国防电子信息技术丛书

天　　线

（第三版）（英文版）

Antennas: For All Applications

Third Edition

[美] John D. Kraus
Ronald J. Marhefka 著

電子工業出版社
Publishing House of Electronics Industry
北京 · BEIJING

内 容 简 介

本书是关于天线的经典著作，全面覆盖了有关天线的各方面内容。前12章介绍了天线的各种基础知识，后12章详细展开了这些内容。书中包括大量实例和习题，便于读者实践掌握。全书图文并茂，更有助于读者的直观理解。书中加入了包括无线革命在内的许多全新的现代应用，对参考文献列表也进行了相应的更新。本书概念清晰，层次分明。无论读者需要的是仅介绍天线基础知识的一个学期的课程，还是计划更深入地进一步学习，本书都能够提供切实的帮助。

本书可作为相关专业高等院校的本科生和研究生的教学用书，以及工程技术人员的参考用书。

John D. Kraus, Ronald J. Marhefka: Antennas: For All Applications, Third Edition
ISBN: 0-07-123201-X
Copyright © 2003 by The McGraw-Hill Companies, Inc.
Original language published by The McGraw-Hill Companies, Inc. All rights reserved. No part of this publication may be reproduced or distributed in any means, or stored in a database or retrieval system, without the prior written permission of the publisher.
English Reprint edition published by McGraw-Hill Education (Asia) Co. and Publishing House of Electronics Industry.
Copyright © 2008.

本书英文影印版由美国麦格劳-希尔教育出版（亚洲）公司授予电子工业出版社。此版本仅限在中国大陆（不包括香港、澳门特别行政区以及台湾地区）发行与销售。未经许可之出口，视为违反著作权法，将受法律之制裁。未经出版者预先书面许可，不得以任何方式复制或抄袭本书的任何部分。
本书封面贴有 McGraw-Hill 公司激光防伪标签，无标签者不得销售。

版权贸易合同登记号　图字：01-2008-2755

图书在版编目（CIP）数据

天线：第3版 = Antennas，3rd Edition：英文/（美）克劳斯（Kraus, J. D.）等著.
北京：电子工业出版社，2008.6
（国防电子信息技术丛书）
ISBN 978-7-121-06878-2

Ⅰ.天...　Ⅱ.克...　Ⅲ.天线-英文　Ⅳ. TN82

中国版本图书馆 CIP 数据核字（2008）第 083433 号

责任编辑：余　义
印　　刷：北京市天竺颖华印刷厂
装　　订：三河市金马印装有限公司
出版发行：电子工业出版社
　　北京市海淀区万寿路 173 信箱　邮编：100036
开　　本：787 × 980　1/16　　印张：60.75　　字数：1652 千字
印　　次：2008 年 6 月第 1 次印刷
定　　价：108.00 元

凡所购买电子工业出版社的图书有缺损问题，请向购买书店调换；若书店售缺，请与本社发行部联系。联系及邮购电话：（010）88254888。
质量投诉请发邮件至 zlts@phei.com.cn，盗版侵权举报请发邮件至 dbqq@phei.com.cn。
服务热线：（010）88258888。

前　言

本书新版较前两版更好地陈述了天线的基本要点，并扩充了最新的无线通信应用的内容。全书结构可按前12章和后12章分成两门课程，具有多种选配组合的灵活性。

第1章“引论”介绍有关符号和记号的知识，并附列了一些有用的图表。第2章“天线基础”包含基本的天线概念和天线语言。在第3章中将简述从偶极子到贴片等20余种天线。第4章介绍点源及其场强、功率、相位的波瓣图。点源的边射阵和端射阵则在第5章（上）中介绍。接着，第6章讨论了电偶极子和直天线及其阵列。各种环天线及其特性在第7章中叙述。第8章（上）是螺旋聚束天线和八木-宇田天线的引论，第8章（下）则是对螺旋天线的详细讨论。第9章介绍缝隙天线、贴片天线和喇叭天线。第10章阐述平板和夹角反射器以及抛物面反射镜天线。第11章讨论宽频带和非频变天线。第12章介绍天线温度、遥感和雷达截面。以上是第一门天线课程的建议章节。

第二门天线课程的建议章节可包含第5章（下）对点源阵内容的拓展，第8章（下）对螺旋天线的深入理解，以及第9章对缝隙天线和喇叭天线的更多知识。然后顺序介绍自阻抗与互阻抗（第13章）、柱型天线（第14章）、口径分布与远场波瓣图之间的傅里叶变换关系（第15章）、偶极子阵和口径阵（第16章）以及透镜天线（第17章）。第18章阐述频率选择表面和周期结构，第19章是对实用大型口径天线的设计考虑。第20章给出若干大型或独特天线的实例。第21章介绍了很多特殊用途的天线，包括移动电话天线、仪表着陆系统（ILS，Instrument Landing System）天线、近地轨道（LEO，Low Earth Orbit）卫星天线，等等。第22章详细介绍了物理尺寸很小的亚毫米波天线。第23章介绍关于平衡-非平衡转换器、变换器、陷波器等的有用知识。最后，第24章给出了当代天线测量的完整知识。附录中包含许多有用的表格、参考文献和计算机程序。

本书的特色是配有大量辅助图示、透彻而富有挑战性的习题集以及丰富的参考文献。下面附有建议的课程设置。当然，也可以有多种不同的选配方案。可以将靠后的一些章节加入第一门天线课程，如第21章中的某些内容可适时地用做第一门天线课程的实例。请读者访问本书的网页www.antennas3.com，以制订出特定的课程计划。

虽然本书是按教科书来编写的，但其中包含的丰富知识，即使是对于从事实践的工程师来说，也仿佛是一座真正的“金矿”。因为书中包含了数以百计的工作实例，有助于将理论转化为实际。

作者们对众多同仁的协助深表感谢，他们是：

Prof. Ben A. Munk, ElectroScience Laboratory, Ohio State University, for his chapters on *Frequency Selective Surfaces* and *Baluns, etc.*

Profs. Arto Lehto and Pertti Vainikainen of the Helsinki University of Technology Radio Laboratory for their chapter on *Measurements*.

Prof. Pertti Vainikainen for his section on *Antennas for Terrestrial Mobile Communication.*

Dr. Edward H. Newman, ElectroScience Laboratory. Ohio State University for his section on *Self-impedance, Radar Cross Section and Mutual Impedance of Short Dipoles by the Method of Moments.*

Prof. Warren Perger, Michigan Technological University for class testing the new manuscript.

Dr. Spencer Webb, AntennaSys, Inc. for assistance on patch and other antennas.

Prof. Jonathan Young, ElectroScience Laboratory, Ohio State University, for assistance on *Ultra Wide Band Antennas for Digital Applications.*

Dr. Brian Baertlein, ElectroScience Laboratory, Ohio State University, for assistance on the *600 THz Antenna.*

Dr. Steven Ellingson, ElectroScience Laboratory, Ohio State University for assistance on the *Argus Array*.

Prof. Richard McFarland, School of Avionics, Ohio University, for assistance on *Instrument Landing System Antennas.*

Dr. Fred J. Dietrich, FD Engineering and Globalstar, for assistance on *Low Earth Orbit Satellite Antennas.*

Dr. Edward E. Altshuler, Air Force Research Laboratory, for assistance on *Genetic Algorithm Antennas*.

Prof. Christopher Walker, Dept. of Astronomy, University of Arizona for sections on *Terahertz Antennas.*

Dr. Eric Walton, ElectroScience Laboratory, Ohio State University for assistance on *Instrument Landing System Antennas.*

Dr. Richard Mallozzi, General Electric Research Laboratory, Schenectady, N.Y., for assistance on many topics.

Drs. James C. Logan and John W. Rockway, EM Scientific, for contributing computer programs available on the book's web site.

感谢本书评阅者对手稿的有益建议，他们是 McGraw-Hill 的项目组：主编 Catherine Fields，执行主编 Emily Lupash 和项目经理 Marilyn Rothenberger。

感谢Jerry Ehman博士和Erich Pacht博士认真的编辑加工。作者们还要分别感谢他们各自的妻子Alice Kraus 和 Deborah Marhefka 以耐心和关爱所给予的支持。

John Kraus & Ronald Marhefka

附言 1：虽经极其仔细的审校,书中谬误在所难免，笔者竭诚欢迎指正。可给Ronald Marhefka发电子邮件至 marhefka.1@osu.edu 。

附言 2：据统计，本书（英文原版）超出 900 页，其中包含 1200 多幅图、130 个例题、75 个表格、300 多道习题、近 900 篇参考文献以及一套多于 2200 词条的索引可供快速接入主题。建议的课程设置

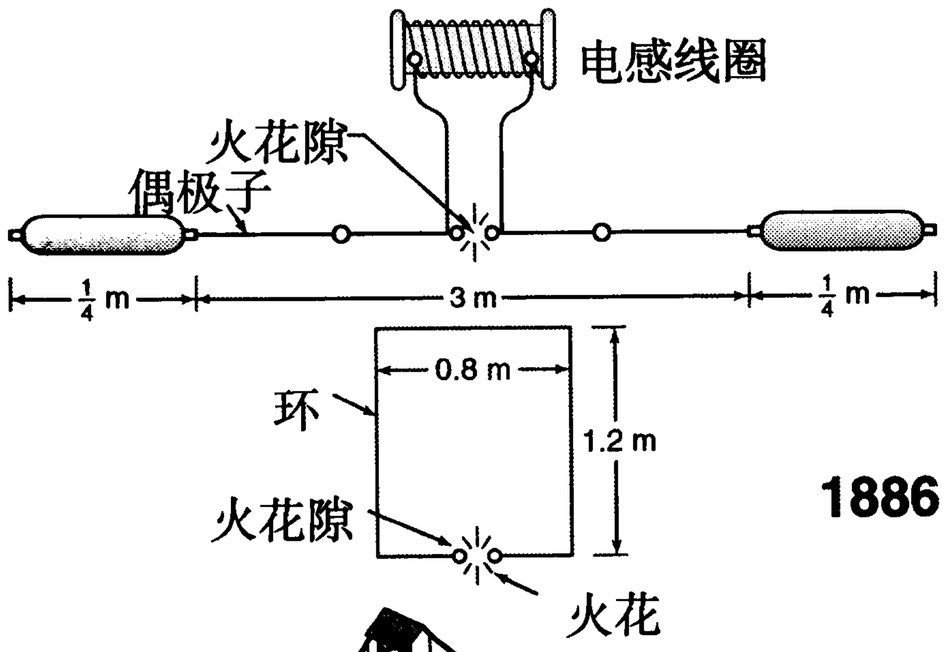
建议的课程设置

每节课平均 8 页

第一学期	1/2 年的学期	1/4 年的学期
章次	节数	节数
1	1	1
2	5	5
3	2	2
4	2	2
5 (上)	4	4
测验	1	1
6	4	2
7	3	3
8 (上)	4	3
9	4	3
10	4	2
测验	1	
11	2	1
12	3	1
测验		合计 1
15	2	30
21	2	
测验	合计 1	
	45	

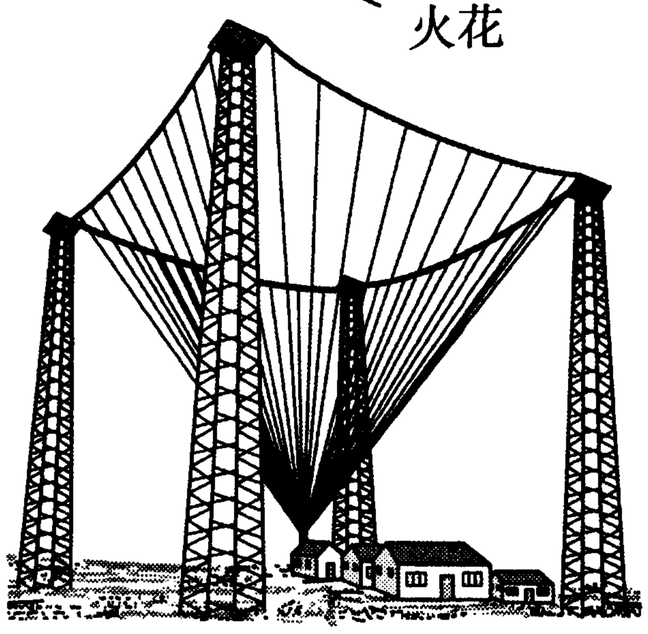
第二学期	1/2 年的学期	1/4 年的学期
章次	节数	节数
5 (下)	4	3
8 (下)	2	1
9	2	1
13	3	2
14	4	3
测验	1	1
15	3	2
16	5	4
17	2	1
18	2	1
19	2	1
20	1	1
测验	1	
21	5	4
22	1	1/2
23	1	1/2
24	5	3
测验	合计 1	合计 1
	45	30

从赫兹天线到手持天线的一个世纪



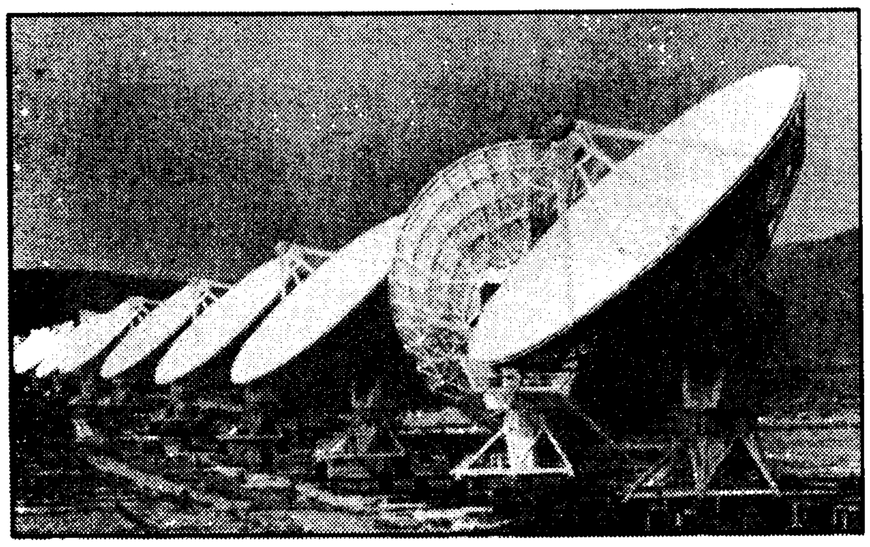
1886

海因里希·赫兹(Heinrich Hertz)的终端加载偶极子发射天线和半波谐振环接收天线工作于8 m波长。在德国卡尔斯洛的赫兹实验室中，电感线圈在偶极子的间隙中所产生的火花，导致相距数米远的环隙中也产生了火花。这就是首个无线电链路，也是最早的用于无线电的偶极子天线和环天线。



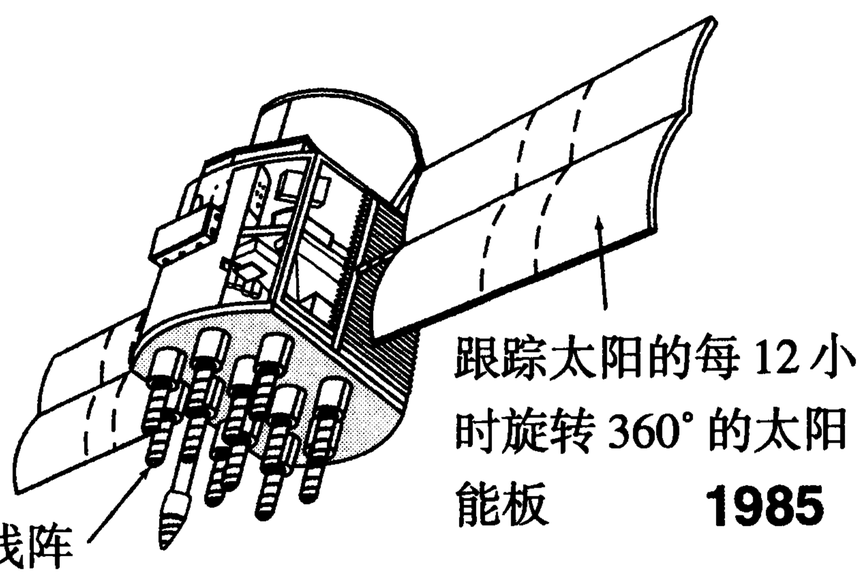
1905

古利莫·马可尼(Guglielmo Marconi) 在英格兰波尔多架设的方锥天线，发射波长为1000 m的信号。



1980

工作在厘米波长的甚大阵（VLA）由27个直径各为25 m的可转向抛物面碟形天线组成，用于观测数亿光年以远的射电源。该阵列位于新墨西哥州索可洛附近的(美国)国立射电天文台(National Radio Astronomy Observatory)。



1985

20 000 km的中高度地球轨道上24颗全球定位卫星（GPS，Global Position Satellite)之一所装载的螺旋天线阵。工作在20 cm波长，为地面或空中的客户提供所处位置（经纬度和高度）的信息，其精确度优于1 m。

2000

到处可见的带有半波天线的手持移动电话，工作在30 cm波长，可与任何人通话。

CONTENTS

Chapter 1 Introduction 1
引论

1-1 Introduction 1
引言

1-2 Dimensions and Units 3
量纲和单位

1-3 Fundamental and Secondary Units 3
基本单位和派生单位

1-4 How to Read the Symbols and Notation 4
如何阅读符号和记号

1-5 Equation and Problem Numbering 6
公式和习题的编号

1-6 Dimensional Analysis 6
量纲分析

1-7 The Electromagnetic Spectrum; Radio-Frequency Bands 9
电磁频谱与无线电频段

References 10
参考文献

Chapter 2 Antenna Basics 11
天线基础

2-1 Introduction 11
引言

2-2 Basic Antenna Parameters 12
基本天线参量

2-3 Patterns 14
波瓣图

2-4 Beam Area (or Beam Solid Angle) Ω_A 19
波束范围或波束立体角 Ω_A

2-5 Radiation Intensity 22
辐射强度

2-6 Beam Efficiency 22
波束效率

2-7 Directivity *D* and Gain *G* 23
定向性 *D* 和增益 *G*
2-8 Directivity and Resolution 26
定向性与分辨率
2-9 Antenna Apertures 27
天线口径
2-10 Effective Height 30
有效高度
2-11 The Radio Communication Link 36
无线电通信线路
2-12 Fields from Oscillating Dipole 37
振荡偶极子产生的场
2-13 Antenna Field Zones 39
天线的场区
2-14 Shape-Impedance Considerations 40
形状 – 阻抗的讨论
2-15 Linear, Elliptical and Circular Polarization 44
线极化、椭圆极化和圆极化
2-16 Poynting Vector for Elliptically and Circularly Polarized Waves 47
椭圆和圆极化波的坡印廷矢量
2-17 The Polarization Ellipse and the Poincaré Sphere 47
极化椭圆和庞加莱球
References 52
参考文献
Problems 52
习题

Chapter 3 The Antenna Family 57
天线家族
3-1 Introduction 57
引言
3-2 Loops, Dipoles and Slots 57
环形、偶极子和缝隙天线
3-3 Opened-Out Coaxial-Line Antennas 60
张开的同轴线天线
3-4 Opened-Out 2-Conductor (Twin-Line) Antennas 61
张开的双导线天线
3-5 Opened-Out Waveguide Antennas (Aperture Types) 62
张开的波导天线（口径类）

3-6 Flat-Sheet Reflector Antennas 63
平板反射器天线
3-7 Parabolic Dish and Dielectric Lens Antennas 64
抛物面和介质透镜天线
3-8 End-Fire Antennas 66
端射天线
3-9 Broad Bandwidth Antennas: The Conical Spiral and the Log-Periodic 67
宽频带天线：圆锥螺蜷和对数周期
3-10 The Patch Antenna, Patch Array, and the Grid Array 67
贴片天线、贴片阵列和栅格阵列
References 70
参考文献
Problems 70
习题

Chapter 4 Point Sources 72
点源
4-1 Introduction. Point Source Defined 72
引言：点源的定义
4-2 Power Patterns 73
功率波瓣图
4-3 A Power Theorem and Its Application to an Isotropic Source 75
功率定理及其应用于个别各向同性源
4-4 Radiation Intensity 75
辐射强度
4-5 Examples of Power Patterns 76
功率波瓣图举例
4-6 Field Patterns 81
场波瓣图
4-7 Phase Patterns 86
相位波瓣图
Problems 88
习题

Chapter 5 Arrays of Point Sources, Part I 90
（上）点源阵
5-1 Introduction 90
引言

5-2 Arrays of Two Isotropic Point Sources 90
两个各向同性点源的阵
5-3 Nonisotropic but Similar Point Sources and the Principle of Pattern Multiplication 98
非各向同性的相似点源和波瓣图乘法原理
5-4 Example of Pattern Synthesis by Pattern Multiplication 102
用图乘法综合波瓣图举例
5-5 Nonisotropic and Dissimilar Point Sources 106
非各向同性的非相似点源
5-6 Linear Arrays of *n* Isotropic Point Sources of Equal Amplitude and Spacing 109
n 个各向同性点源的等幅等间距直线阵
5-7 Null Directions for Arrays of *n* Isotropic Point Sources of Equal Amplitude and Spacing 116
n 个各向同性点源的等幅等间距阵的零方向
Problems 120
习题

Chapter 5 Arrays of Point Sources, Part II 126
（下）点源阵
5-8 Linear Broadside Arrays with Nonuniform Amplitude Distributions. General Considerations 126
非均匀幅度分布的直线边射阵的一般性讨论
5-9 Linear Arrays with Nonuniform Amplitude Distributions. The Dolph-Tchebyscheff Optimum Distribution 129
非均匀幅度分布的直线阵与道尔夫 - 切比雪夫最优分布
5-10 Example of Dolph-Tchebyscheff (D-T) Distribution for an Array of Eight Sources 137
道尔夫 – 切比雪夫分布的八源阵举例
5-11 Comparison of Amplitude Distributions for Eight-Source Arrays 139
八源阵幅度分布的比较
5-12 Continuous Arrays 140
连续阵
5-13 Huygens' Principle 144
惠更斯原理
5-14 Huygens' Principle Applied to the Diffraction of a Plane Wave Incident on a Flat Sheet. Physical Optics 147
惠更斯原理应用于平面波投射到平板的绕射与物理光学
5-15 Rectangular-Area Broadside Arrays 150
矩形面积的边射阵
5-16 Arrays with Missing Sources and Random Arrays 152
缺源阵和随机阵

5-17 Broadside versus End-Fire Arrays. Turns versus Dipoles and Three-Dimensional Arrays 154
边射 – 端射阵与环圈 – 偶极子的三维阵
5-18 Directions of Maxima for Arrays of *n* Isotropic Point Sources of Equal Amplitude and Spacing 159
n 个各向同性点源的等幅等间距阵的最（极）大方向
References 162
参考文献
Problems 162
习题

Chapter 6 The Electric Dipole and Thin Linear Antennas 165
电偶极子和细直天线
6-1 The Short Electric Dipole 165
短电偶极子
6-2 The Fields of a Short Dipole 166
短电偶极子的场
6-3 Radiation Resistance of Short Electric Dipole 175
短电偶极子的辐射电阻
6-4 The Thin Linear Antenna 177
细直天线
6-5 Radiation Resistance of λ/2 Antenna 181
半波天线的辐射电阻
6-6 Radiation Resistance at a Point Which is not a Current Maximum 182
非电流最大点的辐射电阻
6-7 Two Half-Wave Dipoles: Broadside and End-Fire (The W8JK Array) 183
两个半波偶极子：边射和端射
6-8 Fields of a Thin Linear Antenna with a Uniform Traveling Wave 187
载有均匀行波的细直天线的场
References 193
参考文献
Problems 193
习题

Chapter 7 The Loop Antenna 197
环天线
7-1 The Small Loop 197
电小环

7-2 Comparison of Far Fields of Small Loop and Short Dipole 199
小环与短偶极子的远场比较
7-3 The Loop Antenna. General Case 200
环天线的一般情况
7-4 Far-Field Patterns of Circular Loop Antennas with Uniform Current 202
载有均匀电流的圆环天线的远场波瓣图
7-5 The Small Loop as a Special Case 205
作为特例的电小环
7-6 Radiation Resistance of Loops 206
环的辐射电阻
7-7 Directivity of Circular Loop Antennas with Uniform Current 209
载有均匀电流的圆环天线的定向性
7-8 Table of Loop Formulas 212
环的公式列表
7-9 Square Loops 212
方环
7-10 Radiation Efficiency, Q, Bandwidth and Signal-to-Noise Ratio 214
辐射效率、品质因数、频带宽度和信噪比
References 220
参考文献
Problems 220
习题

Chapter 8 End-Fire Antennas: The Helical Beam Antenna and the Yagi-Uda Array, Part I 222
（上）端射天线：螺旋聚束天线和八木－宇田天线
8-1 The Helical Beam Antenna Story *by John Kraus* 222
螺旋聚束天线的故事
8-2 Helical Geometry 227
螺旋的几何表示
8-3 Practical Design Considerations for the Monofilar Axial-Mode Helical Antenna 231
单绕轴向模螺旋天线的实际设计考虑
8-4 Helical Beam Antenna with Loops Replacing the Ground Plane 242
用环取代接地平面的螺旋聚束天线
8-5 Dipole Arrays with Parasitic Elements 242
含有寄生单元的偶极子阵
8-6 The Yagi-Uda Array Story 246
八木－宇田阵的故事
Problems 249
习题

Chapter 8 The Helical Antenna: Axial and Other Modes, Part II 250
（下）螺旋天线：轴向模和其他模
8-7 Introduction 250
引言
8-8 Axial-Mode Patterns and the Phase Velocity of Wave Propagation on Monofilar Helices (Kraus-4) 251
轴向模波瓣图和波沿单绕螺旋的传播相速
8-9 Monofilar Axial-Mode Single-Turn Patterns of Square Helix 258
单绕轴向模单圈方螺旋的波瓣图
8-10 Complete Axial-Mode Patterns of Monofilar Helices 261
单绕螺旋的完整轴向模波瓣图
8-11 Axial Ratio and Conditions for Circular Polarization of Monofilar Axial-Mode Helical Antennas 263
单绕轴向模螺旋天线的轴比和圆极化条件
8-12 Wideband Characteristics of Monofilar Helical Antennas Radiating in the Axial Mode 268
单绕螺旋天线轴向模辐射的宽频带特性
8-13 Table of Pattern, Beamwidth, Gain, Impedance and Axial Ratio Formulas 270
波瓣图、频带宽度、增益、阻抗和轴比公式列表
8-14 Radiation from Linear Periodic Structures with Traveling Waves with Particular Reference to the Helix as a Periodic Structure Antenna 271
载有行波的线形周期性结构的辐射，将螺旋视为周期性结构天线
8-15 Arrays of Monofilar Axial-Mode Helical Antennas 280
单绕轴向模螺旋天线阵
8-16 The Monofilar Axial-Mode Helix as a Parasitic Element and Polarizer (see Fig. 8-57) 284
单绕轴向模螺旋用做寄生单元和极化器
8-17 The Monofilar Axial-Mode Helical Antenna as a Phase and Frequency Shifter 285
单绕轴向模螺旋天线用做移相器和移频器
8-18 Linear Polarization with Monofilar Axial-Mode Helical Antennas 286
单绕轴向模螺旋天线形成线极化
8-19 Monofilar Axial-Mode Helical Antennas as Feeds 287
单绕轴向模螺旋天线用做馈源
8-20 Tapered and Other Forms of Axial-Mode Helical Antennas 289
渐变及其他形式的轴向模螺旋天线
8-21 Multifilar Axial-Mode Helical Antennas 291
多绕轴向模螺旋天线
8-22 Monofilar and Multifilar Normal-Mode Helical Antennas 292
单绕和多绕法向模螺旋天线

8-23 Axial-Mode Helix Terminations 297
轴向模螺旋终端
8-24 Antenna Rotation Experiments 297
天线旋转实验
8-25 Bifilar and Quadrifilar Axial-Mode Helices 299
双绕和四绕轴向模螺旋
8-26 Genetic Algorithm Five-Segment Helix versus Circular Helix 299
五直线段螺旋逼近圆螺旋的遗传算法
References 300
参考文献
Additional References on Helical Antennas 303
螺旋天线的附加参考资料
Problems 303
习题

Chapter 9 Slot, Patch and Horn Antennas 304
缝隙天线、贴片天线和喇叭天线
9-1 Slot Antennas 304
缝隙天线
9-2 Patterns of Slot Antennas in Flat Sheets. Edge Diffraction 307
平板缝隙天线的波瓣图和边缘绕射
9-3 Babinet's Principle and Complementary Antennas 311
巴比涅原理和互补天线
9-4 The Impedance of Complementary Screens 314
互补屏的阻抗
9-5 The Impedance of Slot Antennas 317
缝隙天线的阻抗
9-6 Slotted Cylinder Antennas 321
柱面缝隙天线
9-7 Patch or Microstrip Antennas 322
贴片或微带天线
9-8 Horn Antennas 330
喇叭天线
9-9 The Rectangular Horn Antenna 333
矩形喇叭天线
9-10 Beamwidth Comparison 338
波束宽度的比较
9-11 Conical Horn Antennas 339
圆锥喇叭天线

9-12 Ridge Horns 339
加脊喇叭
9-13 Septum Horns 340
隔膜喇叭
9-14 Corrugated Horns 341
皱纹喇叭
9-15 Aperture-Matched Horn 343
口径匹配喇叭
References 344
参考文献
Problems 345
习题

Chapter 10 Flat Sheet, Corner and Parabolic Reflector Antennas 347
平板和夹角反射器、抛物面反射镜天线
10-1 Introduction 347
引言
10-2 Flat Sheet Reflectors 349
平板反射器
10-3 Corner Reflectors 352
夹角反射器
10-4 Passive (Retro) Corner Reflector 364
无源（返回式）夹角反射器
10-5 The Parabola. General Properties 365
抛物线的一般性质
10-6 A Comparison between Parabolic and Corner Reflectors 367
抛物面反射镜与夹角反射器的比较
10-7 The Paraboloidal Reflector 368
旋转抛物面反射镜
10-8 Patterns of Large Circular Apertures with Uniform Illumination 372
均匀照射大型圆口径的波瓣图
References 375
参考文献
Problems 375
习题

Chapter 11 Broadband and Frequency-Independent Antennas 378
宽频带和非频变天线
11-1 Broadband Basics 378
宽频带基础

11-2 Infinite and Finite Biconical Antennas 380
无限长和有限长的双锥天线
11-3 Directional Biconicals, Conicals, Disk Cones and Bow Ties 382
定向的双锥、圆锥、盘锥和领结天线
11-4 The Frequency-Independent Concept: Rumsey's Principle 386
非频变概念：拉姆塞原理
11-5a The Illinois Story 387
伊利诺依的故事
11-5b The Frequency-Independent Planar Log-Spiral Antenna 387
非频变平面对数螺蜷天线
11-6 The Frequency-Independent Conical-Spiral Antenna 391
非频变圆锥螺蜷天线
11-7 The Log-Periodic Antenna 392
对数周期天线
11-8 The Composite Yagi-Uda-Corner-Log-Periodic (YUCOLP) Array 399
YUCOLP 复合阵
References 399
参考文献
Problems 400
习题

Chapter 12 Antenna Temperature, Remote Sensing and Radar Cross Section 401
天线温度、遥感和雷达截面
12-1 Antenna Temperature 401
天线温度
12-2 System Temperature 409
系统温度
12-3 SNR (Signal-to-Noise Ratio) 411
信噪比
12-4 Passive Remote Sensing 414
被动遥感
12-5 Radar and Radar Cross Section 417
雷达和雷达截面
References 427
参考文献
Problems 427
习题

Chapter 13 Self and Mutual Impedances 438
阻抗与互阻抗
13-1 Introduction 438
引言
13-2 Reciprocity Theorem for Antennas 439
天线的互易性定理
13-3 The 73-Ω Radiation Resistance of a λ/2 Antenna 441
半波长天线的 73 Ω 辐射电阻
13-4 Radiation Resistance at a Point Which is Not a Current Maximum 444
非电流最大点的辐射电阻
13-5 Self-Impedance of a Thin Linear Antenna 446
细线天线的自阻抗
13-6 Mutual Impedance of Two Parallel Linear Antennas 448
两平行线天线的互阻抗
13-7 Mutual Impedance of Parallel Antennas Side-by-Side 450
边靠边平行天线的互阻抗
13-8 Mutual Impedance of Parallel Collinear Antennas 454
平行共线天线的互阻抗
13-9 Mutual Impedance of Parallel Antennas in Echelon 454
梯式平行天线的互阻抗
13-10 Mutual Impedance of Other Configurations 455
其他布置天线的互阻抗
13-11 Mutual Impedance in Terms of Directivity and Radiation Resistance 457
用定向性和辐射电阻表示的互阻抗
References 459
参考文献
Problems 459
习题

Chapter 14 The Cylindrical Antenna and the Moment Method (MM) 461
柱形天线和矩量法
14-1 Introduction 461
引言
14-2 Outline of the Integral-Equation Method 462
积分方程法概述
14-3 Current Distributions 463
电流分布

14-4 Input Impedance 465
输入阻抗
14-5 Patterns of Cylindrical Antennas 470
圆柱天线的波瓣图
14-6 The Thin Cylindrical Antenna 471
细圆柱天线
14-7 Cylindrical Antennas with Conical Input Sections 472
带有圆锥形输入段的圆柱天线
14-8 Antennas of Other Shapes. The Spheroidal Antenna 472
其他形状的天线，类球体天线
14-9 Current Distributions on Long Cylindrical Antennas 473
长圆柱天线上的电流分布
14-10 Integral Equations and the Moment Method (MM) in Electrostatics 477
静电学的积分方程和矩量法
14-11 The Moment Method (MM) and Its Application to a Wire Antenna 482
矩量法及其应用于线天线
14-12 Self-Impedance, Radar Cross Section and Mutual Impedance of Short Dipoles by the Method of Moments *by Edward H. Newman* 490
用矩量法求短偶极子的自阻抗、雷达截面和互阻抗
References 499
参考文献
Problems 500
习题

Chapter 15 The Fourier Transform Relation between Aperture Distribution and Far-Field Pattern 501
口径分布与远场波瓣图之间的傅里叶变换关系
15-1 Continuous Aperture Distribution 501
连续口径分布
15-2 Fourier Transform Relations between the Far-Field Pattern and the Aperture Distribution 503
远场波瓣图与口径分布之间的傅里叶变换关系
15-3 Spatial Frequency Response and Pattern Smoothing 506
空间频率响应和波瓣图平滑处理
15-4 The Simple (Adding) Interferometer 509
简单（加法）干涉仪
15-5 Aperture Synthesis and Multiaperture Arrays 518
口径合成和多口径阵

15-6 Grating Lobes 520
栅瓣
15-7 Two-Dimensional Aperture Synthesis 521
二维口径合成
15-8 Phase Closure or Self-Calibration for "Clean" Maps 523
"干净"图形的相位闭合或自校准
References 526
参考文献
Problems 526
习题

Chapter 16 Arrays of Dipoles and of Apertures 529
偶极子阵和口径阵
16-1 Introduction 529
引言
16-2 Array of Two Driven λ/2 Elements. Broadside Case 530
两根受激 λ/2 单元的边射阵
16-3 Array of 2 Driven λ/2 Elements. End-Fire Case 539
两根受激 λ/2 单元的端射阵
16-4 Array of 2 Driven λ/2 Elements. General Case with Equal Currents of Any Phase Relation 544
两根受激 λ/2 单元的等幅任意相位阵
16-5 Closely Spaced Elements, Radiating Efficiency and *Q*. The W8JK Array 548
密距单元的辐射效率和 *Q* 值，W8JK 阵
16-6 Array of *n* Driven Elements 553
n 根受激单元阵
16-7 Horizontal Antennas above a Plane Ground 555
地平面上的水平天线
16-8 Vertical Antennas above a Plane Ground 566
地平面上的铅垂天线
16-9 Yagi-Uda Modifications 570
八木 – 宇田天线的变种
16-10 Phased Arrays 572
相控阵
16-11 Frequency-Scanning Arrays 576
频率扫描阵
16-12 Retroarrays. The Van Atta Array 581
返回式阵和范阿塔阵

16-13 Adaptive Arrays and Smart Antennas 582
自适应阵和智能阵
16-14 Microstrip Arrays 584
微带阵
16-15 Low-Side-Lobe Arrays 585
低旁瓣阵
16-16 Long-Wire Antennas 585
长导线天线
16-17 Curtain Arrays 591
帘幕形阵
16-18 Location and Method of Feeding Antennas 592
天线的馈电位置和方式
16-19 Folded Dipole Antennas 593
折合偶极子天线
16-20 Modifications of Folded Dipoles 596
折合偶极子的变种
16-21 Argus Type All-Sky Surveillance Antennas 597
百眼巨人型全空监视天线
References 599
参考文献
Additional References on Adaptive Arrays 601
自适应阵的补充参考资料
Problems 602
习题

Chapter 17 Lens Antennas 607
透镜天线
17-1 Introduction 607
引言
17-2 Nonmetallic Dielectric Lens Antennas. Fermat's Principle (Equality of Path Length) 609
非金属介质透镜天线，费马原理（射径等同性）
17-3 Artificial Dielectric Lens Antennas 616
人造介质透镜天线
17-4 *E*-Plane Metal-Plate Lens Antennas 619
E- 面金属板透镜天线
17-5 Tolerances on Lens Antennas 625
透镜天线的容差

17-6 *H*-Plane Metal-Plate Lens Antennas 627
H- 面金属板透镜天线
17-7 Reflector-Lens Antenna 628
反射器 – 透镜天线
17-8 Polyrods 629
介质杆
17-9 Multiple-Helix Lenses 631
多螺旋透镜
17-10 Luneburg and Einstein Lenses 632
龙伯透镜和爱因斯坦透镜
References 634
参考文献
Problems 635
习题

Chapter 18 Frequency-Selective Surfaces and Periodic Structures By Ben A. Munk 636
频率选择表面和周期结构
18-1 Introduction. Definition of FSS 636
引言：频率选择表面的定义
18-2 The Halfwave Dielectric Radome 638
半波长厚的介质天线罩
18-3 Slotted Metal Radome 639
开缝的金属天线罩
18-4 The Simple Hybrid Radome 641
简单的混合天线罩
18-5 The Ideal Stealth Radome 642
理想的隐形天线罩
18-6 Transmission and Reflection Properties of Simple Periodic Surfaces of Wires 642
简单周期性导线表面的传输和反射特性
18-7 Complementary Surfaces and Babinet's Principle 646
互补表面与巴比涅原理
18-8 Oblique Angle of Incidence 646
斜入射角
18-9 On the Shape and Development of Elements 647
单元的形状及其演化
18-10 Controlling Bandwidth with Angle of Incidence and Polarization 651
控制入射角和极化的频带宽度
18-11 Other Applications 654
其他应用

References 658
参考文献
Problems 659
习题

Chapter 19 Practical Design Considerations of Large Aperture Antennas 660
实用大型口径天线的设计考虑
19-1 Aperture Distributions and Efficiencies 660
口径分布与口径效率
19-2 Surface Irregularities and Gain Loss 673
表面不规则性和增益损失
19-3 Off-Axis Operation of Parabolic Reflectors 678
抛物面反射镜的偏轴工作
19-4 Cassegrain Feed, Shaped Reflectors, Spherical Reflectors and Offset Feed 680
卡塞格伦馈电、赋形反射镜、球面反射镜和偏馈
19-5 Low-Side-Lobe Considerations 684
低旁瓣的考虑
References 687
参考文献
Additional References for Further Reading 688
深入阅读的补充参考资料
Problems 689
习题

Chapter 20 Some Examples of Large or Unique Antennas 691
大型或独特的天线实例
References 707
参考文献

Chapter 21 Antennas for Special Applications 708
特殊用途的天线
21-1 Introduction 708
引言
21-2 Electrically Small Antennas 709
电小天线
21-3 Physically Small Antennas 712
物理上的小天线
21-4 Antenna Siting and the Effect of Typical (Imperfect) Ground 713
天线架设和典型（非理想）地面的影响

21-5 Ground-Plane Antennas 719
带接地面的天线
21-6 Sleeve Antennas 721
套筒天线
21-7 Turnstile Antenna 722
旋转场天线
21-8 Superturnstile Antenna 724
超旋转场天线
21-9 Other Omnidirectional Antennas 726
其他全向天线
21-10 Circularly Polarized Antennas 727
圆极化天线
21-11 The High-Gain Omni 729
高增益全向天线
21-12 Submerged Antennas 729
埋地天线
21-13 Surface-Wave and Leaky-Wave Antennas 734
表面波和漏波天线
21-14 Antenna Design Considerations for Satellite Communication 742
设计卫星通信天线的考虑
21-15 Receiving versus Transmitting Considerations 746
用做接收和用做发射的考虑
21-16 Bandwidth Considerations 748
频带的考虑
21-17 Architecturally Acceptable Antennas 748
建筑上可接受的天线
21-18 ILS (Instrument Landing System) Antennas 750
用于仪表着陆系统（ILS）的天线
21-19 The “Sugar Scoop Antenna” and the 3K Cosmic Sky Background Story 755
糖铲形天线和3 K宇宙背景的故事
21-20 LEO Satellite Link Antennas 760
用于近地轨道（LEO）卫星线路的天线
21-21 Asteroid Detection Antenna 769
检测小行星的天线
21-22 Leaky Transmission Lines as Antennas 771
泄漏传输线用做天线
21-23 Artistic Antennas (Fractals) 772
艺术性天线（分形）

21-24 Cell-Tower trees 772
蜂窝 – 塔树
21-25 Antennas for Terrestrial Mobile Communications Systems *by Pertti Vainikainen* 775
用于陆上移动通信系统的天线
21-26 Antennas for Ground Penetrating Radar (GPR): Pulse Bandwidth 781
探地雷达（GPR）的天线：脉冲的频带宽度
21-27 Embedded Antennas 784
植入式天线
21-28 Ultra-Wide-Band (UWB) Antennas for Digital Applications 785
数字式应用的特宽频带天线
21-29 The Plasma Antenna 788
等离子态天线
References 789
参考文献
Problems 792
习题

Chapter 22 Terahertz Antennas 796
太赫频率的天线
22-1 Introduction 796
引言
22-2a Pyramidal Horn Cavity with Dipole 797
带偶极子的棱锥喇叭腔
22-2b Corner Reflector Array 798
夹角反射器阵
22-2c Bow-Tie Dipole 798
领结形偶极子
22-3 A 600-THz Antenna? 798
一种 600 THz 的天线
22-4 Planar Antenna Structures on Dielectric Lenses 799
介质透镜的平面天线结构
22-5 THz Waveguide Structures 800
太赫频率的波导结构
References 802
参考文献

Chapter 23 Baluns, etc. By Ben A. Munk 803
平衡 – 非平衡转换器等
23-1 Introduction 803
引言

23-2 Balun Types I, II and III and Choke Baluns 803
I型，II型，III型和扼流式巴仑
23-3 Bandwidth 806
频带宽度
23-4a Sleeve-Dipole Baluns 808
套筒－偶极子式巴仑
23-4b Modifications 808
变种结构
23-5 Mast Balun 810
支杆式巴仑
23-6 Printed Baluns 810
印刷巴仑
23-7 The λ/2 Bypass Balun 813
半波长旁路式巴仑
23-8 Balanced Transformer 815
平衡变换器
23-9 Cutaway Baluns 818
截割式巴仑
23-10 Natural Baluns 821
天然巴仑
23-11 Folded Dipole to J-Match 822
折合偶极子到J形匹配天线
23-12 Matching Stubs 823
匹配短截线
23-13 Traps 823
陷波器
23-14 Conclusion 824
结论
References 825
参考文献
Additional References 825
补充参考资料
Problems 825
习题

Chapter 24 Antenna Measurements. By Arto Lehto and Pertti Vainikainen 827
天线测量
24-1 Introduction 827
引言

24-2 Basic Concepts 828
基本概念
24-3 Typical Sources of Error in Antenna Measurements 835
天线测量的典型误差源
24-4 Measurement Ranges 838
测量场地
24-5 Measurement of Different Antenna Parameters 854
不同天线参量的测量
24-6 Miscellaneous Topics 873
各种特殊测量
References 883
参考文献
Problems 885
习题

Appendix A Tables for References 887
参考表格
A-1 Table of Antena and Antenna System Relations 887
天线及天线系统的关系式
A-2 Formulas for Input Impedance of Terminated Transmission Lines 889
端接负载传输线的输入阻抗公式
A-3 Reflection and Transmission Coefficients and VSWR 889
反射系数、传输系数与电压驻波比 VSWR
A-4 Characteristic Impedance of Coaxial, 2-Wire and MicroStrip Transmission Lines 890
同轴、双导线及微带传输线的特性阻抗
A-5 Characteristic Impedance of Transmission Lines in Terms of Distributed Parameters 890
用分布参量表示的传输线特性阻抗
A-6 Material Constants (Permittivity, Conductivity and Dielectric Strength) 891
材料参数（介电常数、电导率和介质强度）
A-7 Permittivity Relations 892
介电常数的关系式
A-8 Maxwell' Equations 893
麦克斯韦方程组

Appendix B Books, Tapes and Articles 894
参考书、录像带和文章
B-1 Books 894
参考书

B-2 Video Tapes 897
录像带
B-3 Selected Articles for Further Reading 898
进一步选读的文章

Appendix C Computer Programs (Codes) 902
计算机程序（代码）
C-1 Introduction 902
引言
C-2 Software Modeling Philosophy 902
软件建模的原理
C-3 Codes Illustrating Results in *Antennas* 903
天线软件说明
C-4 Expert MININEC for Students 904
学生版 Expert MININEC
C-5 Other Useful Codes 904
其他有用的软件
References 905
参考文献
References for Further Reading on Antenna Computer Simulation 905
关于天线计算机仿真的深入阅读参考资料

Appendix D Absorbing Materials 906
吸波材料
References 910
参考文献

Appendix E Measurement Error 911
测量误差

Appendix F Answers to Starred Problems 912
带 * 号习题的答案

The List of Glossary 918
术语表

CHAPTER 1

Introduction
引论

Heinrich Hertz

Guglielmo Marconi

1-1 INTRODUCTION（引言）

Since Hertz and Marconi, antennas have become increasingly important to our society until now they are indispensable. They are everywhere: at our homes and workplaces, on our cars and aircraft, while our ships, satellites and spacecraft bristle with them. Even as pedestrians, we carry them.

Although antennas may seem to have a bewildering, almost infinite variety, all operate according to the same basic principles of electromagnetics. The aim of this book is to explain these principles in the simplest possible terms and illustrate them with many practical examples. In some situations intuitive approaches will suffice while in others complete rigor is needed. The book provides a blend of both with selected examples illustrating when to use one or the other. But first a bit of history to provide a perspective.

The other sections of this chapter cover:

- Dimensions and units（量纲和单位）
- Fundamental and secondary units（基本单位和派生单位）
- Symbols and notation（符号和记号）
- Equations and problem numbering（方程和习题的编号）
- Dimensional analysis（量纲分析）
- The electromagnetic spectrum; radio-frequency bands（电磁频谱，无线电频段）

A Short History of Antennas[1]

Antennas are our electronic eyes and ears on the world. They are our links with space. They are an essential, integral part of our civilization.

---

[1] See the list at the end of the chapter for further reading on the history of antennas. See also the book's web site: www.antennas3.com.

Antennas have been around for a long time, millions of years, as the organ of touch or feeling of animals, birds, and insects. But in the last 100 years they have acquired a new significance as the connecting link between a radio system and the outside world.[1]

The first radio antennas were built by Heinrich Hertz, a professor at the Technical Institute in Karlsruhe, Germany. In 1886, he assembled apparatus we would now describe as a complete radio system operating at meter wavelengths with an end-loaded dipole as the transmitting antenna and a resonant square-loop antenna as receiver. He also experimented with a parabolic reflector antenna.

Although Hertz was the pioneer and father of radio, his invention remained a laboratory curiosity until 20-year-old Guglielmo Marconi of Bologna, Italy, went on to add tuning circuits, big antenna and ground systems for longer wavelengths, and was able to signal over large distances. In mid-December 1901 he startled the world by receiving signals at St. Johns, Newfoundland, from a transmitting station he had constructed at Poldhu in Cornwall, England. A year later he began regular transatlantic message service in spite of a suit by the Cable Company for infringing on its monopoly of transatlantic messaging.

Rarely has an invention captured the public imagination as Marconi's wireless did at the beginning of the 20th century. After its value at sea had been dramatized by the *S.S. Republic* and *S.S. Titanic* disasters, Marconi was regarded with universal awe and admiration seldom matched. Before wireless, complete isolation enshrouded a ship at sea. Disaster could strike without anyone on the shore or nearby ships being aware that anything had happened. Marconi became the Wizard of Wireless.

With the advent of radar during World War II, centimeter wavelengths became popular and the entire radio spectrum opened up to wide usage. Thousands of communication satellites bristling with antennas now circle the earth in low, medium, and geostationary orbits. The geostationary satellites form a ring around the earth similar to the rings around Saturn. Your hand-held Global Position Satellite (GPS) receiver gives your latitude, longitude and elevation to centimeter accuracy anywhere on or above the earth day or night, cloudy or clear.

Our probes with their arrays of antennas have visited the planets of the solar system and beyond, responding to our commands and sending back photographs and data at centimeter wavelengths even though it may take over 5 hours for the signals to travel one way. And our radio telescope antennas operating at millimeter to kilometer wavelengths receive signals from objects so distant that it has taken more than 10 billion years for the signals to arrive.

Antennas are the essential communication link for aircraft and ships. Antennas for cellular phones and all types of wireless devices link us to everyone and everything. With mankind's activities expanding into space, the need for antennas will grow to an unprecedented degree. Antennas will provide the vital links to and from everything out there. The future of antennas reaches to the stars.

A word about *wireless*. After Heinrich Hertz first demonstrated radiation from antennas, it was called *wireless* (German: *drahtlos,* French: *sans fils*). And wireless it was until broadcasting

---

[1]The plural of the zoological antenna is "antennae" but the plural of the radio antenna is "antennas." A century ago, antennas were commonly called "aerials" and this term is still used in some countries. In Japanese, the three characters (空中線) for antenna mean literally, "middle sky wire." These are the characters on the cover of the Japanese translation of the first edition of this book.

began about 1920 and the word *radio* was introduced. Now *wireless* is back to describe the many systems that operate without wires as distinguished from *radio,* which to most persons now implies AM or FM.

1–2 DIMENSIONS AND UNITS（量纲和单位）

A *dimension* defines some physical characteristic. For example, length, mass, time, velocity, and force are dimensions. The dimensions of length, mass, time, electric current, temperature, and luminous intensity are considered as the *fundamental dimensions* since other dimensions can be defined in terms of these six. This choice is arbitrary but convenient. Let the letters L, M, T, I, $\mathcal{T}$, and $\mathcal{I}$ represent the dimensions of length, mass, time, electric current, temperature, and luminous intensity. Other dimensions are then secondary dimensions. For example, area is a secondary dimension which can be expressed in terms of the fundamental dimension of length squared (L^2). As other examples, the fundamental dimensions of velocity are L/T and of force are ML/T^2.

A *unit* is a standard or reference by which a dimension can be expressed numerically. Thus, the meter is a unit in terms of which the dimension of length can be expressed, and the kilogram is a unit in terms of which the dimension of mass can be expressed. For example, the length (dimension) of a steel rod might be 2 meters, and its mass (dimension) 5 kilograms.

1–3 FUNDAMENTAL AND SECONDARY UNITS（基本单位和派生单位）

The units for the fundamental dimensions are called the *fundamental* or *base units*. In this book the *metric system* or more precisely the International System of Units, abbreviated SI, is used.[1] In this system the *meter, kilogram, second, ampere, kelvin,* and *candela* are the base units for the six fundamental dimensions of length, mass, time, electric current, temperature, and luminous intensity. The definitions for these fundamental units are:

Meter (m). Equal to the path length traveled by light in vacuum in a time $t = 1/299{,}792{,}458$ second.

Kilogram (kg). Equal to mass of international prototype kilogram, a cylinder of platinum-iridium alloy kept at Sèvres, France. This standard kilogram is the only artifact among the SI base units.

Second (s). Equal to duration of 9,192,631,770 periods of radiation corresponding to the transition between two hyperfine levels of the ground state of cesium-133. The second was formerly defined as 1/86,400 part of a mean solar day. The earth's rotation rate is gradually slowing down, but the atomic (cesium-133) transition is much more constant and is now the standard. The two standards differ by about 1 second per year. Atomic clocks are accurate to about 1 microsecond per year. Distant fast rotating (1000 rps) pulsars may soon replace atomic clocks as a still better standard (nanoseconds per year accuracy).

---

[1]The International System of Units is the modernized version of the metric system. The abbreviation SI is from the French name *Système Internationale d'Unités*. For the complete official description of the system see U.S. National Bureau of Standards (now National Institute of Standards and Technology) *Spec. Pub.* 330, 1971.

Ampere (A). Equal to the electric current flowing in each of two infinitely long parallel wires in vacuum separated by 1 meter which produces a force of 200 nanonewtons per meter of length (200 nN $m^{-1} = 2 \times 10^{-7}$ N m^{-1}).

Kelvin (K). Temperature equal to 1/273.16 of the triple point of water (or triple point of water equals 273.16 kelvins). Note that the degree sign (°) is not used with kelvins. Thus, the boiling temperature of water = 100°C = 373 K.[1]

Candela (cd). Luminous intensity equal to that of 1/600,000 square meter of a perfect radiator at the temperature of freezing platinum at a pressure of 1 standard atmosphere.

The units for other dimensions are called *secondary* or *derived* units and are based on these fundamental units.

The material in this book deals almost exclusively with the four fundamental dimensions *length, mass, time,* and *electric current* (dimensional symbols L, M, T, and I). The four fundamental units for these dimensions are the basis of what was formerly called the *meter-kilogram-second-ampere* (mksa) *system,* now a subsystem of the SI.

The complete SI involves not only units but also other recommendations, one of which is that multiples and submultiples of the SI units be stated in steps of 10^3 or 10^{-3}. Thus, the kilometer (1 km = 10^3 m) and the millimeter (1 mm = 10^{-3} m) are preferred units of length. For example, the proper SI designation for the width of motion-picture film is 35 mm, not 3.5 cm. For a list of the preferred units see Table 1–1. This table also gives the pronunciation, abbreviation, and derivation of these units.

In this book *rationalized* SI units are used. The rationalized system has the advantage that the factor 4π does not appear in Maxwell's equations, although it does appear in certain other relations. A complete table of units in this system is given in Appendix A–1. The table lists dimensions, or quantities, alphabetically under each of the following headings: Fundamental, Mechanical, Electrical, and Magnetic. For each quantity the mathematical symbol (as used in equations), description, SI unit and abbreviation, equivalent units, and fundamental dimensions are listed.

It is a good idea to refer to the table as each new quantity and unit is discussed to become familiar with its fundamental dimensions.

1–4 HOW TO READ THE SYMBOLS AND NOTATION（如何阅读符号和记号）

In this book *quantities,* or *dimensions,* which are scalars, like charge Q, mass M, or resistance R, are always in italics. Quantities which may be vectors *or* scalars are boldface as vectors and italics as scalars, e.g., electric field $\mathbf{E}$ (vector) or E (scalar). Unit vectors are always boldface with a hat (circumflex) over the letter, e.g., $\hat{\mathbf{x}}$ or $\hat{\mathbf{r}}$.[2]

---

[1]Note that the symbol for degrees is not used with kelvins. Thus, the boiling temperature of water (100°C) is 373 kelvins (373 K), *not* 373°K. However, the degree sign is retained with degrees Celsius.

[2]In longhand notation a vector may be indicated by a bar over the letter and hat (^) over the unit vector. Also, no distinction is usually made between quantities (italics) and units (roman). However, it can be done by placing a bar under the letter to indicate italics or by writing the letter with a distinct slant.

Table 1–1 Metric prefixes

Numerical value	Prefix	Pronunciation	Symbol	As used in this book (U.S. meaning)	Meaning in other countries
$1\,000\,000\,000\,000\,000\,000 = 10^{18}$	exa	(ex a)	E	one quintillion	trillion
$1\,000\,000\,000\,000\,000 = 10^{15}$	peta	(pet a)	P	one quadrillion	thousand billion
$1\,000\,000\,000\,000 = 10^{12}$	tera	(tare a)	T	one trillion	billion
$1\,000\,000\,000 = 10^{9}$	giga	(jig a)	G	one billion	milliard
$1\,000\,000 = 10^{6}$	mega	(meg a)	M	one million	
$1\,000 = 10^{3}$	kilo	(key lo)	k	one thousand	
$100 = 10^{2}$	hecto	(hek toe)	h	one hundred	
$10 = 10$	deka	(dek a)	da	ten	
$0.1 = 10^{-1}$	deci	(dec i)	d	one tenth	
$0.01 = 10^{-2}$	centi	(cent i)	c	one hundredth	
$0.001 = 10^{-3}$	milli	(mill i)	m	one thousandth	
$0.000\,001 = 10^{-6}$	micro	(my kro)	μ	one millionth	
$0.000\,000\,001 = 10^{-9}$	nano	(nan o)	n	one billionth	milliardth
$0.000\,000\,000\,001 = 10^{-12}$	pico	(pee ko)	p	one trillionth	billionth
$0.000\,000\,000\,000\,001 = 10^{-15}$	femto	(fem toe)	f	one quadrillionth	thousand billionth
$0.000\,000\,000\,000\,000\,001 = 10^{-18}$	atto	(at o)	a	one quintillionth	trillionth

Examples: 1 kilometer (1 km) = 1000 meters (m)
1 micrometer (1 μm) = one millionth of a meter
1 kilowatt (1 kW) = 1000 watts (W)
1 milliwatt (1 mW) = one thousandth of a watt
Note that whereas billion and trillion (and billionth and trillionth) may be ambiguous, giga, tera, nano, and pico are not.

Units are in roman type, i.e., *not* italic; for example, H for henry, s for second, or A for ampere. The abbreviation for a unit is capitalized if the unit is derived from a proper name; otherwise it is lowercase (small letter). Thus, we have C for coulomb but m for meter. Note that when the unit is written out, it is always lowercase even though derived from a proper name. *Prefixes* for units are also roman, like n in nC for nanocoulomb or M in MW for megawatt.

EXAMPLE 1–4.1

$$\mathbf{D} = \hat{\mathbf{x}}\, 200\ \text{pC m}^{-2}$$

means that the electric flux density **D** is a vector in the positive x direction with a magnitude of 200 picocoulombs per square meter ($= 2 \times 10^{-10}$ coulombs per square meter).

EXAMPLE 1–4.2

$$V = 10\ \text{V}$$

means that the voltage V (a scalar) equals 10 volts. Distinguish carefully between V (italics) for voltage, V (roman) for volts, **v** (lowercase, boldface) for velocity (a vector), and v (lowercase, italics) for volume.

EXAMPLE 1–4.3

$$S = 4 \text{ W m}^{-2}\text{Hz}^{-1}$$

means that the flux density S (a scalar) equals 4 watts per square meter per hertz. This can also be written $S = 4$ W/m^2 Hz or 4 W/(m^2 Hz), but the form W m^{-2} Hz^{-1} is more direct and less ambiguous. However, the slash form (W/m^2 Hz) is easier to write and is used in the problem statements.

Note that for conciseness, prefixes are used where appropriate instead of exponents. Thus, the velocity of light is given as $c = 300$ Mm s^{-1} (300 megameters per second) and *not* 3×10^8 m s^{-1}. However, in solving a problem the exponential form (3×10^8 m s^{-1}) is used.

The metric prefixes are in steps of 10^{-3} or 10^3 and go from *atto* (10^{-18}) to *exa* (10^{18}), a ratio or range of 10^{36}. These are adequate for most purposes. Outside this range the exponential form is used. Thus, there are 10^{79} atoms in the universe.

The modernized metric (SI) units and the conventions used herein combine to give a concise, exact, and unambiguous notation, and if one is attentive to the details, it will be seen to possess both elegance and beauty.

1–5 EQUATION AND PROBLEM NUMBERING（公式和习题的编号）

Important equations and those referred to in the text are numbered consecutively beginning with each section. When reference is made to an equation in a different section, its number is preceded by the chapter and section number. Thus, (2–8–4) refers to Chap. 2, Sec. 8, Eq. (4). A reference to this same equation within Sec. 8 of Chap. 2 would read simply (4). Note that the chapter number and name, and section number and name are printed at the top of each left-hand and right-hand page, respectively.

Problems are numbered according to the section of the book which is relevant. Thus, a problem numbered 1–5–2 is the second problem involving the subject matter of Sec. 1–5.

1–6 DIMENSIONAL ANALYSIS（量纲分析）

It is a necessary condition for correctness that every equation be balanced dimensionally. For example, consider the hypothetical formula

$$\frac{M}{L} = DA$$

where

M = mass
L = length
D = density (mass per unit volume)
A = area

The dimensional symbols for the left side are M/L, the same as those used. The dimensional symbols for the right side are

$$\frac{M}{L^3}L^2 = \frac{M}{L}$$

Therefore, both sides of this equation have the dimensions of mass per length, and the equation is balanced dimensionally. This is not a guarantee that the equation is correct; i.e., it is not a *sufficient* condition for correctness. It is, however, a *necessary* condition for correctness, and it is frequently helpful to analyze equations in this way to determine whether or not they are dimensionally balanced.

Such *dimensional analysis* is also useful for determining what the dimensions of a quantity are. For example, to find the dimensions of force, we make use of Newton's second law that

$$\text{Force} = \text{mass} \times \text{acceleration}$$

Since acceleration has the dimensions of length per time squared, the dimensions of force are

$$\frac{\text{Mass} \times \text{length}}{\text{Time}^2}$$

or in dimensional symbols

$$\text{Force} = \frac{ML}{T^2}$$

Such *dimensional analysis* is also useful for determining the dimensions of a quantity. In free space, wavelength and frequency are related by the velocity of light. Thus,

$$\boxed{f\lambda = c = 3 \times 10^8 \text{ m s}^{-1}} \tag{1}$$

where c = velocity of light, m s^{-1}

EXAMPLE 1–6.1

What is the frequency of a 400-m wave?

■ **Solution**

From (1)

$$f = \frac{c}{\lambda} = 3 \times 10^8/400 = 750 \text{ kHz} \qquad \textit{Ans.}$$

EXAMPLE 1–6.2

What is the wavelength of a 2-GHz wave?

■ **Soiution**

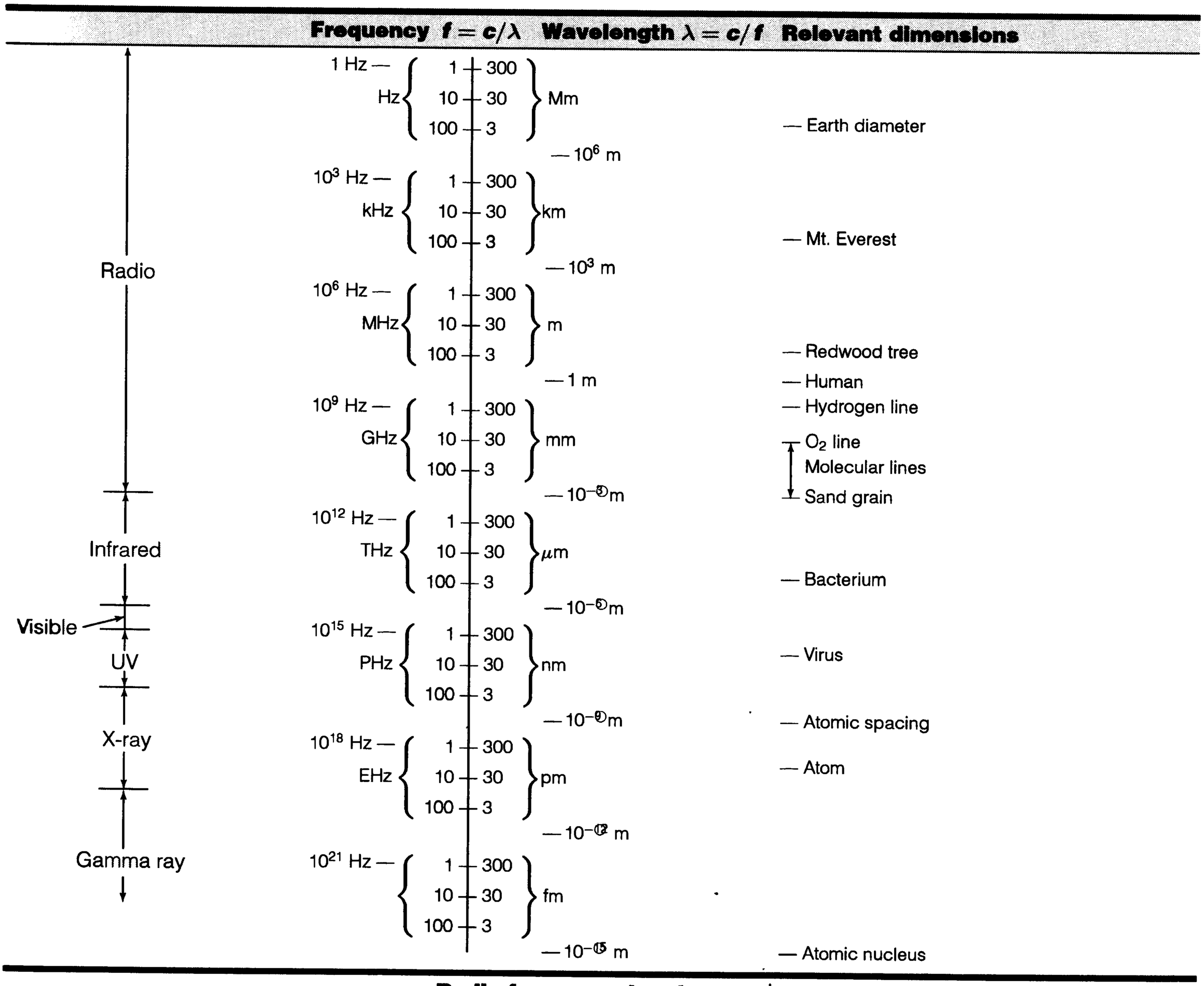
From (1)

$$\lambda = \frac{c}{f} = 3 \times 10^8 / 2 \times 10^9 = 15 \text{ cm} \qquad \textit{Ans.}$$

Confirm both of these answers using the chart of Table 1–2.

1–7 THE ELECTROMAGNETIC SPECTRUM; RADIO-FREQUENCY BANDS
（电磁频谱与无线电频段）

Table 1–2 Frequencies and wavelengths of the electromagnetic spectrum from almost dc to gamma rays



Radio-frequency band names†

Name	Frequency	Principal use
ELF†	3–30 Hz	
SLF	30–300 Hz	Power grids
ULF	300–3000 Hz	
VLF	3–30 kHz	Submarines
LF	30–300 kHz	Beacons
MF	300–3000 kHz	AM broadcast
HF	3–30 MHz	Shortwave broadcast
VHF	30–300 MHz	FM, TV
UHF	300–3000 MHz	TV, LAN, cellular, GPS
SHF	3–30 GHz	Radar, GSO satellites, data
EHF	30–300 GHz	Radar, automotive, data

Microwave bands

"Old"	"New"	Frequency
L	D	1–2 GHz
S	E, F	2–4 GHz
C	G, H	4–8 GHz
X	I, J	8–12 GHz
Ku	J	12–18 GHz
K	J	18–26 GHz
Ka	K	26–40 GHz

†ELF = Extremely Low Frequency, SLF = Super-Low Frequency, VLF = Very Low Frequency, MF = Medium Frequency, HF = High Frequency, UHF = Ultrahigh Frequency, etc.

REFERENCES（参考文献）

Bose, Jagadis Chandra: *Collected Physical Papers,* Longmans, Green, 1927.

Bose, Jagadis Chandra: "On a Complete Apparatus for the Study of the Properties of Electric Waves," *Elect. Engr. (Lond.),* October 1896.

Brown, George H.: "Marconi," *Cosmic Search,* **2,** 5–8, Spring 1980.

Dunlap, Orrin E.: *Marconi—The Man and His Wireless,* Macmillan, 1937.

Faraday, Michael: *Experimental Researches in Electricity,* B. Quaritch, London, 1855.

Gundlach, Friedrich Wilhelm: "Die Technik der kürzesten elecktromagnetischen Wellen seit Heinrich Hertz," *Elektrotech. Zeit. (ETZ),* **7,** 246, 1957.

Hertz, Heinrich Rudolph: "Über Strahlen elecktrischer Kraft," *Wiedemanns Ann. Phys.,* **36,** 769–783, 1889.

Hertz, Heinrich Rudolph: *Electric Waves,* Macmillan, London, 1893; Dover, 1962.

Hertz, Heinrich Rudolph: *Collected Works,* Barth Verlag, 1895.

Hertz, Heinrich Rudolph: *The Work of Hertz and His Successors—Signalling through Space without Wires,* Electrician Publications, 1894, 1898, 1900, 1908.

Hertz, Johanna: *Heinrich Hertz,* San Francisco Press, 1977 (memoirs, letters, and diaries of Hertz).

Kraus, John D.: *Big Ear II,* Cygnus-Quasar, 1995.

Kraus, John D.: "Karl Jansky and His Discovery of Radio Waves from Our Galaxy," *Cosmic Search,* **3,** no. 4, 8–12, 1981.

Kraus, John D.: "Grote Reber and the First Radio Maps of the Sky," *Cosmic Search,* **4,** no. 1, 14–18, 1982.

Kraus, John D.: "Karl Guthe Jansky's Serendipity, Its Impact on Astronomy and Its Lessons for the Future," in K. Kellermann and B. Sheets (eds.), *Serendipitous Discoveries in Radio Astronomy,* National Radio Astronomy Observatory, 1983.

Kraus, John D.: "Antennas Since Hertz and Marconi," *IEEE Trans. Ants. Prop.,* **AP-33,** 131–137, February 1985 (Centennial Plenary Session Paper).

Kraus, John D.: *Radio Astronomy,* 2d ed., Cygnus-Quasar, 1986; Sec. 1–2 on Jansky, Reber, and early history.

Kraus, John D.: "Heinrich Hertz—Theorist and Experimenter," *IEEE Trans. Microwave Theory Tech. Hertz Centennial Issue,* **MTT-36,** May 1988.

Lodge, Oliver J.: *Signalling through Space without Wires,* Electrician Publications, 1898.

Marconi, Degna: *My Father Marconi,* McGraw-Hill, 1962.

Maxwell, James Clerk: *A Treatise on Electricity and Magnetism,* Oxford, 1873, 1904.

Newton, Isaac: *Principia,* Cambridge, 1687.

Poincaré, Henri, and F. K. Vreeland: *Maxwell's Theory and Wireless Telegraphy,* Constable, London, 1905.

Ramsey, John F.: "Microwave Antenna and Waveguide Techniques before 1900," *Proc. IRE,* **46,** 405–415, February 1958.

Rayleigh, Lord: "On the Passage of Electric Waves through Tubes or the Vibrations of Dielectric Cylinders," *Phil. Mag.,* **43,** 125–132, February 1897.

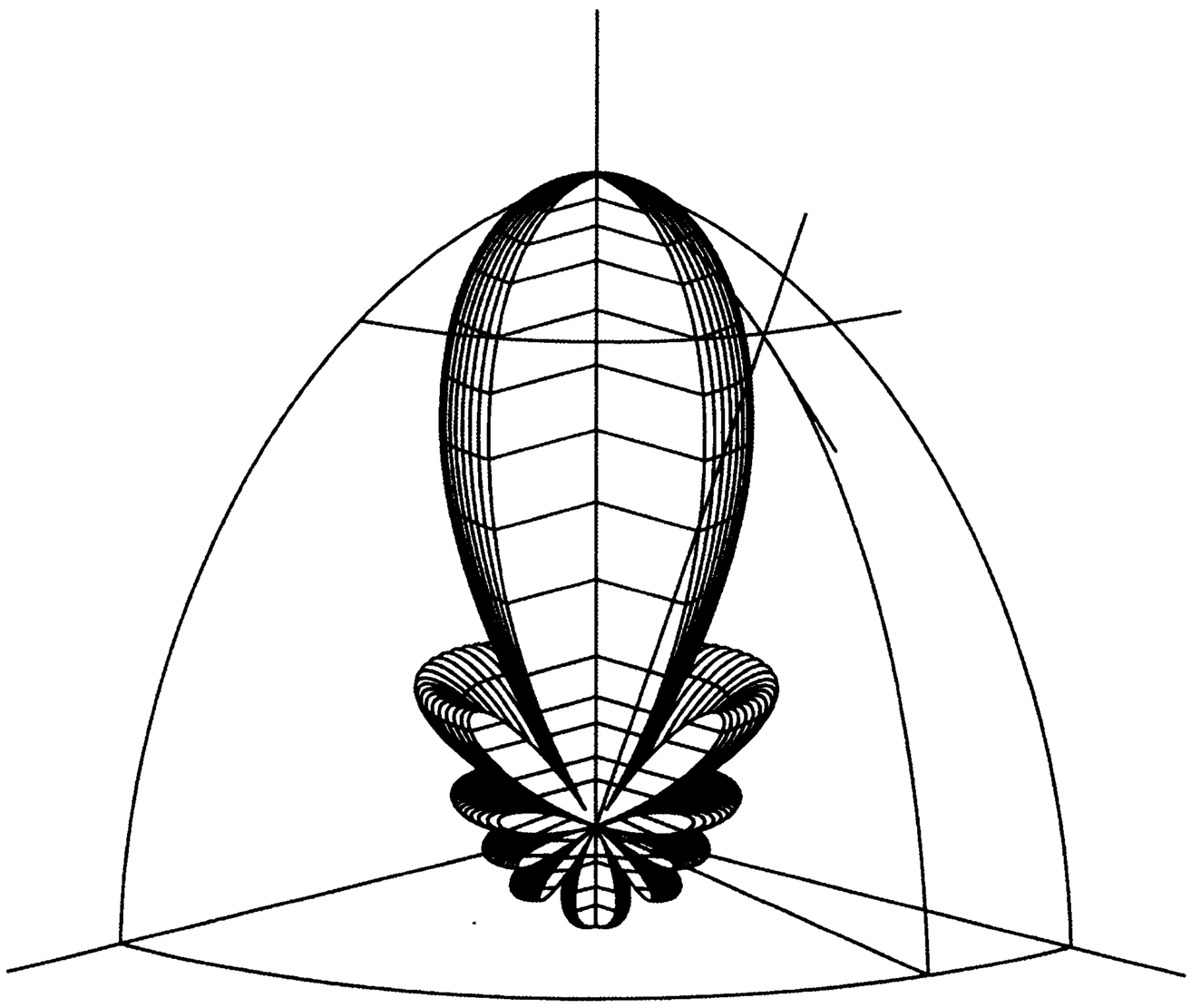
Righi, A.: *L'Ottica della Oscillazioni Elettriche,* Zanichelli, Bologna, 1897.

Rothe, Horst: "Heinrich Hertz, der Entdecker der elektromagnetischen Wellen," *Elektrotech. Zeit. (ETZ),* **7,** 247–251, 1957.

Wolf, Franz: "Heinrich Hertz, Leben and Werk," *Elektrotech. Zeit. (ETZ),* **7,** 242–246, 1957.

CHAPTER 2

Antenna Basics
天线基础



2–1 INTRODUCTION（引言）

Welcome to the wonderful world of antennas, its language and culture; to the aperture family (effective and scattering), the lobe family (main, side, back, and grating); to beamwidths, directivity, and gain.

Antennas are three-dimensional and live in a world of beam area, steradians, square degrees, and solid angle. Antennas have impedances (self and mutual). They couple to all of space and have a temperature measured in kelvins. Antennas have polarizations: linear, elliptical, and circular.

This chapter will make you fluent in the language of antennas and comfortable in its culture. The topics of this chapter include:

- Basic parameters（基本参量）
- Patterns（波瓣图）
- Beam area（波束范围）
- Beam efficiency（波束效率）
- Directivity and gain（定向性和增益）
- Physical and effective apertures（物理口径和有效口径）
- Scattering aperture and radar cross section（散射口径和雷达截面）
- The radio link (Friis formula)［无线电线路（弗里斯公式）］
- Apertures of dipoles and $\lambda/2$ antennas（偶极子和 $\lambda/2$ 天线的口径）
- Radiation resistance（辐射电阻）
- Antenna impedance（天线阻抗）
- Antenna duality（天线的对偶性）
- Sources of radiation（辐射源）
- Field zones（场的分区）
- Shape-impedance considerations（形状 – 阻抗的考虑）
- Polarization（极化）

2–2 BASIC ANTENNA PARAMETERS（基本天线参量）

A *radio antenna* may be defined as the structure associated with the region of transition between a guided wave and a free-space wave, or vice versa. Antennas convert electrons to photons, or vice versa.[1]

Regardless of antenna type, all involve the same basic principle that radiation is produced by accelerated (or decelerated) charge. The *basic equation of radiation* may be expressed simply as

$$\boxed{\dot{I}L = Q\dot{v} \qquad (\text{A m s}^{-1})\ \textbf{\textit{Basic radiation equation}}} \text{（基本辐射方程）} \tag{1}$$

where

$\dot{I}$ = time-changing current, A s^{-1}
L = length of current element, m
Q = charge, C
$\dot{v}$ = time change of velocity which equals the acceleration of the charge, m s^{-2}
L = length of current element, m

Thus, ***time-changing current radiates and accelerated charge radiates.***（时变电流辐射即加速电荷辐射）For steadystate harmonic variation, we usually focus on current. For transients or pulses, we focus on charge.[2] The radiation is perpendicular to the acceleration, and the radiated power is proportional to the square of IL or $Q\dot{v}$.

The two-wire transmission line in Fig. 2–1*a* is connected to a radio-frequency generator (or transmitter). Along the uniform part of the line, energy is guided as a plane Transverse ElectroMagnetic Mode (TEM) wave with little loss. The spacing between wires is assumed to be a small fraction of a wavelength. Further on, the transmission line opens out in a tapered transition. As the separation approaches the order of a wavelength or more, the wave tends to be radiated so that the opened-out line acts like an antenna which launches a free-space wave. The currents on the transmission line flow out on the antenna and end there, but the fields associated with them keep on going.

The transmitting antenna in Fig. 2–1*a* is a region of transition from a guided wave on a transmission line to a free-space wave. The receiving antenna (Fig. 2–1*b*) is a region of transition from a space wave to a guided wave on a transmission line. Thus, ***an antenna is a transition device, or transducer, between a guided wave and a free-space wave, or vice-versa.***（天线是一种导行波与自由空间波之间的转换器件或换能器）The antenna is a device which interfaces a circuit and space.

From the circuit point of view, the antennas appear to the transmission lines as a resistance R_r, called the *radiation resistance*. It is not related to any resistance in the antenna itself but is a resistance coupled from space to the antenna terminals.

In the transmitting case, the radiated power is absorbed by objects at a distance: trees, buildings, the ground, the sky, and other antennas. In the receiving case, passive radiation from distant objects or active radiation from other antennas raises the apparent temperature

---

[1] A photon is a quantum unit of electromagnetic energy equal to hf, where h = Planck's constant ($= 6.63 \times 10^{-34}$ J s) and f = frequency (Hz).

[2] A pulse radiates with a broad bandwidth (the shorter the pulse the broader the bandwidth). A sinusoidal variation results in a narrow bandwidth (theoretically zero at the frequency of the sinusoid if it continues indefinitely).

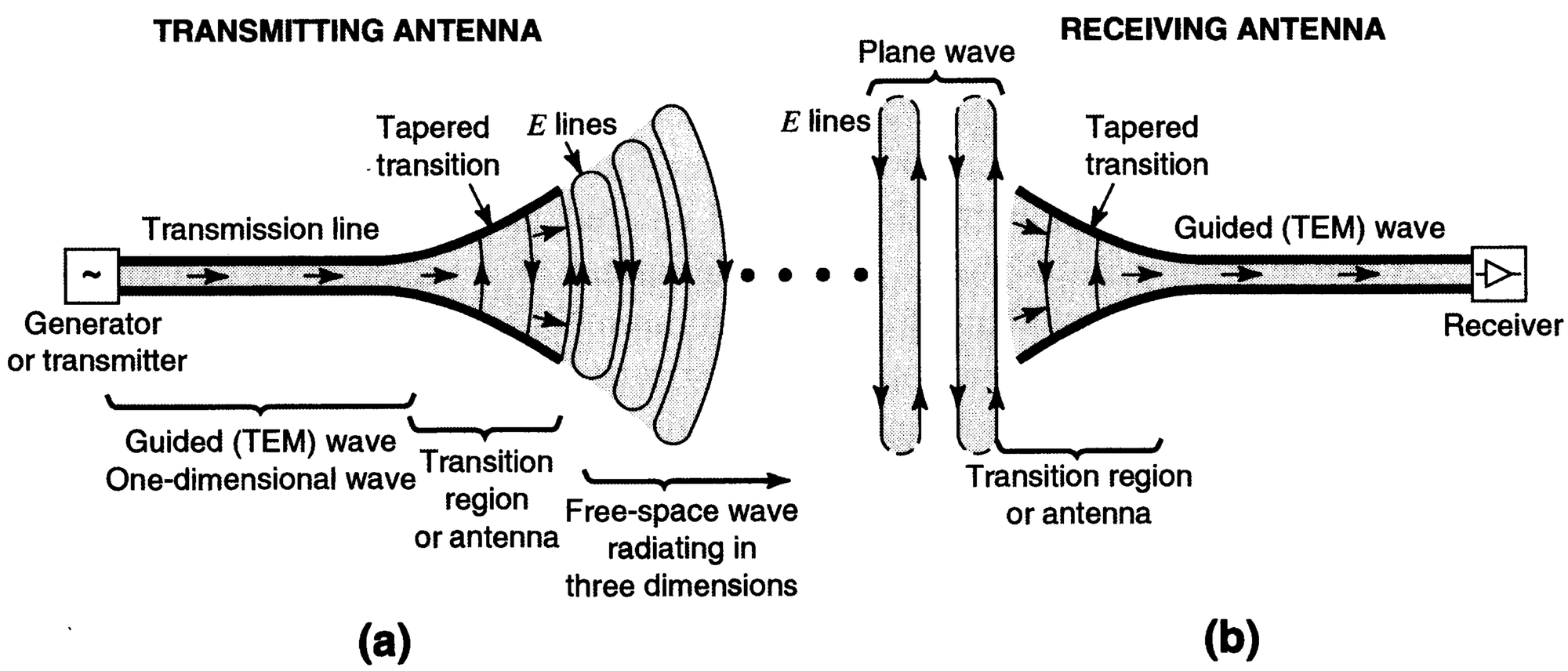


Figure 2–1
(*a*) Radio (or wireless) communication link with transmitting antenna and (*b*) receiving antenna. The receiving antenna is remote from the transmitting antenna so that the spherical wave radiated by the transmitting antenna arrives as an essentially plane wave at the receiving antenna.

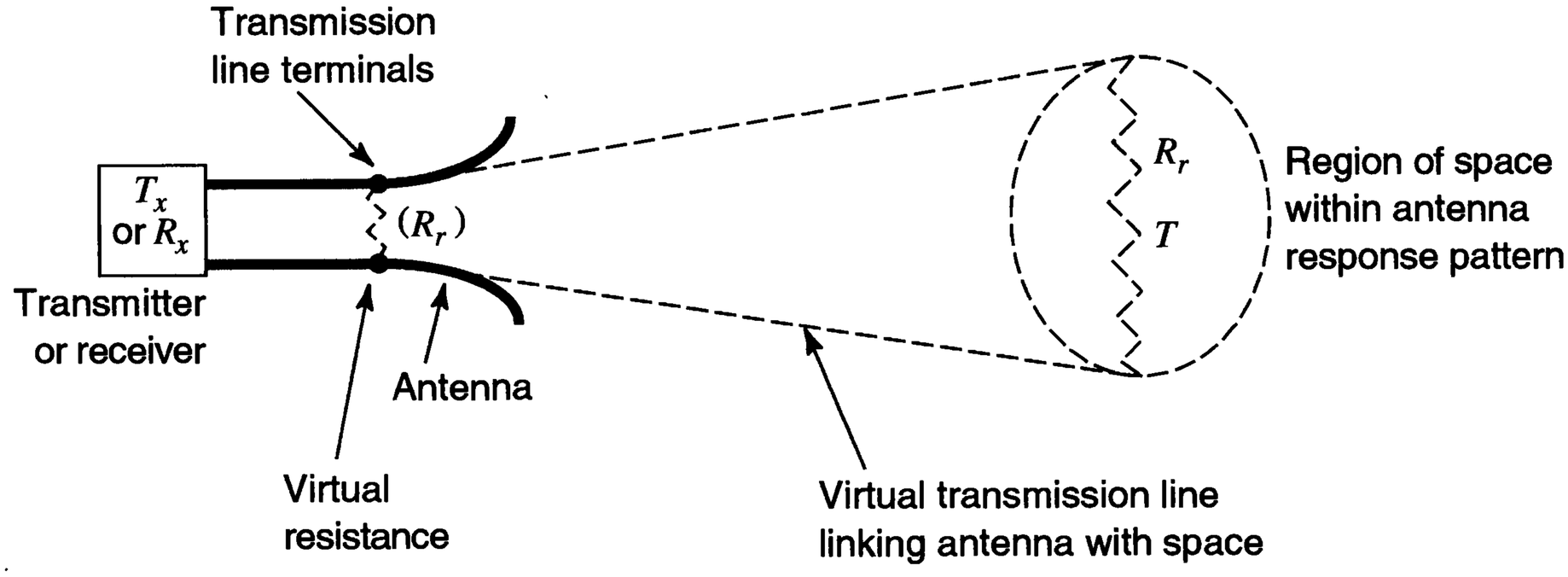


Figure 2–2
Schematic representation of region of space at temperature T linked via a virtual transmission line to an antenna.

of R_r. For lossless antennas this temperature has nothing to do with the physical temperature of the antenna itself but is related to the temperature of distant objects that the antenna is "looking at," as suggested in Fig. 2–2. In this sense, a receiving antenna (and its associated receiver) may be regarded as a remote-sensing temperature-measuring device.

As pictured schematically in Fig. 2–2, the radiation resistance R_r may be thought of as a "virtual" resistance that does not exist physically but is a quantity coupling the antenna to distant regions of space via a "virtual" transmission line.[1]

[1]It is to be noted that the radiation resistance, the antenna temperature, and the radiation patterns are functions of the frequency. In general, the patterns are also functions of the distance at which they are measured, but at distances which are large compared to the size of the antenna and large compared to the wavelength, the pattern is independent of distance. Usually the patterns of interest are for this far-field condition.

2–3 PATTERNS（波瓣图）

Both the radiation resistance R_r, and its temperature T_A are simple scalar quantities. The radiation patterns, on the other hand, are three-dimensional quantities involving the variation of field or power (proportional to the field squared) as a function of the spherical coordinates θ and ϕ. Figure 2–3 shows a three-dimensional field pattern with pattern radius r (from origin to pattern boundary at the dot) proportional to the field intensity in the direction θ and ϕ. The pattern has its *main lobe* (maximum radiation) in the z direction ($\theta = 0$) with *minor lobes* (side and back) in other directions.

To completely specify the radiation pattern with respect to field intensity and polarization requires three patterns:

1. The θ component of the electric field as a function of the angles θ and ϕ or $E_\theta(\theta, \phi)$ (V m^{-1}) as in Figs. 2–3 and 2–4.

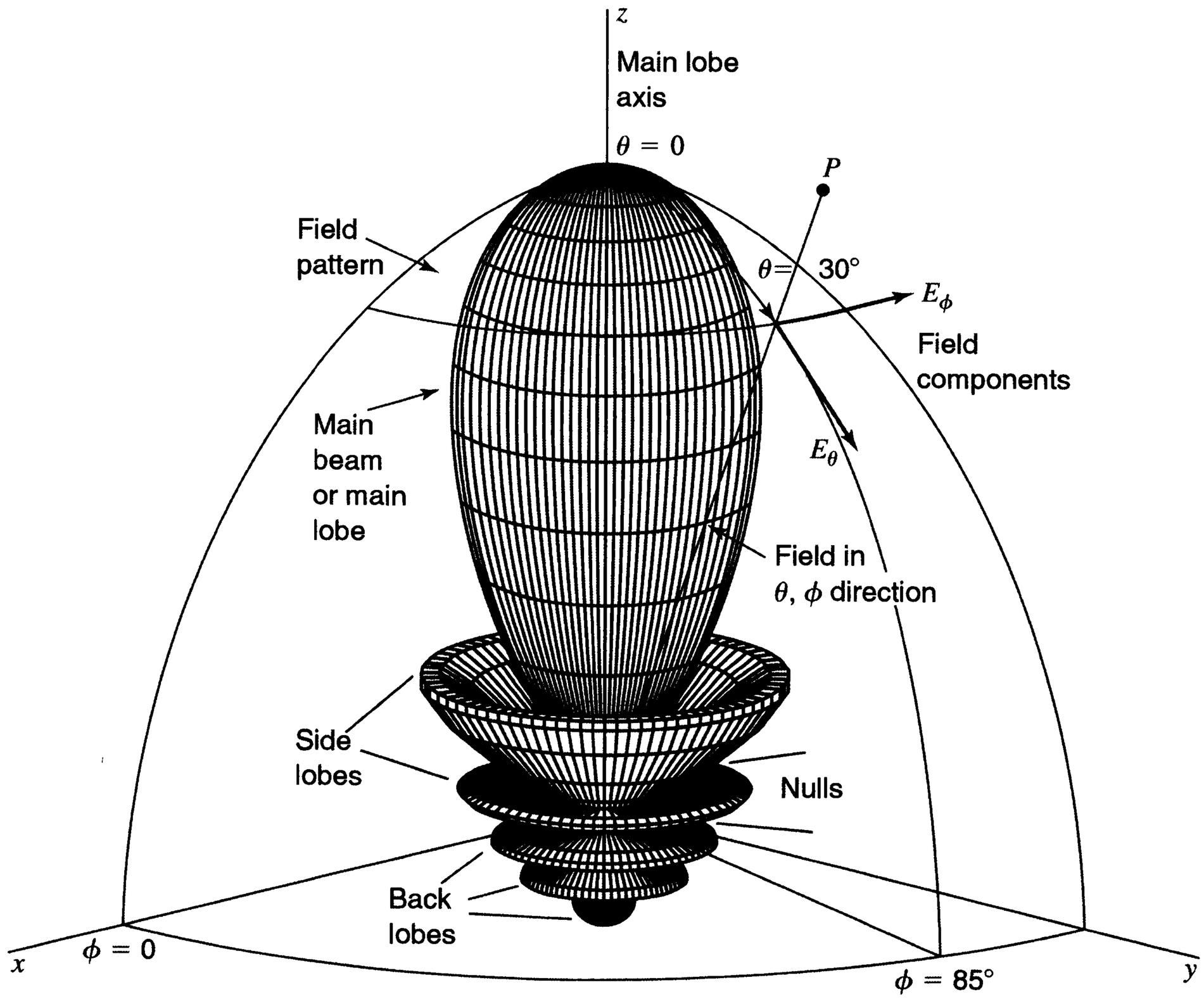


Figure 2–3
Three-dimensional field pattern of a directional antenna with maximum radiation in z-direction at $\theta = 0°$. Most of the radiation is contained in a *main beam* (or *lobe*) accompanied by radiation also in *minor lobes* (*side* and *back*). Between the lobes are *nulls* where the field goes to zero. The radiation in any direction is specified by the angles θ and ϕ. The direction of the point P is at the angles $\theta = 30°$ and $\phi = 85°$. This pattern is symmetrical in ϕ and a function only of θ.

2. The ϕ component of the electric field as a function of the angles θ and ϕ or $E_\phi(\theta, \phi)$ (V m^{-1}).
3. The phases of these fields as a function of the angles θ and ϕ or $\delta_\theta(\theta, \phi)$ and $\delta_\phi(\theta, \phi)$ (rad or deg).

Any field pattern can be presented in three-dimensional spherical coordinates, as in Fig. 2–3, or by plane cuts through the main-lobe axis. Two such cuts at right angles, called the *principal plane patterns* (as in the xz and yz planes in Fig. 2–3) may be required but if the pattern is symmetrical around the z axis, one cut is sufficient.

Figures 2–4*a* and 2–4*b* are principal plane field and power patterns in polar coordinates. The same pattern is presented in Fig. 2–4*c* in rectangular coordinates on a logarithmic, or decibel, scale which gives the minor lobe levels in more detail.

The angular beamwidth at the half-power level or *half-power beamwidth* (HPBW) (or −3-dB beamwidth) and the *beamwidth between first nulls* (FNBW) as shown in Fig. 2–4, are important pattern parameters.

Dividing a field component by its maximum value, we obtain a *normalized or relative field pattern* which is a dimensionless number with maximum value of unity. Thus, the normalized field pattern (Fig. 2–4*a*) for the electric field is given by

(归一化场波瓣图)
$$\boxed{\textbf{\textit{Normalized field pattern}} = E_\theta(\theta, \phi)_n = \frac{E_\theta(\theta, \phi)}{E_\theta(\theta, \phi)_{\max}} \quad \text{(dimensionless)}} \tag{1}$$

The half-power level occurs at those angles θ and ϕ for which $E_\theta(\theta, \phi)_n = 1/\sqrt{2} = 0.707$.

At distances that are large compared to the size of the antenna and large compared to the wavelength, the shape of the field pattern is independent of distance. Usually the patterns of interest are for this *far-field* condition.

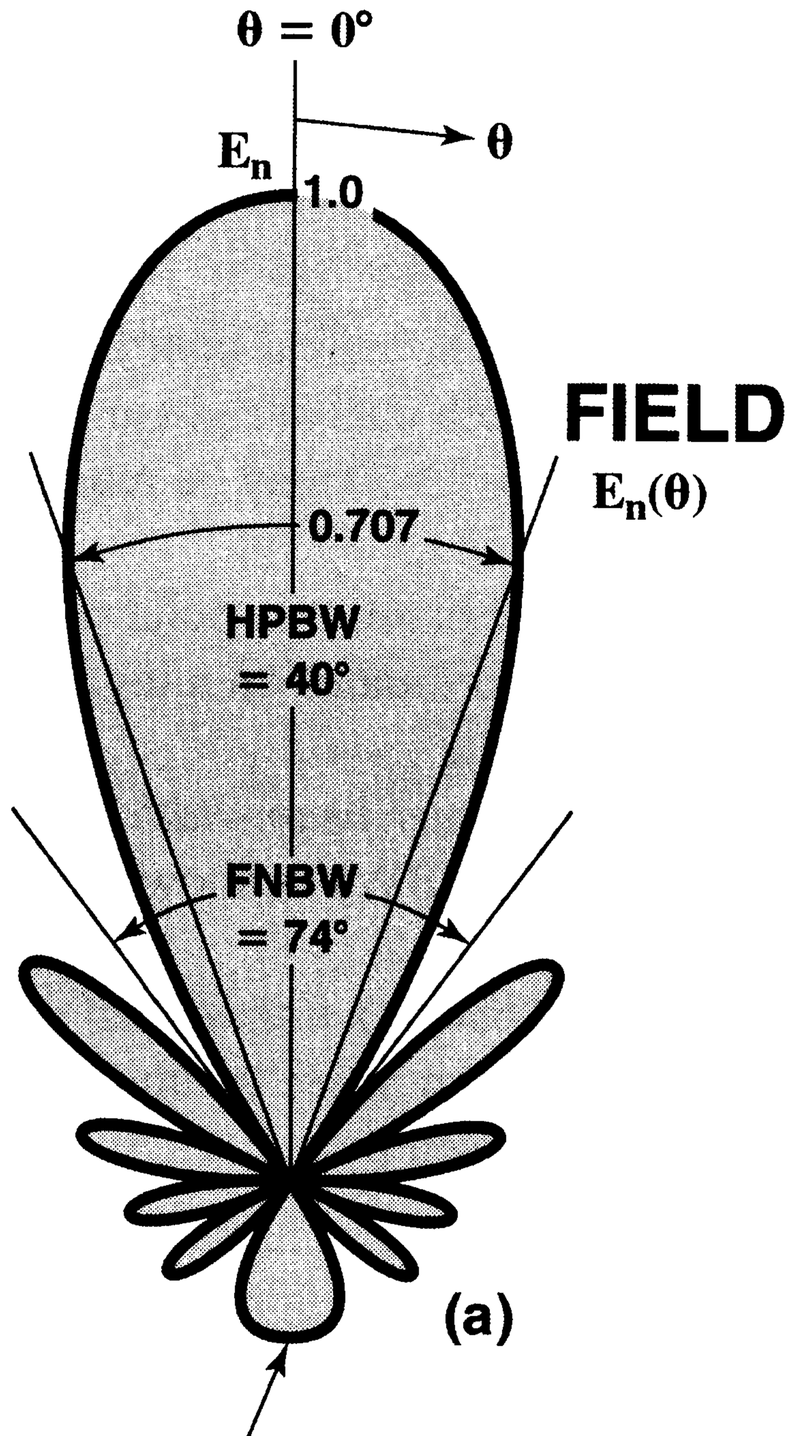
Patterns may also be expressed in terms of the *power per unit area* [or Poynting vector $S(\theta, \phi)$].[1] Normalizing this power with respect to its maximum value yields a *normalized power pattern* as a function of angle which is a dimensionless number with a maximum value of unity. Thus, the normalized power pattern (Fig. 2–4*b*) is given by

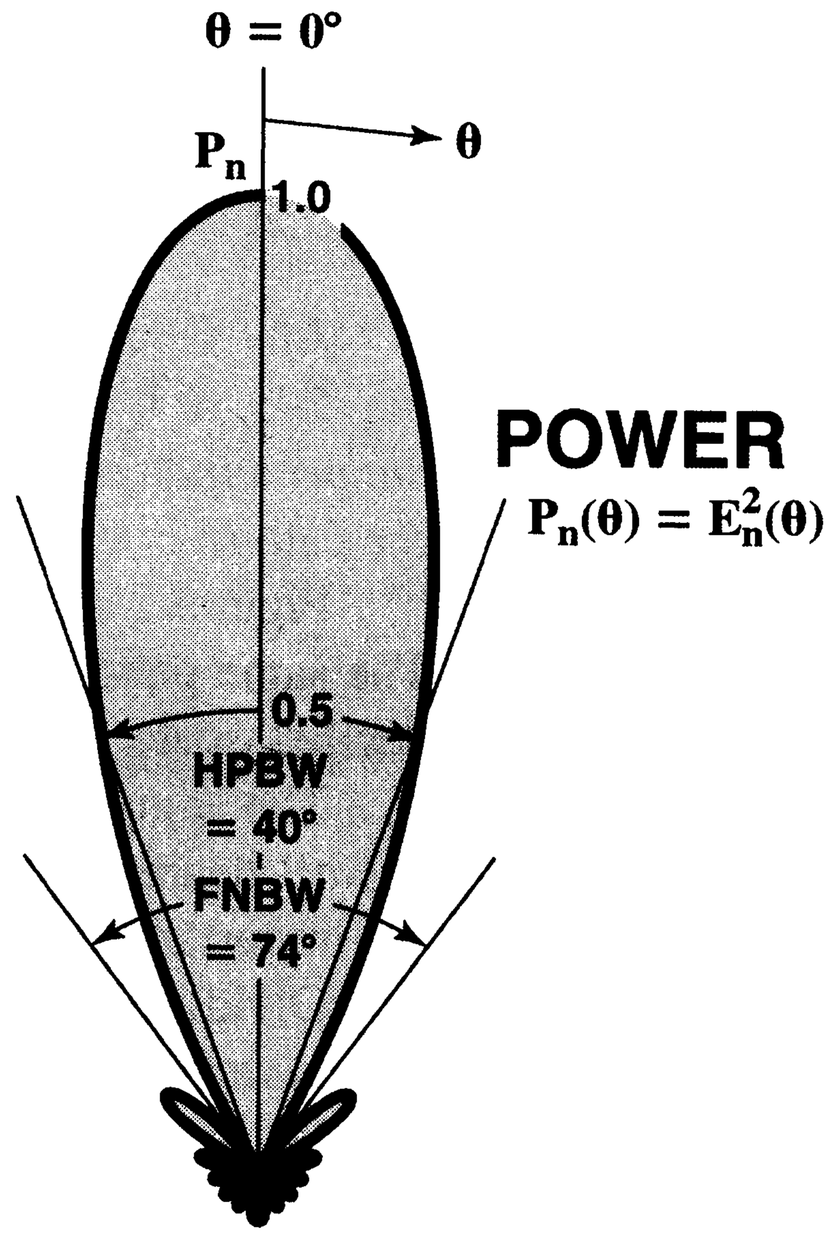
(归一化功率波瓣图)
$$\boxed{\textbf{\textit{Normalized power pattern}} = P_n(\theta, \phi)_n = \frac{S(\theta, \phi)}{S(\theta, \phi)_{\max}} \quad \text{(dimensionless)}} \tag{2}$$

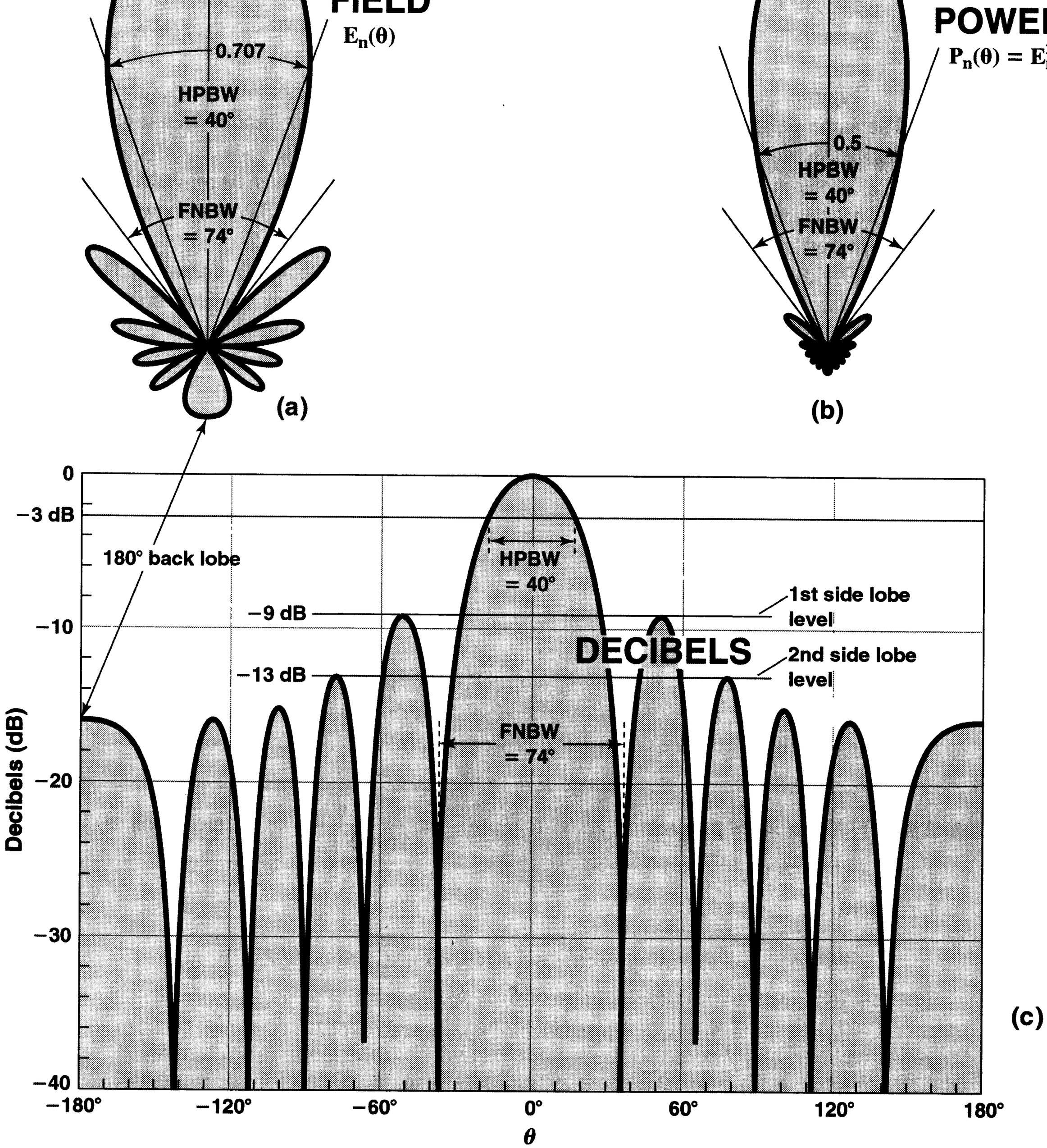
where

$$\begin{aligned} S(\theta, \phi) &= \text{Poynting vector} = \left[E_\theta^2(\theta, \phi) + E_\phi^2(\theta, \phi)\right]/Z_0, \ \text{W m}^{-2} \\ S(\theta, \phi)_{\max} &= \text{maximum value of } S(\theta, \phi), \ \text{W m}^{-2} \\ Z_0 &= \text{intrinsic impedance of space} = 376.7\ \Omega \end{aligned}$$

[1] Although the Poynting vector, as the name implies, is a vector (with magnitude and direction), we use here its magnitude; its direction is the far field is radially outward.


θ = 0°
θ
E_n
1.0
0.707
HPBW
= 40°
FNBW
= 74°
FIELD
$E_n(\theta)$
(a)


θ = 0°
θ
P_n
1.0
0.5
HPBW
= 40°
FNBW
= 74°
POWER
$P_n(\theta) = E_n^2(\theta)$
(b)


0
−3 dB
−10
−20
−30
−40
Decibels (dB)
180° back lobe
HPBW
= 40°
−9 dB
−13 dB
1st side lobe
level
2nd side lobe
level
DECIBELS
FNBW
= 74°
−180°
−120°
−60°
0°
60°
120°
180°
θ
(c)

The decibel level is given by

$$\text{dB} = 10 \log_{10} P_n(\theta, \phi) \tag{3}$$

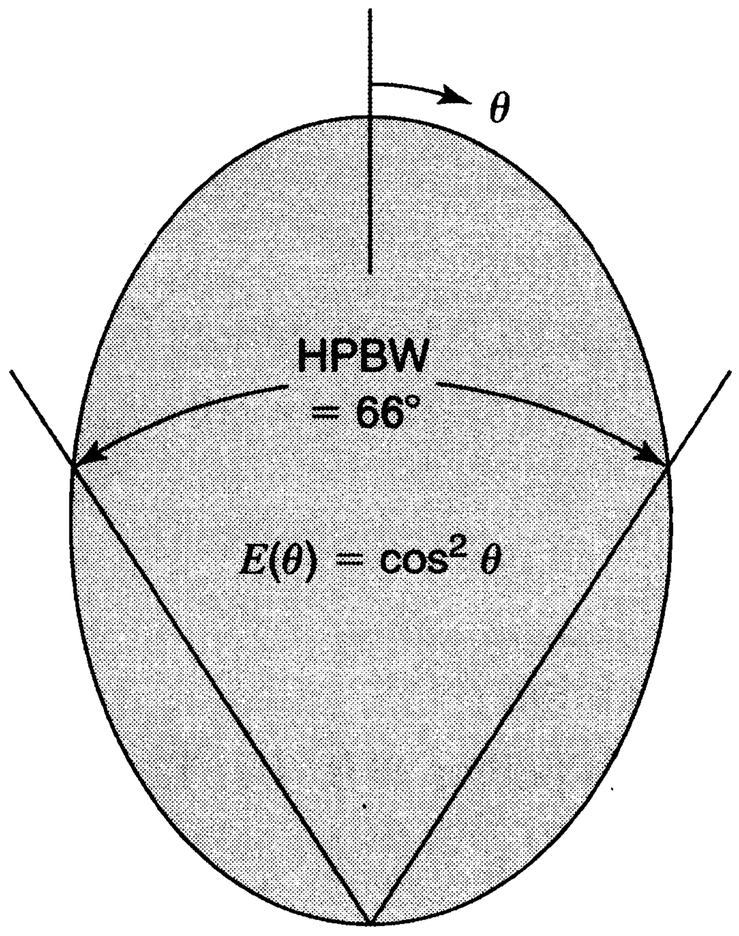
where $P_n(\theta, \phi)$ is as given by (2).

EXAMPLE **2–3.1 Half-Power Beamwidth**

An antenna has a field pattern given by

$$E(\theta) = \cos^2 \theta \quad \text{for } 0° \leq \theta \leq 90°$$

Find the half-power beamwidth (HPBW).



■ **Solution**

$E(\theta)$ at half power $= 0.707$. Thus $0.707 = \cos^2 \theta$ so $\cos \theta = \sqrt{0.707}$ and $\theta = 33°$

$$\text{HPBW} = 2\theta = 66° \qquad \textit{Ans.}$$

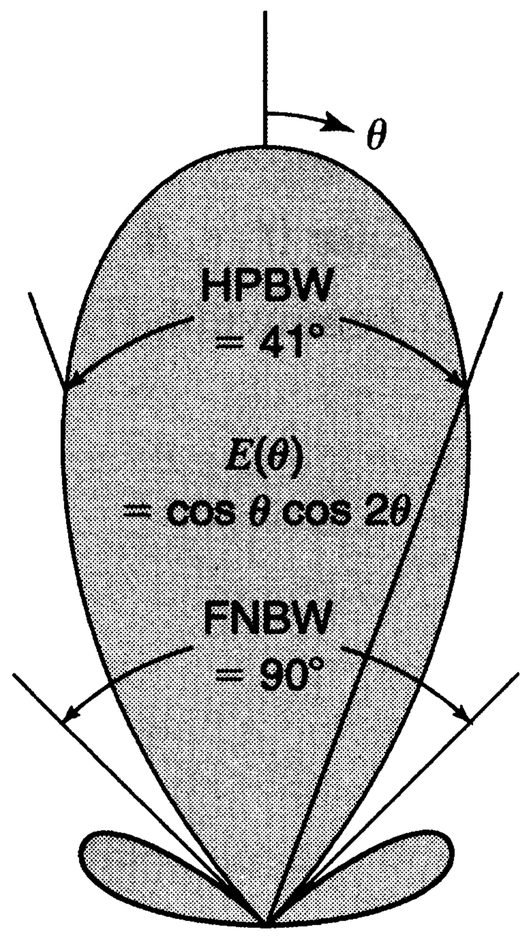
Figure 2–4
Two-dimensional field, power and decibel plots of the 3-D antenna pattern of Fig. 2–3. Taking a slice through the middle of the 3-dimensional pattern of Figure 2–3 results in the 2-dimensional pattern at (*a*). It is a field pattern (proportional to the electric field E in V/m) with normalized relative field $E_n(\theta) = 1$ at $\theta = 0°$. The half-power beam width (HPBW) $= 40°$ is measured at the $E = 0.707$ level.

The pattern at (*b*) is a power plot of (*a*) (proportional to E^2) with relative power $P_n = 1$ at $\theta = 0°$ and with HPBW $= 40°$ as before and measured at the $P_n = 0.5$ level.

A decibel (dB) plot of (*a*) is shown at (*c*) with HPBW $= 40°$ as before and measured at the -3 dB level. The first side lobes are shown at the -9 dB and second side lobes at -13 dB. Decibel plots are useful for showing minor lobe levels.

EXAMPLE **2–3.2 Half-Power Beamwidth and First Null Beamwidth**

An antenna has a field pattern given by $E(\theta) = \cos\theta \cos 2\theta$ for $0° \leq \theta \leq 90°$. Find (*a*) the half-power beamwidth (HPBW) and (*b*) the beamwidth between first nulls (FNBW).



■ **Solution**

(*a*) $E(\theta)$ at half power = 0.707. Thus $0.707 = \cos\theta \cos 2\theta = 1/\sqrt{2}$.

$$\cos 2\theta = \frac{1}{\sqrt{2}\cos\theta} \qquad 2\theta = \cos^{-1}\left(\frac{1}{\sqrt{2}\cos\theta}\right) \qquad \text{and}$$

$$\theta = \frac{1}{2}\cos^{-1}\left(\frac{1}{\sqrt{2}\cos\theta'}\right)$$

Iterating with $\theta' = 0$ as a first guess, $\theta = 22.5°$. Setting $\theta' = 22.5°$, $\theta = 20.03°$, etc., until after next iteration $\theta = \theta' = 20.47° \cong 20.5°$ and

$$\text{HPBW} = 2\theta = 41° \qquad \textit{Ans.}\ (a)$$

(*b*) $0 = \cos\theta \cos 2\theta$, so $\theta = 45°$ and

$$\text{FNBW} = 2\theta = 90° \qquad \textit{Ans.}\ (b)$$

Although the radiation pattern characteristics of an antenna involve three-dimensional vector fields for a full representation, several simple single-valued scalar quantities can provide the information required for many engineering applications. These are:

- Half-power beamwidth, HPBW
- Beam area, Ω_A
- Beam efficiency, ε_M

- Directivity D or gain G
- Effective aperture A_e

The half-power beamwidth was discussed above. The others follow.

2–4 BEAM AREA (OR BEAM SOLID ANGLE) Ω_A (波束范围或波束立体角 Ω_A)

In polar two-dimensional coordinates an incremental area dA on the surface of a sphere is the product of the length $r\,d\theta$ in the θ direction (latitude) and $r\sin\theta\,d\phi$ in the ϕ direction (longitude), as shown in Fig. 2–5.

Thus,

$$dA = (r\,d\theta)(r\sin\theta\,d\phi) = r^2\,d\Omega \tag{1}$$

where

$d\Omega$ = *solid angle* expressed in steradians (sr) or square degrees ($^\square$)
$d\Omega$ = solid angle subtended by the area dA

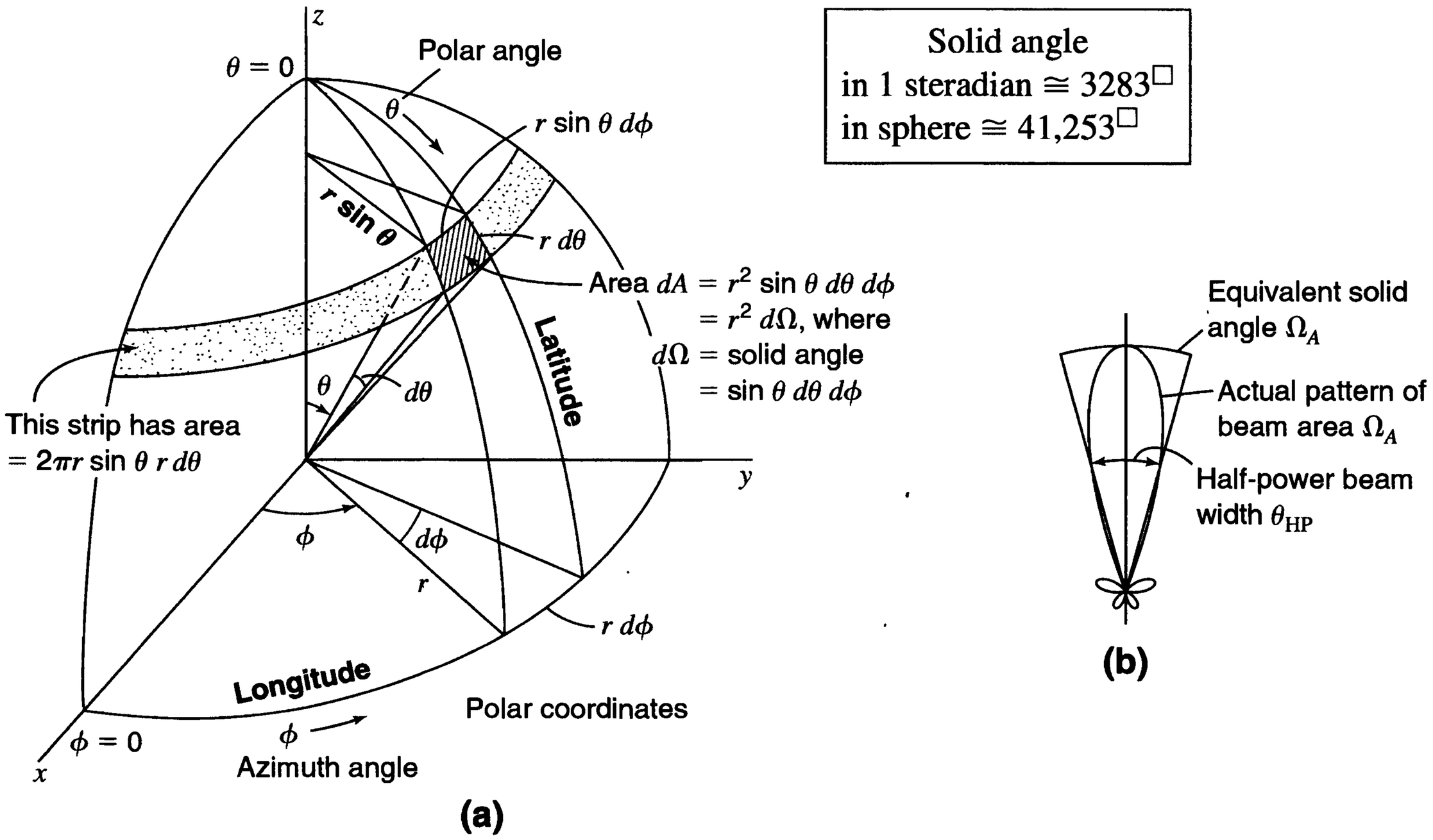


Figure 2–5
Polar coordinates showing incremental solid angle $dA = r^2\,d\Omega$ on the surface of a sphere of radius r where $d\Omega$ = solid angle subtended by the area dA. (*b*) Antenna power pattern and its equivalent solid angle or beam area Ω_A.

The area of the strip of width $r\,d\theta$ extending around the sphere at a constant angle θ is given by $(2\pi r \sin\theta)(r\,d\theta)$. Integrating this for θ values from 0 to π yields the area of the sphere. Thus,

$$\text{Area of sphere} = 2\pi r^2 \int_0^{\pi} \sin\theta\, d\theta = 2\pi r^2[-\cos\theta]_0^{\pi} = 4\pi r^2 \tag{2}$$

where 4π = solid angle subtended by a sphere, sr

Thus,

$$1 \text{ steradian} = 1 \text{ sr} = (\text{solid angle of sphere})/(4\pi)$$

$$= 1 \text{ rad}^2 = \left(\frac{180}{\pi}\right)^2 (\text{deg}^2) = 3282.8064 \text{ square degrees} \tag{3}$$

Therefore,

$$4\pi \text{ steradians} = 3282.8064 \times 4\pi = 41{,}252.96 \cong 41{,}253 \text{ square degrees} = 41{,}253^{\square}$$

$$= \text{solid angle in a sphere} \tag{4}$$

The beam area or *beam solid angle* or Ω_A of an antenna (Fig. 2–5*b*) is given by the integral of the normalized power pattern over a sphere (4π sr)

$$\Omega_A = \int_{\phi=0}^{\phi=2\pi} \int_{\theta=0}^{\theta=\pi} P_n(\theta, \phi) \sin\theta\, d\theta\, d\phi \tag{5a}$$

and

$$\boxed{\Omega_A = \iint_{4\pi} P_n(\theta, \phi)\, d\Omega \qquad \text{(sr)} \qquad \textbf{\textit{Beam area}}} \quad (\text{波束范围}) \tag{5b}$$

where $d\Omega = \sin\theta\, d\theta\, d\phi$, sr.

The beam area Ω_A is the solid angle through which all of the power radiated by the antenna would stream if $P(\theta, \phi)$ maintained its maximum value over Ω_A and was zero elsewhere. Thus the power radiated = $P(\theta, \phi)\Omega_A$ watts.

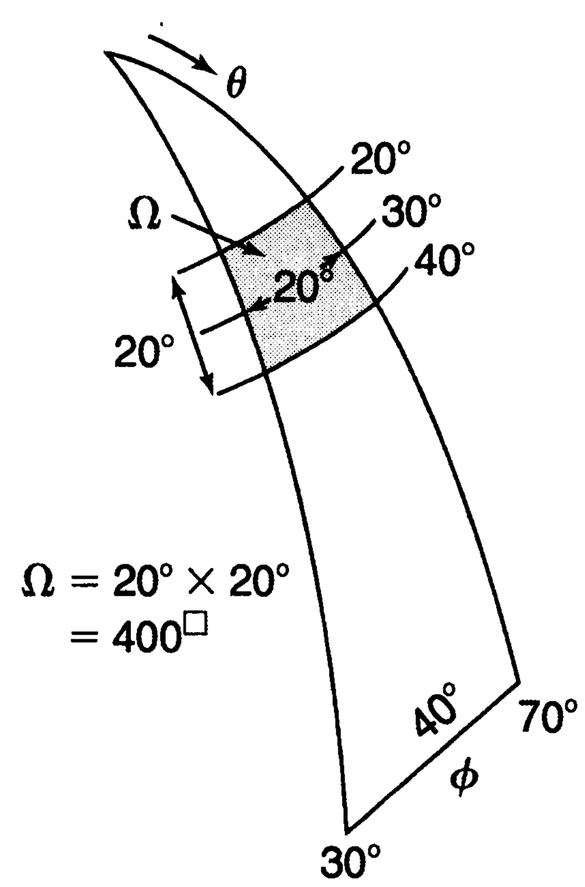
The *beam area* of an antenna can often be described *approximately* in terms of the angles subtended by the *half-power points* of the main lobe in the two principal planes. Thus,

(波束范围) $\boxed{\textbf{\textit{Beam area}} \cong \Omega_A \cong \theta_{\text{HP}}\phi_{\text{HP}} \qquad \text{(sr)}}$ (6)

where θ_{HP} and ϕ_{HP} are the *half-power beamwidths* (HPBW) in the two principal planes, minor lobes being neglected.

EXAMPLE **2–4.1 Solid Angle of Area in Square Degrees**

Find the number of square degrees in the solid angle Ω on a spherical surface that is between $\theta = 20°$ and $\theta = 40°$ (or 70° and 50° north latitude) and between $\phi = 30°$ and $\phi = 70°$ (30° and 70° east longitude).



■ **Solution**

From (1)

$$\Omega = \int_{30°}^{70°} d\phi \int_{20°}^{40°} \sin\theta \, d\theta = \frac{40}{360} 2\pi [-\cos\theta]_{20}^{40}$$

$$= 0.222\pi \times 0.173 = 0.121 \text{ steradians} \qquad \text{(sr)}$$

$$= 0.121 \times 3283 = 397 \text{ square degrees} = 397^{\square} \qquad \textit{Ans.}$$

The solid angle Ω shown in the sketch may be *approximated* as the product of two angles $\Delta\theta = 20°$ and $\Delta\phi = 40° \sin 30° = 40° \times 0.5 = 20°$ where 30° is the median θ value of latitude. Thus, $\Omega = \Delta\theta \, \Delta\phi = 20° \times 20° = 400^{\square}$, which is within 3/4% of the answer given above.

EXAMPLE **2–4.2 Beam Area Ω_A of Antenna with $\cos^2\theta$ Pattern**

An antenna has a field pattern given by $E(\theta) = \cos^2\theta$ for $0° \le \theta \le 90°$. This is the same pattern of Example 2–3.1. Find the beam area of this pattern.

■ **Solution**

From (5)

$$\Omega_A = \int_0^{2\pi} \int_0^{\pi} \cos^4\theta \sin\theta \, d\theta \, d\phi$$

$$= -2\pi \left[\frac{1}{25} \cos^5\theta \right]_0^{\pi/2} = \frac{2\pi}{5} = 1.26 \text{ sr} \qquad \textit{Ans.}$$

From (6) an *approximate* relation for the beam area

$$\Omega_A \cong \theta_{HP}\phi_{HP} \qquad \text{(sr)}$$

where θ_{HP} and ϕ_{HP} are the *half-power beamwidths* (HPBW) in the two principal planes. From Example 2–3.1, $\theta_{HP} = \phi_{HP} = 66°$, so

$$\Omega_A \cong \theta_{HP}\phi_{HP} = 66^2 = 4356 \text{ sq deg} = 4356^{\square}$$

From (3), one square radian = 3283 sq deg so

$$\text{Beam area } \Omega_A = 4356/3282 = 1.33 \text{ sr} \qquad \textit{Approx. Ans.}$$

a difference of 6%.

2–5 RADIATION INTENSITY（辐射强度）

The power radiated from an antenna per unit solid angle is called the *radiation intensity U* (watts per steradian or per square degree). The normalized power pattern of the previous section can also be expressed in terms of this parameter as the ratio of the radiation intensity $U(\theta, \phi)$, as a function of angle, to its maximum value. Thus,

$$P_n(\theta, \phi) = \frac{U(\theta, \phi)}{U(\theta, \phi)_{\max}} = \frac{S(\theta, \phi)}{S(\theta, \phi)_{\max}} \tag{1}$$

Whereas the Poynting vector S depends on the distance from the antenna (varying inversely as the square of the distance), the radiation intensity U is independent of the distance, assuming in both cases that we are in the far field of the antenna (see Sec. 2–13).

2–6 BEAM EFFICIENCY（波束效率）

The (total) *beam area* Ω_A (or *beam solid angle*) consists of the main beam area (or solid angle) Ω_M plus the minor-lobe area (or solid angle) Ω_m.[1] Thus,

$$\Omega_A = \Omega_M + \Omega_m \tag{1}$$

The ratio of the main beam area to the (total) beam area is called the (main) *beam efficiency* ε_M. Thus,

（波束效率）
$$\boxed{\textbf{\textit{Beam efficiency}} = \varepsilon_M = \frac{\Omega_M}{\Omega_A} \qquad \text{(dimensionless)}} \tag{2}$$

The ratio of the minor-lobe area (Ω_m) to the (total) beam area *is* called the *stray factor*. Thus,

$$\varepsilon_m = \frac{\Omega_m}{\Omega_A} = \text{stray factor} \tag{3}$$

It follows that

$$\varepsilon_M + \varepsilon_m = 1 \tag{4}$$

[1] If the main beam is not bounded by a deep null, its extent becomes an arbitrary act of judgment.

2–7 DIRECTIVITY *D* AND GAIN *G* (定向性 *D* 和增益 *G*)

The directivity D and the gain G are probably the most important parameters of an antenna.

The *directivity* of an antenna is equal to the ratio of the maximum power density $P(\theta, \phi)_{\max}$ (watts/m^2) to its average value over a sphere as observed in the far field of an antenna. Thus,

$$D = \frac{P(\theta, \phi)_{\max}}{P(\theta, \phi)_{\text{av}}} \qquad \textit{Directivity from pattern} \quad \text{(来自波瓣图的定向性)} \tag{1}$$

The directivity is a dimensionless ratio ≥ 1.

The average power density over a sphere is given by

$$\begin{aligned} P(\theta, \phi)_{\text{av}} &= \frac{1}{4\pi} \int_{\phi=0}^{\phi=2\pi} \int_{\theta=0}^{\theta=\pi} P(\theta, \phi) \sin\theta \, d\theta \, d\phi \\ &= \frac{1}{4\pi} \iint_{4\pi} P(\theta, \phi) \, d\Omega \qquad (\text{W sr}^{-1}) \end{aligned} \tag{2}$$

Therefore, the directivity

$$D = \frac{P(\theta, \phi)_{\max}}{(1/4\pi) \iint_{4\pi} P(\theta, \phi) \, d\Omega} = \frac{1}{(1/4\pi) \iint_{4\pi} [P(\theta, \phi)/P(\theta, \phi)_{\max}] \, d\Omega} \tag{3}$$

and

$$D = \frac{4\pi}{\iint_{4\pi} P_n(\theta, \phi) \, d\Omega} = \frac{4\pi}{\Omega_A} \qquad \textit{Directivity from beam area } \Omega_A \quad \text{(来自波束范围} \Omega_A \text{的定向性)} \tag{4}$$

where $P_n(\theta, \phi) \, d\Omega = P(\theta, \phi)/P(\theta, \phi)_{\max}$ = normalized power pattern

Thus, the directivity is the ratio of the area of a sphere (4π sr) to the beam area Ω_A of the antenna (Fig. 2–5*b*).

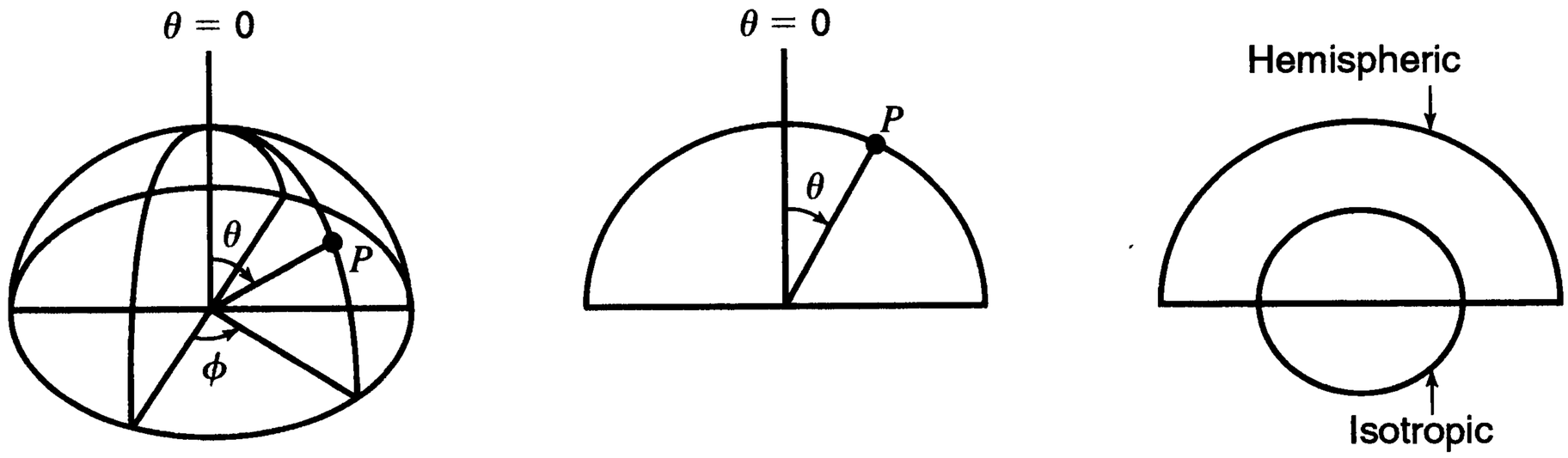
The smaller the beam area, the larger the directivity D. For an antenna that radiates over only half a sphere the beam area $\Omega_A = 2\pi$ sr (Fig. 2–6) and the directivity is

$$D = \frac{4\pi}{2\pi} = 2 \qquad (= 3.01 \text{ dBi}) \tag{5}$$

where dBi = decibels over isotropic

Note that the idealized *isotropic antenna* ($\Omega_A = 4\pi$ sr) has the lowest possible directivity $D = 1$. All actual antennas have directivities greater than 1 ($D > 1$). The simple short dipole has a beam area $\Omega_A = 2.67\pi$ sr and a directivity $D = 1.5$ ($= 1.76$ dBi).

The *gain* G of an antenna is an actual or realized quantity which is less than the directivity D due to ohmic losses in the antenna or its radome (if it is enclosed). In transmitting, these losses involve power fed to the antenna which is not radiated but heats the antenna structure. A mismatch in feeding the antenna can also reduce the gain. The ratio of the gain



(a) (b) (c)

Figure 2–6
Hemispheric power patterns, (*a*) and (*b*), and comparison with isotropic pattern (*c*).

to the directivity is the *antenna efficiency factor*. Thus,

$$G = kD \tag{6}$$

where k = efficiency factor ($0 \leq k \leq 1$), dimensionless.

In many well-designed antennas, k may be close to unity. In practice, G is always less than D, with D its maximum idealized value.

Gain can be measured by comparing the maximum power density of the Antenna Under Test (AUT) with a reference antenna of known gain, such as a short dipole. Thus,

$$\text{Gain} = G = \frac{P_{\max}(\text{AUT})}{P_{\max}(\text{ref. ant.})} \times G(\text{ref. ant.}) \tag{7}$$

If the half-power beamwidths of an antenna are known, its directivity

$$D = \frac{41{,}253^{\square}}{\theta^{\circ}_{\text{HP}}\phi^{\circ}_{\text{HP}}} \tag{8}$$

where

$41{,}253^{\square}$ = number of square degrees in sphere $= 4\pi(180/n)^2$ square degrees ($^{\square}$)
$\theta^{\circ}_{\text{HP}}$ = half-power beamwidth in one principal plane
ϕ°_{HP} = half-power beamwidth in other principal plane

Since (8) neglects minor lobes, a better approximation is a

$$\boxed{D = \frac{40{,}000^{\square}}{\theta^{\circ}_{\text{HP}}\phi^{\circ}_{\text{HP}}} \qquad \textit{Approximate directivity}} \text{ (近似定向性)} \tag{9}$$

If the antenna has a main half-power beamwidth (HPBW) = 20° in both principal planes, its directivity

$$D = \frac{40{,}000^{\square}}{400^{\square}} = 100 \text{ or } 20\,\text{dBi} \tag{10}$$

which means that the antenna radiates 100 times the power in the direction of the main beam as a nondirectional, isotropic antenna.

The *directivity-beamwidth product* $40{,}000^{\square}$ is a rough approximation. For certain types of antennas other values may be more accurate, as discussed in later chapters.

If an antenna has a main lobe with both half-power beamwidths (HPBWs) = 20°, its directivity from (8) is *approximately*

$$D = \frac{4\pi\,(\text{sr})}{\Omega_A\,(\text{sr})} \cong \frac{41{,}253(\text{deg}^2)}{\theta^\circ_{\text{HP}}\phi^\circ_{\text{HP}}} = \frac{41{,}253(\text{deg}^2)}{20^\circ \times 20^\circ}$$

$$\cong 103 \cong 20 \text{ dBi (dB above isotropic)}$$

which means that the antenna radiates a power in the direction of the main-lobe maximum which is about 100 times as much as would be radiated by a nondirectional (isotropic) antenna for the same power input.

EXAMPLE **2–7.1 Gain of Directional Antenna with Three-Dimensional Field Pattern of Fig. 2–3**

The antenna is a lossless end-fire array of 10 isotropic point sources spaced $\lambda/4$ and operating with increased directivity. See Sec. 5–6. The normalized field pattern (see Fig. 2–4*a*) is

$$E_n = \sin\left(\frac{\pi}{2n}\right)\frac{\sin(n\psi/2)}{\sin(\psi/2)} \tag{11}$$

where

$$\psi = d_r(\cos\phi - 1) - \frac{\pi}{n}$$

$$d_r = \pi/2$$

$$n = 10$$

Since the antenna is lossless, gain = directivity.

(*a*) Calculate the gain G.

(*b*) Calculate the gain from the approximate equation (9).

(*c*) What is the difference?

■ **Solution**

(*a*) From (4)

$$\text{Gain } G = \frac{4\pi}{\displaystyle\iint_{4\pi} P_n(\theta,\phi)\,d\Omega}\,d\theta\,d\phi \tag{12}$$

where $P_n(\theta,\phi)$ = normalized power pattern = $[E_n(\theta,\phi)]^2$.
Introducing the given parameters into (11) and (12),

$$G = 17.8 \text{ or } 12.5 \text{ dB} \qquad \textit{Ans.}\ (a)$$

(*b*) From (9) and HPBW = 40° in Fig. 2–4,

$$\text{Gain} = \frac{40{,}000^\square}{(40^\circ)^2} = 25 \text{ or } 14 \text{ dB} \qquad \textit{Ans.}\ (b)$$

(*c*) $\Delta G = 25/17.8 = 1.40$ or 1.5 dB *Ans.* (*c*)

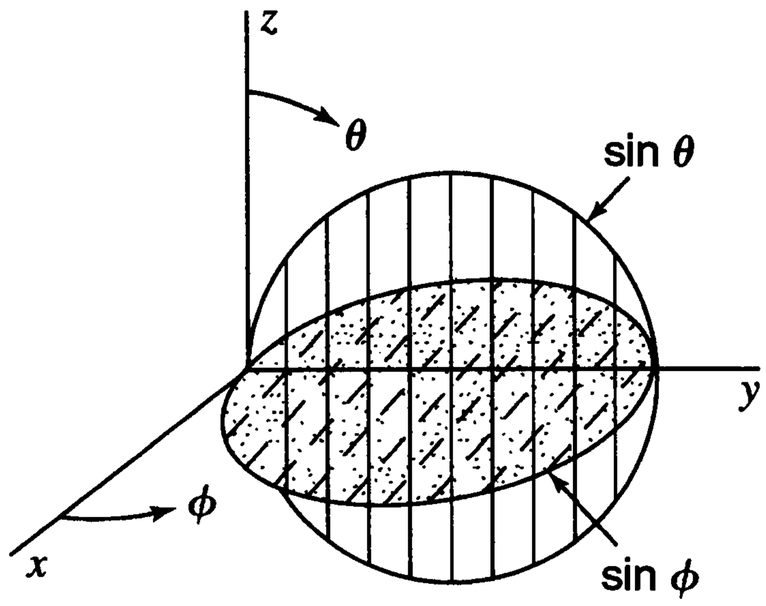
The difference is mostly due to the large minor lobes of the pattern. Changing the formula to

$$\text{Gain} = \frac{28{,}000^{\square}}{(40^{\circ})^2} = 17.5 \text{ or } 12.4 \text{ dB}$$

the gain is much closer to that in (*a*). This approximation is considered more appropriate for end-fire arrays with increased directivity.

EXAMPLE **2–7.2 Directivity**

The normalized field pattern of an antenna is given by $E_n = \sin\theta \sin\phi$, where θ = zenith angle (measured from z axis) and ϕ = azimuth angle (measured from x axis) (see figure). E_n has a value only for $0 \le \theta \le \pi$ and $0 \le \phi \le \pi$ and is zero elsewhere (pattern is unidirectional with maximum in $+y$ direction). Find (*a*) the exact directivity, (*b*) the approximate directivity from (8), and (*c*) the decibel difference.



Unidirectional $\sin\theta$ and $\sin\phi$ field patterns.

■ **Solution**

$$D = \frac{4\pi}{\int_0^{\pi}\int_0^{\pi} \sin^3\theta \sin^2\phi \, d\theta \, d\phi} = \frac{4\pi}{2\pi/3} = 6 \qquad \textit{Ans. (a)}$$

$$D \cong \frac{41{,}253^{\square}}{90^{\circ} \times 90^{\circ}} = 5.1 \qquad \textit{Ans. (b)}$$

$$10 \log \frac{6.0}{5.1} = 0.7 \text{ dB} \qquad \textit{Ans. (c)}$$

2–8 DIRECTIVITY AND RESOLUTION（定向性与分辨率）

The resolution of an antenna may be defined as equal to half the beamwidth between first nulls (FNBW)/2,[1] for example, an antenna whose pattern FNBW = 2° has a resolution of 1° and, accordingly, should be able to distinguish between transmitters on two adjacent

[1]Often called the *Rayleigh resolution*. See J. D. Kraus, *Radio Astronomy,* 2d ed., pp. 6–19, Cygnus-Quasar, 1986.

satellites in the Clarke geostationary orbit separated by 1°. Thus, when the antenna beam maximum is aligned with one satellite, the first null coincides with the adjacent satellite.

Half the beamwidth between first nulls is approximately equal to the half-power beamwidth (HPBW) or

$$\frac{\text{FNBW}}{2} \cong \text{HPBW} \tag{1}$$

Thus, from (2–4–6) the product of the FNBW/2 in the two principal planes of the antenna pattern is a measure of the antenna beam area.[1] Thus,

$$\Omega_A = \left(\frac{\text{FNBW}}{2}\right)_\theta \left(\frac{\text{FNBW}}{2}\right)_\phi \tag{2}$$

It then follows that the number N of radio transmitters or point sources of radiation distributed uniformly over the sky which an antenna can resolve is given approximately by

$$N = \frac{4\pi}{\Omega_A} \tag{3}$$

where Ω_A = beam area, sr

However, from (2–7–4),

$$D = \frac{4\pi}{\Omega_A} \tag{4}$$

and we may conclude that *ideally* the number of point sources an antenna can resolve is numerically equal to the directivity of the antenna or

$$\boxed{D = N} \tag{5}$$

Equation (4) states that the directivity is equal to the number of beam areas into which the antenna pattern can subdivide the sky and (5) gives the added significance that the ***directivity is equal to the number of point sources in the sky that the antenna can resolve*** under the assumed ideal conditions of a uniform source distribution.[2]

2-9 ANTENNA APERTURES（天线口径）

The concept of aperture is most simply introduced by considering a receiving antenna. Suppose that the receiving antenna is a rectangular electromagnetic horn immersed in the field of a uniform plane wave as suggested in Fig. 2–7. Let the Poynting vector, or power density, of the plane wave be S watts per square meter and the area, or physical aperture of the horn, be A_p square meters. If the horn extracts all the power from the wave over its entire physical aperture, then the total power P absorbed from the wave is

$$P = \frac{E^2}{Z} A_p = S A_p \qquad \text{(W)} \tag{1}$$

---

[1] Usually FNBW/2 is slightly greater than the HPBW and (2) is actually a better approximation to Ω_A than $\Omega_A = \theta_{\text{HP}}\phi_{\text{HP}}$ as given by (2–4–6).

[2] A strictly regular distribution of points on a sphere is only possible for 4, 6, 8, 12, and 20 points corresponding to the vertices of a tetrahedron, cube, octahedron, icosahedron and dodecahedron.

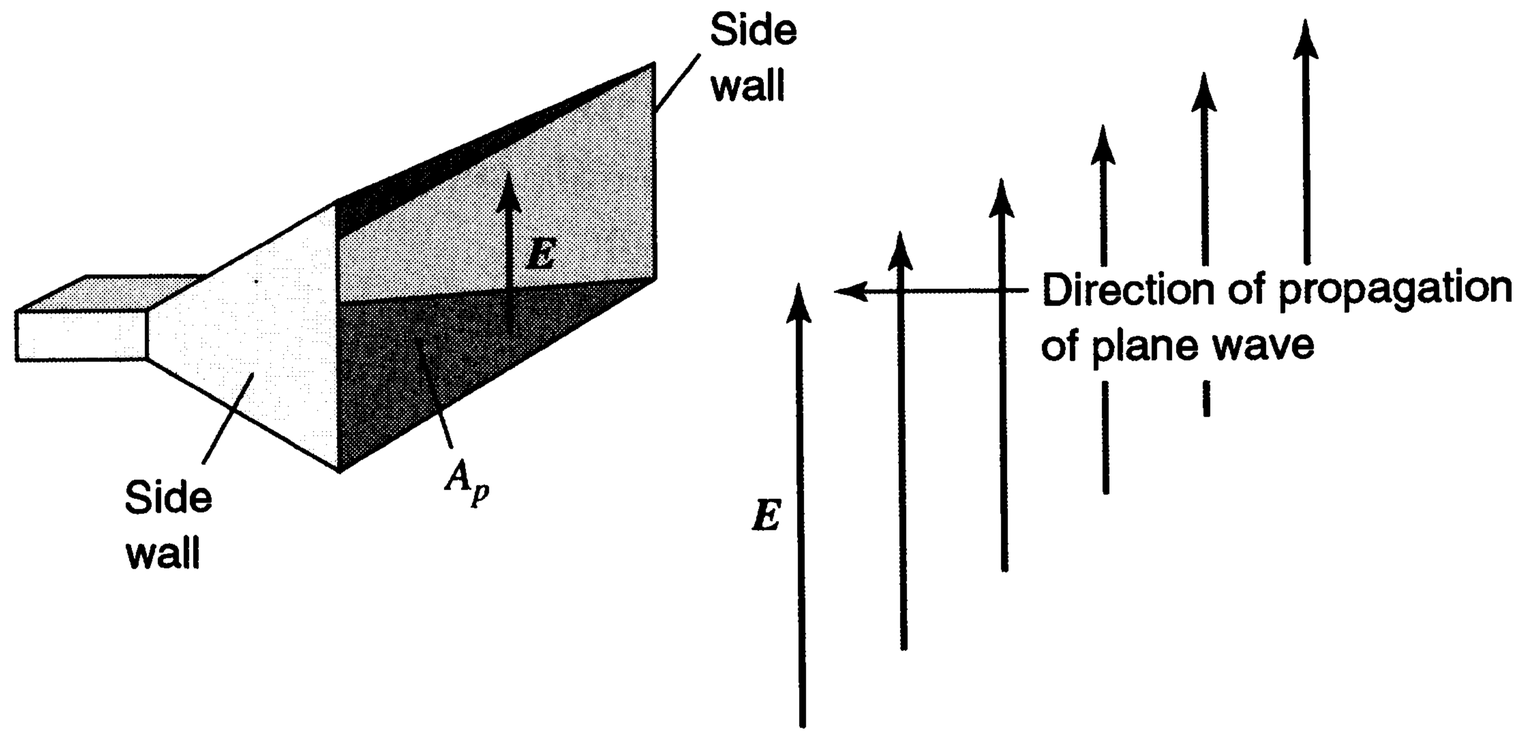


Figure 2–7
Plane wave incident on electromagnetic horn of physical aperture A_p.

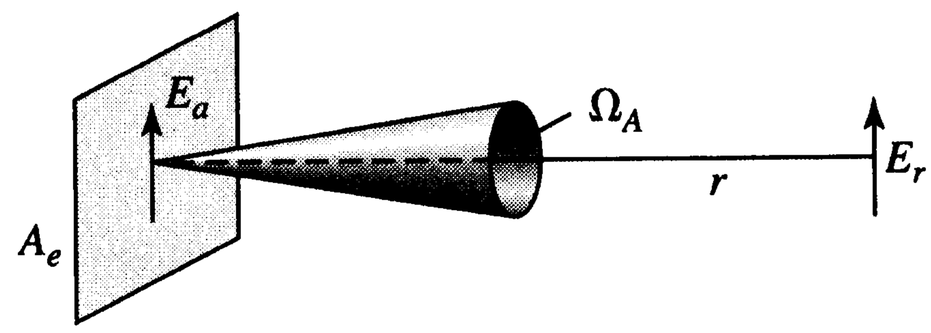


Figure 2–8
Radiation over beam area Ω_A from aperture A_e.

Thus, the electromagnetic horn may be regarded as having an aperture, the total power it extracts from a passing wave being proportional to the aperture or area of its mouth.

But the field response of the horn is NOT uniform across the aperture A because E at the sidewalls must equal zero. Thus, the *effective aperture* A_e of the horn is less than the *physical aperture* A_p as given by

$$\varepsilon_{ap} = \frac{A_e}{A_p} \qquad \text{(dimensionless)} \qquad \textbf{\textit{Aperture efficiency}} \quad (\text{口径效率}) \tag{2}$$

where ε_{ap} = aperture efficiency.

For horn and parabolic reflector antenna, aperture efficiencies are commonly in the range of 50 to 80% ($0.5 \leq \varepsilon_{ap} \leq 0.8$). Large dipole or patch arrays with uniform field to the edges of the physical aperture may attain higher aperture efficiencies approaching 100%. However, to reduce sidelobes, fields are commonly tapered toward the edges, resulting in reduced aperture efficiency.

Consider now an antenna with an *effective aperture* A_e, which radiates all of its power in a conical pattern of beam area Ω_A, as suggested in Fig. 2–8. Assuming a uniform field E_a over the aperture, the power radiated is

$$P = \frac{E_a^2}{Z_0} A_e \qquad \text{(W)} \tag{3}$$

where Z_0 = intrinsic impedance of medium (377 Ω for air or vacuum).

Assuming a uniform field E_r in the far field at a distance r, the power radiated is also given by

$$P = \frac{E_r^2}{Z_0} r^2 \Omega_A \qquad \text{(W)} \tag{4}$$

Equating (3) and (4) and noting that $E_r = E_a A_e / r\lambda$ yields the aperture–beam-area relation

$$\lambda^2 = A_e \Omega_A \qquad (\text{m}^2) \qquad \textit{Aperture–beam-area relation} \quad (\text{口径面积 – 波束范围的关系式}) \tag{5}$$

where Ω_A = beam area (sr).

Thus, if A_e is known, we can determine Ω_A (or vice versa) at a given wavelength. From (5) and (2–7–4) it follows that the directivity

$$D = 4\pi \frac{A_e}{\lambda^2} \qquad \textit{Directivity from aperture} \quad (\text{来自口径的定向性}) \tag{6}$$

All antennas have an effective aperture which can be calculated or measured. Even the hypothetical, idealized isotropic antenna, for which $D = 1$, has an effective aperture

$$A_e = \frac{D\lambda^2}{4\pi} = \frac{\lambda^2}{4\pi} = 0.0796\lambda^2 \tag{7}$$

All lossless antennas must have an effective aperture equal to or greater than this. By reciprocity the effective aperture of an antenna is the same for receiving and transmitting.

Three expressions have now been given for the *directivity* D. They are

$$D = \frac{P(\theta, \phi)_{\max}}{P(\theta, \phi)_{\text{av}}} \qquad (\text{dimensionless}) \qquad \textit{Directivity from pattern} \quad (\text{来自波瓣图的定向性}) \tag{8}$$

$$D = \frac{4\pi}{\Omega_A} \qquad (\text{dimensionless}) \qquad \textit{Directivity from pattern} \quad (\text{来自波束范围的定向性}) \tag{9}$$

$$D = 4\pi \frac{A_e}{\lambda^2} \qquad (\text{dimensionless}) \qquad \textit{Directivity from aperture} \quad (\text{来自口径的定向性}) \tag{10}$$

When the antenna is receiving with a load resistance R_L matched to the antenna radiation resistance R_r ($R_L = R_r$), as much power is reradiated from the antenna as is delivered to the load. This is the condition of *maximum power transfer* (antenna assumed lossless).

In the circuit case of a load matched to a generator, as much power is dissipated in the generator as is delivered to the load. Thus, for the case of the dipole antenna in Fig. 2–9 we have a *load power*

$$P_{\text{load}} = SA_e \qquad (\text{W}) \tag{11}$$

where

S = power density at receiving antenna, W/m^2
A_e = effective aperture of antenna, m^2

and a *reradiated power*

$$P_{\text{rerad}} = \frac{\text{Power reradiated}}{4\pi \text{ sr}} = SA_r \qquad (\text{W})$$

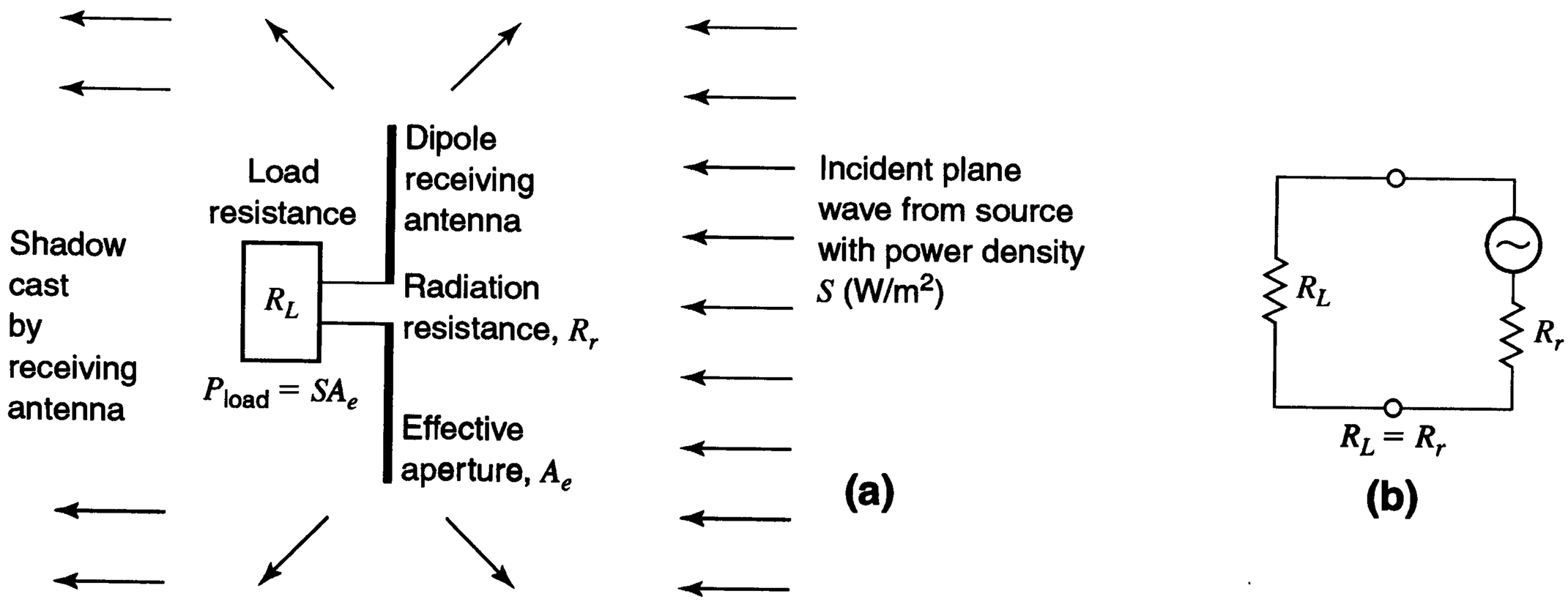


Figure 2–9
(*a*) The receiving antenna matched to a load ($R_r = R_L$) reradiates a power that is equal to the power delivered to the load. More generally, the reradiated and scattered power from any antenna or object yields a *radar cross-section* (RCS) which is proportional to the back-scattered power received at a radar at a distance r, as discussed in Chapter 12. (*b*) Equivalent circuit.

where A_r = reradiating aperture = A_e, m^2 and

$$P_{rerad} = P_{load}$$

The above discussion is applicable to a single dipole ($\lambda/2$ or shorter). However, it does not apply to all antennas. In addition to the reradiated power, an antenna may scatter power that does not enter the antenna-load circuit. Thus, the reradiated plus scattered power may exceed the power delivered to the load. See Sec. 21–15 for a discussion that includes both receiving and transmitting conditions.

2–10 EFFECTIVE HEIGHT（有效高度）

The *effective height* h (meters) of an antenna is another parameter related to the aperture. Multiplying the effective height by the incident field E (volts per meter) of the same polarization gives the voltage V induced. Thus,

$$V = hE \tag{1}$$

Accordingly, the effective height may be defined as the ratio of the induced voltage to the incident field or

$$h = \frac{V}{E} \quad \text{(m)} \tag{2}$$

Consider, for example, a vertical dipole of length $l = \lambda/2$ immersed in an incident field E, as in Fig. 2–9–1(*a*). If the current distribution of the dipole were uniform, its effective height would be l. The actual current distribution, however, is nearly sinusoidal with an *average value* $2/\pi = 0.64$ (of the maximum) so that its effective height $h = 0.64\,l$. It is assumed that the antenna is oriented for maximum response.

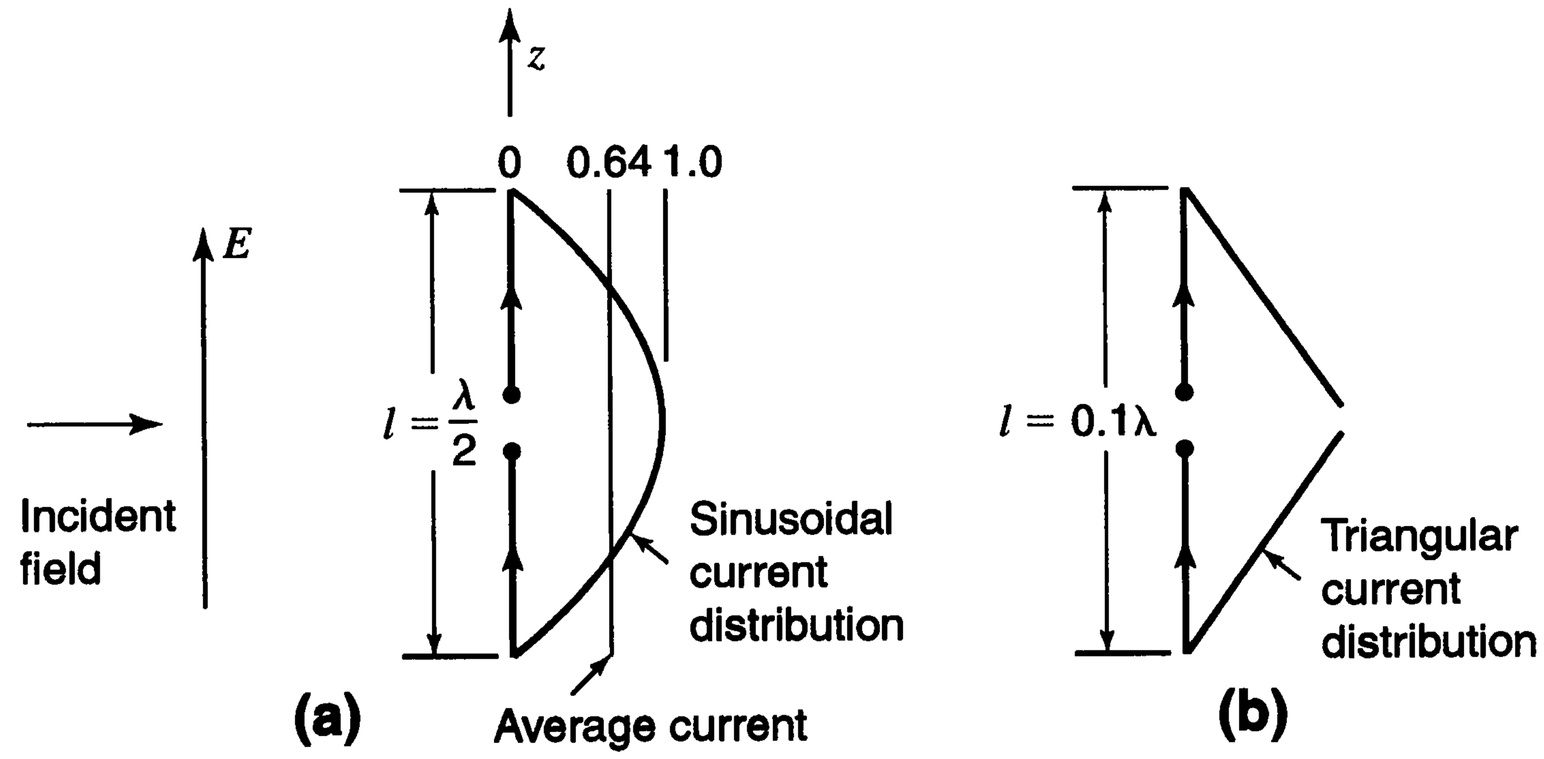


Figure 2–9–1
(*a*) Dipole of length $l = \lambda/2$ with sinusoidal current distribution. (*b*) Dipole of length $l = 0.1\lambda$ with triangular current distribution.

If the same dipole is used at a longer wavelength so that it is only 0.1λ long, the current tapers almost linearly from the central feed point to zero at the ends in a triangular distribution, as in Fig. 2–9–1(*b*). The average current is 1/2 of the maximum so that the effective height is $0.5l$.

Thus, another way of defining effective height is to consider the transmitting case and equate the effective height to the physical height (or length l) multiplied by the (normalized) average current or

$$h_e = \frac{1}{I_0}\int_0^{h_p} I(z)\,dz = \frac{I_{\text{av}}}{I_0}h_p \qquad \text{(m)} \tag{3}$$

where

h_e = effective height, m
h_p = physical height, m
I_{av} = average current, A

It is apparent that *effective height* is a useful parameter for transmitting tower-type antennas.[1] It also has an application for small antennas. The parameter *effective aperture* has more general application to all types of antennas. The two have a simple relation, as will be shown.

For an antenna of radiation resistance R_r matched to its load, the power delivered to the load is equal to

$$P = \frac{1}{4}\frac{V^2}{R_r} = \frac{h^2E^2}{4R_r} \qquad \text{(W)} \tag{4}$$

---

[1]Effective height can also be expressed more generally as a vector quantity. Thus (for linear polarization) we can write

$$V = \mathbf{h}_e \cdot \mathbf{E} = h_e E \cos\theta$$

where

$\mathbf{h}_e$ = effective height and polarization angle of antenna, m
$\mathbf{E}$ = field intensity and polarization angle of incident wave, V m^{-1}
θ = angle between polarization angles of antenna and wave, deg

In a still more general expression (for any polarization state), θ is the angle between polarization states on the Poincaré sphere (see Sec. 2–17).

In terms of the effective aperture the same power is given by

$$P = SA_e = \frac{E^2 A_e}{Z_0} \qquad \text{(W)} \tag{5}$$

where Z_0 = intrinsic impedance of space (= 377 Ω)

Equating (4) and (5), we obtain

$$h_e = 2\sqrt{\frac{R_r A_e}{Z_0}} \qquad \text{(m)} \qquad \text{and} \qquad A_e = \frac{h_e^2 Z_0}{4R_r} \qquad \text{(m}^2\text{)} \tag{6}$$

Thus, effective height and effective aperture are related via radiation resistance and the intrinsic impedance of space.

To summarize, we have discussed the space parameters of an antenna, namely, field and power patterns, beam area, directivity, gain, and various apertures. We have also discussed the circuit quantity of radiation resistance and alluded to antenna temperature, which is discussed further in Sec. 12–1. Figure 2–10 illustrates this ***duality*** of an antenna.

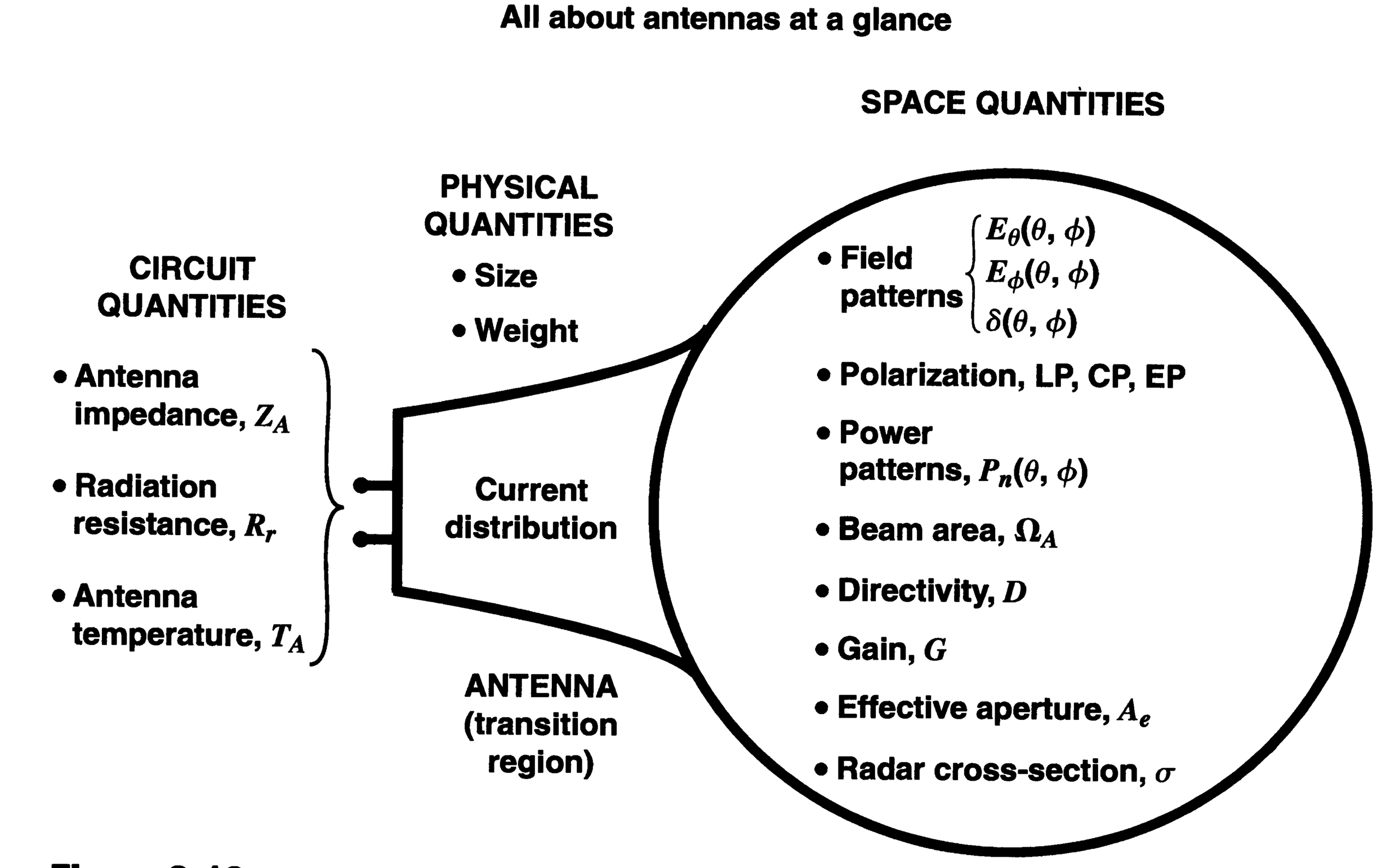


Figure 2–10
The parameters or terminology of antennas illustrating their *duality* as a *circuit device* (with resistance and temperature) on one hand and a *space device* (with patterns, polarization, beam area, directivity, gain, aperture and radar cross-section) on the other. Other antenna quantities are its physical size and bandwidth (involving impedance, Q and pattern).

EXAMPLE **2–10.1 Effective Aperture and Directivity of a Short Dipole Antenna**

A plane wave is incident on a short dipole as in Fig. 2–11. The wave is assumed to be linearly polarized with E in the y direction. The current on the dipole is assumed constant and in the same phase over its entire length, and the terminating resistance R_T is assumed equal to the dipole radiation resistance R_r. The antenna loss resistance R_L is assumed equal to zero. What is (*a*) the dipole's maximum effective aperture and (*b*) its directivity?

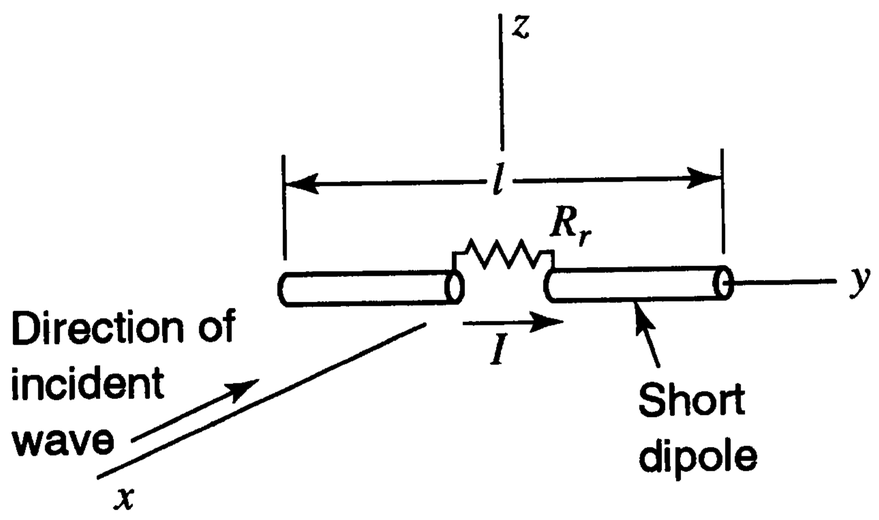


Figure 2–11
Short dipole with uniform current induced by incident wave.

■ **Solution**

(*a*) The maximum effective aperture of an antenna is

$$A_{em} = \frac{V^2}{4SR_r} \tag{7}$$

where the effective value of the induced voltage V is here given by the product of the effective electric field intensity at the dipole and its length, that is,

$$V = El \tag{8}$$

The radiation resistance R_r of a short dipole of length l with uniform current will be shown later to be

$$R_r = \frac{80\pi^2 l^2}{\lambda^2}\left(\frac{I_{\text{av}}}{I_0}\right)^2 = 790\left(\frac{I_{\text{av}}}{I_0}\right)^2\left(\frac{l}{\lambda}\right)^2 \quad (\Omega) \tag{9}$$

where

λ = wavelength
I_{av} = average current
I_0 = terminal current

The power density, or Poynting vector, of the incident wave at the dipole is related to the field intensity by

$$S = \frac{E^2}{Z} \tag{10}$$

where Z = intrinsic impedance of the medium.

In the present case, the medium is free space so that $Z = 120\pi\ \Omega$. Now substituting (8), (9), and (10) into (7), we obtain for the maximum effective aperture of a short dipole (for $I_{av} = I_0$)

$$A_{em} = \frac{120\pi E^2 l^2 \lambda^2}{320\pi^2 E^2 l^2} = \frac{3}{8\pi}\lambda^2 = 0.119\lambda^2 \qquad \textit{Ans. (a)}$$

(*b*)

$$D = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi \times 0.119\lambda^2}{\lambda^2} = 1.5 \qquad \textit{Ans. (b)}$$

A typical short dipole might be $\lambda/10$ long and $\lambda/100$ in diameter for a physical cross-sectional aperture of $0.001\lambda^2$ as compared to the $0.119\lambda^2$ effective aperture of Example 2–10.1. Thus, a single dipole or linear antenna may have a physical aperture that is smaller than its effective aperture. However, a *broadside array* of many dipoles or linear antennas has an *overall* physical aperture that, like horns and dishes, is larger than its effective aperture. On the other hand, an end-fire array of dipoles, as in a Yagi-Uda antenna, has an *end-on* physical cross section that is smaller than the antenna's effective aperture. Thus, depending on the antenna, physical apertures may be larger than effective apertures, or vice versa.

EXAMPLE **2–10.2 Effective Aperture and Directivity of Linear $\lambda/2$ Dipole**

A plane wave incident on the antenna is traveling in the negative x direction as in Fig. 2–12*a*. The wave is linearly polarized with E in the y direction. The equivalent circuit is shown in Fig. 2–12*b*. The antenna has been replaced by an equivalent or Thévenin generator. The infinitesimal voltage dV of this generator due to the voltage induced by the incident wave in an infinitesimal element of length dy of the antenna is

$$dV = E\,dy\cos\frac{2\pi y}{\lambda} \tag{11}$$

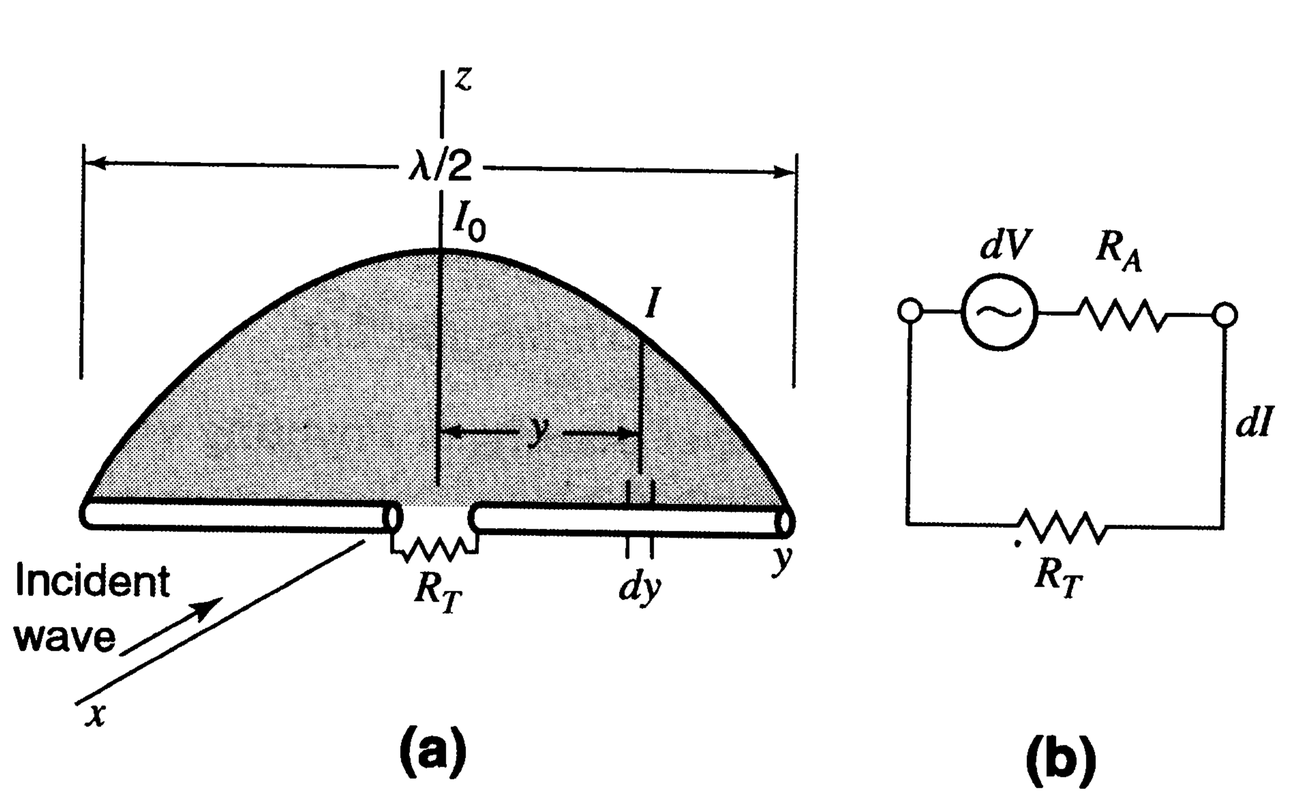


Figure 2–12
Linear $\lambda/2$ antenna in field of electromagnetic wave (*a*) and equivalent circuit (*b*).

It is assumed that the infinitesimal induced voltage is proportional to the current at the infinitesimal element as given by the current distribution (11). Find (*a*) the effective aperture and (*b*) the directivity of the $\lambda/2$ dipole.

■ Solution

(*a*) The total induced voltage V is given by integrating (11) over the length of the antenna. This may be written as

$$V = 2\int_0^{\lambda/4} E\cos\frac{2\pi y}{\lambda}\,dy \tag{12}$$

Performing the integration in (12) we have

$$V = \frac{E\lambda}{\pi} \tag{13}$$

The value of the radiation resistance R_r of the linear $\lambda/2$ antenna will be taken as 73 Ω. The terminating resistance R_T is assumed equal to R_r. Thus, we obtain, for the maximum effective aperture of a linear $\lambda/2$ antenna,

$$A_{em} = \frac{120\pi E^2\lambda^2}{4\pi^2 E^2 \times 73} = \frac{30}{73\pi}\lambda^2 = 0.13\lambda^2 \qquad \textit{Ans. (a)}$$

$$(b)\ D = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi \times 0.13\lambda^2}{\lambda^2} = 1.63 \qquad \textit{Ans. (b)}$$

The maximum effective aperture of the linear $\lambda/2$ antenna is about 10 percent greater than that of the short dipole.

The maximum effective aperture of the $\lambda/2$ antenna is approximately the same as an area 1/2 by $1/4\lambda$ on a side, as illustrated in Fig. 2–13*a*. This area is $0.125\lambda^2$. An elliptically shaped aperture of $0.13\lambda^2$ is shown in Fig. 2–13*b*. The physical significance of these apertures is that power from the incident plane wave is absorbed over an area of this size and is delivered to the terminal resistance or load.

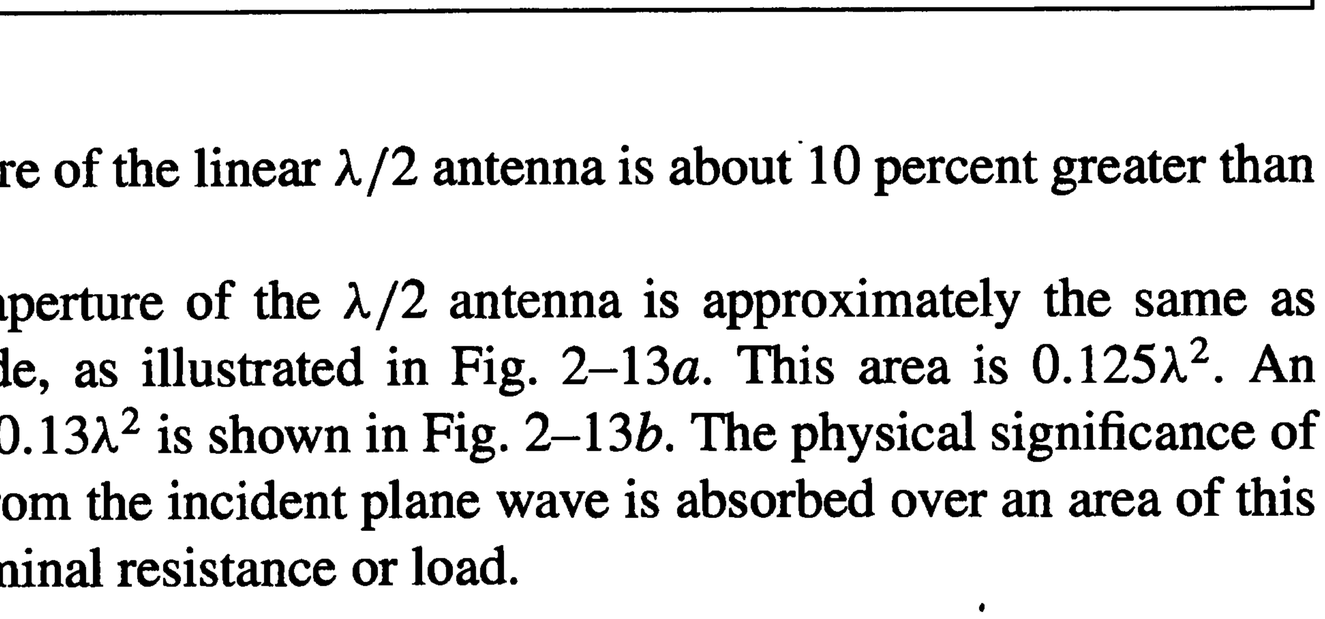


Figure 2–13
(*a*) Maximum effective aperture of linear $\lambda/2$ antenna is approximately represented by rectangle 1/2 by $1/4\lambda$ on a side. (*b*) Maximum effective aperture of linear $\lambda/2$ antenna represented by elliptical area of $0.13\lambda^2$.

Although the radiation resistance, effective aperture, and directivity are the same for both receiving and transmitting, the current distribution is, in general, not the same. Thus, a plane wave incident on a receiving antenna excites a different current distribution than a localized voltage applied to a pair of terminals for transmitting.

2–11 THE RADIO COMMUNICATION LINK（无线电通信线路）

The usefulness of the aperture concept is well illustrated by using it to derive the important Friis transmission formula published in 1946 by Harald T. Friis (1) of the Bell Telephone Laboratory.

Referring to Fig. 2–14, the formula gives the power received over a radio communication link. Assuming lossless, matched antennas, let the transmitter feed a power P_t to a transmitting antenna of effective aperture A_{et}. At a distance r a receiving antenna of effective aperture A_{er} intercepts some of the power radiated by the transmitting antenna and delivers it to the receiver R. Assuming for the moment that the transmitting antenna is isotropic, the power per unit area available at the receiving antenna is

$$S_r = \frac{P_t}{4\pi r^2} \qquad \text{(W)} \tag{1}$$

If the antenna has gain G_t, the power per unit area available at the receiving antenna will be increased in proportion as given by

$$S_r = \frac{P_t G_t}{4\pi r^2} \qquad \text{(W)} \tag{2}$$

Now the power collected by the lossless, matched receiving antenna of effective aperture A_{er} is

$$P_r = S_r A_{er} = \frac{P_t G_t A_{er}}{4\pi r^2} \qquad \text{(W)} \tag{3}$$

The gain of the transmitting antenna can be expressed as

$$G_t = \frac{4\pi A_{et}}{\lambda^2} \tag{4}$$

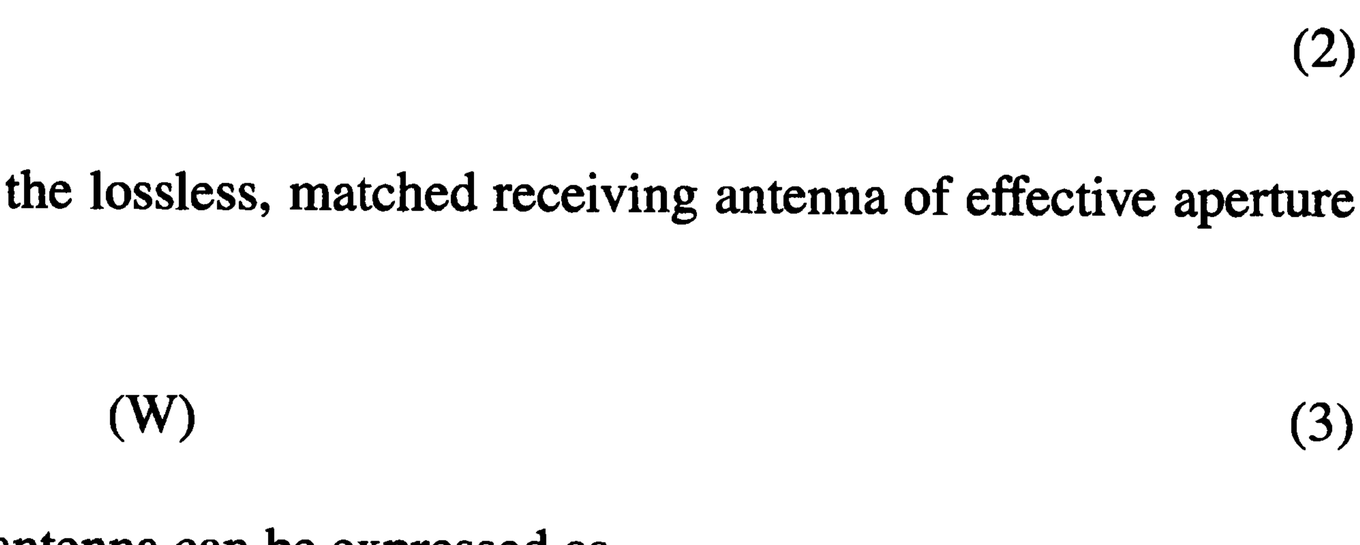


Figure 2–14
Communication circuit with waves from transmitting antenna arriving at the receiving antenna by a direct path of length r.

Substituting this in (3) yields the *Friis transmission formula*

$$\boxed{\frac{P_r}{P_t} = \frac{A_{er}A_{et}}{r^2\lambda^2} \quad \text{(dimensionless)} \quad \textbf{\textit{Friis transmission formula}}} \quad \text{(弗里斯传输公式)} \tag{5}$$

where

P_r = received power, W
P_t = transmitted power, W
A_{et} = effective aperture of transmitting antenna, m^2
A_{er} = effective aperture of receiving antenna, m^2
r = distance between antennas, m
λ = wavelength, m

EXAMPLE **2–11.1 Radio Communication Link**

A radio link has a 15-W transmitter connected to an antenna of 2.5 m^2 effective aperture at 5 GHz. The receiving antenna has an effective aperture of 0.5 m^2 and is located at a 15-km line-of-sight distance from the transmitting antenna. Assuming lossless, matched antennas, find the power delivered to the receiver.

■ **Soiution**

From (5)

$$P = P_t\frac{A_{et}A_{er}}{r^2\lambda^2} = 15\frac{2.5 \times 0.5}{15^2 \times 10^6 \times 0.06^2} = 23\ \mu\text{W} \qquad \textit{Ans.}$$

2–12 FIELDS FROM OSCILLATING DIPOLE（振荡偶极子产生的场）

Although a charge moving with uniform velocity along a straight conductor does not radiate, a charge moving back and forth in simple harmonic motion along the conductor is subject to acceleration (and deceleration) and radiates.

To illustrate radiation from a dipole antenna, let us consider that the dipole of Fig. 2–15 has two equal charges of opposite sign oscillating up and down in harmonic motion with instantaneous separation l (maximum separation l_0) while focusing attention on the electric field. For clarity only a single electric field line is shown.

At time $t = 0$ the charges are at maximum separation and undergo maximum acceleration $\dot{v}$ as they reverse direction (Fig. 2–15*a*). At this instant the current I is zero. At an $\frac{1}{8}$-period later, the charges are moving toward each other (Fig. 2–15*b*) and at a $\frac{1}{4}$-period they pass at the midpoint (Fig. 2–15*c*). As this happens, the field lines detach and new ones of opposite sign are formed. At this time the equivalent current I is a maximum and the charge acceleration is zero. As time progresses to a $\frac{1}{2}$-period, the fields continue to move out as in Fig. 2–15*d* and *e*.

An oscillating dipole with more field lines is shown in Fig. 2–16 at four instants of time.

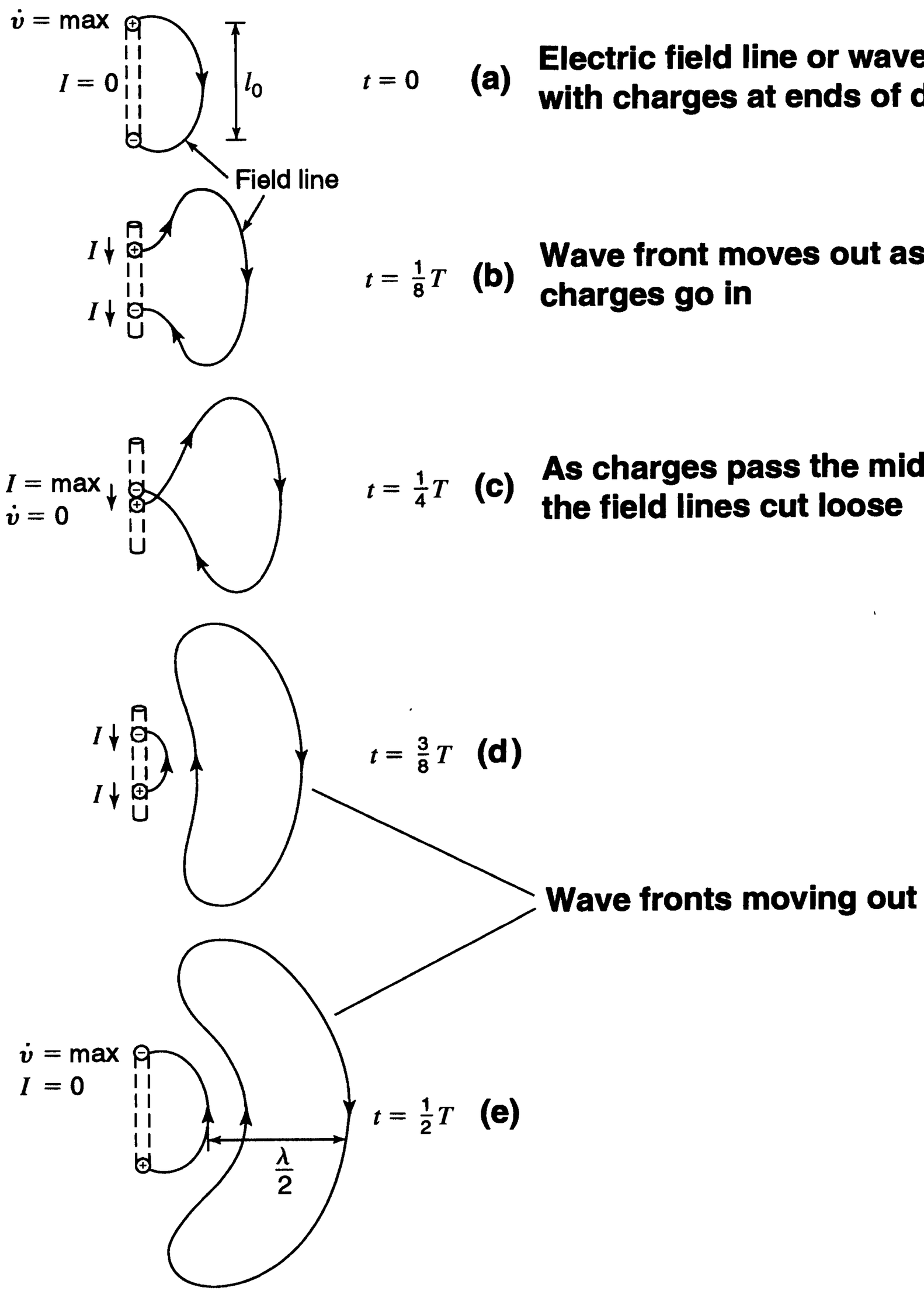


Figure 2–15
Oscillating electric dipole consisting of two electric charges in simple harmonic motion, showing propagation of an electric field line and its detachment (radiation) from the dipole. Arrows next to the dipole indicate current (I) direction.

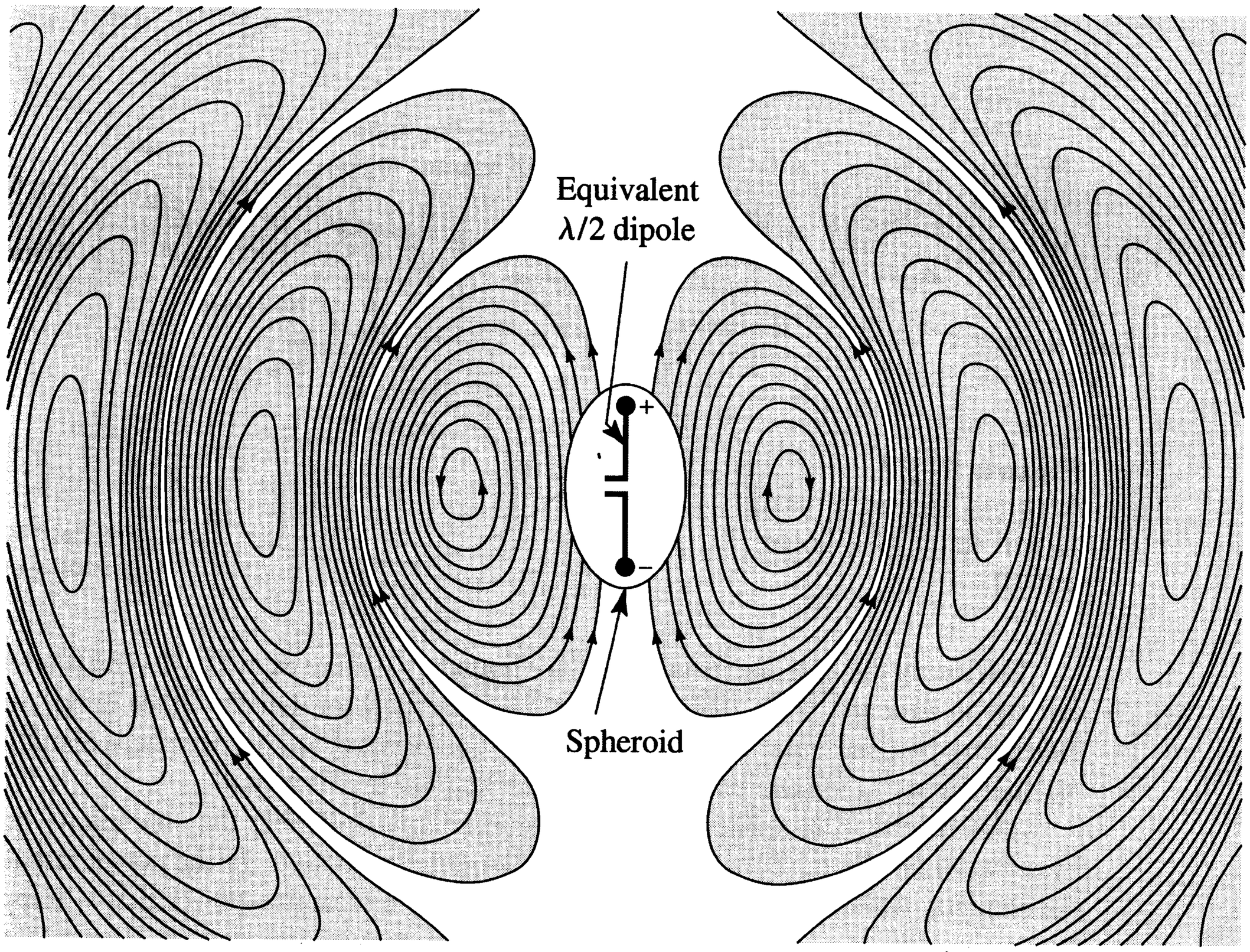


Figure 2–16
Electric field lines of the radiation moving out from λ/2 dipole antenna. (*Produced by Edward M. Kennaugh, courtesy of John D. Cowan, Jr.*)

2–13 ANTENNA FIELD ZONES（天线的场区）

The fields around an antenna may be divided into two principal regions, one near the antenna called the *near field* or *Fresnel zone* and one at a large distance called the *far field* or *Fraunhofer zone*. Referring to Fig. 2–17, the boundary between the two may be arbitrarily taken to be at a radius

$$R = \frac{2L^2}{\lambda} \qquad \text{(m)} \tag{1}$$

where

L = maximum dimension of the antenna, m
λ = wavelength, m

In the far or Fraunhofer region, the measurable field components are transverse to the radial direction from the antenna and all power flow is directed radially outward. In the far field the shape of the field pattern is independent of the distance. In the near or Fresnel region, the longitudinal component of the electric field may be significant and power flow is not entirely radial. In the near field, the shape of the field pattern depends, in general, on the distance.

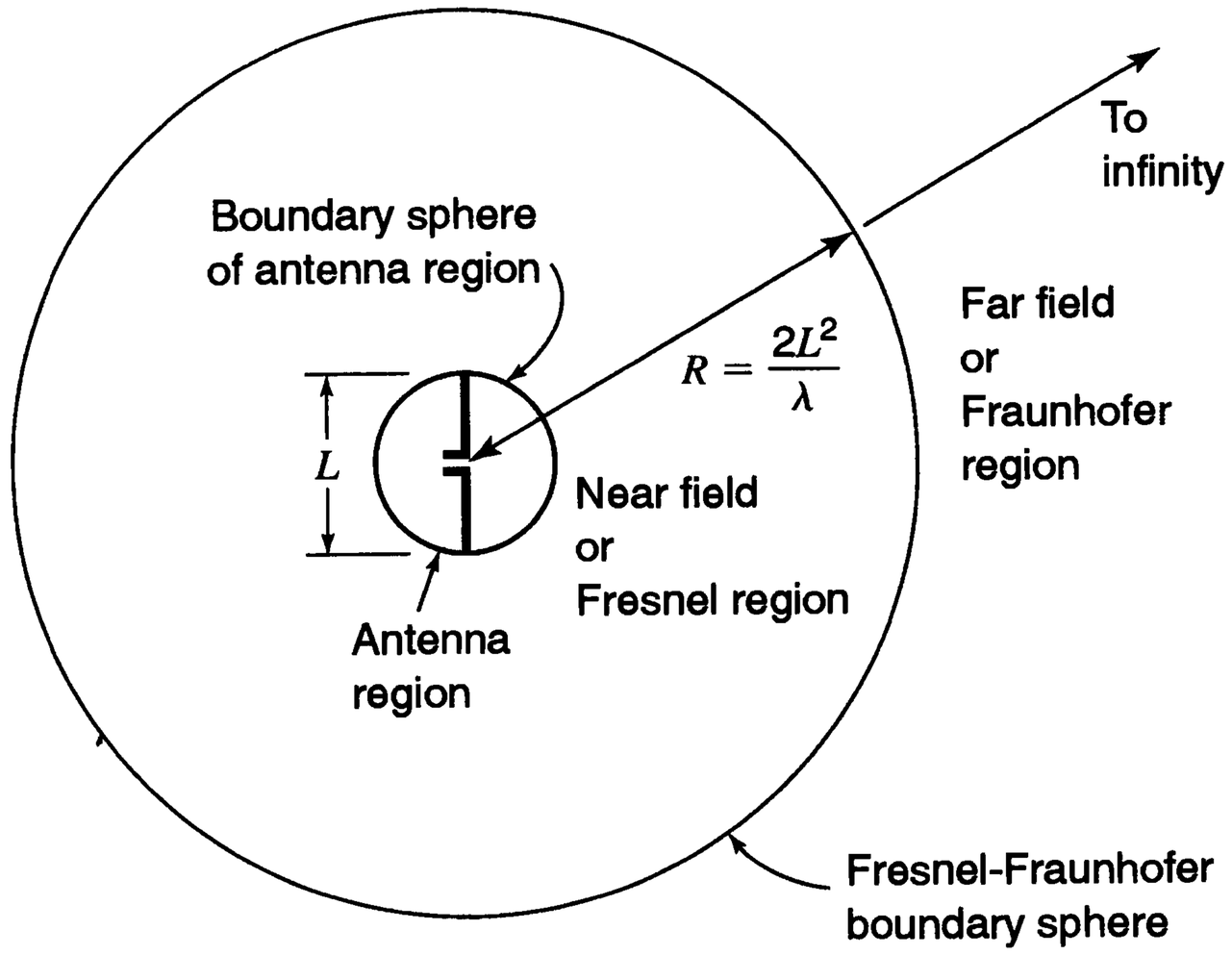


Figure 2–17
Antenna region, Fresnel region and Fraunhofer region.

Enclosing the antenna in an imaginary boundary sphere as in Fig. 2–18*a* it is as though the region near the poles of the sphere acts as a reflector. On the other hand, the waves expanding perpendicular to the dipole in the equatorial region of the sphere result in power leakage through the sphere as if partially transparent in this region.

This results in reciprocating (oscillating) energy flow near the antenna accompanied by outward flow in the equatorial region. The outflow accounts for the power radiated from the antenna, while the reciprocating energy represents reactive power that is trapped near the antenna like in a resonator. This oversimplified discussion accounts in a qualitative way for the field pattern of the $\lambda/2$ dipole antenna as shown in Fig. 2–18*b*. The energy picture is discussed in more detail in Sec. 6–2 and displayed in Fig. 6–6.

For a $\lambda/2$ dipole antenna, the energy is stored at one instant of time in the electric field, mainly near the ends of the antenna or maximum charge regions, while a $\frac{1}{2}$-period later the energy is stored in the magnetic field mainly near the center of the antenna or maximum current region.

Note that although the term *power flow* is sometimes used, it is actually *energy* which flows, power being the time rate of energy flow. A similar loose usage occurs when we say we pay a power bill, when, in fact, we are actually paying for electric energy.

2–14 SHAPE-IMPEDANCE CONSIDERATIONS（形状－阻抗的讨论）

It is possible in many cases to deduce the qualitative behavior of an antenna from its shape. This may be illustrated with the aid of Fig. 2–19. Starting with the opened-out two-conductor transmission line of Fig. 2–19*a*, we find that, if extended far enough, a nearly constant impedance will be provided at the input (left) end for $d \ll \lambda$ and $D \geq \lambda$.

In Fig. 2–19*b* the curved conductors are straightened into regular cones and in Fig. 2–19*c* the cones are aligned collinearly, forming a biconical antenna. In Fig. 2–19*d* the cones degenerate into straight wires. In going from Fig. 2–19*a* to *d*, the bandwidth of relatively constant impedance tends to decrease. Another difference is that the antennas of Fig. 2–19*a*

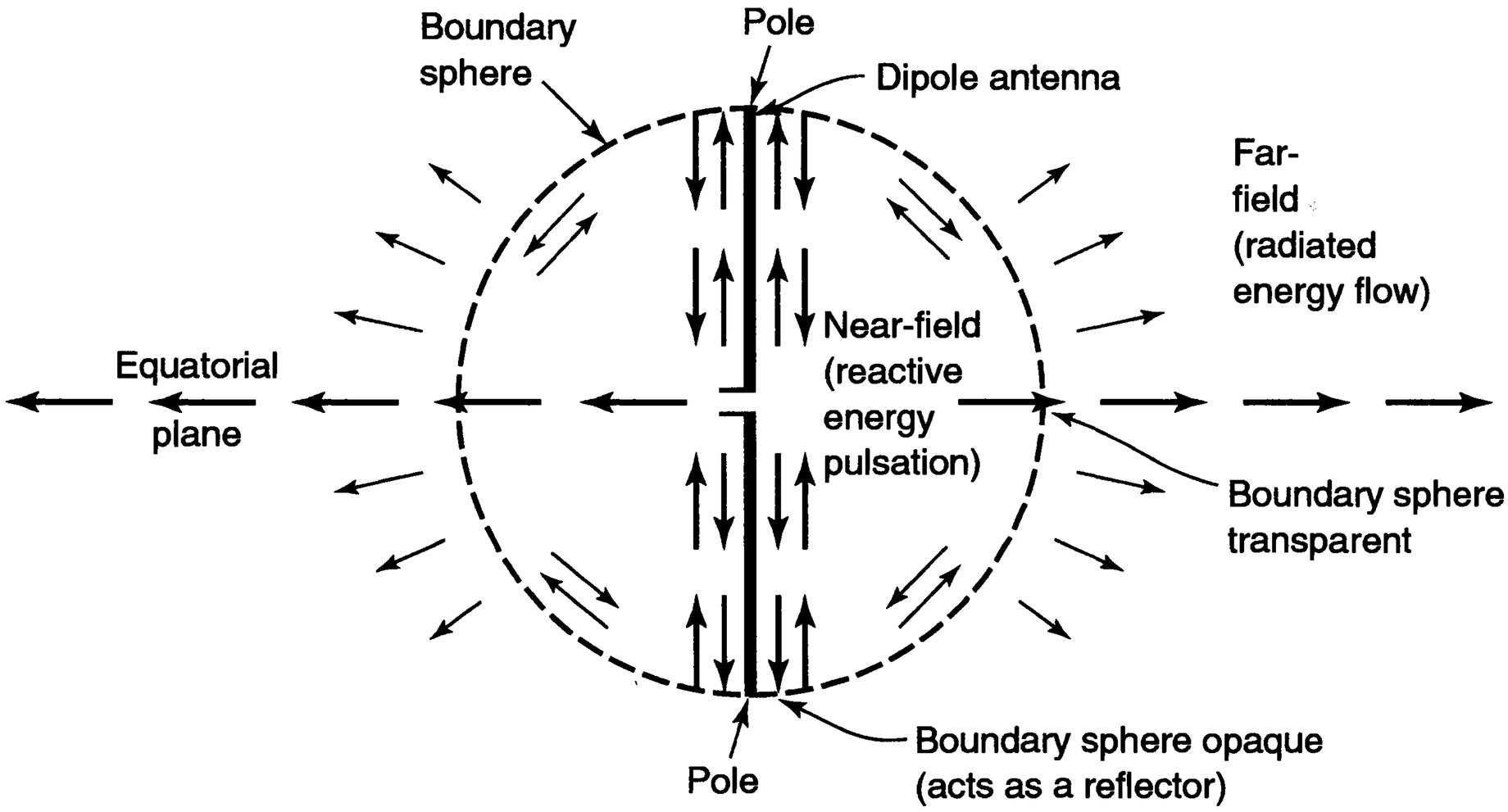


Figure 2–18
Energy flow near a dipole antenna (*a*) and radiation field pattern (*b*). The radius vector *r* is proportional to the field radiated in that direction.

and *b* are unidirectional with beams to the right, while the antennas of Fig. 2–19*c* and *d* are omnidirectional in the horizontal plane (perpendicular to the wire or cone axes).

A different modification is shown in Fig. 2–19*e*. Here the two conductors are curved more sharply and in opposite directions, resulting in a spiral antenna with maximum radiation broadside (perpendicular to the page) and with polarization which rotates clockwise. This antenna, like the one in Fig. 2–19*a*, exhibits very broadband characteristics (see Chap. 11).

The dipole antennas of Fig. 2–19 are balanced, i.e., they are fed by two-conductor (balanced) transmission lines. Figure 2–20 illustrates a similar evolution of monopole antennas, i.e., antennas fed from coaxial (unbalanced) transmission lines.

By gradually tapering the inner and outer conductors or a coaxial transmission line, a very wide band antenna with an appearance reminiscent of a volcanic crater and puff of smoke is obtained, as suggested in the cutaway view of Fig. 2–20*a*.

In Fig. 2–20*b* the volcano form is modified into a double dish and in Fig. 2–20*c* into two wide-angle cones. All of these antennas are omnidirectional in a plane perpendicular

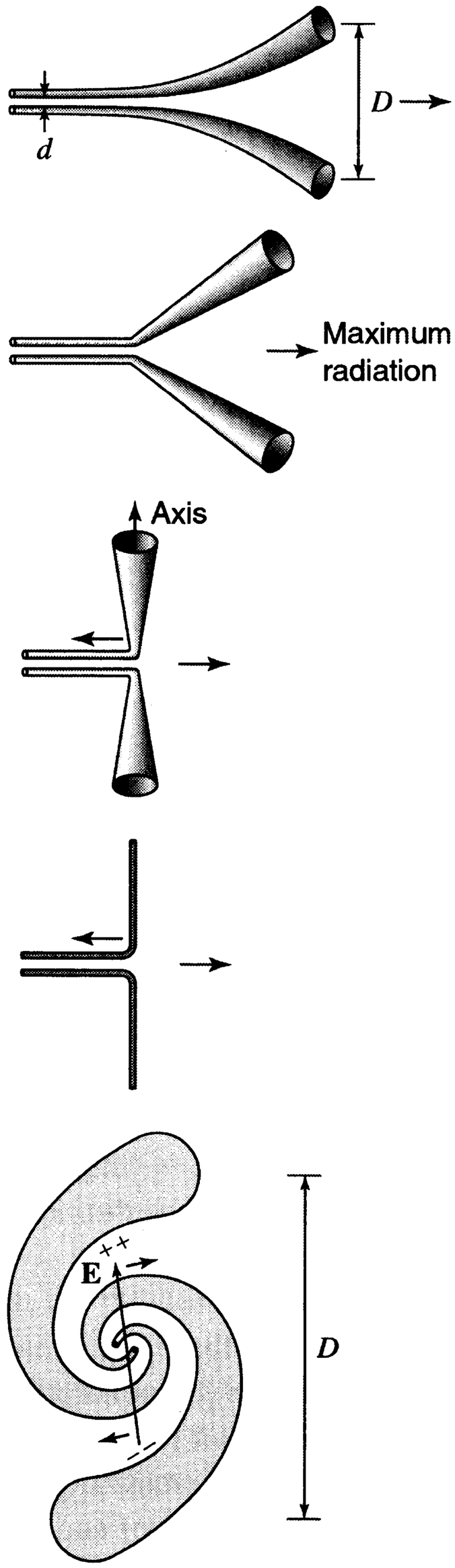


Figure 2–19
Evolution of a thin cylindrical antenna (*d*) from an opened-out twin line (*a*). Curving the conductors as in (*e*) results in the spiral antenna.

to their axes and all have a wide bandwidth. For example, an actual biconical antenna, as in Fig. 2–20*c*, with a full cone angle of 120° has an omnidirectional pattern and nearly constant 50-Ω input impedance (power reflection less than 1 percent or VSWR < 1.2) over a 6 to 1 bandwidth with cone diameter $D = \lambda$ at the lowest frequency.

Increasing the lower cone angle to 180° or into a flat ground plane while reducing the upper cone angle results in the antenna of Fig. 2–20*d*. Collapsing the upper cone into a thin stub, we arrive at the extreme modification of Fig. 2–20*e*. If the antenna of Fig. 2–20*a* is regarded as the most basic form, the stub type of Fig. 2–20*e* is the most degenerate form, with a relatively narrow bandwidth.

Antennas with large and abrupt discontinuities have large reflections and act as reflectionless transducers only over narrow frequency bands where the reflections cancel.

Antennas with discontinuities that are small and gradual have small reflections and are, in general, relatively reflectionless transducers over wide frequency bands.

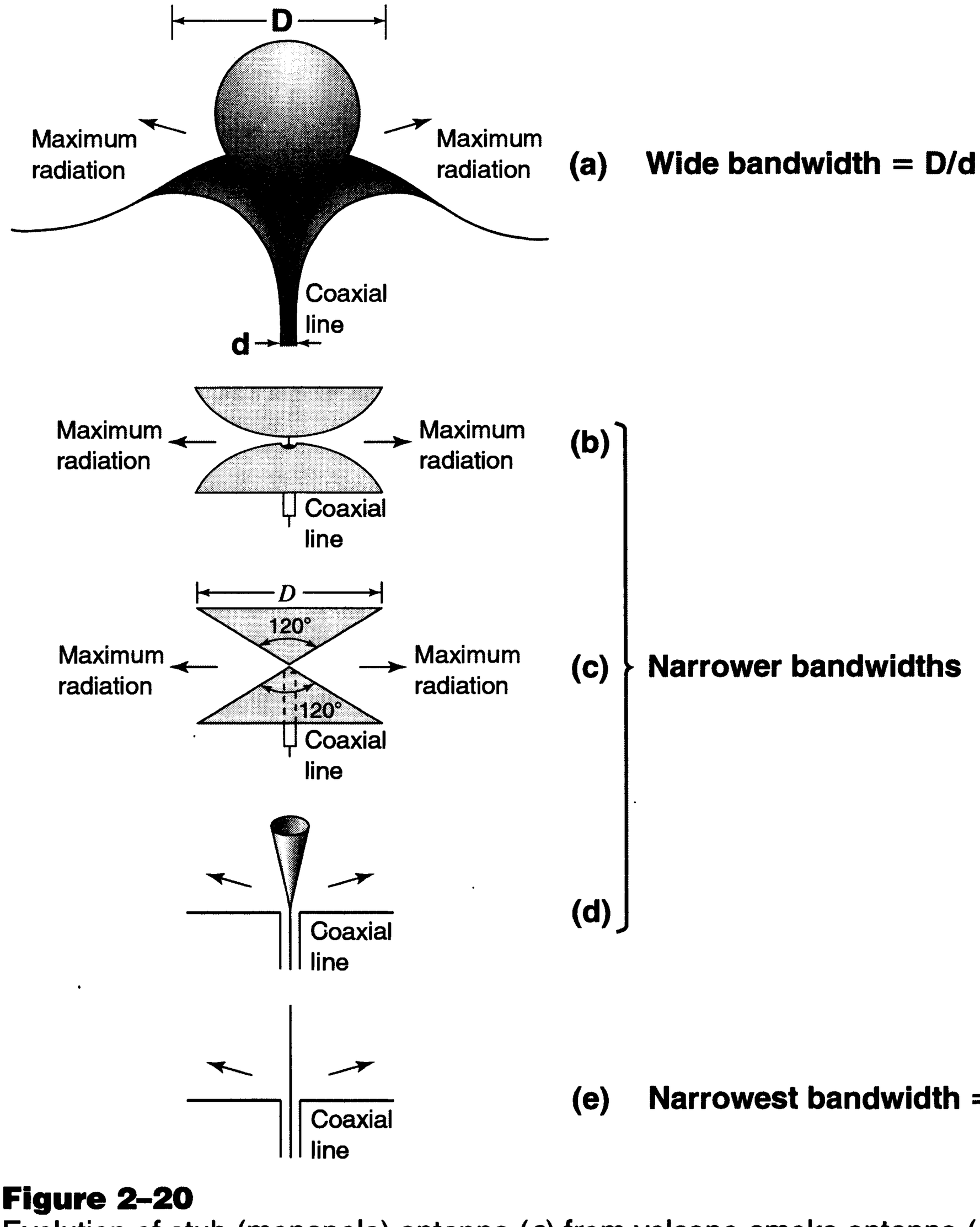


Figure 2–20
Evolution of stub (monopole) antenna (*e*) from volcano-smoke antenna (*a*). Fed from coaxial (unbalanced) transmission lines.

As we depart further from the basic type, the discontinuity in the transmission line becomes more abrupt at what eventually becomes the junction of the ground plane and the coaxial line. This discontinuity results in some energy being reflected back into the line. The reflection at the end of the antenna also increases for thinner antennas. At some frequency the two reflections may compensate, but the bandwidth of compensation is narrow.

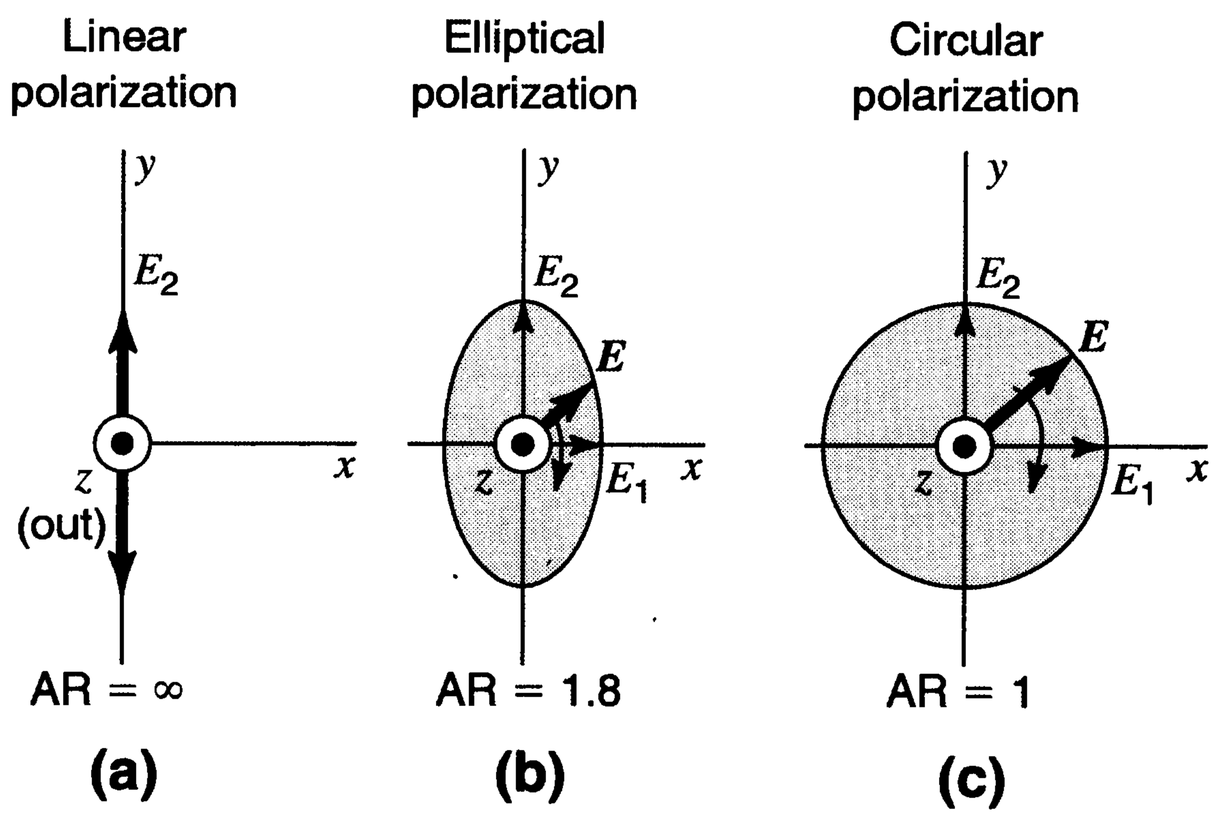


Figure 2–21
(*a*) Linear, (*b*) elliptical, and (*c*) circular polarization for left-circularly polarized wave approaching.

2–15 LINEAR, ELLIPTICAL AND CIRCULAR POLARIZATION[1]
（线极化、椭圆极化和圆极化）

Consider a plane wave traveling out of the page (positive z direction), as in Fig. 2–21*a*, with the electric field at all times in the y direction. This wave is said to be *linearly polarized* (in the y direction). As a function of time and position, the electric field is given by

$$E_y = E_2 \sin(\omega t - \beta z) \tag{1}$$

In general, the electric field of a wave traveling in the z direction may have both a y component and an x component, as suggested in Fig. 2–21*b*. In this more general situation, with a phase difference δ between the components, the wave is said to be *elliptically polarized*. At a fixed value of z the electric vector **E** rotates as a function of time, the tip of the vector describing an ellipse called the ***polarization ellipse***.（椭圆极化）The ratio of the major to minor axes of the polarization ellipse is called the ***Axial Ratio***（轴比） wave Fig. 2–21*b*, AR $= E_2/E_1$. Two extreme cases of elliptical polarization correspond (AR). Thus, for the to *circular polarization,* as in Fig. 2–21*c*, and *linear polarization,* as in Fig. 2–21*a*. *For circular polarization* $E_1 = E_2$ and AR $= 1$, while *for linear polarization* $E_1 = 0$ and AR $= \infty$.

In the most general case of elliptical polarization, the polarization ellipse may have any orientation, as suggested in Fig. 2–22. The elliptically polarized wave may be expressed in terms of two linearly polarized components, one in the x direction and one in the y direction. Thus, if the wave is traveling in the positive z direction (out of the page), the electric field components in the x and y directions are

$$E_x = E_1 \sin(\omega t - \beta z) \tag{2}$$
$$E_y = E_2 \sin(\omega t - \beta z + \delta) \tag{3}$$

---

[1] A much more detailed and complete discussion of wave polarization is given by J. D. Kraus, *Radio Astronomy,* 2d ed., Cygnus-Quasar, Powell, Ohio, 1986.

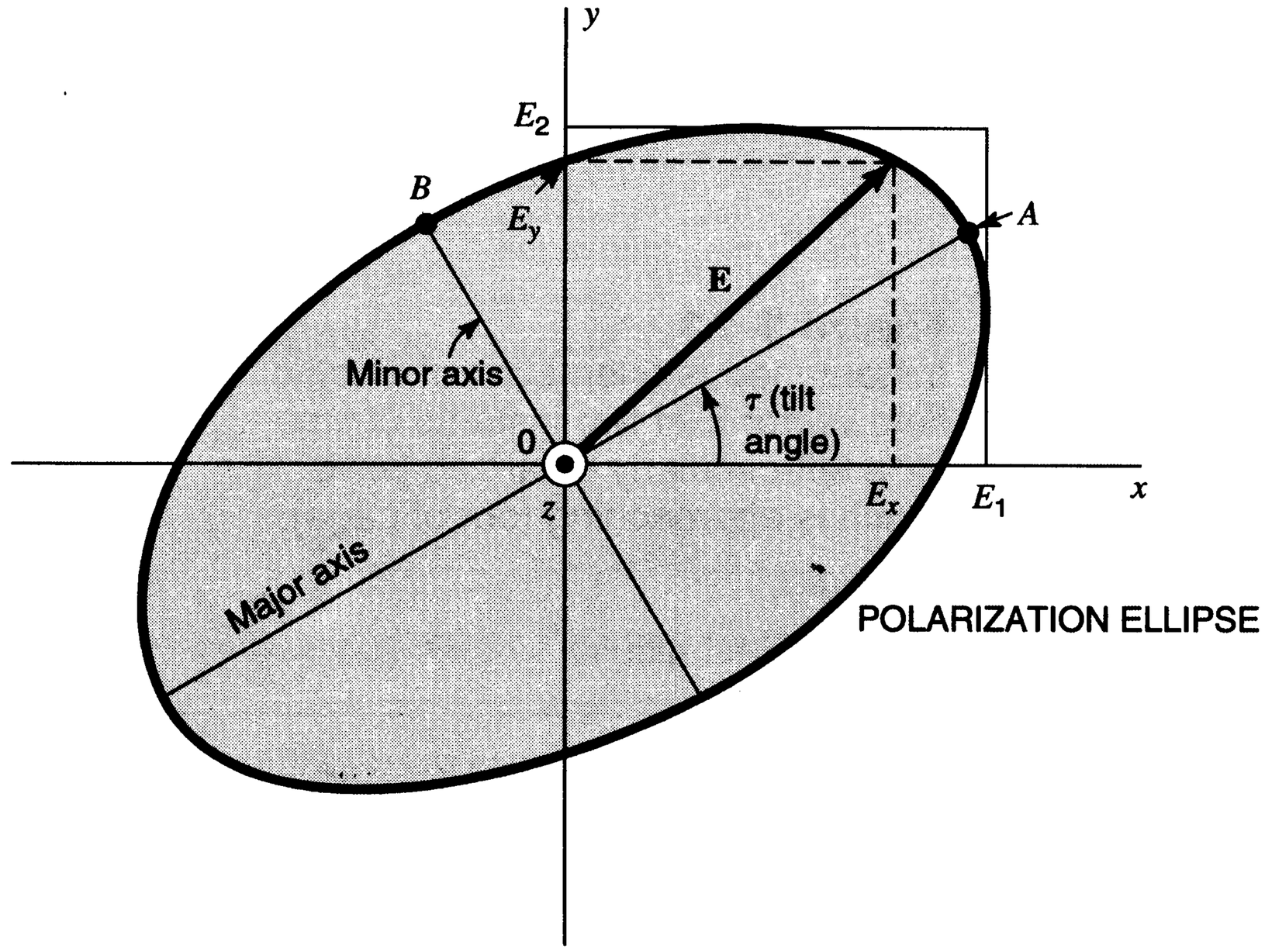


Figure 2–22
Polarization ellipse at tilt angle τ showing instantaneous components E_x and E_y and amplitudes (or peak values) E_1 and E_2.

where

E_1 = amplitude of wave linearly polarized in x direction
E_2 = amplitude of wave linearly polarized in y direction
δ = time-phase angle by which E_y leads E_x

Combining (2) and (3) gives the instantaneous total vector field **E**:

$$\mathbf{E} = \hat{\mathbf{x}}E_1 \sin(\omega t - \beta z) + \hat{\mathbf{y}}E_2 \sin(\omega t - \beta z + \delta) \tag{4}$$

At $z = 0$, $E_x = E_1 \sin \omega t$ and $E_y = E_2 \sin(\omega t + \delta)$. Expanding E_y yields

$$E_y = E_2(\sin \omega t \cos \delta + \cos \omega t \sin \delta) \tag{5}$$

From the relation for E_x we have $\sin \omega t = E_x/E_1$ and $\cos \omega t = \sqrt{1 - (E_x/E_1)^2}$. Introducing these in (5) eliminates ωt, and, on rearranging, we obtain

$$\frac{E_x^2}{E_1^2} - \frac{2E_xE_y \cos \delta}{E_1E_2} + \frac{E_y^2}{E_2^2} = \sin^2 \delta \tag{6}$$

or

$$aE_x^2 - bE_xE_y + cE_y^2 = 1 \tag{7}$$

where

$$a = \frac{1}{E_1^2 \sin^2 \delta} \qquad b = \frac{2 \cos \delta}{E_1E_2 \sin^2 \delta} \qquad c = \frac{1}{E_2^2 \sin^2 \delta}$$

Equation (7) describes a (polarization) ellipse, as in Fig. 2–22. The line segment *OA* is the semimajor axis, and the line segment *OB* is the semiminor axis. The tilt angle of the ellipse

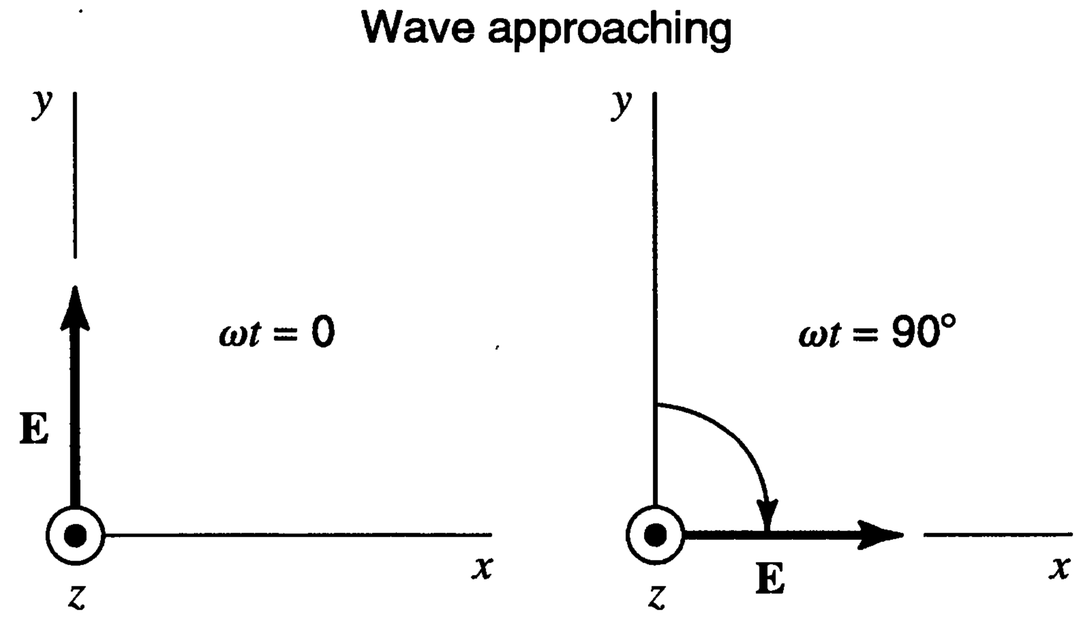


Figure 2–23
Instantaneous orientation of electric field vector **E** at two instants of time for a left-circularly polarized wave which is approaching (out of page).

is τ. The axial ratio is

$$\text{AR} = \frac{OA}{OB} \qquad (1 \leq \text{AR} \leq \infty) \qquad \textit{Axial Ratio} \tag{8}$$

If $E_1 = 0$, the wave is linearly polarized in the y direction. If $E_2 = 0$, the wave is linearly polarized in the x direction. If $\delta = 0$ and $E_1 = E_2$, the wave is also linearly polarized but in a plane at an angle of 45° with respect to the x axis ($\tau = 45°$).

If $E_1 = E_2$ and $\delta = \pm 90°$, the wave is circularly polarized. When $\delta = +90°$, the wave is *left circularly polarized,* and when $\delta = -90°$, the wave is *right circularly polarized*. For the case $\delta = +90°$ and for $z = 0$ and $t = 0$, we have from (2) and (3) that $\mathbf{E} = \hat{\mathbf{y}}E_2$, as in Fig. 2–23*a*. One-quarter cycle later ($\omega t = 90°$), $\mathbf{E} = \hat{\mathbf{x}}E_1$, as in Fig. 4–23*b*. Thus, at a fixed position ($z = 0$) the electric field vector rotates clockwise (viewing the wave approaching). According to the IEEE definition, this corresponds to left circular polarization. The opposite rotation direction ($\delta = -90°$) corresponds to right circular polarization.

If the wave is viewed receding (from negative z axis in Fig. 2–23), the electric vector appears to rotate in the opposite direction. Hence, clockwise rotation of **E** with the wave approaching is the same as counterclockwise rotation with the wave receding. Thus, unless the wave direction is specified, there is a possibility of ambiguity as to whether the wave is left- or right-handed. This can be avoided by defining the polarization with the aid of an axial-mode helical antenna. Thus, a right-handed helical antenna radiates (or receives) right circular (IEEE) polarization.[1] A right-handed helix, like a right-handed screw, is right-handed regardless of the position from which it is viewed. There is no possibility here of ambiguity.

The Institute of Electrical and Electronics Engineers (IEEE) definition is opposite to the classical optics definition which had been in use for centuries. The intent of the IEEE Standards Committee was to make the IEEE definition agree with the classical optics definition, but it got turned around so now we have two definitions. In this book we use the IEEE definition, which has the advantage of agreement with helical antennas as noted above.

---

[1] A left-handed helical antenna radiates (or receives) left circular (IEEE) polarization.

2-16 POYNTING VECTOR FOR ELLIPTICALLY AND CIRCULARLY POLARIZED WAVES（椭圆和圆极化波的坡印廷矢量）

In complex notation the Poynting vector is

$$\mathbf{S} = \frac{1}{2}\mathbf{E} \times \mathbf{H}^* \tag{1}$$

The average Poynting vector is the real part of (1), or

$$\mathbf{S}_{av} = \text{Re}\,\mathbf{S} = \frac{1}{2}\text{Re}\,\mathbf{E} \times \mathbf{H}^* \tag{2}$$

We can also write

$$\boxed{\mathbf{S}_{av} = \frac{1}{2}\hat{\mathbf{z}}\frac{E_1^2 + E_2^2}{Z_0} = \frac{1}{2}\hat{\mathbf{z}}\frac{E^2}{Z_0} \qquad \textit{Average Poynting vector}} \text{（平均坡印廷矢量）} \tag{3}$$

where $E = \sqrt{E_1^2 + E_2^2}$ is the amplitude of the total **E** field.

EXAMPLE **2–16.1 Elliptically Polarized Wave Power**

An elliptically polarized wave traveling in the positive z direction in air has x and y components:

$$E_x = 3\sin(\omega t - \beta x) \qquad (\text{V m}^{-1})$$
$$E_y = 6\sin(\omega t - \beta x + 75°) \qquad (\text{V m}^{-1})$$

Find the average power per unit area conveyed by the wave.

■ **Solution**

The average power per unit area is equal to the average Poynting vector, which from (3) has a magnitude

$$\mathbf{S}_{av} = \frac{1}{2}\frac{E^2}{Z} = \frac{1}{2}\frac{E_1^2 + E_2^2}{Z}$$

From the stated conditions, the amplitude $E_1 = 3\,\text{V m}^{-1}$ and the amplitude $E_2 = 6\,\text{V m}^{-1}$. Also for air $Z = 377\,\Omega$. Hence

$$\mathbf{S}_{av} = \frac{1}{2}\frac{3^2 + 6^2}{377} = \frac{1}{2}\frac{45}{377} \approx 60\,\text{mW m}^{-2} \qquad \textit{Ans.}$$

2-17 THE POLARIZATION ELLIPSE AND THE POINCARÉ SPHERE（极化椭圆和庞加莱球）

In the Poincaré sphere representation of wave polarization, the *polarization state* is described by a point on a sphere where the longitude and latitude of the point are related to parameters of the polarization ellipse (see Fig. 2–24) as follows:

$$\begin{aligned} \text{Longitude} &= 2\tau \\ \text{Latitude} &= 2\varepsilon \end{aligned} \tag{1}$$

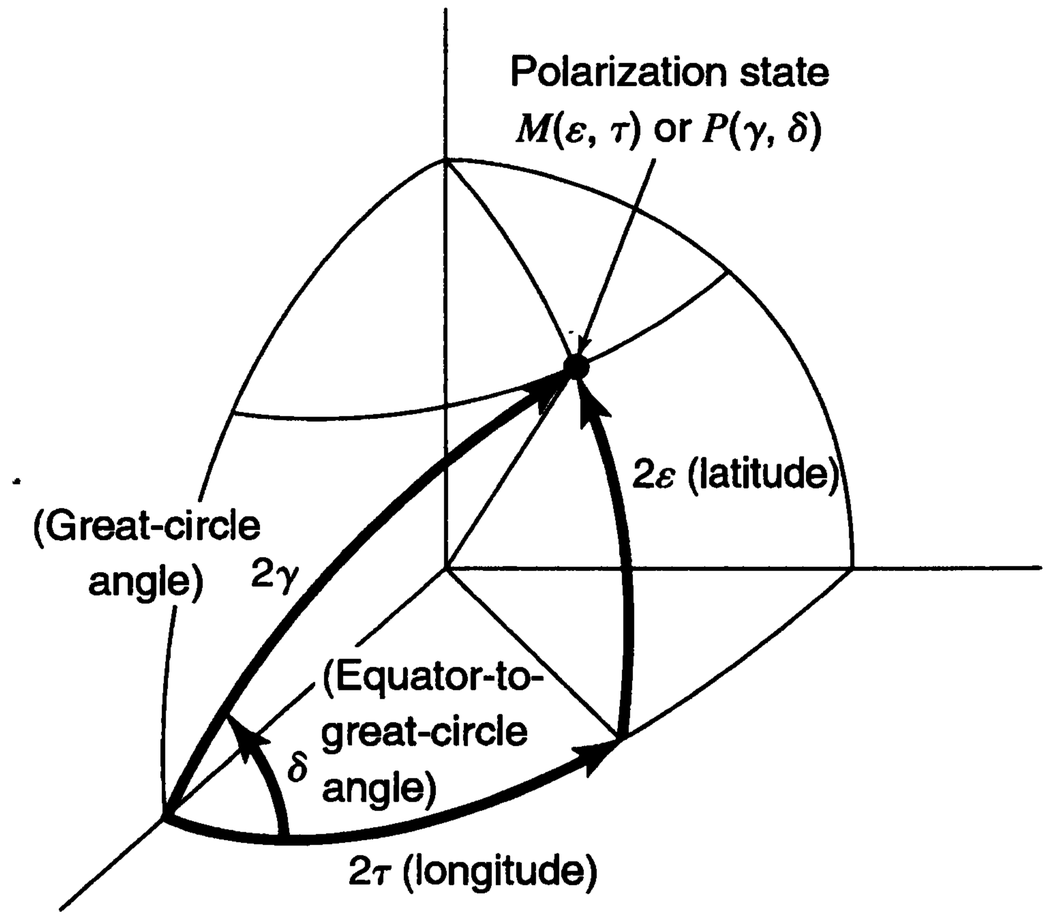


Figure 2–24
Poincaré sphere showing relation of angles ε, τ, δ, γ.

where τ = tilt angle, $0° \leq \tau \leq 180°$,[1] and $\varepsilon = \tan^{-1}(1/\mp \text{AR})$, $-45° \leq \varepsilon \leq +45°$. The Axial Ratio (AR) and angle ε are negative for right-handed and positive for left-handed (IEEE) polarization (Poincaré-1, Deschamps-1).

The polarization state described by a point on a sphere here can also be expressed in terms of the angle subtended by the great circle drawn from a reference point on the equator and the angle between the great circle and the equator (see Fig. 2–24) as follows:

$$\begin{aligned} \text{Great-circle angle} &= 2\gamma \\ \text{Equator-to-great-circle angle} &= \delta \end{aligned} \tag{2}$$

where $\gamma = \tan^{-1}(E_2/E_1)$, $0° \leq \gamma \leq 90°$, and δ = phase difference between E_y and E_x, $-180° \leq \delta \leq +180°$.

The geometric relation of τ, ε, and γ to the polarization ellipse is illustrated in Fig. 2–25. The trigonometric interrelations of τ, ε, γ, and δ are as follows:[2]

$$\boxed{\begin{aligned} \cos 2\gamma &= \cos 2\varepsilon \cos 2\tau \\ \tan\delta &= \frac{\tan 2\varepsilon}{\sin 2\tau.} \\ \tan 2\tau &= \tan 2\gamma \cos\delta \\ \sin 2\varepsilon &= \sin 2\gamma \sin\delta \end{aligned}} \quad \textit{Polarization parameters} \quad (\text{极化参量}) \tag{3}$$

Knowing ε and τ, one can determine γ and δ or vice versa. It is convenient to describe the ***polarization state*** by either of the two sets of angles (ε, τ) or (γ, δ) which describe a point on the Poincaré sphere (Fig. 2–24). Let the polarization state as a function of ε and τ be

---

[1]Note that when the Greek letter τ is used in this section to denote the *tilt angle,* it is associated with the incident, reflected, or transmitted wave (that is, τ_i, τ_r, or τ_t), but when it is used to denote the *transmission coefficient,* it is associated with either a parallel or perpendicular case.

[2]These relations involve spherical trigonometry (see Born-1).

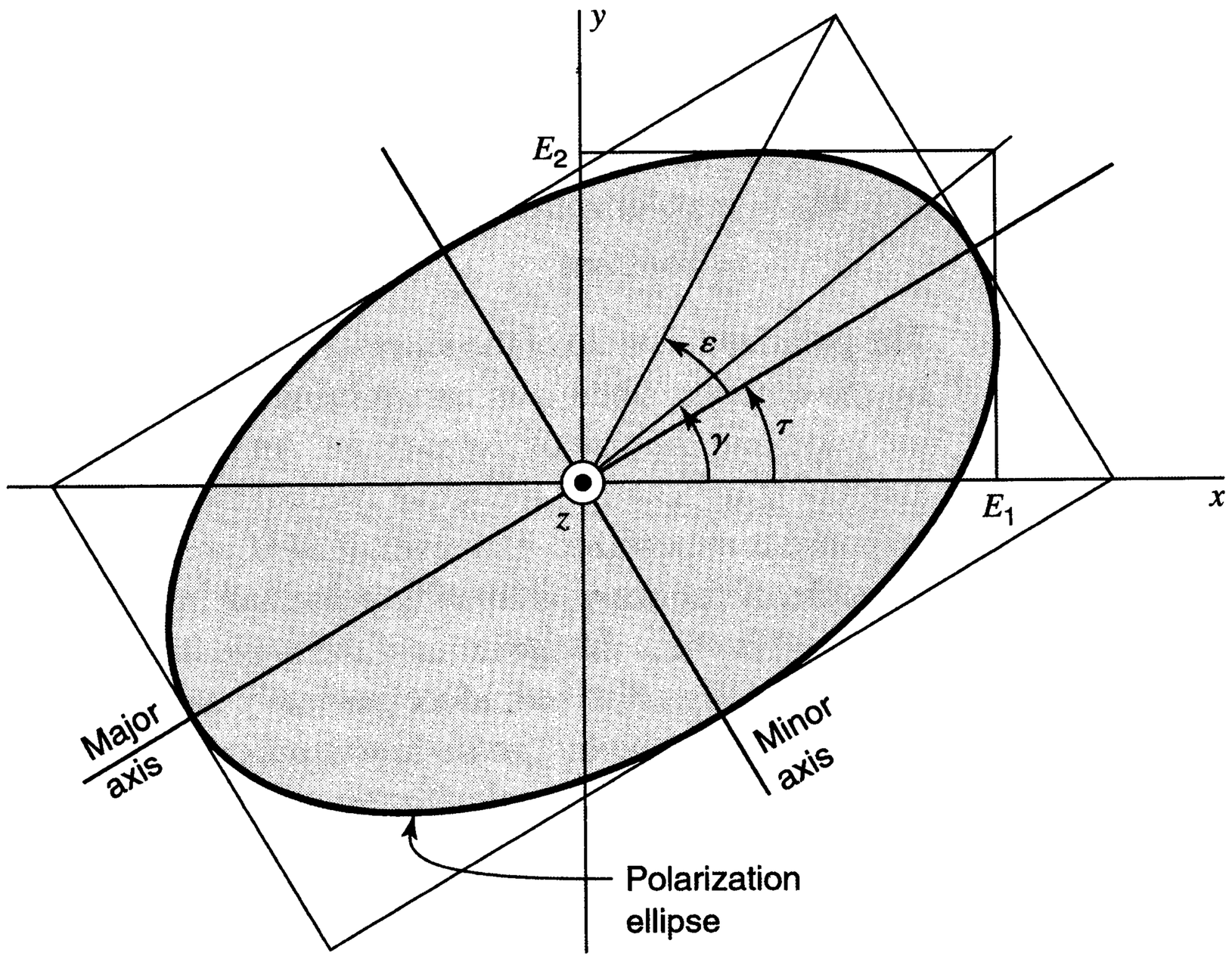


Figure 2–25
Polarization ellipse showing relation of angles ε, γ, and τ.

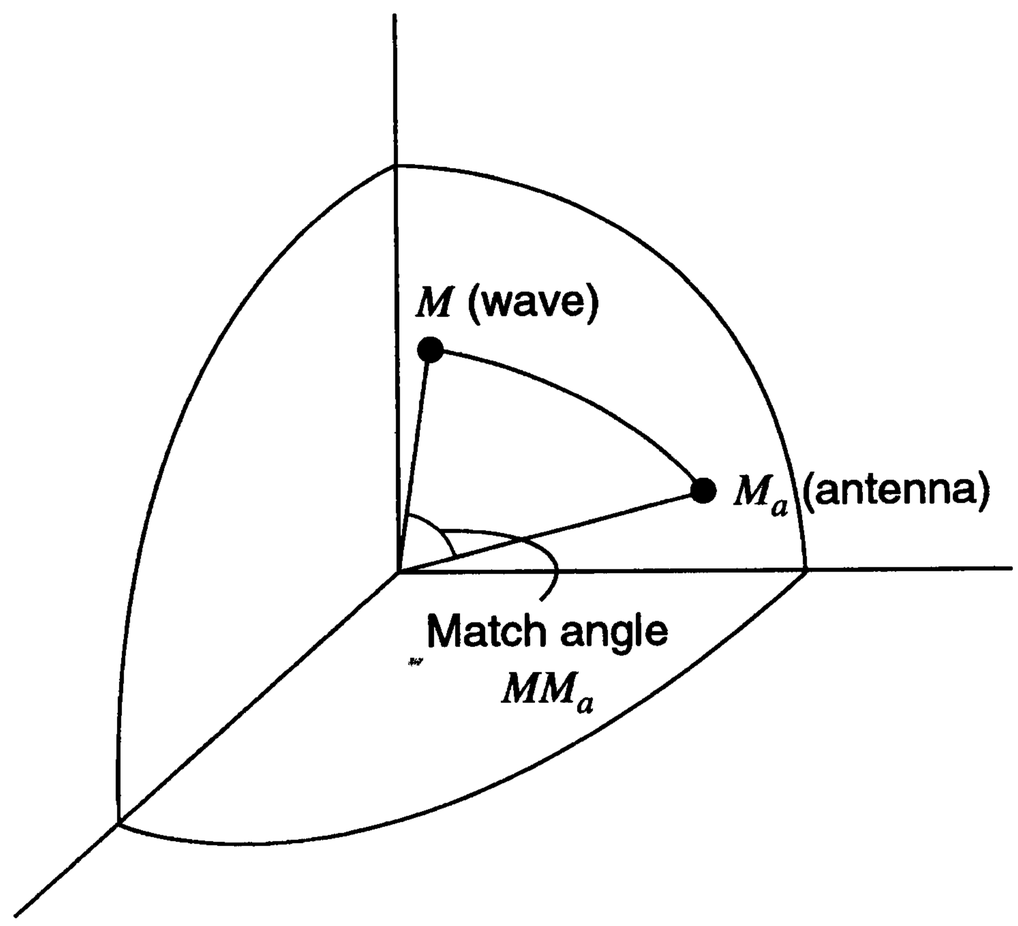


Figure 2–26
The match angle MM_a between the polarization state of wave (M) and antenna (M_a). For $MM_a = 0°$, the match is perfect. For $MM_a = 180°$, the match is zero.

designated by $M(\varepsilon, \tau)$, or simply M, and the polarization state as a function of γ and δ be designated by $P(\gamma, \delta)$, or simply P, as in Fig. 2–25.

As an application of the Poincaré sphere representation (see Fig. 2–26) it may be shown that the voltage response V of an antenna to a wave of arbitrary polarization is given by (Sinclair-1):

$$V = k \cos \frac{MM_a}{2} \qquad \textit{Antenna voltage response} \quad (\text{天线电压响应}) \tag{4}$$

where

MM_a = angle subtended by great-circle line from polarization state M to M_a
M = polarization state of wave
M_a = polarization state of antenna
k = constant

The polarization state of the antenna is defined as the polarization state of the wave radiated by the antenna when it is transmitting. The factor k in (4) involves the field strength of the wave and the size of the antenna. An important result to note is that, if $MM_a = 0°$, the antenna is matched to the wave (polarization state of wave same as for antenna) and the response is maximized. However, if $MM_a = 180°$, the response is zero. This can occur, for example, if the wave is linearly polarized in the y direction while the antenna is linearly polarized in the x direction; or if the wave is left circularly polarized while the antenna is right circularly polarized. More generally we may say that *an antenna is blind to a wave of opposite (or antipodal) polarization state.*

Referring to (4), a *polarization matching factor* F (for power) is given by

$$F = \cos^2 \frac{MM_a}{2} \tag{5}$$

Thus, for a perfect match the match angle $MM_a = 0°$ and $F = 1$ (states of wave and antenna the same). For a complete mismatch the match angle $MM_a = 180°$ and $F = 0$ (Fig. 2–26).

For *linear polarization,* $MM_a/2 = \Delta\tau$ and (5) reduces to

$$F = \cos^2 \Delta\tau \tag{6}$$

where $\Delta\tau$ = difference between the tilt angles of wave and antenna.

In the above discussion we have assumed a completely polarized wave, that is, one where E_x, E_y, and δ are constants. In an unpolarized wave they are not. Such a wave results when the vertical component is produced by one noise generator and the horizontal component by a different noise generator. Most cosmic radio sources are unpolarized and can be received equally well with an antenna of any polarization. If the wave is completely unpolarized, $F = \frac{1}{2}$ regardless of the state of polarization of the antenna. For a more general discussion, see J. D. Kraus, *Radio Astronomy,* 2d ed., Cygnus-Quasar, P.O. Box 85, Powell, OH 43065, 1986, Sec. 4–4.

EXAMPLE **2–17.1 Polarization Matching**

Find the polarization matching factor F for a left elliptically polarized wave (w) with AR = 4 and $\tau = 15°$ incident on a right elliptically polarized antenna (a) with AR = −2 and $\tau = 45°$.

■ **Solution**

From (1), $2\varepsilon(w) = 28.1°$ and $2\varepsilon(a) = -53.1°$. Thus, the wave polarization state M is at latitude +28.1° and longitude 30° while the antenna polarization state M_a is at latitude −53.1° and longitude 90°. Locate these positions on a globe and measure MM_a with

a string. The globe and string give not only the total great-circle angle MM_a but also illustrate the geometry. Then compare this result with an analytical one as follows: From proportional triangles obtain $2\tau(w) = 20.7°$ along the equator and $2\tau(a) = 39.3°$ further along the equator. Next from (3), obtain $2\gamma(w) = 34.3°$ and $2\gamma(a) = 62.4°$. Thus, the total great-circle angle $MM_a = 2\gamma(w) + 2\gamma(a) = 96.7°$ and the polarization matching factor

$$F = \cos^2\left(\frac{96.7}{2}\right) = 0.44$$

or the received power is 44 percent of the maximum possible value. *Ans.*

Although the radiation resistance, effective aperture, effective height, and directivity are the same for both receiving and transmitting, the current distribution is, in general, not the same. Thus, a plane wave incident on a receiving antenna excites a different current distribution than a localized voltage applied to a pair of terminals for transmitting.

Table summarizing important relations of Chapter 2

Wavelength-frequency	$\lambda = \frac{v}{f}$ (m)
Beam area	$\Omega_A = \iint P_n(\theta, \phi)\, d\Omega$ (sr or deg^2)
Beam area (approx.)	$\Omega_A \simeq \theta_{\text{HP}}\phi_{\text{HP}}$ (sr or deg^2)
Beam efficiency	$\varepsilon_M = \frac{\Omega_M}{\Omega_A}$ (dimensionless)
Directivity	$D = \frac{U(\theta, \phi)_{\max}}{U_{\text{av}}} = \frac{S(\theta, \phi)_{\max}}{S_{\text{av}}}$ (dimensionless)
Directivity	$D = \frac{4\pi}{\Omega_A}$ (dimensionless)
Directivity	$D = \frac{4\pi A_e}{\lambda^2}$ (dimensionless)
Directivity (approx.)	$D \simeq \frac{4\pi}{\theta_{\text{HP}}\phi_{\text{HP}}} \simeq \frac{41{,}000}{\theta^\circ_{\text{HP}}\phi^\circ_{\text{HP}}}$ (dimensionless)
Gain	$G = kD$ (dimensionless)
Effective aperture and beam area	$A_e\Omega_A = \lambda^2$ (m^2)
Aperture efficiency	$\varepsilon_{ap} = \frac{A_e}{A_p}$ (dimensionless)
Friis transmission formula	$P_r = P_t \frac{A_{et}A_{er}}{r^2\lambda^2}$ (W)
Current-charge continuity relation	$Il = q\dot{v}$ (A m s^{-1})
Radiation power	$P = \frac{\mu^2 q^2 \dot{v}^2}{6\pi Z}$ (W)
Near-field–far-field boundary	$R = \frac{2L^2}{\lambda}$ (m)
Average power per unit area of elliptically polarized wave in air	$\mathbf{S}_{\text{av}} = \frac{1}{2}\hat{\mathbf{z}}\frac{E_1^2 + E_2^2}{Z_0}$ (W m^{-2})

Table of effective aperture, directivity, effective height, and other parameters for dipoles and loops

Antenna	Radiation resistance* R_r, Ω	Maximum effective aperture A_{em}, λ^2	Effective height, maximum value, h, m	Sphere filling factor	Directivity D	Directivity D, dBi
Isotropic		$\frac{1}{4\pi} = 0.079$		1	1	0
Short dipole,† length l	$80\left(\frac{\pi l / l_{av}}{\lambda / l_0}\right)^2$	$\frac{3}{8\pi} = 0.119$	$\frac{l / l_{av}}{l_0}$	$\frac{2}{3}$	$\frac{3}{2}$	1.76
Short dipole,† $l = \lambda/10$ ($l_{av} = l_0$)	7.9	0.119	$\lambda/10$	$\frac{2}{3}$	$\frac{3}{2}$	1.76
Short dipole,† $l = \lambda/10$ ($l_{av} = \frac{1}{2} l_0$)	1.98	0.119	$\lambda/20$	$\frac{2}{3}$	$\frac{3}{2}$	1.76
Linear, $\lambda/2$ dipole (sinusoidal current distribution)	73	$\frac{30}{73\pi} = 0.13$	$\frac{\lambda}{\pi} = \frac{2l}{\pi}$	0.61	1.64	2.15
Small loop‡ (single turn), any shape	$31{,}200\left(\frac{A}{\lambda^2}\right)^2$	$\frac{3}{8\pi} = 0.119$	$2\pi\frac{A}{\lambda}$	$\frac{2}{3}$	$\frac{3}{2}$	1.76
Small square loop‡ (single turn), side length = l Area $A = l^2 = (\lambda/10)^2$	3.12	$\frac{3}{8\pi} = 0.119$	$\frac{2\pi\lambda}{100}$	$\frac{2}{3}$	$\frac{3}{2}$	1.76

*See Chaps. 5 and 8.

†Length $l \leq \lambda/10$.

‡Area $A \leq \lambda^2/100$, see Sec. 7–9. For n-turn loop, multiply R_r by n^2 and h by n.

REFERENCES（参考文献）

Born, M. (1), and E. Wolf: *Principles of Optics,* pp. 24–27, Macmillan, New York, 1964.

Deschamps, G. A. (1): "Geometrical Representation of the Polarization of a Plane Electromagnetic Wave," *Proc. IRE,* **39,** 540, May 1951.

Friis, H. T. (1): "A Note on a Simple Transmission Formula," *Proc. IRE,* **34,** 254–256, 1946.

Poincaré, H. (1): *Théorie mathématique de la lumière,* G. Carré, Paris, 1892.

Sinclair, G. (1): "The Transmission and Reception of Elliptically Polarized Waves," *Proc. IRE,* **38,** 151, 1950.

PROBLEMS（习题）

2-6-1 **Main-beam efficiency.** For an antenna with field pattern
（主波束效率）

$$E_n = \frac{\sin\theta}{\theta}\frac{\sin\phi}{\phi}$$

where θ = zenith angle (radians) and ϕ = azimuth angle (radians), (*a*) plot the normalized power pattern as a function of θ; (*b*) using your graph, estimate the main beam efficiency of this antenna.

2-7-1 (定向性) **Directivity.** Show that the directivity D of an antenna may be written

$$D = \frac{\dfrac{E(\theta,\phi)_{\max} E^*(\theta,\phi)_{\max}}{Z} r^2}{\dfrac{1}{4\pi}\displaystyle\iint_{4\pi} \frac{E(\theta,\phi)E^*(\theta,\phi)}{Z} r^2\, d\Omega}$$

2-7-2 (近似定向性) **Approximate directivities.** Calculate the approximate directivity from the half-power beamwidths of a unidirectional antenna if the normalized power pattern is given by: (*a*) $P_n = \cos\theta$, (*b*) $P_n = \cos^2\theta$, (*c*) $P_n = \cos^3\theta$, and (*d*) $P_n = \cos^n\theta$. In all cases these patterns are unidirectional ($+z$ direction) with P_n having a value only for zenith angles $0° \le \theta \le 90°$ and $P_n = 0$ for $0° \le \theta \le 180°$. The patterns are independent of the azimuth angle ϕ.

2-7-3 (近似定向性) **Approximate directivities.** Calculate the approximate directivities from the half-power beamwidths of the three unidirectional antennas having power patterns as follows:

$$P(\theta,\phi) = P_m \sin\theta \sin^2\phi$$

$$P(\theta,\phi) = P_m \sin\theta \sin^3\phi$$

$$P(\theta,\phi) = P_m \sin^2\theta \sin^3\phi$$

$P(\theta,\phi)$ has a value only for $0 \le \theta \le \pi$ and $0 \le \phi \le \pi$ and is zero elsewhere.

2-7-4 (定向性和增益) **Directivity and gain.** (*a*) Estimate the directivity of an antenna with $\theta_{HP} = 2°$, $\phi_{HP} = 1°$, and (*b*) find the gain of this antenna if efficiency $k = 0.5$.

2-8-1 (定向性和口径) **Directivity and apertures.** Show that the directivity of an antenna may be expressed as

$$D = \frac{4\pi}{\lambda^2} \frac{\displaystyle\iint_{A_p} E(x,y)\,dx\,dy \iint_{A_p} E^*(x,y)\,dx\,dy}{\displaystyle\iint_{A_p} E(x,y)E^*(x,y)\,dx\,dy}$$

where $E(x, y)$ is the aperture field distribution.

2-8-2 (有效口径和波束范围) **Effective aperture and beam area.** What is the maximum effective aperture (approximately) for a beam antenna having half-power widths of 30° and 35° in perpendicular planes intersecting in the beam axis? Minor lobes are small and may be neglected.

2-8-3 (有效口径和定向性) **Effective aperture and directivity.** What is the maximum effective aperture of a microwave antenna with a directivity of 900?

2-11-1 (接收功率和弗里斯公式) **Received power and the Friis formula.** What is the maximum power received at a distance of 0.5 km over a free-space 1-GHz circuit consisting of a transmitting antenna with a 25-dB gain and a receiving antenna with a 20-dB gain? The gain is with respect to a lossless isotropic source. The transmitting antenna input is 150 W.

2-11-2 (航天飞机间的 100 Mm 线路) **Spacecraft link over 100 Mm.** Two spacecraft are separated by 100 Mm. Each has an antenna with $D = 1000$ operating at 2.5 GHz. If craft A's receiver requires 20 dB over 1 pW, what transmitter power is required on craft B to achieve this signal level?

2-11-3 (航天飞机间的 3 Mm 线路) **Spacecraft link over 3 Mm.** Two spacecraft are separated by 3 Mm. Each has an antenna with $D = 200$ operating at 2 GHz. If craft A's receiver requires 20 dB over 1 pW, what transmitter power

2–11–4 (地球至火星或木星的线路) **Mars and Jupiter links.** (*a*) Design a two-way radio link to operate over earth-Mars distances for data and picture transmission with a Mars probe at 2.5 GHz with a 5-MHz bandwidth. A power of 10^{-19} W Hz^{-1} is to be delivered to the earth receiver and 10^{-17} W Hz^{-1} to the Mars receiver. The Mars antenna must be no larger than 3 m in diameter. Specify effective aperture of Mars and earth antennas and transmitter power (total over entire bandwidth) at each end. Take earth-Mars distance as 6 light-minutes. (*b*) Repeat (*a*) for an earth-Jupiter link. Take the earth-Jupiter distance as 40 light-minutes.

***2–11–5 (地球–月球的线路)** **Moon link.** A radio link from the moon to the earth has a moon-based 5λ-long right-handed monofilar axial-mode helical antenna (see Eq. (8–3–7)) and a 2-W transmitter operating at 1.5 GHz. What should the polarization state and effective aperture be for the earth-based antenna in order to deliver 10^{-14} W to the receiver? Take the earth-moon distance as 1.27 light-seconds.

2–13–1 (最大相位误差) **Maximum phase error.** What is the phase difference between a point on the Fresnel-Fraunhofer boundary sphere and two points on the antenna, one, R, from the origin to the sphere perpendicular to the antenna shown in Fig. 2–17, and the other, F, from the antenna tip, $L/2$?

2–15–1 (半长轴和半短轴) **Semimajor and semiminor axes.** By rotating the coordinates of the polarization ellipse given in Eq. (2–15–7), (*a*) show that the tilt angle τ is given by

$$\tau = \frac{1}{2}\tan^{-1}\left(\frac{2E_1E_2\cos\delta}{E_1^2 - E_2^2}\right)$$

and (*b*) show that

$$OA = \left[(E_1\cos\tau + E_2\cos\delta\sin\tau)^2 + E_2^2\sin^2\delta\sin^2\tau\right]^{1/2} \quad \text{and}$$

$$OB = \left[(E_1\sin\tau + E_2\cos\delta\cos\tau)^2 - E_2^2\sin^2\delta\cos^2\tau\right]^{1/2}$$

2–16–1 (接近月球的宇宙飞船) **Spaceship near moon.** A spaceship at lunar distance from the earth transmits 2-GHz waves. If a power of 10 W is radiated isotropically, find (*a*) the average Poynting vector at the earth, (*b*) the rms electric field **E** at the earth and (*c*) the time it takes for the radio waves to travel from the spaceship to the earth. (Take the earth-moon distance as 380 Mm.) (*d*) How many photons per unit area per second fall on the earth from the spaceship transmitter?

2–16–2 (更多的圆极化波功率) **More power with CP.** Show that the average Poynting vector of a circularly polarized wave is twice that of a linearly polarized wave if the maximum electric field **E** is the same for both waves. This means that a medium can handle twice as much power before breakdown with circular polarization (CP) than with linear polarization (LP).

2–16–3 (圆极化波的坡印廷矢量) **PV constant for CP.** Show that the instantaneous Poynting vector (PV) of a plane circularly polarized traveling wave is a constant.

***2–16–4 (椭圆极化波的功率)** **EP wave power.** An elliptically polarized wave in a medium with constants $\sigma = 0$, $\mu_r = 2$, $\varepsilon_r = 5$ has H-field components (normal to the direction of propagation and normal to each other) of amplitudes 3 and 4 A m^{-1}. Find the average power conveyed through an area of 5 m^2 normal to the direction of propagation.

2–17–1 (交叉偶极子用于圆极化和其他极化态) **Crossed dipoles for CP and other states.** Two $\lambda/2$ dipoles are crossed at 90°. If the two dipoles are fed with equal currents, what is the polarization of the radiation perpendicular to the plane of the dipoles if the currents are (*a*) in phase, (*b*) phase quadrature (90° difference in phase), and (*c*) phase octature (45° difference in phase)?

2-17-2 (两个线极化波的极化)

Polarization of two LP waves. A wave traveling normally out of the page (toward the reader) has two linearly polarized components

$$E_x = 2\cos\omega t$$
$$E_y = 3\cos(\omega t + 90°)$$

(*a*) What is the axial ratio of the resultant wave?
(*b*) What is the tilt angle τ of the major axis of the polarization ellipse?
(*c*) Does **E** rotate clockwise or counterclockwise?

2-17-3 (两个椭圆极化波的叠加)

Superposition of two EP waves. A wave traveling normally outward from the page (toward the reader) is the resultant of two elliptically polarized waves, one with components of **E** given by

$$E'_y = 2\cos\omega t$$
$$E'_x = 6\cos\left(\omega t + \frac{\pi}{2}\right)$$

and the other with components given by

$$E''_y = 1\cos\omega t$$
$$E''_x = 3\cos\left(\omega t - \frac{\pi}{2}\right)$$

(*a*) What is the axial ratio of the resultant wave?
(*b*) Does **E** rotate clockwise or counterclockwise?

2-17-4 (两个线极化分量)

Two LP components. An elliptically polarized plane wave traveling normally out of the page (toward the reader) has linearly polarized components E_x and E_y. Given that $E_x = E_y = 1\ \text{V m}^{-1}$ and that E_y leads E_x by 72°,

(*a*) Calculate and sketch the polarization ellipse.
(*b*) What is the axial ratio?
(*c*) What is the angle τ between the major axis and the x axis?

2-17-5 (两个线极化分量和庞加莱球)

Two LP components and Poincaré sphere. Answer the same questions as in Prob. 2–17–4 for the case where E_y leads E_x by 72° as before but $E_x = 2\ \text{V m}^{-1}$ and $E_y = 1\ \text{V m}^{-1}$.

2-17-6 (两个圆极化波)

Two CP waves. Two circularly polarized waves intersect at the origin. One (y wave) is traveling in the positive y direction with **E** rotating clockwise as observed from a point on the positive y axis. The other (x wave) is traveling in the positive x direction with **E** rotating clockwise as observed from a point on the positive x axis. At the origin, **E** for the y wave is in the positive z direction at the same instant that **E** for the x wave is in the negative z direction. What is the locus of the resultant **E** vector at the origin?

2-17-7 (圆极化波)

CP waves. A wave traveling normally out of the page is the resultant of two circularly polarized components $E_{\text{right}} = 5e^{j\omega t}$ and $E_{\text{left}} = 2e^{j(\omega t+90°)}$ (V m^{-1}). Find (*a*) the axial ratio AR, (*b*) the tilt angle τ, and (*c*) the hand of rotation (left or right).

2-17-8 (椭圆极化波)

EP wave. A wave traveling normally out of the page (toward the reader) is the resultant of two linearly polarized components $E_x = 3\cos\omega t$ and $E_y = 2\cos(\omega t + 90°)$. For the resultant wave find (*a*) the axial ratio AR, (*b*) the tilt angle τ, and (c) the hand of rotation (left or right).

2-17-9 (圆极化波)

CP waves. Two circularly polarized waves traveling normally out of the page have fields given by $E_{\text{left}} = 2e^{-j\omega t}$ and $E_{\text{right}} = 3e^{j\omega t}$ (V m^{-1}) (rms). For the resultant wave find (*a*) AR, (*b*) the hand of rotation, and (*c*) the Poynting vector.

2-17-10
（椭圆极化波）

EP waves. A wave traveling normally out of the page is the resultant of two elliptically polarized (EP) waves, one with components $E_x = 5\cos\omega t$ and $E_y = 3\sin\omega t$ and another with components $E_r = 3e^{j\omega t}$ and $E_l = 4e^{-j\omega t}$. For the resultant wave, find (*a*) AR, (*b*) τ, and (*c*) the hand of rotation.

***2-17-11**
（圆极化波）

CP waves. A wave traveling normally out of the page is the resultant of two circularly polarized components $E_r = 2e^{j\omega t}$ and $E_l = 4e^{-j(\omega t+45°)}$. For the resultant wave, find (*a*) AR, (*b*) τ, and (*c*) the hand of rotation.

2-17-12
（圆－消极化比）

Circular-depolarization ratio. If the axial ratio of a wave is AR, show that the circular-depolarization ratio of the wave is given by

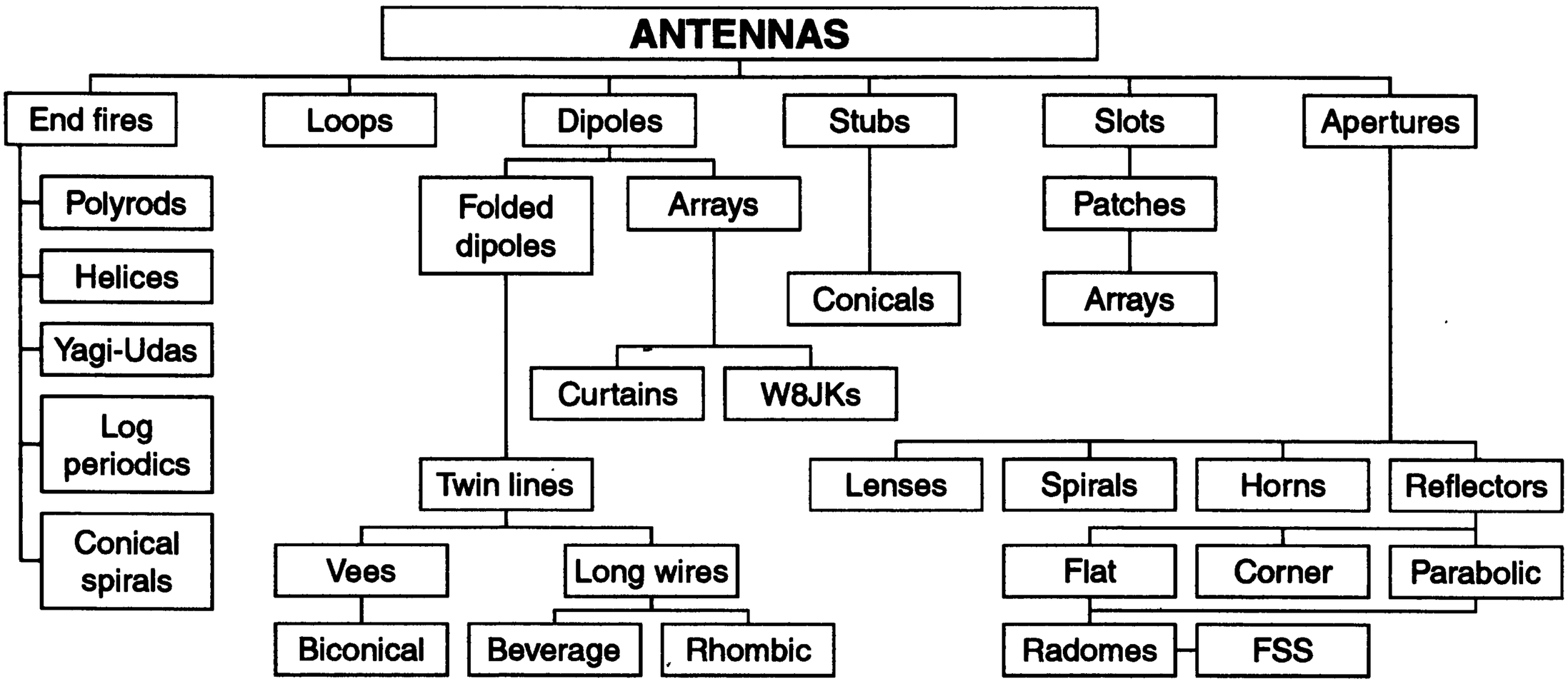
$$R = \frac{AR - 1}{AR + 1}$$

Thus, for pure circular polarization $AR = 1$ and $R = 0$ (no depolarization) but for linear polarization $AR = \infty$ and $R = 1$.

For computer programs see Appendix C.

CHAPTER 3

The Antenna Family
天线家族



3–1 INTRODUCTION（引言）

In this chapter we are introduced to 24 types of antennas as a preview to more detailed treatments in later chapters.

The antennas are grouped into:

- **Basic types**（基本类型）
- **Loops**, dipoles and slots（环形、偶极子和缝隙）
- **Opened**-out coaxial, twin-line and waveguide
  （张开的同轴线、双线和波导）
- Reflector and aperture types（反射镜与口径类型）
- End-fire and broadband types（端射与宽频带类型）
- Flat-panel, slot and grid arrays
  （镶板式、缝隙和栅格阵列）

The coaxial and two-wire (twin-line) types are arranged as an evolutionary sequence from broad to narrow bandwidth. Directivities, bandwidths and field patterns are indicated. These, with dimensions given, are sufficient in many cases to construct an antenna and determine its approximate gain and beamwidth.

The presentation relates the antennas in a genus-species classification that aids understanding how one type carries over or evolves into another.[1]

3–2 LOOPS, DIPOLES AND SLOTS（环形、偶极子和缝隙天线）

The small horizontal loop antenna at (*a*) in Fig. 3–1 may be regarded as the magnetic counterpart of the short vertical dipole at (*b*). Both loop and dipole have identical field

[1]For most antennas, losses are small so that the *gain* is nearly equal to the *directivity*. Thus, as a practical matter, gain and directivity are often used interchangeably.

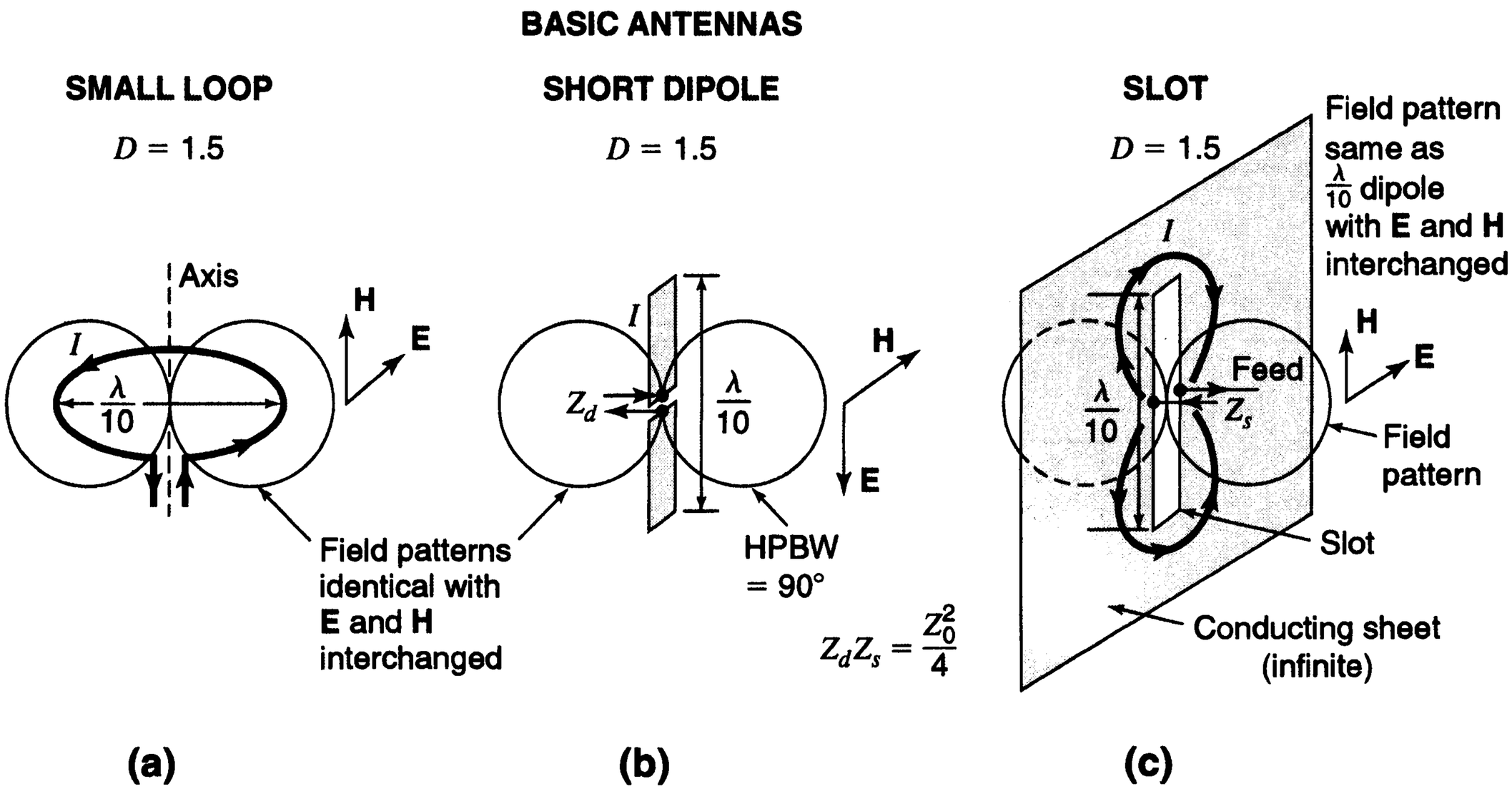


Figure 3–1
Three basic types of antennas: the small loop (*a*), short dipole (*b*), and slot antenna (*c*). The small loop and short dipole (loop axis parallel to dipole) have identical field patterns with **E** and **H** interchanged. The slot and dipole have the same field patterns with **E** and **H** interchanged. Directivities D are identical, as indicated.

patterns but with **E** and **H** interchanged. Thus, the horizontal loop is horizontally polarized and the vertical loop is vertically polarized. Both small loop and short dipole have the same directivity $D = 1.5$. To qualify as a small loop or short dipole, the dimensions should be $\lambda/10$ or less.

EXAMPLE **3–2.1 Loop and Dipole for Circular Polarization**

By placing the short dipole inside the small loop on its axis (Fig. 3–1), the pattern is omnidirectional in the horizontal plane with a null on the vertical axis the same as for the individual loop and dipole patterns.

(*a*) If the loop diameter is $\lambda/15$ and the loop and dipole are fed in-phase with equal power, what is the polarization of the radiation? (*b*) If the loop current is counterclockwise as viewed from above when the dipole current is up, is the polarization left- or right-handed?

From Sec. 2–15, the polarization is right-handed circular. *Ans.* (*a*) and (*b*).

If the dipole has been cut from a metal sheet, leaving a slot as at (*c*), the dipole and slot are said to be *complementary*. The field patterns of the dipole and slot are the same but with **E** and **H** interchanged. Furthermore, the terminal impedance Z_d of the dipole and the terminal impedance Z_s of the slot are related to the intrinsic impedance of space $Z_0 (= 377\ \Omega)$ as given by

$$Z_d Z_s = \frac{Z_0^2}{4} \tag{1}$$

from which

$$\boxed{Z_s = \frac{Z_0^2}{4Z_d} = \frac{Z_0^2}{4} Y_d \qquad \textit{slot impedance}} \quad (缝隙阻抗) \tag{2}$$

so that the *slot impedance* Z_s is proportional to the *dipole admittance* Y_d. If the dipole requires inductance for a match, the complementary slot requires capacitance. Thus, knowing the properties of the dipole enables us to predict the properties of the complementary slot. To be completely complementary, the sheet containing the slot should be large (ideally infinite) and perfectly conducting.

Slot antennas are typically $\lambda/2$ long, and these are then complementary to a $\lambda/2$ dipole antenna (Fig. 3–1). Note that the field patterns in Fig. 3–1 are *not* field lines. However, the directions of **E** and **H** at a point are shown.

EXAMPLE **3–2.2 One-Wavelength Dipole and Slot Antenna**

In Fig. 3–2 a λ-long cylindrical dipole and its complementary slot are compared. The actual length $L = 0.925\lambda$. The dipole cylinder has a diameter $D = L/28 = 0.033\lambda$ and a terminal impedance $Z_d = 710 + j0\ \Omega$. The complementary slot has a width $w = 2D$. Find the terminal impedance Z_s of the slot.

■ **Solution**

From (2),

$$Z_s = \frac{377^2}{4 \times 710} = 50\ \Omega \qquad \textit{Ans.}$$

for an impedance match to a 50-Ω coaxial line as indicated in Fig. 3–2*b*.

Radiation from a patch antenna (Fig. 3–11) occurs as though from two slots, so this example is useful for an understanding of patch antennas.

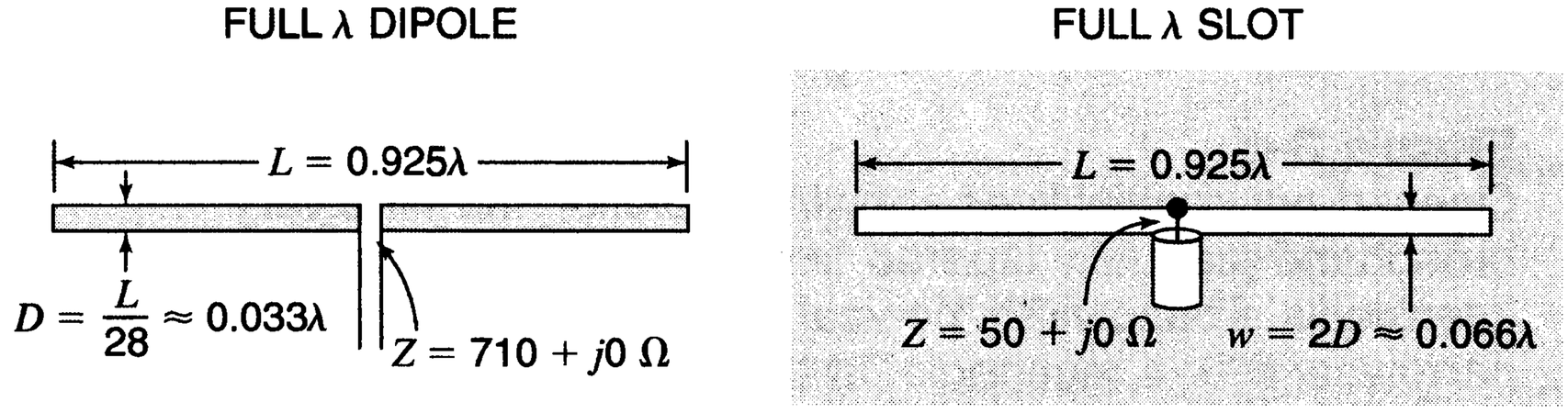


Figure 3–2
Comparison of impedances of cylindrical dipole antenna with complementary slot antenna. The slot in (*b*) matches directly to the 50-Ω coaxial line.

3–3 OPENED-OUT COAXIAL-LINE ANTENNAS（张开的同轴线天线）

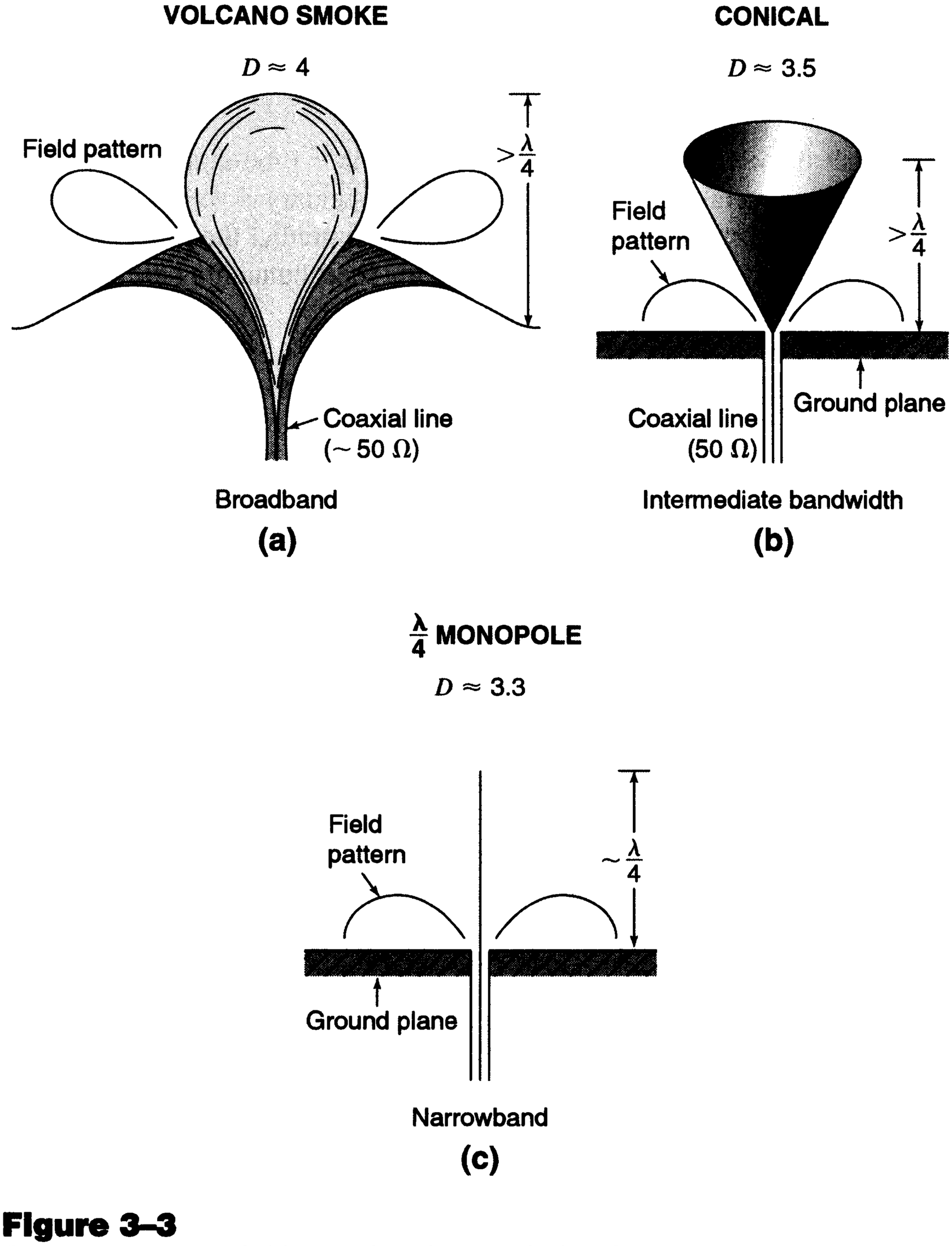


Figure 3–3
Opened-out coaxial lines showing evolution from the very broadband gradually tapered "volcano smoke" at top (*a*) through the intermediate bandwidth "conical antenna" at center (*b*) to the narrowband "$\lambda/4$ monopole" at the bottom (*c*).

All of the antennas in Fig. 3–3 are omnidirectional in the horizontal plane with a null in the vertical (zenith) direction. The directivity D is indicated for each antenna. Going from smooth, gradual transitions to more abrupt ones results in narrower bandwidths.

3–4 OPENED-OUT 2-CONDUCTOR (TWIN-LINE) ANTENNAS（张开的双导线天线）

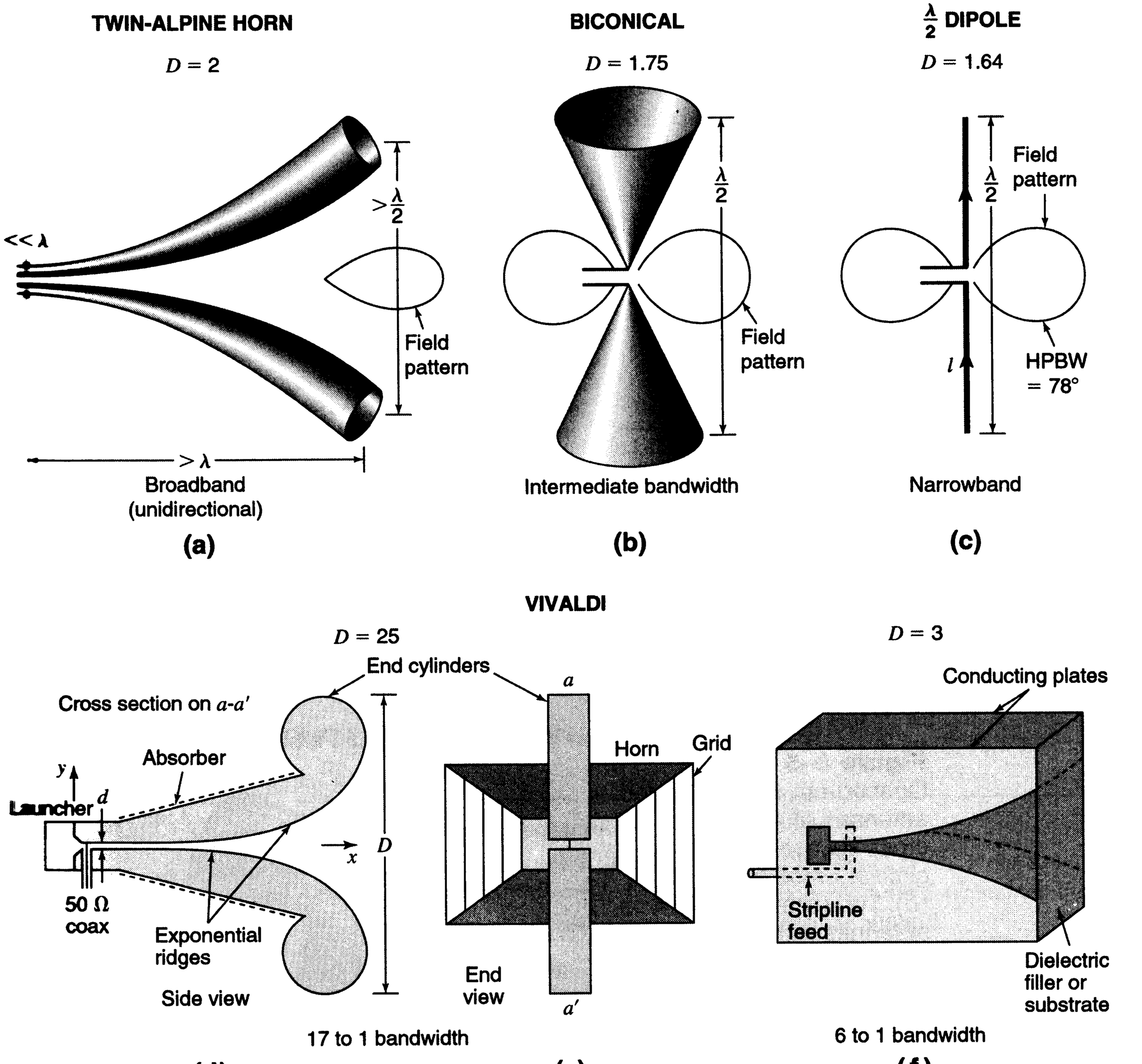


Figure 3–4
Opened-out two-conductor antennas showing the evolution from the very broadband "twin-Alpine-horn antenna" at the top (*a*) through the intermediate bandwidth "biconical antenna" at the center (*b*) to the narrow bandwidth "λ/2 dipole" (*c*) and the compact versions (*d*), (*e*), and (*f*). Whereas the "twin-Alpine horn" is unidirectional, the biconical and dipole antennas are omnidirectional in the horizontal plane. The directivity *D* is indicated for each antenna.

A compact version of the twin-Alpine horn, shown in Fig. 3–4*d* and *e*, has a double-ridge waveguide as the launcher on an exponentially flaring two-conductor balanced transmission line. The design in Fig. 3–4*d* and *e* incorporates features used by Kerr (1) and by Baker (1) and Van der Neut. The exponential taper is of the form $y = k_1 e^{k_2 x}$ where k_1 and k_2 are constants. The exact curvature is not critical provided it is gradual. Another compact version shown in Fig. 3–4*f* is called variously a *slot-vee sandwich* or a *Vivaldi* antenna after the 17th century composer Antonio Vivaldi (Shin-1).

3–5 OPENED-OUT WAVEGUIDE ANTENNAS (APERTURE TYPES)
[张开的波导天线（口径类）]

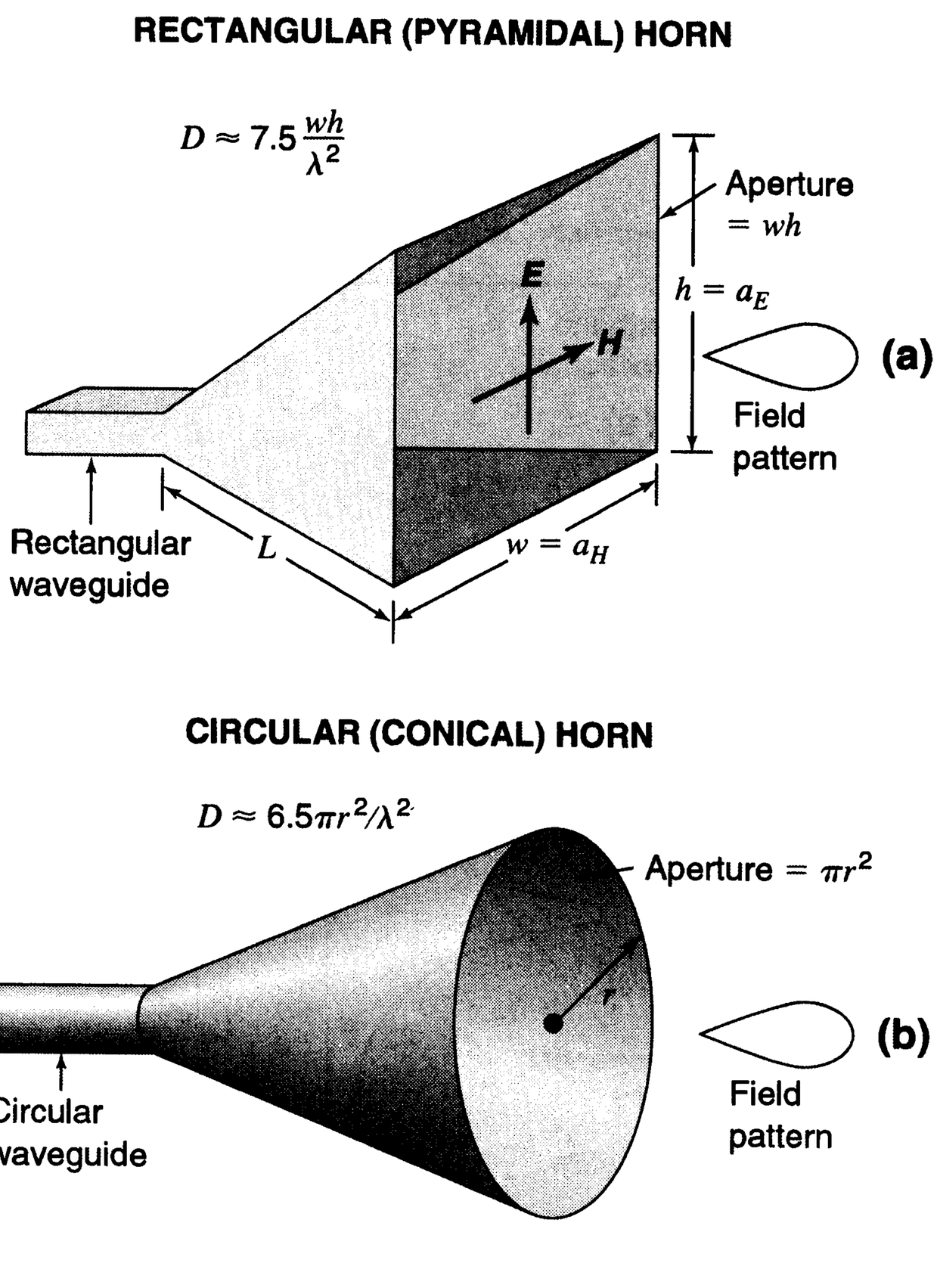


Figure 3–5
Opened-out waveguide-fed antennas of rectangular cross section at top (*a*) and circular cross section at (*b*). They are *aperture-type* antennas with effective aperture A_e and directivity D proportional to the area of the horn opening.

EXAMPLE **3–5.1 Optimum Pyramidal Horn**

Ideally the phase of the field across the horn mouth should be a constant. This requires a very long horn. However, for practical convenience the horn should be as short as possible. Referring to Fig. 3–6, an *optimum horn* is a compromise in which the difference in the path length δ along the edge and the center of the horn is made 0.25λ or less in the E plane. However, in the H plane, δ can be larger, since the field goes to zero at the horn edges (boundary condition, $E_t = 0$ satisfied). From Fig. 3–6 the horn flare angle θ is

given by

$$\theta = 2\cos^{-1}\frac{L}{L+\delta} \tag{1}$$

For a horn with $L = 10\lambda$, find the largest flare angle for which $\delta = 0.25\lambda$.

■ **Solution**

From (1), $\theta = 2\cos^{-1}\dfrac{10}{10.25} = 25.4°$ *Ans.*

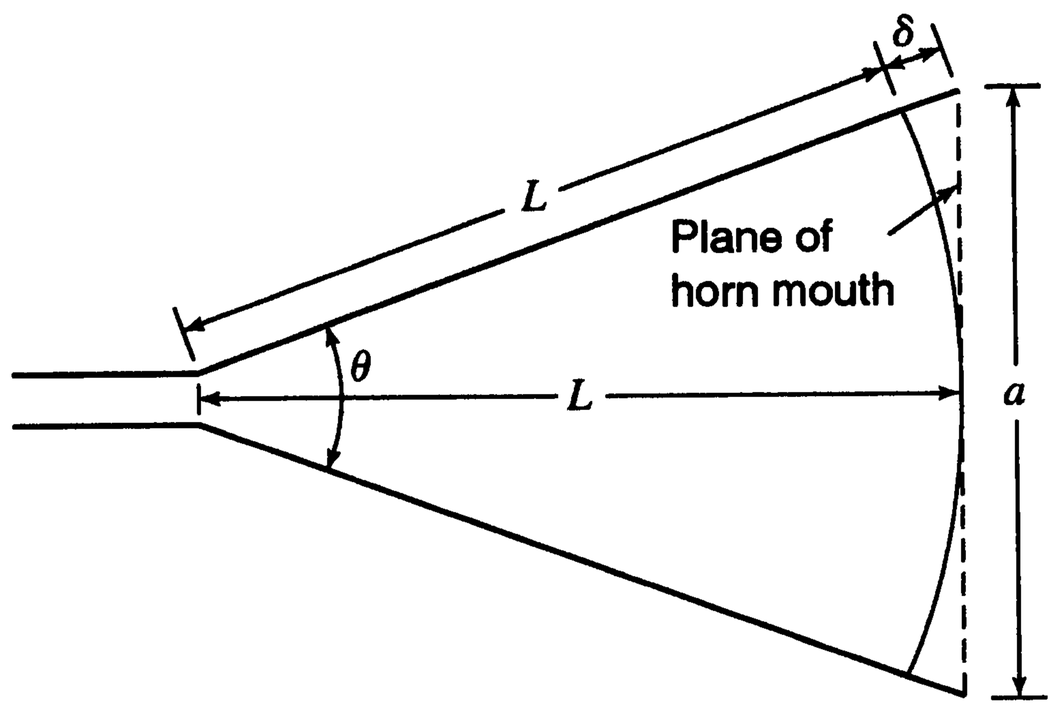


Figure 3–6
Cross section of pyramidal horn with dimensions used in example. The diagram can be used for either *E*-plane or *H*-plane cross sections. For the *E* plane the flare angle is θ_E and aperture dimension is a_E. For the *H* plane the flare angle is θ_H and the aperture dimension is a_H. See Fig. 3–5.

3–6 FLAT-SHEET REFLECTOR ANTENNAS（平板反射器天线）

EXAMPLE **3–6.1 Power Received by Square-Corner Reflector**

A U.S. channel 35 (599 MHz) TV station produces a field strength of 1 μV m^{-1} at a square-corner receiving antenna, as in Fig. 3–7*c*, with optimum dimensions for this channel. Find the power delivered to the receiver assuming it is matched to the antenna.

■ **Solution**

$$\lambda = \frac{c}{f} = \frac{3\times10^8 \text{ m s}^{-1}}{599\times10^6 \text{ Hz}} = 0.501 \text{ m}$$

From Fig. 3–7*c*, $D = 20$, so the effective aperture

$$A_e = \frac{D\lambda^2}{4\pi} = \frac{20\times0.251}{4\pi} = 0.4 \text{ m}^2$$

and the received power is

$$\frac{E^2}{Z_0}A_e = \frac{(10^{-6})^2}{377}0.4 = 1.06\times10^{-15} \text{ W} = 1.06 \text{ fW} \qquad \textit{Ans.}$$

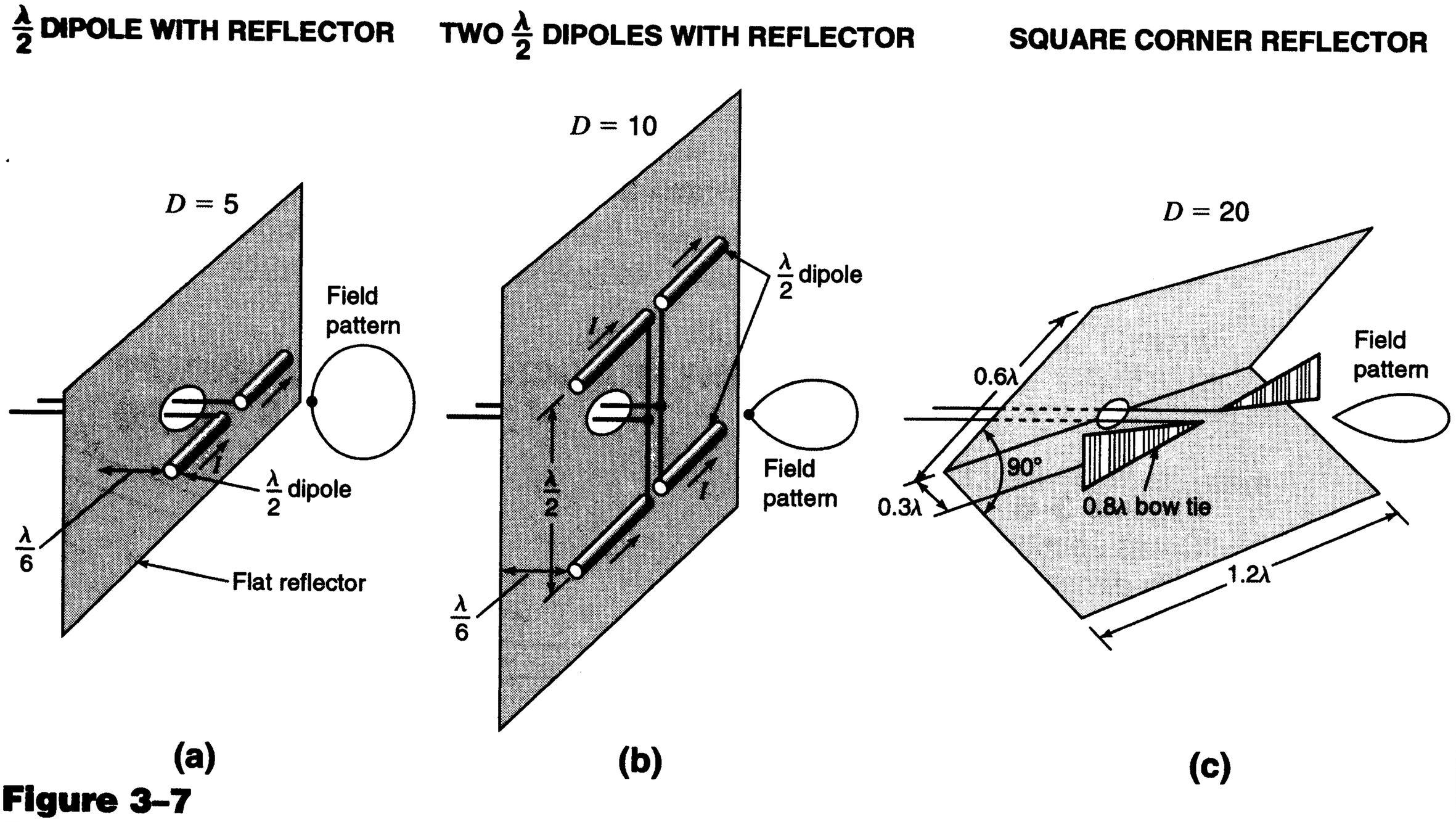


Figure 3–7
Flat reflector antennas with one and two $\lambda/2$ dipoles (*a* and *b*) and 90° or square-corner reflector (*c*). The directivity of a $\lambda/2$ dipole can be increased by placing it in front of a flat conducting reflector as in (*a*). An array of two $\lambda/2$ dipoles in front of a flat reflector as in (*b*) produces higher directivity. Even more directivity is obtained by folding the flat reflector into a 90° or square corner as at (*c*). To reduce wind resistance and the amount of metal required, the reflectors can be replaced by grids of parallel wires spaced $\lambda/10$ or closer. The directivity approximately doubles going from (*a*) to (*b*) and doubles again going to (*c*).

3–7 PARABOLIC DISH AND DIELECTRIC LENS ANTENNAS
（抛物面和介质透镜天线）

EXAMPLE **3–7.1 Parabolic Dish Design**

The directivity (or gain) of a parabolic dish antenna depends on many factors:

1. The pattern of the feed antenna. If its pattern is too broad and spills over the edge of the dish, the gain is reduced. On the other hand, if the pattern is too narrow, the dish is not fully "illuminated" by the feed and the aperture is not fully utilized.
2. The accuracy of the dish surface relative to an ideal parabola. For example, if the surface departs a distance $\delta = \lambda/4$ (or 90° electrical degrees) from the parabolic curve, the reflected field is phase shifted 180°, which reduces the aperture efficiency. See dish surface in Fig. 3–8*a*.
3. Many other factors are also involved. The aperture efficiency varies widely depending on the specific design.

Assuming an aperture efficiency of 70 percent, what is the directivity of a parabolic dish antenna as a function of its radius?

■ **Solution**

$$D = \varepsilon_{ap}\frac{4\pi A_p}{\lambda^2} = 0.7\frac{4\pi\pi r^2}{\lambda^2} = 8.8\frac{\pi r^2}{\lambda^2} = 28\left(\frac{r}{\lambda}\right)^2 \qquad Ans.$$

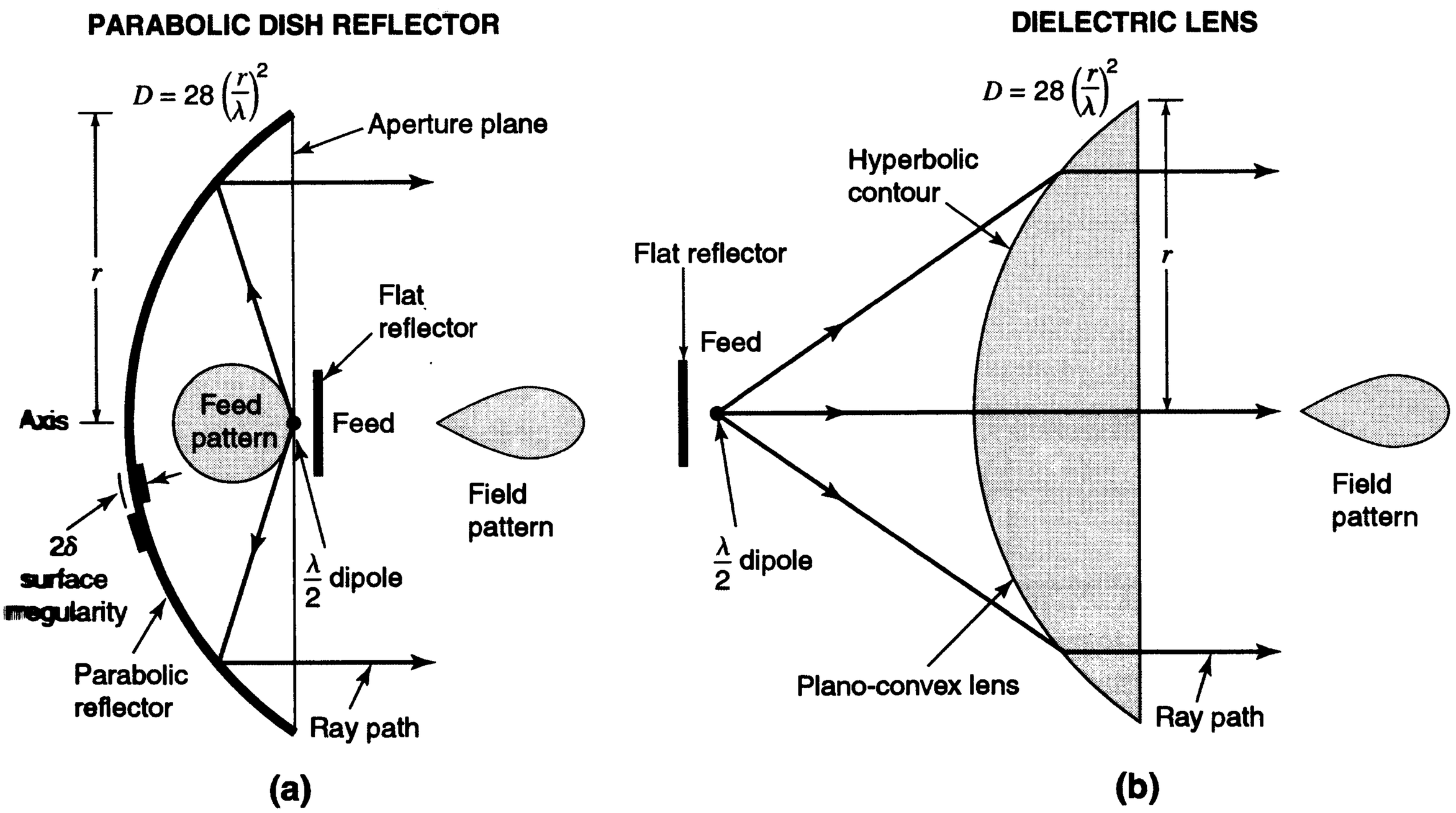
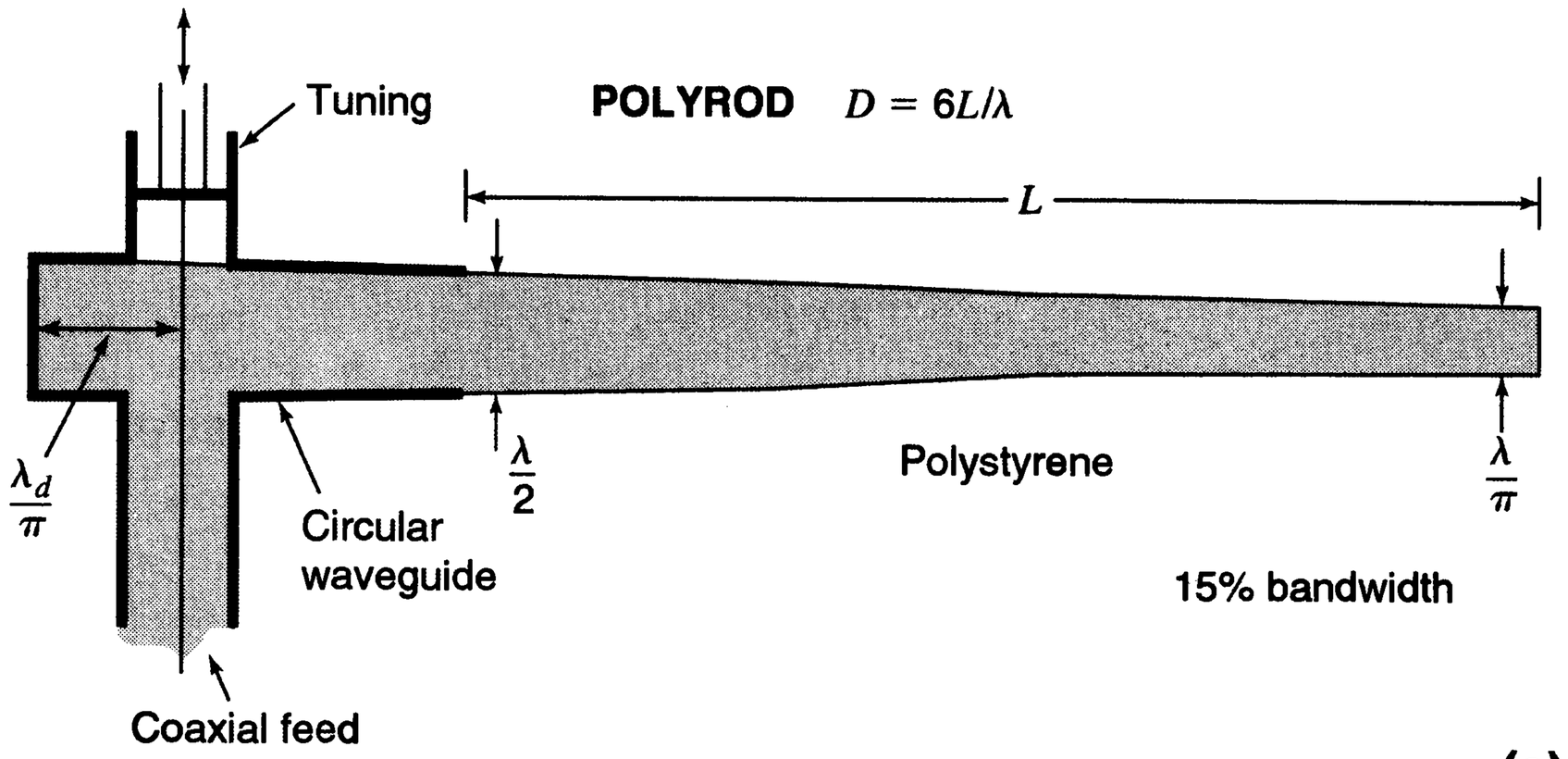
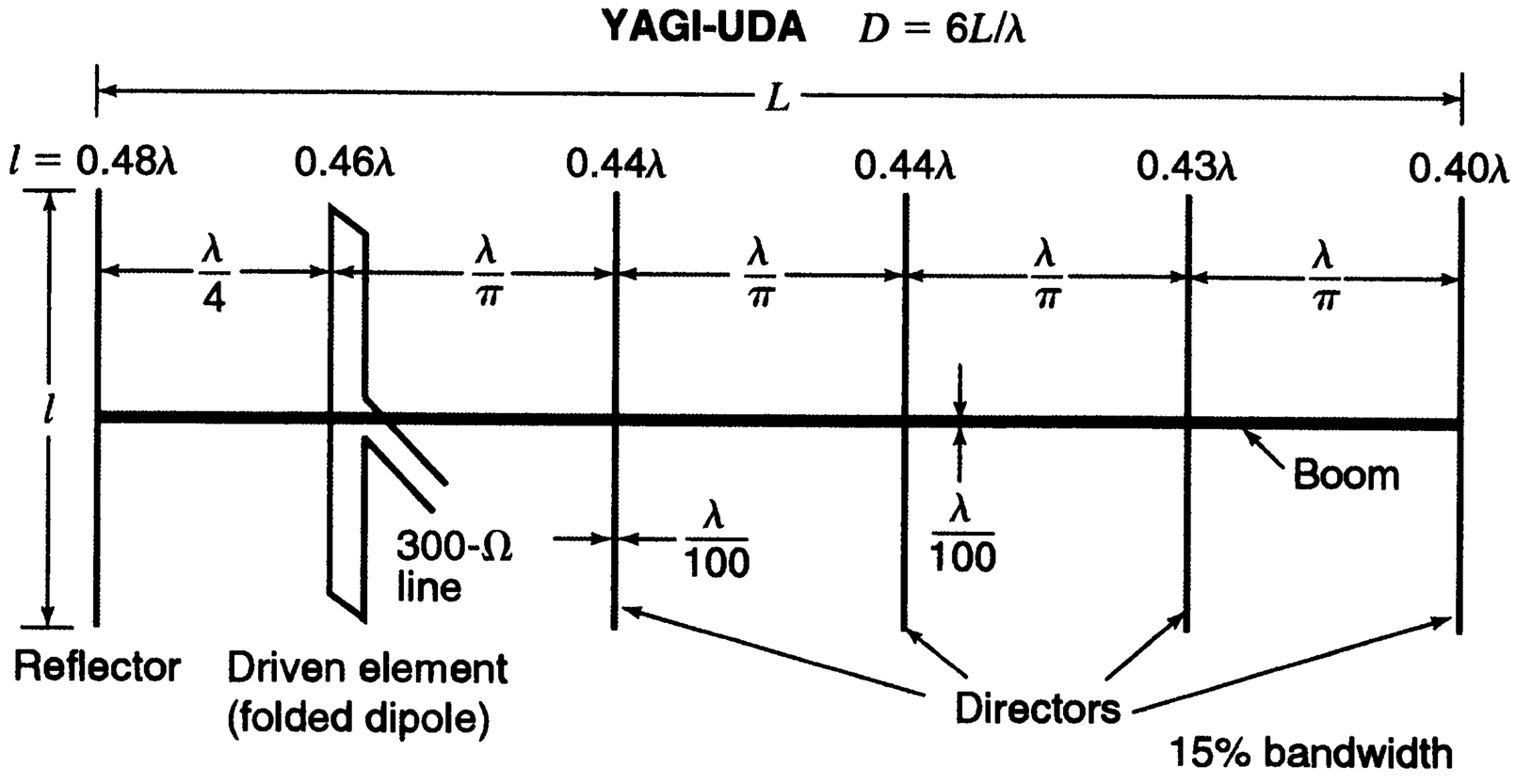


Figure 3–8
A parabolic dish-shaped reflector can provide a high directivity (proportional to its aperture) but for efficient operation requires a suitable feed antenna as shown in (*a*). By contrast a simple dipole is adequate to feed a corner reflector (Fig. 3–7*c*). The dielectric lens antenna at (*b*) is analogous to its optical counterpart. Like the parabolic dish antenna, the lens requires a suitable feed. The directivity D of both dish and lens antenna is proportional to their apertures. Both parabolic-dish and lens antennas require ray theory or optics in their design. Arrows in the figures trace ray paths. The feeds of both types radiate a spherical wave. The parabola converts the spherical wave into a plane wave by *reflection* while the lens does it by *refraction*.

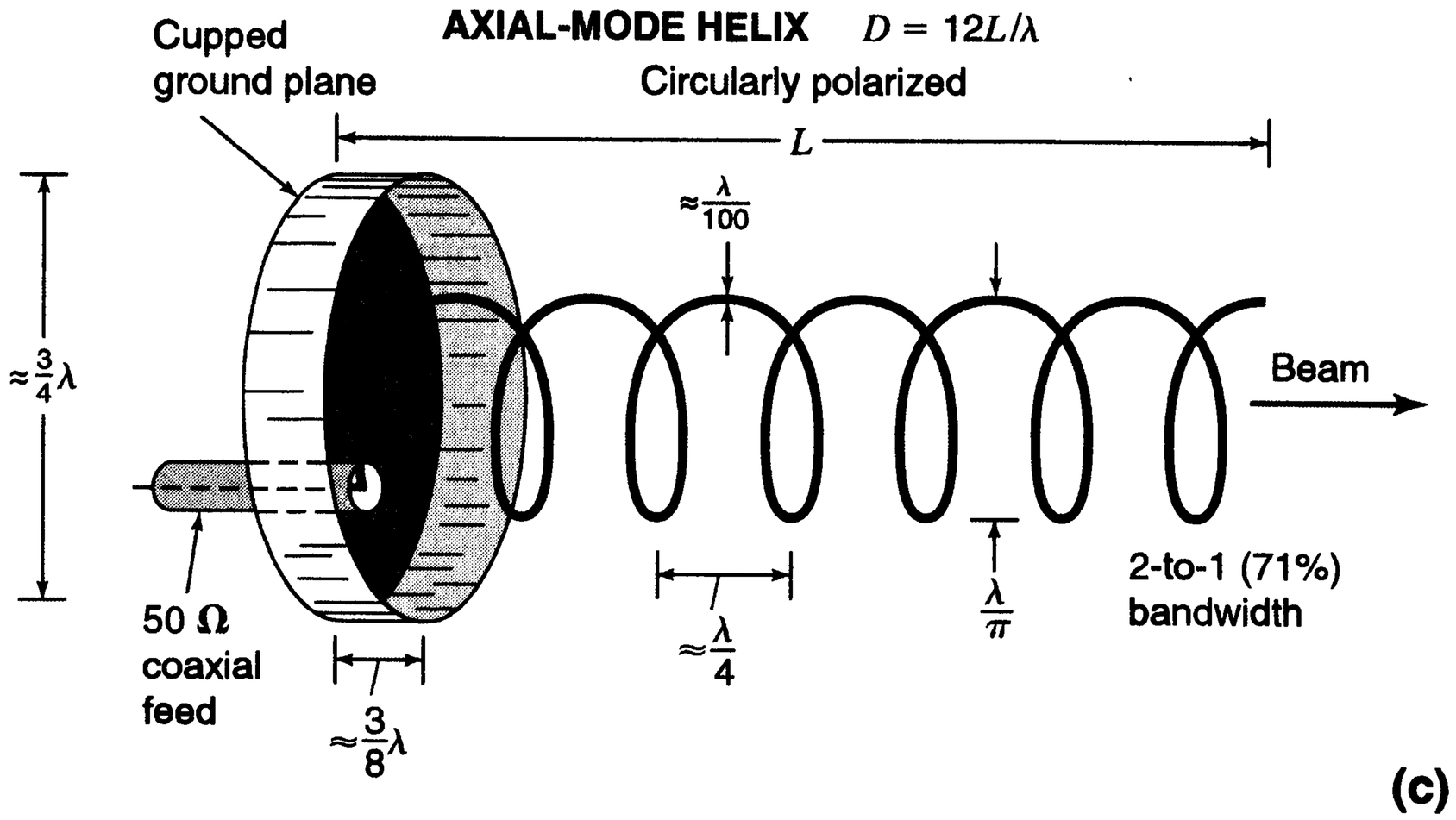
3–8 END-FIRE ANTENNAS（端射天线）



(a)



(b)



(c)

Three types of end-fire antennas are shown in Fig. 3–9. Because of its high directivity, circular polarization, wide bandwidth and noncritical dimensions, the axial-mode helical antenna (Fig. 3–9*c*) is widely employed in space applications. With linear polarization, a change in orientation of the satellite antenna could result in cross polarization and loss of signal. With circular polarization this problem does not occur, provided the satellite and earth-station antennas are of the same hand.

3–9 BROAD BANDWIDTH ANTENNAS: THE CONICAL SPIRAL AND THE LOG-PERIODIC（宽频带天线：圆锥螺蜷和对数周期）

The conical spiral Fig. 3–10*a* may be regarded as a flat spiral which has been wrapped around a dielectric cone. The conical spiral is fed by a coaxial cable bonded to one conducting strip, with its inner conductor joined to the other strip at the apex, as indicated in the sketch of the flat spiral. The lower-frequency limit of the conical spiral occurs when the base diameter is $\lambda/2$. The high-frequency limit occurs when the apex diameter is $\lambda/4$. Thus, the bandwidth is in the ratio $\frac{1}{2}$ base diameter to apex diameter, which, for the cone in Fig. 3–10*a*, is about 7 to 1. The bandwidth of the log-periodic antenna depends on the ratio of the next-to-longest to next-to-shortest dipoles, which for the array in Fig. 3–10*b* is about 4 to 1.

3–10 THE PATCH ANTENNA, PATCH ARRAY, AND THE GRID ARRAY（贴片天线、贴片阵列和栅格阵列）

The Patch Antenna

The "patch" is a low-profile, low-gain, narrow-bandwidth antenna. Aerodynamic considerations require low-profile antennas on aircraft and many kinds of vehicles.

Figure 3–11 shows a patch antenna with its dielectric substrate partially cut away to show the feed point. Typically a patch consists of a thin conducting sheet about 1 by $\frac{1}{2}\lambda_0$ mounted on the substrate.

Radiation from the patch is like radiation from two slots, at the left and right edges of the patch. The "slot" is the narrow gap between the patch and the ground plane. The patch-to-ground-plane spacing is equal to the thickness t of the substrate and is typically about $\lambda_0/100$, as indicated in Fig. 3–11.

Figure 3–9
The polyrod antenna at (*a*), the Yagi-Uda at (*b*), and the axial-mode helical antenna at (*c*) are all end-fire or traveling-wave antennas. The directivity D of each is proportional to its length L with a higher directivity for the helical antenna because it operates in the increased-directivity mode. The retina of the human eye has 100 million rods and cones analogous to the polyrod antenna (*a*). Dimensions, directivities, and bandwidths are indicated. The patterns of all three antennas can be calculated to a good approximation as an array of isotropic sources spaced $\lambda/4$ with 90° phasing for the polyrod and Yagi-Uda and increased directivity phasing for the helical antenna. All three end-fire antennas may be regarded as rudimentary lens antennas which collect energy over an aperture much larger than the physical cross section. In the Yagi-Uda antenna only one element is driven (fed by transmission line), the rest being parasitic elements energized by mutual coupling, the reflector having a lagging phase and the directors leading phases.

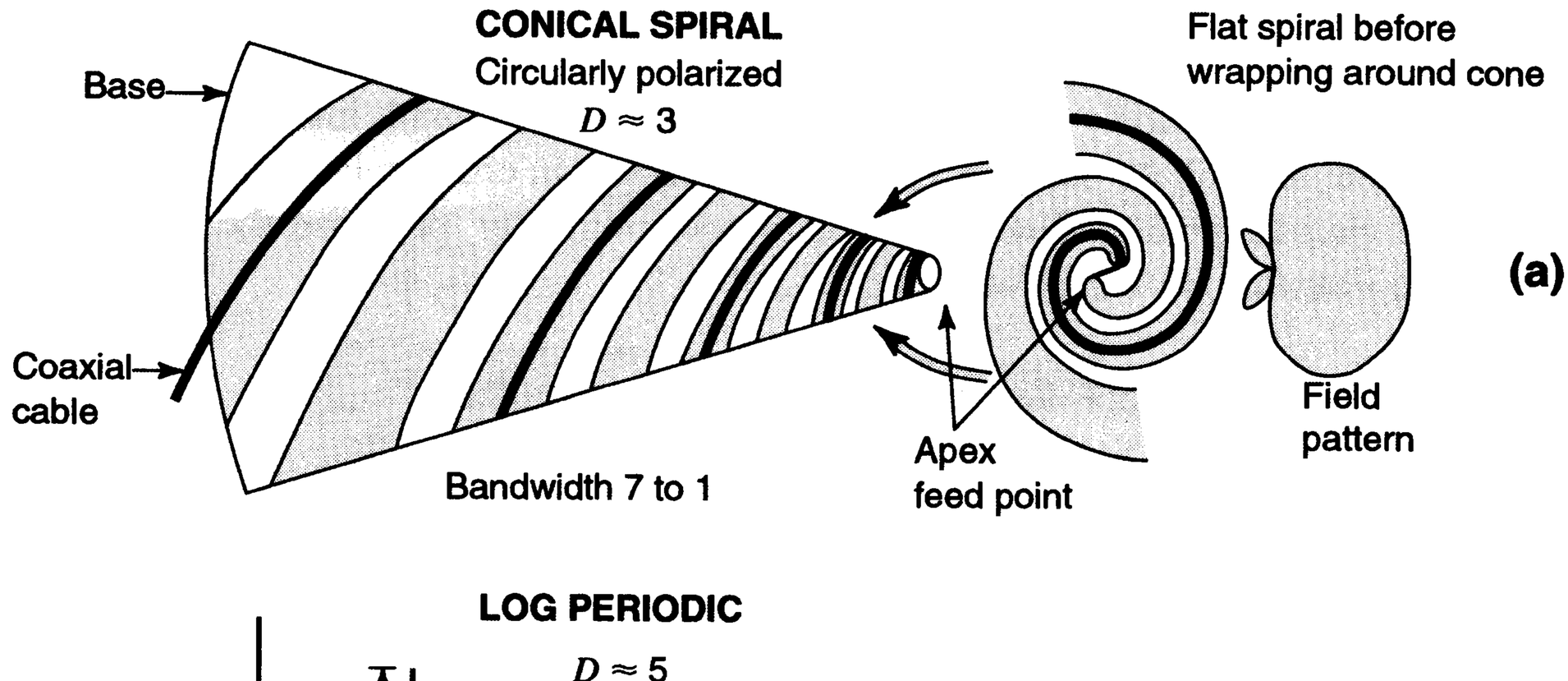


Figure 3–10
The conical spiral and log periodic are very broad bandwidth types with moderate gain.

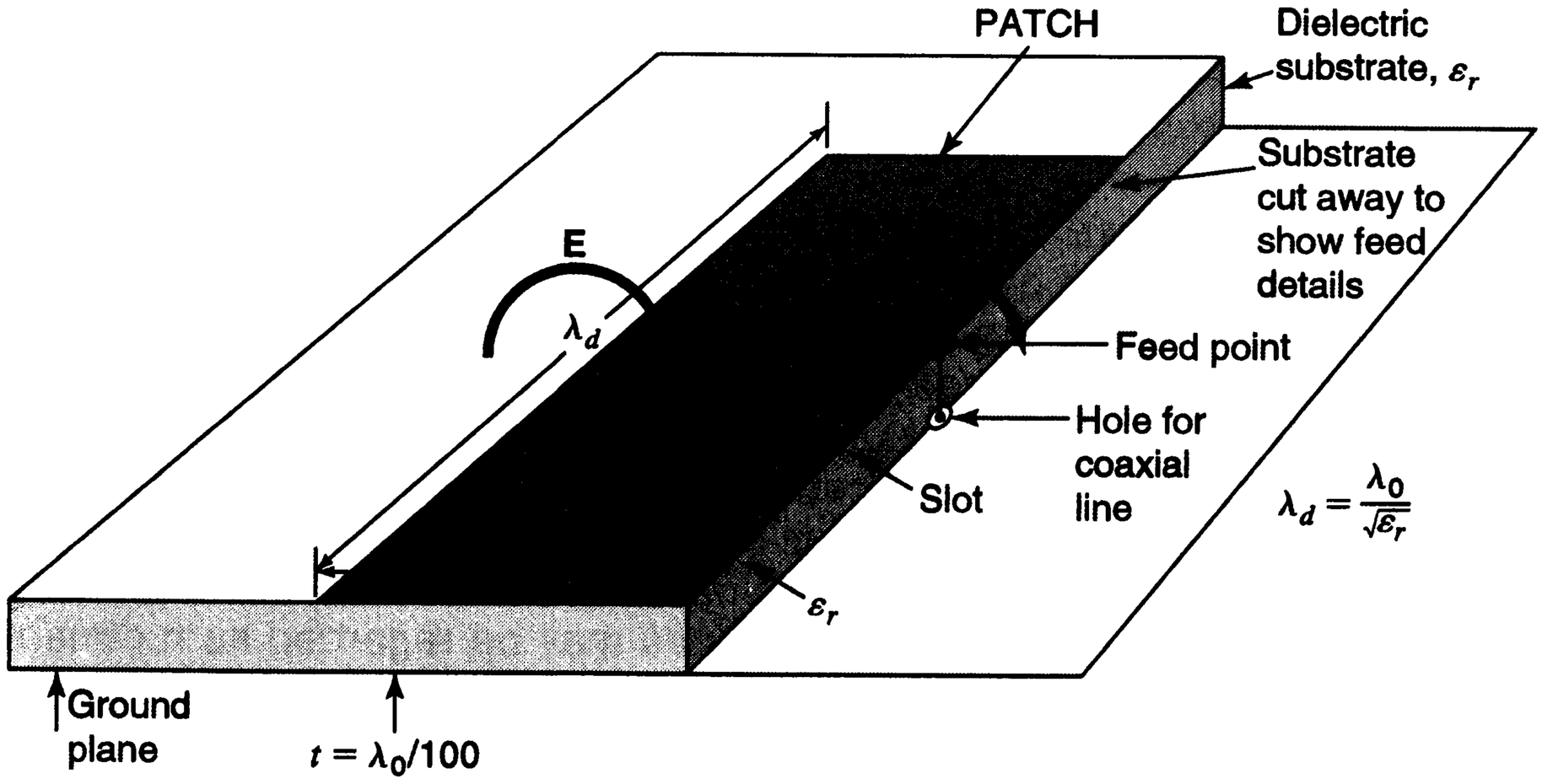


Figure 3–11
Single patch antenna. Arrows show the direction of **E** at the slots.

The Patch Array

EXAMPLE **3–10.1 Four-Patch Array**

This four-patch array shown in Fig. 3–12 is fed in-phase by the matching network shown. Find (*a*) the directivity and (*b*) the beam area.

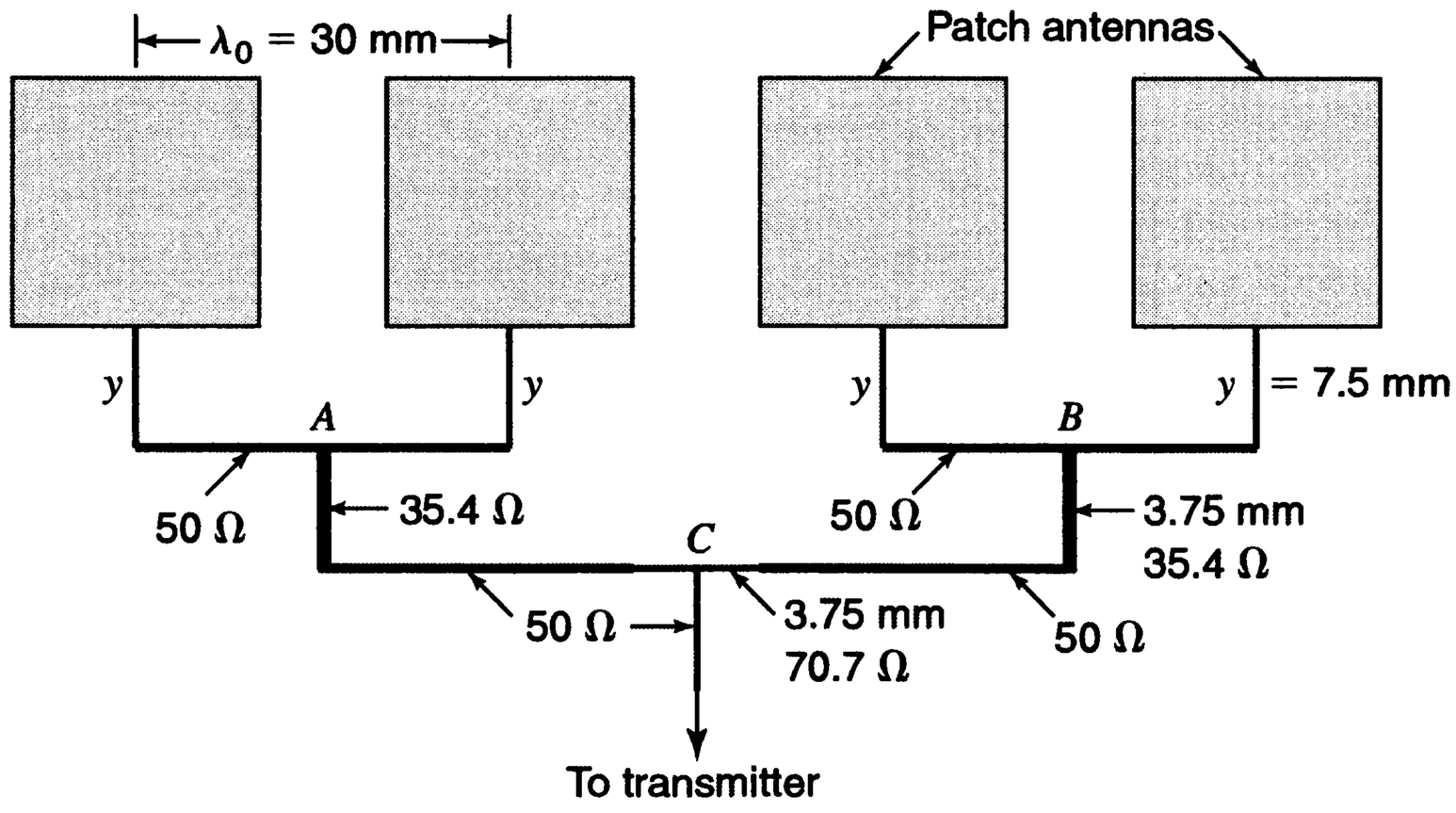


Figure 3–12
Four-patch antenna array.

■ **Solution**

$$A_e \cong 4\lambda \times \frac{\lambda}{2} = 2\lambda^2 \qquad D = \frac{4\pi A_e}{\lambda^2} = 8\pi \cong 25 \text{ (14 dBi)} \qquad \textit{Ans. (a)}$$

$$\Omega_A = \frac{4\pi}{D} = \frac{4\pi}{25} \cong 0.50 \text{ sr} \qquad \textit{Ans. (b)}$$

The Grid Flat-Panel Array

The grid array (Fig. 3–13), measuring $4 \times 2\frac{1}{2}\lambda$, has a directivity $D = 70$ (18.5 dBi) at 75 percent aperture efficiency. It has low side lobes and a VSWR < 1.5 over a 10 percent bandwidth.

The entire array is fed very simply at one point (F) by a 50-Ω coaxial cable with inner conductor through a hole in the ground plane and with the outer conductor bonded to the ground plane.

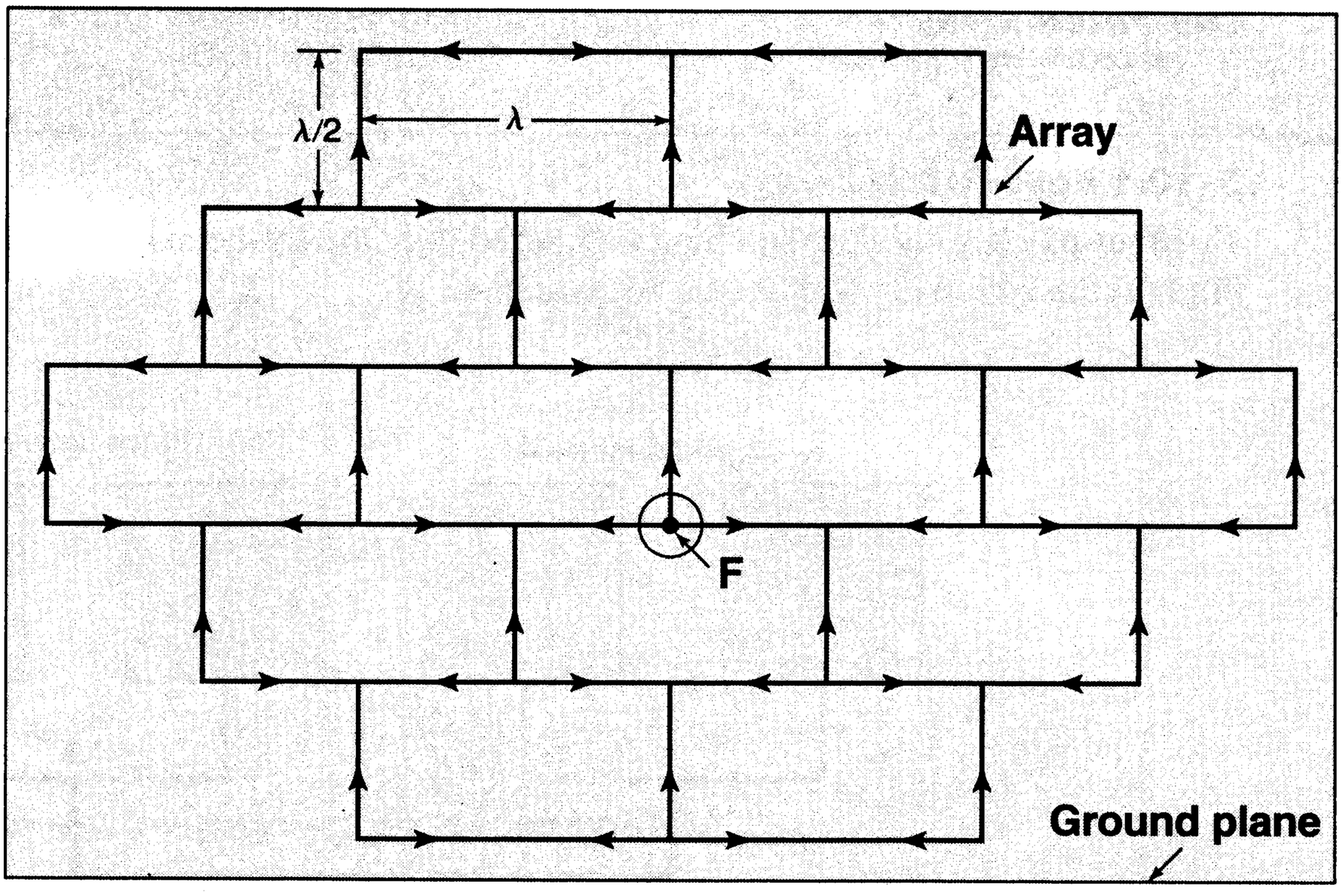


Figure 3–13
Grid flat-panel array with arrows showing the instantaneous current distribution. The fields of the 19 vertical half-wave conductors all add in-phase whereas the fields of the horizontal conductors cancel. The array is mounted on a dielectric substrate $\lambda/4$ above a flat ground plane.

REFERENCES（参考文献）

Baker, D. E. (1), and C. A. Van der Neut: "A Compact, Broadband, Balanced Transmission Line Antenna from Double-Ridged Waveguide," *IEEE. Ant. Prop. Soc. Symp.*, pp. 568–570. Albuquerque, 1982.

Kerr, J. L. (1): "Short Axial Length Broad Band Horns," *IEEE Trans. Ants. Prop.*, **AP-21,** 710–714, September 1973.

Shin, J. (1), and D. H. Schaubert: "A Parametric Study of Stripline-Fed Vivaldi Notch-Antenna Arrays," *IEEE Trans. Ant. Prop.*, **47,** 879–886, May 1999.

PROBLEMS（习题）

***3–2–1**（缝隙尺寸与阻抗）
Slot dimensions and impedance. A thin (diameter $\to 0$) dipole has a terminal impedance of $73 + j53.6\ \Omega$. What are (*a*) the dimensions of the complementary slot, (*b*) the slot impedance, and (*c*) the directivity?

3–4–1（阿尔卑斯喇叭天线）
Alpine-horn antenna. Referring to Fig. 3–4*a*, the low-frequency limit occurs when the open-end spacing $> \lambda/2$ and the high-frequency limit when the transmission-line spacing $d \approx \lambda/4$. If $d = 2$ mm and the open-end spacing $= 1000d$, what is the bandwidth?

***3–4–2**（阿尔卑斯喇叭天线）
Alpine-horn antenna. If $d =$ transmission-line spacing, what open-end spacing is required for a 200-to-1 bandwidth?

***3–5–1**（喇叭天线）
Horn antenna. (*a*) Find the physical aperture of a pyramidal horn antenna with $L = 10\lambda$, δ (E plane) $= 0.25\lambda$, δ (H plane) $= 0.45\lambda$; (*b*) Assuming a hypothetical 100 percent aperture

efficiency, find the directivity. This value is an upper limit. (*c*) Realistically, the aperture efficiency might be 60 percent, resulting in what directivity?

3-5-2 (矩形喇叭天线) **Rectangular horn antenna.** What is the required aperture area for an optimum rectangular horn antenna operating at 2 GHz with 16 dBi gain?

3-5-3 (圆锥形喇叭天线) **Conical horn antenna.** What is the required diameter of a conical horn antenna operating at 3 GHz with 14 dBi gain?

3-6-1 (夹角反射器) **Corner reflector.** Assuming a directivity $D = 20$ for the corner reflector of Fig. 3–7*c*, what are the HPBWs? The H-plane field is given by Eq. 10–3–6,

$$\cos(S_r \cos\phi) - \cos(S_r \sin\phi)$$

where $S_r = 2\pi S/\lambda$ and $S =$ dipole to corner spacing. *Hint:* Use Eq. 2–7–9.

3-7-1 (直播卫星家用抛物面) **Direct broadcast satellite home parabolic dish.** What is the directivity of a 460-mm circular parabolic dish at 12.5 GHz assuming an aperture efficiency of 75 percent?

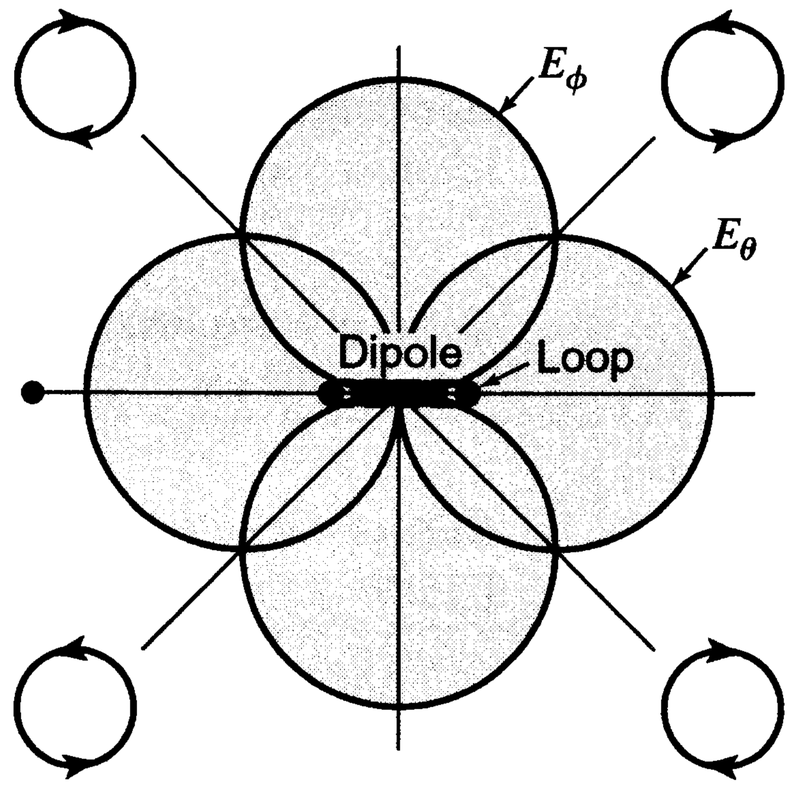
3-7-2 (波束宽度与定向性) **Beamwidth and directivity.** For most antennas, the half-power beamwidth (HPBW) may be estimated as HPBW $= \kappa\lambda/D$, where λ is the operating wavelength, D is the antenna dimension in the plane of interest, and κ is a factor which varies from 0.9 to 1.4, depending on the field amplitude taper across the antenna. Using this approximation, find the directivity and gain for the following antennas: (*a*) circular parabolic dish with 2 m radius operating at 6 GHz, (*b*) elliptical parabolic dish with dimensions of 1 m × 10 m operated at 1 GHz. Assume $\kappa = 1$ and 50 percent efficiency in each case.

For computer programs, see Appendix C.

CHAPTER 4

Point Sources

点源



Topics in this chapter include:

- Point source radiators（点源辐射器）
- Power patterns（功率波瓣图）
- Isotropic sources（各向同性源 ）
- Radiation intensity（辐射强度）
- Examples of power patterns（功率波瓣图举例）
- Field patterns（场波瓣图）
- Phase patterns（相位波瓣图）

4–1 INTRODUCTION. POINT SOURCE DEFINED（引言：点源的定义）

At a sufficient distance in the far field of an antenna, the radiated fields of the antenna are transverse and the power flow or Poynting vector (W m^{-2}) is radial as at the point O at a distance R on the observation circle in Fig. 4–1. It is convenient in many analyses to assume that the fields of the antenna are everywhere of this type. In fact, we may assume, by extrapolating inward along the radii of the circle, that the waves originate at a fictitious volumeless emitter, or *point source,* at the center O of the observation circle. The actual field variation near the antenna, or "near field," is ignored, and we describe the source of the waves only in terms of the "far field" it produces. Provided that our observations are made at a sufficient distance, any antenna, regardless of its size or complexity, can be represented in this way by a single point source.

Instead of making field measurements around the observation circle with the antenna fixed, the equivalent effect may be obtained by making the measurements at a fixed point Q on the circle and rotating the antenna around the center O. This is usually the more convenient procedure if the antenna is small.

In Fig. 4–1*a*, the center O of the antenna coincides with the center of the observation circle. If the center of the antenna is displaced from O, even to the extent that O lies outside the antenna as in Fig. 4–1*b*, the distance d between the two centers has a negligible effect on the field patterns at the observation circle, provided $R \gg d$, $R \gg b$, and $R \gg \lambda$. However, the phase patterns will generally differ, depending on d. If $d = 0$, the phase shift

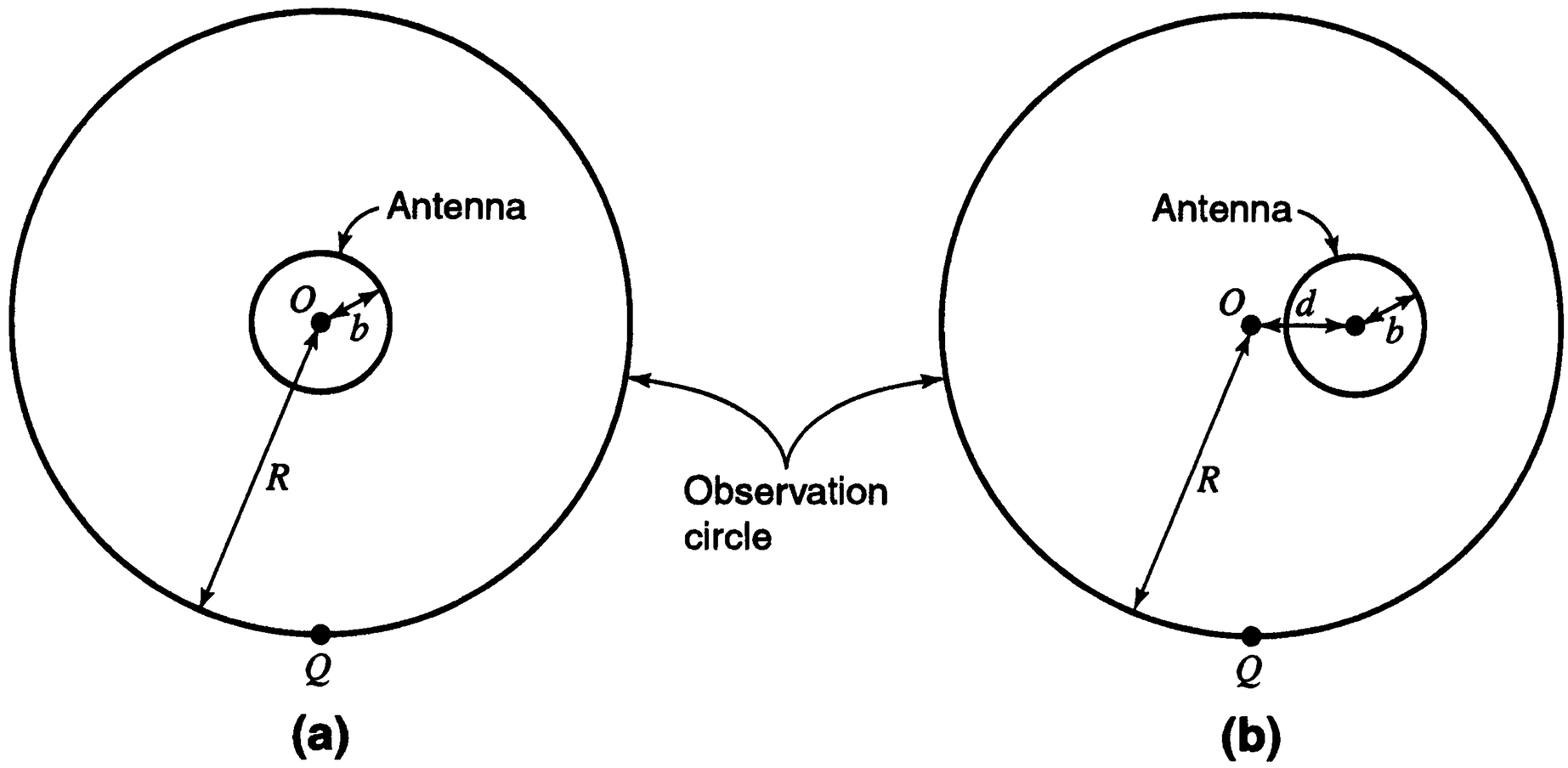


Figure 4–1
Antenna and observation circle.

around the observation circle is usually a minimum. As d is increased, the observed phase shift becomes larger.

As discussed in Sec. 2–3, a complete description of the far field of a source requires three patterns: two patterns of orthogonal field components as a function of angle [$E_\theta(\theta, \phi)$ and $E_\phi(\theta, \phi)$] and one pattern of the phase difference of these fields as a function of angle [$\delta(\theta, \phi)$]. For many purposes, however, such a complete knowledge is not necessary. It may suffice to specify only the variation with angle of the power density or Poynting vector magnitude (power per unit area) from the antenna [$S_r(\theta, \phi)$]. In this case the vector nature of the field is disregarded, and the radiation is treated as a scalar quantity. This is done in Sec. 4–2. The vector nature of the field is recognized later in the discussion on the magnitude of the field components. Although the cases considered as examples in this chapter are hypothetical, they could be approximated by actual antennas.

4–2 POWER PATTERNS（功率波瓣图）

Let a transmitting antenna in free space be represented by a point-source radiator located at the origin of the coordinates in Fig. 4–2 (see also Fig. 2–5). The radiated energy streams from the source in radial lines. The time rate of energy flow per unit area is the *Poynting vector,* or *power density* (watts per square meter). For a point source (or in the far field of any antenna), the Poynting vector **S** has only a radial component S_r with no components in either the θ or ϕ directions ($S_\theta = S_\phi = 0$). Thus, the magnitude of the Poynting vector, or power density, is equal to the radial component ($|\mathbf{S}| = S_r$).

A source that radiates energy uniformly in all directions is an *isotropic source.* For such a source the radial component S_r of the Poynting vector is independent of θ and ϕ. A graph of S_r at a constant radius as a function of angle is a Poynting vector, or power-density, pattern, but is usually called a *power pattern.* The three-dimensional power pattern for an isotropic source is a sphere. In two dimensions the pattern is a circle (a cross section through the sphere), as suggested in Fig. 4–3.

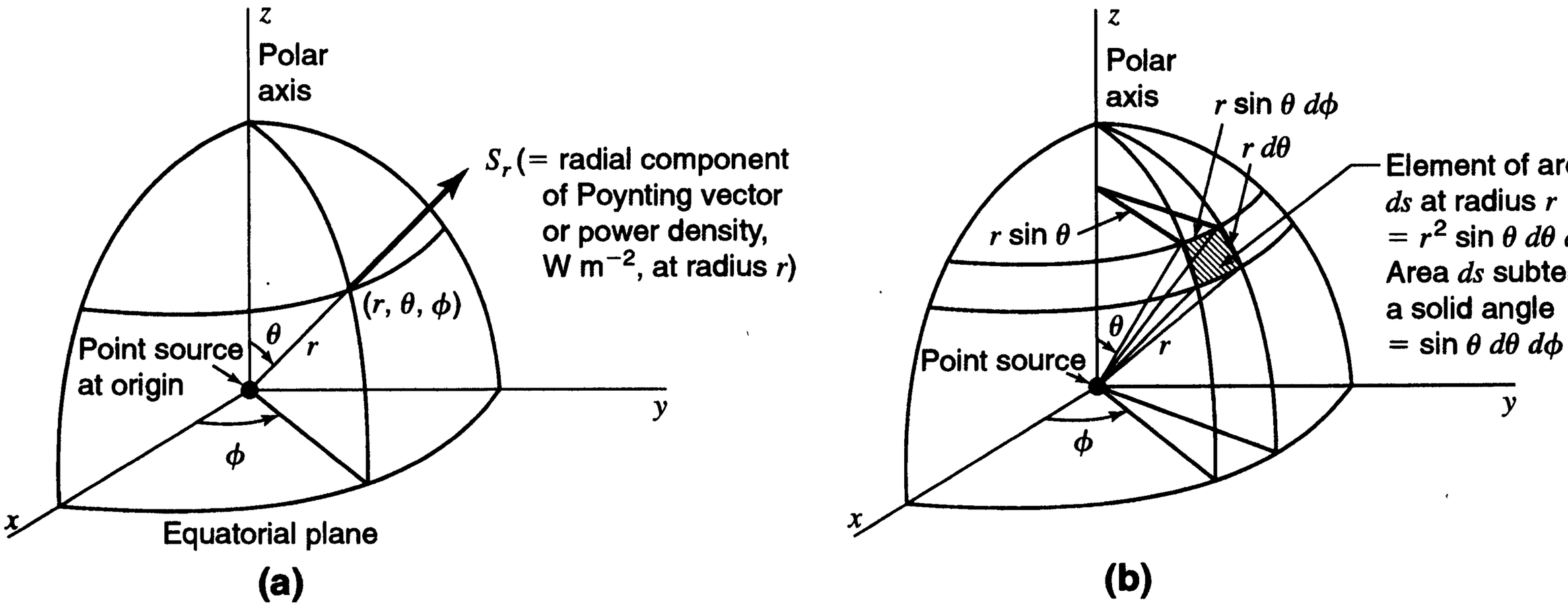


Figure 4–2
Spherical coordinates for a point source of radiation in free space.

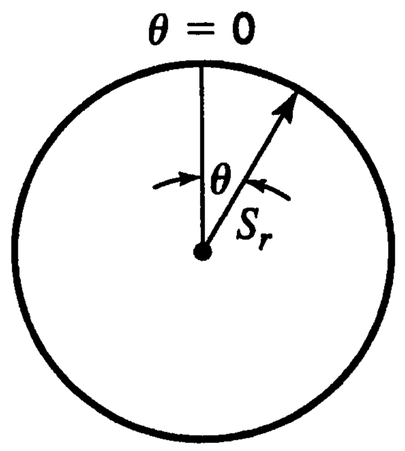


Figure 4–3
Polar power pattern of isotropic source.

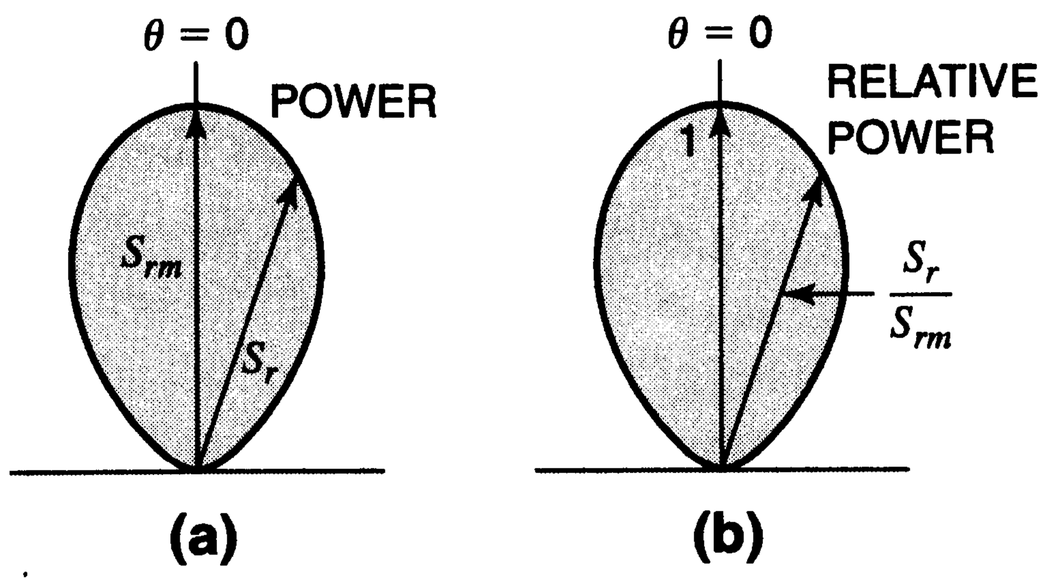


Figure 4–4
(*a*) Power pattern and (*b*) relative power pattern for same source. Both patterns have the same shape. The relative power pattern is normalized to a maximum of unity (1).

Although the isotropic source is convenient in theory, it is not a physically realizable type. Even the simplest antennas have directional properties, i.e., they radiate more energy in some directions than in others. In contrast to the isotropic source, they might be called *anisotropic* sources. As an example, the power pattern of such a source is shown in Fig. 4–4*a* where S_{rm} is the maximum value of S_r.

If S_r is expressed in watts per square meter, the graph is an *absolute power pattern.* On the other hand, if S_r is expressed in terms of its value in some reference direction, the graph is a *relative power pattern.* It is customary to take the reference direction such that S_r is a maximum. Thus, the pattern radius for relative power is S_r/S_{rm} where S_{rm} is the maximum value of S_r. The maximum value of the relative power pattern is unity, as shown in Fig. 4–4*b*. A pattern with a maximum of unity is also called a *normalized pattern.*

4–3 A POWER THEOREM[1] AND ITS APPLICATION TO AN ISOTROPIC SOURCE（功率定理及其应用于个别各向同性源）

If the Poynting vector is known at all points on a sphere of radius r from a point source in a lossless medium, *the total power radiated by the source is the integral over the surface of the sphere of the radial component S_r of the average Poynting vector.* Thus,

$$P = \oiint S \cdot ds = \oiint S_r \, ds \tag{1}$$

where

P = power radiated, W
S_r = radial component of average Poynting vector, W m^{-2}
ds = infinitesimal element of area of sphere (see Fig. 3–2*b*)
$= r^2 \sin\theta \, d\theta \, d\phi$, m^2

For an *isotropic source,* S_r is independent of θ and ϕ so

$$P = S_r \oiint ds = S_r \times 4\pi r^2 \qquad \text{(W)} \tag{2}$$

and

$$S_r = \frac{P}{4\pi r^2} \qquad (\text{W m}^{-2}) \tag{3}$$

Equation (3) indicates that the magnitude of the Poynting vector varies inversely as the square of the distance from a point-source radiator. This is a statement of the well-known law for the variation of power per unit area as a function of the distance.

4–4 RADIATION INTENSITY（辐射强度）

As discussed in Sec. 2–5, the radiation intensity **U** is expressed in watts per unit solid angle (W sr^{-1}). The radiation intensity is independent of radius. It is power per steradian. From (4–3–3) we have

$$r^2 S_r = P/4\pi = U \qquad \text{(W/sr)} \tag{1}$$

---

[1]This theorem is a special case of a more general relation for the complex power flow through any closed surface as given by

$$P = \tfrac{1}{2} \oiint (\mathbf{E} \times \mathbf{H}^*) \cdot ds \tag{1}$$

where P is the total complex power flow and **E** and **H*** are complex vectors representing the electric and magnetic fields, **H*** being the complex conjugate of **H**. The average Poynting vector is

$$S = \tfrac{1}{2} \operatorname{Re}(\mathbf{E} \times \mathbf{H}^*) \tag{2}$$

Now the power flow in the far field is entirely real; hence, taking the real part of (1) and substituting (2), we obtain the special case of (3).

Thus, the power theorem may be restated as follows:

> ***The total power radiated is given by the integral of the radiation intensity over a solid angle of 4π steradians.***
>
> **[点源所辐射的总功率是其辐射强度在 4π 立体角（立体弧度）上的积分]**

As already mentioned in Sec. 2–5, power patterns can be expressed in terms of either the Poynting vector (power density) or the radiation intensity. A power pattern in terms of U is the same as in Fig. 4–4*a* with the maximum Poynting vector (S_m) replaced by the maximum radiation intensity (U_m) and the Poynting vector as a function of r (S_r) replaced by the radiation intensity as a function of $r(U_r)$. The maximum value of U_m is in the $\theta = 0$ direction. Relative Poynting vector (or power density) and relative radiation intensity patterns are identical.

Applying (1) to an isotropic source gives

$$P = 4\pi U_0 \qquad \text{(W)}$$

where U_0 = radiation intensity of isotropic source, W sr^{-1}.

4–5 EXAMPLES OF POWER PATTERNS（功率波瓣图举例）

EXAMPLE **4–5.1 Source with Unidirectional Cosine Power Pattern**

A source has a cosine radiation-intensity pattern, that is,

$$U = U_m \cos\theta \tag{1}$$

where U_m = maximum radiation intensity

The radiation intensity U has a value only in the upper hemisphere ($0 \le \theta \le \pi/2$ and $0 \le \phi \le 2\pi$) and is zero in the lower hemisphere. The radiation intensity is a maximum at $\theta = 0$. The pattern is shown in Fig. 4–5. The space pattern is a figure of revolution of this circle around the polar axis. Find the directivity.

To find the total power radiated by the cosine source, we apply (4–3–1) and integrate only over the upper hemisphere. Thus

$$P = \int_0^{2\pi} \int_0^{\pi/2} U_m \cos\theta \sin\theta \, d\theta \, d\phi = \pi U_m \tag{2}$$

If the power radiated by the unidirectional cosine source is the same as for an isotropic source, then (2) and (1) in Sec. 4–4 may be set equal, yielding

$$\pi U_m = 4\pi U_0$$

or

$$\text{Directivity} = \frac{U_m}{U_0} = 4 = D \qquad \textit{Ans.} \tag{3}$$

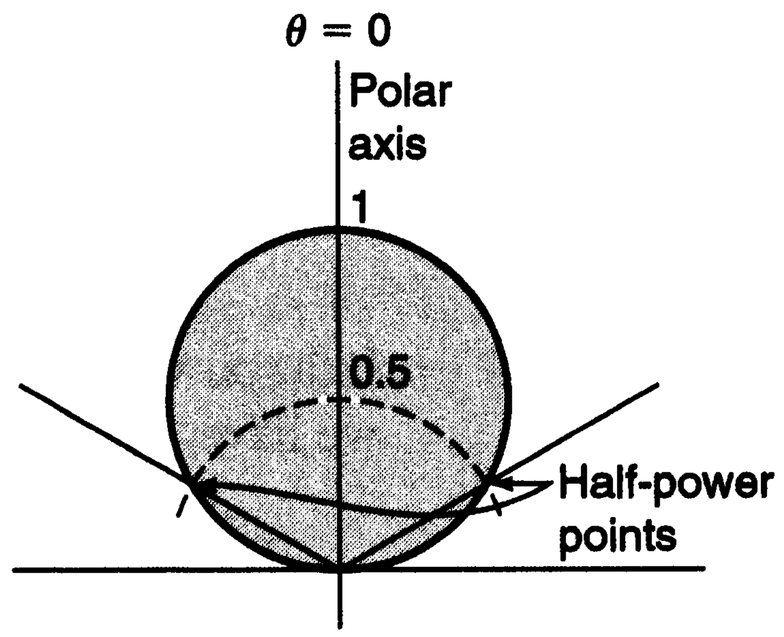


Figure 4–5
Unidirectional cosine power pattern.

Thus, the maximum radiation intensity U_m of the unidirectional cosine source (in the direction $\theta = 0$) is 4 times the radiation intensity U_0 from an isotropic source radiating the same total power. The power patterns for the two sources are compared in Fig. 4–6 for the same total power radiated by each.

EXAMPLE **4–5.2 Source with Bidirectional Cosine Power Pattern**

A source has a cosine power pattern that is bidirectional. Find the directivity. With radiation in two hemispheres instead of one; the maximum radiation intensity is half its value in Example 4–5.1. Thus, from (3),

$$D = 4/2 = 2 \qquad Ans.$$

EXAMPLE **4–5.3 Source with Sine (Doughnut) Power Pattern**

A source has a radiation intensity pattern given by

$$U = U_m \sin\theta \tag{4}$$

The pattern is shown in Fig. 4–6. The space pattern is a figure-of-revolution of this pattern around the polar axis and has the form of a doughnut. Find D.

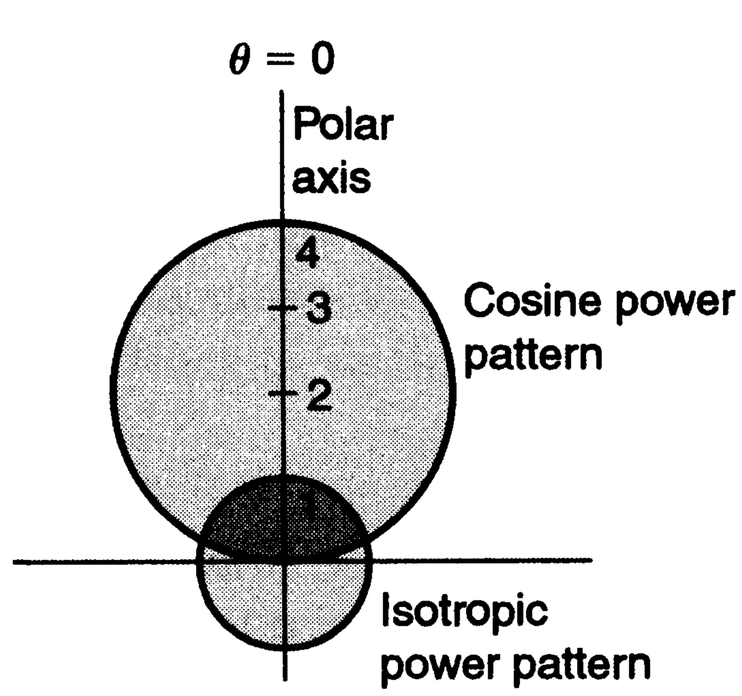


Figure 4–6
Power patterns of cosine and isotropic sources.

■ **Solution**

Applying (4–3–3) the total power radiated is

$$P = U_m \int_0^{2\pi} \int_0^{\pi} \sin^2 \theta \, d\theta \, d\phi = \pi^2 U_m \tag{5}$$

If the power radiated by this source is the same as for an isotropic source taken as reference, we have

$$\pi^2 U_m = 4\pi U_0 \tag{6}$$

and

$$\text{Directivity} = \frac{U_m}{U_0} = \frac{4}{\pi} = 1.27 = D \qquad \textit{Ans.} \tag{7}$$

EXAMPLE **4–5.4 Source with Sine-Squared (Doughnut) Power Pattern**

A source has a sine-squared radiation-intensity power pattern. The radiation-intensity pattern is given by

$$U = U_m \sin^2 \theta \tag{8}$$

The power pattern is shown in Fig. 4–7*a*. This type of pattern is of considerable interest because it is the pattern produced by a short dipole coincident with the polar ($\theta = 0$) axis in Fig. 4–7*a*. Applying (4–3–3), the total power radiated is

$$P = U_m \int_0^{2\pi} \int_0^{\pi} \sin^3 \theta \, d\theta \, d\phi = \frac{8}{3}\pi U_m \tag{9}$$

If P is the same as for the isotropic source,

$$\frac{8}{3}\pi U_m = 4\pi U_0$$

and

$$\text{Directivity} = \frac{U_m}{U_0} = \frac{3}{2} = 1.5 = D \qquad \textit{Ans.} \tag{10}$$

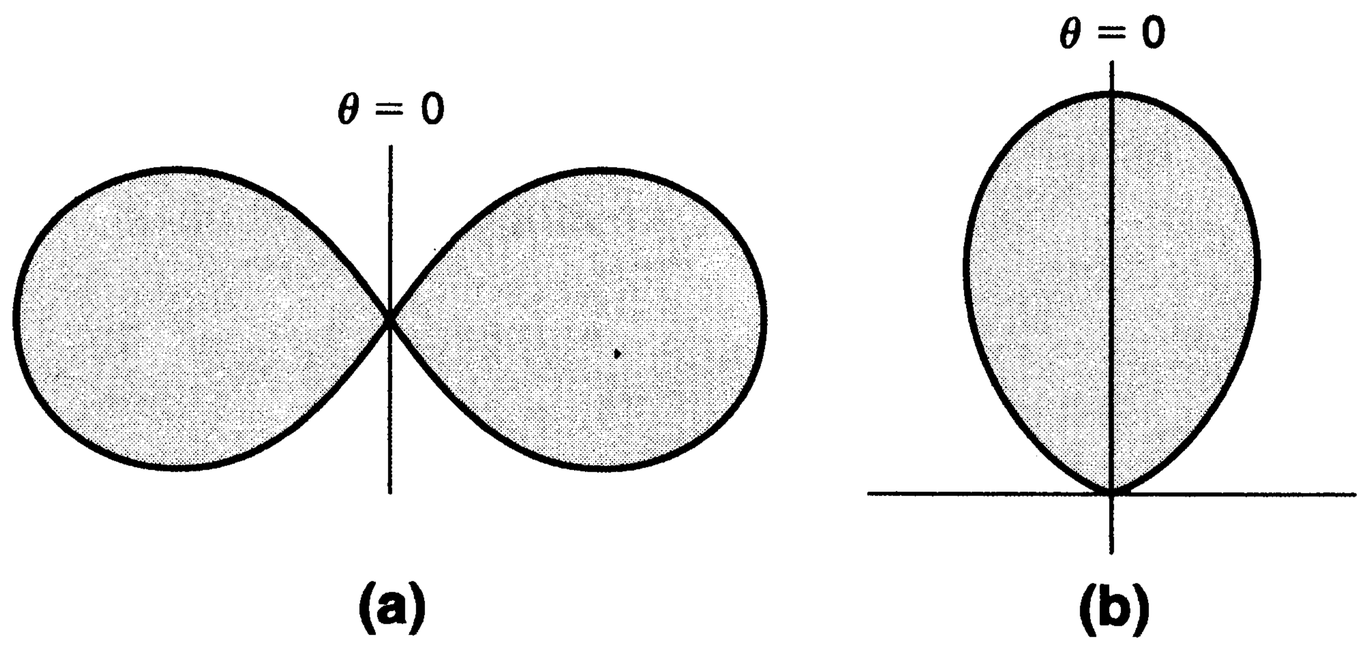


Figure 4–7
(*a*) Sine-squared power pattern and (*b*) unidirectional cosine-squared power pattern.

EXAMPLE **4–5.5 Source with Unidirectional Cosine-Squared Power Pattern**

A source with a unidirectional cosine-squared radiation-intensity pattern is given by

$$U = U_m \cos^2 \theta \tag{11}$$

The radiation intensity has a value only in the upper hemisphere as in Fig. 4–7*b*. The three-dimensional or space pattern is a figure-of-revolution of this pattern around the polar ($\theta = 0$) axis. Find the directivity.

■ **Solution**

The total power radiated is

$$P = U_m \int_0^{2\pi} \int_0^{\pi/2} \cos^2 \theta \sin \theta \, d\theta \, d\phi = \frac{2}{3} \pi U_m \tag{12}$$

If P is the same as radiated by an isotropic source,

$$\frac{2}{3} \pi U_m = 4\pi U_0$$

and

$$\text{Directivity} = \frac{U_m}{U_0} = 6 = D \qquad \textit{Ans.} \tag{13}$$

Thus, the maximum power per unit solid angle (at $\theta = 0$) from the source with the cosine-squared power pattern is six times the power per unit solid angle from an isotropic source radiating the same power.

Directivities are summarized in Table 4–1.

Table 4–1 Directivities of the Point Source Patterns in Examples 4–5.1 to 4–5.5.

Pattern	Directivity
Unidirectional cosine	4
Bidirectional cosine	2
Sine doughnut	1.27
Sine-squared doughnut	1.5
Unidirectional cosine squared	6

Example 4–5.6 provides some valuable insights into the effect minor lobes have on directivity or gain. Without minor lobes the gain of this antenna would be 91.4 or 19.6 dBi as compared to a gain of 18.0 or 12.6 dBi with minor lobes. The minor lobes have large beam or solid angles because they extend 360° in the azimuth or ϕ direction at large sin θ values (θ near 90°). The main lobe, on the other hand, is at small θ angles so the $P_n(\theta)$ sin θ product is small, in fact, zero at $\theta = 0°$.

EXAMPLE **4–5.6 Pencil Beam with Minor Lobes**

As shown in Fig. 4–8, the pattern has pencil beam (symmetrical around the $\theta = 0$ axis) with a main-lobe HPBW of approximately 22° and four minor lobes. Find the directivity.

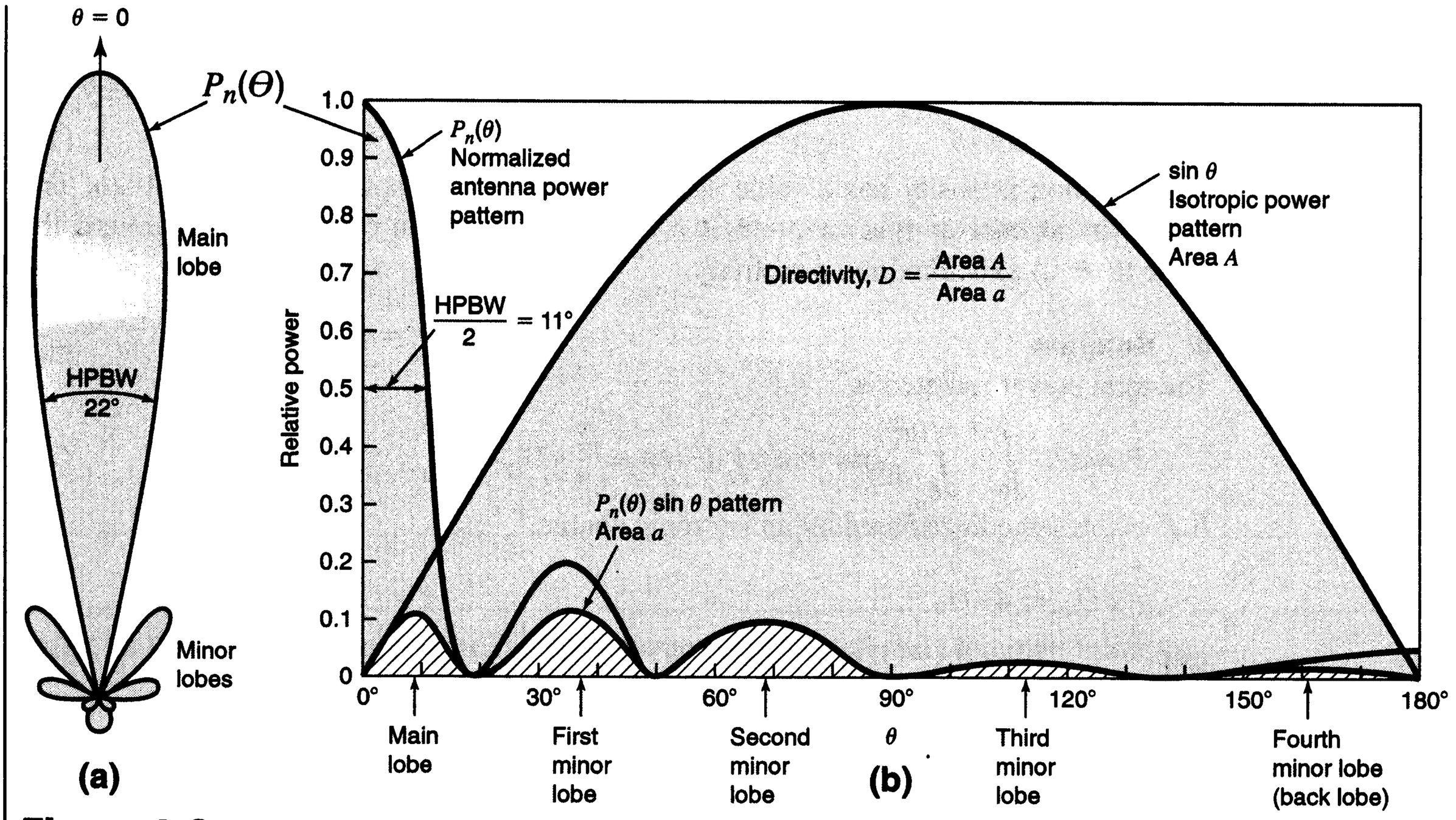


Figure 4–8
Power patterns of beam antenna in polar plot at (*a*) and in rectangular plot at (*b*). The large shaded area *A* of (*b*) is for an isotropic source while the antenna area *a* appears as a series of small areas. The directivity $D = A/a$.

■ **Solution**

The directivity is given by

$$D = \frac{4\pi}{\int_0^{2\pi} \int_0^{\pi} P_n(\theta) \sin\theta \, d\theta \, d\phi} \tag{14}$$

where the denominator equals the total beam area Ω_A.

Since the pattern is symmetrical (no variation with ϕ), the integral with respect to ϕ yields 2π and (14) reduces to

$$D = \frac{4\pi}{2\pi \int_0^{\pi} P_n(\theta) \sin\theta \, d\theta} \tag{15}$$

We have only the pattern graph available (no analytical expression), so let us divide the pattern (Fig. 4–8) into 36 steps of 5° each. The approximate value of the integral in the first ($m = 1$) 5° section ($= \pi/36$ rad) is given by

$$\frac{\pi}{36} P_n(\theta_1)_{\text{av}} \sin\theta_1 = \frac{\pi}{36} \frac{1.0 + 0.93}{2} \sin 2.5° \tag{16}$$

and the approximate directivity is then given by the summation of all 36 sections or by

$$D \simeq \frac{4\pi}{2\pi(\pi/36) \sum_{m=1}^{m=36} P_n(\theta_m)_{\text{av}} \sin\theta_m} \tag{17}$$

Completing the summation, we obtain

$$D = \frac{4\pi}{\Omega_A} \simeq \frac{4\pi}{2\pi(\pi/36)(0.25 + 0.37 + 0.46 + 0.12 + 0.07)} = \frac{72}{1.27\pi} = 18.0 \quad (18)$$

Main lobe; First minor lobe; Second minor lobe; Third minor lobe; Fourth minor lobe (back lobe)

or $D \simeq 12.6$ dBi

It is noteworthy that the second minor lobe contributes most to the total beam area, the first minor lobe almost as much, and the main lobe less than either. Thus, the directivity is greatly affected by the minor lobes, which is a common situation with actual antennas. For this antenna pattern the beam efficiency is given by

$$\varepsilon_M = \frac{0.25}{1.27} = 0.20 \quad (19)$$

If the second minor lobe were eliminated, the directivity would increase to 14.5 dBi (up 1.9 dB) and if both first and second minor lobes were eliminated, the directivity would increase to 17.1 dBi (up 4.5 dB).

The directivity obtained in Example 4–5.6 is approximate. By sufficiently reducing the step size (5° in the example), the summation can be made as precise as the available data will allow. Computation of this numerical integration can be facilitated by using a computer.

The half-power beamwidth of the pattern in the example is about 22°. Taking $k_p = 1$ and ε_M as in (19), the approximate directivity is then

$$D \simeq \frac{41{,}000\varepsilon_M}{k_p \times \text{HPBW}^2} = \frac{41{,}000 \times 0.2}{(22°)^2} = 16.9 \text{ or } 12.3 \text{ dBi} \quad (20)$$

which is 0.3 dB less than obtained by the 36-step summation.

The beam area of an isotropic source equals 4π steradians. In Fig. 4–8*b* this corresponds to the area A under the $\sin\theta$ curve. The beam area of the source in Example 4–5.6 corresponds to the area a under the $P_n(\theta)\sin\theta$ curve. Thus, the directivity is simply A/a or the ratio of the area of the isotropic source to the area of the source being measured. Hence,

$$D = \frac{4\pi}{\Omega_A} = \frac{A}{a} \quad (21)$$

If the areas A and a are cut from a lead sheet of uniform thickness, the directivity equals the ratio of the weight of A to the weight of a.

4–6 FIELD PATTERNS（场波瓣图）

The discussion in the preceding sections is based on considerations of power. This has afforded a simplicity of analysis, since the power flow from a point source has only a radial

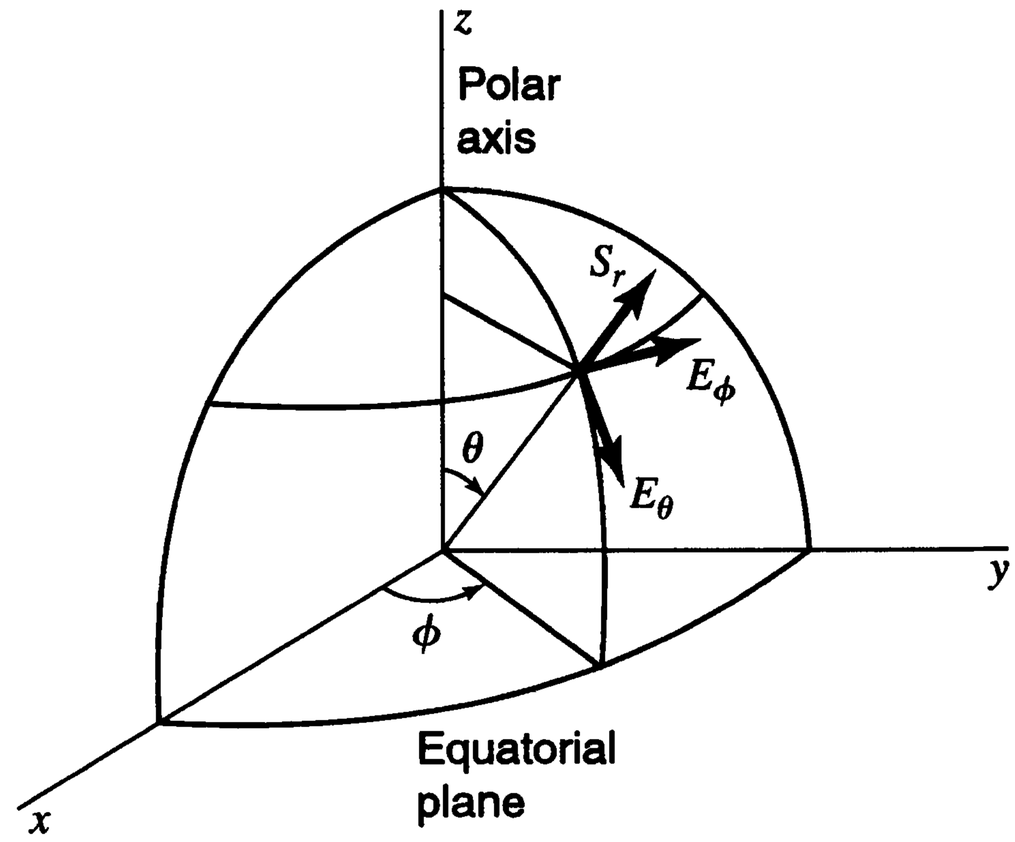


Figure 4–9
Relation of the Poynting vector S and the two electric field components of the far field.

component which can be considered as a scalar quantity. To describe the field of a point source more completely, we need to consider the electric field **E** and/or the magnetic field **H** (both vectors). For point sources we deal entirely with *far fields* so **E** and **H** are both entirely transverse to the wave direction, are perpendicular to each other, are in-phase, and are related in magnitude by the intrinsic impedance of the medium ($\mathbf{E}/\mathbf{H} = Z = 377\ \Omega$ for free space). For our purposes it suffices to consider only one field vector, and we arbitrarily choose the electric field **E**.

Since the Poynting vector around a point source is everywhere radial, it follows that the electric field is entirely transverse, having only E_θ and E_ϕ components. The relation of the radial component S_r of the Poynting vector and the electric field components is illustrated by the spherical coordinate diagram of Fig. 4–9. The conditions characterizing the far field are then:

1. Poynting vector radial (S_r component only)
2. Electric field transverse (E_θ and E_ϕ components only)

The Poynting vector and the electric field at a point of the far field are related in the same manner as they are in a plane wave, since, if r is sufficiently large, a small section of the spherical wave front may be considered as a plane.

The relation between the average Poynting vector and the electric field at a point of the far field is

$$S_r = \frac{1}{2}\frac{E^2}{Z} \tag{1}$$

where Z_0 = intrinsic impedance of medium and

$$E = \sqrt{E_\theta^2 + E_\phi^2} \tag{2}$$

where

E = amplitude of total electric field intensity
E_θ = amplitude of θ component
E_ϕ = amplitude of ϕ component

The field may be elliptically, linearly or circularly polarized.

If the field components are rms values, rather than amplitudes, the Poynting vector is twice that given in (1).

A pattern showing the variation of the electric field intensity at a constant radius r as a function of angle (θ, ϕ) is called a *field pattern*. In presenting information concerning the far field of an antenna, it is customary to give the field patterns for the two components, E_θ and E_ϕ, of the electric field since the total electric field **E** can be obtained from the components by (2), but the components cannot be obtained from a knowledge of only **E**.

When the field intensity is expressed in volts per meter, it is an *absolute field pattern*.[1] On the other hand, if the field intensity is expressed in units relative to its value in some reference direction, it is a *relative field pattern*. The reference direction is usually taken in the direction of maximum field intensity. The relative pattern of the E_θ component is then given by

$$\frac{E_\theta}{E_{\theta m}} \tag{3}$$

and the relative pattern of the E_ϕ component is given by

$$\frac{E_\phi}{E_{\phi m}} \tag{4}$$

where

$E_{\theta m}$ = maximum value of E_θ
$E_{\phi m}$ = maximum value of E_ϕ

The magnitudes of both the electric field components, E_θ and E_ϕ, of the far field vary inversely as the distance from the source. However, they may be different functions, F_1 and F_2, of the angular coordinates, θ and ϕ. Thus, in general,

$$E_\theta = \frac{1}{r} F_1(\theta, \phi) \tag{5}$$

$$E_\phi = \frac{1}{r} F_2(\theta, \phi) \tag{6}$$

Since $S_{rm} = E_m^2/2Z$, where E_m is the maximum value of E, it follows on dividing this into (1) that the relative total power pattern is equal to the square of the relative total field

[1]The magnitude depends on the radius, varying inversely as the distance ($E \propto 1/r$).

pattern. Thus,

$$P_n = \frac{S_r}{S_{rm}} = \frac{U}{U_m} = \left(\frac{E}{E_m}\right)^2 \tag{7}$$

EXAMPLE **4–6.1 Source with Cosine Field Pattern**

An antenna's far field has only an E_ϕ component in the equatorial plane, the E_θ component being zero in this plane. The relative equatorial-plane pattern of the E_ϕ component (that is, E_ϕ as a function of ϕ for $\theta = 90°$) is given by

$$\frac{E_\phi}{E_{\phi m}} = \cos\phi \tag{8}$$

This pattern is illustrated at the left of Fig. 4–10. The length of the radius vector in the diagram is proportional to E_ϕ. A pattern of this form could be produced by a short dipole coincident with the y axis. Find D.

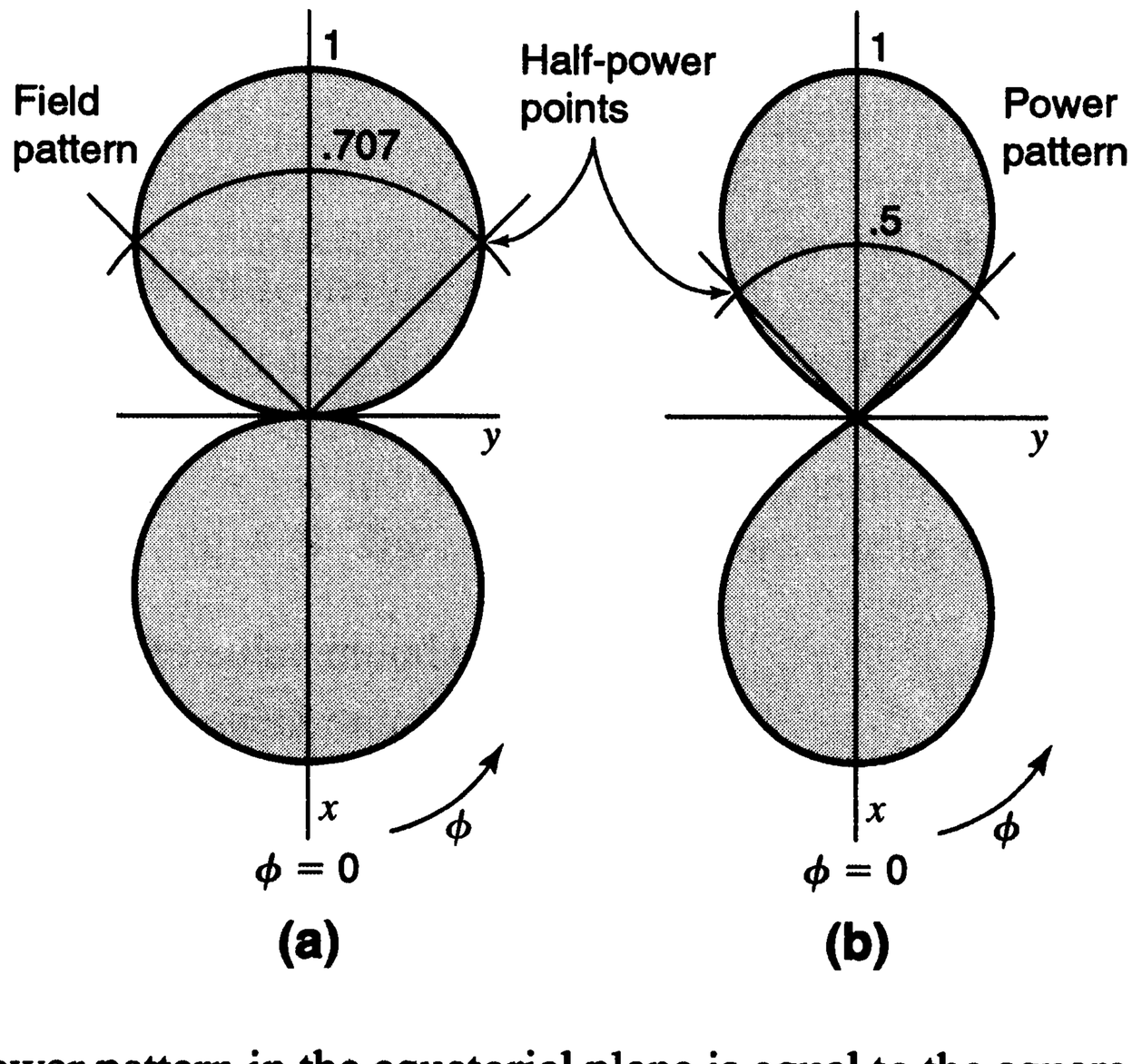


Figure 4–10
(*a*) Relative E_ϕ pattern of Example 4–6.1 and (*b*) the relative power pattern.

The relative (normalized) power pattern in the equatorial plane is equal to the square of the relative field pattern. Thus

$$P_n = \frac{S_r}{S_{rm}} = \frac{U}{U_m} = \left(\frac{E_\phi}{E_{\phi m}}\right)^2 \tag{9}$$

and substituting (8) into (9) we have

$$P_n = \cos^2\phi$$

This pattern is illustrated at the right of Fig. 4–10.

EXAMPLE **4–6.2 Source with Sine Field Pattern**

An antenna has a far field that has only an E_θ component in the equatorial plane, the E_ϕ component being zero in this plane. Assume that the relative equatorial-plane pattern of the E_θ component (that is, E_θ as a function of ϕ for $\theta = 90°$) for this antenna is given by

$$\frac{E_\theta}{E_{\theta m}} = \sin\phi \tag{10}$$

This pattern is illustrated by Fig. 4–11*a* and could be produced by a small loop antenna, the axis of the loop coincident with the x axis. Find D.

The relative (normalized) power pattern in the equatorial plane is

$$P_n = \sin^2\phi \tag{11}$$

This pattern is shown by Fig. 4–11.

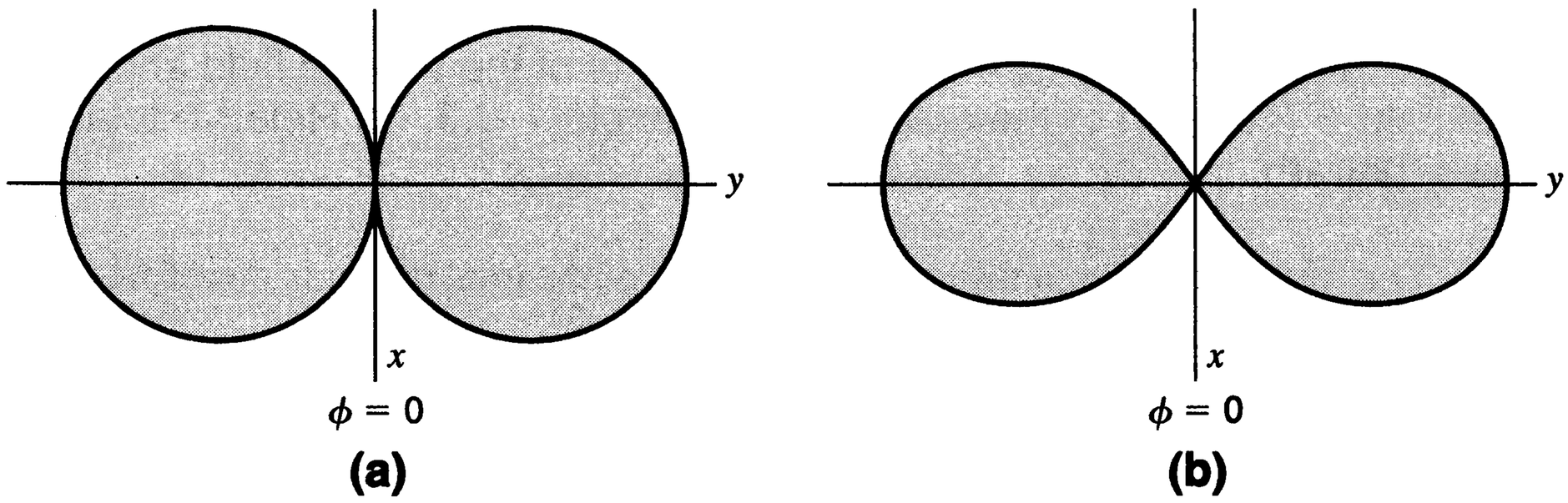


Figure 4–11
(*a*) Relative E_θ pattern of Example 4–6.2 and (*b*) the relative power pattern.

EXAMPLE **4–6.3 Short Dipole and Loop Patterns**

An antenna's far field has both E_θ and E_ϕ components in the equatorial plane ($\theta = 90°$). Suppose that this antenna is a composite of the two antennas we have just considered in Examples 4–6.1 and 4–6.2 and that equal power is radiated by each antenna. If both patterns are of identical shape in three dimensions as well as in the xy plane, as from a short dipole and a small loop, it then follows that at a radius r from the composite antenna, $E_{\theta m} = E_{\phi m}$. The individual patterns for the E_θ and E_ϕ components as given by (10) and (8) may then be shown to the same scale by one diagram, as in Fig. 4–12*a*. The relative pattern of the total field E is

$$\frac{E}{E_m} = \sqrt{\sin^2\phi + \cos^2\phi} = 1 \tag{12}$$

which is a circle as indicated by the dashed line in Fig. 4–12*a*. Find D.

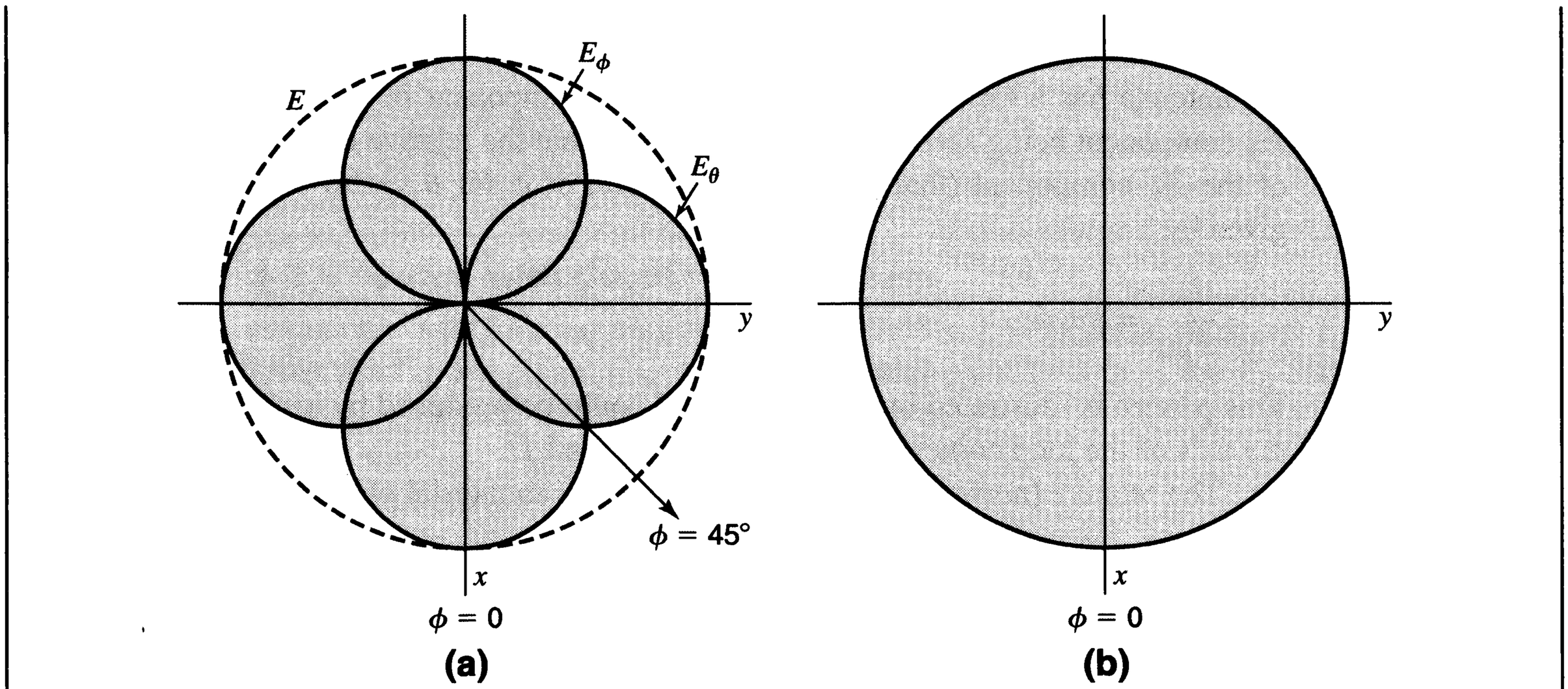


Figure 4–12
(*a*) Relative patterns of E_θ and E_ϕ components of the electric field and the total field E (dashed) of the antenna of Example 4–6.3. (*b*) Relative total power pattern.

The relative pattern in the equatorial plane for the total power is therefore a circle of radius unity as illustrated by Fig. 4–12*b*.

We note in Fig. 4–12*a* that at $\phi = 45°$ the magnitudes of the two field components, E_θ and E_ϕ, are equal. Depending on the time phase between E_θ and E_ϕ, the field in this direction could be plane, elliptically or circularly polarized, but regardless of phase the power is the same. To determine the type of polarization requires that the phase angle between E_θ and E_ϕ be known. This is discussed in the next section.

4–7 PHASE PATTERNS（相位波瓣图）

Assuming that the field varies harmonically with time and that the frequency is known, the far field in all directions from a source may be completely specified by a knowledge of the following four quantities:

1. Amplitude of the polar component E_θ of the electric field as a function of r, θ, and ϕ
2. Amplitude of the azimuthal component E_ϕ of the electric field as a function of r, θ, and ϕ
3. Phase lag δ of E_ϕ behind E_θ as a function of θ and ϕ
4. Phase lag η of either field component behind its value at a reference point as a function of r, θ, and ϕ

Since we regard the field of a point source as a far field everywhere, the above four quantities can be considered as those required for a complete knowledge of the field of a point source.

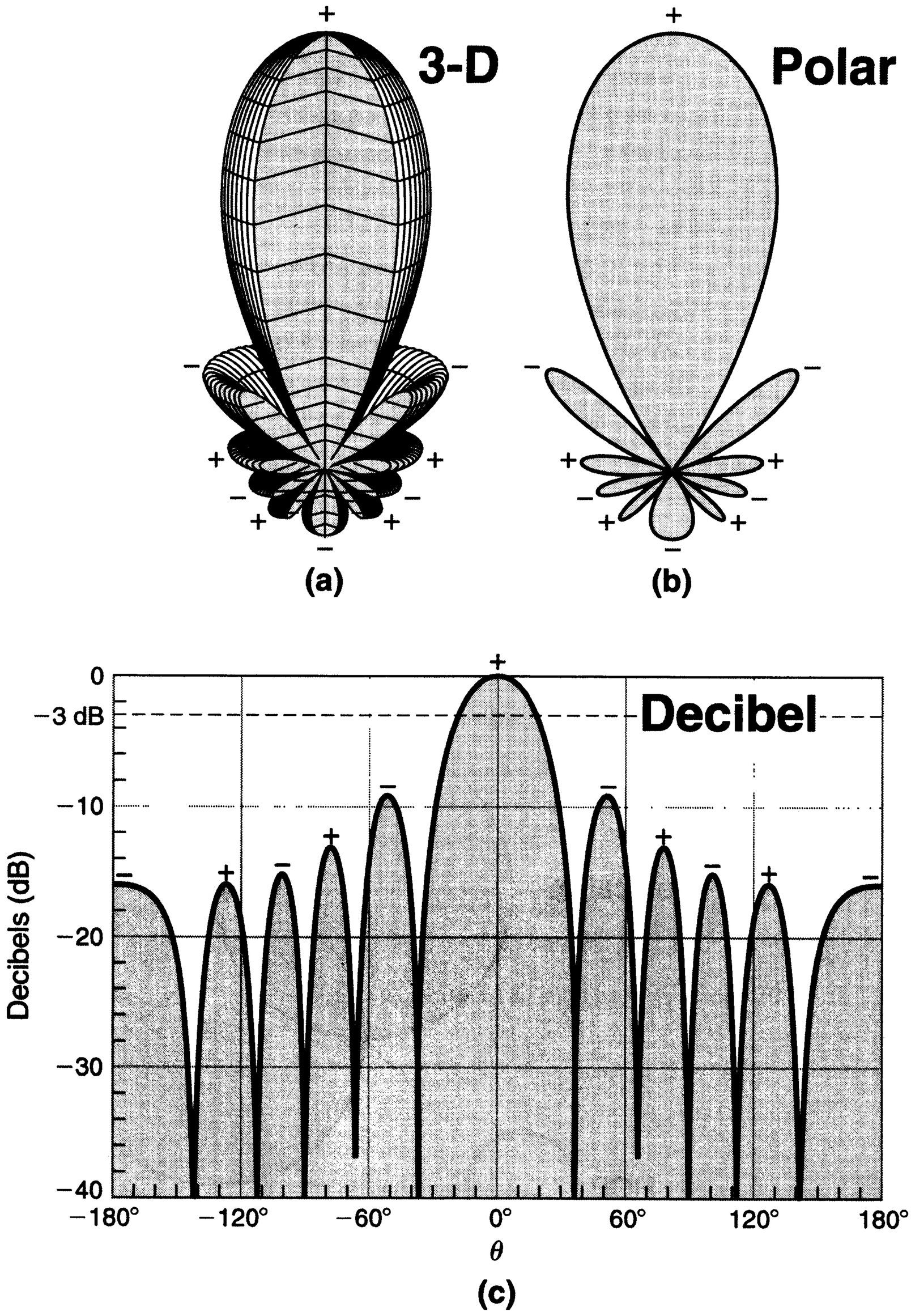


Figure 4–13
Three-dimensional field pattern at (*a*), polar pattern at (*b*), and decibel pattern at (*c*) showing alternate phasing (+ and −) of pattern lobes.

If the amplitudes of the field components are known at a particular radius from a point source in free space, their amplitudes at all distances are known from the inverse-distance law. Thus, it is usually sufficient to specify E_θ and E_ϕ as a function only of θ and ϕ as, for example, by a set of field patterns.

Figure 4–13 shows the pattern of Fig. 2–3 in three-dimensional, polar and decibel displays. Note that the polarity of the lobes alternate (+ and −). Thus, when the magnitude of the field of one lobe (+) and the adjacent lobe (−) are equal, the total field goes to zero, producing a null.

EXAMPLE **4–7.1 Field of Dipole and Loop in Phase Quadrature**

A short dipole is situated inside a small loop as in Fig. 4–14. The magnitude of the fields of both dipole and loop are equal. If the dipole and loop are fed in quadrature or 90° phasing, what are the fields that are observed as a function of azimuth in the plane of the page?

■ **Solution**

The fields north and south are horizontally polarized (in the plane of the page). The fields east and west are vertically polarized. At 45° or NE the field is right circularly polarized (RCP). At 135° or SE the field is left circularly polarized (LCP). At 225° or SW the field is again right circularly polarized. Finally, at 315° or NW the field is again left circularly polarized. At intermediate angles the field is elliptically polarized.

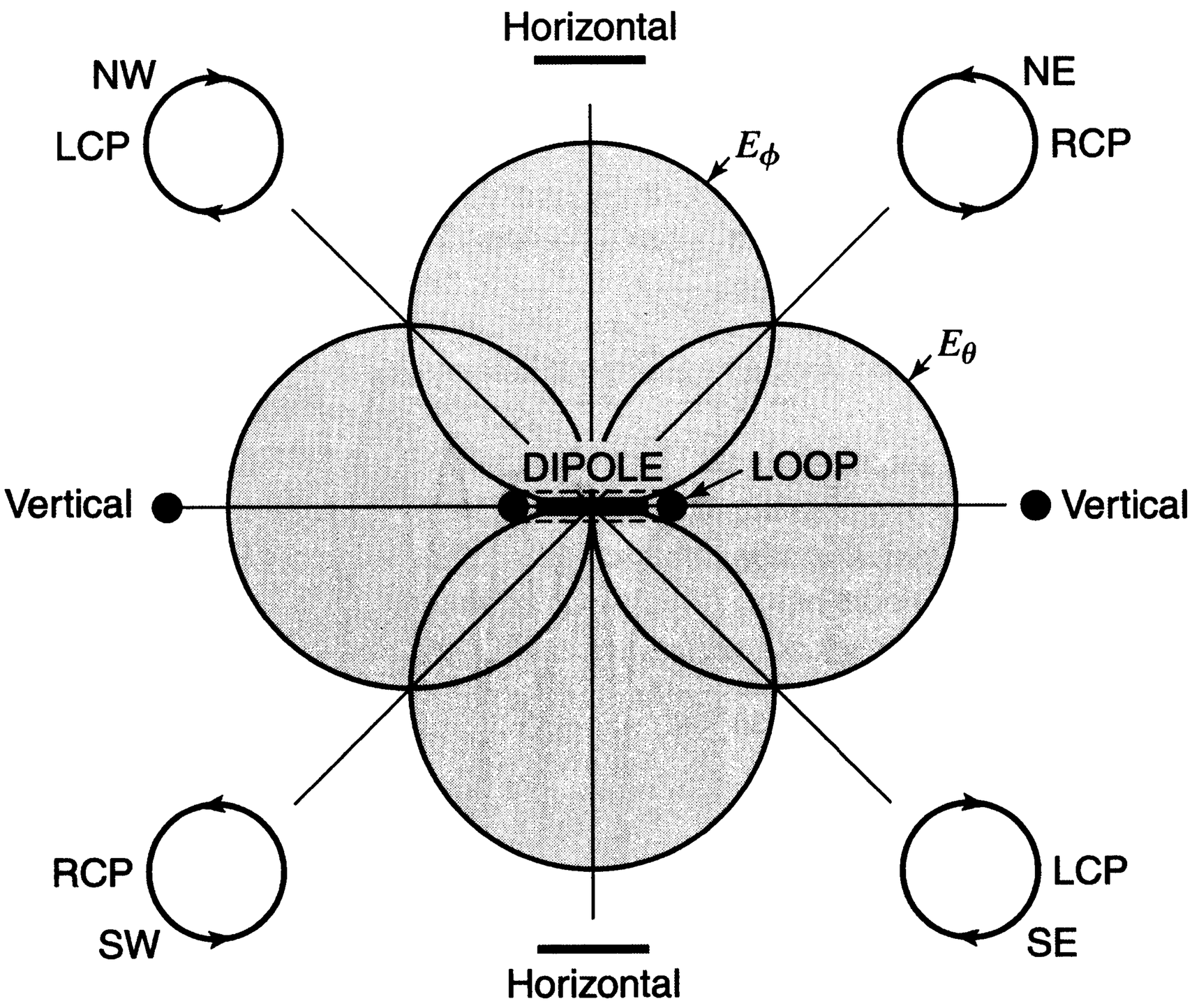


Figure 4–14
Fields of short dipole and small loop of equal magnitude and in phase quadrature (90°).

PROBLEMS（习题）

***4–3–1** （太阳功率）

Solar power. The earth receives from the sun 2.2 g cal min^{-1} cm^{-2}.

(*a*) What is the corresponding Poynting vector in watts per square meter?

(*b*) What is the power output of the sun, assuming that it is an isotropic source?

(*c*) What is the rms field intensity at the earth due to the sun's radiation, assuming all the sun's energy is at a single frequency?

Note: 1 watt = 14.3 g cal min^{-1}, distance earth to sun = 149 Gm.

4–5–1 (近似定向性) **Approximate directivities.** (*a*) Show that the directivity for a source with at unidirectional power pattern given by $U = U_m \cos^n \theta$ can be expressed as $D = 2(n + 1)$. U has a value only for $0° \leq \theta \leq 90°$. The patterns are independent of the azimuth angle ϕ. (*b*) Compare the exact values calculated from (*a*) with the approximate values for the directivities of the antennas found in Prob. 2–7–2 and find the dB difference from the exact values.

4–5–2 (正确/近似定向性) **Exact versus approximate directivities.** (*a*) Calculate the exact directivities of the three unidirectional antennas having power patterns as follows:

$$P(\theta, \phi) = P_m \sin\theta \sin^2\phi$$

$$P(\theta, \phi) = P_m \sin\theta \sin^3\phi$$

$$P(\theta, \phi) = P_m \sin^2\theta \sin^3\phi$$

$P(\theta, \phi)$ has a value only for $0 \leq \theta \leq \pi$ and $0 \leq \phi \leq \pi$ and is zero elsewhere.

(*b*) Compare the exact values in (*a*) with the approximate values found in Prob. 2–7–3.

4–5–3 (定向性与旁瓣) **Directivity and minor lobes.** Prove the following theorem: If the minor lobes of a radiation pattern remain constant as the beamwidth of the main lobe approaches zero, then the directivity of the antenna approaches a constant value as the beamwidth of the main lobes approaches zero.

4–5–4 (定向性积分) **Directivity by integration.** (*a*) Calculate by graphical integration or numerical methods the directivity of a source with a unidirectional power pattern given by $U = \cos\theta$. Compare this directivity value with the exact value from Prob. 4–5–1. U has a value only for $0° \leq \theta \leq 90°$ and $0° \leq \phi \leq 360°$ and is zero elsewhere.

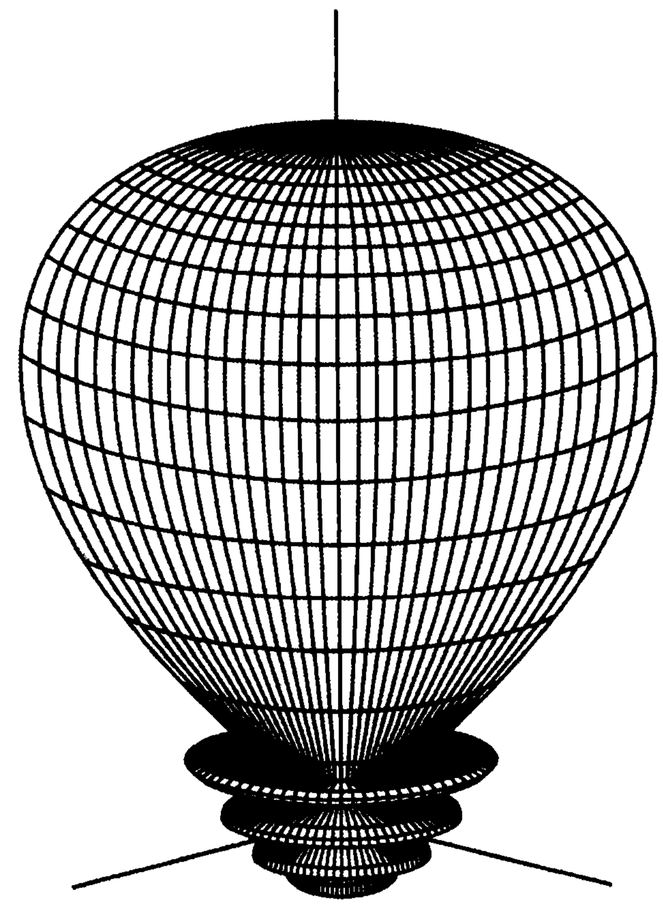
(*b*) Repeat for a unidirectional power pattern given by $U = \cos^2\theta$.

(*c*) Repeat for a unidirectional power pattern given by $U = \cos^3\theta$.

4–5–5 (定向性) **Directivity.** Calculate the directivity for a source with relative field pattern $E = \cos 2\theta \cos\theta$.

For computer programs, see Appendix C.

CHAPTER 5



Arrays of Point Sources, Part I

（上）点源阵

Topics in this chapter include:

- Examples of arrays of two point sources（两个点源的阵）
- Pattern multiplication（波瓣图乘法）
- Pattern synthesis（波瓣图综合）
- Nonisotropic sources（非各向同性源）
- Linear arrays of *n* point sources（*n* 个点源的直线阵）
- Broadside and end-fire arrays（边射阵和端射阵）
- Null directions and beamwidths（零方向和波束宽度）

5–1 INTRODUCTION（引言）

In Chap. 2 an antenna was treated as an aperture. In Chap. 4 an antenna was considered as a single point source. In this chapter we continue with the point-source concept, but extend it to a consideration of arrays of point sources. This approach is of great value since ***the pattern of any antenna can be regarded as produced by an array of point sources.*** （任何天线的波瓣图都能可认为是由点源阵产生的）Much of the discussion will concern arrays of isotropic point sources which may represent different kinds of antennas.

With the information of this chapter and the computer programs on the book's web site, you will be able to design arrays producing almost any pattern you may desire.

5–2 ARRAYS OF TWO ISOTROPIC POINT SOURCES（两个各向同性点源的阵）

Let us introduce the subject of arrays of point sources by considering the simplest situation, namely, that of two isotropic point sources. As illustrations, five cases involving two isotropic point sources are discussed.

Case 1. Two Isotropic Point Sources of Same Amplitude and Phase

The first case we shall analyze is that of two isotropic point sources having equal amplitudes and oscillating in the same phase. Let the two point sources, 1 and 2, be separated by a distance d and located symmetrically with respect to the origin of the coordinates as shown

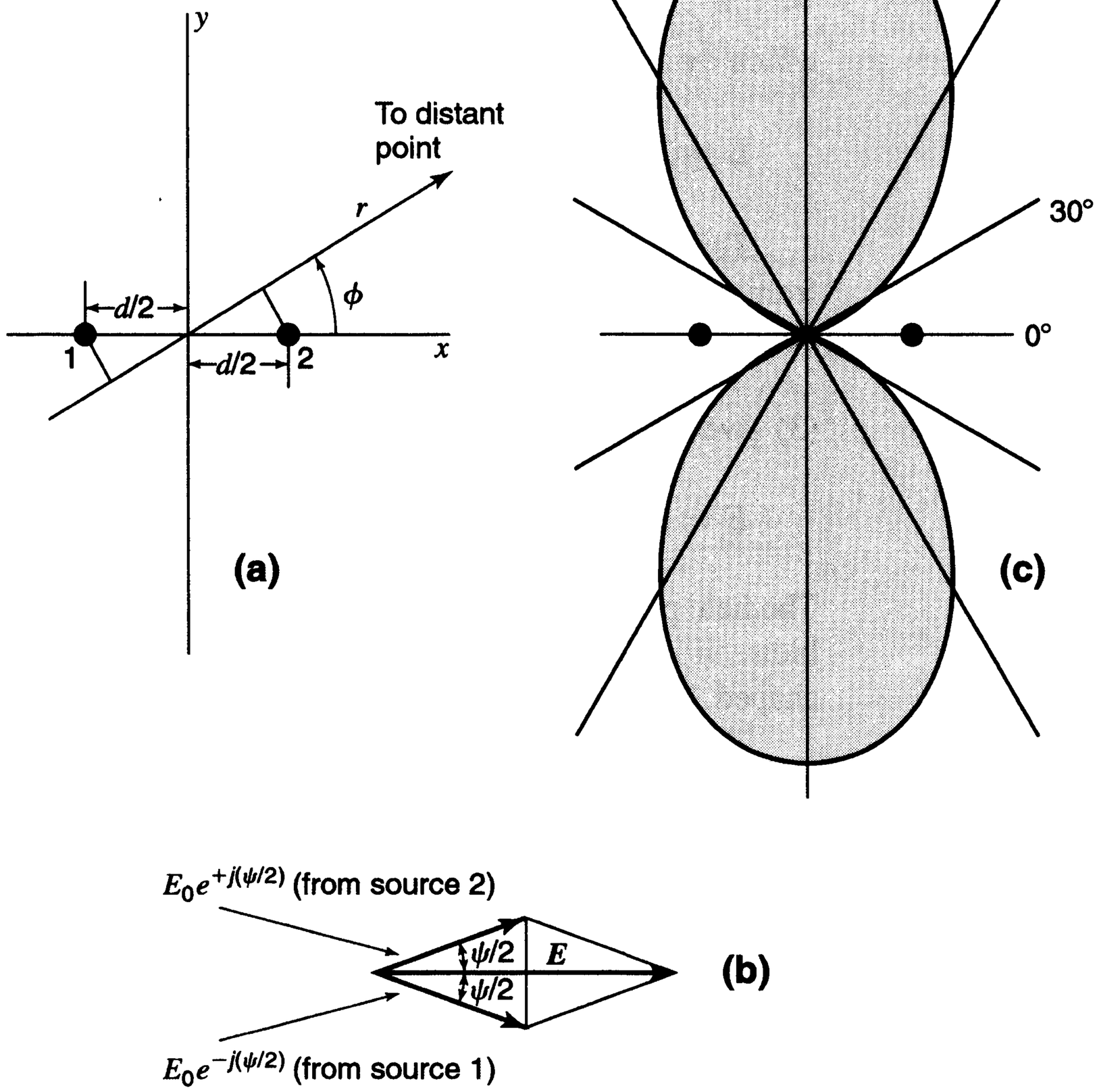


Figure 5–1
(*a*) Relation to coordinate system of two isotropic point sources separated by a distance *d*. (*b*) Vector addition of the fields from two isotropic point sources of equal amplitude and same phase located as in (*a*). (*c*) Field pattern of two isotropic point sources of equal amplitude and same phase located as in (*a*) for the case where the separation $d = \lambda/2$.

in Fig. 5–1*a*. The angle ϕ is measured counterclockwise from the positive x axis. The origin of the coordinates is taken as the reference for phase. Then at a distant point in the direction ϕ the field from source 1 is retarded by $\frac{1}{2}d_r \cos\phi$, while the field from source 2 is advanced by $\frac{1}{2}d_r \cos\phi$, where d_r is the distance between the sources expressed in radians; that is,

$$d_r = \frac{2\pi d}{\lambda} = \beta d$$

The total field at a large distance r in the direction ϕ is then

$$E = E_0 e^{-j\psi/2} + E_0 e^{+j\psi/2} \tag{1}$$

where $\psi = d_r \cos\phi$ and the amplitude of the field components at the distance r is given by E_0.

The first term in (1) is the component of the field due to source 1 and the second term the component due to source 2. Equation (1) may be rewritten

$$E = 2E_0 \frac{e^{+j\psi/2} + e^{-j\psi/2}}{2} \tag{2}$$

which by a trigonometric identity is

$$E = 2E_0 \cos\frac{\psi}{2} = 2E_0 \cos\left(\frac{d_r}{2}\cos\phi\right) \tag{3}$$

This result may also be obtained with the aid of the vector diagram[1] shown in Fig. 5–1*b*, from which (3) follows directly. We note in Fig. 5–1*b* that the phase of the total field E does not change as a function of ψ. To normalize (3), that is, make its maximum value unity, set $2E_0 = 1$. Suppose further that d is $\lambda/2$. Then $d_r = \pi$. Introducing these conditions into (3) gives

$$E = \cos\left(\frac{\pi}{2}\cos\phi\right) \tag{4}$$

The field pattern of E versus ϕ as expressed by (4) is presented in Fig. 5–1*c*. The pattern is a bidirectional figure-of-eight with maxima along the y axis. The space pattern is doughnut-shaped, being a figure-of-revolution of this pattern around the x axis.

The same pattern can also be obtained by locating source 1 at the origin of the coordinates and source 2 at a distance d along the positive x axis as indicated in Fig. 5–2*a*. Taking now the field from source 1 as reference, the field from source 2 in the direction ϕ is advanced by $d_r \cos\phi$. Thus, the total field E at a large distance r is the vector sum of the fields from the two sources as given by

$$E = E_0 + E_0 e^{+j\psi} \tag{5}$$

where $\psi = d_r \cos\phi$.

The relation of these fields is indicated by the vector diagram of Fig. 5–2*b*. From the vector diagram the magnitude of the total field is

$$E = 2E_0 \cos\frac{\psi}{2} = 2E_0 \cos\frac{d_r \cos\phi}{2} \tag{6}$$

as obtained before in (3). The phase of the total field E is, however, not constant in this case but is $\psi/2$, as also shown by rewriting (5) as

$$E = E_0(1 + e^{j\psi}) = 2E_0 e^{j\psi/2}\left(\frac{e^{j\psi/2} + e^{-j\psi/2}}{2}\right) = 2E_0 e^{j\psi/2}\cos\frac{\psi}{2} \tag{7}$$

Normalizing by setting $2E_0 = 1$, (7) becomes

$$E = e^{j\psi/2}\cos\frac{\psi}{2} = \cos\frac{\psi}{2}\,\underline{/\psi/2} \tag{8}$$

---

[1] It is to be noted that the quantities represented here by vectors are not true space vectors but merely vector representations of the time phase (i.e., phasors).

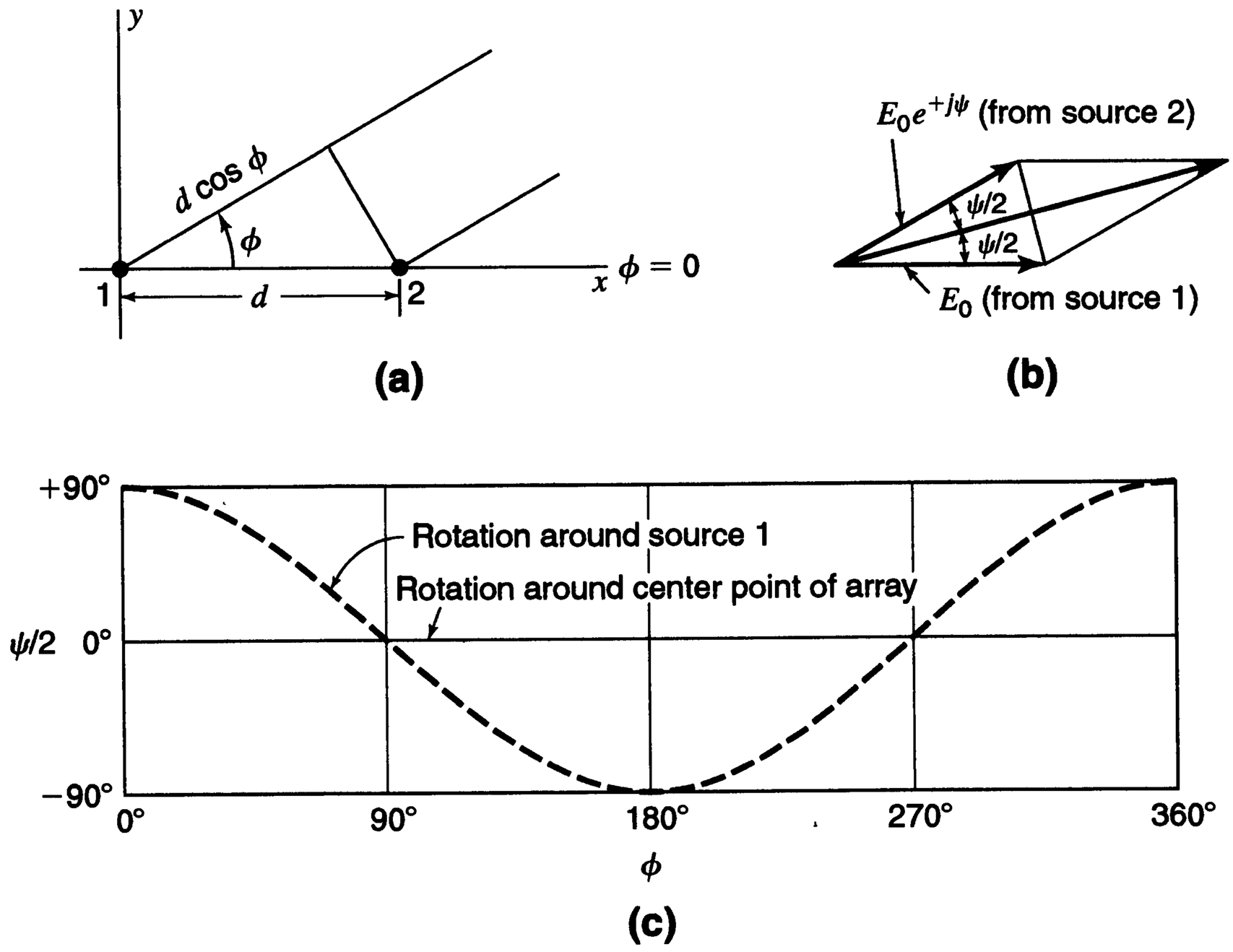


Figure 5–2
(*a*) Two isotropic point sources with the origin of the coordinate system coincident with one of the sources. (*b*) Vector addition of the fields from two isotropic point sources of equal amplitude and same phase located as in (*a*). (*c*) Phase of total field as a function of ϕ for two isotropic point sources of same amplitude and phase spaced $\lambda/2$ apart. The phase change is zero when referred to the center point of the array but is $\psi/2$ as shown by the dashed curve when referred to source 1.

In (8) the cosine factor gives the amplitude variation of E, and the exponential or angle factor gives the phase variation *with respect to source* 1 as the reference. The phase variation for the case of $\lambda/2$ spacing ($d_r = \pi$) is shown by the dashed line in Fig. 5–2*c*. Here the phase angle with respect to the phase of source 1 is given by $\psi/2 = (\pi/2)\cos\phi$. The magnitude variation for this case has already been presented in Fig. 5–1*c*. When the phase is referred to the point midway between the sources (Fig. 5–1*a*), there is no phase change around the array as shown by the solid line in Fig. 5–2*c*. Thus, an observer at a fixed distance observes no phase change when the array is rotated (with respect to ϕ) around its midpoint, but a phase change (dashed curve of Fig. 5–2*c*) is observed if the array is rotated with source 1 as the center of rotation.

Case 2. Two Isotropic Point Sources of Same Amplitude but Opposite Phase

This case is identical with the one we have just considered except that the two sources are in opposite phase instead of in the same phase. Let the sources be located as in Fig. 5–1*a*.

Then the total field in the direction ϕ at a large distance r is given by

$$E = E_0 e^{+j\psi/2} - E_0 e^{-j\psi/2} \tag{9}$$

from which

$$E = 2jE_0 \sin\frac{\psi}{2} = 2jE_0 \sin\left(\frac{d_r}{2}\cos\phi\right) \tag{10}$$

Whereas in Case 1 (3) involves the cosine of $\psi/2$, (10) for Case 2 involves the sine. Equation (10) also includes an operator j, indicating that the phase reversal of one of the sources in Case 2 results in a 90° phase shift of the total field as compared with the total field for Case 1. This is unimportant here. Thus, putting $2jE_0 = 1$ and considering the special case of $d = \lambda/2$, (10) becomes

$$E = \sin\left(\frac{\pi}{2}\cos\phi\right) \tag{11}$$

The directions ϕ_m of maximum field are obtained by setting the argument of (11) equal to $\pm(2k+1)\pi/2$. Thus,

$$\frac{\pi}{2}\cos\phi_m = \pm(2k+1)\frac{\pi}{2} \tag{11a}$$

where $k = 0, 1, 2, 3 \ldots$. For $k = 0$, $\cos\phi_m = \pm 1$ and $\phi_m = 0°$ and 180°.

The null directions ϕ_0 are given by

$$\frac{\pi}{2}\cos\phi_0 = \pm k\pi \tag{11b}$$

For $k = 0$, $\phi_0 = \pm 90°$.

The half-power directions are given by

$$\frac{\pi}{2}\cos\phi = \pm(2k+1)\frac{\pi}{4} \tag{11c}$$

For $k = 0$, $\phi = \pm 60°, \pm 120°$.

The field pattern given by (11) is shown in Fig. 5–3. The pattern is a relatively broad figure-of-eight with the maximum field in the same direction as the line joining the sources (x axis). The space pattern is a figure-of-revolution of this pattern around the x axis. The two sources, in this case, may be described as a simple type of "end-fire" array. In contrast to this pattern, the in-phase point sources produce a pattern with the maximum field normal to the line joining the sources, as shown in Fig. 5–1*c*. The two sources for this case may be described as a simple "broadside" type of array.

Case 3. Two Isotropic Point Sources of the Same Amplitude and In-Phase Quadrature

Let the two point sources be located as in Fig. 5–1*a*. Taking the origin of the coordinates as the reference for phase, let source 1 be retarded by 45° and source 2 advanced by 45°. Then the total field in the direction ϕ at a large distance r is given by

$$E = E_0 \exp\left[+j\left(\frac{d_r\cos\phi}{2} + \frac{\pi}{4}\right)\right] + E_0 \exp\left[-j\left(\frac{d_r\cos\phi}{2} + \frac{\pi}{4}\right)\right] \tag{12}$$

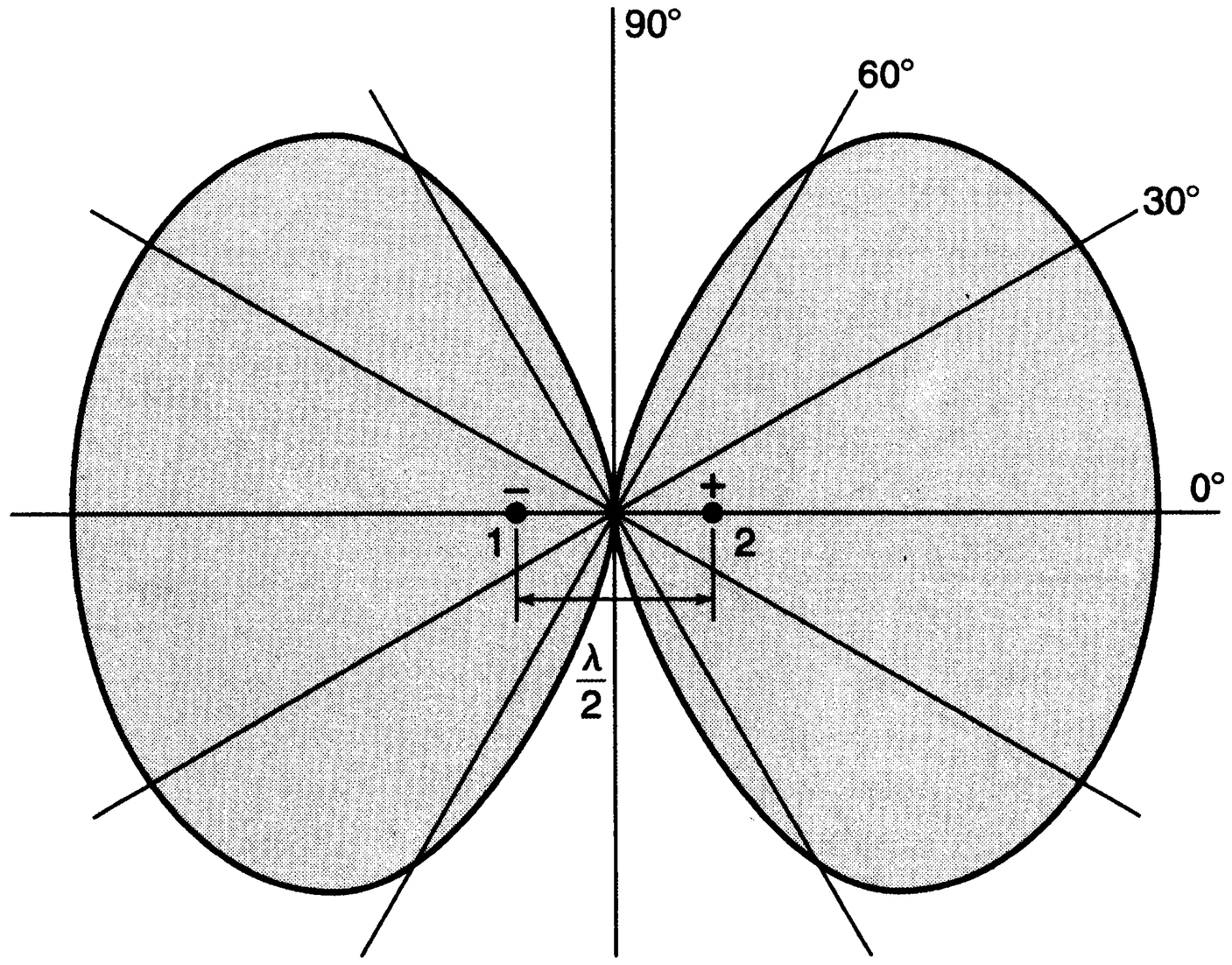


Figure 5–3
Relative field pattern for two isotropic point sources of the same amplitude but opposite phase, spaced $\lambda/2$ apart.

From (12) we obtain

$$E = 2E_0 \cos\left(\frac{\pi}{4} + \frac{d_r}{2}\cos\phi\right) \tag{13}$$

Letting $2E_0 = 1$ and $d = \lambda/2$, (13) becomes

$$E = \cos\left(\frac{\pi}{4} + \frac{\pi}{2}\cos\phi\right) \tag{14}$$

The field pattern given by (14) is presented in Fig. 5–4. The space pattern is a figure-of-revolution of this pattern around the x axis. Most of the radiation is in the second and third quadrants. It is interesting to note that the field in the direction $\phi = 0°$ is the same as in the direction $\phi = 180°$. The directions ϕ_m of maximum field are obtained by setting the argument of (14) equal to $k\pi$, where $k = 0, 1, 2, 3 \ldots$. In this way we obtain

$$\frac{\pi}{4} + \frac{\pi}{2}\cos\phi_m = k\pi \tag{15}$$

For $k = 0$,

$$\frac{\pi}{2}\cos\phi_m = -\frac{\pi}{4} \tag{16}$$

and

$$\phi_m = 120° \text{ and } 240° \tag{17}$$

If the spacing between the sources is reduced to $\lambda/4$, (13) becomes

$$E = \cos\left(\frac{\pi}{4} + \frac{\pi}{4}\cos\phi\right) \tag{18}$$

The field pattern for this case is illustrated by Fig. 5–5*a*. It is a cardioid-shaped, unidirectional pattern with maximum field in the negative x direction. The space pattern is a figure-of-revolution of this pattern around the x axis.

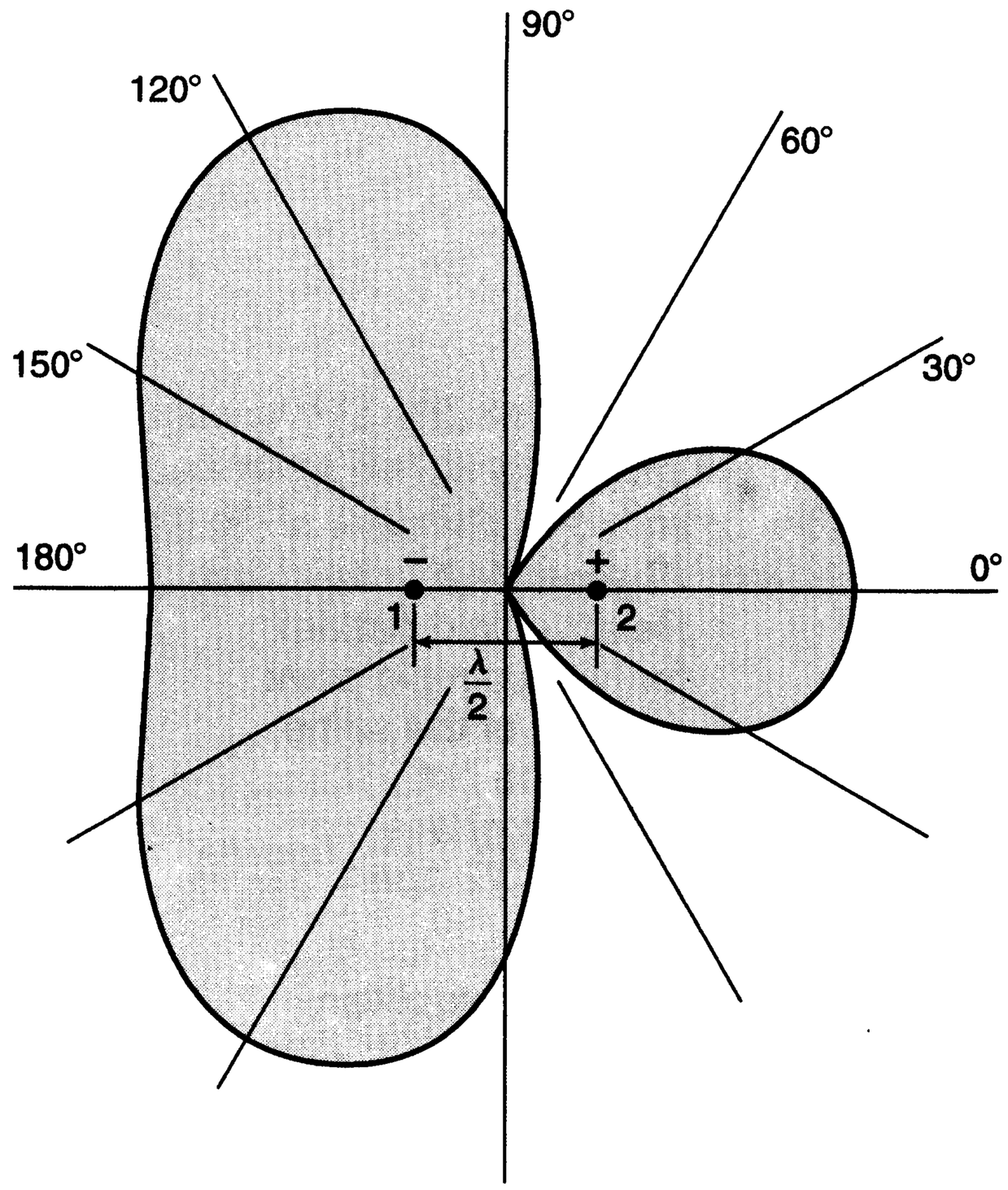


Figure 5–4
Relative field pattern of two isotropic point sources of the same amplitude and in phase quadrature for a spacing of $\lambda/2$. The source to the right leads that to the left by 90°.

A simple method of determining the direction of maximum field is illustrated by Fig. 5–5*b*. As indicated by the vectors, the phase of source 2 is 0° (vector to right) and the phase of source 1 is 270° (vector down). Thus, source 2 leads source 1 by 90°.

To find the field radiated to the left, imagine that we start at source 2 (phase 0°) and travel to the left, riding with the wave (phase 0°) like a surfer rides a breaker. The phase of the wave we are riding is 0° and does not change but by the time we have traveled $\lambda/4$ and arrived at source 1, a $\frac{1}{4}$-period has elapsed so the current in source 1 will have advanced 90° (vector rotated ccw) from 270° to 0°, making its phase the same as that of the wave we are riding, as in the middle diagram of Fig. 5–5*b*. Thus, the field of the wave from source 2 reinforces that of the field of source 1, and the two fields travel to the left together in phase producing a maximum field to the left which is twice the field of either source alone.

Now imagine that we start at source 1 with phase 270° (vector down) and travel to the right. By the time we arrive at source 2 the phase of its field has advanced from 0 to 90° so it is in phase opposition and cancels the field of the wave we are riding, as in the bottom diagram in Fig. 5–5*b*, resulting in zero radiation to the right.

Case 4. General Case of Two Isotropic Point Sources of Equal Amplitude and Any Phase Difference

Proceeding now to a more general situation, let us consider the case of two isotropic point sources of equal amplitude but of any phase difference δ. The total phase difference ψ between the fields from source 2 and source 1 at a distant point in the direction ϕ (see

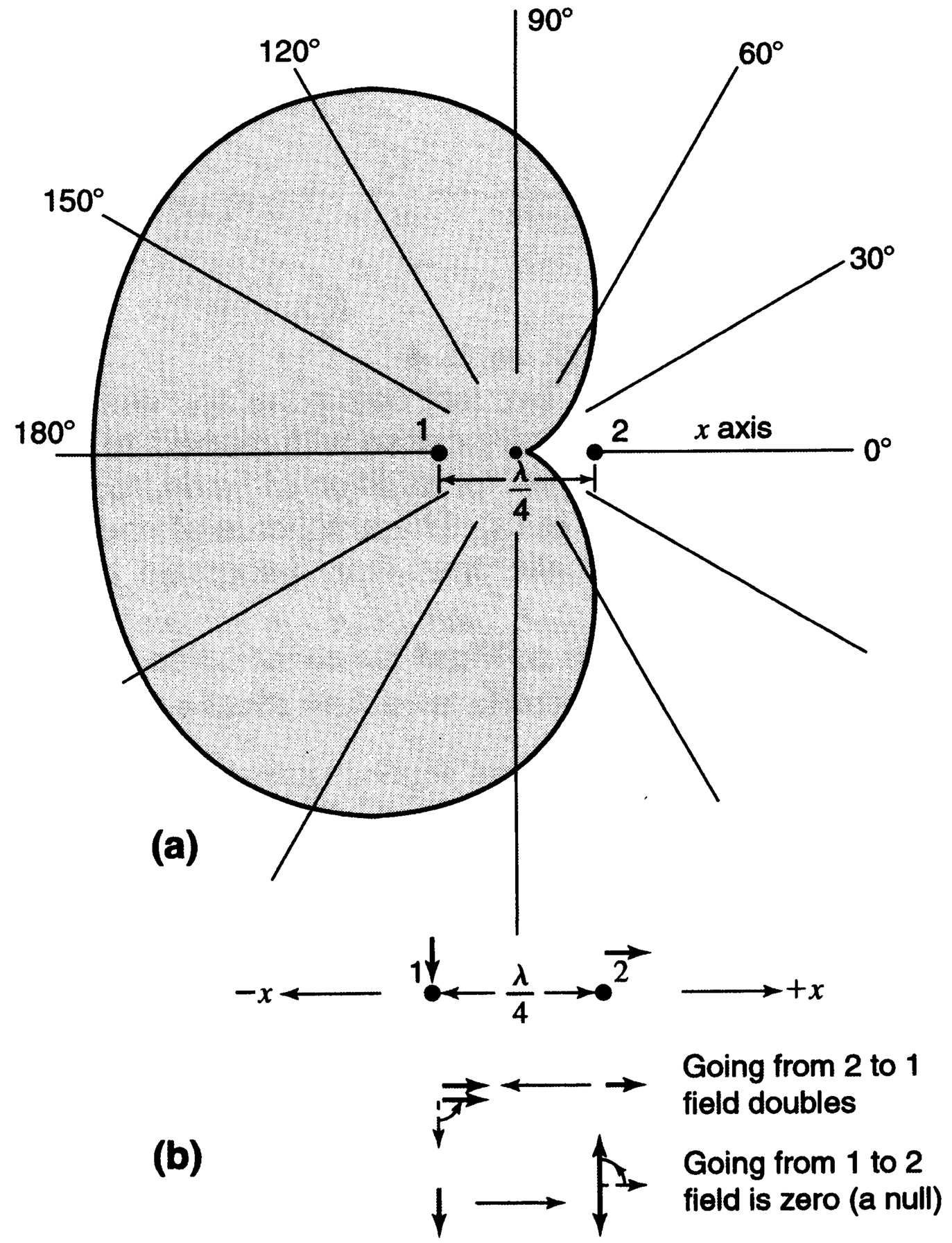


Figure 5–5
(*a*) Relative field pattern of two isotropic sources of same amplitude and in-phase quadrature for a spacing of $\lambda/4$. Source 2 leads source 1 by 90°. (*b*) Vector diagrams illustrating field reinforcement in the $-x$ direction and field cancellation in the $+x$ direction.

Fig. 5–2*a*) is then

$$\psi = d_r \cos\phi + \delta \tag{19}$$

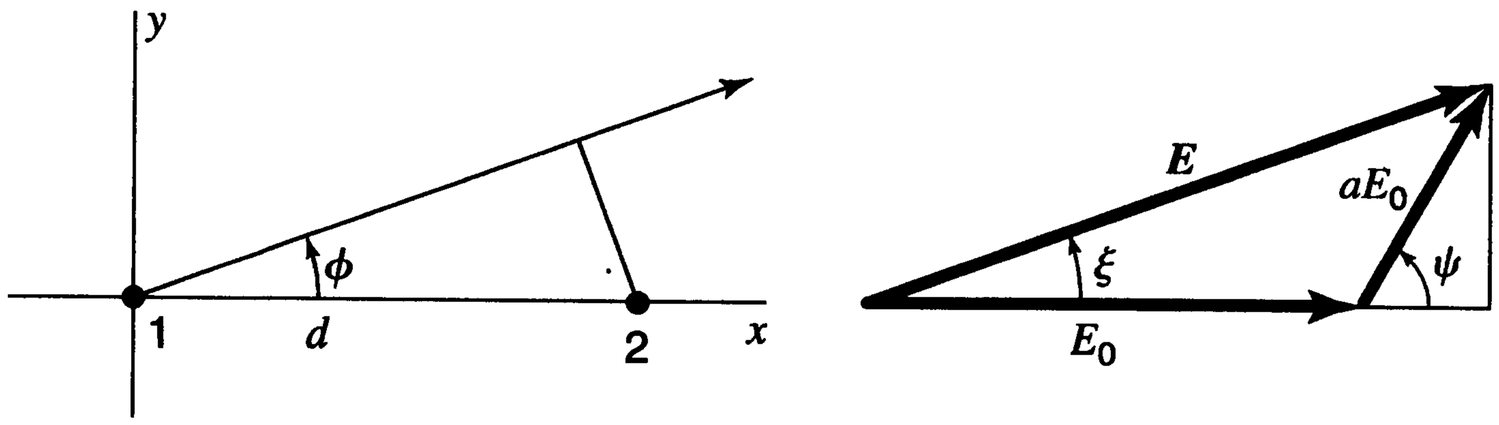
Taking source 1 as the reference for phase, the positive sign in (19) indicates that source 2 is advanced in phase by the angle δ. A minus sign would be used to indicate a phase retardation. If, instead of referring the phase to source 1, it is referred to the centerpoint of the array, the phase of the field from source 1 at a distant point is given by $-\psi/2$ and that from source 2 by $+\psi/2$. The total field is then

$$E = E_0(e^{j\psi/2} + e^{-j\psi/2}) = 2E_0 \cos\frac{\psi}{2} \tag{20}$$

Normalizing (20), we have the general expression for the field pattern of two isotropic sources of equal amplitude and arbitrary phase,

$$E = \cos\frac{\psi}{2} \tag{21}$$

where ψ is given by (19). The three cases we have discussed are obviously special cases of (21). Thus, Cases 1, 2, and 3 are obtained from (21) when $\delta = 0°$, 180°, and 90° respectively.



(a) (b)

Figure 5–6
(*a*) Two isotropic point sources of unequal amplitude and arbitrary phase with respect to the coordinate system. (*b*) Vector addition of fields from unequal sources arranged as in (*a*). The amplitude of source 2 is assumed to be smaller than that of source 1 by the factor *a*.

Case 5. Most General Case of Two Isotropic Point Sources of Unequal Amplitude and Any Phase Difference

A still more general situation, involving two isotropic point sources, exists when the amplitudes are unequal and the phase difference is arbitrary. Let the sources be situated as in Fig. 5–6*a* with source 1 at the origin. Assume that the source 1 has the larger amplitude and that its field at a large distance r has an amplitude of E_0. Let the field from source 2 be of amplitude aE_0 $(0 \leq a \leq 1)$ at the distance r. Then, referring to Fig. 5–6*b*, the magnitude and phase angle of the total field E is given by

$$E = E_0\sqrt{(1 + a\cos\psi)^2 + a^2\sin^2\psi}\ \underline{/\arctan[a\sin\psi/(1 + a\cos\psi)]} \tag{22}$$

where $\psi = d_r\cos\phi + \delta$ and the phase angle ($\angle$) is referred to source 1. This is the phase angle ξ shown in Fig. 5–6*b*.

5–3 NONISOTROPIC BUT SIMILAR POINT SOURCES AND THE PRINCIPLE OF PATTERN MULTIPLICATION（非各向同性的相似点源和波瓣图乘法原理）

The cases considered in the preceding section all involve *isotropic* point sources. These can readily be extended to a more general situation in which the sources are *nonisotropic but similar*.

The word *similar* is here used to indicate that the variation with absolute angle ϕ of both the amplitude and phase of the field is the same.[1] The maximum amplitudes of the individual sources may be unequal. If, however, they are also equal, the sources are not only similar but are *identical*.

As an example, let us reconsider Case 4 of Sec. 5–2 in which the sources are identical, with the modification that both sources 1 and 2 have field patterns given by

$$E_0 = E_0'\sin\phi \tag{1}$$

Patterns of this type might be produced by short dipoles oriented parallel to the x axis as suggested by Fig. 5–7. Substituting (1) in (5–2–20) and normalizing by setting $2E_0' = 1$

[1]The patterns not only must be of the same shape but also must be oriented in the same direction to be called "similar."

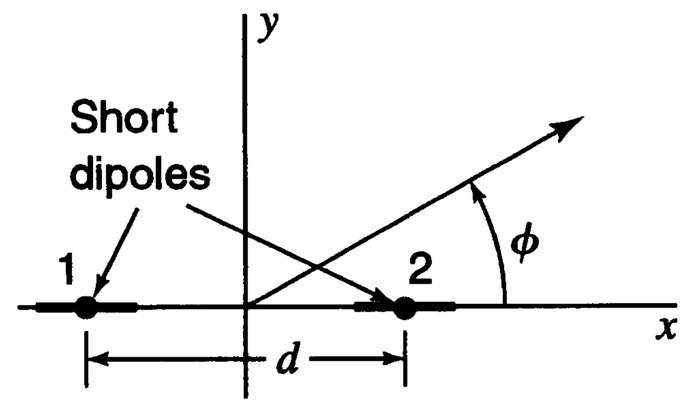


Figure 5–7
Two nonisotropic sources with respect to the coordinate system.

gives the field pattern of the array as

$$E = \sin\phi \cos\frac{\psi}{2} \tag{2}$$

where $\psi = d_r \cos\phi + \delta$

This result is the same as obtained by multiplying the pattern of the individual source ($\sin\phi$) by the pattern of two isotropic point sources ($\cos\psi/2$).

If the similar but unequal point sources of Case 5 (Sec. 5–2) have patterns as given by (1), the total normalized pattern is

$$E = \sin\phi\sqrt{(1 + a\cos\psi)^2 + a^2\sin^2\psi} \tag{3}$$

Here again, the result is the same as that obtained by multiplying the pattern of the individual source by the pattern of an array of isotropic point sources.

These are examples illustrating the ***principle of pattern multiplication,*** (**波瓣图乘法原理**) which may be expressed as follows:

> ***The field pattern of an array of nonisotropic but similar point sources is the product of the pattern of the individual source and the pattern of an array of isotropic point sources having the same locations, relative amplitudes, and phase as the nonisotropic point sources.***
> **(非各向同性而相似的点源阵场波瓣图是其个别源波瓣图与该阵列中具有相同的位置、相对幅度和相位的各向同性点源阵波瓣图之乘积)**

This principle may be applied to arrays of any number of sources provided only that they are similar. The individual nonisotropic source or antenna may be of finite size but can be considered as a point source situated at the point in the antenna to which phase is referred. This point is said to be the "phase center."

The above discussion of pattern multiplication has been concerned only with the field pattern or magnitude of the field. If the field of the nonisotropic source and the array of isotropic sources vary in phase with space angle, i.e., have a phase pattern which is not a constant, the statement of the principle of pattern multiplication may be extended to include this more general case as follows:

> The total field pattern of an array of nonisotropic but similar sources is the product of the individual source pattern and the pattern of an array of isotropic point sources each located at the phase center of the individual source and having the same relative amplitude and phase, while the total phase pattern is the sum of the phase patterns of the individual source and the array of isotropic point sources.

The total phase pattern is referred to the phase center of the array. In symbols, the total field E is then

$$E = \underbrace{f(\theta,\phi)F(\theta,\phi)}_{\text{Field pattern}} \underbrace{\angle f_p(\theta,\phi) + F_p(\theta,\phi)}_{\text{Phase pattern}} \tag{4}$$

where

$f(\theta, \phi)$ = field pattern of individual source
$f_p(\theta, \phi)$ = phase pattern of individual source
$F(\theta, \phi)$ = field pattern of array of isotropic sources
$F_p(\theta, \phi)$ = phase pattern of array of isotropic sources

The patterns are expressed in (4) as a function of both polar angles to indicate that the principle of pattern multiplication applies to space patterns as well as to the two-dimensional cases we have been considering.

To illustrate the principle, let us apply to it two special modifications of Case 1 (Sec. 5–2).

EXAMPLE 5–3.1

Assume two identical point sources separated by a distance d, each source having the field pattern given by (1) as might be obtained by two short dipoles arranged as in Fig. 5–7. Let $d = \lambda/2$ and the phase angle $\delta = 0$. Then the total field pattern is

$$E = \sin\phi \cos\left(\frac{\pi}{2}\cos\phi\right) \tag{5}$$

This pattern is illustrated by Fig. 5–8*c* as the product of the individual source pattern ($\sin\phi$) shown at (*a*) and the array pattern $\{\cos[(\pi/2)\cos\phi]\}$ as shown at (*b*). The pattern is sharper than it was in Case 1 (Sec. 5–2) for the isotropic sources. In this instance, the maximum field of the individual source is in the direction $\phi = 90°$, which coincides with the direction of the maximum field for the array of two isotropic sources.

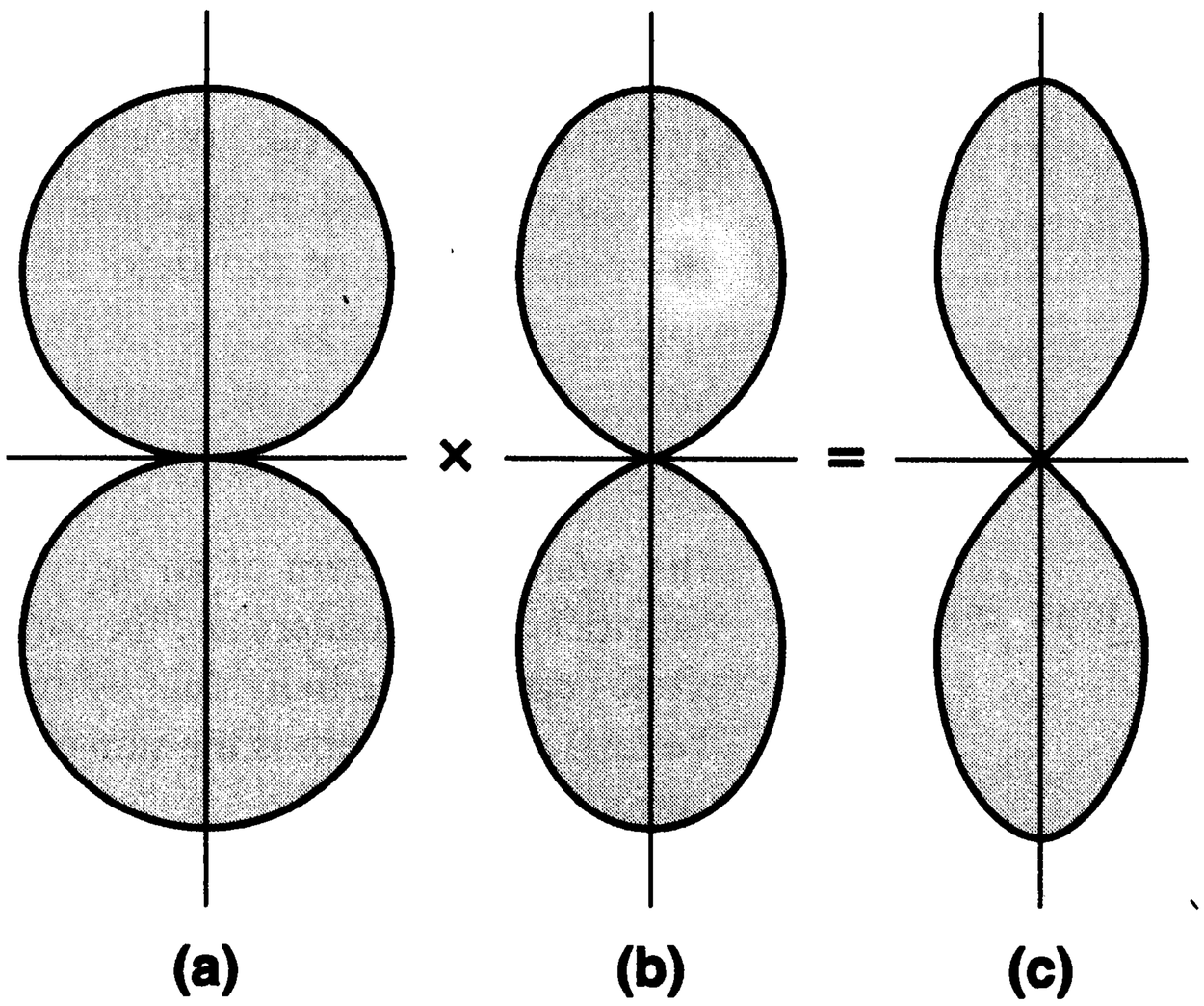


Figure 5–8
Example of pattern multiplication. Two nonisotropic but identical point sources of the same amplitude and phase, spaced $\lambda/2$ apart and arranged as in Fig. 5–7, produce the pattern shown at (*c*). The individual source has the pattern shown at (*a*), which, when multiplied by the pattern of an array of two isotropic point sources (of the same amplitude and phase) as shown at (*b*), yields the total array pattern of (*c*).

EXAMPLE 5–3.2

Let us consider next the situation in which $d = \lambda/2$ and $\delta = 0$ as in Example 5–3.1 but with individual source patterns given by

$$E_0 = E_0' \cos\phi \tag{6}$$

This type of pattern might be produced by short dipoles oriented parallel to the y axis as in Fig. 5–9. Here the maximum field of the individual source is in the direction ($\phi = 0$) of a null from the array, while the individual source has a null in the direction ($\phi = 90°$) of the pattern maximum of the array. By the principle of pattern multiplication the total normalized field is

$$E = \cos\phi \cos\left(\frac{\pi}{2}\cos\phi\right) \tag{7}$$

The total array pattern in the xy plane as given by (7) is illustrated in Fig. 5–10c as the product of the individual source pattern ($\cos\phi$) shown at (a) and the array pattern $\{\cos[(\pi/2)\cos\phi]\}$ shown at (b). The total array pattern in the xy plane has four lobes with nulls at the x and y axes.

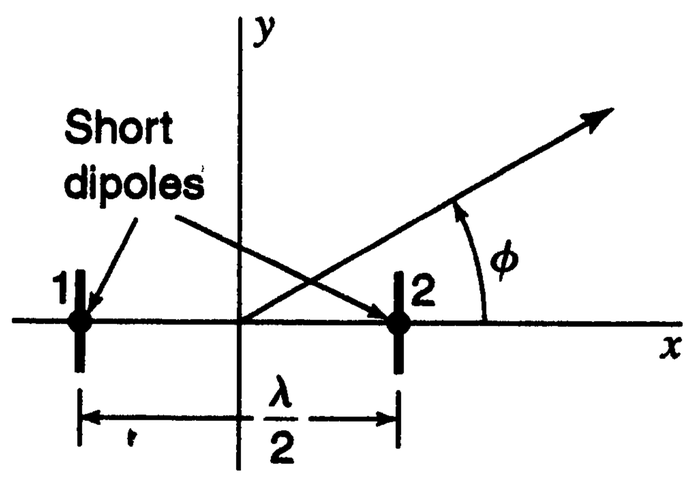


Figure 5–9
Array of two nonisotropic sources with respect to the coordinate system.

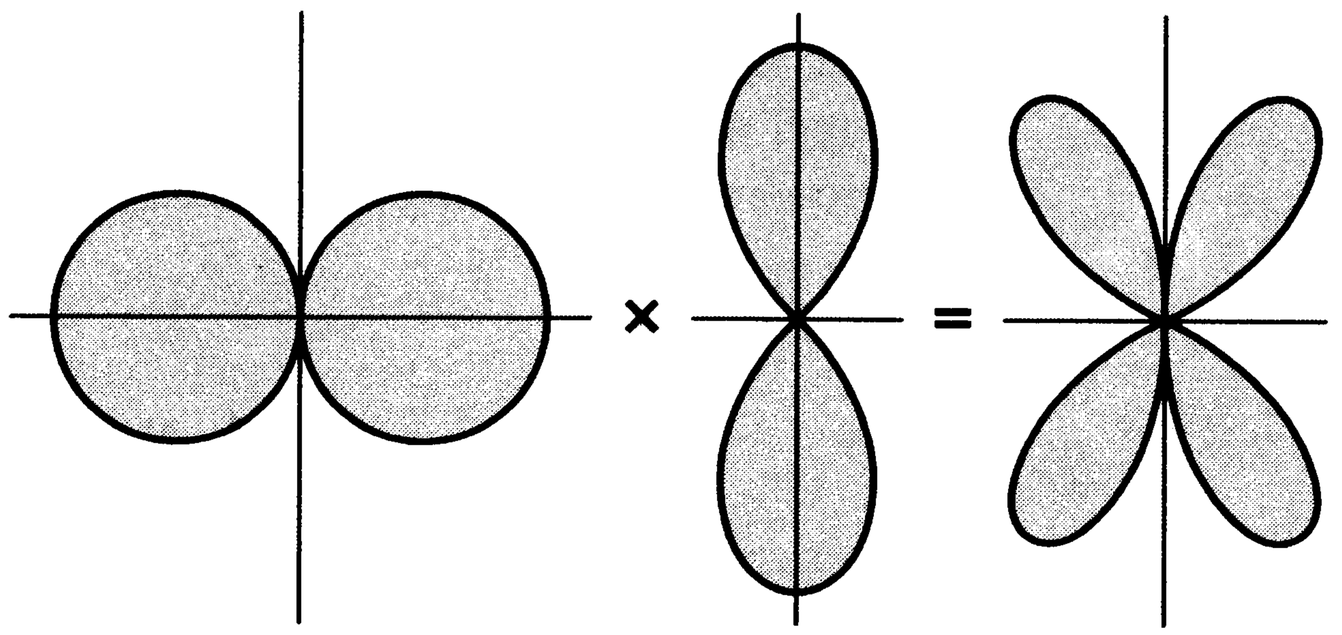


Figure 5–10
Example of pattern multiplication. Total array pattern (c) as the product of pattern (a) of individual nonisotropic source and pattern (b) of array of two isotropic sources. The pattern (b) for the array of two isotropic sources is identical with that of Fig. 5–8b, but the individual source pattern (a) is rotated through 90° with respect to the one in Fig. 5–8a.

The above examples illustrate two applications of the principle of pattern multiplication to arrays in which the source has a simple pattern. However, in the more general case the individual source may represent an antenna of any complexity provided that the amplitude and phase of its field can be expressed as a function of angle, that is to say, provided that

the field pattern and the phase pattern with respect to the phase center are known. If only the total field pattern is desired, phase patterns need not be known provided that the individual sources are identical.

If the arrays in the above examples are parts of still larger arrays, the smaller arrays may be regarded as nonisotropic point sources in the larger array—another application of the principle of pattern multiplication yielding the complete pattern. In this way the principle of pattern multiplication can be applied n times to find the patterns of arrays of arrays of arrays.

5–4 EXAMPLE OF PATTERN SYNTHESIS BY PATTERN MULTIPLICATION
（用图乘法综合波瓣图举例）

The principle of pattern multiplication, discussed in the preceding section, is of great value in pattern synthesis. By pattern synthesis is meant the process of finding the source or array of sources that produces a desired pattern. Theoretically an array of isotropic point sources can be found that will produce any arbitrary pattern. This process is not always simple and may yield an array that is difficult or impossible to construct. A simpler, less elegant approach to the problem of antenna synthesis is by the application of pattern multiplication to combinations of practical arrays, the combination which best approximates the desired pattern being arrived at by a trial-and-error process.

To illustrate this application of pattern multiplication, let us consider the following hypothetical problem. A broadcasting station (in the 500- to 1500-kHz frequency band) requires a pattern in the horizontal plane fulfilling the conditions indicated in Fig. 5–11*a*. The maximum field intensity, with as little variation as possible, is to be radiated in the 90° sector between northwest and northeast. No nulls in the pattern can occur in this sector. However, nulls may occur in any direction in the complementary 270° sector, but, as an additional requirement, nulls must be present in the due east and the due southwest directions in order to prevent interference with other stations in these directions. An idealized sector-shaped pattern fulfilling those requirements is illustrated in Fig. 5–11*b*. The antenna producing this pattern is to consist of an array of four vertical towers. The currents in all towers are to be equal in magnitude, but the phase may be adjusted to any relationship. There is also no restriction on the spacing or geometrical arrangement of the towers.

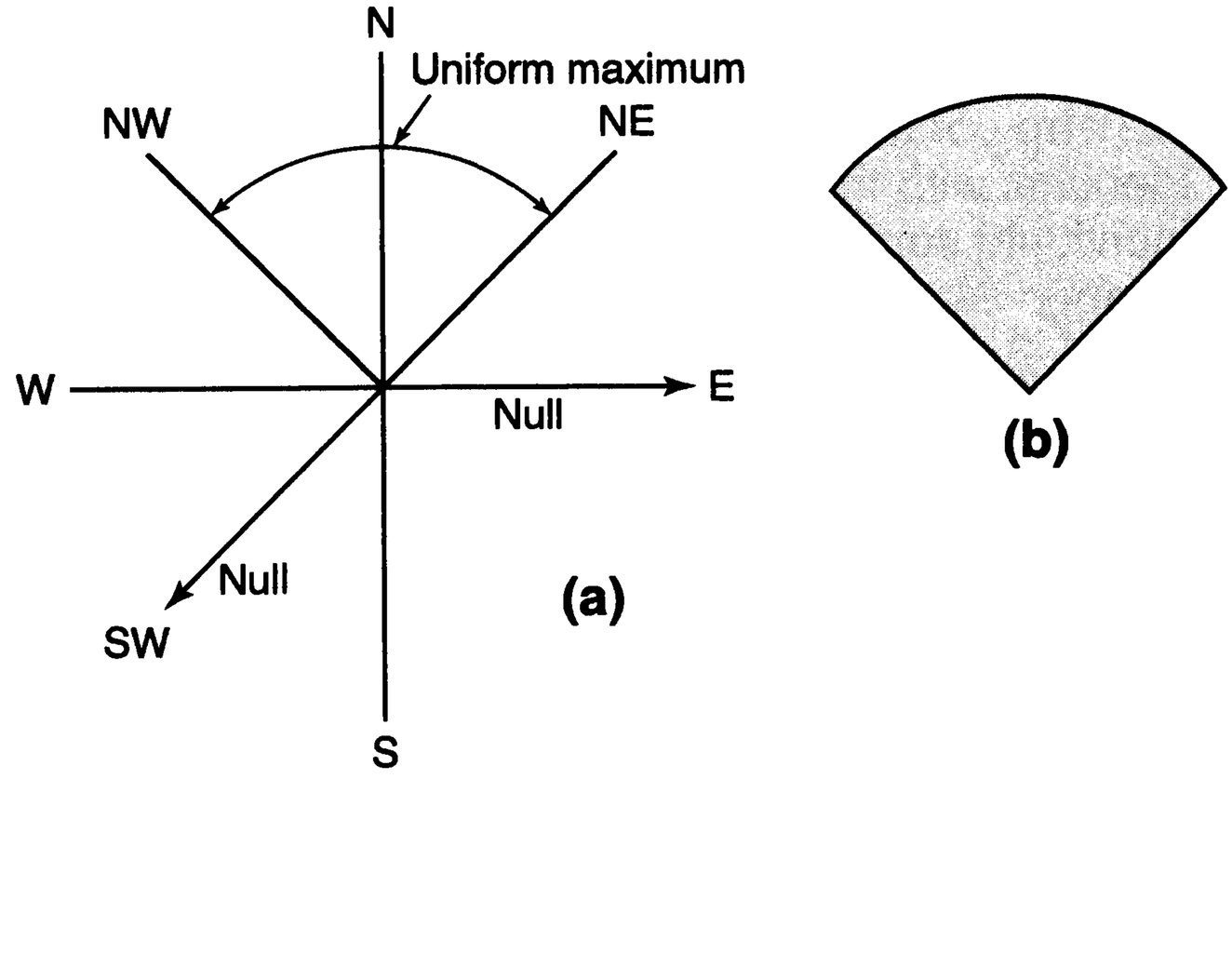


Figure 5–11
(*a*) Requirements for pattern of broadcast station and (*b*) idealized pattern fulfilling them.

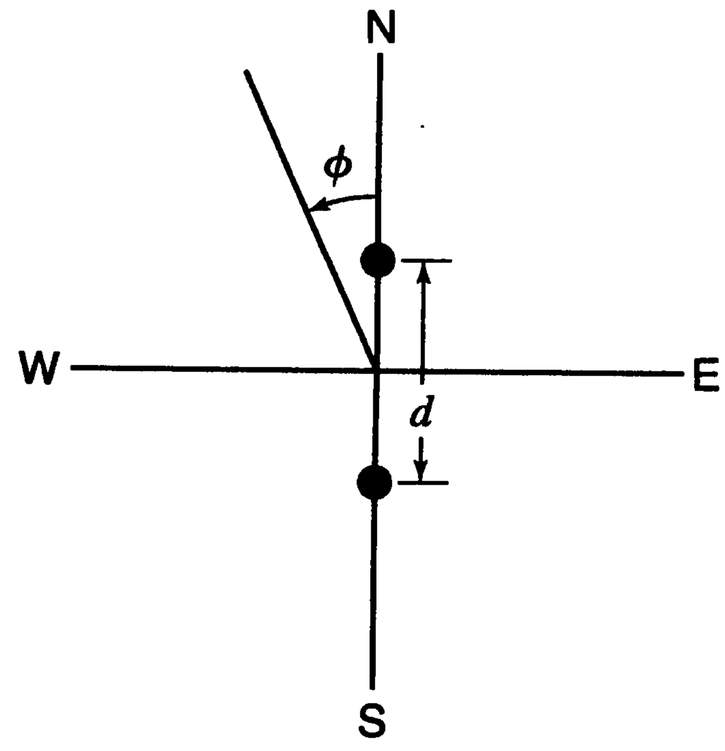


Figure 5–12
Arrangement of two isotropic point sources for both primary and secondary arrays.

Since we are interested only in the horizontal plane pattern, each tower may be considered as an isotropic point source. The problem then becomes one of finding a space and phase relation of four isotropic point sources located in the horizontal plane which fulfills the above requirements.

The principle of pattern multiplication will be applied to the solution of this problem by seeking the patterns of two pairs of isotropic sources which yield the desired pattern when multiplied together. First let us find a pair of isotropic sources whose pattern fulfills the requirements of a broad lobe of radiation with maximum north and a null southwest. This will be called the "primary" pattern.

Two isotropic sources phased as an end-fire array can produce a pattern with a broader major lobe than when phased as a broadside array (for example, compare Figs. 5–1*c* and 5–5). Since a broad lobe to the north is desired, an end-fire arrangement of two isotropic sources as shown in Fig. 5–12 will be tried. From a consideration of pattern shapes as a function of separation and phase, a spacing between $\lambda/4$ and $3\lambda/8$ appears suitable (see Fig. 16–11). (See Brown-1, Terman-1, and Smith-1.) Accordingly, let $d = 0.3\lambda$. Then the field pattern for the array is

$$E = \cos\frac{\psi}{2} \tag{1}$$

where

$$\psi = 0.6\pi \cos\phi + \delta \tag{2}$$

For there to be a null in the pattern of (1) at $\phi = 135°$, it is necessary that[1]

$$\psi = (2k + 1)\pi \tag{3}$$

where $k = 0, 1, 2, 3, \ldots$

[1]The azimuth angle ϕ (Fig. 5–12) is measured counterclockwise (ccw) from the north. This is consistent with the engineering practice of measuring positive angles in a counterclockwise sense. However, it should be noted that the *geodetic azimuth angle* of a point is measured in the opposite, or clockwise (cw), sense from the reference direction, which is sometimes taken as south and sometimes as north.

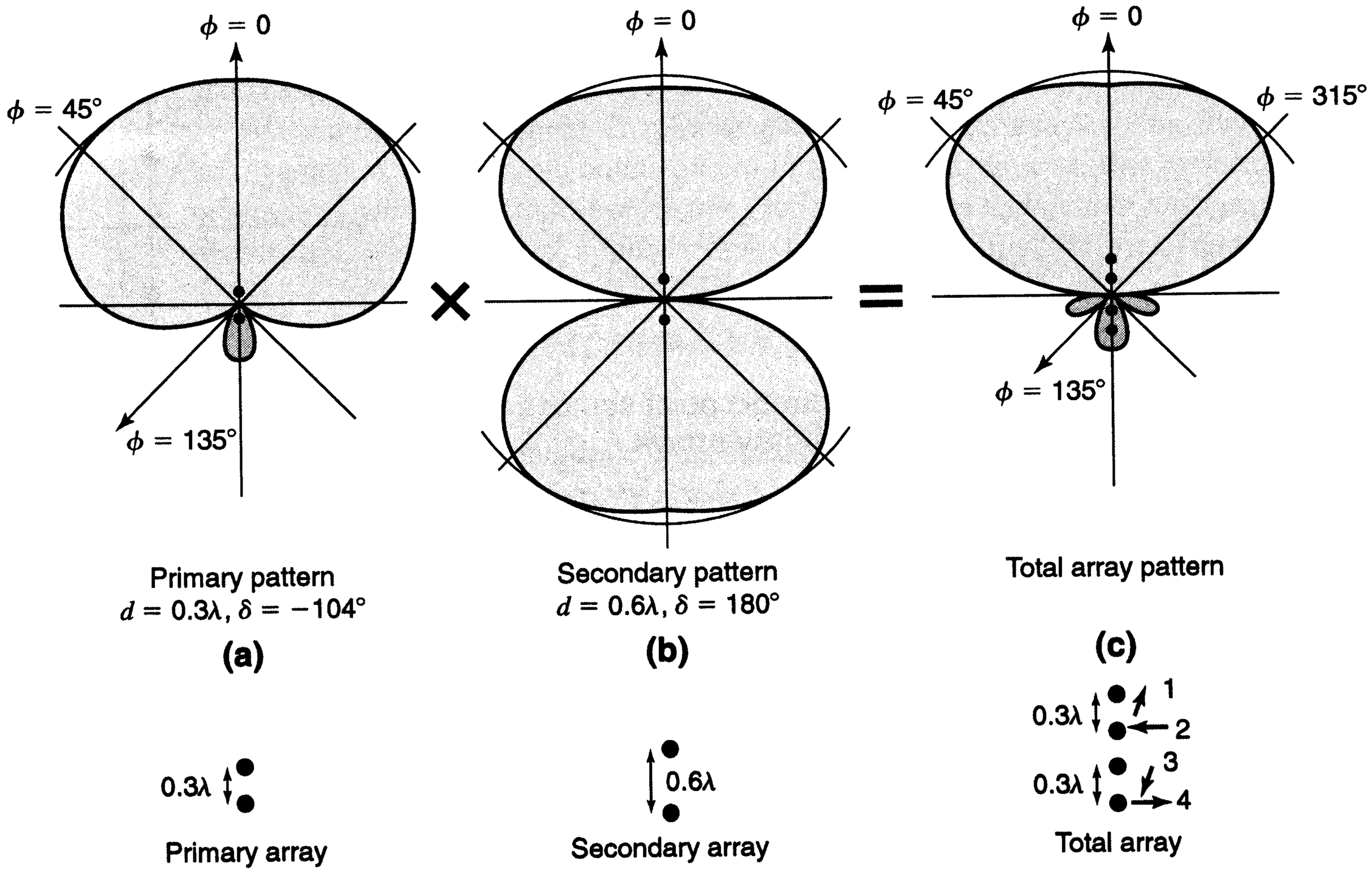


Figure 5–13
Field patterns of primary and secondary arrays of two isotropic sources which multiplied together give pattern of total array of four isotropic sources.

Equating (2) and (3) then gives

$$-0.6\pi\frac{1}{\sqrt{2}}+\delta=(2k+1)\pi \tag{4}$$

or

$$\delta=(2k+1)\pi+0.425\pi \tag{5}$$

For $k=0$, $\delta=-104°$. The pattern for this case ($d=0.3\lambda$ and $\delta=-104°$) is illustrated by Fig. 5–13*a*.

Next, let us find the array of two isotropic point sources that will produce a pattern that fulfills the requirements of a null at $\phi = 270°$ and that also has a broad lobe to the north. This will be called the "secondary" pattern. This pattern multiplied by the primary array pattern will then yield the total array pattern. If the secondary isotropic sources are also arranged as in Fig. 5–12 and have a phase difference of 180°, there is a null at $\phi = 270°$. Let the spacing $d = 0.6\lambda$. Then the secondary pattern is given by (1) where

$$\psi = 1.2\pi\cos\phi+\pi \tag{6}$$

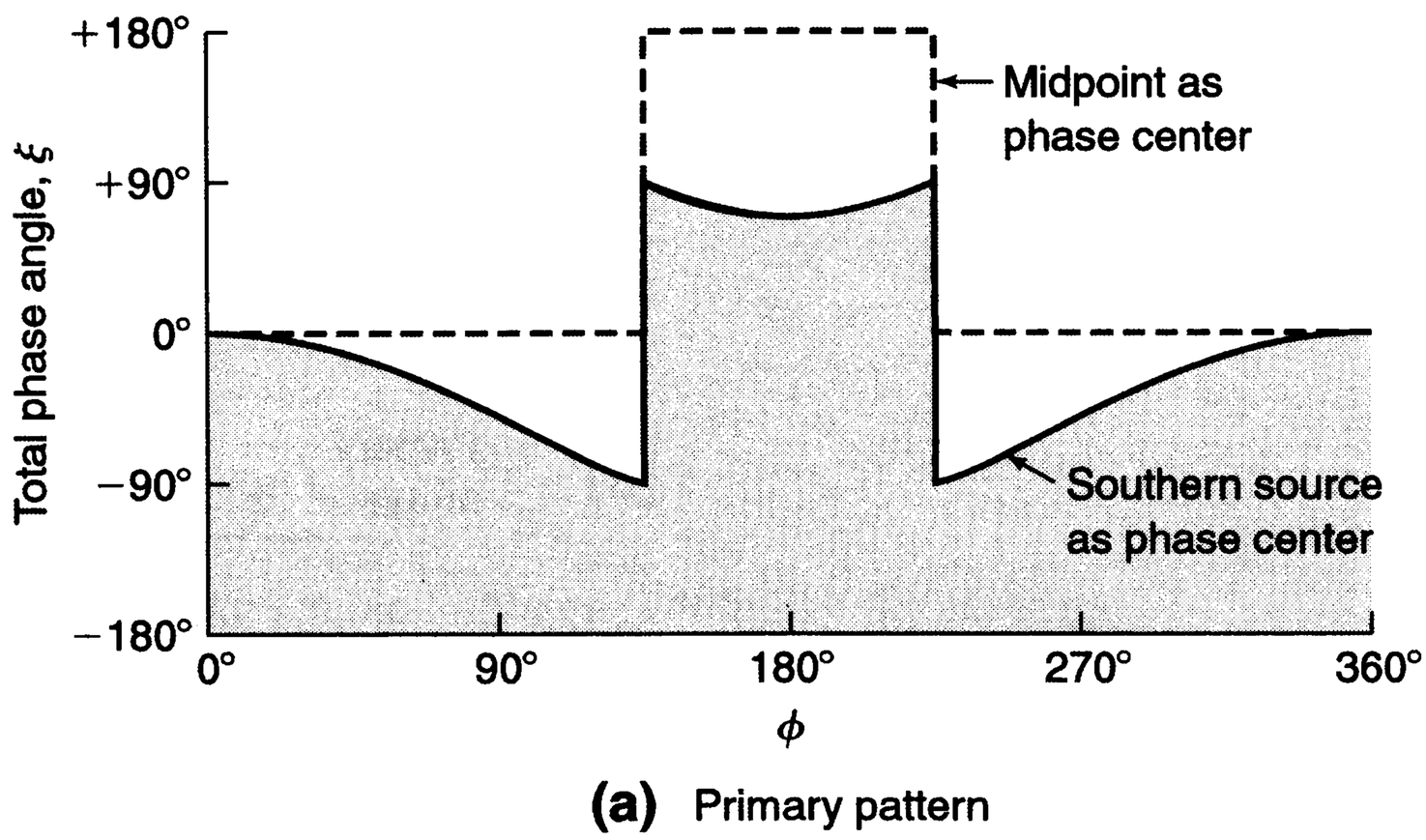
The pattern is illustrated by Fig. 5–13*b*. By the principle of pattern multiplication, the total array pattern is the product of this pattern and the primary array pattern, or

$$E=\cos(54°\cos\phi-52°)\cos(108°\cos\phi+90°) \tag{7}$$

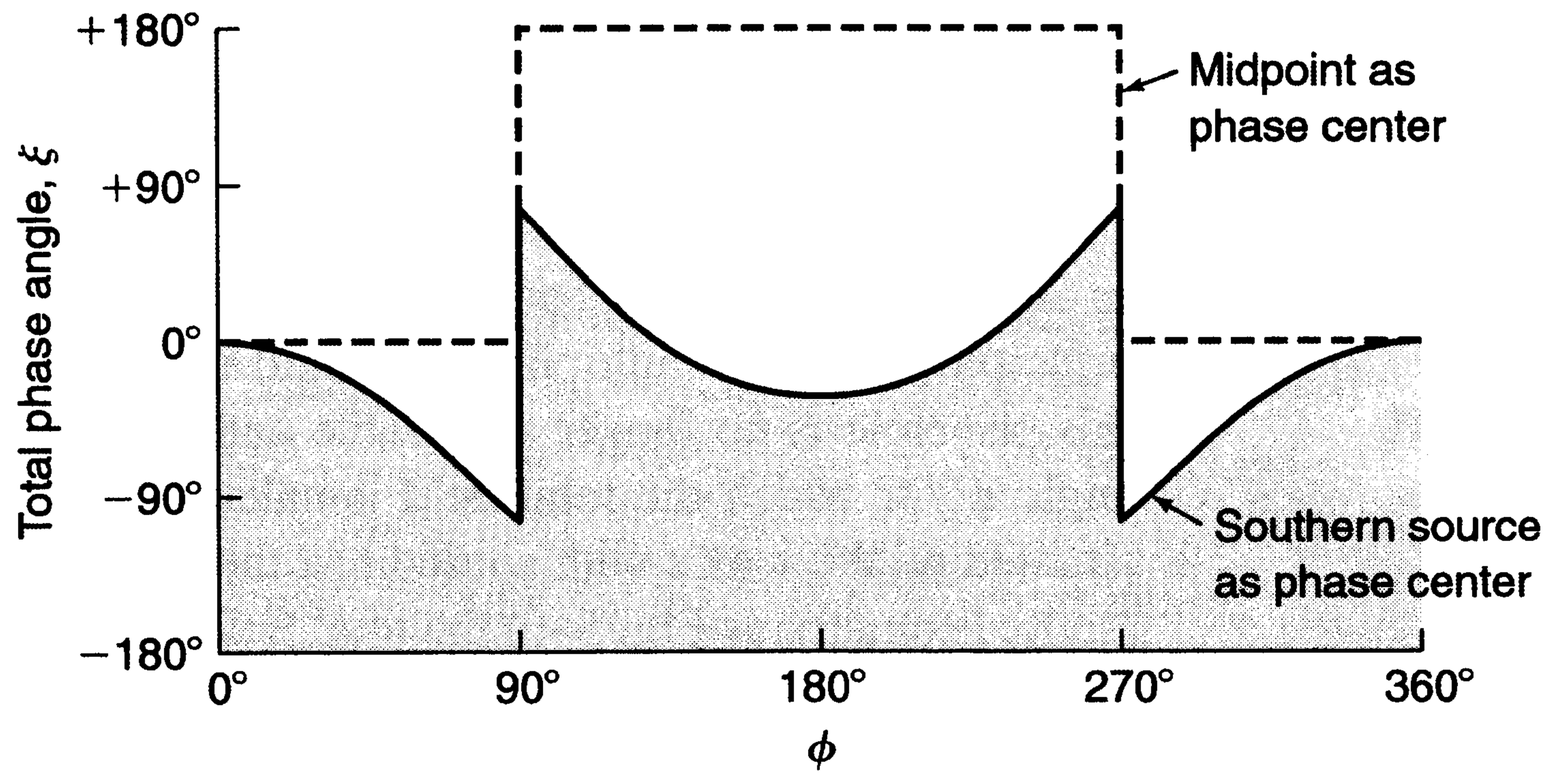
This pattern, which is illustrated by Fig. 5–13*c*, satisfies the pattern requirements. The complete array is obtained by replacing each of the isotropic sources of the secondary pattern by the two-source array producing the primary pattern. The midpoint of each primary array is its phase center, so this point is placed at the location of a secondary source. The complete antenna is then a linear array of four isotropic point sources as shown in the lower part of Fig. 5–13, where now each source represents a single vertical tower. All towers carry the same current. The current of tower 2 leads tower 1 and the current of tower 4 leads tower 3 by 104°, while the current in towers 1 and 3 and 2 and 4 are in phase opposition. The relative phase of the current is illustrated by the vectors in the lower part of Fig. 5–13*c*.

The solution obtained is only one of an infinite number of possible solutions involving four towers. It is, however, a satisfactory and practical solution to the problem.

The phase variation ξ around the primary, secondary, and total arrays is shown in Fig. 5–14*a*, *b*, and *c* with the phase center at the centerpoint of each array and also at the southernmost source. The arrangement of the arrays with their phase centers is illustrated in Fig. 5–14*d* for both cases.



(a) Primary pattern



(b) Secondary pattern

Figure 5–14
Phase patterns of primary, secondary and total arrays having the field patterns shown in Fig. 5–13. Phase patterns are given for the phase center at the midpoint of the array and at the southernmost source, the arrangement of the arrays and the phase centers being shown at (*d*). The phase angle ξ is adjusted to zero at $\phi = 0$ in all cases.

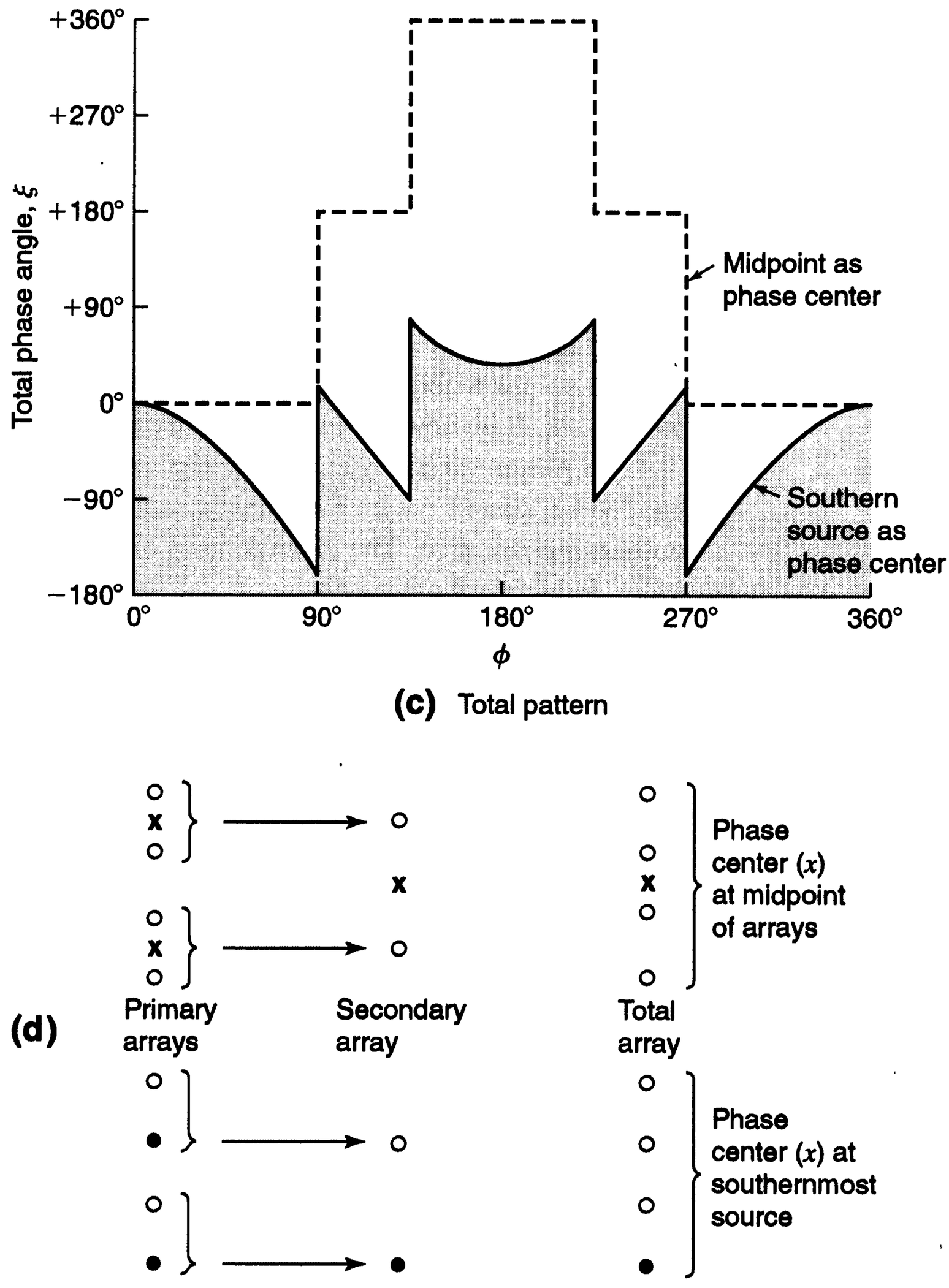


Figure 5–14
Continued.

5–5 NONISOTROPIC AND DISSIMILAR POINT SOURCES
（非各向同性的非相似点源）

In Sec. 5–3 nonisotropic but similar point sources were discussed, and it was shown that the principle of pattern multiplication could be applied. However, if the sources are dissimilar, this principle is no longer applicable and the fields of the sources must be added at each angle ϕ for which the total field is calculated. Thus, for two dissimilar sources 1 and 2 situated on the x axis with source 1 at the origin and the sources separated by a distance d (same geometry as Fig. 5–6) the total field is in general

$$E = E_1 + E_2 = E_0\sqrt{[f(\phi) + aF(\phi)\cos\psi]^2 + [aF(\phi)\sin\psi]^2}$$
$$\underline{/f_p(\phi) + \arctan[aF(\phi)\sin\psi/(f(\phi) + aF(\phi)\cos\psi)]} \qquad (1)$$

where the field from source 1 is taken as

$$E_1 = E_0 f(\phi) \angle f_p(\phi) \tag{2}$$

and from source 2 as

$$E_2 = aE_0 F(\phi) \angle F_p(\phi) + d_r \cos\phi + \delta \tag{3}$$

where

E_0 = constant
a = ratio of maximum amplitude of source 2 to source 1 ($0 \leq a \leq 1$)
ψ = $d_r \cos\phi + \delta - f_p(\phi) + F_p(\phi)$, where
δ = relative phase of source 2 with respect to source 1
$f(\phi)$ = relative field pattern of source 1
$f_p(\phi)$ = phase pattern of source 1
$F(\phi)$ = relative field pattern of source 2
$F_p(\phi)$ = phase pattern of source 2

In (1) the phase angle ($\angle$) is referred to the phase of the field from source 1 in some reference direction ($\phi = \phi_0$).

In the special case where the field patterns are identical but the phase patterns are not, $a = 1$, and

$$f(\phi) = F(\phi) \tag{4}$$

from which

$$E = 2E_0 f(\phi) \cos\frac{\psi}{2} \angle f_p(\phi) + \psi/2 \tag{5}$$

where phase is again referred to source 1 in some reference direction ϕ_0.

As an illustration of nonisotropic, dissimilar point sources, let us consider an example in which the field from source 1 is given by

$$E_1 = \cos\phi \angle 0 \tag{6}$$

and from source 2 by

$$E_2 = \sin\phi \angle \psi \tag{7}$$

where $\psi = d_r \cos\phi + \delta$

The relation of the two sources to the coordinate system and the individual field patterns is shown in Fig. 5–15. Source 1 is located at the origin. The total field E is then the vector

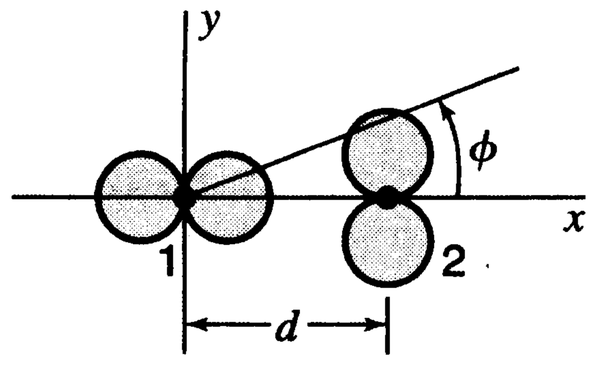


Figure 5–15
Relation of two nonisotropic dissimilar sources to coordinate system.

sum of E_1 and E_2, or

$$E = \cos\phi + \sin\phi \angle\psi \tag{8}$$

Let us consider the case for $\lambda/4$ spacing ($d = \lambda/4$) and phase quadrature of the sources ($\delta = \pi/2$). Then

$$\psi = \frac{\pi}{2}(\cos\phi + 1) \tag{9}$$

The calculation for this case is easily carried out by graphical vector addition. The resulting field pattern for the total field E of the array is presented in Fig. 5–16, and the resulting phase pattern for the angle ξ is given in Fig. 5–17. The angle ξ is the phase angle between the total field and the field of source 1 in the direction $\phi = 0$.

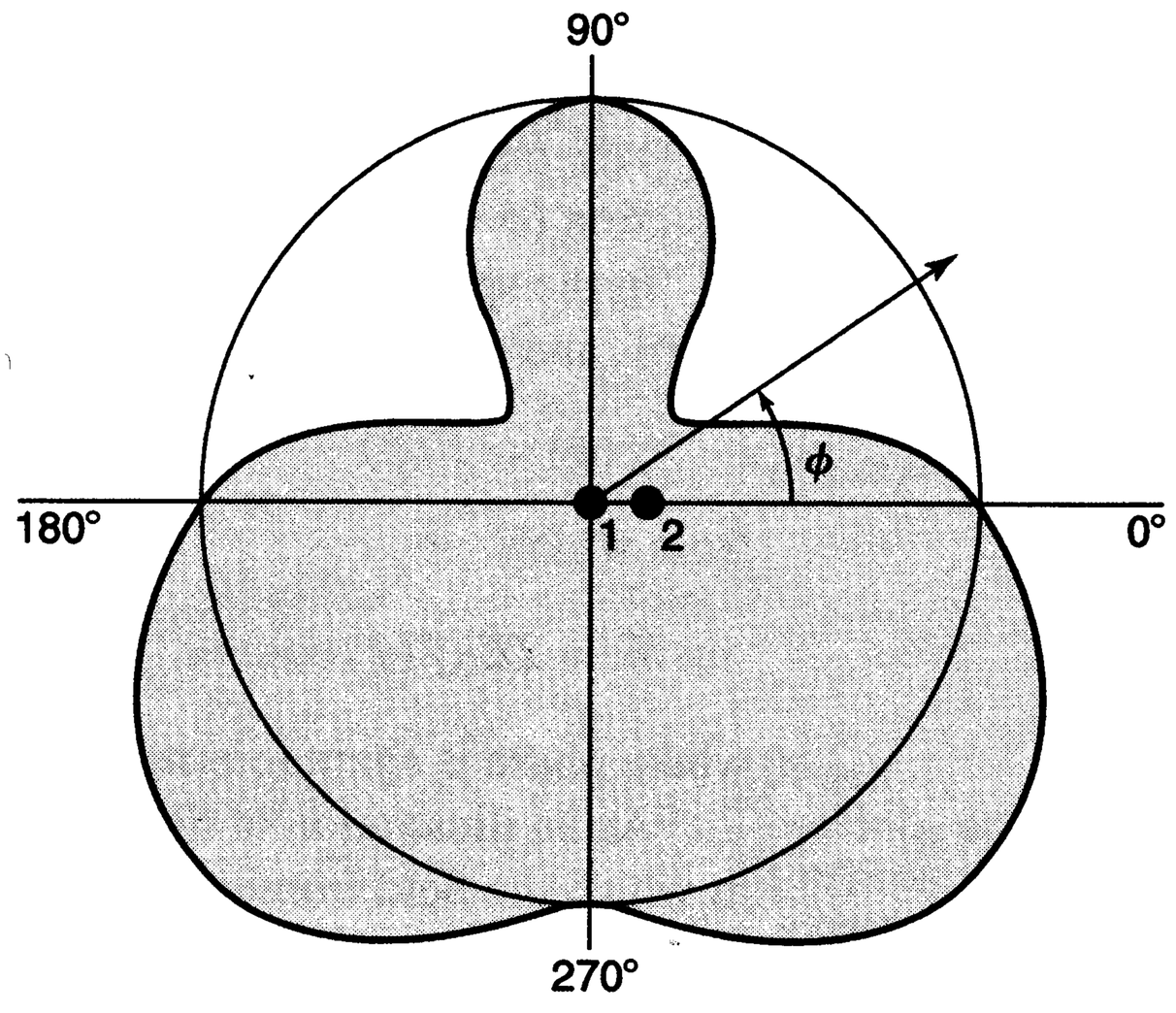


Figure 5–16
Field pattern of array of two nonisotropic dissimilar sources of Fig. 5–15 for $d = \lambda/4$ and $\delta = 90°$.

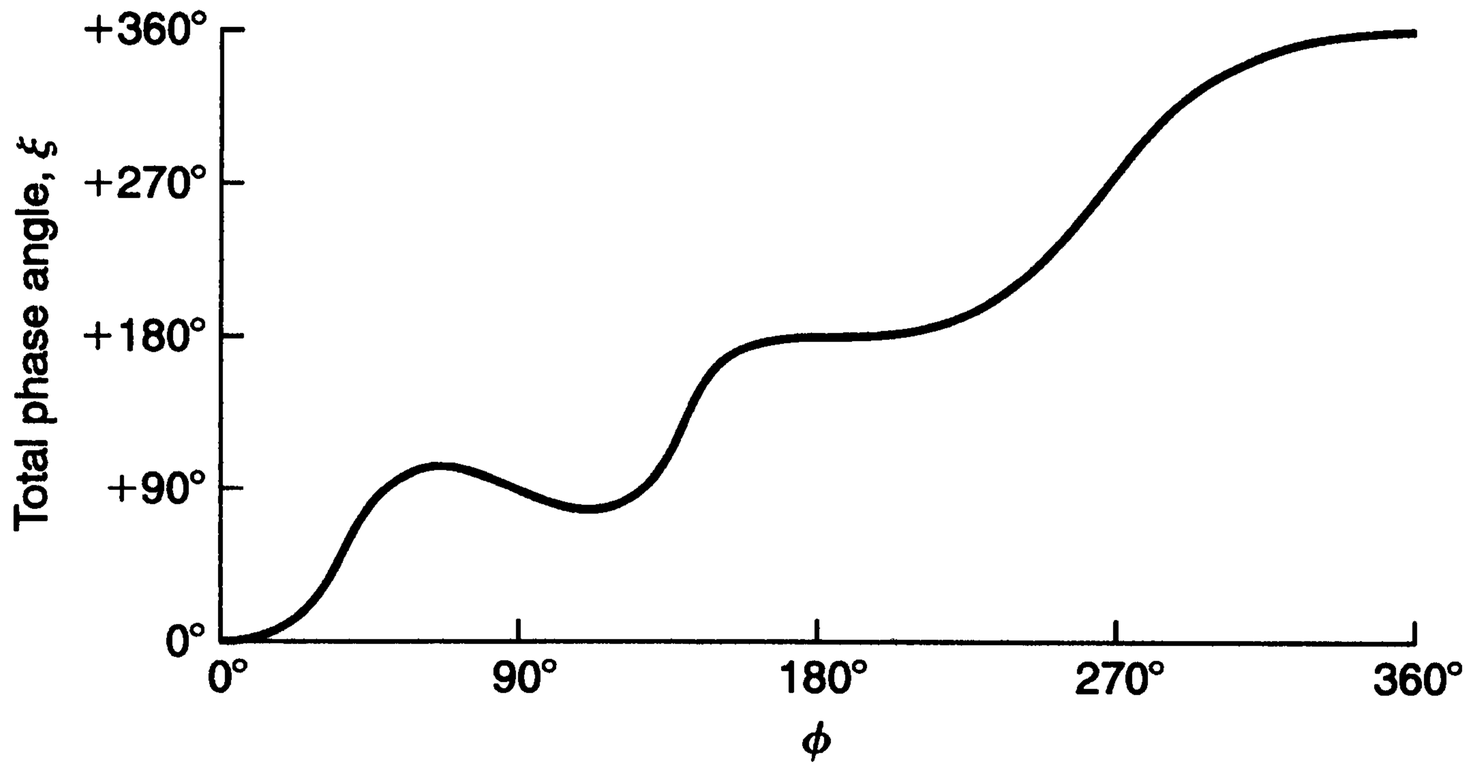


Figure 5–17
Phase pattern of array having field pattern of Fig. 5–16. The phase angle ξ is with respect to source 1 as phase center.

5–6 LINEAR ARRAYS OF *n* ISOTROPIC POINT SOURCES OF EQUAL AMPLITUDE AND SPACING（*n*个各向同性点源的等幅等间距直线阵）

Introduction

Let us now proceed to the case of n isotropic point sources of equal amplitude and spacing arranged as a linear array, as indicated in Fig. 5–18, where n is any positive integer. The total field E at a large distance in the direction ϕ is given by

$$E = 1 + e^{j\psi} + e^{j2\psi} + e^{j3\psi} + \cdots + e^{j(n-1)\psi} \tag{1}$$

where ψ is the total phase difference of the fields from adjacent sources as given by

$$\psi = \frac{2\pi d}{\lambda} \cos\phi + \delta = d_r \cos\phi + \delta \tag{2}$$

where δ is the phase difference of adjacent sources, i.e., source 2 with respect to 1, 3 with respect to 2, etc. (Schelkunoff-1, Stratton-1).

The amplitudes of the fields from the sources are all equal and taken as unity. Source 1 (Fig. 5–18) is the phase reference. Thus, at a distant point in the direction ϕ the field from source 2 is advanced in phase with respect to source 1 by ψ, the field from source 3 is advanced in phase with respect to source 1 by 2ψ, etc.

Equation (1) is a geometric series. Each term represents a phasor, and the amplitude of the total field E and its phase angle ξ can be obtained by phasor (vector) addition as in Fig. 5–19. Analytically, E can be expressed in a simple trigonometric form which we now develop as follows:

Multiply (1) by $e^{j\psi}$, giving

$$Ee^{j\psi} = e^{j\psi} + e^{j2\psi} + e^{j3\psi} + \cdots + e^{jn\psi} \tag{3}$$

Now subtract (3) from (1) and divide by $1 - e^{j\psi}$, yielding

$$E = \frac{1 - e^{jn\psi}}{1 - e^{j\psi}} \tag{4}$$

Equation (4) may be rewritten as

$$E = \frac{e^{jn\psi/2}}{e^{j\psi/2}} \left(\frac{e^{jn\psi/2} - e^{-jn\psi/2}}{e^{j\psi/2} - e^{-j\psi/2}} \right) \tag{5}$$

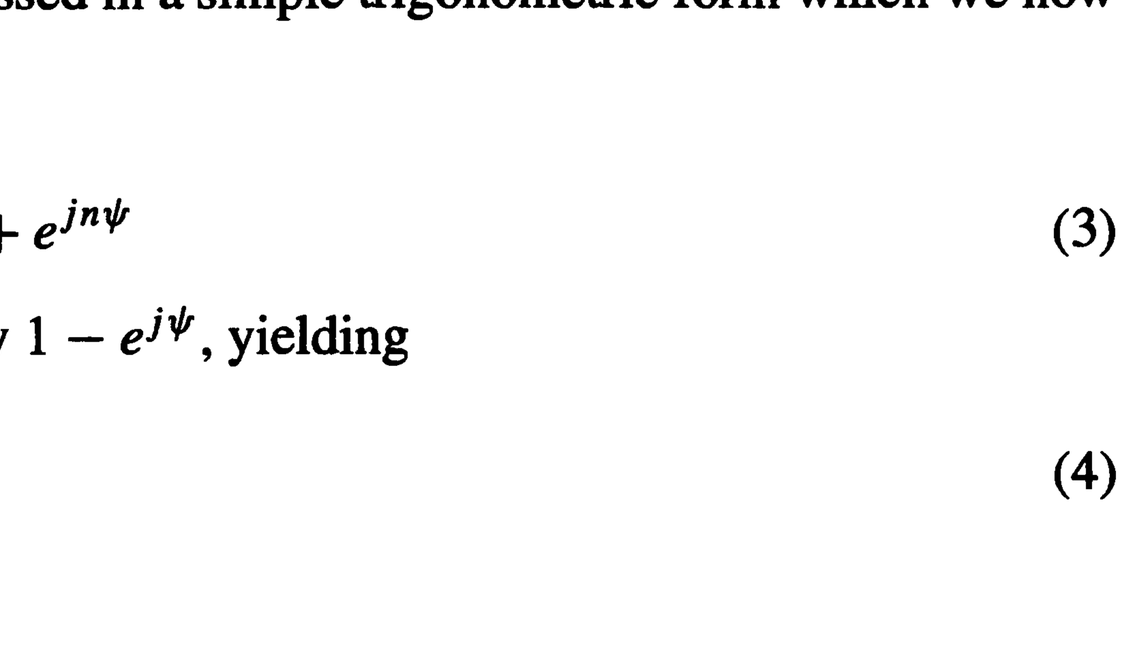


Figure 5–18
Arrangement of linear array of n isotropic point sources.

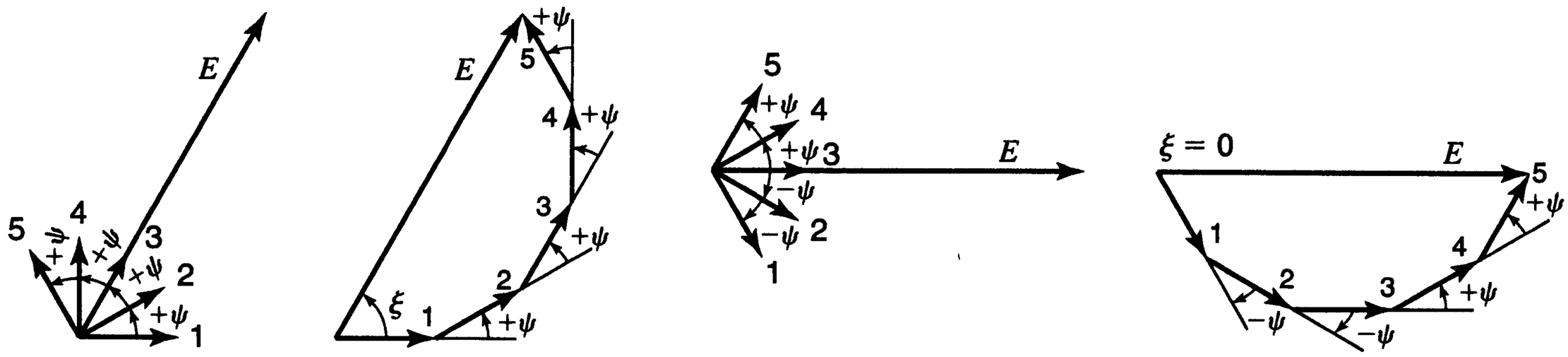


Figure 5–19
(*a*) Vector addition of fields at a large distance from the linear array of five isotropic point sources of equal amplitude with source 1 as the phase center (reference for phase). (*b*) Same, but with midpoint of array (source 3) as phase center.

from which

$$E = e^{j\xi}\frac{\sin(n\psi/2)}{\sin(\psi/2)} = \frac{\sin(n\psi/2)}{\sin(\psi/2)}\angle\xi \tag{6}$$

where ξ is referred to the field from source 1. The value of ξ is given by

$$\xi = \frac{n-1}{2}\psi \tag{7}$$

If the phase is referred to the centerpoint of the array, (6) becomes

$$E = \frac{\sin(n\psi/2)}{\sin(\psi/2)} \tag{8}$$

In this case the phase pattern is a step function as given by the sign of (8). The phase of the field is constant wherever E has a value but changes sign when E goes through zero.

When $\psi = 0$, (6) or (8) is indeterminate so that for this case E must be obtained as the limit of (8) as ψ approaches zero. Thus, for $\psi = 0$ we have the relation that

$$E = n \tag{8a}$$

This is the maximum value that E can attain. Hence, the normalized value of the total field for $E_{\max} = n$ is

$$E = \frac{1}{n}\frac{\sin(n\psi/2)}{\sin(\psi/2)} \tag{9}$$

The field as given by (9) will be referred to as the "array factor." Values of the array factor as obtained from (9) for various numbers of sources are presented in Fig. 5–20. If ψ is known as a function of ϕ, then the field pattern can be obtained directly from Fig. 5–20.

We may conclude from the above discussion that the field from the array will be a maximum in any direction ϕ for which $\psi = 0$. Stated in another way, the fields from the sources all arrive at a distant point in the same phase when $\psi = 0$. In special cases, ψ may not be zero for any value of ϕ, and in this case the field is usually a maximum at the minimum value of ψ.

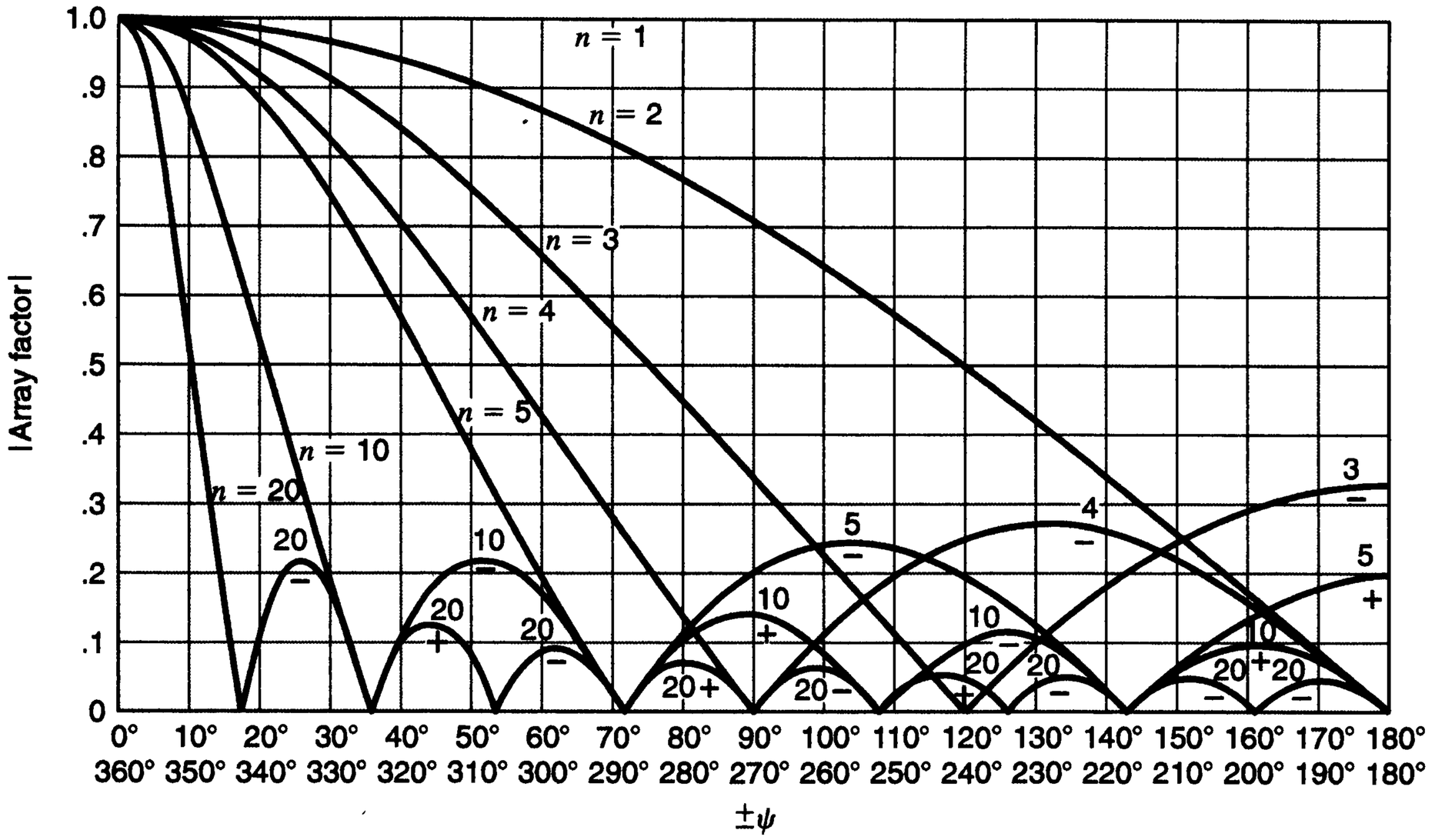


Figure 5–20
Universal field-pattern chart for arrays of various numbers n of isotropic point sources of equal amplitude and spacing.

To illustrate some of the properties of linear arrays (9) will now be applied to several special cases. See programs on the book's web site involving these different cases. See also discussion in Appendix C.

Case 1. Broadside Array (Sources in Phase)

The first case is a linear array of n isotropic sources of the same amplitude and phase. Therefore, $\delta = 0$ and

$$\psi = d_r \cos\phi \tag{10}$$

To make $\psi = 0$ requires that $\phi = (2k + 1)(\pi/2)$, where $k = 0, 1, 2, 3, \ldots$. The field is, therefore, a maximum when

$$\phi = \frac{\pi}{2} \qquad \text{and} \qquad \frac{3\pi}{2} \tag{10a}$$

That is, the maximum field is in a direction normal to the array. Hence, this condition, which is characterized by in-phase sources ($\delta = 0$), results in a "broadside" type of array.

As an example, the pattern of a broadside array of four in-phase isotropic point sources of equal amplitude is shown in Fig. 5–21*a*. The spacing between sources is $\lambda/2$.[1] The field pattern in rectangular coordinates and the phase patterns for this array are presented in Fig. 5–21*b*.

[1] If the spacing between elements exceeds λ, sidelobes appear which are equal in amplitude to the main (center) lobe. These are called *grating lobes* (see Sec. 15–6)

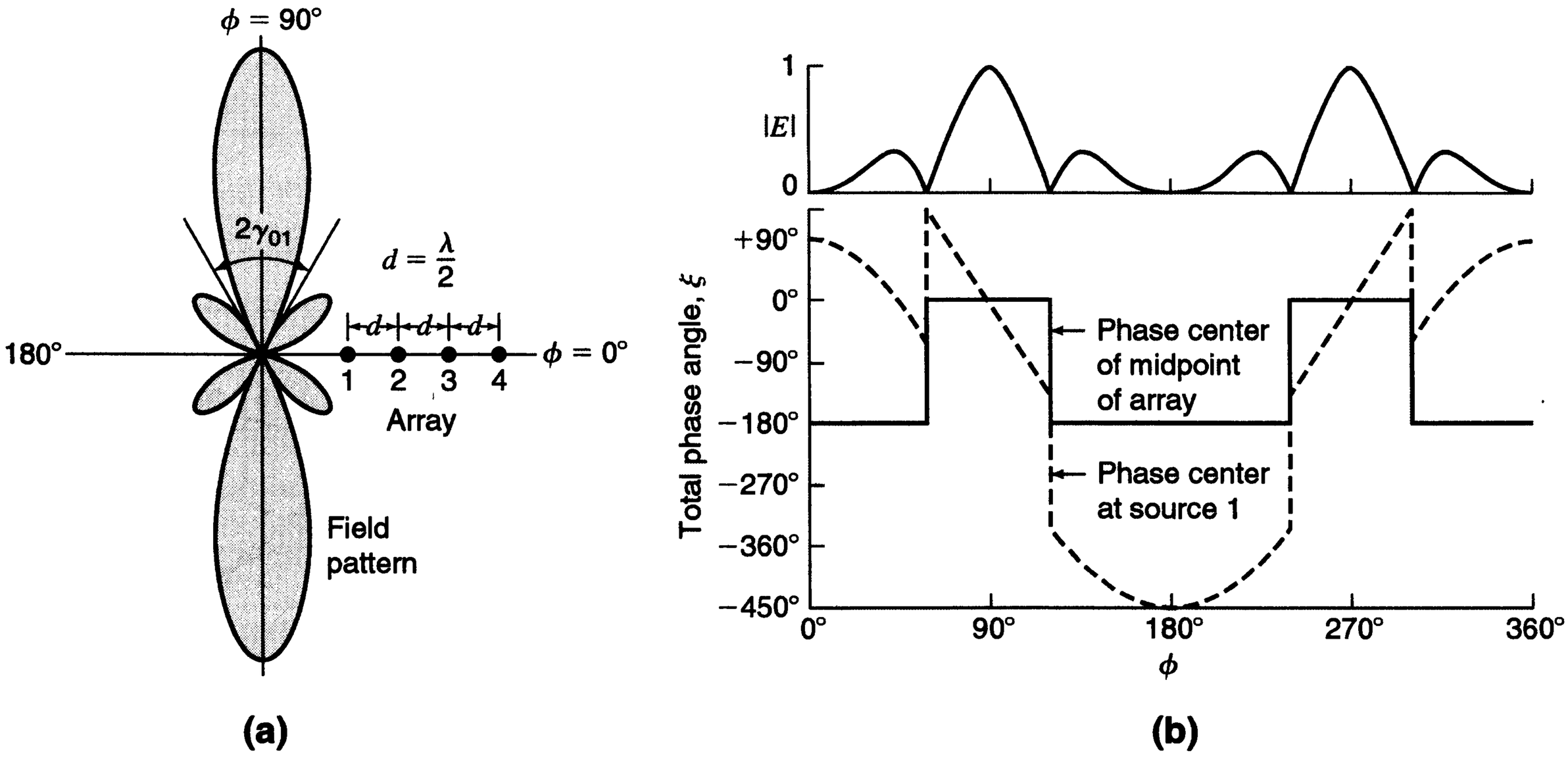


Figure 5–21
(*a*) Field pattern of broadside array of four isotropic point sources of the same amplitude and phase. The spacing between sources is $\lambda/2$. (*b*) Field pattern in rectangular coordinates and phase patterns of same array with phase center at midpoint and at source 1. The reference direction for phase is at $\phi = 90°$.

Case 2. Ordinary End-Fire Array

Let us now find the phase angle between adjacent sources that is required to make the field a maximum in the direction of the array ($\phi = 0$). An array of this type may be called an "end-fire" array. For this we substitute the conditions $\psi = 0$ and $\phi = 0$ into (2), from which

$$\delta = -d_r \tag{11}$$

Hence, for an end-fire array, the phase between sources is retarded progressively by the same amount as the spacing between sources in radians. Thus, if the spacing is $\lambda/4$, source 2 in Fig. 5–18 should lag source 1 by 90°, source 3 should lag source 2 by 90°, etc.

As an example, the field pattern of an end-fire array of four isotropic point sources is presented in Fig. 5–22*a*. The spacing between sources is $\lambda/2$ and $\delta = -\pi$. The field pattern in rectangular coordinates and the phase patterns are shown in Fig. 5–22*b*. The same shape of field pattern is obtained in this case if $\delta = +\pi$ since, with $d = \lambda/2$, the pattern is bidirectional. However, if the spacing is less than $\lambda/2$, the maximum radiation is in the direction $\phi = 0$ when $\delta = -d_r$ and in the direction $\phi = 180°$ when $\delta = +d_r$.

Case 3. End-Fire Array with Increased Directivity

The situation discussed in Case 2, namely, for $\delta = -d_r$, produces a maximum field in the direction $\phi = 0$ but does not give the maximum directivity. It has been shown by Hansen (1)

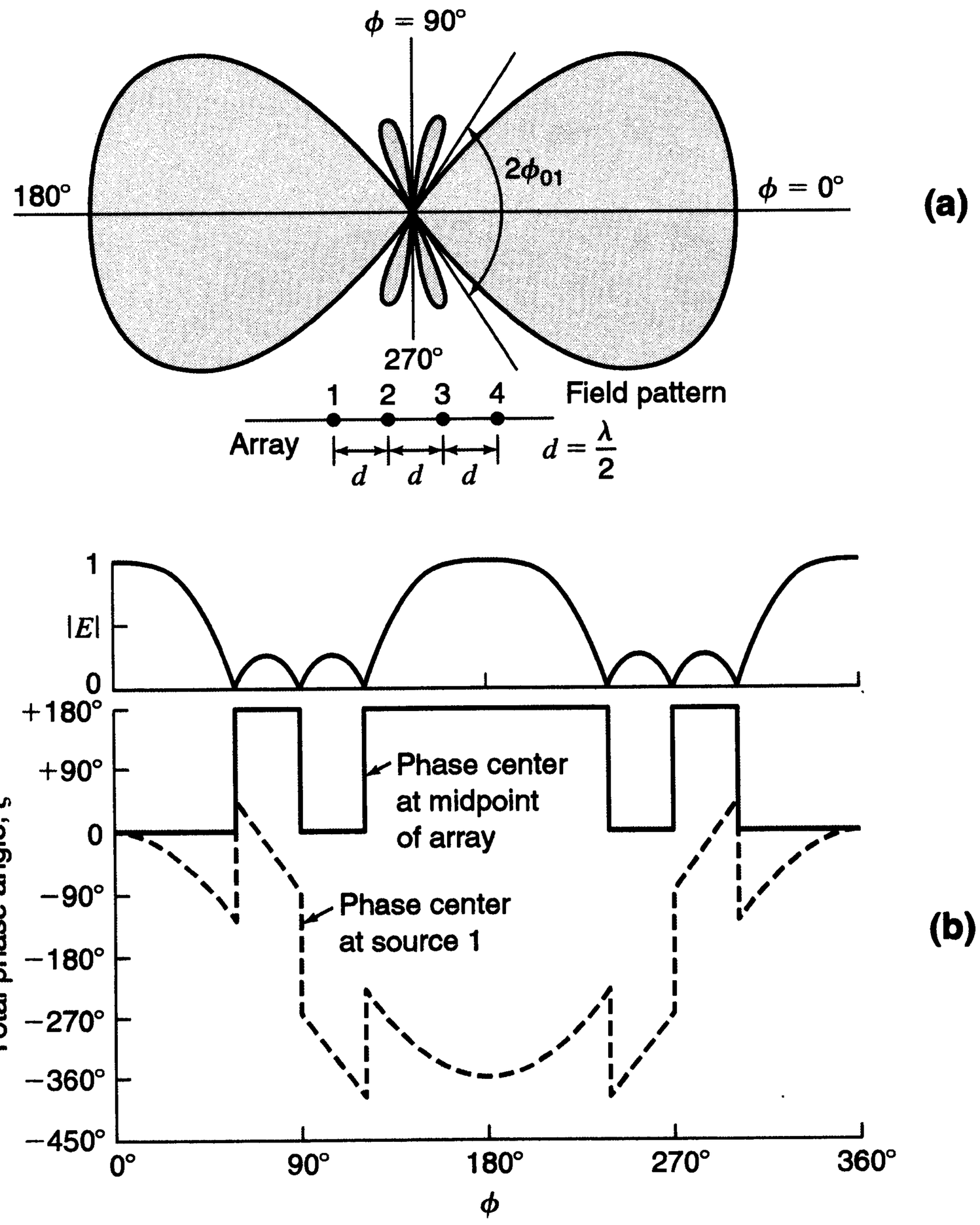


Figure 5–22
(*a*) Field pattern of ordinary end-fire array of four isotropic point sources of same amplitude. Spacing is $\lambda/2$ and the phase angle $\delta = -\pi$. (*b*) Field pattern in rectangular coordinates and phase patterns of same array with phase center at midpoint and at source 1. The reference direction for phase is at $\phi = 0$.

and Woodyard that a larger directivity is obtained by increasing the phase change between sources so that

$$\delta = -\left(d_r + \frac{\pi}{n}\right) \tag{12}$$

This condition will be referred to as the condition for "increased directivity." Thus for the phase difference of the fields at a large distance we have

$$\psi = d_r(\cos\phi - 1) - \frac{\pi}{n} \tag{13}$$

As an example, the field pattern of an end-fire array of four isotropic point sources for this case is illustrated in Fig. 5–23. The spacing between sources is $\lambda/2$, and therefore $\delta = -(5\pi/4)$. Hence, the conditions are the same as for the array with the pattern of Fig. 5–22, except that the phase difference between sources is increased by $\pi/4$. Comparing the field patterns of Figs. 5–22*a* and 5–23, it is apparent that the additional phase difference yields a considerably sharper main lobe in the direction $\phi = 0$. However, the

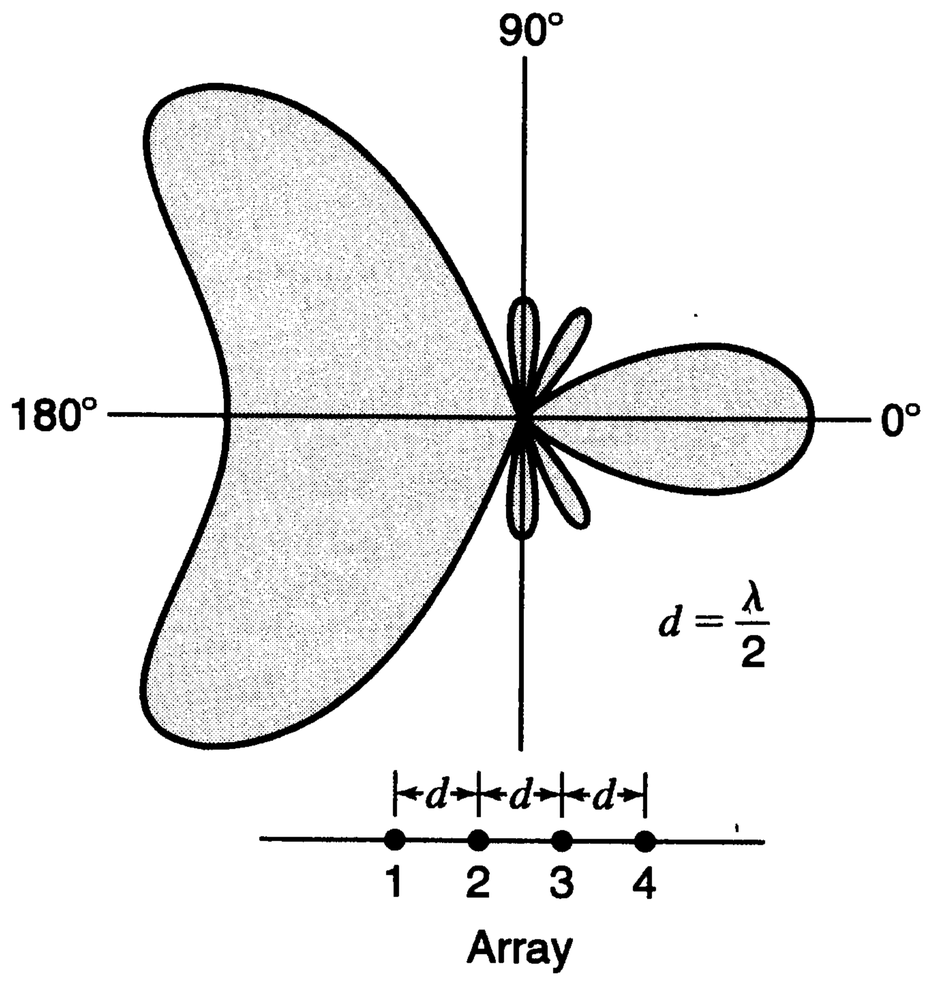


Figure 5–23
Field pattern of end-fire array of four isotropic point sources of equal amplitude spaced $\lambda/2$ apart. The phasing is adjusted for increased directivity ($\delta = -\frac{5}{4}\pi$).

back lobes in this case are excessively large because the large value of spacing results in too great a range in ψ.

To realize the directivity increase afforded by the additional phase difference requires that $|\psi|$ be restricted in its range to a value of π/n at $\phi = 0$ and a value in the vicinity of π at $\phi = 180°$. This can be fulfilled if the spacing is reduced. For example, the field pattern of an end-fire array of 10 isotropic point sources of equal amplitude and spaced $\lambda/4$ apart is presented in Fig. 5–24*a* for the phase condition giving increased directivity ($\delta = -0.6\pi$). In contrast to this pattern, one is presented in Fig. 5–24*b* for the identical antenna with the phasing of an ordinary end-fire array ($\delta = -0.5\pi$). Both patterns are plotted to the same maximum. The increased directivity is apparent from the greater sharpness of the pattern. Integrating the pattern, including the minor lobes, the directivity is found to be about 19 and of the ordinary endfire about 11. The beamwidths and directivities for the two cases are compared in Table 5–1.

The maximum of the field pattern of Fig. 5–24*a* occurs at $\phi = 0$ and $\psi = -\pi/n$. In general, any increased directivity end-fire array, with maximum at $\psi = -\pi/n$, has a normalized field pattern given by

$$E = \sin\left(\frac{\pi}{2n}\right)\frac{\sin(n\psi/2)}{\sin(\psi/2)} \tag{14}$$

Case 4. Array with Maximum Fieid in an Arbitrary Direction. Scanning Array

Let us consider the case of an array with a field pattern having a maximum in some arbitrary direction ϕ_1 not equal to $k\pi/2$ where $k = 0, 1, 2,$ or 3. Then (2) becomes

$$0 = d_r \cos\phi_1 + \delta \tag{15}$$

By specifying the spacing d_r, the required phase difference δ is then determined by (15). Conversely, by changing δ the beam direction ϕ_1 can be shifted or scanned.

Phase shift 180° (0.6π) versus 90° (0.5π)

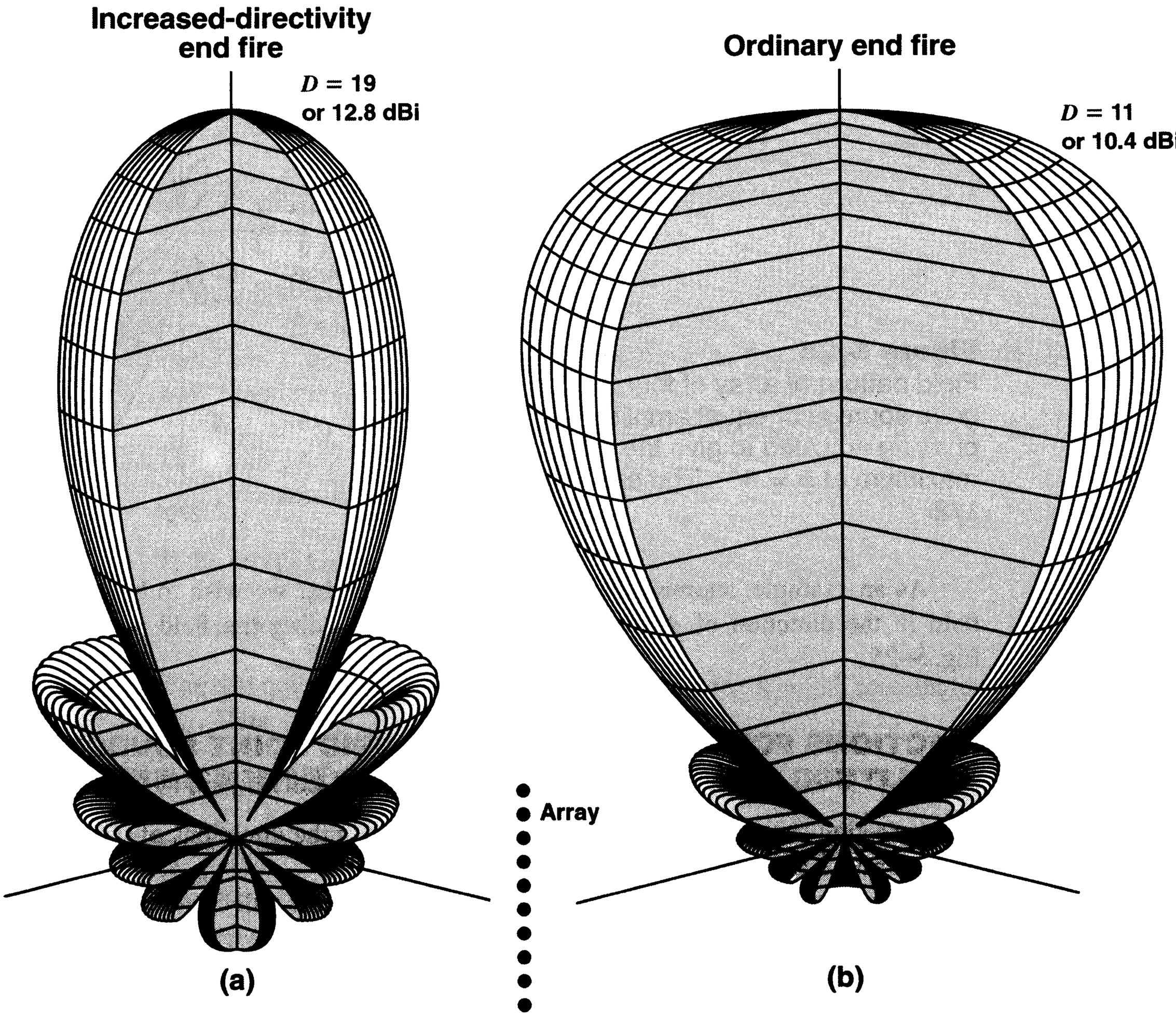


Figure 5–24
Field patterns of end-fire arrays of 10 isotropic point sources of equal amplitude spaced $\lambda/4$ apart. The pattern at (*a*) has the phase adjusted for increased directivity ($\delta = -0.6\pi$), while the pattern at (*b*) has the phasing of an ordinary end-fire array ($\delta = -0.5\pi$).

Table 5–1 Comparison of end-fire arrays

	Ordinary end-fire array	End-fire array with increased directivity
Beamwidth between half-power points	69°	38°
Beamwidth between first nulls	106°	74°
Directivity	11	19

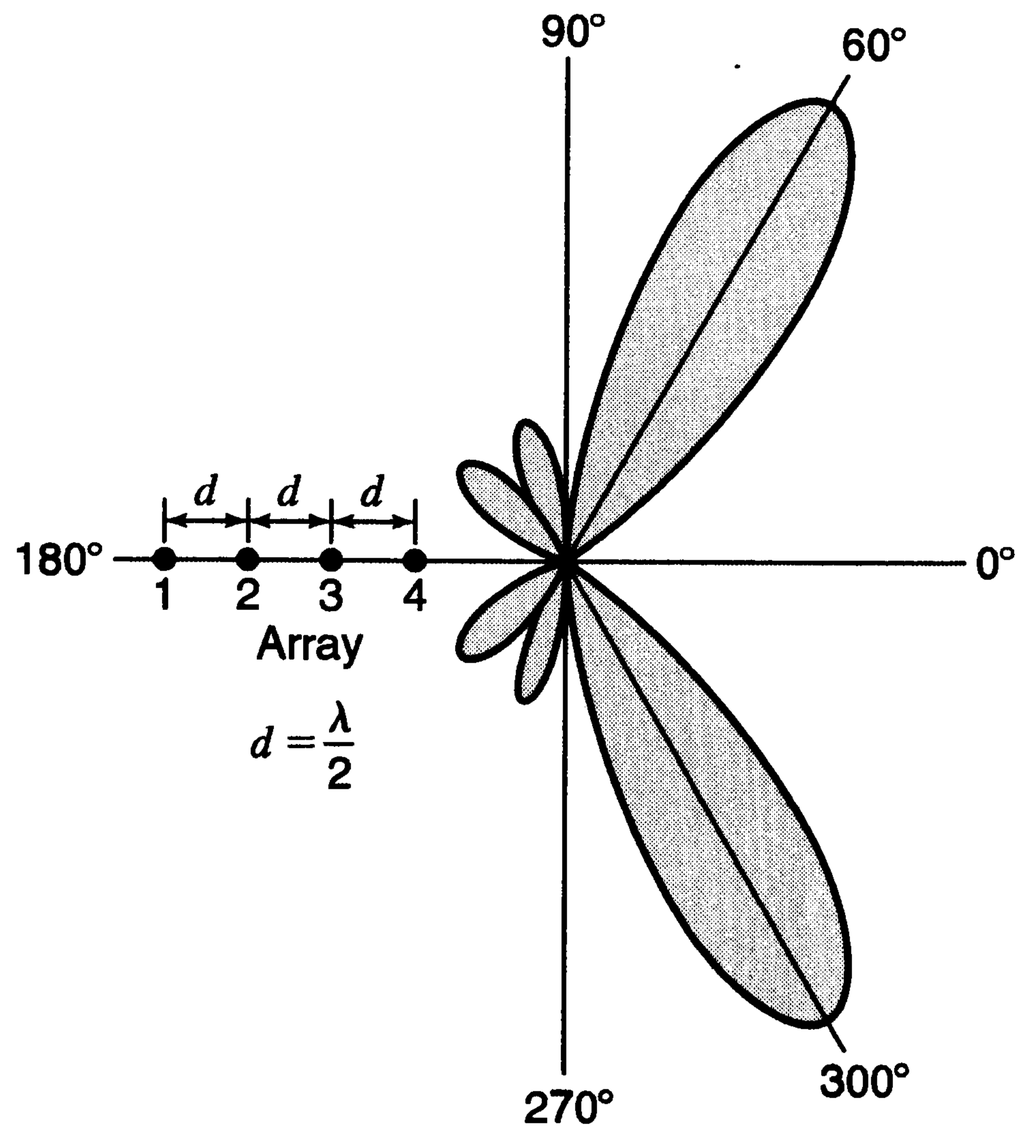


Figure 5–25
Field pattern of array of four isotropic point sources of equal amplitude with phasing adjusted to give the maximum at $\phi = 60°$. The spacing is $\lambda/2$.

As an example, suppose that $n = 4$, $d = \lambda/2$, and that we wish to have a maximum field in the direction of $\phi = 60°$. Then $\delta = -\pi/2$, yielding the field pattern shown in Fig. 5–25.

5–7 NULL DIRECTIONS FOR ARRAYS OF *n* ISOTROPIC POINT SOURCES OF EQUAL AMPLITUDE AND SPACING（*n*个各向同性点源的等幅等间距阵的零方向）

In this section simple methods are discussed for finding the directions of the pattern nulls of the arrays considered in Sec. 5–6.

Following the procedure given by Schelkunoff (2, 3) the null directions for an array of n isotropic point sources of equal amplitude and spacing occur when $E = 0$ or, provided that the denominator of Eq. (5–6–4) is not zero, when

$$e^{jn\psi} = 1 \tag{1}$$

Equation (1) requires that

$$n\psi = \pm 2K\pi \tag{2}$$

where $K = 1, 2, 3, \ldots$

Equating the value of ψ in (2) to its value in Eq. (5–6–2) gives

$$\psi = d_r \cos\phi_0 + \delta = \pm\frac{2K\pi}{n} \tag{3}$$

Thus,

$$\phi_0 = \arccos\left[\left(\pm\frac{2K\pi}{n} - \delta\right)\frac{1}{d_r}\right] \tag{4}$$

where ϕ_0 gives the direction of the pattern nulls. Note that values of K must be excluded

for which $K = mn$, where $m = 1, 2, 3, \ldots$. Thus, if $K = mn$, (2) reduces to $\psi = \pm 2m\pi$ and the denominator of Eq. (5–6–4) equals zero so that the null condition of (1), that the numerator of Eq. (5–6–4) be zero, is insufficient.

In a broadside array $\delta = 0$, so that for this case (4) becomes

$$\phi_0 = \arccos\left(\pm\frac{2K\pi}{nd_r}\right) = \arccos\left(\pm\frac{K\lambda}{nd}\right) \tag{5}$$

As an example, the field pattern of Fig. 5–21 ($n = 4, d = \lambda/2, \delta = 0$) has the null directions

$$\phi_0 = \arccos\left(\pm\frac{K}{2}\right) \tag{6}$$

For $K = 1, \phi_0 = \pm 60°$ and $\pm 120°$, and for $K = 2, \phi_0 = 0°$ and $180°$. These are the six null directions for this array.

If ϕ_0 in (3) is replaced by its complementary angle γ_0 (see Fig. 5–18), then (5) becomes

$$\gamma_0 = \arcsin\left(\pm\frac{K\lambda}{nd}\right) \tag{7}$$

If the array is long, so that $nd \gg K\lambda$,

$$\gamma_0 \simeq \pm\frac{K\lambda}{nd} \tag{8}$$

The first nulls either side of the maximum occur for $K = 1$. These angles will be designated γ_{01}. Thus,

$$\gamma_{01} \simeq \pm\frac{\lambda}{nd} \tag{9}$$

and the total beamwidth of the main lobe between first nulls *for a long broadside array* is then

$$2\gamma_{01} \simeq \frac{2\lambda}{nd} \tag{10}$$

For the field pattern in Fig. 5–21 this width is exactly 60°, while as given by (10) it is 1 rad, or 57.3°. This pattern is for an array 2λ long. The agreement would be better with longer arrays.

Turning next to *end-fire arrays,* the condition for an ordinary end-fire array is that $\delta = -d_r$. Thus, for this case (3) becomes

$$\cos\phi_0 - 1 = \pm\frac{2K\pi}{nd_r} \tag{11}$$

from which we obtain

$$\frac{\phi_0}{2} = \arcsin\left(\pm\sqrt{\frac{K\pi}{nd_r}}\right) \tag{12}$$

or

$$\phi_0 = 2\arcsin\left(\pm\sqrt{\frac{K\lambda}{2nd}}\right) \tag{13}$$

As an example, the field pattern of Fig. 5–22 ($n = 4, d = \lambda/2, \delta = -\pi$) has the null directions

$$\phi_0 = 2 \arcsin\left(\pm\sqrt{\frac{K}{4}}\right) \tag{14}$$

For $K = 1$, $\phi_0 = \pm 60°$; for $K = 2$, $\phi_0 = \pm 90°$, etc.

If the array is long, so that $nd \gg K\lambda$, (13) becomes

$$\phi_0 \simeq \pm\sqrt{\frac{2K\lambda}{nd}} \tag{15}$$

The first nulls either side of the main lobe occur for $K = 1$. These angles will be designated ϕ_{01}. Thus,

$$\phi_{01} \simeq \pm\sqrt{\frac{2\lambda}{nd}} \tag{16}$$

and the total beamwidth of the main lobe between first nulls *for a long ordinary end-fire array* is then

$$2\phi_{01} \simeq 2\sqrt{\frac{2\lambda}{nd}} \tag{17}$$

For the field pattern in Fig. 5–22 this width is exactly 120°, while as given by (17) it is 2 rad, or 115°.

For end-fire arrays with increased directivity as proposed by Hansen (1) and Woodyard, the condition is that $\delta = -(d_r + \pi/n)$. Thus, for this case (3) becomes

$$d_r(\cos\phi_0 - 1) - \frac{\pi}{n} = \pm 2\frac{K\pi}{n} \tag{18}$$

from which

$$\frac{\phi_0}{2} = \arcsin\left[\pm\sqrt{\frac{\pi}{2nd_r}(2K-1)}\right] \tag{19}$$

or

$$\phi_0 = 2\arcsin\left[\pm\sqrt{\frac{\lambda}{4nd}(2K-1)}\right] \tag{20}$$

If the array is long, so that $nd \gg K\lambda$, (20) becomes

$$\phi_0 \simeq \pm\sqrt{\frac{\lambda}{nd}(2K-1)} \tag{21}$$

The first nulls either side of the main lobe, ϕ_{01}, occur for $K = 1$. Thus,

$$\phi_{01} \simeq \pm\sqrt{\frac{\lambda}{nd}} \tag{22}$$

and the total beamwidth of the main lobe between first nulls *for a long end-fire array with increased directivity* is then

$$2\phi_{01} \simeq 2\sqrt{\frac{\lambda}{nd}} \tag{23}$$

This width is $1/\sqrt{2}$, or 71 percent, of the width of the ordinary end-fire array. As an example, the ordinary end-fire array pattern of Fig. 5–24*b* has a beamwidth between first nulls of 106°. The width of the pattern in Fig. 5–24*a* for the array with increased directivity is 74°, or 70 percent as much.

Table 5–2 lists the formulas for null directions and beamwidths for the different arrays considered above. The null directions in column 2 apply to arrays of any length. The formulas in the third and fourth columns are approximate and apply only to long arrays.

The formulas in Table 5–2 have been used to calculate the curves presented in Fig. 5–26. These curves show the beamwidth between first nulls as a function of nd_λ for three types of arrays: broadside, ordinary end-fire, and end-fire with increased directivity. The quantity

Table 5–2 Null directions and beamwidths between first nulls for linear arrays of n isotropic point sources of equal amplitude and spacing. (For $n \geq 2$. The angles in columns 3 and 4 are expressed in radians. To convert to degrees, multiply by 57.3)

Type of array	Null directions (array any length)	Null directions (long array)	Beamwidth between first nulls (long array)
General case	$\phi_0 = \arccos\left[\left(\frac{\pm 2K\pi}{n} - \delta\right)\frac{1}{d_r}\right]$		
Broadside	$\gamma_0 = \arcsin\left(\pm\frac{K\lambda}{nd}\right)$	$\gamma_0 \simeq \pm\frac{K\lambda}{nd}$	$2\gamma_{01} \simeq \frac{2\lambda}{nd}$
Ordinary end-fire	$\phi_0 = 2\arcsin\left(\pm\sqrt{\frac{K\lambda}{2nd}}\right)$	$\phi_0 \simeq \pm\sqrt{\frac{2K\lambda}{nd}}$	$2\phi_{01} \simeq 2\sqrt{\frac{2\lambda}{nd}}$
End-fire with increased directivity (Hansen and Woodyard)	$\phi_0 = 2\arcsin\left[\pm\sqrt{\frac{\lambda}{4nd}(2K-1)}\right]$	$\phi_0 \simeq \pm\sqrt{\frac{\lambda}{nd}(2K-1)}$	$2\phi_{01} \simeq 2\sqrt{\frac{\lambda}{nd}}$

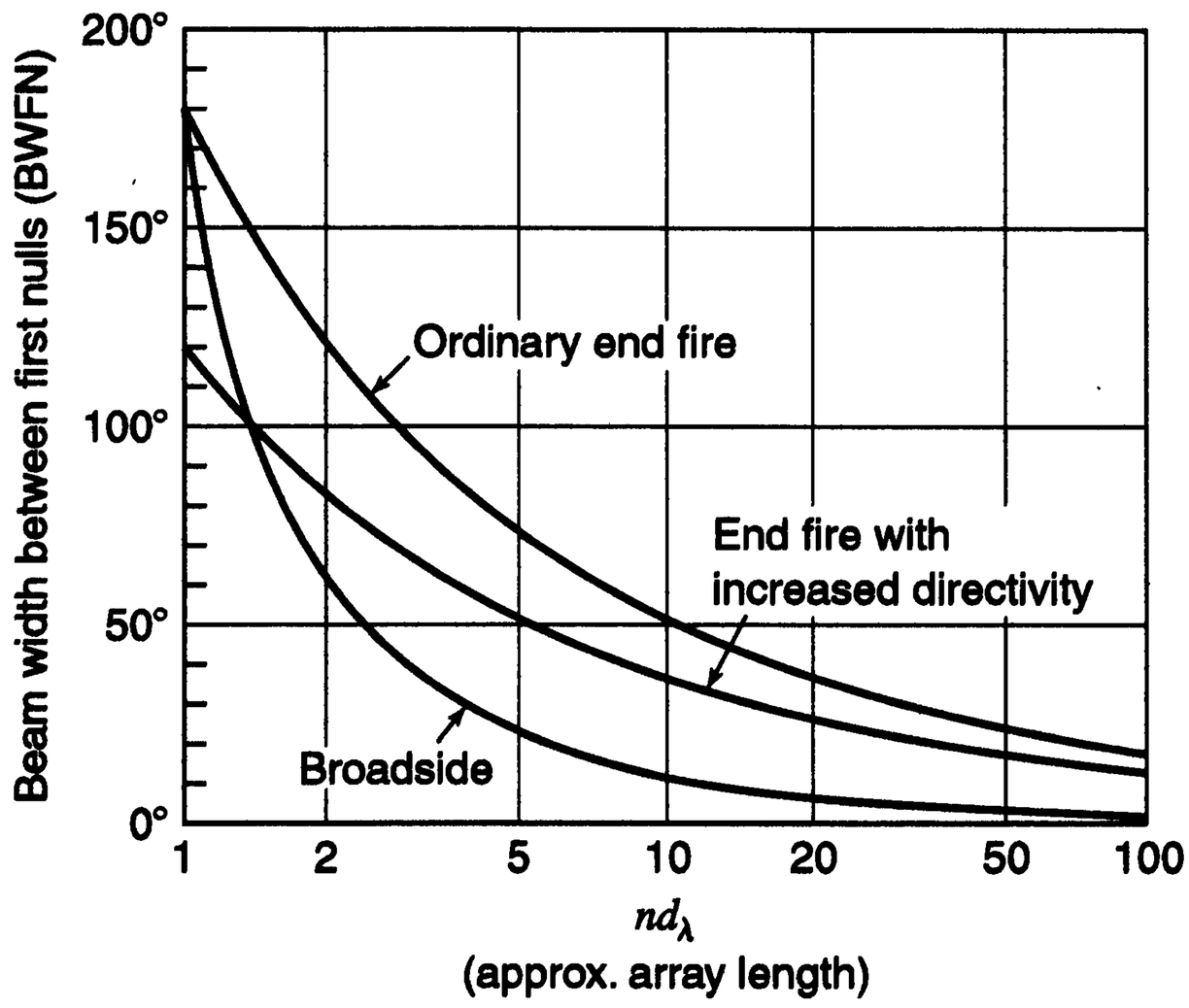


Figure 5–26
Beamwidth between first nulls as a function of nd_λ for arrays of n isotropic point sources of equal amplitude. For long arrays, nd_λ is approximately equal to the array length.

$nd_\lambda (=nd/\lambda)$ is approximately equal to the length of the array in wavelengths for long arrays. The exact value of the array length is $(n-1)d_\lambda$.

The beamwidth of long broadside arrays is inversely proportional to the array length, whereas the beamwidth of long end-fire types is inversely proportional to the square root of the array length. Hence, the beamwidth in the plane of a long linear broadside array is much smaller than for end-fire types of the same length as shown by Fig. 5–26. It should be noted, however, that the broadside array has a disk-shaped pattern with a narrow beamwidth in a plane through the array axis but a circular pattern (360° beamwidth) in the plane normal to the array axis. On the other hand, the end-fire array has a cigar-shaped pattern with the same beamwidth in all planes through the array axis.

PROBLEMS（习题）

***5–2–1**
（两个点源）

Two point sources. (*a*) Show that the relative $E(\phi)$ pattern of an array of two identical isotropic in-phase point sources arranged as in Fig. P5–2–1 is given by $E(\phi) = \cos[(d_r/2)\sin\phi]$, where $d_r = 2\pi d/\lambda$.

(*b*) Show that the maxima, nulls, and half-power points of the pattern are given by the following relations:

Maxima:

$$\phi = \arcsin\left(\pm\frac{k\lambda}{d}\right)$$

Nulls:

$$\phi = \arcsin\left[\pm\frac{(2k+1)\lambda}{2d}\right]$$

Half-power points:

$$\phi = \arcsin\left[\pm\frac{(2k+1)\lambda}{4d}\right]$$

where $k = 0, 1, 2, 3, \ldots$.

(*c*) For $d = \lambda$ find the maxima, nulls, and half-power points, and from these points and any additional points that may be needed plot the $E(\phi)$ pattern for $0° \le \phi \le 360°$. There are four maxima, four nulls, and eight half-power points.

(*d*) Repeat for $d = 3\lambda/2$.

(*e*) Repeat for $d = 4\lambda$.

(*f*) Repeat for $d = \lambda/4$. Note that this pattern has two maxima and two half-power points but no nulls. The half-power points are minima.

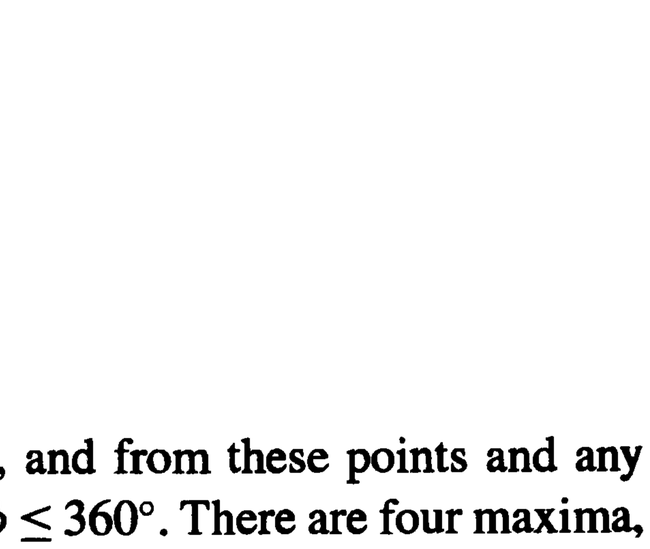


Figure P5–2–1
Two point sources.

5-2-2 (两个点源)

Two point sources. (*a*) What is the expression for $E(\phi)$ for an array of two point sources arranged as in the figure for Prob. 5–2–1? The spacing d is $3\lambda/8$. The amplitude of source 1 in the ϕ plane is given by $|\cos\phi|$ and the phase by ϕ. The amplitude of the field of source 2 is given by $|\cos(\phi - 45°)|$ and the phase of the field by $\phi - 45°$.

(*b*) Plot the normalized amplitude and the phase of $E(\phi)$ referring the phase to the center point of the array.

5-2-3 (两源边射阵)

Two-source broadside array. (*a*) Calculate the directivity of a broadside array of two identical isotropic in-phase point sources spaced $\lambda/2$ apart along the polar axis, the relative field pattern being given by

$$E = \cos\left(\frac{\pi}{2}\cos\theta\right)$$

where θ is the polar angle.

(*b*) Show that the directivity for a broadside array of two identical isotropic in-phase point sources spaced a distance d is given by

$$D = \frac{2}{1 + (\lambda/2\pi d)\sin(2\pi d/\lambda)}.$$

5-2-4 (两源端射阵)

Two-source end-fire array. (*a*) Calculate the directivity of an end-fire array of two identical isotropic point sources in phase opposition, spaced $\lambda/2$ apart along the polar axis, the relative field pattern being given by

$$E = \sin\left(\frac{\pi}{2}\cos\theta\right)$$

where θ is the polar angle.

(*b*) Show that the directivity for an ordinary end-fire array of two identical isotropic point sources spaced a distance d is given by

$$D = \frac{2}{1 + (\lambda/4\pi d)\sin(4\pi d/\lambda)}.$$

5-2-5 (两源波瓣图)

Two-source patterns. Calculate and plot the field and phase patterns for an array of two isotropic sources of the same amplitude and phase, for two cases:

(*a*) $d = 3/4\lambda$

(*b*) $d = 3/2\lambda$

Plot the field pattern in polar coordinates and phase pattern in rectangular coordinates with:

1. Phase center at source 1

2. Phase center at midpoint

5-2-6 (两源阵)

Two-source array. Show that for a two-source array the field patterns

$$E = \frac{\sin(n\psi/2)}{\sin(\psi/2)} \quad \text{and} \quad E = 2\cos\frac{\psi}{2}$$

are equivalent.

5-2-7 (两不等源振幅列)

Two-unequal-source array. In Case 5 (Sec. 5–2) for two isotropic point sources of unequal amplitude and any phase difference, show that the phase angle of the total field with midpoint of the array as phase center is given by

$$\tan^{-1}\left(\frac{a-1}{a+1}\tan\psi\right)$$

5–2–8
（四源方阵）

Four sources in square array. (*a*) Derive an expression for $E(\phi)$ for an array of four identical isotropic point sources arranged as in Fig. P5–2–8. The spacing d between each source and the center point of the array is $3\lambda/8$. Sources 1 and 2 are inphase, and sources 3 and 4 in opposite phase with respect to 1 and 2.

(*b*) Plot, approximately, the normalized pattern.

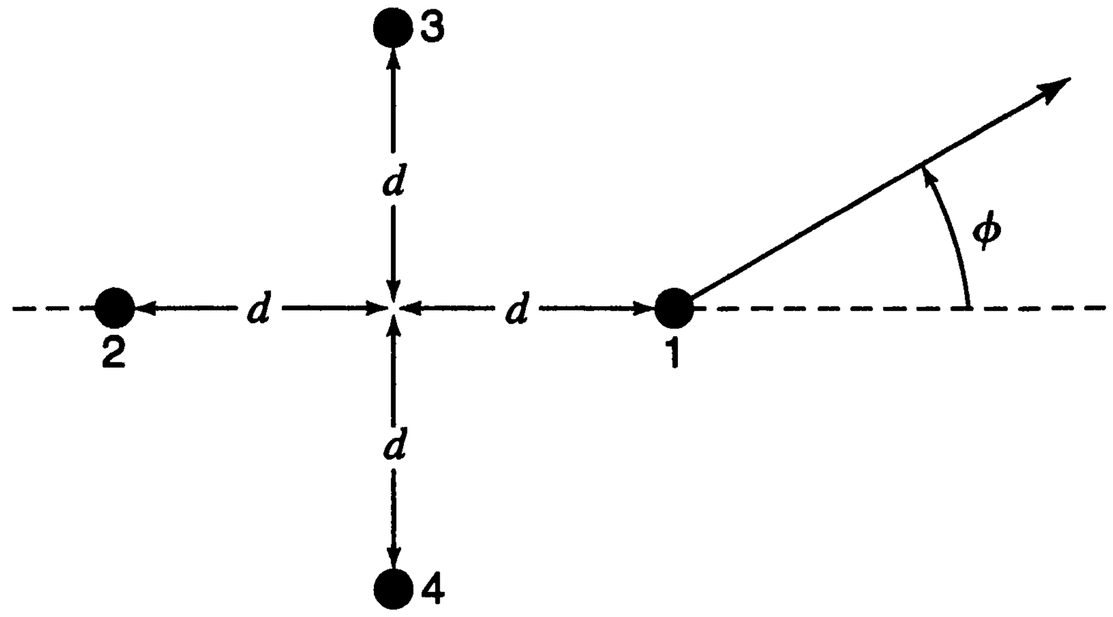


Figure P5–2–8
Four sources in square array.

5–3–1
（各向同性点源阵的波瓣图）

Array patterns, isotropic sources. Plot the field amplitude and phase patterns for two isotropic point sources with equal amplitude and phase at spacing of (*a*) $\lambda/16$, (*b*) $\lambda/8$, (*c*) $\lambda/4$, and (*d*) λ.

5–3–2
（偶极子阵的波瓣图）

Array patterns, dipoles. Repeat Prob. 5–3–1 if the sources are short dipoles with $E(\phi) = \cos\phi$.

5–3–3
（各种单元的阵列波瓣图）

Array patterns, various elements. Repeat Prob. 5–3–1 if the sources have field patterns given by (*a*) $\cos^2\phi$, (*b*) $\cos^3\phi$, (*c*) $(\sin\phi)/\phi$.

5–4–1
（四塔广播阵）

Four-tower broadcast array. A broadcasting station requires the horizontal plane pattern indicated in Fig. P5–4–1. The maximum filed intensity is to be radiated northeast with as little decrease as possible in field intensity in the 90° sector between north and east. No nulls are permitted in this sector. Nulls may occur in any direction in the complementary 270° sector. However, it is required that nulls must be present for the directions due west and due southwest, in order to prevent interference with other stations in these directions.

Design a four-vertical-tower array to fulfill these requirements. The currents are to be equal in magnitude in all towers, but the phase may be adjusted to any relationship. There is also no restriction on the spacing or geometrical arrangements of the towers. Plot the field pattern.

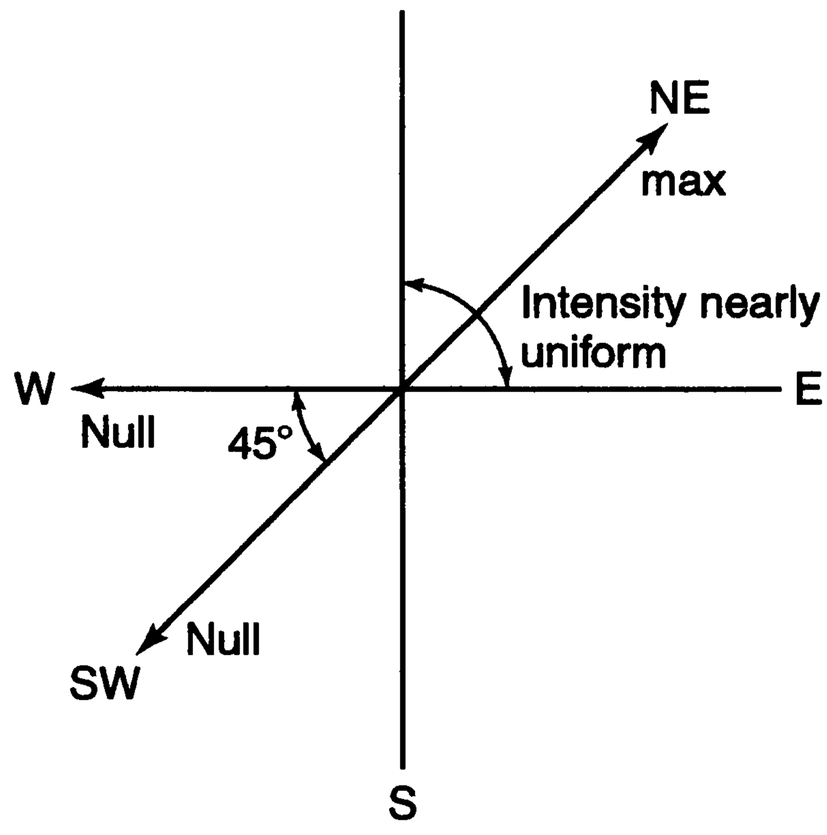


Figure P5–4–1
Four-tower BC array requirements.

5-4-2 (两塔广播阵)
Two-tower broadcast array. A broadcast array of two vertical towers with equal currents is to have a horizontal plane pattern with a broad maximum of field intensity to the north and a null at an azimuth angle of 131° measured counterclockwise from the north. Specify the arrangement of the towers, their spacing and phasing. Calculate and plot the field pattern in the horizontal plane.

5-4-3 (三塔广播阵)
Three-tower broadcast array. A broadcast array with three vertical towers arranged in a straight horizontal line is to have a horizontal-plane pattern with a broad maximum of field intensity to the north and nulls at azimuth angles of 105°, 147°, and 213° measured counterclockwise from the north. The towers need not have equal currents. For the purpose of analysis the center tower (2) may be regarded as two towers, $2a$ and $2b$, $2a$ belonging to an array of itself and tower 1 and $2b$ to an array of itself and tower 3. Specify the arrangement of towers, their spacing, currents and phasing. Calculate and plot the field pattern in the horizontal plane.

5-4-4 (四塔广播阵)
Four-tower broadcast array. A broadcast array of four vertical towers with equal currents is to have a symmetrical four-lobed pattern in the horizontal plane with maximum field intensity to the north, east, south, and west and reduced field intensity to the northeast, southeast, southwest, and northwest equal to half the maximum. Specify the array arrangement, orientation, spacing, and phasing. Calculate and plot the field pattern in the horizontal plane.

5-5-1 (场/相位[illegible])
Field and phase patterns. Calculate and plot the field and phase patterns of an array of two nonisotropic dissimilar sources for which the total field is given by

$$E = \cos\phi + \sin\phi \angle\psi$$

where $\psi = d\cos\phi + \delta = \frac{\pi}{2}(\cos\phi + 1)$

Take source 1 as the reference for phase. See Fig. P5–5–1.

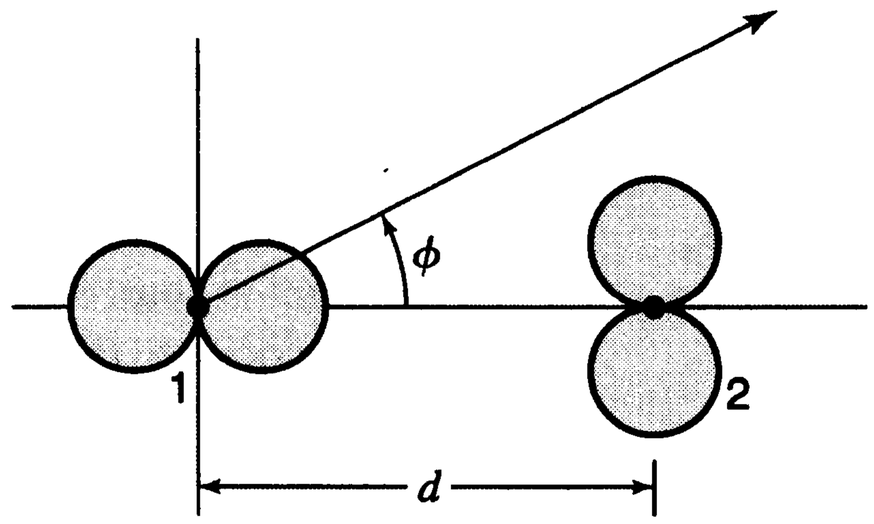


Figure P5–5–1
Field and phase patterns.

5-6-1 (n源阵)
***n*-source array.** (a) Derive an expression for $E(\psi)$ for an array of n identical isotropic point sources where $\psi = f(\phi, d, \delta)$. ϕ is the azimuthal position angle with $\phi = 0$ in the direction of the array. δ is the phase lag between sources as one moves along the array in the $\phi = 0°$ direction, and d is the spacing.

(b) Plot the normalized field as ordinate and ψ as abscissa for $n = 2, 4, 6, 8, 10$, and 12 for $0° \le \psi \le 180°$.

5-6-2 (十源端射阵)
Ten-source end-fire array. Plot $E(\phi)$ for an end-fire array of $n = 10$ identical isotropic point sources spaced $3\lambda/8$ apart with $\delta = -3\pi/4$.

(b) Repeat with $\delta = -\pi[(3/4) + (1/n)]$.

5-6-3 (场/相位波瓣图)
Field and phase patterns. Calculate and plot the field and phase patterns for the cases of Figs. 5–21 and 5–22 and compare with the curves shown.

***5–6–4** (八源端射阵)

Eight-source end-fire array. (*a*) Calculate and plot the field pattern of a linear array of eight isotropic point sources of equal amplitude spaced 0.2λ apart for the ordinary end-fire condition.

(*b*) Repeat, assuming that the phasing satisfies the Hansen and Woodyard increased-directivity condition.

(*c*) Calculate the directivity in both cases by graphical or numerical integration of the entire pattern.

5–6–5 (十二源端射阵)

Twelve-source end-fire array. (*a*) Calculate and plot the field pattern of a linear end-fire array of 12 isotropic point sources of equal amplitude spaced $\lambda/4$ apart for the ordinary end-fire condition.

(*b*) Calculate the directivity by graphical or numerical integration of the entire pattern. Note that it is the power pattern (square of field pattern) which is to be integrated. It is most convenient to make the array axis coincide with the polar or z axis of Fig. 2–5 so that the pattern is a function of θ.

(*c*) Calculate the directivity by the approximate half-power beamwidth method and compare with that obtained in (*b*).

***5–6–6** (十二源边射阵)

Twelve-source broadside array. (*a*) Calculate and plot the pattern of a linear broadside array of 12 isotropic point sources of equal amplitude spaced $\lambda/4$ apart with all the sources in the same phase.

(*b*) Calculate the directivity by graphical or numerical integration of the entire pattern and compare with the directivity obtained in Prob. 5–6–5 for the same-sized array operating end-fire.

(*c*) Calculate the directivity by the approximate half-power beamwidth method and compare with that obtained in (*b*).

5–6–7 (十二源增强定向性端射阵)

Twelve-source end-fire with increased directivity. (*a*) Calculate and plot the pattern of a linear end-fire array of 12 isotropic point sources of equal amplitude spaced $\lambda/4$ apart and phased to fulfill the Hansen and Woodyard increased-directivity condition.

(*b*) Calculate the directivity by graphical or numerical integration of the entire pattern and compare with the directivity obtained in Probs. 5–6–5 and 5–6–6.

(*c*) Calculate the directivity by the approximate half-power beamwidth method and compare with that obtained in (*b*).

5–6–8 (*n*源阵，可变相速)

***n*-source array. Variable phase velocity.** Referring to Fig. 5–18, assume that the uniform array of n isotropic point sources is connected by a transmission system extending along the array with the feed point at source 1 so that the phase of source 2 lags 1 by $\omega d/v$, 3 lags 1 by $2\omega d/v$, etc.; v is the phase velocity to the right along the transmission system. Show that the field is given by Eq. (5–6–8) where $\psi = d_r[\cos\phi - (1/p)]$, where p is the relative phase velocity, that is, $p = v/c$ where c is the velocity of light. Show also that $p = \infty$ for the broadside case, $p = 2$ for the maximum field at $\phi = 60°$, $p = 1$ for the ordinary end-fire case, and $p = 1/[1 + (1/2nd_\lambda)]$ for the increased-directivity end-fire case.

5–6–9 (常规端射阵的定向性)

Directivity of ordinary end-fire array. Show that the directivity of an ordinary end-fire array may be expressed as

$$D = \frac{n}{1 + (\lambda/2\pi nd)\sum_{k=1}^{n-1}[(n-k)/k]\sin(4\pi kd/\lambda)}$$

Note that

$$\left[\frac{\sin(n\psi/2)}{(\psi/2)}\right]^2 = n + \sum_{k=1}^{n-1} 2(n-k)\cos 2k\frac{\psi}{2}$$

5-6-10 **Directivity of broadside array.** Show that the directivity of a broadside array may be expressed as

$$D = \frac{n}{1 + (\lambda/\pi nd)\sum_{k=1}^{n-1}[(n-k)/k]\sin(2\pi kd/\lambda)}$$

5-6-11 **Phase Center.** Show that the phase center of a uniform array is at its centerpoint.

5-6-12 **Ordinary and Increased-directivity end-fire array gains.** (*a*) Use ARRAYPATGAIN to calculate the directivity of ordinary end-fire arrays with $\lambda/4$ spacing and number of sources $n = 5$, 10, 15, and 20. In the formula directivity $D = fn$, what is the value of f?

(*b*) Repeat for increased-directivity end-fire arrays. What is the value of f for this case?

(*c*) What is the ratio of f for the two cases?

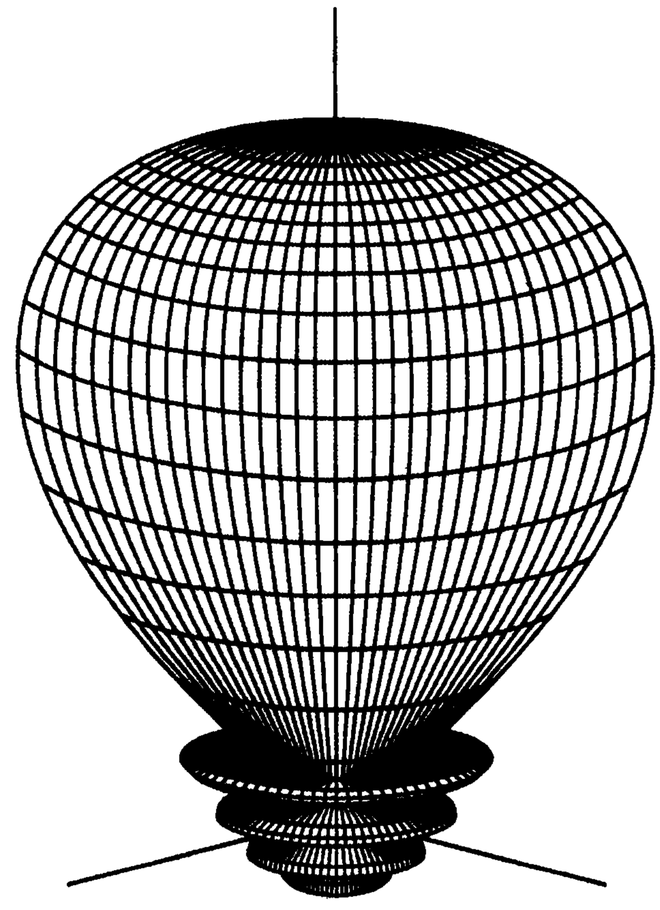
(*d*) Find f in the formula $D = f(L/\lambda)$ where $L = (n-1)\lambda/4$ = length of array.

(*e*) Make a table comparing the above cases.

For computer programs, see Appendix C.

CHAPTER 5

Arrays of Point Sources, Part II
（下）点源阵



Linear arrays of *equal* amplitude isotropic sources were discussed in Chap. 5, Part I. In this part we extend our discussion to the more general case of *nonuniform* distributions. The topics in this part include:

- Nonuniform amplitude distributions (非均匀幅度分布)
- D-T (Dolph-Tschebycheff) or optimum distribution（最优（D-T）分布）
- A comparison of amplitude distributions（幅度分布的比较）
- Continuous arrays（连续阵）
- Huygens' principle（惠更斯（Huygens）原理）
- Diffraction by a flat sheet（平板的绕射）
- Rectangular broadside arrays（矩形边射阵）
- Random arrays（随机阵）
- Broadside versus end-fire arrays（边射－端射阵）
- Directions of array maxima（阵列的主向）

5–8 LINEAR BROADSIDE ARRAYS WITH NONUNIFORM AMPLITUDE DISTRIBUTIONS. GENERAL CONSIDERATIONS
（非均匀幅度分布的直线边射阵的一般性讨论）

Let us begin by comparing the field patterns of four amplitude distributions, namely, *uniform, binomial, edge,* and *optimum*. To be specific, we will consider a linear array of five isotropic point sources with $\lambda/2$ spacing. If the sources are in phase and all equal in amplitude, we may calculate the pattern as discussed in Sec. 5–6 the result being as shown in Fig. 5–27 by the pattern designated *uniform. A uniform distribution yields the maximum directivity* or gain. The pattern has a half-power beamwidth of 23°, but the side lobes are relatively large. The amplitude of the first side lobe is 24 percent of the main-lobe maximum. In some applications this minor-lobe amplitude may be undesirably large.

To reduce the Side-Lobe Level (SLL) of linear in-phase broadside arrays, John Stone Stone (1) proposed that the sources have amplitudes proportional to the coefficients of a binomial series of the form

$$(a+b)^{n-1} = a^{n-1} + (n-1)a^{n-2}b + \frac{(n-1)(n-2)}{2!}a^{(n-3)}b^2 + \cdots \tag{1}$$

where n is the number of sources. Thus, for arrays of three to six sources the relative

Table 5–3

n	Relative amplitudes (Pascal's triangle)
3	1 2 1
4	1 3 3 1
5	1 4 6 4 1
6	1 5 10 10 5 1

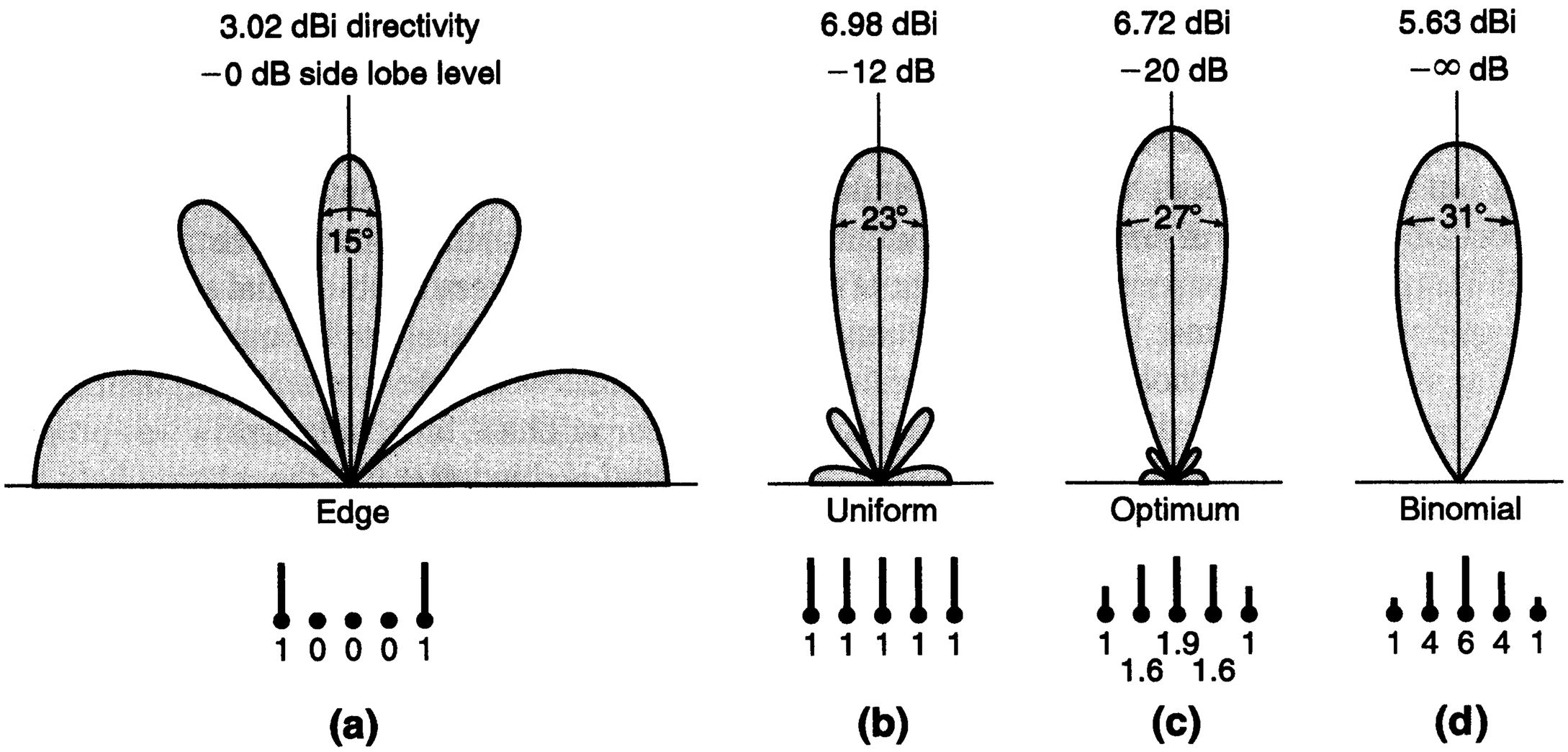


Figure 5–27
Normalized field patterns of broadside arrays of five isotropic point sources spaced $\lambda/2$ apart. All sources are in the same phase, but the relative amplitudes have four different distributions: edge, uniform, optimum, and binomial. Only the upper half of the pattern is shown. The relative amplitudes of the five sources are indicated in each case by the array below the pattern, the height of the line at each source being proportional to its amplitude. All patterns are adjusted to the same maximum amplitude.

amplitudes are given by Table 5–3, where the amplitudes are arranged as in *Pascal's triangle* (any inside number is equal to the sum of the adjacent numbers in the row above).

Applying the binomial distribution to the array of five sources spaced $\lambda/2$ apart, the sources have the relative amplitudes 1, 4, 6, 4, 1. The resulting pattern, designated *binomial,* is shown in Fig. 5–27. Methods of calculating such patterns are discussed in the next section. The pattern has no minor lobes, but this has been achieved at the expense of an increased beamwidth (31°). For spacings of $\lambda/2$ or less between elements, the minor lobes are eliminated by Stone's binomial distribution. However, the increased beamwidth and the large ratio of current amplitudes required in large arrays are disadvantages.

At the other extreme from the binomial distribution, we might try an edge distribution in which only the end sources of the array are supplied with power, the three central sources being either omitted or inactive. The relative amplitudes of the five-source array

Table 5–4

Type of distribution	Half-power beamwidth	Minor-lobe amplitude (% of major lobe)
Binomial	31°	0
Edge	15°	100

are, accordingly, 1, 0, 0, 0, 1. The array has, therefore, degenerated to two sources 2λ apart and has the field pattern designated as *edge* in Fig. 5–27. The beamwidth between half-power points of the "main" lobe (normal to the array) is 15°, but "minor" lobes are the same amplitude as the "main" lobe.

Comparing the binomial and edge distributions for the five-source array with $\lambda/2$ spacing, we have Table 5–4.

Although for most applications it would be desirable to combine the 15° beamwidth of the edge distribution with the zero side-lobe level of the binomial distribution, this combination is not possible. However, if the distribution is between the binomial and the edge type, a compromise between the beamwidth and the side-lobe level can be made; i.e., the side-lobe level will not be zero, but the beamwidth will be less than for the binomial distribution. An amplitude distribution of this nature for linear in-phase broadside arrays was proposed by Dolph (1) which has the further property of ***optimizing*** (**最优化**) the relation between beamwidth and side-lobe level; i.e., ***if the side-lobe level is specified, the beamwidth between first nulls is minimized;*** or, conversely, ***if the beamwidth between first nulls is specified, the sidelobe level is minimized.*** (**对于指定的旁瓣电平，其第一零点波束宽度为最窄；反之，对于指定的第一零点波束宽度，其旁瓣电平为最低**) Dolph's distribution is based on the properties of the Tchebyscheff polynomials and accordingly will be referred to as the Dolph-Tchebyscheff or optimum distribution.

Applying the Dolph-Tchebyscheff distribution to our array of five sources with $\lambda/2$ spacing, let us specify a side-lobe level 20 dB below the main lobe, i.e., a minor-lobe amplitude 10 percent of the main lobe. The relative amplitude distribution for this side-lobe level is 1, 1.6, 1.9, 1.6, 1 and yields the pattern designated *optimum* in Fig. 5–27. Methods of calculating the distribution and pattern are discussed in the next section. The beamwidth between half-power points is 27°, which is less than for the binomial distribution. Smaller beamwidths can be obtained only by raising the side-lobe level. The Dolph-Tchebyscheff distribution includes all distributions between the binomial and the edge. In fact, the binomial and edge distributions are special cases of the Dolph-Tchebyscheff distribution, the binomial distribution corresponding to an infinite ratio between main- and side-lobe levels and the edge distribution to a ratio of unity. The uniform distribution is, however, not a special case of the Dolph-Tchebyscheff distribution.

Referring to Fig. 5–27, we may draw a number of general conclusions regarding the relation between patterns and amplitude distributions. We note that if the amplitude tapers to a small value at the edge of the array (binomial distribution), minor lobes can be eliminated. On the other hand, if the distribution has an inverse taper with maximum amplitude at the edges and none at the center of the array (edge distribution), the minor lobes are accentuated, being in fact equal to the "main" lobe. From this we may quite properly conclude that the side-lobe level is closely related to the abruptness with which the amplitude distribution ends at the edge of the array. An abrupt discontinuity in the distribution results in large minor

lobes, while a gradually tapered distribution approaching zero at the edge minimizes the discontinuity and the minor-lobe amplitude. In the next section, we shall see that the abrupt discontinuity produces large higher "harmonic" terms in the Fourier series representing the pattern. On the other hand, these higher harmonic terms are small when the distribution tapers gradually to a small value at the edge. There is an analogy between this situation and the Fourier analysis of wave shapes. Thus, a square wave has relatively large higher harmonics, whereas a pure sine wave has none, the square wave being analogous to the uniform array distribution while the pure sine wave is analogous to the binomial distribution.

The preceding discussion has been concerned with arrays of discrete sources separated by finite distances. However, the general conclusions concerning amplitude distributions which we have drawn can be extended to large arrays of continuous distributions of an infinite number of point sources, such as might exist in the case of a continuous current distribution on a metal sheet or in the case of a continuous field distribution across the mouth of an electromagnetic horn or across a parabolic reflector antenna. If the amplitude distribution follows a Gaussian error curve, which is similar to a binomial distribution for discrete sources, then minor lobes are absent but the beamwidth is relatively large. An increase of amplitude at the edge reduces the beamwidth but results in minor lobes, as we have seen. Thus, in the case of a high-gain parabolic reflector type of antenna, the illumination of the reflector by the primary antenna is usually arranged to taper toward the edge of the parabola. However, a compromise is generally made between beamwidth and side-lobe level so that the illumination is not zero at the edge but has an appreciable value as in a Dolph-Tchebyscheff distribution.

5–9 LINEAR ARRAYS WITH NONUNIFORM AMPLITUDE DISTRIBUTIONS. THE DOLPH-TCHEBYSCHEFF OPTIMUM DISTRIBUTION

（非均匀幅度分布的直线阵与道尔夫－切比雪夫最优分布）

In this section linear in-phase arrays with nonuniform amplitude distributions are analyzed, and the development and application of the Dolph-Tchebyscheff distribution are discussed. It is shown that the far-field pattern of a linear array of isotropic point sources can be expressed as a finite Fourier series of N terms. Then Dolph's procedure is described for matching the terms of the Fourier polynomial with the terms of like degree of a Tchebyscheff polynomial. This then yields the optimum source amplitude distribution for a specified Side-Lobe Level (SLL) with all side lobes of the same level. To apply these results to an array, one can give this section a quick scan and proceed directly to the worked example of Sec. 5–10.

We will consider a linear array of an even number n_e of isotropic point sources of uniform spacing d arranged as in Fig. 5–28*a*. All sources are in the same phase. The direction $\theta = 0$ is taken normal to the array with the origin at the center of the array as shown. The individual sources have the amplitudes A_0, A_1, A_2, etc., as indicated, the amplitude distribution being symmetrical about the center of the array. The total field E_{n_e} from the *even number of sources* at a large distance in a direction θ is then the sum of the fields of the symmetrical pairs of sources, or

$$E_{n_e} = 2A_0 \cos\frac{\psi}{2} + 2A_1 \cos\frac{3\psi}{2} + \cdots + 2A_k \cos\left(\frac{n_e - 1}{2}\psi\right) \tag{1}$$

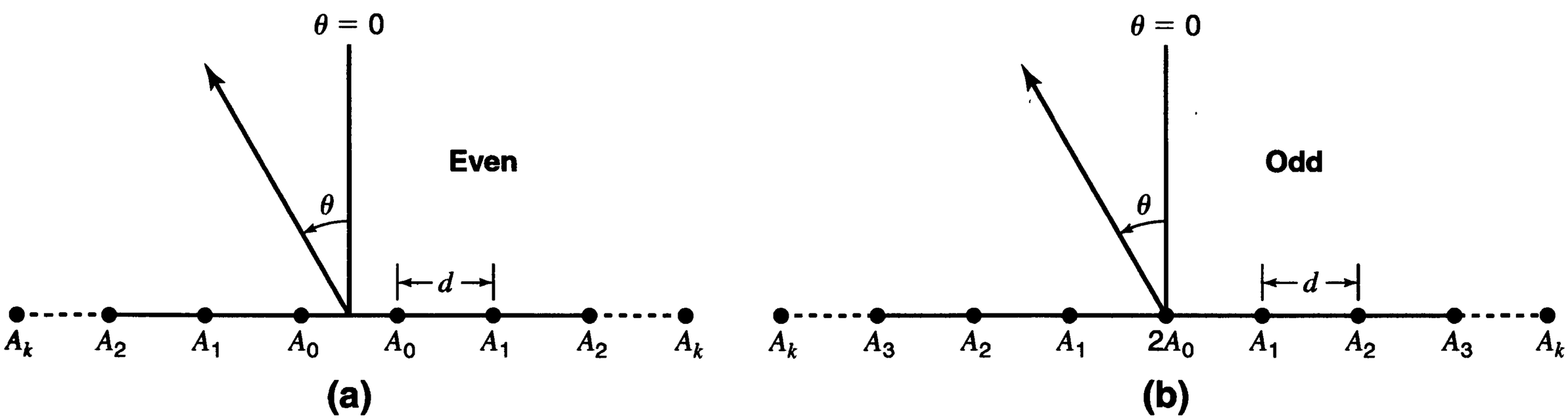


Figure 5–28
Linear broadside arrays of *n* isotropic sources with uniform spacing for *n* even (*a*) and *n* odd (*b*).

where

$$\psi = \frac{2\pi d}{\lambda} \sin\theta = d_r \sin\theta \tag{2}$$

Each term in (1) represents the field due to a symmetrically disposed pair of the sources.

Now let

$$2(k+1) = n_e$$

where $k = 0, 1, 2, 3, \ldots$ so that

$$\frac{n_e - 1}{2} = \frac{2k+1}{2}$$

Then (1) becomes

$$\boxed{E_{n_e} = 2\sum_{k=0}^{k=N-1} A_k \cos\left(\frac{2k+1}{2}\psi\right) \qquad \textit{Even number Fourier series}} \quad (\text{偶数傅里叶级数}) \tag{3}$$

where $N = n_e/2$

Next let us consider the case of a linear array of an odd number n_o of isotropic point sources of uniform spacing arranged as in Fig. 5–28*b*. The amplitude distribution is symmetrical about the center source. The amplitude of the center source is taken as $2A_0$, the next as A_1, the next as A_2, etc. The total field E_{n_o} from the odd number of sources at a large distance in a direction θ is then

$$E_{n_o} = 2A_0 + 2A_1 \cos\psi + 2A_2 \cos 2\psi + \cdots + 2A_k \cos\left(\frac{n_o - 1}{2}\psi\right) \tag{4}$$

Now for this case let

$$2k + 1 = n_o$$

where $k = 0, 1, 2, 3, \ldots$. Then (4) becomes

$$\boxed{E_{n_o} = 2\sum_{k=0}^{k=N} A_k \cos\left(2k\frac{\psi}{2}\right) \qquad \textit{Odd number Fourier series}} \quad (\text{奇数傅里叶级数}) \tag{5}$$

where $N = (n_o - 1)/2$

The series expressed by (4) or by (5) may be recognized as a **finite Fourier series** of N terms (Wolff-1). For $k = 0$ we have a constant term $2A_0$ representing the contribution of the center source. For $k = 1$ we have the term $2A_1 \cos \psi$ representing the contribution of the first pair of sources on either side of the center source. For each higher value of k we have a higher harmonic term which in each case represents the contribution of a pair of symmetrically disposed sources. Thus, the total field pattern is simply the sum of a series of terms of increasing order in the same way that the waveform of an alternating current can be represented as a Fourier series involving, in general, a constant term, a fundamental term, and higher harmonic terms. The field pattern of an even number of sources as given by (1) or (3) is also a finite Fourier series but one which has no constant term and only odd harmonics. The coefficients $A_0, A_1, \ldots$ in both series are arbitrary and express the amplitude distribution.

To illustrate the Fourier nature of the field-pattern expression, let us consider the simple example of an array of nine isotropic point sources spaced $\lambda/2$ apart, having the same amplitude and phase. Hence, the coefficients are related as follows: $2A_0 = A_1 = A_2 = A_3 = A_4 = \frac{1}{2}$. The number of sources is odd; hence the expression for the field pattern is then given by (5) as

$$E_9 = \tfrac{1}{2} + \cos \psi + \cos 2\psi + \cos 3\psi + \cos 4\psi \tag{6}$$

The first term ($k = 0$) is a constant so that the field pattern is a circle of amplitude $\frac{1}{2}$ as shown in Fig. 5–29*a*. The second term ($k = 1$) may be regarded as the fundamental term of the Fourier series and gives the pattern of the two sources (A_1 in Fig. 5–29*b*) on either side of the center. This pattern has four lobes of maximum amplitude of unity, as illustrated in Fig. 5–29*b*. The next term ($k = 2$) may be regarded as the second harmonic term and gives the pattern of the next pair of sources (A_2 in Fig. 5–29*b*). This pattern has eight lobes as shown by Fig. 5–29*c*. The last two terms represent the third and fourth harmonics, and

Resolution of pattern (*f*) into five Fourier components

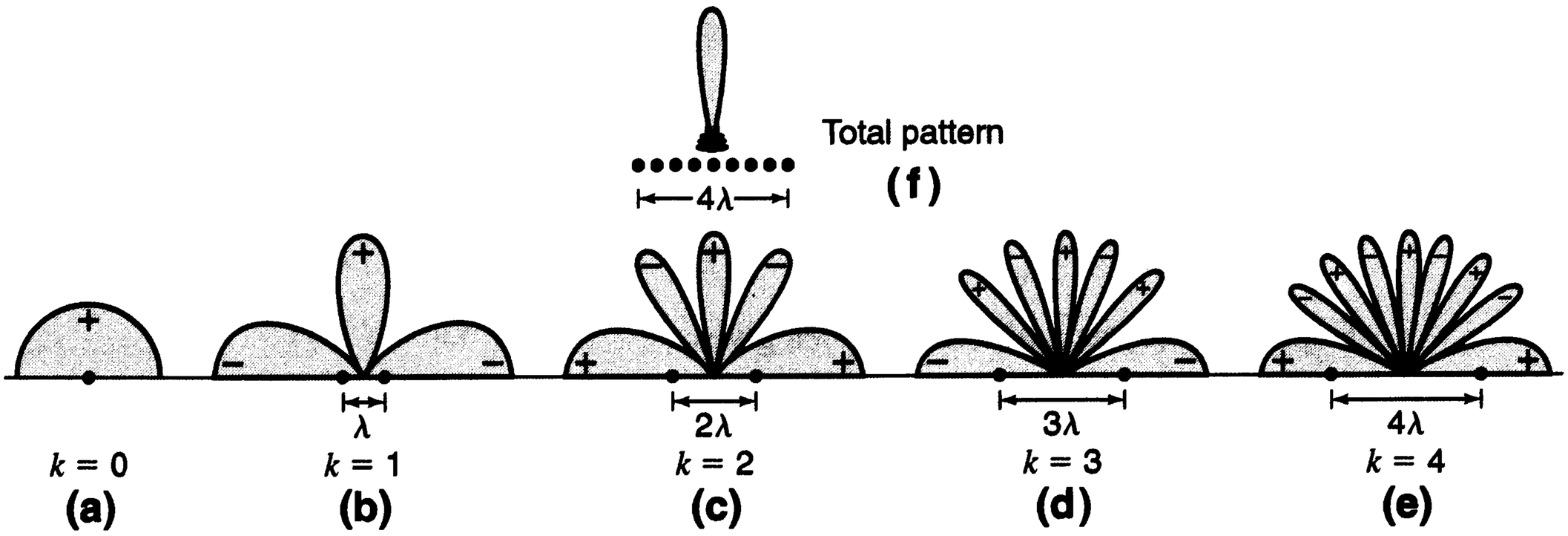


Figure 5–29
Resolution of total pattern of array of nine isotropic sources into Fourier components due to center source and pairs of symmetrically disposed sources. The relative field pattern of the entire array is shown by (*f*). The lower halves of patterns are not shown. (Note that the end-fire lobes are wider than the broadside lobes.)

Table 5–5

k	Sources	Spacing	Fourier term	Pattern
0	1	0	Constant	Circle
1	2	1λ	Fundamental	4 lobes
2	2	2λ	Second harmonic	8 lobes
3	2	3λ	Third harmonic	12 lobes
4	2	4λ	Fourth harmonic	16 lobes

the patterns have 12 and 16 lobes, respectively, as indicated by Fig. 5–29*d* and *e*. The above relations may be summarized as in Table 5–5.

The algebraic sum of the patterns given by the five terms is the total far-field pattern of the array which is presented in Fig. 5–29*f*. If the middle source of the array has zero amplitude or is omitted, the total pattern is then the sum of the four terms for which $k = 1, 2, 3$, and 4. If in addition the pair of sources A_1 is omitted, the total pattern is the sum of three terms for which $k = 2, 3$, and 4. Since these are higher harmonic terms, we may properly expect that in this case the minor lobes of the total pattern will be accentuated. It is apparent from the above discussion that the field pattern of any symmetrical amplitude distribution can be expressed as a series of the form of (3) or (5).

Proceeding now to the Dolph-Tchebyscheff amplitude distribution, it will be shown that the coefficients of the pattern series[1] can be uniquely determined so as to produce a pattern of minimum beamwidth for a specified side-lobe level. The first step in the development of the Dolph-Tchebyscheff distribution is to show that (3) and (5) can be regarded as polynomials of degree $n_e - 1$ and $n_o - 1$, that is, polynomials of degree equal to the number of sources less 1. In the present discussion we shall consider only the case of the broadside type of array, i.e., where $\delta = 0$. Thus,

$$\psi = d_r \sin\theta \tag{7}$$

Now by de Moivre's theorem,

$$e^{jm\psi/2} = \cos m\frac{\psi}{2} + j \sin m\frac{\psi}{2} = \left(\cos\frac{\psi}{2} + j \sin\frac{\psi}{2}\right)^m \tag{8}$$

On taking real parts of (8) we have

$$\cos m\frac{\psi}{2} = \mathrm{Re}\left(\cos\frac{\psi}{2} + j \sin\frac{\psi}{2}\right)^m \tag{9}$$

Expanding (9) as a binomial series gives

$$\cos m\frac{\psi}{2} = \cos^m\frac{\psi}{2} - \frac{m(m-1)}{2!}\cos^{m-2}\frac{\psi}{2}\sin^2\frac{\psi}{2} + \frac{m(m-1)(m-2)(m-3)}{4!}\cos^{m-4}\frac{\psi}{2}\sin^4\frac{\psi}{2} - \cdots \tag{10}$$

[1]Equations (1), (3), (4) and (5).

Putting $\sin^2(\psi/2) = 1 - \cos^2(\psi/2)$, and substituting particular values of m, (10) then reduces to the following:

$$\left.\begin{aligned} m &= 0 \qquad \cos m\frac{\psi}{2} = 1 \\ m &= 1 \qquad \cos m\frac{\psi}{2} = \cos\frac{\psi}{2} \\ m &= 2 \qquad \cos m\frac{\psi}{2} = 2\cos^2\frac{\psi}{2} - 1 \\ m &= 3 \qquad \cos m\frac{\psi}{2} = 4\cos^3\frac{\psi}{2} - 3\cos\frac{\psi}{2} \\ m &= 4 \qquad \cos m\frac{\psi}{2} = 8\cos^4\frac{\psi}{2} - 8\cos^2\frac{\psi}{2} + 1 \\ &\text{etc.} \end{aligned}\right\} \tag{11}$$

Now let

$$x = \cos\frac{\psi}{2} \tag{12}$$

whereupon the equations of (11) become

$$\left.\begin{aligned} \cos m\frac{\psi}{2} &= 1 \qquad\qquad \text{when } m = 0 \\ \cos m\frac{\psi}{2} &= x \qquad\qquad \text{when } m = 1 \\ \cos m\frac{\psi}{2} &= 2x^2 - 1 \qquad \text{when } m = 2 \\ &\text{etc.} \end{aligned}\right\} \tag{13}$$

The polynomials of (13) are called Tchebyscheff polynomials, which may be designated in general by

$$T_m(x) = \cos m\frac{\psi}{2} \tag{14}$$

For particular values of m, the first eight Tchebyscheff polynomials are

$$\left.\begin{aligned} T_0(x) &= 1 \\ T_1(x) &= x \\ T_2(x) &= 2x^2 - 1 \\ T_3(x) &= 4x^3 - 3x \\ T_4(x) &= 8x^4 - 8x^2 + 1 \\ T_5(x) &= 16x^5 - 20x^3 + 5x \\ T_6(x) &= 32x^6 - 48x^4 + 18x^2 - 1 \\ T_7(x) &= 64x^7 - 112x^5 + 56x^3 - 7x \end{aligned}\right\} \tag{15}$$

We note in (15) that the degree of the polynomial is the same as the value of m.

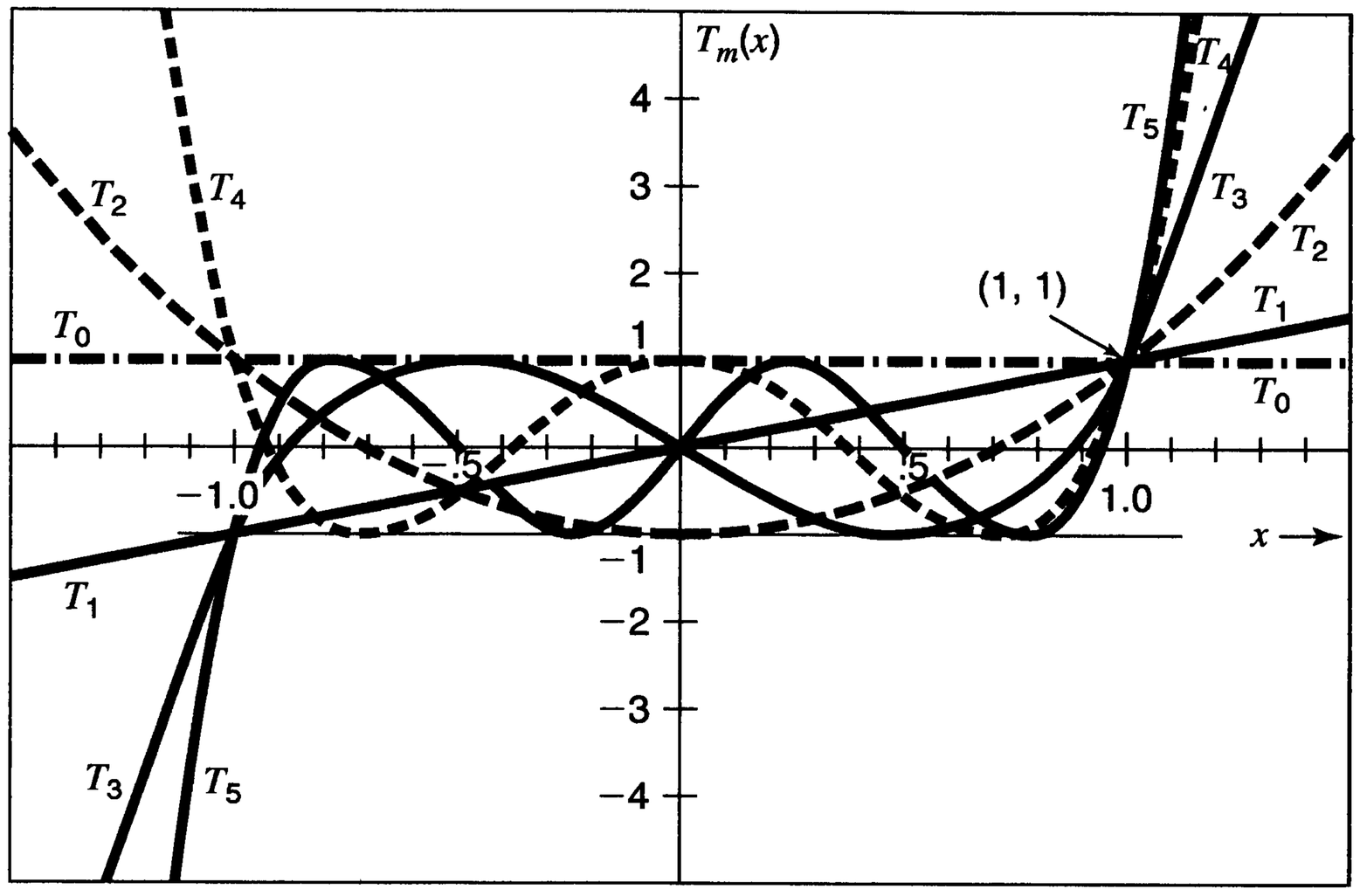


Figure 5–30
Tchebyscheff polynomials of degree $m = 0$ through $m = 5$.

The roots of the polynomials occur when $\cos m(\psi/2) = 0$ or when

$$m\frac{\psi}{2} = (2k - 1)\frac{\pi}{2} \tag{16}$$

where $k = 1, 2, 3, \ldots$

The roots of x, designated x', are thus

$$x' = \cos\left[(2k - 1)\frac{\pi}{2m}\right] \tag{17}$$

We have shown that $\cos m(\psi/2)$ can be expressed as a polynomial of degree m. Thus, (3) and (5) are expressible as polynomials of degree $2k + 1$ and $2k$, respectively, since each is the sum of cosine polynomials of the form $\cos m(\psi/2)$. For an even number n_e of sources $2k + 1 = n_e - 1$, while for an odd number n_o, $2k = n_o - 1$. Therefore, (3) and (5), which express the field pattern of a symmetric in-phase equispaced linear array of n isotropic point sources, are polynomials of degree equal to the number of sources less 1. If we now set the array polynomial as given by (3) or (5) equal to the Tchebyscheff polynomial of like degree ($m = n - 1$) and equate the array coefficients to the coefficients of the Tchebyscheff polynomial, then the amplitude distribution given by these coefficients is a Tchebyscheff distribution and the field pattern of the array corresponds to the Tchebyscheff polynomial of degree $n - 1$.

The Tchebyscheff polynomials of degree $m = 0$ through $m = 5$ are presented in Fig. 5–30. Referring to Fig. 5–30, the following properties of the polynomials are worthy of note:

1. All pass through the point (1, 1).
2. For values of x in the range $-1 \leq x \leq +1$, the polynomials all lie between ordinate values of $+1$ and -1. All roots occur between $-1 \leq x \leq +1$, and all maximum values in this range are ± 1.

We may now describe Dolph's method of applying the Tchebyscheff polynomial to obtain an optimum pattern. Suppose that we have an array of 6 sources. The field pattern is then a polynomial of degree 5. If this polynomial is equated to the Tchebyscheff polynomial

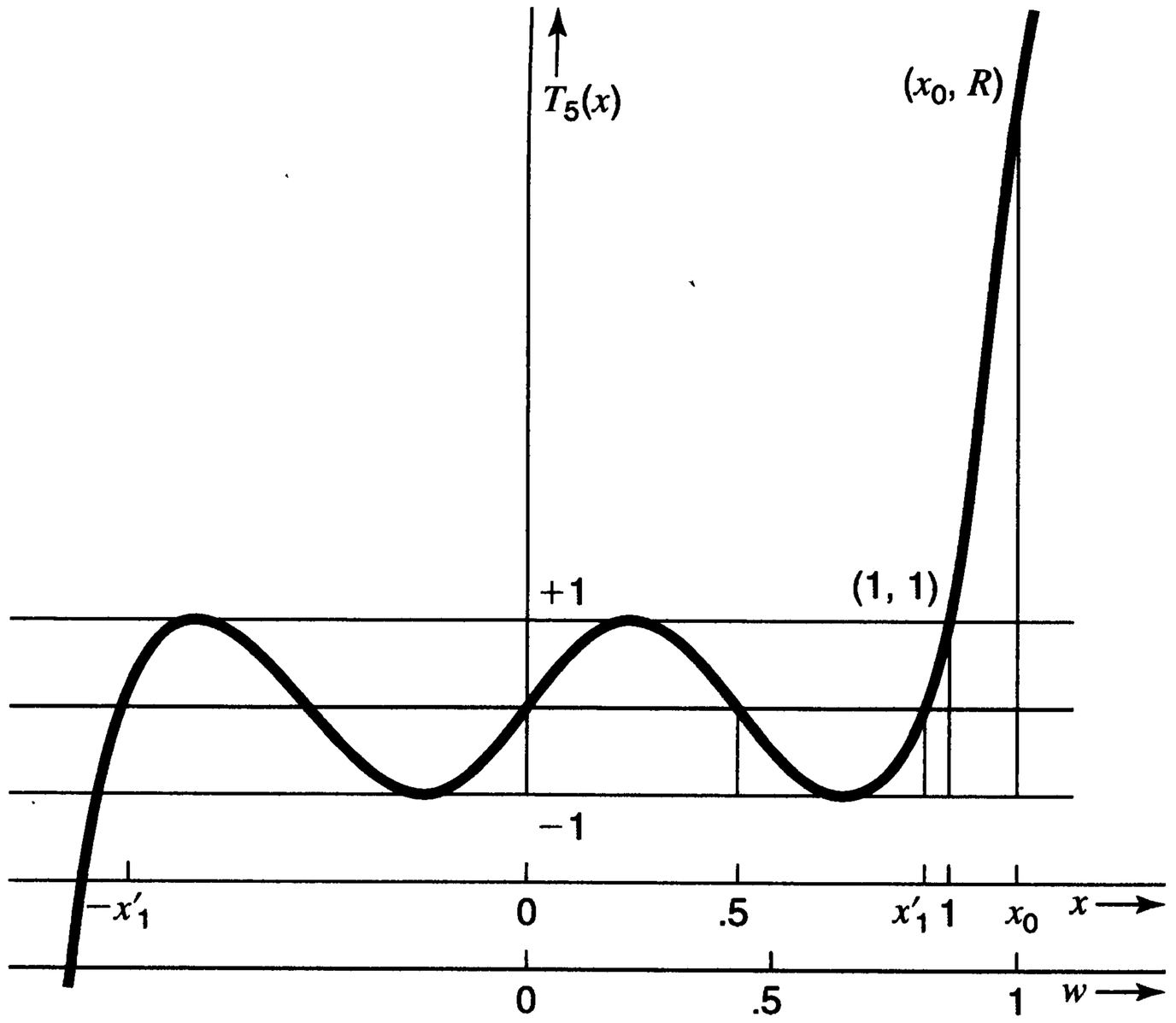


Figure 5–31
Tchebyscheff polynomial of fifth degree with relation to coordinate scales.

of degree 5, shown in Fig. 5–31, then the optimum pattern may be derived as follows: Let the ratio of the main-lobe maximum to the minor-lobe level be specified as R; that is,

$$R = \frac{\text{main-lobe maximum}}{\text{side-lobe level}}$$

The point (x_0, R) on the $T_5(x)$ polynomial curve then corresponds to the main-lobe maximum, while the minor lobes are confined to a maximum value of unity. The roots of the polynomial correspond to the nulls of the field pattern. ***The important property of the Tchebyscheff polynomial is that if the ratio R is specified, the beamwidth to the first null ($x = x'_1$) is minimized. The corollary also holds that if the beamwidth is specified, the ratio R is maximized (side-lobe level minimized).*** [切比雪夫多项式的重要性质是：若指定比值 R（即指定旁瓣电平），则第一零点（$x = x'_1$）的波束宽度为最窄；若指定波束宽度，则比值 R 为最大（旁瓣电平最低）]。

The procedure will now be summarized. Let us write (3) and (5) again. It is to be noted that they are functions of $\psi/2$. Thus,

$$E_{n_e} = 2\sum_{k=0}^{k=N-1} A_k \cos\left[(2k+1)\frac{\psi}{2}\right] \qquad (n \text{ even}) \tag{18}$$

and

$$E_{n_o} = 2\sum_{k=0}^{k=N} A_k \cos\left(2k\frac{\psi}{2}\right) \qquad (n \text{ odd}) \tag{19}$$

Since we are usually interested only in the relative field pattern, the factor 2 before the summation sign in (18) and (19) may be dropped.

For an array of n sources, the first step is to select the Tchebyscheff polynomial of the same degree as the array polynomial, (3) or (5). This is given by

$$T_{n-1}(x) \tag{20}$$

where n is the number of sources and $m = n - 1$. Next we choose R and solve

$$T_m(x_0) = R \tag{21}$$

for x_0. Referring to Fig. 5–31, we note that, for $R > 1$, x_0 is also greater than unity. This presents a difficulty since, according to (12), x must be restricted to the range $-1 \leq x \leq +1$. If, however, a change of scale is made by introducing a new abscissa w (Fig. 5–31) where

$$w = \frac{x}{x_0} \tag{22}$$

then the restriction of (12) can be fulfilled by putting

$$w = \cos\frac{\psi}{2} \tag{23}$$

where now the range of w is restricted to $-1 \leq w \leq +1$. The pattern polynomial, (18) or (19), may now be expressed as a polynomial in w. The final step is to equate the Tchebyscheff polynomial of (20) and the array polynomial obtained by substituting (23) into (18) or (19). Thus,

$$T_{n-1}(x) = E_n \tag{24}$$

The coefficients of the array polynomial are then obtained from (24), yielding the Dolph-Tchebyscheff amplitude distribution, which is an optimum for the side-lobe level specified.

As a proof of the optimum property of the Tchebyscheff polynomial, let us consider any other polynomial $P(x)$ of degree 5 which passes through (x_0, R) in Fig. 5–31 and the highest root x_1' and for all smaller values of x lies between $+1$ and -1. If the range in ordinate of $P(x)$ is less than ± 1, then this polynomial would give a smaller side-lobe level for this same beamwidth, and $T_5(x)$ would not be optimum. Since $P(x)$ lies between ± 1 in the range $-x_1' \leq x \leq +x_1'$ it must intersect the curve $T_5(x)$ in at least $m + 1 = 6$ points, including (x_0, R). Two polynomials of the same degree m which intersect in $m + 1$ points must be the same polynomial,[1] so that

$$P(x) = T_5(x)$$

and the $T_5(x)$ polynomial is therefore the optimum.

If the spacing between sources exceeds $\lambda/2$, it should be noted that as the spacing approaches λ a large lobe develops at $\theta = \pm 90°$ which equals the main lobe when $d = \lambda$. However, if the individual sources of the array are nonisotropic, i.e., are directional with the maximum at $\theta = 0$ and with little or no radiation at $\theta = \pm 90°$, then by pattern multiplication the lobes of the total pattern at $\theta = \pm 90°$ can be made small.

---

[1] This follows from the fact that a polynomial of degree m has $m + 1$ arbitrary constants. Further, if $m + 1$ points on the polynomial's curve are specified, $m + 1$ independent equations with $m + 1$ unknowns can be written and the $m + 1$ constants thereby determined.

5–10 EXAMPLE OF DOLPH-TCHEBYSCHEFF (D-T) DISTRIBUTION FOR AN ARRAY OF EIGHT SOURCES（道尔夫－切比雪夫分布的八源阵举例）

To illustrate the method for finding the Dolph-Tchebyscheff distribution, let us work the following problem.

An array of $n = 8$ in-phase isotropic sources, spaced $\lambda/2$ apart, is to have a side-lobe level 26 dB below the main-lobe maximum. Find: (*a*) the amplitude distribution fulfilling this requirement that produces the minimum beamwidth between first nulls, (*b*) the HPBW, and (*c*) the gain.

Since

$$\text{Side-lobe level in dB below main-lobe maximum} = 20 \log_{10} R \tag{1}$$

it follows that

$$R = 20 \tag{2}$$

The Tchebyscheff polynomial of degree $n - 1$ is $T_7(x)$. Thus, we set

$$T_7(x_0) = 20 \tag{3}$$

The value of x_0 may be determined by trial and error from the $T_7(x)$ expansion as given in (5–9–15) or x_0 may be calculated from

$$x_0 = \tfrac{1}{2}[(R + \sqrt{R^2 - 1})^{1/m} + (R - \sqrt{R^2 - 1})^{1/m}] \tag{4}$$

Substituting $R = 20$ and $m = 7$ in (4) yields

$$x_0 = 1.15 \tag{5}$$

Now substituting (5–9–23) in (5–9–18) and dropping the factor 2, we have

$$E_8 = A_0 w + A_1(4w^3 - 3w) + A_2(16w^5 - 20w^3 + 5w) + A_3(64w^7 - 112w^5 + 56w^3 - 7w) \tag{6}$$

But $w = x/x_0$, so making this substitution in (6) and grouping terms of like degree,

$$E_8 = \frac{64A_3}{x_0^7}x^7 + \frac{16A_2 - 112A_3}{x_0^5}x^5 + \frac{4A_1 - 20A_2 + 56A_3}{x_0^3}x^3 + \frac{A_0 - 3A_1 + 5A_2 - 7A_3}{x_0}x \tag{7}$$

The Tchebyscheff polynomial of like degree is

$$T_7(x) = 64x^7 - 112x^5 + 56x^3 - 7x \tag{8}$$

Now equating (7) and (8),

$$E_8 = T_7(x) \tag{9}$$

For (9) to be true requires that the coefficients of (7) equal the coefficients of the terms of like degree in (8). Therefore,

$$\frac{64A_3}{x_0^7} = 64 \tag{10}$$

or

$$A_3 = x_0^7 = 1.15^7 = 2.66 \tag{11}$$

In a similar way we find that

$$\left.\begin{aligned} A_2 &= 4.56 \\ A_1 &= 6.82 \\ A_0 &= 8.25 \end{aligned}\right\} \tag{12}$$

The relative amplitudes of the eight sources are then

$$1, 1.7, 2.6, 3.1, 3.1, 2.6, 1.7, 1$$

To obtain the field pattern given by the Dolph-Tchebyscheff distribution, we recall that $\psi/2 = (d_r \sin\theta)/2$, $\cos(\psi/2) = w$, and $w = x/x_0$, from which

$$x = x_0 \cos\frac{d_r \sin\theta}{2} \tag{13}$$

The value of x corresponding to a given value of θ, as obtained from (13), is then introduced in the appropriate Tchebyscheff polynomial, in this case $T_7(x)$, or scaled from a graph of this polynomial, as shown in Fig. 5–32. The value of the polynomial for this value of x is then the relative field strength in the direction θ. In general, as θ ranges from $-\pi/2$ to $+\pi/2$, the variables $\psi/2$, w and x range as indicated by Table 5–6. Thus, in general, as θ ranges from $-\pi/2$ to 0 to $+\pi/2$, x ranges from some point, such as a in Fig. 5–32, to x_0 and back again to a, the ordinate value giving the relative field intensity.

(*continued*)

(*continued*)

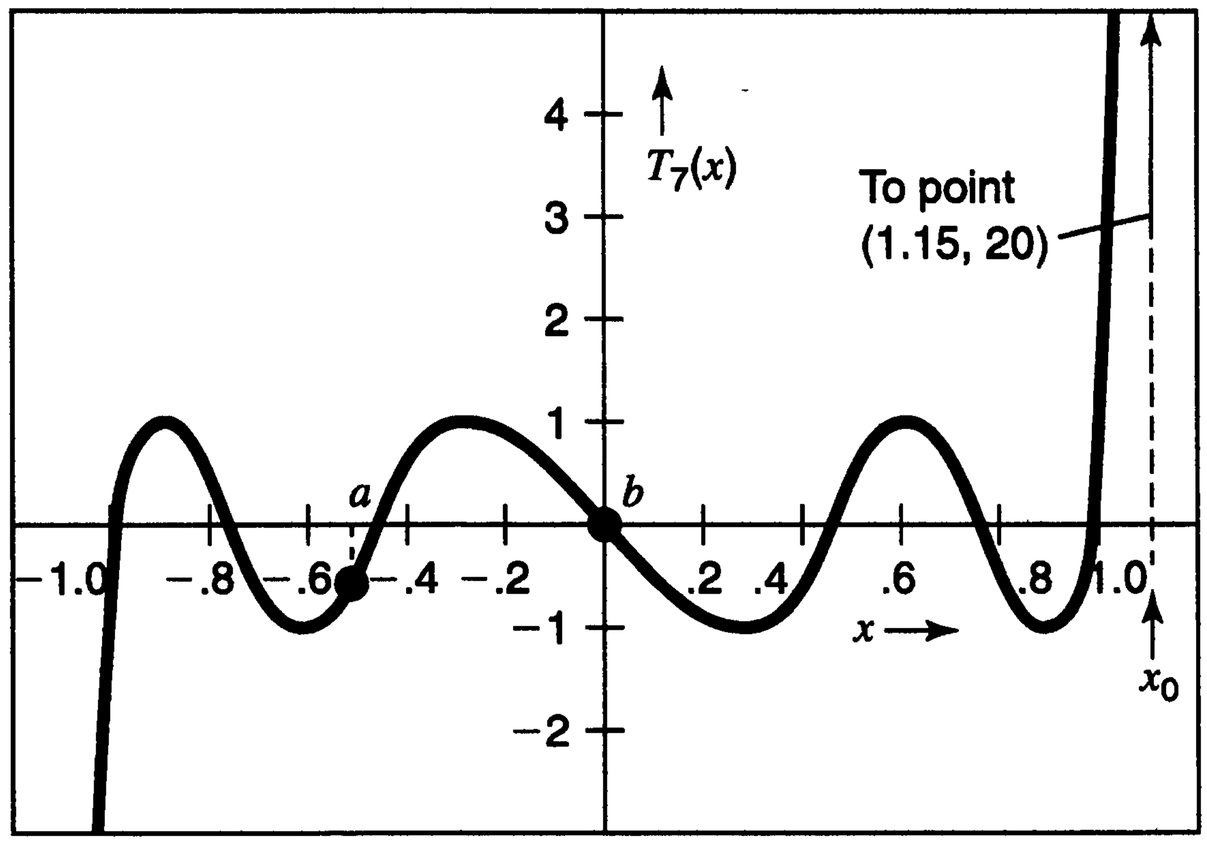


Figure 5–32
Tchebyscheff polynomial of the seventh degree.

In our problem, $d_r = \pi$ and $x_0 = 1.15$, so that the range of x is as shown in Table 5–6. Hence, at $\theta = -90°$ we start at the origin in Fig. 5–32 (point b), and as θ approaches 0° we proceed to the right along the polynomial curve, reaching the point (x_0, $R = 1.15, 20$) when $\theta = 0°$. As θ continues to increase, we retrace the polynomial curve, reaching the origin when $\theta = 90°$. Thus, the pattern is symmetrical about the $\theta = 0°$ direction.

As a preliminary step to plotting the field pattern, it is usually helpful to make a plot of x versus θ from (13). Then, knowing the values of x for the nulls and maxima of the $T_m(x)$ curve, the corresponding values of θ may be determined. As many intermediate points as are needed may also be obtained in the same manner. Following this procedure, the field pattern for our problem of the eight-source array is presented in Fig. 5–33*a* in rectangular coordinates and in Fig. 5–33*b* in polar coordinates.

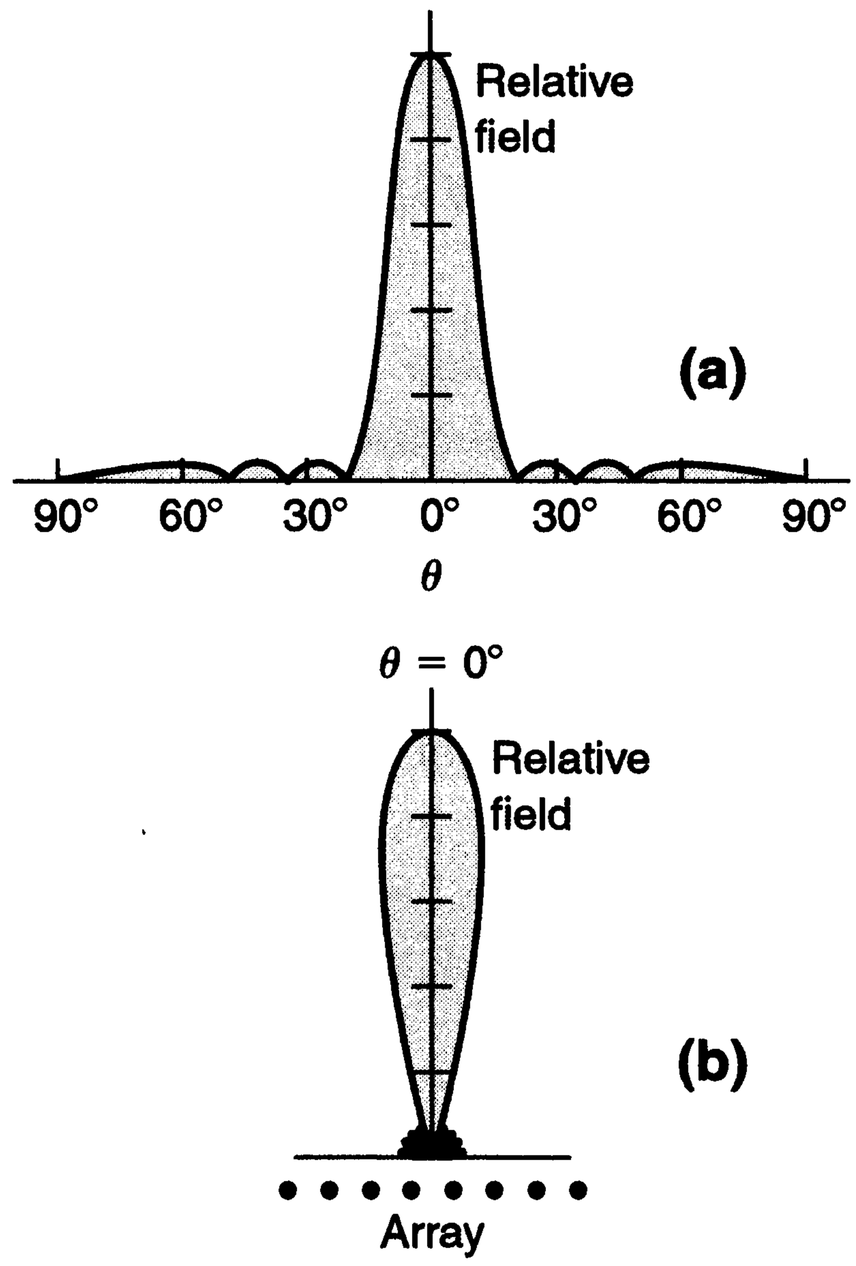


Figure 5–33
Relative field pattern of broadside array of eight isotropic sources spaced $\lambda/2$ apart. The D-T amplitude distribution gives a minimum beamwidth for a side-lobe level $\frac{1}{20}$ of the main lobe. The pattern is shown in rectangular coordinates at (*a*) and in polar at (*b*). Both diagrams show the pattern only from −90° to +90°, the other half of the pattern being identical.

Table 5–6

Variable	Range		
θ	$-\frac{\pi}{2}$	0	$+\frac{\pi}{2}$
$\frac{\psi}{2}$	$-\frac{d_r}{2}$	0	$+\frac{d_r}{2}$
w	$\cos\frac{d_r}{2}$	1	$\cos\frac{d_r}{2}$
x	$x_0\cos\frac{d_r}{2}$	x_0	$x_0\cos\frac{d_r}{2}$

5-11 COMPARISON OF AMPLITUDE DISTRIBUTIONS FOR EIGHT-SOURCE ARRAYS（八源阵幅度分布的比较）

In the problem worked in the preceding section, the side-lobe level was 26 dB below the maximum of the main beam ($R = 20$). It is of interest to compare the amplitude for this case with the distributions for other side-lobe levels. This is done in Fig. 5–34 for a uniform distribution and three optimum (D-T) distributions with side-lobe levels −20 dB, −40 dB, and $-\infty$ dB below the main beam maximum. The infinite decibel case corresponds to $R = \infty$ (zero side-lobe level) and is identical with Stone's binomial distribution. The relative

Table 5–7

Variable	Range		
θ	$-\frac{\pi}{2}$	0	$+\frac{\pi}{2}$
x	0	1.15	0

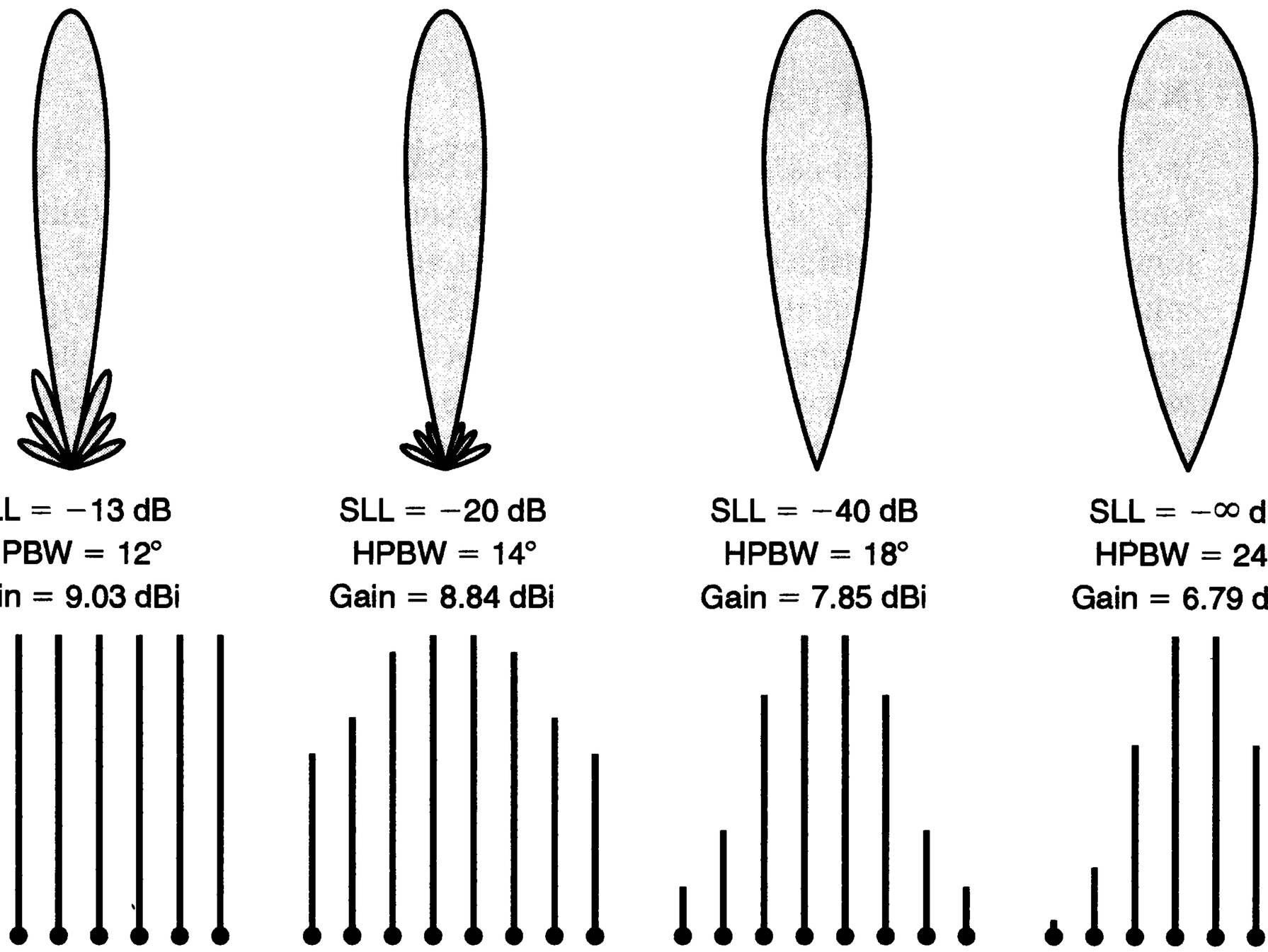


Figure 5–34
Uniform and three optimum (D-T) source distributions for eight in-phase isotropic sources spaced $\lambda/2$ with field patterns. The distributions result in Side-Lobe Levels (SLLs) ranging from −13 dB for the uniform array to $-\infty$ for the binomial array. Note that as the SLL is reduced, the distribution is more tapered, the HPBW is larger, and the gain is less. Thus, if low SLL is required, the gain is reduced. Conversely, if maximum gain is desired, a larger SLL must be tolerated. This is the designer's dilemma.

amplitudes for this case are 1, 7, 21, 35, 35, 21, 7, 1 (Riblet-1). The ratio of amplitudes of the center sources to the edge sources is 35 to 1. Such a large ratio might be difficult to achieve in practice. *Both the binomial and edge distributions are special cases of the Dolph-Tchebyscheff (D-T) distribution, but the uniform amplitude distribution is not.*

The D-T optimum amplitude distribution, as discussed in the preceding sections, *is optimum only if* $d \geq \lambda/2$, which covers the cases of most interest for broadside arrays. By a generalization of the method, however, cases with smaller spacings can also be optimized.

In conclusion, it should be pointed out that the properties of the Tchebyscheff polynomials may be applied not only to antenna patterns as discussed above but also to other situations. It is necessary, however, that the function to be optimized be expressible as a polynomial.

The D-T (Dolph-Tchebyscheff) source amplitude distribution for linear arrays on N sources are given by the computer program ARRAYPATGAIN on the book's web site antennas3.com. The program also plots the pattern and gives the HPBW and gain as discussed in App. C–3*a*.

5–12 CONTINUOUS ARRAYS（连续阵）

In the preceding sections, the discussion has been restricted to arrays of discrete point sources, i.e., to arrays of a finite number of sources separated by finite distances. We now proceed to a consideration of *continuous* arrays of point sources, i.e., arrays of an infinite number of sources separated by infinitesimal distances. By Huygens' principle, a continuous array of point sources is equivalent to a continuous field distribution. In this way, our discussion of continuous arrays can be extended to include the radiation patterns of field distributions across apertures, as, for example, the pattern of an electromagnetic horn where the field distribution across the mouth of the horn is known.

We shall now develop an expression for the far field of a continuous array of point sources of uniform amplitude and of the same phase. Let the array of length a be parallel to the y axis with its center at the origin as indicated in Fig. 5–35. Then the field dE at a distant point in the direction θ due to the point sources in the infinitesimal length dy at a

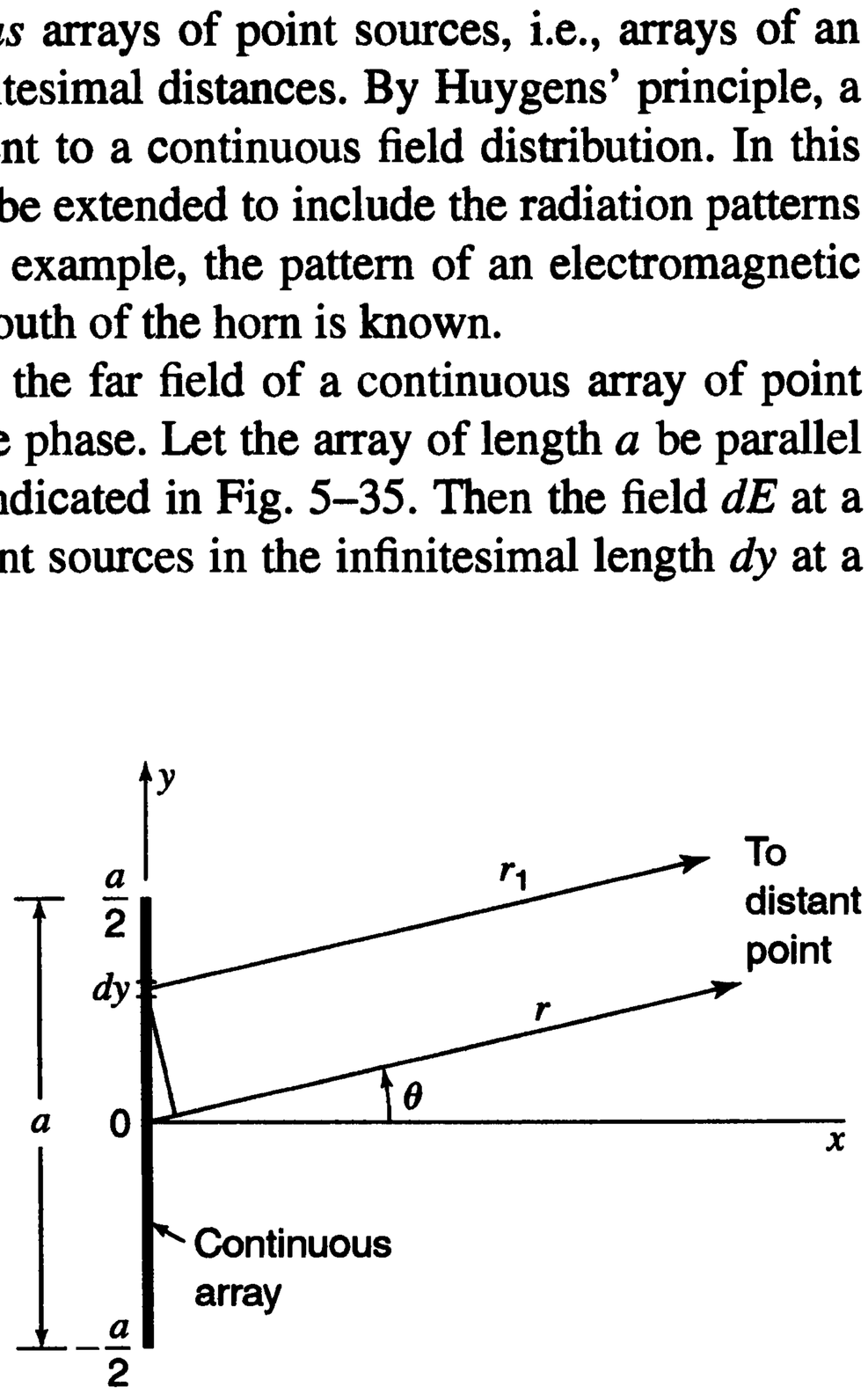


Figure 5–35
Continuous broadside array of point sources of length *a*.

distance y from the origin is

$$dE = \frac{A}{r_1} e^{j\omega[t-(r_1/c)]}\, dy = \frac{A}{r_1} e^{j(\omega t - \beta r_1)}\, dy \tag{1}$$

where $\beta = \omega/c = 2\pi/\lambda$ and A is a constant involving amplitude. The total field E at the distant point is then the integrated value of (1) over the array of length a as given by

$$E = \int_{-a/2}^{a/2} \frac{A}{r_1} e^{j(\omega t - \beta r_1)}\, dy \tag{2}$$

Both A and the time factor may be taken outside the integral, and r_1 may also be if $r_1 \gg a$. Thus,

$$E = \frac{Ae^{j\omega t}}{r_1} \int_{-a/2}^{a/2} e^{-j\beta r_1}\, dy \tag{3}$$

However, referring to Fig. 5–35,

$$r_1 = r - y \sin\theta \tag{4}$$

Substituting (4) in (3) and taking the constant factor $e^{-j\beta r}$ outside the integral, we have

$$E = A' \int_{-a/2}^{a/2} e^{j\beta y \sin\theta}\, dy \tag{5}$$

where

$$A' = \frac{Ae^{j(\omega t - \beta r)}}{r_1} \tag{6}$$

Integrating (5) yields

$$E = \frac{2A'}{\beta \sin\theta} \frac{e^{j(\beta a/2)\sin\theta} - e^{-j(\beta a/2)\sin\theta}}{2j} \tag{7}$$

which may be written as

$$E = \frac{2A'}{\beta \sin\theta} \sin\left(\frac{\beta a}{2} \sin\theta\right) \tag{8}$$

Let

$$\psi' = \beta a \sin\theta = a_r \sin\theta \tag{9}$$

where $a_r = \beta a = 2\pi a/\lambda =$ array length, rad

Then

$$E = \frac{2A'}{\beta \sin\theta} \sin\frac{\psi'}{2} \tag{10}$$

However, from (9),

$$\beta \sin\theta = \frac{\psi'}{a}$$

so that (10) becomes

$$E = aA' \frac{\sin(\psi'/2)}{\psi'/2} \tag{11}$$

Normalizing (11) gives finally

$$E = \frac{\sin(\psi'/2)}{\psi'/2} \tag{12}$$

Equation (12) expresses the far field, or Fraunhofer diffraction pattern, of a continuous broadside array of length a having uniform amplitude and phase. For n discrete, equally spaced sources, it was previously shown by (5–6–9) that the normalized value of the total field is

$$E = \frac{\sin(n\psi/2)}{n\sin(\psi/2)} \tag{13}$$

where $\psi = d\cos\phi + \delta$

For in-phase sources, $\delta = 0$. Comparing Figs. 5–18 and 5–35, we note that $\phi = \theta + \pi/2$, so that

$$\psi = -d_r \sin\theta = -\beta d \sin\theta \tag{14}$$

For small values of ψ, which occur for small values of θ, d, or both, (13) can be expressed as

$$E = \frac{\sin(n\psi/2)}{n\psi/2} = \frac{\sin[(\beta nd/2)\sin\theta]}{(\beta nd/2)\sin\theta} \tag{15}$$

The length a of the array of discrete sources is

$$a = d(n-1) \tag{16}$$

where n = number of sources

d = spacing

If $n \gg 1$, $a \simeq nd$ and (15) becomes

$$E = \frac{\sin[(\beta a/2)\sin\theta]}{(\beta a/2)\sin\theta} = \frac{\sin[(a_r/2)\sin\theta]}{(a_r/2)\sin\theta} \tag{17}$$

where $a_r = \beta a = 2\pi a/\lambda$

By (9) this can now be expressed as

$$E = \frac{\sin(\psi'/2)}{\psi'/2} \tag{18}$$

which is identical with the value obtained in (12) for the continuous array. Thus, the field pattern for an array of many discrete sources ($n \gg 1$) and for small values of ψ is the same as the pattern of a continuous array of the same length. If the array is long, that is, if $nd \gg \lambda$, the main beam and the first minor lobes are confined to small values of θ. It therefore follows that the main features of the pattern of a large array are the same, whether the array has many discrete sources or is a continuous distribution of sources. Many of the conclusions derived in previous sections concerning amplitude distributions

for arrays of discrete sources can also be applied to continuous arrays provided that the arrays are large.

The null directions θ_0 of the continuous array pattern are given by

$$\frac{\psi'}{2} = \pm K\pi \tag{19}$$

where $K = 1, 2, 3, \ldots$

Thus,

$$\theta_0 = \arcsin\left(\pm\frac{K\lambda}{a}\right) \tag{20}$$

For a long array (20) can be expressed as

$$\theta_0 \simeq \pm\frac{K}{a_\lambda}(\text{rad}) \simeq \pm\frac{57.3K}{a_\lambda}(\text{deg}) \tag{21}$$

where $a_\lambda = a/\lambda$

The beamwidth between first nulls ($K = 1$) for a long array is then

(第一零点波束宽度)

$$\boxed{\textbf{\textit{Beamwidth between first nulls}} \quad 2\theta_{01} \simeq \frac{2}{a_\lambda}(\text{rad}) \simeq \frac{115}{a_\lambda}(\text{deg}) \qquad \textbf{\textit{long array}}\ (\text{长阵})} \tag{22}$$

It is to be noted that (20), (21), and (22) are identical with the expressions given for the broadside array of discrete sources, if nd is replaced by a (see Table 5–2). Therefore, the null locations for long arrays of either discrete or continuous sources are the same provided only that $n \gg 1$.

The field patterns of the main beam of continuous arrays of point sources 5, 10, and 50λ long are compared in Fig. 5–36. It may be noted that the beamwidth between half-power points, θ_{HP}, of a long, uniform broadside array is given approximately by

$$\theta_{HP} = 0.9\theta_{01} = \frac{0.9}{a_\lambda} \qquad (\text{rad}) \tag{23}$$

or

$$\theta_{HP} = \frac{51}{a_\lambda} \qquad (\text{deg}) \tag{24}$$

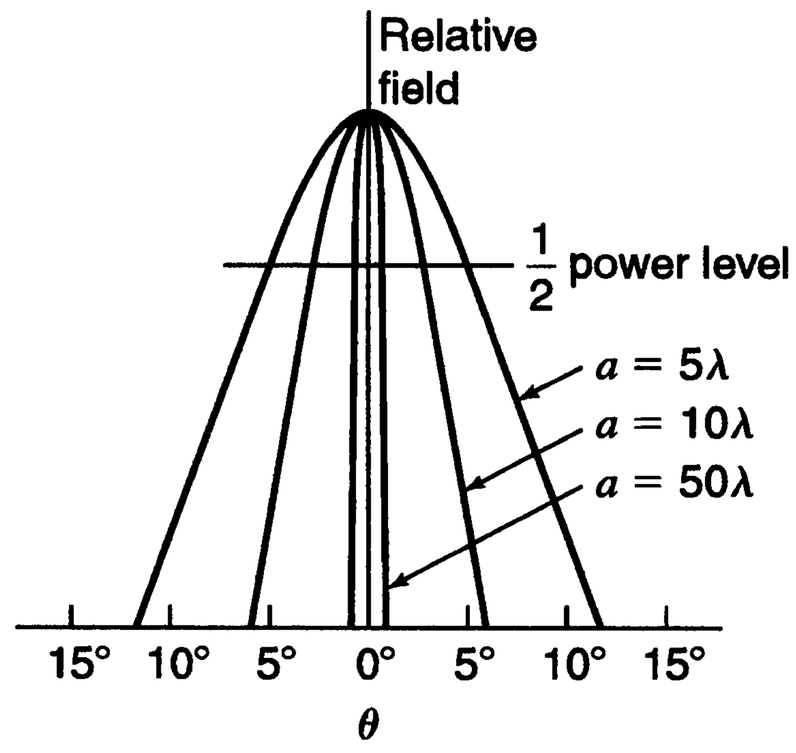


Figure 5–36
Main-lobe field patterns of continuous uniform broadside arrays 5, 10, and 50λ long.

5–13 HUYGENS' PRINCIPLE（惠更斯原理）

The principle proposed by Christian Huygens (1629–1695) has been of fundamental importance to the development of wave theory (Huygens-1; Born-1; Sommerfeld-1).

Huygens' principle states that each point on a primary wave front can be considered to be a new source of a secondary spherical wave and that a secondary wave front can be constructed as the envelope of these secondary waves, as suggested in Fig. 5–37.

（惠更斯原理说明，某个初始波阵面上的每一点都应视为新的次级球面波的源，迷些次级波的包络构成了次级波阵面，见图 5.37）

Thus, a spherical wave from a single point source propagates as a spherical wave as indicated in Fig. 5–37*a*, while an infinite plane wave continues as a plane wave as suggested in Fig. 5–37*b*. This principle of physical optics can be used to explain the apparent bending of electromagnetic waves around obstacles, i.e., the diffraction of waves, a *diffracted* ray being one that follows a path that cannot be interpreted as either reflection or refraction.

Let us consider the situation shown in Fig. 5–38*a* in which an infinite plane electromagnetic wave is incident on an infinite flat sheet which is opaque to the waves. The sheet has a slot of width a and of infinite length in the direction normal to the page. The field everywhere to the right of the sheet is the result of the section of the wave that passes through the slot. If a is many wavelengths, the field distribution across the slot may be assumed, in the first approximation, to be uniform, as shown in Fig. 5–38*b*. By Huygens' principle the field everywhere to the right of the sheet is the same as though each point in the plane of the slot is the source of a new spherical wave. Each of these point sources is of equal amplitude and phase. Thus, by Huygens' principle the slotted sheet with a uniform field across the opening can be replaced by a continuous array of point sources which just fills the opening. The field pattern in the xy plane (Fig. 5–38*a*) is then calculated in the same way as for a continuous linear array of point sources of length a oriented parallel to the y axis.

The far field, or Fraunhofer diffraction pattern, of such an array was shown in the preceding section to be given by

$$E = \frac{\sin(\psi'/2)}{\psi'/2} \tag{1}$$

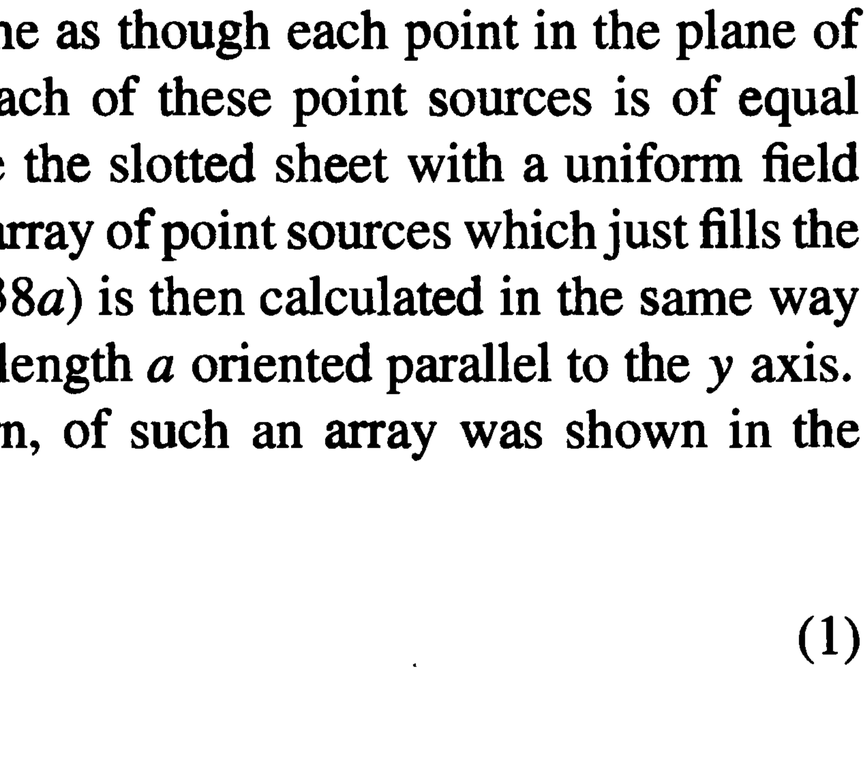


Figure 5–37
Spherical and plane wavefronts with secondary waves of Huygens.

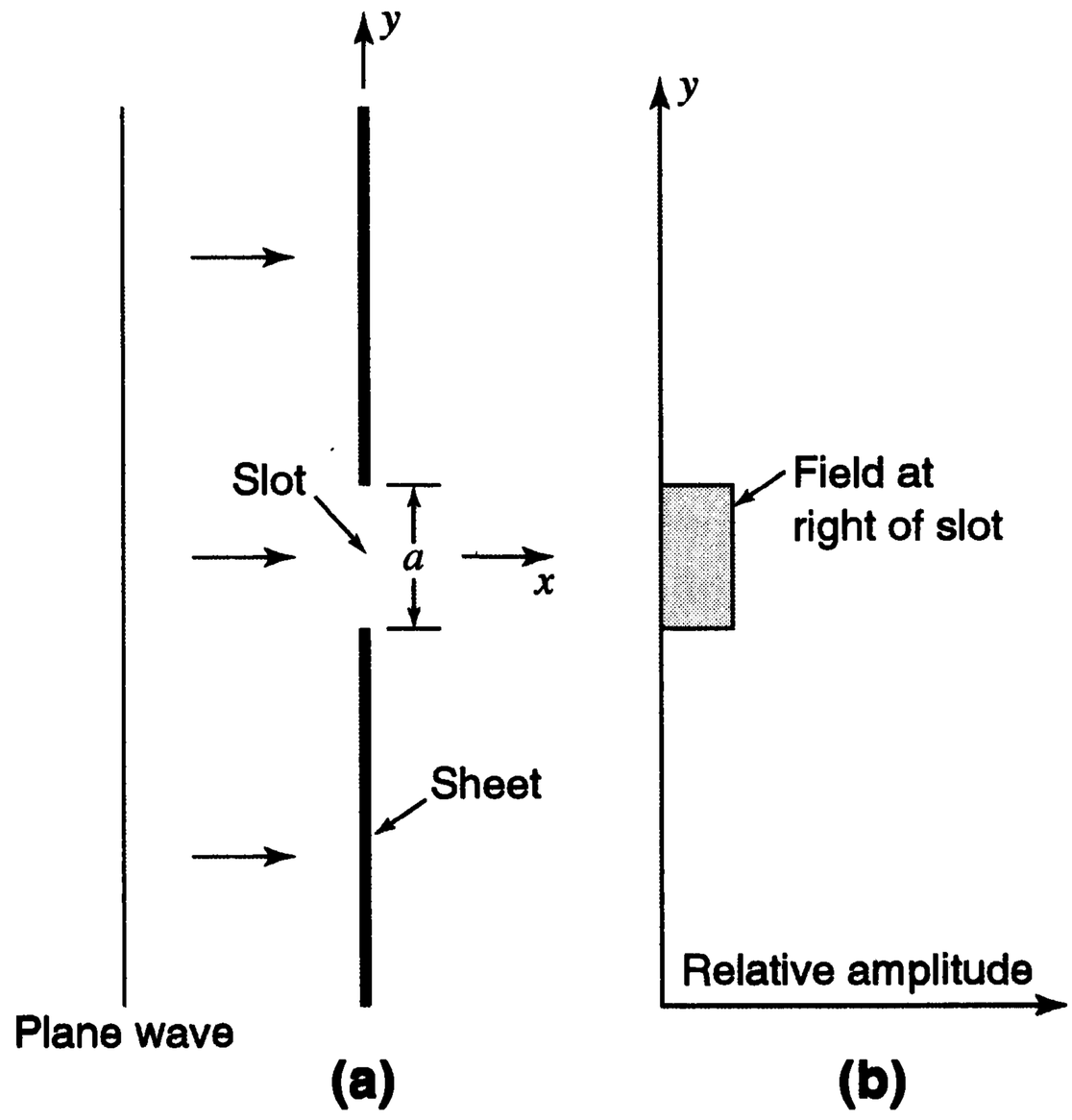


Figure 5–38
Plane wave incident on opaque sheet with slot of width *a*.

where $\psi' = (2\pi a/\lambda)\sin\theta$ and where θ is in the xy plane (Fig. 5–35). This pattern, in the xy plane, is independent of the extent of the array in the z direction (normal to the page).

In deriving (1), that is, (5–12–12), the total field at a point was obtained by integrating the contributions from a continuous array of sources distributed over a length a. For points at a great distance from the array the integral can be simplified, and the integration is straightforward, as demonstrated in the preceding section. For points near to the array, however, the integral does not simplify in this way but can be reduced to the form of Fresnel's integrals. The field variation near the slot as obtained in this way is commonly called a Fresnel diffraction pattern. Along a straight line parallel to the slot and a short distance from it, the field variation is as suggested at (*a*) in Fig. 5–39, the variation approximating the uniform distribution of field at the slot as shown in Fig. 5–38*b*. As the distance x from the slot is increased, the Fresnel patterns change through a series of transitional forms, such as suggested at (*b*) in Fig. 5–39, until at large distances we enter the Fraunhofer region and the pattern assumes a form as suggested by (*c*) in Fig. 5–39. Ordinarily the Fraunhofer pattern is obtained by rotating the slot around its center so that the field is observed at a constant radius rather than at a constant distance x. The resulting field pattern in polar coordinates is then as suggested at (*d*) in Fig. 5–39. Once we have entered the Fraunhofer region, this pattern is the same at all greater distances. For a point to be in the Fraunhofer region, it must be at a sufficient distance from the slot so that we can make the assumption that lines extending from the edges of the slot to the point are parallel. This is commonly assumed to be the case when the point is at a distance r from the slot given by

$$r \geq \frac{2a^2}{\lambda} \tag{2}$$

where a is the width or aperture of the slot, which is assumed to be large. Thus, the larger the aperture or the shorter the wavelength, the greater must be the distance at which the

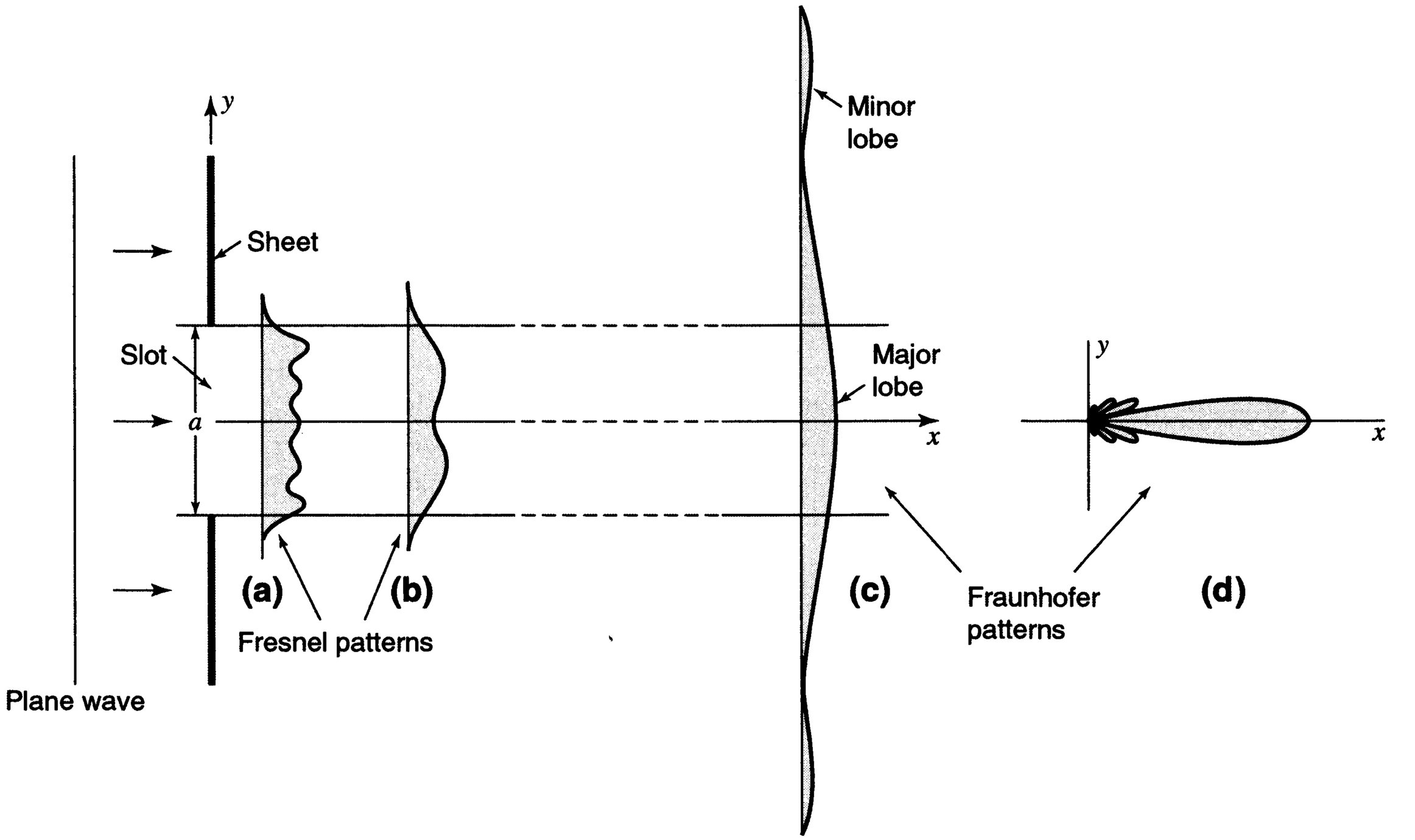


Figure 5–39
Fresnel and Fraunhofer patterns of a slot of width *a*.

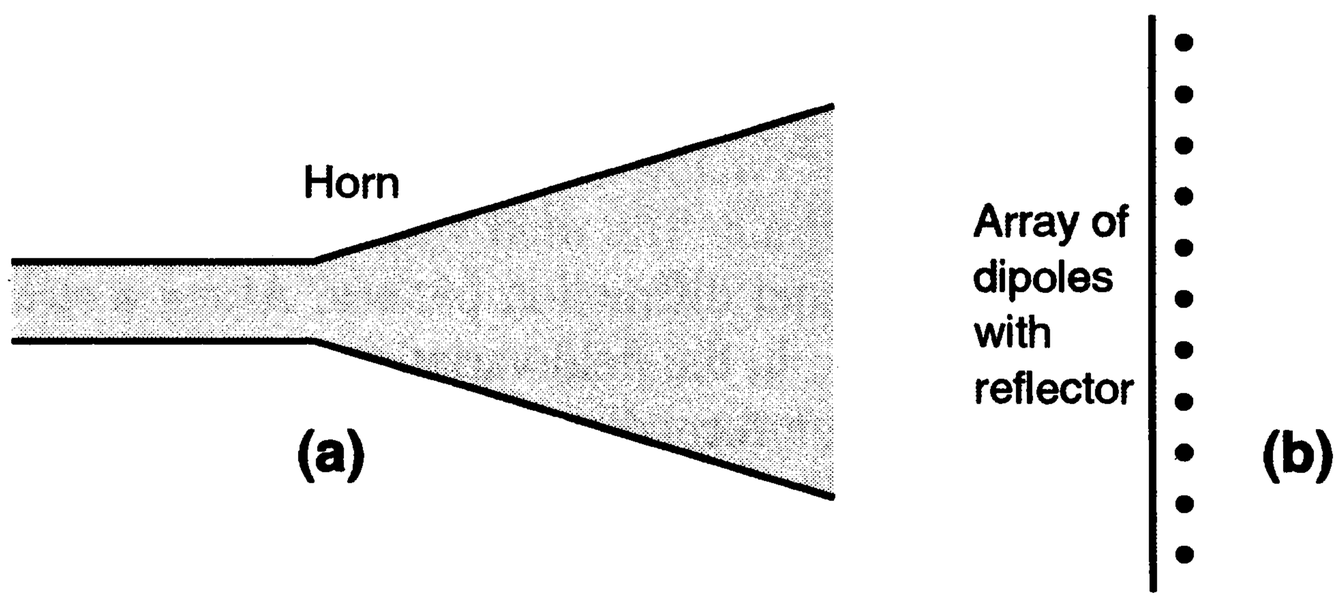


Figure 5–40
Electromagnetic horn antenna and array of dipoles with reflector.

pattern is measured if we wish to avoid the effects of Fresnel diffraction. This is discussed further in Sec. 24–2–2.

A nearly uniform type of field distribution across an aperture such as discussed above in connection with Figs. 5–38 and 5–39 occurs in optics when a beam of light is incident on a slit. It also may be realized by the field distribution across the mouth of a long electromagnetic horn antenna as in Fig. 5–40*a*. Since the pattern of a uniform field distribution is the same as the pattern of a uniform distribution of point sources of equal extent, another form of antenna equivalent to the optical slit or electromagnetic horn is a uniform current sheet. This can be approximated by a "billboard" type of array, as in Fig. 5–40*b*, having many dipole antennas carrying equal currents. The expressions which have been developed can thus be applied to a calculation of the Fraunhofer diffraction pattern of an optical slit

or the far field of a horn or uniform current sheet. If the field or current distribution across the slit or antenna aperture is not uniform, the form factor for the distribution will appear in the integral for the field expression. If the aperture is large, the relations developed for amplitude distributions of arrays of discrete sources can be applied to the case of continuous arrays of sources.

Huygens' principle is not without its limitations. Thus, it neglects the vector nature of the electromagnetic field. It also neglects the effect of currents which flow at the edge of the slot, as in Figs. 5–38 and 5–39, or at the edge of the horn, as in Fig. 5–40*a*. However, if the aperture is sufficiently large and we confine our attention to directions roughly normal to aperture, the scalar theory of Huygens' principle gives satisfactory results.

5–14 HUYGENS' PRINCIPLE APPLIED TO THE DIFFRACTION OF A PLANE WAVE INCIDENT ON A FLAT SHEET. PHYSICAL OPTICS

（惠更斯原理应用于平面波投射到平板的绕射与物理光学）

Consider a uniform plane wave incident on a perfectly conducting half-plane, as in Fig. 5–41*a* (Kraus-1). We want to calculate the electric field at point P at a distance r behind the plane. By Huygens' principle,

$$E = \int_{\substack{\text{over} \\ x \text{ axis}}} dE \tag{1}$$

where dE is the electric field at P due to a point source at a distance x from the origin, as in Fig. 5–41*b*. Thus,

$$dE = \frac{E_0}{r} e^{-j\beta(r+\delta)}\, dx \tag{2}$$

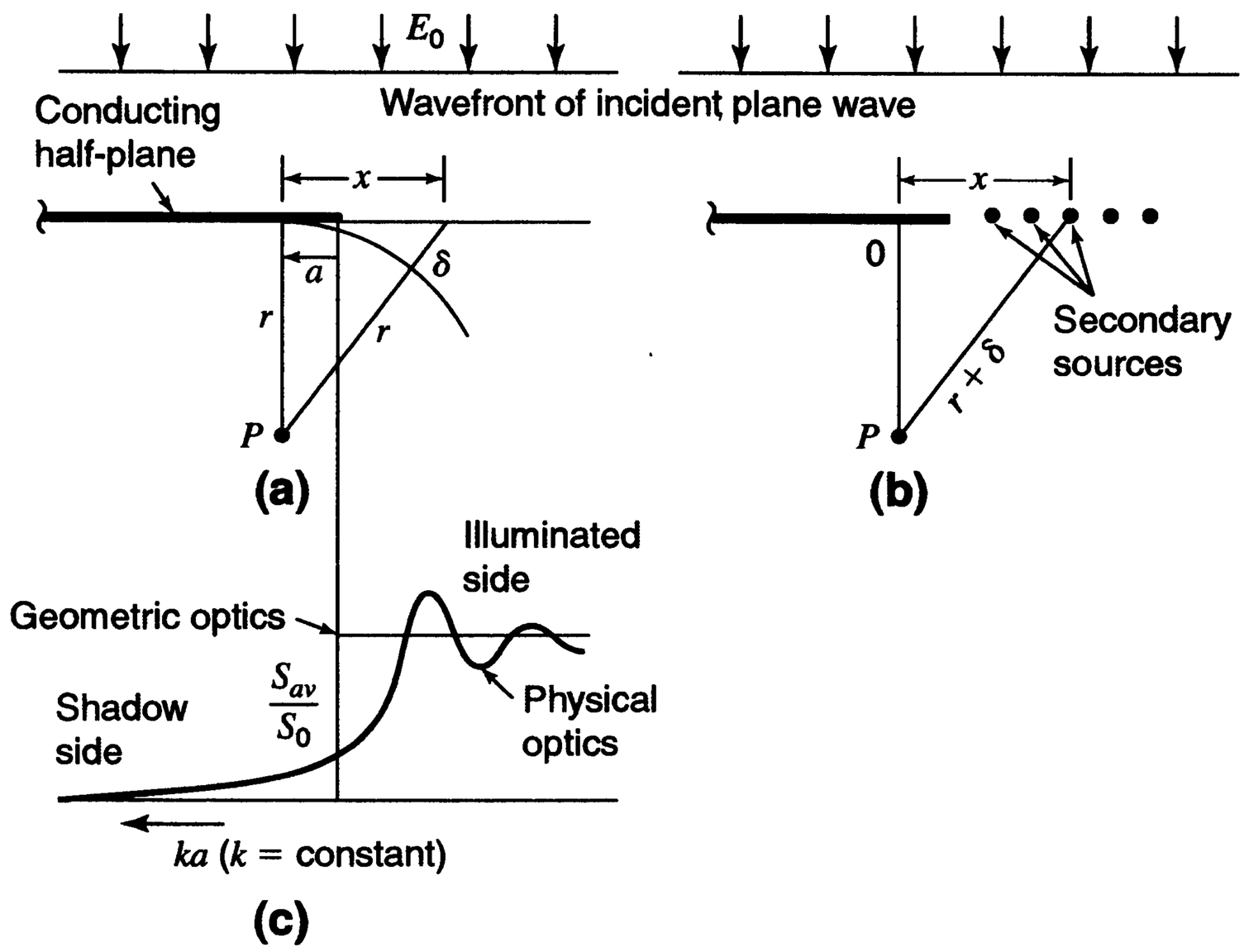


Figure 5–41
Plane wave incident from above onto a conducting half-plane with resultant power-density variation below the plane as obtained by physical optics.

so that

$$E = \frac{E_0}{r} e^{-j\beta r} \int_a^\infty e^{-j\beta\delta}\, dx \tag{3}$$

If $\delta \ll r$, it follows that

$$\delta = \frac{x^2}{2r} \tag{4}$$

When we let $2/r\lambda = k^2$ and $kx = u$, (3) becomes

$$E = \frac{E_0}{kr} e^{-j\beta r} \int_{ka}^\infty e^{-j\pi u^2/2}\, du \tag{5}$$

which can be rewritten as

$$E = \frac{E_0}{kr} e^{-j\beta r} \int_0^\infty e^{-j\pi u^2/2}\, du - \int_0^{ka} e^{-j\pi u^2/2}\, du \tag{6}$$

The integrals in (6) have the form of Fresnel integrals so (6) can be written

$$E = \frac{E_0}{kr} e^{-j\beta r} \left\{ \frac{1}{2} + \frac{1}{2} j - [C(ka) + jS(ka)] \right\} \tag{7}$$

where

$$C(ka) = \int_0^{ka} \cos \frac{\pi u^2}{2}\, du = \text{Fresnel cosine integral} \tag{8}$$

$$S(ka) = \int_0^{ka} \sin \frac{\pi u^2}{2}\, du = \text{Fresnel sine integral} \tag{9}$$

where $ka = \sqrt{\frac{2}{r\lambda}}\, a$, dimensionless.

A graph of $C(ka)$ and $S(ka)$ yields the Cornu spiral (Fig. 5–42). Since $C(-ka) = -C(ka)$ and $S(-ka) = -S(ka)$, the spiral for negative values of ka is in the third quadrant and is symmetrical with respect to the origin for the spiral in the first quadrant.

The power density as a function of ka is then

$$S_{av} = \frac{EE^*}{Z} = S_0 \frac{1}{2} \left\{ \left[\frac{1}{2} - C(ka) \right]^2 + \left[\frac{1}{2} - S(ka) \right]^2 \right\} \qquad (\text{W m}^{-2}) \tag{10}$$

where

$$S_0 = \frac{E_0^2 \lambda}{2Zr} \qquad (\text{W m}^{-2}) \tag{11}$$

The power density variation of (10) as a function of ka (with r, λ, and k constant) is shown in Fig. 5–41*c*. Assuming that the plane wave originates from a distant source, we have

1. For no obscuration, $ka = -\infty$ and $S_{av} = S_0$.
2. For source, observer, and edge of obscuring plane in line, $ka = 0$ and $S_{av} = \frac{1}{4} S_0$.
3. For complete obscuration, $ka = +\infty$ and $S_{av} = 0$.

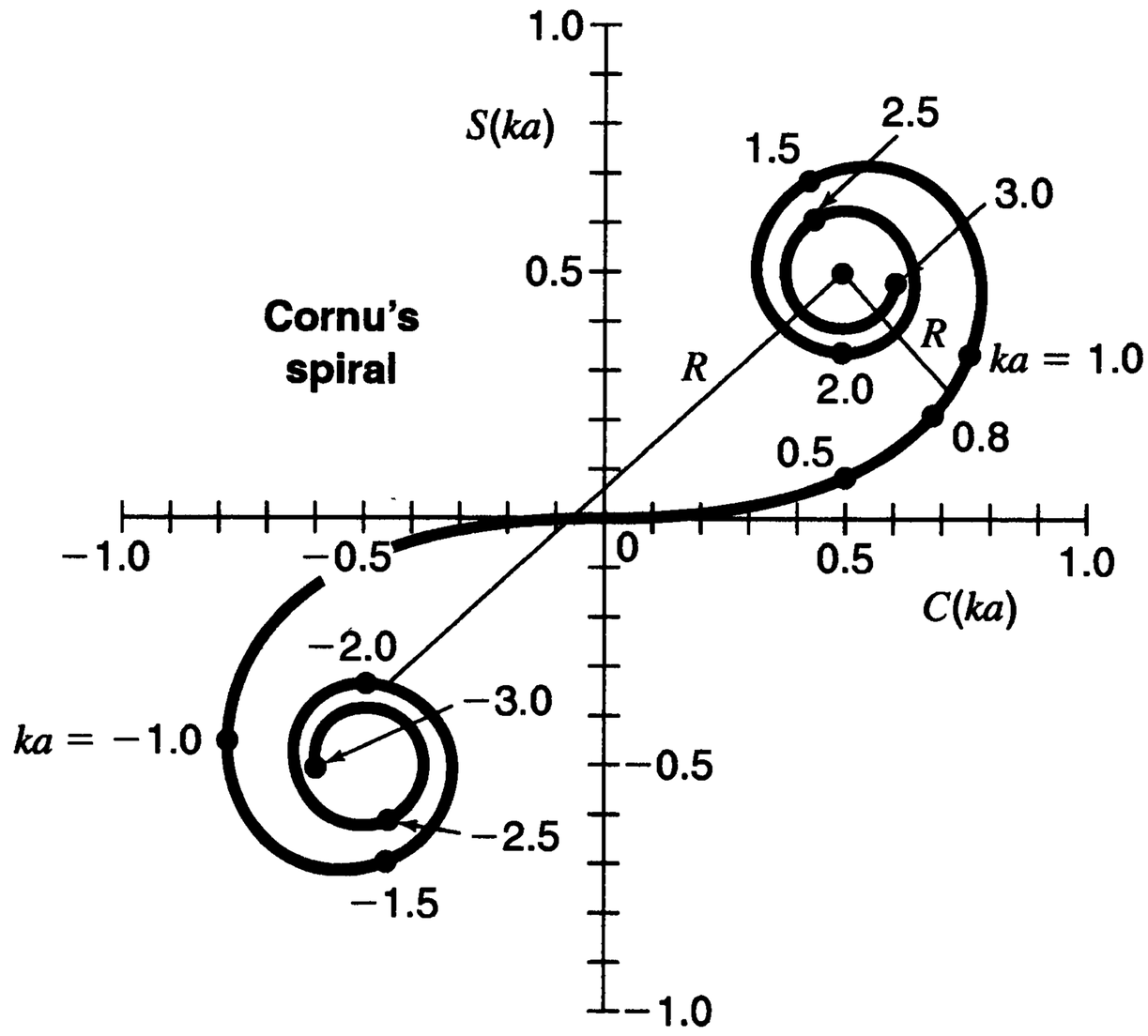


Figure 5–42
Cornu spiral showing $C(ka)$ and $S(ka)$ as a function of ka values along the spiral. For example, when $ka = 1.0$, $C(ka) = 0.780$ and $S(ka) = 0.338$. When $ka = \infty$, $C(ka) = S(ka) = \frac{1}{2}$.

Thus, the power density does not go to zero abruptly as the point of observation goes from the illuminated side ($ka < 0$) to the shadow side ($ka > 0$); rather, there are fluctuations followed by a gradual decrease in power density.

From (10) and (11) the relative power density as a function of ka is

$$S_{\text{av}}(\text{relative}) = \frac{S_{\text{av}}}{S_0} = \frac{1}{2}\left\{\left[\frac{1}{2} - C(ka)\right]^2 + \left[\frac{1}{2} - S(ka)\right]^2\right\} \tag{12}$$

The relative power density (12) is equal to $\frac{1}{2}R^2$, where R is the distance from a ka value on the Cornu spiral to the point $(\frac{1}{2}, \frac{1}{2})$ (see Fig. 5–42). For large positive values of ka, R approaches $1/\pi ka$, so that (12) reduces approximately to

$$S_{\text{av}}(\text{relative}) = \frac{1}{2}\left(\frac{1}{\pi ka}\right)^2 = \frac{r\lambda}{4\pi^2 a^2} \tag{13}$$

where

r = distance from obstacle (conducting half-plane), m
λ = wavelength, m
a = distance into shadow region, m

Equation (13) gives the relative power density for large ka (>3) (well into the shadow region). For this condition it is apparent that the power flux density (Poynting vector) due to diffraction increases with wavelength and with distance (from edge) but decreases as the square of the distance a into the shadow region.

EXAMPLE 5–14.1

A vertical conducting wall 25 m high extends above a flat ground plane. A $\lambda = 10$-cm transmitter is situated 25 m above the ground plane at a large distance to one side of the vertical wall and a receiver is located on the ground plane 100 m to the other side of the wall. Find the signal level at the receiver due to diffraction over the wall as compared to the direct path signal without the wall.

■ **Solution**

The constant $k = \sqrt{2/r\lambda} = \sqrt{2/100 \times 0.1} = 0.44$ and $a = 25$ m, so $ka = 11$, which is greater than 3. Thus, (13) is applicable and

$$S_{av}(\text{relative}) = \frac{r\lambda}{4\pi^2 a^2} = \frac{100 \times 0.1}{4\pi^2 \times 25^2} \simeq \frac{1}{2500} \text{ or } -34 \text{ dB} \qquad \textit{Ans.}$$

Thus, the vertical wall causes 34 dB of attenuation as compared to a direct path signal.

If the half-plane in Fig. 5–41 is replaced by a strip of width D and length $\gg D$, diffraction occurs from both edges, scattering radiation into the shadow region behind the strip. On the centerline of the strip, diffraction fields from both edges are equal in magnitude and of the same phase since the path lengths from both edges are equal. Thus, the diffracted field has a maximum or central peak on the centerline.

If the strip is replaced by a disk of diameter D, there is diffraction around its entire edge and all diffracted fields arrive in phase on the centerline behind the disk, producing a larger central peak. In optics this peak is called the *axial bright spot*. In a similar way, the diffracted fields from the feed system at the focus of a parabolic dish reflector can produce a back lobe on the axis of the parabola. See additional discussions on diffraction in Secs. 9–2, 10–2, 12–5, and 24–4*d*.

5–15 RECTANGULAR-AREA BROADSIDE ARRAYS（矩形面积的边射阵）

The method of obtaining the field patterns of linear arrays discussed in the preceding sections can be easily extended to the case of rectangular broadside arrays, i.e., arrays of sources which occupy a flat area of rectangular shape, as in Fig. 5–43. For such a rectangular array, the field pattern in the xy plane (as a function of θ) depends only on the y-dimension a of the array, while the field pattern in the xz plane (as a function of ϕ) depends only on the z-dimension b of the array. The assumption is made that the field or current distribution across the array in the y direction is the same for any values of z between $\pm b/2$. Likewise, it is assumed that the amplitude distribution across the array in the z direction is the same for all values of y between $\pm a/2$. Therefore, the field pattern in the xy plane is calculated as though the array consists only of a single linear array of height a coincident with the y axis (y array). In the same way, the pattern in the xz plane is obtained by calculating the pattern of a single linear array of length b coincident with the z axis (z array). If the array also has depth in the x direction, i.e., has end-fire directivity, then the pattern in the xy plane is the product of the patterns of the single linear x and y arrays, while the pattern in the xz plane is the product of the patterns of the x and z arrays.

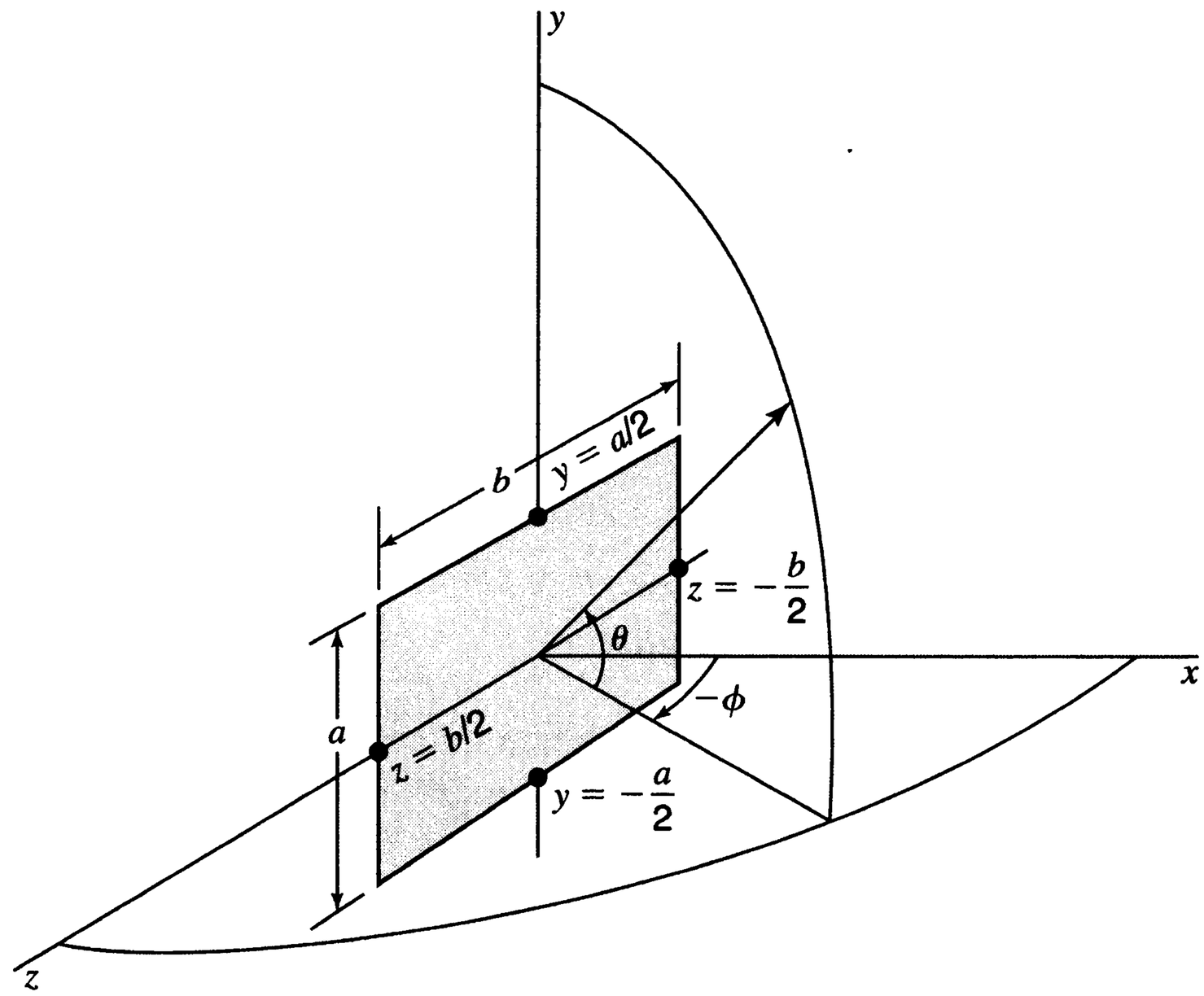


Figure 5–43
Rectangular broadside array of height *a* and length *b* with relation to coordinates.

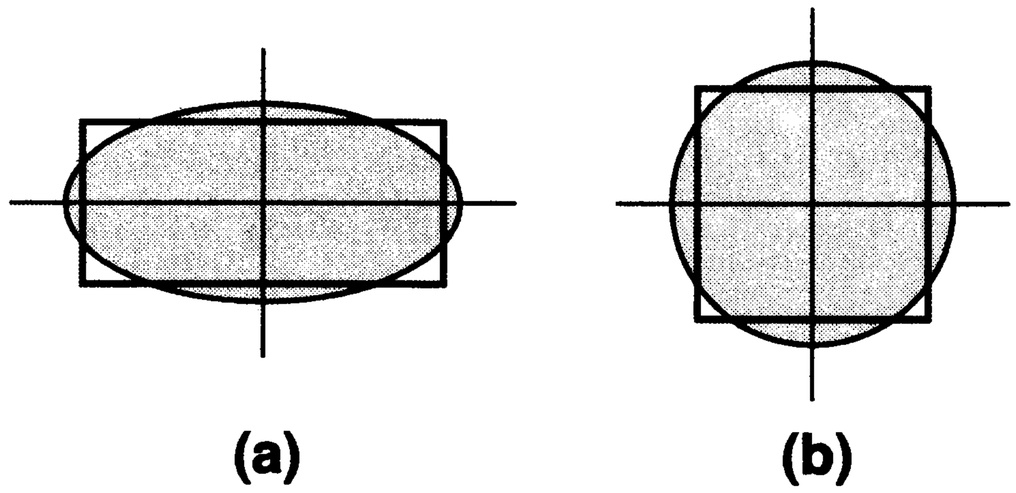


Figure 5–44
Elliptical array with equivalent rectangular array (*a*) and circular array with equivalent square array (*b*).

If the area occupied by the array is not rectangular in shape, the above principles do not hold. However, the approximate field patterns may be obtained in the case of an array of elliptical area, for example, by assuming that it is a rectangular area as in Fig. 5–44*a*, or in the case of a circular area by assuming that it is square as in Fig. 5–44*b*.

From the field patterns in two planes (xy and xz) of a rectangular array the beamwidths between half-power points can be obtained. If the minor lobes are not large, the directivity D is then given approximately by

$$D = \frac{41{,}000}{\theta_1^\circ \phi_1^\circ} \tag{1}$$

where θ_1° and ϕ_1° are the half-power beamwidths in degrees in the xy and xz planes, respectively. The limitations of (1) are discussed following (2–7–9).

An expression for the directivity of a large rectangular broadside array of height a and width b (Fig. 5–43) and with a uniform amplitude distribution may also be derived

rigorously as follows. By (2–7–3), the directivity of an antenna is given by

$$D = \frac{4\pi f(\theta, \phi)_{\max}}{\int\int f(\theta, \phi) \sin\theta \, d\theta \, d\phi} \tag{2}$$

where $f(\theta, \phi)$ is the space power pattern, which varies as the square of the space field pattern. From (5–12–17), the space field pattern of a large rectangular array is

$$E(\theta, \phi) = \frac{\sin[(a_r \sin\theta)/2]}{(a_r \sin\theta)/2} \frac{\sin[(b_r \sin\phi)/2]}{(b_r \sin\phi)/2} \tag{3}$$

where

$$a_r = 2\pi a/\lambda$$
$$b_r = 2\pi b/\lambda$$

The main beam maximum is in the direction $\theta = \phi = 0$ in Fig. 5–43. In (3), $\theta = 0$ at the equator, while in (2), $\theta = 0$ at the zenith. For large arrays and relatively sharp beams we can therefore replace $\sin\theta$ and $\sin\phi$ in (3) by the angles, while $\sin\theta$ in (2) can be set equal to unity. Assuming that the array is unidirectional (no field in the $-x$ direction), the integral in the denominator of (2) then becomes

$$\int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} \frac{\sin^2(\pi a\theta/\lambda)}{(\pi a\theta/\lambda)^2} \frac{\sin^2(\pi b\phi/\lambda)}{(\pi b\phi/\lambda)^2} \, d\theta \, d\phi \tag{4}$$

Making the limits of integration $-\infty$ to $+\infty$ instead of $-\pi/2$ to $+\pi/2$, (4) may be evaluated as λ^2/ab. Therefore, the approximate directivity D of a large unidirectional rectangular broadside array with a uniform amplitude distribution is

$$D = \frac{4\pi ab}{\lambda^2} = 12.6\frac{ab}{\lambda^2} \tag{5}$$

As an example, the directivity of a broadside array of height $a = 10\lambda$ and length $b = 20\lambda$ is, from (5), equal to 2520, or 34 dB.

5–16 ARRAYS WITH MISSING SOURCES AND RANDOM ARRAYS
(缺源阵和随机阵)

A linear array of five isotropic point sources with $\lambda/2$ spacing is discussed in Sec. 5–10 for several amplitude distributions including uniform, binomial, and Dolph-Tchebyscheff. Let us consider this five-source array again with all sources of equal amplitude with pattern as in Fig. 5–45*a* (same as Fig. 5–29*a*) and note what happens to the pattern if one of the sources is turned off (amplitude reduced to zero). If the off source is next to the edge source, the beamwidth is essentially unchanged but the minor-lobe level is up and nulls are filled as shown in Fig. 5–45*b*. When the center source is off, the beamwidth is reduced but the minor-lobe level is higher, as indicated in Fig. 5–45*c*. If an edge source is off, the array is identical with a uniform array of four sources and, as shown in Fig. 5–45*d*, has a larger beamwidth with the minor-lobe level slightly higher than for the uniform source array of Fig. 5–45*a* (compare curves for $n = 4$ and 5 in Fig. 5–20).

If the amplitude distribution of the array is tapered (binomial or Dolph-Tchebyscheff) as in Fig. 5–27*b* and *c*, turning off a source at the edge will have less effect than in the uniform-amplitude case.

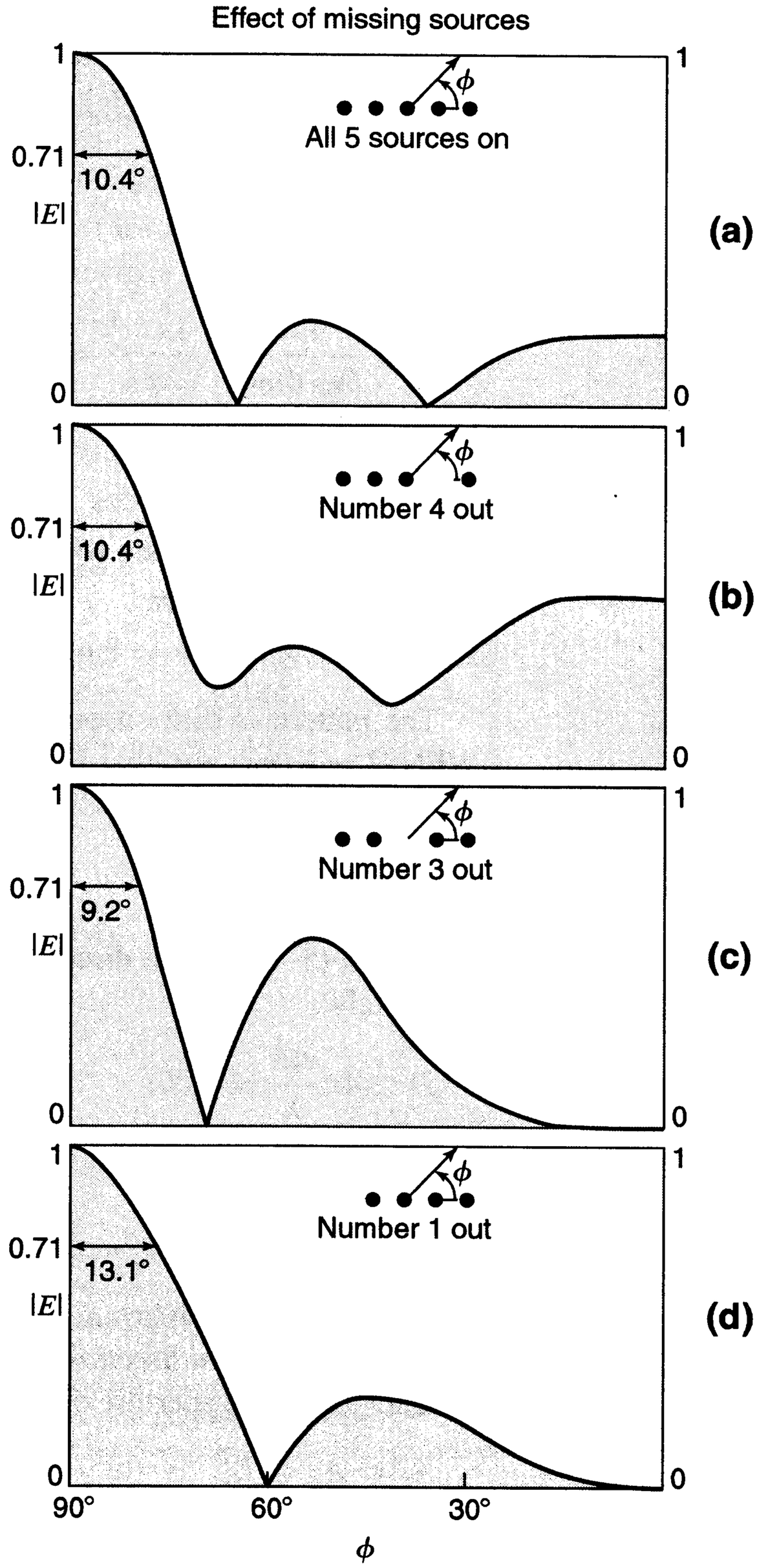


Figure 5–45
Field patterns of linear array of five isotropic point sources of equal amplitude and $\lambda/2$ spacing: (*a*) all five sources on, (*b*) one source (next to the edge) off, (*c*) one source (at the center) off, and (*d*) one source (at the edge) off.

In an array of a large number of sources, it is of interest to know what happens if one or more sources are turned off either intentionally or inadvertently. Also to reduce cost, a designer would like to know how many (and which) sources can be omitted without appreciably affecting the performance characteristics. It has been shown by Lo (1) and Maher (1) and Cheng that if sources are positioned with random instead of uniform spacing in a large array ($L \gg \lambda$) the number n of sources can be reduced without affecting the beamwidth appreciably. The gain, however, is proportional to n, and to keep the largest side lobes below a certain level, a minimum n is required.

5–17 BROADSIDE VERSUS END-FIRE ARRAYS. TURNS VERSUS DIPOLES AND THREE-DIMENSIONAL ARRAYS

（边射－端射阵与环圈－偶极子的三维阵）

Assuming that the half-power beamwidth (HPBW) is half the beamwidth between first nulls (BWFN) (an approximation), we have from (2–7–8) and (5–7–10) that the directivity of a *linear broadside array* is given approximately by

$$D \simeq \frac{4\pi}{\theta_{\text{HP}}\,\phi_{\text{HP}}} = \frac{4\pi nd}{2\pi\lambda} = 2L_\lambda \tag{1}$$

where

n = number of sources

d = spacing between sources, m

λ = wavelength, m

$L_\lambda = nd/\lambda = L/\lambda$ = length of array in wavelengths, dimensionless

The pattern is disk-shaped so $\theta = 360°$, or 2π radians. It is assumed that $\phi_{\text{HP}} = \text{BWFN}/2 = \lambda/nd$; also that $L_\lambda \gg 1$ so that $L \simeq nd$.

From (5–7–17), the directivity of an *ordinary end-fire array* is approximately

$$D \simeq 4\pi\frac{nd}{2\lambda} = 2\pi L_\lambda \tag{2}$$

while from (5–7–23) the directivity of an *end-fire array with increased directivity* is approximately

$$D \simeq 4\pi\frac{nd}{\lambda} = 4\pi L_\lambda \tag{3}$$

A general expression for directivity, as given by (2–22–9), is

$$D = \frac{4\pi A_e}{\lambda^2} \tag{4}$$

where A_e = effective aperture, m^2

For a *square unidirectional broadside array or aperture* with 100 percent aperture efficiency (uniform aperture distribution) we have

$$A_e = A_p = L^2 \qquad (\text{m}^2)$$

and

$$D = 4\pi L_\lambda^2 \tag{5}$$

For 100 percent aperture efficiency (uniform aperture distribution) the directivity of a *circular unidirectional broadside array or aperture* is

$$D = \pi^2 d_\lambda^2 \tag{6}$$

where $d_\lambda = d/\lambda$ = diameter of array or aperture in wavelengths, dimensionless

These directivity relations are summarized in Table 5–8 which also gives half-power beamwidths and numerical directivities for array lengths (or diameters) of 1 to 1000 as measured in wavelengths.

Table 5–8 Directivities and beamwidths of arrays and apertures†

Array (or aperture)‡	Directivity formula	Directivity for L_λ or d_λ equal to 1	10	100	1000	Half-power beamwidths
Linear broadside array of length L_λ	$2L_\lambda$	2	20	200	2000	$\frac{50.8°}{L_\lambda} \times 360°$
Ordinary end-fire array of length L_λ	$2\pi L_\lambda$	6.3	63	630	6300	$\frac{108°}{\sqrt{L_\lambda}}$
Increased-directivity end-fire array of length L_λ	$4\pi L_\lambda$	12.6	126	1,260	12,600	$\frac{52°}{\sqrt{L_\lambda}}$
Square broadside aperture with side length L_λ	$4\pi L_\lambda^2$	12.6	1260	126,000	1.26×10^7	$\frac{50.8°}{L_\lambda} \times \frac{50.8°}{L_\lambda}$
Circular broadside aperture with diameter d_λ	$\pi^2 d_\lambda^2$	9.9	990	99,000	9.9×10^6	$\frac{58°}{d_\lambda}$
Flat array (length L_λ) of ordinary end-fire antennas (length $L_{\lambda E}$)§	$\pi L_\lambda \sqrt{8L_{\lambda E}}$					
Same but square ($L_{\lambda E} = L_\lambda$)§	$\pi \sqrt{8L_\lambda^3}$	8.9	281	8,900	281,000	
Flat array (length L_λ) of increased-directivity end-fire antennas (length $L_{\lambda E'}$)§	$4\pi L_\lambda \sqrt{L_{\lambda E'}}$					
Same but square ($L_{\lambda E'} = L_\lambda$)§	$4\pi \sqrt{L_\lambda^3}$	12.6	398	12,600	398,000	

†The directivities for the arrays (broadside and end-fire) are approximate while for the apertures (square and circular) the directivities are exact. Note that if the directivity of the square or circular aperture is calculated using the approximation $41{,}000/\theta^\circ_{HP}\phi^\circ_{HP}$ the result is larger than the directivity given in the table. See discussion regarding this approximation in connection with (2–7–8) and (2–7–9).

Arrays or apertures are assumed to be large compared to the wavelength and to have uniform amplitude distribution. Specifically, the square and circular apertures are assumed to have 100 percent aperture efficiency. See text for other assumptions involved.

The directivities for the square and circular apertures are identical in form to the more general exact relation given by (4), which applies to an aperture of any shape.

‡L_λ = array length, wavelengths
d_λ = array diameter, wavelengths
$L_{\lambda E}$ = ordinary end-fire array length, wavelengths
$L_{\lambda E'}$ = increased-directivity end-fire array length, wavelengths
§Beam off edge, *not* perpendicular to flat side.

For array dimensions of the order of λ, the increased directivity end-fire array and square broadside array have comparable directivities. We note, however, that the end-fire directivity is proportional to the length L_λ while the directivity of the broadside square array is proportional to the *square* of the side length L_λ. Hence, for a high directivity an end-fire array must have a much greater dimension than a square broadside array. For example, a square array with 1000λ on a side has a directivity (see Table 5–8) of 12.6 million. To equal this directivity an increased-directivity end-fire array must be 1 million λ long and an ordinary end-fire array 2 million λ long. Even if all the sources or elements of such a long end-fire array could be fed with equal amplitude and in proper phase, the great length of the array is a severe disadvantage. Thus, it is apparent why broadside arrays or apertures are invariably used for high-directivity (high-gain) applications. The broadside aperture may consist of an array of $\lambda/2$ dipoles or it may be the aperture of a parabolic reflector antenna with single feed point or, as discussed next, a broadside array of intermediate-length end-fire antennas.

With uniform amplitude of all $\lambda/2$ dipole elements across a large aperture (and dipole spacing no more than $\lambda/2$), there is nothing to be gained by replacing each $\lambda/2$ dipole by a more directional end-fire antenna. However, if each end-fire antenna is used to *replace several* $\lambda/2$ dipoles, then some benefits may accrue as described in the following example involving a broadside array of end-fire antennas. An array of this type may be called a *volume* or *three-dimensional array*.

Figure 5–46*a* is an edge view of a 12×12 broadside array of 144 point sources with $\lambda/2$ spacing. The figure shows 12 elements as seen from one edge. Let each source consist of a $\lambda/2$ dipole element as suggested in partial broadside view in Fig. 5–46*e*. The array of Fig. 5–46*a* is bidirectional but if backed at $\lambda/4$ spacing by an identical array fed with equal amplitude currents and 90° phase difference, each pair of dipoles has a unidirectional pattern (as in Fig. 5–5) and the entire $12 \times 12 \times 2$ array is unidirectional as suggested in Fig. 5–46*b*. An alternative, simpler arrangement is to replace the second array with a conducting flat-sheet reflector or *ground plane* at an appropriate spacing, as in Fig. 5–46*c*.

A $\lambda/2$ dipole has a directivity $D = 1.64$, and taking the directivity of dipole with reflector as twice this value or 3.28 ($=2 \times 1.64$), the equivalent effective aperture is

$$A_e = \frac{D\lambda^2}{4\pi} = \frac{3.28}{4\pi}\lambda^2 = 0.26\lambda^2 \tag{7}$$

or approximately $\lambda^2/4$, as suggested in Fig. 5–46*e*.

The 12×12 array has 144 feed points. By substituting an end-fire array of appropriate directivity and effective aperture for a group of $\lambda/2$ dipole elements, the number of elements and feed points can be reduced. Thus, let us replace 9 $\lambda/2$ dipoles by a single end-fire array. The 9 $\lambda/2$ elements have an effective aperture of

$$A_e \simeq \left(\frac{3}{2}\right)^2 \lambda^2 = \frac{9}{4}\lambda^2 = 2.25\lambda^2 \tag{8}$$

For an increased-directivity end-fire array to provide an effective aperture of $2.25\lambda^2$ requires an array directivity

$$D(\text{required}) = \frac{4\pi A_e}{\lambda^2} = 4\pi \times 2.25 = 28.3 \tag{9}$$

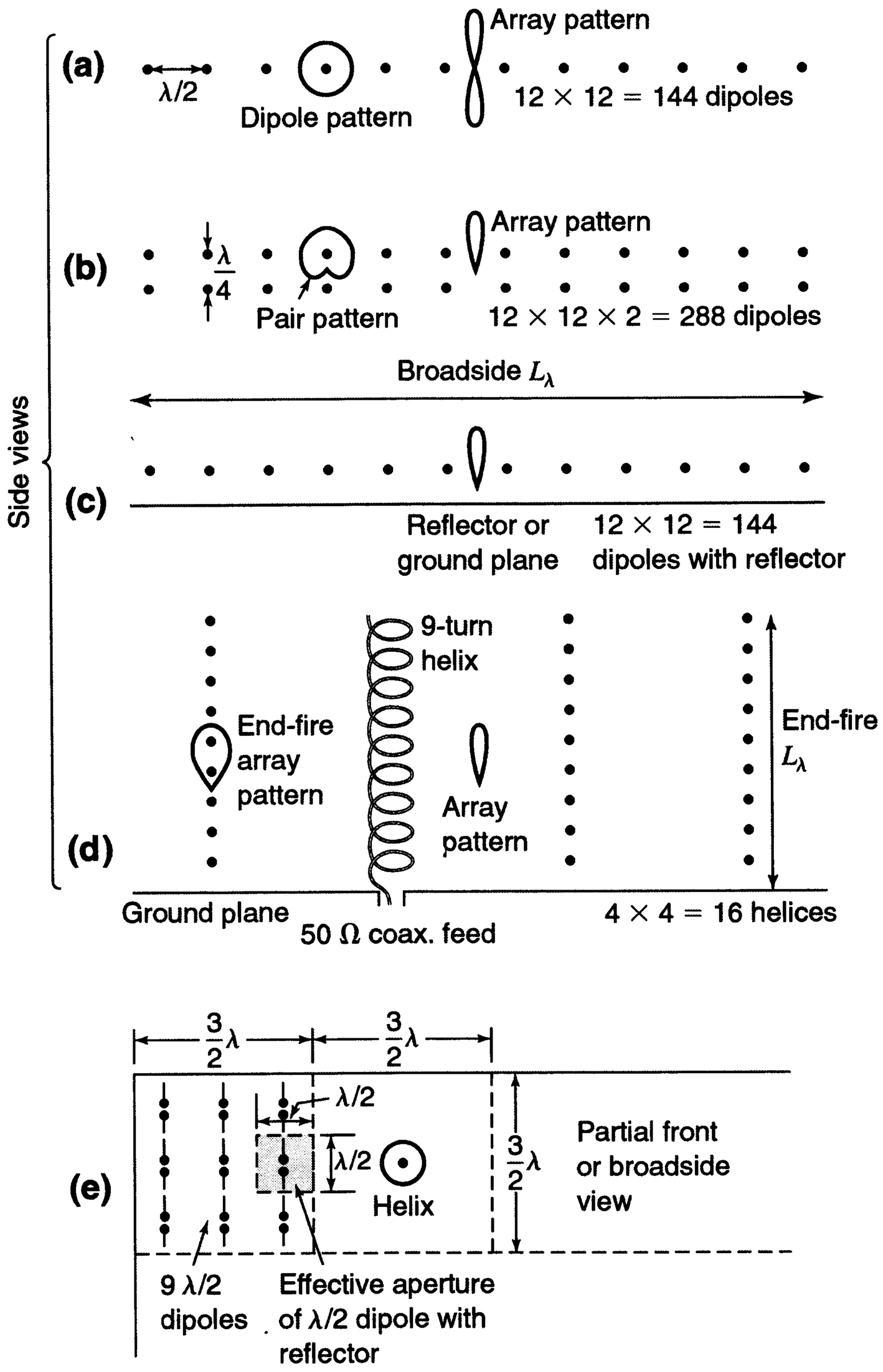


Figure 5–46
Equivalence of broadside array of $\lambda/2$ dipoles and three-dimensional broadside-end-fire array. Parts *a*, *b*, *c*, and *d* are edge or side views while *e* is a front or broadside view.

The required length of the increased-directivity end-fire array is then

$$L(\text{required}) = \frac{D(\text{required})\lambda}{4\pi} = \frac{4\pi \times 2.25\lambda}{4\pi} = 2.25\lambda \tag{10}$$

An effective end-fire array which meets the above requirements is a nine-turn monofilar axial-mode helical antenna (see Chap. 8) with λ/π diameter and $\lambda/4$ spacing between turns making the length $L = 9 \times 0.25\lambda = 2.25\lambda$. This end-fire antenna has the remarkable properties of increased directivity, wide bandwidth (over 2 to 1), and very small mutual coupling between adjacent helices. It is fed from one end through the ground plane by a coaxial transmission line (as in Fig. 5–27*d*) which may be 50 Ω or other convenient impedance.

With each nine-turn helix replacing 9 $\lambda/2$ dipoles the broadside array of 144 $\lambda/2$ dipoles is replaced by 16 (=144/9) helices, which reduces the complexity of the feed system and provides an array readily capable of wide bandwidth operation.

The effective aperture of the array is $6\lambda \times 6\lambda = 36\lambda^2$ for a directivity

$$D = 4\pi \times 36 = 452 \text{ (or 26.6 dBi)} \tag{11}$$

As a further step, the 16 nine-turn helices could be replaced by four 36-source end-fire arrays (36-turn helices), and as a final step, the four helices by a single 144-source end-fire array (l44-turn helix) 36λ long. Such a long helix is not practical but, even if it were, its great length is a disadvantage as compared to a more compact array with a number of shorter helices.

In the above example each turn of a helix (as an end-fire source) has a directivity equal to a $\lambda/2$ dipole with reflector (as a broadside source), but polarizations differ (helix circularly polarized and dipole linearly polarized). In summary, we considered the following cases:

1. 12×12 array of reflector-backed $\lambda/2$ dipoles: $12 \times 12 = 144$ sources
2. 4×4 array of 9-turn helical end-fire antennas: $4 \times 4 \times 9 = 144$ sources
3. 2×2 array of 36-turn helical end-fire antennas: $2 \times 2 \times 36 = 144$ sources
4. Single 144-turn helical end-fire antenna: $1 \times 144 = 144$ sources

Thus, for constant directivity and effective aperture, the *number of sources is a constant* whether the sources are arranged in a flat broadside array or in three-dimensional broadside-end-fire configurations, all sources having uniform amplitude. Although the constant directivity and effective aperture in the above example may not apply fully for all three-dimensional arrays, the example illustrates the principle that for a given number of sources, various configurations may produce (ideally) similar, if not identical, directivities and effective apertures.

Another broadside-end-fire combination is a *flat (or planar) array* consisting of a linear array with only a single row of end-fire antennas (helices) as in Fig. 5–46*d* (no other arrays stacked perpendicular to the page). The beamwidth in one plane is determined by the broadside length L_λ and the beamwidth in the other plane by the end-fire length $L_{\lambda E}$ (for ordinary end-fire) or length $L_{\lambda E'}$ (for increased directivity).

The directivity for the ordinary end-fire case is

$$D(\text{ordinary}) = \pi L_\lambda \sqrt{8 L_{\lambda E}} \tag{12}$$

and if $L_\lambda = L_{\lambda E}$ (square flat array),

$$D(\text{ordinary}) = \pi \sqrt{8 L_\lambda^3} \tag{13}$$

The directivity for the increased directivity case is

$$D(\text{increased directivity}) = 4\pi L_\lambda \sqrt{L_{\lambda E'}} \tag{14}$$

and if $L_\lambda = L_{\lambda E'}$ (square flat array),

$$D(\text{increased directivity}) = 4\pi \sqrt{L_\lambda^3} \tag{15}$$

These relations are summarized in Table 5–8.

An early application of end-fire antennas in a large three-dimensional array, which Kraus designed and built in 1951, is shown in Fig. 8–4. It had 96 helices of 11 turns each in a 4×24 configuration. Equivalent to an array of 1056 $(=11 \times 96)\lambda/2$ dipoles with reflectors, it had a wide bandwidth (over 2 to 1) and was simple to feed.

5–18 DIRECTIONS OF MAXIMA FOR ARRAYS OF *n* ISOTROPIC POINT SOURCES OF EQUAL AMPLITUDE AND SPACING

[*n*个各向同性点源的等幅等间距阵的最（极）大方向]

Let us now proceed to a discussion of the methods for locating the positions of the pattern maxima. The major-lobe maximum usually occurs when $\psi = 0$. This is the case for the broadside or ordinary end-fire array. The main lobes of the broadside array are then at $\phi = 90°$ and 270°, while for the ordinary end-fire array the main lobe is at 0° or 180° or both. For the end-fire array with increased directivity the main-lobe maximum occurs at a value of $\psi = \pm\pi/n$ with the main lobe at 0° or 180°. Referring to Fig. 5–24*a* the main-lobe maximum (first maximum) for this case occurs at the first maximum of the numerator of (5–6–8).

The maxima of the minor lobes are situated between the first- and higher-order nulls. It has been pointed out by Schelkunoff that these maxima occur approximately whenever the numerator of (8) is a maximum, i.e., when

$$\sin\frac{n\psi}{2} = 1 \tag{1}$$

Referring to Fig. 5–47, we note that the numerator of (5–6–8) varies as a function of ψ more rapidly than the denominator $\sin(\psi/2)$. This is especially true when n is large. Thus, although the nulls occur exactly where $\sin(n\psi/2) = 0$, the maxima occur approximately where $\sin(n\psi/2) = 1$. This condition requires that

$$\frac{n\psi}{2} = \pm(2K+1)\frac{\pi}{2} \tag{2}$$

where $K = 1, 2, 3, \ldots$

Substituting the value of ψ from (2) into (5–6–2) gives

$$d_r \cos\phi_m + \delta = \frac{\pm(2K+1)\pi}{n} \tag{3}$$

Therefore

$$\phi_m \simeq \arccos\left\{\left[\frac{\pm(2K+1)\pi}{n} - \delta\right]\frac{1}{d_r}\right\} \tag{4}$$

where ϕ_m = direction of the minor-lobe maxima

For a *broadside array,* $\delta = 0$ so that (4) becomes

$$\phi_m \simeq \arccos\frac{\pm(2K+1)\lambda}{2nd} \tag{5}$$

As an example, the field pattern of Fig. 5–21 ($n = 4$, $d = \lambda/2$, $\delta = 0$) has the minor-lobe maxima at

$$\phi_m \simeq \arccos\frac{\pm(2K+1)}{4} \tag{6}$$

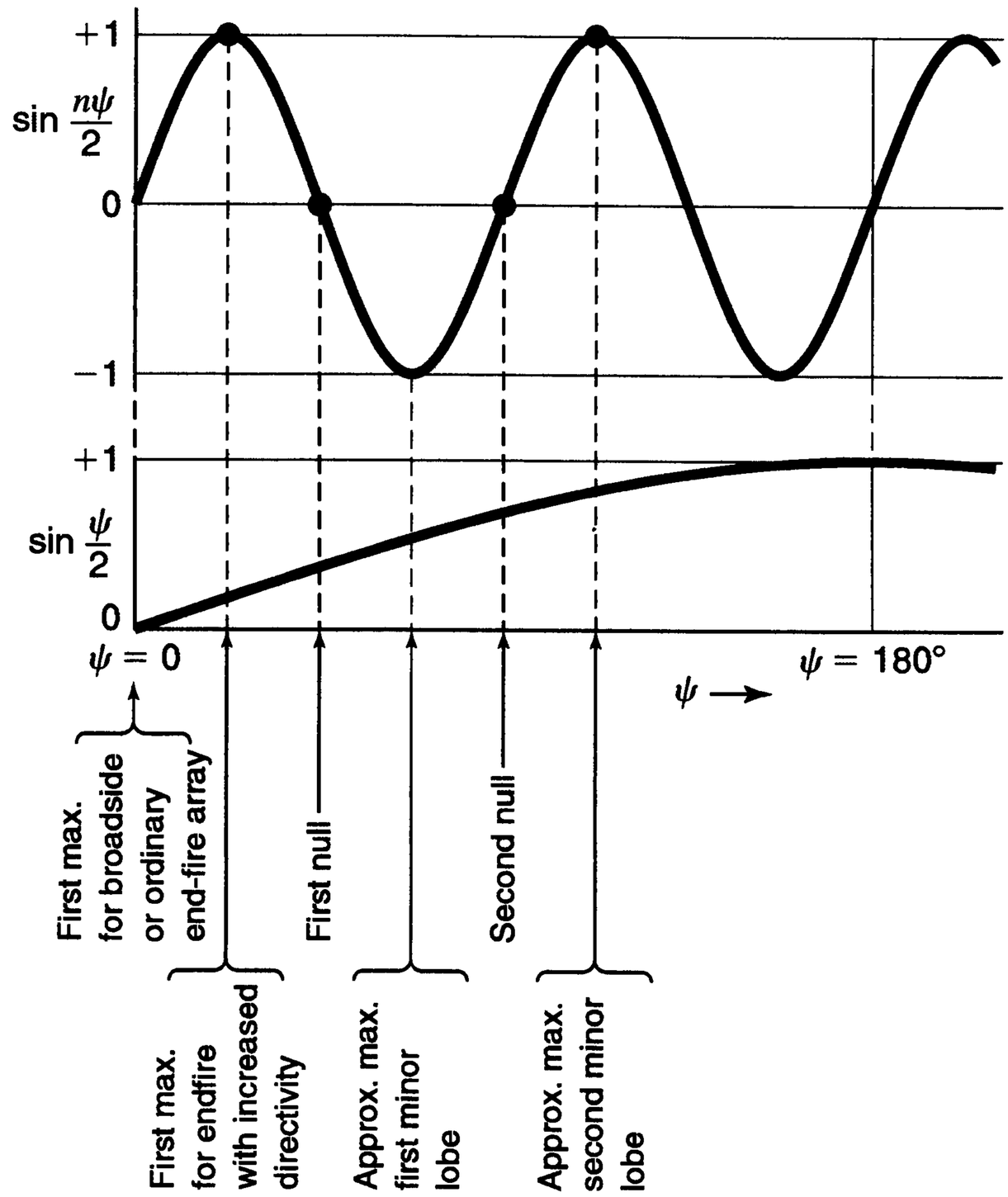


Figure 5–47
Graphs of the numerator ($\sin n\psi/2$) and denominator ($\sin \psi/2$) of the array factor as functions of ψ, showing the values of ψ corresponding to maxima and nulls of a field pattern for the case $n = 8$.

For $K = 1, \phi_m = \pm 41.4°$ and $\pm 138.6°$. These are the approximate directions for the maxima of the four minor lobes of this pattern.

For an *ordinary end-fire array,* $\delta = -d_r$ so that (4) becomes

$$\phi_m \simeq \arccos\left[\frac{\pm(2K+1)\lambda}{2nd} + 1\right] \tag{7}$$

while, for an *end-fire array with increased directivity,* $\delta = -(d_r + \pi/n)$ and

$$\phi_m \simeq \arccos\left\{\frac{\lambda}{2nd}[1 \pm (2K+1)] + 1\right\} \tag{8}$$

The above formulas for the approximate location of the minor-lobe maxima are listed in Table 5–9 ($K = 1$ for first minor lobe, $K = 2$ for second minor lobe, etc.).

The amplitudes of the field at the minor-lobe maxima are also of interest. It has been shown by Schelkunoff that since the numerator of (5–6–9) is approximately unity at the maximum of a minor lobe, the relative amplitude of a minor-lobe maximum E_{ML} is given by

$$E_{ML} \simeq \frac{1}{n \sin(\psi/2)} \tag{9}$$

Introducing the value of ψ from (2) into (9) yields

$$E_{ML} \simeq \frac{1}{n \sin[(2K+1)\pi/2n]} \tag{10}$$

Table 5–9 Directions of minor-lobe maxima for linear arrays of n isotropic point sources of equal amplitude and spacing

Type of array	Directions of minor-lobe maxima
General	$\phi_m \simeq \arccos\left\{\left[\frac{\pm(2K+1)\pi}{n} - \delta\right]\frac{1}{d_r}\right\}$
Broadside	$\phi_m \simeq \arccos\frac{\pm(2K+1)\lambda}{2nd}$
Ordinary end-fire	$\phi_m \simeq \arccos\left[\frac{\pm(2K+1)\lambda}{2nd} + 1\right]$
End-fire with increased directivity (Hansen and Woodyard)	$\phi_m \simeq \arccos\left\{\frac{\lambda}{2nd}[1 \pm (2K+1)] + 1\right\}$
where $K = 1, 2, 3\ldots$	

When $n \gg K$, that is, for the first few minor lobes of an array of a large number of sources, we have the further approximation

$$E_{\mathrm{ML}} \simeq \frac{2}{(2K+1)\pi} \tag{11}$$

Thus, for arrays of a large number of sources the relative amplitude of the first few minor lobes is given by (11) for $K = 1, 2, 3$, etc. In a broadside or ordinary end-fire array, the major-lobe maximum is unity so that the relative amplitudes of the maximum and first five minor lobes for arrays of these types and many sources are 1, 0.21, 0.13, 0.09, 0.07, and 0.06. From the curve for $n = 20$ in Fig. 5–20 we have the corresponding relative amplitudes given by 1, 0.22, 0.13, 0.09, 0.07, and 0.06. For an end-fire array with increased directivity the maximum for $\phi = 0$ and $n = 20$ occurs at $\psi = \pi/20 = 9°$. At this value of ψ the array factor is 0.63. Putting the maximum equal to unity then makes the relative amplitudes 1, 0.35, 0.21, 0.14, 0.11, and 0.09. It is interesting to note in (10) that the maximum amplitude of the smallest minor lobe occurs for $2K + 1 = n$. Then

$$\sin\left[\frac{(2K+1)\pi}{2n}\right] = 1 \tag{12}$$

and

$$E_{\mathrm{ML}} \simeq \frac{1}{n} \tag{13}$$

The condition $2K + 1 = n$ is exactly fulfilled when n is odd for the minor-lobe maximum at $\psi = 180°$ (see Fig. 5–26). When n is even, the condition is approximately fulfilled by the minor lobes nearest $\psi = 180°$. Thus, the maximum amplitude of the smallest minor lobe of the field pattern of any array of n isotropic point sources of equal amplitude and spacing will never be less than $1/n$ of the major-lobe maximum. An exception to this is where the range of ψ ends after a null in the array factor has been passed but before the next maximum has been reached. In this case the maximum of the smallest minor lobe may be arbitrarily small.

REFERENCES（参考文献）

Born, Max (1): *Optik,* Springer-Verlag, 1933.

Brown, G. H. (1): "Directional Antennas," *Proc. IRE,* **25,** January 1937.

Dolph, C. L. (1): "A Current Distribution for Broadside Arrays Which Optimizes the Relationship between Beam Width and Side-Lobe Level," *Proc. IRE,* **34,** no. 6, 335–348, June 1946.

Hansen, W. W. (1), and J. R. Woodyard: "A New Principle in Directional Antenna Design," *Proc. IRE,* **26,** 333–345, March 1938.

Huygens, C. (1): *Traité de la Lumière,* Leyden, 1690.

Kraus, J. D. (1): *Radio Astronomy,* 2d ed., Cygnus-Quasar Books, 1986.

Lo, Y. T. (1): "A Probabilistic Approach to the Design of Large Antenna Arrays," *IEEE Trans. Ants. Prop.,* **AP-11,** 95–96, 1963.

Maher, T. M. (1), and D. K. Cheng: "Random Removal of Radiators from Large Linear Arrays," *IEEE Trans. Ants. Prop.,* **AP-11,** 106–112, 1963.

Riblet, H. J. (1): *Proc. IRE,* no. 5, 489–492, May 1947.

Schelkunoff, S. A. (1): *Electromagnetic Waves,* p. 342, Van Nostrand, New York, 1943.

Schelkunoff, S. A. (2): *Electromagnetic Waves,* p. 343, Van Nostrand, New York, 1943.

Schelkunoff, S. A. (3): "A Mathematical Theory of Arrays," *Bell System Tech. J.,* **22,** 80–107, January 1943*c*.

Smith, C. E. (1): *Directional Antennas,* Cleveland Institute of Radio Electronics, Cleveland, Ohio, 1946.

Sommerfeld, Arnold (1): "Theorie der Beugung," in Frank and von Mises (eds.), *Differential und Integralgleichungen der Mechanik und Physik,* Vieweg, 1935.

Stone, John Stone (1): U.S. Patents 1,643,323 and 1,715,433.

Stratton, J. A. (1): *Electromagnetic Theory,* p. 451, McGraw-Hill, New York, 1941.

Terman, F. E. (1): *Radio Engineers' Handbook,* p. 804, McGraw-Hill, New York, 1943.

Wolff, Irving (1): "Determination of the Radiating System Which Will Produce a Specified Directional Characteristic," *Proc. IRE,* **25,** 630–643, May 1937.

PROBLEMS（习题）

5-8-1（三个不等幅源） **Three unequal sources.** Three isotropic in-line sources have $\lambda/4$ spacing. The middle source has three times the current of the end sources. If the phase of the middle source is 0°, the phase of one end source +90°, and the phase of the other end source −90°, make a graph of the normalized field pattern.

5-8-2（长边射阵） **Long broadside array.** Show that the HPBW of a long uniform broadside array is given (without approximation) by $50.8°/L_\lambda$, where $L_\lambda = L/\lambda =$ length of array in wavelengths.

5-8-3（两源阵的相位中心） **Phase center of two-source array.** An array consists of two isotropic point sources, one at the origin and one at a distance of $\lambda/2$ in the x direction. If the source at the origin has twice the amplitude (field) of the other source, find the position of the phase center of the array.

***5-8-4**（四源边射阵） **Four sources in broadside array.** (*a*) Derive an expression for $E(\phi)$ for a linear in-phase broadside array of four identical isotropic point sources. Take $\phi = 0$ in the broadside direction. The spacing between sources is $5\lambda/8$.

(*b*) Plot, approximately, the normalized field pattern ($0° \leq \phi \leq 360°$).

(*c*) Repeat parts (*a*) and (*b*) with the changed condition that the amplitudes of the four sources are proportional to the coefficients of the binomial series for $(a + b)^{n-1}$.

5–8–5 (二项式分布)
Binomial distribution. Use the principle of pattern multiplication to show that a linear array with binomial amplitude distribution has pattern with no minor lobes.

5–8–6 (二项式阵波瓣图)
Binomial array pattern. (*a*) Compute and plot the field pattern of a seven-source binomial array of isotropic sources, (*b*) find the HPBW of the array, and (*c*) estimate the directivity of this array using (2–7–9).

5–8–7 (杂散因子和定向增益)
Stray factor and directive gain. The ratio of the main beam solid angle Ω_M to (total) beam solid angle Ω_A is called the *main beam efficiency*. The ratio of the minor-lobe solid angle Ω_m to the (total) beam solid angle Ω_A is called the *stray factor*. It follows that $\Omega_M/\Omega_A + \Omega_m/\Omega_A = 1$. Show that the *average directivity gain* over the minor lobes of a highly directive antenna is nearly equal to the stray factor. The directive gain is equal to the directivity multiplied by the normalized power pattern $[=DP_n(\theta, \phi)]$, making it a function of angle with the maximum value equal to D.

5–9–1 (切比雪夫 $T_3(x)$和 $T_6(x)$)
Tchebyscheff $T_3(x)$ and $T_6(x)$. (*a*) Calculate and plot $\cos\theta$ as x and $\cos 3\theta$ as y, for $-1 \le x \le +1$. Compare with the curve for $T_3(x)$.

(*b*) Calculate and plot $\cos\theta$ as x and $\cos 6\theta$ as y, for $-1 \le x \le +1$. Compare with the curve for $T_6(x)$.

***5–9–2** (三源阵)
Three-source array. The center source of a three-source array has a (current) amplitude of unity. For a side-lobe level 0.1 of the main-lobe maximum field, find the Dolph-Tchebyscheff value of the amplitude of the end sources. The source spacing $d = \lambda/2$.

***5–9–3** (五源道尔夫–切比雪夫分布)
Five-source Dolph-Tchebyscheff (D-T) distribution. (*a*) Find the Dolph-Tchebyscheff current distribution for the minimum beamwidth of a linear in-phase broadside array of five isotropic point sources. The spacing between sources is $\lambda/2$ and the side-lobe level is to be 20 dB down. Take $\phi = 0$ in the broadside direction.

(*b*) Locate the nulls and maxima of the minor lobes.

(*c*) Plot, approximately, the normalized field pattern ($0° \le \phi \le 360°$).

(*d*) What is the half-power beamwidth?

5–9–4 (八源道尔夫–切比雪夫分布)
Eight source D-T distribution. (*a*) Find the Dolph-Tchebyscheff current distribution for the minimum beamwidth of a linear in-phase broadside array of eight isotropic sources. The spacing between the elements is $\lambda/4$ and the side-lobe level is to be 40 dB down. Take $\phi = 0$ in the broadside direction.

(*b*) Locate the nulls and the maxima of the minor lobes.

(*c*) Plot, approximately, the normalized field pattern ($0° \le \phi \le 360°$).

(*d*) What is the half-power beamwidth?

***5–9–5** (道尔夫–切比雪夫六源阵)
D-T Six-Source Array. Calculate the Dolph-Tchebyscheff distribution of a six-source broadside array for $R = 5$, 7 and 10. Explain the variation.

5–9–6
Beamwidth and gain of uniform arrays. Use ARRAYPATGAIN to obtain (*a*) the HPBW and (*b*) the gain in dBi of uniform linear arrays of 4, 16, and 48 isotropic point sources spaced $\lambda/2$ apart for a SLL of -15 dB. Six answers are required.

***5–9–7** (五源阵)
Five-source array. (*a*) What is an expression for the field pattern of an array of five identical isotropic point sources arranged in line and spaced a distance d ($\le\lambda/2$) apart? The phase lead of source 2 over 1, 3 over 2, etc., is δ.

(*b*) What value should δ have to make the array a broadside type? For this broadside case, what are the relative current magnitudes of the sources for:

1. Maximum directivity?
2. No side lobes?
3. Side lobes equal in magnitude to "main" lobe?

***5–12–1** (矩形电流片) **Rectangular current sheet.** Calculate and plot the patterns in both planes perpendicular to rectangular sheet carrying a current of uniform density and everywhere of the same direction and phase if the sheet measures 10 by 20λ. What is the approximate directivity?

5–12–2 (连续阵，可变相速) **Continuous array. Variable phase velocity.** Consider that the array of discrete sources in Fig. 5–18 is replaced by a continuous array of length L and assume that it is energized like the array of Prob. 5–6–8. Show that the far field for the general case of any phase lag δ' per unit distance along the continuous array is given by (5–12–18) where $\psi' = L_r \cos\phi - \delta' L = L_r[\cos\phi - (1/p)]$, where $p = v/c$ as in Prob. 5–6–8. Show also that in the four cases considered in Prob. 5–6–8 the p values are the same except for the increased-directivity end-fire case where $p = 1/[1 + (1/2L_\lambda)]$.

***5–17–1** (24 源端射阵) **24-source end-fire array.** A uniform linear array has 24 isotropic point sources with a spacing of $\lambda/2$. If the phase difference $\delta = -\pi/2$ (ordinary end-fire condition), calculate exactly (*a*) the HPBW, (*b*) the first side-lobe level, (*c*) the beam solid angle, (*d*) the beam efficiency, (*e*) the directivity, and (*f*) the effective aperture.

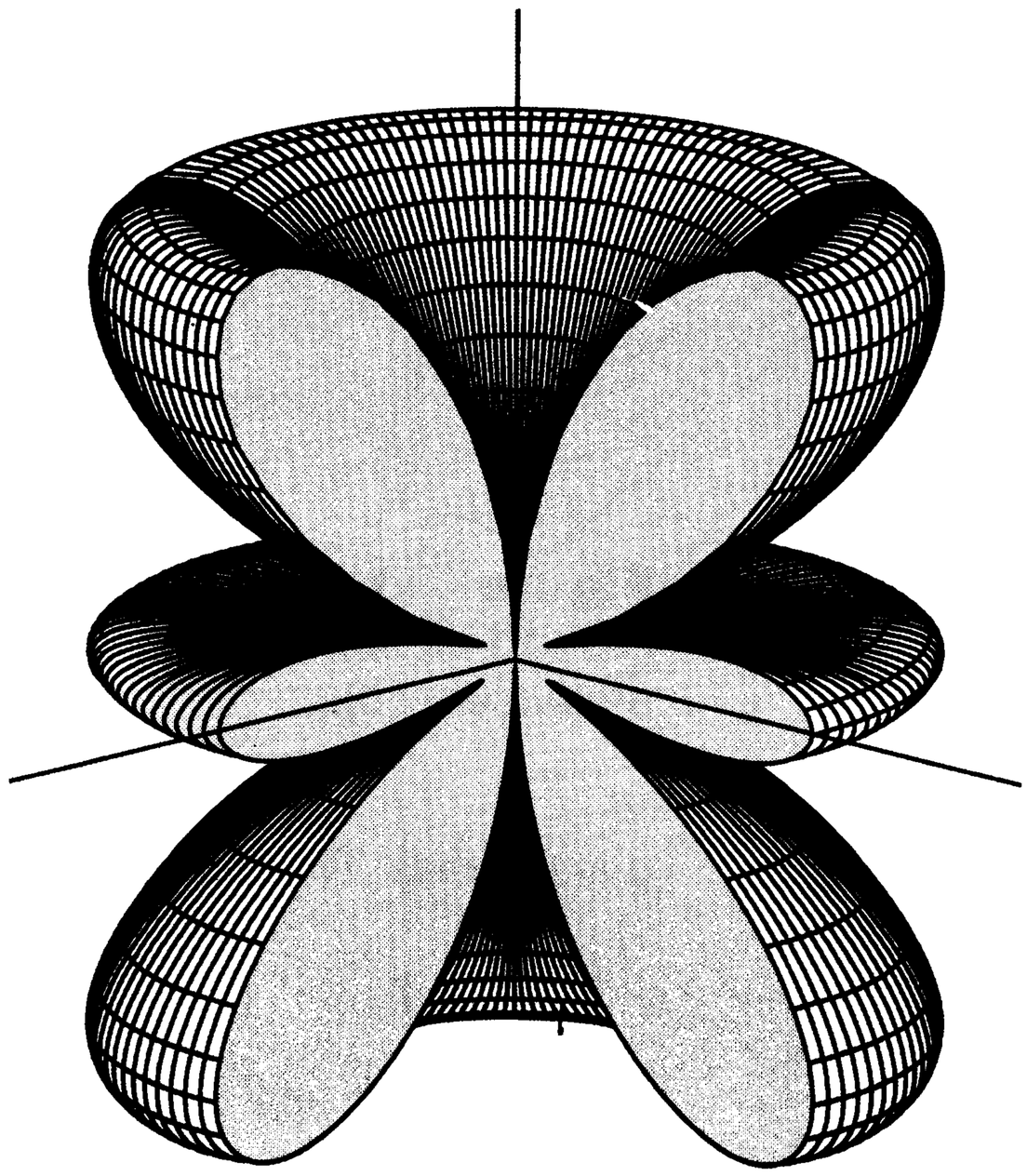
***5–18–1** (两个同相源) **Two sources in phase.** Two isotropic point sources of equal amplitude and same phase are spaced 2λ apart. (*a*) Plot a graph of the field pattern. (*b*) Tabulate the angles for maxima and nulls.

5–18–2 (两个反相源) **Two sources in opposite phase.** Two isotropic sources of equal amplitude and opposite phase have 1.5λ spacing. Find the angles for all maxima and nulls.

5–18–3 (均匀阵的波束宽度与增益) **Beamwidth and gain of uniform arrays.** Use ARRAYPATGAIN to obtain (*a*) the HPBW and (*b*) the gain in dBi of uniform linear arrays of 4, 16, and 48 isotropic point sources spaced $\lambda/2$ for a SLL of −15 dB. Six answers are required. For computer programs, see Appendix C.

CHAPTER 6

The Electric Dipole and Thin Linear Antennas
电偶极子和细直天线



Topics in this chapter include:

- Fields of short dipole（短电偶极子的场）
- Radiation resistance of short dipole（短电偶极子的辐射电阻）
- Thin linear antenna: $\lambda/2$, λ, and $3\lambda/2$（细直天线：$\lambda/2$，λ，$3\lambda/2$）
- Radiation resistance of $\lambda/2$ dipole（$\lambda/2$ 偶极子的辐射电阻）
- Arrays of two $\lambda/2$ dipoles（两个 $\lambda/2$ 偶极子的阵列）
- Broadside, end-fire and close-spaced arrays（边射、端射和密排阵）
- Radiation resistance at a point which is not a current maximum（非电流最大点的辐射电阻）
- Traveling wave antennas（行波天线）

6–1 THE SHORT ELECTRIC DIPOLE（短电偶极子）

Since any linear antenna may be considered as consisting of a large number of very short conductors connected in series, it is of interest to examine first the radiation properties of short conductors. From a knowledge of the properties of short conductors, we can then proceed to a study of long linear conductors such as are commonly employed in practice.

A short linear conductor is often called a *short dipole*. In the following discussion, a short dipole is always of finite length even though it may be very short. If the dipole is vanishingly short, it is an infinitesimal dipole.

Let us consider a short dipole such as shown in Fig. 6–1*a*. The length L is very short compared to the wavelength ($L \ll \lambda$). Plates at the ends of the dipole provide capacitive loading. The short length and the presence of these plates result in a uniform current I along the entire length L of the dipole. The dipole may be energized by a balanced transmission line, as shown. It is assumed that the transmission line does not radiate and, therefore, its presence will be disregarded. Radiation from the end plates is also considered to be negligible. The diameter d of the dipole is small compared to its length ($d \ll L$). Thus, for purposes of analysis we may consider that the short dipole appears as in Fig. 6–1*b*. Here it consists simply of a thin conductor of length L with a uniform current I and point charges

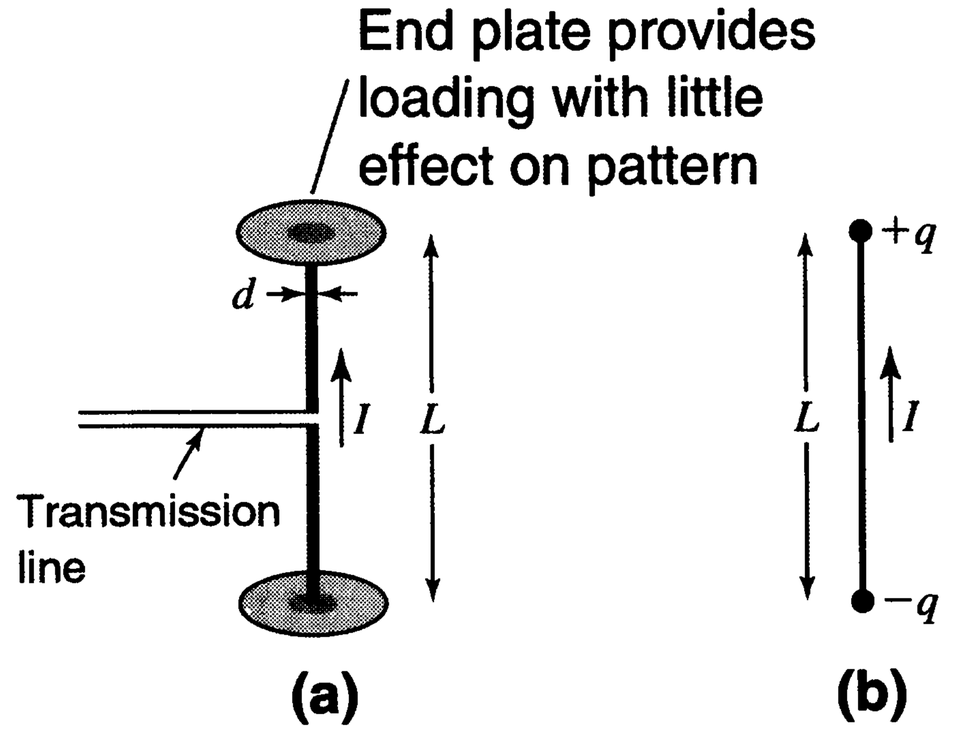


Figure 6–1
A short dipole antenna (*a*) and its equivalent (*b*).

q at the ends. The current and charge are related by

$$\frac{dq}{dt} = I \tag{1}$$

6–2 THE FIELDS OF A SHORT DIPOLE（短电偶极子的场）

Let us now proceed to find the fields everywhere around a short dipole. Let the dipole of length L be placed coincident with the z axis and with its center at the origin as in Fig. 6–2. The relation of the electric field components, E_r, E_θ and E_ϕ, is then as shown. It is assumed that the medium surrounding the dipole is air or vacuum.

In dealing with antennas or radiating systems, the propagation time is a matter of great importance. Thus, if a current is flowing in the short dipole of Fig. 6–3, the effect of the current is not felt instantaneously at the point P, but only after an interval equal to the time required for the disturbance to propagate over the distance r. We have already recognized this in Chap. 5 in connection with the pattern of arrays of point sources, but here we are more explicit and describe it as a *retardation* effect.

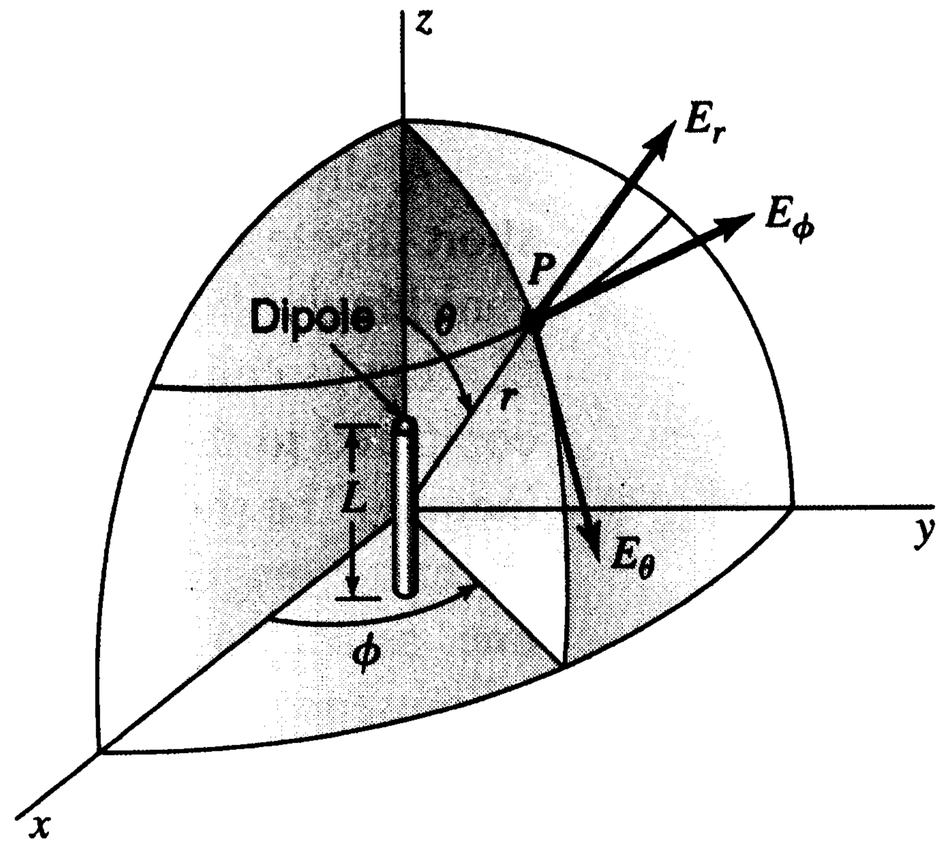


Figure 6–2
Relation of dipole to coordinates.

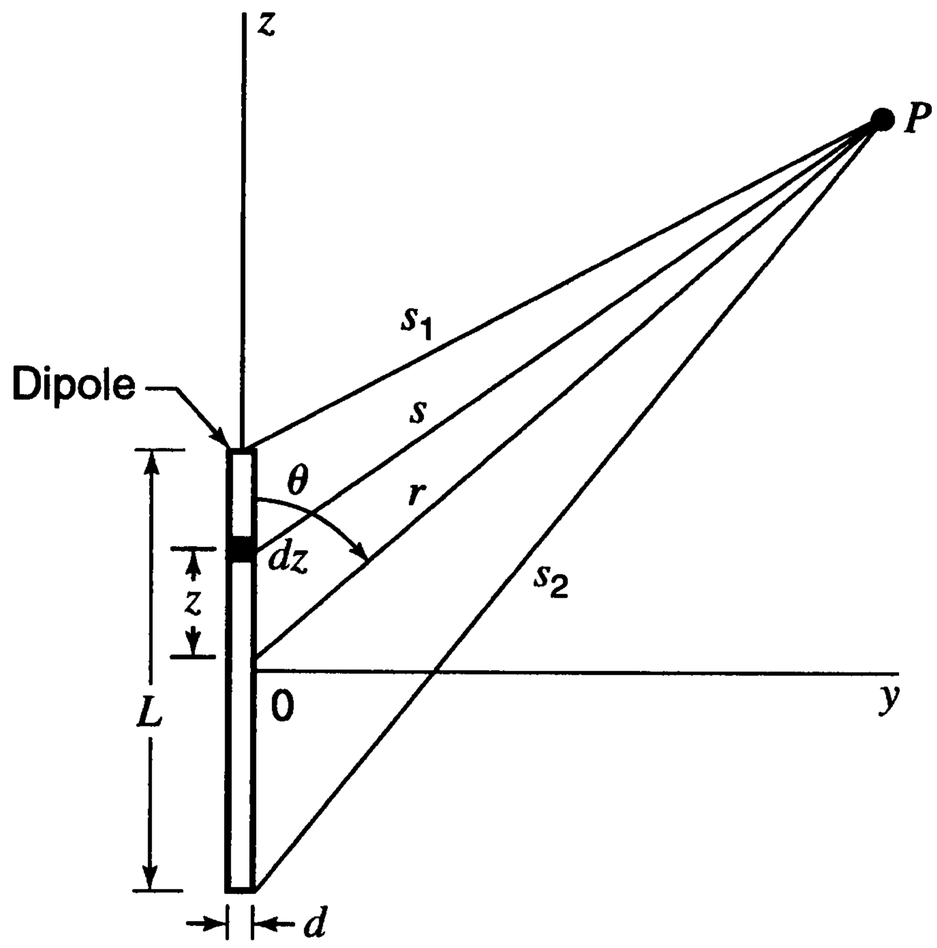


Figure 6–3a
Geometry for short dipole.

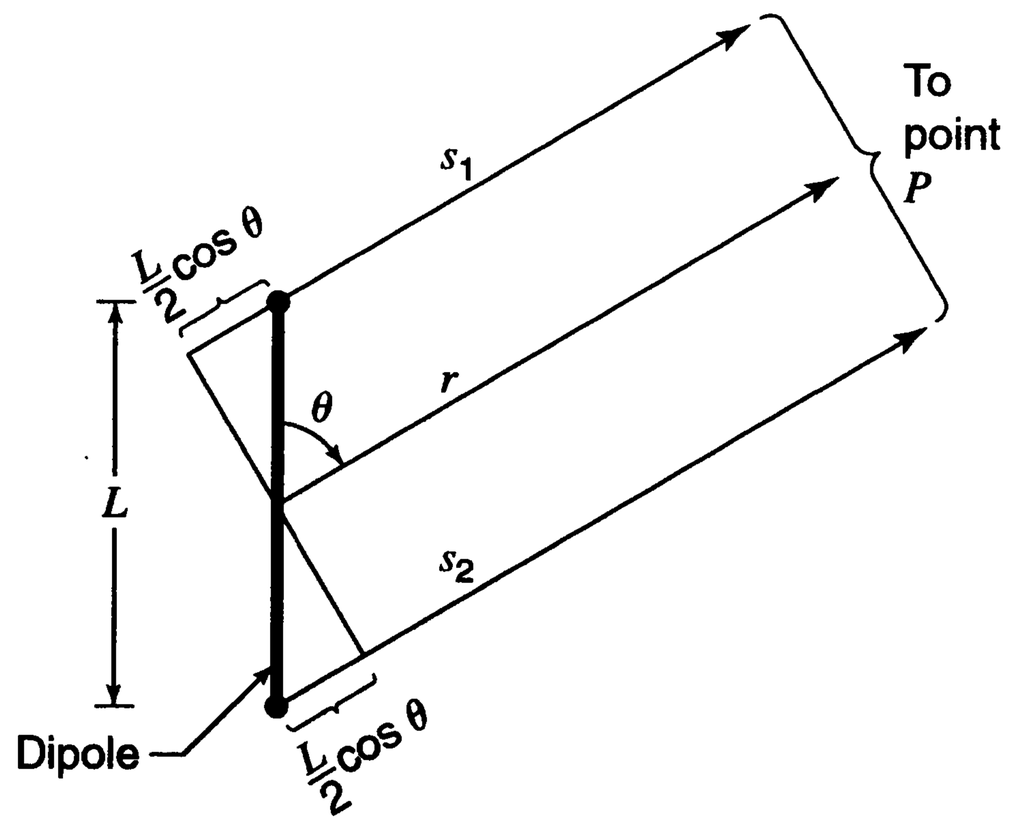


Figure 6–3b
Relations for short dipole when $r \gg L$.

Accordingly, instead of writing the current I as[1]

$$I = I_0 e^{j\omega t} \tag{1}$$

which implies instantaneous propagation of the effect of the current, we introduce the propagation (or retardation) time as done by Lorentz and write

$$[I] = I_0 e^{j\omega[t-(r/c)]} \tag{2}$$

where $[I]$ is called the *retarded current*. Specifically, the retardation time r/c results in a *phase retardation* $\omega r/c = 2\pi f r/c$ radians $= 360°\ fr/c = 360°\ t/T$, where $T = 1/f =$ time of one period or cycle (seconds) and $f =$ frequency (hertz, Hz = cycles per second). The brackets may be added as in (2) to indicate explicitly that the effect of the current is retarded.

---

[1]It is assumed that we take either the real ($\cos \omega t$) or imaginary ($\sin \omega t$) part of $e^{j\omega t}$.

Equation (2) is a statement of the fact that the disturbance at a time t and at a distance r from a current element is caused by a current $[I]$ that occurred at an earlier time $t - r/c$. The time difference r/c is the interval required for the disturbance to travel the distance r, where c is the velocity of light ($= 300$ Mm s^{-1}).

Electric and magnetic fields can be expressed in terms of vector and scalar potentials. Since we will be interested not only in the fields near the dipole but also at distances which are large compared to the wavelength, we must use *retarded potentials,* i.e., expressions involving $t - r/c$. For a dipole located as in Fig. 6–2 or Fig. 6–3a, the retarded vector potential of the electric current has only one component, namely, A_z. Its value is

$$A_z = \frac{\mu_0}{4\pi} \int_{-L/2}^{L/2} \frac{[I]}{s}\, dz \tag{3}$$

where $[I]$ is the retarded current given by

$$[I] = I_0 e^{j\omega[t-(s/c)]} \tag{3a}$$

In (3) and (3a),

z = distance to a point on the conductor

I_0 = peak value in time of current (uniform along dipole)

μ_0 = permeability of free space $= 4\pi \times 10^{-7}$ H m^{-1}

If the distance from the dipole is large compared to its length ($r \gg L$) and if the wavelength is large compared to the length ($\lambda \gg L$), we can put $s = r$ and neglect the phase differences of the field contributions from different parts of the wire. The integrand in (3) can then be regarded as a constant, so that (3) becomes

$$A_z = \frac{\mu_0 L I_0 e^{j\omega[t-(r/c)]}}{4\pi r} \tag{4}$$

The retarded scalar potential V of a charge distribution is

$$V = \frac{1}{4\pi\varepsilon_0} \int_V \frac{[\rho]}{s}\, d\tau \tag{5}$$

where $[\rho]$ is the retarded charge density given by

$$[\rho] = \rho_0 e^{j\omega[t-(s/c)]} \tag{6}$$

and $d\tau$ = infinitesimal volume element

ε_0 = permittivity or dielectric constant of free space $= 8.85 \times 10^{-12}$ F m^{-1}

Since the region of charge in the case of the dipole being considered is confined to the points at the ends as in Fig. 6–1b, (5) reduces to

$$V = \frac{1}{4\pi\varepsilon_0} \left\{ \frac{[q]}{s_1} - \frac{[q]}{s_2} \right\}. \tag{7}$$

From (6–1–1) and (3a),

$$[q] = \int [I]\, dt = I_0 \int e^{j\omega[t-(s/c)]}\, dt = \frac{[I]}{j\omega} \tag{8}$$

Substituting (8) into (7),

$$V = \frac{I_0}{4\pi\varepsilon_0 j\omega}\left[\frac{e^{j\omega[t-(s_1/c)]}}{s_1} - \frac{e^{j\omega[t-(s_2/c)]}}{s_2}\right] \tag{9}$$

Referring to Fig. 6–3*b*, when $r \gg L$, the lines connecting the ends of the dipole and the point P may be considered as parallel so that

$$s_1 = r - \frac{L}{2}\cos\theta \tag{10}$$

and

$$s_2 = r + \frac{L}{2}\cos\theta \tag{11}$$

Substituting (10) and (11) into (9), it may be shown that the fields of a short electric dipole are:

（短电偶极子的电场）

Electric fields of short dipole

$$E_r = \frac{I_0 L\cos\theta e^{j\omega[t-(r/c)]}}{2\pi\varepsilon_0}\left(\frac{1}{cr^2} + \frac{1}{j\omega r^3}\right) \quad \text{（一般情况）} \textit{General case} \tag{12}$$

$$E_\theta = \frac{I_0 L\sin\theta e^{j\omega[t-(r/c)]}}{4\pi\varepsilon_0}\left(\frac{j\omega}{c^2 r} + \frac{1}{cr^2} + \frac{1}{j\omega r^3}\right) \tag{13}$$

In obtaining (12) and (13) the relation was used that $\mu_0\varepsilon_0 = 1/c^2$, where c = velocity of light.

Turning our attention now to the ***magnetic field,*** （磁场） this may be calculated from curl of A as follows:

$$\nabla \times \mathbf{A} = \frac{\hat{\mathbf{r}}}{r\sin\theta}\left[\frac{\partial(\sin\theta)A_\phi}{\partial\theta} - \frac{\partial(A_\theta)}{\partial\phi}\right] + \frac{\hat{\theta}}{r\sin\theta}\left[\frac{\partial A_r}{\partial\phi} - \frac{\partial(r\sin\theta)A_\phi}{\partial r}\right] + \frac{\hat{\phi}}{r}\left[\frac{\partial(rA_\theta)}{\partial r} - \frac{\partial A_r}{\partial\theta}\right] \tag{14}$$

Since $A_\phi = 0$, the first and fourth terms of (14) are zero, since A_r and A_θ are independent of ϕ, so that the second and third terms of (14) are also zero. Thus, only the last two terms contribute, so that $\nabla \times \mathbf{A}$, and hence also $\mathbf{H}$, have only a ϕ component. Thus,

（短电偶极子的磁场）

Magnetic fields of short dipole

$$|\mathbf{H}| = H_\phi = \frac{I_0 L\sin\theta e^{j\omega[t-(r/c)]}}{4\pi}\left(\frac{j\omega}{cr} + \frac{1}{r^2}\right) \quad \text{（一般情况）} \textit{General case} \tag{15}$$

$$H_r = H_\theta = 0 \tag{16}$$

Thus, the fields from the dipole have only three components E_r, E_θ and H_ϕ. The components E_ϕ, H_r and H_θ are everywhere zero.

When r is very large, the terms in $1/r^2$ and $1/r^3$ in (12), (13), and (15) can be neglected in favor of the terms in $1/r$. Thus, in the *far field* E_r is negligible, and we have effectively

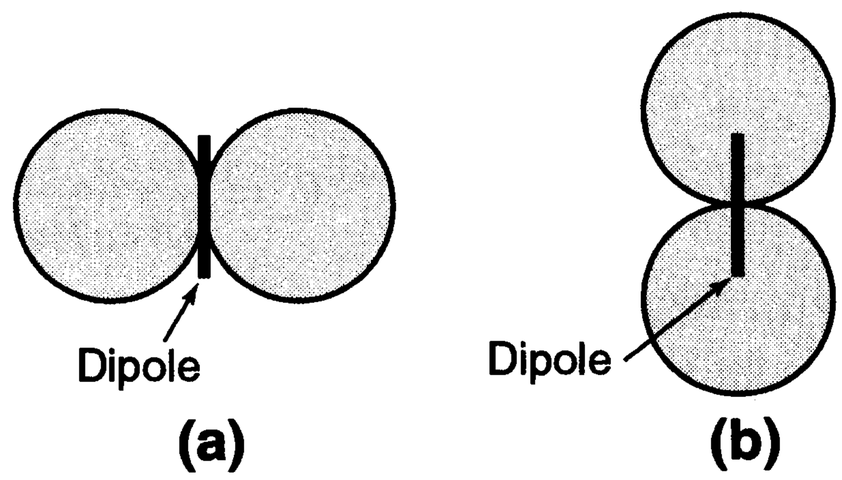


Figure 6–4
Near- and far-field patterns of E_θ and H_ϕ components for short dipole (*a*) and near-field pattern of E_r component (*b*).

only two field components, E_θ and H_ϕ, given by

（电场）（短电偶极子的磁场）

Electric and magnetic fields of short dipole

$$E_\theta = \frac{j\omega I_0 L \sin\theta e^{j\omega[t-(r/c)]}}{4\pi\varepsilon_0 c^2 r} = j\frac{I_0\beta L}{4\pi\varepsilon_0 c r}\sin\theta e^{j\omega[t-(r/c)]} \quad (17)$$

$$H_\phi = \frac{j\omega I_0 L \sin\theta e^{j\omega[t-(r/c)]}}{4\pi c r} = j\frac{I_0\beta L}{4\pi r}\sin\theta e^{j\omega[t-(r/c)]} \quad (18)$$

Far-field case（远场情况）

Taking the ratio of E_θ to H_ϕ as given by (17) and (18), we obtain

$$\frac{E_\theta}{H_\phi} = \frac{1}{\varepsilon_0 c} = \sqrt{\frac{\mu_0}{\varepsilon_0}} = \mathbf{376.7\ \Omega} \quad (19)$$

Impedance of space（空间阻抗）

This is the ***intrinsic impedance of free space***（自由空间的本征阻抗）(a pure resistance). It is a very important constant.

Comparing (17) and (18) we note that E_θ and H_ϕ are in time phase in the far field. We note also that the field patterns of both are proportional to $\sin\theta$. The pattern is independent of ϕ, so that the space pattern is doughnut-shaped, being a figure-of-revolution of the pattern in Fig. 6–4*a* about the axis of the dipole. Referring to the near-field expressions given by (12), (13) and (15), we note that for a small r the electric field has two components E_r and E_θ, which are both in time-phase quadrature with the magnetic field, as in a resonator. At intermediate distances, E_θ and E_r can approach time-phase quadrature so that the total electric field vector rotates in a plane parallel to the direction of propagation, thus exhibiting the phenomenon of *cross-field*. For the E_θ and H_ϕ components, the near-field patterns are the same as the far-field patterns, being proportional to $\sin\theta$ (Fig. 6–4*a*). However, the near-field pattern for E_r is proportional to $\cos\theta$ as indicated by Fig. 6–4*b*. The space pattern for E_r is a figure-of-revolution of this pattern around the dipole axis.

Let us now consider the situation at very low frequencies. This will be referred to as the *quasi-stationary,* or dc case. Since from

$$[I] = I_0 e^{j\omega[t-(r/c)]} = j\omega[q] \quad (20)$$

(12) and (13) can be rewritten as

$$E_r = \frac{[q]L\cos\theta}{2\pi\varepsilon_0}\left(\frac{j\omega}{cr^2} + \frac{1}{r^3}\right) \quad (21)$$

and

$$E_\theta = \frac{[q]L\sin\theta}{4\pi\varepsilon_0}\left(-\frac{\omega^2}{c^2 r} + \frac{j\omega}{cr^2} + \frac{1}{r^3}\right) \quad (22)$$

The magnetic field is given by (15) as

$$H_\phi = \frac{[I]L\sin\theta}{4\pi}\left(\frac{j\omega}{cr}+\frac{1}{r^2}\right) \tag{23}$$

At low frequencies, ω approaches zero so that the terms with ω in the numerator can be neglected. As $\omega \to 0$, we also have

$$[q] = q_0 e^{j\omega[t-(r/c)]} = q_0 \tag{24}$$

and

$$[I] = I_0 \tag{25}$$

Thus, for the quasi-stationary, or dc, case, the field components become from (21), (22) and (23)

（短电偶极子的电场和磁场）

Electric and magnetic fields of short dipole

$$E_r = \frac{q_0 L\cos\theta}{2\pi\varepsilon_0 r^3} \tag{26}$$

$$E_\theta = \frac{q_0 L\sin\theta}{4\pi\varepsilon_0 r^3} \tag{27}$$

$$H_\phi = \frac{I_0 L\sin\theta}{4\pi r^2} \tag{28}$$

Low-frequency case
（低频情况）

The restriction that $r \gg L$ still applies.

The expressions for the electric field, (26) and (27), are identical to those obtained in electrostatics for the field of two point charges, $+q_0$ and $-q_0$, separated by a distance L. The relation for the magnetic field, (28), may be recognized as the Biot-Savart relation for the magnetic field of a short element carrying a steady or slowly varying current. Since in the expressions for the quasi-stationary case the fields decrease as $1/r^2$ or $1/r^3$, the fields are confined to the vicinity of the dipole and there is negligible radiation. In the general expressions for the fields, (21), (22) and (23), it is the $1/r$ terms which are important in the far field and hence take into account the radiation.

The expressions for the fields from a short dipole developed above are summarized in Table 6–1.

Table 6–1 Fields of a short electric dipole†

Component	General expression	Far field	Quasi-stationary
E_r	$\frac{[I]L\cos\theta}{2\pi\varepsilon_0}\left(\frac{1}{cr^2}+\frac{1}{j\omega r^3}\right)$	0	$\frac{q_0 L\cos\theta}{2\pi\varepsilon_0 r^3}$
E_θ	$\frac{[I]L\sin\theta}{4\pi\varepsilon_0}\left(\frac{j\omega}{c^2 r}+\frac{1}{cr^2}+\frac{1}{j\omega r^3}\right)$	$\frac{[I]Lj\omega\sin\theta}{4\pi\varepsilon_0 c^2 r} = \frac{j60\pi[I]\sin\theta}{r}\frac{L}{\lambda}$	$\frac{q_0 L\sin\theta}{4\pi\varepsilon_0 r^3}$
H_ϕ	$\frac{[I]L\sin\theta}{4\pi}\left(\frac{j\omega}{cr}+\frac{1}{r^2}\right)$	$\frac{[I]Lj\omega\sin\theta}{4\pi cr} = \frac{j[I]\sin\theta}{2r}\frac{L}{\lambda}$	$\frac{I_0 L\sin\theta}{4\pi r^2}$

†The restriction applies that $r \gg L$ and $\lambda \gg L$. The quantities in the table are in SI units, that is, E in volts per meter, H in amperes per meter, I in amperes, r in meters, etc. $[I]$ is as given by (20). Three of the field components of an electric dipole are everywhere zero, that is,

$$E_\phi = H_r = H_\theta = 0$$

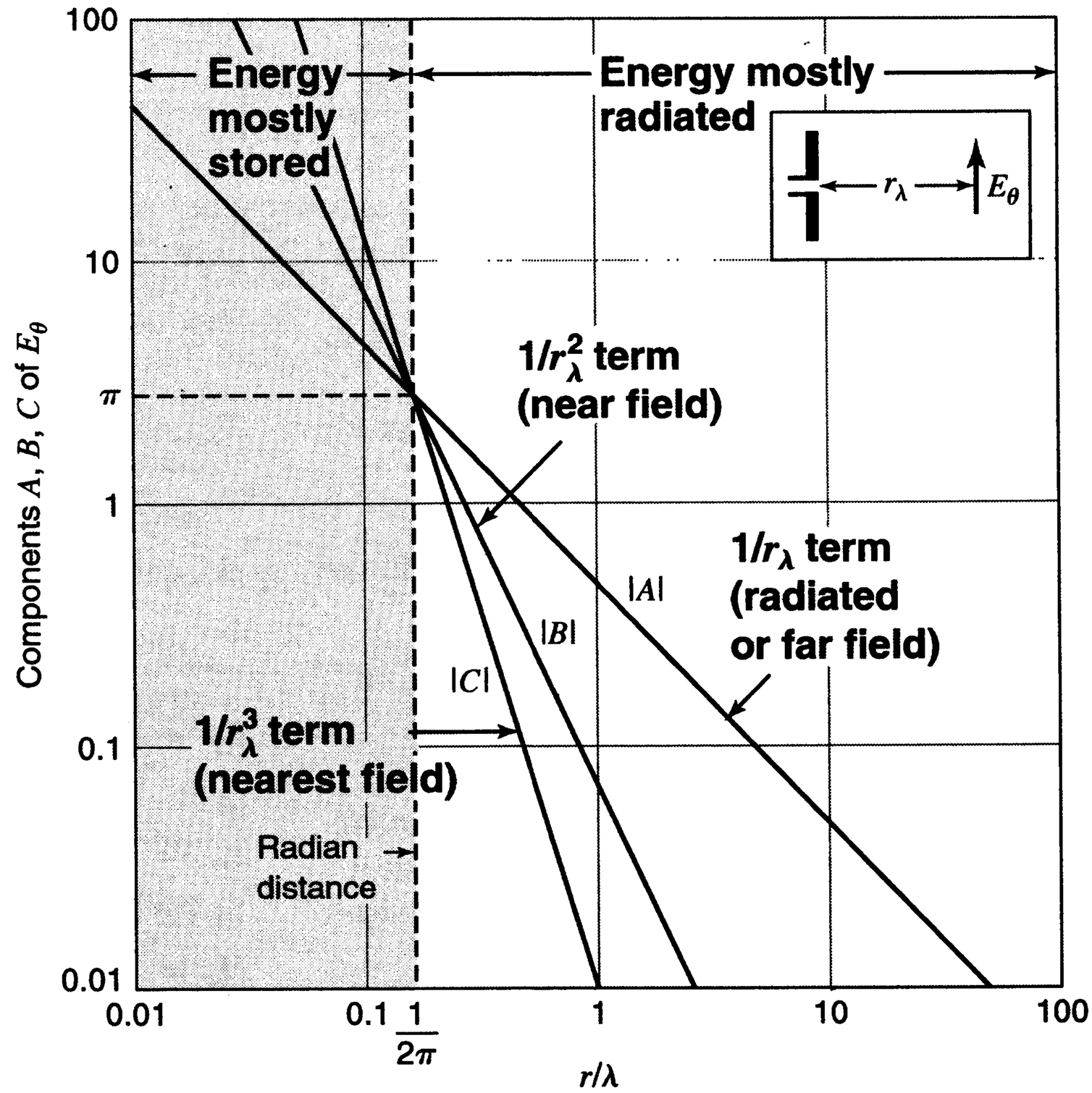


Figure 6–5
Variation of the magnitudes of the components of E_θ of a short electric dipole as a function of distance (r/λ). The magnitudes of all components equal π at the radian distance $1/(2\pi)$. At larger distances energy is mostly radiated, at smaller distances mostly stored.

Setting

$$|A| = \frac{1}{2r_\lambda}$$

$$|B| = \frac{1}{4\pi r_\lambda^2}$$

$$|C| = \frac{1}{8\pi^2 r_\lambda^3}$$

for the three components of E_θ, their variation with distance is as shown in Fig. 6–5. For r_λ greater than the radian distance $[1/(2\pi)]$, component A of the electric field is dominant, for r_λ less than the radian distance component C of the electric field is dominant, while *at* the radian distance only B contributes ($= \pi$) because although $|A| = |B| = |C| = \pi$, A and C are in phase opposition and cancel.

For the special case where $\theta = 90°$ (perpendicular to the dipole in the xy plane of Fig. 6–2) and at $r_\lambda \gg 1/(2\pi)$,

$$|H_\phi| = \frac{I_0 L_\lambda}{2r} \qquad (\text{A m}^{-1}) \tag{29}$$

while at $r_\lambda \ll 1/(2\pi)$,

$$|H_\phi| = \frac{I_0 L}{4\pi r^2} \tag{30}$$

which is identical to the relation for the magnetic field perpendicular to a short linear conductor carrying direct current as given by (28).

The magnetic field *at any distance r* from an infinite linear conductor with direct current is given by

$$H_\phi = \frac{I_0}{2\pi r} \tag{31}$$

which is *Ampere's law*.

Remarkably, the magnitude of the magnetic field in the equatorial plane ($\theta = 90°$) *in the far field of an oscillating* $\lambda/2$ *dipole* is identical to (31) (Ampere's law). It is assumed that the current distribution on the $\lambda/2$ dipole is sinusoidal. This is discussed in more detail in Sec. 6–4.

Rearranging the three field components of Table 6–1 for a short electric dipole, we have

$$E_r = \frac{[I]L_\lambda Z\cos\theta}{\lambda}\left[\frac{1}{2\pi r_\lambda^2} - j\frac{1}{4\pi^2 r_\lambda^3}\right] \tag{32}$$

$$E_\theta = \frac{[I]L_\lambda Z\sin\theta}{\lambda}\left[j\frac{1}{2r_\lambda} + \frac{1}{4\pi r_\lambda^2} - j\frac{1}{8\pi^2 r_\lambda^3}\right] \tag{33}$$

$$H_\phi = \frac{[I]L_\lambda \sin\theta}{\lambda}\left[j\frac{1}{2r_\lambda} + \frac{1}{4\pi r_\lambda^2}\right] \tag{34}$$

We note that the constant factor in each of the terms in brackets differs from the factors of adjacent terms by a factor of 2π.

At *the radian distance* ($r_\lambda = 1/2\pi$) the fields of (32), (33) and (34) reduce to

$$E_r = \frac{2\sqrt{2}\pi[I]L_\lambda Z\cos\theta}{\lambda}\underline{/-45°} \tag{35}$$

$$E_\theta = \frac{\pi[I]L_\lambda Z\sin\theta}{\lambda} \tag{36}$$

$$H_\phi = \frac{\sqrt{2}\pi[I]L_\lambda \sin\theta}{\lambda}\underline{/45°} \tag{37}$$

The magnitude of the *average power flux* or *Poynting vector* in the θ direction is given by

$$S_\theta = \tfrac{1}{2}\,\mathrm{Re}\,E_r H_\phi^* = \tfrac{1}{2}E_r H_\phi\,\mathrm{Re}\,1\underline{/-90°} = \tfrac{1}{2}E_r H_\phi\cos(-90°) = 0 \tag{38}$$

indicating that *no power is transmitted. However, the product* $E_r H_\phi$ *represents imaginary or reactive energy that oscillates back and forth from electric to magnetic energy twice per cycle.*

In like manner the magnitude of the power flux or Poynting vector in the r direction is given by

$$S_r = \tfrac{1}{2}E_\theta H_\phi\cos(-45°) = \frac{1}{2\sqrt{2}}E_\theta H_\phi \tag{39}$$

indicating *energy flow in the r direction.*

Much closer to the dipole [$r_\lambda \ll 1/(2\pi)$], (32), (33) and (34) reduce approximately to

$$E_r = -j\frac{[I]L_\lambda Z\cos\theta}{4\pi^2\lambda r_\lambda^3} \tag{40}$$

$$E_\theta = -j\frac{[I]L_\lambda Z\sin\theta}{8\pi^2\lambda r_\lambda^3} \tag{41}$$

$$H_\phi = \frac{[I]L_\lambda\sin\theta}{4\pi\lambda r_\lambda^2} \tag{42}$$

From these equations it is apparent that $S_r = S_\theta = 0$. However, the products $E_r H_\phi$ and $E_\theta H_\phi$ represent imaginary or reactive energy oscillating back and forth but not going anywhere. Thus, *close to the dipole there is a region of almost complete energy storage.*

Remote from the dipole [$r_\lambda \gg 1/(2\pi)$], (32), (33) and (34) reduce approximately to

$$E_r = 0 \tag{43}$$

$$E_\theta = j\frac{[I]L_\lambda Z\sin\theta}{2\lambda r_\lambda} \tag{44}$$

$$H_\phi = j\frac{[I]L_\lambda\sin\theta}{2\lambda r_\lambda} \tag{45}$$

Since $E_r = 0$, there is no energy flow in the θ direction ($S_\theta = 0$). However, *since $E_\theta H_\phi$ are in time phase, their product represents real power flow in the outward radial direction. This power is radiated.*

Many antennas behave like the dipole with large energy storage close to the antenna.

The region near the dipole is one of stored energy (reactive power) while regions remote from the dipole are ones of radiation. The radian sphere at $r_\lambda = 1/(2\pi)$ marks a zone of transition from one region to the other with a nearly equal division of the imaginary and real (radiated) power.

The region close to the dipole may be likened to a spherical resonator within which pulsating energy is trapped, but with some leakage which is radiated. There is no exact boundary to this resonator region, but if we arbitrarily put it at the radian distance a qualitative picture may be sketched as in Fig. 6–6.

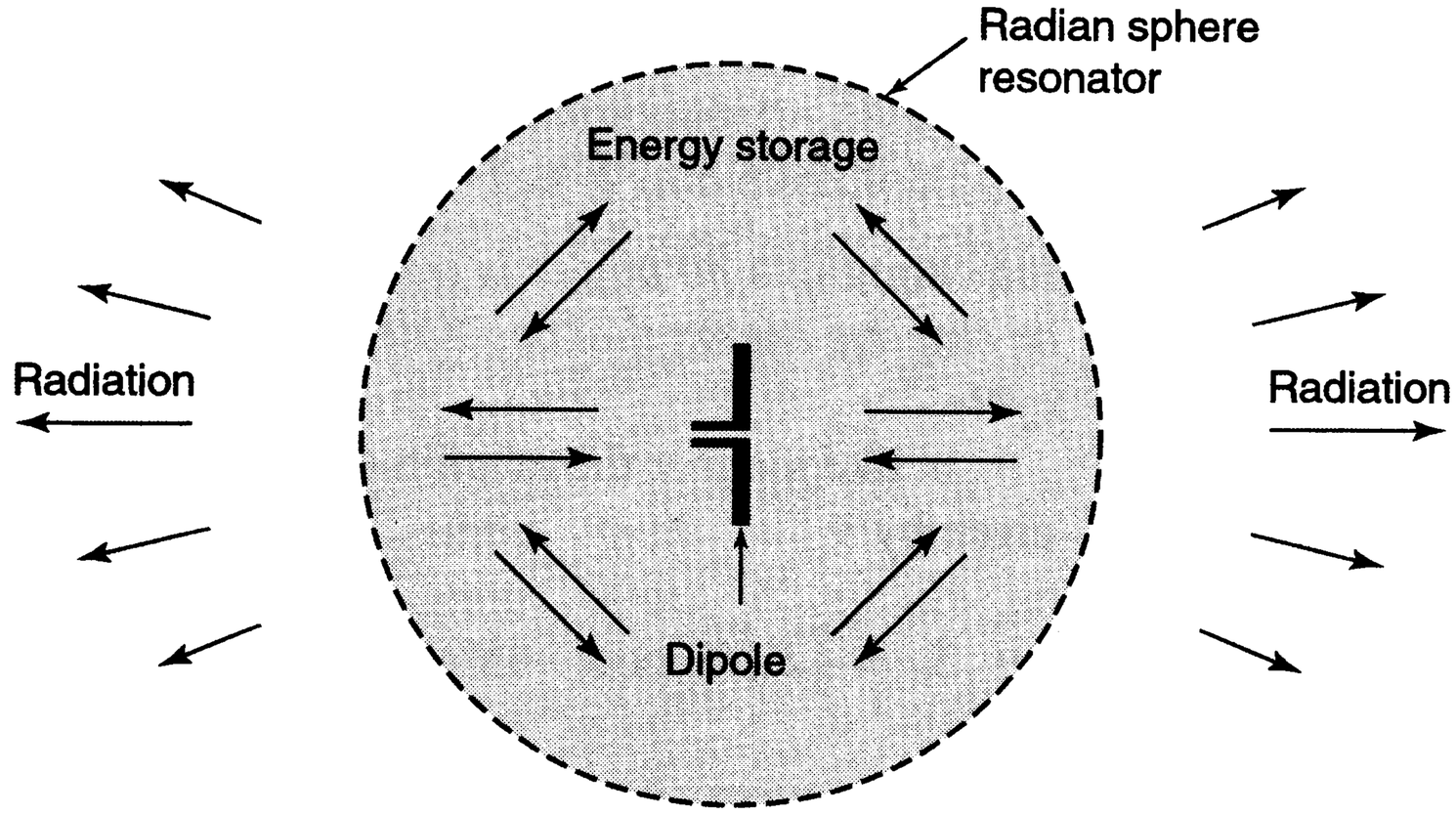


Figure 6–6
Sketch suggesting that within the radian sphere at $r = \lambda/2\pi = 0.16\lambda$ the situation is like that inside a resonator with high-density pulsating energy accompanied by leakage which is radiated.

6–3 RADIATION RESISTANCE OF SHORT ELECTRIC DIPOLE
(短电偶极子的辐射电阻)

Let us now calculate the radiation resistance of the short dipole of Fig. 6–1*b*. This may be done as follows. The Poynting vector of the far field is integrated over a large sphere to obtain the total power radiated. This power is then equated to I^2R where I is the rms current on the dipole and R is a resistance, called the radiation resistance of the dipole.

The *average* Poynting vector is given by

$$\boldsymbol{S} = \tfrac{1}{2}\,\mathrm{Re}(\boldsymbol{E} \times \boldsymbol{H}^*) \tag{1}$$

The far-field components are E_θ and H_ϕ so that the radial component of the Poynting vector is

$$S_r = \tfrac{1}{2}\,\mathrm{Re}\, E_\theta H_\phi^* \tag{2}$$

where E_θ and H_ϕ^* are complex.

The far-field components are related by the intrinsic impedance of the medium. Hence,

$$E_\theta = H_\phi Z = H_\phi \sqrt{\frac{\mu}{\varepsilon}} \tag{3}$$

Thus, (2) becomes

$$S_r = \tfrac{1}{2}\,\mathrm{Re}\, Z H_\phi H_\phi^* = \tfrac{1}{2}|H_\phi|^2\,\mathrm{Re}\, Z = \tfrac{1}{2}|H_\phi|^2 \sqrt{\frac{\mu}{\varepsilon}} \tag{4}$$

The total power P radiated is then

$$P = \int\int S_r\, ds = \frac{1}{2}\sqrt{\frac{\mu}{\varepsilon}} \int_0^{2\pi} \int_0^{\pi} |H_\phi|^2 r^2 \sin\theta\, d\theta\, d\phi \tag{5}$$

where the angles are as shown in Fig. 6–2 and $|H_\phi|$ is the absolute value of the magnetic field, which from (6–2–18) is

$$|H_\phi| = \frac{\omega I_0 L \sin\theta}{4\pi c r} \tag{6}$$

Substituting this into (5), we have

$$P = \frac{1}{32}\sqrt{\frac{\mu}{\varepsilon}} \frac{\beta^2 I_0^2 L^2}{\pi^2} \int_0^{2\pi} \int_0^{\pi} \sin^3\theta\, d\theta\, d\phi \tag{7}$$

The double integral equals $8\pi/3$ and (7) becomes

$$P = \sqrt{\frac{\mu}{\varepsilon}} \frac{\beta^2 I_0^2 L^2}{12\pi} \tag{8}$$

This is the *average* power or rate at which energy is streaming out of a sphere surrounding the dipole. Hence, it is equal to the power radiated. Assuming no losses, it is also equal to the power delivered to the dipole. Therefore, P must be equal to the square of the rms current I flowing on the dipole times a resistance R_r called the *radiation resistance* of the dipole. Thus,

$$\sqrt{\frac{\mu}{\varepsilon}} \frac{\beta^2 I_0^2 L^2}{12\pi} = \left(\frac{I_0}{\sqrt{2}}\right)^2 R_r \tag{9}$$

Solving for R_r,

$$R_r = \sqrt{\frac{\mu}{\varepsilon}} \frac{\beta^2 L^2}{6\pi} \tag{10}$$

For air or vacuum $\sqrt{\mu/\varepsilon} = \sqrt{\mu_0/\varepsilon_0} = 377 = 120\pi\,\Omega$ so that (10) becomes[1]

(均匀电流的偶极子)

Dipole with uniform current $R_r = 80\pi^2 \left(\frac{L}{\lambda}\right)^2 = 80\pi^2 L_\lambda^2 = 790 L_\lambda^2 \quad (\Omega)$ ***Radiation resistance*** (辐射电阻) (11)

As an example suppose that $L_\lambda = \frac{1}{10}$. Then $R_r = 7.9\ \Omega$. If $L_\lambda = 0.01$, then $R_r = 0.08\ \Omega$. Thus, the radiation resistance of a short dipole is small.

In developing the field expressions for the short dipole, which were used in obtaining (11), the restriction was made that $\lambda \gg L$. This made it possible to neglect the phase difference of field contributions from different parts of the dipole. If $L_\lambda = \frac{1}{2}$ we violate this assumption, but, as a matter of interest, let us find what the radiation resistance of a $\lambda/2$ dipole is, when calculated in this way. Then for $L_\lambda = \frac{1}{2}$, we obtain $R_r = 197\ \Omega$. The correct value is 168 Ω (see Prob. 6–5–1), which indicates the magnitude of the error introduced by violating the restriction that $\lambda \gg L$ to the extent of taking $L = \lambda/2$.

It has been assumed that with end loading (see Fig. 6–1*a*) the dipole current is uniform. However, with no end loading the current must be zero at the ends and, if the dipole is short, the current tapers almost linearly from a maximum at the center to zero at the ends, as in Fig. 2–12, with an average value of $\frac{1}{2}$ of the maximum. Modifying (8) for the general case where the current is not uniform on the dipole, the *radiated power* is

$$P = \sqrt{\frac{\mu}{\varepsilon}} \frac{\beta^2 I_{\text{av}}^2 L^2}{12\pi} \qquad \text{(W)} \tag{12}$$

where I_{av} = amplitude of *average current* on dipole (peak value in time)

The *power delivered* to the dipole is, as before,

$$P = \tfrac{1}{2} I_0^2 R_r \qquad \text{(W)} \tag{13}$$

where I_0 = amplitude of *terminal current* of center-fed dipole (peak value in time). Equating the *power radiated* (12) to the *power delivered* (13) yields, for free space ($\mu = \mu_0$ and $\varepsilon = \varepsilon_0$), a radiation resistance

$$R_r = 790 \left(\frac{I_{\text{av}}}{I_0}\right)^2 L_\lambda^2 \qquad (\Omega) \tag{14}$$[2]

For a short dipole without end loading, we have $I_{\text{av}} = \frac{1}{2} I_0$, as noted above, and (14) becomes

$$R_r = 197 L_\lambda^2 \qquad (\Omega) \tag{15}$$

---

[1] $\sqrt{\mu_0/\varepsilon_0} = 376.73\ \Omega$. 377 and 120π are convenient approximations.

[2] As already given by (2–10–9).

6–4 THE THIN LINEAR ANTENNA（细直天线）

In this section expressions for the far-field patterns of thin linear antennas will be developed. It is assumed that the antennas are symmetrically fed at the center by a balanced two-wire transmission line. The antennas may be of any length, but it is assumed that the current distribution is sinusoidal. Current-distribution measurements indicate that this is a good assumption provided that the antenna is thin, i.e., when the conductor diameter is less than, say, $\lambda/100$. Thus, the sinusoidal current distribution approximates the natural distribution on thin antennas. Examples of the approximate natural-current distributions on a number of thin, linear center-fed antennas of different length are illustrated in Fig. 6–7. The currents are in phase over each $\lambda/2$ section and in opposite phase over the next.

Referring to Fig. 6–8, let us now proceed to develop the far-field equations for a symmetrical, thin, linear, center-fed antenna of length L. The retarded value of the current at any point z on the antenna referred to a point at a distance s is

$$[I] = I_0 \sin\left[\frac{2\pi}{\lambda}\left(\frac{L}{2} \pm z\right)\right] e^{j\omega[t-(r/c)]} \tag{1}$$

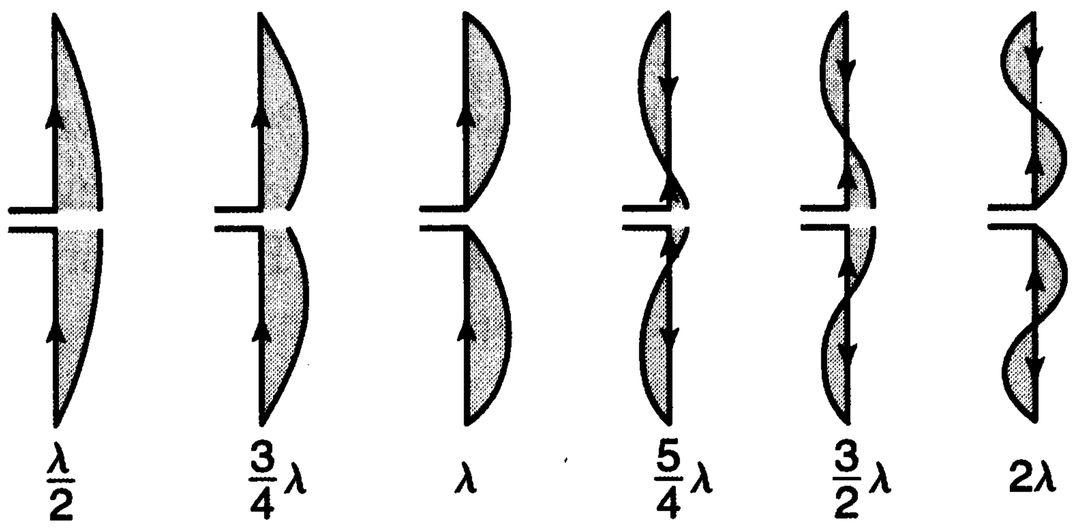


Figure 6–7
Approximate natural-current distribution for thin, linear, center-fed antennas of various lengths.

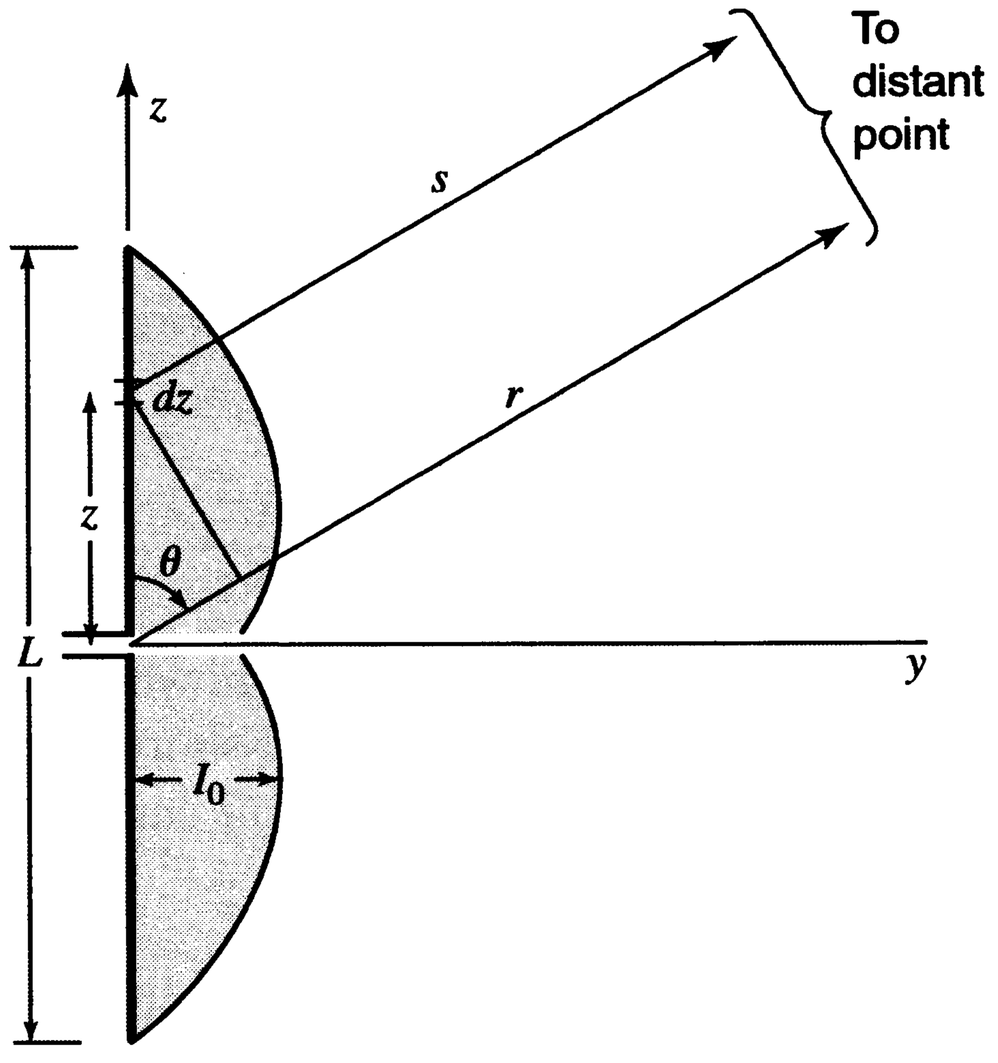


Figure 6–8
Relations for symmetrical, thin, linear, center-fed antenna of length L.

In (1) the function

$$\sin\left[\frac{2\pi}{\lambda}\left(\frac{L}{2}\pm z\right)\right]$$

is the form factor for the current on the antenna. The expression $(L/2)+z$ is used when $z<0$ and $(L/2)-z$ is used when $z>0$. By regarding the antenna as made up of a series of infinitesimal dipoles of length dz, the field of the entire antenna may then be obtained by integrating the fields from all of the dipoles making up the antenna with the result[1]

（中馈偶极子的远场）

Far fields of center-fed dipole

$$H_\phi = \frac{j[I_0]}{2\pi r}\left[\frac{\cos[(\beta L\cos\theta)/2]-\cos(\beta L/2)}{\sin\theta}\right] \tag{2}$$

$$E_\theta = \frac{j60[I_0]}{r}\left[\frac{\cos[(\beta L\cos\theta)/2]-\cos(\beta L/2)}{\sin\theta}\right] \tag{3}$$

where $[I_0]=I_0e^{j\omega[t-(r/c)]}$ and

$$E_\theta = 120\pi H_\phi \tag{3a}$$

Equations (2), (3) and (3*a*) give the far fields H_ϕ and E_θ of a *symmetrical, center-fed, thin linear antenna of length L*. The shape of the far-field pattern is given by the factor in the brackets. The factors preceding the brackets in (2) and (3) give the instantaneous magnitude of the fields as functions of the antenna current and the distance r. To obtain the rms value of the field, we let $[I_0]$ equal the rms current at the location of the current maximum. There is no factor involving phase in (2) or (3), since the center of the antenna is taken as the phase center. Hence any phase change of the fields as a function of θ will be a jump of 180° when the pattern factor changes sign.

As examples of the far-field patterns of linear center-fed antennas, three antennas of different lengths will be considered. Since the amplitude factor is independent of the length, only the relative field patterns as given by the pattern factor will be compared.

EXAMPLE **6–4.1 λ/2 Antenna**

When $L=\lambda/2$, the pattern factor becomes

$$E=\frac{\cos[(\pi/2)\cos\theta]}{\sin\theta} \tag{4}$$

This pattern is shown in Fig. 6–9*a*. It is only slightly more directional than the pattern of an infinitesimal or short dipole which is given by $\sin\theta$. The beamwidth between half-power points of the $\lambda/2$ antenna is 78° as compared to 90° for the short dipole.

[1]For complete development, see the second edition of this book, pp. 220–221.

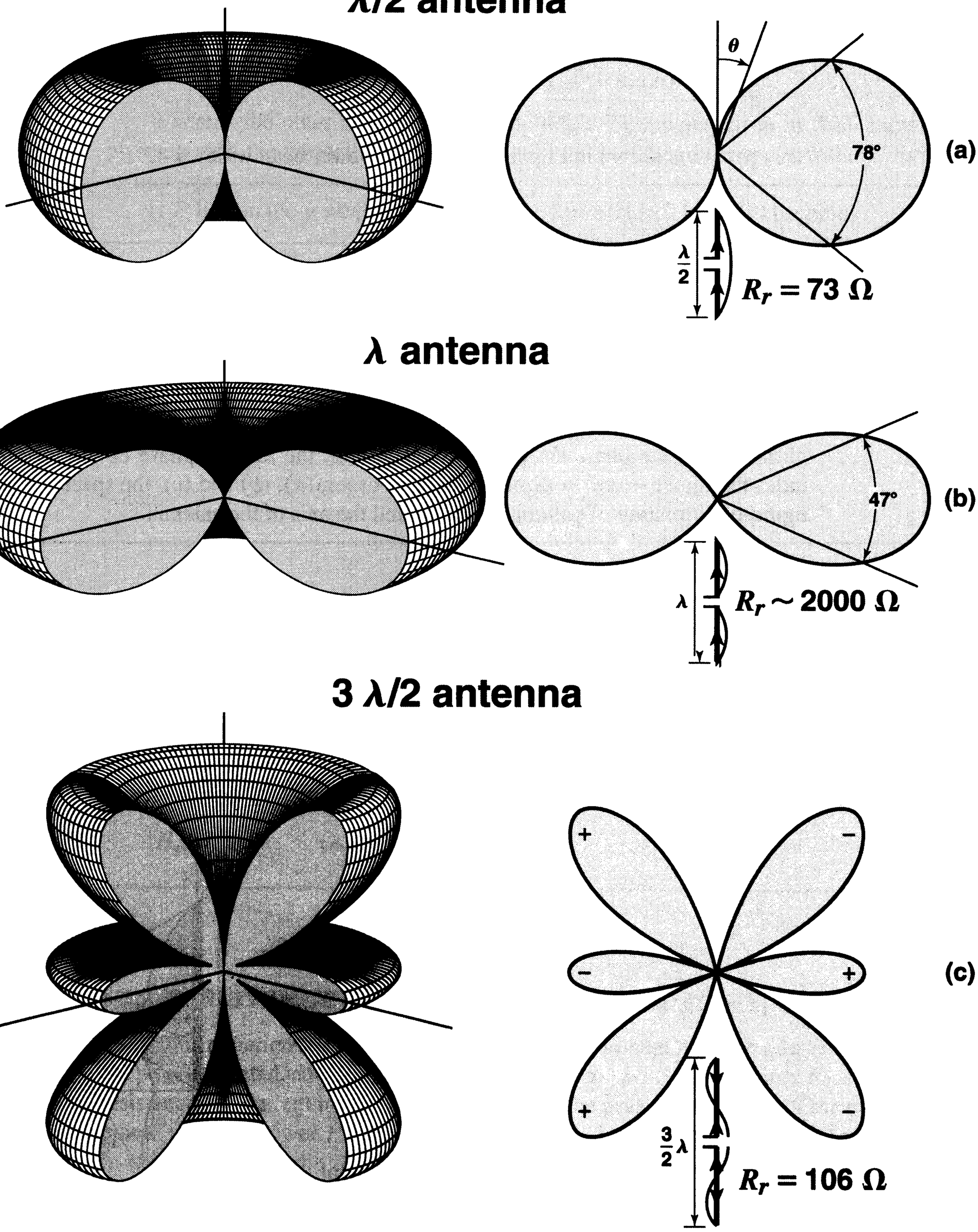


Figure 6–9
Three-dimensional and polar plots of the patterns of λ/2, λ, and 3 λ/2 antennas. The antennas are center-fed with current distributions assumed sinusoidal as indicated.

EXAMPLE **6–4.2 Full-Wave (λ) Antenna**

When $L = \lambda$, the pattern factor becomes

$$E = \frac{\cos(\pi \cos\theta) + 1}{\sin\theta} \tag{5}$$

This pattern is shown in Fig. 6–9*b*. The half-power beamwidth is 47°.

EXAMPLE **6–4.3 Three Half-Wave (3 $\lambda/2$) Antenna**

When $L = 3\,\lambda/2$, the pattern factor is

$$E = \frac{\cos\left(\frac{3}{2}\pi \cos\theta\right)}{\sin\theta} \tag{6}$$

The pattern for this case is presented in Fig. 6–9*c*. With the midpoint of the antenna as phase center, the phase shifts 180° at each null, the relative phase of the lobes being indicated by the + and − signs. In all three cases, (*a*), (*b*) and (*c*), the space pattern is a figure-of-revolution of pattern shown around the axis of the antenna.

EXAMPLE **6–4.4 Field at Any Distance from Center-Fed Dipole**

The geometry for the field at the point P from a symmetrical center-fed dipole of length L with sinusoidal current distribution is presented in Fig. 6–10. The maximum current is I_0. It may be shown that the z component of the electric field at the point P is given by

$$E_z = \frac{-jI_0Z}{4\pi}\left[\frac{e^{-j\beta s_1}}{s_1} + \frac{e^{-j\beta s_2}}{s_2} - 2\cos\frac{\beta L}{2}\frac{e^{-j\beta r}}{r}\right] \tag{7}$$

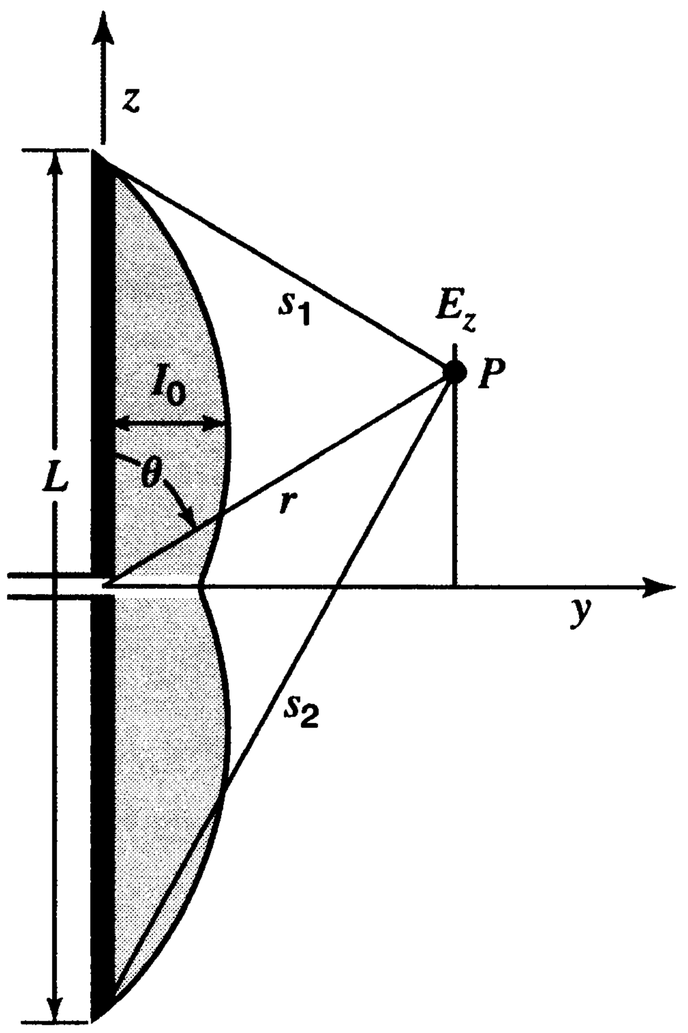


Figure 6–10
Symmetrical center-fed dipole with sinusoidal current distribution. The field component E_z at any distance can be expressed as the sum of three components radiating from the ends and the center of the dipole.

The ϕ component of the magnetic field at the point P (Fig. 6–10) is given by

$$H_\phi = \frac{jI_0}{4\pi r \sin\theta}\left(e^{-j\beta s_1} + e^{-j\beta s_2} - 2\cos\frac{\beta L}{2}e^{-j\beta r}\right) \tag{8}$$

Whereas the other field equations for oscillating dipoles given in this chapter apply only with the restrictions of $\lambda \gg L$ and $r \gg L$, (7) and (8) apply without distance restrictions.

If P lies on the y axis ($\theta = 90°$) and the dipole is $\lambda/2$ long, (7) becomes

$$E_z = \frac{I_0 Z}{2\pi\lambda\sqrt{\frac{1}{16} + r_\lambda^2}} \angle -360°\sqrt{\frac{1}{16} + r_\lambda^2} - 90° \qquad (\text{V m}^{-1}) \tag{9}$$

and (8) becomes

$$H_\phi = \frac{I_0}{2\pi r} \angle -360°\sqrt{\frac{1}{16} + r_\lambda^2} + 90° \qquad (\text{A m}^{-1}) \tag{10}$$

where

$r_\lambda = r/\lambda$

$Z = 377\ \Omega$

$I_0 =$ maximum current $=$ terminal current

At a large distance the ratio of E_z as given by (9) to H_ϕ as given by (10) is

$$\frac{E_z}{H_\phi} = Z = 377\ \Omega = \text{intrinsic impedance (resistance) of space} \tag{11}$$

The magnitude of H_ϕ is

$$|H_\phi| = \frac{I_0}{2\pi r} \qquad (\text{A m}^{-1}) \tag{12}$$

6–5 RADIATION RESISTANCE OF λ/2 ANTENNA（半波天线的辐射电阻）

To find the radiation resistance, the Poynting vector is integrated over a large sphere yielding the power radiated, and this power is then equated to $(I_0/\sqrt{2})^2 R_0$, where R_0 is the radiation resistance at a current maximum point and I_0 is the peak value in time of the current at this point. The total power P radiated was given in (6–3–5)[1] in terms of H_ϕ for a short dipole. In (6–3–5), $|H_\phi|$ is the absolute value. Hence, the corresponding value of H_ϕ for a linear antenna is obtained from (6–4–2) by putting $|j[I_0]| = I_0$. Substituting this into 6–3–5,

[1] $P = \iint \boldsymbol{S}\cdot d\boldsymbol{s} = \frac{1}{2}\sqrt{\mu/\varepsilon}\iint |H_\phi|^2\, ds$

we obtain

$$P = \frac{15I_0^2}{\pi} \int_0^{2\pi} \int_0^{\pi} \frac{\{\cos[(\beta L/2)\cos\theta] - \cos(\beta L/2)\}^2}{\sin\theta} \, d\theta \, d\phi \tag{1}$$

$$= 30I_0^2 \int_0^{\pi} \frac{\{\cos[(\beta L/2)\cos\theta] - \cos(\beta L/2)\}^2}{\sin\theta} \, d\theta \tag{2}$$

Equating the radiated power as given by (2) to $I_0^2 R_0/2$ we have

$$P = \frac{I_0^2 R_0}{2} \tag{3}$$

and

$$R_0 = 60 \int_0^{\pi} \frac{\{\cos[(\beta L/2)\cos\theta] - \cos(\beta L/2)\}^2}{\sin\theta} \, d\theta \tag{4}$$

where the radiation resistance R_0 is referred to the current maximum. In the case of a $\lambda/2$ antenna this is at the center of the antenna or at the terminals of the transmission line (see Fig. 6–7).

Proceeding with the evaluation of (4) with the aid of the sine integral, $\mathrm{Si}(x)$, and the cosine integral, $\mathrm{Cin}(x)$, it may be shown that the radiation resistance of the $\lambda/2$ antenna is

$$\boxed{R_r = 30\,\mathrm{Cin}(2\pi) = 30 \times 2.44 = \mathbf{73\ \Omega}} \tag{5}$$

This is the well-known value for the radiation resistance of a thin, linear, center-fed, $\lambda/2$ antenna with sinusoidal current distribution. The terminal impedance also includes some inductive reactance as discussed in Chap. 13. That is,

$$\boxed{Z = 73 + j42.5\ \Omega} \tag{6}$$

To make the reactance zero, that is, to make the antenna resonant, requires that the antenna be shorted a few percent less than $\lambda/2$. This shortening also results in a reduction in the value of the radiation resistance to about 65 Ω.

6–6 RADIATION RESISTANCE AT A POINT WHICH IS NOT A CURRENT MAXIMUM（非电流最大点的辐射电阻）

If we calculate, for example, the radiation resistance of a $3\lambda/4$ antenna (see Fig. 6–7) by the above method, we obtain its value at a current maximum. This is not the point at which the transmission line is connected. Neglecting antenna losses, the value of radiation resistance so obtained is the resistance R_0 which would appear at the terminals of a transmission line connected at a current maximum in the antenna, *provided* that the current distribution on the antenna is the same as when it is center-fed as in Fig. 6–7. Since a change of the feed point from the center of the antenna may change the current distribution, the radiation resistance R_0 is not the value which would be measured on a $3\lambda/4$ antenna or on any symmetrical antenna whose length is not an odd number of $\lambda/2$. However, R_0 can be easily transformed to the value which would appear across the terminals of the transmission line connected at the center of the antenna.

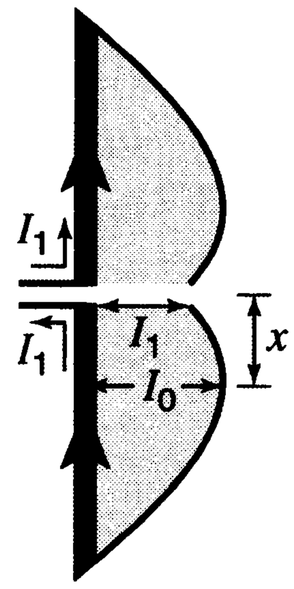


Figure 6–11
Relation of current I_1 at transmission-line terminals to current I_0 at current maximum.

This may be done by equating (6–5–3) to the power supplied by the transmission line, given by $I_1^2 R_1/2$, where I_1 is the current amplitude at the terminals and R_1 is the radiation resistance at this point (see Fig. 6–11). Thus,

$$\frac{I_1^2}{2} R_1 = \frac{I_0^2}{2} R_0 \tag{1}$$

where R_0 is the radiation resistance calculated at the current maximum. Thus, the radiation resistance appearing at the terminals is

$$R_1 = \left(\frac{I_0}{I_1}\right)^2 R_0 \tag{2}$$

The current I_1 at a distance x from the nearest current maximum, as shown in Fig. 6–11, is given by

$$I_1 = I_0 \cos \beta x \tag{3}$$

where

I_1 = terminal current
I_0 = maximum current

Therefore, (2) can be expressed as

$$R_1 = \frac{R_0}{\cos^2 \beta x} \tag{4}$$

When $x = 0$, $R_1 = R_0$; but when $x = \lambda/4$, $R_1 = \infty$ if $R_0 \neq 0$. However, the radiation resistance measured at a current minimum ($x = \lambda/4$) is not infinite as would be calculated from (4), since an actual antenna is not infinitesimally thin and the current at a minimum point is not zero. Nevertheless, the radiation resistance at a current minimum may in practice be very large, i.e., thousands of ohms.

6–7 TWO HALF-WAVE DIPOLES: BROADSIDE AND END-FIRE (THE W8JK ARRAY) (两个半波偶极子：边射和端射)

As an introduction to larger arrays in later chapters, we consider here two simple arrays each using two $\lambda/2$ thin, linear dipoles.

First, the case of two horizontal $\lambda/2$ dipoles fed with equal *in-phase* currents and stacked one above the other, as in Fig. 6–12*a*, at a spacing of $\lambda/2$, the *broadside array*.

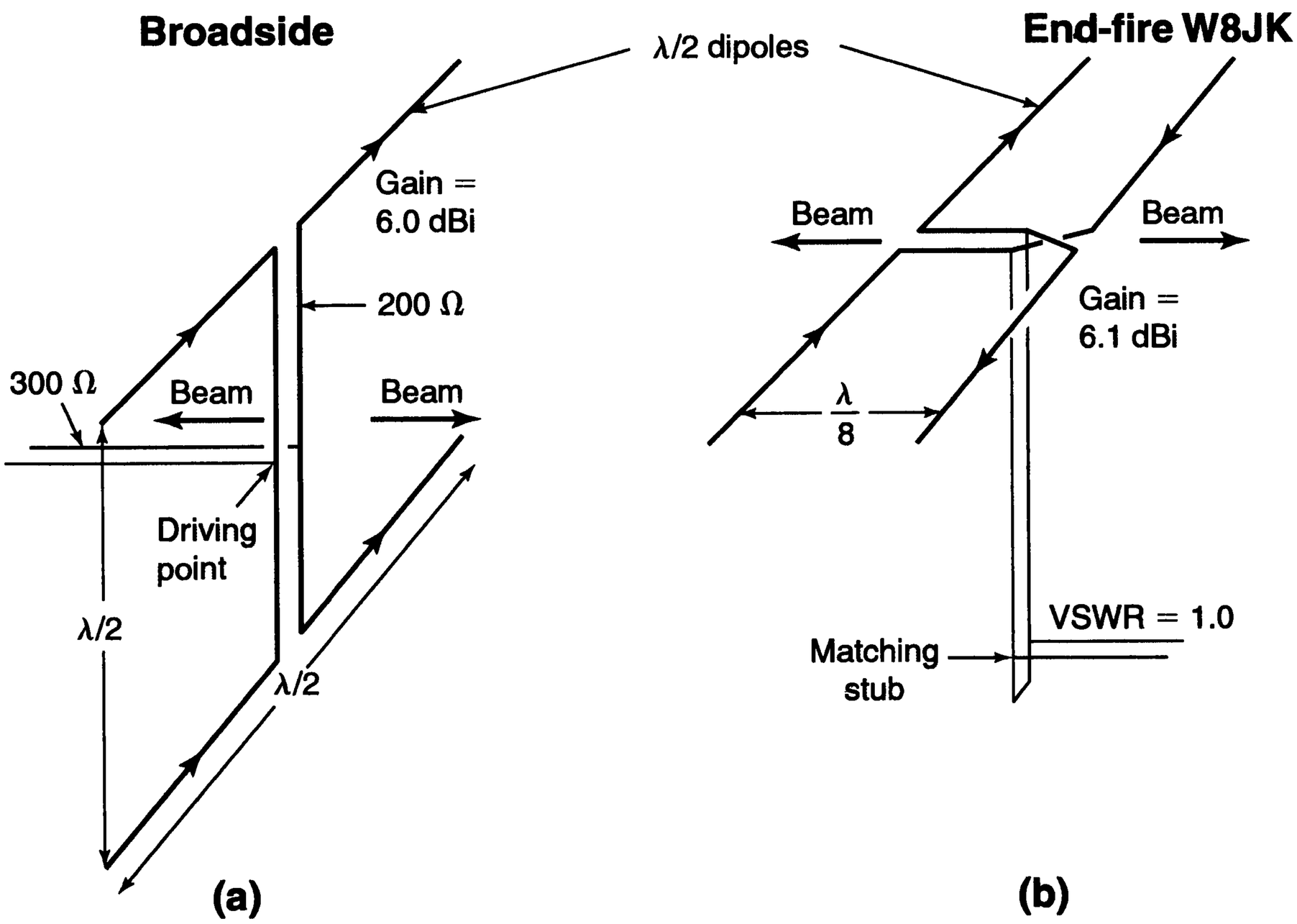


Figure 6–12
Arrays of two $\lambda/2$ dipoles arranged broadside in (*a*) and end-fire in (*b*).

Second, the case of two horizontal dipoles fed with equal *opposite*-phase currents situated side by side, as in Fig. 6–12*b*, at a spacing of $\lambda/8$, the *end-fire (W8JK) array*.

Case 1. Broadside Array.

In this configuration the array is fed by a twin-line which joins the two dipoles, as in Fig. 6–12*a*. The pattern in the horizontal (x-y) plane is that of a single dipole.

The self-impedance of an isolated $\lambda/2$ dipole is given by (13–5–7) as

$$Z = 73 + j43\ \Omega$$

due to the presence of the other dipole there is also a mutual impedance given by

$$Z_m = -13 - j29\ \Omega$$

so that the total impedance is

$$Z = 73 - 13 + j(43 - 29) = 60 + j14\ \Omega$$

Tuning out the reactance with a series capacitor, the 60 Ω is transformed by a $\lambda/4$ of 200 Ω twin-line to an impedance

$$Z = 200^2/60 = 667\ \Omega$$

This appears at the junction halfway between the dipoles. With the lines from both dipoles connected in parallel, the junction or *driving point impedance* is $667/2 = 333\ \Omega$.

EXAMPLE **6–7.1 VSWR and Gain of Broadside Array**

If the junction point (Fig. 6–12*a*) connects to a 300-Ω twin-line going to the transmitter or receiver, find: (*a*) VSWR on the 300-Ω line, (*b*) the gain of the array over a single $\lambda/2$ dipole and (*c*) the gain over an isotropic source.

■ **Solution**

The reflection coefficient is

$$\rho_v = (333 - 300)/(333 + 300) = 0.052$$

and

$$\text{VSWR} = (1 + 0.052)/(1 - 0.052) = 1.11 \qquad \textit{Ans. (a)}$$

The field gain over $\lambda/2$ dipole $= \sqrt{(2 \times 73)/(73 - 13)} = 1.56$ or 3.9 dB *Ans. (b)*
Gain over isotropic source $= 1.56^2 \times 1.64 = 4.0$ or 6.0 dBi *Ans. (c)*

Instead of feeding the array with 300-Ω twin-line, the array could be fed with a coaxial line of convenient impedance via a balun (see Fig. 23–20).

THE CLOSELY-SPACED END-FIRE (W8JK) ARRAY STORY

by John Kraus

In 1932 when I joined the Institute of Radio Engineers (IRE), the fore-runner of the IEEE, I was completing my Ph.D. degree on the propagation of 60 MHz waves in and around Ann Arbor, Michigan. FM and TV were still many years in the future and no one except an occasional amateur was using such "high frequencies." The following year I received my Ph.D. degree from the University of Michigan, publishing my dissertation in the *Proceedings of the Institute of Radio Engineers*.

I read every issue with interest. On opening the January 1937 *Proceedings* I delved into a monumental treatise on "Directional Antennas" by George H. Brown (1) of RCA. Buried deep in the article was, to me, an astonishing calculation which indicated that parallel linear dipoles with spacings of $\lambda/8$ or less had higher gains than the customary larger spacings.

Within one week of the time I received my *Proceedings* I had designed and built an array of 4 close-spaced $\lambda/2$ dipoles at my amateur station W8JK. Operating at a wavelength of 20 m, the array was phenomenally effective. I published the design and in subsequent articles extended it to an entire family of close-spaced arrays. The antennas outperformed all others. I called them "flat-top beams" but everyone else called them W8JK arrays after my amateur station call sign. They were soon in use by thousands of amateur and commercial short-wave stations world-wide (Kraus-1, 6).

In 1937 close spacing was a new and revolutionary concept. In George Brown's autobiography he states (Brown-2):

> Ironically, the particular portion of my paper which John Kraus used so effectively was a small paper which I submitted to the *Proceedings* in 1932 only to have it rejected by a reviewer who denied its validity. When I prepared "Directional Antennas" I tucked this older material into the middle of this bulky manuscript on the assumption that the reviewer would not notice it.

George Brown's ruse worked and the world finally learned of his idea but only after it had languished in obscurity for five years! Brown's new concept was to calculate gain on the basis of *constant power* to the antenna instead of assuming constant currents.

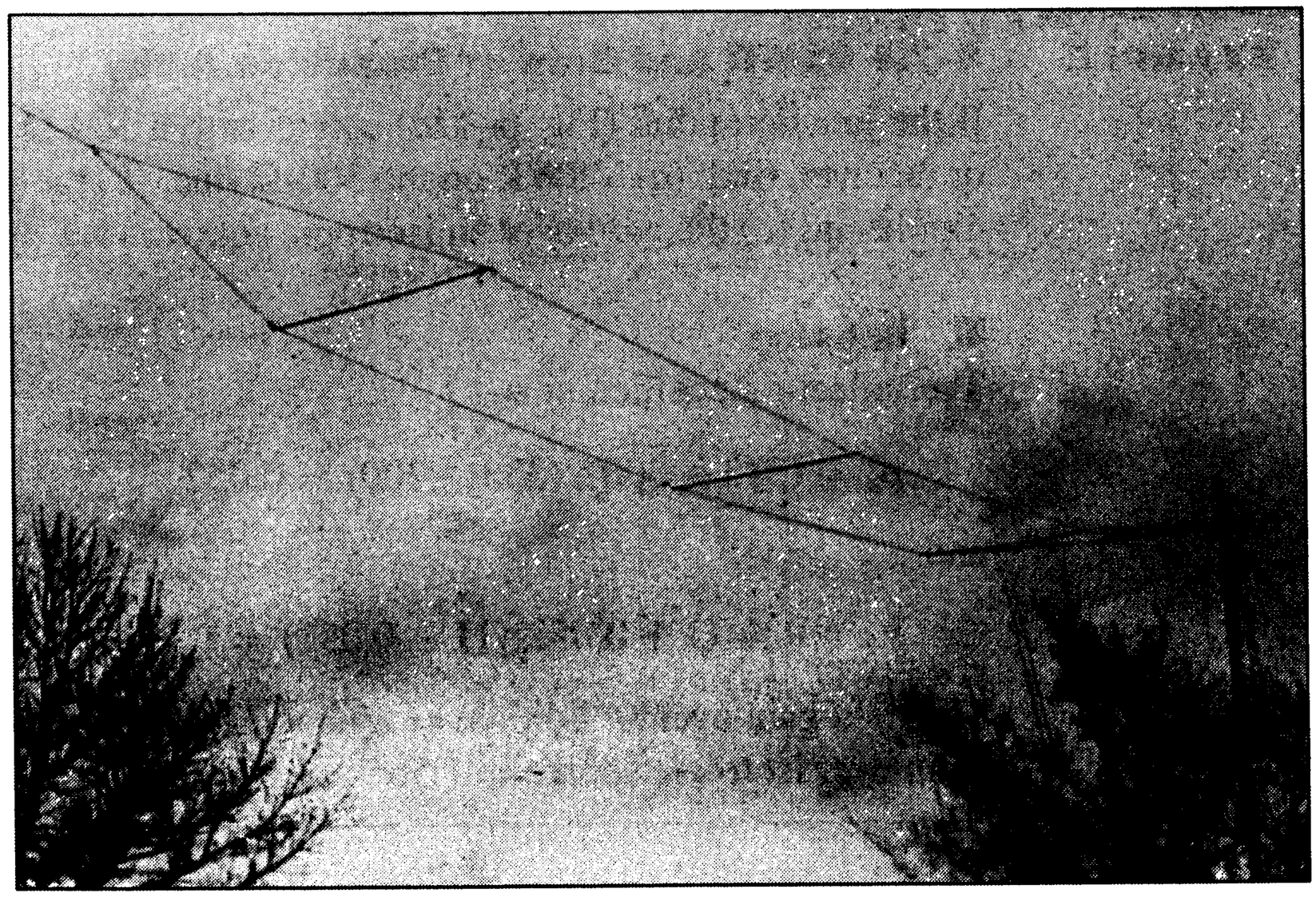


Figure 6–13
The first W8JK array (1937) with two full-wavelength (4 $\lambda/2$) dipoles suspended between two 15-m-tall poles for 14-MHz (20-m) operation. The array had a gain of 8.0 dBi. It proved especially effective for long-distance (DX) communication.

Case 2. End-Fire (W8JK) Array

With two $\lambda/2$ dipoles arranged as in Fig. 6–12*b* and fed equal currents in opposite phase the gain is 4.4 dBi when the dipoles are spaced $\lambda/2$ apart. However, with the spacing reduced to $\lambda/8$ the gain is greater, as will be shown.

The mutual impedance of the two dipoles at $\lambda/8$ spacing, as given in Chap. 13, is

$$Z_m = 64.4 - j5\ \Omega$$

so that the total dipole impedance is

$$Z = 73 + j43 - (64.4 - j5) = 9 + j48\ \Omega \tag{1}$$

The array is fed from the center point of a twin-line which has a cross-over so that the dipoles are fed in opposite phase.

EXAMPLE **6–7.2 Gain of End-Fire (W8JK) Array**

Find the gain of the $\lambda/8$ spacing W8JK array.

■ **Solution**

From (16–5–8), field gain over $\lambda/2$ dipole,

$$D = \sqrt{(2 \times 73)/(73 - 64.4)}\sin(\pi/8) = 1.57 \text{ or } 3.9\text{ dB}$$

$$\text{Gain over isotropic source} = 1.57^2 \times 1.64 = 4.04 \text{ or } 6.1\text{ dBi} \qquad \textit{Ans.}$$

This gain assumes 100% efficiency. As the dipole spacing is reduced, the dipole currents increase for a constant power input so that *efficiency* becomes a consideration (see Sec. 16–5). Also with smaller spacing the Q of the array increases, which reduces the usable bandwidth. However, when properly constructed, the W8JK array has proven highly effective for many applications.

The W8JK array may be fed with twin-lines of convenient value and a matching stub as shown in Fig. 6–12*b*. Thus, the array can be matched to the line going to the transmitter or receiver with a VSWR = 1.0. It may also be fed by coaxial line via a balun (see Fig. 23–20).

For a loss resistance of 1 Ω, the gain is 0.3 dB less, but in a well-constructed array using copper wire of suitable gauge, the loss resistance should be much less than an ohm.

With both broadside and end-fire (W8JK) arrays at the same maximum height as in Fig. 6–12, the W8JK array is $\lambda/4$ higher above ground than the center of the broadside array. This results in a lower angle of maximum radiation for the W8JK array, making it especially effective for long-distance communication. The first one constructed is shown in Fig. 6–13.

6–8 FIELDS OF A THIN LINEAR ANTENNA WITH A UNIFORM TRAVELING WAVE（载有均匀行波的细直天线的场）

A sinusoidal current distribution may be regarded as the standing wave produced by two uniform (unattenuated) traveling waves of equal amplitude moving in opposite directions along the antenna. If, however, only one such wave is present on the antenna, the current distribution is uniform. The amplitude is a constant along the antenna, and the phase changes linearly with distance as suggested by Fig. 6–14.

The condition of a uniform traveling wave on an antenna is one of considerable importance, as this condition may be approximated in a number of antenna systems. For example, a single-wire antenna terminated in its characteristic impedance, as in Fig. 6–15*a*, may have essentially a uniform traveling wave.[1] This type of antenna is often referred to as a Beverage or wave antenna. A terminated rhombic antenna (Fig. 6–15*b*) may also have essentially a single traveling wave. The Beverage and rhombic antennas are discussed further in Chap. 16. Other types of antennas that have, in the first approximation, a single outgoing traveling wave, are a long monofilar axial-mode helical antenna and a long, thick linear antenna as illustrated in Fig. 6–15*c* and *d*. These antennas have no terminating impedance but behave in a similar way to terminated antennas. Thus, the thick linear conductor has a current distribution similar to a thin terminated linear conductor, and the patterns are similar if the conductor diameter is not too large. The results for a traveling wave on a linear

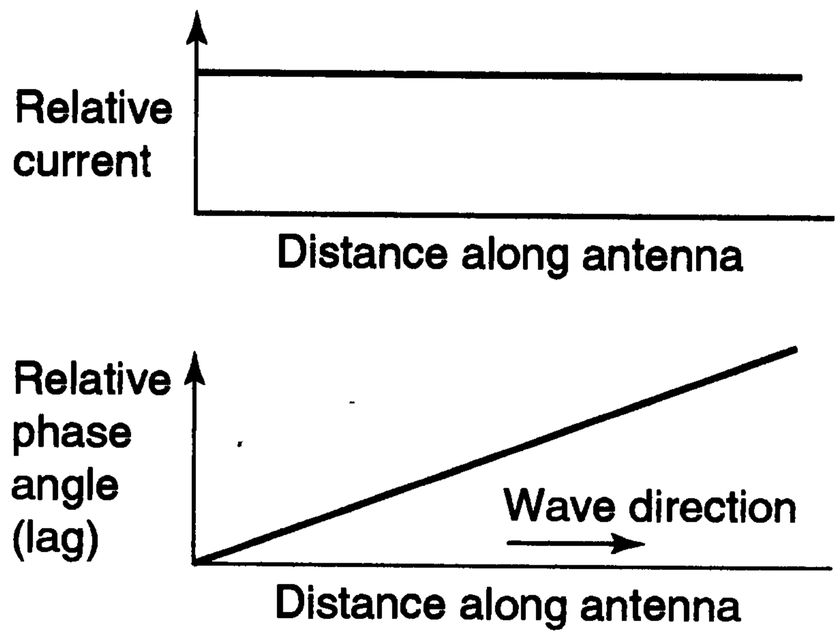


Figure 6–14
Current amplitude and phase relations along an antenna carrying a single uniform traveling wave.

[1] Since the fields of an antenna are not confined to the immediate vicinity of the antenna, it is not possible to provide a nonreflecting termination with a lumped impedance. However, a lumped impedance may greatly reduce reflections at the termination.

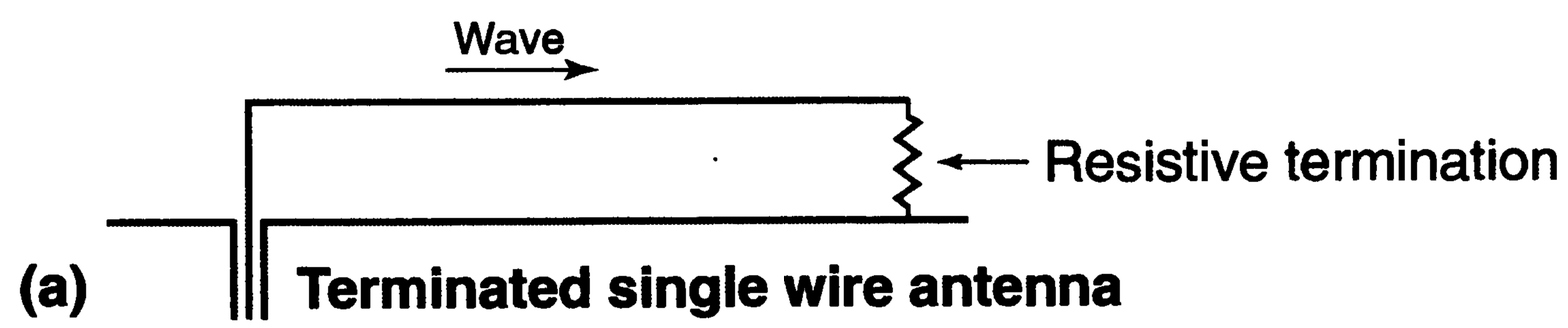


Figure 6–15
Various antennas having essentially a single traveling wave.

conductor can be applied to a helix, as shown in Chap. 8, by considering that the helix consists of a number of short linear segments. On the linear antennas, the phase velocity of the traveling wave is substantially equal to the velocity of light. However, the phase velocity along the conductor of a monofilar axial-mode helical antenna may differ appreciably from the velocity of light. Hence, to make the results applicable to any of the antenna types shown in Fig. 6–15, the fields from an antenna with a traveling wave will be developed for the general case where the phase velocity v of the wave along the conductor may have any arbitrary value (Alford-1; Kraus-7; Grosskopf-1).

Proceeding now to find the field radiated by a traveling wave on a thin linear conductor, let us consider a conductor of length b coincident with the z axis and with one end at the origin of a cylindrical coordinate system (ρ, ξ, z) as in Fig. 6–16. It is assumed that a single, uniform traveling wave is moving to the right along the conductor.

Since the current is entirely in the z direction, the magnetic field has but one component H_ξ. The ξ direction is normal to the page at P in Fig. 6–16, and its positive sense is outward from the page. The magnetic field H_ξ can be obtained from the Hertz vector $\mathbf{\Pi}$. Since the current is entirely in the z direction, the Hertz vector has only a z component. Thus,

$$H_\xi = j\omega\varepsilon(\nabla \times \mathbf{\Pi})_\xi = -j\omega\varepsilon \frac{\partial \Pi_z}{\partial \rho} \tag{1}$$

where Π_z is the z component of the retarded Hertz vector at the point P, as given by

$$\Pi_z = \frac{1}{4\pi j\omega\varepsilon} \int_0^b \frac{|I|}{r}\, dz_1 \tag{2}$$

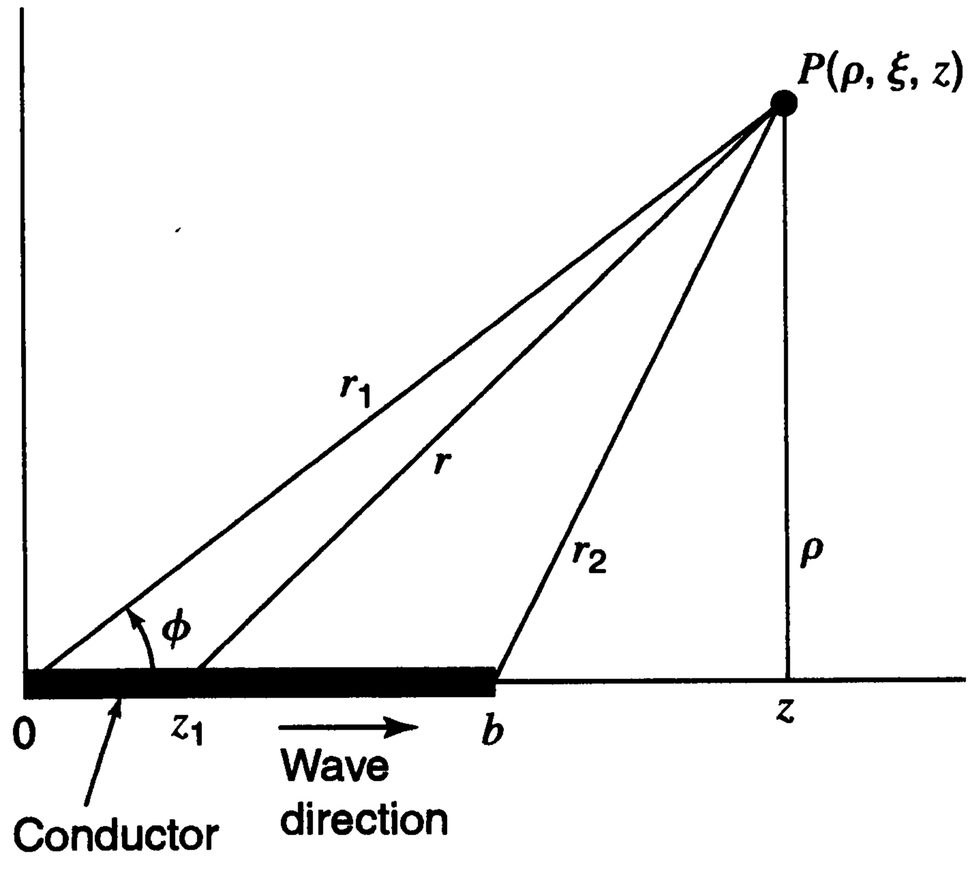


Figure 6–16
Relation of conductor of length b with single traveling wave to cylindrical coordinate system.

where

$$[I] = I_0 \sin \omega\left(t - \frac{r}{c} - \frac{z_1}{v}\right) \tag{3}$$

where z_1 = a point on the conductor and

$$v = pc \qquad \text{or} \qquad p = \frac{v}{c} \tag{4}$$

In (4), p is the ratio of the velocity along the conductor v to the velocity of light c. This ratio will be called the *relative phase velocity*.

All the conditions required for calculating the magnetic field due to a single traveling wave on the linear conductor are contained in the relations (1) through (4). That is, if $[I]$ in (3) is substituted into (2) and Π_z from this equation into (1) and the indicated operations performed, we obtain the field H_ξ, as follows:

$$H_\xi = \frac{I_0 p}{2\pi r_1}\left\{\frac{\sin\phi}{1 - p\cos\phi}\sin\left[\frac{\omega b}{2pc}(1 - p\cos\phi)\right]\right\} \Bigg/ \left[\omega\left(t - \frac{r_1}{c}\right) - \frac{\omega b}{2pc}(1 - p\cos\phi)\right] \tag{5}$$

Equation (5) gives the instantaneous magnetic field at large distances from the linear antenna carrying a single traveling wave of amplitude I_0, in terms of the distance r_1, direction angle ϕ, relative phase velocity p, radian frequency ω, conductor length b, time t and velocity of light c. The distant or far electric field E_ϕ is obtained from H_ξ by $E_\phi = H_\xi Z$, where $Z = 377\ \Omega$.

In (5) the shape of the field pattern is given by the expression in the braces { }. The expression indicated as an angle $\angle$ gives the phase of the field referred to the origin of the coordinates (see Fig. 6–16) as the phase center. The relative phase pattern at a constant distance is given by the right-hand term, $[\omega b/(2pc)](1 - p\cos\phi)$.

Several examples will now be considered to illustrate the nature of the field patterns obtained on linear conductors carrying a uniform traveling wave.

EXAMPLE **6–8.1 Linear λ/2 Antenna**

Let us consider a linear antenna, $\lambda/2$ long as measured in free-space wavelengths. Thus, assuming that $p = 1$, the phase velocity along the antenna is equal to that of light and the pattern calculated from (5) is as shown by Fig. 6–17*a*. The difference between this pattern and that for a linear $\lambda/2$ antenna with a sinusoidal current distribution or standing wave (Fig. 6–9*a*) is striking. The lobes are sharper and also tilted forward in the case of the traveling wave antenna (Fig. 6–17*a*). The tilt is in the direction of the traveling wave. The *tilt angle* τ of the direction of maximum radiation is 25° and the beamwidth between half-power points is about 60°. This is in contrast to $\tau = 0$ and a beamwidth of 78° for the $\lambda/2$ antenna with a sinusoidal current distribution or standing wave.

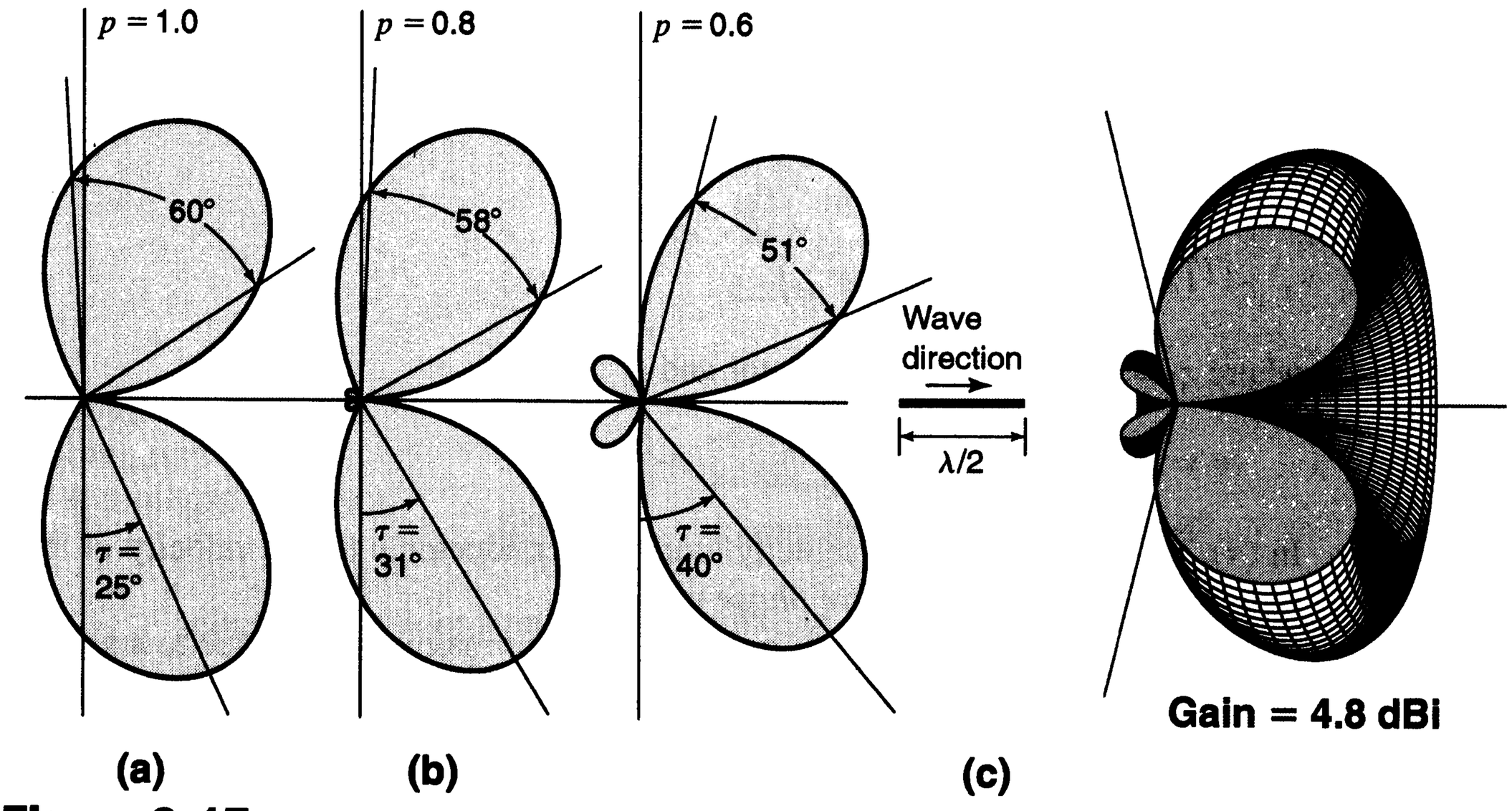


Figure 6–17
Far-field patterns of linear $\lambda/2$ antenna carrying a uniform traveling wave (to right) for three conditions of relative phase velocity ($p = 1.0$, 0.8 and 0.6). The *tilt angle* τ and the half-power beamwidths are indicated for each pattern.

As the phase velocity of the traveling wave on the $\lambda/2$ antenna is reduced, the tilt angle is increased and the beamwidth reduced further, as illustrated by the patterns of Fig. 6–17*b* and *c*, which are for the cases of $p = 0.8$ and $p = 0.6$, respectively.

EXAMPLE **6–8.2 Linear Antenna 5λ Long**

The field pattern for a 5λ linear antenna with a single traveling wave is presented in Fig. 6–18 for the case where $p = 1$ (that is, $v = c$). This pattern is typical of those for long, terminated antennas, the radiation being beamed forward in a cone having the antenna as its axis. The tilt angle for this antenna is about 68°.

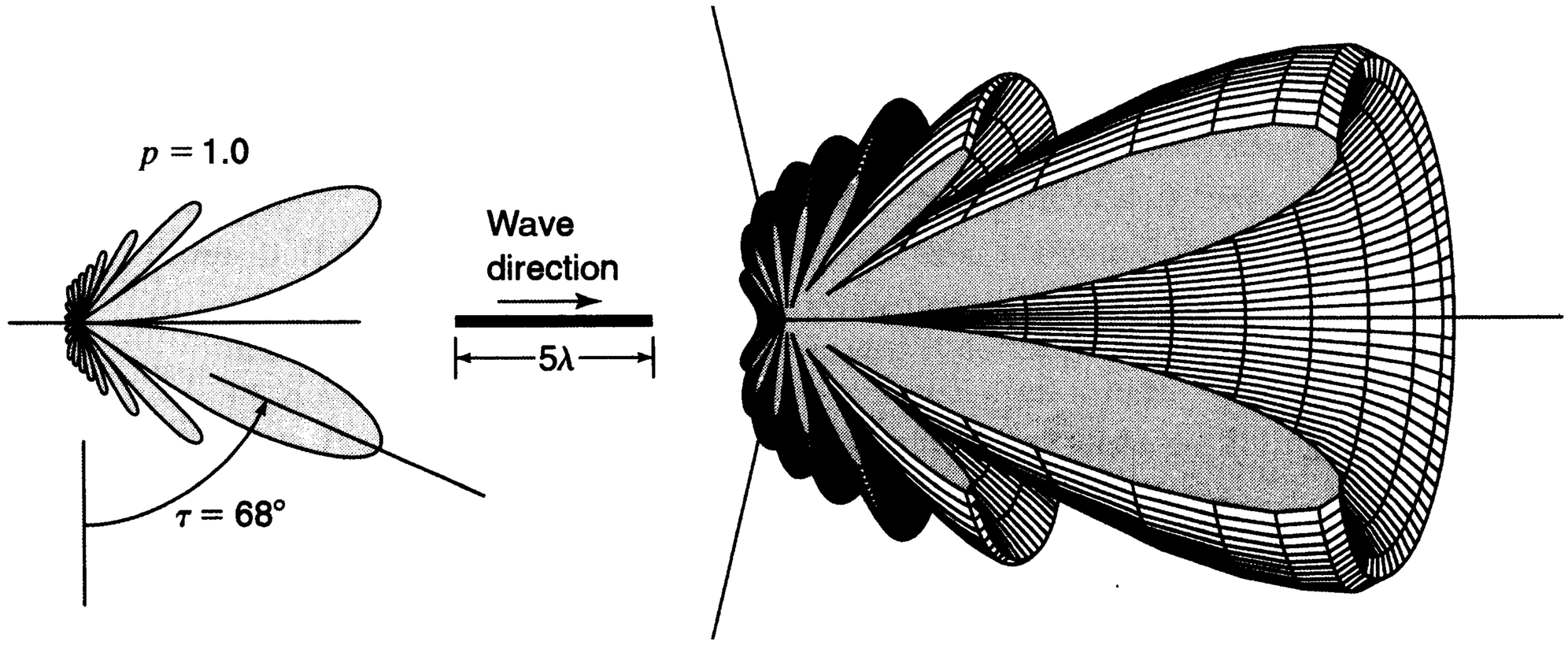


Figure 6–18
Far-field pattern of 5λ antenna carrying a uniform traveling wave ($\rho = 1$). Polar plot at left, three-dimensional at right. Gain = 10.7 dBi.

EXAMPLE **6–8.3 Linear Antenna λ/2 to 25λ Long**

As the length of the antenna is increased the tilt angle increases further, reaching about 78° (12° from antenna) when the length is 20λ for $p = 1$. The variation of the angle of the conical beam from the antenna is shown in Fig. 6–19 as a function of the antenna length for a wave traveling at the velocity of light ($p = 1$). Note that α in Fig. 6–19 is the complement of the tilt angle τ, that is, $\alpha = 90° - \tau$.

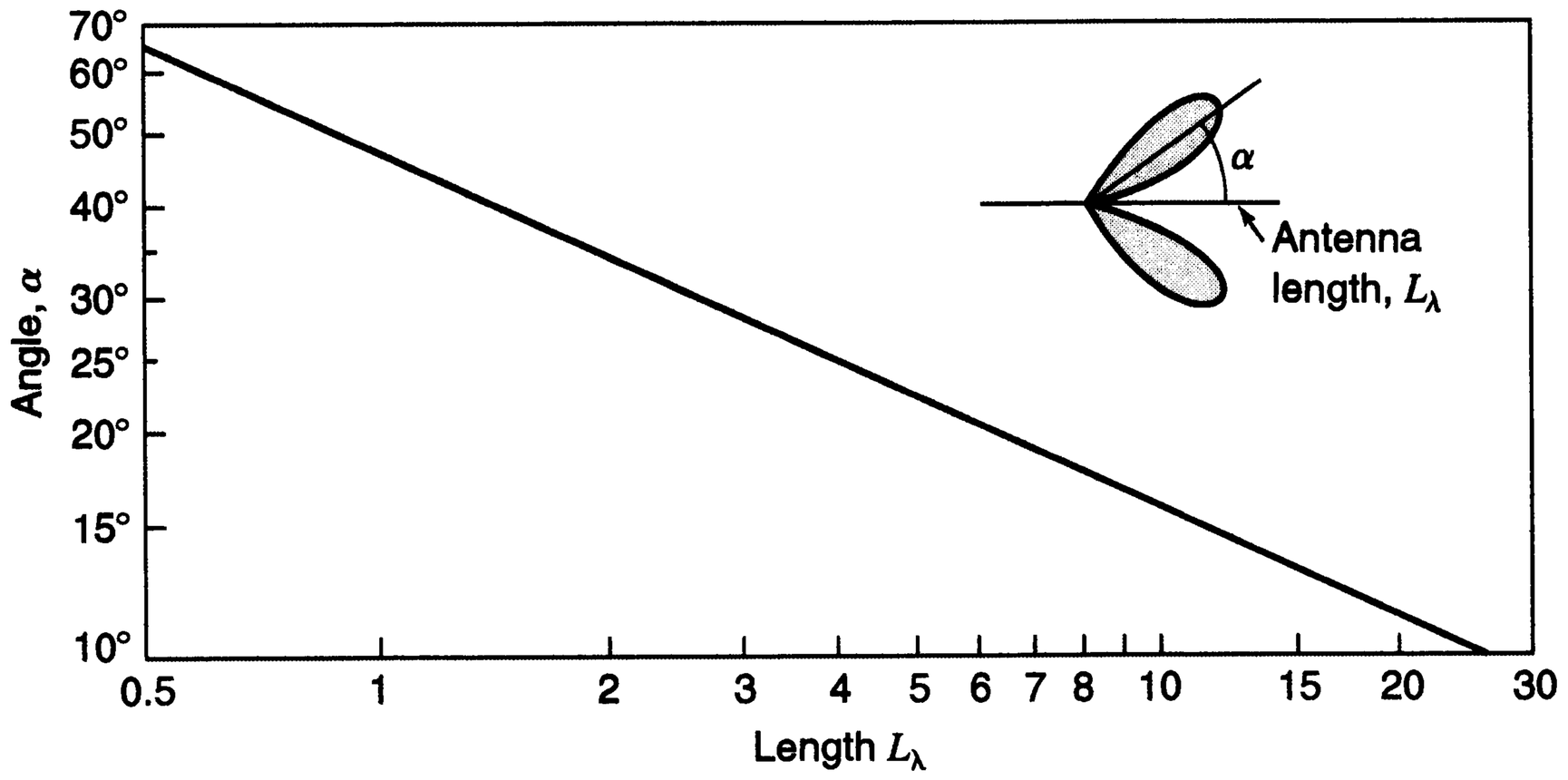


Figure 6–19
Angle α of main beam maximum from a linear traveling wave antenna as a function of antenna length in wavelengths (L_λ) with $v = c$ ($p = 1$) for antennas λ/2 to 25λ long.

More on traveling wave antennas is given in Chap. 17. Table 6–2 presents patterns, directivities and gains (dBi) for basic antenna types and simple arrays (Kraus-8).

Table 6–2 Basic antenna types (left). Arrays of two (center) and of four λ/2 dipoles (right). Note that the highest gain for two λ/2 dipoles is obtained with the most compact configureation (used in the W8JK antenna).

Antenna	Field pattern	Directivity	Gain over isotropic (dBi)
Isotropic .hypothetical/		1.00	0
Short dipole HPBW = 90°		1.50	1.76
λ/2 dipole HPBW = 78°		1.64	2.15
Hemispherical radiator		2.00	3.0
Short monopole		3.00	4.8
λ/4 monopole		3.28	5.2
λ/2 monopole		4.80	6.8
Small loop HPBW = 90°		1.50	1.76
Square loop (circumference = λ)		1.9	2.8

Basic types

Antenna	Field pattern	Directivity	Gain over one element (dB)	Gain over isotropic (dBi)
Two λ/2 dipoles collinear in-phase δ = 0°	HPBW = 47°	2.40	1.7	3.8
Two λ/2 dipoles S = λ/2 δ = 0°	S, HPBW = 57°	1.56	3.8	6.0
Two λ/2 dipoles S = λ/8 δ = 180° W8JK	HPBW = 90°	1.57	3.9	6.1
Two λ/2 dipoles S = λ/4 δ = 180°		1.51	3.6	5.7
Two λ/2 dipoles S = λ/4 δ = 90°		1.41	3.0	5.2
Two λ/2 dipoles S = λ/2 δ = 180°	S	1.30	2.3	4.5

Two λ/2 dipoles

Antenna	Field pattern	Directivity	Gain over one element (dB)	Gain over isotropic (dBi)
Four λ/2 dipoles collinear in-phase δ = 0°		4.38	4.3	6.4
Four λ/2 dipoles S = λ/2 δ = 0°	S	8.47	7.1	9.3
Four λ/2 dipoles S = λ/4 δ = 90°	ORDINARY END-FIRE	5.59	5.3	7.5
Four λ/2 dipoles S = λ/4 δ = 45°	INCREASED DIRECTIVITY END-FIRE	9.29	7.5	9.7

Four λ/2 dipoles

REFERENCES（参考文献）

Alford, A. (1): "A Discussion of Methods Employed in Calculations of Electromagnetic Fields of Radiating Conductors, "*Elec. Commun.*, **15,** 70–88, July 1936. Treats case where velocity is equal to light.

Brown, G. H. (1): "Directional Antennas," *Proc. IRE,* **25,** 78–145, January 1937.

Brown, G. H. (2): "And Part of Which I Was," Angus Cupar, 117 Hunt Drive, Princeton, NJ 08540, 1982.

Grosskopf, J. (1): "Über die Verwendung zweier Lösungsansätze der Maxwellschen Gleichungen bei der Berechnung der electromagnetischen Felder strahlender Leiter," *Hochfrequenztechnik und Electroakustik,* **49,** 205–211, June 1937. Treats case where velocity is equal to light.

Kraus, J. D. (1): "A Small but Effective Flat Top Beam Antenna," *Radio,* no. 213, 56–58, March 1937 and no. 216, 10–16, June 1937.

Kraus, J. D. (2): "Rotary Flat Top Beam Antennas," *Radio,* no. 222, 11–16, December 1937.

Kraus, J. D. (3): "Directional Antennas with Closely-Spaced Elements," *QST,* **22,** 21–23, January 1938.

Kraus, J. D. (4): "Characteristics of Antennas with Closely-Spaced Elements," *Radio,* no. 236, 9–19, February 1939.

Kraus, J. D. (5): "Antenna Arrays with Closely-Spaced Elements," *Proc. IRE,* **28,** 76–84, February 1940.

Kraus, J. D. (6): "W8JK 5-Band Rotary Beam Antenna," *QST,* **54,** 11–14, July 1970.

Kraus, J. D. (7), and J. C. Williamson: "Characteristics of Helical Antennas Radiating in the Axial Mode," *J. App. Phys.*, **19,** 87–96, January 1948. Treats general case.

Kraus, J. D. (8): "Antennas, Our Electronic Eyes and Ears," *Microwave Jour.*, 77–92, January 1989.

PROBLEMS（习题）

***6–2–1** （电偶极子）

Electric dipole. (*a*) Two equal static electric charges of opposite sign separated by a distance L constitute a static electric dipole. Show that the electric potential at a distance r from such a dipole is given by

$$V = \frac{QL\cos\theta}{4\pi\varepsilon r^2}$$

where Q is the magnitude of each charge and θ is the angle between the radius r and the line joining the charges (axis of dipole). It is assumed that r is very large compared to L. (*b*) Find the vector value of the electric field E at a large distance from a static electric dipole by taking the gradient of the potential expression in part (*a*).

***6–2–2** （短偶极子的场）

Short dipole fields. A dipole antenna of length 5 cm is operated at a frequency of 100 MHz with terminal current $I_o = 120$ mA. At time $t = 1$ s, angle $\theta = 45°$, and distance $r = 3$ m, find (*a*) E_r, (*b*) E_θ, and (*c*) H_ϕ.

6–2–3 （短偶极子远场）

Short dipole far fields. For the dipole antenna of Prob. 6–2–2, at a distance $r = 1$ m, use the general expressions of Table 6–1 to find (*a*) E_r, (*b*) E_θ, and (*c*) H_ϕ. Compare these results to those obtained using the far-field expressions of Table 6–1.

***6–2–4** （短偶极子似稳场）

Short dipole quasi-stationary fields. For the dipole antenna of Prob. 6–2–2, at a distance $r =$ 100 m, use the general expressions of Table 6–1 to find (*a*) E_r, (*b*) E_θ, and (*c*) H_ϕ. Compare these results to those obtained using the far-field expressions of Table 6–1.

6–2–5 （短偶极子的场）

Short dipole fields. At what distance from a 1-m dipole antenna operated at 15 MHz are the amplitudes of the fields E_θ and H_ϕ within 1 percent of their far-field values?

***6–3–1** （各向同性天线，辐射电阻）

Isotropic antenna. Radiation resistance. An omnidirectional (isotropic) antenna has a field pattern given by $E = 10I/r$ (V m^{-1}), where I = terminal current (A) and r = distance (m). Find the radiation resistance.

*6-3-2 (短偶极子的功率) **Short dipole power.** (*a*) Find the power radiated by a 10-cm dipole antenna operated at 50 MHz with an average current of 5 mA. (*b*) How much (average) current would be needed to radiate power of 1 W?

6-3-3 (短偶极子的辐射电阻) **Short dipole radiation resistance.** (*a*) Find the radiation resistance of a 10-m dipole antenna operated at 500 kHz (assume $I_{av} = \frac{1}{2} I_o$). (*b*) How long must this antenna be for a radiation resistance of 1 Ω?

6-3-4 (短偶极子) **Short dipole.** For a thin center-fed dipole $\lambda/15$ long find (*a*) directivity D, (*b*) gain G, (*c*) effective aperture A_e, (*d*) beam solid Ω_A and (*e*) radiation resistance R_r. The antenna current tapers linearly from its value at the terminals to zero at its ends. The loss resistance is 1 Ω.

*6-3-5 (圆锥波瓣图) **Conical pattern.** An antenna has a conical field pattern with uniform field for zenith angles (θ) from 0 to 60° and zero field from 60 to 180°. Find exactly (*a*) the beam solid angle and (*b*) directivity. The pattern is independent of the azimuth angle (ϕ).

6-3-6 (圆锥波瓣图) **Conical pattern.** An antenna has a conical field pattern with uniform field for zenith angles (θ) from 0 to 45° and zero field from 45 to 180°. Find exactly (*a*) the beam solid angle, (*b*) directivity and (*c*) effective aperture. (*d*) Find the radiation resistance if the $E = 5$ V m^{-1} at a distance of 50 m for a terminal current $I = 2$ A (rms). The pattern is independent of the azimuth angle (ϕ).

*6-3-7 (θ和ϕ的定向波瓣图) **Directional pattern in θ and ϕ.** An antenna has a uniform field pattern for zenith angles (θ) between 45 and 90° and for azimuth (ϕ) angles between 0 and 120°. If $E = 3$ V m^{-1} at a distance of 500 m from the antenna and the terminal current is 5 A, find the radiation resistance of the antenna. $E = 0$ except within the angles given above.

*6-3-8 (θ和ϕ的定向波瓣图) **Directional pattern in θ and ϕ.** An antenna has a uniform field $E = 2$ V m^{-1} (rms) at a distance of 100 m for zenith angles between 30 and 60° and azimuth angles ϕ between 0 and 90° with $E = 0$ elsewhere. The antenna terminal current is 3 A (rms). Find (*a*) directivity, (*b*) effective aperture and (*c*) radiation resistance.

*6-3-9 (带有后瓣的定向波瓣图) **Directional pattern with back lobe.** The field pattern of an antenna varies with zenith angle (θ) as follows: E_n ($= E_{\text{normalized}}$) $= 1$ between $\theta = 0°$ and $\theta = 30°$ (main lobe), $E_n = 0$ between $\theta = 30°$ and $\theta = 90°$ and $E_n = 1/3$ between $\theta = 90°$ and $\theta = 180°$ (back lobe). The pattern is independent of azimuth angle (ϕ). (*a*) Find the exact directivity. (*b*) If the field equals 8 V m^{-1} (rms) for $\theta = 0°$ at a distance of 200 m with a terminal current $I = 4$ A (rms), find the radiation resistance.

6-3-10 (短偶极子) **Short dipole.** The radiated field of a short-dipole antenna with uniform current is given by $|E| = 30\beta l(I/r) \sin\theta$, where l = length, I = current, r = distance and θ = pattern angle. Find the radiation resistance.

6-3-11 (辐射电阻与波束范围的关系) **Relation of radiation resistance to beam area.** Show that the radiation resistance of an antenna is a function of its beam area Ω_A as given by

$$R_r = \frac{Sr^2}{I^2} \Omega_A$$

where

S = Poynting vector at distance r in direction of pattern maximum
I = terminal current.

*6-3-12 (辐射电阻) **Radiation Resistance.** An antenna measured at a distance of 500 m is found to have a far-field pattern of $|E| = E_o(\sin\theta)^{1.5}$ with no ϕ dependence. If $E_o = 1$ V/m and $I_o = 650$ mA, find the radiation resistance of this antenna.

6–4–1
（载有驻波的全波天线）

1λ antenna with standing wave. Calculate the field pattern in the plane of the 1λ antenna shown in Fig. P6–4–1. Assume that the current distribution on each wire is sinusoidal and that all currents are in-phase. Plot the pattern.

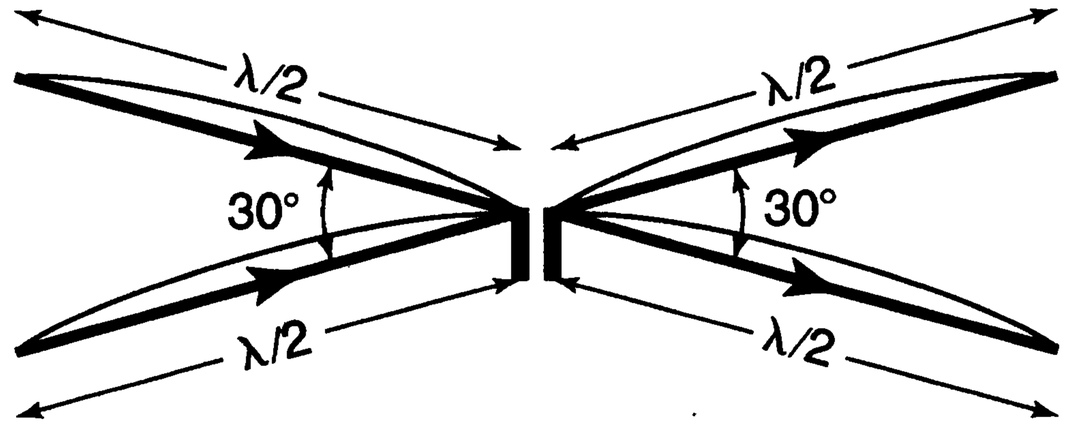


Figure P6–4–1
1λ antenna with standing wave.

***6–5–1**
（半波天线）

λ/2 antenna. Assume that the current is of uniform magnitude and in-phase along the entire length of a λ/2 thin linear element.
(*a*) Calculate and plot the pattern of the far field.
(*b*) What is the radiation resistance?
(*c*) Tabulate for comparison:

1. Radiation resistance of part (*b*) above
2. Radiation resistance at the current loop of a λ/2 thin linear element with sinusoidal in-phase current distribution
3. Radiation resistance of a λ/2 dipole calculated by means of the short dipole formula

(*d*) Discuss the three results tabulated in part (*c*) and give reasons for the differences.

6–6–1
（倍波天线）

2λ antenna. The instantaneous current distribution of a thin linear center-fed antenna 2λ long is sinusoidal as shown in Fig. P6–6–1.
(*a*) Calculate and plot the pattern of the far field.
(*b*) What is the radiation resistance referred to a current loop?
(*c*) What is the radiation resistance at the transmission-line terminals as shown?
(*d*) What is the radiation resistance λ/8 from a current loop?

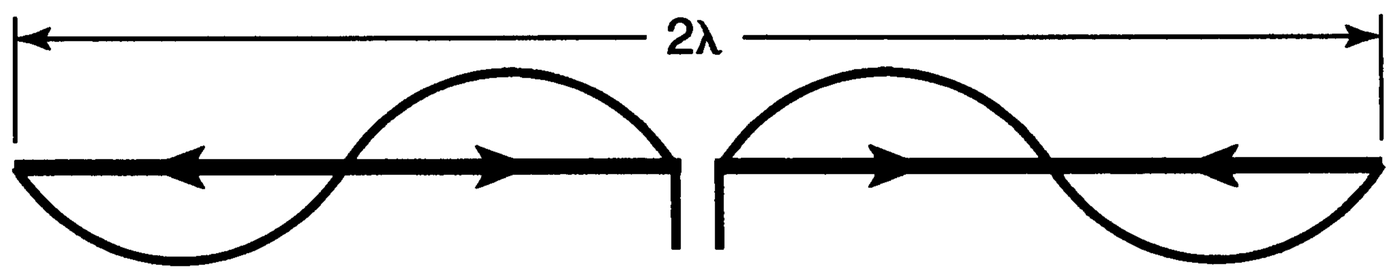


Figure P6–6–1
2λ antenna.

6–7–1
（梯次排列的两根半波天线）

λ/2 antennas in echelon. Calculate and plot the radiation-field pattern in the plane of two thin linear λ/2 antennas with equal in-phase currents and the spacing relationship shown in Fig. P6–7–1. Assume sinusoidal current distributions.

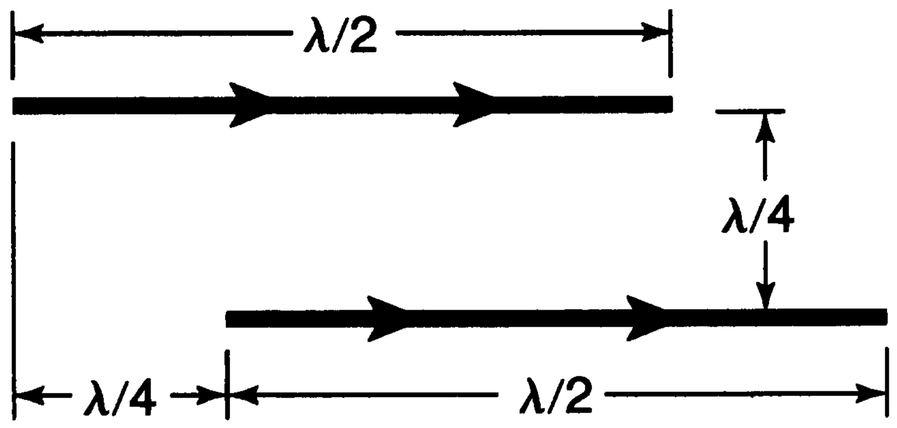


Figure P6–7–1
λ/2 antennas in echelon.

***6-8-1**
（载有行波的全波天线和10波长天线）

1λ and 10λ antennas with traveling waves. (*a*) Calculate and plot the far-field pattern in the plane of a thin linear element 1λ long, carrying a single uniform traveling wave for two cases of the relative phase velocity $p = 1$ and 0.5. (*b*) Repeat for the single case of an element 10λ long and $p = 1$.

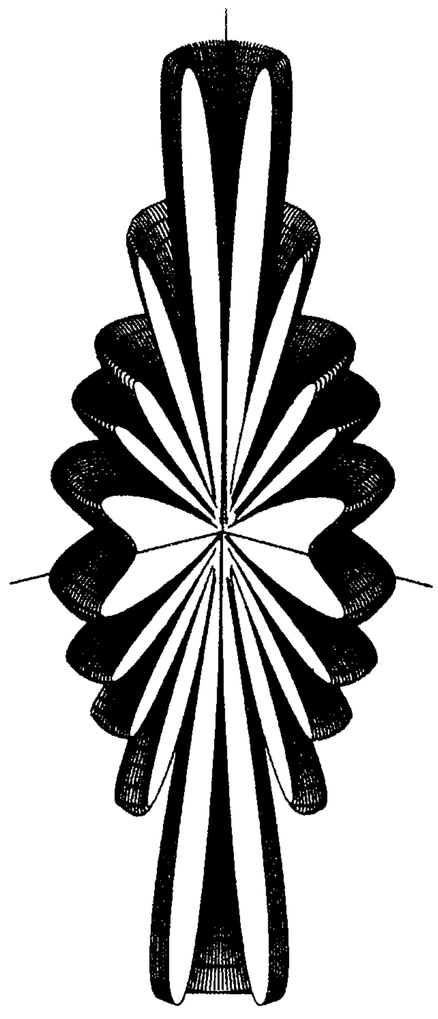
6-8-2
（波瓣图因子的等效）

Equivalence of pattern factors. Show that the field pattern of an ordinary end-fire array of a large number of collinear short dipoles as given by Eq. (5–6–8), multiplied by the dipole pattern $\sin\phi$, is equivalent to Eq. (6–8–5) for a long linear conductor with traveling wave for $p = 1$.

For computer programs, see Appendix C.

CHAPTER 7

The Loop Antenna
环天线



The topics in this chapter include:

- The small loop（小环）
- Small loop and short dipole compared（小环和短偶极子的比较）
- The loop antenna: General case（环天线的一般情况）
- Patterns of loop antennas（环天线的波瓣图）
- Radiation resistance and directivity of loops（环的辐射电阻和定向性）
- Square loops（方环）
- Loop efficiency, Q, bandwidth and SNR
  （环的效率、Q值、频带宽度和信噪比）
- The ferrite-loaded loop
  （铁氧体加载的环）

7–1 THE SMALL LOOP（电小环）

The field pattern of a small circular loop of radius a may be determined very simply by considering a square loop of the same area, that is,

$$d^2 = \pi a^2 \tag{1}$$

where d = side length of square loop as shown in Fig. 7–1.

It is assumed that the loop dimensions are small compared to the wavelength. It will be shown that the far-field patterns of circular and square loops of the same area are the same when the loops are small but different when they are large in terms of the wavelength.

If the loop is oriented as in Fig. 7–2, its far electric field has only an E_ϕ component. To find the far-field pattern in the yz plane, it is only necessary to consider two of the four small linear dipoles (2 and 4). A cross section through the loop in the yz plane is presented in Fig. 7–3. Since the individual small dipoles 2 and 4 are nondirectional in the yz plane, the field pattern of the loop in this plane is the same as that for two isotropic point sources as treated in Sec. 5–2. Thus,

$$E_\phi = -E_{\phi 0}\, e^{j\psi/2} + E_{\phi 0}\, e^{-j\psi/2} \tag{2}$$

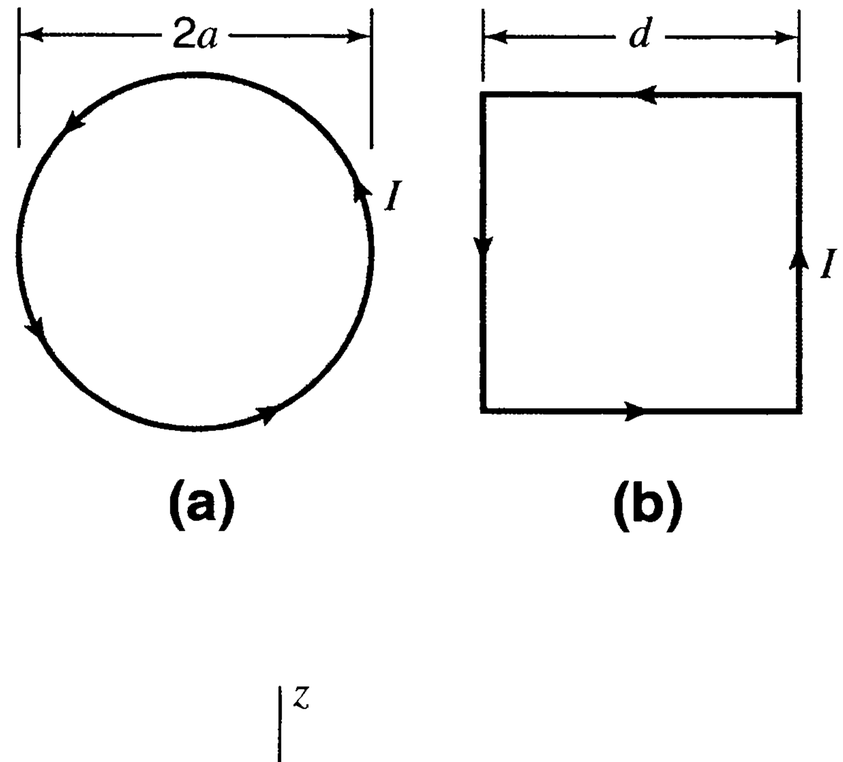


Figure 7–1
Circular loop (*a*) and square loop (*b*) of equal area.

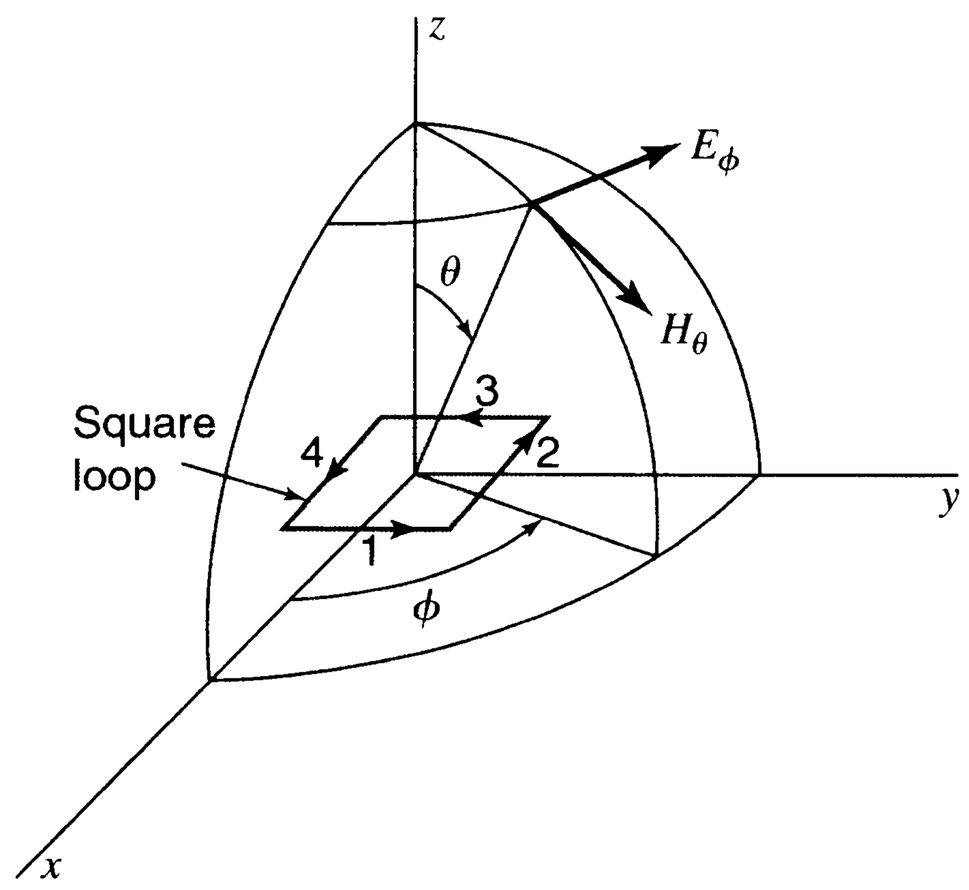


Figure 7–2
Relation of square loop to coordinates.

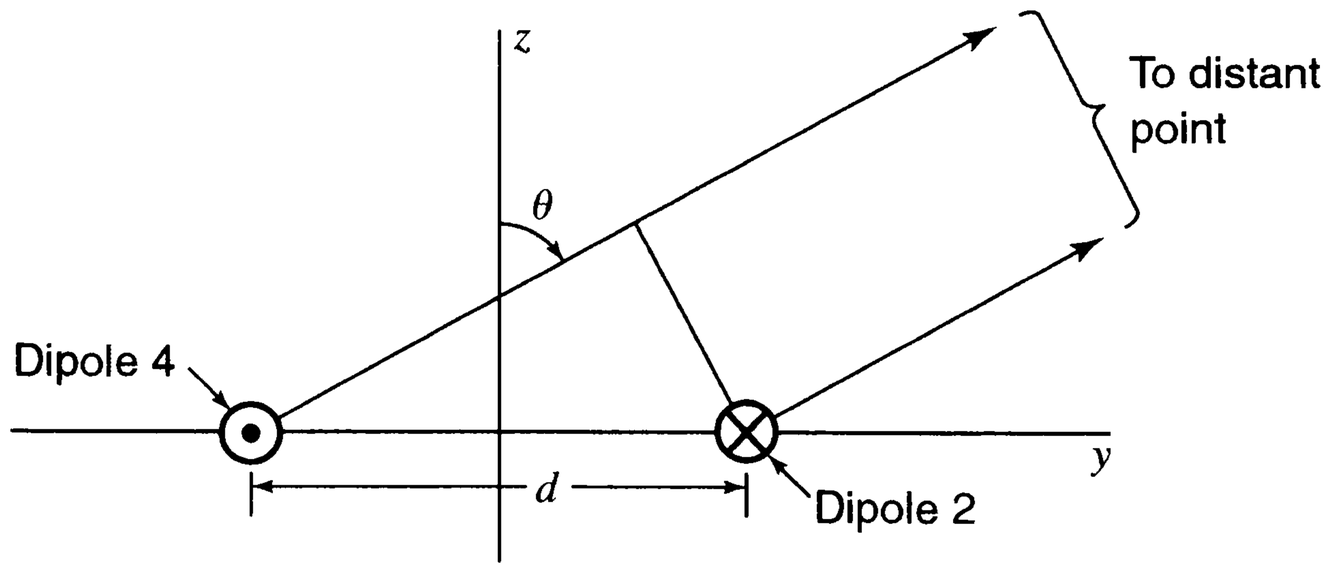


Figure 7–3
Construction for finding far field of dipoles 2 and 4 of square loop.

where $E_{\phi 0}$ = electric field from individual dipole and

$$\psi = \frac{2\pi d}{\lambda} \sin\theta = d_r \sin\theta \tag{3}$$

It follows that

$$E_\phi = -2jE_{\phi 0} \sin\left(\frac{d_r}{2} \sin\theta\right) \tag{4}$$

The factor j in (4) indicates that the total field E_ϕ is in phase quadrature with the field $E_{\phi 0}$ of the individual dipole. This may be readily seen by a vector construction of the type of

Fig. 5–1*b* of Chap. 5. Now if $d \ll \lambda$, (4) can be written

$$E_\phi = -jE_{\phi 0} d_r \sin\theta \tag{5}$$

The far field of the individual dipole was developed in Chap. 6, being given in Table 6–1. In developing the dipole formula, the dipole was in the z direction, whereas in the present case it is in the x direction (see Figs. 7–2 and 7–3). The angle θ in the dipole formula is measured from the dipole axis and is 90° in the present case. The angle θ in (5) is a different angle with respect to the dipole, being as shown in Figs. 7–2 and 7–3. Thus, we have for the far field $E_{\phi 0}$ of the individual dipole

$$E_{\phi 0} = \frac{j60\pi[I]L}{r\lambda} \tag{6}$$

where $[I]$ is the retarded current on the dipole and r is the distance from the dipole. Substituting (6) in (5) then gives

$$E_\phi = \frac{60\pi[I]Ld_r \sin\theta}{r\lambda} \tag{7}$$

However, the length L of the short dipole is the same as d, that is, $L = d$. Noting also that $d_r = 2\pi d/\lambda$ and that the area A of the loop is d^2, (7) becomes

（小环） ***Small loop*** $E_\phi = \dfrac{120\pi^2[I]\sin\theta}{r}\dfrac{A}{\lambda^2}$ ***Far E_ϕ field*** （远场） (8)

This is the instantaneous value of the E_ϕ component of the far field of a small loop of area A. The peak value of the field is obtained by replacing $[I]$ by I_0, where I_0 is the peak current in time on the loop. The other component of the far field of the loop is H_θ, which is obtained from (8) by dividing by the intrinsic impedance of the medium, in this case, free space. Thus,

$$H_\theta = \frac{E_\phi}{120\pi} = \frac{\pi[I]\sin\theta}{r}\frac{A}{\lambda^2} \tag{9}$$

7–2 COMPARISON OF FAR FIELDS OF SMALL LOOP AND SHORT DIPOLE
（小环与短偶极子的远场比较）

It is of interest to compare the far-field expressions for a small loop with those for a short electric dipole. The comparison is made in Table 7–1. The presence of the operator j in

Table 7–1 Far fields of small electric dipoles and loops

Field	Electric dipole	Loop
Electric	$E_\theta = \dfrac{j60\pi[I]\sin\theta}{r}\dfrac{L}{\lambda}$	$E_\phi = \dfrac{120\pi^2[I]\sin\theta}{r}\dfrac{A}{\lambda^2}$
Magnetic	$H_\phi = \dfrac{j[I]\sin\theta}{2r}\dfrac{L}{\lambda}$	$H_\theta = \dfrac{\pi[I]\sin\theta}{r}\dfrac{A}{\lambda^2}$

the dipole expressions and its absence in the loop equations indicate that the fields of the electric dipole and of the loop are in time-phase quadrature, the current I being in the same phase in both the dipole and loop. This quadrature relationship is a fundamental difference between the fields of loops and dipoles. See Prob. 7–2–1.

The formulas in Table 7–1 apply to a loop oriented as in Fig. 7–2 and a dipole oriented parallel to the polar or z axis. The formulas are exact only for vanishingly small loops and dipoles. However, they are good approximations for loops up to $\lambda/10$ in diameter and dipoles up to $\lambda/10$ long.

7–3 THE LOOP ANTENNA. GENERAL CASE（环天线的一般情况）

The general case of a loop antenna with uniform, in-phase current will now be discussed. The size of the loop is not restricted to a small value compared to the wavelength as in the preceding sections but may assume any value. The method of treatment follows that given by Foster (1) and also Glinski (1) for small radii loops.

Let the loop of radius a be located with its center at the origin of the coordinates as in Fig. 7–4. The current I is uniform and in phase around the loop. Although this condition is readily obtained when the loop is small, it is not a natural condition for large loops energized at a point. For loops with perimeters of about $\lambda/4$ or larger, phase shifters of some type must be introduced at intervals around the periphery in order to approximate a uniform, in-phase current on the loop. Assuming that the current is uniform and in phase, the far-field expressions will be derived with the aid of the vector potential of the electric current. The vector potential will first be developed for a pair of short, diametrically opposed

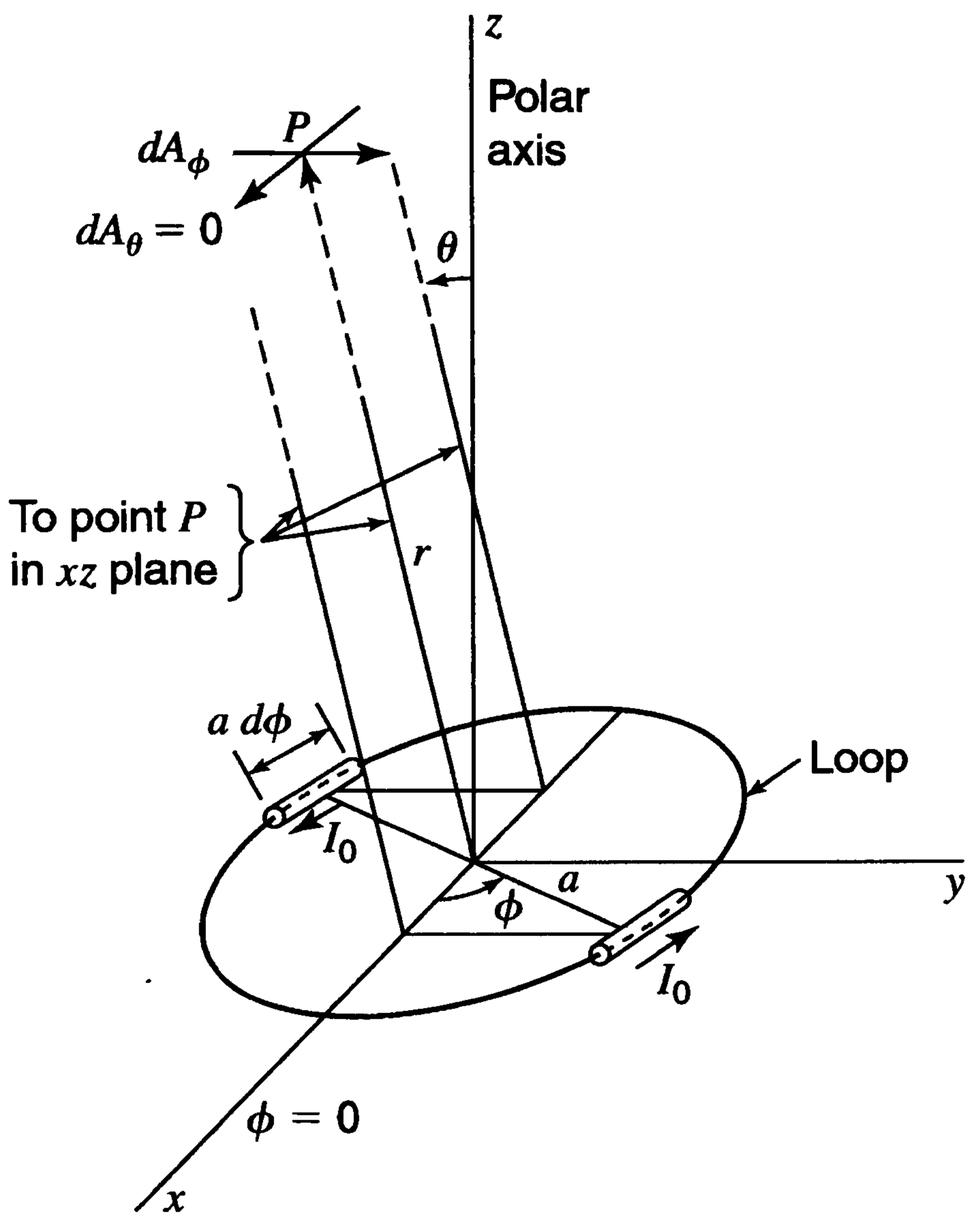


Figure 7–4
Loop of any radius *a* with relation to coordinates.

electric dipoles of length $a\,d\phi$, as in Fig. 7–4. Then integrating over the loop, the total vector potential is obtained, and from this the far-field components are derived.

Since the current is confined to the loop, the only component of the vector potential having a value is A_ϕ. The other components are zero: $A_\theta = A_r = 0$. The infinitesimal value at the point P of the ϕ component of A from two diametrically opposed infinitesimal dipoles is

$$dA_\phi = \frac{\mu\, dM}{4\pi r} \tag{1}$$

where dM is the current moment due to one pair of diametrically opposed infinitesimal dipoles of length, $a\,d\phi$. In the $\phi = 0$ plane (Fig. 7–4) the ϕ component of the retarded current moment due to one dipole is

$$[I]a\,d\phi\cos\phi \tag{2}$$

where $[I] = I_0\, e^{j\omega[t-(r/c)]}$ and I_0 is the peak current in time on the loop.

Figure 7–5 is a cross section through the loop in the xz plane of Fig. 7–4. Referring again to Fig. 7–5, the resultant moment $d\,M$ at a large distance due to a pair of diametrically opposed dipoles is

$$dM = 2j[I]a\,d\phi\cos\phi\sin\frac{\psi}{2} \tag{3}$$

where $\psi = 2\beta a\cos\phi\sin\theta$ radians

Introducing this value for ψ into (3), we have

$$dM = 2j[I]a\cos\phi[\sin(\beta a\cos\phi\sin\theta)]\,d\phi \tag{4}$$

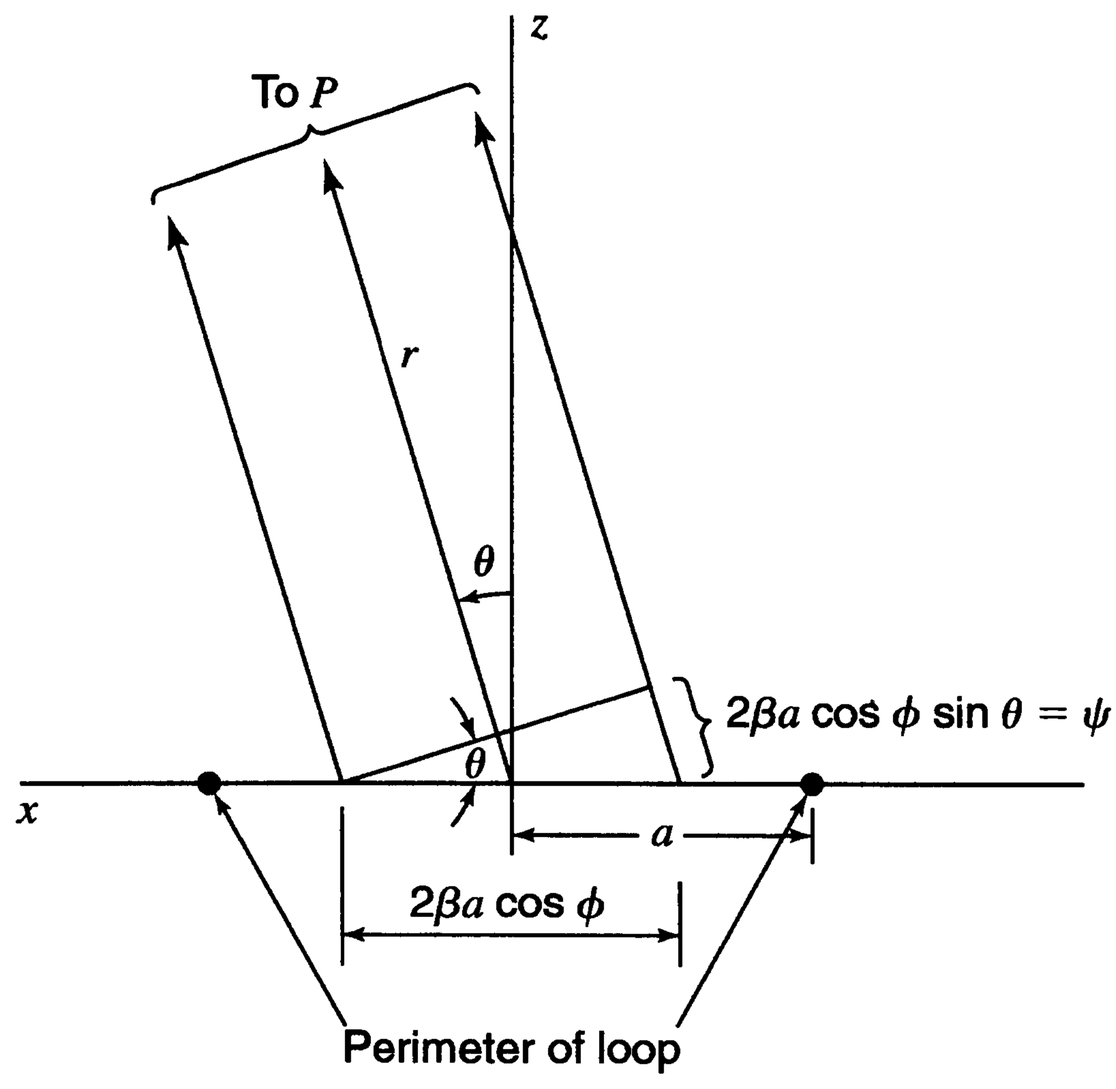


Figure 7–5
Cross section in xz-plane through loop of Fig. 7–4.

Now substituting (4) into (1) and integrating,

$$A_\phi = \frac{j\mu[I]a}{2\pi r}\int_0^\pi \sin(\beta a\cos\phi\sin\theta)\cos\phi\, d\phi \tag{5}$$

or

$$A_\phi = \frac{j\mu[I]a}{2r}J_1(\beta a\sin\theta) \tag{6}$$

where J_1 is a Bessel function of the first order and of argument $(\beta a\sin\theta)$. The integration of (5) is performed on equivalent dipoles which are all situated at the origin but have different orientations with respect to ϕ. The retarded current $[I]$ is referred to the origin and, hence, is constant in the integration.

The far electric field of the loop has only a ϕ component given by

$$E_\phi = -j\omega A_\phi \tag{7}$$

Substituting the value of A_ϕ from (6) into (7) yields

(对所有环) ***For all loops***

$$E_\phi = \frac{\mu\omega[I]a}{2r}J_1(\beta a\sin\theta) \tag{8}$$

or

$$E_\phi = \frac{60\pi\beta a[I]}{r}J_1(\beta a\sin\theta) \tag{9}$$

Far fields (远场)

This expression gives the instantaneous electric field at a large distance r from a loop of any radius a. The peak value of E_ϕ is obtained by putting $[I] = I_0$, where I_0 is the peak value (in time) of the current on the loop. The magnetic field H_θ at a large distance is related to E_ϕ by the intrinsic impedance of the medium, in this case, free space. Thus,

$$H_\theta = \frac{\beta a[I]}{2r}J_1(\beta a\sin\theta) \tag{10}$$

This expression gives the instantaneous magnetic field at a large distance r from a loop of any radius a.

7–4 FAR-FIELD PATTERNS OF CIRCULAR LOOP ANTENNAS WITH UNIFORM CURRENT（载有均匀电流的圆环天线的远场波瓣图）

The far-field patterns for a loop of any size are given by (7–3–9) and (7–3–10). For a loop of a given size, βa is constant and the shape of the far-field pattern is given as a function of θ by

$$J_1(C_\lambda\sin\theta) \tag{1}$$

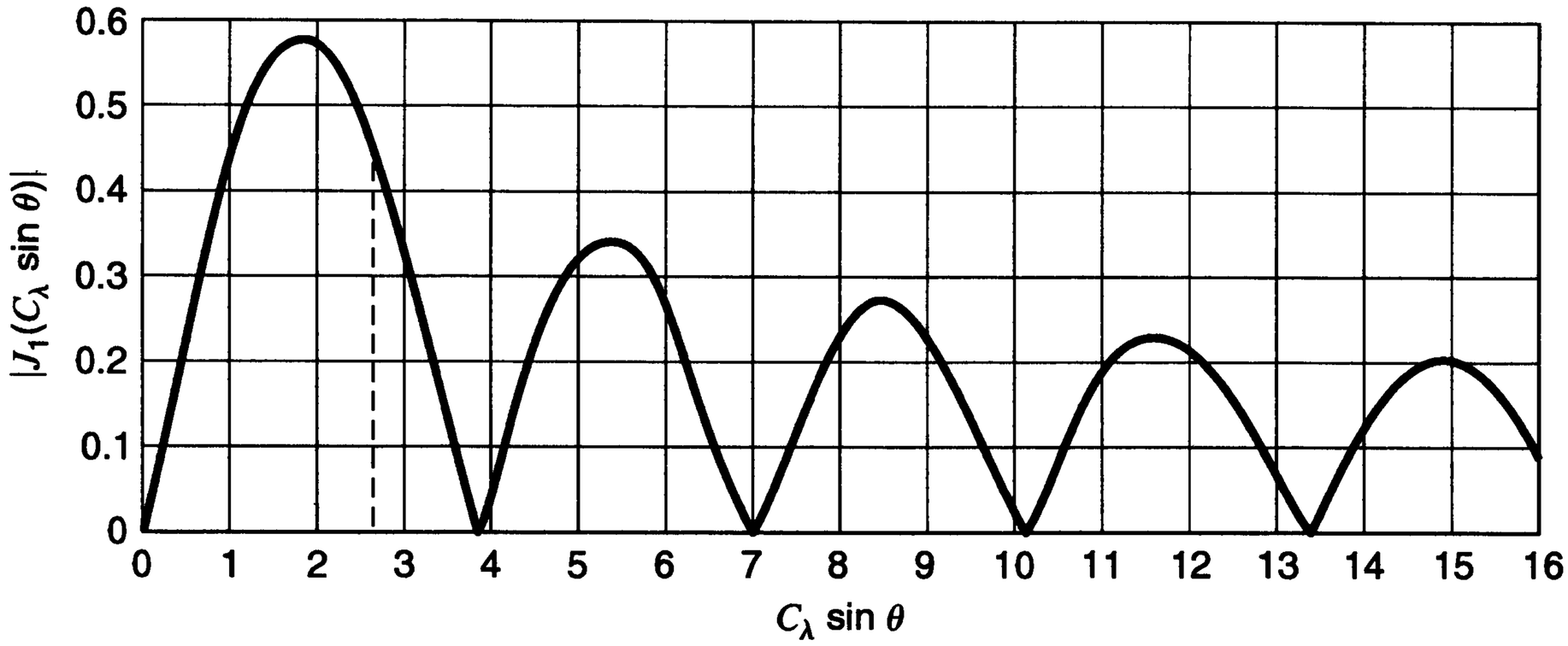


Figure 7–6
Rectified first-order Bessel curve for patterns of loops.

where C_λ is the circumference of the loop in wavelengths. That is,

$$C_\lambda = \frac{2\pi a}{\lambda} = \beta a \tag{2}$$

The value of $\sin\theta$ as a function of θ ranges in magnitude between zero and unity. When $\theta = 90°$, the relative field is $J_1(C_\lambda)$, and as θ decreases to zero, the values of the relative field vary in accordance with the J_1 curve from $J_1(C_\lambda)$ to zero. This is illustrated by Fig. 7–6 in which a rectified first-order Bessel curve is shown as a function of $C_\lambda \sin\theta$.

EXAMPLE **7–4.1 Feed System for 1λ Diameter Loop**

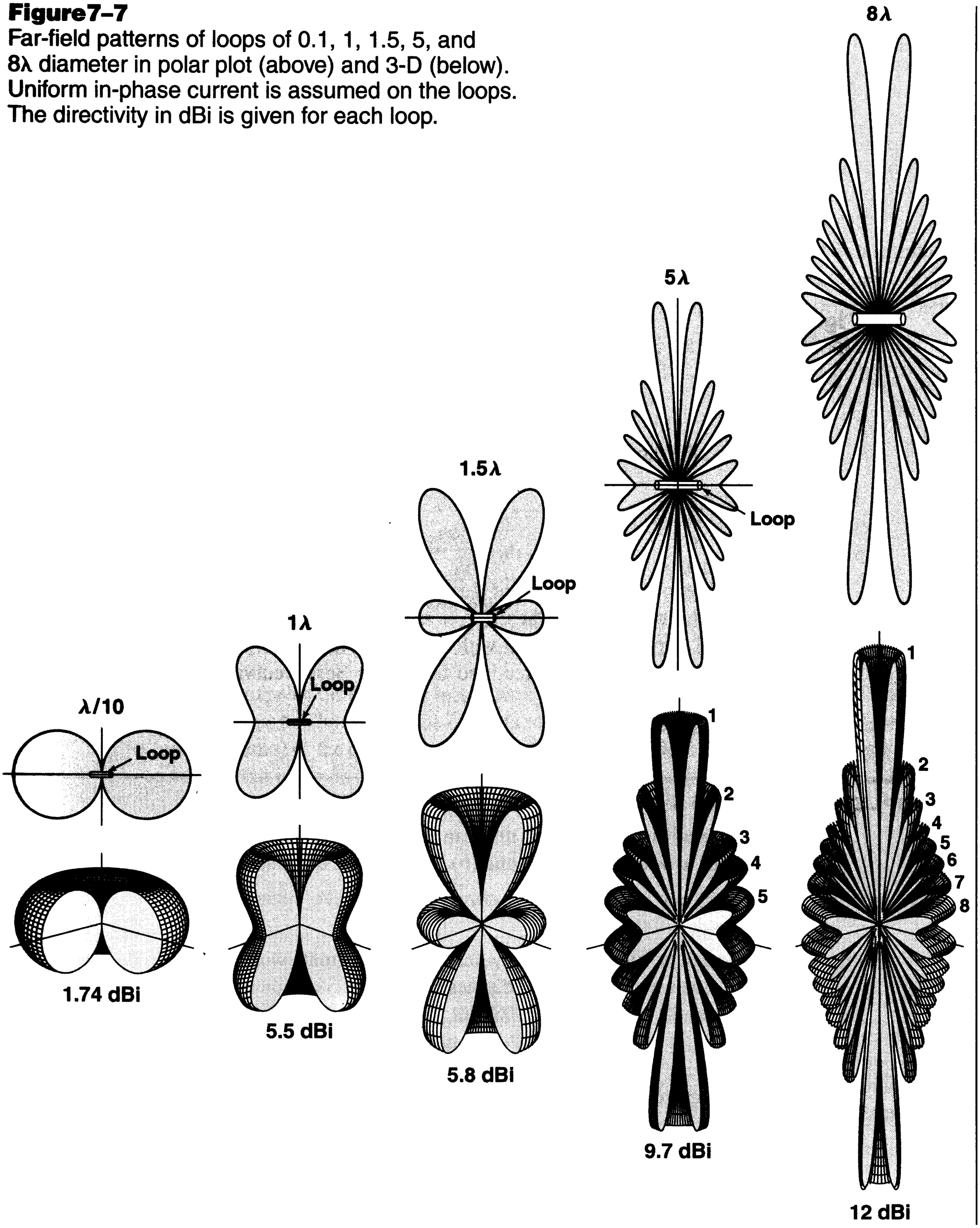
The patterns of Fig. 7–7 assume uniform in-phase currents. (*a*) Design a feed system that does this for a 1λ diameter loop and (*b*) for a 0.212λ diameter loop.

■ **Solution**

(*a*) The design of Figure 7–8*a* provides in-phase currents with the amplitude variation of a rectified sine wave. All six $\lambda/2$ dipoles are fed at a common point. Although this system does not provide uniform current, it is a partial solution.

(*b*) The design of Fig. 7–8*b* provides in-phase currents having a constant amplitude with a + and − 25 percent ripple. It is a better approximation to the complete requirement. A loop fed at one point, as in Fig. 7–9, will have in-phase essentially uniform current only for diameters of 0.1λ or less. For larger loops, feed systems such as illustrated in Fig. 7–8*a* and *b* are required to approximate the desired condition.

Figure7–7
Far-field patterns of loops of 0.1, 1, 1.5, 5, and 8λ diameter in polar plot (above) and 3-D (below). Uniform in-phase current is assumed on the loops. The directivity in dBi is given for each loop.



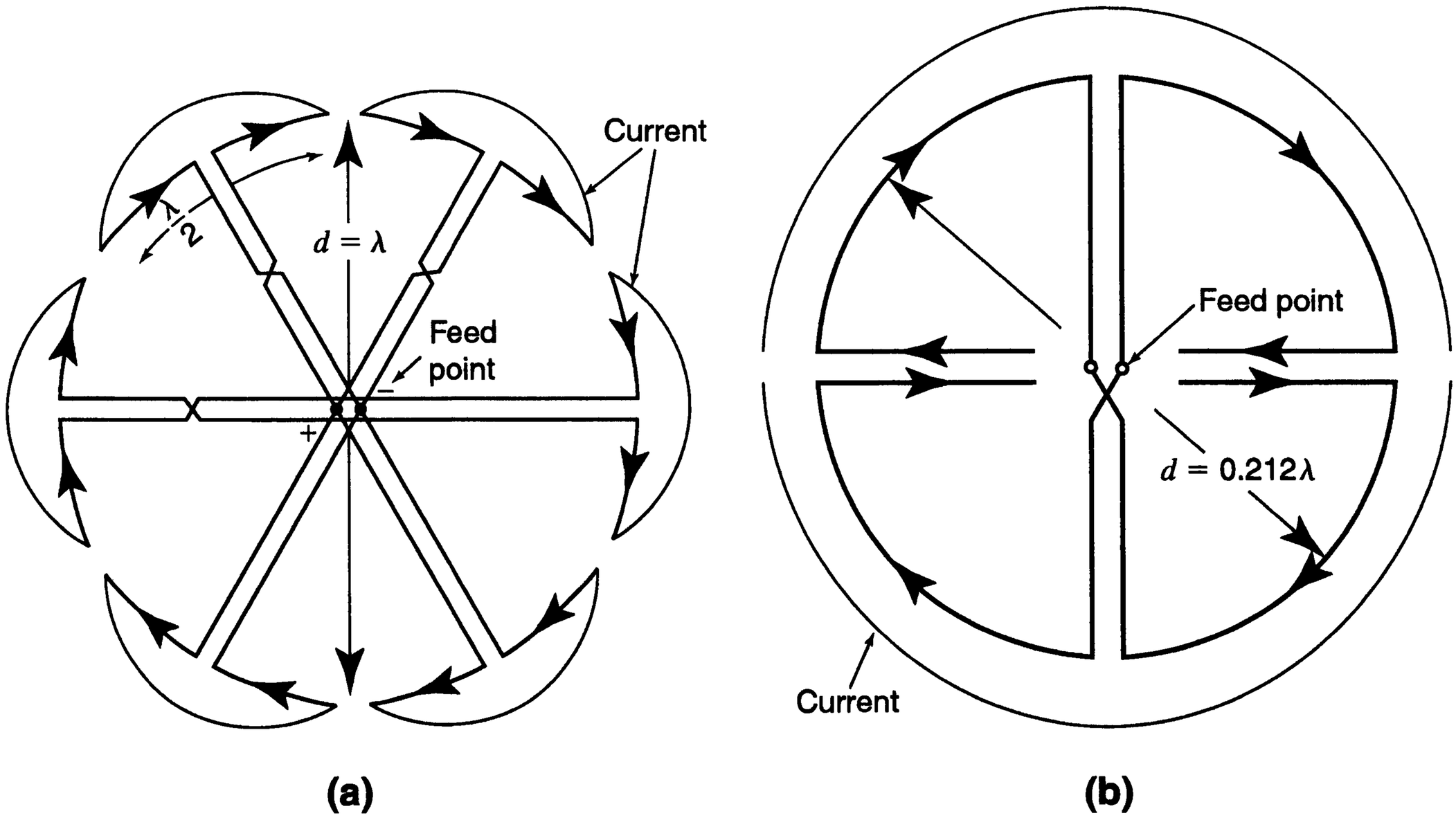


Figure 7–8
(*a*) Feed system for providing a 1λ diameter loop with in-phase currents. (*b*) Feed system for providing a 0.212λ diameter loop with in-phase currents having a constant amplitude with a + and – 25 percent ripple.

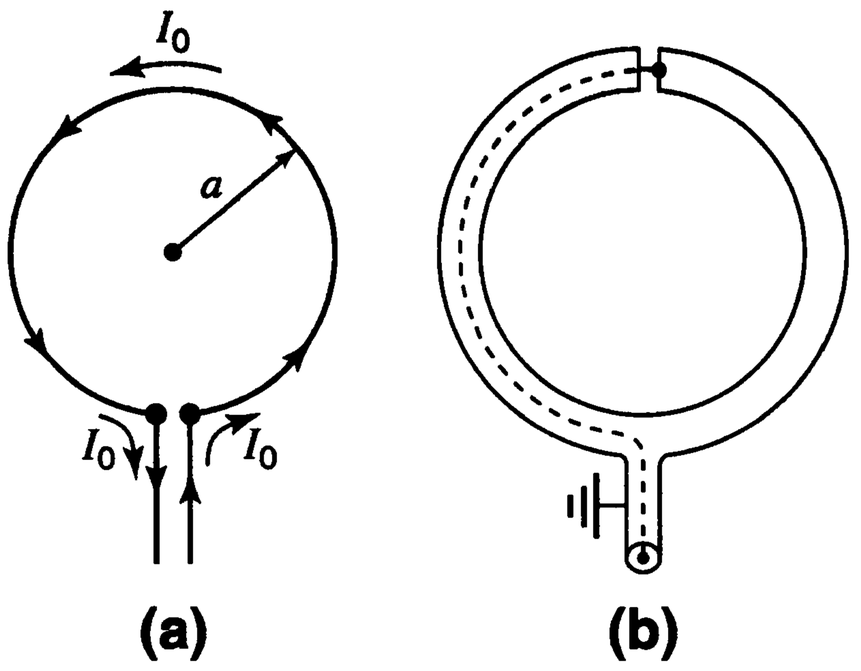


Figure 7–9
(*a*) Loop fed by twin-line. (*b*) grounded loop fed with coaxial cable.

7–5 THE SMALL LOOP AS A SPECIAL CASE（作为特例的电小环）

The relations of (7–3–9) and (7–3–10) apply to loops of any size. It will now be shown that for the special case of a small loop, these expressions reduce to the ones obtained previously.

For small arguments of the first-order Bessel function, the following approximate relation can be used:[1]

$$J_1(x) = \frac{x}{2} \tag{1}$$

where x is any variable. When $x = \frac{1}{3}$, the approximation of (1) is about 1 percent in error. The relation becomes exact as x approaches zero. Thus, if the perimeter of the loop is $\lambda/3$ or less ($C_\lambda < \frac{1}{3}$), (1) may be applied to (7–3–9) and (7–3–10) with an error which is about 1 percent or less. Equations (7–3–9) and (7–3–10) then become

(小环) ***Small loop***

$$E_\phi = \frac{60\pi\beta a[I]\beta a \sin\theta}{2r} = \frac{120\pi^2[I]\sin\theta}{r}\ \frac{A}{\lambda^2} \tag{2}$$

$$H_\theta = \frac{\beta a[I]\beta a \sin\theta}{4r} = \frac{\pi[I]\sin\theta}{r}\ \frac{A}{\lambda^2} \tag{3}$$

Far fields (远场)

These far-field equations for a small loop are identical with those obtained in earlier sections (see Table 7–1).

7–6 RADIATION RESISTANCE OF LOOPS (环的辐射电阻)

To find the radiation resistance of a loop antenna, the Poynting vector is integrated over a large sphere yielding the total power P radiated. This power is then equated to the square of the effective current on the loop times the radiation resistance R_r (Foster-1).

$$P = \frac{I_0^2}{2} R_r \tag{1}$$

where I_0 = peak current in time on the loop. The radiation resistance so obtained is the value which would appear at the loop terminals connected to the twin-line, as in Fig. 7–9*a* and coaxial line as in Fig. 7–9*b*. It is assumed that the current is uniform and in phase for any radius a, this condition being obtained by means of phase shifters, multiple feeds or other devices (see Fig. 7–8). The average Poynting vector of a far field is given by

$$S_r = \tfrac{1}{2}|H|^2 \operatorname{Re} Z \tag{2}$$

where $|H|$ is the absolute value of the magnetic field and Z is the intrinsic impedance of the medium, which in this case is free space. Substituting the absolute value of H_θ from (7–3–10) for $|H|$ in (2) yields

$$S_r = \frac{15\pi(\beta a I_0)^2}{r^2} J_1^2(\beta a \sin\theta) \tag{3}$$

The total power radiated P is the integral of S_r over a large sphere; that is,

$$P = \iint S_r\, ds = 15\pi(\beta a I_0)^2 \int_0^{2\pi}\int_0^{\pi} J_1^2(\beta a \sin\theta)\sin\theta\, d\theta\, d\phi \tag{4}$$

---

[1]For small arguments, the J_1 curve is nearly linear (see Fig. 7–6). The general relation for a Bessel function of any order n is $J_n(x) \simeq x^n/n!2^n$, where $|x| \ll 1$.

or

$$P = 30\pi^2(\beta a I_0)^2 \int_0^{\pi} J_1^2(\beta a \sin\theta) \sin\theta \, d\theta \tag{5}$$

In the case of a loop that is small in terms of wavelengths, the approximation of (7–5–1) can be applied. Thus (5) reduces to

$$P = \frac{15}{2}\pi^2(\beta a)^4 I_0^2 \int_0^{\pi} \sin^3\theta \, d\theta = 10\pi^2\beta^4 a^4 I_0^2 \tag{6}$$

Since the area $A = \pi a^2$, (6) becomes

$$P = 10\beta^4 A^2 I_0^2 \tag{7}$$

Assuming no antenna losses, this power equals the power delivered to the loop terminals as given by (1). Therefore,

$$R_r \frac{I_0^2}{2} = 10\beta^4 A^2 I_0^2 \tag{8}$$

and

(小环) ***Small loop*** ***Radiation resistance*** (辐射电阻)

$$R_r = 31{,}171 \left(\frac{A}{\lambda^2}\right)^2 = 197 C_\lambda^4 \quad (\Omega) \tag{9}$$

or

$$R_r \simeq 31{,}200 \left(\frac{A}{\lambda^2}\right)^2 \quad (\Omega) \tag{10}$$

This is the radiation resistance of a small single-turn loop antenna, circular or square, with uniform in-phase current. The relation is about 2 percent in error when the loop perimeter is $\lambda/3$. A circular loop of this perimeter has a diameter of about $\lambda/10$. Its radiation resistance by (10) is nearly 2.5 Ω.

The radiation resistance of a small loop consisting of one or more turns is given by (Alford-1),

(小环) ***Small loop*** ***Radiation resistance*** (辐射电阻)

$$R_r = 31{,}200 \left(n\frac{A}{\lambda^2}\right)^2 \quad (\Omega) \tag{10a}$$

where n = number of turns

Let us now proceed to find the radiation resistance of a circular loop of any radius a. To do this we must integrate (5). However, the integral of (5) may be reexpressed. Thus, in general (Watson-1),

$$\int_0^{\pi} J_1^2(x \sin\theta) \sin\theta \, d\theta = \frac{1}{x} \int_0^{2x} J_2(y) \, dy \tag{11}$$

where y is any function

Applying (11) to (5) we obtain

$$P = 30\pi^2 \beta a I_0^2 \int_0^{2\beta a} J_2(y)\, dy \tag{12}$$

Equating (12) and (1) and putting $\beta a = C_\lambda$ yields

$$R_r = 60\pi^2 C_\lambda \int_0^{2C_\lambda} J_2(y)\, dy \qquad (\Omega) \tag{13}$$

This is the radiation resistance as given by Foster for a single-turn circular loop with uniform in-phase current and of any circumference C_λ.

When the loop is large ($C_\lambda \geq 5$), we can use the approximation

$$\int_0^{2C_\lambda} J_2(y)\, dy \simeq 1 \tag{14}$$

so that (13) reduces to

(大环) ***Large loop*** $C_\lambda \geq 5$ $\quad R_r = 60\pi^2 C_\lambda = 592 C_\lambda = 3720 \frac{a}{\lambda}$ $\quad$ ***Radiation resistance*** (辐射电阻) (15)

For a loop of 10λ perimeter, the radiation resistance by (15) is nearly 6000 Ω.

For values of C_λ between $\frac{1}{3}$ and 5 the integral in (13) can be evaluated using the transformation

$$\int_0^{2C_\lambda} J_2(y)\, dy = \int_0^{2C_\lambda} J_0(y)\, dy - 2J_1(2C_\lambda) \tag{16}$$

where the expressions on the right of (16) are tabulated functions (Lowan-1).

For perimeters of over 5λ ($C_\lambda > 5$) one can also use the asymptotic development,

$$\int_0^{2x} J_2(y)\, dy \simeq 1 - \frac{1}{\sqrt{\pi x}} \left[\sin\left(2x - \frac{\pi}{4}\right) + \frac{11}{16x} \cos\left(2x - \frac{\pi}{4}\right) \right] \tag{17}$$

where $x = \beta a = C_\lambda$.

For small values of x, one can use a series obtained by integrating the ascending power series for J_2. Thus,

$$\int_0^{2x} J_2(y)\, dy = \frac{x^3}{3} \left(1 - \frac{x^2}{5} + \frac{x^4}{56} - \frac{x^6}{1080} + \frac{x^8}{31{,}680} - \cdots \right) \tag{18}$$

When $x = C_\lambda = 2$ (perimeter of 2λ), the result with four terms is about 2 percent in error. This same percentage error is obtained with one term when the perimeter is about λ/3.

A graph showing the radiation resistance of single-turn loops with uniform current as a function of the circumference in wavelengths is presented in Fig. 7–10. The data for the curve are based on Foster's formulas as given above. Curves for the approximate formulas of small and large loops are shown by the dashed lines.

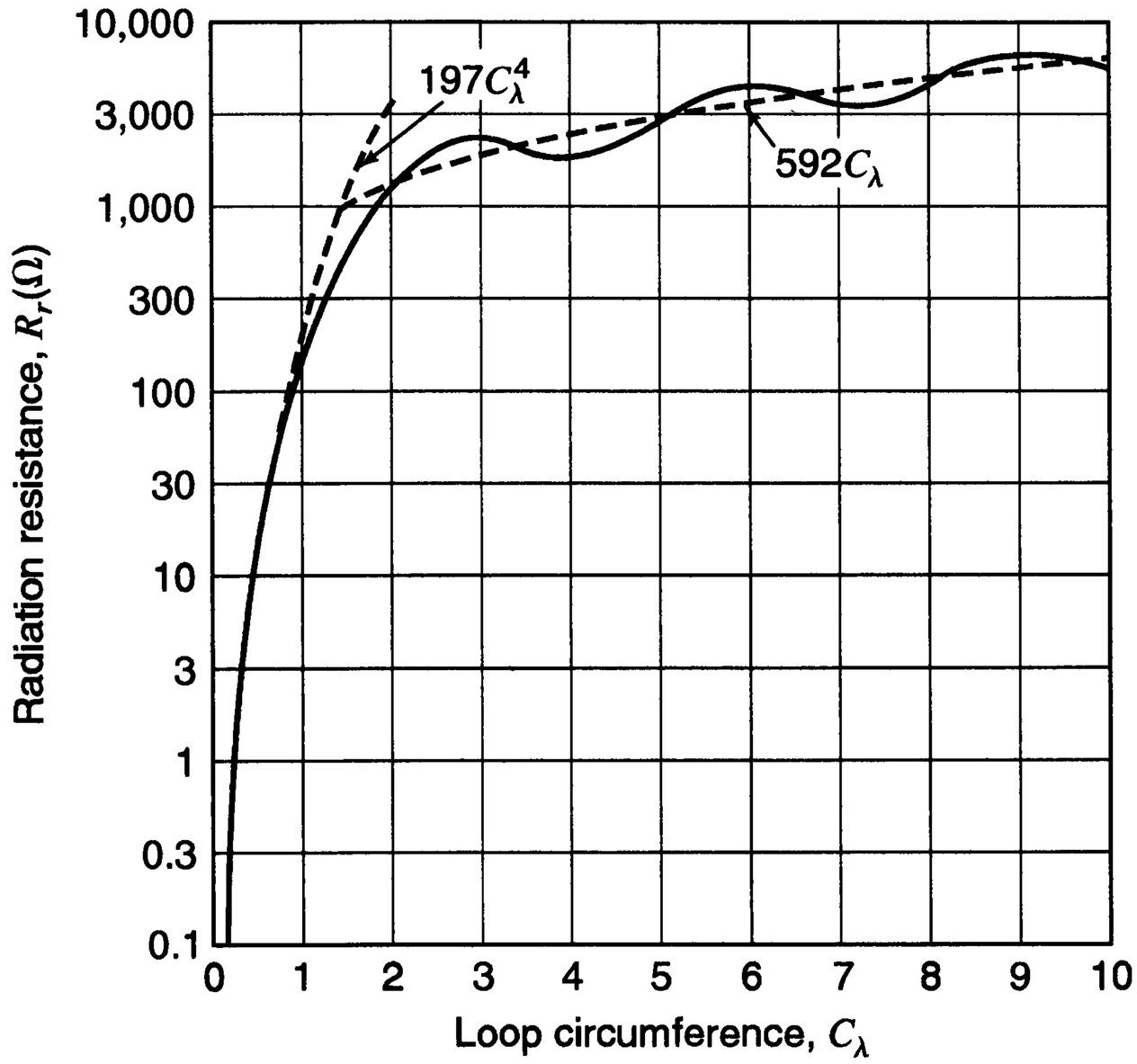


Figure 7–10
Radiation resistance of single-turn circular loop with uniform, in-phase current as a function of the loop circumference in wavelengths, C_λ.

7–7 DIRECTIVITY OF CIRCULAR LOOP ANTENNAS WITH UNIFORM CURRENT
（载有均匀电流的圆环天线的定向性）

The directivity D of an antenna is shown in the table at the end of Chap. 2 as the ratio of maximum radiation intensity to the average radiation intensity. The maximum radiation intensity for a loop antenna is given by r^2 times (7–6–3). The average radiation intensity is given by (7–6–5) divided by 4π. Thus, the directivity of a loop is

$$D =: \frac{2C_\lambda \left[J_1^2(C_\lambda \sin\theta)\right]_{\max}}{\int_0^{2C_\lambda} J_2(y)\, dy} \tag{1}$$

This is Foster's expression for the directivity of a circular loop with uniform in-phase current of any circumference C_λ. The angle θ in (1) is the value for which the field is a maximum.

For a small loop ($C_\lambda < \frac{1}{3}$), the directivity expression reduces to

（小环） ***Small loop*** $C_\lambda \le 1/3$ $\qquad D = \frac{3}{2}\sin^2\theta = \frac{3}{2}$ $\qquad$ ***Directivity*** （定向性） (2)

since the field is a maximum at $\theta = 90°$. The value of $\frac{3}{2}$ is the same as for a short electric dipole. This is to be expected since the pattern of a short dipole is the same as for a small loop.

For a large loop ($C_\lambda > 5$), (1) reduces to

$$D = 2C_\lambda J_1^2(C_\lambda \sin\theta) \tag{3}$$

From Fig. 7–6 we note that for any loop with $C_\lambda \ge 1.84$, the maximum value of $J_1(C_\lambda \sin\theta)$ is 0.582. Thus, the directivity expression of (3) for a large loop becomes

（大环） ***Large loop*** $C_\lambda \ge 2$ $\qquad D = 0.68C_\lambda$ $\qquad$ ***Directivity*** （定向性） (4)

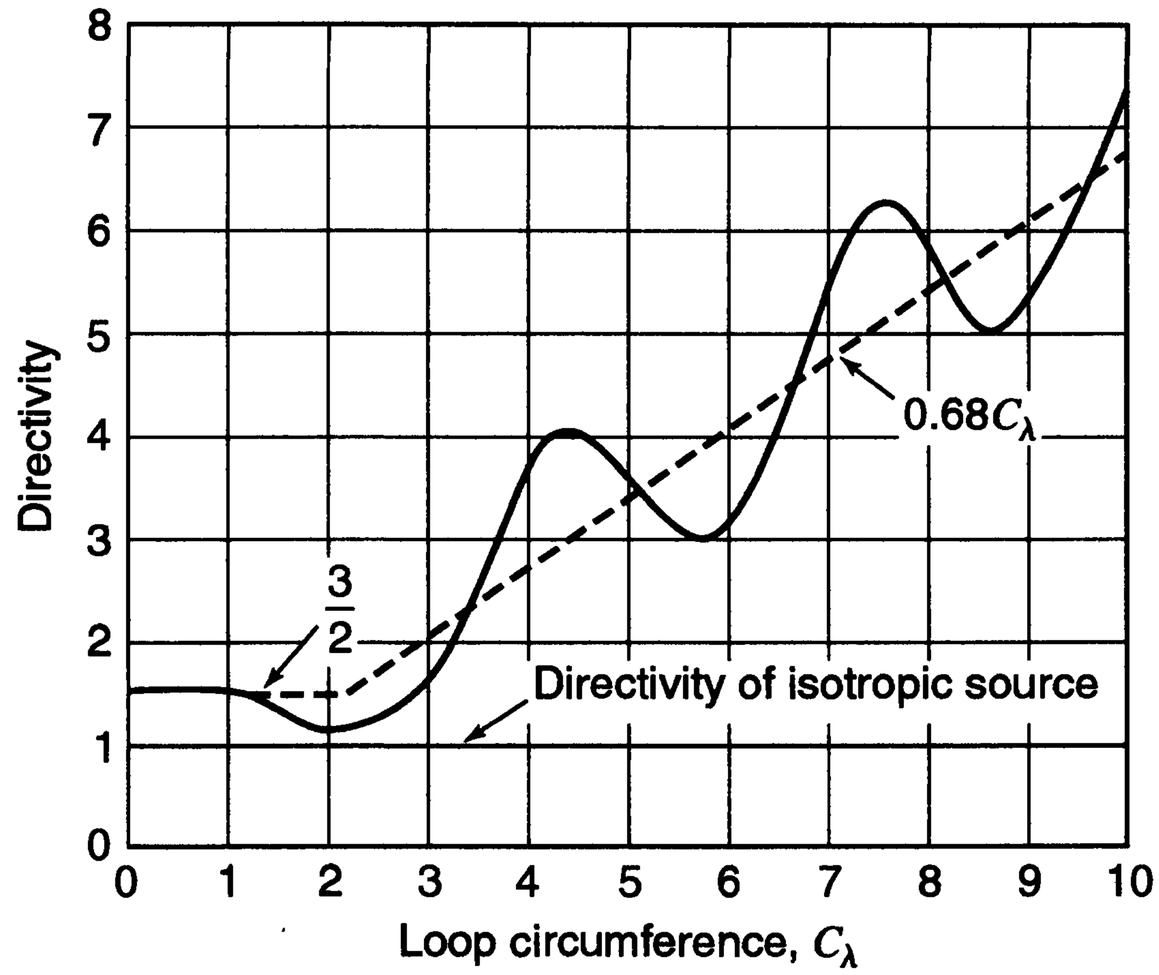


Figure 7–11
Directivity of circular loop antenna with uniform, in-phase current as a function of loop circumference in wavelengths, C_λ. (Foster-1)

The directivity of a loop antenna as a function of the loop circumference C_λ is presented in Fig. 7–11. Curves based on the approximate relations of (2) and (4) for small and large loops are indicated by dashed lines.

EXAMPLE **7–7.1 $\lambda/10$ DIAMETER LOOP**

Although small loops are commonly used for direction finding, they have the disadvantage that the two nulls of the pattern result in a 180° ambiguity. However, with a ground plane the ambiguity disappears. Thus, the $\lambda/10$ diameter loop of Fig. 7–12*a* spaced $\lambda/10$ from the ground plane has the field pattern of Fig. 7–12*c* provided the ground plane is big enough ($\simeq \lambda$).

(*a*) What is the directivity or gain of the loop alone assuming no losses?

(*b*) What is the directivity or gain of the loop with ground plane assuming no losses?

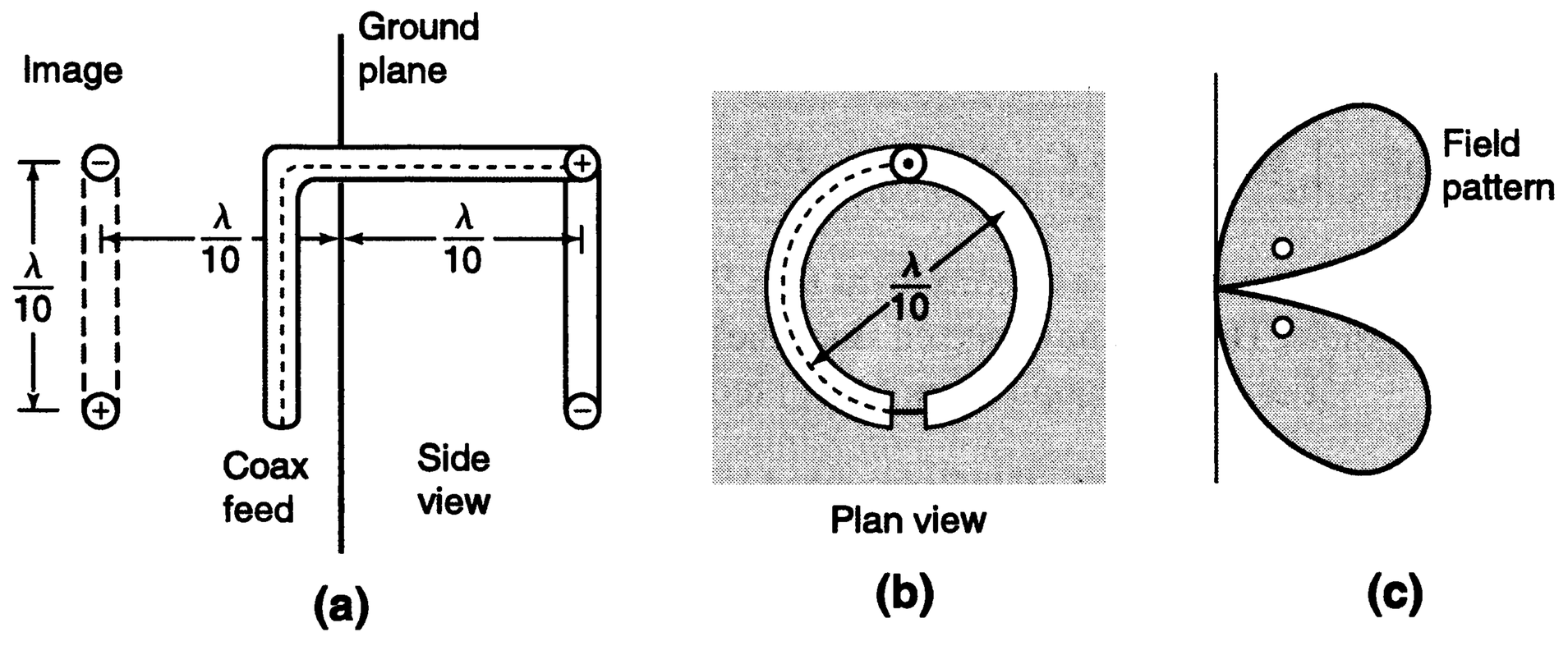


Figure 7–12
(*a*) $\lambda/10$ diameter loop spaced $\lambda/10$ from ground plane. (*b*) Feed arrangment (like that for loop of Fig. 7–8*b*). (*c*) Field in polar plot.

■ **Solution**

(*a*) For the loop alone we have from (2), $D = 1.5$ or 1.76 dBi. *Ans.* (*a*)

(*b*) For the loop with ground plane we have from formula (1) of Table 10–1 that

$$D = \left[2\sqrt{R_r/(R_r - R_m)}\sin(2\pi/10)\right]^2 \times 1.5 \tag{5}$$

where

R_r = self-resistance of loop, Ω

R_m = mutual resistance of the two loops, Ω

From (7–6–9), $R_r = 197C_\lambda^4 = 197(0.1\pi)^4 = 1.92\ \Omega$

It may be shown that $R_m = 1.60\ \Omega$

Introducing these values in (1), we have

$$D = \left[2\sqrt{1.92/(1.92 - 1.60)}\sin 36°\right]^2 \times 1.5 \doteq 8.3 \times 1.5 = 12.45 \text{ or } 11 \text{ dBi}$$

Ans. (*b*)

EXAMPLE **7–7.2 λ/π DIAMETER LOOP**

A loop with a diameter of λ/π or circumference of 1 λ is of particular interest. Figure 7–13 shows the loop at (*a*) with its square or *quad* antenna equivalent at (*b*) and its *folded* $\lambda/2$ *dipole* equivalent at (*c*). All of these antennas are close to resonance (reactance nearly zero) and have a radiation resistance R_r of 200 to 300 Ω. Furthermore, when the loop is $\lambda/10$ from a ground plane its radiation resistance $R_r \simeq 50\ \Omega$ and its reactance $R_x \simeq 0$, making it easy to feed with 50-Ω coaxial cable as in Fig. 7–14.

Question: What is the directivity of this loop with ground plane?

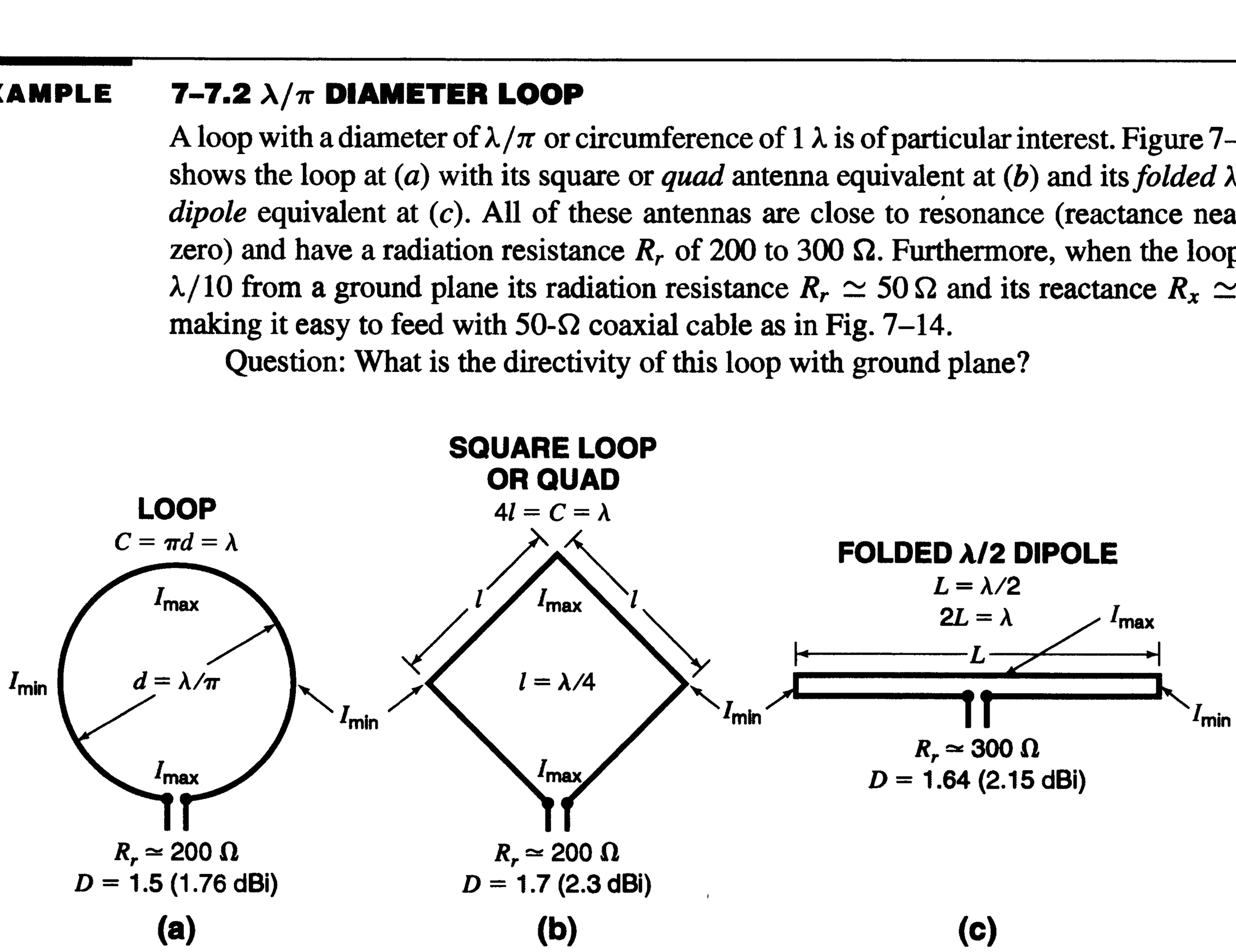


Figure 7–13

Loop at (*a*), quad at (*b*) and folded $\lambda/2$ dipole at (*c*) all with total conductor length of 1λ.

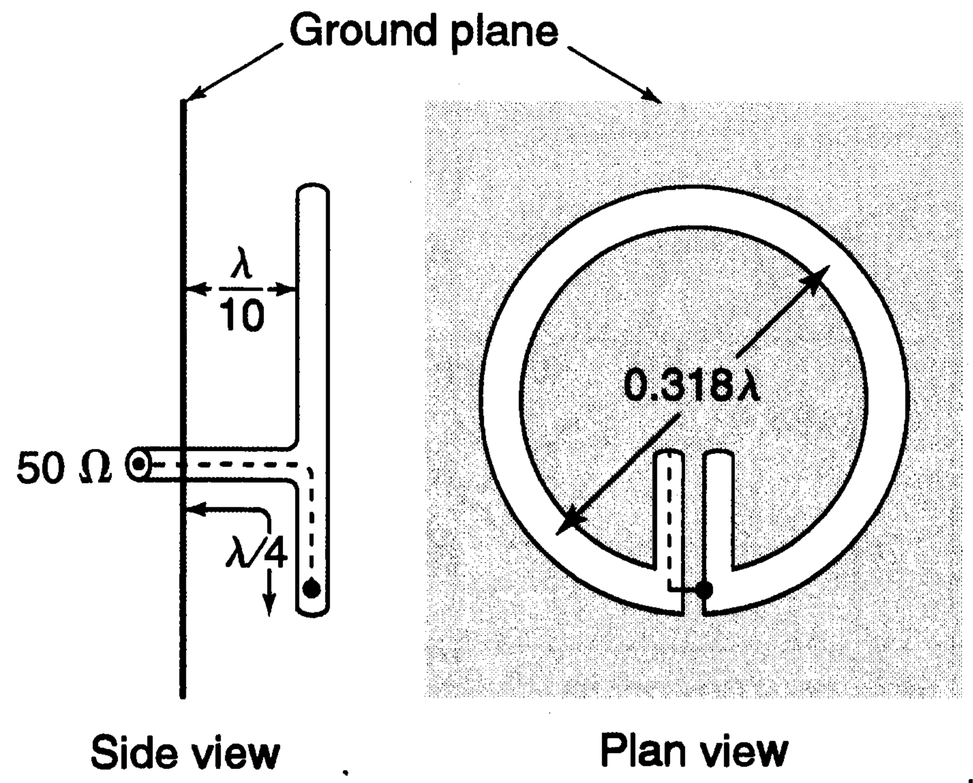


Figure 7–14
1λ circumference loop (diameter = 0.318 λ) mounted λ/10 from ground plane and fed by 50-Ω coaxial cable.

■ **Soiution**

From formula (1) of Table 10–1

$$D = \left[2\sqrt{R_r/(R_r - R_m)}\sin(2\pi/10)\right]^2 \times 1.5 \tag{6}$$

From (7–6–9), $R_r = 197\ \Omega$.

It may be shown that $R_m = 157\ \Omega$.

Substituting these values in (10), we have

$$D = \left(2\sqrt{197/40}\sin 36°\right)^2 \times 1.5 = 10.2 \quad \text{or} \quad 10.1\ \text{dBi.} \qquad \textit{Ans.}$$

Even though this 0.318 λ diameter of loop has 1 dB less gain than the 0.1-λ-diameter loop of Example 7–7.1, its much larger radiation resistance gives it a smaller Q, a broader bandwidth and less susceptibility to losses.

7–8 TABLE OF LOOP FORMULAS（环的公式列表）

The relations developed in the preceding sections are summarized in Table 7–2. The general and large loop formulas are based on Foster's results.

7–9 SQUARE LOOPS*（方环）

It was shown in Sec. 7–3 that the far-field patterns of square and circular loops of the same area are identical when the loops are small ($A < \lambda^2/100$). As a generalization, we may say that the properties depend only on the area and that the shape of the loop has no effect when the loop is small. However, this is not the case when the loop is large. The pattern of a circular loop of any size is independent of the angle ϕ but is a function of θ (see Fig. 7–2). On the other hand, the pattern of a large square loop is a function of both θ and ϕ.

*Loops of 1 λ circumference or perimeter and many shapes are the basic elements used by Ben Munk in Chap. 18 on Frequency-Selective Surfaces.

Table 7–2 Formulas for circular loops with uniform current

Quantity	General expression (any size loop)	Small loop† $A < \lambda^2/100, C_\lambda < \frac{1}{3}$	Large loop $C_\lambda > 5$
Far E_ϕ	$\dfrac{60\pi[I]C_\lambda J_1(C_\lambda \sin\theta)}{r}$	$\dfrac{120\pi^2[I]\sin\theta}{r}\dfrac{A}{\lambda^2}$	Same as general
Far H_θ	$\dfrac{[I]C_\lambda J_1(C_\lambda \sin\theta)}{2r}$	$\dfrac{\pi[I]\sin\theta}{r}\dfrac{A}{\lambda^2}$	Same as general
Radiation resistance, Ω	$60\pi^2 C_\lambda \int_0^{2C_\lambda} J_2(y)\,dy$	$31{,}200\left(\dfrac{A}{\lambda^2}\right)^2 = 197C_\lambda^4$	$3720\dfrac{a}{\lambda} = 592C_\lambda$
Directivity	$\dfrac{2C_\lambda J_1^2(C_\lambda \sin\theta)}{\int_0^{2C_\lambda} J_2(y)\,dy}$	$\dfrac{3}{2}$	$4.25\dfrac{a}{\lambda} = 0.68C_\lambda$

A = area of loop; C_λ = circumference of circular loop, wavelengths.

†The small loop formulas apply not only to circular loops but also to square loops of area A and in fact to small loops of any shape having an area A. The formula involving C_λ applies, of course, only to a circular loop.

Referring to Fig. 7–15, the pattern in a plane normal to the plane of the loop and parallel to two sides (1 and 3), as indicated by the line AA', is simply the pattern of two point sources representing sides 2 and 4 of the loop. The pattern in a plane normal to the plane of the loop and passing through diagonal corners, as indicated by the line BB', is different. The complete range in the pattern variation as a function of ϕ is contained in this 45° interval between AA' and BB' in Fig. 7–15.

An additional difference of large circular and square loops is in the θ patterns. For instance, Fig. 7–7 (5λ) shows the pattern as a function of θ for a circular loop 5λ in diameter. By way of comparison, the pattern for a square loop of the same area is presented in Fig. 7–16. The square loop is 4.44λ on a side. The pattern is in a plane perpendicular to the plane of the loop and parallel to the sides (plane contains AA' in Fig. 7–15). Comparing

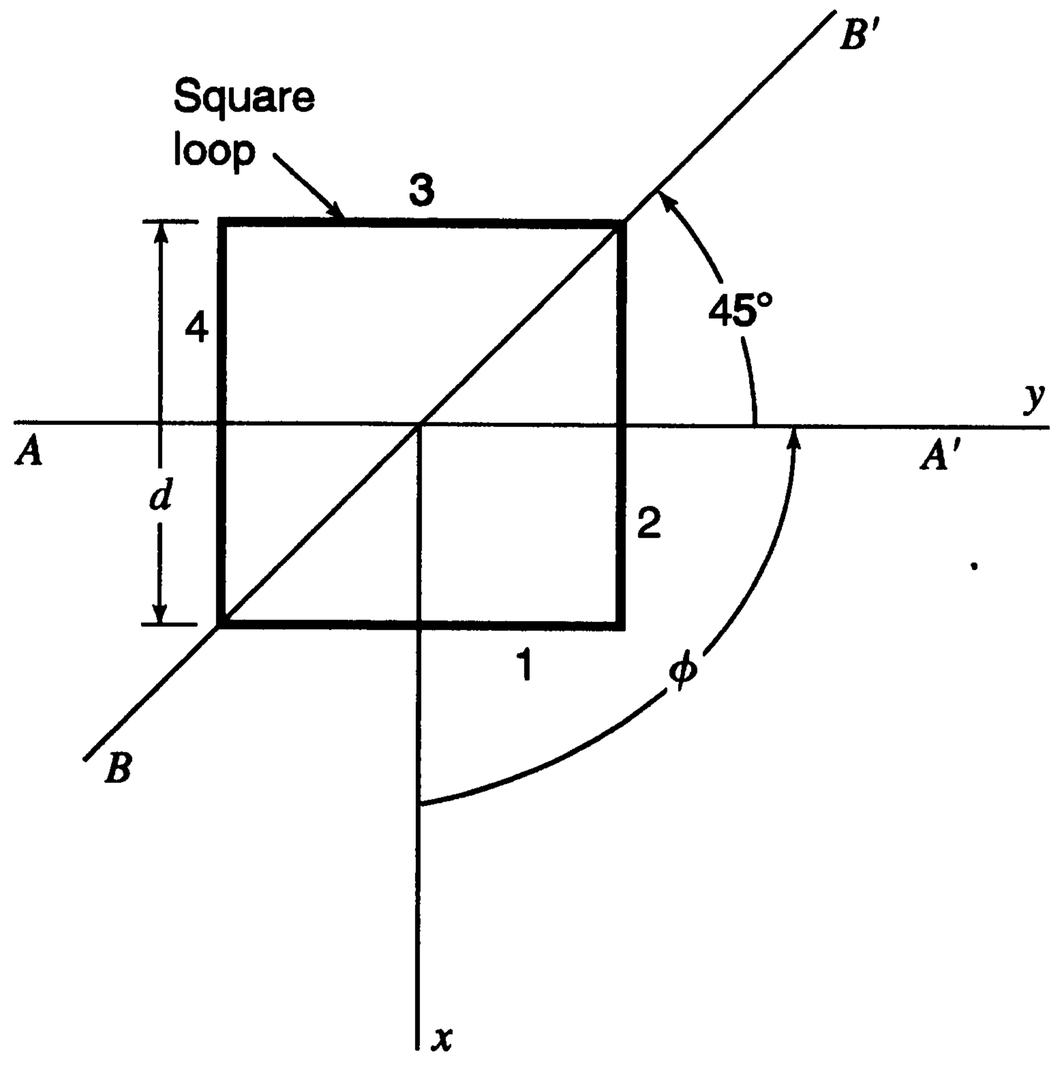


Figure 7–15
Large square loop.

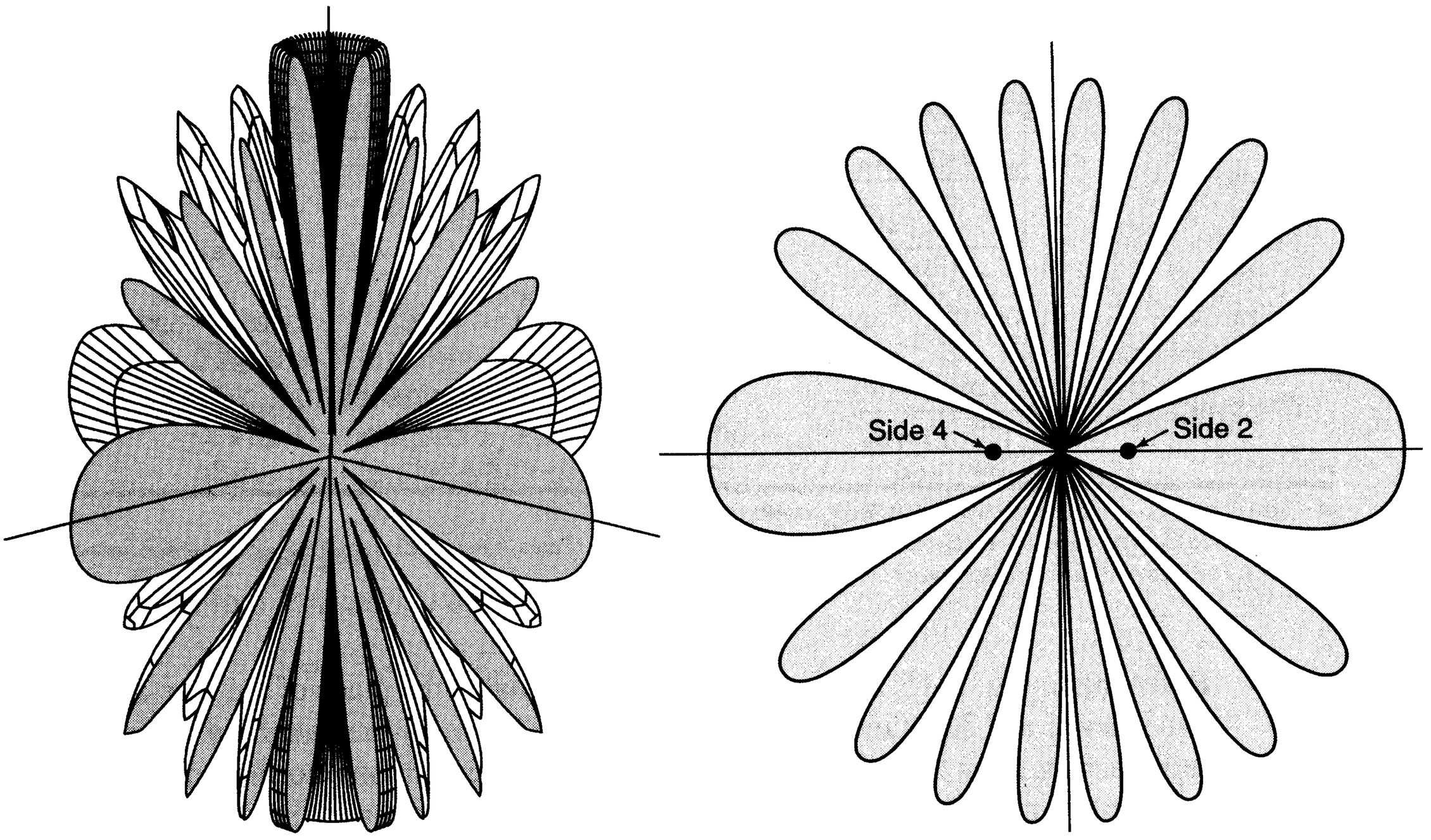


Figure 7–16
Three-dimensional and polar plots of square loop with uniform, in-phase current. The loop is 4.44 λ on a side. Gain = 9.5 dBi. The polar plot is in a plane normal to the plane of the loop and through the line AA'' of Fig. 7–15.

Fig. 7–7 (5λ loop) and Fig. 7–16 we note that the pattern lobes of the circular loop decrease in magnitude as θ approaches 90° while the lobes of the square loop are of equal magnitude. This illustrates the difference of the Bessel function pattern of the circular loop and the trigonometric function pattern of the square loop. In the above discussion, uniform in-phase currents are assumed.

7–10 RADIATION EFFICIENCY, *Q*, BANDWIDTH AND SIGNAL-TO-NOISE RATIO
（辐射效率、品质因数、频带宽度和信噪比）

In Sec. 2–7 we noted that the gain G of an antenna with respect to an isotropic source is identical with the antenna's directivity D provided no losses other than radiation are present. For the more general case we write as in (2–7–6) that

$$G = kD \tag{1}$$

where k = efficiency factor ($0 \leq k \leq 1$), dimensionless

For a lossless antenna, $k = 1$, but with ohmic losses k is less than 1.

If an antenna has a radiation resistance R_r and a loss resistance R_L, then its (radiation) efficiency factor

$$k = \frac{R_r}{R_r + R_L} \tag{2}$$

and the gain

$$G = \frac{R_r}{R_r + R_L} \frac{4\pi A_{em}}{\lambda^2} = \frac{4\pi A_{em} k}{\lambda^2} \tag{3}$$

For antennas which are small compared to the wavelength, the radiation resistance R_r is small and, if ohmic losses R_L are significant, radiation efficiency is reduced. Thus, short dipoles and small loops may be inefficient radiators when losses are present. For example, when $R_L = R_r$ the radiation efficiency is 50 percent; only half of the power input to the antenna is radiated, the other half being dissipated as heat in the antenna structure.

An rf wave entering a conductor attenuates to $1/e$ of its surface value in a distance δ given by (Kraus-1),

$$\delta = \frac{1}{\sqrt{f \pi \mu \sigma}} = \text{depth of penetration} \tag{4}$$

where

f = frequency, Hz
μ = permeability of medium, H m^{-1}
σ = conductivity of medium, $\mho$ m^{-1}

It is assumed that $\sigma \gg \omega\varepsilon$. The induced current density in the conductor also attenuates in the same way. This means that the current density associated with a wave traveling along a conductor is greatest close to the surface, the so-called *skin effect*. The quantity δ is referred to as the *1/e depth of penetration*. It follows that the rf resistance of a round wire or solid cylindrical conductor is equivalent to the dc resistance of a hollow tube of the same material of wall thickness δ. It is assumed that the wire or conductor diameter is much larger than δ. Thus, assuming that the perimeter or circumference L is much smaller than the wavelength so that the current is essentially uniform around the loop, the *ohmic* (or loss resistance) of a small loop antenna is given by

$$R_L = \frac{L}{\sigma \pi d \delta} = \frac{L}{d} \sqrt{\frac{f \mu_0}{\pi \sigma}} \qquad (\Omega) \tag{5}$$

where

L = loop length (perimeter or circumference), m
d = wire or conductor diameter, m

From (7–6–9) the radiation resistance of a small loop is

$$R_r \simeq 31{,}200 \left(\frac{A}{\lambda^2}\right)^2 \simeq 197 C_\lambda^4 \tag{6}$$

where

A = loop area (square or circular), m^2
$C_\lambda = C/\lambda$, where C = circumference of circular loop

Assuming that the loop's inductive reactance is balanced by a capacitor, the terminal impedance will be resistive and equal to

$$R_T = R_r + R_L \tag{7}$$

and the radiation efficiency, or ratio of power radiated to input power, will be

$$k = \frac{1}{1 + (R_L/R_r)} \tag{8}$$

For a 1-turn copper-conductor circular loop (perimeter $L = C$) in air ($\sigma = 5.7 \times 10^7$ ℧ m^{-1}, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}),

$$\frac{R_L}{R_r} = \frac{3430}{C^3 f_{\text{MHz}}^{3.5} d} \tag{9}$$

where

C = circumference of loop, m
f_{MHz} = frequency, MHz
d = wire (or conductor) diameter, m

For small square loops of side length l ($L = 4l$), we may take $C = 3.5l$.

EXAMPLE 7–10.1

Find the radiation efficiency of a 1-m-diameter loop ($C = \pi$ m) of 10-mm-diameter copper wire at (*a*) 1 MHz and (*b*) 10 MHz.

■ **Solution**

(*a*) From (9),

$$\frac{R_L}{R_r} = \frac{3430}{\pi^3 \times 1 \times 10^{-2}} = 11{,}000 \tag{10}$$

and the radiation efficiency

$$k = \frac{1}{1 + 11{,}000} = 9 \times 10^{-5} \text{ (or } -40.5 \text{ dB)} \tag{10a}$$

(*b*) At 10 MHz we have

$$k = 0.22 \text{ (or } -6.6 \text{ dB)} \tag{11}$$

The radiation efficiency as a function of frequency for a small single-turn copper loop in air is shown in Fig. 7–17. It is assumed that the loop is small compared to the wavelength ($C \ll \lambda$) and that the wire or conductor diameter is small compared to the loop circumference ($d \ll C$). Dielectric losses are neglected.

In spite of the low efficiency of a small loop, there are many applications where such loops are useful in receiving applications provided the received signal-to-noise ratio is acceptable as discussed later in this section [see (19)].

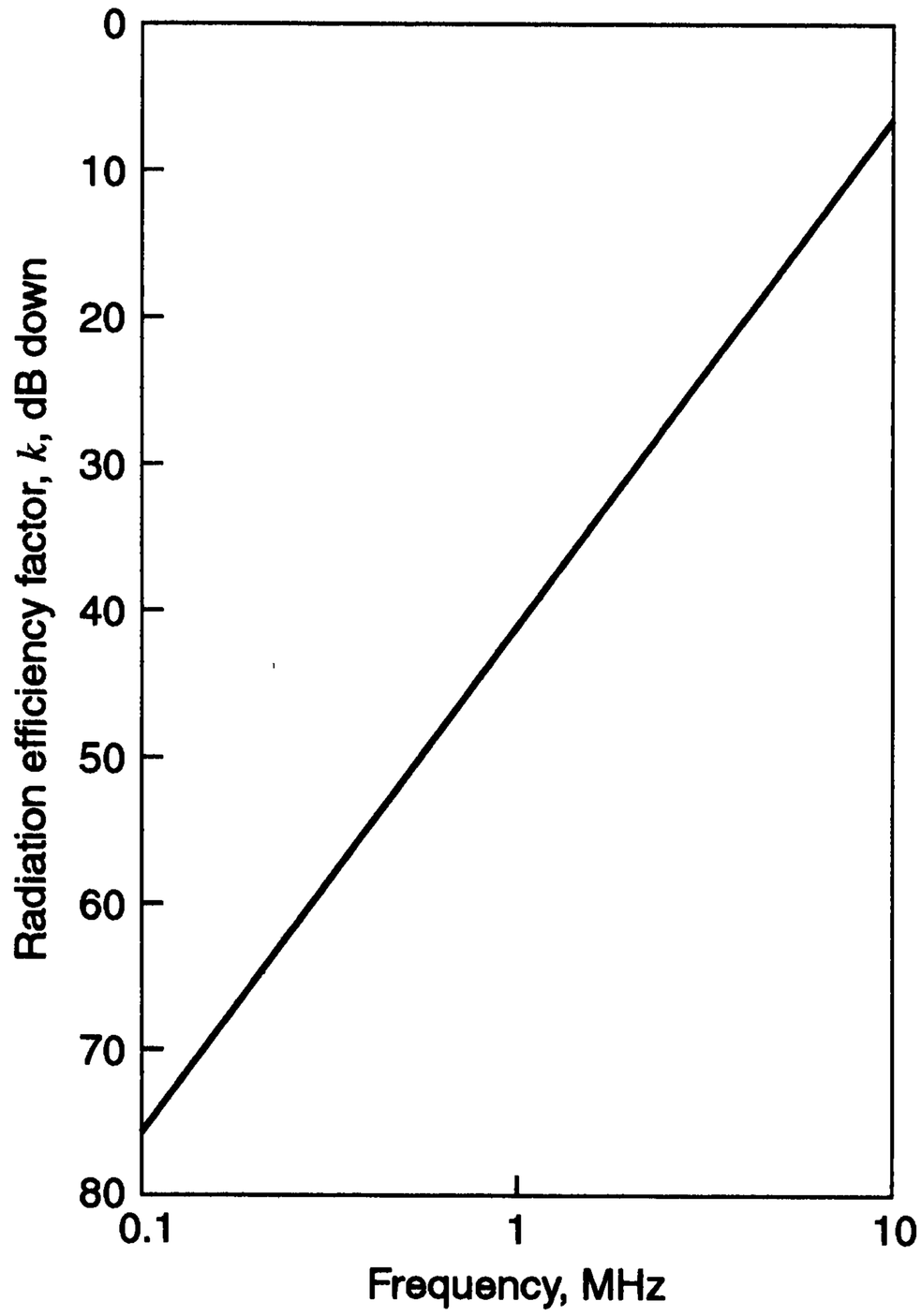


Figure 7–17
Radiation efficiency factor as a function of frequency for a 1-m-diameter single-turn copper loop in air ($C = \pi$ m, $d = 10$ mm).

For loops with n turns, R_r increases in proportion to n^2 while R_L increases in proportion to n. Hence, for multiturn loops (9) becomes

$$\frac{R_L}{R_r} = \frac{3430}{C^3 f_{\text{MHz}}^{3.5} nd} \tag{12}$$

and the radiation efficiency k is increased by a factor which approaches n if R_L/R_r is large. In (12) the effect of capacitance between turns has been neglected but if the turns are spaced sufficiently and are few in number, (12) can be a useful approximation.

The radiation efficiency of a multiturn loop or coil antenna can be increased by introducing a *ferrite rod* into the coil as in Fig. 7–18. Here the coil (horizontal to receive vertical polarization) serves the function of both an antenna and also (with a series capacitor) of the resonant circuit for the first (mixer) stage of a broadcast receiver (500 to 1600 kHz).

The radiation resistance of a *ferrite loaded loop or coil* is given by

（辐射电阻）

Radiation resistance

$$R_r = 31{,}200\mu_{er}^2 n^2 \left(\frac{A}{\lambda^2}\right)^2 = 197\mu_{er}^2 n^2 C_\lambda^4 \quad (\Omega)$$

（铁氧体加载环）***Ferrite loaded loop*** (13)

and the loss resistance (due to the ferrite rod) by

$$R_f = 2\pi f \mu_{er} \frac{\mu_r''}{\mu_r'} \mu_0 n^2 \frac{a}{l} \quad (\Omega) \tag{14}$$

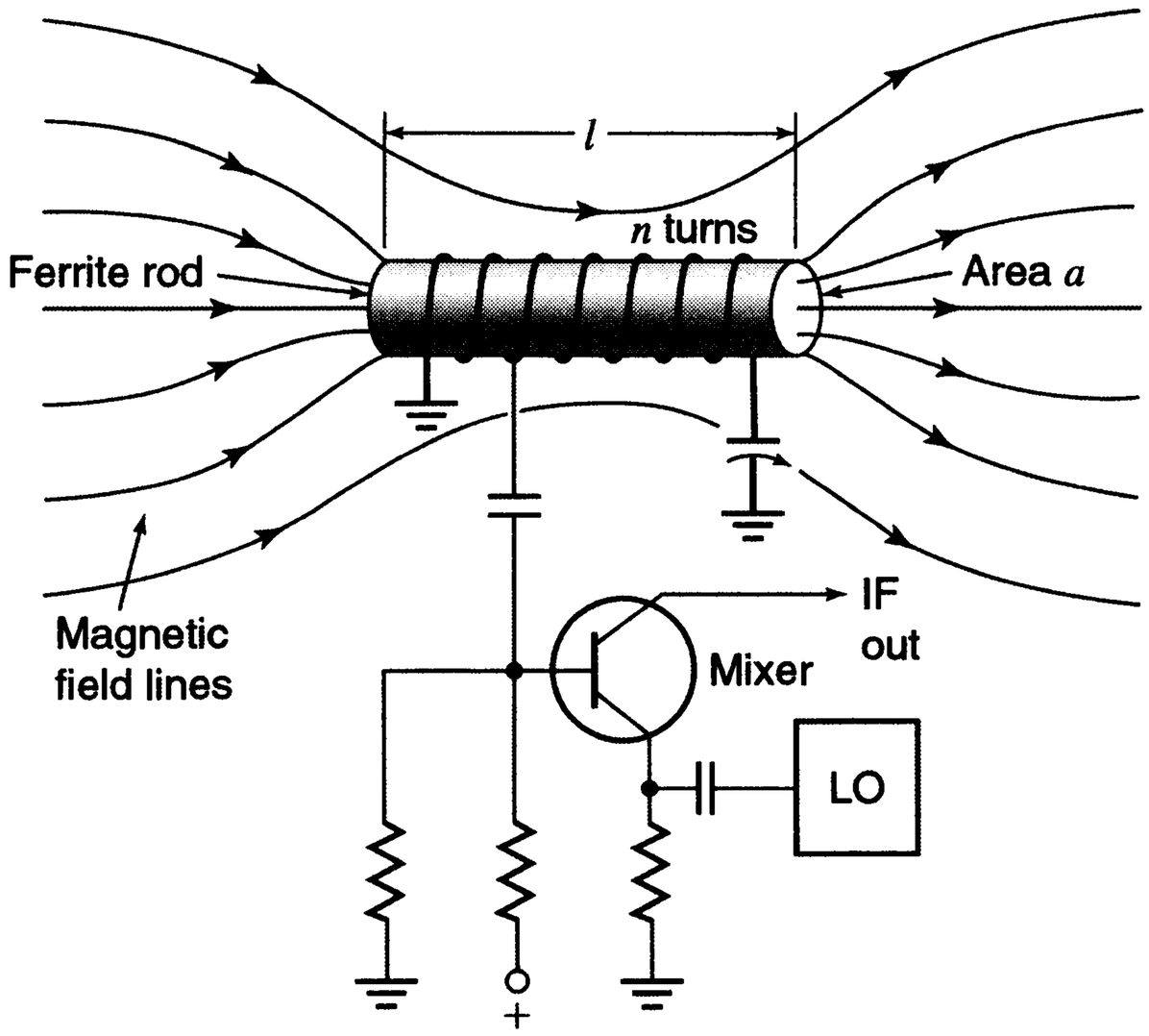


Figure 7–18
Ferrite rod antenna and associated tuned circuit of receiver front-end mixer stage.

where

f = frequency, Hz
μ_{er} = effective relative permeability of ferrite rod, dimensionless
μ'_r = real part of relative permeability of ferrite material, dimensionless
μ''_r = imaginary part of relative permeability of ferrite material, dimensionless
$\mu_0 = 4\pi \times 10^{-7}$, H m^{-1}
n = number of turns
a = ferrite rod cross-sectional area, m^2
l = length of ferrite rod, m

Because of its open geometry (as contrasted to a closed core or ring) a ferrite rod with a relative permeability μ_r will have a smaller *effective relative permeability* μ_{er} (due to demagnetization effect). Typically for a rod with $\mu_r = 250$ and a length-diameter ratio of 10, the effective relative permeability is about 50.

The ohmic loss resistance R_L of the coil is as given by (5). The radiation efficiency factor for the ferrite rod coil antenna is then

$$k = \frac{R_r}{R_r + R_L + R_f} = \frac{1}{1 + [(R_L + R_f)/R_r]} \tag{15}$$

Dielectric loss is neglected.

Knowing the total resistance $(R_r + R_L + R_f)$ of the ferrite rod antenna, one can calculate the Q and bandwidth of the tuned circuit of which it is a part. The Q (ratio of energy stored to energy lost per cycle) is given by

$$Q = \frac{2\pi f_0 L}{R_r + R_L + R_f} = \frac{f_0}{\Delta f_{\text{HP}}} \tag{16}$$

where

f_0 = center frequency, Hz
L = $\mu_{er} n^2 a \mu_0 / l$ = inductance, H[1]
Δf_{HP} = bandwidth at half-power, Hz

EXAMPLE 7–10.2

The multiturn ferrite rod antenna of a broadcast receiver has 10 turns of 1-mm-diameter enameled copper wire wound on a ferrite rod 1 cm in diameter and 10 cm long. The ferrite rod $\mu_r = \mu_r' - \mu_r'' = 250 - j2.5$. Take $\mu_{er} = 50$. At 1 MHz find (*a*) the radiation efficiency, (*b*) the Q and (*c*) the half-power bandwidth.

■ **Solution**

(*a*) From (13), $R_r = 1.91 \times 10^{-4}\ \Omega$. From (14), $R_f = 0.31\ \Omega$. From (4), $\delta = 7 \times 10^{-5}$ m. Thus, the ratio $d/\delta = 14.3$ so we can use (5) (times n), which makes $R_L = 0.026\ \Omega$. Accordingly, $(R_L + R_f)/R_r = 1790$ and $k = 1/1790 = 5.6 \times 10^{-4}$. (Dielectric losses are neglected.)

(*b*) From (16), $Q = 162$.

(*c*) From (16), $\Delta f_{HP} = 6.170$ kHz.

Although 6.17 kHz is adequate front-end *selectivity* for the 10-kHz channel spacing of the broadcast band, the low aperture efficiency of less than 0.06 percent makes it uncertain whether the *sensitivity* is adequate. To determine this, a calculation of the *signal-to-noise* ratio for a typical application is required. From the Friis transmission formula (2–11–5), the power received from a transmitter of power P_t at a distance r is

$$P_r = \frac{P_t A_{et} A_{er}}{r^2 \lambda^2} \quad \text{(W)} \tag{17}$$

where

A_{et} = effective aperture of transmitting antenna, m^2
A_{er} = effective aperture of receiving antenna, m^2

For a small loop receiving antenna, $D = \frac{3}{2}$ so

$$A_{er} = \frac{1.5\lambda^2 k}{4\pi} \quad (\text{m}^2) \tag{18}$$

where k = radiation efficiency factor

The signal-to-noise ratio (S/N) is given by

$$\frac{\text{S}}{\text{N}} = \frac{P_r}{N} \quad \text{(dimensionless)} \tag{19}$$

Table 7–3 lists formulas for loops.

[1]Distinguish between L for inductance in (16) and L for length in (5).

Table 7–3 Radiation resistance, loss resistance and radiation efficiency of small loop antennas and also formulas for Q, bandwidth and signal-to-noise ratio†

Quantity	Formula	Reference equation for units
Radiation resistance, single turn	$R_r = 31{,}200\left(\frac{A}{\lambda^2}\right)^2\ (\Omega)$	(7–10–6)
Radiation resistance, single turn	$R_r = 197C_\lambda^4\ (\Omega)$	(7–10–6)
Loss resistance, n-turn	$R_L = \frac{nL}{d}\sqrt{\frac{f\mu_0}{\pi\sigma}}\ (\Omega)$	(7–10–5)
Radiation efficiency, n-turn	$k = \frac{1}{1+(R_L/R_r)}$	(7–10–8)
R_L/R_r ratio, n-turn copper conductor	$\frac{R_L}{R_r} = \frac{3430}{C^3 f_{\text{MHz}}^{3.5} nd}$	(7–10–9)
Loss resistance, n-turn ferrite rod antenna	$R_f = 2\pi f\mu_{er}\frac{\mu_r''}{\mu_r'}\mu_0 n^2\frac{a}{l}\ (\Omega)$	(7–10–14)
Radiation efficiency, n-turn ferrite rod antenna	$k = \frac{1}{1+[(R_L+R_f)/R_r]}$	(7–10–15)
Q	$Q = \frac{2\pi f_0 L}{R_r + R_L + R_f}$	(7–10–16)
Bandwidth	$\Delta f_{\text{HP}} = \frac{f_0}{Q}$	(7–10–16)
Signal-to-noise ratio	$\frac{S}{N} = \frac{P_r}{N} = \frac{P_t A_{et} A_{er}}{r^2\lambda^2 kT_s \Delta f_{\text{HP}}}$	(7–10–17) and (7–10–19)

† L in third row for loss resistance is perimeter length, whereas L in eighth row for Q is inductance [see (7–10–16)].

REFERENCES（参考文献）

Alford, A. (1), and A. G. Kandoian: "Ultrahigh-Frequency Loop Antennas," *Trans. AIEE,* **59,** 843–848, 1940.

Foster, Donald (1): "Loop Antennas with Uniform Current," *Proc. IRE,* **32,** 603–607, October 1944.

Glinski, G. (1): "Note on Circular Loop Antennas with Nonuniform Current Distribution," *J. Appl. Phys.,* **18,** 638–644, July 1947.

Kraus (1) and Fleisch: *Electromagnetics with Applications,* 5th ed., McGraw-Hill, New York, p. 183.

Lowan, A. N. (1), and M. Abramowitz: *J. Math. Phys.,* **22,** 2–12, May 1943. The integral involving J_0 for the interval $0 \le x \le 5$ (where $x = C_\lambda$) is given; and given also by *Natl. Bur. Standards Tech. Memo* 20.

Watson, G. N. (1): *A Treatise on the Theory of Bessel Functions,* Cambridge University Press, London, 1922.

PROBLEMS（习题）

7–2–1（圆极化的环/偶极子组合） **Loop and dipole for circular polarization.** If a short electric dipole antenna is mounted inside a small loop antenna (on polar axis, Fig. 7–3) and both dipole and loop are fed in phase with equal. power, show that the radiation is everywhere circularly polarized with a pattern as in Fig. 7–7 for the 0.1 λ diameter loop.

7–4–1 (直径为 3λ/4 的环) **The 3λ/4 diameter loop.** Calculate and plot the far-field pattern normal to the plane of a circular loop 3λ/4 in diameter with a uniform in-phase current distribution.

***7–6–1** (环的辐射电阻) **Radiation resistance of loop.** What is the radiation resistance of the loop of Prob. 7–4–1?

7–6–2 (小环的电阻) **Small-loop resistance.** (*a*) Using a Poynting vector integration, show that the radiation resistance of a small loop is equal to $320\pi^4(A/\lambda^2)^2\ \Omega$ where A = area of loop (m^2). (*b*) Show that the effective aperture of an isotropic antenna equals $\lambda^2/4\pi$.

7–7–1 (直径为 λ/10 的环) **The λ/10 diameter loop.** What is the maximum effective aperture of a thin loop antenna 0.1 λ in diameter with a uniform in-phase current distribution?

7–8–1 (环的波瓣图、辐射电阻和定向性) **Pattern, radiation resistance and directivity of loops.** A circular loop antenna with uniform in-phase current has a diameter d. What is (*a*) the far-field pattern (calculate and plot), (*b*) the radiation resistance and (*c*) the directivity for each of three cases where (1) $d = \lambda/4$, (2) $d = 1.5\lambda$ and (3) $d = 8\lambda$?

***7–8–2** (圆环) **Circular loop.** A circular loop antenna with uniform in-phase current has a diameter d. Find (*a*) the far-field pattern (calculate and plot), (*b*) the radiation resistance and (*c*) the directivity for the following three cases: (1) $d = \lambda/3$, (2) $d = 0.75\lambda$ and (3) $d = 2\lambda$.

***7–9–1** (边长 1λ 的方环) **The 1 λ square loop.** Calculate and plot the far-field pattern in a plane normal to the plane of a square loop and parallel to one side. The loop is 1 λ on a side. Assume uniform in-phase currents.

7–9–2 (小方环) **Small square loop.** Resolving the small square loop with uniform current into four short dipoles, show that the far-field pattern in the plane of the loop is a circle.

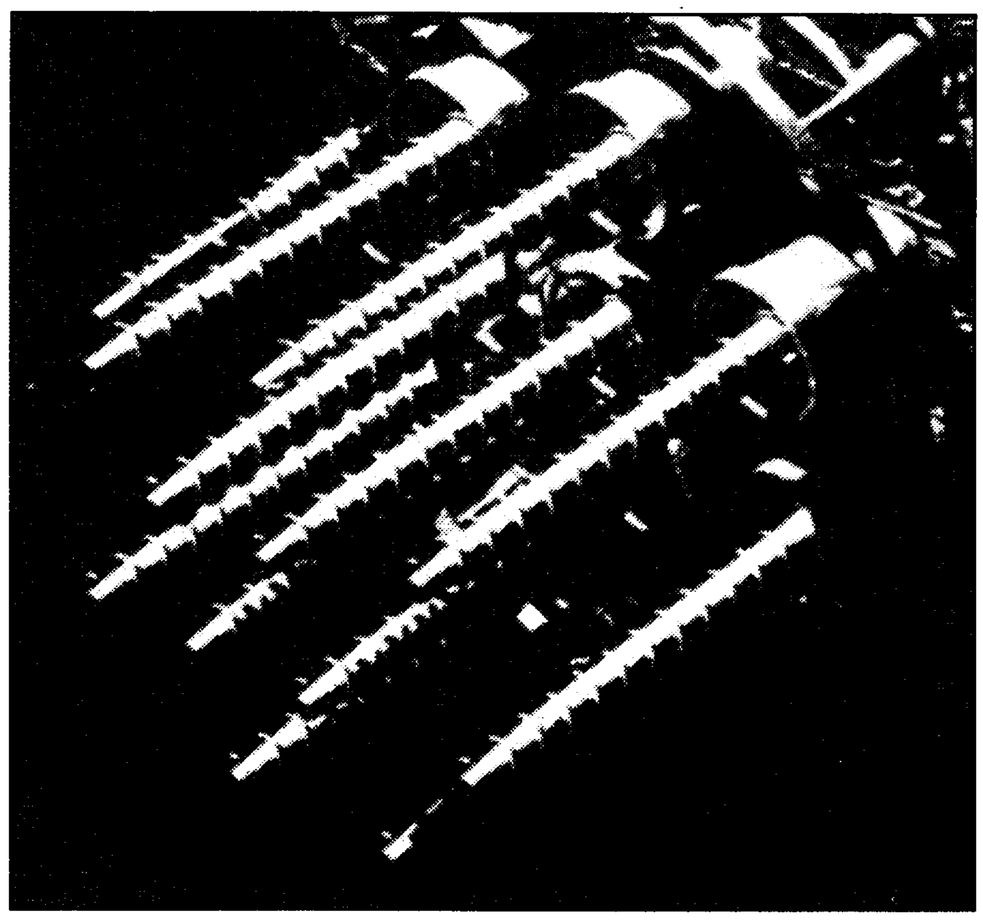
7–10–1 (波长 2 m 的测向天线) **Direction-finding antenna for 2 m.** An amateur radio operator would like to join some fellow hams for a direction-finding hunt with the transmitter operating at 146 MHz with a bandwidth of 0.1 MHz. The transmitting antenna will be a half-wave dipole with only 10 W of power. (*a*) Design a loop antenna for the amateur to use to find the transmitter. Note that direction finding is based on minimizing the signal in the antennas null and that a loop needs to be small in circumference to have a null on axis. How close to the transmitter will he have to be before he can detect it, assuming he needs a 10 dB SNR? (*b*) With this in mind, what is the main drawback of using a loop or similar antenna for direction finding? (See Prob. 21–9–3.)

For computer programs, see Appendix C.

CHAPTER 8

End-Fire Antennas: The Helical Beam Antenna[1] and the Yagi-Uda Array, Part I

（上）端射天线：螺旋聚束天线和八木 – 宇田天线



Courtesy Aerospace Corp.

Topics in this chapter include:

- The helical beam antenna story（螺旋聚束天线的故事）
- Helical geometry（螺旋的几何表示）
- Helical antenna design（螺旋天线设计）
- Helix without ground plane（不带接地板的螺旋）
- Dipole arrays with parasitic elements（含有寄生单元的偶极子阵）
- The Yagi-Uda array story（八木 – 宇田阵的故事）
- Yagi-Uda array theory（八木 – 宇田阵的理论）

8–1 THE HELICAL BEAM ANTENNA STORY *by John Kraus*
（螺旋聚束天线的故事）

In 1946, a few months after joining the faculty at Ohio State University, I attended an afternoon lecture on traveling-wave tubes by a famous scientist who was visiting the campus. In these tubes an electron beam is fired down the inside of a long wire helix for amplification of waves traveling along the helix. The helix is only a small fraction of a wavelength in diameter and acts as a guiding structure. After the lecture I asked the visitor if he thought a helix could be used as an antenna. "No," he replied, "I've tried it and it doesn't work." The finality of his answer set me thinking. If the helix were larger in diameter than in a traveling-wave tube, I felt that it would have to radiate in some way, but how, I did not know. I determined to find out.

That evening in the basement of my home I wound a seven-turn helical coil of wire 1 λ in circumference and fed it via coaxial line and ground plane from my 12-cm oscillator (Fig. 8–1). I was thrilled to find that it produced a sharp beam of circularly polarized radiation off its open end.

---

[1] Also called the "axial-mode helical antenna." The normal, bifilar, quadrifilar, four-lobed and other modes are discussed later, in this chapter.

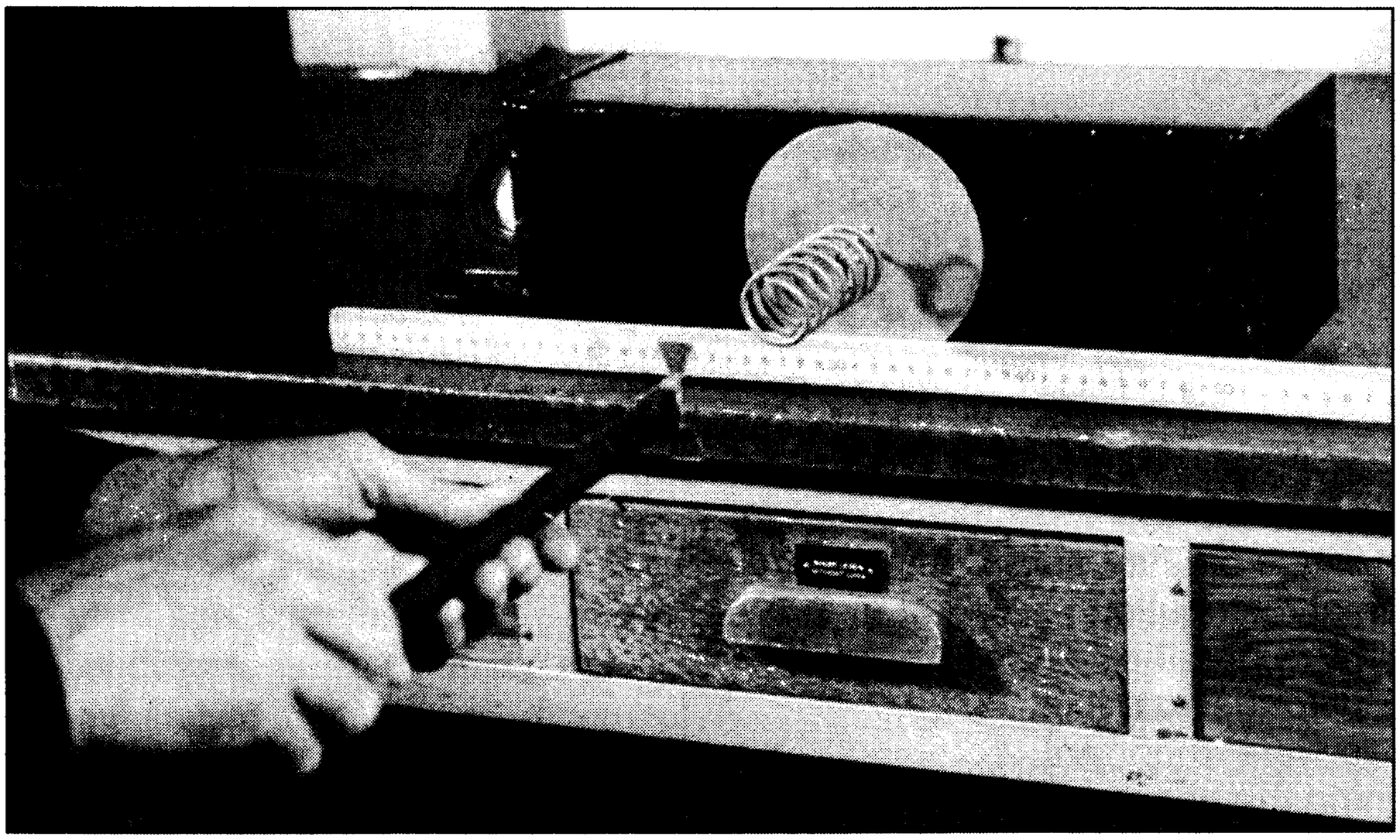


Figure 8–1
The first helical-beam antenna (1946). When I rotated the hand-held dipole probe, there was no change in response, indicating circular polarization.

Next I wound other helices with larger and smaller diameters, noting little change in behavior. Adding more turns, however, resulted in sharper beams. Although my invention/discovery had come quickly, I realized then that much work would be required to understand this remarkable antenna. Actually it took years of extensive measurements and calculations. My first publication entitled "The Helical Beam Antenna" appeared in the April 1947 issue of *Electronics*. I followed with many articles, a few with students to whom I had assigned studies of specific properties of the antenna (Kraus-1, 5; Glasser-1; Tice-1).

The steps taken to unravel the mystery of the helix went something like this. The input impedance was measured and found to be essentially resistive and constant over a wide bandwidth. This suggested that the helix behaved like a terminated (matched) transmission line. This was hard to understand because the open end of the helix was completely unterminated. New insights came when we measured the current distribution along the helix. This we did by rotating a helix and its ground plane while holding a small loop (current probe) under the helical conductor (Fig. 8–2). At a low frequency (helix circumference about $\lambda/2$) there was an almost pure standing wave (VSWR $\rightarrow \infty$) all along the helix (outgoing and reflected waves nearly equal) (Fig. 8–3*a*), but as the frequency increased, the distribution changed dramatically. For a helix circumference of about 1 λ three regions appeared: near the input end the current decayed exponentially, near the open end there was a standing wave over a short distance, while between the ends there was a relatively uniform current amplitude (small VSWR) which extended over most of the helix (Fig. 8–3*b*). The decay at the input end could be understood as a transition between a helix-to-ground-plane mode and a pure helix mode. The reflection of the outgoing wave at the open end also decayed exponentially to a much smaller reflected wave, leaving the outgoing wave dominant over most

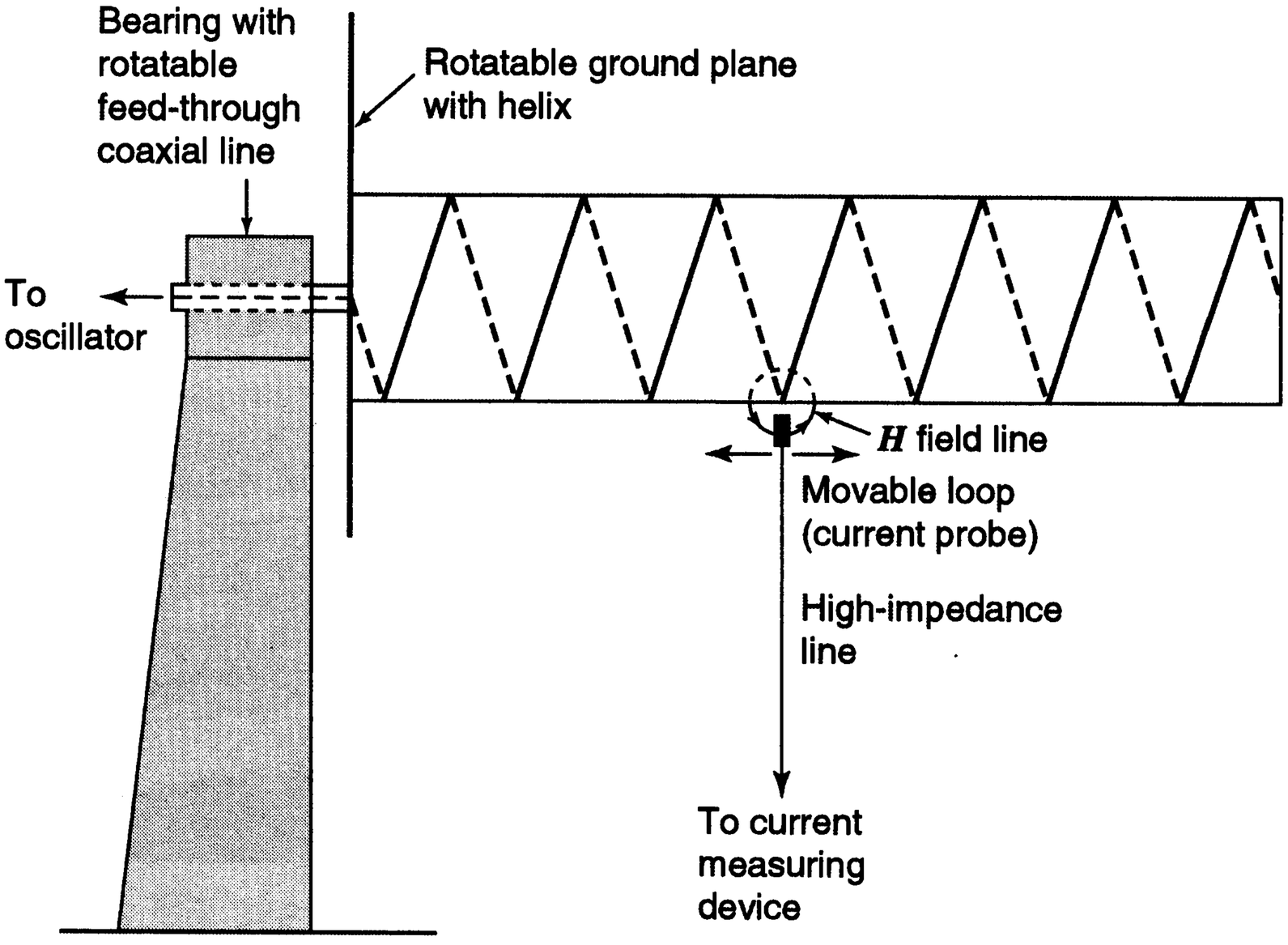


Figure 8–2
Helix-and-ground-plane mounted to rotate on the helix axis for current distribution measurements along the helical conductor using a loop probe. These were important for determining the phase velocity along the helix. (*After Kraus (2) and Williamson.*)

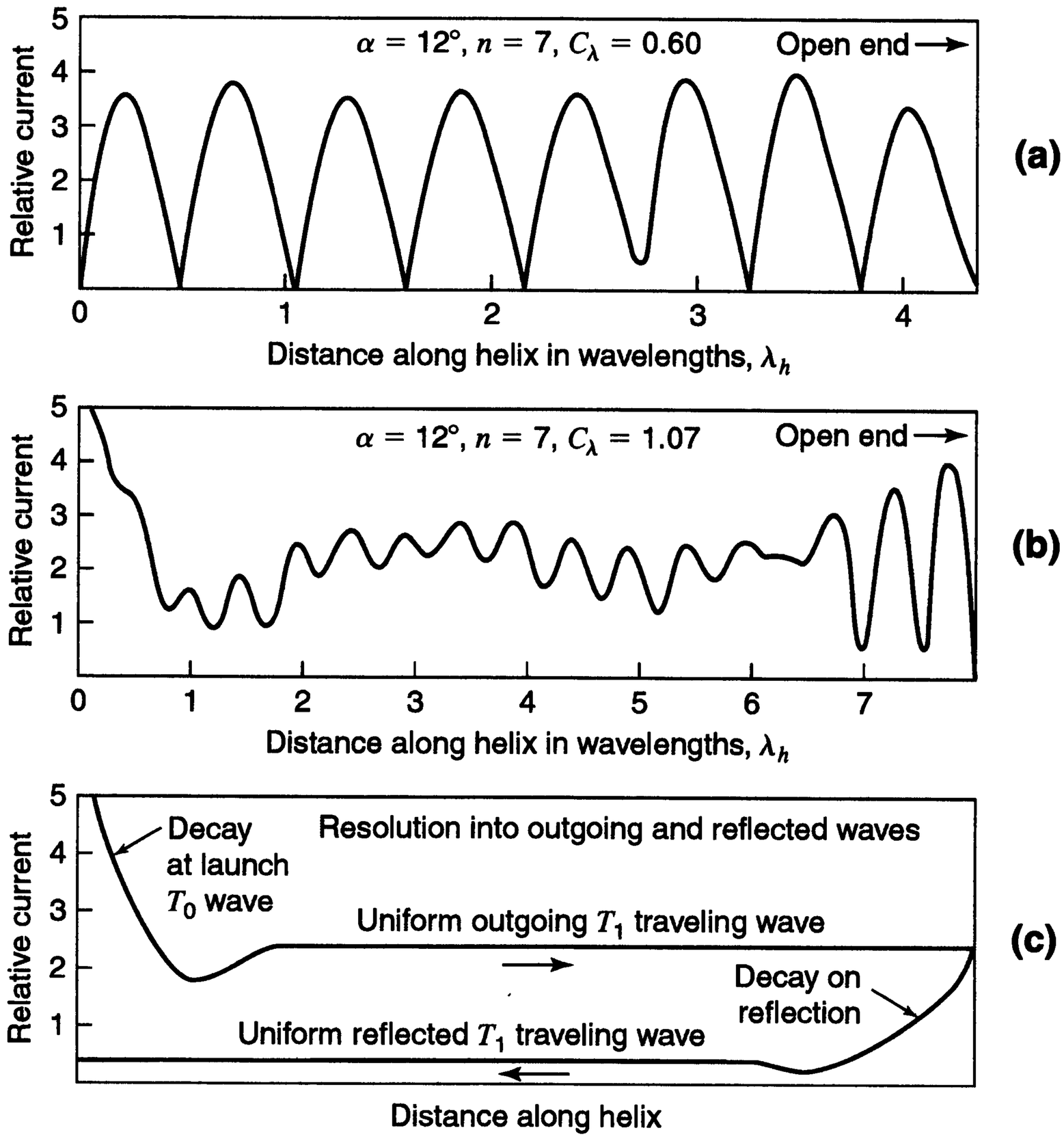


Figure 8–3
Typical measured current distribution (*a*) at a frequency below the axial mode of operation and (*b*) at a frequency near the center of the axial-mode region. (*c*) Resolution of currents into outgoing and reflected waves. (*After Kraus (2) and Williamson.*)

of the helix (VSWR small). The small VSWR ripple was sufficient, however, to measure the relative phase velocity ($= \lambda_h/\lambda_0$) along the helix, which was useful for an understanding of the radiation patterns. The current distribution resolved into outgoing and reflected waves is shown in Fig. 8–3*c* (Kraus (2) and Williamson).

Our extensive pattern measurements showed that the end-fire beam mode persists over a frequency range of about 2 to 1 centered on the frequency for which the circumference is 1 λ. Thus, the diameter I had chosen for the first helix I tried was optimum!

Although the helix is continuous, it can also be regarded as a periodic structure. Thus, assuming that an *n*-turn helix is an end-fire array of *n* sources, I calculated the pattern using the formula (5–6–9) for the ordinary end-fire condition. Surprisingly, the measured helix patterns were much sharper. Could the helix be operating in the increased-directivity condition? I calculated patterns for this condition using the formula (5–6–14) and obtained good agreement with the measured patterns. Furthermore, this condition persisted over a wide bandwidth, indicating that the phase velocity on the helix changes by just the right amount to maintain the increased-directivity condition. The phase velocity measurements we had made also confirmed this. Thus, the helix locks onto the increased-directivity condition and automatically stays locked over the full bandwidth.

Not only does the helix have a nearly uniform resistive input over a wide bandwidth but it also operates as a "supergain" end-fire array over the same bandwidth! Furthermore, it is noncritical with respect to conductor size and turn spacing. It is also easy to use in arrays because of almost negligible mutual impedance.

[螺旋天线不仅在宽频带上具有近乎一致的电阻性输入阻抗，而且在同样的频带上按"超增益"(super gain)端射阵的波瓣图工作！进一步说，它的性能对导线尺寸和螺旋节距不敏感；它的互阻抗几乎可以忽略，因此很容易用来组阵。]

The helix immediately found wide application. I employed it in an array of 96 11-turn helices in a radio telescope I designed and built with my students in 1951 (Fig. 8–4). Operating at frequencies of 200 to 300 MHz, the array measured 50 m in length and had a gain of 35 dB. With it we produced some of the first and most extensive maps of the radio sky (Kraus-6). Others employed the helix over a wide range of frequencies, some at frequencies as low as 10 MHz.

Following Sputnik the helical antenna became the workhorse of space communications for telephone, television and data, being employed both on satellites and at ground stations. Many U.S. satellites, including its weather satellites, Comsat, Fleetsatcom (Fig. 8–6), GOES (global environmental satellites), Leasat, GPS (Global Position Satellites) (Fig. 8–5), Westar and tracking and data-relay satellites, all have helical antennas, the latter with arrays of 30. Russian satellites also have helical antennas, each of the Ekran class satellites being equipped with an array of 96 helicals. The helical antenna has been carried to the Moon and Mars (Fig. 8–7). It is also on many other probes of planets and comets, being used alone, in arrays or as feeds for parabolic reflectors, its circular polarization, high gain and simplicity making it especially attractive for space applications.[1]

This short account provides a brief introduction to the helix in which some of the experimental and analytical steps taken to understand its behavior are outlined. Specifically, the helix can be described as a *monofilar (one-wire) axial-mode helical antenna*.

---

[1] A more detailed personal account of his early work on the helical antenna is given by John Kraus in "Big Ear II," 1995, Cygnus-Quasar Books.

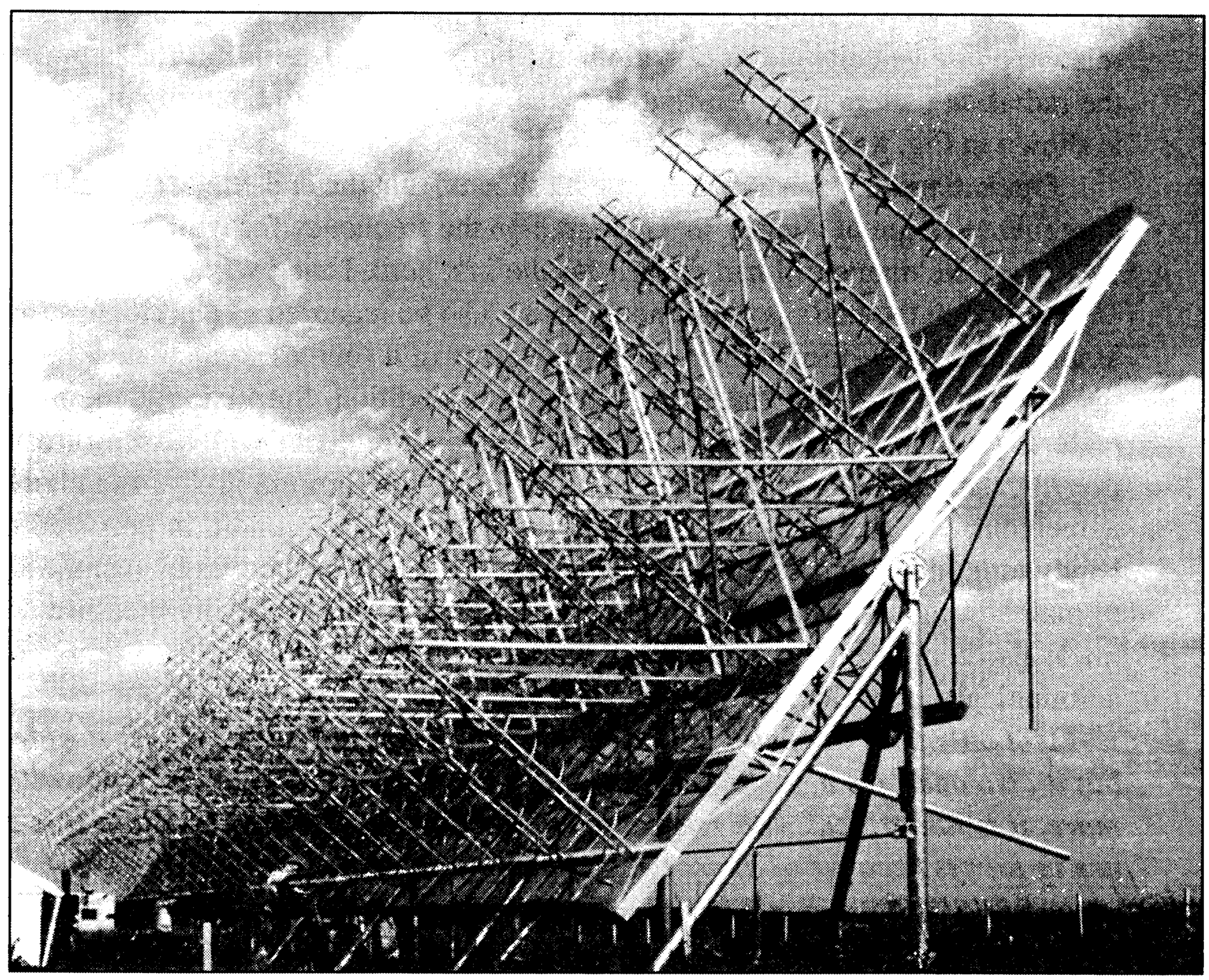


Figure 8–4
Radio telescope at the Ohio State University Radio Observatory with array of 96 11-turn monofilar axial-mode helical antennas mounted on a tiltable ground plane 50 m long. This array was used to make some of the first and most extensive maps of the radio sky.

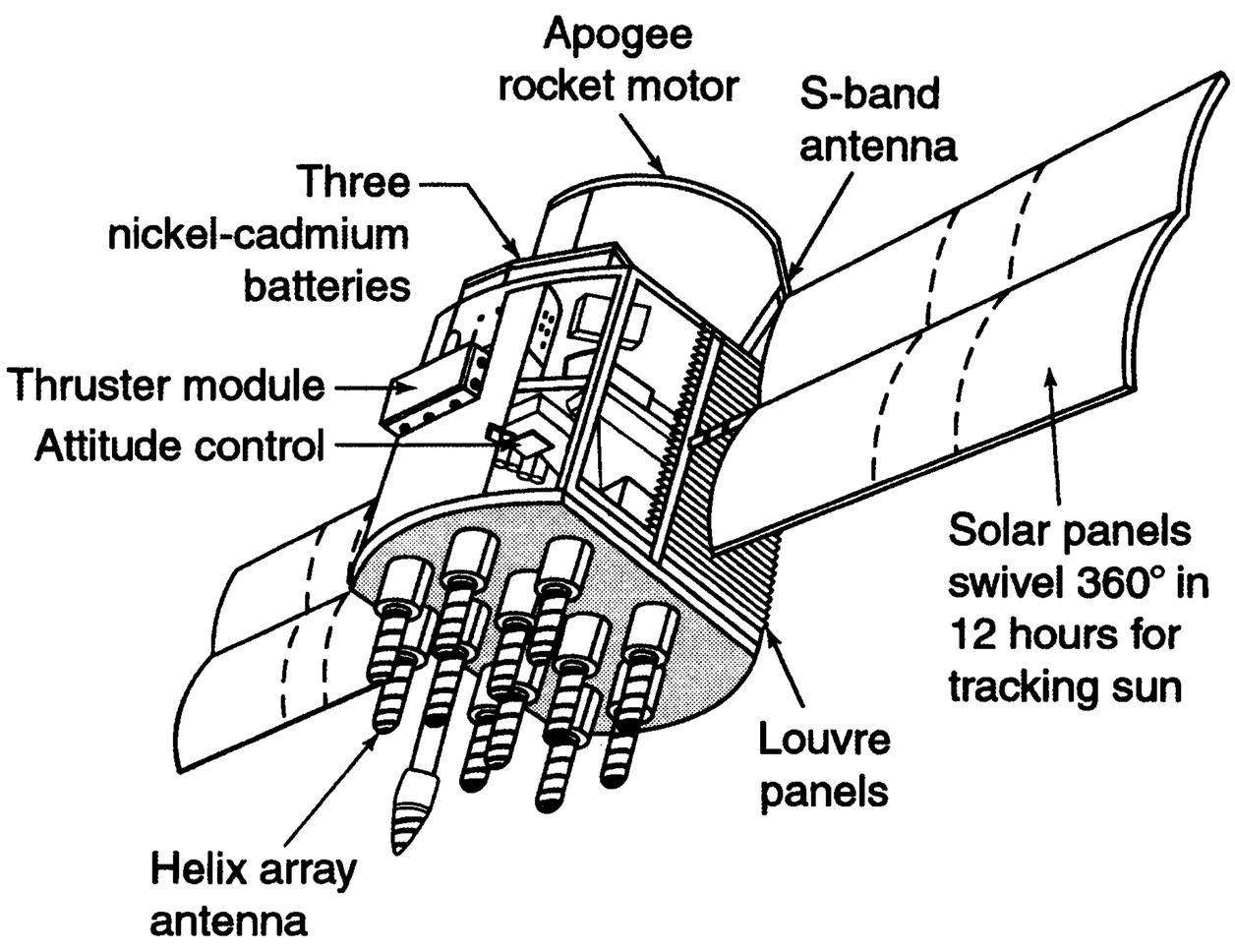


Figure 8–5
GPS or Global-Position Satellite with array of 12 helical antennas. Twenty four of these satellites are in elliptical orbits around the earth. From them one can determine one's absolute position anywhere on the earth (latitude, longitude and altitude), and at any time and in any weather, to a precision of a few centimeters and relative position to a few millimeters.

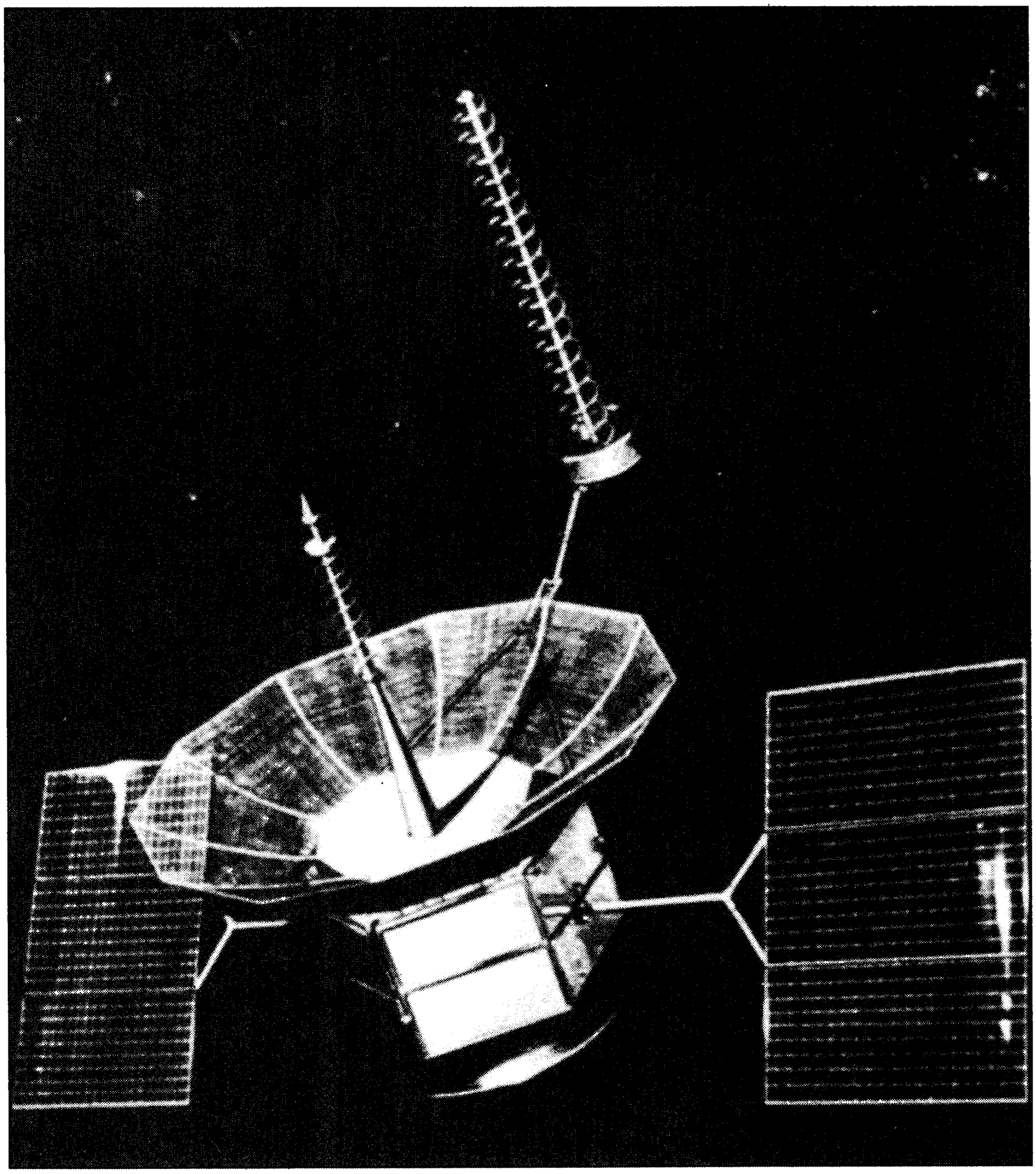


Figure 8–6
Fleetsatcom geostationary relay satellite with monofilar axial-mode helical antennas for transmission and reception with one as the feed for a dish. These satellites provide global communications for the U.S. government. (*Courtesy TRW Corp., King-1.*)

8–2 HELICAL GEOMETRY（螺旋的几何表示）

The helix is a basic three-dimensional geometric form. A helical wire on a uniform cylinder becomes a straight wire when unwound by rolling the cylinder on a flat surface. Viewed end-on, a helix projects as a circle. Thus, a helix combines the geometric forms of a straight line, a circle and a cylinder. In addition a helix has handedness; it can be either left- or right-handed.

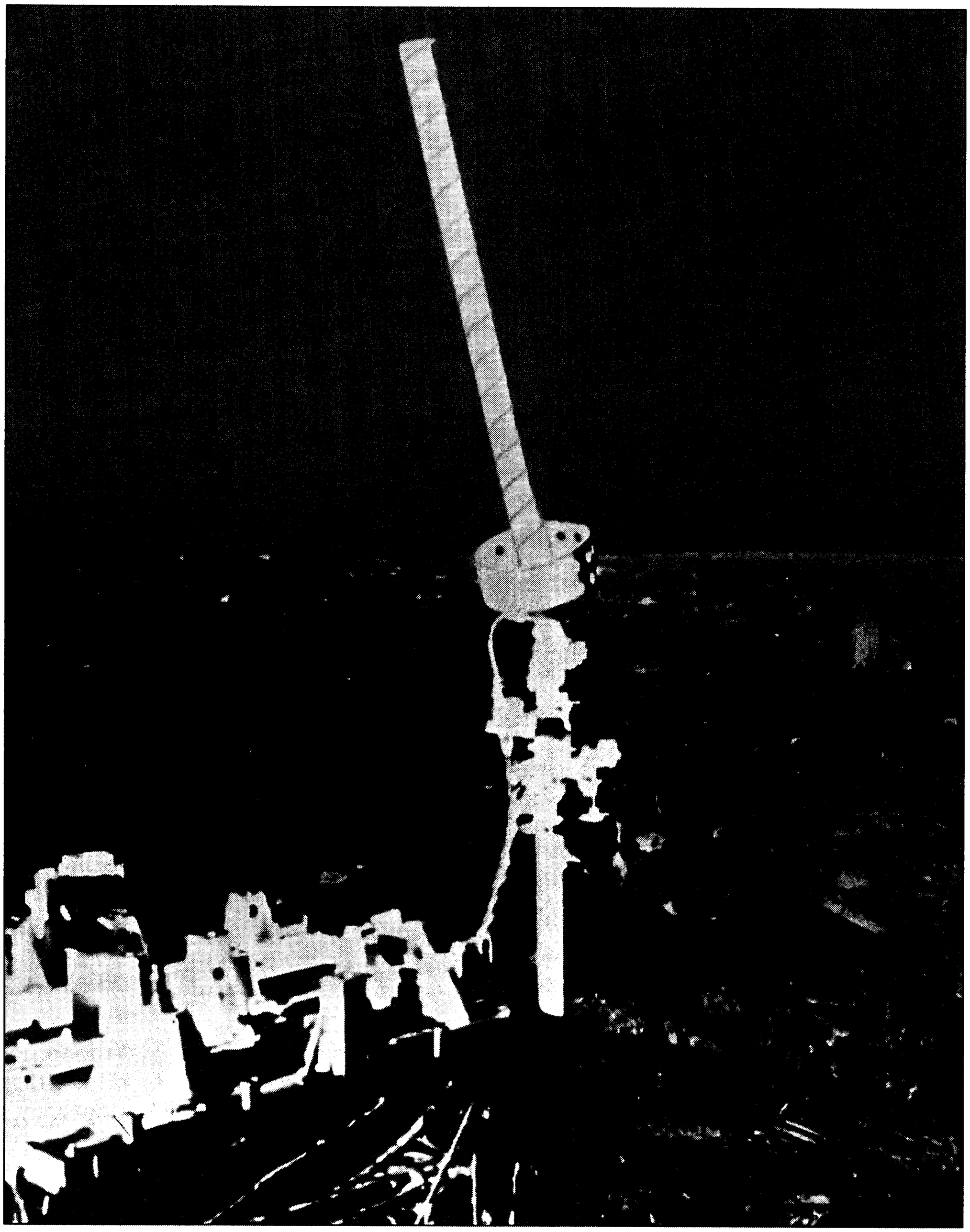


Figure 8–7
Helical beam antenna placed on the Fra Mauro highlands of the moon by the Apollo 14 astronauts Alan Shepard, Jr., and Edgar Mitchell for radioing information back to earth about conditions there. The helix wire is wound on a thin plastic tube.

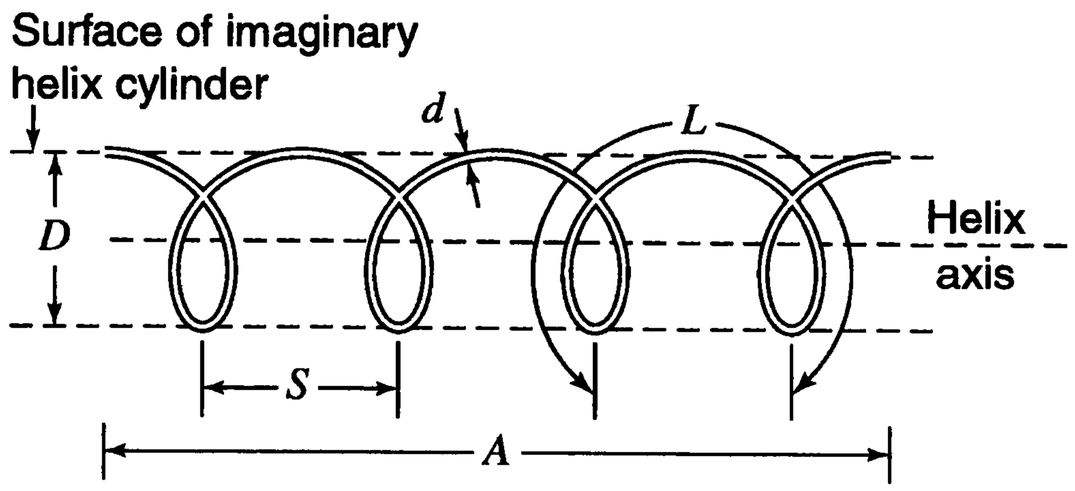


Figure 8–8
Helix and its dimensions.

The following symbols are used to describe a helix (Fig. 8–8).

D = diameter of helix (center to center)
C = circumference of helix = πD
S = spacing between turns (center to center)
α = pitch angle = arctan $S/\pi D$
L = length of 1 turn
n = number of turns
A = axial length = nS
d = diameter of helix conductor

The diameter D and circumference C refer to the imaginary cylinder whose surface passes through the centerline of the helix conductor. A subscript λ signifies that the dimension is measured in *free-space wavelengths*. For example, D_λ is the helix diameter in free-space wavelengths.

If 1 turn of a circular helix is unrolled on a flat plane, the relation between the spacing S, circumference C, turn length L and pitch angle α is as illustrated by the triangle in Fig. 8–9.

The dimensions of a helix are conveniently represented by a diameter-spacing chart or, as in Fig. 8–10, by a circumference-spacing chart. On this chart the dimensions of a helix may be expressed either in rectangular coordinates by the spacing S_λ and circumference C_λ or in polar coordinates by the length of 1 turn L_λ and the pitch angle α. When the spacing is zero, $\alpha = 0$ and the helix becomes a loop. On the other hand, when the diameter is zero, $\alpha = 90°$ and the helix becomes a linear conductor. Thus, in Fig. 8–10 the ordinate axis represents loops while the abscissa axis represents linear conductors. The entire area between the two axes represents the general case of the helix.

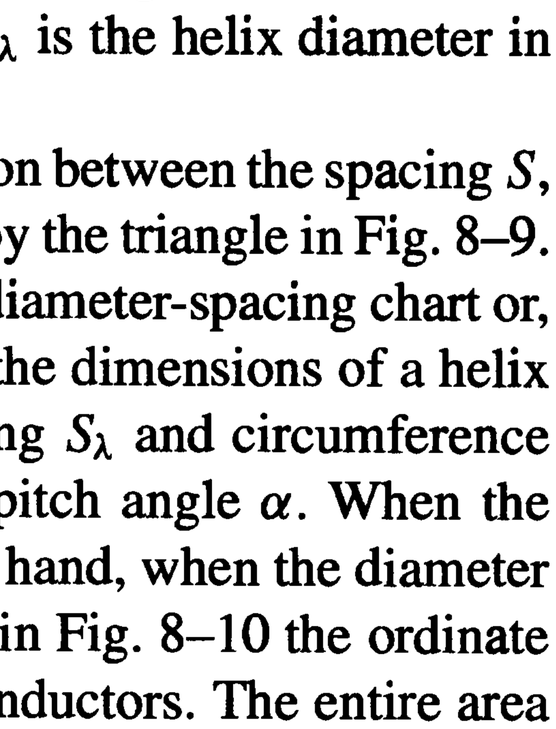


Figure 8–9
Relation between circumference, spacing, turn length and pitch angle of a helix.

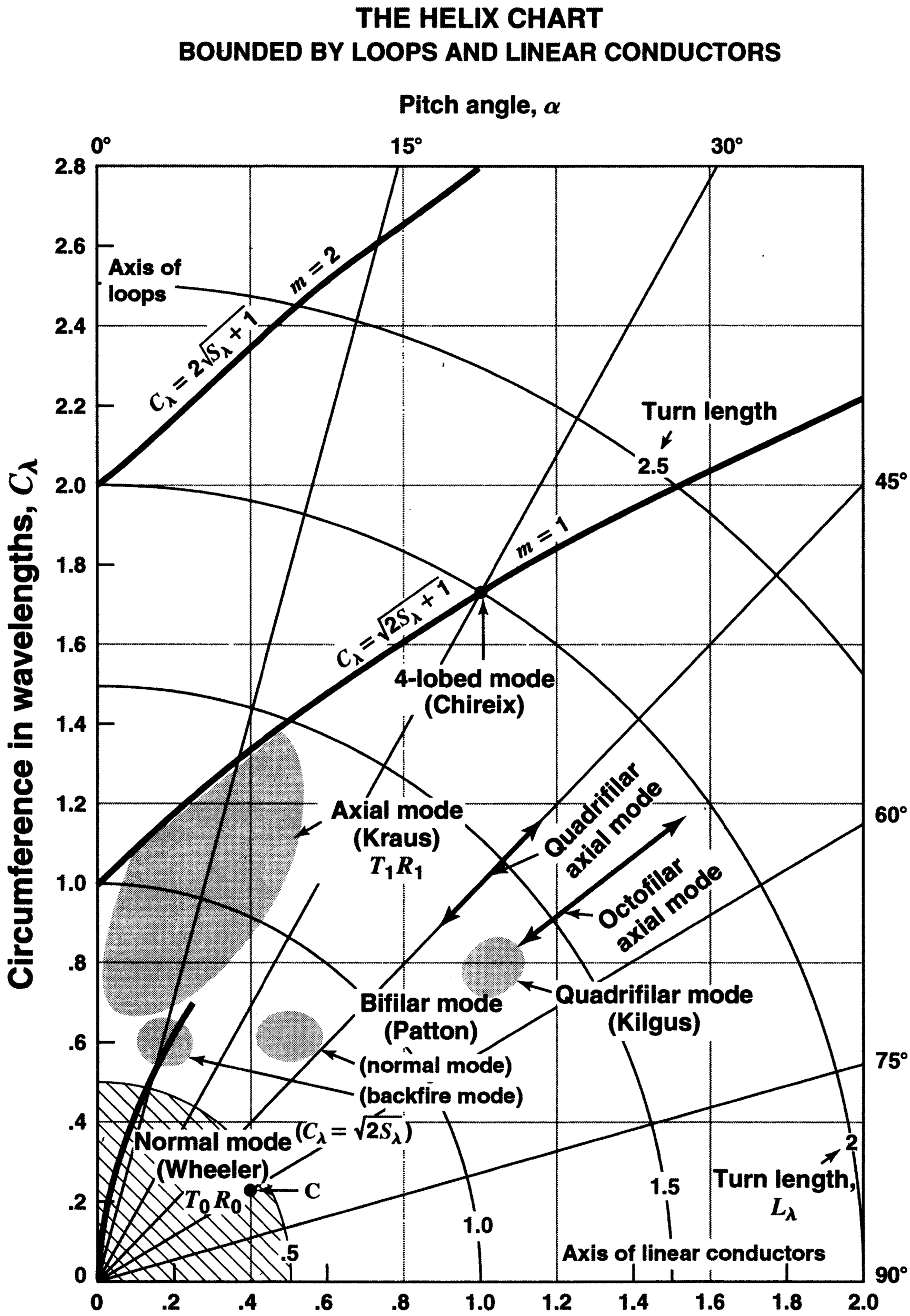


Figure 8–10
Helix chart showing the location of different modes of operation as a function of the helix dimensions (diameter, spacing and pitch angle). As a function of frequency the helix moves along a line of constant pitch angle. *Along the vertical axis the helices become loops and along the horizontal axis they become linear conductors.* The 4-lobed, quadrifilar, bifilar and normal modes are discussed in Chap. 8, Part II. Point *C* is a bifilar mode location (see Sec. 8–25).

Suppose that we have a 1-turn helix with a turn length of 1 λ ($L_\lambda = 1$). When $\alpha = 0$, the helix is a loop of 1 λ circumference or of diameter equal to 1 λ/π. As the pitch angle α increases, the circumference decreases and the dimensions of the helix move along the $L_\lambda = 1$ curve in Fig. 8–10 until, when $\alpha = 90°$, the "helix" is a straight conductor 1 λ long.

The helix dimensions for various operational modes are indicated in Fig. 8–10. Reference to these will be made as we proceed through the chapter.

8–3 PRACTICAL DESIGN CONSIDERATIONS FOR THE MONOFILAR AXIAL-MODE HELICAL ANTENNA（单绕轴向模螺旋天线的实际设计考虑）

Before analyzing the many facets of the antenna individually, an overall picture will be given by describing the performance of some practical designs.

The monofilar axial-mode helical antenna is very noncritical and one of the easiest of all antennas to build. Nevertheless, attention to details can maximize its performance.

The important parameters are:

1. Beamwidth
2. Gain
3. Impedance
4. Axial ratio

Gain and *beamwidth,* which are interdependent [$G \propto (1/\text{HPBW}^2)$], and the other parameters are all functions of the number of turns, the turn spacing (or pitch angle) and the frequency. For a given number of turns the behavior of the *beamwidth, gain, impedance* and *axial ratio* determines the useful *bandwidth*. The nominal center frequency of this bandwidth corresponds to a helix circumference of about 1 λ ($C_\lambda = 1$). For a given bandwidth to be completely useful, all 4 parameters must be satisfactory over the entire bandwidth.

The parameters are also functions of the ground plane size and shape, the helical conductor diameter, the helix support structure and the feed arrangement. The ground plane may be flat (either circular or square) with a diameter or side dimension of at least $3\lambda/4$ or the ground plane (launching structure) may be cup-shaped forming a shallow cavity (Fig. 8–11) or replaced by loops (Sec. 8–4).

A two-turn flush-mounted design described by Bystrom (1) and Bernsten for aircraft applications is shown in Fig. 8–11*c*. These authors found that two turns are required to obtain satisfactory pattern and impedance characteristics but that no significant improvement is obtained with a deeper cavity and a larger number of turns since the size of the aperture opening remains the same (like an open-ended cylindrical waveguide).

The deep conical arrangement of Fig. 8–11*d* is effective in reducing the side- and back-lobe radiation [Carver (1–2)]. Launching a wave on the helix may also be done without a ground plane using loops (Sec. 8–4) or to produce a back fire beam for a dish feed as in Fig. 8–6.

Conductor size is not critical (Tice-1) and may range from 0.005 λ or less to 0.05 λ or more (Fig. 8–12). The helix may be supported by a few radial insulators mounted on an axial dielectric or metal rod or tube whose diameter is a few hundredths of a wavelength, by one or more longitudinal dielectric rods mounted peripherally (secured directly to the helical conductor) or by a thin-wall dielectric tube on which the helix is wound. With the

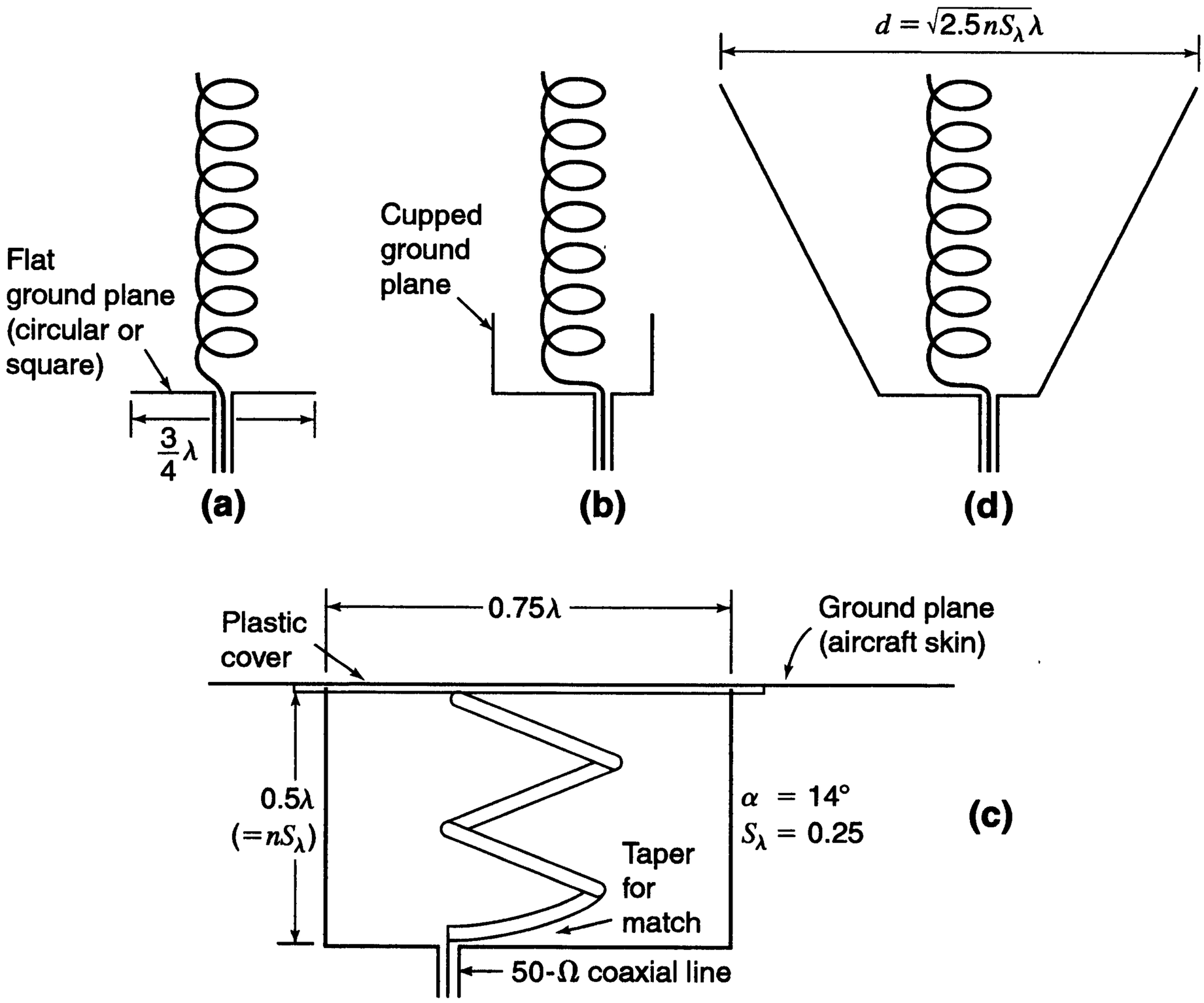


Figure 8–11
(*a*) Monofilar axial-mode helical antenna on flat ground plane and (*b*) in shallow cupped ground plane (see also Fig. 8–16*c*). (*c*) General-purpose flush-mounted two-turn monofilar axial-mode helical antenna with taper feed for matching to a 50-Ω coaxial line (*after Bystrom (1) and Bernsten*) (see also Fig. 8–16*a* and *b*). (*d*) Deep conical ground-plane enclosure for reducing side and back lobes. (*After K. R. Carver (1, 2).*)

latter arrangement the operating bandwidth is shifted to lower frequencies so that for a given frequency the antenna is smaller. Several of these mounting arrangements are illustrated in Fig. 8–13.

The helix may be fed axially, peripherally or from any convenient location on the ground-plane launching structure with the inner conductor of the coaxial line connected to the helix and the outer conductor bonded to the ground plane.

With axial feed the terminal impedance (resistive) is given within 20 percent by

$$R = 140C_\lambda \qquad (\Omega) \tag{1}$$

while with peripheral feed Baker (1) gives its value within 10 percent as

$$R = \frac{150}{\sqrt{C_\lambda}} \qquad (\Omega) \tag{2}$$

These relations have the restrictions that $0.8 \leq C_\lambda \leq 1.2$, $12° \leq \alpha \leq 14°$ and $n \geq 4$.

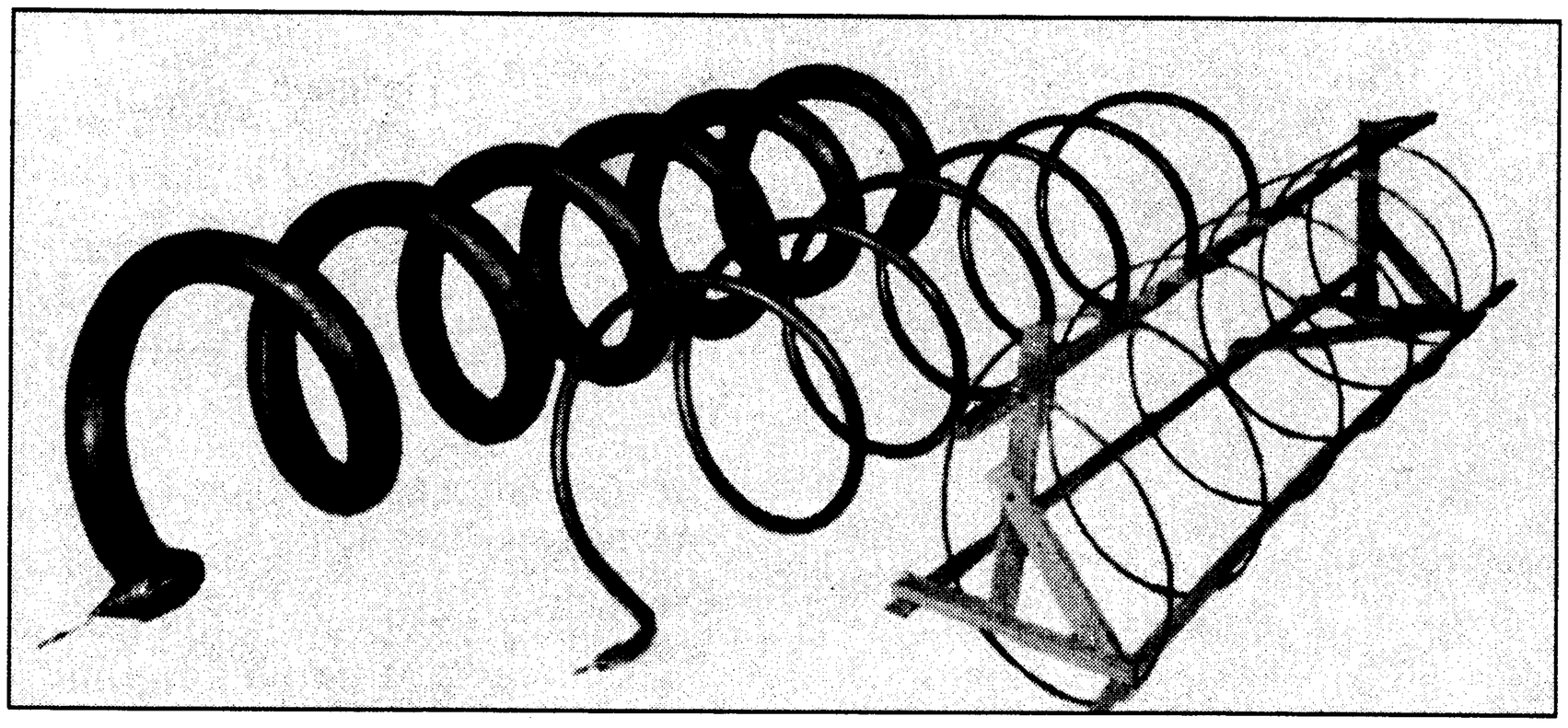


Figure 8–12
Peripherally fed monofilar axial-mode helical antennas with helix conductors of 0.055, 0.017 and 0.0042 λ diameter at center frequency of 400 MHz for determining effect of conductor diameter on helix performance. Only minor differences were measured. (*After T. E. Tice (1) and J. D. Kraus.*) The 0.055 λ diameter tubing (4.1 cm diameter) is about the largest size which could be bent to the radius of 11 cm ($=\lambda/2\pi$).

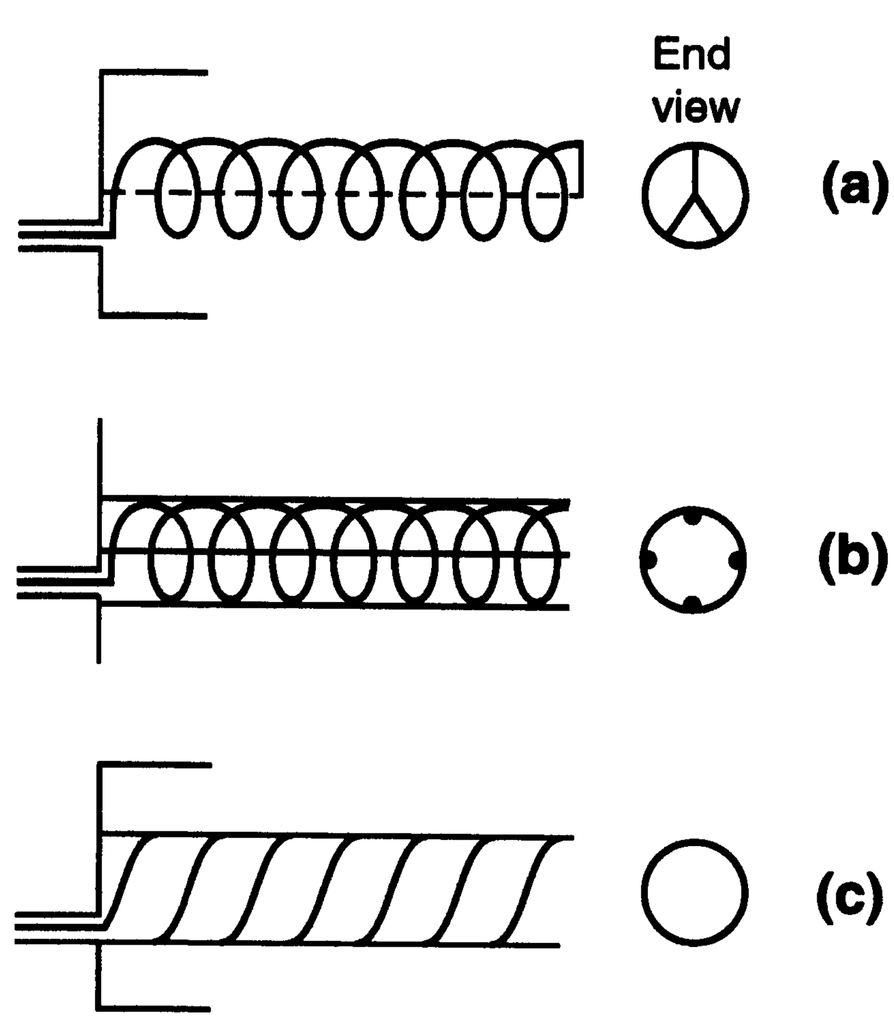


Figure 8–13
Monofilar axial-mode helical antenna supported by axial metal or dielectric rod (or tube) with radial insulators (*a*), by four peripheral dielectric rods secured to the helix (*b*) and by a dielectric tube on which the helix is wound (*c*).

With a suitable matching section the terminal impedance (resistive) can be made any desired value from less than 50 Ω to more than 150 Ω. Thus, by bringing the last $\frac{1}{4}$-turn of the helix parallel to the ground plane in a gradual manner, a tapered transition between the 140- or 150-Ω helix impedance and a 50-Ω coaxial line can be readily accomplished. This can be done with either axially or peripherally fed helices but is more convenient with a peripheral feed. Details of a suitable arrangement are shown in Fig. 8–14*a* and *b*.

As the helix tubing is brought close to the ground plane, it is gradually flattened until it is completely flat at the termination, where it is spaced from the ground plane by a dielectric

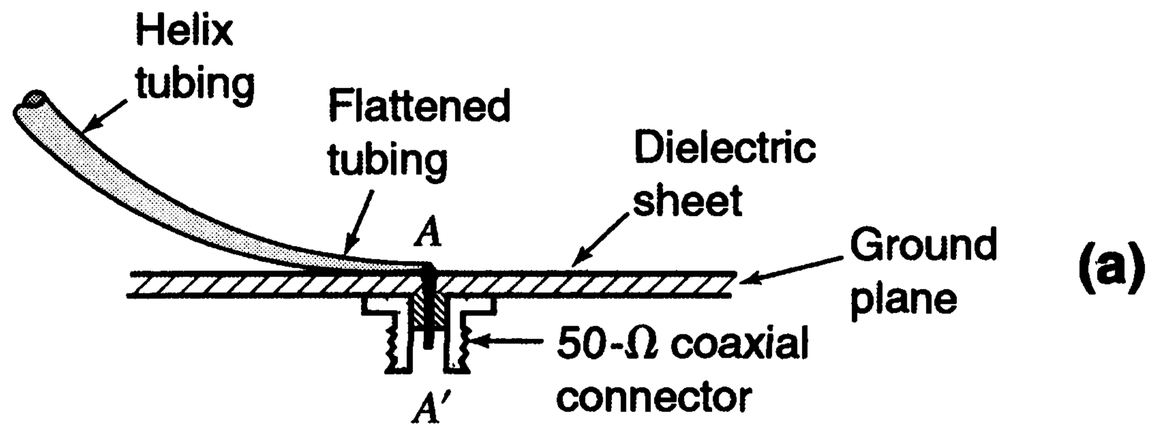


Figure 8–14
(*a*) Gradually tapered transition from helix to coaxial line with detailed cross section at (*b*).

sheet (or slab). The appropriate height h (or thickness of the sheet) is given by

$$h = \frac{w}{\left[377/\left(\sqrt{\varepsilon_r} Z_0\right)\right] - 2} \tag{3}$$

where

w = width of conductor at termination
h = height of conductor above ground plane (or thickness of dielectric sheet) in same units as w
ε_r = relative permittivity of dielectric sheet
Z_0 = characteristic impedance of dielectric sheet

EXAMPLE 8–3.1

If the flattened tubing width is 5 mm, find the required thickness of a polystyrene sheet ($\varepsilon_r = 2.7$) for matching to a 50-Ω coaxial transmission line.

■ **Solution**

From (3),

$$h = \frac{5}{\left[377/\left(\sqrt{2.7} \times 50\right)\right] - 2} = 1.9 \text{ mm}$$

A typical peripherally fed monofilar axial-mode helical antenna with cup ground-plane launcher matched to a 50-Ω line, as in Fig. 8–14*a* and *b*, is shown in Fig. 8–14*c* with dimensions given in wavelengths at the center frequency for which $C_\lambda = 1$. Support may be an axial rod with radial insulators or one or more peripheral rods (Fig. 8–13*a* and *b*).

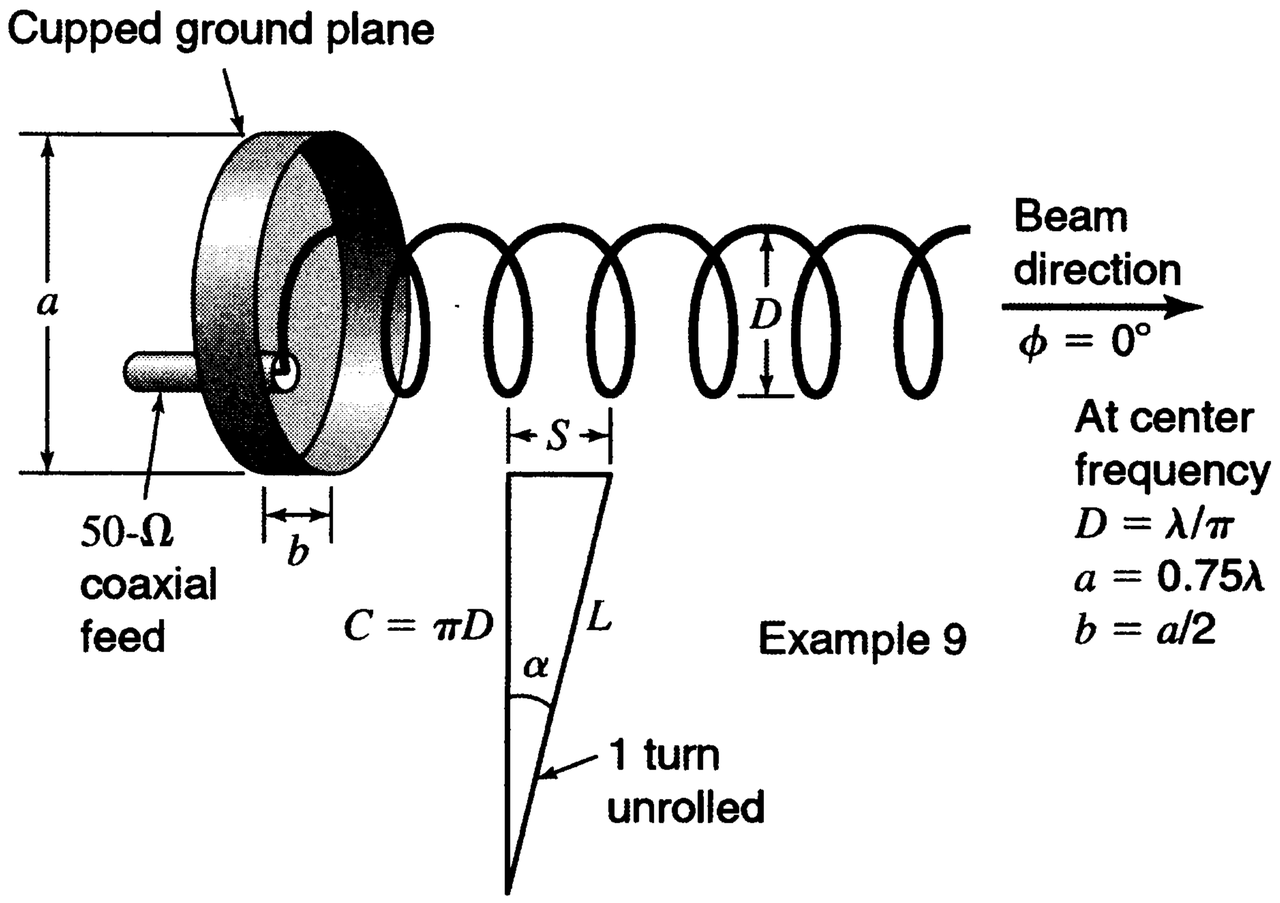


Figure 8–14c
Typical peripherally fed monofilar axial-mode helical antenna with cupped ground plane matched to a 50-Ω coaxial transmission line as in Fig. 8–14*a* and *b*. The turn spacing $S = 0.225\ \lambda$ and the circumference $C = \lambda$ at the center frequency. The relative phase velocity p changes automatically by just the right amount to lock onto end-fire ($\phi = 0°$) with supergain over a frequency range of about one octave. Typical dimensions of the cupped ground plane are $a = 0.75\ \lambda$ and $b = a/2$ at the center frequency.

A monofilar axial-mode helical antenna with flat circular ground plane and supported by dielectric members is shown in Fig. 8–15 and one with cupped ground plane supported by a dielectric cylinder is illustrated in Fig. 8–16.

Measured patterns of a 6-turn helix as a function of frequency are presented in Fig. 8–17 and patterns at the center frequency ($C_\lambda = 1$) as a function of length (number of turns) are shown in Fig. 8–18.

Based on a large number of such pattern measurements Kraus made during 1948 and 1949, the beamwidths were found to be given by the following quasi-empirical relations.

$$\text{HPBW (half-power beamwidth)} \simeq \frac{52}{C_\lambda \sqrt{n S_\lambda}} \quad \text{(deg)} \tag{4}$$

$$\text{BWFN (beamwidth between first nulls)} \simeq \frac{115}{C_\lambda \sqrt{n S_\lambda}} \quad \text{(deg)} \tag{5}$$

The HPBW as given by (4) is shown graphically in Fig. 8–19. Dividing the square of (4) into the number of square degrees in a sphere (41,253) yields an *approximate* directivity relation:[1]

$$D \simeq 15 C_\lambda^2 n S_\lambda \tag{6}$$

This calculation disregards the effect of minor lobes and the details of the pattern shape. A more realistic relation is

(定向性) ***Directivity*** $D \simeq 12 C_\lambda^2 n S_\lambda$ (7)

Restrictions are that (4) to (7) apply only for $0.8 < C_\lambda < 1.15$, $12° < \alpha < 14°$ and $n > 3$.

The measured gains of King (1) and Wong for 12.8° monofilar axial-mode helical antennas are presented in Fig. 8–20 as a function of helix length ($L_\lambda = n S_\lambda$) and frequency.

[1]It is assumed that the patterns of both field components are of the same shape and are figures-of-revolution around the helix axis.

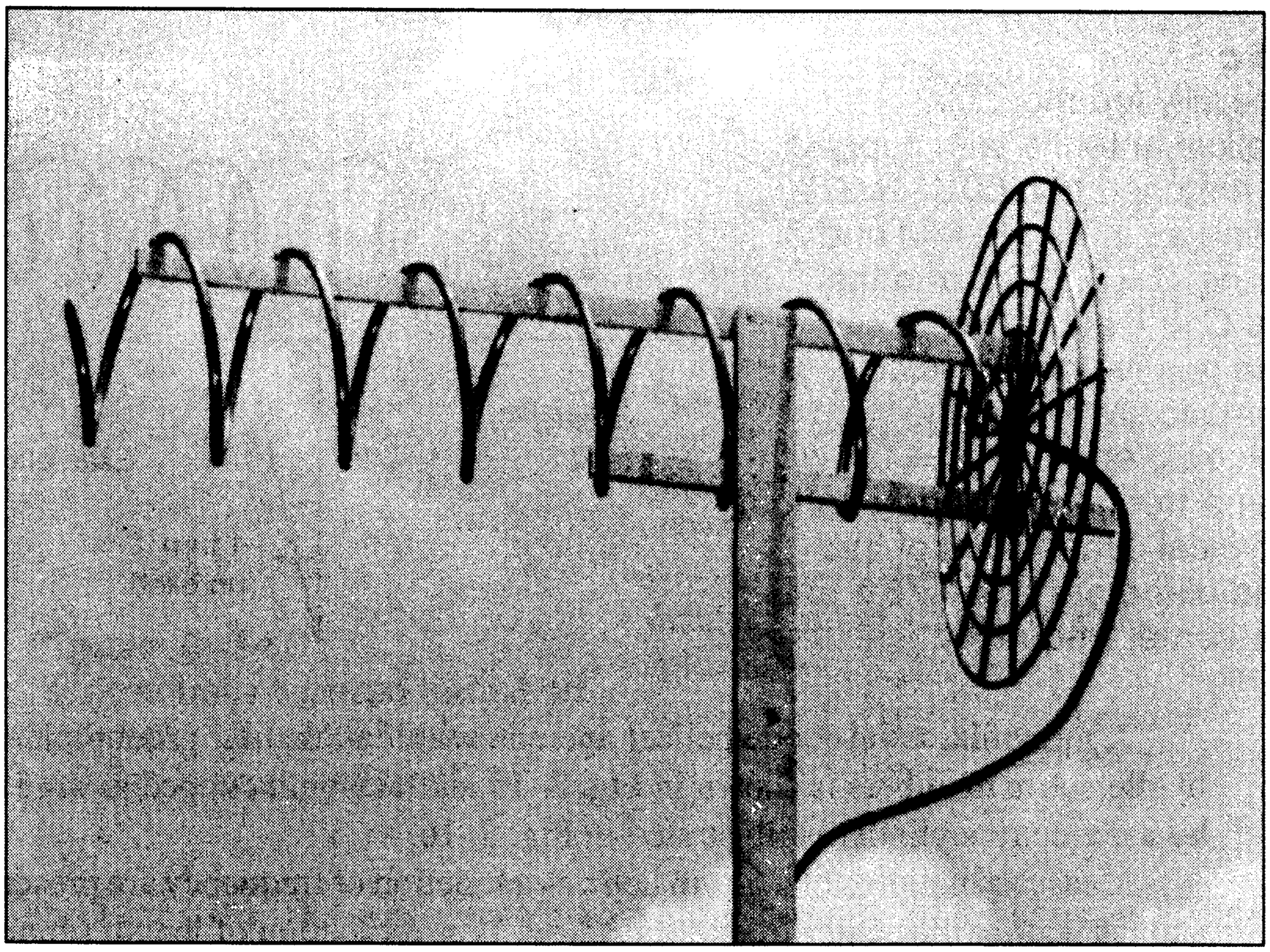


Figure 8–15
Dielectric-member-supported monofilar axial-mode helical antenna with flat circular ground plane. The pitch angle is 12.5°. The axial feed is directly from a 150-Ω coaxial cable (no matching section required). The conductor tubing is 0.02 λ in diameter. Note that the open grid of the ground plane has both circular and radial conductors. Both are essential. (Built by Kraus.)

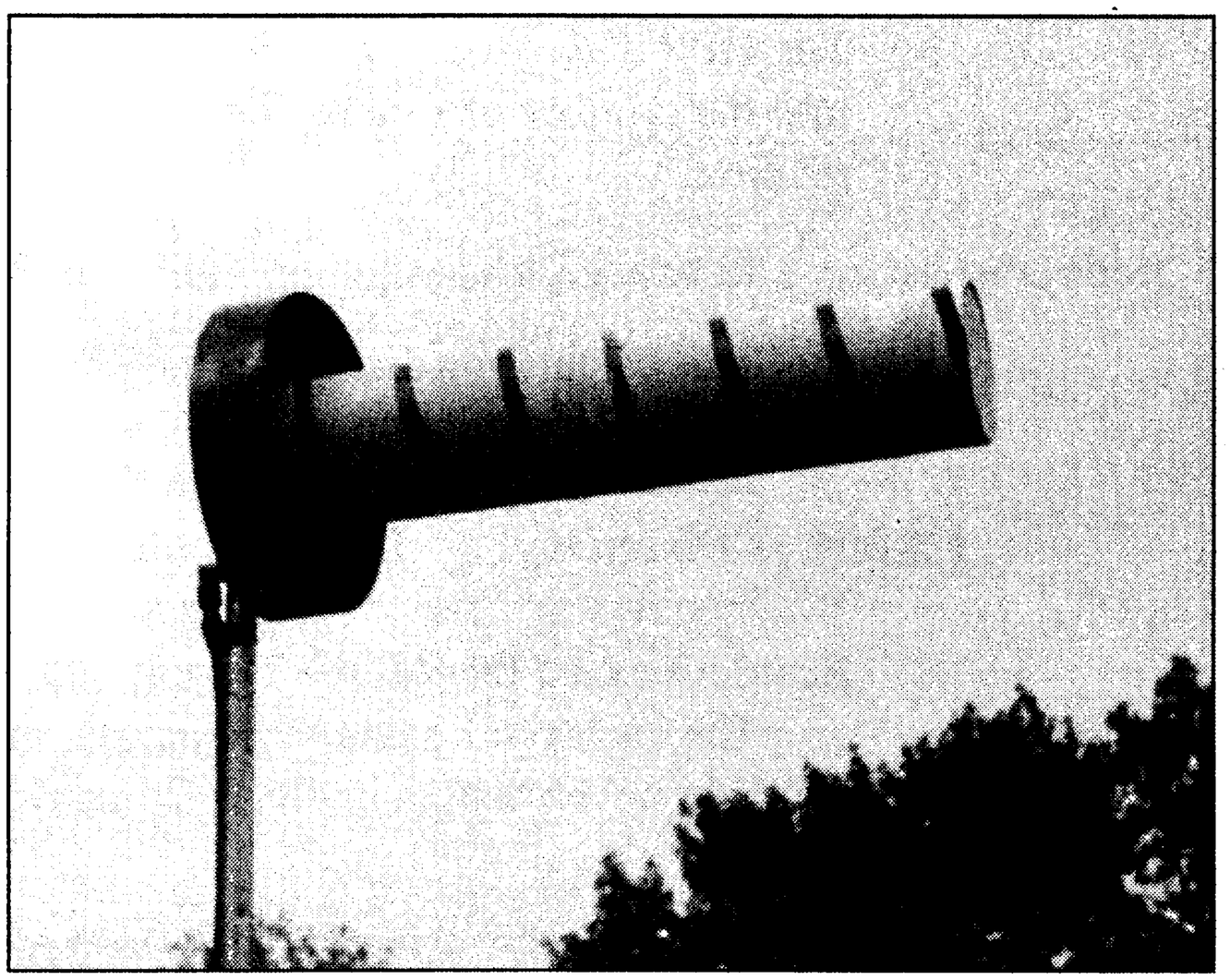


Figure 8–16
Thin-wall plastic-cylinder-supported $6\frac{1}{2}$-turn monofilar axial-mode helical antenna with solid metal cupped ground plane. Feeding is via a matching transition from a 50-Ω cable connected through a fitting mounted on the back of the cup ground plane at a point between the plastic cylinder and the lip of the cup. The helix is a flat metal strip bonded to the plastic cylinder. The strip width is 0.03 λ (equivalent to a 0.015 λ diameter round conductor). The pitch angle is 12.8°. Built by the author for UHF TV band operation with VSWR < 2 from channel 25 to 83 (524 to 890 MHz) and less than 1.2 from channel 27 to 75 (548 to 842 MHz).

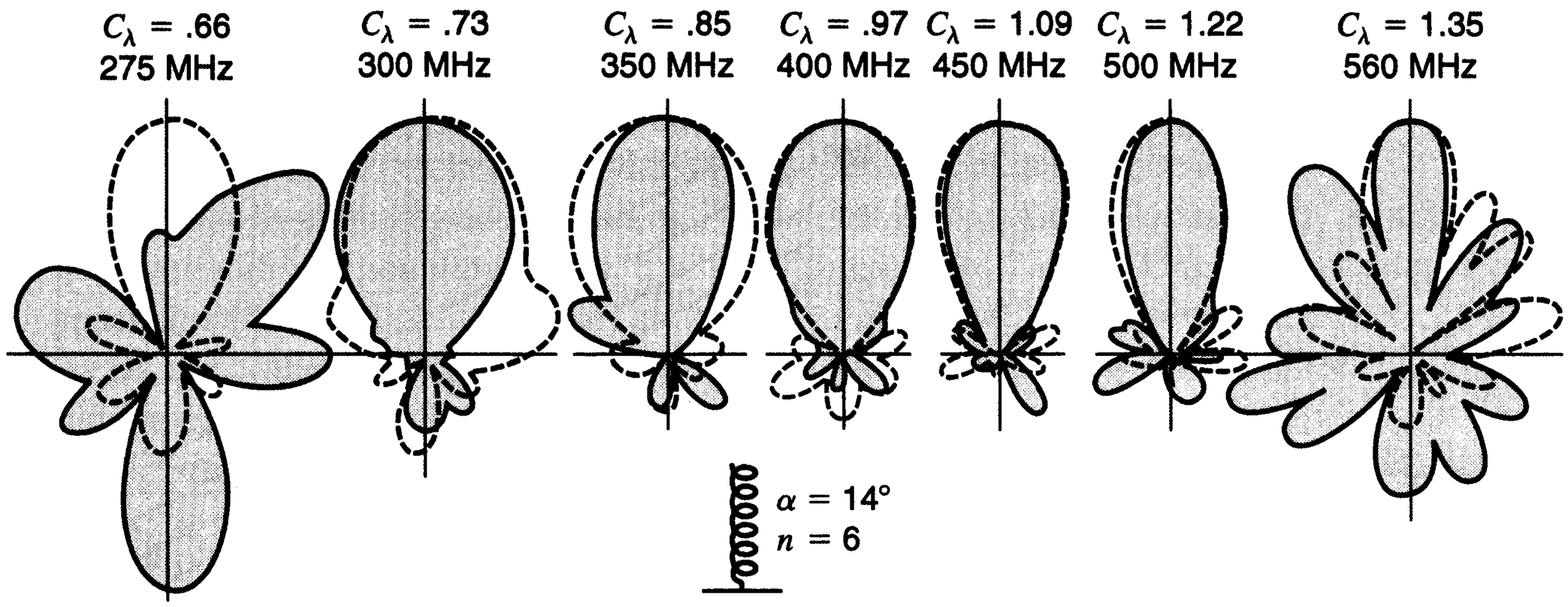


Figure 8–17
Measured field patterns of monofilar axial-mode helical antenna of 6 turns and 14° pitch angle. Patterns are characteristic of the axial mode of radiation over a range of circumferences from about 0.73 to 1.22 λ. Both the circumference and the frequency (in megahertz) are indicated. The solid patterns are for the horizontally polarized field component (E_ϕ) and the dashed for the vertically polarized (E_θ). Both are adjusted to the same maximum. (*After Kraus.*)

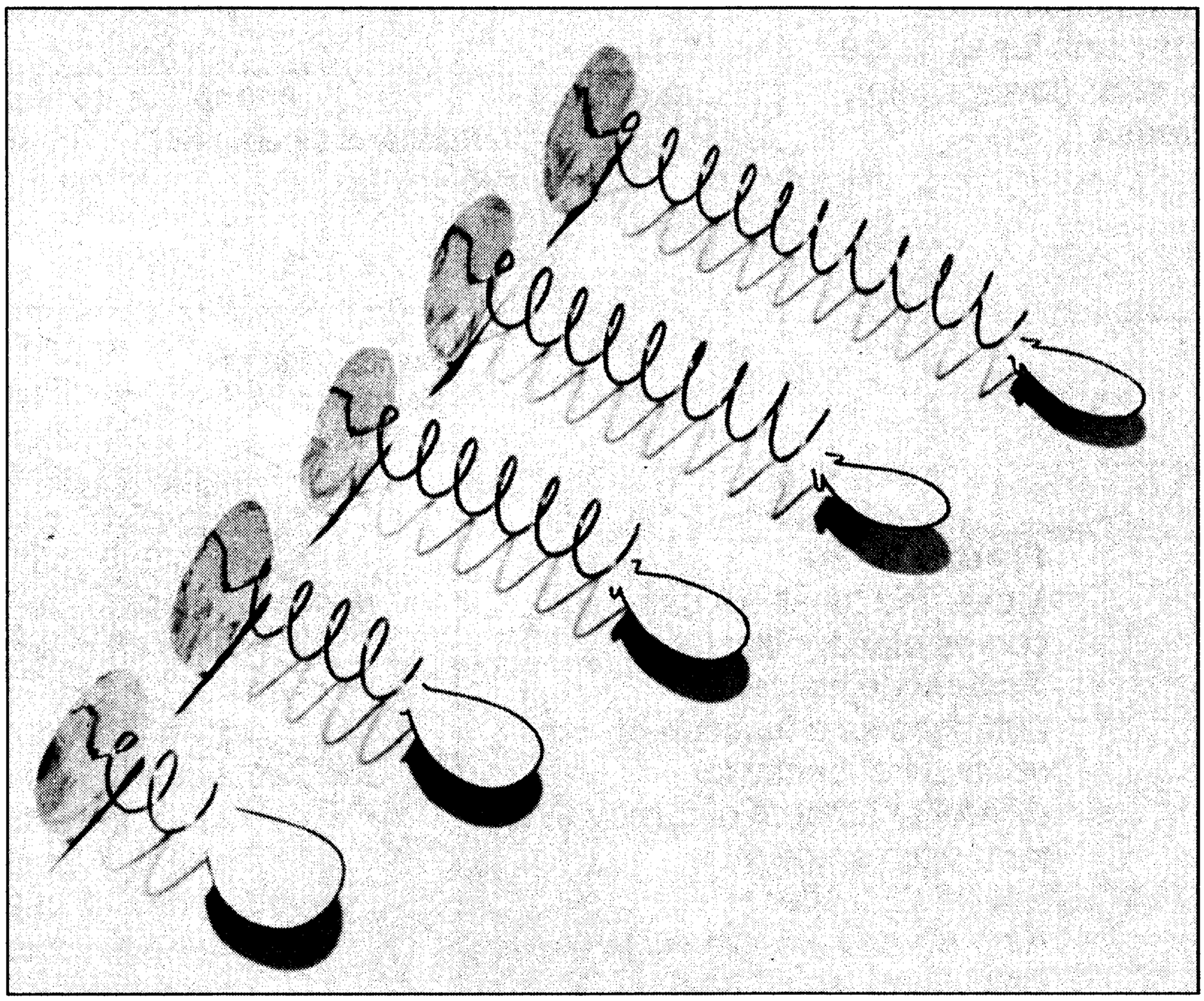


Figure 8–18
Effect of number of turns on measured field patterns. Helices have 12.2° pitch angle and 2, 4, 6, 8, 10 turns. Patterns shown are average of measured E_θ and E_ϕ patterns. (*After Kraus.*)

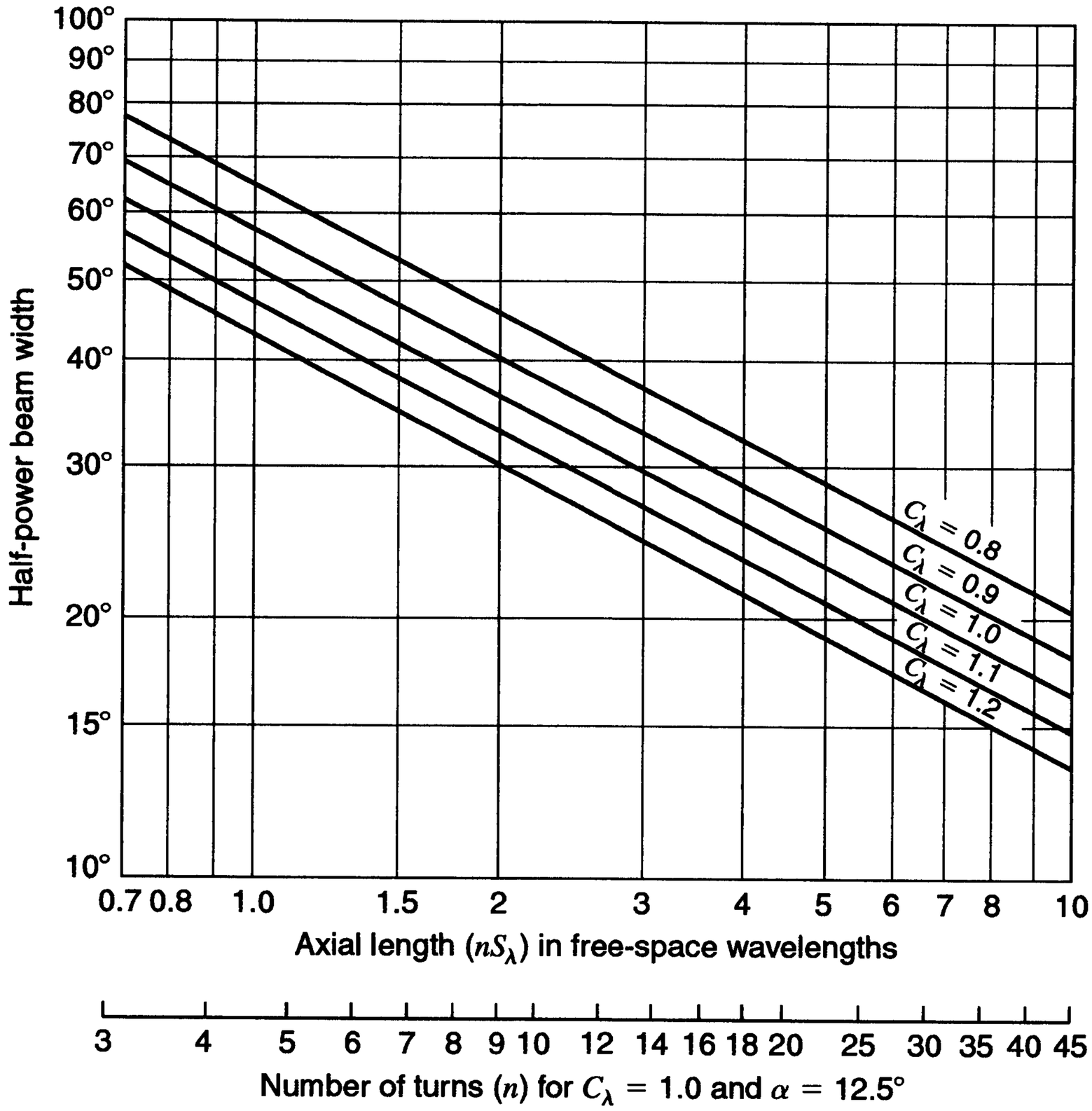


Figure 8–19
Half-power beamwidth of monofilar axial-mode helical antenna as a function of the axial length and circumference in free-space wavelengths and also as a function of the number of turns for $C_\lambda = 1.0$ and $\alpha = 12.5°$ (lower scale). (*After Kraus.*)

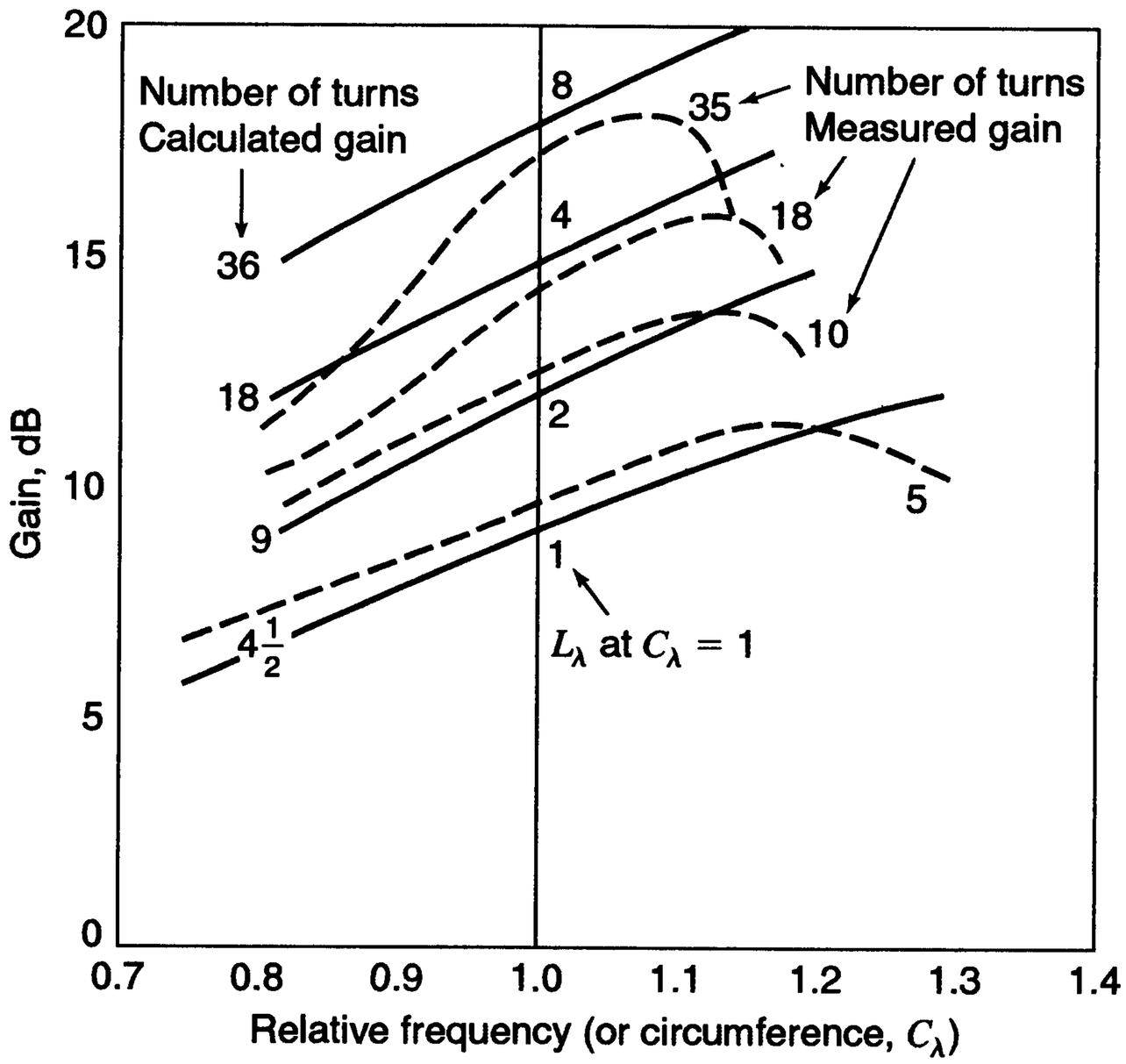


Figure 8–20
Measured (dashed) gain curves of monofilar axial-mode helical antennas as a function of relative frequency for different numbers of turns for a pitch angle of $\alpha = 12.8°$. (*After H. E. King (2) and J. L. Wong.*) Calculated (solid) gain curves are also shown for different numbers of turns.

Although higher gains are obtained by an increased number of turns, the bandwidth tends to become smaller. Highest gains occur at 10 to 20 percent above the center frequency for which $C_\lambda = 1$. Although the gains in Fig. 8–20 tend to be less than calculated from (7), they were measured on helices with 0.08λ diameter axial metal tubes.

Although pitch angles as small as 2°, as noted by MacLean (1) and Kouyournjian, and as large as 25°, as noted by Kraus, can be used, angles of 12° to 14° (corresponding to turn spacings of 0.21 to 0.25 at $C_\lambda = 1$) are optimum. King and Wong found that on helices with metal axial tubes, smaller pitch angles (near 12°) resulted in a slightly higher (1 dB) gain but a narrower bandwidth than larger angles (near 14°).

Turning to other parameters, the pattern, axial ratio and impedance (VSWR) performance as a function of frequency for a 6-turn, 14° pitch angle monofilar axial-mode helical antenna are summarized in Fig. 8–21. This is the same antenna for which the patterns

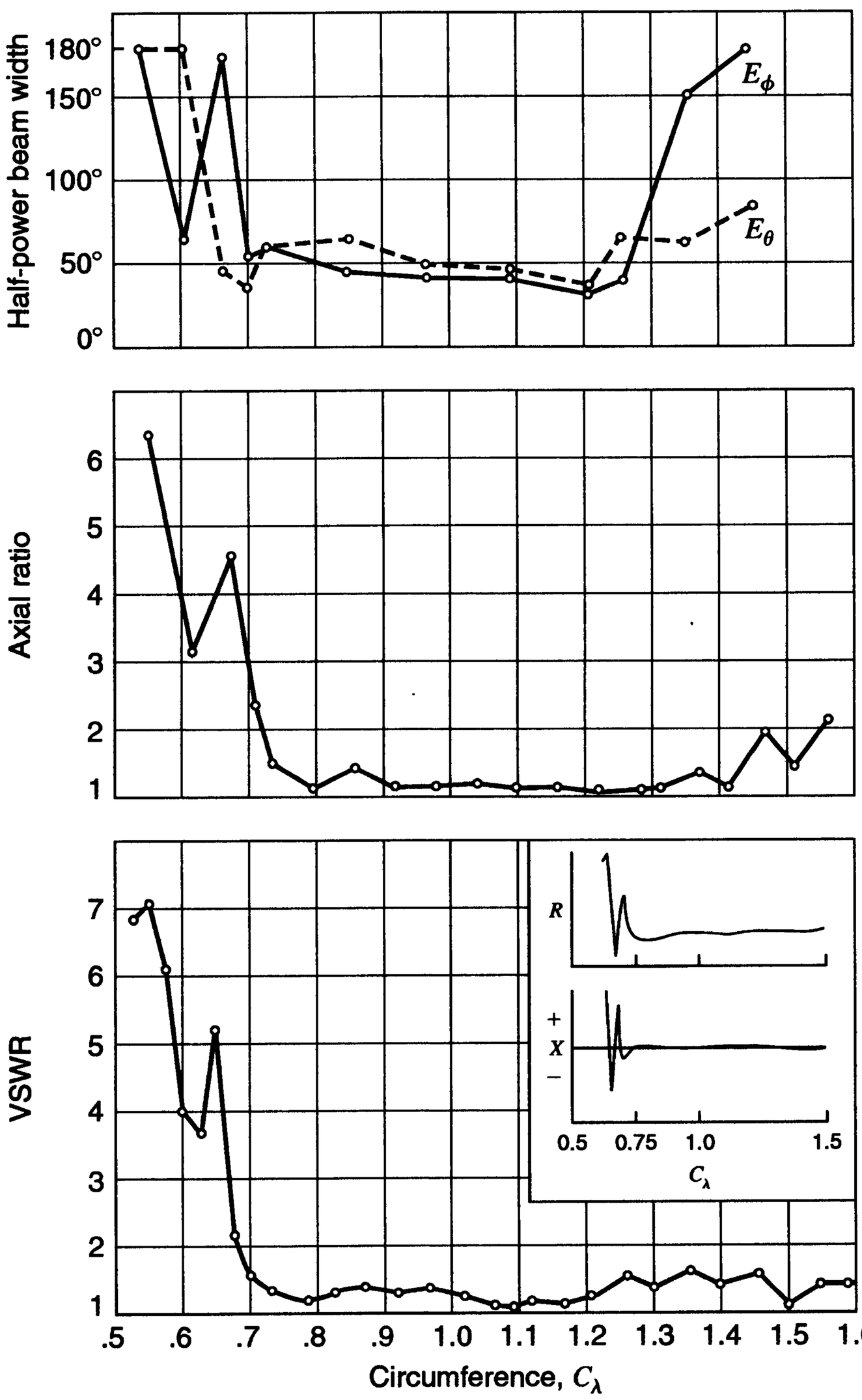


Figure 8–21
Summary of measured performance of 6-turn, 14° monofilar axial-mode helical antenna. The curves show the HPBW for both field components, the axial ratio and the VSWR on a 53-Ω line as a function of the relative frequency (or circumference C_λ). Trends of (relative) resistance *R* and reactance *X* are shown in the VSWR inset. Note the relatively constant *R* and small *X* for $C_\lambda > 0.7$. (*After Kraus.*)

are shown in Fig. 8–17. The half-power beamwidth is taken between half-power points regardless of whether these occur on the major lobe or on minor lobes. This definition is arbitrary but is convenient to take into account a splitting up of the pattern into many lobes of large amplitude at frequencies outside the beam mode. Beamwidths of 180° or more are arbitrarily plotted as 180°. The axial ratio is the value measured in the direction of the helix axis. The standing-wave ratio is the value measured on a 53-Ω coaxial line. A transformer section $\lambda/2$ long at the center frequency is located at the helix terminals to transform the terminal resistance of approximately 130 to 53 Ω. Considered altogether, these pattern, polarization and impedance characteristics represent remarkably good performance over a wide frequency range for a circularly polarized beam antenna.

The onset of the axial mode at a relative frequency of about 0.7 is very evident with axial-mode operation, extending from this frequency over at least an octave for VSWR and axial ratio and almost an octave for pattern.

(轴比) $$\boxed{\textit{Axial ratio} = (2n + 1)/2n} \tag{8}$$

where n = number of turns

EXAMPLE **8–3.2 16-Turn Helical Beam Antenna**

A 16-turn helical beam antenna (Fig. 8–22) has a circumference of λ, and turn spacing of $\lambda/4$. What is (*a*) HPBW, (*b*) axial ratio, (*c*) gain and (*d*) power pattern?

■ **Solution**

The helix can be represented by an end-fire array of point sources spaced $\lambda/4$ with one source for each turn. For an ordinary end-fire array the phasing between sources is −90° and the fields from all sources add in-phase on axis. The helical beam antenna or axial-mode helix, however, "self-programs" the phasing to $\delta = -101.25°$ and the fields do NOT add in-phase BUT the beam is sharper and the GAIN IS HIGHER. Furthermore, this improvement is maintained over an almost 2-to-1 bandwidth.

(*a*) From (8–3–4) HPBW = 26° *Ans.*

(*b*) From (8–3–8), AR = 33/32 = 1.03 or only 3 percent from perfect circular polarization.

(*c*) Gain = 15.4 dBi from pattern integration.

(*d*) The pattern is shown in Fig. 8–22.

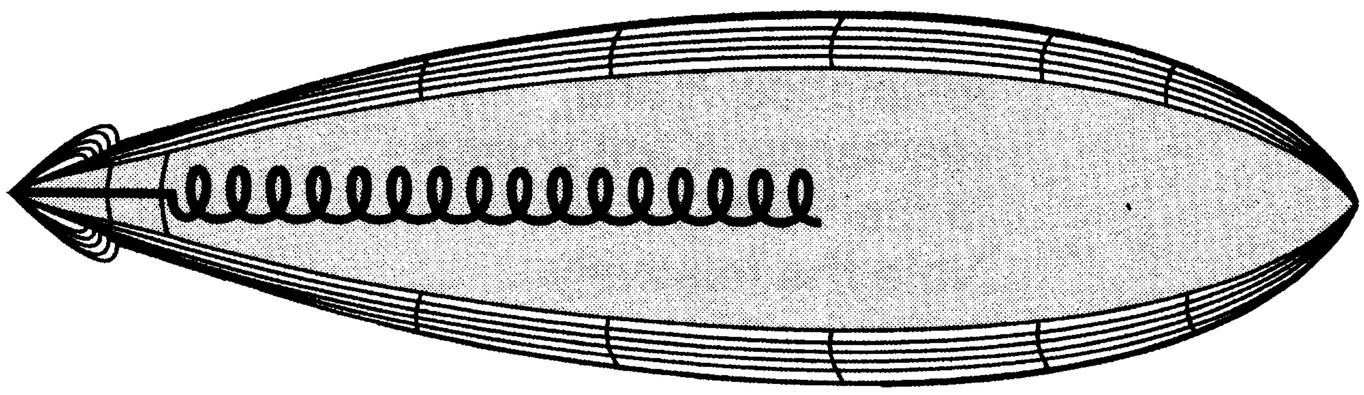


Figure 8–22
16-turn helical beam antenna and its power pattern.

EXAMPLE **8–3.3 Design of Quad-Helix Earth Station Antenna**

Figure 8–23 shows an array of four right-handed axial-mode helical antennas for communication with satellites. Since the fields hug the helixes, there is minimal coupling or "cross talk" between adjacent helixes and the terminal impedance of each helix is approximately 50 Ω in the array, the same as when used alone.

Determine (*a*) the best spacing based on the effective apertures of the helixes, (*b*) the directivity of the array, and (*c*) connections for feeding all helixes equally and in phase.

■ **Solution**

The directivity of an axial-mode helix with circumference equal to λ at the center frequency is approximately

$$D = 12nS_\lambda$$

where n = number of turns and S_λ = spacing between turns in wavelengths.

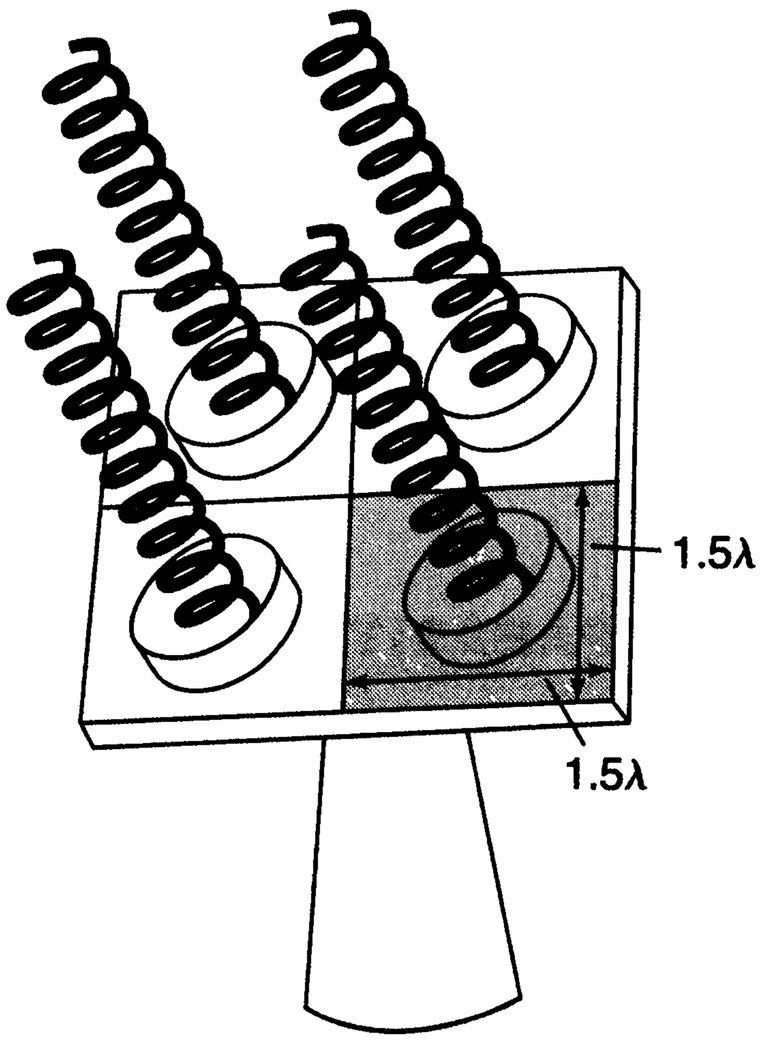
For each helix in the array, $n = 10$ and $S_\lambda = 0.236$. Thus,

$$D = 12 \times 10 \times 0.236 = 28.3$$

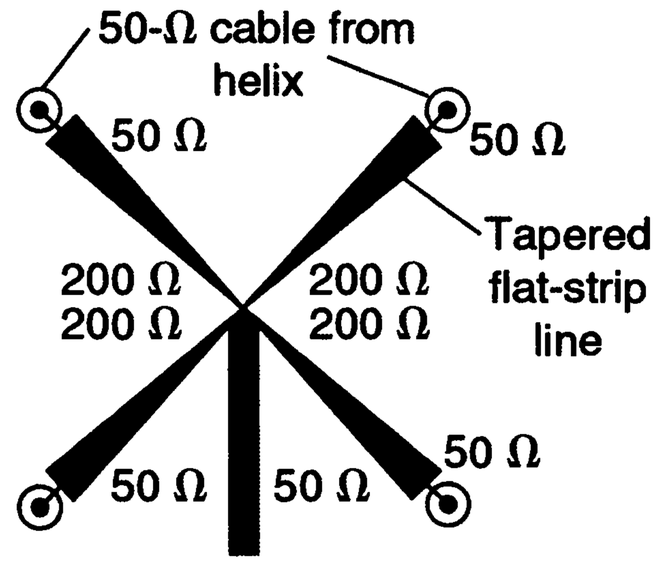
The effective aperture of each helix is then

$$A_e = \frac{D\lambda^2}{4\pi} = \frac{28.3}{4\pi}\lambda^2 = 2.25\lambda^2$$

This is the area of a square equal to $\sqrt{2.25} = 1.5\ \lambda$ on a side. Therefore, a spacing between helixes of 1.5 λ is appropriate. For a smaller spacing, the effective apertures of adjacent helixes overlap, decreasing the gain. A larger spacing does not increase the total aperture or the gain and may introduce grating lobes. *Ans.* (*a*)



(a)



(b)

Figure 8–23
(*a*) Earth station with quad-helix array. (*b*) Strip-line feed.

At 1.5 λ spacing the four helixes have a total effective aperture of $2.25 \times 4 = 9\lambda^2$ for an array directivity of

$$D = \frac{4\pi A_e}{\lambda^2} = 113 \text{ (20.5 dBi)} \qquad \textit{Ans. (b)}$$

A strip-line feed system is shown in Fig. 8–23*b*, which has 50- to 200-Ω tapered transitions joined to match a 50-Ω line. The tapered transitions have wide bandwidths which preserve the wide 2-to-1 bandwidth of the helixes. *Ans.* (*c*)

8–4 HELICAL BEAM ANTENNA WITH LOOPS REPLACING THE GROUND PLANE（用环取代接地平面的螺旋聚束天线）

Although the helical beam antenna or axial-mode helix is conveniently fed with a ground plane, it may be fed by two loops (replacing the ground plane) as shown in Fig. 8–24 (Kraus-7).

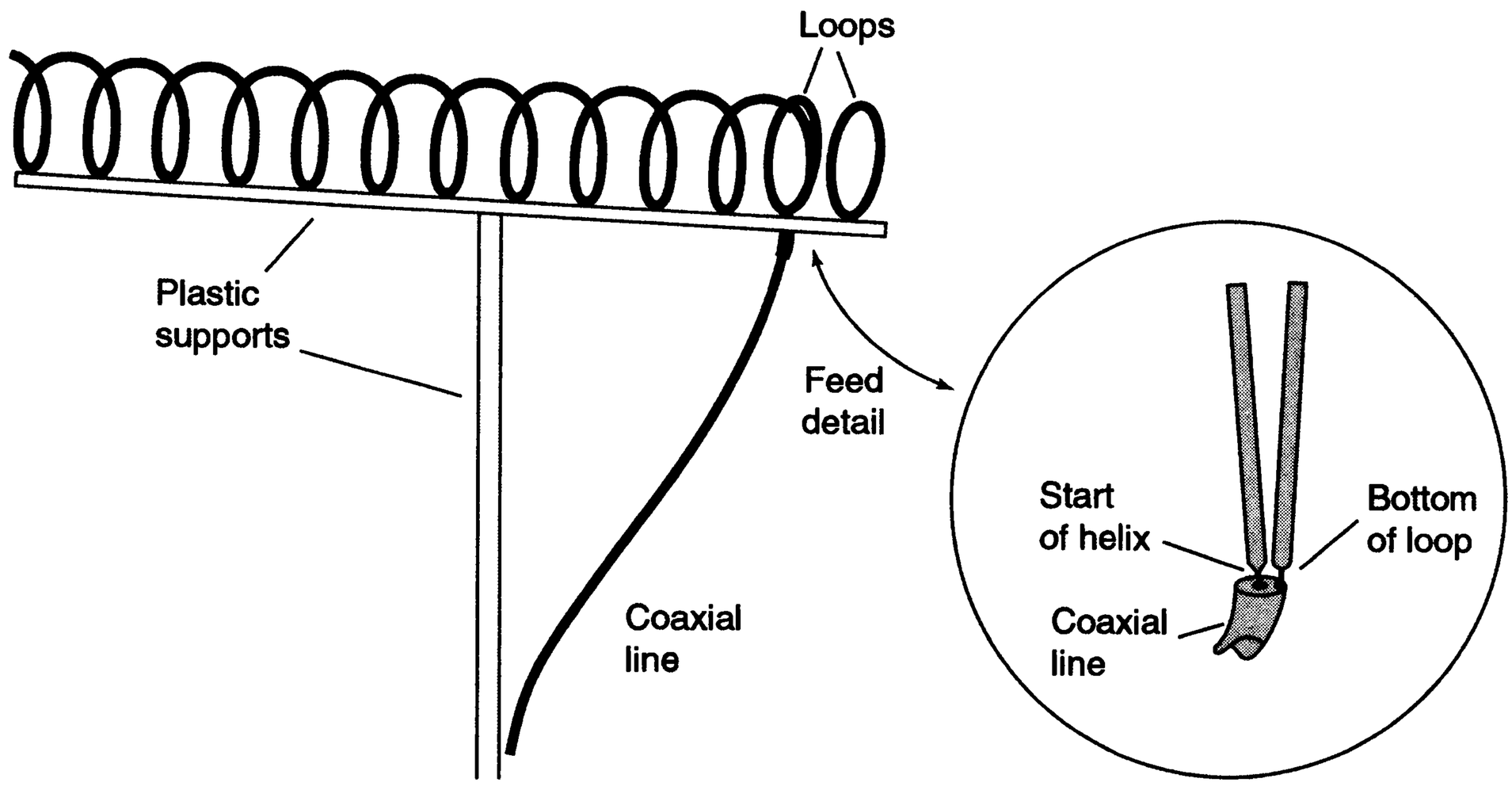


Figure 8–24
Ten-turn axial-mode helix with two loops instead of ground plane. Loop 1 is at the feed point (see inset). Loop 2 is λ/3 to λ/2 from the feed point.

8–5 DIPOLE ARRAYS WITH PARASITIC ELEMENTS（含有寄生单元的偶极子阵）

In all of the arrays we have considered earlier, except for the loop reflector with the helix antenna of Sec. 8–4, all elements were driven, that is, supplied with power by means of a transmission line. Antennas can also be constructed with "*parasitic elements*" in which currents are induced by the fields from a driven element. Such elements have no transmission line connection.

Although the helix can be used as a parasitic element (Sec. 17–9) , this section discusses dipoles as the parasitic elements.

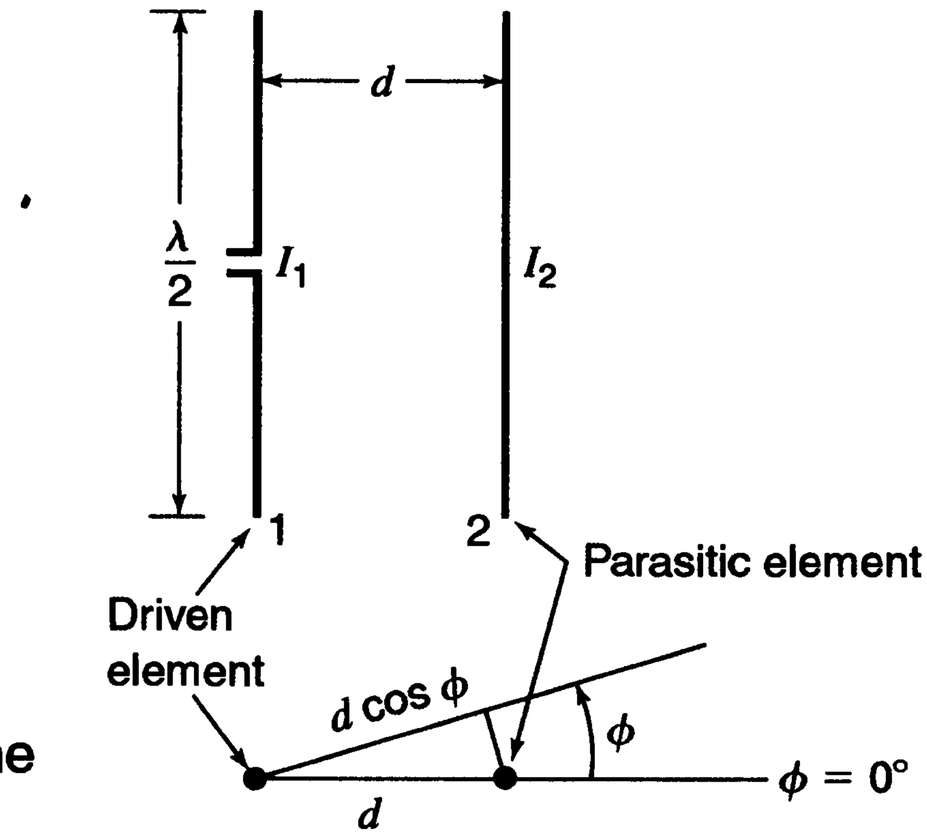


Figure 8–25
Array with one driven dipole element and one parasitic element.

Let us consider the case of an array in free space consisting of one driven $\lambda/2$ dipole element (element 1) and one parasitic element (element 2), as in Fig. 8–25. The procedure follows that used by Brown (1). Suppose that both elements are vertical so that the azimuth angle ϕ is as indicated. The circuit relations for the elements are

$$V_1 = I_1 Z_{11} + I_2 Z_{12} \tag{1}$$

$$0 = I_2 Z_{22} + I_1 Z_{12} \tag{2}$$

From (2) the current in element 2 is

$$I_2 = -I_1 \frac{Z_{12}}{Z_{22}} = -I_1 \frac{|Z_{12}| \angle \tau_m}{|Z_{22}| \angle \tau_2} = -I_1 \left| \frac{Z_{12}}{Z_{22}} \right| \angle \tau_m - \tau_2 \tag{3}$$

or

$$I_2 = I_1 \left| \frac{Z_{12}}{Z_{22}} \right| \angle \xi \tag{4}$$

where $\xi = \pi + \tau_m - \tau_2$, in which

$$\tau_m = \arctan \frac{X_{12}}{R_{12}}$$

$$\tau_2 = \arctan \frac{X_{22}}{R_{22}}$$

where

$R_{12} + jX_{12} = Z_{12}$ = mutual impedance of elements 1 and 2, Ω
$R_{22} + jX_{22} = Z_{22}$ = self-impedance of the parasitic element, Ω

The electric field intensity at a large distance from the array as a function of ϕ is

$$E(\phi) = k\left(I_1 + I_2 \angle d_r \cos \phi\right) \tag{5}$$

where $d_r = \beta d = \dfrac{2\pi}{\lambda} d$

Substituting (4) for I_2 in (5),

$$E(\phi) = kI_1\left(1 + \left|\frac{Z_{12}}{Z_{22}}\right| \underline{/\xi + d_r \cos\phi}\right) \tag{6}$$

Solving (1) and (2) for the driving-point impedance Z_1 of the driven element, we get

$$Z_1 = Z_{11} - \frac{Z_{12}^2}{Z_{22}} = Z_{11} - \frac{|Z_{12}^2| \underline{/2\tau_m}}{|Z_{22}| \underline{/\tau_2}} \tag{7}$$

The real part of Z_1 is

$$R_1 = R_{11} - \left|\frac{Z_{12}^2}{Z_{22}}\right| \cos(2\tau_m - \tau_2) \tag{8}$$

Adding a term for the effective loss resistance, if any is present, we have

$$R_1 = R_{11} + R_{1L} - \left|\frac{Z_{12}^2}{Z_{22}}\right| \cos(2\tau_m - \tau_2) \tag{9}$$

For a power input P to the driven element,

$$I_1 = \sqrt{\frac{P}{R_1}} = \sqrt{\frac{P}{R_{11} + R_{1L} - |Z_{12}^2/Z_{22}| \cos(2\tau_m - \tau_2)}} \tag{10}$$

and substituting (10) for I_1 in (6) yields the electric field intensity at a large distance from the array as a function of ϕ. Thus,

$$E(\phi) = k\sqrt{\frac{P}{R_{11} + R_{1L} - |Z_{12}^2/Z_{22}| \cos(2\tau_m - \tau_2)}}\left(1 + \left|\frac{Z_{12}}{Z_{22}}\right| \underline{/\xi + d_r \cos\phi}\right) \tag{11}$$

For a power input P to a single vertical $\lambda/2$ element the electric field intensity at the same distance is

$$E_{\mathrm{HW}}(\phi) = kI_0 = k\sqrt{\frac{P}{R_{00} + R_{0L}}} \qquad (\mathrm{V\ m^{-1}}) \tag{12}$$

where

R_{00} = self-resistance of single $\lambda/2$ element, Ω

R_{0L} = loss resistance of single $\lambda/2$ element, Ω

The gain in field intensity (as a function of ϕ) of the array with respect to a single $\lambda/2$ antenna with the same power input is the ratio of (11) to (12). Since $R_{00} = R_{11}$ and letting $R_{0L} = R_{1L}$, we have

$$G_f(\phi)\left[\frac{A}{\mathrm{HW}}\right] = \sqrt{\frac{R_{11} + R_{1L}}{R_{11} + R_{1L} - |Z_{12}^2/Z_{22}| \cos(2\tau_m - \tau_2)}}\left(1 + \left|\frac{Z_{12}}{Z_{22}}\right| \underline{/\xi + d_r \cos\phi}\right) \tag{13}$$

If Z_{22} is made very large by detuning the parasitic element (i.e., by making X_{22} large), (13)

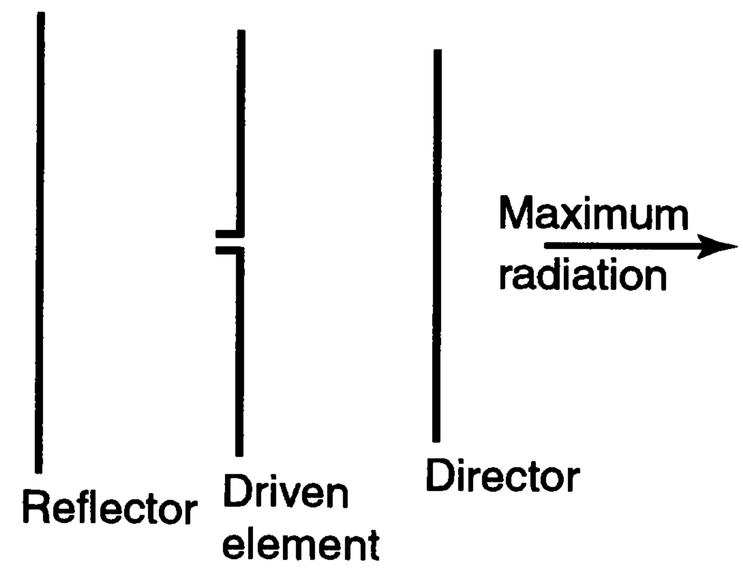


Figure 8–26
Three-element array.

reduces to unity, that is to say, the field of the array becomes the same as the single $\lambda/2$ dipole comparison antenna.

By means of a relation equivalent to (13), Brown (1) analyzed arrays with a single parasitic element for various values of parasitic element reactance (X_{22}) and was the first to point out that spacings of less than $\lambda/4$ were desirable.

The magnitude of the current in the parasitic element and its phase relation to the current in the driven element depends on its tuning. The parasitic element may have a fixed length of $\lambda/2$, the tuning being accomplished by inserting a lumped reactance in series with the antenna at its center point. Alternatively, the parasitic element may be continuous and the tuning accomplished by adjusting the length. This method is often simpler in practice but is more difficult of analysis. ***When the $\lambda/2$ parasitic element is inductive (longer than its resonant length) it acts as a reflector. When it is capacitive (shorter than its resonant length) it acts as a director.***[1] [当 λ/2 的寄生单元为电感性（长度大于其谐振长度）时，起反射器的作用；为电容性（长度小于其谐振长度）时，起引向器的作用。]

Arrays may be constructed with both a reflector and a director. A three-element array of this type is shown in Fig. 8–26, one parasitic element acting as a reflector and the other as a director. The analysis for the three-element array is more complex than for the two-element type treated above.

Experimentally measured field patterns of a horizontal three-element array situated 1 λ above a square horizontal ground plane about 13 λ on a side are presented in Fig. 8–27. The element lengths and spacings are as indicated. The gain at $\alpha = 15°$ for this array at a height of 1 λ is about 5 dB with respect to a single $\lambda/2$ dipole antenna at the same height.[2] The vertical plane pattern is shown in Fig. 8–27*a*. It is interesting to note that because of the finite size of the ground plane there is radiation at negative elevation angles. This phenomenon is characteristic of antennas with finite ground planes, the radiation at negative angles being largely the result of currents on the edges of the ground plane or beneath it. The azimuthal patterns at elevation angles $\alpha = 10$, 15 and 20° are shown in Fig. 8–27*b*. A parasitic array of this type with closely spaced elements has a small driving-point radiation resistance and a relatively narrow bandwidth.

---

[1]From Fig. 14–5, we note that the reactance of a thin linear element varies rapidly as a function of frequency when its length is about $\lambda/2$, going from positive (inductive) reactance through zero reactance (resonance) to negative (capacitive) reactance values as the length is reduced.

[2]Note that it is necessary to specify both the height and elevation angle at which the comparison is made. In comparing one antenna with another, the gain as a function of elevation angle at a given height (or as a function of height at a given elevation angle) may, in general, range from zero to infinity.

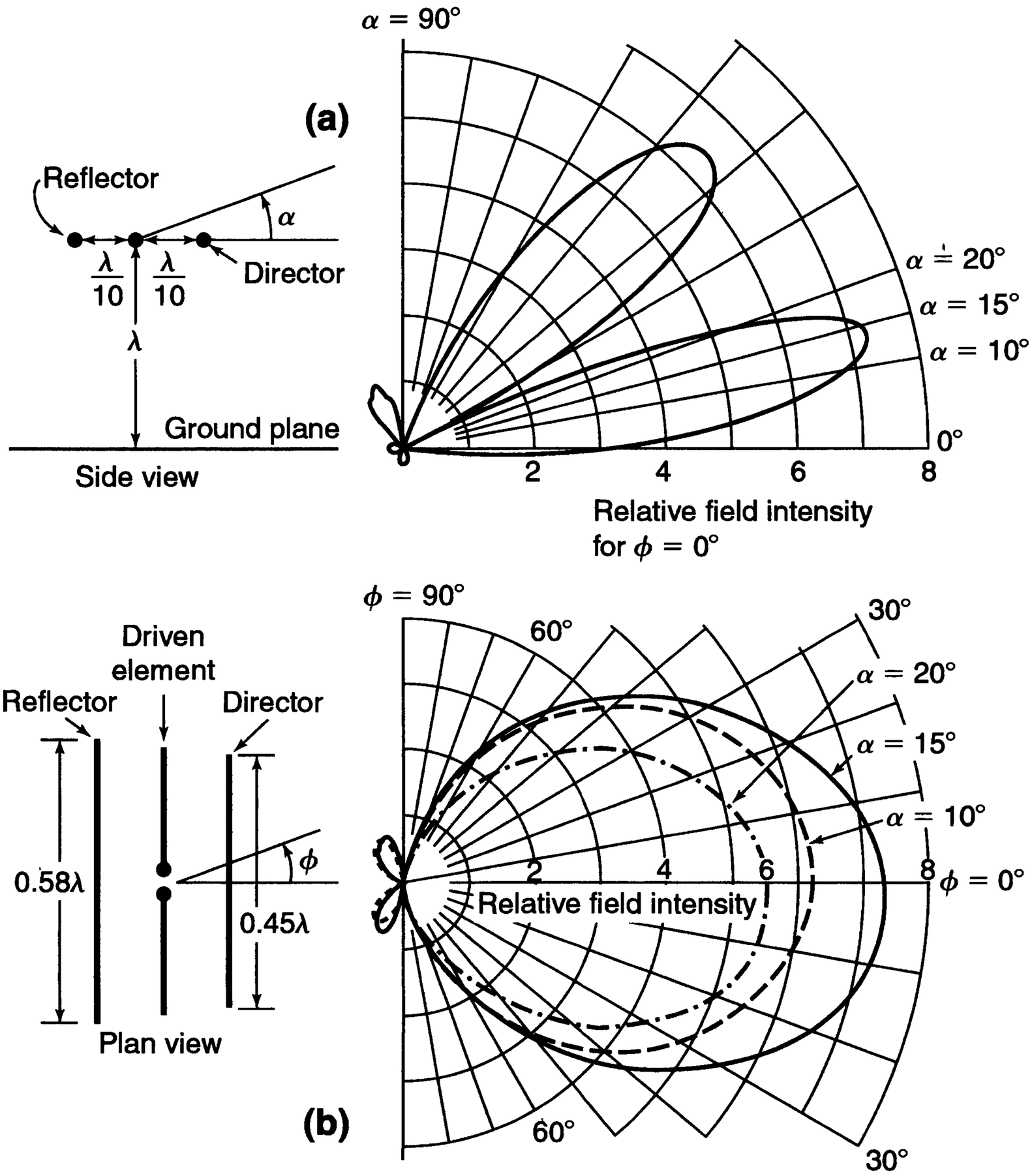


Figure 8–27
Measured vertical plane pattern (*a*) and horizontal plane patterns (*b*) at three elevation angles for a three-element array located 1 λ above a large ground plane. (*Patterns by D. C. Cleckner, Ohio State University.*)

8–6 THE YAGI-UDA ARRAY STORY（八木－宇田阵的故事）

Shintaro Uda, an assistant professor at Tohoku University, had not turned 30 when he conducted experiments on the use of parasitic reflector and director elements in 1926. This led to his publication of a series of 11 articles (from March 1926 to July 1929) in the *Journal of the Institute of Electrical Engineers of Japan* titled "On the Wireless Beam of Short Electric Waves." (Uda-1). He measured patterns and gains with a single parasitic reflector, a single parasitic director and with a reflector and as many as 30 directors. One of his many experimental arrays is shown in Fig. 8–28. He found the highest gain with the reflector about $\lambda/2$ in length and spaced about $\lambda/4$ from the driven element, while the best director lengths were about 10 percent less than $\lambda/2$ with optimum spacings about $\lambda/3$. Even though many patterns were measured in the near field, these lengths and spacings agree remarkably well with optimum values determined since then by further experimental and computer techniques. After George H. Brown demonstrated the advantages of close spacing, the reflector-to-driven-element spacings were reduced.

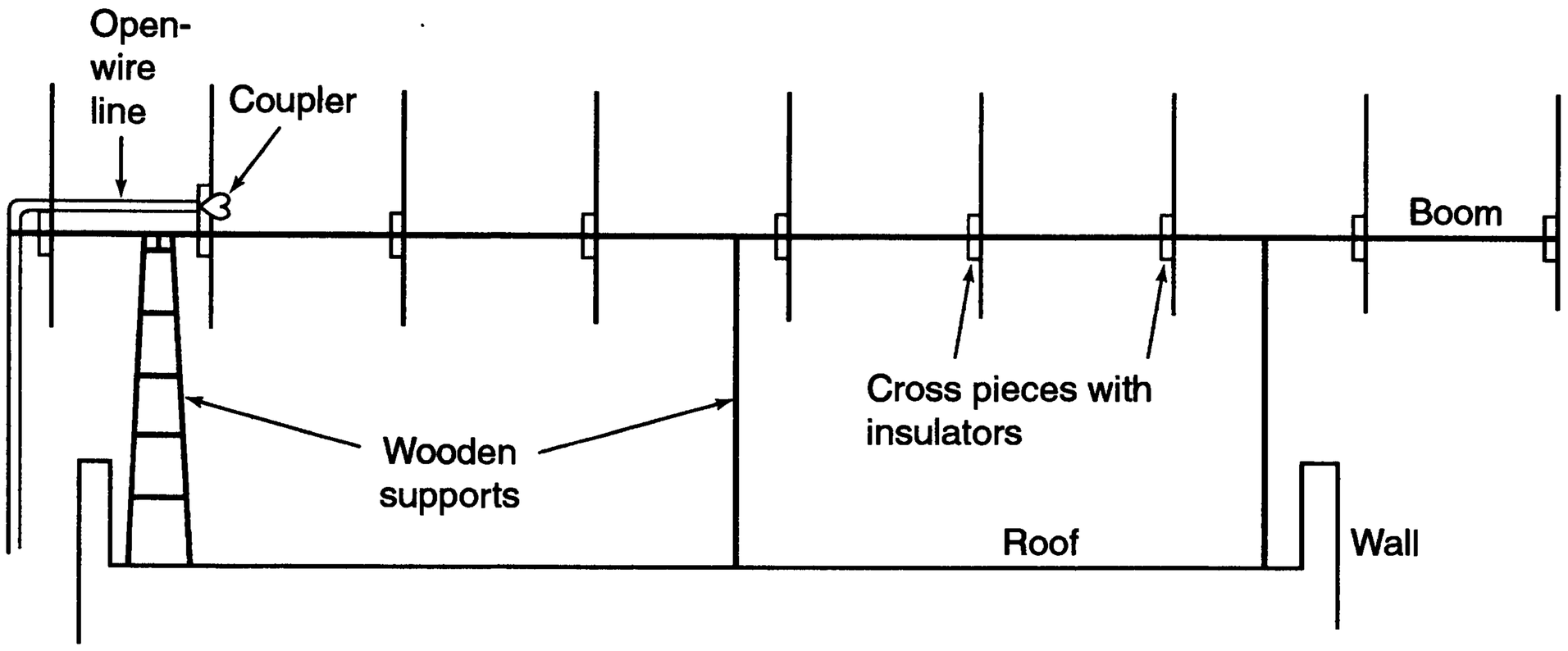


Figure 8–28
Shintaro Uda's experimental antenna with 1 reflector and 7 directors on the roof of his laboratory at Tohoku University for vertically polarized transmission tests during 1927 and 1928 over land and sea paths up to 135 km using a wavelength $\lambda = 4.4$ m. The horizontal wooden boom supporting the array elements is 15 m long.

Hidetsugu Yagi, professor of electrical engineering at Tohoku University and 10 years Uda's senior, presented a paper with Uda at the Imperial Academy on the "Projector of the Sharpest Beam of Electric Waves" in 1926, and in the same year they both presented a paper before the Third Pan-Pacific Congress in Tokyo titled "On the Feasibility of Power Transmission by Electric Waves." The narrow beams of short waves produced by the guiding action of the multidirector periodic structure, which they called a "wave canal," had encouraged them to suggest using it for short-wave power transmission, an idea now being considered for beaming solar power to the earth from a space station or from earth to a satellite.

It is reported that Professor Yagi had received a substantial grant from Sendai businessman Saito Zenuemon which supported the antenna research done by Uda with Yagi's collaboration. Then in 1928 Yagi toured the United States presenting talks before Institute of Radio Engineers sections in New York, Washington and Hartford, and in the same year Yagi published his now famous article on "Beam Transmission of Ultra Short Waves" in the *Proceedings of the IRE* (Yagi-1). Although Yagi noted that Uda had already published 9 papers on the antenna and acknowledged that Uda's ingenuity was mainly responsible for its successful development, the antenna soon came to be called "a Yagi." In deference to Uda's contributions, we refer to the array as a *Yagi-Uda antenna,* a practice now becoming common. Uda has summarized his researches on the antenna in two data-packed books (Uda-2, 3).

A typical modern-version 6-element Yagi-Uda antenna is shown in Fig. 8–29. It consists of a driven element (folded $\lambda/2$ dipole) fed by a 300-Ω 2-wire transmission line (twin line), a reflector and 4 directors. Dimensions (lengths and spacings) are indicated on the figure. The antenna provides a gain of about 10 dBi (maximum) with a bandwidth at half-power of 10 percent. By adjusting lengths and spacings appropriately (tweeking), the dimensions can

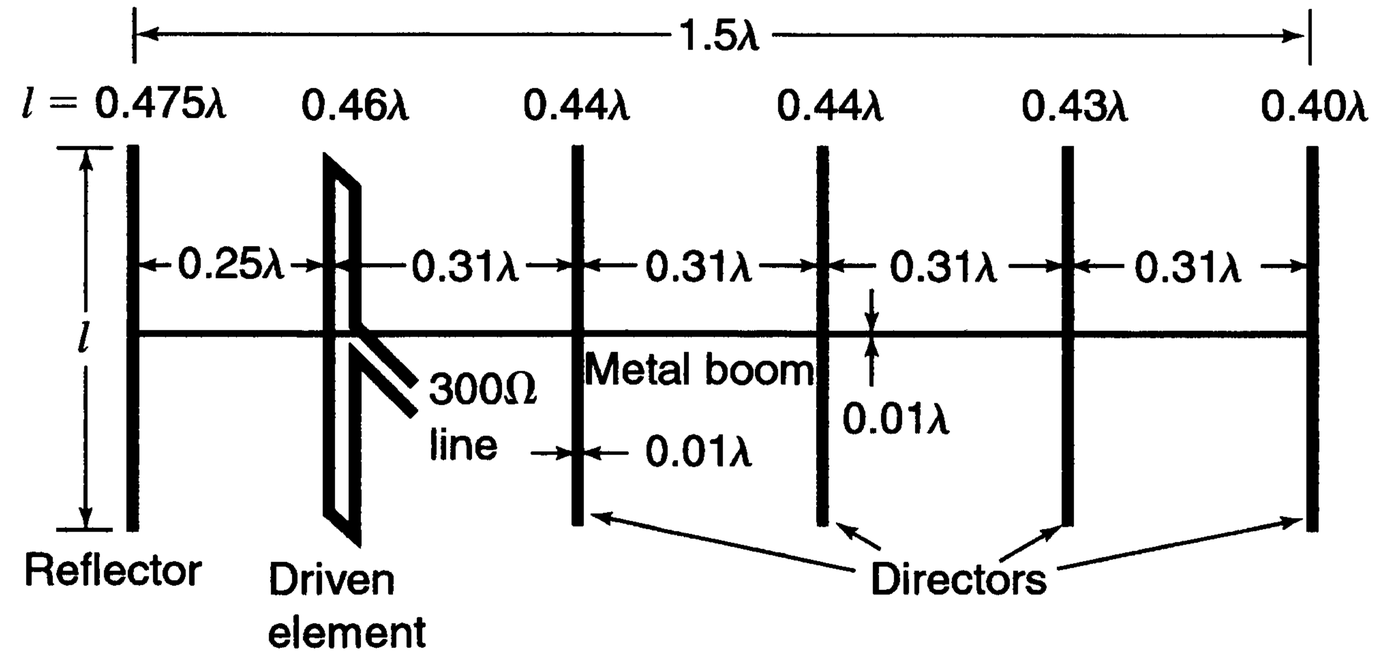


Figure 8–29
Modern-version 6-element Yagi-Uda antenna with dimensions. It has a maximum directivity of about 12 dBi at the center of a bandwidth of 10 percent at half-power.

be optimized, producing an increase in gain of another decibel (Chen-1, 2; Viezbicke-1). However, the dimensions are critical.

The inherently narrow bandwidth of the Yagi-Uda antenna can be broadened to 1.5 to 1 by lengthening the reflector to improve operation at low frequencies and shortening the directors to improve high-frequency operation. However, this is accomplished at a sacrifice in gain of as much as 5 dB.

EXAMPLE **8–6.1 Yagi-Uda 1.5 λ Array**

For the 1.5 λ array of Fig. 8–29, what is (*a*) HPBW, (*b*) axial ratio, (*c*) gain and (*d*) the pattern?

■ **Solution**

(*a*) HPBW = 44° in plane of elements, HPBW = 64° in plane perpendicular to elements (from pattern).

(*b*) AR = infinite (pure linear polarization).

(*c*) Gain = 9.4 dBi by pattern integration.

(*d*) Pattern is shown in Fig. 8–30.

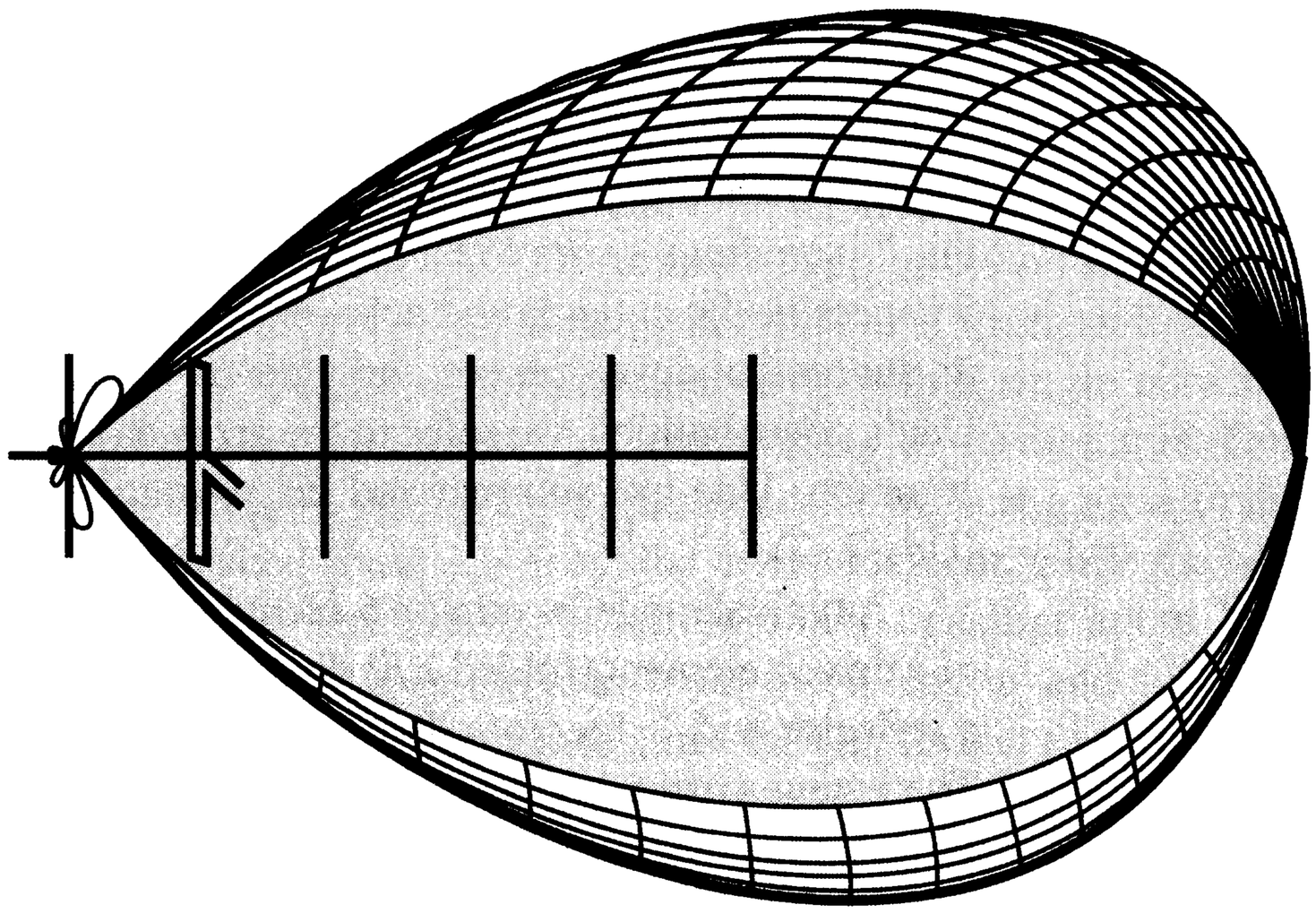


Figure 8–30
Power pattern of the Yagi-Uda array of Fig. 8–29. The narrower pattern is in the plane of the elements.

The polyrod antenna, as shown in Fig. 3–9, is another type of end-fire antenna. It may be regarded as an extended lens and is discussed in Chap. 17 with lens antennas.

PROBLEMS（习题）

8–3–1
（10 圈的螺旋）

A 10-turn helix. A right-handed monofilar helical antenna has 10 turns, 100 mm diameter and 70 mm turn spacing. The frequency is 1 GHz. (*a*) What is the HPBW? (*b*) What is the gain? (*c*) What is the polarization state? (*d*) Repeat the problem for a frequency of 300 MHz.

8–3–2
（30 圈的螺旋）

A 30-turn helix. A right-handed monofilar axial-mode helical antenna has 30 turns, $\lambda/3$ diameter and $\lambda/5$ turn spacing. Find (*a*) HPBW, (*b*) gain and (*c*) polarization state.

8–3–3
（左旋 / 右旋的螺旋）

Helices, left and right. Two monofilar axial-mode helical antennas are mounted side by side with axes parallel (in the x direction). The antennas are identical except that one is wound left-handed and the other right-handed. What is the polarization state in the x direction if the two antennas are fed (*a*) in phase and (*b*) in opposite phase?

***8–3–4**
（轴向模螺旋天线）

Axial-mode helical antenna. Design a right circularly polarized (RCP) axial-mode helical antenna with 17 dBi gain for operation at 1600 MHz with turn spacing λ/π. Find (*a*) number of turns, (*b*) turn diameter, and (*c*) axial ratio.

***8–5–1**
（八木 – 宇田天线）

Yagi-Uda antenna. Design a Yagi-Uda six-element antenna for operation at 500 MHz with a folded dipole feed. What are the lengths of (*a*) reflector element, (*b*) driven element, and (*c*) four-director element? What is the spacing (*d*) between reflector and driven element and (*e*) between director elements? What is (*f*) frequency bandwidth (upper and lower frequencies) and (*g*) gain?

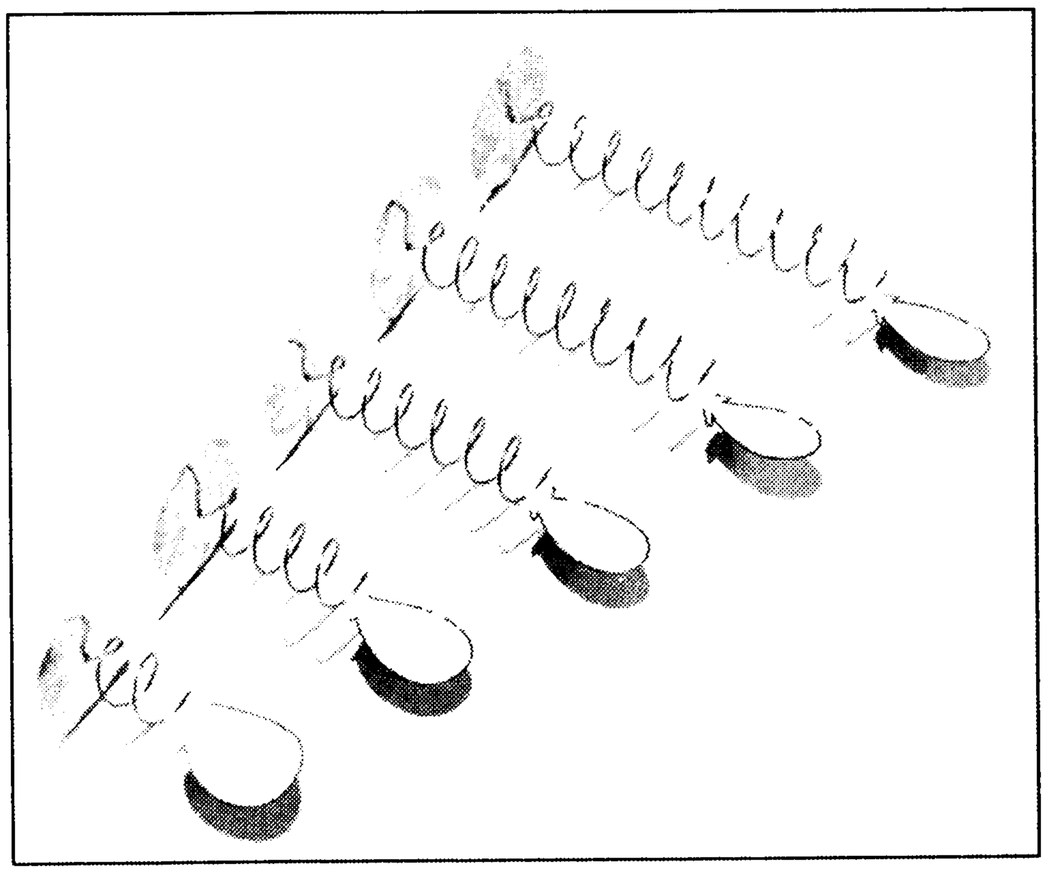
For computer programs, see Appendix C.

See Table 8–2, p. 270, for a summary of important formulas for axial-mode helical antennas.

CHAPTER 8

The Helical Antenna: Axial and Other Modes, Part II

（下）螺旋天线：轴向模和其他模



The topics in this part include:

- Axial-mode patterns and phase velocity（轴向模的波瓣图和相速）
- The square helix（方螺旋）
- Axial ratios（轴比）
- Mutual impedances（互阻抗）
- Wide-band properties（宽频带特性）
- The axial-mode helix as a periodic structure（轴向模螺旋作为周期性结构）
- The axial-mode helix as a phase and frequency shifter（轴向模螺旋作为移相器和移频器）
- The axial-mode helix as a polarizer and parasitic element in 7 applications（轴向模螺旋作为极化器以及寄生单元的 7 种应用）
- The axial-mode helix as a parabolic dish feed（轴向模螺旋作为抛物面的馈源）
- Arrays of 2 to 96 helices（2 ~ 96 支螺旋的阵列）
- Axial-mode helix on dielectric cylinder（介质柱上的轴向模螺旋）
- Conical-taper and flat-spiral terminations（锥形渐变螺旋和平面螺蜷终端）
- Bifilar, quadrifilar, 4-lobed and normal modes（双绕、四绕、四瓣和法向模）
- Genetic algorithm 5-segment helix（5 直线段螺旋的遗传算法）

8–7 INTRODUCTION（引言）

Many of the basic properties of the axial-mode helix were described in Part I. In this part more aspects of axial-mode operation are considered including pattern, phase velocity, axial ratio and mutual impedance. The axial-mode helix as a frequency shifter, polarizer and parasitic element is described for a number of applications. Its use as a feed antenna and in arrays of many helices is discussed. Finally, the bifilar, quadrifilar, four-lobed and normal modes of helix operation are described.

8–8 AXIAL-MODE PATTERNS AND THE PHASE VELOCITY OF WAVE PROPAGATION ON MONOFILAR HELICES (KRAUS-4)

（轴向模波瓣图和波沿单绕螺旋的传播相速）

As a first approximation, a monofilar (single conductor) helical antenna radiating in the axial mode may be assumed to have a single traveling wave of uniform amplitude along its conductor. By the principle of pattern multiplication, the far-field pattern of a helix is the product of the pattern for 1 turn and the pattern for an array of n isotropic point sources as in Fig. 8–31. The number n equals the number of turns. The spacing S between sources is equal to the turn spacing. When the helix is long (say, $nS_\lambda > 1$), the array pattern is much sharper than the single-turn pattern and hence largely determines the shape of the total far-field pattern. Hence, the approximate far-field pattern of a long helix is given by the array pattern. Assuming now that the far-field variation is given by the array pattern or factor and that the phase difference between sources of the array is equal to the phase shift over 1 turn length L_λ for a single traveling wave, it is possible to obtain a simple, approximate expression for the phase velocity required to produce axial-mode radiation. This value of phase velocity is then used in pattern calculations.

The array pattern or array factor E for an array of n isotropic point sources arranged as in Fig. 8–31 is given by (5–6–8). Thus,

$$E = \frac{\sin(n\psi/2)}{\sin(\psi/2)} \tag{1}$$

where n = number of sources and

$$\psi = S_r \cos\phi + \delta \tag{2}$$

where $S_r = 2\pi S/\lambda$

In the present case, (2) becomes

$$\psi = 2\pi\left(S_\lambda \cos\phi - \frac{L_\lambda}{p}\right) \tag{3}$$

where $p = v/c$ = relative phase velocity of wave propagation along the helical conductor, v being the phase velocity along the helical conductor and c being the velocity of light in free space.

If the fields from all sources are in phase at a point on the helix axis ($\phi = 0$), the radiation will be in the axial mode. For the fields to be in phase (ordinary end-fire condition) requires that

$$\psi = -2\pi m \tag{4}$$

where $m = 0, 1, 2, 3, \ldots$.

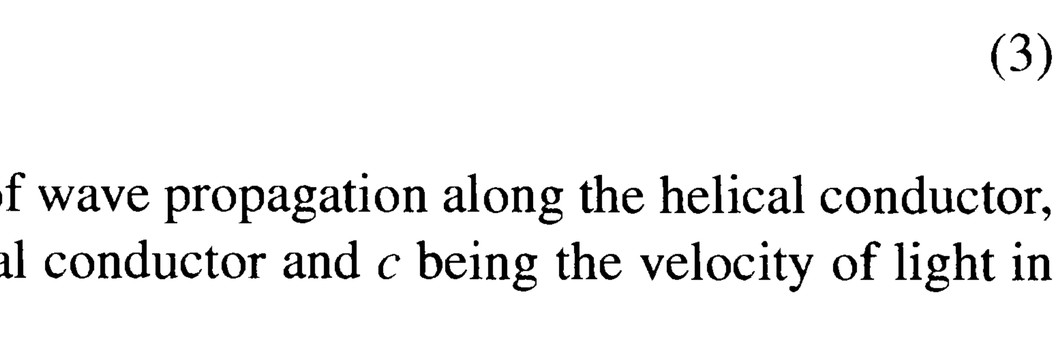


Figure 8–31
Array of isotropic sources, each source representing 1 turn of the helix.

The minus sign in (4) results from the fact that the phase of source 2 is retarded by $2\pi L_\lambda / p$ with respect to source 1. Source 3 is similarly retarded with respect to source 2, etc.

Now putting $\phi = 0$ and equating (3) and (4), we have

$$\frac{L_\lambda}{p} = S_\lambda + m \tag{5}$$

When $m = 1$ and $p = 1$, we have the relation

$$L_\lambda - S_\lambda = 1 \qquad \text{or} \qquad L - S = \lambda \tag{6}$$

This is an approximate relation between the turn length and spacing required for a helix radiating in the axial mode. Since for a helix $L^2 = \pi^2 D^2 + S^2$, (6) can be rewritten as

$$D_\lambda = \frac{\sqrt{2S_\lambda + 1}}{\pi} \qquad \text{or} \qquad C_\lambda = \sqrt{2S_\lambda + 1} \tag{7}$$

Equation (7) is shown graphically by the curve marked $C_\lambda = \sqrt{2S_\lambda + 1}$ in Fig. 8–10. The curve defines approximately the upper limit of the axial- or beam-mode region.

When $m = 1$, (5) is appropriate for a helix operating in the first-order (T_1) transmission mode. When $m = 2$, (5) is appropriate for the T_2 transmission mode, etc. A curve for $m = 2$ is shown in Fig. 8–10 by the line marked $C_\lambda = 2\sqrt{S_\lambda + 1}$. Hence, m corresponds to the order of the transmission mode on a helix radiating a maximum field in the axial direction. The case of particular interest here is where $m = 1$.

The case where $m = 0$ does not represent a realizable condition, unless p exceeds unity, since when $m = 0$ and $p = 1$ in (5) we have $L = S$. This is the condition for an end-fire array of isotropic sources excited by a straight wire connecting them ($\alpha = 90°$). However, the field in the axial direction of a straight wire is zero so that there can be no axial mode of radiation in this case.

Returning now to a consideration of the case where $m = 1$ and solving (5) for p, we have

$$p = \frac{L_\lambda}{S_\lambda + 1} \tag{8}$$

From the triangle of Fig. 8–9, (8) can also be expressed as

$$p = \frac{1}{\sin\alpha + [(\cos\alpha)/C_\lambda]} \tag{9}$$

Equation (9) gives the required variation in the relative phase velocity p as a function of the circumference C_λ for in-phase fields in the axial direction. The variation for helices of different pitch angles is illustrated in Fig. 8–32. These curves indicate that when a helix is radiating in the axial mode ($\frac{3}{4} < C_\lambda < \frac{4}{3}$) the value of p may be considerably less than unity. This is borne out by direct measurements of the phase velocity. In fact, the observed phase velocity is found to be slightly less than called for by (8) or (9). Calculating the array pattern for a 7-turn helix using values of p from (8) and (9) yields patterns much broader than observed. The p value of (8) or (9) corresponds to the ordinary end-fire condition discussed in Chap. 5. If the increased directivity condition of Hansen and Woodyard is

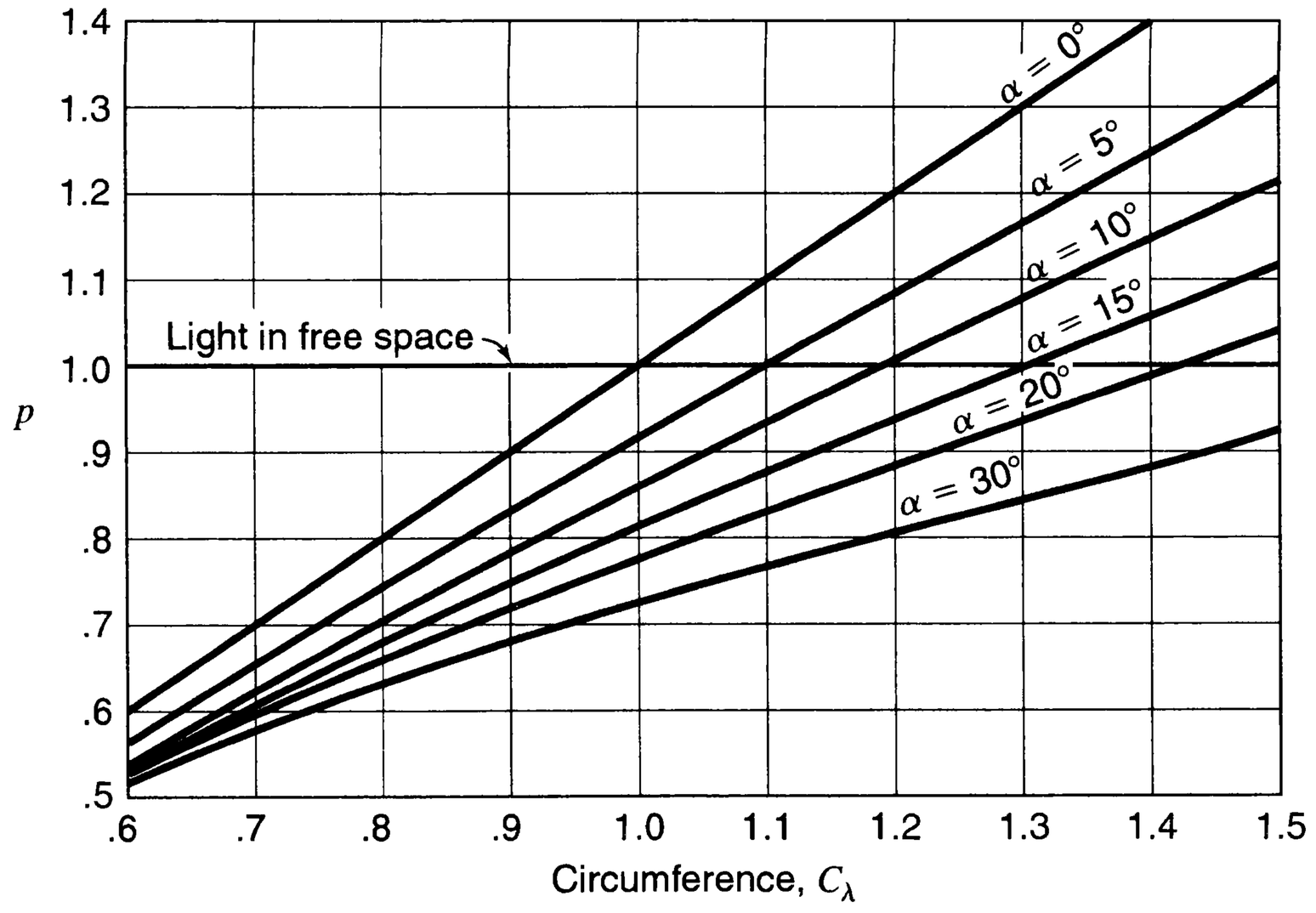


Figure 8–32
Relative phase velocity p for different pitch angles as a function of the helix circumference C_λ for the condition of in-phase fields in the axial direction.

presumed to exist, (4) becomes

$$\psi = -\left(2\pi m + \frac{\pi}{n}\right) \tag{10}$$

Now equating (10) and (3), putting $\phi = 0$ and solving for p we have

$$p = \frac{L_\lambda}{S_\lambda + m + (1/2n)} \tag{11}$$

For the case of interest $m = 1$ and

$$p = \frac{L_\lambda}{S_\lambda + [(2n+1)/2n]} \tag{12}$$

For large values of n, (12) reduces to (8). Equation (12) can also be expressed as[1]

$$p = \frac{1}{\sin\alpha + [(2n+1)/2n][(\cos\alpha)/C_\lambda]} \tag{13}$$

Using p as obtained from (12) or (13) to calculate the array factor yields patterns in good agreement with measured patterns. The p value from (12) or (13) also is in closer agreement with measured values of the relative phase velocity. Hence, it appears that the increased directivity condition is approximated as a natural condition on helices radiating in the axial mode.[2]

Another method of finding the relative phase velocity p on helical antennas radiating in the axial mode is by measuring the angle ϕ_0 at which the first minimum or null occurs

---

[1] It is to be noted that, as n becomes large, this relation (13) for increased directivity reduces to (9).

[2] The axial mode region is shown by the shaded ($T_1 R_1$) area in Fig. 8–10. Helices with dimensions in this region radiate in the axial mode, and (9), or more properly (13), applies. Outside this region these equations generally do not apply.

in the far-field pattern. This corresponds to the first null in the array factor, which is at ψ_0 (see Fig. 5–20). Then in this case (4) becomes

$$\psi = -(2\pi m + \psi_0) \tag{14}$$

Now equating (14) and (3) and putting $m = 1$ and solving for p, we have

$$p = \frac{L_\lambda}{S_\lambda \cos\phi_0 + 1 + (\psi_0/2\pi)} \tag{15}$$

Three relations for the relative phase velocity p have been discussed for helices radiating in the axial mode with transmission in the T_1 mode. These are given by (9), (13) and (15).

A fourth relation for p appropriate to the T_1 and higher-order transmission modes on infinite helices has been obtained by Bagby (1) by applying boundary conditions approximating a helical conductor to a solution of the general wave equation expressed in a new coordinate system he called "helicoidal cylindrical coordinates." Bagby's solution is obtained by applying boundary conditions to the two points c and d in Fig. 8–33. His value of the relative phase velocity is given by

$$p = \frac{C_\lambda}{m\cos\alpha + hR\sin\alpha} \tag{16}$$

where

$$hR = \tan\alpha \frac{m J_m^2(kR)}{J_{m-1}(kR)\, J_{m+1}(kR)} \tag{17}$$

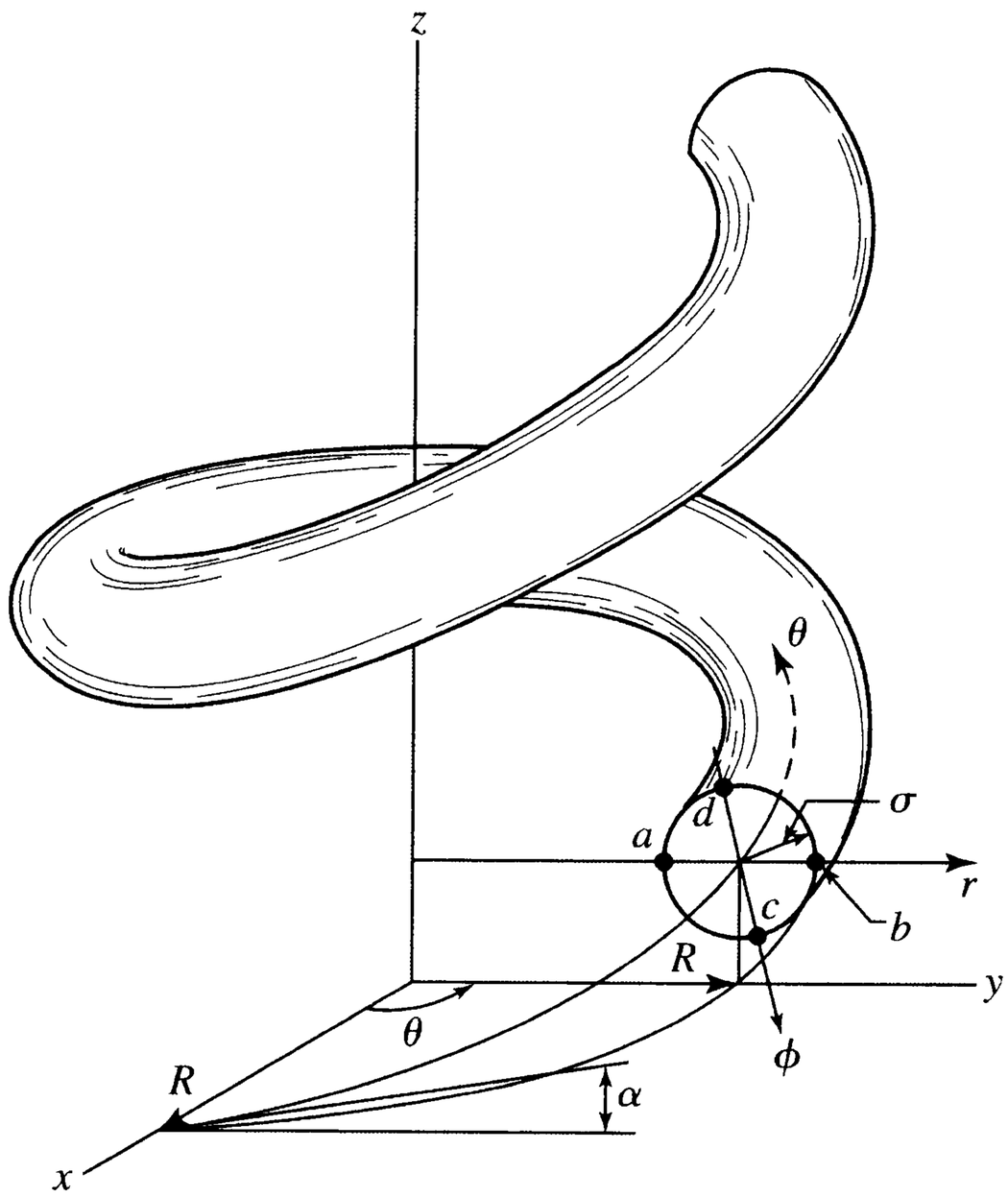
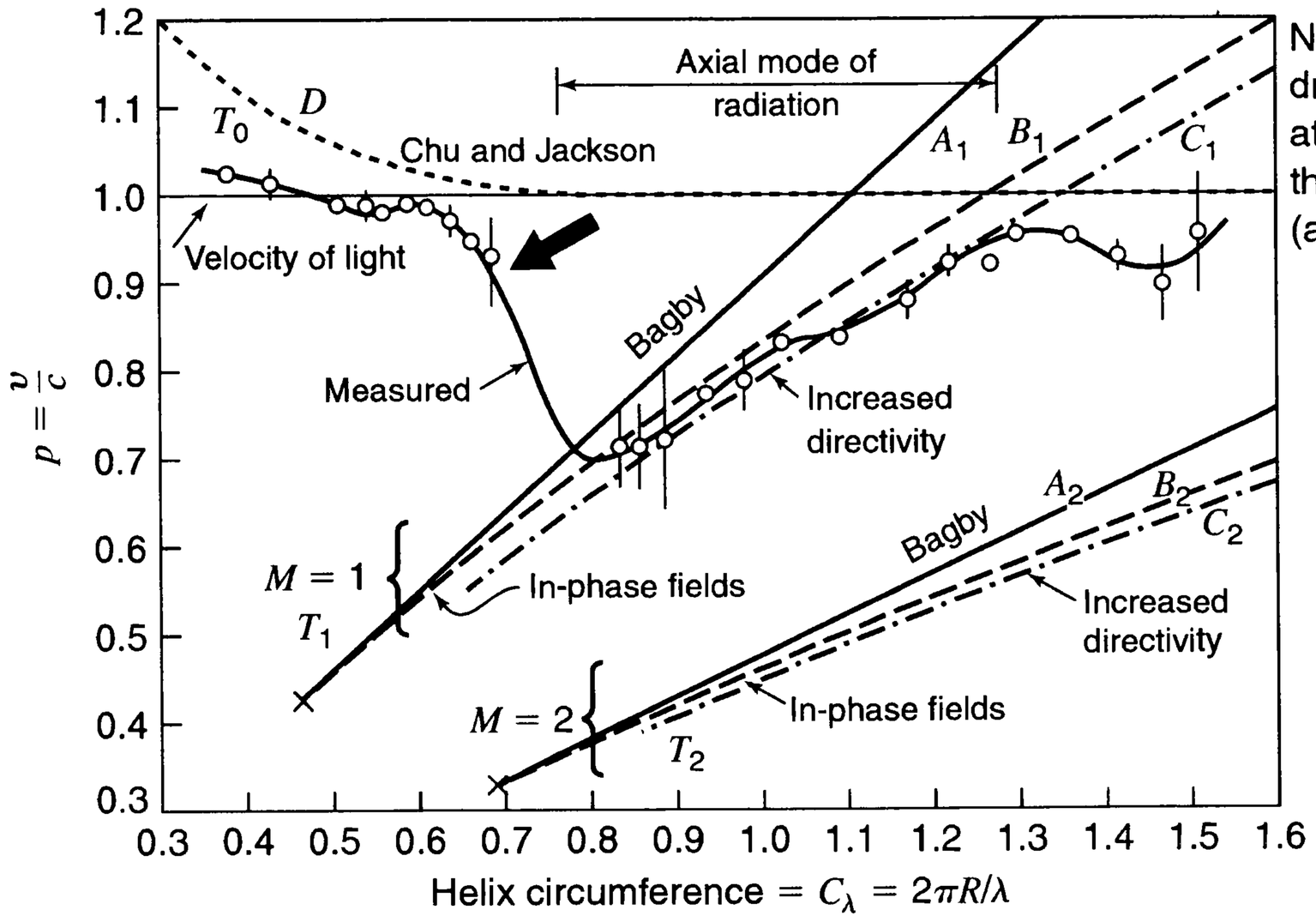


Figure 8–33
Helix showing points c and d at the conductor surface.



Note the abrupt drop in velocity at the onset of the axial mode (arrow).

Figure 8–34
Relative phase velocity p as a function of the helix circumference C_λ for 13° helices. The solid curve is measured on a 13°, 7-turn helix. Curves A_1 and A_2 are as calculated by Bagby for T_1 and T_2 transmission modes on an infinite 13° helix. Curves B_1 and B_2 are calculated for in-phase fields and curves C_1 and C_2 for increased directivity for T_1 and T_2 transmission modes. Curve D is from data by Chu and Jackson as calculated for the T_0 transmission mode. (*After Kraus.*)

where

m = order of transmission mode (= 1, 2, 3, . . .) ($m \neq 0$)
R = radius of helix cylinder
$kR = \sqrt{C_\lambda^2 - (hR)^2}$
h = constant
J = Bessel function of argument kR

The variation of p as a function of C_λ for a 13° helix as calculated by (16) and (17) for the case $m = 1$ is illustrated by the curve A_1 in Fig. 8–34. A curve for the T_1 transmission mode ($m = 1$) as calculated for the in-phase condition from (9) is shown by B_1. A curve for the increased directivity condition on a 13°, 7-turn helix, with $m = 1$, is presented by C_1.

Curves for the T_2 transmission mode for each of the three cases considered above are also presented in Fig. 8–34. In addition, a curve of the measured relative phase velocity on a 13°, 7-turn helix is shown for circumferences between about 0.4 and 1.5 λ. It is to be noted that in the circumference range where the helix is radiating in the axial mode ($\frac{3}{4} < C_\lambda < \frac{4}{3}$), the increased directivity curve, of the three calculated curves, lies closest to the measured curve.[1] The measured curve gives the value of the total or resultant phase

[1]The increased directivity curve is the only curve calculated for a helix of 7 turns. The in-phase field curve refers to no specific length while Bagby's curve is for an infinite helix.

velocity owing to all modes present (T_0, T_1, etc.) as averaged over the region of the helix between the third and sixth turns from the feed end. The vertical lines indicate the spread, if any, in values observed at one frequency. In general, each transmission mode propagates with a different velocity so that when waves of more than one transmission mode are present the resultant phase velocity becomes a function of position along the helix and may vary over a considerable range of values [Marsh (1)]. When $\frac{3}{4} < C_\lambda < \frac{4}{3}$ the phase velocity as measured in the region between the third and sixth turns corresponds closely to that of the T_1 transmission mode. The T_0 mode is also present on the helix but is only important near the ends. When the circumference $C_\lambda < \frac{2}{3}$, the T_0 mode may be obtained almost alone over the entire helix and the measured phase velocity approaches that for a pure T_0 mode indicated by curve D in Fig. 8–34, based on data given by Chu (1) and Jackson. This curve indicates that at small circumferences the relative velocity of a pure T_0 mode wave attains values considerably greater than that of light in free space. At $C_\lambda = \frac{2}{3}$, curve D has decreased to a value of nearly unity, and if no higher-order transmission mode were permissible, the phase velocity would approach that of light for large circumferences. However, higher-order modes occur, and, when C_λ exceeds about $\frac{2}{3}$, the resultant velocity drops abruptly, as shown by the measured curve in Fig. 8–34. This change corresponds to a transition from the T_0 to the T_1 transmission mode. For a circumference in the transition region, such as 0.7 λ, both T_0 and T_1 modes are of about equal importance.

When C_λ is about $\frac{3}{4}$ or somewhat more, the measured phase velocity approaches a value associated with the T_1 mode. As C_λ increases further, the relative phase velocity increases in an approximately linear fashion, agreeing most closely with the theoretical curve for the increased directivity condition (curve C_1). When C_λ reaches about $\frac{4}{3}$, a still higher order transmission mode (T_2) appears to become partially effective, causing further dips in the measured curve. However, the radiation may no longer be in the axial mode.

The formulas given for helical antennas operating in the first-order transmission mode ($m = 1$) are summarized in Table 8–1.

As mentioned above, the approximate far-field pattern of a monofilar helix radiating in the axial mode is given by the array factor for n isotropic point sources, each source replacing a single turn of the helix (see Fig. 8–31).

Table 8–1 Relative phase velocities for first-order transmission mode on helical antennas

Condition	Relative phase velocity
In-phase fields (ordinary end-fire)	$p = \dfrac{L_\lambda}{S_\lambda + 1} = \dfrac{1}{\sin\alpha + [(\cos\alpha)/C_\lambda]}$
Increased directivity	$p = \dfrac{L_\lambda}{S_\lambda + [(2n+1)/2n]} = \dfrac{1}{\sin\alpha + [(2n+1)/2n][(\cos\alpha)/C_\lambda]}$
From first null of measured field pattern	$p = \dfrac{L_\lambda}{S_\lambda \cos\phi_0 + (\psi_0/2\pi) + 1}$
Helicoidal cylindrical coordinate solution	$p = \dfrac{C_\lambda}{\cos\alpha + hR\sin\alpha}$ where hR is as given by (17)

The normalized array factor is

$$E = \sin\frac{\pi}{2n}\frac{\sin(n\psi/2)}{\sin(\psi/2)} \tag{18}$$

where $\psi = 2\pi[S_\lambda \cos\phi - (L_\lambda/p)]$

The normalizing factor is $\sin(\pi/2n)$ instead of $1/n$ since the increased directivity end-fire condition is assumed to exist (see Sec. 5–6, Case 3). For a given helix, S_λ and L_λ are known and p can be calculated from (12) or (13), ψ is then obtained as a function of ϕ. From (18), these values of ψ give the field pattern.

As an illustration, the calculated array factor patterns for a 7-turn, 12° helix with $C_\lambda = 0.95$ are shown in Fig. 8–35 for p values corresponding to increased directivity and also in-phase fields and for $p = 1$, 0.9 and 0.725. A measured curve (average of E_ϕ and E_θ) is shown for comparison. It is apparent that the pattern calculated for the increased-directivity condition ($p = 0.76$) agrees most closely with the measured pattern. The measured pattern was taken on a helix mounted on a ground plane 0.88 λ in diameter. The calculated patterns

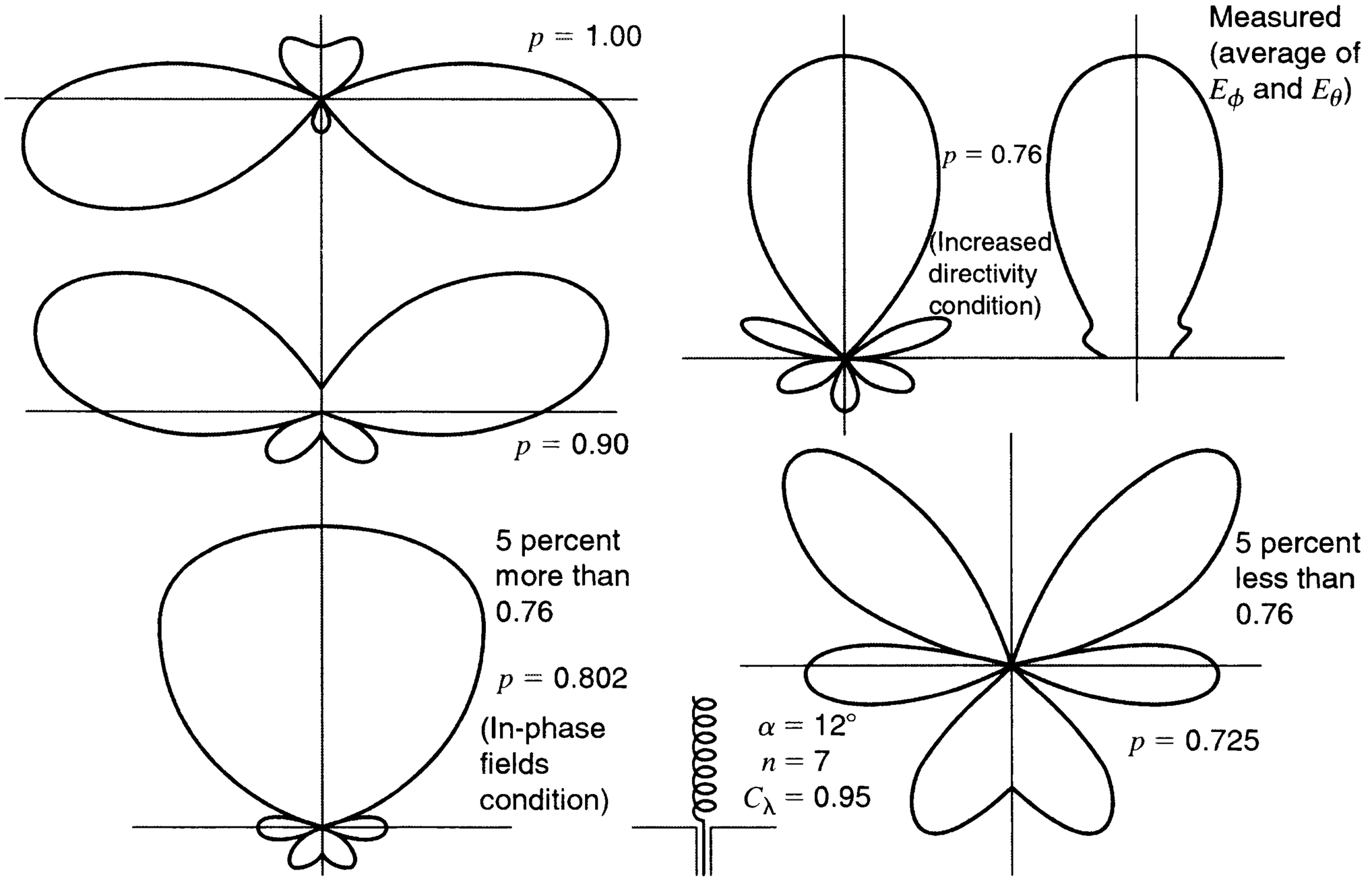


Figure 8–35
Array factor patterns for 12°, 7-turn helix with $C_\lambda = 0.95$. Patterns are shown for $p = 1, 0.9, 0.802$ (in-phase fields or ordinary end-fire condition), 0.76 (increased directivity) and 0.725. A measured curve is also presented. All patterns are adjusted to the same maximum. The sensitivity of the pattern to phase velocity is evident. A change of as little as 5 percent produces a drastic change in pattern, as may be noted by comparing the pattern for $p = 0.802$ (5 percent high) and the one for $p = 0.725$ (5 percent low) with the one for $p = 0.76$ which matches the measured pattern.

neglect the effect of a ground plane. This effect is small if the back lobe is small compared to the front lobe, as it is for $p = 0.802$ and $p = 0.76$.

The sensitivity of the pattern to the phase velocity is very apparent from Fig. 8–35. In particular, we note that as little as a 5 percent difference in phase velocity from that required for the increased directivity condition ($p = 0.76$) produces marked changes as shown by the patterns for $p = 0.802$ (5 percent high) and $p = 0.725$ (5 percent low).

8–9 MONOFILAR AXIAL-MODE SINGLE-TURN PATTERNS OF SQUARE HELIX
（单绕轴向模单圈方螺旋的波瓣图）

In this section expressions will be developed for the far-field patterns from a single turn of a monofilar helix radiating in the axial mode. It is assumed that the single turn has a uniform traveling wave along its entire length. The product of the single-turn pattern and the array factor then gives the total helix pattern.

A circular helix may be treated approximately by assuming that it is of square cross section. The total field from a single turn is then the resultant of the fields of four short, linear antennas as shown in Fig. 8–36*a*. A helix of square cross section can, of course, be treated exactly by this method. Measurements indicate that the difference between helices of circular and square cross section is small.

Referring to Fig. 8–37, the far electric field components, $E_{\phi T}$ and $E_{\theta T}$, in the xz plane will be calculated as a function of ϕ for a single-turn helix.

Let the area of the square helix be equal to that of the circular helix so that

$$g = \frac{\sqrt{\pi}\, D}{2} \tag{1}$$

where g is as shown in Fig. 8–36*a*.

The far magnetic field for a linear element with a uniform traveling wave is given by (6–8–5). Multiplying (6–8–5) by the intrinsic impedance Z of free space, putting

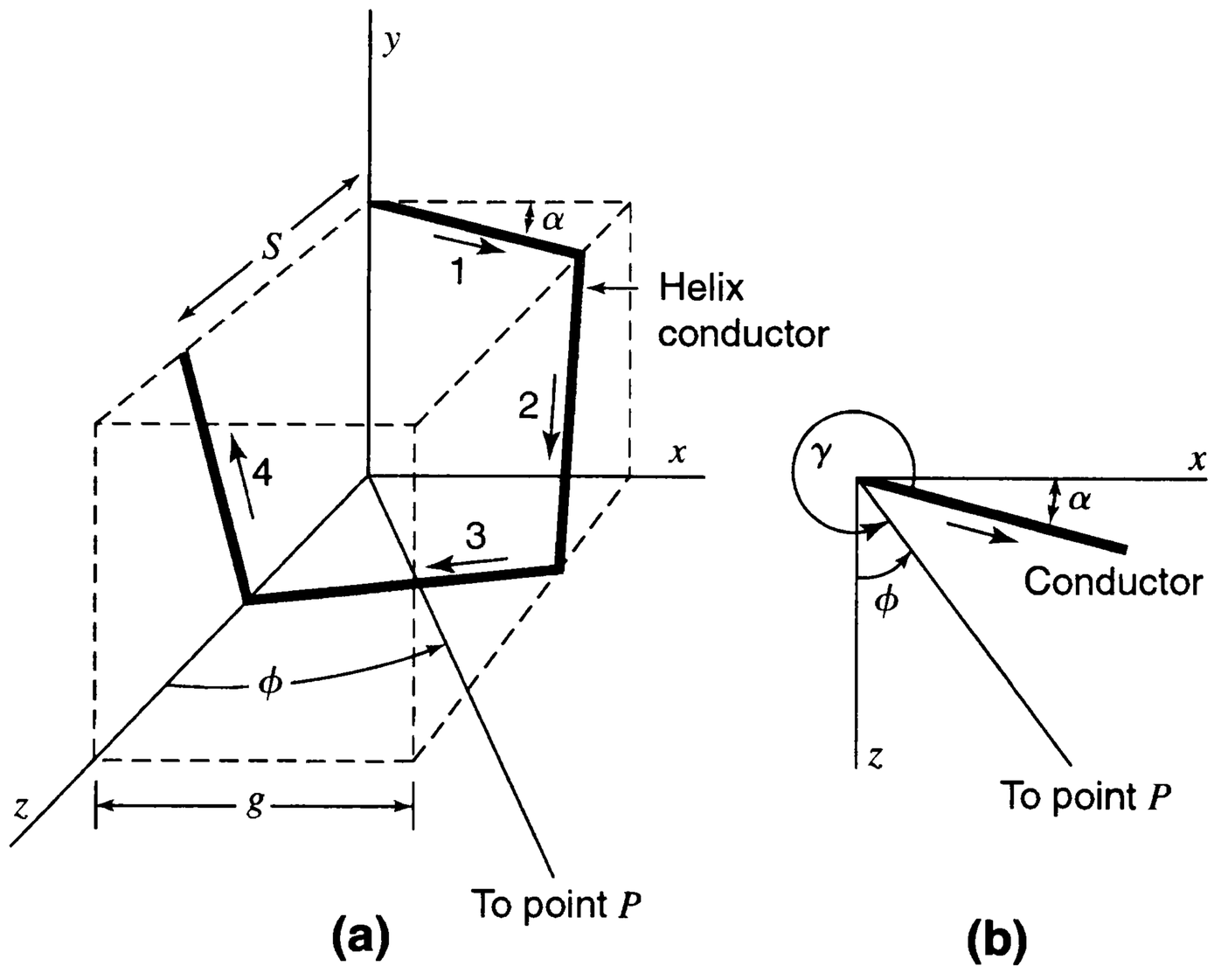


Figure 8–36
Square helix used in calculating single-turn pattern.

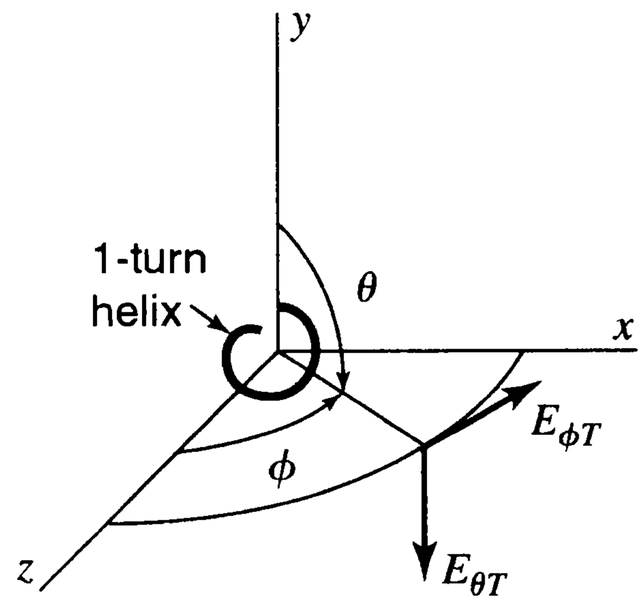


Figure 8–37
Field components with relation to single-turn helix.

$\gamma = (3\pi/2) + \alpha + \phi$, $t = 0$ and $b = g/\cos\alpha$, we obtain the expression for the ϕ component $E_{\phi 1}$ of the far field in the xz plane due to element 1 of the square helix as follows:

$$E_{\phi 1} = k\frac{\sin\gamma}{A}\sin BA \Big/ \left(-\frac{\omega r_1}{c} - BA\right) \tag{2}$$

where

$$k = \frac{I_0 pZ}{2\pi r_1}$$
$$A = 1 - p\cos\gamma$$
$$B = \frac{\omega g}{2pc\cos\alpha}$$

The expressions for $E_{\phi 2}$, $E_{\phi 3}$, etc., due to elements 2, 3 and 4 of the square turn are obtained in a similar way. Since the elements are all dissimilar sources, the total ϕ component, $E_{\phi T}$, from a single square turn is obtained by adding the fields from the four elements at each angle ϕ for which the total field is calculated. The sum of the fields from the four elements is then

$$E_{\phi T} = k\frac{\sin\gamma}{A}\sin BA \Big/ \left(-BA - \frac{\omega r_1}{c}\right)$$
$$+ k\frac{\sin BA''\sin\alpha\sin\phi}{A''} \Big/ \left[-BA'' - \frac{L\omega}{4pc} + \frac{\omega}{c}\left(\frac{S\cos\phi}{4} + g\sin\phi - r_1\right)\right]$$
$$+ k\frac{\sin\gamma'}{A'}\sin BA' \Big/ \left[-BA' - \frac{L\omega}{2pc} + \frac{\omega}{c}\left(\frac{S\cos\phi}{2} + g\sin\phi - r_1\right)\right]$$
$$+ k\frac{\sin BA''\sin\alpha\sin\phi}{A''} \Big/ \left[-BA'' - \frac{3L\omega}{4pc} + \frac{\omega}{c}\left(\frac{3S\cos\phi}{4} - r_1\right)\right] \tag{3}$$

where

$$\gamma = \frac{3\pi}{2} + \alpha + \phi, \qquad \gamma' = \frac{\pi}{2} - \alpha + \phi, \qquad \gamma'' = \arccos(\sin\alpha\cos\phi)$$
$$A = 1 - p\cos\gamma, \qquad A' = 1 - p\cos\gamma', \qquad A'' = 1 - p\cos\gamma''$$

When a helix of circular cross section is being calculated, $L = \pi D/\cos\alpha$ in (3), while for a helix of square cross section $L = 4b$.

If the contributions of elements 2 and 4 are neglected, which is a good approximation when both α and ϕ are small, the expression for $E_{\phi T}$ is considerably simplified. Making this approximation, letting $k = 1$ and $r_1 =$ constant, we obtain

$$E_{\phi T} = \frac{\sin\gamma}{A}\sin BA \angle(-BA) + \frac{\sin\gamma'}{A'}\sin BA' \angle[-BA' - 2\sqrt{\pi}B + \pi(S_\lambda \cos\phi + \sqrt{\pi}D_\lambda \sin\phi)] \quad (4)$$

Equation (4) applies specifically to helices of circular cross section, so that B in (4) is

$$B = \frac{D_\lambda \pi^{3/2}}{2p\cos\alpha} \quad (5)$$

Equation (4) gives the approximate pattern of the ϕ component of the far field in the xz plane from a single-turn helix of circular cross section.

In the case of the θ component of the far field in the xz plane, only elements 2 and 4 of the square turn contribute. Putting $k = 1$, the magnitude of the approximate θ pattern of the far field of a single-turn helix of circular cross section can be shown to be

$$|E_{\theta T}| = 2\frac{\sin\gamma''\sin BA''\cos\alpha}{A''(1-\sin^2\alpha\cos^2\phi)^{1/2}}\sin\tfrac{1}{2}\left[\pi\left(S_\lambda\cos\phi - \sqrt{\pi}D_\lambda\sin\phi\right) - 2\sqrt{\pi}B\right] \quad (6)$$

where B is as given by (5) and γ'' and A'' are as in (3).

As an example, the $E_{\phi T}$ and $E_{\theta T}$ patterns for a single-turn 12° helix with $C_\lambda = 1.07$ have been calculated and are presented in Fig. 8–38. Although the two patterns are of different form, both are broad in the axial direction ($\phi = 0$).

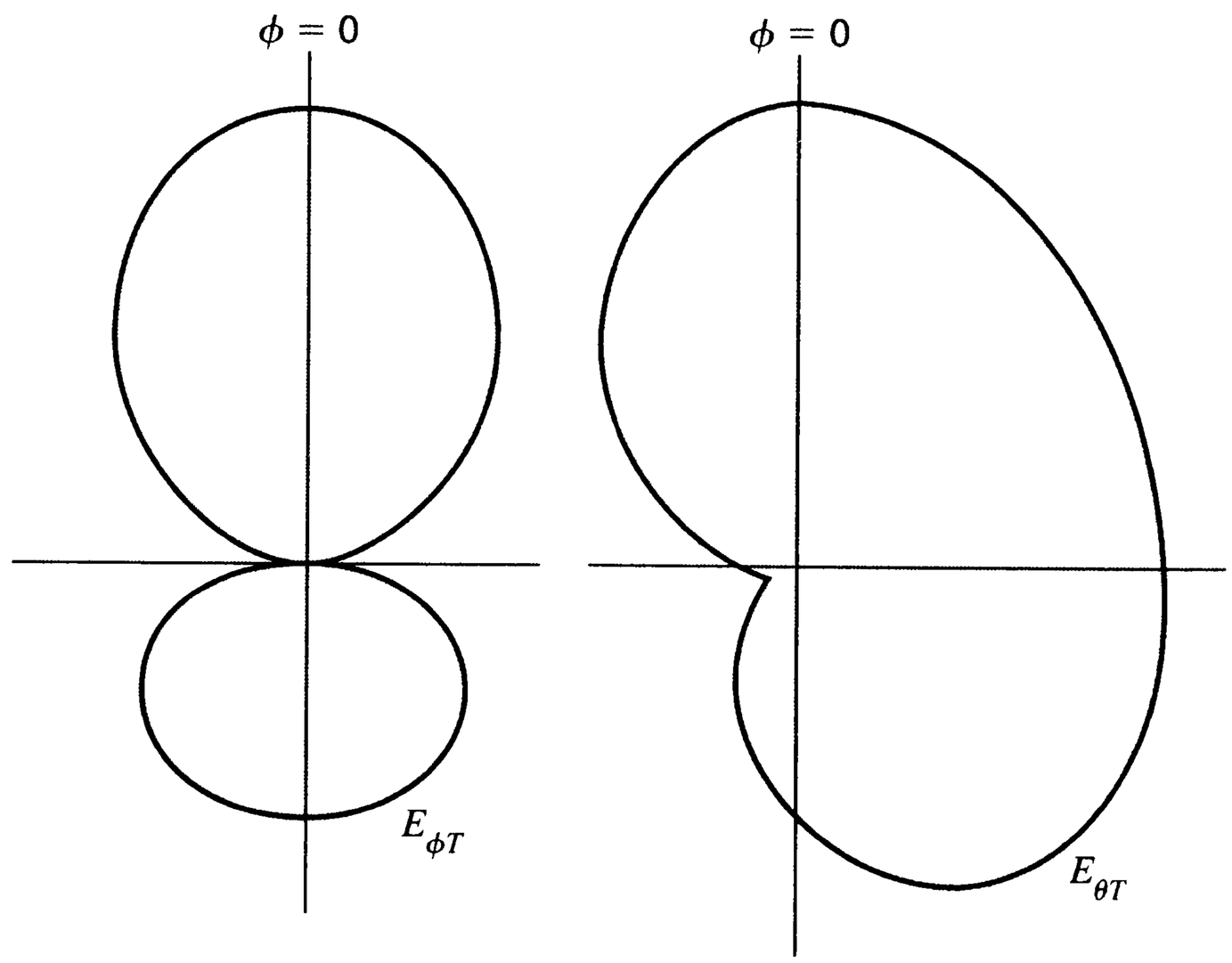


Figure 8–38
Calculated patterns for $E_{\phi T}$ and $E_{\theta T}$ fields of single turn of a 12° helix.

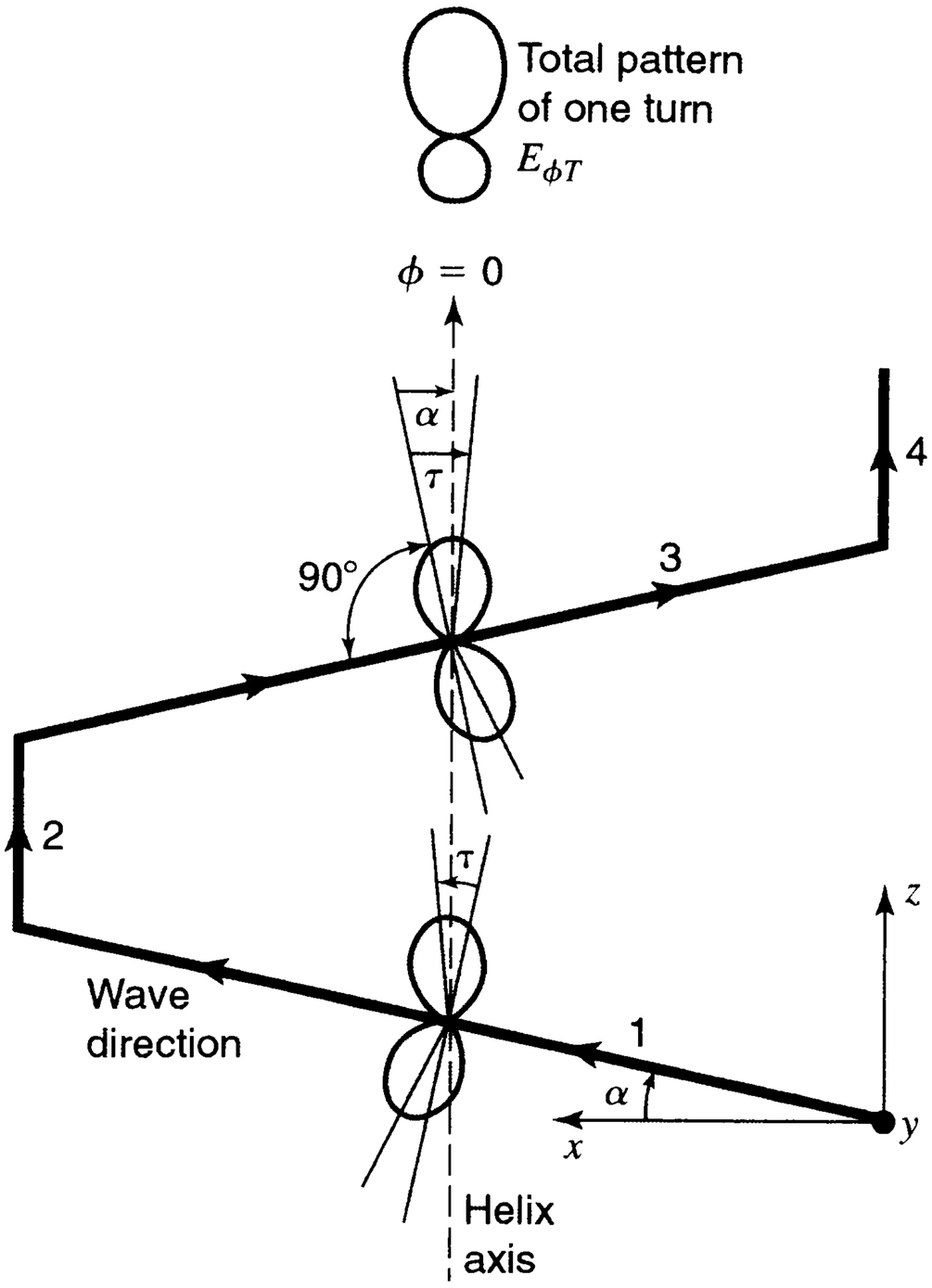


Figure 8–39
Individual E_ϕ patterns of elements 1 and 3 and total pattern of single turn, $E_{\phi T}$. The single turn is shown in plan view (in the xz plane of Fig. 8–36). The single turn and coordinate axes have been rotated around the y axis so that the z direction ($\phi = 0$) is toward the top of the page.

The individual E_ϕ patterns of elements 1 and 3 of the single turn are as suggested in Fig. 8–39. One lobe of each pattern is nearly in the axial direction, the tilt angle τ being nearly equal to the pitch angle α. The individual patterns add to give the $E_{\phi T}$ pattern for the single turn as shown (see also Fig. 8–38).

8–10 COMPLETE AXIAL-MODE PATTERNS OF MONOFILAR HELICES
(单绕螺旋的完整轴向模波瓣图)

By the principle of pattern multiplication, the total far-field pattern of a helix radiating in the axial mode is the product of the single-turn pattern and the array factor E. Thus, the total ϕ component E_ϕ of the distant electric field of a helix of circular cross section is the product of (8–9–4) and (8–8–18) or

$$E_\phi = E_{\phi T} E \tag{1}$$

The total θ component E_θ is the product of (8–9–6) and (8–8–18) or

$$E_\theta = E_{\theta T} E \tag{2}$$

As examples, the approximate E_ϕ and E_θ patterns, as calculated by the above procedure, for a 12°, 7-turn uniform helix of circular cross section with $C_\lambda = 1.07$ are presented in Fig. 8–40 at (a) and (c). The helix is shown at (e), with E_ϕ in the plane of the page and E_θ normal to the page. The array factor is shown at (b). The single-turn patterns are as presented in Fig. 8–38. The value of p used in these calculations is approximately that for the increased directivity condition. The product of the single-turn patterns (Fig. 8–38) and

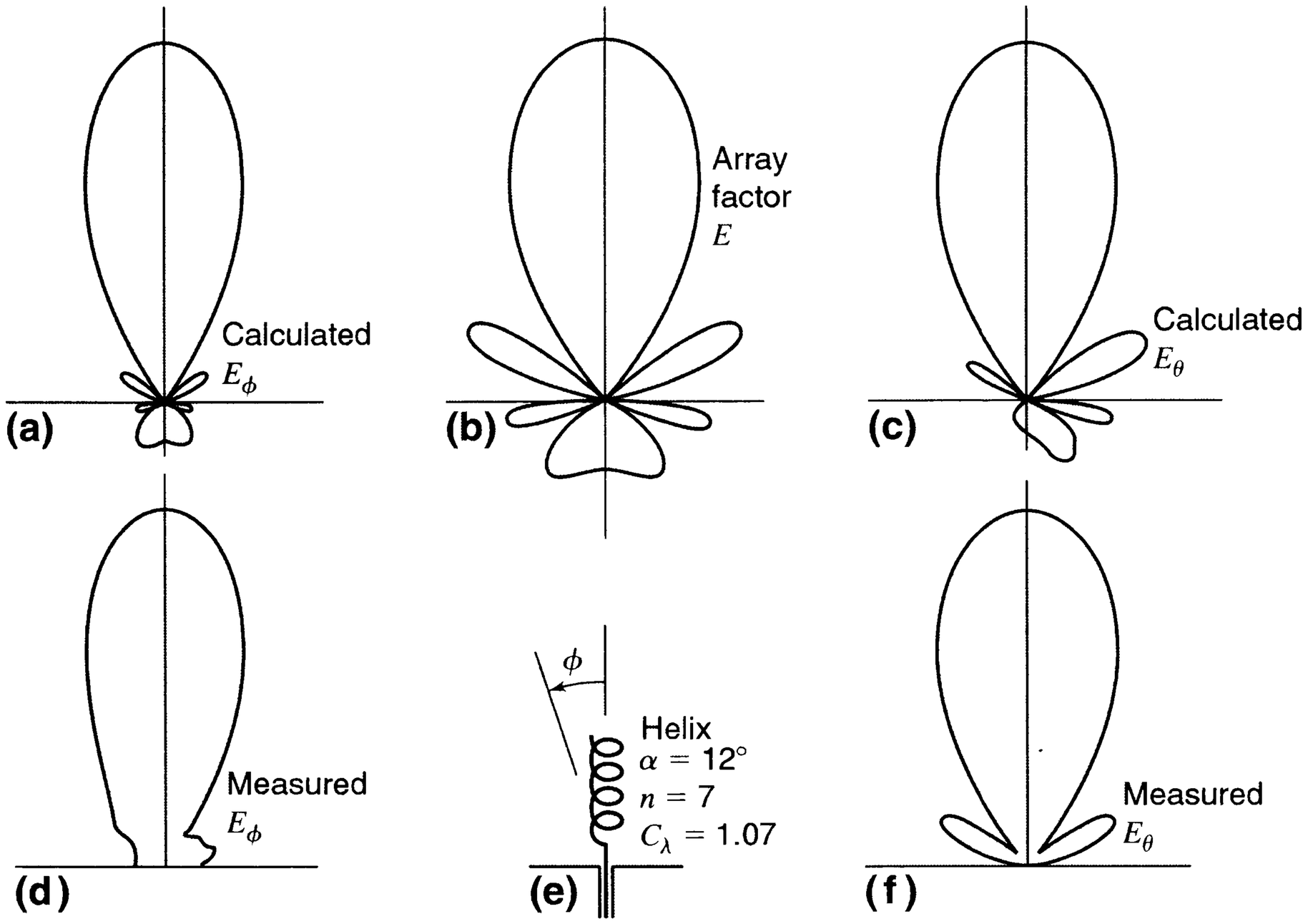


Figure 8–40
Comparison of complete calculated patterns (product of single-turn pattern and array factor) with measured patterns for a 12°, 7-turn helix with $C_\lambda = 1.07$ radiating in the axial mode. Agreement is satisfactory.

the array factor pattern at (*b*) yields the total patterns at (*a*) and (*c*). The agreement with the measured patterns shown at (*d*) and (*f*) is satisfactory.

Comparing the patterns of Figs. 8–38 and 8–40, it is to be noted that the array factor is much sharper than the single-turn patterns. Thus, the total E_ϕ and E_θ patterns (*a*) and (*c*) (Fig. 8–40) are nearly the same, in spite of the difference in the single-turn patterns. Furthermore, the main lobes of the E_ϕ and E_θ patterns are very similar to the array factor pattern. For long helices (say, $nS_\lambda > 1$) it is, therefore, apparent that a calculation of only the array factor suffices for an approximate pattern of any field component of the helix. Ordinarily the single-turn pattern need not be calculated except for short helices.

The far-field patterns of a helix radiating in the axial mode can, thus, be calculated to a good approximation from a knowledge of the dimensions of the helix and the wavelength. The value of the relative phase velocity used in the calculations may be computed for the increased-directivity condition from the helix dimensions and number of turns.

The effect of the ground plane on the axial-mode patterns is small if there are at least a few turns, since the returning wave on the helix and also the back lobe of the outgoing wave are both small. Hence, the effect of the ground plane may be neglected unless the helix is very short ($nS_\lambda < \frac{1}{2}$).

The approximate pattern of an axial-mode helix can be calculated very simply, while including the approximate effect of the single-turn pattern, by assuming that the single-turn

pattern is given by $\cos\phi$. Then the normalized total radiation pattern is expressed by

$$E = \left(\sin\frac{90^\circ}{n}\right)\frac{\sin(n\psi/2)}{\sin(\psi/2)}\cos\phi \tag{3}$$

where n = number of turns and

$$\psi = 360^\circ[S_\lambda(1-\cos\phi) + (1/2n)] \tag{4}$$

The value of ψ in (4) is for the increased-directivity condition and is obtained by substituting (8–8–12) in (8–8–3) and simplifying. The first factor in (3) is a normalizing factor, i.e., makes the maximum value of E unity.

8–11 AXIAL RATIO AND CONDITIONS FOR CIRCULAR POLARIZATION OF MONOFILAR AXIAL-MODE HELICAL ANTENNAS[1]

（单绕轴向模螺旋天线的轴比和圆极化条件）

In this section the axial ratio in the direction of the helix axis will be determined, and also the conditions necessary for circular polarization in this direction will be analyzed.

Consider the helix shown in Fig. 8–41. Let us calculate the electric field components E_ϕ and E_θ, as shown, at a large distance from the helix in the z direction. The helix is assumed to have a single uniform traveling wave as indicated. The relative phase velocity is p. The diameter of the helix is D and the spacing between turns is S. Unrolling the helix in the xz plane, the relations are as shown in Fig. 8–42. The helix as viewed from a point on the z axis is as indicated in Fig. 8–43. The angle ξ is measured from the xz plane. The coordinates of a point Q on the helix can be specified as r, ξ, z. The point Q is at a distance l from the terminal point T as measured along the helix. From the geometry of Figs. 8–42

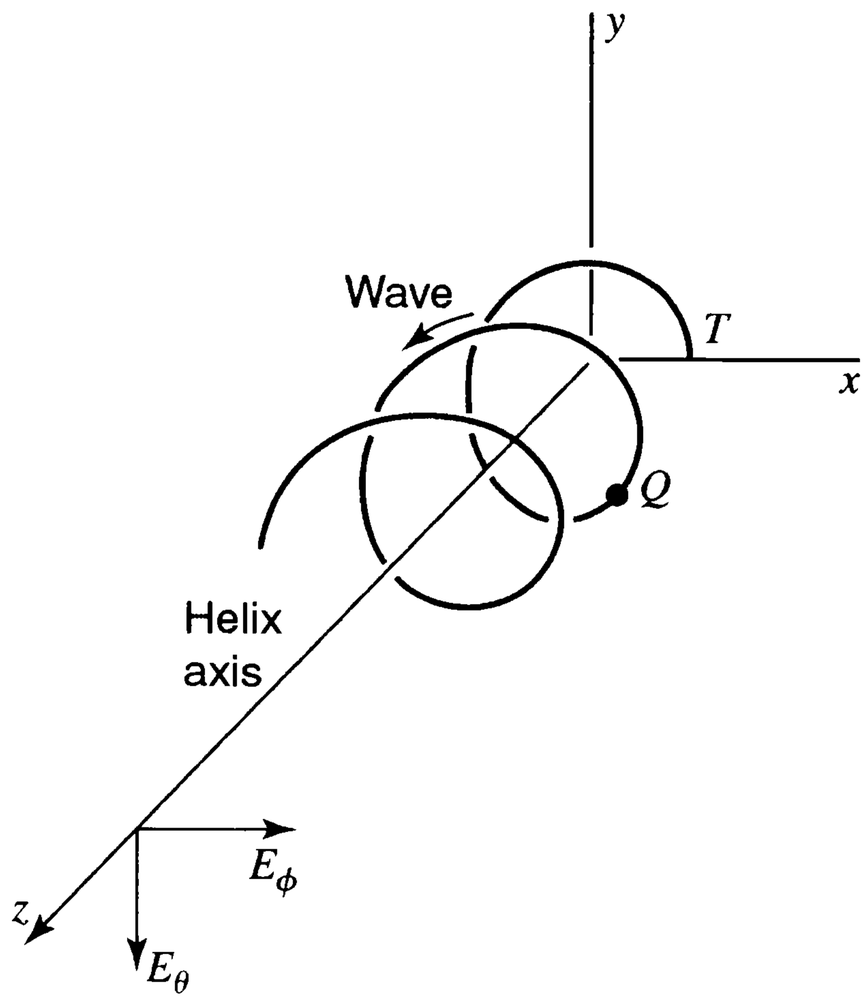


Figure 8–41
Field components as viewed from the helix axis.

[1]For a general discussion of elliptical and circular polarization see Secs. 2–15, 2–16 and 2–17; also see Kraus (6), Chap. 4.

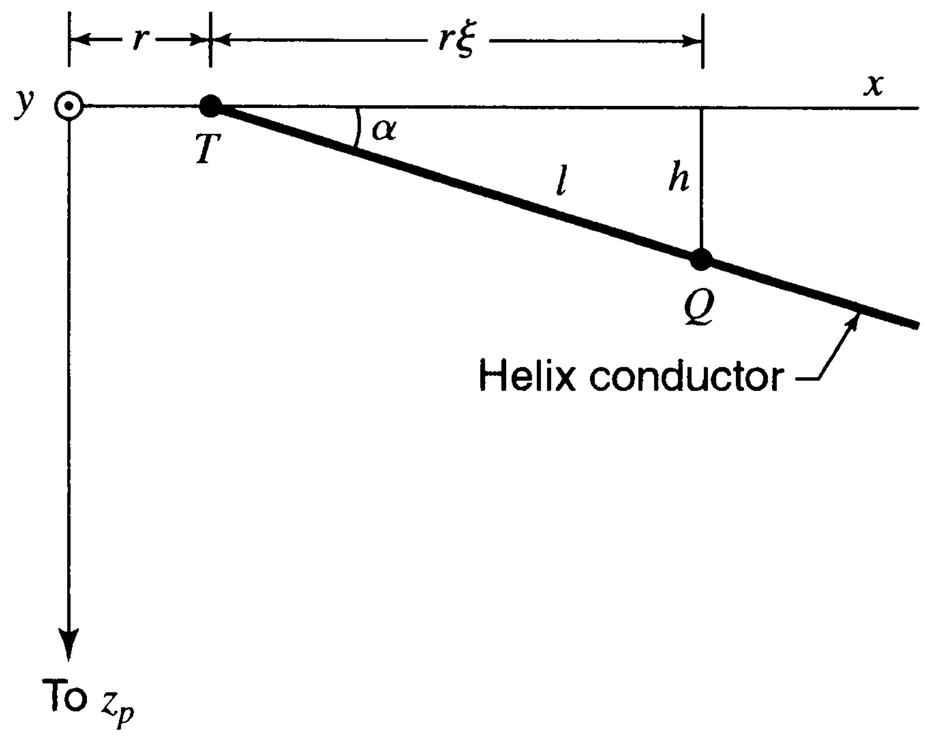


Figure 8–42
Geometry for calculating fields in the z direction.

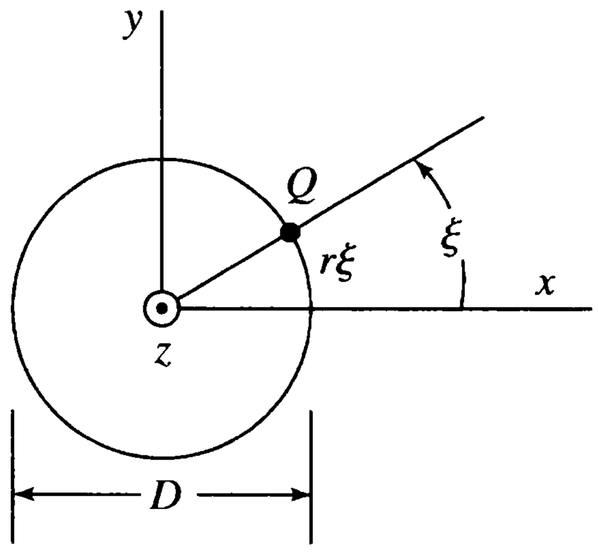


Figure 8–43
Helix of Fig. 8–41 as viewed from the positive z axis.

and 8–43, we can write

$$\left.\begin{aligned} h &= l\sin\alpha \\ z_p - h &= z_p - l\sin\alpha \\ \alpha &= \arctan\frac{S}{\pi D} = \arccos\frac{r\xi}{l} \\ r\xi &= l\cos\alpha \end{aligned}\right\} \tag{1}$$

where z_p is the distance from the origin to the distant point P on the z axis.

At the point P the ϕ component E_ϕ of the electric field for a helix of an integral number of turns n is

$$E_\phi = E_0\int_0^{2\pi n}\sin\xi\exp\left[j\omega\left(t - \frac{z_p}{c} + \frac{l\sin\alpha}{c} - \frac{l}{pc}\right)\right]d\xi \tag{2}$$

where E_0 is a constant involving the current magnitude on the helix.

From (1) the last two terms of the exponent in (2) may be rewritten. Thus,

$$\frac{l\sin\alpha}{c} - \frac{l}{pc} = \frac{r\xi}{c}\left(\tan\alpha - \frac{1}{p\cos\alpha}\right) = \frac{r\xi q}{c} \tag{3}$$

where

$$q = \tan\alpha - \frac{1}{p\cos\alpha} \tag{4}$$

When $\alpha = 0$, the helix becomes a loop and $q = -1/p$. The relation being obtained is, thus, a general one, applying not only to helices but also to loops as a special case. Equation (2)

now reduces to

$$E_\phi = E_0\, e^{j(\omega t - \beta s_p)} \int_0^{2\pi n} \sin\xi e^{jk\xi}\, d\xi \tag{5}$$

where quantities independent of ξ have been taken outside the integral and where

$$\beta = \frac{\omega}{c} = \frac{2\pi}{\lambda}$$

and

$$k = \beta r q = L_\lambda \left(\sin\alpha - \frac{1}{p} \right) \tag{6}$$

On integration (5) becomes

$$E_\phi = \frac{E_1}{k^2 - 1}(e^{j2\pi nk} - 1) \tag{7}$$

where $E_1 = E_0\, e^{j(\omega t - \beta z_p)}$

In a similar way we have for the θ component E_θ of the electric field at the point P,

$$E_\theta = E_0 \int_0^{2\pi n} \cos\xi \exp\left[j\omega \left(t - \frac{z_p}{c} + \frac{l\sin\alpha}{c} - \frac{l}{pc} \right) \right] d\xi \tag{8}$$

Making the same substitutions as in (2), we obtain from (8)

$$E_\theta = \frac{jE_1 k}{k^2 - 1}(e^{j2\pi nk} - 1) \tag{9}$$

The condition for circular polarization in the direction of the z axis is

$$\frac{E_\phi}{E_\theta} = \pm j \tag{10}$$

The ratio of (7) to (9) gives

$$\frac{E_\phi}{E_\theta} = \frac{1}{jk} = -\frac{j}{k} \tag{11}$$

Accordingly, for circular polarization in the axial direction of a helix of an integral number of turns, k must equal ± 1.

Equation (11) indicates that E_ϕ and E_θ are in time-phase quadrature. Therefore, the axial ratio AR is given by the magnitude of (11) or

$$AR = \frac{|E_\phi|}{|E_\theta|} = \left| \frac{1}{jk} \right| = \frac{1}{k} \tag{12}$$

The axial ratio will be restricted here to values between unity and infinity. Hence, if (12) is less than unity, its reciprocal is taken.

Substituting the value of k from (6) into (12) yields

$$AR = \frac{1}{|L_\lambda[\sin\alpha - (1/p)]|} \tag{13}$$

or

$$AR = \left| L_\lambda \left(\sin\alpha - \frac{1}{p} \right) \right| \tag{14}$$

Either (13) or (14) is used so that $1 \leq AR \leq \infty$.

From (13) and (14), it appears that the axial ratio can be calculated from the turn length L_λ and pitch angle α of the helix, and the relative phase velocity p. If we introduce the value of p for the condition of in-phase fields (see Table 8–1), it is found that $AR = 1$. In other words, the in-phase field condition is also the condition for circular polarization in the axial direction.

This may also be shown by noting that (11) satisfies the condition for circular polarization when $k = -1$, or

$$L_\lambda \left(\sin\alpha - \frac{1}{p} \right) = -1 \tag{15}$$

Solving (15) for p, we obtain

$$p = \frac{L_\lambda}{S_\lambda + 1} \tag{16}$$

which is identical with the relation for in-phase fields (ordinary end-fire condition).

Our previous discussion on phase velocity indicated that p followed more closely the relation for increased directivity than the relation for in-phase fields. Thus, introducing p in (14) for the condition of increased directivity, we obtain

$$AR \text{ (on axis)} = \frac{2n + 1}{2n} \tag{17}$$

where n is the number of turns of the helix. If n is large the axial ratio approaches unity and the polarization is nearly circular.[1]

When John Kraus first derived (17) in 1947, it came as a pleasant surprise to him that the axial ratio could be given by such a simple expression.

As an example, let us consider the axial ratio in the direction of the helix axis for a 13°, 7-turn helix. The axial ratio is unity if the relative velocity for the condition of in-phase fields is used. By (17) the axial ratio for the condition of increased directivity is $15/14 = 1.07$. This axial ratio is independent of the frequency or circumference C_λ as shown by the dashed line in Fig. 8–44. In this figure, the axial ratio is presented as a function of the helix circumference C_λ in free-space wavelengths.

If the axial ratio is calculated from (13) or (14), using the measured value of p shown in Fig. 8–44, an axial ratio variation is obtained as indicated by the solid curve in Fig. 8–44. This type of axial ratio versus circumference curve is typical of ones measured on helical beam antennas. Usually, however, the measured axial ratio increases more sharply as C_λ decreases to values less than about $\frac{3}{4}$. This difference results from the fact that the calculation of axial ratio by (13) or (14) neglects the effect of the back wave on the helix. This is usually small when the helix is radiating in the axial mode but at lower frequencies

---

[1] With circularly polarized feed an $AR = 1$ can be obtained on the axis for a helix of any length according to R. G. Vaughan (1) and J. B. Andersen. They also deduce the axial ratio as a function of the off-axis angle.

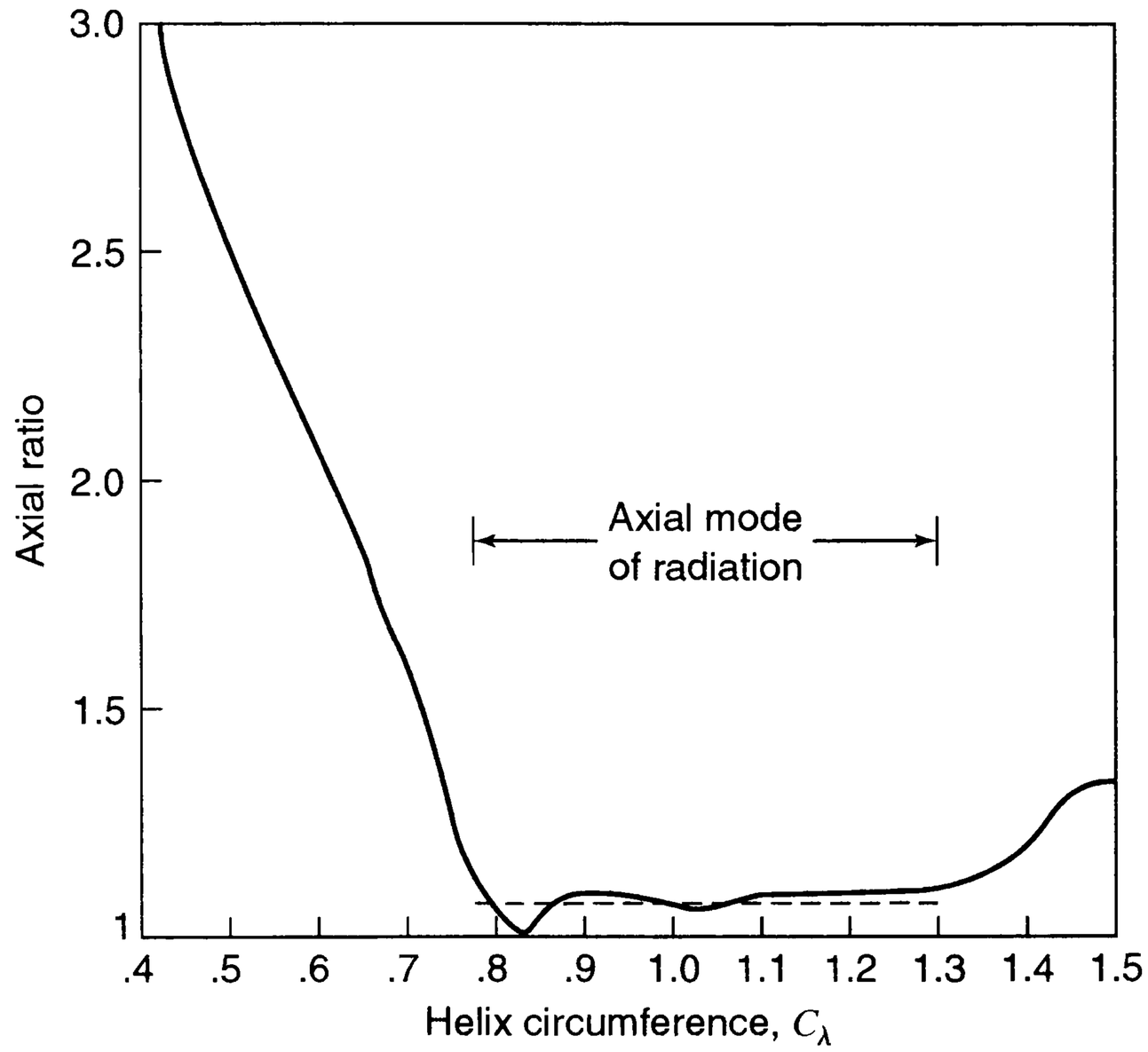


Figure 8–44
Axial ratio as a function of helix circumference C_λ for a 13°, 7-turn monofilar axial-mode helical antenna. The dashed curve is from (17). (*After Kraus.*)

or smaller circumferences ($C_\lambda < \frac{3}{4}$) the back wave is important. The back wave on the helix produces a wave reflected from the ground plane having the opposite direction of field rotation to that produced by the outgoing traveling wave on the helix. This causes the axial ratio to increase more rapidly than indicated in Fig. 8–44.

The foregoing discussion applies to helices of an integral number of turns. Let us now consider a long helix where the number of turns may assume nonintegral values. Hence, the length of the helical conductor will be specified as ξ_1 instead of $2\pi n$. It is further assumed that k is nearly unity. Thus, (5) becomes

$$E_\phi = \frac{E_1}{2j} \int_0^{\xi_1} \left(e^{j(k+1)\xi} - e^{j(k-1)\xi} \right) d\xi \tag{18}$$

Since $k \simeq -1, k + 1 \simeq 0$, and it follows that

$$e^{j(k+1)\xi_1} \simeq 1 + j(k+1)\xi_1 \tag{19}$$

Now integrating (18) and introducing the condition that k is nearly equal to -1 and the approximation of (19), we have

$$E_\phi = -\frac{E_1}{2} \left(j\xi_1 - \frac{e^{j(k-1)\xi_1} - 1}{k-1} \right) \tag{20}$$

Similarly the θ component E_θ of the electric field is

$$E_\theta = +\frac{E_1}{2j} \left(j\xi_1 + \frac{e^{j(k-1)\xi_1} - 1}{k-1} \right) \tag{21}$$

When the helix is very long

$$\xi_1 \gg 1$$

and (20) and (21) reduce to

$$E_\phi = -j\frac{E_1\xi_1}{2} \qquad \text{and} \qquad E_\theta = +\frac{E_1\xi_1}{2} \tag{22}$$

Taking the ratio of E_ϕ to E_θ,

$$\frac{E_\phi}{E_\theta} = -j \tag{23}$$

which fulfills the condition for circular polarization.

Still another condition resulting in circular polarization is obtained when $(k \pm 1)\xi_1 = 2\pi m$, where m is an integer. This condition is satisfied when either the positive or negative sign in $k \pm 1$ is chosen but not for both.

To summarize, the important conditions for circular polarization are as follows:

1. The radiation in the axial direction from a helical antenna of any pitch angle and of an integral number of 1 or more turns will be circularly polarized if $k = -1$ (in-phase fields or ordinary end-fire condition).
2. The radiation in the axial direction from a helical antenna of any pitch angle and a large number of turns, which are not necessarily an integral number, is nearly circularly polarized if k is nearly -1.

8–12 WIDEBAND CHARACTERISTICS OF MONOFILAR HELICAL ANTENNAS RADIATING IN THE AXIAL MODE（单绕螺旋天线轴向模辐射的宽频带特性）

The helical beam antenna[1] has inherent broadband properties, possessing desirable pattern, impedance and polarization characteristics over a relatively wide frequency range. The natural adjustment of the phase velocity so that the fields from each turn add nearly in phase in the axial direction accounts for the persistence of the axial mode of radiation over a nearly 2 to 1 range in frequency. If the phase velocity were constant as a function of frequency, the axial-mode patterns would be obtained only over a narrow frequency range. The terminal impedance is relatively constant over the same frequency range because of the large attenuation of the wave reflected from the open end of the helix. The polarization is nearly circular over the same range in frequency because the condition of fields in phase is also the condition for circular polarization.

As shown in Fig. 8–45*a*, the dimensions of a helix in free-space wavelengths move along a constant pitch-angle line as a function of frequency. If F_1 is the lower frequency limit of the axial mode of radiation and F_2 the upper frequency limit of this mode, then the range in dimensions for a 10° helix would be as suggested by the heavy line on the diameter-spacing chart of Fig. 8–45*a*. The center frequency F_0 is arbitrarily defined as $F_0 = (F_1 + F_2)/2$.

The properties of a helical beam antenna are a function of the pitch angle. The angle resulting in a maximum frequency range $F_2 - F_1$ of the axial mode of radiation is said to

[1] Or monofilar axial-mode helical antenna.

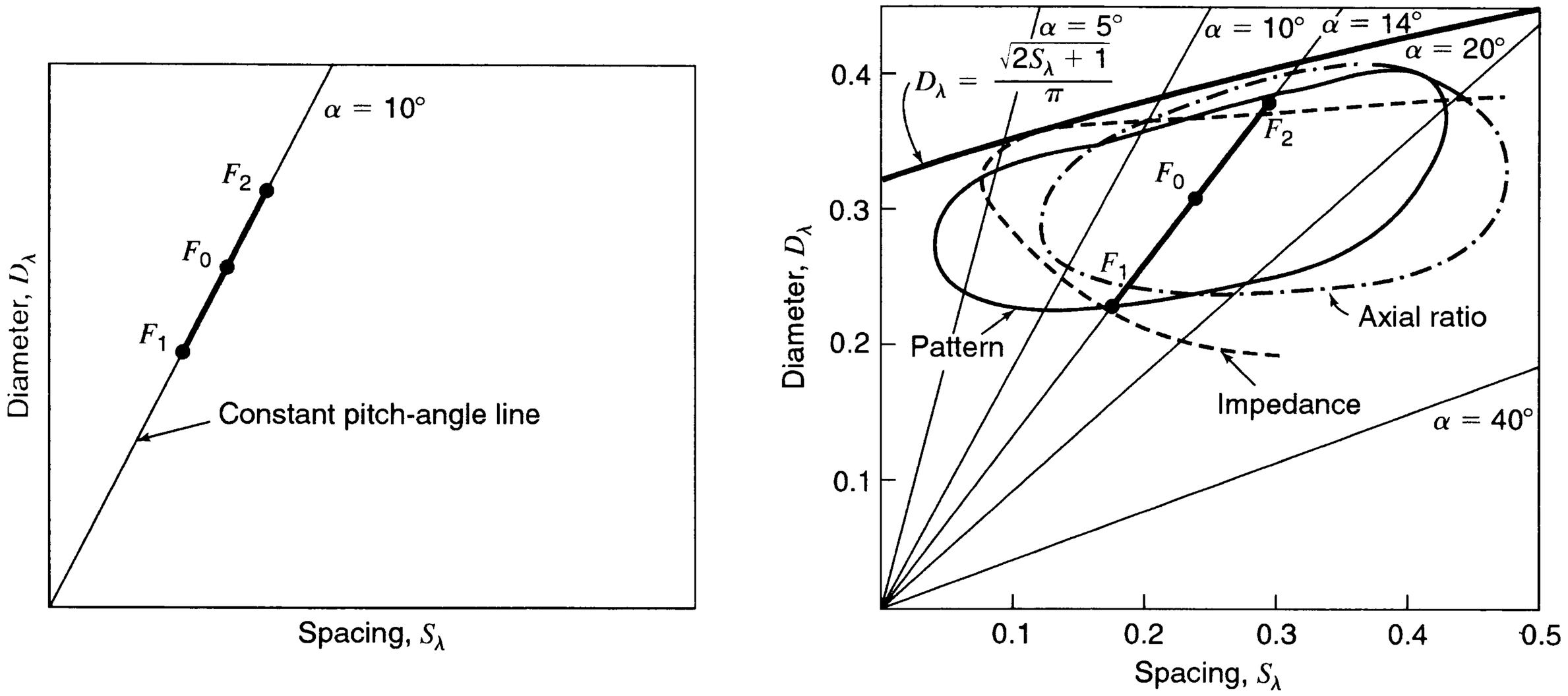


Figure 8–45
Diameter-spacing charts for monofilar helices with measured performance contours (*b*) for the axial mode of radiation.

be an "optimum" pitch angle. To determine an optimum angle, the pattern, impedance and polarization characteristics of helical antennas may be compared on a diameter-spacing chart as in Fig. 8–45*b*. The three contours indicate the region of satisfactory pattern, impedance and polarization values as determined by measurements on helices of various pitch angle as a function of frequency. The axial length of the helices tested is about 1.6 λ at the center frequency. The pattern contour in Fig. 8–45*b* indicates the approximate region of satisfactory patterns. A satisfactory pattern is considered to be one with a major lobe in the axial direction and with relatively small minor lobes. Inside the pattern contour, the patterns are of this form and have half-power beamwidths of less than 60° and as small as 30°. Inside the impedance contour in Fig. 8–45*b* the terminal impedance is relatively constant and is nearly a pure resistance of 100 to 150 Ω. Inside the axial ratio contour, the axial ratio in the direction of the helix axis is less than 1.25. Note that all contours lie below the line for which $D_\lambda = \sqrt{2S_\lambda + 1}/\pi$. This line may be regarded as an upper limit for the beam mode. It is apparent that the frequency range $F_2 - F_1$ is small if the pitch angle is either too small or too large. A pitch angle of about 12 or 14° would appear to be "optimum" for helices about 1.6 λ long at the center frequency. Since the properties of the helix change slowly in the vicinity of the optimum angle, there is nothing critical about this value. The contours are arbitrary but are suitable for a general-purpose beam antenna of moderate directivity. The exact values of the frequency limits, F_1 and F_2, are also arbitrary but are relatively well defined by the close bunching of the contours near the frequency limits.

Based on the above conclusions, John Kraus constructed a 14°, 6-turn helix in 1948 and measured its properties. The helix has a diameter of 0.31 λ at the center frequency

(400 MHz). The diameter of the conductor is about 0.02 λ. Conductor diameters of 0.005 to 0.05 λ can be used with little difference in the properties of this helix in the frequency range of the beam mode (Kraus-5).

The measured patterns between 275 and 560 MHz are presented in Fig. 8–17. It is apparent that the patterns are satisfactory over a frequency range from 300 MHz ($C_\lambda = 0.73$) to 500 MHz ($C_\lambda = 1.22$).

A summary of the characteristics of this antenna are given in Fig. 8–21 in which the half-power beamwidth, axial ratio and standing-wave ratio are shown as a function of the helix circumference.

8–13 TABLE OF PATTERN, BEAMWIDTH, GAIN, IMPEDANCE AND AXIAL RATIO FORMULAS（波瓣图、频带宽度、增益、阻抗和轴比公式列表）

Expressions developed in the preceding sections for calculating the pattern, beamwidth, directivity, terminal resistance and axial ratio for axial-mode helical antennas are summarized in Table 8–2. These relations apply to helices for $12^\circ < \alpha < 15^\circ$, $\frac{3}{4} < C_\lambda < \frac{4}{3}$ and $n > 3$ or to the more specific restrictions listed in the footnote to the table.

Table 8–2 Formulas for monofilar axial-mode helical antennas

Pattern	$\begin{cases} E = \left(\sin\dfrac{90^\circ}{n}\right)\dfrac{\sin(n\psi/2)}{\sin(\psi/2)}\cos\phi \\ \text{where } \psi = 360^\circ\left[S_\lambda(1-\cos\phi)+\dfrac{1}{2n}\right] \end{cases}$
Beamwidth (half-power) See restrictions	$\text{HPBW} = \dfrac{52^\circ}{C_\lambda\sqrt{nS_\lambda}}$
Beamwidth (first nulls) See restrictions	$\text{BWFN} = \dfrac{115^\circ}{C_\lambda\sqrt{nS_\lambda}}$
Directivity (or gain†) See restrictions	$D = 12C_\lambda^2 nS_\lambda$
Terminal resistance	$R = 140C_\lambda\,\Omega$ (axial feed)
See restrictions	$R = 150/\sqrt{C_\lambda}\,\Omega$ (peripheral feed)
Axial ratio (on axis)	$AR = \dfrac{2n+1}{2n}$ (increased directivity)
Axial ratio (on axis)	$AR = \left\lvert L_\lambda\left(\sin\alpha - \dfrac{1}{p}\right)\right\rvert$ (p unrestricted)

n = number of turns of helix
C_λ = circumference in free-space wavelengths
S_λ = spacing between turns in free-space wavelengths
L_λ = turn length in free-space wavelengths
α = pitch angle
p = relative phase velocity
ϕ = angle with respect to helix axis

Restriction for beamwidth and directivity $0.8 < C_\lambda < 1.15$; $12^\circ < \alpha < 14^\circ$; $n > 3$

Restriction for terminal resistance $0.8 < C_\lambda < 1.2$; $12^\circ < \alpha < 14^\circ$; $n \geq 4$

† Assuming no losses.

8–14 RADIATION FROM LINEAR PERIODIC STRUCTURES WITH TRAVELING WAVES WITH PARTICULAR REFERENCE TO THE HELIX AS A PERIODIC STRUCTURE ANTENNA

（载有行波的线形周期性结构的辐射，将螺旋视为周期性结构天线）

Radiation from continuous linear antennas carrying a traveling wave was discussed in Sec. 6–8. Although the helical beam antenna consists of a continuous conductor carrying a traveling wave, it is also a periodic structure with period equal to the turn spacing as considered in Sec. 8–8.

Now let us develop the periodic structure approach in a more general way which illustrates the relation of helical antennas to other periodic-structure (dipole) antennas.[1]

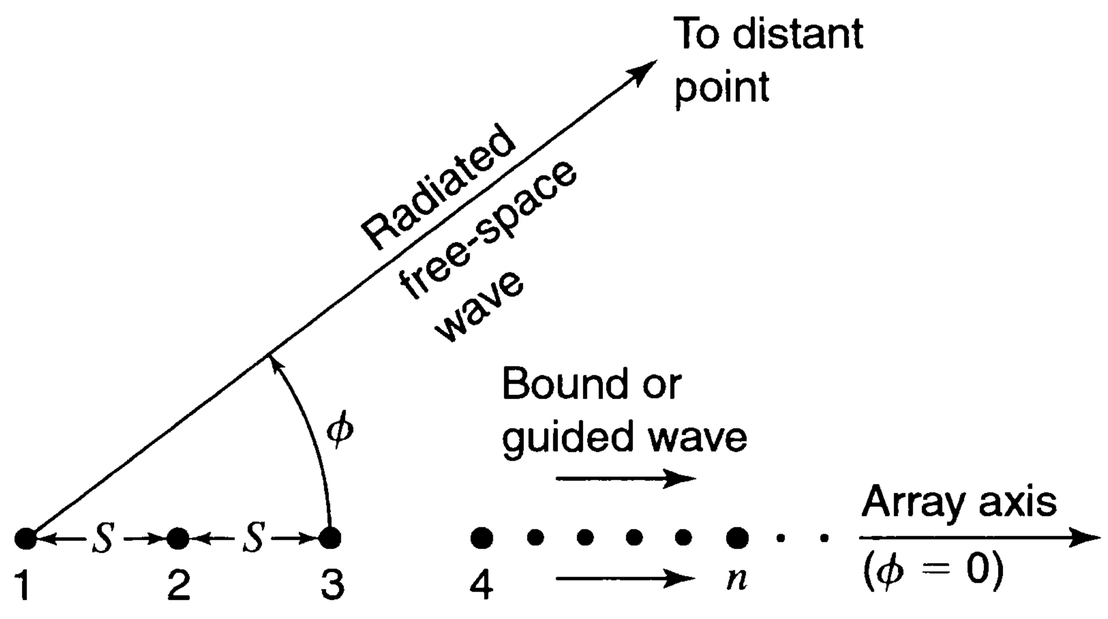


Figure 8–46
Linear array of n isotropic point sources of equal amplitude and spacing, S, representing a linear periodic structure carrying a traveling wave.

A linear array of n isotropic point sources of equal amplitude and spacing is shown in Fig. 8–46 representing a linear periodic structure carrying a traveling wave. As discussed previously the phase difference of the fields from adjacent sources as observed at a distance point is given by

$$\psi = \frac{2\pi}{\lambda_0} S \cos\phi - \delta \tag{1}$$

where

S = spacing between sources, m
λ_0 = free-space wavelength, m
ϕ = angle between array axis and direction of distant point, rad or deg
δ = phase difference of source 2 with respect to source 1, 3 with respect to 2, etc., rad or deg

We assume that the array is fed by a wave traveling along it from left to right via a guiding structure which may, for example, be an open-wire transmission line, a waveguide

[1] Although the helix is a periodic structure, it is continuous. The dipole arrays are discontinuous.

or a helix.[1] The *phase constant* of the traveling wave is given by

$$\beta = \frac{2\pi}{\lambda_0 p} \text{ (rad m}^{-1}) = \frac{360^\circ}{\lambda_0 p} \text{ (deg m}^{-1}) \tag{2}$$

where

λ_0 = free-space wavelength, m
p = v/c = relative phase velocity, dimensionless
v = wave velocity, m s^{-1}
c = velocity of light, m s^{-1}

The *phase difference* between sources is given by

$$\delta = \frac{2\pi}{\lambda_0 p} S \text{ (rad)} = \frac{360^\circ}{\lambda_0 p} S \text{ (deg)} \tag{3}$$

where S = spacing between sources, m

In general, for the fields from the n sources to be in-phase at a distant point requires that

$$\psi = 2\pi m \qquad \text{(rad or deg)} \tag{4}$$

where m = mode number = 0, ±1, ±2, etc.

Introducing (3) and (4) into (1) yields

$$2\pi m = \frac{2\pi}{\lambda_0} S \cos\phi - \frac{2\pi}{\lambda_0 p} S \tag{5}$$

or

$$2\pi m = \beta_0 S \cos\phi - \beta S \tag{6}$$

where $\beta_0 = \dfrac{2\pi}{\lambda_0}$ = phase constant of free-space wave, rad m^{-1}

For *mode number* $m = 0$, the phase difference of the fields from adjacent sources at a distant point is zero; for $m = 1$, the phase difference is 2π; for $m = 2$, the phase difference is 4π; etc.

$\beta = \dfrac{2\pi}{\lambda_0 p}$ = phase constant of guided wave, rad m^{-1}
$\beta_0 S$ = electrical distance between sources for a free-space wave, rad
βS = electrical distance between sources for the guided wave, rad
$\beta_0 S \cos\phi$ = electrical distance between sources for a free-space wave in direction of distant point

[1] We assume a uniform traveling wave. Although such a wave is approximated with a monofilar axial-mode helix, it is not necessarily realized with the dipole arrays presented in this section without the addition of suitable impedance matching networks (not shown). See Sec. 16–11 on Phased Arrays and Sec. 21–22 on Leaky Wave Antennas.

From (6) we have

$$\beta_0 \cos\phi = \beta + \frac{2\pi m}{S} \tag{7}$$

or

$$\cos\phi = \frac{\beta}{\beta_0} + \frac{2\pi m}{\beta_0 S} = \frac{1}{p} + \frac{m}{S/\lambda_0} \tag{8}$$

Let us now consider several examples.

EXAMPLE **8–14.1 Mode Number $m = 0$**

For different relative phase velocities p, the beam angles ϕ, as given by (8), are as tabulated:

Relative phase velocity, p	$\cos\phi$	ϕ	Beam direction
1 ($v = c$)	1	0°	End-fire
∞	0	90°	Broadside
−1 ($v = c$ with wave right-to-left)	−1	180°	Back-fire
<1	>1	Imaginary	No beam

For the last entry ($p < 1$), ϕ is imaginary. This implies that all of the wave energy is bound to the array (guided along it) and that there is no radiation (no beam).

■ **Summary**

For p values from $+1$ to $+\infty$ and $-\infty$ to -1, the beam swings from end-fire ($\phi = 0°$) through broadside ($\phi = 90°$) to back-fire ($\phi = 180°$). For p values between -1 and $+1$ ($-1 < p < 1$) ϕ is imaginary (no radiation). We note that these results are independent of the spacing S.

EXAMPLE **8–14.2 Mode Number $m = -1$. *Relative phase velocity* $p = 1$ ($v = c$)**

Fields from adjacent sources have 2π (= 360°) phase difference at a distant point in the direction of the beam maximum. For different spacings S, the beam angles ϕ, as given by (8), are as tabulated:

Spacing S	$\cos\phi$	ϕ	Beam direction
λ_0	0	90°	Broadside†
$\lambda_0/2$	−1	180°	Back-fire‡

† For this case, there are also equal beams end-fire ($\phi = 0°$) and back-fire ($\phi = 180°$)

‡ For this case, there is an equal end-fire lobe ($\phi = 0°$)

■ **Summary**

For spacings between $\lambda_0/2$ and λ_0 the beam swings between 90 and 180°. Larger spacings are required to swing the beam to angles less than 90° but other lobes also appear.

EXAMPLE **8–14.3 Mode Number $m = -1$. *Relative phase velocity $p = \frac{1}{2}$ (slow wave)***

For different spacings S, the beam angles ϕ, as given by (8), are as tabulated:

Spacing S	$\cos\phi$	ϕ	Beam direction
λ_0	1	0°, 90°, 180°	End-fire, broadside and backfire
$\lambda_0/2$	0	90°	Broadside
$\lambda_0/3$	−1	180°	Back-fire

■ **Summary**

For spacings between $\lambda_0/3$ and λ_0 the beam swings from back-fire (180°) through broadside (90°) to end-fire (0°), but for $S = \lambda_0$ broadside and back-fire lobes also appear. For spacings greater than λ_0 or less than $\lambda_0/3$, ϕ is imaginary (no beam).

EXAMPLE **8–14.4 Mode Number $m = -1$. *Relative phase velocity $p = \frac{1}{5}$ (slow wave)***

For different spacings S, the beam angles ϕ, as given by (8), are as tabulated:

Spacing S	$\cos\phi$	ϕ	Beam direction
$\lambda_0/4$	1	0°	End-fire
$\lambda_0/5$	0	90°	Broadside
$\lambda_0/6$	−1	180°	Back-fire

■ **Summary**

For spacings between $\lambda_0/4$ and $\lambda_0/6$ the beam swings from end-fire, through broadside to back-fire. For spacings less than $\lambda_0/6$ or more than $\lambda_0/4$, ϕ is imaginary (no beam).

The results of Examples 2, 3 and 4 are shown graphically in Fig. 8–47.

Another way of analyzing an array, and periodic structures in general, is to plot the electrical spacing $\beta_0 S$ of the free-space wave (as ordinate) versus the electrical spacing βS of the guided wave traveling along the array (as abscissa). Dividing both coordinates by 2π, we obtain S/λ_0 as ordinate and $S/(p\lambda_0) = S/\lambda$ as abscissa (where $\lambda = p\lambda_0$ = wavelength

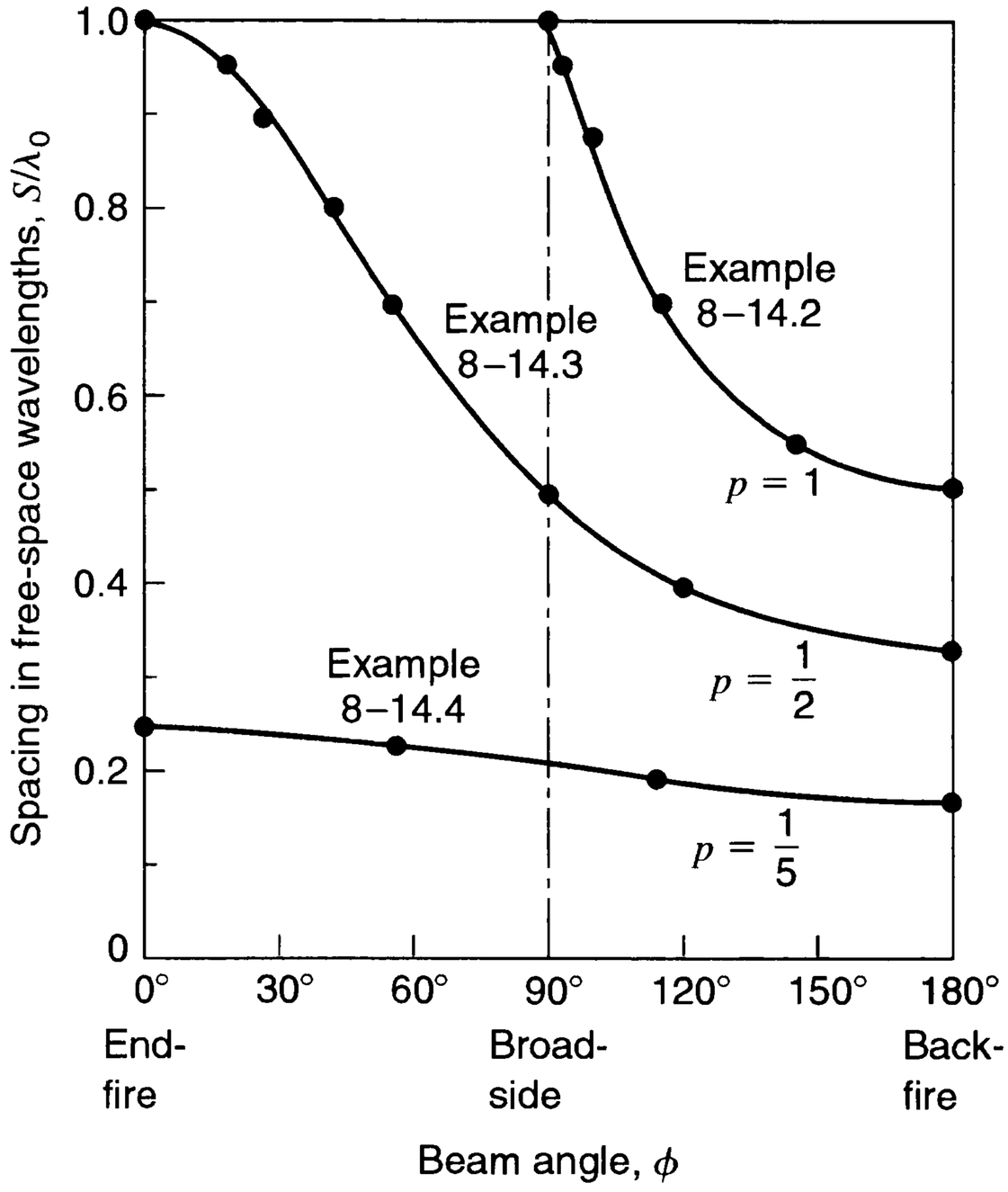


Figure 8–47
Relation of spacing and beam angle for linear arrays with traveling waves of relative phase velocities $p = 1$, $\frac{1}{2}$ and $\frac{1}{5}$ (Examples 8–14.2, 8–14.3, and 8–14.4) all with mode number $m = -1$.

on the array). This type of S-S diagram[1] is presented in Fig. 8–48, illustrating the three arrays of Examples 2, 3 and 4 (shown also in the S-ϕ diagram of Fig. 8–47).

For a relative phase velocity $p = 1$, $S/\lambda = S/\lambda_0$, and the array operates along the $p = 1$ line (at 45° to the axes). Back-fire occurs at $S/\lambda_0 = \frac{1}{2}$ and broadside at $S/\lambda_0 = 1$ (Example 8–14.2).

For a relative phase velocity $p = \frac{1}{2}$, the array operates along the $p = \frac{1}{2}$ line with back-fire at $S/\lambda_0 = \frac{1}{3}$, broadside at $S/\lambda_0 = \frac{1}{2}$ and end-fire at $S/\lambda_0 = 1$ at right edge of diagram (Example 8–14.3).

For a relative phase velocity $p = \frac{1}{5}$, the array operates along the $p = \frac{1}{5}$ line with back-fire at $S/\lambda_0 = \frac{1}{6}$, broadside at $S/\lambda_0 = \frac{1}{5}$ and end-fire at $S/\lambda_0 = \frac{1}{4}$ (Example 8–14.4). For a higher mode number $m = -2$, the array again produces beams swinging from back-fire at $S/\lambda_0 = \frac{1}{3}$, through broadside at $S/\lambda_0 = \frac{2}{5}$ to end-fire at $S/\lambda_0 = \frac{1}{2}$ (off diagram at right).

Let us consider several more examples of traveling-wave periodic-structure arrays. On an S-S diagram each type of array occupies a unique location or niche which is characteristic of the antenna's behavior. Differences between arrays are clearly evident from their locations on the diagram.

---

[1] Some authors use k for β_0 and refer to the graph as a k-β diagram, also called a *Brillouin diagram* after Leon Brillouin.

Array chart showing regions for broad-side, end-fire and back-fire operation with excursion distances for monofilar, bifilar, quadrifilar, octafilar, axial-mode helices and for log periodics

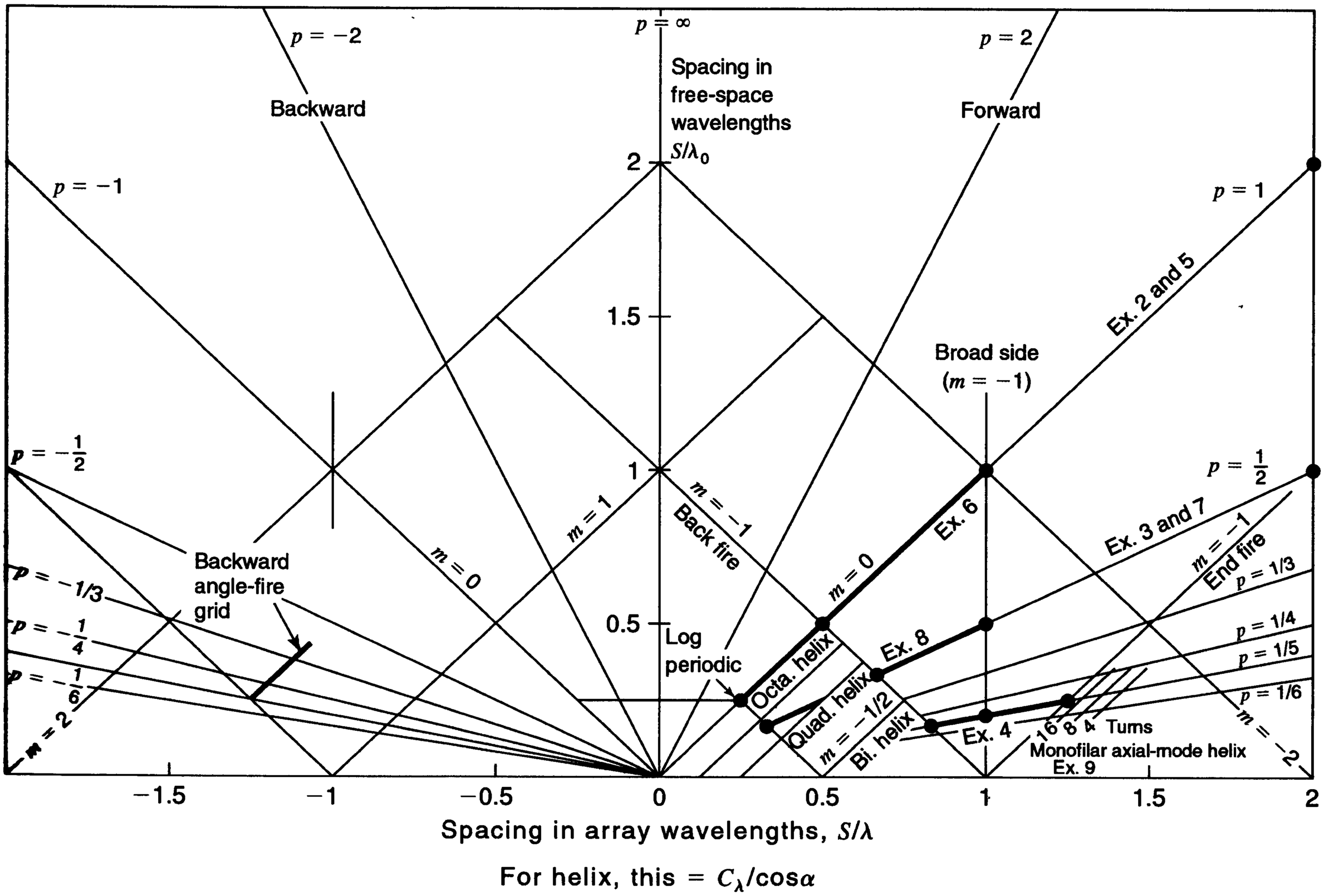


Figure 8–48
S-S diagram for linear traveling-wave antennas showing regions of operation for different types of arrays including Examples 8–14.2 through 8–14.9. Note that for a helix, S/λ here corresponds to the turn length L_λ.

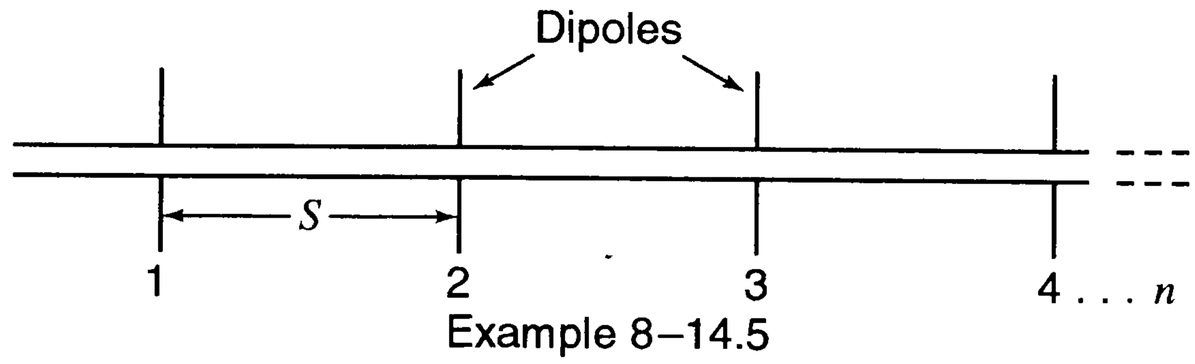
EXAMPLE **8–14.5 Scanning Array of Dipoles with $S = \lambda_0$ at Center Frequency (Fig. 8–49)**

Beam scanning is by shifting frequency. Physical element spacing is constant. Array is fed from the left end with a two-wire transmission line ($p = 1$). The mode number $m = -1$, so (8) becomes

$$\cos\phi = \frac{1}{p} - \frac{1}{S/\lambda_0} = 1 - \frac{1}{S/\lambda_0} \tag{9}$$

At the center frequency $S = \lambda_0$ and from (9) $\cos\phi = 0$ and $\phi = 90°$ (beam broadside). Halving the frequency makes $S = \lambda_0/2$ and $\phi = 180°$ (beam back-fire). Doubling the frequency makes $S = 2\lambda_0$ and $\phi = 60°$ (only 30° beyond broadside). To swing the beam further toward end-fire requires a further increase in frequency. The position of

Figure 8–49
Scanning array of dipoles with $S = \lambda_0$ at center frequency, relative phase velocity $p = 1$ ($v = c$) and mode number $m = -1$. The beam angle ϕ (with respect to dipole 1) is outward from the page. (Example 8–14.5.)



Example 8–14.5

this scanning array is shown in Fig. 8–48 (line labeled Examples 2 and 5). It is to be noted that there are other lobes present not given by (9).

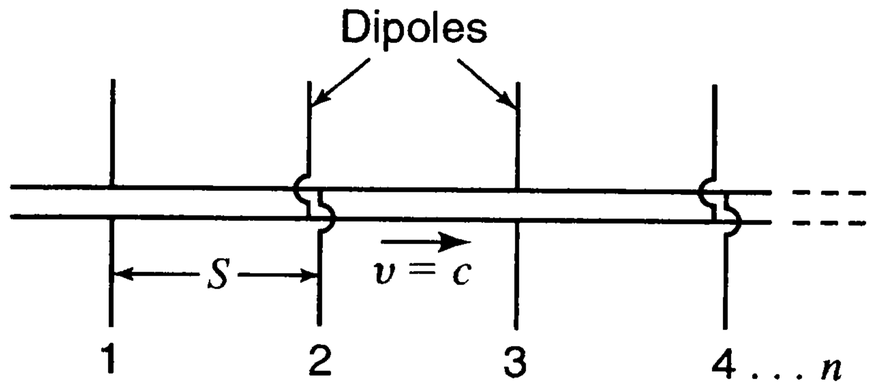
EXAMPLE **8–14.6 Scanning Array of Alternately Reversed Dipoles with $S = \lambda_0/2$ at Center Frequency (Fig. 8–50)**

Beam scanning is by shifting frequency. Physical element spacing is constant. Array is fed from the left end with a two-wire transmission line ($p = 1$). The mode number $m = -\frac{1}{2}$, so (8) becomes

$$\cos\phi = 1 - \frac{1}{2S/\lambda_0} \tag{10}$$

At the center frequency $S = \lambda_0/2$ and from (10) $\cos\phi = 0$ and $\phi = 90°$ (beam broadside). Halving the frequency makes $S = \lambda_0/4$ and $\phi = 180°$ (beam back-fire). Doubling the frequency makes $S = \lambda_0$, swinging the beam to $\phi = 60°$. The position of this scanning array is shown in Fig. 8–48.

Figure 8–50
Scanning array of alternately reversed dipoles with $S = \lambda_0/2$ at center frequency, relative phase velocity $p = 1$ and mode number $m = -\frac{1}{2}$. The beam angle ϕ (with respect to dipole 1) is outward from the page. (Example 8–14.6.)



Example 8–14.6

EXAMPLE **8–14.7 Scanning Array of Dipoles with $S = \lambda_0/2$ at Center Frequency with Slow Wave ($p = \frac{1}{2}$) (Fig. 8–51)**

Beam scanning is by shifting frequency. Physical element spacing is constant. Array is fed from the left end by a two-wire transmission line with line length $2S$ between dipoles so that $p = \frac{1}{2}$. The mode number $m = -1$, so (8) becomes

$$\cos\phi = 2 - \frac{1}{S/\lambda_0} \tag{11}$$

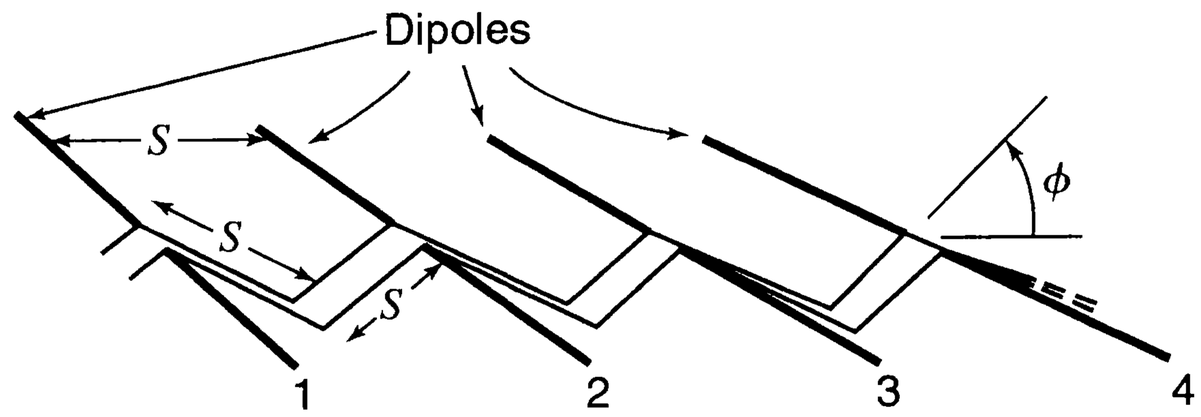


Figure 8–51
Scanning array of dipoles with $S = \lambda_0/2$ at center frequency, relative phase velocity $p = \frac{1}{2}$ (slow wave) and mode number $m = -1$. (Example 8–14.7.)

At the center frequency $S = \lambda_0/2$ and from (11) $\phi = 90°$ (beam broadside). Reducing the frequency so $S = \lambda_0/3$ makes $\phi = 180°$ (beam back-fire). Doubling the frequency makes $S = \lambda_0$ and $\phi = 0°$ (beam end-fire). The position of this scanning array is shown in Fig. 8–48 (line labeled Examples 3 and 7).

EXAMPLE **8–14.8 Scanning Array of Alternately Reversed Dipoles with $S = \lambda_0/4$ at Center Frequency with Slow Wave ($p = \frac{1}{2}$) (Fig. 8–52)**

Beam scanning is by shifting frequency. Physical element spacing is constant. Array is fed from the left end by a two-wire transmission line with line length $2S$ between dipoles so that $p = \frac{1}{2}$. The mode number $m = -\frac{1}{2}$, so (8) becomes

$$\cos\phi = 2 - \frac{1}{2S/\lambda_0} \tag{12}$$

At the center frequency $S = \lambda_0/4$ and from (12) $\phi = 90°$ (beam broadside). Decreasing the frequency to $\frac{1}{6}$ makes $S = \lambda_0/6$ and $\phi = 180°$ (beam back-fire). Doubling the frequency makes $S = \lambda_0/2$ and $\phi = 0°$ (beam end-fire). The position of this scanning array is also shown in Fig. 8–48.

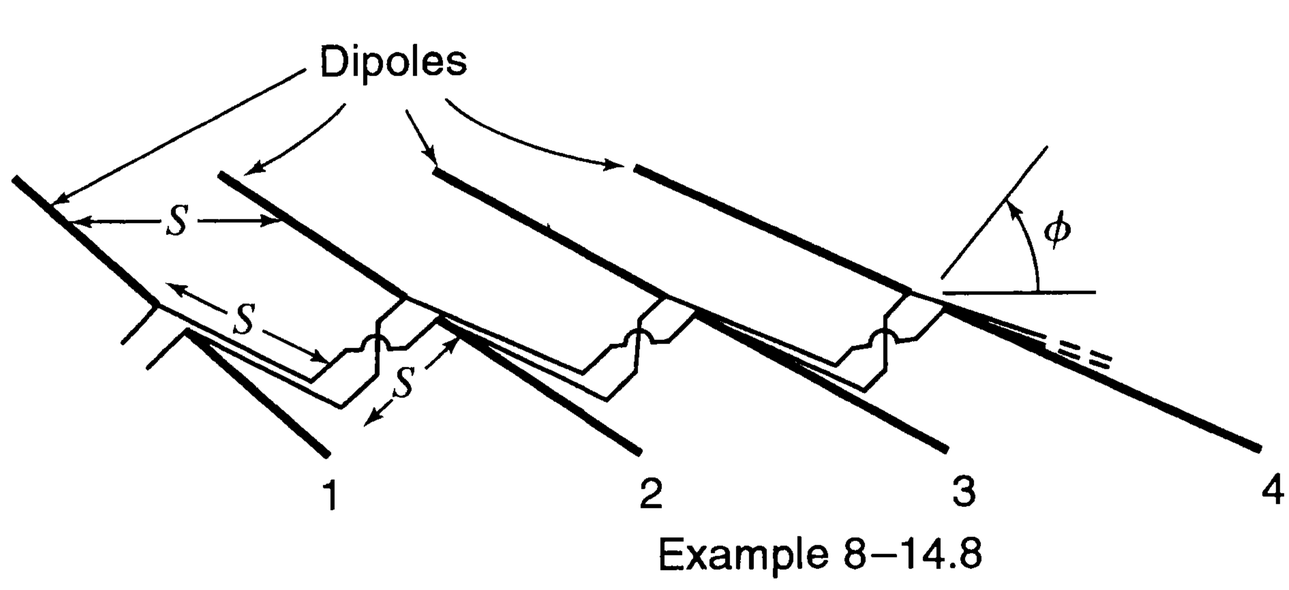


Figure 8–52
Scanning array of alternately reversed dipoles with $S = \lambda_0/4$ at center frequency, relative phase velocity $p = \frac{1}{2}$ (slow wave) and mode number $m = -\frac{1}{2}$. (Example 8–14.8.)

Now let us consider the monofilar helical antenna in comparison to the above examples of other periodic structure arrays with traveling waves.

EXAMPLE

8–14.9 Monofilar Axial-Mode Helical Antenna with Turn-Spacing $S = \lambda_0/4$ and Circumference $C = \lambda_0$ at Center Frequency (Fig. 8–53)

As discussed in Sec. 8–2, the monofilar axial-mode helical antenna operates in the increased-directivity condition resulting in a supergain end-fire beam ($\phi = 0°$). The mode number $m = -1$ and from (5) we have for $\phi = 0°$ that

$$-2\pi - \frac{\pi}{n} = \frac{2\pi}{\lambda_0} S - \frac{2\pi}{\lambda_0} \frac{S}{p} \tag{13}$$

where

n = number of turns

$p = (v/c) \sin\alpha$ = relative phase velocity of wave *in direction of helix axis* (not along the helical conductor as in Sec. 8–2)

α = pitch angle

Rearranging (13) we obtain

$$p = \frac{1}{1 + \dfrac{2n+1}{2n} \dfrac{1}{S/\lambda_0}} \tag{14}$$

For large n,

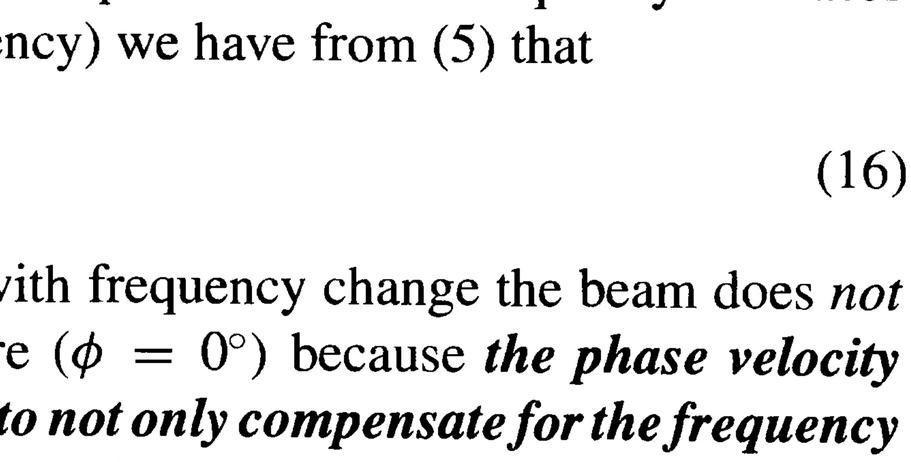
$$p \simeq \frac{1}{1 + [1/(S/\lambda_0)]} \tag{15}$$

Thus, end-fire ($\phi = 0°$) with $S = \lambda_0/4$ requires that $p = 0.20$. If the frequency decreases so that $S = \lambda_0/5$ (a 25 percent change in frequency) we have from (5) that

$$\cos\phi = \frac{1}{p} - \frac{1}{S/\lambda_0} = \frac{1}{0.20} - 5 = 0 \tag{16}$$

making $\phi = 90°$ (beam broadside). However, with frequency change the beam does *not* swing broadside but remains locked on end-fire ($\phi = 0°$) because ***the phase velocity changes automatically by just the right amount to not only compensate for the frequency change but also to provide increased directivity and supergain.*** (相速自动地改变到不仅补偿频变、而且恰能提供增强定向性或超增益的正确值。) This is one of the very remarkable properties of the monofilar axial-mode helical antenna.

The increased-directivity condition involves not only the turn spacing S but also the number of turns n. However, for large n the difference in p for the two end-fire conditions is small. Thus, for large n we find from (15) that p values range from about 0.25 for



Example 8–14.9

Figure 8–53
Monofilar axial-mode helical antenna which locks on end-fire ($\phi = 0°$) with increased directivity over more than an octave frequency range by an automatic shift in relative phase velocity p (see Fig. 8–48). The mode number $m = -1$.

$S = \lambda_0/3$ through 0.20 for $S = \lambda_0/4$ to 0.167 for $S = \lambda_0/5$, locking the beam on end-fire with supergain over a frequency range of about 2 to 1, which is in marked contrast to the beam-swinging of the scanning arrays discussed above. The position of the monofilar helical antenna over a 2 to 1 frequency range is shown in Fig. 8–48 for $n = 4, 8$ and 16 turns. For very large n, the position moves along the end-fire line which intersects $S/\lambda = 1$ on the $S/\lambda_0 = 0$ axis. We note that while other arrays move along constant p lines as the frequency changes, the *monofilar axial-mode helical antenna moves along a constant beam-angle (end-fire) line, cutting across lines of constant p value.* [而单绕轴向模螺旋天线则是按相同波束指向角（端射）线移动，从而切割各等 p 值线。]

Another remarkable property and great advantage of the monofilar helical antenna is that the input impedance is an almost constant resistance over an octave bandwidth, the resistance being easily set at any convenient value from 50 to 150 Ω. This is in contrast to the large impedance fluctuations of the above dipole arrays with change in frequency.

8–15 ARRAYS OF MONOFILAR AXIAL-MODE HELICAL ANTENNAS
（单绕轴向模螺旋天线阵）

With arrays of monofilar axial-mode helical antennas the designer must strike a balance between the number and length of helices needed to achieve a desired gain. The choice is between more lower-gain antennas and fewer higher-gain antennas appropriately spaced. As an illustration consider the following problem.

EXAMPLE 8–15.1

Design a circularly polarized antenna using one or more end-fire elements to produce a gain of 24 dB for operation at a given wavelength λ.

■ Solution

The highest end-fire gain is obtained with the increased-directivity condition which is automatic with monofilar axial-mode helical antennas. From (8–3–7) for $\alpha = 12.7°$ and $C_\lambda = 1.05$, the required length of a single helix is

$$L = nS = \frac{252}{12 \times 1.05^2} = 19\lambda \tag{1}$$

requiring an 80-turn helix (Fig. 8–54*a*).

A more compact configuration results if four 20-turn helices are used in a broadside array. Assuming uniform aperture distribution, the effective aperture of each helix is

$$A_e = \frac{D\lambda^2}{4\pi} = \frac{63\lambda^2}{4\pi} = 5.0\lambda^2 \tag{2}$$

Assuming a square aperture, the side length is 2.24λ $(= \sqrt{5.0})$. With each helix placed at the center of its aperture area, the spacing between helices is 2.24λ (Fig. 8–54*b*).

A third configuration results if nine 9-turn helices are used in a broadside array (Fig. 8–54*c*).

A fourth possible configuration results if sixteen 5-turn helices are used in a broadside array with a spacing of 1.12 λ between helices (Fig. 8–54*d*).

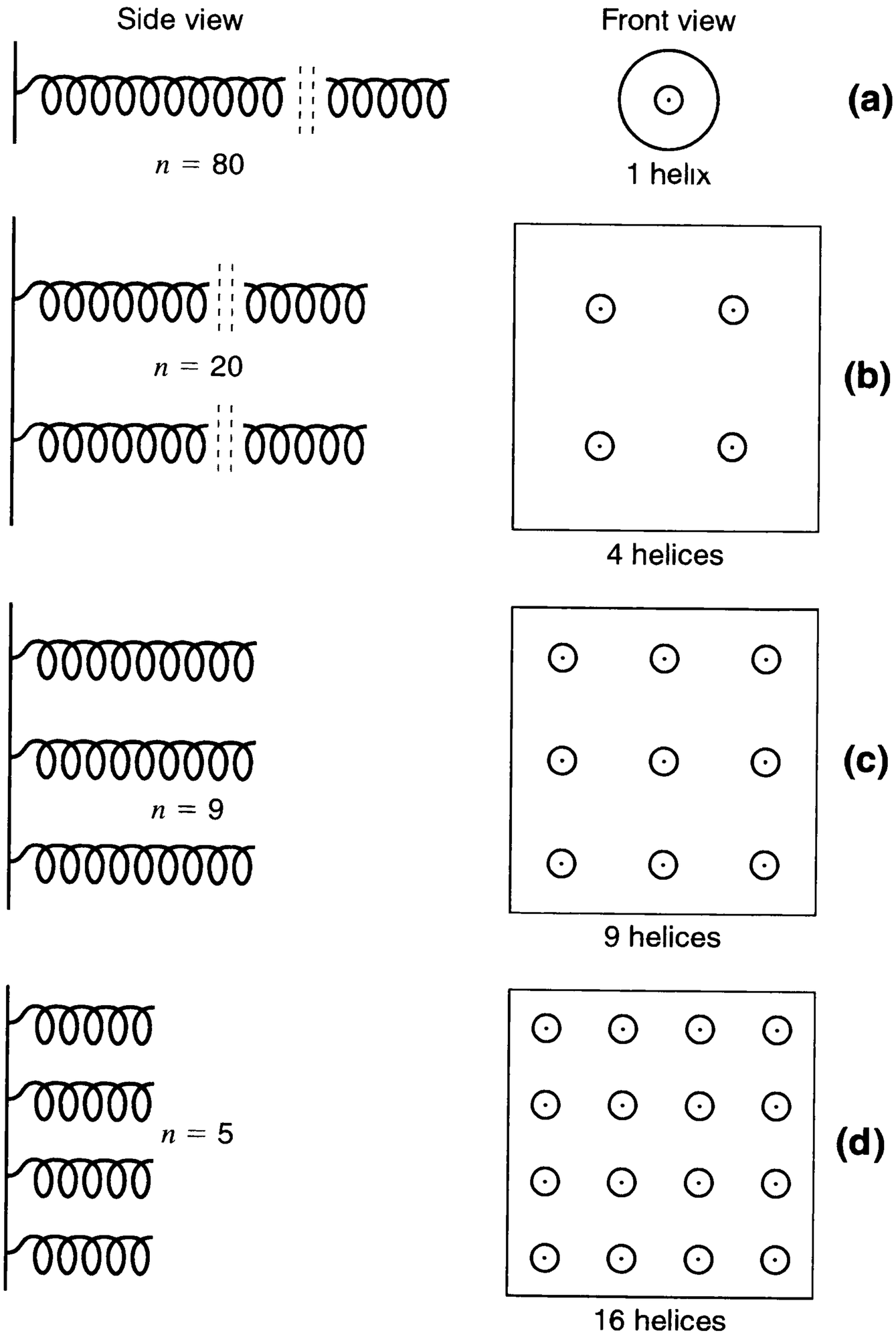


Figure 8–54
Single monofilar axial-mode helix with 80 turns (*a*) compared with an array of four 20-turn helices (*b*), an array of nine 9-turn helices (*c*) and sixteen 5-turn helices (*d*) for worked example. All have 24 dB gain. Note also that the product of the number of helices and number of turns for each array equals 80 (±1). See Sec. 5–17.

Which of the above configurations should be used? The decision will depend on considerations of support structure and feed connections. The single helix has a single feed point and a small ground plane but is very long. The other configurations have larger ground planes but are more compact. The 4- and 16-helix configurations have the advantage that the helices can be fed by a symmetrical corporate structure. Also the 16-helix configuration can be operated as a phased array.

With the multiple-helix arrays the mutual impedance of adjacent helices is a consideration. Figure 8–55 shows the resistive and reactive components of the mutual impedance of a pair of same-handed 8-turn monofilar axial-mode helical antennas as a function of the separation distance in wavelengths ($C_\lambda = 1, \alpha = 12°$) as measured by Blasi (1). At spacings of a wavelength or more, as is typical in helix arrays, the mutual impedance is only

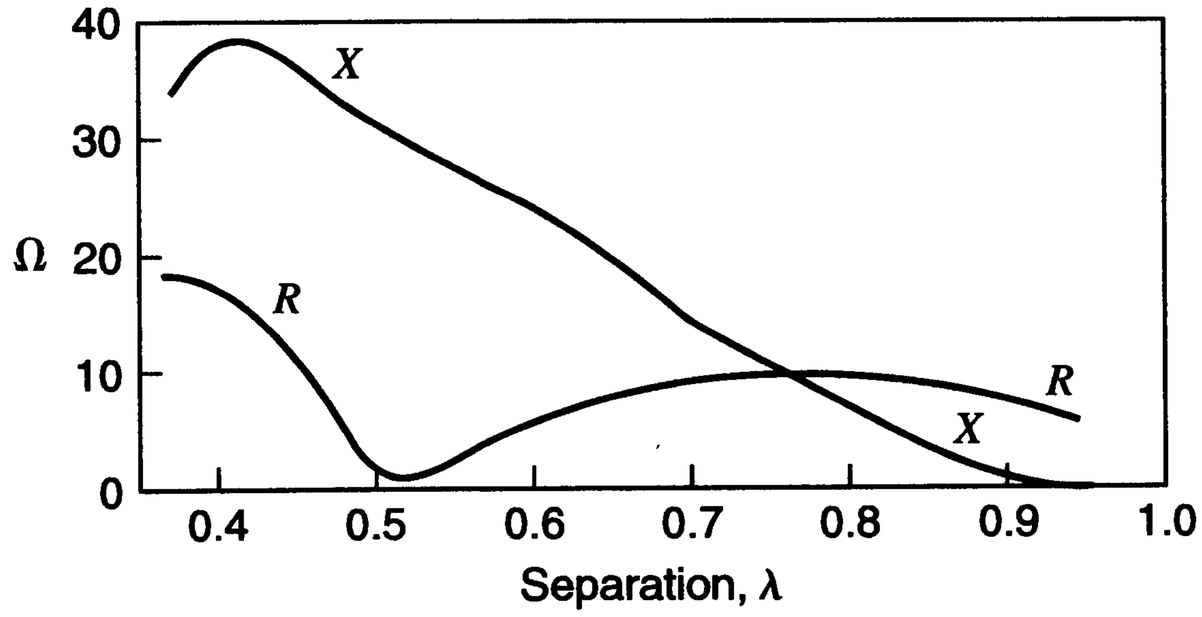


Figure 8–55
Resistive (R) and reactive (X) components of the mutual impedance of a pair of same-handed 8-turn monofilar axial-mode helical antennas of 12° pitch angle as a function of the separation distance (center-to-center) in wavelengths. Helix circumference $C_\lambda = 1$. Conductor diameter is 0.016 λ. Self-impedance $Z_{11} \simeq 140 + j0\ \Omega$. (*After E. A. Blasi-1.*)

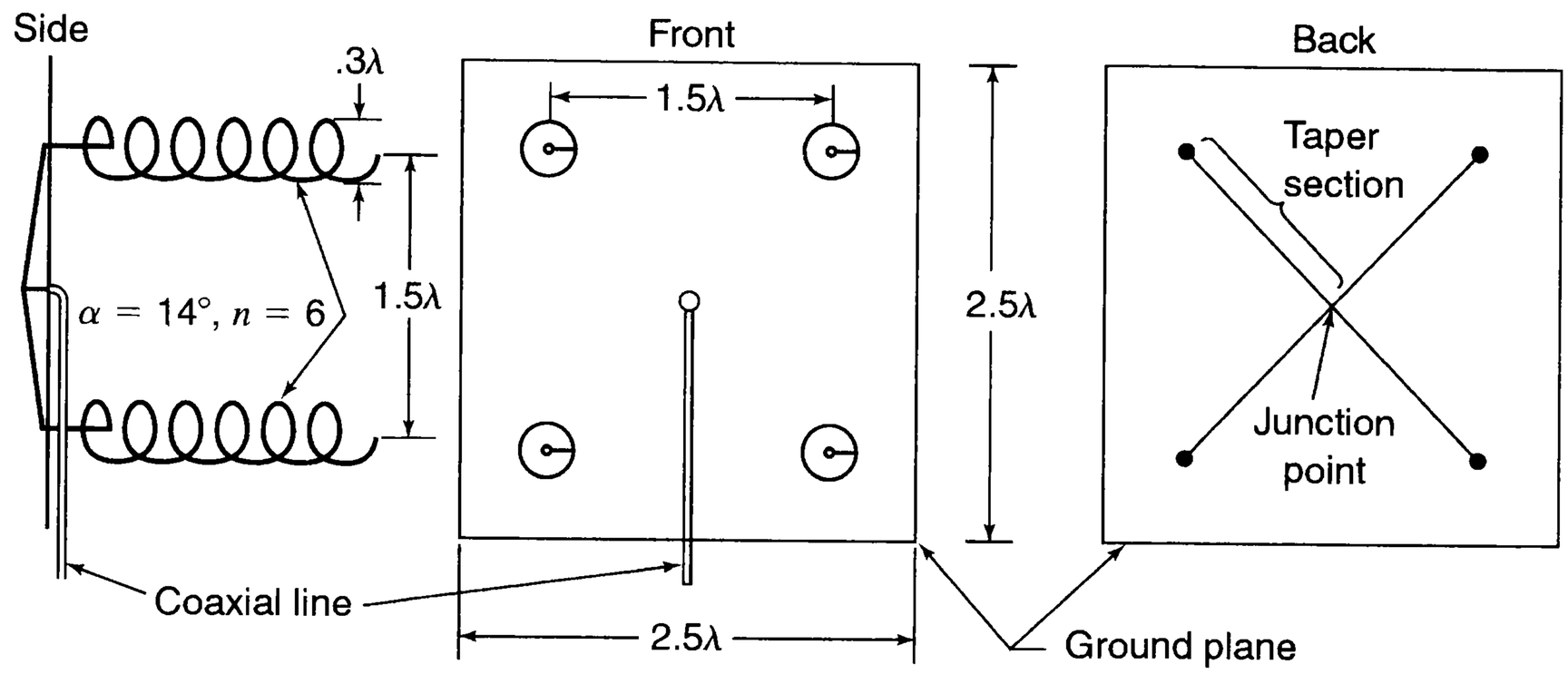
a few percent or less of the helix self-impedance (140 Ω resistive). Thus, in designing the feed connections for a helix array the effect of mutual impedance can often be neglected without significant consequences. As examples, let us consider the feed systems for two helix arrays, one with 4 helices and the other with 96 helices.

Array of Four Monofilar Axial-Mode Helical Antennas

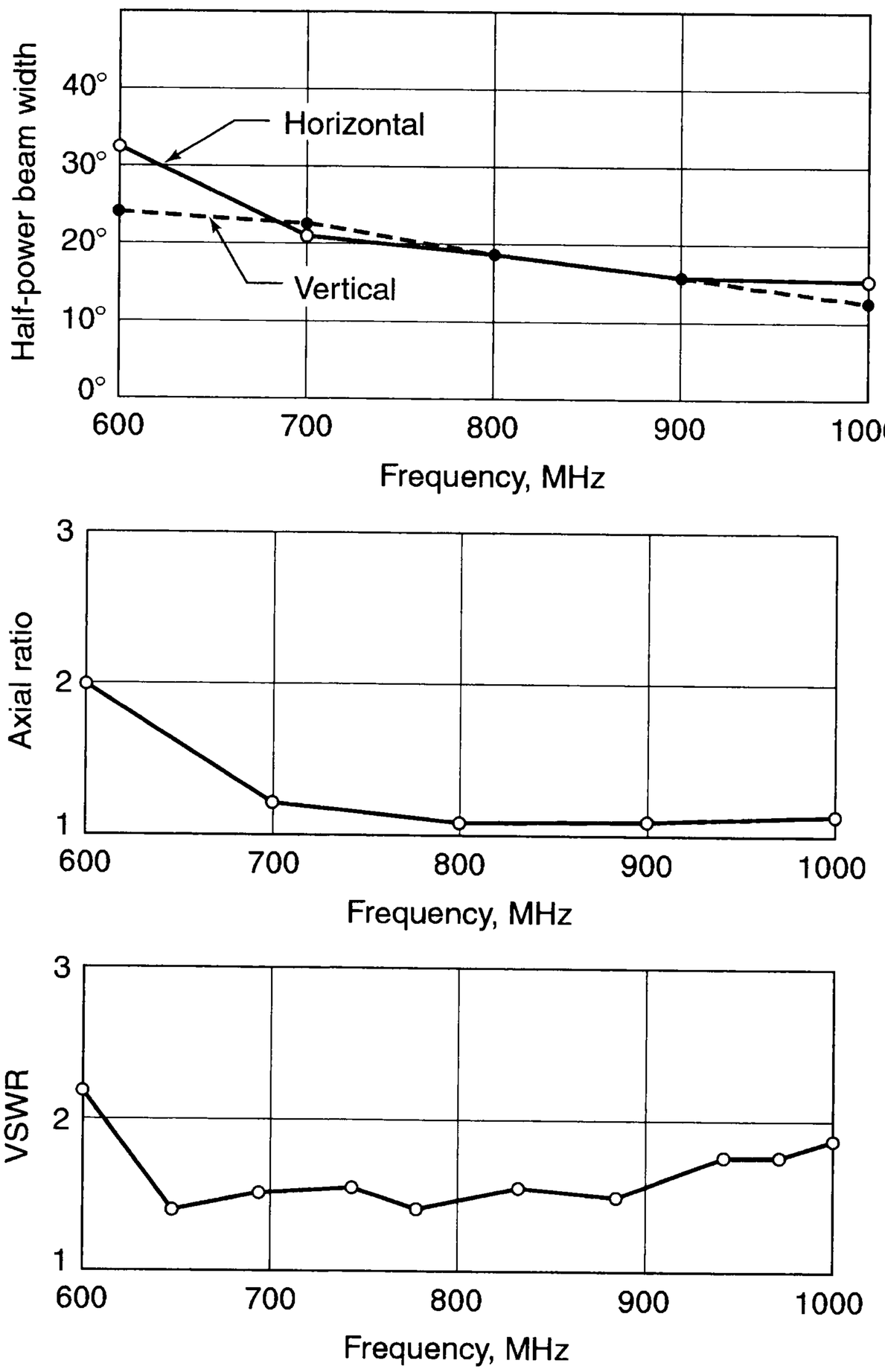
This array, shown in Fig. 8–56*a*, built by John Kraus in 1947, has four 6-turn, 14° pitch angle helices mounted with 1.5 λ spacing on a 2.5 × 2.5 λ square ground plane (Kraus-3). The helices are fed axially through an insulated fitting in the ground plane. Each feed point is connected on the back side of the ground plane to a junction point at the center of the ground plane. Each conductor acts as a single-wire versus ground-plane transmission line with a spacing between wire and ground plane which tapers gradually so that the approximately 140 Ω at each helix is transformed to 200 Ω at the junction. The four 200-Ω lines in parallel yield 50 Ω at the junction which is fed through an insulated connector to a 50-Ω coaxial fitting on the front (helix) side of the ground plane. Since the taper sections are about 1 λ long, the arrangement provides a low VSWR on the 50-Ω line connected to the junction over a wide bandwidth. Beamwidth, axial ratio and VSWR performance of the array are presented in Fig. 8–56*b*. The array gain at the center frequency (800 MHz) is about 18.5 dB and at 1000 MHz about 21.5 dB. Compare with the quad-helix array of Fig. 8–54.

Array of 96 Monofilar Axial-Mode Helical Antennas

This array, shown in Fig. 8–4, which John Kraus (8) designed and constructed in 1951, has 96 11-turn 12.5° pitch angle helices mounted on a tiltable flat ground plane 40 λ long by 5 λ wide for operation at a center frequency of 250 MHz. Each helix is fed by a 150-Ω coaxial cable. Equal-length cables from each helix of one bay or group of 12 helices are connected in parallel to one end of a 2 λ long tapered transition section which transforms the 12.5-Ω (= 150/12) resistive impedance to 50 Ω. Equal-length 50-Ω coaxial cables then connect the transition section of each bay to a central location resulting in a uniform in-phase aperture distribution with low VSWR over a wide bandwidth. The array produces a gain of about 35 dB at the center frequency of 250 MHz (λ = 1.2 m) and increased gain at higher frequencies. At 250 MHz the beam is fan-shaped with half-power beamwidths of 1 by 8°.



(a)



(b)

Figure 8–56
(*a*) Constructional details for broadside array of four 6-turn, 14° monofilar axial-mode helical antennas. Dimensions are in wavelengths at the center frequency. (*b*) Measured performance of 4-helix array of (*a*) showing beamwidths, axial ratio and VSWR on a 53-Ω line as a function of frequency. (*After Kraus-3.*)

8–16 THE MONOFILAR AXIAL-MODE HELIX AS A PARASITIC ELEMENT AND POLARIZER (SEE FIG. 8–57)（单绕轴向模螺旋用做寄生单元和极化器）

Helix-helix (Fig. 8–57*a*). If the conductor of a 6-turn monofilar axial-mode helical antenna is cut at the end of the second turn, the antenna continues to operate with the first 2 turns launching the wave and the remaining 4 turns acting as a parasitic director.

Polyrod-helix (Fig. 8–57*b*). By slipping a parasitic helix of several turns over a linearly polarized polyrod antenna, it becomes a circularly polarized antenna.

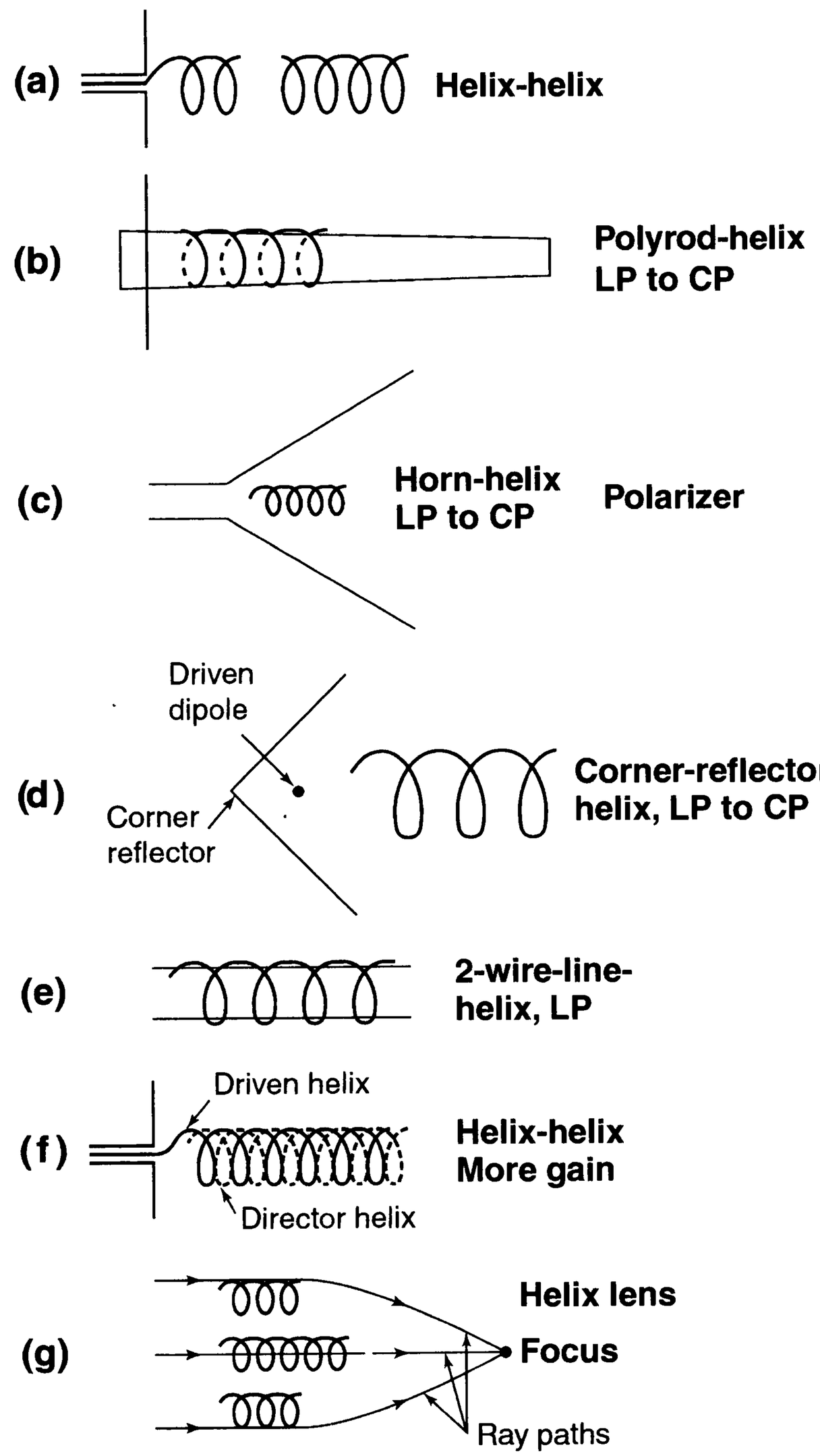


Figure 8–57
The monofilar axial-mode helical antenna in seven applications as a polarizer parasitic element.

Horn-helix (Fig. 8–57*c*). By placing a parasitic helix of several turns in the throat of a linearly polarized pyramidal horn antenna without touching the horn walls, the horn radiation becomes circularly polarized.

Corner-helix (Fig. 8–57*d*). A parasitic helix in front of a corner-reflector antenna results in a circularly polarized antenna.

The 2-wire-line-helix (Fig. 8–57*e*). If a parasitic helix of many turns is slipped over a 2-wire transmission line without touching it (helix diameter slightly greater than line spacing), it is reported that the combination becomes a linearly polarized end-fire antenna with **E** parallel to the plane of the 2-wire line [Broussaud-1].

Helix-helix (Fig. 8–57*f*). If a parasitic helix is wound between the turns of a driven monofilar axial-mode helical antenna without touching it (diameters the same), Nakano *et al.* report that the combination gives an increased gain of about 1 dB without an increase in the axial length of the antenna. The increased gain occurs for helices of any number of turns between 8 and 20. The parasitic helix may be regarded as a director for the driven helix [Nakano-1, 2].

Helix lens (Fig. 8–57*g*). A monofilar axial-mode helical antenna (or, for that matter, any end-fire antenna) acts as a lens. An array of parasitic helices of appropriate length arranged in a broadside configuration can operate as a large aperture-lens antenna.

8–17 THE MONOFILAR AXIAL-MODE HELICAL ANTENNA AS A PHASE AND FREQUENCY SHIFTER（单绕轴向模螺旋天线用做移相器和移频器）

The monofilar axial-mode helical antenna is a simple, beautiful device for changing phase or frequency. Thus, if the monofilar axial-mode helical antenna in Fig. 8–58 is transmitting at a frequency F, rotating the helix on its axis by 90° will advance the phase of the radiated wave by 90° (or retard it, depending on the direction of rotation). Rotating the helix continuously f times per second results in radiation at a frequency $F \pm f$ depending on the direction of rotation (see also Sec. 8–24).

As an application, consider the 3-helix lobe-sweeping antenna of Fig. 8–59. All helices are of the same hand. By rotating helix 1 clockwise and helix 3 counterclockwise with helix 2 at the center stationary, a continuously swept lobe is obtained as suggested in the figure. In operation a small lobe appears about 30° to the left, then grows in amplitude while sweeping to the right, reaching a maximum at 0° (at right angles to the array). Sweeping

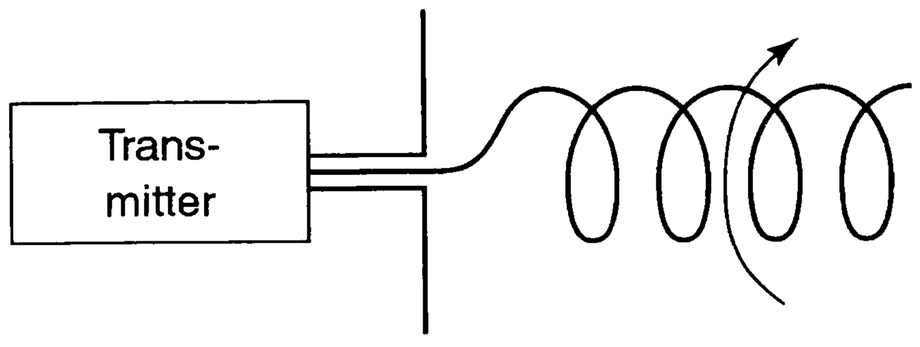
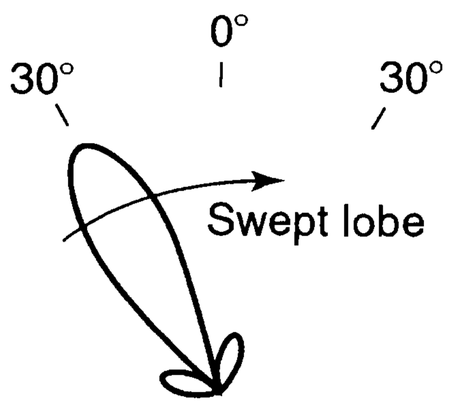


Figure 8–58
The monofilar axial-mode helical antenna as a phase-shifting device.



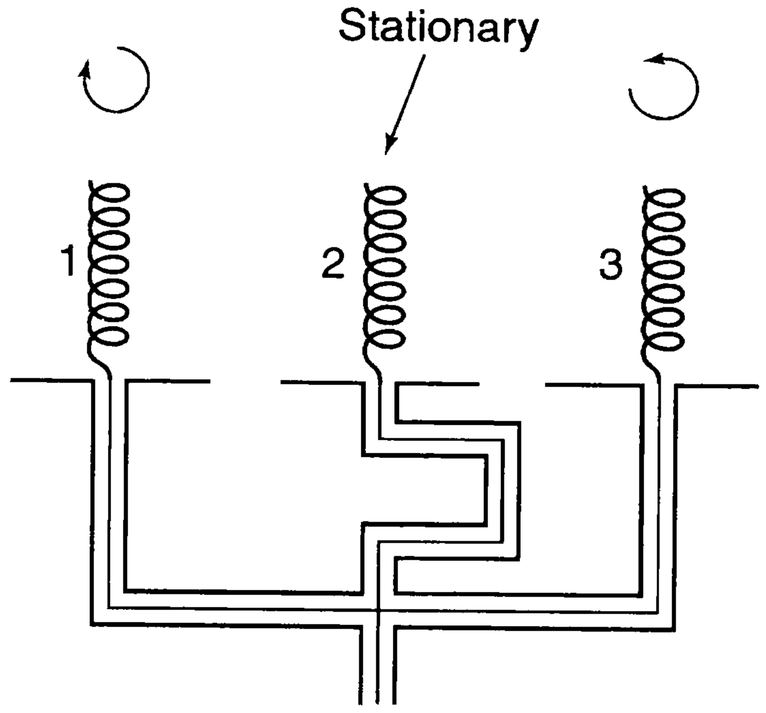


Figure 8–59
Array of 3 right-handed monofilar axial-mode helical antennas with outer 2 rotating in opposite directions to produce a continuously sweeping lobe.

further to the right, the lobe decreases to a small amplitude at an angle of about 30° and simultaneously a small lobe appears at 30° to the left and the process is repeated, giving a continuously sweeping lobe (left to right) which crosses the 0° direction *n* times per minute for a helix rotation speed of *n* revolutions per minute. By using more helices, the beamwidth of the swept lobe can be made arbitrarily small.

John Kraus (9) and Richard McFarland (2) operated this 3-helix lobe-sweeping array in 1957 for observing radio emissions from the planet Jupiter at frequencies of 25 to 35 MHz. Each helix had 3 turns and was 3 m in diameter. The sweeping lobe allowed one to identify celestial objects moving at sidereal or near-sidereal rates and distinguish them from terrestrial or near-earth objects. See Fig. 16–41 for a photograph of the helix array.

Another application of phase-shifting with a helix is discussed in the next section in connection with helices for linear polarization.

8–18 LINEAR POLARIZATION WITH MONOFILAR AXIAL-MODE HELICAL ANTENNAS（单绕轴向模螺旋天线形成线极化）

If two monofilar axial-mode helical antennas are mounted side by side and fed equal power, the radiation on axis will be linearly polarized provided the helices are of opposite hand but otherwise identical (Fig. 8–60*a*). With switch 1 right, switch 2 left and switch 3 up, polarization is linear. Rotating one of the helices on its axis 90° rotates the plane of linear polarization by 45°. Rotating one helix through 180° rotates the plane of linear polarization 90°. With switches 1 and 3 left, as in the figure, the polarization is LCP (left-handed circular polarization). With switches 2 and 3 right, the polarization is RCP (right-handed circular polarization). Thus, the two helices can provide either left or right circular polarization or any plane of linear polarization.

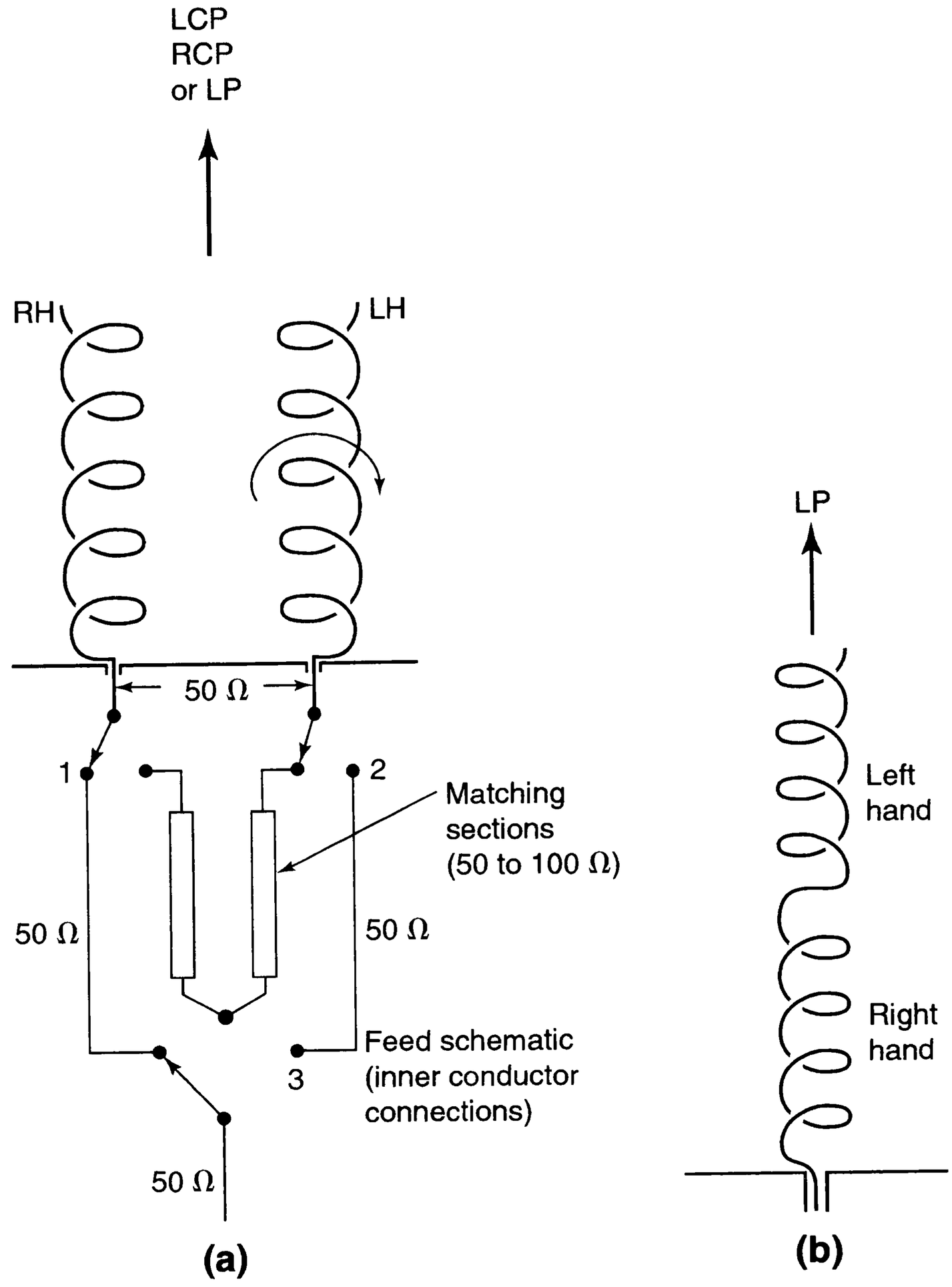


Figure 8–60
(*a*) Arrangement for producing left circular polarization (LCP), right circular polarization (RCP) or any plane of linear polarization (LP). (*b*) Two helices of opposite hand in series for producing linear polarization (LP).

Another method of obtaining linear polarization is to connect a left- and a right-handed helix in series as in Fig. 8–60*b*.

A third method has already been discussed in Fig. 8–57*e* (2-wire-line-helix).

Elliptical polarization approaching linear polarization can be obtained by flattening a helix so that its cross section is elliptical instead of circular.

8–19 MONOFILAR AXIAL-MODE HELICAL ANTENNAS AS FEEDS
(单绕轴向模螺旋天线用做馈源)

Figure 8–61 shows a driven helix feeding an array of crossed dipoles acting as directors for producing circular polarization. Although this arrangement has less gain and bandwidth than a full helix of the same length the crossed dipoles may be simpler to support than a long helix. The feed connections for the helix are also simpler than for a pair of crossed Yagi-Uda antennas, which require equal power with voltages in phase quadrature.

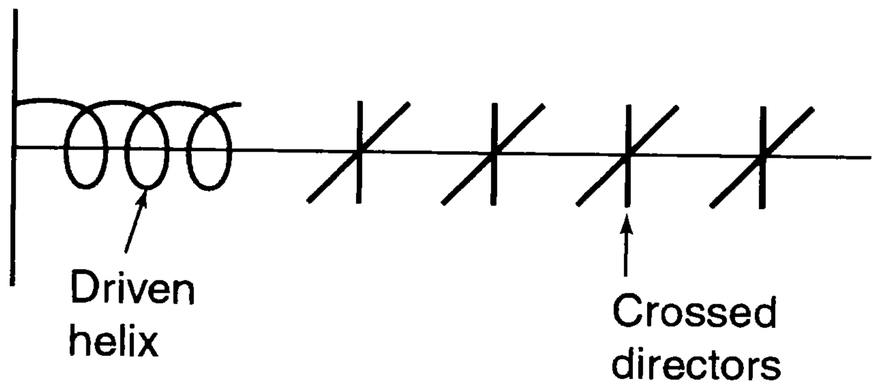


Figure 8–61
Monofilar axial-mode helical antenna as feed for an end-fire array of crossed dipoles.

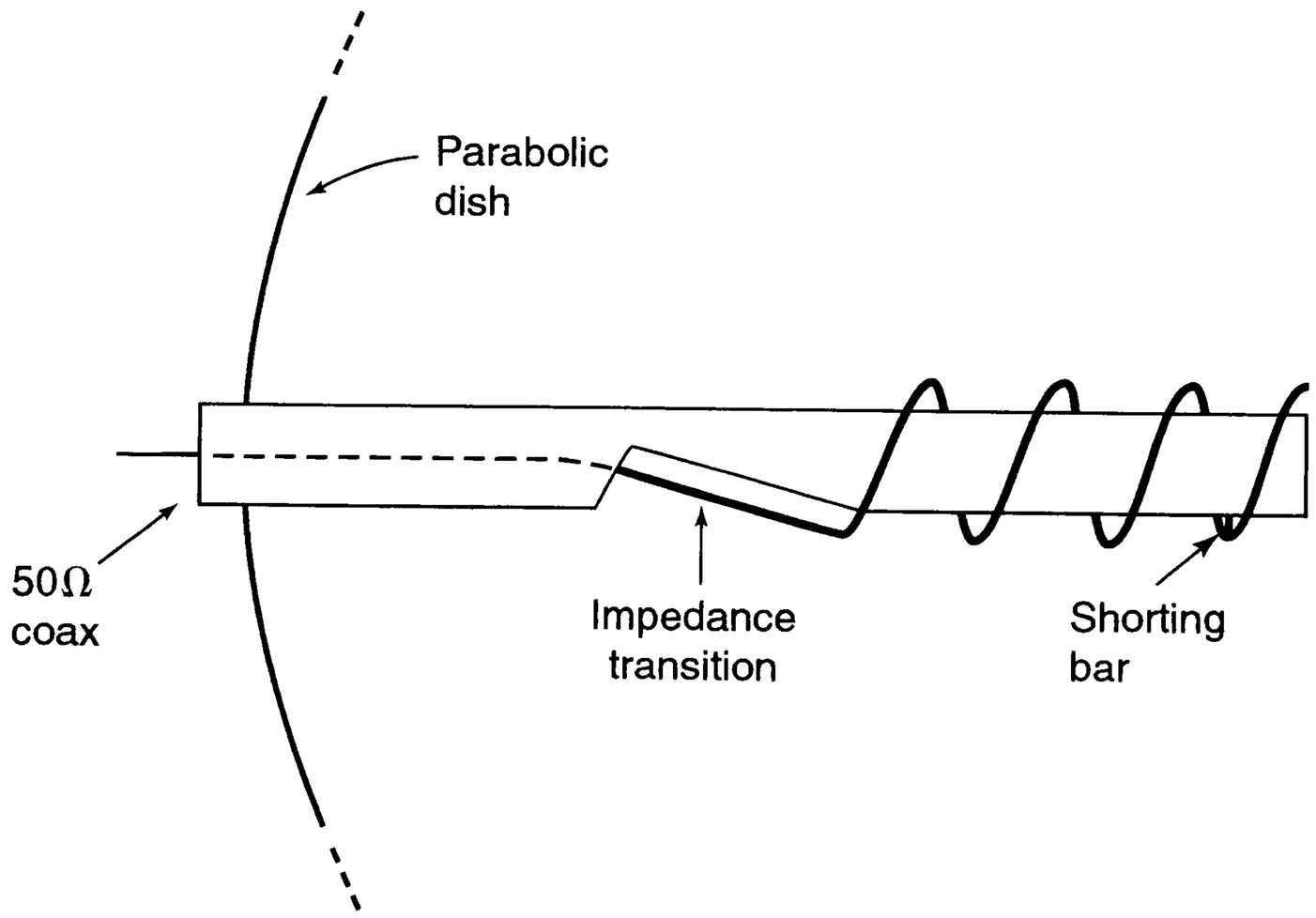


Figure 8–62
Short back-fire monofilar axial-mode helix as high-power (200 kW) circularly polarized feed for a parabolic dish antenna. (*After R. C. Johnson (1) and R. B. Cotton.*)

Helices are useful as feed elements for parabolic dish antennas. An example constructed by Johnson and Cotton (1) is shown in Fig. 8–62 in which a 3.5-turn monofilar axial-mode helix operates in the back-fire mode as a high-power unpressurized (200 kW) circularly polarized feed element for a parabolic dish reflector. Without a ground plane the helix naturally radiates in the backward axial direction. Another example of a back-fire helix feed is shown in Fig. 8–6 (King-1). A monofilar back-fire helical antenna was also constructed by Patton (1) for comparison with bifilar back-fire helical antennas.

Short monofilar axial-mode end-fire helices of a few turns with cupped ground plane are also useful as feeds for parabolic dish reflectors for producing sharp beams of circularly polarized radiation. Short conical helices (α constant, D and S increasing) are also useful because of their broad patterns for short focal-length dishes (see the helix in Fig. 8–65).

For dish feeds covering a frequency range greater than provided by a single helix, two or more helices can be mounted coaxially inside each other with phase centers coincident as shown in Fig. 8–63. This combination is superior to a log-periodic antenna as the feed since the phase center of a log-periodic antenna moves with frequency, resulting in defocusing of the parabolic reflector system.

Holland (1) has built a feed of this type with a larger helix for the L band and a smaller helix for the S band. The number of turns required for the helix feed antennas depends on the beamwidth desired. For the pattern to be 10 dB down at the edge of the parabolic dish reflector, the required number of turns is approximately given by

$$n \simeq \frac{8400}{\phi^2 S_\lambda} \tag{1}$$

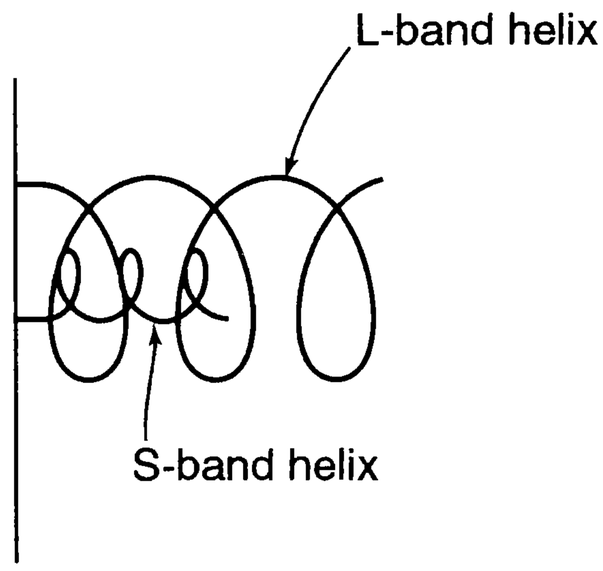


Figure 8–63
Coaxially mounted peripherally fed monofilar axial-mode helical antennas of same hand as parabolic dish feeds with same stationary phase centers for covering a 5 to 1 frequency range. (*After J. Holland-1.*)

where

ϕ = 10 dB beamwidth
S_λ = turn spacing in wavelengths

Thus, if $S_\lambda = 0.21$ ($\alpha = 12°$) and the required value of $\phi = 115°$ we have from (1) that $n = 3$.

To reduce mutual coupling of the helices, Holland placed the peripheral feed points of the two helices on opposite sides of the axis, as suggested in Fig. 8–63, obtaining 2-port fixed-phase-center operation over a 5 to 1 bandwidth.

8–20 TAPERED AND OTHER FORMS OF AXIAL-MODE HELICAL ANTENNAS

（渐变及其他形式的轴向模螺旋天线）

In this section a number of variants of the uniform (constant diameter, constant pitch) monofilar axial-mode helical antenna are discussed. Some of these forms are shown in Figs. 8–64, 8–65 and 8–66 which are reproduced here without changes from the first edition of *Antennas* (1950). In Fig. 8–64 we recognize in (*a*) a uniform helix with ground plane and in (*c*) a uniform helix without ground plane (the same configuration as in Fig. 8–62 for a back-fire feed). The double winding in (*i*) is similar to the one of Nakano (1) (Fig. 8–57*f*) except that both windings in (*i*) are driven while in Nakano one is parasitic.

Figure 8–65 shows 9 forms of tapered monofilar axial-mode helical antennas grouped into 3 classes: (1) pitch angle α constant but turn spacing S and diameter D variable, (2) diameter D constant but pitch angle α and turn spacing S variable and (3) turn spacing S constant but pitch angle α and diameter D variable. Many of these forms have been investigated—the class (2) form by Day (1).

Day measured patterns of monofilar axial-mode helical antennas of 6 turns with the diameter constant but α increasing or decreasing (D constant but α and S variable as in Fig. 8–65*d*, *e* and *f* or the middle row of Fig. 8–65). The helix conductor diameter was 0.02λ. Pitch angles were varied on a given helix from 1 to 20°, 5 to 17° or 9 to 15°, both increasing and decreasing. These were compared with a constant pitch angle of 12.5° at helix circumferences C_λ of 0.6, 0.8, 1.0, 1.2 and 1.4—a total of 35 cases. For pitch angle tapers between 5 and 17° and $0.8 \leq C_\lambda \leq 1.2$, the pattern variations are minor. However, at $C_\lambda = 1.2$ and with the pitch angle decreasing from 17° at the feed end to 5° at the open end, the gain is 1 dB more than for $C_\lambda = 1.2$ and $\alpha = 12.5°$ (constant pitch). This is a significant improvement since the highest gain for a uniform 6-turn helix occurs when α is

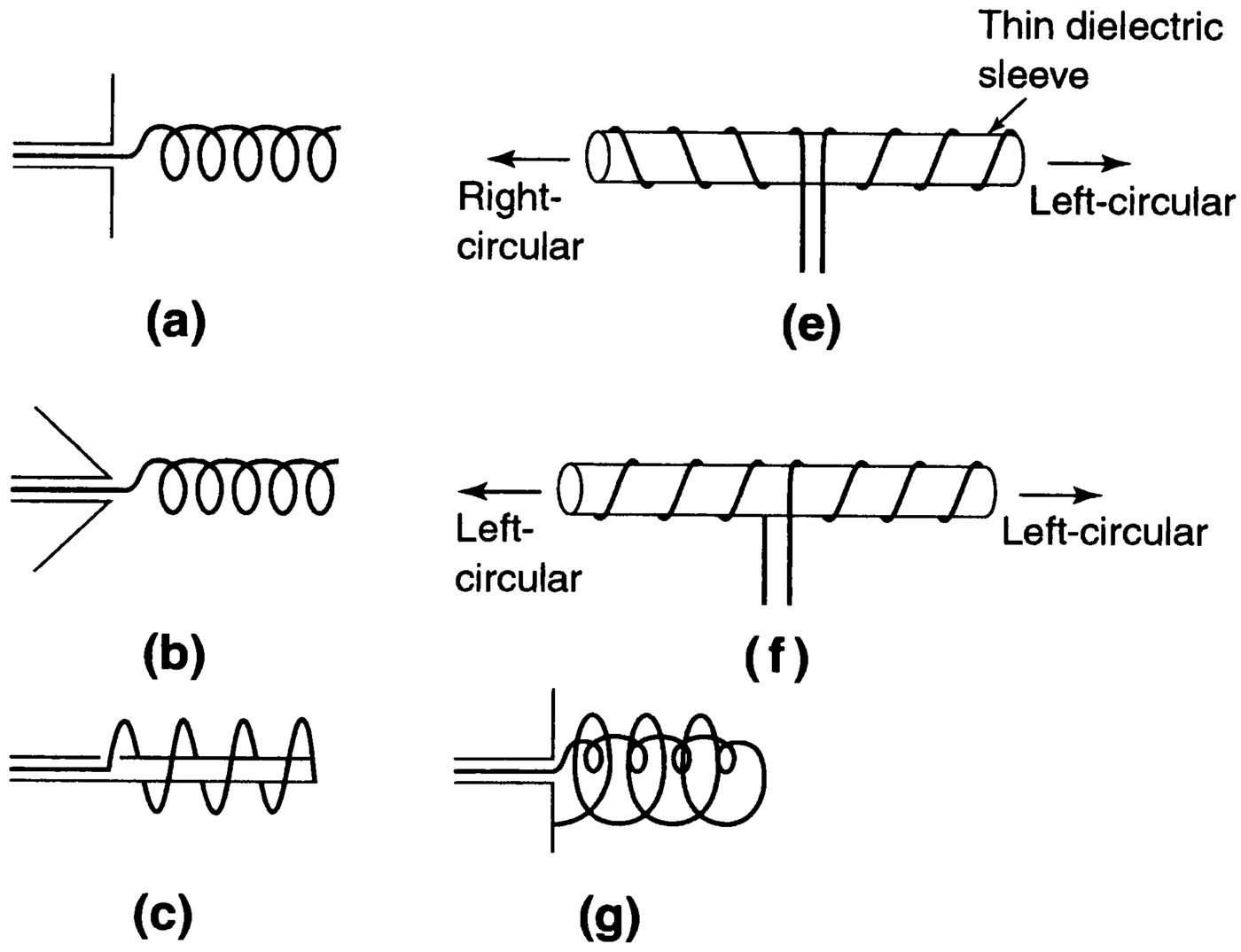


Figure 8–64
Axial-mode helices showing various constructional and feed arrangements. (*After Kraus.*)

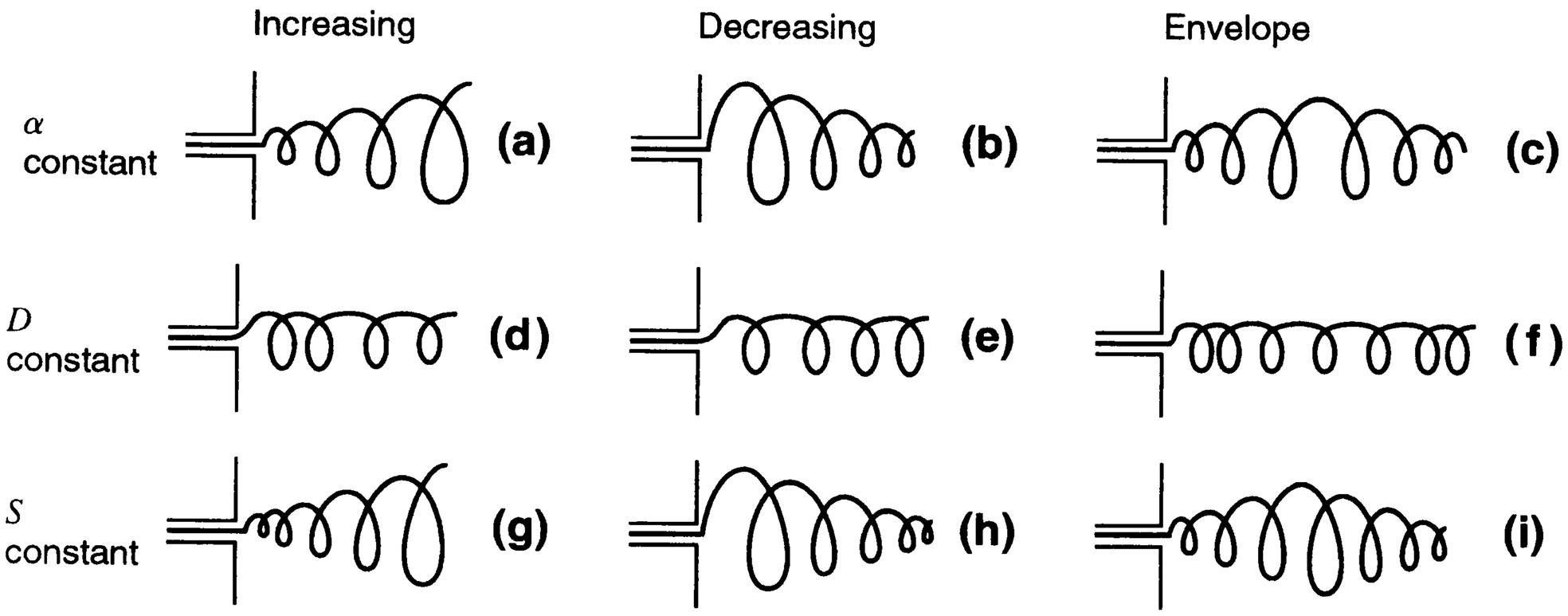


Figure 8–65
Types of tapered monofilar axial-mode helical antennas. (*After Kraus-1.*)

approximately 12.5° and the circumference C_λ approximately 1.2. Thus, the center helix (D constant, α and S decreasing), of the 9 shown in Fig. 8–65 (i.e., Fig. 8–65*e*), is a useful variant of the uniform helix.

The conical helix in Fig. 8–65*a* for which α is constant and D (or C) and S are increasing has been investigated by Chatterjee (1), Nakano (3), Mikawa and Yamauchi and others. With small pitch angles Chatterjee found that very broad patterns can be obtained

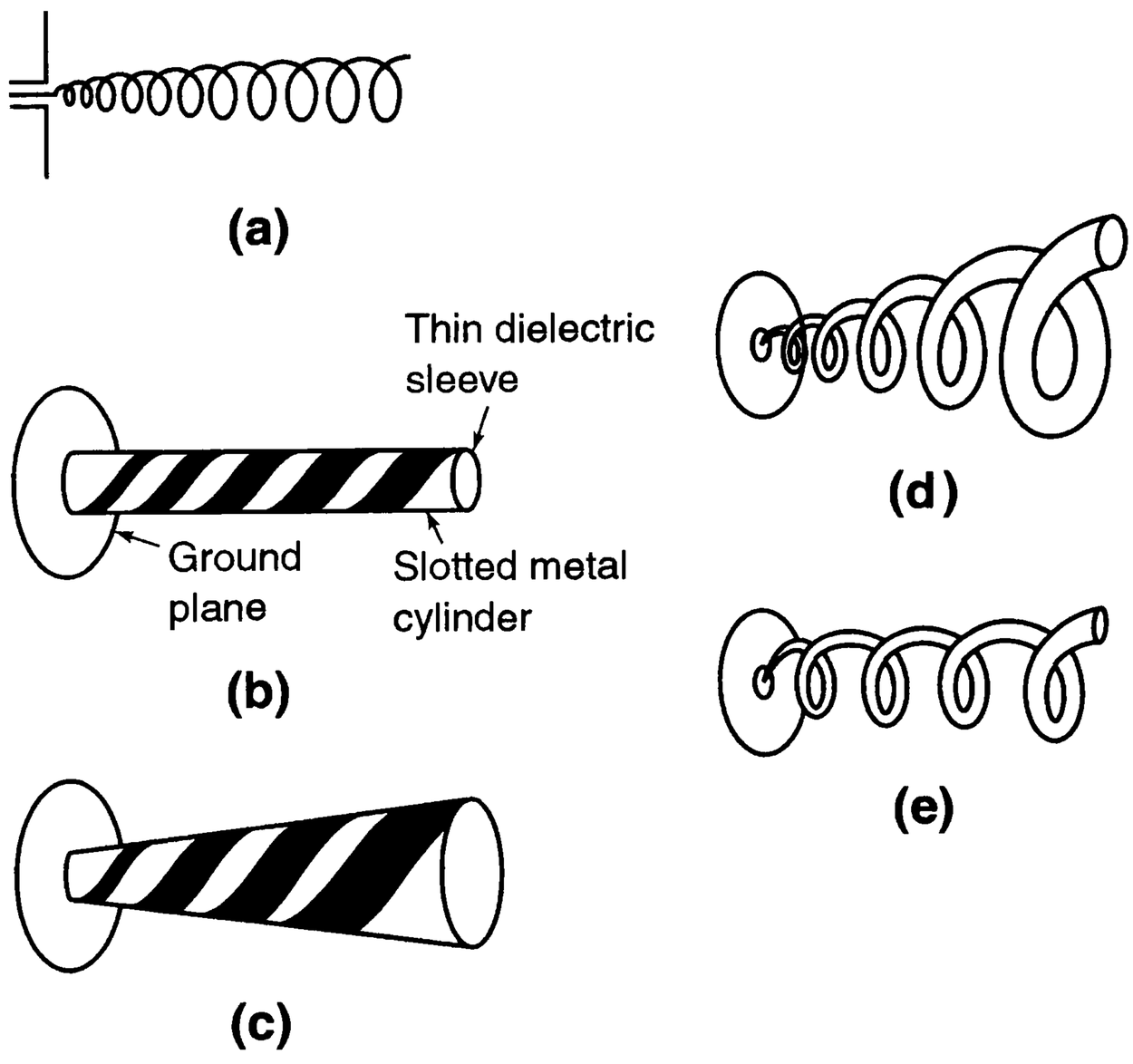


Figure 8–66
Types of tapered monofilar axial-mode helical antennas including ones in which conductor size is tapered. (*After Kraus.*)

over a 5 to 1 bandwidth. According to Nakano (3), Mikama and Yamauchi, the currents involved are those of the attenuating wave-launching region close to the feed point (see Fig. 8–3*b* and *c*).

Additional tapered types are shown in Fig. 8–66. The one at (*a*) has a tapered and uniform section but reversed in order from the uniform-to-taper type of Wong (1) and King and others. The other designs shown in Fig. 8–66 involve variation in the diameter of the helical conductor d or the width w of a flat strip conductor. Thus, there are 4 quantities which can be varied, α, D, S and d (or w). Since the characteristics of the monofilar axial-mode helical antenna are relatively insensitive to moderate changes in dimensions, the effect of moderate departures from uniformity is, in general, not large. However, some changes may produce significant increases in gain, as discussed above, and significant decreases in axial ratio and VSWR.

8–21 MULTIFILAR AXIAL-MODE HELICAL ANTENNAS（多绕轴向模螺旋天线）

Four wires, each $\lambda/2$ long and forming $\frac{1}{2}$-turn of a helix as in Fig. 8–67, produce a cardioid-shaped back-fire circularly polarized pattern (HPBW $\simeq 120°$) when the two pairs are fed in phase quadrature. This coil described by Kilgus (1) has two $\frac{1}{2}$-turn bifilar helices or one $\frac{1}{2}$-turn quadrifilar helix. The antenna is resonant and the bandwidth is narrow (about 4 percent). The 4 wires can also be $\lambda/4$ or λ long. For these lengths the lower ends are open-circuited instead of short-circuited as for the $\lambda/2$ wires of Fig. 8–67. Each bifilar helix can be balun-fed at the top from a coaxial line extending to the top along the central axis. By increasing the number of turns Kilgus (2) reports that shaped-conical patterns can be obtained which may be more useful for some applications than a cardioid (heart-shaped) pattern.

A bifilar helix end-fed by a balanced 2-wire transmission line produces a back-fire beam when operated above the cutoff frequency of the principal mode of the helical waveguide.

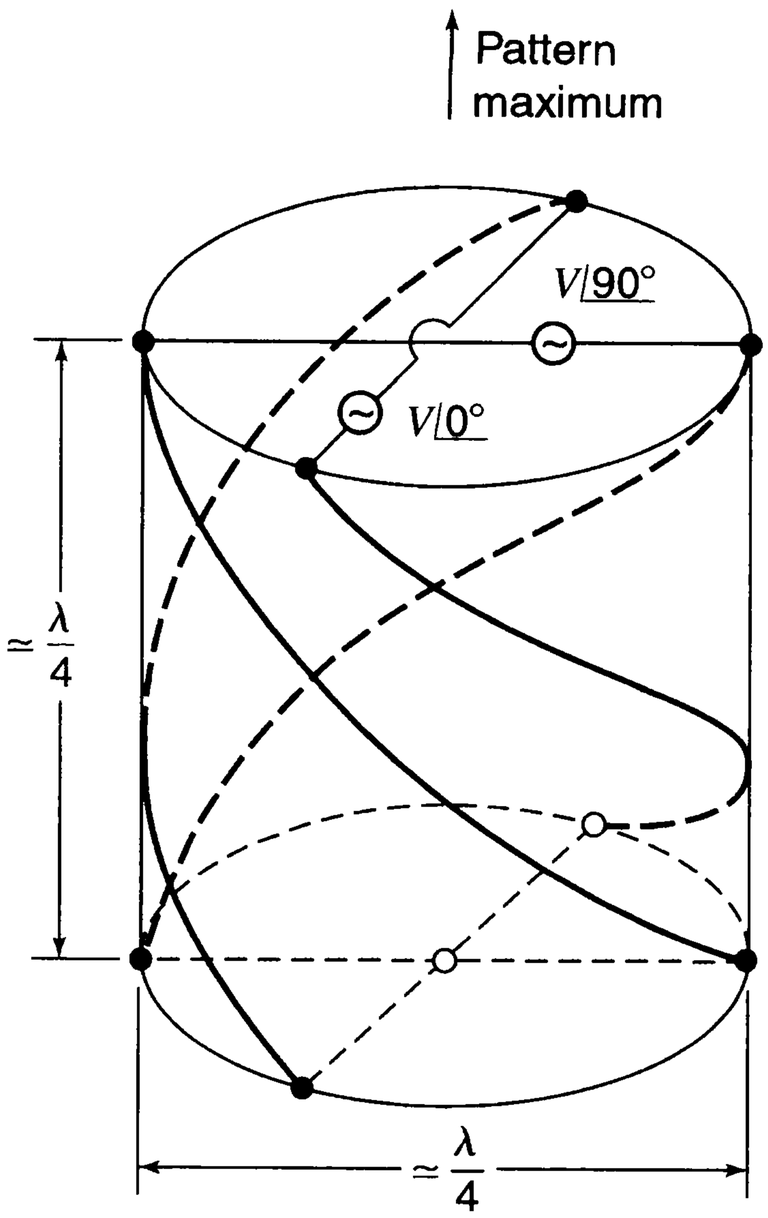


Figure 8–67
Resonant narrowband back-fire quadrifilar coil for very broad circularly polarized pattern. (*After C. C. Kilgus-1.*) Wires are situated in space as though wrapped around a cylinder as suggested in the figure.

The maximum directivity of this coil described by Patton (1) occurs slightly above the cutoff frequency. The pattern broadens with increasing frequency and at pitch angles of about 45° the back-fire beam splits and scans toward side-fire.

Below the cutoff frequency, there is a standing-wave current distribution along the helical conductor. Above cutoff, the standing wave gives way to a gradually decaying traveling wave. With a further increase in frequency the rate of decay increases and a low-level standing wave appears, indicating the existence of a higher-order helical waveguide mode. This establishes the upper frequency limit of the bifilar helix back-fire radiation.

Quadrifilar and octofilar axial-mode forward end-fire circularly polarized helical antennas using large pitch angles (30 to 60°) have been investigated by Gerst (1) and Worden and Adams et al. (1).

8–22 MONOFILAR AND MULTIFILAR NORMAL-MODE HELICAL ANTENNAS
（单绕和多绕法向模螺旋天线）

The previous sections deal with *axial-mode* helical antennas with maximum radiation in the direction of the helix axis. The radiation may be (forward) end-fire or back-fire. In this section the *normal mode* of radiation is discussed, *normal* being used in the sense of *perpendicular to* or at *right angles to* the helix axis. This radiation with its maximum normal to the helix axis may also be described as *side-fire* or *broadside*.

When the helix circumference is approximately a wavelength the axial mode of radiation is dominant, but when the circumference is much smaller the normal mode is dominant. Figure 8–68*a* and *c* shows helices radiating in both modes while Fig. 8–68*b* shows a 4-lobed mode helix of Chireix with the relative sizes for producing the modes being indicated.

Now let us examine the requirements for normal-mode radiation in more detail. Consider a helix oriented with axis coincident with the polar or z axis as in Fig. 8–69*a*. *If*

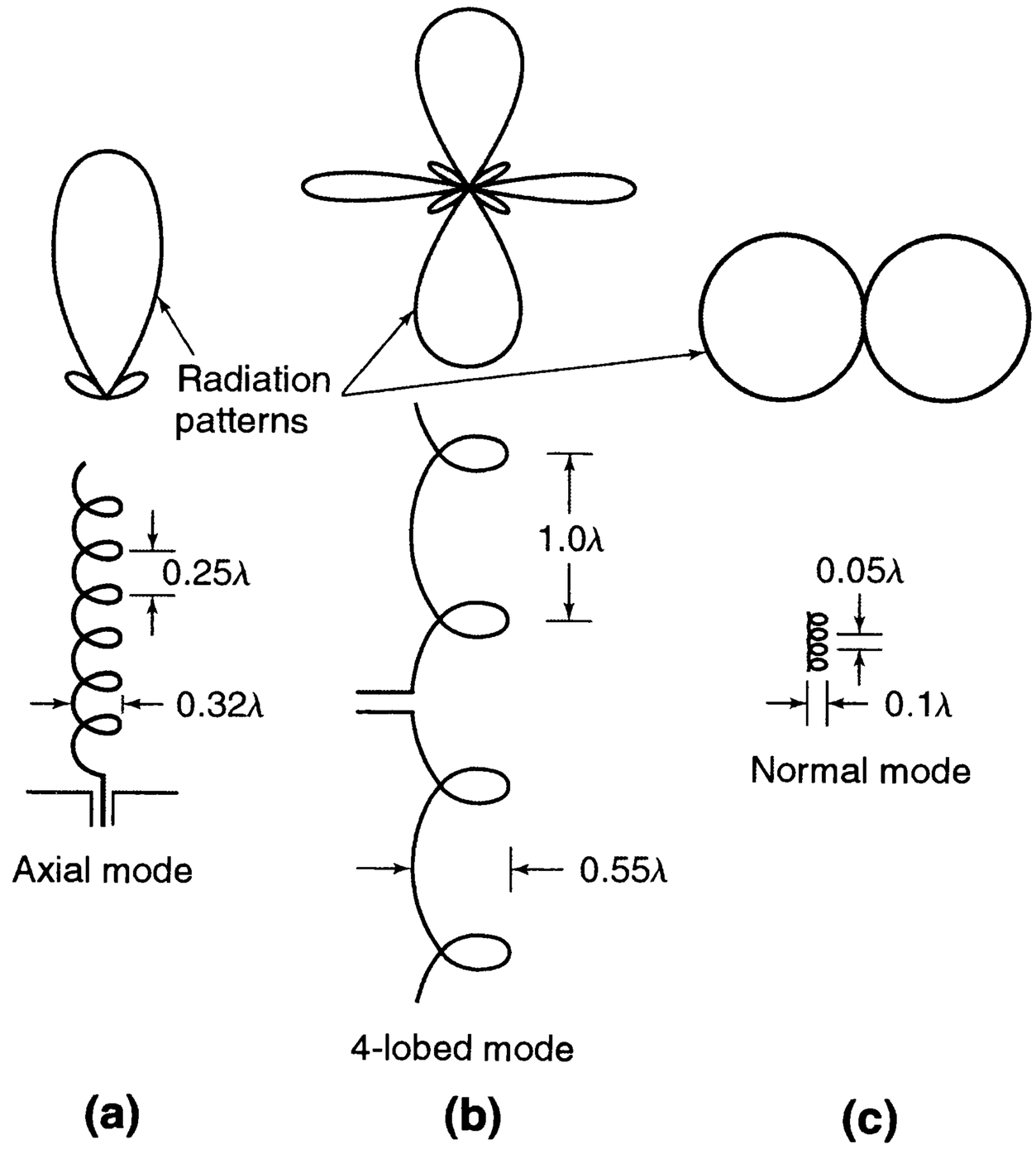


Figure 8–68
Field patterns of axial, 4-lobed and normal radiation modes of helical antennas with relative size indicated.

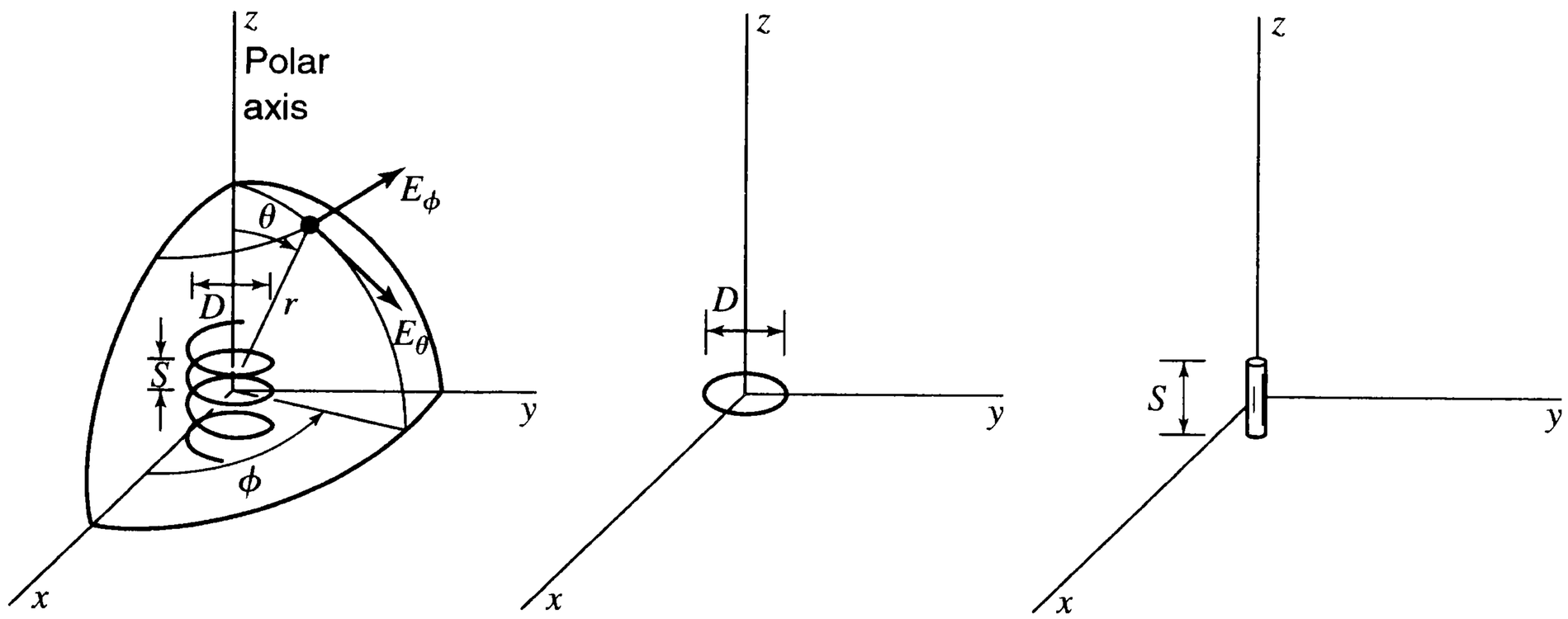


Figure 8–69
Dimensions for helix, loop and dipole.

the dimensions are small ($nL \ll \lambda$), the maximum radiation is in the xy plane for a helix oriented as in Fig. 8–69*a*, with zero field in the z direction.

When the pitch angle is zero, the helix becomes a loop as in Fig. 8–69*b*. When the pitch angle is 90°, the helix straightens out into a linear antenna as in Fig. 8–69*c*, the loop and straight antenna being limiting cases of the helix.

The far field of the helix may be described by two components of the electric field, E_ϕ and E_θ, as shown in Fig. 8–69*a*. Let us now develop expressions for the far-field patterns of

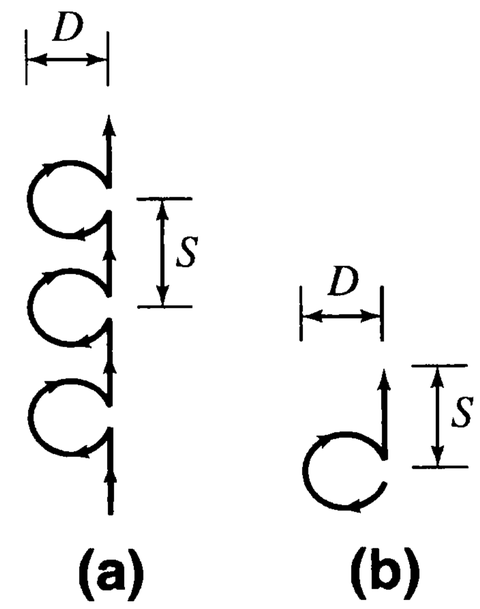


Figure 8–70
Modified helix for normal-mode calculations.

these components for a small short helix. The development is facilitated by assuming that the helix consists of a number of small loops and short dipoles connected in series as in Fig. 8–70*a*. The diameter D of the loops is the same as the helix diameter, and the length of the dipoles S is the same as the spacing between turns of the helix. Provided that the helix is small, the modified form of Fig. 8–70*a* is equivalent to the true helix of Fig. 8–69*a*. The current is assumed to be uniform in magnitude and in phase over the entire length of the helix. Since the helix is small, the far-field pattern is independent of the number of turns. Hence, it suffices to calculate the far-field patterns of a single small loop and one short dipole as illustrated in Fig. 8–70*b*.

The far field of the small loop has only an E_ϕ component. Its value is given in Table 7–1 as

$$E_\phi = \frac{120\pi^2[I]\sin\theta}{r}\frac{A}{\lambda^2} \tag{1}$$

where the area of the loop $A = \pi D^2/4$

The far field of the short dipole has only an E_θ component. Its value is given in the same table as

$$E_\theta = j\frac{60\pi[I]\sin\theta}{r}\frac{S}{\lambda} \tag{2}$$

where S has been substituted for L as the length of the dipole.

Comparing (1) and (2), the j operator in (2) and its absence in (1) indicates that E_ϕ and E_θ are in phase quadrature. The ratio of the magnitudes of (1) and (2) then gives the axial ratio of the polarization ellipse of the far field. Hence, dividing the magnitude of (2) by (1) we obtain for the axial ratio:

$$\text{AR} = \frac{|E_\theta|}{|E_\phi|} = \frac{S\lambda}{2\pi A} = \frac{2S\lambda}{\pi^2 D^2} = \frac{2S_\lambda}{C_\lambda^2} \tag{3}$$

Three special cases of the polarization ellipse are of interest. (1) When $E_\phi = 0$, the axial ratio is infinite and the polarization ellipse is a vertical line indicating linear vertical polarization. The helix in this case is a vertical dipole. (2) When $E_\theta = 0$, the axial ratio is zero[1] and the polarization ellipse is a horizontal line indicating linear horizontal polarization. The helix in this case is a horizontal loop. (3) The third special case of interest occurs when

[1]The axial ratio is here allowed to range from 0 to ∞, instead of from 1 to ∞ as customarily, in order to distinguish between linear vertical and linear horizontal polarization.

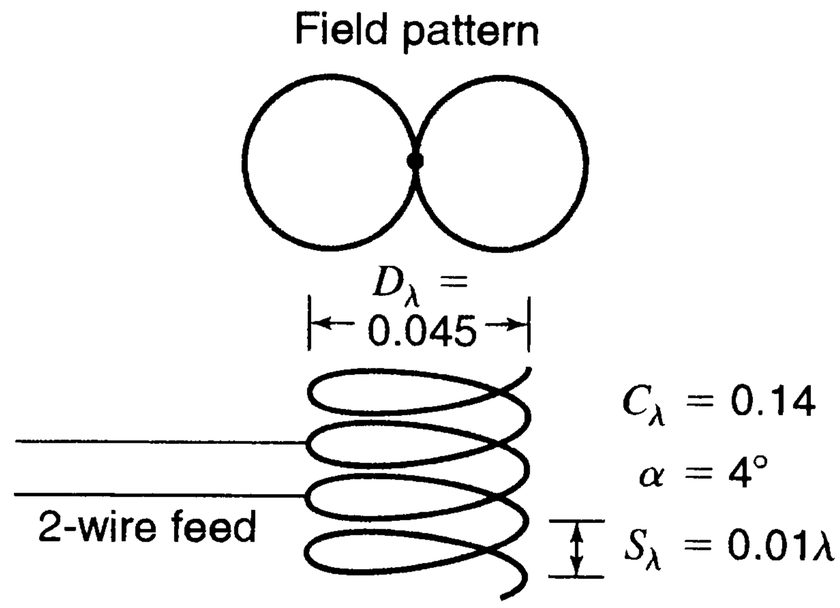


Figure 8–71
Resonant narrowband circularly polarized monofilar normal-mode of Wheeler. Pattern is that of a short dipole.

$|E_\theta| = |E_\phi|$. For this case the axial ratio is unity and the polarization ellipse is a circle, indicating circular polarization. Thus, setting (3) equal to unity yields

$$\pi D = \sqrt{2S\lambda} \qquad \text{or} \qquad C_\lambda = \sqrt{2S_\lambda} \tag{4}$$

This relation was first obtained by Wheeler (1) in an equivalent form. The radiation is circularly polarized in all directions in space but with zero field on axis (z direction, Fig. 8–69*a*). A monofilar normal-mode helix or *Wheeler coil* fulfilling condition (4) is shown in Fig. 8–71. It is a resonant, narrowband antenna.

We have considered three special cases of the polarization ellipse involving linear and circular polarization. In the general case, the radiation is elliptically polarized. Therefore, the radiation from a helix of constant turn-length changes progressively through the following forms as the pitch angle is varied. When $\alpha = 0$, we have a loop (Fig. 8–69*b*) and the polarization is linear and horizontal. As α increases, let us consider the helix dimensions as we move along a constant L_λ line (circle with center at origin, Fig. 8–10). As α increases from zero, the polarization becomes elliptical with the major axis of the polarization ellipse horizontal. When α reaches a value such that $C_\lambda = \sqrt{2S_\lambda}$ the polarization is circular. With the aid of Fig. 8–9, this value of α is given by

$$\alpha = \arcsin \frac{-1 + \sqrt{1 + L_\lambda^2}}{L_\lambda} \tag{5}$$

As α increases still further, the polarization again becomes elliptical but with the major axis of the polarization ellipse vertical. Finally, when α reaches 90°, we have a dipole (Fig. 8–69*c*) and the polarization is linear and vertical. Wheeler's relation for circular polarization from a helix radiating in the normal mode as given by (4) or (5) is shown in Fig. 8–10 by the curve marked $C_\lambda = \sqrt{2S_\lambda}$.

In the preceding discussion on the normal mode of radiation, the assumption is made that the current is uniform in magnitude and in phase over the entire length of the helix. This condition could be approximated if the helix is very small ($nL \ll \lambda$) and is end-loaded. However, the bandwidth of such a small helix is very narrow, and the radiation efficiency is low. The bandwidth and radiation efficiency could be increased by increasing the size of the helix, but to approximate the uniform, in-phase current distribution requires that some type of phase shifter be placed at intervals along the helix. This may be inconvenient or impractical. Hence, the production of the normal mode of radiation from a helix has practical limitations.

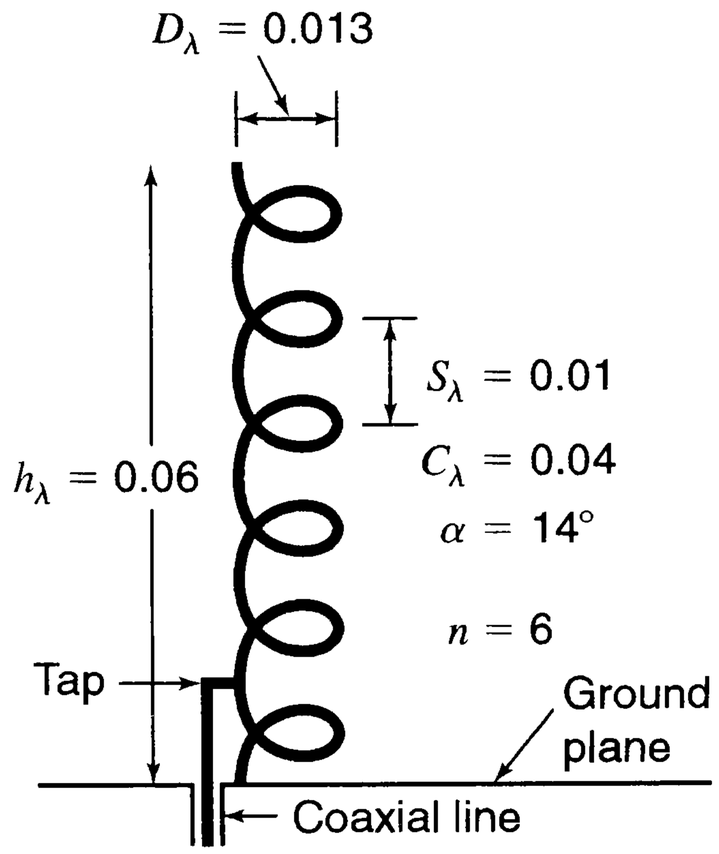


Figure 8–72
Short resonant narrowband monofilar normal-mode helical antenna mounted over a ground plane as substitute for a $\lambda/4$ stub.

An antenna having four slanting dipoles that is suggestive of a fractional-turn quadrifilar helix radiating in the normal mode had been built by Brown (2) and Woodward (see Fig. 21–19*f*). Their arrangement is based on a design described by Lindenblad (1).

Resonant monofilar normal-mode helical antennas are useful as short, essentially vertically polarized, radiators. Referring to Fig. 8–72, the helix mounted on a ground plane with axis vertical acts as a resonant narrowband substitute for a $\lambda/4$ vertical stub or monopole above a ground plane. The helix in Fig. 8–72 is 0.06 λ in height or about $\frac{1}{4}$-height of a $\lambda/4$ stub. From (3) the axial ratio of the helix is given by

$$\text{AR} = \frac{2S_\lambda}{C_\lambda^2} = \frac{2 \times 0.01}{(0.04)^2} = 12.5 \tag{6}$$

with the major axis of the polarization ellipse vertical. The polarization is, thus, essentially linear and vertical with an omnidirectional pattern in the horizontal plane (plane of ground plane). The radiation resistance is nearly the same as for a short monopole of height h_λ above the ground plane where $h_\lambda = nS_\lambda$, which from (2–10–9) or (6–3–11) (for a short dipole) is given by

$$R_s = \tfrac{1}{2} \times 790 \left(\frac{I_{\text{av}}}{I_0} \right)^2 h_\lambda^2 \qquad (\Omega) \tag{7}$$

Assuming a sinusoidal current distribution (maximum current at ground plane, zero at open end),

$$R_s = 395 \times \left(\frac{2}{\pi} \right)^2 \times 0.06^2 = 0.6\ \Omega \tag{8}$$

This is the radiation resistance between the base of the helix and ground. Connection to a coaxial line would require an impedance transformer, but with the shunt feed of Fig. 8–72 the helix can be matched directly to a coaxial line by adjusting the tap point on the helix. With such a small radiation resistance, any loss resistance can reduce efficiency. The advantage of the helix over a straight stub or monopole is that its inductance can resonate the antenna.

A center-fed monofilar helix ($\alpha = 30°$) with $S_\lambda = 1$, $L_\lambda = 2$ and $C_\lambda = \sqrt{3}$ has a 4-lobed pattern, 1 lobe each way on axis and 2 lobes normal to the axis. Its location is indicated on the $m = 1$ line of Fig. 8–10 where the $L_\lambda = 2$ and $\alpha = 30°$ lines intersect, for which also $C_\lambda = 1.73$ and $S_\lambda = 1$ (Chireix design, see also Fig. 8–68*b*).

Patton (1) has demonstrated that a bifilar helix end-fed by a balanced 2-wire transmission line can produce circularly polarized omnidirectional side-fire radiation when pitch angles of about 45° are used.

Some other monofilar and multifilar normal mode (side-fire) helices for omnidirectional FM and TV broadcasting are described by King (3) and Wong and by DuHamel (1).

8–23 AXIAL-MODE HELIX TERMINATIONS（轴向模螺旋终端）

A dielectric tube supporting a helical conductor may significantly affect performance. The magnitude of the effect depends on the dielectric's properties and its geometry, especially the thickness of the tubing wall. For a peripherally fed helix supported on a polyvinyl chloride (PVC) tube with $\varepsilon_r = 2.70$, Baker (1), using the VSWR as a criterion, found that the relative frequency for the onset of the axial mode shifted from 0.72 without the tube to 0.625 with the tube for a ratio of 1.15. Thus, the effective relative permittivity ε_{eff} (with tube) = 1.32 ($= 1.15^2$), making the terminal resistance at the center frequency 130 Ω ($= 150/\sqrt{1.32}$). A precision matching section designed by Baker converts this to 50 Ω with measured VSWR < 1.2 ($\rho_v < -20$ dB) over a 1.7 to 1 bandwidth. The helix wire is wound in a groove of half the wall thickness machined with a computer-controlled lathe. All dimensions of the helix, matching section and supporting structure are specified in Baker's design.

Several arrangements have been proposed to reduce the axial ratio and VSWR to even lower values. These include a conical end-taper section by Wong (1) and King, Donn (1), Angelakor (3) and Kajfez and Jamwal (1) and Vakil, and a flat spiral termination by Baker (1). The flat spiral adds no axial length to the helix. The reflection coefficient (or VSWR) as measured by Baker for a 10-turn peripherally fed monofilar axial-mode helical antenna with and without the spiral termination is presented in Fig. 8–73. The improvement occurs at relative frequencies above 1.1 which, however, is a region where the gain is decreasing.

8–24 ANTENNA ROTATION EXPERIMENTS（天线旋转实验）

Consider the radio circuit shown in Fig. 8–74*a* in which both the transmitting and receiving antennas are linearly polarized. If either of the antennas is rotated about its axis at a frequency f (r/s), the received signal is *amplitude modulated* at this frequency. The direction of rotation is immaterial.

Consider next the radio circuit shown in Fig. 8–74*b* in which one antenna is circularly polarized and the other is linearly polarized. If one of the antennas is rotated about its axis at a frequency f (r/s), the received signal is *shifted* to $F \pm f$, where F is the transmitter frequency. This experiment may also be conducted with 2 circularly polarized antennas of the same type. The frequency f is added or subtracted from F depending on the direction of antenna rotation relative to the rotation direction of **E** (or hand of circular polarization).

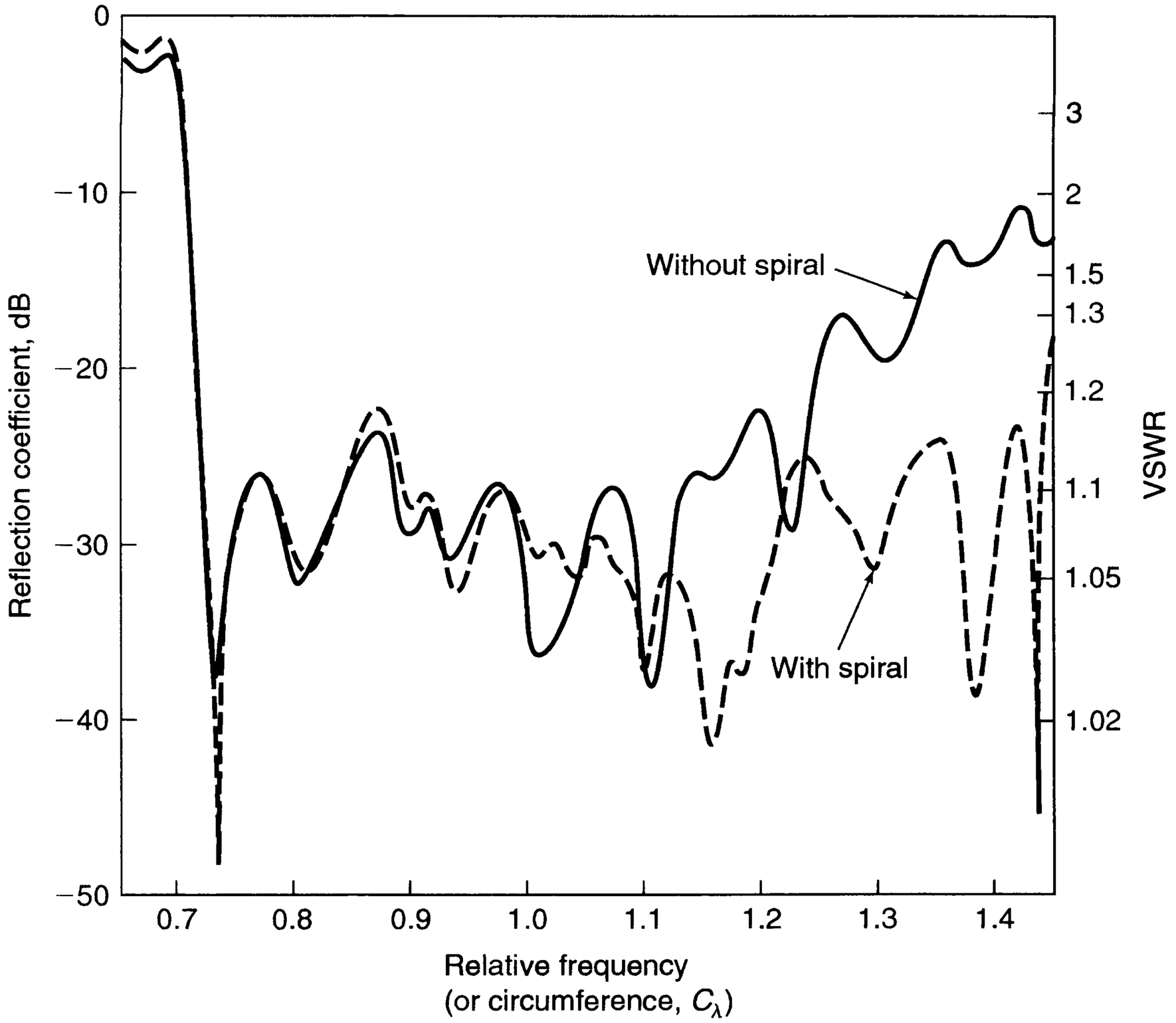


Figure 8–73
Reflection coefficient and VSWR for an impedance-matched peripherally fed 10-turn, 13.8° monofilar axial-mode helical antenna as a function of relative frequency (or circumference C_λ) without spiral termination (solid) and with it (dashed). (*After D. E. Baker-1.*)

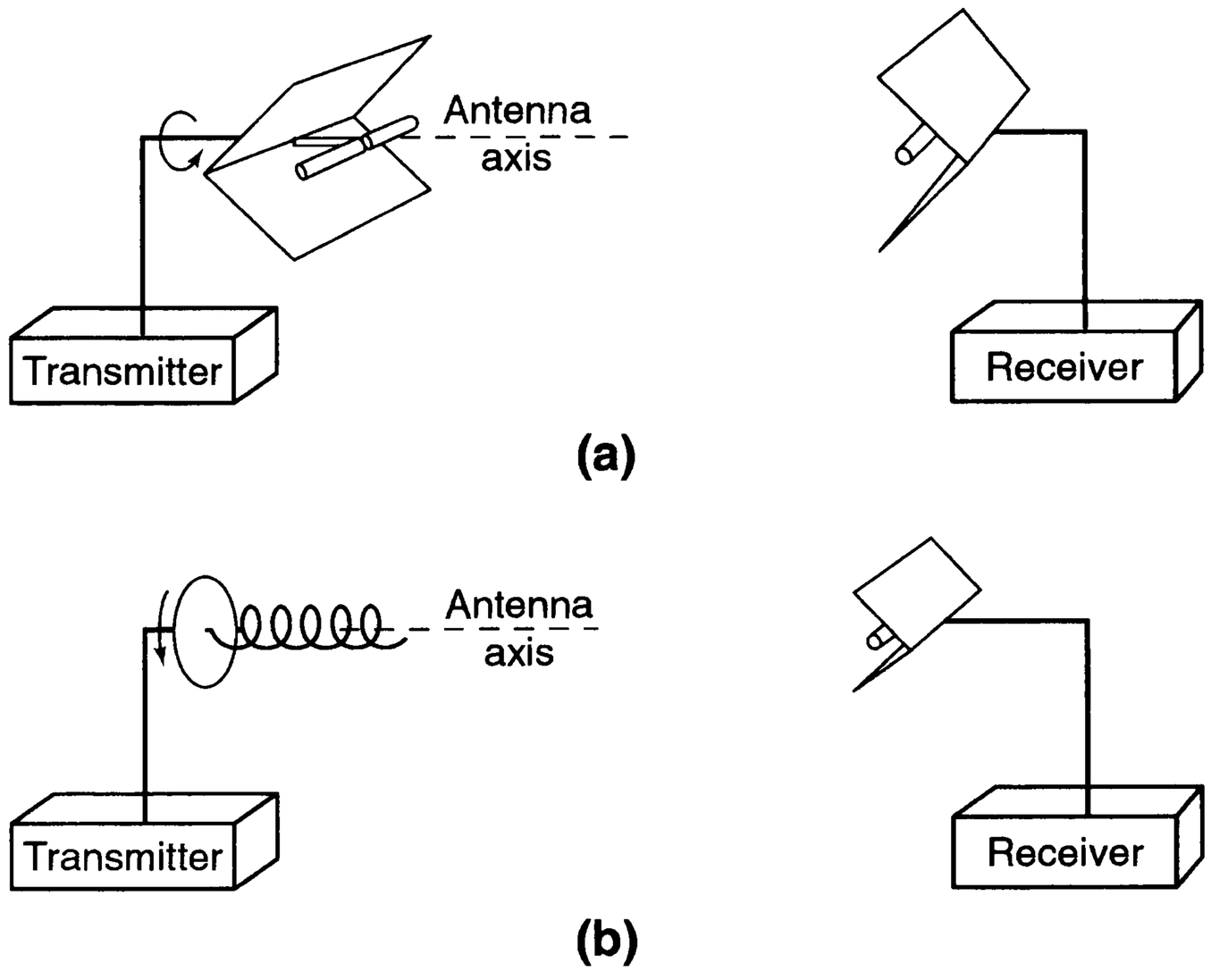


Figure 8–74
Arrangements for antenna rotation experiments. (*a*) Antenna rotation produces amplitude modulation. (*b*) Rotating the monofilar axial-mode helix increases or decreases the signal frequency by the rotation rate.

8–25 BIFILAR AND QUADRIFILAR AXIAL-MODE HELICES
（双绕和四绕轴向模螺旋）

Intertwining two 13° pitch-angle helices fed in phase opposition, as in Fig. 8–75, results in an improved pattern (smaller minor lobes) according to Holtum (1). Intertwining four helices fed in progressive phase-quadrature steps results in an approximation to a continuous helical current sheet with reduced bandwidth due to the feed network. These helices operate with the reduced phase velocity of the monofilar helix in region *A* of Fig. 8–10.

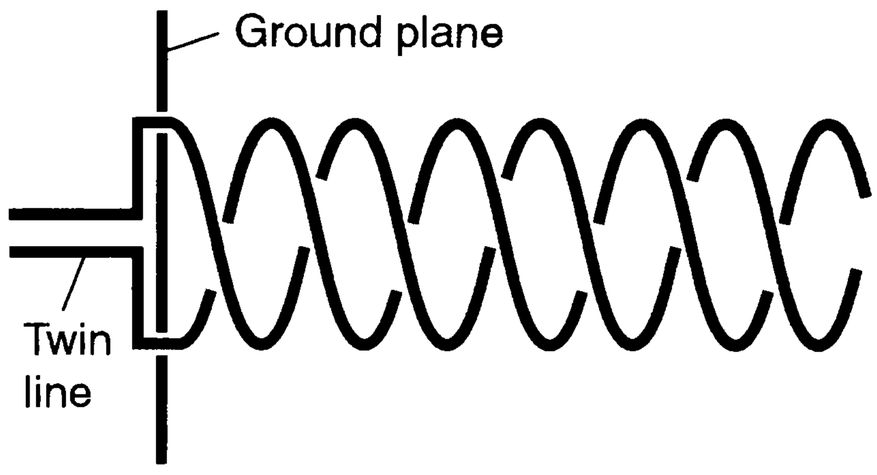


Figure 8–75
Intertwined axial-mode helices fed in phase opposition from a twin line.

A bifilar helix with large (68°) pitch angle and small diameter (<0.1 λ) has a pattern that scans in angle as a function of frequency [Zimmerman (1), Nakano (3)]. The phase velocity $v = c$. See point *C* of Fig. 8–10.

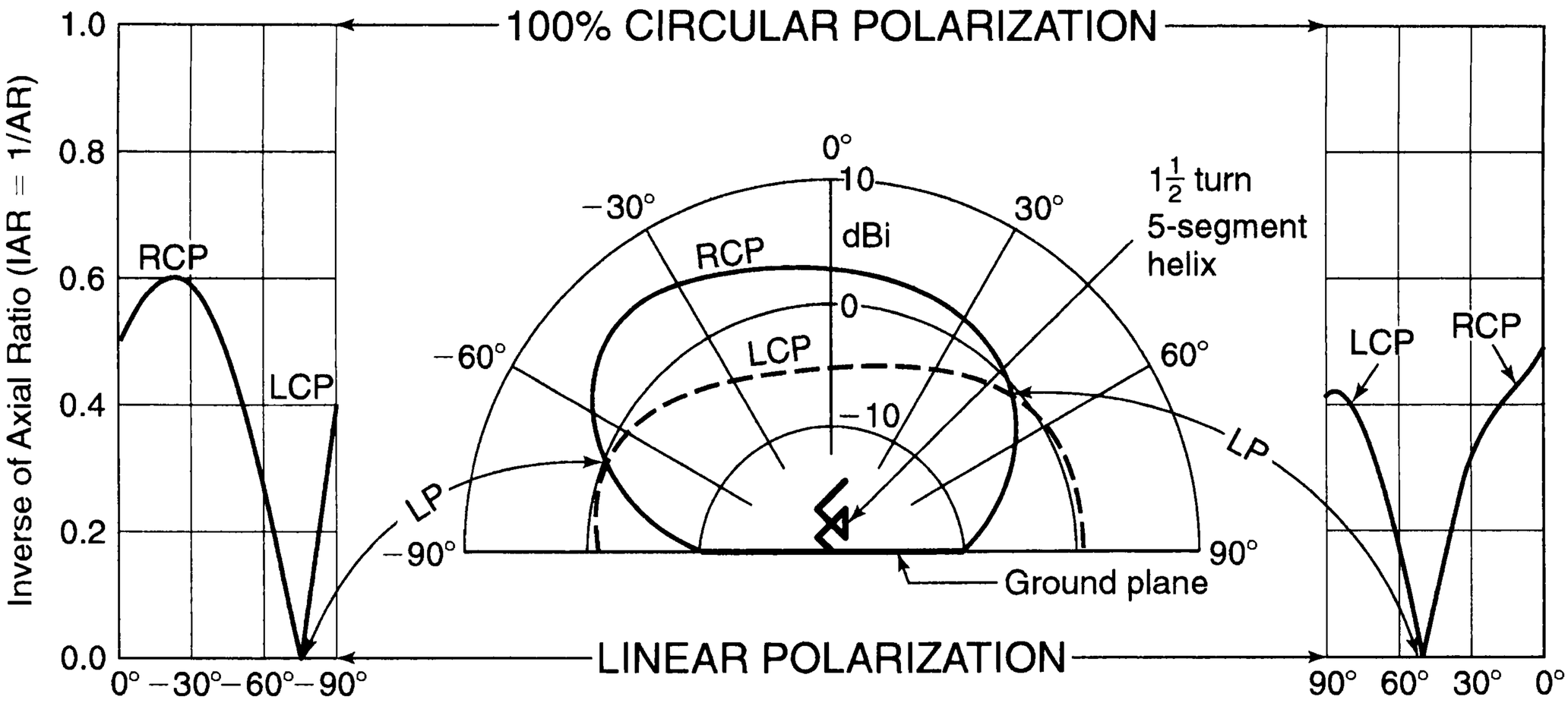
8–26 GENETIC ALGORITHM FIVE-SEGMENT HELIX VERSUS CIRCULAR HELIX
（五直线段螺旋逼近圆螺旋的遗传算法）

Using a genetic algorithm, Altshuler (1) has designed an antenna intended for use as the vehicular terminal of an LEO or MEO satellite link. It is a wire with five straight segments of irregular length bent in the form of a $1\frac{1}{2}$-turn helix. It has a circumference of ~λ and a turn spacing of ~0.5 λ. Its position on the chart of Fig. 8–10 is shown by the point *G*. It closely resembles the $1\frac{1}{4}$ turn "curl antenna" of Nakano (5). See Fig. 21–47–1.

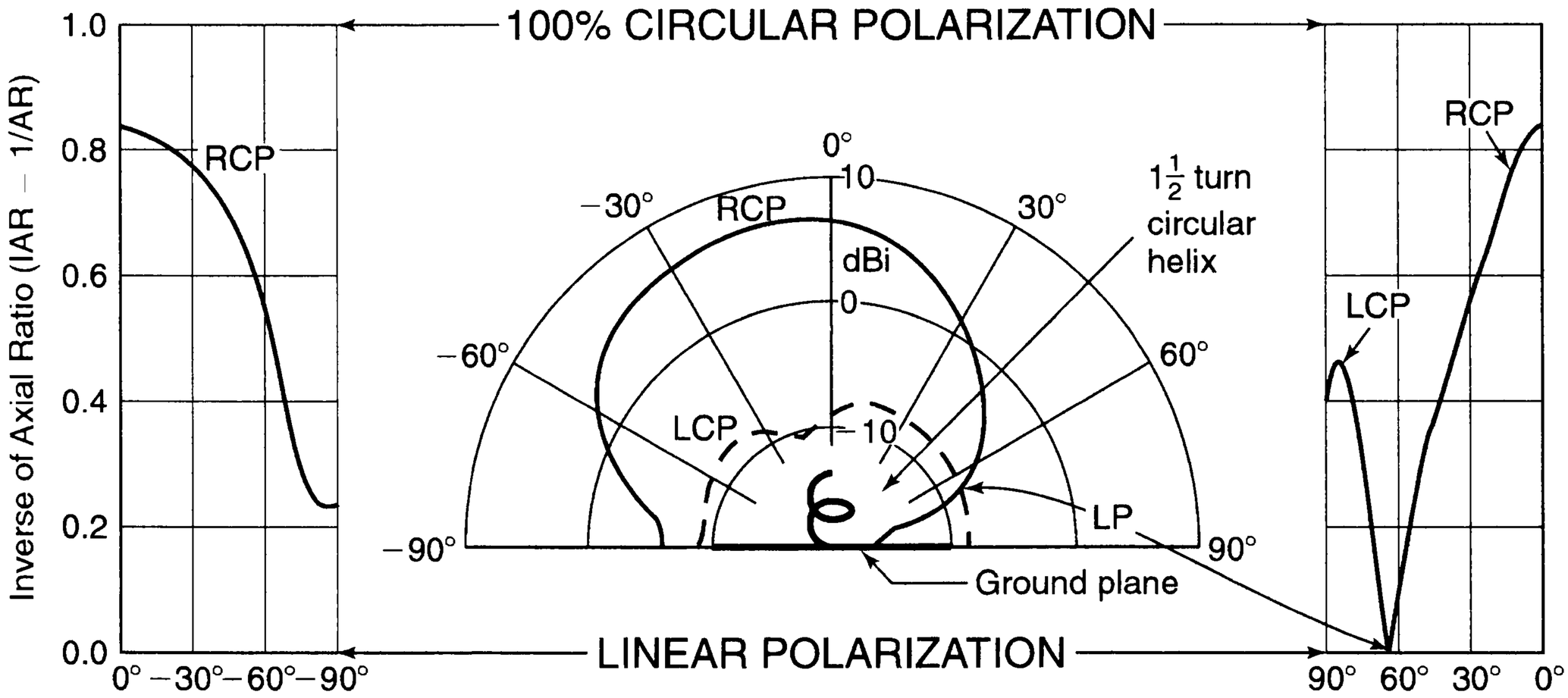
An antenna of Altshuler's dimensions has been measured and compared with measurements for a regular circular $1\frac{1}{2}$-turn helix of the same circumference and length. The results are shown in Fig. 8–76 with patterns in decibels above isotropic (dBi) and polarization in inverse axial ratio (IAR = 1/AR).

The 5-segment helix is 0.5 IAR right-circularly polarized (RCP) on axis (0°) with a gain of 2 dBi changing to linear polarization (LP) at −75° and 50° and then to left-circular polarization (LCP) of 0.4 IAR at the ground plane (±90°).

The circular helix is 0.84 IAR RCP on axis with a gain of 7.5 dBi changing to 0.25 IAR RCP at −90°, LP at 70° and 0.4 IAR LCP at 90°. Comparing the two antennas, it is apparent that the 5-segment helix approximates a circular helix. In this case, the circular helix has more gain on axis (7.5 dBi vs. 2.0 dBi), more RCP purity on axis (0.84 vs. 0.5) and a wider RCP angle of 160° vs. 120°. The genetic-algorithm helix has more gain at large angles away from the axis but with impure polarization. Trends are similar for other cuts through the pattern. The 5-segment helix measurements are similar to Altschuler's when one takes into account that his measurements were made with an RCP horn antenna and the field was not resolved into right- and left-handed components as done above.



(a)



(b)

Figure 8–76
Patterns in decibels above isotropic (dBi) of 5-segment helix (*a*) and of circular helix (*b*) flanked at left and right by the inverse of the axial ratio (IAR = 1/AR). At 1.0 the field is 100 percent circularly polarized while at 0.0 it is linearly polarized. Measurements were made at 1575 MHz on a 1.1 λ square ground plane. (*Measurements by Kevin Sickles at the ElectroScience Laboratory of the Ohio State University.*)

REFERENCES（参考文献）

Adams, A. A. (1), R. K. Greenough, R. F. Wallenberg, A. Mendelovicz and C. Lumjiak: "The Quadrifilar Helix Antenna," *IEEE Trans. Ants. Prop.,* **AP-22,** 173–178, March 1974.

Altshuler, E. E. (1), "Design of a Vehicular Antenna for GPS/IRIDIUM Using a Genetic Algorithm," *IEEE Trans. Ants. Prop.,* 968–971, June 2000.

Angelakor, D. J. (3), and D. Kajfez: "Modifications on the Axial-Mode Helical Antenna," *Proc. IEEE,* 558–559, April 1967.

Bagby, C. K. (1): "A Theoretical Investigation of Electro-magnetic Wave Propagation on the Helical Beam Antenna," Master's thesis, Electrical Engineering Department, Ohio State University, 1948.

Baker, D. E. (1): "Design of a Broadband Impedance Matching Section for Peripherally Fed Helical Antennas," *Antenna Applications Symposium,* University of Illinois, September 1980.

Blasi, E. A. (1): "Theory and Application of the Radiation Mutual Coupling Factor," M.S. thesis, Electrical Engineering Department, Ohio State University, 1952.

Broussaud, G. (1), and E. Spitz: "Endfire Antennas," *Proc. IRE,* **49,** 515–516, February 1961.

Brown, G. H. (1): "Directional Antennas," *Proc. IRE,* **25,** 78–145, January 1937.

Brown, G. H. (2), and O. M. Woodward: "Circularly Polarized Omnidirectional Antenna," *RCA Rev.,* **8,** 259–269, June 1947.

Bystrom, A., Jr. (1), and D. G. Bernsten: "An Experimental Investigation of Cavity-Mounted Helical Antennas," *IRE Trans. Ants. Prop.,* **AP-4,** 53–58, January 1956.

Carver, K. R. (1): "The Helicone: A Circularly Polarized Antenna with Low Side Lobe Level," *Proc. IEEE,* **55,** 559, April 1967.

Carver, K. R. (2), and B. M. Potts: "Some Characteristics of the Helicone Antenna," *Antennas and Propagation International Symposium,* 1970, pp. 142–150.

Chatterjee, J. S. (1): "Radiation Field of a Conical Helix," *J. Appl. Phys.,* **24,** 550–559, May 1953; **26,** 331–335, March 1955.

Chen, C. A. (1), and D. K. Cheng: "Optimum Spacings for Yagi-Uda Arrays," *IEEE Trans. Ants. Prop.,* **AP-21,** 615–623, September 1973.

Chen, C. A. (2), and D. K. Cheng: "Optimum Element Lengths for Yagi-Uda Arrays," *IEEE Trans. Ants. Prop.,* **AP-23,** 8–15, 1975.

Chu, L. J. (1), and J. D. Jackson: "Field Theory of Traveling Wave Tubes," *Proc. IRE,* **36,** 853–863, July 1948.

Day, P. C. (1): "Some Characteristics of Tapered Helical Beam Antennas," M.S. thesis, Ohio State University, 1950.

Donn, C. (1): "A New Helical Antenna Design for Better On-and-Off Boresight Axial Ratio Performance," *IEEE Trans. Ants. Prop.,* **AP-28,** 264–267, March 1980.

DuHamel, R. H. (1), and A. R. Mahnad: Chap. 28 in R. C. Johnson (ed.), *Radio Engineering Handbook,* 3d ed., McGraw-Hill, New York, 1993.

Gerst, C. W. (1), and R. A. Worden: "Helix Antennas Take a Turn for the Better;" *Electronics,* 100–110, Aug. 22, 1966.

Glasser, O. J. (1), and J. D. Kraus: "Measured Impedances of Helical Beam Antennas," *J. Appl. Phys.,* **19,** 193–197, February 1948.

Holland, J. (1): "Multiple Feed Antenna Covers *L*, *S* and *C*-Band Segments," *Microwave J.,* 82–85, October 1981.

Holtum Jr., A. G. (1): "Improving the Helical Beam Antenna," *Electronics,* Apr. 29, 1960.

Jamwal, K. K. S. (1), and R. Vakil: "Design Analysis of Gain-Optimized Helix Antennas for X-band Frequencies," *Microwave J.,* 177–183, September 1985.

Johnson, R. C. (1), and R. B. Cotton: "A Backfire Helical Feed," Georgia Institute of Technology, Engineering Experimental Station Report, 1982.

Kilgus, C. C. (1): "Resonant Quadrifilar Helix," *IEEE Trans. Ants. Prop.,* **AP-17,** 349–351, May 1969.

Kilgus, C. C. (2): "Shaped-Conical Radiation Pattern Performance of the Backfire Quadrifilar Helix," *IEEE Trans. Ants. Prop.,* **AP-23,** 392–397, May 1975.

King, H. E. (1), and J. L. Wong: "Antenna System for the FleetSatCom Satellites," *IEEE International Symposium on Antennas and Propagation,* pp. 349–352, 1977.

King, H. E. (2), and J. L. Wong: "Characteristics of 1 to 8 Wavelength Uniform Helical Antennas," *IEEE Trans. Ants. Prop.,* **AP-28,** 291, March 1980.

King H. E. (3), and J. L. Wong: Chap. 13 in R. C. Johnson (ed.), *Radio Engineering Handbook,* 3d ed., McGraw-Hill, New York, 1993.

Kraus, J. D. (1): "Helical Beam Antenna," *Electronics,* **20,** 109–111, April 1947.

Kraus, J. D. (2), and J. C. Williamson: "Characteristics of Helical Antennas Radiating in the Axial Mode," *J. Appl. Phys.,* **19,** 87–96, January 1948.

Kraus, J. D. (3): "Helical Beam Antennas for Wide-Band Applications," *Proc. IRE,* **36,** 1236–1242, October 1948.

Kraus, J. D. (4): "The Helical Antenna," *Proc. IRE,* **37,** 263–272, March 1949.

Kraus, J. D. (5): "Helical Beam Antenna Design Techniques," *Communications,* **29,** 6–9, 34–35, September 1949.

Kraus, J. D. (6): *Radio Astronomy,* 2d ed., Cygnus-Quasar, 1986, p. 8-2.

Kraus, J. D. (7): "A 50-ohm Input Impedance for Helical Beam Antennas," *IEEE Trans. Ants. Prop.,* **AP-25,** 913, November 1977.

Kraus, J. D. (8): "The Ohio State Radio Telescope," *Sky and Tel.,* **12,** 157–159, April 1953.

Kraus, J. D. (9): "Planetary and Solar Radio Emission at 11 Meters Wavelength," *Proc. IRE,* **46,** 266–274, January 1958.

Lindenblad, N. E. (1): "Antennas and Transmission Lines at the Empire State Television Station," *Communications,* **21,** 10–14, 24–26, April 1941.

MacLean, T. S. M. (1), and R. G. Kouyournjian: "The Bandwidth of Helical Antennas," *IRE Trans. Ants. Prop.,* **AP-7,** S379–386, December 1959.

Marsh, J. A. (1): "Measured Current Distributions on Helical Antennas," *Proc. IRE,* **39,** 668–675, June 1951.

McFarland, R. M. (1): "A Lobe-Sweeping Antenna for the Reception of Radio Waves of Extraterrestrial Origin," M.S. thesis, Ohio State University, 1957.

Nakano, H. (1), T. Yamane, J. Yamauchi and Y. Samada: "Helical Antenna with Increased Power Gain," *IEEE AP-S Int. Symp.,* **1,** 417–420, 1984.

Nakano, H. (2), Y. Samada and J. Yamauchi: "Axial Mode Helical Antennas," *IEEE Trans. Ants. Prop.,* **AP-34,** 1143–1148, September 1986.

Nakano, H. (3), T. Mikawa and J. Yamauchi: "Numerical Analysis of Monofilar Conical Helix," *IEEE AP-S Int. Symp.,* **1,** 177–180, 1984.

Nakano, H. (4), H. M. Mimaki and J. Yamauschi: "Loaded Bifilar Helical Antenna with Small Radius and Large Pitch Angle," *Electronics Lett.,* **27,** 1568–1569, 1991.

Nakano, H. (5), S. Okuzawa, K. Ohishi, H. Mimaki and J. Yamauchi: "A Curl Antenna," *IEEE Trans. Ants. Prop.,* pp. 1570–1575, November 1993.

Patton, W. T. (1), "The Backfire Bifilar Helical Antenna," Ph.D. thesis, University of Illinois, 1963.

Tice, T. E. (1), and J. D. Kraus: "The Influence of Conductor Size on the Properties of Helical Beam Antennas," *Proc. IRE,* **37,** 1296, November 1949.

Uda, S. (1): "On the Wireless Beam of Short Electric Waves," *JIEE (Japan),* March 1926. pp. 273–282 (I); April 1926 (II); July 1926 (III); January 1927 (IV); January 1927 (V); April 1927 (VI); June 1927 (VII); October 1927 (VIII); November 1927, pp. 1209–1219 (IX); April 1928 (X); July 1929 (XI).

Uda, S. (2), and Y. Mushiake: *Yagi-Uda Antenna,* Maruzden, Tokyo, 1954.

Uda, S. (3): *Short Wave Projector; Historical Records of My Studies in the Early Days,* published by Uda, 1974.

Vaugm, R. G. (1), and J. B. Anderson: "Polarization Properties of the Axial-Mode Helix Antenna," *IEEE Trans. Ants. Prop.,* **AP-33,** 10–20, January 1985.

Viezbicke, P. P. (1): "Yagi Antenna Design," NBS Technical Note 688, December 1968.

Wheeler, H. A. (1): "A Helical Antenna for Circular Polarization," *Proc. IRE.,* **35,** 1484–1488, December 1947.

Wong, J. L. (1), and H. E. King: "Broadband Quasi-Taper Helical Antennas," *IEEE Trans. Ants. Prop.,* **AP-27,** 72–78, January 1979.

Yagi, H. (1): "Beam Transmission of Ultra Short Waves," *Proc. IRE.,* **16,** 715–740, June 1928.

Zimmerman, R. K. Jr. (1): "Traveling-wave Analysis of a Bifilar Scanning Helical Antenna," *IEEE Trans. Ants. Prop.,* 1007–1009, June 2000.

ADDITIONAL REFERENCES ON HELICAL ANTENNAS（螺旋天线的附加参考资料）

Dunham, Mark E., Max Light and Daniel N. Holden: "Broad-Band Pulse Performance of Short Helices," *IEEE Transactions on Antennas and Prop.*, vol. 43, p. 1017, October 1995. "The helix has proven to be the radiator of choice for many radio astronomy and space science applications."

Emerson, Darrel: "The Gain of the Axial-Mode Helix Antenna," *Antenna Compendium,* vol. 4, pp. 64–68, pub. by ARRL. "The axial-mode helix antenna is probably the most widely used circularly polarized antenna, either in space or on the ground." Discusses gains.

Krause, L. O.: "Sidefire Helix UHF TV Transmitting Antenna," *Electronics,* p. 107, August 1951.

Nakano, H.: *Helical and Spiral Antennas—A Numerical Approach,* Wiley, New York, 1987.

Smith, H. G.: "High-Gain Side Firing Helical Antennas," *AIEE Trans. Comm. and Electronics,* vol. 73, pp. 135–138, May 1954.

PROBLEMS（习题）

***8–8–1**（8圈螺旋）

An 8-turn helix. A monofilar helical antenna has $\alpha = 12°$, $n = 8$, $D = 225$ mm. (*a*) What is p at 400 MHz for (1) in-phase fields and (2) increased directivity? (*b*) Calculate and plot the field patterns for $p = 1.0$, 0.9, and 0.5 and also for p equal to the value for in-phase fields and increased directivity. Assume each turn is an isotropic point source. (*c*) Repeat (*b*) assuming each turn has a cosine pattern.

8–9–1（6圈螺旋）

A 6-turn helix. A monofilar axial-mode helical antenna has 6 turns, 231 mm diameter and 181 mm turn spacing. Neglect the effect of the ground plane. Assume that the relative phase velocity p along the helical conductor satisfies the increased-directivity condition. Calculate and plot the following patterns as a function of ϕ (0 to 360°) in the $\theta = 90°$ plane at 400 MHz. Use the square helix approximation. (*a*) $E_{\phi T}$ for a single turn and E_ϕ for the entire helix. (*b*) Repeat (*a*) neglecting the contribution of sides 2 and 4 of the square turn. (*c*) $E_{\theta T}$ for a single turn and E_θ for the entire helix.

***8–11–1**（法向模螺旋）

Normal-mode helix. (*a*) What is the approximate relation required between the diameter D and height H of an antenna having the configuration shown in Fig. P8–11–1, in order to obtain a circularly polarized far field at all points at which the field is not zero? The loop is circular and is horizontal, and the linear conductor of length H is vertical. Assume D and H are small compared to the wavelength, and assume the current is of uniform magnitude and in phase over the system. (*b*) What is the pattern of the far circularly polarized field?

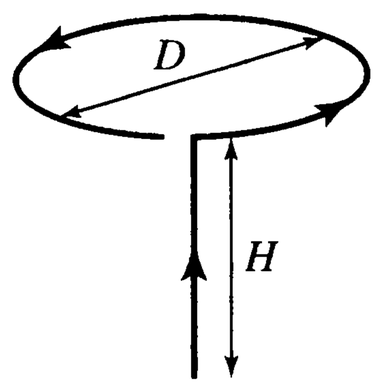


Figure P8–11–1
Normal mode helix.

8–15–1（四螺旋地球站天线的设计）

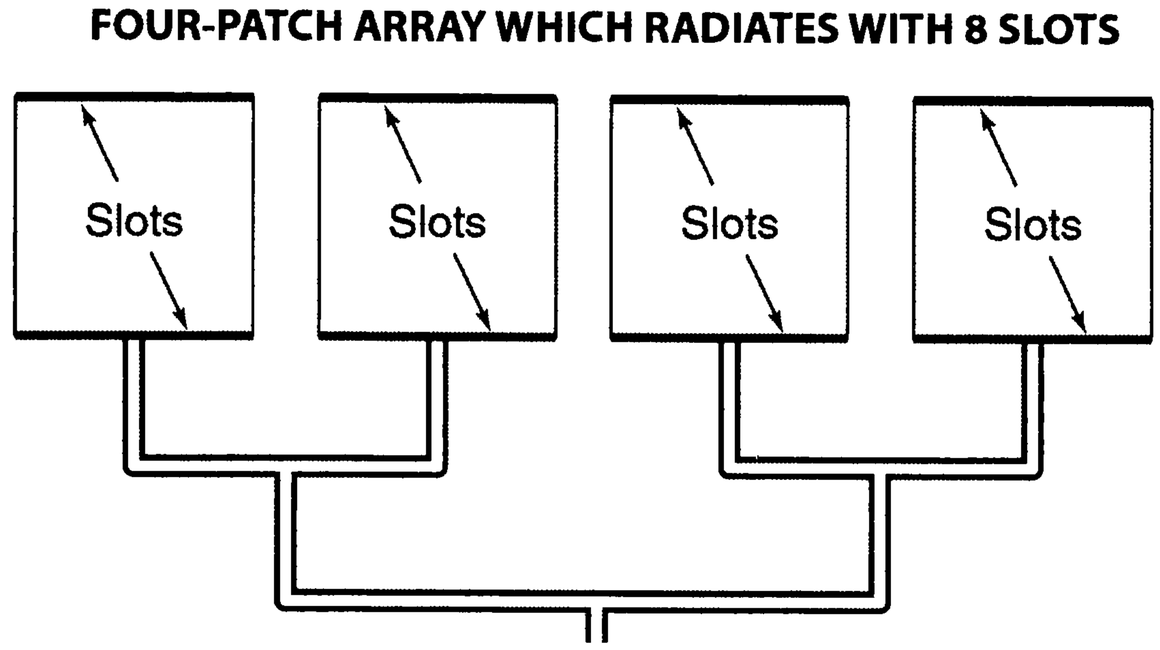
Design of quad-helix earth station antenna. An array of four right-handed axial-mode helical antennas, shown in Fig. 8–54, can be used for communications with satellites. Determine (*a*) the best spacing based on the effective apertures of the helices, (*b*) the directivity of the array. Assume the number of turns is 20 and the spacing between turns is 0.25 λ.

For computer programs, see Appendix C.

CHAPTER 9

Slot, Patch and Horn Antennas

缝隙天线、贴片天线和喇叭天线



The topics in this chapter include:

- Slot antennas (缝隙天线)
- Slot antenna patterns (缝隙天线的波瓣图)
- Slots and their complementary forms (Babinet's principle) (缝隙及其互补形式（巴比涅原理))
- Impedance of slot antennas (缝隙天线的阻抗)
- Slotted cylinder antennas (开缝柱形天线)
- Patch or microstrip antennas (贴片或微带天线)
- Impedance of patch antennas (贴片天线的阻抗)
- Horn antennas (Fermat's principle) (喇叭天线（费马原理))
- Rectangular horns (矩形喇叭)
- Conical and biconical horns (圆锥和双锥喇叭)
- Ridge horns (加脊喇叭)
- Septum horns (隔膜喇叭)
- Corrugated horns (皱纹喇叭)
- Aperture matched horns (口径匹配喇叭)

9–1 SLOT ANTENNAS (缝隙天线)

Slot antennas are useful in many applications, especially where low-profile or flush mountings are required as, for example, on high-speed aircraft. Any slot has its complementary form in wires or strips, so that pattern and impedance data of these forms can be used to predict the patterns and impedances of the corresponding slots. The discussion is based largely on a generalization and extension of Babinet's (Ba-bi-naý's) principle by Henry Booker (1).

The antenna shown in Fig. 9–1*a*, consisting of two resonant $\lambda/4$ stubs connected to a 2-wire transmission line, is an inefficient radiator. The long wires are closely spaced ($w \ll \lambda$) and carry currents of opposite phase so that their fields tend to cancel. The end wires carry currents in the same phase, but they are too short to radiate efficiently. Hence, enormous currents may be required to radiate appreciable amounts of power.

The antenna in Fig. 9–1*b*, on the other hand, is a very efficient radiator. In this arrangement a $\lambda/2$ slot is cut in a flat metal sheet. Although the width of the slot is small ($w \ll \lambda$), the currents are not confined to the edges of the slot but spread out over the sheet. This is a simple type of slot antenna. Radiation occurs equally from both sides of

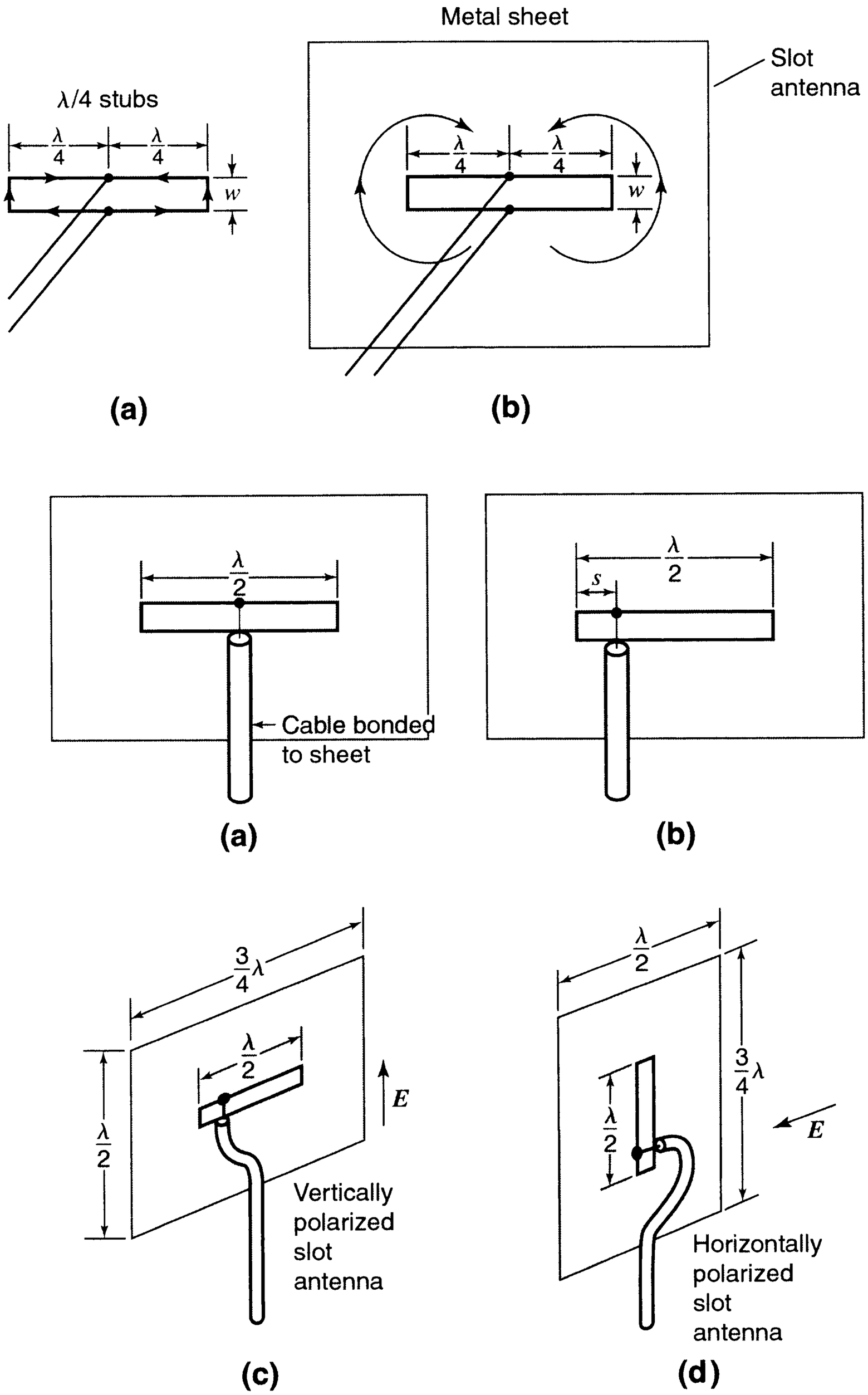


Figure 9–1
Whereas the stubs of (*a*) are a poor radiator, the slot of (*b*) is a good, efficient radiator because the currents can spread out on the metal sheet.

Figure 9–2
Slot antennas fed by coaxial transmission lines.

the sheet. If the slot is horizontal, as shown, the radiation normal to the sheet is vertically polarized.

A slot antenna may be conveniently energized with a coaxial transmission line as in Fig. 9–2*a*. The outer conductor of the cable is bonded to the metal sheet. Since the terminal resistance at the center of a resonant $\lambda/2$ slot in a large sheet is about 500 Ω and the characteristic impedance of coaxial transmission lines is usually much less, an off-center feed such

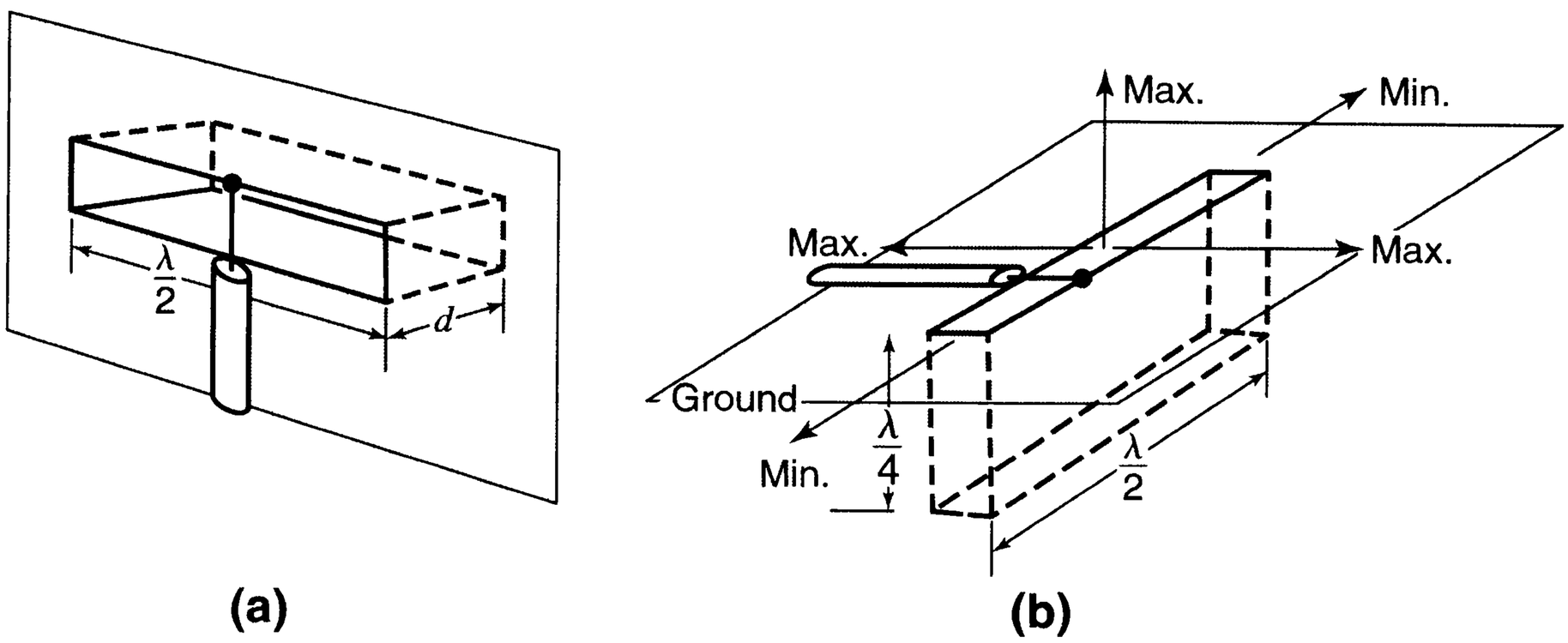


Figure 9–3
Boxed-in slot antenna (*a*) and application to provide flush radiator (*b*).

as shown in Fig. 9–2*b* may be used to provide a better impedance match. For a 50-Ω coaxial cable the distance s should be about $\lambda/20$. Slot antennas fed by a coaxial line in this manner are illustrated in Fig. 9–2*c* and *d*. The radiation normal to the sheet with the horizontal slot (Fig. 9–2*c*) is vertically polarized while radiation normal to the sheet with the vertical slot (Fig. 9–2*d*) is horizontally polarized. The slot may be $\lambda/2$ long, as shown, or more.

A flat sheet with a $\lambda/2$ slot radiates equally on both sides of the sheet. However, if the sheet is very large (ideally infinite) and boxed in as in Fig. 9–3*a*, radiation occurs only from one side. If the depth d of the box is appropriate ($d \sim \lambda/4$ for a thin slot), no appreciable shunt susceptance appears across the terminals. With such a zero susceptance box, the terminal impedance of the resonant $\lambda/2$ slot is nonreactive and approximately twice its value without the box, or about 1000 Ω.

The boxed-in slot antenna might be applied even at relatively long wavelengths by using the ground as the flat conducting sheet and excavating a trench $\lambda/2$ long by $\lambda/4$ deep as suggested in Fig. 9–3*b*. The absence of any structure above ground level might make this type of antenna attractive, for example, in applications near airports. To improve the ground conductivity, the walls of the trench and the ground surrounding the slot can be covered with copper sheet or screen. Radiation is maximum in all directions at right angles to the slot and is zero along the ground in the directions of the ends of the slot, as suggested in Fig. 9–3*b*. The radiation along the ground is vertically polarized.

Radiation from only one side of a large flat sheet may also be achieved by a slot fed with a waveguide as in Fig. 9–4*a*. With transmission in the guide in the TE_{10} mode, the direction of the electric field **E** is as shown. The width L of the guide must be more than $\lambda/2$ to transmit energy, but it should be less than 1λ to suppress higher-order transmission modes. With the slot horizontal, as shown, the radiation normal to the sheet is vertically polarized. The slot opening constitutes an abrupt termination to the waveguide. It has been found that the resulting impedance mismatch is least over a wide frequency band if the ratio L/w is less than 3.

A compact wideband method for feeding a boxed-in slot is illustrated in Fig. 9–4*b*. In this T-fed arrangement the bar compensates the impedance characteristics so as to provide a VSWR on a 50-Ω feed line of less than 2 over a frequency range of nearly 2 to 1. The ratio L/w of the length to width of the slot is about 3 (Dorne-1).

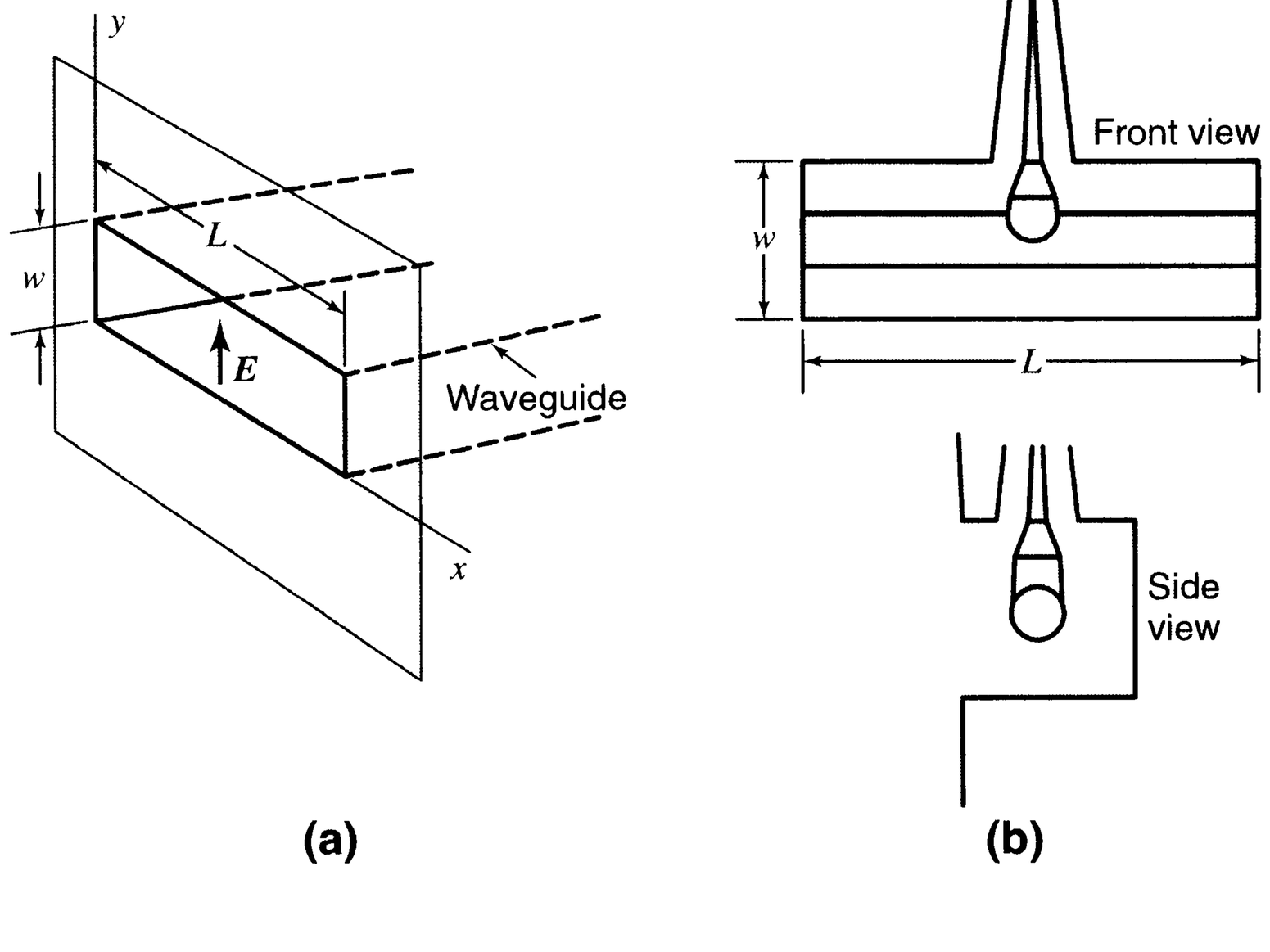


Figure 9–4
Waveguide-fed slot (*a*) and T-fed slot (*b*).

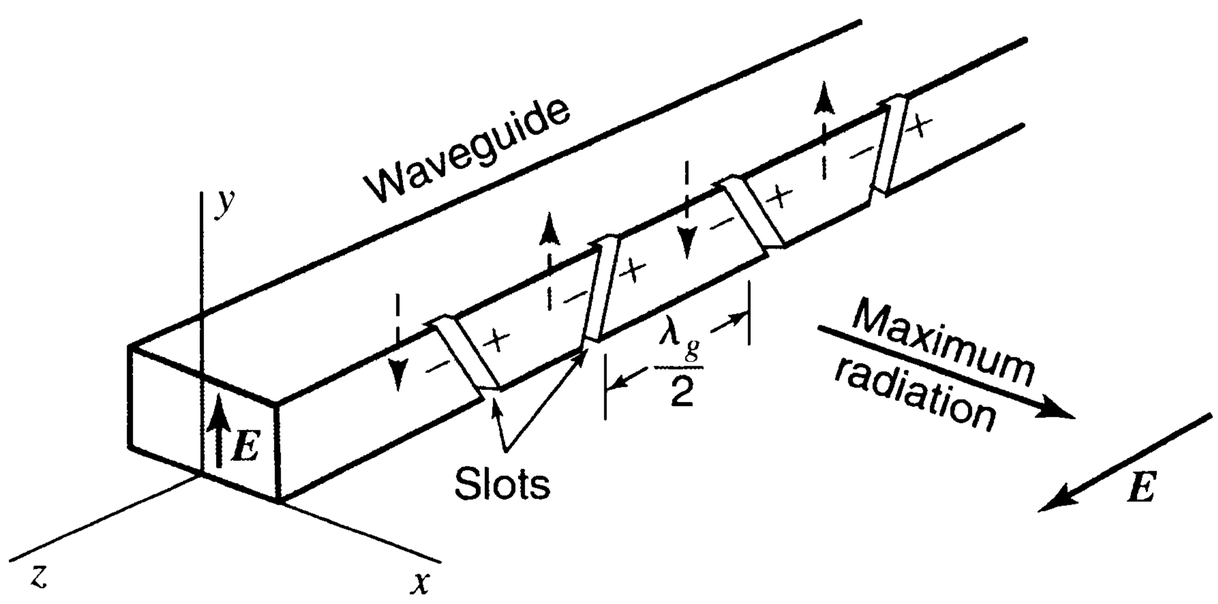


Figure 9–5
Broadside array of slots in waveguide.

Dispensing with the flat sheet altogether, an array of slots may be cut in the waveguide as in Fig. 9–5 so as to produce a directional radiation pattern (Watson-1). With transmission in the guide in the TE_{10} mode, the instantaneous direction of the electric field **E** inside the guide is as indicated by the dashed arrows. By cutting inclined slots as shown at intervals of $\lambda_g/2$ (where λ_g is the wavelength in the guide), the slots are energized in phase and produce a directional pattern with maximum radiation broadside to the guide. If the guide is horizontal and **E** inside the guide is vertical, the radiated field is horizontally polarized as suggested in Fig. 9–5.

9–2 PATTERNS OF SLOT ANTENNAS IN FLAT SHEETS. EDGE DIFFRACTION

（平板缝隙天线的波瓣图和边缘绕射）

Consider the horizontal $\lambda/2$ slot antenna of width w in a perfectly conducting flat sheet of infinite extent, as in Fig. 9–6*a*. The sheet is energized at the terminals *FF*. It has been postulated by Booker (1) that the radiation pattern of the slot is the same as that of the

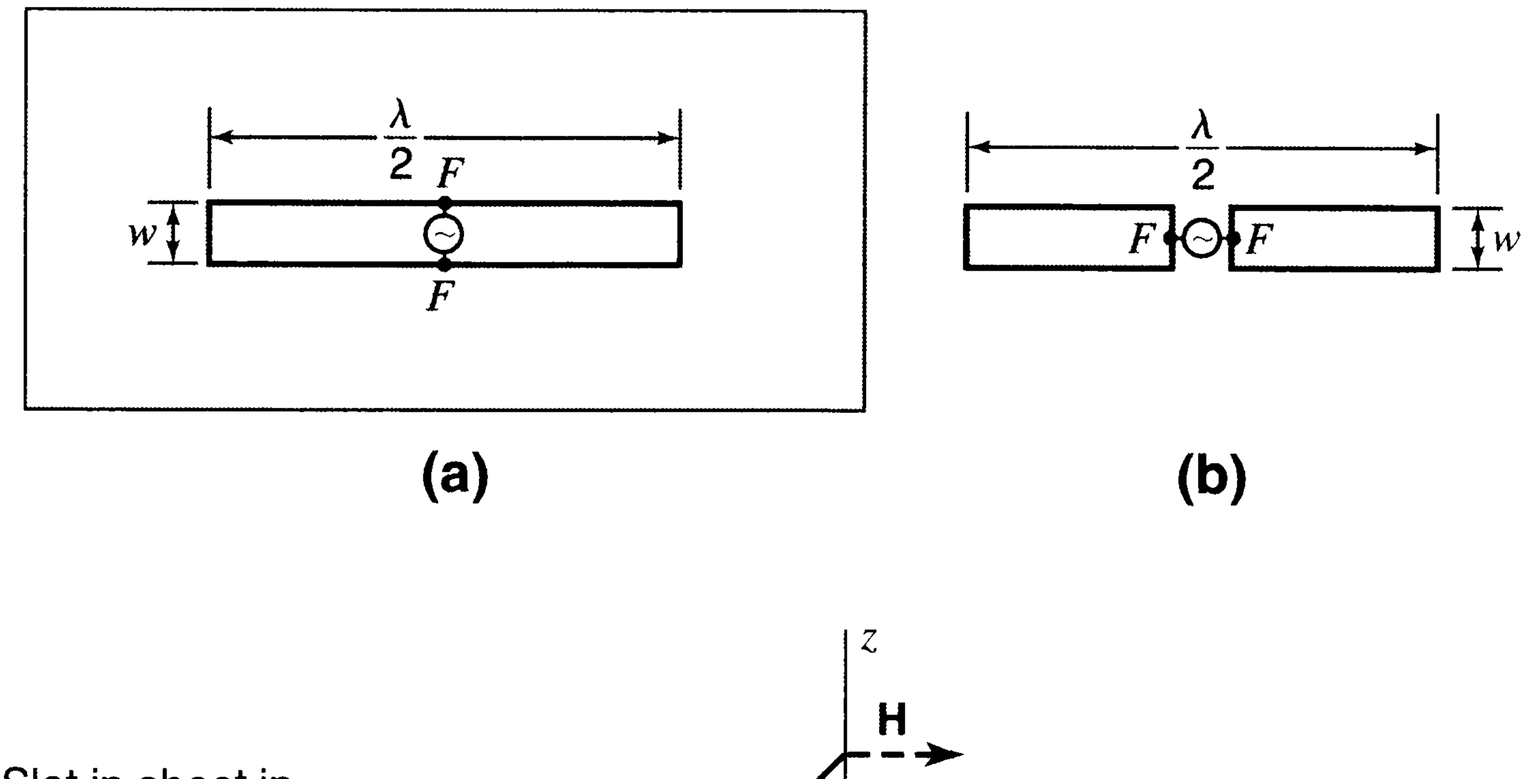


Figure 9–6
A $\lambda/2$ slot in an infinite flat sheet (*a*) and a complementary $\lambda/2$ dipole antenna (*b*).

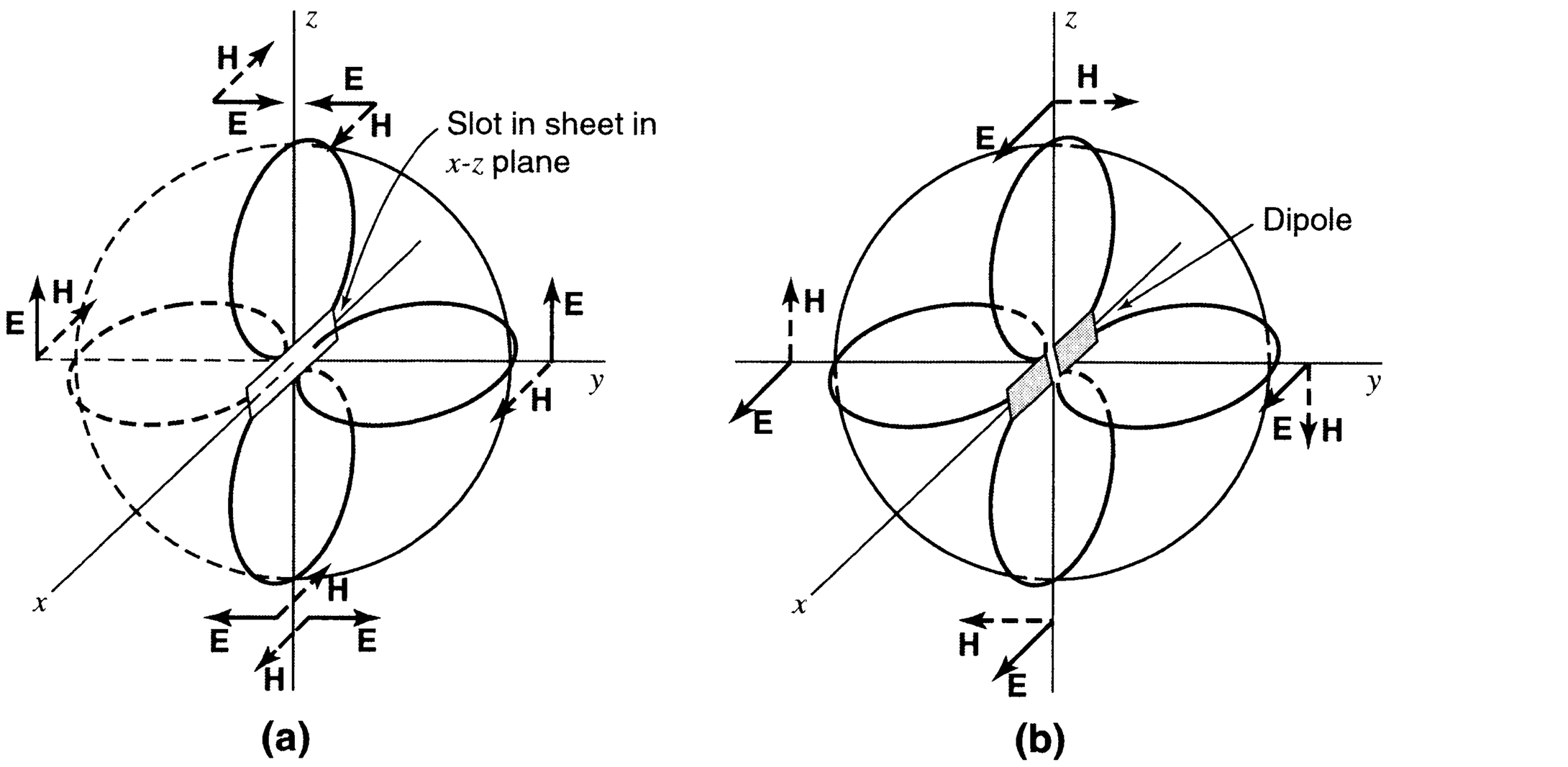


Figure 9–7
Radiation-field patterns of slot in an infinite sheet (*a*) and of complementary dipole antenna (*b*). The patterns have the same shape but with **E** and **H** interchanged.

complementary horizontal $\lambda/2$ dipole consisting of a perfectly conducting flat strip of width w and energized at the terminals FF, as indicated in Fig. 9–6*b*, but with two differences. These are (1) that the electric and magnetic fields are interchanged and (2) that the component of the electric field of the slot normal to the sheet is discontinuous from one side of the sheet to the other, the direction of the field reversing. The tangential component of the magnetic field is, likewise, discontinuous.

The patterns of the $\lambda/2$ slot and the complementary dipole are compared in Fig. 9–7. The infinite flat sheet is coincident with the xz plane, and the long dimension of the slot is in the x direction (Fig. 9–7*a*). The complementary dipole is coincident with the x axis (Fig. 9–7*b*). The radiation-field patterns have the same doughnut shape, as indicated, but the directions of **E** and **H** are interchanged. The solid arrows indicate the direction of the electric field **E** and the dashed arrows the direction of the magnetic field **H**.

If the xy plane is horizontal and the z axis vertical as in Fig. 9–7*a*, the radiation from the horizontal slot is vertically polarized everywhere in the xy plane. Turning the slot to a vertical position (coincident with the z axis) rotates the radiation pattern through 90° to the

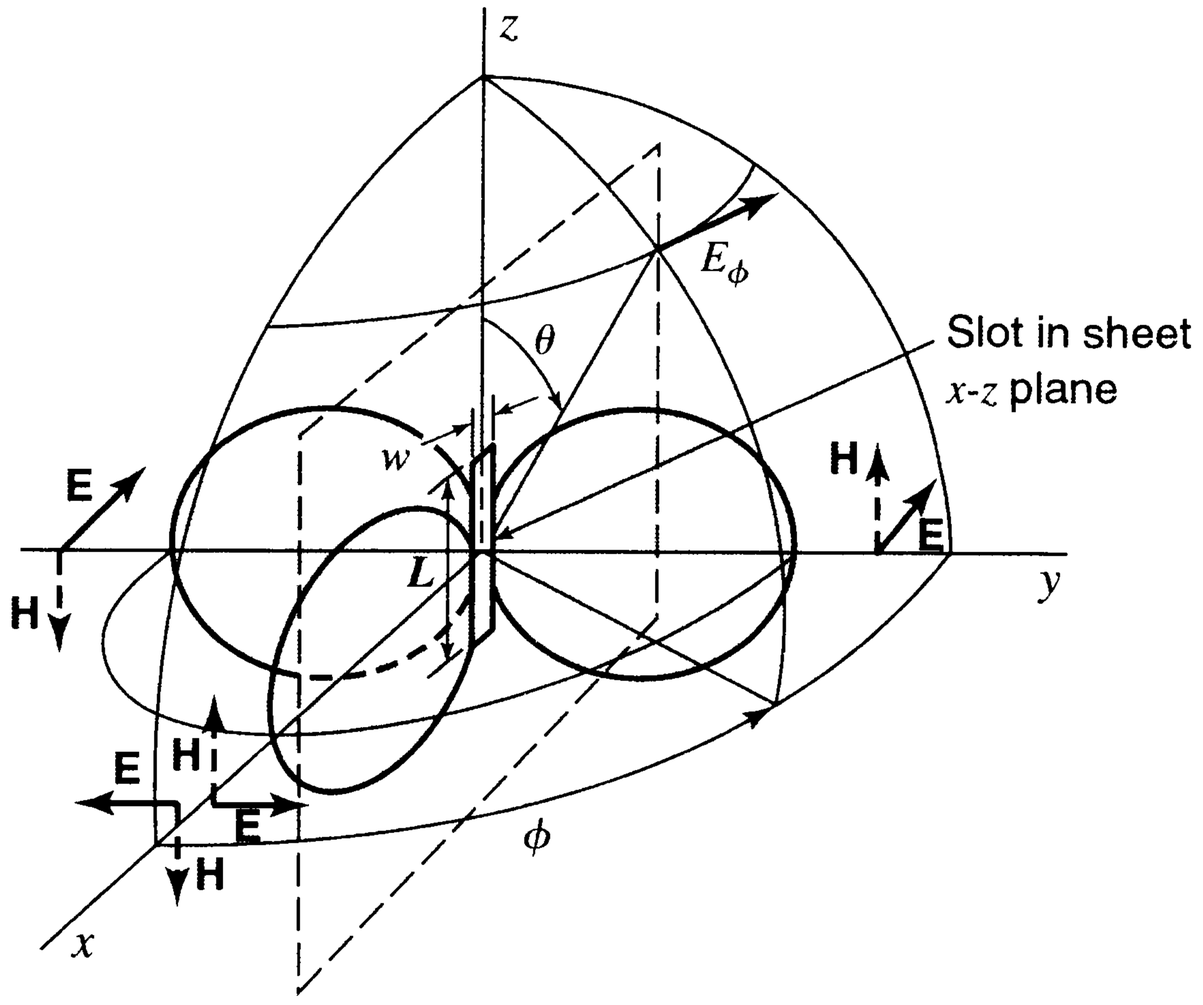


Figure 9–8
Radiation pattern of vertical slot in an infinite flat sheet.

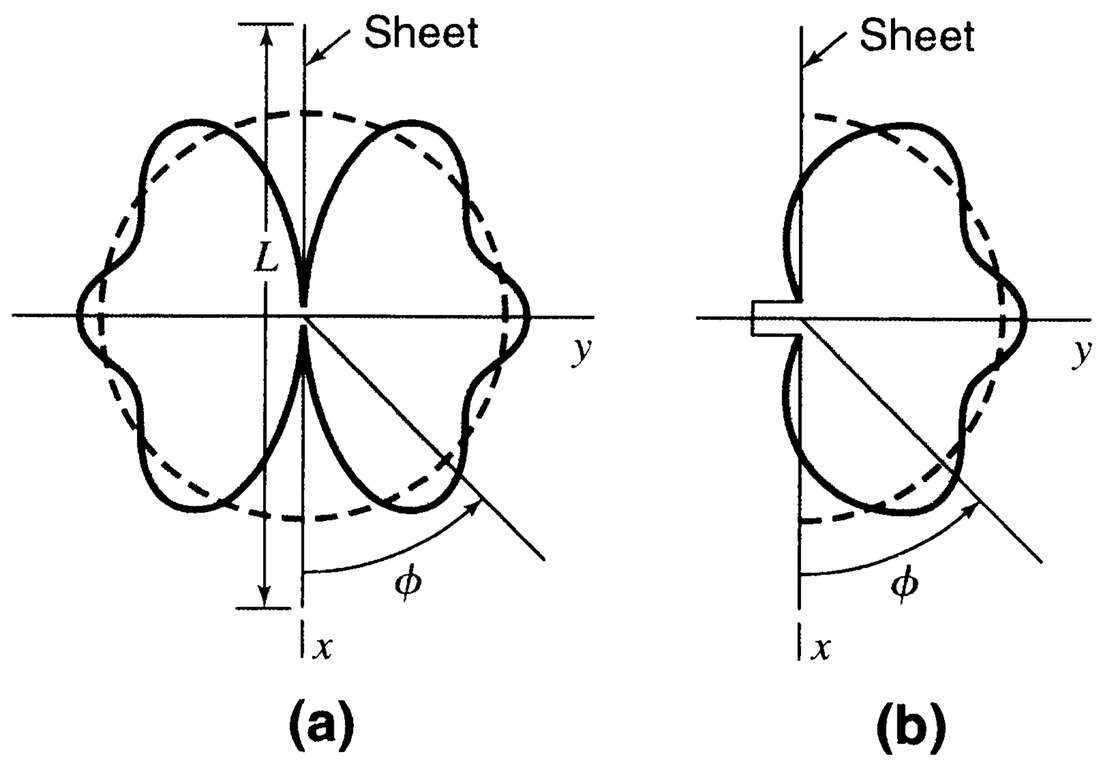


Figure 9–9
Solid curves show patterns in *xy* plane of Fig. 9–8 for slot in finite sheet of length *L*. Slot is open on both sides in (*a*) and closed on left side in (*b*). Dashed curves show pattern for infinite sheet. All patterns idealized.

position shown in Fig. 9–8. The radiation in this case is everywhere horizontally polarized; i.e., the electric field has only an E_ϕ component. If the slot is very thin ($w \ll \lambda$) and $\lambda/2$ long ($L = \lambda/2$), the variation of E_ϕ as a function of θ is given by

$$E_\phi(\theta) = \frac{\cos[(\pi/2)\cos\theta]}{\sin\theta} \tag{1}$$

Assuming that the sheet is perfectly conducting and infinite in extent, the magnitude of the field component E_ϕ remains constant as a function of ϕ for any value of θ. Thus,

$$E_\phi(\phi) = \text{constant} \tag{2}$$

Consider now the situation where the slot is cut in a sheet of finite extent as suggested by the dashed lines in Fig. 9–8. This change produces relatively little effect on the $E_\phi(\theta)$ pattern given by (1). However, there must be a drastic change in the $E_\phi(\phi)$ pattern since in the x direction, for example, the fields radiated from the two sides of the sheet are equal in magnitude but opposite in phase so that they cancel. Hence, there is a null in all directions in the plane of the sheet. For a sheet of given length L in the x direction, the field pattern in the xy plane might then be as indicated by the solid curve in Fig. 9–9*a*. The dashed curve

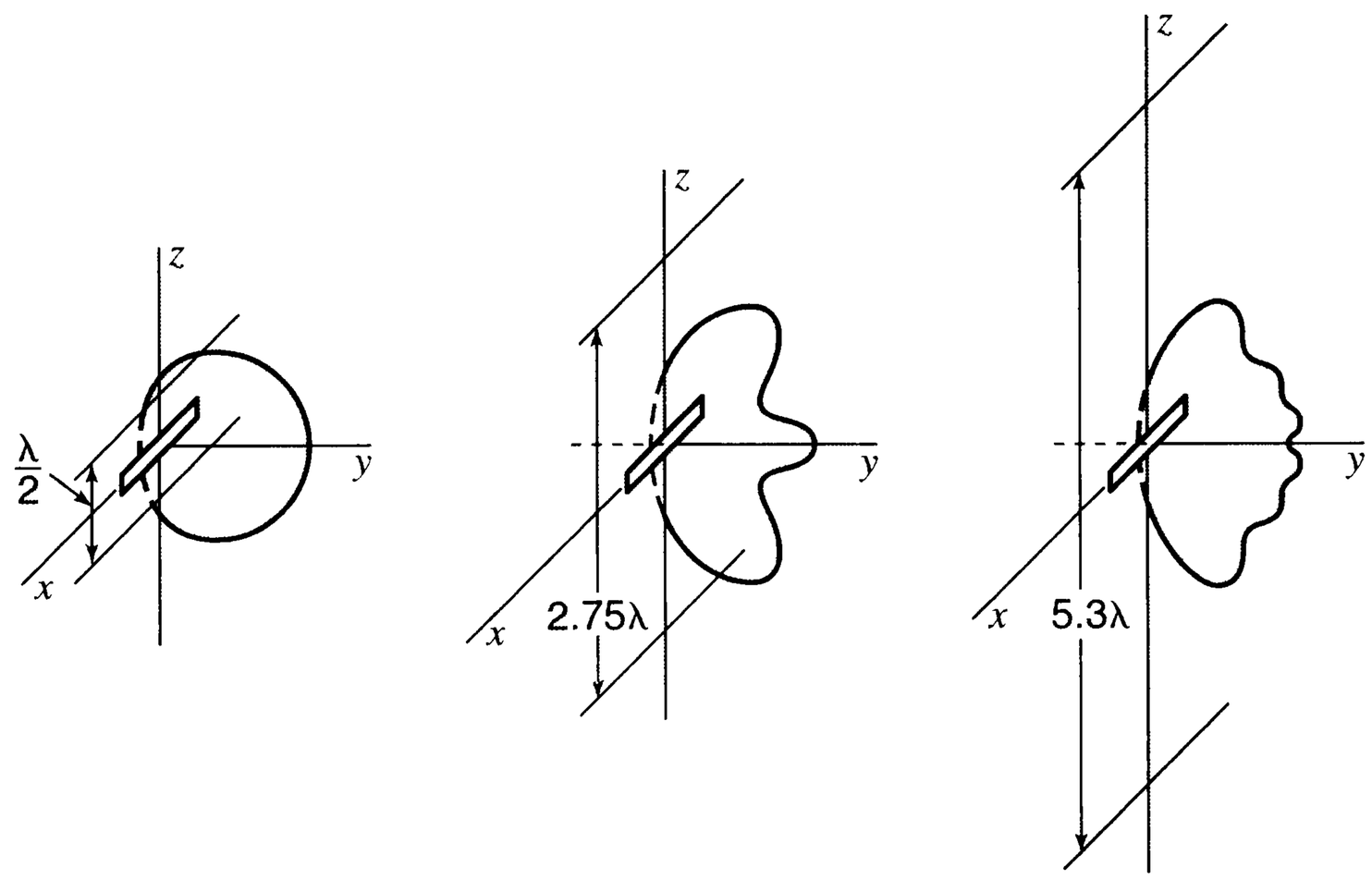


Figure 9–10
Measured ϕ-plane patterns of $\lambda/2$ boxed-in slot antennas in finite sheets of three lengths $L = 0.5$, 2.75 and 5.3λ. The width of the slots is 0.1λ. (*After Dorne (1) and Lazarus*).

is for an infinite sheet ($L = \infty$). If one side of the slot is boxed in, there is radiation in the plane of the sheet as suggested by the pattern in Fig. 9–9*b*.[1] With a finite sheet the pattern usually exhibits a scalloped or undulating characteristic, as suggested in Fig. 9–9. As the length L of the sheet is increased, the pattern undulations become more numerous but the magnitude of the undulations decreases, so that for a very large sheet the pattern conforms closely to a circular shape. Measured patterns illustrating this effect are shown in Fig. 9–10 for 3 values of L. A method due to Andrew Alford for locating the angular positions of the maxima and minima is described by Dorne (1) and Lazarus. In this method the assumption is made that the far field is produced by three sources (see Fig. 9–11), one (1) at the slot of strength $1 \sin \omega t$ and two (2 and 3) at the edges of the sheet (edge diffraction effect) with a strength $k \sin(\omega t - \delta)$, where $k \ll 1$ and δ gives the phase difference of the edge sources with respect to the source (1) at the slot. At the point P at a large distance in the direction ϕ, the relative field intensity is then

$$E = \sin \omega t + k \sin(\omega t - \delta - \varepsilon) + k \sin(\omega t - \delta + \varepsilon) \tag{3}$$

where $\varepsilon = (\pi/\lambda) L \cos \phi$

By trigonometric expansion and rearrangement,

$$E = (1 + 2k \cos \delta \cos \varepsilon) \sin \omega t - (2k \sin \delta \cos \varepsilon) \cos \omega t \tag{4}$$

and the modulus of E is

$$|E| = \sqrt{(1 + 2k \cos \delta \cos \varepsilon)^2 + (2k \sin \delta \cos \varepsilon)^2} \tag{5}$$

Squaring and neglecting terms with k^2, since $k \ll 1$, (5) reduces to

$$|E| = \sqrt{1 + 4k \cos \delta \cos \varepsilon} \tag{6}$$

---

[1] According to H. G. Booker (1), the energy density in the $\phi = 0$ or 180° directions is $\frac{1}{2}$ that for an infinite sheet, or the field intensity is 0.707 that for an infinite sheet.

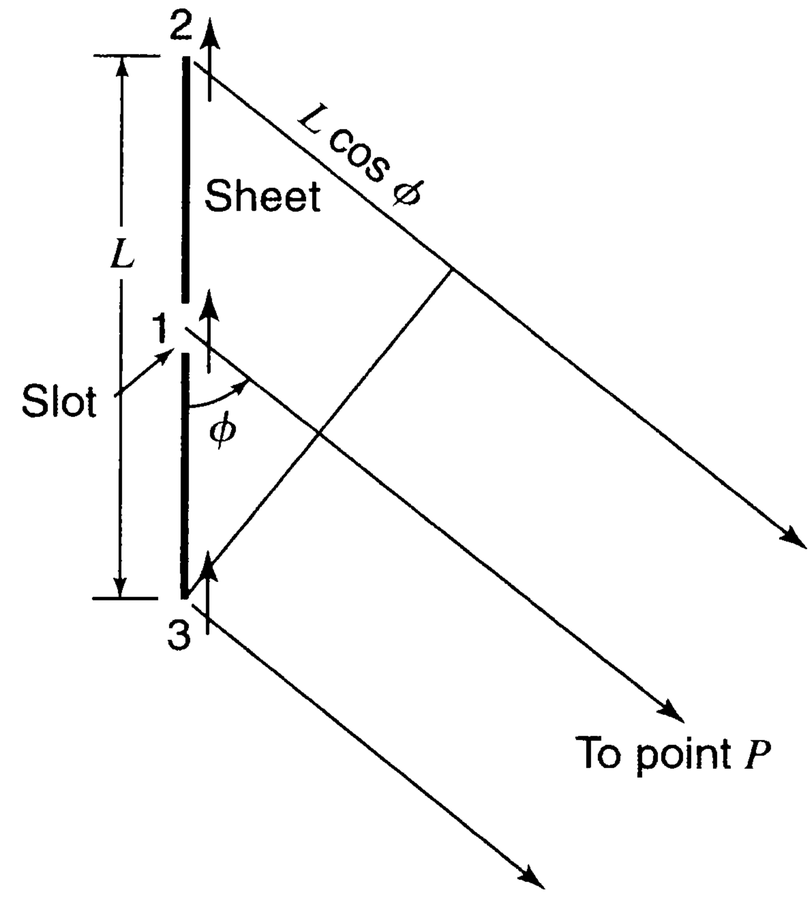


Figure 9–11
Construction for locating maxima and minima of ϕ pattern for slot in a finite sheet.

The maximum and minimum values of $|E|$ as a function of ε occur when $\varepsilon = n\pi$, so that

$$\varepsilon = \frac{\pi}{\lambda} L \cos\phi = n\pi \tag{7}$$

where n is an integer. Thus

$$\cos\phi = \frac{n\lambda}{L} \qquad \text{and} \qquad \phi = \arccos\frac{n\lambda}{L} \tag{8}$$

The values of ϕ for maxima and minima in the ϕ pattern are given by (8). These locations are independent of k and δ. If cos δ is positive, then the maxima correspond to even values of n and the minima to odd values of n.

9–3 BABINET'S PRINCIPLE AND COMPLEMENTARY ANTENNAS
（巴比涅原理和互补天线）

By means of Babinet's (Ba-bi-naý's) principle many of the problems of slot antennas can be reduced to situations involving complementary linear antennas for which solutions have already been obtained. In optics Babinet's principle (Born-1) may be stated as follows:

The field at any point behind a plane having a screen, if added to the field at the same point when the complementary screen is substituted, is equal to the field when no screen is present. （位于屏障所在平面后方任意点处的场，加上用互补屏障替换后在同点处的场，等于全无屏障时该点处的场。）

The principle may be illustrated by considering an example with 3 cases. Let a source and 2 imaginary planes, plane of screens A and plane of observation B, be arranged as in Fig. 9–12. As Case 1, let a perfectly absorbing screen be placed in plane A. Then in plane B there is a region of shadow as indicated. Let the field behind this screen be some function f_1 of x, y and z. Thus,

$$F_s = f_1(x, y, z) \tag{1}$$

As Case 2 let the first screen be replaced by its complementary screen and the field behind it be given by

$$F_{cs} = f_2(x, y, z) \tag{2}$$

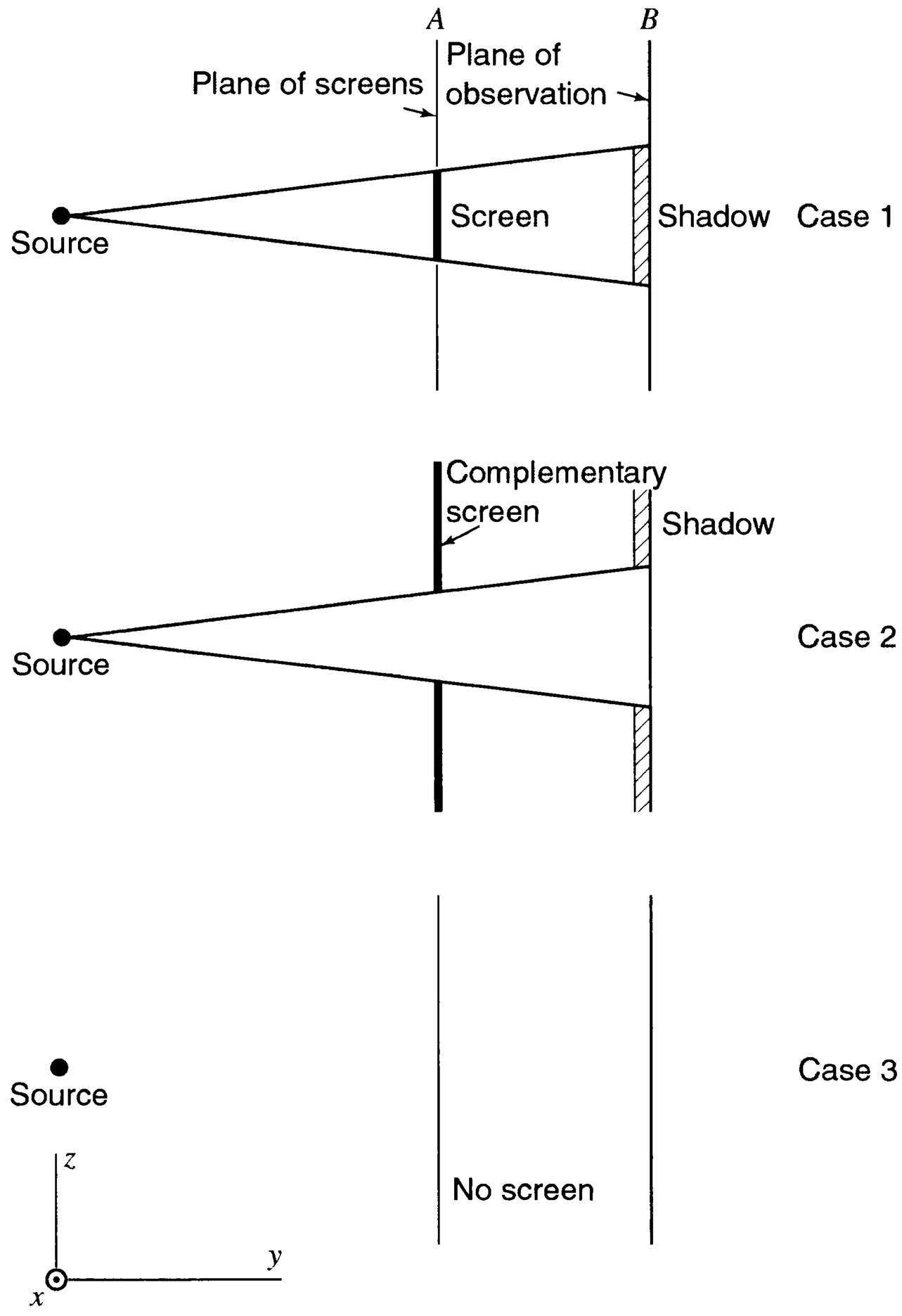


Figure 9–12
Illustration of Babinet's principle.

As Case 3 with no screen present the field is

$$F_0 = f_3(x, y, z) \tag{3}$$

Then Babinet's principle asserts that at the same point x_1, y_1, z_1,

$$F_s + F_{cs} = F_0 \tag{4}$$

The source may be a point as in the above example or a distribution of sources. The principle applies not only to points in the plane of observation B as suggested in Fig. 9–12 but also to any point behind screen A. Although the principle is obvious enough for the simple shadow case above, it also applies where diffraction is considered.

Babinet's principle has been extended and generalized by Booker (1) to take into account the vector nature of the electromagnetic field. In this extension it is assumed that the screen is plane, perfectly conducting and infinitesimally thin. Furthermore, if one screen is perfectly conducting ($\sigma = \infty$), the complementary screen must have infinite permeability ($\mu = \infty$). Thus, if one screen is a perfect conductor of electricity, the complementary

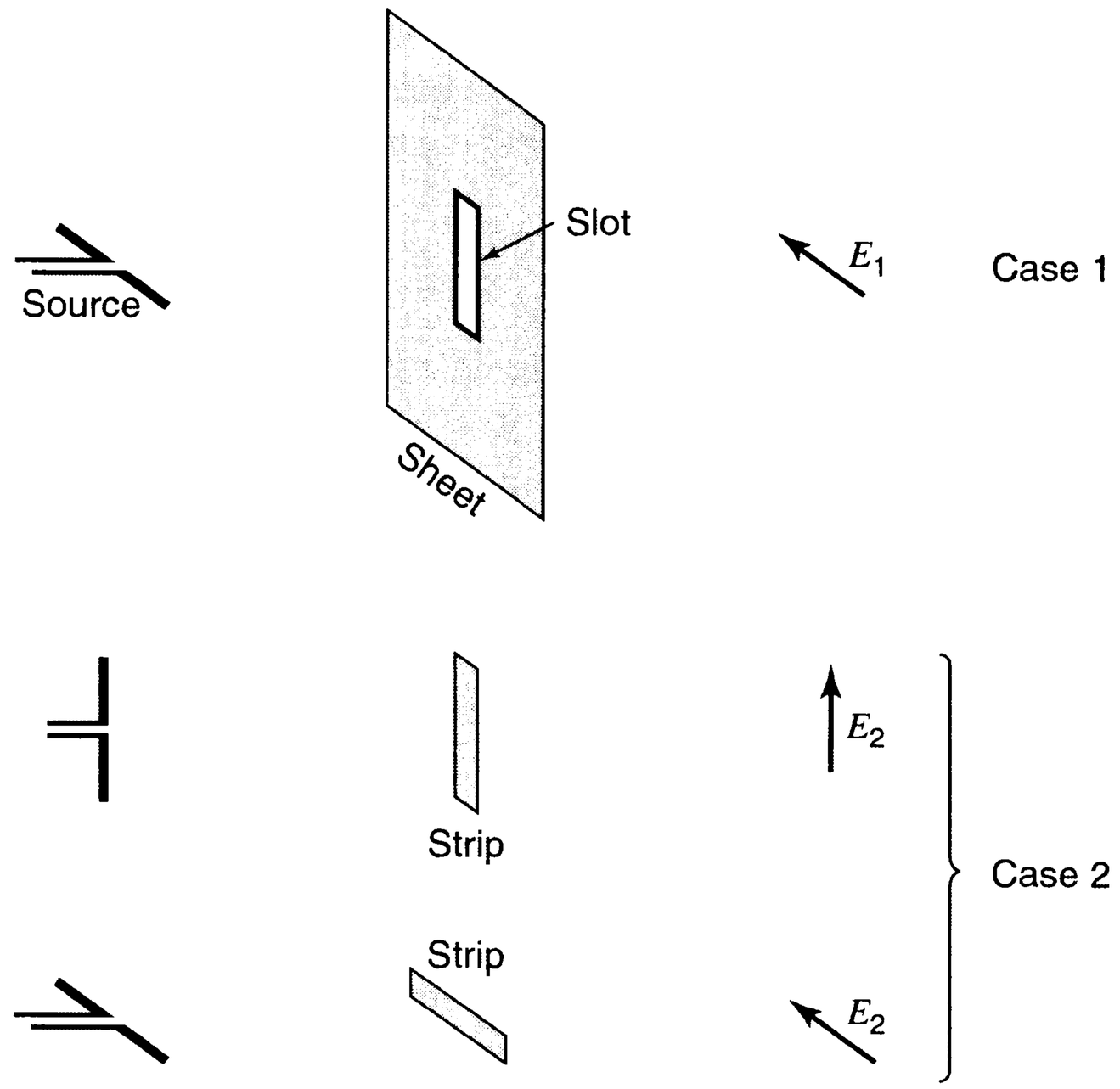


Figure 9–13
Illustration of Babinet's principle applied to a slot in an infinite metal sheet and the complementary metal strip.

screen is a perfect "conductor" of magnetism. No infinitely permeable material exists, but the equivalent effect may be obtained by making both the original and complementary screens of perfectly conducting material and interchanging electric and magnetic quantities everywhere. The only perfect conductors are *superconductors* which soon may be available at ordinary temperatures for antenna applications. However, many metals, such as silver and copper, have such high conductivity that we may assume the conductivity is infinite with a negligible error in most applications.

As an illustration of Booker's extension of Babinet's principle, consider the cases in Fig. 9–13. The source in all cases is a short dipole, theoretically an infinitesimal dipole. In Case 1 the dipole is horizontal and the original screen is an infinite, perfectly conducting, plane, infinitesimally thin sheet with a vertical slot cut out as indicated. At a point P behind the screen the field is E_1. In Case 2 the original screen is replaced by the complementary screen consisting of a perfectly conducting, plane, infinitesimally thin strip of the same dimensions as the slot in the original screen. In addition, the dipole source is turned vertical so as to interchange **E** and **H**. At the same point P behind the screen the field is E_2. As an alternative situation for Case 2 the dipole source is horizontal and the strip is also turned horizontal. Finally, in Case 3 no screen is present and the field at point P is E_0. Then, by

Babinet's principle,

$$E_1 + E_2 = E_0 \tag{5}$$

or

$$\frac{E_1}{E_0} + \frac{E_2}{E_0} = 1 \tag{6}$$

Babinet's principle may also be applied to points in front of the screens. In the situation of Case 1 (Fig. 9–13) a large amount of energy may be transmitted through the slot so that the field E_1 may be about equal to the field E_0 with no intermediate screen (Case 3). In such a situation the complementary dipole acts like a reflector and E_2 is very small. (See Chap. 18 on frequency-sensitive surfaces.) Since a metal sheet with a $\lambda/2$ slot or, in general, an orifice of at least 1λ perimeter may transmit considerable energy, slots or orifices of this size should be assiduously avoided in sheet reflectors such as described in Chap. 10 when **E** is not parallel to the slot.

9–4 THE IMPEDANCE OF COMPLEMENTARY SCREENS（互补屏的阻抗）

In this section Babinet's principle is applied with the aid of a transmission-line analogy to finding the relation between the surface impedance Z_1 of a screen and the surface impedance Z_2 of the complementary metal screen (Booker-1).

Consider the infinite transmission line shown in Fig. 9–14*a* of characteristic impedance Z_0 or characteristic admittance $Y_0 = 1/Z_0$. Let a shunt admittance Y_1 be placed across the line. An incident wave traveling to the right of voltage V_i is partly reflected at Y_1 as a wave of voltage V_r and partly transmitted beyond Y_1 as a wave of voltage V_t. The voltages are measured across the line.

This situation is analogous to a plane wave of field intensity E_i incident normally on a plane screen of infinite extent with a surface admittance, or admittance per square, of Y_1; that is, the admittance measured between the opposite edges of any square section of the sheet as in Fig. 9–14*c* is Y_1. Neglecting the impedance of the leads the admittance

$$Y_1 = \frac{I}{V} \qquad (\mho \text{ per square}) \tag{1}$$

The value of Y is the same for *any square section* of the sheet. Thus, the section may be 1 cm square or 1 meter square. Hence, (1) has the dimensions of admittance rather than of admittance per length squared and is called a *surface admittance,* or *admittance per square.*

The field intensities of the waves reflected and transmitted normally to the screen are E_r and E_t. Let the medium surrounding the screen be free space. It has a characteristic admittance Y_0 which is a pure conductance G_0. Thus,

$$Y_0 = \frac{1}{Z_0} = \frac{1}{377} = G_0 \qquad (\mho) \tag{2}$$

The ratio of the magnetic to the electric field intensity of any plane traveling wave in free space has this value. Hence,

$$Y_0 = \frac{H_i}{E_i} = -\frac{H_r}{E_r} = \frac{H_t}{E_t} \qquad (\mho) \tag{3}$$

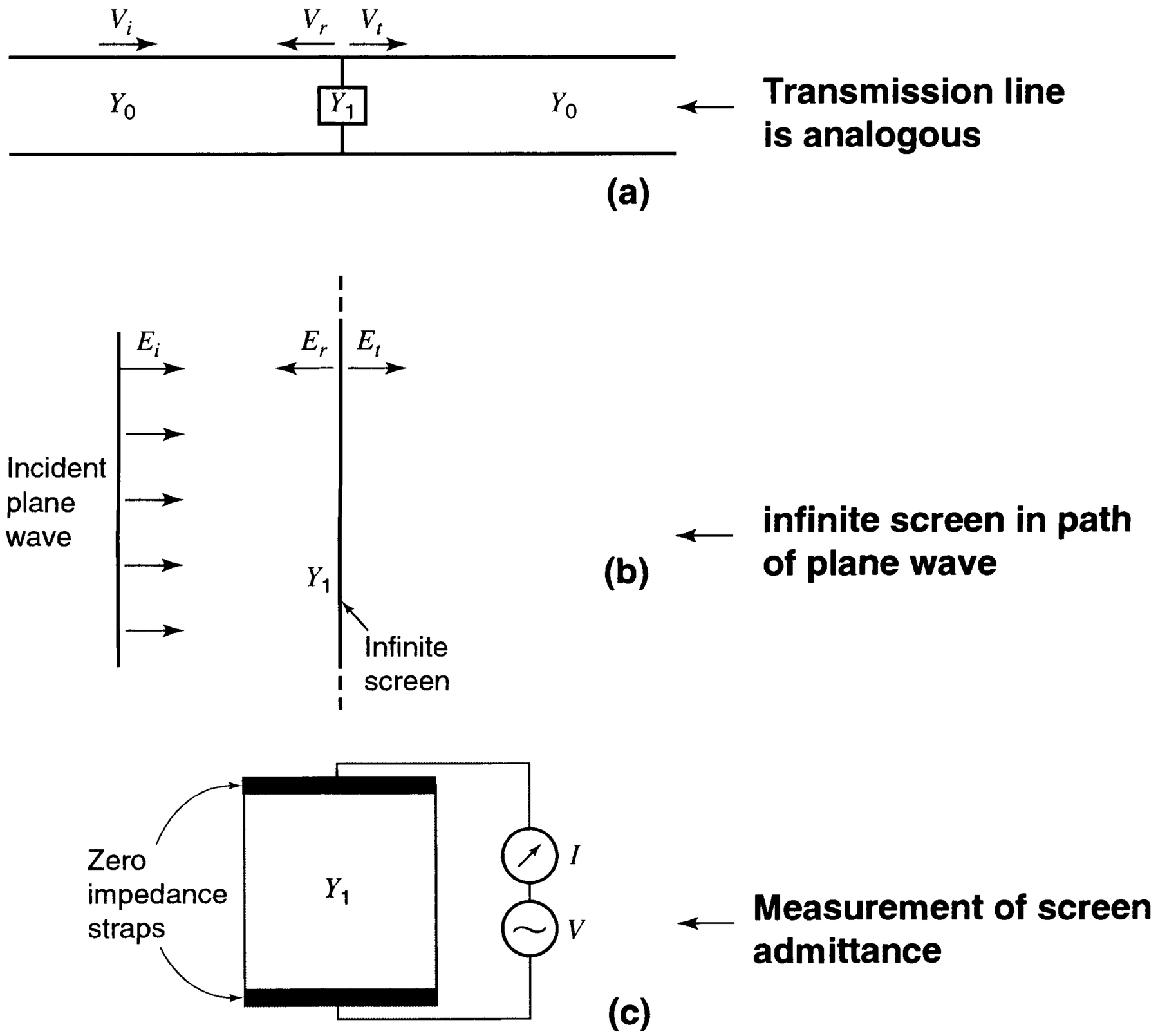


Figure 9–14
Shunt admittance across transmission line (*a*) is analogous to infinite screen in path of plane wave (*b*). Method of measuring surface admittance of screen is suggested in (*c*).

where H_i, H_r and H_t are the magnetic field intensities of the incident, reflected and transmitted waves respectively.

The transmission coefficient for voltage τ_v of the transmission line (Fig. 9–14*a*) is (Schelkunoff-1):

$$\tau_v = \frac{V_t}{V_i} = \frac{2Y_0}{2Y_0 + Y_1} \tag{4}$$

By analogy the transmission coefficient for the electric field (Fig. 9–14*b*) is

$$\tau_E = \frac{E_t}{E_i} = \frac{2Y_0}{2Y_0 + Y_1} \tag{5}$$

If now the original screen is replaced by its complementary screen with an admittance per square of Y_2, the new transmission coefficient is the ratio of the new transmitted field E_t' to the incident field. Thus,

$$\tau_E' = \frac{E_t'}{E_i} = \frac{2Y_0}{2Y_0 + Y_2} \tag{6}$$

Applying Babinet's principle, we have from (9–3–6) that

$$\frac{E_t}{E_i} + \frac{E_t'}{E_i} = 1 \tag{7}$$

or

$$\tau_E + \tau_E' = 1 \tag{8}$$

Therefore,

$$\frac{2Y_0}{2Y_0 + Y_1} + \frac{2Y_0}{2Y_0 + Y_2} = 1 \tag{9}$$

and we obtain Booker's result that

$$Y_1 Y_2 = 4Y_0^2 \tag{10}$$

Since $Y_1 = 1/Z_1$, $Y_2 = 1/Z_2$ and $Y_0 = 1/Z_0$,

$$Z_1 Z_2 = \frac{Z_0^2}{4} \qquad \text{or} \qquad \sqrt{Z_1 Z_2} = \frac{Z_0}{2} \tag{11}$$

Thus, the geometric mean of the impedances of the two screens equals $\frac{1}{2}$ the intrinsic impedance of the surrounding medium. Since, for free space, $Z_0 = 376.7\ \Omega$,

$$Z_1 = \frac{35{,}476}{Z_2} \qquad (\Omega) \tag{12}$$

If screen 1 is an infinite grating of narrow parallel metal strips as in Fig. 9–15*a*, then the complementary screen (screen 2) is an infinite grating of narrow slots as shown in Fig. 9–15*b*. Suppose that a low-frequency plane wave is incident normally on screen 1 with the electric field parallel to the strips. Then the grating acts as a perfectly reflecting screen and zero field penetrates to the rear. Thus $Z_1 = 0$ and, from (12), $Z_2 = \infty$, so that the complementary screen of slots (screen 2) offers no impediment to the passage of the wave. If the frequency is increased sufficiently, screen 1 begins to transmit part of the incident wave. If at the frequency F_0 screen 1 has a surface impedance $Z_1 = j188\ \Omega$ per square, the impedance Z_2 of screen 2 is $-j188\ \Omega$ per square, so that both screens transmit equally well. If screen 1 becomes more transparent (Z_1 larger) as the frequency is further increased, screen 2 will become more opaque (Z_2 smaller). At any frequency the sum of the fields

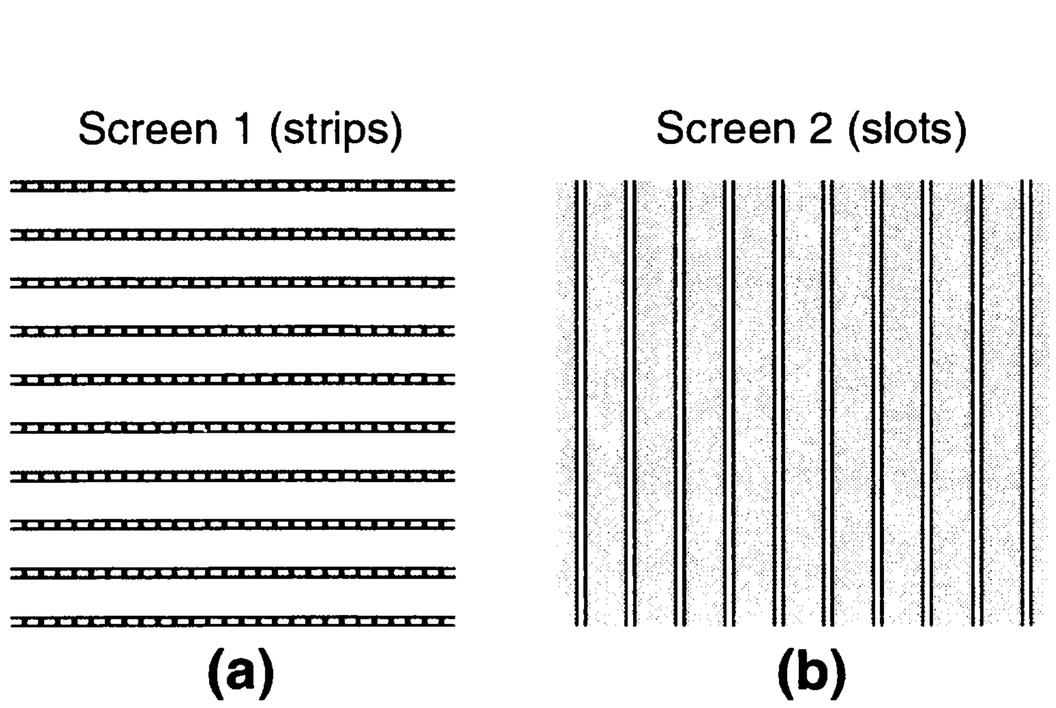


Figure 9–15
Screen of parallel strips (*a*) and complementary screen of slots (*b*).

transmitted through screen 1 and through screen 2 is a constant and equal to the field without any screen present.

9–5 THE IMPEDANCE OF SLOT ANTENNAS（缝隙天线的阻抗）

In this section a relation is developed for the impedance Z_s of a slot antenna in terms of the impedance Z_d of the complementary dipole antenna (Booker-1). Knowing Z_d for the dipole, the impedance Z_s of the slot can then be determined.

Consider the slot antenna shown in Fig. 9–16*a* and the complementary dipole antenna shown in Fig. 9–16*b*. The terminals of each antenna are indicated by *FF*, and it is assumed that they are separated by an infinitesimal distance. It is assumed that the dipole and slot are cut from an infinitesimally thin, plane, perfectly conducting sheet.

Let a generator be connected to the terminals of the slot. The driving-point impedance Z_s at the terminals is the ratio of the terminal voltage V_s to the terminal current I_s. Let $\mathbf{E}_s$ and $\mathbf{H}_s$ be the electric and magnetic fields of the slot at any point P. Then the voltage V_s at the terminals *FF* of the slot is given by the line integral of $\mathbf{E}_s$ over the path C_1 (Fig. 9–16*a*) as C_1 approaches zero. Thus,

$$V_s = \lim_{C_1 \to 0} \int_{C_1} \mathbf{E}_s \cdot d\mathbf{l} \tag{1}$$

where $d\mathbf{l}$ = infinitesimal vector element of length $d\mathbf{l}$ along the contour or path C_1

The current I_s at the terminals of the slot is

$$I_s = 2 \lim_{C_2 \to 0} \int_{C_2} \mathbf{H}_s \cdot d\mathbf{l} \tag{2}$$

The path C_2 is just outside the metal sheet and parallel to its surface. The factor 2 enters because only $\frac{1}{2}$ of the closed line integral is taken, the line integral over the other side of the sheet being equal by symmetry.

Turning our attention to the complementary dipole antenna, let a generator be connected to the terminals of the dipole. The driving-point impedance Z_d at the terminals is the ratio

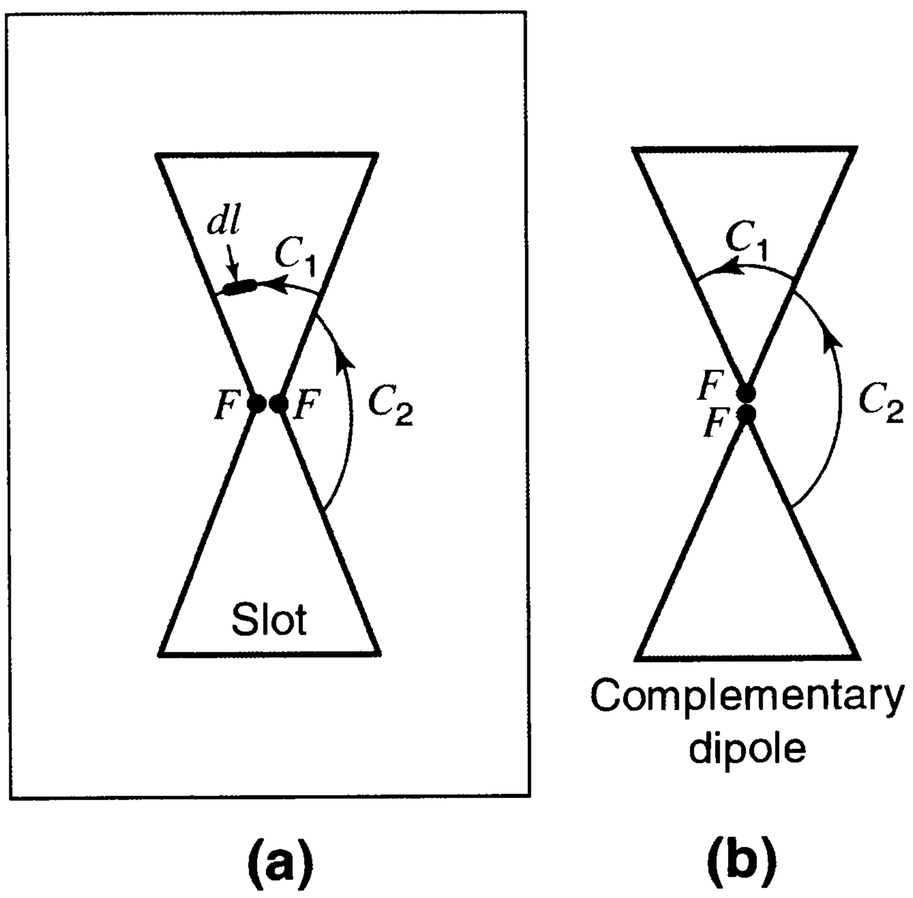


Figure 9–16
Slot antenna (*a*) and complementary dipole antenna (*b*).

of the terminal voltage V_d to the terminal current I_d. Let $\mathbf{E}_d$ and $\mathbf{H}_d$ be the electric and magnetic fields of the dipole at any point P. Then the voltage at the dipole terminals is

$$V_d = \lim_{C_2 \to 0} \int_{C_2} \mathbf{E}_d \cdot d\mathbf{l} \tag{3}$$

and the current is

$$I_d = 2 \lim_{C_1 \to 0} \int_{C_1} \mathbf{H}_d \cdot d\mathbf{l} \tag{4}$$

However,

$$\lim_{C_2 \to 0} \int_{C_2} \mathbf{E}_d \cdot d\mathbf{l} = Z_0 \lim_{C_2 \to 0} \int_{C_2} \mathbf{H}_s \cdot d\mathbf{l} \tag{5}$$

and

$$\lim_{C_1 \to 0} \int_{C_1} \mathbf{H}_d \cdot d\mathbf{l} = \frac{1}{Z_0} \lim_{C_1 \to 0} \int_{C_1} \mathbf{E}_s \cdot d\mathbf{l} \tag{6}$$

where Z_0 is the intrinsic impedance of the surrounding medium. Substituting (3) and (2) in (5) yields

$$V_d = \frac{Z_0}{2} I_s \tag{7}$$

Substituting (4) and (1) in (6) gives

$$V_s = \frac{Z_0}{2} I_d \tag{8}$$

Multiplying (7) and (8) we have

$$\frac{V_s}{I_s} \frac{V_d}{I_d} = \frac{Z_0^2}{4} \tag{9}$$

or

$$Z_s Z_d = \frac{Z_0^2}{4} \qquad \text{or} \qquad Z_s = \frac{Z_0^2}{4 Z_d} \tag{10}$$

Thus, we obtain Booker's result that the terminal impedance Z_s of a slot antenna is equal to $\frac{1}{4}$ of the square of the intrinsic impedance of the surrounding medium divided by the terminal impedance Z_d of the complementary dipole antenna. For free space $Z_0 = 376.7\ \Omega$, so[1]

(互补 – 偶极子的阻抗)

$$\boxed{\textit{Slot-dipole impedances} \qquad Z_s = \frac{Z_0^2}{4 Z_d} = \frac{35{,}476}{Z_d} \qquad (\Omega)} \tag{11}$$

[1] If the intrinsic impedance Z_0 of free space were unknown, (11) provides a means of determining it by measurements of the impedance Z_s of a slot antenna and the impedance Z_d of the complementary dipole antenna. The impedance Z_0 is twice the geometric means of Z_s and Z_d or

$$Z_0 = 2\sqrt{Z_s Z_d} \tag{12}$$

The impedance of the slot is proportional to the admittance of the dipole, or vice versa. Since, in general, Z_d may be complex, we may write

$$Z_s = \frac{35{,}476}{R_d + jX_d} = \frac{35{,}476}{R_d^2 + X_d^2}(R_d - jX_d) \tag{13}$$

where R_d and X_d are the resistive and reactive components of the dipole terminal impedance Z_d. Thus, if the dipole antenna is inductive, the slot is capacitive, and vice versa. Lengthening a $\lambda/2$ dipole makes it more inductive, but lengthening a $\lambda/2$ slot makes it more capacitive.

Let us now consider some numerical examples, proceeding from known dipole types to the complementary slot types.

EXAMPLE **9–5.1 Thin $\lambda/2$ Dipole**

The impedance of an *infinitesimally thin $\lambda/2$ antenna* ($L = 0.5\lambda$ and $L/D = \infty$) is $73 + j42.5\ \Omega$ (see Chap. 13). Therefore, the terminal impedance of an *infinitesimally thin $\lambda/2$ slot antenna* ($L = 0.5\lambda$ and $L/w = \infty$) is

$$Z_1 = \frac{35{,}476}{73 + j42.5} = 363 - j211\ \Omega \tag{14}$$

See Fig. 9–17*a*.

EXAMPLE **9–5.2 Resonant $\lambda/2$ Dipole**

As more practical example, a cylindrical antenna with a length-diameter ratio of 100 ($L/D = 100$) is resonant when the length is about 0.475λ ($L = 0.475\lambda$). The terminal impedance is resistive and equal to about 67 Ω. The terminal resistance of the complementary slot antenna is then

$$Z_1 = \frac{35{,}476}{67} = 530 + j0\ \Omega \tag{15}$$

See Fig. 9–17*b*.

The complementary slot has a length $L = 0.475\lambda$, the same as for the dipole, but the width of the slot should be twice the diameter of the cylindrical dipole. A flat strip of width w is equivalent to a cylindrical conductor of diameter D provided that $w = 2D$. Thus, in this example, the width of the complementary slot is

$$w = 2D = \frac{2L}{100} = \frac{2 \times 0.475\lambda}{100} \simeq 0.01\lambda \tag{16}$$

EXAMPLE **9–5.3 Fullwave Dipole**

As a third example, a cylindrical dipole with an L/D ratio of 28 and length of about 0.925λ has a terminal resistance of about $710 + j0\ \Omega$. The terminal resistance of the

complementary slot is then about $50 + j0\ \Omega$ so that an impedance match will be provided to a 50 Ω coaxial line. See Fig. 9–17*c*.

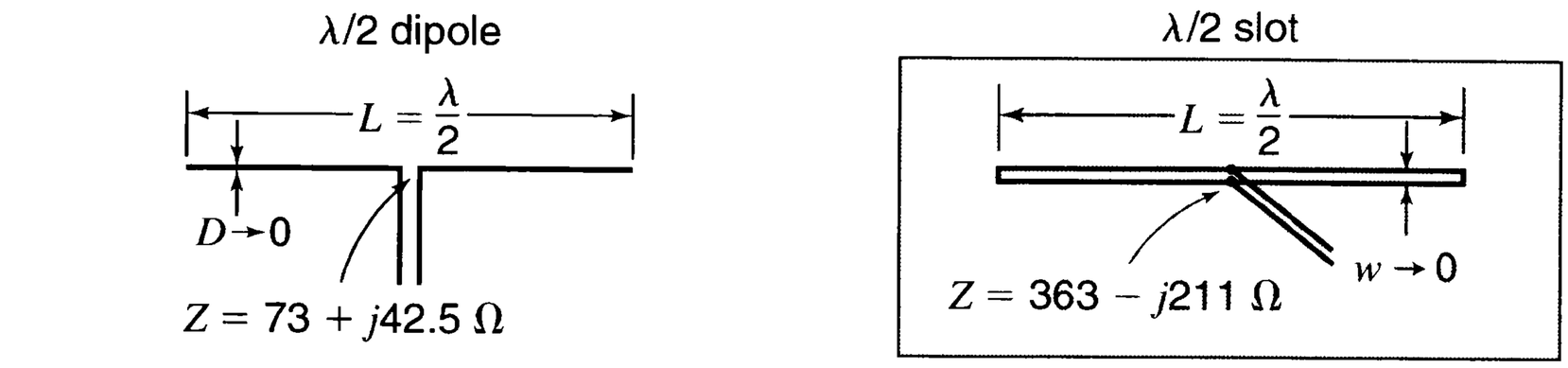


Figure 9–17
Comparison of impedances of cylindrical dipole antennas with complementary slot antennas. The slot in (*c*) matches directly to the 50 Ω coaxial line.

If the slots in these examples are enclosed on one side of the sheet with a box of such size that zero susceptance is shunted across the slot terminals, due to the box, the impedances are doubled.

The bandwidth or selectivity characteristics of a slot antenna are the same as for the complementary dipole. Thus, widening a slot (smaller L/w ratio) increases the bandwidth of the slot antenna, the same as increasing the thickness of a dipole antenna (smaller L/D ratio) increases its bandwidth.

The above discussion of this section applies to slots in sheets of infinite extent. If the sheet is finite, the impedance values are substantially the same provided that the edge of the sheet is at least a wavelength from the slot. However, the measured slot impedance is sensitive to the nature of the terminal connections.

9–6 SLOTTED CYLINDER ANTENNAS（柱面缝隙天线）

A slotted sheet antenna is shown in Fig. 9–18*a* (Alford-1, 2; Jordan-1; Sinclair-1). By bending the sheet into a U-shape as in (*b*) and finally into a cylinder as in (*c*), we arrive at a slotted cylinder antenna. The impedance of the path around the circumference of the cylinder may be sufficiently low so that most of the current tends to flow in horizontal loops around the cylinder as suggested. If the diameter D of the cylinder is a sufficiently small fraction of a wavelength, say, less than $\lambda/8$, the vertical slotted cylinder radiates a horizontally polarized field with a pattern in the horizontal plane which is nearly circular. As the diameter of the cylinder is increased, the pattern in the horizontal plane tends to become more unidirectional with the maximum radiation from the side of the cylinder with the slot. For resonance, the length L of the slot is greater than $\lambda/2$. This may be explained as follows. Referring to Fig. 9–19*a*, the 2-wire transmission line is resonant when it is $\lambda/2$ long. However, if this line is loaded with a series of loops of diameter D as at (*b*), the phase velocity of wave transmission on the line can be increased, so that the resonant frequency is raised. With a sufficient number of shunt loops the arrangement of (*b*) becomes equivalent to a slotted cylinder of diameter D. Typical slotted cylinder dimensions for resonance are $D = 0.125\lambda$, $L = 0.75\lambda$ and the slot width about 0.02λ.

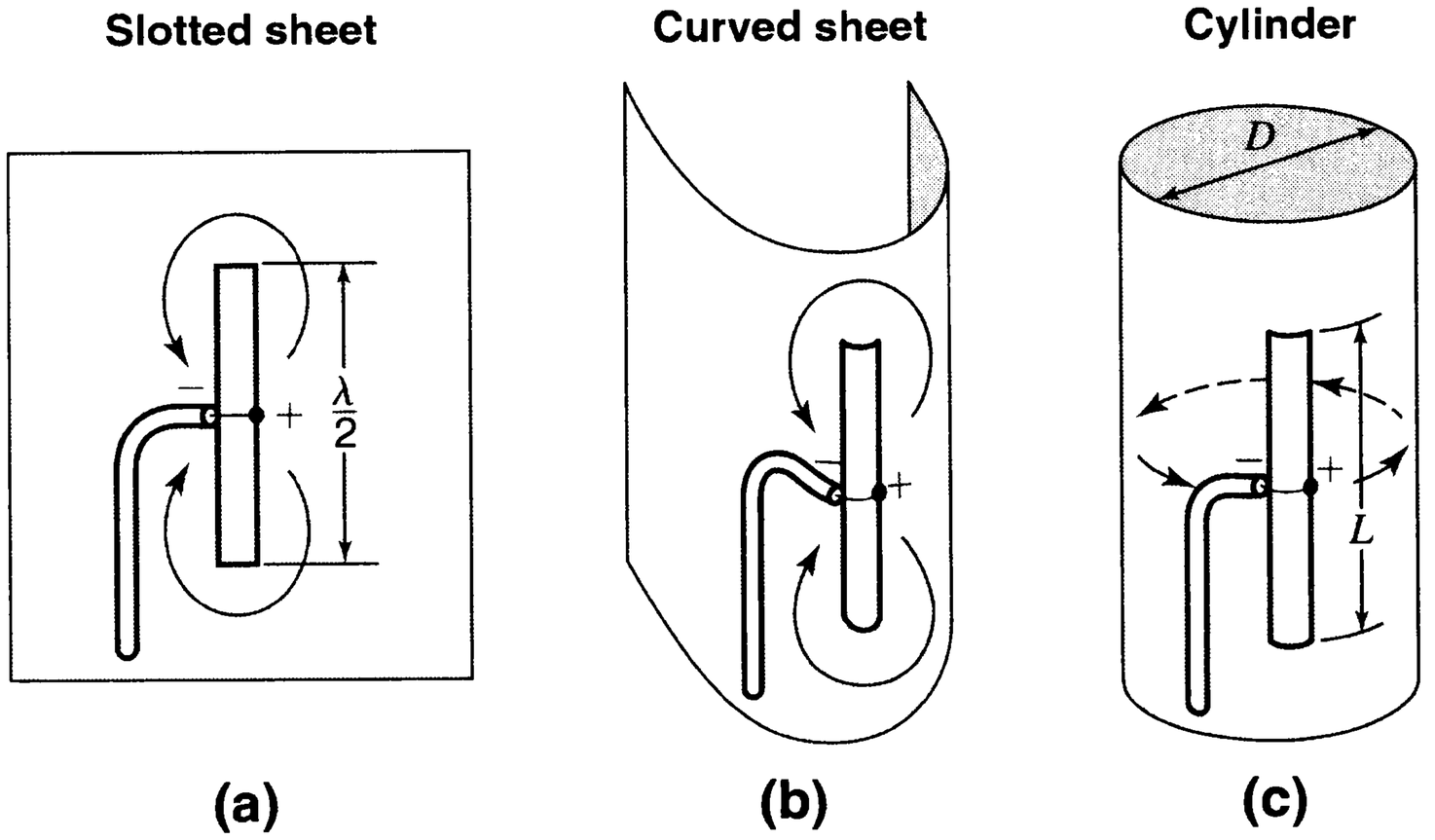


Figure 9–18
Evolution of slotted cylinder from slotted sheet.

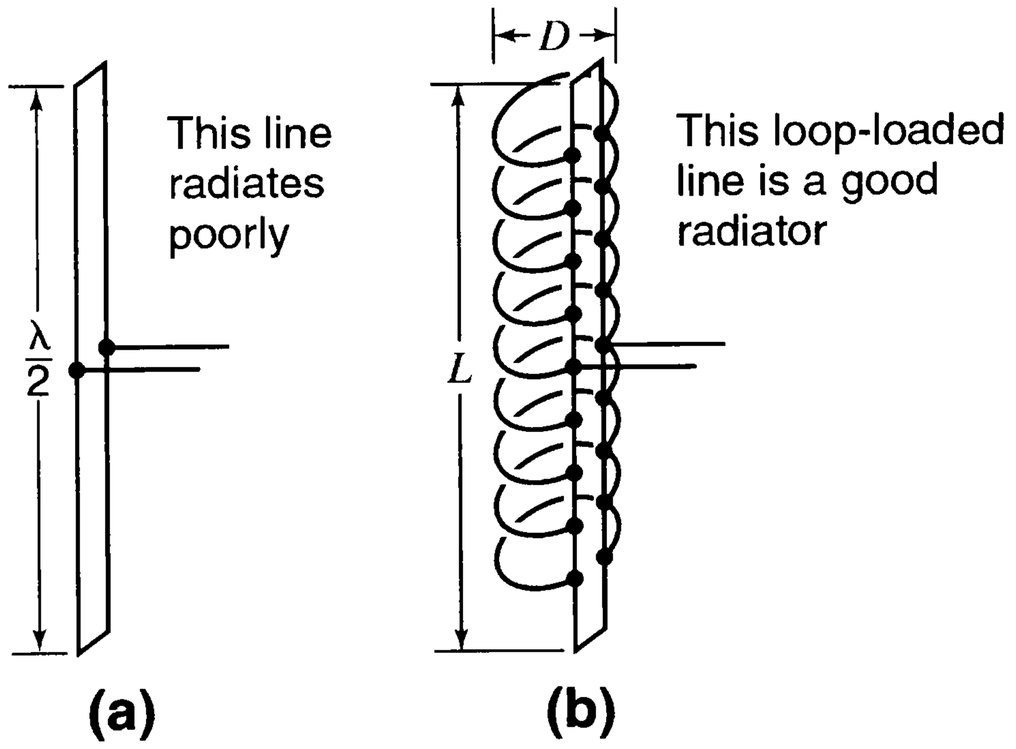


Figure 9–19
Slotted cylinder as a loop-loaded transmission line.

This type of antenna, pioneered by Andrew Alford, has found considerable application for television broadcasting of a horizontally polarized wave with an omnidirectional or circular pattern in the horizontal plane. Vertical-plane directivity may be increased by using stacked collinear slots in a long vertical cylinder.

9–7 PATCH OR MICROSTRIP ANTENNAS（贴片或微带天线）

These antennas are popular for low-profile applications at frequencies above 100 MHz ($\lambda_0 < 3$ m). They commonly consist of a rectangular or square metal patch on a thin layer of dielectric (called the substrate) on a ground plane. Patches may be photoetched making them adapted for low-cost, mass production.

Typical patch dimensions of length L, width W and thickness t are indicated in Fig. 9–20 with feed from a coaxial line at the center of the left edge. The horizontal components of the electric fields at the left and right edges are in the same direction, giving in-phase linearly polarized radiation with a maximum broadside to the patch (up in Fig. 9–20*b*).

The patch acts as a resonant $\lambda/2$ parallel-plate microstrip transmission line with characteristic impedance equal to the reciprocal of the number n of parallel field-cell transmission lines (Kraus (1) and Fleisch). Each field-cell transmission line has a characteristic impedance equal to the intrinsic impedance Z_i of the medium where

$$Z_i = \sqrt{\frac{\mu}{\varepsilon}} = Z_0\sqrt{\frac{\mu_r}{\varepsilon_r}} \qquad (\Omega) \tag{1}$$

For air, $\mu_r = \varepsilon_r = 1$ and $Z_i = Z_0 = 377\,\Omega$.

An end view of the patch (from the left side) is shown in Fig. 9–21. The cross section (as drawn) has 10 parallel field-cell transmission lines so that for $\varepsilon_r = 2$ the microstrip

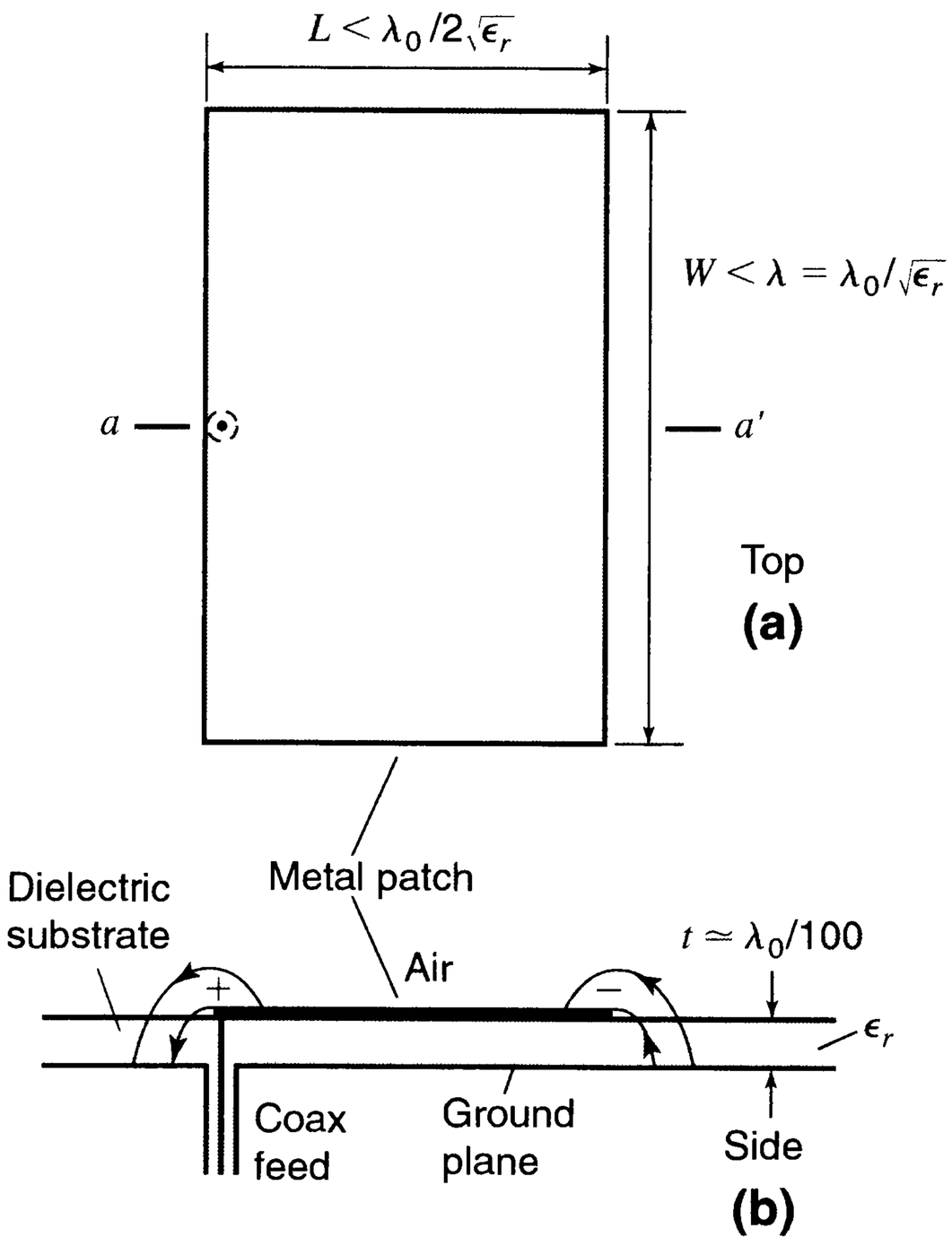


Figure 9–20
Top view (*a*) and side (cross-section) view (*b*) of patch antenna fed by coaxial line at left edge.

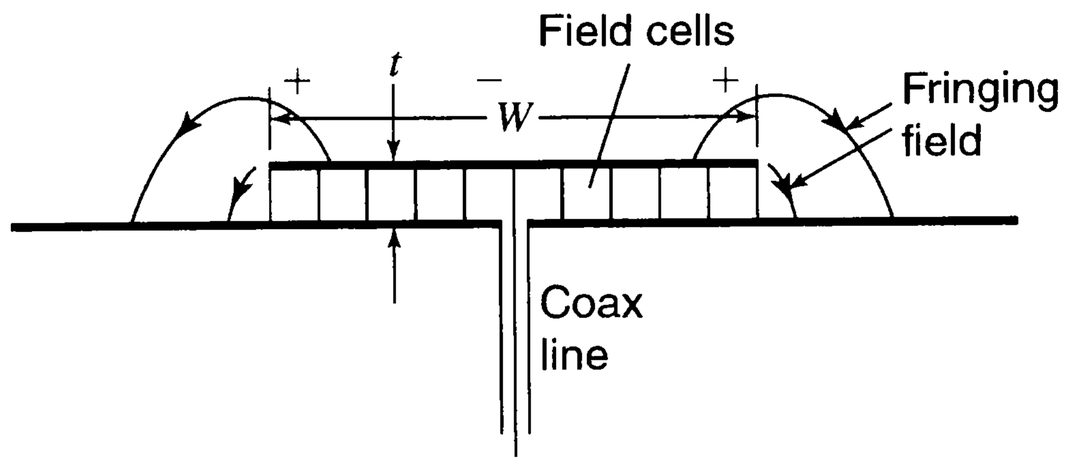


Figure 9–21
View from coax feed side of patch of Fig. 9–20 with gap (or slot) divided into (square) field cells.

characteristic impedance Z_c is given by

$$Z_c = \frac{Z_0}{n\sqrt{\varepsilon_r}} = \frac{377}{10\sqrt{2}} = 26.7\,\Omega \tag{2}$$

Since $n = W/t$, a more general relation is

$$Z_c = \frac{Z_0 t}{W\sqrt{\varepsilon_r}} \tag{3}$$

This relation neglects fringing of the field at the edges. Since W is typically even much larger than the t value in this example, this effect is small for a patch. However, for a microstrip transmission line where the ratio W/t is smaller, the fringing effect can be accounted for approximately by adding 2 cells, giving a more accurate formula for microstrip line impedance as

$$Z_c = \frac{Z_0}{\sqrt{\varepsilon_r}[(W/t) + 2]} \tag{4}$$

The resonant length L of the patch is critical and typically is a couple of percent less than $\lambda/2$, where λ is the wavelength in the dielectric ($\lambda = \lambda_0/\sqrt{\varepsilon_r}$).

Radiation from the patch occurs as though from 2 slots, at the left and right in Fig. 9–20. Let us calculate their impedance for the case where the dielectric substrate is air ($\varepsilon_r = 1$).

From (9–5–11) the impedance Z_s of a slot antenna is given by

$$Z_s = \frac{Z_0^2}{4Z_d} = \frac{35{,}476}{Z_d} \qquad (\Omega) \tag{5}$$

where

Z_0 = intrinsic impedance of empty space = 377 Ω

Z_d = impedance of complementary dipole, Ω

For a slot λ long and a few $\lambda/100$ wide, the complementary dipole impedance is approximately $700 + j0\ \Omega$. Thus, the approximate slot impedance is

$$Z_s \simeq \frac{35{,}476}{700} \simeq 50\,\Omega \tag{6}$$

This is the impedance of a slot in a conducting sheet open on both sides.

The $\lambda/2$ microstrip is equivalent to a $\lambda/2$ section of transmission line shunted with a radiation (and loss) resistance at each end. The centerpoint of the line is at low potential and can be grounded with little effect on operation. Thus, the slot at each end is effectively boxed-in, doubling its impedance. However, the 2 patch "slots" act in parallel, which halves the impedance resulting in an input resistance

$$R_{\text{in}} \simeq 50\,\Omega$$

for $W \simeq \lambda_0$, which is a typical value even with $\varepsilon_r > 1$. If W is smaller, R_{in} increases proportionately.

The radiation pattern of the patch is broad. Typically, the beam area Ω_A is $\frac{1}{2}$ of a half-space or about π sr. The resulting patch directivity D is then given by

贴片定向性（近似）

$$\textit{Patch directivity (approximate)} \qquad D = \frac{4\pi}{\Omega_A} \simeq \frac{4\pi}{\pi} \simeq 4 \text{ (or 6 dBi)} \tag{7}$$

The impedance bandwidth of a patch antenna is usually much narrower than its pattern bandwidth. The impedance bandwidth is proportional to the thickness t of the dielectric substrate. Since t is small, the bandwidth is small, typically a few percent.

According to Jackson (1) and Alexopolus, the radiation resistance of a resonant half-wave patch ($L = 0.49\lambda_0/\sqrt{\varepsilon_r}$), as in Fig. 9–20, is

$$R_r = 90\left(\varepsilon_r^2/(\varepsilon_r - 1)\right)(L/W)^2\ \Omega \tag{7.1}$$

and

$$X_r = 0$$

An empirical formula by Jackson and Alexopolus for the bandwidth (VSWR < 2) is

$$\text{Bandwidth } = 3.77\left((\varepsilon_r - 1)/\varepsilon_r^2\right)(W/L)(t/\lambda_0) \tag{8}$$

appropriate for small values of t/λ_0, where λ_0 = free space wavelength, m.

Note that the bandwidth is a linear function of the substrate thickness t. Increasing t to increase the bandwidth, however, results in greater surface waves, spurious radiation and reduced directivity.

It is interesting to evaluate the effective height h_e for a typical patch antenna. The effective height h_e of an antenna is given by

$$h_e = \sqrt{\frac{2R_r A_e}{Z_0}} \tag{9}$$

where

R_r = radiation resistance, Ω

A_e = effective aperture, λ^2

Z_0 = intrinsic impedance of space, Ω

If we take $D = 4$ and $R_r = 50\ \Omega$ for a typical patch we have as its effective aperture

$$A_e = \frac{D\lambda_0^2}{4\pi} = \frac{\lambda_0^2}{\pi} \tag{10}$$

and as its effective height

$$h_e = \sqrt{\frac{2 \times 50\lambda_0^2}{377\pi}} \simeq 0.3\lambda_0 \tag{11}$$

It is interesting that an antenna which extends only $\lambda_0/100$ above a flat ground plane has an "effective height" 30 times as much.[1]

[1] However, a problem with patches is that radiation does not break away cleanly due to surface modes.

The dimensions of a patch are not electrically small and at, say, 2 GHz ($\lambda = \lambda_0/\sqrt{2} =$ 106 mm) a patch is also not physically very small.

Although the above discussion is considerably simplified, it outlines some of the important properties of patch antennas. Many other shapes and configurations are possible.

EXAMPLE **9–7.1 Rectangular Patch**

A rectangular patch with dimensions as given Fig. 9–22*a* has a dielectric substrate of relative permittivity $\varepsilon_r = 2.27$ and a thickness $t = \lambda_0/100$. Find: (*a*) impedance, (*b*) bandwidth, (*c*) directivity, (*d*) length and impedance of $\lambda/4$ section of microstrip transmission line to match the patch to a 50-Ω line, (*e*) width of microstrip lines.

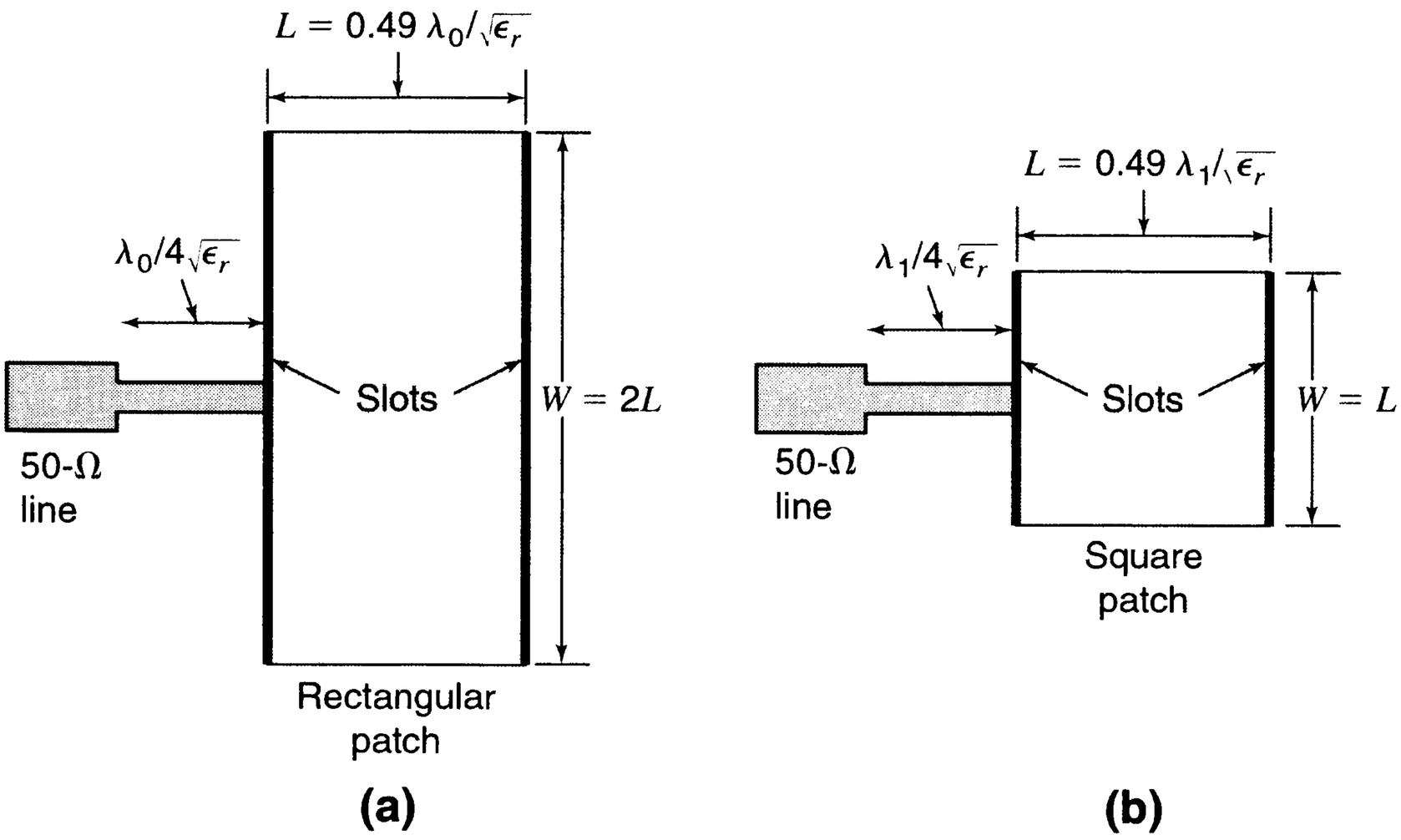


Figure 9–22
Rectangular and square patches for Examples 9–7.1 and 9–7.2.

■ **Solution**

(*a*) From (7–1) $Z = 91 + 0\ \Omega$

(*b*) From (8), bandwidth = 0.019 or 1.9%

(*c*) In the *E*-plane, parallel to *L*, the pattern approximates that of 2 equal in-phase point sources spaced $0.49\lambda_0/\sqrt{2.27} = 0.325\lambda_0$. Using ARRAYPATGAIN for the pattern, the HPBW = 180°. In the *H*-plane, parallel to *W*, the pattern is that of the complementary full-wave dipole of Fig. 9–17 with HPBW = 47°. Thus, from (2–7–9)

$$\text{Directivity} = 40{,}000^{\square}/180^\circ \times 47^\circ = 4.7 \quad \text{or} \quad 6.7\ \text{dBi}$$

(*d*) To match the patch to a 50-Ω, microstrip line the impedance of the $\lambda/4$ section is given by

$Z = \sqrt{91 \times 50} = 67.5\ \Omega$. Its length should be $\lambda_0/4\sqrt{2.27} = 0.166\lambda_0$

(*e*) From (9–7–4), the width of microstrip line is given by

$$W = \left(\left(377/Z_{\text{line}}\sqrt{\varepsilon_r}\right) - 2\right)t = ((377/67.5 \times 1.5) - 2)\frac{\lambda_0}{100} = 0.017\lambda_0$$

for the $\lambda/4$ matching line and $0.030\lambda_0$ for the 50-Ω line.

EXAMPLE **9–7.2 Square Patch**

A square patch with dimensions as given in Fig. 9–22*b* has the same substrate constants as for the rectangular patch of Example 9–7.1. Find (*a*) impedance, (*b*) bandwidth, (*c*) directivity and (*d*) impedance and length of $\lambda/4$ matching section.

■ **Solution**

(*a*) From (7–1), $Z = 365 + j0\ \Omega$

(*b*) From (8), bandwidth = 0.01 or 1%

(*c*) In the *E*-plane, parallel to *L*, the pattern approximates that of 2 equal in-phase point sources with $0.325\lambda_0$ spacing. From ARRAYPATGAIN the HPBW = 180°. In the *H*-plane, parallel to *W*, the pattern is that of the complementary $\lambda/2$ dipole of Fig. 9–17. From (6–4–4), the HPBW = 78°. Thus, from (2–7–9)

$$\text{Directivity} = 40{,}000^{\square}/180^\circ \times 78^\circ = 2.85 \quad \text{or} \quad 4.5\ \text{dBi}$$

(*d*) To match the patch to a 50-Ω microstrip line requires a $\lambda/4$ section of impedance $50 \times 365 = 135\ \Omega$ that has a length of $\lambda_0/4\sqrt{2.27} = 0.166\lambda_0$.

Patches also lend themselves to microstrip arrays in which the patches are fed by microstrip transmission lines. The configuration of a 4-patch array is shown in Fig. 9–23*a*. See also Fig. 16–49 showing an array of 896 microstrip elements.

The literature on patch or microstrip antennas is extensive. Some basic references are those of Munson (1, 2), Derneryd (1), Carver and Mink (1), Pozar (1) and Brookner (1).

Monolithic Microwave Integrated Circuits (MMICs) combine microstrip antennas and associated circuitry in very compact form with applications at frequencies from 50 MHz to 100 GHz (Pozar-2; Levine-1).

EXAMPLE **9–7.3 Microstrip Array of 4 Patches**

An array of 4 $\lambda/2$ patches is shown in Fig. 9–23*a*. The spacing is *d* and the substrate relative permittivity is 2.27. Find the directivity when (*a*) $d = \lambda_0$ and (*b*) when $d = \lambda_0/\sqrt{2.27}$.

■ **Solution**

In the *E*-plane, up in Fig. 9–23*a* the pattern is that of two equal in-phase point sources as in Examples 9–7.1 and 9–7.2, with the HPBW = 180°.

In the *H*-plane, horizontal in Fig. 9–23*a* the pattern is that of 4 equal in-phase point sources with spacing *d*.

(*a*) For $d = \lambda_0$ the pattern from ARRAYPATGAIN is as shown in Fig. 9–23*b*. It has large minor or grating lobes parallel to the ground plane that are equal in magnitude to the "main" lobe. This is not satisfactory.

(*b*) For $d = \lambda_0\sqrt{2.27}$ the pattern from ARRAYPATGAIN is as shown in Fig. 9–23*c*. This is the *array* pattern with HPBW = 28°. The total pattern is the product of this pattern and that of the $\lambda/2$ slot (or complementary $\lambda/2$ dipole) with HPBW = 78°. This pattern is also shown in Fig. 9–23*c*. The product of this pattern and the array pattern results in a

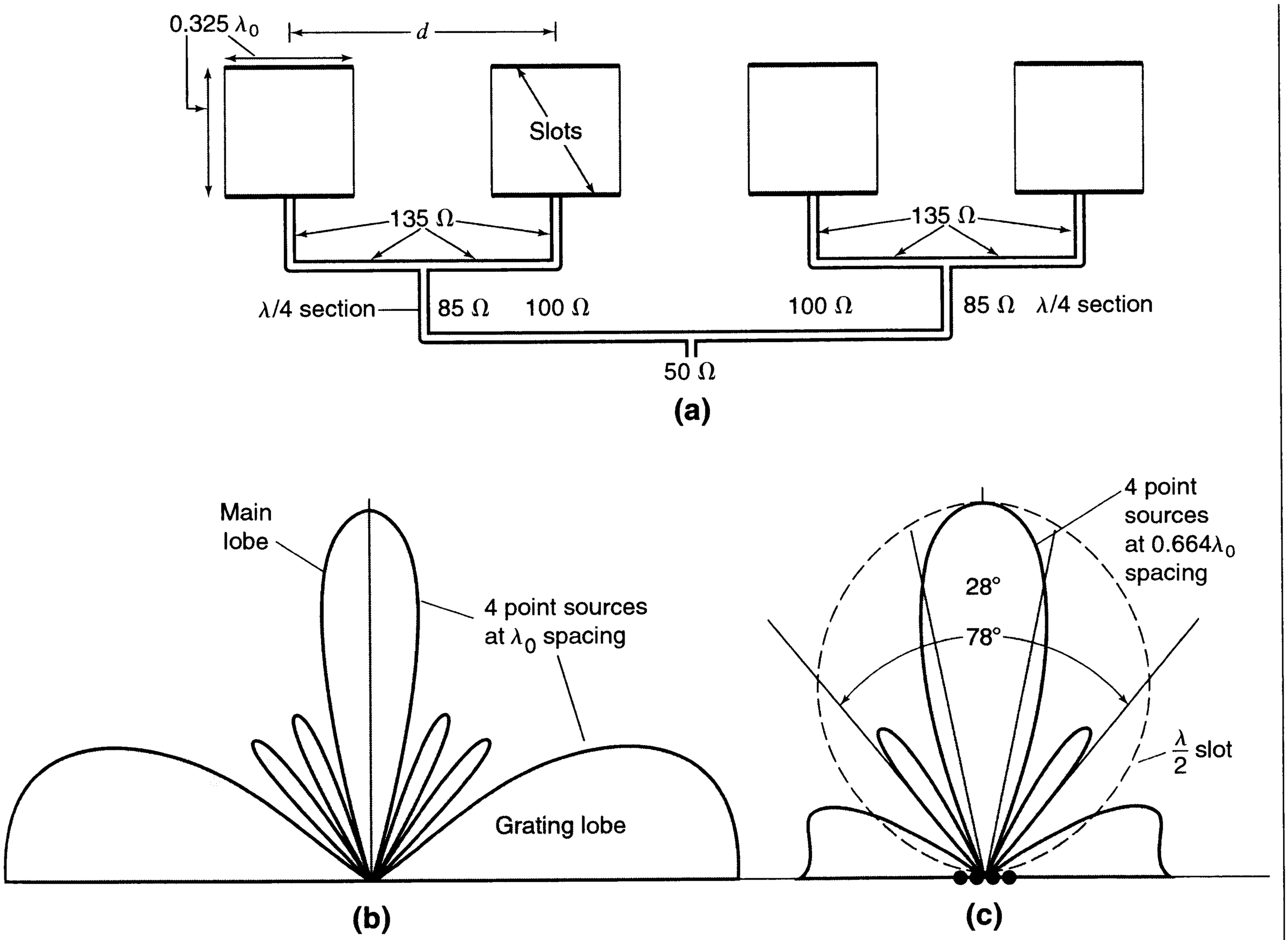


Figure 9–23
(*a*) Array of 4 $\lambda/2$ patches with spacing d connected by microstrip transmission lines to a 50-Ω source. (*b*) Field pattern when $d = \lambda_0$. (*c*) Field pattern when $d = \lambda_0/\sqrt{2.27}$. The $\lambda/2$ slot field pattern is also shown.

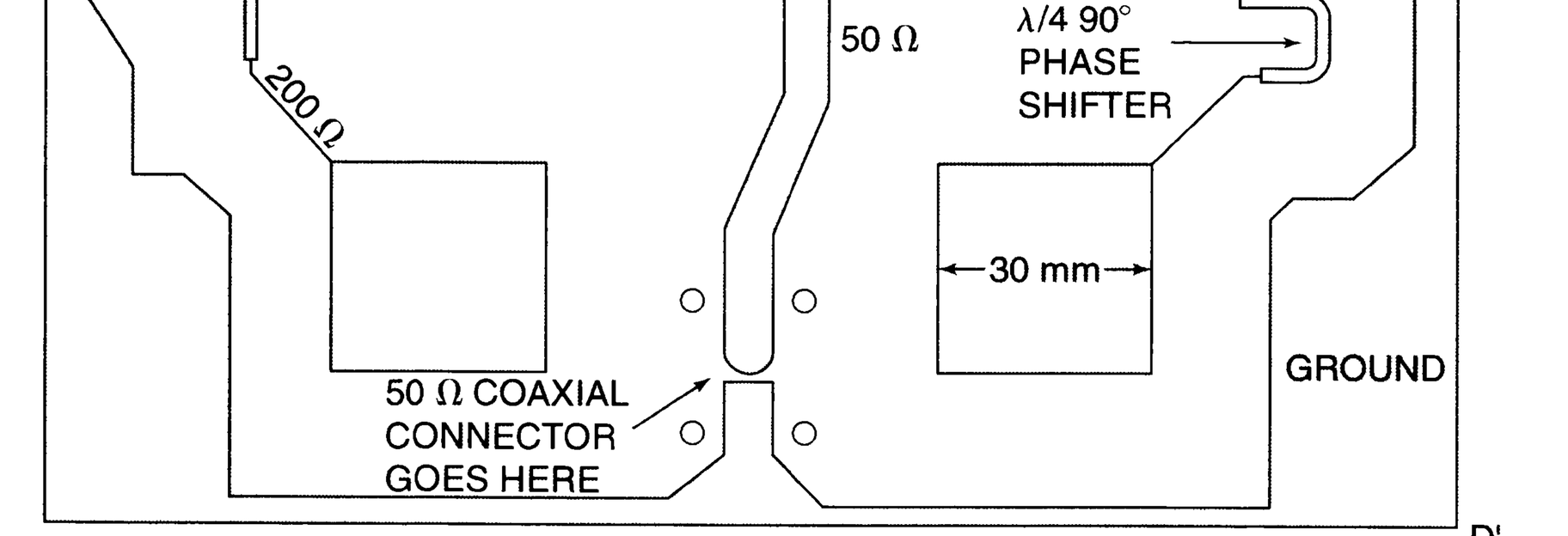
total HPBW $= 26°$. Thus, the

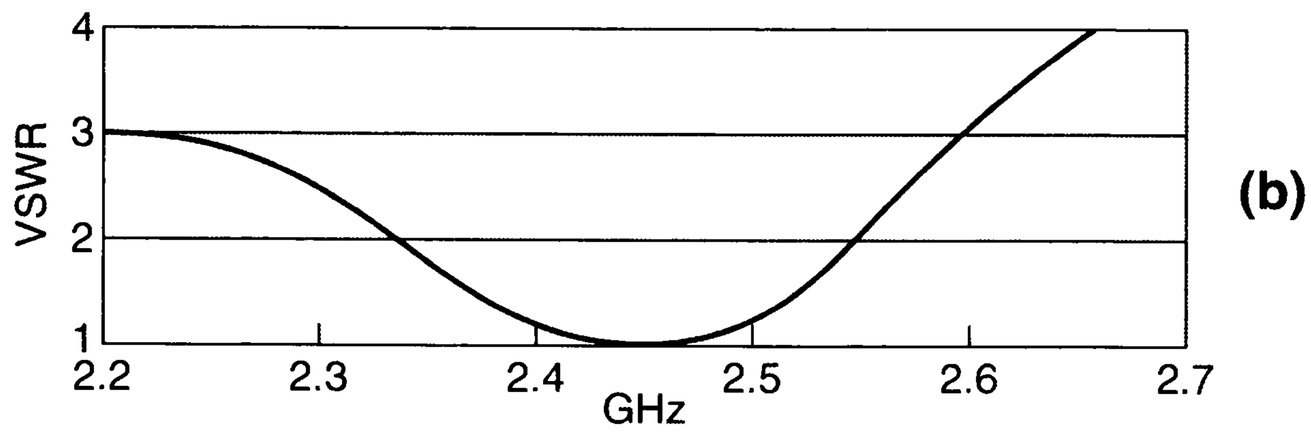
$$\text{Directivity} = 40{,}000^{\square}/180° \times 26° = 8.6 \quad \text{or} \quad 9.3 \text{ dBi}$$

A 4-patch circularly polarized array for 2.45 GHz ($\lambda = 122.4$ mm) is shown in Fig. 9–23.1(*a*). The equivalent dielectric substrate $\varepsilon_r = 4$ so the patch dimension is given by $0.49 \times 122.4/4^{1/2} = 30$ mm. The measured VSWR vs. frequency is shown at (*b*) with match to 50 Ω (VSWR = 1) at the design frequency of 2.45 GHz with a measured gain of 11.0 dBi (circular). The measured RCP field patterns are shown at (*c*), (*d*) and (*e*) with patterns measured through planes perpendicular to CC', DD' and EE'. The commercial model of this patch antenna has a plastic cover or "radome." With this in place, a substrate with relative permittivity less than 4 is used. However, the combination of this substrate and the radome is equivalent to a substrate with $\varepsilon_r = 4$ and no radome as assumed in the worked example. The measurements were made on the commercial model with radome.

EXAMPLE **9–7.4 Comparison of 4-Patch CP Array with Axial-Mode Helix**

(*a*) What are the helix dimensions required to obtain the $D = 12.6$ (11.0 dBi) circularly polarized gain of the 4-patch array of Fig. 9–23–1?





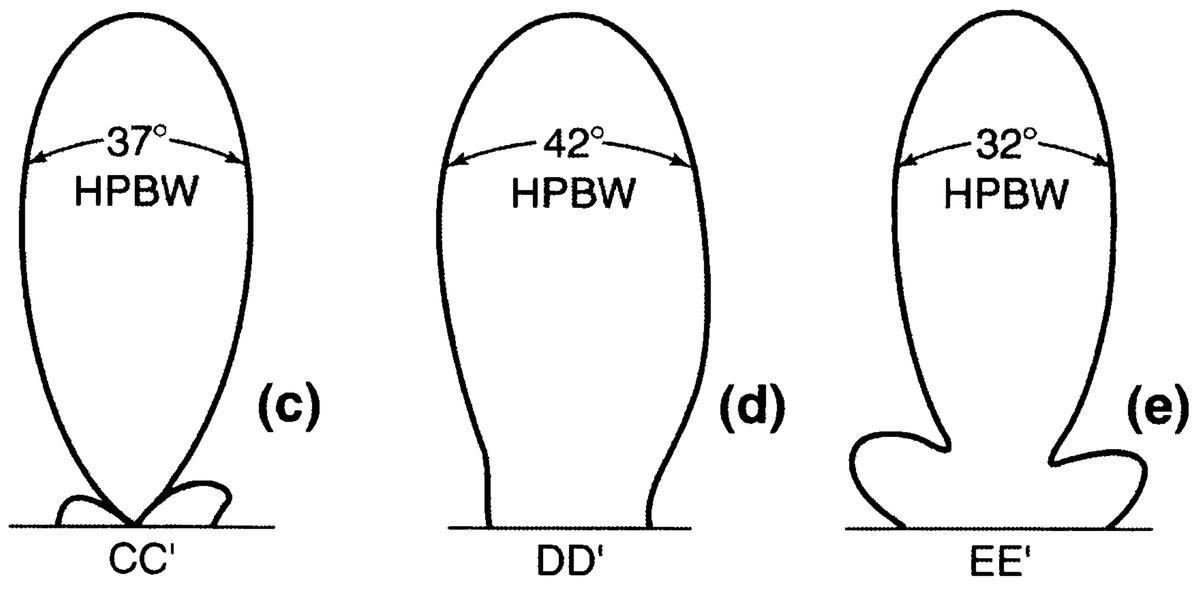


Figure 9–23–1
Four-patch RCP array (*a*) with measured VSWR (*b*) and measured field patterns (*c*), (*d*) and (*e*). (*Array courtesy DTC Communications, Inc. Measurements by Çağatay Tokgöz of ElectroScience Laboratory, Ohio State University.*)

(*b*) Compare impedance bandwidths.
(*c*) Compare axial ratios.
(*d*) Compare patterns.

■ **Solution**

(*a*) For a helix with turn spacing 0.2λ, we have from (8–3–7) that the number of turns $n = 12.6/12 \times 0.2 = 5.25$. Since (8–3–7) is only approximate, we make $n = 6$. The helix length is $6 \times 0.2\lambda = 1.2\lambda$ or $122.4 \times 1.2 = 127$ mm. The helix diameter $= 122.4/\pi =$ 39 mm. For flush mounting the helix is in a metal cup forming a recessed cavity as in Fig. 9–23–2.

(*b*) From Fig. 8–21 the impedance bandwidth for VSWR < 1.5 is 1.72 to 3.06 GHz for the helix as compared to the measured bandwidth of 2.27 to 2.53 GBz for the 4-patch array. Thus, the helix bandwidth is $3.06 - 1.72/2.53 - 2.27 = 8.4$ times wider than for the patch array.

(*c*) The axial ratio (AR) of the helix from (8–3–8) is $= (2 \times 6) + 1/2 \times 6 = 1.1$ as compared to a measured axial ratio of 2 for the patch array. (AR $= 1$ for pure circular polarization).

(*d*) From Fig. 8–19, the helix HPBWs are a few degrees more than the HPBWs of the patch patterns of Fig. 9–23–1(*c*) but the helix gain is equivalent because of the better polarization purity of the helix.

In summary, the helix has equivalent gain but with much wider bandwidth and more polarization purity than the 4-patch array. The advantage of the patch array is that it can be flush mounted whereas the helix requires a recessed cavity.

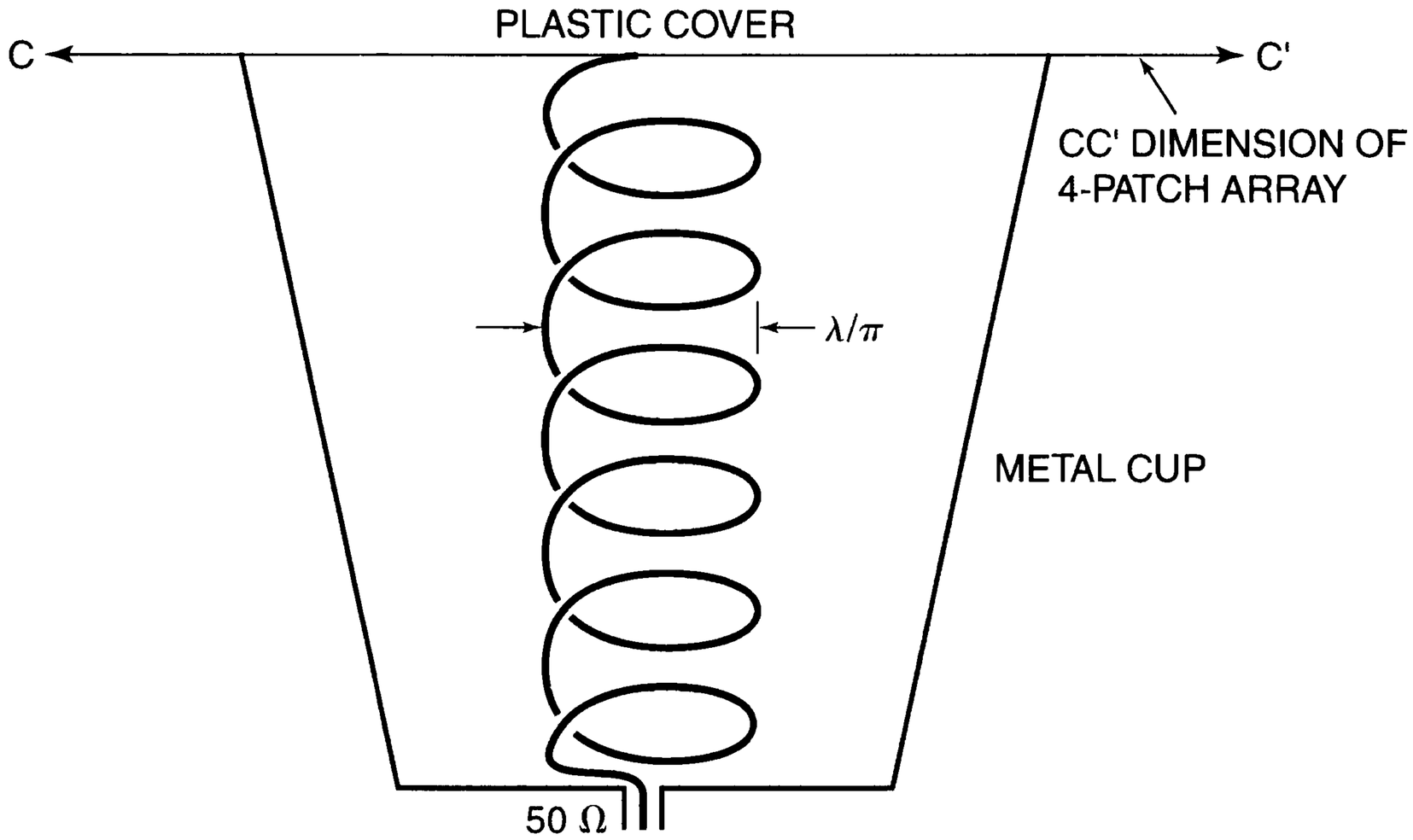


Figure 9–23–2
Six-turn 11.0 dBi (circular) gain axial-mode helix in cavity. The dimension *CC′* of the patch array is shown for size comparison.

9–8 HORN ANTENNAS（喇叭天线）

A horn antenna may be regarded as a flared-out (or opened-out) waveguide. The function of the horn is to produce a uniform phase front with a larger aperture than that of the waveguide and hence greater directivity. Horn antennas are not new. Jagadis Chandra Bose constructed a pyramidal horn in 1897.

Several types of horn antennas are illustrated in Fig. 9–24. Those in the left column are rectangular horns. All are energized from rectangular waveguides. Those in the right column are circular types. To minimize reflections of the guided wave, the transition region or horn between the waveguide at the throat and free space at the aperture could be given a gradual exponential taper as in Fig. 9–24*a* or *e*. However, it is the general practice to make horns

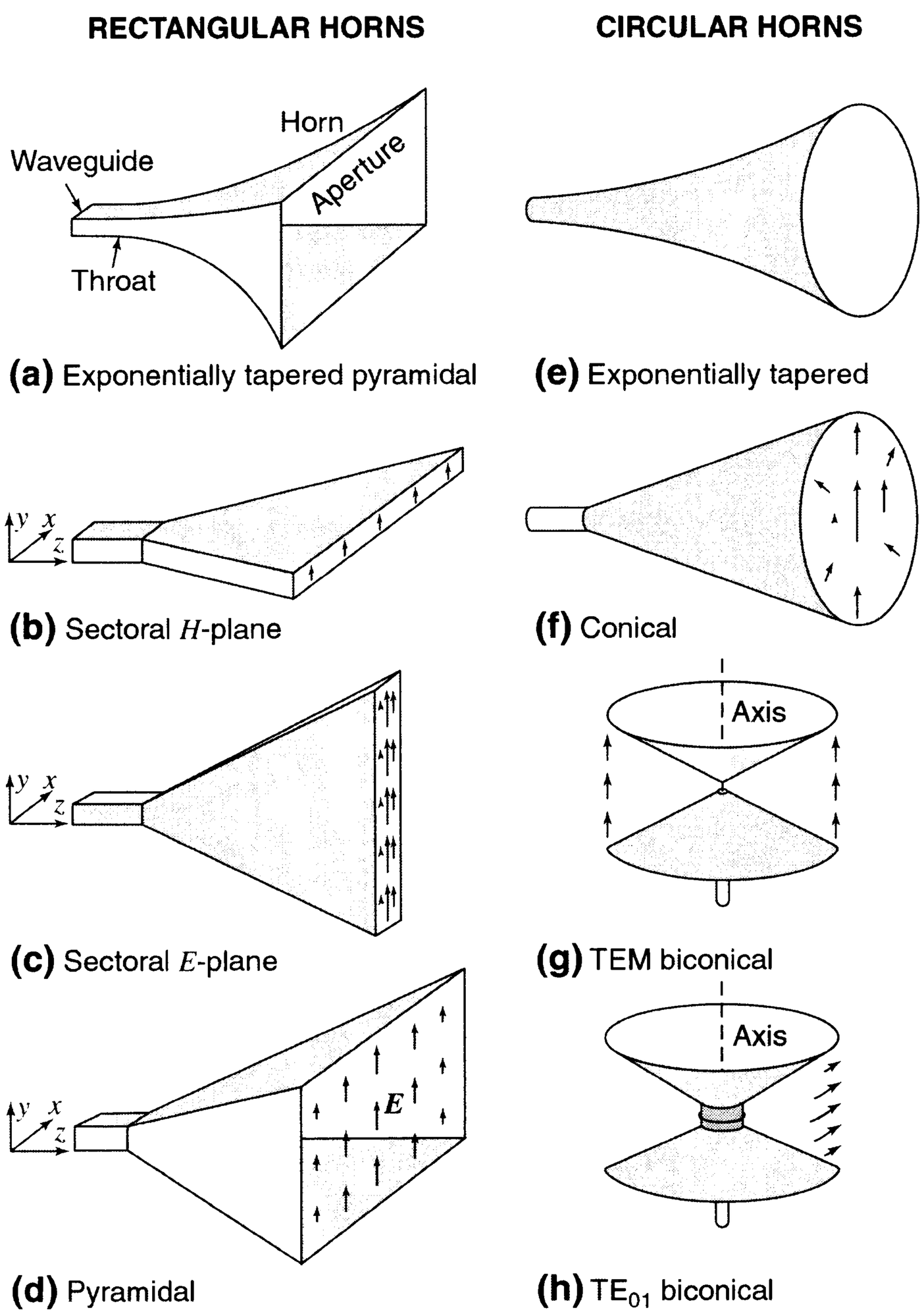


Figure 9–24
Types of rectangular and circular horn antennas. Arrows indicate **E**-field directions.

with straight flares as suggested by the other types in Fig. 9–24.[1] The types in Fig. 9–24*b* and *c* are sectoral horns. They are rectangular types with a flare in only one dimension. Assuming that the rectangular waveguide is energized with a TE_{10} mode wave electric field (**E** in the *y* direction), the horn in Fig. 9–24*b* is flared out in a plane perpendicular to **E**. This is the plane of the magnetic field **H**. Hence, this type of horn is called a sectoral horn flared in the *H* plane or simply *an H-plane sectoral horn*. The horn in Fig. 9–24*c* is flared out in the plane of the electric field **E**, and, hence, is called an *E-plane sectoral horn*. A rectangular horn with flare in both planes, as in Fig. 9–24*d*, is called a *pyramidal horn*. With a TE_{10} wave in the waveguide the magnitude of the electric field is quite uniform in the *y* direction across the apertures of the horns of Fig. 9–24*b*, *c* and *d* but tapers to zero in the *x* direction across the apertures. This variation is suggested by the arrows at the apertures in Fig. 9–24*b*, *c* and *d*. The arrows indicate the direction of the electric field **E**, and their length gives an approximate indication of the magnitude of the field intensity. For small flare angles the field variation across the aperture of the rectangular horns is similar to the sinusoidal distribution of the TE_{10} mode across the waveguide.

The horn shown in Fig. 9–24*f* is a *conical type*. When excited with a circular guide carrying a TE_{11} mode wave, the electric field distribution at the aperture is as shown by the arrows. The horns in Fig. 9–24*g* and *h* are *biconical* types. The one in Fig. 9–24*g* is excited in the TEM mode by a vertical radiator while the one in Fig. 9–24*h* is excited in the TE_{01} mode by a small horizontal loop antenna. These biconical horns are nondirectional in the horizontal plane. The biconical horn of Fig. 9–24*g* is like the one shown in Fig. 2–20*c*.

Neglecting edge effects, the radiation pattern of a horn antenna can be determined if the aperture dimensions and aperture field distribution are known. For a given aperture the directivity is maximum for a uniform distribution. Variations in the magnitude or phase of the field across the aperture decrease the directivity. Since the *H*-plane sectoral horn (Fig. 9–24*b*) has a field distribution over the *x* dimension which tapers to zero at the edges of the aperture, one would expect a pattern in the *xz* plane relatively free of minor lobes as compared to the *yz* plane pattern of an *E*-plane sectoral horn (Fig. 9–24*c*) for which the magnitude of **E** is quite constant over the *y* dimension of the aperture. This is borne out experimentally.

The *principle of equality of path length (Fermat's principle)* is applicable to the horn design but with a different emphasis. Instead of requiring a constant phase across the horn mouth, the requirement is relaxed to one where the phase may deviate, but by less than a specified amount δ, equal to the path length difference between a ray traveling along the side and along the axis of the horn. From Fig. 9–25,

$$\cos\frac{\theta}{2} = \frac{L}{L+\delta} \tag{1}$$

$$\sin\frac{\theta}{2} = \frac{a}{2(L+\delta)} \tag{2}$$

$$\tan\frac{\theta}{2} = \frac{a}{2L} \tag{3}$$

[1] Horns with a straight flare tend to have a *constant phase center* while those with a taper do not.

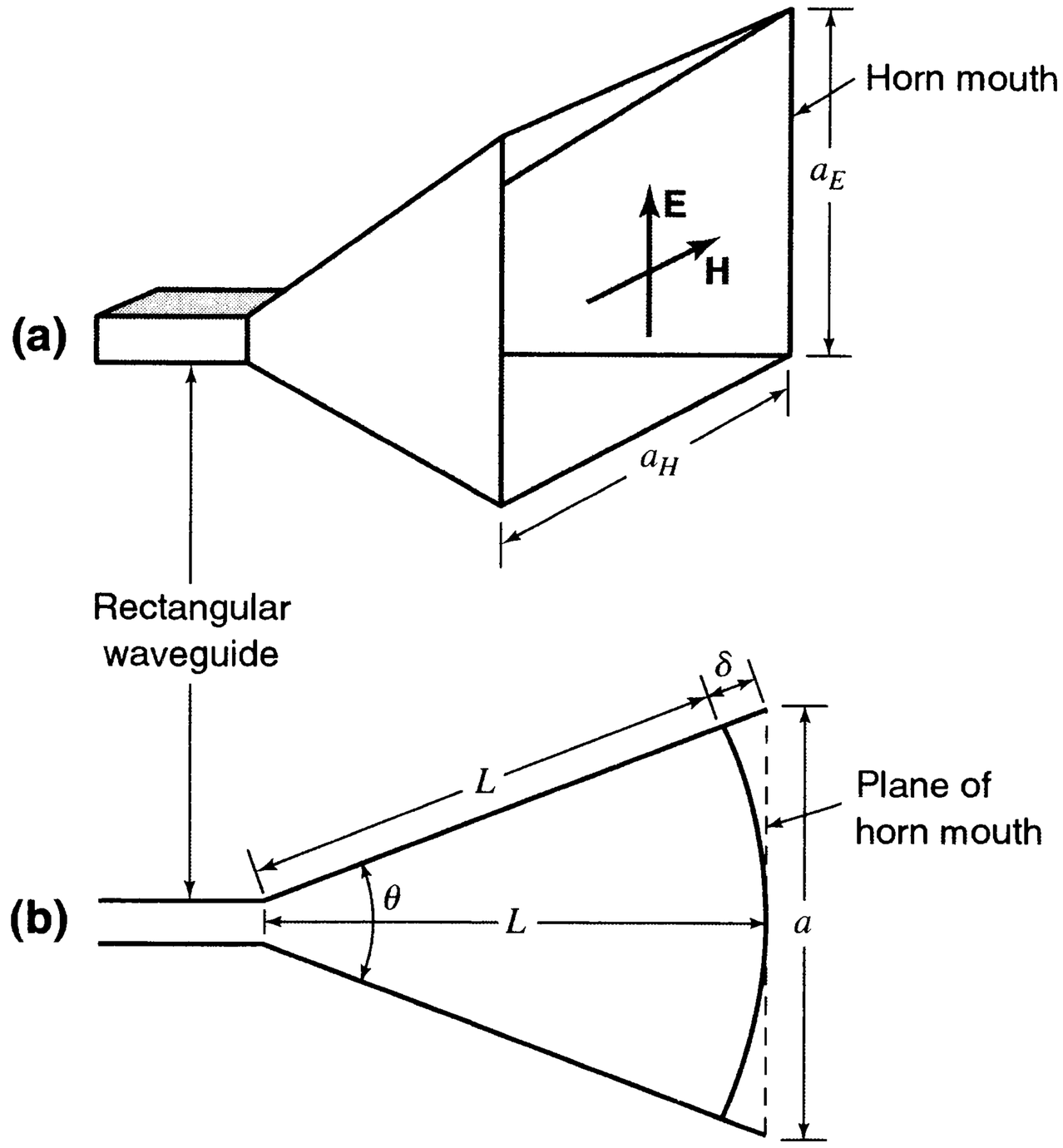


Figure 9–25
(*a*) Pyramidal horn antenna. (*b*) Cross section with dimensions used in analysis. The diagram can be for either *E*-plane or *H*-plane cross sections. For the *E* plane the flare angle is θ_E and aperture a_E. For the *H* plane the flare angle is θ_H and the aperture a_H. See Fig. 9–26.

where

θ = flare angle (θ_E for E plane, θ_H for H plane), deg
a = aperture (a_E for E plane, a_H for H plane), m
L = horn length, m
δ = path length difference, m

From the geometry we have also that

$$L = \frac{a^2}{8\delta} \qquad (\delta \ll L) \tag{4}$$

and

$$\theta = 2\tan^{-1}\frac{a}{2L} = 2\cos^{-1}\frac{L}{L+\delta} \tag{5}$$

In the E plane of the horn, δ is usually held to 0.25λ or less. However, in the H plane, δ can be larger, or about 0.4λ, since **E** goes to zero at the horn edges (boundary condition, $E_t = 0$ satisfied).

To obtain as uniform an aperture distribution as possible, a very long horn with a small flare angle is required. However, from the standpoint of practical convenience the horn should be as short as possible. An *optimum horn* is between these extremes and has the minimum beamwidth without excessive side-lobe level (or most gain) for a given length.

If δ is a sufficiently small fraction of a wavelength, the field has nearly uniform phase over the entire aperture. For a constant length L, the directivity of the horn increases (beamwidth decreases) as the aperture a and flare angle θ are increased. However, if the aperture and flare angle become so large that δ is equivalent to 180 electrical degrees, the field at the edge of the aperture is in phase opposition to the field on the axis. For all but very large flare angles the ratio $L/(L+\delta)$ is so nearly unity that the effect of the additional path length δ on the distribution of the field magnitude can be neglected. However, when $\delta = 180°$, the phase reversal at the edges of the aperture reduces the directivity (increases side lobes). It follows that the maximum directivity occurs at the largest flare angle for which δ does not exceed a certain value (δ_0). Thus, from (1) the *optimum horn dimensions* can be related by

(最优喇叭尺寸)

Optimum horn dimensions

$$\delta_0 = \frac{L}{\cos(\theta/2)} - L = \text{optimum } \delta \tag{6}$$

or

$$L = \frac{\delta_0 \cos(\theta/2)}{1 - \cos(\theta/2)} = \text{optimum length} \tag{7}$$

It turns out that the value of δ_0 must usually be in the range of 0.1 to 0.4 free-space wavelength.[1] Suppose that for an optimum horn $\delta_0 = 0.25\lambda$ and that the axial length $L = 10\lambda$. Then from (5), $\theta = 25°$. This flare angle then results in the maximum directivity for a 10λ horn.

The path length, or δ effect, discussed above is an inherent limitation of all horn antennas of the conventional type.[2] The relations of (1) through (7) can be applied to all the horns of Fig. 9–24 to determine the optimum dimensions. However, the appropriate value of δ_0 may differ as discussed in the following sections. Another limitation of horn antennas is that for the most uniform aperture illumination higher modes of transmission in the horn must be suppressed. It follows that the width of the waveguide at the throat of the horn must be between $\lambda/2$ and 1λ, or if the excitation system is symmetrical, so that even modes are not energized, the width must be between $\lambda/2$ and $3\lambda/2$.

9–9 THE RECTANGULAR HORN ANTENNA[3] (矩形喇叭天线)

Provided that the aperture in both planes of a rectangular horn exceeds 1λ, the pattern in one plane is substantially independent of the aperture in the other plane. Hence, in general, the H-plane pattern of an H-plane sectoral horn is the same as the H-plane pattern of a pyramidal horn with the same H-plane cross section. Likewise, the E-plane pattern of an E-plane sectoral horn is the same as the E-plane pattern of a pyramidal horn with the same

---

[1] At a given frequency the wavelength in the horn λ_h is always equal to or greater than the free-space wavelength λ. Since λ_h depends on the horn dimensions, it is more convenient to express δ_0 in free-space wavelengths λ.

[2] In the lens-compensated type of horn antenna (see Fig. 17–18*b* and Fig. 9–34) the velocity of the wave is increased near the edge of the horn with respect to the velocity at the axis in order to equalize the phase over the aperture.

[3] Barrow (1, 2), Chu (1), Terman (1), Risser (1), Stavis (1).

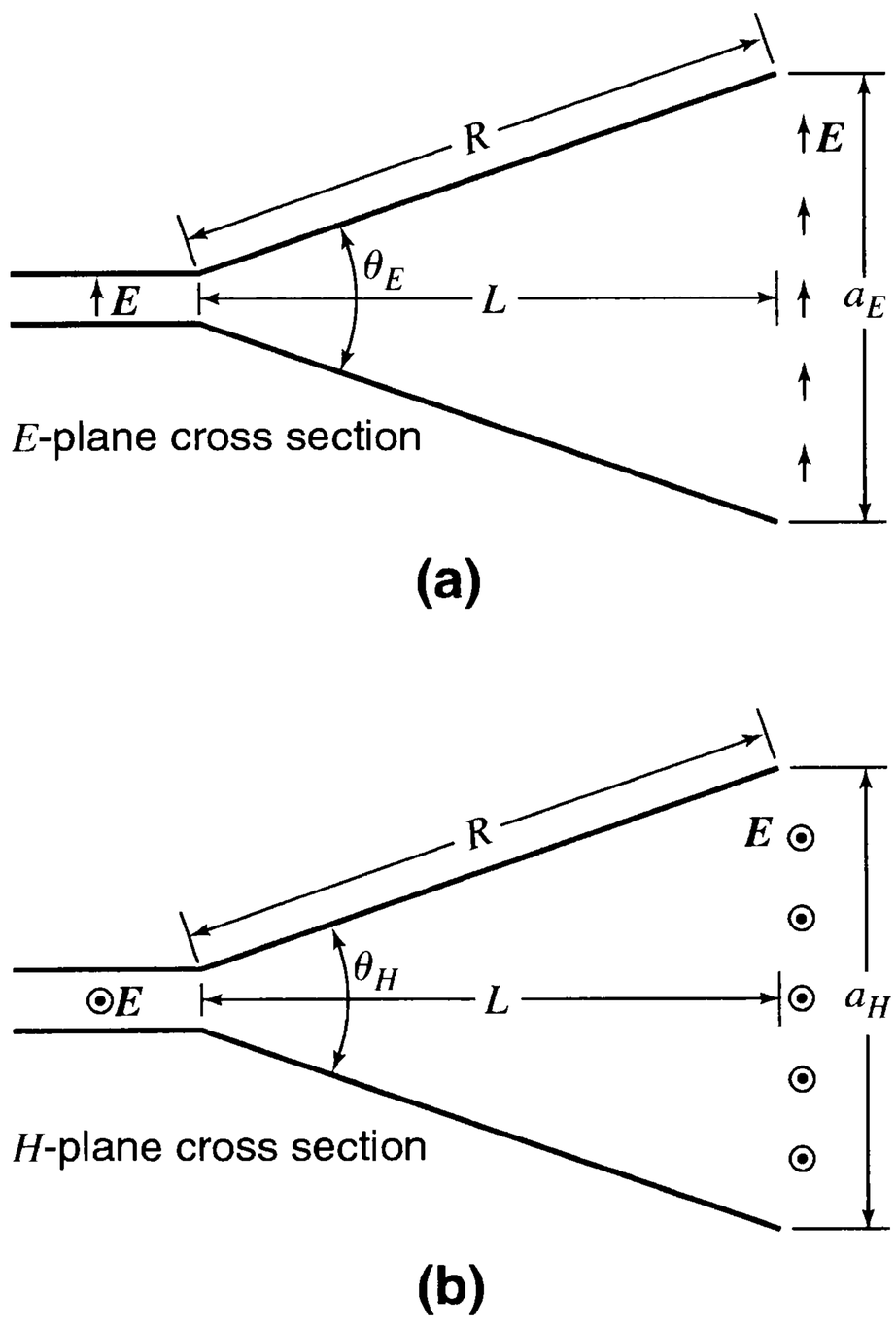


Figure 9–26
E-plane and *H*-plane cross sections.

E-plane cross section. Referring to Fig. 9–26, the total flare angle in the E plane is θ_E and the total flare angle in the H plane is θ_H. The axial length of the horn from throat to aperture is L and the radial length is R. Patterns measured by Donald Rhodes (1) are shown in Fig. 9–27. In (a) the patterns in the E plane and H plane are compared as a function of R. Both sets are for a flare angle of 20°. The E-plane patterns have minor lobes whereas the H-plane patterns have practically none. In (b) measured patterns for horns with $R = 8\lambda$ are compared as a function of flare angle. In the upper row E-plane patterns are given as a function of the E-plane flare angle θ_E and in the lower row H-plane patterns are shown as a function of the H-plane flare angle θ_H. For a flare angle $\theta_E = 50°$ the E-plane pattern is split, whereas for $\theta_H = 50°$ the H-plane pattern is not. This is because a given phase shift at the aperture in the E-plane horn has more effect on the pattern than the same phase shift in the H-plane horn. In the H-plane horn the field goes to zero at the edges of the aperture, so the phase near the edge is relatively less important. Accordingly, we should expect the value of δ_0 for the H plane to be larger than for the E plane. This is illustrated in Fig. 9–28 and discussed in the next paragraph.

From Rhodes's experimental patterns, optimum dimensions were selected for both E- and H-plane flare as a function of flare angle and horn length L. These optimum dimensions are indicated by the solid lines in Fig. 9–28. The corresponding half-power beamwidths and apertures in wavelengths are also indicated. The dashed curves show the calculated dimensions for a path length $\delta_0 = 0.25\lambda$ and $\delta_0 = 0.4\lambda$. The value of 0.25λ gives a curve close to the experimental curve for E-plane flare, while the value of 0.4λ gives a curve close to the experimental one for H-plane flare over a considerable range of horn length. Thus,

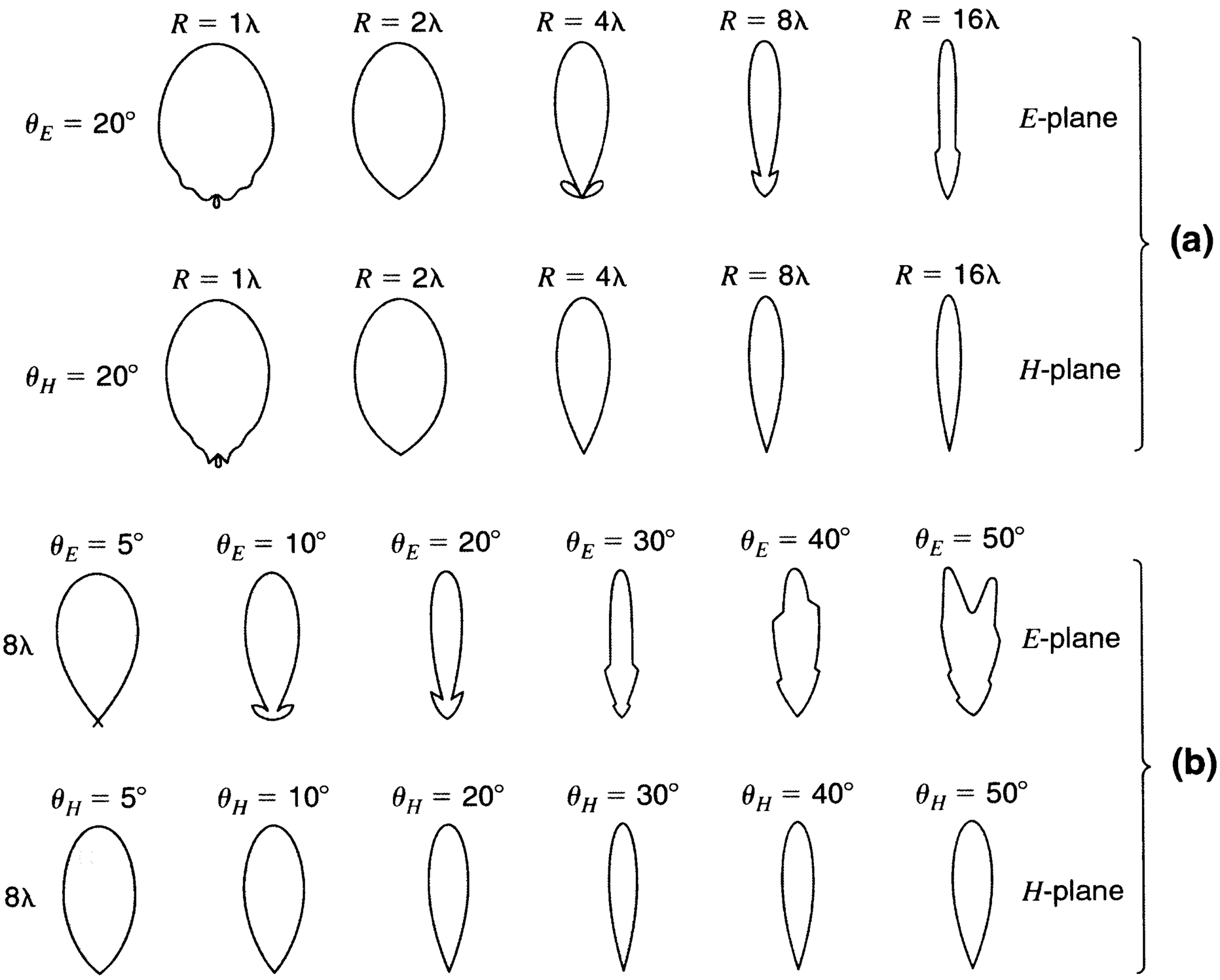


Figure 9–27
Measured E- and H-plane field patterns of rectangular horns as a function of flare angle and horn length. (*After Rhodes-1.*)

the tolerance in path length is greater for H-plane flare than for E-plane flare, as indicated above.

Suppose we wish to construct an optimum horn with length $L = 10\lambda$. From Fig. 9–28 we note that for this length the HPBW (E plane) = 11° and the HPBW (H plane) = 13°, the E-plane aperture $a_E = 4.5\lambda$ and the H-plane aperture $a_H = 5.8\lambda$. Thus, although the E-plane aperture is not so large as the H-plane aperture, the beamwidth is less (but minor lobes larger) because the E-plane aperture distribution is more uniform. For horn operation over a frequency band it is desirable to determine the optimum dimensions for the highest frequency to be used, since δ as measured in wavelengths is largest at this highest frequency.

The directivity (or gain, assuming no loss) of a horn antenna can be expressed in terms of its effective aperture. Thus,

$$D = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi \varepsilon_{ap} A_p}{\lambda^2} \tag{1}$$

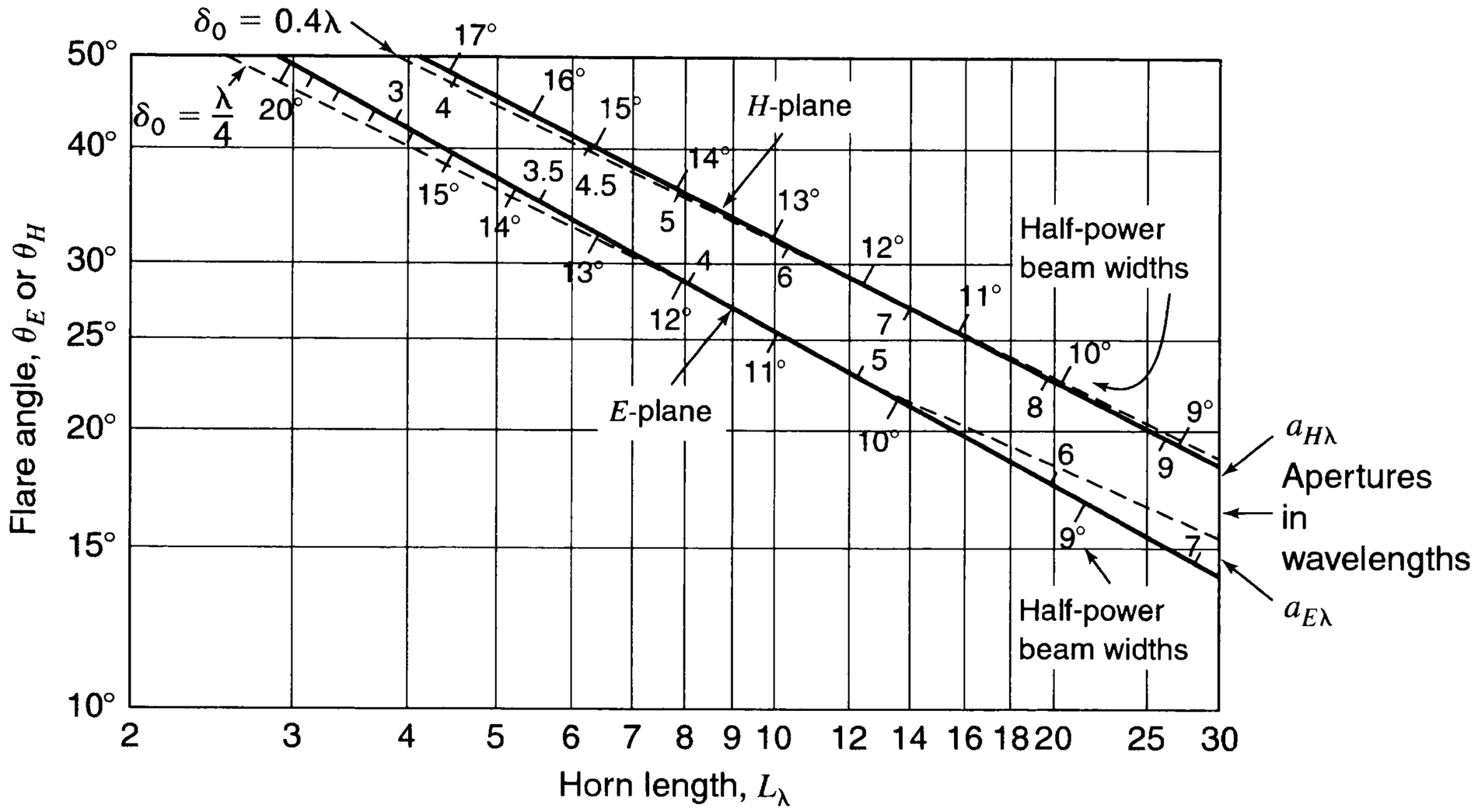


Figure 9–28
Experimentally determined optimum dimensions for rectangular horn antennas. Solid curves give relation of flare angle θ_E in E plane and flare angle θ_H in H plane to horn length (see Fig. 9–26). The corresponding half-power beamwidths and apertures in wavelengths are indicated along the curves. Dashed curves show calculated dimensions for $\delta_0 = 0.25$ and 0.4λ.

where

A_e = effective aperture, m^2
A_p = physical aperture, m^2
ε_{ap} = aperture efficiency = A_e/A_p
λ = wavelength, m

For a rectangular horn $A_p = a_E a_H$ and for a conical horn $A_p = \pi r^2$, where r = aperture radius. It is assumed that a_E, a_H or r are all at least 1λ. Taking $\varepsilon_{ap} \simeq 0.6$, (1) becomes

$$D \simeq \frac{7.5A_p}{\lambda^2} \tag{2}$$

or

$$D \simeq 10\log\left(\frac{7.5A_p}{\lambda^2}\right) \quad \text{(dBi)} \tag{3}$$

For a pyramidal (rectangular) horn (3) can also be expressed as

$$D \simeq 10\log(7.5a_{E\lambda}a_{H\lambda}) \tag{4}$$

where

$a_{E\lambda}$ = E-plane aperture in λ
$a_{H\lambda}$ = H-plane aperture in λ

EXAMPLE 9–9.1

(*a*) Determine the length L, H-plane aperture and flare angles θ_E and θ_H (in the E and H planes, respectively) of a pyramidal horn as in Fig. 9–24*d* for which the E-plane aperture $a_E = 10\lambda$. The horn is fed by a rectangular waveguide with TE_{10} mode. Let $\delta = 0.2\lambda$ in the E plane and 0.375λ in the H plane. (*b*) What are the beamwidths? (*c*) What is the directivity?

■ **Solution**

Taking $\delta = \lambda/5$ in the E plane, we have from (9–8–4) that the required horn length

$$L = \frac{a^2}{8\delta} = \frac{100\lambda}{8/5} = 62.5\lambda \tag{5}$$

and from (9–8–5) that the flare angle in the E plane

$$\theta_E = 2\tan^{-1}\frac{a}{2L} = 2\tan^{-1}\frac{10}{125} = 9.1^\circ \tag{6}$$

Taking $\delta = 3\lambda/8$ in the H plane we have from (9–8–5) that the flare angle in the H plane

$$\theta_H = 2\cos^{-1}\frac{L}{L+\delta} = 2\cos^{-1}\frac{62.5}{62.5+0.375} = 12.52^\circ \tag{7}$$

and from (9–8–5) that the H-plane aperture

$$a_H = 2L\tan\frac{\theta_H}{2} = 2\times 62.5\lambda\tan 6.26^\circ = 13.7\lambda \tag{8}$$

From Table 9–1,

$$\text{HPBW } (E \text{ plane}) = \frac{56^\circ}{a_{E\lambda}} = \frac{56^\circ}{10} = 5.6^\circ \tag{9a}$$

$$\text{HPBW } (H \text{ plane}) = \frac{67^\circ}{a_{H\lambda}} = \frac{67^\circ}{13.7} = 4.9^\circ \tag{9b}$$

From (3),

$$D \simeq 10\log\left(\frac{7.5A_p}{\lambda^2}\right) = 10\log(7.5\times 10\times 13.7) = 30.1 \text{ dBi} \tag{10}$$

The δ values used in this example are conservative. For an optimum horn, the δ values are larger, resulting in a considerably shorter horn but at the expense of slightly less gain (because fields are less uniform across the aperture of an optimum horn).

Figure 9–29*a* shows the optimum dimensions for pyramidal (rectangular) horns versus gain (or directivity, if no loss) (Schrank-1). For a given desired gain, the graph gives the dimensions for the length L_λ, E-plane aperture $a_{E\lambda}$ and H-plane aperture $a_{H\lambda}$, all in

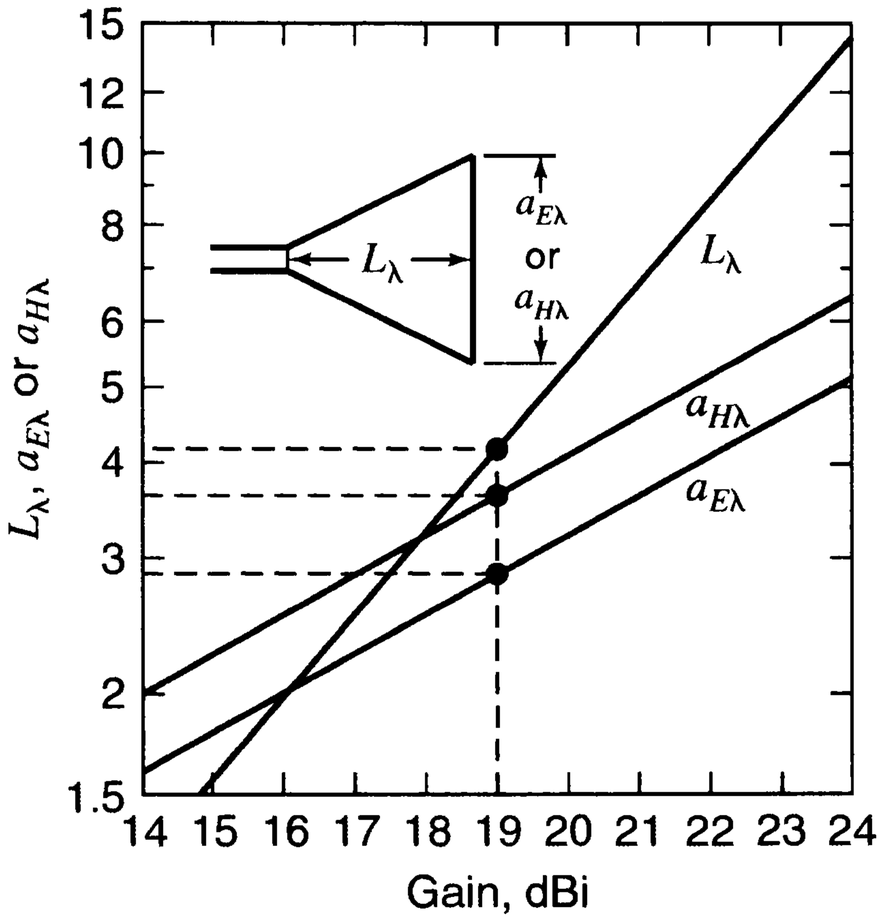


Figure 9–29a
Dimensions of rectangular (pyramidal) horns (in wavelengths) versus directivity (or gain, if no loss). Thus, noting the dashed lines, a gain of 19 dBi requires a horn length $L_\lambda = 4.25$, an H-plane aperture $a_{H\lambda} = 3.7$ and an E-plane aperture $a_{E\lambda} = 2.9$. These are inside dimensions. It is assumed that δ (E plane) $= 0.25\lambda$ and δ (H plane) $= 0.4\lambda$, making the dimensions close to optimum. It is also assumed that $\varepsilon_{ap} = 0.6$.

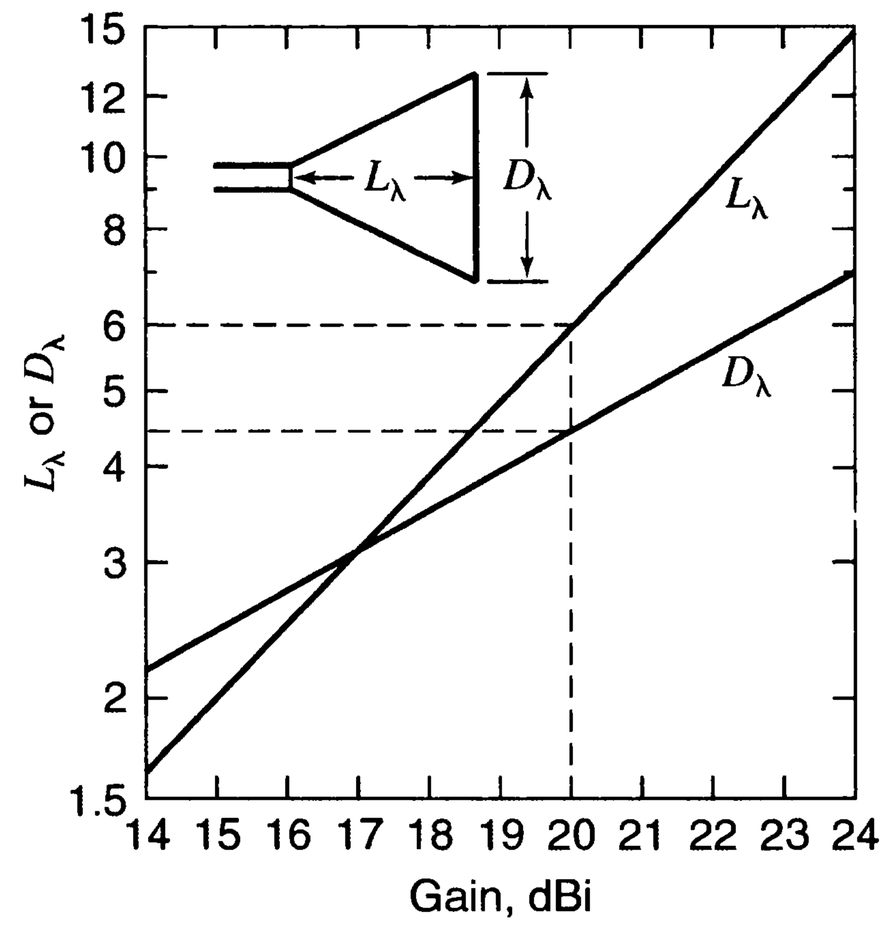


Figure 9–29b
Dimensions of conical horn (in wavelengths) versus directivity (or gain, if no loss). Thus, noting the dashed lines, a gain of 20 dBi requires a horn length $L_\lambda = 6.0$ and a diameter $D_\lambda = 4.3$. These (inside) dimensions are close to optimum.

wavelengths. For a given length, the graph also gives the appropriate apertures and the gain. The dimensions are close to optimum. Such dimensions are only of importance on large horns (many wavelengths long) where it is desired that the length be a minimum. For small (short) horns, optimization is usually unwarranted.

9–10 BEAMWIDTH COMPARISON（波束宽度的比较）

It is interesting to compare the beamwidth between first nulls and between half-power points for uniformly illuminated rectangular and circular apertures obtained in previous chapters with those for optimum rectangular horn antennas (sectoral or pyramidal). This is done in Table 9–1. In general, the relations apply to apertures that are at least several wavelengths long. The beamwidths between nulls for the horns are calculated and the half-power beamwidths are empirical (Stavis-1).

Table 9–1

Type of aperture	Beamwidth, deg: Between first nulls	Beamwidth, deg: Between half-power points
Uniformly illuminated rectangular aperture or linear array	$\frac{115}{L_\lambda}$	$\frac{51}{L_\lambda}$
Uniformly illuminated circular aperture	$\frac{140}{D_\lambda}$	$\frac{58}{D_\lambda}$
Optimum *E*-plane rectangular horn	$\frac{115}{a_{E\lambda}}$	$\frac{56}{a_{E\lambda}}$
Optimum *H*-plane rectangular horn	$\frac{172}{a_{H\lambda}}$	$\frac{67}{a_{H\lambda}}$

L_λ = length of rectangular aperture or linear array in free-space wavelengths
D_λ = diameter of circular aperture in free-space wavelengths
$a_{E\lambda}$ = aperture in E plane in free-space wavelengths
$a_{H\lambda}$ = aperture in H plane in free-space wavelengths

9–11 CONICAL HORN ANTENNAS（圆锥喇叭天线）

The conical horn (Fig. 9–24*f*) can be directly excited from a circular waveguide. Dimensions can be determined from (9–8–5), (9–8–6) and (9–8–7) by taking $\delta_0 = 0.32\lambda$ (Southworth-1; King-1). For optimum conical horns King gives half-power beamwidths of $60/a_{E\lambda}$ in the E plane and $70/a_{H\lambda}$ in the H plane. Those are about 6 percent more than the values for a rectangular horn as given in Table 9–1.

The biconical horns (Fig. 9–24*g* and *h*) have patterns that are nondirectional in the horizontal plane (axis of horns vertical). These horns may be regarded as modified pyramidal horns with a 360° flare angle in the horizontal plane. The optimum vertical-plane flare angle is about the same as for a sectoral horn of the same cross section excited in the same mode.

Figure 9–29*b* shows optimum dimensions for conical horns versus directivity (or gain if no loss) as adapted from King. For a given desired gain, the graph gives the length L_λ and diameter D_λ, or for a given length, the graph gives the appropriate aperture and gain.

9–12 RIDGE HORNS（加脊喇叭）

A central ridge loads a waveguide and increases its useful bandwidth by lowering the cutoff frequency of the dominant mode (Cohn-1; Chen-1). A rectangular guide with single ridge is shown in Fig. 9–30*a* and with a double ridge in Fig. 9–30*b*. A very thin ridge or fin is also effective in producing the loading of a central ridge. It may consist of a metal-clad ceramic sheet which facilitates the installation of shunt circuit elements as suggested in Fig. 9–30*c* (Meier-1).

Of course, the cutoff frequency can be lowered by placing dielectric material in the waveguide, but this does not increase the bandwidth and it may increase losses.

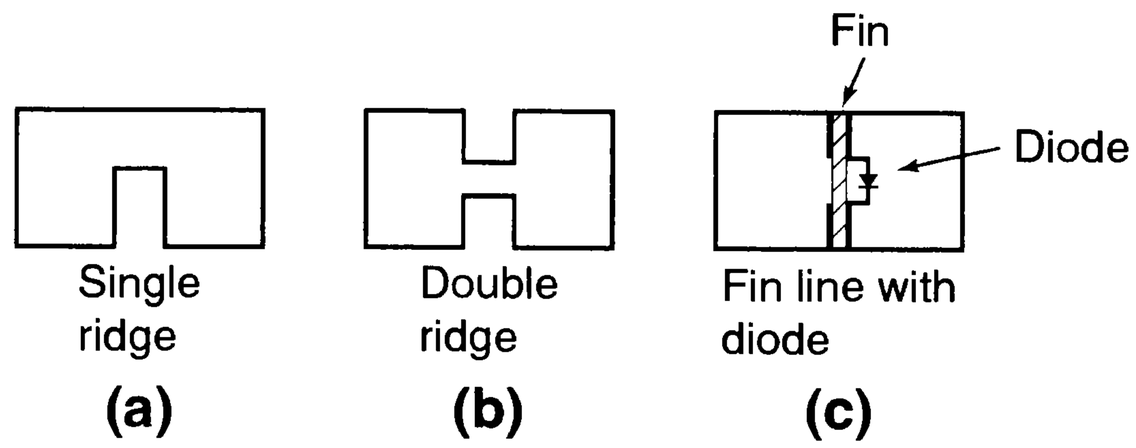


Figure 9–30
Single- and double-ridge rectangular waveguide and fin-line with diode.

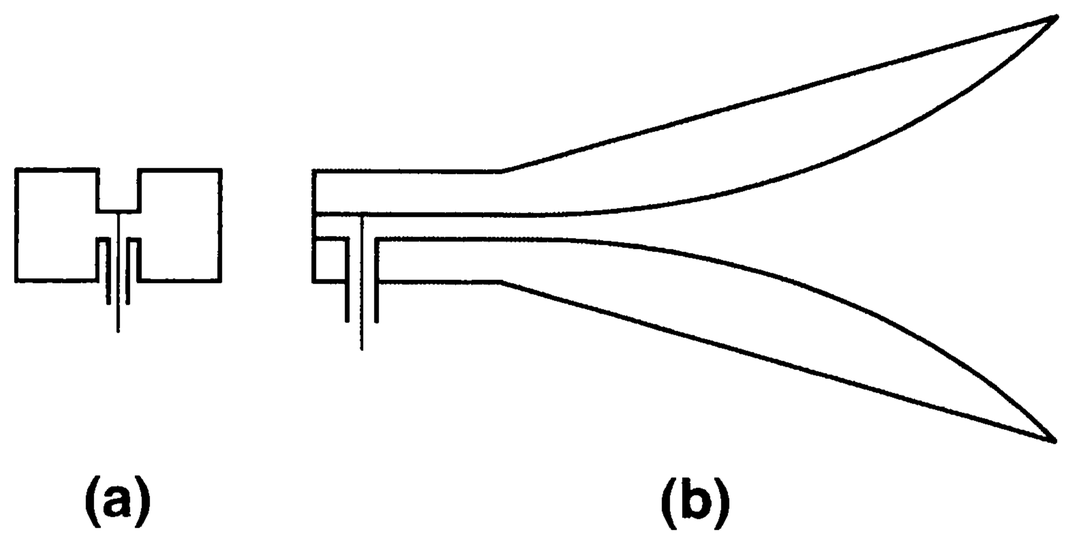


Figure 9–31
Double-ridge or vivaldi horn with coaxial feed. The view at (*a*) is a cross section at the feed point.

By continuing a double-ridge structure from a waveguide into a pyramidal horn as suggested in Fig. 9–31, the useful bandwidth of the horn can be increased manyfold (Walton-1). A quadruple-ridge horn connected to a dual-fed quadruple-ridge square waveguide can provide dual orthogonal linear polarizations over bandwidths of more than 6 to 1.

9–13 SEPTUM HORNS（隔膜喇叭）

Although the electric field in the H plane of a pyramidal horn tends to zero at the edges, resulting in a tapered distribution and reduced side lobes, the electric field in the E plane may be close to uniform in amplitude to the edges, resulting in significant side lobes. By introducing septum plates bonded to the horn walls, a stepped-amplitude distribution can be achieved in the E plane with a reduction in E-plane side lobes. Typically, the first side lobes of a uniform amplitude distribution are down about 13 dB. With a 2-septum horn Peace (1) and Swartz were able to achieve a side-lobe level more than 30 dB down, which is lower than the side-lobe level in the H plane.

A cosine field distribution is approximated with 1 : 2 : 1 stepped amplitude distribution with apertures also in the ratio 1 : 2 : 1 as suggested in Fig. 9–32. To achieve this distribution the septums must be appropriately spaced at the throat of the horn.

Figure 9–33 shows a stepped-amplitude septum horn with metal-plate lens for feeding "Big Ear," the Ohio State University 110-m radio telescope. The dual-feed stacked twin horn with metal-plate E-plane lens, as in Fig. 9–32, provides a compact arrangement $\frac{1}{4}$ the length of a single horn. This twin horn is one of a pair used for radio astronomy observations at 1 to 2 GHz (Nash-1, 2; Kraus-1). Both the aperture and field distributions have the ratios 1 : 2 : 1 (binomial series ratio).

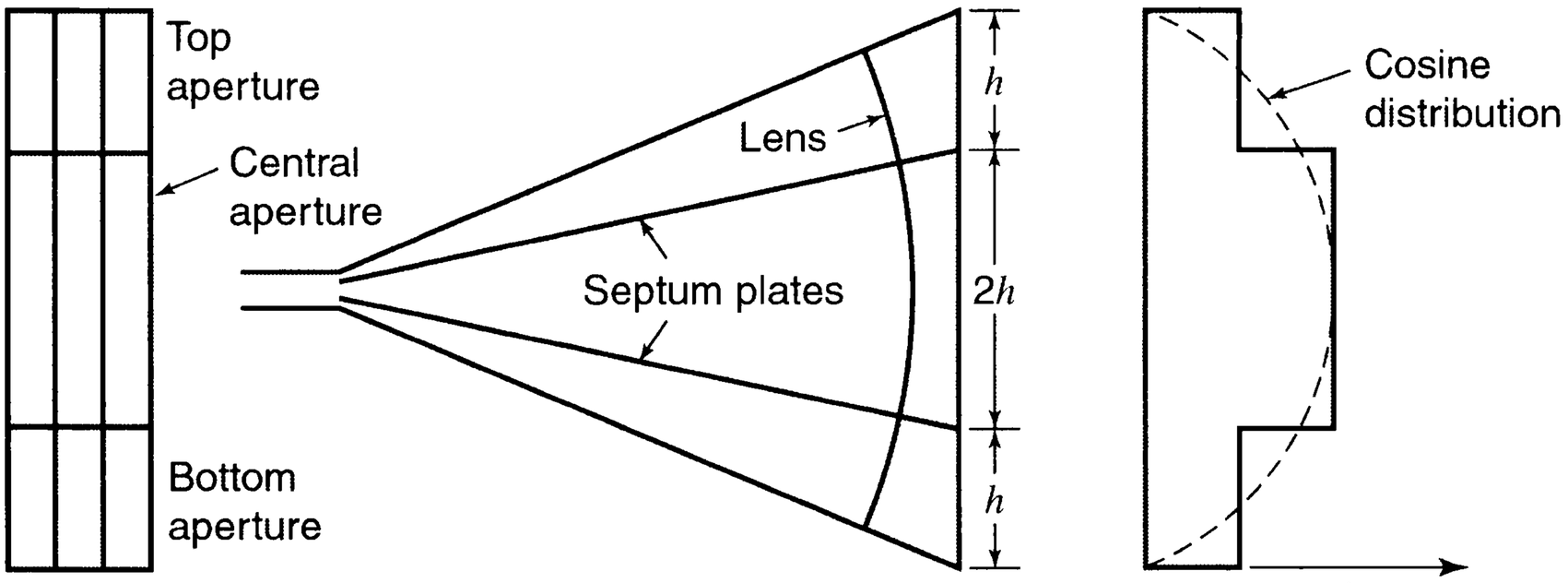


Figure 9–32
Two-septum horn with 1 : 2 : 1 stepped amplitude distribution in field intensity at mouth of horn (approximating a cosine distribution).

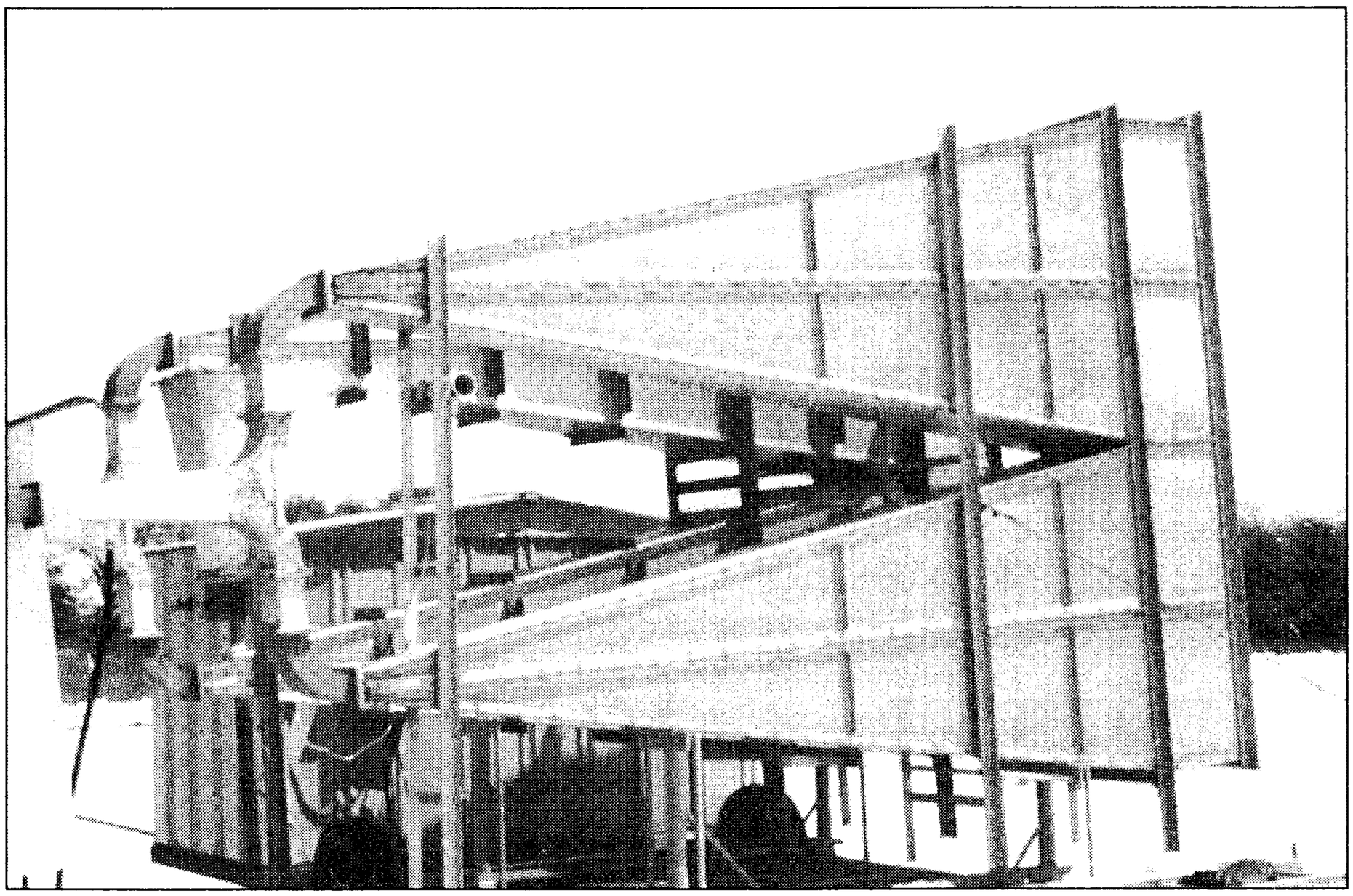


Figure 9–33
Stacked septum horns with *E*-plane metal-plate lenses for feeding 110-m Ohio State University radio telescope. The aperture height (= 4*h* in Fig. 9–32) is 3 m.

9–14 CORRUGATED HORNS（皱纹喇叭）

Corrugated horns can provide reduced edge diffraction, improved pattern symmetry and reduced cross-polarization (less **E** field in the **H** plane).

Corrugations on the horn walls acting as $\lambda/4$ chokes are used to reduce **E** to very low values at all horn edges for all polarizations. These prevent waves from diffracting around

the edges of the horn (or surface currents flowing around the edge and over the outside) (Kay-1).

Consider the corrugations of width w and depth d shown in Fig. 9–34. A square cross section (w by w), as indicated in the figure, constitutes the open end of a short-circuited field-cell transmission line of length d with a characteristic impedance $Z = 377\,\Omega$. The reactance at the open end is given approximately by

$$X \simeq 377 \tan\left(\frac{2\pi d}{\lambda}\right) \qquad (\Omega) \tag{1}$$

where d = depth, λ

The reactance for any square area of the corrugations (such as $3w \times 3w$) is also as given by (1). Thus, this is the surface reactance in ohms per square. It is assumed that the corrugations are air-filled and that the wall thickness is small enough to be neglected.

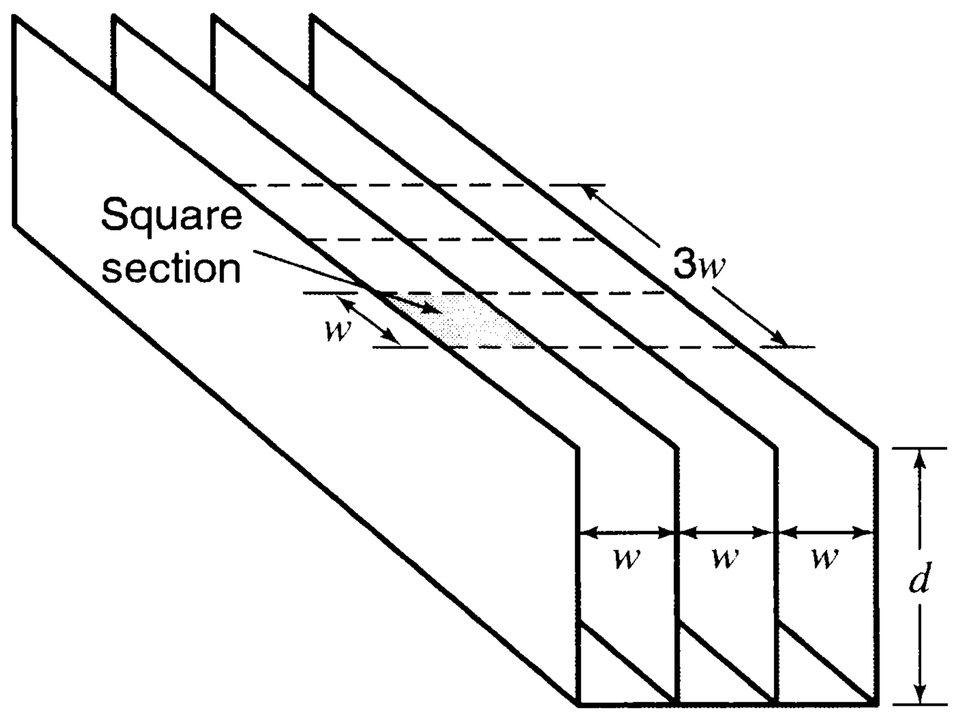


Figure 9–34
Corrugations of width w and depth d.

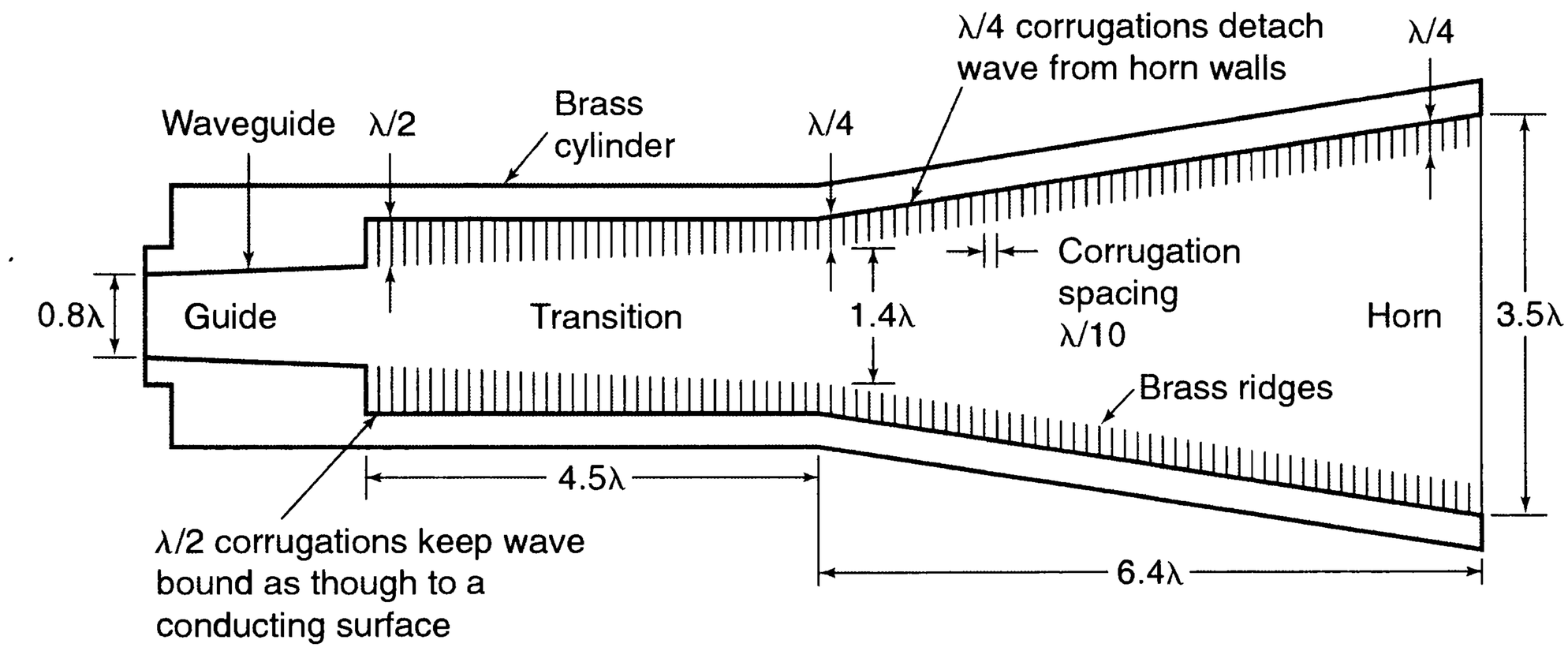


Figure 9–35
Cross section of circular waveguide-fed corrugated horn with corrugated transition. Corrugations with depth of $\lambda/2$ at waveguide act like a conducting surface while corrugations with $\lambda/4$ depth in horn present a high impedance. (*After Chu-1.*)

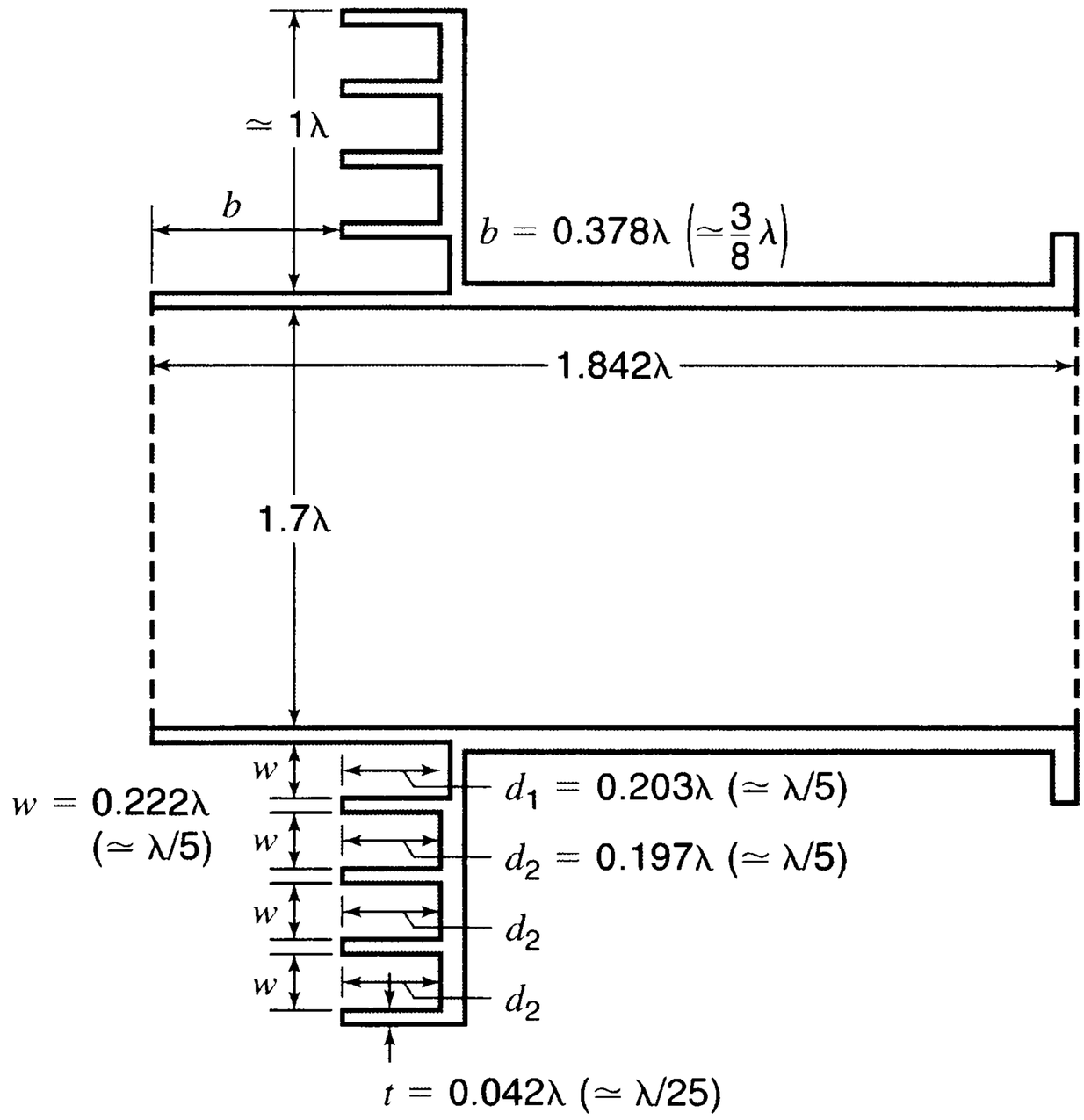


Figure 9–36
Cross section of circular waveguide with flange and 4 chokes for wide-beamwidth high-efficiency feed of low *F/D* parabolic reflectors. (*After Wohlleben-1*).

When $d = \lambda/4$, X becomes infinite, while when $d = \lambda/2$, $X = 0$ and, assuming no radiation or loss, $R = 0$ and, hence, $Z = 0$.

As an example, a circular waveguide-fed corrugated horn with a corrugated transition is shown in Fig. 9–35. In the transition section the corrugation depth changes from $\lambda/2$, where the corrugations act like a conducting surface, to $\lambda/4$, where the corrugations present a high impedance. The corrugation spacing or width $w = \lambda/10$. The corrugations are air-filled. This corrugated horn was developed by Chu et al. (1) for feeding a 7-m millimeter-wave reflector antenna of the Bell Telephone Laboratories.

A simpler feed with choke corrugations was developed for deep dishes with F/D ratios of less than 0.35 by Wohlleben, Mattes and Lochner (1) for use on the Bonn 100-m radiotelescope. It consists of a circular waveguide equipped with a disk (or flange) projecting 1λ beyond the guide and 4 chokes $\lambda/5$ deep as shown in Fig. 9–36. The location of the disk with chokes $3\lambda/8$ behind the waveguide opening gives a broad 130° 10-dB beamwidth with a steep edge taper. This results in high aperture efficiency.

9–15 APERTURE-MATCHED HORN (口径匹配喇叭)

By attaching a smooth curved (or rolled) surface section to the outside of the aperture edge of a horn, Burnside (1) and Chuang have achieved a significant improvement in the pattern, impedance and bandwidth characteristics. This arrangement, shown in Fig. 9–37 is an attractive alternative to a corrugated horn. The shape of the rolled edge is not critical but its radius of curvature should be at least $\lambda/4$.

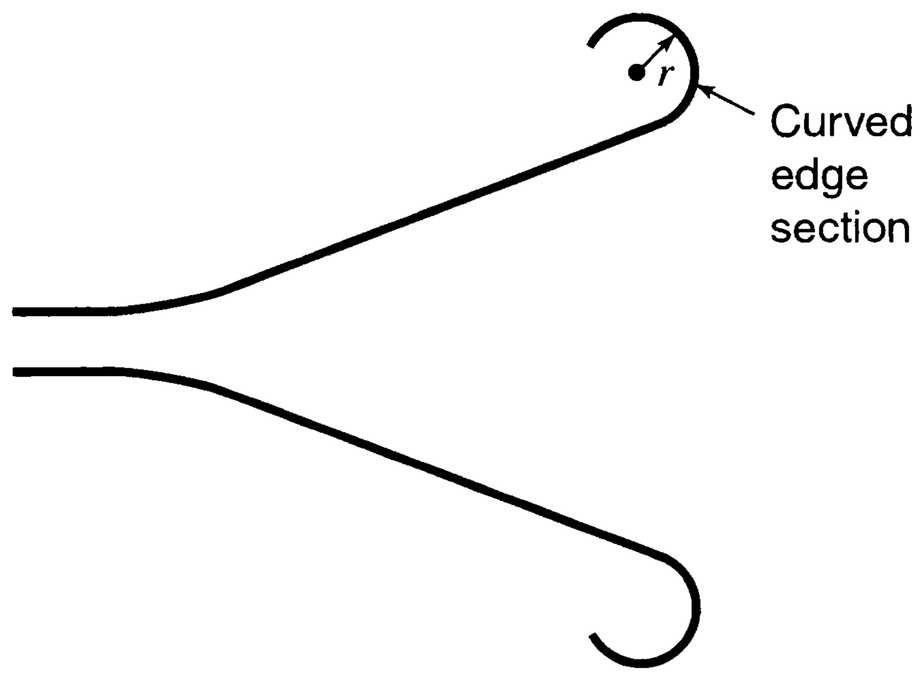


Figure 9–37
Cross section of Burnside and Chuang's aperture-matched horn. The radius of curvature r of the rolled edge should be at least $\lambda/4$.

REFERENCES（参考文献）

Alford, A. (1), "Long Slot Antennas," *Proc. Natl. Electronics Conf.,* 1946, p. 143.

Alford, A. (2), "Antenna for F-M Station WGHF," *Communications,* 26, 22, February 1946.

Barrow, W. L. (1), and F. D. Lewis, "The Sectoral Electromagnetic Horn," *Proc. IRE,* **27,** 41–50, January 1939.

Barrow, W. L. (2), and L. J. Chu, "Theory of the Electromagnetic Horn," *Proc. IRE,* **27,** 51–64, January 1939.

Booker, H. G. (1), "Slot Aerials and Their Relation to Complementary Wire Aerials," *JIEE (Lond.),* **93,** pt. IIIA, no. 4, 1946.

Born, Max (1), *Optik,* Verlag Julius Springer, Berlin, 1933, p. 155.

Brookner, E. (1), "Array Radars: An Update," *Microwave J.,* **30,** 134–138, February 1987, and 167–174, March 1987.

Burnside, W. D. (1), and C. W. Chuang, "An Aperture-Matched Horn Design," *IEEE Trans. Ants. Prop.,* **AP-30,** 790–796, July 1982.

Carver, K. R. (1), and J. W. Mink, "Microstrip Antenna Technology," *IEEE Trans. Ants. Prop.,* **AP-29,** 25–38, January 1981.

Chen, T-S (1), "Calculation of the Parameters of Ridge Waveguides," *IRE Trans. Microwave Theory Tech.,* **MTT-26,** 726–732, October 1978.

Chu, L. J. (1), and W. L. Barrow, "Electromagnetic Horn Design," *Trans AIEE,* **58,** 333–337, July 1939.

Chu, T. S. (1), R. W. Wilson, R. W. England, D. A. Gray and W. E. Legg, "Crawford Hill 7-m Millimeter Wave Antenna," *Bell Sys. Tech. J.,* **57,** 1257–1288, May–June 1978.

Cohn, S. B. (1), "Properties of the Ridge Wave Guide," *Proc. IRE,* **35,** 783–789, August 1947.

Derneryd, A. (1), "A Theoretical Investigation of the Rectangular Microstrip Antenna Element," *IEEE Trans. Ants. Prop.,* **AP-26,** 532–535, July 1978.

Dorne, A. (1), and D. Lazarus, in *Very High Frequency Techniques,* Radio Research Laboratory Staff, McGraw-Hill, New York, 1947, Chap. 7.

Jordan, E. C. (1), and W. E. Miller, "Slotted Cylinder Antennas," *Electronics,* **20,** 90–93, February 1947.

Kay, A. F. (1), "The Scalar Feed," AFCRL Rept., 64–347, March 1964.

King, A. P. (1), "The Radiation Characteristics of Conical Horn Antennas," *Proc. IRE,* **38,** 249–251, March 1950.

Kraus, J. D. (1), and D. A. Fleisch, *Electromagnetics with Applications,* 5th ed., McGraw-Hill, 1999, pp. 127–133.

Kraus, J. D. (2), *Radio Astronomy,* 2d ed., Cygnus-Quasar, 1986.

Levine, E. (1), J. Ashenasy and D. Treves, "Printed Dipole Arrays on a Cylinder," *Microwave J.,* **30,** 85–92, March 1987.

Meier, P. J., "Integrated Fin-Line Millimeter Components," *IEEE Trans. Microwave Theory Tech.,* **MTT-22,** 1209–1216, December 1974.

Munson, R. E. (1), "Conformal Microstrip Antennas and Microstrip Phased Arrays," *IEEE Trans. Ants. Prop.*, **AP-22,** 74–78, January 1974.

Munson, R. E. (2), "Microstrip Antennas," *Antenna Engineering Handbook,* McGraw-Hill, 1984, Chap. 7.

Nash, R. T. (1), "Stepped-Amplitude Distributions," *IEEE Trans. Ants. Prop.,* **AP-12,** July 1964.

Nash, R. T. (2), "Beam Efficiency Limitations of Large Antennas," *IEEE Trans. Ants. Prop.,* **AP-12,** 918–923, December 1964.

Peace, G. M. (1), and E. E. Swartz, "Amplitude Compensated Horn Antenna," *Microwave J.,* **7,** 66–68, February 1964.

Pozar, D. M. (1), "An Update on Microstrip Antenna Theory and Design Including Some Novel Feeding Techniques," *IEEE Ants. Prop. Soc. Newsletter,* **28,** 5–9, October 1986.

Pozar, D. M. (2), "Considerations for Millimeter Wave Printed Antennas," *IEEE Trans. Ants. Prop.,* **31,** 740–747, September 1983.

Rhodes, D. R. (1), "An Experimental Investigation of the Radiation Patterns of Electromagnetic Horn Antennas," *Proc. IRE,* **36,** 1101–1105, September 1948.

Risser, J. R. (1), in S. Silver (ed.), *Microwave Antenna Theory and Design,* McGraw-Hill, New York, 1949, Chap. 10, pp. 349–365.

Schelkunoff, S. A. (1), *Electromagnetic Waves,* Van Nostrand, New York, 1943, p. 212.

Schrank, H. (1), "Optimum Horns," *Ant. Prop. Soc. Newsletter,* **27,** 13–14, February 1985.

Sinclair, George (1), "The Patterns of Slotted Cylinder Antennas," *Proc. IRE,* **36,** 1487–1492, December 1948.

Southworth, G. C. (1), and A. P. King, "Metal Horns as Directive Receivers of Ultrashort Waves," *Proc. IRE,* **27,** 95–102, February 1939.

Stavis, G. (1), and A. Dome, in *Very High Frequency Techniques,* Radio Research Laboratory Staff, McGraw-Hill, New York, 1947, Chap. 6.

Terman, F. E. (1), *Radio Engineers' Handbook,* McGraw-Hill, New York, 1943, pp. 824–837 (this reference includes a summary of design data on horns).

Walton, K. L. (1), and V. C. Sundberg, "Broadband Ridge Horn Design," *Microwave J.,* **7,** 96–101, March 1964.

Watson, W. H. (1), *The Physical Principles of Wave Guide Transmission and Antenna Systems,* Oxford University Press, London, 1947.

Wohlleben, R. (1), H. Mattes and O. Lochner, "Simple, Small, Primary Feed for Large Opening Angles and High Aperture Efficiency," *Electronics Letters,* **8,** 19, September 1972.

PROBLEMS（习题）

9–2–1
（两条λ/2缝隙）

Two λ/2 slots. Two λ/2-slot antennas are arranged end-to-end in a large conducting sheet with a spacing of 1λ between centers. If the slots are fed with equal in-phase voltages, calculate and plot the far-field pattern in the 2 principal planes. Note that the *H* plane coincides with the line of the slots.

***9–5–1**
（盒式缝隙的阻抗）

Boxed-slot impedance. What is the terminal impedance of a slot antenna boxed to radiate only in one half-space whose complementary dipole antenna has a driving-point impedance of $Z = 150 + j0\ \Omega$? The box adds no shunt susceptance across the terminals.

***9–5–2**
（盒式缝隙）

Boxed slot. The complementary dipole of a slot antenna has a terminal impedance $Z = 90 + j10\ \Omega$. If the slot antenna is boxed so that it radiates only in one half-space, what is the terminal impedance of the slot antenna? The box adds no shunt susceptance at the terminals.

9–5–3
（敞开缝隙的阻抗）

Open-slot impedance. What dimensions are required of a slot antenna in order that its terminal impedance be $75 + j0\ \Omega$? The slot is open on both sides.

9–7–1
（50 Ω和100 Ω的贴片）

50 and 100 Ω patches. What value of the patch length W results in (*a*) a 50-Ω and (*b*) a 100-Ω input resistance for a rectangular patch as in Fig. 9–22*a*?

9–7–2 **Radiation from microstrip feed lines.** What prevents the microstrip feed lines of Examples 9–7.1, 2 and 3 from radiating like the patches that they feed?

9–7–3 **Microstrip line.** For a polystyrene substrate ($\varepsilon_r = 2.7$) what width-substrate thickness ratio results in a 50-Ω microstrip transmission line?

*9–9–1 **Optimum horn gain.** What is the approximate maximum power gain of an optimum horn antenna with a square aperture 9λ on a side?

9–9–2 **Horn pattern.** (*a*) Calculate and plot the *E*-plane pattern of the horn of Prob. 9–9–1, assuming uniform illumination over the aperture.

(*b*) What is the half-power beamwidth and the angle between first nulls?

9–9–3 **Rectangular horn antenna.** What is the required aperture area for an optimum rectangular horn antenna operating at 2 GHz with 12 dBi gain?

9–9–4 **Conical horn antenna.** What is the required diameter of a conical horn antenna operating at 2 GHz with a 12 dBi gain?

9–9–5 **Pyramidal horn.** (*a*) Determine the length L, aperture a_H and half-angles in E and H planes for a pyramidal electromagnetic horn for which the aperture $a_E = 8\lambda$. The horn is fed with a rectangular waveguide with TE_{10} mode. Take $\delta = \lambda/10$ in the E plane and $\delta = \lambda/4$ in the H plane.

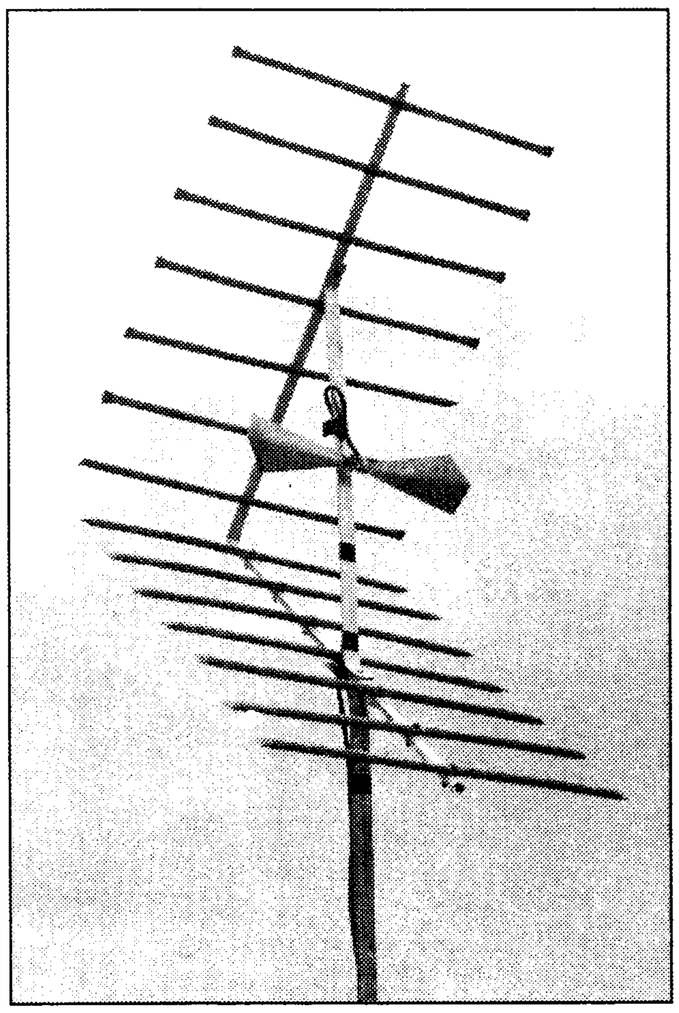
(*b*) What are the HPBWs in both E and H planes?

(*c*) What is the directivity?

(*d*) What is the aperture efficiency?

For computer programs, see Appendix C.

CHAPTER 10



Flat Sheet, Corner and Parabolic Reflector Antennas

平板和夹角反射器、抛物面反射镜天线

Topics in this chapter include:

- Flat-sheet reflectors（平板反射器）
- Corner reflectors（夹角反射器）
- Passive corner reflectors（无源夹角反射器）
- Parabolic reflectors（抛物面反射镜）
- Patterns of large circular apertures（大圆口径的波瓣图）

10–1 INTRODUCTION（引言）

Reflectors are widely used to modify the radiation pattern of a radiating element. For example, the backward radiation from an antenna may be eliminated with a plane sheet reflector of large enough dimensions. In the more general case, a beam of predetermined characteristics may be produced by means of a large, suitably shaped, and illuminated reflector surface. The characteristics of antennas with sheet reflectors or their equivalent are considered in this chapter.

Several reflector types are illustrated in Fig. 10–1. The arrangement in Fig. 10–1*a* has a large, flat sheet reflector near a linear dipole antenna to reduce the backward radiation (to the left in the figure). With small spacings between the antenna and sheet this arrangement also yields a substantial gain in the forward radiation. The desirable properties of the sheet reflector may be largely preserved with the reflector reduced in size as in Fig. 10–1*b* and even in the limiting case of Fig. 10–1*c*. Here the sheet has degenerated into a thin reflector element. Whereas the properties of the large sheet are relatively insensitive to small frequency changes, the thin reflector element is highly sensitive to frequency changes. The case of a $\lambda/2$ antenna with parasitic reflector element was treated in Sec. 8–5.

With two flat sheets intersecting at an angle α ($<180°$) as in Fig. 10–1*d*, a sharper radiation pattern than from a flat sheet reflector ($\alpha = 180°$) can be obtained. This arrangement, called an *active corner reflector antenna,* is most practical where apertures of 1 or 2λ are of convenient size. A corner reflector without an exciting antenna can be used as a *passive reflector* or target for radar waves. In this application the aperture may be many wavelengths, and the corner angle is *always 90°*. Reflectors with this angle have the property

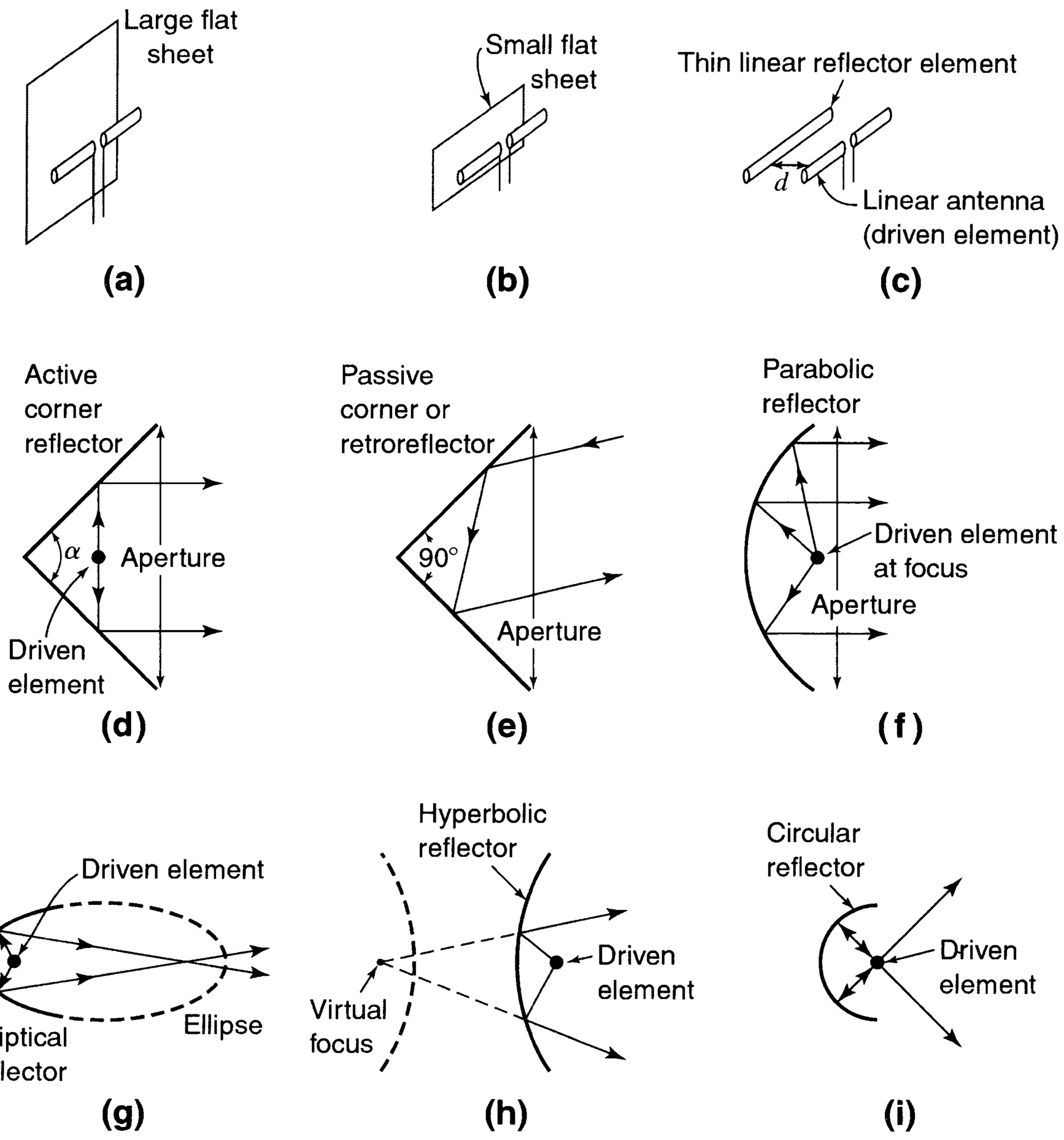


Figure 10–1
Reflectors of various shapes.

that an incident wave is reflected back toward its source as in Fig. 10–1*e*, the corner acting as a *retroreflector*.

When it is feasible to build antennas with apertures of many wavelengths, parabolic reflectors can be used to provide highly directional antennas. A parabolic reflector antenna is shown in Fig. 10–1*f*. The parabola reflects the waves originating from a source at the focus into a parallel beam, the parabola transforming the curved wave front from the feed antenna at the focus into a plane wave front. Many other shapes of reflectors can be employed for special applications. For instance, with an antenna at one focus, the elliptical reflector (Fig. 10–1*g*) produces a diverging beam with all reflected waves passing through the second focus of the ellipse. Examples of reflectors of other shapes are the hyperbolic (Stavis-1) and the circular reflectors (Ashmead-1) shown in Fig. 10–1*h* and *i*.

The plane sheet reflector, the corner reflector, the parabolic reflector and other reflectors are discussed in more detail in the following sections. Feed systems, aperture blockage,

aperture efficiency, diffraction, surface irregularities and frequency-selective surfaces are considered in later chapters.

10–2 FLAT SHEET REFLECTORS（平板反射器）

The problem of an antenna at a distance S from a perfectly conducting plane sheet reflector of infinite extent is readily handled by the method of images (Brown-1). In this method the reflector is replaced by an image of the antenna at a distance $2S$ from the antenna, as in Fig. 10–2. This situation is identical with a horizontal antenna above ground. If the antenna is a $\lambda/2$ dipole this in turn reduces to the problem of the W8JK antenna discussed in Sec. 6–7. Assuming zero reflector losses, the gain in field intensity of a $\lambda/2$ dipole antenna at a distance S from an infinite plane reflector is

$$G_f(\phi) = 2\sqrt{\frac{R_{11} + R_L}{R_{11} + R_L - R_{12}}}\,|\sin(S_r \cos\phi)| \tag{1}$$

where $S_r = 2\pi S/\lambda$.

The gain in (1) is expressed relative to a $\lambda/2$ antenna in free space with the same power input. The field patterns of $\lambda/2$ antennas at distances $S = \lambda/4$, $\lambda/8$ and $\lambda/16$ from the flat sheet reflector are shown in Fig. 10–3. These patterns are calculated from (1) for the case where $R_L = 0$.

The gain as a function of the spacing S is presented in Fig. 10–4 for assumed antenna loss resistances $R_L = 0$, 1 and 5 Ω. These curves are calculated from (1) for $\phi = 0$. It is apparent that very small spacings can be used effectively provided that losses are small. However, the bandwidth is narrow for small spacings. With wide spacings the gain is less, but the bandwidth is larger. Assuming an antenna loss resistance of 1 Ω, a spacing of 0.125λ yields the maximum gain.

A large flat sheet reflector can convert a bidirectional antenna array into a unidirectional system. An example is shown in Fig. 10–5*a*. Here a broadside array of 16 in-phase $\lambda/2$ elements spaced $\lambda/2$ apart is backed up by a large sheet reflector so that a unidirectional beam is produced. The feed system for the array is indicated, equal in-phase voltages being

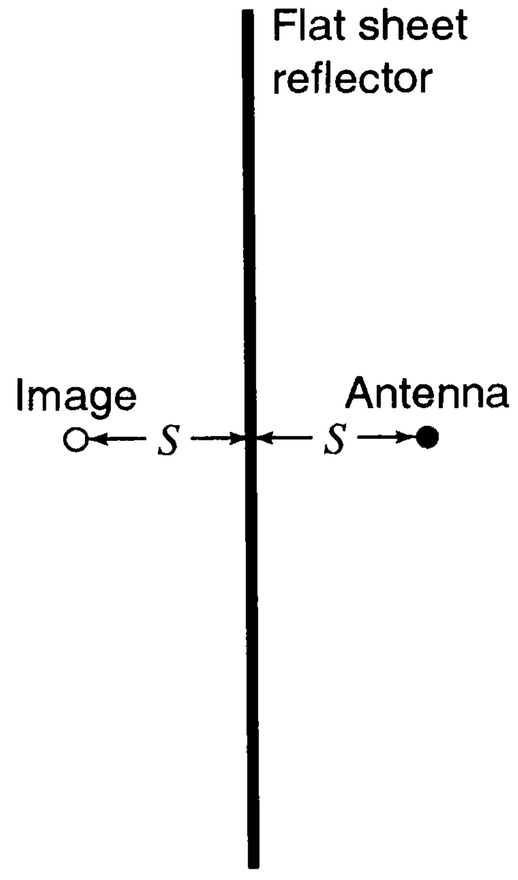


Figure 10–2
Antenna with flat sheet reflector.

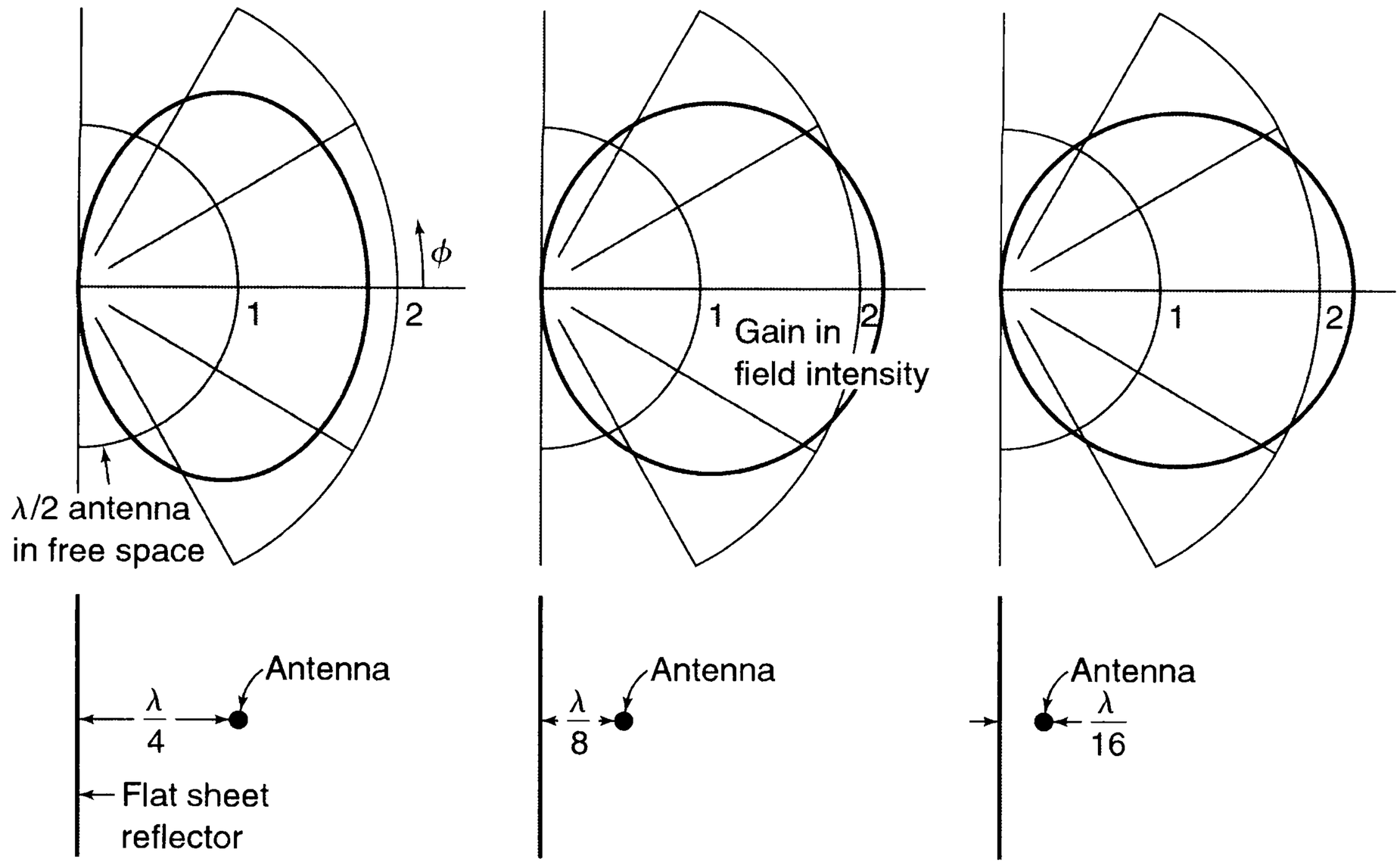


Figure 10–3
Field patterns of a $\lambda/2$ antenna at spacings of $\frac{1}{4}$, $\frac{1}{8}$ and $\frac{1}{16}\lambda$ from an infinite flat sheet reflector. Patterns give gain in field intensity over a $\lambda/2$ antenna in free space with same power input. For $\lambda/8$ spacing, the gain is 2.2 (= 6.7 dB = 8.9 dBi).

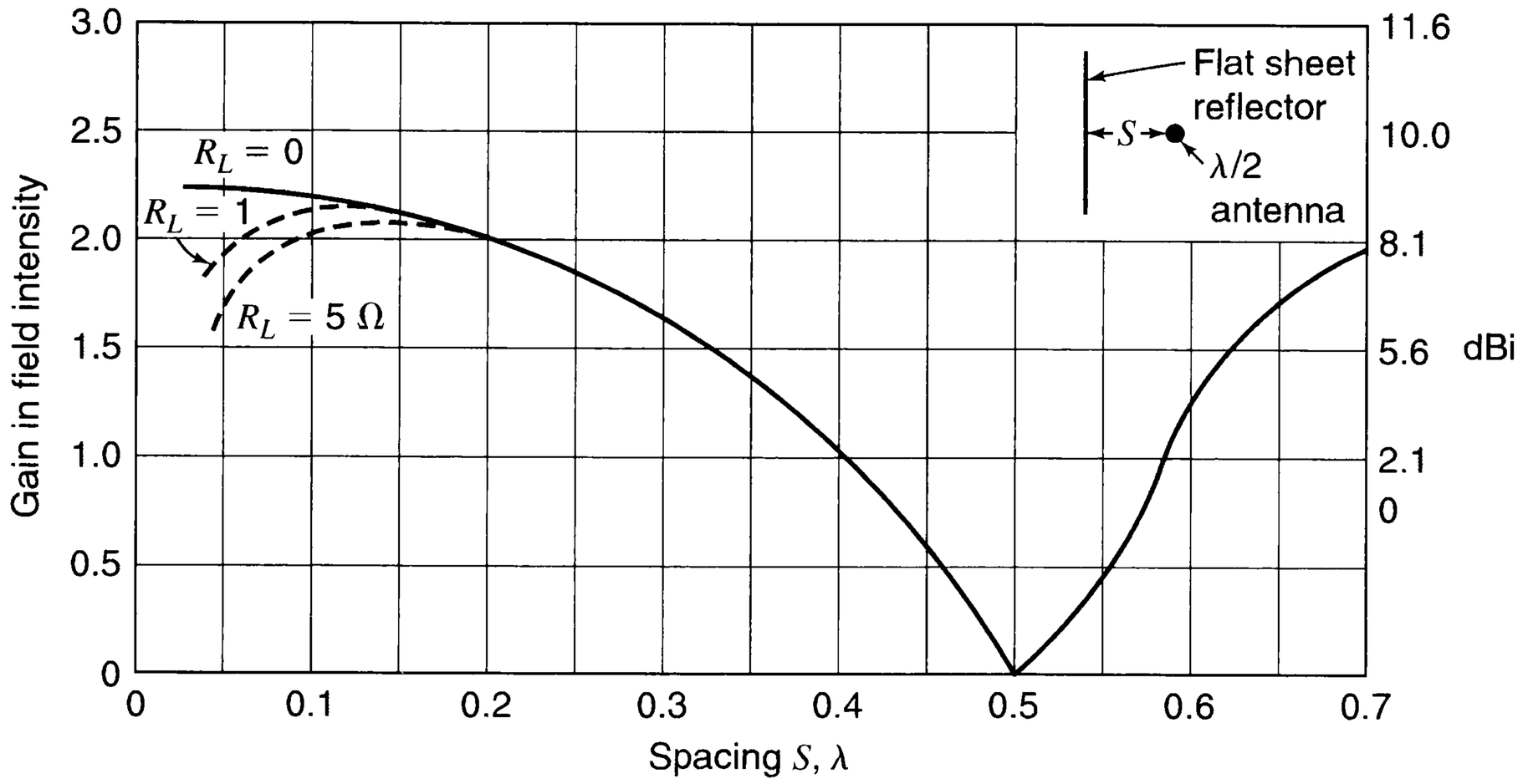
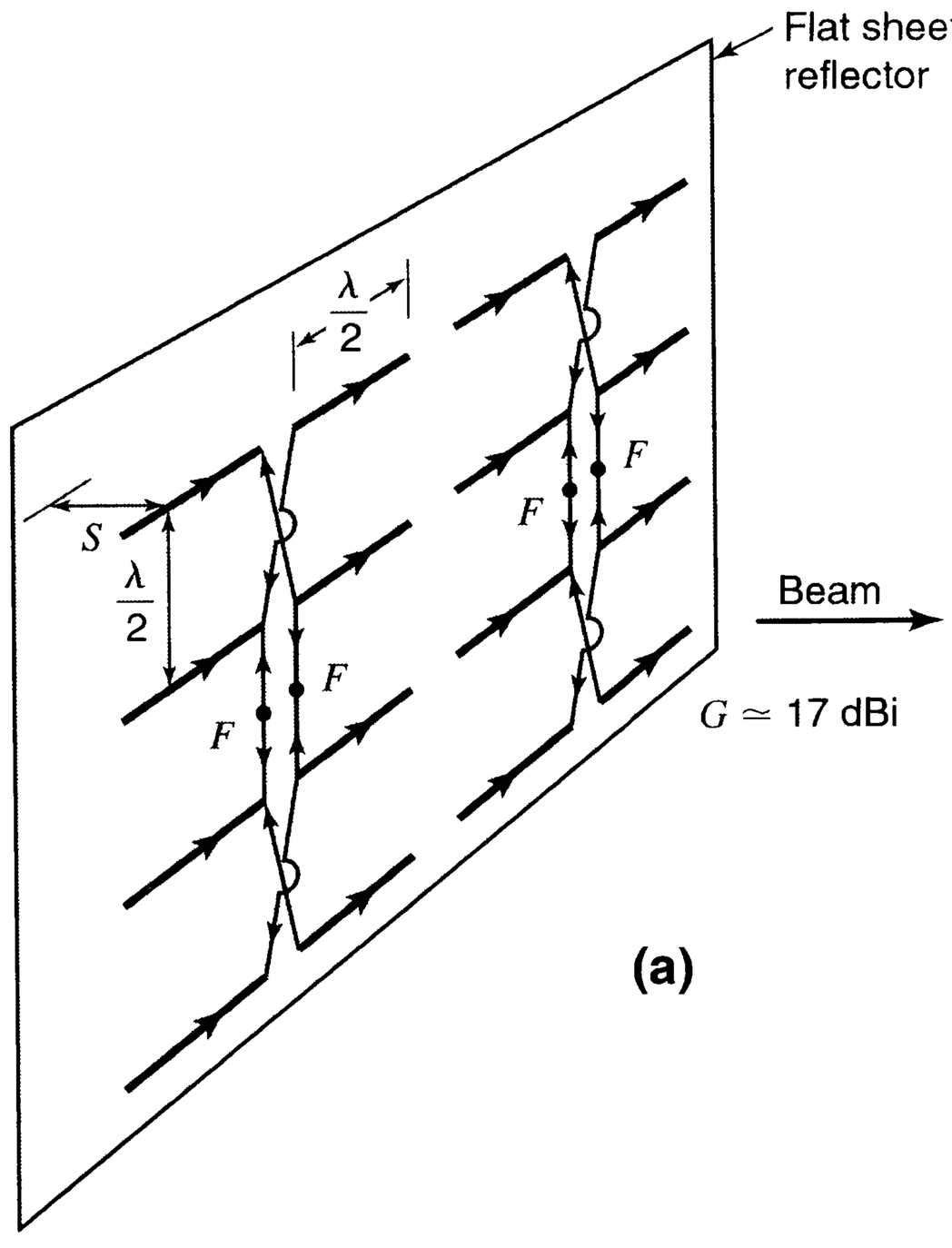


Figure 10–4
Gain in field intensity of $\lambda/2$ dipole antenna at distance S from flat sheet reflector. Gain is relative to $\lambda/2$ dipole antenna in free space with the same power input. Gain in dBi is also shown. Gain is in direction $\phi = 0$ and is shown for assumed loss resistances $R_L = 0.1$ and 5 Ω.



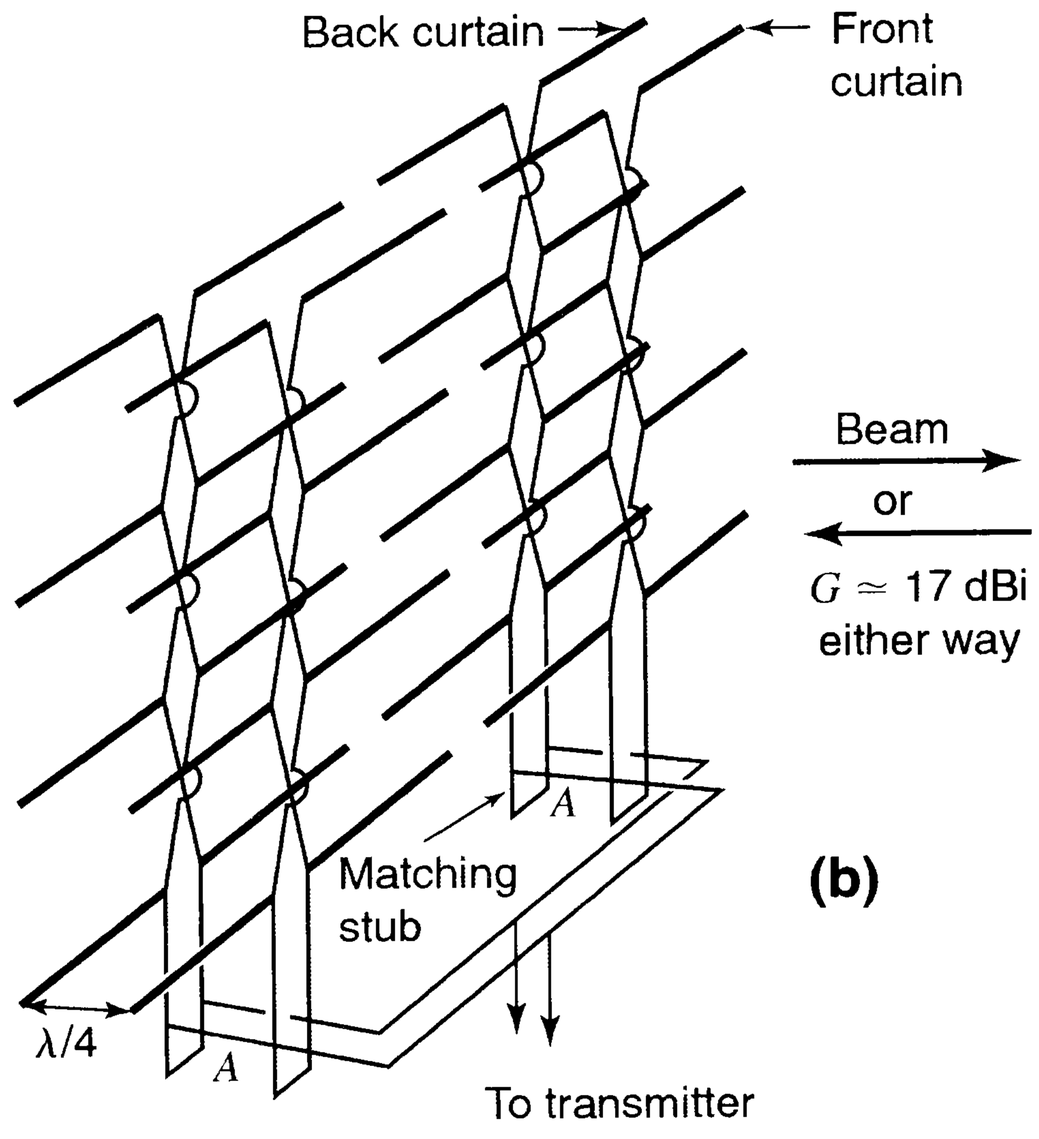


Figure 10–5
(*a*) Array of 16 $\lambda/2$ dipoles with flat-sheet reflector (billboard antenna). (*b*) Flat-sheet reflector replaced by a back curtain of 16 $\lambda/2$ dipoles for reduced wind resistance. Also by reversing the transmission lines at *A*, the beam direction can be reversed.

applied at the 2 pairs of terminals *F*-*F*. If the edges of the sheet extend some distance beyond the array, the assumption that the flat sheet is infinite in extent is a good first approximation. The choice of the spacing *S* between the array and the sheet usually involves a compromise between gain and bandwidth. If a spacing of $\lambda/8$ is used, the radiation resistance of the elements of a large array remains about the same as with no reflector present (Wheeler-1). This spacing also has the advantage over wider spacings of reduced interaction between elements. On the other hand, a spacing such as $\lambda/4$ provides a greater bandwidth, and the precise value of *S* is less critical in its effect on the element impedance.

Many shortwave broadcast stations operating at 15 to 50 m wavelengths and beaming worldwide use large curtain arrays; as in Fig. 10–5*b*, supported by tall towers. The array in Fig. 10–5*b* has a gain of about 17 dBi. Adding another set of curtain arrays (both front and back) alongside the array of Fig. 10–5*b* for a total of 64 dipoles increases the gain to about 20 dBi. Doubling again to 128 dipoles, the gain is about 23 dBi and doubling again to 256 the gain is about 26 dBi. Doubling once more to 512 dipoles the gain is about 29 dBi.

EXAMPLE **10–2.1 Short-Wave Station with 512 $\lambda/2$-Dipole Array**

This large array is equal in size to 16 of the arrays of Fig. 10–5*b* placed side-by-side.

(*a*) If the station transmitter delivers 25,000 W to the 512-dipole array what is the effective radiated power (ERP)?

(*b*) What is the field strength at a distance of 7500 km assuming that the signal is trapped between the ionosphere at a height of 250 km and the earth?

■ **Solution**

(*a*) antilog 2.9 = 794

$$794 \times 25{,}000 = 19.9 \text{ MW} \qquad \textit{Ans. (a)}$$

(*b*) $\text{ERP}/2\pi rh = 19.9 \times 10^6/2\pi 7.5 \times 10^6 \times 250 \times 10^3 = 1.68 \times 10^{-6}$ W/m^2

$E = \sqrt{WZ/\text{m}^2} = \sqrt{1.68 \times 10^{-6} \times 377} = 25$ mV/m, a very strong signal. *Ans. (b)*

When the reflecting sheet is reduced in size, the analysis is less simple. The situation is shown in Fig. 10–6*a*. There are 3 principal angular regions:

Region 1 (above or in front of the sheet). In this region the radiated field is given by the resultant of the *direct field* of the dipole and the *reflected field* from the sheet.

Region 2 (above and below at the sides of the sheet). In this region there is only the direct field from the dipole. This region is in the shadow of the reflected field.

Region 3 (below or behind the sheet). In this region the sheet acts as a shield, producing a full shadow (no direct or reflected fields, only diffracted fields).

If the sheet is 1 or 2λ in width and the dipole is close to it, image theory accounts adequately for the radiation pattern in region 1. In region 2, the distant field is dominated by the direct ray from the dipole. In the full shadow behind the sheet (region 3) the Geometrical Theory of Diffraction (GTD) must be used.[1] The pattern in this region is effectively that of 2 weak line sources, one along each edge. The fields in the 3 regions are shown in Fig. 10–6*b* for the case of the sheet width dimension $D = 2.25\lambda$ and the dipole spacing $d = 3\lambda/8$. It is assumed that the sheet is very long perpendicular to the page ($\gg D$).

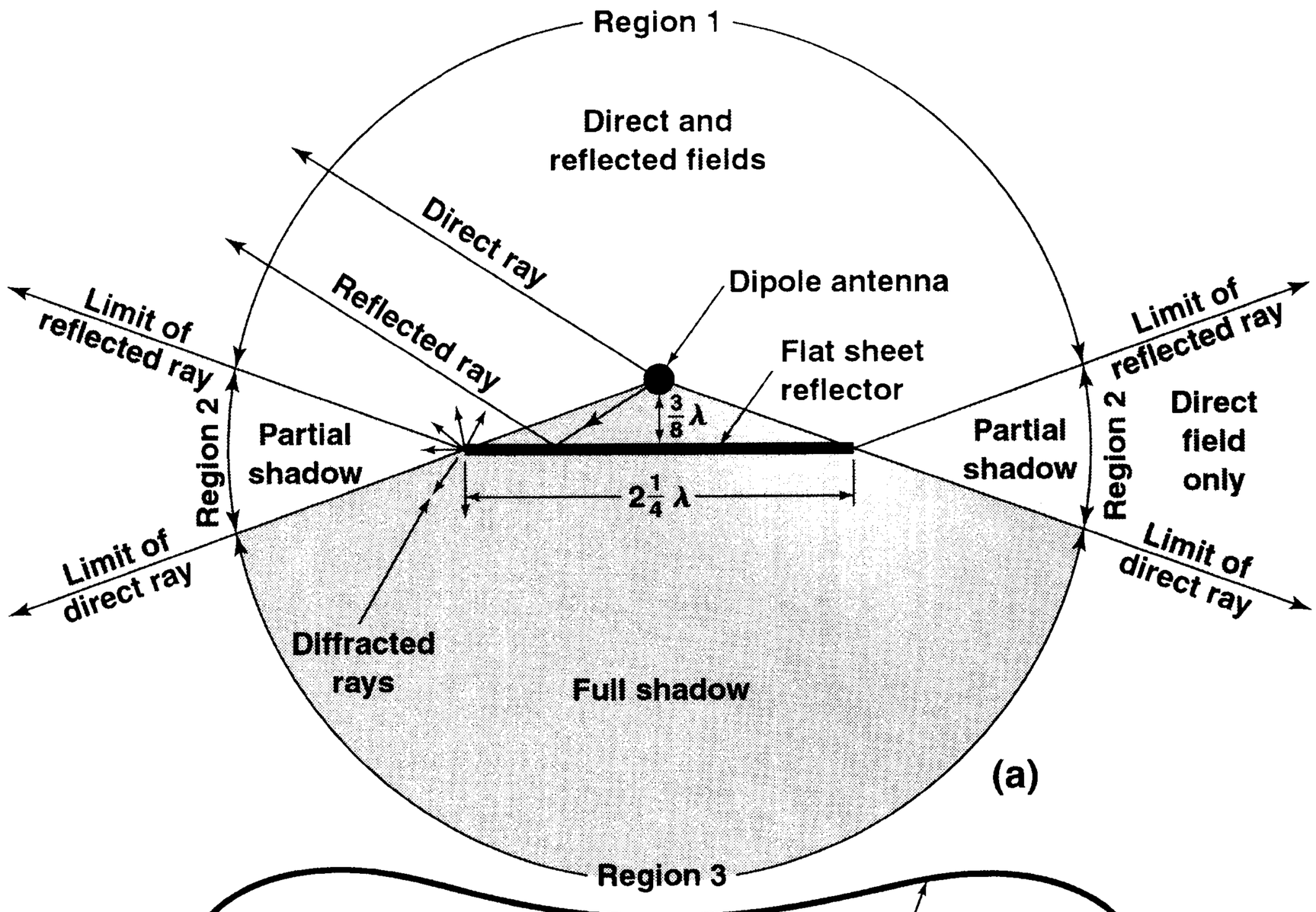
Narrower reflecting sheets result in more radiation into region 3 but this diffracted radiation can be minimized by using a rolled edge (radius of curvature $>\lambda/4$) and absorbing material, as suggested in Fig. 10–6*c*.

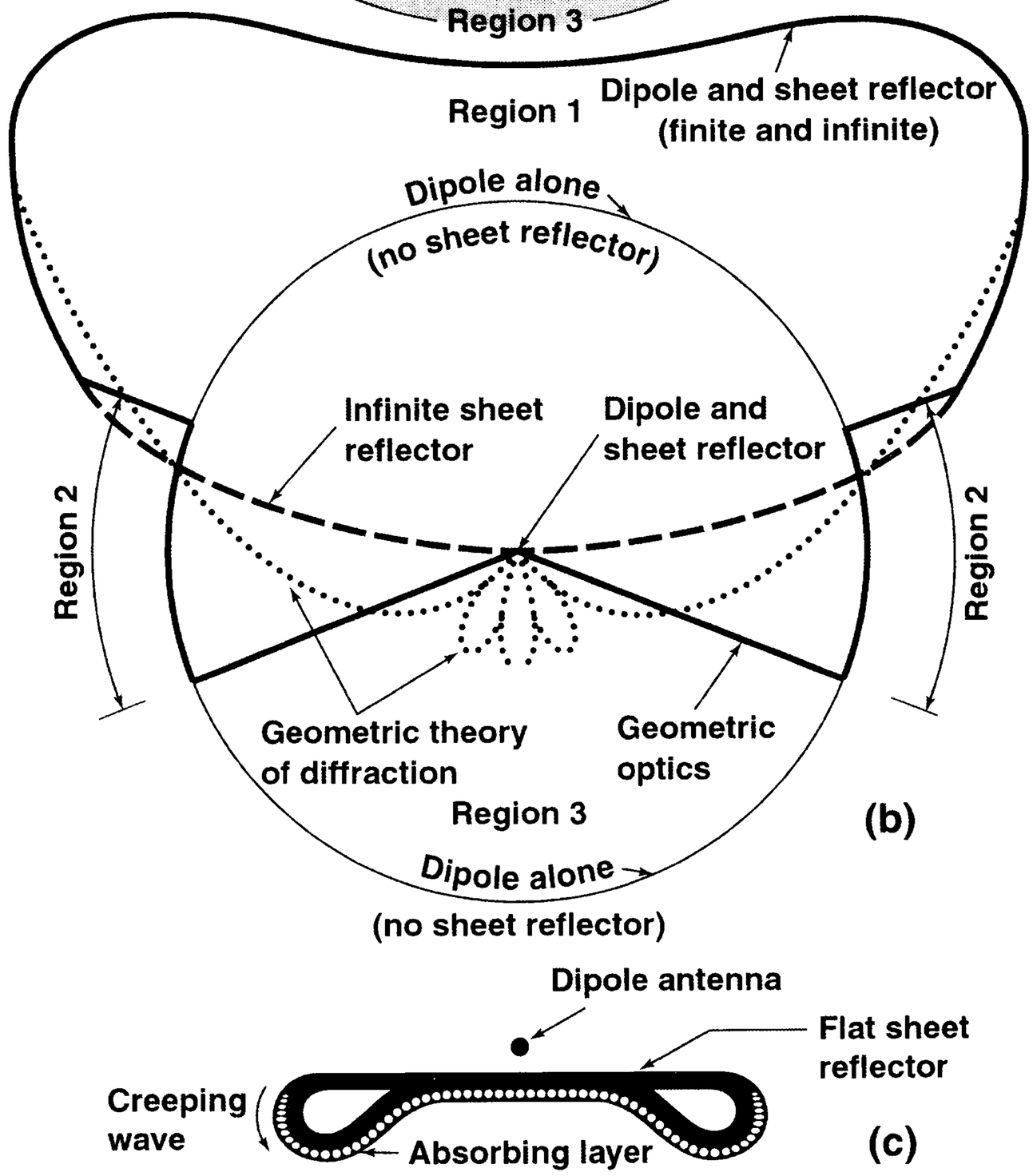
10–3 CORNER REFLECTORS（夹角反射器）

10–3a Introduction by John Kraus［引言(John Kraus)］

In 1938 while analyzing the radiation from a dipole parallel to and closely spaced from a flat reflecting sheet, I realized that when the sheet is replaced by its image, the dipole and its image form a W8JK array. When the flat sheet (180° included angle) is folded into a square (90°) corner the theory calls for 3 images, and my calculations showed correspondingly higher gain. Thus, the corner reflector developed as an extension of my analysis of the W8JK array. I immediately constructed several corner reflectors to obtain experimental

[1] Diffraction is also discussed in Secs. 5–14 and 9–2. For more details on the principles of physical and geometrical diffraction theories see Kraus-1.





The Geometric Theory of Diffraction (GTD) in a nutshell.

Figure 10–6
(*a*) Dipole antenna with 2.25λ flat sheet reflector with 3 regions of radiation according to geometric optics. (*b*) Field pattern of dipole and sheet reflector according to geometric optics (heavy solid line) and according to geometric theory of diffraction (dotted line). The solid circle indicates the field from the dipole alone (in free space) and the dashed line gives the pattern for dipole with an infinite sheet reflector. (*c*) Modification of edges of sheet to reduce diffracted back-side radiation (in region 3).

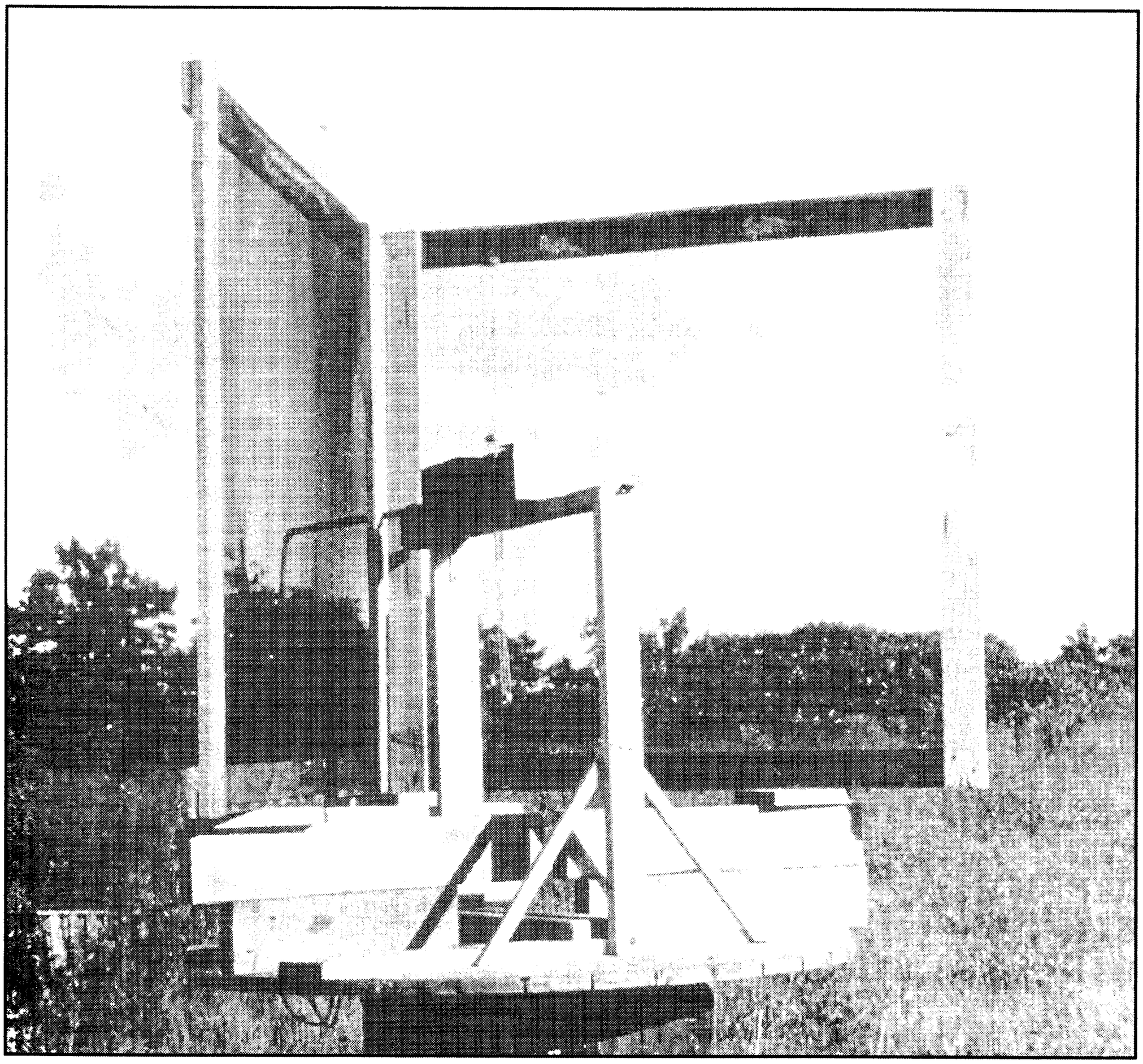


Figure 10–7
90-degree or square-corner reflector on turntable for measurements at $\lambda = 1$ m outside my home in 1939. The reflector is made of copper window screen. The screen reflector is 70 cm square.

confirmation. I tried parallel-wire grid reflectors, modifying both the spacing and length of the reflector wires to determine the limiting dimensions required. Figure 10–7 shows a 90° corner for $\lambda = 1$ m operation I built in 1938 with patterns measured by rotating the antenna on a turntable.

The following discussion is based mainly on my articles in the *Proceedings of the I.R.E.,* in *Radio* and a patent filed for me by the *Radio Corporation of America (RCA)* (Kraus-2, 3, 4).

10–3b Corner Reflector Design（夹角反射器的设计）

Two flat reflecting sheets intersecting at an angle or corner as in Fig. 10–8 form an effective directional antenna. When the corner angle $\alpha = 90°$, the sheets intersect at right angles, forming a square-corner reflector. Corner angles both greater or less than 90° can be used although there are practical disadvantages to angles much less than 90°. A corner reflector

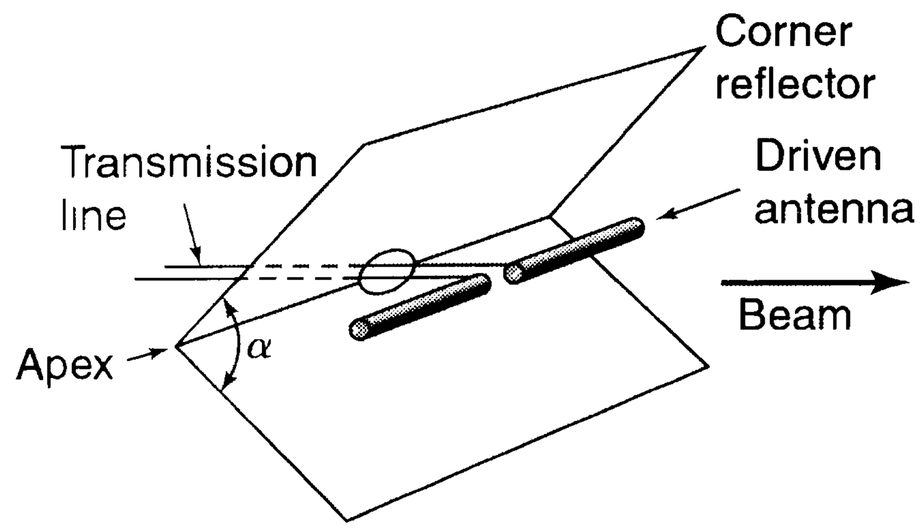


Figure 10–8
Corner reflector antenna.

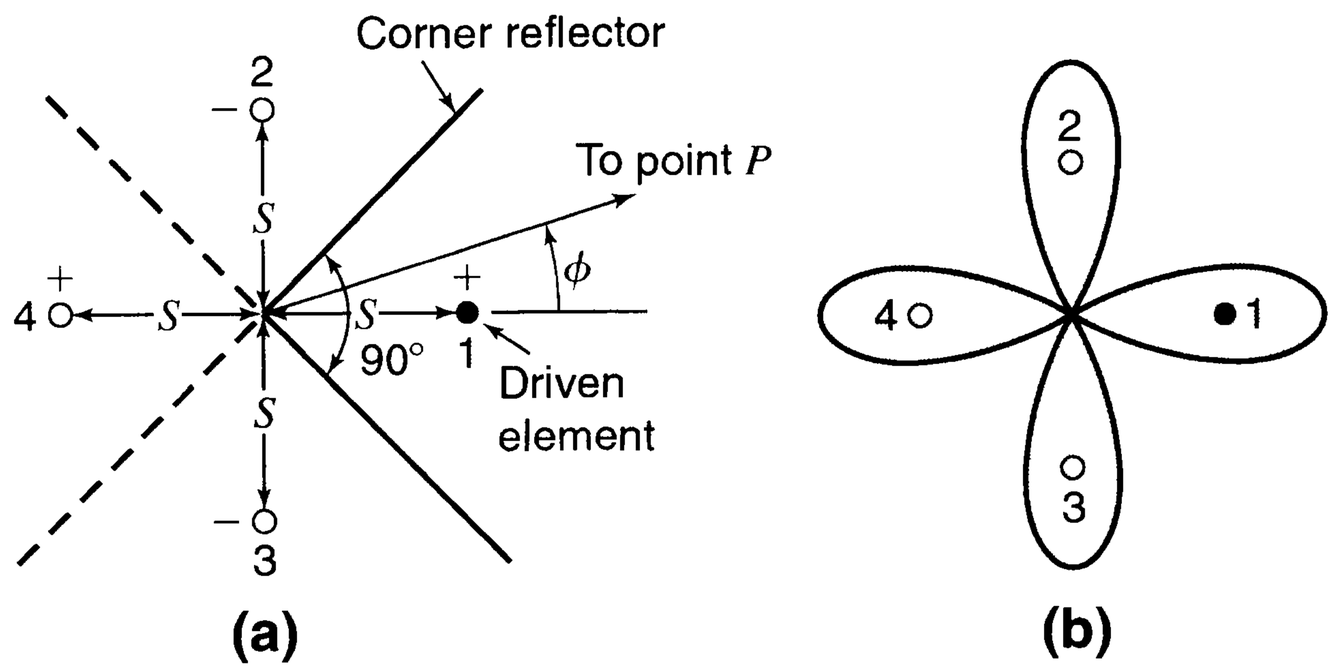


Figure 10–9
Square-corner reflector with images used in analysis (*a*) and 4-lobed pattern of driven element and images (*b*).

with $\alpha = 180°$ is equivalent to a flat sheet reflector and may be considered as a limiting case of the corner reflector. This case has been treated in Sec. 10–2.

Assuming perfectly conducting reflecting sheets of infinite extent, the *method of images* can be applied to analyze the corner reflector antenna for angles $\alpha = 180°/n$, where n is any positive integer. This method of handling corners is well known in electrostatics (Jeans-1). Corner angles of 180° (flat sheet), 90°, 60°, etc., can be treated in this way. Corner reflectors of intermediate angle cannot be determined by this method but can be interpolated approximately from the others.

In the analysis of the 90° corner reflector there are 3 image elements, 2, 3 and 4, located as shown in Fig. 10–9*a*. The driven antenna 1 and the 3 images have currents of equal magnitude. The phase of the currents in 1 and 4 is the same. The phase of the currents in 2 and 3 is the same but 180° out of phase with respect to the currents in 1 and 4. All elements are assumed to be $\lambda/2$ long.

At the point P at a large distance D from the antenna, the field intensity is

$$E(\phi) = 2kI_1|[\cos(S_r \cos\phi) - \cos(S_r \sin\phi)]| \tag{1}$$

where

I_1 = current in each element
S_r = spacing of each element from the corner, rad
$= 2\pi(S/\lambda)$
k = constant involving the distance D, etc.

For arbitrary corner angles, analysis involves integrations of cylindrical functions which can be approximated by infinite sums as shown by Klopfenstein (1). The emf V_1 at the

terminals at the center of the driven element is

$$V_1 = I_1 Z_{11} + I_1 R_{1L} + I_1 Z_{14} - 2I_1 Z_{12} \tag{2}$$

where

Z_{11} = self-impedance of driven element
R_{1L} = equivalent loss resistance of driven element
Z_{12} = mutual impedance of elements 1 and 2
Z_{14} = mutual impedance of elements 1 and 4

Similar expressions can be written for the emf's at the terminals of each of the images. Then if P is the power delivered to the driven element (power to each image element is also P), we have from symmetry that

$$I_1 = \sqrt{\frac{P}{R_{11} + R_{1L} + R_{14} - 2R_{12}}} \tag{3}$$

Substituting (3) in (1) yields

$$E(\phi) = 2k\sqrt{\frac{P}{R_{11} + R_{1L} + R_{14} - 2R_{12}}}\,|[\cos(S_r \cos\phi) - \cos(S_r \sin\phi)]| \tag{4}$$

The field intensity at the point P at a distance D from the driven $\lambda/2$ element with the reflector removed is

$$E_{HW}(\phi) = k\sqrt{\frac{P}{R_{11} + R_{1L}}} \tag{5}$$

where k = the same constant as in (1) and (4)

This is the relation for field intensity of a $\lambda/2$ dipole antenna in free space with a power input P and provides a convenient reference for the corner reflector antenna. Thus, dividing (4) by (5), we obtain the gain in field intensity of a square-corner reflector antenna over a single $\lambda/2$ antenna in free space with the same power input, or

$$G_f(\phi) = \frac{E(\phi)}{E_{HW}(\phi)}$$

$$= 2\sqrt{\frac{R_{11} + R_{1L}}{R_{11} + R_{1L} + R_{14} - 2R_{12}}}\,|[\cos(S_r \cos\phi) - \cos(S_r \sin\phi)]| \tag{6}$$

where the expression in brackets is the pattern factor and the expression included under the radical sign is the coupling factor. The pattern shape is a function of both the angle ϕ and the antenna-to-corner spacing S. The pattern calculated by (6) has 4 lobes as shown in Fig. 10–9*b*. However, only one of the lobes is real.

Expressions for the gain in field intensity of corner reflectors with corner angles of 60°, 45°, etc., can be obtained in a similar manner. The driven element is a $\lambda/2$ dipole. For the 60° corner the analysis requires a total of 6 elements, 1 actual antenna and 5 images

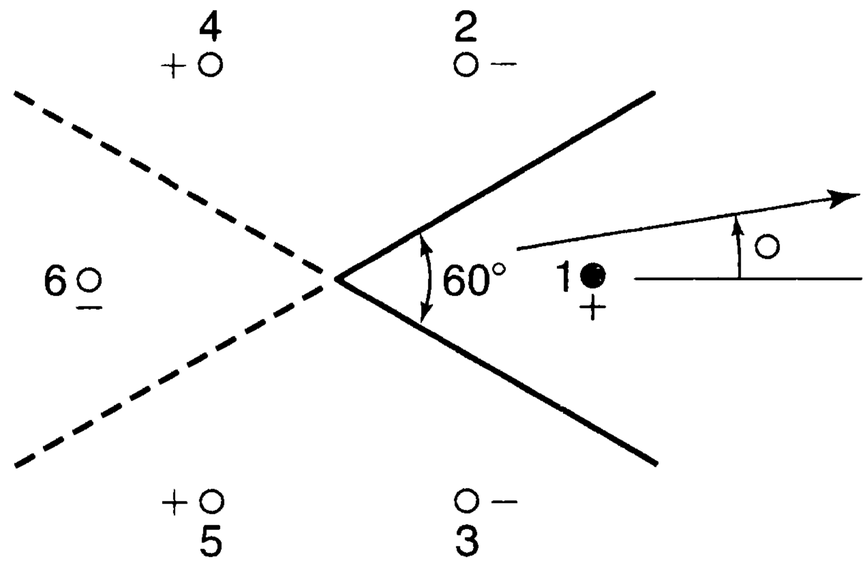


Figure 10–10
A 60° corner reflector with images used in analysis.

Table 10–1 Gain-pattern formulas for corner reflector antennas

Corner angle, deg	Number of elements in analysis	Gain in field intensity over $\lambda/2$ antenna in free space with same power input
180	2	$2\sqrt{\dfrac{R_{11}+R_{1L}}{R_{11}+R_{1L}-R_{12}}}\sin(S_r\cos\phi)$
90	4	$2\sqrt{\dfrac{R_{11}+R_{1L}}{R_{11}+R_{1L}+R_{14}-2R_{12}}}\lvert[\cos(S_r\cos\phi)-\cos(S_r\sin\phi)]\rvert$
60	6	$2\sqrt{\dfrac{R_{11}+R_{1L}}{R_{11}+R_{1L}+2R_{14}-2R_{12}-R_{16}}}\times\lvert\{\sin(S_r\cos\phi)-\sin[S_r\cos(60^\circ-\phi)]-\sin[S_r\cos(60^\circ+\phi)]\}\rvert$

as in Fig. 10–10. Gain-pattern expressions for corner reflectors of 90 and 60° are listed in Table 10–1. The expression for a 180° "corner" or flat sheet is also included.

In the formulas of Table 10–1 it is assumed that the reflector sheets are perfectly conducting and of infinite extent. Curves of gain versus spacing calculated from these relations are presented in Fig. 10–11. The gain given is in the direction $\phi = 0$. Two curves are shown for each corner angle. The solid curve in each case is computed for zero losses ($R_{1L} = 0$), while the dashed curve is for an assumed loss resistance $R_{1L} = 1\ \Omega$. It is apparent that for efficient operation too small a spacing should be avoided. A small spacing is also objectionable because of narrow bandwidth. On the other hand, too large a spacing results in less gain.[1]

The calculated pattern of a 90° corner reflector with antenna-to-corner spacing $S = 0.5\lambda$ is shown in Fig. 10–12*a*. The gain is nearly 10 dB over a reference $\lambda/2$ antenna or 12 dBi. This pattern is typical if the spacing S is not too large. If S exceeds a certain value, a multilobed pattern may be obtained. For example, a square-corner reflector with $S = 1.0\lambda$ has a 2-lobed pattern as in Fig. 10–12*b*. If the spacing is increased to 1.5, the pattern shown in Fig. 10–12*c* is obtained with the major lobe in the $\phi = 0$ direction but with minor lobes present. This pattern may be considered as belonging to a higher-order radiation mode of the antenna. The gain over a single $\lambda/2$ dipole antenna is 12.9 dB ($\simeq$15 dBi).

[1]Displacing the driven dipole from the bisector of the corner angle shifts (squints) the beam direction to the other side of the bisector (see Prob. 10–3–13).

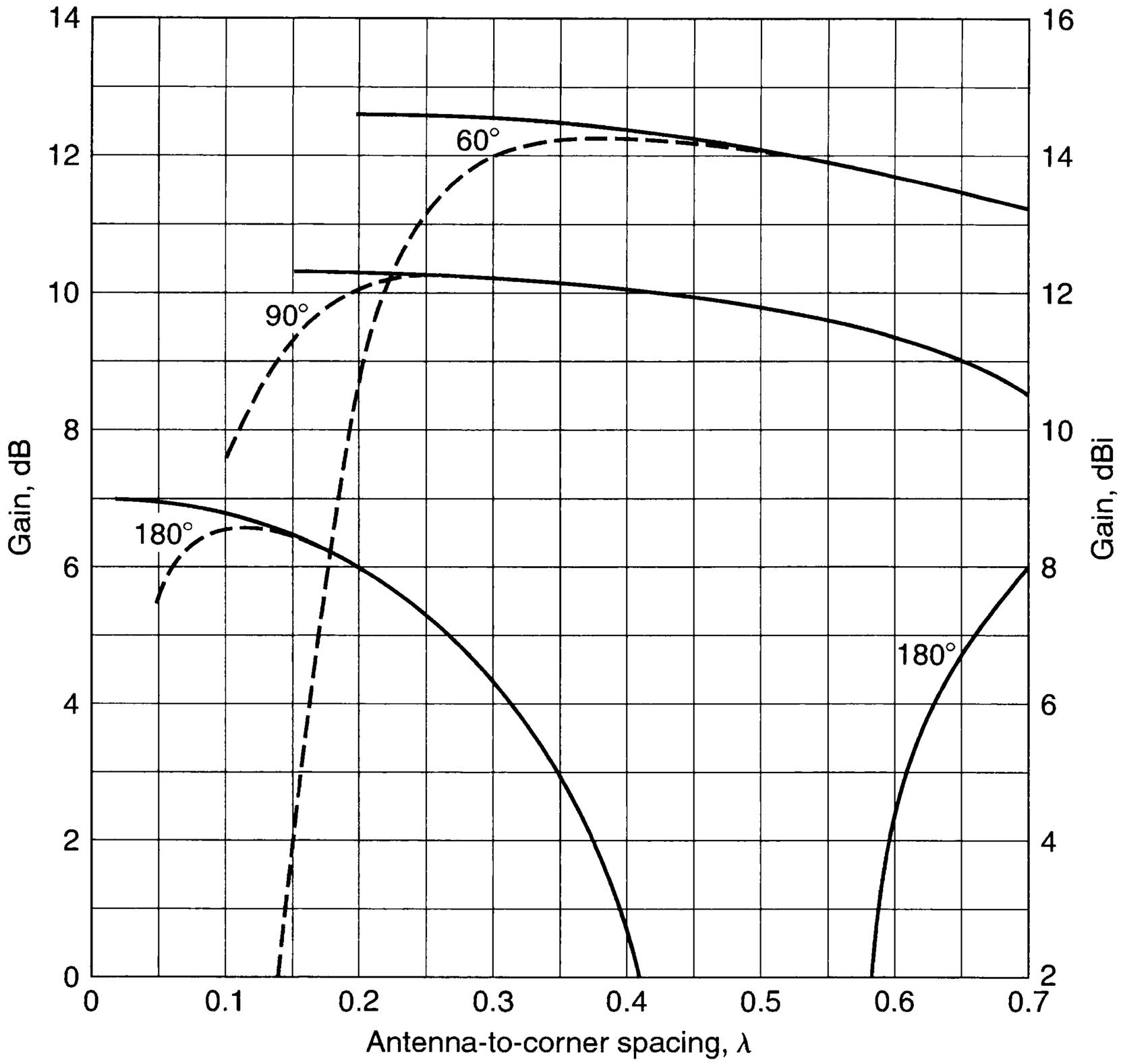


Figure 10–11
Gain of corner reflector antennas over a $\lambda/2$ dipole antenna in free space with the same power input as a function of the antenna-to-corner spacing. Gain is in the direction $\phi = 0$ and is shown for zero loss resistance (solid curves) and for an assumed loss resistance of 1 Ω ($R_{1L} = 1\ \Omega$) (dashed curves). (*After Kraus.*)

Restricting patterns to the lower-order radiation mode (no minor lobes), it is generally desirable that S lie between the following limits:

Corner angle, α	Corner-to-dipole spacing, S
90°	0.25–0.7λ
180° (flat sheet)	0.1–0.3λ

The terminal resistance R_T of the driven antenna is obtained by dividing (2) by I_1 and taking the real parts of the impedances. Thus,

$$R_T = R_{11} + R_{1L} + R_{14} - 2R_{12} \tag{7}$$

If $R_{1L} = 0$, the terminal resistance is all radiation resistance. The variation of the terminal radiation resistance of the driven element is presented in Fig. 10–13*a* as a function of the

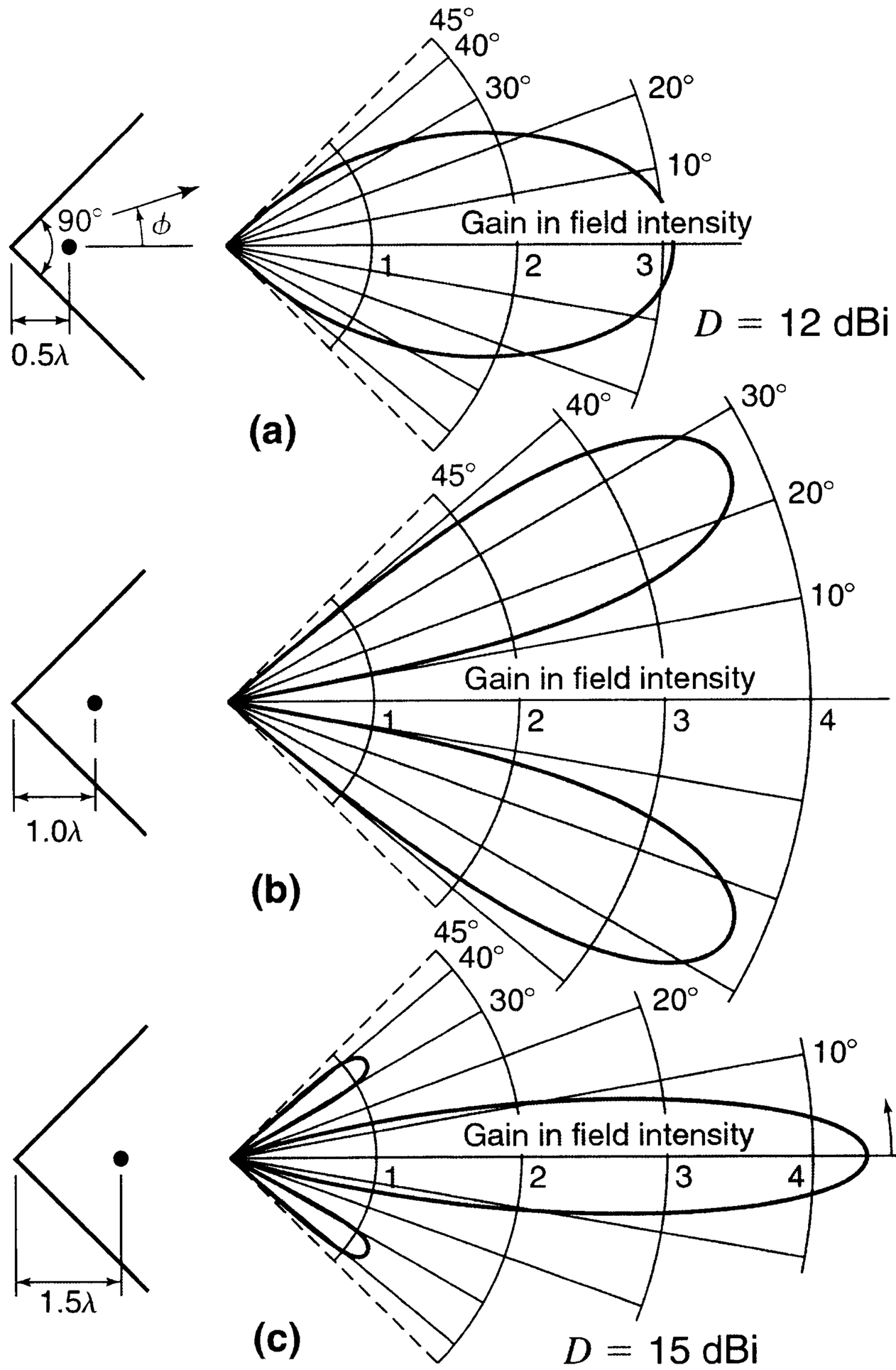


Figure 10–12
Calculated patterns of square-corner reflector antennas with antenna-to-corner spacings of (*a*) 0.5λ, (*b*) 1.0λ and (*c*) 1.5λ. Patterns give gain relative to the λ/2 dipole antenna in free space with the same power input.

spacing S for corner angles $\alpha = 180, 90$ and $60°$. We note that for $\alpha = 90°$ and $S = 0.35\lambda$, the resistance of the driven $\lambda/2$ dipole is the same as for a $\lambda/2$ dipole in free space.

In the above analysis it is assumed that the reflectors are perfectly conducting and of infinite extent, with the exception that the gains with a finitely conducting reflector may be approximated with a proper choice of R_{1L}. The analysis provides a good first approximation to the gain-pattern characteristics of actual corner reflectors with finite sides provided that the sides are not too small. Neglecting edge effects, a suitable value for the length of sides may be arrived at by the following line of reasoning. An essential region of the reflector is that near the point at which a wave from the driven antenna is reflected parallel to the axis. For example, this is the point A of the square-corner reflector of Fig. 10–13*b*. This point is at a distance of $1.41S$ from the corner C, where S is the antenna-to-corner spacing. If the reflector ends at the point B at a distance $L = 2S$ from the corner, as in Fig. 10–13*b*, the reflector extends approximately $0.6S$ beyond A. With the reflector ending at B, it is to be noted that the only waves reflected from infinite sides, but not from finite sides, are those

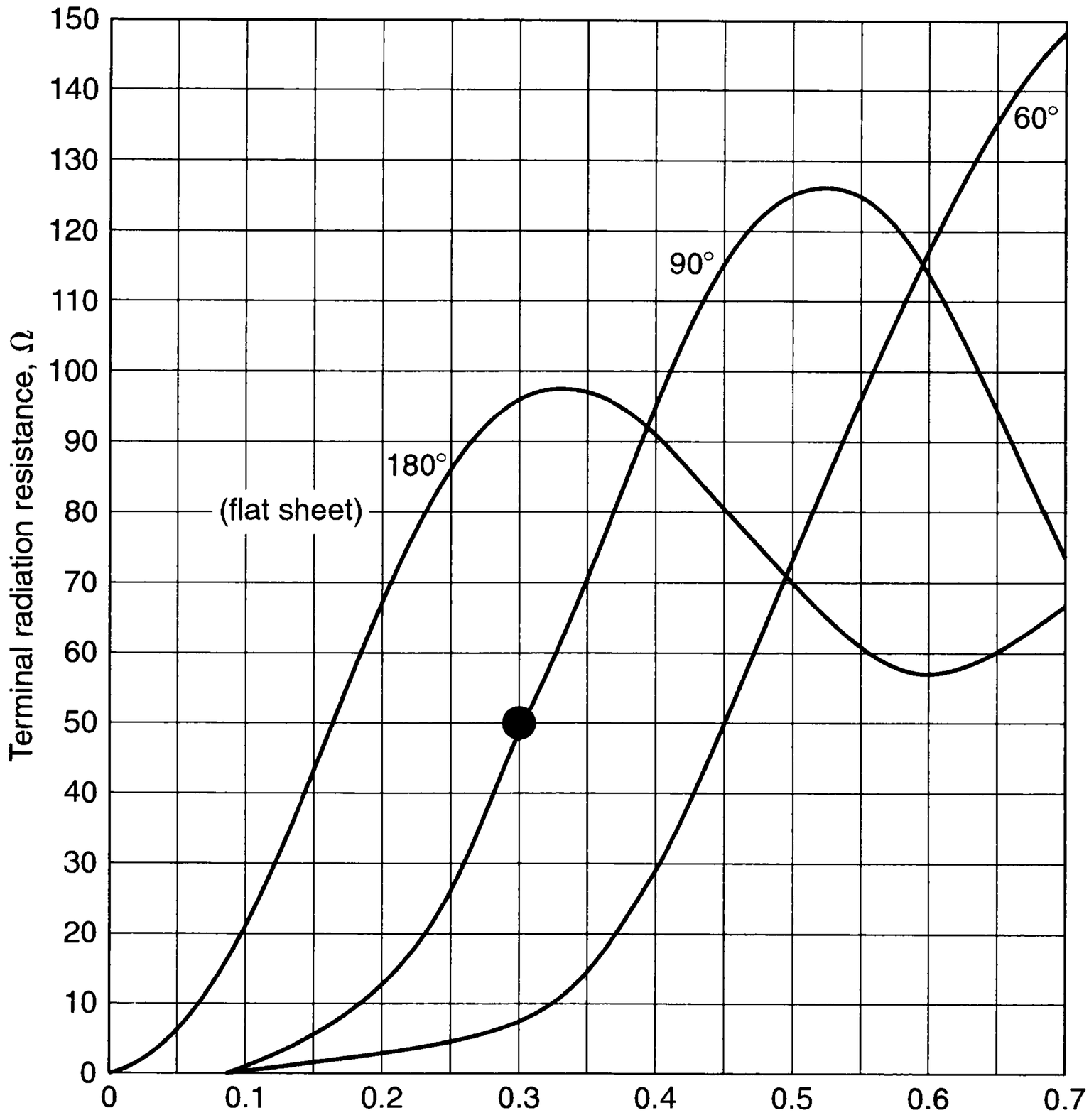


Figure 10–13a
Radiation resistance of driven λ/2 dipole antenna as a function of the dipole-to-corner spacing for 60°, 90° and 180° corner reflectors (*after Kraus*). Note the convenient 50-Ω radiation resistance for the 90° corner with dipole-to-corner spacing of 0.3 λ (large dot).

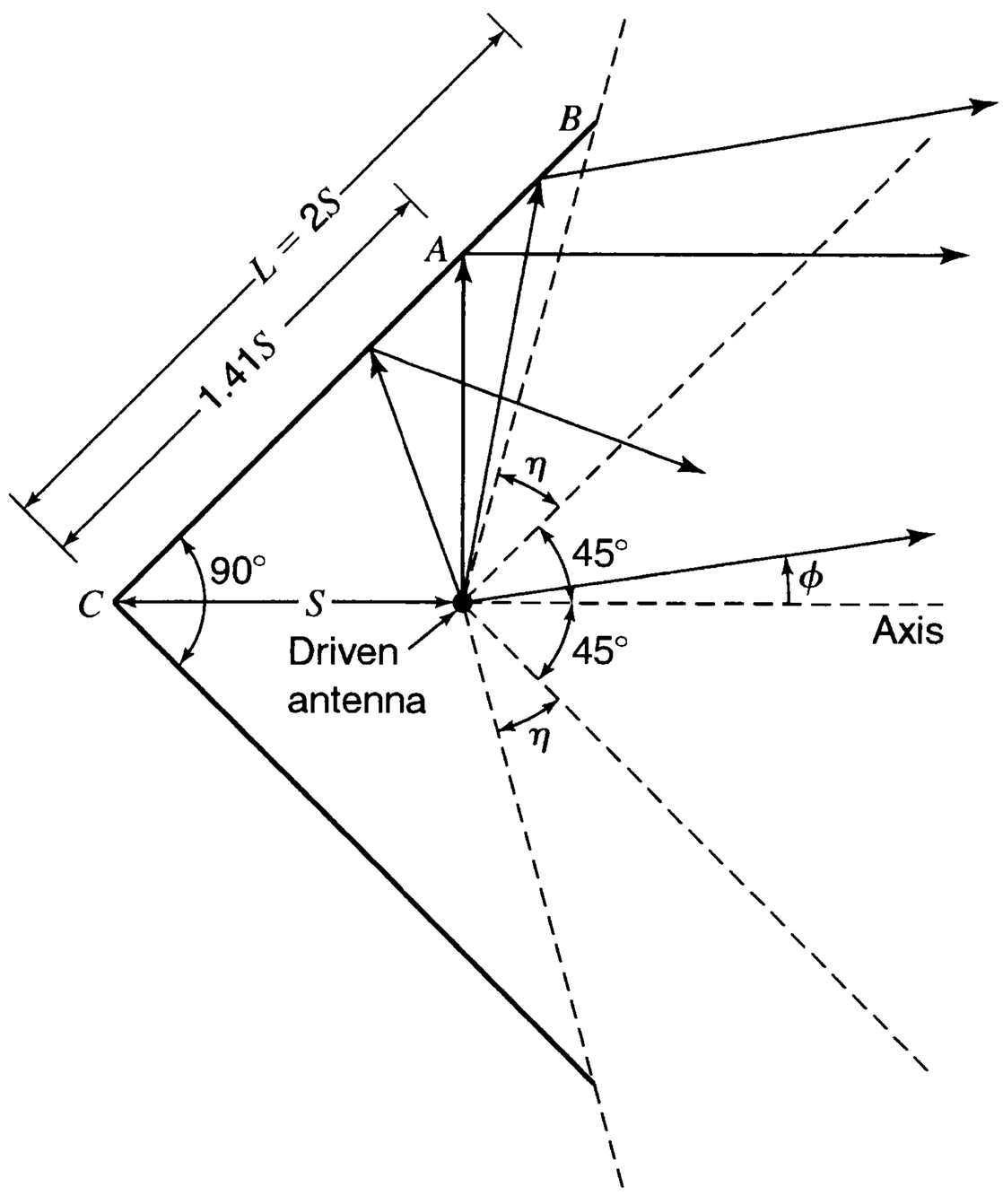


Figure 10–13b
Square-corner reflector with sides of length *L* equal to twice the antenna-to-corner spacing *S*.

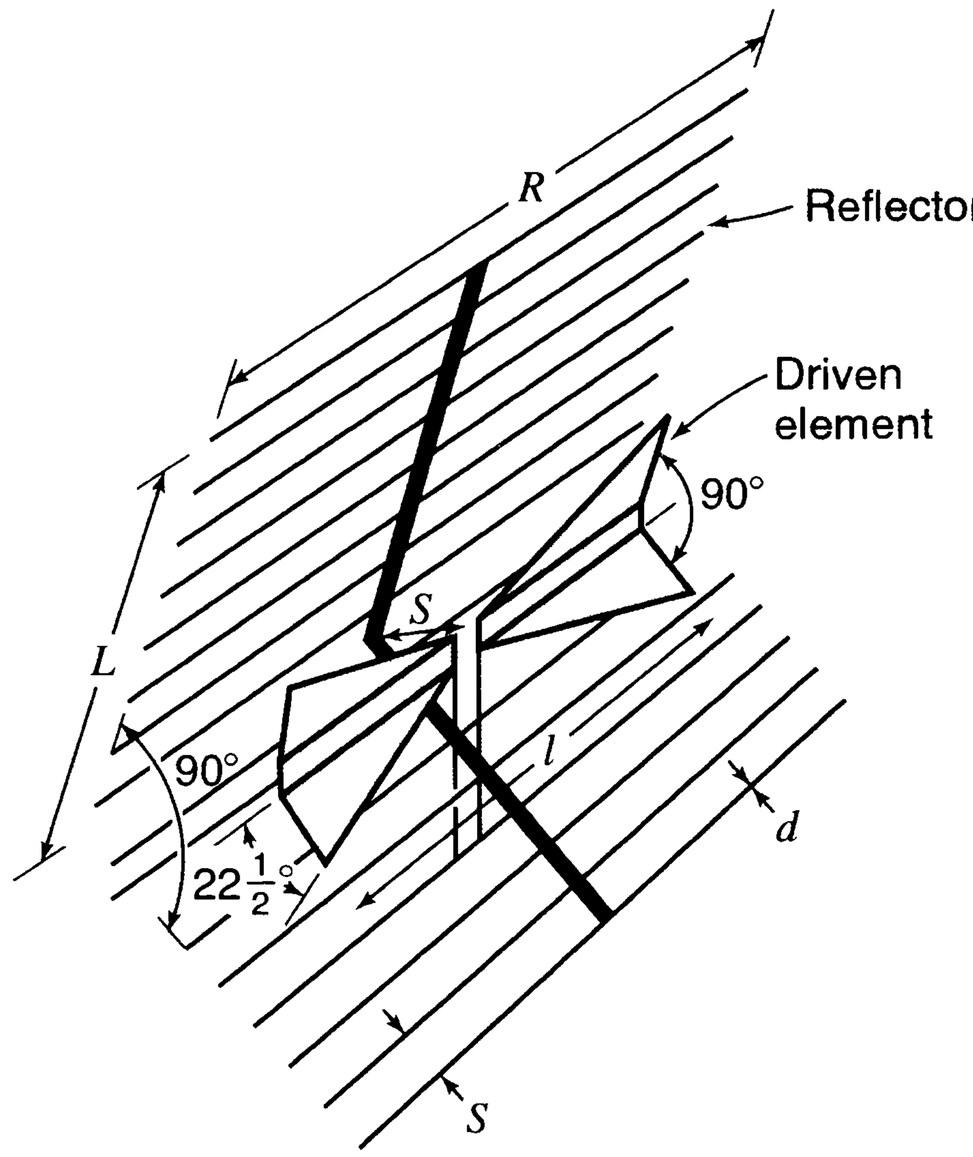


Figure 10–14
Square-corner (grid) reflector with bow-tie dipole for wideband operation (see Table 10–2).

radiated in the sectors η. Furthermore, these waves are reflected with infinite sides into a direction that is at a considerable angle ϕ with respect to the axis. Hence, the absence of the reflector beyond B should not have a large effect. It should also have relatively little effect on the driving-point impedance. The most noticeable effect with finite sides is that the measured pattern is appreciably broader than that calculated for infinite sides and a null does not occur at $\phi = 45°$ but at a somewhat larger angle. If this is not objectionable, a side length of twice the antenna-to-corner spacing ($L = 2S$) is a practical minimum value for square-corner reflectors.[1]

Although the gain of a corner reflector with infinite sides can be increased by reducing the corner angle, it does not follow that the gain of a corner reflector with finite sides of fixed length will increase as the corner angle is decreased. To maintain a given efficiency with a smaller corner angle requires that S be increased. Also on a 60° reflector, for example, the point at which a wave is reflected parallel to the axis is at a distance of $1.73S$ from the corner as compared to $1.41S$ for the square-corner type. Hence, to realize the increase in gain requires that the length of the reflector sides be much larger than for a square-corner reflector designed for the same frequency. Usually this is a practical disadvantage in view of the relatively small increase to be expected in gain.

To reduce the wind resistance offered by a solid reflector, a grid of parallel wires or conductors can be used as in Fig. 10–14. The supporting member joining the midpoints of the reflector conductors may be either a conductor or an insulator. In general the spacing s between reflector conductors should be equal to or less than $\lambda/8$. With a $\lambda/2$ driven element the length R of the reflector conductors should be equal to or greater than 0.7λ. If the length R is reduced to values of less than 0.6λ, radiation to the sides and rear tends to increase

[1] See App. C for UTD code.

Table 10–2 Design data for wideband 90° corner reflector with bow-tie dipole (see Figs. 10–14 and 10–15)

Dipole-to-corner spacing, $S = \lambda/4$ at lowest frequency
Length of reflector, $L = 3\lambda/4$ at lowest frequency
Reflector rod length, $R = 4\lambda/5$ at lowest frequency
Reflector rod spacing, $s = \lambda/8$ at highest frequency
Reflector rod diameter, $d = \lambda/50$ at highest frequency
Bow-tie dipole length, $l = 4\lambda/5$ at mid frequency

	Lowest frequency f_1	**Mid frequency f_2**	**Highest frequency f_3**	**Units**
Dipole-to-corner spacing, S	0.27	0.40	0.54	λ
Length of reflector, L	0.75	1.13	1.50	λ
Reflector rod length, R	0.81	1.20	1.62	λ
Reflector rod spacing, s	0.061	0.092	0.122	λ
Reflector rod diameter, d	0.01	0.015	0.02	λ
Bow-tie dipole length, l	0.53	0.80	1.06	λ
Gain	11.0	13.0	14.0	dBi

and the gain decreases. When R is decreased to as little as 0.3λ, the strongest radiation is no longer forward and the "reflector" acts as a director.

The square-corner reflector is a simple, practical, inherently wideband antenna producing substantial gains (11 to 14 dBi). Typical design data for a 90° (square) corner reflector with bow-tie dipole for wideband (2 to 1 frequency range) operation are given in Table 10–2.

The driven element is a 45° Brown-Woodward (bow-tie) dipole (see Fig. 11–8–3) bent at 90° as suggested in Fig. 10–14, so that its flat sides are parallel to the reflector. The dipole can be fed by a 300- or 400-Ω twin line with low VSWR over the 2 to 1 frequency range. None of the corner reflector dimensions are critical. A moderate increase or decrease in the reflector dimensions L and R from the values in Table 10–2 results in only small changes in gain. Thus, a 10 percent increase in L and R can increase the gain by a decibel or less while a 10 percent decrease in L and R can decrease the gain by a decibel or more. Also the (center-to-center) spacing s between reflector rods can be increased with only a small gain reduction. However, an increase in the rod diameter d can compensate for the larger spacing, keeping the gain essentially constant (Wong-1).

For operation at a single frequency the dimensions for f_1, f_2 or f_3 may be used depending on the gain desired. However, for f_1 or f_2 values of S, L, R and l, f_3 values may be used for s and d, resulting in fewer reflector elements.

The corner reflector is often used as a gain standard. Corner reflectors for THz (terahertz) frequencies are described in Chap. 22.

The composite or hybrid corner-reflector-Yagi-Uda antenna of Fig. 10–16 is a popular design for the U.S. UHF TV band. The lengths and spacings of the directors are appropriate for the high-frequency end of the band. With fewer reflector elements the corner reflector is still effective at the low and mid-frequencies while with more director elements the Yagi-Uda array is effective at the high frequencies resulting in a gain of 7 to 8 dBi over a bandwidth of nearly 2-to-1. This is adequate for the UHF band. Although less than the gain of a full-element corner reflector antenna, the hybrid has less wind resistance.

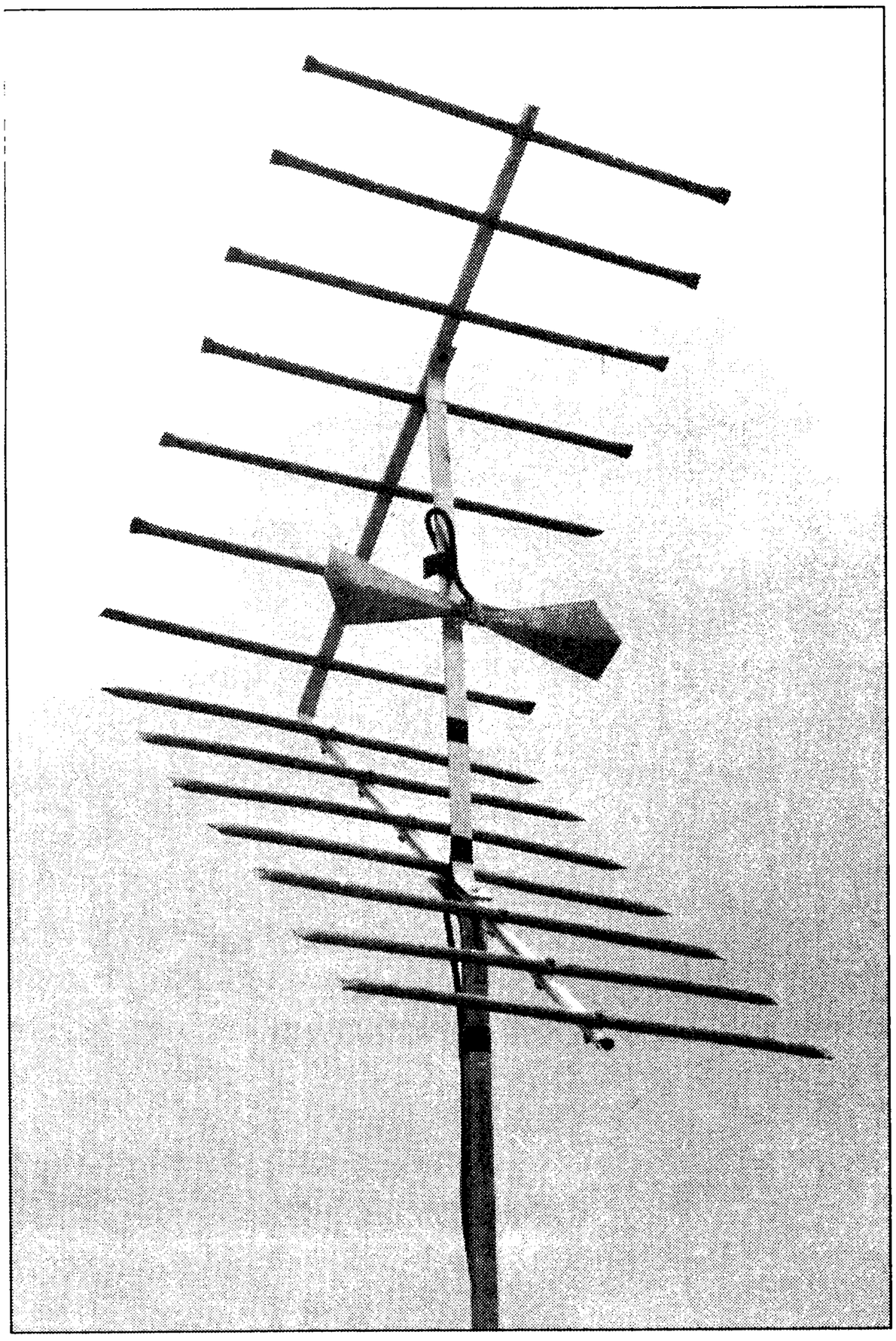


Figure 10–15
Square-corner grid-reflector bow-tie dipole antenna used by the million for UHF TV reception. It has an 11 to 14 dBi gain over a 2-to-1 bandwidth.

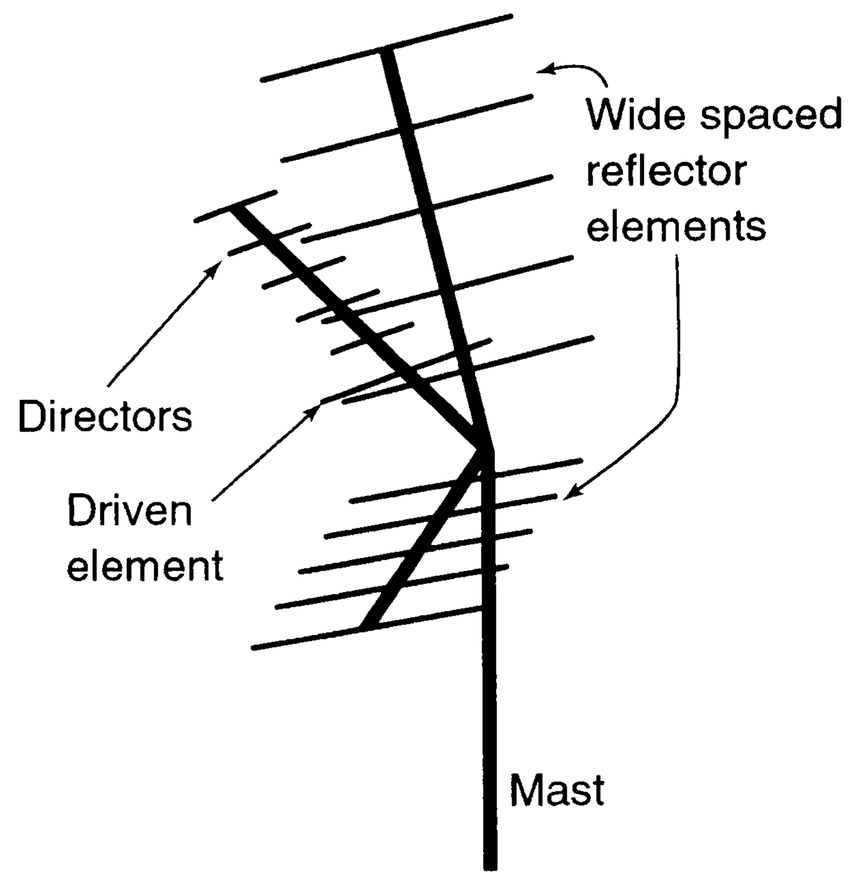


Figure 10–16
Square-corner-Yagi-Uda hybrid antenna. It has an average directivity of about 8 dBi over a 2 to 1 bandwidth.

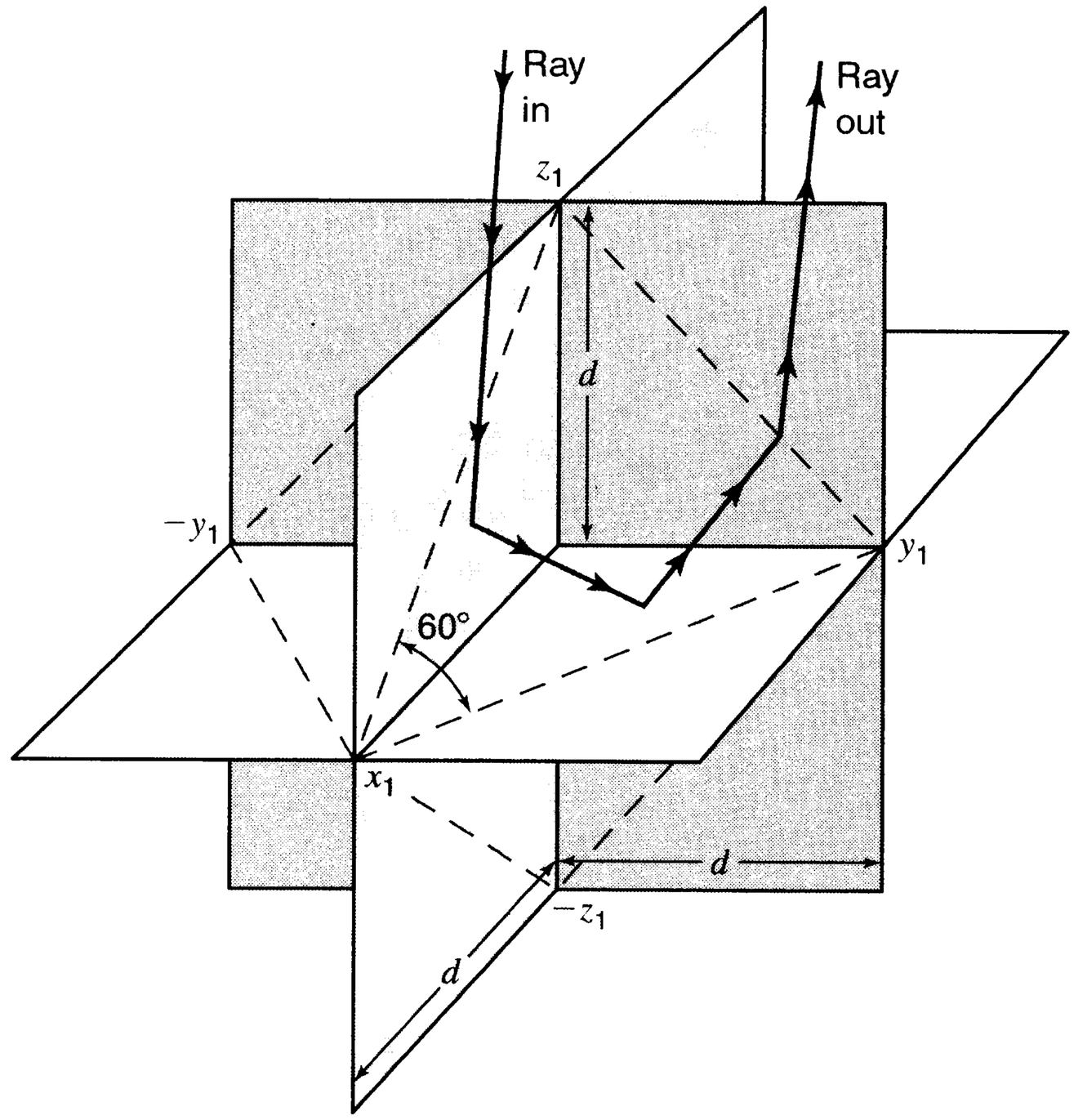


Figure 10–17
Retroreflector of 8 square corners for reflecting back waves from any direction. Path of ray returning via triple bounce is shown.

Compare with the wide-bandwidth corner reflector of Fig. 10–14 and also the Yagi-Uda-corner-log-periodic (YUCOLP) array described in Sec. 11–8.

If a 3-dimensional square corner (see Fig. 10–17) is ***driven*** (受激) by a $\lambda/2$ to $3\lambda/4$ ***monopole*** (单极子) spaced 0.9λ from the corner with $d = 2\lambda$, a beam is obtained in a direction making approximately equal angles with the 3 coordinate axes, and Inagaki (1) reports a gain of 17 dBi with higher gains for smaller corner angles. By placing a cylindrical surface of constant radius from the apex of a corner reflector, Elkamchouchi (1) reports improved performance.

To distinguish the corner reflector with dipole (driven element) discussed in this section from corners without dipoles (or monopoles), the ones with the driven element may be called ***active*** (有源) corners and the others ***passive*** (无源) (retro) corners. While the active corners may have any included angle, with 90° the most common, the passive corners are ***always*** (总是) square (90°). These corners are discussed further in the next section.

10–4 PASSIVE (RETRO) CORNER REFLECTOR [无源(返回式)夹角反射器]

A passive (retro) corner consisting of 2 flat reflecting sheets is shown in Fig. 10–1*e*.

With 3 mutually perpendicular reflecting sheets, as in Fig. 10–17, each sheet extending a distance d ($= \pm x_1 = \pm y_1 = \pm z_1$) from the origin, we have a cluster of eight ***3-dimensional square-corner retroreflectors.*** (三维夹直角的返回式反射器) Each square corner occupies one octant ($4\pi/8$ sr $= \pi/2$ sr $= 5157$ square deg). Any ray or wave incident within this solid angle is reflected back in the same direction, as suggested in the figure. Together the

cluster of 8 square corners form a retroreflector for waves from *any direction* within a full sphere solid angle ($=4\pi$ sr $= 41.253$ square deg) with the equivalent (normal incidence) flat-sheet reflecting area being a function of the angle of incidence. Its maximum value is $\sqrt{3}d^2$ except that in the 6 directions, exactly on the 3 axes (x, y and z), the area is $4d^2$. Just off these directions the area approaches zero.

The enhanced reflection of radar signals from such passive corner clusters makes them useful for many applications. For example, small water-craft commonly carry one (usually with reflecting surfaces of wire mesh) on a tall mast to make the craft's presence more visible on radar screens. To be most effective, the reflector dimensions should be many λ and the periphery of the mesh hole $<\lambda/2$. The surface should also be flat to better than $\lambda/16$ and, to increase the probability that the radar echo will be noticed, the reflector can be rotated to avoid a persistent null condition.

Truncating the sides of the reflecting sheets along the diagonal lines (dashed lines in Fig. 10–17 results in 8 truncated corners, each bounded by an equilateral triangle (included angle 60°) with an aperture area of $\frac{3}{4}d^2$. If the surface of a sphere is divided into triangles (in the manner of a Buckminster Fuller geodesic dome) and a truncated corner inserted in each triangular area, it is possible to array dozens of corner reflectors over the sphere and obtain a more uniform echo area as a function of angle of incidence over 4π sr, but at the expense of a smaller maximum value.

Retroreflectors can also be constructed in other ways. Thus, a Luneburg lens (Sec. 17–10) with a reflecting cap over, say π sr, acts as a retroreflector over this angle. The Van Atta array (Sec. 16–12) is another example.

10–5 THE PARABOLA. GENERAL PROPERTIES（抛物线的一般性质）

Suppose that we have a point source and that we wish to produce a plane-wave front over a large aperture by means of a sheet reflector. Referring to Fig. 10–18*a*, it is then required that the distance from the source to the plane-wave front via path 1 and 2 be equal or[1]

$$2L = R(1 + \cos\theta) \tag{1}$$

and

$$R = \frac{2L}{1 + \cos\theta} \tag{2}$$

This is the equation for the required surface contour. It is the equation of a parabola with the focus at F.

Referring to Fig. 10–18*b*, the parabolic curve may be defined as follows. The distance from any point P on a parabolic curve to a fixed point F, called the *focus,* is equal to the perpendicular distance to a fixed line called the *directrix*. Thus, in Fig. 10–18*b*, $PF = PQ$. Referring now to Fig. 10–18*c*, let AA' be a line normal to the axis at an arbitrary distance QS from the directrix. Since $PS = QS - PQ$ and $PF = PQ$, it follows that the distance

[1]This is an application of the *principle of equality of path length (Fermat's principle)* to the special case where all paths are in the same medium. For the more general situation involving more than one medium see Chap. 17.

from the focus to S is

$$PF + PS = PF + QS - PQ = QS \tag{3}$$

Thus, a property of a parabolic reflector is that all waves from an isotropic source at the focus that are reflected from the parabola arrive at a line AA' with equal phase. The "image" of the focus is the directrix, and the reflected field along the line AA' appears as though it originated at the directrix as a plane wave. The plane BB' (Fig. 10–18*c*) at which a reflector is cut off is called the *aperture plane*.

A cylindrical parabola converts a cylindrical wave radiated by an in-phase line source at the focus, as in Fig. 10–19*a*, into a plane wave at the aperture, or a paraboloid-of-revolution converts a spherical wave from an isotropic source at the focus, as in Fig. 10–18*b*, into a uniform plane wave at the aperture. Confining our attention to a single ray or wave path, the paraboloid has the property of directing or collimating radiation from the focus into a beam parallel to the axis (see Fig. 10–19).

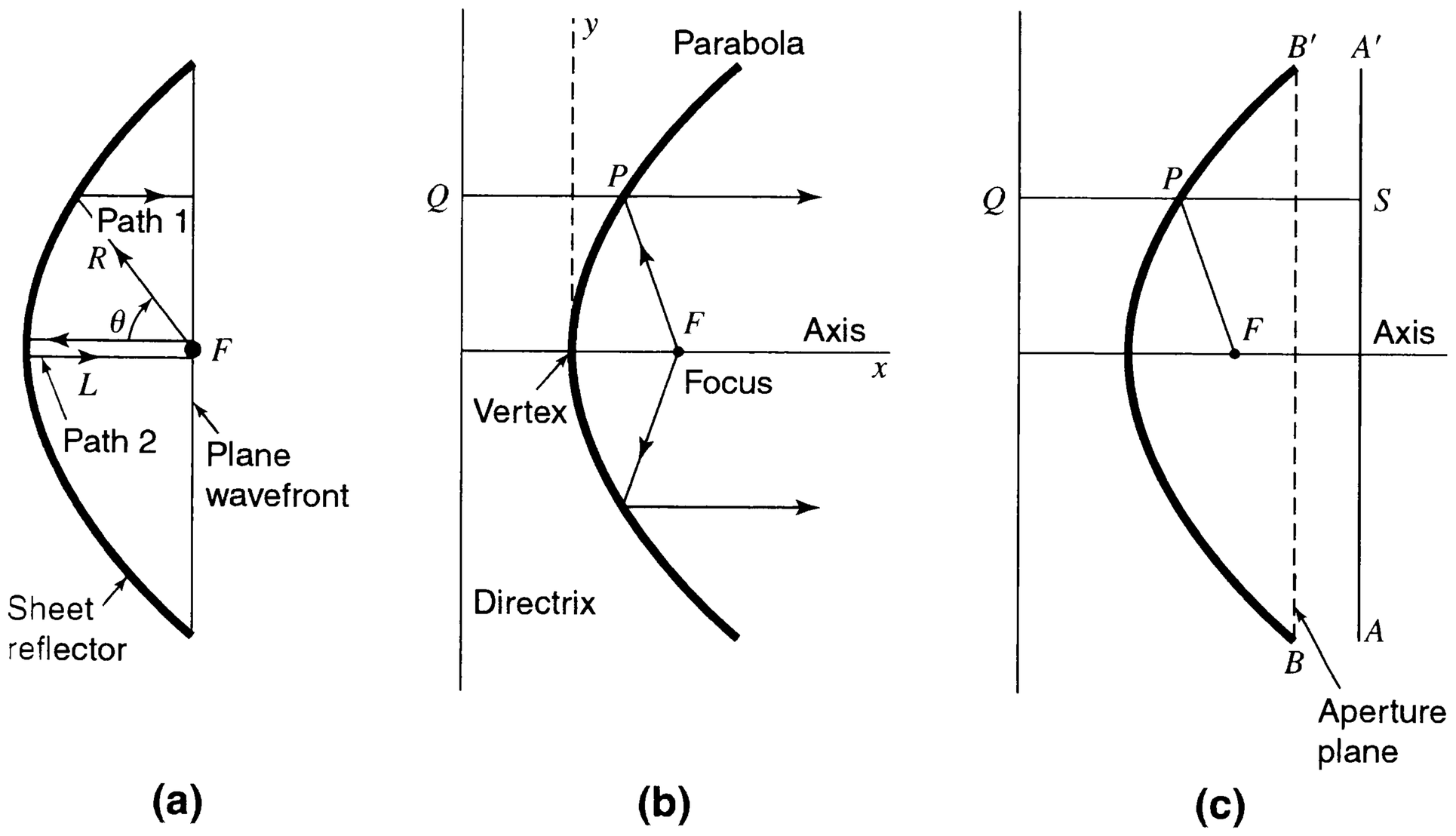


Figure 10–18
Parabolic reflectors.

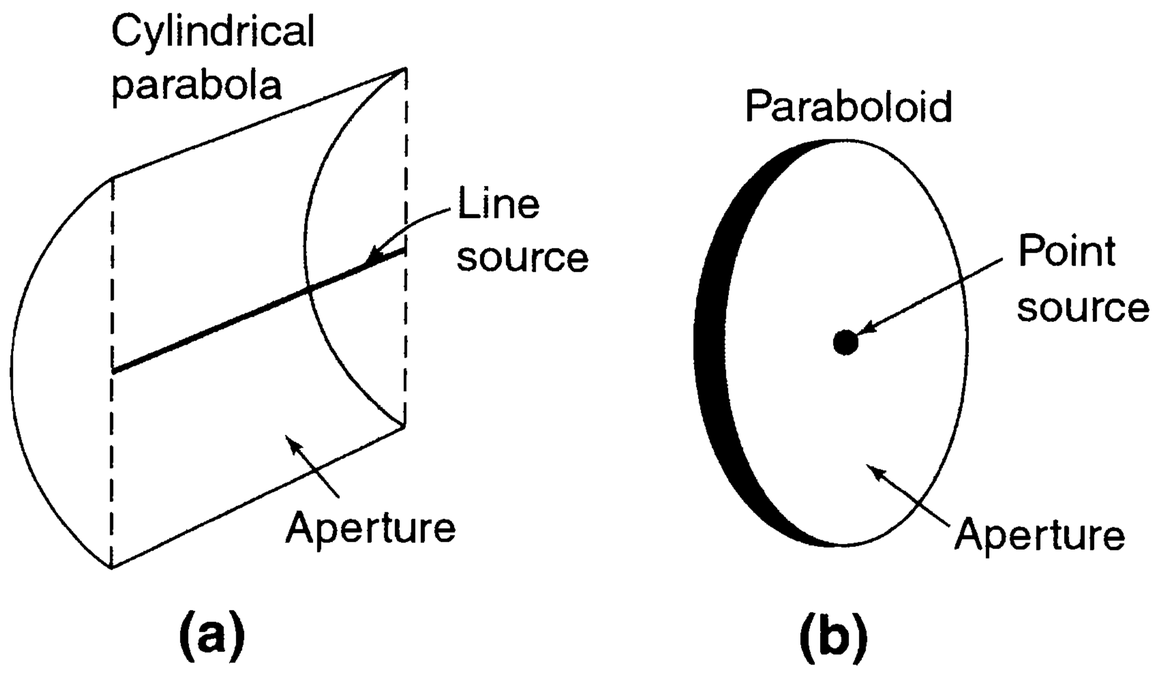


Figure 10–19
Line source and cylindrical parabolic reflector (*a*) and point source and paraboloidal reflector (*b*).

10–6 A COMPARISON BETWEEN PARABOLIC AND CORNER REFLECTORS
（抛物面反射镜与夹角反射器的比较）

Referring to Fig. 10–20 any radiation from the primary source or feed antenna at the focus of the parabola which is not directed into the parabola is not collimated but is radiated by direct paths over a large solid angle. This is not only inefficient but the distributed radiation can degrade the pattern of the radiation from the parabola. Thus, ***it is essential that a parabolic reflector have a directional feed which radiates all or most of the energy into the parabola. A corner reflector, on the other hand, does not require a directional feed since the direct and reflected waves are properly combined (image theory). Furthermore, a corner reflector has no specific focal point***. [抛物面反射镜必须具有定向的馈源，以将全部或大部分能量射向抛物面。另一方面，夹角反射器却并不要求有定向的馈源，因为其直射与反射波总是合成相加的（镜像原理）。进一步说，夹角反射器不存在特定的焦点。]

Practical aperture dimensions for a square-corner reflector are 1 to 2λ. For larger apertures parabolic reflectors should be used, and for a large parabola of many λ aperture, a practical choice for the feed can be a corner reflector with a corner angle of 90 to 180° depending on the F/D ratio of the parabola (see Sec. 10–7). Corner angles of about 120° have the advantage that the beamwidths are approximately equal in both principal planes.

Although the corner reflector differs in principle from the parabolic reflector, there are situations in which the two may be nearly equivalent. This may be illustrated with the aid of Fig. 10–20. Let a linear antenna be located at the focus F of a cylindrical parabolic reflector, and let this arrangement be compared with a square-corner reflector of the same aperture and with an antenna-to-corner spacing AF. The parabolic and corner reflectors are superimposed for comparison in Fig. 10–20. A wave radiated in the positive y direction from F is reflected at 0 by the corner reflector and at 0'' by the cylindrical parabolic reflector. Hence, this wave travels a shorter distance in the corner reflector by an amount 00′. If $AF = 2\lambda$, the electrical length of 00′, is about 180° so that a marked difference would be expected in the field patterns of the two reflectors. However, if $AF = 0.35\lambda$ the electrical length of 00′ is only about 30°, and this will cause only a slight difference in the field patterns. It follows that if AF is small in terms of the wavelength the exact shape of the reflector is not of great importance. The practical advantage of the corner reflector is the simplicity and ease of construction of the flat sides.

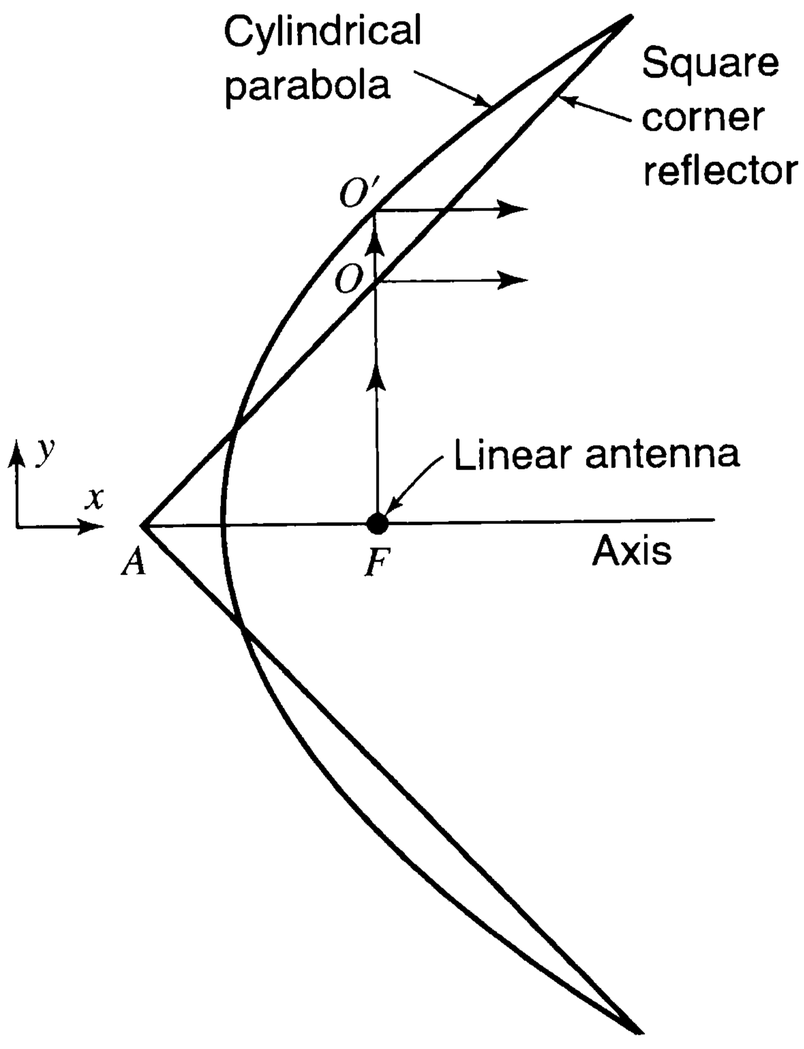


Figure 10–20
Cylindrical parabolic reflector compared with square-corner reflector.

10–7 THE PARABOLOIDAL REFLECTOR（旋转抛物面反射镜）

The surface generated by the revolution of a parabola around its axis is called a *paraboloid* or a *parabola of revolution* (Friis-1, Silver-1, Cutler-1, Slater-1). If an isotropic source is placed at the focus of a paraboloidal reflector as in Fig. 10–21*a*, the portion *A* of the source radiation that is intercepted by the paraboloid is reflected as a plane wave of circular cross section provided that the reflector surface deviates from a true parabolic surface by no more than a small fraction of a wavelength.

If the distance L between the focus and vertex of the paraboloid is an even number of $\lambda/4$, the direct radiation in the axial direction from the source will be in opposite phase and will tend to cancel the central region of the reflected wave. However, if

$$L = \frac{n\lambda}{4} \tag{1}$$

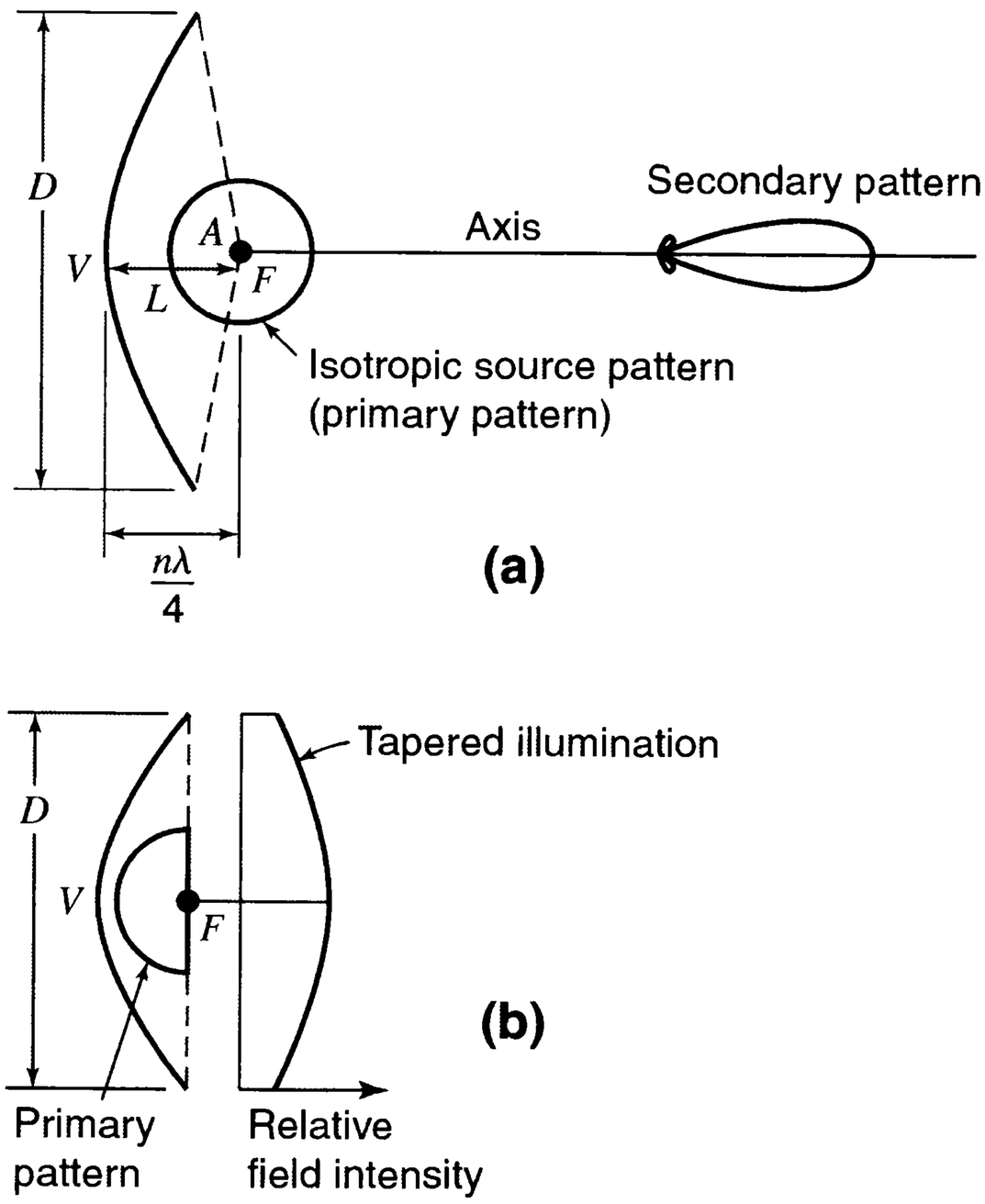


Figure 10–21
Parabolic reflectors of different focal lengths (L) and with sources of different patterns.

where $n = 1, 3, 5, \ldots$, the direct radiation in the axial direction from the source will be in the same phase and will tend to reinforce the central region of the reflected wave.

Direct radiation from the source can be eliminated by means of a directional source or primary antenna[1] as in Fig. 10–21*b* and *c*. A primary antenna with the idealized hemispherical pattern shown in Fig. 10–21*b* (solid curve) results in a wave of uniform phase over the reflector aperture. However, the amplitude is tapered as indicated. To obtain a more uniform aperture field distribution or illumination. it is necessary to make θ_1 small, as suggested in Fig. 10–21*c* by increasing the focal length L while keeping the reflector diameter D constant.[2] If the source pattern is uniform between the angles $\pm\theta_1$ (solid curve), the aperture illumination is more nearly uniform (solid curve) but not entirely so. The path length from F to the edges (at P_1 and P_2) is greater than from F to V (at the vertex). Although ray paths via the edges (at P_1 and P_2) and via the vertex V are of equal total length to the plane wave front (see Fig. 10–18*a*), giving phase equality, there is more path length to the edges in a spherical $(1/r)$ attenuating wave. Thus, the field at the edges is weaker, as will be calculated. To make the field completely uniform across the aperture would require a feed pattern with *inverse* taper.

A typical pattern for a directional source as indicated by the dashed curve in Fig. 10–21*c* (left) gives a more tapered aperture distribution as shown by the dashed curve in Fig. 10–21*c* (right). The greater amount of taper with resultant reduction in edge illumination may be desirable in order to reduce the minor-lobe level, this being achieved, however, at some sacrifice in directivity.

The arrangement of Fig. 10–21*b* illustrates the case of a small focal ratio. The arrangement in Fig. 10–21*c* illustrates the case of a larger ratio.

Suitable directional patterns may be obtained with various types of primary antennas. As examples, a $\lambda/2$ antenna with a small ground plane is shown in Fig. 10–22*a* and a small horn antenna in Fig. 10–22*b*.

The presence of the primary antenna in the path of the reflected wave, as in the above examples, has 2 principal disadvantages. These are, first, that waves reflected from the parabola back to the primary antenna produce interaction and mismatching.[3] Second, the primary antenna acts as an obstruction, blocking out the central portion of the aperture and increasing the minor lobes. To avoid both effects, a portion of the paraboloid can be used and the primary antenna displaced as in Fig. 10–22*c*. This is called an *offset feed*.

Let us next develop an expression for the field distribution across the aperture of a parabolic reflector. Since the development is simpler for a cylindrical parabola, this case is treated first, as an introduction to the case for a paraboloid. Consider a *cylindrical parabolic reflector* with line source as in Fig. 10–23*a*. The line source is isotropic in a plane perpendicular to its axis (plane of page). For a unit distance in the z direction (normal to page in

---

[1] It is convenient to refer to the pattern of the source or primary antenna as the *primary pattern* and the pattern of the entire antenna as the *secondary pattern*.

[2] That is, by increasing the ratio of L to D. This ratio is called the *F ratio,* the *F/D ratio* or the *focal ratio* $(= L/D = FV/D)$.

[3] This may be greatly reduced by using a circularly polarized primary antenna, such as an axial-mode helix. If the primary antenna radiation is right-circularly polarized, the wave reflected from the parabola is mostly left-circularly polarized and the primary antenna is insensitive to this polarization.

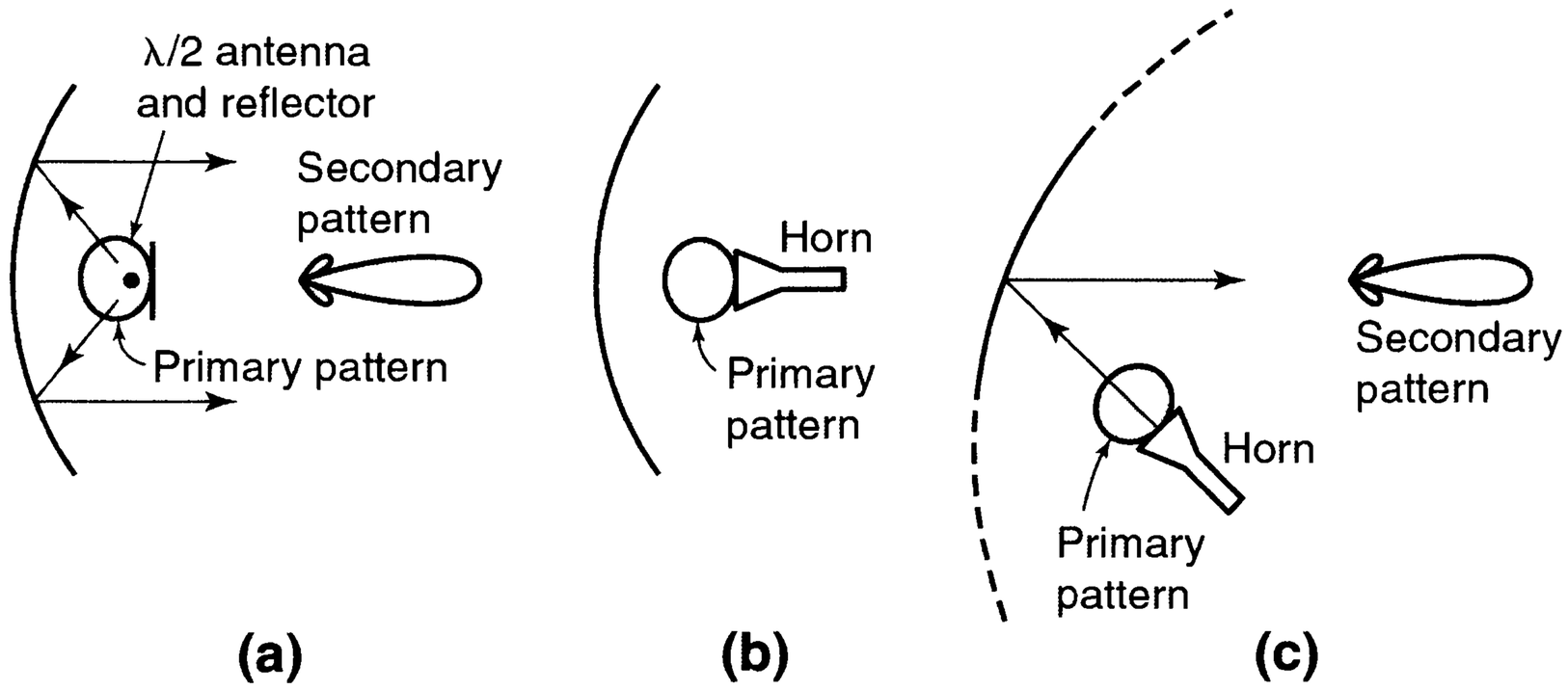


Figure 10–22
Parabolic reflectors (*a* and *b*) and partial reflector with offset feed (*c*).

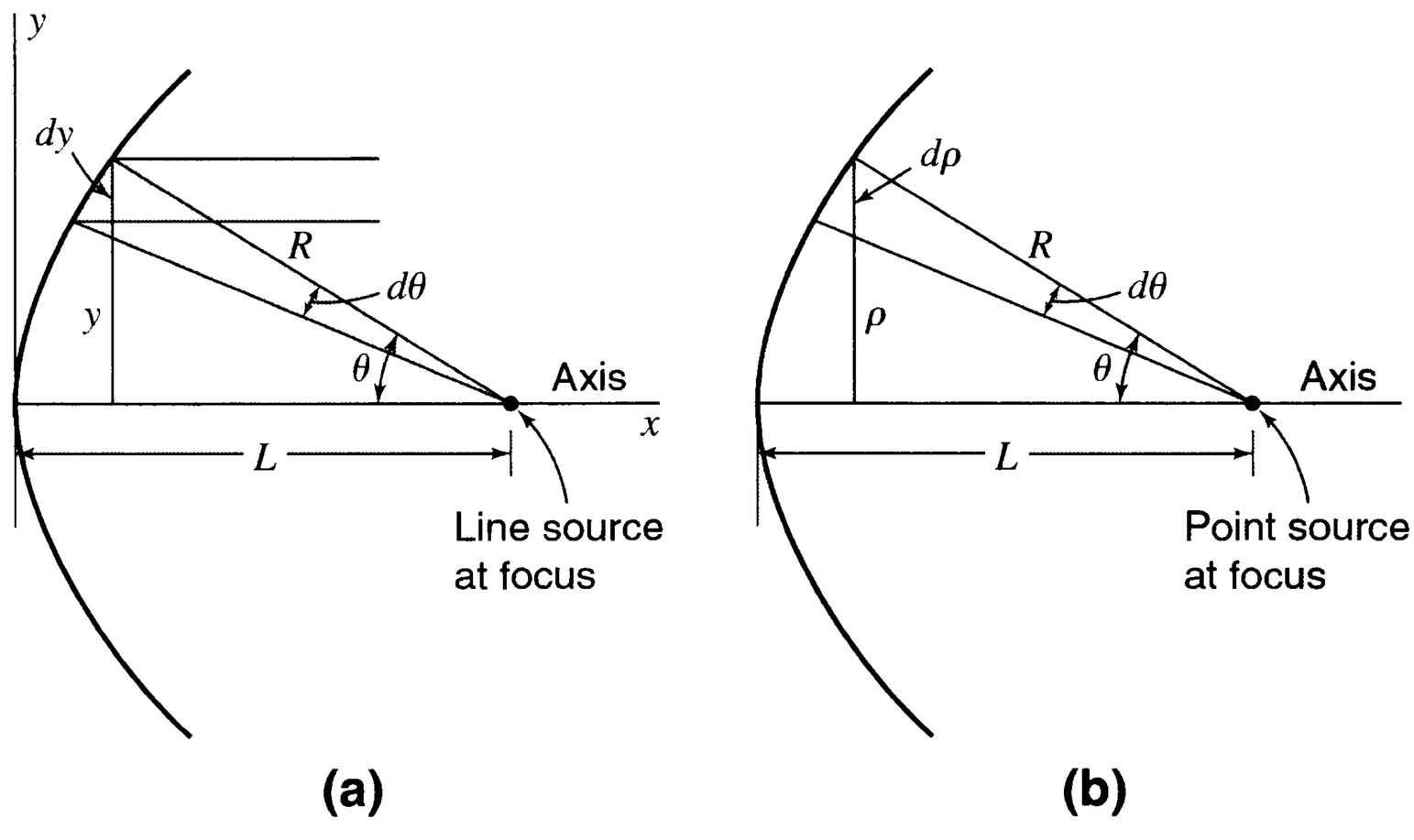


Figure 10–23
Cross sections of cylindrical parabola (*a*) and of paraboloid-of-revolution (*b*).

Fig. 10–23*a*) the power P in a strip of width dy is

$$P = dy\, S_y \tag{2}$$

where S_y = the power density at y, W m^{-2}

However, we also have that

$$P = d\theta\, U' \tag{3}$$

where U' = the power per unit angle per unit length in the z direction

Thus,

$$S_y\, dy = U'\, d\theta \tag{4}$$

and

$$\frac{S_y}{U'} = \frac{1}{(d/d\theta)(R \sin\theta)} \tag{5}$$

where

$$R = \frac{2L}{1 + \cos\theta} \tag{6}$$

This yields

$$S_y = \frac{1 + \cos\theta}{2L} U' \tag{7}$$

The ratio of the power density S_θ at θ to the power density S_0 at $\theta = 0$ is then given by the ratio of (7) when $\theta = \theta$ to (7) when $\theta = 0$, or

$$\frac{S_\theta}{S_0} = \frac{1 + \cos\theta}{2} \tag{8}$$

The field-intensity ratio in the aperture plane is equal to the square root of the power ratio or

$$\frac{E_\theta}{E_0} = \sqrt{\frac{1 + \cos\theta}{2}} \tag{9}$$

where E_θ / E_0 is the relative field intensity at a distance y from the axis as given by $y = R \sin\theta$.

Turning now to the case of a *paraboloid-of-revolution* with an isotropic point source as in Fig. 10–23 the total power P through the annular section of radius ρ and width $d\rho$ is

$$P = 2\pi\rho \, d\rho \, S_\rho \tag{10}$$

where S_ρ = the power density at a distance ρ from the axis, W m^{-2}

This power must be equal to the power radiated by the isotropic source over the solid angle $2\pi \sin\theta \, d\theta$. Thus,

$$P = 2\pi \sin\theta \, d\theta \, U \tag{11}$$

where U = the radiation intensity, W sr^{-1}

Then

$$\rho \, d\rho \, S_\rho = \sin\theta \, d\theta \, U \tag{12}$$

or

$$\frac{S_\rho}{U} = \frac{\sin\theta}{\rho(d\rho/d\theta)} \tag{13}$$

where $\rho = R \sin\theta = \frac{2L \sin\theta}{1 + \cos\theta}$

This yields

$$S_\rho = \frac{(1 + \cos\theta)^2}{4L^2} U \tag{14}$$

The ratio of the power density S_θ at the angle θ to the power density S_0 at $\theta = 0$ is then

$$\frac{S_\theta}{S_0} = \frac{(1 + \cos\theta)^2}{4} \tag{15}$$

The field-intensity ratio in the aperture plane is equal to the square root of the power ratio or

$$\frac{E_\theta}{E_0} = \frac{1 + \cos\theta}{2} \tag{16}$$

where E_θ / E_0 is the relative field intensity at a radius ρ from the axis as given by $\rho = R \sin\theta$.

10–8 PATTERNS OF LARGE CIRCULAR APERTURES WITH UNIFORM ILLUMINATION（均匀照射大型圆口径的波瓣图）

The radiation from a large paraboloid with uniformly illuminated aperture is essentially equivalent to that from a circular aperture of the same diameter D in an infinite metal plate with a uniform plane wave incident on the plate as in Fig. 10–24. The radiation-field pattern for such a uniformly illuminated aperture can be calculated (Slater-2, Silver-1) by applying Huygens' principle in a similar way to that for a rectangular aperture in Chap. 5. The normalized field pattern $E(\phi)$ as a function of ϕ and D is

$$E(\phi) = \frac{2\lambda}{\pi D} \frac{J_1[(\pi D/\lambda)\sin\phi]}{\sin\phi} \tag{1}$$

where

D = diameter of aperture, m
λ = free-space wavelength, m
ϕ = angle with respect to the normal to the aperture (Fig. 10–24)
J_1 = first-order Bessel function

The angles ϕ_0 to the first nulls of the radiation pattern are given by

$$\frac{\pi D}{\lambda}\sin\phi_0 = 3.83 \tag{2}$$

since $J_1(x) = 0$ when $x = 3.83$. Thus,

$$\phi_0 = \arcsin\frac{3.83\lambda}{\pi D} = \arcsin\frac{1.22\lambda}{D} \tag{3}$$

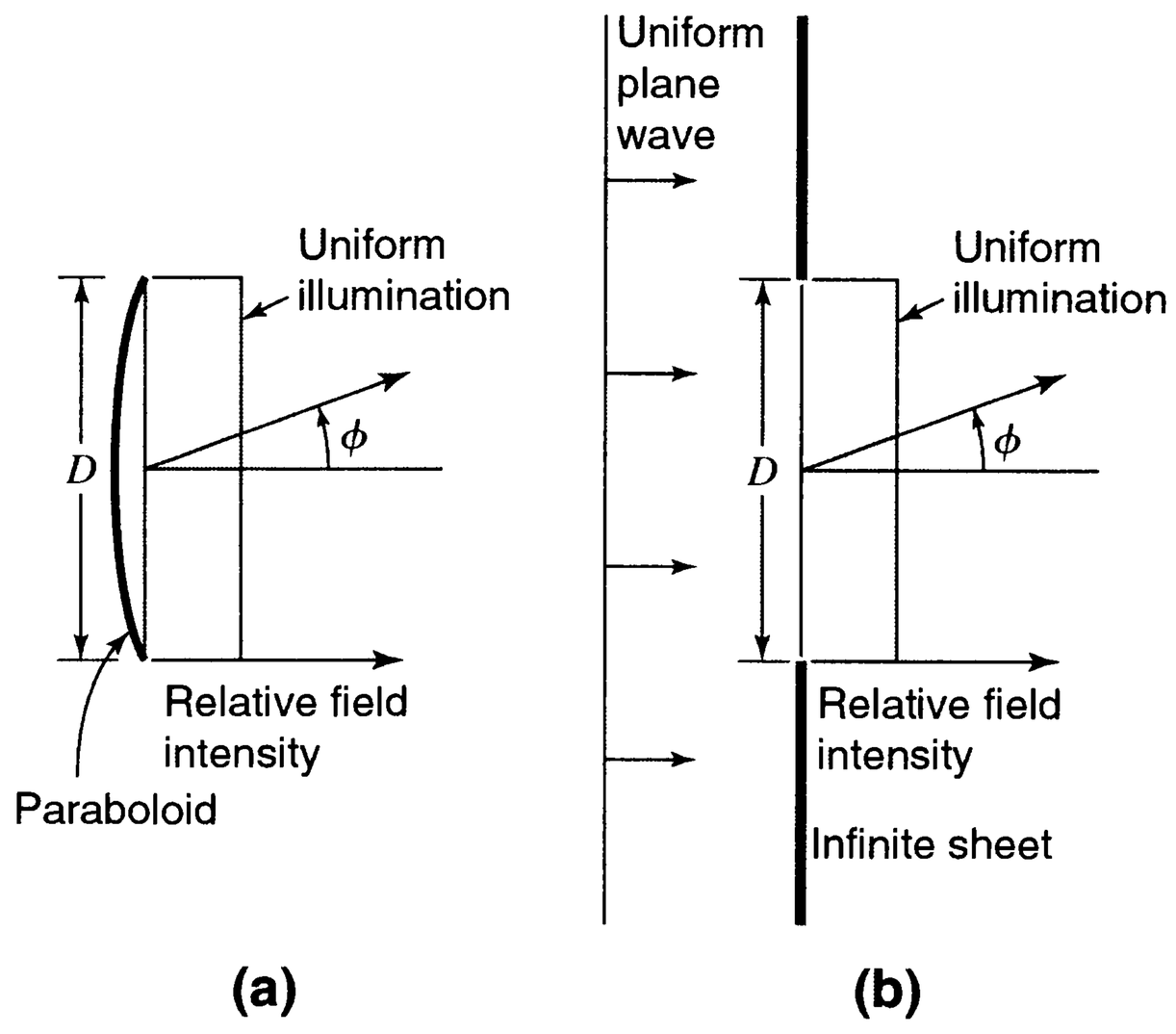


Figure 10–24
Large paraboloid with uniformly illuminated aperture (*a*) and equivalent uniformly illuminated aperture of same diameter D in infinite flat sheet (*b*).

When ϕ_0 is very small (aperture large)

$$\phi_0 \simeq \frac{1.22}{D_\lambda} \text{ (rad)} = \frac{70}{D_\lambda} \text{ (deg)} \tag{4}$$

where $D_\lambda = D/\lambda$ = diameter of aperture, λ

The beamwidth between first nulls is twice this. Hence for large *circular apertures,* the *beamwidth between first nulls*

$$\text{BWFN} = \frac{140}{D_\lambda} \quad \text{(deg)} \tag{5}$$

By way of comparison, the beamwidth between first nulls for a large uniformly illuminated *rectangular aperture* or a long linear array is

$$\text{BWFN} = \frac{115}{L_\lambda} \quad \text{(deg)} \tag{6}$$

where L_λ = length of aperture, λ

The *beamwidth between half-power points* for a large circular aperture is (Silver-1)

$$\text{HPBW} = \frac{58}{D_\lambda} \quad \text{(deg)} \tag{7}$$

The directivity D of a large *uniformly illuminated aperture* is given by

$$D = 4\pi \frac{\text{area}}{\lambda^2} \tag{8}$$

For a *circular aperture*

$$D = 4\pi \frac{\pi D^2}{4\lambda^2} = 9.87 D_\lambda^2 \tag{9}$$

where D_λ = the diameter of the aperture, λ

The power gain G of a circular aperture over a $\lambda/2$ dipole antenna is

$$G = 6 D_\lambda^2 \tag{10}$$

For example, an antenna with a uniformly illuminated circular aperture 10λ in diameter has a gain of 600 or nearly 28 dB with respect to a $\lambda/2$ dipole antenna ($\simeq$ 30 dBi).

For a *square aperture,* the directivity is

$$D = 4\pi \frac{L^2}{\lambda^2} = 12.6 L_\lambda^2 \tag{11}$$

and the power gain over a $\lambda/2$ dipole is

$$G = 7.7 L_\lambda^2 \tag{12}$$

where L_λ = the length of a side, λ

For example, an antenna with a square aperture 10λ on a side has a gain of 770 or nearly 29 dB over a $\lambda/2$ dipole (=31 dBi).

The above directivity and gain relations are for uniformly illuminated apertures at least several wavelengths across. If the illumination is tapered, the directivity and gain are less.

The patterns for a square aperture 10λ on a side and for a circular aperture 10λ in diameter are compared in Fig. 10–25. In both cases the field is assumed to be uniform in

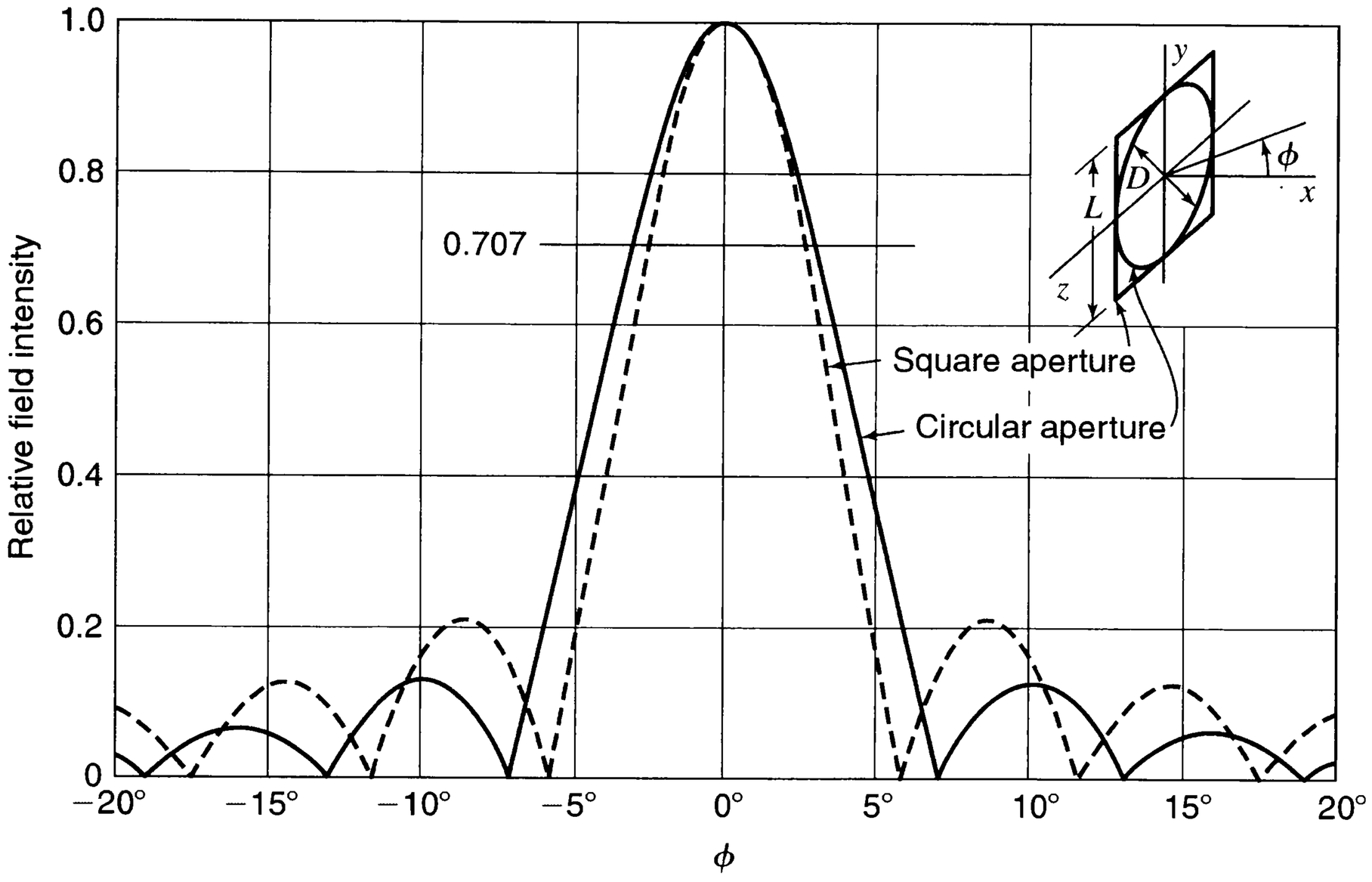


Figure 10–25
Relative radiation patterns of circular aperture of diameter $D = 10\lambda$ and of square aperture of side length $L = 10\lambda$.

both magnitude and phase across the aperture. The patterns are given as a function of ϕ in the xy plane. The patterns in the xz plane are identical to those in the xy plane. Although the beamwidth for the circular aperture is greater than for the square aperture, the side-lobe level for the circular aperture is smaller. A similar effect could be produced with the square aperture by tapering the illumination.

Beamwidths, directivities and gains are summarized in Table 10–3. Beamwidths are compared with horn antennas in Table 9–1.

Table 10–3 Beamwidths, directivities and gains of circular and rectangular apertures with uniform aperture distributions†

	Aperture	
	Circular	**Rectangular**
Half-power beamwidth	$\frac{58^\circ}{D_\lambda}$	$\frac{51^\circ}{L_\lambda}$
Beamwidth between first nulls	$\frac{140^\circ}{D_\lambda}$	$\frac{115^\circ}{L_\lambda}$
Directivity (gain over isotropic source)	$9.9D_\lambda^2$	$12.6L_\lambda L'_\lambda$
Gain over $\lambda/2$ dipole	$6D_\lambda^2$	$7.7L_\lambda L'_\lambda$

where
D_λ = diameter, λ
L_λ = side length, λ
L'_λ = length of other side (if aperture is square, $L'_\lambda = L_\lambda$)

†Apertures are assumed to be large compared to λ. With tapered distributions beamwidths are larger, and directivities, gains and minor lobes are less

REFERENCES（参考文献）

Ashmead J. (1), and A. B. Pippard, "The Use of Spherical Reflectors as Microwave-Scanning Aerials," *JIEE* (Lond.), **93,** pt. IIIA, no. 4, 627–632. 1946

Brown, G. H. (1), "Directional Antennas." *Proc. IRE.* **25,** 122, January 1937.

Cutler, C. C. (1), "Parabolic Antenna Design for Microwaves," *Proc. IRE,* **37,** 1284–1294, November 1947.

Elkamchouchi, H. M. (1), "Cylindrical and Three-Dimensional Corner Reflector Antennas," *IEEE Trans. Ants. Prop.,* **AP-31,** 451–455, May 1983.

Friis, H. T. (1), and W. D. Lewis, "Radar Antennas." *Bell System Tech. J.,* **26,** 219–317, April 1947.

Inagaki, N. (1), "Three Dimensional Corner Reflector Antennas," *IEEE Trans. Ants. Prop.,* **AP-22,** 580, July 1974.

Jeans, Sir James (1), *Mathematical Theory of Electricity and Magnetism,* 5th ed., Cambridge University Press, London, p. 188.

Klopfenstein (1), "Corner Reflector Antennas with Arbitrary Dipole Orientation and Apex Angle," *IRE Trans. Ants. Prop.,* **AP-5,** 297–305, July 1957.

Kraus, J. D. (1), *Radio Astronomy,* 2d ed., Cygnus-Quasar, 1986, pp. 642–646.

Kraus, J. D. (2), "The Corner Reflector Antenna," *Proc. IRE,* **28,** 513–519, November 1940.

Kraus, J. D. (3), "The Square-Corner Reflector," *Radio,* no. 237, 19–24, March 1939.

Kraus, J. D. (4), "Corner Reflector Antenna," U.S. Patent 2,270,314, granted Jan. 20, 1942.

Silver, S. (ed.) (1), *Microwave Antenna Theory and Design,* McGraw-Hill, New York, 1949.

Slater J. C. (1), *Microwave Transmission,* McGraw-Hill, New York, 1942, pp. 272–276.

Slater J. C. (2), and N. H. Frank, *Introduction to Theoretical Physics,* McGraw-Hill, New York, 1933, p. 325.

Stavis, G. (1), and A. Dome, in *Very High Frequency Techniques,* Radio Research Laboratory Staff, McGraw-Hill, New York, 1947, Chap. 6.

Wheeler, H. A. (1), "The Radiation Resistance of an Antenna in an Infinite Array or Waveguide," *Proc. IRE,* **36,** 478–487, April 1948.

Wong, J. L. (1), and H. E. King, "A Wide-Band Corner-Reflector Antenna for 240 to 400 MHz," *IEEE Trans. Ants. Prop.,* **AP-33,** 891–892, August 1985.

PROBLEMS（习题）

10–2–1 平板反射器）

Flat sheet reflector. Calculate and plot the radiation pattern of a $\lambda/2$ dipole antenna spaced 0.15λ from an infinite flat sheet for assumed antenna loss resistance $R_L = 0$ and 5 Ω. Express the patterns in gain over a $\lambda/2$ dipole antenna in free space with the same power input (and zero loss resistance).

10–2–2 接地平面的绕射）

Diffraction by ground plane. Using the UTD program calculate the field pattern for a $\lambda/2$ dipole $\lambda/4$ above a 1λ square flat-sheet reflector. Obtain the pattern in all 3 regions of Fig. 10–6. (*a*) How many dB down is the maximum field in region 2 from the maximum in region 1? (*b*) How many dB down is the maximum in region 3 from the maximum in region 1?

10–2–3 大接地平面的绕射）

Diffraction by a larger ground plane. Repeat Prob. 10–2–2 for a ground plane 2λ square.

10–3–1 夹直角反射器）

Square-corner reflector. A square-corner reflector has a driven $\lambda/2$ dipole antenna space $\lambda/2$ from the corner. Assume perfectly conducting sheet reflectors of infinite extent (ideal reflector). Calculate and plot the radiation pattern in a plane at right angles to the driven element.

10–3–2 夹直角反射器）

Square-corner reflector. (*a*) Show that the relative field pattern in the plane of the driven $\lambda/2$ element of a square-corner reflector is given by

$$E = [1 - \cos(S_r \sin\theta)]\frac{\cos(90^\circ \cos\theta)}{\sin\theta}$$

where θ is the angle with respect to the element axis. Assume that the corner-reflector sheets are perfectly conducting and of infinite extent.

(*b*) Calculate and plot the field pattern in the plane of the driven element for a spacing of $\lambda/2$ to the corner. Compare with the pattern at right angles (Prob. 10–3–1).

10–3–3 (矢直角反射器)

Square-corner reflector. Calculate and plot the pattern of an ideal square-corner reflector with $\lambda/2$ driven antenna spaced $\lambda/2$ from the corner but with the antenna displaced 20° from the bisector of the corner angle. The pattern to be calculated is in a plane perpendicular to the antenna and to the reflecting sides.

10–3–4 (矢直角反射器)

Square-corner reflector. (*a*) Calculate and plot the pattern of a 90° corner reflector with a thin center-fed $\lambda/2$ driven antenna spaced 0.35λ from the corner. Assume that the corner reflector is of infinite extent.

(*b*) Calculate the radiation resistance of the driven antenna.

(*c*) Calculate the gain of the antenna and corner reflector over the antenna alone. Assume that losses are negligible.

10–3–5 (矢直角反射器对应的镜像单元阵)

Square-corner reflector versus array of its image elements. Assume that the corner reflector of Prob. 10–3–4 is removed and that in its place the three images used in the analysis are present physically, resulting in 4-element driven array.

(*a*) Calculate and plot the pattern of this array.

(*b*) Calculate the radiation resistance at the center of one of the antennas.

(*c*) Calculate the gain of the array over one of the antennas alone.

10–3–6 (矢直角反射器阵)

Square-corner reflector array. Four 90° corner-reflector antennas are arranged in line as a broadside array. The corner edges are parallel and side-by-side as in Fig. P10–3–6. The spacing between corners is 1λ. The driven antenna in each corner is a $\lambda/2$ element spaced 0.4λ from the corner. All antennas are energized in phase and have equal current amplitude. Assuming that the properties of each corner are the same as if its sides were of infinite extent, what is (*a*) the gain of the array over a single $\lambda/2$ antenna and (*b*) the half-power beamwidth in the *H* plane?

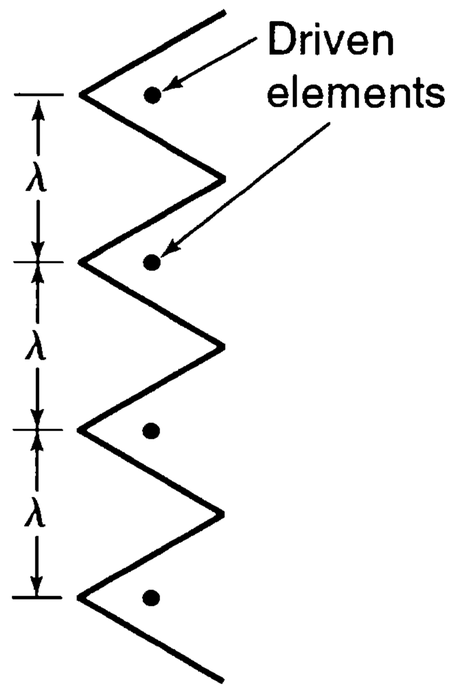


Figure P10–3–6
Square-corner reflector array.

10–3–7 (受激单元相距 $\lambda/4$ 的夹角反射器)

Corner reflector. $\lambda/4$ to the driven element. A square-corner reflector has a spacing of $\lambda/4$ between the driven $\lambda/2$ element and the corner. Show that the directivity $D = 12.8$ dBi.

10–3–8 (受激单元相距 $\lambda/2$ 的夹角反射器)

Corner reflector. $\lambda/2$ to the driven element. A square-corner reflector has a driven $\lambda/2$ element $\lambda/2$ from the corner.

(*a*) Calculate and plot the far-field pattern in both principal planes.

(*b*) What are the HPBWs in the two principal planes?

(*c*) What is the terminal impedance of the driven element?

(*d*) Calculate the directivity in two ways: (1) from impedances of driven and image dipoles and (2) from HPBWs, and compare. Assume perfectly conducting sheet reflectors of infinite extent.

10–3–9 带有领结形偶极子的夹角反射器）

Corner reflector with bow-tie dipole. A 90-degree corner reflector with 0.75λ bow-tie dipole as in Fig. 10–15 has 20 reflector rods 1.2λ long spaced 0.1λ. (*a*) Draw the field pattern and (*b*) calculate the directivity.

10–3–10 夹角反射器的接收功率）

Power received by corner reflector. Find the maximum power delivered to a 50-Ω matched load by a 1 μVm^{-1} field from a U.S. channel 36 (602–608 MHz) station into a 90° corner reflector with 11 dBi gain.

10–3–11 多径条件下天线的最优架高）

Optimum antenna height under multipath conditions. A U.S. channel 44 (650–656 MHz) station transmits a horizontally polarized signal from a 300-m tower. What is the optimum height for a horizontally polarized corner reflector antenna to receive the station at a distance of 10 km? Assume flat, perfectly conducting ground.

10–3–12 三面夹直角反射器天线）

Trihedral square-corner reflector antenna. Three flat mutually perpendicular conducting sheets intersecting at a point make a trihedral square-corner reflector like one of the corners of Fig. 10–17. If a λ/4 stub monopole is place at a distance *S* from the corner find: (*a*) the value of *S* which provides a match to a 50-Ω coaxial feed line and (*b*) the directivity of the antenna. Assume infinite perfectly conducting sheets.

10–3–13 夹直角单脉冲雷达天线）

Square-corner monopulse radar antenna. Design a square-corner reflector antenna with 2 off-axis feeds to operate as a monopulse radar. See footnote preceding Equation (10–3–7).

10–7–1 旋转抛物面反射镜）

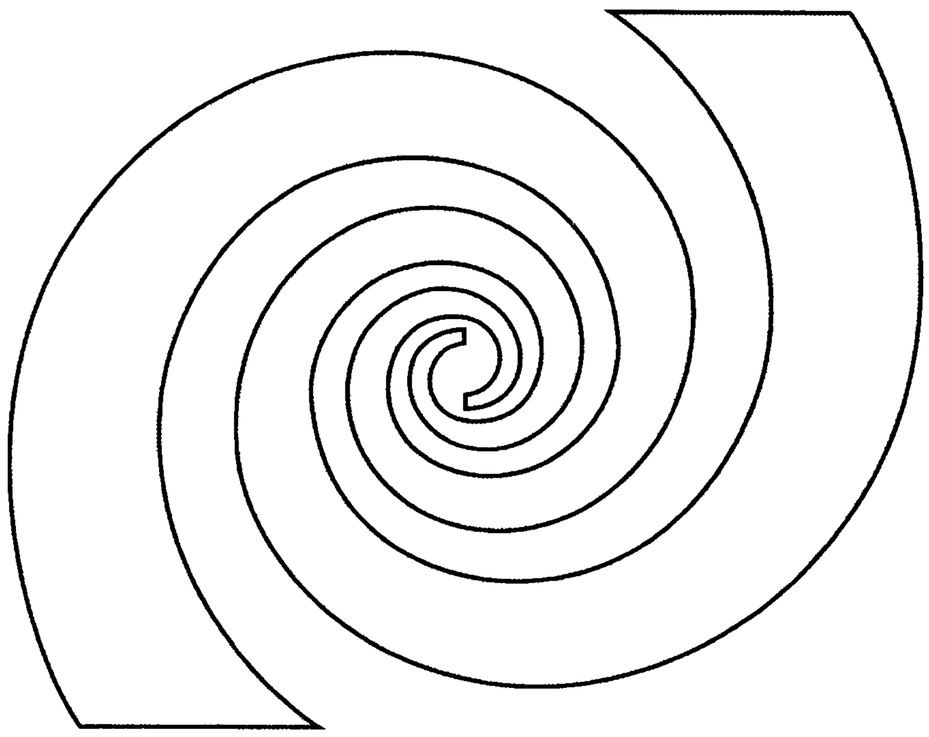
Paraboloidal reflector. Calculate and plot the radiation patterns of a paraboloidal reflector with uniformly illuminated aperture when the diameter is 8λ and when the diameter is 16λ.

***10–7–2** 带有缺损扇形片的抛物面反射镜）

Parabolic reflector with missing sector. A circular parabolic dish antenna has an effective aperture of 100 m^2. If one 30° sector of the parabola is removed, find the new effective aperture. The rest of the antenna, including the feed, is unchanged.

For computer programs, see Appendix C.

CHAPTER 11



Broadband and Frequency-Independent Antennas

宽频带和非频变天线

Topics in this chapter include:

- Broadband basics（宽频带基础）
- Infinite and finite biconical antennas（无限和有限的双锥天线）
- Directional biconicals or vees, conicals, disk cones and bow ties（定向的双锥或V形、圆锥、盘锥和领结天线）
- The frequency-independent concept, Rumsey's principle（非频变概念、拉姆塞(Rumsey)原理）
- The Illinois story（伊利诺伊(Illinois)的故事）
- Planar log spirals（平面对数螺蜷）
- Conical spirals（圆锥螺蜷）
- Log-periodic antennas（对数周期天线）
- YUCOLP composite antenna（YUCOLP复合天线）

11-1 BROADBAND BASICS（宽频带基础）

What is the difference between a broadband and a narrowband antenna? The curved-biconical constant-impedance, twin-line vee antenna of Fig. 11–1*a* is a broadband type. It is basically a transmission line of constant impedance Z_k (ratio of conductor spacing S to conductor radius r constant). If the length L is a wavelength or more, most of the energy of the outgoing wave is radiated with very little energy reflected. The antenna is a nonresonant low-Q radiator with an input impedance that remains essentially constant over a wide frequency range. The antenna is well-matched to space providing a smooth transition from the guided wave on the input transmission line to a free-space wave.

The arrows in Fig. 11–1*a* indicate the direction and magnitude of the energy flow, most being radiated with little reflected back.

By contrast, the short dipole of Fig. 11–1*b* has an abrupt transition from a guided wave on the transmission line to space with a large reflection of energy which oscillates back and forth near the dipole like in a resonator before being radiated. The dipole is a resonant relatively high-Q antenna with an input impedance that changes rapidly with frequency resulting in a narrow bandwidth. The arrows in Fig. 11–1*b* indicate the large energy storage before radiation.

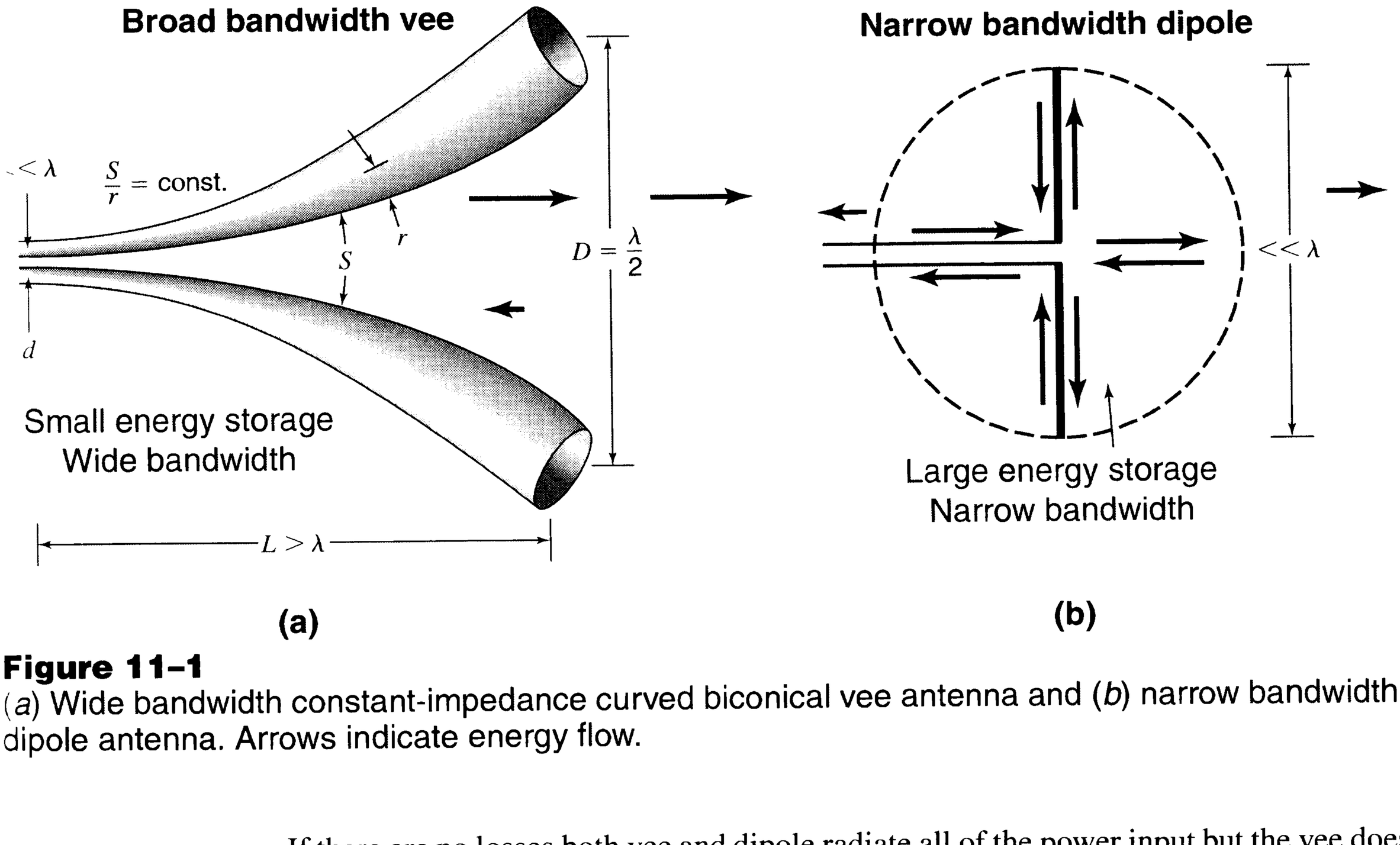


Figure 11–1
(*a*) Wide bandwidth constant-impedance curved biconical vee antenna and (*b*) narrow bandwidth dipole antenna. Arrows indicate energy flow.

If there are no losses both vee and dipole radiate all of the power input but the vee does it with a smooth transition.

EXAMPLE **11–1.1 Bandwidth of Curved Biconical Vee Antenna of Fig. 11–1*a***

Referring to Fig. 11–1*a*, what is the bandwidth if $d = 4$ mm and $D = 100$ mm?

■ **Solution**

Assuming that the spacing $d = \lambda/10$, the shortest $\lambda = 4 \times 10 = 40$ mm. For $D = 100$ mm, the longest $\lambda = 100 \times 2 = 200$ mm.

Thus, the approximate bandwidth $= 200/40 = 5$ to 1 *Ans.*

The VSWR may be less than 2 over this bandwidth. By contrast the VSWR of the dipole may be less than 2 over a bandwidth of only a few percent.

This discussion is oversimplified but illustrates some aspects of bandwidth behavior of *individual* antenna elements. Due to mutual coupling the behavior in an array, however, may be different.

The constant-impedance curved biconical "V" (Fig. 11–1*a*) is a traveling-wave antenna. The axial-mode helical antenna (Fig. 11–2) is also a traveling-wave antenna. Little energy is reflected from the open end so the input impedance remains essentially constant over a wide bandwidth. Typically, the VSWR ≤ 1.5 over a 2 to 1 bandwidth. And this behavior is maintained even in an array of many helices because of the small mutual coupling of helices.

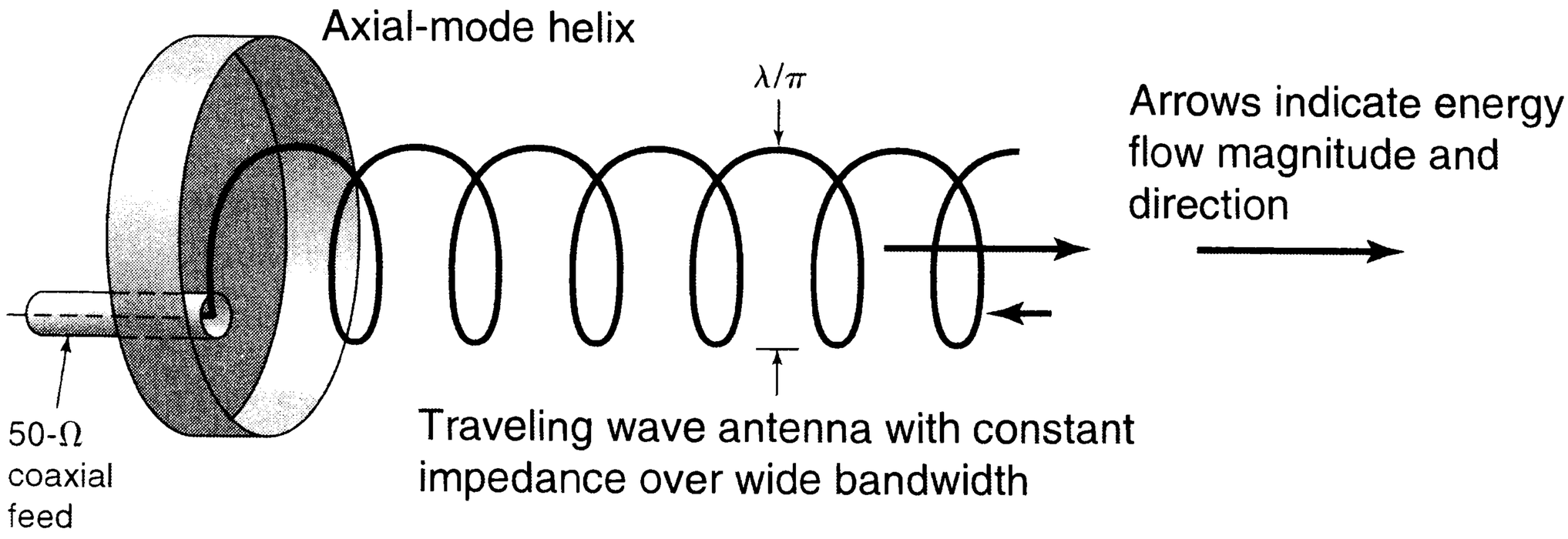


Figure 11–2
Wideband helical-beam antenna.

11-2 INFINITE AND FINITE BICONICAL ANTENNAS（无限长和有限长的双锥天线）

An infinite biconical antenna acts as a guide for a traveling outgoing spherical wave in the same way that a uniform transmission line acts as a guide for a traveling plane wave.

The two situations are compared in Fig. 11–3. They both have a constant characteristic impedance Z_k and since they are infinite the input impedance $Z_i = Z_k$. These values are

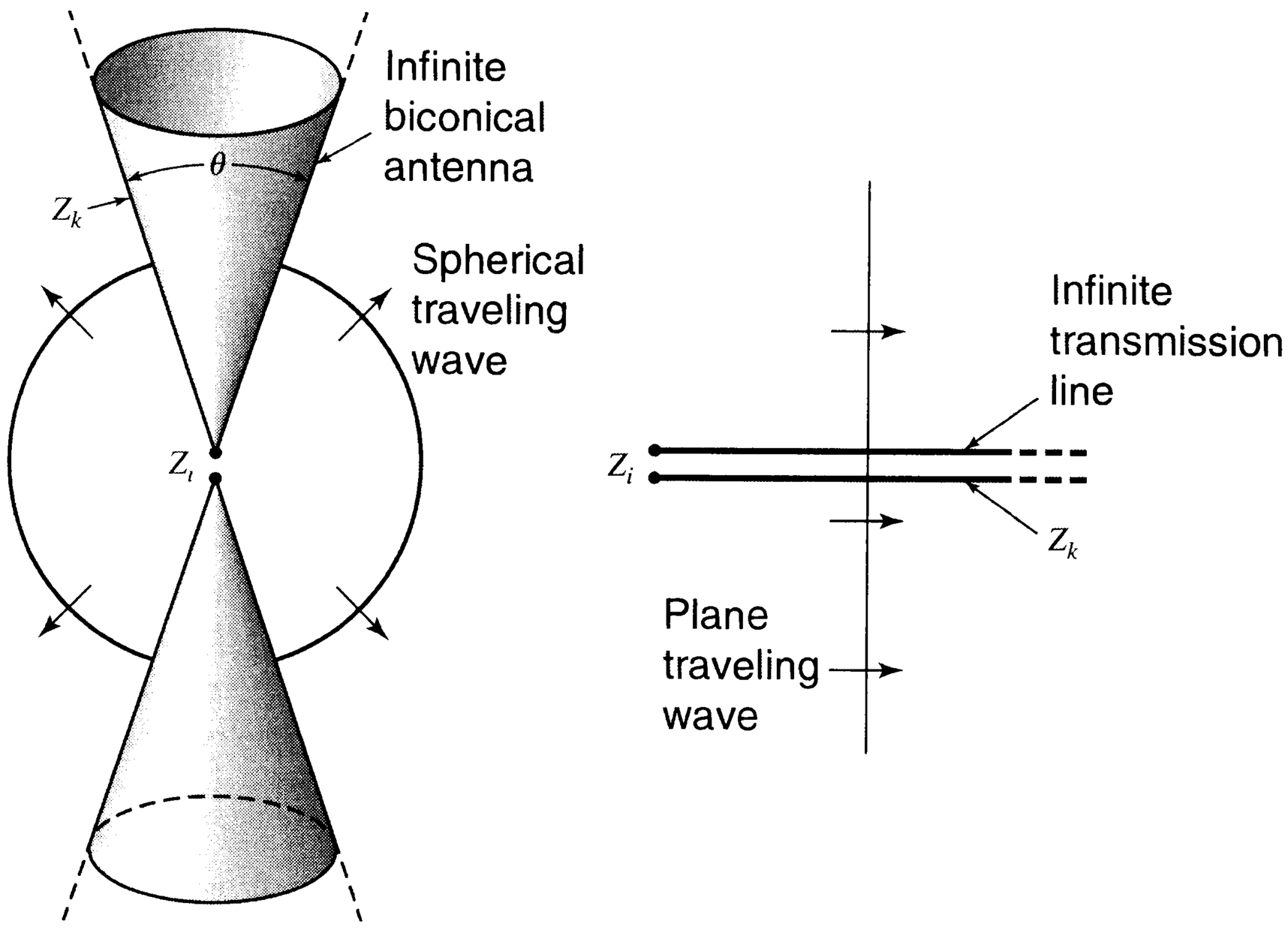


Figure 11–3
An infinite biconical antenna (*a*) is analogous to an infinite uniform transmission line (*b*).

purely resistive so the input resistance

$$R_i = Z_i - Z_k \tag{1}$$

For the infinite biconical antenna

$$R_i = 120 \ln \cot (\theta/4) \tag{2}$$

where θ = cone angle

The solid line of Fig. 11–4 shows the variation of the input resistance R_i as a function of cone angle θ. If the lower cone is replaced by a large ground plane (see insert in Fig. 11–4) the resistance is $\frac{1}{2}$ the value given by (2) as shown by the dashed line in Fig. 11–4. Note that a single cone of 90° angle has an input resistance of about 50 Ω.

With the infinite biconical antenna as an introduction, let us now consider the practical case of a biconical antenna of finite radius r (Fig. 11–5). When the outgoing spherical wave reaches a radius r part of the energy is reflected, resulting in energy storage. The remaining energy is radiated, with more radiated perpendicular to the axis than close to the cones as suggested in Fig. 11–5.

The input impedance Z_i is given by

$$Z_i = Z_k \frac{Z_k + jZ_m \tan \beta r}{Z_m + jZ_k \tan \beta r} \tag{3}$$

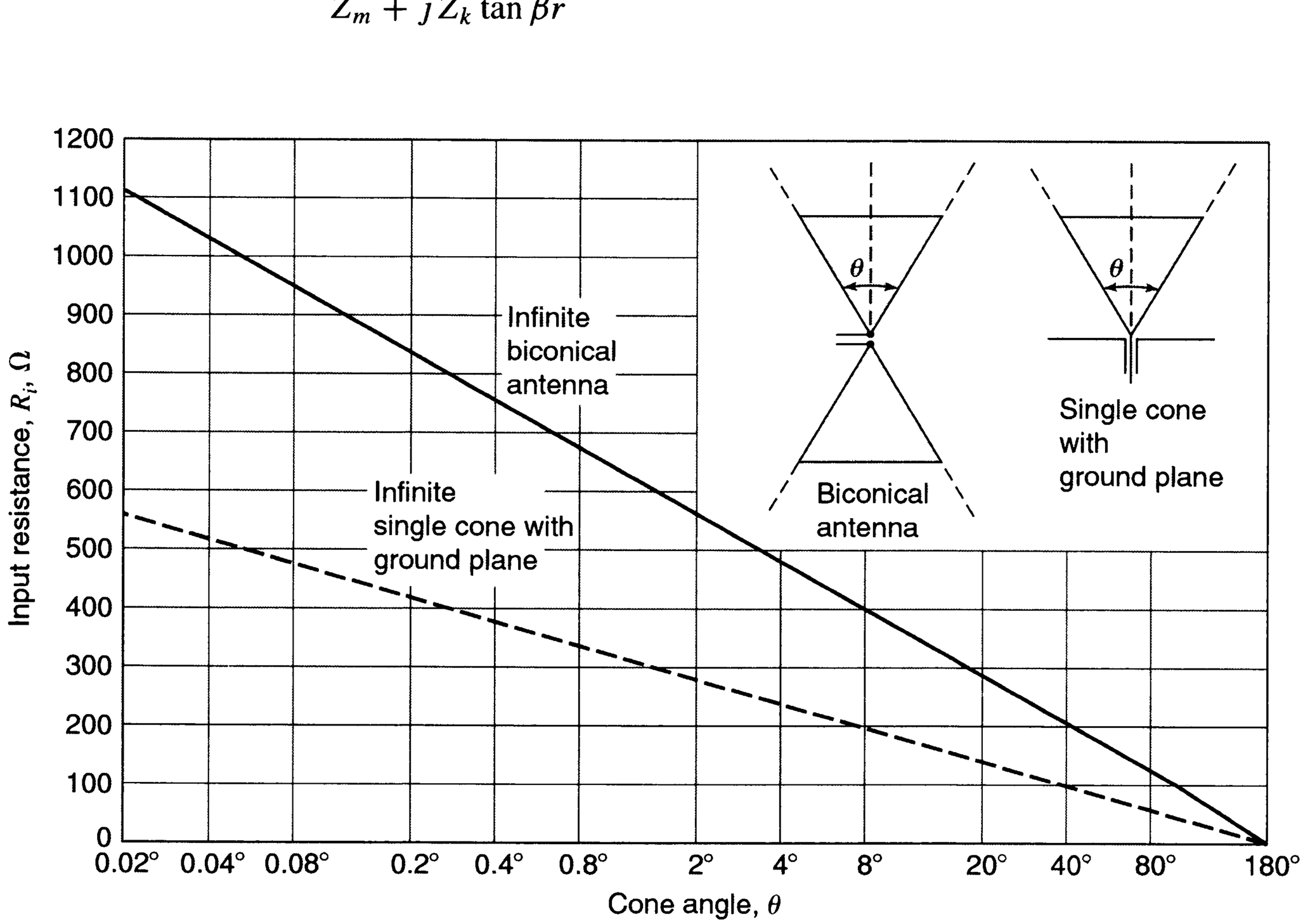


Figure 11–4
Characteristic resistance R_k of infinite biconical and single cone antennas. Since the antenna is infinitely long, the input resistance $R_i = R_k$.

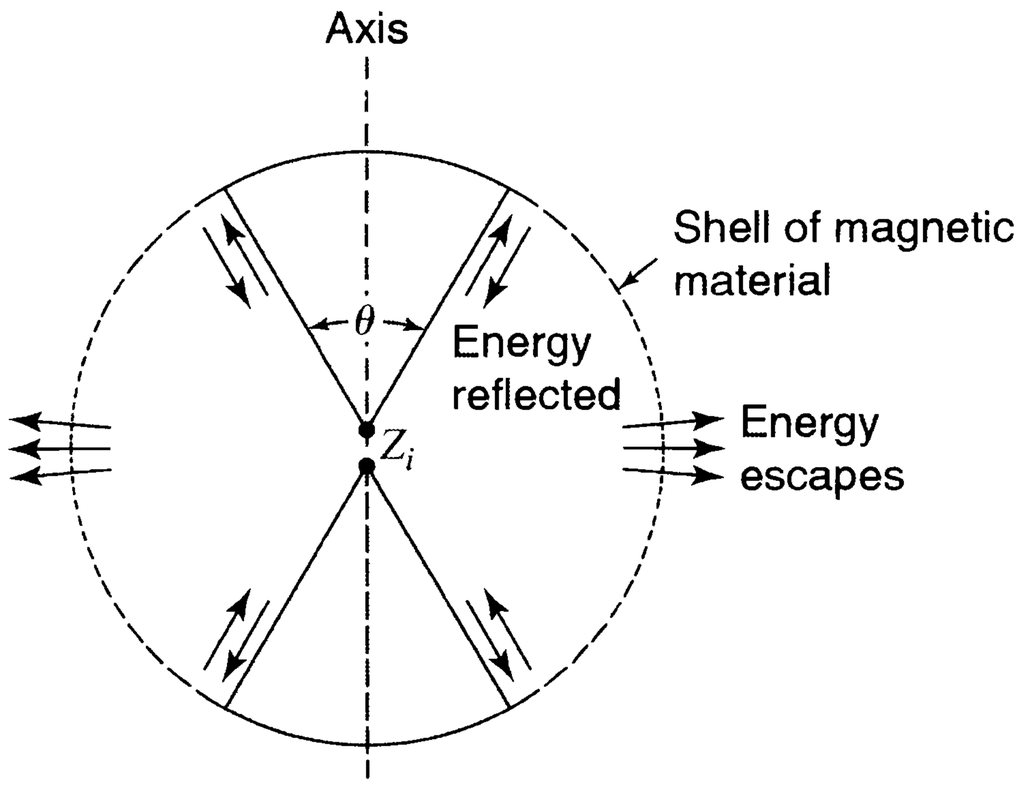


Figure 11–5
Finite biconical antenna enclosed in hypothetical sphere where energy flowing near the cone is reflected but with energy escaping perpendicular to the axis in the equatorial region.

where

r = cone length, m
$\beta = 2\pi/\lambda$
$Z_k = 120 \ln \cot(\theta/4)$
$Z_m = R_m + jX_m$

The R_m and X_m values are given by Schellkunoff (1) for thin cones ($\theta < 5°$) by

$$R_m = 60\,\text{Cin}\,2\beta l + 30(0.577 + \ln \beta l - 2\,\text{Ci}\,2\beta l + \text{Ci}\,4\beta l)\cos 2\beta l + 30(\text{Si}\,4\beta l - 2\,\text{Si}\,2\beta l)\sin 2\beta l \quad (\Omega) \tag{4}$$

$$X_m = 60\,\text{Si}\,2\beta l + 30(\text{Ci}\,4\beta l - \ln \beta l - 0.577)\sin 2\beta l - 30(\text{Si}\,4\beta l)\cos 2\beta l \quad (\Omega) \tag{5}$$

Measured values by RRL (1) of the VSWR for larger cone angles over a 2 to 1 bandwidth are:

Cone angle	VSWR
20°	<5
40°	<3
60°	<2

Thus, the lowest VSWR over a given bandwidth is obtained with the largest cone angle.

11–3 DIRECTIONAL BICONICALS, CONICALS, DISK CONES AND BOW TIES
（定向的双锥、圆锥、盘锥和领结天线）

Figure 11–6 shows a progression of V-type antennas from the simple V at (a) to the biconical V at (b) to the curved biconical V at (c). The characteristic impedance Z_k of the simple V at (a) is not constant and the V has a narrow bandwidth. Its pattern is bidirectional.

The biconical V at (b) has a constant characteristic impedance Z_k and the antenna has a wider bandwidth. It also tends to be unidirectional.

The curved biconical antenna (c) also has a constant characteristic impedance Z_k, is more unidirectional and has a still wider bandwidth.

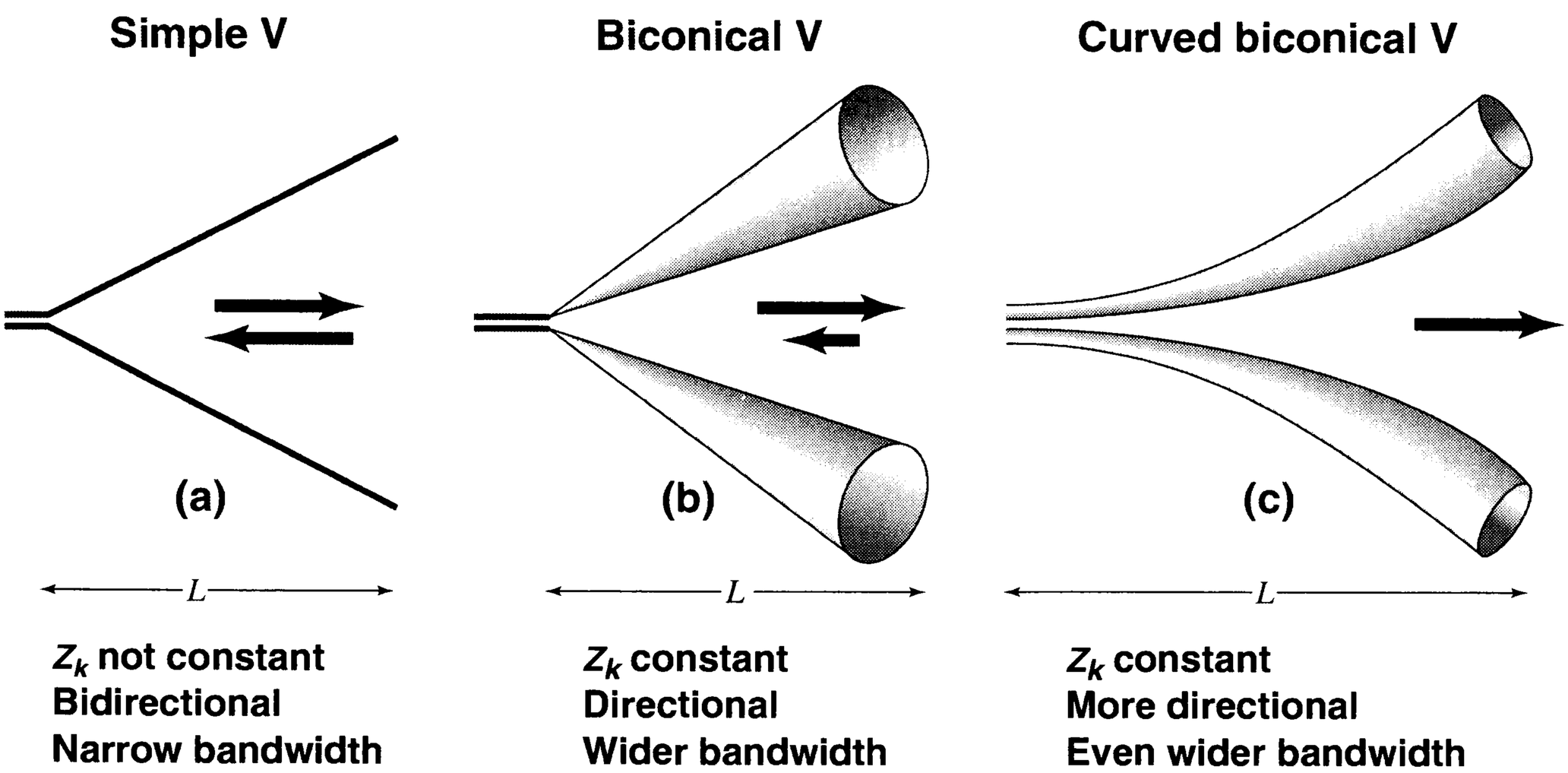


Figure 11–6
(*a*) Simple V, (*b*) biconical V and (*c*) curved biconical V antennas having increasing bandwidth. The arrows indicate that the simple V is bidirectional with the others more nearly unidirectional to the right. The length *L* is a wavelength or more.

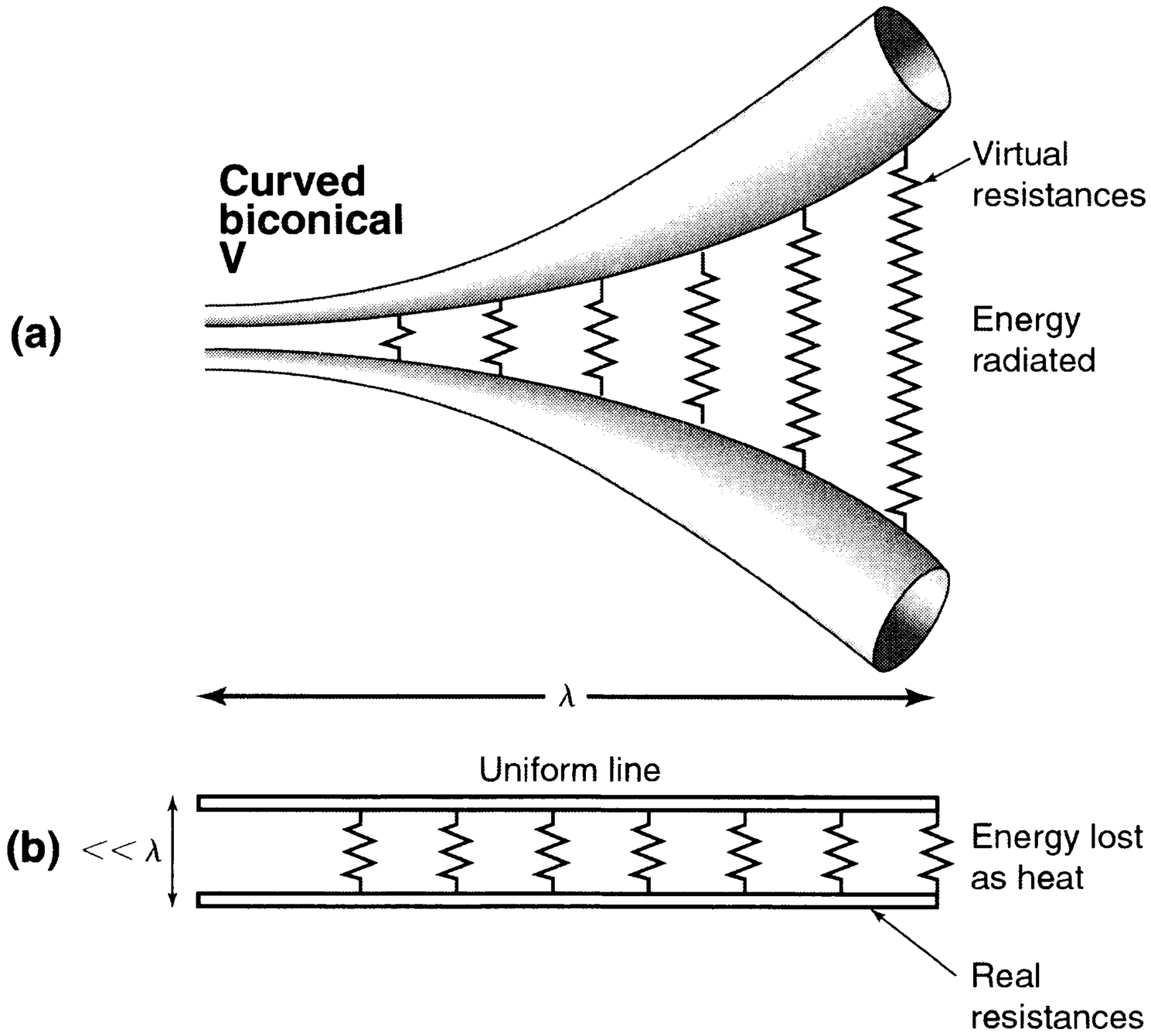


Figure 11–6–1
Comparison of curved biconical V antenna with a resistance-loaded uniform transmission line. Both have similar current distributions, decreasing from left to right. The virtual resistances of the biconical antenna radiate energy whereas the real resistances of the uniform line absorb energy which is lost as heat.

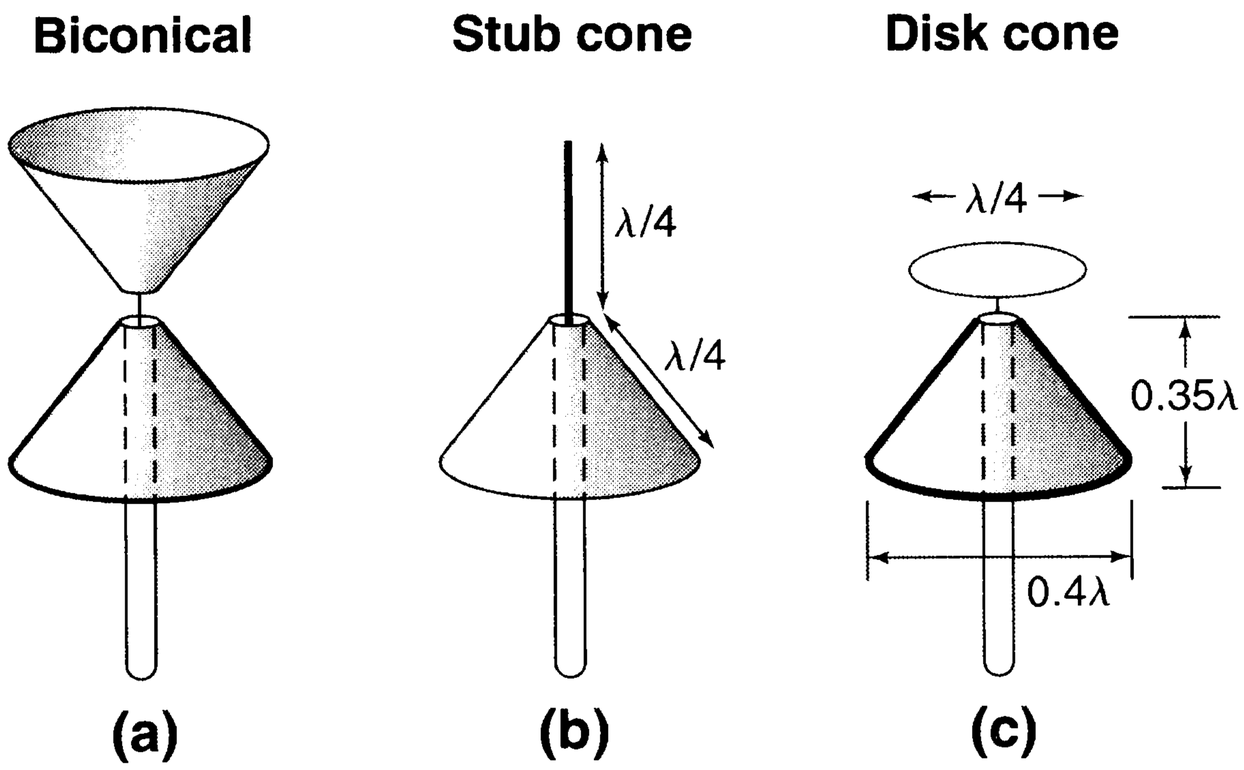


Figure 11–7
Coaxial fed conical antennas for mounting on masts with omnidirectional radiation in the horizontal plane.

Figure 11–7 shows a progression of conical antennas with coax feed. The one at (*a*) is a full biconical. The one at (*b*) has the upper cone replaced by a thin stub, while at (*c*) the cone is replaced by a disk. The coax feed makes these antennas convenient for mounting on masts. Radiation is a maximum in the horizontal plane.

A bow-tie antenna (Brown (1) and Woodward) is a flat-plane version of the biconical antenna. The 60° bow tie of Fig. 11–8*a* provides a VSWR < 2 over a 2 to 1 bandwidth for $L = 0.8\lambda$ at the center frequency. An open-wire version of the biconical antenna is shown in Fig. 11–8*b* with properties nearly the same as with solid surface cones.

Figure 11–8–1 shows Brown and Woodward's results for conical and triangular antennas operating against a ground plane as a function of the length l_λ (or height) for flare angles θ of 30, 60 and 90°. Although the conical measurements were made with open-ended cones, Brown and Woodward found no significant difference in impedance values for a 60° cone with and without end caps. The gain of conical dipoles of length $2l_\lambda$ with respect to a $\lambda/2$ dipole is shown in Fig. 11–8–2. The gains are calculated from measured patterns.

Although the conical antennas have a smaller resistance fluctuation with frequency than the triangular antennas, the flat geometry of the triangles is attractive. The measured performance of a Brown-Woodward (bow-tie) antenna 34 cm long connected to a 300-Ω twin line for frequencies between 480 and 900 MHz (UHF TV channels 15 to 83) is presented in Fig. 11–8–3.

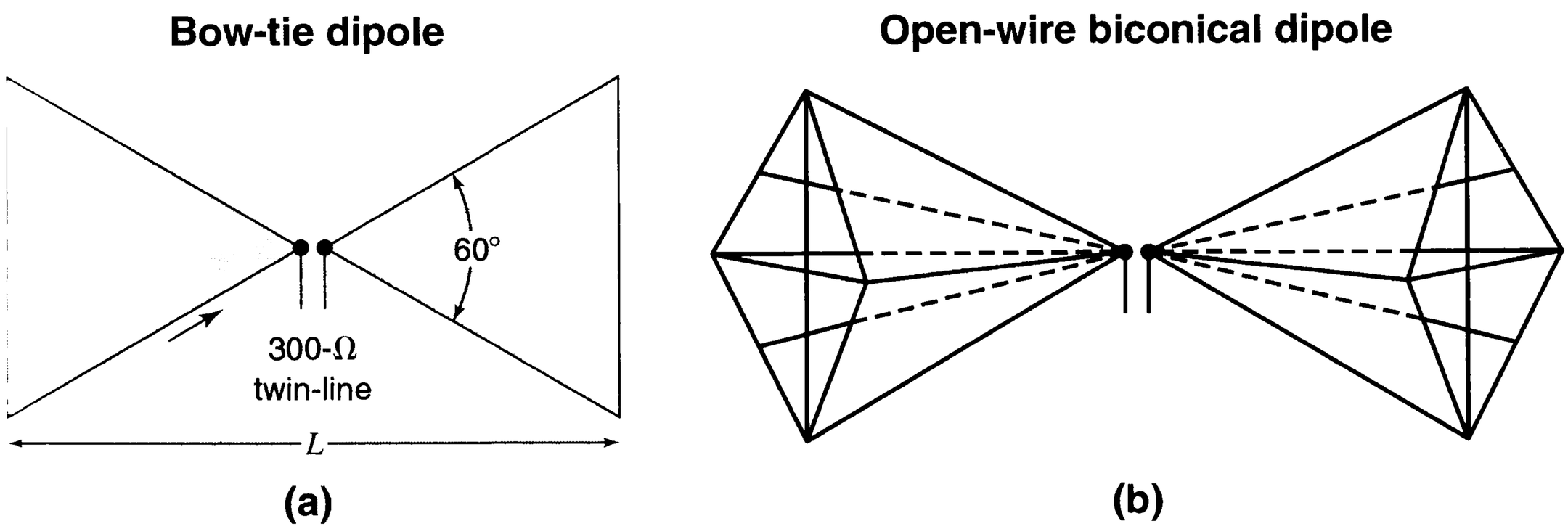


Figure 11–8
(*a*) Flat-plane bow-tie antenna and (*b*) open-wire biconical dipole antenna.

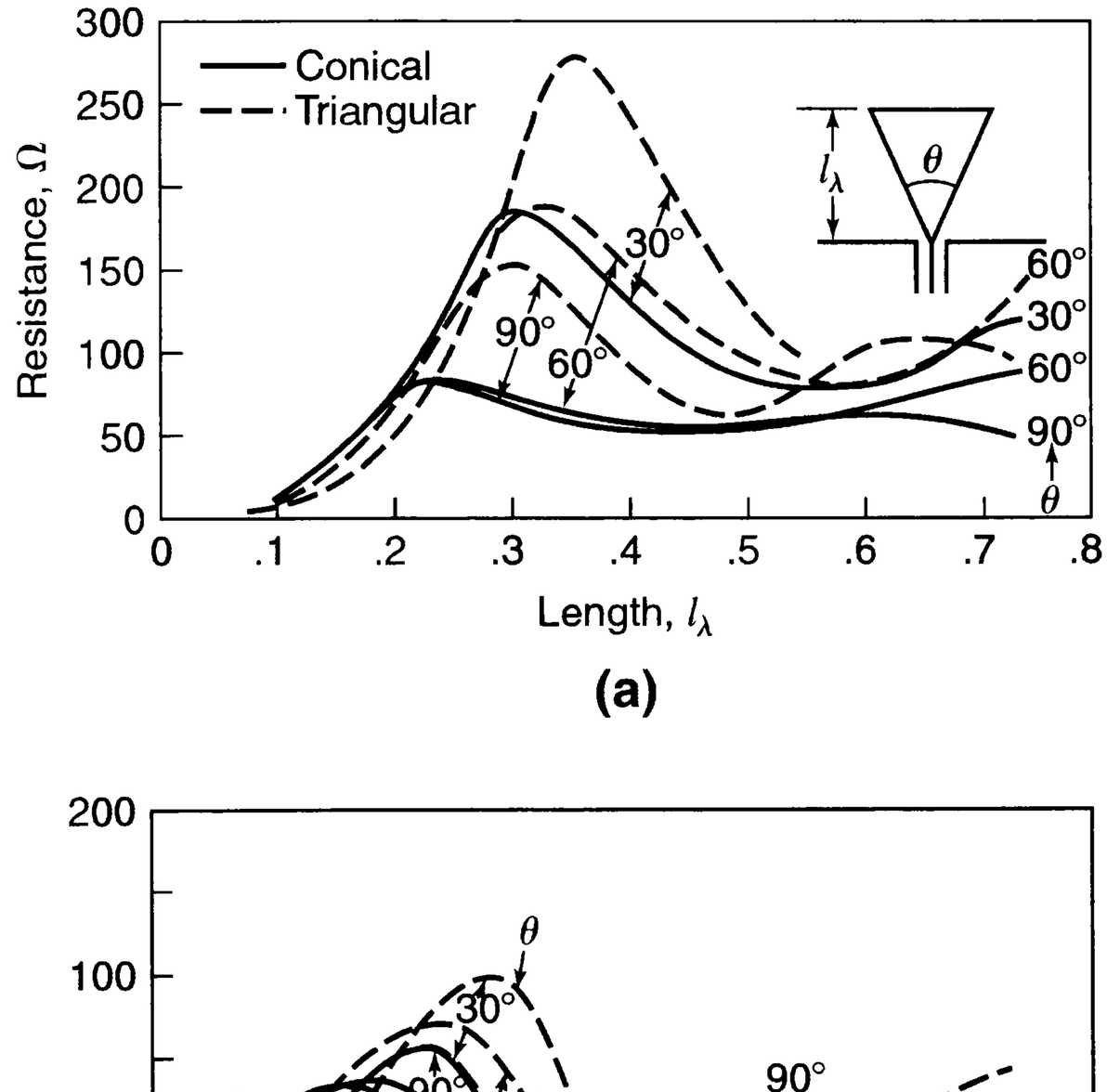


Figure 11–8–1
Measured resistance (*a*) and reactance (*b*) values for monoconical and monotriangular (flat sheet) antennas as a function of length l_λ for flare angles θ of 30, 60 and 90°. (*Brown (1) and Woodward.*)

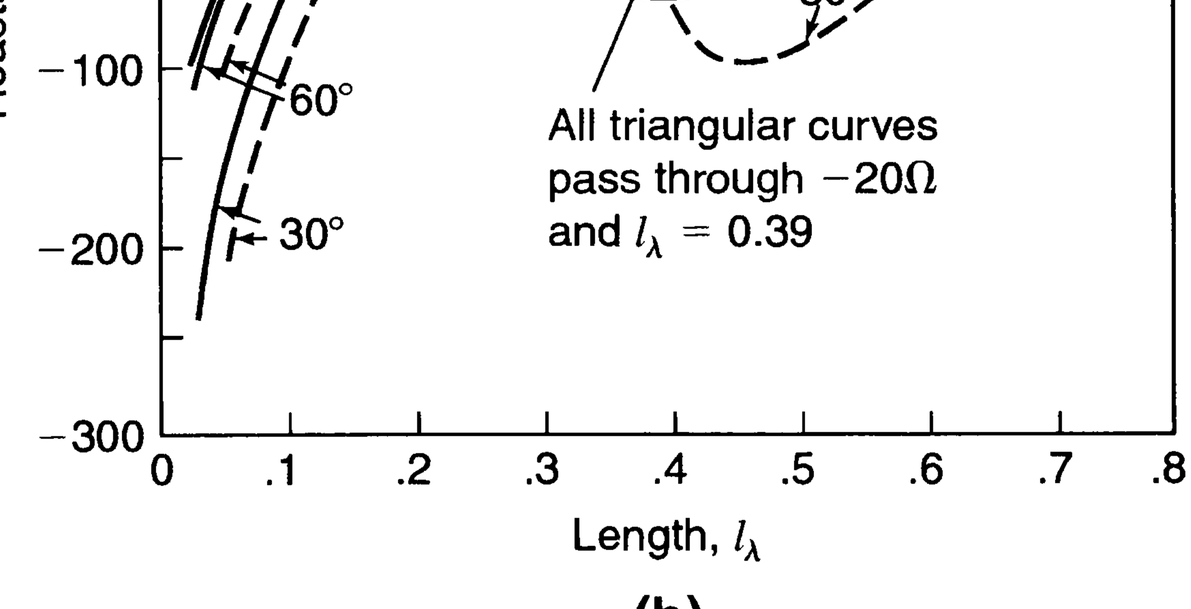
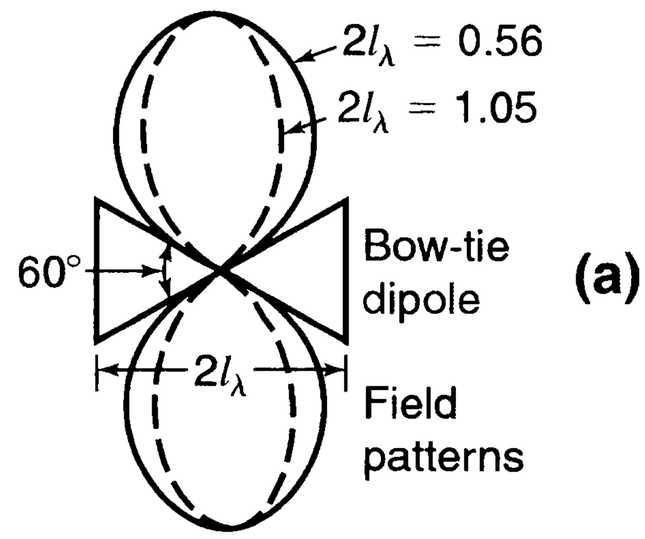


Figure 11–8–2
Gain of biconical antennas with respect to a $\lambda/2$ dipole as a function of length $2l_\lambda$ for full cone (flare) angles θ of 30, 60 and 90°. (*Brown (1) and Woodward.*)



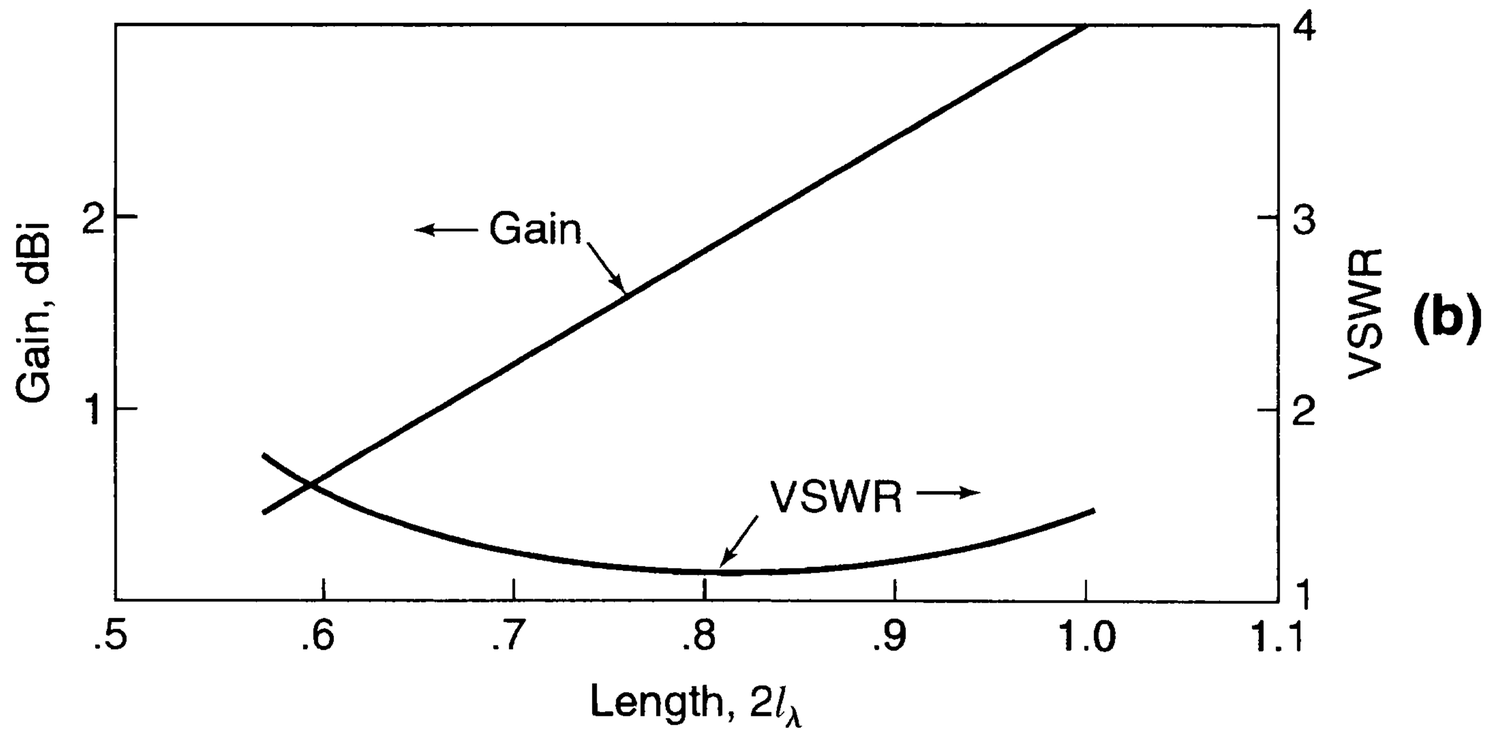


Figure 11–8–3
Gain in dBi and VSWR of UHF Brown-Woodward (bow-tie) antenna with 60° flare angle as a function of the length $2l_\lambda$. (*Brown and Woodward.*) The field patterns shown in (*a*) are actually those with the plane of the bow tie perpendicular to the page (rotated 90° on its axis) instead of with the plane of the bow tie parallel to the page as drawn.

11–4 THE FREQUENCY-INDEPENDENT CONCEPT: RUMSEY'S PRINCIPLE
（非频变概念：拉姆塞原理）

A true frequency-independent antenna is physically fixed in size and operates on an instantaneous basis over a wide bandwidth with relatively constant impedance, pattern, polarization and gain. These kinds of antennas are discussed in subsequent sections.

Beginning while at the Ohio State University in the early 1950s, continuing from 1954 to 1957 at the University of Illinois, and later at the University of California, first at Berkeley and subsequently at San Diego, Victor H. Rumsey (1) developed and introduced a new way of looking at antennas and their operation as a function of the frequency (Jordan-1; Mayes-1).

Rumsey was intrigued with Mushiake's observation in 1949 that *self-complementary antennas*[1] have a constant impedance of $Z_0/2$, or half the intrinsic impedance of space, at all frequencies (Mushiake-1; Uda-1). This is remarkable since there is an infinity of self-complementary shapes. A self-complementary planar antenna has a metal area congruent to the open area; i.e., the two areas can be brought into coincidence by a rigid motion. Three examples of self-complementary antennas are shown in Fig. 11–9. The metal and open areas are congruent since a rotation of either brings both into coincidence.

The slot and complementary dipole antennas of Chap. 9 are similarly related but usually require a translation for coincidence. Mushiake's $Z_0/2$ result comes directly from Booker's relation for complementary slots and dipoles as given by (9–5–11).

---

[1]The concept of complementary antennas applies strictly only to infinitesimally thin planar, perfectly conducting shapes of infinite extent.

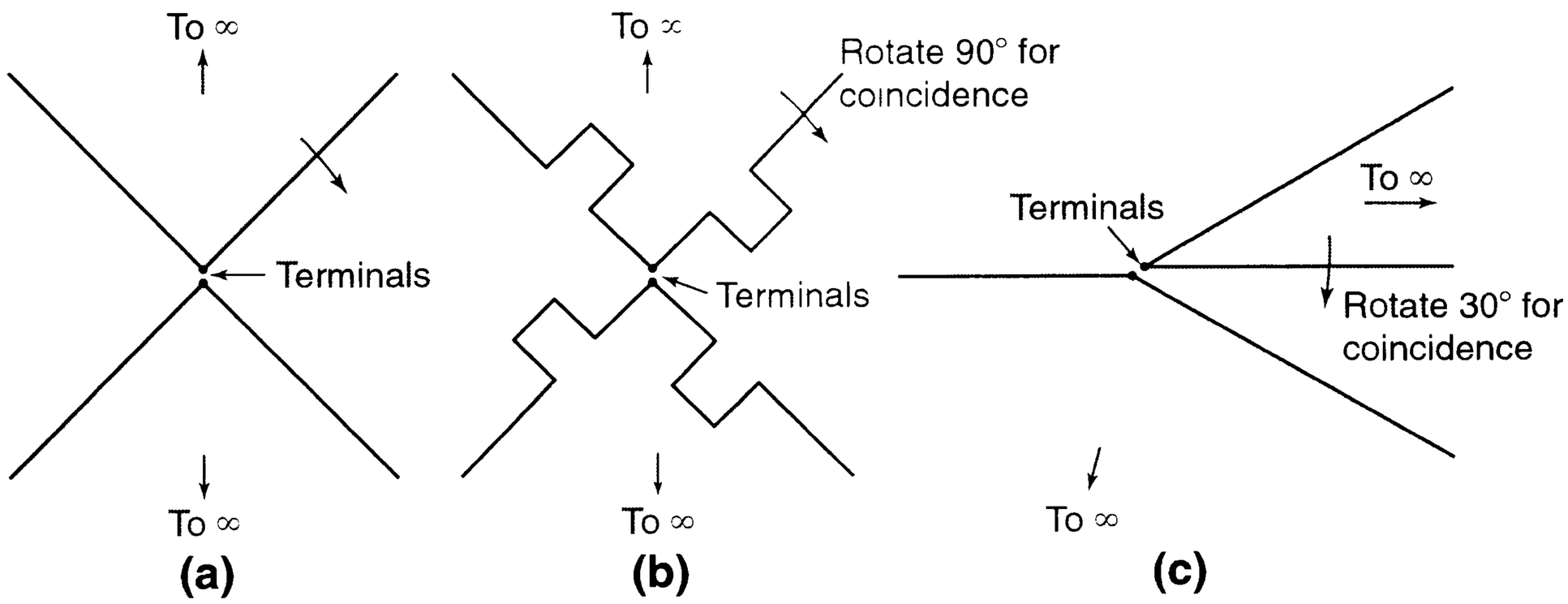


Figure 11–9
Three self-complementary planar antennas. Theoretical terminal impedance is 188 Ω.

Rumsey's principle is that the impedance and pattern properties of an antenna will be frequency-independent if the antenna shape is specified only in terms of angles.
(拉姆塞原理：若天线的形状仅由角度来决定，则该天线具有非频变的阻抗和波瓣图特性。)

Thus, an infinite logarithmic spiral should meet the requirement.

The biconical antenna of Secs. 11–2 and 11–3 are examples of an antenna that can be specified only in terms of the included cone angle, but it is frequency-independent only if it is infinitely long. When truncated (without a matched termination) there is a reflected wave from the ends of the cones which results in modified impedance and pattern characteristics.

To meet the frequency-independent requirement in a finite structure requires that the current attenuate along the structure and be negligible at the point of truncation. For radiation and attenuation to occur, as stated in Chap. 2, charge must be accelerated (or decelerated) and this happens when a conductor is curved or bent normally to the direction in which the charge is traveling. Thus, the curvature of a spiral results in radiation and attenuation so that, even when truncated, the spiral provides frequency-independent operation over a wide bandwidth.

11–5a THE ILLINOIS STORY (伊利诺依的故事)

Rumsey's principle was implemented experimentally by John D. Dyson at the University of Illinois, who constructed the first practical frequency-independent spiral antennas in 1958, first the bidirectional planar spiral and then the unidirectional conical spiral. These two types are described in this and the next section (11–6). Further work on spirals at Illinois was done by Dyson, Paul Mayes and G. A. DeSchamps. Concurrently Raymond DuHamel and Dwight Isbell were developing the log-periodic dipole array, and in 1961 Mayes and Robert Carrel described the log-periodic array with V-dipole elements (Sec. 11–7).

11–5b THE FREQUENCY-INDEPENDENT PLANAR LOG-SPIRAL ANTENNA (非频变平面对数螺蜷天线)

The equation for a *logarithmic (or log) spiral* is given by

$$r = a^{\theta} \tag{1}$$

or

$$\ln r = \theta \ln a \tag{2}$$

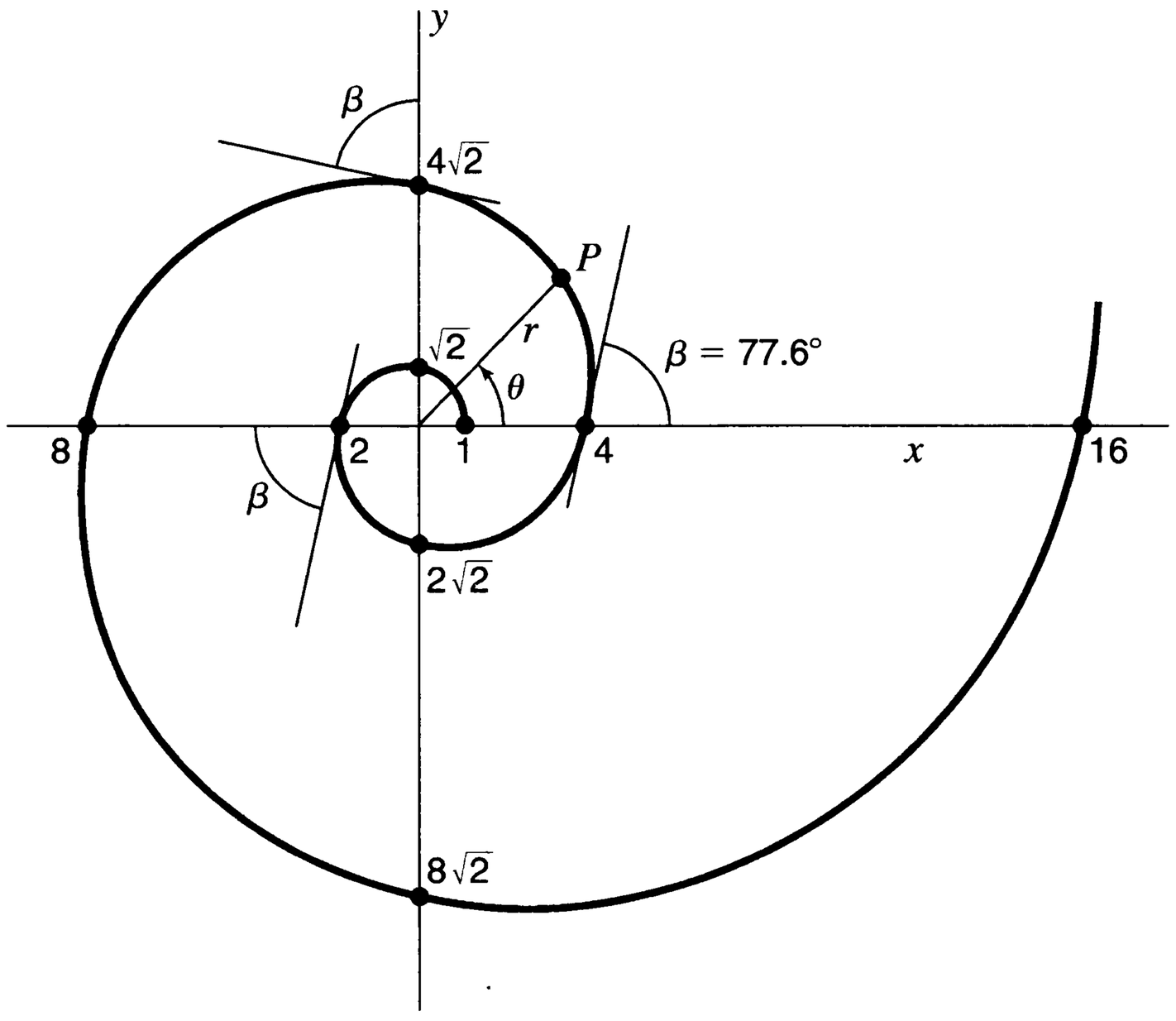


Figure 11-10
Logarithmic or log spiral.

where, referring to Fig. 11–10,

r = radial distance to point P on spiral

θ = angle with respect to x axis

a = a constant

From (1), the rate of change of radius with angle is

$$\frac{dr}{d\theta} = a^{\theta} \ln a = r \ln a \tag{3}$$

The constant a in (3) is related to the angle β between the spiral and a radial line from the origin as given by

$$\ln a = \frac{dr}{r\,d\theta} = \frac{1}{\tan \beta} \tag{4}$$

Thus, from (4) and (2),

$$\theta = \tan \beta \ln r \tag{5}$$

The log spiral in Fig. 11–10 is constructed so as to make $r = 1$ at $\theta = 0$ and $r = 2$ at $\theta = \pi$. These conditions determine the value of the constants a and β. Thus, from (4) and (5), $\beta = 77.6°$ and $a = 1.247$. Thus, the shape of the spiral is determined by the angle β which is the same for all points on the spiral.[1]

---

[1] Although it is a broadband antenna, the *Archimedes spiral* is not regarded as completely frequency-independent. The angle β for an Archimedes spiral as given by $r = a\theta$ is not a constant but is a function of position along the spiral. However, remote from the origin on a tight Archimedes spiral, β approaches a nearly constant angle and the Archimedes spiral becomes a close approximation of a tightly wound log spiral (Bawer-1; Mayes-1).

Let a second log spiral, identical in form to the one in Fig. 11–10, be generated by an angular rotation δ so that (1) becomes

$$r_2 = a^{\theta-\delta} \tag{6}$$

and a third and fourth spiral given by

$$r_3 = a^{\theta-\pi} \tag{7}$$

and

$$r_4 = a^{\theta-\pi-\delta} \tag{8}$$

Then, for a rotation $\delta = \pi/2$ we have 4 spirals at 90° angles. Metalizing the areas between spirals 1 and 4 and 2 and 3, with the other areas open, self-complementary and congruence conditions are satisfied. Connecting a generator or receiver across the inner terminals, we obtain Dyson's frequency-independent planar spiral antenna of Fig. 11–11 (Dyson-1).

The arrows indicate the direction of the outgoing waves traveling along the conductors resulting in right-circularly polarized (RCP) radiation (IEEE definition) outward from the page and left-circularly polarized radiation into the page. The high-frequency limit of operation is determined by the spacing d of the input terminal and the low-frequency limit by the overall diameter D. The ratio D/d for the antenna of Fig. 11–11 is about 25 to 1. If we take $d = \lambda/10$ at the high-frequency limit and $D = \lambda/2$ at the low-frequency limit,

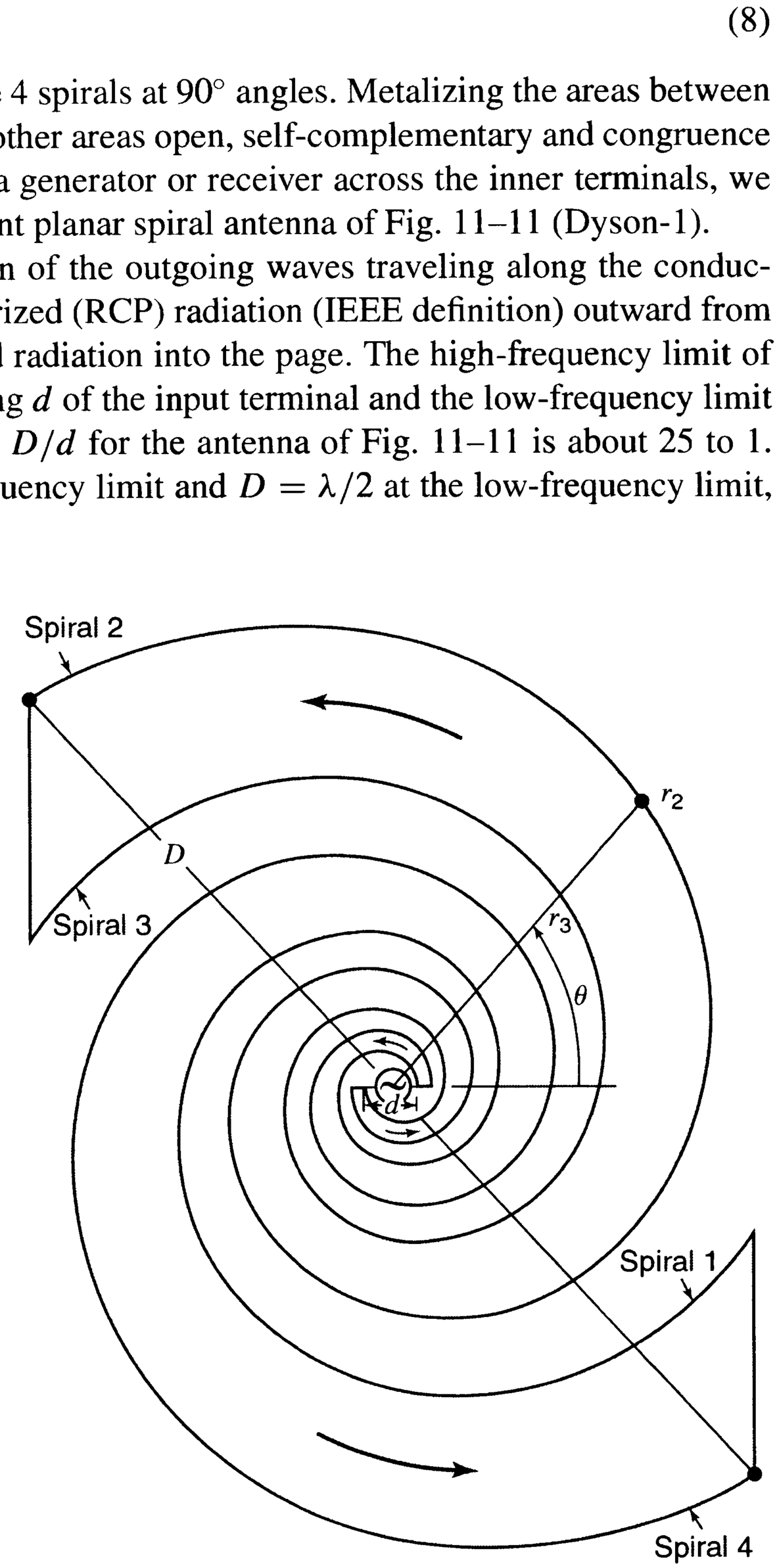


Figure 11–11
Frequency-independent planar spiral antenna.

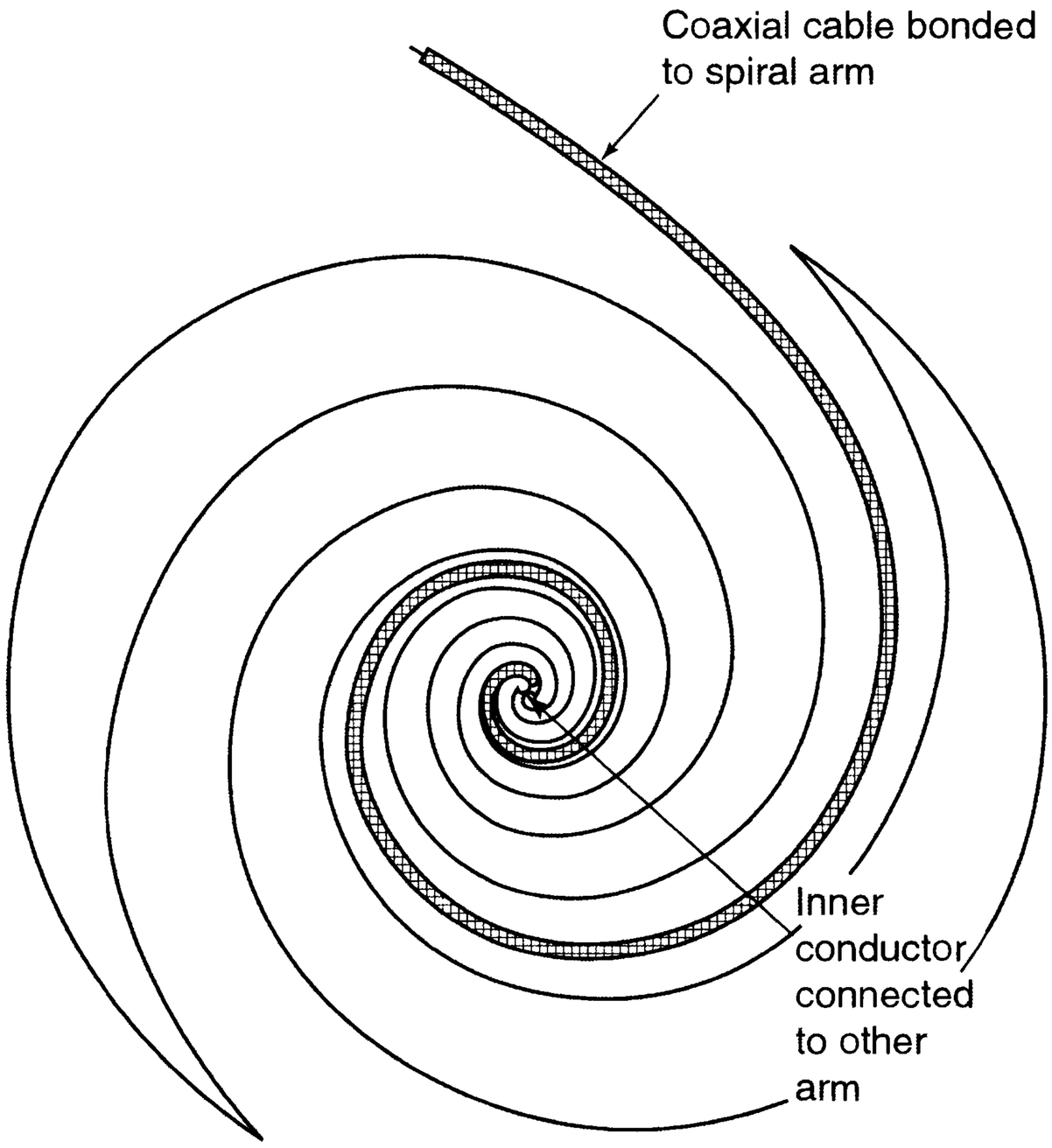


Figure 11–12
Frequency-independent planar spiral antenna cut from large ground plane.

the antenna bandwidth is 5 to 1. The spiral should be continued to a smaller radius but, for clarity, the terminal separation shown in Fig. 11–11 is larger than it should be. Halving it doubles the bandwidth.

In practice, it is more convenient to cut the slots for the antenna from a large ground plane, as done by Dyson, and feed the antenna with a coaxial cable bonded to one of the spiral arms as in Fig. 11–12 the spiral acting as a balun.[1] A dummy cable may be bonded to the other arm for symmetry but is not shown.

Radiation for the antennas of Figs. 11–11 and 11–12 is bidirectional broadside to the plane of the spiral. The patterns in both directions have a single broad lobe so that the gain is only a few dBi. The input impedance depends on the parameters δ and a and the terminal separation. According to Dyson, typical values are in the range 50 to 100 Ω or considerably less than the theoretical 188 Ω ($=Z_0/2$). The smaller measured values are apparently due to the finite thickness of spirals.

Referring to Fig. 11–11, the ratio K of the radii across any arm, such as between spirals 2 and 3, is given by the ratio of (7) to (6), or

$$K = \frac{r_3}{r_2} = a^{-\pi+\delta} \tag{9}$$

[1] Balance-to-unbalance transformer. See Chap. 23.

For the antenna of Fig. 11–11, $\delta = \pi/2$ so

$$K = \frac{r_3}{r_2} = a^{-\pi/2} = 0.707\ (=1/\sqrt{2}) \tag{10}$$

This is seen to be the ratio of the radial distances to the spiral of Fig. 11–10 at successive 90° intervals.

11–6 THE FREQUENCY-INDEPENDENT CONICAL-SPIRAL ANTENNA

（非频变圆锥螺蜷天线）

A tapered helix is a conical-spiral antenna and these were described and investigated extensively in the years following 1947. In the first article on the helical antenna which Kraus (1) published in 1947, he described a tapered helix which he constructed and measured. Figure 11–13*a* and *b* show tapered helical or conical spiral antennas in which the pitch angle is constant with diameter and turn spacing variable. These figures appeared in the first and second editions of this book. These tapered helix or conical spirals were investigated by Springer (1) (1950) and by Chatterjee (1) (1953, 1955) and others, and more recently by Nakano (1), Mikawa and Yamauchi, and found capable of bandwidths of 5 to 1 or more. Chatterjee also described a planar spiral antenna.

However, it was not until 1958 that John D. Dyson (2) at the University of Illinois made the tapered helix or conical spiral fully frequency-independent by wrapping or projecting multiple planar spirals onto a conical surface.

A typical balanced 2-arm Dyson conical spiral is shown in Fig. 11–14. The conical spiral retains the frequency-independent properties of the planar spiral while providing broad-lobed unidirectional circularly polarized radiation off the small end or apex of the cone. As with the planar spiral, the two arms of the conical spiral are fed at the centerpoint or apex from a coaxial cable bonded to one of the arms, the spiral acting as a balun. For symmetry a dummy cable may be bonded to the other arm, as suggested in Fig. 11–14. In some models the metal straps are dispensed with and the cables alone used as the spiral conductors. According to Dyson, the input impedance is between 100 and 150 Ω for a pitch angle $\alpha = 17°$ and full cone angles of 20 to 60°. The smaller cone angles (30° or less) have higher front-to-back ratios of radiation. The bandwidth, as with the planar spiral, depends on the ratio of the base diameter ($\sim\lambda/2$ at the lowest frequency) to the truncated apex diameter ($\sim\lambda/4$ at the highest frequency). This ratio may be made arbitrarily large.

Conical and planar spirals with more than 2 arms are also possible and have been investigated by Mayes (2) and Dyson and by Deschamps (1), all at the University of Illinois, and also by Atia (1) and Mei.

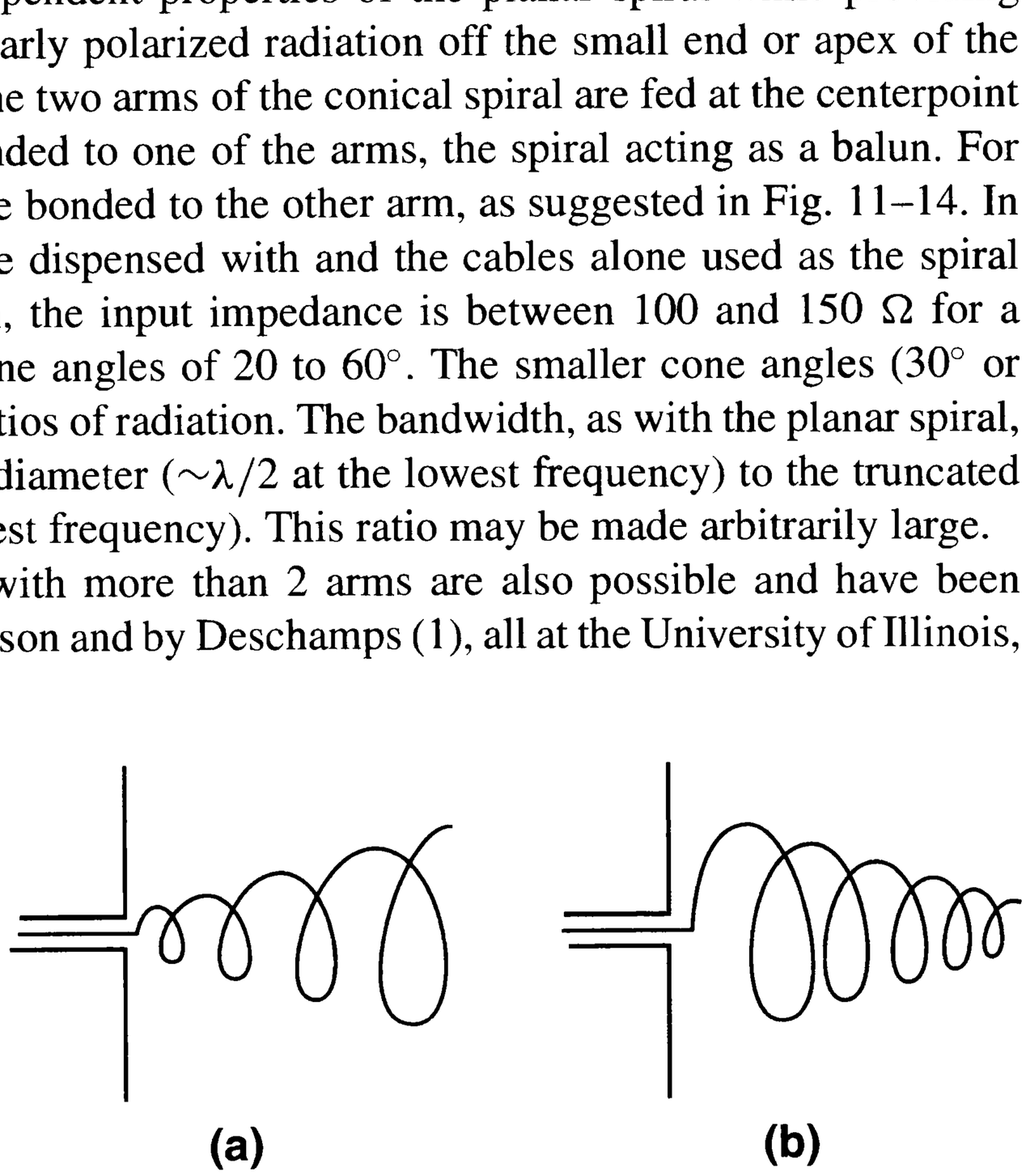


Figure 11–13
Tapered helical or conical-spiral (forward-fire) CP antennas.

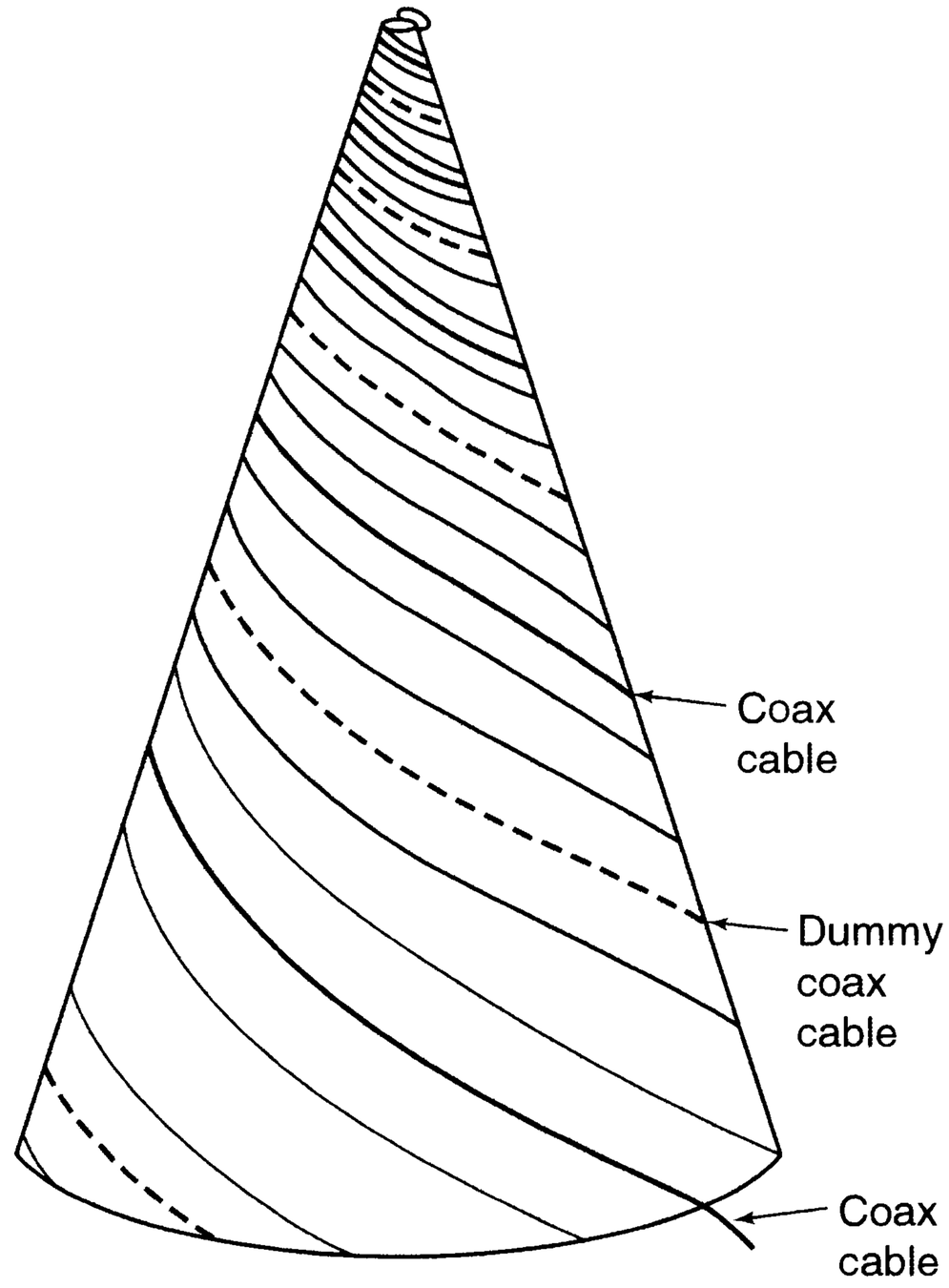


Figure 11–14
Dyson 2-arm balanced conical-spiral (backward-fire) antenna. Polarization is RCP. Inner conductor of coax connects to dummy at apex.

11–7 THE LOG-PERIODIC ANTENNA（对数周期天线）

While the planar and conical spirals were being developed, Raymond DuHamel (1) and Dwight Isbell, also at the University of Illinois, created a new type of frequency-independent antenna with a self-complementary toothed structure as suggested in Fig. 11–15. In an alternative version, the metal and slot areas of Fig. 11–15 are interchanged. Since $\beta_1 + \beta_2 = 90°$ the self-complementary condition is fulfilled. The expansion parameter

$$k_1 = \frac{R_n + 1}{R_n} \tag{1}$$

and the tooth-width parameter

$$k_2 = \frac{r_n}{R_n} \tag{2}$$

Further work at the University of Illinois showed that the self-complementary condition was not required, and by 1960 Dwight Isbell (1) had demonstrated the first *log-periodic dipole array.*[1] The basic concept is that *a gradually expanding periodic structure array* radiates most effectively when the array elements (dipoles) are near resonance so that with change in frequency the active (radiating) region moves along the array. This *expanding* structure array differs from the *uniform* arrays considered in Sec. 11–5*b*.

---

[1] So called because the structure repeats periodically with the logarithm of the frequency. Put another way, the structure doubles for each doubling of the wavelength.

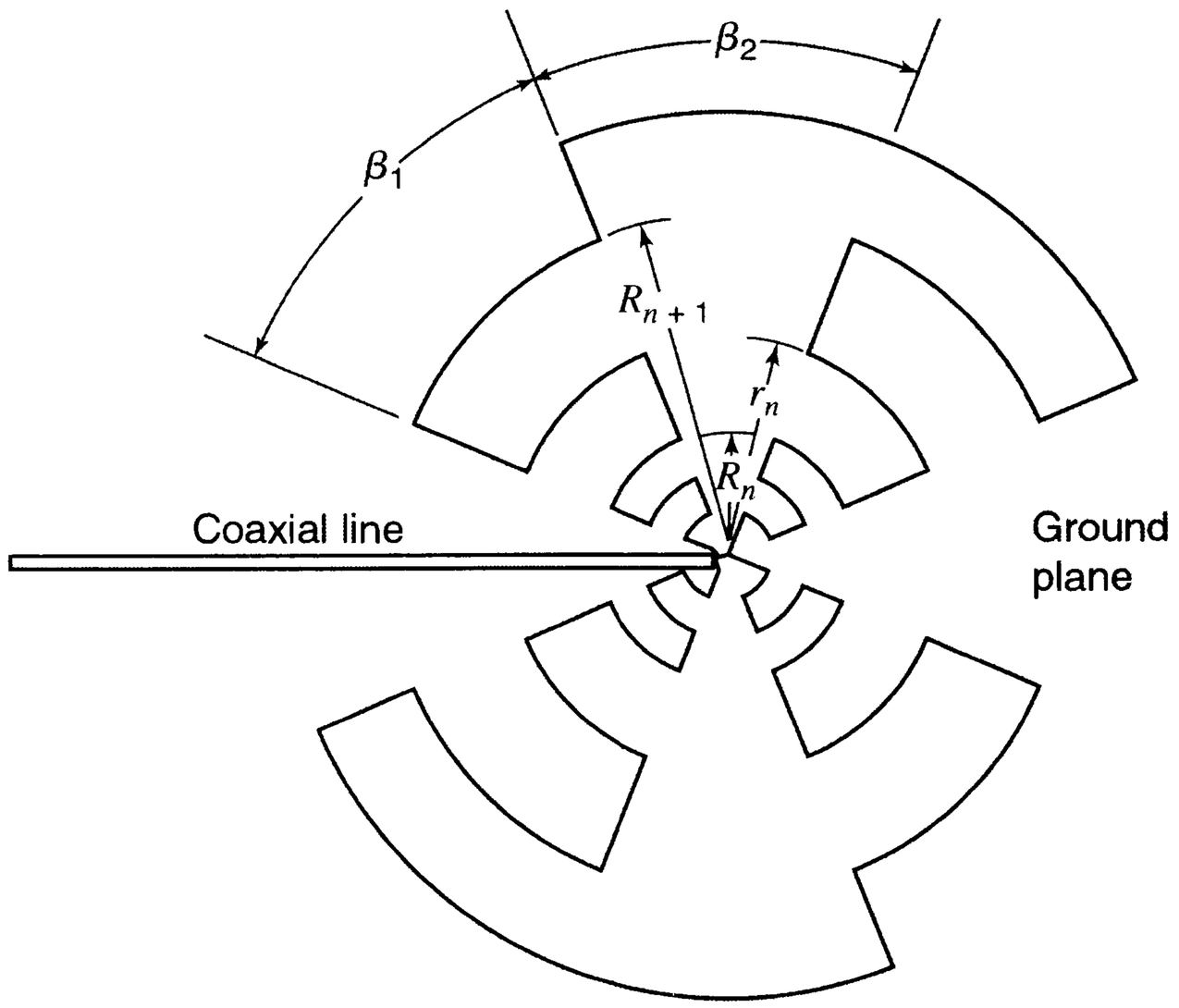


Figure 11–15
Self-complementary toothed log-periodic antenna of DuHamel and Isbell.

MECHANICALLY TUNABLE WIDEBAND DIPOLE

The arrangement of Fig. 11–16 consists of an adjustable $\lambda/2$ dipole made of two drum-type (roll-up) pocket rulers. If L is adjusted to approximately $\lambda/2$ at the frequency of operation, the impedance and pattern remain the same. Strictly speaking, the element thickness or width w and the size of the drum housing should also be adjusted, but if these dimensions remain small compared to λ, this effect is small and for many purposes may be negligible. This simple antenna illustrates the requirement that the antenna should expand or contract in proportion to the wavelength in order to be frequency-independent, or if the antenna-structure is not mechanically adjustable as above, ***the size of the active or radiating region should be proportional to the wavelength.*** (则其有效辐射区的尺寸应正比于波长)

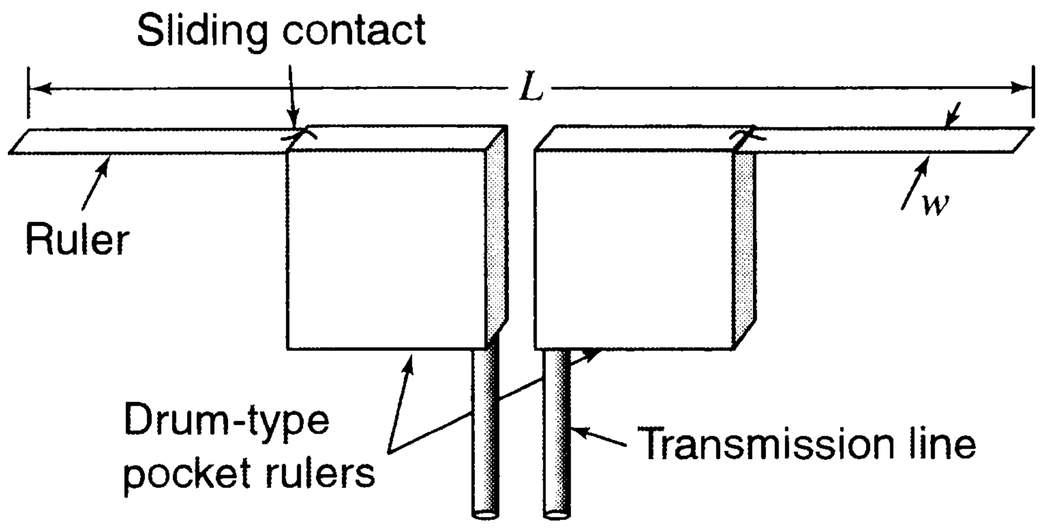


Figure 11–16
Adjustable $\lambda/2$ dipole of 2 drum-type rulers illustrates the requirement that to be frequency-independent an antenna must expand or contract in proportion to the wavelength.

This mechanically adjustable antenna can provide wideband operation for a scanning receiver if the dipole length is adjusted in synchronism with the receiver wavelength. This adjustment could be continuous or in incremental steps, as, for example, 11 steps. In a comparable log-periodic dipole array (Fig. 11–17) wideband operation is achieved with an array of 11 dipoles of graduated lengths mounted herringbone fashion along a transmission line. It has been proposed that computer controlled *laser beams* replace the rulers (or other antenna elements) to provide instantaneous frequency control with a **"plasma antenna."** (等离子体天线) See Sec. 21–29.

(*continued*)

(*Continued*)

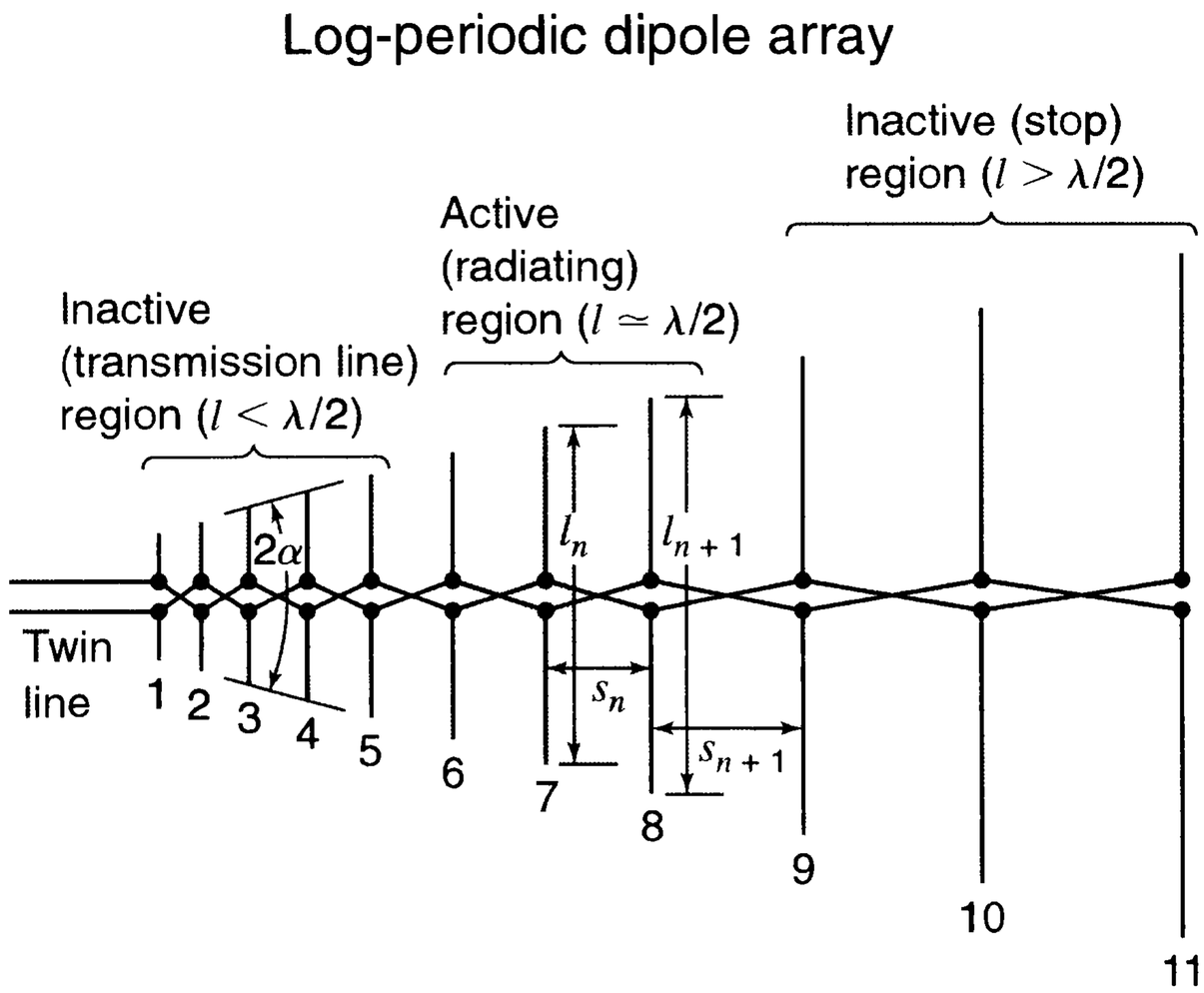


Figure 11–17
Isbell log-periodic frequency-independent type of dipole array of 7 dBi gain with 11 dipoles showing active central region and inactive regions (left and right ends).

The log-periodic dipole array is a popular design. Referring to Fig. 11–17, the dipole lengths increase along the antenna so that the included angle α is a constant, and the lengths l and spacings s of adjacent elements are scaled so that

$$\frac{l_{n+1}}{l_n} = \frac{s_{n+1}}{s_n} = k \tag{3}$$

where k is a constant. At a wavelength near the middle of the operating range, radiation occurs primarily from the central region of the antenna, as suggested in Fig. 11–17. The elements in this active region are about $\lambda/2$ long.

Elements 9, 10 and 11 are in the neighborhood of 1λ long and carry only small currents (they present a large inductive reactance to the line). The small currents in elements 9, 10 and 11 mean that the antenna is effectively truncated at the right of the active region. Any small fields from elements 9, 10 and 11 also tend to cancel in both forward and backward directions. However, some radiation may occur broadside since the currents are approximately in phase. The elements at the left (1, 2, 3, etc.) are less than $\lambda/2$ long and present a large capacitive reactance to the line. Hence, currents in these elements are small and radiation is small.

Thus, at a wavelength λ, radiation occurs from the middle portion where the dipole elements are $\sim\lambda/2$ long. When the wavelength is increased the radiation zone moves to the right and when the wavelength is decreased it moves to the left with maximum radiation toward the apex or feed point of the array.

At any given frequency only a fraction of the antenna is used (where the dipoles are about $\lambda/2$ long). At the short-wavelength limit of the bandwidth only 15 percent of the length may be used, while at the long-wavelength limit a larger fraction is used but still less than 50 percent.

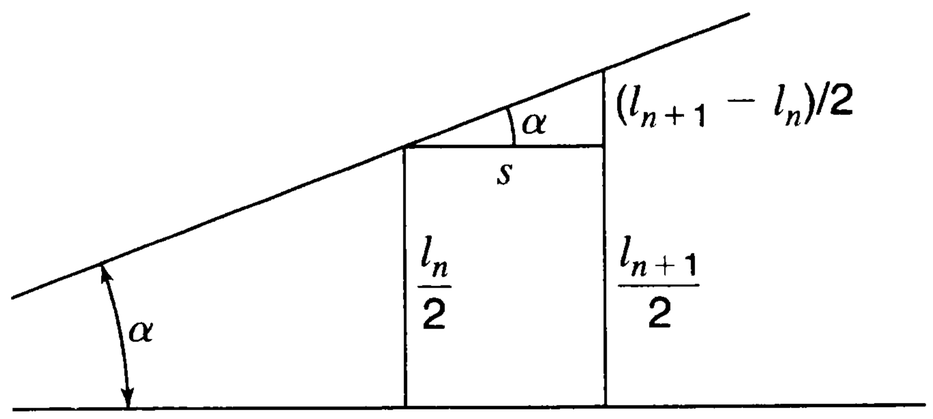


Figure 11–18
Log-periodic array geometry for determining the relation of parameters.

From the geometry of Fig. 11–18 for a section of the array, we have

$$\tan\alpha = \frac{(l_{n+1} - l_n)/2}{s} \tag{4}$$

or from (3),

$$\tan\alpha = \frac{[1 - (1/k)](l_{n+1}/2)}{s} \tag{5}$$

Taking $l_{n+1} = \lambda/2$ (when active) we have

$$\tan\alpha = \frac{1 - (1/k)}{4s_\lambda} \tag{6}$$

where

α = apex angle
k = scale factor
s_λ = spacing in wavelengths shortward of $\lambda/2$ element

Specifying any 2 of the 3 parameters α, k and s_λ determines the third. The relationship of the 3 parameters is displayed in Fig. 11–19 with the optimum design line (maximum gain for a given value of scale factor k) and gain along this line from calculations of Carrel (1), Cheong (1) and King, De Vito (1) and Stracca and Butson (1) and Thomson.

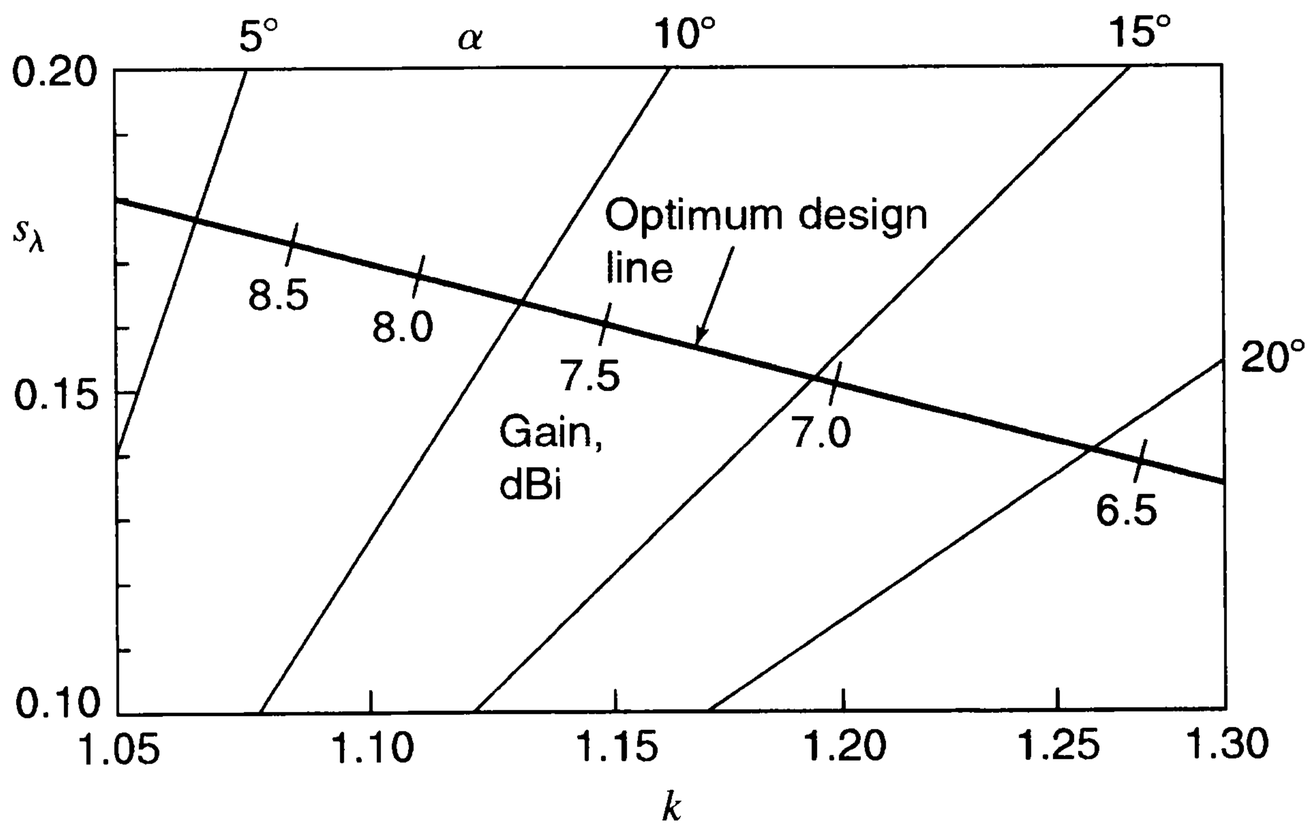


Figure 11–19
Relation of log-periodic array parameters of apex angle α, scale factor k and spacing s_λ [from (6)] with optimum design line and gain values according to Carrel and others (see text for details).

The length l (and spacing s) for any element $n+1$ is k^n greater than for element 1, or

$$\frac{l_{n+1}}{l_1} = k^n = F \tag{7}$$

where F = frequency ratio or bandwidth

Thus, if $k = 1.19$ and $n = 4$, $F = k^4 = 1.19^4 = 2$ and element $5(=n+1)$ is twice the length l_1 of element 1. Thus, with 5 elements and $k = 1.19$, the frequency ratio is 2 to 1.

EXAMPLE 11–7.1

Design a log-periodic dipole array with 7 dBi gain and a 4 to 1 bandwidth. Specify apex angle α, scale constant k and number of elements.

■ **Solution**

From Fig. 11–19, the 7 dBi point on the maximum gain line corresponds to the apex angle $\alpha = 15°$ and $k = 1.2$. (We also note that $s_\lambda = 0.15$.) From (7),

$$k^n = F \qquad \text{or} \qquad n \ln k = \ln F \tag{8}$$

and

$$n = \frac{\ln F}{\ln k} = \frac{\ln 4}{\ln 1.2} = \frac{1.386}{0.182} = 7.6 \tag{9}$$

Taking $n = 8, n+1 = 9$. Adding 2 more elements for a conservative design brings the total to 11.[1]

The array in Fig. 11–17 corresponds to the parameters of the above example. The E-plane HPBW $\simeq 60°$. The H-plane beamwidth is a function of the gain as given by

$$\text{HPBW } (H \text{ plane}) \leq \frac{41{,}000}{D \times 60°} \qquad \text{(deg)} \tag{10}$$

For the antenna of the example $D = 5$, since $\log_{10} 5 = 7$ dBi, so

$$\text{HPBW } (H \text{ plane}) \leq \frac{41{,}000}{5 \times 60°} = 137° \tag{11}$$

Details of construction and feeding are shown in Fig. 11–20. The arrangement in (a) is fed with coaxial cable, the one at (b) with twin line.

To obtain more gain than with a single log-periodic dipole array, 2 arrays may be stacked. However, for frequency-independent operation, Rumsey's principle requires that the locations of all elements be specified by angles rather than distances. This means that both log-periodic arrays must have a common apex, and, accordingly, the beams of the 2 arrays point in different directions. For a stacking angle of 60°, the situation is as suggested

[1] The number of extra elements needed depends, for example, on the design gain. Thus, for a high-gain design the active region requires more elements than for a low-gain design so that the bandwidth is less than given by k^n.

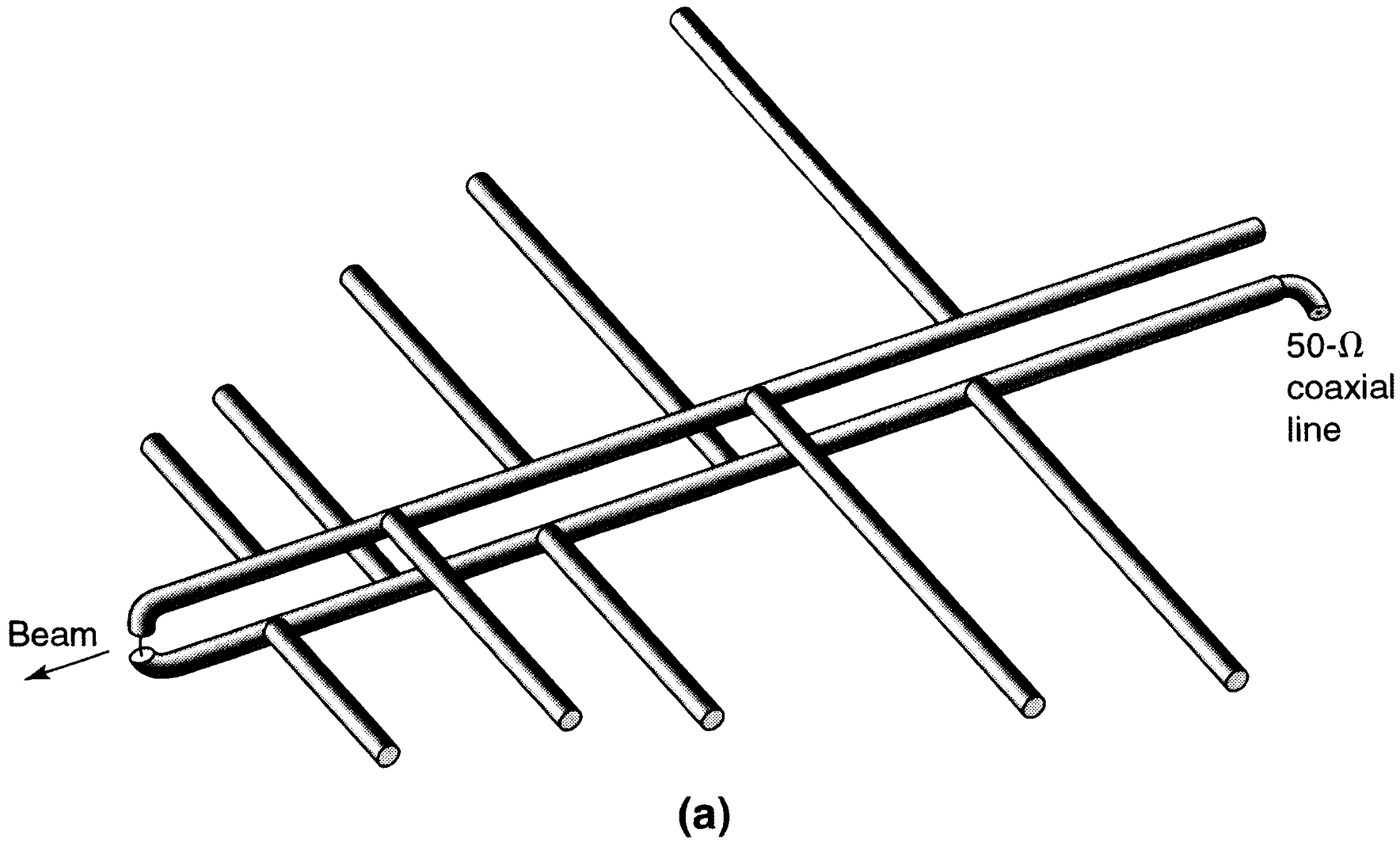


Figure 11–20
Construction and feed details of log-periodic dipole array. Arrangement at (*a*) has a 50- or 75-Ω coaxial feed. The one at (*b*) has criss-crossed open-wire line for 300-Ω twin-line feed.

in Fig. 11–21 for dipole arrays of the type shown in Figs. 11–17 and 11–20. The array in Fig. 11–21*b* is a skeleton-tooth or edge-fed trapezoidal type. Wires supported by a central boom replace the teeth of the antenna of Fig. 11–15.

For very wide bandwidths the log-periodic array must be correspondingly long. To shorten the structure, Paul Mayes (3) and Robert Carrel, of the University of Illinois, developed a more compact V-dipole array which can operate in several modes. In the

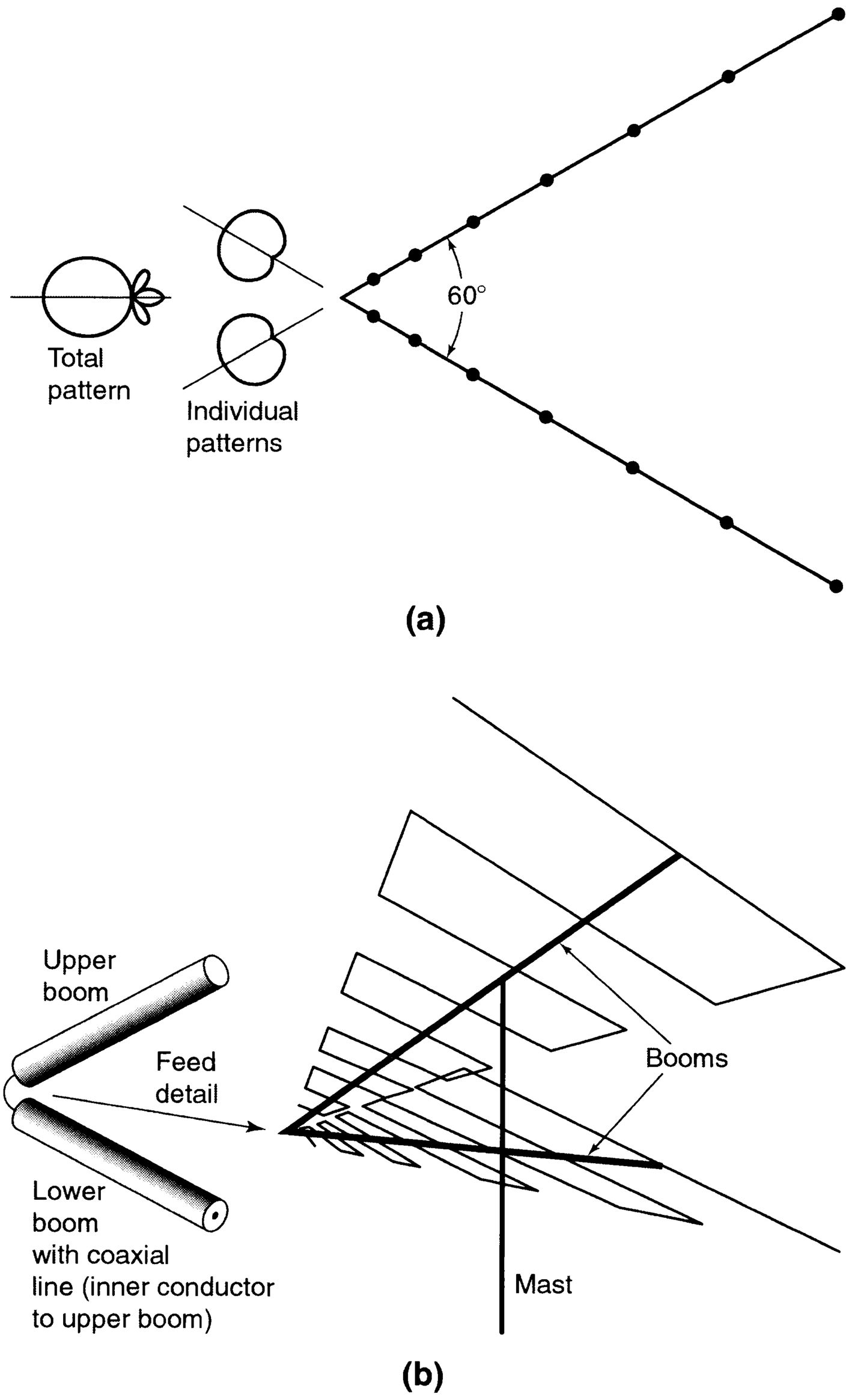


Figure 11–21
Stacked log-periodic arrays, with dipole type, as in Figs. 11–17 and 11–18 at (*a*) and trapezoidal or wedge-fed type at (*b*).

lowest mode, with the central region dipoles $\sim\lambda/2$ long, operation is as already described. However, as the frequency is increased to the point where the shortest elements are too long to give $\lambda/2$ resonance, the longest elements become active at $3\lambda/2$ resonance. As the frequency is increased further, the active region moves to the small end in the $3\lambda/2$ mode. With still further increase in frequency the large end becomes active in still higher-order modes. The forward tilt of the V-dipoles has little effect on the $\lambda/2$ mode but in the higher modes provides essential forward beaming. An example of a V-dipole array is shown in Fig. 11–21.

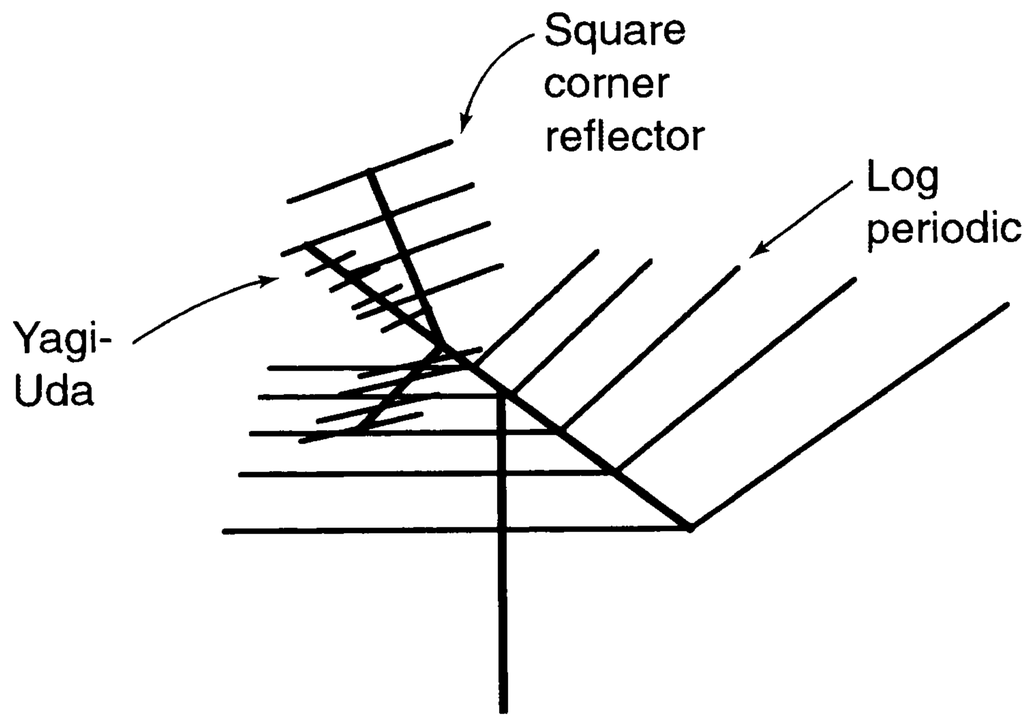


Figure 11–22
YUCOLP (Yagi-Uda–Corner–Log-Periodic) hybrid array for covering U.S. VHF TV and FM bands and UHF TV band. The YU–corner combination provides higher gain for the UHF TV band than an extension of the LP dipole array.

11–8 THE COMPOSITE YAGI-UDA–CORNER–LOG-PERIODIC (YUCOLP) ARRAY
(YUCOLP 复合阵)

For ultimate compactness and gain, to cover the 54 to 890 MHz U.S. TV and FM bands, a hybrid composite YUCOLP (Yagi-Uda–Corner–Log-Periodic) array is a popular design. A typical model, shown in Fig. 11–22, has an $\alpha = 43°$, $k = 1.3$ LP array of 5 V-dipoles to cover the 54 to 108 MHz TV and FM bands with a 6 dBi gain in the $\lambda/2$ mode, the 174 to 216 MHz band with 8 or 9 dBi gain in the $3\lambda/2$ mode and a square-corner–YU array to cover the 470 to 890 UHF TV band with a 7 to 10 dBi gain. The total included angle of the V-dipoles is 120°. The corner–YU array is similar in design to the one in Fig. 11–17. As frequency increases, the active region moves from the large to the small end of the LP array in the $\lambda/2$ mode, then from the large to the small end in the $3\lambda/2$ mode, next to the corner reflector and finally to the YU array. The corner–YU array provides more gain for the UHF band than possible with a high-frequency extension of the LP array.

REFERENCES (参考文献)

Atia, A. E. (1), and K. K. Mei, "Analysis of Multiple Arm Conical Log-Spiral Antennas," *IEEE Trans. Ants. Prop.*, **AP-19,** 320–331, May 1971.

Bawer, R. (1), and J. J. Wolfe, "A Printed Circuit Balun for Use with Spiral Antennas," *IRE Trans. Microwave Th. Tech.*, **MTT-8,** 319–325, May 1960.

Brown G. H. (1), and O. M. Woodward, "Experimentally Determined Radiation Characteristics of Conical and Triangular Antennas," *RCA Rev.*, **13,** 425–452, December 1952.

Butson, P. O. (1), and G. T. Thomson, "A Note on the Calculation of the Gain of Log-Periodic Dipole Antennas," *IEEE Trans. Ants. Prop.*, **AP-24,** 105–106, January 1976.

Carrel, R. L. (1), "The Design of Log-Periodic Dipole Antennas," *IRE Int. Conv. Rec.*, **1,** 61–75, 1961.

Chatterjee, J. S. (1), "Radiation Field of a Conical Helix," *J. Appl. Phys.*, **24,** 550–559, May 1953; "Radiation Characteristics of a Conical Helix of Low Pitch Angle," *J. Appl. Phys.*, **26,** 331–335, March 1955.

Cheong, W. M. (1), and R. W. P. King, "Log Periodic Dipole Antenna," *Radio Sci.*, **2,** 1315–1325, November 1967.

Deschamps, G. A. (1), "Impedance Properties of Complementary Multiterminal Planar Structures," *IRE Trans. Ants. Prop.*, **AP-7,** S371–378, December 1959.

DuHamel, R. H. (1), and D. E. Isbell, "Broadband Logarithmically Periodic Antenna Structures," *IRE Natl. Conv. Rec.*, pt. 1, 119–128, 1957.

Dyson, J. D. (1), "The Equiangular Spiral Antenna," *IRE Trans. Ants. Prop.*, **AP-7,** 181–187, April 1959.

Dyson, J. D. (2), "The Unidirectional Spiral Antenna," *IRE Trans. Ants. Prop.*, **AP-7,** 329–334, October 1959.

Dyson, J. D. (3), and P. E. Mayes, "New Circularly-Polarized Frequency-Independent Antennas with Conical Beam or Omnidirectional Patterns," *IRE Trans. Ants. Prop.*, **AP-9,** 334–342, July 1961.

Isbell, D. E. (1), "Log Periodic Dipole Arrays," *IRE Trans. Ants. Prop.*, **AP-8,** 260–267, May 1960.

Jordan, E. C. (1), G. A. Deschamps, J. D. Dyson and P. E. Mayes, "Developments in Broadband Antennas," *IEEE Spectrum,* **1,** 58–71, April 1964.

Kraus, J. D. (1), "Helical Beam Antenna," *Electronics,* **20,** 109–111, April 1947.

Mayes, P. E. (1), "Frequency-Independent Antennas: Birth and Growth of an Idea," *IEEE Ants. Prop. Soc. Newsletter,* **24,** 5–8, August 1982.

Mayes, P. E. (2), and J. D. Dyson, "A Note on the Difference between Equiangular and Archimedes Spiral Antennas," *IRE Trans. Microwave Th. Tech.*, **MTT-9,** 203–205, March 1961.

Mayes, P. E. (3), and R. L. Carrel, "Log Periodic Resonant-V Arrays," San Francisco Wescon Conf., August 1961.

Mushiake, Y. (1), *J. IEE Japan,* **69,** 1949.

Nakano, H. (1), T. Mikawa and J. Yamauchi, "Numerical Analysis of Monofilar Conical Helix," *IEEE AP-S Int. Symp.*, **1,** 177–180, 1984.

RRL (1), Radio Research Laboratory Staff, *Very High Frequency Techniques,* McGraw-Hill, 1947, Chap. 4.

Rumsey, V. H. (1), *Frequency Independent Antennas,* Academic Press, 1966.

Schellkunoff, S. A. (1), Electromagnetic Waves, Van Nostrand, New York, 1943, Chap. 11.

Springer, P. S. (1), "End-Loaded and Expanding Helices as Broad-Band Circularly Polarized Radiators," Electronic Subdiv. Tech. Rept. 6104, Wright-Patterson AFB, 1950.

Uda, S. (1), and Y. Mushiake, "Input Impedance of Slit Antennas," Tech. Rept. Tohoku University 14, pp. 46–59, 1949.

Vito, G. De (1), and G. B. Stracca, "Comments on the Design of Log-Periodic Dipole Antennas," *IEEE Trans. Ants. Prop.*, **AP-21,** 303–308, May 1973; **AP-22,** 714–718, September 1974.

PROBLEMS（习题）

11-2-1 三锥和接地平面）

Single cone and ground plane. Prove that the characteristic impedance of Z_k for a single cone and ground plane is half Z_k for a biconical antenna.

11-2-2 2°的圆锥）

The 2° cone. Calculate the terminal impedance of a conical antenna of 2° total angle operating against a very large ground plane. The length l of the cone is $3\lambda/8$.

11-5-1 对数螺蜷）

Log spiral. Design a planar log-spiral antenna of the type shown in Fig. 11–11 to operate at frequencies from 1 to 10 GHz. Make a drawing with dimensions in millimeters.

11-7-1 对数周期）

Log-periodic. Design an "optimum" log-periodic antenna of the type shown in Fig. 11–17 to operate at frequencies from 100 to 500 MHz with 11 elements. Give (*a*) length of longest element, (*b*) length of shortest element, and (*c*) gain.

11-7-2 三叠对数周期）

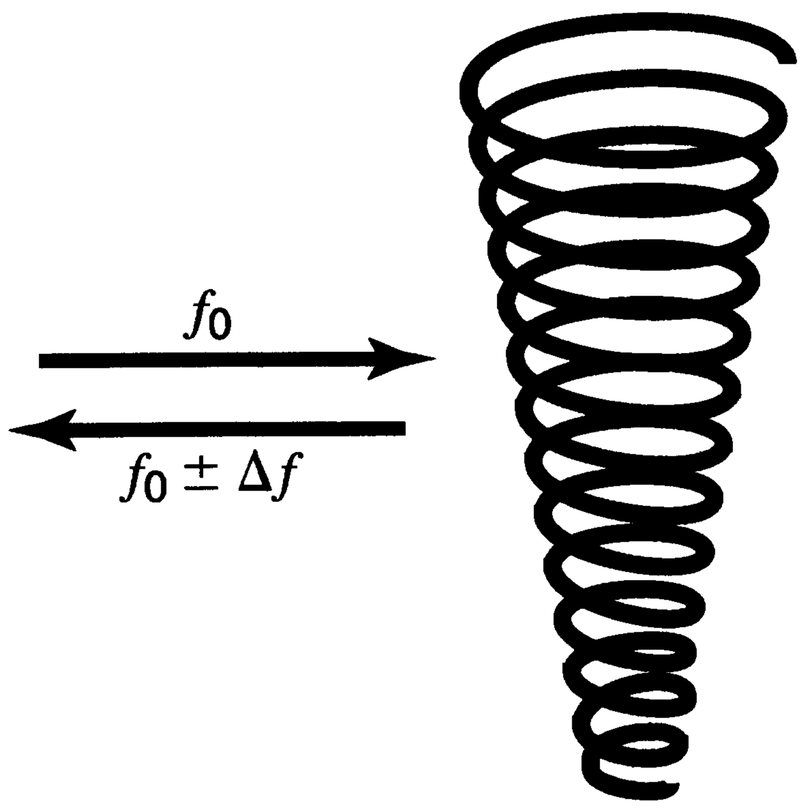
Stacked LPs. Two LP arrays like in the worked example of Sec. 11–7 are stacked as in Fig. 11–21*a*.

(*a*) Calculate and plot the vertical plane field pattern. Note that pattern multiplication cannot be applied.

(*b*) What is the gain?

For computer programs, see Appendix C.

CHAPTER 12



Antenna Temperature, Remote Sensing and Radar Cross Section
天线温度、遥感和雷达截面

Topics in this chapter include:

- Antenna temperature（天线温度）
- System temperature[1]（系统温度）
- SNR (Signal-to-Noise Ratio)（信号噪声比）
- Passive remote sensing（被动遥感）
- Radar (active remote sensing)（雷达（主动遥感））
- Doppler radar（多普勒雷达）
- RCS (Radar Cross Section)（雷达截面）

12–1 ANTENNA TEMPERATURE（天线温度）

The *noise power per unit bandwidth* available at the terminals of a resistor of resistance R at a temperature T_r (Fig. 12–1*a*) is given by the Nyquist (1) relation as

$$p = kT_r \qquad (\text{W Hz}^{-1}) \tag{1}$$

where

p = power per unit bandwidth, W Hz^{-1}
k = Boltzmann's constant = 1.38×10^{-23} J K^{-1}
T_r = absolute temperature, K

If the resistor R is replaced by a *lossless* antenna of radiation resistance R in an anechoic chamber at temperature T_c (Fig. 12–1*b*), the noise power per unit bandwidth available at the antenna terminals is the same as in (1) provided $T_c = T_r$.

Now if the antenna is removed from the anechoic chamber and pointed at a sky of temperature T_s (Fig. 12–1*c*), the noise power per unit bandwidth is again the same as in (1)

[1]For a more detailed treatment, see J. D. Kraus, *Radio Astronomy,* 2d ed., Cygnus-Quasar, Powell, Ohio, 1986, pp. 3–39 to 3–45.

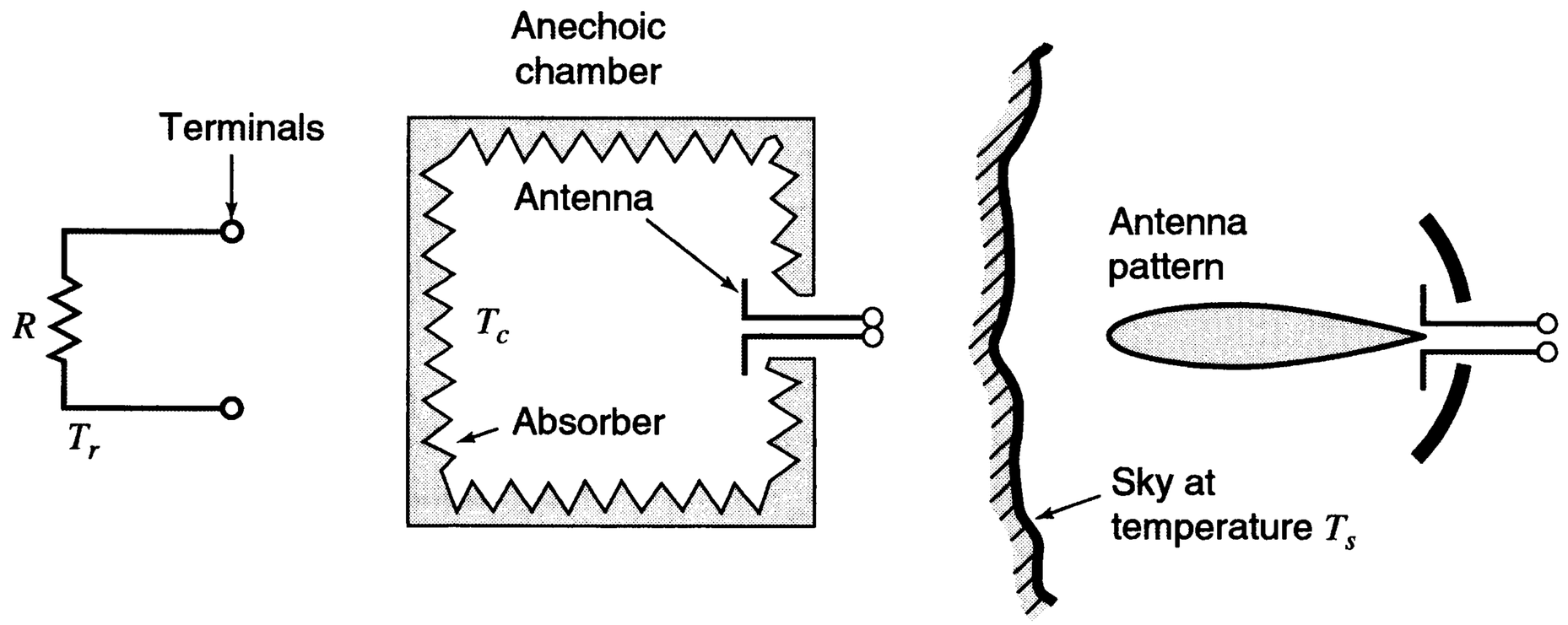


Figure 12–1
(*a*) Resistor at temperature T_r. (*b*) Antenna in an anechoic chamber at temperature T_c. (*c*) Antenna observing sky at temperature T_s. The same noise power per unit bandwidth is available at the terminals in all three cases if $T_r = T_c = T_s$.

(provided $T_s = T_r$), and we can say that the antenna has a *noise temperature* T_A equal to the sky temperature T_s.[1]

Using an antenna to measure a distant temperature in this way is *passive remote sensing,* and the antenna in this application may be called a **radio telescope**. (射电望远镜) This passive remote sensing is in contrast to the *active remote sensing* of radar.

To measure the distant or sky temperature T_s, the antenna noise temperature may be compared with that of a resistor at an adjustable temperature T_r by alternately connecting antenna (directed at sky) and resistor to a receiver. When the receiver detects no difference, $T_A = T_s = T_r$.

The noise temperature T_A of the antenna (assumed *lossless*) is equal to the sky temperature T_s, and *not* the physical temperature of the antenna. This contrasts to the resistor of Fig. 12–1*a*, which is completely lossy and, therefore, has a noise temperature equal to its physical temperature. Thus, for a radio-telescope antenna the noise power per unit bandwidth is given by

$$p = kT_A \qquad (\text{W Hz}^{-1}) \tag{2}$$

where T_A is the *antenna* (*noise temperature*) or temperature of the antenna's radiation resistance, determined by the sky temperature at which the antenna beam is directed.

Hence, a radio telescope antenna (and receiver) may be regarded as a radiometer (or temperature-measuring device) for remote sensing the temperature of distant regions coupled to the system through the radiation resistance of the antenna. An extreme view is to imagine that with a radio telescope we can, in effect, stretch the wires from the terminals in Fig. 12–1*a*, until the resistor R comes into contact with the distant regions.

---

[1] It is assumed that the entire antenna pattern "sees" the sky of temperature T_s.

In the above we have assumed that the antenna has no thermal losses and also that all of its pattern is encompassed by the region being observed (negligible side and back lobes).

Multiplying (2) by the bandwidth B, we obtain the total power available as

$$\boxed{P = kT_A B \qquad \text{(W)}} \tag{3}$$

where B = receiver bandwidth, Hz.

It is often convenient to express the received power per unit bandwidth in terms of a *flux density* S. Thus, dividing (2) by the effective aperture A_e of the antenna, we have

$$S = \frac{p}{A_e} = \frac{kT_A}{A_e} \qquad (\text{W m}^{-2}\,\text{Hz}^{-1}) \tag{4}$$

In the above development we have assumed a single celestial source whose extent is greater than the antenna beam. In practice the antenna temperature may include contributions from several sources, or the source under observation may be superimposed on a background temperature region. To measure the temperature of a source under these circumstances, the radio telescope beam is moved onto and then off of the source, and an incremental or difference temperature ΔT_A is measured. Thus, (4) for the *source flux density* can be rewritten as

$$\boxed{S = \frac{k\Delta T_A}{A_e} \qquad (\text{W m}^{-2}\,\text{Hz}^{-1}) \qquad \textbf{\textit{Source flux density}}} \quad (\text{源通量密度}) \tag{5}$$

Note that the units for the flux density S (W m^{-2} Hz^{-1}) are the same as for the Poynting vector per unit bandwidth, so we may regard the flux density as a measure of the Poynting vector (per unit bandwidth) received from distant regions. In radio astronomy observations, flux densities are very small, and the unit of convenience is the jansky (Jy) $= 10^{-26}$ W m^{-2} Hz^{-1}, after Karl G. Jansky who made the first radio astronomy observations in 1933.

If the remote source is small compared to the antenna beamwidth, all of ΔT_A is due to the source, and (5) gives the correct flux density, but ΔT_A is *not* equal to the source temperature. However, if the source solid angle Ω_s and the antenna beam solid angle Ω_A (Fig. 12–2) are known, the source temperature is given very simply by

$$T_s = \frac{\Omega_A}{\Omega_s}\Delta T_A \qquad \text{(K)} \tag{6}$$

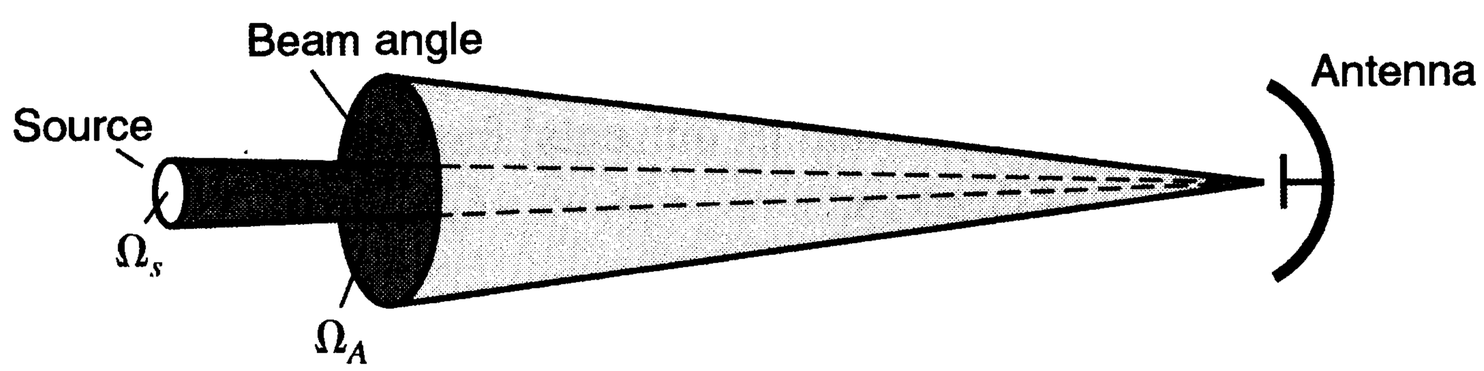


Figure 12–2
Situation where source Ω_s is smaller than the beam area Ω_A.

where

T_s = source temperature, K
ΔT_A = incremental antenna (noise) temperature, K
Ω_s = source solid angle (see figure), sr
Ω_A = antenna beam solid angle (see figure), sr

It is important to note that the antenna temperature has nothing to do with the physical temperature of the antenna provided the antenna is lossless.

Let us now apply (6), by way of an example, to a classic, historic remote-sensing observation.

EXAMPLE **12–1.1 Mars Temperature**

The incremental antenna temperature for the planet Mars measured with the U.S. Naval Research Laboratory 15-m radio telescope antenna at 31.5-mm wavelength was found to be 0.24 K (Mayer-1). Mars subtended an angle of 0.005° at the time of the measurement. The antenna HPBW = 0.116°. Find the average temperature of Mars at 31.5-mm wavelength.

■ **Solution**

Assuming that Ω_A is given by the solid angle within the HPBW, we have from (6) that the Mars temperature

$$T_s = \frac{\Omega_A}{\Omega_s}\Delta T_A \approx \frac{0.116^2}{\pi(0.005^2/4)}0.24 = 164 \text{ K} \qquad \textit{Ans.}$$

This temperature is less than the infrared temperature measured for the sunlit side (250 K), implying that the 31.5-mm radiation may originate farther below the Martian surface than the infrared radiation. This is an example of *passive remote sensing* of the surface of another planet from the earth.

The source temperature in the above discussion and example is an *equivalent temperature*. It may represent the physical temperature of a planetary surface, as in the example, but, on the other hand, a celestial plasma cloud with oscillating electrons which is at a physical temperature close to absolute zero may generate radiation with an equivalent temperature of thousands of kelvins. The temperatures we are discussing are thermal (noise) temperatures like those of a perfect emitting-absorbing object called a *blackbody*. A hot object filling the beam of a receiving antenna will ideally produce an antenna temperature equal to its thermometer-measured temperature.[1] However, the oscillating currents of a transmitting antenna can produce an equivalent temperature of millions of degrees (K) even though the antenna structure is at normal outdoor temperature. It may be said that the antenna (and its currents) have an equivalent blackbody (or noise) temperature of millions of degrees.

[1] Assuming the object's intrinsic impedance = 377 Ω.

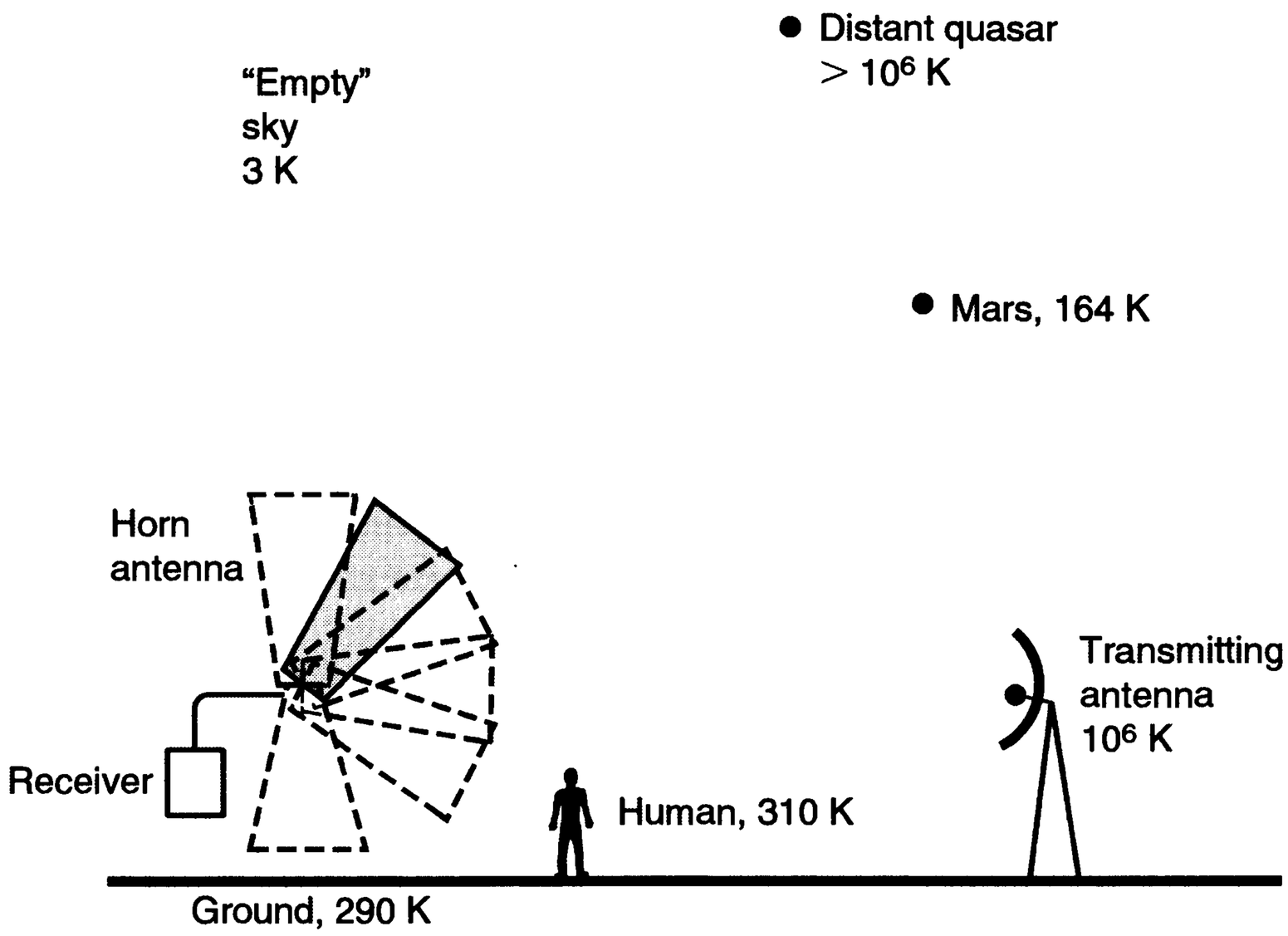


Figure 12–3
Horn antenna directed at various objects senses different temperatures as suggested.

All objects not at absolute zero produce radiation which, in principle, may be detected with a radio antenna-receiver. A few objects are shown in Fig. 12–3 with the equivalent temperatures measured when the horn antenna is pointed at them. Thus, the temperature of a distant quasar is over 10^6 K, of Mars 164 K, of a transmitter on the earth 10^6 K, of a man 310 K, of the ground 290 K,[1] while the empty sky at the zenith is 3 K. This temperature, called the 3 K sky background, is the residual temperature of the primordial fireball which created the universe and is the *minimum possible temperature* of any antenna looking at the sky.

An assumption was made in the above discussion which requires comment. It was assumed that the antenna and source polarizations were matched (same polarization states on the Poincaré sphere). Although this is possible for the transmitting antenna in Fig. 12–3, it is not possible for the other sources because their radiation is *unpolarized* and *any* antenna, whether linearly or circularly polarized, receives only half of the available power. Hence, for such sources the flux density of the source at the antenna is given by twice (4) or

$$S = \frac{2k\Delta T_A}{A_e} \qquad (\text{W m}^{-2}\,\text{Hz}^{-1}) \qquad \textbf{\textit{Source flux density}} \quad (\text{源通量密度}) \tag{7}$$

We have considered two extreme cases, one where the source extent is much broader than the antenna beamwidth and one where the source extent is much less than the beamwidth. Let us consider now the general situation for any source beamwidth–size relation. For this general situation the *total antenna temperature* is

$$T_A = \frac{1}{\Omega_A}\int_0^{\pi}\int_0^{2\pi} T_s(\theta,\phi)\,P_n(\theta,\phi)\,d\Omega \qquad (\text{K}) \tag{8}$$

[1] Due to reflection, more realistic values for a human and the ground might be less.

where

T_A = total antenna temperature (not ΔT_A), K
$T_s(\theta, \phi)$ = brightness temperature of source or sources as a function of angle, K
$P_n(\theta, \phi)$ = normalized antenna power pattern, dimensionless
Ω_A = antenna beam solid angle, sr
$d\Omega$ = $\sin\theta \, d\theta \, d\phi$ = infinitesimal element of solid angle, sr

Note that T_A in (8) is the *total antenna temperature* including not only contributions from a particular source in the main beam but from sources of radiation in all directions in proportion to the pattern response. Note also that the temperatures are in *kelvins*, K (= Celsius degrees above absolute zero).

EXAMPLE **12–1.2 Antenna Temperature**

A circular reflector antenna of 500 m^2 effective aperture operating at $\lambda = 20$ cm is directed at the zenith. What is the total antenna temperature assuming the sky temperature is uniform and equal to 10 K? Take the ground temperature equal to 300 K and assume that half the minor-lobe beam area is in the back direction (toward the ground).[1] The beam efficiency is 0.7 ($= \Omega_M / \Omega_A$).

■ **Solution**

Assuming that the antenna aperture efficiency is 50 percent, its physical aperture is 1000 m^2 and its diameter 35.7 m ($= 2\sqrt{1000/\pi}$ m). At $\lambda = 0.2$ m the diameter is 179λ, implying that the HPBW $\simeq 0.4°$ ($= 70°/179$). Thus, the antenna is highly directional with the main beam directed entirely at the sky (close to the zenith).

Since the (main) beam efficiency is 0.7, 70 percent of the beam area Ω_A is directed at the 10 K sky, half of the remainder or 15 percent at the sky and the other half of the remainder or 15 percent at the 300 K ground. Thus, integrating (8) in 3 steps, we have

$$\text{Sky contribution} = \frac{1}{\Omega_A}(10 \times 0.7\,\Omega_A) = 7 \text{ K}$$

$$\text{Side-lobe contribution} = \frac{1}{\Omega_A}\left(10 \times \frac{1}{2} \times 0.3\,\Omega_A\right) = 1.5 \text{ K}$$

$$\text{Back-lobe contribution} = \frac{1}{\Omega_A}\left(300 \times \frac{1}{2} \times 0.3\,\Omega_A\right) = 45 \text{ K}$$

and

$$T_A = 7 + 1.5 + 45 = 53.5 \text{ K}$$

Note that 45 of the 53.5 K, or 84 percent, of the total antenna temperature results from the back-lobe pickup from the ground. With no back lobes the antenna temperature could ideally be only 10 K, so that in this example the back lobes are very detrimental to

[1] Since much sky radiation reaches the antenna via reflection from the ground, a more realistic ground temperature might be less than 300 K.

the system sensitivity (see Sec. 12–2). It is for this reason that radio telescope and space communication antennas are usually designed to reduce back- and side-lobe response to a minimum.

The information given regarding aperture and wavelength is relevant to the problem only to the extent that it indicates that the main beam is directed entirely at the sky.

In contrast to the above example, let us recall the temperature measurements made by Arno Penzias (1) and Robert Wilson in 1965 at 4 GHz on their 6.2-m horn-reflector antenna which resulted in their discovery of the 3-K sky background. When directed at regions of "empty" sky near the zenith, Penzias and Wilson measured a total antenna temperature $T_A = 6.7$ K.

Contributions to this temperature were measured as

$$\begin{aligned} &2.3 \pm 0.3 \text{ K due to the atmosphere} \\ &0.8 \pm 0.4 \text{ K due to ohmic losses} \\ &\underline{<0.1 \text{ K due to back lobes into the ground}} \\ &3.2 \pm 0.5 \end{aligned}$$

The difference, $6.7 - 3.2 = 3.5$ K, they attributed to the sky background. Theirs was the first measurement of the residual temperature of the primordial (Big Bang) fireball which created the universe, and sets a lower limit to the temperature of any antenna looking at the sky. See Sec. 21–19.

The 0.1-K ground pickup by the antenna of Penzias and Wilson is one of the smallest values ever measured for an antenna. Note also that, in their analysis, they attributed 0.8 K to *ohmic losses* in the antenna and rotary joint.

The antenna noise temperature from the sky as a function of frequency (and wavelength) is presented in Fig. 12–4. A beam angle (HPBW) of less than a few degrees and 100 percent (main) beam efficiency are assumed. Curves are given for beam angles from the zenith (complementary to elevation angles). At lower frequencies the temperature is dominated by radiation from the galaxy. At higher frequencies the atmosphere introduces noise due to absorption. Above the earth's atmosphere (in space) this noise is avoided, but there is a universal photon or quantum noise temperature limit at still higher frequencies given by the photon energy hf divided by Boltzmann's constant, or

$$T = \frac{hf}{k} \qquad \text{(K)} \tag{9}$$

where

$$\begin{aligned} h &= \text{Planck's constant} = 6.63 \times 10^{-34} \text{ J s} \\ f &= \text{frequency, Hz} \\ k &= \text{Boltzmann's constant} = 1.38 \times 10^{-23} \text{ J K}^{-1} \end{aligned}$$

Across the spectrum between these sources of noise there is the noise background or floor of 3 K (or more precisely 2.7 K) due to radiation from the primordial fireball. The low-noise region between galactic radiation and atmospheric absorption defines an *earth-based radio window,* while the region between galactic radiation and quantum limit establishes a *cosmic radio window*.

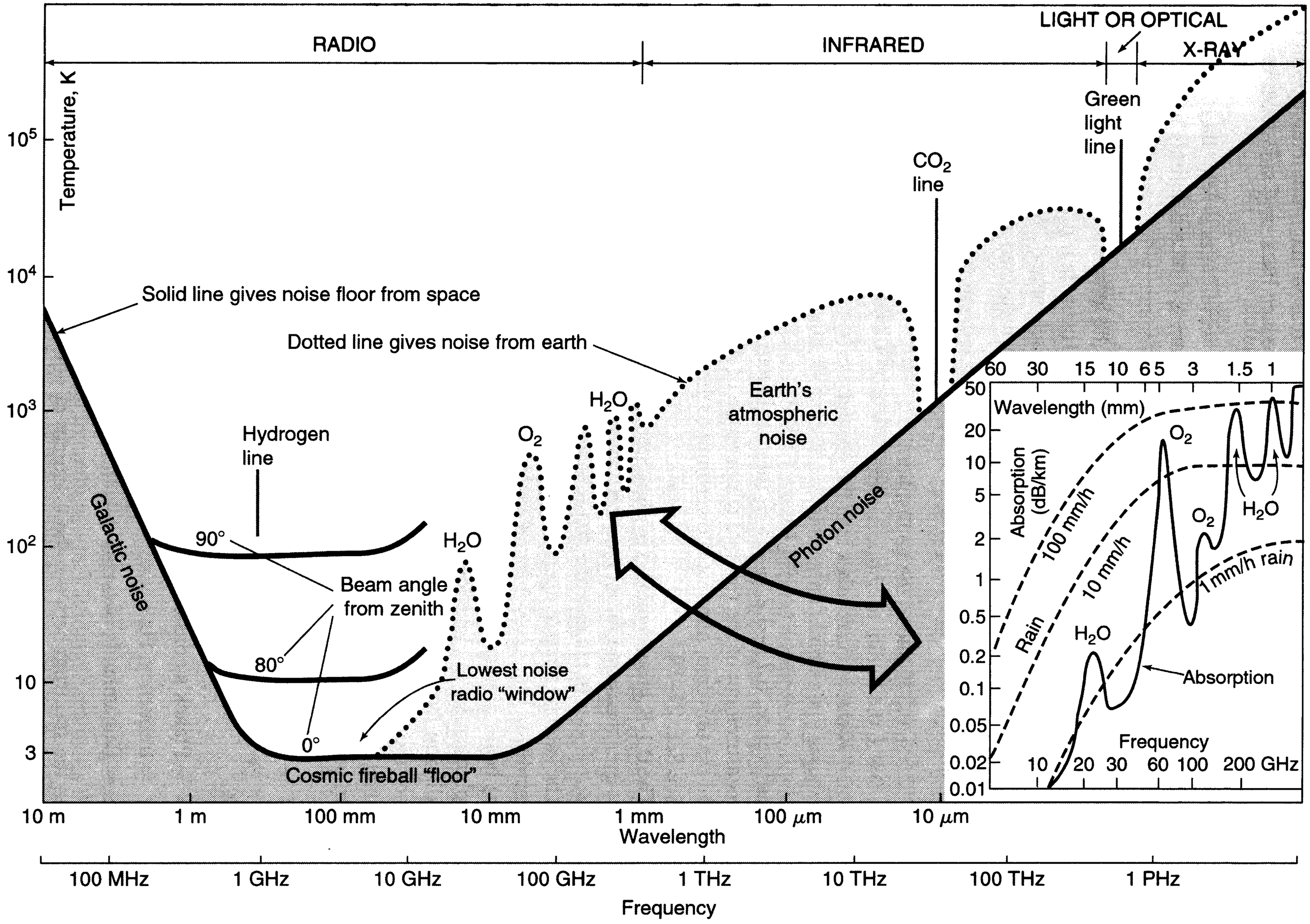


Figure 12–4
Sky noise temperature from radio to x-rays. Inset gives attenuation by the atmosphere and by rain. This figure also appears as Fig. 21–44 but is duplicated here for convenience.

12–2 SYSTEM TEMPERATURE（系统温度）

An antenna is part of a receiving system consisting, in general, of an antenna, a receiver and a transmission line which connects them. The temperature of the system, or *system temperature,* is a critical factor in determining the sensitivity and signal-to-noise ratio of a receiving system.

Let us consider a receiving system as shown schematically in Fig. 12–5, with an antenna, a receiver and a transmission line (or waveguide) connecting them.

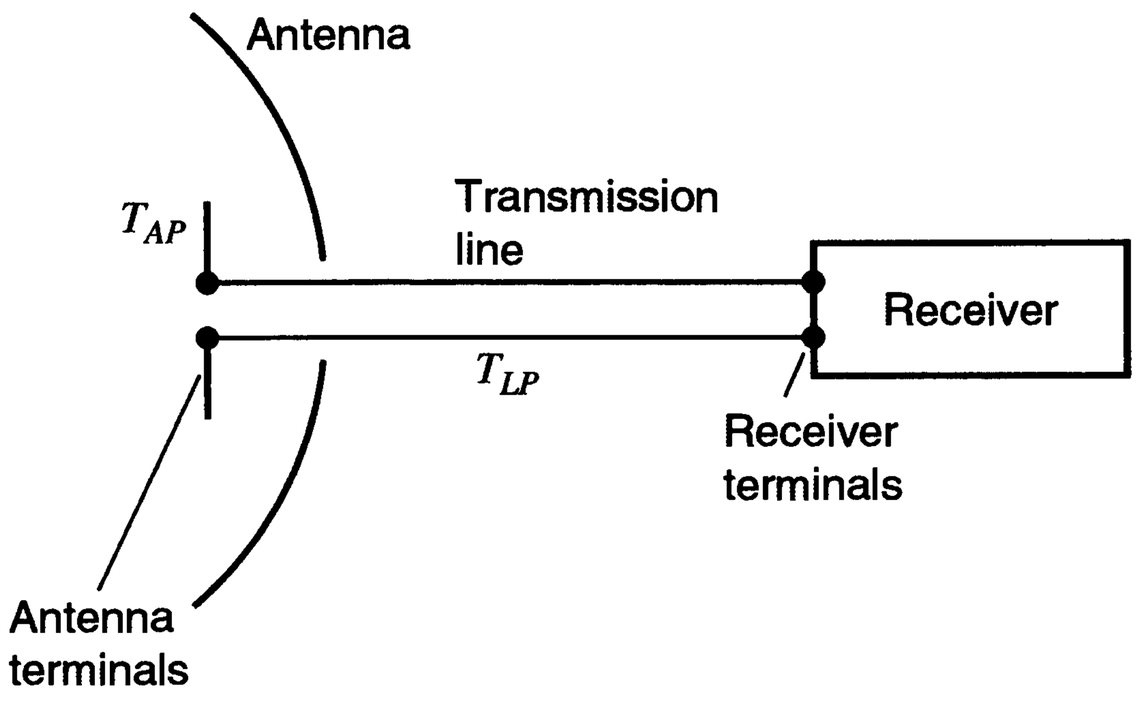


Figure 12–5
Antenna, transmission line and receiver for system temperature determination.

The system temperature depends on the noise temperature of the sky, the ground and antenna environs, the antenna pattern, the antenna thermal efficiency, the receiver noise temperature and the efficiency of the transmission line (or waveguide) between the antenna and receiver. The *system temperature at the antenna terminals* is given by

$$T_{\text{sys}} = T_A + T_{AP}\left(\frac{1}{\varepsilon_1} - 1\right) + T_{LP}\left(\frac{1}{\varepsilon_2} - 1\right) + \frac{1}{\varepsilon_2}T_R \tag{1}$$

where

T_A = antenna noise temperature [as given by (12–1–8)], K
T_{AP} = antenna physical temperature, K
ε_1 = antenna (thermal) efficiency ($0 \leq \varepsilon_1 \leq 1$), dimensionless
T_{LP} = line physical temperature, K
ε_2 = line efficiency ($0 \leq \varepsilon_2 \leq 1$),[1] dimensionless
T_R = receiver noise temperature (see next paragraph), K

The *receiver noise temperature* is given by

$$T_R = T_1 + \frac{T_2}{G_1} + \frac{T_3}{G_1 G_2} + \cdots \tag{2}$$

where

T_1 = noise temperature of first stage of receiver, K
T_2 = noise temperature of second stage, K

[1] $\varepsilon_2 = e^{-\alpha l}$, where α = attenuation constant (Np m^{-1}) and l = length of line (m).

T_3 = noise temperature of third stage, K
G_1 = power gain of first stage
G_2 = power gain of second stage

Terms for additional stages may be required if the temperatures are sufficiently high and the gains sufficiently low.

EXAMPLE **12–2.1 System Temperature**

A receiving system has an antenna with a total noise temperature of 50 K, a physical temperature of 300 K and an efficiency of 99 percent, a transmission line at a physical temperature of 300 K and an efficiency of 90 percent, and a receiver with the first 3 stages all of 80 K noise temperature and 13 dB gain. Find the system temperature.

■ **Solution**

From (2) the receiver noise temperature is

$$T_R = 80 + \frac{80}{20} + \frac{80}{20^2} = 80 + 4 + 0.2 = 84.2 \text{ K}$$

From (1) the system temperature is

$$T_{\text{sys}} = 50 + 300\left(\frac{1}{0.99} - 1\right) + 300\left(\frac{1}{0.9} - 1\right) + \frac{1}{0.9}84.2$$
$$= 50 + 3 + 33.3 + 93.6 \simeq 180 \text{ K}$$

Note that due to losses in the antenna, its physical temperature contributes 3 K. The line contributes about 33 K and the receiver about 94 K.

The sensitivity, or *minimum detectable temperature,* ΔT_{min}, of a receiving system is equal to the rms noise temperature ΔT_{rms} of the system, as given by

$$\Delta T_{\text{min}} = \frac{k' T_{\text{sys}}}{\sqrt{\Delta f t}} = \Delta T_{\text{rms}} \tag{3}$$

where

k' = system constant (order of unity), dimensionless
T_{sys} = system temperature [sum of antenna, line and receiver temperatures as given by (1)], K
T_{rms} = rms noise temperature = ΔT_{min}, K
Δf = predetection bandwidth of receiver, Hz
t = postdetection time constant, s

The *criterion of detectability* is that the incremental antenna temperature ΔT_A due to a radio source be equal to or exceed ΔT_{min}, that is,

$$\Delta T_A \geq \Delta T_{\text{min}} \tag{4}$$

and the *signal-to-noise* (S/N) *ratio* is then[1]

$$\frac{S}{N} = \frac{\Delta T_A}{\Delta T_{\min}} \tag{5}^2$$

Many space communication systems, radio telescopes and remote sensing systems operate at such high sensitivity (low signal levels) that a low system temperature is essential.

EXAMPLE **12–2.2 Minimum Detectable Flux Density**

The Ohio State University 110 by 21 m radio telescope antenna (see Fig. 20–6) had a physical aperture of 2208 m^2, and at 1415 MHz an aperture efficiency of 54 percent and a system temperature of 50 K. The rf bandwidth was 100 MHz, the output time constant 10 s and the system constant was 2.2. Find the minimum detectable flux density.

■ **Solution**

From (3) the minimum detectable temperature was

$$\Delta T_{\min} = \frac{k' T_{\text{sys}}}{\sqrt{\Delta f t}} = \frac{2.2 \times 50}{\sqrt{100 \times 10^6 \times 10}} = 0.0035 \text{ K}$$

The effective aperture $A_e = A_p \varepsilon_{ap} = 2208 \times 0.545 = 1203 \text{ m}^2$. From (12–1–7) the minimum detectable flux density is

$$\Delta S_{\min} = \frac{2k\Delta T_{\min}}{A_e} = \frac{2 \times 1.38 \times 10^{-23} \times 0.0035}{1203}$$

$$= 8.1 \times 10^{-29} \text{ W m}^{-2} \text{ Hz}^{-1} \simeq 8 \text{ mJy}^*$$

By repeating observations and averaging, the minimum can be further reduced ($\alpha \sqrt{1/n}$, where n = number of observations).

In a large sky survey at 1415 MHz, about 20,000 radio sources were detected and cataloged at flux densities above 180 mJy. Thus, the signal-to-noise ratio for these cataloged sources is

$$\frac{S}{N} = \frac{180 \text{ mJy}}{8.0 \text{ mJy}} = \frac{\Delta T_A}{\Delta T_{\min}} = \frac{0.0785 \text{ K}}{0.0035 \text{ K}} \simeq 22.5$$

12–3 SNR (SIGNAL-TO-NOISE RATIO)（信噪比）

Let us consider next the signal-to-noise ratio for a receiving system which is part of a *communication* link. If a transmitter radiates a power P_t isotropically and uniformly over a bandwidth Δf_t, it produces a flux density at a distance r of $P_t/(4\pi r^2 \Delta f_t)$. A receiving

---

[1]Distinguish between S here for *signal* and S elsewhere for *flux density* or *Poynting vector*.

[2]The *minimum detectable signal* is given by $S = N$ or $S/N = 1$. The ratio is sometimes expressed $(S + N)/N$ and for the minimum detectable signal case this ratio equals 2.

*8 millijansky, where 1 Jy $= 10^{-26}$ W m^{-2} Hz^{-1}.

antenna of effective aperture A_{er} at a distance r can collect a power

$$P_r = \frac{P_t A_{er}}{4\pi r^2} \frac{\Delta f_r}{\Delta f_t} \quad \text{(W)} \tag{1}$$

where

P_t = radiated transmitter power, W
A_{er} = effective aperture of receiving antenna, m^2
Δf_r = receiver bandwidth, Hz
Δf_t = transmitter bandwidth, Hz
r = distance between transmitter and receiver, m

It is assumed that $\Delta f_r \leq \Delta f_t$.

With a transmitting antenna of directivity $D = 4\pi A_{et}/\lambda^2$, the received power becomes

$$P_r = \frac{P_t A_{er} A_{et}}{r^2 \lambda^2} \frac{\Delta f_r}{\Delta f_t} \quad \text{(W)} \tag{2}$$

where

λ = wavelength, m
A_{et} = transmitter effective aperture, m^2

For $\Delta f_r = \Delta f_t$ (bandwidths matched), (2) is the Friis transmission formula.

The sensitivity of a receiving system (antenna and receiver) depends not only on the antenna temperature T_A but also on the temperature or noise contribution of the receiver and the transmission line connecting the antenna to the receiver. The resultant of these temperatures is called the *system temperature* T_{sys}, which is a factor in the *signal-to-noise ratio* (S/N or *SNR*) for a *radio link* as given by

$$\frac{S}{N} = \frac{P_t A_{et} A_{er}}{r^2 \lambda^2 B k T_{\text{sys}}} \quad \text{(dimensionless)} \quad \textbf{\textit{Signal-to-noise ratio SNR of radio link}} \tag{3}$$

（无线电线路的信噪比 SNR）

where

P_t = transmitter power, W
A_{et} = effective aperture of transmitting antenna, m^2
A_{er} = effective aperture of receiving antenna, m^2
r = distance between transmitter and receiver, m
λ = wavelength, m
B = bandwidth, Hz
k = Boltzmann's constant = 1.38×10^{-23} J K^{-1}
T_{sys} = system temperature, K

In (3) matched polarizations and bandwidths are assumed.

EXAMPLE **12–3.1 Downlink Signal-to-Noise Ratio**

The criterion of detectability of a signal is the *signal-to-noise* ratio (S/N) (3). For a transmitter power of 1 W and an isotropic (nondirectional) antenna, the signal-to-noise ratio of a lossless line-of-sight radio link is given by

$$\frac{S}{N} = \frac{\lambda^2}{16\pi^2 r^2 k T_{\text{sys}} B}$$

where

λ = wavelength, m
r = distance from transmitter to receiver, m
k = Boltzmann's constant = 1.38×10^{-23} J K^{-1}
T_{sys} = system temperature, K
B = bandwidth, Hz

For a Clarke-orbit geostationary satellite C-band transponder downlink to an earth station (Fig. 12–6), the transponder power = 5 W, distance = 36,000 km, λ = 7.5 cm, and antenna gain = 30 dB. If the earth station antenna has 38-dB gain and the earth station receiver a system temperature of 100 K, find the earth station S/N. The bandwidth B = 30 MHz for frequency-modulated (FM) TV signals.

Note: The S/N ratio as used in this example actually represents the carrier-to-noise (C/N) ratio, defined as the ratio of the power in the unmodulated carrier to the noise power. For FM video signals as employed by the Clarke-orbit satellites, the S/N at the output of the FM demodulator can exceed the carrier-to-noise ratio (C/N) by 35 dB or more, provided that the C/N is above the "FM threshold" of approximately 10 dB. The amount of enhancement depends on the signal's *modulation index,* the ratio of the peak frequency deviation to the modulation frequency. In practice, satellite designers strive for a C/N of 13 to 20 dB, corresponding to 3 to 10 dB of *link margin* to allow for antenna misalignment, attenuation along the propagation path, attenuation due to water or snow in the earth-station dish, transponder power variations, and demodulator inefficiency.

■ **Solution**

For 1 W and isotropic antenna,

$$\frac{S}{N} = \frac{\lambda^2}{16\pi^2 r^2 k T_{\text{sys}} B}$$

$$= \frac{0.075^2}{16\pi^2 (36^2 \times 10^{12})(1.38 \times 10^{-23})(100)(30 \times 10^6)}$$

$$= 6.64 \times 10^{-7} = -61.8 \text{ dB}$$

For transponder antenna gain = 30 dB and transponder power = 5 W = 7 dB,

ERP (effective radiated power) of transponder = 30 + 7

= 37 dB (over 1 W isotropic)

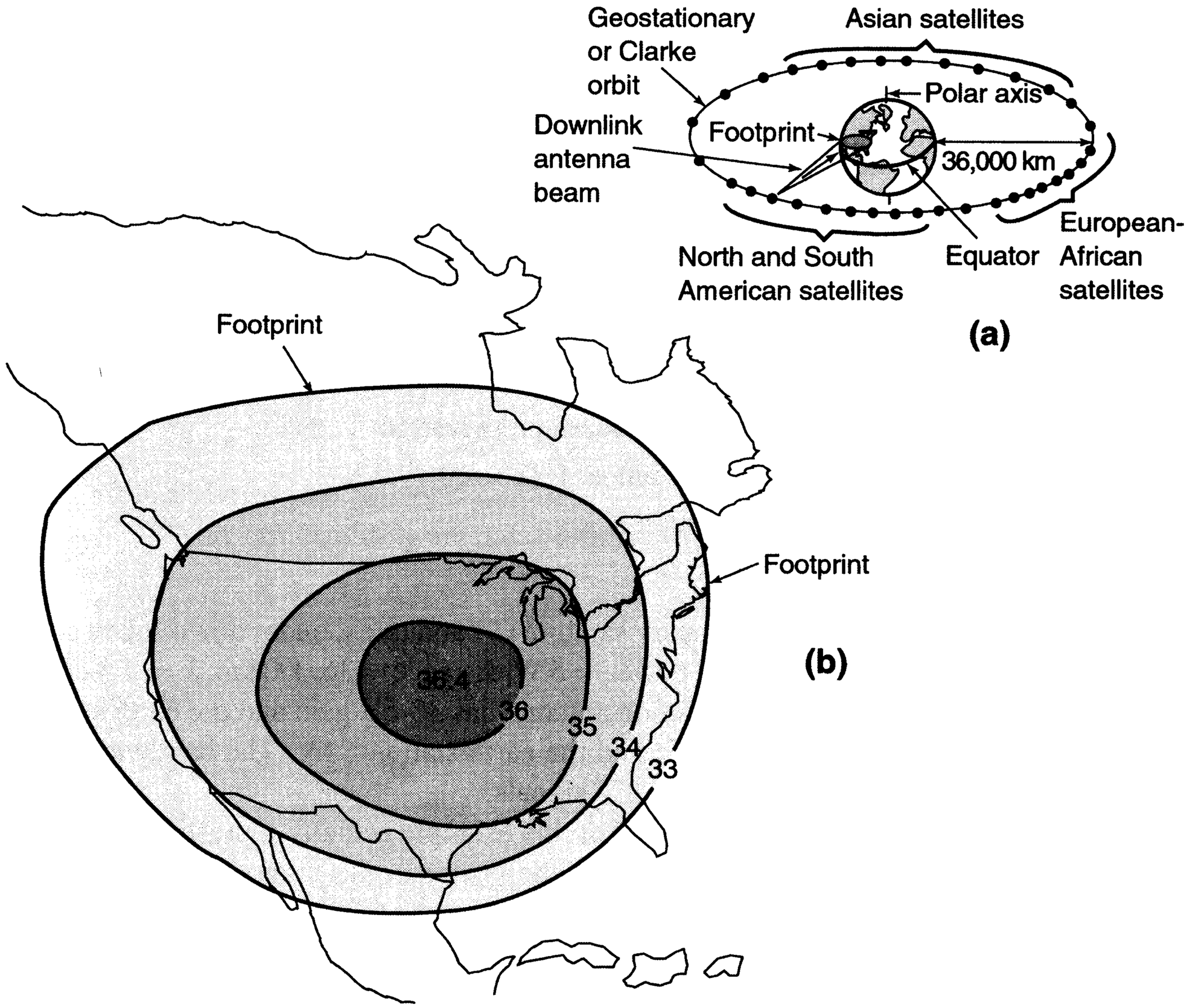


Figure 12–6
(*a*) Geostationary TV and communication relay satellites in Clarke orbit around the earth. (*b*) Effective radiated power contours, or footprint, in decibels above 1-W isotropic of typical transponder downlink over North America. Note that the footprint gives radiation contours over a spherical surface (the earth's) and, in most cases, at an oblique angle. Only when the satellite beams directly down on the earth's equator with a relatively narrow pattern will the footprint contours approximate the true antenna pattern contours.

For earth station gain = 38 dB,

$$S/N \text{ (downlink)} = 37 + 38 - 61.8 = 13.2 \text{ dB} \qquad \textit{Ans.}$$

This is 3.2 dB more than the minimum S/N ratio considered to be acceptable.

12–4 PASSIVE REMOTE SENSING（被动遥感）

A radio telescope is a remote sensing device whether it is earth-based and pointed at the sky for observing celestial objects or on an aircraft or satellite and pointed at the earth. In this section we consider the case where the radiation detected or sensed by the

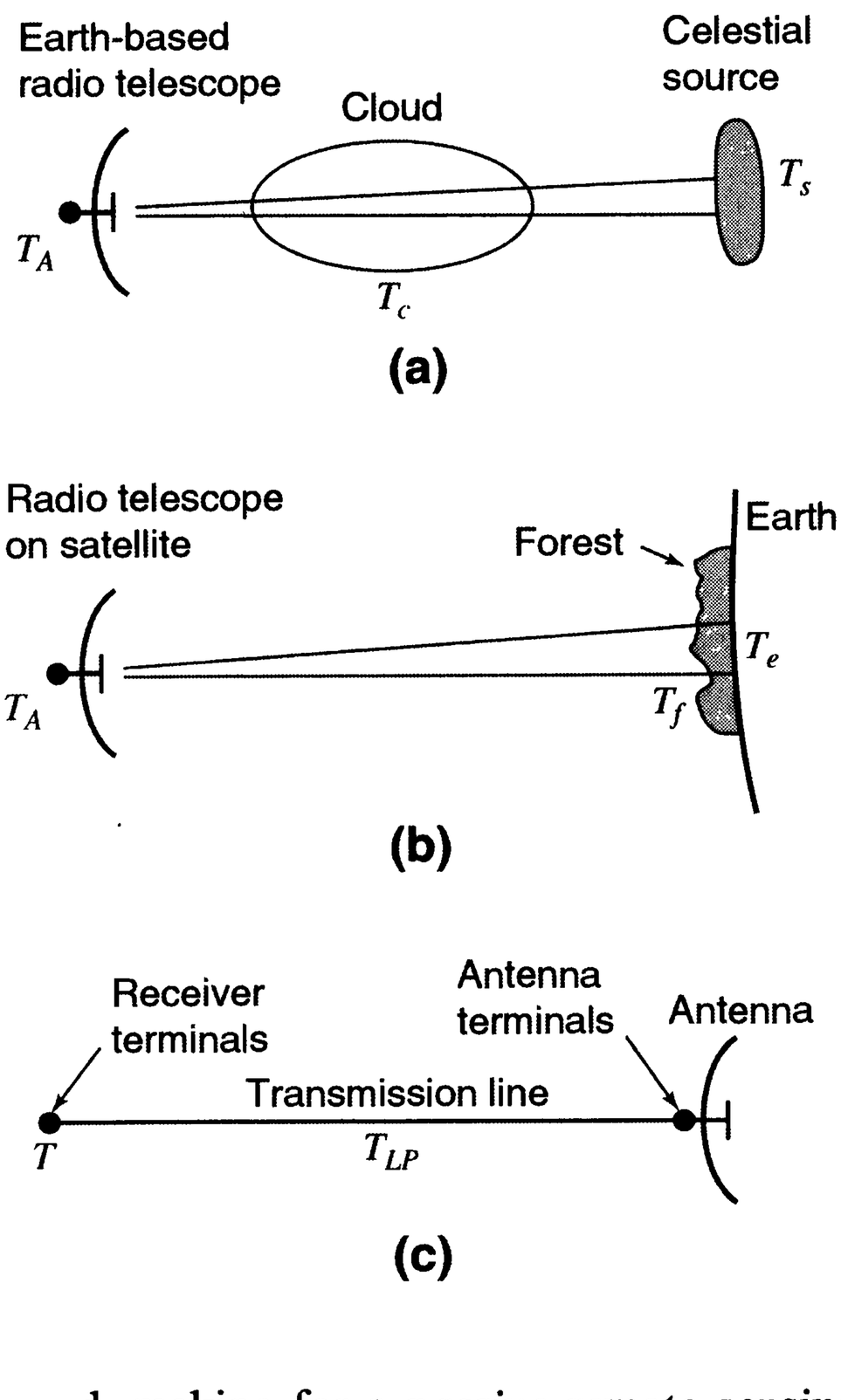


Figure 12–7
(*a*) Earth-based radio telescope remote-sensing celestial source through intervening interstellar cloud. (*b*) Radio telescope on satellite remote-sensing the earth through forest. (*c*) Receiver detecting antenna output through transmission line. The cloud, forest and transmission line have analogous emitting-absorbing properties.

telescope originates in the objects being observed, making for a *passive remote sensing* system in distinction to *radar or active remote sensing* where signals are transmitted and their reflections observed and analyzed. The active case is discussed in the next section.

Consider the situation of Fig. 12–7*a*, in which the earth-based radio telescope antenna beam is completely subtended by a celestial source of temperature T_s with an intervening absorbing-emitting cloud of temperature T_c. *With no cloud present,* the incremental antenna temperature $\Delta T_A = T_s$, but *with the cloud* it may be shown that the observed antenna temperature

$$\Delta T_A = T_c(1 - e^{-\tau_c}) + T_s\, e^{-\tau_c} \qquad \text{(K)} \tag{1}$$

where τ_c = absorption coefficient of the cloud[1] (= 0 for no absorption and = ∞ for infinite absorption). Thus, knowing T_s and τ_c, the cloud's equivalent blackbody temperature T_c can be determined.

Now, referring to Fig. 12–7*b*, let us reverse the situation and put the radio telescope on an orbiting satellite for observing the surface of the earth at temperature T_e with the antenna beam completely subtended by a large forest at a temperature T_f. The incremental satellite antenna temperature is then

$$\Delta T_A = T_f(1 - e^{-\tau_f}) + T_e\, e^{-\tau_f} \qquad \text{(K)} \tag{2}$$

[1] Astronomers call τ_c the "optical depth." The quantity $e^{-\tau_c}$ is equivalent to the efficiency factor ε in (12–2–1).

where τ_f = absorption coefficient of the forest. Knowing T_e and τ_f, the temperature of the forest can be determined, or knowing T_e and T_f, the absorption coefficient can be deduced. It is by such a technique that the whole earth can be surveyed and much information obtained about the temperatures of land and water areas, and from absorption coefficients about the nature of the surface cover.

EXAMPLE **12–4.1 Forest Temperature by Remote Sensing**

The remote-sensing antenna of a 3-GHz orbiting satellite measures a temperature $\Delta T_A = 300$ K when directed at a tropical forest region having an absorption coefficient $\tau_f = 0.693$ at vertical incidence. If the earth temperature $T_e = 305$ K, find the temperature of the forest.

■ **Solution**

Since $\tau_f = 0.693$, $e^{-\tau_f} = 0.5$, so from (2),

$$T_f = \frac{\Delta T_A - T_e e^{-\tau_f}}{1 - e^{-\tau_f}} = \frac{300 - 305 \times 0.5}{1 - 0.5} = 295 \text{ K}$$

If the antenna-transmission line-receiver system is viewed from the receiver terminals as in Fig. 12–7*c* (instead of from the antenna terminals as in Fig. 12–5), we note that the analogy between the remote-sensing situations discussed above extends here to the transmission line. Thus, the emitting-absorbing transmission line of Fig. 12–7*c* is like the emitting-absorbing cloud of Fig. 12–7*a* and like the emitting-absorbing forest of Fig. 12–7*b*. The analogy may be emphasized by comparing (1) and (2) with the temperature as seen from the receiver terminals, so that the equations for the 3 situations have identical form as follows:

Antenna looking at celestial source (see Fig. 12–7*a*):

$$\Delta T_A = T_c(1 - e^{-\tau_c}) + T_s e^{-\tau_c} \qquad \text{(K)} \tag{3}$$

Antenna looking at earth from satellite (see Fig. 12–7*b*):

$$\Delta T_A = T_f(1 - e^{-\tau_f}) + T_e e^{-\tau_f} \qquad \text{(K)} \tag{4}$$

Receiver looking at antenna (see Fig. 12–7*c*):

$$T = T_{LP}(1 - e^{-\alpha l}) + T_A e^{-\alpha l} \qquad \text{(K)} \tag{5}$$

where

ΔT_A = incremental antenna temperature, K
T_c = cloud temperature, K
τ_c = cloud absorption coefficient (optical depth), dimensionless
T_s = celestial source temperature, K

T_f = forest temperature, K
τ_f = forest absorption coefficient, dimensionless
T_e = temperature of earth, K
T_{LP} = transmission line physical temperature, K
α = transmission line attenuation constant, Np m^{-1}
l = length of transmission line, m

Note that the *system temperature* should be referred to the antenna terminals as in (12–2–1) and *not* to the receiver terminals as in (5). Thus, if the line is completely lossy ($e^{-\alpha l} = \varepsilon_2 = 0$), (12–2–1) gives an infinite system temperature which is correct, meaning that the system has no sensitivity whatever. However, with this condition, a "system temperature" viewed from the receiver terminals, as in (5), would equal the temperature T_{LP} of the line plus the receiver temperature, a completely misleading result since it indicates that the system still has sensitivity.

The Mars measurement is an example of *passive remote sensing* in which the temperature of a very distant object is determined with a radio telescope. As another example of passive remote sensing, many satellites circling the earth carry radio telescopes which are directed down at the earth for measuring the temperature of the earth's surface in great detail in the same way that the telescope in the worked example was used to determine the surface temperature of Mars. By contrast, radar detection is *active remote sensing* since a signal is transmitted and an echo received.

12–5 RADAR AND RADAR CROSS SECTION（雷达和雷达截面）

Consider now the situation shown in Fig. 12–8 where either a transmitter or a receiver can be connected to an antenna. Now, with transmitter connected to the antenna, a pulse is sent out that strikes a passive reflecting-scattering object as in Fig. 12–8. The power intercepted by the object is given by the Friis transmission formula (12–3–2) as

$$P_{\text{int}} \text{ (by object)} = \frac{P_t A_{et}}{r^2 \lambda^2} \sigma \qquad \text{(W)} \tag{1}$$

where σ = *radar cross section* of the object, m^2.

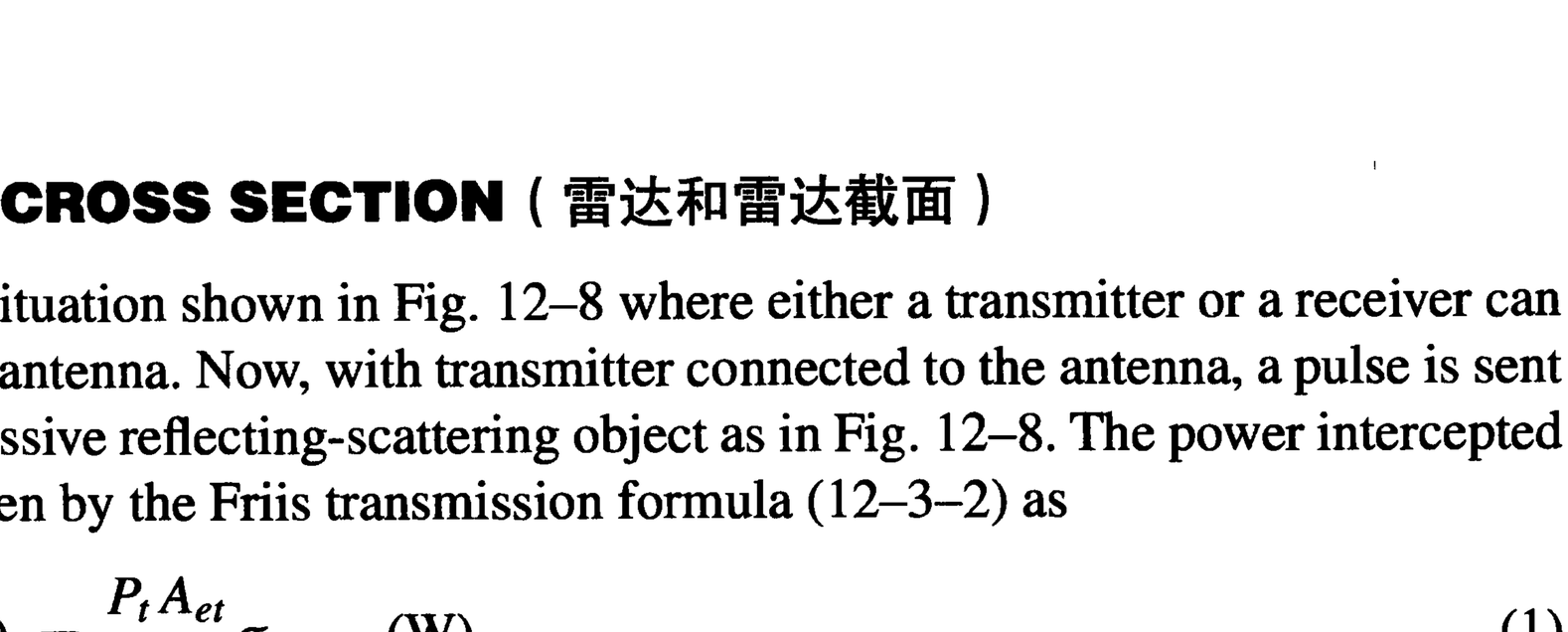


Figure 12–8
Double-path geometry (wave out and scattered wave back) used in obtaining radar equation. The antenna can be switched between transmitter and receiver.

The term *radar* is an acronym for *ra*dio *d*irection *a*nd *r*ange. Assuming that the object scatters isotropically ($D = 1$), its effective aperture from (2–9–7) is $\lambda^2/4\pi$, so the scattered power from the object which is *received back at* the transmitter location is, by another application of the Friis transmission formula (2–11–5), given by

$$P_r \text{ (Power received by antenna)} = \frac{P_{\text{int}} \text{ (Power intercepted by object)} A_{et}}{r^2\lambda^2} \frac{\lambda^2}{4\pi} \quad \text{(W)} \tag{2}$$

Now, connecting the antenna to the receiver (before the backscattered pulse or echo arrives), the ratio of the backscattered power collected by the antenna to the transmitted power yields the *radar equation* as given by

$$\boxed{\frac{P_r \text{ (by antenna)}}{P_t} = \frac{A^2\sigma}{4\pi r^4\lambda^2} \quad \text{(dimensionless)} \quad \textbf{\textit{Radar equation}}} \quad \text{(雷达方程)} \tag{3}$$

where $A = A_{et}$ = effective aperture of antenna (same for transmitting and receiving), m^2.

It is assumed in (3) that polarizations are matched (no cross-polarized component of the backscattered wave), i.e., $F = 1$ or $MM_a = 0$ and also that the antenna and receiver are matched. If polarizations are not matched, then (3) should be multiplied by F from (2–17–5).

From (3), the radar cross section (RCS) is

$$\sigma = \frac{P_r \text{ (by antenna)} 4\pi r^4\lambda^2}{P_t A^2} \tag{4}$$

or (雷达截面)

$$\boxed{\sigma = \frac{S_r}{S_{\text{inc}}/4\pi r^2} = \frac{4\pi r^2 S_r}{S_{\text{inc}}} = \frac{\text{scattered power}}{\text{incident power density}} \quad (\text{m}^2) \quad \textbf{\textit{Radar cross section}}} \tag{5}$$

where

S_r = isotropically backscattered power density at distance r
$= [P_r \text{ (by antenna)}]/A, \text{ W m}^{-2}$
S_{inc} = power density incident on object
$= (P_t A)/(r^2\lambda^2), \text{ W m}^{-2}$

In words, the *radar cross section* σ of an object is its effective area intercepting the incident power density S_{inc} which, if scattered isotropically, would result in the backscattered power density S_r.

For a large perfectly reflecting metal sphere of radius a, the radar cross section is equal to the physical cross section πa^2. For imperfectly reflecting spheres, the radar cross section is smaller. For example, at meter wavelengths, the radar cross section of the moon is about 0.1 of its physical cross section.

To measure accurate far-field values of the *Radar Cross Section* (RCS) of objects, the radar antenna should be at a distance well in excess of w^2/λ, where w = maximum width (or height) of the object being measured, in order that the radar wave front be sufficiently planar. At high frequencies and/or for big objects, this may require an impractically large

distance. A solution is to use a parabola to convert the spherical wave front from the radar to a plane wave. This greatly reduces the distance required and results in what is called a "compact range." The Ohio State University 110-m radio telescope doubled as a compact range for RCS measurements of large objects. See Figs. 20–6 and 20–8.

Notice in (3) that the backscattered power at the radar receiver is inversely proportional to the fourth power of the distance. This means that if you have a receiver as sensitive and an antenna as large as the radar's, you should be able to detect the radar at greater distances than it can detect you. The signal needs to cover the path to you only once ($1/r^2$ attenuation), but it must cover it twice to get back to the radar ($1/r^4$ attenuation).

In *pulse radar* the antenna is connected to the transmitter while the pulse is sent. It is then switched to the receiver which listens for the echoes. The greater the range being observed, the longer the time needed between pulses. In *doppler radar* the antenna is connected continuously to both transmitter and receiver through a circulator which isolates the receiver from the transmitter, and the transmitter is on all the time. The term CW (continuous wave) *doppler* is used to describe this mode of operation, which measures only velocity. By measuring both time delay and frequency shift of the echoes from a pulse radar, we have what is called *pulse doppler radar* which measures both distance and velocity.

Radars have wide application for ship and aircraft navigation; for harbor, airport and highway surveillance; for weather forecasting (storms, rain, hail, etc.); for terrain mapping; for measuring the distance to the moon or the rotation of Venus (whose surface is hidden by clouds); for monitoring the speed of a pitcher's fast ball; or determining the velocity of a hummingbird.

The radar cross sections of several objects are listed in Table 12–1 where it is assumed that the objects are large compared to the wavelength. Note that, whereas the RCS of a sphere is equal to its cross-sectional area, the RCS of a plate or sheet is larger than its area (see Prob. 12–5–7). See also the RCS data in Fig. 12–9.

In *pulse radar* the time Δt between the transmission of the pulse and reception of its echo gives the *distance d* of the object as

$$d = \frac{1}{2} c \Delta t \qquad \text{(m)} \tag{6}$$

where $c = 300$ Mm s^{-1} (for air) and Δt = delay time of echo, s.

Table 12–1 Radar cross sections†

Object	Radar cross section σ
Sphere, radius a	πa^2
Flat plate, area A	$4\pi A^2/\lambda^2$
Cylinder, radius a, length L	$2\pi aL^2/\lambda$

†Objects perfectly conducting and large compared to the wavelength (a and $L \gg \lambda$). Plate and cylinder at normal incidence. Cylinder length L parallel to plane of polarization of radar wave. For the general case where the object's dimensions may also be smaller than the wavelength, see R. J. Kouyoumjian (1).

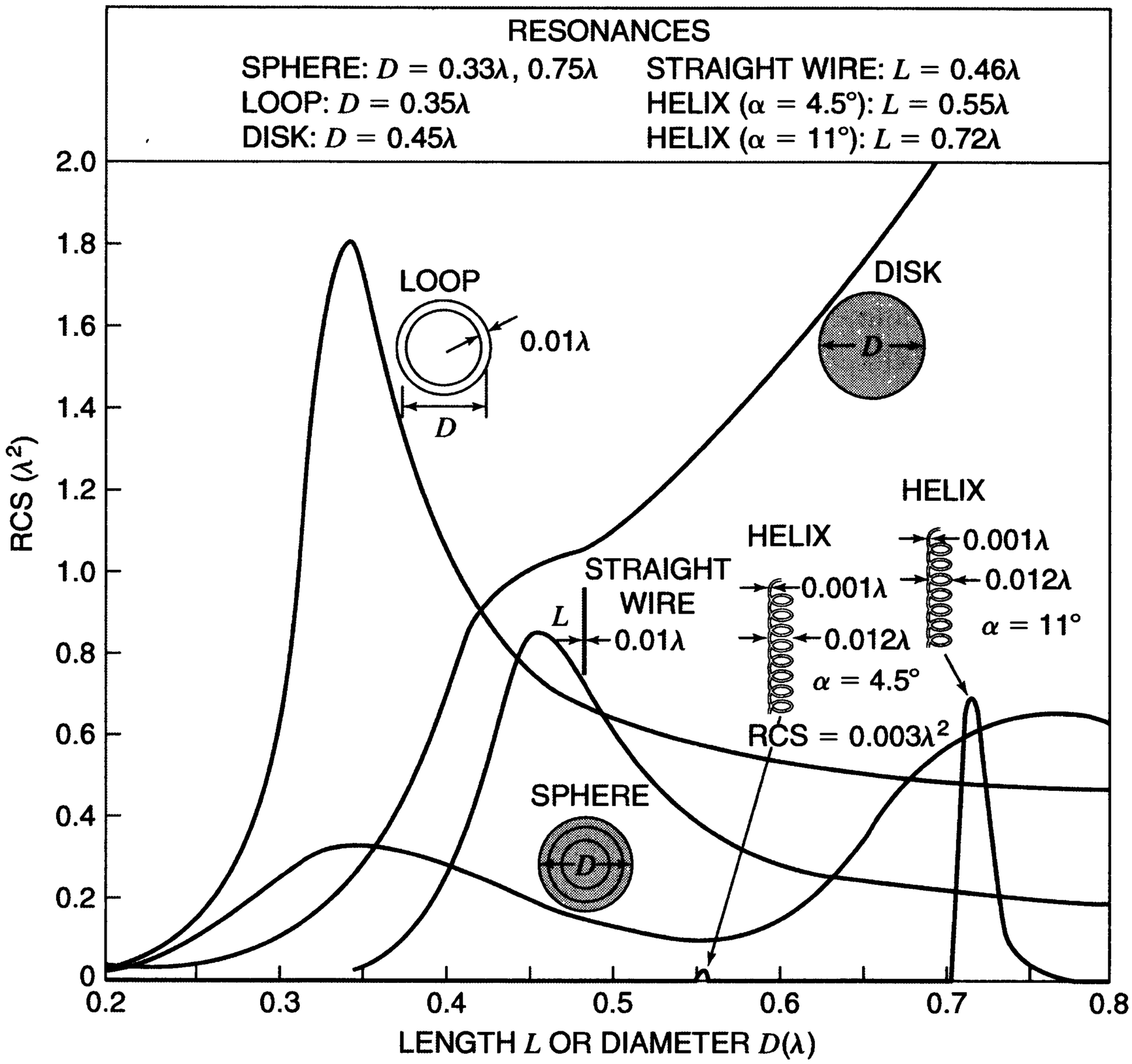


Figure 12–9
RCS (Radar Cross Section) values of wire, loop, disk, sphere and helices. The wire, loop, and disk values are from Kouyoumjian (1), the sphere from Mie (1) while the helix values are from measurements by Kraus (1).

In *doppler radar* the change in frequency Δf of the echo with respect to the transmitted frequency f_0 gives the *velocity* v of the object as

$$v = \frac{1}{2}\frac{\Delta f}{f_0}c \qquad (\text{m s}^{-1}) \tag{7}$$

where

Δf = (doppler) shift of frequency, Hz
f_0 = transmitting frequency, Hz
c = 300 Mm s^{-1} (for air)

For positive values of Δf (increase in echo frequency), the object is approaching; for negative values (decrease in echo frequency), it is receding (going away).

By moving or scanning the antenna beam, a radar can provide information on the *direction, distance,* and *velocity* of objects within its view. Also, since a returning pulse carries

a characteristic signature, different objects may be recognized, a very short pulse response being the Fourier transform of the object's frequency response.

Pulse Doppler Weather Radar

Coherent pulse doppler radars measure both the amplitude and the phase of radar echoes. The phase of the return is related to the distance r to the scattering object and the wavelength λ by

$$\phi = \frac{4\pi r}{\lambda} \tag{8}$$

This differs from the one-way phase shift $2\pi r/\lambda$ by a factor of 2, since radar scattering is a two-way process.

If the source of the radar echo is moving, the phase of the return changes with time as

$$\phi(t) = \frac{4\pi(r + v_r t)}{\lambda} \tag{9}$$

where v_r = velocity component along the radar line of sight.

The time derivative of this changing phase is an angular frequency $\Delta\omega$:

$$\Delta\omega = \frac{\partial\phi}{\partial t} = \frac{4\pi v_r}{\lambda} \tag{10}$$

The doppler frequency shift due to motion of the scatterer is, therefore,

$$\Delta f = \frac{\Delta\omega}{2\pi} = \frac{2v_r}{\lambda} \qquad \text{(Hz)} \tag{11}$$

In a pulse doppler radar, both the distance to the echo source and its velocity are measured by transmitting a series of pulses and measuring the phase of the return for each pulse. From the Nyquist requirement of 2 samples per cycle, the maximum doppler frequency shift which may be unambiguously measured is

$$\Delta f_{\max} = \frac{1}{2T} \qquad \text{(Hz)} \tag{12}$$

where T = pulse repetition interval (PRI), s.

If a total of N pulses is used to determine the velocity, the total measurement time is NT seconds. Two frequencies may be resolved in this time if one of them undergoes an additional cycle of phase change in the observation period. Thus, the frequency resolution of a pulse doppler radar is

$$\Delta f_{\min} = \frac{1}{NT} \tag{13}$$

where N = number of pulses observed and T = pulse repetition interval, s.

Pulse doppler weather radar uses scattering from water droplets and fluctuations in refractive index to measure the intensity of rain and wind velocity.

EXAMPLE **12–5.1 Weather Radar**

For an X-band (10-GHz) weather radar, find: (*a*) the minimum pulse repetition frequency (PRF = 1/PRI) which may be used to unambiguously measure the wind velocity in a tornado with a wind speed of 350 km h^{-1} (Fig. 12–10). (*b*) At this PRF, how many pulses must be sampled to resolve in frequency two portions of the tornado with a differential velocity of 1 km h^{-1}?

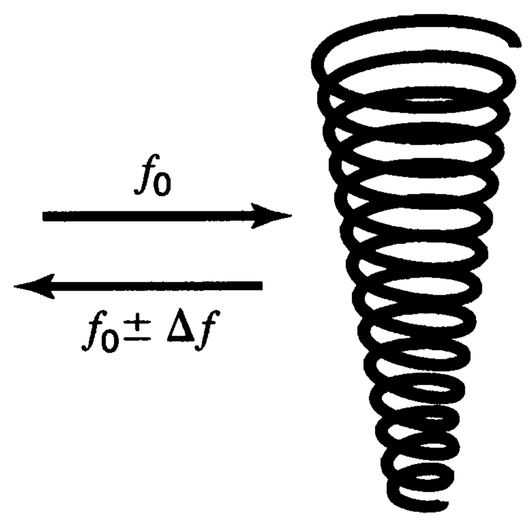


Figure 12–10

■ **Solution**

(*a*) Using (11), the doppler shift for a velocity of 350 km h^{-1} is

$$\Delta f = \frac{2v_r}{\lambda} = \frac{2\left(\dfrac{3.5 \times 10^5}{3600}\right)}{0.03} = 6.5 \text{ kHz}$$

From (12),

$$\Delta f_{\max} = \frac{1}{2T}$$

$$T = \frac{1}{2\Delta f_{\max}} = \frac{1}{2(6.5 \times 10^3)} = 7.7 \times 10^{-5} = 77\ \mu\text{s}$$

$$\text{PRF} = \frac{1}{T} = \frac{1}{7.7 \times 10^{-5}} = 13 \text{ kHz} \qquad \textit{Ans. (a)}$$

(*b*) Rearranging (13) to solve for N gives

$$N = \frac{1}{\Delta f_{\min} T}$$

For two scatterers with $\Delta v_r = 1$ km h^{-1}, we have from (11)

$$\Delta f_1 - \Delta f_2 = \frac{2v_{r1}}{\lambda} - \frac{2v_{r2}}{\lambda} = \frac{2(v_{r1} - v_{r2})}{\lambda} = \frac{2\left(\dfrac{1000}{3600}\right)}{0.03} = 18.5 \text{ Hz}$$

Therefore,

$$N = \frac{1}{\Delta f_{\min} T} = \frac{1}{(18.5)(7.7 \times 10^{-5})} = 702 \text{ pulses} \qquad \textit{Ans. (b)}$$

The Corner Reflector (Dihedral and Trihedral)

A square-corner dihedral (or two-surface) reflector, as in Fig. 12–11*a* with internal angle = 90°, acts as a retroreflector over a wide, almost 90° angle, as suggested in Fig. 12–11*b* and *c*.

To distinguish this type of corner reflector from the one in Fig. 10–8, this one may be called a *passive corner* and the one of Fig. 10–8 an *active corner.*

The measured radar field strength return from a square-corner reflector is shown in Fig. 12–12. The corner surfaces are large compared to the wavelength. There is strong reflection from the front side of the reflector over a broad angle (almost 90°). There are also four strong but narrow spikes of reflection from the flat sides at normal incidence. But there is only a tiny spike return from the sharp edge of the back side of the corner. Thus, the front sides of corners and flat surfaces at normal incidence are good to *enhance* detection, but they should be avoided to *escape* detection.

Adding a third side to the *dihedral* corner results in a *trihedral* corner. Clustering eight of them together as in Fig. 12–13 a retroreflector is obtained that provides a strong reflection over almost 4π sr. Such reflectors are widely used to enhance radar return. See Fig. 10–17.

For example, small watercraft commonly carry one (usually with reflecting surfaces of wire mesh) on a tall mast to make the craft's presence more visible on radar screens and reduce chances of being rammed in a fog. See the yacht in Fig. 12–15. To be most effective, the reflector dimensions should be many wavelengths and the periphery of the mesh hole less than $\lambda/2$. The surfaces should be flat to better than $\lambda/12$. And, to increase the probability that the radar echo will be noticed, the reflector can be rotated to avoid a persistent low return in the directions of the three planes of the reflector. Such a retroreflector may not give an echo like the Queen Elizabeth II but it can make a little watercraft appear like a sizable ship. Aircraft also need large RCS values to be easily detected and tracked by airport radars.

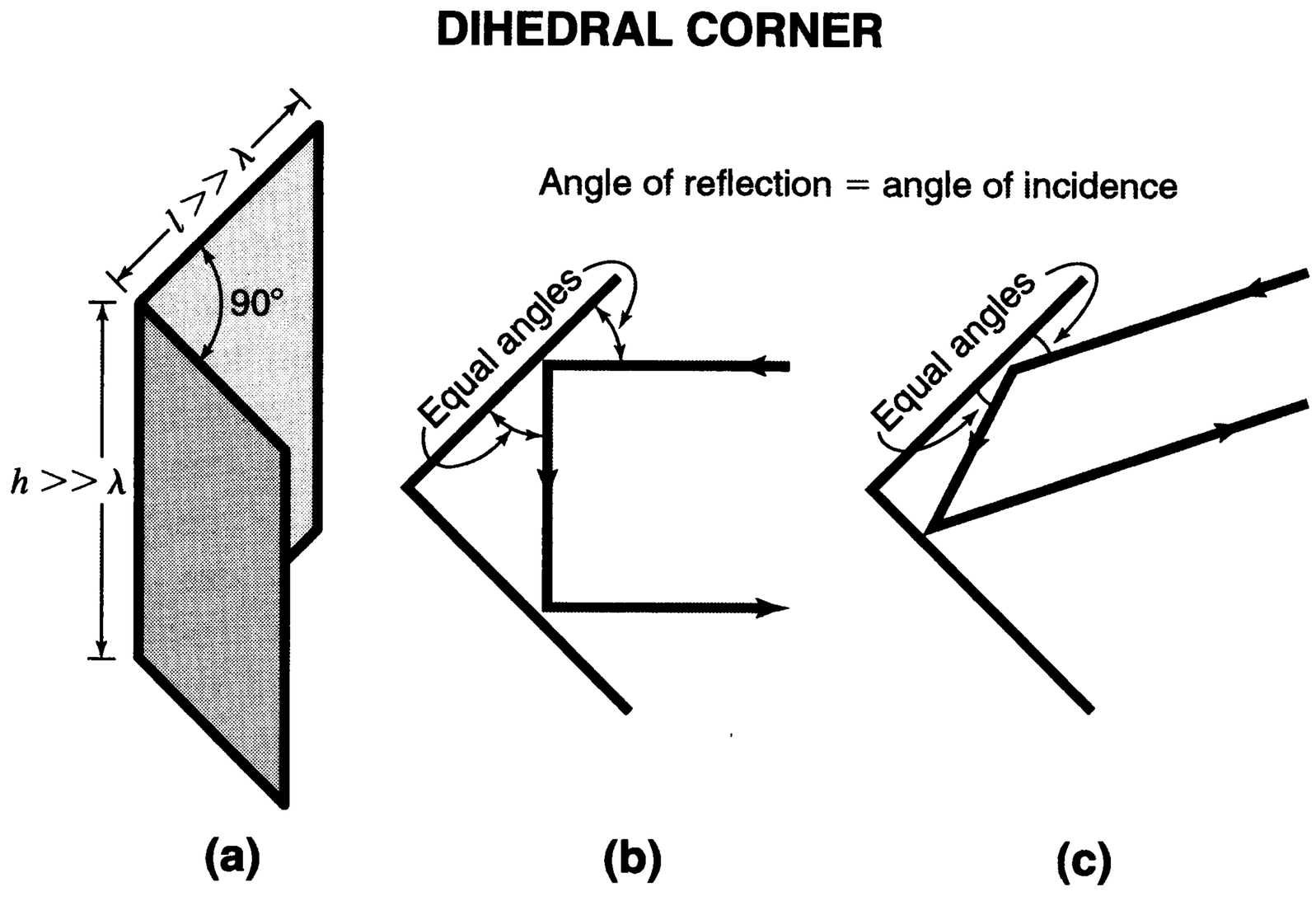


Figure 12–11
Square-corner reflector at (*a*) with ray diagrams at (*b*) and (*c*) showing retroreflective action at different angles of incidence.

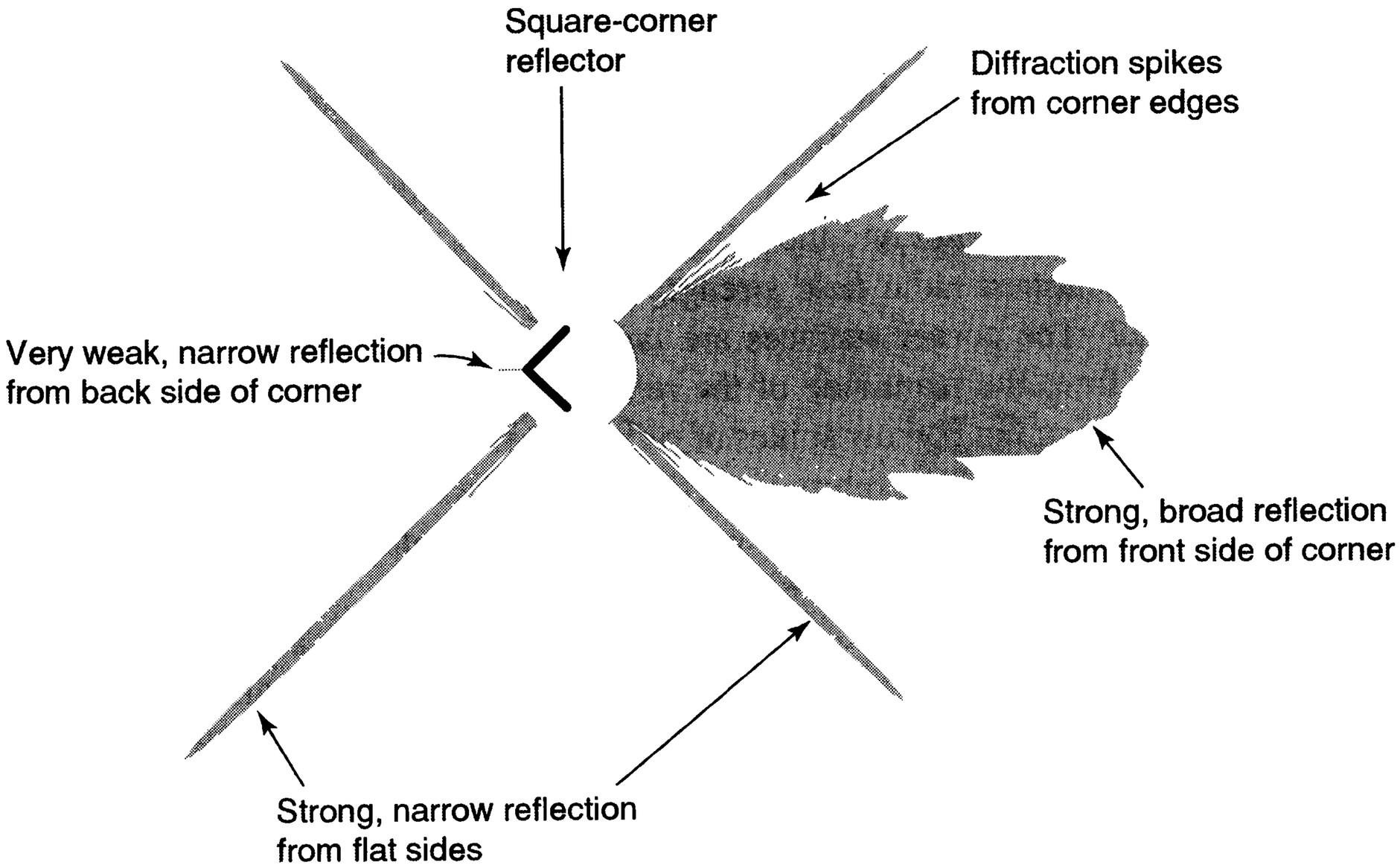


Figure 12–12
Measured radar field strength return from corner reflector. The corner dimensions are much larger than the wavelength.

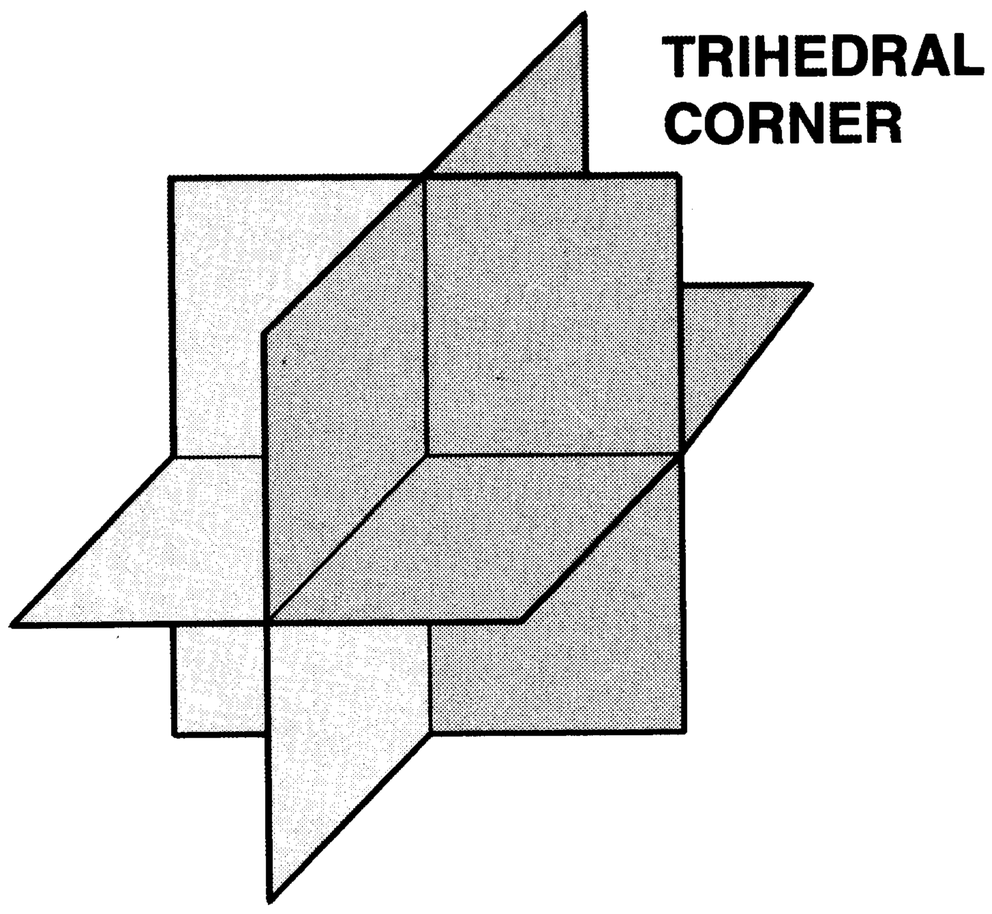


Figure 12–13
Cluster of eight trihedral corner reflectors providing retroreflective action over almost 4π sr.

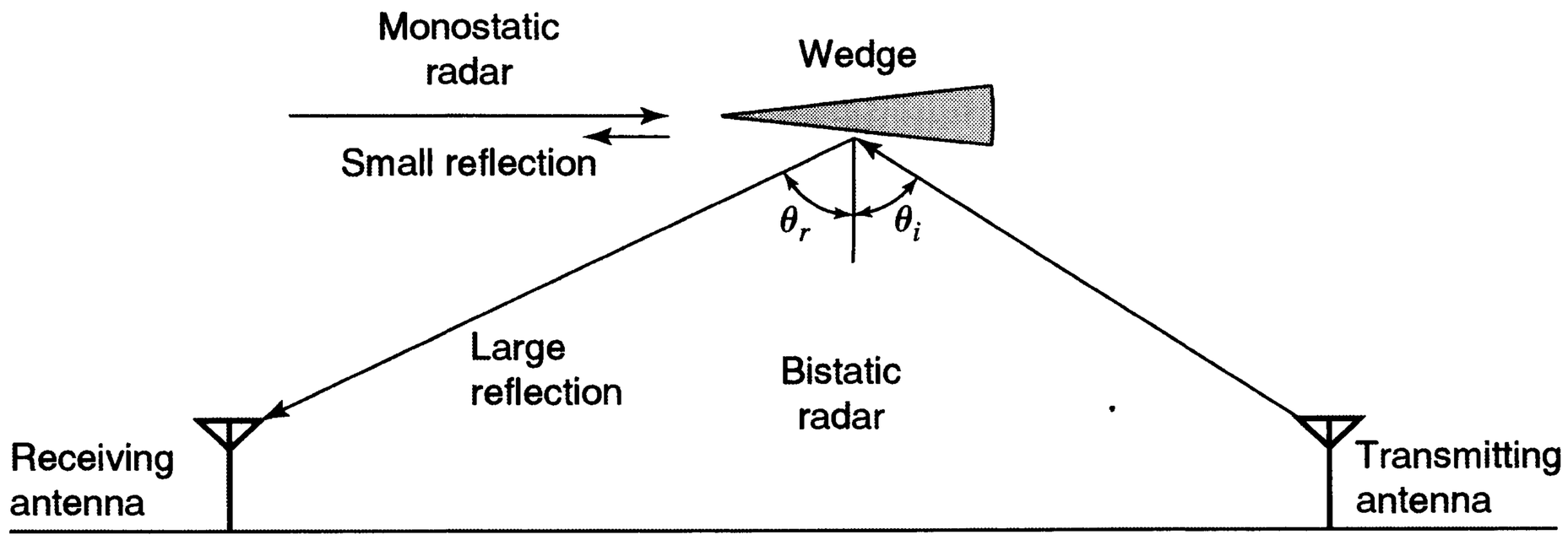


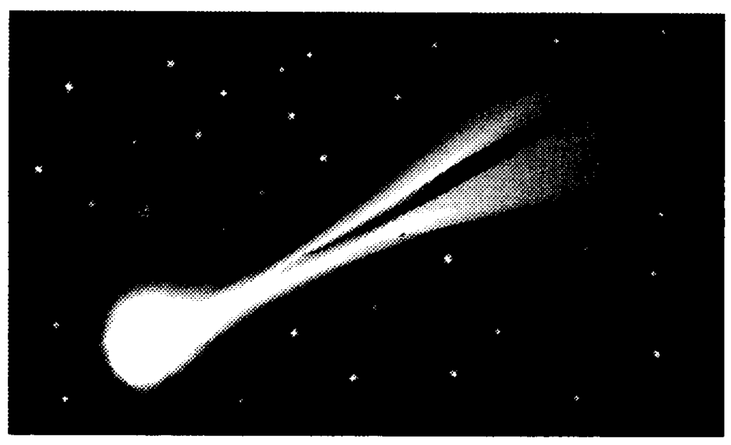
Figure 12–14
Monostatic and bistatic reflections from wedge. The monostatic reflection from the edge is small but the bistatic reflection from the flat side is large with the angle of reflection equal to the angle of incidence ($\theta_r = \theta_i$).

Bistatic Radar

The *monostatic* radar we have been discussing uses the same antenna for both transmitting and receiving (or if not the same antenna, both are at the same location). However, in *bistatic* radar the transmitting and receiving antennas are at different locations as suggested in Fig. 12–14. The two may, or may not, be coordinated. Thus, the transmitting antenna may be any of 100s of radio antennas transmitting in the area (short-wave to microwave). For example, the signal from a weak distant FM station may momentarily increase with the passage of an object like the wedge in Fig. 12–14. If the station's location is known, this gives the object's direction with its distance likely to be about halfway to the station. Although this informal bistatic system is less precise than the monostatic radar, it may be useful.

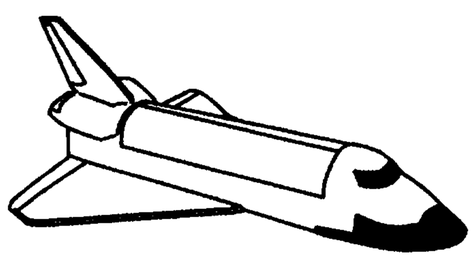
The space shuttle or other objects in low-earth orbit can often be detected bistatically from the magnetohydrodynamic (MHD) ripples they produce in the ionosphere. If the undisturbed electron density is just below the critical value, the electron density in the crests of the ripples may exceed the critical value permitting bistatic reflection between transmitting and receiving antennas separated by 100s or 1000s of km.

Typical RCS values that might be expected for a variety of objects are displayed in Fig. 12–15 for a radar wavelength of 3 cm (frequency = 10 GHz). The values are nominal maxima. An exception is the yacht which has both a maximum and a minimum value depending on the yacht's orientation. Whereas the yacht's RCS may vary, the RCS of the trihedral corner is essentially constant over 360°. Another exception is the space shuttle where the large value is from the ionospheric disturbance caused by the shuttle. More accurate RCS values require a consideration of the polarization, angles of incidence and observation, and the geometry and composition of the object. For more detailed information see Kouyoumjian (1) and Skolnik (1).

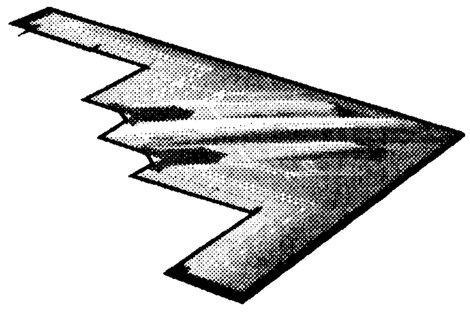


Comet Hyakataki 10^{18}

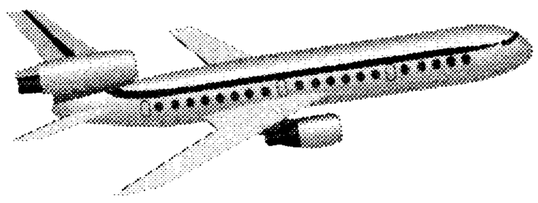
- Mars 10^{30}



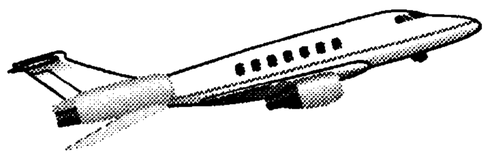
Space shuttle 10 to 10^{7}



Stealth aircraft $< 10^{-4}$



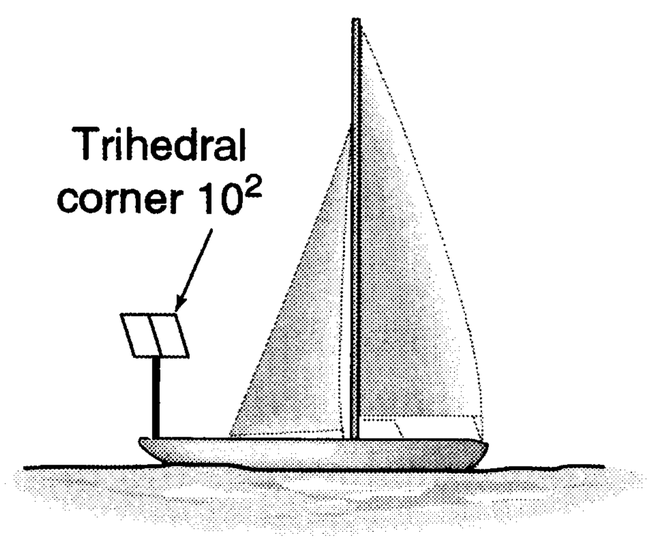
Jet liner 10^{3}



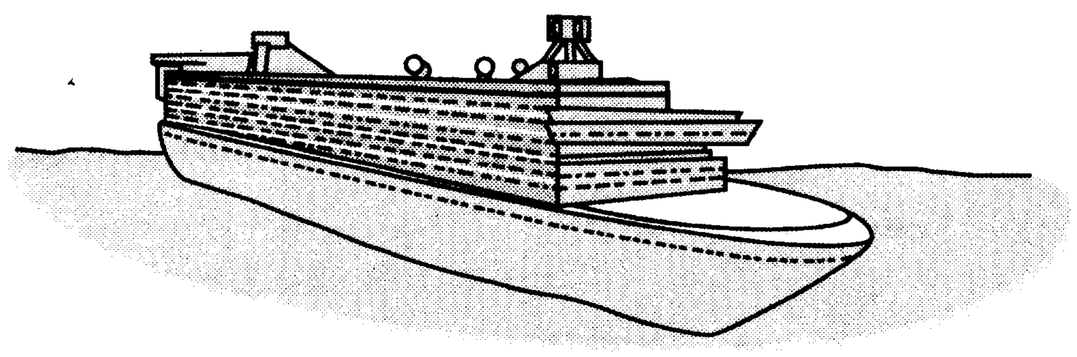
Executive jet 10^{2}

Barn swallow 10^{-2}

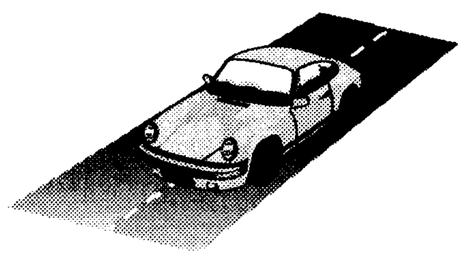
Mosquito 10^{-4}



Yacht 1 to 10

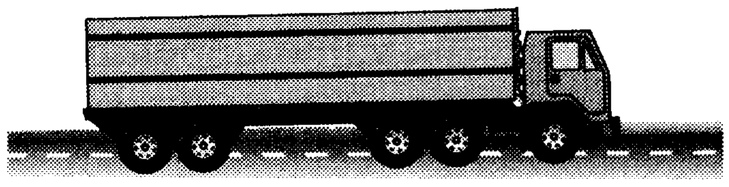


Cruise ship 10^{6}



Automobile 10^{2}

Adult human 1



Big truck 10^{4}

Figure 12–15
Land, sea, air and space objects with RCS values in square meters at 10 GHz.

REFERENCES（参考文献）

Gordon, W. E. (1), "Radar Backscatter from Earth's Ionosphere," *IEEE Trans. Ants. Prop.*, **AP-12,** 873–876, December 1964.

Kouyoumjian, R. J. (1), in *Antennas,* Kraus, 2d ed., pp. 791–797.

Kraus, J. D. (1), "Antennas Our Electronic Eyes and Ears," *Microwave Jour.,* Jan. 1989, pp. 77–92.

Mayer, C. H. (1), T. P. McCullough and R. M. Sloanaker, "Observations of Mars and Jupiter at a Wavelength of 3.15 cm," *Astrophys. J.,* **127,** 11–16, January 1958.

Mie, G. (1), *Ann. Phys.,* **25,** 377–446, 1908.

Nyquist, H. (1), Thermal Agitation of Electric Charge in Conductors, *Phys. Rev.,* **32:**110–113 (1928).

Penzias, A. A. (1), and R. W. Wilson, "A Measurement of Excess Antenna Temperature at 4080 MHz," *Astrophys. J.,* **142,** 419–421, 1965.

Shannon, C. E. (1), "Communication in the Presence of Noise," *Proceedings of the IRE,* **37** (January 1949).

Skolnik, M. I. (1), *Radar Systems,* McGraw-Hill, 1980.

PROBLEMS（习题）

***12–2–1**（天线温度）

Antenna temperature. An end-fire array is directed at the zenith. The array is located over flat nonreflecting ground. If $0.9\Omega_A$ is within 45° of the zenith and $0.08\Omega_A$ between 45° and the horizon calculate the antenna temperature. The sky brightness temperature is 5 K between the zenith and 45° from the zenith, 50 K between 45° from the zenith and the horizon and 300 K for the ground (below the horizon). The antenna is 99 percent efficient and is at a physical temperature of 300 K.

***12–2–2**（地球站的天线温度）

Earth-station antenna temperature. An earth-station dish of 100 m^2 effective aperture is directed at the zenith. Calculate the antenna temperature assuming that the sky temperature is uniform and equal to 6 K. Take the ground temperature equal to 300 K and assume that 1/3 of the minor-lobe beam area is in the back direction. The wavelength is 75 mm and the beam efficiency is 0.8.

12–3–1（信噪比）

Signal-to-noise ratio. Show that the S/N ratio for a radio link with 1 W transmitter and isotropic antennas is

$$\frac{S}{N} = \frac{\lambda^2}{16\pi^2 r^2 k T_{\text{sys}} \Delta f}$$

where symbols are as given in Eqs. (12–3–1) and (12–3–3).

***12–3–2**（信噪比）

***S/N* ratio.** What is the S/N ratio of a communications link operating with 50 MHz bandwidth (for 5 TV channels at 10 MHz each) over a distance of 1500 km if the parabolic dish antennas are 1 m diameter operating at 3 GHz? The transmitter power is 10 W and the receiver system temperature is 200 K.

***12–3–3**（来自太阳系背后"先驱者 10"的信号）

Signals from Pioneer 10 beyond the solar system. In March 1992 Pioneer 10 was 8 billion kilometers (8×10^{12} m) distant, well outside the solar system into deep space. Launched by NASA 20 years earlier, on March 1972, it was the first man-made object to escape from the solar system. Signals from its 8 W transmitter were still being received, informing us, among other things, that the solar wind was still observable. Pioneer 10 gave us our first close-up views of Jupiter and has become our first interstellar probe. But even at its velocity of 40,000 km/h it will be nearly a million years before it reaches the nearest stars. Although its radio may presently fail, the craft carries a gold-anodized aluminum plaque with an engraved symbolic message from us. If Pioneer 10 antenna gain is 36 dBi and the NASA earth-station antenna gain is 66 dBi, find the following as of the year 2000 assuming that Pioneer 10 travels at a constant speed away from the earth and that the transmitter

is still operating: (*a*) the signal time delay (Pioneer 10 to earth), (*b*) the received power, and (*c*) the maximum bandwidth for a signal-to-noise ratio of 7 (or 8.5 dB) at a wavelength of 10 cm and system temperature equal to 20 K.

***12–3–4** (卫星电视下传) **Satellite TV downlink.** A transmitter (transponder) on a Clarke orbit satellite produces an effective radiated power (ERP) at an earth station of 35 dB over 1 W isotropic. (*a*) Determine the S/N ratio (dB) if the earth station antenna diameter is 3m, the antenna temperature 25 K, the receiver temperature 75 K and the bandwidth 30 MHz. Take the satellite distance as 36,000 km. Assume the antenna is a parabolic reflector (dish-type) of 50 percent efficiency. (See Example 12–3.1.) (*b*) If a 10-dB S/N ratio is acceptable, what is the required diameter of the earth station antenna?

***12–3–5** (系统温度) **System temperature.** The digital output of a 1.4 GHz radio telescope gives the following values (arbitrary units) as a function of the sidereal time while scanning a uniform brightness region. The integration time is 14 s, with 1 s idle time for printout. The output units are proportional to power.

Time	Output	Time	Output
$31^m\ 30^s$	234	$32^m\ 45^s$	229
31 45	235	33 00	236
32 00	224	33 15	233
32 15	226	33 30	230
32 30	239	33 45	226

If the temperature calibration gives 170 units for 2.9 K applied, find (*a*) the rms noise at the receiver, (*b*) the minimum detectable temperature, (*c*) the system temperature and (*d*) the minimum detectable flux density. The calibration signal is introduced at the receiver. The transmission line from the antenna to the receiver has 0.5 dB attenuation. The antenna effective aperture is 500 m^2. The receiver bandwidth is 7 MHz. The receiver constant $k' = 2$.

12–3–6 (系统温度) **System temperature.** Find the system temperature of a receiving system with 15 K antenna temperature, 0.95 transmission-line efficiency, 300 K transmission-line temperature, 75 K receiver first-stage temperature, 100 K receiver second-stage temperature and 200 K receiver third-stage temperature. Each receiver stage has 16 dB gain.

***12–3–7** (太阳对地球站的干扰) **Solar interference to earth station.** Twice a year the sun passes through the apparent declination of the geostationary Clarke-orbit satellites, causing solar-noise interference to earth stations. A typical forecast notice appearing on U.S. satellite TV screens reads:

ATTENTION CHANNEL USERS:
WE WILL BE EXPERIENCING
SOLAR OUTAGES FROM
OCTOBER 15 TO 26
FROM 12:00 TO 15:00 HOURS

(*a*) If the equivalent temperature of the sun at 4 GHz is 50,000 K, find the sun's signal-to-noise ratio (in decibels) for an earth station with a 3-m parabolic dish antenna at 4 GHz. Take the sun's diameter as 0.5° and the earth-station system temperature as 100 K. (*b*) Compare this result with that for the carrier-to-noise ratio calculated in Example 12–3.1 for a typical Clarke-orbit TV transponder. (*c*) How long does the interference last? Note that the relation $\Omega_A = \lambda^2/A_e$ gives the solid beam angle in

steradians and not in square degrees. (*d*) Why do the outages occur between October 15 and 26 and not at the autumnal equinox around September 20 when the sun is crossing the equator? (*e*) How can satellite services work around a solar outage?

***12–3–8**
(接近海王星的"旅行者2号"(Voyager 2))

Voyager 2 at Neptune. On August 24, 1989, Voyager 2 made a close encounter with Neptune sending back close-up pictures of the planet. With its 2.5-m-diameter parabolic dish antenna and 10 W, 10 GHz transmitter, what maximum earth station system temperature is permissible to provide a signal-to-noise (S/N) ratio of 5 dB for reception of a picture with 3×10^6 pixels (picture elements) in 3 minutes if the earth station antenna diameter is 70 m? Assume aperture efficiencies of 70 percent. The earth-Neptune distance = 4 light-hours.

***12–3–9**
[临界频率和最大可用频率(MUF)]

Critical frequency. MUF. Layers may be said to exist in the earth's ionosphere where the ionization gradient is sufficient to refract radio waves back to earth. [Although the wave actually may be bent gradually along a curved path in an ionized region of considerable thickness, a useful simplification for some situations is to assume that the wave is reflected as though from a horizontal perfectly conducting surface situated at a (*virtual*) *height h*.] The highest frequency at which this layer reflects a vertically incident wave back to the earth is called the *critical frequency* f_o. Higher frequencies at the vertical incidence pass through. For waves at oblique incidence ($\phi > 0$ in Fig. P12–3–9) the *maximum usable frequency* (MUF) for point-to-point communication on the earth is given by MUF $= f_o/\cos\phi$, where ϕ = angle of incidence. The critical frequency $f_o = 9\sqrt{N}$, where N = electron density (number m^{-3}). N is a function of solar irradiation and other factors. Both f_o and h vary with time of day, season, latitude and phase of the 11-year sunspot cycle. Find the MUF for (*a*) a distance $d = 1.3$ Mm by F_2-layer ($h = 325$ km) reflection with F_2-layer electron density $N = 6 \times 10^{11}$ m^{-3}; (*b*) a distance $d = 1.5$ Mm by F_2-layer ($h = 275$ km) reflection with $N = 10^{12}$ m^{-3}; and (*c*) a distance $d = 1$ Mm by sporadic E-layer ($h = 100$ km) reflection $N = 8 \times 10^{11}$ m^{-3}. Neglect earth curvature.

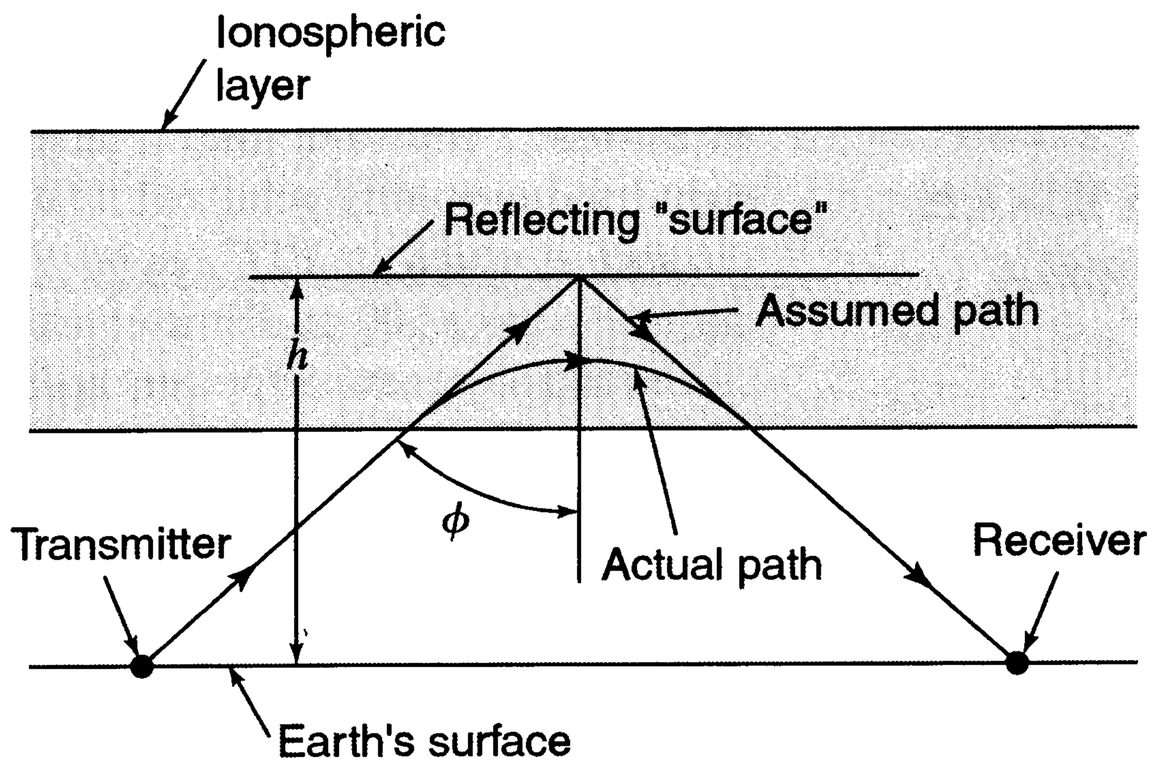


Figure P12–3–9
Communication path via reflection from ionospheric layer.

12–3–10
[克拉克轨道卫星的最小可用频率(mUF)]

mUF for Clarke-orbit satellites. Stationary communication (relay) satellites are placed in the Clarke orbit at heights of about 36 Mm. This is far above the ionosphere, so that the transmission path passes completely through the ionosphere twice, as in Fig. P12–3–10. Since frequencies of 2 GHz and above are usually used the ionosphere has little effect. The high frequency also permits wide bandwidths. If the ionosphere consists of a layer 200 m thick between heights of 200 and 400 km with a uniform electron density $N = 10^{12}$ m^{-1}, find the lowest frequency (or *minimum usable frequency*, mUF) which can be used with a communication satellite (*a*) for vertical incidence and (*b*) for paths 30° from the zenith. (*c*) For an earth station on the equator, what is the mUF for a satellite 15° above the eastern or western horizon?

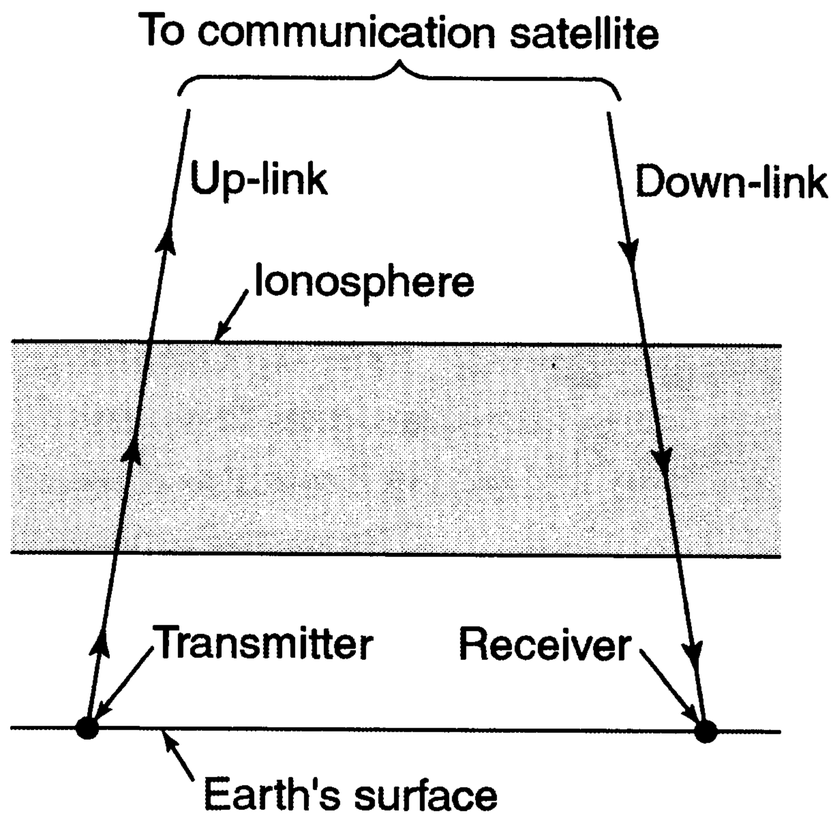


Figure P12–3–10
Communication path via geostationary Clarke-orbit relay satellite.

***12–3–11** (最小可检测温度) **Minimum detectable temperature.** A radio telescope has the following characteristics: antenna noise temperature 50 K, receiver noise temperature 50 K, transmission-line between antenna and receiver 1 dB loss and 270 K physical temperature, receiver bandwidth 5 MHz, receiver integration time 5 s, receiver (system) constant $k' = \pi/\sqrt{2}$ and antenna effective aperture 500 m^2. If two records are averaged, find (*a*) the minimum detectable temperature and (*b*) the minimum detectable flux density.

12–3–12 (最小可检测温度) **Minimum detectable temperature.** A radio telescope operates at 2650 MHz with the following parameters: system temperature 150 K, predetection bandwidth 100 MHz, postdetection time constant 5 s, system constant $k' = 2.2$ and effective aperture of antenna 800 m^2. Find (*a*) the minimum detectable temperature and (*b*) the minimum detectable flux density. (*c*) If four records are averaged, what change results in (*a*) and (*b*)?

***12–3–13** (星际无线电线路) **Interstellar wireless link.** If an extraterrestrial civilization (ETC) transmits 10^6 W, 10 s pulses of right-hand circularly polarized 5 GHz radiation with a 100-m-diameter dish, what is the maximum distance at which the ETC can be received with an SNR = 37? Assume the receiving antenna on the earth also has a 100-m-diameter antenna responsive to right circular polarization, that both antennas (theirs and ours) have 50 percent aperture efficiency, and that the earth station has a system temperature of 10 K and bandwidth of 0.1 Hz.

***12–3–14** (背包企鹅) **Backpacking penguin.** This penguin (Fig. P12–3–14) participated in a study of Antarctic penguin migration habits. Its backpack radio with $\lambda/4$ antenna transmitted data on its body temperature and its

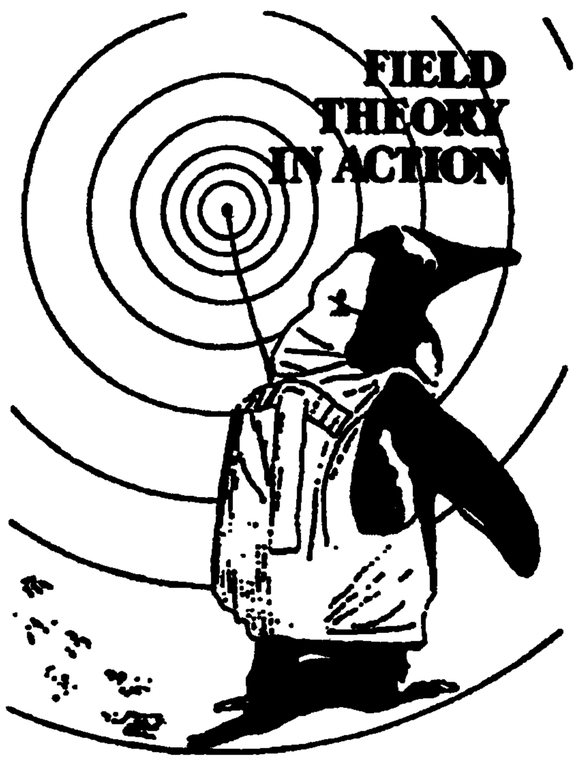


Figure P12–3–14
Antarctic backpacking penguin.

heart and respiration rates. It also provided information on its location as it moved with its flock across the ice cap. The backpack operated at 100 MHz with a peak power of 1 W and a bandwidth of 1 kHz of tone-modulated data signals. If $T_{sys} = 1000$ K and SNR = 30 dB, what is the maximum range? The transmitting and receiving antennas are $\lambda/4$ stubs.

12–3–15 (卫星的载噪比) **Satellite carrier-to-noise ratio.** Since the signal-to-noise ratio as defined Eq. (12–3–3) depends on the bandwidth of the receiver, which in turn depends on the modulation applied to the signal, communications satellite engineers use a related quantity called the *carrier-to-noise* or C/N. This represents the ratio of the power in the carrier signal to the noise power per hertz of bandwidth and is given by

$$\frac{C}{N} = \frac{P_t A_{et} A_{er}}{r^2 \lambda^2 k T_{sys}}$$

where C = power density of carrier (W/Hz) and N = noise power density (W/Hz). Find the carrier-to-noise ratio for the system described in Example 12–3.1.

12–3–16 (全路径载噪比) **Full-path C/N.** Show that the full-path or "circuit" (up- and downlink) carrier-to-noise ratio is given by

$$(C/N)^{-1}_{\text{circuit}} = (C/N)^{-1}_{\text{up}} + (C/N)^{-1}_{\text{down}}$$

where $(C/N)_{\text{up}}$ and $(C/N)_{\text{down}}$ are the uplink and downlink carrier-to-noise ratios.

***12–3–17 (近地轨道通信卫星)** **Low earth orbit communications satellite.** A communications satellite in low earth orbit (LEO) has $r_{\text{up}} = 1500$ km and $r_{\text{down}} = 1000$ km, with an uplink frequency of 14.25 GHz and a downlink frequency of 12 GHz. Find the full-circuit C/N if the transmitting earth station ERP is 60 dBW and the satellite ERP is 25 dBW. Assume the satellite receiver G/T is 5 dB/K and the earth-station G/T is 30 dB/K.

***12–3–18 [直播卫星(DBS)]** **Direct broadcast satellite (DBS).** Direct broadcast satellite services provide CD quality audio to consumers via satellites in geosynchronous orbit. The World Administrative Radio Conference (WARC) has established these requirements for such services.

Frequency band	11.7 to 12.2 GHz (K_u band)
Channel bandwidth	27 MHz
Minimum power flux density	-103 dBW/m^2
Receiver figure of merit (G/T)	6 dB/K
Minimum carrier-to-noise ratio	14 dB

(*a*) Find the effective radiated power (ERP) over 1 W isotropic needed to produce the specified flux density at the earth's surface from a DBS satellite in a 36,000-km orbit. (*b*) If the satellite has a 100-W transmitter and is operated at 12 GHz, what size circular parabolic dish antenna must be used to achieve the required ERP? Assume 50 percent efficiency. (*c*) Does a consumer receiver with circular 1 m dish antenna with 50 percent efficiency and system noise temperature of 1000 K meet the specified G/T? (*d*) By how much does the system specified in parts (*a*) through (*c*) exceed the required carrier-to-noise ratio?

12–3–19 (载噪比的简化表达式) **Simplified expression for C/N.** The expression for C/N provided in P12–3–15 may be simplified by making the following substitutions:

$$\text{Effective isotropic radiated power} = \text{ERP} = P_t G_t \text{ (W)}$$

$$\text{Link path loss} = L_{\text{link}} = 4\pi r^2/\lambda^2$$

The carrier-to-noise ratio may then be written as

$$\frac{C}{N} = \text{ERP}\frac{1}{L_{\text{link}}}\frac{1}{k}\frac{G_r}{T_{\text{sys}}}$$

where G_r/T_{sys} is the receive antenna gain divided by the system noise temperature. This ratio, referred to as "*G over T*," is commonly used as a figure of merit for satellite and earth station receivers. Find the C/N ratio for the uplink to a satellite at the Clarke orbit ($r = 36{,}000$ km) equipped with a 1 m parabolic dish antenna with efficiency of 50 percent and a receiver with noise temperature of 1500 K. Assume that the transmitting earth station utilizes a 1 kW transmitter and a 50 percent efficient 10 m dish antenna and operates at a frequency of 6 GHz.

12-3-20 (载噪比与最大数据率) **Carrier-to-noise and maximum data rate.** The importance of carrier-to-noise density ratio (C/N_d) in communication links was established in 1949 by C. E. Shannon (1). According to Shannon's theorem for the *information capacity* of a communication channel, the maximum data rate of a channel with bandwidth B is given by

$$M = B\log_2\left(1 + \frac{C}{N_d B}\right)$$

where M = channel capacity (bits/s), B = channel bandwidth (Hz), C = carrier signal power (W), and N_d = noise power density (W/Hz). Show that the maximum data rate for any channel, even if infinite bandwidth is used, is

$$M \underset{B\to\infty}{=} \frac{C}{N_d}\log_2(e) = 1.44\frac{C}{N_d}$$

Hint: Let $x = C/(N_d B)$ and use the relation

$$\lim_{x\to\infty}(1+x)^{1/x} = e$$

12-3-21 (卫星通信中继系统) **Satellite communications relay system.** A proposed global communication system uses a ring of medium earth orbit (MEO) satellites to relay signals from one side of the earth to the other and to relay data from distant spacecraft to ground stations. The ground stations use receivers with G/T of 20 dB/K and have transmit power of 10 W. The uplink frequency is 14 GHz, downlink frequency is 12 GHz, and intersatellite frequency is 100 GHz (since there is no atmospheric absorption between satellites). Each satellite has an ERP of 5000 W and receives G/T of 5 dB/K. (*a*) Find the full-link carrier-to-noise ratio and maximum data rate if three satellites are used to link two earth stations. (*b*) Find the full-link carrier-to-noise ratio and maximum data rate between a probe at Saturn (distance = 1.5×10^9 km) with ERP of 500 W if a single satellite is used to relay the signal to the receiving ground station.

12-3-22 (不协调的伽利略天线) **Galileo's uncooperative antenna.** When the Galileo spacecraft arrived at Jupiter in 1995, ground controllers had been struggling for 3 years to open the spacecraft's 5-m high-gain communications dish, which was to operate at 10 GHz (X band). Unable to deploy this antenna because of prelaunch loss of lubricant, a low-directivity ($G = 10$ dB) S-band antenna operating at 2 GHz had to be used to relay all pictures and data from the spacecraft to the earth. For a spacecraft transmit power of 20 W, distance to earth of 7.6×10^{11} m, and 70 m dish with 50 percent efficiency at the receiving station, find (*a*) the maximum achievable data rate if the 5 m X-band antenna had deployed and (*b*) the maximum data rate using the low-gain 1-m S-band antenna.

***12–4–1**
（有吸收云时的天线温度）

Antenna temperature with absorbing cloud. A radio source is occulted by an intervening emitting and absorbing cloud of unity optical depth and brightness temperature 100 K. The source has a uniform brightness distribution of 200 K and a solid angle of 1 square degree. The radio telescope has an effective aperture of 50 m^2. If the wavelength is 50 cm, find the antenna temperature when the radio telescope is directed at the source. The cloud is of uniform thickness and has an angular extent of 5 square degrees. Assume that the antenna has uniform response over the source and cloud.

12–4–2
（被动遥感天线）

Passive remote-sensing antenna. Design a 3-GHz antenna for an earth-resource passive remote-sensing satellite to measure earth-surface temperature with 1 km^2 resolution from a 300-km orbital height.

12–4–3
（树林的吸收）

Forest absorption. An earth-resource satellite passive remote-sensing antenna directed at the Amazon River Basin measures a night-time temperature $T_A = 21°C$. If the earth temperature $T_e = 27°C$ and the Amazon forest temperature $T_f = 15°C$, find the forest absorption coefficient τ_f.

***12–4–4**
（木星信号）

Jupiter signals. Flux densities of 10^{-20} W m^{-2} Hz^{-1} are commonly received from Jupiter at 20 MHz. What is the power per unit bandwidth radiated at the source? Take the earth-Jupiter distance as 40 light-minutes and assume that the source radiates isotropically.

12–4–5
（红移，功率）

Red shifts. Powers. Some radio sources have been identified with optical objects and the Doppler or red shift z $(= \Delta\lambda/\lambda)$ measured from an optical spectrum. The distance-red shift or Hubble relation is

$$R = \frac{v}{H_o} = \frac{m-1}{m+1}\frac{c}{H_o}$$

where R = distance in megaparsecs (1 megaparsec = 1 Mpc = 3.26×10^6 light-years), v = velocity of recession of object in m s^{-1}, $m = (z+1)^2$, c = velocity of light and H_o = Hubble's constant = 75 km s^{-1} Mpc^{-1}. Determine the distance R in light-years to the following radio sources: (*a*) Cygnus A (prototype radio galaxy), $z = 0.06$; (*b*) 3C273 (quasistellar radio source, or quasar), $z = 0.16$; and (*c*) OQ172 (distant quasar), $z = 3.53$. The above sources have flux densities as follows at 3 GHz: Cygnus A, 600 Jy; 3C273, 30 Jy; OQ172, 2 Jy (1 Jy = 10^{-26} W m^{-2} Hz^{-1}). (*d*) Determine the radio power per unit bandwidth radiated by each source. Assume that the source radiates isotropically.

12–5–1
（雷达检测）

Radar detection. A radar receiver has a sensitivity of 10^{-12} W. If the radar antenna effective aperture is 1 m^2 and the wavelength is 10 cm, find the transmitter power required to detect an object with 5 m^2 radar cross section at a distance of 1 km.

12–5–2
（金星和月球雷达）

Venus and moon radar. (*a*) Design an earth-based radar system capable of delivering 10^{-15} W of peak echo power from Venus to a receiver. The radar is to operate at 2 GHz and the same antenna is to be used for both transmitting and receiving. Specify the effective aperture of the antenna and the peak transmitter power. Take the earth-Venus distance as 3 light-minutes, the diameter of Venus as 12.6 Mm and the radar cross section of Venus as 10 percent of a sphere. (*b*) If the system of (*a*) is used to observe the moon, what will the received power be? Take the moon diameter as 3.5 Mm and the moon radar cross section as 10 percent of a sphere.

***12–5–3**
（电子的雷达截面）

RCS of electron. The alternating electric field of a passing electromagnetic wave causes an electron (initially at rest) to oscillate (Fig. P12–5–3). This oscillation of the electron makes it equivalent to a short dipole antenna with $D = 1.5$. Show that the ratio of the power scattered per steradian to the incident Poynting vector is given by $(\mu_o e^2 \sin\theta/4\pi m)^2$, where e and m are the charge and mass of the electron and θ is the angle of the scattered radiation with respect to the direction of the

electric field **E** of the incident wave. This ratio times 4π is the radar cross section of the electron. Such reradiation is called *Thompson scatter*.

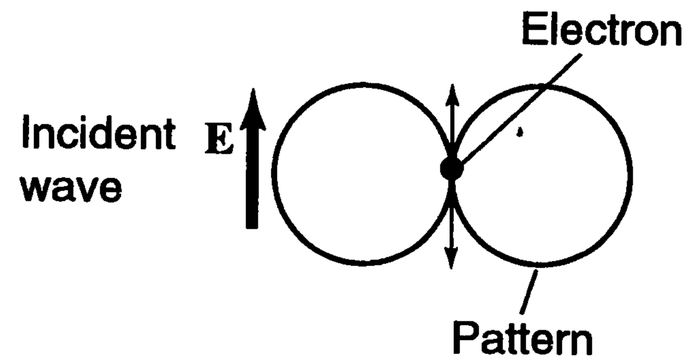


Figure P12–5–3

12–5–4 (汤姆孙散射雷达) **Thompson-scatter radar.** A ground-based vertical-looking radar can be used to determine electron densities in the earth's ionosphere by means of Thompson scatter (see Prob. 12–5–3). The scattered power radar return is proportional to the electron density. If a short pulse is transmitted by the radar, the backscattered power as a function of time is a measure of the electron density as a function of height. Design a Thompson-scatter radar operating at 430 MHz capable of measuring ionospheric electron densities with 1 km resolution in height and horizontal position to heights of 1 Mm. The radar should also be capable of detecting a minimum of 100 electrons at a height of 1 Mm. The design should specify radar peak power, pulse length, antenna size and receiver sensitivity (Gordon-1).

***12–5–5 (10 km 处单个电子的检测)** **Detecting one electron at 10 km.** If the Arecibo ionospheric 300-m-diameter antenna operates at 100 MHz, how much power is required to detect a single electron at a height (straight up) of 10 km with an SNR −0 dB? See Fig. P12–5–5. The bandwidth is 1 Hz, $T_{sys} = 100$ K and the aperture efficiency = 50 percent.

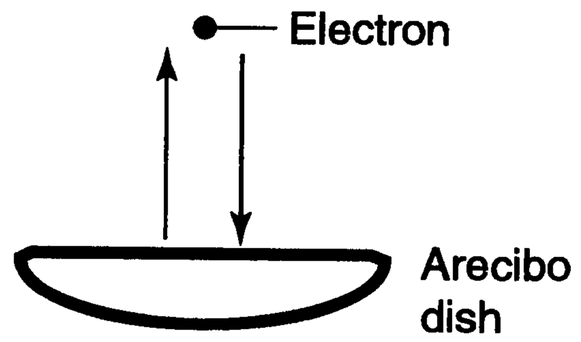


Figure P12–5–5

12–5–6 (短偶极子雷达截面的谐振效应) **Effect of resonance on radar cross section of short dipoles.** (*a*) Calculate the radar cross section of a lossless resonant dipole ($Z_L = -jX_A$) with length = $\lambda/10$ and diameter = $\lambda/100$. (See Secs. 2–9 and 14–12.) (*b*) Calculate the radar cross section of the same dipole from $34L^6/\lambda^4[\ln(L/2b) - 1]^2$, where L is the dipole length and b is its radius. (*c*) Compare both values with the maximum radar cross section shown in Fig. 12–9. Comment on results.

***12–5–7 (圆球、平盘和圆柱的雷达散射截面)** **Radar cross section of sphere, disk and cylinder.** What are the radar cross sections at $\lambda = 10$ cm for (*a*) a 1-m-diameter sphere, (*b*) a 1-m^2-flat disk, and (*c*) a cylinder 1 m long and 4 mm in diameter with L parallel to the plane of polarization. All are perfectly conducting and at normal incidence. (*d*) Same as (*c*) but cross polarized.

***12–5–8 (雷达散射截面)** **Radar cross section.** What is the radar cross section of an object at a distance of 10 km which gives a return of 1 nW for a transmitted power of 1 kW? The transmit-receive antenna is a 1-m-diameter dish operating at 3 GHz. Take aperture efficiency as 50 percent.

***12–5–9** (龙卷风雷达)

Tornado radar. For a maximum doppler shift of 5.3 kHz with the weather radar of Example 12–5.1 scanning a tornado, find (*a*) the maximum wind speed in the tornado funnel. (*b*) If the lowest doppler shift observable is 920 Hz, how fast is the tornado approaching or receding from the weather radar antenna?

12–5–10 (鸟腿箍(bird band)的雷达截面)

RCS of bird band. (*a*) What is the maximum RCS of a bird leg band that is 2 mm wide by 3 mm in diameter made of aluminum 0.1 mm thick? (*b*) What is the frequency for the maximum RCS? (*c*) How far can the band and bird be tracked with a 100-W peak power radar using a 100 wavelength diameter dish antenna for a 5 dB SNR?

12–5–11 (雷达作用距离)

Radar range. Find the maximum range at which a flat plate of area $A = 1$ m^2 can be detected with a 3 cm radar having a peak power of 10 kW and a 3-m-diameter dish antenna with 70% aperture efficiency for a SNR = 10 dB. The receiver system temperature = 100 K and the bandwidth $B = 1$ MHz.

***12–5–12** (投球速度)

Fastball velocity. A 20 GHz radar measures a Doppler shift of 6 kHz on a baseball pitcher's fastball. What is the fastball's velocity?

***12–5–13** (投球测量的雷达功率)

Radar power for fastball measurement. To measure the velocity of the fastball of Prob. 12–5–12 with the 20-GHz radar at a distance of 100 m, what power is required for an SNR = 30 dB? The radar uses a conical horn with diameter = 8 cm and aperture efficiency $\varepsilon_{ap} = 0.5$. The ball diameter = 7 cm and it has a radar cross section (RCS) half that of a perfectly conducting sphere of the same diameter.

***12–5–14** (防撞雷达)

Anticollision radar. To provide anticollision warnings, forward-looking radars on automobiles, trucks and other vehicles (see Fig. P12–5–14) can alert the driver of vehicles ahead that are decelerating too fast or have stopped. The brake light on the vehicle ahead may not be working or it may be obscured by poor visibility. To warn of clear-distance decrease rates of 9 m/s or more, what doppler shift must a 20-GHz radar be able to detect?

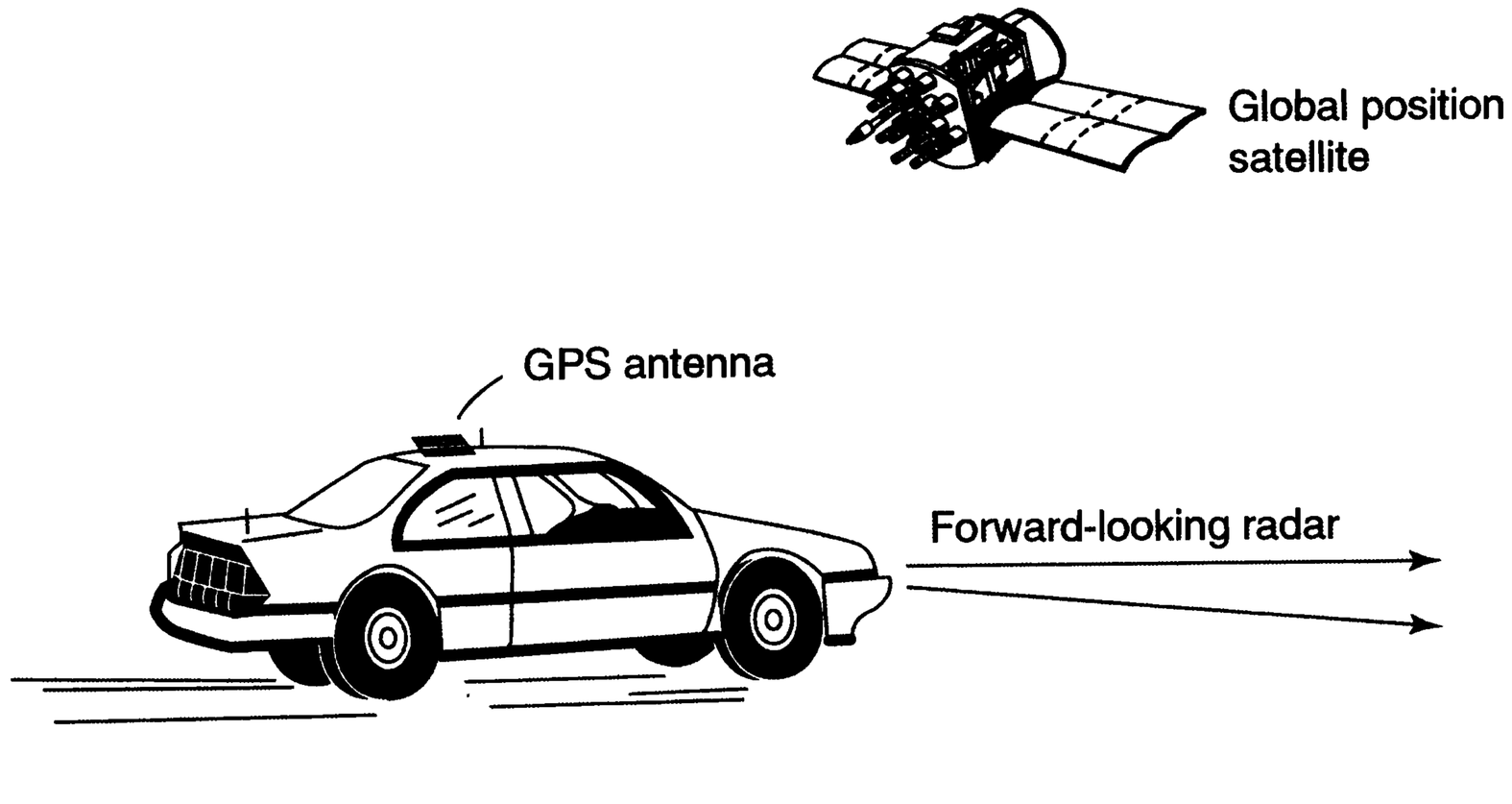


Figure P12–5–14

12–5–15 (防撞雷达的频带宽度和功率)

Beamwidth and power of anticollision radar. For the anticollision 20-GHz radar of Prob. 12–5–14 to avoid false warnings from parked cars and bridge abutments along the side of the road, the beamwidth between first nulls should be 10 m at a range of 250 m. To avoid returns from bridges crossing the highway, the vertical beamwidth should be one-half as much. Assume that the radar antenna is a flat panel broadside array of patch antennas with uniform aperture distribution. (*a*) What are the antenna dimensions? (*b*) To detect a 1 m^2 object at a range of 300 m with an SNR = 30 dB, what power is required?

12–5–16 (防 CFIT 雷达)

Anti-CFIT radar. For aircraft to avoid flying into mountainous terrain during normal flight in poor visibility, called *controlled flight into terrain* (CFIT), a forward-looking radar is required that is similar to that in Prob. 12–5–14 for highway vehicles but with different range and closure rates. Thus, for a range of 15 km and rate of distance closure of 100 m/s or more, find (*a*) pulse rate and (*b*) power required for a 10-GHz radar with 33-dB gain antenna and SNR = 30 dB.

12–5–17 (雷达高度计)

Radar altimeter. For a 10-GHz pulse radar altimeter to measure altitudes of 100 m to 10 km, (*a*) what power and (*b*) pulse rate are required? The antenna is a 3 × 30 cm downward-looking flat panel patch array with uniform aperture distribution. A frequency of 10 GHz has been chosen to avoid the water absorption band at 20 GHz. Although an altimeter is a valuable navigation aid, an anti-CFIT radar is also needed to warn of rapidly rising or steep terrain. The optimum system is to have both altimeter and CFIT radar working in conjunction with the aircraft's navigational radar and a GPS display that indicates the aircraft's position and the elevation of the terrain in the vicinity.

***12–5–18** (警用雷达)

Police radar. A pulsed speed measuring radar must be able to resolve the returns from two cars separated by 30 m. Find the maximum pulse width that can be used to prevent overlapping of the returns from the two vehicles. Note that it takes $t = 2R/c$ seconds for a signal to travel from the transmitter to a target and return, where t is the time, R is the range and c is the velocity in the media. If τ is the pulse width, then it can be shown that the *range resolution* (the minimum range difference between objects for which the returns do not overlap in time) is given by

$$\Delta R = R_2 - R_1 = \frac{c\tau}{2}$$

where the subscripts denote the different objects.

12–5–19 (透地雷达的分辨率)

Ground penetrating radar resolution. For a ground penetrating radar operating in sandy soil with $\varepsilon_r = 8$ and $\mu_r = 0.01$, what pulse width is required to resolve two buried objects at depths of 10 and 15 m?

***12–5–20** (海杂波)

Sea clutter. Search-and-rescue aircraft using radar to locate lost vessels must contend with backscatter from the surface of the ocean. The amplitude of these returns (know as *sea clutter*) depends on the frequency and polarization of the radar waveform, the size of the illuminated patch on the surface, the angle of incidence, and the sea state. The scattering geometry is shown in Fig. P12–5–20. To characterize sea cluster independently on the radar footprint on the surface, the scattering cross section of the ocean may be specified per unit area. This parameter, designated σ_0, has dimensions of square meters of dB above a square meter (dBsm). The total RCS of a patch of ocean surface is found by

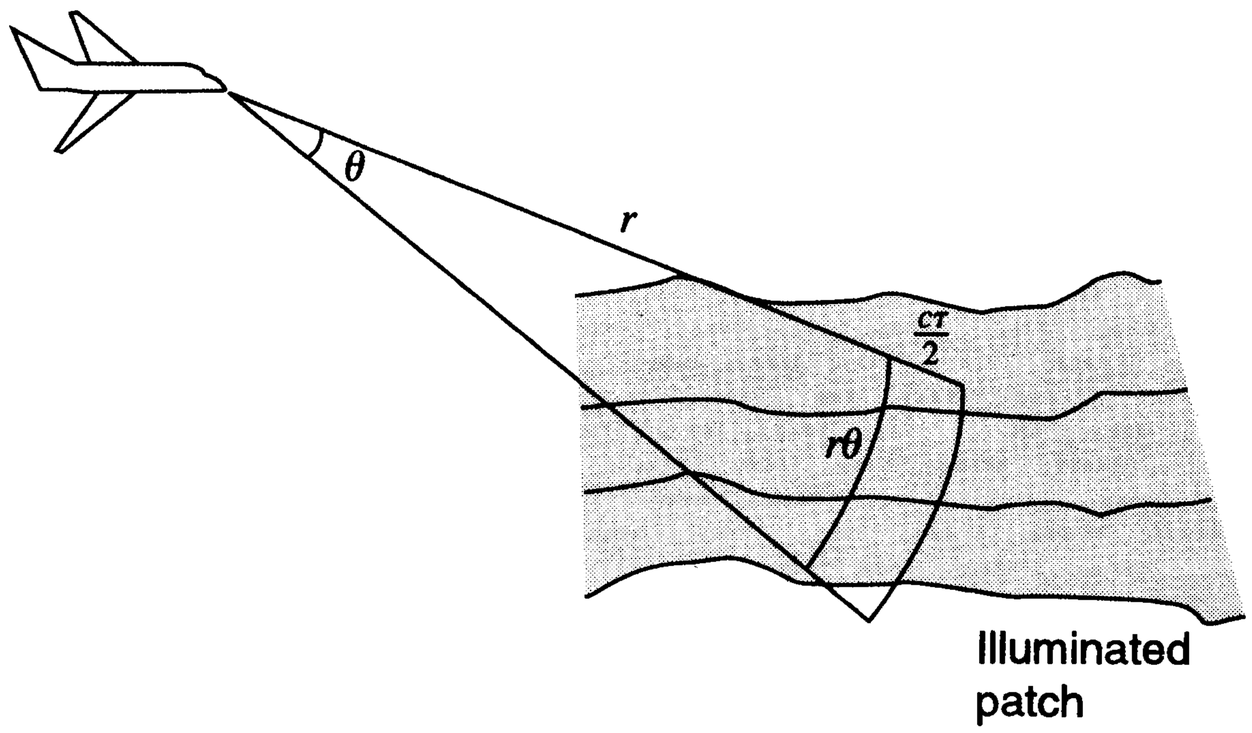


Figure P12–5–20
Sea search-and-rescue geometry.

multiplying σ_0 by the area of the patch. For area scattering, the radar equation is written

$$P_r = P_t \frac{A_e^2 \lambda^2}{4\pi r^4} \sigma_0 A_{\text{patch}}$$

For a pulsed radar with pulse width τ and 3 dB antenna beamwidth of θ rad, the illuminated area for low grazing angle is approximately $(c\tau/2)(r\theta)$. The radar equation may therefore be written as

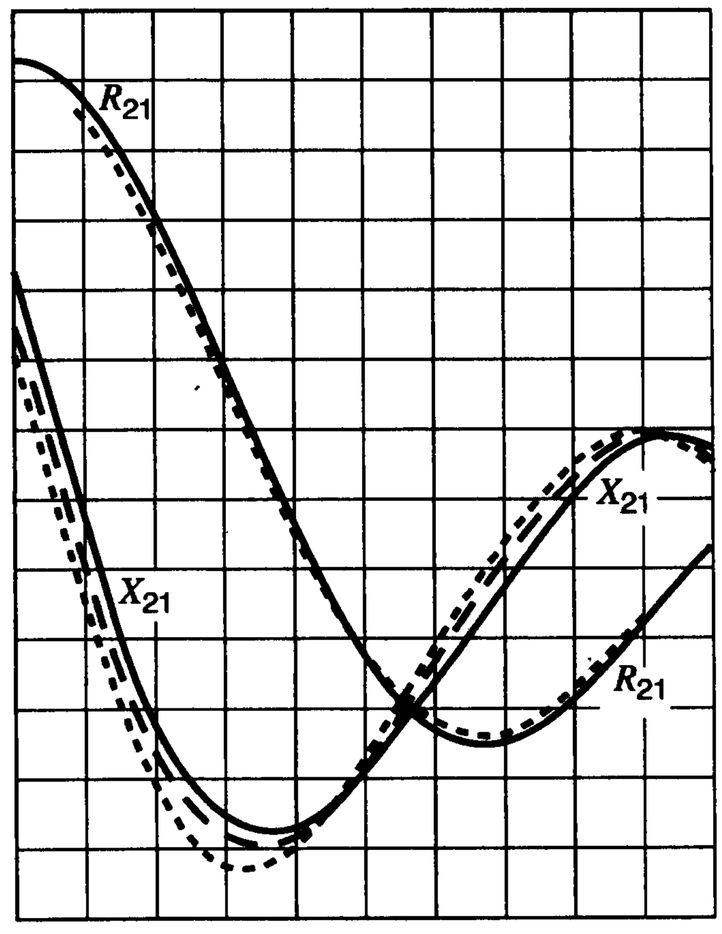
$$P_r = P_t \frac{A_e^2 \lambda^2}{4\pi r^4} \sigma_0 \left(\frac{c\tau}{2}\right)(r\theta) = P_r \frac{A_e^2 \lambda^2 \sigma_0 c\tau\theta}{8\pi r^3}$$

(a) Determine the received power from sea clutter at a range of 10 km for a monostatic pulsed radar transmitting a 1 μs pulse with 1 kW peak power at a frequency of 6 GHz. Assume a 1.5 m circular dish antenna with 50 percent efficiency and sea state 4 ($\sigma_0 = -30$ dBsm/m^2 at C band). (b) For a receiver bandwidth of 10 kHz and noise figure of 3.5 dB, find the receiver noise power. (c) If this radar is used to search for a ship with RCS = 33 dBsm under these conditions, what signal-to-noise and signal-to-clutter ratios can be expected?

For computer programs, see Appendix C.

CHAPTER 13

Self and Mutual Impedances
阻抗与互阻抗



The topics in this chapter include:

- Reciprocity theorem for antennas（天线的互易性定理）
- The 73 Ω radiation resistance of a λ/2 antenna（半波长天线的 73 Ω 辐射电阻）
- Radiation resistance of antenna with off-center current maximum（电流最大点偏离中心的天线辐射电阻）
- Self-impedance of thin linear antennas（细线天线的自阻抗）
- Mutual impedance of two parallel linear antennas（两平行线天线的互阻抗）
- Mutual impedance of parallel side-by-side antennas（边靠边平行天线的互阻抗）
- Mutual impedance of parallel collinear antennas（平行共线天线的互阻抗）
- Mutual impedance of antennas in echelon（梯式平行天线的互阻抗）
- Mutual impedance of other configurations（其他布置天线的互阻抗）
- Mutual impedance in terms of directivity and radiation resistance（用定向性和辐射电阻表示的互阻抗）

13–1 INTRODUCTION（引言）

The impedance presented by an antenna to a transmission line can be represented by a 2-terminal network. This is illustrated in Fig. 13–1 in which the antenna is replaced by an equivalent impedance Z connected to the terminals of the transmission line. In designing a transmitter and its associated transmission line, it is convenient to consider that the antenna is simply a 2-terminal impedance. This impedance into which the transmission line operates is called the ***terminal*** or ***driving-point impedance.***（馈端阻抗或激励点阻抗）If the antenna is isolated, i.e., remote from the ground or other objects, and is lossless,[1] its terminal impedance is the same as the ***self-impedance***（自阻抗）of the antenna. This impedance

[1]By lossless is meant that there is no Joule heating associated with the antenna. There may, of course, be radiation. If the antenna is not lossless, an equivalent loss resistance appears at the terminals in series with the self-resistance or radiation resistance.

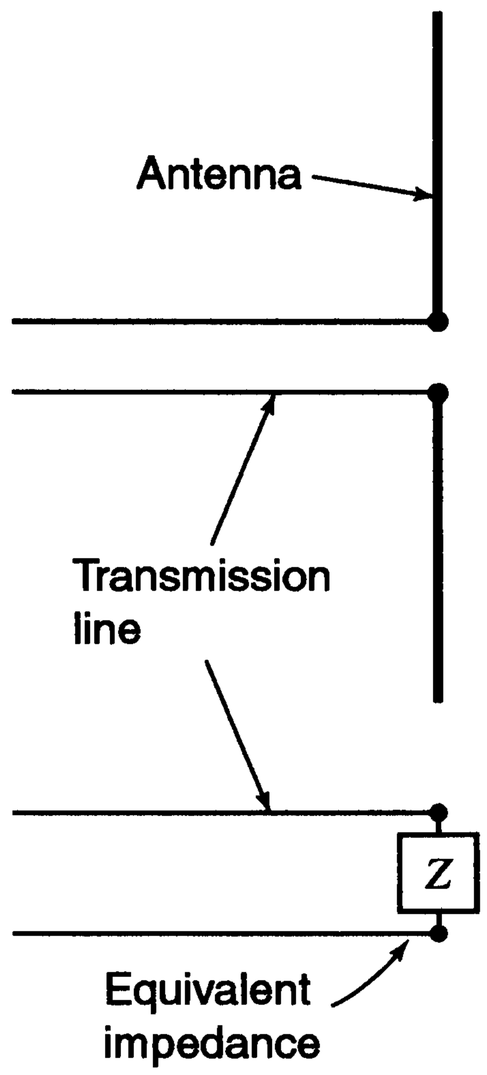


Figure 13–1
Transmission line with antenna and with equivalent impedance.

has a real part called the ***self-resistance*** (自电阻) (radiation resistance) and an imaginary part called the ***self-reactance.*** (自电抗) The self-impedance is the same for reception as for transmission.

In case there are nearby objects, say several other antennas, the terminal impedance can still be replaced by a 2-terminal network. However, its value is determined not only by the self-impedance of the antenna but also by the mutual impedances between it and the other antennas and the currents flowing on them. The terminal impedance is the same for both transmission and reception (see next section).

In developing the subject of antenna impedance, an important and much-used theorem is that of *reciprocity*. Accordingly, this topic is discussed first and then applied to the impedance problem.

13–2 RECIPROCITY THEOREM FOR ANTENNAS (天线的互易性定理)

The Rayleigh-Helmholtz reciprocity theorem has been generalized by Carson to include continuous media (Rayleigh-1; Carson-1, 2; Ballantine-1). This theorem as applied to antennas may be stated as follows:

If an emf is applied to the terminals of an antenna A and the current measured at the terminals of another antenna B, then an equal current (in both amplitude and phase) will be obtained at the terminals of antenna A if the same emf is applied to the terminals of antenna B. [若在天线 *A* 的馈端上施加电动势，在天线 *B* 的馈端处测得电流，则对应于在天线 B 的馈端施加相同电动势的情况，在天线 *A* 的馈端处也将得到相等 (幅度和相位) 的电流。]

It is assumed that the emfs are of the same frequency and that the media are linear, passive and also isotropic. An important consequence of this theorem is the fact that under these conditions the transmitting and receiving patterns of an antenna are the same. Also, for matched impedances, the power flow is the same either way.

As an illustration of the reciprocity theorem for antennas, consider the following two cases.

Case 1. Let an emf V_a be applied to the terminals of antenna A as in Fig. 13–2a. This antenna acts as a transmitting antenna, and energy flows from it to antenna B, which may be considered as a receiving antenna, producing a current I_b at its terminals.[1] It is assumed that the generator supplying the emf and the ammeter for measuring the current have zero impedance or, if not zero, that the generator and ammeter impedances are equal.

Case 2. If an emf V_b is applied to the terminals of antenna B, then it acts as a transmitting antenna and energy flows from it to antenna A as in Fig. 13–2b, producing a current I_a at its terminals.

Now if $V_b = V_a$, then by the reciprocity theorem $I_a = I_b$.

[1] Although the emf V_a and the current I_b are scalar space quantities, they are complex or vector quantities with respect to time phase. The term "phasor" is sometimes used to distinguish such a quantity from a true space vector.

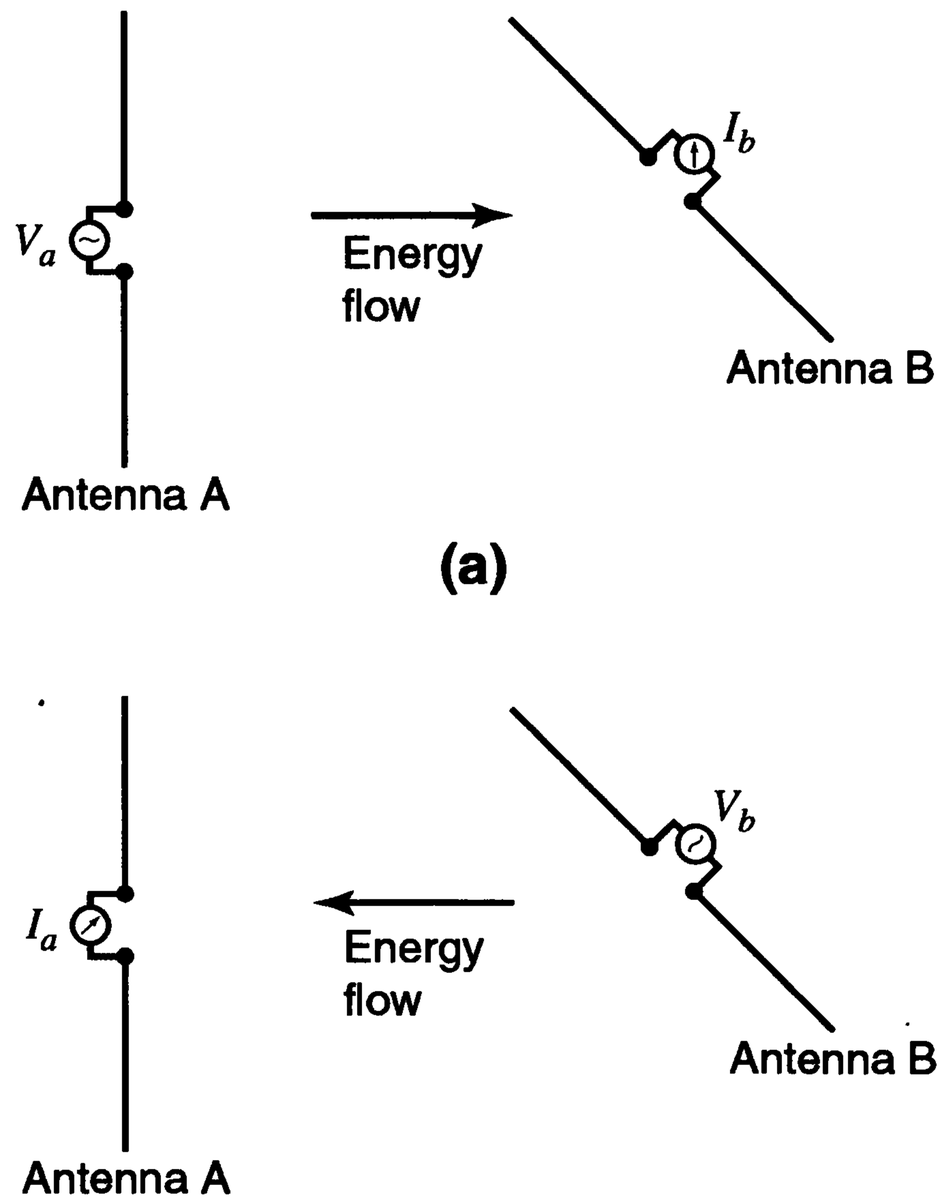


Figure 13–2
Illustrations for reciprocity theorem.

The ratio of an emf to a current is an impedance. In Case 1, the ratio of V_a to I_b may be called the *transfer impedance* Z_{ab}, and in Case 2 the ratio V_b to I_a may be called the transfer impedance Z_{ba}. Then, by the reciprocity theorem it follows that these impedances are equal. Thus,

$$\frac{V_a}{I_b} = Z_{ab} = Z_{ba} = \frac{V_b}{I_a} \tag{1}$$

In order to prove the reciprocity theorem for antennas, let the antennas and the space between them be replaced by a network of linear, passive, bilateral impedances. Since any 4-terminal network can be reduced to an equivalent T section,[1] the antenna arrangement of Case 1 (see Fig. 13–2a) can be replaced by the network of Fig. 13–3a.

The current through the meter is

$$I_b = I_1 \frac{Z_3}{Z_2 + Z_3} \tag{2}$$

where

$$I_1 = \frac{V_a}{Z_1 + [Z_2 Z_3/(Z_2 + Z_3)]} = \frac{V_a(Z_2 + Z_3)}{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1} \tag{3}$$

[1] This is true insofar as the amplitude and phase of the input voltage and output current are concerned.

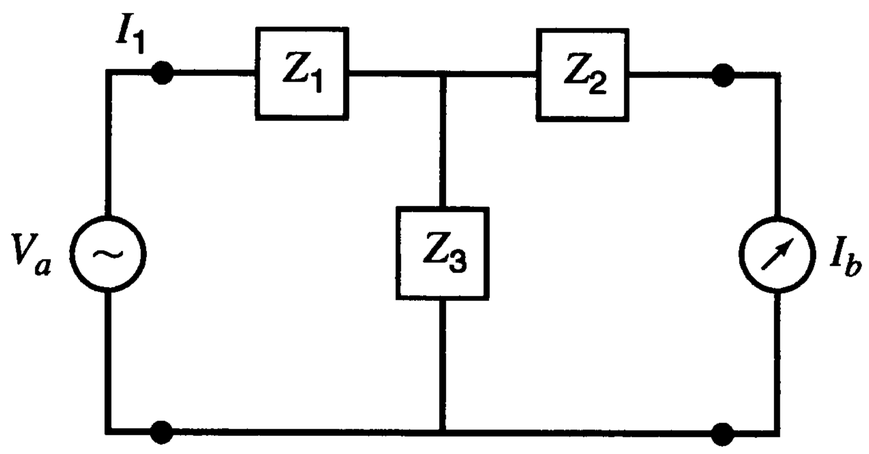


Figure 13–3
Equivalent circuits used in proof of reciprocity theorem.

Introducing (3) into (2) yields the current through the meter in terms of the emf V_a and the network impedances. Thus,

$$I_b = \frac{V_a Z_3}{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1} \tag{4}$$

If the locations of the emf and current meter are interchanged, as in Fig. 13–3*b*, we obtain

$$I_a = \frac{V_b Z_3}{Z_1 Z_2 + Z_2 Z_3 + Z_3 Z_1} \tag{5}$$

Comparing (4) and (5), it follows that if $V_a = V_b$ then $I_a = I_b$, proving the theorem.

13–3 THE 73 Ω RADIATION RESISTANCE OF A λ/2 ANTENNA
（半波长天线的 73 Ω 辐射电阻）

To find the radiation resistance or real part of the self-impedance, the Poynting vector is integrated over a large sphere yielding the power radiated, and this power is then equated to $(I_0/\sqrt{2})^2 R_0$, where R_0 is the radiation resistance at a current maximum point and I_0 is the peak value in time of the current at this point. The total power P radiated was given in (6–3–5) in terms of H_ϕ for a short dipole. In (6–3–5), $|H_\phi|$ is the absolute value. Hence, the corresponding value of H_ϕ for a linear antenna is obtained from (6–4–2) by putting $|j[I_0]| = I_0$. Substituting this into (6–3–5), we obtain

$$P = \frac{15 I_0^2}{\pi} \int_0^{2\pi} \int_0^{\pi} \frac{\{\cos[(\beta L/2)\cos\theta] - \cos(\beta L/2)\}^2}{\sin\theta} \, d\theta \, d\phi \tag{1}$$

$$= 30 I_0^2 \int_0^{\pi} \frac{\{\cos[(\beta L/2)\cos\theta] - \cos(\beta L/2)\}^2}{\sin\theta} \, d\theta \tag{2}$$

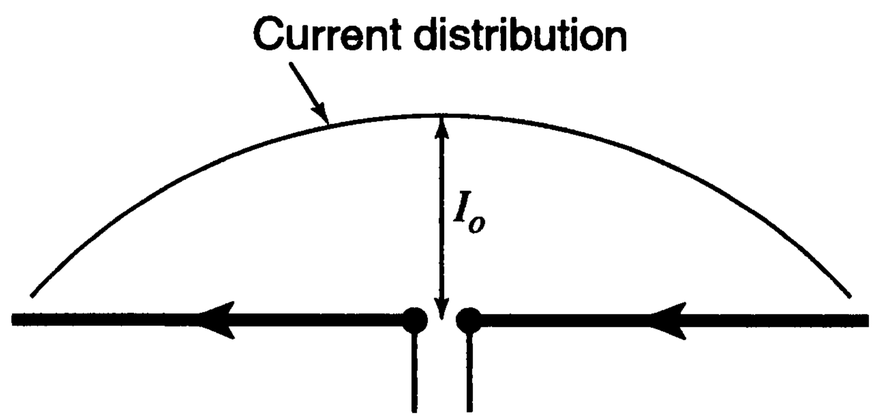


Figure 13–4
$\lambda/2$ antenna with maximum current I_0 at terminals of transmission line.

Equating the radiated power as given by (2) to $I_0^2 R_0/2$ we have

$$P = \frac{I_0^2 R_0}{2} \tag{3}$$

and

$$R_0 = 60 \int_0^{\pi} \frac{\{\cos[(\beta L/2)\cos\theta] - \cos(\beta L/2)\}^2}{\sin\theta}\, d\theta \tag{4}$$

where the radiation resistance R_0 is referred to the current maximum. In the case of a $\lambda/2$ antenna this is at the center of the antenna or at the terminals of the transmission line (see Fig. 13–4).

Proceeding now to evaluate (4), let

$$u = \cos\theta \qquad \text{and} \qquad du = -\sin\theta\, d\theta \tag{5}$$

by which (4) is transformed to

$$R_0 = 60 \int_{-1}^{+1} \frac{[\cos(\beta L u/2) - \cos(\beta L/2)]^2}{1 - u^2}\, du \tag{6}$$

However,

$$\frac{1}{1-u^2} = \frac{1}{(1+u)(1-u)} = \frac{1}{2}\left(\frac{1}{1+u} + \frac{1}{1-u}\right) \tag{7}$$

Also putting $k = \beta L/2$, (6) becomes

$$R_0 = 30 \int_{-1}^{+1} \left[\frac{(\cos ku - \cos k)^2}{1+u} + \frac{(\cos ku - \cos k)^2}{1-u}\right] du \tag{8}$$

This integral gives the radiation resistance for a thin linear antenna of any length L. For the special case being considered where $L = \lambda/2$, we have $k = \pi/2$. Thus, in the case of a thin $\lambda/2$ antenna, (8) reduces to

$$R_0 = 30 \int_{-1}^{+1} \left[\frac{\cos^2(\pi u/2)}{1+u} + \frac{\cos^2(\pi u/2)}{1-u}\right] du \tag{9}$$

Now in the first term let

$$1 + u = \frac{v}{\pi} \qquad \text{and} \qquad du = \frac{dv}{\pi} \tag{10}$$

and in the second term let

$$1 - u = \frac{v'}{\pi} \qquad \text{and} \qquad du = -\frac{dv'}{\pi} \tag{11}$$

Noting also that $(v - \pi)/2 = (\pi - v')/2$, Eq. (9) becomes

$$R_0 = 60 \int_0^{2\pi} \frac{\cos^2[(v - \pi)/2]}{v}\, dv \tag{12}$$

But $\cos^2(x/2) = \frac{1}{2}(1 + \cos x)$ so that

$$R_0 = 30 \int_0^{2\pi} \frac{1 + \cos(v - \pi)}{v}\, dv = 30 \int_0^{2\pi} \frac{1 - \cos v}{v}\, dv \tag{13}$$

The last integral in (13) is often designated as $\mathrm{Cin}(x)$. Thus,

$$\mathrm{Cin}(x) = \int_0^x \frac{1 - \cos v}{v}\, dv = \ln \gamma x - \mathrm{Ci}(x)$$

$$= 0.577 + \ln x - \mathrm{Ci}(x) \tag{14}$$

where $\gamma = e^c = 1.781$ or $\ln \gamma = c = 0.577 =$ Euler's constant

The part of this integral given by

$$\mathrm{Ci}(x) = \ln \gamma x - \mathrm{Cin}(x) \tag{15}$$

is called the cosine integral. The value of this integral is given by

$$\mathrm{Ci}(x) = \int_\infty^x \frac{\cos v}{v}\, dv = \ln \gamma x - \frac{x^2}{2!2} + \frac{x^4}{4!4} - \frac{x^6}{6!6} + \cdots \tag{16}$$

When x is small ($x < 0.2$),

$$\mathrm{Ci}(x) \simeq \ln \gamma x = 0.577 + \ln x \tag{17}$$

When x is large ($x \gg 1$),

$$\mathrm{Ci}(x) = \frac{\sin x}{x} \tag{18}$$

A curve of the cosine integral as a function of x is presented in Fig. 13–5. It is to be noted that $\mathrm{Ci}(x)$ converges around zero at large values of x. From (16) and (14) we obtain $\mathrm{Cin}(x)$ as an infinite series,

$$\mathrm{Cin}(x) = \frac{x^2}{2!2} - \frac{x^4}{4!4} + \frac{x^6}{6!6} - \cdots \tag{19}$$

While discussing $\mathrm{Cin}(x)$ and $\mathrm{Ci}(x)$, mention may be made of another integral which commonly occurs in impedance calculations. This is the sine integral, $\mathrm{Si}(x)$, given by

$$\mathrm{Si}(x) = \int_0^x \frac{\sin v}{v}\, dv = x - \frac{x^3}{3!3} + \frac{x^5}{5!5} - \cdots \tag{20}$$

When x is small ($x < 0.5$),

$$\mathrm{Si}(x) \simeq x \tag{21}$$

When x is large ($x \gg 1$),

$$\mathrm{Si}(x) \simeq \frac{\pi}{2} - \frac{\cos x}{x} \tag{22}$$

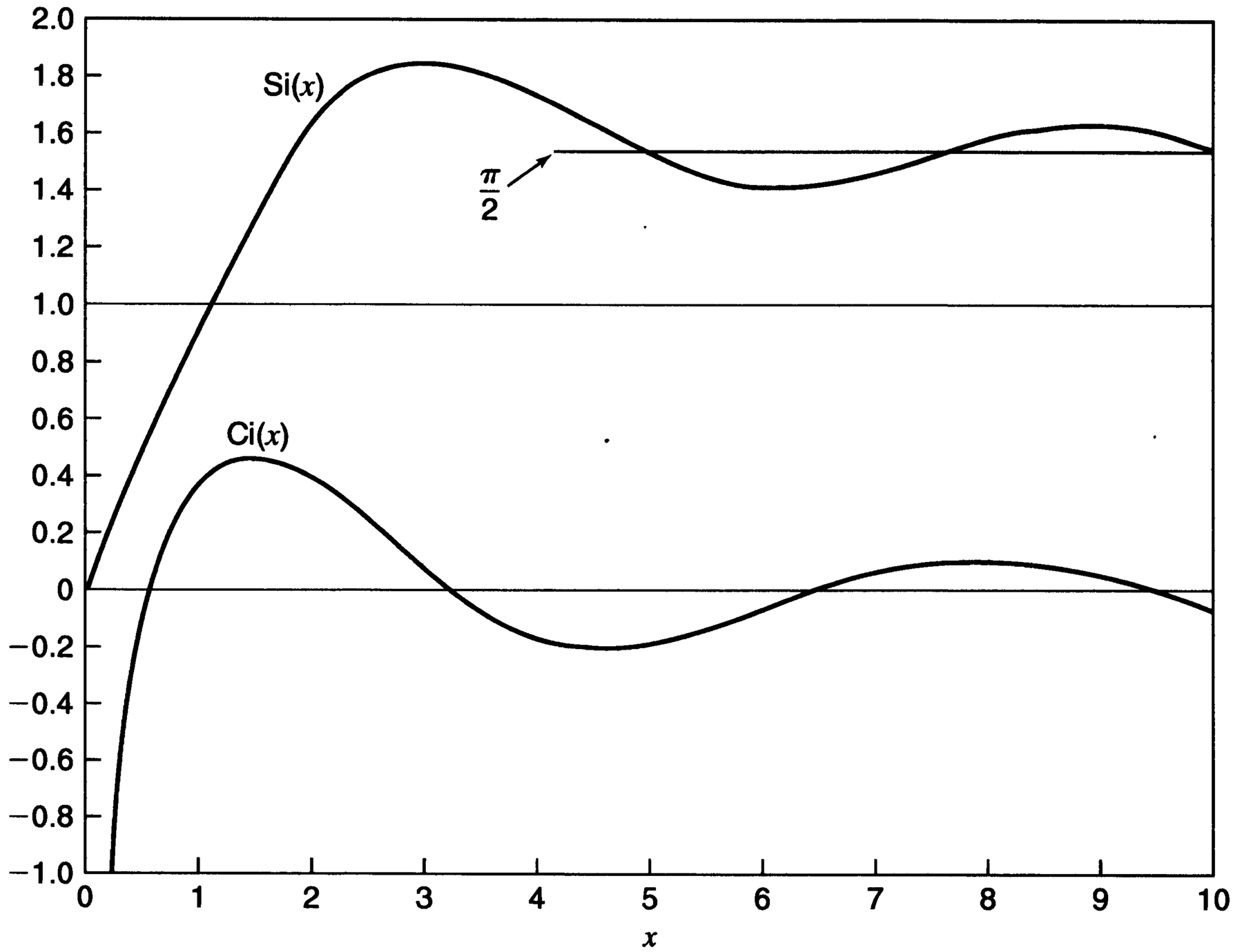


Figure 13–5
The sine integral Si(x) and cosine integral Ci(x). The sine integral converges around $\pi/2$ while the cosine integral converges around zero.

A curve of the sine integral as a function of x is presented in Fig. 13–5. It is to be noted that Si(x) converges around $\pi/2$ at large values of x.

Returning now to (13), this can be written as

$$R_0 = 30\,\text{Cin}(2\pi) = 30 \times 2.44 = 73\ \Omega \qquad \textit{Radiation resistance of } \lambda/2 \textit{ antenna} \tag{23}$$

This is the well-known value for the radiation resistance of a thin, linear, center-fed, $\lambda/2$ antenna with sinusoidal current distribution. The terminal impedance also includes some inductive reactance in series with R_0 (see Sec. 13–5). To make the reactance zero, i.e., to make the antenna resonant, requires that the antenna be a few percent less than $\lambda/2$. This shortening also results in a reduction in the value of the radiation resistance.

13–4 RADIATION RESISTANCE AT A POINT WHICH IS NOT A CURRENT MAXIMUM（非电流最大点的辐射电阻）

If we calculate, for example, the radiation resistance of a $3\lambda/4$ antenna (see Fig. 6–7) by the above method, we obtain its value at a current maximum. This is not the point at which the transmission line is connected. Neglecting antenna losses, the value of radiation resistance so obtained is the resistance R_0 which would appear at the terminals of a transmission

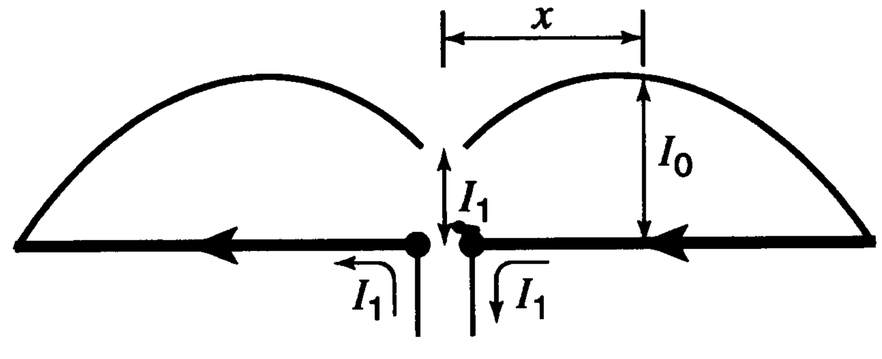


Figure 13–6
Relation of current I_1 at transmission-line terminals to current I_0 at current maximum.

line connected at a current maximum in the antenna, *provided* that the current distribution on the antenna is the same as when it is center-fed as in Fig. 6–7. Since a change of the feed point from the center of the antenna may change the current distribution, the radiation resistance R_0 is not the value which would be measured on a $3\lambda/4$ antenna or on any symmetrical antenna whose length is not an odd number of $\lambda/2$. However, R_0 can be easily transformed to the value which would appear across the terminals of the transmission line connected at the center of the antenna.

This may be done by equating (6–5–3) to the power supplied by the transmission line, given by $I_1^2 R_1/2$, where I_1 is the current amplitude at the terminals and R_1 is the radiation resistance at this point (see Fig. 13–6). Thus,

$$\frac{I_1^2}{2} R_1 = \frac{I_0^2}{2} R_0 \tag{1}$$

where R_0 is the radiation resistance calculated at the current maximum. Thus, the radiation resistance appearing at the terminals is

$$R_1 = \left(\frac{I_0}{I_1}\right)^2 R_0 \tag{2}$$

The current I_1 at a distance x from the nearest current maximum, as shown in Fig. 13–6, is given by

$$I_1 = I_0 \cos \beta x \tag{3}$$

where

I_1 = terminal current
I_0 = maximum current

Therefore, (2) can be expressed as

$$R_1 = \frac{R_0}{\cos^2 \beta x} \tag{4}$$

When $x = 0$, $R_1 = R_0$; but when $x = \lambda/4$, $R_1 = \infty$ if $R_0 \neq 0$. However, the radiation resistance measured at a current minimum ($x = \lambda/4$) is not infinite as would be calculated from (4), since an actual antenna is not infinitesimally thin and the current at a minimum point is not zero. Nevertheless, the radiation resistance at a current minimum may in practice be very large, i.e., thousands of ohms.

13–5 SELF-IMPEDANCE OF A THIN LINEAR ANTENNA（细线天线的自阻抗）

Section 14–11 discusses how the current distribution and self-impedance of a dipole antenna can be obtained using the ***moment method.***（矩量法）Here we use an ***induced emf method.***（感应电动势法）Only the self-impedance can be determined with this method, the current distribution being assumed at the outset. Measurements indicate that the current distribution on thin dipoles is nearly sinusoidal, except near current minima. This distribution results in satisfactory values for dipoles with length-diameter ratios as small as 100, provided the terminals are at a current maximum. Omitting the 50-step derivation, we obtain a self-impedance given by

$$Z_{11} = R_{11} + jX_{11} = 30[\text{Cin}(2\pi n) + j\,\text{Si}(2\pi n)] \qquad (\Omega) \tag{1}$$

or

$$Z_{11} = 30[0.577 + \ln(2\pi n) - \text{Ci}(2\pi n) + j\,\text{Si}(2\pi n)] \qquad (\Omega) \tag{2}$$

The self-resistance is

$$R_{11} = 30\,\text{Cin}(2\pi n) = 30[0.577 + \ln(2\pi n) - \text{Ci}(2\pi n)] \qquad (\Omega) \tag{3}$$

and the self-reactance is

$$X_{11} = 30\,\text{Si}(2\pi n) \qquad (\Omega) \tag{4}$$

These equations give the impedance values for a thin linear center-fed antenna that is an odd number (n) of $\lambda/2$ long. The current distribution is assumed to be sinusoidal (Fig. 13–7*a*). The values are those appearing at the terminals at the center of the antenna.

In the case of a $\lambda/2$ *antenna* as shown in Fig. 13–7*a*, $n = 1$, and we have for the self-resistance and self-reactance

$$R_{11} = 30\,\text{Cin}(2\pi) \qquad (\Omega) \tag{5}$$

and

$$X_{11} = 30\,\text{Si}(2\pi) \qquad (\Omega) \tag{6}$$

The value of (5) is identical with that given for the radiation resistance of a $\lambda/2$ antenna, in Sec. 13–3, Eq. (13–2–23). Evaluating (5) and (6), we obtain for the self-impedance

（半波长天线的自阻抗）

$$\boxed{Z_{11} = R_{11} + jX_{11} = 73 + j42.5\ \Omega \qquad \textit{Self-impedance of } \lambda/2 \textit{ antenna}} \tag{7}$$

Since X_{11} is not zero, an antenna exactly $\lambda/2$ long is not resonant. To obtain a resonant antenna, it is common practice to shorten the antenna a few percent to make $X_{11} = 0$. In this case the self-resistance is somewhat less than 73 Ω.

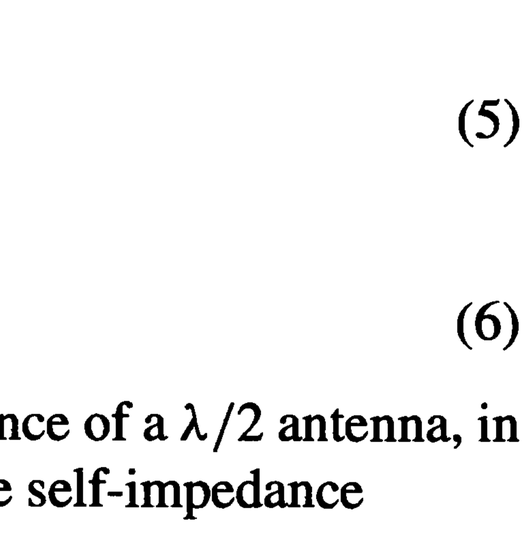


Figure 13–7
$\lambda/2$ and $3\lambda/2$ antennas with current distribution.

For a $3\lambda/2$ *antenna* as shown in Fig. 13–7*b*, $n = 3$ and the self-impedance is

$$Z_{11} = 30[\text{Cin}(6\pi) + j\,\text{Si}(6\pi)]$$

or

$$\boxed{Z_{11} = 105.5 + j45.5\ \Omega \qquad \textit{Self-impedance of } 3\lambda/2 \textit{ antenna}} \quad (3\lambda/2\text{天线的自阻抗}) \tag{8}$$

It is interesting that the self-reactance of center-fed antennas, an exact odd number of $\lambda/2$ long, is always positive since the sine integral $\text{Si}(2\pi n)$ is always positive. For large n the sine integral converges around a value of $\pi/2$ (see Fig. 13–5) which corresponds to a reactance of 47.1 Ω. It should be noted that for antenna lengths not an exact odd number of $\lambda/2$ the reactance may be positive or negative. However, the foregoing analysis of this section is limited to antennas that are an exact odd number of $\lambda/2$ long.

For large n, the self-resistance expression (3) approaches the value

$$R_{11} = 30[0.577 + \ln(2\pi n)] \tag{9}$$

since $\text{Ci}(2\pi n)$ approaches zero. Thus, the self-resistance continues to increase indefinitely with increasing n but at a logarithmic rate.

The more general situation, where the antenna length L is not restricted to an odd number of $\lambda/2$, has also been treated. The antenna is center-fed, and the current distribution is assumed to be sinusoidal. The self-resistance for this case is

$$R_{11} = 30\left[\left(1 - \cot^2\frac{\beta L}{2}\right)\text{Cin}\,2\beta L + 4\cot^2\frac{\beta L}{2}\,\text{Cin}\,\beta L + 2\cot\frac{\beta L}{2}(\text{Si}\,2\beta L - 2\,\text{Si}\,\beta L)\right] \qquad (\Omega) \tag{10}$$

When the length L is small, (10) reduces very nearly to

$$R_{11} = 5(\beta L)^2 \qquad (\Omega) \tag{11}$$

which is the same as (2–10–9) of Example 2–10.1 when $I_{\text{av}} = (1/2)I_0$

For the special case of $L = n\lambda/2$, where $n = 1, 3, 5, \ldots$, (10) reduces to the relation given previously by (3).

The above discussion of this section applies to balanced center-fed antennas. For a thin linear stub antenna of height l perpendicular to an infinite, perfectly conducting ground plane as in Fig. 13–8*a*, the self-impedance is $\frac{1}{2}$ that for the corresponding balanced type. The general formula (10) for Fig. 13–8*b* self-resistance can be converted for a stub antenna above a ground plane by changing the factor 30 to 15 and making the substitution $L = 2l$. The formulas (3) and (4) can be converted for a stub antenna with ground plane where the antenna is an odd number n of $\lambda/4$ long by changing the factor 30 to 15. Thus, for a $\lambda/4$ antenna perpendicular to an infinite, perfectly conducting ground plane, the self-impedance is

$$\boxed{Z_{11} = 36.5 + j21\ \Omega \qquad \textit{Self-impedance of } \lambda/4 \textit{ stub antenna}} \quad (3\lambda/4\text{短桩形天线的自阻抗}) \tag{12}$$

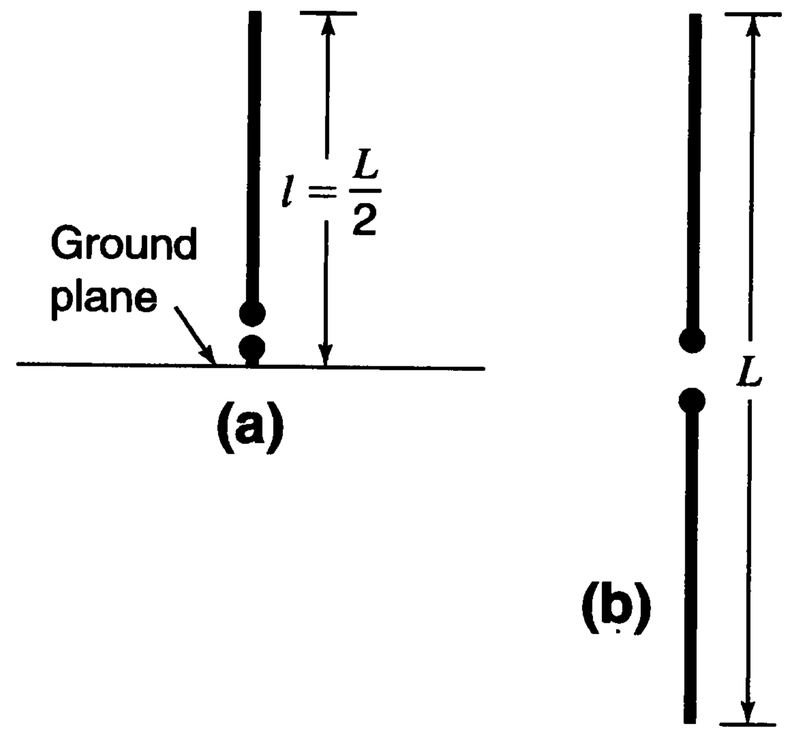


Figure 13–8
Stub antenna of length l at (*a*) and center-fed antenna of length L at (*b*).

13–6 MUTUAL IMPEDANCE OF TWO PARALLEL LINEAR ANTENNAS
(两平行线天线的互阻抗)

The *mutual impedance* of 2 coupled circuits is defined in circuit theory as the negative of the ratio of the emf V_{21} *induced* in circuit 2 to the current I_1 flowing in circuit 1 with circuit 2 open. Consider, for example, the coupled circuit of Fig. 13–9 consisting of the primary and secondary coils of a transformer. The mutual impedance Z_{21} is then

$$Z_{21} = -\frac{V_{21}}{I_1} \tag{1}$$

where V_{21} is the emf induced across the terminals of the open-circuited secondary by the current I_1 in the primary. The mutual impedance, so defined, is not the same as a transfer impedance such as discussed in connection with the reciprocity theorem in Sec. 13–2. In general, a *transfer impedance* is the ratio of an emf *impressed* in one circuit to the resulting current in another with all circuits closed. For example, if the generator in Fig. 13–9 is removed from the primary and is connected to the secondary terminals, the ratio of the emf V *applied* by this generator to the current I_1 in the closed primary circuit is a transfer impedance Z_T. Thus,

$$\frac{V}{I_1} = Z_T \tag{2}$$

This impedance is not the same as the mutual impedance Z_{21} given in (1).

Instead of the coupled circuit of Fig. 13–9, let us consider now the case of 2 coupled antennas 1 and 2 as shown in Fig. 13–10. Suppose a current I_1 in antenna 1 induces an emf V_{21} at the open terminals of antenna 2. Then the ratio of $-V_{21}$ to I_1 is the mutual

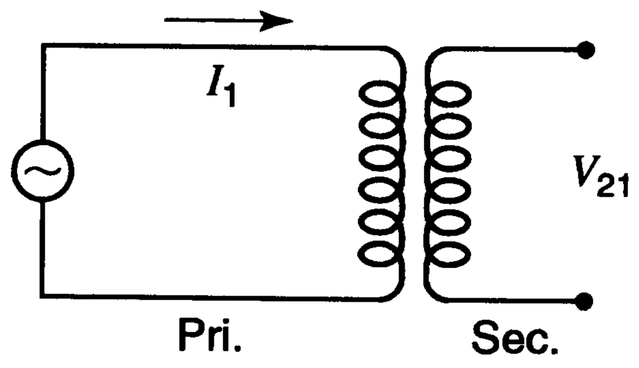


Figure 13–9
Coupled circuit or transformer.

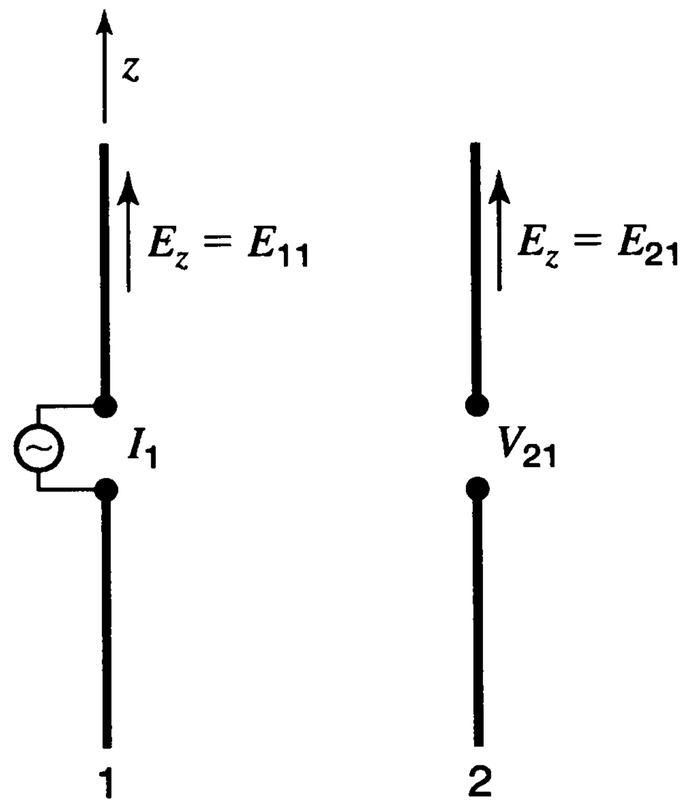


Figure 13–10
Parallel coupled antennas.

impedance Z_{21}. Thus,

$$Z_{21} = \frac{-V_{21}}{I_1} \tag{3}$$

If the generator is moved to the terminals of antenna 2, then by reciprocity the mutual impedance Z_{12} or ratio of $-V_{12}$ to I_2 is the same as before, where V_{12} is the emf induced at the open terminals of antenna 1 by the current I_2 in antenna 2. Thus,

$$\frac{-V_{21}}{I_1} = Z_{21} = Z_{12} = \frac{-V_{12}}{I_2} \tag{4}$$

To calculate the mutual impedance, we need to know V_{21} and I_1. Let the antennas be in the z direction as shown in Fig. 13–10. The emf V_{21} induced at the open terminals of antenna 2 by the current in antenna 1 is given by

$$V_{21} = \frac{1}{I_2} \int_0^L I_z E_{21}\, dz \tag{5}$$

where I_2 is the maximum current and I_z the value at a distance z from the lower end of antenna 2 with its terminals closed, and where E_{21} is the electric field along antenna 2 produced by the current in antenna 1. Assuming that this current distribution is sinusoidal as given by

$$I_z = I_2 \sin \beta z \tag{6}$$

so that (5) becomes

$$V_{21} = \int_0^L E_{21} \sin \beta z\, dz \tag{7}$$

then

$$Z_{21} = \frac{-V_{21}}{I_1} = -\frac{1}{I_1} \int_0^L E_{21} \sin \beta z\, dz \tag{8}$$

This is the general expression for the mutual impedance of two thin linear, parallel, center-fed antennas with sinusoidal current distribution. We will consider first the situation where both antennas are the same length L, where L is an odd number of $\lambda/2$ long

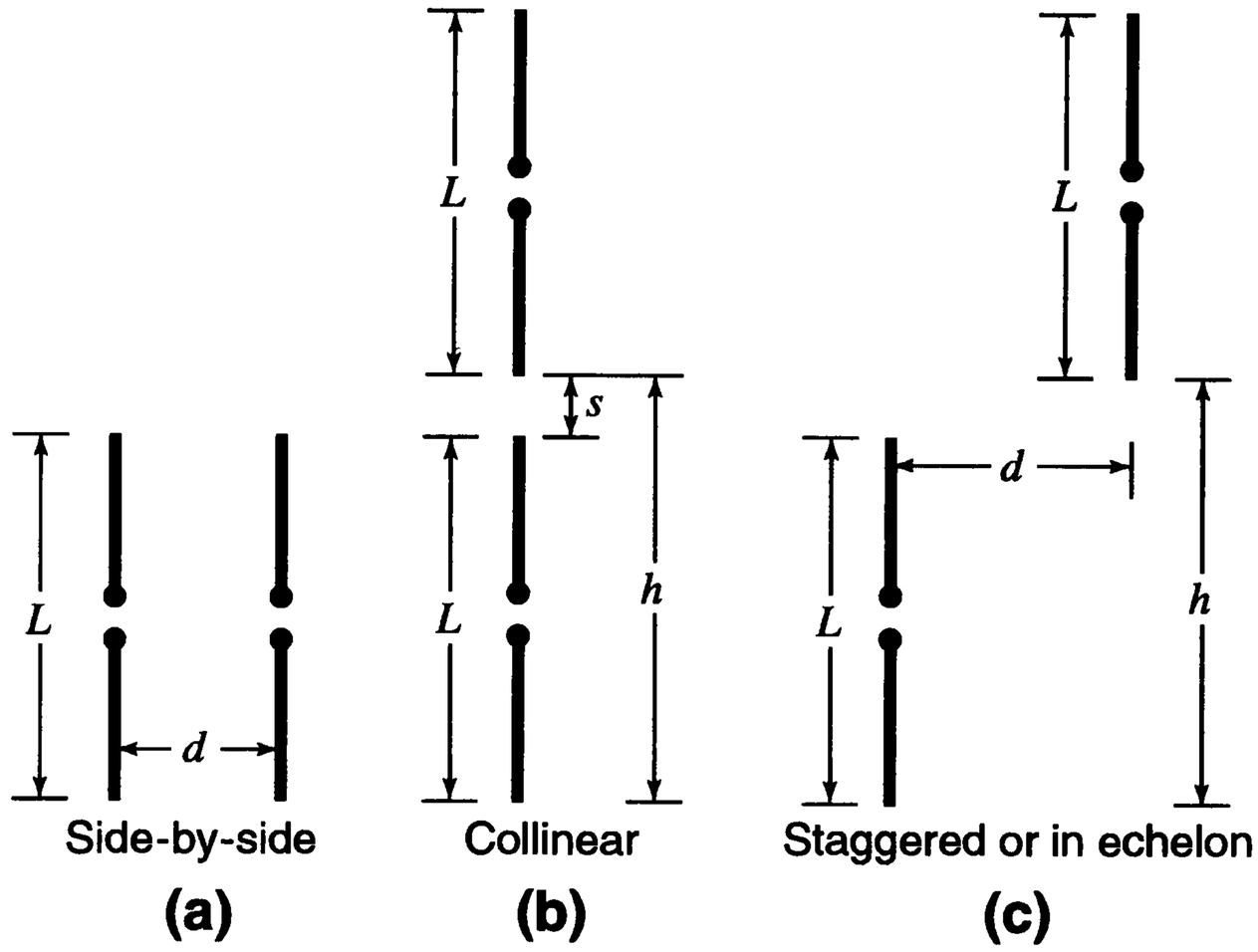


Figure 13–11
Three arrangements of two parallel antennas.

($L = n\lambda/2$; $n = 1, 3, 5, \ldots$). A case of particular interest is where both antennas are $\lambda/2$ long ($n = 1$). The relative positions of the antennas may be divided into three situations: side-by-side, collinear or end-to-end, and staggered or in echelon. These arrangements are illustrated in Fig. 13–11. Mutual impedance expressions for the three arrangements are given in the following sections.

13–7 MUTUAL IMPEDANCE OF PARALLEL ANTENNAS SIDE-BY-SIDE
（边靠边平行天线的互阻抗）

Let d be separation of the antennas. Referring to the arrangement of Fig. 13–11*a* and Fig. 13–12, we have

$$r_1 = \sqrt{d^2 + z^2} \tag{1}$$

and

$$r_2 = \sqrt{d^2 + (L - z)^2} \tag{2}$$

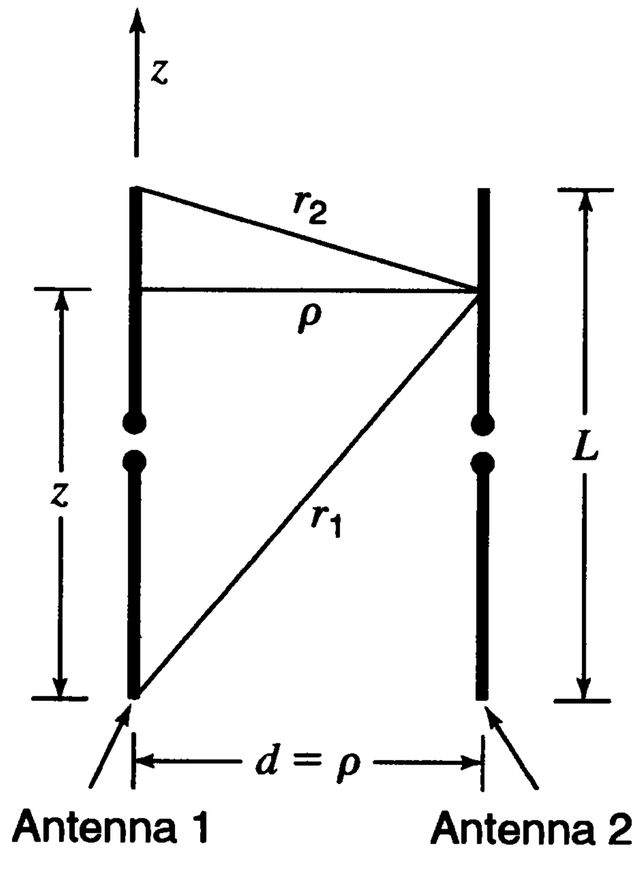


Figure 13–12
Parallel coupled antennas with dimensions.

The mutual impedance then becomes

$$Z_{21} = j30 \int_0^L \left\{ \frac{\exp(-j\beta\sqrt{d^2+z^2})}{\sqrt{d^2+z^2}} + \frac{\exp\left[-j\beta\sqrt{d^2+(L-z)^2}\right]}{\sqrt{d^2+(L-z)^2}} \right\} \sin \beta z \, dz \tag{3}$$

Carter has shown that upon integration of (3),

$$Z_{21} = 30\left\{2\,\mathrm{Ei}(-j\beta d) - \mathrm{Ei}\left[-j\beta\left(\sqrt{d^2+L^2}+L\right)\right] - \mathrm{Ei}\left[-j\beta\left(\sqrt{d^2+L^2}-L\right)\right]\right\} \quad (\Omega) \tag{4}$$

where the exponential integral

$$\mathrm{Ei}(\pm jy) = \mathrm{Ci}(y) \pm j\,\mathrm{Si}(y) \tag{5}$$

Thus, the mutual resistance is

$$R_{21} = 30\left\{2\,\mathrm{Ci}(\beta d) - \mathrm{Ci}\left[\beta\left(\sqrt{d^2+L^2}+L\right)\right] - \mathrm{Ci}\left[\beta\left(\sqrt{d^2+L^2}-L\right)\right]\right\} \quad (\Omega) \tag{6}$$

and the mutual reactance is

$$X_{21} = -30\left\{2\,\mathrm{Si}(\beta d) - \mathrm{Si}\left[\beta\left(\sqrt{d^2+L^2}+L\right)\right] - \mathrm{Si}\left[\beta\left(\sqrt{d^2+L^2}-L\right)\right]\right\} \quad (\Omega) \tag{7}$$

where

$$R_{21} + jX_{21} = Z_{21} = Z_{12} = R_{12} + jX_{12} \tag{8}$$

and where $L = n\lambda/2$ for n odd.

The mutual resistance and reactance calculated by (6) and (7) for the case of $\lambda/2$ antennas ($L = \lambda/2$) are presented by the solid curves in Fig. 13–13 as a function of the spacing d. The mutual resistance R_{21} is also listed in Table 13–1.

An integral-equation method for the calculation of the mutual impedance of linear antennas has been presented by King (1) and Harrison and by Tai (1). The method is related to that discussed in Chap. 14. In this method the diameter of the antenna conductor is a factor. By way of comparison, curves for the mutual resistance and reactance given by Tai are also shown in Fig. 13–12. The dashed curves are for a total length-diameter ratio (L/D) of 11,000 (very thin antenna) and the dotted curves for a ratio of 73.

In Table 13–1 the quantity $R_{11} - R_{21}$, which is important in array calculations, is also tabulated. When d is small, it has been shown by Brown (2) that this quantity is given approximately by the simple relation

$$R_{11} - R_{21} = 60\pi^2\left(\frac{d}{\lambda}\right)^2 = 592.2\left(\frac{d}{\lambda}\right)^2 \quad (\Omega) \tag{9}$$

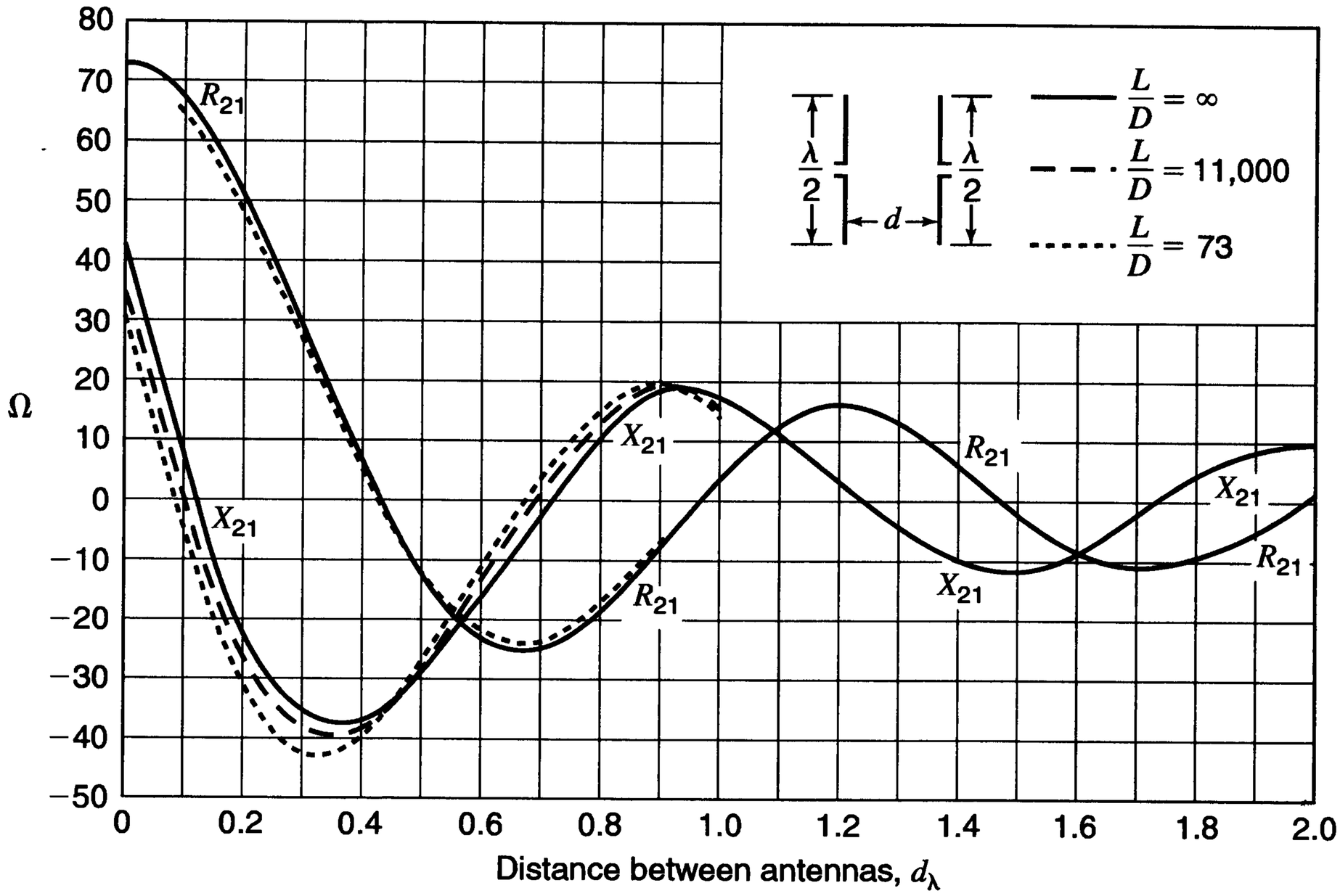


Figure 13–13
Curves of mutual resistance (R_{21}) and reactance (X_{21}) of two parallel side-by-side linear $\lambda/2$ antennas as a function of distance between them. Solid curves are for infinitesimally thin antennas as calculated from Carter's formulas. Dashed and dotted curves between 0 and 1.0λ spacing are from Tai's data for antennas with L/D ratios of 11,000 and 73, respectively.

where λ = the free-space wavelength. This relation is accurate to within 1 percent when $d \leq 0.05\lambda$ and to within about 5 percent when $d \leq 0.1\lambda$.

In the more general situation where the antenna length L is not restricted to an odd number of $\lambda/2$, the mutual resistance and reactance are given by Brown (1) and King as

$$R_{21} = 30\frac{1}{\sin^2(\beta L/2)}\left\{2(2+\cos\beta L)\,\mathrm{Ci}\,\beta d\right.$$

$$-4\cos^2\frac{\beta L}{2}\left[\mathrm{Ci}\,\frac{\beta}{2}\left(\sqrt{4d^2+L^2}-L\right)+\mathrm{Ci}\,\frac{\beta}{2}\left(\sqrt{4d^2+L^2}+L\right)\right]$$

$$+\cos\beta L\left[\mathrm{Ci}\,\beta\left(\sqrt{d^2+L^2}-L\right)+\mathrm{Ci}\,\beta\left(\sqrt{d^2+L^2}+L\right)\right]$$

$$+\sin\beta L\left[\mathrm{Si}\,\beta\left(\sqrt{d^2+L^2}+L\right)-\mathrm{Si}\,\beta\left(\sqrt{d^2+L^2}-L\right)\right.$$

$$\left.\left.-2\,\mathrm{Si}\,\frac{\beta}{2}\left(\sqrt{4d^2+L^2}+L\right)+2\,\mathrm{Si}\,\frac{\beta}{2}\left(\sqrt{4d^2+L^2}-L\right)\right]\right\} \qquad (\Omega) \qquad (10)$$

Table 13–1 Mutual resistance versus spacing for thin center-fed side-by-side $\lambda/2$ antennas ($\beta L = 180°$), with sinusoidal current distribution

Spacing d, λ	Mutual resistance R_{21}, Ω	Self minus mutual resistance $(R_{11} - R_{21})$, Ω
0.00	73.13	0.00
0.01	73.07	0.06
0.05	71.65	1.48
0.10	67.5	5.63
0.125	64.4	8.7
0.15	60.6	12.5
0.20	51.6	21.5
0.25	40.9	32.2
0.3	29.4	43.7
0.4	+6.3	66.8
0.5	−12.7	85.8
0.6	−23.4	96.5
0.7	−24.8	97.9
0.8	−18.6	91.7
0.9	−7.2	80.3
1.0	+3.8	69.3
1.1	+12.1	61.0
1.2	+15.8	57.3
1.3	+12.4	60.7
1.4	+5.8	67.3
1.5	−2.4	75.5
1.6	−8.3	81.4
1.7	−10.7	83.8
1.8	−9.4	82.5
1.9	−4.8	77.9
2.0	+1.1	72.0

Note that at a spacing $d = 0.125\lambda$ between the dipoles, the total resistance (self − mutual) is only 8.7 Ω versus 73 Ω for a single $\lambda/2$ antenna. At smaller spacings the total resistance decreases rapidly. In the presence of any appreciable losses, the result is loss of gain and low efficiency. This is discussed in more detail in Sec. 16–5 on "Closely Spaced Elements and Radiating Efficiency."

and

$$X_{21} = 30\frac{1}{\sin^2(\beta L/2)}\left\{-2(2+\cos\beta L)\,\mathrm{Si}\,\beta d\right.$$
$$+4\cos^2\frac{\beta L}{2}\left[\mathrm{Si}\,\frac{\beta}{2}\left(\sqrt{4d^2+L^2}-L\right)+\mathrm{Si}\,\frac{\beta}{2}\left(\sqrt{4d^2+L^2}+L\right)\right]$$
$$-\cos\beta L\left[\mathrm{Si}\,\beta\left(\sqrt{d^2+L^2}-L\right)+\mathrm{Si}\,\beta\left(\sqrt{d^2+L^2}+L\right)\right]$$
$$+\sin\beta L\left[\mathrm{Ci}\,\beta\left(\sqrt{d^2+L^2}+L\right)-\mathrm{Ci}\,\beta\left(\sqrt{d^2+L^2}-L\right)\right.$$
$$\left.\left.-2\,\mathrm{Ci}\,\frac{\beta}{2}\left(\sqrt{4d^2+L^2}+L\right)+2\,\mathrm{Ci}\,\frac{\beta}{2}\left(\sqrt{4d^2+L^2}-L\right)\right]\right\} \qquad (\Omega) \qquad (11)$$

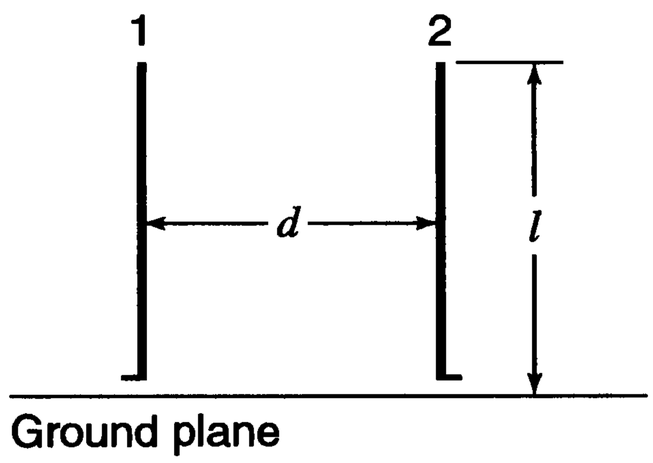


Figure 13–14
Two coupled linear parallel stub antennas.

In the special case of $L = n\lambda/2$, where n is odd, (10) and (11) reduce to the relations given previously by (6) and (7).

The above relations of this section apply to balanced center-fed antennas. The mutual impedance of two stub antennas of height $l = L/2$ above an infinite, perfectly conducting ground plane as in Fig. 13–14 is $\frac{1}{2}$ that given by (6) and (7) or (10) and (11). These relations are converted to the ground-plane case by changing the factor 30 to 15 and making the substitution $L = 2l$.

13–8 MUTUAL IMPEDANCE OF PARALLEL COLLINEAR ANTENNAS
（平行共线天线的互阻抗）

Let each antenna be an odd number of $\lambda/2$ long and arranged as in Fig. 13–11*b*. For the case where h is greater than L, Carter (1) gives the mutual resistance and reactance as

$$R_{21} = -15\cos\beta h\left[-2\,\mathrm{Ci}\,2\beta h + \mathrm{Ci}\,2\beta(h-L) + \mathrm{Ci}\,2\beta(h+L) - \ln\left(\frac{h^2-L^2}{h^2}\right)\right] + 15\sin\beta h[2\,\mathrm{Si}\,2\beta h - \mathrm{Si}\,2\beta(h-L) - \mathrm{Si}\,2\beta(h+L)] \quad (\Omega) \tag{1}$$

and

$$X_{21} = -15\cos\beta h[2\,\mathrm{Si}\,2\beta h - \mathrm{Si}\,2\beta(h-L) - \mathrm{Si}\,2\beta(h+L)] + 15\sin\beta h \times \left[2\,\mathrm{Ci}\,2\beta h - \mathrm{Ci}\,2\beta(h-L) - \mathrm{Ci}\,2\beta(h+L) - \ln\left(\frac{h^2-L^2}{h^2}\right)\right] \quad (\Omega) \tag{2}$$

Curves for R_{21} and X_{21} of parallel collinear $\lambda/2$ antennas ($L = \lambda/2$) are presented in Fig. 13–15 as a function of the spacing s where $s = h - L$ (see Fig. 13–11*b*).

13–9 MUTUAL IMPEDANCE OF PARALLEL ANTENNAS IN ECHELON
（梯式平行天线的互阻抗）

For this case the antennas are staggered or in echelon as in Fig. 13–11*c*. Each antenna is an odd number of $\lambda/2$ long. The mutual resistance and reactance of two such antennas are given by Carter (1) and King (1) as

$$R_{21} = -15\cos\beta h(-2\,\mathrm{Ci}\,A - 2\,\mathrm{Ci}\,A' + \mathrm{Ci}\,B + \mathrm{Ci}\,B' + \mathrm{Ci}\,C + \mathrm{Ci}\,C') + 15\sin\beta h(2\,\mathrm{Si}\,A - 2\,\mathrm{Si}\,A' - \mathrm{Si}\,B + \mathrm{Si}\,B' - \mathrm{Si}\,C + \mathrm{Si}\,C') \quad (\Omega) \tag{1}$$

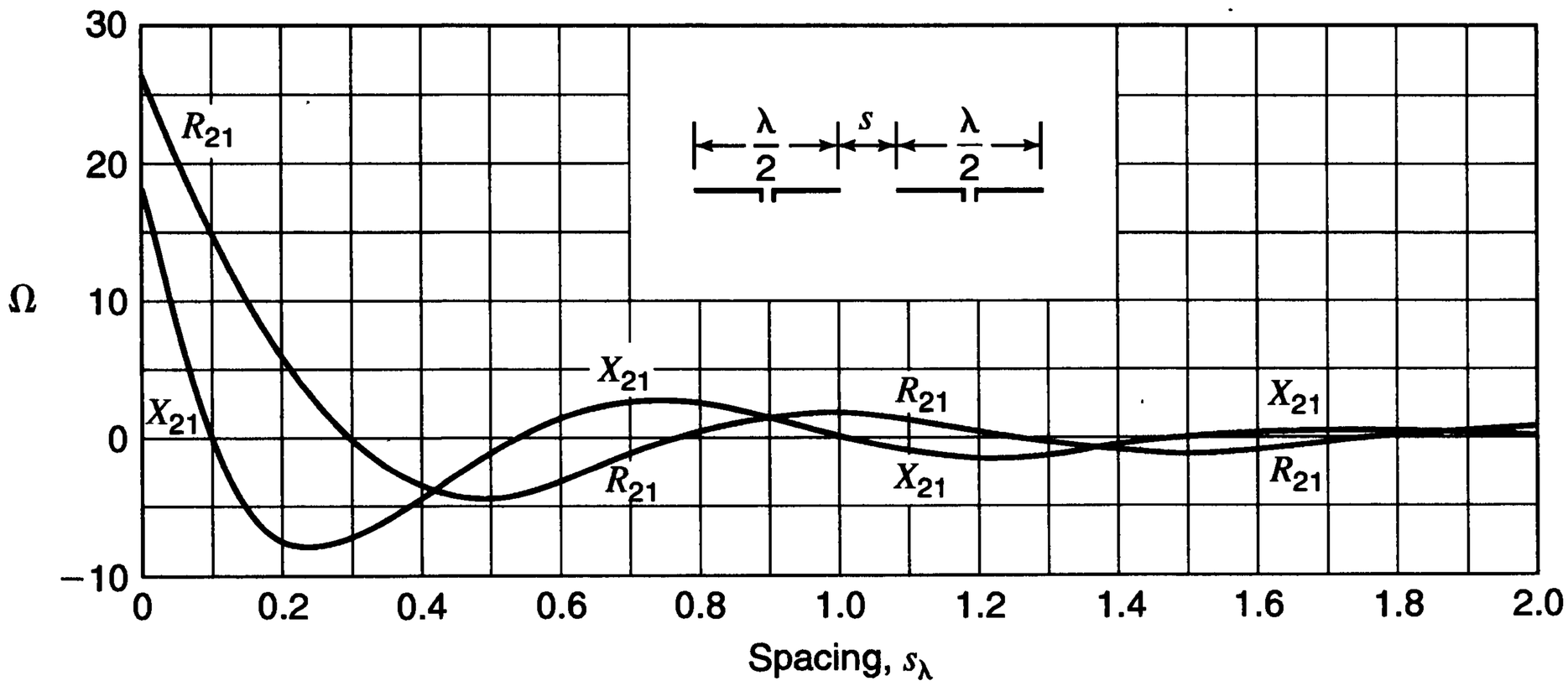


Figure 13–15
Curves of mutual resistance (R_{21}) and reactance (X_{21}) of two parallel collinear infinitesimally thin $\lambda/2$ antennas as a function of the spacing s between adjacent ends.

and

$$X_{21} = -15\cos\beta h(2\,\mathrm{Si}\,A + 2\,\mathrm{Si}\,A' - \mathrm{Si}\,B - \mathrm{Si}\,B' - \mathrm{Si}\,C - \mathrm{Si}\,C')$$
$$+ 15\sin\beta h(2\,\mathrm{Ci}\,A - 2\,\mathrm{Ci}\,A' - \mathrm{Ci}\,B + \mathrm{Ci}\,B' - \mathrm{Ci}\,C + \mathrm{Ci}\,C') \qquad (\Omega) \qquad (2)$$

where

$$A = \beta\left(\sqrt{d^2 + h^2} + h\right)$$
$$A' = \beta\left(\sqrt{d^2 + h^2} - h\right)$$
$$B = \beta\left[\sqrt{d^2 + (h - L)^2} + (h - L)\right]$$
$$B' = \beta\left[\sqrt{d^2 + (h - L)^2} - (h - L)\right]$$
$$C = \beta\left[\sqrt{d^2 + (h + L)^2} + (h + L)\right]$$
$$C' = \beta\left[\sqrt{d^2 + (h + L)^2} - (h + L)\right]$$

Values of the mutual resistance in ohms as calculated from (1) are listed in Table 13–2 (Pistolkors-1) as a function of d and h for the case where the antennas are $\lambda/2$ long ($L = \lambda/2$), as indicated in Fig. 13–16.

The staggered or echelon arrangement is the more general situation of which the side-by-side position (Sec. 13–7) and the collinear position (Sec. 13–8) are special cases.

13–10 MUTUAL IMPEDANCE OF OTHER CONFIGURATIONS
（其他布置天线的互阻抗）

There are many other antenna configurations for which the mutual impedance may be of interest. The variety is enormous, but two will be mentioned and references given which the reader may consult for further information.

1. *Parallel antennas of unequal height.* This case has been treated by Cox (1). His data apply specifically to stub antennas perpendicular to an infinite, perfectly conducting

Table 13–2 Mutual resistance as a function of *d* and *h* (Fig. 13–16) for thin $\lambda/2$ antennas in echelon

	Spacing *h*						
Spacing *d*	**0.0λ**	**0.5λ**	**1.0λ**	**1.5λ**	**2.0λ**	**2.5λ**	**3.0λ**
0.0λ	+73.1	+26.4	−4.1	+1.8	−1.0	+0.6	−0.4
0.5λ	−12.7	−11.8	−0.8	+0.8	−1.0	+0.5	−0.3
1.0λ	+3.8	+8.8	+3.6	−2.9	+1.1	−0.4	+0.1
1.5λ	−2.4	−5.8	−6.3	+2.0	+0.6	−1.0	+0.9
2.0λ	+1.1	+3.8	+6.1	+0.2	−2.6	+1.6	−0.5
2.5λ	−0.8	−2.8	−5.7	−2.4	+2.7	−0.3	−0.1
3.0λ	+0.4	+1.9	+4.5	+3.2	−2.1	−1.6	+1.7
3.5λ	−0.3	−1.5	−3.9	−3.8	+0.7	+2.7	−1.0
4.0λ	+0.2	+1.1	+3.1	+3.7	+0.5	−2.5	−0.1
4.5λ	−0.2	−0.9	−2.5	−3.4	−1.3	+2.0	+1.1
5.0λ	+0.2	+0.7	+2.1	+3.1	+1.8	−1.4	−1.9
5.5λ	−0.1	−0.6	−1.8	−2.9	−2.2	+0.5	+1.8
6.0λ	+0.1	+0.5	+1.6	+2.6	+2.3	−0.1	−2.0
6.5λ	−0.1	−0.5	−1.2	−2.3	−2.3	−0.5	+1.7
7.0λ	+0.1	+0.4	+1.1	+2.1	+2.3	+0.9	−1.3
7.5λ	0.0	−0.3	−1.0	−1.9	−2.1	−1.0	+0.7

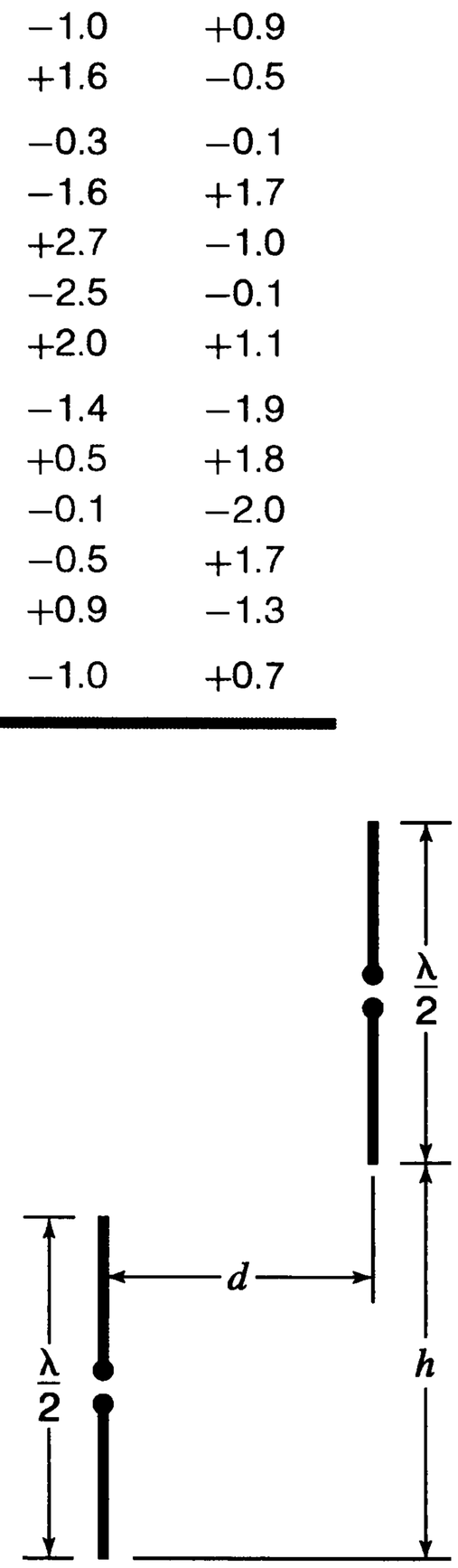


Figure 13–16
Two parallel linear $\lambda/2$ antennas in echelon.

ground, but can be used with symmetrical center-fed antennas of twice the length by multiplying the resistance and reactance values by 2 (see also Howard E. King-1).

2. *V or skew antennas.* Some antenna systems involve nonparallel linear radiators. The mutual impedances of such inclined antennas are readily calculated by the moment method, as, for example, by J. H. Richmond's FORTRAN IV program, ASIS-NAPS Document NAPS-02223.

For MININEC programs, see the book's web site "antennas3.com."
For two short dipoles, however, a simple, useful relation can be derived as follows.

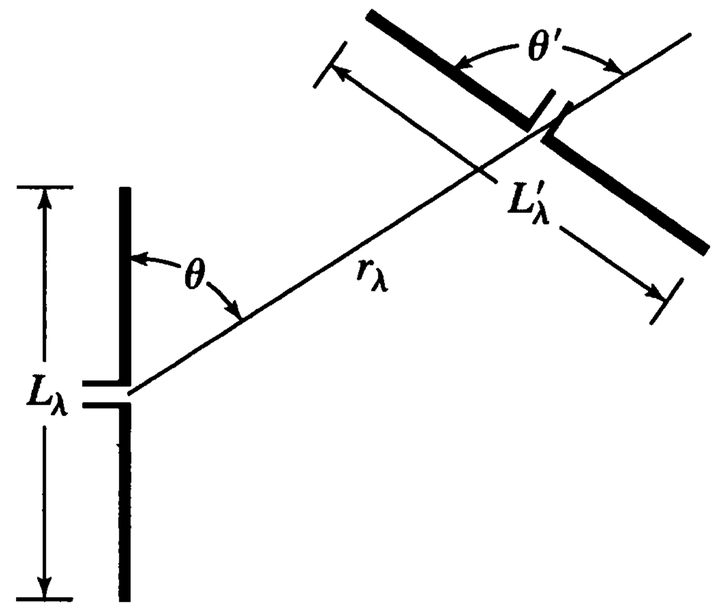


Figure 13–17
Two short center-fed dipoles of length L_λ and L'_λ with separation r_λ and orientation angles θ and θ' for mutual impedance equation.

Referring to Fig. 13–17, consider the 2 short center-fed dipoles 1 and 2 of length L_λ and L'_λ separated by a distance r_λ with orientation angles θ and θ' as indicated. The mutual impedance Z_{21} is given by the ratio of the voltage V_{21} induced in dipole 2 by the current I_1 flowing in dipole 1. Then for $L_\lambda \ll 1$, $L'_\lambda \ll 1$ and $r_\lambda \gg 1$, we have, from (6–2–17),

$$Z_{21} = \frac{V_{21}}{I_1} = \frac{60\pi L_\lambda}{r_\lambda} \sin\theta\, e^{-j\beta r} L'_\lambda \sin\theta' \tag{1}$$

or

Magnitude Orientation Periodic function

$$Z_{21} = \frac{60\pi L_\lambda L'_\lambda}{r_\lambda} (\sin\theta \sin\theta')(\sin 2\pi r_\lambda + j \cos 2\pi r_\lambda) \tag{2}$$

with maximum value

$$Z_{21}(\text{max}) = \frac{60\pi L_\lambda L'_\lambda}{r_\lambda} \qquad (\Omega) \tag{3}$$

We note in (2) that there are 3 factors: the first is a *magnitude factor* involving the lengths of the dipoles and their separation, the second involves their mutual *orientation,* while the third factor is a *periodic or complex function* of unit magnitude giving the phase as a function of the separation distance. The mutual impedance for antennas, in general, involves these 3 factors.

13–11 MUTUAL IMPEDANCE IN TERMS OF DIRECTIVITY AND RADIATION RESISTANCE（用定向性和辐射电阻表示的互阻抗）

Consider a transmitting antenna and a receiving antenna separated by a distance r. The power delivered to the receiving antenna load under matched conditions is

$$P_r = \frac{(V_r/2)^2}{R_r} \qquad (\text{W}) \tag{1}$$

where

V_r = voltage induced at terminals of receiving antenna, V
R_r = radiation resistance of receiving antenna, Ω

The power transmitted is given by

$$P_t = I_t^2 R_t \qquad \text{(W)} \tag{2}$$

where

I_t = current at terminals of transmitting antenna, A
R_t = radiation resistance of transmitting antenna, Ω

From (1) and (2), the magnitude of the mutual impedance of the antennas is

$$|Z_m| = \frac{V_r}{I_t} = \frac{2\sqrt{P_r R_r}}{\sqrt{P_t/R_t}} = 2\sqrt{\frac{P_r}{P_t}}\sqrt{R_r R_t} \tag{3}$$

From the Friis transmission formula (2–11–5),

$$\frac{P_r}{P_t} = \frac{A_{er} A_{et}}{r^2 \lambda^2} \tag{4}$$

where

A_{er} = effective aperture of receiving antenna, m^2
A_{et} = effective aperture of transmitting antenna, m^2

Introducing (4) in (3), the maximum mutual impedance becomes

$$Z_m(\text{max}) = \frac{2\sqrt{A_{er} A_{et} R_r R_t}}{r\lambda} \tag{5}$$

However, the directivity $D = 4\pi A_e/\lambda^2$, so the *maximum mutual impedance* can be re-expressed as

$$|Z_m|(\text{max}) = \frac{\sqrt{D_r D_t R_r R_t}}{2\pi r_\lambda} \qquad (\Omega) \tag{6}$$

where

D_r = directivity of receiving antenna, dimensionless
D_t = directivity of transmitting antenna, dimensionless
R_r = radiation resistance of receiving antenna, Ω
R_t = radiation resistance of transmitting antenna, Ω
r_λ = separation, wavelengths

Thus, the mutual impedance of two antennas is a function of their directivities (or apertures) and radiation resistances. In the above, the restriction applies that the separation distance is large compared to the antenna dimensions and also that $r \gg \lambda$.

EXAMPLE 13–11.1

Calculate the maximum mutual impedance of 2 center-fed dipole antennas 0.1λ long separated by 10λ. Assume uniform current distribution on the dipoles. The receiving antenna is terminated for maximum power transfer.

■ **Solution**

From (9) of Example 2–10.1

$$|Z_m|(\max) = \frac{1.5 \times 0.8\pi^2}{2\pi \times 10} = 0.1885\ \Omega$$

From (13–10–3)

$$Z_{21}(\max) = \frac{60\pi \times 0.1^2}{10} = 0.1885\ \Omega \qquad \textit{Ans.}$$

Although (6) and (13–10–3) bear little resemblance, except for the r_λ in the denominator, both yield an identical impedance!

REFERENCES（参考文献）

Ballantine, Stuart (1), "Reciprocity in Electromagnetic, Mechanical, Acoustical, and Interconnected Systems," *Proc. IRE,* **17,** 929–951, June 1929.

Brown, G. H. (1), and R. King, "High Frequency Models in Antenna Investigations," *Proc. IRE,* **22,** 457–480, April 1934.

Brown, G. H. (2), personal communication to John Kraus, June 16, 1938.

Carson, J. R. (1), "A Generalization of the Reciprocal Theorem," *Bell System Tech. J.,* **3,** 393–399, July 1924.

Carson, J. R. (2), "Reciprocal Theorems in Radio Communication," *Proc. IRE,* **17,** 952–956, June 1929.

Carter, P. S. (1), "Circuit Relations in Radiating Systems and Applications to Antenna Problems," *Proc. IRE,* **20,** 1004–1041, June 1932.

Cox, C. R. (1), "Mutual Impedance between Vertical Antennas of Unequal Heights," *Proc. IRE,* **35,** 1367–1370, November 1947.

King, H. E. (1), "Mutual Impedance of Unequal Length Antennas in Echelon," *IEEE Trans. Ants. Prop.,* **AP-5,** 306–313, July 1957.

King, R. (1), and C. W. Harrison, Jr., "Mutual and Self Impedance for Coupled Antennas," *J. Appl. Phys.,* 481–495, June 1944.

Labus, J. (1), "Recherische Ermittlung der Impedanz von Antennen," *Hochfrequenztechnik und Electroakustik,* **17,** January 1933.

Pistolkors, A. A. (1), "The Radiation Resistance of Beam Antennas," *Proc. IRE,* **17,** 562–579, March 1929.

Rayleigh, Lord (1), *The Theory of Sound,* The Macmillan Company, New York, vol. 1 (1877, 1937); pp. 98 and 150–157, and vol. 2 (1878, 1929), p. 145.

Tai, C. T. (1), "Coupled Antennas," *Proc. IRE,* **36,** 487–500, April 1948.

PROBLEMS（习题）

***13–4–1 (5λ/2 天线)** **A $5\lambda/2$ antenna.** Calculate the self-resistance and self-reactance of a thin, symmetrical center-fed linear antenna $5\lambda/2$ long.

13–6–1 (平行边靠边 λ/2 天线) **Parallel side-by-side λ/2 antennas.** Calculate the mutual resistance and mutual reactance for two parallel side-by-side thin linear $\lambda/2$ antennas with a separation of 0.15λ.

13–6–2 (Brown 公式) **Brown's equation.** Prove Brown's relation $R_{11} - R_{21} = 60\pi^2(d/\lambda)^2$ given in (13–7–9).

***13–6–3** (三根边靠边天线) **Three side-by-side antennas.** Three antennas are arranged as shown in Fig. P13–6–3. The currents are of the same magnitude in all antennas. The currents are in-phase in (*a*) and (*c*), but the current in (*b*) is in antiphase. The self-resistance of each antenna is 100 Ω, while the mutual resistances are: $R_{ab} = R_{bc} = 40\ \Omega$ and $R_{ac} = -10\ \Omega$. What is the radiation resistance of each of the antennas? The resistances are referred to the terminals, which are in the same location in all antennas.

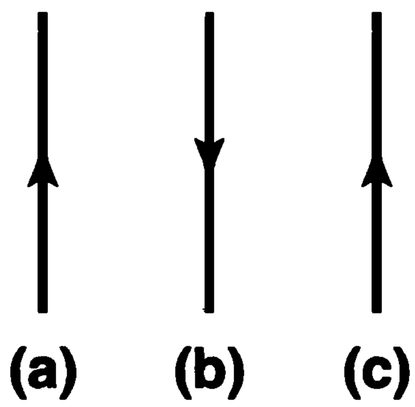


Figure P13–6–3
Three side-by-side antennas.

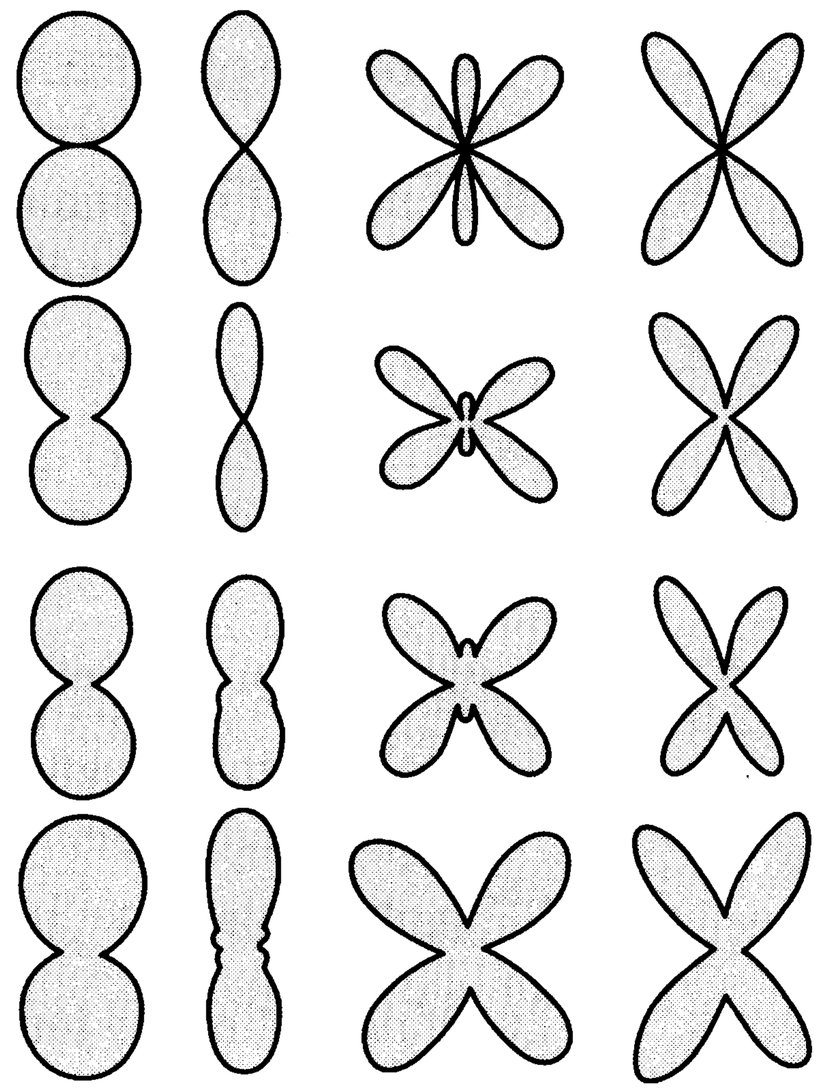
13–8–1 (两根梯式 λ/2 天线) **Two λ/2 antennas in echelon.** Calculate the mutual resistance and reactance of two parallel thin linear $\lambda/2$ antennas in echelon for the case where $d = 0.25\lambda$ and $h = 1.25\lambda$ (see Fig. 13–16).

13–9–1 (自电阻和互电阻) **Self-resistance and mutual resistance.** Explain why the mutual resistance of two antennas can be both positive and negative but the self-resistance of a single antenna can only be positive.

13–10–1 (馈端阻抗) **Terminal Impedance.** (*a*) Show by means of an equivalent network that at the terminals of a receiving antenna, the equivalent or Thévenin generator has an impedance $Z_{22} - (Z_m^2/Z_{11})$ and an emf $V_1 Z_m/Z_{11}$, where Z_{11} = self-impedance of transmitting antenna, Z_{22} = self-impedance of receiving antenna, Z_m = mutual impedance and V_1 = emf applied to terminals of transmitting antenna. (*b*) What load impedance connected to the terminals of the receiving antenna results in the maximum power transfer?

For computer programs, see Appendix C.

CHAPTER 14



The Cylindrical Antenna and the Moment Method (MM)

柱形天线和矩量法

The topics in this chapter include:

- Integral-equation method（积分方程法）
- Current distributions on cylindrical antennas（圆柱天线上的电流分布）
- Input impedance of thick and thin cylinders（粗柱和细柱的输入阻抗）
- Patterns（波瓣图）
- Conical or tapered input sections（圆锥或锥削的输入段）
- Spheroidal antenna（类球体天线）
- Current distribution on long cylindrical antennas（长圆柱天线上的电流分布）
- Charge distribution by the Moment Method [矩量法（MOM）求电荷分布]
- Current distribution and impedance by the Moment Method（矩量法求电流分布和阻抗）
- Self-impedance, mutual impedance and radar cross section using the Moment Method by Edward H. Newman（用矩量法求自阻抗、互阻抗和雷达截面）

14–1 INTRODUCTION（引言）

In previous chapters, the assumption is made that the current distribution on a finite antenna is sinusoidal. This assumption is a good one provided that the antenna is very thin. In this chapter, a method for calculating the current distribution of a cylindrical center-fed antenna is discussed, taking into account the thickness of the antenna conductor.

This is a boundary-value problem. The antenna as a boundary-value problem was treated many years ago by Abraham (1), who obtained an exact solution for a freely oscillating elongated ellipsoid of revolution. However, the earliest treatments of the cylindrical center-driven antenna as a boundary-value problem are those of Hallén (1) and L. V. King (1).

Hallén's method leads to an integral equation, approximate solutions of which yield the current distribution. Knowing the current distribution and the voltage applied at the input terminals, the input impedance is then obtained as the ratio of the voltage to the current at the terminals.

An outline of Hallén's integral-equation method follows in the next section with results given in Sec. 14–3. Later in the chapter the Moment Method (MM) is applied to the solution of the current distribution, impedance and radar cross section of a short dipole antenna.

More details on Hallén's integral-equation method are presented in earlier editions of this book.

14–2 OUTLINE OF THE INTEGRAL-EQUATION METHOD（积分方程法概述）

The *objective* of the method is twofold:

1. To obtain the current distribution of a cylindrical center-fed antenna in terms of its length and diameter.
2. To obtain the input impedance.

An *outline of the procedure* is given by the following steps. These are treated more fully in the sections which follow.

1. The field **E** inside the conductor is expressed in terms of the current and skin effect resistance.
2. The field **E** outside the conductor is expressed in terms of the vector potential.
3. The tangential components of **E** are equated, obtaining a wave equation in the vector potential **A**.
4. The wave equation in **A** is solved as the sum of a complementary function and a particular integral given as

$$A_z = \frac{-j}{c}(C_1 \cos \beta z + C_2 \sin \beta z) + \frac{jZ}{c} \int_0^z I(s) \sin \beta(z - s)\, ds$$

5. The constant C_2 in the solution is evaluated in terms of the conditions at the input terminals.
6. The vector potential **A** is expressed in terms of the antenna current I.
7. The value of C_2 from 5 and of **A** from 6 are inserted in the solution 4, obtaining Hallén's integral equation. This is an integral equation in the current I.
8. A partial solution for the current I is then obtained by evaluating one of the integrals so that the current is expressed as the sum of several terms, some of which also involve I.
9. Neglecting certain terms in I, an approximate (zero-order) solution is obtained for I.
10. This value of I is substituted back in the current equation, obtaining a first-order approximation for the current. This process of iteration can be continued, yielding second-order and higher-order solutions.
11. The constant C_1 is evaluated and an asymptotic expansion obtained for the current; that is,

$$I_z = \frac{jV_T}{60\Omega'} \left[\frac{\sin \beta(l - |y|) + (b_1/\Omega') + (b_2/\Omega'^2) + \cdots}{\cos \beta l + (d_1/\Omega') + (d_2/\Omega'^2) + \cdots} \right] \tag{1}$$

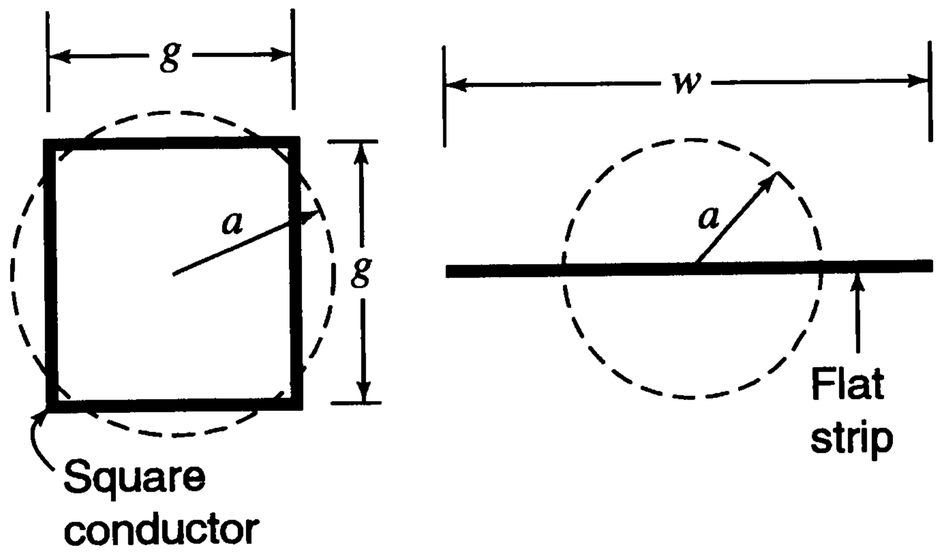


Figure 14–1
Conductors of square and flat cross section with equivalent circular conductors of radius a.

where $\Omega' = 2\ln(2l/a)$, where l is the half-length of the antenna and a the radius. The first-order approximation involves terms only as high as b_1/Ω' and d_1/Ω'. A second-order approximation involves b_2/Ω'^2 and d_2/Ω'^2, etc.

12. The input impedance is then obtained as the ratio of the input terminal voltage V_T to the current at the input terminals I_T. This is discussed in Sec. 14–4.

It is assumed that $l \gg a$ and $\beta a \ll l$. The condition that $l \gg a$ will be arbitrarily taken to mean that

$$\frac{l}{a} \geq 60$$

The ratio l/a equals the ratio of the total length of the cylindrical antenna to the diameter. Thus,

$$\frac{\text{Total length}}{\text{Diameter}} = \frac{2l}{2a} = \frac{l}{a}$$

The above discussion deals with uniform cylindrical antennas, i.e., antennas of circular cross section (radius $= a$). According to Hallén (3), uniform antennas with noncircular cross section can also be treated by taking an equivalent radius. For *square cross sections* of side length g (Fig. 14–1), the equivalent radius is

$$a = 0.59g$$

while for *thin flat strips* of width w the equivalent radius is

$$a = 0.25w$$

For any shape of cross section there exists equivalent radius. In all cases it is assumed that the cross section is uniform over the entire length of the antenna.

14–3 CURRENT DISTRIBUTIONS（电流分布）

The amplitude and phase of the current along cylindrical antennas of three lengths and two values of the total length-diameter ratio (l/a) are presented in Figs. 14–2, 14–3 and 14–4. Figure 14–2 is for a $\lambda/2$ antenna ($2l = \lambda/2$), Fig. 14–3 for a full-wavelength antenna ($2l = \lambda$) and Fig. 14–4 for a $5\lambda/4$ antenna ($2l = 5\lambda/4$). For each length the relative amplitude and phase of the current are presented for $\Omega' = 10$ and $\Omega' = \infty$ corresponding to total length-diameter ratios (l/a) of 75 and ∞. The amplitude curves are adjusted to the

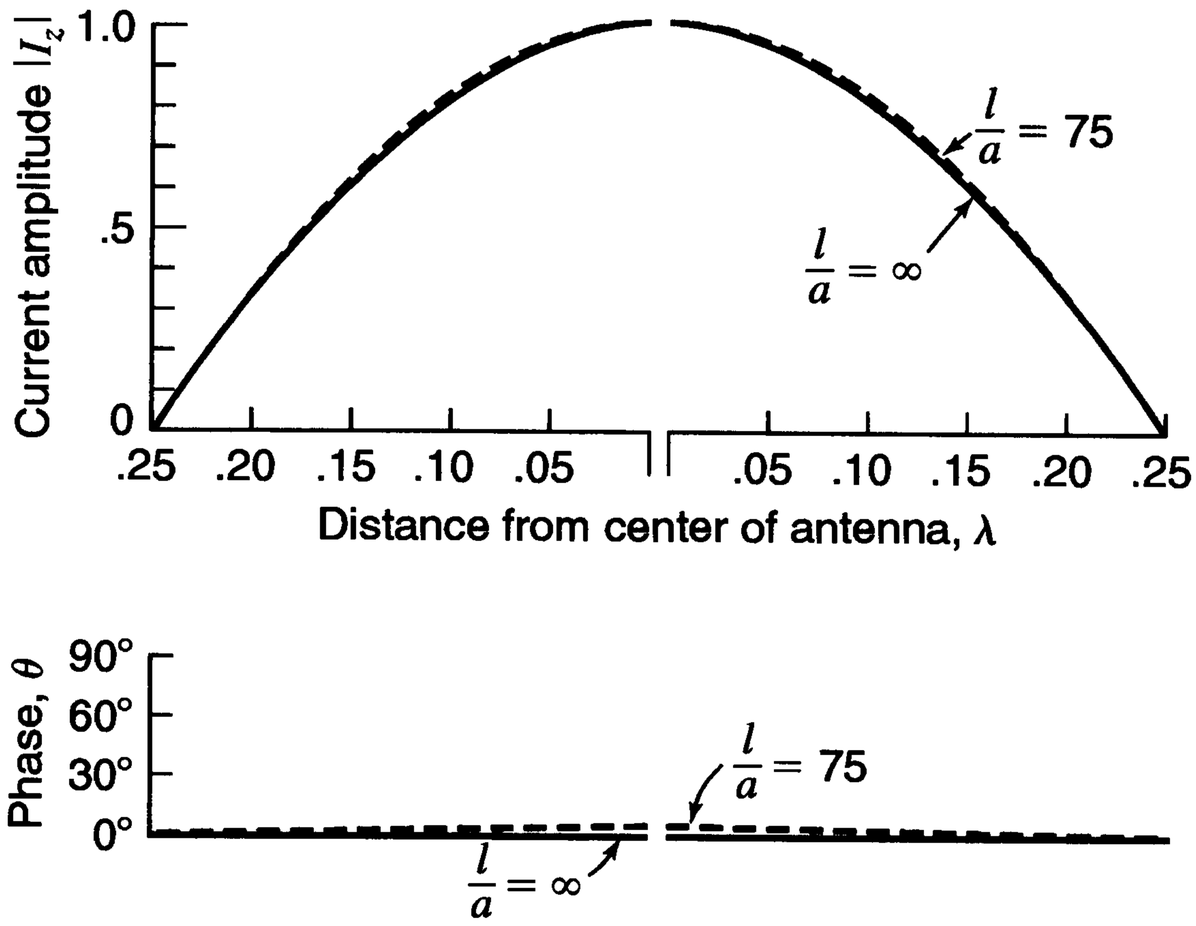


Figure 14–2
Relative current amplitude and phase along a center-fed $\lambda/2$ cylindrical antenna ($2l = \lambda/2$) for total length-diameter ratios (l/a) of 75 and infinity. (*R. King (1) and C. W. Harrison, Jr.*) Distance from the center of the antenna is expressed in wavelengths.

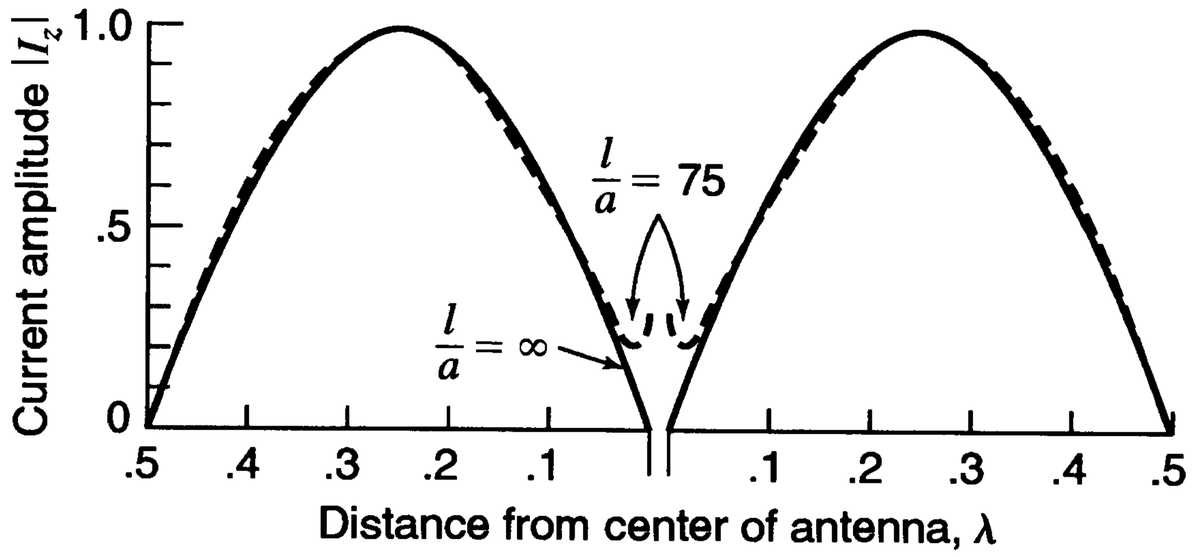
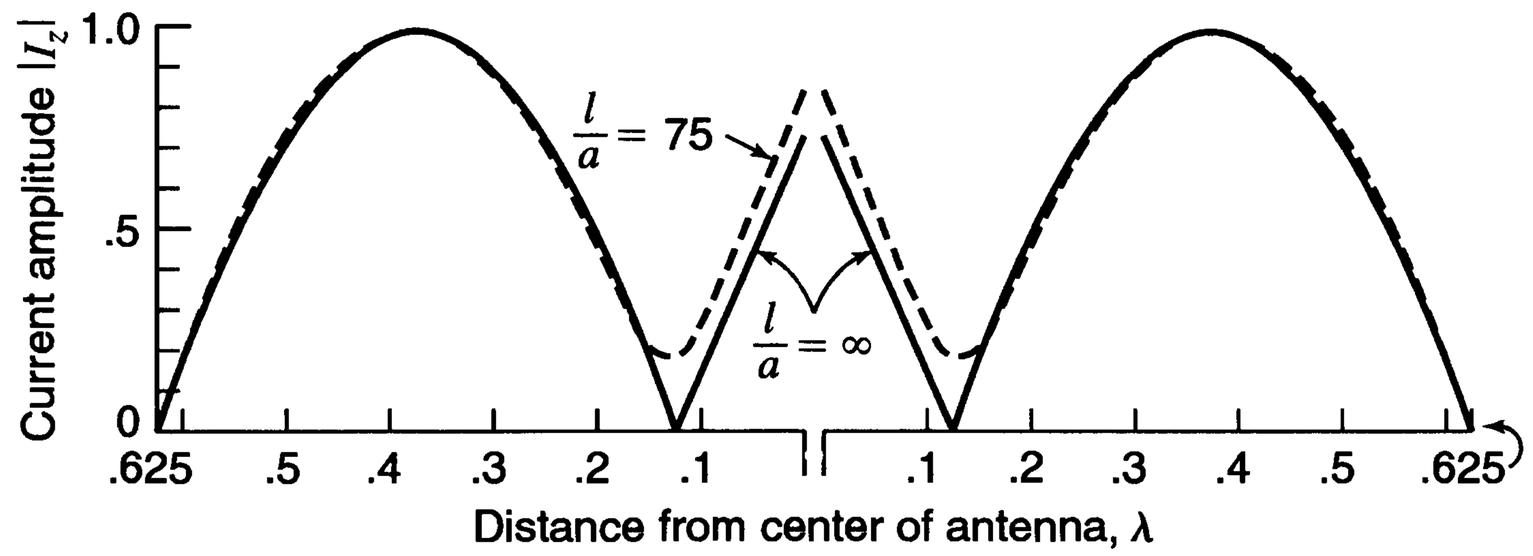


Figure 14–3
Relative current amplitude and phase along a center-fed full-wavelength cylindrical antenna ($2l = \lambda$) for total length-diameter ratios (l/a) of 75 and infinity. (*After R. King (1) and C. W. Harrison, Jr.*) Distance from the center of the antenna is expressed in wavelengths.

same maximum value, and all phase curves are adjusted to the same value at the ends of the antenna.

It is generally assumed that the current distribution of an infinitesimally thin antenna ($l/a = \infty$) is sinusoidal, and that the phase is constant over a $\lambda/2$ interval, changing abruptly by 180° between intervals. This behavior is illustrated by the solid curves in Figs. 14–2, 14–3 and 14–4.

The dashed curves illustrate the current amplitude and phase variation for $l/a = 75$ ($\Omega' = 10$). The difference between these curves and the solid curves ($l/a = \infty$) is not large but is appreciable. The dashed curves ($l/a = 75$) are from the distributions given by King (1) and Harrison, the current being expressed in terms of its amplitude and the phase



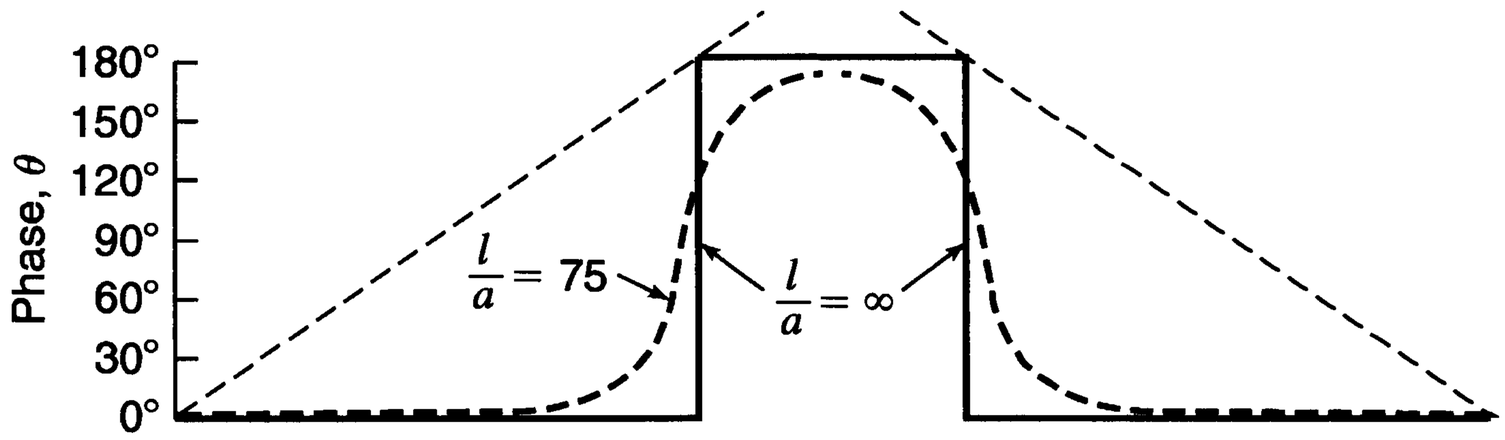


Figure 14–4
Relative current amplitude and phase along a center-fed $5\lambda/4$ cylindrical antenna ($2l = 5\lambda/4$) for total length-diameter ratios (l/a) of 75 and infinity. (*R. King (1) and C. W. Harrison, Jr.*) Distance from the center of the antenna is expressed in wavelengths.

angle relative to a reference point. Thus,

$$I_z = |I_z| \angle \theta \tag{1}$$

The effect of the length-thickness ratio on the current amplitude is well illustrated by Fig. 14–3 for a full-wavelength antenna. When the antenna is infinitesimally thin, the current is zero at the center. As the antenna becomes thicker, the current minimum increases and at the same time shifts slightly toward the end of the antenna.

The effect of the length-thickness ratio on the phase variation is well illustrated by Fig. 14–4 for a $5\lambda/4$ antenna. When the antenna is infinitesimally thin, the phase varies as a step function, being constant over $\lambda/2$ and changing by 180° at the end of the $\lambda/2$ interval (solid line, Fig. 14–4). This type of phase variation is observed in a pure standing wave. As the antenna becomes thicker, the phase shift at the end of the $\lambda/2$ interval tends to become less abrupt (dashed curve for $l/a = 75$). For still thicker antennas ($l/a < 75$), it might be expected that this trend would continue and for very thick antennas would tend to approach that of a pure traveling wave, as indicated by the straight dashed lines in Fig. 14–4.

14–4 INPUT IMPEDANCE（输入阻抗）

The input impedance Z_T of a center-fed cylindrical antenna is found by taking the ratio of the input or terminal voltage V_T and the current I_T at the input terminals; that is,

$$Z_T = \frac{V_T}{I_T} = R_T + jX_T \tag{1}$$

where

$$I_T = I_z(0)$$
$$R_T = \text{terminal resistance}$$
$$X_T = \text{terminal reactance}$$

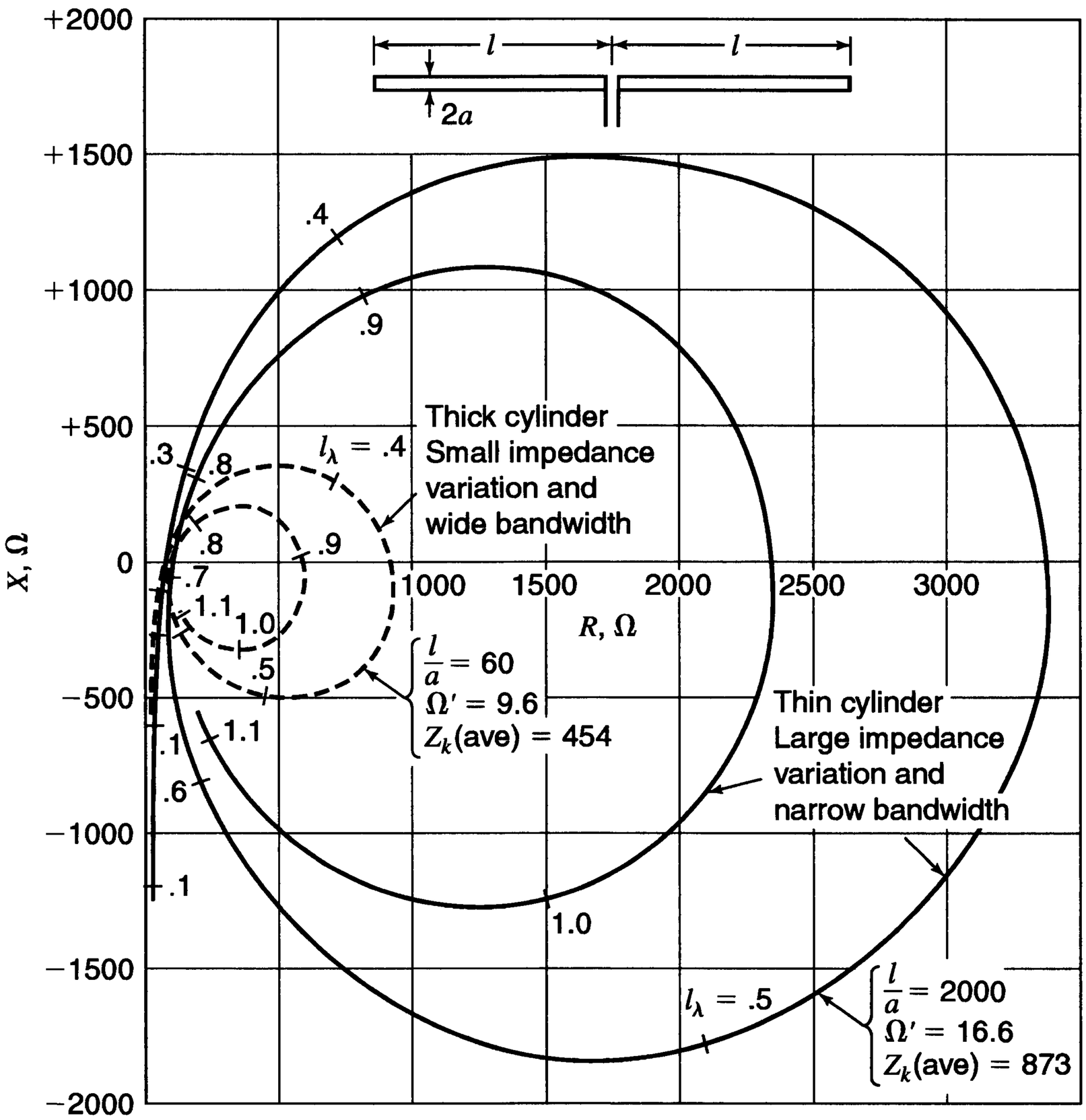


Figure 14–5
Calculated input impedance ($R + jX$, Ω') for cylindrical center-fed antennas with ratios of total length to diameter ($2l/2a$) of 60 and 2000 as a function of l_λ (along spiral). (*After E. Hallén-2.*)

Therefore, inserting this value of current in (14–2–1) yields Hallén's relation for the input impedance,

$$Z_T = -j60\Omega' \left[\frac{\cos \beta l + (d_1/\Omega')}{\sin \beta l + (b_1/\Omega')} \right] \qquad (2)^*$$

This is a first-order approximation for the input impedance. If the second-order terms are included, Hallén's input-impedance expression has the form

$$Z_T = -j60\Omega' \left[\frac{\cos \beta l + (d_1/\Omega') + (d_2/\Omega'^2)}{\sin \beta l + (b_1/\Omega') + (b_2/\Omega'^2)} \right] \qquad (3)^*$$

This relation has been evaluated by Hallén (1), who has also presented the results in chart form (Hallén-2). Impedance spirals based on Hallén's data are presented in Fig. 14–5 for center-fed cylindrical antennas with ratios of total length to diameter (l/a) of 60 and 2000. The half-length l of the antenna is given along the spirals in free-space wavelengths. The impedance variation is that which would be obtained as a function of frequency for

---

*$b_1 = F_1(z) \sin \beta l - F_1(l) \sin \beta|z| + G_1(l) \cos \beta z - G_1(z) \cos \beta l$
$d_1 = F_1(l)$ b_2, d_2 are higher-order terms. The quantities b_1 and d_1 have been calculated in terms of real and imaginary functions by King (1) and Harrison for several values of l and curves given.

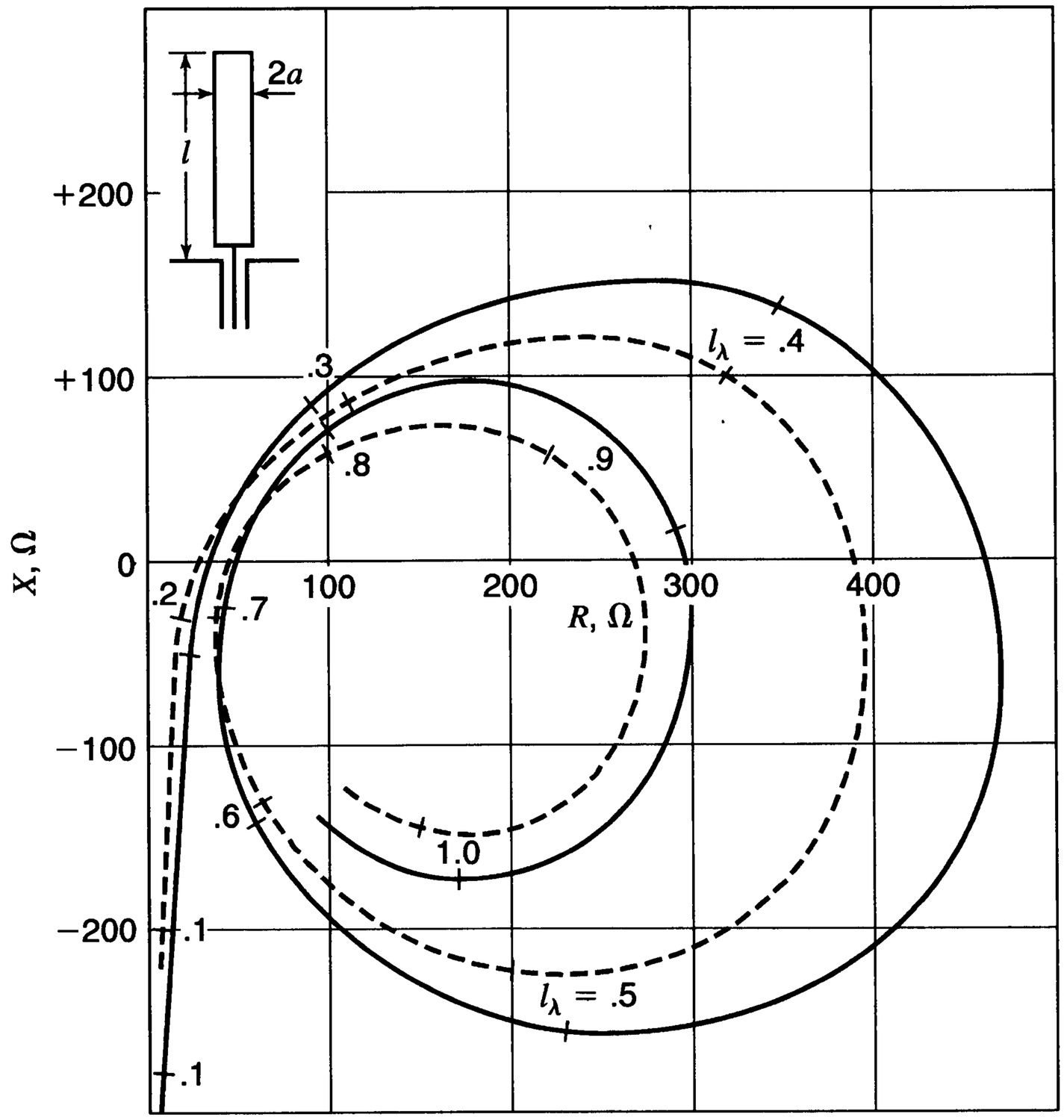


Figure 14–6
Comparison of calculated (solid curve) and measured (dashed curve) input impedance ($R + jX$, Ω') for cylindrical stub antenna with ground plane for the length-radius ratio (l/a) of 60 as a function of l_λ (along spiral).

an antenna of fixed physical dimensions. The difference in the impedance behavior of the thinner antenna ($l/a = 2000$) and of the thicker antenna ($l/a = 60$) is striking, the variation in impedance with frequency of the thicker antenna being much less than that of the thinner antenna.

The impedance, given by (2) or (3), applies to center-fed cylindrical antennas of total length $2l$ and diameter $2a$. To obtain the impedance of a cylindrical stub antenna of length l and diameter $2a$ operating against a very large perfectly conducting ground plane, (2) and (3) are divided by 2. The impedance curve based on Hallén's calculations for a cylindrical stub antenna with an l/a ratio of 60 is given by the solid spiral in Fig. 14–6. The length l of the stub is indicated in free-space wavelengths along the spiral. The measured impedance variation of the same type of antenna ($l/a = 60$) as given by Dorne (1) is also shown in Fig. 14–6 by the dashed spiral. The agreement is good considering the fact that the measured curve includes the effect of the shunt capacitance at the gap and the small but finite antenna terminals (Brown-1; King, D. D.-1; Tai-1; King, R.-1).

The measured input impedance of a cylindrical stub antenna with an l/a ratio of 20 is shown in Fig. 14–7. Comparing this curve with the dashed curve of Fig. 14–6, it is apparent that the trend toward decreased impedance variation with smaller l/a ratio (increased thickness) suggested by Fig. 14–5 is continued to smaller l/a ratios. A measured impedance curve for $l/a = 472$ is also included in Fig. 14–7.[1]

[1]The curves in Fig. 14–7 are based on data presented by Dorne (1).

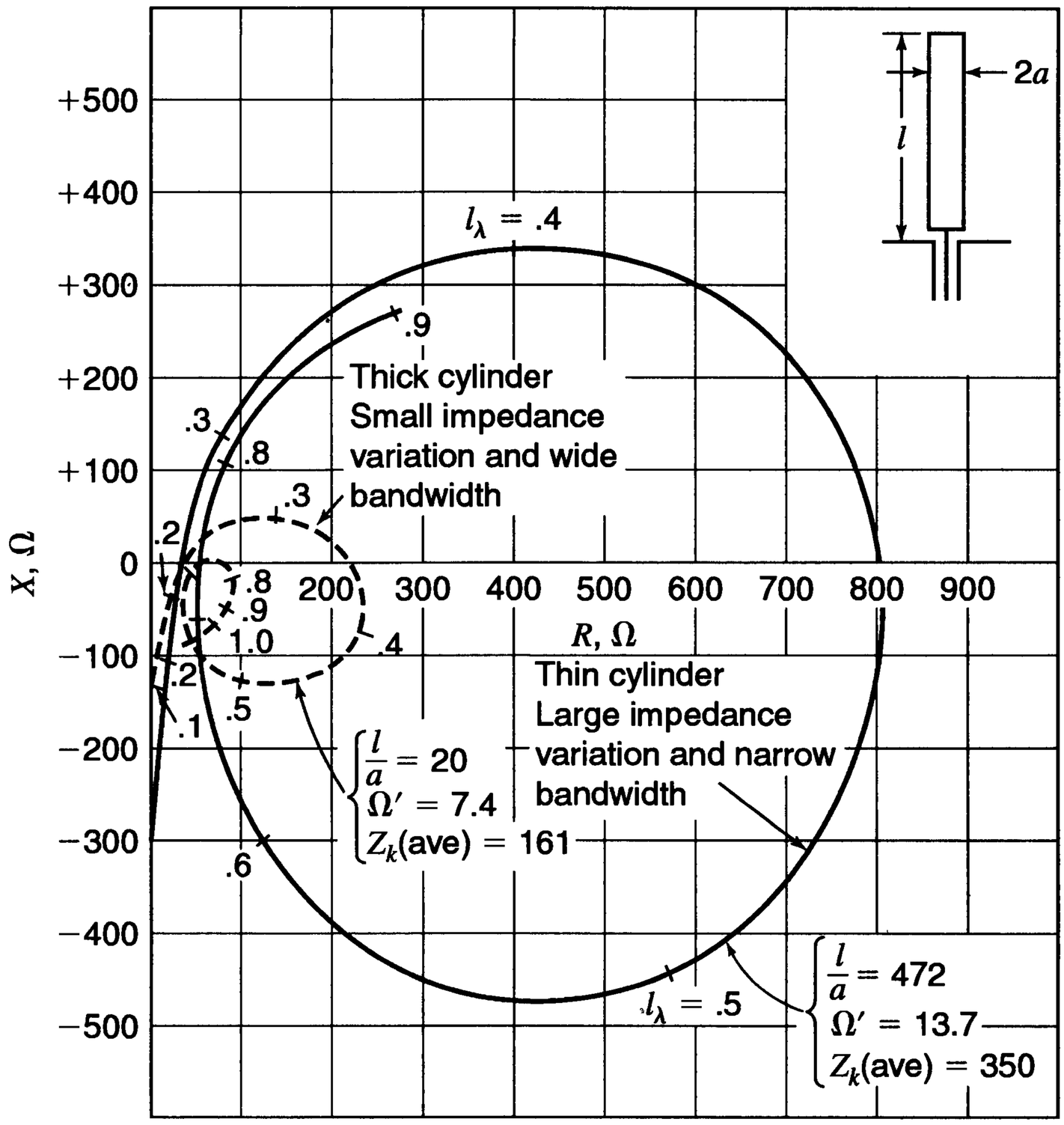


Figure 14–7
Measured input impedance ($R + jX$, Ω') of cylindrical stub antenna with ground plane for the length-radius ratio (l/a) of 20 and 472 as a function of l_λ (along spiral).

An antenna is said to be resonant when the input impedance is a pure resistance. On the impedance diagrams of Figs. 14–5, 14–6 and 14–7 resonance occurs where the spirals cross the $X = 0$ axis. At zero frequency all the impedance spirals start at $R = 0$ and $X = -\infty$. As the frequency increases, the reactance decreases and the resistance also increases, although more slowly. The first resonance occurs when the length l of the antenna is about $\lambda/4$. The resistance at the first resonance is designated R_1. As the frequency is increased, the length of the antenna becomes greater and the second resonance occurs when the length l is about $\lambda/2$. The resistance at the second resonance is designated R_2. At the third resonance (resistance = R_3), the antenna length l is about $3\lambda/4$ and at the fourth resonance (resistance = R_4) l is about 1λ. As the frequency is increased indefinitely, an infinite number of such resonances can be obtained except where the impedance stays reactive.

Since it is common practice to operate antennas at or near resonance, the values of the resonant resistances are of interest. Curves based on Hallén's calculated graphs are presented in Fig. 14–8 for the first four resonances of a cylindrical stub antenna with large ground plane as a function of the length-radius ratio (l/a). The lowest value of l/a for which Hallén gives data is 60, since the accuracy of (3) (Hallen-3) tends to deteriorate for smaller l/a values. Thus, the solid part of the curves ($l/a > 60$) are according to Hallén's calculated values. The dashed parts of the curves are extrapolations to smaller values of l/a. The extrapolation is without theoretical basis but is probably not much in error. A few measured values of resonant resistance from Dorne's data (Dorne-1) are shown as points in Fig. 14–8, the dotted lines indicating to which resonant resistance the points correspond.

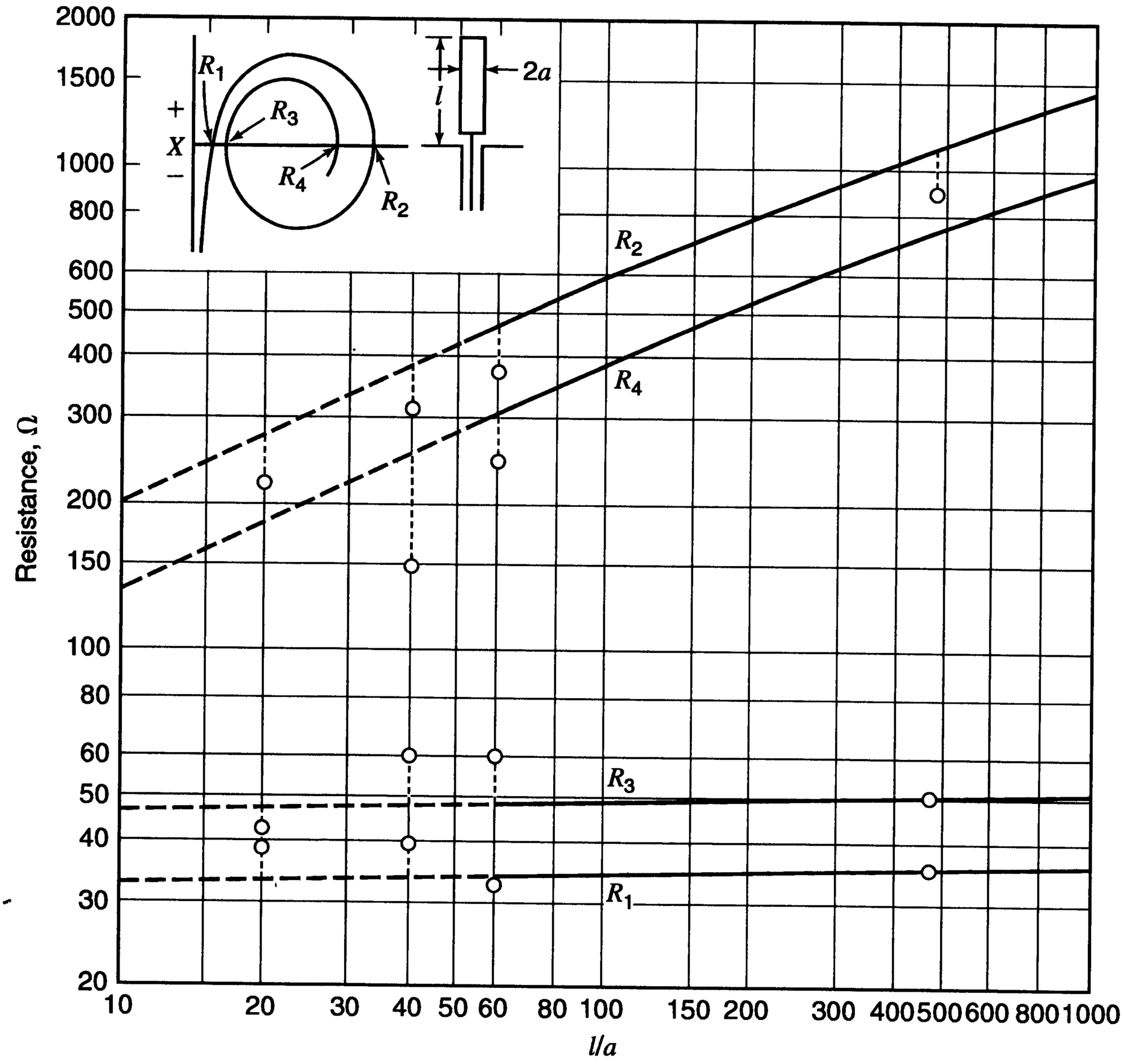


Figure 14–8
Resonant resistance of cylindrical stub antenna with ground plane as a function of the length-radius ratio (l/a). Curves are shown for the first four resonances. For cylindrical center-fed antennas (total length $2l$) multiply the resistance by 2.

Figure 14–8 illustrates the difference in the effect of antenna thickness on the resistance at odd and even resonances. The resistance at odd resonances (R_1, R_3, etc.) is nearly independent of the antenna thickness. The first resonant resistance R_1 is about 35 Ω and the third resonant resistance R_3 is about 50 Ω over a large range of l/a ratios. On the other hand, the antenna thickness has a large effect on the resistance at even resonances (R_2, R_4, etc.). The thicker the antenna, the smaller the resistance. For example, the second resonant resistance R_2 is about 200 Ω when $l/a = 10$ and increases to about 1500 Ω at $l/a = 1000$. The fourth resonant resistance behaves in a similar fashion, the values being somewhat less.

The difference in the resistance behavior at odd and even resonances is related to the current distribution. Thus, at odd resonances the antenna length l is an odd number of $\lambda/4$ (approximately), and a current maximum appears at or near the input terminals. At even resonances the antenna length l is an even number of $\lambda/4$ (approximately), and a current minimum appears at or near the input terminals. As indicated by the current distribution curves

of Figs. 14–3 and 14–4, one of the most noticeable effects of an increase in antenna thickness is the increase of the current at current minima. Thus, when a current minimum is at or near the input terminals, an increase in the antenna thickness raises the input current I_T for a constant input voltage V_T so that the resonant resistance given by the ratio V_T / I_T is reduced.

14–5 PATTERNS OF CYLINDRICAL ANTENNAS（圆柱天线的波瓣图）

Formulas for calculating the far-field patterns of thin linear antennas were developed in Chap. 6. Although these relations apply strictly to infinitesimally thin conductors, they provide a first approximation to the pattern of even a relatively thick cylindrical antenna. This is illustrated by the patterns in Fig. 14–9 for center-fed linear cylindrical antennas of total length $2l$ equal to $\lambda/2$, 1λ, $3\lambda/2$ and 2λ. The calculated patterns for infinitesimally thin antennas are shown in the top row. Three of these patterns were given previously in Fig. 6–9. In the next three rows patterns measured by Dorne (1) are given for l/a ratios of

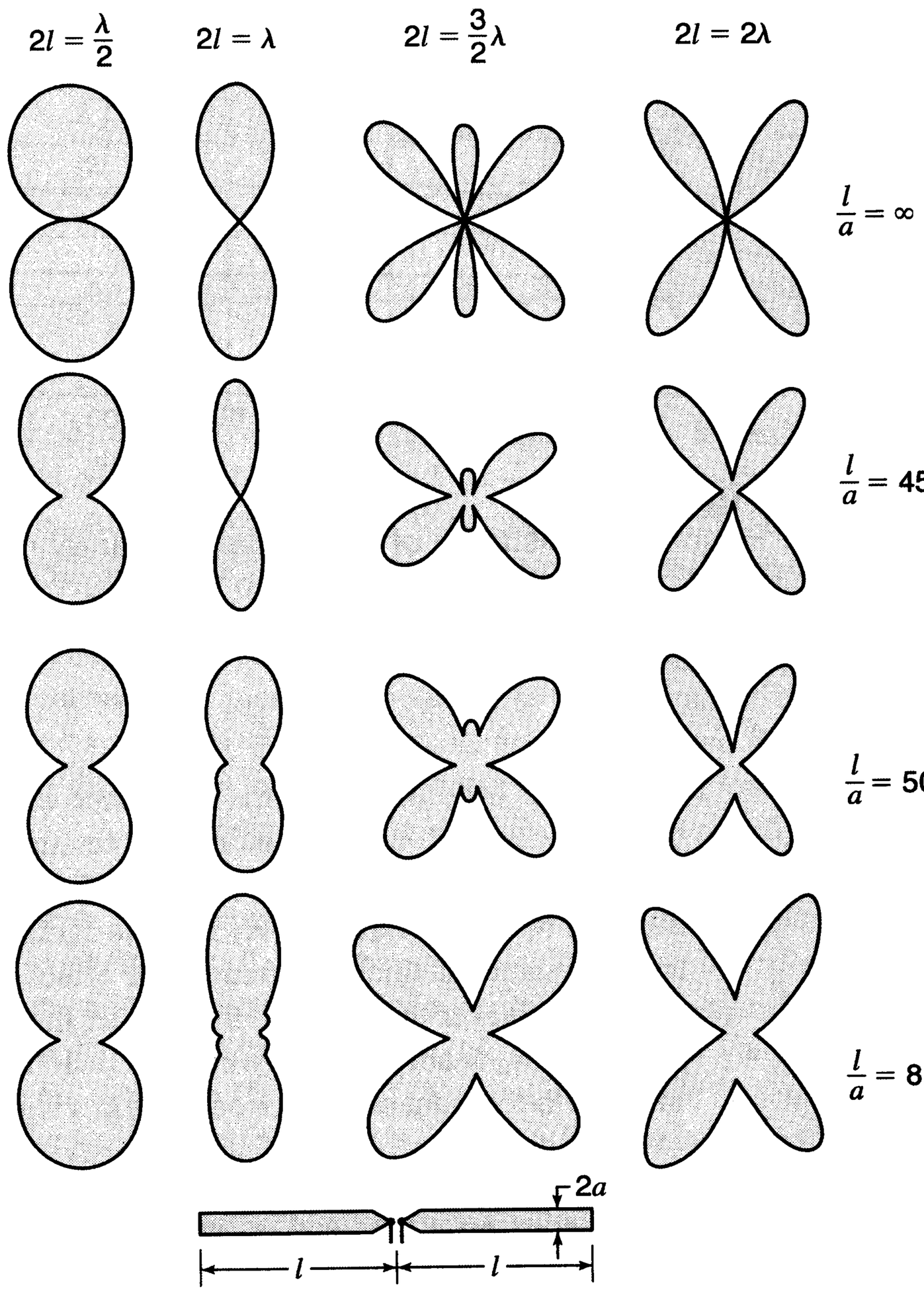


Figure 14–9
Field patterns of cylindrical center-fed linear antennas of total length $2l$ as a function of the total length-diameter ratio (l/a) and also as a function of the total length ($2l$) in wavelengths.

450, 50 and 8.7. The principal effect of increased antenna thickness appears to be that some of the pattern nulls are filled in and that some minor lobes are obliterated (note the patterns in the third column for $2l = 3\lambda/2$).

14–6 THE THIN CYLINDRICAL ANTENNA（细圆柱天线）

If the assumption is made that the cylindrical antenna is infinitesimally thin ($\Omega' \to \infty$), the current distribution is given

$$I_z = \frac{jV_T}{60\Omega'} \frac{\sin \beta(l - |z|)}{\cos \beta l} \tag{1}$$

Although Ω' approaches infinity, the ratio V_T / Ω' may be maintained constant by also letting V_T approach infinity. According to (1), the shape of the current distribution is sinusoidal; that is,

$$I_z = k \sin \beta(l - |z|) \tag{2}$$

where k = a constant

The input impedance Z_T is the ratio V_T/I_T where I_T is the current at the terminals ($z = 0$). Thus from (1),

$$Z_T = \frac{V_T}{I_T} = -j60\Omega' \cot \beta l \tag{3}$$

In (3) we may regard Ω' as large but finite. The terminal impedance Z_T according to (3) is a pure reactance X_T. Equation (3) is identical to the relation for the input impedance of an open-circuited lossless transmission line of length l provided that $60\Omega'$ is taken equal to the characteristic impedance of the line. If, by analogy, $60\Omega'$ is taken equal to the average characteristic impedance Z_k (ave) of the center-fed cylindrical antenna then,

$$Z_k \text{ (ave)} = 60\Omega' = 120 \ln \frac{2l}{a} \tag{4}$$

This relation is of the same form as Schelkunoff's (1) expression for the characteristic impedance Z_k of a thin biconical antenna since for small cone angles $\theta_{hc} = a/l$ so

$$Z_k = 120 \ln \frac{2l}{a} \tag{5}$$

where a = end radius of the cone as shown in Fig. 14–10.

The average characteristic impedance of a center-fed cylindrical antenna as given by Schelkunoff is

$$Z_k \text{ (ave)} = 120\left(\ln \frac{2l}{a} - 1\right) \tag{6}$$

The average impedance of a cylindrical stub antenna with a large ground plane is $\frac{1}{2}$ the value of (6).

As $l/a \to \infty$, (6) reduces to the form given in (4). However, for finite values of l/a, the average characteristic impedance of a cylindrical antenna is the same as for a biconical antenna of the same length l but with an end radius a which is larger than the radius of the

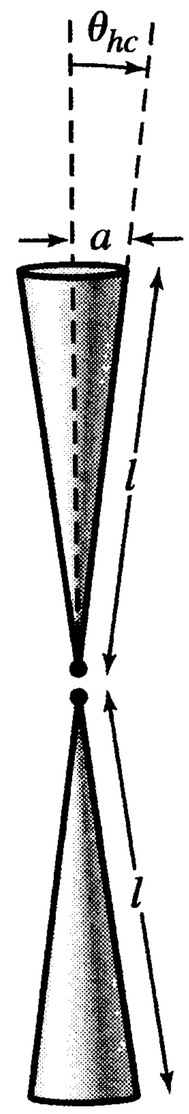


Figure 14–10
Biconical antenna of end radius a and length l.

cylindrical conductor. For example, a cylindrical antenna with an l/a ratio of 500 has an average characteristic impedance equal to that of a biconical antenna of the same length with an end radius 2.8 times larger than the radius of the cylindrical conductor.

In Fig. 14–5 the calculated input impedance is presented for cylindrical center-fed antennas with total length-diameter ratios ($2l/2a = l/a$) of 60 and 2000. The average characteristic impedance of these antennas by (6) is 454 and 873 Ω, respectively. The curve for the l/a ratio of 60 [Z_k (ave) = 454 Ω] has approximately the same form as the calculated impedance spiral for a 2.7° half-angle biconical antenna ($Z_k = 450$ Ω).

In Fig. 14–7 the measured input impedance is shown for cylindrical stub antennas with l/a ratios of 20 and 472. The average characteristic impedance of these antennas as given by $\frac{1}{2}$ of (6) is 161 and 350 Ω, respectively.

14–7 CYLINDRICAL ANTENNAS WITH CONICAL INPUT SECTIONS
（带有圆锥形输入段的圆柱天线）

It is common practice to construct cylindrical antennas with short conical sections at the input terminals as indicated at the bottom of Fig. 14–9. If the cylinders are of large cross section, the conical sections are particularly desirable in order to reduce the shunt capacitance at the gap. Since the measured impedance of an antenna includes the effect of the gap capacitance and the small but finite terminals, the measured impedances will differ more or less from the theoretical values. It is to be expected that measured values will agree better with calculated ones when end cones are used rather than when the ends of the cylinders are butted close together.

14–8 ANTENNAS OF OTHER SHAPES. THE SPHEROIDAL ANTENNA
（其他形状的天线，类球体天线）

The solution of a boundary-value problem may be facilitated if the boundary can be specified by one coordinate of an appropriate coordinate system. A spherical antenna or one in the shape of an elongated ellipsoid of revolution (prolate spheroid), as in Fig. 14–11, is amenable

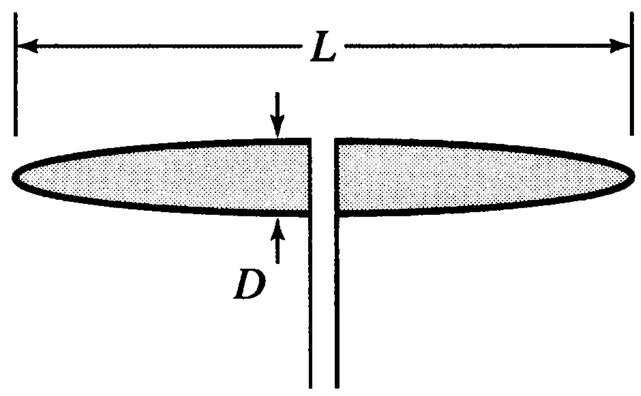


Figure 14–11
Prolate spheroidal antenna.

to such treatment since the surface of the spheroid corresponds to a particular value of one coordinate of a spheroidal coordinate system. By varying the eccentricity of the ellipsoid, one may study the properties of the sphere at the one extreme of eccentricity and of a long thin conductor at the other extreme. This problem has been treated at length by Stratton (1) and Chu and by Page (1) and Adams. Stratton and Chu give admittance and impedance curves for various length-diameter (L/D) ratios (see Fig. 14–11). For long, thin ellipsoids the impedance characteristics are similar to those deduced by other methods. The current distribution for thin $\lambda/2$ spheroids is also found to be nearly sinusoidal.

A point of interest is that for spheroids of the order of $\lambda/2$ long, the impedance variation with frequency decreases with decreasing L/D ratios (thicker spheroids); that is to say, resonance with thick spheroids is broader than with thin ones. This is in agreement with the well-known fact that thick antennas have broader band impedance characteristics than thin ones.

14–9 CURRENT DISTRIBUTIONS ON LONG CYLINDRICAL ANTENNAS
（长圆柱天线上的电流分布）

On a matched lossless transmission line an outgoing wave has a uniform current magnitude and a linear phase change with distance (Fig. 14–12). If the line is mismatched and the reflected (returning) wave is $\frac{1}{2}$ the magnitude of the outgoing wave, a standing wave appears on the line with VSWR given by[1]

$$\text{VSWR} = \frac{I_{\max}}{I_{\min}} = \frac{I_0 + I_1}{I_0 - I_1} = \frac{1 + \frac{1}{2}}{1 - \frac{1}{2}} = 3 \tag{1}$$

where

I_0 = current magnitude of outgoing wave
I_1 = current magnitude of returning wave

The phase change is also nonuniform (fluctuating with distance), as indicated in Fig. 14–12.

When the line is completely mismatched (open- or short-circuited), so that the returning wave equals the outgoing wave, the VSWR = ∞ and the phase changes in a stepwise fashion (Fig. 14–12).

[1]Properly this should be ISWR for the current standing-wave ratio. However, the VSWR = ISWR, although their standing-wave patterns are displaced in position.

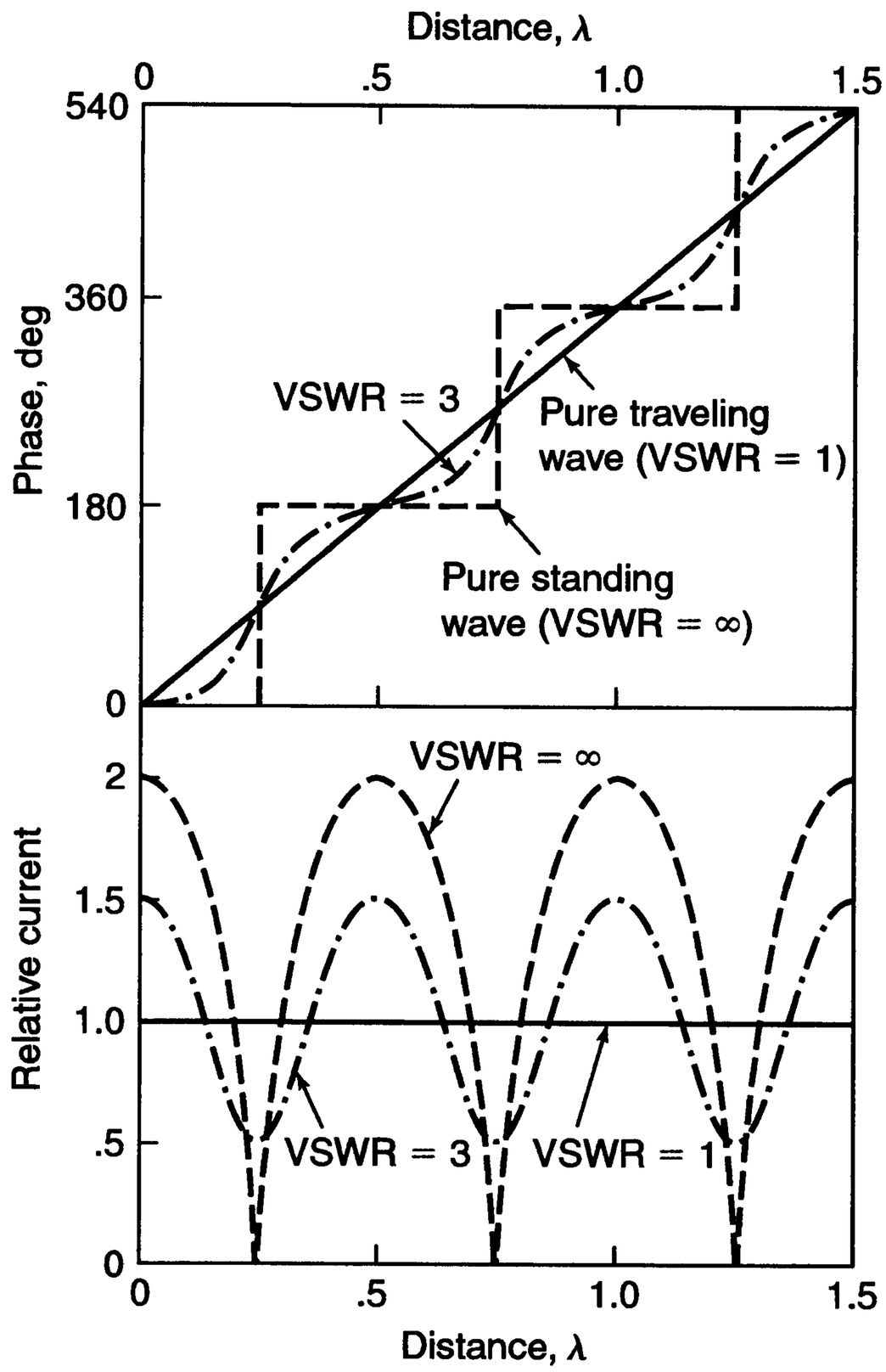


Figure 14–12
Current distribution and phase variation (1) of single uniform traveling wave (solid lines), (2) of two waves traveling in opposite directions with magnitudes 1 and $\frac{1}{2}$ (dash-dot lines) and (3) of 2 waves of equal magnitudes (dashed lines). The last case represents a full (pure) standing wave.

The phase velocity of a wave on the line is given by

$$v = \frac{\omega}{\beta(x)} = \frac{\omega}{d\phi/dx} \tag{2}$$

where

$\omega = 2\pi f$, Hz

ϕ = phase, rad or deg

$\beta = 2\pi/\lambda$, rad (or deg) λ^{-1}

In general, for two opposite current waves of unequal magnitude and velocity, the phase velocity is (Marsh-1)

$$v = \omega\{I_0^2 + I_1^2 + 2I_0I_1\cos[(\beta_1 + \beta_0)x + \gamma]\} \Big/ \left\{ I_0^2\beta_0 - I_1^2\beta_1 \right.$$

$$\left. + I_0I_1(\beta_0 - \beta_1)\cos[(\beta_1 - \beta_0)x + \gamma] + \sin[(\beta_1 + \beta_0)x + \gamma]\left(I_0\frac{dI_1}{dx} - I_1\frac{dI_0}{dx}\right)\right\} \tag{3}$$

where

$\beta_0 = 2\pi/\lambda_0$
$\beta_1 = 2\pi/\lambda_1$
λ_0 = wavelength for outgoing wave, m
λ_1 = wavelength for returning wave, m
γ = phase difference of two waves, rad or deg

For the case where $\gamma = 0$, the velocity of both waves is the same ($\beta_1 = \beta_0$), I_0 and I_1 are constant with distance and (3) reduces to

$$v = \frac{\omega(I_0^2 + I_1^2 + 2I_0I_1 \cos 2\beta x)}{\beta_0(I_0^2 - I_1^2)} \tag{4}$$

Dividing by c ($=\omega/\beta_0$) yields the *relative phase velocity* p. The *ratio* of the maximum to minimum relative phase velocity is

$$\frac{p_{\max}}{p_{\min}} = \frac{(I_0 + I_1)^2}{(I_0 - I_1)^2} \tag{5}$$

Comparing (5) and (1) we note that

$$\frac{p_{\max}}{p_{\min}} = \frac{(I_0 + I_1)^2}{(I_0 - I_1)^2} = \left(\frac{I_{\max}}{I_{\min}}\right)^2 = \text{VSWR}^2 \tag{6}$$

The variation of p ($= v/c$), phase (ϕ) and current magnitude ($|I|$) for $I_1 = \frac{1}{2}I_0$ are presented in Fig. 14–13 over a distance of 2λ. We note that the VSWR = 3 and the relative phase velocity ratio equals 9, and also that the relative phase velocity p is a maximum where I is a maximum.

The current and phase measured by Marsh (1) along a 5λ open-ended cylindrical conductor 0.2λ in diameter are shown in Fig. 14–14, as well as the deduced relative phase velocity variation and the magnitudes of the outgoing and reflected waves. The attenuation of the outgoing and reflected waves is evident.

It is interesting to compare the current distribution of Fig. 14–14 for the long (5λ), thick (0.2λ diameter open-ended) cylindrical conductor with the distributions of Fig. 8–3 for a 7-turn helix with much thinner conductor (0.02λ diameter at $C_\lambda = 1.07$). When $C_\lambda = 0.60$, the helix has a nearly uniform standing wave, indicating outgoing and returning T_0 mode waves of almost equal amplitude. When $C_\lambda = 1.07$ the outgoing T_0 mode wave attenuates rapidly with energy transferred to a nearly uniform T_1 mode wave over the rest of the helix. At the open end, a reflected or returning T_0 mode wave is excited which attenuates rapidly while transforming into a small nearly uniform returning T_1 mode wave.

On the cylindrical conductor (Fig. 14–14) a T_0 mode wave attenuates gradually over the length of the conductor and on reflection from the open end excites a gradually attenuating returning wave.

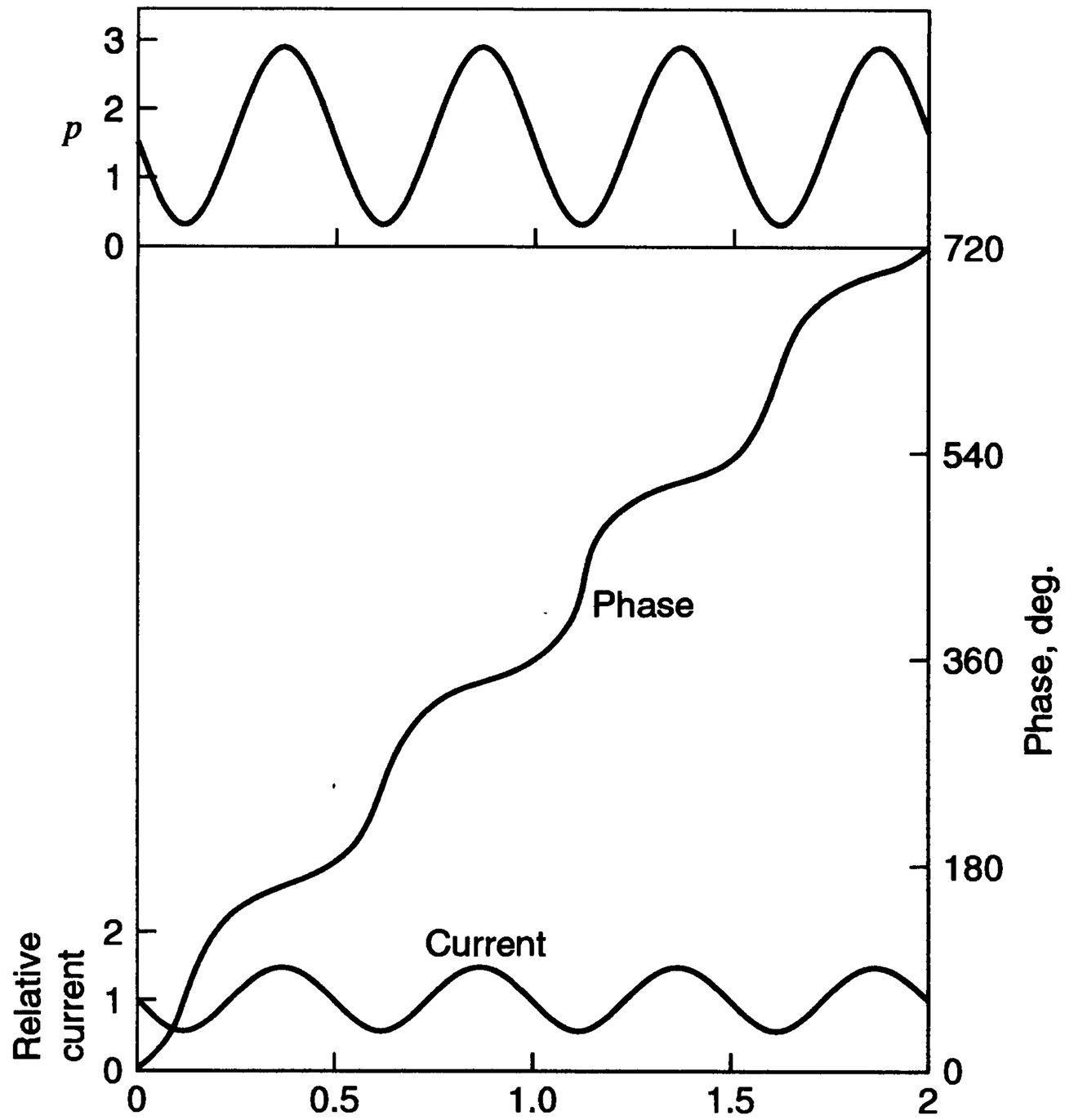


Figure 14–13
Current, phase and relative phase velocity p for two waves traveling in opposite directions with magnitudes 1 and $\frac{1}{2}$ (VSWR = 3 and relative phase velocity ratio = 9).

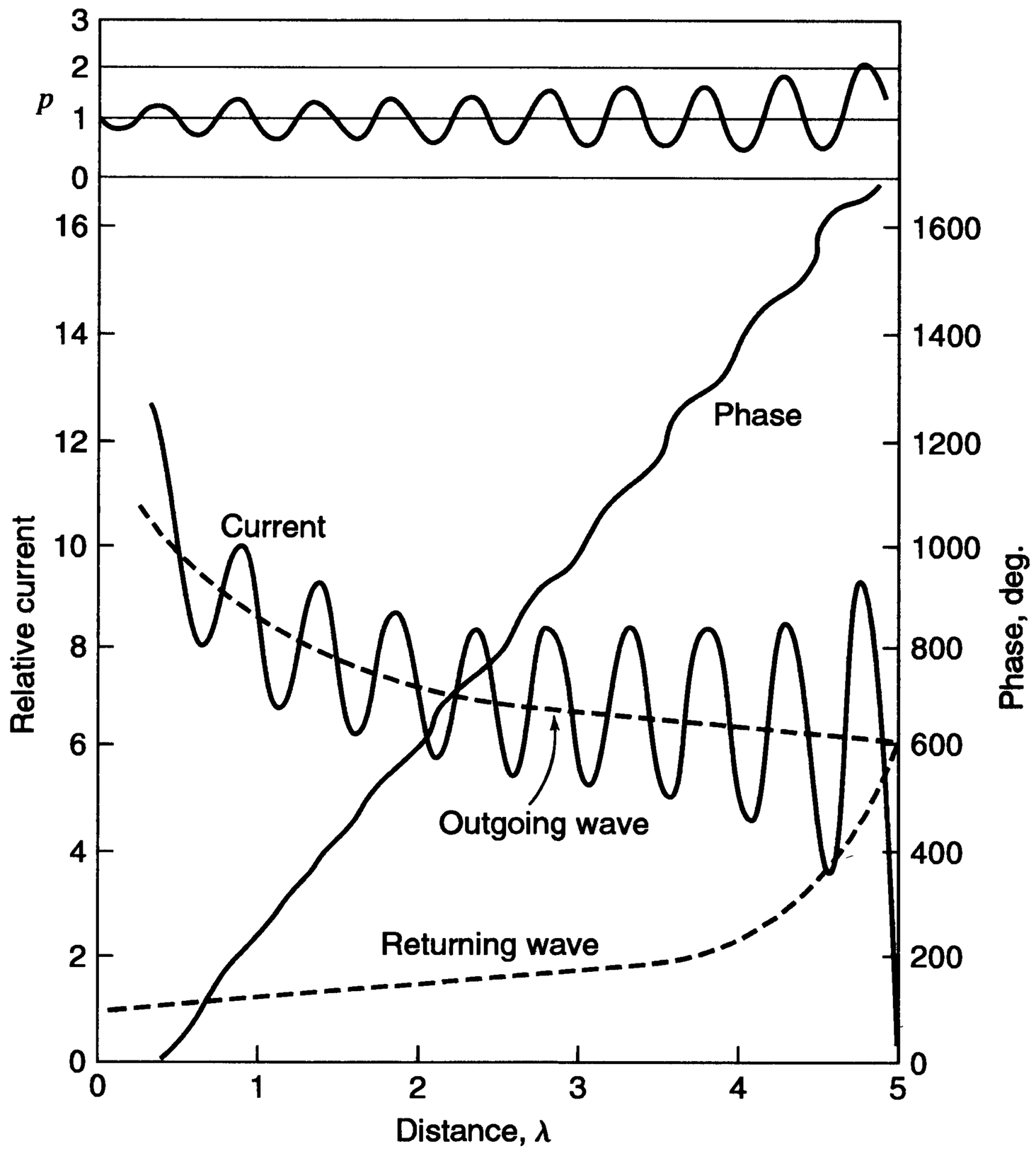


Figure 14–14
Measured current distribution on a long (5λ), thick (0.2λ diameter) cylindrical conductor with attendant phase and relative phase velocity (p) of the total wave. (*J. A. Marsh-1.*) Resolution into outgoing and returning (reflected) waves is indicated.

Table 14–1 Comparison of currents on long, thick cylindrical antenna and on helical antennas

Antenna	Mode	Relative phase velocity, $p (= v/c)$†	Remarks
Cylindrical conductor	All T_0	1	Gradual attenuation of outgoing and returning T_0 waves
Helix, $C_\lambda = 0.6$	All T_0	>1	Uniform equal outgoing and returning T_0 waves
Helix, $C_\lambda = 1.07$ (axial mode)	T_0 at ends, T_1 over remainder	>1 <1 (~0.8)	T_0 waves attenuate rapidly T_1 waves uniform

†For single traveling wave.

The behavior of the two antennas is summarized in Table 14–1.

14–10 INTEGRAL EQUATIONS AND THE MOMENT METHOD (MM) IN ELECTROSTATICS（静电学的积分方程和矩量法）

As an introduction to the moment method, let us consider its application to an electrostatics problem.

In calculus we deal with *differential equations* as, for example,

$$\frac{dx}{dt} = v \tag{1}$$

or the rate of change of distance (x) with time (t) equals the velocity (v). It is implied that we know x as a function of t or how x varies with time [$x(t)$].

On the other hand, suppose we know how the velocity varies as a function of time [$v(t)$]. Then the distance is given by the *integral* of the velocity with respect to time or

$$x = \int_0^{t_1} v(t)\, dt \tag{2}$$

If v is constant, then (2) becomes

$$x = vt_1 \tag{3}$$

or x is the distance traveled at velocity v in time t_1. Now suppose that x is known at $t = 0$ and $t = t_1$ but v is not known during this period of time. Then (2) is an *integral equation* with the problem being to obtain a solution for the velocity as a function of time [$v(t)$].

Referring to Fig. 14–15, a basic relation of electrostatics is that the electric potential V at a point P due to a charge Q is given by

$$V = \frac{Q}{4\pi\varepsilon r} \qquad \text{(V)} \tag{4}$$

where

Q = charge, C
r = distance from P to the charge Q, m
ε = permittivity of medium, F m^{-1}

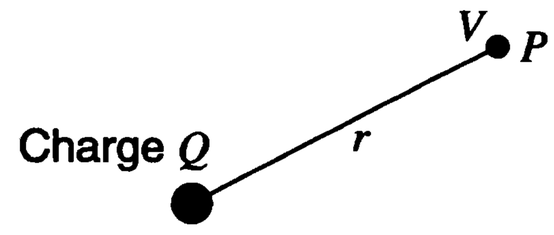


Figure 14–15
Electric potential V at point P is inversely proportional to the distance r from charge Q.

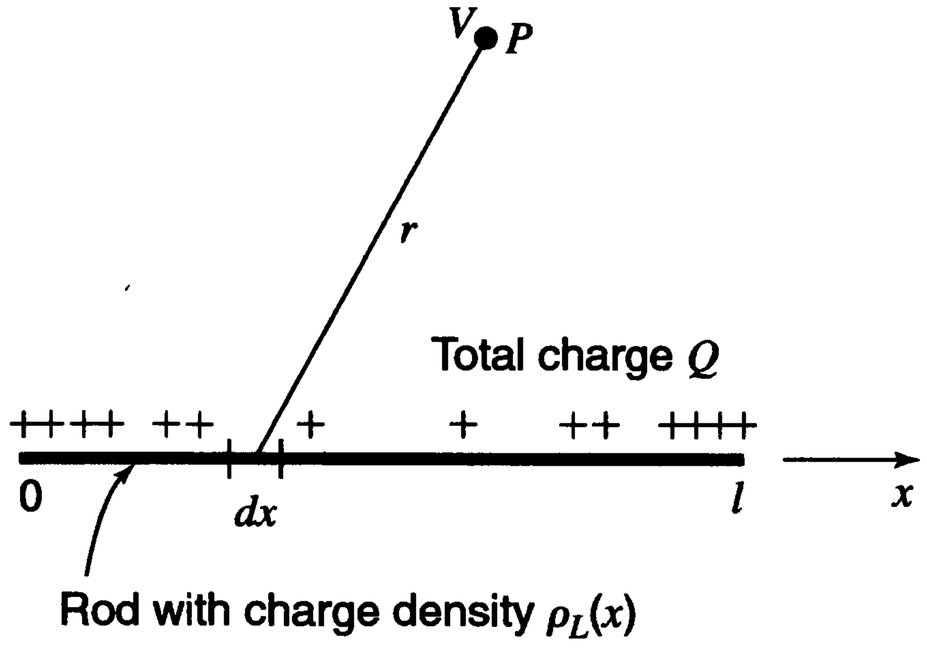


Figure 14–16
Electric potential V at point P due to a rod with charge density $\rho_L(x)$ which is a function of position (x).

For a line of charge of density ρ_L (C m^{-1}) as in Fig. 14–16, then V at some observation point P is given by the integral of (4) over the length l of the line or

$$V = \frac{1}{4\pi\varepsilon}\int_0^l \frac{\rho_L(x)}{r}\,dx \tag{5}$$

where $\rho_L(x)$ = charge per unit of length of line as a function of x, C m^{-1}

If $\rho_L(x)$ is known as a function of x, then (5) can be integrated in a straightforward manner. However, if $\rho_L(x)$ is not known, (5) represents *an integral equation* with the problem being to find a solution for $\rho_L(x)$.

EXAMPLE **14–10.1 Charge Distribution on Wire**

Let the line be an isolated conducting rod or wire of radius a and length $l = 8a$ on which a total charge $+Q$ has been placed. Since like charges repel, it may be anticipated that the charge will tend to separate and pile up near the ends of the rod, making the charge distribution along the rod nonuniform. The problem is to determine this charge distribution $\rho_L(x)$ using an incremental numerical technique or moment method as an introduction to integral equations.

■ **Solution**

First, let us divide the rod of length l into 4 segments or increments with each segment of length $2a$ as in Fig. 14–17 ($l = 8a$). Let the total charge on segment 1 be Q_1 and on segment 2 be Q_2. By symmetry, the charge on segment 3 is the same as on segment 2, or Q_2. Likewise, the charge on segment 4 is equal to Q_1. Specifically, our problem is to find the ratio of Q_1 to Q_2 [a first step or approximation in solving for $\rho_L(x)$].

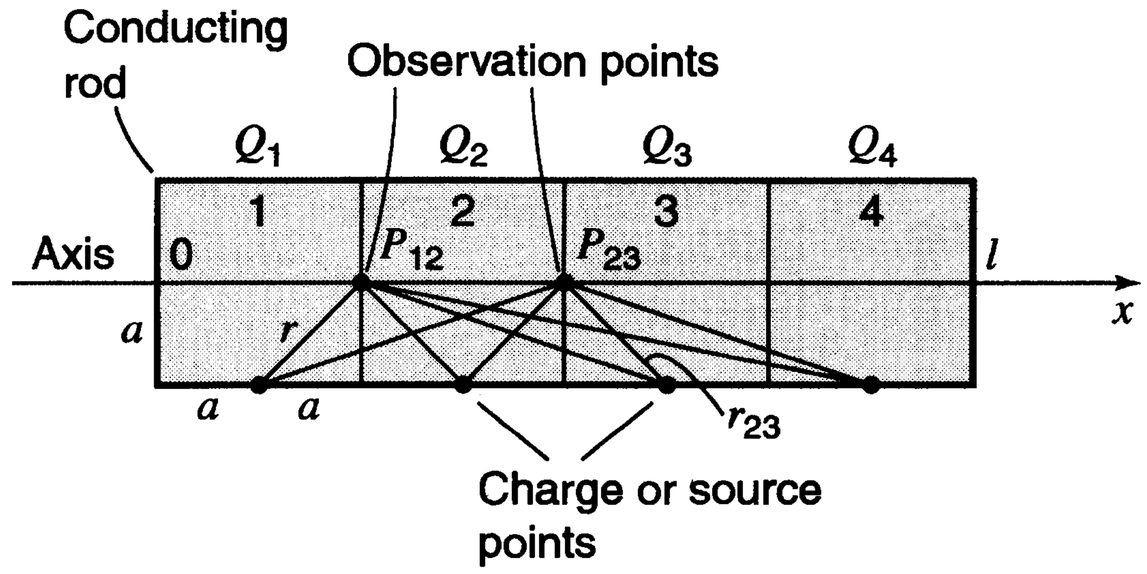


Figure 14–17
Charged rod of radius a divided into 4 segments of equal length ($2a$) for calculation of charge ratios.

Let us assume that all of the charge on each segment is concentrated on a circle on the surface of the segment around its midpoint with the observation or test points on the wire axis. Since the distance r from an observation point to any point on the circle of charge is constant, we may consider that all of the charge of a segment is at one point (charge or source point), as in Fig. 14–17. The situation may now be regarded as one with 4 points of charge (source points) in empty space with the potential at observation or test points on the axis to be determined. Thus, from (4) the potential at point P_{12} is given by

$$V(P_{12}) = \frac{1}{4\pi\varepsilon}\left[\frac{Q_1}{\sqrt{a^2+a^2}} + \frac{Q_2}{\sqrt{a^2+a^2}} + \frac{Q_2}{\sqrt{a^2+9a^2}} + \frac{Q_1}{\sqrt{a^2+25a^2}}\right] \tag{6}$$

Likewise at point P_{23} the potential is given by

$$V(P_{23}) = \frac{1}{4\pi\varepsilon}\left[\frac{Q_1}{\sqrt{a^2+9a^2}} + \frac{Q_2}{\sqrt{a^2+a^2}} + \frac{Q_2}{\sqrt{a^2+a^2}} + \frac{Q_1}{\sqrt{a^2+9a^2}}\right] \tag{7}$$

A *boundary condition* is that (even though the charge density varies along the rod) the potential is constant. Therefore, $V(P_{12}) = V(P_{23})$ so that equating (6) and (7) we find that

$$Q_1 = 1.45Q_2 \tag{8}$$

Thus, the charge (or average charge density) for the outer segments is 45 percent greater than for the inner segments and we can write

$$Q_1 : Q_2 = 1.45 : 1.00 \tag{9}$$

Dividing the rod into 6 segments and proceeding as above results in the ratios

$$Q_1 : Q_2 : Q_3 = 1.84 : 1.03 : 1.00 \tag{10}$$

The charge density distribution along the rod is shown by the step or pulse functions in Fig. 14–18 for the cases of 4, 6 and 8 segments. A smooth curve is also drawn through the centers of the pulse functions.

To simplify the above calculations we neglected the effect of the end surfaces of the rod. For 4 segments the cylindrical area of a segment is 4 times the end of the rod so that the effect of neglecting the ends is not large. However, with more segments the effect becomes greater, especially for the charge on the end segments.

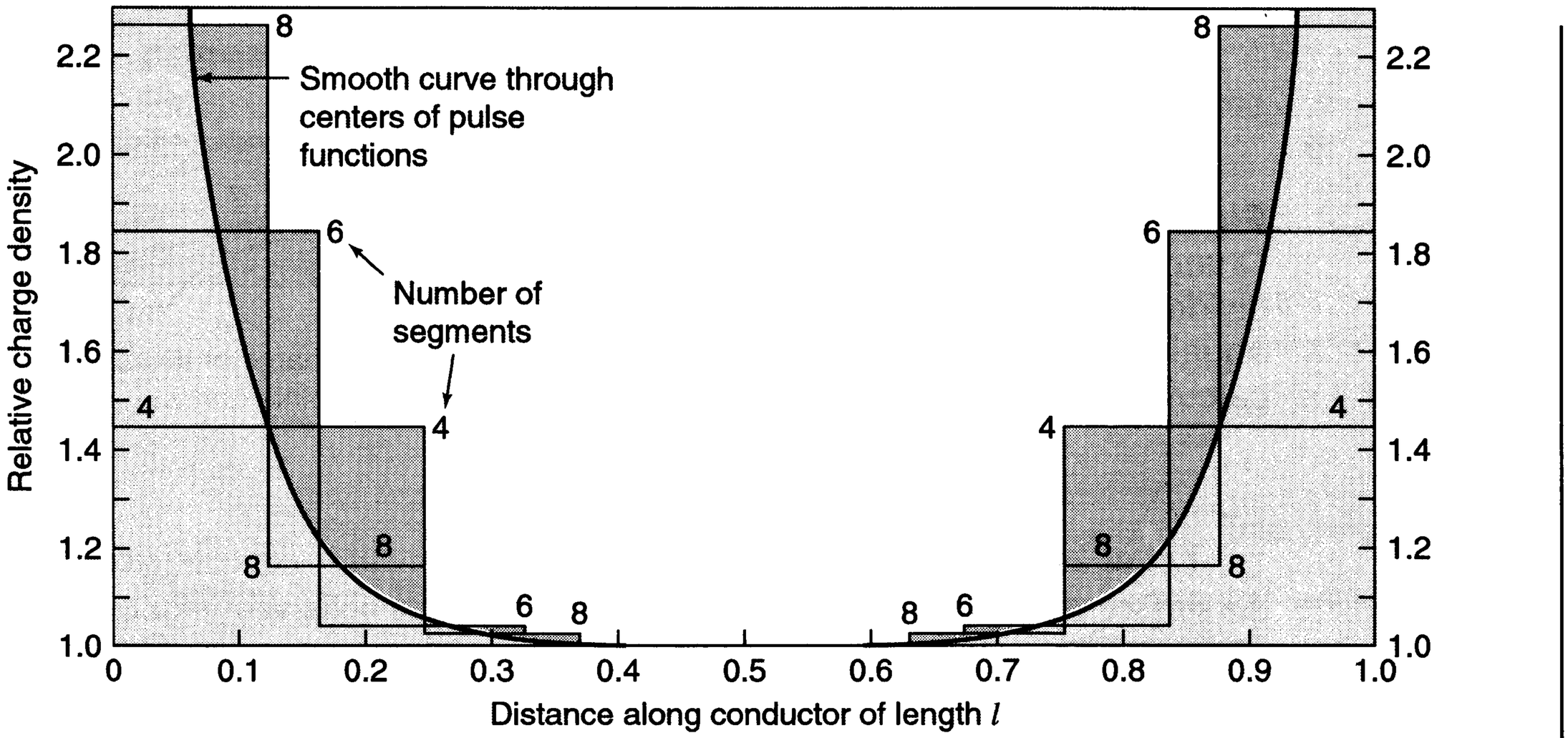


Figure 14–18
Relative charge density along straight conducting rod of radius a and length $8a$ as calculated by the moment method using 4, 6 and 8 segments.

Let us now discuss the problem more formally. From (5),

$$V = \frac{1}{4\pi\varepsilon}\int_0^l \frac{\rho_L(x)}{r}\,dx \tag{11}$$

Referring to Fig. 14–19, let the line be divided into N segments of equal length Δx with average charge density $\rho_L(x)_n$ for segment Δx_n. Then the charge on segment n is given by

$$Q_n = \overline{\rho_L(x)_n}\,\Delta x_n, \qquad n = 1, 2, 3, \ldots, N \tag{12}$$

and the total charge on the wire by

$$Q = \sum_{n=1}^{N} Q_n \tag{13}$$

Equation (11) can now be written as

$$\sum_{n=1}^{N} l_{mn} Q_n = V_m \tag{14}$$

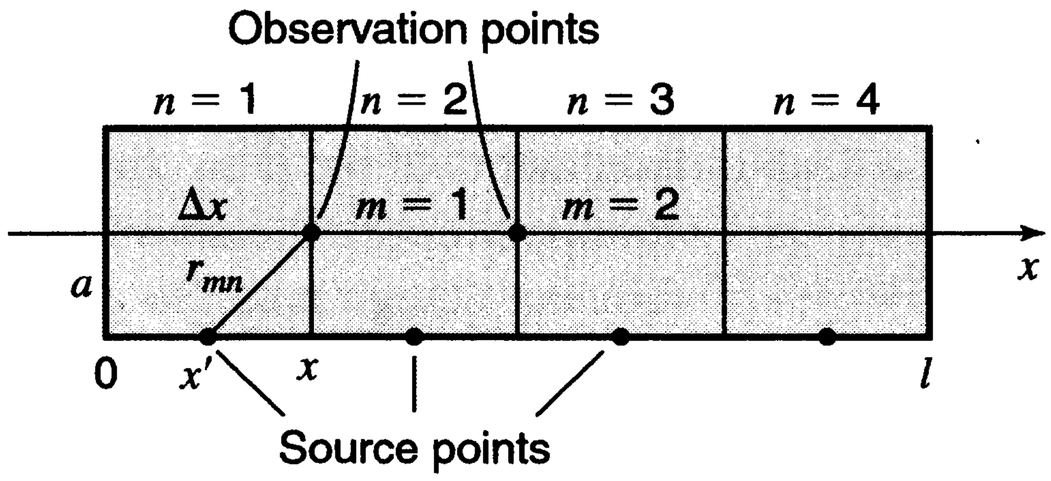


Figure 14–19
Charged rod for calculation of charge density distribution.

where

$$l_{mn} = \frac{1}{4\pi\varepsilon r_{mn}}, \qquad m = 1, 2, 3, \ldots, M \tag{15}$$

$$r_{mn} = \sqrt{a^2 + (x - x')^2}$$

a = rod radius

x = axial distance of observation or test point m

x' = axial distance of source point at middle of segment n

In matrix notation (14) is

$$[l_{mn}][Q_n] = [V_m] \tag{16}$$

or

$$\begin{bmatrix} l_{11} & l_{12} & l_{13} & \cdots & l_{1N} \\ l_{21} & l_{22} & l_{23} & \cdots & l_{2N} \\ l_{31} & l_{32} & l_{33} & \cdots & l_{3N} \\ \vdots & \vdots & \vdots & & \vdots \\ l_{M1} & l_{M2} & l_{M3} & \cdots & l_{M4} \end{bmatrix} \begin{bmatrix} Q_1 \\ Q_2 \\ Q_3 \\ \vdots \\ Q_N \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ \vdots \\ V_M \end{bmatrix} \tag{17}$$

Thus, the integral equation (11) has been transformed into a set of N simultaneous linear algebraic equations (17) where l_{mn} represents a known function (the inverse distance relation), V_m represents potentials determined by the boundary conditions and Q_n represents the N unknown charges whose values are sought.

In the charged rod example we have from symmetry that $M = N/2$ and from the boundary condition that

$$V_1 = V_2 = V_3 = \cdots \tag{18}$$

For 4 segments ($m = 2, n = 4$), (17) then reduces to

$$\begin{bmatrix} l_{11} & l_{12} & l_{13} & l_{14} \\ l_{21} & l_{22} & l_{23} & l_{24} \end{bmatrix} \begin{bmatrix} Q_1 \\ Q_2 \\ Q_2 \\ Q_1 \end{bmatrix} = V \tag{19}$$

Introducing (15) for l_{mn}, (19) is identical in form with (6) and (7) for increments of $\Delta x = 2a$, where m designates the test or observation points (P_{12} for $m = 1$ and P_{23} for $m = 2$ in Fig. 14–17) and r_{mn} is equal to the distance between the test point m and the source point of segment n. Thus, r_{23} is the distance between test point $m = 2$ (P_{23} in Fig. 14–17) and the center of segment 3.

In the above we have used a pulse or step-function approximation in which the boundary condition (V = constant) is not enforced everywhere along the rod but only at certain observation or test points. In between observation points the boundary condition may not

be satisfied. However, as the number of segments and observation points increase, the boundary condition is enforced at more points (the solution converging) and the accuracy of the results should improve. In the sense that the residual discrepancies or moments should vanish with a sufficient number of properly selected pulse functions, the procedure we have discussed may be called a *moment method.*

14–11 THE MOMENT METHOD (MM) AND ITS APPLICATION TO A WIRE ANTENNA（矩量法及其应用于线天线）

As discussed in the previous section, an integral equation can be transformed into a set of simultaneous linear algebraic equations (or matrix equation) which may then be solved by numerical techniques. Roger Harrington (1) has unified the various procedures into a general moment method (MM) now widely used with powerful computers for solving electromagnetic field problems.

In this section the method will be developed for a wire antenna and applied to an example for a short dipole.

Consider a cylindrical current-carrying conductor (or wire) of radius a isolated in free space (Fig. 14–20). Let its conductivity $\sigma = \infty$ so that we can consider the radio-frequency current to be entirely on the surface ($1/e$ depth $= 0$). The total current at point z' on the conductor is

$$I(z') = K(z')2\pi a \tag{1}$$

where $K(z')$ = surface current density at z' (A m^{-1})

All of the current is at a distance a from the conductor axis (z axis), and we will consider it as flowing in empty space along an infinitesimally thin filament parallel to the z axis at a distance a, as in Fig. 14–21, with the conductor no longer present.

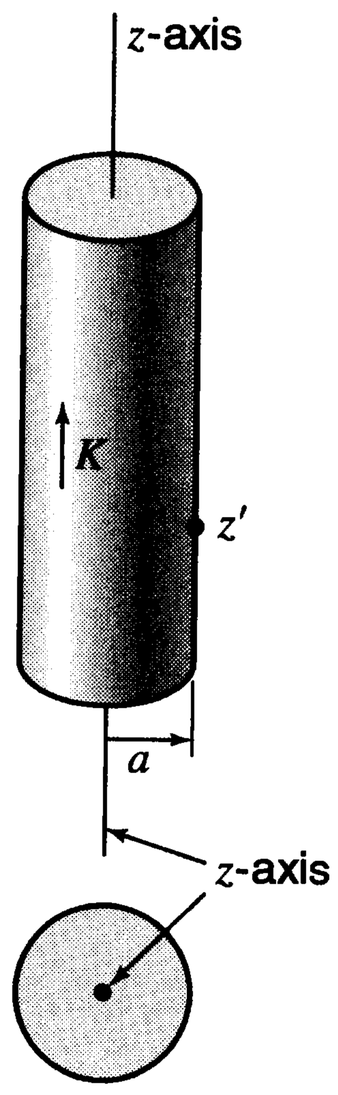


Figure 14–20
Cylindrical conductor of radius a with surface current density K (A m^{-1}).

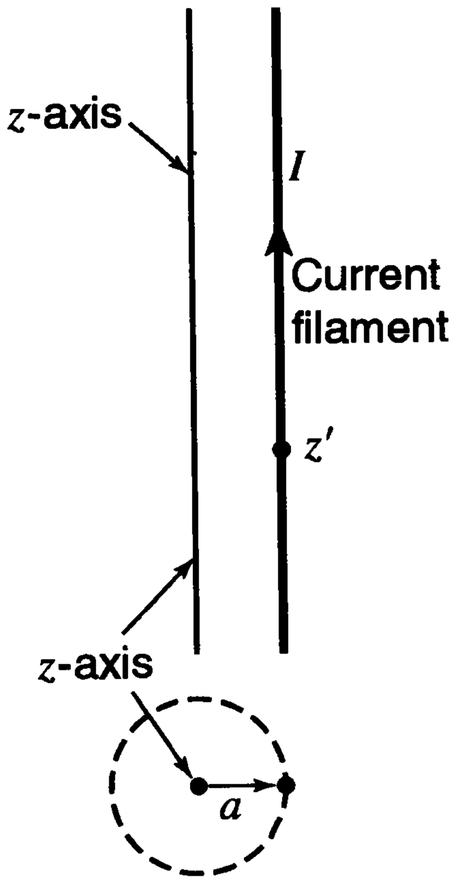


Figure 14–21
Conductor replaced by current filament $I = 2\pi aK$ (A) at distance a from the z-axis.

The electric field of charges and current is given by

$$\mathbf{E} = -j\omega\mu_0\mathbf{A} - \nabla V \tag{2}$$

where

$\mathbf{A}$ = vector potential
V = scalar potential

For current only in the z direction (2) becomes

$$E_z = -j\omega\mu_0 A_z - \frac{\partial V}{\partial z} \tag{3}$$

We also have

$$\frac{\partial A_z}{\partial z} = -j\omega\varepsilon_0 V \tag{4}$$

and its z derivative

$$\frac{\partial^2 A_z}{\partial z^2} = -j\omega\varepsilon_0 \frac{\partial V}{\partial z} \tag{4a}$$

or

$$-\frac{\partial V}{\partial z} = \frac{1}{j\omega\varepsilon_0}\frac{\partial^2 A_z}{\partial z^2} \tag{5}$$

Introducing (5) into (3),

$$E_z = \frac{1}{j\omega\varepsilon_0}\frac{\partial^2 A_z}{\partial z^2} - j\omega\mu_0 A_z \tag{6}$$

$$= \frac{1}{j\omega\varepsilon_0}\left(\frac{\partial^2 A_z}{\partial z^2} + \omega^2\mu_0\varepsilon_0 A_z\right) \tag{7}$$

$$= \frac{1}{j\omega\varepsilon_0}\left(\frac{\partial^2 A_z}{\partial z^2} + \beta^2 A_z\right) \tag{8}$$

For a current element dz, the vector potential

$$dA_z = \frac{I(z')e^{-j\beta r}\,dz}{4\pi r} \tag{9}$$

where

$e^{-j\beta r}/r = G_{zz'}$ = free space Green's function
$r = \sqrt{(z-z')^2 + a^2}$
z = observation point
z' = source point (see Fig. 14–22)

The field from this current element is then

$$dE_z = \frac{I(z')}{j\omega\varepsilon_0 4\pi}\left(\frac{\partial^2}{\partial z^2}G_{zz'} + \beta^2 G_{zz'}\right)dz' \tag{10}$$

For a conductor of length L, the total field is given by the integral of (10), known as *Pocklington's equation* (Pocklington-1):

$$E_z = \frac{1}{4\pi j\omega\varepsilon_0}\int_{-L/2}^{L/2}\left(\frac{\partial^2}{\partial z^2}G_{zz'} + \beta^2 G_{zz'}\right)I(z')\,dz' \tag{11}$$

E_z is the radiated field due to the current $I(z')$, resulting from an impressed or source field E_z, from, for example, a voltage applied at the antenna terminals or from an incident plane wave (scattering case). On (and inside) the conductor the sum of these fields must vanish ($\sigma = \infty$) so

$$E_z = -E_{z'} \tag{12}$$

Richmond (1) has differentiated and rearranged (11) in a more convenient form as follows:

$$-E_{z'} = \frac{\lambda Z_0}{8\pi^2 j}\int_{-L/2}^{L/2}\frac{e^{-j\beta r}}{r^5}[(1+j\beta r)(2r^2-3a^2)+\beta^2 a^2 r^2]I(z')\,dz' \qquad (\text{V m}^{-1}) \tag{13}$$

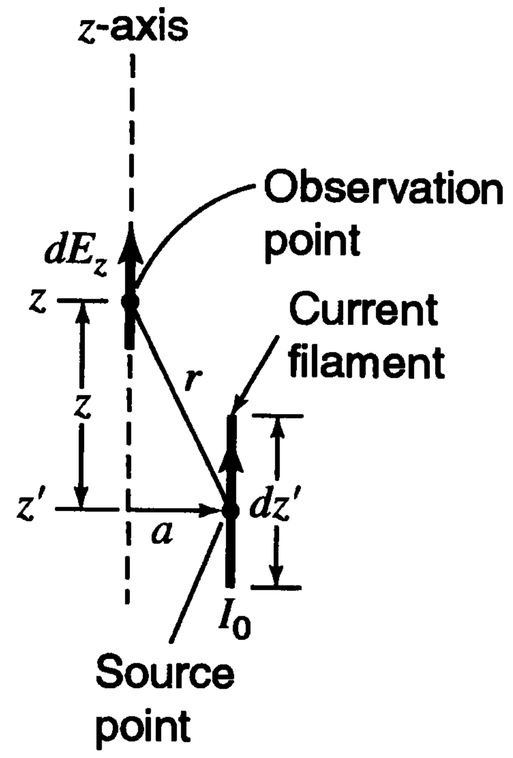


Figure 14–22
Source point on current filament with field dE_z at distance r on the z-axis.

where

r = distance between source and observation points
$= \sqrt{(z' - z)^2 + a^2}$, m
$Z_0 = 377\ \Omega$

With parameters in *dimensionless form,* (13) becomes

$$-V = -\Delta z\, E(z')$$
$$= -j\frac{Z_0\, \Delta z_\lambda}{8\pi^2} \int_{-L/2}^{L/2} \frac{e^{-j2\pi r_\lambda}}{r_\lambda^3} \left\{(1 + j2\pi r_\lambda)\left[2 - 3\left(\frac{a}{r}\right)^2\right] + 4\pi^2 a_\lambda^2\right\} I(z')\, dz_\lambda \quad \text{(V)} \tag{14}$$

where

$r_\lambda = r/\lambda$, dimensionless
V = voltage developed by $E(z')$ over Δz_λ, V

For brevity let $-E(z')$ in (14) be written as

$$-E(z') = \int_{-L/2}^{L/2} I(z')G(r_{mn})\, dz' \qquad (\text{V m}^{-1}) \tag{15}$$

where

$$G(r_{mn}) = \frac{-jZ_0}{8\pi^2\lambda^2}\left(\frac{e^{-j2\pi r_\lambda}}{r_\lambda^3}\right)\left\{(1 + j2\pi r_\lambda)\left[2 - 3\left(\frac{a}{r}\right)^2\right] + 4\pi^2 a_\lambda^2\right\} \qquad (\Omega\ \text{m}^{-2}) \tag{16}$$

$r = r_{mn}$
m = observation point
n = source point

Approximating the current with a series expansion we let

$$I(z') = \sum_{n=1}^{N} I_n F_n(z') \tag{17}$$

where $F_n(z')$ is a pulse function (equal to zero or unity) for incremental segments $\Delta z'_n$. (Other functions are possible, e.g., overlapping segments, each with a triangular or piecewise sinusoidal current distribution.)[1] For the mth segment we have

$$-E_{z'}(z_m) = \sum_{n=1}^{N} I_n \int_{\Delta z'_n} G(r_{mn})\, dz' \qquad (\text{V m}^{-1}) \tag{18}$$

[1]The piecewise sinusoidal distribution (section of a sine curve) is assumed to be somewhat more appropriate than a strictly triangular distribution. A piecewise sinusoidal distribution is used in Sec. 14–12. The differences between distributions are most significant for a small number of segments. With a large number of segments the different distribution functions should all give equivalent results.

Putting

$$G_{mn} = \int_{\Delta z'_n} G(r_{mn})\, dz' \simeq G(r_{mn})\Delta z'_n \qquad (\Omega\ \mathrm{m}^{-1}) \tag{19}$$

(18) becomes approximately

$$-E_{z'}(z_m) = I_1 G_{m1} + I_2 G_{m2} + \cdots + I_n G_{m3} + \cdots + I_N G_{m4} \tag{20}$$

and the antenna equation now takes the form of a network equation. Writing (20) for each of the N segments ($m = 1, 2, 3, \ldots, N$), we obtain a set of equations:

$$\begin{aligned} I_1 G_{11} + I_2 G_{12} + \cdots + I_N G_{1N} &= -E_{z'}(z_1) \\ I_1 G_{21} + I_2 G_{22} + \cdots + I_N G_{2N} &= -E_{z'}(z_2) \\ \vdots \qquad\quad \vdots \qquad\qquad \vdots \qquad\quad & \vdots \\ I_1 G_{N1} + I_2 G_{N2} + \cdots + I_N G_{NN} &= -E_{z'}(z_N) \end{aligned} \tag{21}$$

which may be expressed in matrix form as

$$\begin{bmatrix} G_{11} & G_{12} & \cdots & G_{1N} \\ G_{21} & G_{22} & \cdots & G_{2N} \\ \vdots & \vdots & & \vdots \\ G_{N1} & G_{N2} & \cdots & G_{NN} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_N \end{bmatrix} = \begin{bmatrix} -E_{z'}(z_1) \\ -E_{z'}(z_2) \\ \vdots \\ -E_{z'}(z_N) \end{bmatrix} \tag{22}$$

and in compact notation by

$$[G_{mn}][I_n] = -[E_m] \qquad (\mathrm{V}\ \lambda^{-1} \text{ or } \mathrm{V}\ \mathrm{m}^{-1}) \tag{23}$$

where

$$\begin{aligned} m &= 1, 2, 3, \ldots, N \\ n &= 1, 2, 3, \ldots, N \end{aligned}$$

Multiplying both sides of the equation by the distance Δz,

$$\Delta z[G_{mn}][I_n] = -\Delta z[E_m] \qquad (\mathrm{V}) \tag{24}$$

we have on the left the product of an impedance (Z) and current (I) and on the right a voltage (V) as in an electric circuit equation (Ohm's law):

$$[Z_{mn}][I_n] = -[V_m] \qquad (\mathrm{V}) \tag{25}$$

EXAMPLE **14–11.1 Current Distribution and Impedance of a Short Dipole by the Moment Method**

To illustrate the application of the moment method (circuit equation) technique, use (16), (19) and (21) to calculate the current distribution and input impedance of a perfectly conducting center-fed cylindrical dipole 0.1λ long with a radius of 0.001λ. We presume no knowledge of what the impedance or current distribution should be except that the current distribution is symmetrical.

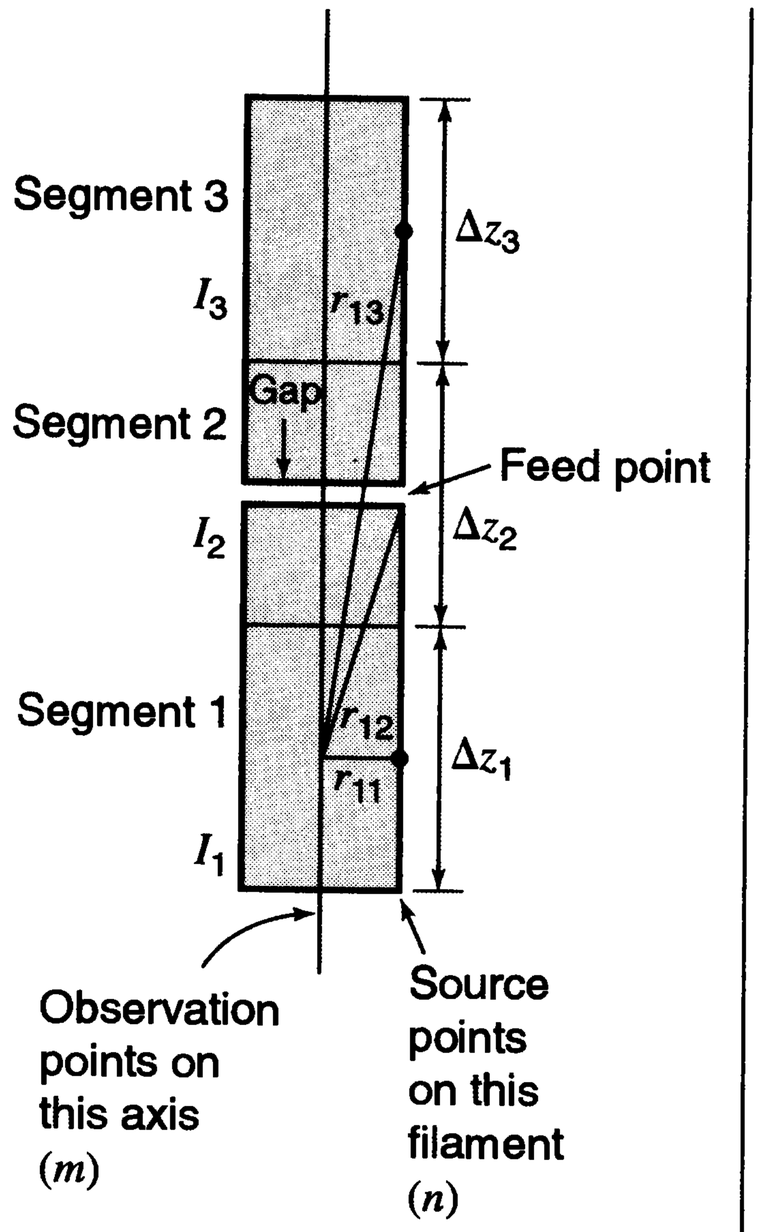


Figure 14–23
Short center-fed dipole 0.1λ long divided into 3 segments $\Delta z_1 = \Delta z_2 = \Delta z_3 = 0.033\lambda$ long.

■ **Solution**

Referring to Fig. 14–23, the dipole is divided into 3 segments $\Delta z' = 0.033\lambda$ long and each assumed to have a uniform current over each segment (pulse function) given by I_1, I_2 and I_3. Thus, $N = 3$ and (21) becomes

$$\begin{aligned} I_1 G_{11} + I_2 G_{12} + I_3 G_{13} &= -E(z_1') \\ I_1 G_{21} + I_2 G_{22} + I_3 G_{23} &= -E(z_2') \\ I_1 G_{31} + I_2 G_{32} + I_3 G_{33} &= -E(z_3') \end{aligned} \tag{26}$$

The upper and lower halves of the dipole are symmetrical so that $r_{12} = r_{21} = r_{23} = r_{32}$ and $r_{13} = r_{31}$. Also $r_{11} = r_{22} = r_{33}$. Accordingly, we need to evaluate (16) for only 3 distances, r_{11}, r_{12} and r_{13}, between the source and observation points (point matching).

Introducing numerical values for $r_{12} = 0.066\lambda$, $a = 0.001\lambda$ and $\Delta z = 0.033\lambda$ into (16),

$$\begin{aligned} G_{13} &= \frac{-j377 \times 0.033}{8\pi^2 \times 0.066^3}(\cos 2\pi \times 0.066 - j \sin 2\pi \times 0.066) \\ &\quad \times \left\{(1 + j2\pi \times 0.066)\left[2 - 3\left(\frac{0.001}{0.066}\right)^2\right] + 4\pi^2 \times 0.001^2\right\} \\ &= -25.8 - j1184 \qquad (\Omega\,\lambda^{-1}) \end{aligned} \tag{27}$$

Similarly,

$$G_{11} = -20.0 + j52{,}700 \qquad (\Omega\,\lambda^{-1}) \tag{28}$$

$$G_{12} = -25.6 - j12{,}800 \qquad (\Omega\,\lambda^{-1}) \tag{29}$$

Since the integrand of (16) is sensitive to small changes in r when r is small (particularly in the region where $r = \sqrt{1.5}a$ or less), G_{11} was divided into 5 subsegments, as indicated in Fig. 14–24. The segment with the smallest r contributes negligible resistance to the dipole impedance but a large negative reactance. For larger r's there is a contribution to the resistance and the reactance becomes positive. The change in sign of the reactance occurs at $r = \sqrt{1.5}a$. To perform the integration [by summation of (19)], the contributions of the subsegments are added to obtain the value in (28). Likewise, G_{12} was subdivided into 2 segments and the contributions added to obtain (29).

Introducing (27), (28) and (29) in (26) and multiplying by $\Delta z\,(=0.033\lambda)$ we obtain

	Amperes	Ohms	Volts	
$I_1(0.66 - j1739) + I_2(0.85 + j422)$	$+ I_3(0.85 + j39)$		$= V_1$	(30)
$I_1(0.85 + j422) + I_2(0.66 - j1739)$	$+ I_3(0.85 + j422)$		$= V_2$	(31)
$I_1(0.85 + j39) + I_2(0.85 + j422)$	$+ I_3(0.66 - j1739)$		$= V_3$	(32)

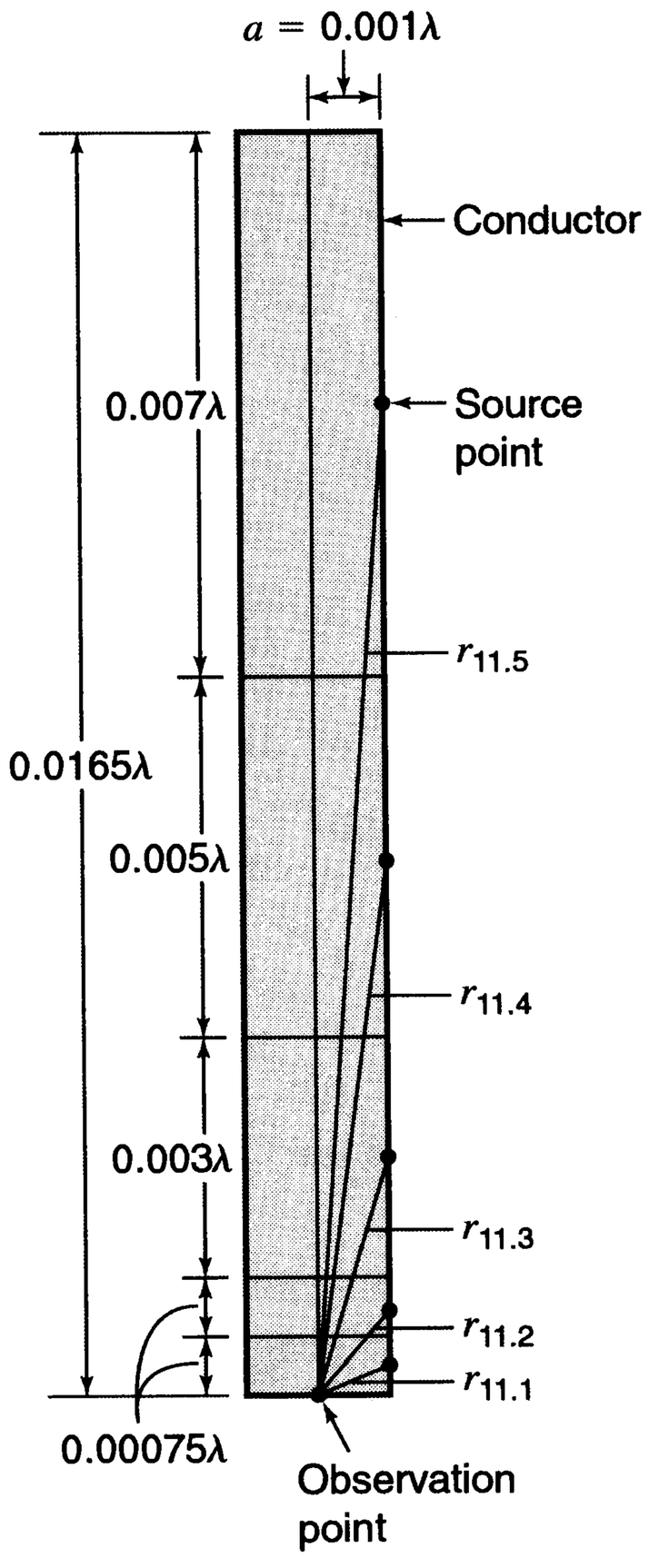


Figure 14–24
Half of one dipole segment divided into 5 subsegments for calculations of G_{11} in the example.

By symmetry $I_1 = I_3$; also for a center-fed dipole $V_1 = V_3 = 0$ so (30) and (32) are identical. Introducing the condition that $V_1 = 0$ in (30) yields a *current ratio* of

$$\frac{I_1}{I_2} = 0.25 + j0.0002 \qquad (Ans.)$$

Putting this ratio in (31), dividing by I_2 and setting $V_2 = 1$ V yields an *input impedance* of

$$Z = \frac{V_2}{I_2} = R + jX = 1.1 - j1528\ \Omega \qquad (Ans.) \tag{33}$$

For a short center-fed dipole the current distribution is nearly triangular so that the pulse function current ratio I_1/I_2 should be $\frac{1}{3}$ as suggested in Fig. 14–25. Ideally, the moment method should yield this value. If this ratio ($\frac{1}{3}$) is substituted in (31) we obtain

$$Z = 1.2 - j1458\ \Omega \tag{34}$$

For a dipole of the same length (0.1λ) but only $\frac{1}{10}$ the radius (0.0001λ), Richmond's convergence value of $Z = 1.852 - j1895\ \Omega$.

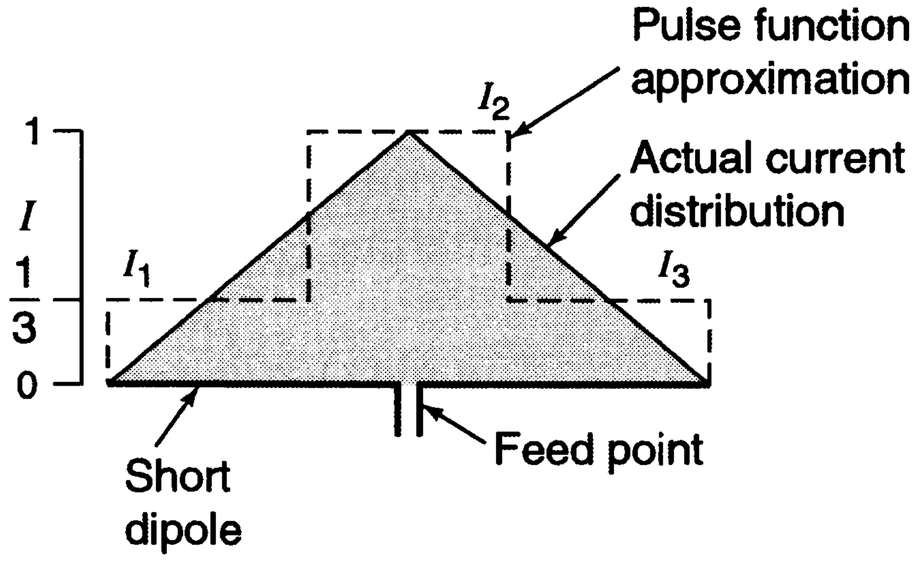


Figure 14–25
Stepped (pulse function) approximation and actual (triangular) current distribution on short center-fed dipole antenna.

It is not expected with the few segments used and the assumptions made that high accuracy would be obtained. The purpose of the example is to illustrate the moment method with actual numerical values for a simple case which can be solved with a small pocket calculator. It took the author an hour or so using one. However, such calculations if done frequently are most appropriately accomplished with a computer, especially one programmed to accommodate more segments and a range of dipole parameters such as length and diameter. The literature describing programs for doing this is extensive. The book *Antenna Design Using Personal Computers* by David M. Pozar (1) includes a number of programs on wire antennas. One of these called DIPOLE uses a standard moment method solution in a 254-step BASIC program which can handle a wide range of dipole dimensions.

The use of personal computers for antenna problems has been presented by Miller (1) and Burke with discussions of the Numerical Electromagnetics Code (NEC) and its subset MININEC for wire antennas. Many additional references on computer programs are listed at the end of this chapter and in App. C.

For MININEC programs, see also the book's website.

14–12 SELF-IMPEDANCE, RADAR CROSS SECTION AND MUTUAL IMPEDANCE OF SHORT DIPOLES BY THE METHOD OF MOMENTS *By Edward H. Newman*[1]

（用矩量法求短偶极子的自阻抗、雷达截面和互阻抗）

In this section the moment method (MM) with piecewise sinusoidal current modes is used to calculate the current distribution, input impedance and radar cross section of a short dipole. The mutual coupling of two adjacent short dipoles is also calculated. Using a simplification of the mutual impedance equations of Howard E. King (King, H. E.-1) the calculations can be done on a scientific hand calculator in an hour or so. Thus, the MM can be illustrated without resorting to a digital computer and taking the time and effort to write an appropriate computer program.

As in (14–11–17), the dipole current can be approximated by the series expansion[2]

$$I(z') = \sum_{n=1}^{N} I_n F_n(z') \tag{1}$$

where $F_n(z')$ is a piecewise sinusoidal mode. For example, the dipole may be divided into 4 equal segments of length $d = l/4$ as shown in Fig. 14–26. Segment n extends from z_n to z_{n+1}. The piecewise sinusoidal modes are placed on the dipole in an overlapping fashion, with mode n existing on segment n and $n+1$. Mode n has endpoints z_n and z_{n+2}, and center or terminals at z_{n+1}. F_n is a filament of electric current, located a radius a from the wire centerline (i.e., on the surface of the wire) and with current

$$F_n(z) = \frac{\sin\beta(d - |z - z_{n+1}|)}{\sin\beta d} \qquad \text{(A)} \tag{2}$$

where $\beta = 2\pi/\lambda$

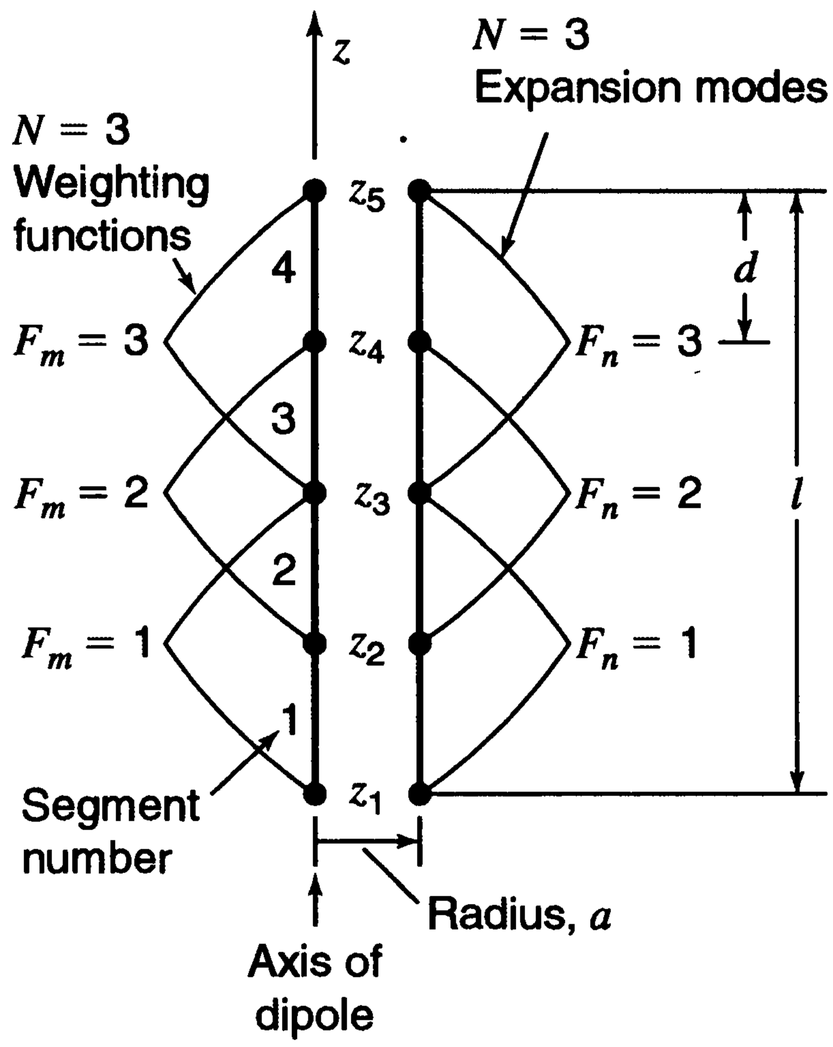


Figure 14–26
Three piecewise sinusoidal dipole modes on a dipole divided into 4 equal segments for Example 14–12.1.

[1]ElectroScience Laboratory, Ohio State University.

[2]We assume that the dipole is perfectly conducting and that the surrounding medium is free space.

$F_n(z)$ is zero at its endpoints and rises sinusoidally to a maximum at its center with terminal current of $I_{n0} = F_n(z_{n+1}) = 1$ A. Note that the piecewise sinusoidal modes produce a current which is continuous and also zero at the dipole endpoints. Except at the dipole endpoints, the dipole current of (1) at z_{n+1} is I_n amperes. Equation (1) produces a sinusoidal interpolation of the current values at the $N + 2$ points.

We require that the radiated and impressed fields satisfy (14–11–12). Substituting (1) into (14–11–12) we have

$$-\sum_{n=1}^{N} I_n E_{zn} \simeq E_{z'} \tag{3}$$

where

E_{zn} = free-space z component of the electric field of F_n

$E_{z'}$ = z component of the incident field

E_{zn} is available in terms of simple functions given by Schelkunoff (1) and Friis and H. E. King (1).

The weighting functions in the MM solution are chosen identical to the expansion functions, except that they are located along the centerline of the dipole. This is because we enforce (14–11–12) on the centerline. Then multiplying both sides of (3) by the sequence of N weighting functions, F_m ($m = 1, 2, \ldots, N$), (3) becomes an $N \times N$ system of simultaneous linear algebraic equations which can be written compactly in matrix form as

$$[Z]I = V \tag{4}$$

Here I is the current column vector whose N components contain the I_n of (1). $[Z]$ is the $N \times N$ impedance matrix whose typical term is

$$Z_{mn} = -\int_m E_{zn} F_m \, dz \tag{5}$$

In general $[Z]$ is dependent on the geometry and material composition of the scatterer, but not on the incident fields. A typical element of the right-hand-side or voltage vector V is given by

$$V_m = \int_m E_{z'} F_m \, dz \tag{6}$$

The integration in (5) and (6) is on the dipole centerline, and over the extent of F_m, that is, from $z = z_m$ to z_{m+2}. The dimensions of the elements of $[Z]$ and V are volt-amperes (VA), while the elements of I are dimensionless. If the Z_{mn} were divided by $I_{m0} I_{n0}$, then the Z_{mn} would have dimensions of ohms (Ω). Since in our case the modal terminal currents are $I_{n0} = 1$ A, Z_{mn} can be considered to have the dimensions of ohms. In any case, the $[Z]$ matrix is usually referred to as an impedance matrix and V as a voltage vector since the matrix equation (4) resembles an N-port generalization of Ohm's law.

The major problem in an MM solution is usually the evaluation of the elements in the impedance matrix. Typically this involves numerical integrations and/or the evaluation of special functions. As a result, most MM solutions are done on a digital computer and require a great deal of programming time and effort. For this reason, most MM solutions

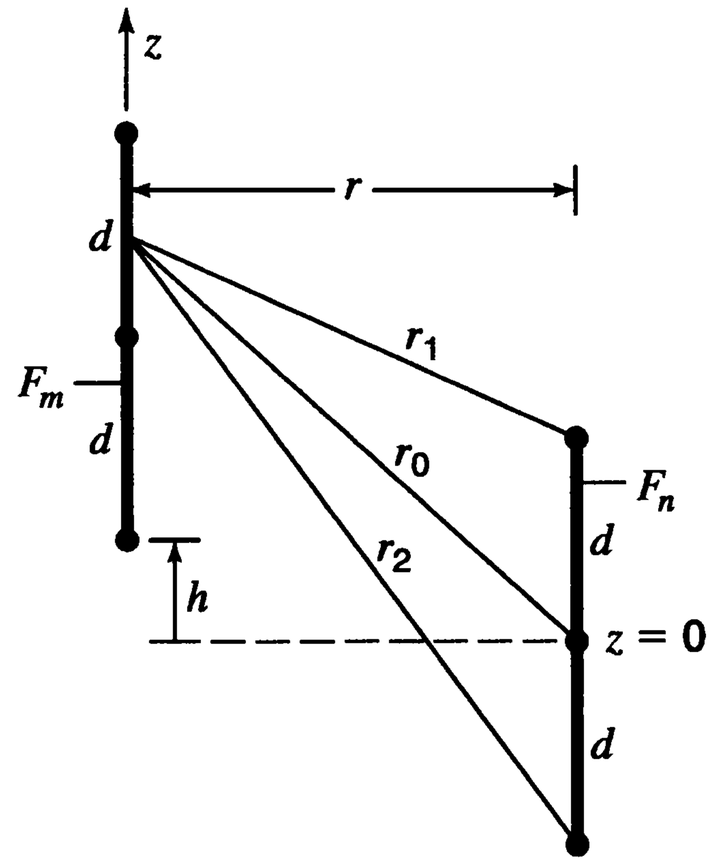


Figure 14–27
Geometry for the mutual impedance of 2 parallel piecewise sinusoidal dipoles.

are not suitable as a simple example problem which can be accomplished in about an hour using only a hand calculator. Relatively simple expressions for the elements in the dipole MM impedance matrix are presented here, thus eliminating the need for a digital computer to carry out the MM solution to the examples given.

For the dipole antenna, the elements in the impedance matrix, as given by (5), are the mutual impedances between parallel piecewise sinusoidal dipole modes. Figure 14–27 shows two parallel piecewise sinusoidal dipole modes of length $2d$. The bottom of weighting mode m is located a distance h above the center of expansion mode n, and the modes are staggered by the distance r. For convenience, the expansion mode has its center at $z = 0$. Exact expressions for the mutual impedance between these modes has been given by H. E. King (1). King's expressions are very lengthy and also require the evaluation of sine and cosine integrals. In order to simplify King's expressions we assume that the modes are electrically small and electrically close. Specifically, if we assume that βd, βh and βr are all $\ll 1$, it may be shown that King's expressions for the mutual impedance between modes m and n reduce to[1]

$$Z_{mn} = R_{mn} + jX_{mn} \tag{7}$$

where

$$R_{mn} = 20(\beta d)^2 \tag{8}$$

and

$$\begin{aligned} X_{mn} = -\frac{30}{\beta d^2}[&-4A + 6B - 4C + D + E + 4h \ln(2A + 2h) \\ &- 6(d + h) \ln(2B + 2h + 2d) + 4(2d + h) \ln(2C + 2h + 4d) \\ &+ (d - h) \ln(2D + 2h - 2d) - (3d + h) \ln(2E + 2h + 6d)] \end{aligned} \tag{9}$$

[1]I appreciate the assistance of Linda Bingham in obtaining these simplified expressions.

where

$$A = \sqrt{r^2 + h^2}$$
$$B = \sqrt{r^2 + (d+h)^2}$$
$$C = \sqrt{r^2 + (2d+h)^2}$$
$$D = \sqrt{r^2 + (d-h)^2}$$
$$E = \sqrt{r^2 + (3d+h)^2}$$

Equations (8) and (9) are suitable for a hand calculator, since they involve no operations more complicated than a logarithm and square root. Note that R_{mn} is the well-known formula for the radiation resistance of a short dipole, and is independent of mode separation. Equation (9) can be further simplified if we assume $r = a \ll d$ (wire radius much less than the segment length) and also consider certain special values of h. For self-impedance terms, $m = n$, and

$$X_{mm}(h = -d) = -\frac{30}{\beta d}\left[-4 + 4\ln\left(\frac{d}{a}\right)\right] \tag{10}$$

For adjacent modes with one overlapping segment, $|m - n| = 1$ and

$$X_{mn}(h = 0) = -\frac{30}{\beta d}\left[1 + 2\ln\left(\frac{1.54a}{d}\right)\right] \tag{11}$$

If $|m - n| = 2$, then the modes share a single point and

$$X_{mn}(h = d) = -\frac{30}{\beta d}[-0.68] \tag{12}$$

If $|m - n| \geq 3$, then the modes are not touching and

$$X_{mn}(h \geq 2d) = -\frac{30}{\beta d}\left[\frac{h}{d}\ln\frac{h^4(2d+h)^4}{(d+h)^6(h-d)(3d+h)} + \ln\frac{(2d+h)^8(h-d)}{(d+h)^6(3d+h)^3}\right] \tag{13}$$

Now consider the evaluation of the right-hand-side vector V. As seen in (6), V is dependent upon the z component of the incident electric field.

First, consider the case where the dipole is excited by a *voltage generator*. The simplest, and probably the most commonly used, model for a voltage generator is the so-called *delta-gap model* (Stutzman (1) and Thiele). A delta-gap generator is one that creates an extremely large, but highly localized, electric field polarized parallel to the wire centerline. A delta-gap generator located at $z = z'$ has the incident field

$$E_{z'} = v\,\delta(z - z') \tag{14}$$

where

v = generator voltage
$\delta(z)$ = unit area Dirac delta function

Normally the generators are placed at the center or terminals of the piecewise sinusoidal modes. Thus, referring to Fig. 14–26 for a dipole with $N = 3$ modes, the generator could be placed at $z = z_2$, z_3 or z_4, which would be at the terminals of modes 1, 2 or 3, respectively. Inserting the incident field from (14) into (6) shows that, if a delta-gap generator of v_m volts

is placed at the terminals of mode m, then

$$V_m = v_m \tag{15}$$

Element m of V is nonzero only if a nonzero generator is placed at the terminals of mode m.

Now consider the effect of placing a *lumped load* in the wire. A lumped load of Z_{lm} ohms, placed at the terminals of mode m, will produce a voltage of $-I_m Z_{lm}$ volts at these terminals. If we treat this voltage as a dependent delta-gap generator, then according to (15) we should add $-I_m Z_{lm}$ to V_m. However, this is an unknown voltage, since initially I_m is unknown. Since it is conventional to write all unknowns on the left-hand side of the matrix equation, we add $I_m Z_{lm}$ to both sides of row m of the matrix equation. Thus, it can be seen that a lumped load of Z_{lm} ohms placed at the terminals of mode m simply results in Z_{mm} being replaced by $Z_{mm} + Z_{lm}$.

There is no physical break or gap in the wire where a generator or load is placed. Thus, the current is continuous through generators and loads. However, there is a slope discontinuity, or jump in the derivative of the current, at the generator or load. Note that the piecewise sinusoidal modes account for this behavior by enforcing continuity of current on the wire and by allowing a slope discontinuity at their terminals.

Next consider the situation where the wire is excited by a normally incident plane wave. If a z-polarized plane wave is incident from the $+x$ axis with magnitude E_0, then

$$E_{z'} = E_0 e^{j\beta x} \tag{16}$$

Inserting (16) into (6) and integrating yields

$$V_m = \frac{2E_0}{\beta} \tan\left(\frac{\beta d}{2}\right) \tag{17}$$

EXAMPLE **14–12.1 Dipole Current Distribution and Input Impedance**

Compute the current distribution and input impedance of a center-fed dipole antenna. The dipole length $l = \lambda/10$ and radius $a = \lambda/10{,}000$.

■ **Solution**

As illustrated in Fig. 14–26 we use $N = 3$ piecewise sinusoidal modes on the dipole and segment the dipole into $N + 1 = 4$ equal segments of length $d = l/4 = 0.025\lambda$. In this case the 3×3 MM matrix (4) can be explicitly written as

$$\begin{bmatrix} Z_{11} & Z_{12} & Z_{13} \\ Z_{21} & Z_{22} & Z_{23} \\ Z_{31} & Z_{32} & Z_{33} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} \tag{18}$$

Since the current on the center-fed dipole is symmetric, $I_1 = I_3$. In this case, we can add column 3 of the matrix equation to column 1 and reduce the order 3 matrix equation to the order 2 matrix equation

$$\begin{bmatrix} (Z_{11} + Z_{13}) & Z_{12} \\ (Z_{21} + Z_{23}) & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \tag{19}$$

Table 14–2 Elements of the [Z] matrix (VA) for Example 14–12.1

Element	Approximate	Exact
Z_{11}	$0.4935 - j3454$	$0.4944 - j3426$
Z_{12}	$0.4935 + j1753$	$0.4945 + j1576$
Z_{13}	$0.4935 + j129.9$	$0.4885 + j132.2$

Reducing the order of the matrix equation from 3 to 2 greatly reduces the effort in the hand calculations required to solve the matrix equation.

Although [Z] in (18) and (19) contain 9 elements, only 3 are distinct, since from the symmetry of the dipole,

$$Z_{11} = Z_{22} = Z_{33} \tag{20}$$

$$Z_{12} = Z_{21} = Z_{23} = Z_{32} \tag{21}$$

$$Z_{13} = Z_{31} \tag{22}$$

The real part of each Z_{mn} is given by (8) as

$$R_{mn} = 0.4935 \text{ VA} \tag{23}$$

The imaginary part of the Z_{mn} can be computed from (9); however, here we choose to use the simpler forms of (10), (11) and (12). Table 14–2 shows the elements in the first row of the [Z] matrix of (18) computed by (7), (8), (10), (11) and (12) and by King's exact expressions. Note that the approximate values of [Z] are within 11 percent of the exact values.

If the approximate values of [Z] from Table 14–2 are substituted into (19) we obtain

$$\begin{bmatrix} 0.9869 - j3324 & 0.4935 + j1753 \\ 0.9869 + j3506 & 0.4935 - j3454 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{24}$$

where we have set $V_2 = 1$ VA since there is a 1-V generator at the terminals of mode 2. Equation (24) can be easily solved using Cramer's rule. The results for the elements in the dimensionless current vector are

$$I_1 = I_3 = 0.000\,328\,6 \angle 89.892° \tag{25}$$

$$I_2 = 0.000\,623\,0 \angle 89.926° \tag{26}$$

Dividing (26) by (25), the current ratio I_2/I_1 is very nearly equal to 1.9, indicating a nearly triangular current distribution on the dipole.

The dipole current in amperes can now be obtained by inserting these coefficients into (1) with $N = 3$. Thus, the dipole input or terminal current is I_2 amperes. The input impedance is given by the ratio of the input voltage to the input current; that is,

$$Z_{\text{in}} = \frac{1}{I_2} = 2.083 - j1605\ \Omega \qquad (\textit{Ans.}) \tag{27}$$

By contrast, if we were to use the exact values of $[Z]$ from Table 14–2, then the results for the current distribution and input impedance would be

$$I_1 = I_3 = 0.000\,249\,8 \angle 90.0^\circ \tag{28}$$

$$I_2 = 0.000\,521\,9 \angle 89.9^\circ \tag{29}$$

$$Z_{\text{in}} = 1.892 - j1916\ \Omega \qquad (Ans.) \tag{30}$$

EXAMPLE **14–12.2 Scattering from a Short Dipole**

Compute the radar cross section[1] of the same dipole considered in Example 14–12.1 for a wave at normal (broadside) incidence with the dipole terminated in a (conjugate) matched load.

■ **Solution**

To terminate the dipole in its conjugate matched load, we place a lumped load of Z_{in}^* at the center of the dipole, i.e., at the terminals of mode 2. Using the value of Z_{in} from (27),

$$Z_{l2} = Z_{\text{in}}^* = 2.083 + j1605\ \Omega \tag{31}$$

The impedance matrix for the loaded dipole is identical to that of the unloaded dipole except that we add Z_{l2} to the self-impedance of mode 2 to obtain

$$Z_{22} = (0.4935 - j3454) + (2.083 + j1605) = 2.576 - j1849\ \text{VA} \tag{32}$$

If the incident electric field is a unit amplitude z-polarized plane wave incident from the $+x$ axis, then the elements of the right-hand-side vector are identical and given by (17) with $E_0 = 1$:

$$V_m = 0.025\,05\ \text{VA}, \qquad m = 1, 2, 3 \tag{33}$$

Since the excitation and loading of the antenna are symmetric with respect to the center of the dipole, the current on the dipole remains symmetric. Thus, the current vector can still be computed from the order 2 matrix (19):

$$\begin{bmatrix} 0.9869 - j3324 & 0.4935 + j1753 \\ 0.9869 + j3506 & 2.576 - j1849 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 0.025\,05 \\ 0.025\,05 \end{bmatrix} \tag{34}$$

Equation (34) can be solved using Cramer's rule. The results for the elements in the dimensionless current vector are

$$I_1 = I_3 = 0.006\,515 \angle 0.581^\circ \tag{35}$$

$$I_2 = 0.012\,35 \angle 0.548^\circ \tag{36}$$

The scattered field is given by

$$E_s = \sum_{n=1}^{N} I_n E_{zn} \tag{37}$$

[1] See Sec. 12–5.

where E_{zn} is the free-space electric field of expansion mode F_n as in (3). For a field point on the $+x$ axis (i.e., in the backscatter direction) and in the far zone of the dipole, the electric field of F_n will be z polarized and given by

$$E_{zn} = j60 \tan\left(\frac{\beta d}{2}\right)\frac{e^{-j\beta x}}{x} \qquad (\text{V m}^{-1}) \tag{38}$$

Using (38) and the above values for the I_n, the far-zone backscattered electric field is

$$E_s = 0.1199 \underline{/90.6^\circ}\,\frac{e^{-j\beta x}}{x} \qquad (\text{V m}^{-1}) \tag{39}$$

The *radar cross section* of the dipole is

$$\sigma = 4\pi x^2 \frac{|E_s|^2}{|E_0|^2} = 0.1806\lambda^2 \qquad (Ans.) \tag{40}$$

By contrast, if we were to repeat this example with the exact value of $[Z]$ from Table 14–2, the results would be

$$I_1 = I_3 = 0.006\,199 \underline{/0.0^\circ} \tag{41}$$

$$I_2 = 0.012\,95 \underline{/0.0^\circ} \tag{42}$$

$$\sigma = 0.1801\lambda^2 \qquad (Ans.) \tag{43}$$

From Example 2–10.1 the maximum effective aperture of a short matched lossless dipole is given by

$$A_{em} = \frac{3}{8\pi}\lambda^2 = 0.119\lambda^2 \tag{44}$$

provided that the dipole length $l \ll \lambda$. The total scattering aperture equals A_{em} and the radar cross section is this value times the short dipole directivity D (=1.5) or

$$\sigma = DA_{em} = \frac{1.5 \times 3}{8\pi}\lambda^2 = 0.179\lambda^2 \tag{45}$$

as compared to 0.1806 and $0.1801\lambda^2$ above.

EXAMPLE **14–12.3 Mutual Impedance of 2 Short Dipoles**

Compute the mutual impedance of 2 short side-by-side dipoles separated by $\lambda/100$ as in Fig. 14–28. The dipoles are identical to the ones in Examples 14–12.1 and 14–12.2.

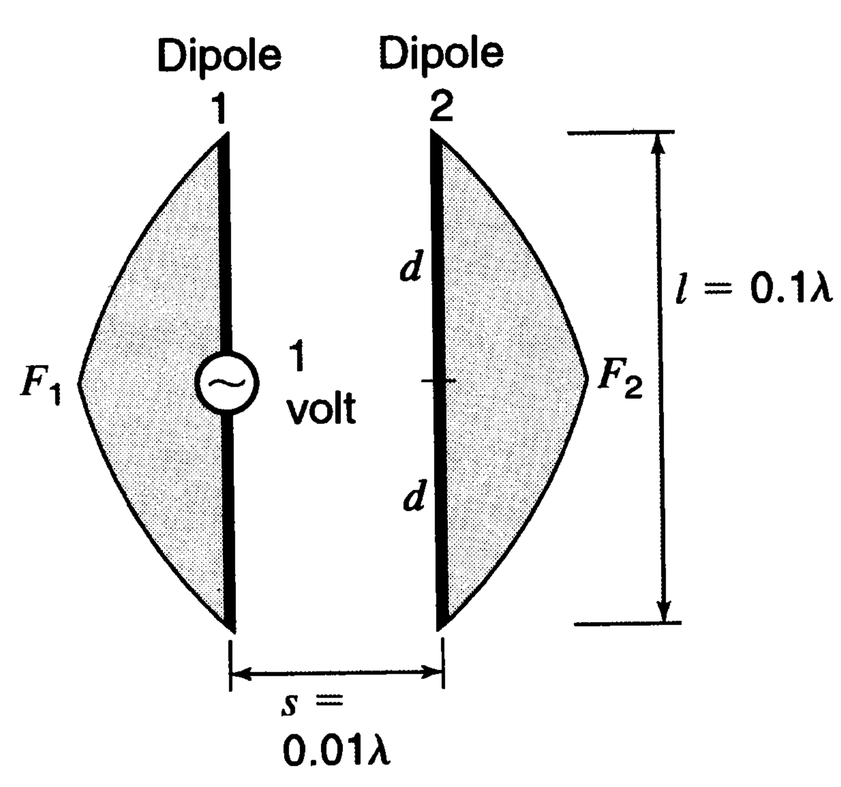


Figure 14–28
Geometry for 2 coupled dipoles of Example 3. The radius of each dipole is 0.0001λ.

Table 14–3 Elements of the [Z] matrix (VA) for Example 14–12.3

Element	Approximate	Exact
Z_{11}	$1.9739 - j1992$	$2.0000 - j1921$
Z_{12}	$1.9739 - j232.8$	$1.9971 - j325.1$

■ **Solution**

To simplify the computations, we place only one piecewise sinusoidal mode on each dipole. Thus, the order $N = 2$ matrix equation for this example is

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \tag{46}$$

Only Z_{11} and Z_{12} need be computed, since from the symmetry of the dipoles $Z_{11} = Z_{22}$ and $Z_{12} = Z_{21}$. Z_{11} is evaluated from (7), (8) and (10) with $d = -h = l/2 = 0.05\lambda$ and $r = a = 0.0001\lambda$. Z_{12} is evaluated from the same equations with a replaced by $s = 0.01\lambda$, since $h = -d$. The results for Z_{11} and Z_{12} are shown in Table 14–3 where they are compared with the exact values of H. E. King (1) and of Richmond (2). For this simple one mode per dipole solution, Z_{12} is the *mutual impedance* between the two dipoles.

Inserting the approximate values from Table 14–3 into (46) we obtain

$$\begin{bmatrix} 1.9739 - j1992 & 1.9739 - j232.8 \\ 1.9739 - j232.8 & 1.9739 - j1992 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \tag{47}$$

In (47) we set $V_1 = 1$, since to compute the input impedance of dipole 1, we place a 1-V generator at the terminals of mode 1. Equation (47) has the solution

$$I_1 = 0.000\,509\,0 \angle 89.955^\circ \tag{48}$$

$$I_2 = 0.000\,059\,49 \angle -89.616^\circ \tag{49}$$

The input impedance is then

$$Z_{\text{in}} = \frac{1}{I_1} = 1.539 - j1964\ \Omega \tag{50}$$

The $N = 1$ mode solution for the input impedance of dipole 1 alone is

$$Z_{\text{in}} = Z_{11} = 1.974 - j1992\ \Omega \tag{51}$$

If the exact values of Table 14–4 are used, the impedance of dipole 1 in the presence of dipole 2 is

$$Z_{\text{in}} = 1.382 - j1822\ \Omega \tag{52}$$

Table 14–4 gives a summary of self-impedance, mutual impedance and radar cross section for short dipoles.

Table 14–4 Summary of self-impedance, mutual impedance and radar cross section for short dipoles†

	Modes	Equation
$Z_s = 1.974 - j1992\ \Omega$	$N = 1$ (approx. King)*	(51)
$= 2.000 - j1921\ \Omega$	$N = 1$ (exact King)*	Z_{11} (exact), Table 14–3
$= 2.083 - j1605\ \Omega$	$N = 3$ (approx. King)*	(27)
$= 1.892 - j1916\ \Omega$	$N = 3$ (exact King)*	(30)
$= 1.864 - j1905\ \Omega$	$N = 5$ (exact King)*	J. H. Richmond (2)
$= 1.856 - j1899\ \Omega$	$N = 7$ (exact King)*	FORTRAN IV program (see References below)
$Z_m = 1.974 - j233\ \Omega$	$N = 1$ (approx. King)*	Z_{12} (approx.), Table 14–3
$= 1.997 - j325\ \Omega$	$N = 1$ (exact King)*	Z_{12} (exact), Table 14–3
$Z_\Delta = Z_s - Z_m = 0.000 - j1759\ \Omega$	$N = 1$ (approx. King)*	(51) $- Z_{12}$ (approx.)
$= 0.003 - j1596\ \Omega$	$N = 1$ (exact King)*	Z_{11} (exact) $- Z_{12}$ (exact)
$\sigma = 0.1806\lambda^2$ (approx. King)*		(40)
$= 0.1801\lambda^2$ (exact King)*		(43)
$= 0.179\lambda^2$ (short dipole theory)	*H. E. King (1)	(45)

† Z_s = self-impedance = input impedance of isolated dipole, Ω
Z_m = mutual impedance, Ω
σ = radar cross section, λ^2
Dipole length, $l = \lambda/10$
Dipole radius, $a = \lambda/10{,}000$
Dipole separation, $s = \lambda/100$ (for Z_m)
Note that Z_s should approach the true value as N increases provided the solution converges, also that the self-mutual difference $Z_\Delta \to 0$ as $s \to 0$ and $Z_\Delta \to Z_s$ as $s \to \infty$ ($Z_m \to 0$).

REFERENCES（参考文献）

Abraham, M. (1), "Die electrischen Schwingungen um einen stabförmigen Leiter, behandelt nach der Maxwellschen Theorie," *Ann. Physik,* **66,** 435–472, 1898.

Brown, G. H. (1), and O. M. Woodward, "Experimentally Determined Impedance Characteristics of Cylindrical Antennas," *Proc. IRE,* **33,** 257–262, April 1945.

Dorne, A. (1), in *Very High Frequency Techniques,* by Radio Research Laboratory Staff, McGraw-Hill, New York, 1947, Chap. 4.

Hallén, Erik (1), "Theoretical Investigations into the Transmitting and Receiving Qualities of Antennae," *Nova Acta Regiae Soc. Sci. Upsaliensis,* ser. IV, **11,** no. 4, 1–44, 1938.

Hallén, Erik (2), "On Antenna Impedances," *Trans. Roy. Inst. Technol.,* Stockholm, no. 13, 1947.

Hallén, Erik (3), "Admittance Diagrams for Antennas and the Relation between Antenna Theories," Cruft Laboratory Tech. Rept. **46,** Harvard University, 1948.

Harrington, R. F. (1), *Field Computation by Moment Methods,* Macmillan, 1968.

King, D. D. (1), "The Measured Impedance of Cylindrical Dipoles," *J. Appl. Phys.,* **17,** 844–852, October 1946.

King, H. E. (1), "Mutual Impedance of Unequal Length Antennas in Echelon," *IEEE Trans. Ants. Prop.,* **AP-5,** 306–313, July 1957.

King, L. V. (1), "On the Radiation Field of a Perfectly Conducting Base-Insulated Cylindrical Antenna over a Perfectly Conducting Plane Earth, and the Calculation of the Radiation Resistance and Reactance," *Phil. Trans. R. Soc. (Lond.),* **236,** 381–422, 1937.

King, Ronold (1), and C. W. Harrison, Jr., "The Distribution of Current along a Symmetrical Center-Driven Antenna," *Proc. IRE,* **31,** 548–567, October 1943.

Marsh, J. A. (1), "A Study of Phase Velocity on Long Cylindrical Conductors," Ph.D. thesis, Ohio State University, 1949.

Miller, E. K. (1), and G. J. Burke, "Personal Computer Applications in Electromagnetics," *IEEE Ants. Prop. Soc. Newsletter,* 5–9, August 1983.

Page, L. (1), and N. I. Adams, "The Electrical Oscillations of a Prolate Spheroid," *Phys. Rev.,* **53,** 819–831, 1938.

Pocklington, H. C. (1), "Electrical Oscillations in Wires," *Camb. Phil. Soc. Proc.,* **9,** 324–332, 1897.

Pozar, D. M. (1), *Antenna Design Using Personal Computers,* Artech House, 1985.

Richmond, J. H. (1), "Digital Computer Solutions of the Rigorous Equations for Scattering Problems," *Proc. IEEE,* **53,** 796–804, August 1965.

Richmond, J. H. (2), "Radiation and Scattering by Thin-Wire Structures in a Homogeneous Conducting Medium," *IEEE Trans. Ants. Prop.,* **AP-22,** 365, March 1974.

Schelkunoff, S. A. (1), and H. T. Friis, *Antennas Theory and Practice,* Wiley, New York, 1952.

Stratton, J. A. (1), and L. J. Chu, "Steady State Solutions of Electromagnetic Field Problems," *J. Appl. Phys.,* **12,** 230–248, March 1941..

Stutzman, W. L. (1), and G. A. Thiele, *Antenna Theory and Design,* 2d ed., Wiley, New York, 1998, Chap. 10.

Tai, C-T. (1), "Dipoles and Monopoles," in *Antenna Engineering Handbook,* McGraw-Hill, New York, 1984, Chap. 4.

PROBLEMS（习题）

14–6–1（偶极子阻抗） **Dipole impedance.** Calculate the input impedance of a cylindrical center-fed dipole antenna $\lambda/12$ long with a length-diameter ratio of 25.

14–10–1（电荷分布） **Charge distribution.** Determine the electrostatic charge distribution on a cylindrical conducting rod with a length-diameter ratio of 6.

14–11–1（短偶极子的输入阻抗） **Input impedance of short dipole.** Calculate the current distribution and input impedance of a center-fed dipole antenna $\lambda/15$ long with a length-diameter ratio of 200 using MM.

14–12–1（短偶极子的互阻抗） **Mutual impedance of short dipoles.** Calculate the mutual impedance of 2 side-by-side dipole antennas separated by $\lambda/25$ with each dipole $\lambda/8$ long and $\lambda/100$ diameter using MM.

14–12–2（λ/10 偶极子阻抗） **$\lambda/10$ dipole impedance.** Show that the convergence or true value of the self-impedance Z_s of the dipole of Table 14–4 is $1.852 - j1895\ \Omega$.

For computer programs, see Appendix C.

CHAPTER 15

The Fourier Transform Relation between Aperture Distribution and Far-Field Pattern[1]

口径分布与远场波瓣图之间的傅里叶变换关系

*See footnote.

The topics in this chapter include:

- Continuous aperture distribution（连续口径分布）
- Aperture distribution versus field pattern（口径分布与场波瓣图）
- Spatial frequency response and pattern smoothing（空间频率响应和波瓣图平滑处理）
- Simple adding interferometer（简单加法干涉仪）
- Aperture synthesis and multiple aperture arrays（口径合成和多口径阵）
- Grating lobes（栅瓣）
- Two-dimensional aperture synthesis（二维口径合成）
- Phase closure and self-calibration; "clean" maps（相位闭合和自校准，"干净"图形）

15–1 CONTINUOUS APERTURE DISTRIBUTION（连续口径分布）

Consider a continuous-current sheet or field distribution over an aperture as in Fig. 15–1. Assuming a current or field perpendicular to the page (y direction) that is uniform with respect to y, the electric field at a distance r from an elemental aperture $dx\,dy$ is

$$dE = -j\omega\, dA_y = -\frac{j\omega\mu}{4\pi r}\frac{E(x)}{Z}\, e^{-j\beta r}\, dx\, dy \qquad (1)^2$$

*If we could travel 10,000,000 light-years out into space and look back, this is how our galaxy might appear. The arrow points to where we probably live inside a spiral arm but we couldn't see our sun from this distance. Our solar system is so minuscule that it is smaller than the tiniest dot in the picture and invisible among the 100,000,000 other stars of our galaxy which turns slowly like a great wheel 100,000 light-years in diameter. This photograph is of the beautiful spiral galaxy Messier 81 which is believed to closely resemble our own. (Photo courtesy K. A. Strand).

[1]This chapter is taken from Chap. 6 of J. D. Kraus, *Radio Astronomy,* 2d ed., Cygnus-Quasar, 1986.

[2]Note that $E(x)/Z = E_y(x)/Z = H_x = J_y z$, where z (up in Fig. 15–1) is the thickness of the current sheet. Also $dE = dE_y$.

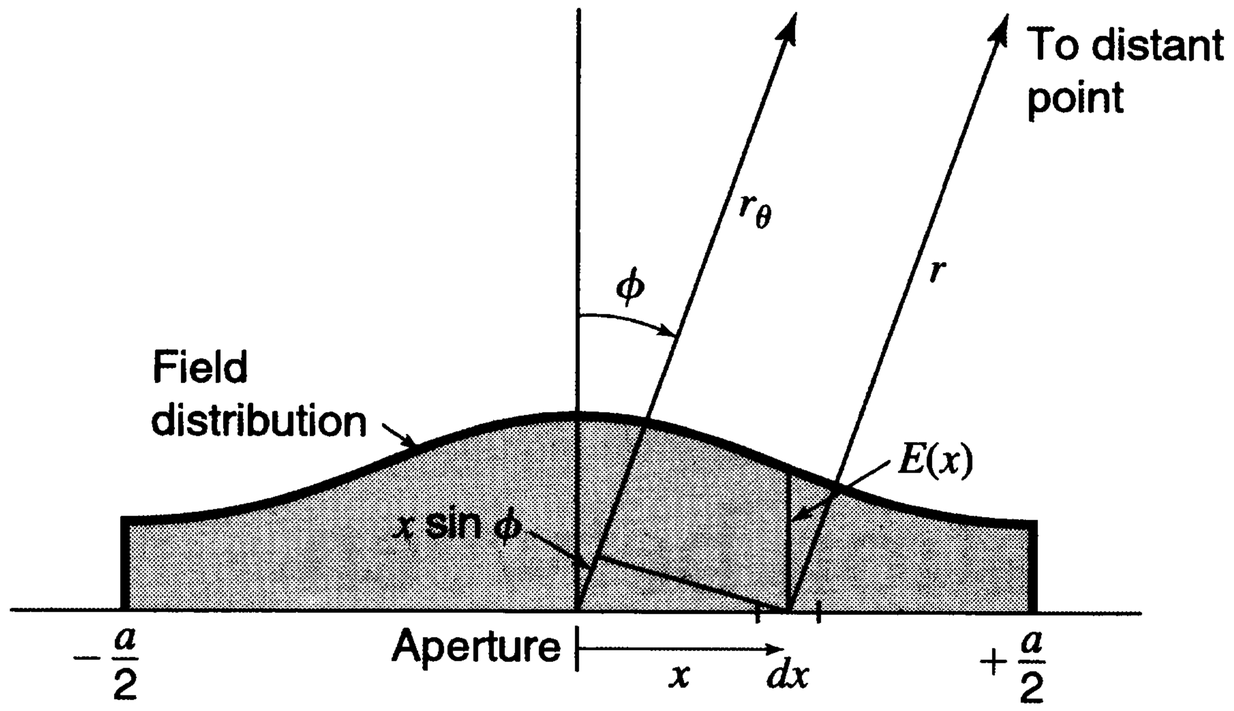


Figure 15–1
Aperture of width a and amplitude distribution $E(x)$.

where

$$\begin{aligned}
\mathcal{A}_y &= \text{vector potential } \left(= \frac{\mu}{4\pi} \iiint \frac{J_y}{r}\, dv, \text{ in general}\right), \text{ V s m}^{-1} \\
J_y &= \text{current density, A m}^{-2} \\
E(x) &= \text{aperture electric-field distribution, V m}^{-1} \\
Z &= \text{intrinsic impedance of medium, } \Omega \text{ square}^{-1} \\
\omega &= 2\pi f \ (f = \text{frequency}), \text{ rad s}^{-1} \\
\mu &= \text{permeability of medium, H m}^{-1}
\end{aligned}$$

For an aperture with a uniform dimension y_1 perpendicular to the page and with the field distribution over the aperture a function only of x, the electric field as a function of ϕ at a large distance from the aperture ($r \gg a$) is, from (1),

$$E(\phi) = \frac{-j\omega\mu y_1 e^{-j\beta r_0}}{4\pi r_0 Z} \int_{-a/2}^{+a/2} E(x)\, e^{j\beta x \sin\phi}\, dx \tag{2}$$

The magnitude of $E(\phi)$ is then

$$|E(\phi)| = \frac{y_1}{2r_0\lambda} \int_{-a/2}^{+a/2} E(x)\, e^{j\beta x \sin\phi}\, dx \tag{3}$$

where $\beta = 2\pi/\lambda$. For a uniform aperture distribution [$E(x) = E_a$], (3) reduces to

$$|E(\phi)| = \frac{y_1 E_a}{2r_0\lambda} \int_{-a/2}^{+a/2} e^{j\beta x \sin\phi}\, dx \tag{4}$$

and on axis ($\phi = 0$) we have

$$|E(\phi)| = \frac{E_a a y_1}{2r_0\lambda} = \frac{E_a A}{2r_0\lambda} \tag{5}$$

where

$$\begin{aligned}
A &= \text{aperture area } (= ay_1) \\
E_a &= \text{electric field in aperture plane}
\end{aligned}$$

For unidirectional radiation from the aperture (in direction $\phi = 0$ but not in direction $\phi = 180°$), $|E(\phi)|$ is twice the value given in (5).

Integration of (4) gives

$$|E(\phi)| = k_0 \frac{\sin[(\beta a/2)\sin\phi]}{(\beta a/2)\sin\phi} \tag{6}$$

where

$$k_0 = \frac{AE_a}{2r_0\lambda} \tag{7}$$

From (5–12–17) the field of a long array of n discrete sources of spacing d is

$$E = nE_0 \frac{\sin[(\beta a'/2)\sin\phi]}{(\beta a'/2)\sin\phi} \tag{8}$$

where the length of the long array is $a' = (n-1)d \simeq nd$ and E_0 = field of one source. It is also assumed in (8) that ϕ is restricted to small angles. This is not an undue restriction if the array is large and only the main lobe and first side lobes are of interest. Under these conditions it is clear that the field pattern (8) of the long array of discrete sources is the same as the pattern (6) for the continuous array of the same length ($a = a'$) (as already noted in Sec. 5–12).

15–2 FOURIER TRANSFORM RELATIONS BETWEEN THE FAR-FIELD PATTERN AND THE APERTURE DISTRIBUTION

（远场波瓣图与口径分布之间的傅里叶变换关系）

According to Booker (1) and Clemmow a 1-dimensional aperture distribution $E(x_\lambda)$ and its far-field distribution $E(\sin\phi)$ are reciprocal Fourier transforms as given by (Booker-1; Bracewell-1)

$$E(\sin\phi) = \int_{-\infty}^{\infty} E(x_\lambda)\, e^{j2\pi x_\lambda \sin\phi}\, dx_\lambda \tag{1}$$

and

$$E(x_\lambda) = \int_{-\infty}^{\infty} E(\sin\phi)\, e^{-j2\pi x_\lambda \sin\phi}\, d(\sin\phi) \tag{2}$$

where $x_\lambda = x/\lambda$. For real values of ϕ, $|\sin\phi| \leq 1$, the field distribution represents radiated power, while for $|\sin\phi| > 1$ it represents reactive or stored power (Rhodes-1). The field distribution $E(\sin\phi)$, or angular spectrum, refers to an angular distribution of plane waves. Except for $|\sin\phi| > 1$ the angular spectrum for a finite aperture is the same as the far-field pattern $E(\phi)$ (the far-field condition $r \gg a$ does not hold for an infinite aperture, i.e., where $a = \infty$). Thus, for a finite aperture the Fourier integral representation of (1) may be written

$$E(\phi) = \int_{-a_\lambda/2}^{+a_\lambda/2} E(x_\lambda)\, e^{j2\pi x_\lambda \sin\phi}\, dx_\lambda \tag{3}$$

This is identical with (15–1–2) except for constant factors. Equation (15–1–2) is an absolute relation, whereas (3) is relative. Examples of the far-field patterns $E(\phi)$ for several aperture distributions $E(x_\lambda)$ of the same extent are presented in Fig. 15–2.

FOURIER-TRANSFORM APERTURE-FAR-FIELD RELATIONS FOR RECTANGULAR APERTURES

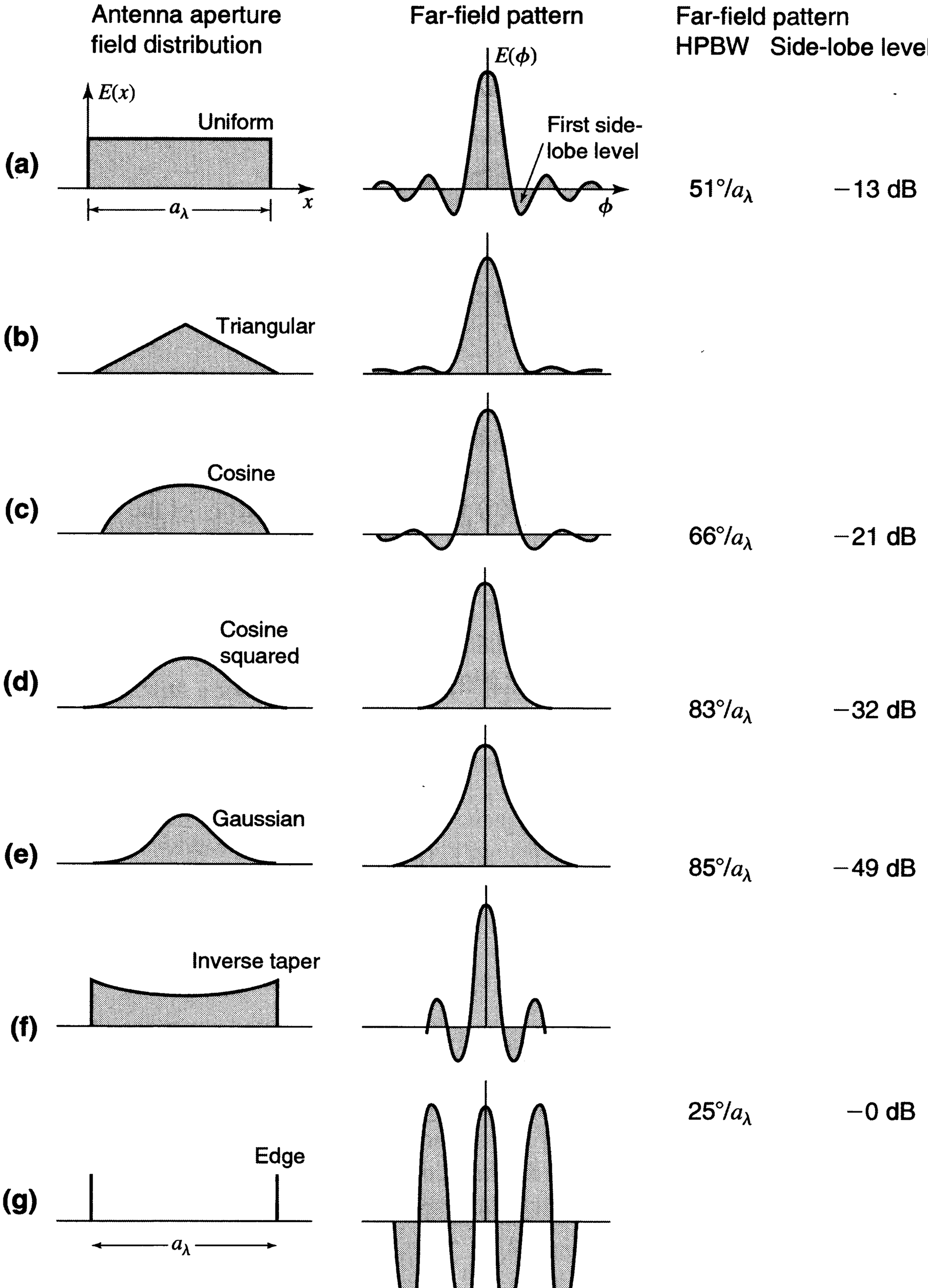


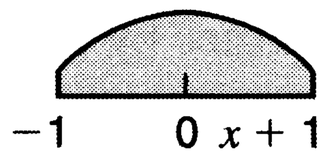
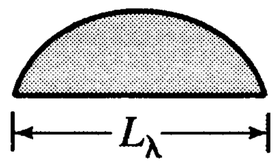
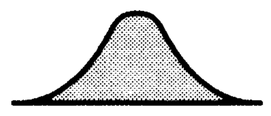
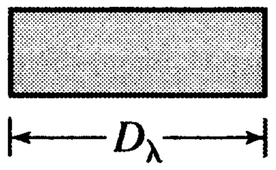
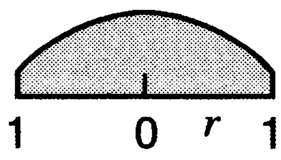
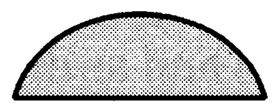
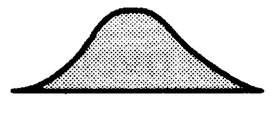
Figure 15–2
The near-field distributions across linear or rectangular aperture antenna (at left). Fourier transforms to the far-field patterns (at right) with the half-power beamwidths (HPBWs) and first side-lobe levels indicated. Thus, measuring the near field with a probe can give you the far-field pattern.

Taking the uniform distribution as reference, the more tapered distributions (triangular and cosine) have larger beamwidths and smaller minor lobes, while the most gradually tapered distributions (cosine squared and Gaussian) have still larger beamwidths but no minor lobes. On the other hand, an inverse taper (less amplitude at the center than at the edge), such as shown in Fig. 15–2*f*, yields a smaller beamwidth but larger minor lobes than for the uniform distribution. Such an inverse taper might inadvertently result from aperture blocking due to a feed structure in front of the aperture. Carrying the inverse taper to its extreme limit results in the edge distribution of Fig. 15–2*g*. This distribution is equivalent to that of a 2-element interferometer and has a beamwidth $\frac{1}{2}$ that of the uniform distribution but side lobes equal in amplitude to the main lobe.

A uniform line source or rectangular aperture distribution (Fig. 15–2*a*) produces the highest directivity. However, the first side lobe is only about 13 dB down. Thus, *aperture distributions used in practice are a trade-off or compromise between desired directivity (or gain) and side-lobe level,* as already discussed.

For circular dish antennas the relation between aperture, beamwidth, and first side-lobe level is shown in Table 15–1.

Table 15–1 Beamwidth and side-lobe level for rectangular and circular aperture distributions†

Aperture field distribution Rectangular or linear apertures $E(x)$		Half-power beamwidth	Level of first side lobe, dB
Tapered to $\frac{1}{3}$ at edge (~10 dB down) $E(x) = 1 - 2x^2/3$		$\frac{59°}{L_\lambda}$	−19
Tapered to zero at edge $E(x) = 1 - x^2 \simeq \cos(\pi x/2)$		$\frac{66°}{L_\lambda}$	−21
Tapered to zero at edge $E(x) = \cos^2(\pi x/2)$		$\frac{83°}{L_\lambda}$	−32
Circular apertures $E(r)$			
Uniform		$\frac{58°}{D_\lambda}$	−18
Tapered to $\frac{1}{3}$ at edge (~10 dB down) $E(r) = 1 - 2r^2/3$		$\frac{66°}{D_\lambda}$	−23
Tapered to zero at edge $E(r) = 1 - r^2$		$\frac{73°}{D_\lambda}$	−25
Tapered to zero at edge $E(r) = (1 - r^2)^2$		$\frac{84°}{D_\lambda}$	−31

†$L_\lambda = D/\lambda$, $D_\lambda = D/\lambda$. It is assumed that $L_\lambda \gg 1$ and $D_\lambda \gg 1$. For a uniform rectangular or linear aperture HPBW $\simeq 51°/D_\lambda$ with first side lobe −13 dB.

Theoretically, the directivity of a uniform aperture distribution can be exceeded (supergain condition) by large field fluctuations near the edges of the aperture. However, according to Rhodes (2), to obtain a $\frac{1}{2}$ dB increase in directivity from a 5λ aperture requires that the field near the edges be at least 20 times the field for a uniform aperture. The currents to produce these fields would have I^2R losses which offset any gain unless all conductors were perfectly conducting and the surrounding media were completely lossless. Furthermore, the antenna would have an enormous Q and an extremely narrow bandwidth, making supergain attempts impractical.

The situation here is reminiscent of the W8JK array which theoretically has a 4-dB gain over a single dipole even when the spacing between elements approaches zero; however, enormous currents would be required. With appreciable losses and for usable bandwidths a minimum spacing of $\lambda/8$ is a practical limit.

A useful property of (3) is that the distribution may be taken as the sum of 2 or more component distributions, $E_1(x_\lambda)$, $E_2(x_\lambda)$, etc., the resulting pattern being the sum of the transforms of these distributions. Thus,

$$E_1(\phi) + E_2(\phi) + \cdots = \int_{-a_\lambda/2}^{+a_\lambda/2} E_1(x_\lambda)\, e^{j2\pi x_\lambda \sin\phi}\, dx_\lambda + \int_{-a_\lambda/2}^{+a_\lambda/2} E_2(x_\lambda)\, e^{j2\pi x_\lambda \sin\phi}\, dx_\lambda + \cdots \tag{4}$$

15–3 SPATIAL FREQUENCY RESPONSE AND PATTERN SMOOTHING

（空间频率响应和波瓣图平滑处理）

It has been shown further by Booker (1) and Clemmow that the Fourier transform of the antenna power pattern is proportional to the complex autocorrelation function of the aperture distribution. Thus,

$$\overline{P}(x_{\lambda_0}) \propto \int_{-\infty}^{\infty} E(x_\lambda - x_{\lambda_0}) E^*(x_\lambda)\, dx_\lambda \tag{1}$$

where

$\overline{P}(x_{\lambda_0})$ = Fourier transform of antenna power pattern $P_n(\phi) \propto$ autocorrelation function of aperture distribution
$E(\phi)$ = field pattern
$E(x_\lambda)$ = aperture distribution
x_λ = x/λ = distance, λ
x_{λ_0} = x/λ_0 = displacement, λ

The autocorrelation function involves displacement x_{λ_0}, multiplication and integration. The situation for a uniform aperture distribution is illustrated by Fig. 15–3. The aperture distribution is shown at (*b*) and as displaced by x_{λ_0} at (*a*). The autocorrelation function, as shown at (*c*), is proportional to the area under the product curve of the upper 2 distributions or, in this case, to the area of overlap. It is apparent that the autocorrelation function is zero for values of x_{λ_0} greater than the aperture width a_λ since $\overline{P}(x_{\lambda_0}) = 0$ for $|x_{\lambda_0}| > a_\lambda$.

If a source moves through the beam of an antenna (or if the source is fixed and the antenna is rotated) the *observed response* of this scanning process is proportional to the

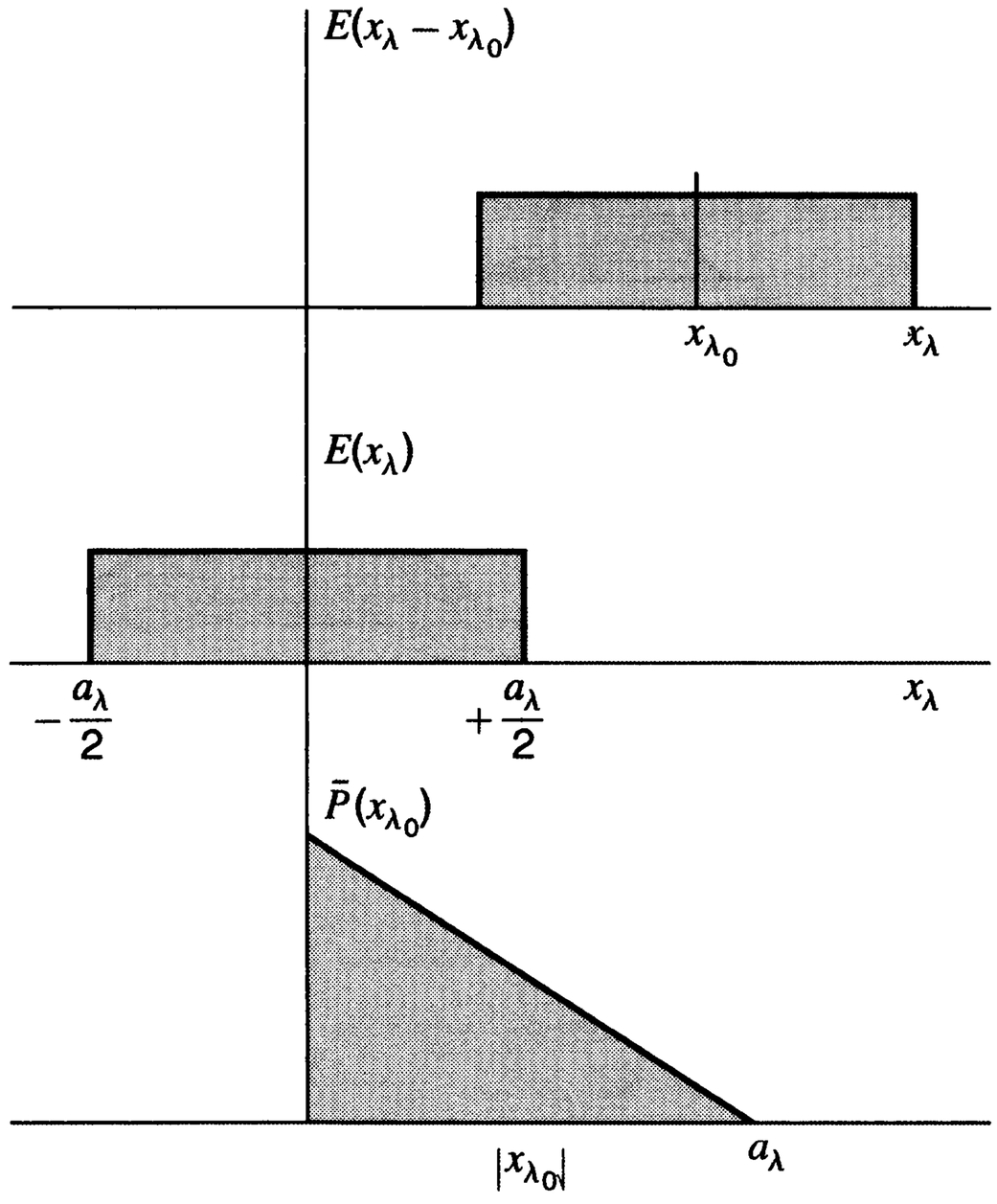


Figure 15–3
The autocorrelation function of the aperture distribution yields the Fourier transform of the antenna pattern.

convolution of the antenna power pattern and the source brightness distribution. Thus,

$$S(\phi_0) = \int_{-\infty}^{\infty} B(\phi)\tilde{P}_n(\phi_0 - \phi)\, d\phi \tag{2}$$

where

$S(\phi_0)$ = observed power distribution, W m^{-2} Hz^{-1}
$B(\phi)$ = true source brightness distribution, W m^{-2} Hz^{-1} sr^{-1}
$\tilde{P}_n(\phi)$ = mirror image of normalized antenna power pattern
ϕ_0 = displacement angle (scan angle)

It follows that

$$\overline{S}(x_\lambda) = \overline{B}(x_\lambda)\overline{P}(x_{\lambda_0}) \tag{3}$$

where the bars mean the Fourier transform. Since $\overline{P}(x_{\lambda_0})$ varies as the autocorrelation function of the aperture distribution, it follows that $\overline{S}(x_\lambda)$ and $S(\phi_0)$ are zero where $\overline{P}(x_{\lambda_0}) = 0$. This means that there is a cutoff for all values of x_{λ_0} greater than a_λ (Bracewell-2). The quantity x_{λ_0} is called the *spatial frequency* (wavelengths per aperture) and a_λ its *cutoff value*. Thus,

$$x_{\lambda_c} = a_\lambda \tag{4}$$

where x_{λ_c} = spatial-frequency cutoff

The reciprocal of x_{λ_c} gives an angle

$$\phi_c = \frac{1}{a_\lambda} \text{ rad} = \frac{57.3}{a_\lambda} \text{ deg} \tag{5}$$

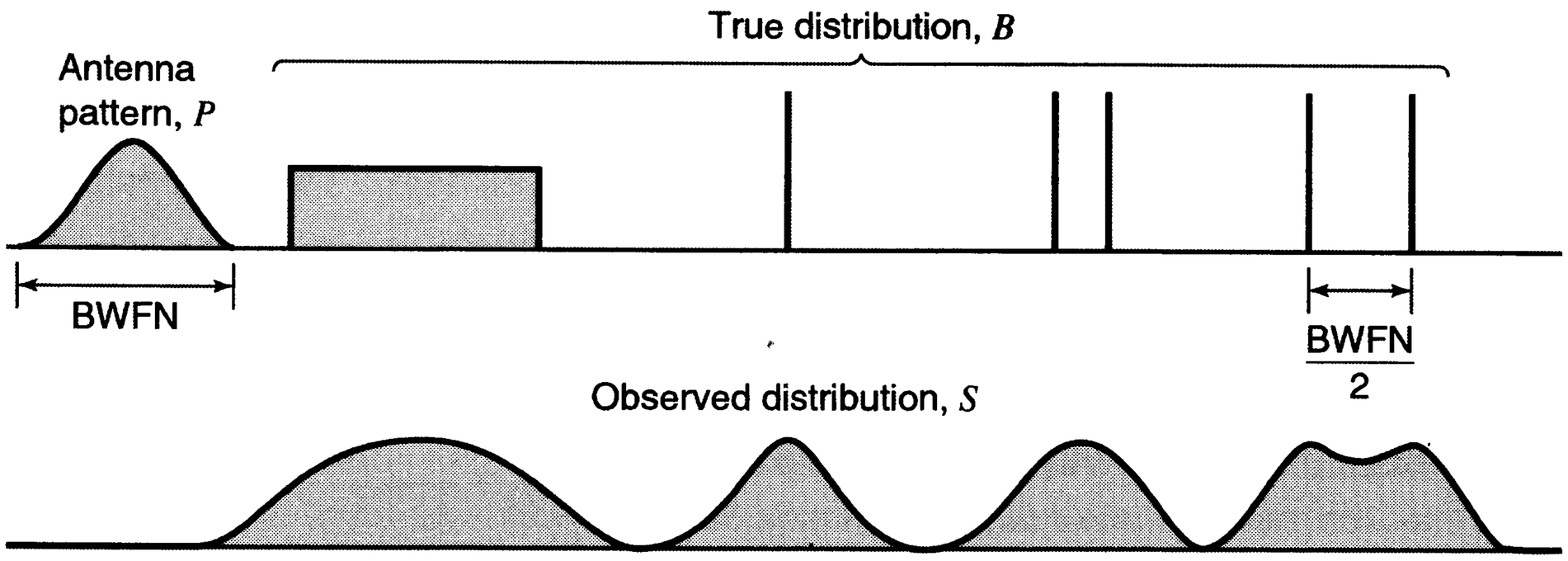


Figure 15–4
Smoothed distribution *S* observed with antenna pattern *P*.

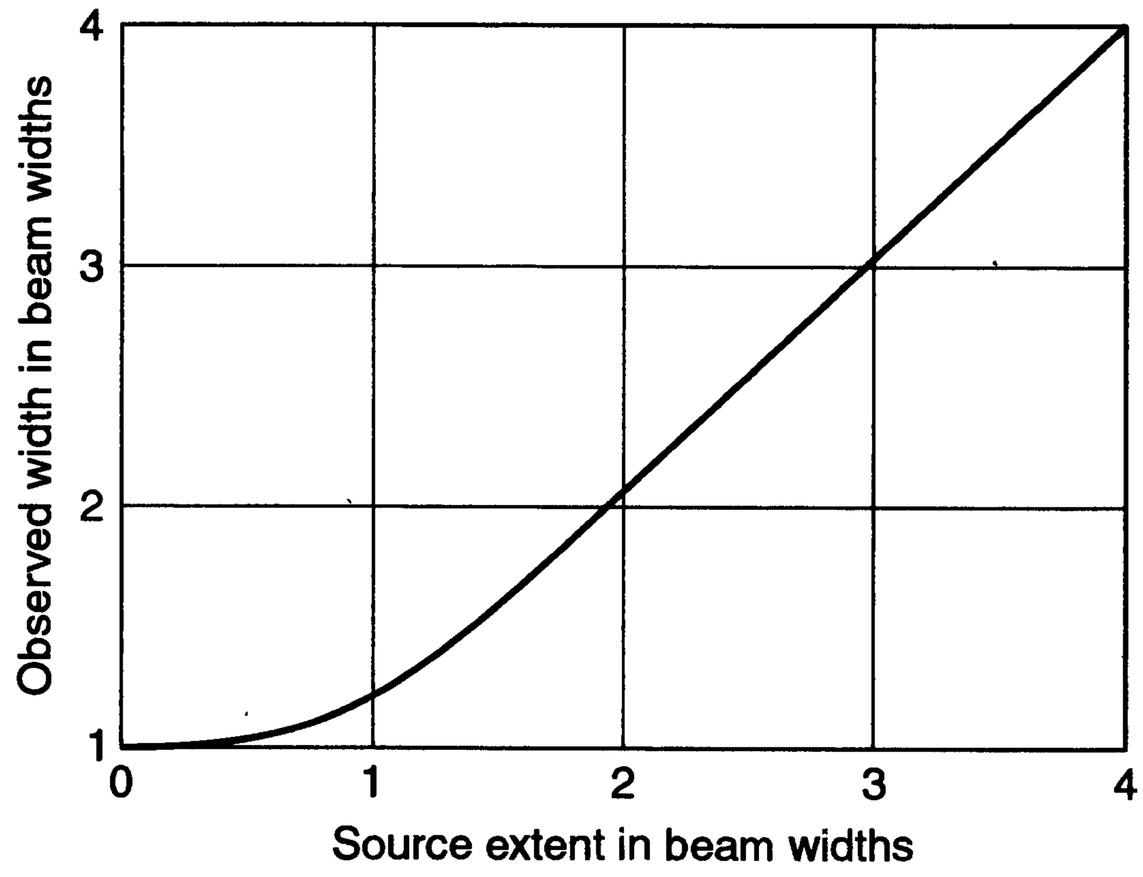


Figure 15–5
Observed half-power width as a function of source width in half-power beamwidths for a large uniform linear-aperture antenna and a uniform 1-dimensional source.

It follows that this (cutoff) angle ϕ_c is equal to $\frac{1}{2}$ the Beamwidth between First Nulls (BWFN) for a uniform aperture distribution ($\phi_c = \text{BWFN}/2$), and is 12 percent greater than the beamwidth at half-power ($\phi_c = 1.12$ HPBW). The significance of ϕ_c is that structure in the source distribution having a period of less than BWFN/2 will not appear in the observed response. Thus, the antenna tends to smooth the true brightness distribution (Bracewell-2). This is illustrated in Fig. 15–4. Half of the beamwidth between first nulls (BWFN/2) is equal to the *Rayleigh resolution*.[1] Thus, 2 point sources separated by this distance will be just resolved, as indicated at the right in Fig. 15–4.

The observed half-power width as a function of the source width in half-power beamwidths for a large uniform linear-aperture antenna and a uniform 1-dimensional source is shown in Fig. 15–5. A source of half-power width equal to the antenna half-power beamwidth produces about 20 percent beam broadening, or an observed width of 1.2 beamwidths. For larger source widths the observed width approaches the actual source width. Thus, from the amount of broadening an estimate may be made of the equivalent source extent.

---

[1]The *Rayleigh resolution* can, in principle, be improved upon in the signal-processing domain of an adaptive array.

15–4 THE SIMPLE (ADDING) INTERFEROMETER [简单（加法）干涉仪]

The resolution of an antenna or of a radio telescope can be improved, for example, by increasing the aperture a. However, this may not be economically feasible. A common expedient is to use two antennas spaced a distance s apart, as in Fig. 15–6.

If each antenna has a uniform aperture distribution of width a, the resulting autocorrelation function is as shown in Fig. 15–7. It is apparent that by making observations with spacings out to s_λ it is possible to obtain higher spatial-frequency components in the observed pattern to a cutoff

$$x_{\lambda_c} = s_\lambda + a_\lambda \tag{1}$$

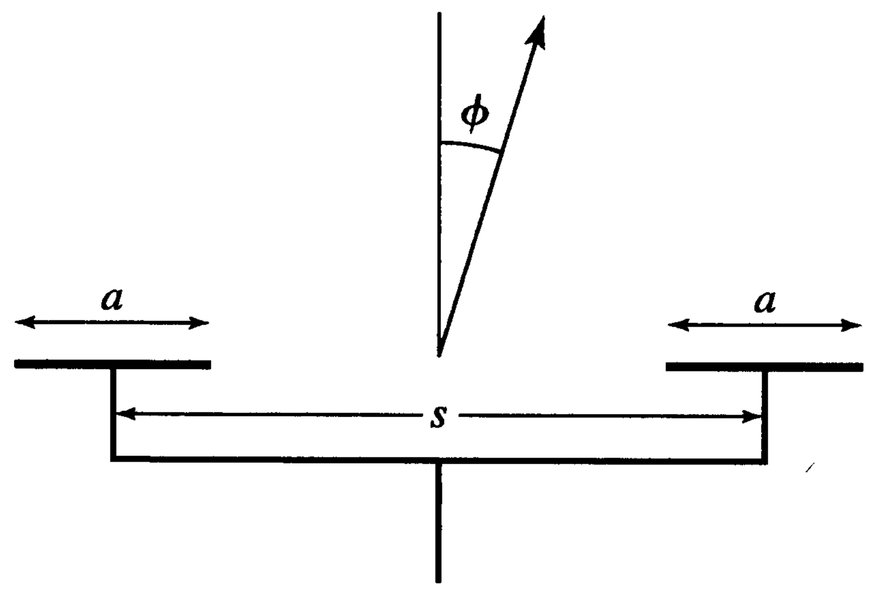


Figure 15–6
Simple 2-element interferometer.

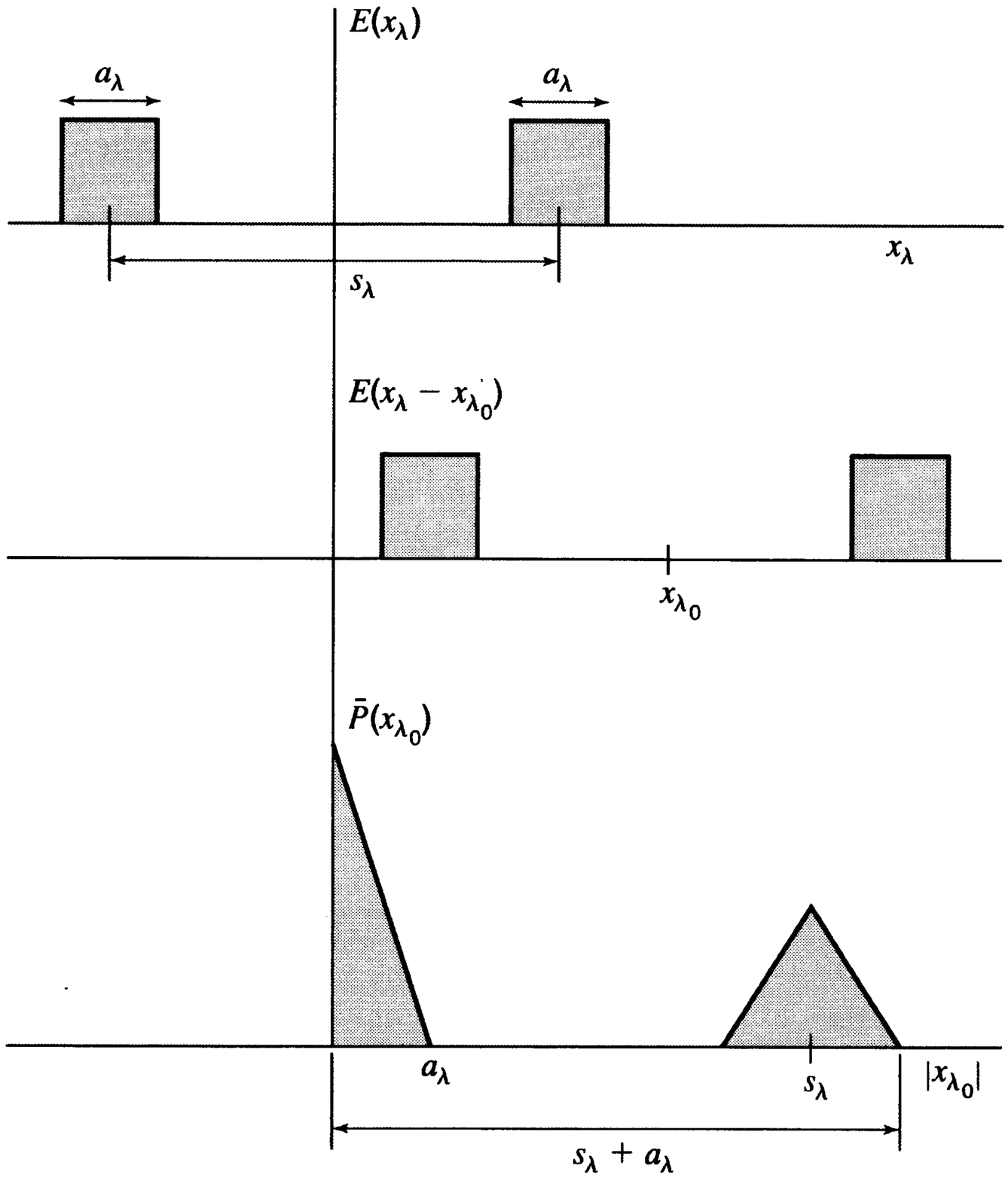


Figure 15–7
Autocorrelation function of aperture distribution of simple 2-element interferometer.

and a smaller resolution angle

$$\phi_c = \frac{1}{s_\lambda + a_\lambda} \quad \text{(rad)} = \frac{57.3}{s_\lambda + a_\lambda} \quad \text{(deg)} \tag{2}$$

In the following analysis it will be shown that if observations are made to sufficiently large spacings, it is possible, in principle, to deduce the true source distribution.

The normalized far-field pattern of the 2-element array is

$$E(\phi) = E_n(\phi) \cos \frac{\psi}{2} \tag{3}$$

where

$E_n(\phi)$ = normalized field pattern of individual array element
$\psi \quad = 2\pi s_\lambda \sin \phi$

The relative power pattern is equal to the square of $|E(\phi)|$, or

$$P(\phi) = |E(\phi)|^2 = |E_n(\phi)|^2 \cos^2 \frac{\psi}{2} = |E_n(\phi)|^2 \frac{1 + \cos \psi}{2} \tag{4}$$

For large spacings the pattern has many lobes, which, in optics, are referred to as fringes. The first null occurs when $\psi = \pi$, from which the beamwidth between first nulls, or *fringe spacing,* is

$$\text{BWFN} = \frac{1}{s_\lambda} \quad \text{(rad)} \quad = \frac{57.3}{s_\lambda} \quad \text{(deg)} \tag{5}$$

This is the BWFN/2 value for a continuous array of aperture width $a_\lambda = s_\lambda$ or a large array of discrete sources of the same length ($L_\lambda = s_\lambda$).

The pattern maxima occur when $\psi = 2\pi n$, where n (= 0, 1, 2, 3, . . .) is the *fringe order*. Thus,

$$\phi_{\max} = \frac{n}{s_\lambda} \quad \text{(rad)} \quad = \frac{57.3n}{s_\lambda} \quad \text{(deg)} \tag{6}$$

Referring to Fig. 15–8 the first factor in (4) represents the individual-element pattern, as shown in (*a*), and the second factor the pattern of the array of 2 elements, as in (*b*). The product of the 2 factors gives the interferometer pattern, as indicated in (*c*). In these patterns a point source is implied. In the general case, for a source of angular extent α the observed flux density is the convolution of the true source distribution and the antenna power pattern. Assuming that the source extent is small compared to the individual-element pattern, so that $|E_n(\phi)|$ is essentially constant across the source, we have in the 1-dimensional case that

$$\begin{aligned} S(\phi_0, s_\lambda) &= |E_n(\phi)|^2 \int_{-\alpha/2}^{+\alpha/2} B(\phi)\{1 + \cos[2\pi s_\lambda \sin(\phi_0 - \phi)]\}\, d\phi \\ &= |E_n(\phi)|^2 \left\{ \int_{-\alpha/2}^{+\alpha/2} B(\phi)\, d\phi + \int_{-\alpha/2}^{+\alpha/2} B(\phi) \cos[2\pi s_\lambda \sin(\phi_0 - \phi)]\, d\phi \right\} \\ &= |E_n(\phi)|^2 \left\{ S_0 + \int_{-\alpha/2}^{+\alpha/2} B(\phi) \cos[2\pi s_\lambda \sin(\phi_0 - \phi)]\, d\phi \right\} \end{aligned} \tag{7}$$

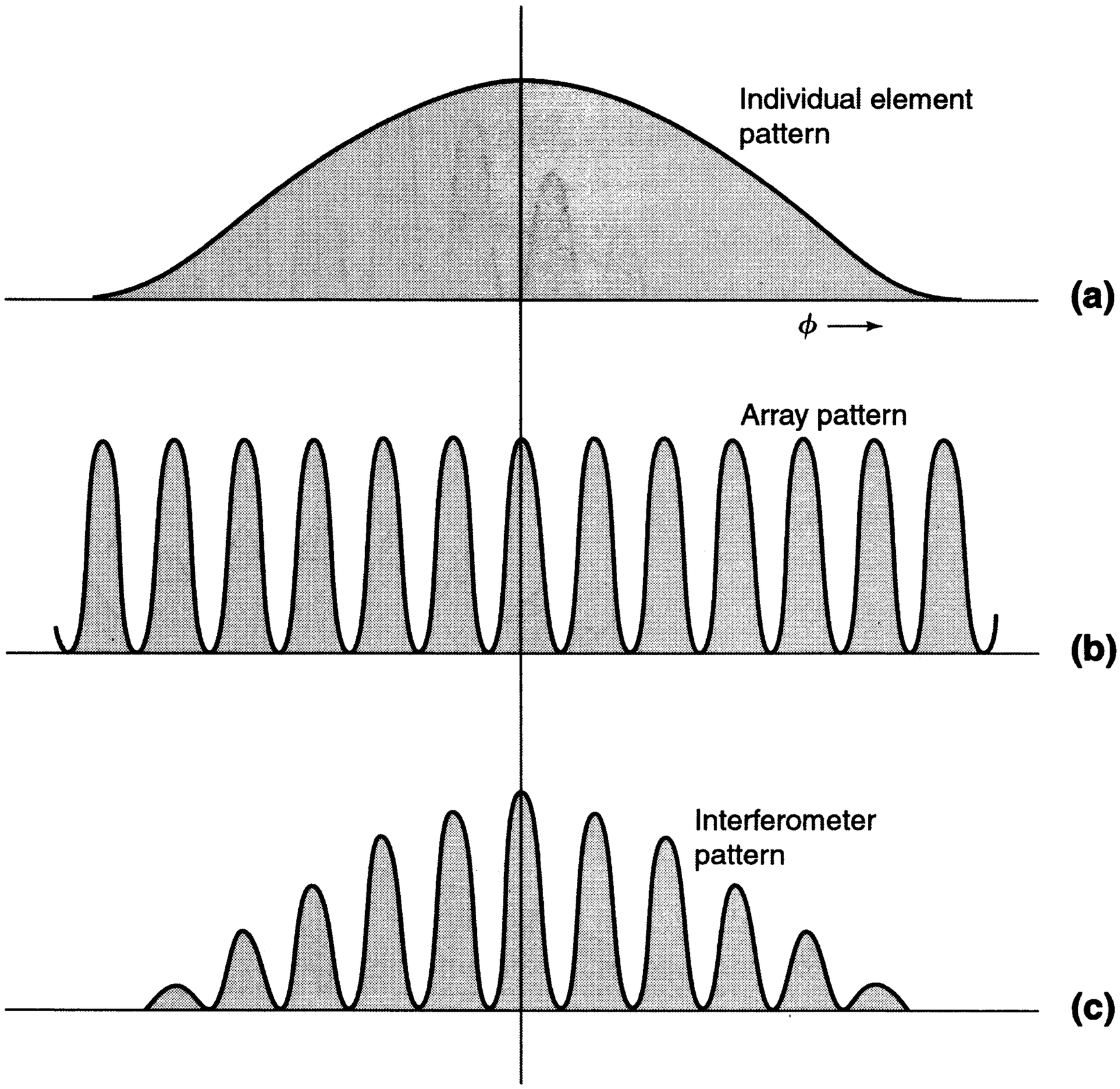


Figure 15–8
(*a*) Individual-element pattern, (*b*) array pattern and (*c*) the resultant interferometer pattern for the case of a point source.

where

$S(\phi_0, s_\lambda)$ = observed flux-density distribution, W m^{-2} Hz^{-1}
$B(\phi)$ = true source (brightness) distribution, W m^{-2} Hz^{-1} sr^{-1}
ϕ_0 = displacement angle (= hour angle), rad
α = source extent, rad
$s_\lambda = s/\lambda$ (where s = interferometer element spacing)
S_0 = flux density of source, W m^{-2} Hz^{-1}

The observed distribution as a function of displacement or scan angle is shown in Fig. 15–9 for 3 cases: Fig. 15–9*a* source extent very small compared with the lobe spacing ($\alpha \ll 1/s_\lambda$), the same as in Fig. 15–8; Fig. 15–9*b* source extent comparable to, but smaller than, the lobe spacing ($\alpha < 1/s_\lambda$); and Fig. 15–9*c* source distribution uniform and equal in extent to the lobe spacing ($\alpha = 1/s_\lambda$).

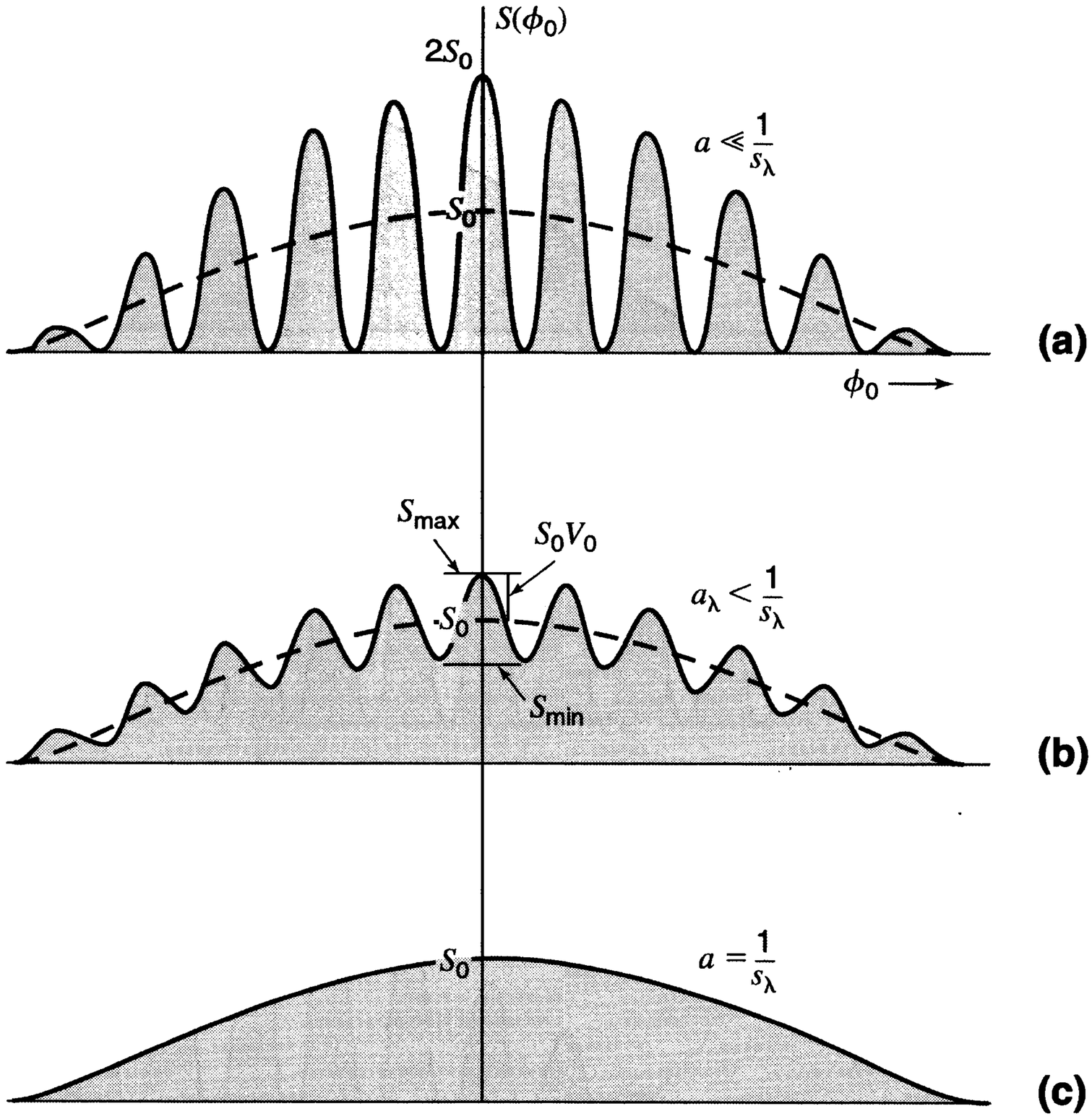


Figure 15–9
Interferometer pattern (*a*) for point source, (*b*) for a uniform extended source of angle $\alpha < 1/s_\lambda$ and (*c*) for a uniform extended source of angle $\alpha = 1/s_\lambda$.

Assuming that the observations are made broadside to the array or that the source is tracked by the individual array elements, so that $|E_n(\phi)|^2 = 1$, (7) becomes

$$S(\phi_0, s_\lambda) = S_0 + \int_{-\alpha/2}^{+\alpha/2} B(\phi) \cos[2\pi s_\lambda \sin(\phi_0 - \phi)]\, d\phi \tag{8}$$

If the source is small, so that $\phi_0 - \phi \ll \pi$, we may write

$$S(\phi_0, s_\lambda) = S_0 + \cos 2\pi s_\lambda \phi_0 \int_{-\alpha/2}^{+\alpha/2} B(\phi) \cos 2\pi s_\lambda \phi\, d\phi$$

$$+ \sin 2\pi s_\lambda \phi_0 \int_{-\alpha/2}^{+\alpha/2} B(\phi) \sin 2\pi s_\lambda \phi\, d\phi \tag{9}$$

$S(\phi_0, s_\lambda)$ may also be expressed as the sum of a constant term and a variable term (sum of 2 terms). Thus,

$$S(\phi_0, s_\lambda) = S_0[1 + V(\phi_0, s_\lambda)] \tag{10}$$

where

$$V(\phi_0, s_\lambda) = \frac{1}{S_0} \cos 2\pi s_\lambda \phi_0 \int_{-\alpha/2}^{+\alpha/2} B(\phi) \cos 2\pi s_\lambda \phi\, d\phi$$

$$+ \frac{1}{S_0} \sin 2\pi s_\lambda \phi_0 \int_{-\alpha/2}^{+\alpha/2} B(\phi) \sin 2\pi s_\lambda \phi\, d\phi \tag{11}$$

The variable term may also be expressed as a cosine function with a displacement $\Delta\phi_0$. Thus,

$$V(\phi_0, s_\lambda) = V_0(s_\lambda)\cos[2\pi s_\lambda(\phi_0 - \Delta\phi_0)] \tag{12}$$

or

$$V(\phi_0, s_\lambda) = V_0(s_\lambda)(\cos 2\pi s_\lambda\phi_0 \cos 2\pi s_\lambda\Delta\phi_0 + \sin 2\pi s_\lambda\phi_0 \sin 2\pi s_\lambda\Delta\phi_0) \tag{13}$$

The quantity $V_0(s_\lambda)$ represents the amplitude of the observed lobe pattern, i.e., the *fringe amplitude*. It is also called the *fringe visibility* or simply the *visibility*. As a function of s_λ, it may be referred to as the *visibility function*. The angle $\Delta\phi_0$ represents the fringe displacement from the position with a point source. Hence,

$$V_0(s_\lambda)\cos 2\pi s_\lambda\Delta\phi_0 = \frac{1}{S_0}\int_{-\alpha/2}^{+\alpha/2} B(\phi)\cos 2\pi s_\lambda\phi\, d\phi \tag{14}$$

and

$$V_0(s_\lambda)\sin 2\pi s_\lambda\Delta\phi_0 = \frac{1}{S_0}\int_{-\alpha/2}^{+\alpha/2} B(\phi)\sin 2\pi s_\lambda\phi\, d\phi \tag{15}$$

It follows that

$$V_0(s_\lambda)\, e^{j2\pi s_\lambda\Delta\phi_0} = \frac{1}{S_0}\int_{-\alpha/2}^{+\alpha/2} B(\phi)\, e^{j2\pi s_\lambda\phi}\, d\phi \tag{16}$$

The quantity $V_0(s_\lambda)e^{j2\pi s_\lambda\Delta\phi_0}$ is called the *complex visibility function*. If the source is contained within a small angle, the limits can be extended to infinity without appreciable error, giving

$$V_0(s_\lambda)\, e^{j2\pi s_\lambda\Delta\phi_0} = \frac{1}{S_0}\int_{-\infty}^{+\infty} B(\phi)\, e^{j2\pi s_\lambda\phi}\, d\phi \tag{17}$$

According to (17), the complex visibility function is equal to the Fourier transform of the source brightness distribution (times $1/S_0$). By the inverse Fourier transform we obtain

$$B(\phi_0) = S_0\int_{-\infty}^{+\infty} V_0(s_\lambda)\, e^{j2\pi s_\lambda\Delta\phi_0}\, e^{-j2\pi s_\lambda\phi_0}\, ds_\lambda \tag{18}$$

or

$$B(\phi_0) = S_0\int_{-\infty}^{+\infty} V_0(s_\lambda)\, e^{-j2\pi s_\lambda(\phi_0-\Delta\phi_0)}\, ds_\lambda \tag{19}$$

According to (18) and (19), the true brightness distribution of a source may be obtained, in principle, as the Fourier transform of the complex visibility function (an observable quantity).

To do this in practice requires observations at suitable intervals out to sufficiently large spacings, a high source signal-to-noise ratio and no other (confusing) sources of significant power in the individual-element response pattern. Thus, there are practical limits to the detail with which the source distribution can be determined. According to Bracewell (3), the spacing interval need be no smaller than $1/\alpha$, where α is the full source extent.

Referring to Fig. 15–9*b*, the visibility may be read from the observed record as

$$V_0(s_\lambda) = \frac{S_{\max} - S_{\min}}{S_{\max} + S_{\min}} = \text{visibility} \tag{20}$$

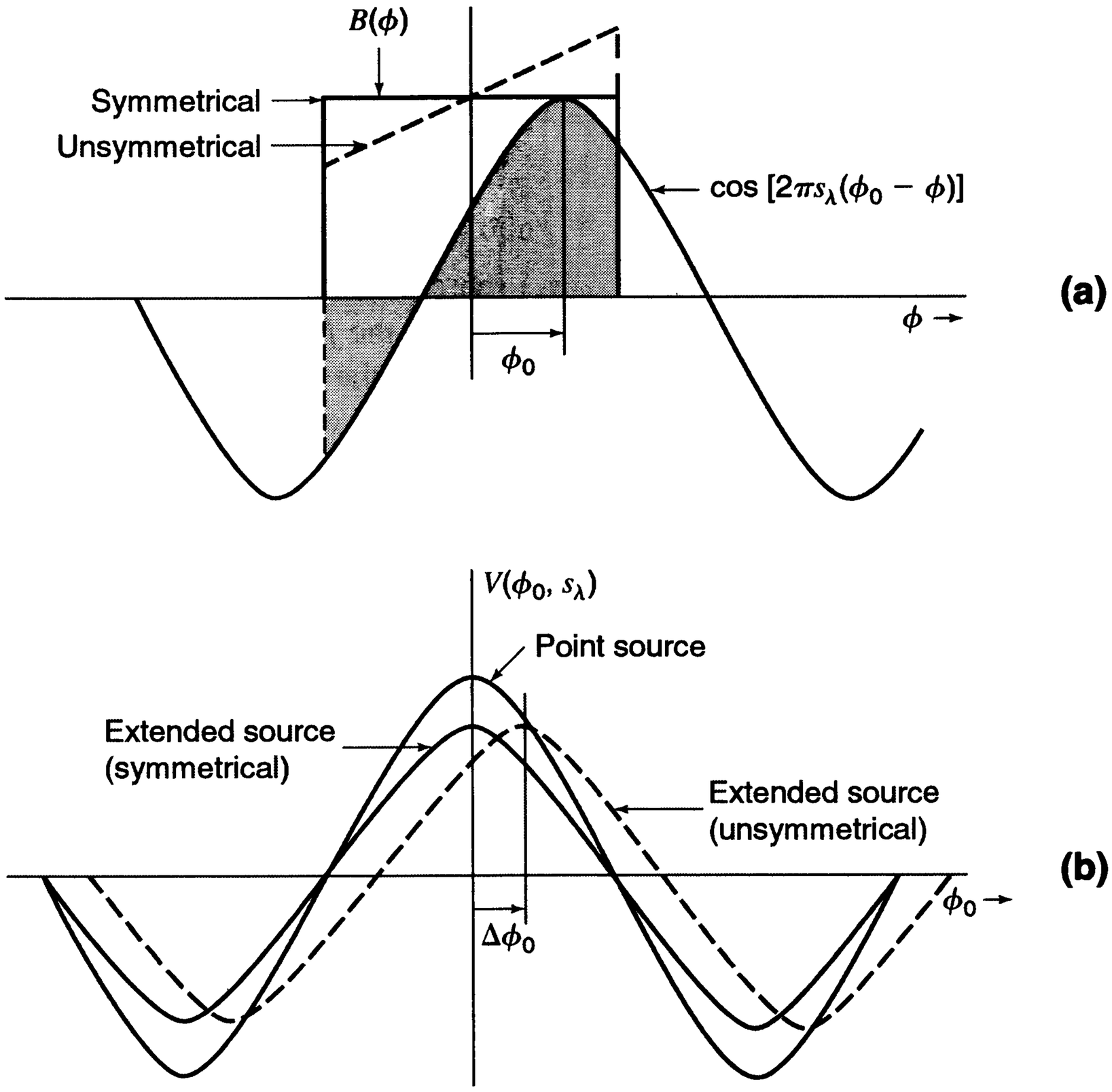


Figure 15–10
Interferometer patterns for symmetrical and unsymmetrical source distributions.

where

$$V_0(s_\lambda) = \text{visibility } [0 \leq V_0(s_\lambda) \leq 1]$$

Referring to Fig. 15–10*a* the value of the integral in (8) is proportional to the net shaded area, the areas above the ϕ axis being positive and the areas below negative. This integral (times $1/S_0$) is the variable quantity $V(\phi_0, s_\lambda)$, and its variation with respect to ϕ_0 for a fixed s_λ is a cosine function, as suggested by the solid curves in Fig. 15–10*b*, one for a point source ($\alpha \rightarrow 0$) and the other for an extended source. For symmetrical source distributions (even functions) the fringe displacement is zero or $\frac{1}{2}$-fringe ($\Delta\phi_0 = \frac{1}{2}s_\lambda$). For unsymmetrical sources, such as the one shown by the dashed lines in Fig. 15–10*a*, the fringes will have a displacement $\Delta\phi_0$, as suggested in Fig. 15–10*b*.

For symmetrical sources the visibility is, from (14),

$$V_0(s_\lambda) = \pm\frac{1}{S_0}\int_{-\alpha/2}^{+\alpha/2} B(\phi)\cos 2\pi s_\lambda \phi \, d\phi \tag{21}$$

For a uniform source [$B(\phi) =$ constant] and noting that $\alpha B(\phi) = S_0$, (21) reduces to

$$V_0(s_\lambda) = \pm\frac{\sin 2\pi s_\lambda(\alpha/2)}{2\pi s_\lambda(\alpha/2)} \tag{22}$$

A graph of the visibility $V_0(s_\lambda)$ as a function of s_λ is presented in Fig. 15–11 for the case

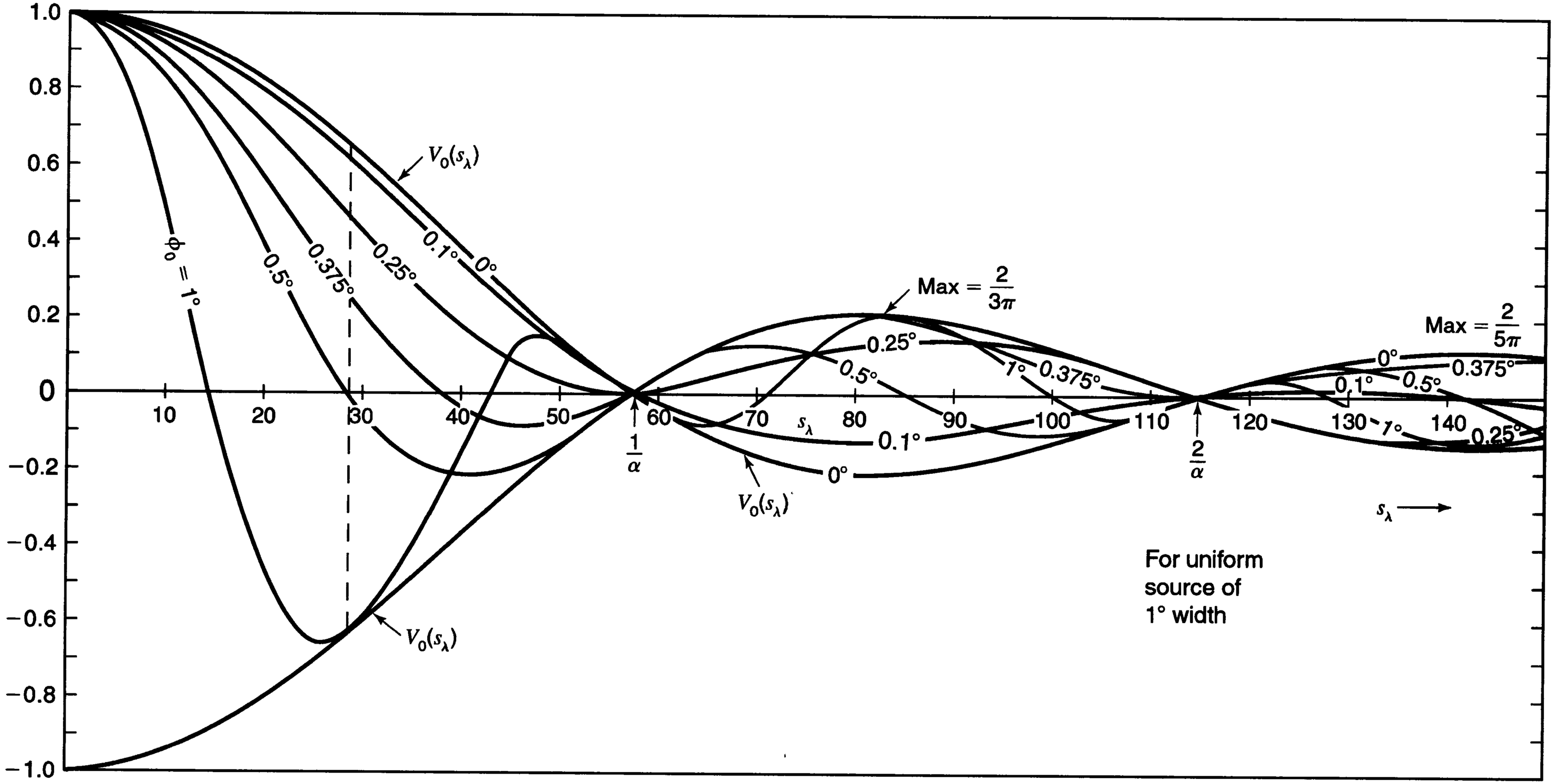


Figure 15–11
Visibility functions versus interferometer spacing s_λ for a uniform source of 1° width. At a fixed spacing s_λ, $V(\phi_0, s_\lambda)$ varies between the maximum and minimum curves $V_0(s_\lambda)$ as a function of the fringe drift or displacement ϕ_0. Thus, for $s_\lambda = 28.6$ (dashed line), $V(\phi_0, s_\lambda)$ is about 0.65 for $\phi_0 = 0°$, about 0.25 for $\phi_0 = 0.375°$, 0.0 for $\phi_0 = 0.5°$ and about -0.65 for $\phi_0 = 1°$.

where the source is uniform and 1° in width. As the source extent becomes small compared to the fringe spacing ($\alpha \ll 1/s_\lambda$), the visibility $V_0(s_\lambda)$ approaches unity, but as the fringe spacing becomes very small compared to the source extent ($\alpha \gg 1/s_\lambda$), $V_0(s_\lambda)$ tends to zero. The visibility is also zero (for a uniform source) when the source extent is equal to the fringe spacing ($1/s_\lambda$) or integral multiples thereof. For symmetrical sources we have from (19) that the source power distribution is given by the Fourier cosine transform of the visibility function, or

$$B(\phi_0) = S_0 \int_{-\infty}^{+\infty} V_0(s_\lambda) \cos 2\pi s_\lambda \phi_0 \, ds_\lambda \tag{23}$$

Also, from (13),

$$V(\phi_0, s_\lambda) = V_0(s_\lambda) \cos 2\pi s_\lambda \phi_0 \tag{24}$$

so that another form for (23) is

$$B(\phi_0) = S_0 \int_{-\infty}^{+\infty} V(\phi_0, s_\lambda) \, ds_\lambda = 2S_0 \int_0^{+\infty} V(\phi_0, s_\lambda) \, ds_\lambda \tag{25}$$

Curves of $V(\phi_0, s_\lambda)$ as a function of s_λ for several values of ϕ_0 are also shown in Fig. 15–11 for the uniform 1° source.

EXAMPLE **15–4.1 Source Observation with 2-Element Interferometer**

A uniform source of unknown extent is observed with a simple 2-element interferometer at spacings up to 100λ for determining the source visibility function. Find the source extent.

■ **Solution**

Substituting (22) in (23),

$$B(\phi_0) = 2S_0 \int_0^{\infty} \cos 2\pi s_\lambda \phi_0 \frac{\sin 2\pi s_\lambda (\alpha/2)}{2\pi s_\lambda (\alpha/2)} \, ds_\lambda \tag{26}$$

This equation has the form

$$B(\phi_0) = \frac{2S_0}{\alpha\pi} \int_0^{\infty} \cos mx \frac{\sin x}{x} \, dx \tag{27}$$

where

$$x = 2\pi s_\lambda \alpha/2$$
$$m = 2\phi_0/\alpha$$

The definite integral (27) is well known and yields

$$B(\phi_0) = \frac{2S_0}{\alpha\pi} \frac{\pi}{2} = \frac{S_0}{\alpha} \qquad (\mathrm{W\,m^{-2}\,Hz^{-1}\,sr^{-1}})$$

for $-1 < m < +1$ and $B(\phi_0) = 0$ for $m < -1$ and $m > +1$.

Suppose that the visibility function is like the one in Fig. 15–11 with the visibility going through zero at an interferometer spacing of 57.3λ ($s_\lambda = 57.3$) or

$$\alpha \text{ (source width)} = \frac{1 \text{ rad}}{s_\lambda} = \frac{57.3^\circ}{57.3} = 1^\circ$$

with the resulting source brightness

$$B_0 = \frac{S_0}{\alpha} = 57.3 S_0 \qquad (\text{W m}^{-2}\,\text{Hz}^{-1}\,\text{rad}^{-1})$$

where S_0 = source power density.

The source has this brightness in the range $-\frac{1}{2}^\circ < \phi_0 < +\frac{1}{2}^\circ$ $(-1 < m < +1)$ and is zero outside, as indicated in Fig. 15–12.

By scanning a source with an interferometer over a range of spacings the visibility function was obtained. Then by taking the Fourier transform of the visibility a source brightness distribution was reconstructed (Fig. 15–12) which is the same one discussed earlier in our analysis. Thus, in this hypothetical example we have gone full circle.

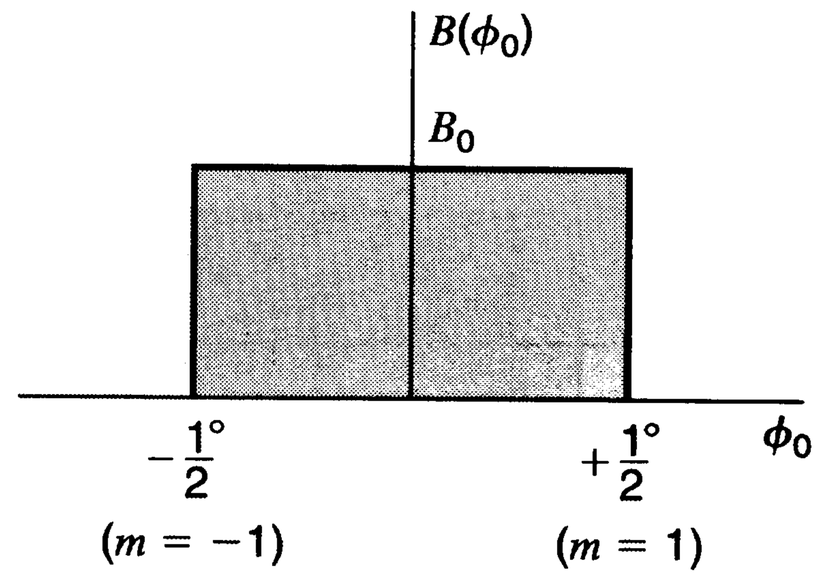


Figure 15–12
Source distribution reconstructed from visibility function.

Examples of visibility functions for other source distributions are given in Fig. 15–13. The case of a uniform source of extent α with a symmetrical hole of extent β is shown in Fig. 15–13*a* for several cases of hole width. When β approaches α, the distribution approaches that of 2 point sources with a separation α. When $\beta = 0$, the source distribution is uniform. In Fig. 15–13*b* the visibility function is presented for a uniform source with a bright center of 4 times the side brightness.

It is to be noted that the visibility functions of Fig. 15–13*a* can be obtained as the visibility for a uniform source distribution of width α minus the visibility of a uniform source of width β, while in Fig. 15–13*b* the result can be obtained as the sum of 2 uniform distributions of widths α and β.

Comparing (17) with (15–2–1) and (15–2–3), it is apparent that the complex visibility function is related to the source brightness distribution in the same manner that the far-field pattern of an antenna is related to the antenna aperture distribution. Accordingly, the graphs of Fig. 15–2 may also be interpreted as giving the visibility functions for various source distributions and the graphs of Fig. 15–13 the field patterns for various aperture distributions. The restriction holds that the source extent is small and the aperture extent large.

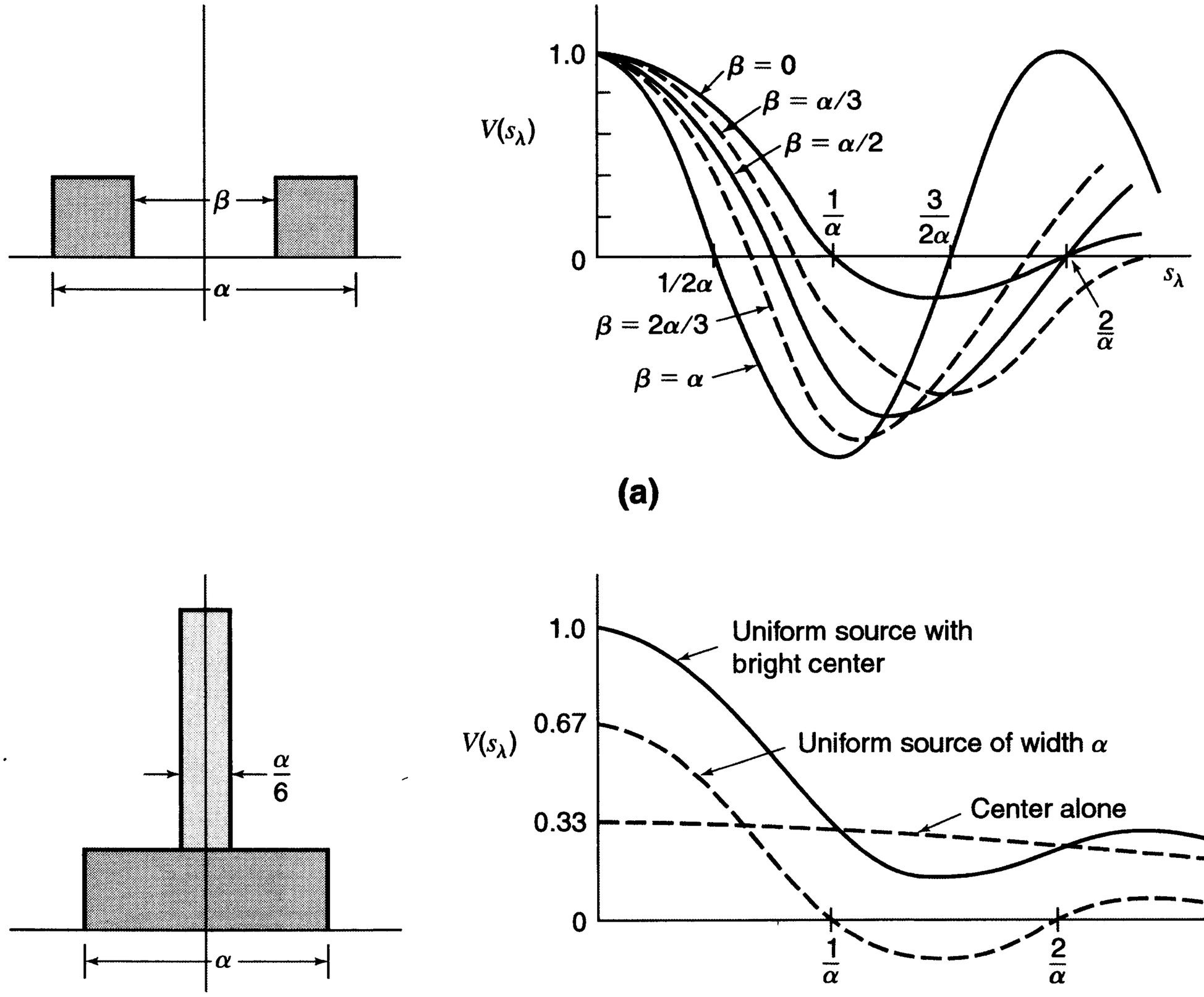


Figure 15–13
(*a*) Visibility functions for a source of uniform brightness with holes of various widths β; (*b*) visibility function for a uniform source with a bright center.

For simplicity only 1-dimensional distributions have been considered. This case is of considerable practical importance. The principles may be extended also to the more general 2-dimensional case. A thorough treatment of the 2-dimensional problem is given by Bracewell (3) and Thompson (1).

In the above discussion monochromatic radiation at a single frequency is assumed. If the antennas and receiver respond uniformly over a bandwidth $f_0 \pm \Delta f/2$ with the radiation considered to be made up of mutually incoherent monochromatic components, the result for a point source is similar in form to the one above for a uniform source of width α, but with $\alpha/2$ replaced by $\Delta f/2$. A result of too wide a bandwidth is that the higher-order fringes may be obliterated.

15–5 APERTURE SYNTHESIS AND MULTI-APERTURE ARRAYS
（口径合成和多口径阵）

The example of the preceding section is illustrative of 1-dimensional aperture synthesis. To obtain resolution of the 1° uniform source of the example to a Rayleigh resolution (=BWFN/2) of 0.1° requires a single aperture 573λ in length [= 114.6°/(2 × 0.1°)]. An

interferometer, on the other hand, can produce the same result with two small (say, 5λ) apertures operated at various spacings provided that the source is strong enough to give a satisfactory signal-to-noise (S/N) ratio with the small apertures. Thus, the interferometer can synthesize a large continuous aperture—hence the term ***aperture synthesis.***

Instead of using only 2 apertures and moving one or both of them, a number of apertures can be employed with unequal spacings in order to provide data points on the visibility curve.

In radio astronomy observations, the baseline separation of the apertures and the angle of the baseline changes with the earth's rotation providing more visibility data, making it possible to do 2-dimensional aperture synthesis and produce 2-dimensional maps of the source distribution. Spreading the apertures over a plane (instead of all in-line) also improves the data. Typically, a celestial object is tracked (as the earth rotates) by a number of radio telescope antennas with each antenna-pair combination producing a *complex visibility function* (amplitude *and* phase) as a function of displacement (hour) angle. A map or image of the object or sky region is then constructed as a Fourier transform of the complex visibility.

The largest and most elaborate aperture synthesis array as of 2000 is the VLA (Very Large Array) of the National Radio Astronomy Observatory built at a cost of $78M (1975 $) on the Plains of San Augustin near Socorro, New Mexico. This array consists of 27 Cassegrain-type reflector antennas (apertures), shown in Fig. 15–14 each with a diameter of 25 m. Each dish can be moved along one of three radial railroad tracks 21 km long so

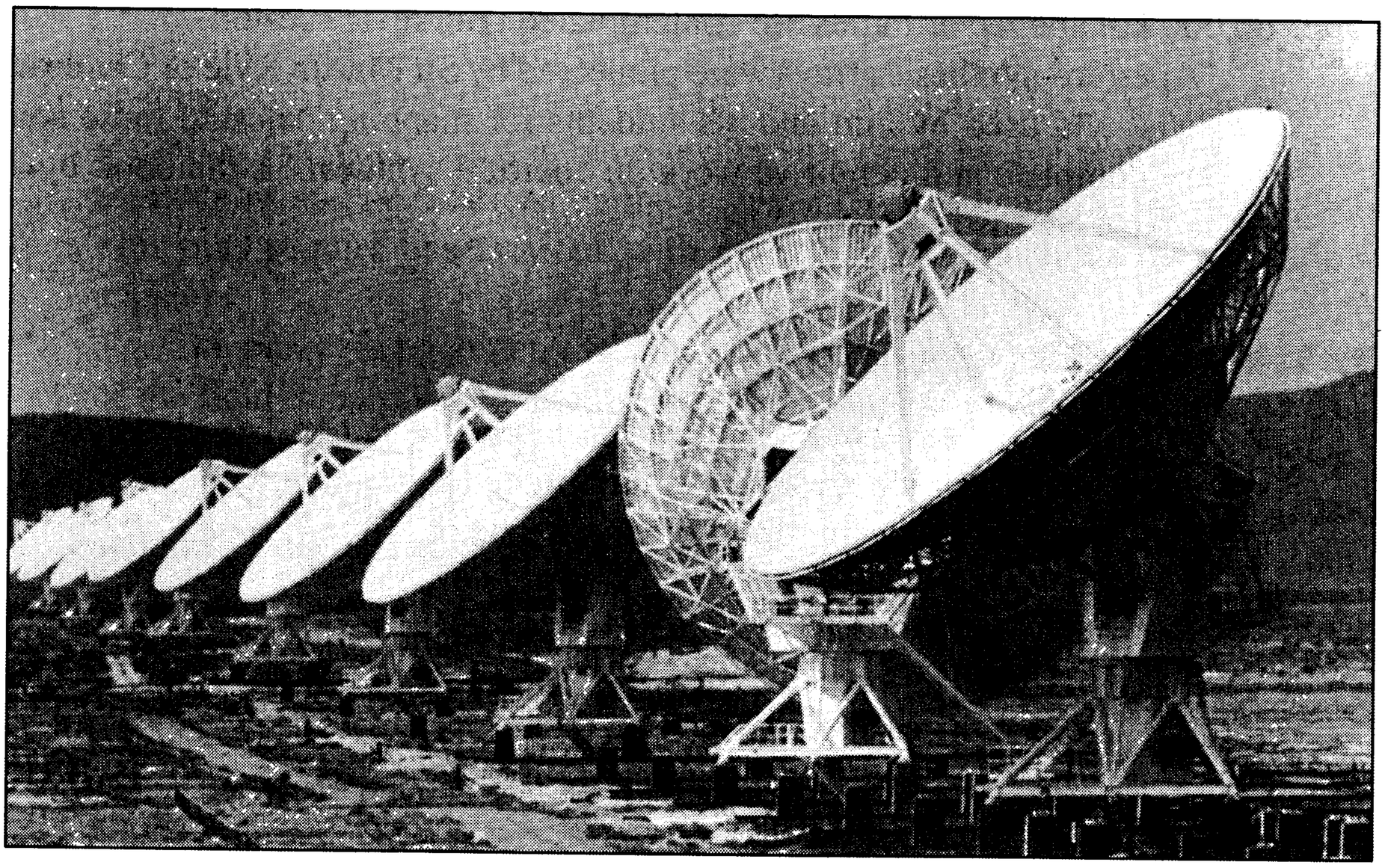


Figure 15–14
Antennas of the Very Large Array (VLA) of the National Radio Astronomy Observatory in compact configuration. The array, on the Plains of San Augustin, New Mexico, has 27 steerable Cassegrain-type reflector antennas 25 m in diameter mounted on 3 rail tracks (radials at 120°), each 21 km long for deployment in a variety of aperture spacings. Where the tracks cross highways, signs warn motorists: "Caution, Radio Telescope Crossing." (*National Radio Astronomy Observatory.*)

that the dishes can be deployed in a Y-shaped configuration with spacings of 0.6 to 36 km in order to maximize the visibility data with the earth's rotation. In the photograph 9 dishes are shown in their most compact, close-in configuration along one track. Synthesized sky maps of 10^7 pixels (picture elements) can be produced with sensitivities of 1 mJy or better. At $\lambda = 6$ cm the resolution is about $\frac{1}{2}$ arcsecond.

Radio telescope antennas separated by intercontinental distances have also been operated as interferometers at even higher resolution. With antenna apertures separated by 10,000 km, milliarcsecond resolution is possible at $\lambda = 6$ cm. In this Very Long Baseline Interferometry (VLBI), signals are downconverted and the IF signals taped and sent to a central location. By synchronizing the tapes using atomic standards, a real-time comparison can be simulated. Such observations require a high degree of coordination between participating observatories and being time-consuming have only been performed on an intermittent basis. A dedicated full-time Very Long Baseline Array (VLBA) has 10 antennas (apertures) with locations coast-to-coast in the continental United States and in Puerto Rico and Hawaii (Kellermann-1).

A next step is to add an orbiting antenna to the array, increasing the resolution and visibility data resulting in maps with better detail and dynamic range. The rapid change of baseline distances for an orbiting antenna also reduces the mapping time. Ultimately with 2 or more orbiting antennas, all of the interferometry could be done above the earth's atmosphere obtaining higher phase stability. High elliptical or circular orbits of 10,000 to 60,000 km apogee or radius are contemplated (Burke-1; Preston-1).

Aperture-synthesis arrays were pioneered by Sir Martin Ryle of Cambridge University, England, and used by him and his students for mapping celestial radio sources (Ryle-1). A very complete and definitive work on aperture synthesis is the book by Thompson (1), Moran and Swenson.

15–6 GRATING LOBES (柵瓣)

If a uniform linear array of n elements has a spacing d_λ between elements exceeding unity, side lobes appear which are equal in amplitude to the main (center) lobe. These so-called *grating lobes* have a spacing from the main lobe of

$$\phi_G = \sin^{-1} \frac{m}{d_\lambda} \text{ (rad)} \qquad \text{where } m = 1, 2, 3, \ldots \tag{1}$$

If $d_\lambda \gg 1$, this reduces approximately to

$$\phi_G = \frac{m}{d_\lambda} \qquad \text{(rad)} \tag{2}$$

with the first side lobe ($m = 1$) at $1/d_\lambda$ as given by the solid line (array factor) in Fig. 15–15 [individual array elements are nondirectional (isotropic)]. To suppress all grating lobes including the first requires directional elements, each with an aperture of approximately d_λ. This puts the first null of the individual element pattern on the first grating lobe, but the array is now equivalent to a continuous aperture. With less directivity (smaller aperture) elements, with pattern as suggested by the upper dashed line in Fig. 15–15 the resultant (total) pattern is as suggested by the lower dashed line with some grating lobe suppression but not elimination.

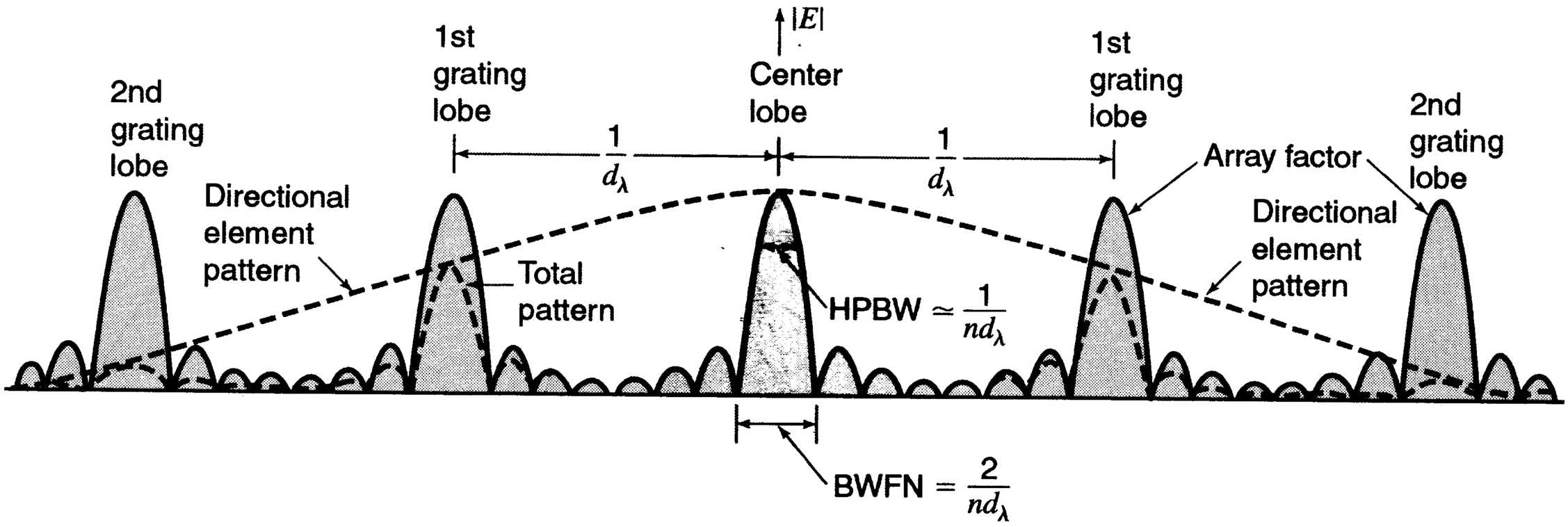


Figure 15–15
Grating lobes with array of *n* widely spaced elements. Solid line: pattern with isotropic elements (array factor). Dashed line: total pattern with directional elements, each with pattern of dotted line.

15–7 TWO-DIMENSIONAL APERTURE SYNTHESIS[1] (二维口径合成)

In 2-dimensional aperture synthesis with a multielement array, a celestial object is tracked by the system (as the earth rotates) so $P_n(\theta, \phi) = 1$ with each antenna-pair combination producing a complex visibility (amplitude and phase) as a function of hour angle. A map or image of the object or field is then constructed as a Fourier transform of the complex visibility. For good maps it is necessary that there be adequate coverage of antenna-pair separations and orientations on the *uv* plane (perpendicular to the direction of the object). This may require observations with many pair combinations over a wide range of hour angles.

The observing geometry is shown in Fig. 15–16 with the *z*-axis directed to the center of the field to be mapped and the *x*-*y* plane tangent to the celestial sphere at the center of the field. Parallel to the *xyz* system of coordinates is a *uvw* system as shown.

For simplicity the field to be mapped is shown at the moment it is on the meridian and with only one pair of antennas of the synthesis array. The actual (ground distance)

---

[1]Regarding the term *aperture synthesis,* A. Richard Thompson (1985) comments as follows: "Some people, myself included, feel that the term 'aperture synthesis' is an unfortunate choice of terminology when applied to instruments like the VLA. This is because, in general, there is no equivalent aperture for such an instrument. The power reception pattern of an aperture antenna is the Fourier transform of the autocorrelation function of the field distribution. For an instrument like the VLA, the Fourier transform of the power reception pattern is the function that represents the spatial spectral sensitivity in the u, v plane, which is an ensemble of elliptical arcs. This function should correspond to the autocorrelation of the equivalent aperture, but, in fact, there exists no aperture distribution which when correlated will result in a series of disconnected elliptical arcs. So it cannot be said that the VLA synthesises an aperture in any precise manner. One can only say that it produces a beam of width equal to that of some aperture of dimensions comparable to the size of the array. A further objection to the term aperture synthesis is that it suggests that there is something fundamentally desirable about a large aperture. However, interferometry theory shows that to obtain high resolution we need to sample the visibility function (or mutual coherence function) in a uniform manner over a wide range of spacings. Use of a large aperture is only one way, and not the most efficient way, of obtaining such information."

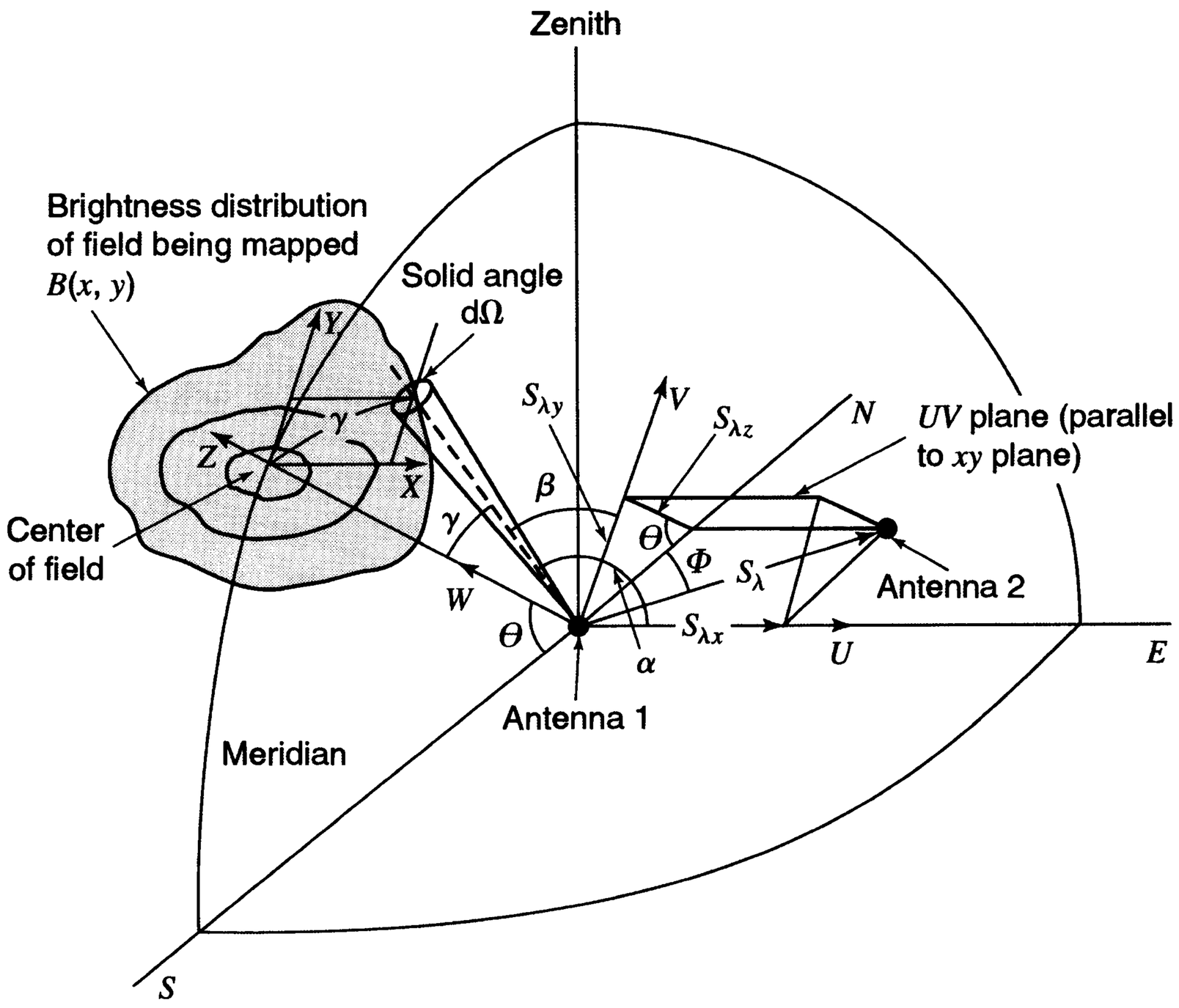


Figure 15–16
Geometry for 2-dimensional aperture synthesis relations.

separation of the antennas (1 and 2) in wavelengths is s_λ and the projections of this distance on the uv plane are $s_{\lambda x}$ and $s_{\lambda y}$ with the axial displacement distance $s_{\lambda z}$ ($= s_\lambda \cos\phi \cos\theta$).

With the field center at the zenith, $s_{\lambda z} = 0$ and the visibility function (amplitude and phase of the interferometer pair or correlator) is an exact 2-dimensional Fourier transform of the brightness distribution of the object. But if $s_{\lambda z} \neq 0$, the output of the interferometer is

$$V(u, v, w)\, e^{j2\pi w} = \int_{-1}^{+1} \int_{-1}^{+1} \frac{B(x, y)}{\sqrt{1 - x^2 - y^2}}\, e^{j2\pi\left(ux+vy+\sqrt{1-x^2-y^2}\,w\right)} dx\, dy \qquad (1)$$

where

$V(u, v, w)$ = visibility function

$B(x, y)$ = brightness distribution of object, W m^{-2} Hz^{-1} sr^{-1}

$u = s_{\lambda x}$ = antenna separation in x direction $= s_x/\lambda$

$v = s_{\lambda y}$ = antenna separation in y direction $= s_y/\lambda$

$w = s_{\lambda z}$ = antenna separation in z direction $= s_z/\lambda$

$x = \cos\alpha$ = direction cosine with respect to x direction

$y = \cos\beta$ = direction cosine with respect to y direction

$z = \cos\gamma$ = direction cosine with respect to z direction

and the factor $e^{j2\pi w}$ on the left side represents the fringe pattern. Only x and y need to be specified since x, y and z are related by

$$x^2 + y^2 + z^2 = 1 \qquad \text{and} \qquad z = \sqrt{1 - x^2 - y^2}$$

For small x and y, $\sqrt{1-x^2-y^2} \simeq z \simeq 1$ and we get

$$V(u, v, w) = \int_{-1}^{-1} \int_{-1}^{-1} \frac{B(x, y)}{\sqrt{1-x^2-y^2}} e^{j2\pi(ux-vy)}\, dx\, dy \tag{2}$$

Thus, for small x and y, V is independent of w and we can replace $V(u, v, w)$ by $V(u, v)$. The inverse Fourier transform yields the object or field brightness distribution as

$$B(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} V(u, v)\, e^{-j2\pi(ux+vy)}\, du\, dv \tag{3}$$

but with restrictions on the size of the field that can be accurately mapped (x and y small). Alternatively, (3) can be written

$$B(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} V(s_{\lambda x}, s_{\lambda y})\, e^{-j2\pi(s_{\lambda x}x+s_{\lambda y}y)}\, dx_\lambda\, ds_{\lambda y} \tag{4}$$

15–8 PHASE CLOSURE OR SELF-CALIBRATION FOR "CLEAN" MAPS
("干净"图形的相位闭合或自校准)

If an interferometer or aperture-synthesis system has phase errors in its images introduced by the atmosphere, by the ionosphere or by its own instabilities, its maps or images are degraded. R. C. Jennison (1) has described a method of phase closure or self-calibration by which such phase errors can, in principle, be eliminated. Thus, consider 3 antennas, A, B, and C, of a switched interferometer which form 3 interferometer pairs AB, BC and AC.

Let antenna A be the reference for phase. Next let θ_{AB}, θ_{BC} and θ_{AC} represent the phases of the complex Fourier transforms of the 3 interferometer pairs due to the source structure; ω_B and ω_C the time-variable phase rotation of the interferometer at antennas B and C; ϕ_B and ϕ_C the phase angle produced by the position of the source relative to B and C; and δ_B and δ_C the atmospheric and/or internal interferometer instability errors at B and C. Then the AB phase output is given by

$$\theta_{AB} + \omega_B + \phi_B + \delta_B \tag{1}$$

while the BC phase output is given by

$$\theta_{BC} + \omega_C - \omega_B + \phi_C - \phi_B + \delta_C - \delta_B \tag{2}$$

and the AC phase output is given by

$$\theta_{AC} + \omega_C + \phi_C + \delta_C \tag{3}$$

Adding (1) and (2) we obtain

$$\theta_{AB} + \theta_{BC} + \omega_C + \phi_C + \delta_C \tag{4}$$

Subtracting (3) from (4) yields

$$\theta_{AB} + \theta_{BC} - \theta_{AC} \tag{5}$$

The rotation, position and error phase angles cancel, *leaving only information due to the actual source structure*. With more antennas and interferometer pairs, additional error-free

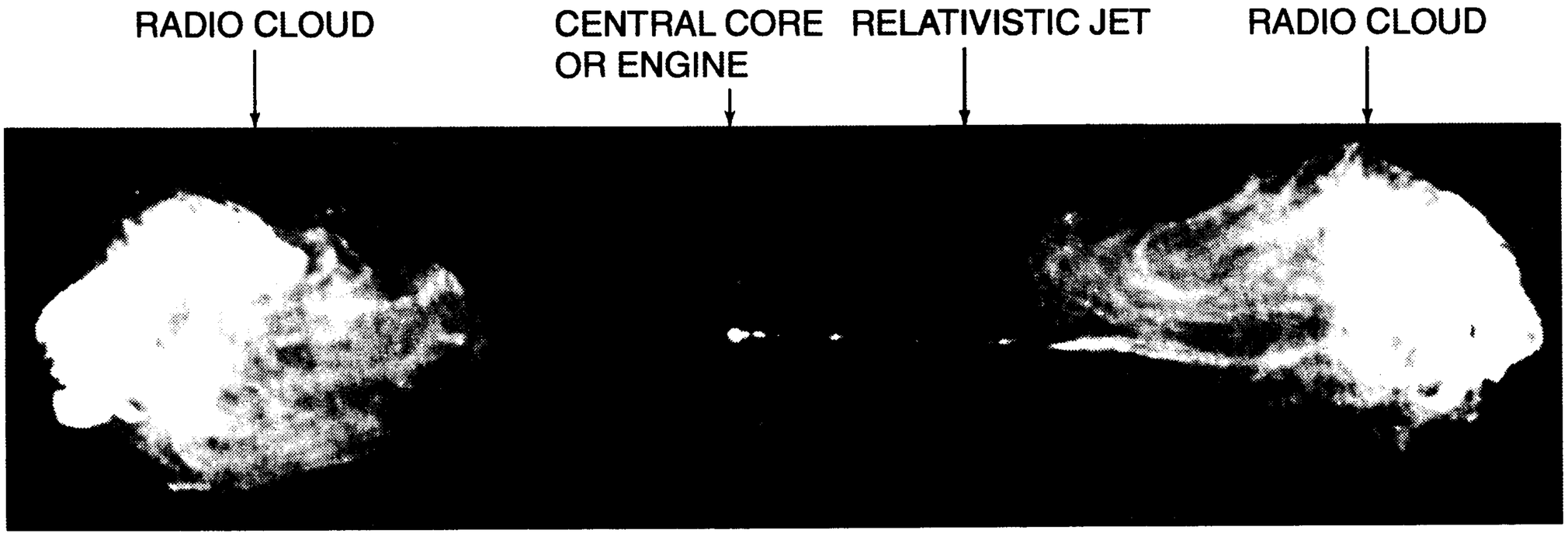


Figure 15–17
Radio picture of the exploded galaxy Cygnus A at a distance far outside our galaxy as observed with the 27 dish array of the VLA (Fig. 15–14) at 5 GHz using aperture synthesis. For its distance see Example 15–8.1. (Source: *Data enhancement by JPL*. Photograph Courtesy of National Radio Astronomy Observatory.)

information about the source structure can be obtained and used in the construction of an improved "clean" image. It is noteworthy that the errors can be removed by a computer *after* the observational data has been recorded. Real-time corrections are not required.

The above technique has been expanded in scope by Readhead (1) and Wilkinson, Cornwell (1) and Wilkinson and others and applied to maps made with the VLA, Westerbork, MERLIN (University of Manchester) and various VLBI arrays. An excellent discussion of the technique and its historical development is given by Ekers (1).

A spectacular example of the application of these techniques to VLA maps is presented in Fig. 15–17 (Cygnus A).

These VLA observations by Perley (1), Dreher and Cowan revealed for the first time the relativistic jets from the central core or engine to the radio clouds as well as a wealth of detailed filamentary structure as shown in Fig. 15–17.

EXAMPLE **15–8.1 The Very Distant Superpowerful Radio Galaxy Cygnus A**

(*a*) If the overall extent of the galaxy shown in Fig. 15–17 is 500,000 light-years, what is the distance of the galaxy?

(*b*) If the maximum VLA spacing is 36 km, what is the resolution or pixel (picture element) size?

(*c*) Approximately how many pixels does Fig. 15–17 contain?

(*d*) If the average flux density between 1 and 2 GHz is 2.5×10^{-23} W/m^2, what is the radio power of the galaxy assuming isotropic radiation?

■ **Solution**

(*a*) From Fig. 15–17 the overall extent of the galaxy $\alpha = 0.032°$ so

$$\begin{aligned}\text{Distance} &= \text{galaxy extent}/\sin\alpha \\ &= 4 \times 10^5/\sin 0.032° \\ &= 7 \times 10^8 \text{ or } 700 \text{ million light-years} \qquad \textit{Ans.}\end{aligned}$$

(*b*) Assuming the maximum resolution of a uniform aperture distribution, the pixel size $= 51°/a_\lambda$ where $a_\lambda = a/\lambda = 36 \times 10^3$ m/60 mm

Thus, resolution or pixel size $= 85 \times 10^{-6}$ degrees

$= 85 \times 3600 \times 10^{-6}$ arc seconds

$= 0.31$ arc seconds *Ans.*

(*c*) The area of Fig. 15–17 is approximately $0.008° \times 0.03°$

$= 24 \times 10^{-3}$ square degrees

$= 3.1 \times 10^5$ arc sec

Thus, the number of pixels $= 3.1 \times 10^5/0.31^2 = 3.2 \times 10^6$

or more than 3 million pixels *Ans.*

(*d*) Cygnus A radio power

$=$ flux density $\times 4\pi$(distance)$^2 \times$ bandwidth

$= 2.5 \times 10^{-23} \times 4\pi(7 \times 10^8$ light-years$)^2 \times 10^9$ watts

where 1 light-year $= 365 \times 24 \times 3600$ seconds $\times 3 \times 10^8$ m/s

$= 1.4 \times 10^{37}$ W *Ans.*

This is more than a trillion, trillion, trillion watts where 1 trillion $= 10^{12}$. By comparison the earth's total electric power consumption is 10 trillion watts.

EXAMPLE **15–8.2 100-m Dish for Deep-Space Communication**

The aperture distribution of a 100-m-diameter dish antenna is tapered to one-third at the edge or about 10 dB down (a typical value). See Table 15–1. At 10 GHz, what is its (*a*) HPBW, (*b*) gain from beamwidth, (*c*) gain from effective aperture, and (*d*) first side-lobe level?

■ **Solution**

From Table 15–1, HPBW $= 66°/D_\lambda = 66° \times 0.03/100 = 0.02°$. *Ans.* (*a*)

From (2–7–9),

$$\text{Gain} = \frac{41{,}000}{\text{HPBW}^2} = 1.05 \times 10^8 \text{ or } 80 \text{ dBi} \qquad \textit{Ans. } (b)$$

From (2–9–2) and (2–9–6), for gain from effective aperture,

$$\text{Gain} = \frac{4\pi A_p \varepsilon_{ap}}{\lambda^2} = \frac{4\pi^2(50)^2(0.725)}{(0.03)^2}$$

$$= 8.0 \times 10^7 = 79 \text{ dBi} \qquad \textit{Ans. } (c)$$

First side-lobe level: -23 dB. *Ans.* (*d*)

REFERENCES（参考文献）

Booker, H. G. (1), and P. C. Clemmow, "The Concept of an Angular Spectrum of Plane Waves and Its Relation to That of Polar Diagram and Aperture Distribution," *Proc. Instn. Elec. Engrs. London,* ser. 3, **97,** 11–17, January 1950.

Bracewell, R. N. (1), "Radio Astronomy Techniques," in S. Flügge (ed.), *Handbuch der Physik,* **54,** Springer-Verlag OHG, Berlin, 1962, pp. 42–129 (a more general 2-dimensional case).

Bracewell, R. N. (2), and J. A. Roberts, "Aerial Smoothing in Radio Astronomy," *Australian J. Phys.,* **7,** 615–640, 1954.

Bracewell, R. N. (3), "Radio Interferometry of Discrete Sources," *Proc. IRE,* **46,** 97–105, January 1958.

Burke, B. F. (1), "Orbiting VLBI: A Survey," in R. Fanti, K. Kellermann and G. Setti (eds.), *VLBI* and *Compact Radio Sources,* Reidel, 1984.

Cornwell, T. J. (1), and P. N. Wilkinson, "A New Method of Making Maps with Unstable Radio Interferometers," *Mon. Not. Roy. Ast. Soc.,* vol. 196, pp. 1067–1086, 1981.

Ekers, R. D. (1), "The Almost Serendipitous Discovery of Self Calibration," pp. 154–159, in "Serendipitous Discoveries in Radio Astronomy," K. Kellermann and B. Sheets, eds., NRAO, Green Bank, 1983.

Jennison, R. C. (1), *Fourier Transforms and Convolutions for the Experimentalist,* Pergamon Press, New York, 1961.

Kellermann, K. I. (1), and A. R. Thompson, "The Very Long Baseline Array," *Science,* **229,** 123–130, 1985.

Napier, P. J. (1), and A. R. Thompson and R. D. Ekers, "The Very Large Array: Design and Performance of a Modern Synthesis Radio Telescope," *Proc. IEEE,* **71,** 1295–1320, November 1983 (includes 155 references).

Perley, R. A. (1), J. W. Dreher and J. J. Cowan, "The Jets and Filaments in Cygnus A," *Astrophys. J.,* **285,** L335–38, 1984.

Preston, R. A. (1), B. F. Burke, R. Doxsey, J. F. Jordan, S. H. Morgan, D. H. Roberts and I. I. Shapiro, "The Future of VLBI Observations in Space," in F. Biraud (ed.), *Very Long Baseline Interferometry Techniques,* Cepadues, 1983.

Readhead, A. C. S. (1), and P. N. Wilkinson, "Mapping of Compact Radio Sources from VLBI Data," *Astrophys. J.,* vol. 223, pp. 25–36, 1978.

Rhodes, D. R. (1), "The Optimum Line Source for the Best Mean Square Approximation to a Given Radiation Pattern," *IEEE Trans. Ants. Prop.,* **AP-11,** 440–446, July 1963.

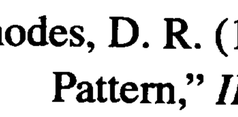
Rhodes, D. R. (2),"On an Optimum Line Source for Maximum Directivity," *IEEE Trans. Ants. Prop.,* **AP-19,** 485–492, July 1971.

Ryle, M. (1), A. Hewish and J. R. Shakeshaft, "The Synthesis of Large Radio Telescopes by the Use of Radio Interferometers," *IRE Trans. Ants. Prop.,* **AP-7,** S120–124, December 1959.

Thompson, A. R. (1), J. M. Moran and G. W. Swenson, Jr., *Interferometry and Synthesis in Radio Astronomy,* Wiley-Interscience, New York, 1986.

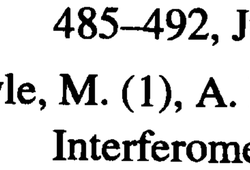
PROBLEMS（习题）

15-2-1（四种口径分布） **Four aperture distributions.** For the following aperture distributions show that the far-field patterns are given as:

(*a*) 

Stepped

$$E(\phi) = \frac{2}{3}\frac{\sin\psi}{\psi} + \frac{1}{3}\frac{\sin\psi/2}{\psi/2} \qquad \text{where } \psi = \pi L_\lambda \sin\phi$$

(*b*) 
Circular

$$E(\phi) = \frac{2J_1(\psi)}{\psi} \qquad \text{where } \psi = 2\pi L_\lambda \sin\phi$$

(*c*) Triangular

$$E(\phi) = \frac{\sin^2 \psi}{\psi^2} \qquad \text{where } \psi = \frac{\pi}{2} L_\lambda \sin \phi$$

(*d*) Triangular asymmetric

$$E(\phi) = \frac{\sin \psi}{\psi} + j\left(\frac{\cos \psi}{\psi} - \frac{\sin \psi}{\psi^2}\right) \qquad \text{where } \psi = \pi L_\lambda \sin \phi$$

15–2–2 (傅里叶变换) **Fourier transform.** Apply the Fourier transform method to obtain the far-field pattern of an array of 2 equal in-phase isotropic point sources with a separation d. Reduce the expression to its simplest trigonometric form.

15–2–3 (笔状波束波瓣图) **Pencil-beam patterns.** For symmetrical circular aperture pencil-beam patterns (function only of θ) show that the main beam solid angle Ω_M is given by

$1.13\ \theta_{HP}^2$ for a Gaussian pattern

$0.988\ \theta_{HP}^2$ for a $\sin(x)/x$ pattern

$1.008\ \theta_{HP}^2$ for a Bessel pattern

where θ_{HP} is the half-power beamwidth. Also show that Ω_M is given by $1.036\ \theta_{HP}\phi_{HP}$ for a $\sin(u)/u$ (square aperture) pattern.

***15–3–1 (波瓣图的平滑化)** **Pattern smoothing.** An idealized antenna pattern-brightness distribution is illustrated by the 1-dimensional diagram in Fig. P15–3–1. The brightness distribution consists of a point source of flux density S and a uniform source 2° wide, also of flux density S. The point source is 2° from the center of the 2° source. The antenna pattern is triangular (symmetrical) with a 2° beamwidth between zero points and with zero response beyond.

(*a*) Draw an accurate graph of the observed flux density as a function of angle from the center of the 2° source.

(*b*) What is the maximum ratio of the observed to the actual total flux density ($2S$)?

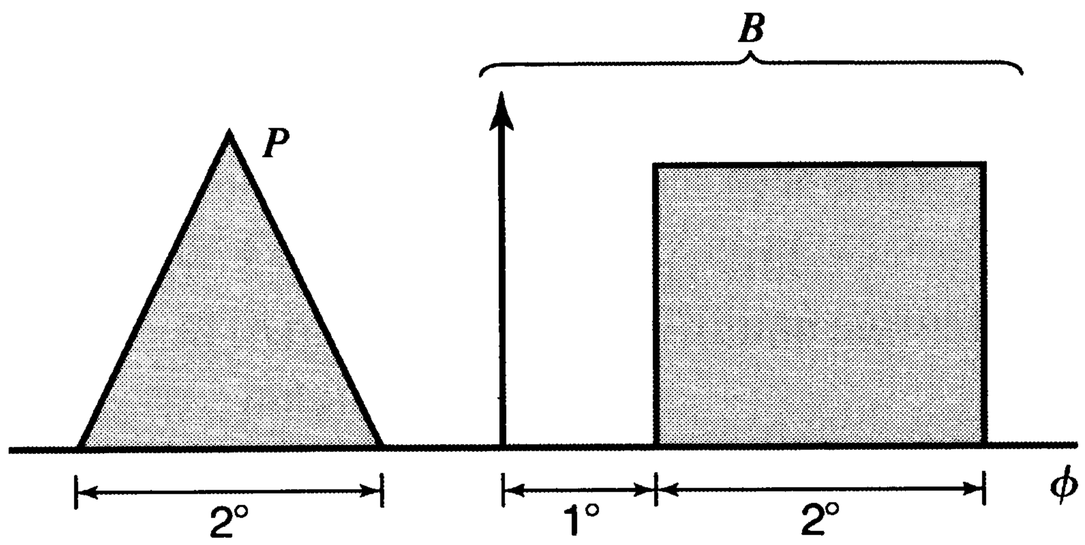


Figure P15–3–1
Pattern smoothing.

15–4–1 (干涉仪，波瓣图乘法) **Interferometer. Pattern multiplication.** An interferometer antenna consists of 2 square broadside in-phase apertures with uniform field distribution.

(*a*) If the apertures are 10λ square and are separated 60λ on centers, calculate and plot the far-field pattern to the first null of the single aperture pattern.

(*b*) How many lobes are contained between first nulls of the aperture pattern?

(*c*) What is the effective aperture?

(*d*) What is the HPBW of the central interferometer lobe?

(*e*) How does this HPBW compare with the HPBW for the central lobe of two isotropic in-phase point sources separated 60λ?

15–4–2 (可见度函数) **Visibility funotion.** Show that the visibility function observed with a simple interferometer of spacing s_λ for a uniform source of width α with a symmetrical uniform bright center of width $\beta = \alpha/6$ is

$$V(s_\lambda) = \frac{\sin(\pi s_\lambda \alpha) + 3\sin(\pi s_\lambda \alpha/6)}{3\pi s_\lambda \alpha/2}$$

if the center brightness is 4 times the side brightness (see Fig. 15–13*b*).

15–4–3 (可见度函数) **Visibility function.** Show that the visibility function observed with a simple interferometer of spacing s_λ for 2 equal uniform sources of width $\alpha/6$ spaced between centers by $5\alpha/6$ is

$$V(s_\lambda) = \frac{\sin(\pi s_\lambda \alpha)}{\pi s_\lambda \alpha} - \frac{2}{3}\frac{\sin(2\pi s_\lambda \alpha/3)}{2\pi s_\lambda \alpha/3}$$

15–4–4 (干涉仪的输出) **Interferometer output.** Show that the output of a simple interferometer of spacing s and bandwidth $f_0 \pm \Delta f/2$ for a point source is given by

$$S_0 \Delta f \left(1 + \frac{\sin\left(\frac{2\pi s}{c}\frac{\Delta f}{2}\phi\right)}{\frac{2\pi s}{c}\frac{\Delta f}{2}\phi} \cos\left(\frac{2\pi s}{c} f_0 \phi\right) \right)$$

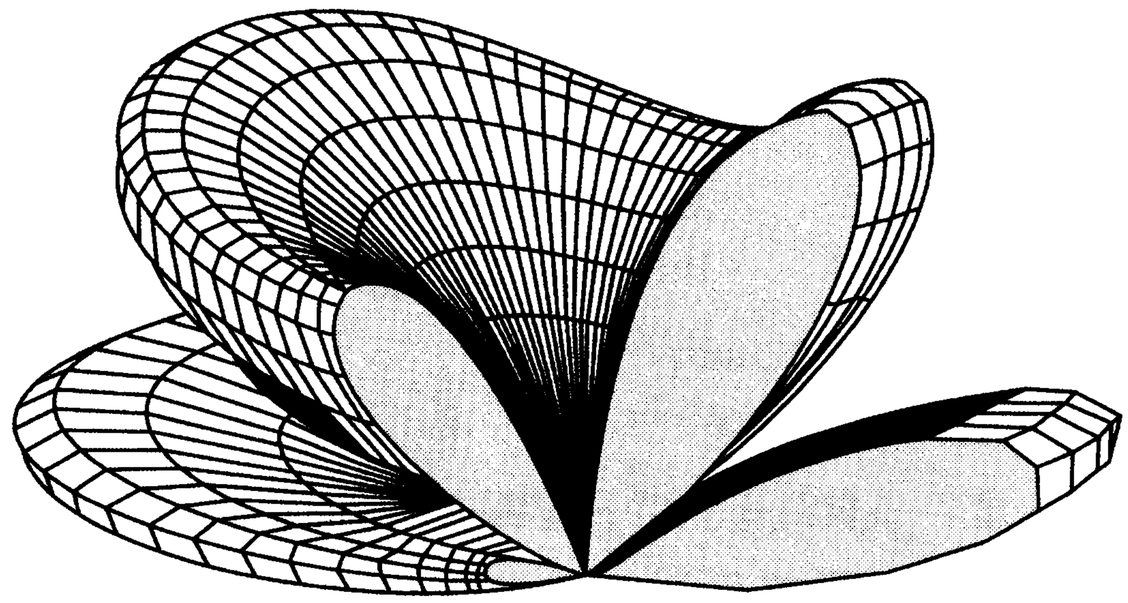
where c = velocity of light.

15–4–5 (干涉仪的频带) **Interferometer bandwidth.** Show that for an interferometer with bandwidth $f_0 \pm \Delta f/2$ (Prob. 15–4–4), the condition $\Delta f/f_0 \ll 1/n$, where n = fringe order, must hold in order that the fringe amplitude not be decreased.

15–6–1 (阵元的数量) **Number of elements.** In Fig. 15–15 how many elements n have been assumed?

For computer programs, see Appendix C.

CHAPTER 16



Arrays of Dipoles and of Apertures

偶极子阵和口径阵

Topics in this chapter include:

- Arrays of 2 driven elements; broadside and end-fire (两根受激单元的边射阵和端射阵)
- Patterns, driving point impedance, and gain (波瓣图、激励点阻抗和增益)
- Arrays of 2 driven elements; general case (两根受激单元阵的一般情况)
- Closely spaced elements (密距单元)
- Radiating efficiency (辐射效率)
- Arrays of *n* driven elements (*n* 根受激单元的阵)
- Horizontal and vertical antennas above ground (地面上的水平和铅垂天线)
- Shaped dipole array (赋形偶极子的阵)
- Phased arrays (相控阵)
- Grid and chain arrays (栅格形阵和链式阵)
- Retroarrays (返回式阵)
- Digital beam-forming or adaptive or smart arrays (数字波束形成：自适应阵和智能阵)
- Microstrip arrays (微带阵)
- Low-side-lobe arrays (低旁瓣阵)
- Long-wire antennas: V, rhombic and Beverage (长导线天线：V 形、菱形和贝弗瑞天线)
- Curtain arrays (帘幕形阵)
- Array feed points (阵列馈电点)
- Folded dipoles (折合偶极子)
- Argus array: The ultimate surveillance antenna (百眼巨人型阵的全空监视天线)

16–1 INTRODUCTION (引言)

Essential background for this chapter is covered in Chap. 5 on arrays of point sources, Chap. 6 on linear antennas, and Chap. 13 on self and mutual impedances. The heading for this chapter could appropriately be "Arrays of Antenna Elements" where an *element* refers to the basic unit of which an array is constructed. In the first part of this chapter the "elements" are mostly thin linear *dipoles* while in the latter sections of the chapter the array "elements" are *apertures* in general, which may be helices, horns, big reflectors or arrays of dipoles (arrays of arrays).

The *far-* or *radiation field pattern,* the *driving point impedance* and the *array gain* are first derived in that order for several different arrays of dipoles. The method of analysis is general and is applicable to other dipole arrays, the specific types discussed serving merely as examples. Array gain is calculated by treating the dipoles as circuit elements having self and mutual impedances. Although direct pattern integration could be used to determine the gain, the circuit approach is simpler provided impedance values are available (patterns having been utilized in the impedance calculations).

16–2 ARRAY OF TWO DRIVEN λ/2 ELEMENTS. BROADSIDE CASE
（两根受激λ/2单元的边射阵）

Consider two center-fed λ/2 elements (dipoles) arranged side by side with a spacing d as in Fig. 16–1. Two special cases will be considered: the *broadside case*[1] treated in this section, in which the two elements are fed with equal *in-phase* currents, and the *end-fire case*[2] (Sec. 16–3), in which the two elements are fed with equal currents in *opposite phase*. The more general case where the currents are equal in magnitude but in any phase relation is treated in Sec. 16–4.

16–2a Field Patterns（场波瓣图）

The first part of the analysis will be to determine the absolute far-field patterns. It is convenient to obtain two pattern expressions, one for the horizontal plane and one for the vertical plane. Ordinarily, the relative patterns would be sufficient. However, the absolute patterns will be needed in gain calculations. Let the elements be vertical as shown in Fig. 16–2*a*. It is assumed that the array is in free space, i.e., at an infinite distance from the ground or other objects. The field intensity $E_1(\phi)$ from a single element as a function of ϕ and at a large

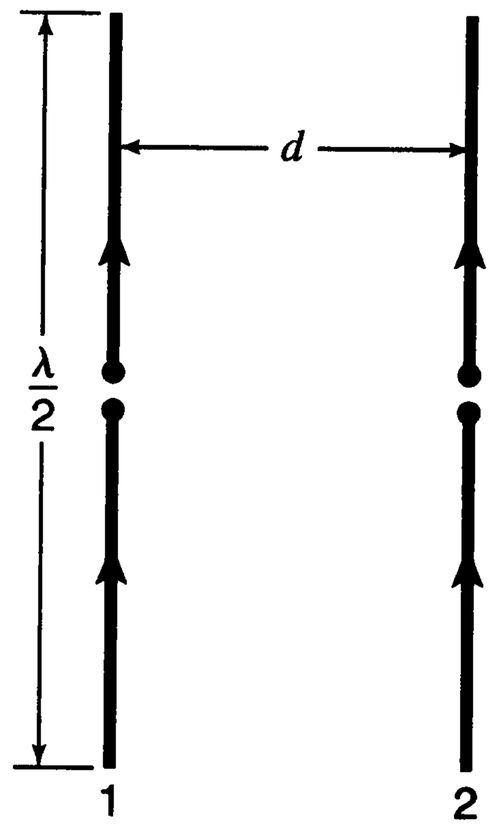
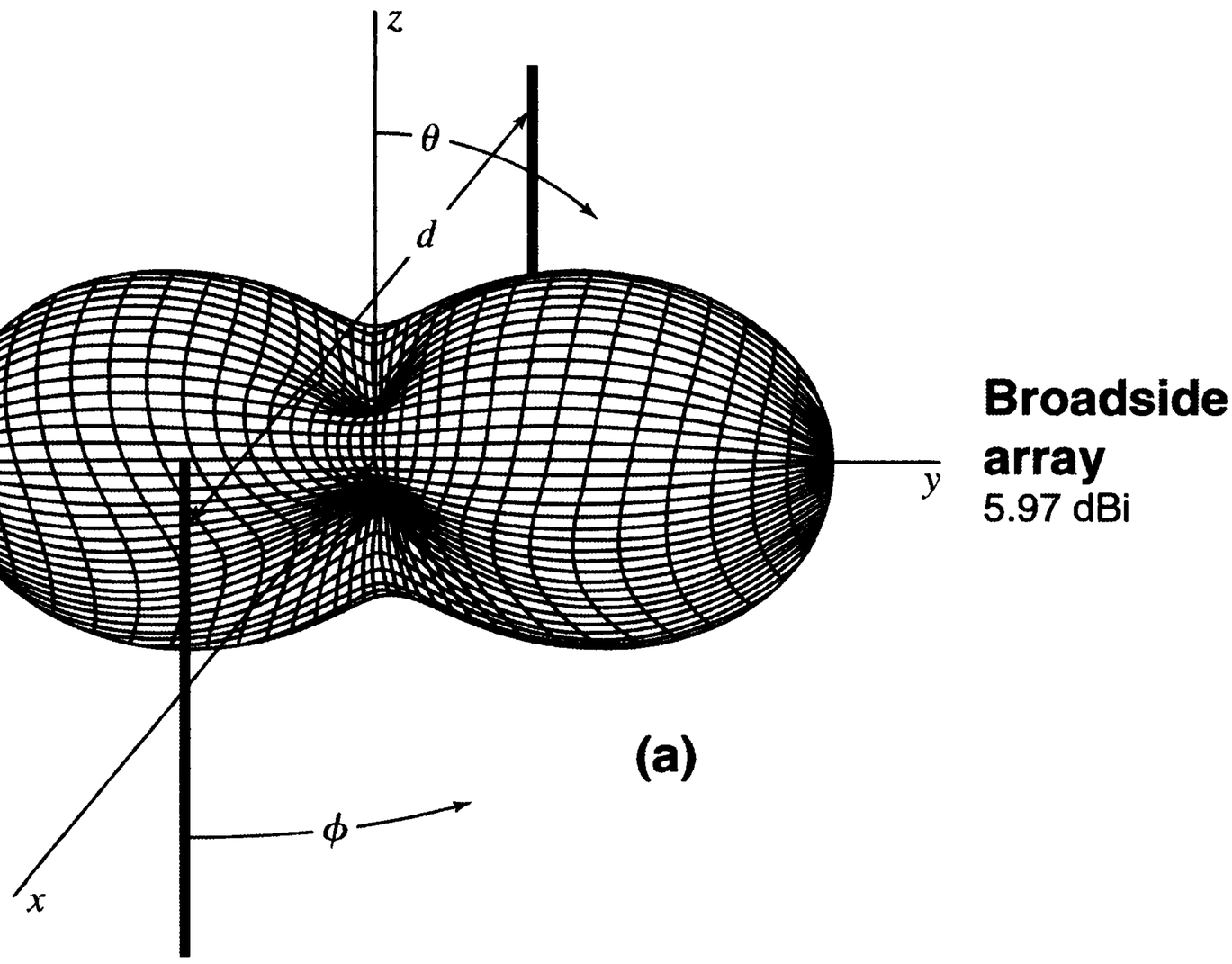
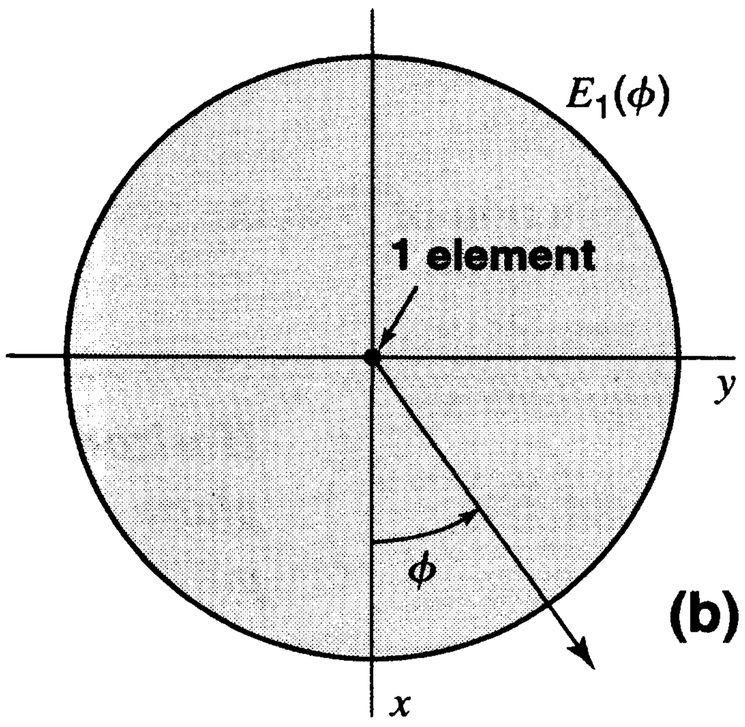


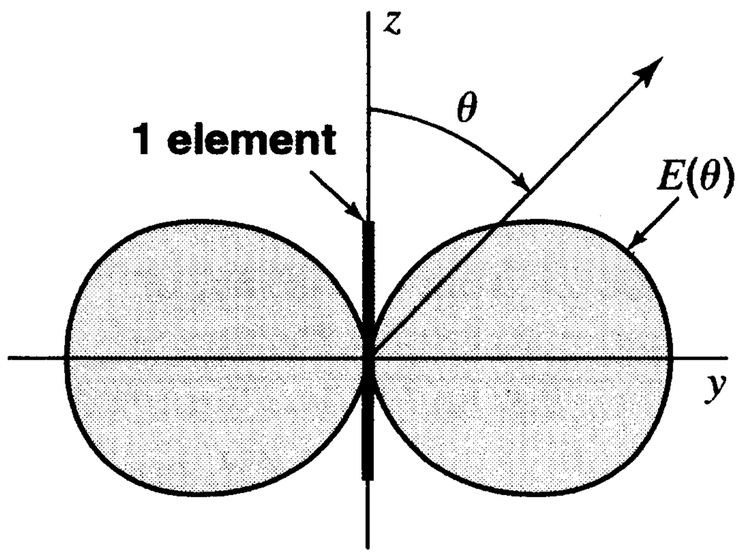
Figure 16–1
Broadside array of 2 in-phase λ/2 elements.

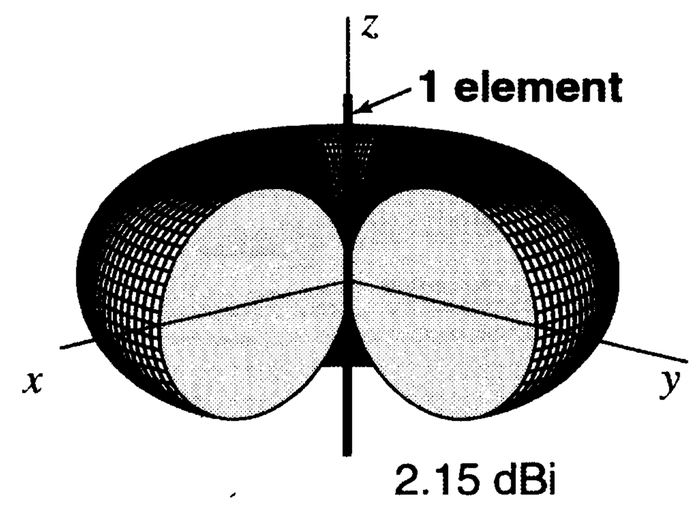
[1] In the so-called "broadside case" there is always a major lobe of radiation broadside to the array, although at large spacings there may be an end-fire lobe of equal magnitude (as, for example, when the spacing is 1λ).

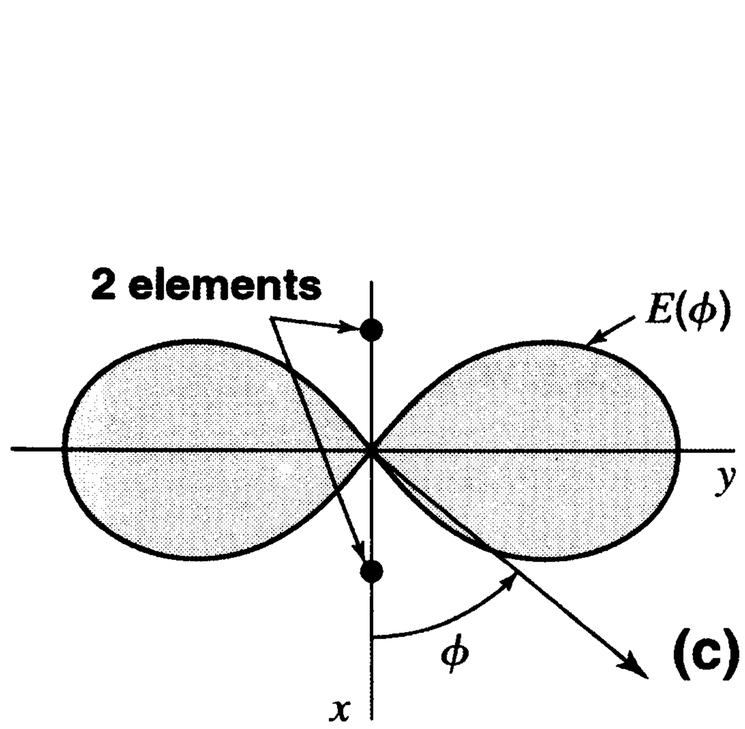
[2] In the so-called "end-fire case" the pattern always has zero radiation broadside. The maximum radiation is always end-fire if the spacing is λ/2 or less. However, for greater spacings the maximum radiation is, in general, not end-fire. Since spacings of λ/2 or less are of principal interest, the array may be referred to as an end-fire type.

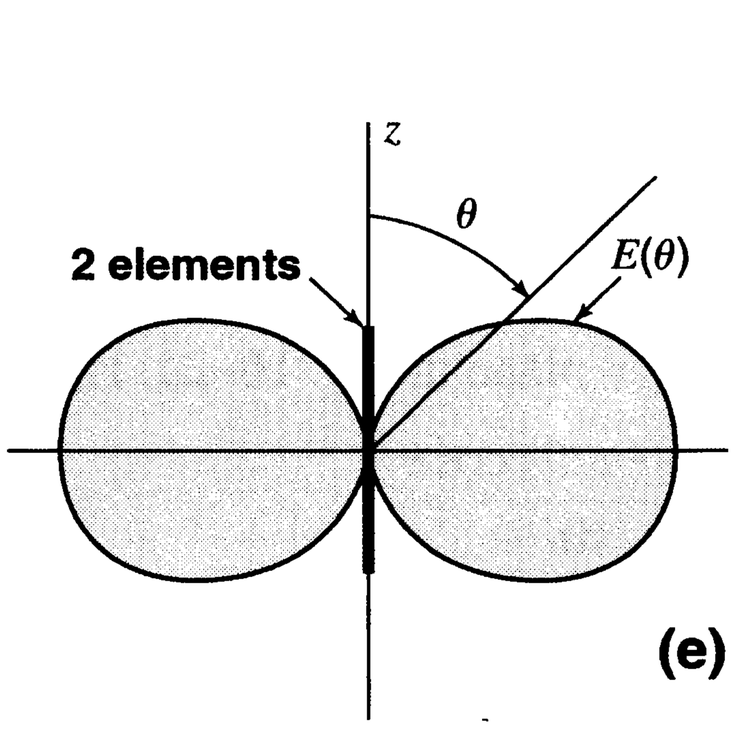












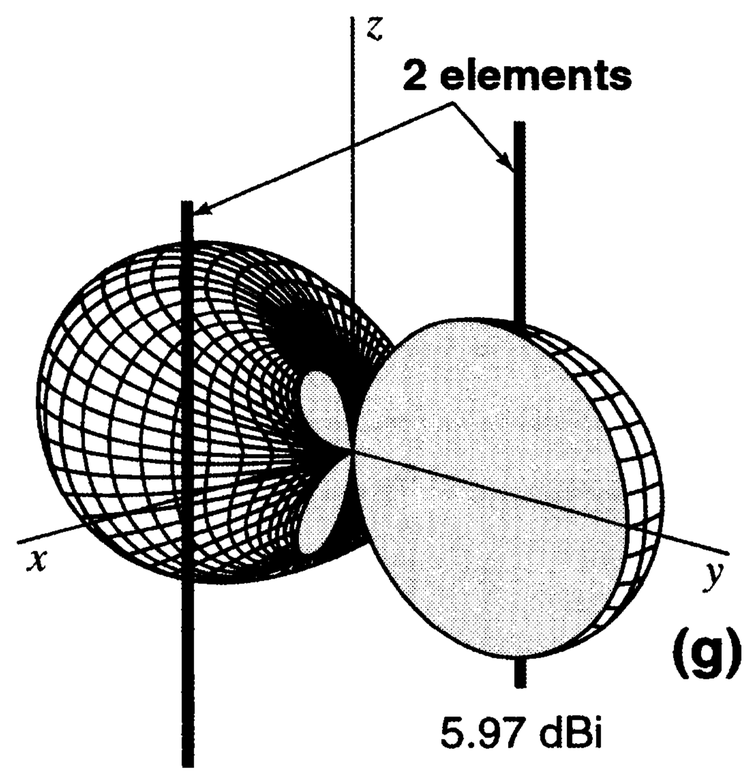


Figure 16–2
Field patterns for broadside array of 2 linear in-phase $\lambda/2$ elements with spacing $d = \lambda/2$ shown in 3-dimensional (3-D) pattern, at (*a*). Figures (*b*) and (*d*) show the 2-dimensional (2-D) field pattern of one element while figures (*c*) and (*e*) show the 2-D field pattern for the 2-element array. Figure (*f*) is a 3-D field pattern of one element and Figure (*g*) is a 3-D field pattern of the 2-element array. Note the minor lobes at 45° in (*g*) which do not appear in the 2-D patterns.

distance D ($D \gg d$) in a horizontal plane ($\theta = 90°$ or xy plane in Fig. 16–2*a*) is

$$E_1(\phi) = kI_1 \tag{1}$$

where k is a constant ($\Omega\ \mathrm{m}^{-1}$) involving the distance D and I_1 is the terminal current. Equation (1) is the absolute field pattern in the horizontal plane. It is independent of ϕ so that the relative pattern is a circle as indicated in Fig. 16–2*b*.

Next let the elements be replaced by isotropic point sources of equal amplitude. The pattern $E_{\text{iso}}(\phi)$ as a function of ϕ in the horizontal plane for two such isotropic in-phase point sources is given by (5–2–6) as

$$E_{\text{iso}}(\phi) = 2E_0 \cos\left(\frac{d_r \cos\phi}{2}\right) \tag{2}$$

where d_r is the distance between sources expressed in radians; that is,

$$d_r = \frac{2\pi d}{\lambda} \tag{3}$$

Applying the principle of pattern multiplication, we may consider that E_0 is the field intensity from a single element at a distance D. Thus,

$$E_0 = E_1(\phi) = kI_1 \tag{4}$$

Introducing (4) into (2) yields the field intensity $E(\phi)$ as a function of ϕ in the horizontal plane at a large distance D from the array, or

（两 λ/2 单元边射阵）

2 λ/2 broadside array $E(\phi) = E_1(\phi)2\cos\left(\frac{d_r \cos\phi}{2}\right) = 2kI_1 \cos\left(\frac{d_r \cos\phi}{2}\right)$ ***Horizontal plane pattern***

（水平面波瓣图）

(5)

This expression may be called the ***absolute field pattern in the horizontal plane***. The electric field at points in this plane is everywhere vertically polarized. The shape of this pattern is illustrated in Fig. 16–2*c*, and also partially in Fig. 16–2*a*, for the case where $d = \lambda/2$. The maximum field intensity is at $\phi = 90°$ or broadside to the array.

The field intensity $E_1(\theta)$ as a function of θ from a single $\lambda/2$ element at a distance D in the vertical plane (yz plane in Fig. 16–2*a*) is, from (6–4–4), given by

$$E_1(\theta) = kI_1 \frac{\cos[(\pi/2)\cos\theta]}{\sin\theta} \tag{6}$$

The shape of this pattern is shown in Fig. 16–2*d*. It is independent of the angle ϕ. The pattern $E_{\text{iso}}(\theta)$ in the vertical plane for two isotropic sources in place of the two elements is

$$E_{\text{iso}}(\theta) = 2E_0 \tag{7a}$$

Applying the principle of pattern multiplication, we put

$$E_0 = E_1(\theta) \tag{7b}$$

so that the field intensity $E(\theta)$ in the vertical plane at a distance D from the array is

$$E(\theta) = 2kI_1 \frac{\cos[(\pi/2)\cos\theta]}{\sin\theta} \quad \textit{Vertical plane pattern} \quad \text{(铅垂面波瓣图)} \tag{8}$$

This may be called the ***absolute field pattern in the vertical plane.*** (绝对场波瓣图) This pattern has the same shape as the pattern for a single element in the vertical plane and is independent of the spacing. The relative pattern is presented in Fig. 16–2*e* and also partially in Fig. 16–2*a*. The relative 3-dimensional field variation for the case where $d = \lambda/2$ is suggested in Fig. 16–2*a*.

16–2b Driving-Point Impedance (激励点阻抗)

Suppose that the array is energized by the transmission-line arrangement shown in Fig. 16–3. Two transmission lines of equal length l join at P to a third line extending to a transmitter. Let us find the driving-point impedance presented to the third line at the point P (Brown-1, 2). This will be called the driving point for the array.

Let V_1 be the emf applied at the terminals of element 1. Then

$$V_1 = I_1 Z_{11} + I_2 Z_{12} \tag{9}$$

where I_1 is the current in element 1, I_2 the current in element 2, Z_{11} is the self-impedance of element 1 and Z_{12} is the mutual impedance between the two elements. Likewise, if V_2 is the emf applied at the terminals of element 2,

$$V_2 = I_2 Z_{22} + I_1 Z_{12} \tag{10}$$

where Z_{22} = the self-impedance of element 2

The currents are equal and in phase so

$$I_1 = I_2 \tag{11}$$

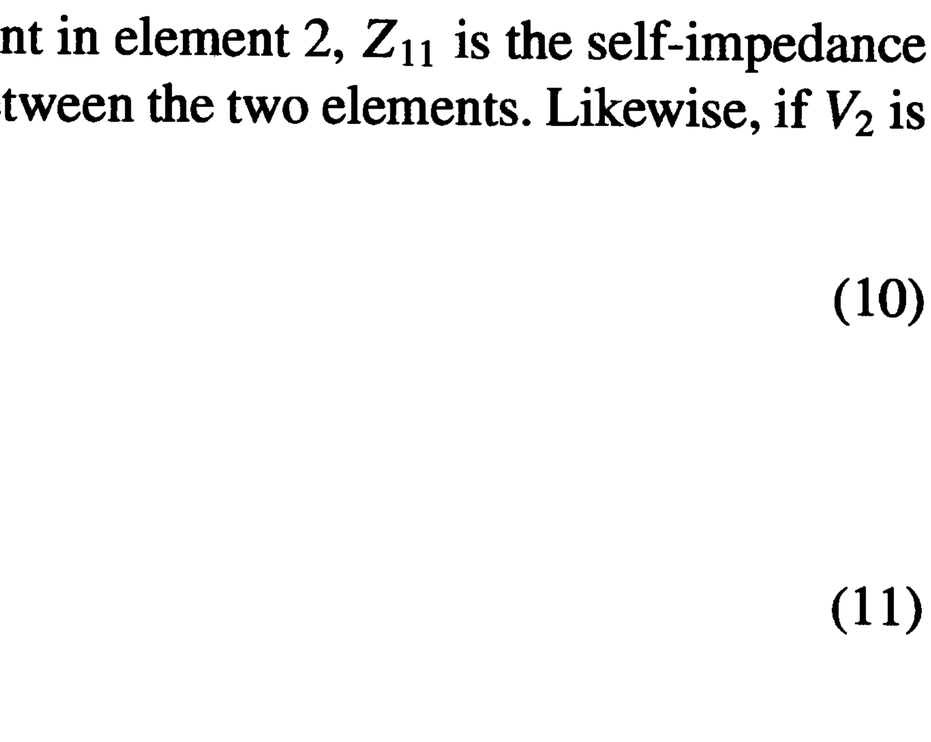


Figure 16–3
Broadside array of 2 linear $\lambda/2$ elements with arrangement for driving elements with equal in-phase currents.

Therefore, (9) and (10) become

$$V_1 = I_1(Z_{11} + Z_{12}) \tag{12}$$

and

$$V_2 = I_2(Z_{22} + Z_{12}) \tag{13}$$

The terminal impedance Z_1 of element 1 is

$$Z_1 = \frac{V_1}{I_1} = Z_{11} + Z_{12} \tag{14}$$

and the terminal impedance Z_2 for element 2 is

$$Z_2 = \frac{V_2}{I_2} = Z_{22} + Z_{12} \tag{15}$$

Since the elements are identical

$$Z_{22} = Z_{11} \tag{16}$$

Therefore, the terminal impedances given by (14) and (15) are equal; that is,

$$Z_1 = Z_2 = Z_{11} + Z_{12} \tag{17}$$

Since $Z_1 = Z_2$ and $I_1 = I_2$ it is necessary that the emf V_1 applied at the terminals of element 1 be equal and in phase with respect to the emf V_2 applied at the terminals of element 2.

For the case where the spacing d is $\lambda/2$, the terminal impedance Z_1 of each element is

$$\begin{aligned} Z_1 = Z_{11} + Z_{12} &= R_{11} + R_{12} + j(X_{11} + X_{12}) \\ &= 73 - 13 + j(43 - 29) \\ &= 60 + j14\ \Omega = \text{impedance of each element} \end{aligned} \tag{18}$$

Suppose that the reactance of 14 Ω is tuned out at the terminals by a series capacitance.[1] The terminal impedance then becomes a pure resistance of 60 Ω. If the length l of each transmission line between the antenna terminals and P is $\lambda/2$, the driving-point impedance of the array at P is a pure resistance of 30 Ω. This value is independent of the characteristic impedance of the $\lambda/2$ lines. However, a resistance of 30 Ω is too low to be matched readily by an open-wire transmission line. Therefore, a more practical arrangement would be to make l equal to $\lambda/4$. Suppose that we wish to have a driving-point resistance of 600 Ω. To do this, we let the characteristic impedance of each $\lambda/4$ line be $\sqrt{1200 \times 60} = 269\ \Omega$.[2] Each line transforms the 60 Ω to 1200 Ω and since two such lines are connected in parallel at P, the **driving-point impedance for the array** (激励点阻抗) equals $600 + j0\ \Omega$. This is the impedance presented to the line to the transmitter. For an impedance match this line should have a characteristic impedance of $600 + j0\ \Omega$.

---

[1] It is often simpler to resonate the elements by shortening them slightly. This modifies the resistive component of the impedance and also alters the $E(\theta)$ field pattern, but to a first approximation these effects can usually be neglected.

[2] For the special case of a $\lambda/4$ line, the general transmission-line formula reduces to $Z_{\text{in}} = Z_0^2/Z_L$ where Z_{in} is the input impedance, Z_0 the characteristic impedance and Z_L the load impedance. Thus, $Z_0 = \sqrt{Z_{\text{in}} Z_L}$.

16–2c Gain in Field Intensity（场强增益）

As the last part of the analysis of the array, let us determine the gain in field intensity for the array. This could be done by pattern integration as in Chap. 4, but with self- and mutual-impedance values available a shorter method is as follows.

Let the total power input (real power) to the array be P.[1] Assuming no heat losses, the power P_1 in element 1 is

$$P_1 = I_1^2(R_{11} + R_{12}) \tag{19}$$

and the power P_2 in element 2 is

$$P_2 = I_2^2(R_{22} + R_{12}) \tag{20}$$

where I_1 and I_2 are rms currents. However, $R_{22} = R_{11}$ and $I_2 = I_1$. Making these substitutions and adding (19) and (20) to obtain the total power P, we have

$$P = P_1 + P_2 = 2I_1^2(R_{11} + R_{12}) \tag{21}$$

and

$$I_1 = \sqrt{\frac{P}{2(R_{11} + R_{12})}} \tag{22}$$

Suppose that we express the gain with respect to a single $\lambda/2$ element as the reference antenna. Let the same power P be supplied to this antenna. Then assuming no heat losses, the current I_0 at its terminals is

$$I_0 = \sqrt{\frac{P}{R_{00}}} \tag{23}$$

where R_{00} is the self-resistance of the reference antenna ($= R_{11}$).

In general, the *gain in field intensity*[2] of an array over a reference antenna is given by the ratio of the field intensity from the array to the field intensity from the reference antenna when both are supplied with the same power P. The comparison is, of course, made in the same direction from both the array and the reference antenna. In the present case it will be convenient to obtain two gain expressions, one for the horizontal plane and the other for the vertical plane.

In the horizontal plane the field intensity $E_{HW}(\phi)$, as a function of ϕ, at a distance D from a single vertical center-fed $\lambda/2$ reference antenna is of the form of (1). Thus,

$$E_{HW}(\phi) = kI_0 \tag{24}$$

---

[1] It is important that the antenna power P be considered constant. Most transmitters are essentially constant power devices which can be coupled to a wide range of antenna impedance. Until the antenna power was considered constant by G. H. Brown (2) the advantages of closely spaced elements were not apparent. Prior to this time the antenna current had usually been considered constant.

[2] The *power gain* discussed in Chap. 2 is equal to the square of the *gain in field intensity*. The power gain is the ratio of the radiation intensities (power per unit solid angle) for the array and reference antennas, the radiation intensity being proportional to the square of the field intensity.

where I_0 is the terminal current and "HW" indicates "Half-Wavelength ($\lambda/2$) antenna." Substituting the value of I_0 from (23), we obtain

$$E_{\mathrm{HW}}(\phi) = k\sqrt{\frac{P}{R_{00}}} \tag{25}$$

The field intensity $E(\phi)$ in the horizontal plane at a distance D from the array is given by (5). Introducing the value of the terminal current I_1 from (22) into (5) yields

$$E(\phi) = k\sqrt{\frac{2P}{R_{11} + R_{12}}}\cos\left(\frac{d_r \cos\phi}{2}\right) \tag{26}$$

The ratio of (26) to (25) gives the gain in field intensity of the array (as a function of ϕ in the horizontal plane) with respect to a vertical $\lambda/2$ reference antenna with the same power input. This gain will be designated by the symbol $G_f(\phi)[A/\mathrm{HW}]$ where the expression in the brackets is by way of explanation that it is the gain in field of the *array* (A) with respect to a *half-wavelength reference antenna* (HW)[1] in the same direction from both array and reference antenna. Thus,

（图 16.3 中两根 λ/2 单元阵列的增益）

$$G_f(\phi)\left[\frac{A}{\mathrm{HW}}\right] = \frac{E(\phi)}{E_{\mathrm{HW}}(\phi)} = \sqrt{\frac{2R_{00}}{R_{11} + R_{12}}}\left|\cos\left(\frac{d_r \cos\phi}{2}\right)\right| \quad \textit{Gain of 2 } \lambda/2 \textit{ element array of Fig. 16–3} \tag{27}$$

The absolute value bars || are introduced so that the gain will be confined to positive values (or zero) regardless of the values of d_r and ϕ. A negative gain would merely indicate a phase difference between the fields of the array and the reference antenna.

If the gain is the ratio of the *maximum* field of the array to the *maximum* field of the reference antenna, it is designated by G_f (not a function of angle).

The self-resistances $R_{00} = R_{11} = 73\ \Omega$. For the case where the spacing is $\lambda/2$, $d_r = \pi$ and $R_{12} = -13\ \Omega$ so that (27) becomes

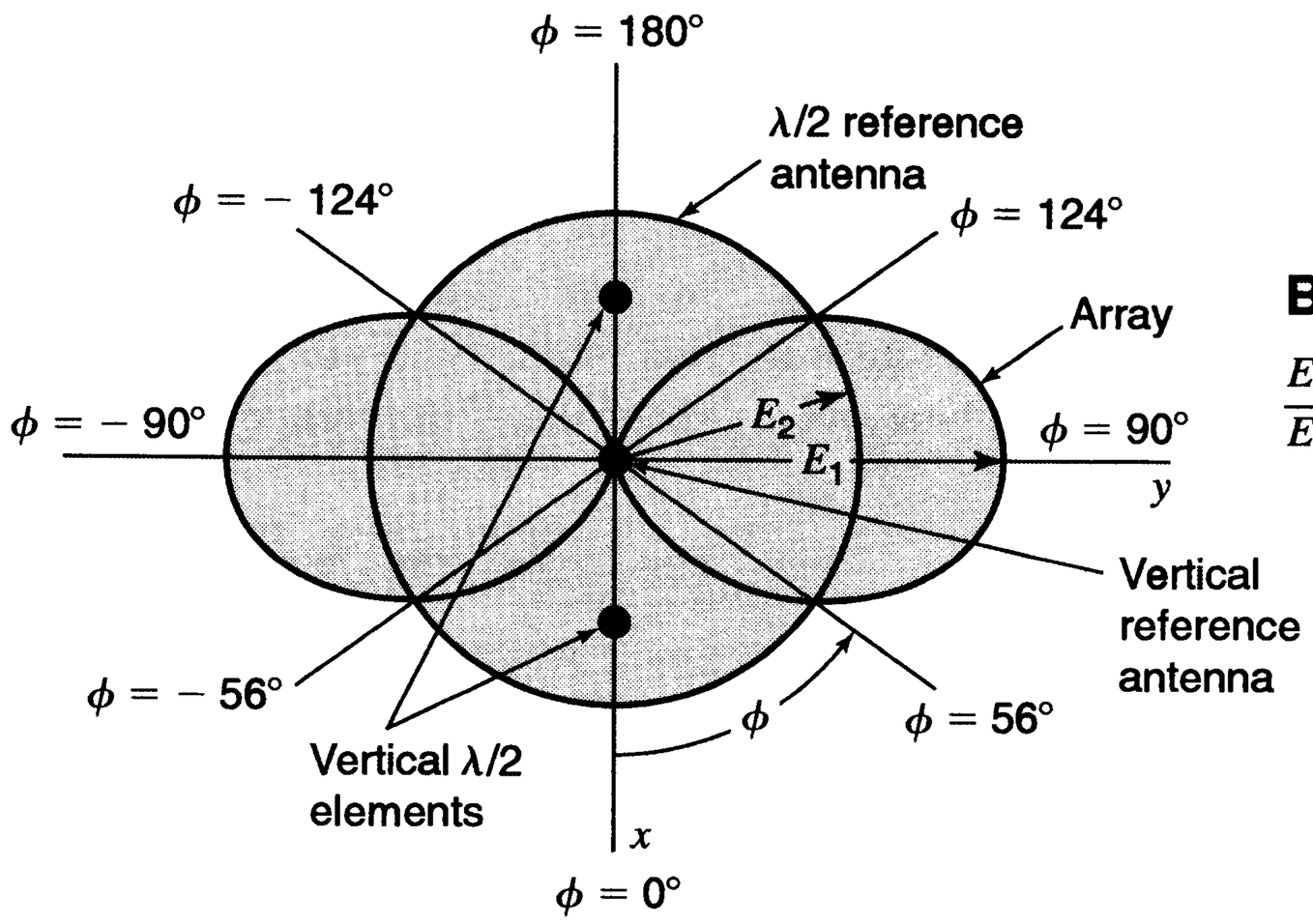
$$G_f(\phi)\left[\frac{A}{\mathrm{HW}}\right] = 1.56\cos\left(\frac{\pi}{2}\cos\phi\right) \tag{28}$$

In the broadside direction ($\phi = \pi/2$), the pattern factor becomes unity. The gain is then 1.56. This is the ratio of the maximum field of the array to the maximum field of the reference antenna (see Fig. 16–4). Hence, $G_f = 1.56$.

It is also of interest to find the angle ϕ_0 for which the gain is unity. For this condition (28) becomes

$$\cos\left(\frac{\pi}{2}\cos\phi_0\right) = 0.64 \tag{29}$$

[1]Both the array and the $\lambda/2$ reference antenna are assumed to be in free space. Thus, to be more explicit, the expression $G_f(\phi)$[AFS/HWFS], meaning the gain in field intensity of the *Array in Free Space* (AFS) with respect to a *Half-Wavelength reference antenna in Free Space* (HWFS), might be used. However, to simplify the notation, the letters "FS" will be omitted when *both* antennas are in free space.



BROADSIDE ARRAY

$\frac{E_1}{E_2} = 1.56 = G_f = 3.86$ dB or 6.0 dBi

Figure 16–4
Horizontal plane field patterns of broadside array of 2 vertical in-phase $\lambda/2$ elements spaced $\lambda/2$. The pattern of a single vertical $\lambda/2$ reference antenna with the same power input is shown for comparison.

or

$$\phi_0 = \pm 56° \qquad \text{or} \qquad \pm 124°$$

These angles are shown in Fig. 16–4. The array has a gain of greater than unity in both broadside directions over an angle of 68°.

The gain as a decibel ratio is given by the relation

$$\text{Gain} = 20 \log_{10} G_f \qquad \text{(dB)}$$

where G_f = gain in field intensity

Thus, a field-intensity gain of 1.56 is equal to 3.86 dB.

Turning our attention now to the gain in the vertical plane (yz plane of Fig. 16–2a), the field intensity $E_{\text{HW}}(\theta)$ as a function of θ in this vertical plane at a distance D from a single vertical $\lambda/2$ reference antenna with the same power input is of the form of (6). Thus,

$$E_{\text{HW}}(\theta) = k I_0 \frac{\cos[(\pi/2)\cos\theta]}{\sin\theta} \tag{30}$$

where I_0 = the terminal current

Substituting its value from (23), we get

$$E_{\text{HW}}(\theta) = k \sqrt{\frac{P}{R_{00}}} \frac{\cos[(\pi/2)\cos\theta]}{\sin\theta} \tag{31}$$

The field intensity $E(\theta)$ as a function of θ in the vertical plane at a distance D from the array is given by (8). Introducing the value of the terminal current I_1 from (22) into (8),

we have

$$E(\theta) = k\sqrt{\frac{2P}{R_{11} + R_{12}}}\ \underset{\sin\theta}{\cos[(\pi/2)\cos\theta]} \tag{32}$$

The ratio of (32) to (31) gives the gain in field intensity, $G_f(\theta)[A/\text{HW}]$, of the array as a function of θ in the vertical plane over a vertical $\lambda/2$ reference antenna with the same power input. Thus,

（图 16.3 中两根 λ/2 单元阵列的增益）

（边射阵）

$$\boxed{\textit{Broadside array} \quad G_f(\theta)\left[\frac{A}{\text{HW}}\right] = \frac{E(\theta)}{E_{\text{HW}}(\theta)} = \sqrt{\frac{2R_{00}}{R_{11} + R_{12}}} \quad \textit{Gain of 2 } \lambda/2 \textit{ element array of Fig. 16–3}} \tag{33}$$

The gain is a constant, being independent of the angle θ. For the case where the spacing is $\lambda/2$, (33) becomes

$$G_f(\theta)\left[\frac{A}{\text{HW}}\right] = 1.56 \text{ (or } 3.86 \text{ dB)} \tag{34}$$

The shape of the pattern for the array and for the $\lambda/2$ reference antenna is the same as shown in Fig. 16–5 but the ratio of the radius vectors in a given direction is a constant equal to 1.56.

If the reference antenna is an isotropic source instead of a $\lambda/2$ antenna, the gain in the vertical plane is a function of the angle θ. The maximum gain in field intensity of the array over an isotropic source with the same power input is $\sqrt{1.64}$ times greater than the voltage gain over a $\lambda/2$ reference antenna [$D(\lambda/2) = 1.64$, see table at end of Chap. 2]. Thus, when the spacing is $\lambda/2$, the *maximum gain in field intensity of the array with respect to an*

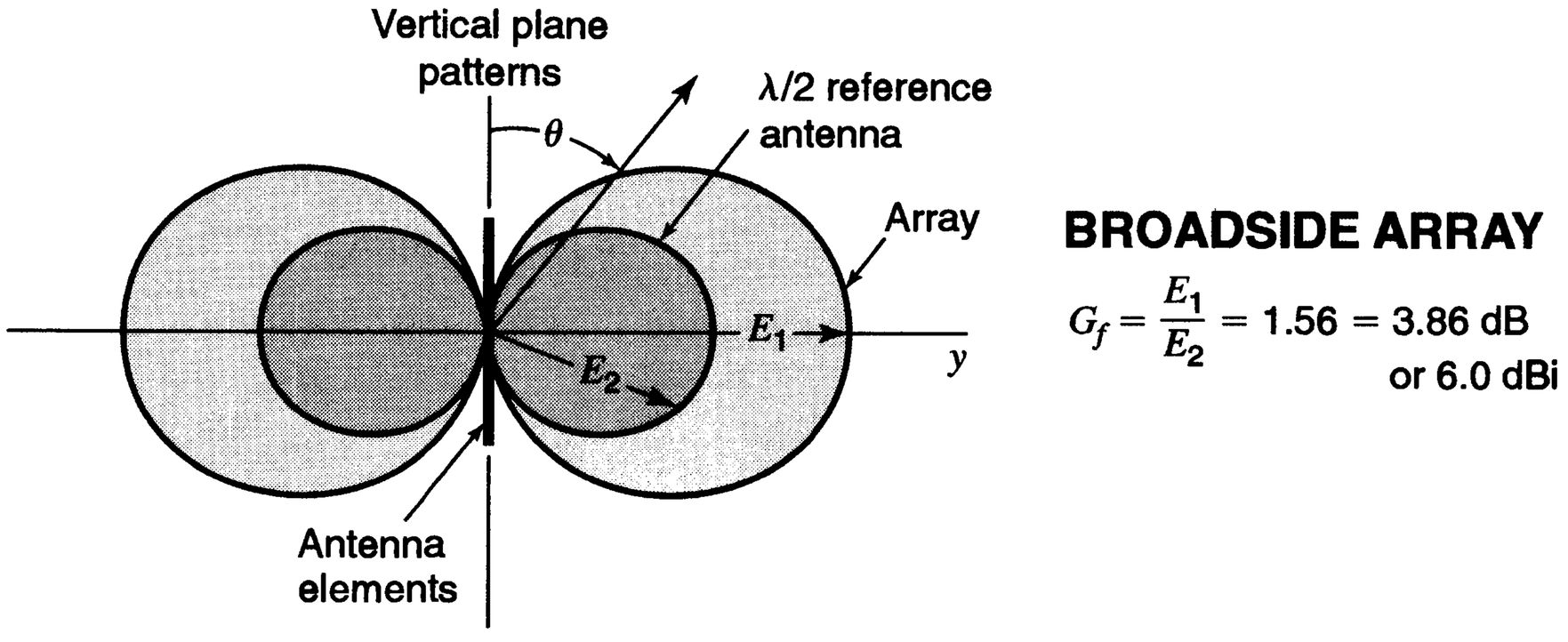


Figure 16–5
Vertical-plane pattern of broadside array of 2 vertical in-phase $\lambda/2$ elements spaced $\lambda/2$. The pattern of a single vertical $\lambda/2$ reference antenna with the same power input is shown for comparison.

isotropic source is

$$G_f\left[\frac{A}{\text{iso}}\right] = 1.56 \times \sqrt{1.64} = 2.0 \text{ (or } 6.0 \text{ dBi)}^1 \tag{35}$$

This value is in the broadside direction ($\phi = \theta = 90°$).

16–3 ARRAY OF 2 DRIVEN $\lambda/2$ ELEMENTS. END-FIRE CASE

(两根受激 λ/2 单元的端射阵)

Consider an array of 2 center-fed vertical $\lambda/2$ elements (dipoles) in free space arranged side by side with a spacing d and equal currents in opposite phase as in Fig. 16–6. The only difference between this case and the one discussed in Sec. 16–2 is that the currents in the elements are taken to be in the opposite phase instead of in the same phase. As in Sec. 16–2, the analysis will be divided into 3 subsections on the field patterns, driving-point impedance and gain in field intensity.

16–3a Field Patterns (场波瓣图)

The field intensity $E_1(\phi)$ as a function of ϕ at a distance D in a horizontal plane (xy or ϕ plane in Fig. 16–7*a*) from a single element is

$$E_1(\phi) = kI_1$$

where

k = a constant involving the distance D
I_1 = the terminal current

Replacing the elements by isotropic point sources of equal amplitude, the pattern $E_{\text{iso}}(\phi)$ in the horizontal plane for two such isotropic out-of-phase sources is given

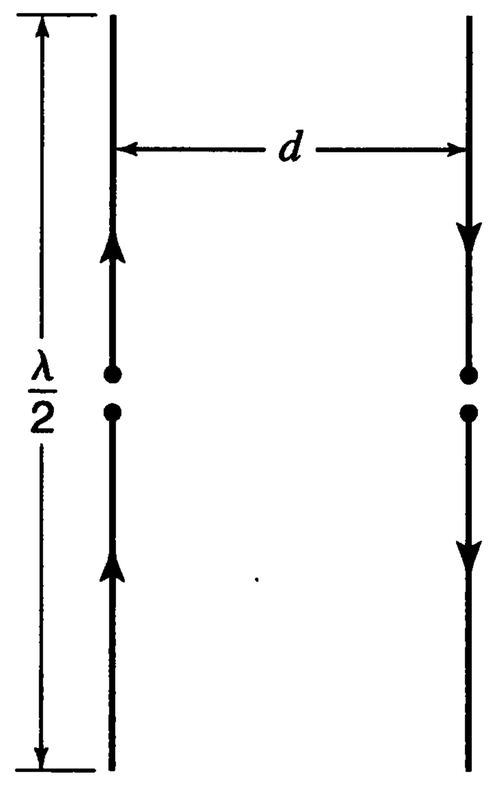


Figure 16–6
End-fire array of 2 linear $\lambda/2$ elements with currents of equal magnitude but opposite phase.

[1]Distinguish between "dB" for gain with respect to a *reference antenna* ($\lambda/2$ dipole in the present case) and "dBi" for gain with respect to an *isotropic source*.

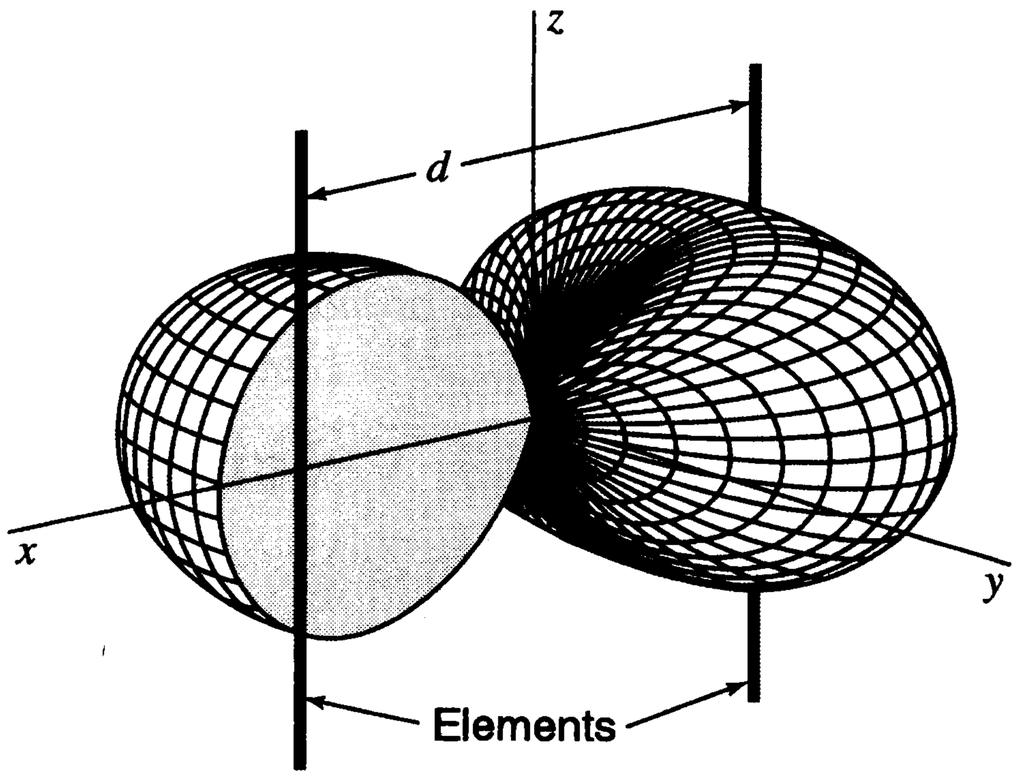


Figure 16–7
Field patterns for end-fire array of 2 linear out-of-phase $\lambda/2$ elements with spacing $d = \lambda/2$.

by (5–2–10) as

$$E_{\text{iso}}(\phi) = 2E_0 \sin\left(\frac{d_r \cos\phi}{2}\right) \tag{1}$$

Applying the principle of pattern multiplication, we may consider that E_0 is the field intensity from a single element at a large distance D. Thus

$$E_0 = E_1(\phi) = kI_1 \tag{2}$$

and the field intensity $E(\phi)$ as a function of ϕ in the horizontal plane at a large distance D from the array is

$$E(\phi) = 2kI_1 \sin\left(\frac{d_r \cos\phi}{2}\right) \tag{3}$$

This is the absolute field pattern in the horizontal plane. The electric field at points in this plane is everywhere vertically polarized. The relative pattern for the case where the spacing d is $\lambda/2$ is shown in Fig. 16–7*b* and also partially in Fig. 16–7*a*. The maximum field intensity is at $\phi = 0°$ and $\phi = 180°$. Hence, the array is commonly referred to as an "end-fire" type.

The field intensity $E_1(\theta)$ as a function of θ from a single $\lambda/2$ element at a distance D in the vertical plane (xz plane in Fig. 16–7*a*) is, from (6–4–4), given by

$$E_1(\theta) = kI_1 \frac{\cos[(\pi/2)\cos\theta]}{\sin\theta} \tag{4}$$

The pattern $E_{\text{iso}}(\theta)$ as a function of θ in the vertical plane for two isotropic sources in place of the two elements is, from (5–2–10),

$$E_{\text{iso}}(\theta) = 2E_0 \sin\left(\frac{d_r \sin\theta}{2}\right) \tag{5}$$

Note that θ is complementary to ϕ in (5–2–10), so $\cos\phi = \sin\theta$.

Putting $E_0 = E_1(\theta)$, the field intensity $E(\theta)$ as a function of θ in the vertical plane at a large distance D from the array is

（水平面波瓣图）

（两 λ/2 单元边射阵）

2 λ/2 element end-fire array $E(\theta) = 2kI_1 \dfrac{\cos[(\pi/2)\cos\theta]}{\sin\theta} \sin\left(\dfrac{d_r \sin\theta}{2}\right)$ ***Vertical plane pattern*** (6)

This is the absolute field pattern in the vertical plane. The relative pattern is illustrated in Fig. 16–7*c*, and also partially in Fig. 16–7*a*, for the case where the spacing is $\lambda/2$. The relative 3-dimensional field variation for this case ($d = \lambda/2$) is suggested in Fig. 16–7*a*.

16–3b Driving-Point Impedance（激励点阻抗）

Let V_1 be the emf applied to the terminals of element 1. Then

$$V_1 = I_1 Z_{11} + I_2 Z_{12} \tag{7}$$

Likewise, if V_2 is the emf applied to the terminals of element 2,

$$V_2 = I_2 Z_{22} + I_1 Z_{12} \tag{8}$$

The currents are equal in magnitude but opposite in phase so

$$I_2 = -I_1 \tag{9}$$

Therefore, (7) and (8) become

$$V_1 = I_1(Z_{11} - Z_{12}) \tag{10}$$

and

$$V_2 = I_2(Z_{22} - Z_{12}) \tag{11}$$

The terminal impedance Z_1 of element 1 is

$$Z_1 = \frac{V_1}{I_1} = Z_{11} - Z_{12} \tag{12}$$

and the terminal impedance Z_2 of element 2 is

$$Z_2 = \frac{V_2}{I_2} = Z_{22} - Z_{12} \tag{13}$$

Therefore,

$$Z_1 = Z_2 = Z_{11} - Z_{12} \tag{14}$$

or

$$\frac{V_1}{I_1} = \frac{V_2}{I_2} \tag{15}$$

Since $I_2 = -I_1$ it follows from (15) that $V_2 = -V_1$. This means that the 2 elements must be energized with emfs which are equal in magnitude and opposite in phase. This may be done by means of a crossover in the transmission line from the driving point P to one of the elements as shown in Fig. 16–8. The length l of each line is the same.

For the case where the spacing between elements is $\lambda/2$, the terminal impedance of each element is

$$\begin{aligned} Z_1 &= R_{11} - R_{12} + j(X_{11} - X_{12}) \\ &= 86 + j72\ \Omega = \text{impedance of each element} \end{aligned} \tag{16}$$

Consider that the reactance of 72 Ω is tuned out by a series capacitance at the terminals of each element. The terminal impedance is then a pure resistance of 86 Ω. To obtain a ***driving-point resistance of 600 Ω,*** (600 Ω的激励点电阻) let the length l of the line from P to each element be $\lambda/4$ with a line impedance of $\sqrt{1200 \times 86} = 321\ \Omega$. For an impedance match, the line from the driving point P to the transmitter should have a characteristic impedance of 600 Ω.

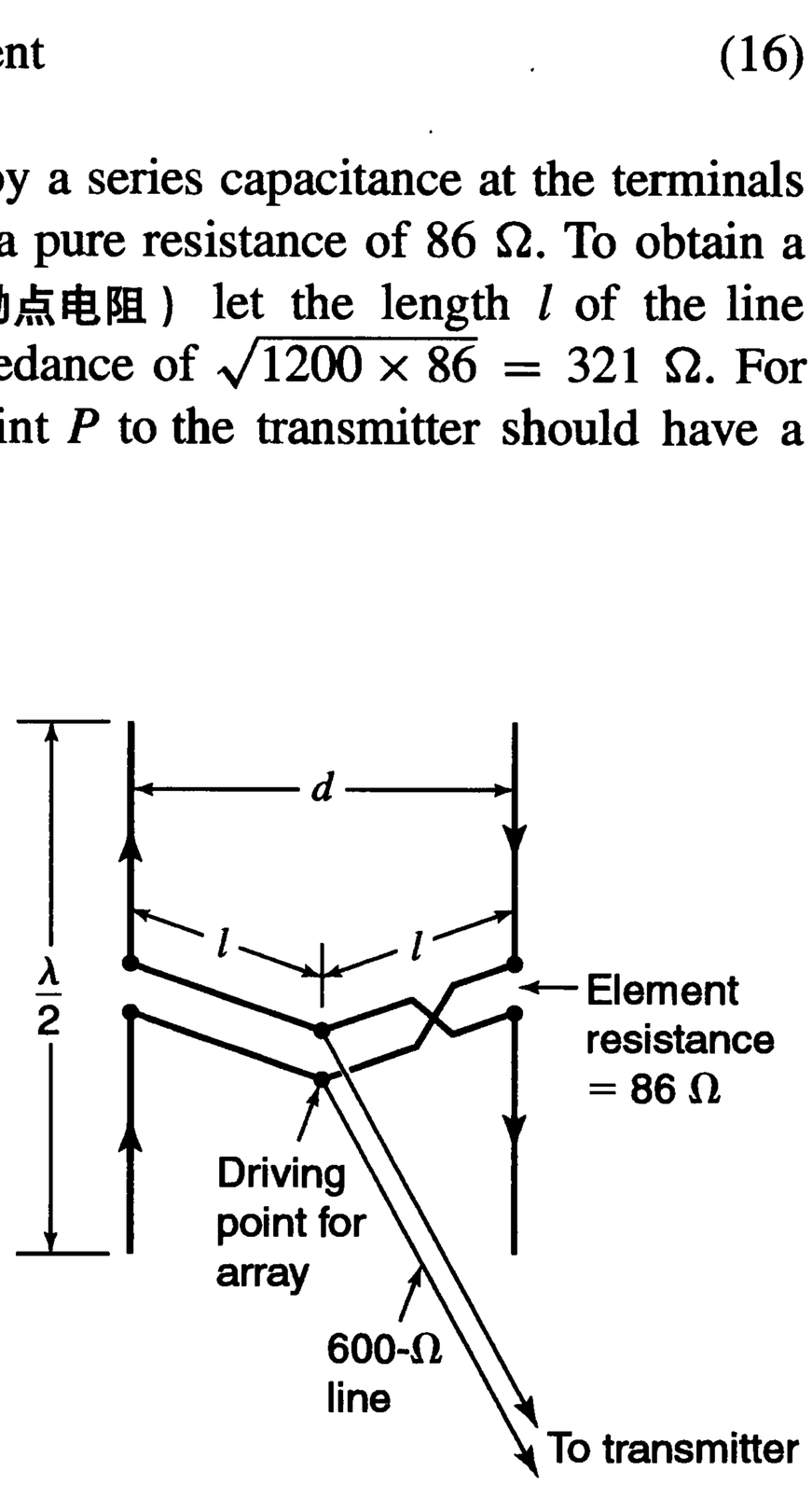


Figure 16–8
End-fire array of 2 linear $\lambda/2$ elements with arrangement for driving elements, with currents of equal magnitude but opposite phase.

16–3c Gain in Field Intensity（场强增益）

Using the same method as in Sec. 16–2*c*, the current I_1 in each element for a power input P to the array is given by

$$I_1 = \sqrt{\frac{P}{2(R_{11} - R_{12})}} \tag{17}$$

It is assumed that there are no heat losses. The current I_0 in a single $\lambda/2$ reference antenna is given by (16–2–23). The gain in field intensity $G_f(\phi)[A/\text{HW}]$ as a function of ϕ in the horizontal plane with respect to a $\lambda/2$ reference antenna is obtained by substituting (17) in (3) and taking the ratio of this result to (16–2–25). This yields

$$G_f(\phi)\left[\frac{A}{\text{HW}}\right] = \sqrt{\frac{2R_{00}}{R_{11} - R_{12}}}\left|\sin\left(\frac{d_r \cos\phi}{2}\right)\right| \tag{18}$$

For a spacing of $\lambda/2$, (18) reduces to

$$G_f(\phi)\left[\frac{A}{\text{HW}}\right] = 1.3\left|\sin\left[\left(\frac{\pi}{2}\right)\cos\phi\right]\right| \tag{19}$$

In the end-fire directions ($\phi = 0°$ and $180°$) the pattern factor becomes unity, and the gain is 1.3 or 2.3 dB. This is the gain G_f (see Fig. 16–9).

The gain in field intensity $G_f(\theta)[A/\text{HW}]$ as a function of θ in the vertical plane (xz plane of Fig. 16–7*a*) with respect to a $\lambda/2$ reference antenna is found by substituting (17) in (6) and taking the ratio of this result to (16–2–31), obtaining

（图 16.6 中两根 λ/2 单元阵列的增益）

（端射阵）

End-fire array $$G_f(\theta)\left[\frac{A}{\text{HW}}\right] = \sqrt{\frac{2R_{00}}{R_{11} - R_{12}}}\left|\sin\left(\frac{d_r \sin\theta}{2}\right)\right|$$ ***Gain of 2 λ/2 elements of Fig. 16–9*** (20)

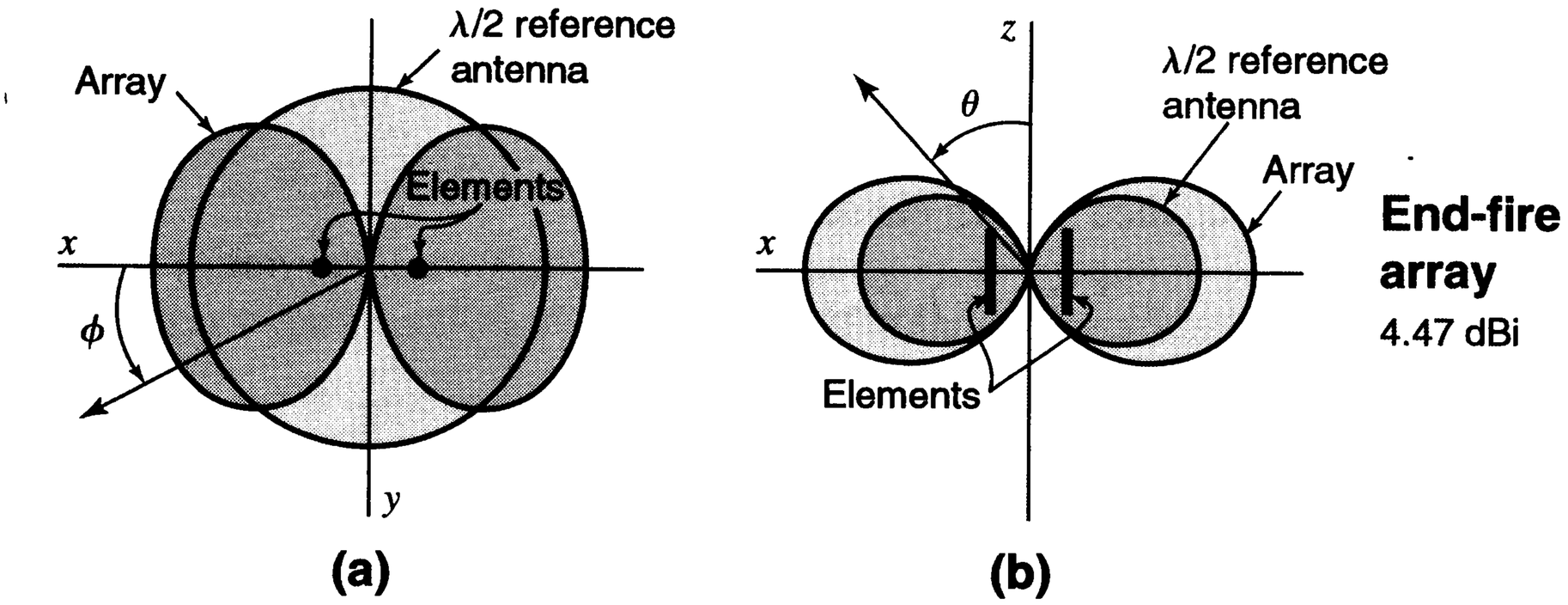


Figure 16–9
Horizontal plane field pattern (*a*) and vertical plane field pattern (*b*) of end-fire array of 2 vertical $\lambda/2$ elements with $\lambda/2$ spacing. The patterns of a vertical $\lambda/2$ reference antenna with the same power input are shown for comparison.

which is of the same form as the gain expression (18) for the horizontal plane (note that maximum radiation is in a direction $\theta = 90°, \phi = 0°$).

The gain in field intensity G_f of the array over an isotropic source with the same power input is $1.3 \times \sqrt{1.64} = 1.66$ (or 4.4 dBi).

16–4 ARRAY OF 2 DRIVEN λ/2 ELEMENTS. GENERAL CASE WITH EQUAL CURRENTS OF ANY PHASE RELATION[1]

（两根受激 λ/2 单元的等幅任意相位阵）

In the preceding sections 2 special cases of an array of two $\lambda/2$ driven elements have been treated. In one case the currents in the elements are in phase (phase difference = 0°) and in the other the currents are in opposite phase (phase difference = 180°). In this section the more general case is considered where the phase difference may have any value. As in the preceding cases, the two $\lambda/2$ elements are arranged side by side with a spacing d and are driven with currents of equal magnitude.

For the general-phase case the radiation-field pattern in the horizontal plane (xy plane of Fig. 16–7*a*) is, from (5–2–20), given by

$$E(\phi) = 2kI_1 \cos\frac{\psi}{2} \tag{1}$$

where ψ is the total phase difference between the fields from element 1 and element 2 at a large distance in the direction ϕ (see Fig. 16–10). Thus,

$$\psi = d_r \cos\phi + \delta \tag{2}$$

where δ = the phase difference of the currents in the elements

A positive sign in (2) indicates that the current in element 2 of Fig. 16–10 is advanced in phase by an angle δ with respect to the current in element 1; that is,

$$I_2 = I_1 \angle\delta$$

or

$$I_1 = I_2 \angle -\delta \tag{3}$$

The voltages applied at each element are

$$V_1 = I_1 Z_{11} + I_2 Z_{12} = I_1 (Z_{11} + Z_{12} \angle\delta) \tag{4}$$

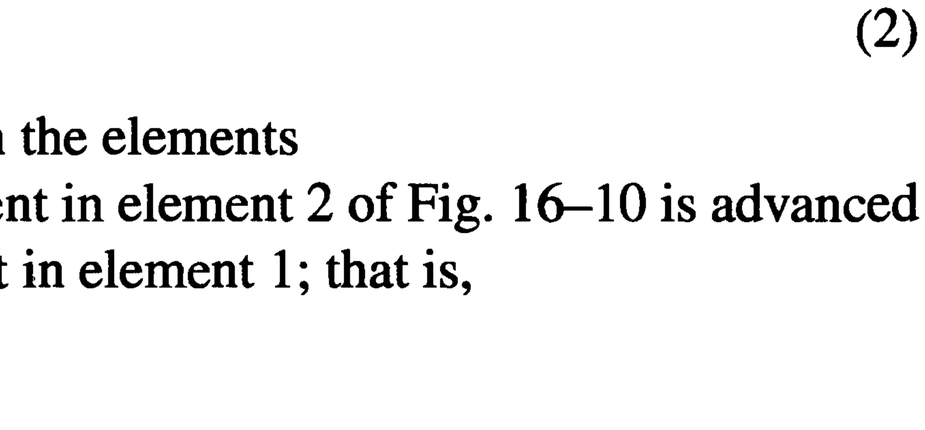


Figure 16–10
Array of 2 side-by-side $\lambda/2$ elements normal to plane of page.

[1]For a more detailed discussion of this case and also of the most general case where the current amplitudes are unequal, see G. H. Brown (2).

and

$$V_2 = I_2 Z_{22} + I_1 Z_{12} = I_2(Z_{22} + Z_{12}\angle -\delta) \tag{5}$$

The driving-point impedances of the elements are then

$$Z_1 = \frac{V_1}{I_1} = Z_{11} + Z_{12}\angle \delta \tag{6}$$

and

$$Z_2 = \frac{V_2}{I_2} = Z_{22} + Z_{12}\angle -\delta \tag{7}$$

The real parts of the driving-point resistances are

$$R_1 = R_{11} + |Z_{12}|\cos(\tau + \delta) \tag{8}$$

and

$$R_2 = R_{22} + |Z_{12}|\cos(\tau - \delta) \tag{9}$$

where

τ = the phase angle of the mutual impedance Z_{12} (that is, $\tau = \arctan X_{12}/R_{12}$ where $Z_{12} = R_{12} + jX_{12}$)

Therefore, the power P_1 in element 1 is

$$P_1 = |I_1|^2 R_1 = |I_1|^2[R_{11} + |Z_{12}|\cos(\tau + \delta)] \tag{10}$$

and the power P_2 in element 2 is

$$P_2 = |I_2|^2[R_{22} + |Z_{12}|\cos(\tau - \delta)] \tag{11}$$

Since $R_{11} = R_{22}$, the total power P is

$$\begin{aligned} P = P_1 + P_2 &= |I_1|^2\{2R_{11} + |Z_{12}|[\cos(\tau + \delta) + \cos(\tau - \delta)]\} \\ &= 2|I_1|^2(R_{11} + |Z_{12}|\cos\tau\cos\delta) \\ &= 2|I_1|^2(R_{11} + R_{12}\cos\delta) \end{aligned} \tag{12}$$

It follows that the gain in field intensity as a function of ϕ in the horizontal plane[1] of the array over a single $\lambda/2$ element with the same power input is

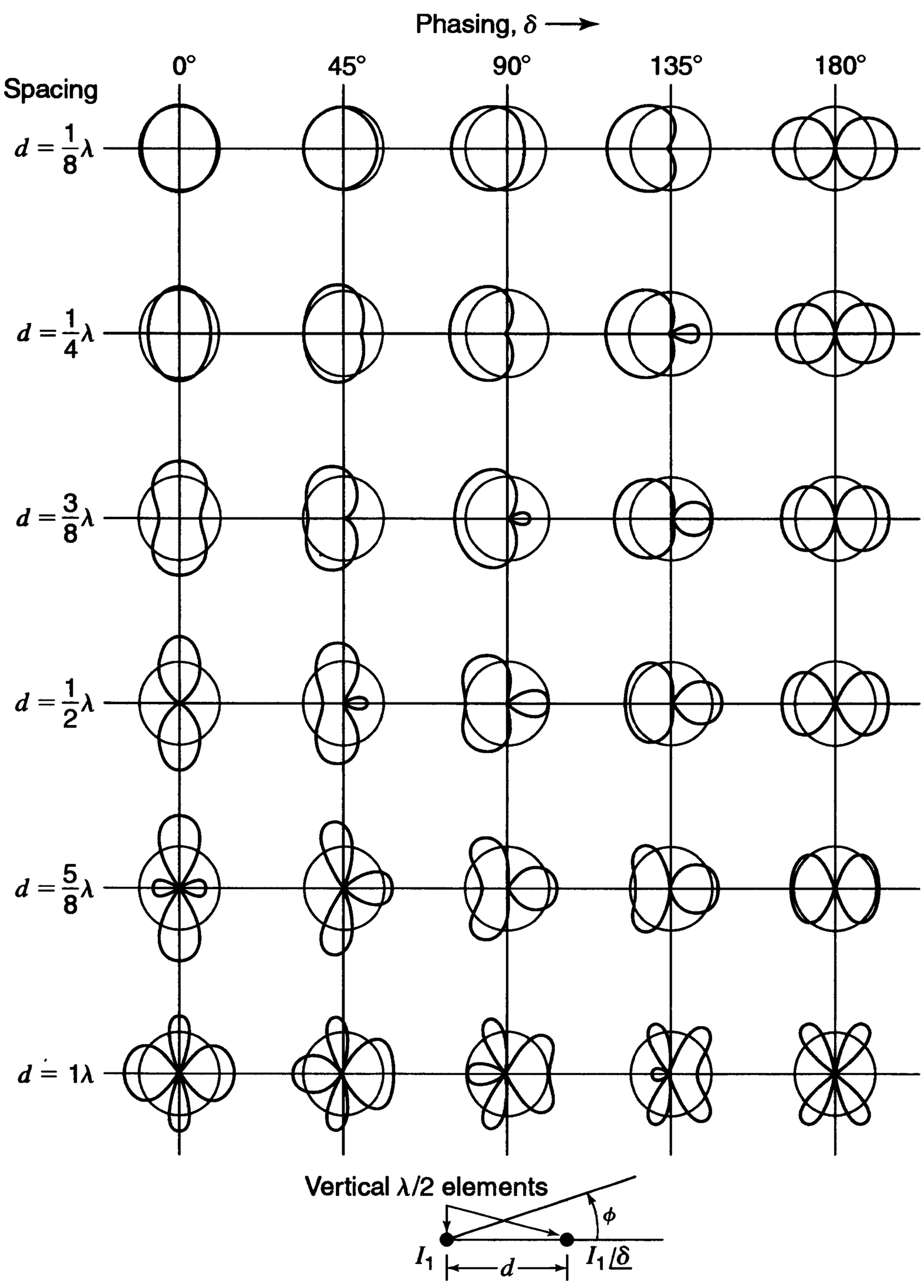
（水平面内场强增益）

Gain in field intensity in horizontal plane

$$G_f(\phi)\left[\frac{A}{\text{HW}}\right] = \sqrt{\frac{2R_{11}}{R_{11} + R_{12}\cos\delta}}\left|\cos\left(\frac{d_r\cos\phi + \delta}{2}\right)\right| \tag{13}$$

[1]This is the plane of the page in Fig. 16–10.

A polar plot of (13) with respect to the azimuth angle ϕ gives the radiation-field pattern of the array in the horizontal plane, the ratio of the magnitude of the radius vector to a unit radius indicating the gain over a reference $\lambda/2$ antenna. Brown (2) has calculated such patterns as a function of phase difference δ and spacing d_r. Examples of these are shown in Fig. 16–11.



Horizontal plane field patterns of 2 vertical λ/2 elements with spacings d and phasing δ. See sketch at bottom of figure

Figure 16–11
Horizontal-plane field patterns of 2 vertical elements as a function of the phase difference δ and spacing d. (*After G. H. Brown-2*). Both elements are the same length and have currents of equal magnitude. The circles indicate the field intensity of a single reference element of the same length with the same power input.

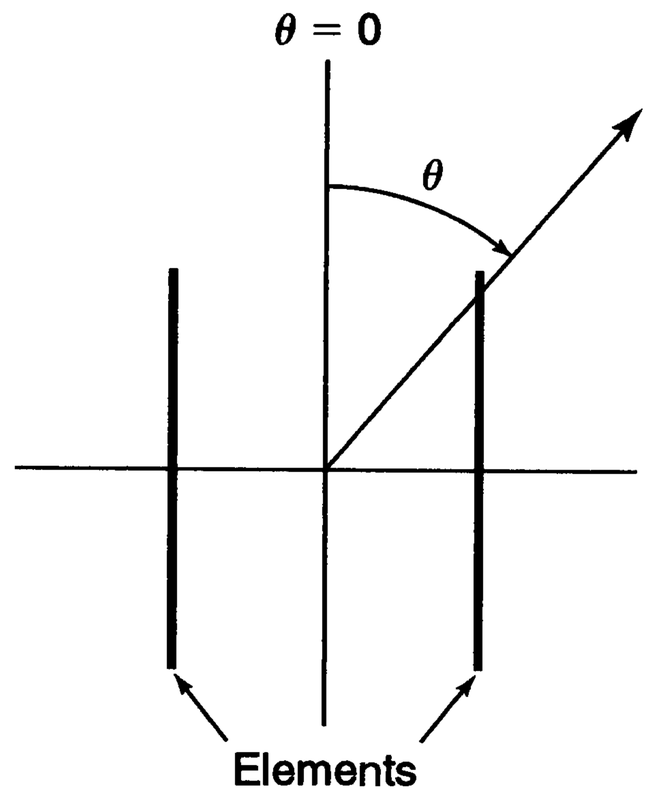


Figure 16–12
Relation of polar angle θ in the plane of the elements.

The radiation-field pattern in the vertical plane containing the elements (in the plane of the page of Fig. 16–12) is

$$E(\theta) = 2kI_1 \cos\left(\frac{d_r \sin\theta + \delta}{2}\right) \frac{\cos[(\pi/2)\cos\theta]}{\sin\theta} \tag{14}$$

Thus, the pattern in the vertical plane has the shape of the patterns of Fig. 16–11 multiplied by the pattern of a single $\lambda/2$ antenna. The gain in the vertical plane over a vertical $\lambda/2$ reference antenna with the same power input is then

$$G_f(\theta)\left[\frac{A}{\text{HW}}\right] = \sqrt{\frac{2R_{11}}{R_{11} + R_{12}\cos\delta}}\left|\cos\left(\frac{d_r \sin\theta + \delta}{2}\right)\right| \tag{15}$$

It is often convenient to refer the gain to an isotropic source with the same power input. Since the power gain of a $\lambda/2$ antenna over an isotropic source is 1.64 [$D(\lambda/2) = 1.64$, see table at end of Chap. 2], the gain in field intensity as a function of θ in the vertical plane of a vertical $\lambda/2$ antenna in free space over an isotropic source is

$$G_f(\theta)\left[\frac{\text{HW}}{\text{iso}}\right] = \sqrt{1.64}\left|\frac{\cos[(\pi/2)\cos\theta]}{\sin\theta}\right| \tag{16}$$

The gain in field intensity in the vertical plane of the array over the isotropic source is then the product of (15) and (16) or

（铅垂面内场强增益）

$$\boxed{\begin{aligned} G_f(\theta)\left[\frac{A}{\text{iso}}\right] &= G_f(\theta)\left[\frac{A}{\text{HW}}\right] G_f(\theta)\left[\frac{\text{HW}}{\text{iso}}\right] \qquad \textit{Gain in field intensity in vertical plane} \\ &= \sqrt{\frac{3.28R_{11}}{R_{11} + R_{12}\cos\delta}}\left|\cos\left(\frac{d_r \sin\theta + \delta}{2}\right)\frac{\cos[(\pi/2)\cos\theta]}{\sin\theta}\right| \end{aligned}} \tag{17}$$

EXAMPLE **16–4.1 Gain in dBi of Array of 2 $\lambda/2$ In-Phase Antennas**

Find the gain of this array, (*a*) if $s = 0.4\ \lambda$, (*b*) if $s = 0.5\ \lambda$ and (*c*) if $s = 0.6\ \lambda$.

■ **Solution**

(*a*) From Table 13–1 the mutual resistance $R_{21} = +6.3\ \Omega$. Thus, from (13)

Gain in field over half-wave antenna $= (2 \times 73.13/79.4)^{1/2} = 1.84^{1/2}$. So power gain over isotropic $= 1.84 \times 1.64 = 3.02$ or 4.8 dBi *Ans.*

(*b*) Similarly from (13), power gain over isotropic $= 2.42 \times 1.64 = 3.97$ or 5.98 dBi *Ans.*

(*c*) Similarly from (13), power gain over isotropic $= 2.94 \times 1.64 = 4.83$ or 6.84 dBi *Ans.*

■ **Question**

Is there a spacing which gives even more gain? See Prob. 16–4–1.

16–5 CLOSELY SPACED ELEMENTS, RADIATING EFFICIENCY AND *Q*. THE W8JK ARRAY（密距单元的辐射效率和 *Q* 值，W8JK 阵）

The end-fire array of two side-by-side, out-of-phase $\lambda/2$ dipole elements discussed in Sec. 16–3 can produce substantial gains when the spacing is decreased to small values. As indicated by the $R_L = 0$ curve in the gain-versus-spacing graph of Fig. 16–13*a*, the gain approaches 3.9 dB at small spacings. At $\lambda/2$ spacing the gain is 2.3 dB. This curve is calculated from (16–3–18) for $\phi = 0°$ or (16–3–20) for $\theta = 90°$. As the spacing d approaches zero, the coupling factor becomes infinite, but at the same time the pattern factor approaches zero. The product of the two or gain stays finite, leveling off at a value of about 3.9 dB (6.0 dBi) for small spacings, as illustrated by Fig. 16–13*b*. The fact that increased gain is associated with small spacings makes this arrangement attractive for many applications.

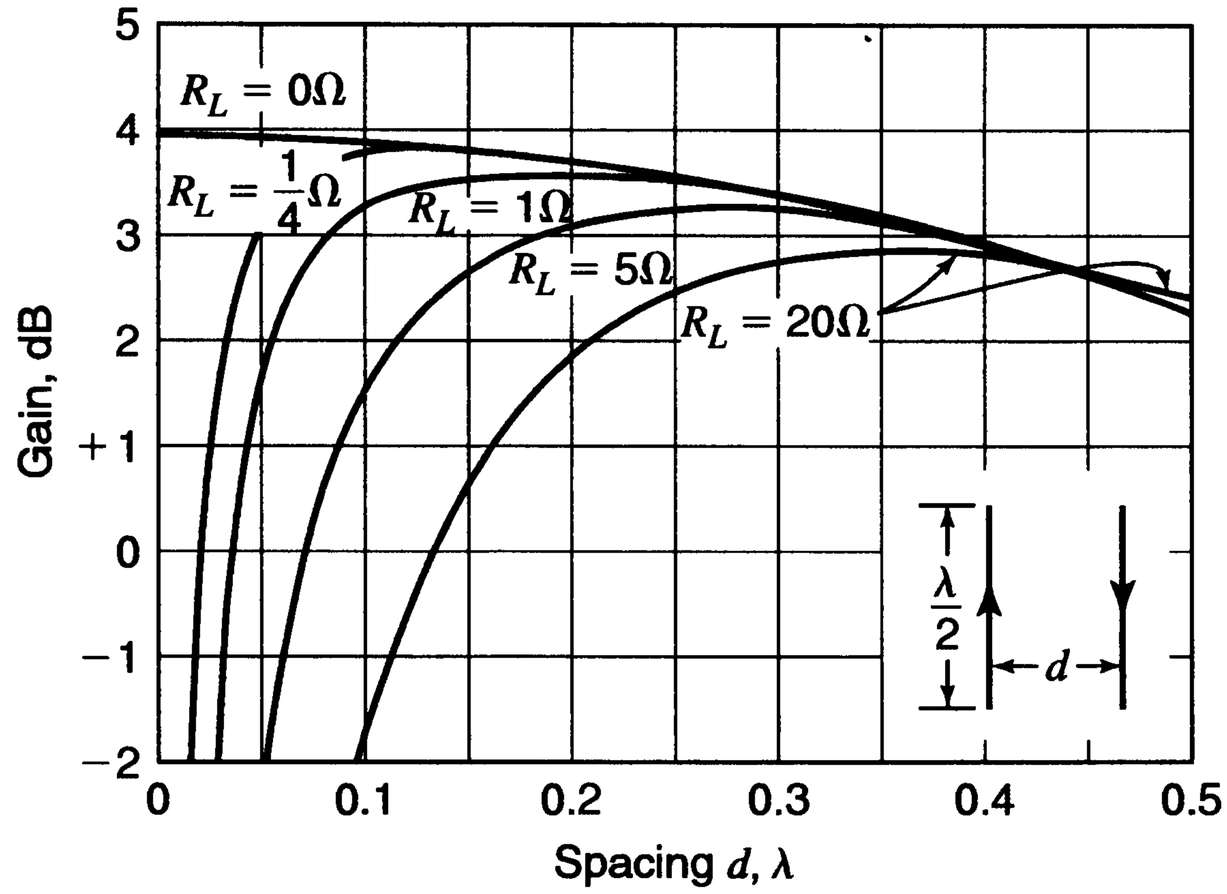
Thus far it has been assumed that there are no heat losses in the antenna system. In many antennas such losses are small and can be neglected. However, in the W8JK antenna such losses may have considerable effect on the gain (Fig. 16–13*b*). Therefore, the question of losses and of radiating efficiency will be treated in this section in connection with a discussion of arrays of 2 closely spaced, out-of-phase elements. The term "closely spaced" will be taken to mean that the elements are spaced $\lambda/4$ or less.

A transmitting antenna is a device for radiating radio-frequency power. Let the *radiating efficiency* be defined as the ratio of the power radiated to the power input of the antenna. The real power delivered to the antenna that is not radiated is dissipated in the *loss resistance* and appears chiefly in the form of heat in the antenna conductor, in the insulators supporting the antenna, etc. An antenna with a total terminal resistance R_{1T} may be considered to have a terminal radiation resistance R_1 and an equivalent terminal loss resistance R_{1L} (see Sec. 2–10) such that

$$R_{1T} = R_1 + R_{1L} \tag{1}$$

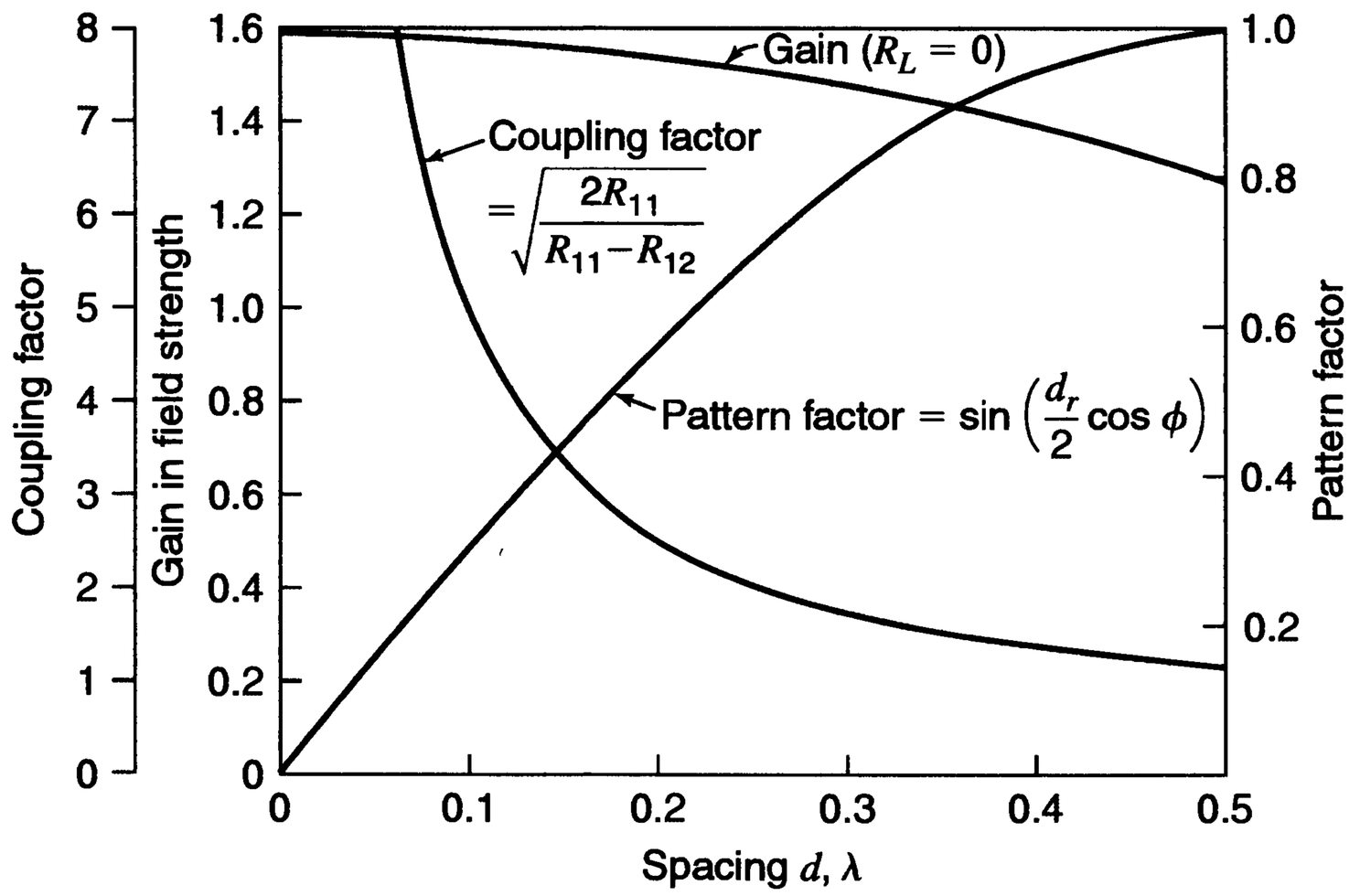
It follows that

$$\text{Radiating efficiency (\%)} = \frac{R_1}{R_1 + R_{1L}} \times 100 \tag{2}$$



(a)

Gain in dB of two λ/2 antennas with different spacings d and loss resistance R_L. For gain in dBi add 2.1 dB



(b)

Gain factors

Figure 16–13
(*a*) Gain of end-fire array of 2 out-of-phase λ/2 elements (W8JK array) with respect to a λ/2 reference antenna as a function of the spacing for 5 values of the loss resistance R_L. (*b*) Gain curve for $R_L = 0$ with variation of its component factors, the coupling factor and the pattern factor, for $\phi = 0$.

Since many types of high-frequency antennas have radiation resistances that are large compared to any loss resistance, the efficiencies are high. In an array with closely spaced, out-of-phase elements, however, the radiation resistance may be relatively small and the antenna current very large, as illustrated by Fig. 16–14. Hence, *a considerable reduction in radiating efficiency may result from the presence of any loss resistance*. The radiating efficiency may also be small for low-frequency antennas which are very short compared to the wavelength. *Although the effect of loss resistance will be discussed specifically for an*

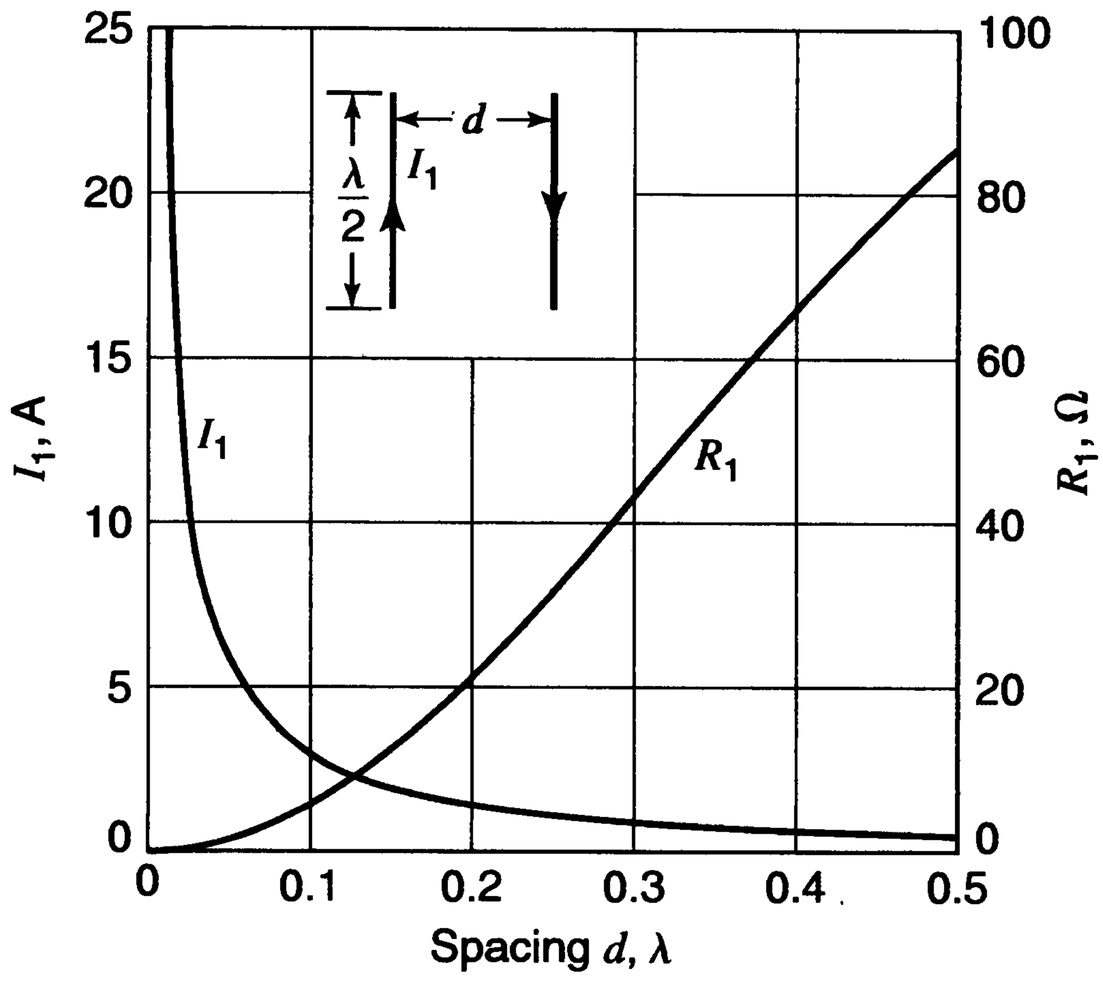


Figure 16–14
Current I_1 and radiation resistance R_1 in each element of a W8JK antenna as a function of the spacing. The current is calculated for a constant input power of 100 W to the array.

array of 2 closely spaced λ/2 elements, the method is general and may be applied to any type of antenna.

Let the equivalent loss resistance at the terminals of each element be R_{1L}. The elements are center-fed and are arranged side by side with a spacing d. The total terminal resistance R_{1T} is as given by (1). The terminal radiation resistance R_1 is given by

$$R_1 = R_{11} - R_{12} \tag{3}$$

Substituting (3) in (1) the total terminal resistance for each element is then

$$R_{1T} = R_{11} + R_{1L} - R_{12} \tag{4}$$

If a power P is supplied to the 2-element array, the current I_1 in each element is

$$I_1 = \sqrt{\frac{P}{2(R_{11} + R_{1L} - R_{12})}} \tag{5}$$

The total terminal resistance R_{0T} of a single, center-fed λ/2 reference antenna is

$$R_{0T} = R_{00} + R_{0L} \tag{6}$$

where R_{00} is the self-resistance and R_{0L} the loss resistance of the reference antenna. The current I_0 at the terminals of the reference antenna is then

$$I_0 = \sqrt{\frac{P}{R_{00} + R_{0L}}} \tag{7}$$

With the array elements vertical, the gain in field intensity as a function of ϕ in the horizontal plane (xy plane in Fig. 16–7*a*) is obtained by substituting (5) in (16–3–3), (7) in (16–2–24) and taking the ratio which gives

$$G_f(\phi)\left[\frac{A}{\text{HW}}\right] = \sqrt{\frac{2(R_{00} + R_{0L})}{R_{11} + R_{1L} - R_{12}}}\left|\sin\left(\frac{d_r \cos\phi}{2}\right)\right| \tag{8}$$

This expression reduces to (16–3–18) if the loss resistances are zero ($R_{0L} = R_{1L} = 0$).

In a similar way the gain in field intensity as a function of θ in the vertical plane (xz plane in Fig. 16–7*a*) is

$$G_f(\theta)\left[\frac{A}{\text{HW}}\right] = \sqrt{\frac{2(R_{00}+R_{0L})}{R_{11}+R_{1L}-R_{12}}}\left|\sin\left(\frac{d_r\sin\theta}{2}\right)\right| \qquad \textit{Gain in field intensity including effect of loss resistance, } R_L$$

（包含损耗电阻 R_L 效应的场强增益） (9)

This reduces to (16–3–20) if the loss resistances are zero.

The effect of loss resistance on the gain of a closely spaced array of 2 out-of-phase $\lambda/2$ elements over a $\lambda/2$ reference antenna is illustrated by the curves in Fig. 16–13*a*. The gain presented is the maximum gain which occurs in the directions of maximum radiation from the array ($\phi = 0$ and 180°; $\theta = 90°$). The top curve is for zero loss resistance ($R_{0L} = R_{1L} = 0$). The lower curves are for 4 different values of assumed loss resistance: $\frac{1}{4}$, 1, 5 and 20 Ω. The assumption is made that the loss resistance R_{1L} of each element of the array is the same as the loss resistance R_{0L} of the reference $\lambda/2$ antenna (that is, $R_{1L} = R_{0L}$). It is apparent from the curves that a loss resistance of only 1 Ω seriously limits the gain at spacings of less than $\lambda/10$, and larger loss resistances cause reductions in gain at considerably greater spacings. If the loss resistance is taken to be 1 Ω (a not unlikely value for a typical high-frequency antenna), the gain is almost constant (within 0.1 dB) for spacings between $\lambda/8$ and $\lambda/4$. Smaller spacings result in reduced gain because of decreased efficiency while larger spacings also give reduced gain, not because of decreased efficiency but because of the decrease in the coupling factor. A spacing of $\lambda/8$ has the advantage that the physical size of the antenna is less. However, resonance is sharper for this spacing than for wider spacings. Hence, a spacing of $\lambda/4$ is to be preferred if a wide bandwidth is desired. In some situations an intermediate or compromise spacing is indicated.

The Q of an antenna, like the Q of any resonant circuit, is proportional to the ratio of the energy stored to the energy lost (in heat or radiation) per cycle. For a constant power input to the closely spaced array *the Q is nearly proportional to the square of the current I in each element*. Referring to Fig. 16–14, it is apparent that the current for $\lambda/8$ spacing is about twice the value for $\lambda/4$ spacing. Hence the Q for $\lambda/8$ spacing is about 4 times the Q for $\lambda/4$ spacing. A large Q indicates a large amount of stored energy near the antenna in proportion to the energy radiated per cycle. This also means that the antenna acts like a sharply tuned circuit. Since the bandwidth (if it is narrow) is inversely proportional to the Q, a spacing of $\lambda/4$ provides about 4 times the bandwidth obtained with $\lambda/8$ spacing. Although the efficiency of an array with closely spaced, out-of-phase elements might be increased, e.g., by using a large-diameter conductor for each element, any substantial increase in bandwidth requires an increase in the spacing between the elements. This increase also raises the radiating efficiency.

A single-section horizontally polarized W8JK closely spaced array consists of two side-by-side, out-of-phase $\lambda/2$ elements as indicated in Fig. 16–15. A single-section array is also shown in Fig. 16–16*a*. Five other examples of W8JK antennas are shown in Fig. 16–16 with arrows located at current maxima, indicating the instantaneous current directions. The type at Fig. 16–16*b* has an additional collinear $\lambda/2$ section, the 2 sections being energized

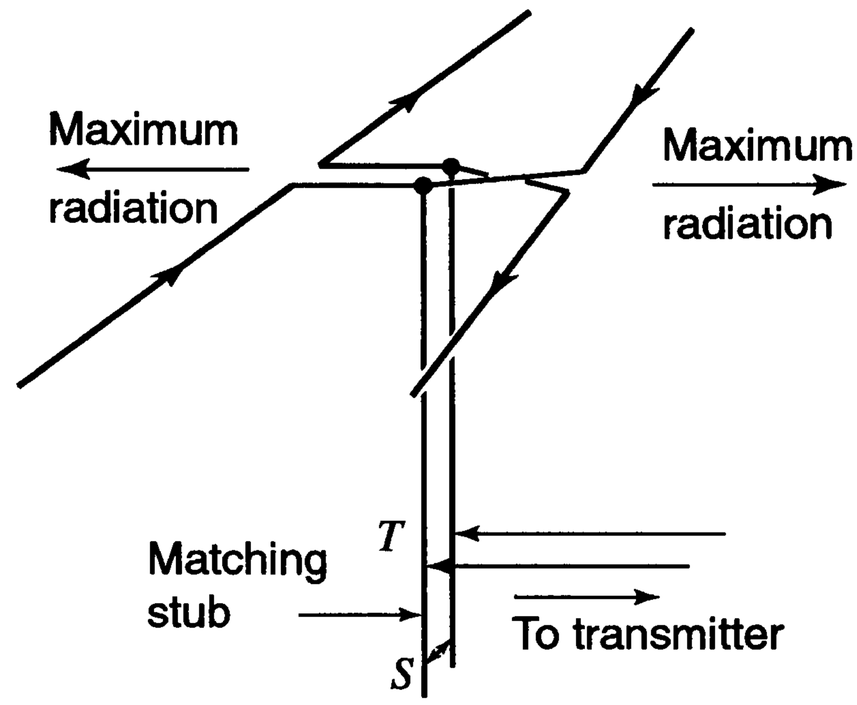


Figure 16–15
Horizontally polarized W8JK array with closely spaced elements carrying equal out-of-phase currents.

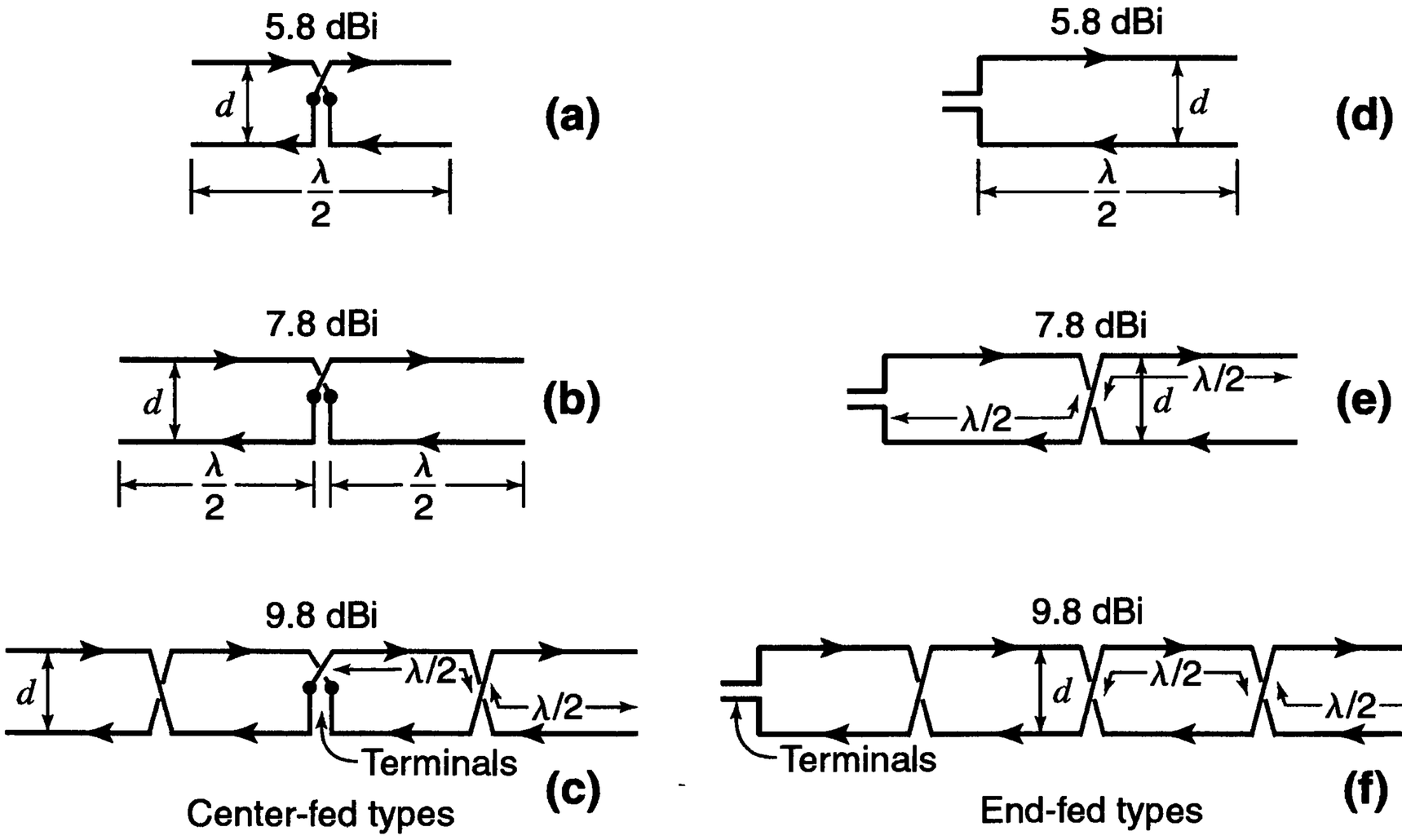


Figure 16–16
Six types of W8JK antennas with gains in dBi for $d = \lambda/8$ and $R_L = 1/2\ \Omega$.

from the center. A 4-section center-fed array is illustrated in Fig. 16–16*c*. The additional sections yield a higher gain by virtue of the sharper beam in the plane of the elements. The antennas of Fig. 16–16*d*, *e* and *f* are end-fed types corresponding to the enter-fed arrays in the left-hand group. The spacing d is usually between $\lambda/8$ and $\lambda/4$.

Referring to the matching stub in the W8JK array in Fig. 16–15, the location of the short circuit S on the vertical line is adjusted for resonance (total length from open end of horizontal dipole to S an odd number of $\lambda/4$ approximately). The distance of the tap point T above S is then adjusted for minimum VSWR on the line to the transmitter (or receiver). The other types in Fig. 16–16 can also be matched in the same way. An adjustable balun for matching a W8JK array to a coaxial line is shown in Fig. 23–20. A W8JK array fed by constant-impedance lines without tuners is shown in Fig. 16–57. See also Table 6–2.

16–6 ARRAY OF *n* DRIVEN ELEMENTS（*n*根受激单元阵）

The field pattern of an array of many elements can often be obtained by an application of the principle of pattern multiplication. As an example, consider the volume array of Fig. 16–17 consisting of 16 $\lambda/2$ dipole elements with equal currents. In the y direction the spacing between elements is d, in the x direction the spacing is a and in the z direction the spacing is h. Let the y-direction and z-direction arrays be broadside types and the x-direction array an end-fire type such that the maximum radiation of the entire volume array is in the positive x-direction. Let $d = h = \lambda/2$ and $a = \lambda/4$. Consider that the currents in all elements are equal in magnitude and that the currents in the front 8 elements are in phase but retarded by 90° with respect to the currents in the rear 8 elements. By the principle of pattern multiplication the pattern of the array is given by the pattern of a single element multiplied by the pattern of a volume array of point sources, where the point sources have the same space distribution as the elements. In general, the field pattern $E(\theta, \phi)$ of a volume array as a function of θ and ϕ is

$$E(\theta, \phi) = E_s(\theta, \phi)E_x(\theta, \phi)E_y(\theta, \phi)E_z(\theta, \phi) \tag{1}$$

where

$E_s(\theta, \phi)$ = pattern of single element
$E_x(\theta, \phi)$ = pattern of linear array of point sources in x direction
$E_y(\theta, \phi)$ = pattern of linear array of point sources in y direction
$E_z(\theta, \phi)$ = pattern of linear array of point sources in z direction

The product of the last 3 terms in (1) is the pattern of a volume array of point sources. If, for instance, we wish to obtain the pattern of the entire array $E(\phi)$ as a function of ϕ in the xy plane ($\theta = 90°$), we introduce the appropriate pattern expression in this plane for each component array in (1). For the example being considered the normalized pattern becomes

$$E(\phi) = \frac{\sin(2\pi \sin\phi)}{4\sin[(\pi/2)\sin\phi]} \cos\left[\frac{\pi}{4}(1 - \cos\phi)\right] \tag{2}$$

Only the $E_y(\phi)$ broadside pattern and the $E_x(\phi)$ end-fire pattern contribute to the array pattern in the xy plane, since in this plane the $E_s(\phi)$ pattern of a single element and the $E_z(\phi)$ broadside pattern are uniform.

The impedance relations for an array of any number n of identical elements are derived by an extension of the analysis used in the special cases in the preceding sections (Carter-1). Thus, for n driven elements we have

$$\left.\begin{aligned}
V_1 &= I_1Z_{11} + I_2Z_{12} + I_3Z_{13} + \cdots + I_nZ_{1n} \\
V_2 &= I_1Z_{21} + I_2Z_{22} + I_3Z_{23} + \cdots + I_nZ_{2n} \\
V_3 &= I_1Z_{31} + I_2Z_{32} + I_3Z_{33} + \cdots + I_nZ_{3n} \\
&\vdots \quad\quad \vdots \quad\quad \vdots \quad\quad \vdots \quad\quad\quad\quad \vdots \\
V_n &= I_1Z_{n1} + I_2Z_{n2} + I_3Z_{n3} + \cdots + I_nZ_{nn}
\end{aligned}\right\} \tag{3}$$

where

V_n = terminal voltage of the nth element
I_n = terminal current of the nth element
Z_{1n} = mutual impedance between element 1 and the nth element
Z_{nn} = self-impedance of the nth element

In matrix form (3) can be expressed as

$$[V_n] = [I_n][Z_{nn}] \tag{4}$$

The driving-point or terminal impedance of one of the elements, say element 1, is then

$$Z_1 = \frac{V_1}{I_1} = Z_{11} + \frac{I_2}{I_1}Z_{12} + \frac{I_3}{I_1}Z_{13} + \cdots + \frac{I_n}{I_1}Z_{1n} \tag{5}$$

If the currents in the elements and the self and mutual impedances are known, the driving-point impedance Z_1 can be evaluated.

The voltage gain of an array of n elements over a single element can be determined in the same manner as outlined for the special cases considered in the previous sections. For instance, the gain in field intensity as a function of ϕ in the xy plane ($\theta = 90°$) for the array of Fig. 16–17 with respect to a single vertical $\lambda/2$ element with the same power input is

$$G_f(\phi)\left[\frac{A}{\text{HW}}\right] = \sqrt{\frac{R_{11} + R_{1L}}{R_{11} + R_{1L} + R_{13} + R_{15} + R_{17} + \frac{3}{2}(R_{12} + R_{16}) + \frac{1}{2}(R_{14} + R_{18})}} \times \frac{\sin(2\pi \sin\phi)}{\sin[(\pi/2)\sin\phi]} \cos\left[\frac{\pi}{4}(1 - \cos\phi)\right] \tag{6}$$

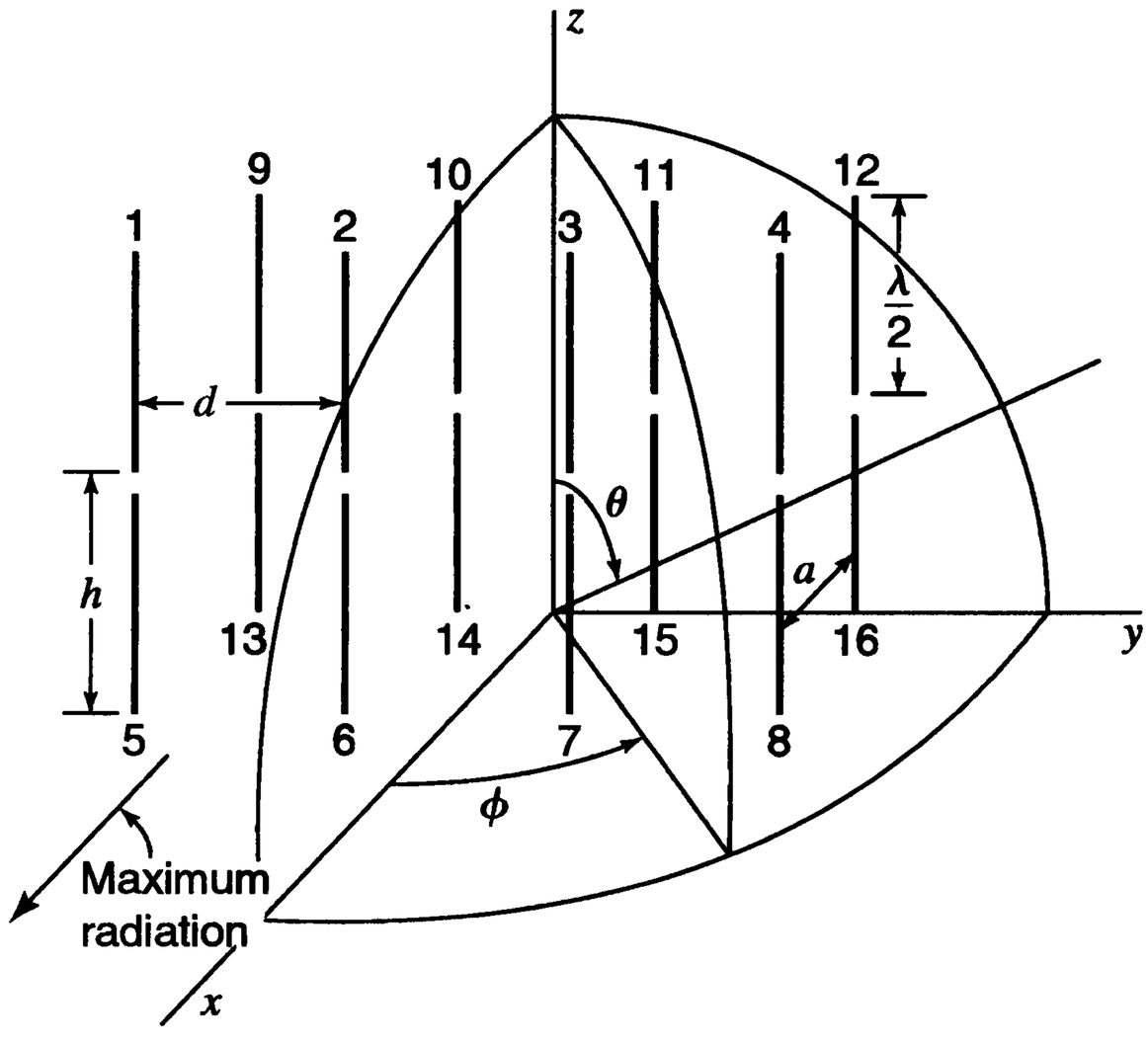


Figure 16–17
Array of 16 $\lambda/2$ dipole elements.

where

R_{11} = self-resistance of one element
R_{1L} = loss resistance of one element
R_{12} = mutual resistance between element 1 and element 2
R_{13} = mutual resistance between element 1 and element 3, etc.

The numbering of the elements is as indicated in Fig. 16–17. It is assumed that $d = h = \lambda/2$ and $a = \lambda/4$ and that the current magnitudes are equal, the currents in the front 8 elements being all in the same phase but retarded 90° with respect to the currents in the rear 8 elements.

16–7 HORIZONTAL ANTENNAS ABOVE A PLANE GROUND
（地平面上的水平天线）

In the previous discussions it has been assumed that the antenna is in free space, i.e., infinitely remote from the ground. Although the fields near elevated microwave antennas may closely approximate this idealized situation, the fields of most antennas are affected by the presence of the ground. The change in the pattern from its free-space shape is of primary importance. The impedance relations may also be different than when the array is in free space, especially if the array is very close to the ground. In this section the effect of the ground on horizontal antennas is discussed. In Sec. 16–8 the effect of the ground is analyzed for vertical antennas. A number of special cases are treated in each section, these being limited to single elements or to simple arrays of several elements. Perfectly conducting ground is assumed.

16–7a Horizontal $\lambda/2$ Antenna above Ground（地面上方的水平 λ/2 天线）

Consider the horizontal $\lambda/2$ antenna shown in Fig. 16–18 at a height h above a plane ground of infinite extent. Owing to the presence of the ground, the field at a distant point P is the resultant of a direct wave and a wave reflected from the ground as in Fig. 16–19. Assuming that the ground is perfectly conducting, the tangential component of the electric field must vanish at its surface. To fulfill this boundary condition, the reflected wave must suffer a phase reversal of 180° at the point of reflection.

To obtain the field at a distant point P, it is convenient to transform the problem by the *method of images*. In this method the ground is replaced by an image of the antenna situated a distance h below the ground plane. By taking the current in the image equal in magnitude but reversed in phase by 180° with respect to the antenna current, the condition of zero

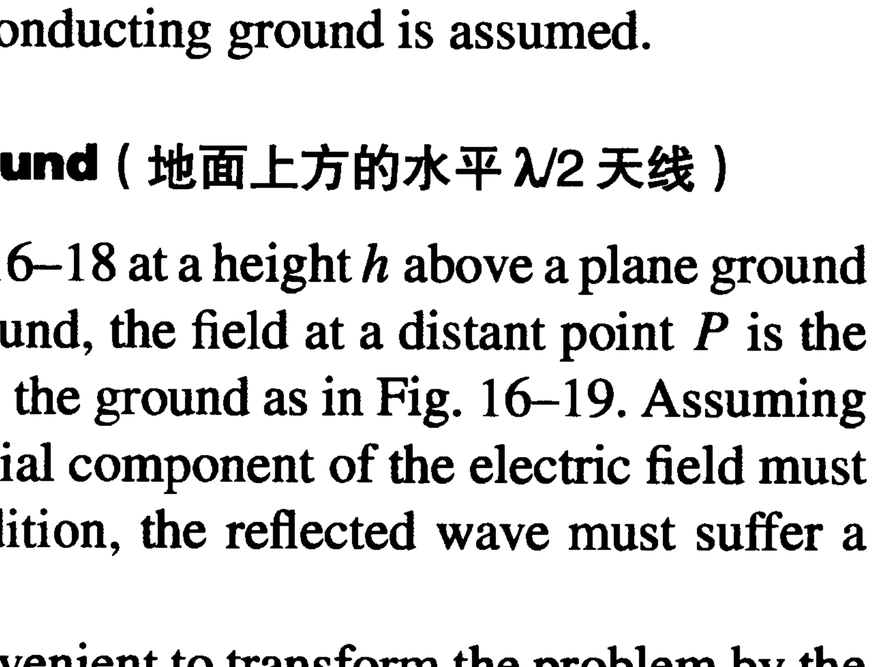


Figure 16–18
Horizontal $\lambda/2$ antenna at height h above ground with image at equal distance below ground.

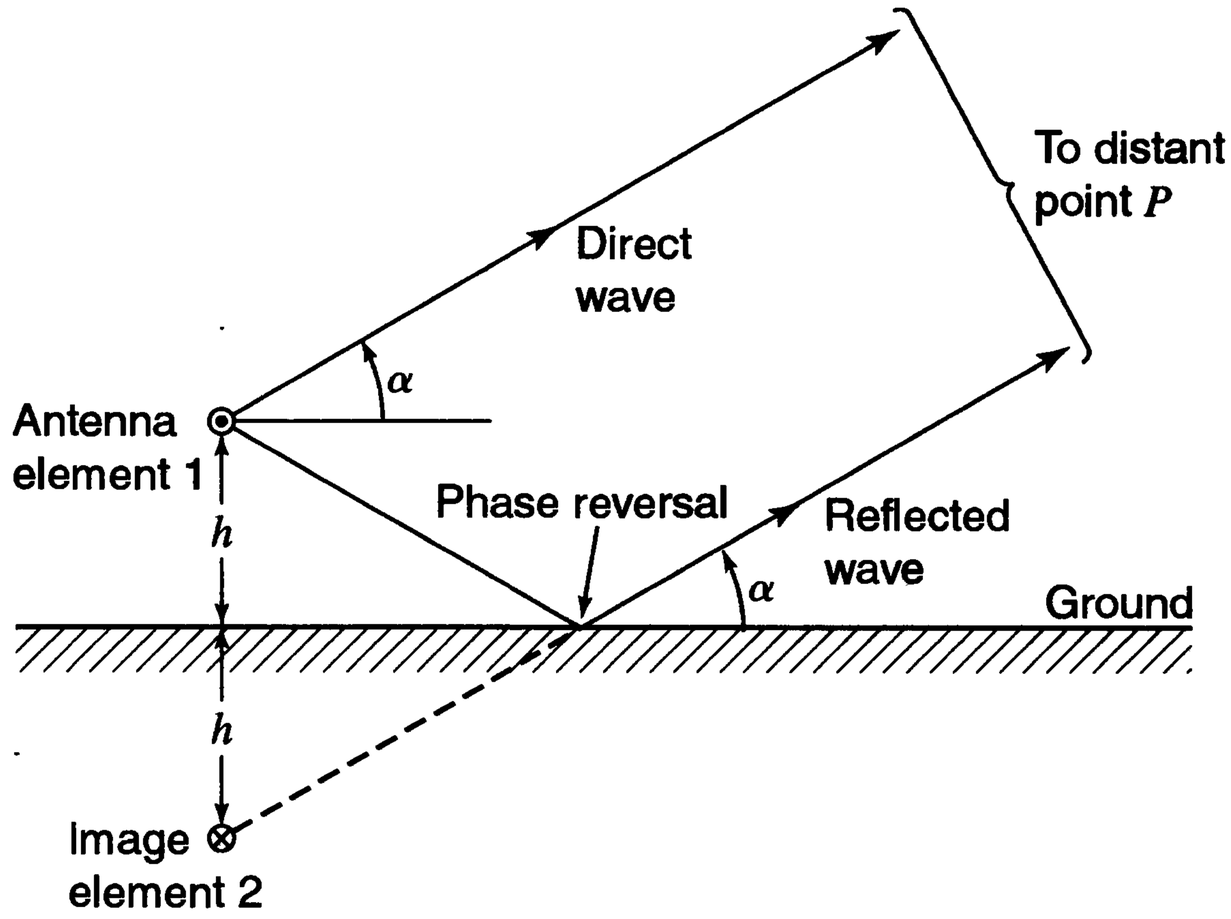


Figure 16–19
Antenna above ground with image showing direct and reflected waves.

tangential electric field is met at all points along a plane everywhere equidistant from the antenna and the image. This is the plane of the ground which the image replaces. In this way, the problem of a horizontal antenna above a perfectly conducting ground[1] of infinite extent can be transformed into the problem already treated in Sec. 16–3 of a so-called end-fire array. One point of difference is that in developing the gain expression it is assumed that if a power P is delivered to the antenna, an equal power is also supplied to the image. Hence, a total power $2P$ is furnished to the "end-fire array" consisting of the antenna and its image.

Owing to the presence of the ground, the driving-point impedance of the antenna is, in general, different from its free-space value. Thus, the applied voltage at the antenna terminals is

$$V_1 = I_1 Z_{11} + I_2 Z_m \tag{1}$$

where

I_1 = the antenna current
I_2 = the image current
Z_{11} = the self-impedance of the antenna
Z_m = the mutual impedance of the antenna and its image at a distance of $2h$

Since $I_2 = -I_1$, the driving- or feed-point impedance of the antenna is

$$Z_1 = \frac{V_1}{I_1} = Z_{11} - Z_m \tag{2}$$

The real part of (2) or driving-point radiation resistance is

$$R_1 = R_{11} - R_m \tag{3}$$

---

[1]It is also possible to apply the method of images to the case of a ground of infinite extent but of finite conductivity σ and of dielectric constant ε by properly adjusting the relative magnitude and phase of the image current with respect to the antenna current.

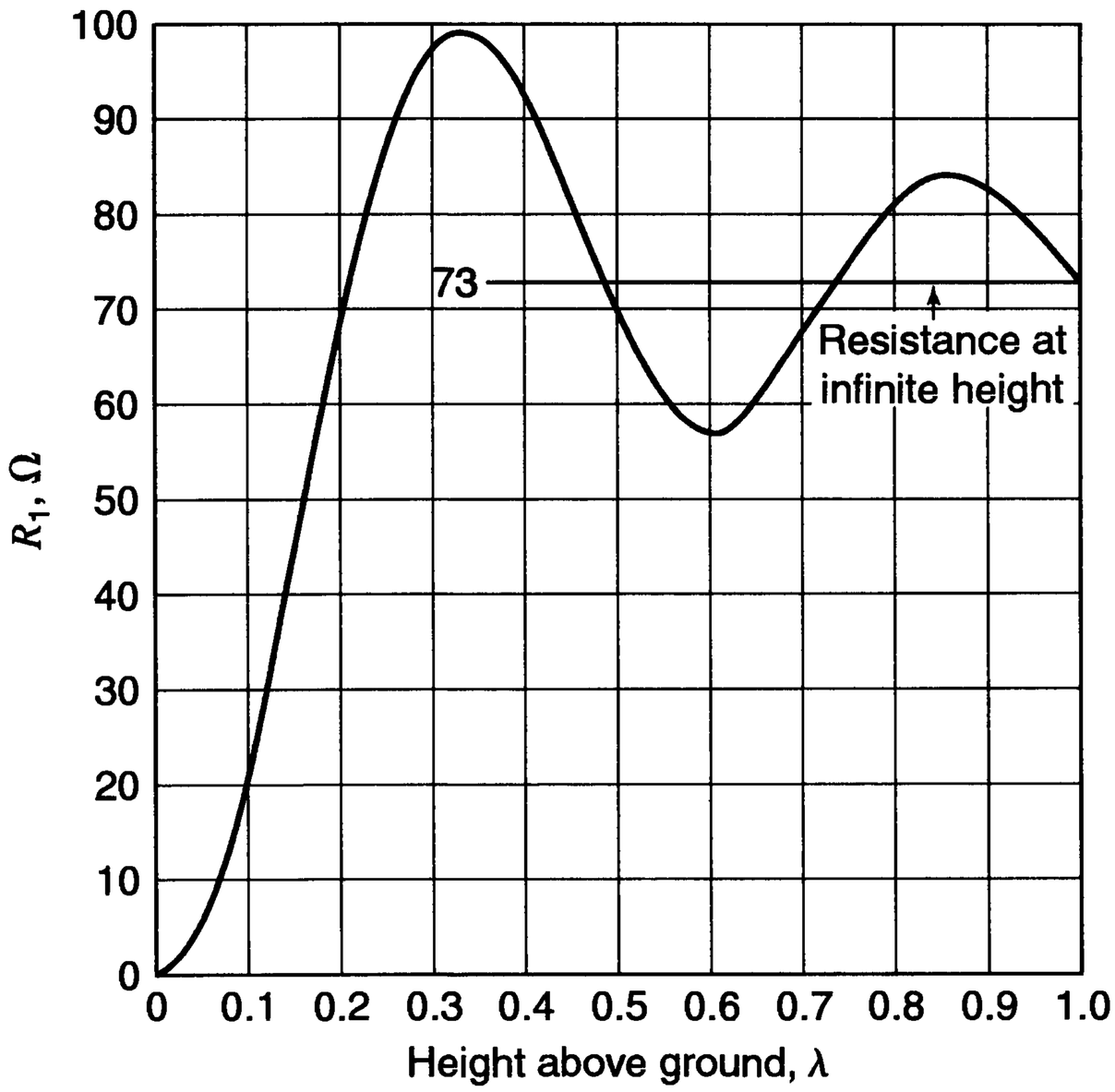


Figure 16–20
Driving- or feed-point resistance R_1 at the center of a horizontal $\lambda/2$ dipole antenna as a function of its height above a perfectly conducting ground.

The variation of this resistance at the center of the $\lambda/2$ antenna is shown in Fig. 16–20 as a function of the antenna height h above the ground. As the height becomes very large, the effect of the image on the resistance decreases, the radiation resistance approaching its free-space value.

Since the antenna and image have currents of equal magnitude but opposite phase, there is zero radiation in the horizontal plane, i.e., in the direction for which the elevation angle α is zero (see Fig. 16–19). If the height h is $\lambda/4$ or less, the maximum radiation is always in the vertical direction ($\alpha = 90°$). For larger heights the maximum radiation is, in general, at some elevation angle between 0 and 90°.

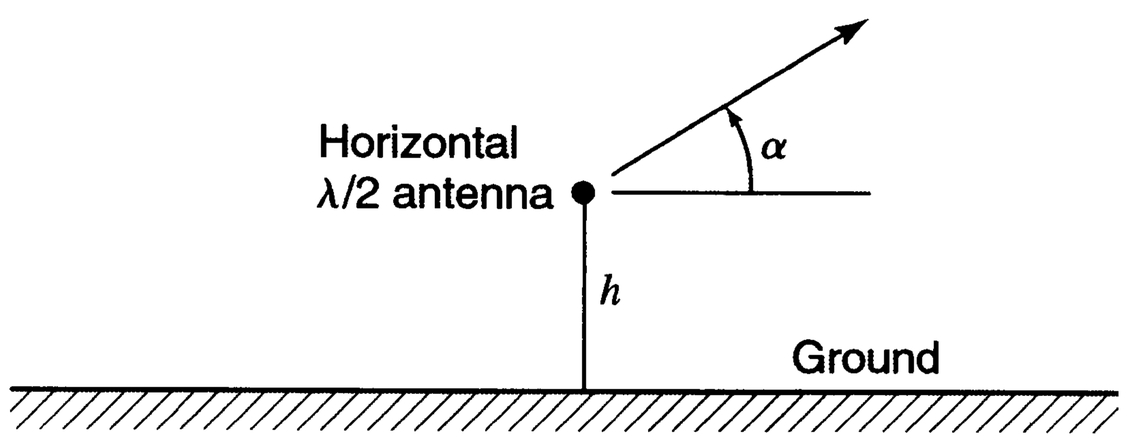
It is convenient to compare the horizontal $\lambda/2$ antenna at a height h above ground with respect to a $\lambda/2$ antenna in free space with the same power input. At a large distance the gain in field intensity of the "Half-Wavelength antenna Above Ground" (HWAG) with respect to the "Half-Wavelength antenna in Free Space" (HWFS) is given by

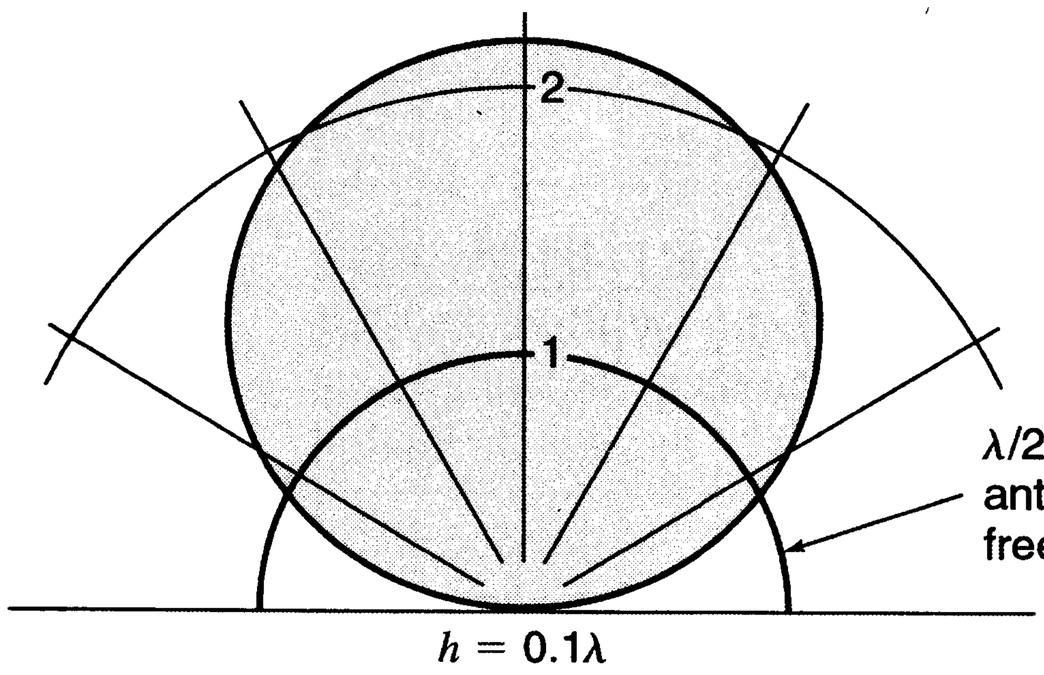
$$G_f(\alpha)\left[\frac{\text{HWAG}}{\text{HWFS}}\right] = \sqrt{\frac{R_{11} + R_{1L}}{R_{11} + R_{1L} - R_m}}\,|2\sin(h_r \sin\alpha)| \tag{4}$$

where

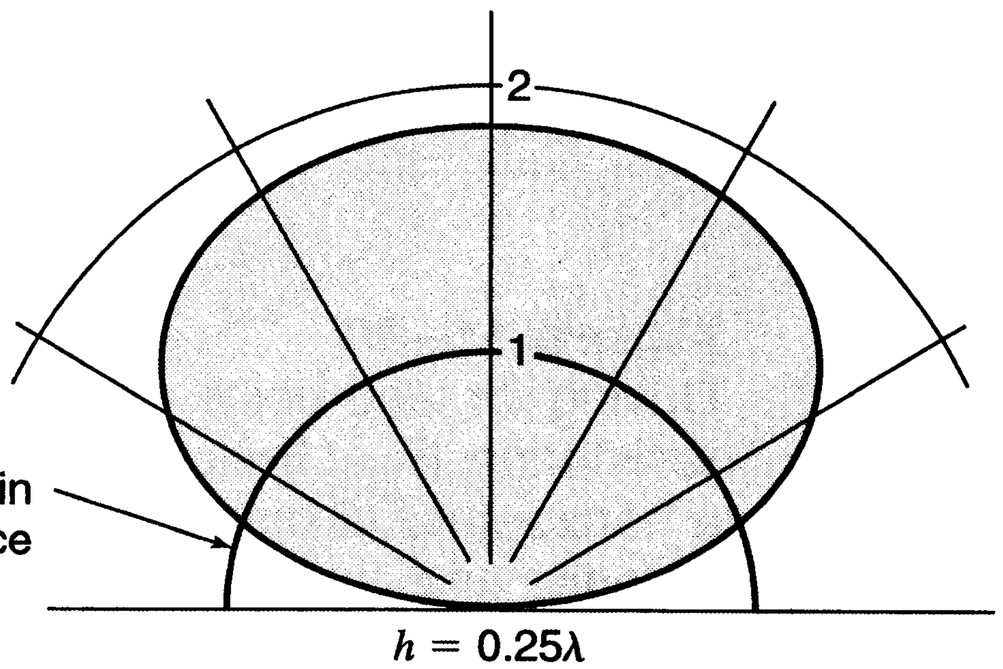
$h_r = (2\pi/\lambda)h$
R_{11} = self-resistance of $\lambda/2$ antenna
R_{1L} = loss resistance of $\lambda/2$ antenna
R_m = mutual resistance of $\lambda/2$ antenna and its image at a distance of $2h$

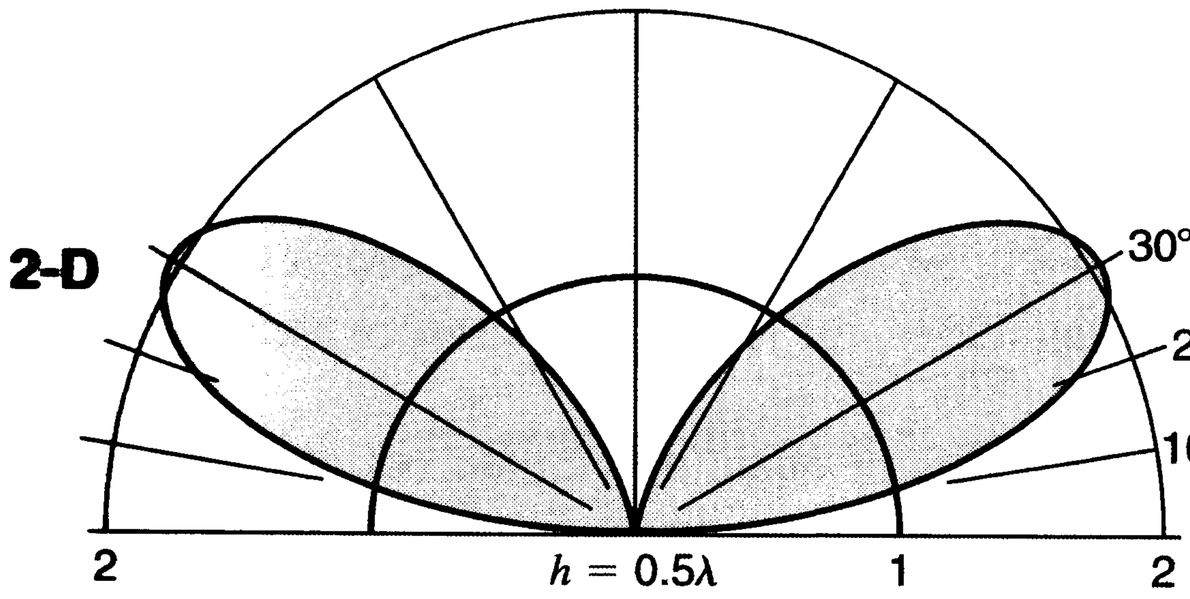
Equation (4) gives the gain in the vertical plane normal to the antenna as a function of α (see Fig. 16–21).

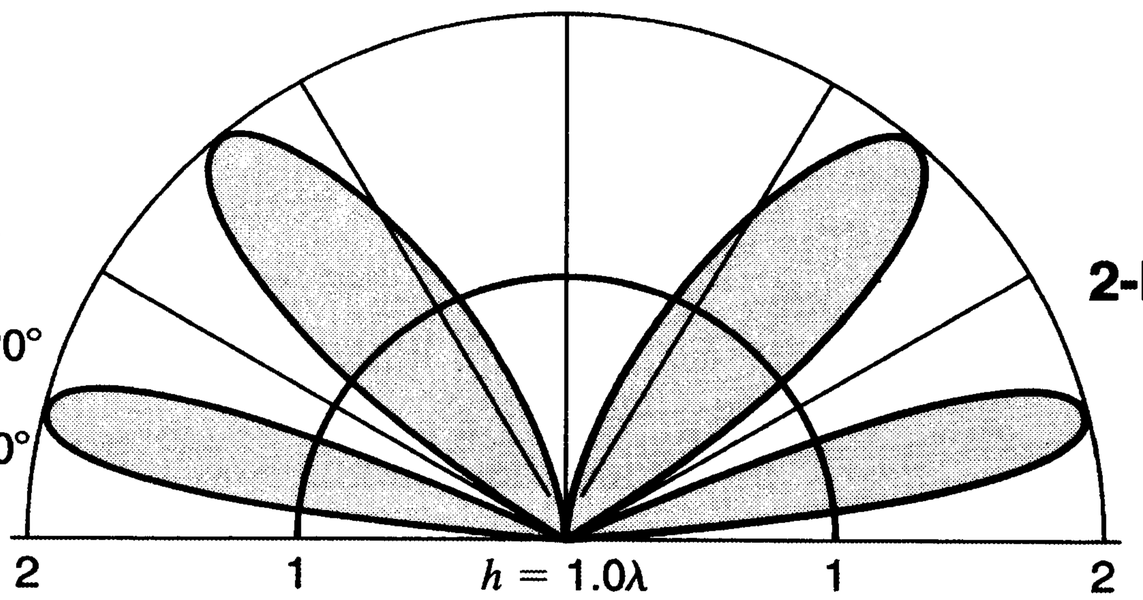


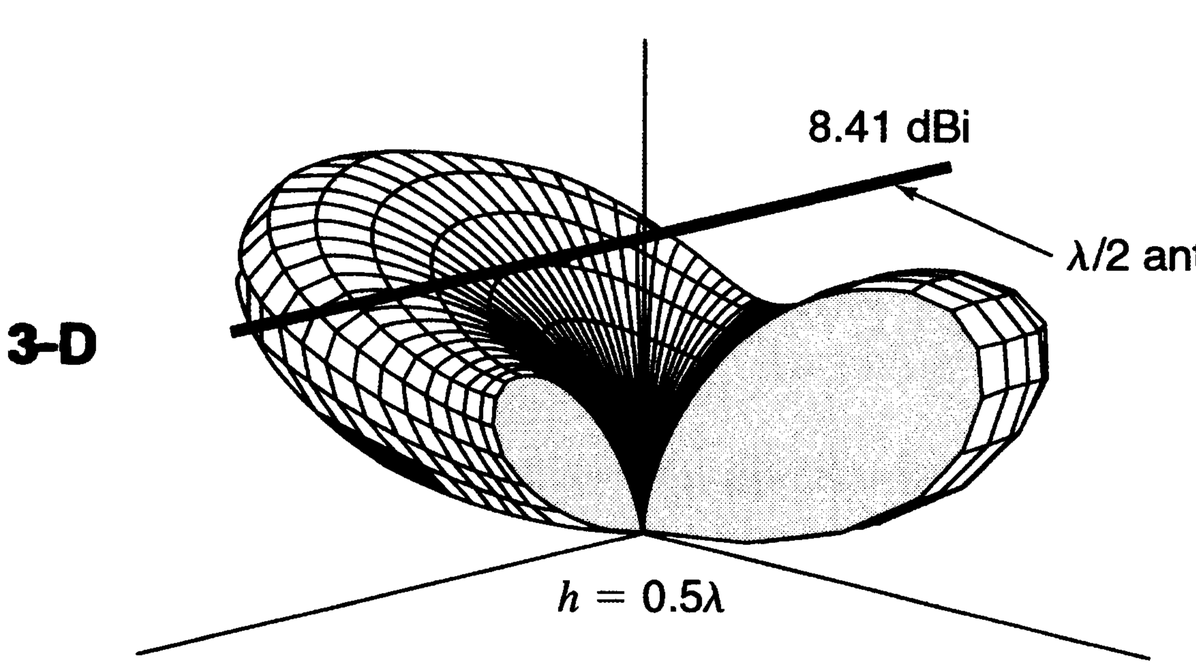


λ/2 antenna in free space









λ/2 antenna

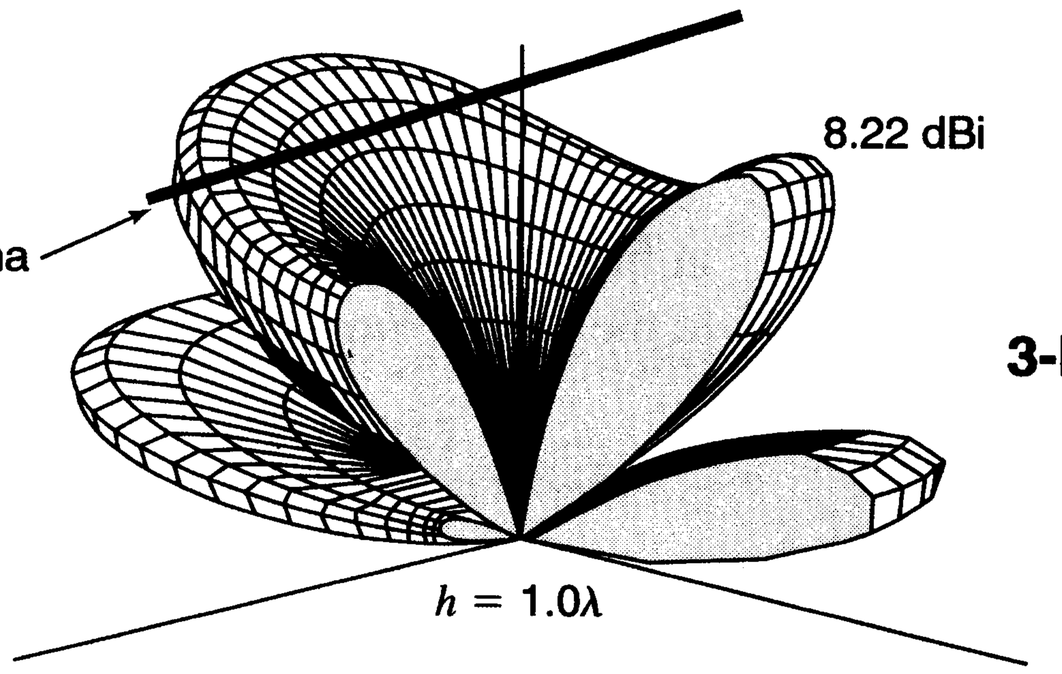


Figure 16–21
Vertical-plane field patterns of a horizontal $\lambda/2$ dipole at various heights h above a perfectly conducting ground as calculated from (4) for $R_{1L} = 0$. Patterns give gain in field intensity over a $\lambda/2$ dipole with the same power input. Note that the presence of the ground increases the field by approximately 6 dB or more in certain directions.

The vertical-plane patterns of a horizontal $\lambda/2$ antenna are shown in Fig. 16–21 for heights $h = 0.1, 0.25, 0.5$ and 1.0λ. The circular pattern is for a $\lambda/2$ antenna in free space (i.e., with the ground removed) with the same power input. It is assumed that loss resistances are zero.

It is also of interest to calculate the field pattern as a function of the azimuth angle ϕ for a constant elevation angle α. The radius vector to the distant point P then sweeps out a cone as suggested in Fig. 16–22. To find this field pattern, let us first consider the field pattern of a horizontal antenna in free space as in Fig. 16–23. The xy plane is horizontal. The field intensity at a large distance in the direction of α and ϕ is then given by the length OA between the origin and the point of intersection of a cone of elevation angle α and the surface

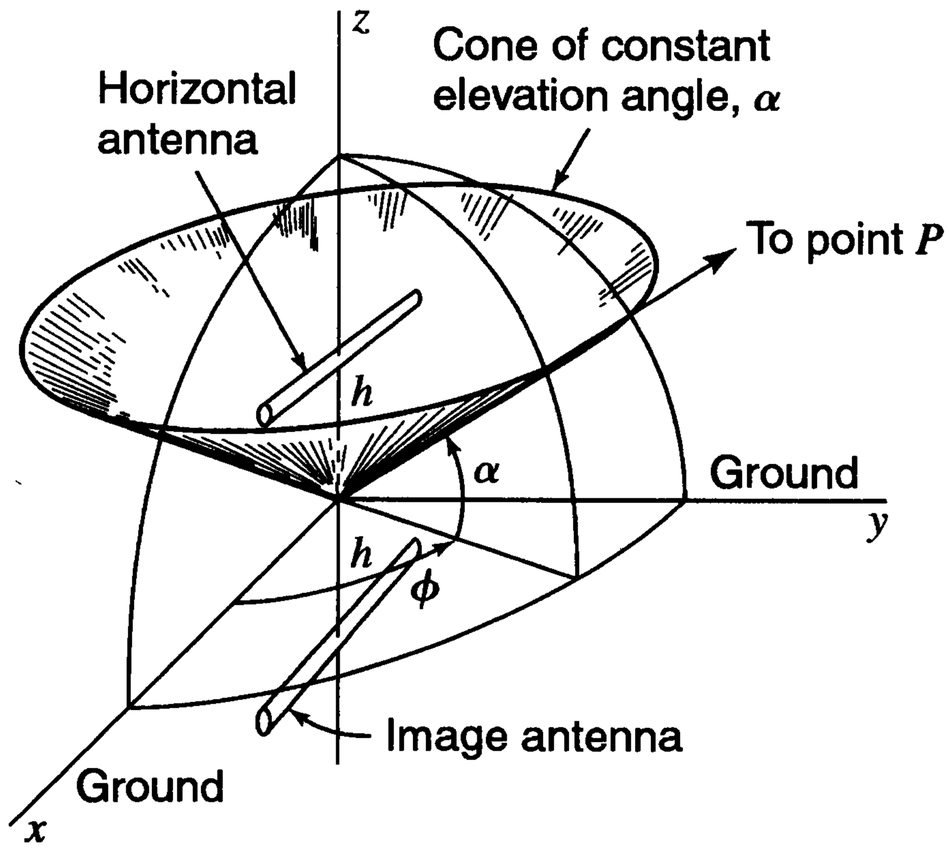


Figure 16–22
Horizontal antenna at height h above ground (xy plane) showing azimuth angle ϕ and elevation angle α for a distant point P.

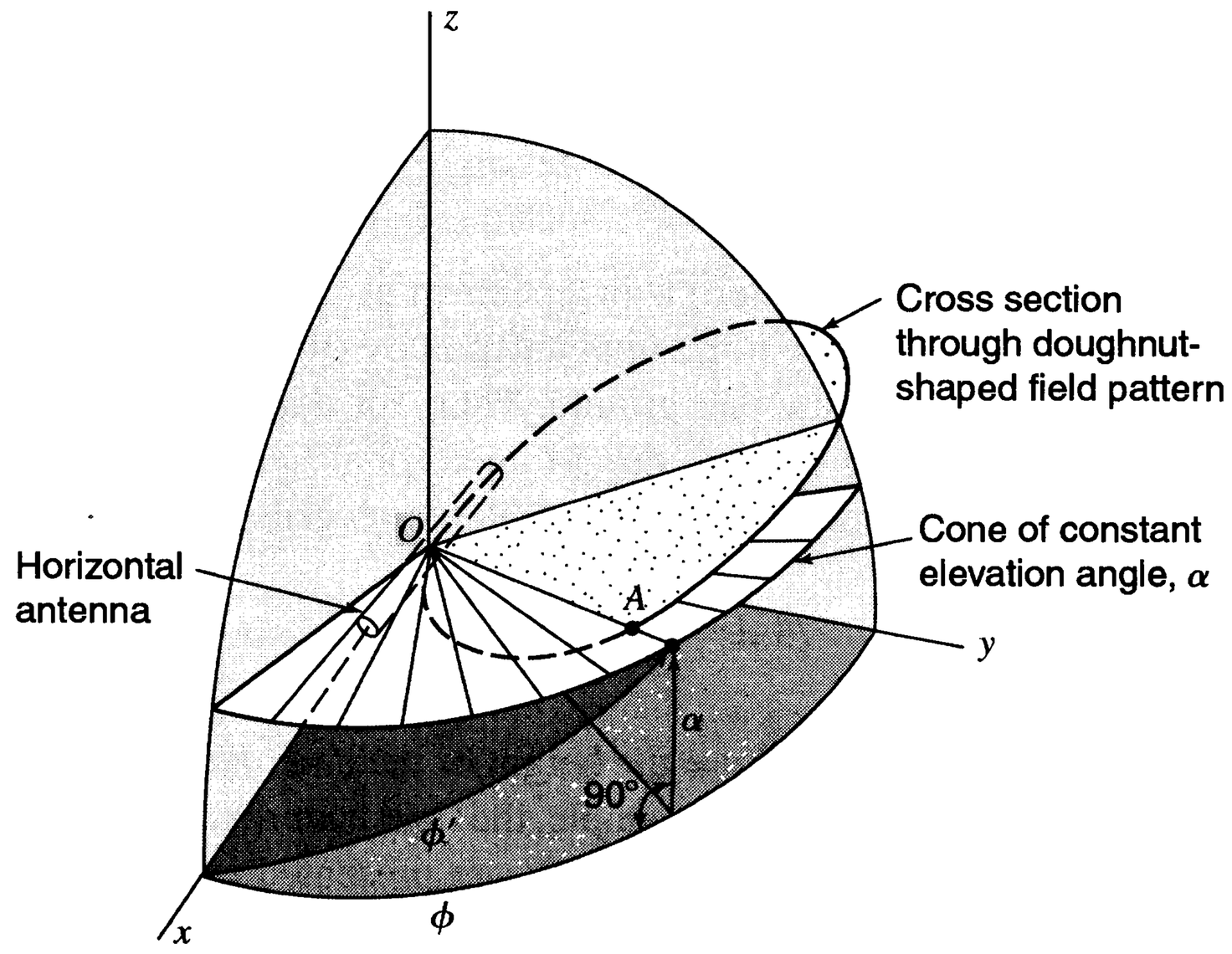


Figure 16–23
Geometrical construction for finding the field intensity at a constant elevation angle α in direction of line OA.

of the 3-dimensional doughnut field pattern of the antenna as suggested in Fig. 16–23. This length is obtained from the field-pattern formula of the antenna in free space by expressing the polar angle ϕ' from the antenna axis in terms of α and ϕ. For the spherical right triangle in Fig. 16–23 we have

$$\cos\phi' = \cos\phi\cos\alpha \tag{5}$$

or

$$\sin\phi' = \sqrt{1 - \cos^2\phi\cos^2\alpha} \tag{6}$$

Substituting these relations in the pattern formula, we get the field intensity in the direction α, ϕ. For example, by substituting (5) and (6) into (6–4–2), noting that ϕ' in (5) and (6) equals θ in (6–4–2), we obtain for the field of a $\lambda/2$ horizontal dipole antenna

$$E(\alpha, \phi) = \frac{\cos[(\pi/2)\cos\phi\cos\alpha]}{\sqrt{1 - \cos^2\phi\cos^2\alpha}} \tag{7}$$

Then the relative field pattern of the horizontal $\lambda/2$ dipole antenna in free space as a function of ϕ at a fixed elevation angle α_0 is given by

$$E(\phi) = \frac{\cos[(\pi/2)\cos\phi\cos\alpha_0]}{\sqrt{1 - \cos^2\phi\cos^2\alpha_0}} \tag{8}$$

To obtain the field pattern of the antenna when situated at a height h above a perfectly conducting ground, we multiply the above free-space relations by the pattern of 2 isotropic point sources of equal amplitude but opposite phase. The sources are separated by a distance $2h$ along the z axis. From (5–2–10) the pattern of the isotropic sources becomes in the present case

$$E_{\text{iso}} = \sin(h_r \sin\alpha) \tag{9}$$

where h_r is the height of the antenna above ground in radians; that is,

$$h_r = \frac{2\pi h}{\lambda}$$

The pattern is independent of the azimuth angle ϕ. Multiplying the free-space field pattern of any horizontal antenna by (9) yields the field pattern for the antenna above a perfectly conducting ground. Thus, for a horizontal $\lambda/2$ dipole antenna above a perfectly conducting ground, the 3-dimensional field pattern as a function of both α and ϕ is obtained by multiplying (7) and (9) which gives

$$E = \frac{\cos[(\pi/2)\cos\phi\cos\alpha]}{\sqrt{1 - \cos^2\phi\cos^2\alpha}} \sin(h_r \sin\alpha) \tag{10}$$

where h_r = the height of the antenna above ground, rad

As an example, the field patterns as a function of the azimuth angle ϕ at elevation angles $\alpha = 10, 20$ and $30°$ are presented in Fig. 16–24 as calculated from (10) for a horizontal $\lambda/2$ antenna at a height of $\lambda/2$ ($h_r = \pi$) above a perfectly conducting ground of infinite

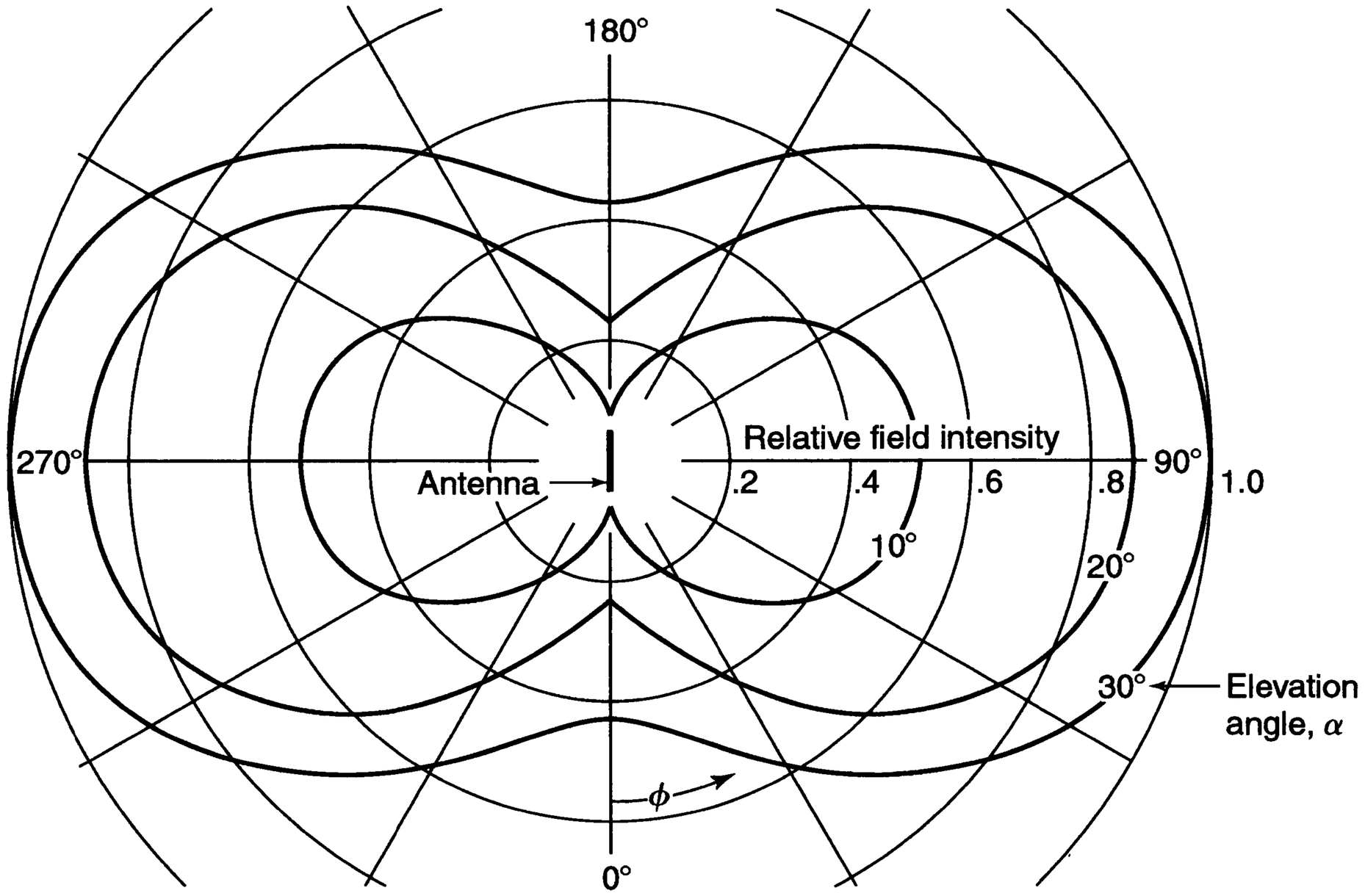


Figure 16–24
Azimuthal field patterns of horizontal $\lambda/2$ antenna $\lambda/2$ above ground at elevation angles $\alpha = 10$, 20 and 30°.

extent. The relative magnitudes of these patterns at $\phi = 90$ or 270° are seen to correspond to the field intensities at $\alpha = 10$, 20 and 30° in the vertical-plane pattern of Fig. 16–21 for $h = 0.5\lambda$. It should be noted that the field is horizontally polarized at $\phi = 90$ or 270° and is vertically polarized at $\phi = 0°$ and $\phi = 180°$. At intermediate azimuth angles the field is linearly polarized at a slant angle.

16–7b W8JK Antenna above Ground（地面上方的 W8JK 天线）

In this section the case of 2 horizontal, closely spaced, out-of-phase $\lambda/2$ elements or W8JK antenna above a perfectly conducting ground is discussed. Referring to Fig. 16–25, let the $\lambda/2$ elements be at a height h above the ground and separated by a distance d. The gain in field intensity of this antenna relative to a $\lambda/2$ antenna in free space with the same power input is given by (Kraus-2)

$$G_f(\alpha)\left[\frac{\text{AAG}}{\text{HWFS}}\right]^{\dagger} = \sqrt{\frac{R_{11}+R_{1L}}{2(R_{11}+R_{1L}+R_{14}-R_{12}-R_{13})}} \times |[1 - 1\angle(d_r\cos\alpha) - 1\angle(2h_r\sin\alpha) + 1\angle(d_r\cos\alpha + 2h_r\sin\alpha)]| \tag{11}$$

†The symbols in the brackets are by way of explanation that the gain in field intensity is for the "Array Above Ground with respect to a Half-Wavelength antenna in Free Space."

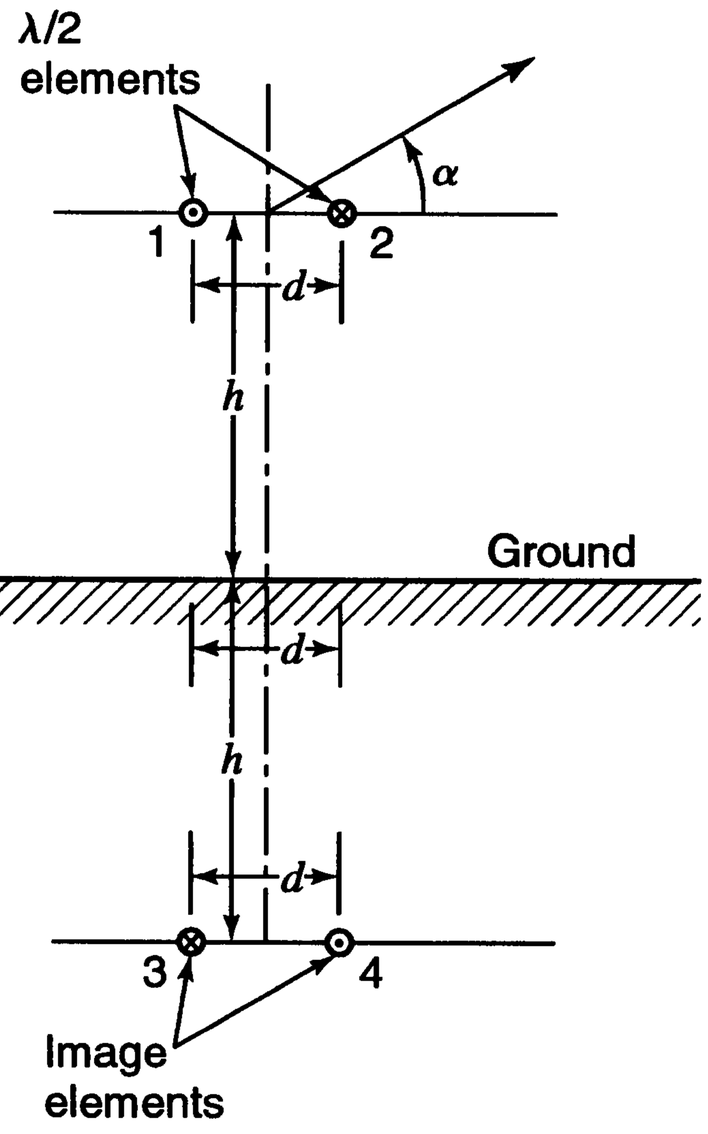


Figure 16–25
W8JK antenna above ground.

where

d_r = spacing of elements, rad $= 2\pi d/\lambda$
h_r = height of element above ground, rad $= 2\pi h/\lambda$
R_{11} = self-resistance of a single element
R_{1L} = loss resistance of a single element
R_{12} = mutual resistance of elements 1 and 2
R_{13} = mutual resistance of elements 1 and 3, etc.

where the elements are numbered as in Fig. 16–25. The gain in (11) is expressed as a function of α in the vertical plane normal to the elements.

Polar plots calculated by (11) for the gain in field intensity of a W8JK antenna consisting of two $\lambda/2$ elements spaced $\lambda/8$ apart are presented by the solid curves in Fig. 16–26 for antenna heights of $\lambda/2$ and $3\lambda/4$ above ground. Patterns of a single $\lambda/2$ antenna at the same heights above ground and with the same power input are shown for comparison (dashed curves). The gain in field intensity is expressed relative to a $\lambda/2$ antenna in free space with the same power input.

In Fig. 16–27 the gain is given as a function of height above ground for several elevation angles. Curves are shown for both a 2-element W8JK and a single horizontal $\lambda/2$ antenna. It is assumed that loss resistances are zero. If, for example, the effective elevation angle at a particular time on a certain short-wave circuit (transmission via ionospheric reflections) is 30°, we note from Fig. 16–27 that the optimum height for a 2-element W8JK beam is 0.5λ. For a single $\lambda/2$ antenna the optimum height is about 0.57λ.

It is interesting to consider the effect of tilting the plane of the W8JK elements by an angle γ as in Fig. 16–28. Results calculated by an extension of the above analysis are illustrated in Figs. 16–29 and 16–30 for 2-element arrays at average heights of $\lambda/2$ and

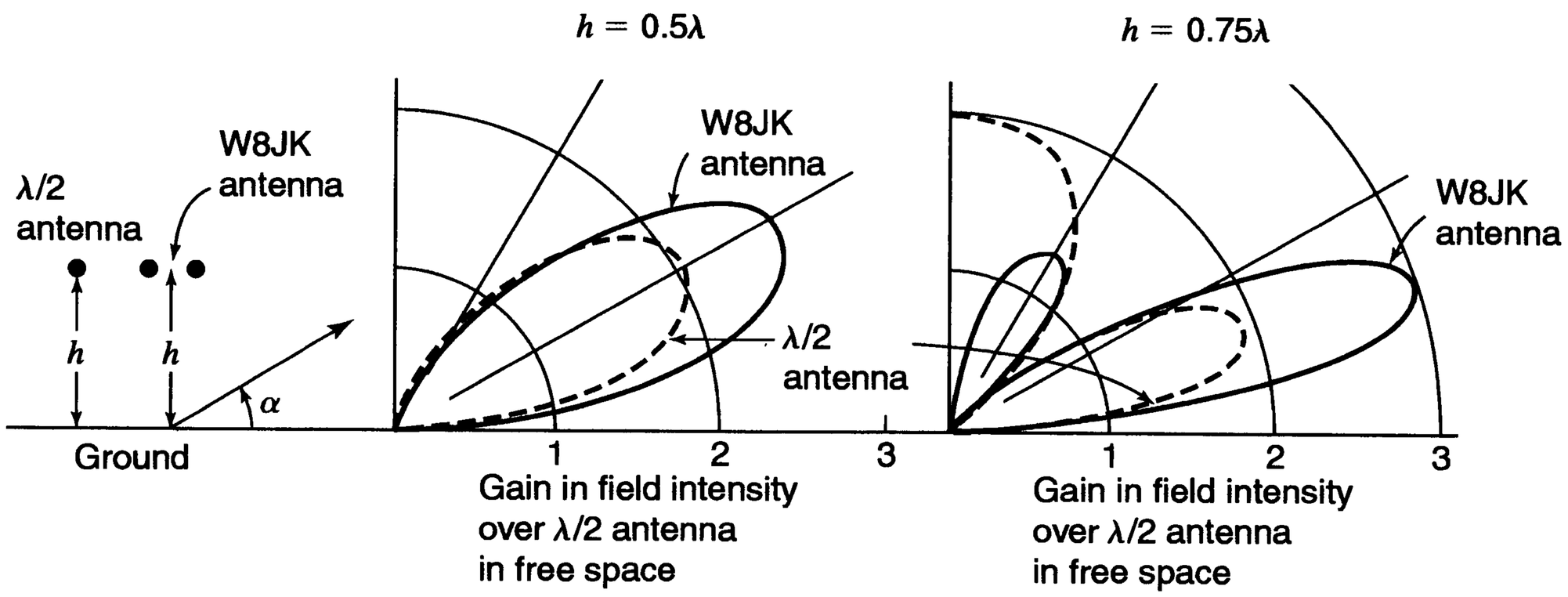


Figure 16–26
Vertical-plane patterns (solid curves) of 2-element W8JK antenna with $\lambda/8$ spacing at heights of $\lambda/2$ and $3\lambda/4$ above ground. The patterns are plotted relative to a $\lambda/2$ antenna in free space with the same power input. The vertical plane patterns of a single $\lambda/2$ antenna at the same heights above ground and with the same power input are shown for comparison by the dashed curves. The left-hand quadrants of the vertical planes are mirror images. Since the W8JK has zero radiation vertically and maximum horizontally, it is very effective for low-angle long-distance communication.

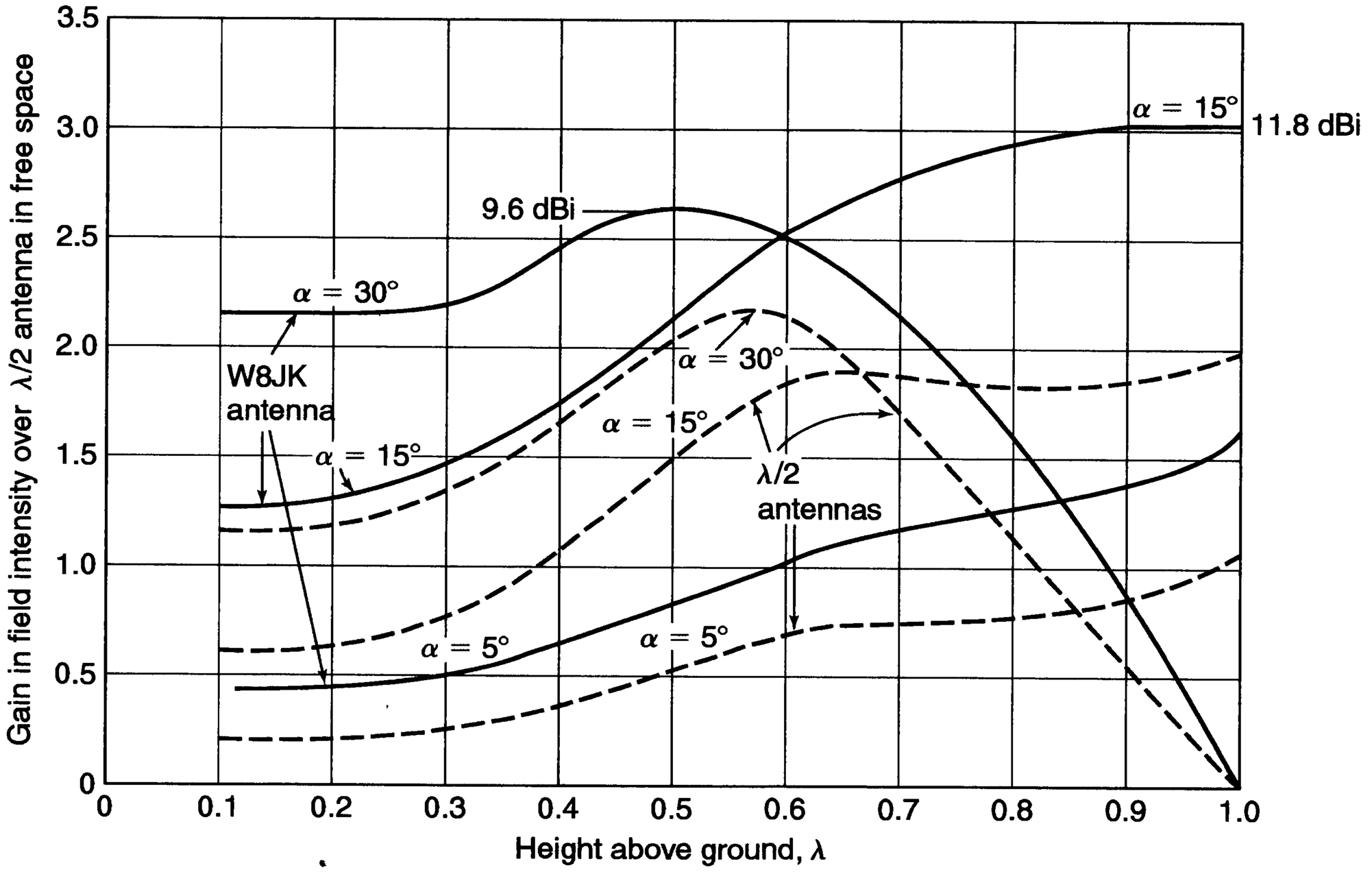


Figure 16–27
Gain in field intensity of 2-element W8JK antenna with $\lambda/8$ spacing (solid curves) and of a single $\lambda/2$ antenna (dashed curves) as a function of the height above a perfectly conducting ground. Gains are relative to a single $\lambda/2$ antenna in free space with the same power input. Curves are given for elevation angles $\alpha = 5$, 15 and 30°. We note that the gain of a 1-section (2-element) W8JK antenna at $\alpha = 15°$ and $h = \lambda$ exceeds 3 (=11.8 dBi). For a 4-section W8JK antenna at this angle and height the gain is about 16 dBi.

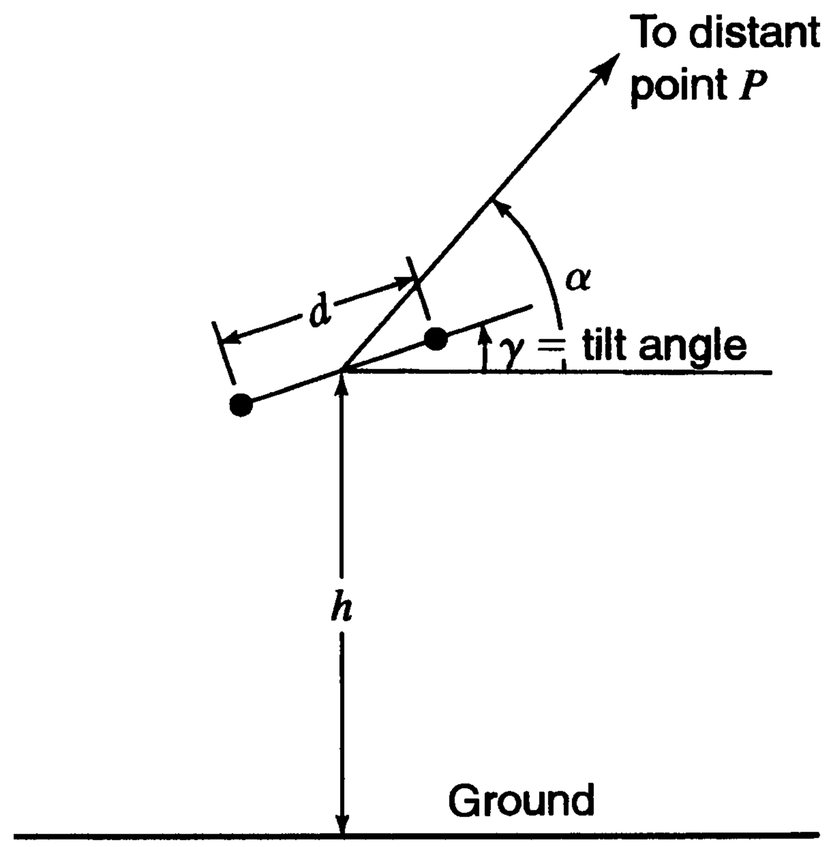


Figure 16–28
Tilted W8JK antenna.

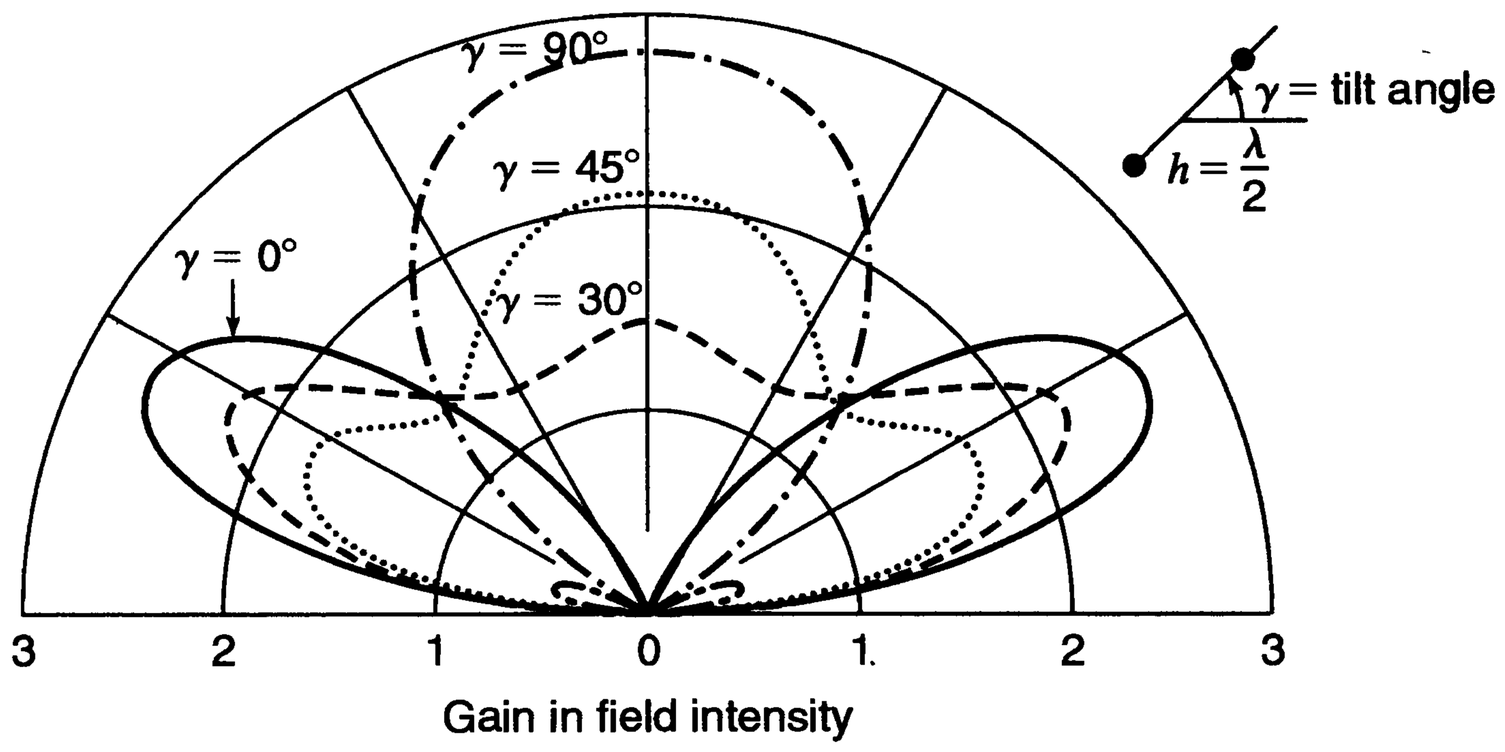


Figure 16–29
Vertical-plane patterns for horizontal 2-element W8JK antenna with $\lambda/8$ spacing at an average height of $\lambda/2$ above ground for tilt angles $\gamma = 0$, 30, 45 and 90°. Patterns give gain in field intensity over a single $\lambda/2$ antenna in free space with the same power input.

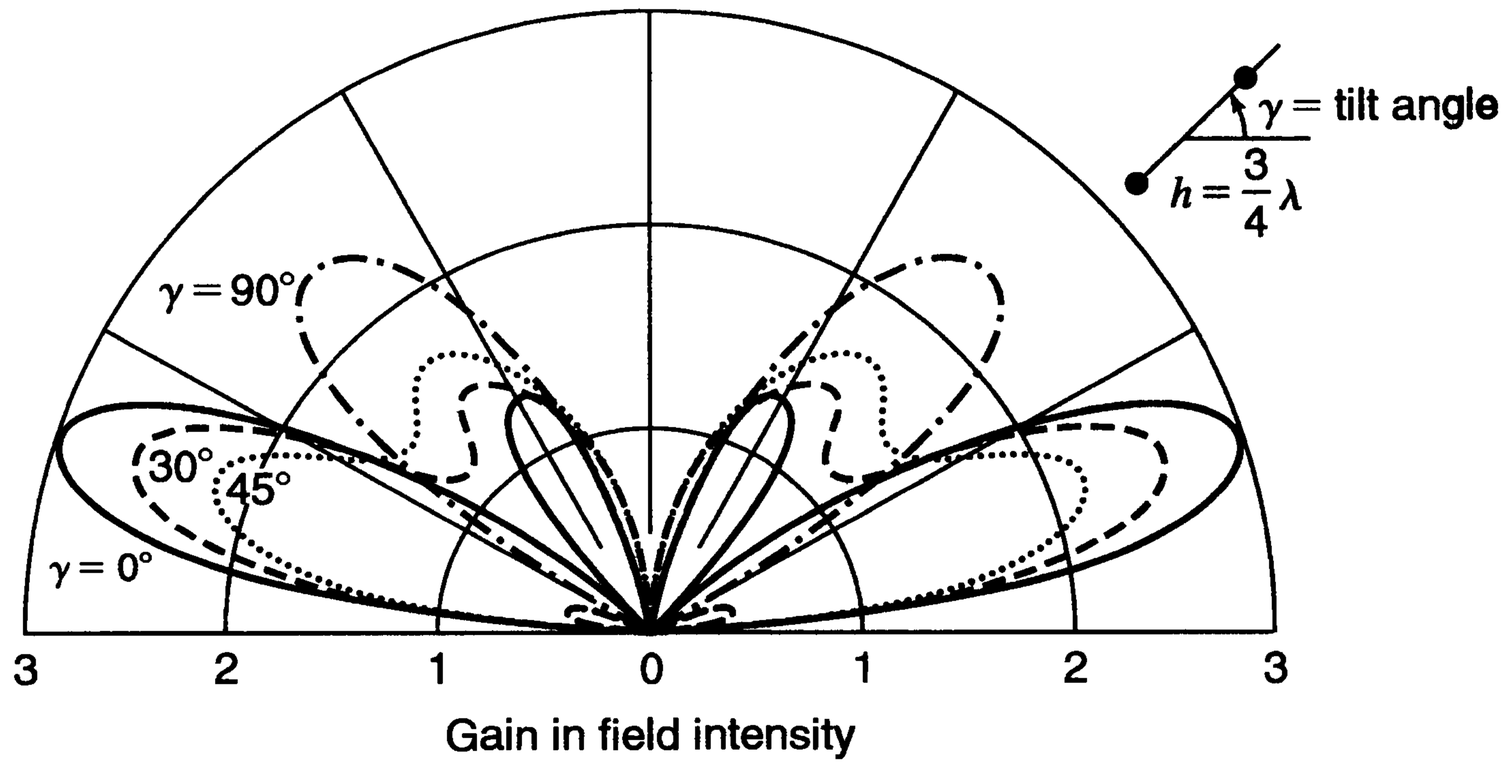


Figure 16–30
Same as for Fig. 16–29 but with array elements at average height of $3\lambda/4$ above ground.

$3\lambda/4$ above a perfectly conducting ground (Kraus-3). Patterns are shown for tilt angles $\gamma = 0, 30, 45$ and $90°$. *In all cases the effect of the tilt is to increase the field intensity at large elevation angles and to decrease it at small angles.*

16–7c Stacked Horizontal $\lambda/2$ Antennas above Ground
(地面上方的层叠水平λ/2天线)

Consider the case of two horizontal $\lambda/2$ elements stacked in a vertical plane above a perfectly conducting ground of infinite extent. The elements have equal in-phase currents. The arrangement of the elements and their images is shown in Fig. 16–31. The height of the upper element above ground is h. Let the spacing between elements be $\lambda/2$ so that the height of the lower element above ground is $h - \lambda/2$. The gain in field intensity of this array over a single $\lambda/2$ dipole antenna in free space with the same power input is

$$G_f(\alpha)\left[\frac{\text{AAG}}{\text{HWFS}}\right] = \sqrt{\frac{R_{00} + R_{0L}}{2(R_{11} + R_{1L} + R_{12} - R_{13}) - R_{23} - R_{14}}} \times 2|\{\sin(h_r \sin\alpha) + \sin[(h_r - \pi)\sin\alpha]\}| \tag{12}$$

where R_{12} is the mutual resistance between elements 1 and 2, R_{13} the mutual resistance between elements 1 and 3, etc. The elements are numbered as in Fig. 16–31. This expression gives the gain as a function of h and of the elevation angle α in the vertical plane normal to the plane of the elements. As an example, the gain in field intensity for two stacked in-phase horizontal $\lambda/2$ elements over a free-space $\lambda/2$ antenna with the same power input is presented in Fig. 16–32 as a function of the height h above ground, for an elevation angle $\alpha = 20°$. The gains at $\alpha = 20°$ for a 2-element W8JK antenna and a single horizontal $\lambda/2$ antenna are also shown as a function of height for comparison. At all heights less than 0.9λ the W8JK antenna produces the highest gain. *This gain improvement at low heights and angles is an important advantage and has been emphasized by Regier (1).*

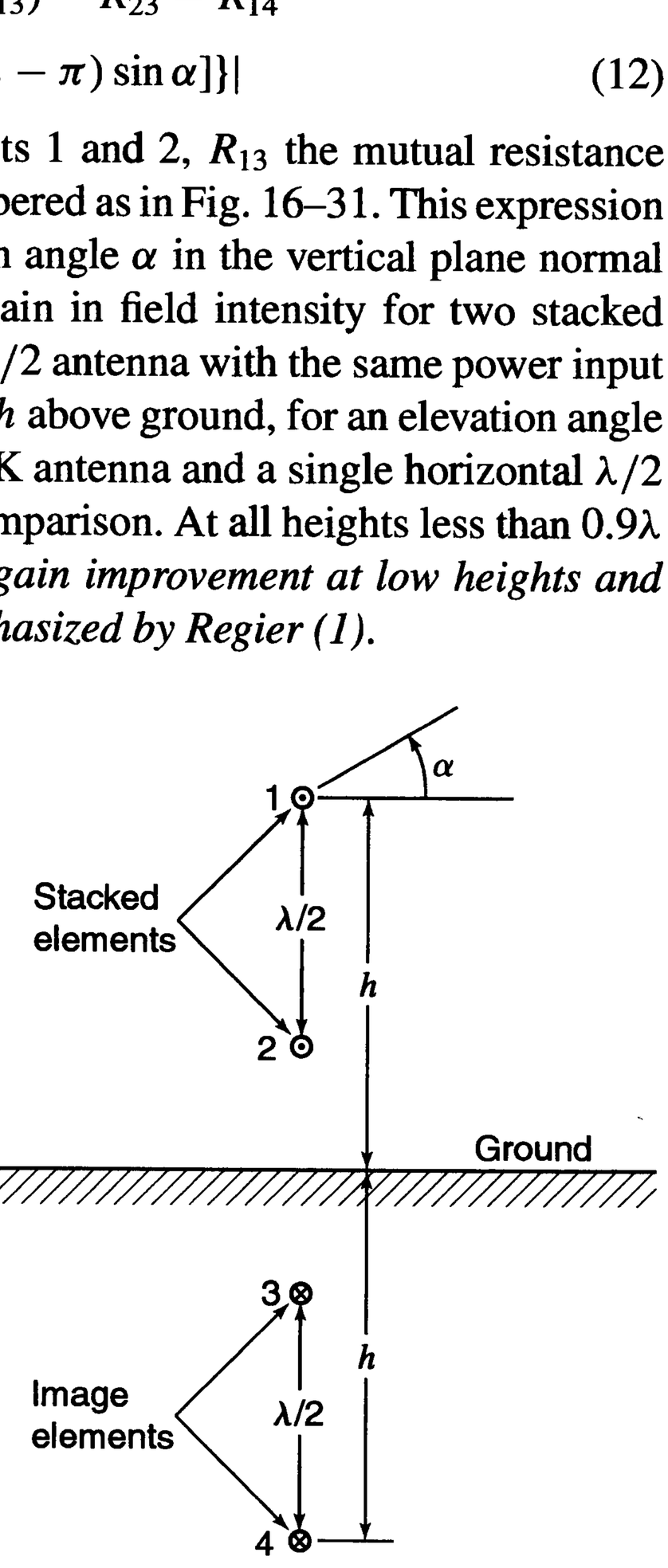


Figure 16–31
Array of stacked horizontal $\lambda/2$ elements.

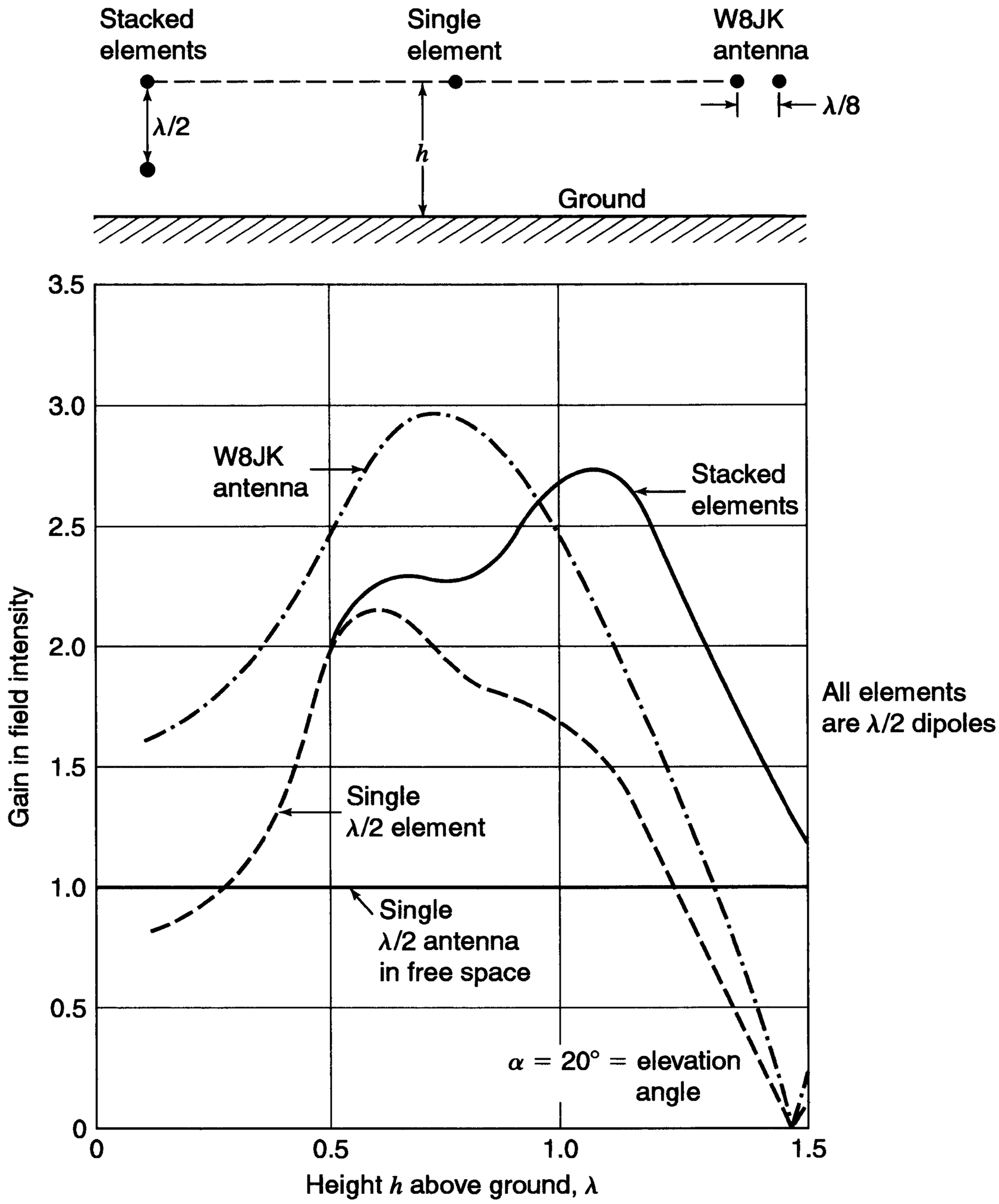


Figure 16–32
Gain in field intensity of array of 2 stacked horizontal $\lambda/2$ elements as a function of the height above ground of the upper element for an elevation angle of 20°. The elements are stacked $\lambda/2$ apart. The gain is relative to a single $\lambda/2$ dipole antenna in free space with the same power input. Gains of a 2-element W8JK antenna and single $\lambda/2$ antenna as a function of the height above ground are also shown for comparison at the same elevation angle. At all heights less than 0.9λ the W8JK antenna produces the highest gain.

16–8 VERTICAL ANTENNAS ABOVE A PLANE GROUND（地平面上的铅垂天线）

Consider a vertical stub antenna of length l above a plane horizontal ground of infinite extent and perfect conductivity as in Fig. 16–33. By the method of images the ground may be replaced by an image antenna of length l with sinusoidal current distribution and instantaneous current direction as indicated. The problem of the stub antenna above ground

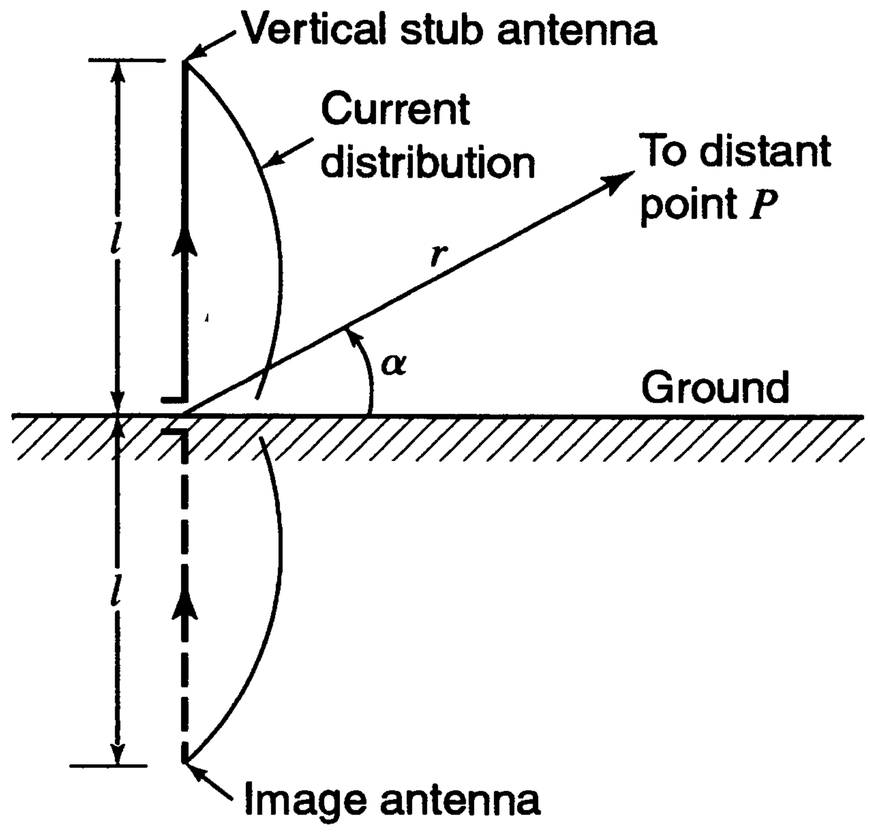


Figure 16–33
Vertical stub antenna above a ground plane.

then reduces to the problem already treated in Chap. 6 of a linear center-fed antenna with symmetrical current distribution. The electric field intensity as a function of the elevation angle α and distance r may be derived from (6–4–3), obtaining

$$E(\alpha, r) = \frac{60}{r}\sqrt{\frac{P}{R_{11} + R_{1L}}}\frac{\cos(l_r \sin\alpha) - \cos l_r}{\cos\alpha} \qquad (\text{V m}^{-1}) \tag{1}$$

where

l_r $= \beta l = (2\pi/\lambda)l$, dimensionless
R_{11} = self-resistance of a vertical stub antenna of length l referred to the point of current maximum, Ω
R_{1L} = effective loss resistance of antenna referred to same point, Ω
P = power input, W
r = distance, m

Values of the self-resistance referred to the current loop of a vertical stub antenna above a perfectly conducting ground have been given by Brown (1) and by Labus (1). These values are presented as a function of antenna length in Fig. 16–34. Using these values of self-resistance, or radiation resistance, the field intensity of a vertical stub antenna of any length l and power input P can be calculated by (1) at any elevation angle α and distance r. Thus, the field intensity by (1) along the ground ($\alpha = 0$) for a $\lambda/4$ vertical antenna ($l_r = \pi/2$) with a power input $P = 1$ W at a distance of 1609 m is 6.5 mV m^{-1}. The value of R_{11} for a $\lambda/4$ stub antenna is 36.5 Ω, and R_{1L} is assumed to be zero.

Vertical stub antennas, singly or in directional arrays, are very widely used for AM broadcasting. In this application the field intensity along the ground ($\alpha = 0$) is of particular interest. It is also customary to compare field intensities at some standard distance, say 1.61 km, and for some standard input such as 1 kW. For this case (1) reduces to

$$E = \frac{1.18(1 - \cos l_r)}{\sqrt{R_{11} + R_{1L}}} \qquad (\text{V m}^{-1}) \tag{2}$$

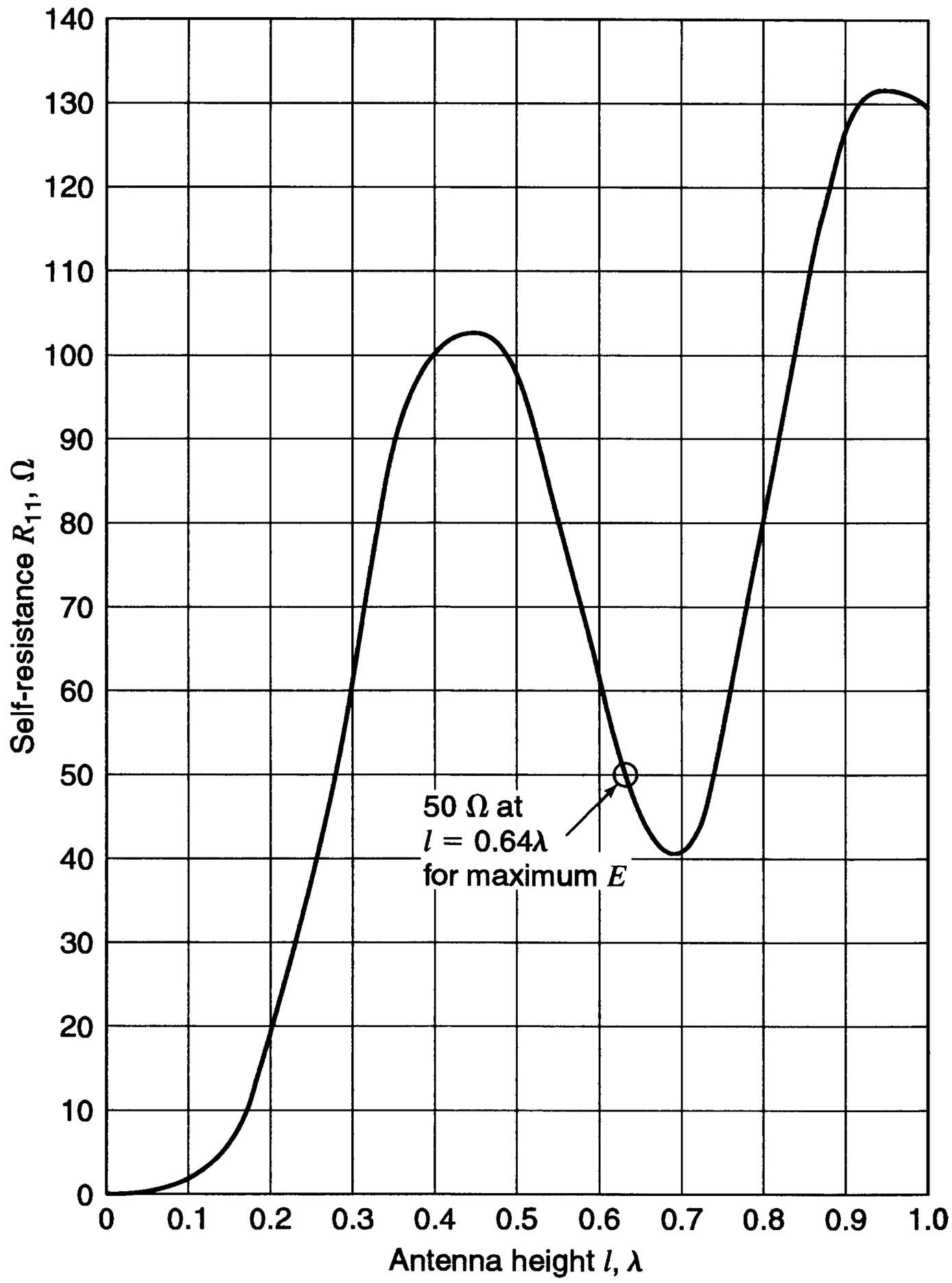


Figure 16–34
Radiation resistance at the current maximum of a thin vertical antenna as a function of the height *l* of the antenna. (*After G. H. Brown-1 and also J. Labus-1.*)

where E is the field intensity along the ground at a distance of 1.61 km for a power input of 1 kW. The variation of E as given by (1) is presented in Fig. 16–35 as a function of antenna length. The vertical-plane patterns calculated by (1) as a function of the elevation angle α for vertical antennas of various heights are presented in Fig. 16–36 (Brown-2; Smith-1). A length of about 0.64λ yields the greatest field intensity along the ground, but as pointed out by Brown (1) the large high-angle radiation (at $\alpha = 60°$) for this length reduces the nonfading range at broadcast frequencies (500 to 1500 kHz) as compared, for example, with an antenna about $\lambda/2$ long. The nonfading range is largest for an antenna height of 0.528λ. It is assumed that the loss resistance $R_{1L} = 0$, that is, the entire input to the antenna is radiated. The small amount of high-angle radiation, which is an important factor in reducing fading, is apparent for the $l = 0.528\lambda$ antenna as compared to other lengths (see Fig. 16–36*b* and *c*).

The analysis of arrays of several vertical stub antennas can be reduced in a similar fashion to arrays of symmetrical center-fed antennas. Many of these have been treated in previous sections. In this case it is often convenient to compare the pattern and refer the gain

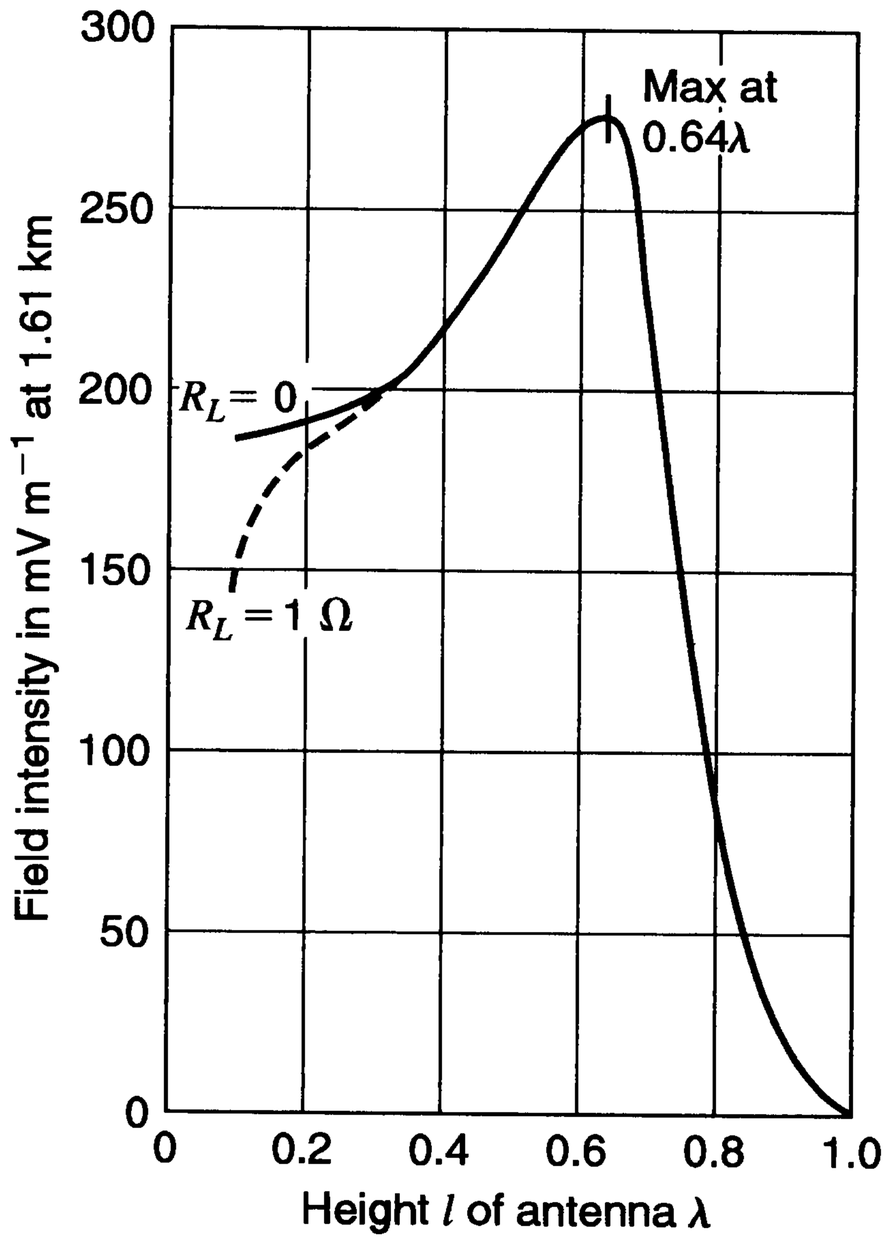


Figure 16–35
Field intensity at the ground (zero elevation angle) at a distance of 1.6 km from a vertical antenna with 1 kW input as a function of its height *l*. Perfectly conducting ground is assumed. The solid curve is for an assumed loss resistance $R_L = 0$ and the dashed curve for $R_L = 1\ \Omega$.

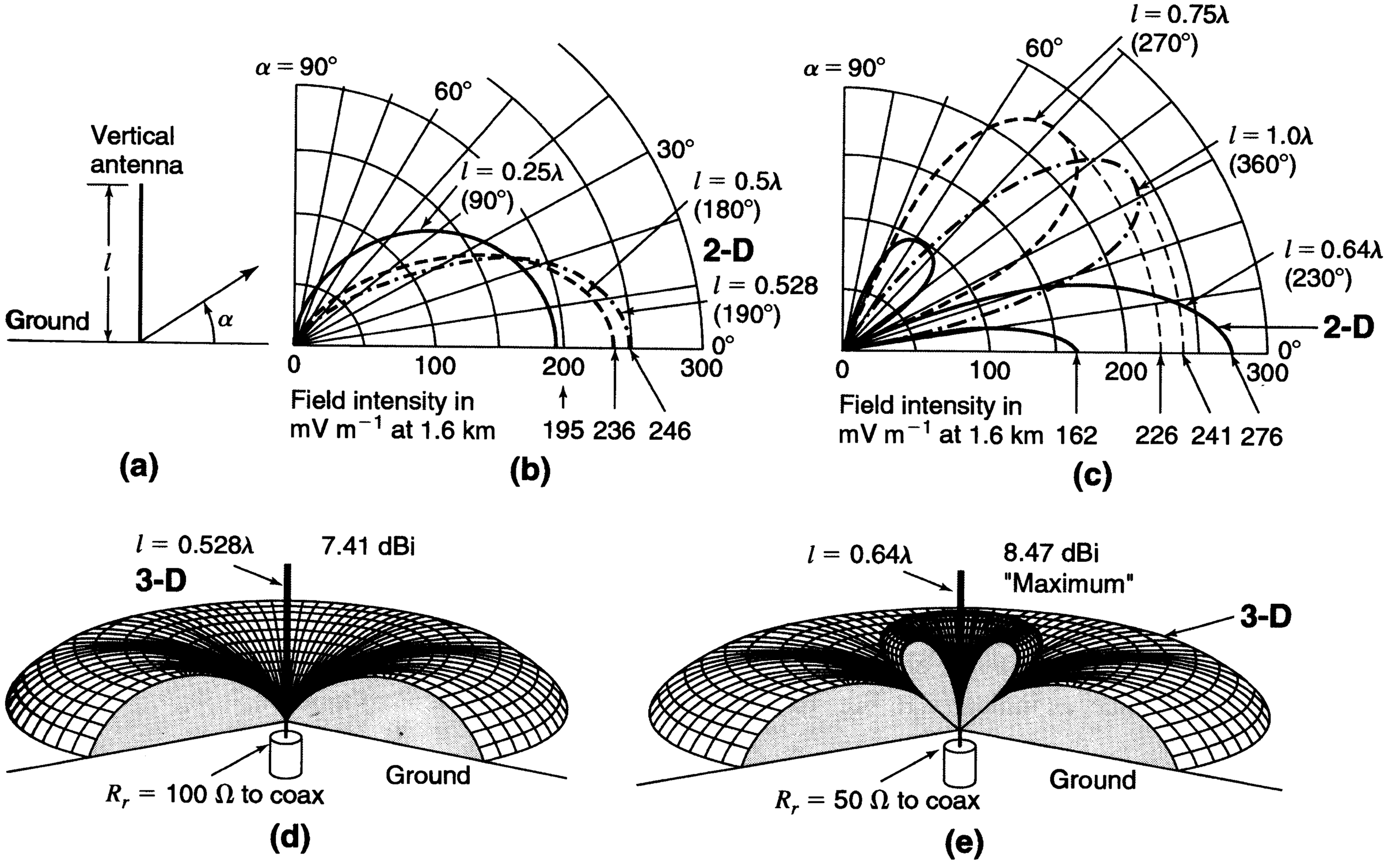


Figure 16–36
Vertical-plane field patterns of vertical antennas for several values of antenna height *l*. The field intensity is expressed in millivolts per meter at a distance of 1.6 km for 1 kW input. Perfectly conducting ground and zero loss resistance are assumed.

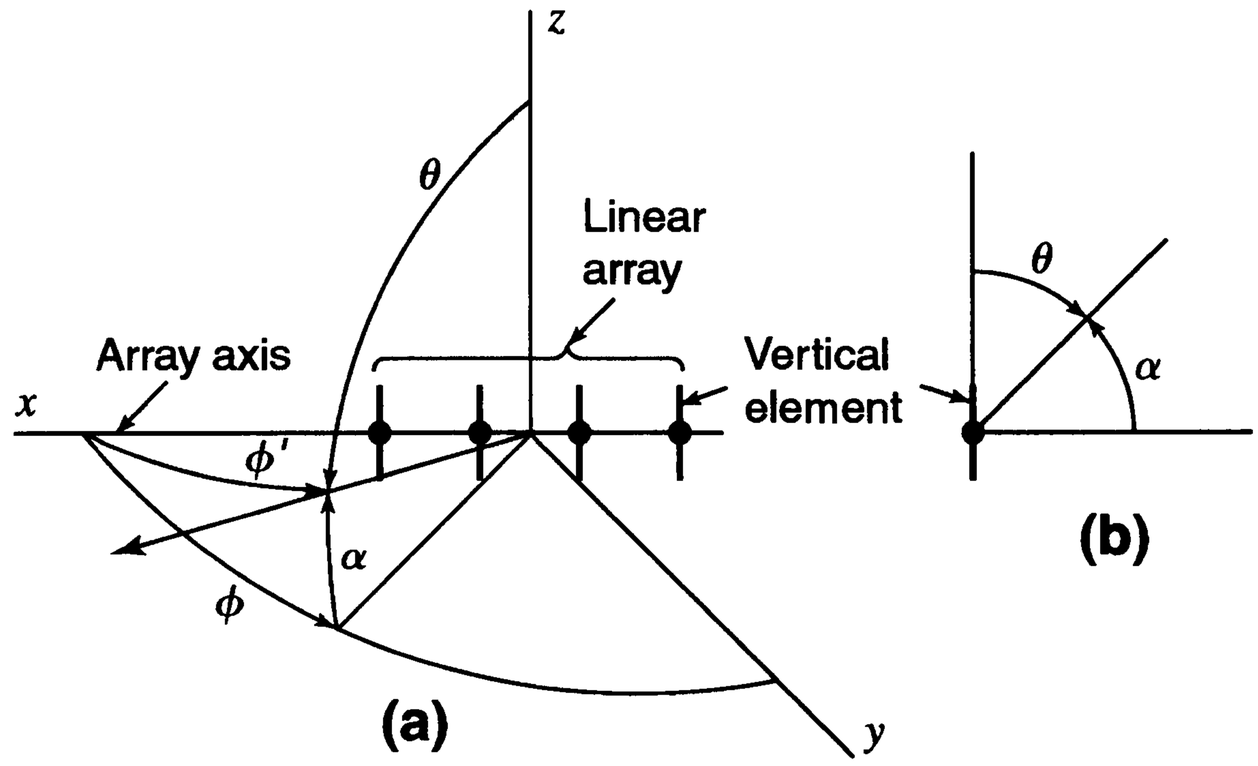


Figure 16–37
Geometrical construction for finding field intensity of a linear array of vertical elements at a constant elevation angle α.

to a single vertical stub antenna with the same power input. The situation of a symmetrical center-fed vertical antenna with its lower end some distance above the ground can also be treated by the method of images. In this case the antenna is reduced to a collinear array.

For the case of a linear array of vertical elements of equal height and of the same current distribution, the pattern $E(\phi)$ as a function of the azimuth angle ϕ at a constant elevation angle α is given by

$$E(\phi) = E_{\text{iso}}(\phi') \times E_1 \tag{3}$$

where

$E_{\text{iso}}(\phi')$ = relative pattern of array of isotropic point sources used to replace elements
E_1 = relative field intensity of a single vertical element at the elevation angle α

The angle ϕ' in the pattern formula of the array of isotropic sources is the angle with respect to the array axis or x axis in Fig. 16–37*a*. Before inserting this formula into (3), it is necessary to express ϕ' in terms of the azimuth angle ϕ and elevation angle α (Fig. 16–37*a*). This is done by the substitutions

$$\cos\phi' = \cos\phi\cos\alpha \tag{4}$$

and

$$\sin\phi' = \sqrt{1 - \cos^2\phi\cos^2\alpha} \tag{5}$$

If the relative field intensity formula E_1 of a single vertical element is given in terms of the polar angle θ, the elevation angle α is introduced by means of the substitution $\theta = 90° - \alpha$, since, as indicated in Fig. 16–37*b*, θ and α are complementary angles.

16–9 YAGI-UDA MODIFICATIONS（八木－宇田天线的变种）

16–9a Circular Polarization with a Yagi-Uda Antenna （借助八木－宇田天线的圆极化）

To produce circular polarization, 2 Yagi-Uda antennas can be crossed (elements at right angles on the same boom) with the driven elements fed in phase quadrature, or both driven

elements can be fed in phase but with one array displaced $\lambda/4$ along the boom with respect to the other. Another alternative is to feed the crossed director pairs with a monofilar axial-mode helical antenna (see Fig. 8–61). An advantage of this arrangement is that it can be fed by a single coaxial transmission line.

16–9b The Landsdorfer Shaped-Dipole Array (Landsdorfer 赋形偶极子阵列)

The gain of a ***Yagi-Uda*** antenna can be increased by adding more directors and increasing the length of the array. As with all end-fire arrays, twice the gain (3 dB improvement) requires a 4-fold increase in length. Another method of obtaining a 3 dB improvement is to stack 2 arrays. As yet another alternative, Landsdorfer (1, 2, 3) has demonstrated that higher gain can be obtained by extending and shaping the conductors of a 3-element close-spaced Yagi-Uda antenna.

Consider the center-fed $3\lambda/2$ dipole shown in Fig. 16–38*a*. Assuming a sinusoidal current distribution, the field pattern is as indicated in (*b*) (see also Fig. 6–8). There are small broadside lobes and also large lobes at an angle. If the center $\lambda/2$ section is folded into a $\lambda/4$ stub as in (*c*) the antenna reduces to an in-phase 1λ dipole with a bidirectional broadside field pattern as in (*d*). Now pulling the stub apart and shaping it and the $\lambda/2$ sections, it can be arranged with a similarly shaped director and reflector as done by Landsdorfer and shown in (*e*) with the unidirectional pattern at (*f*). The overall length is 1.3λ with gain reported to be about 11.5 dBi. This compares to about 8.5 dBi for a close-spaced 3-element array of $\lambda/2$ dipoles.

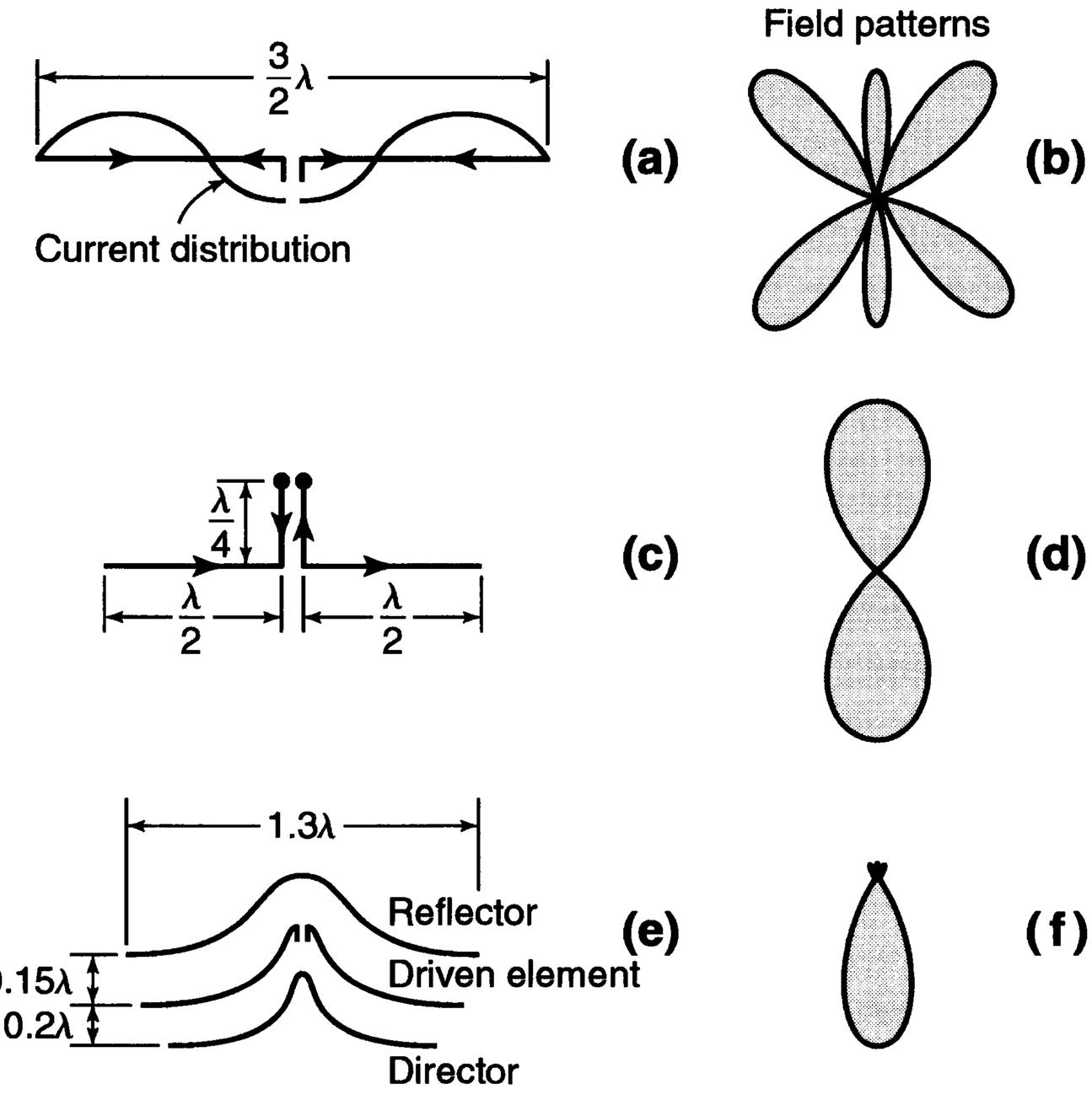


Figure 16–38
(*a*) A $3\lambda/2$ center-fed dipole and (*b*) its field pattern. (*c*) Center section folded into a $\lambda/4$ stub reducing length to 1λ and (*d*) the field pattern. (*e*) Further shaping and addition of shaped director and reflector forms a Landsdorfer array with the field pattern (*f*).

16–10 PHASED ARRAYS（相控阵）

16–10a Introduction（引言）

Although the elements of any antenna array must be phased in some manner, the term *phased array* has come to mean an array of many elements with the phase (and also, in general, the amplitude) of each element being a variable, providing *control of the beam direction* and pattern shape including side lobes. These arrays are discussed in the second part of this section.

Specialized phased arrays given different names are the frequency scanning array, the retroarray and the adaptive array.

In the scanning array, phase change is accomplished by varying the frequency. These *frequency scanning arrays* are among the simplest phased arrays since no phase control is required at each element. Several of these arrays were discussed in Sec. 8–14. Additional ones are described in the next section (16–11).

A *retroarray* is one which automatically reflects an incoming signal back toward its source. This type of array is considered in Sec. 16–12.

Adaptive arrays have an awareness of their environment and adjust to it in a desired fashion. Thus, an *adaptive array* can automatically steer its beam toward a desired signal while steering a null toward an undesired or interfering signal. In a more versatile adaptive array the output of each element is sampled, digitized and processed by a computer which can be programmed to accomplish tasks limited mainly by the sophistication of the computer program and the available computer power. Such an array may be called a *smart antenna.* These arrays are described in Sec. 16–13.

16–10b Phased Array Designs（相控阵设计）

An objective of a phased array is to accomplish beam steering without the mechanical and inertial problems of rotating the entire array. In principle, the beam steering of a phased array can be instantaneous and, with suitable networks, all beams can be formed simultaneously. However, the look angle or field of view of a planar phased array may be more restricted than for a steerable array (although a phased array on a curved surface may cover as much solid angle). Also the beam of a rotatable array maintains its shape with change in direction whereas a phased array beam may not.

Another objective of the phased array is to provide beam control at a fixed frequency or at any number of frequencies within a certain bandwidth in a *frequency-independent manner.*

In its most simplistic form, beam steering of a phased array can be done by mechanical switching. Thus, consider the case of the rudimentary 3-element array of Fig. 16–39*a*. Let each element be a $\lambda/2$ dipole (seen end-on in the figure). An incoming wave arriving broadside as in (*a*) will induce voltages in the transmission lines (or cables) in the same phase so that if all cables are of the same length ($l_1 = l_2 = l_3$) the voltages will be in phase at the (dashed) in-phase line. By bringing all 3 cables to a common point as in (*b*), the 3-element array will operate as a broadside array. For an impedance match, the cable to the receiver (or transmitter) should be $\frac{1}{3}$ the impedance of the 3 cables, or a 3 to 1

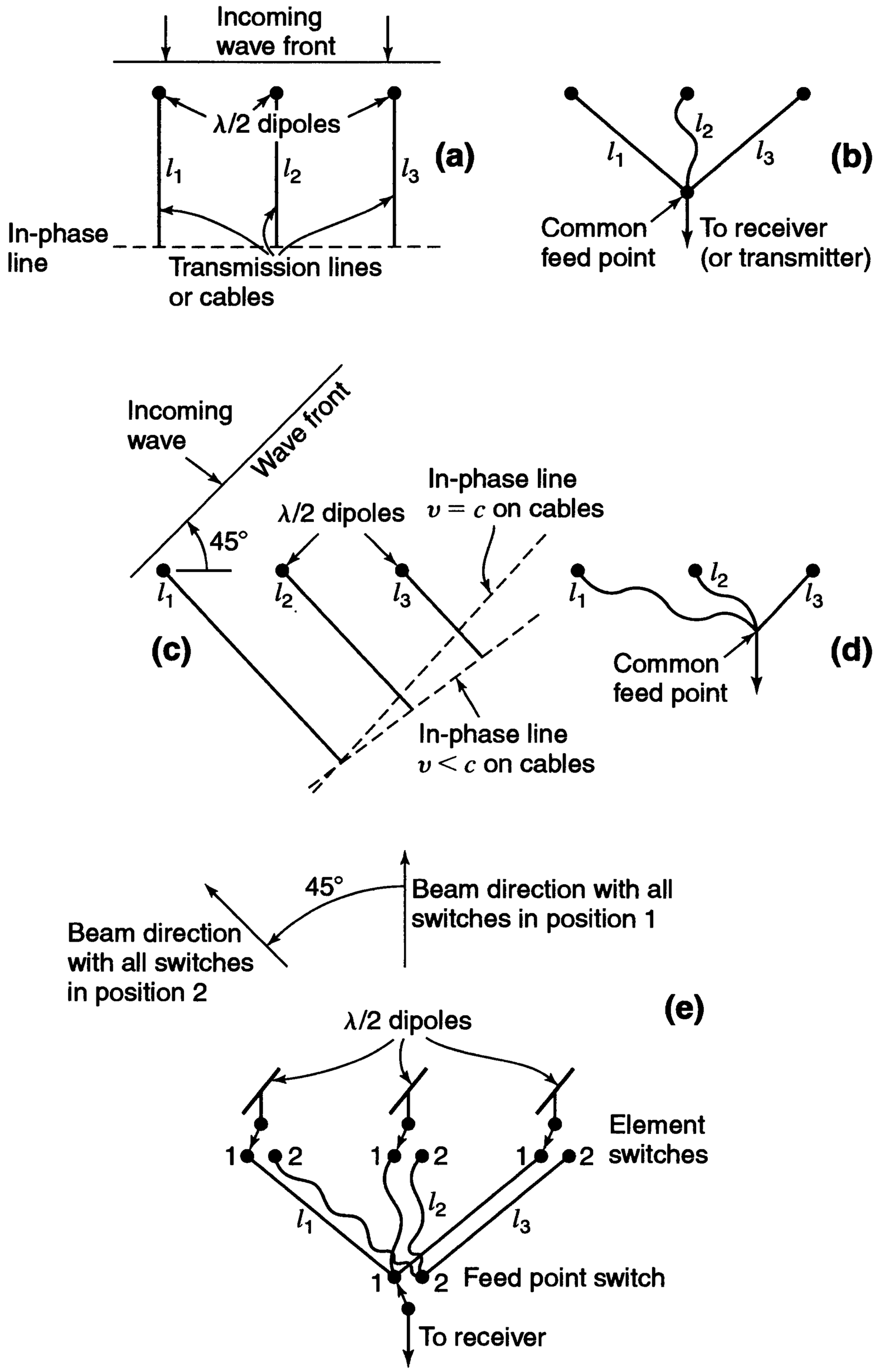


Figure 16–39
(*a*) Array of three $\lambda/2$ dipoles (seen end-on) with incoming wave broadside. (*b*) Equal length cables joined. (*c*) Incoming wave at 45° from broadside. (*d*) Appropriate lengths of cable joined. (*e*) Switches for shifting from broadside to 45° reception.

impedance transformer can be inserted at the common junction point with all cables of the same impedance.

Now consider a wave arriving at an angle of 45° from broadside as in Fig. 16–39*c*. If the wave velocity $v = c$ on the cables, the (dashed) in-phase line is parallel to the wave front of the incoming wave, as suggested in (*c*). However, if $v < c$, the lengths l_2 and l_3 must be increased as suggested in order for all phases to be the same (the in-phase condition). Then, if cables of these lengths are joined as in (*d*) the 3-element array will have its beam 45° from broadside.

By installing a switch at each antenna element and one at the common feed point as in Fig. 16–39*e* and mechanically ganging all switches together, the beam can be shifted from broadside to 45° by operating the ganged switch.

By adding more switch points and more cables of appropriate length the beam can be steered to an arbitrarily large number of directions. With more elements, narrower beams can be formed. With diodes (PIN type[1]) in place of mechanical switches, control can be electronic. However, even with these modifications, it is obvious that for precision beam steering with many antenna elements, the required number of interconnecting cables can become astronomic. Many schemes have been proposed to reduce the required number of cables. One of these, called a *Butler* (1, 2) *matrix,* is a cable-connected matrix which is the hard-wire equivalent of a discrete fast Fourier transform. For an N element, N output-port matrix, forming N simultaneous beams, the number of required cables is reduced from N^2 to $N \ln N$, resulting in a significant economy for large values of N. Computers can do the same thing by appropriate programming of sampled signals.

Instead of controlling the beam by switching cables, a phase shifter can be installed at each element. Phase shifting may be accomplished by a ferrite device. The same effect may be produced by the insertion of sections of cable (delay line) by electronic switching. Thus, insertion of cables of $\lambda/4$, $\lambda/2$, $3\lambda/4$ (and no cable) provides phase increments of 90°. For more precise phasing, cables with smaller incremental differences are used.

Figure 16–40*a* is the schematic of a phased array with a phase shifter and attenuator at each element. The feed cables are all of equal length in a *corporate structure*[2] arrangement. Figure 16–40*b* shows an end-fed phased array, also with individual element phase shifter and attenuator. Since a progressive phase shift is introduced between elements with a frequency change, the phase shifters must introduce opposing phase changes to compensate, in addition to making the desired phase changes.

Figure 16–40*c* shows a 4-element end-fed phased receiving array with each element fed from a transmission line via a *directional coupler*. The transmission line has a matched termination (zero reflection) so that (ideally) there is a pure traveling wave on the line. Phasing is accomplished by physically sliding the directional couplers along the line. Element amplitude is controlled by changing the closeness of coupling. Reduction of amplitude can control or eliminate minor lobes as with a 1 : 3 : 3 : 1 (binomial) amplitude distribution.

The literature on phased arrays is extensive. Robert Mailloux (1) gives a very comprehensive overview of the subject, updating an earlier review article by L. Stark (1).

16–10c Rotatable Helix Phased Array[3] (可旋转的螺旋相控阵)

With monofilar axial-mode helices as elements of the array, phasing can be accomplished by rotating the helices on their axes, a rotation of 90° providing a 90° shift in phase of the (circularly polarized) wave. For example, with 3 helices of the same hand connected as in

---

[1]PIN (Positive Intrinsic Negative): high open-circuit impedance, low short-circuit impedance.

[2]Named after the organizational *structure of a corporation* with president over 2 vice presidents each over 2 subordinates, etc., the diagram in Fig. 16–40*a* being an upside-down version.

[3]Described in Sec. 8–1.

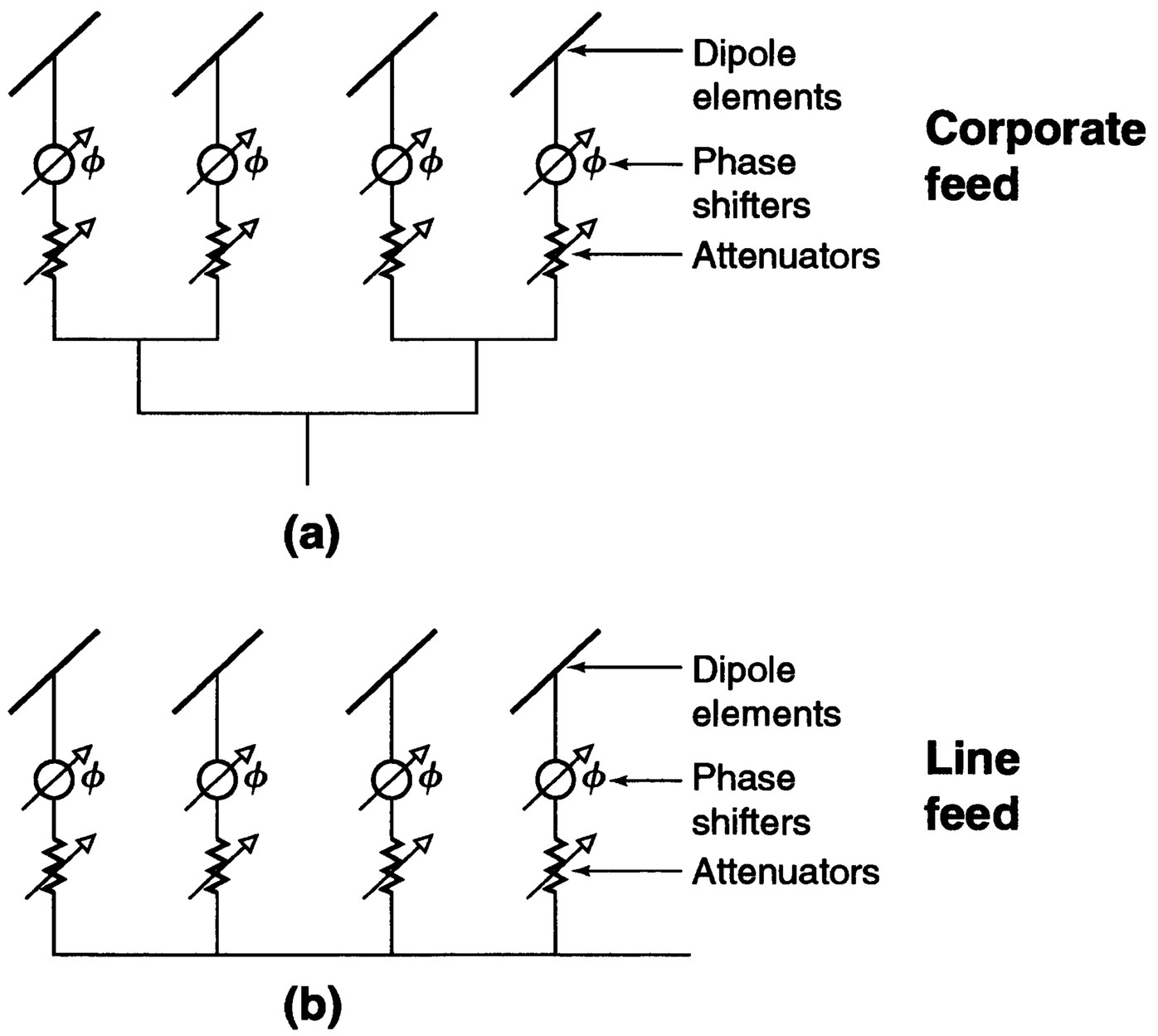


Figure 16–40
(*a*) Schematic of phased array fed by corporate structure and (*b*) end-fed. (*c*) Phased array with each element fed from a matched transmission line via a directional coupler.

Fig. 8–59 the beam direction can be steered by rotation of the outer helices (1 and 3) with helix 2 stationary. Thus, continuously rotating helix 1, clockwise and helix 3 counterclockwise will result in a continuously sweeping beam between two angular extremes. In this type of operation a lobe appears at the left extreme of the sweep angle, grows in amplitude as it sweeps to the right, reaching maximum amplitude at broadside. It then becomes smaller as it sweeps further to the right. After reaching the extreme right of the sweep angle, a new lobe appears at the left extreme and the process repeats. The *angle of sweep* is determined by the pattern of a single helix.

John Kraus designed and built a 3-helix beam-sweeping array of this type in 1958 for operation at 25 to 35 MHz at the Ohio State University Radio Observatory for planetary (Jupiter) and solar radio observations shown in Fig. 16–41 (Kraus-1, 2). The helices were 3 m in diameter, the outer 2 rotating in opposite directions. Each helix had 3 turns so that the beamwidth between first nulls (and sweep angle) was about 130°. With helix rotation at 3 revolutions per hour, the equatorial zone of the sky was swept or scanned from east to west every 20 min.

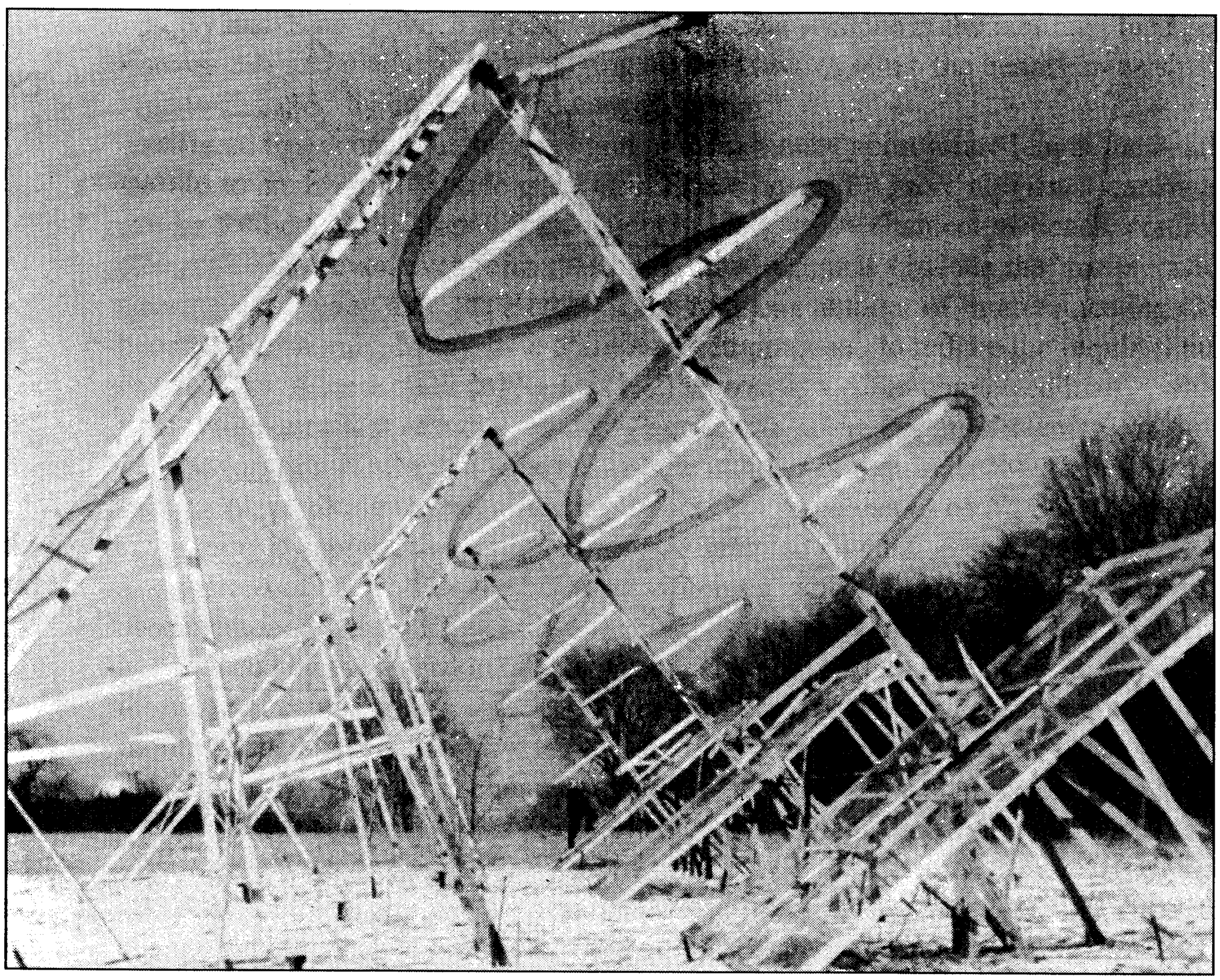


Figure 16–41
Three-helix lobe-sweeping array with 2 outer helices rotating in opposite directions producing a continuously swept beam at the O.S.U. Radio Observatory site. Note man on frame of far helix for scale. See Fig. 8–59 for feed arrangement and additional information.

16–11 FREQUENCY-SCANNING ARRAYS（频率扫描阵）

16–11a Frequency-Scanning Line-Fed Array（端馈式频率扫描阵）

Consider a line-fed array of uniformly spaced elements (dipoles) with a receiver connected at the right end of the line as suggested by the schematic of Fig. 16–42. Each element is fed from the transmission line via a directional coupler. This arrangement is similar to that in Fig. 16–40*c* but in the present case the coupler positions are fixed, with beam sweeping or scanning done by changing the frequency. The transmission line is matched to eliminate reflections and ensure an essentially pure traveling wave on the line. From (8–14–8),

$$\cos\phi = \frac{1}{p} + \frac{m}{d/\lambda_0} \tag{1}$$

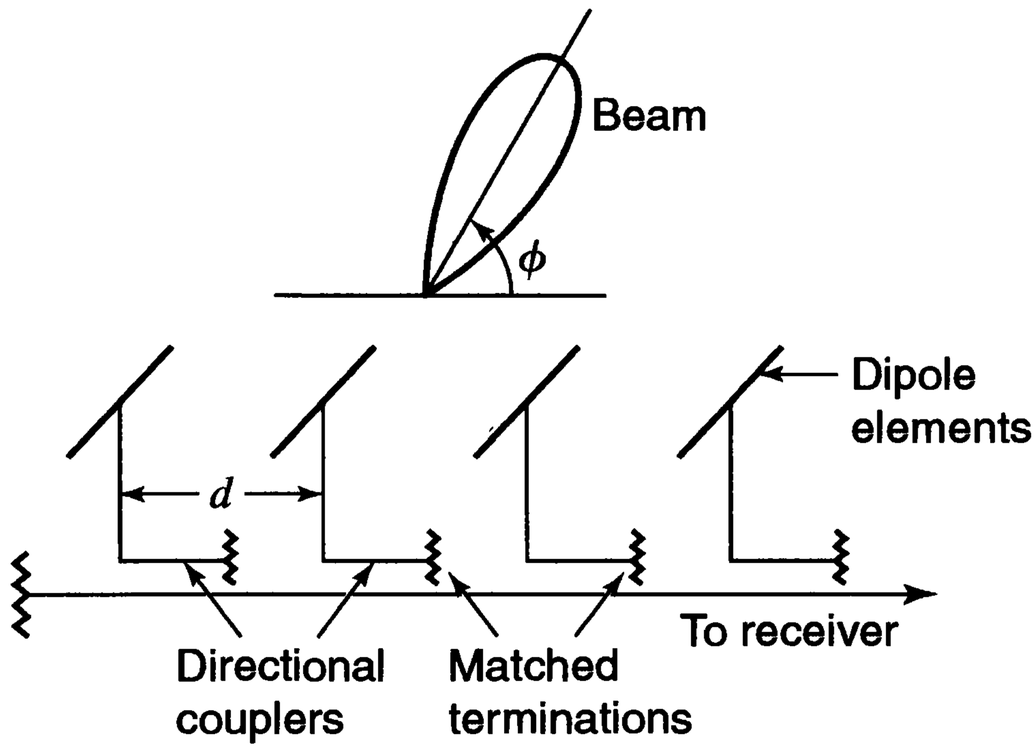


Figure 16–42
Frequency-scanning line-fed array of uniformly spaced elements with tunable receiver at left end. Beam angle ϕ is a function of the frequency.

where

ϕ = beam angle from array axis, rad or deg
p = phase velocity on transmission line = v/c, dimensionless
m = mode number, dimensionless
d = element spacing, m
λ_0 = free-space wavelength at center frequency of array operation, m

For $p = 1$ and $m = 0$, $\phi = 0°$ (beam fixed at end-fire independent of λ_0).

Consider now the situation for $p = 1$, $m = -1$, $d = 1$ m and $\lambda_0 = 1$ m, or

$$\cos\phi = 1 - 1 = 0 \tag{2}$$

and $\phi = 90°$ (beam broadside).

Suppose next that the frequency is increased so that the wavelength is $0.9\lambda_0$ or 0.9 m; then

$$\cos\phi = 1 - 0.9 = 0.1 \tag{3}$$

and $\phi = 84.3°$, or 5.7° right of broadside. Shifting to a lower frequency so that the wavelength is $1.1\lambda_0$ or 1.1 m,

$$\cos\phi = 1 - 1.1 = -0.1$$

and $\phi = 95.7°$, or 5.7° left of broadside.

Thus, a ±10 percent shift in wavelength (or frequency) swings the beam ±5.7° from broadside (total scan 11.4°) with larger frequency shifts resulting in larger scan angles. To eliminate or reduce beams at $\phi = -90°$ (mirror image), and also end-fire and back-fire beams, the $\lambda/2$ dipoles can be replaced by identical unidirectional elements, the scan angle then being restricted to the beamwidth of the individual element.

This frequency-scanning array has no moving parts, no phase shifters and no switches, making it one of the simplest types of phased arrays.

16–11b Frequency-Scanning Backward Angle-Fire Grid and Chain Arrays
（频率扫描的背射栅格形阵和链式阵）

The current on folded-wire antennas usually assumes a sinusoidal (standing-wave) distribution. In the 1930s John Kraus constructed and used folded-wire antennas of this kind, such as the Bruce curtain (Fig. 16–53*b*, and became familiar with their operation. A broadside array of these antennas (for increased gain) may require many interconnecting transmission lines to feed them. In thinking about these arrays during the winter of 1961–1962 he wondered if it would be possible to construct a continuous wire grid as a broadside array and feed it at a single centrally located point.

The basic arrangement he tried is illustrated in Fig. 16–43. The dimensions of each mesh of the grid are $l \simeq \lambda$ by $s \simeq \lambda/2$. Assuming standing waves, the instantaneous current distribution would be as indicated by the arrows, one located at each current maximum point. Currents on all of the short sides ($\lambda/2$ long, horizontal in the figure) would be in phase, while on the long sides of the meshes (λ long, vertical in the figure) there are as many current maxima in one direction as in the opposite so that radiation broadside from the long sides should (ideally) be zero. Thus, the array should produce a linearly polarized (horizontal in the figure) beam broadside to the array with a gain proportional to the number of $\lambda/2$ sides (or elements) (31 in the figure).

Kraus constructed an array similar to the one in Fig. 16–43, mounted it approximately $\lambda/4$ from a flat conducting ground plane and fed it with a balanced transmission line at the

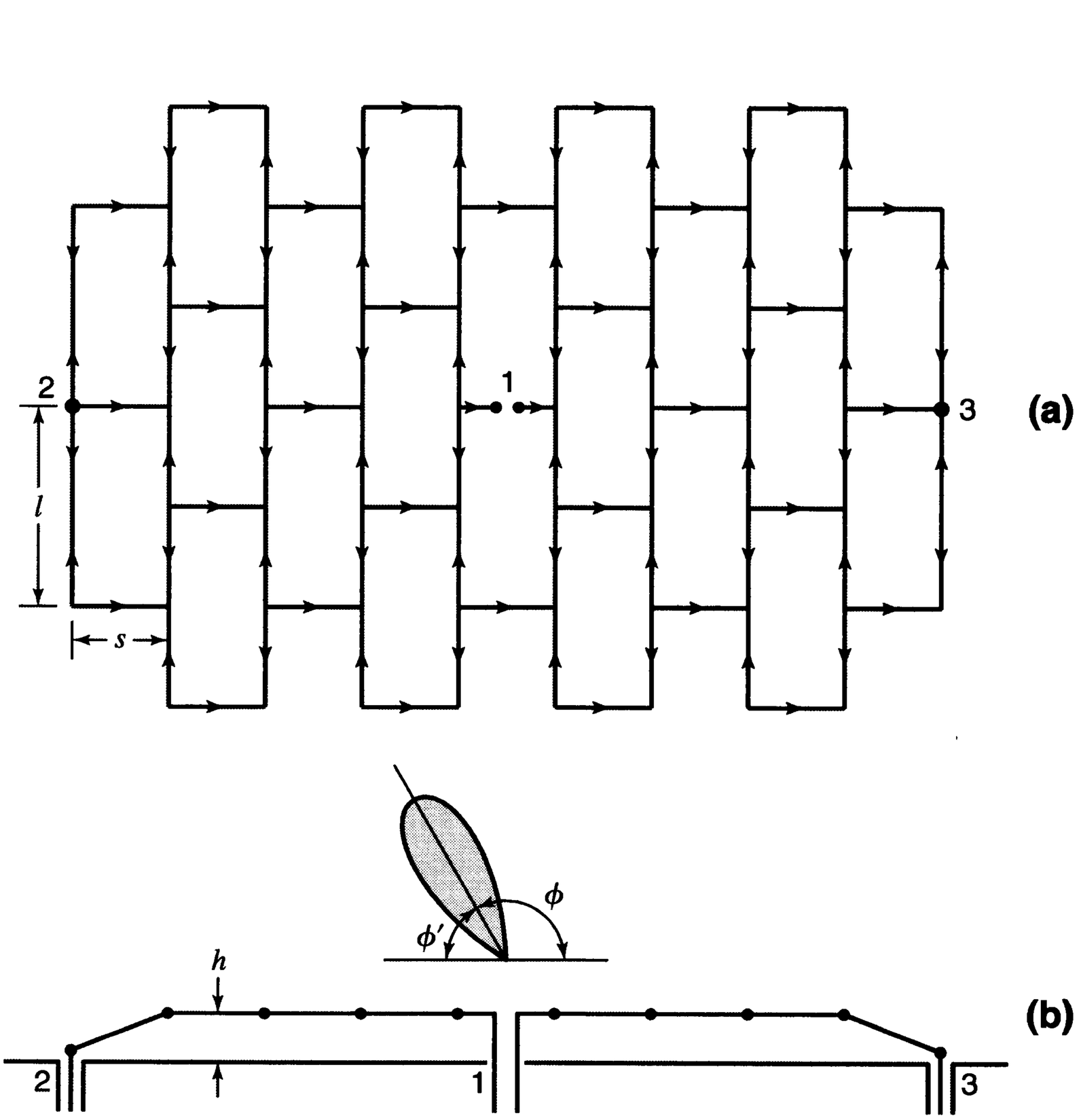


Figure 16–43
Frequency-scanning Kraus grid array. When fed at point 2 (with terminals at point 1 short-circuited and a matched load connected at point 3) the beam is at an angle ϕ which is a function of the frequency. A frequency shift of ±14 percent swings the beam angle ϕ through about 75°. Switching the feed point to 3 and load to 2 puts the beam in the right-hand quadrant, making the total scan angle 150°. Typically, $l \simeq \lambda$, $s < \lambda/2$, $h < \lambda/4$.

central point (1 in the figure). To his surprise, he found that the radiation was not in a single broadside beam but was split into 2 equal lobes, one left and one right of broadside.

It was apparent that the current distribution was not that of a resonant standing wave but rather of two traveling waves, one to the left and one to the right from point 1. Accordingly, in order to have only one (left-to-right) traveling wave across the entire grid he short-circuited the terminals at point 1 and fed the grid with a coaxial line at the left edge (point 2). This resulted in a single beam in the back-fire direction (opposite to the traveling wave) as indicated in Fig. 16–43, with the beam angle ϕ a function of the frequency. Thus, in this mode of operation, the antenna is a nonresonant frequency-scanning array with the long sides of the meshes behaving essentially as transmission-line sections and the short sides as both radiating and transmission-line sections. Although the terminals at the other edge of the array (point 3) may be left open, connecting a matched load reduces any reflected wave which could degrade the desired condition of a single left-to-right traveling wave. Thus, the antenna may be regarded as a traveling-wave-fed array of discrete radiating elements (Kraus-6, 7).

Feeding the array at point 2 (terminals at 1 short-circuited and 3 matched), the beam direction ϕ as a function of the grid parameters is given by

$$\frac{2\pi s}{\lambda}\cos\phi - \frac{\pi l}{\lambda p_x} - \frac{2\pi s}{\lambda p_y} = -2\pi m \tag{4}$$

where

s = length of short side, m
l = length of long side, m
ϕ = beam angle from array axis
p_x = relative phase velocity along short side = v_x/c
p_y = relative phase velocity along long side = v_y/c
m = mode number = integer

For $m = 1$, $p_x = p_y = 1$, (4) becomes

$$-2\pi = \frac{2\pi}{\lambda}\left(s\cos\phi - \frac{l}{2} - s\right) \tag{5}$$

or typically ($l = 2.75s$)

$$\cos\phi = 2.37 - \frac{l}{s/\lambda} \tag{6}$$

The beam direction ϕ' (complement to ϕ in Fig. 16–43) varies from about 15 to 90° for changes in s from 0.3 to 0.42λ. A wavelength (or frequency) change of ±14 percent swings the beam through a scan angle of 75°. By switching the feed point to 3 and matched load to point 2 the beam can be placed in the right quadrant, increasing the total scan angle to 150°. Although this analysis is oversimplified it illustrates the basic relations. Comparing calculations with measurements indicates that the relative phase velocity p ($= p_x = p_y$) is a function of the frequency, as suggested by Fig. 16–44, and not a constant (= 1) as assumed above. Additional measurements suggest further that $p_x \neq p_y$ (Kraus-8).

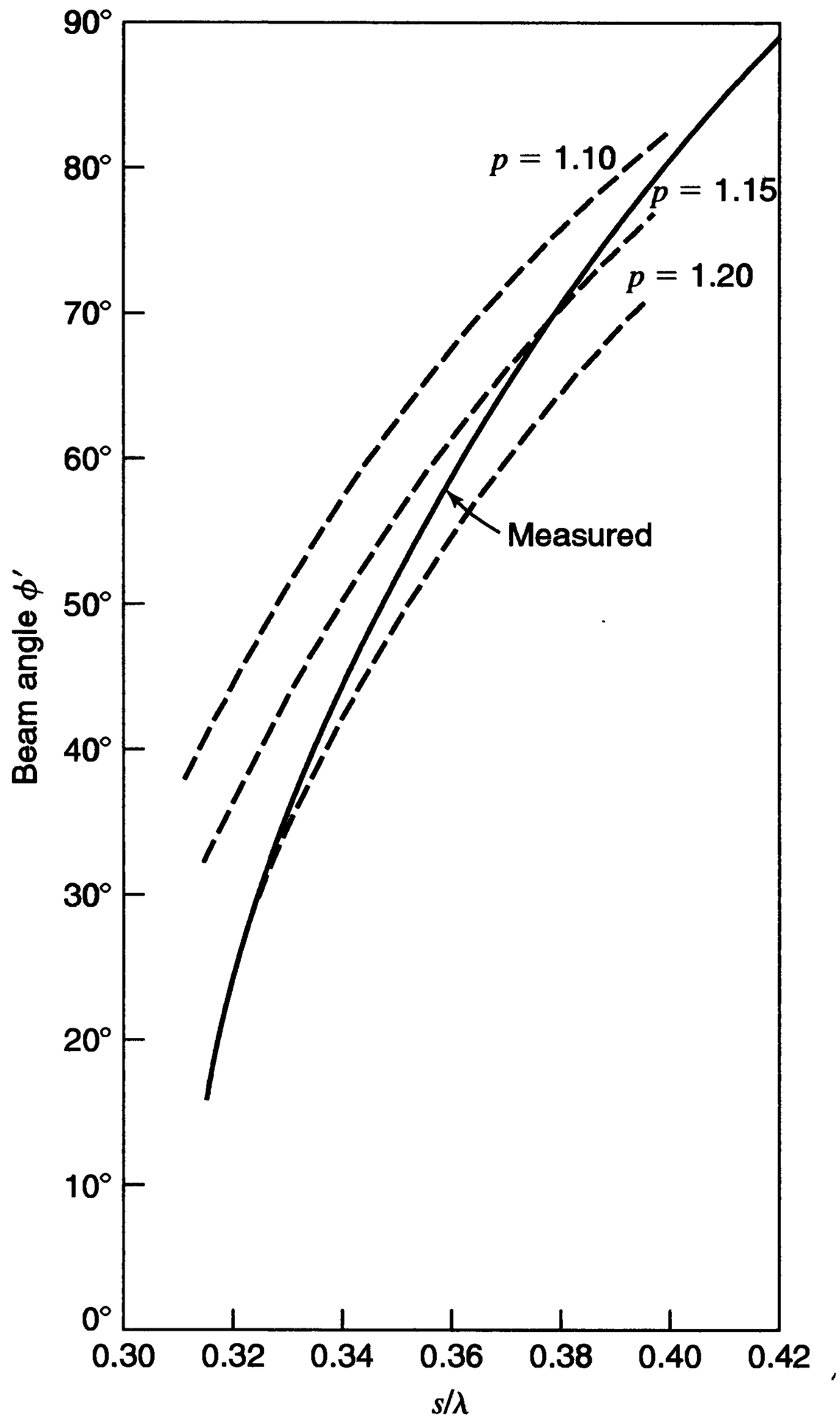


Figure 16–44
Beam angle ϕ' as a function of frequency expressed in terms of s/λ. The dashed curves are calculated for different values of relative phase velocity p. The solid curve is measured, suggesting that p is a (weak) function of the frequency.

The experimentally determined "best" average value of s $(=0.36\lambda)$ corresponds to an average value for the long sides of $l = 2.75s \simeq \lambda$. Thus, in practice the long sides of the meshes are $\sim\lambda$ long, as envisioned in the initial design, but the short sides are less than $\lambda/2$.

Considering the array as a transmission line, the extra path length $\lambda/2$ between radiating elements reduces the effective phase velocity (left to right) in the ratio $s/[s + (\lambda/2)]$. Typically, $l = 2.75s$ so that the effective phase velocity is about $0.4c$, making the grid a *slow-wave* structure. The average gain of a grid array as in Fig. 16–43 is about 17 dBi.

A microstrip version of the array is described by Conti (1).

The *Kraus grid* array principle has been extended by Tiuri (1), Tallqvist and Urpo and by Tallqvist (1) to an array in which the meshes are divided into parallel matched chains, with a typical configuration as in Fig. 16–45. This *Tiuri chain* design is also well adapted to microstrip or printed circuit construction. The array shown has an endpoint input impedance of 50 Ω for an impedance of 300 Ω for the individual chains. The current attenuation from input to matched output is about 10 dB, which is considered optimum. Higher attenuation

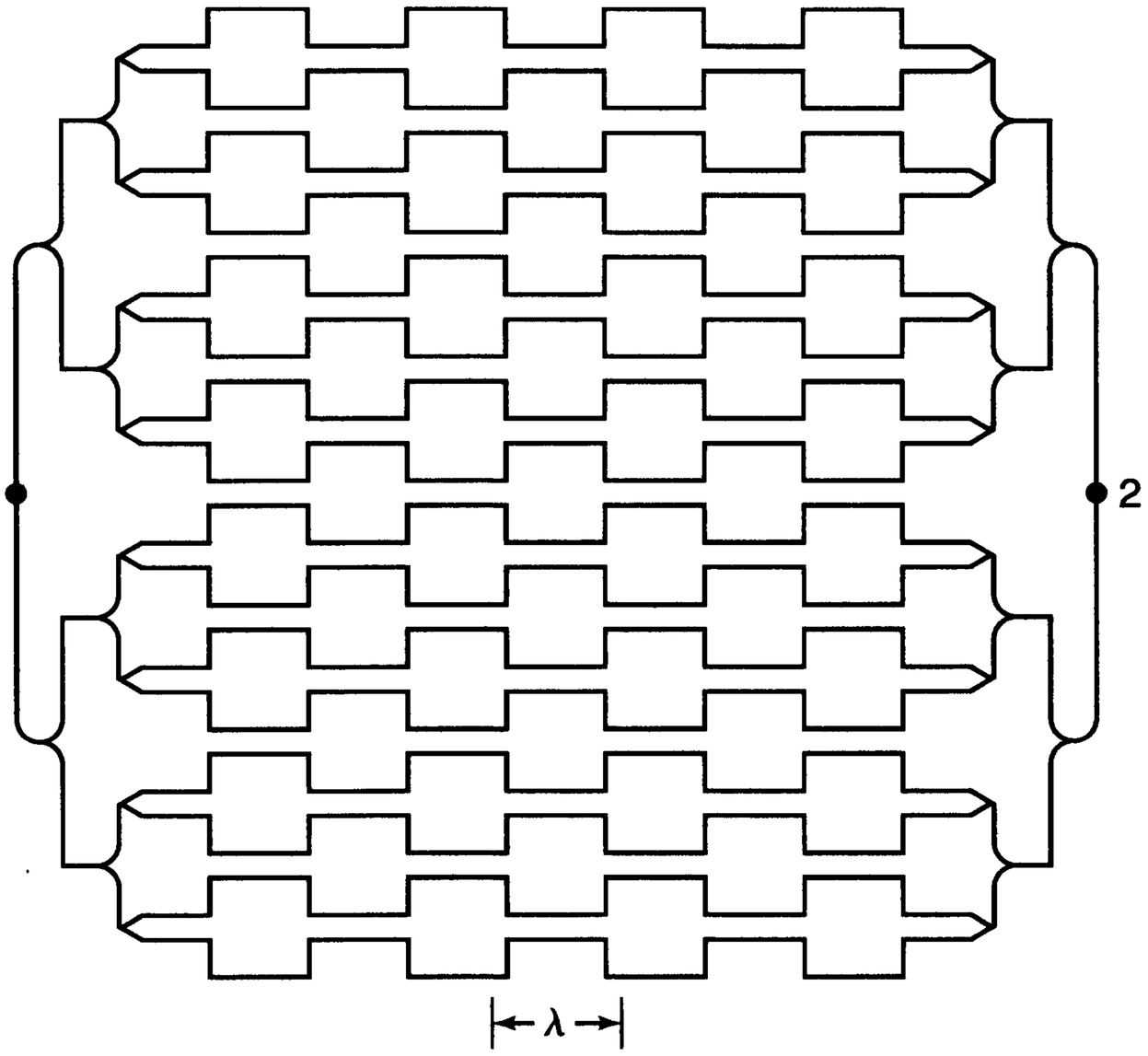


Figure 16–45
Frequency-scanning Tiuri chain array for feeding at one end (point 1) with matched load at point 2, or vice versa. With array mounted about 0.15λ from a flat conducting ground plane, the impedance of a single chain is typically about 300 Ω and the terminal impedance (at points 1 or 2) typically 50 Ω. Polarization is horizontal. (*After S. Tallqvist-1.*)

reduces the gain due to the larger taper in current distribution while lower attenuation lowers the gain because more power is lost in the matched load. Average aperture efficiencies are typically about 50 percent.

By bending the chain elements, Hendriksson (1), Markus and Tiuri have developed a circularly polarized chain array.

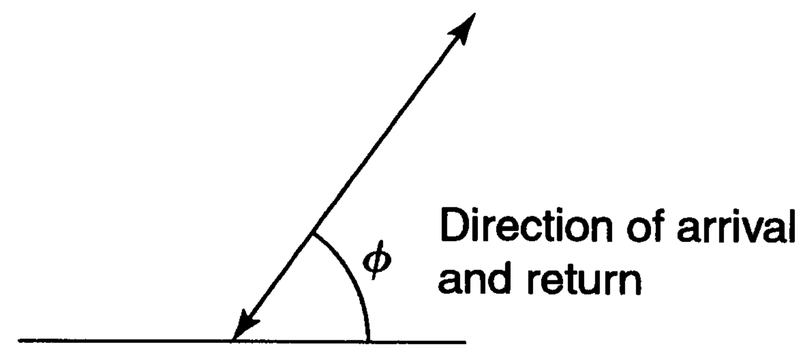
16–11c Grid Array with Broadside Beam（边射波束的栅格形阵）

By reducing the size of the grid array of Fig. 16–43 to 4 × 2 1/2 wavelengths, Hildebrand and McNamara found that the array provides a ***single broadside beam*** with 18.5 dBi gain (Hildebrand-1). The array which is shown in Fig. 3–13 is printed on a λ/4 thick plastic sheet mounted on a ground plane. It is fed very simply by a 50-Ω coaxial cable at a central point. The entire array is encapsulated in hard plastic providing a durable flat-panel array which has found wide use.

16–12 RETROARRAYS. THE VAN ATTA ARRAY（返回式阵和范阿塔阵）

If a wave incident on an array is received and transmitted back in the same direction, the array acts as a *retroreflector* or *retroarray*. The passive square-corner reflector (Sec. 10–4) does the same thing.

In general, each element of a retroarray reradiates a signal which is the conjugate of the received signal. The Van Atta array of Fig. 16–46 is an example (Van Atta-1). The 8 identical elements may be λ/2 dipoles, shown in end view in the figure. With element pairs (1 and 8, 2 and 7, 3 and 6, and 4 and 5) connected by identical equal-length cables, as indicated, a wave arriving at any angle ϕ is transmitted back in the same direction. The



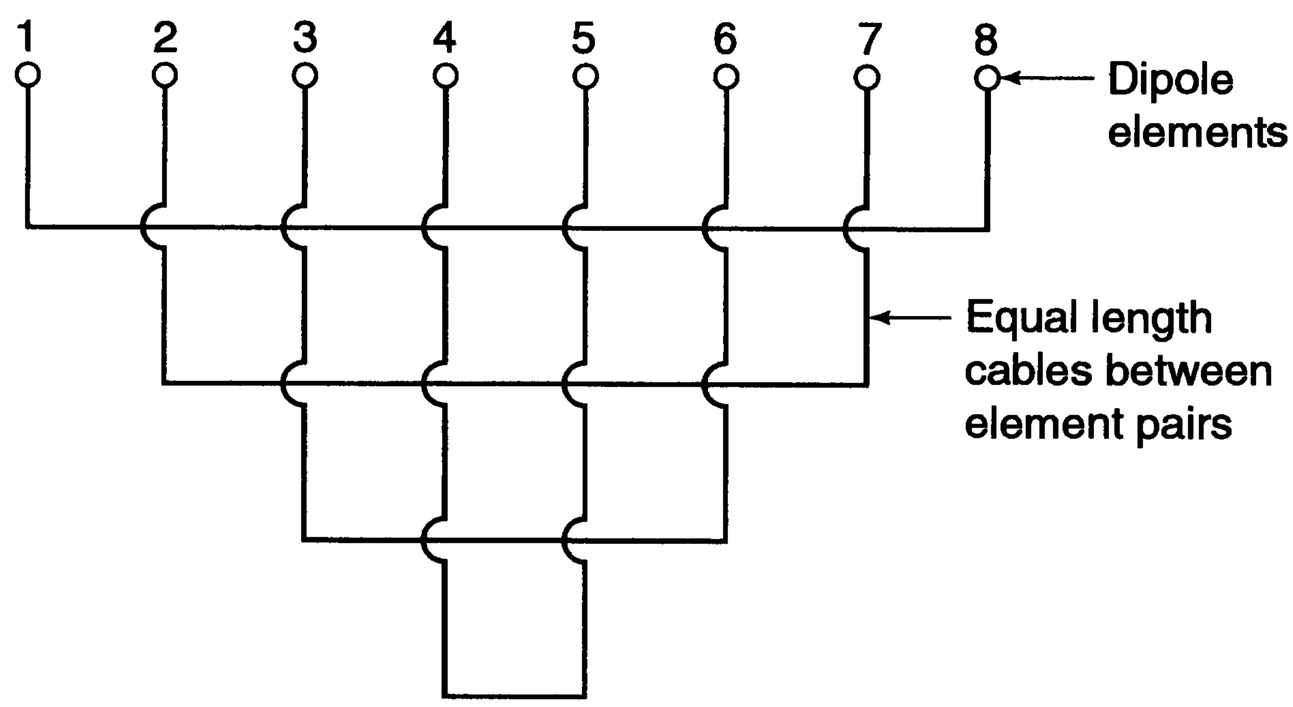


Figure 16–46
Eight-element Van Atta retroarray. Element pairs are connected by equal length lines.

array shown in Fig. 16–46, like the square-corner reflector, is passive. An adaptive (active) array (Sec. 16–13) can also be made retrodirective by using a mixer to produce a conjugate phase shift for each element (Pon-1). An advantage of an active array is that the elements need not be arranged in a line or, in a 2-dimensional case, in a plane. Active retroarrays can also incorporate amplifiers (Andre-1).

16–13 ADAPTIVE ARRAYS AND SMART ANTENNAS（自适应阵和智能阵）

The antenna elements and their transmission-line interconnections discussed so far produce a beam or beams in predetermined directions. Thus, when receiving, these arrays look in a given direction regardless of whether any signals are arriving from that direction or not. However, by processing the signals from the individual elements, an array can become active and react intelligently to its environment, steering its beam toward a desired signal while simultaneously steering a null toward an undesired, interfering signal and thereby maximizing the signal-to-noise ratio of the desired signal. The term *adaptive array* is applied to this kind of antenna.

Also, by suitable signal processing, performance may be further enhanced, giving *simulated patterns*[1] of higher resolution and lower side lobes. In addition, by appropriate sampling and digitizing the signals at the terminals of each element and processing them with a computer, a very intelligent or *smart antenna* can, in principle, be built. For a given number of elements, such an antenna's capabilities are limited, mainly by the ingenuity of the programmer and the available computer power. Thus, for example, multiple beams may

[1] Simulated patterns are ones that exist only in the signal-processing domain.

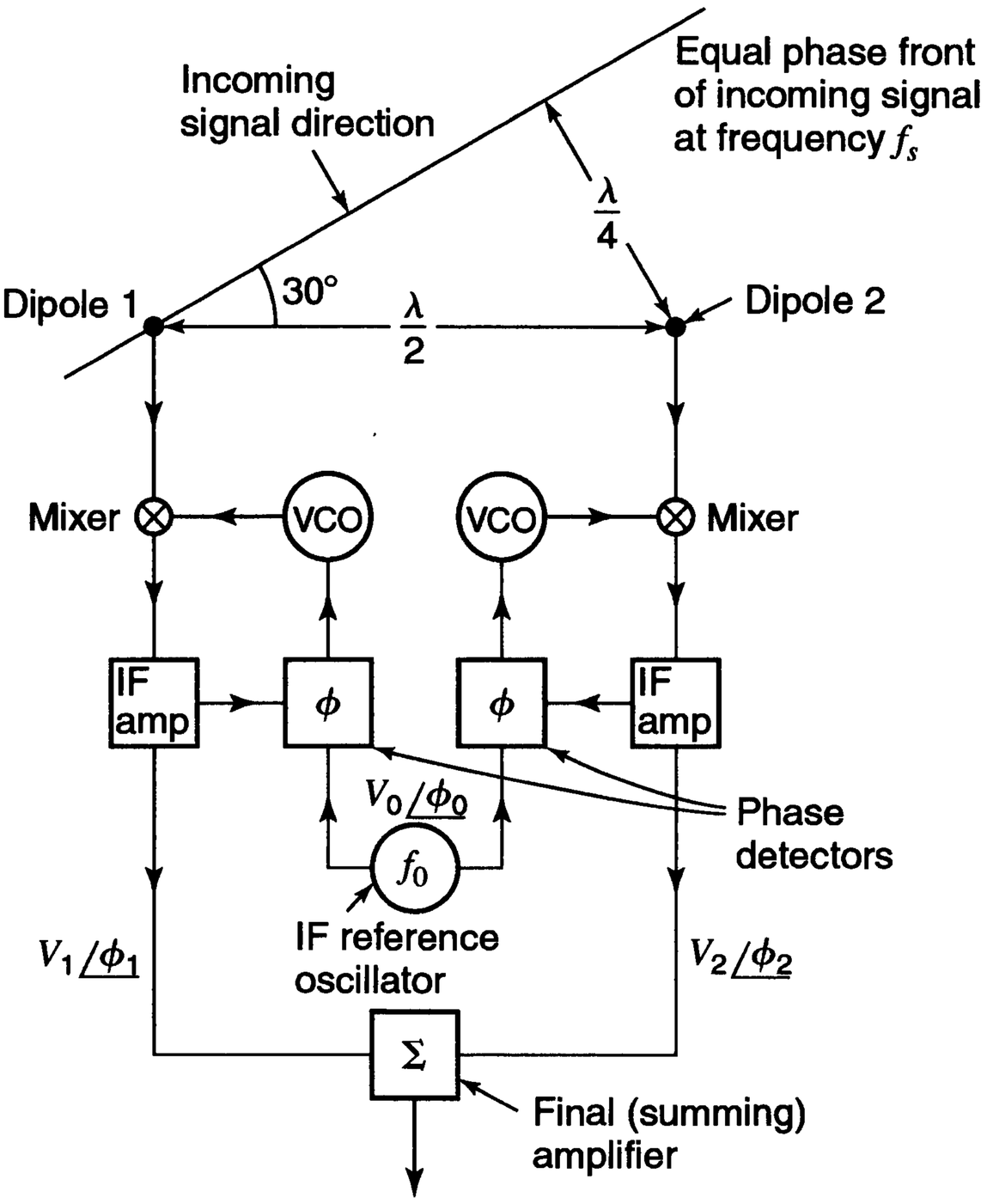


Figure 16–47
Two-element adaptive array with signal-processing circuitry.

be simultaneously directed toward many signals arriving from different directions within the field of view of the antenna (ideally $= 2\pi$ sr for a planar array). These antennas are sometimes called *Digital Beam Forming (DBF) antennas* (Steyskal-1).

As a rudimentary example of an adaptive array, a simple 2-element system is shown in Fig. 16–47 with $\lambda/2$ spacing between the elements at the signal frequency f_s. Let each element be a $\lambda/2$ dipole seen end-on in Fig. 16–47 so that the patterns of the elements are uniform in the plane of the page. With elements operating in phase, the beam is broadside (up in the figure).

Consider now the case of a signal at 30° from broadside as suggested in Fig. 16–47 so that the wave arriving at element 2 travels $\lambda/4$ farther than to element 1, thus retarding the phase of the signal by 90° at element 2. Each element is equipped with its own mixer, Voltage-Controlled Oscillator (VCO), intermediate frequency amplifier and phase detector. An oscillator at the intermediate frequency f_0 is connected to each phase detector as reference. The phase detector compares the *phase of the downshifted signal* with the *phase of the reference oscillator* and produces a voltage proportional to the phase difference. This voltage, in turn, advances or retards the phase of the VCO output so as to reduce the phase difference to zero (phase locking). The voltage for the VCO of element 1 would ideally be equal in magnitude but of opposite sign to the voltage for the VCO of element 2 so that the downshifted signals from both elements are locked in phase, making

$$\phi_1 = \phi_2 = \phi_0 \tag{1}$$

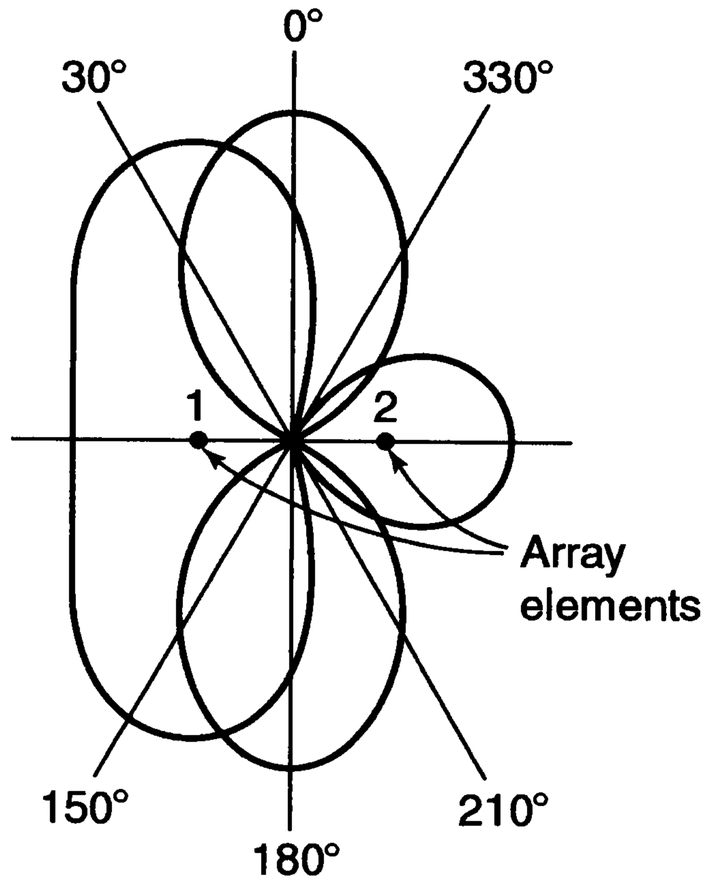


Figure 16–48
Patterns of 2-element adaptive array for signals from 0 and 30° directions. For the 0° signal, nulls are at 90 and 270° while for the 30° signal, nulls are at 210 and 330°. These patterns are identical with those of Figs. 5–1 and 5–4.

where

ϕ_1 = phase of downshifted signal from element 1
ϕ_2 = phase of downshifted signal from element 2
ϕ_0 = phase of reference oscillator

With equal gain from both IF amplifiers the voltages V_1 and V_2 from both elements should be equal so that

$$V_1 \angle \phi_1 = V_2 \angle \phi_2 \tag{2}$$

making the voltage from the summing amplifier proportional to $2V_1$ ($= 2V_2$) and maximizing the response of the array to the incoming signal by steering the beam onto the incoming signal. In our example, 45° phase corrections of opposite sign would be required by the VCOs (+ for element 1, − for element 2).

In our rudimentary 2-element example, the beam will be in the 0° direction for a signal from the 0° direction and at 30° for a signal from that direction, as shown by the patterns in Fig. 16–48. If interfering signals are arriving from the 210 and 330° directions when the main signal is at 30°, the nulls at 210 and 330° will suppress the interference. However, an interfering signal at 150° would be at a pattern maximum, the same as the desired signal at 30°. To provide more effective adaptation to its environment, an array with more elements and more sophisticated signal processing is required. For example, the main beam may be steered toward the desired signal by changing the progressive phase difference between elements, while, independently, one or more nulls are steered toward interfering signals by modifying the array element amplitudes with digitally controlled attenuators.

Many references on adaptive arrays are listed at the end of the chapter.

16–14 MICROSTRIP ARRAYS（微带阵）

Printed circuit and microstrip techniques facilitate the construction of multielement arrays for microwave frequencies (Oltman-1). The grid and chain arrays (Sec. 16–11*b*) are examples. The 896-element microstrip antenna for space research shown in Fig. 16–49 is another.

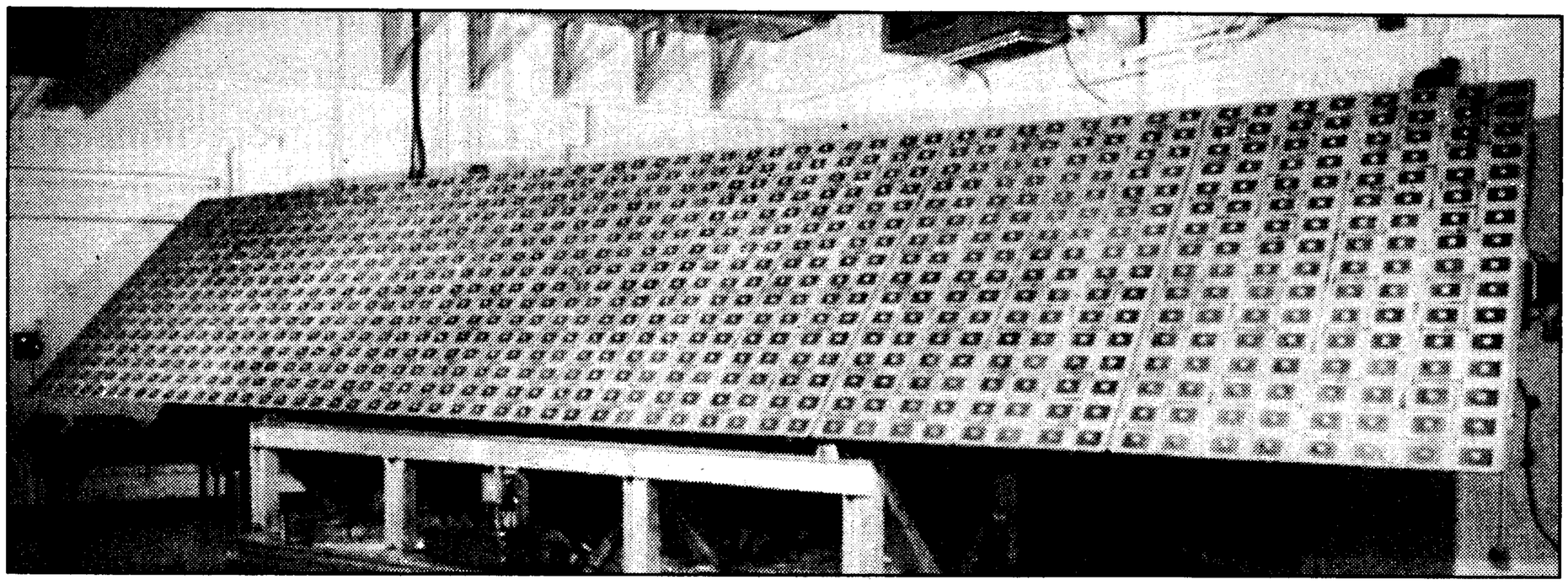


Figure 16–49
An 896-element microstrip array for remote sensing the earth from space. The 9.5 × 2.4 m array has 34 dBi gain at $\lambda \simeq 30$ cm. (*Courtesy Robert Munson, Ball Aerospace, Boulder, Colorado.*)

All elements are photoetched from one side of a printed circuit board. The corporate structure feed has amplitude taper in the narrow dimension. The 9.5 × 2.4 m array has a 34-dBi gain at $\lambda \simeq 30$ cm.

Although the microstrip element is inherently narrowband, log-periodic patch arrays have achieved bandwidths of 4 to 1 (Hall-1). Patch (or microstrip) antennas are discussed in Chap. 9.

16–15 LOW-SIDE-LOBE ARRAYS（低旁瓣阵）

Ideally it may be desirable for an antenna to have a narrow, well-defined beam with no side lobes, or at least none above a certain prescribed level. A prime factor affecting the side-lobe level is the aperture distribution (Chapter 19). Although some distributions yield (theoretically) zero side lobes, they may be more difficult of realization on reflector antennas than on phased arrays. The typical large parabolic reflector antenna may also have significant side lobes due to other causes such as aperture blocking and diffraction from its prime focus feed or Cassegrain reflector, from struts of the supporting structure, from feed spillover, from surface irregularities and from the edge of the parabola. The edge diffraction may be reduced by using a rolled edge (see Sec. 24–4–4). Although offset feeds or integral horn reflectors may have significantly lower minor lobes, accurate construction techniques providing precise amplitude and phase control of all elements of a phased array have resulted in phased array designs with side-lobe levels down 50 dB or more. This is as good or better than has been achieved to date with reflector antennas. However, phased arrays are inherently narrower band than reflector antennas.

16–16 LONG-WIRE ANTENNAS（长导线天线）

Most of the preceding parts of this chapter deal with arrays of individual, discrete elements (usually $\lambda/2$ long) interconnected by transmission lines. A linear wire antenna, many wavelengths long, may also be regarded as an array of $\lambda/2$ elements but connected in a continuous

linear fashion with each element serving as both a radiator and a transmission line. The long-wire antennas discussed in this section are the V, rhombic and Beverage types. The V antenna may be either unterminated (with standing wave) or terminated (with traveling wave). The rhombic and Beverage antennas are almost always terminated (with traveling wave).

16–16a V Antennas（V形天线）

By assuming a sinusoidal (standing-wave) current distribution, the pattern of a long thin wire antenna can be calculated as described in Chap. 6 (Carter-1, 2). A typical pattern is shown in Fig. 16–50*a* for a wire 2λ long. The main lobes are at an angle $\beta = 36°$ with respect to the wire. By arranging two such wires in a V with an included angle $\gamma = 72°$ as in Fig. 16–50*b* a bidirectional pattern can be obtained. This pattern is the sum of the patterns of the individual wires or legs. Although an included angle $\gamma = 2\beta$ results in the

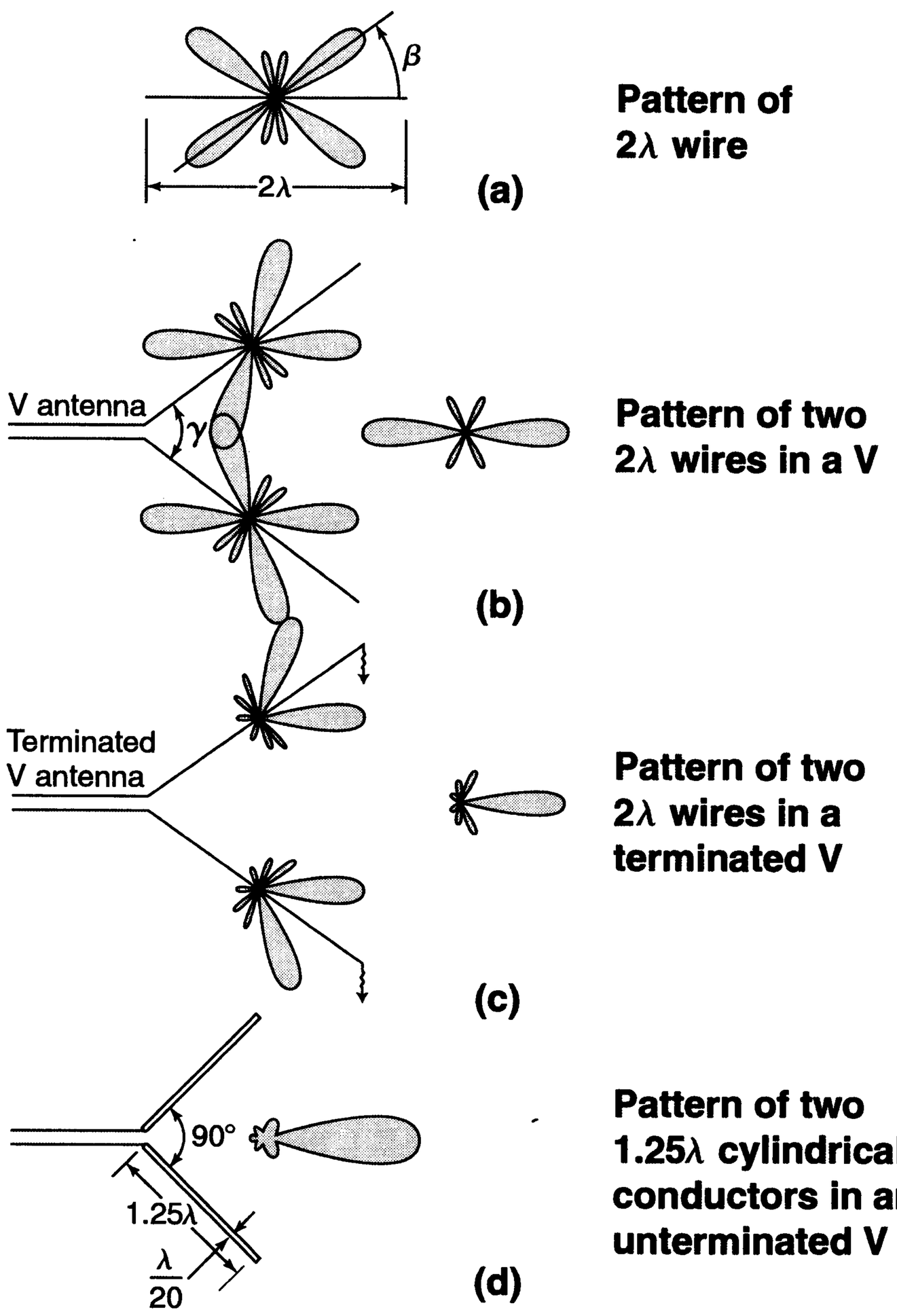


Figure 16–50
(*a*) Calculated pattern of 2λ wire with standing wave, (*b*) V antenna of two such wires, (*c*) terminated V antenna with legs 2λ long and (*d*) V antenna of cylindrical conductors 1.25λ long with measured pattern.

alignment of the major lobes at zero elevation angle (wires horizontal) and in free space, it is necessary to make γ somewhat less than 2β in order to obtain alignment at elevation angles greater than zero (ARRL-1). This is because the space pattern of a single wire is conical, being obtained by revolving the pattern of Fig. 16–50*a*, for example, with the wire as the axis.

If the legs of the thin-wire V antenna are terminated in their characteristic impedance, as in Fig. 16–50*c*, so that the wires carry only an outgoing traveling wave, the back-radiation is greatly reduced. The patterns of the individual wires can be calculated, assuming a single traveling wave as done in Chap. 6.

A similar effect may be produced without terminations by the use of V conductors of considerable thickness. The reflected wave on such a conductor may be small compared to the outgoing wave, and a condition approaching that of a single traveling (outgoing) wave may result. For example, a V antenna consisting of two cylindrical conductors 1.25λ long and $\lambda/20$ diameter with an included angle $\beta = 90°$ has the highly unidirectional pattern of Fig. 16–50*d* (Dorne-1).

16–16b Rhombic Antennas（菱形天线）

A rhombic antenna may be regarded as a double-V type. The wires at the end remote from the feed end are in close proximity, as in Fig. 16–51*a*. A terminating resistance, usually 600 to 800 Ω, can be conveniently connected at this location so that there is substantially a single outgoing traveling wave on the wires. The length of each leg is L, and half of the

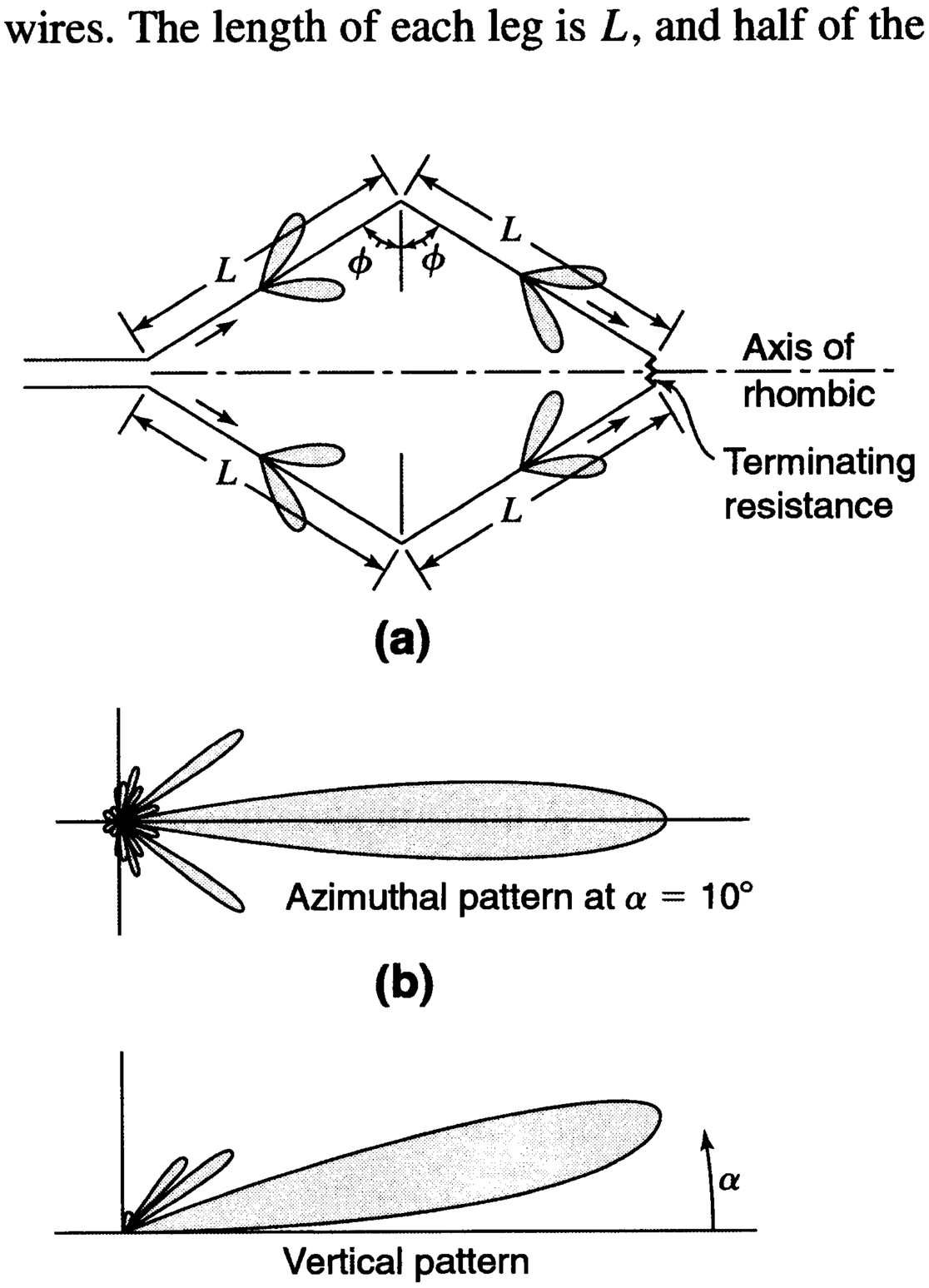


Figure 16–51
Terminated rhombic antenna (*a*) with azimuthal pattern (*b*) and vertical plane pattern (*c*) for a rhombic 6λ long on each leg, $\phi = 70°$, and at a height of 1.1λ above a perfectly conducting ground. (*After A. E. Harper-1.*)

included side angle is ϕ. The calculated patterns of a terminated rhombic with legs 6λ long are shown in Fig. 16–51*b* and *c* (Harper-1). The rhombic is assumed to be 1.1λ above a perfectly conducting ground, and $\phi = 70°$ (Bruce-1, Foster-1).

In designing a rhombic antenna, the angle ϕ, the leg length and the height above ground may be so chosen that (1) the maximum of the main lobe coincides with the desired elevation angle α (*alignment design*) or (2) the maximum relative field intensity E for a constant antenna current is obtained at the desired elevation angle α (*maximum E design*) (Bruce-2).

If the height above ground is less than that required for these designs, alignment may be obtained by increasing the leg length. If the height is maintained but the leg length is reduced, alignment may be obtained by changing the angle ϕ. As a third possibility, if both the height and the leg length are reduced, the angle ϕ can be changed to produce alignment. Any of these 3 modifications results in a so-called *compromise design* having reduced gain. If moderate departures from optimum performance are acceptable, a rhombic antenna can be operated without adjustment over a frequency band of the order of 2 to 1.

The pattern of a rhombic antenna may be calculated as the sum of the patterns of four tilted wires each with a single outgoing traveling wave. The effect of a perfectly conducting ground may be introduced by the method of images. For a horizontal rhombic of perfectly conducting wire above a perfectly conducting plane ground, Bruce (2), Beck and Lowry give the relative field intensity E in the vertical plane coincident with the rhombic axis[1] as a function of α, ϕ, L_λ and H_λ as

$$E = \frac{(\cos\phi)[\sin(H_r \sin\alpha)][\sin(\psi L_r)]^2}{\psi} \tag{1}$$

where

α = elevation angle with respect to ground
ϕ = half included side angle of rhombic antenna
$H_\lambda = H/\lambda$ = height of rhombic antenna above ground
$L_\lambda = L/\lambda$ = leg length
$H_r = 2\pi H_\lambda = 2\pi(H/\lambda)$
$L_r = 2\pi L_\lambda = 2\pi L/\lambda$
$\psi = (1 - \sin\phi\cos\alpha)/2$

A uniform antenna current is assumed and mutual coupling is neglected.

Following the procedure of Bruce, Beck and Lowry, the various parameters may be determined as follows (Bruce-2). For the maximum E condition, E is maximized with respect to H_λ, that is, we make

$$\frac{\partial E}{\partial H_\lambda} = 0 \tag{2}$$

---

[1]The radiation in this plane is horizontally polarized. However, in other planes the polarization is not, in general, horizontal.

which yields

$$\cos(2\pi H_\lambda \sin\alpha) = 0$$

which is satisfied when

$$2\pi H_\lambda \sin\alpha = n\frac{\pi}{2}$$

where $n = 1, 3, 5, \ldots$

For the lowest practical height, $n = 1$. Therefore,

$$H_\lambda = \frac{1}{4\sin\alpha} \tag{3}$$

Equation (3) gives the height H_λ for the antenna. To find the leg length, E is maximized with respect to L_λ, obtaining

$$L_\lambda = \frac{1}{2(1 - \sin\phi\cos\alpha)} \tag{4}$$

Finally, by maximizing E with respect to ϕ and introducing the condition of (4),

$$\phi = 90^\circ - \alpha \tag{5}$$

Substituting (5) back into (4) yields

$$L_\lambda = \frac{1}{2\sin^2\alpha} \tag{6}$$

Equations (3), (5) and (6) then give the height in wavelengths H_λ, the half-side angle ϕ and the leg length in wavelengths L_λ, for *maximum* E at the desired elevation angle α. This is for a uniform antenna current. It does not follow that the field intensity at the desired elevation angle is a maximum for a given power input to the antenna. However, it is probably very close to this condition. It is also of interest that for the maximum E condition the maximum point of the main lobe of radiation is not, in general, aligned with the desired elevation angle.

In the *alignment design* the maximum point of the main lobe of radiation is aligned with the desired elevation angle α. For this condition, E at α is slightly less than for the maximum E condition. Alignment is accomplished by maximizing E with respect to α and introducing the condition of (3). This gives

$$L_\lambda = \frac{0.371}{1 - \sin\phi\cos\alpha} \tag{7}$$

Substituting (7) in (1) and maximizing the resulting relation for the field with respect to ϕ gives

$$\phi = 90^\circ - \alpha \tag{8}$$

as before. Finally substituting (8) in (7) we obtain

$$L_\lambda = \frac{0.371}{\sin^2\alpha} \tag{9}$$

Equations (3), (8) and (9) then give H_λ, ϕ and L_λ for *alignment* of the maximum point of the main lobe of radiation with the desired elevation angle α. Only the length is

Table 16–1 Design formulas for terminated rhombic antennas†

Type of rhombic antenna	Formulas
Maximum E at elevation angle α	$H_\lambda = \dfrac{1}{4\sin\alpha}$ $\phi = 90° - \alpha$ $L_\lambda = \dfrac{0.5}{\sin^2\alpha}$
Alignment of major lobe with elevation angle α	$H_\lambda = \dfrac{1}{4\sin\alpha}$ $\phi = 90° - \alpha$ $L_\lambda = \dfrac{0.371}{\sin^2\alpha}$
Reduced height H' Compromise design for alignment at elevation angle α	$\phi = 90° - \alpha$ $L_\lambda = \dfrac{\tan[(\pi L_\lambda)\sin^2\alpha]}{\sin\alpha}\left[\dfrac{1}{2\pi\sin\alpha} - \dfrac{H'_\lambda}{\tan(H'_r\sin\alpha)}\right]$ where $H'_\lambda = \dfrac{H'}{\lambda}$ and $H'_r = 2\pi\dfrac{H'}{\lambda}$
Reduced length L' Compromise design for alignment at elevation angle α	$\dot{H}_\lambda = \dfrac{1}{4\sin\alpha}$ $\phi = \arcsin\left[\dfrac{L'_\lambda - 0.371}{L'_\lambda\cos\alpha}\right]$ where $L'_\lambda = L'/\lambda$
Reduced height H' and length L' Compromise design for alignment at elevation angle α	Solve this equation for ϕ: $\dfrac{H'_\lambda}{\sin\phi\tan\alpha\tan(H'_r\sin\alpha)} = \dfrac{1}{4\pi\psi} - \dfrac{L'_\lambda}{\tan(\psi L'_r)}$ where $\psi = \dfrac{1-\sin\phi\cos\alpha}{2}$ and $L'_r = 2\pi\dfrac{L'}{\lambda}$

†After Bruce-2.

different in the alignment design, being $0.371/0.5 = 0.74$ of the value for the maximum E design.

The above design relations are summarized in Table 16–1 together with design formulas for 3 kinds of compromise designs.

An end-to-end receiving array of a number of rhombics may be so connected as to provide an electrically controllable vertical plane pattern which can be adjusted to coincide with the optimum elevation angle of downcoming waves. This Multiple Unit Steerable Antenna, or MUSA, is a vertically steerable system of this kind for long-distance short-wave reception of horizontally polarized downcoming waves (Friis-1).

16–16c Beverage Antennas（贝弗瑞天线）

The electric field of a wave traveling along a perfectly conducting surface is perpendicular to the surface as in Fig. 16–52*a*. However, if the surface is an imperfect conductor, such as the earth's surface or ground, the electric field lines have a forward tilt near the surface as in Fig. 16–52*b*. Hence, the field at the surface has a vertical component E_y and a horizontal

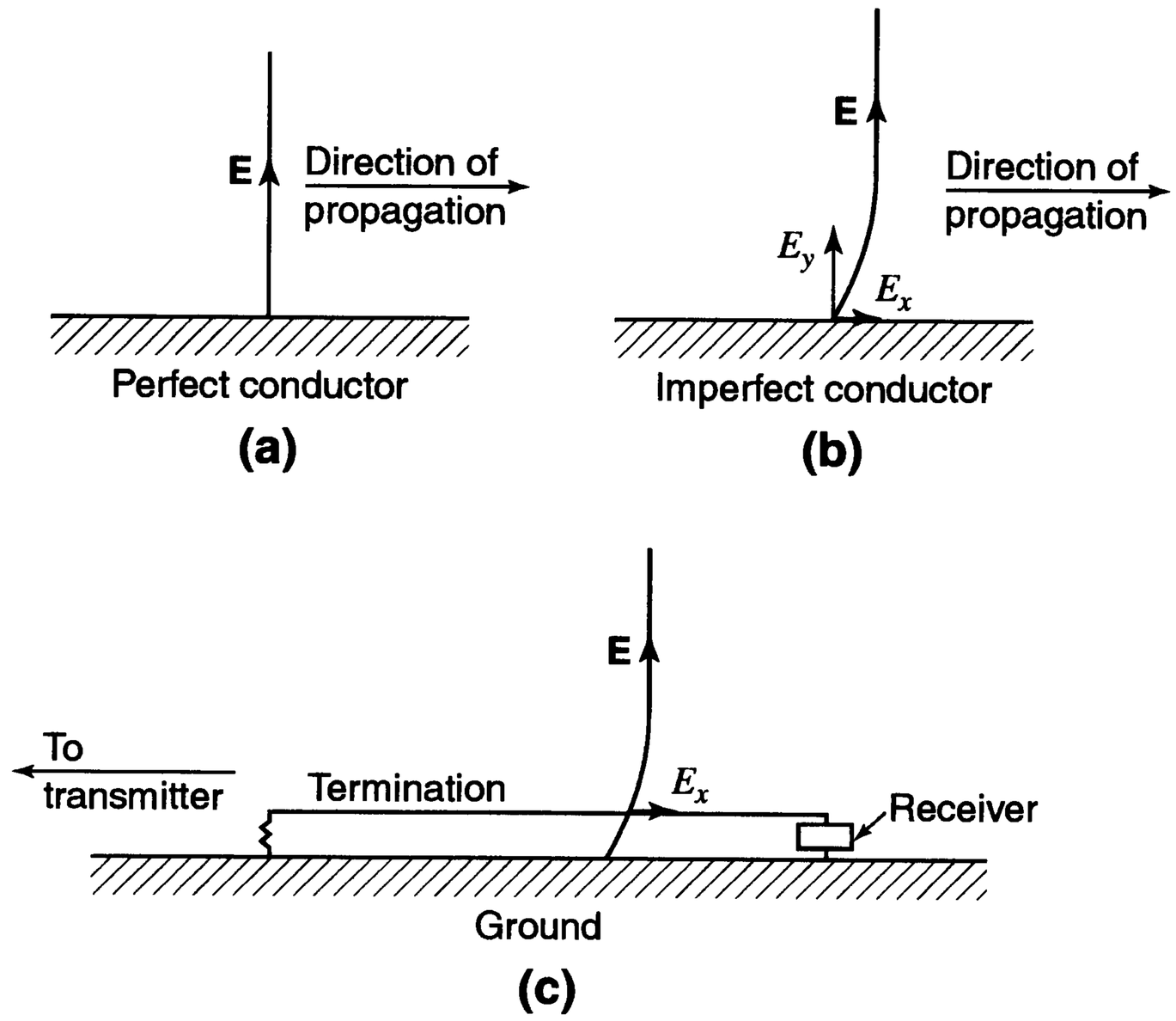


Figure 16–52
(*a*) Wave front over a perfect conductor. (*b*) Wave front over imperfect conductor. (*c*) Beverage antenna.

component E_x.[1] The component E_x is associated with that part of the wave that enters the surface and is dissipated as heat. The E_y component continues to travel along the surface.

The fact that a horizontal component E_x exists is applied in the wave antenna of Beverage (1), Rice and Kellogg for receiving vertically polarized waves. This antenna consists of a long horizontal wire terminated in its characteristic impedance at the end toward the transmitting station as in Fig. 16–52*c*. The ground acts as the imperfect conductor. The emfs induced along the antenna by the E_x component, as the wave travels toward the receiver, all add up in the same phase at the receiver. Energy from a wave arriving from the opposite direction is largely absorbed in the termination. Hence, the antenna exhibits a directional pattern in the horizontal plane with maximum response in the direction of the termination (to the left in Fig. 16–52*c*). The Beverage antenna finds application as a receiving antenna in the low- and medium-frequency range.

16–17 CURTAIN ARRAYS（帘幕形阵）

In short-wave communications the curtain type of array finds many applications. As an example, a curtain type is illustrated in Fig. 16–53 that consists of an array of $\lambda/2$ dipoles with a similar curtain at a distance of about $\lambda/4$ acting as a reflector (Brückmann-1). If the array is large in terms of wavelengths, the reflector curtain is nearly equivalent to a large sheet reflector. See also the array of Fig. 20–15 (photograph).

Several other examples of curtain arrays are the *Bruce type* (Bruce-1) of Fig. 16–53*b*, the *Sterba curtain* (Sterba-1) of Fig. 16–53*c* and the *Chireix-Mesny type* (Chireix-1) of

[1] Actually the wave exhibits elliptical cross-field, i.e., the electric vector describes an ellipse whose plane is parallel to the direction of propagation. However, the axial ratio of this ellipse is usually very large.

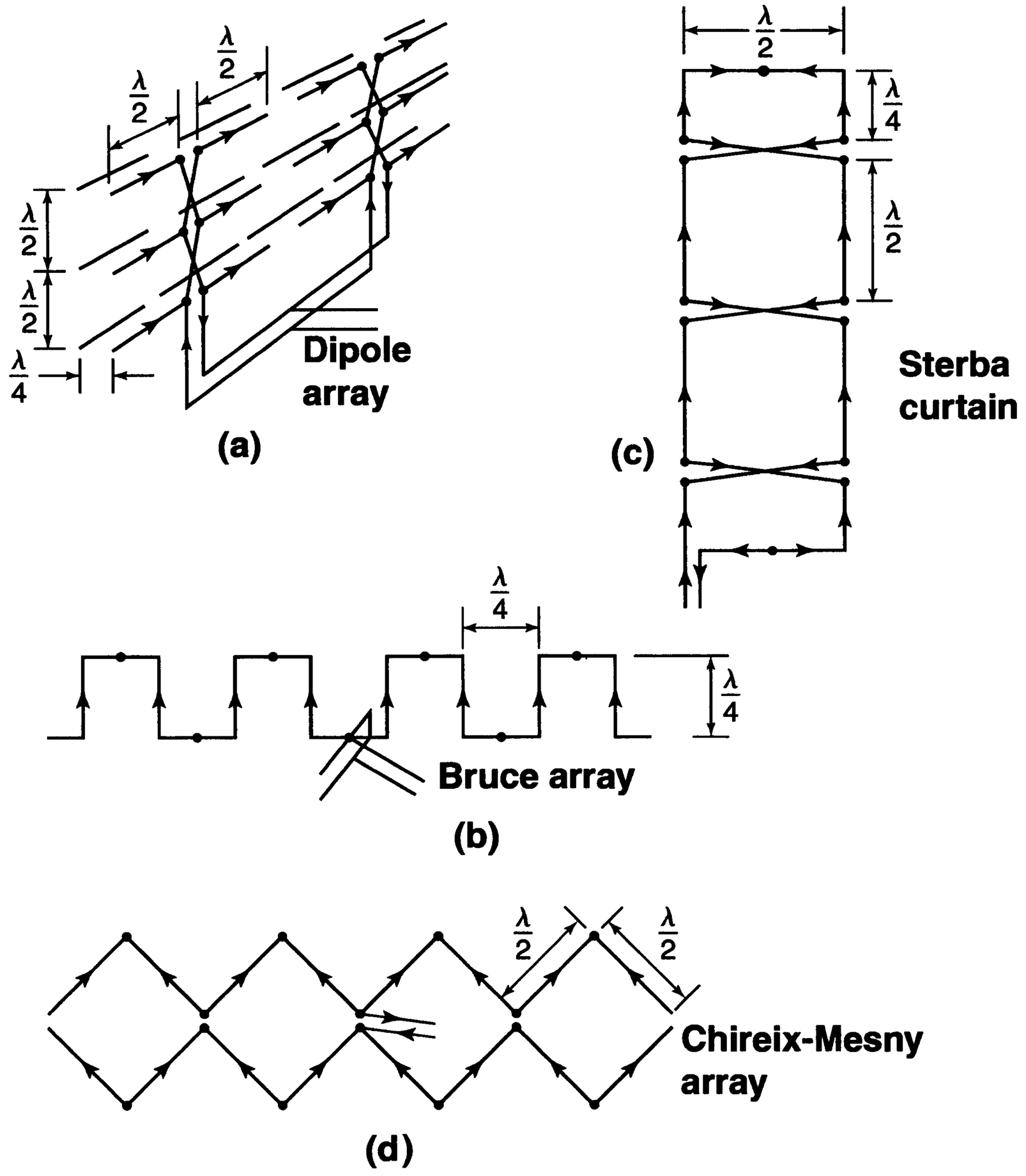


Figure 16–53
(*a*) Array of $\lambda/2$ dipoles with reflectors, (*b*) symmetrical Bruce antenna, (*c*) Sterba curtain array and (*d*) Chireix-Mesny array. Arrows indicate instantaneous current directions and dots indicate current minimum points.

Fig. 16–53*d*. The arrows are located at or near current maxima and indicate the instantaneous current direction. The small dots indicate the locations of current minima.

16–18 LOCATION AND METHOD OF FEEDING ANTENNAS
（天线的馈电位置和方式）

It is interesting to note the effect that the method and location of feeding has on the characteristics of an antenna. As illustrations, let us consider the following cases.

If an antenna is fed with a balanced 2-wire line, equal out-of-phase currents must flow at the feed point. Thus, a square loop 1λ in perimeter and fed at the bottom as in Fig. 16–54*a* must have the current distribution indicated. The arrows indicate the instantaneous current directions and the dots the locations of current minima. The radiation normal to this loop is horizontally polarized.

Consider now the situation shown in Fig. 16–54*b*. Here the loop is fed at the same location. However, the loop is continuous and is fed at a point by an unbalanced line. In this case, the antenna currents flowing to the feed point are equal and in phase, so that the

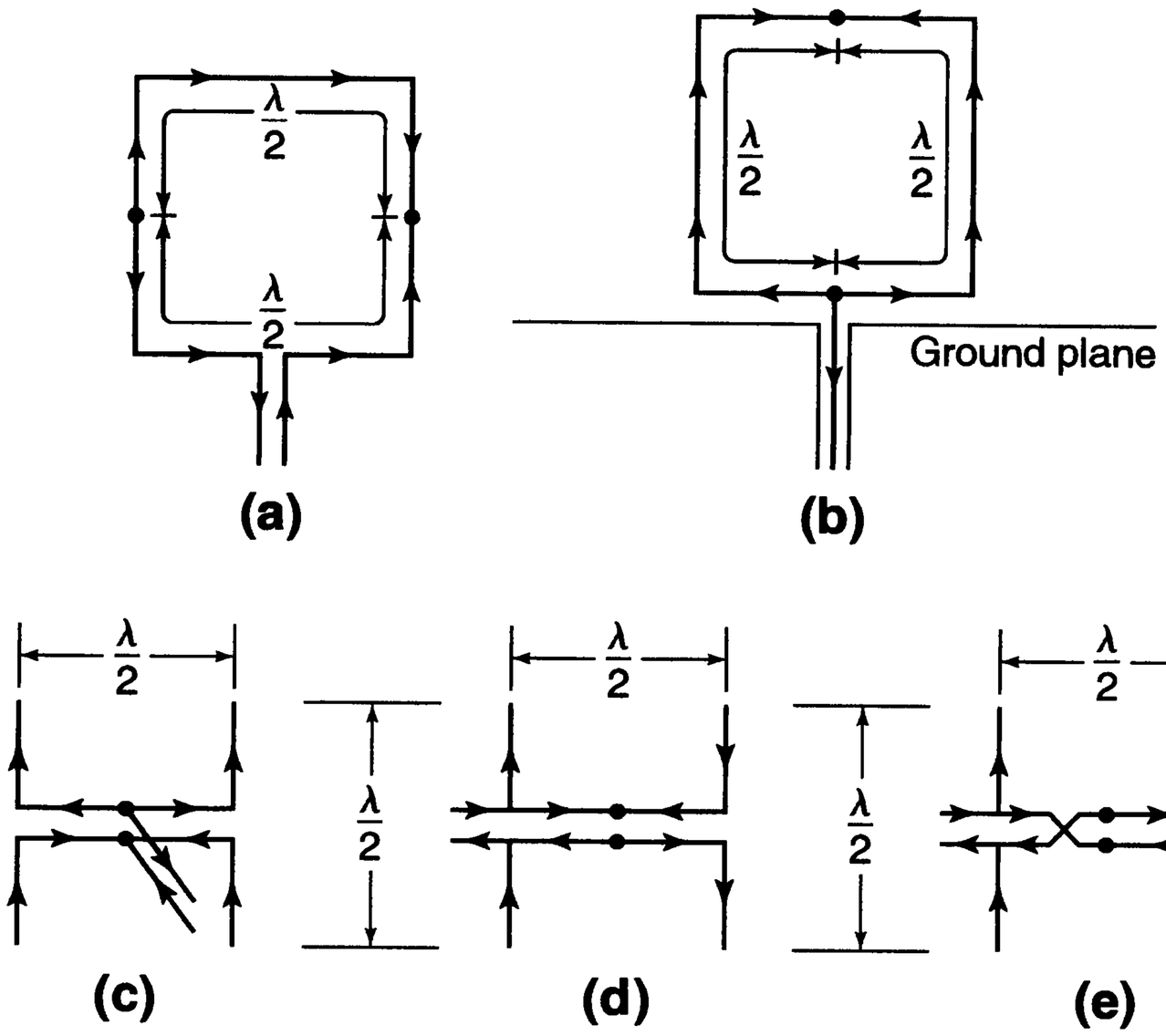


Figure 16–54
(*a*) Loop with 2-wire feed for horizontal polarization, (*b*) loop with 1-wire feed from coaxial line for vertical polarization, (*c*) center-fed broadside array of two $\lambda/2$ dipoles, (*d*) end-fed end-fire array of two $\lambda/2$ dipoles and (*e*) end-fed broadside array of two $\lambda/2$ dipoles. Arrows indicate instantaneous current directions and dots indicate current minimum points.

current distribution on the antenna must be as indicated. The radiation normal to this loop is vertically polarized.

The location at which an antenna is energized also may be important. For example, two $\lambda/2$ elements have in-phase currents when symmetrically fed as in Fig. 16–54*c* but out-of-phase currents when fed from one end as in Fig. 16–54*d*. For the currents to be in phase when the array is fed from one end requires that the line between the elements be transposed as in Fig. 16–54*e*.

16–19 FOLDED DIPOLE ANTENNAS（折合偶极子天线）

A simple $\lambda/2$ dipole has a terminal resistance of about 70 Ω so that an impedance transformer is required to match this antenna to a 2-wire line of 300 to 600 Ω characteristic impedance. However, the terminal resistance of the modified $\lambda/2$ dipole shown in Fig. 16–55 is nearly 300 Ω so that it can be directly connected to a 2-wire line having a characteristic impedance of the same value. This "ultra close-spaced type of array" is called a *folded dipole*. More specifically the one in Fig. 16–55*a* is a 2-wire folded $\lambda/2$ dipole. The antenna consists of 2 closely spaced $\lambda/2$ elements connected together at the outer ends. The currents in the elements are substantially equal and in phase.

Assuming that both conductors of the dipole have the same diameter, the approximate value of the terminal impedance may be deduced very simply as follows (King-1; Roberts-1). Let the emf V applied to the antenna terminals be divided between the 2 dipoles as in Fig. 16–55*b*. Then

$$\frac{V}{2} = I_1 Z_{11} + I_2 Z_{12} \tag{1}$$

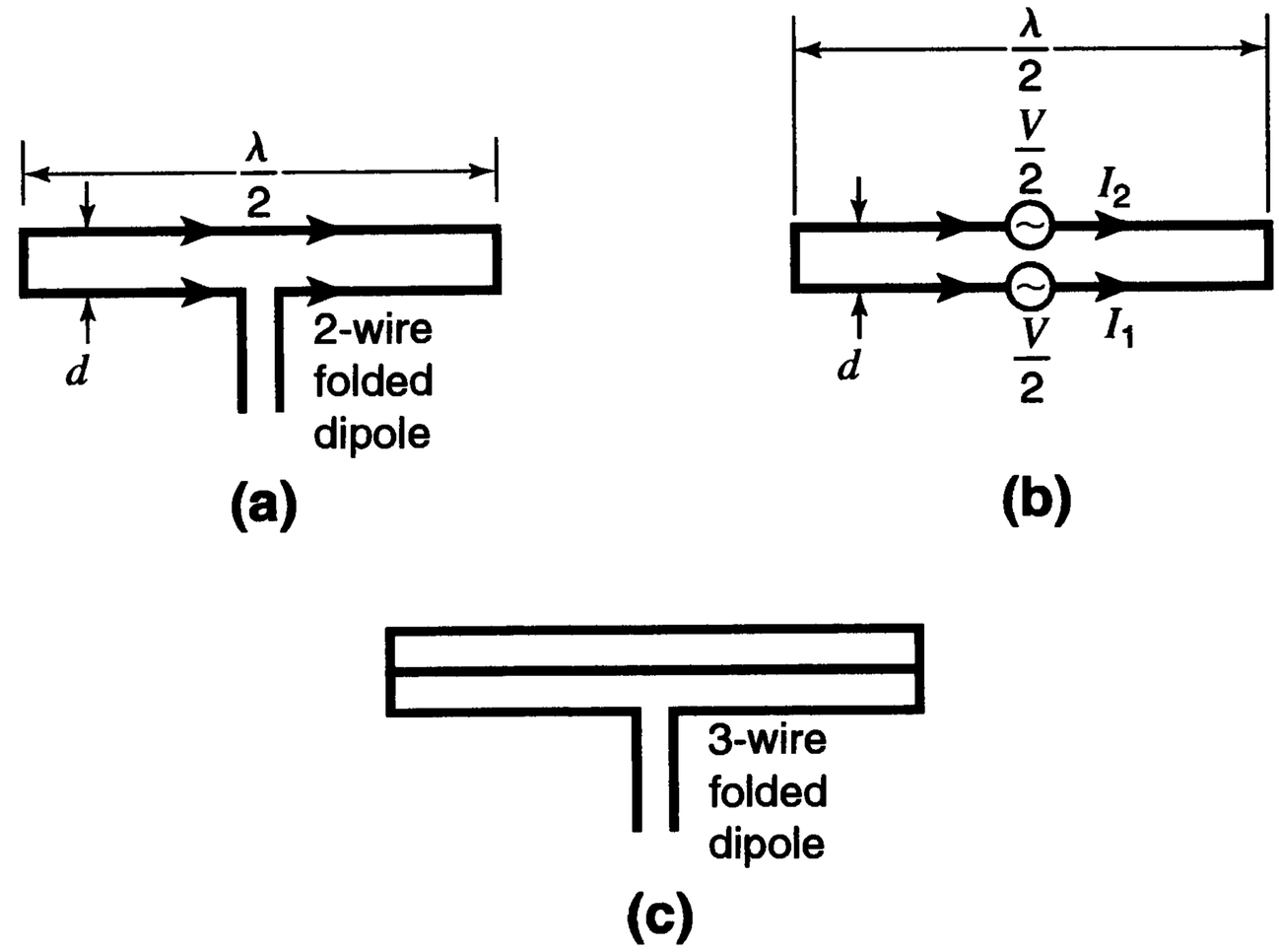


Figure 16–55
Folded dipoles.

where

I_1 = current at terminals of dipole 1
I_2 = current at terminals of dipole 2
Z_{11} = self-impedance of dipole 1
Z_{12} = mutual impedance of dipoles 1 and 2

Since $I_1 = I_2$, (1) becomes

$$V = 2I_1(Z_{11} + Z_{12}) \tag{2}$$

Further, since the 2 dipoles are close together, usually d is of the order of $\lambda/100$, $Z_{12} \simeq Z_{11}$. Thus, the terminal impedance Z of the antenna is given by

$$Z = \frac{V}{I_1} \simeq 4Z_{11} \tag{3}$$

Taking $Z_{11} \simeq 70 + j0\ \Omega$ for a $\lambda/2$ dipole, the terminal impedance of the 2-wire folded dipole becomes

$$Z \simeq 280\ \Omega$$

For a 3-wire folded $\lambda/2$ dipole as in Fig. 16–55*c* the terminal resistance calculated in this way is $9 \times 70 = 630\ \Omega$. In general, for a folded $\lambda/2$ dipole of N wires, the terminal resistance is $70N^2\ \Omega$. Equal currents in all wires are assumed.

Several other types of folded-wire antennas (Kraus-9) are shown in Fig. 16–56. The one at (*a*) is a 3-wire type which differs from the one in Fig. 16–55*c* in that there are no closed loops. The measured terminal resistance of this antenna is about 1200 Ω. The antenna at (*b*) is a 4-wire type with a measured terminal resistance of about 1400 Ω. Thus far, all the folded dipoles discussed have been $\lambda/2$ types. The total current distribution for these types is nearly sinusoidal, the same as for a simple $\lambda/2$ dipole. Folded dipoles of length other than $\lambda/2$ are illustrated in Fig. 16–56*c* and *d*. The one at (*c*) is a 2-wire type $3\lambda/4$ long and that at

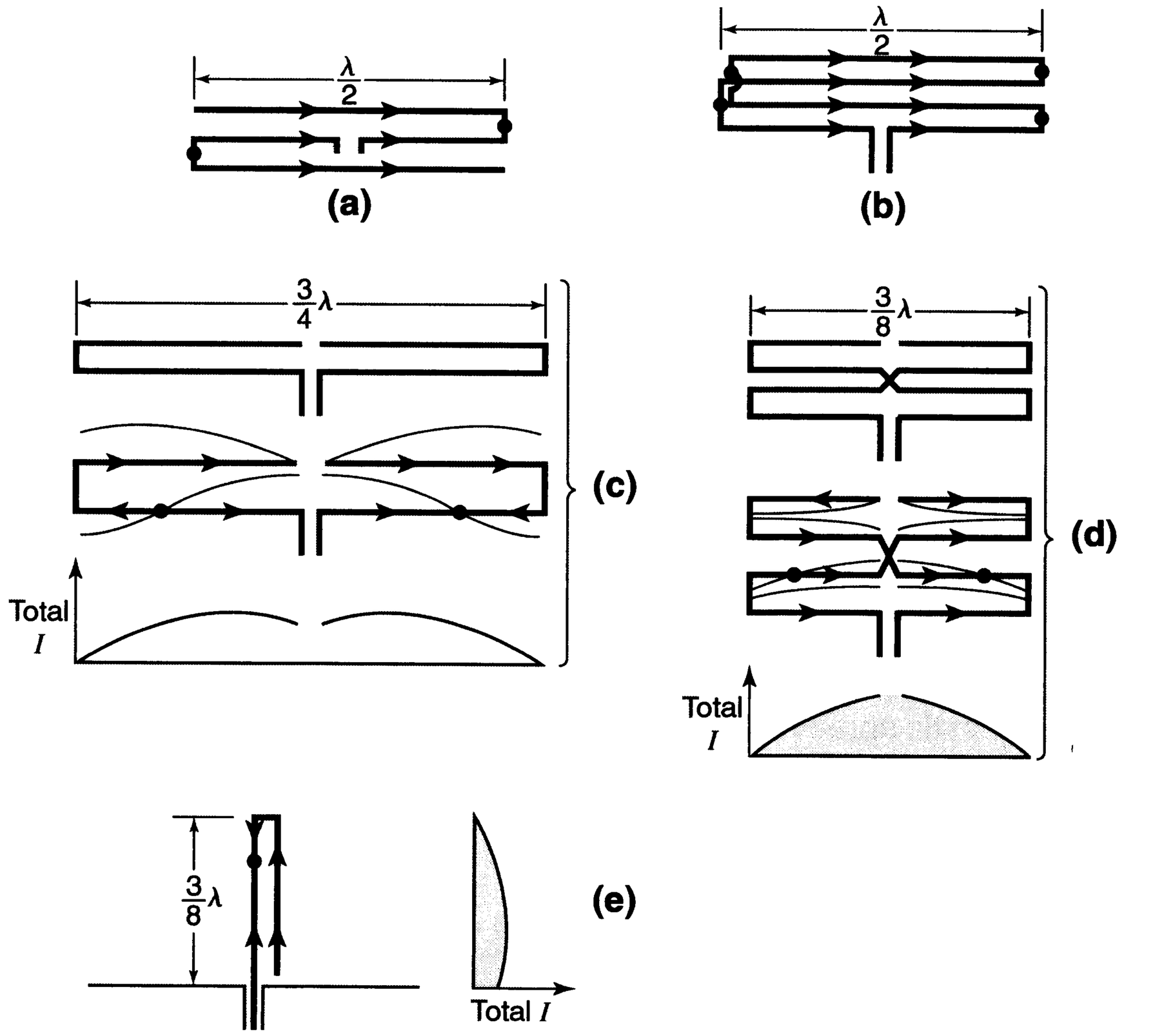


Figure 16–56
(*a*) Three-wire folded $\lambda/2$ dipole, (*b*) 4-wire folded $\lambda/2$ dipole, (*c*) 2-wire $3\lambda/4$ antenna, (*d*) 4-wire $3\lambda/8$ antenna and (*e*) 2-wire $3\lambda/8$ stub antenna. Arrows indicate instantaneous current directions and dots indicate current minimum points. (*After Kraus-9.*)

(*d*) is a 4-wire type $3\lambda/8$ long. The instantaneous current directions, the current distribution on the individual conductors and the total current distribution are also indicated. Half of the 2-wire $3\lambda/4$ dipole can be operated with a ground plane as in Fig. 16–56, yielding the $3\lambda/8$ stub antenna with total current distribution shown. The measured terminal resistance of the 2-wire $3\lambda/4$ dipole is about 450 Ω, of the 4-wire $3\lambda/8$ dipole about 225 Ω and of the 2-wire $3\lambda/8$ stub antenna about 225 Ω.

An application of the 3-wire folded $\lambda/2$ dipole of Fig. 16–56*a* to a W8JK array with $\lambda/5$ spacing is shown in Fig. 16–57 (Kraus-10). The impedance of each folded dipole in free space is about 1200 Ω (resistive) but in the array is reduced to 300 Ω, which transforms via a $\lambda/4$ 600-Ω line to 1200 Ω. At the junction of the two transformers the impedance is half 1200 Ω, or 600 Ω, matching a 600-Ω line to the transmitter or receiver. Thus, the W8JK array is fed entirely by lines of constant impedance (600 Ω) with no resonant stubs or tuning adjustments required.

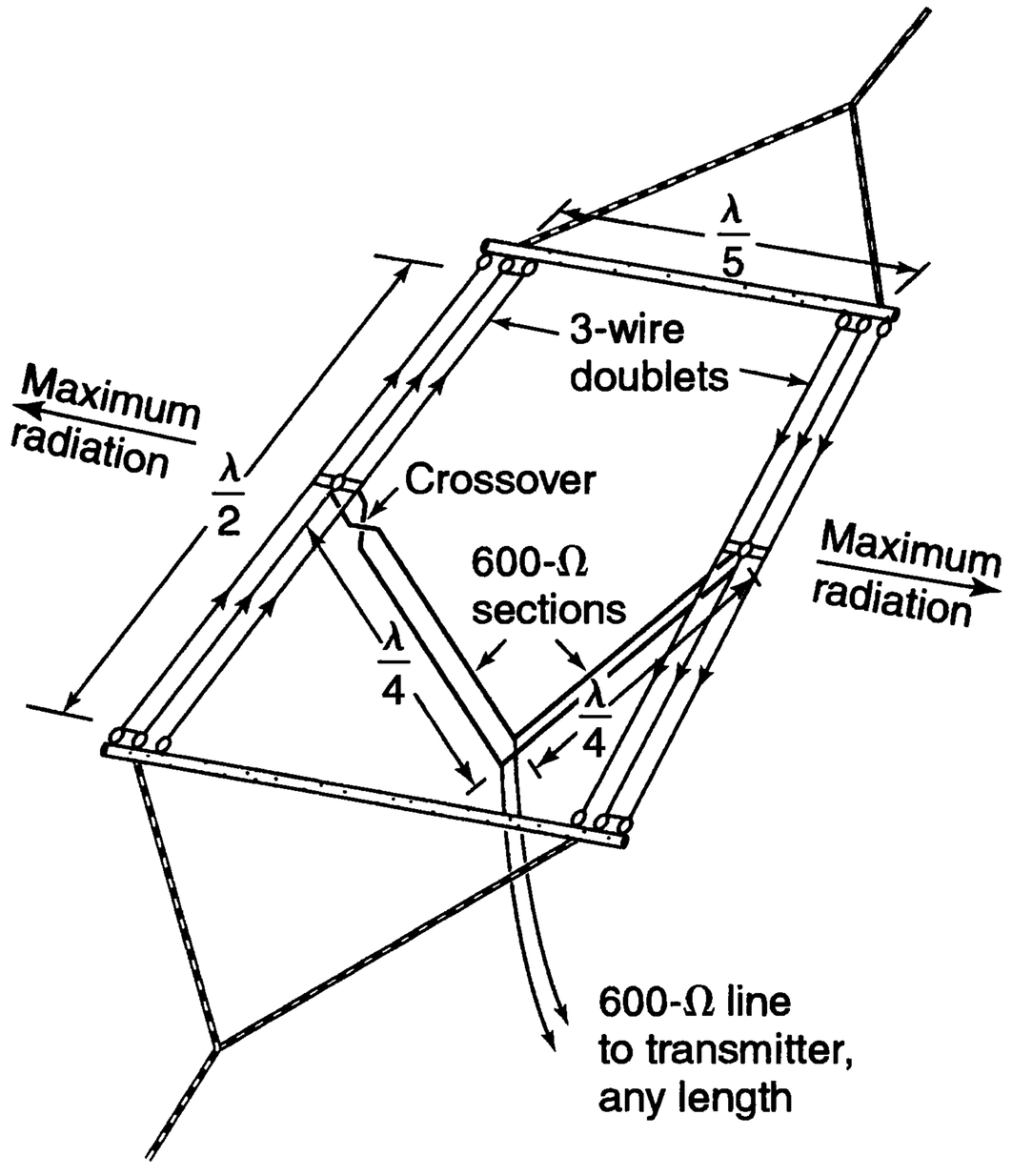


Figure 16–57
W8JK array with 3-wire folded dipole elements fed by transmission lines of constant impedance. The dipoles are separated by wooden or plastic spreaders and supported by nylon rope.

16–20 MODIFICATIONS OF FOLDED DIPOLES（折合偶极子的变种）

Consider a 2-wire folded dipole shown in Fig. 16–58*a*. The terminal resistance is approximately 300 Ω. By modifying the dipole to the general form shown in Fig. 16–58*b*, a wide range of terminal resistances can be obtained, depending on the value of *D*. This arrangement is called a *T-match antenna* (Kraus (11) and Sturgeon). Dimensions in wavelengths for providing an impedance match to a 600-Ω line are shown in Fig. 16–58*c*. See also Fig. 23–19.

A 2-wire folded λ/2 dipole is also shown in Fig. 16–59*a*. The arrows indicate the instantaneous current direction and the small dots indicate the locations of current minima.

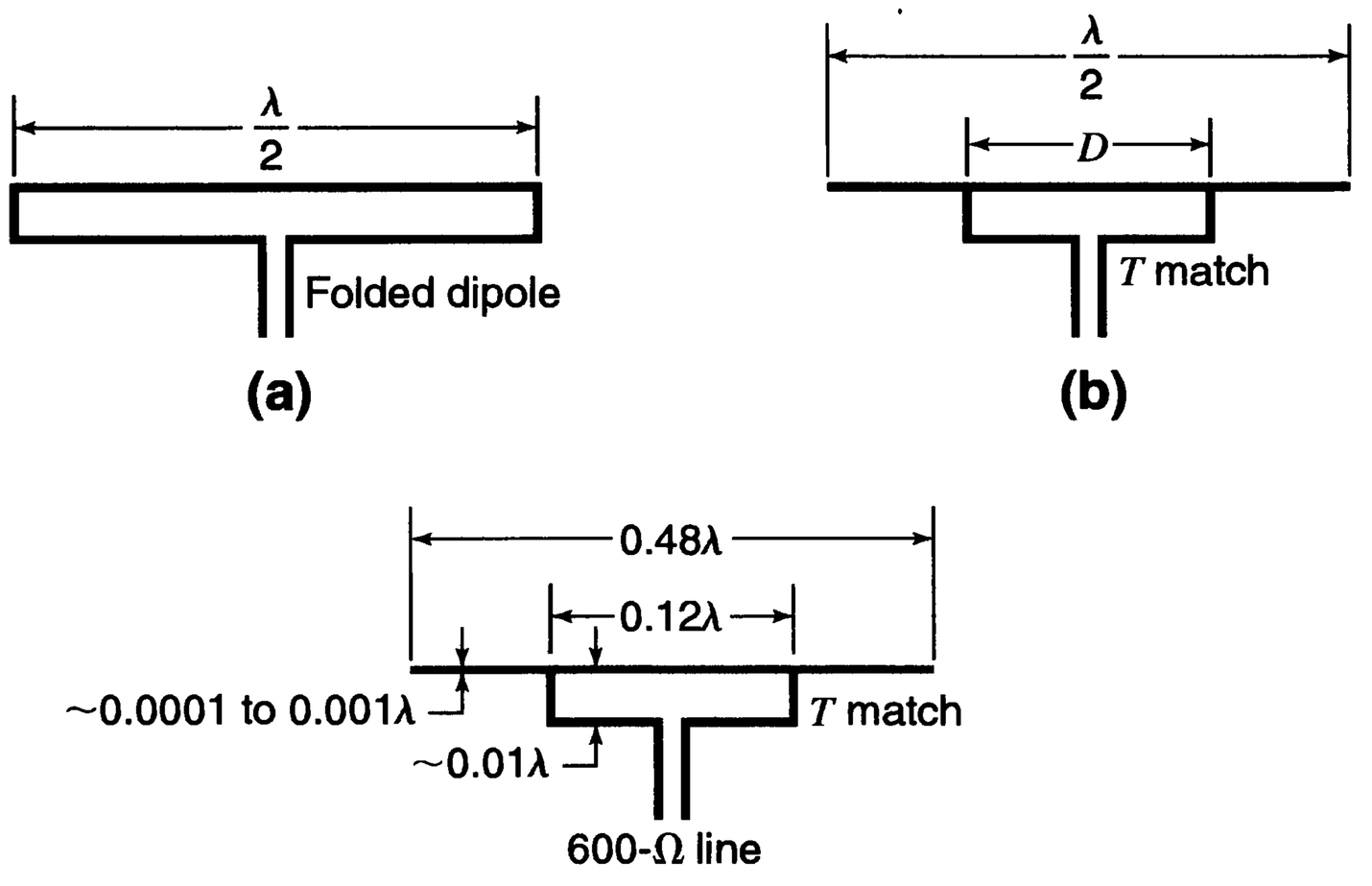


Figure 16–58
Folded dipole and T-match antennas.

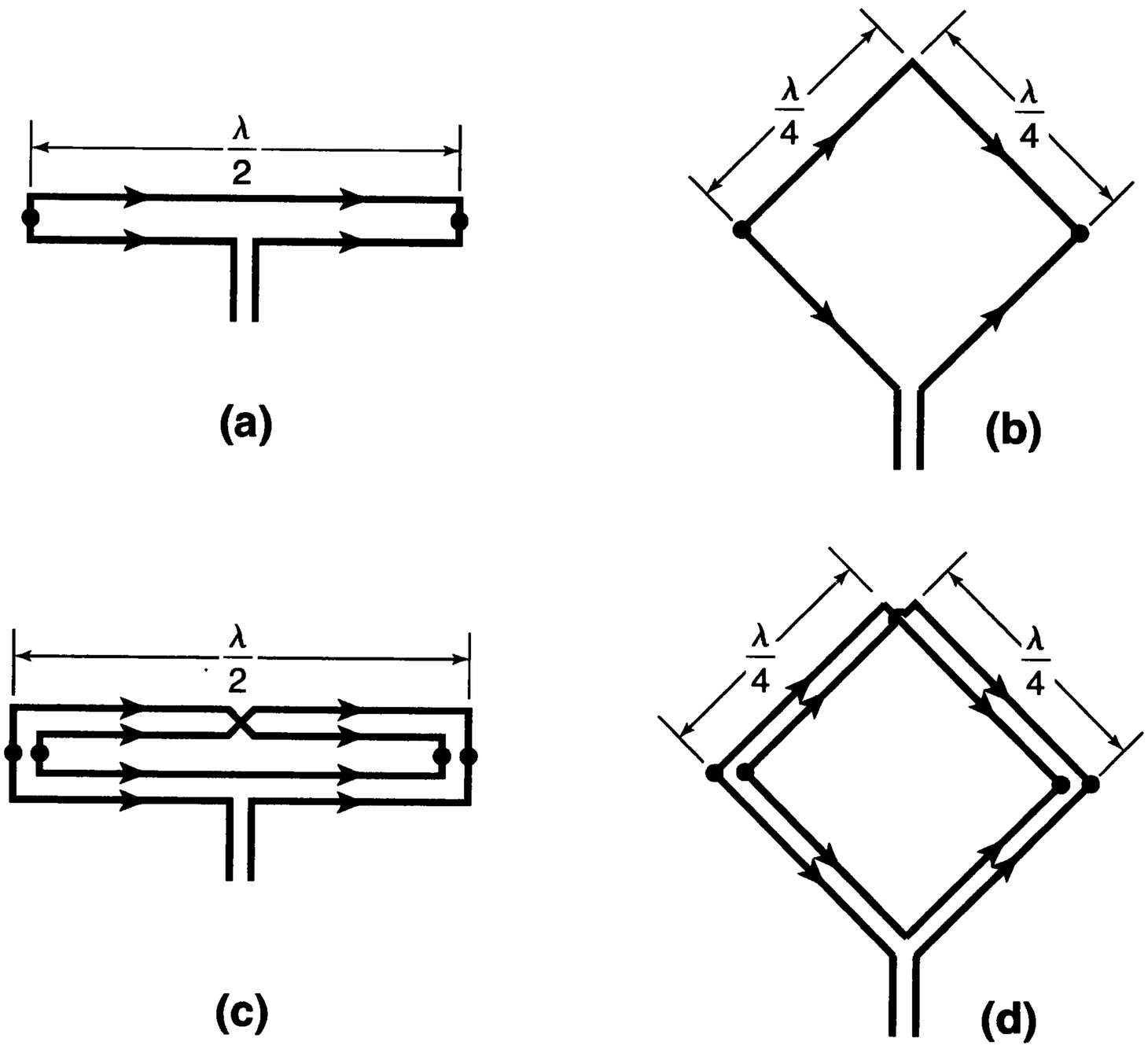


Figure 16–59
(*a*) Two-wire folded dipole and (*b*) as modified to form single-turn loop. (*c*) Four-wire folded dipole and (*d*) as modified to form 2-turn loop.

By pulling the dipole wires apart at the center, the single-turn loop antenna of Fig. 16–59*b* is obtained. The length of each side is $\lambda/4$. The loop has a lower terminal resistance than the folded dipole.

A 4-wire folded $\lambda/2$ dipole is shown in Fig. 16–59*c*. This dipole is the same type as shown in Fig. 16–56*b*. It is, however, sketched in a different manner. By pulling this dipole apart at the center the 2-turn loop or *quad antenna* of Fig. 16–59*d* results.

The directivity of all the types shown in Fig. 16–59 is nearly the same as for a simple $\lambda/2$ dipole, although the types at (*b*) and (*d*) are equivalent to 2 horizontal dipoles stacked $\sim 0.18\lambda$ giving a small increase. With the loop types vertical and the terminals at the lowest corner, the radiation normal to the plane of the loops is horizontally polarized.

16–21 ARGUS TYPE ALL-SKY SURVEILLANCE ANTENNAS
（百眼巨人型全空监视天线）

Argus is the ultimate surveillance antenna with a continuous real-time image of the entire sky over a wide bandwidth. Depending on the surveillance requirements, an Argus antenna array might consist of 16 to 1,000,000 individual antenna units. Each antenna unit contains a receiver and analog-to-digital converter with all outputs connected to a computer.

Argus could replace huge, expensive dish antennas that can only look in one direction at a time with an array of small, low-cost, mass-produced antenna units that monitor the entire sky in real time. The Argus array is analogous to the photographic film of a camera where each photosensitive crystal is connected to a computer. Instead of forming beams, Argus measures the electric field incident on its aperture and applies the appropriate

signal-processing algorithm. By comparing instantaneous sky images with a time-average sky image, even short bursts of signal from anywhere in the sky could be noted.[1]

Argus differs from the usual digital-beam-forming array (DBA) in that it provides continuous all-sky coverage (equivalent to multiple simultaneous beams) as compared to DBA's more limited coverage.

Argus, named for the see-all, many-eyed guardian of Greek mythology, is the ultimate surveillance antenna. The SETI Institute is interested in using an Argus array to Search for ExtraTerrestrial Intelligence (SETI) at frequencies between 1 and 2 GHz where the galactic noise background is the lowest (~3 K, see Fig. 21–44). Another application is envisioned by the Netherlands Foundation for Research in Astronomy with a Square Kilometer Array (SKA) of one million square meters of collecting aperture covering a frequency range of 200 to 2000 MHz (Braun-1). There would be 32 clusters, each with thousands of individual wideband antenna units. Most of these would be deployed within an inner area of 50 km diameter (for gain) and the rest within larger areas of 300 km diameter (for resolution). Thus, there would be sufficient sensitivity to observe weak sources and applying aperture-synthesis techniques to achieve angular resolutions of 1/10 arcsecond on stronger sources. It would have the potential of producing high-resolution sky images instantaneously as compared to the slow process of moving the large VLA dishes (Fig. 15–14) to different configurations, taking days or weeks to achieve the same results.

An Argus prototype was constructed at the Ohio State University in 1987 by Bolinger (1). Robert Dixon of O.S.U. has been an ardent proponent of the Argus concept (Dixon-1, 2). A prototype at O.S.U. has 64 spiral antennas deployed within an area 7.5 m in diameter (Bartlein-1). The array is too small for detailed sky observations but would be useful as a companion to large communication, earth station or radio astronomy installations to detect and locate interfering signals and provide real-time nulling for interference mitigation (Ellingson-1).

The Netherlands Foundation for Research in Astronomy (NFRA) is developing a Thousand Element Array (THEA) (see Fig. 16–60) which consists of 1024 receiving wide-band antenna elements for detecting (strong) radio sources in the frequency band from 600 to 1700 MHz in the presence of strong RF interfering signals. THEA is built up of 16 tiles (or basic units) each consisting of 64 antenna elements. The THEA system has special features, such as multibeaming (32 simultaneous beams on the sky), adaptive RFI nulling, no moving parts and on-line interference monitoring. In addition THEA is set up as a flexible system;

---

[1] In many ways Argus is analogous to the human eye-brain combination. The eye has more than 100 million rods and cones acting as the antennas. These are connected via bipolar cells, dendrites and axon transmission lines to the brain for signal processing and image recognition. The brain, with its hundred billion neurons or nerve cells surrounded by a trillion supporting cells, acts as the computer.

With peripheral vision, the eye has nearly hemispherical coverage and very high, one-arcminute, resolution or sharpness of vision over most of it. And, it does this over millions of channels covering a two-to-one bandwidth from red to blue.

Argus works at wavelengths almost a million times longer than the light wavelengths the eye detects. An eye scaled up to work at radio wavelengths would be one kilometer across! So think of it this way, a human has two microminature Argus arrays peering out of its head with many microminature super-Cray supercomputers inside. One may say that Nature anticipated Argus by millions of years.

There is one point of difference, however, between Argus and the eye. The eye has a lens whereas Argus does not.

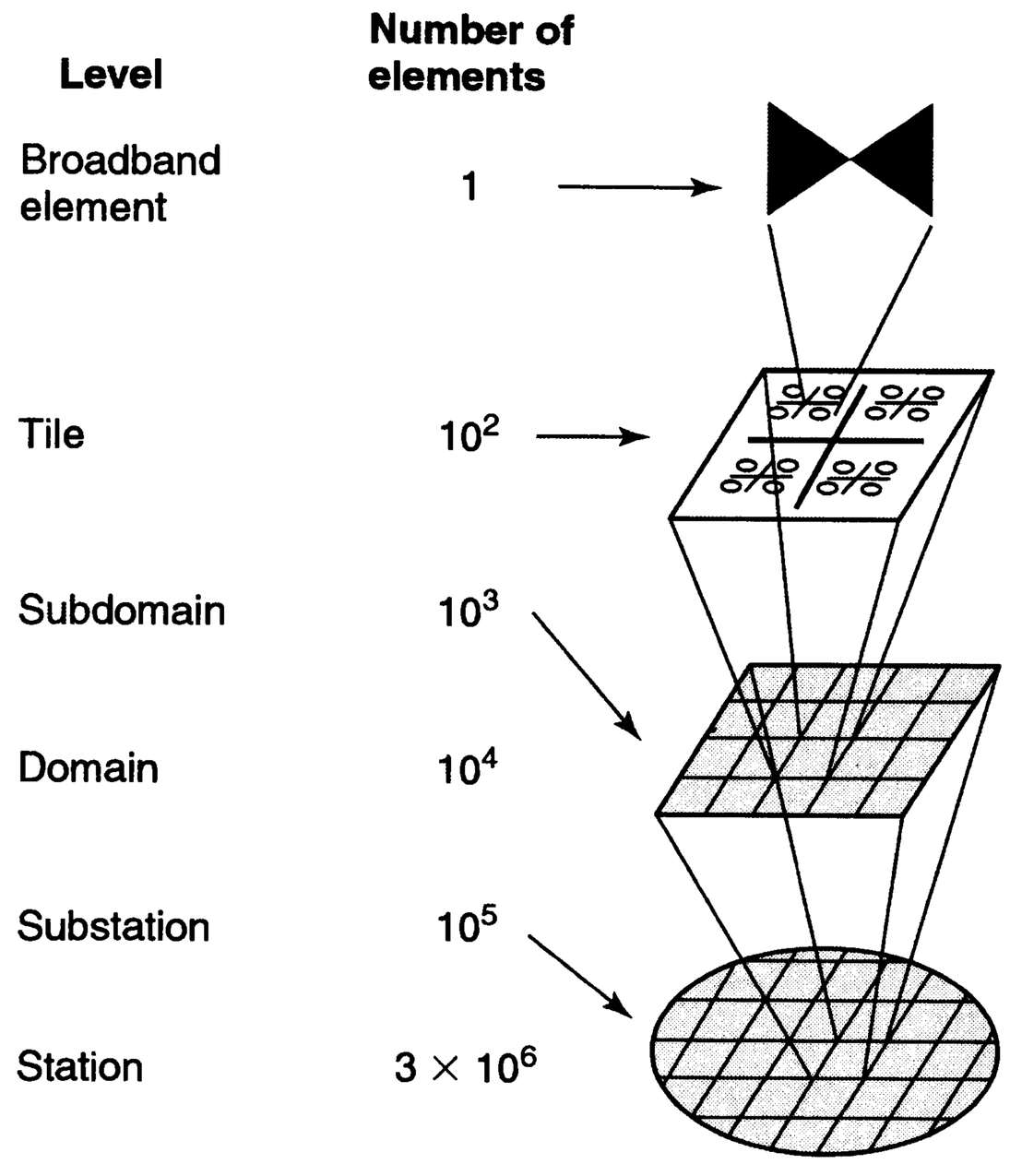


Figure 16–60
Hierarchy of elements that make up a station in the Square Kilometer Array (SKA). (*After NFRA Newsletter, September 1998.*)

e.g., the location of the tiles can be changed and the very powerful digital back end can be easily reconfigured (Smolders-1).

The Low Frequency Array (LOFAR) is a telescope that will operate at the lowest radio frequencies that are accessible from the earth. The current plan is that LOFAR will work in the range from 10 to 150 MHz, although a possible extension to higher frequencies is also under study. The telescope is being developed by NFRA, located in Dwingeloo (the Netherlands), the Naval Research Laboratory in Washington (United States) and MIT Haystack Observatory (United States) and in collaboration with Fokker Space in Leiden (Bregman-1).

REFERENCES（参考文献）

Andre, S. N. (1), and D. J. Leonard, "An Active Retrodirective Array for Satellite Communications," *IEEE Trans. Ants. Prop.*, **AP-12,** 181–186, March 1964.

ARRL-1, *The A.R.R.L. Antenna Book,* American Radio Relay League, West Hartford, Conn., 1984, p. 7–4.

Bartlein, B. A. (1), and J. Brinkman, "Argus Antenna Design," private communication.

Beverage, H. H. (1), C. W. Rice and E. W. Kellogg, "The Wave Antenna, a New Type of Highly Directive Antenna," *Trans. AIEE,* **42,** 215, 1923.

Bolinger, J. L. (1), "A Simultaneous Multi-beam Phased Array Using Digital Processing Technique," M.S.E.E. thesis, Ohio State University, 1988.

Braun, R. (1), "The Concept of the Square Kilometer Array Interferometer," in High Sensitivity Radio Astronomy, M. Jackson and R. J. Davies (eds.), Cambridge University Press, 1997, pp. 260–268.

Bregman, J. D. (1), "Concept Design for a Low Frequency Array," SPIE Proceedings Astronomical Telescopes and Instrumentation 2000, Radio Telescopes, vol. 4051, 2000.

Brown, G. H. (1), "A Critical Study of the Characteristics of Broadcast Antennas as Affected by Antenna Current Distribution," *Proc. IRE,* **24,** 48–81, January 1936.

Brown, G. H. (2), "Directional Antennas," *Proc. IRE,* **25,** 78–145, January 1937.

Bruce, E. (1), "Development in Short-wave Directive Antennas," *Proc. IRE,* **19,** 1406–1433, August 1931.

Bruce, E. (2), A. C. Beck and L. R. Lowry, "Horizontal Rhombic Antennas," *Proc. IRE,* **23,** 24–46, January 1935.

Brückmann, H. (1), *Antennen, ihre Theorie und Technik,* S. Hirzel, Leipzig, 1939, p. 300.

Butler, J. (1), and R. Lowe, "Beamforming Matrix Simplifies Design of Electronically Scanned Antennas," *Elect. Des.,* **9,** 170–173, 1961.

Butler, J. (2), in R. C. Hansen (ed.), *Microwave Scanning Antennas,* vol. 3, Academic Press, 1966, Chap. 3 (includes 80 references).

Carter, P. S. (1), "Circuit Relations in Radiating Systems and Applications to Antenna Problems," *Proc. IRE,* **20,** 1007, June 1932.

Carter, P. S. (2), C. W. Hansell and N. E. Lindenblad, "Development of Directive Transmitting Antennas by R. C. A. Communications, Inc.," *Proc. IRE,* **19,** 1773–1842, October 1931.

Chireix, H. (1), "French System of Directional Aerials for Transmission on Short Waves," *Exp. Wireless and Wireless Engr.,* **6,** 235, May 1929.

Conti, R. (1), J. Toth, T. Dowling and J. Weiss, "The Wire Grid Microstrip Antenna," *IEEE Trans. Ants. Prop.,* **AP-29,** 157–166, January 1981.

Dixon, R. S. (1), "Argus: A Next-Generation Omnidirectional Powerful Radiotelescope," *Acta Astronautica,* **35,** no. 9–11, 1995, p. 745.

Dixon, R. S. (2), "Argus: A Future SETI Telescope," Progress in the Search for Extraterrestrial Life, Astronomical Society of the Pacific Series 74, vol. 74, 1995, pp. 355–366.

Dorne, A. (1), in *Very High Frequency Techniques,* Radio Research Laboratory Staff, McGraw-Hill, New York, 1947, Chap. 4, p. 115.

Ellingson, S. W. (1), and G. A. Hampson, "A Subspace-Tracking Approach to Interference Nulling for Phased Array-Based Radio Telescopes," *IEEE Trans. Ants. Prop.,* in press.

Foster, Donald (1), "Radiation from Rhombic Antennas," *Proc. IRE,* **25,** 1327–1353, October 1937.

Friis, H. T. (1), and C. B. Feldman, "A Multiple Unit Steerable Antenna for Short-Wave Reception," *Proc. IRE,* **25,** 841–917, July 1937.

Hall, P. S. (1), "Microstrip Antenna Array with Multi-Octave Bandwidth," *Microwave J.,* **29,** 133–139, March 1986.

Harper, A. E. (1), *Rhombic Antenna Design,* Van Nostrand, New York, 1941.

Hendriksson, J. (1), K. Markus and M. Tiuri, "A Circularly Polarized Traveling-Wave Chain Antenna," *European Microwave Conf.,* Brighton, September 1979.

Hildebrand, L. T. (1), and D. A. McNamara, "Experimental Verification of Spacial Domain Integral Equation Analysis of Microstrip Wire-Grid Arrays," Dept. of Electrical and Electronic Engineering, University of Pretoria.

King, R. W. P. (1), H. R. Mimno and A. H. Wing, *Transmission Lines, Antennas and Wave Guides,*" McGraw-Hill, New York, 1945, p. 224.

Kraus, J. D. (1), "Antennas, Our Electronic Eyes and Ears," *Microwave Jour.,* January 1989, pp. 77–92.

Kraus, J. D. (2), "Antenna Arrays with Closely Spaced Elements," *Proc. IRE,* **28,** 76–84, February 1940.

Kraus, J. D. (3), "Characteristics of Antennas with Closely Spaced Elements," *Radio,* 9–19, February 1939.

Kraus, J. D. (4), "Planetary and Solar Radio Emission at 11 m Wavelength," *Proc. IRE,* **46,** 266–274, January 1958.

Kraus, J. D. (5), *Radio Astronomy,* 2d ed., Cygnus-Quasar, 1986, p. 6–40 (1st ed., 1966, p. 193).

Kraus, J. D. (6), "Backward Angle-Fire Array Antenna," *IEEE Trans. Ants. Prop.,* **AP-12,** 48–50, January 1964.

Kraus, J. D. (7), "Backward Angle Traveling Wave Wire Mesh Antenna Array," U.S. Patent 3,290,688, applied for June 11, 1962, granted Dec. 6, 1966.

Kraus, J. D. (8), and K. R. Carver, "Wave Velocities on the Grid-Structure Backward Angle-Fire Antenna," *IEEE Trans. Ants. Prop.,* **AP-12,** 509–510, July 1964.

Kraus, J. D. (9), Multi-wire Dipole Antennas," *Electronics,* **13,** 26–27, January 1940.

Kraus, J. D. (10), "Twin-Three Flat-Top Beam Antenna," *Radio,* no. 243, 10–16, November 1939.

Kraus, J. D. (11), and S. S. Sturgeon, "The T-Matched Antenna," *QST,* **24,** 24–25, September 1940.

Labus, J. (1), "Rechnerische Ermittlung der Impedanz von Antennen," *Hochfrequenztechnik und Electroakustik,* **41,** 17–23, January 1933.

Landsdorfer, F. M. (1), "Zur Optimalen Form von Linearantennen," *Frequenz,* **30,** 344–349, 1976.

Landsdorfer, F. M. (2), "A New Type of Directional Antenna," *Ant. Prop. Soc. Int. Symp. Digest,* 169–172, 1976.

Landsdorfer, F. M. (3), and R. R. Sacher, *Optimization of Wire Antennas,* Wiley, New York, 1985.

Mailloux, R. J. (1), "Phased Array Theory and Technology," *Proc. IEEE,* **70,** 246–291, March 1982 (includes 174 references).

Oltman, H. G. (1), and D. A. Huebner, "Electromagnetically Coupled Microstrip Dipoles," *IEEE Trans. Ants. Prop.,* **AP-29,** 151–157, January 1981.

Pon, C. Y. (1), "Retrodirective Array Using the Heterodyne Method," *IEEE Trans. Ants. Prop.,* **AP-12,** 176–180, March 1964.

Regier, F. (1), "A New Look at the W8JK Antenna," *Ham Radio,* 60–63, July 1981.

Roberts, W. V. B. (1), "Input Impedance of a Folded Dipole," *RCA Rev.,* **8,** 289–300, June 1947, which treats folded dipoles with conductors of equal diameter and also unequal diameter.

Schrank, H. E. (1), "Low Sidelobe Phased Array Antennas," *IEEE Ant. Prop. Soc. Newsletter,* **25,** 5–9, April 1983.

Smith, C. E. (1), *Directional Antennas,* Cleveland Institute of Radio Electronics, Cleveland, Ohio, 1946.

Smolders, A. (1), and D. Kant, "System Specification THEA," NFRA-document SKA-development-140, 1999.

Stark, L. (1), "Microwave Theory of Phased Array Antennas—A Review," *Proc. IEEE,* **62,** 1661–1701, 1974.

Sterba, E. J. (1), "Theoretical and Practical Aspects of Directional Transmitting Systems," *Proc. IRE,* **19,** 1184–1215, July 1931.

Steyskal, H. (1), "Digital Beamforming Antennas," *Microwave J.,* **30,** 107–124, January 1987.

Tallqvist, S. (1), "Theory of the Meander Type Chain and Grid Antenna," Lic. thesis, Helsinki University of Technology, 1977.

Tiuri, M. (1), S. Tallqvist and S. Urpo, "Chain Antenna," *IEEE Ant. Prop. Soc. Int. Symp. Digest,* June 10, 1974.

Van Atta, L. C. (1), "Electromagnetic Reflector," U.S. Patent 2,909,002, Oct. 6, 1959.

ADDITIONAL REFERENCES ON ADAPTIVE ARRAYS（自适应阵的补充参考资料）

There is an extensive literature on adaptive arrays. Three special issues of the *IEEE Transactions on Antennas and Propagation* have been devoted to adaptive arrays. They are: vol. AP-12, March 1964; vol. AP-24, September 1976; and vol. AP-34, March 1986. Additional references are as follows:

Bickmore, R. W., "Time Versus Space in Antenna Theory," in R. C. Hansen (ed.), *Microwave Scanning Antennas,* vol. 3, Academic Press, 1966, pp. 289–339.

Blank, S., "An Algorithm for the Empirical Optimization of Antenna Arrays," *IEEE Trans. Ants. Prop.,* **AP-31,** 685–689, July 1983.

Butler, J. L., "Digital, Matrix, and Intermediate Frequency Scanning," in R. C. Hansen (ed.), *Microwave Scanning Antennas,* vol. 3, Academic Press, 1966, pp. 217–288.

Dinger, R. J., "A Computer Study of Interference Nulling by Reactively Steered Adaptive Arrays," *Ant. Prop. Soc. Int. Symp. Proc.,* **2,** 807–810, 1984.

Einarsson, O., Optimization of Planar Arrays, *IEEE Trans. Ants. Prop.,* **AP-27,** 86–92, January 1979.

Ersoy, O., "Real Discrete Fourier Transform," *IEEE Trans. Acoustics, Speech, and Signal Processing,* **ASSP-33,** 880–882, August 1985.

Fan, H., E. I. El-Masry and W. K. Jenkins, "Resolution Enhancement of Digital Beamformers," *IEEE Trans. Acoustics, Speech, and Signal Processing,* **ASSP-32,** 1041–1052, October 1984.

Griffiths, L. J., and C. W. Jim, "An Alternative Approach to Linearly Constrained Adaptive Beamforming," *IEEE Trans. Ants. Prop.,* **AP-30,** January 1982.

Gupta, I. J., and A. A. Ksienski, "Effect of Mutual Coupling on the Performance of Adaptive Arrays," *IEEE Trans. Ants. Prop.,* **AP-31,** 785–791, September 1983.

Hansen, R. C., "Gain Limitations of Large Antennas," *IRE Trans. Ants. Prop.,* **AP-8,** 491–495, September 1960.

Hatcher, B. R., "Granularity of Beam Positions in Digital Phased Arrays," *Proc. IEEE,* **56,** November 1968.

Johnson, H. W., and C. S. Burrus, "The Design of Optimal DFT Algorithms Using Dynamic Programming," *IEEE Trans. Acoustics, Speech, and Signal Processing,* **ASSP-31,** 378–387, April 1983.

Laxpati, S. R., "Planar Array Synthesis with Prescribed Pattern Nulls," *IEEE Trans. Ants. Prop.,* **AP-30,** 1176–1183, November 1982.

Mucci, R. A., "A Comparison of Efficient Beamforming Algorithms," *IEEE Trans. Acoustics, Speech, and Signal Processing,* **ASSP-32,** 548–558, June 1984.

Ricardi, L. J., "Adaptive Antennas," in R. C. Johnson and H. Jasik (eds.), *Antenna Engineering Handbook,* McGraw-Hill, 1984, Chap. 22.

Shelton, J. P., and K. S. Kelleher, "Multiple Beams from Linear Arrays," *IRE Trans. Ants. Prop.,* **AP-9,** 154–161, March 1961.

Sorenson, H. V., M. T. Heideman and C. S. Burrus, "On Computing the Split-Radix FFT," *IEEE Trans. Acoustics, Speech, and Signal Processing,* **ASSP-34,** 152–156, February 1986.

Steinberg, B. D., "Design Approach for a High-Resolution Microwave Imaging Radio Camera," *J. Franklin Inst.,* **296,** 415–432, December 1973.

Volakis, J. L., and J. D. Young, "Phase Linearization of a Broad-Band Antenna Response in Time Domain," *IEEE Trans. Ants. Prop.,* **AP-30,** 309–313, March 1982.

Vu, T. B., "Null Steering by Controlling Current Amplitudes Only," *Ant. Prop. Soc. Int. Symp. Proc.,* **2,** 811–814, 1984.

Waldron, T. P., S. K. Chin and R. J. Naster, "Distributed Beamsteering Control of Phased Array Radars," *Microwave J.,* **29,** 133–146, September 1986.

Weber, M. E., and R. Heisler, "A Frequency-Domain Beamforming Algorithm for Wideband Coherent Signal Processing," *J. Acoustic Society of America,* **76,** October 1984.

Wheeler, H. A., "The Grating-Lobe Series for the Impedance Variation in a Planar Phased-Array Antenna," *IEEE Trans. Ants. Prop.,* **AP-14,** 707–714, November 1966.

PROBLEMS（习题）

***16–2–1**
（两根λ/2单元的边射阵）

Two λ/2-element broadside array. (*a*) Calculate and plot the gain of a broadside array of 2 side-by-side λ/2 elements in free space as a function of the spacing *d* for values of *d* from 0 to 2λ. Express the gain with respect to a single λ/2 element. Assume all elements are 100 percent efficient. (*b*) What spacing results in the largest gain? (*c*) Calculate and plot the radiation field patterns for λ/2 spacing. Show also the patterns of the λ/2 reference antenna to the proper relative scale.

16–3–1
（两根λ/2单元的端射阵）

Two λ/2-element end-fire array. A 2-element end-fire array in free space consists of 2 vertical side-by-side λ/2 elements with equal out-of-phase currents. At what angles in the horizontal plane is the gain equal to unity: (*a*) When the spacing is λ/2? (*b*) When the spacing is λ/4?

***16–3–2**
（两单元阵的阻抗和增益）

Impedance and gain of 2-element array. Two thin center-fed λ/2 antennas are driven in phase opposition. Assume that the current distributions are sinusoidal. If the antennas are parallel and spaced 0.2λ, (*a*) Calculate the mutual impedance of the antennas. (*b*) Calculate the gain of the array in free space over one of the antennas alone.

16–4–1
（两根同相λ/2天线的最大增益）

Maximum gain of 2 in-phase half-wave antennas. What spacing of two in-phase side-by-side half-wave antennas produces the maximum gain? What is the gain in dBi? See Example 16–4.1.

16–4–2 (两单元的铅垂极化阵) **Two-element VP array.** Calculate and plot the field and phase patterns of the far field for an array of 2 vertical side-by-side $\lambda/2$ elements in free space with $\lambda/4$ spacing when the elements are: (*a*) in phase and (*b*) 180° out of phase. For the in-phase case also include on the graph the patterns in both the yz or vertical plane and xy or horizontal plane of Fig. 16–2*a*. For the out-of phase case do the same for the pattern in both the xz or vertical plane and xy or horizontal plane of Fig. 16–7*a*.

16–4–3 (不等电流的两单元阵) **Two-element array with unequal currents.** (*a*) Consider two $\lambda/2$ side-by-side vertical elements spaced a distance d with currents related by $I_2 = aI_1/\delta$. Develop the gain expression in a plane parallel to the elements and the gain normal to the elements, taking a vertical $\lambda/2$ element with the same power input as reference ($0 \leq a \leq 1$). Check that these reduce to (16–4–15) and (16–4–13) when $a = 1$. (*b*) Plot the field patterns in both planes and also show the field pattern of the reference antenna in proper relative proportion for the case where $d = \lambda/4$, $a = 1/2$ and $\delta = 120°$.

16–5–1 (W8JK 阵) **W8JK array.** Calculate the vertical and horizontal plane free space field patterns of a W8JK antenna consisting of two horizontal out-of-phase $\lambda/2$ elements spaced $\lambda/8$. Assume a loss resistance of 1 Ω and show the relative patterns of $\lambda/2$ reference antenna with the same power input.

***16–6–1** (道尔夫 – 切比雪夫阵的阻抗) **Impedance of D-T array.** (*a*) Calculate the driving-point impedance at the center of each element of an in-phase broadside array of 6 side-by-side $\lambda/2$ elements spaced $\lambda/2$ apart. The currents have a Dolph-Tchebyscheff distribution such that the minor lobes have 1/5 the field intensity of the major lobe. (*b*) Design a feed system for the array.

16–6–2 (三角形阵) **Triangle array.** Three isotropic point sources of equal amplitude are arranged at the corners of an equilateral triangle, as in Fig. P16–6–2. If all sources are in phase, determine and plot the far-field pattern.

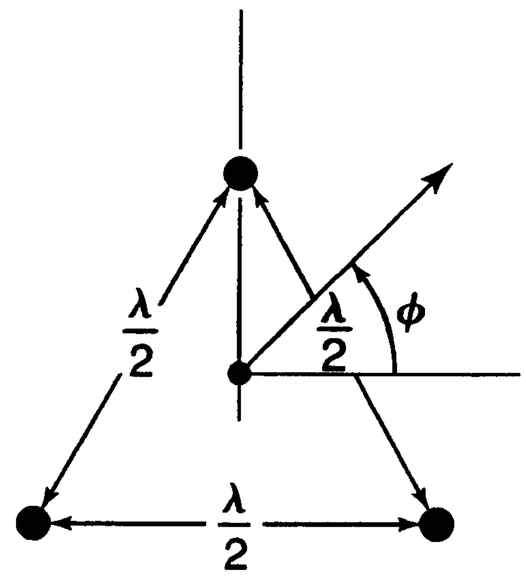


Figure P16–6–2
Triangle array.

16–6–3 (方形阵) **Square array.** Four isotropic point sources of equal amplitude are arranged at the corners of a square, as in Fig. P16–6–3. If the phases are as indicated by the arrows, determine and plot the far-field patterns.

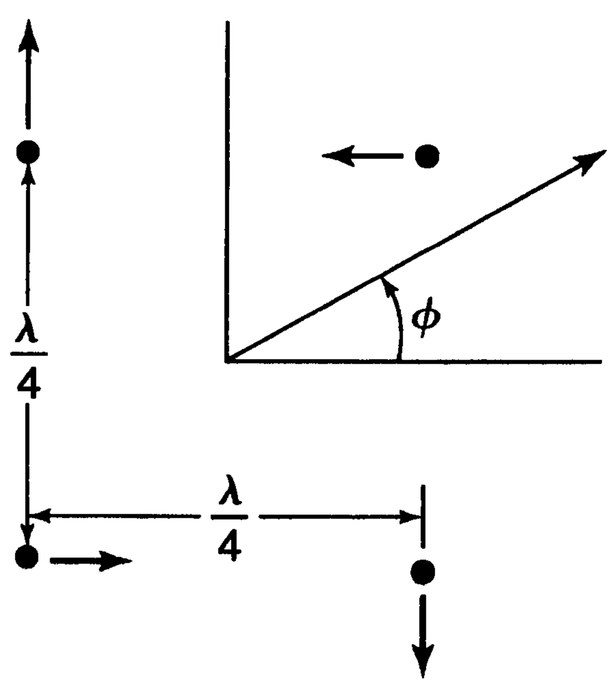


Figure P16–6–3
Square array.

***16–6–4** (七根短偶极子)

Seven short dipoles. 4-dB angle. A linear broadside (in-phase) array of 7 short dipoles has a separation of 0.35λ between dipoles. Find the angle from the maximum field for which the field is 4 dB (to nearest 0.1°).

16–6–5 (方形阵)

Square array. Four identical short dipoles (perpendicular to page) are arranged at the corners of a square $\lambda/2$ on a side. The upper left and lower right dipoles are in the same phase while the 2 dipoles at the other corners are in the opposite phase. If the direction to the right (x direction) corresponds to $\phi = 0°$, find the angles ϕ for all maxima and minima of the field pattern in the plane of the page.

16–6–6 (三根λ/2偶极子的共线阵)

Collinear array of three $\lambda/2$ dipoles. An antenna array consists of 3 in-phase collinear thin $\lambda/2$ dipole antennas, each with sinusoidal current distribution and space $\lambda/2$ apart. The current in the center dipole is twice the current in the end dipoles (binomial array). (*a*) Calculate and plot the far-field pattern. (*b*) What is the HPBW? (*c*) How does this HPBW value compare with the HPBW for a binomial array of 3 isotropic point sources spaced $\lambda/2$ apart?

***16–6–7** (16个点源的边射阵)

Sixteen-source broadside array. A uniform linear array has 16 isotropic in-phase point sources with a spacing $\lambda/2$. Calculate exactly (*a*) the half-power beamwidth, (*b*) the level of the first side lobe, (*c*) the beam solid angle, (*d*) the beam efficiency, (*e*) the directivity and (*f*) the effective aperture.

16–6–8 (笔状波束波瓣图)

Pencil-beam patterns. For symmetrical circular pencil-beam patterns (function only of θ) show that the main beam solid angle Ω_M is given by

$1.13\,\theta_{\text{HP}}^2$ for a Gaussian pattern
$0.998\,\theta_{\text{HP}}^2$ for a $(\sin x)/x$ pattern
$1.008\,\theta_{\text{HP}}^2$ for a Bessel pattern

where θ_{HP}^2 is the half-power beam width. Also show that Ω_M is given by

$1.036\,\theta_{\text{HP}}\phi_{\text{HP}}$ for a $(\sin u)/u$ (square aperture) pattern

where $u = x = y$. Note that for a Gaussian pattern $\theta_{\text{HP}}^2 = 0.88\Omega_A$ so $\Omega_M = \Omega_A$.

16–7–1 (16元阵和W8JK阵的增益公式)

Sixteen-element and W8JK array gain equations. Confirm (16–6–6) and (16–7–11).

16–7–2 (地面上的λ/2层叠单元和W8JK阵)

Staked $\lambda/2$ elements and W8JK array above ground. (*a*) Develop (16–7–12). (*b*) Calculate and plot from (16–7–12) the gain in field intensity for an array of 2 in-phase horizontal $\lambda/2$ elements stacked $\lambda/2$ apart (as in Fig. 16–31) over a $\lambda/2$ antenna in free space with the same power input as a function of h up to $h = 1.5\lambda$ for an elevation angle $\alpha = 10°$. Also calculate and plot for comparison on the same graph the gains at $\alpha = 10°$ for a 2-element horizontal W8JK antenna over a single horizontal $\lambda/2$ antenna as a function of the height above ground from $h = 0$ to $h = 1.5\lambda$. Note difference of these curves and those for $\alpha = 20°$ in Fig. 16–32.

16–8–1 (两塔广播阵)

Two-tower BC array. A broadcast-station antenna array consists of two vertical $\lambda/4$ towers spaced $\lambda/4$ apart. The currents are equal in magnitude and in phase quadrature. Assume a perfectly conducting ground and zero loss resistance. Calculate and plot the azimuthal field pattern in millivolts (rms) per meter at 1.6 km with 1 kW input for vertical elevation angle $\alpha = 0, 20, 40, 60$ and 80°. The towers are series fed at the base. Assume that the towers are infinitesimally thin.

16–8–2 (两塔广播阵)

Two-tower BC array. Calculate and plot the relative field pattern in the vertical plane through the axis of the 2-tower broadcast array fulfilling the requirements of Prob. 5–4–2 if the towers are $\lambda/4$ high and are series fed at the base. Assume that the towers are infinitesimally thin and that the ground is perfectly conducting.

16–8–3 (三塔广播阵)

Three-tower BC array. Calculate and plot the relative field pattern in the vertical plane though the axis of the 3-tower broadcast array fulfilling the requirements of Prob. 5–4–3 if the towers are $3\lambda/8$

high and are series fed at the base. Assume that the towers are infinitesimally thin and that the ground is perfectly conducting.

16–8–4 (α = 30° 为零点的广播阵)

BC array with null at $\alpha = 30°$. Design a broad cast-station antenna array of 2 vertical base-fed towers $\lambda/4$ high and spaced $3\lambda/8$ which produce a broad maximum of field intensity to the north in the horizontal plane and a null at an elevation angle $\alpha = 30°$ and azimuth angle $\phi = 135°$ measured ccw from north to reduce interference via ionospheric bounce. Assume that the towers are infinitesimally thin, that the ground is perfectly conducting and that the base currents of the two towers are equal. Specify the orientation and phasing of the towers. Calculate and plot the azimuthal field pattern at $\alpha = 10°$ and $\alpha = 30°$ and also the pattern in the vertical plane through $\phi = 135°$. The suggested procedure is as follows. Solve (16–8–4) for ϕ' at the null. Then set ϕ in the pattern factor in (16–4–13) equal to ϕ' and solve for the value δ which makes the pattern factor zero. The relative field intensity at any angle (ϕ, α) is then given by (16–4–14) where $\sin\theta = \cos\phi' = \cos\alpha\cos\phi$ in the first pattern factor and $\theta = 90° - \alpha$ in the second pattern factor.

16–8–5 (西向所有α都为零的广播阵)

BC array with null to west at all α. Design a broadcast-station array of 2 vertical base-fed towers $\lambda/4$ high that produces a broad maximum of field intensity to the north in the horizontal plane and a null at all vertical angles to the west. Assume that the towers are infinitesimally thin and that the ground is perfectly conducting. Specify the spacing, orientation and phasing of the towers. Calculate and plot the azimuthal relative field pattern at elevation angles of $\alpha = 0, 30$ and 60°.

***16–8–6** (四塔广播阵)

Four-tower broadcast array. A broadcast array has 4 identical vertical towers arranged in an east-west line with a spacing d and progressive phase shift δ. Find (*a*) d and (*b*) δ so that there is a maximum field at $\phi = 45°$ (northeast) and a null at $\phi = 90°$ (north). There can be other nulls and maxima, but no maximum can exceed the one at 45°. The distance d must be less than $\lambda/2$.

16–10–1 (8个点源的扫描阵)

Eight-source scanning array. A linear broadside array has 8 sources of equal amplitude and $\lambda/2$ spacing. Find the progressive phase shift required to swing the beam (*a*) 5°, (*b*) 10° and (*c*) 15° from the broadside direction. (*d*) Find BWFN when all sources are in phase.

16–10–2 (24根偶极子的扫描阵)

A 24-dipole scanning array. A linear array consists of an in-line configuration of 24 $\lambda/2$ dipoles spaced $\lambda/2$. The dipoles are fed with equal currents but with an arbitrary progressive phase shift δ between dipoles. What value of δ is required to put the main-lobe maximum (*a*) perpendicular to the line of the array (broadside condition), (*b*) 25° from broadside, (*c*) 50° from broadside and (*d*) 75° from broadside? (*e*) Calculate and plot the four field patterns in polar coordinates. (*f*) Discuss the feasibility of this arrangement for a scanning array by changing feed-line lengths to change δ or by keeping the array physically fixed but changing the frequency. What practical limits occur in both cases?

16–10–3 (三螺旋扫描阵)

Three-helix scanning array. Three 4-turn right-handed monofilar axial-mode helical antennas spaced 1.5λ apart are arranged in a broadside array as in Fig. 16–41. The pitch angle $\alpha = 12.5°$ and the circumference $C = \lambda$. (*a*) If the outer two helices rotate on their axes in opposite directions while the center helix is fixed, determine the angle ϕ of the main lobe with respect to the broadside direction and describe how ϕ varies as the helices rotate. (*b*) What is the maximum scan angle ϕ? (*c*) What is the main-lobe HPBW as a function of ϕ?

***16–16–1** (端接负载的V形行波天线)

Terminated V. Traveling wave. (*a*) Calculate and plot the far-field pattern of a terminated-V antenna with 5λ legs and 45° included angle. (*b*) What is the HPBW?

***16–16–2** (最大E型菱形天线)

E-type rhombic. Design a maximum E-type rhombic antenna for an elevation angle $\alpha = 17.5°$.

16–16–3 (瞄准型菱形天线)

Alignment rhombic. Design an alignment-type rhombic antenna for an elevation angle $\alpha = 17.5°$.

***16–16–4** (折中型菱形天线) **Compromise rhombic.** Design a compromise-type rhombic antenna for an elevation angle $\alpha = 17.5°$ at a height above ground of $\lambda/2$.

16–16–5 (折中型菱形天线) **Compromise rhombic.** Design a compromise-type rhombic antenna for an elevation angle $\alpha = 17.5°$ with leg length of 3λ.

***16–16–6** (折中型菱形天线) **Compromise rhombic.** Design a compromise-type rhombic antenna for an elevation angle $\alpha = 17.5°$ at a height above ground of $\lambda/2$ and a leg length of 3λ.

16–16–7 (菱形天线波瓣图) **Rhombic patterns.** Calculate the relative vertical plane patterns in the axial direction for the rhombics of Probs. 16–16–2, 16–16–3, 16–16–4, 16–16–5 and 16–16–6. Compare the patterns with the main lobes adjusted to the same maximum value.

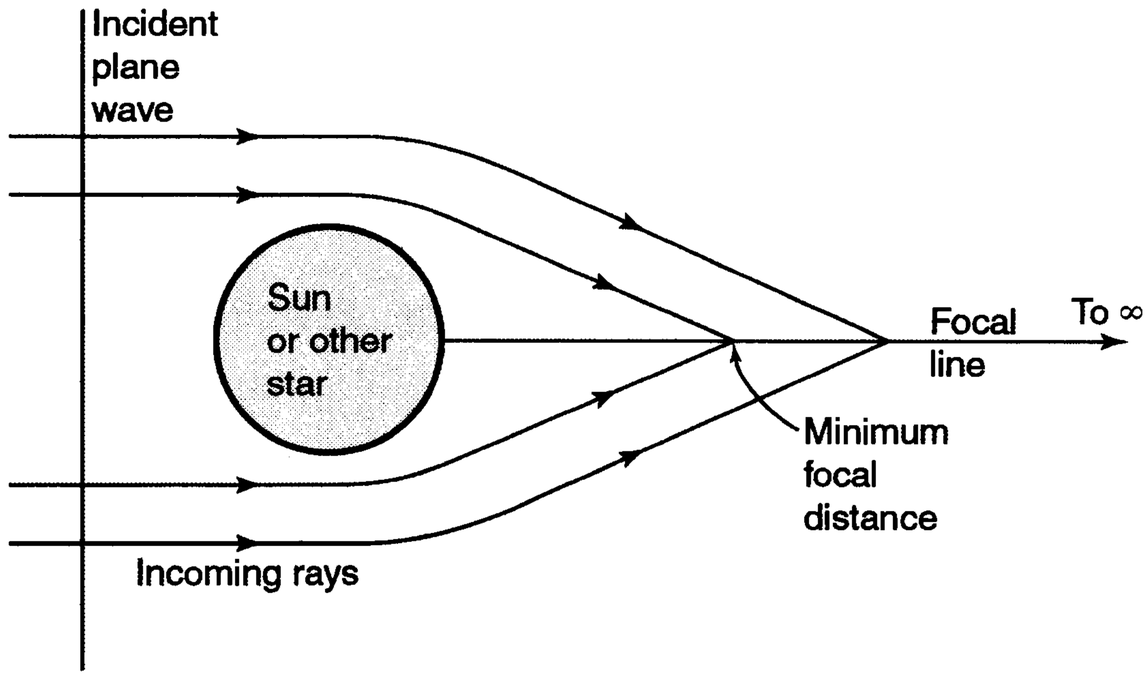
16–16–8 (菱形天线公式) **Rhombic equation.** Derive (16–16–1) for the relative field intensity of a horizontal rhombic antenna above a perfectly conducting ground.

16–16–9 (瞄准型菱形天线公式) **Alignment rhombic equation.** Verify (16–16–3), (16–16–8) and (16–16–9) for the alignment design rhombic antenna.

For computer programs, see Appendix C.

CHAPTER 17

Lens Antennas
透镜天线



Topics in this chapter include:

- Dielectric lens antennas（介质透镜天线）
- Fermat's principle（费马原理）
- Artificial dielectric lens antennas（人造介质透镜天线）
- *E*-plane metal-plate lens antennas（*E*- 面金属板透镜天线）
- Tolerances on lens antennas（透镜天线的容差）
- *H*-plane metal-plate lens antennas（*H*- 面金属板透镜天线）
- Reflector lens antennas（反射镜 – 透镜天线）
- Polyrod antennas（介质杆天线）
- Multiple-helix lens antennas（多螺旋透镜天线）
- Luneberg lens antennas（龙伯透镜天线）
- The Einstein gravity lens（爱因斯坦重力透镜）

17–1 INTRODUCTION（引言）

Lens antennas may be divided into two distinct types: (1) delay lenses, in which the electrical path length is increased by the lens medium, and (2) fast lenses, in which the electrical path length is decreased by the lens medium. In delay lenses the wave is retarded by the lens medium. Dielectric lenses and *H*-plane metal-plate lenses are of the delay type. *E*-plane metal-plate lenses are of the fast type. The actions of a dielectric lens and an *E*-plane metal-plate lens are compared in Fig. 17–1.

The dielectric lenses may be divided into two groups:

1. Lenses constructed of nonmetallic dielectrics, such as lucite or polystyrene
2. Lenses constructed of metallic or artificial dielectrics

These types are considered in the next two sections (17–2 and 17–3). *E*-plane metal-plate lenses are discussed in Sec. 17–4, tolerances in Sec. 17–5 and the *H*-plane metal-plate lens in Sec. 17–6. A reflector lens is presented in Sec. 17–7.

All lens antennas of the delay type may be regarded basically as end-fire antennas with the polyrod and monofilar axial-mode helical antennas as rudimentary forms as suggested in Fig. 17–2. Likewise, the director structure of a many-element Yagi-Uda antenna is a rudimentary lens. Polyrods are covered in Sec. 17–8, monofilar axial-mode helical antennas

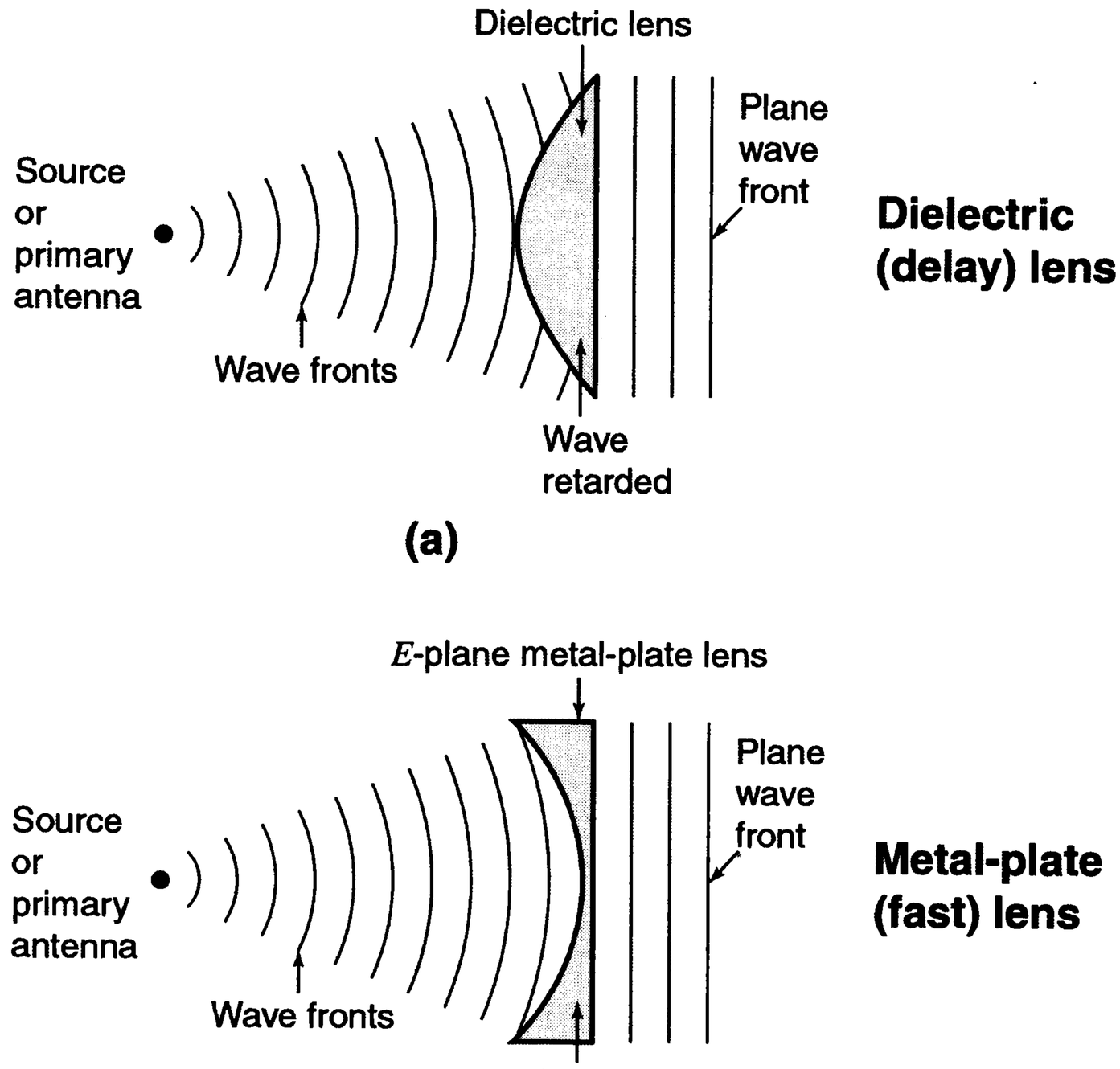


Figure 17–1
Comparison of dielectric (delay) lens and *E*-plane metal-plate (fast) lens actions.

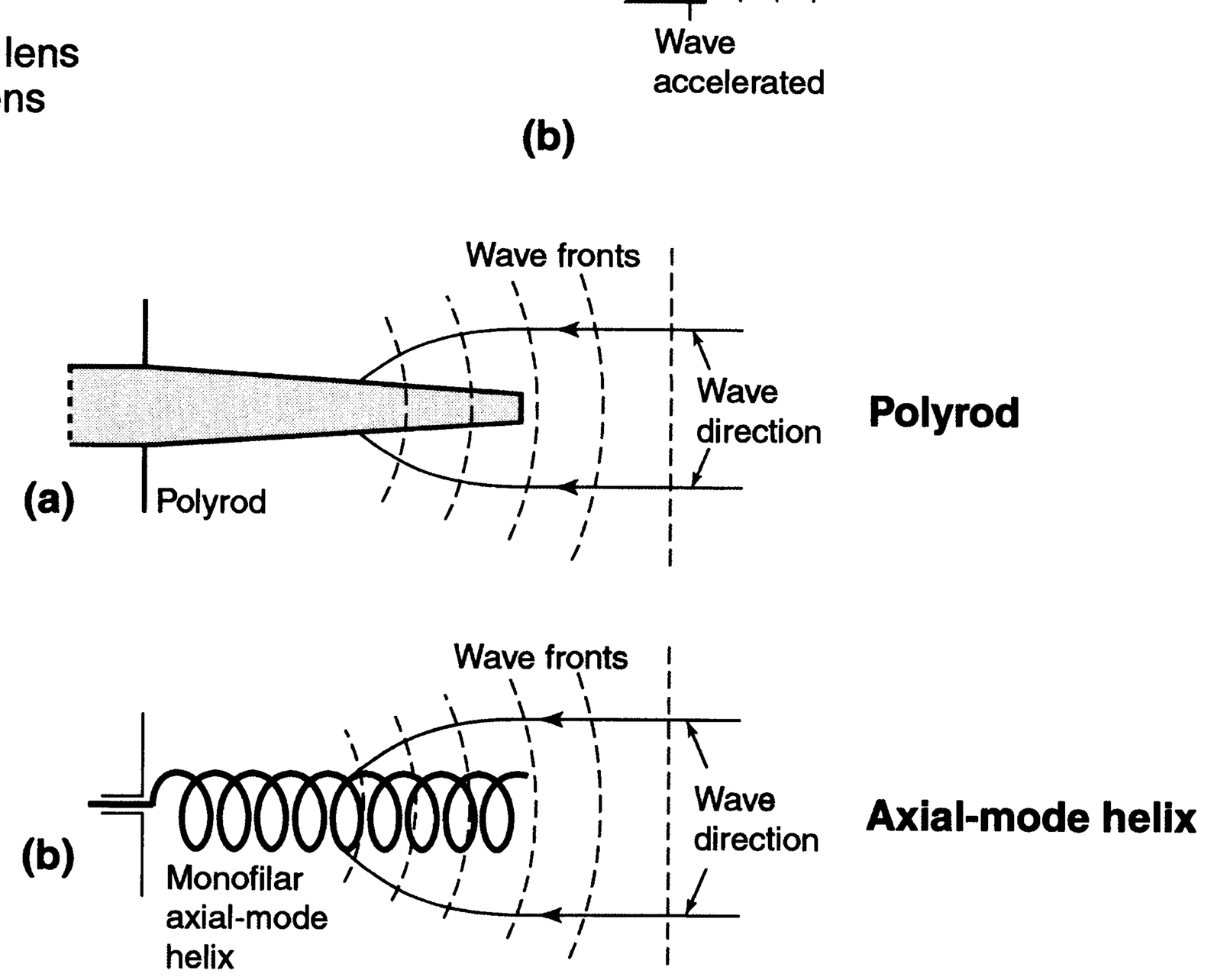


Figure 17–2
Three forms of basic lens antennas.

and Yagi-Uda antennas having already been discussed in Chap. 8. Lenses of multiple helices are also described in Sec. 17–9. The last section of this chapter (17–10) discusses two spherically symmetric lenses of special type, the Luneburg and Einstein lenses, the latter utilizing a large mass as the focusing device.

At millimeter wavelengths low-loss dielectric lens antennas are competitive in weight and performance with reflector antennas (Goldsmith-1, 2, 3).

17–2 NONMETALLIC DIELECTRIC LENS ANTENNAS. FERMAT'S PRINCIPLE (EQUALITY OF PATH LENGTH) [非金属介质透镜天线，费马原理（射径等同性）]

This type is similar to the optical lens (Risser-1). It may be designed by the ray analysis methods of geometrical optics. As an example, let us determine the shape of the plano-convex lens of Fig. 17–1*a* for transforming the spherical wave front from an isotropic point source or primary antenna into a plane wave front.[1] The field over the plane surface can be made everywhere in phase by shaping the lens so that all paths from the source to the plane are of equal electrical length. ***This is the principle of equality of electrical (or optical) path length (Fermat's principle).*** [这是射径的电（或光）长度等同性原理（费马原理）] Thus, in Fig. 17–3, the electrical length of the path OPP' must equal the electrical length of the path $OQQ'Q''$, or more simply OP must equal OQ'. Let $OQ = L$ and $OP = R$, and let the medium surrounding the lens be air or vacuum. Then

$$\frac{R}{\lambda_0} = \frac{L}{\lambda_0} + \frac{R\cos\theta - L}{\lambda_d} \tag{1}$$

where

λ_0 = wavelength in free space (air or vacuum)

λ_d = wavelength in the lens

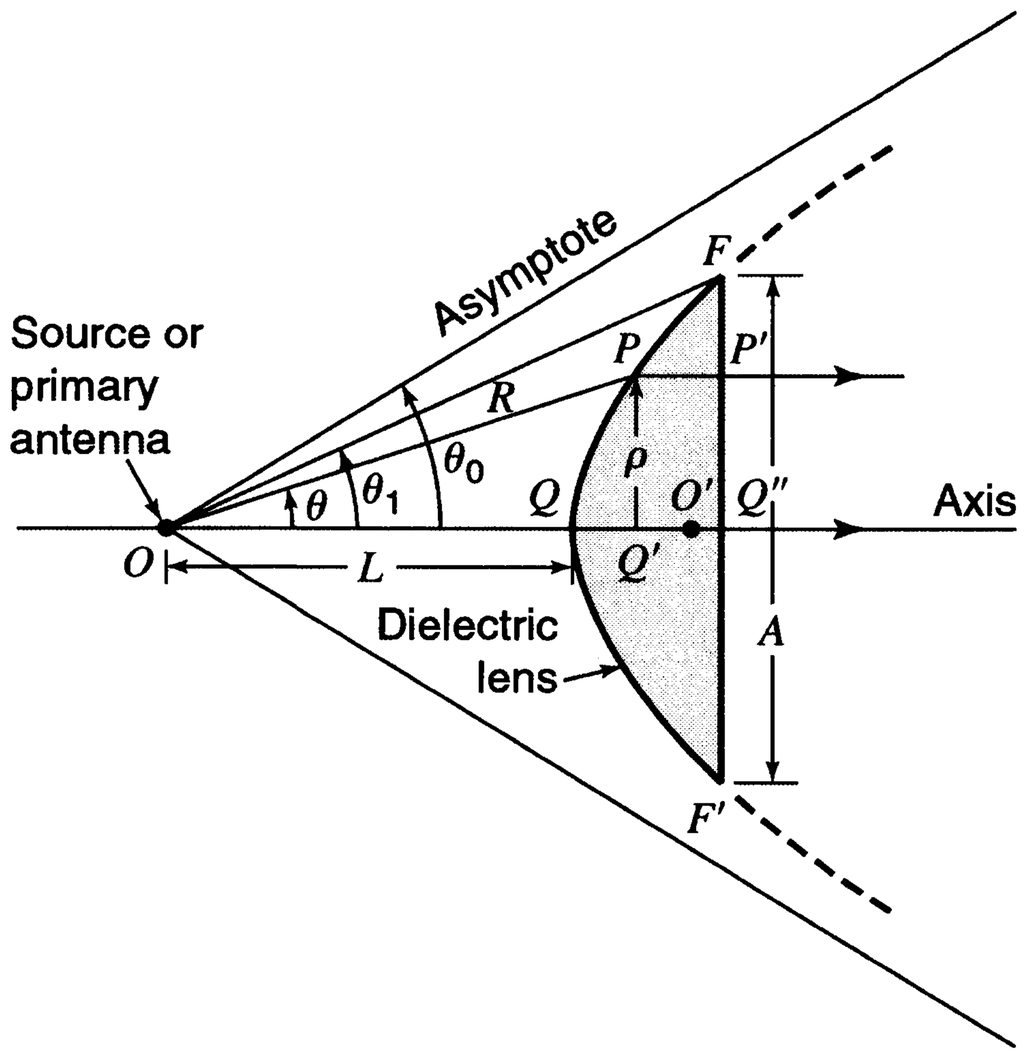


Figure 17–3
Path lengths in dielectric lens.

[1] A wave front is defined as a surface on all points of which the field is in the same phase, i.e., a *phase front*.

Multiplying (1) by λ_0,

$$R = L + n(R\cos\theta - L) \tag{2}$$

where $n = \lambda_0/\lambda_d$ = index of refraction

In general,

$$n = \frac{\lambda_0}{\lambda_d} = \frac{f\lambda_0}{f\lambda_d} = \frac{v_0}{v_d} = \frac{\sqrt{\mu\varepsilon}}{\sqrt{\mu_0\varepsilon_0}} \tag{3}$$

where

f = frequency, Hz
v_0 = velocity in free space, m s^{-1}
v_d = velocity in dielectric, m s^{-1}
μ = permeability of the dielectric medium, H m^{-1}
ε = permittivity of the dielectric medium, F m^{-1}
μ_0 = permeability of free space = $4\pi \times 10^{-7}$, H m^{-1}
ε_0 = permittivity of free space = 8.85×10^{-12}, F m^{-1}

However,

$$\mu = \mu_0\mu_r \tag{4}$$

and

$$\varepsilon = \varepsilon_0\varepsilon_r \tag{5}$$

where

$\mu_r = \dfrac{\mu}{\mu_0}$ = relative permeability of dielectric medium
$\varepsilon_r = \dfrac{\varepsilon}{\varepsilon_0}$ = relative permittivity of dielectric medium

Thus, from (3),

$$n = \sqrt{\mu_r\varepsilon_r} \tag{6}$$

For nonmagnetic materials μ_r is very nearly unity so that

$$n = \sqrt{\varepsilon_r}$$

The index of refraction of dielectric substances is always greater than unity. For vacuum, $\varepsilon_r = 1$ by definition. For air at atmospheric pressure, $\varepsilon_r = 1.0006$, but in most applications it is sufficiently accurate to take $\varepsilon_r = 1$ for air. The relative permittivity (or dielectric constant), index of refraction and power factor for a number of lens materials are listed in Table 17–1 in order of increasing ε_r. Although the *permittivity* of materials may vary with frequency (ε_r for water is 81 at radio frequencies and about 1.8 at optical frequencies), the table values are appropriate at radio wavelengths down to the order of 1 cm. The power factor is also a function of frequency. The values listed merely indicate the order of magnitude at radio frequencies.

Table 17–1 Permittivity and power factor of materials

Material	Relative permittivity ε_r	Index of refraction n	Power factor[§]
Paraffin	2.1	1.4	0.0003
Polyethylene	2.2	1.5	0.0003
Lucite or Plexiglas (methacrylic resin)	2.6	1.6	0.01
Polystyrene	2.5	1.6	0.0004
Flint glass	7	2.5	0.004
Polyglas (TiO_2 or titanate fillers)	4–16[†]	2–4	0.003
Rutile (TiO_2)	85–170[‡]	9–13	0.0006

[†]Depends on composition.
[‡]Depends on orientation of crystal with respect to field.
[§]Equals cosine of angle between conduction and total currents.

Returning now to Eq. (2) and solving for R, we have

$$R = \frac{(n-1)L}{n\cos\theta - 1} \tag{7}$$

This equation gives the required shape of the lens. It is the equation of a hyperbola. Referring to Fig. 17–3, the distance L is the focal length of the lens.[1] The asymptotes of the hyperbola are at an angle θ_0 with respect to the axis. The angle θ_0 may be determined from (7) by letting $R = \infty$. Thus,

$$\theta_0 = \arccos\frac{1}{n} \tag{8}$$

The point O is at one focus of the hyperbola. The other focus is at O'. For a point source at the focus, the 3-dimensional lens surface is a spherical hyperbola obtained by rotating the hyperbola on its axis. For an in-phase line source normal to the page (Fig. 17–3) as the primary antenna, the lens surface is a cylindrical hyperbola obtained by translating the hyperbola parallel to the line source.

Although Eq. (7) for the lens surface was derived without using Snell's laws of refraction,[2] these laws are satisfied by the lens boundary as given by (7).

The plane wave emerging from the right side of the lens produces a secondary pattern with maximum radiation in the direction of the axis. The shape of the secondary pattern is a function of both the aperture A and the type of illumination. This aperture-pattern relation has been discussed in previous chapters.

---

[1]The F number of a lens is the ratio of the focal distance to the diameter A of the lens aperture. Thus, $F = L/A$.

[2]Snell's laws of refraction are (1) that the incident ray, the refracted ray and the normal to the surface lie in a plane and (2) that the ratio of the sine of the angle of incidence to the sine of the angle of refraction equals a constant for any two media. If the medium of the incident wave is air, the constant is the index of refraction n of the medium with the refracted ray. Thus, $\sin\alpha/\sin\beta = n$, where α is the angle between the incident ray in air and the normal to the surface and β is the angle between the refracted ray in the dielectric medium and the normal to the surface.

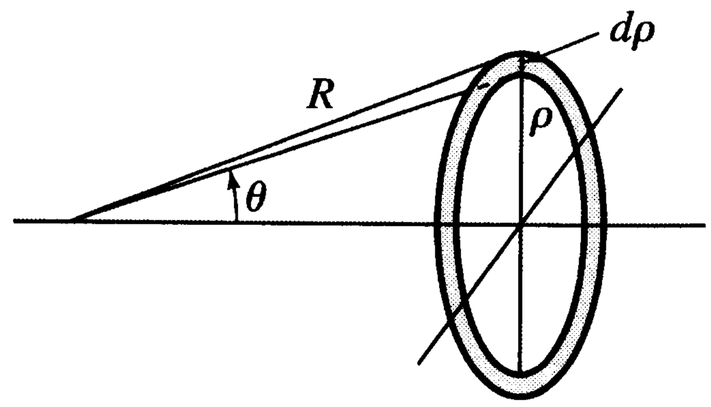


Figure 17–4
Annular zone.

For an isotropic point-source primary antenna and a given focal distance L, the field at the edge of the lens ($\theta = \theta_1$) is less than at the center ($\theta = 0$), the effects of reflections at the lens surfaces and losses in the lens material being neglected. The variation of field intensity in the aperture plane of the spherical lens can be determined by calculating the power per unit area passing through an annular section of the aperture as a function of the radius ρ (Risser-1). Referring to Fig. 17–4, the total power P through the annular section of radius ρ and width $d\rho$ is given by

$$P = 2\pi\rho\, d\rho\, S_\rho \tag{9}$$

where S_ρ = power density or Poynting vector (power per unit area) at radius ρ

This power must be equal to that radiated by the isotropic source over the solid angle $2\pi \sin\theta\, d\theta$. Thus,

$$P = 2\pi \sin\theta\, d\theta\, U \tag{10}$$

where U = radiation intensity of the isotropic source (power per unit solid angle)

Equating (9) and (10),

$$\rho\, d\rho\, S_\rho = \sin\theta\, d\theta\, U \tag{11}$$

and

$$\frac{S_\rho}{U} = \frac{\sin\theta}{\rho(d\rho/d\theta)} \tag{12}$$

However, $\rho = R\sin\theta$, and introducing the value of R from (7),

$$S_\rho = \frac{(n\cos\theta - 1)^3}{(n-1)^2(n-\cos\theta)L^2}U \tag{13}$$

The ratio of the power density S_θ at the angle θ to the power density S_0 at the axis ($\theta = 0$) is given by the ratio of (13) when $\theta = \theta$, to (13) when $\theta = 0$. Thus,

$$\frac{S_\theta}{S_0} = \frac{(n\cos\theta - 1)^3}{(n-1)^2(n-\cos\theta)} \tag{14}$$

In the aperture plane the field-intensity ratio is equal to the square root of (14), or[1]

$$\frac{E_\theta}{E_0} = \sqrt{\frac{S_\theta}{S_0}} = \frac{1}{n-1}\sqrt{\frac{(n\cos\theta - 1)^3}{n - \cos\theta}} \tag{15a}$$

The ratio E_θ/E_0 is the relative field intensity at a radius ρ given by $\rho = R\sin\theta$. For $n = 1.5$,

$$\frac{E_\theta}{E_0} = 0.7 \text{ at } \theta = 20°$$

and

$$\frac{E_\theta}{E_0} = 0.14 \text{ at } \theta = 40°$$

Hence, for nearly uniform aperture illumination an angle θ_1 to the edge of the lens even less than 20° is essential unless the pattern of the primary antenna is an inverted type, i.e., one with less intensity in the axial direction ($\theta = 0$) than in directions off the axis.

Instead of uniform aperture illumination, a tapered illumination may be desired in order to suppress minor lobes. Thus, in the above example with $\theta_1 = 40°$, the field at the edge of the lens is 0.14 its value at the center. The disadvantage of this method of producing a taper is that the lens is bulky (Fig. 17–5*a*). An alternative arrangement, shown in Fig. 17–5*b*, has a lens of smaller θ_1 value with the desired taper obtained with a directional primary antenna at a larger focal distance (relative to the aperture). The lens in this case is less bulky, but the focal distance is larger (F number $= L/A$ larger).

For compactness and mechanical lightness it would be desirable to combine the short focal distance of the lens at (*a*) with the light weight of the lens at (*b*). This combination may be largely achieved with the short focal distance zoned lens of Fig. 17–5*c*. The weight of this lens is reduced by the removal of sections or zones, the geometry of the zones being such that the lens performance is substantially unaffected at the design frequency. Whereas the unzoned lens is not frequency sensitive, the zoned lens is, and this may be a disadvantage. The thickness z of a zone step is such that the electrical length of z in the dielectric is an integral number of wavelengths longer (usually unity) than the electrical length of z in air. Thus, for a 1λ difference,

$$\frac{z}{\lambda_d} - \frac{z}{\lambda_0} = 1 \tag{16}$$

or

$$z = \frac{\lambda_0}{n-1} \tag{17}$$

---

[1]Equation (15*a*) is for a spherical lens. Attenuation in the lens is neglected. For a cylindrical lens the field-intensity ratio is

$$\frac{E_\theta}{E_0} = \frac{n\cos\theta - 1}{\sqrt{(n-1)(n-\cos\theta)}} \tag{15b}$$

where E_θ/E_0 is the relative field intensity at a distance y from the axis given by $y = R\sin\theta$.

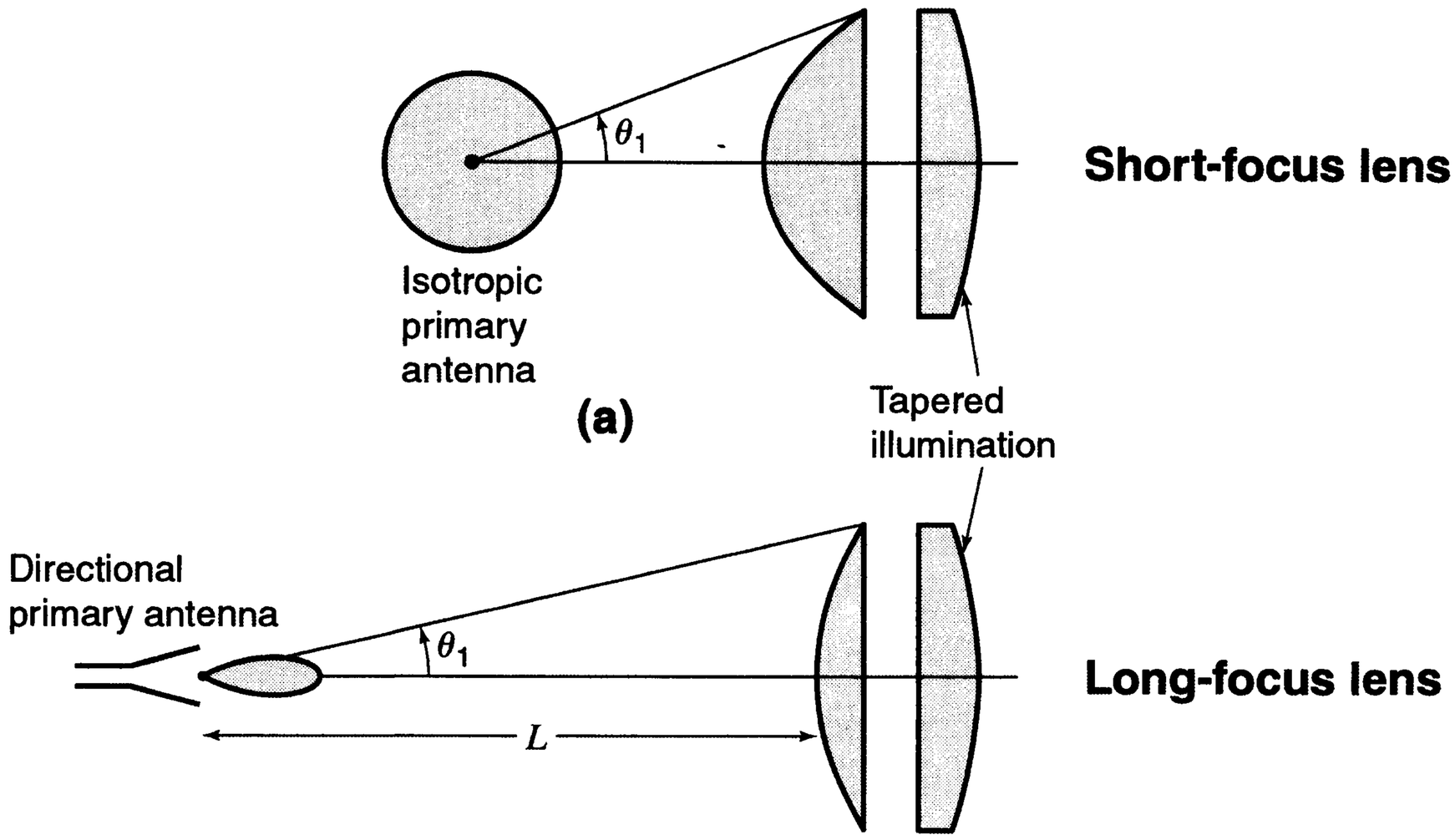


Figure 17–5
Short-focus lens (*a*), long-focus lens (*b*) and zoned lens (*c*).

For a dielectric with index of refraction $n = 1.5$,

$$z = 2\lambda_0$$

that is, each zone step is twice the free-space wavelength. Since $n = \lambda_0/\lambda_d$,

$$z = 3\lambda_d$$

Thus, in this case, the electrical length of z in the dielectric is 3λ, while the electrical length of z in air is 2λ (see Fig. 17–5*c*).

In lens antennas the primary antenna does not interfere with the plane wave leaving the aperture as it does in a symmetrical parabolic reflector antenna (with prime focus feed) (see Chap. 10). However, the energy reflected from the lens surfaces may be sufficient to cause a mismatch of the primary antenna to its feed line or guide. In the lens of Fig. 17–6*a* reflections from the convex surface of the lens do not return to the source except from points at or near the axis. This is not serious, but the wave reflected internally from the plane lens surface is refocused at the primary antenna and may be disturbing. In this case, the wave

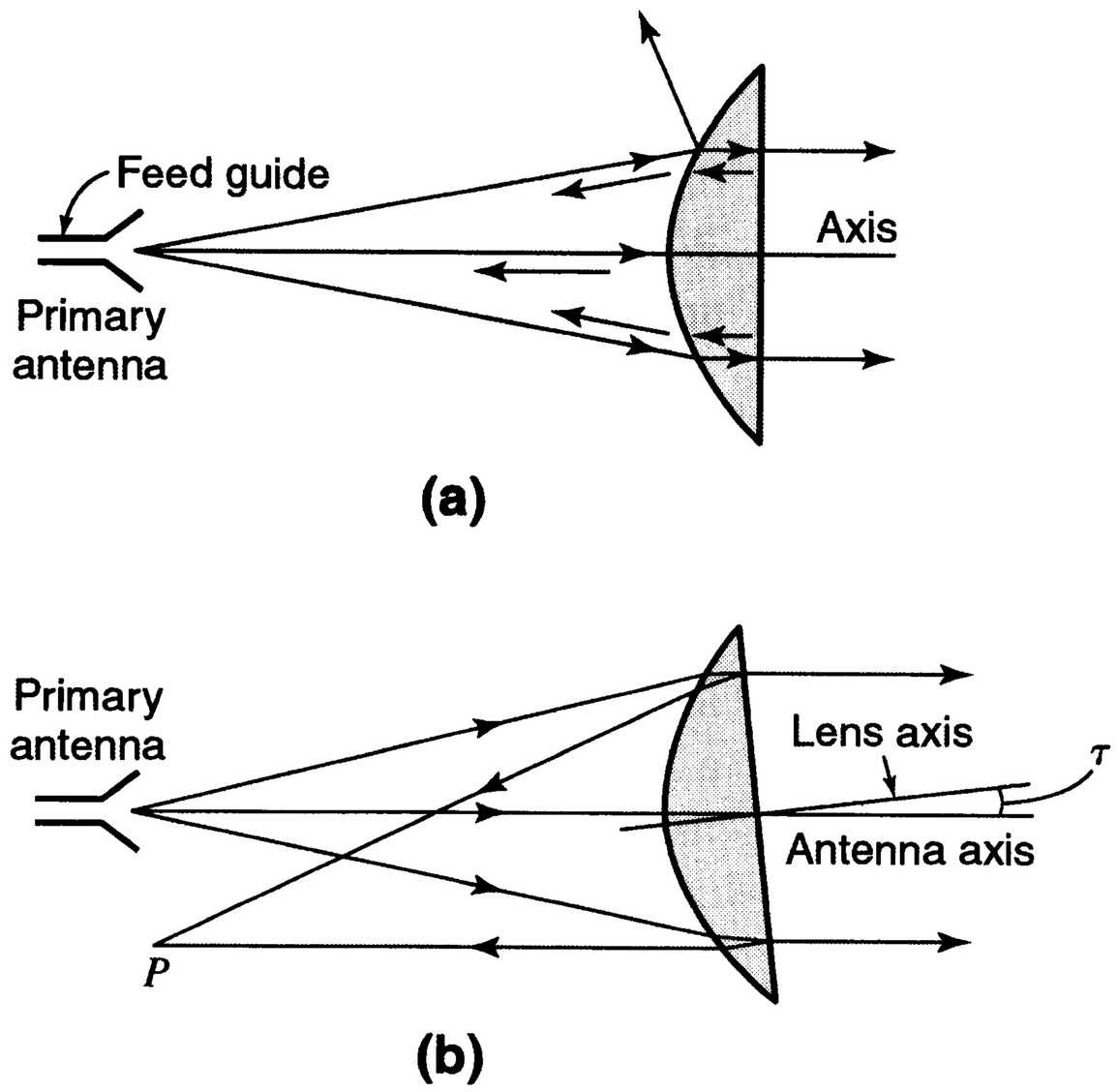


Figure 17–6
Reflected waves entering primary antenna (*a*) and refocused to one side of primary antenna (*b*).

is reflected at normal incidence, and the reflection coefficient is

$$\rho = \frac{Z_0 - Z}{Z_0 + Z} \tag{18}$$

where

Z_0 = intrinsic impedance of free space = $\sqrt{\mu_0/\varepsilon_0}$
Z = intrinsic impedance of dielectric lens material = $\sqrt{\mu/\varepsilon}$

Thus,

$$\rho = \frac{(Z_0/Z) - 1}{(Z_0/Z) + 1} = \frac{n - 1}{n + 1} \tag{19}$$

where n = the index of refraction of the dielectric lens material

For $n = 1.5$, $\rho = 0.2$, while for $n = 4$, $\rho = 0.6$. Hence, for a small reflection a low index of refraction is desirable. The reflection can also be minimized by other methods. For example, a $\lambda/4$ plate can be applied to the plane lens surface with the refractive index of the plate made equal to $\sqrt{n}$, where n is the refractive index of the lens proper.[1] Another method is to tilt the lens slightly as indicated in Fig. 17–6*b* so that the reflected wave refocuses to one side of the primary antenna. Even though the lens is tilted, the antenna beam remains on axis.

---

[1]In general the refractive index of a $\lambda/4$ matching plate between two media should be equal to the geometric mean of the indices of the two media. This is equivalent to saying that the intrinsic impedance Z_p of the plate material is made equal to the geometric mean of the intrinsic impedances Z_1 and Z_2 of the two media. Thus, $Z_p = \sqrt{Z_1 Z_2}$.

17–3 ARTIFICIAL DIELECTRIC LENS ANTENNAS（人造介质透镜天线）

Instead of using ordinary, nonmetallic dielectrics for the lens, Kock (1) has demonstrated that artificial or metallic dielectrics can be substituted, generally with a saving in weight. Whereas the ordinary dielectric consists of molecular particles of microscopic size, the artificial dielectric consists of discrete metal particles of macroscopic size. The size of the metal particles should be small compared to the design wavelength to avoid resonance effects. It is found that this requirement is satisfied if the maximum particle dimension (parallel to the electric field) is less than $\lambda/4$. A second requirement is that the spacing between the particles be less than λ to avoid diffraction effects.

The particles may be metal spheres, disks, strips or rods. For example, a plano-convex lens constructed of metal spheres is illustrated in Fig. 17–7. The spheres are arranged in a 3-dimensional array or lattice structure. Such an arrangement simulates the crystalline lattice of an ordinary dielectric substance but on a much larger scale. The radio waves from the source or primary antenna cause oscillating currents to flow on the spheres. The spheres are, thus, analogous to the oscillating molecular dipoles of an ordinary dielectric.

An artificial dielectric lens can be designed in the same manner as an ordinary dielectric lens (Sec. 17–2). To do this, it is necessary to know the effective index of refraction of the artificial dielectric. This can be measured experimentally with a slab of the material, or it can be calculated approximately by the following method of analysis. Metal disks or strips are generally preferable to spheres because they are lighter in weight. The strips may be continuous in a direction perpendicular to the electric field as indicated in Fig. 17–8. Since, however, the sphere is more readily analyzed, the method will be illustrated for the case of the sphere.

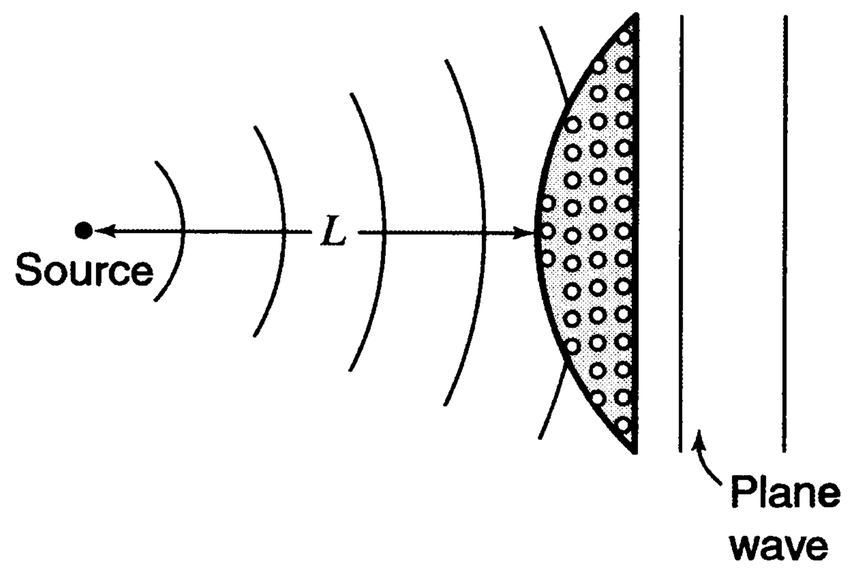


Figure 17–7
Artificial dielectric lens of metal spheres.

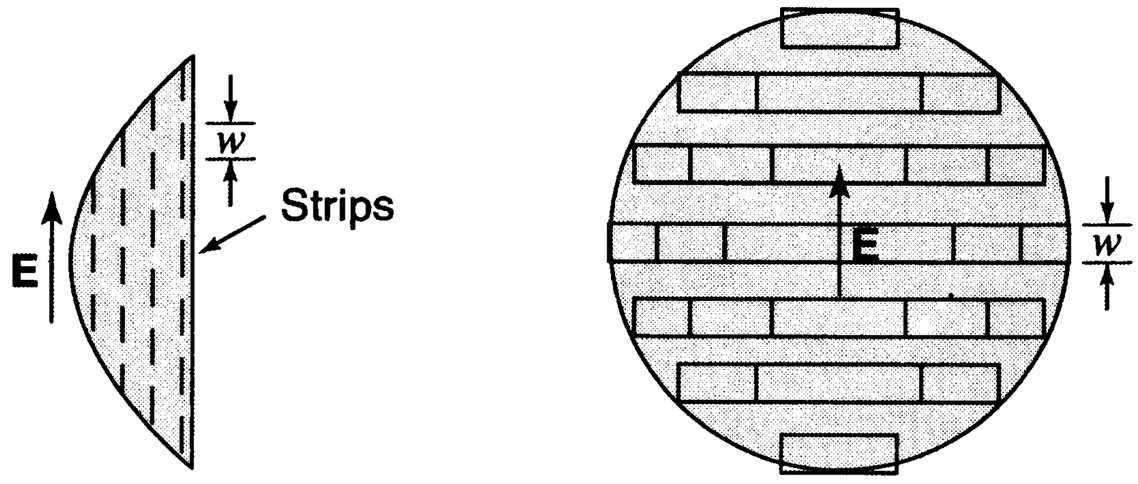


Figure 17–8
Artificial dielectric lens of flat metal strips of width *w*.

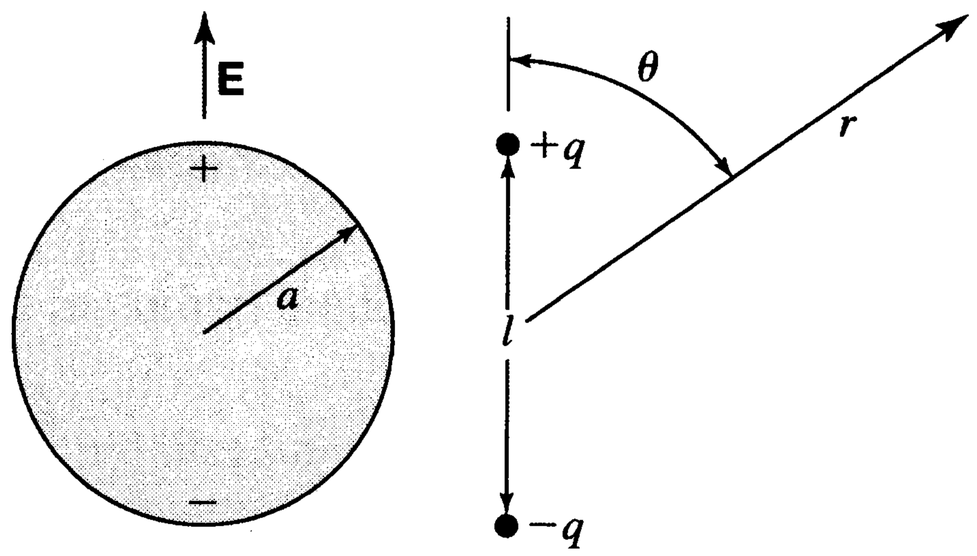


Figure 17–9
Charged sphere and equivalent dipole.

Let an uncharged conducting sphere be placed in an electric field **E** as in Fig. 17–9*a*. The field induces positive and negative charges as indicated. At a distance the effect of these charges may be represented by point charges $+q$ and $-q$ separated by a distance l as in Fig. 17–9*b*. Such a configuration is an *electric dipole of dipole moment ql*. At a distance $r \gg l$ the potential due to the dipole is given by

$$V = \frac{ql \cos\theta}{4\pi\varepsilon_0 r^2} \tag{1}$$

The polarization **P** of the artificial dielectric is given by

$$\mathbf{P} = Nq\mathbf{l} \tag{2}$$

where

N = number of spheres per cubic meter
$\mathbf{l}$ = vector of length l joining the charges q

The electric displacement **D**, the electric field intensity **E** and the polarization **P** are related by

$$\mathbf{D} = \varepsilon\mathbf{E} = \varepsilon_0\mathbf{E} + \mathbf{P} \tag{3}$$

where ε_0 = dielectric constant of free space

Thus, the effective dielectric constant ε of the artificial dielectric medium is

$$\varepsilon = \varepsilon_0 + \frac{\mathbf{P}}{\mathbf{E}} = \varepsilon_0 + N\frac{q\mathbf{l}}{\mathbf{E}} \tag{4}$$

Hence, if the number of spheres per unit volume and the dipole moment of one sphere per unit applied field are known, the effective dielectric constant can be determined. Let us now determine the dipole moment per unit applied field.

We have $\mathbf{E} = -\nabla V$. Then in a uniform field the potential

$$V = -\int_0^r E\cos\theta\, dr = -Er\cos\theta \tag{5}$$

where θ is the angle between the radius vector and the field (see Fig. 17–9*b*). The potential

V_0 outside the sphere placed in an originally uniform field is the sum of (1) and (5), or

$$V_0 = -Er\cos\theta + \frac{ql\cos\theta}{4\pi\varepsilon_0 r^2} \tag{6}$$

At the sphere (radius a) (6) becomes[1]

$$0 = -Ea\cos\theta + \frac{ql\cos\theta}{4\pi\varepsilon_0 a^2}$$

and solving for ql/E we obtain

$$\frac{ql}{E} = 4\pi\varepsilon_0 a^3 \tag{7}$$

Introducing this value for the dipole moment per unit applied field in (4),

$$\varepsilon = \varepsilon_0 + 4\pi\varepsilon_0 N a^3$$

or

$$\varepsilon_r = 1 + 4\pi N a^3 \tag{8}$$

where ε_r = effective relative permittivity of the artificial dielectric

If the effective relative permeability of the artificial dielectric is unity, the index of refraction is given by the square root of (8). However, the lines of magnetic field of a radio wave are deformed around the sphere since high-frequency fields penetrate to only a very small distance in good conductors. The effective relative permeability of an artificial dielectric of conducting spheres is

$$\mu_r = 1 - 2\pi N a^3 \tag{9}$$

The effective index of refraction of the artificial dielectric of conducting spheres is then given by

$$n = \sqrt{\varepsilon_r \mu_r} = \sqrt{(1 + 4\pi N a^3)(1 - 2\pi N a^3)} \tag{10}$$

Equation (10) gives a smaller n than obtained by the square root of (8) alone. According to (10) the index of refraction of an artificial dielectric of conducting spheres can be calculated if the radius a of the sphere (in meters) and the number N of spheres per cubic meter are known. The relative permeability of disk or strip-type artificial dielectrics is more nearly unity so that one can take $\sqrt{\varepsilon_r}$ as their index of refraction. Theoretical values of ε_r, μ_r and n for artificial dielectrics made of conducting spheres, disks and strips are listed in Table 17–2 (Kock-1). According to Kock the table values are reliable only for $\varepsilon_r \leq 1.5$, and only approximate for larger ε_r. For $\varepsilon_r > 1.5$, N becomes sufficiently large that the particles interact because of their close spacing. This effect is neglected by the formulas.

---

[1]The potential of the sphere is zero since there is as much positive as negative charge on its surface.

Table 17–2 Artificial dielectric materials†

Type of particle	Relative permittivity ε_r	Relative permeability μ_r	Index of refraction n
Sphere	$1 + 4\pi Na^3$	$1 - 2\pi Na^3$	$\sqrt{(1 + 4\pi Na^3)(1 - 2\pi Na^3)}$
Disk	$1 + 5.33Na^3$	~ 1	$\sqrt{1 + 5.33Na^3}$
Strip	$1 + 7.85N'w^2$	~ 1	$\sqrt{1 + 7.85N'w^2}$

† N = number of spheres or disks per cubic meter
a = radius of sphere or disk, m
N' = number of strips per square meter in lens cross section (see Fig. 17–8*a*)
w = width of strips, m (see Fig. 17–8)

17–4 *E*-PLANE METAL-PLATE LENS ANTENNAS (*E*-面金属板透镜天线)

Whereas the ordinary and artificial dielectric lens depend for their action on a retardation of the wave in the lens, the *E*-plane metal-plate type of lens depends for its action on an *acceleration* of the wave by the lens (Kock-2). In this type of lens the metal plates are parallel to the *E* plane (or plane of the electric field). Referring to Fig. 17–10, the velocity v of propagation of a TE_{10} wave (**E** as indicated) in the x direction between two parallel conducting plates of large extent is given by Chu (1) as

$$v = \frac{v_0}{\sqrt{1 - (\lambda_0/2b)^2}} \tag{1}$$

where

v_0 = velocity in free space
λ_0 = wavelength in free space
b = spacing of plates or sheets

The plates act as a guide, transmitting the wave for values of $b \geq \lambda_0/2$. The spacing $b = \lambda_0/2$ is the critical spacing since for smaller values of b the guide is opaque and the wave is not

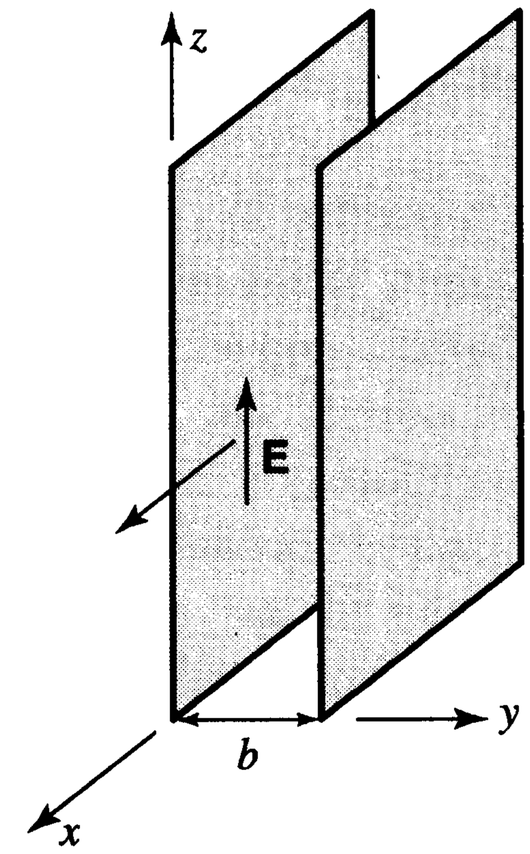


Figure 17–10
Wave between plates in *E*-plane type of metal-plate lens.

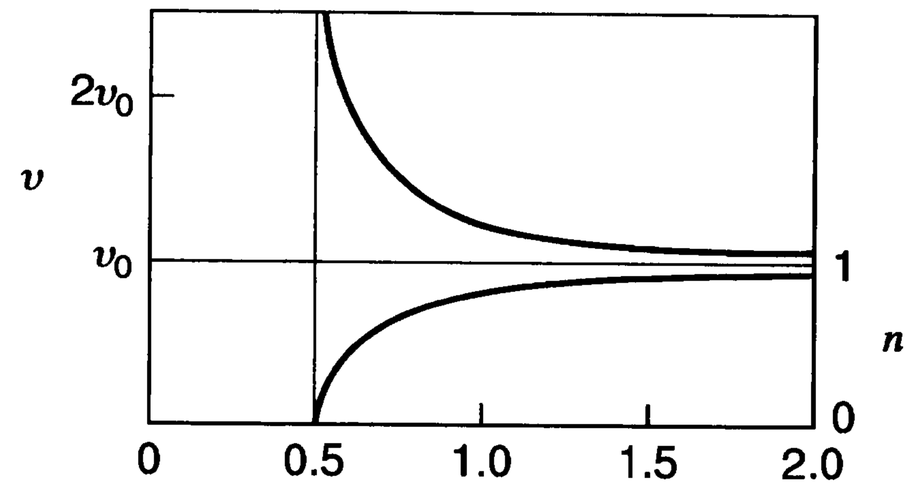


Figure 17–11
Velocity v of wave between parallel plates and equivalent index of refraction n as a function of spacing b between plates.

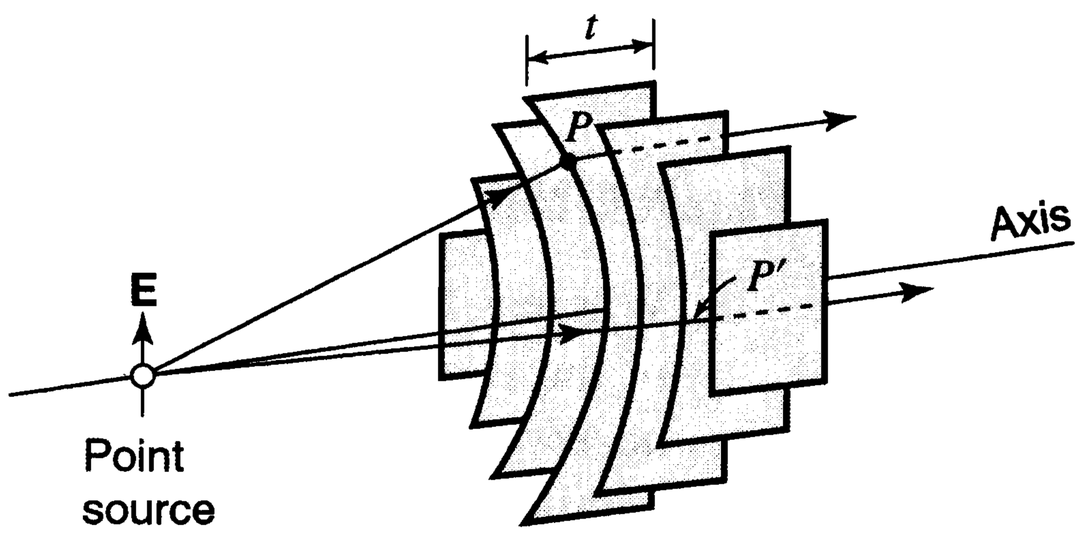


Figure 17–12
E-plane type of metal-plate lens.

transmitted.[1] The variation of the velocity for a fixed wavelength as a function of the plate spacing b is illustrated in Fig. 17–11. The velocity of the wave between the plates is always greater than the free-space velocity v_0. It approaches infinity as b approaches $0.5\lambda_0$, and it approaches v_0 as b becomes infinite.

The equivalent index of refraction of a medium constructed of many such parallel plates with a spacing b is

$$n = \frac{v_0}{v} = \sqrt{1 - \left(\frac{\lambda_0}{2b}\right)^2} \tag{2}$$

The index is always *less* than unity, as shown in Fig. 17–11.

The acceleration of waves between plates has been applied in a metal-plate lens for focusing radio waves (Kock-2). For instance, a metal lens equivalent to the plano-convex dielectric lens of Fig. 17–1*a* or Fig. 17–3 is a plano-concave type as illustrated in Fig. 17–12. The plates are cut from flat sheets, the thickness t at any point being such as to transform the spherical wave from the source into a plane wave on the plane side of the lens. The electric field is parallel to the plates.

The lens plate on the axis of the lens in Fig. 17–12 is shown in Fig. 17–13. The shape of the plate can be determined by the principle of equality of electrical path length (Fermat's principle). Thus, in Fig. 17–13, OPP' must be equal to OQQ' in electrical length, or

$$\frac{L}{\lambda_0} = \frac{R}{\lambda_0} + \frac{L - R\cos\theta}{\lambda_g} \tag{3}$$

[1]However, for typical spacings of $b \simeq 3\lambda/4$ at normal incidence the transmission coefficient is nearly unity.

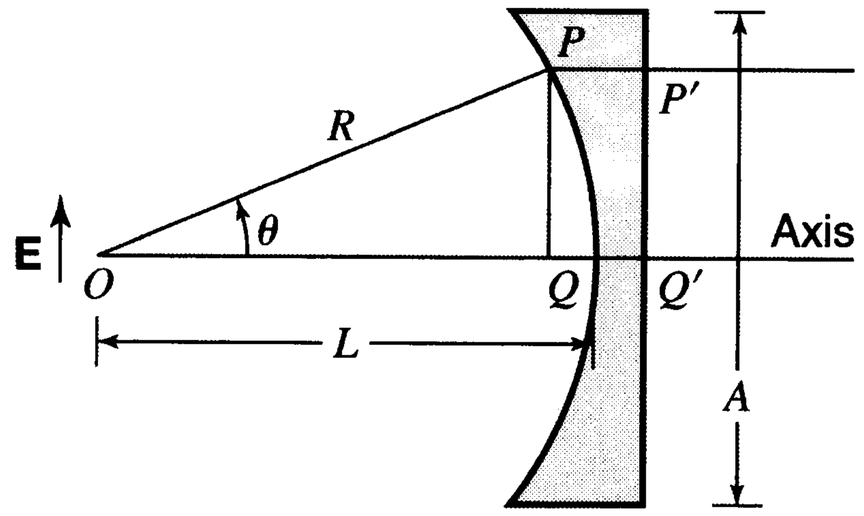


Figure 17–13
Geometry for *E*-plane type of metal-plate lens.

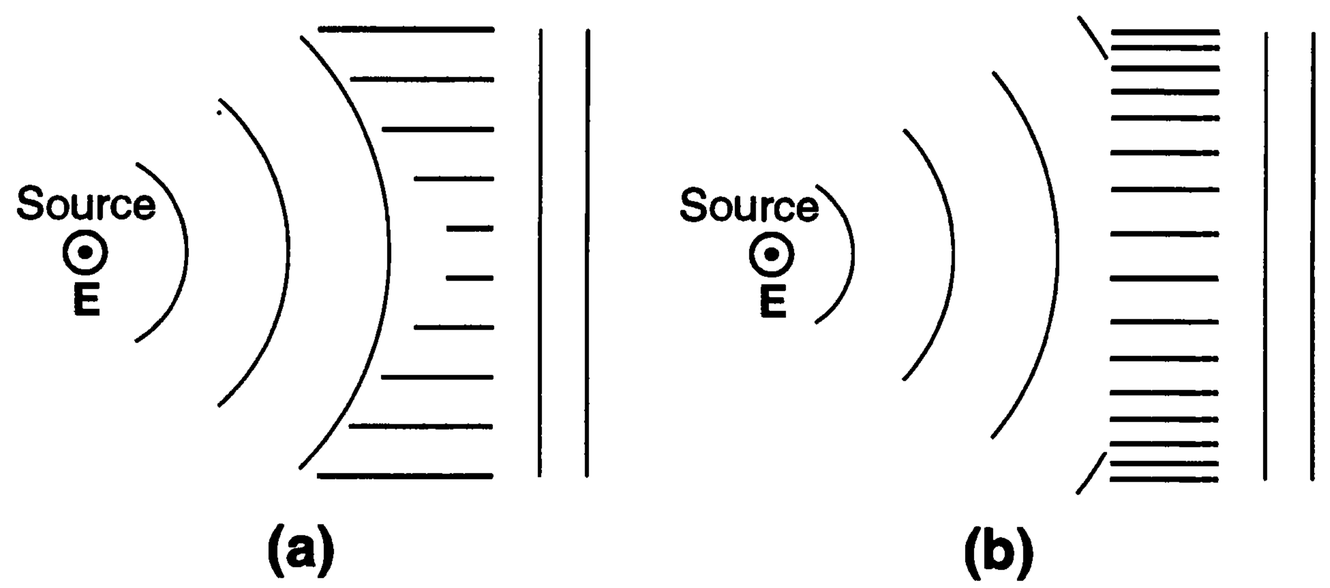


Figure 17–14
Cross sections of constrained types of *E*-plane metal-plate lenses.

where

λ_0 = wavelength in free space

λ_g = wavelength in lens

Then

$$R = \frac{(1-n)L}{1-n\cos\theta} \tag{4}$$

This relation is identical with (17–2–7). However, to keep both numerator and denominator positive (since $n < 1$ in the present case), the numerator and denominator of (17–2–7) should be multiplied by minus 1. With $n < 1$, (4) is the equation of an ellipse. The 3-dimensional concave surface of the lens in Fig. 17–12 would be generated by rotating the contour for the center plate, as given by (4), on the axis. If the primary antenna were a line source perpendicular to the page in Fig. 17–13, all the plates would be identical and the lens surface would be in the form of an elliptical cylinder.

Waves entering the lens of Fig. 17–12 at the point P obey Snell's laws of refraction. However, this is not necessarily the case for waves entering at P' where the metal plates constrain the wave to travel between them. *E*-plane metal-plate lenses may be constructed that have only such constrained refraction. Two types are illustrated in cross section in Fig. 17–14. Both have a line source normal to the page. The electric field **E** is parallel to the source. All lens cross sections perpendicular to the line sources are the same as the ones shown in the figure. In the lens at (*a*) the spacing between plates is uniform, but the width varies from plate to plate. In the lens at (*b*) all plates have the same width, but the spacing varies.

A disadvantage of the *E*-plane metal-plate lens as compared to the dielectric type is that it is frequency-sensitive, i.e., the lens has a relatively small bandwidth. To determine

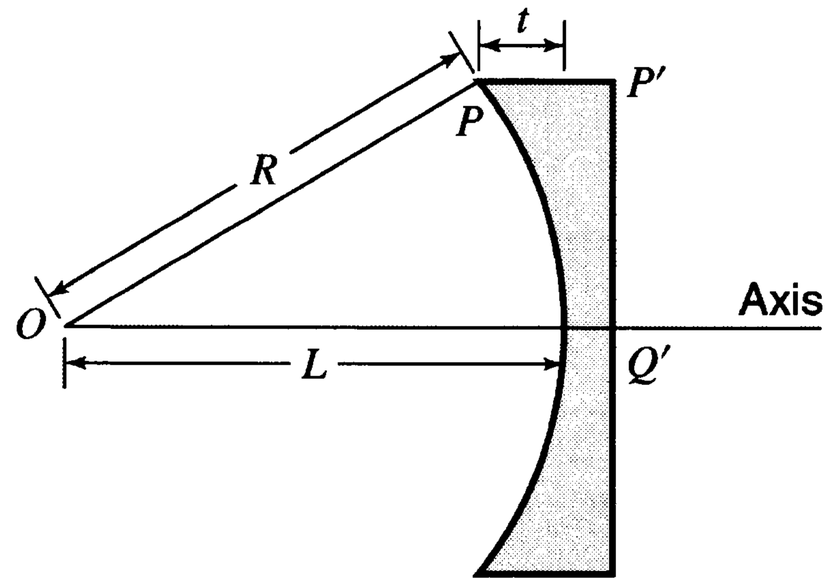


Figure 17–15
Geometry for bandwidth considerations.

the bandwidth, consider the geometry of Fig. 17–15 (Risser-1). At the design frequency f we have from (3) that

$$\frac{L}{\lambda_0} = \frac{R}{\lambda_0} + \frac{t}{\lambda_g} \tag{5}$$

or

$$L = R + nt \tag{6}$$

where n = index of refraction at the design frequency f

At some other frequency f',

$$L + \delta = R + n't \tag{7}$$

where

δ = the difference in electrical path length of OQ and OPP'
n' = index of refraction at the frequency f'

Subtracting (6) from (7),

$$\delta = \Delta n\, t \tag{8}$$

where $\Delta n = n' - n$

For a small wavelength difference $\Delta\lambda_0$,

$$\Delta n = \frac{\partial n}{\partial \lambda_0}\, \Delta\lambda_0 \tag{9}$$

Introducing n from (2) into (9), differentiating and substituting this value of Δn in (8) yields

$$\delta = \frac{n^2 - 1}{n} \frac{\Delta\lambda_0}{\lambda_0} t \tag{10}$$

or

$$\left| \frac{\Delta\lambda_0}{\lambda_0} \right| = \frac{n\delta}{(1 - n^2)t} \tag{11}$$

The total bandwidth B is twice (11) so

$$B = \frac{2n\delta}{(1-n^2)t} = \frac{2n}{1-n^2}\frac{\delta_\lambda}{t_\lambda} \tag{12}$$

where

δ_λ = maximum tolerable path difference in free-space wavelengths

t_λ = thickness of lens plate at edge of lens in free-space wavelengths

If we arbitrarily take $\delta = 0.25\lambda$,

$$B = \frac{50n}{(1-n^2)t_\lambda} \qquad (\%) \tag{13}$$

For $n = 0.5$ and $t = 6\lambda$, the bandwidth

$$B = 5.5\%$$

Thus, the usable frequency band for this antenna is 5.5 percent of the design frequency.[1] Although zoning a dielectric lens introduces frequency sensitivity, the effect of zoning an E-plane metal-plate lens is to decrease the frequency sensitivity. Hence, zoning is desirable with E-plane metal-plate lens, both to save weight and to increase the bandwidth. An E-plane metal-plate lens 40λ square with nine zones is illustrated in Fig. 17–16. The patterns of this lens, fed with a short primary horn antenna, are shown in Fig. 17–17.

The bandwidth of a zoned E-plane metal-plate lens is given approximately by

$$B = \frac{50n}{1+Kn} \qquad (\%) \tag{14}$$

where

n = index of refraction at the design frequency

K = number of zones, the zone on the axis of the lens being counted as the first zone

A zoned lens comparable to the unzoned lens of $n = 0.5$, $t = 6\lambda_0$ and $B = 5.5$ percent has $n = 0.5$ and $K = 3$ since with $n = 0.5$, $K \sim t_\lambda/2$. The bandwidth B of this zoned lens is 10 percent, or nearly double the bandwidth of the unzoned lens.

The aperture efficiency to be expected of large lens antennas is about 0.6 so that the directivity is about the same as for optimum horns of the same size aperture (see Chap. 9).

---

[1] $\frac{2\Delta\lambda}{\lambda} = \frac{\lambda_2-\lambda_1}{\lambda} = \frac{(1/f_2)-(1/f_1)}{(1/f)} = \frac{f(f_1-f_2)}{f_1 f_2} \simeq \frac{f_1-f_2}{f} = \frac{2\Delta f}{f}$ (for $\Delta f \ll f$)

where

λ = design wavelength

f = design frequency

λ_1 = short wavelength limit of band

λ_2 = long wavelength limit of band

f_1 = high-frequency limit of band

f_2 = low-frequency limit of band

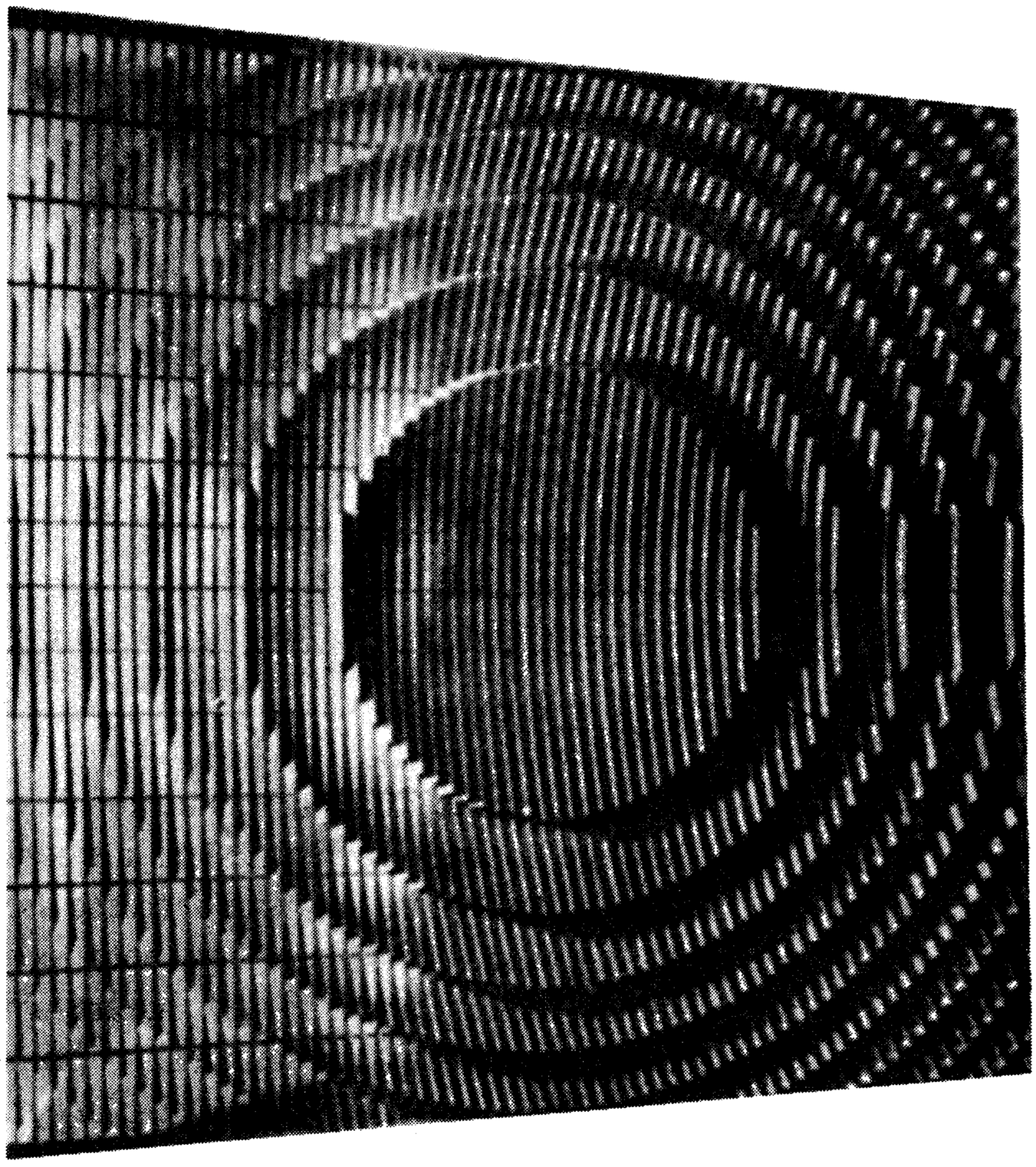


Figure 17–16
Zoned type of *E*-plane metal-plate lens with a square aperture 40λ on a side. (*Courtesy W. E. Kock, Bell Telephone Laboratories.*)

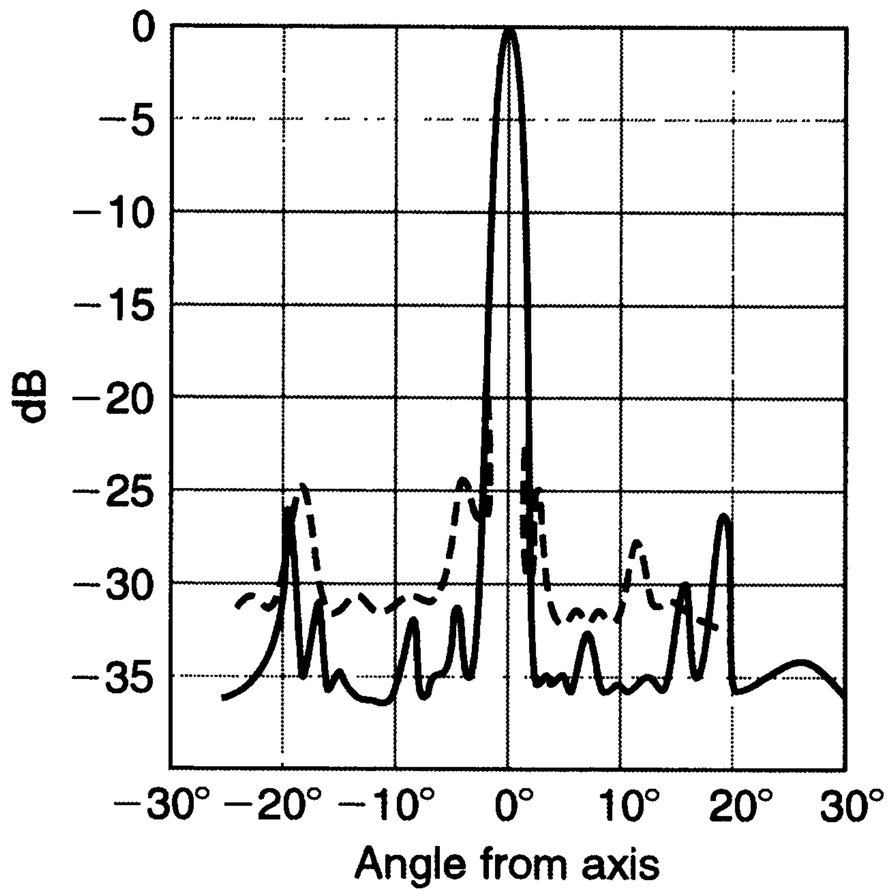
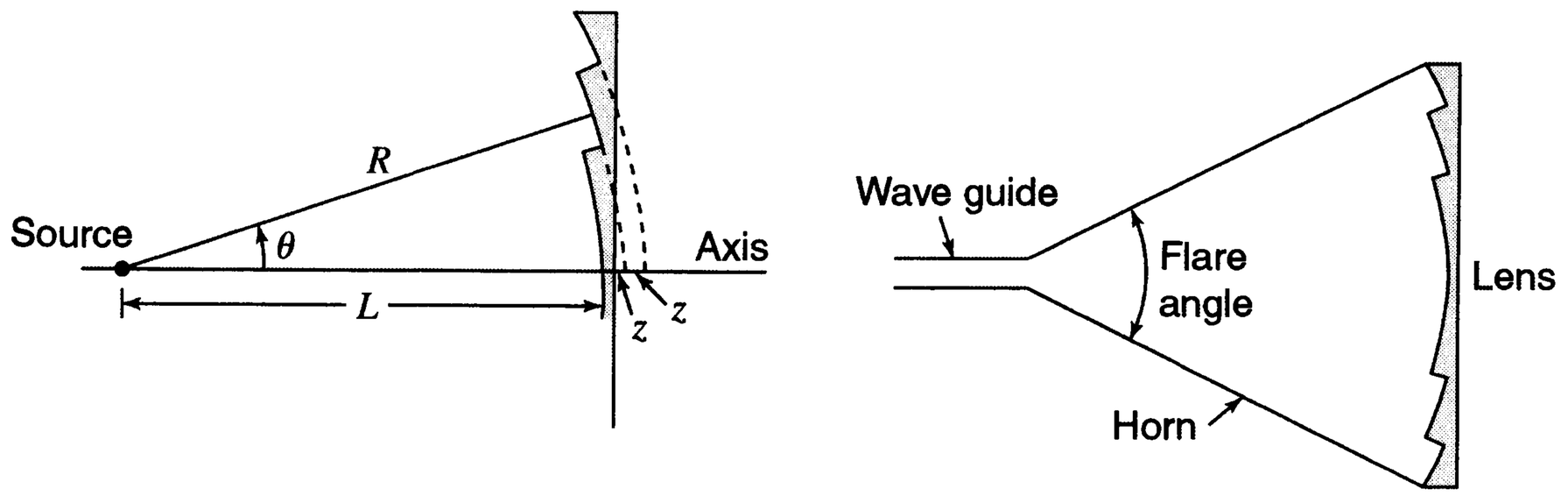


Figure 17–17
E-plane pattern (solid) and *H*-plane pattern (dashed) of 40λ square zoned *E*-plane lens of Fig. 17–16. (*Kock-1, 2*)



(a) (b)

Figure 17–18
(*a*) Zoned lens plate. (*b*) Horn with lens.

Referring to Fig. 17–18*a*, the thickness z of a zone step is given by

$$\frac{z}{\lambda_0} - \frac{z}{\lambda_g} = 1$$

or

$$z = \frac{\lambda_0}{1 - n} \tag{15}$$

The equation for the contour of the zoned lens is the same as (4) for the unzoned lens except that L is replaced by L_k, where

$$L_k = L + \frac{(K-1)\lambda_0}{1-n} = L + (K-1)z \tag{16}$$

For the first zone (on the axis) $L_k = L$, for the second zone $L_k = L + z$, for the third zone $L_k = L + 2z$, etc.

To shield against stray radiation from the source side of a lens, a metallic enclosure may be used as in Fig. 17–18*b*. This enclosure forms an electromagnetic horn of wide flare angle with a lens at the aperture. Without the lens an optimum horn of the same aperture would be much longer (smaller flare angle). The fact that the lens permits a much shorter structure for the same size aperture is, perhaps, the principal advantage of a lens or lens-horn combination over a simple horn antenna.

17–5 TOLERANCES ON LENS ANTENNAS（透镜天线的容差）

In a *dielectric lens,* differences in the path length may be caused by deviations in thickness from the ideal contour and by variations in the index of refraction (Risser-1). Assigning an allowable variation of $\lambda_0/32$ rms to either cause, we have as the *thickness tolerance* that

$$\frac{\Delta t}{\lambda_d} - \frac{\Delta t}{\lambda_0} = \frac{1}{32}$$

or

$$\Delta t = \frac{\lambda_0}{32(n-1)}$$

or

$$\Delta t = \frac{\lambda_0}{32(n-1)} = \frac{0.03\lambda_0}{n-1} \tag{1}$$

For $n = 1.5$,

$$\Delta t = 0.06\lambda_0$$

For the *tolerance on n,*

$$\Delta n\, t = \frac{\lambda_0}{32}$$

or

$$\Delta n = \frac{0.03}{t_\lambda} \tag{2}$$

where t_λ = thickness of lens in free-space wavelengths

Dividing (2) by n,

$$\frac{\Delta n}{n} = \frac{3}{nt_\lambda} \qquad (\%) \tag{3}$$

If $n = 1.5$ and $t = 4\lambda_0$, $\Delta n/n = \frac{1}{2}\%$.

In an ***E-plane metal-plate lens*** (*E*-面金属板透镜中) the path length may be affected by both the thickness of the lens and the spacing b between lens plates. Assigning $\lambda_0/32$ to each cause, we have as the ***thickness tolerance*** (板的间隙容差) that

$$\Delta t = \frac{\lambda_0}{32(1-n)} = \frac{0.03\lambda_0}{1-n} \tag{4}$$

and for the ***tolerance on the spacing b between plates*** (抛物面反射镜)

$$\frac{\Delta b}{b} = \frac{3n}{(1-n^2)t_\lambda} \qquad (\%) \tag{5}$$

It is interesting to compare these tolerances with the surface contour requirement of a *parabolic reflector*. A displacement Δx normal to the surface of the reflector at the vertex (i.e., a displacement in the axial direction) results in a change in wave path of $2\Delta x$. Thus, for the same effect as either Δt or Δn, Δx must be $\frac{1}{2}$ as large.

This is a severe requirement for a large reflector at a small wavelength, placing a strict limitation on the allowable warping or twisting of the reflector. In contrast to this, *the thickness tolerance on a lens refers only to the thickness dimension*. It does not imply that the lens contour be maintained to this accuracy. With a lens, a relatively large amount of warping or twisting can be tolerated, and this is an important advantage of this type of antenna. Furthermore, the lens axis can be tilted a considerable angle τ with respect to the axis through the primary antenna and center of the lens (see Fig. 17–6*b*) without serious effects.[1]

The above-mentioned tolerances are summarized in Table 17–3. Tolerances for zoned lenses are also listed. These are derived from the unzoned lens tolerances by taking the

---

[1]Little difference in radiation-field patterns of an *E*-plane metal-plate lens antenna is revealed for a tilt angle τ as large as 30° according to patterns presented by Friis (1) and Lewis.

Table 17–3 Tolerances on lens and reflector antennas

Type of antenna	Type of tolerance	Amount of tolerance (rms)
Parabolic reflector	Surface contour	$0.016\lambda_0$
Dielectric lens† (unzoned)	Thickness	$\frac{0.03\lambda_0}{n-1}$
	Index of refraction	$\frac{3}{nt_\lambda}\%$
Dielectric lens† (zoned)	Thickness	3%
	Index of refraction	$\frac{3(n-1)}{n}\%$
E-plane metal-plate lens‡ (unzoned)	Thickness	$\frac{0.03\lambda_0}{1-n}$
	Plate spacing	$\frac{3n}{(1-n^2)t_\lambda}\%$
E-plane metal-plate lens‡ (zoned)	Thickness	3%
	Plate spacing	$\frac{3n}{1+n}\%$

n = index of refraction
t = lens thickness
t_λ = lens thickness in free-space wavelengths
†$n > 1$.
‡$n < 1$.

dielectric lens thickness as nearly equal to $\lambda_0/(n-1)$ and the metal-plate lens thickness as nearly equal to $\lambda_0/(1-n)$. All tolerances in the table assume $\lambda_0/32$ rms for the individual lens variations and $\lambda_0/64$ rms for the reflector variation resulting, in each case, in a gain-loss factor $k_g = 0.16$ dB as calculated from (19–2–3), provided the correlation distance $>\lambda$ and other conditions of (19–2–3) are met. For a combination of random variations (as index *and* thickness) the net effective variation is given by the quadrature sum of the individual variations.

17–6 *H*-PLANE METAL-PLATE LENS ANTENNAS（*H*-面金属板透镜天线）

A wave entering a stack of metal plates oriented parallel to the H plane (perpendicular to the E plane) as in Fig. 17–19*a* is little affected in its velocity. However, the wave is constrained to pass between the plates so that, once inside, the path length can be increased if the plates are deformed, as suggested in Fig. 17–19*b*. An increase in path length can also be produced by slanting the plates as at (*c*). The increase of path length is $S - T$. Using the slant-plate method of increasing the path length, an H-plane metal-plate lens can be designed by applying the principle of equality of electrical path length. This type of lens is called an H-plane type since the plates are parallel to the magnetic field (perpendicular to the E-plane) (Kock-3).

Referring to Fig. 17–19*b*, the condition for equality of electrical path length requires that

$$R = L + \frac{R\cos\theta - L}{\cos\xi} \tag{1}$$

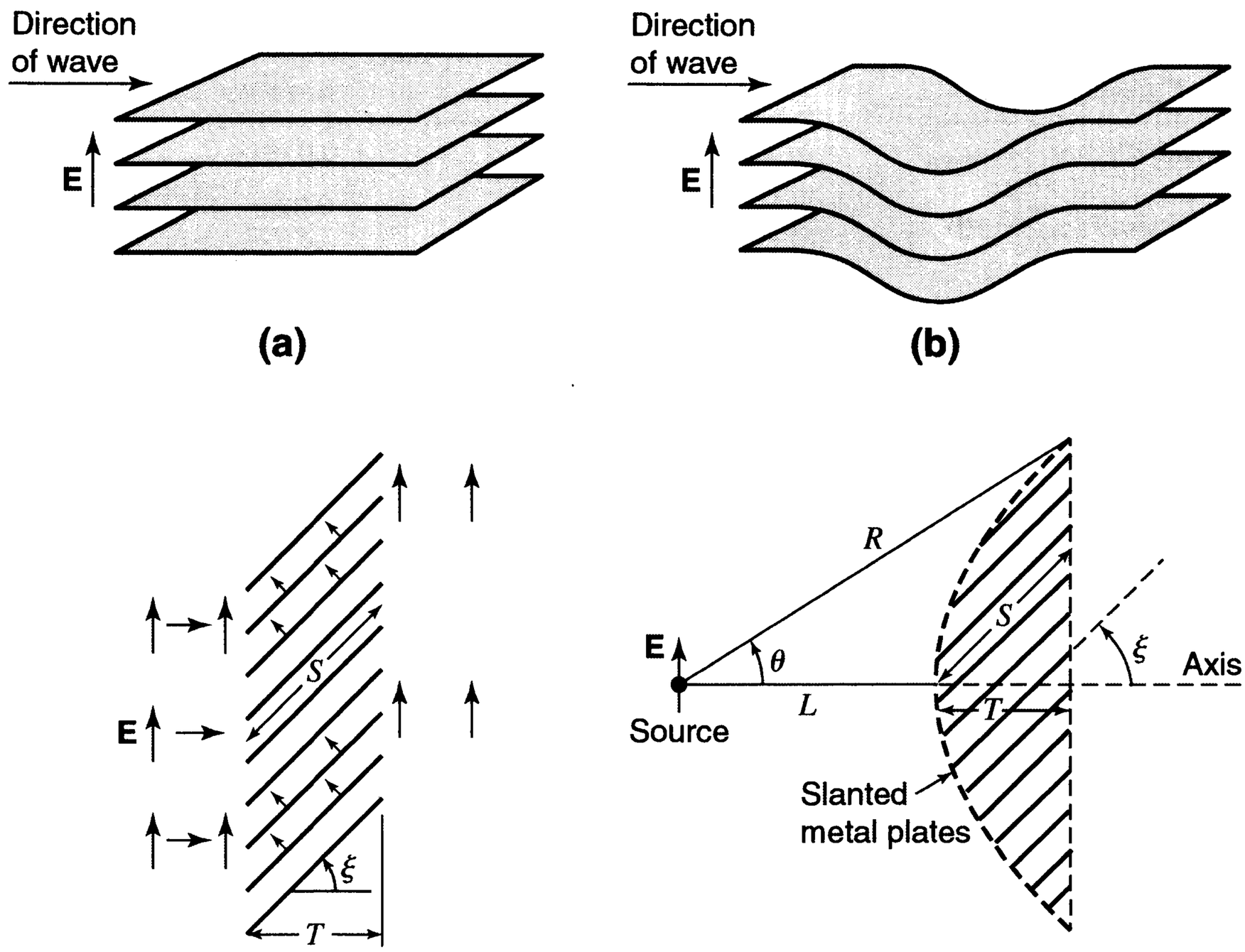


Figure 17–19
(*a*) *H*-plane stack of flat metal plates. (*b*) *H*-plane stack with increased path length. (*c*) Slanted *H*-plane plates. (*d*) *H*-plane metal-plate lens using slanted plate construction.

or

$$R = \frac{(n-1)L}{n\cos\theta - 1} \tag{2}$$

where $n = 1/\cos\xi$ = effective index of refraction of the slant-plate lens medium

In this case the index of refraction is equal to or greater than unity so that (2) is identical with (17–2–7) for a dielectric lens. The index n depends only on the plate slant angle ξ and is not a function of the frequency as in the E-plane type of metal-plate lens. The most critical dimension is the path length S in the lens. This may be affected by a change in T or in ξ. Assuming a maximum allowable variation $\delta = \lambda_0/8$ in electrical path length, the tolerance in S is given by

$$\Delta S = \pm 0.06\lambda_0 \tag{3}$$

A disadvantage of the H-plane metal-plate lens is that this type of construction tends to produce unsymmetrical aperture illumination in the E plane.

17–7 REFLECTOR-LENS ANTENNA（反射器 – 透镜天线）

This antenna combines a dielectric lens with a flat reflecting sheet as suggested in Fig. 17–20 (Kraus-1). An incoming ray traverses the lens twice and is brought to a focus at F. For thin

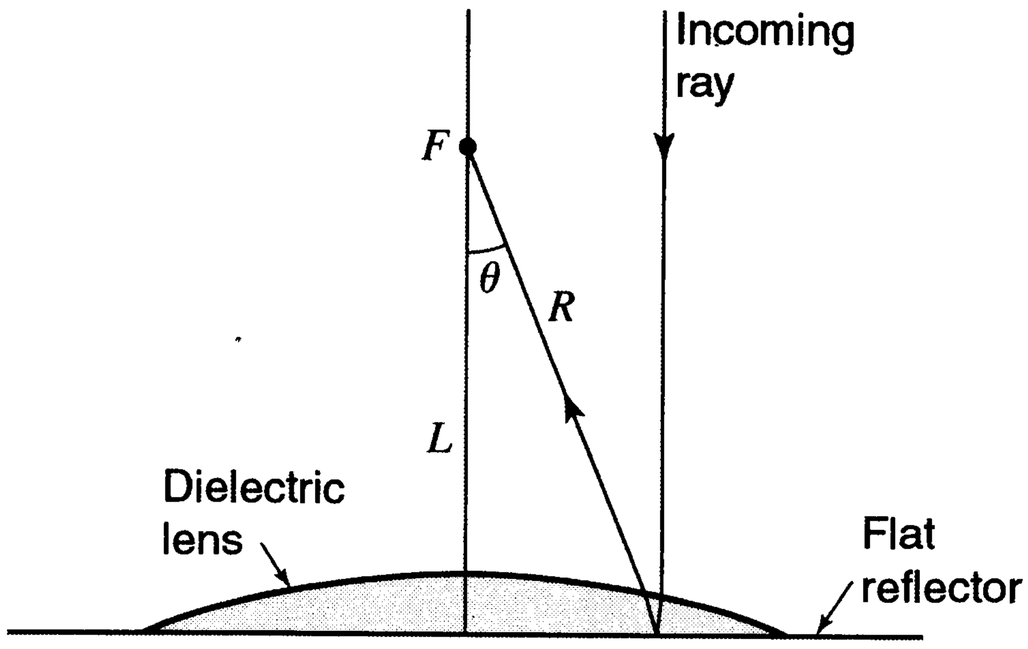


Figure 17–20
Reflector-lens antenna consisting of a plano-convex dielectric lens on a flat reflecting sheet.

lenses the distance R is given approximately by

$$R = \frac{(n-1)2L}{(2n-1)\cos\theta - 1} \tag{1}$$

where

n = index of refraction of lens
L = focal length (in same units as R)

This reflector lens is approximately half the thickness and half the weight of a simple dielectric lens (Figs. 17–1*a* or 17–3) and is an alternative to a parabolic reflector. An interesting, specialized application would be to use a pool of conducting liquid for the flat surface with a long focal length reflector lens situated above as in Fig. 17–20. Beam squinting by horizontal displacements of the feed would allow observations of a region near the zenith. Only the thickness of the lens is critical since the flatness of the reflecting surface is maintained automatically, thanks to gravity. The lens need not be in contact with the flat surface. A meridian-transit millimeter-wave radio astronomy application at low cost is envisioned.

17–8 POLYRODS[1]（介质杆）

A dielectric rod or wire can act as a guide for electromagnetic waves (Hondros-1; Schelkunoff-1; Whitmer-1). The guiding action, however, is imperfect since considerable power may escape through the wall of the rod and be radiated. This tendency to radiate is turned to advantage in the *polyrod antenna* so called because the dielectric rod is usually made of polystyrene (Mueller (1) and Tyrrell). A 6λ long polyrod antenna is shown in cross section in Fig. 17–21*a*. The rod is fed by a short section of cylindrical waveguide which, in

[1]During World War II, George E. Mueller and W. A. Tyrrell of the Bell Telephone Laboratories developed the polyrod for use in arrays of 16 or more rods as a ship-borne radar antenna operating at 20 cm wavelength. Following the war, Dr. Mueller joined the Electrical Engineering faculty of the Ohio State University. He later joined the National Aeronautics and Space Administration (NASA) to direct the Apollo program which put the U.S. astronauts on the moon. As of 2001, he is CEO of the Kisler Aerospace Corporation, Kirkland, Washington.

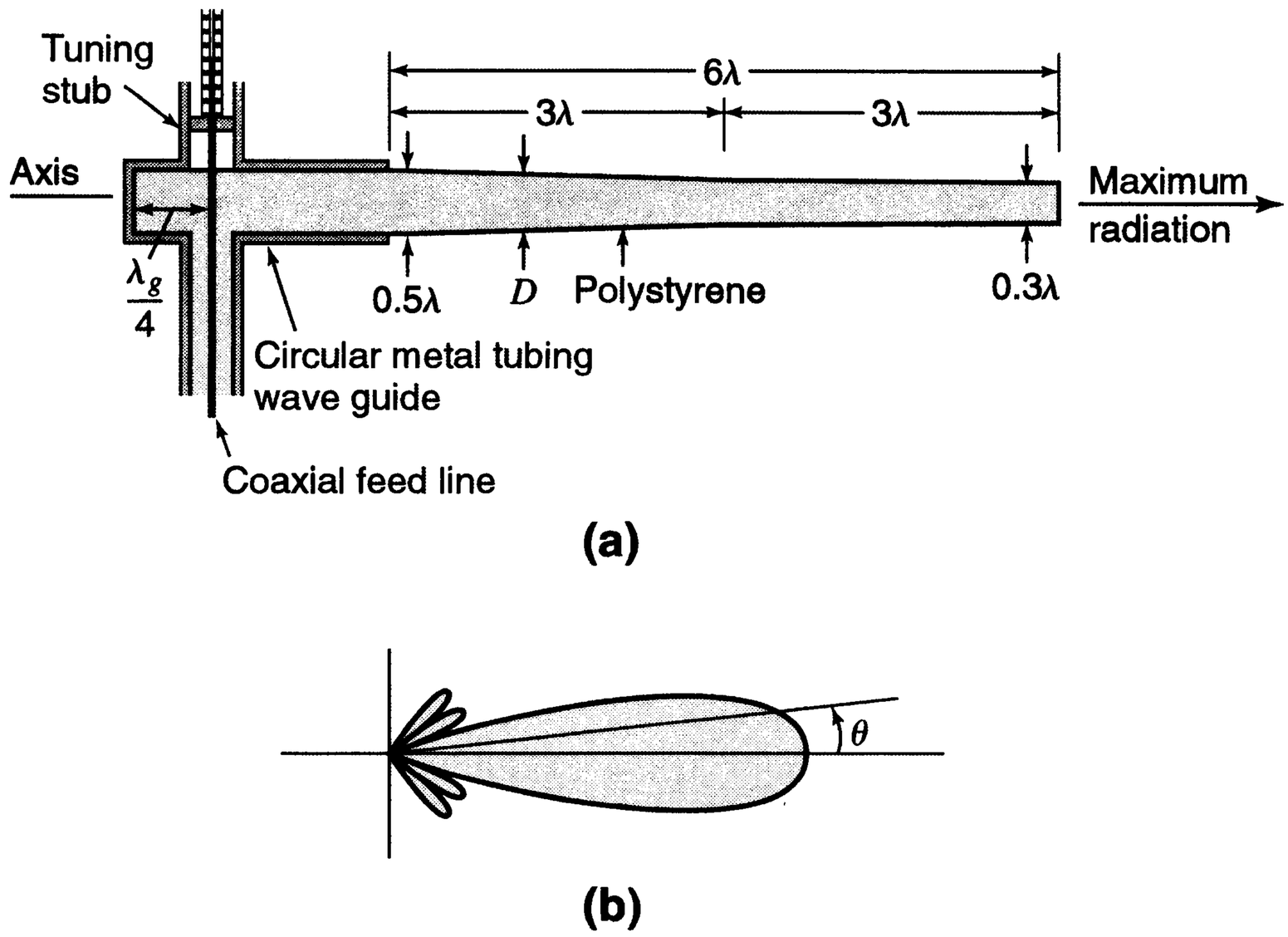


Figure 17–21
(*a*) Cross section of cylindrical polystyrene antenna 6λ long. (*b*) Radiation pattern. (*Mueller (1) and Tyrrell*)

turn, is energized by a coaxial transmission line.[1] This type of polyrod acts as an end-fire antenna (Wilkes-1).

The phase velocity of wave propagation in the rod and also the ratio of the power guided outside the rod to the power guided inside are both functions of the rod diameter D in wavelengths and the dielectric constant[2] of the rod material (Mueller (1) and Tyrrell). For polystyrene rods with $D < \lambda/4$, the rod possesses little guiding effect on the wave, and only a small fraction of the power is confined to the inside of the rod. The phase velocity in the rod is also close to that for the surrounding medium (free space). For diameters of the order of a wavelength, however, most of the power is confined to the rod, and the phase velocity in the rod is nearly the same as in an unbounded medium of polystyrene. For increased directivity operation the diameter D_λ in free-space wavelengths of a uniform rod (length $L_\lambda > 2$ and $2 < \varepsilon_r < 5$) is

$$D_\lambda \simeq \frac{3}{\varepsilon_r^{3/2}\sqrt{1+2L_\lambda}} + 0.2 \tag{1}$$

In practice, polystyrene rod diameters in the range 0.5 to 0.3λ are used.[3] The rod may be uniform or to reduce minor lobes can be tapered as in Fig. 17–21*a*. This polyrod is tapered halfway and is uniform in cross section the remainder of its length. The diameter D is 0.5λ at the butt end and 0.3λ at the far end. The radiation-field pattern for this polyrod as given by Mueller (1) and Tyrrell is shown in Fig. 17–21*b*. The gain is about 16 dBi.

---

[1] An end-fire polyrod antenna may be regarded as a degenerate or rudimentary form of lens antenna with an effective lens cross section of the order of a wavelength. See Gilbert Wilkes (1).

[2] The relative permittivity $\varepsilon_r = 2.5$ for polystyrene. See Table 17–1.

[3] To transmit the lowest (TE_{11}) mode in a circular waveguide, the diameter D of the guide must be at least $0.58\lambda/\sqrt{\varepsilon_r}$, where λ is the free-space wavelength and ε_r is the relative permittivity of the guide. Thus, for a rod of polystyrene ($\varepsilon_r = 2.5$) fed from a circular waveguide as in Fig. 17–21*a*, the guide diameter must be at least 0.37λ to allow transmission in the metal tube.

To a first approximation the radiation pattern of a polyrod antenna excited uniformly along its length may be calculated by assuming that it is a continuous array of isotropic point sources with a phase shift of about 360 $(1+1/2L_\lambda)$ deg/wavelength of antenna, where L_λ is the total length of the antenna in free-space wavelengths.[1] The relative field pattern as a function of the angle θ from the axis is then given by

$$E(\theta) = \frac{\sin(\psi'/2)}{\psi'/2} \tag{2}$$

where $\psi' = 2\pi L_\lambda \cos\theta - 2\pi L_\lambda\left(1+\dfrac{1}{2L_\lambda}\right) = 2\pi\left[L_\lambda(\cos\theta - 1) - \dfrac{1}{2}\right]$

The radiation field could be calculated exactly by applying Schelkunoff's equivalence principle, provided the fields on the surface were known (Schelkunoff-2). By this principle the fields at the rod surfaces are replaced by equivalent electric and fictitious magnetic current sheets, and the radiation field is calculated from these currents. An approximate calculation has been made by assuming a field distribution (Watson-1).

The directivity D of a polyrod antenna is given approximately by Mueller (1) and Tyrrell as

$$D \simeq 8L_\lambda \tag{3}$$

and the half-power beamwidth by

$$\text{HPBW} \simeq \frac{60^\circ}{\sqrt{L_\lambda}} \tag{4}$$

where L_λ = length of polyrod in free-space wavelengths

Polyrod antennas may also be of square or rectangular cross section. Another possibility is to use a dielectric sleeve of circular or square cross section, the interior of the sleeve being air-filled. In this case the appropriate diameter of the sleeve may be of the order of 1λ (Kiely-1).

The rods and cones of the retina of a human eye are similar to polyrod antennas but with unity front-to-back ratio and diameters of 1 to 2λ as compared to the high front-to-back ratio and $\lambda/2$ diameter or less of the polyrod of Fig. 17–21. The retina has an array of more than 100 million "polyrods." Curiously, to an antenna engineer, all of the feed system and connections (axons and dendrites) are *in front* of the "polyrods." However, this is all right because the "wiring" is transparent. For more detail see, for example, Kraus (1) and Fleisch.

17–9 MULTIPLE-HELIX LENSES（多螺旋透镜）

As mentioned earlier, an axial-mode monofilar helical antenna is a rudimentary form of lens antenna (see Fig. 17–2*b*). A multiplicity of parasitic axial-mode monofilar helices can also be arranged as in Fig. 17–22 to form a *phase-controlled lens.* Helices of opposite hand are mounted back-to-back on a spherical shell with the center at the focus. The helices on the outer side of the shell receive the incoming wave which is retransmitted to the focal point by the helices on the inner side of the shell. The lens is focused by rotating each helix pair

[1]This is the Hansen and Woodyard condition for increased directivity of an end-fire array.

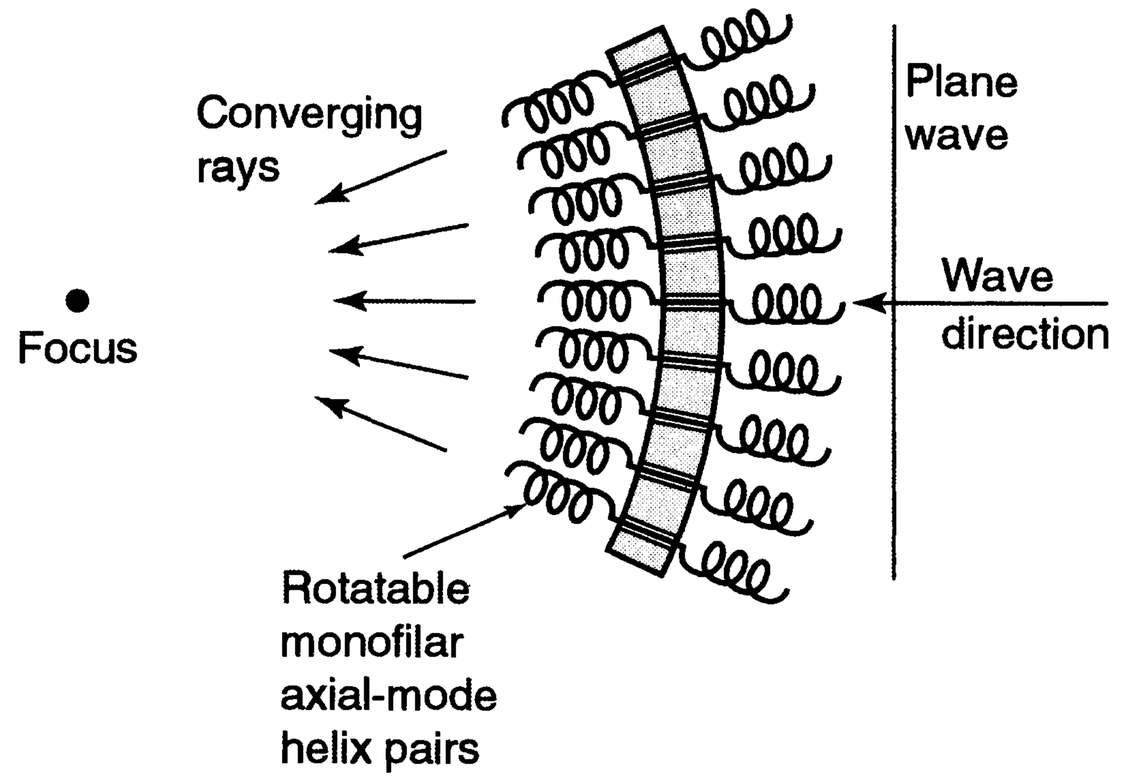


Figure 17–22
Helix lens. Focusing is accomplished by rotating helix pairs.

with respect to a reference pair (at the center) to compensate for path-length differences. Rotation of a helix pair through an angle θ shifts the phase by 2θ. The number of turns n of each helix is given by

$$n = \frac{4\pi A_p}{12 S_\lambda \lambda^2 N} \tag{1}$$

where

A_p = lens area
S_λ = helix turn spacing
N = number of helix pairs

To avoid grating lobes the spacing between helix pairs should be less than λ. The helices and coaxial interconnections should, of course, be matched. A helix lens 1.3 m in diameter with 1213 helix pairs has been constructed for operation at 8.15 GHz ($\lambda = 3.7$ cm). At this wavelength the lens diameter is 35λ producing a half-power beamwidth of about 2°.

17–10 LUNEBURG AND EINSTEIN LENSES（龙伯透镜和爱因斯坦透镜）

The *Luneburg lens* (Luneburg-1) shown in Fig. 17–23 is a spherically symmetric delay-type lens formed of a dielectric with index of refraction n which varies as a function of radius, as given by

$$n = \sqrt{2 - \left(\frac{r}{R}\right)^2} \tag{1}$$

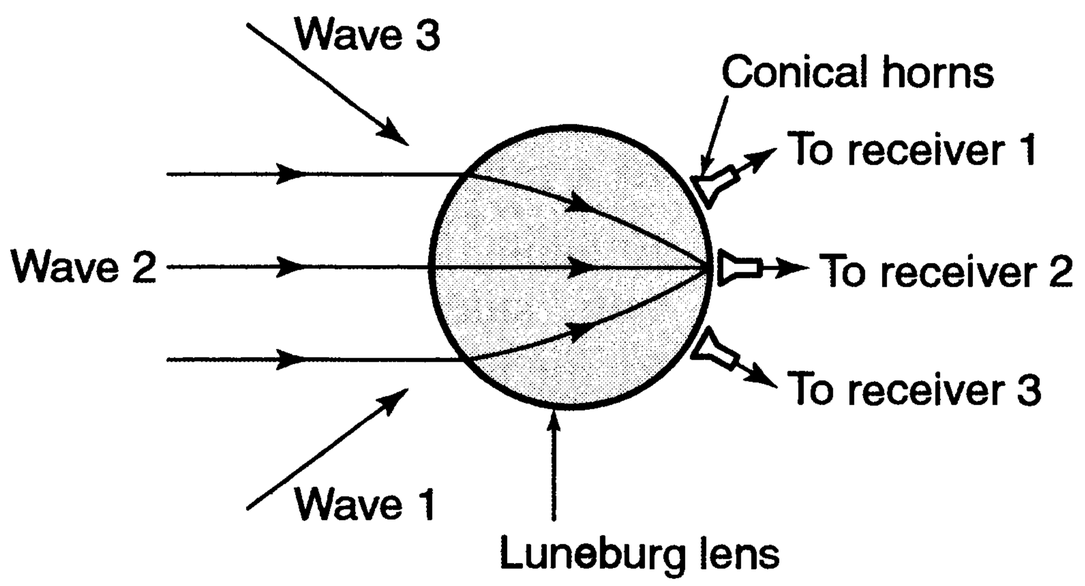


Figure 17–23
Luneburg lens. A plane wave incident on one side is brought to a focus on the opposite side.

where

r = radial distance from center of sphere
R = radius of sphere

When $r = R, n = 1$ while at the center of the sphere ($r = 0$), $n = \sqrt{2}$ (its maximum value). The lens has the property that an incident plane wave is brought to a focus on the opposite side of the sphere as suggested for wave 2 in Fig. 17–23. Simultaneously, waves from other directions will be brought to a focus at a point on the opposite side of the sphere, as suggested in Fig. 17–23 for waves 1 and 3. Thus, signals can be received simultaneously with a Luneburg lens from as many directions as there is space available on the sphere to place feed horns or other receiving devices. For steering a single beam the receiver (or transmitter) can be switched to different feed horns, or a single movable feed horn can be used. The variable index required can be obtained with an artificial dielectric material (Sec. 17–3) or with concentric shells of dielectric of different indices of refraction.

If a Luneburg sphere is cut in half and a reflecting sheet placed on the flat side, a Luneburg reflector-lens antenna results with incoming wave at an angle of incidence θ_i brought to a focus at the corresponding angle of reflection $\theta_r = \theta_i$.

The full spherical Luneburg lens provides beam steering in both polar coordinates (θ and ϕ). For steering in only one coordinate (ϕ), a plane (parallel-sided) section through the center of the sphere can be used. However, the beam is no longer the same in both coordinates due to vignetting in the θ direction. Kelleher gives a good review on the Luneburg lens and its variants (Kelleher-1).

Although the idea had been mentioned earlier, Albert Einstein's brief note in *Science* in 1936 was the first analysis of lens action by the gravitational field of a star such as the sun (Einstein-1). Actually any large mass—the sun, Jupiter, the earth, a neutron star or a black hole—would do. Incident electromagnetic waves passing around a star are deflected through an angle which is proportional to the mass of the star and brought to a focus on the far side, as suggested by the ***Einstein gravity lens*** of Fig. 17–24. There is no single focal point, rather a *focal line* extending from a minimum focal distance to infinity. The gain of the lens is proportional to the mass of the star and inversely proportional to the wavelength.

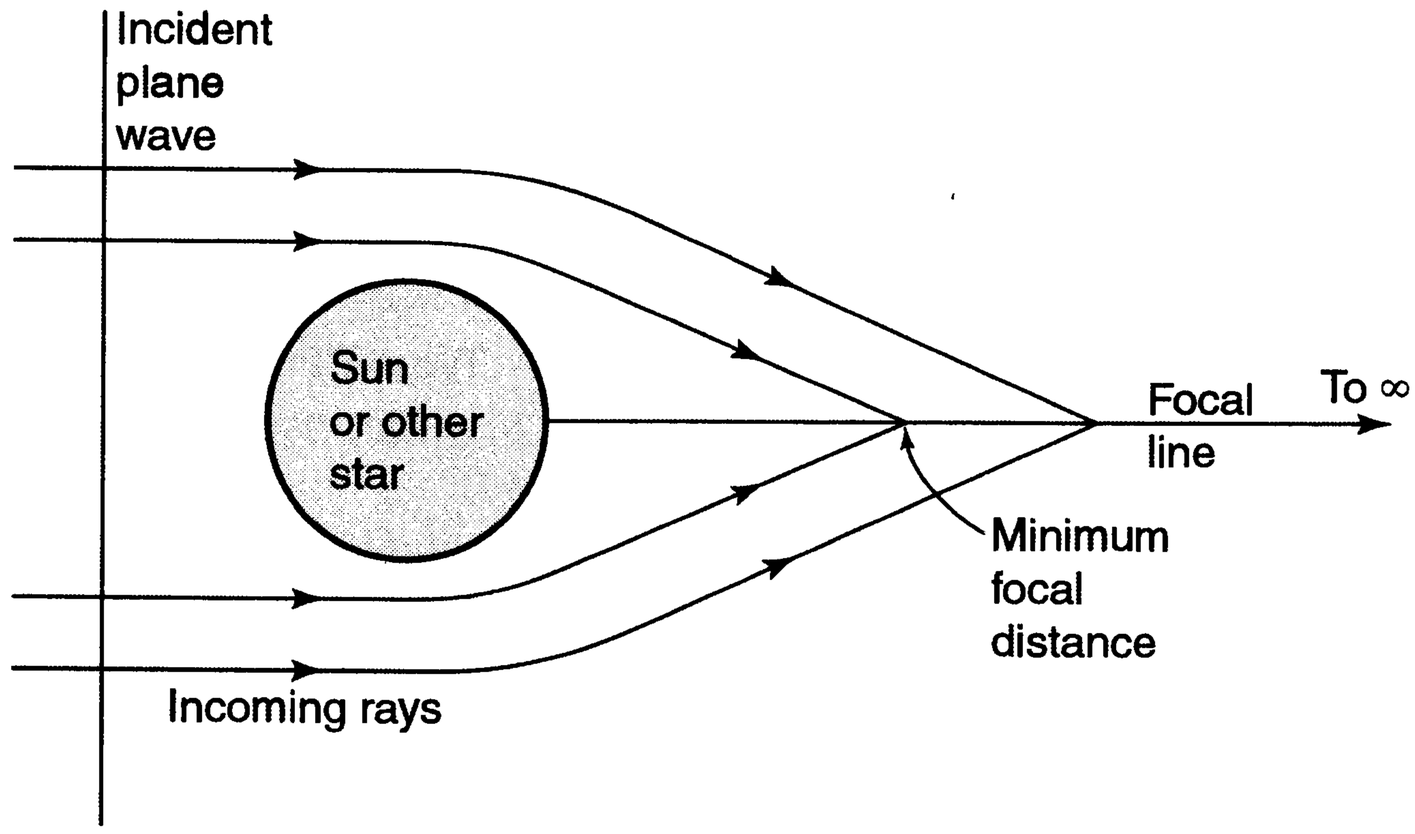


Figure 17–24
Einstein gravity lens. A plane wave incident on a large mass (such as the sun) is brought to a focus along a line extending from a minimum distance to infinity. Large gains are possible.

At $\lambda = 1$ mm a solar lens can, in principle, give a gain of more than 80 dB. If the spacecraft on the focal line has an antenna with 80 dB gain, the total system gain is 160 dB, equivalent to the gain of an array of 100 million 80 dB-gain antennas.

The minimum focal distance in the case of the sun is about a dozen times the distance of Pluto so that we must wait until it is possible to send a properly equipped spacecraft to that distance before the sun can be put to use as a gravity lens.

More details on the Einstein lens with a worked example are given in *Radio Astronomy* (Kraus-3) based on the very extensive treatment of Von R. Eshleman (Eshleman-1). See also Maccone (1).

REFERENCES（参考文献）

Chu, L. J. (1), and W. L. Barrow, "Electromagnetic Waves in Hollow Metal Tubes of Rectangular Cross Section," *Proc. IRE,* **26,** 1520–1555, December 1938.

Einstein, A., "Lens-like Action of a Star by the Deviation of Light in the Gravitational Field," *Science,* **84,** 506, 1936.

Eshleman, V. R. (1), "Gravitational Lens of the Sun: Its Potential for Observations and Communications over Interstellar Distances," *Science,* **205,** 1133, 1979.

Friis, H. T. (1), and W. D. Lewis, "Radar Antennas," *Bell System Tech. J.,* **26,** 270, April 1947.

Goldsmith, P. F. (1), and E. L. Moore, "Gaussian Optics Lens Antennas (GOLAs)," *Microwave J.,* **26,** 153–156, July 1984.

Goldsmith, P. F. (2), "Quasioptical Techniques at Millimeter and Submillimeter Wavelengths," in K. Button (ed.), *Infrared and Millimeter Waves,* vol. 6, Academic Press, 1982, pp. 277–343.

Goldsmith, P. F. (3), and G. J. Gill, "Dielectric Wedge Conical Scanned Gaussian Optics Lens Antenna," *Microwave J.,* **29,** 207–212, September 1986.

Hondros, D. (1), and P. Debye, "Elektromagnetische Wellen an dielektrischen Drahten," *Ann. Physik,* **32,** 465–476, 1910.

Kelleher, K. S. (1), "Electromechanical Scanning Antennas," in R. C. Johnson (ed.), *Antenna Engineering Handbook,* 3d ed., McGraw-Hill, New York, 1993, Chap. 18.

Kiely, D. G. (1), *Dielectric Aerials,* Methuen, 1953.

Kock, W. E. (1), "Metallic Delay Lens," *Bell System Tech. J.,* **27,** 58–82, January 1948.

Kock, W. E. (2), "Metal Lens Antennas," *Proc. IRE,* **34,** 828–836, November 1946.

Kock, W. E. (3), "Path Length Microwave Lenses," *Proc. IRE,* **37,** 852–855; August 1949.

Kraus, J. D. (1), "Some Unique Reflector Antennas," *IEEE Ant. Prop. Soc. Newsletter,* **24,** 10, April 1982.

Kraus, J. D. (2), and D. A. Fleisch, *Electromagnetics,* 5th ed., McGraw-Hill, New York, 1999, pp. 503–506.

Kraus, J. D. (3), *Radio Astronomy,* 2d ed., Cygnus-Quasar, 1986.

Luneburg, R. K. (1), *Mathematical Theory of Optics,* Brown University Press, 1944. (Luneburg has sometimes been misspelled as Luneberg).

Maccone, Claudio (1), *The Sun as a Gravitational Lens: Proposed Space Missions,* IPI Press, 1999.

Mueller, G. E. (1), and W. A. Tyrrell, "Polyrod Antennas," *Bell System Tech. J.,* **26,** 837–851, October 1947.

Nakatami, D. T. (1), and T. S. Ajioka, "Lens Designs Using Rotatable Phasing Elements," *IEEE APS Int. Symp. Dig.,* 357–360, June 1977.

Risser, J. R. (1), in S. Silver (ed.), *Microwave Antenna Theory and Design,* McGraw-Hill, New York, 1949, Chap. 11.

Schelkunoff, S. A. (1), *Electromagnetic Waves,* Van Nostrand, New York, 1943, pp. 425–428.

Schelkunoff, S. A. (2), "Equivalence Theorems in Electromagnetics," *Bell System Tech. J.,* **15,** 92–112, 1936.

Watson, R. B. (1), and C. W. Horton, "The Radiation Patterns of Dielectric Rods—Experiment and Theory," *J. Appl. Phys,* **19,** 661–670, July 1948.

Whitmer, R. M. (1), "Fields in Non-metallic Guides," *Proc. IRE,* **36,** 1105–1109, September 1948.

Wilkes, Gilbert (1), "Wavelength Lens," *Proc. IRE,* **36,** 206–212, February 1948.

PROBLEMS（习题）

17–2–1 （介质透镜）

Dielectric lens. (*a*) Design a plano-convex dielectric lens for 5 GHz with a diameter of 10λ. The lens material is to be paraffin and the F number is to be unity. Draw the lens cross section. (*b*) What type of primary antenna pattern is required to produce a uniform aperture distribution?

17–2–2 （柱形透镜）

Cylindrical lens. Prove (17–2–15*b*).

17–3–1 （人造介质）

Artificial dielectric. Design an artificial dielectric with relative permittivity of 1.4 for use at 3 GHz when the artificial dielectric consists of (*a*) copper spheres, (*b*) copper disks, (*c*) copper strips.

***17–4–1** （未分区的金属板透镜）

Unzoned metal-plate lens. Design an unzoned plano-concave E-plane type of metal plate lens of the unconstrained type with an aperture 10λ square for use with a 3-GHz line source 10λ long. The source is to be 20λ from the lens ($F = 2$). Make the index of refraction 0.6. (*a*) What should the spacing between the plates be? (*b*) Draw the shape of the lens and give dimensions. (*c*) What is the bandwidth of the lens if the maximum tolerable path difference is $\lambda/4$?

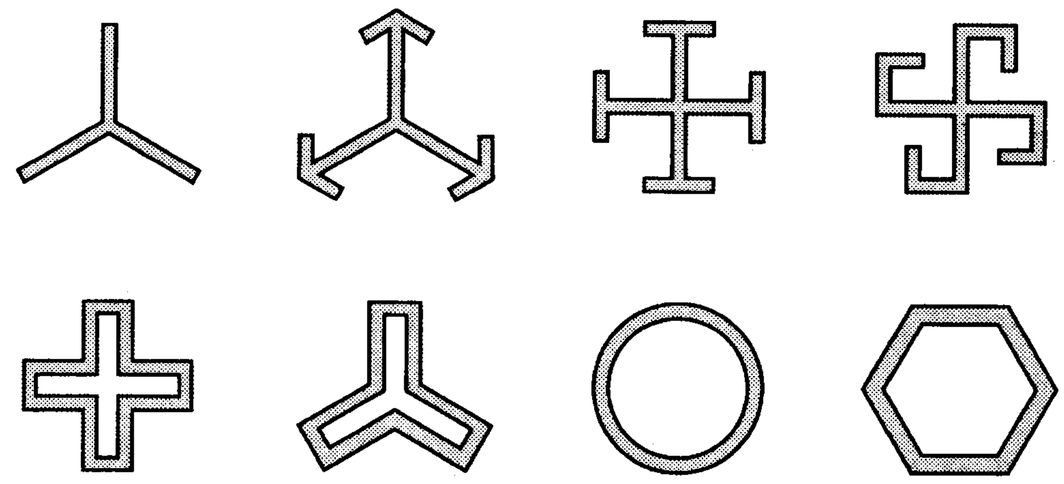
17–9–1 （螺旋透镜）

Helix lens. Confirm (17–9–1).

For computer programs, see Appendix C.

CHAPTER 18

Frequency-Selective Surfaces and Periodic Structures 频率选择表面和周期结构



By Ben A. Munk[1]

Topics in this chapter include:

- Definition of FSS（频率选择表面的定义）
- Basic dipole and slot FSS
  （基本的偶极子或缝隙频率选择表面）
- Dielectric slab FSS for radome
  （介质片频率选择表面与天线罩）
- Slotted metal radome（开缝的金属天线罩）
- Hybrid radome（混合天线罩）
- Ideal stealth radome（理想的隐形天线罩）
- Transmission and reflection properties of simple periodic surfaces of wires
  （简单周期性导线表面的传输和反射特性）
- Complementary surfaces and Babinet's principle（互补表面与巴比涅原理）
- Oblique angles of incidence（斜入射角）
- Element types（单元的型式）
- Controlling bandwidth with angle of incidence and polarization
  （控制入射角和极化的频带宽度）
- Other applications: dichroic surfaces and meanderline polarizers
  （其他应用:分频表面和回折线极化器）

18–1 INTRODUCTION. DEFINITION OF FSS（引言：频率选择表面的定义）

A Frequency-Selective Surface (FSS) is a surface which exhibits different reflection and/or transmission properties as a function of frequency. Two basic types are an array of wires and the array of slots, as shown in Fig. 18–1. This is followed by a dielectric slab as the FSS. A surface which is NOT frequency-selective is a flat perfectly conducting sheet.

These surfaces perform only for linear polarizations with E as shown in Fig. 18–1. To perform for all polarizations, crossed dipoles and crossed slots can be used but only successfully for normal angle of incidence. For a discussion of elements in general see Sec. 18–9.

Frequency-Selective Surfaces (FSSs) have many important applications in today's technology. Perhaps the most interesting is for making radomes that are either improved in their performance or designed to have special features.

---

[1]ElectroScience Laboratory, The Ohio State University.

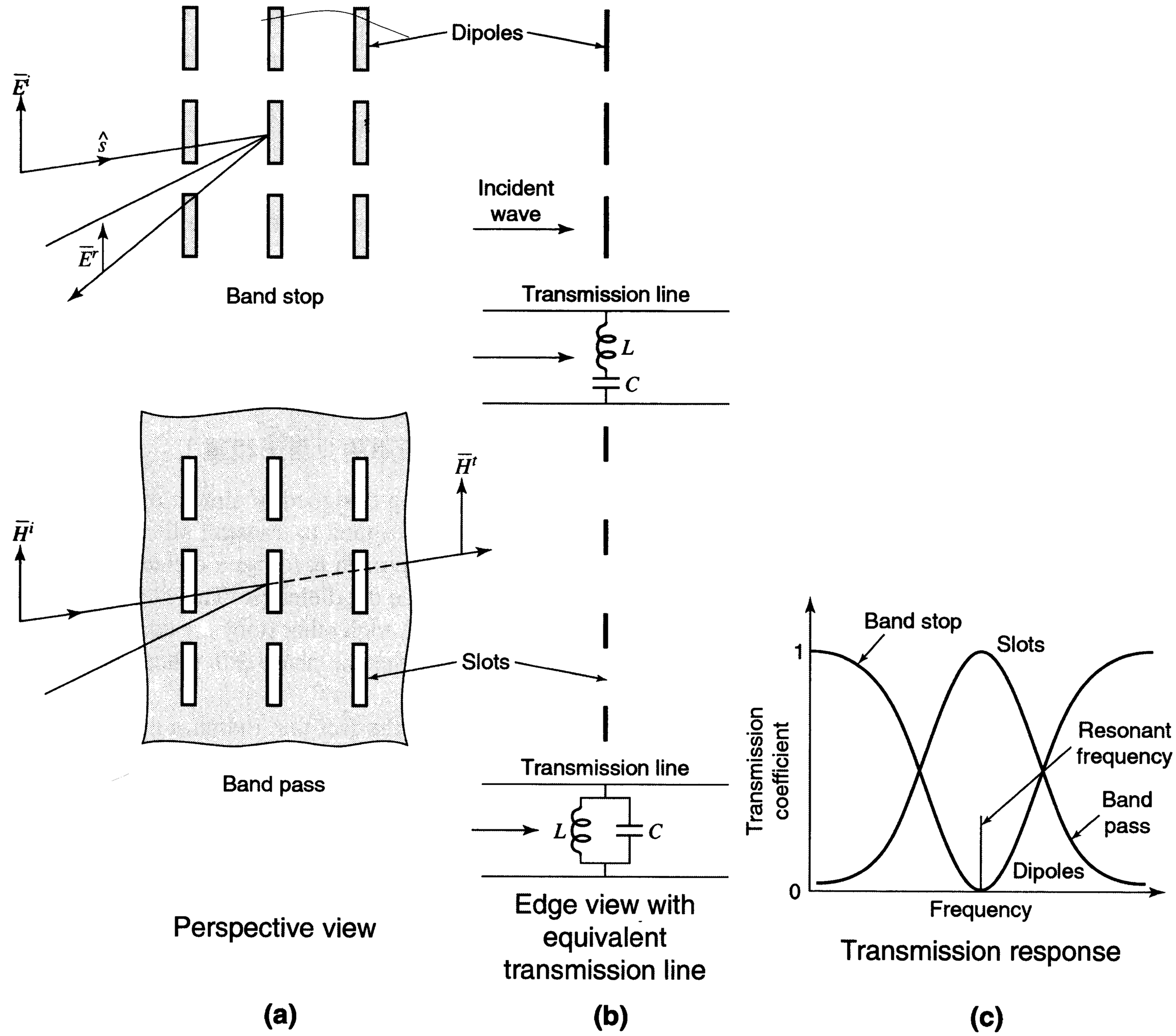


Figure 18–1
Basic frequency selective surfaces: Array of dipoles (above) acts as a band-stop filter while the array of complementary slots in a perfectly conducting sheet (below) acts as a bandpass filter. The equivalent transmission lines are also shown. The transmission coefficient for both cases is shown at (*c*). Note that for the dipoles *E* is vertical but for the slots *E* is horizontal or *H* is vertical.

Basically, a radome can be considered as merely a protective cover placed in front of an antenna. Thus, a radome should be mechanically strong and at the same time transparent at the operating frequency of the antenna. While an exhaustive treatment of the classical radome theory is given elsewhere (Tice-1), we shall in this chapter limit ourselves to very simple radome types that lead to the more exotic combinations of FSS and a stratified dielectric medium usually referred to as a "hybrid radome."

18–1a Basic Dipole and Slot FSS（基本的偶极子或缝隙频率选择表面）

An array of resonant dipoles (Fig. 18–1) acts as a ***band-stop filter,*** passing waves above and below the dipole resonant frequency but not at the resonant frequency. The complementary array of slots acts as a ***bandpass filter,*** passing waves at the resonant frequency of the slots but rejecting them at higher and lower frequencies.

The action of the dipoles is equivalent to that of a series-tuned circuit on a transmission line as suggested in Fig. 18–1*b*, while the action of the slots is analogous to that of a parallel tuned circuit. In each case the inductor and capacitor resonate at the "pass" or "stop" frequencies. *Note:* The equivalent circuit will in general vary with angle of incidence and polarization.

18–2 THE HALFWAVE DIELECTRIC RADOME（半波长厚的介质天线罩）

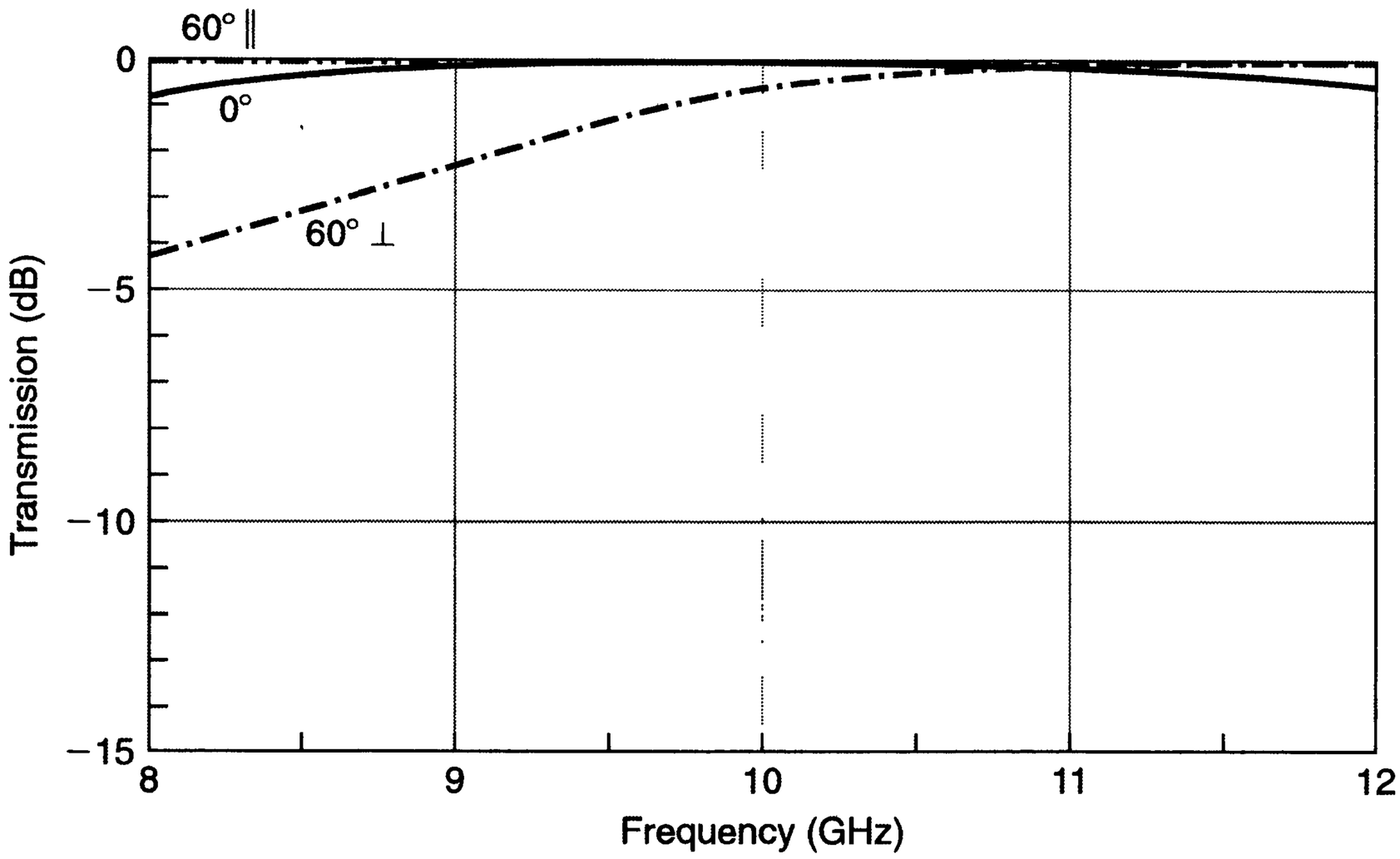
The earliest radomes were often quite primitive, being designed of simple dielectric materials, even plywood. Ideally, a radome should be designed to transmit all of the energy incident upon it and reflect nothing. The simplest approach is to use a dielectric sheet of electrical thickness equal to one-half wavelength inside the dielectric. This simply ensures that the reflections from the front and back will cancel each other (they are equal in amplitude but with opposite phase). A distance of $\lambda_\varepsilon/2$ leaves the phase difference unchanged, where λ_ε equals wavelength in the dielectric.

However, considerable complications arise from the fact that radomes must function for different angles of incidence as well as polarizations. Fundamentally, all stratified media with typical slab thickness d will have electrical thickness equal to $\beta_\varepsilon d \cos \eta_\varepsilon$, where $\beta_\varepsilon = 2\pi/\lambda_\varepsilon$ is the propagation constant in the slab, and η_ε is the angle of propagation inside the slab measured from the normal (see Fig. 18–2, bottom).

Thus, as we increase the angle of incidence η_ε, $\cos \eta_\varepsilon$ will be reduced. In general, the trend for a stratified medium without FSS is that full transmission or resonance is observed at a higher frequency when β_n is higher yielding $\beta_n d_n \cos \eta_n$ about the same as for normal incidence.

A typical set of transmission curves for various angles of incidence as well as polarizations is shown in Fig. 18–2, top. Here we show the case of a dielectric slab of thickness $d \sim 0.75$ cm and dielectric constant $\varepsilon_r = 4.0$. This is seen to give us a perfect transmission around 10 GHz for normal angles of incidence. However, as we increase the angle of incidence we clearly see how the resonant frequency is moving upward as predicted earlier. Only when the incident E-field is parallel to the plane of incidence is this hard to observe. The reason is simply that in this case we will observe the Brewster's angle when perfect transmission is obtained from "dc to THz," i.e., we are washing out any resonance whatsoever.

More specifically we observe a narrowing of the resonant curves for E orthogonal to the plane of incidence and a broadening in the parallel case. This has to do with a change of the characteristic impedances of the equivalent transmission lines, as will be discussed later.



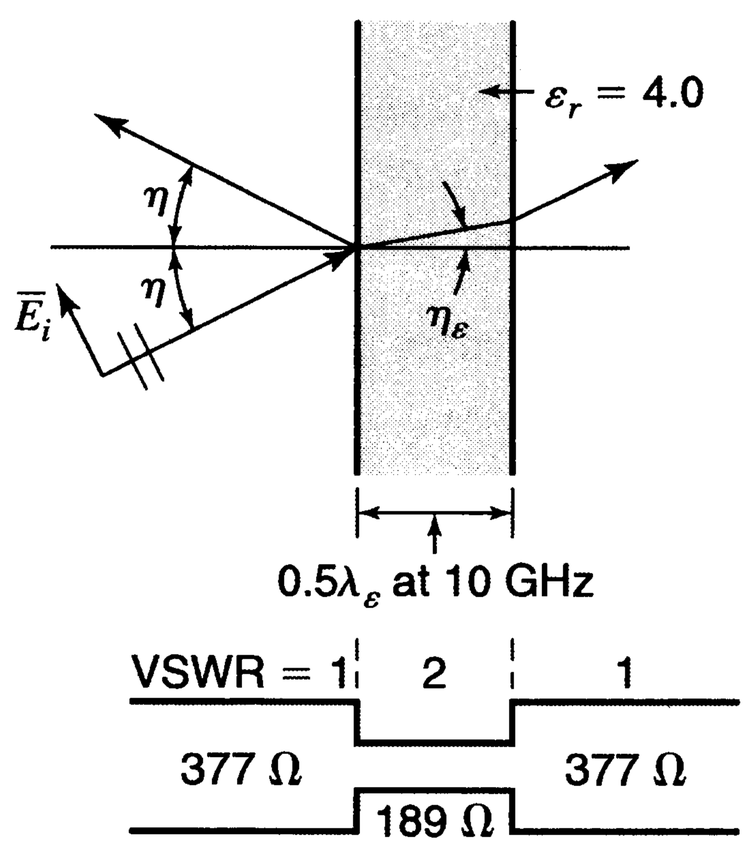
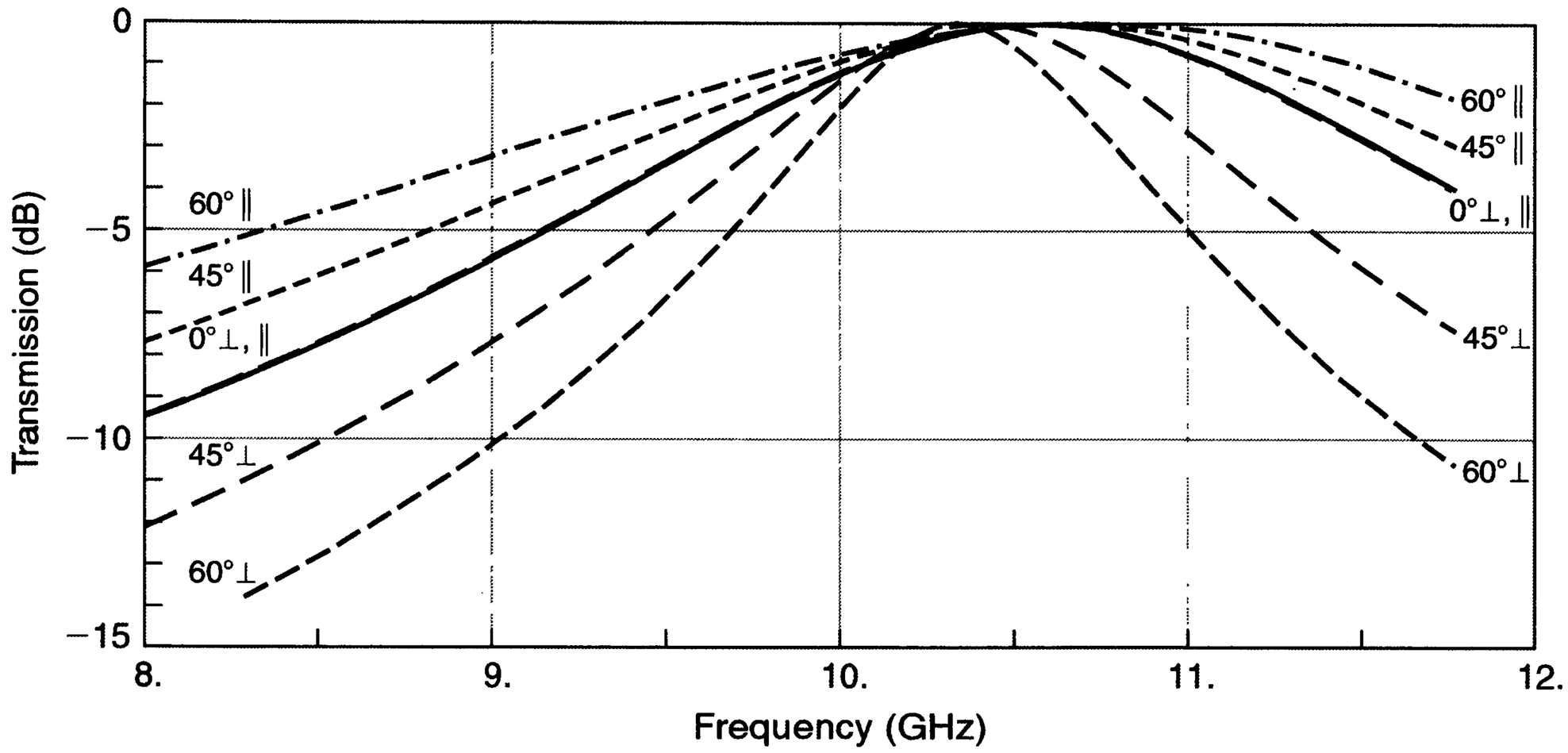


Figure 18–2
Typical transmission curves as a function of frequency for a simple dielectric sheet with $\varepsilon_r = 4$. ($\lambda_\varepsilon/2$ radome). Note: the equivalent circuit at the bottom is valid only at normal angles of incidence.

18–3 SLOTTED METAL RADOME（开缝的金属天线罩）

The mere suggestion of a radome made of metal might at first sound like an absurdity. However, if a conducting surface is provided with small slot elements arranged in a periodic pattern, this arrangement can be perfectly transparent at one frequency. The possible shapes of such slots are discussed later; see Sec. 18–9. We shall here limit ourselves to show a typical example of three-legged loaded elements as shown at the bottom of Fig. 18–3. Note that the bandwidth becomes narrower for higher angles of incidence when the incident



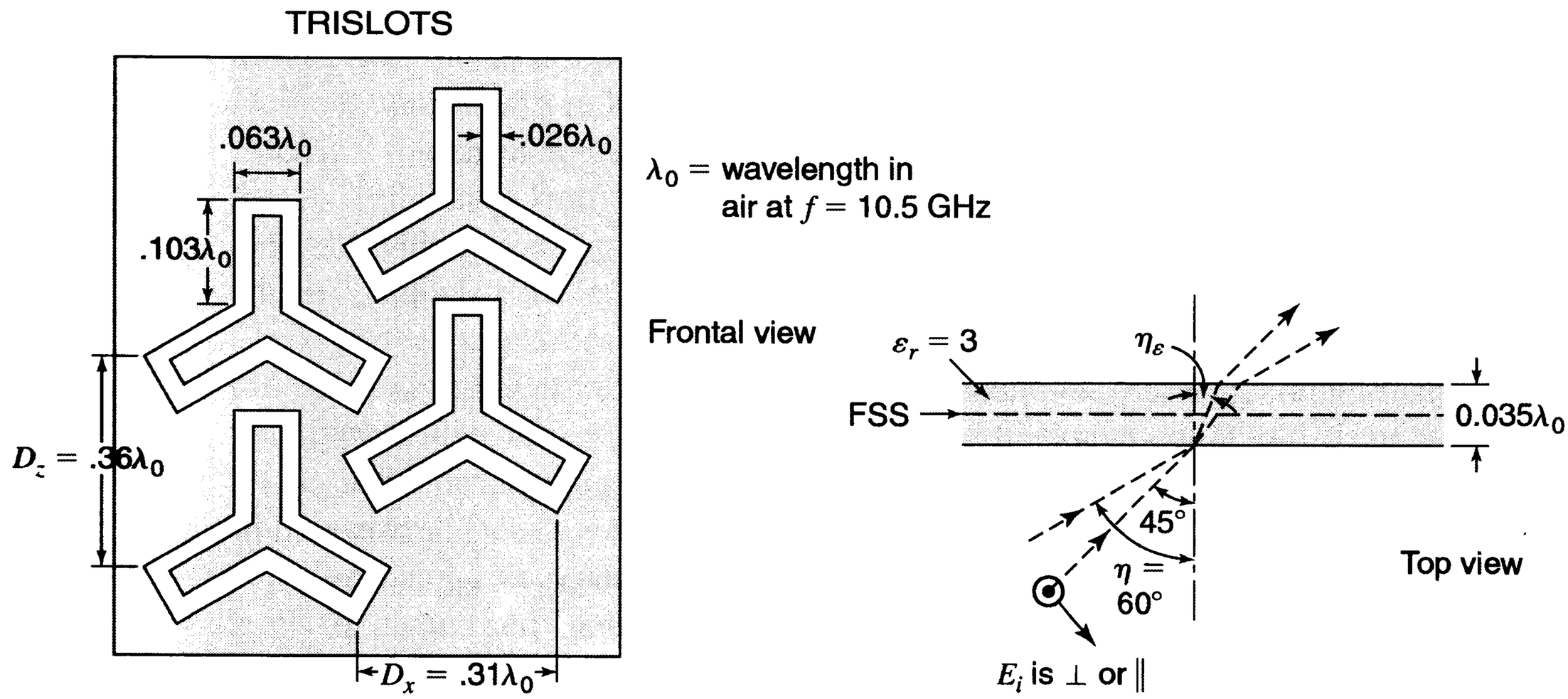


Figure 18–3
Transmission curves as a function of frequency for various angles of incidence on an FSS of loaded trislot elements bounded by dielectric sheets as shown at the bottom. The perimeter of the elements is ~0.81λ in air or ~1.4λ in the dielectric at 10.5 GHz, yielding a circumference of 1.1λ on average. Along the centerline of the slot element the circumference is ~1.0λ on average.

E-field is orthogonal to the plane of incidence in much the same way, as was the case for the purely dielectric case in Fig. 18–2. However, there is a notable difference between the two cases, namely, that while the resonance frequency moves up for oblique angles of incidence in the dielectric case, it moves downward in the dielectric-metal sandwich of Fig. 18–3. This strongly suggests to the creative designer that a combination of dielectric and metallic surfaces could lead to designs with more stable resonant frequency. Such surfaces are usually denoted "Hybrid Periodic Surfaces," as will be discussed next.

18–4 THE SIMPLE HYBRID RADOME（简单的混合天线罩）

The simplest form of a hybrid radome is obtained by combining the slotted surface in Fig. 18–3 with the dielectric slab shown in Fig. 18–2. Typical transmission curves obtained from the PMM program are shown in Fig. 18–4. We observe that the resonant frequency remains fairly constant with angle of incidence as expected, since the metallic and dielectric surfaces went in opposite directions with the angle of incidence. However, the bandwidth still changes dramatically. This feature may not be of any great concern if

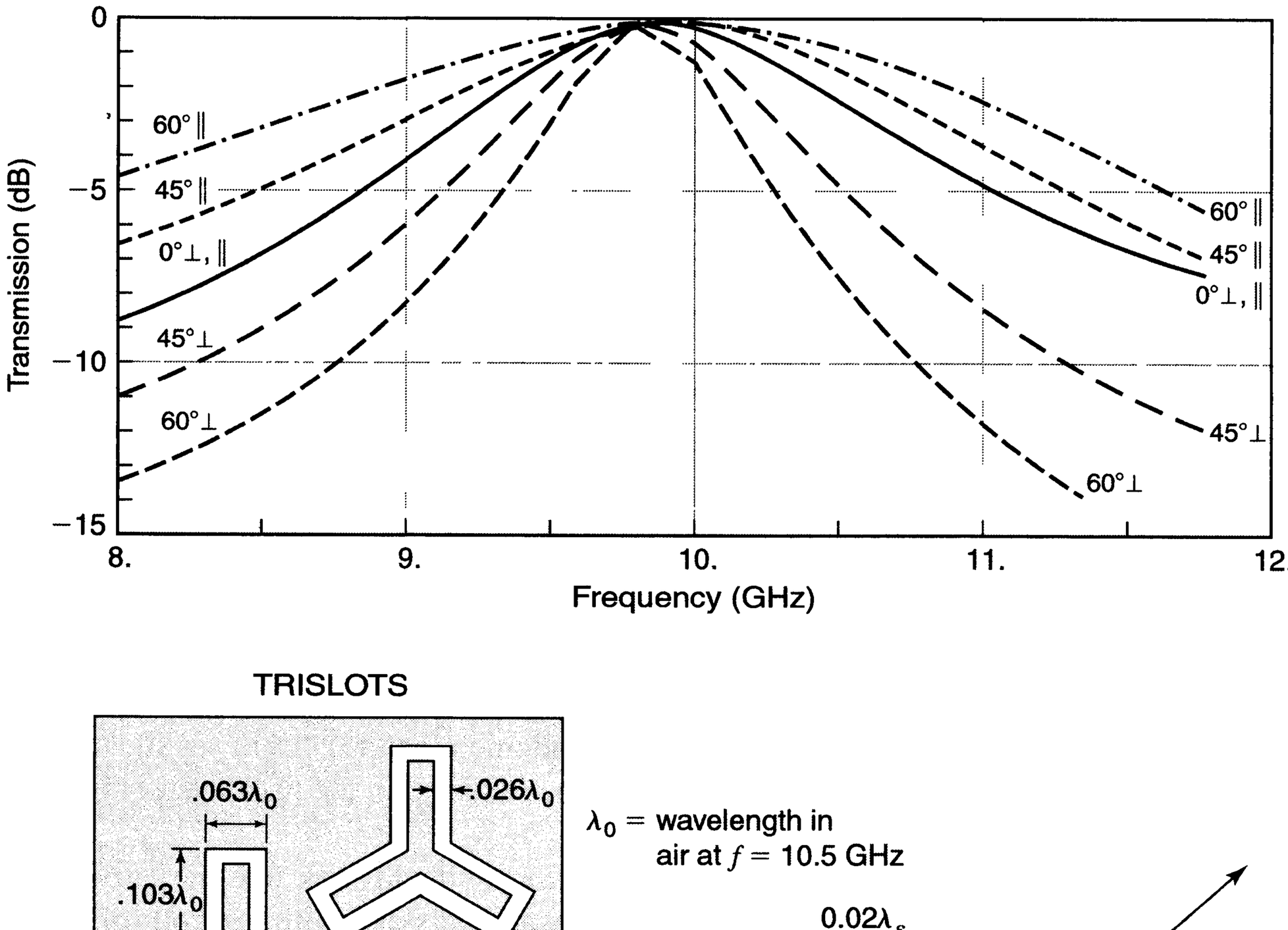


Figure 18–4
Typical transmission curves as a function of frequency for the combination of the $\lambda_\varepsilon/2$ slab in Fig. 18–2 with the FSS design in Fig. 18–3. Such a combination is usually called for a hybrid radome.

the radome merely has to let energy through in a narrow frequency range. However, as shown in the next section, there are important cases when we must require much more than that!

Note also that the resonant frequency of the slotted surface should be very close to the frequency where the dielectric slab is electrically one-half wavelength thick. Otherwise transmission loss will occur. This feature makes this design somewhat difficult to work with in practice.

18–5 THE IDEAL STEALTH RADOME（理想的隐形天线罩）

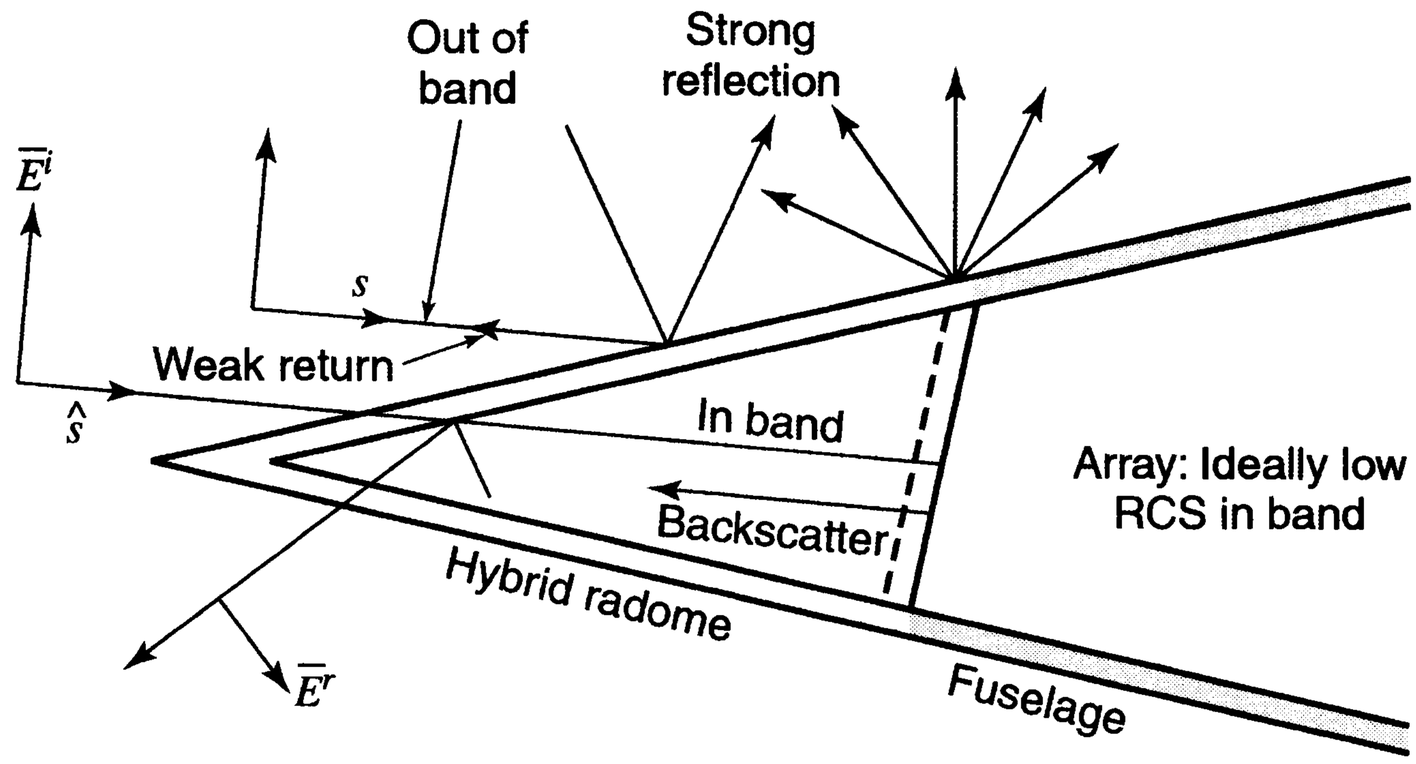
To *see* but not *be seen* is the objective of a "stealth" radome. It is well known that the Radar Cross Section (RCS) of an antenna can be significant. For example, a planar array can have an RCS as high as that of a flat plate with an area equal to the total aperture of the array (for normal angle of incidence and no grating lobes). However, while it never can be higher (!), it can be significantly lower by proper design of the array. In fact, it may for proper matching of the elements and no grating lobes in principle be equal to zero. Unfortunately a good match can only be maintained over a limited bandwidth. Thus, when the antenna is exposed to frequencies outside the matching range of the array, the RCS will increase and eventually be as large as that of an equivalent flat plate as mentioned above. To reduce the RCS of any antenna outside its operating frequency range we may place a radome with bandpass characteristic in front of it as illustrated in Fig. 18–5, top. The concept here is simply that the antenna can receive and transmit freely in the frequency range where the radome is transparent; see the typical transmission curve in Fig. 18–5, bottom. In that range the radome will not affect the RCS of the antenna that consequently bears the major responsibility for attaining a low backscattering signal. However, in the frequency range when the radome is opaque, an incident signal will always be strongly reflected in the bistatic direction(s). If the radome is shaped "right" as shown, for example, in Fig. 18–5, top, only a small signal will be reflected in the backscatter direction; i.e., we obtain low RCS.

Ideally, the bandpass curve for the radome should look like the curve in Fig. 18–5, bottom; i.e., have a flat top inside the passband and skirts with fast roll-off outside toward the opaque region.

Conceptually this sounds like a simple classical network problem. However, in actuality the situation is complicated significantly by the fact that the shape of the bandpass curve has to be maintained for all polarizations and a large range of angles of incidence. This is a complex problem that will be discussed in the next sections.

18–6 TRANSMISSION AND REFLECTION PROPERTIES OF SIMPLE PERIODIC SURFACES OF WIRES（简单周期性导线表面的传输和反射特性）

In this section we will discuss the fundamental physical properties of periodic surfaces of wires. Thus, we show in Fig. 18–6 the simplest FSS, namely, the one comprised of simple straight elements arranged in a rectangular array. However, let us emphasize right at the start that this type of FSS is rarely used. First of all it will only handle one polarization.



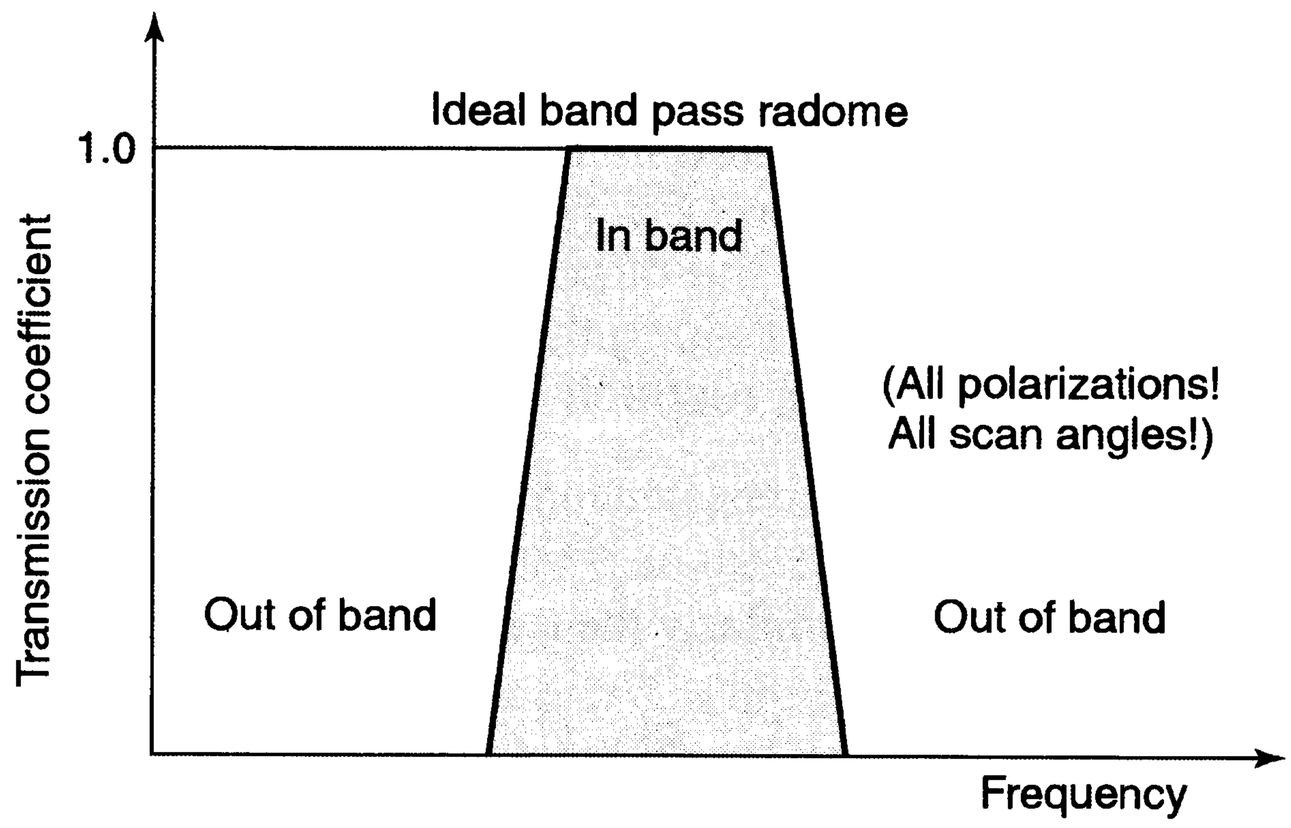


Figure 18–5
Use of a hybrid radome (top) with an ideal bandpass characteristic (bottom) to reduce the antenna RCS out of band. No reduction will be observed in the pass band of the radome.

While this problem could eventually be remedied by using, for example, crossed elements, such a surface will in general exhibit many undesirable features such as a second resonance close to the fundamental and also poor stability of the resonance frequency with angle of incidence. For a survey of more suitable element types, see Sec. 18–9. For an in-depth discussion and actual calculated resonance curves, see Munk (1).

We denote the length of the elements by $2l$ and the interelement spacings by D_x and D_z as shown in Fig. 18–6. While such a configuration is exposed to an incident plane wave with field vector $\overline{E}_i$, a wave $\overline{E}_r$ will be reflected in the specular direction while another part $\overline{E}_t$ of the incident signal will be transmitted through the surface in the forward direction.

We define the specular reflection coefficient as

$$^{E}\Gamma = \frac{E_r}{E_i} \tag{1}$$

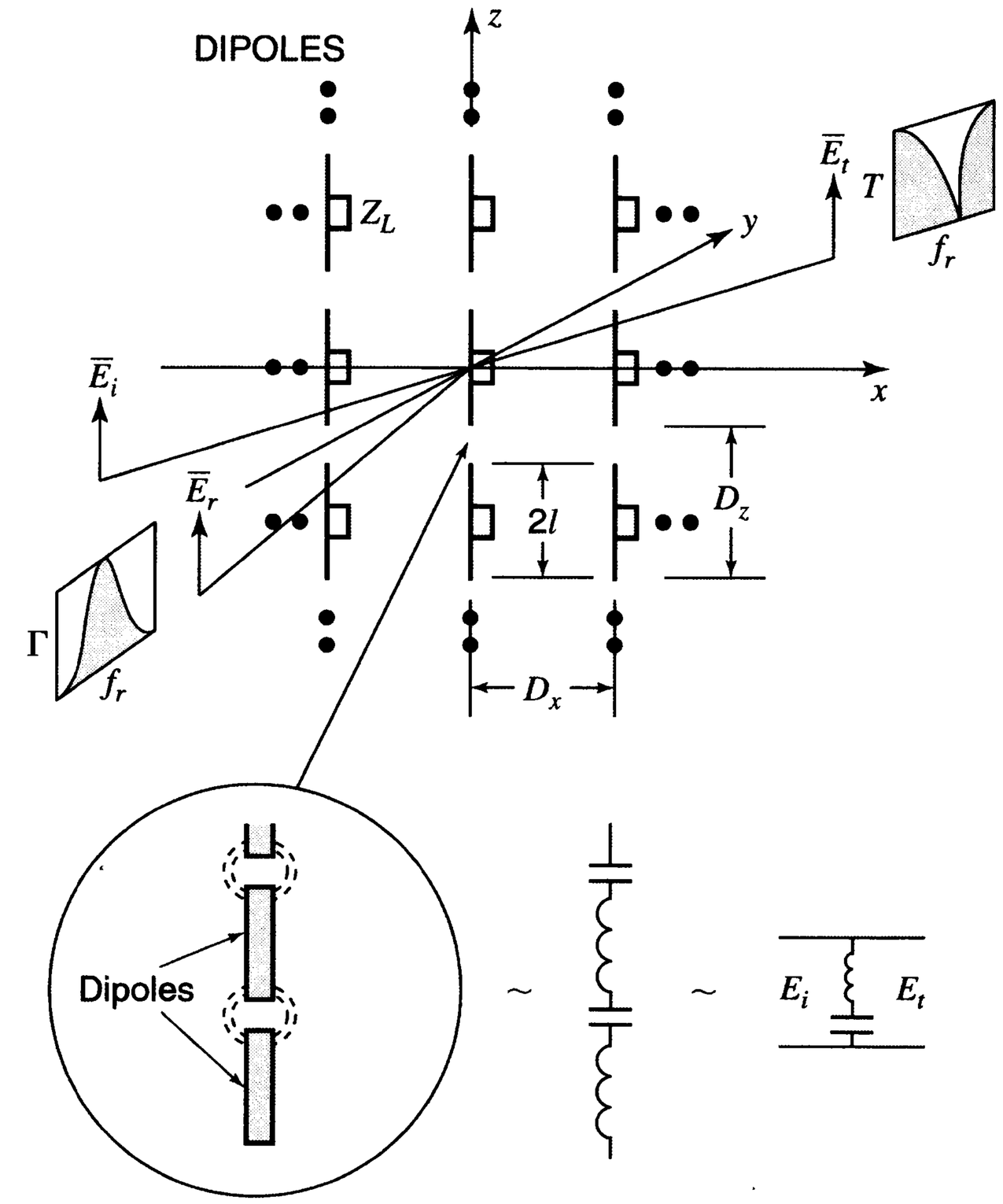


Figure 18–6
A periodic structure of electric conductors ("dipoles") with load impedances Z_L excited by an incident plane wave $\overline{E}_i$ which is being partly reflected in the specular direction ($\overline{E}_r$) and partly transmitted in the forward direction ($\overline{E}_t$). (s_x and s_z denote the directional cosines along the x- and z-axes, respectively.)

and similarly the transmission coefficient in the forward direction as

$$^{E}T = \frac{E^t}{E^i} \tag{2}$$

When we change the frequency we will find that the reflection as well as the transmission coefficient also changes as indicated in the inserts of Fig. 18–6.

A physical explanation is simply that series capacitances are associated with the gaps between the elements while the strips themselves exhibit a series inductance (see Fig. 18–6, bottom). Thus, it is not surprising that a typical equivalent circuit consists of an equivalent transmission line with a series LC circuit in shunt as shown. At the resonant frequency f_r the reflection coefficient for the electrical field $^{E}\Gamma = -1$, and transmission field $^{E}T = 0$, i.e., the periodic structure acts as a groundplane. At other frequencies the periodic structure is partly transparent. Thus, such a structure can be used as a simple filter. It is often called a "Frequency-Selective Surface" (FSS). By cascading two or more surfaces and by proper use of dielectric slabs we can obtain highly sophisticated filters, as will be discussed later.

We further show in Fig. 18–7 an infinite array of slots in an infinite perfectly conducting groundplane. As above we now expose this configuration to an incident plane wave; however, we now require the incident H vector to be parallel to the elements as shown.

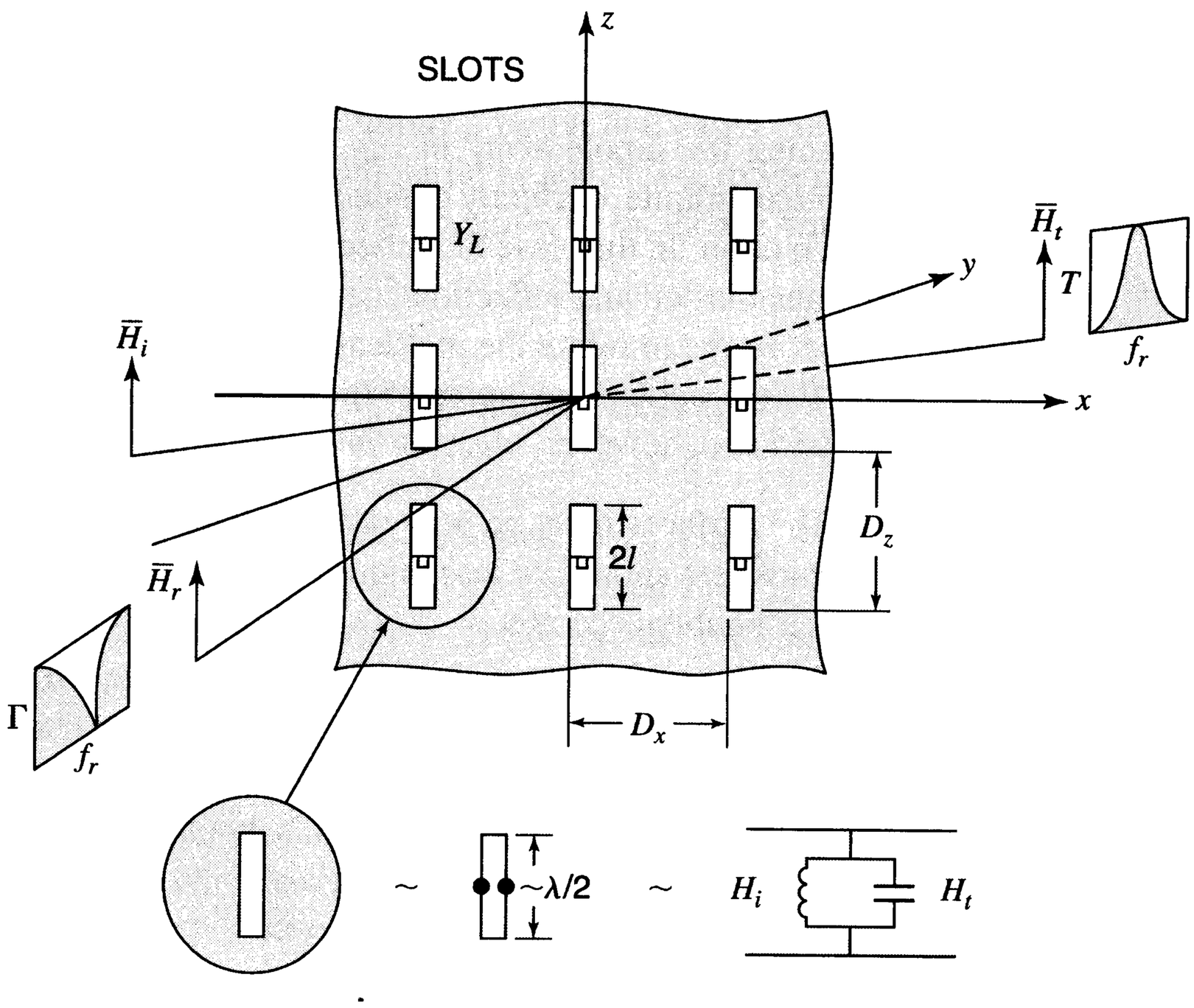


Figure 18–7
A periodic structure of "slots" with load admittance Y_L. Passively excited by an incident plane wave $\overline{H}_i$ being partly reflected in the specular direction ($\overline{H}_r$) and partly transmitted in the forward direction ($\overline{H}_t$).

We define the reflection and transmission coefficients analogous to (1) and (2) above except that we (in general) refer to the H field rather than the E field. This notation leads to a very convenient symmetry between the two cases.

The conducting strips in Fig. 18–6 can be considered as dipoles short-circuited at their center. Thus, this type of periodic surface is often referred to as a dipole or wire case while similarly the case in Fig. 18–7 is referred to as a slot case. Note that both configurations alternatively can be reactively loaded at their centers. As we shall see later, this feature plays an important role in reducing the physical size of the elements which is an absolute must for producing high-quality FSSs. If we, for example, add a series inductor to the dipole case the resonance f will be lower. The slot surface in Fig. 18–7 will exhibit a resonance similar to its counterpart in Fig. 18–6. However, instead of being reflective at that frequency it is completely transparent. A physical explanation is suggested by simply considering the edges of the slots as two twin lead transmission lines of length $\sim\lambda/4$ and in parallel as shown in the insert of Fig. 18–7, bottom. Since the ends of the two transmission-line segments are short-circuited the impedances at the centers of the slots will be infinite and thereby allow perfect transmission of the incident field. The equivalent circuit for a typical slotted surface is also shown in Fig. 18–7, bottom. It consists of a transmission line shunted with a parallel LC circuit. At resonance the incident signal will go through the surface with no attenuation while the reflected signal is zero; see inserts for 1 and 2 in Fig. 18–7. *Note:* The equivalent circuits are only valid for no grating lobes and change with angle of incidence and polarization.

18–7 COMPLEMENTARY SURFACES AND BABINET'S PRINCIPLE (互补表面与巴比涅原理)

If placing the dipole array in Fig. 18–6 on top of the slot array in Fig. 18–7 leads to a complete, infinite, perfectly conducting surface, the two surfaces are called complementary to each other. In that case the reflection and transmission coefficients for one of them equal the transmission and reflection coefficients of the other, respectively, as shown in Fig. 18–1 (do not forget to rotate the incident field vectors E_i and H_i as shown).

This is a simple example of Babinet's principle, which is well known from optics. For it to be valid, however, we must observe certain requirements:

1. The surfaces must be perfectly conducting and "infinitely thin." In practice the screen thickness should be less than $\sim 1/1000\lambda$ in order for Babinet's principle to be reasonably accurate.
2. Adding a thin layer ($< \sim 0.05\lambda$) of dielectric will lead to a lower resonant frequency in both the dipole and the slot cases of approximately the same amount. However, use of thicker slabs leads to quite different results for the two cases.
3. Cascading two or more surfaces behind each other leads to radically different transmission and reflection curves in the two cases.

18–8 OBLIQUE ANGLE OF INCIDENCE (斜入射角)

The discussion in the last section has assumed that the angle of incidence is constant. However, periodic structures should always be carefully examined for variations with the angle of incidence as well as polarization.

A complete and exact treatment of this subject is beyond the scope of this chapter. For an in-depth treatment see Munk (1), Vardaxoglou (1) and Wu (1). It will be sufficient here to say that the changes with angle of incidence essentially are:

1. Change of the resonance frequency, usually downward.
2. Change of bandwidth.

The first problem, namely, change of resonance frequency, can usually be minimized by keeping the interelement spacings D_x and D_z small in terms of wavelength like $< \sim 0.4\lambda$ (this requires either small elements or interlacing them into each other). The second problem is harder to handle. It can be shown (Munk-2) that the bandwidth variation of any periodic surface can be essentially explained by a variation of the characteristic impedance of the equivalent transmission line, shown in Fig. 18–6, bottom right, namely, it should be divided by $\cos\eta$ (η = angle of incidence) when E is orthogonal to the plane of incidence and multiplied by $\cos\eta$ when E is parallel. Elementary network theory then readily shows that this leads to a broadening of the resonance curve for $\overline{E}$ orthogonal and vice versa for $\overline{E}$ parallel.

Consistently we find in the slot case as in Fig. 18–7 that the intrinsic admittance of the equivalent transmission line should be divided and multiplied by $\cos\eta$ when H is orthogonal and parallel to the plane of incidence, respectively. Again, as above in the dipole case, it leads to a broadening of the resonance curve when H is orthogonal and vice versa for the parallel case.

Note how the dipole and slot cases are similar in their behavior with angle of incidence as a result of referring to the E field in the dipole case and to the H field in the slot case as well as working with impedance versus admittance in the two cases. Several actual calculated examples of resonance curves for slotted surfaces are shown in Figs. 18–3 and 18–4. Note the broadening of the resonance curves with angle of incidence when H is orthogonal to the plane of incidence.

18–9 ON THE SHAPE AND DEVELOPMENT OF ELEMENTS
(单元的形状及其演化)

Arriving at various element shapes is often clouded in a mystery. Actually a good starting point is to consider the straight dipole element shown in Fig. 18–8*a*. When it is $\sim\lambda_\varepsilon/2$ long it will resonate and scatter effectively. However, the problem with this element is that the resonance in general may change considerably with angle of incidence because the interelement spacings must be larger than $\lambda_\varepsilon/2$. One remedy is simply to shorten the elements. They then no longer resonate but they can be tuned back to resonance by being "loaded" with a small inductance at its center as shown in Fig. 18–8*b*. Alternatively, this inductance can be replaced by a two-wire transmission line as shown in Fig. 18–8*c*. And finally, it can be made into a bipolarized version called a "four-legged loaded element" as shown in Fig. 18–8*d*. It is interesting that although we start our element development with a straight dipole we are actually ending up with an element that can be considered a loop. It simply resonates when the circumference is equal to λ_ε, leading to relatively compact

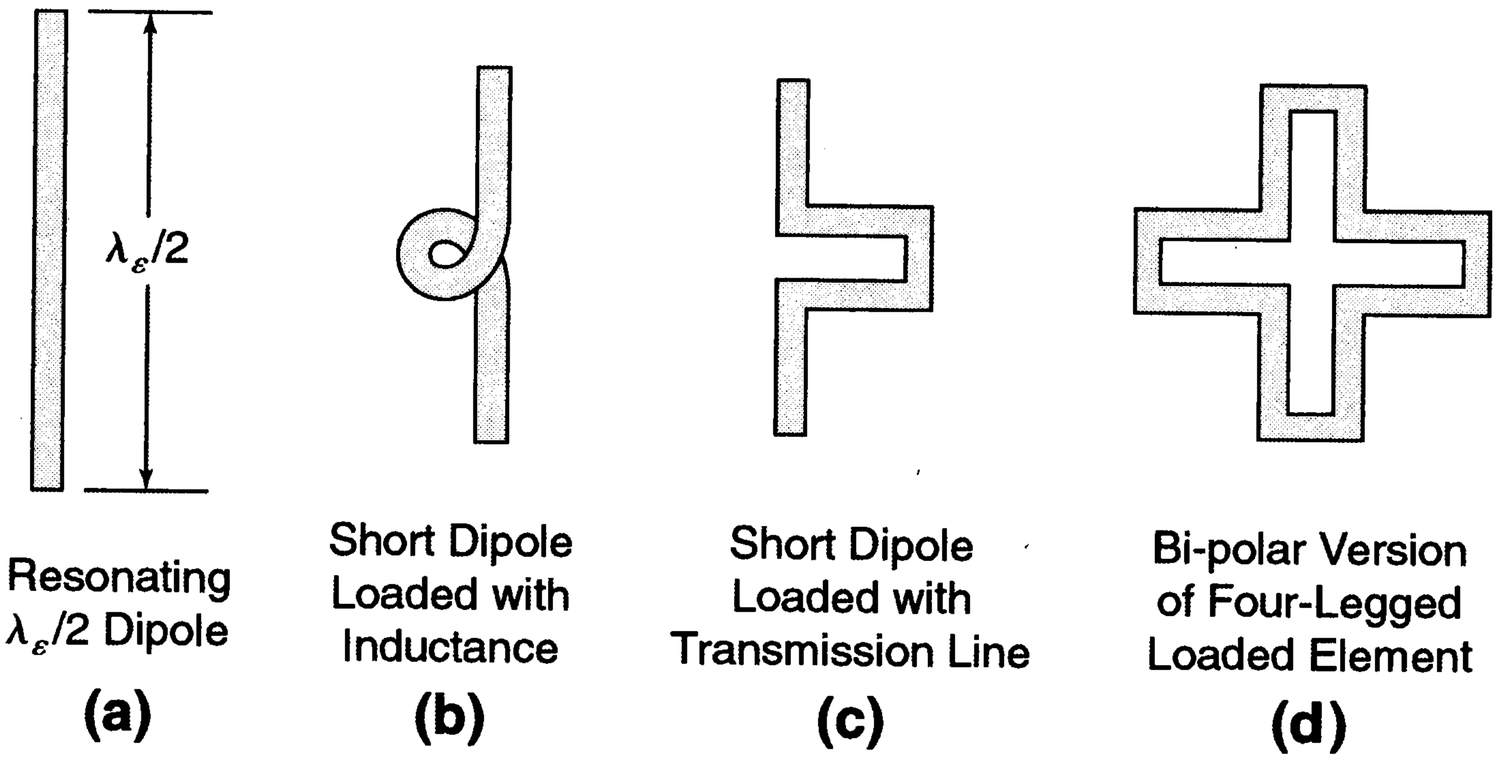


Figure 18–8
The development of the so-called "loop" elements originates with a simple straight $\lambda_\varepsilon/2$ dipole that is shortened and inductively loaded. The loop elements are preferred for making high-quality FSS for oblique angle of incidence. All loop elements resonate when their average circumference is $\sim\lambda_\varepsilon$.

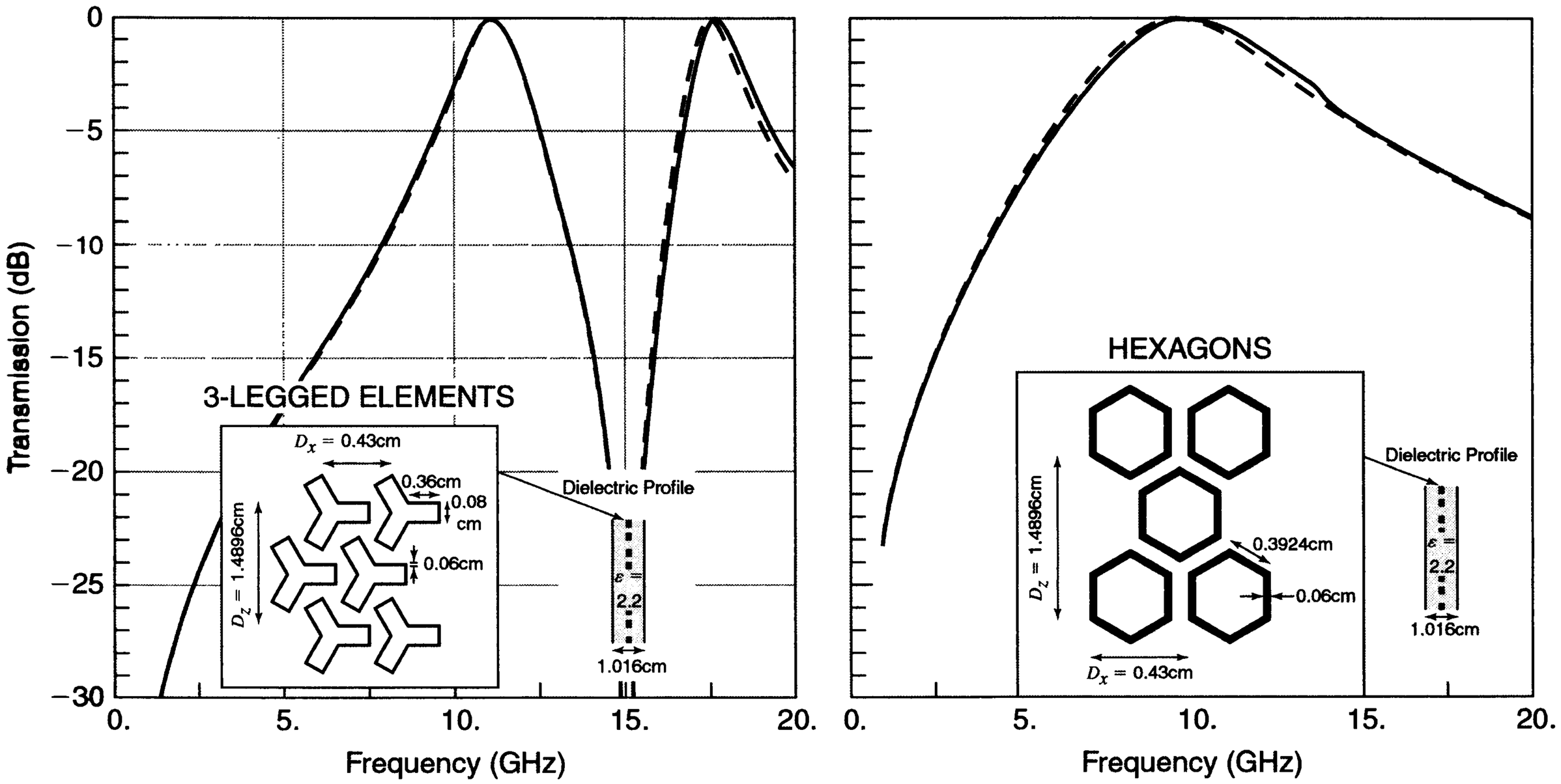


Figure 18–9
By expanding the three-legged loaded element (left) into a hexagon element (right), the bandwidth can be greatly increased. Note: the two arrays have identical interelement spacings as well as element circumferences ($\sim\lambda_\varepsilon$).

elements. They are naturally in a rectangular array. If a triangular array grid is preferred, one should use the "three-legged loaded element" shown in Fig. 18–8*e*. Finally, the sides of the three-legged loaded element can be "pulled" outward into a hexagon element as shown in Fig. 18–8*f* or alternatively into an ordinary loop as shown in Fig. 18–8*g*. All of these elements will resonate when their average circumference is $\sim\lambda_\varepsilon$. Their differences lie in their bandwidth as illustrated in Fig. 18–9. Here, the three-legged loaded elements have been expanded into hexagon elements. The difference in bandwidth is seen to be ~2.5 (actually the bandwidth of the hexagon array could be made considerably larger by reducing the interelement spacings).

Other element types can also be obtained by "shooting" elements out from the center. However, the reader should be warned not to cross two $\lambda/2$ dipoles, as logical as it seems (not shown). First of all, as stated earlier, the $\lambda/2$ dipole is not a particularly good element by itself but when two are crossed (and touching) it can quite often go terribly wrong. Not for normal incidence nor oblique when orthogonally polarized. However, for oblique incidence and parallel polarization we will observe not just one but two resonances close to each other. While this may sound like a minor point, the fact is that we also will obtain a null between the two resonance frequencies. This null can easily occur at the same frequency where resonance is observed for orthogonal polarization. Thus, an FSS with crossed elements is complex and in general not recommended. For an in-depth discussion see Pelton (1) and Munk.

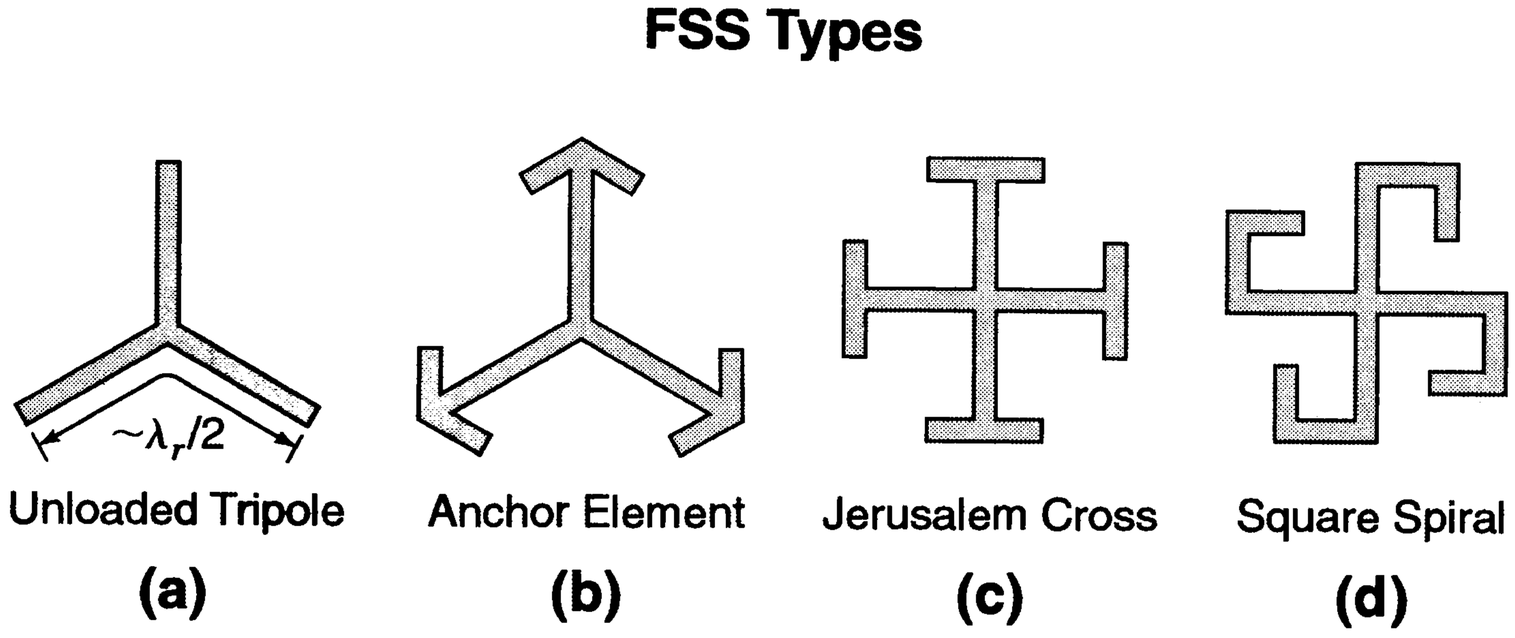


Figure 18–10
The "center connected," or "*N*-pole" elements, all have a "second" resonance as illustrated in Fig. 18–11 for the three-legged unloaded element.

The problem with double resonance can be avoided if we instead use the unloaded tripole element as shown in Fig. 18–10*a*. In that case the two resonances degenerate into one. Alternatively, end capacitances can be added to obtain the anchor element shown in Fig. 18–10*b*. Similarly, end capacitances can be added to a four-legged unloaded element, yielding the Jerusalem cross depicted by Fig. 18–10*c*. The effect of the end capacitance is simply to move the lowest resonance well below the second resonance as well as the null. Thus, this element is OK. Finally we show in Fig. 18–10*d* the so-called square loop element. It is characterized by being very broadbanded, almost as broadbanded as the hexagon element in Fig. 18–8*f*. The elements in Fig. 18–10 are often called "center connected" or "N-poles." They are characterized by having all of their element arms connected at the center. This simply means that not only can an even mode exist on these elements but also an odd mode, for oblique incidence as shown in Fig. 18–11, bottom. Typically the odd mode will resonate at a frequency about twice the even or fundamental. We show an example in Fig. 18–11, top, for an unloaded three-legged element for 45° angle of incidence for both orthogonal and parallel polarizations. The fundamental resonance is at ~10 GHz while the odd (for parallel polarization) is at ~17 GHz. Further, it is a fundamental fact in the theory for periodic surfaces that two resonances always will have a perfect null somewhere between them, in our case at ~15 GHz. This null is often referred to as a ***modal interaction null***. (模式互作用零点) To exist, the periodic structure must be lossless and have no grating lobes. All center-connected elements have at least one modal interaction null. The ***loop types*** (环型单元) are characterized by having their fundamental resonance when the circumference of the elements is about one wavelength in the dielectric surrounding them. This leads to relatively small elements, typically about $\lambda_\varepsilon/3$ to $\lambda_\varepsilon/4$ across. This is smaller than the center-connected elements that typically measure $\lambda_\varepsilon/2$ from tip to tip. Since a small interelement spacing is a must to obtain periodic structure that is stable with angle of incidence, it is clear that the loop types in general will be preferable to the centerconnected.

The elements, in Fig. 18–12, top, are called "solid interior" or "plate" elements. Because of their simplicity, they were among the first elements to be investigated, see (Kieburtz-1; Lee-1; Chen-1). However, they are not very effective resonators, and when they do resonate, it is usually associated with the onset of a grating lobe. In other words, they are not at all

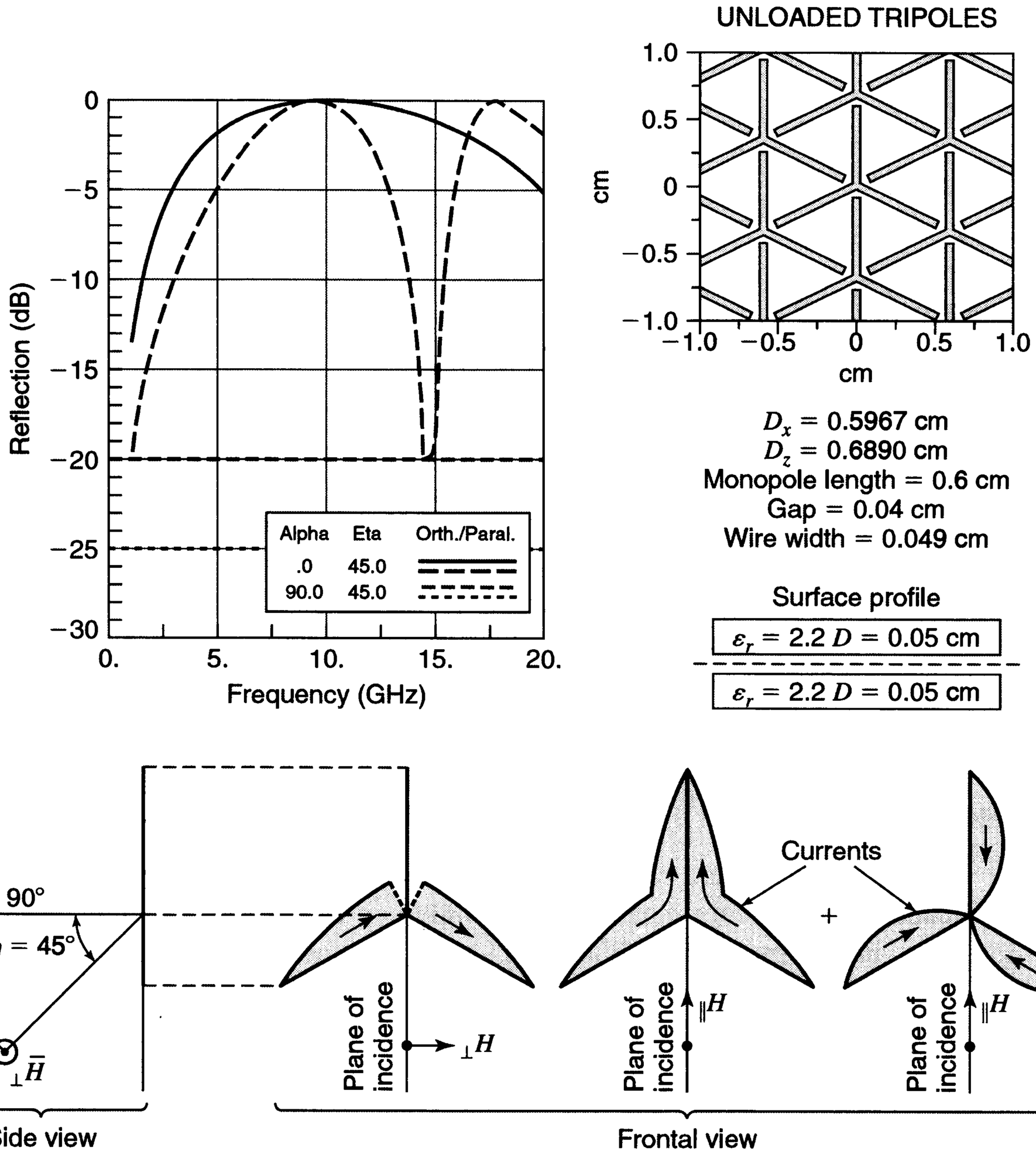


Figure 18–11
Typical reflection curves for an FSS of unloaded tripoles. Note the odd resonance at ~17 GHz for parallel polarization due to excitation of an odd mode as shown below to the far right. Also note the "modal interaction null" at ~15 GHz.

like their similar-looking counterparts in Fig. 18–8. They are, in general, not recommended for precision work.

Finally, there is an endless list of combinations and special-purpose elements as shown in Fig. 18–12, bottom. The first element to the left is merely a four-legged loaded element provided with end capacitors like the Jerusalem cross. Further, the element in the middle and to the right represents types without any "floating" interior such as the case is with the loop elements. These elements have merit if they are produced by electroforming.

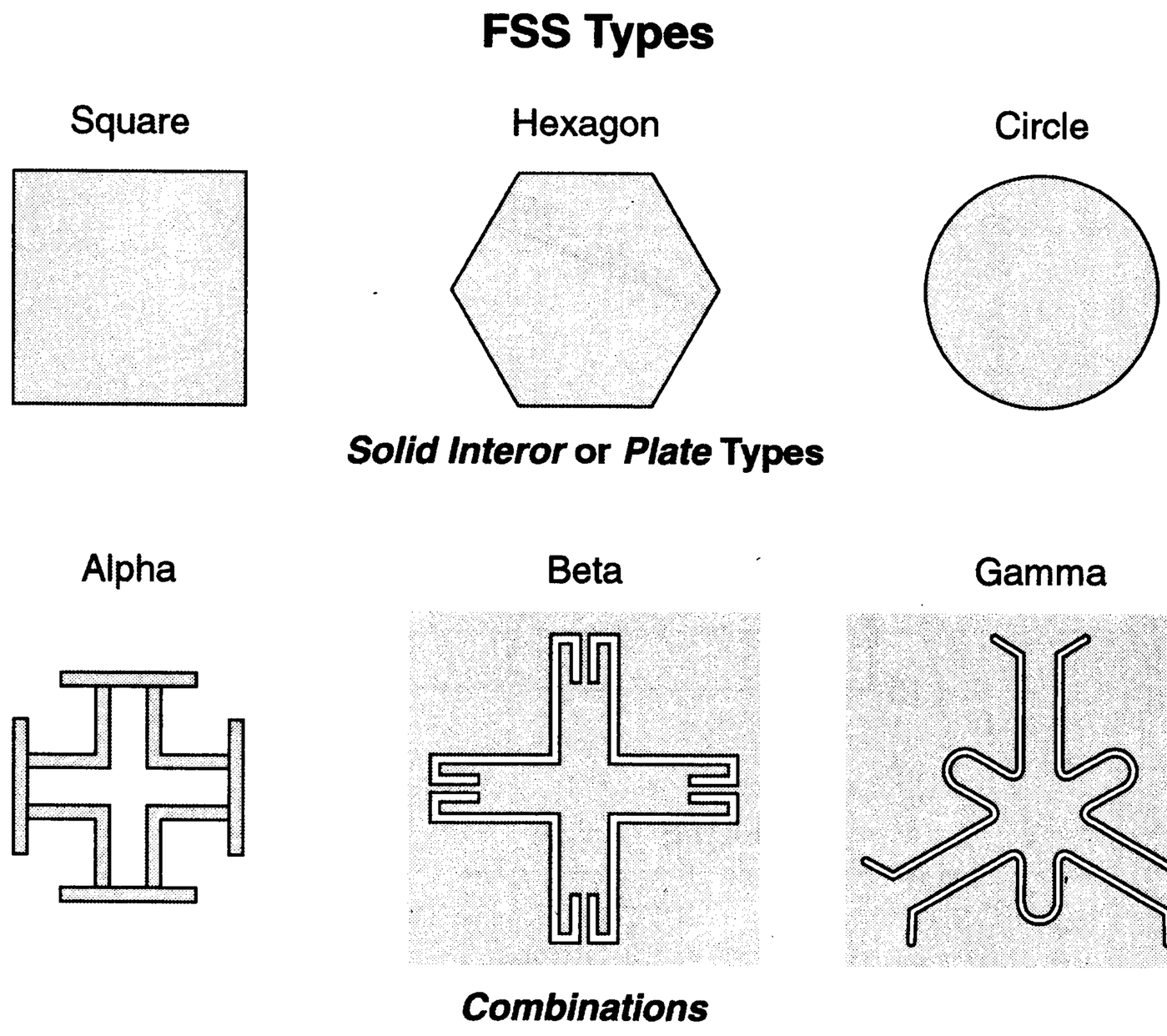


Figure 18–12
Top: the "solid interior" or "plate type" elements. Bottom: some of the many so-called "combination" elements.

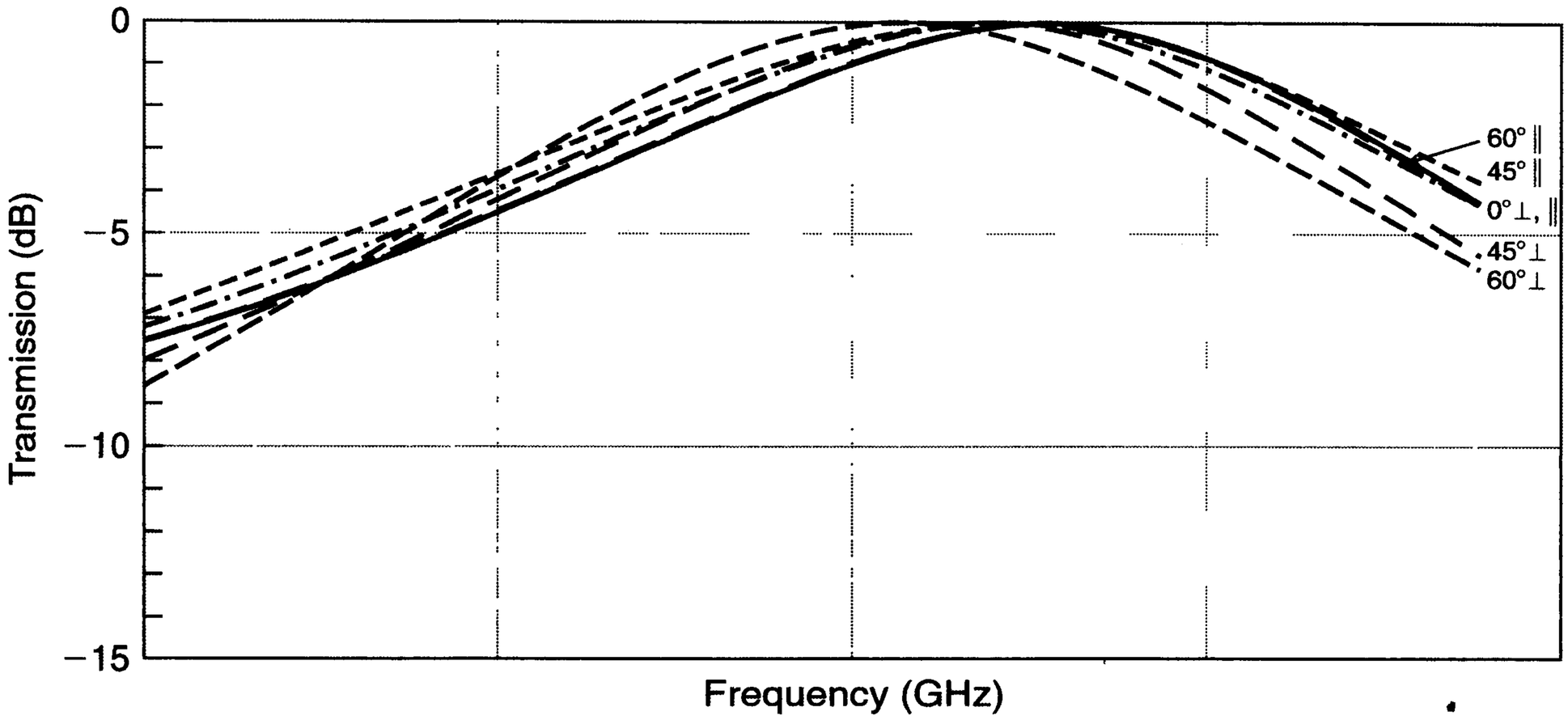
A detailed examination of most of the types shown in Figs. 18–8 to 18–12 is given in Munk (3).

18–10 CONTROLLING BANDWIDTH WITH ANGLE OF INCIDENCE AND POLARIZATION（控制入射角和极化的频带宽度）

Based on the discussions earlier, the bandwidth of a typical FSS will either increase or decrease as $\sim\cos\eta$ depending on the polarization. Thus, since most FSS are exposed to arbitrary polarization, the bandwidth variation typically will be as high as $\sim\cos^2\eta$ or ~4 : 1 for 60° angle of incidence. Although some reduction in this variation can be obtained by choice of elements, this approach is basically academic compared to what is really needed. In fact, it is fairly safe to state that a "superelement" capable of producing constant bandwidth for changing angle of incidence and polarization simply does not exist unless we consider nonplanar elements (see English-1; Andre-1; Kent-1; Ng-1; Schuman-1; Lin-1).

The most practical solution to this problem is simply to use one or more FSSs placed in an appropriate stratified dielectric medium. A couple of practical examples will be shown next.

First, we show in Fig. 18–13, top, the same single FSS as presented in Fig. 18–3 but now provided with additional dielectric slabs on each side as shown at the bottom ($\varepsilon_r > 1.3$). These surfaces are seen to be substantially thicker, namely, slightly more than $\lambda_\varepsilon/4$ as compared to the thin substrate next to the FSS. The purpose of the thicker slabs is to make the bandwidth much more stable with angle of incidence, while the thin ones (often referred



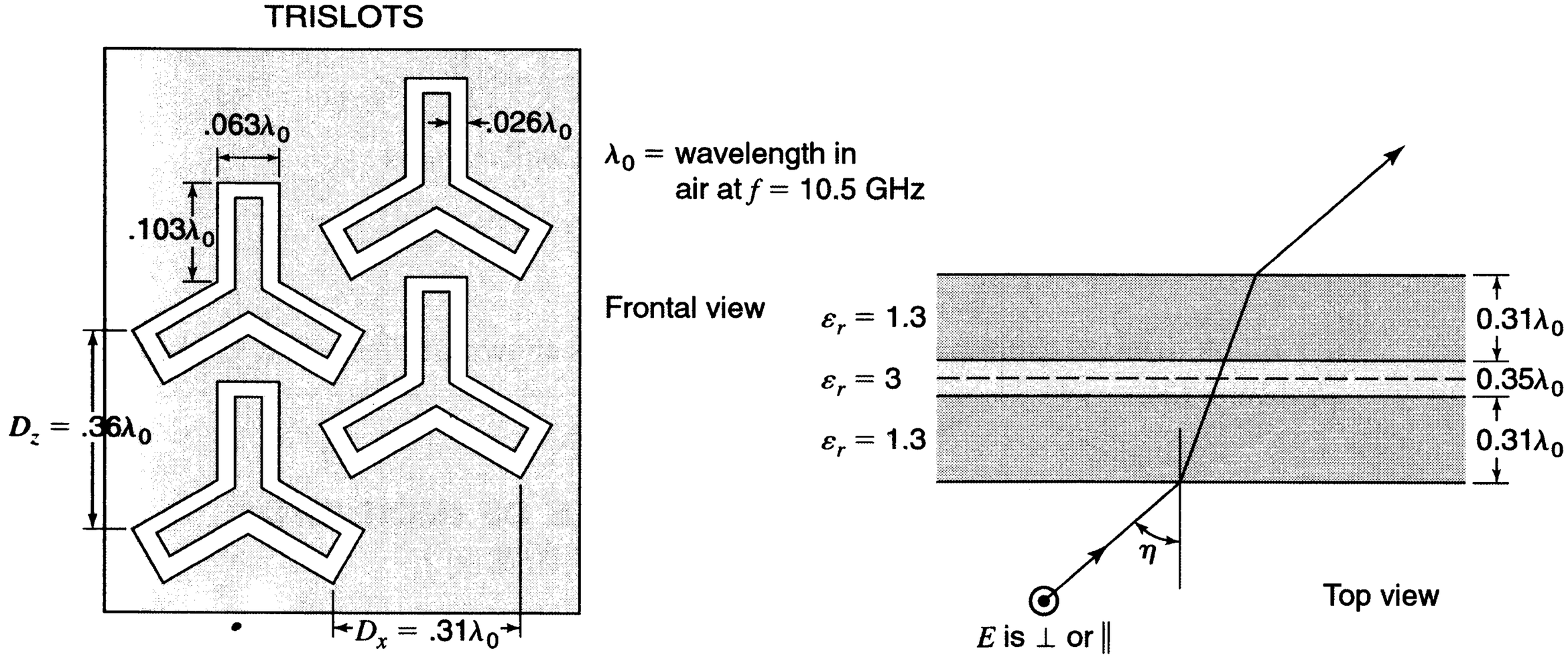
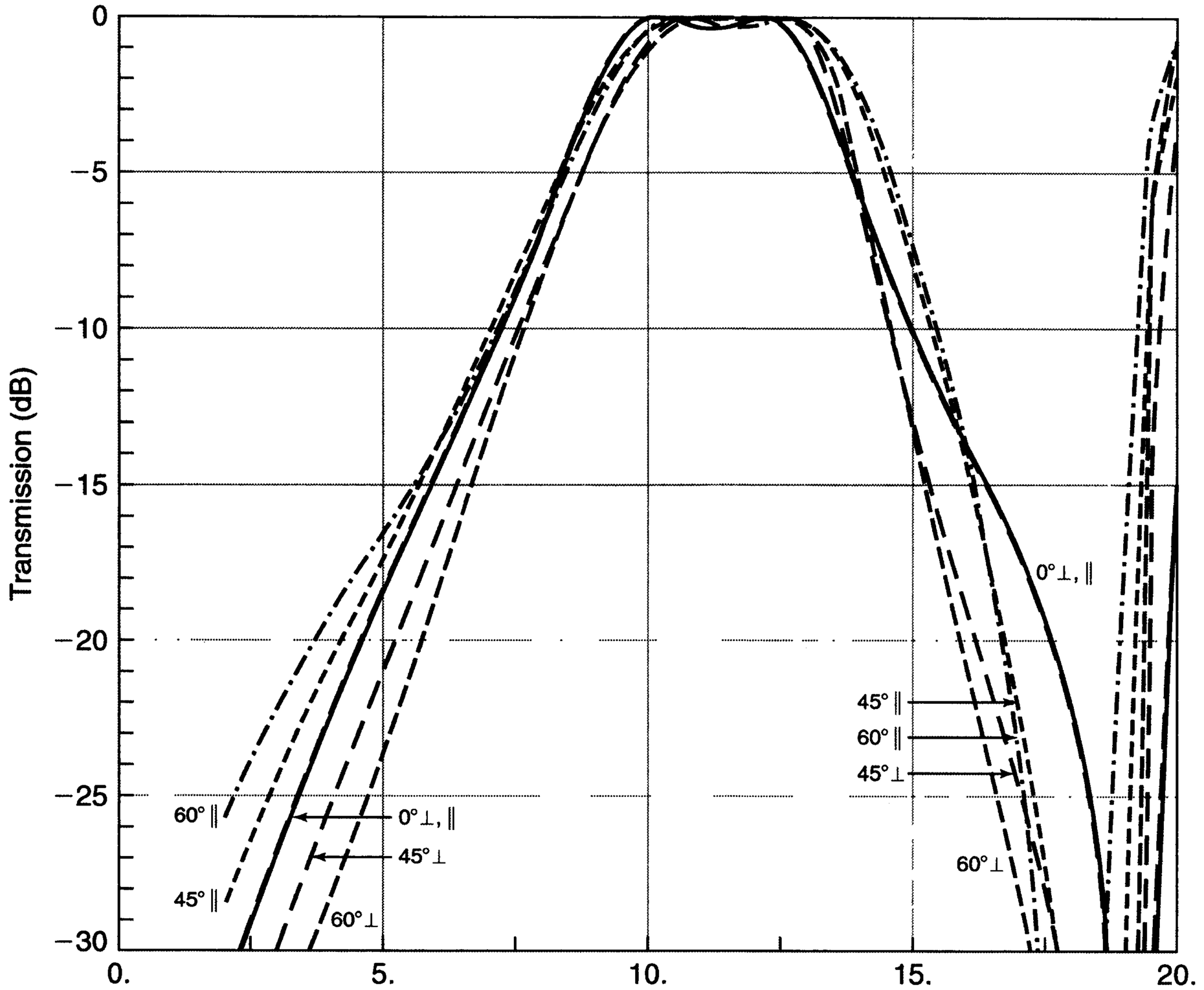


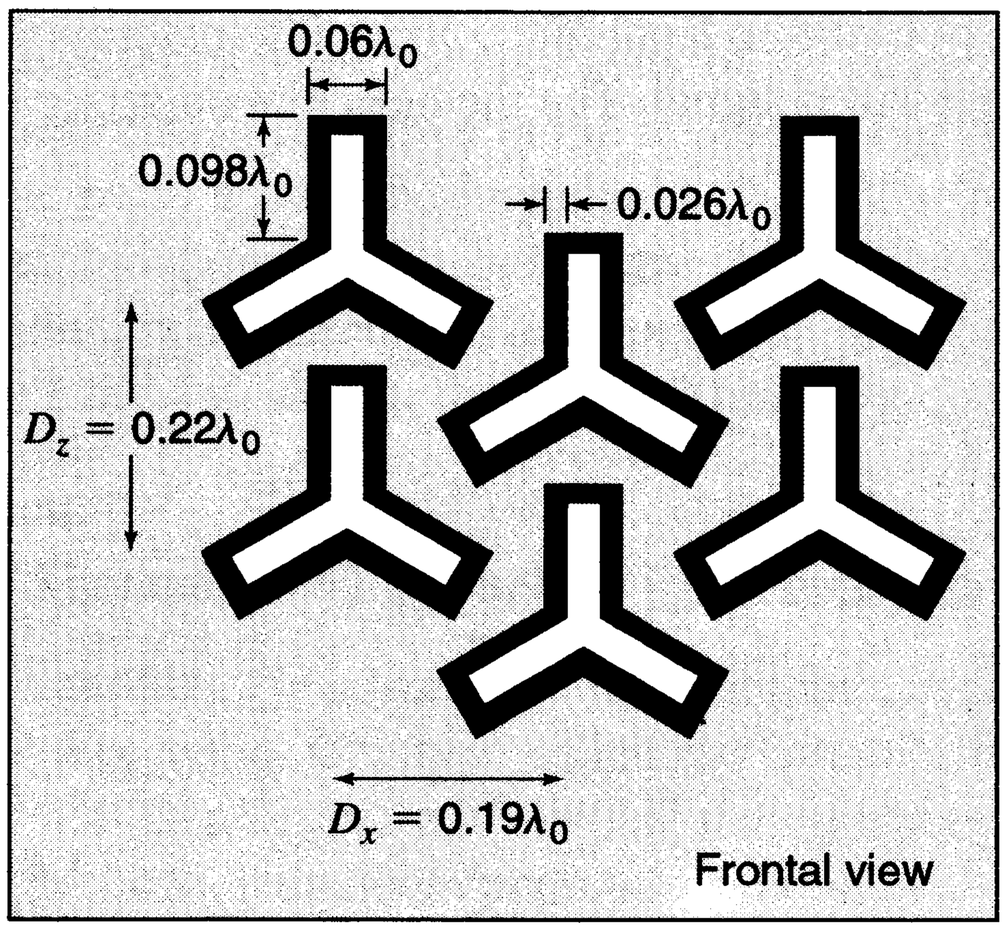
Figure 18–13
By adding dielectric slabs of the proper thickness and dielectric constant to the FSS shown in Fig. 18–3, we can obtain a hybrid radome with a bandwidth that is fairly stable with angle of incidence as well as polarization.

to as "dielectric underwear") essentially provide mechanical protection as well as dielectric loading (Munk-1, 4).

Note that while the bandwidth of the FSS in Fig. 18–3 varies approximately as ~4 : 1 (for 60° angle of incidence) it has been reduced to ~1.4 : 1 in Fig. 18–13. Greater stability of the resonance frequency with angle of incidence is possible.

Although the monoplanar design shown in Fig. 18–13 is satisfactory for some applications, it is often desirable to produce a transmission curve with a flat top (see Fig. 18–5). This is accomplished by cascading two or more surfaces behind each other as shown in Fig. 18–14. We show the transmission curves for two cascaded surfaces of the same type





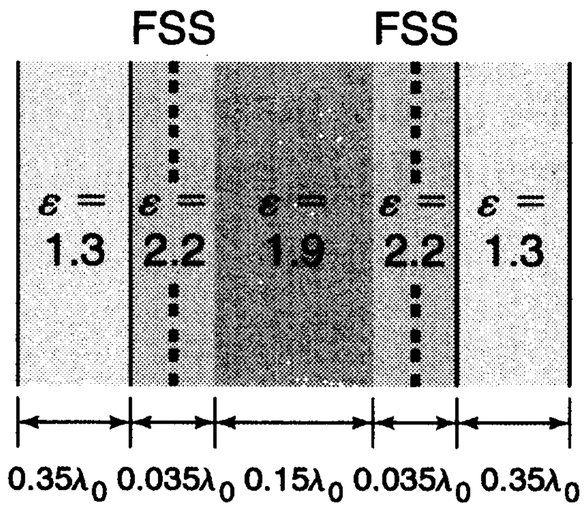


Figure 18–14
The typical transmission curve for a biplanar hybrid radome showing almost constant bandwidth with angle of incidence as well as polarization. Also note the flat tops in contrast to the case in Fig. 18–13.

as used in Fig. 18–3. These two surfaces have been placed in a stratified dielectric medium comprised of three dielectric slabs plus substrates, as shown in the insert. Again the effect of the dielectric slabs in the case at the bottom is spectacular: The bandwidth varied less than $\sim 1.2:1$ as compared to $\sim 4:1$ in the nondielectric case. A closer analysis is give in Munk (4) where it is shown that the outer dielectric slabs are responsible for producing the constant bandwidth while the slab in the middle is responsible for the flatness of the top of the transmission curves.

18–11 OTHER APPLICATIONS（其他应用）

The discussion so far has been directed mostly toward one application, namely, the hybrid radome. However, periodic surfaces have many other uses. We shall present a few.

18–11a Dichroic Surfaces（分频表面）

Dichroic refers to two frequencies, and a typical situation would be where the surface is transparent at one frequency and opaque at another. A common application is shown in Fig. 18–15, where the subreflector in a Cassegrain system has been designed to be transparent at the frequency f_1 and reflective at f_2. Placing feeds operating at these frequencies (or bands) enables us to operate the Cassegrain antenna over a wider frequency band.

In Fig. 18–16, we have made the entire main reflector of dichroic material and placed it in front of an array operating in X-band. This is the frequency band where the reflector is transparent while it is opaque where the feed horn operates. We obtain in this way a very compact design that could be used, for example, in the periscope of a submarine.

Note also the polarizer in front of the array. It transforms the linear signal from the array into a circularly polarized signal. The meanderline polarizer is discussed in Sec. 18–11c.

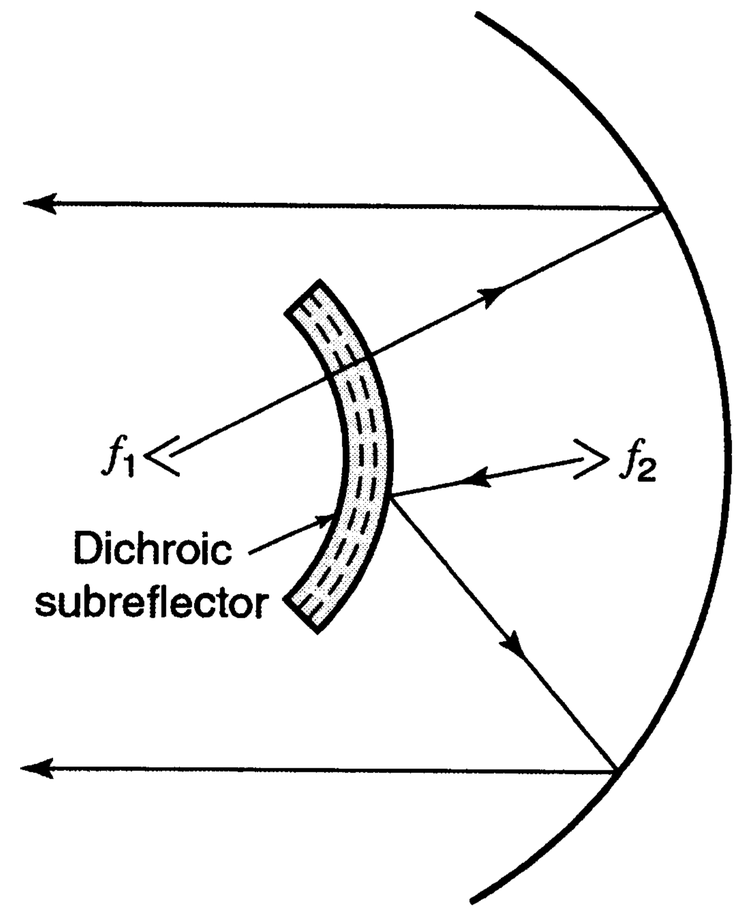


Figure 18–15
Use of a periodic structure as a Cassegrain subreflector.

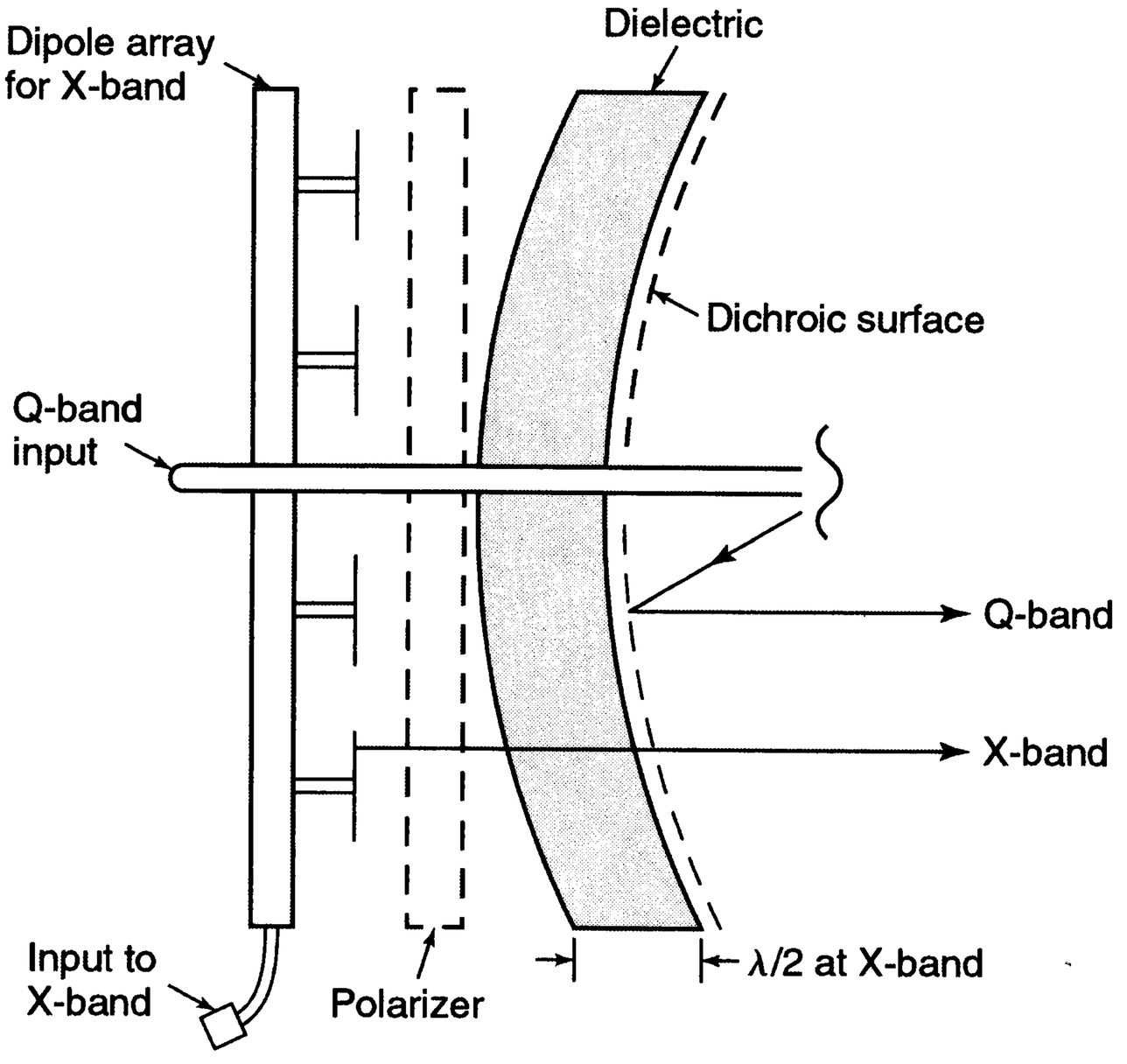


Figure 18–16
An X-band array (left) is looking through a polarizer and a dichroic main reflector reflecting Q-band.

18–11b Circuit Analog Absorbers（电路模拟的吸波器）

Periodic structures with elements made of resistive materials find application as absorbers as shown in Fig. 18–17. A periodic array of crossed lossy elements is placed in front of a groundplane. As explained in Fig. 18–6, the elements provide us with capacitance and inductance in series while the resistivity of the elements adds a series resistance leading to an equivalent circuit as shown in the middle of Fig. 18–17. Finally we show in the Smith chart at the bottom how the impedance (3) at the front of the absorber is obtained as a parallel combination of the groundplane impedance Y_g (1) and the circuit analog sheet Y_a (2). Cascading more sheets can lead to analog absorbers with more than 25 dB attenuation over a 10 : 1 bandwidth. For details see Munk (5).

18–11c Meanderline Polarizer（回折线极化器）

Generally speaking, polarizers are devices that transform a linear polarized signal into a circular or vice versa. One type called the meanderline polarizer is shown in Fig. 18–18, top. It consists of a structure of vertical meanderlines. A plane wave with its E vector $\overline{E}^i$ tilted 45° as shown is incident upon this structure.

The workings of such a polarizer are readily understood if we decompose the incident field $\overline{E}^i$ into a vertical component $\overline{E}^v$ and a horizontal component $\overline{E}^h$. To the first of these components, the meanderlines will act like a shunt inductance while it will act as a shunt capacitor to the horizontal component. Thus, the equivalent circuits for these two components will be equivalent transmission line with shunt inductance or capacitance as shown in Fig. 18–18, bottom. From basic network theory we readily see that the transmitted

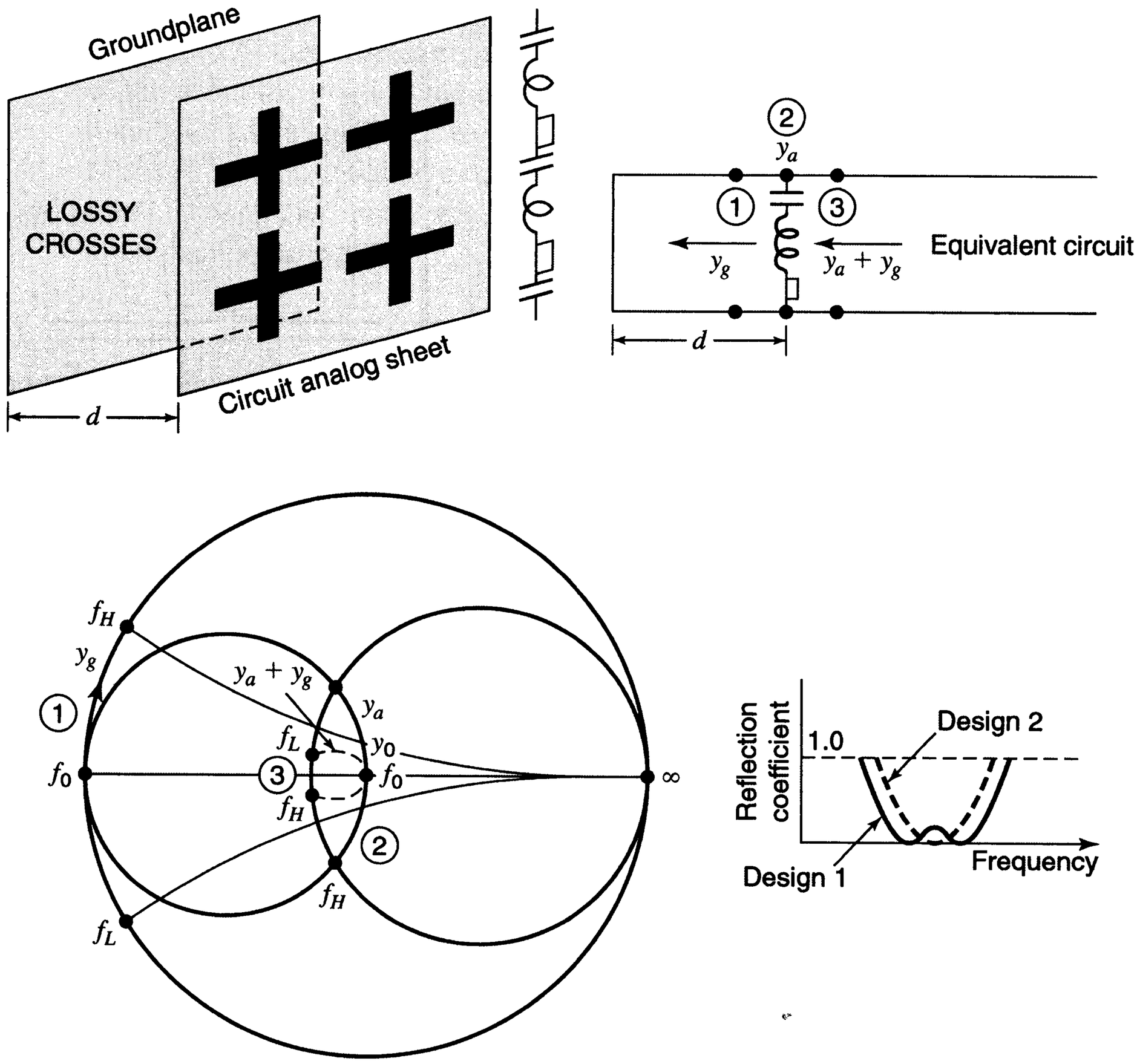


Figure 18–17
Circuit analog absorbers are made of periodic structures of very lossy material. They have more bandwidth per given thickness than most other absorbers. The reflection coefficient of two designs is shown at lower right.

field of the vertical component will be advanced in phase while the horizontal component will be delayed. If the two transmitted signals have the same magnitude and a net phase difference of 90°, the combined transmitted field will be circularly polarized.

It is interesting to note that requiring a 45° advance or delay of a single meanderline sheet invariably leads to a reflection coefficient that is numerically equal to the transmission coefficient (i.e., half the incident signal is reflected).

This problem can be overcome by using two or more meanderline sheets cascaded after each other. In that way we require the phase advances and delays of each sheet to be only half or less than 45°. This again leads to smaller reflections per sheet. And in addition the reflections from the individual sheets can be designed to largely cancel each other. In this way we can obtain polarizers with a bandwidth exceeding an octave. For references see Marino (1).

Meanderline Polarizers

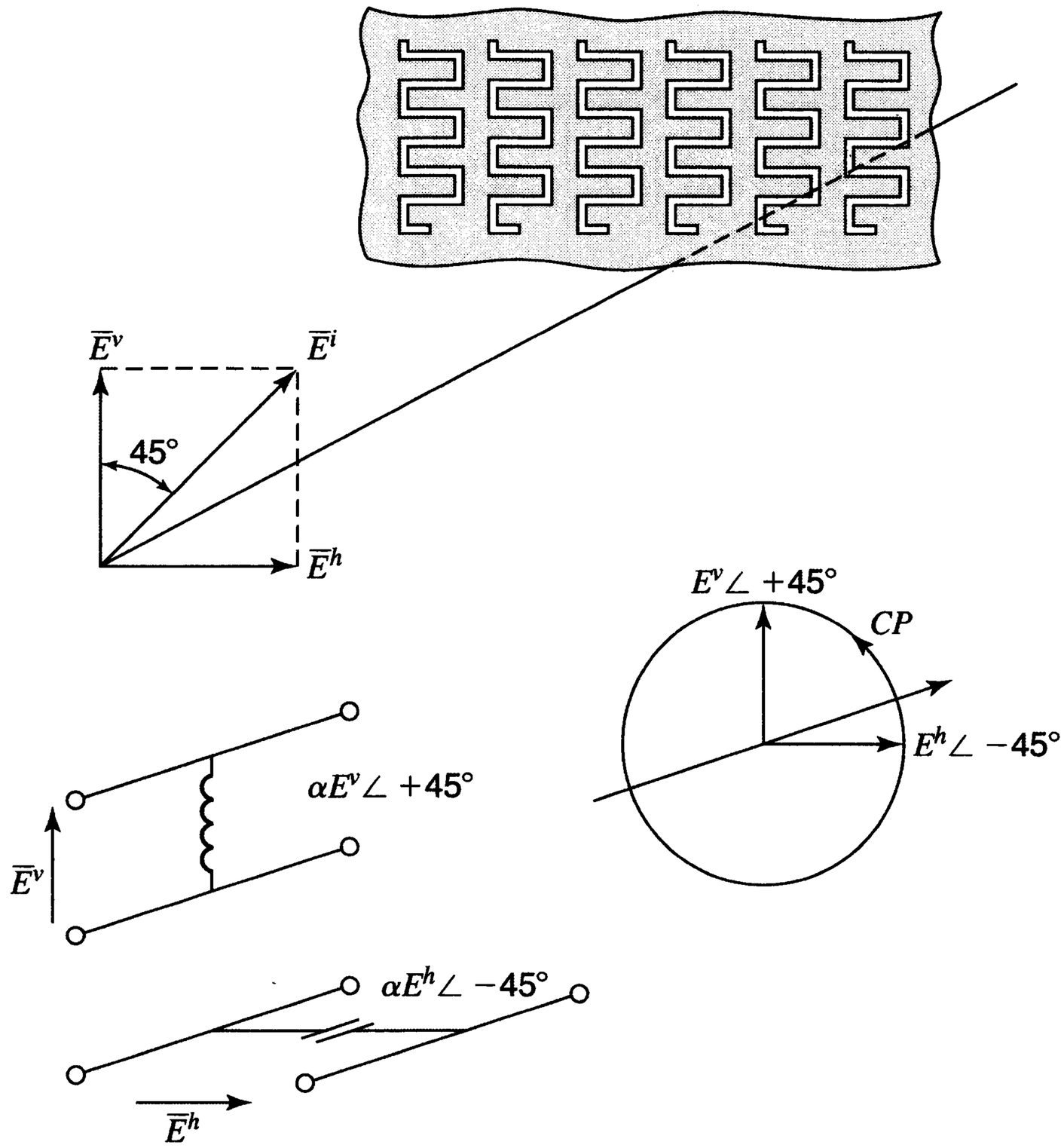


Figure 18–18
A meanderline structure acts like a shunt inductance for vertical polarization and as a shunt capacitance for horizontal. For a net difference of 90° a signal tilted 45° as shown will be transformed into a circularly polarized signal.

EXAMPLE **18–11.1 Sandwich Comparison**

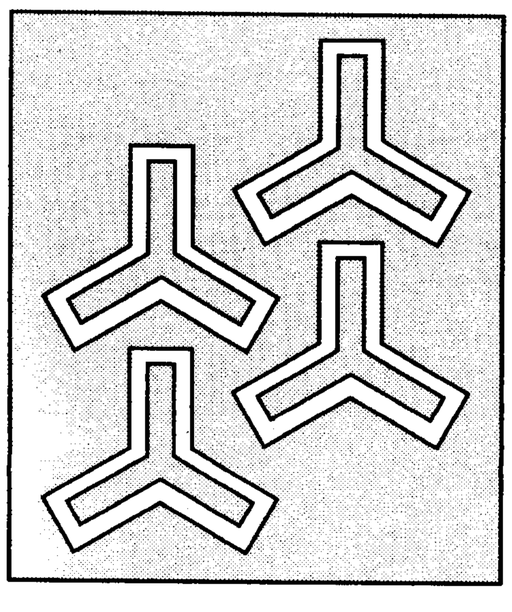
Compare the −3 dB bandwidth of trislots, dual 3-legged elements and hexagons, all in dielectric sandwiches.

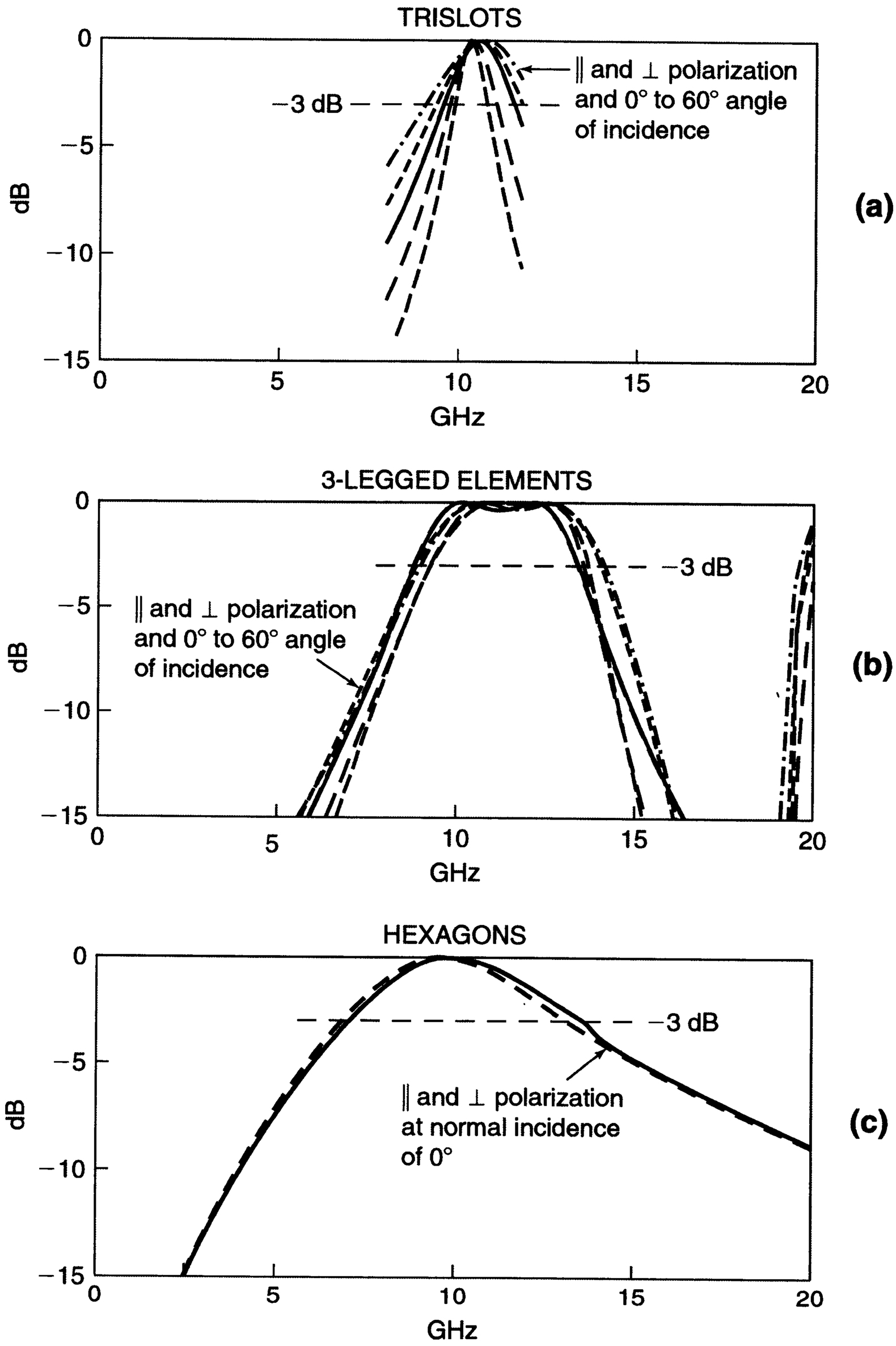
■ **Solution**

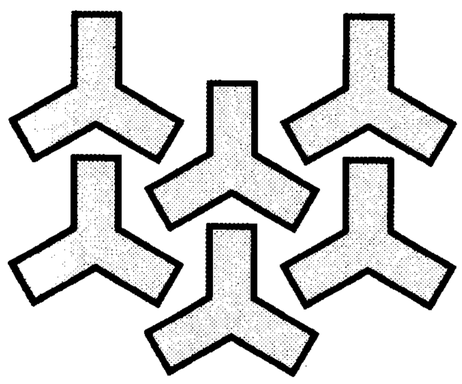
The comparison is shown in Fig. 18–19 with trislots at (*a*), dual 3-legged elements at (*b*) and hexagons at (*c*). The shaded areas in (*a*) and (*b*) include transmission for parallel (∥) and perpendicular (⊥) polarization and for 0° to 60° angles of incidence. Only normal incidence (0°) for both polarizations is shown in (*c*). Note the flat-topped wideband transmission for the dual 3-legged elements in (*b*).

The −3 dB bandwidths are:

	GHz	Ratio	
Trislots	8.5–11.5	1.35	
3-legged elements	9–14	1.55	
Hexagons	7–14	2.00	*Ans.*







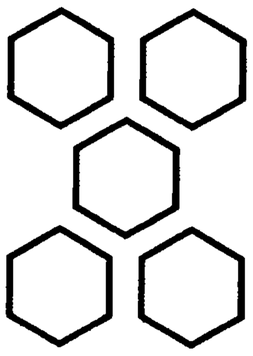


Figure 18–19
Comparison of transmission through dielectric sandwiches with trislots, dual 3-legged elements and hexagons.

REFERENCES（参考文献）

Andre, R. (1), *An Analysis Method for Doubly Periodic Nonplanar Antenna Arrays,* M.Sc. thesis, The Ohio State University, Dept. of Electrical Engineering, Columbus, Ohio, 1985.

Chen, C.-C. (1), "Transmission of Microwave through Perforated Flat Plates of Finite Thickness," *IEEE Trans. Microwave Theory and Techniques,* **MITT-21,** no. 1, January 1973.

English, E. K. (1), *Electromagnetic Scattering from Infinite Periodic Arrays of Arbitrarily Oriented Dipole Elements Embedded in a General Stratified Medium,* Ph.D. dissertation, The Ohio State University, Dept. of Electrical Engineering, Columbus, Ohio, 1973.

Kent, B. M. (1), *Impedance Properties of an Infinite Array of Non-Planar Rectangular Loop Antennas Embedded in a General Stratified Medium,* Ph.D. dissertation, The Ohio State University, Dept. of Electrical Engineering, Columbus, Ohio, 1984.

Kieburtz, R. B. (1), and A. Ishimaru, "Scattering by a Periodically Apertured Conducting Screen," *IRE Trans. Ants. Prop.,* **AP-9,** no. 6, pp. 506–514, November 1961.

Kyriazidou, C. A., R. E. Diaz and N. G. Alexópoulos, "Novel Material with Narrow-Band Transparency Window in the Bulk," *IEEE Trans. Ants, Prop.,* **48,** January 2000, pp. 107–116.

Lee, S. W. (1), "Scattering by Dielectric-Loaded Screen," *IEEE Trans. Ants. Prop.,* **AP-19,** no. 5, pp. 656–665, September 1971.

Lin, S. J. (1), *On the Scan Impedance of an Array of V-Dipoles and the Effect of the Feedlines,* Ph.D. dissertation, The Ohio State University, Dept. of Electrical Engineering, Columbus, Ohio, 1985.

Marino, R. A. (1), "Accurate and Efficient Modeling of Meander-line Polarizers," *Microwave Journal,* November 1998.

Munk, B. A. (1), *Frequency Selective Surfaces, Theory and Design,* Wiley, New York, 2000, Chaps. 2 and 7.

Munk, B. A. (2), FSSTD, pp. 109–112.

Munk, B. A. (3), FSSTD, pp. 26–63.

Munk, B. A. (4), FSSTD, pp. 227–278.

Munk, B. A. (5), FSSTD, pp. 391–394.

Ng, K. T. (1), *Admittance Properties of a Slot Array with Parasitic Wire Arrays in a Stratified Medium,* Ph.D. dissertation, The Ohio State University, Dept. of Electrical Engineering, Columbus, Ohio, 1985.

Pelton, E. L. (1), and B. A. Munk, "Scattering from Arrays of Crossed Dipoles," *IEEE Trans. Ants. Prop.,* **AP-27,** pp. 323–330, May 1979.

Schuman, H. K. (1), D. R. Pflug, L. D. Thompson, "Infinite Planar Array of Arbitrarily Bent Thin Wire Radiators," *IEEE Trans. Ants. Prop.,* **AP-32,** no. 4, pp. 364–477, April 1984.

Tice, T. E. (1), editor-in-chief, *Techniques for Airborne Radome Design,* vols. 1 and 11, McGraw-Hill Technical Writing Services, TR AFAL-TR-66-391, 1966.

Vardaxoglou, J. C. (1), *Frequency Selective Surfaces,* Research Studies Press, Ltd., 1997.

Wu, T. K. (1), *Frequency Selective Surface and Grid Array,* Wiley, New York, 1995.

PROBLEMS（习题）

18–9–1（无载三极子） **Unloaded tripole.** Determine the approximate length of the legs of an unloaded trislot operating at $f = 15$ GHz with (*a*) No dielectric substrate. (*b*) Dielectric substrate $\varepsilon_r = 2.2$ and thickness 0.50 mm located on both sides of the FSS (use arithmetic average of ε_r and ε_o for ε_{eff}). (*c*) Determine D_x just short enough that no grating lobes are present when scanning in the xy-plane for any angle of incidence.

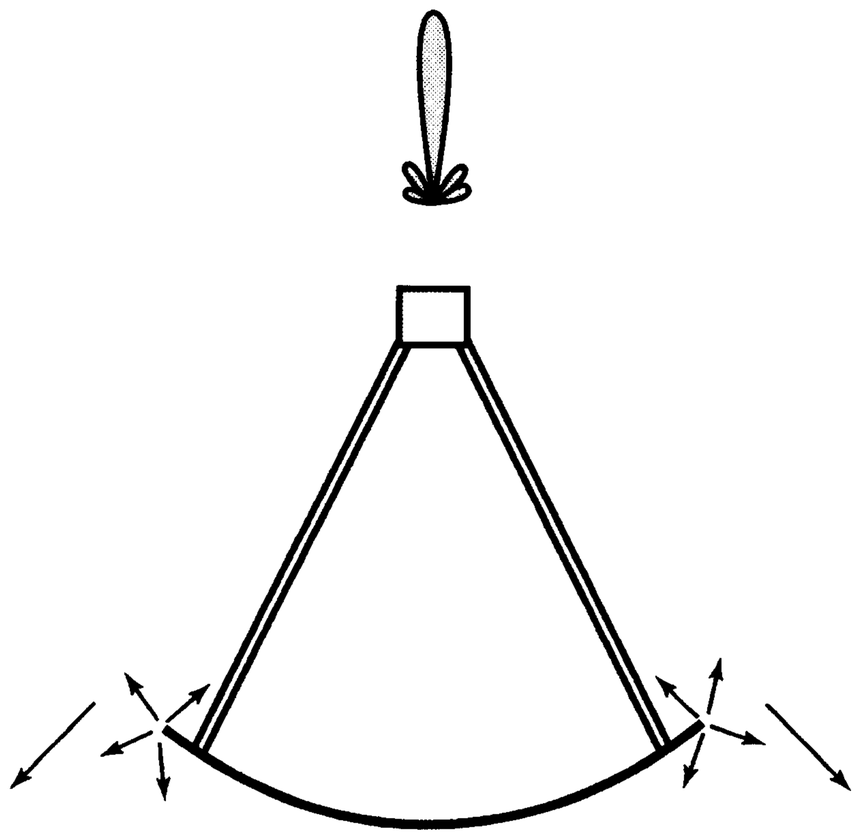
18–9–2（四腿形加载单元） **Four-legged loaded element.** Determine the approximate dimensions for a four-legged loaded element operating at $f = 15$ GHz with (*a*) No dielectric substrate. (*b*) Dielectric substrate $\varepsilon_r = 2.2$ and thickness 0.50 mm located on only one side of the FSS (estimate ε_{eff}). (*c*) Leave a separation of 1 mm between adjacent elements (rectangular grid); determine the lowest onset frequency for grating lobes for any angle of incidence.

For computer programs, see Appendix C.

CHAPTER 19

Practical Design Considerations of Large Aperture Antennas

实用大型口径天线的设计考虑



Topics in this chapter include:

- Aperture distribution and efficiency
  （口径分布与口径效率）
- Surface irregularities and gain loss
  （表面不规则性和增益损失）
- Cassegrain feed; offset feed
  （卡塞格伦馈电；偏馈）
- Low-side-lobe considerations
  （低旁瓣的考虑）

19–1 APERTURE DISTRIBUTIONS AND EFFICIENCIES[1]（口径分布与口径效率）

As an introduction to aperture distributions and efficiencies, a few basic concepts discussed in earlier chapters are reviewed briefly. Then a number of criteria useful in antenna design are developed.

Let a plane wave of power density S (W m^{-2}) be incident on an antenna, as in Fig. 19–1. The power P delivered by the antenna to the receiver is then

$$P = SA_e \tag{1}$$

where A_e = *effective aperture* of the antenna
or

$$A_e = \frac{P}{S} \tag{2}$$

A completely polarized point source is assumed with the antenna matched to the wave.

If ohmic losses are not negligible, as assumed above, we may distinguish between the actual effective aperture (including the effect of ohmic losses) and an effective aperture based entirely on the pattern (losses neglected). Thus, we may write

$$A_e = k_o A_{ep} \tag{3}$$

[1]This section (19–1) and the following sections (19–2 and 19–3) are from J. D. Kraus, *Radio Astronomy,* 2d ed., Cygnus-Quasar, 1986.

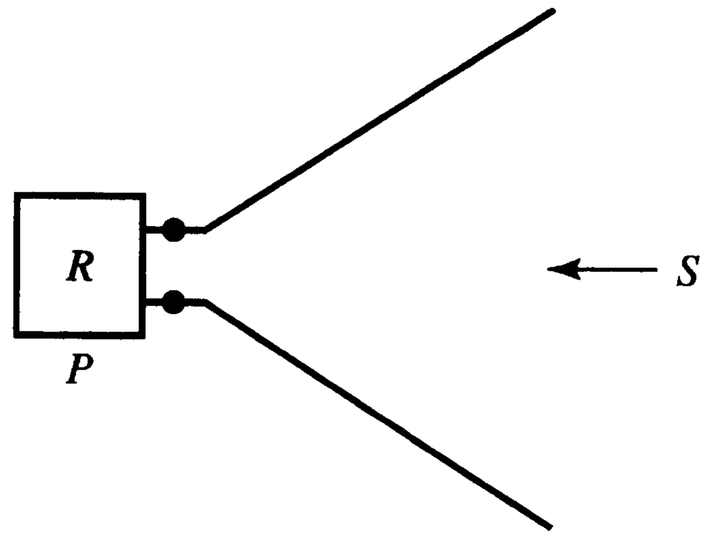


Figure 19–1
Wave of flux density S incident on antenna delivers a power P to the receiver R.

where

A_e = actual effective aperture, m^2
k_o = ohmic-loss factor, dimensionless $(0 \leq k_o \leq 1)$
A_{ep} = effective aperture as determined entirely by *pattern*, m^2

Using these symbols, the directivity of an antenna is given by

$$D = \frac{4\pi}{\Omega_A} = \frac{4\pi}{\lambda^2} A_{ep} \tag{4}$$

from which

$$A_{ep}\Omega_A = \lambda^2 \tag{5}$$

and

$$A_e\Omega_A = k_o\lambda^2 \tag{6}$$

where

Ω_A = antenna beam solid angle, sr
λ = wavelength, m

The *aperture efficiency* is defined as the ratio of the effective aperture to the physical aperture, or

$$\varepsilon_{ap} = \frac{A_e}{A_p} \tag{7}$$

so that the ratio of the aperture and beam efficiencies is

$$\frac{\varepsilon_{ap}}{\varepsilon_M} = \frac{A_e\Omega_A}{A_p\Omega_M} = \frac{k_o\lambda^2}{A_p\Omega_M} \tag{8}$$

where

$\varepsilon_M = \Omega_M/\Omega_A$ = beam efficiency, dimensionless
Ω_M = main beam area, sr
Ω_A = total beam area, sr

Although the physical aperture A_p may not have a unique meaning on some apertures, its value tends to be readily defined on apertures that are large in terms of wavelength.

The directivity of an antenna depends only on the radiation pattern, so that the *directivity* D is given by

$$D = \frac{4\pi}{\lambda^2} A_{ep} \tag{9}$$

The *gain* G is then

$$G = Dk_o = k_o \frac{4\pi}{\lambda^2} A_{ep} \tag{10}$$

The *maximum directivity* D_m will be defined as the directivity obtainable from an antenna (assumed to be large in terms of λ) if the field is uniform over the aperture, i.e., the physical aperture. Hence,

$$D_m = \frac{4\pi}{\lambda^2} A_p \tag{11}$$

In designing an antenna, one may have a certain *design directivity* D_d one wishes to achieve. In general, this will be less than D_m, since, to reduce side lobes, some taper will probably be introduced into the aperture distribution. Thus we may write

$$D_d = \frac{4\pi}{\lambda^2} A_p k_u = D_m k_u \tag{12}$$

where D_d = design directivity

The factor k_u is called the *utilization factor*. It is the ratio of the directivity chosen by design to that obtainable with a uniform aperture distribution, or

$$k_u = \frac{D_d}{D_m} \qquad 0 \leq k_u \leq 1 \tag{13}$$

After designing and building the antenna and measuring its performance, it will probably be found, however, that the actual directivity D is less than the design directivity D_d. It is possible, but unlikely, that D exceeds D_d. The actual directivity may then be expressed as

$$D = D_d k_a = D_m k_u k_a \tag{14}$$

where k_a is the *achievement factor,* which is a measure of how well the objective has been achieved. Thus,

$$k_a = \frac{D}{D_d} \qquad 0 \leq k_a \leq 1 \text{ (usually)} \tag{15}$$

The gain G of the antenna can now be written as

$$G = D_m k_o k_u k_a = \frac{4\pi}{\lambda^2} A_p k_o k_u k_a = \frac{4\pi}{\lambda^2} A_p \varepsilon_{\text{ap}} \tag{16}$$

where

A_p = physical aperture, m
k_o = ohmic efficiency factor, dimensionless
k_u = utilization factor, dimensionless
k_a = achievement factor, dimensionless
ε_{ap} = aperture efficiency, dimensionless

We may also write

$$A_{ep} = k_u k_e A_p \tag{17}$$

$$A_e = k_o k_u k_a A_p \tag{18}$$

$$\varepsilon_{\text{ap}} = \frac{A_e}{A_p} = k_o k_u k_a \tag{19}$$

where

A_{ep} = effective aperture (as determined entirely by pattern)
A_e = actual effective aperture

A basic definition for the directivity of an antenna is that the directivity is equal to the ratio of the maximum to the average radiation intensity from the antenna (assumed transmitting). By reciprocity the directivity will be the same in the receiving case. Hence,

$$D = \frac{U_m}{U_{\text{av}}} \tag{20}$$

where

U_m = maximum radiation intensity, W sr^{-1}
U_{av} = average radiation intensity, W sr^{-1}

The average value may be expressed as the integral of the radiation intensity $U(\theta, \phi)$ over a solid angle of 4π divided by 4π. Thus

$$D = \frac{U_m}{\frac{1}{4\pi}\iint_{4\pi} U(\theta, \phi)\, d\Omega} = \frac{4\pi U_m}{P} \tag{21}$$

where P is the total power radiated. From (21) the effective power (power which would need to be radiated if the antenna were isotropic) is

$$DP = 4\pi U_m \tag{22}$$

Let us now consider an aperture under two conditions. Under Condition 1 the field distribution is assumed to be uniform, so that the directivity is D_m. The power radiated is P'. Under Condition 2 the field has its actual distribution, and the directivity and power radiated are D and P, respectively. If the effective powers are equal in the two conditions,

$$DP = D_m P' \tag{23}$$

and

$$D = D_m \frac{P'}{P} = \frac{4\pi}{\lambda^2} A_p \frac{\frac{E_{\text{av}} E_{\text{av}}^*}{Z} A_p}{\iint_{A_p} \frac{E(x, y) E^*(x, y)}{Z}\, dx\, dy} \tag{24}$$

In (24) the surface of integration has been collapsed over the antenna, with part of the

surface coinciding with the aperture. Further, it is assumed that all the power radiated flows out through the aperture. In (24) Z is the intrinsic impedance of the medium (Ω per square) and E_{av} is the average field across the aperture, as given by

$$E_{\text{av}} = \frac{1}{A_p} \iint_{A_p} E(x, y)\, dx\, dy \tag{25}$$

where $E(x, y)$ is the field at any point (x, y) of the aperture. Rearranging (24), we have for the actual directivity

$$D = \frac{4\pi}{\lambda^2} A_p \frac{1}{\dfrac{1}{A_p} \displaystyle\iint_{A_p} \left[\frac{E(x, y)}{E_{\text{av}}}\right]\left[\frac{E(x, y)}{E_{\text{av}}}\right]^* dx\, dy} \tag{26}$$

This relation was developed originally by Ronald N. Bracewell (1). Following Bracewell's discussion and also elaborating on it by introducing the utilization factor, we can write for the design directivity

$$D_d = \frac{4\pi}{\lambda^2} A_p \frac{1}{\dfrac{1}{A_p} \displaystyle\iint_{A_p} \left[\frac{E'(x, y)}{E'_{\text{av}}}\right]\left[\frac{E'(x, y)}{E'_{\text{av}}}\right]^* dx\, dy} \tag{27}$$

where the primes indicate the design field values. The right-hand factor in (27) may be recognized as the *utilization factor* k_u. Multiplying and dividing (26) by k_u, as given in (27), yields

$$D = \frac{4\pi}{\lambda^2} A_p \frac{1}{\dfrac{1}{A_p} \displaystyle\iint_{A_p} \left[\frac{E'(x, y)}{E'_{\text{av}}}\right]\left[\frac{E'(x, y)}{E'_{\text{av}}}\right]^* dx\, dy} \times \frac{\dfrac{1}{A_p} \displaystyle\iint_{A_p} \left[\frac{E'(x, y)}{E'_{\text{av}}}\right]\left[\frac{E'(x, y)}{E'_{\text{av}}}\right]^* dx\, dy}{\dfrac{1}{A_p} \displaystyle\iint_{A_p} \left[\frac{E(x, y)}{E_{\text{av}}}\right]\left[\frac{E(x, y)}{E_{\text{av}}}\right]^* dx\, dy} \tag{28}$$

We also have, from (14),

$$D = D_m k_u k_a \tag{29}$$

Thus, the last factor in (28) is the *achievement factor,* a result given by Bracewell. Further, as done by Bracewell, let

$$E(x, y) = E_{\text{av}} + \delta E_{\text{av}} \tag{30}$$

or

$$\frac{E(x, y)}{E_{\text{av}}} = 1 + \delta \tag{31}$$

where δ is the *complex deviation (factor)* of the field from its average value. Thus, the denominator of the last factor in (28) may be written as

$$\frac{1}{A_p}\iint_{A_p}(1+\delta)(1+\delta)^*\,dx\,dy \tag{32}$$

Since the average of δ or δ^* over the aperture is zero, (32) simplifies to

$$1+\frac{1}{A_p}\iint_{A_p}\delta\delta^*\,dx\,dy = 1+\operatorname{var}\delta \tag{33}$$

where var δ is used to signify the *variance* of δ or average of $\delta\delta^*$ $(=|\delta|^2)$ over the aperture. Thus, (28) may be written more concisely as

$$D=\frac{4\pi}{\lambda^2}A_p\frac{1}{1+\operatorname{var}\delta'}\frac{1+\operatorname{var}\delta'}{1+\operatorname{var}\delta} \tag{34}$$

where

δ' = design value

δ = actual value

We also have that the *utilization factor*

$$k_u=\frac{1}{1+\operatorname{var}\delta'} \tag{35}$$

and the *achievement factor*

$$k_a=\frac{1+\operatorname{var}\delta'}{1+\operatorname{var}\delta} \tag{36}$$

Turning now to the *beam efficiency* ε_M, or the ratio of the solid angle of the main beam Ω_M to the total beam solid angle Ω_A, we have

$$\varepsilon_M=\frac{\Omega_M}{\Omega_A}=\frac{\displaystyle\iint_{\substack{\text{main}\\ \text{lobe}}}P(\theta,\phi)\,d\Omega}{\displaystyle\iint_{4\pi}P(\theta,\phi)\,d\Omega}\qquad 0\le\varepsilon_M\le 1 \tag{37}$$

where $P(\theta,\phi)$ = antenna power pattern $(=EE^*=|E|^2)$

Let us consider next the effect of the aperture field distribution on the beam and aperture efficiencies. We take first the simplest case of a 1-dimensional distribution, i.e., a rectangular aperture (L_λ by L'_λ) with a uniform distribution in the y direction (aperture L'_λ) and a distribution in the x direction (aperture L_λ) as in Fig. 19–2, given by

$$E(x)=K_1+K_2\left[1-\left(\frac{2x}{L_\lambda}\right)^2\right]^n \tag{38}$$

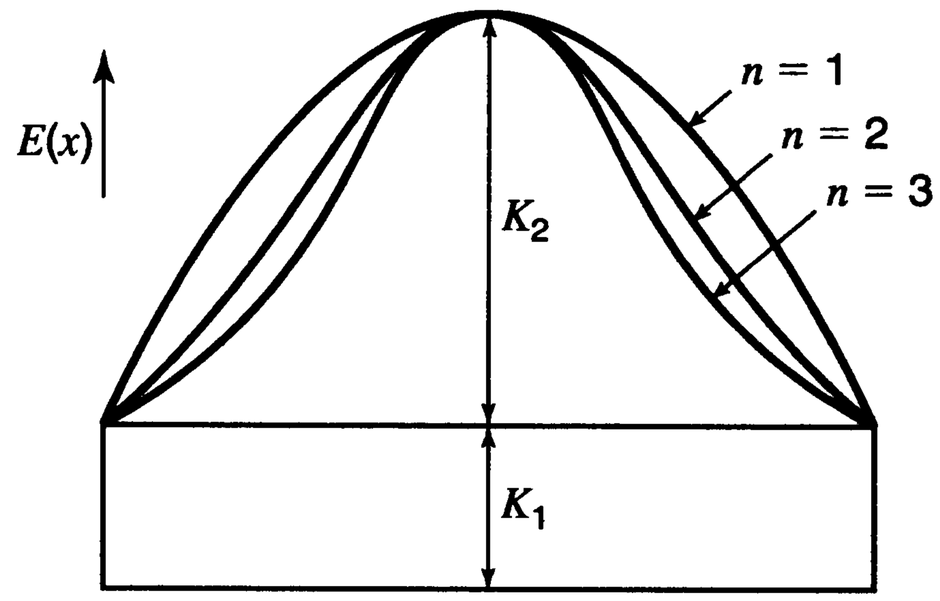


Figure 19–2
Various shapes of aperture distribution.

$-L_\lambda/2$	0	x	$+L_\lambda/2$
$-D_\lambda/2$	0	r	$+D_\lambda/2$

where

$E(x)$ = field distribution
K_1 = constant (see Fig. 19–2)
K_2 = constant (see Fig. 19–2)
L_λ = L/λ = aperture width, λ
n = integer (= 1, 2, 3, . . .)

If $K_2 = 0$, the distribution is uniform. If $K_1 = 0$, the distribution is parabolic for $n = 1$ and more severely tapered toward the edges for larger values of n, as indicated in Fig. 19–2. The beam and aperture efficiency of a 1-dimensional aperture with $n = 1$ has been calculated by Nash (Nash-1) as a function of the ratio $K_1/(K_1 + K_2)$, with the results shown in Fig. 19–3. For $K_1 = 0$, the distribution tapers to zero at the edge (maximum taper). As the abscissa value in Fig. 19–3 [ratio $K_1/(K_1 + K_2)$] increases, the taper decreases until at an abscissa value of 1 ($K_2 = 0$), there is no taper; i.e., the distribution is uniform.[1] The curves of Fig. 19–3 show that the beam efficiency tends to increase with an increase in taper but the aperture efficiency decreases. *Maximum aperture efficiency occurs for a uniform aperture distribution, but maximum beam efficiency occurs for a highly tapered distribution.* In most cases a taper is used that is intermediate between the two extremes of Fig. 19–3 [$K_1/(K_1 + K_2) = 0$ or 1], and a compromise is reached between large beam and aperture efficiencies.

For a 2-dimensional aperture distribution, i.e., a rectangular aperture (L_λ by L'_λ) with the same type of distribution in both the x and y directions, the beam and aperture efficiencies as a function of taper have been calculated by Nash (1). A field distribution as given by (38) with $n = 1$ is assumed (K_2 part of distribution parabolic). Thus, for this case

$$E(x, y) = \left\{K_1 + K_2\left[1 - \left(\frac{2x}{L_\lambda}\right)^2\right]\right\}\left\{K_1 + K_2\left[1 - \left(\frac{2y}{L'_\lambda}\right)^2\right]\right\} \tag{39}$$

[1]The step function or pedestal K_1 is an idealization which is not physically realizable. The actual field cutoff must be more gradual. Rhodes shows that the electric field component perpendicular to the edge of a planar structure must vanish as the first power of the distance from the edge and that the component parallel to the edge must vanish as the second power. See D. R. Rhodes (1).

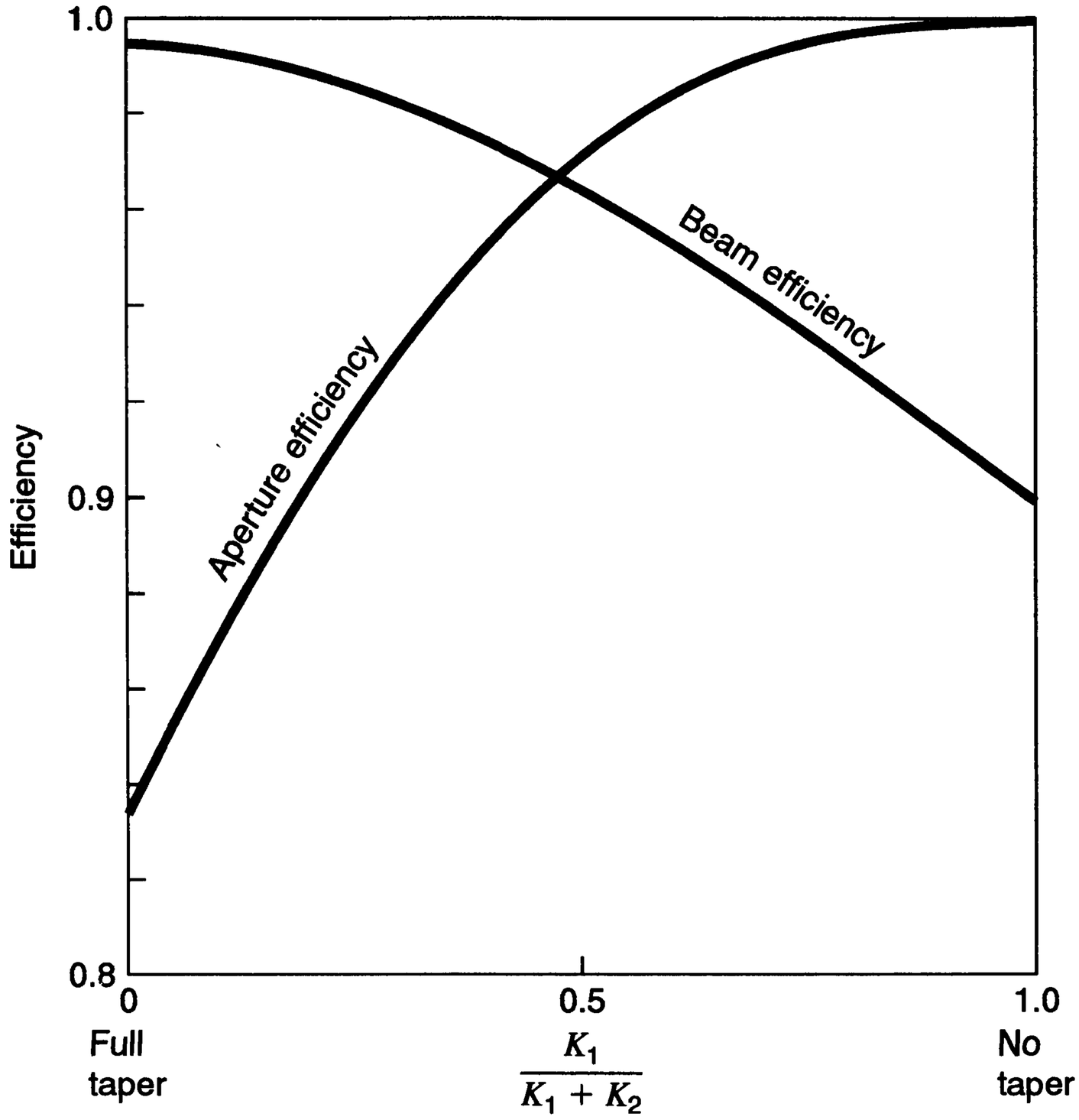


Figure 19–3
Beam and aperture efficiencies for a 1-dimensional aperture as a function of taper. The aperture efficiency is a maximum with no taper, while the beam efficiency is a maximum with full taper. A parabolic distribution for K_2 is assumed (see $n = 1$ in Fig. 19–2). (*Nash-1*)

Nash's data, which also show the effect of random phase errors across the aperture, are presented in Fig. 19–4*a*. There is a family of 4 curves for the aperture efficiency (solid) and another family of 4 curves for the beam efficiency (dashed) for random displacements (or errors) of 0, 0.04, 0.06 and 0.08λ rms deviation. To obtain these curves the aperture and beam efficiencies calculated for the smooth distribution (39) were multiplied by the gain-degradation (or gain-loss) factor of Ruze (1), given by

$$k_g = e^{-(2\pi\delta/\lambda)^2} \tag{40}$$

where δ is the rms phase front displacement from planar over the aperture. It is assumed that the correlation intervals of the deviations are greater than the wavelength. The curves of Fig. 19–4*a* indicate that the controlling effect of the taper on the efficiencies (beam and aperture) tends to decrease as the phase error increases. The efficiencies are also reduced by the presence of the phase error, since such errors tend to scatter radiation into the side-lobe regions. Thus, the phase errors constitute a primary limitation on the antenna efficiency.

The gain-loss formula (40) of Ruze assumes specifically that the deviations from the best-fit paraboloid are random and distributed in a Gaussian manner, that the errors are uniformly distributed, that the region over which the errors are substantially constant is large compared to the wavelength and that the number of such uncorrelated regions is large.

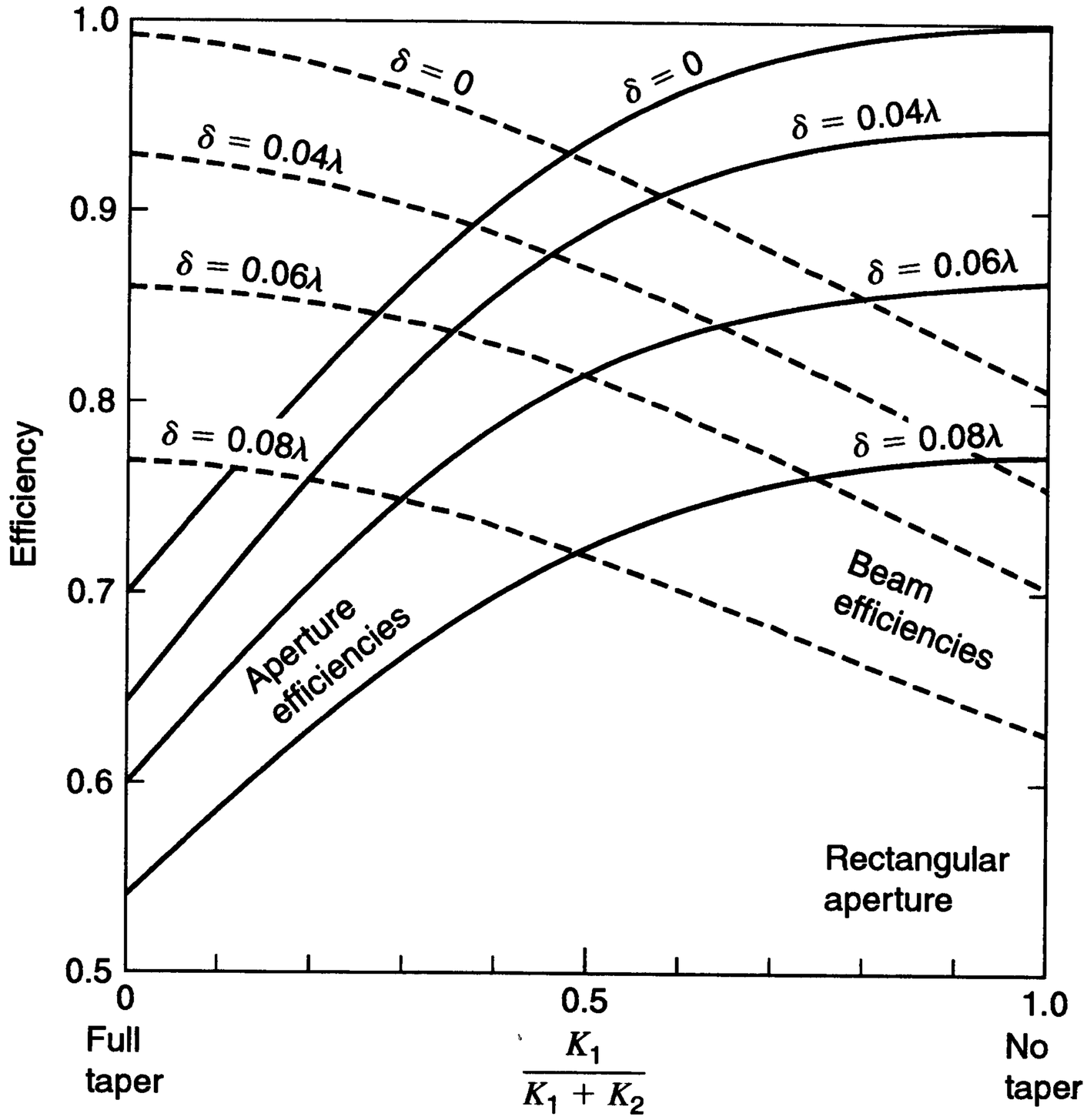


Figure 19–4a
Aperture efficiency (solid) and beam efficiency (dashed) of a rectangular aperture as a function of taper and phase error. A parabolic distribution for K_2 is assumed (see $n = 1$ in Fig. 19–2). (*Nash-1*)

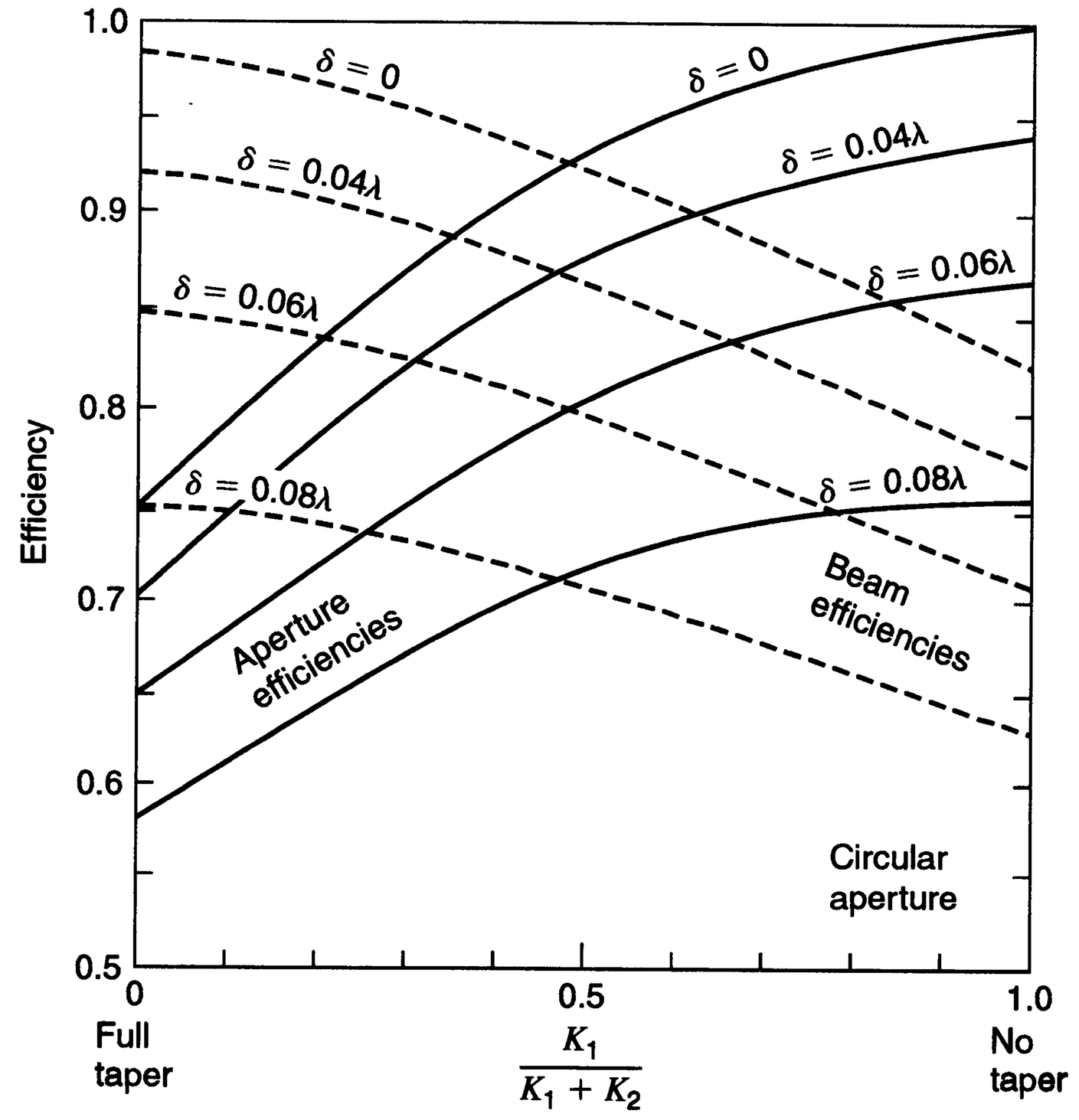


Figure 19–4b
Aperture efficiency (solid) and beam efficiency (dashed) of a circular aperture as a function of taper and phase error. A parabolic distribution for K_2 is assumed (see $n = 1$ in Fig. 19–2). (*Nash-1*)

A circular aperture of diameter D_λ with a distribution as given by (38), where x is replaced by r and L_λ by D_λ, has also been investigated by Nash, with the results shown in Fig. 19–4*b*. Two families of 4 curves each are given for the beam and aperture efficiencies for 4 conditions of random phase error. It is assumed that $n = 1$ (K_2 part of distribution parabolic). The curves of Fig. 19–4*b* (circular aperture) are seen to be very similar to those of Fig. 19–4*a* for the rectangular aperture. For reflector antennas with an rms surface deviation δ' it should be noted that the phase front deviation δ in (40) will be approximately twice as large; that is, $\delta = 2\delta'$.

Another problem to be considered with reflector antennas is the efficiency with which the primary, or feed, antenna illuminates the reflector. This may be defined as the *feed efficiency* ε_f, where

$$\varepsilon_f = \frac{\displaystyle\iint_{\Omega_R} P_f(\theta, \phi)\, d\Omega}{\displaystyle\iint_{4\pi} P_f(\theta, \phi)\, d\Omega} \tag{41}$$

where

$P_f(\theta, \phi)$ = power pattern of feed
Ω_R = solid angle subtended by reflector as viewed from feed point

If the first null of the feed antenna pattern coincides with the edge of the reflector, the feed efficiency given by (41) is identical with the beam efficiency of the feed.[1]

In the general situation with a reflector antenna, the beam efficiency of the system may be expressed as

$$\varepsilon_M = e^{-(4\pi\delta'/\lambda)^2} \frac{\displaystyle\iint_{\substack{\text{main}\\ \text{lobe}}} P(\theta, \phi)\, d\Omega}{\displaystyle\iint_{4\pi} P(\theta, \phi)\, d\Omega} \frac{\displaystyle\iint_{\Omega_R} P_f(\theta, \phi)\, d\Omega}{\displaystyle\iint_{4\pi} P_f(\theta, \phi)\, d\Omega} \tag{42}$$

where

δ' = rms surface error of reflector
$P(\theta, \phi)$ = power pattern of reflector due to aperture distribution produced by the feed assuming no phase error
$P_f(\theta, \phi)$ = power pattern of feed

Surface leakage is neglected. If it is appreciable, another factor is required.

[1]Common practice is to taper the feed illumination by 10 dB or so at the edges of the reflector.

An experimental procedure for determining the *beam efficiency* ε_M of a large antenna using a *celestial source* is as follows. The main-beam solid angle Ω_M is evaluated from

$$\Omega_M = k_p \theta_{\text{HP}}\, \phi_{\text{HP}}, \text{ sr} \tag{43}$$

where

k_p = factor between about 1.0 for a uniform aperture distribution and 1.13 for a Gaussian power pattern
θ_{HP} = half-power beamwidth in θ plane, rad
ϕ_{HP} = half-power beamwidth in ϕ plane, rad

The half-power beamwidths are measured, while k_p may be calculated or estimated from the pattern shape. The half-power beamwidth in right ascension in degrees is given by the observed half-power beamwidth of a drift profile in minutes of time multiplied by $\cos\delta/4$, where δ is the declination of the observed point source; that is,

$$\text{HPBW(deg)} = \frac{\text{HPBW(min)} \cos\delta}{4} \tag{44}$$

The total beam solid angle Ω_A is obtained from the relation

$$\Omega_A = \frac{k_0 \lambda^2}{A_e} \tag{45}$$

where the effective aperture A_e is determined by observing a celestial point source of known flux density;[1] that is,

$$A_e = \frac{2kT_A}{S} \tag{46}$$

Combining relations, the *beam efficiency* is then given by

$$\varepsilon_M = \frac{\Omega_M}{\Omega_A} = \frac{k_p \theta_{\text{HP}}\, \phi_{\text{HP}} 2kT_A}{k_o \lambda^2 S} \tag{47}$$

where

k = Boltzmann's constant ($= 1.38 \times 10^{-23}$ J K^{-1})
T_A = antenna temperature due to radio source (measured value corrected for cable loss), K
S = flux density of radio source, W m^{-2} Hz^{-1}
λ = wavelength, m
k_o = antenna ohmic-loss factor, dimensionless

and where k_p, θ_{HP} and ϕ_{HP} are as defined in (43).

[1] See the table of celestial sources, Sec. 24–5*b*.

A procedure for determining the *aperture efficiency* ε_{ap} of a large antenna is to observe a *celestial source* of known flux density to find A_e by (46), from which

$$\varepsilon_{ap} = \frac{A_e}{A_p} \tag{48}$$

where A_p = physical aperture

It is often of interest to the antenna designer, however, to break ε_{ap} up into a number of factors, as in (19), in order to account as completely as possible for all the causes of efficiency degradation. Thus, from knowledge of the antenna structure and its conductivity the ohmic-loss factor k_o may be determined. The utilization factor k_u can be calculated from design considerations. The achievement factor k_a could be calculated from the relation in (28) if the actual fields across the aperture were accurately known; however, these are rarely measured.[1] As an alternative the achievement factor can be separated into many factors involving the random surface error, feed efficiency, aperture blocking, feed displacement normal to axis (squint or coma), feed displacement parallel to axis (astigmatism), etc., of the (parabolic) reflector antenna. It is assumed that each of these subfactors can be independently calculated or estimated.

The aperture efficiency ε_{ap} may then be expressed as

$$\varepsilon_{ap} = \frac{A_e}{A_p} = k_o k_u k_a = k_o k_u k_1 k_2 \cdots k_N$$

$$= k_o k_u \prod_{n=1}^{N} k_n \tag{49}$$

where

k_o = ohmic-loss factor
k_u = utilization factor (by design)
k_a = achievement factor
$k_1 = e^{-(4\pi\delta'/\lambda)^2}$ = random reflector-surface-error (gain-loss) factor = k_g
$k_2 = \varepsilon_f$ = feed-efficiency factor
k_3 = aperture-blocking factor
k_4 = squint factor
k_5 = astigmatism factor
k_6 = surface-leakage factor

etc.

A close agreement between ε_{ap} as calculated by (49) and as measured by (48) does not necessarily mean that the designer has taken all factors properly into account (some could have been overestimated and others underestimated), but it does provide some confidence in understanding the factors involved. However, if there is significant disagreement between the two methods, the designer knows the analysis is incorrect or incomplete in one or more

[1] See Fig. 24–15 regarding a *holographic measurement.*

respects. This comparative method has been used by Nash in analyzing the performance characteristics of a 110-m radio telescope (Big Ear) (Nash-2). A discussion of tolerances in large antennas is given by Bracewell (1).

For in-phase fields over a lossless aperture, the aperture efficiency is given from (26) by

$$\varepsilon_{ap} = \frac{E_{av}^2}{(E^2)_{av}} \tag{50}$$

where

E = field at any point in aperture
E_{av} = average of E over aperture
$(E^2)_{av}$ = average of E^2 over aperture

EXAMPLE 19–1.1

Determine the aperture efficiency and the beam efficiency $\varepsilon_M = \Omega_M / \Omega_A$ for a 50λ diameter circular aperture with aperture distribution of the form $(1 - r^2)^n$, where $r = 0$ at the center and $r = 1$ at the rim for various values of n. Show the results graphically.

■ **Solution**

The aperture efficiency is obtained from (50). The beam efficiency is obtained by integrating the pattern over 4π for Ω_A and over the main beam for Ω_M, using numerical methods. The results are shown in Fig. 19–5 with efficiency as ordinate and side-lobe level as abscissa, with aperture distribution shape as given by n also indicated. For an $n = 2$ distribution the aperture efficiency is about 56 percent and the first side lobe 31 dB down.

For continuous apertures which are large ($\gg\lambda$) some conclusions are:

1. A uniform amplitude distribution yields the maximum directivity (nonuniform edge-enhanced distributions for supergain being considered impractical).
2. Tapering the amplitude from a maximum at the center to a smaller value at the edges reduces the side-lobe level but results in larger (main-lobe) beamwidth and less directivity.
3. A distribution with an inverse taper (amplitude depression at center) results in smaller (main-lobe) beamwidth but in increased side-lobe level and less directivity. (The amplitude depression at the center might be produced inadvertently by a primary (feed) antenna blocking the center of the aperture.)
4. Maximum aperture efficiency occurs for a uniform aperture distribution, but maximum beam efficiency occurs for a highly tapered distribution.
5. Aperture phase errors are a primary limitation on antenna efficiency.
6. Depending on aperture size (in λ) and phase error, there is a frequency (or λ) for which the gain peaks, rolling off to smaller values as the frequency is raised (see Fig. 19–9.)

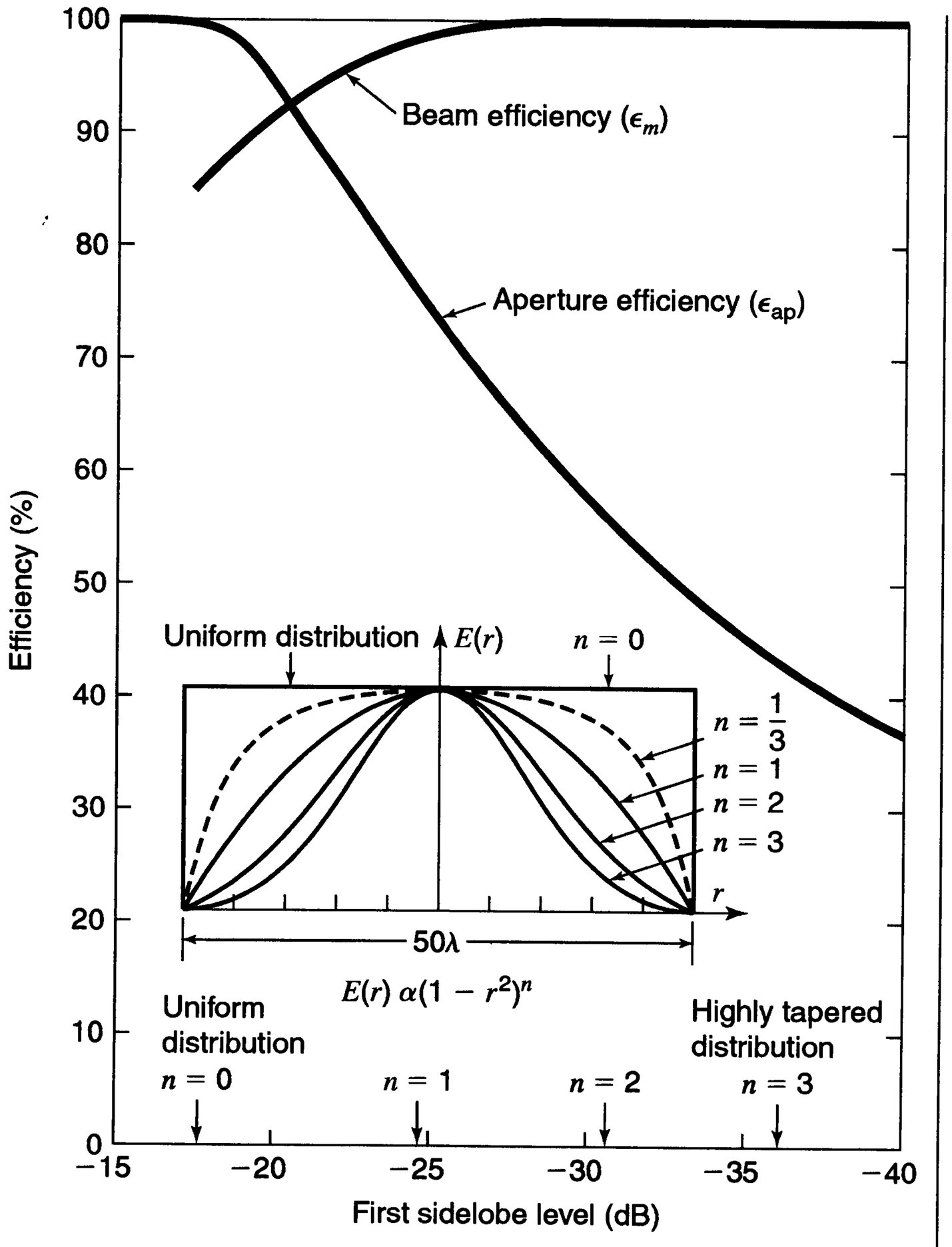


Figure 19–5
Aperture and beam efficiencies for various aperture distributions and side-lobe levels from worked example. Uniform phase is assumed. With phase variation, aperture efficiency decreases and side lobes increase.

19–2 SURFACE IRREGULARITIES AND GAIN LOSS（表面不规则性和增益损失）

Referring to Fig. 19–6, consider the idealized case of a reflecting surface with irregularities which depart a distance δ' above and below the ideal surface (Christiansen (1) and Hogbom). Plane waves reflected from the irregular surface will be advanced or retarded with respect

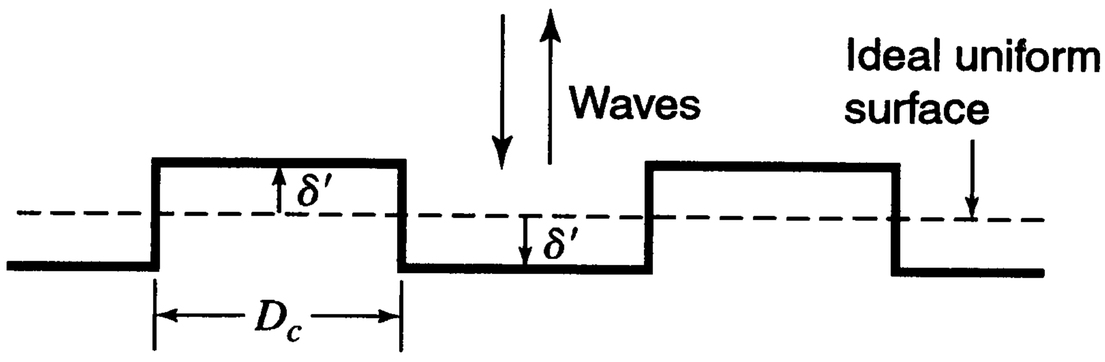


Figure 19–6
Geometry for determining the effect of surface irregularities.

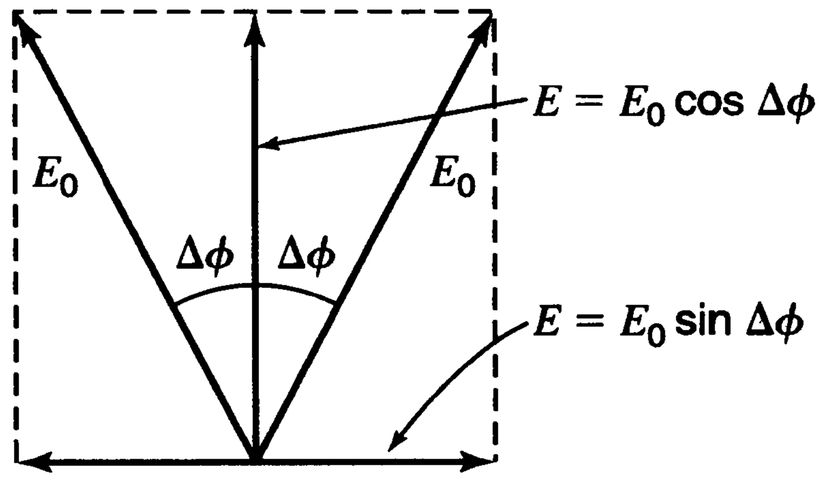


Figure 19–7
Phase variation due to surface irregularities.

to waves reflected from the uniform surface by

$$2\pi \frac{\delta'}{\lambda} \times 2 = 2\pi \frac{\delta}{\lambda} \qquad \text{(rad)} \tag{1}$$

where

δ' = surface deviation (surface error)
δ = twice surface deviation $= 2\delta'$

Referring to the phase diagram of Fig. 19–7, the resultant field

$$E = E_0 \cos \Delta\phi \tag{2}$$

where $\Delta\phi = 4\pi(\delta'/\lambda)$ rad $= 720°(\delta'/\lambda) = 360°(\delta/\lambda) =$ phase error

Assuming that the separation (correlation distance D_c) of the irregularities is at least as large as the wavelength and that there are as many positive as negative irregularities, the (normalized) surface gain-loss factor is

$$k_g = \left(\frac{E_0 \cos \Delta\phi}{E_0}\right)^2 = \cos^2 \Delta\phi = \cos^2\left(720° \frac{\delta'}{\lambda}\right) \tag{3}$$

where $k_g =$ gain-loss factor $(0 \le k_g \le 1)$, dimensionless

The variation of the gain-loss factor as a function of rms surface deviation in wavelengths and degrees as calculated from (3) is given in Fig. 19–8. The gain-loss factor as calculated from the Ruze relation (19–1–40) $(= \exp[-(4\pi\delta'/\lambda)^2])$ is also shown. Correlation distances of at least a wavelength are assumed.

EXAMPLE 19–2.1

A surface has an rms deviation $\delta' = \lambda/20$. Find the reduction in gain.

■ **Solution**

From (3),

$$\Delta\phi = 4\pi \frac{\delta'}{\lambda} \text{ (rad)} = 720° \frac{\delta'}{\lambda} = 720° \frac{1}{20} = 36° \tag{4}$$

and

$$k_g = \cos^2 36° = 0.65 \tag{5}$$

or a gain reduction of 1.8 dB (as may be noted in Fig. 19–8).

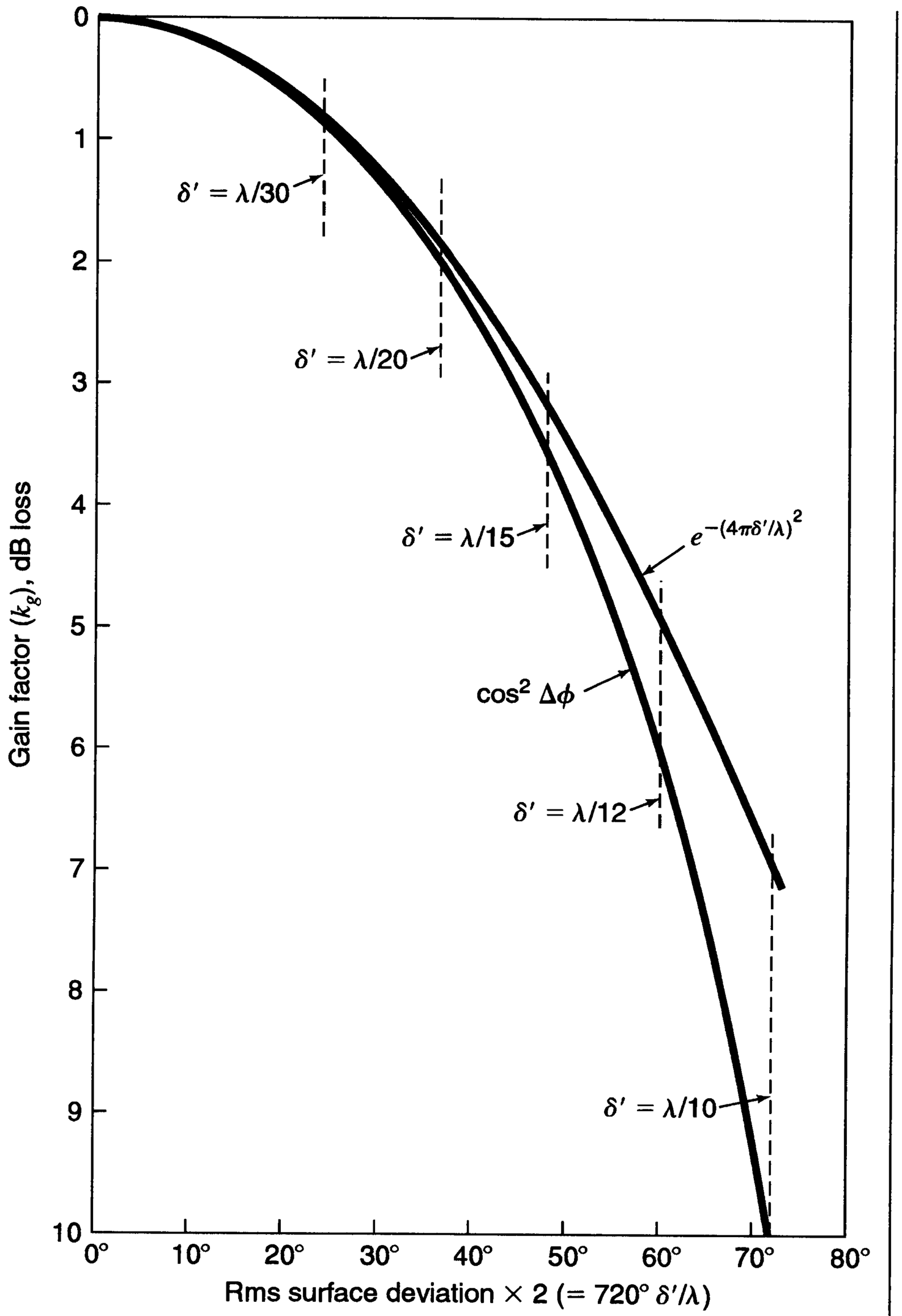


Figure 19–8
Gain-loss factor of reflector antenna as a function of twice the rms surface deviation expressed in degrees of phase angle (of the reflected wave) as calculated from (3) and (19–1–40). The vertical dashed lines correspond to phase angles for rms surface deviations expressed in fractions of a wavelength. See text for assumptions involved.

Assuming 50 percent aperture efficiency ($A_e = \frac{1}{2}A_p$) and a circular dish diameter D,

$$A_e = \frac{1}{2}\pi\left(\frac{D}{2}\right)^2$$

Introducing the gain-loss factor for surface irregularities from (3),

$$G = k_g \frac{4\pi}{\lambda^2}\frac{\pi}{2}\left(\frac{D}{2}\right)^2 = k_g \frac{\pi^2}{2}\left(\frac{D}{\lambda}\right)^2 = \cos^2\left(720°\frac{\delta'}{\lambda}\right)\frac{\pi^2}{2}\left(\frac{D}{\lambda}\right)^2 \tag{6}$$

where δ' = rms surface deviation

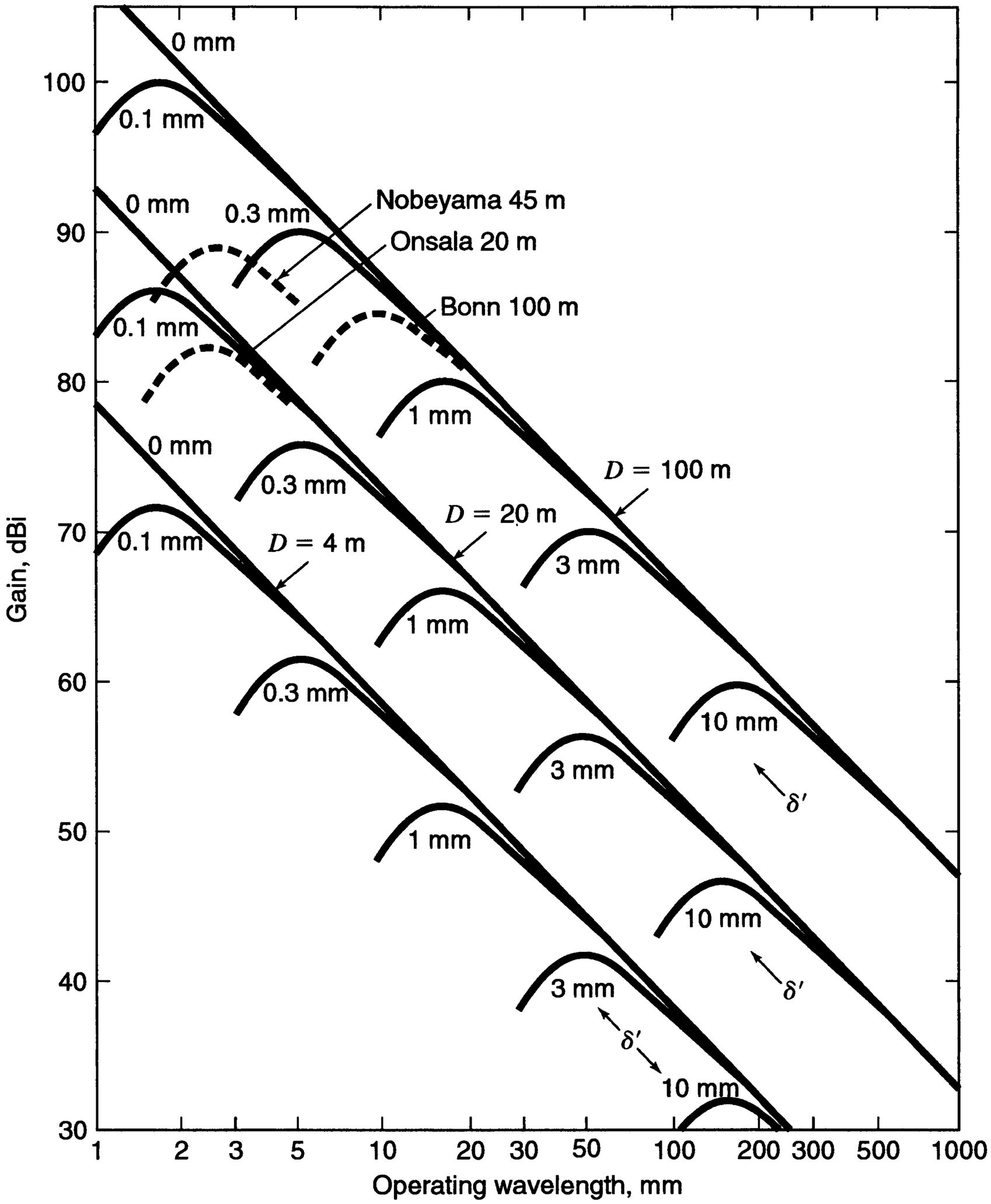


Figure 19–9
Gain peaking and roll-off with decreasing wavelength. The curves show antenna gain as a function of operating wavelength for various rms surface deviations (δ') and antenna diameters (D) equal to 4, 20 and 100 m as calculated from (6). Since scales are logarithmic, other values of D and δ' can be readily interpolated. Curves for Onsala, Nobeyama and Bonn dishes are shown dashed. See text for assumptions involved.

Values of the gain calculated from (6) are shown in Fig. 19–9 for various surface deviations δ' and circular dish diameters $D = 4, 20$ and 100 m. Measured gains of three radio telescope dishes (dashed curves) are shown for comparison. It may be inferred from Fig. 19–8 that the equivalent rms surface deviations of the Onsala and Nobeyama dishes are about 150 μm and of the Bonn dish about 500 μm.

Referring to Fig. 19–7, the quadrature field components are given by

$$E_q = E_0 \sin \Delta\phi \tag{7}$$

For equal positive and negative irregularities these total zero. Suppose, however, that in some other direction they all *add in phase*. The power level of this minor lobe relative to the main lobe is then

$$\left(\frac{E_{\text{minor}}}{E_{\text{major}}}\right)^2 = \left(\frac{NE_0 \sin \Delta\phi}{NE_0 \cos \Delta\phi}\right)^2 = \tan^2 \Delta\phi \tag{8}$$

where N = number of irregularities

Equation (8) gives the maximum level a minor lobe could have due to these irregularities. Consider, on the other hand, that the quadrature fields *add randomly*. Their total is then $\sqrt{N}E_0 \sin \Delta\phi$ and the power level of the resulting minor lobe relative to the main lobe is

$$\left(\frac{E_{\text{minor}}}{E_{\text{major}}}\right)^2 = \left(\frac{\sqrt{N}E_0 \sin \Delta\phi}{NE_0 \cos \Delta\phi}\right)^2 = \frac{1}{N} \tan^2 \Delta\phi \tag{9}$$

EXAMPLE 19–2.2

If the rms phase error is 36° and the number of irregularities $N = 100$, find the maximum and random side-lobe levels.

■ **Solution**

From (8),

$$\text{Maximum side-lobe level} = \tan^2 36° = 0.527$$

or down 2.8 dB from the main lobe. From (9),

$$\text{Side-lobe level (random case)} = \frac{1}{N} \tan^2 36° = \frac{0.527}{100} = 0.00527$$

or down 22.8 dB from the main lobe.

Although the gain-loss factor (3) is not a function of the number N of irregularities, the side-lobe level (9) for the random case does depend on N and decreases as N increases. Small-scale irregularities with correlation distances which are small compared to the wavelength result in fields which smooth out at large distances from the reflecting surface and, accordingly, these small-scale irregularities do not affect the gain or side-lobe level as discussed above.

For a reflector antenna with perfect surface the main lobe and near side lobes are determined principally by the aperture distribution and taper, while the minor lobes at larger angles to 90° or so are influenced by scatter from the feed support structure. Back-radiation is a function of spillover and diffraction around the edge of the reflector. For a circular dish a substantial back lobe may occur on axis (180° from the main lobe) due to all diffracted waves adding in phase (see Fig. 19–10). To reduce diffraction the reflector should have a rolled edge with radius of curvature at least $\lambda/4$ at the longest wavelength of operation (Burnside-1).

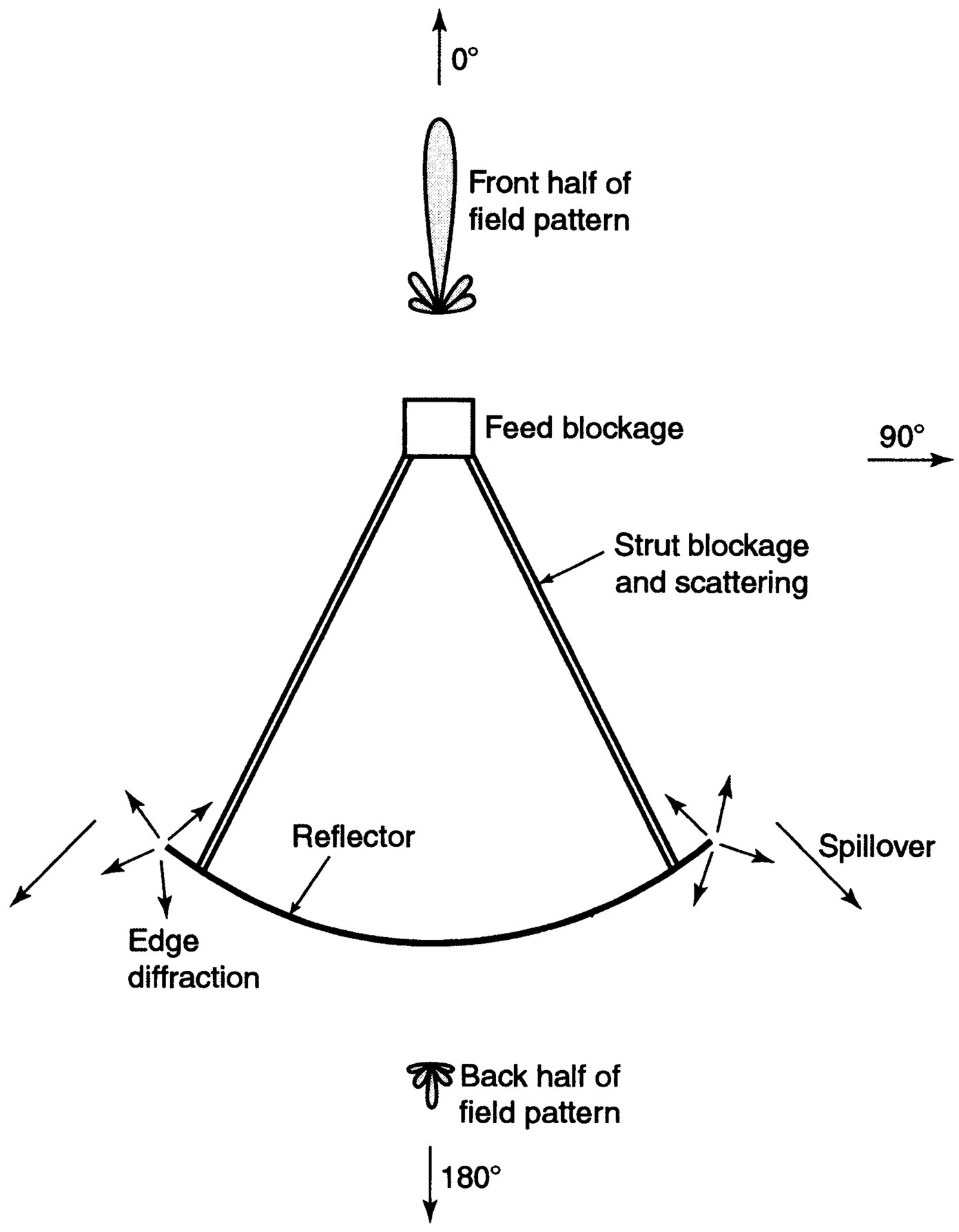


Figure 19–10
Typical parabolic reflector with feed (or subreflector). Front part of field pattern (main lobe and side lobes) is determined by aperture distribution, surface irregularities and scattering from feed and struts. Back half of pattern (axial back lobe and near lobes) is determined by spillover and diffraction around dish.

Due to the force of gravity, a ground-based steerable parabolic reflector deforms as the reflector is tilted so that for a given wavelength there is a maximum diameter which cannot be exceeded by adding metal to the backup structure. However, this limit can be exceeded by a *homologous design* in which one paraboloid deforms into another. The next limit is then imposed by thermal deformation (Von Hoerner-1).

19–3 OFF-AXIS OPERATION OF PARABOLIC REFLECTORS
（抛物面反射镜的偏轴工作）

When the feed of a parabolic reflector antenna is displaced laterally from the focal point the beam is shifted (squinted) off-axis in the opposite direction to the feed displacement. Such squinting is accompanied by gain loss, beam broadening and the appearance of a coma side lobe. The amount of such degradation in performance is a function of the F/D ratio and the feed displacement, as illustrated in Fig. 19–11 (Baker-1).

With $F/D = 1.0$, the gain loss is 0.3 dB for a feed displacement of 10 HPBWs but with $F/D = 0.25$ the loss is 10 dB for the same displacement. Thus, a long focal length parabola is more tolerant of feed displacement than a short focal length parabola.

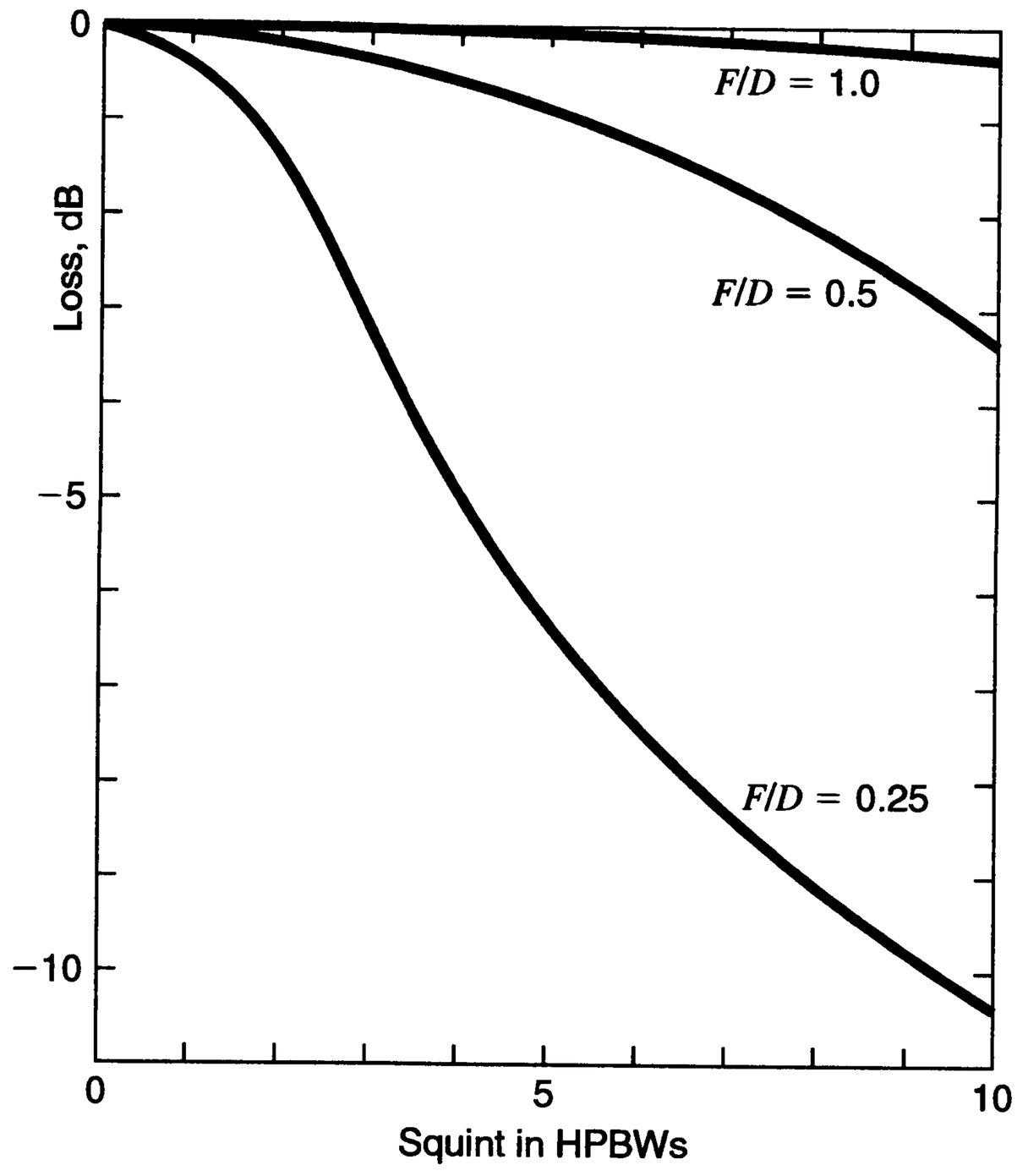


Figure 19–11
Universal squint diagram for parabolic antennas of any size showing the gain loss as a function of the squint displacement in HPBWs for 3 values of F/D ratio. The loss for other F/D ratios may be interpolated. (*Baker-1*)

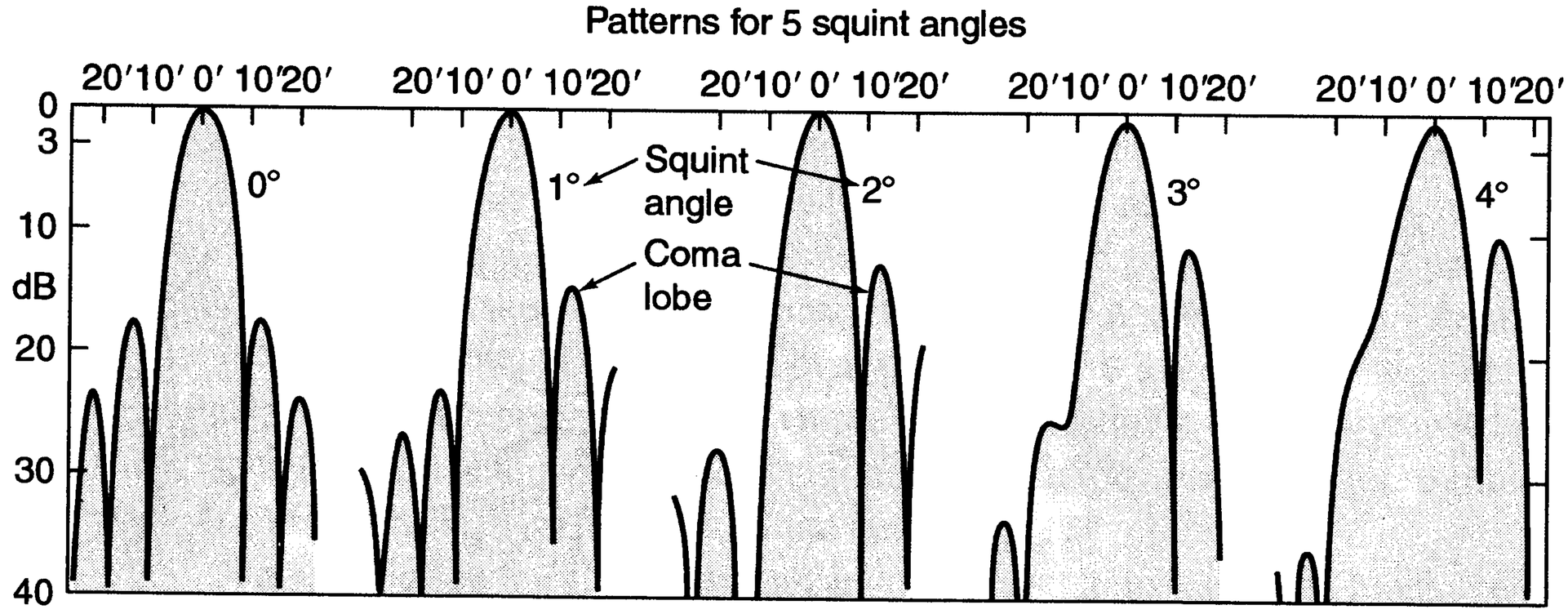


Figure 19–12
Calculated patterns of a 530λ diameter parabola with $F/D = 1.2$ for 5 squint angles. At 4° squint the gain is 1 dB down, the HPBW has increased from 8′ to 8.5′ (arcminutes) and the coma lobe is up 7.1 dB. (*Baker-2*)

Calculated patterns for a 530λ diameter parabola with $F/D = 1.2$ are shown in Fig. 19–12 for 5 values of squint angle as calculated by Baker (2). The aperture distribution is uniform, resulting in symmetrical first side lobes 17.6 dB down as shown for 0° squint. As the squint angle increases, the gain decreases, the beamwidth increases, the pattern becomes asymmetrical and a first side lobe increases (coma lobe). Thus, at a squint

of 4° the gain is down 1 dB, the HPBW has increased from 8′ to 8.5′ and the coma lobe is only 10.5 dB down (up 7.1 dB from the 0° squint condition). Squinting with multiple feeds deployed near the prime focus (or by a movable feed for tracking) is frequently useful in spite of some degradation in performance. More detailed discussions of off-axis operation are given by Baker (1), Lo (1), Rudge (1) and Withers, Rusch (1) and Ludwig, Sandler (1) and von Gniss (1) and Ries.

19–4 CASSEGRAIN FEED, SHAPED REFLECTORS, SPHERICAL REFLECTORS AND OFFSET FEED（卡塞格伦馈电、赋形反射镜、球面反射镜和偏馈）

Subreflectors offer flexibility of design for reflecting telescopes. The classical arrangement introduced by N. Cassegrain of France over 300 years ago uses a subreflector of hyperbolic shape which surrounds the prime focus point of the main parabolic reflector. Referring to Fig. 19–13, we require that all rays from the focal point F form a spherical wave front

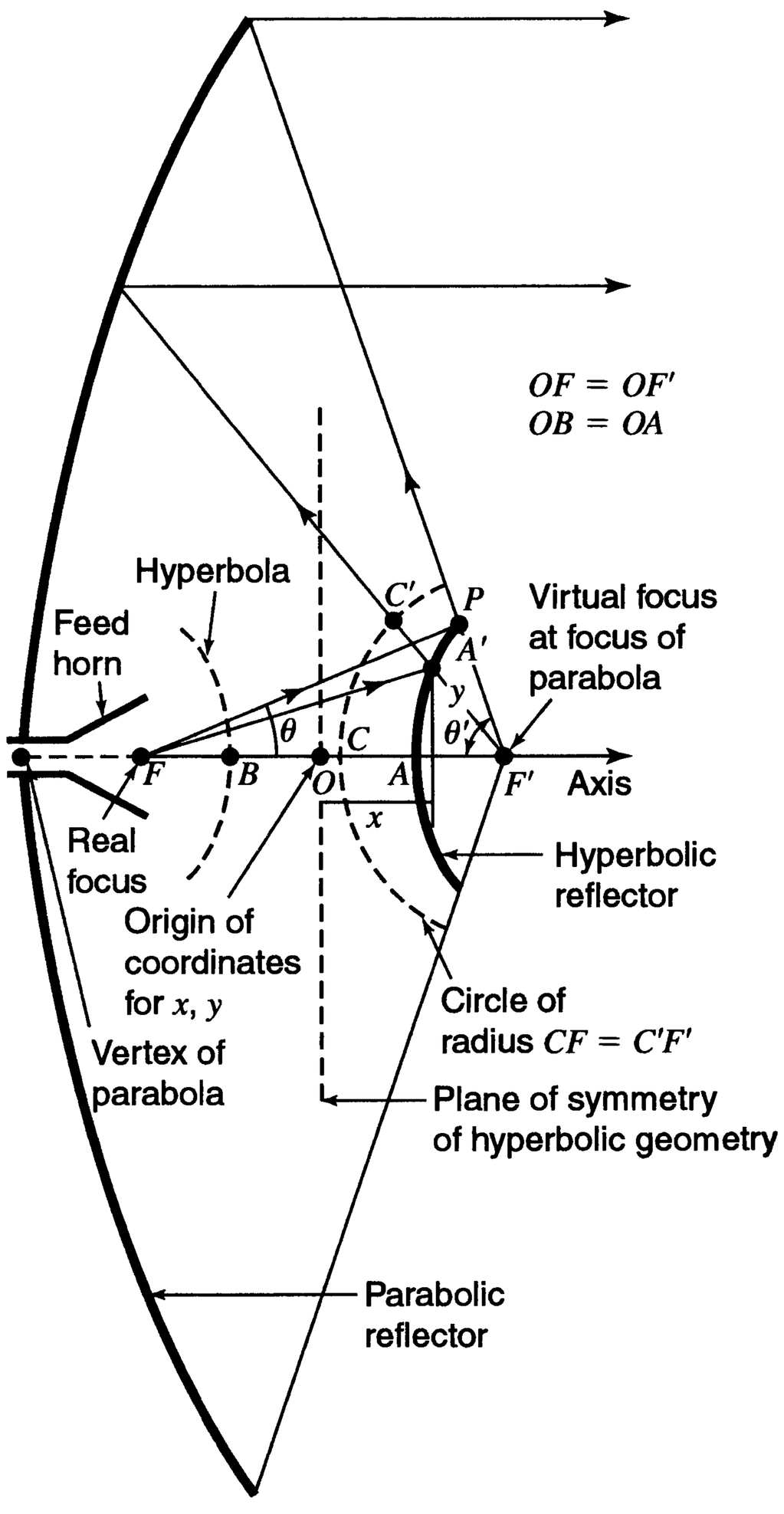


Figure 19–13
Geometry for Cassegrain reflector.

(circle of radius CF') on reflection from the (hyperbolic) subreflector (as though radiating isotropically from the parabola focus F') or by Fermat's principle of equality of path length that

$$C'A' + FA' = CA + FA \tag{1}$$

Noting that $CA = CF' - AF'$ and that $FA - AF' = 2OA$ we obtain

$$FA' - A'F' = 2OA = BA \tag{2}$$

which is the relation for a hyperbola with standard form

$$\frac{x^2}{a^2} - \frac{y^2}{f^2 - a^2} = 1 \tag{3}$$

where $a = OA = OB$, $f = OF' = OF$, and x and y are as shown in Fig. 19–13. Or

$$y^2 = \left(\frac{x^2}{a^2} - 1\right)(f^2 - a^2) \tag{4}$$

The hyperbolic subreflector is then truncated at point P for which a ray reflected from the hyperbola hits the edge of the parabolic reflector. The hyperbolic reflector then subtends an angle θ from the feed location at the focal point F while the (main) parabolic reflector subtends an angle θ' from the focal point F' of the parabola. Thus, the feed horn beam angle is increased in the ratio θ'/θ to fill the parabola aperture.

An advantage of the Cassegrain design is its compactness, with feed and amplifier near the vertex of the parabola. Higher aperture efficiency may also be realized by shaping the subreflector (or modifying the hyperbolic contour) to correct the primary pattern of the feed and produce a more uniform field distribution across the parabola aperture.[1]

Referring to Fig. 19–14, the surface at AA' is deformed to enlarge or restrict the incremental ray bundle BB', thereby decreasing or increasing the watts per steradian in the bundle and finally the watts per square meter in the aperture plane of the parabola. This *shaping technique* may be extended over the entire subreflector and often both subreflector and parabola are shaped. As a result a more uniform aperture distribution and higher aperture efficiency can be achieved but with higher first side lobes and also more rapid gain loss as the feed is moved off-axis to squint the beam.

A constraint on the Cassegrain arrangement is that to minimize blockage the subreflector should be small compared to the parabola, yet the subreflector must be large compared to the wavelength. Details on Cassegrain reflectors are given by Love (1).

In Fig. 19–15 a *parabola* is given by

$$y^2 = 4fx \tag{5}$$

where f = focal distance $= VF$

---

[1] If the rays are allowed to pass through the focus of the parabola instead of being intercepted by the hyperbolic (convex) Cassegrain surface, as in Fig. 19–13, they can be reflected by a concave ellipsoidal surface beyond (to the right of) the focus. This type of subreflector is called *Gregorian* after James Gregory of England who devised it about 1660.

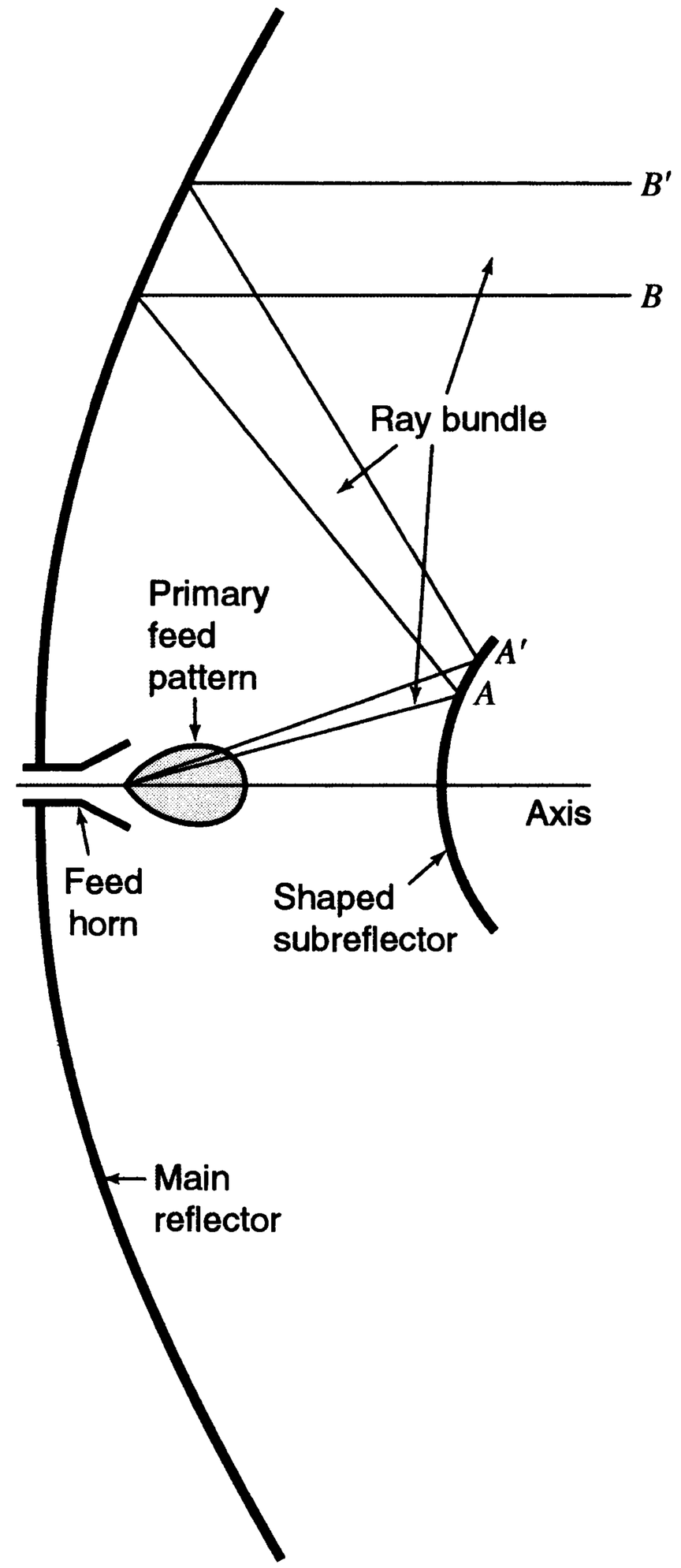


Figure 19–14
Geometry for shaped reflectors.

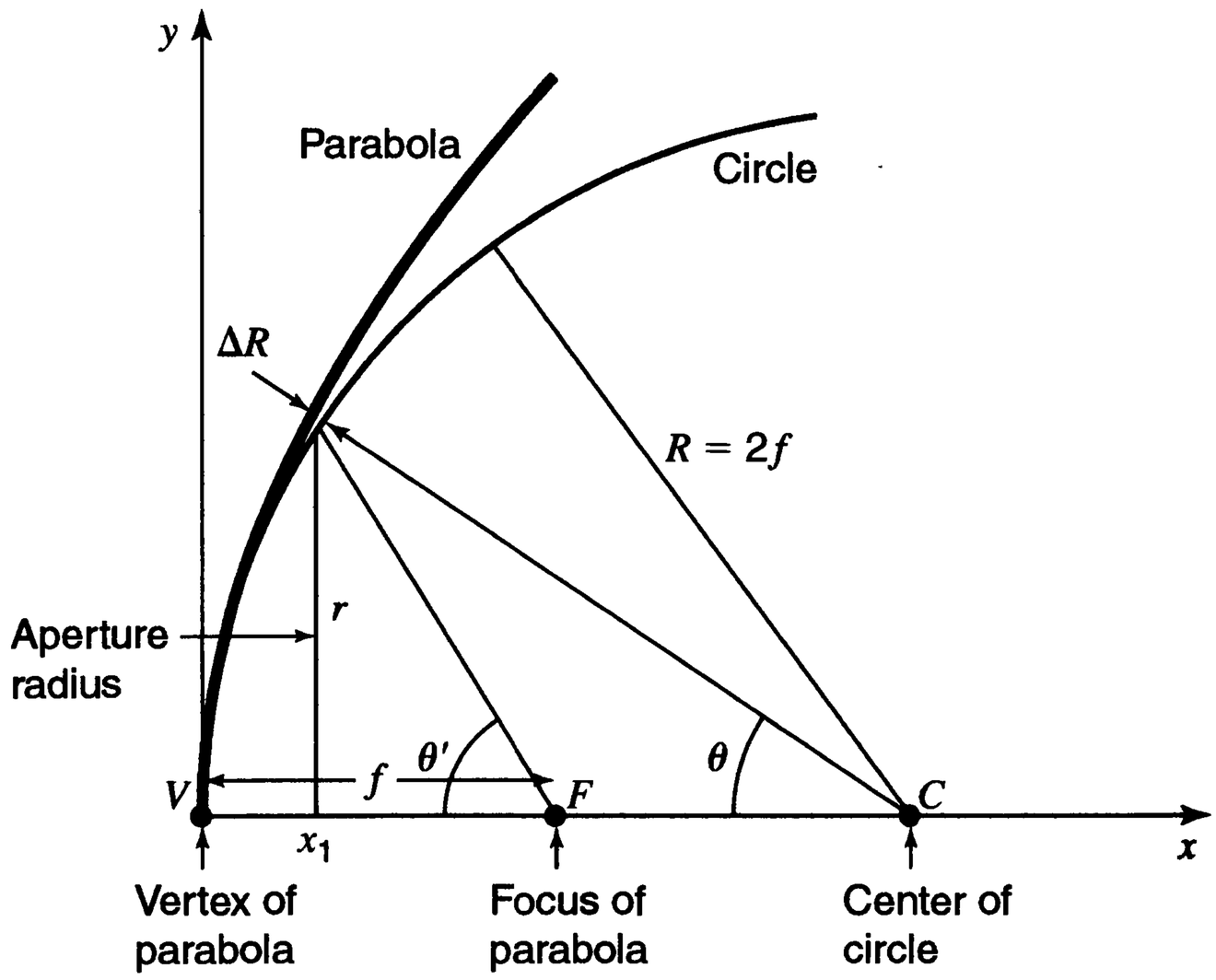


Figure 19–15
Circle and parabola compared, with radius of circle equal to twice the focal length of the parabola.

This parabola is compared with a *circle* of radius $R = VC$. It may be shown that for small values of x, the circle is of nearly the same form as the parabola when

$$R = 2f \tag{6}$$

Over an angle θ and *aperture radius*

$$r = R \sin\theta \tag{7}$$

the circle differs from the parabola by less than ΔR. If $\Delta R \ll \lambda$ (or specifically $<\lambda/16$) the field radiated from a point source at F within an angle θ' and reflected from the circle will be within 45° ($=2 \times 360°/16$) of the phase of a field radiated from F and reflected from the parabola.

Let Fig. 19–15 represent the cross section of a reflecting sphere and paraboloid-of-revolution. Then a feed antenna at the focal point F which illuminates the sphere only within the angle θ' will produce a plane wave over the aperture of diameter $2r$ having a phase deviation of less than 45°, this amount of deviation occurring only near the edge of the aperture.

It is apparent that with a *spherical reflector* of 2π sr solid angle, which remains fixed in position, a beam can be scanned over almost all of this solid angle by moving a suitable feed antenna along a spherical surface of radius $R/2$, as suggested in Fig. 19–16.

Although the spherical reflector is fixed and does not need to be moved, only a fraction of its total aperture (πR^2) is used.[1] If the primary feed beam angle θ' is increased, the phase degradation increases at the edges of the beam aperture. The effects of this spherical aberration can be reduced by suitable corrections of the feed distribution. Ashmead and Pippard considered the necessary corrections when the spherical aberration is small (Ashmead-1). Spencer, Sletten and Walsh (Spencer-1) and Love (1) describe a phase-correcting line source. Diffraction theory of large spherical reflectors is given by Schell and means of correcting

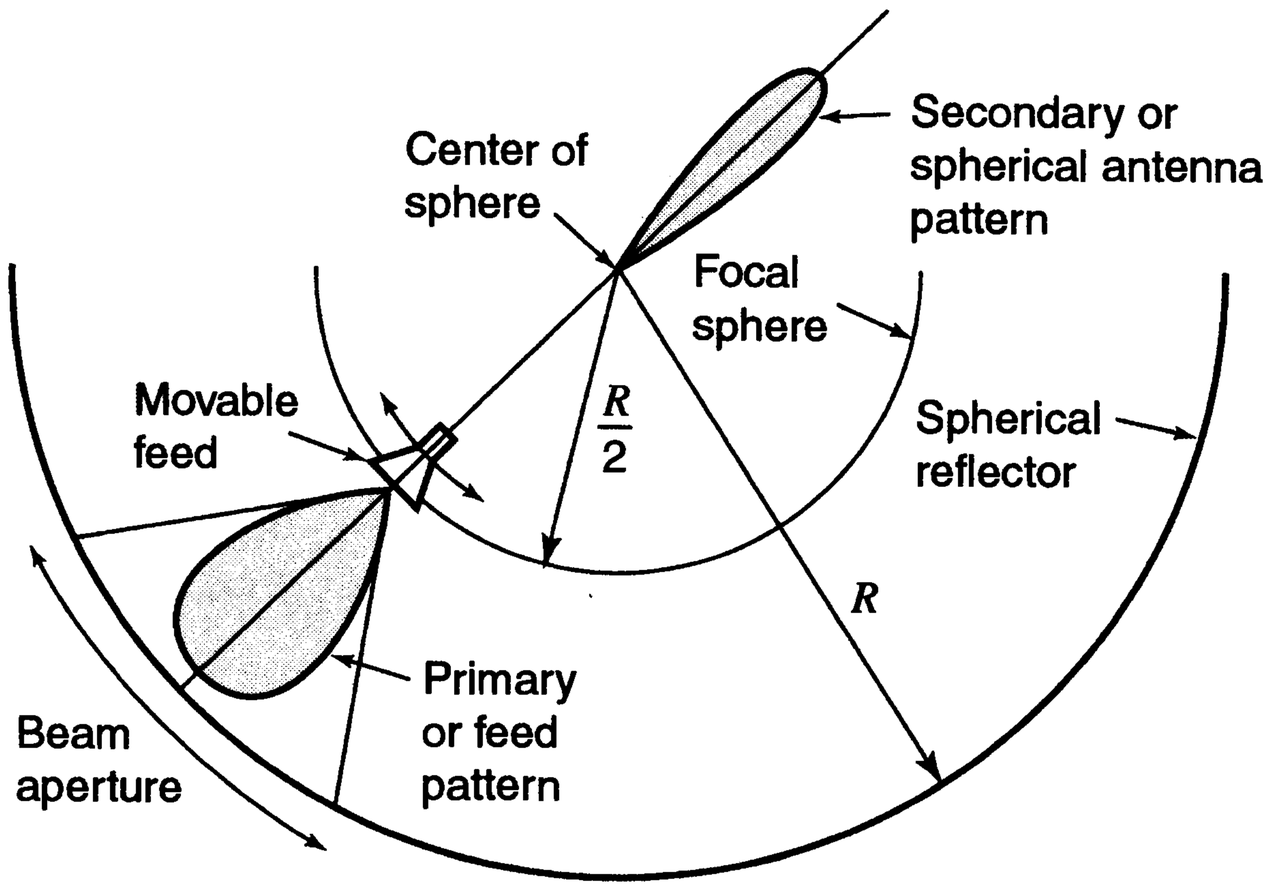


Figure 19–16
Fixed spherical reflector of 2π sr solid angle with movable feed for beam steering over nearly 2π sr.

[1] However, with multiple feeds multiple simultaneous beams could be produced.

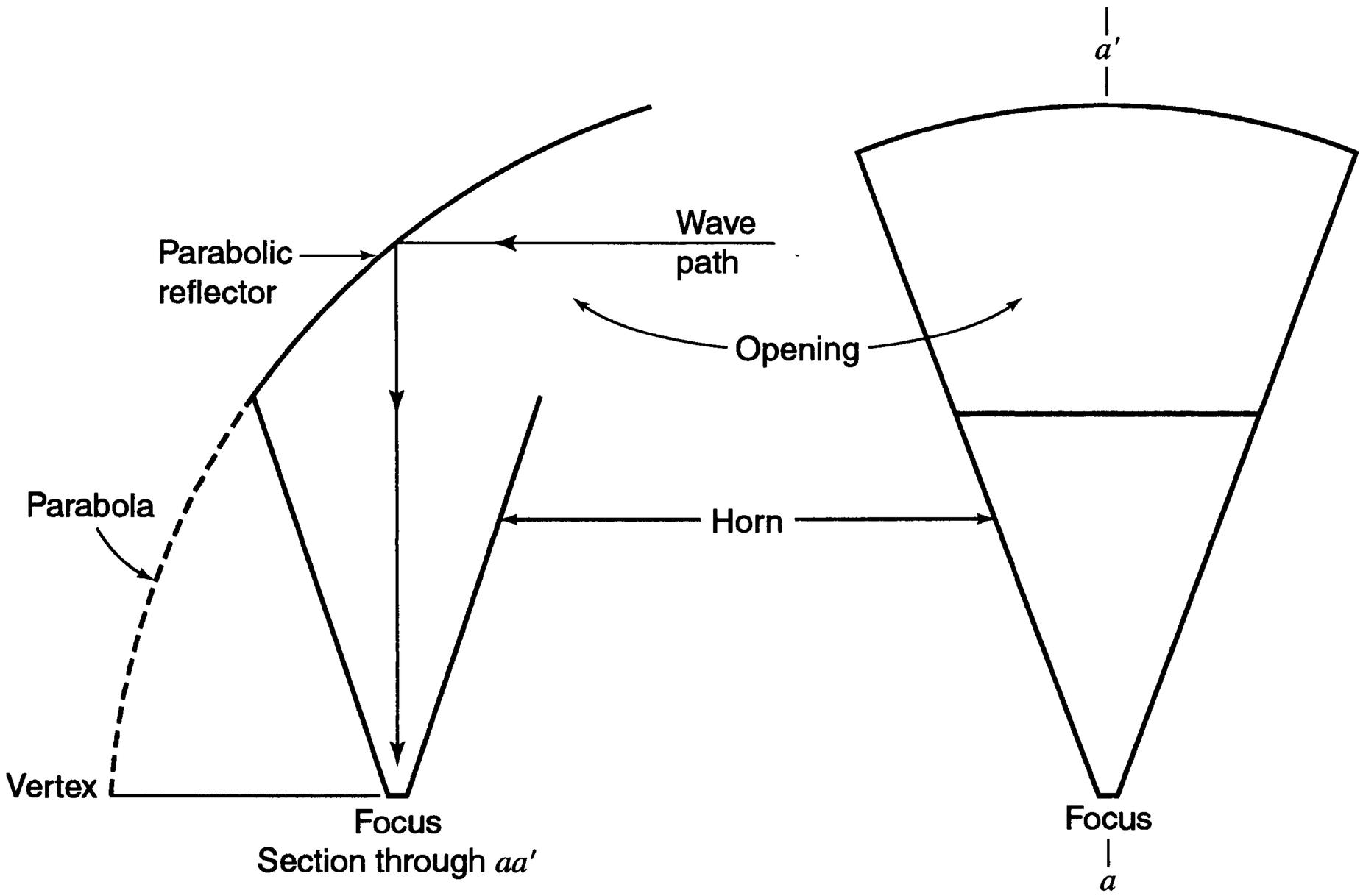


Figure 19–17
"Sugar scoop" low-side-lobe horn-reflector antenna in cross section and front view.

for spherical aberration are discussed by Rumsey (Schell-1; Rumsey-1). See Chap. 20 for discussion of the elaborate feed system of the Arecibo spherical reflector.

The loss in aperture due to feed antenna blockage with attendant scattering and minor lobes can be avoided by the use of an *offset feed* as in Fig. 10–22*c*. Although the asymmetry of the offset feed makes full mechanical steerability more awkward for ground-based antennas, the asymmetry is less of a problem under the weightless conditions of space. However, with the standing-curved-reflector tiltable-flat-reflector (Kraus-type) radio telescope, aperture blockage can be made zero or nearly so (see Chap. 20 and Fig. 20–6). A fully steerable offset-fed paraboloid is also described in Chap. 20 (see Fig. 20–4).

An extreme example of an offset feed is provided by the horn reflector of Fig. 19–17 in which the energy from the feed point is guided inside a horn attached to the reflector. A famous example of this design is the one used by Penzias and Wilson in their discovery of the 3 K cosmic background radiation (see Sec. 12–1). The absence of aperture blocking and the shielding by the horn structure resulted in very low side and back lobes so that the ground contributed negligibly (less than $\frac{3}{100}$ of 1 percent of the nominal 300 K ground temperature) to the antenna temperature.

19–5 LOW-SIDE-LOBE CONSIDERATIONS（低旁瓣的考虑）

Referring to Fig. 19–10, it is to be noted that the side lobes in the forward direction (0 to 90°) of a large parabolic dish reflector are determined (1) by the aperture distribution, (2) by the irregularities of the dish surface, (3) by scattering or diffraction from the feed structure and support struts and (4) by diffraction from the edge of the dish. Effect (3) is absent with

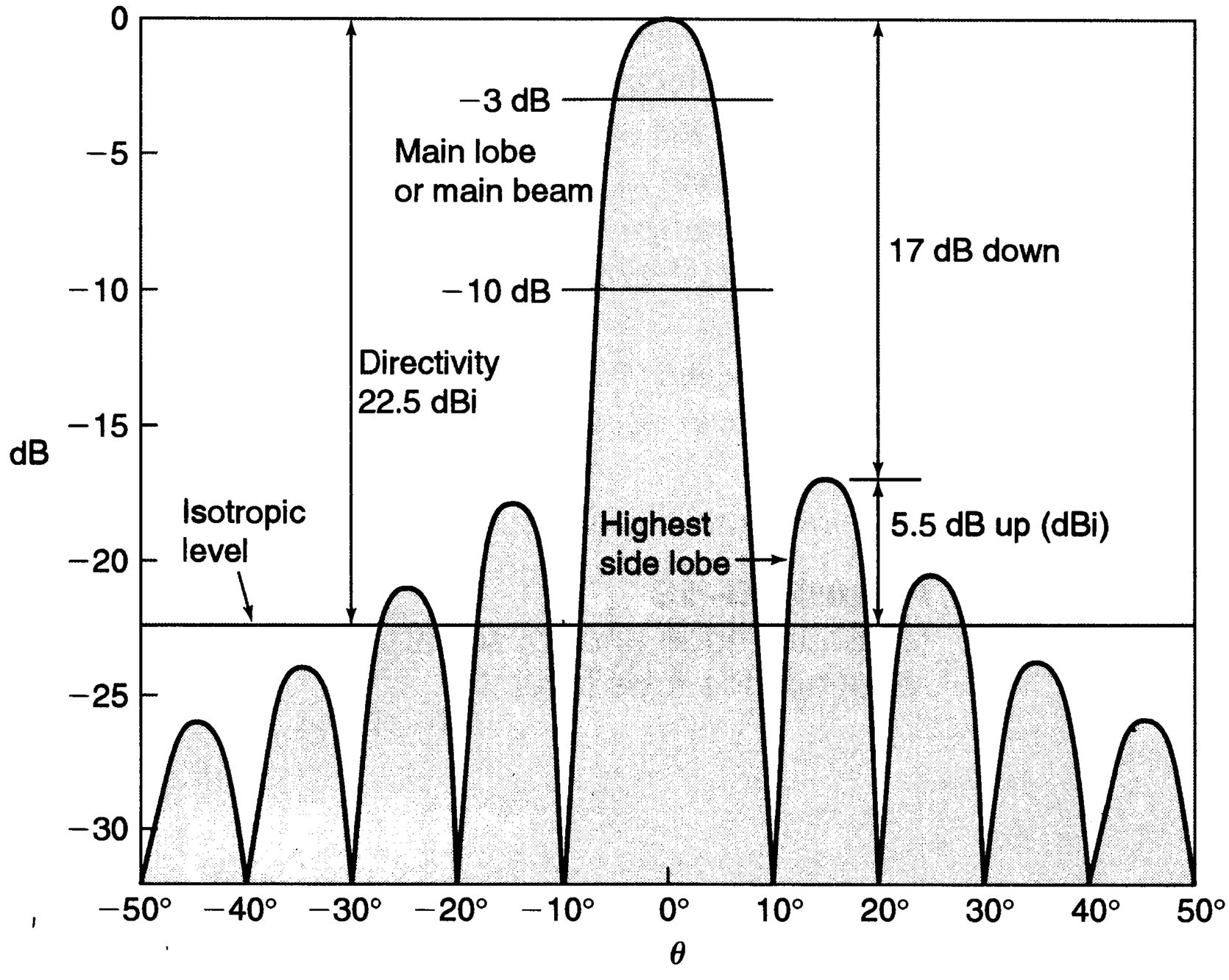


Figure 19–18
Main beam and side lobes with respect to the isotropic level.

offset feeds or with the horn-reflector antenna. The side lobes to the rear (90 to 180°) are determined (1) by the spillover and (2) by diffraction around the edges of the dish. Schrank gives a thorough summary of low-side-lobe reflector antennas (Schrank-1).

For an isotropic source the directivity $D = 1$ (0 dBi) so the *isotropic level* and the relative power level (uniform in all directions) are the same. As D increases, the main lobe rises above the isotropic level in proportion to D. In the hypothetical (but typical) pattern shown in Fig. 19–18, the directivity $D = 22.5$ dBi (= lossless gain) and the highest side lobe is 17 dB *below* (the main-lobe maximum) or 5.5 dB *above* the isotropic level (5.5 dBi). Side-lobe levels are usually referred to the main lobe but sometimes to the isotropic level.

As shown in Fig. 15–2 a triangular or cosine tapered aperture distribution drops to zero field at the edge of the aperture, yet it results in front side lobes. Back side lobes should, in principle, be absent. However, the cosine squared or Gaussian distributions have no side lobes but the HPBWs are greater.

Dish surface irregularities with a regular (periodic) spacing of a wavelength or more are apt to result in side lobes called *grating lobes*.[1] Diffraction from the sharp edge of a dish also contributes to the side lobes both front and back. To randomize the phase of the diffracted rays, the edge may be serrated (saw-tooth edge) as in Fig. 19–19. The tooth dimensions should be of the order of a wavelength or more. The straight-versus-diagonal or sawtooth effect may also be noted with a square dish or ground plane. Thus, as suggested in Fig. 19–20, the side lobes in the plane of the diagonal tend to be less than in the plane of the sides.

[1] Grating lobes are typical for arrays with interelement spacings of λ or more (see Sec. 15–6).

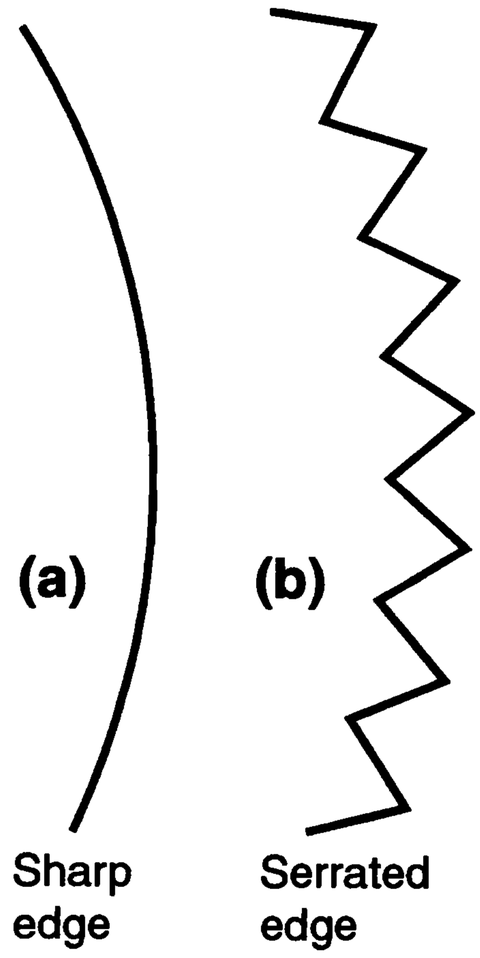


Figure 19–19
(*a*) Sharp edge and (*b*) serrated or sawtooth edge for reduced side-lobe level.

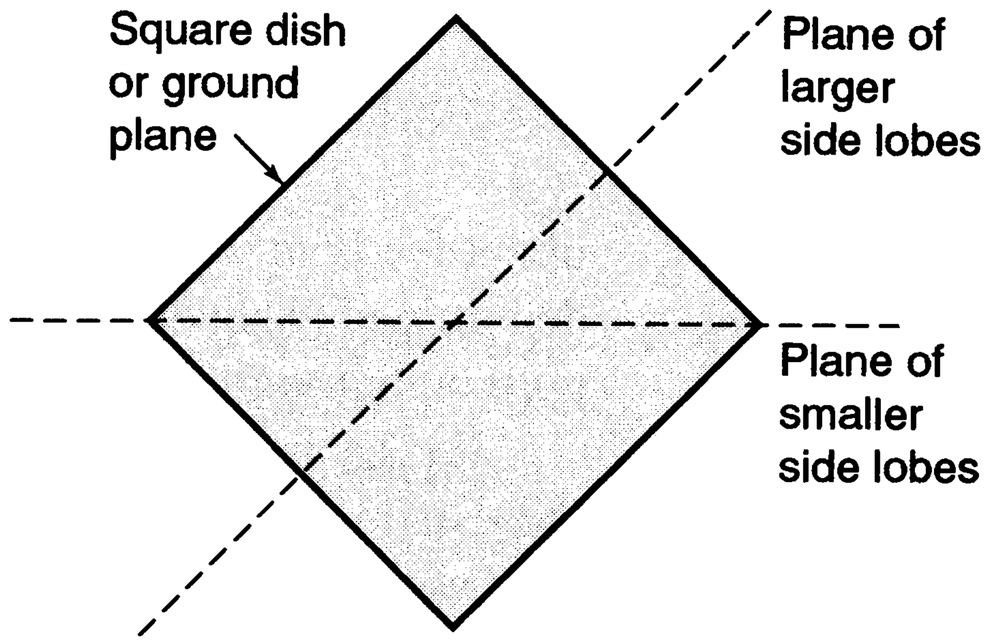


Figure 19–20
Square dish or ground plane with lower side lobes in plane of diagonal.

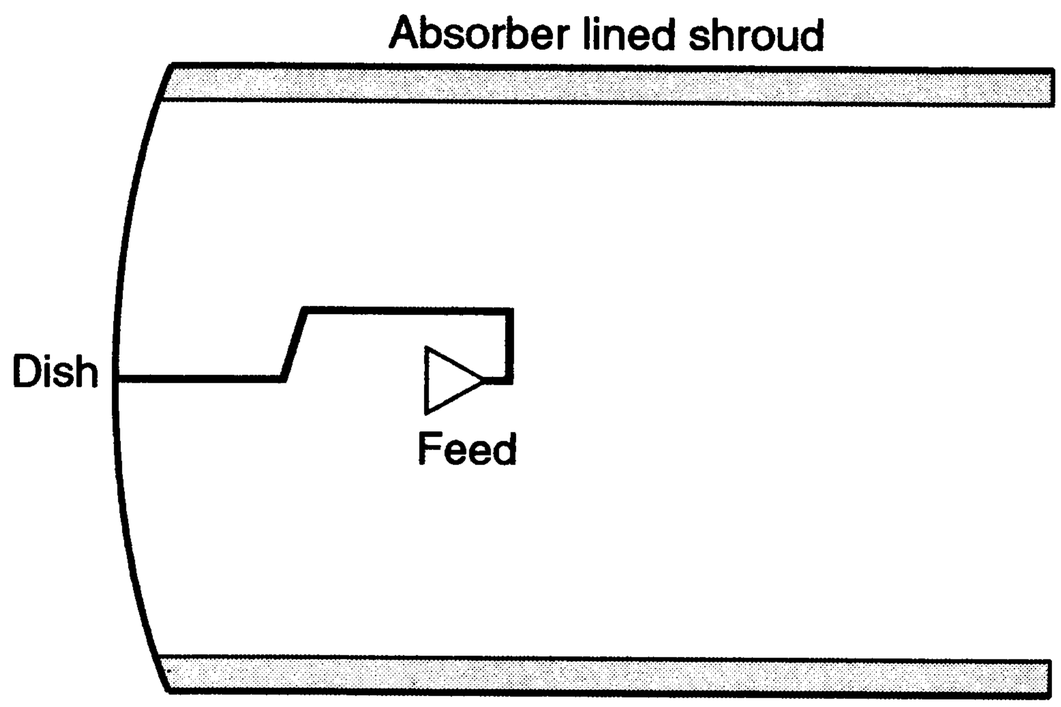


Figure 19–21
Reflector antenna with cylindrical shroud of absorber for reducing far-out side lobes.

Edge diffraction may be reduced by means of a rolled edge with (or without) absorbing material, as illustrated in Fig. 10–6. An oversize parabolic dish with reduced-edge illumination might also be used to reduce edge diffraction but a problem with this approach is that the underilluminated edge and outer regions of the parabola (with its irregularities) still contribute to a diffracted field. However, if the outer region of the dish is curved away from a parabolic contour and blended into a rolled or curved edge, diffraction effects are much reduced (see Sec. 24–4*d*) (Burnside-1).

Although the horn-reflector antenna (Fig. 19–17) has very low wide-angle side lobes, a side lobe tends to appear at $\theta = 90°$, which may be objectionable. Thomas has used a serrated edge (blinder) to reduce it (Thomas-1).

Another side and back lobe suppression technique involves the addition of a cylindrical absorbing shroud attached to the edge of the dish as in Fig. 19–21. The outer surface of the shroud may be metal or dielectric. Dybdal (1) and King found that with a shroud twice as long as the dish diameter the far-out side lobes were 60 to 75 dB down, although the on-axis back lobe ($\theta = 180°$) was only 50 dB down.

REFERENCES（参考文献）

Ashmead, J. (1), and A. B. Pippard, "The Use of Spherical Reflectors as Microwave Scanning Aerials," *JIEE,* **93,** 627–632, 1946.

Baker, D. E. (1), "The Focal Regions of Paraboloidal Reflectors of Arbitrary *F/D* Ratio," Ph.D. thesis, Ohio State University, 1974.

Baker, D. E. (2), personal communication, 1977.

Bracewell, R. N. (1), "Tolerance Theory of Large Antennas," *IRE Trans. Ants. Prop.,* **AP-9,** 49–58, January 1961.

Burnside, W. D. (1), M. C. Gilreath, B. M. Kent and G. C. Clerici, "Curved Edge Modification of Compact Range Reflector," *IEEE Trans. Ants. Prop.,* **AP-35,** 176–182, February 1987.

Christiansen, W. N. (1), and J. A. Hogbom, *Radiotelescopes,* Cambridge University Press, 1985 (gives an excellent, lucid discussion of the effect of surface irregularities on gain which served as an inspiration in preparing parts of this section).

Dybdal, R. B. (1), and H. E. King, "Performance of Reflector Antennas with Absorber-Lined Tunnels," *IEEE Symp. Ants. Prop.,* Seattle, June 1979.

Lo, Y. T. (1), "On the Beam Deviation Factor of a Paraboloidal Reflector," *IRE Trans. Ants. Prop.,* **AP-8,** 347, May 1960.

Love, A. W. (1), "Spherical Reflecting Antennas with Corrected Line Sources," *IRE Trans. Ants. Prop.,* **AP-10,** 529–537, September 1962.

Nash, R. T. (1), "Beam Efficiency Limitations of Large Antennas," *IEEE Trans. Ants. Prop.,* **AP-12,** 918–923, December 1964.

Nash, R. T. (2), "A Multi-reflector Meridian Transit Radio Telescope Antenna for the Observation of Waves of Extra-terrestrial Origin," Ph.D. thesis, Ohio State University, 1961.

Rhodes, D. R. (1), "On a New Condition for Physical Realizability of Planar Antennas," *IEEE Trans. Ants. Prop.,* **AP-19,** 162–166, March 1971.

Rudge, A. W. (1), and M. J. Withers, "New Technique for Beam Steering with Fixed Parabolic Reflectors," *Proc. IEE,* **118,** 857, July 1971.

Rumsey, V. H. (1), "On the Design and Performance of Feeds for Correcting Spherical Aberration," *IEEE Trans. Ants. Prop.,* **AP-18,** 343–351, May 1970.

Rusch, W. V. T. (1), and A. C. Ludwig, "Determination of the Maximum Scan-Gain Contours of a Beam-Scanning Paraboloid and Their Relation to the Petzval Surface," *IEEE Trans. Ants. Prop.*, **AP-21,** 141, March 1973.

Ruze, J. (1), "Physical Limitations of Antennas," *MIT Res. Lab. Electron. Tech. Rept.*, 248, 1952.

Sandler, S. S. (1), "Paraboloidal Reflector Patterns for Off-Axis Feed," *IRE Trans. Ants. Prop.*, **AP-8,** 368, July 1960.

Schell, A. C. (1), "The Diffraction Theory of Large Aperture Spherical Reflector Antennas," *IEEE Trans. Ants. Prop.*, **AP-11,** 428–432, July 1963.

Schrank, H. E. (1), "Low Sidelobe Reflector Antennas," *IEEE Ant. Prop. Soc. Newsletter,* **27,** 5–16, April 1985.

Spencer, R. C. (1), C. J. Sletten and J. E. Walsh, "Correction of Spherical Aberration by a Phased Line Source," *Proc. Natl. Elect. Conf.*, **5,** 320, 1949.

Thomas, D. T. (1), "Design of Multiple-Edge Blinders for Large Horn Reflector Antennas," *IEEE Trans. Ants. Prop.*, **AP-21,** 153–158, March 1973.

von Gniss, H. (1), and G. Ries, "Feld Bild um den Brennpunkt von Parabolreflektoren mit Kleinen *F/D* Verhaltnis," *Archiv der Elek. Ubertrag.*, **23,** 481, October 1969.

Von Hoerner, S. (1), "The Design and Improvement of Tiltable Radio Telescopes," *Vistas in Astron.*, **20,** pt. 4, 411–444, 1977.

ADDITIONAL REFERENCES FOR FURTHER READING（深入阅读的补充参考资料）

Bach, H., and H-H. Viskum, "The SNFGTD (Spherical Near-Field Geometrical Theory of Diffraction) Method and Its Accuracy," *IEEE Trans. Ants. Prop.*, **AP-35,** 169–175, February 1987.

Bennet, J. C., A. P. Anderson, P. A. McInnes and A. J. T. Whitaker, "Microwave Holographic Metrology of Large Reflector Antennas," *IEEE Trans. Ants. Prop.*, **AP-24,** 295–303, 1976.

Burnside, W. D., M. C. Gilreath and B. M. Kent, "Rolled Edge Modification of Compact Range Reflector," *AMTA Symp.*, San Diego, 1984.

Chu, T-S., and R. H. Turrin, "Depolarization Properties of Offset Reflector Antennas," *IEEE Trans. Ants. Prop.*, **AP-21,** 339–345, May 1973.

Chu, T-S., "An Imaging Beam Waveguide Feed," *IEEE Trans. Ants. Prop.*, **AP-31,** 614–619, July 1983.

Chuang, C. W., and W. D. Burnside, "A Diffraction Coefficient for a Cylindrically Truncated Planar Surface (Rolled Edge Effect)," *IEEE Trans. Ants. Prop.*, **AP-28,** 177–182, March 1980.

Crawford, A. B., D. C. Hogg and L. E. Hunt, "A Horn-Reflector Antenna for Space Communication," *BSTJ,* **40,** 1045–1116, July 1961.

Cross, D. C., D. D. Howard and J. W. Titus, "Mirror-Antenna Radar Concept," *Microwave J.*, **29,** 323–335, May 1986.

Dragone, C., "Offset Multireflector Antennas with Perfect Pattern Symmetry and Polarization Discrimination," *BSTJ,* **57,** 2663–2684, September 1978.

Ehrenspeck, H. W., "The Short Backfire Antenna," *Proc. IEEE,* **53,** 1138–1140, August 1965.

Godwin, M. P., A. J. T. Whitaker and A. P. Anderson, "Microwave Diagnostics of the Chilbolton 25 m Antenna Using the OTS Satellite," *Proc. Inst. Elec. Eng. Int. Conf.,* York, England, pp. 232–236, 1981.

James, G. L., *Geometrical Theory of Diffraction for Electromagnetic Waves,* Peregrinus, 1980.

Kaplan, P. D., "Predicting Antenna Sidelobe Performance (Sidelobe Pop-Up Probability)," *Microwave J.*, **29,** 201–206, September 1986.

Keller, J. B., "Geometrical Theory of Diffraction," *J. Opt. Soc. Am.*, **52,** 116–130, February 1962.

Kieburtz, R. B., and A. Ishimaru, "Scattering by a Periodically Apertured Conducting Screen," *IRE Trans. Ants. Prop.*, **AP-9,** 506–514, November 1961.

Kieburtz, R. B., and A. Ishimaru, "Aperture Fields of an Array of Rectangular Apertures," *IRE Trans. Ants. Prop.*, **AP-10,** 663–671, November 1962.

Koffman, I., "Feed Polarization for Parallel Currents in Reflectors Generated by Conic Sections," *IEEE Trans. Ants. Prop.*, **AP-14,** 37–40, January 1966.

Kouyoumjian, R. G., and P. H. Pathak, "A Uniform Geometrical Theory of Diffraction for an Edge in a Perfectly Conducting Surface," *Proc. IEEE,* **62,** 1448–1461, November 1974.

Lamb, J. W., "Analysis of a Cassegrain Antenna Having a Gaussian Illumination Pattern," Helsinki Univ. Tech. Rept. S147, 1983.

Ludwig. A. C., "Low Sidelobe Aperture Distribution for Blocked and Unblocked Circular Apertures," *IEEE Trans. Ants. Prop.*, **30,** 933–946, September 1982.

Mayer, C. E., J. H. Davis, W. L. Peters and W. J. Vogel, "A Holographic Surface Measurement of the Texas 4.9-meter Antenna at 86 GHz," *IEEE Trans. Instrum. Meas.*, **IM-32,** 102–109, 1983.

Nash, R. T., "Stepped Amplitude Distributions," *IEEE Trans. Ants. Prop.*, **AP-12,** July 1964.

Ott, R. H., and J. H. Richmond, "A Linear Equation Solution for Scattering by a Periodic Plane Array of Thin Conducting Plates," ElectroSci. Lab. Rept. 1774-9, 1965.

Ott, R. H., R. G. Kouyoumjian and L. Peters, Jr., "The Scattering by a Two-Dimensional Periodic Narrow Array of Plates," *Radio Sci.,* November 1967.

Peters, L., Jr., and C. E. Ryan, Jr., "Empirical Formulas for E-Plane Creeping Waves on General Smooth Conducting Bodies," *IEEE Trans. Ants. Prop,* **AP-18,** 432–434, May 1970.

Rahmat-Samii, Y., "Surface Diagnosis of Large Reflector Antennas Using Microwave Holographic Metrology—An Iterative Approach," *Radio Sci.* **13,** 1205–1217, September–October 1984.

Rahmat-Samii, Y., "Microwave Holography of Large Reflector Antennas: Simulation Algorithms," *IEEE Trans. Ants. Prop.,* **AP-33,** 1194–1203, November 1985.

Ratnasiri, P. A. J., R. G. Kouyoumjian and P. H. Pathak, "The Wide Angle Side Lobes of Reflector Antennas," OSU Elect. Sci. Lab. Rept. 5635-4, 1970.

Rudge, A. W., and N. A. Adatia, "Offset-Parabolic-Antennas: A Review," *Proc. IEEE,* **66,** 1592–1618, December 1978.

Rudge, A. W., and M. J. Withers, "Design of Flared-Horn Primary Feeds for Parabolic Reflector Antennas," *Proc. IEE,* **117,** 1741–1747, September 1970.

Rudge, A. W., and M. J. Withers, "New Technique for Beam Steering with Fixed Parabolic Reflectors," *Proc. IEE,* **118,** 857–863, July 1971.

Rusch, W. V. T ., Y. Rahmat-Samii and R. A. Shore, "The Equivalent Paraboloid of an Optimized Off-Set Cassegrain Antenna," *Dig. IEEE Int. Symp. Ants. Prop.,* 1986.

Scott, P. F., and M. Ryle, "A Rapid Method for Measuring the Figure of a Radio Telescope Reflector," *Royal Ast. Soc. Mon. Notices,* **178,** 539–545, 1977.

Shore, R. A., "A Simple Derivation of the Basic Design Equation for Offset Dual Reflector Antennas with Rotational Symmetry and Zero Cross Polarization," *IEEE Trans. Ants. Prop.*, **AP-33,** 114–116, January 1985.

Sletten, C. J., W. Payne and W. Shillue, "Offset Dual Reflector Antennas for Very Low Sidelobes," *Microwave J.*, **29,** 221–240, May 1986.

Tanaka, H., and M. Mizuasawa, "Elimination of Cross-Polarization in Offset Dual Reflector Antennas," *Elec. Commun. (Japan),* **58-B,** 71–78, 1975.

Twersky, V., "Multiple Scattering of Waves and Optical Phenomena," *J. Opt. Soc. Am.*, **52,** 145–171, 1962.

PROBLEMS（习题）

19–1–1（柱形抛物面反射镜） **Cylindrical parabolic reflector.** Calculate the radiation pattern of a cylindrical parabolic reflector of square aperture 16λ on a side when the illumination is uniform over the aperture and when the field intensity across the aperture follows a cosine variation with maximum intensity at the center and zero intensity at the edges. Compare the two cases by plotting the normalized curves on the same graph.

19–1–2 (抛物面反射镜的口径分布) **Paraboloidal reflector aperture distribution.** (*a*) Show that the variation of field across the aperture of paraboloidal reflector with an isotropic source is proportional to $1/[1+(\rho/2L)^2]$ where ρ is the radial distance from the axis of the paraboloid. Show that this relation is equivalent to $(1+\cos\theta)/2$. (*b*) If the parabola extends to the focal plane and the feed is isotropic over the hemisphere subtended by the parabola, calculate the aperture efficiency.

***19–1–3** (具有部分锥削分布矩形口径的效率) **Efficiency of rectangular aperture with partial taper.** Calculate the aperture efficiency and directivity of an antenna with rectangular aperture $x_1 y_1$ with a uniform field distribution in the y direction and a cosine field distribution in the x direction (zero at edges, maximum at center) if $x_1 = 20\lambda$ and $y_1 = 10\lambda$.

***19–1–4** (具有完全锥削分布矩形口径的效率) **Efficiency of rectangular aperture with full taper.** Repeat Prob. 19–1–3 for the case where the aperture field has a cosine distribution in both the x and y directions.

19–1–5 (具有相位起伏分布口径的效率) **Efficiency of aperture with phase ripple.** A square unidirectional aperture ($x_1 y_1$) is 10λ on a side and has a design distribution for the electric field which is uniform in the x direction but triangular in the y direction with maximum at the center and zero at the edges. Design phase is constant across the aperture. However, in the actual aperture distribution there is a plus-and-minus-30° sinusoidal phase variation in the x direction with a phase cycle per wavelength. Calculate (*a*) the design directivity, (*b*) the utilization factor, (*c*) the actual directivity, (*d*) the achievement factor, (*e*) the effective aperture and (*f*) the aperture efficiency.

***19–1–6** (单边余弦锥削的矩形口径) **Rectangular aperture. Cosine taper.** An antenna with rectangular aperture $x_1 y_1$ has a uniform field in the y direction and a cosine field distribution in the x direction (zero at edges, maximum at center). If $x_1 = 16\lambda$ and $y_1 = 8\lambda$, calculate (*a*) the aperture efficiency and (*b*) the directivity.

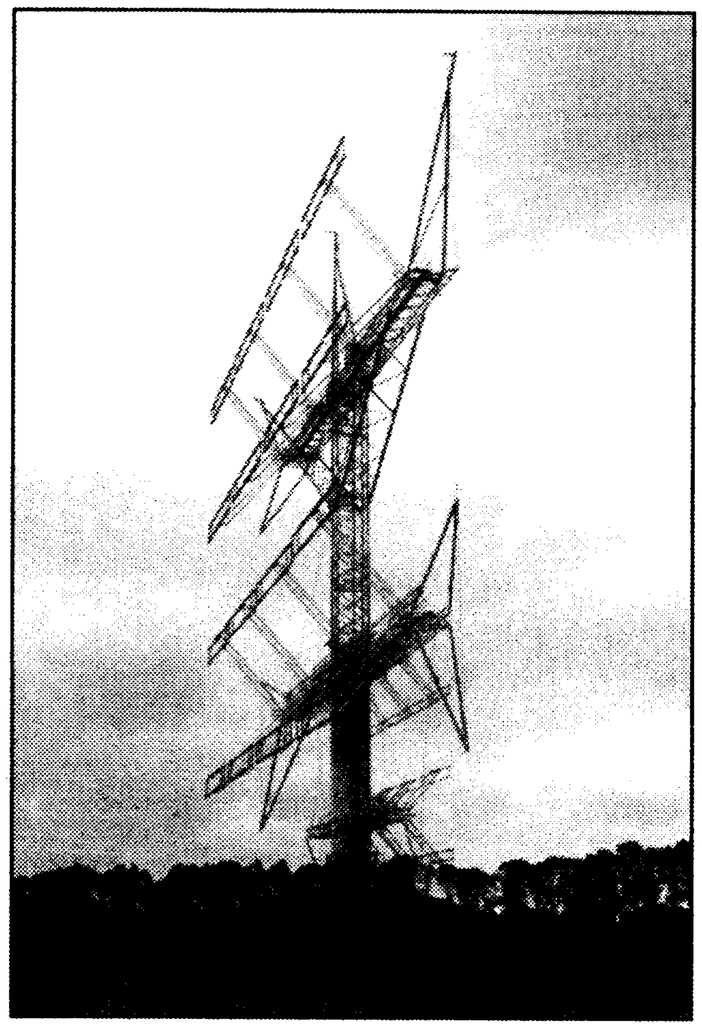
19–1–7 (双边余弦锥削的矩形口径) **Rectangular aperture. Cosine tapers.** Repeat Prob. 19–1–6 for the case where the aperture field has a cosine distribution in both the x and y directions.

***19–1–8** (余弦平方锥削的 20λ 线源) **A 20λ line source. Cosine-squared taper.** (*a*) Calculate and plot the far-field pattern of a continuous in-phase line source 20λ long with cosine-squared field distribution. (*b*) What is the HPBW?

For computer programs, see Appendix C.

CHAPTER 20

Some Examples of Large or Unique Antennas
大型或独特的天线实例



Courtesy of Deutsche Welle

Examples include antennas at:

- Bonn（波恩）
- Arecibo（阿利西波）
- Bell Laboratories（贝尔实验室）
- Nobeyama（野边山）
- Ohio（俄亥俄）
- Gorki（高尔基）
- Five College（五大学）
- Nancay（纳西）
- RATAN-600（拉坦 -600）
- Parkes（帕克斯）
- Manchester（曼彻斯特）
- VLA（甚大口径阵）
- Nauen（诺恩）
- Green Bank（绿滩）

Bonn. The telescope is shown in Fig. 20–1. Standing 40 stories tall in a valley of the Eifel mountains near Bonn, Germany, the 3200-tonne 100-m-diameter parabolic reflector antenna is used for radio astronomy and space research. The dish can be elevated 20° per minute and rotated in azimuth on a circular track 40° per minute. It can operate at wavelengths as short as 1 cm. Feeds may be either prime focus or Cassegrain. See the gain-versus-wavelength curve in Fig. 19–9.

Arecibo. The fixed 305-m-diameter spherical reflector of the National Astronomy and Ionosphere Center at Arecibo, Puerto Rico, is shown in elevation cross section in Fig. 20–2 and from the air in Fig. 20–3.

Designed originally for ionospheric studies, it was adapted for radio astronomy observations. The slotted waveguide feed for longer wavelengths is now supplemented by a Gregorian feed for shorter wavelengths. The Gregorian secondary and tertiary reflectors are housed in a dome. Aperture efficiency with the line feed at the zenith is about 50% at 430 MHz. Both line feed and Gregorian feed can swing through a 20° angle from the zenith allowing beam scanning over a solid angle around the zenith of 6% of the sky above the horizon but with reduced aperture efficiency.

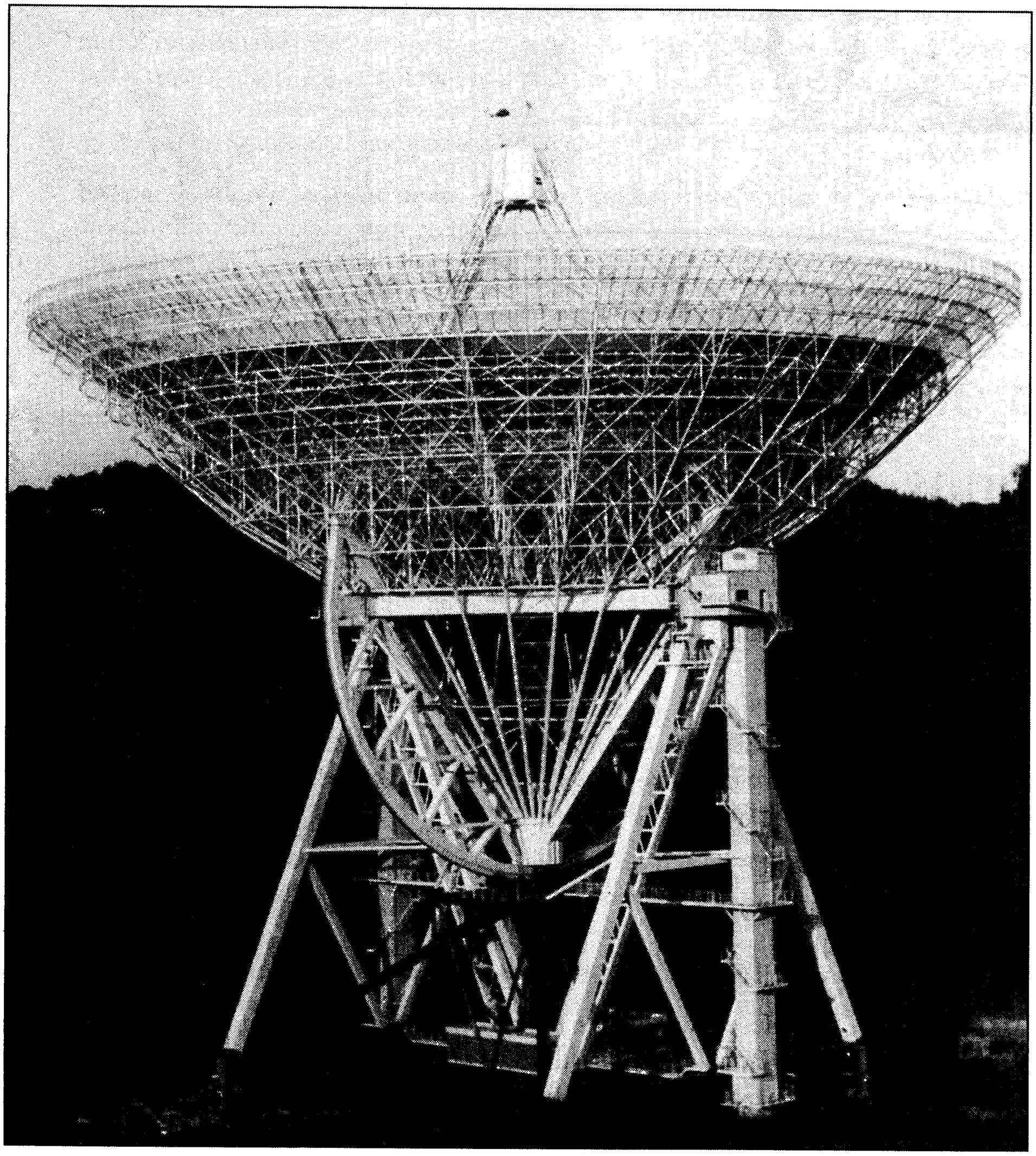


Figure 20–1
Fully steerable 100-m-diameter antenna near Bonn, West Germany, standing nearly 40 stories tall. The moving parts weigh 3200 tonnes. Note the massive backup structure required to provide rigidity to the dish. (*Courtesy Dr. R. Wielebinski, Max Planck Institute for Radioastronomy.*)

At 1.5 GHz at the zenith the effective aperture is 28,000 m^2 for an aperture efficiency of 39%. At 20° from the zenith the aperture efficiency drops to 31%. After upgrading of the reflector, operation is expected to extend to 8 GHz with an aperture efficiency of 20% at the zenith. In many situations only part of the dish is used. Taking this smaller value as the physical aperture gives higher values of aperture efficiency of 50 to 70% instead of 20 to 40% (Davis-1).

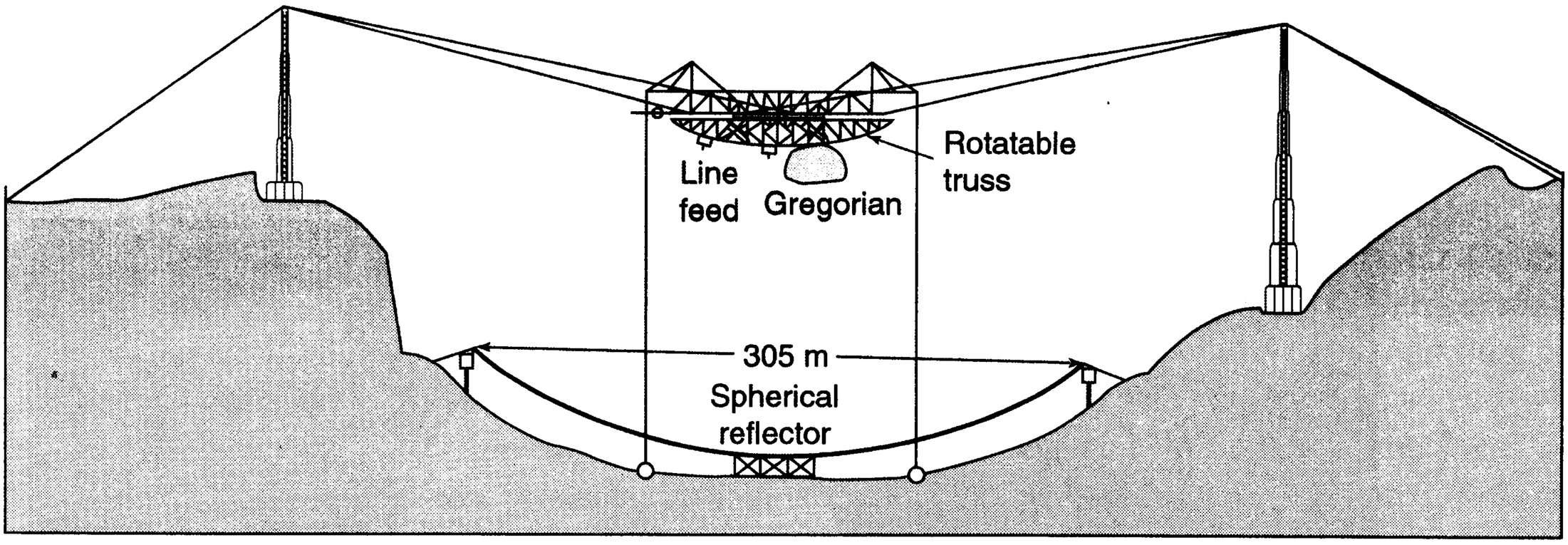


Figure 20–2
Elevation cross section of the 305-m-diameter fixed spherical reflector situated in a mountain valley of Puerto Rico.

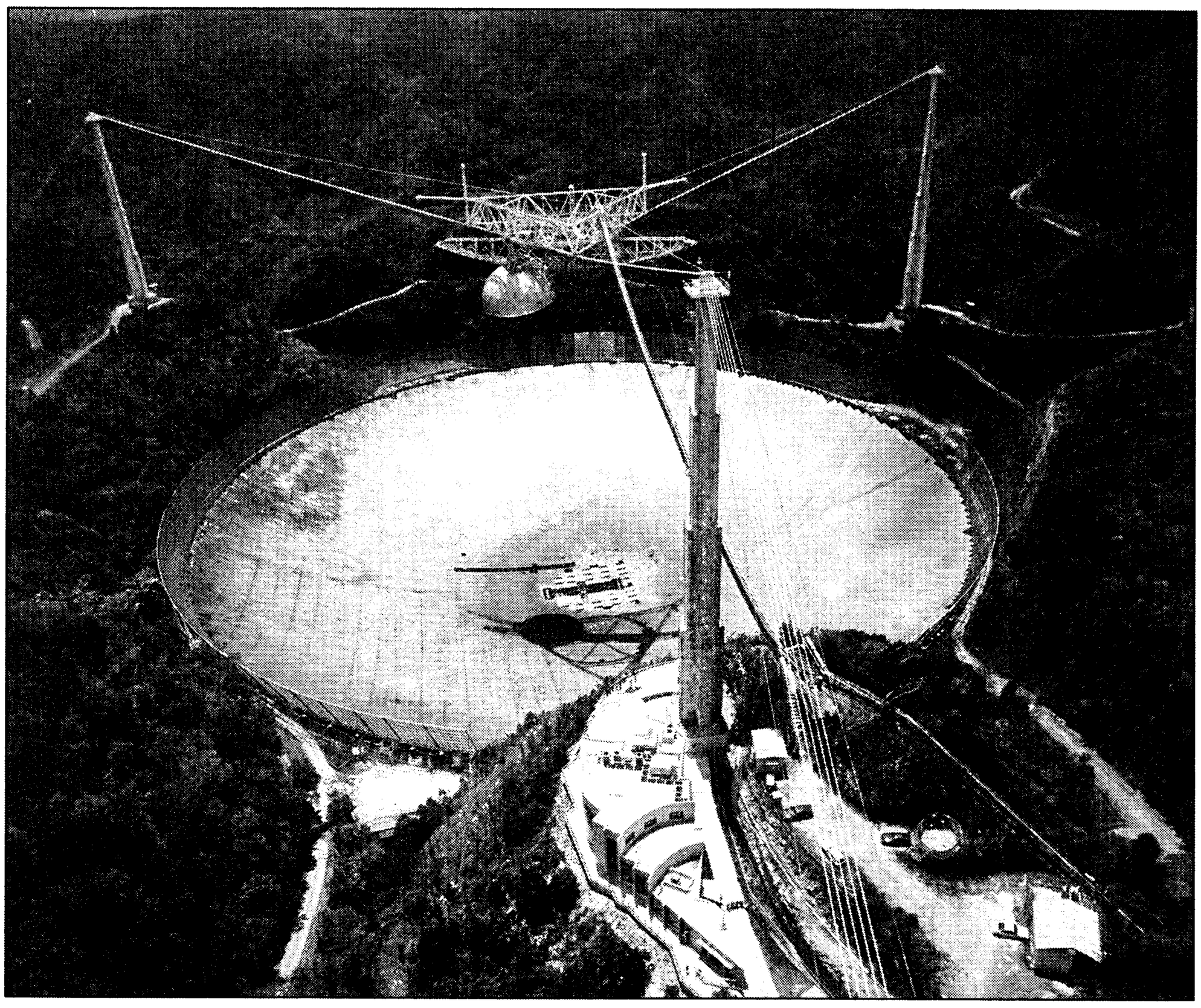


Figure 20–3
Air view of 305-m Arecibo dish with control and administrative buildings in the foreground. The feed structure is supported by cables from 3 towers. Beam steering by moving the feeds allows observations up to 20° from the zenith. Away from the zenith, only part of the reflector can be used resulting in a decrease in aperture efficiency. (*Photo by David Parker. The Arecibo Observatory of the NAIC is operated by Cornell University under a cooperative agreement with the National Science Foundation.*)

Bell Telephone Laboratories. The 7-m-diameter parabolic reflector with offset feed of the Bell Telephone Laboratories is shown in Fig. 20–4. This antenna, located at Crawford Hill, New Jersey, has a 100-μm rms surface accuracy permitting operation at wavelengths as short as 3 mm (100 GHz) with 59 percent aperture efficiency, a gain of 77 dBi and HPBW

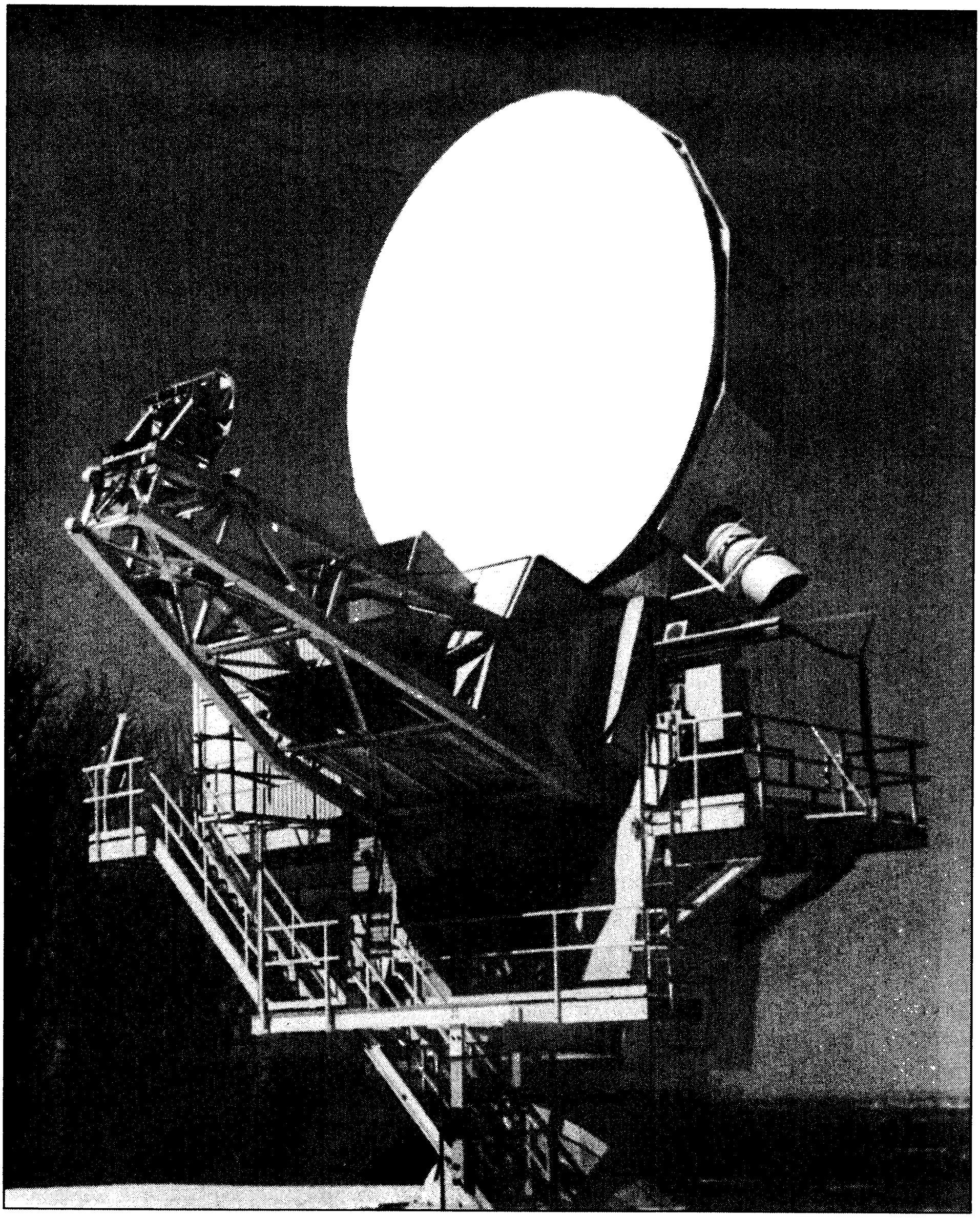


Figure 20–4
The 7-m offset Cassegrain-fed low-side-lobe millimeter-wave antenna of the Bell Telephone Laboratories at Crawford Hill, New Jersey. Key structural parts are covered with insulation to reduce thermal effects. (*Courtesy Dr. T-S. Chu, Bell Telephone Laboratories.*)

of 2 arcminutes (Chu-1). At 1 cm (30 GHz) the aperture efficiency is 69 percent and the gain 66.5 dBi. With a 15-dB feed taper and no aperture blockage the side-lobe level is over 40 dB down 1° off-axis at 30 GHz. The antenna also has very low cross-polarization (over 40 dB down across the main beam). These low levels are achieved using a quasi-optical feed system and conical corrugated horn with hybrid-mode launcher (see Fig. 9–35).

Whereas the atmosphere contributes a temperature of only 2.3 K at 4 GHz (at the zenith), the dry atmosphere temperature at 100 GHz is about 100 K so that the temperature contribution due to spillover (12 K) and mylar waveguide window loss (1 K) add little of significance to the system temperature at 100 GHz (Chu-2).

In radio astronomy observations, receiver noise reflected back into prime focus or Cassegrain feeds can produce spurious spectral lines. This effect is eliminated with an offset feed as in the Crawford Hill antenna.

Nobeyama. The world's largest millimeter-wave dish, shown in Fig. 20–5, is located at the Nobeyama Radio Observatory at a high elevation in the central mountains of Honshu

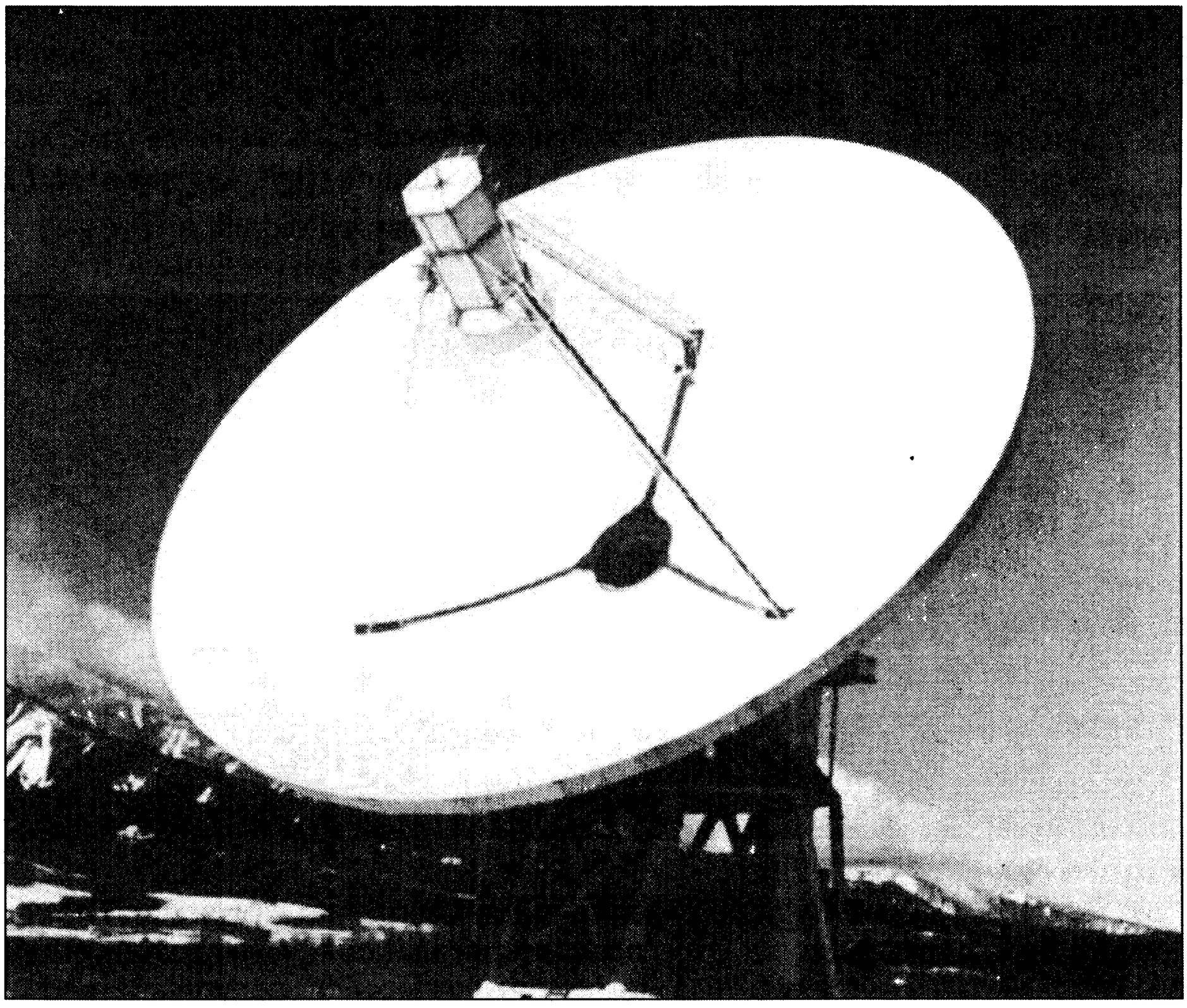


Figure 20–5
The 45-m fully steerable parabolic reflector antenna with gain of nearly 90 dBi at wavelengths of 2 to 3 mm at the Nobeyama Radio Observatory in the Yatsugatake Mountains, Japan. (*NRO photo.*)

Island, Japan. It has a 200-μm rms overall equivalent surface deviation and a pointing accuracy of 2 arcseconds (Akabane-1). Maximum gain is almost 90 dBi at 2- to 3-mm wavelengths. See the gain curve in Fig. 19–9. Thermal effects are minimized by covering the dish structure with insulated panels and circulating air inside the enclosure. Each of the 600 rigid panels of the dish has a surface accuracy of 60 μm rms.

Ohio. The fixed-standing-curved reflector with tiltable-flat reflector, dubbed "Big Ear," of the Ohio State University Radio Observatory is shown in Fig. 20–6. The curved-standing reflector is a rectangular section of a sphere with dimensions of 110 by 21 m. The tiltable flat reflector dimensions are 104 by 31 m (Fig. 20–7). Both reflectors are joined by a flat aluminum ground plane (Kraus-1, 6).

The basic design consideration of the antenna was that of obtaining the maximum aperture per unit of cost. It also had other advantages such as less than $\frac{1}{2}$ of 1 percent effective aperture blocking, reduced susceptibility to terrestrial interference because of the low profile of the feed (right at ground level) and shielding by the large reflectors, and a spacious underground receiver laboratory directly below the prime focus where weight restrictions were not a consideration.

Tilting the flat reflector allows observations between declinations of -36 and $+64°$ (a range of 100°) which gave a coverage of 90 percent of the sky observable from the site. Movement of the feed car permitted beam steering of 15° in azimuth (or tracking of sources for an hour or more in right ascension). The telescope (see Fig. 20–8*a*) has a very long focal length with $F/D = 1.17$ in azimuth (right ascension) and $F/D = 6.0$ in

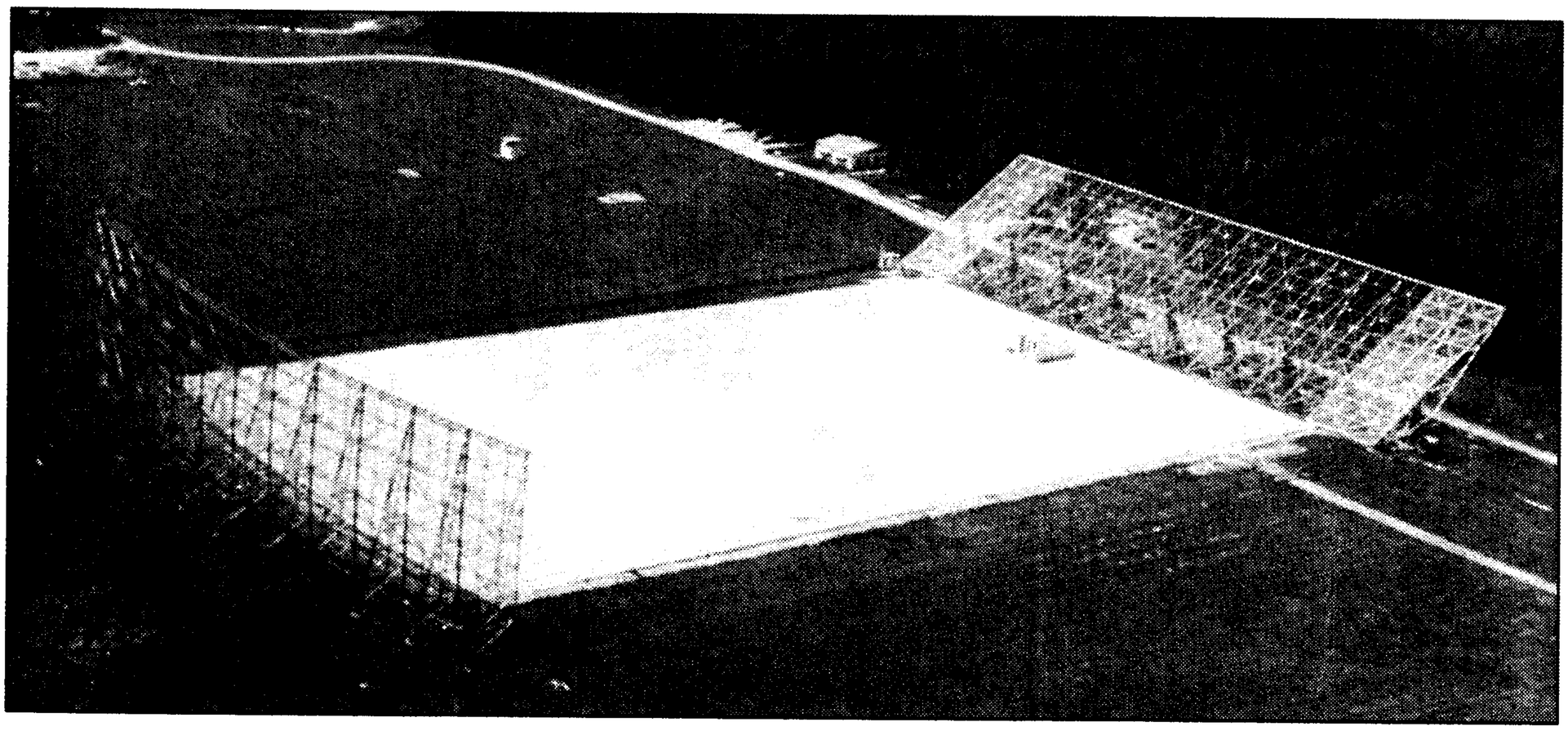


Figure 20–6
"Big Ear," the 110-m telescope antenna at the Ohio State University. The low-profile design features a large aperture per unit cost, negligible aperture blocking, reduced susceptibility to terrestrial interference, convenient feed location, long focal length and extended tracking capability. This telescope was used for the Ohio Sky Survey, in which 20,000 radio sources were cataloged and mapped at 1415 MHz and many unique sources discovered, including the most distant known objects in the universe. (*Tom Root photo.*)

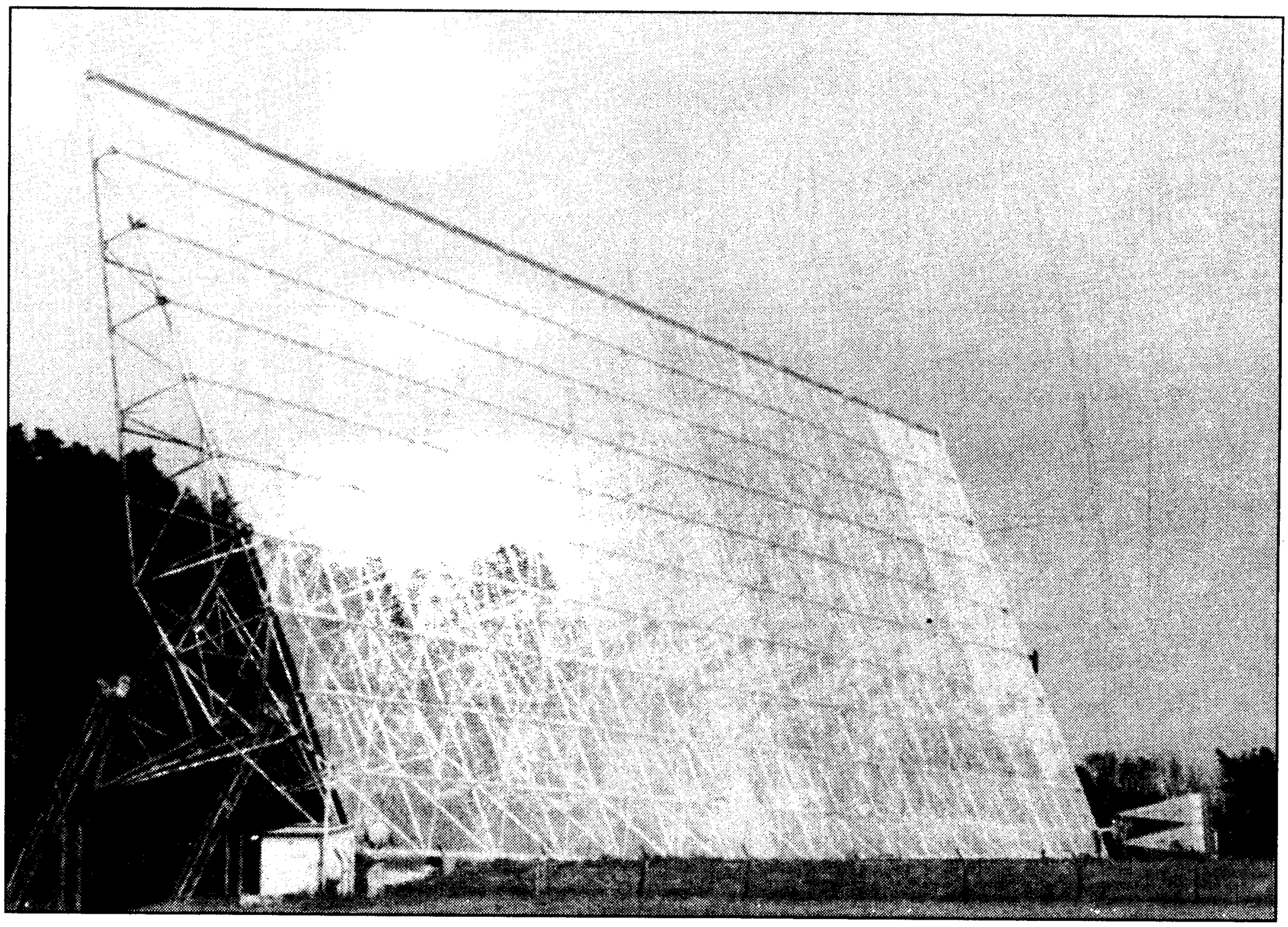


Figure 20–7
The tiltable flat reflector of Big Ear. For scale, note the man halfway up the ladder on the far side.

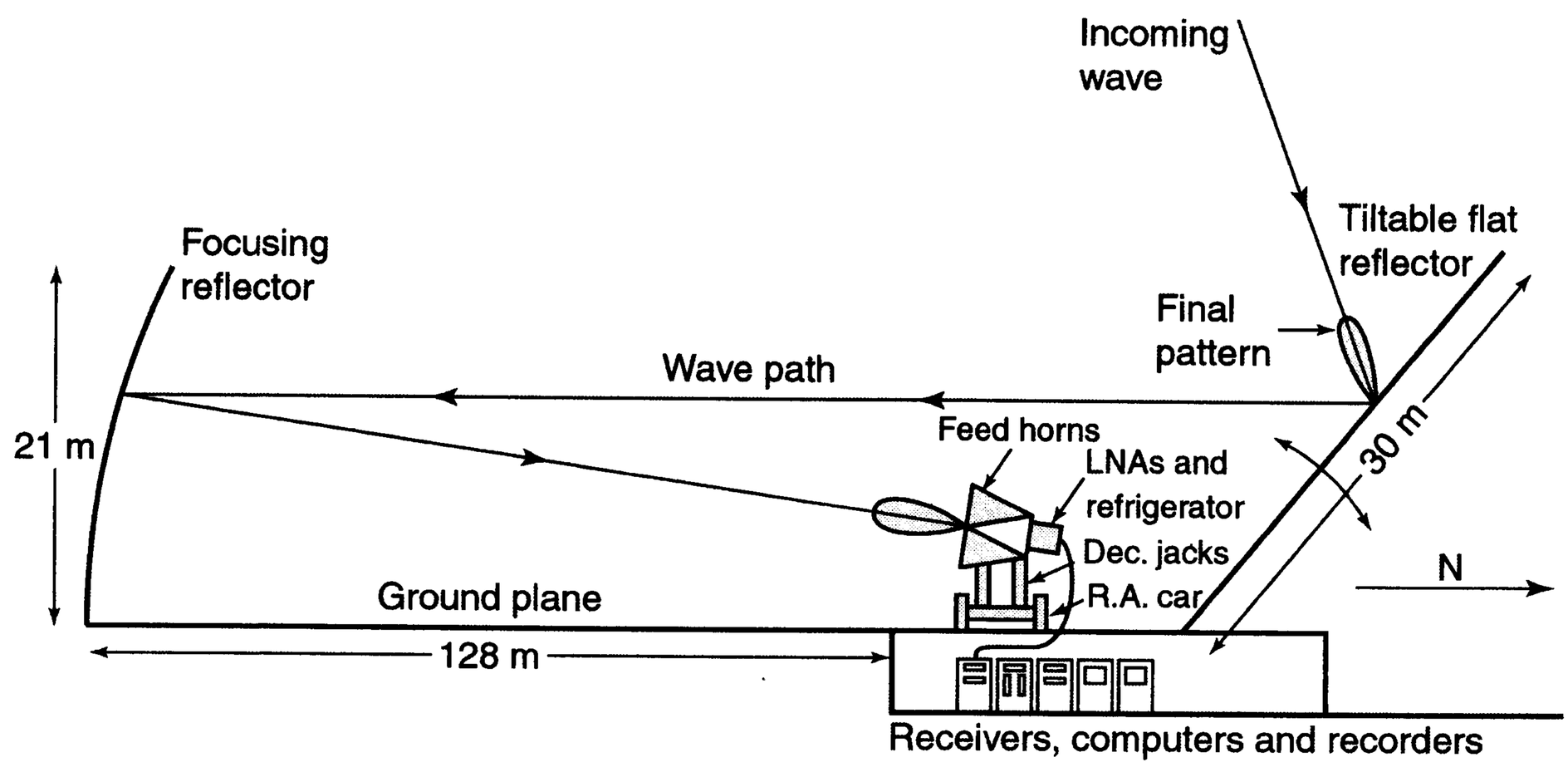


Figure 20–8a
Elevation cross section through the Big Ear radio telescope.

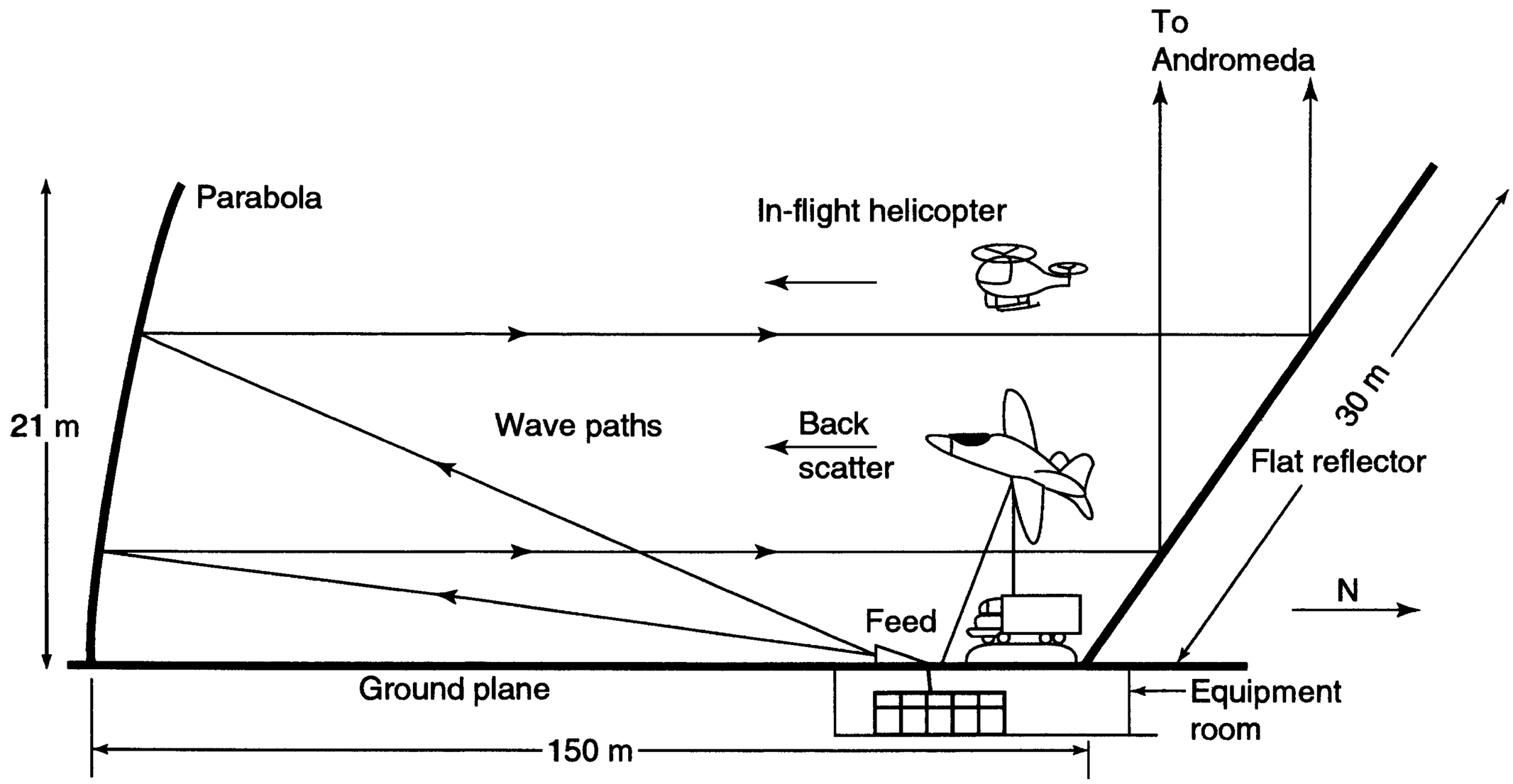


Figure 20–8b
Big Ear radio telescope of the Ohio State University as a compact range for scattering measurements of large objects. Actual full-size vehicles, aircraft and in-flight helicopters could be accommodated. The flat reflector deflected waves into the empty sky, replacing absorbing materials used in standard compact ranges.

elevation (declination) so that it was possible to deploy many feed systems efficiently in the focal region for simultaneous operation. It is noteworthy that because of the long focal length the curved reflector can be described as either a parabola or a sphere since both are almost identical except at the extreme E-W edges where they differ by only a few millimeters. The antenna was operated routinely at frequencies as low as 20 MHz (15 m) and as high as 3 GHz (10 cm). Incoming waves deflected off the flat reflector into the parabola bringing them to a focus at ground level near the base of the flat reflector. Moving the flat reflector though 50° tilted the antenna beam through a 100° range in elevation (or declination).

Another important application of Big Ear was as the world's largest compact range as shown in Fig. 20–8*b*.

In 1998, Big Ear was demolished for a golf course.

Gorki. Albert Kislyakov has constructed a standing-parabola tiltable-flat-reflector antenna near Gorki, USSR, for operation at millimeter wavelengths. Although the basic principle is the same as for Big Ear, Kislyakov has added a unique feature of *self-calibration* by providing that the flat reflector can be set vertically (Kislyakov-1). With flat reflector vertical and with feed displaced laterally off-axis (and transmitting), radiation is brought to a focus on the opposite side of the axis where the signal can be received as suggested by

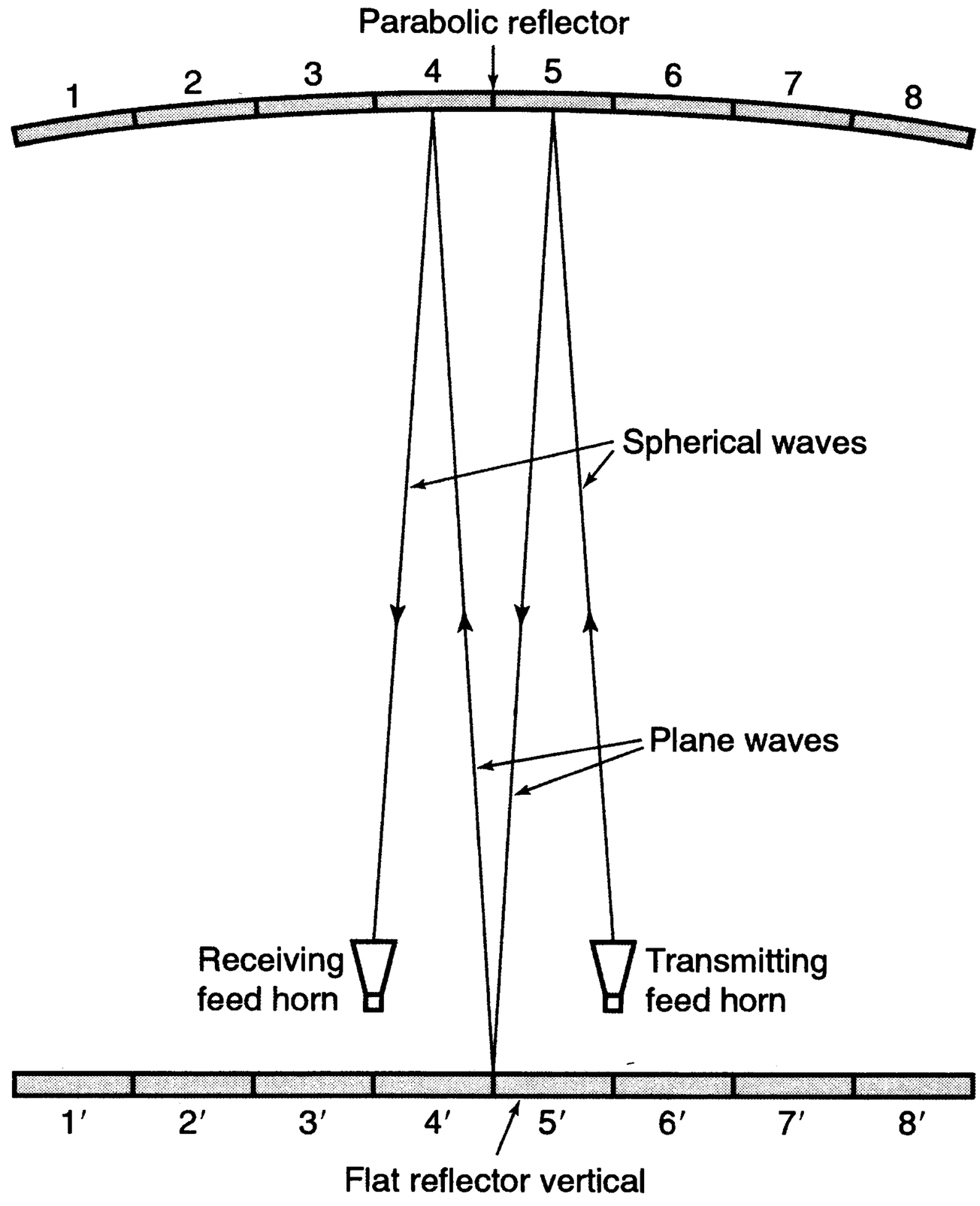


Figure 20–9
Arrangement of standing-parabolic and tiltable-flat reflectors of 25-m millimeter-wave antenna at Gorki, USSR, used by Albert Kislyakov at a wavelength of 1 mm, with flat reflector turned vertically, for self-calibration.

Fig. 20–9. By moving the receiving horn sideways or up and down, the *far-field pattern* of the antenna can be measured with an arbitrarily high S/N ratio.

With the same configuration (flat reflector vertical), panels of the reflectors can be adjusted to maximize the gain. For example, with panels 1, 2, 3, 5, 6 and 7 covered with absorbing material, only panels 4, 4′, 8 and 8′ are operational, and, with 4 and 4′ as reference, panels 8 and 8′ can be adjusted for maximum signal or gain. The same procedure is then repeated with other pairs of panels until all panels are adjusted.

Five College Observatory. The 14-m dish of the Five College Radio Astronomy Observatory is an example of a millimeter-wave antenna enclosed in a Buckminster Fuller triangular-panel geodesic radome. Front and back views of the dish are shown in Fig. 20–10. The reflector is constructed of 72 panels contoured to a section of a paraboloid with a 65-μm rms surface accuracy, or $\lambda/20$ at 1.3 mm. To set the panels accurately a *holographic technique* was used in which the amplitude *and* phase of a 38-GHz satellite beacon signal was measured over an angular extent much larger than the main beam of the antenna. The amplitude and phase of the antenna aperture field distribution was then obtained by a

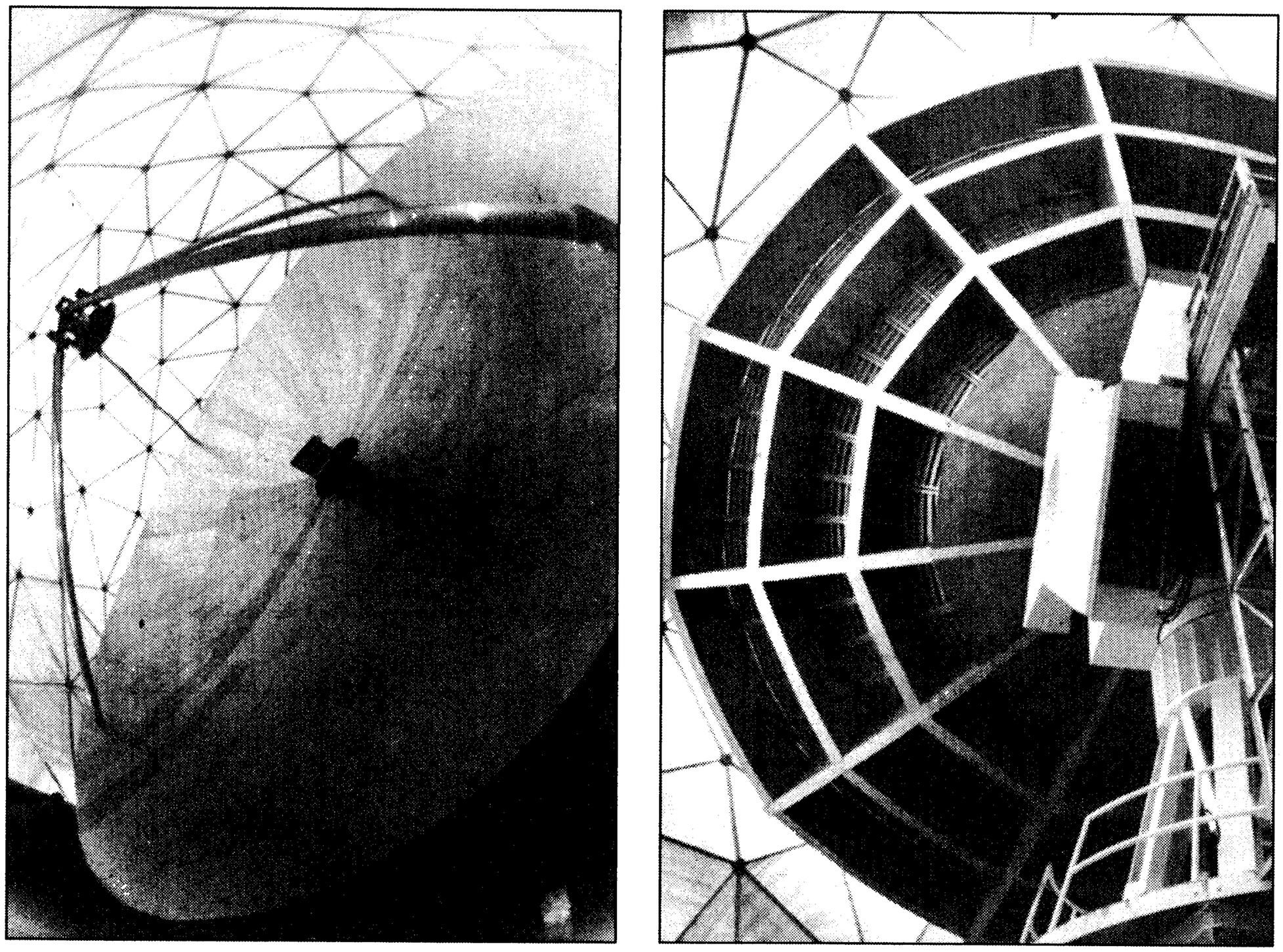


Figure 20–10
Front and back views of the Five College Radio Astronomy Observatory 14-m radome-enclosed radio telescope.

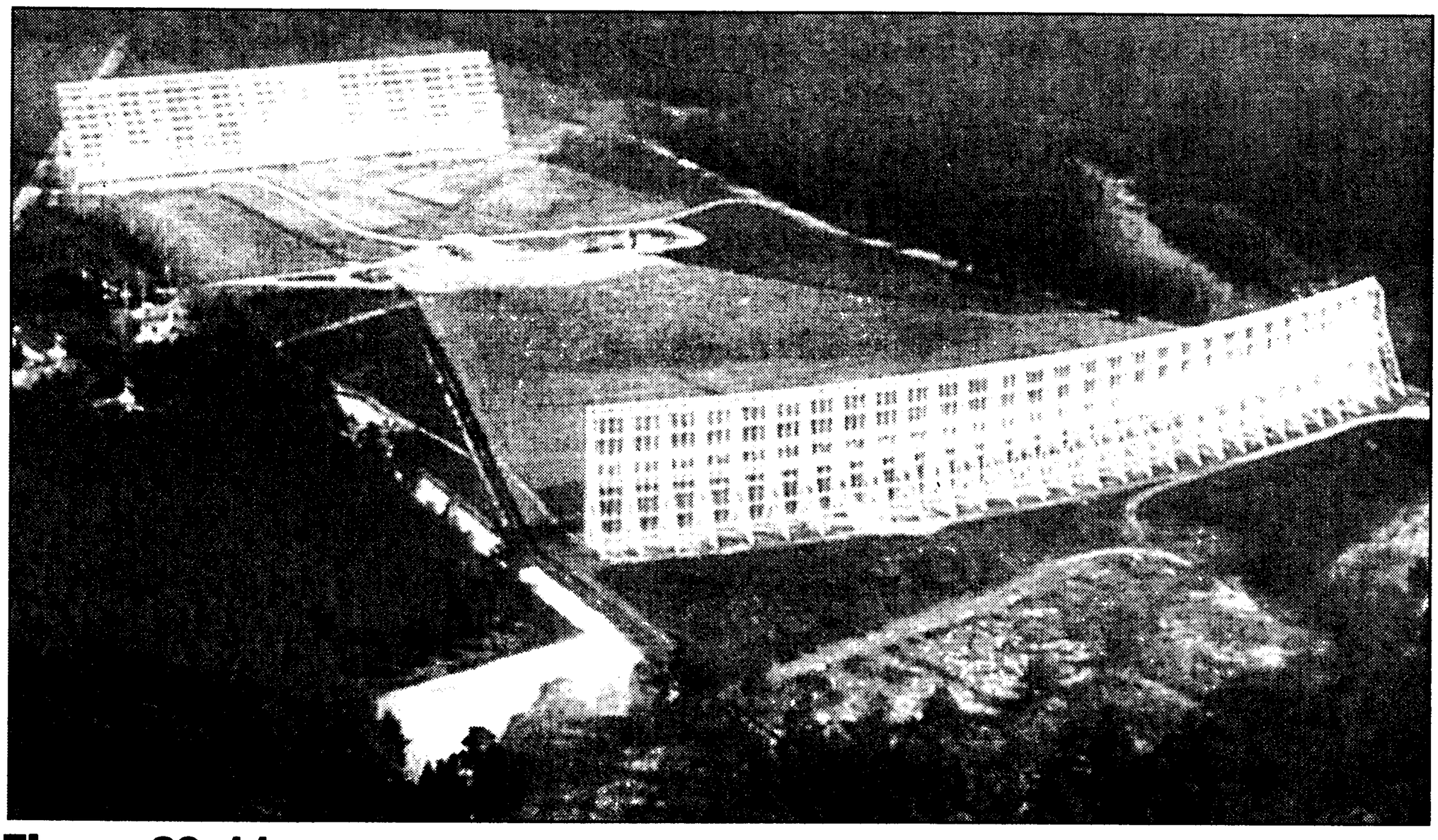


Figure 20–11
Air view of the 300-m Nancay radio telescope antenna.

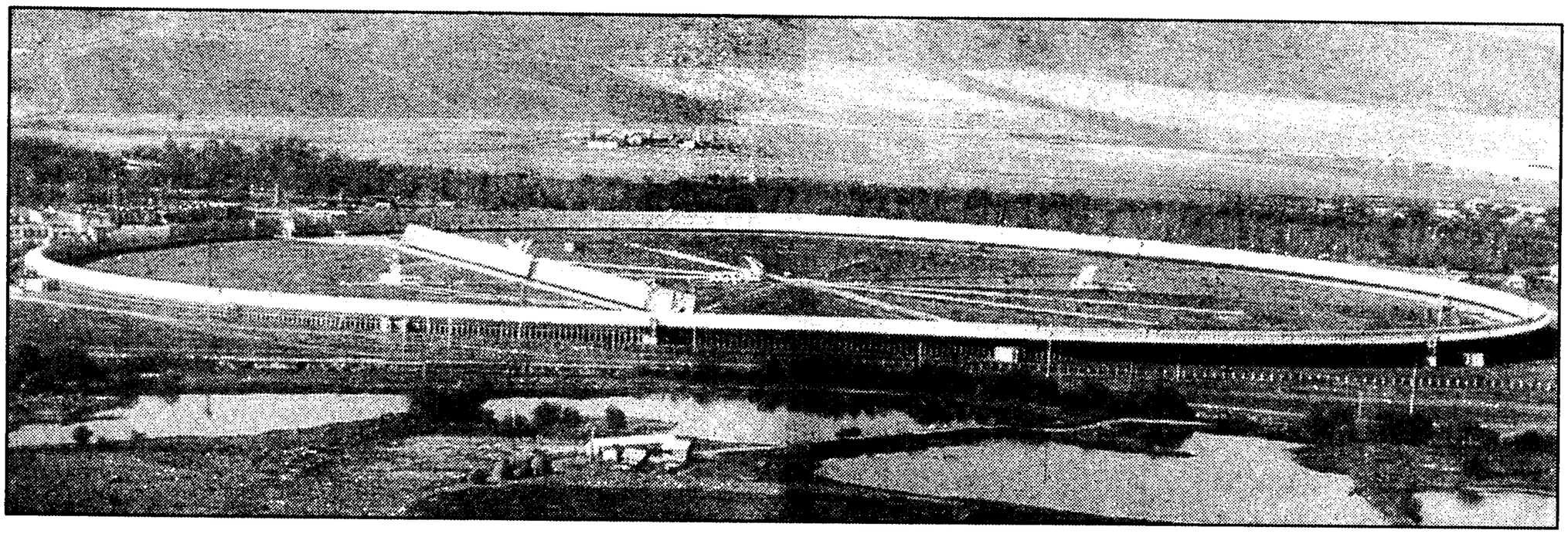


Figure 20–12
The 600-m-diameter RATAN-600 radio telescope near Zelenchukskaya, Russia with tiltable flat and curved reflectors. Although the ring reflector is large enough to hold four Arecibo dishes, the collecting area is less. (*Photo courtesy, Dr. Y. Parijskij.*)

2-dimensional Fourier transform.[1] This indicated a 90-μm rms residual panel position error with the antenna at a 45° elevation angle. At 38 GHz ($\lambda = 7.9$ mm) the gain-loss factor from (19–2–3) is

$$k_g = \cos^2\left(720° \frac{0.09\,\text{mm}}{7.9\,\text{mm}}\right) = 0.98$$

or only 0.09 dB loss from surface error.

At 230 GHz ($\lambda = 1.3$ mm) the loss is larger and irregular lens behavior of the radome fabric becomes significant.

Nancay. A dual reflector antenna, shown in Fig. 20–11, is located at Nancay, France. On completion in 1965, it was dedicated by French President Charles DeGaulle. The basic principle is the same as for the Ohio telescope except that it operates without a ground plane. It is 4 times larger than the Ohio telescope having a 200 by 40 m tiltable flat reflector and a 300 by 35 m standing curved reflector. A movable feed system permits tracking sources for 2 hours or more. It was upgraded in 2000 to work at shorter wavelengths.

RATAN-600. This radio telescope located near Zelenchukskaya, Russia, is shown in Fig. 20–12. It has both tiltable curved and flat reflectors. The curved reflector is 600 m in

[1] A complete description of the procedure is given by P. F. Goldsmith and N. R. Erickson in Sec. 6–22 of J. D. Kraus, *Radio Astronomy,* 2d ed., Cygnus-Quasar, 1986. For additional information on holographic techniques see Bennet *et al.*, Godwin, Whitaker and Anderson, Mayer *et al.* and Rahmat-Samii in the references listed at the end of Chap. 19.

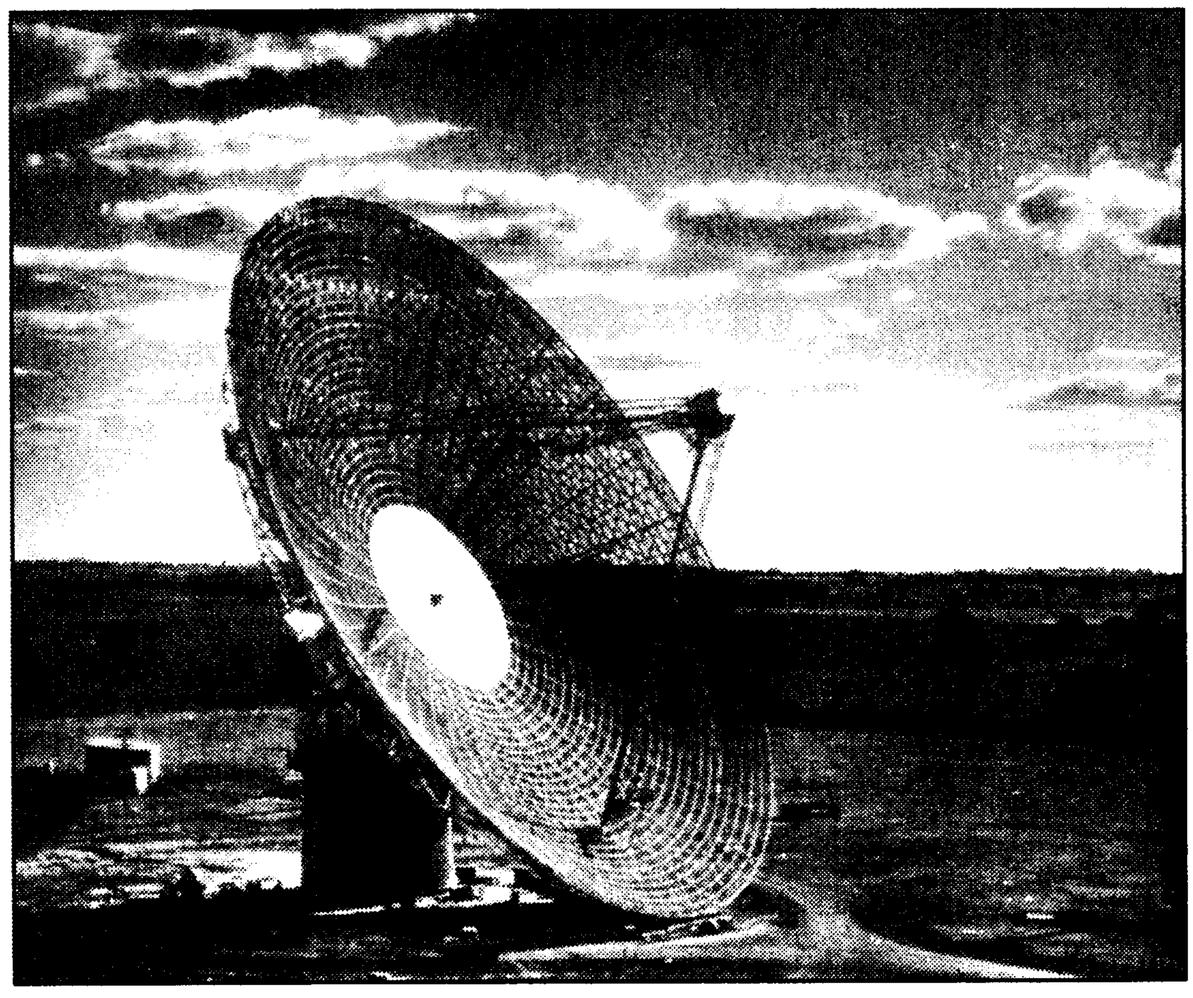


Figure 20–13
The 64-m diameter radio telescope of the CSIRO at Parkes, Australia. (*Courtesy Dr. J. L. Pawsey.*)

diameter and is constructed of 895 movable aluminum panels. Feed systems are situated near the center of the circle. Operation can be as a flat-reflector-spherical-reflector combination as in the Ohio and Nancay telescopes and also in other modes where panels of the circular reflector are used to bring the waves directly to a focus. Although it does not have a filled aperture, the RATAN-600 has the longest continuous aperture of any radio telescope.

Parkes. This 64-m-diameter completely steerable alt-azimuth mounted dish is situated at Parkes, New South Wales, Australia. It is shown in Fig. 20–13. It has been used for surveys of the southern skies and other radio astronomy observations.

Manchester. The 76-m-diameter alt-azimuth mounted parabolic dish antenna shown in Fig. 20–14 was the world's first large fully steerable antenna, rotating in azimuth on a circular track. It was completed in 1957 and is now called "The Lovell Telescope" after Sir Bernard Lovell of the University of Manchester Jodrell Bank Radio Observatory. As of 2000 the "British Government Joint Infrastructure Fund" has provided £2 million for an extensive upgrade that will permit observations at shorter wavelengths.

VLA. This Very Large Aperture array of 27 steerable reflector antennas each 25 m in diameter, shown in part in Fig. 15–14, is situated on the Plains of San Augustin, New Mexico. Operated by the National Radio Astronomy Observatory, it is used for aperture-synthesis mapping of the radio sky. The antennas are movable and can be deployed at stations along railroad tracks having a Y-shaped configuration with each arm of the Y = 21 km long. The page facing page 1 also has a photograph of the VLA.

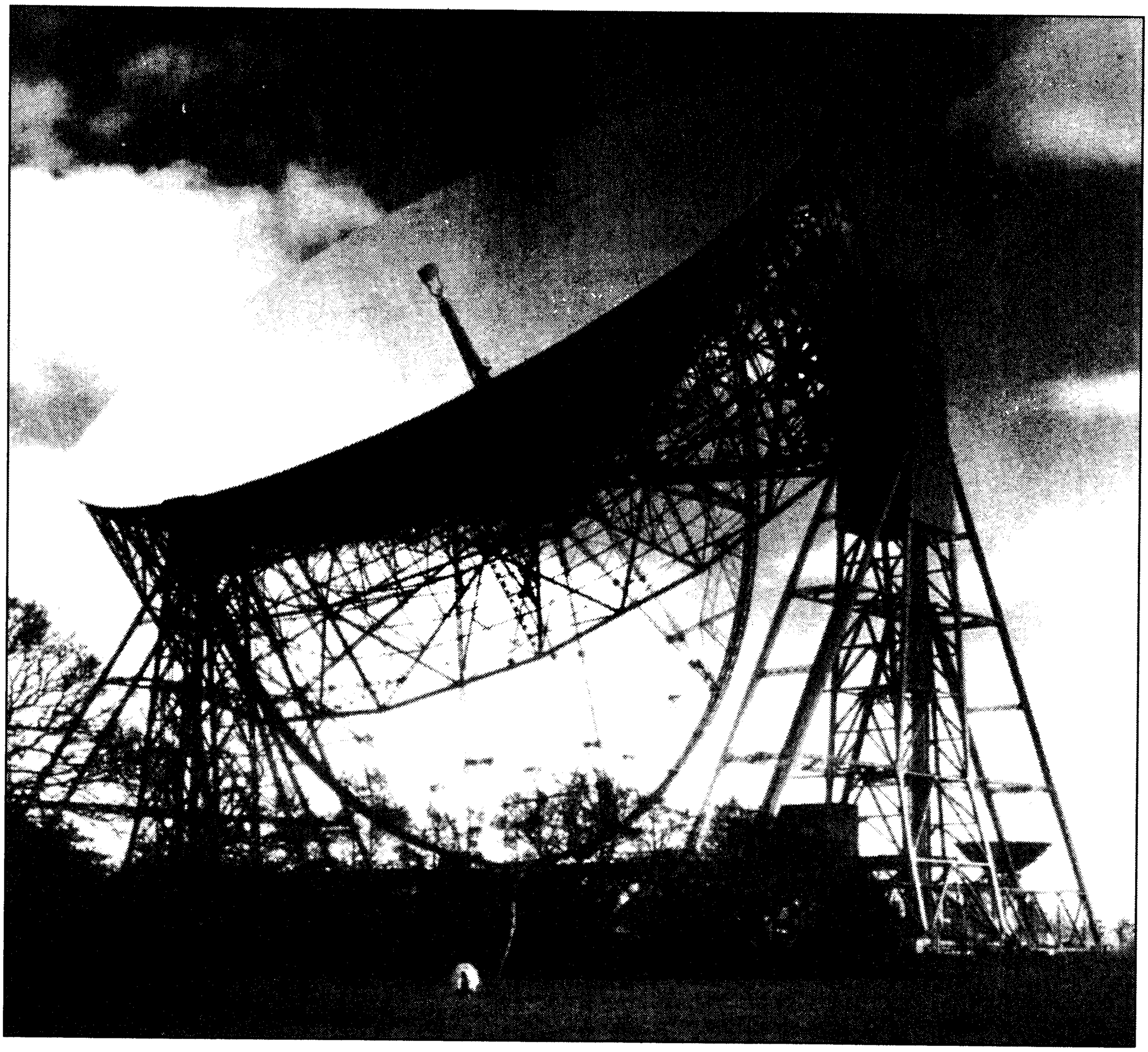


Figure 20–14
The 76-m Lovell radio telescope at the University of Manchester Jodrell Bank Observatory. (*Courtesy Sir Bernard Lovell.*)

Nauen. This antenna system has two arrays, one for $\lambda = 50$ m and the other smaller array for $\lambda = 25$ m with a reflector screen between which is held by outriggers set at an angle. The antenna system can be turned to any compass direction. Each array has 16 $\lambda/2$ dipoles.

EXAMPLE **Deutsche Welle Short Wave Antenna**

(*a*) What is the gain of the Deutsche Welle $\lambda = 50$ m antenna array of Fig. 20–15*a* including the effect of ground if the center of the array is 62.5 m above ground?

(*b*) What is the ground bounce angle?

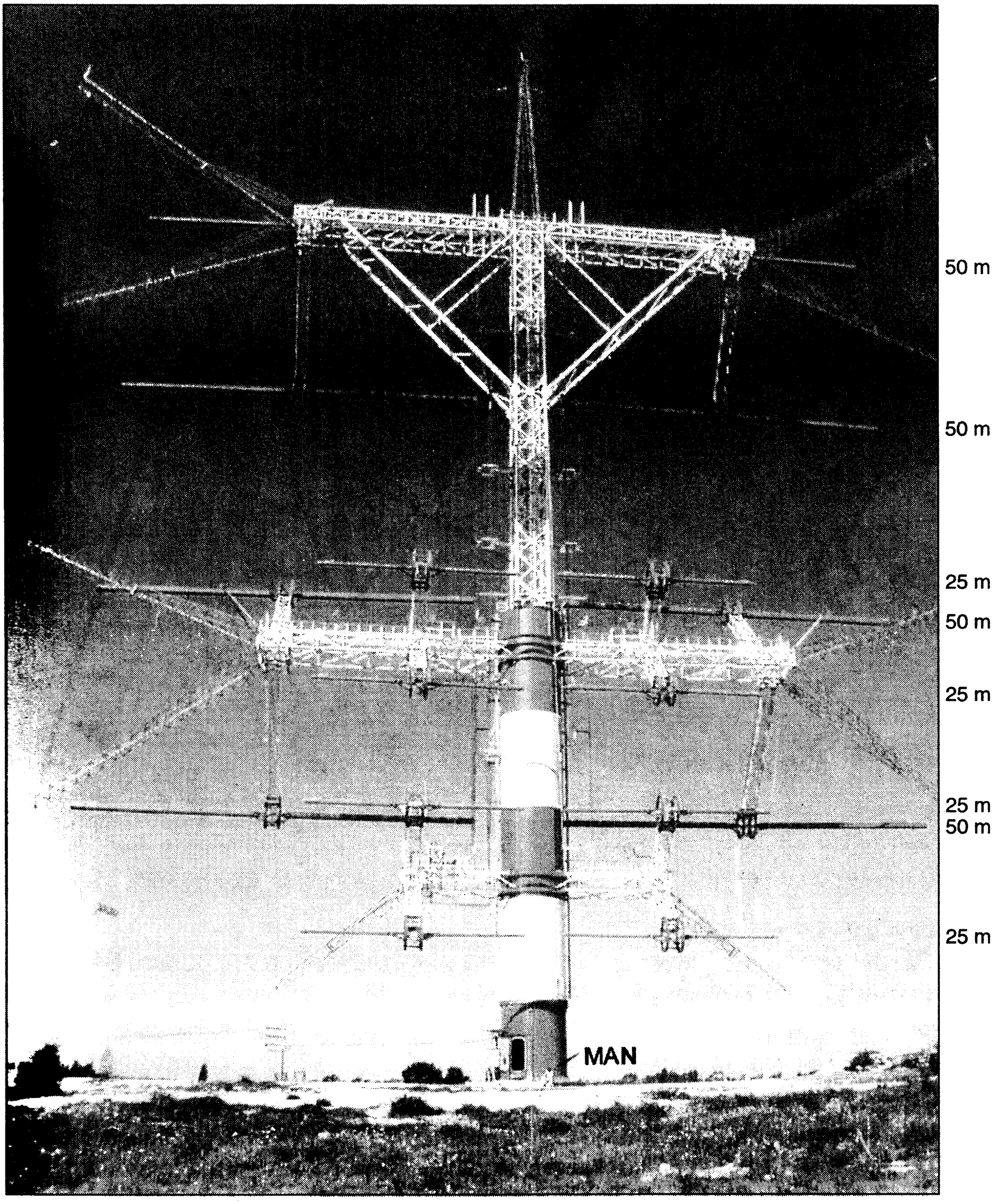


Figure 20–15a
Tannenbaum antenna array of Deutsche Welle at Nauen, Germany, for 50 and 25 m short-wave broadcasting. For scale, note the technician at the base of the tower. The location of the 50 and 25 m collinear dipoles is indicated along the right side of the photograph. Other short-wave antennas (fixed curtain arrays) are in the distance at the lower right. (*Photograph courtesy of Deutsche Welle.*)

■ **Solution**

(*a*) The gain of a single $\lambda/2$ dipole is 2.15 dBi and of 2 collinear in-phase $\lambda/2$ dipoles is 3.8 dBi. The array of 8 such collinear dipoles adds $3+3+3=9$ dB. The reflector screen adds 3 dB more and the ground bounce another 6 dB for a total gain of $3.8+9+3+6=21.8$ dBi or a directivity of 151 *approx.* *Ans.* (*a*)

Thus, for a power input of 10 kW to the antenna the ERP (effective radiated power) = $10^4 \times 151 = 1.5$ MW *approx.*

(*b*) For 2 out-of-phase point sources separated by 2.5λ or 125 m the ground bounce angle $\alpha = \sin^{-1}(0.5\lambda/2.5\lambda) = \sin^{-1}(25\,\text{m}/125\,\text{m}) = 11.5°$ *Ans.* (*b*). See sketch.

The pattern of the two point sources multiplied by the pattern of the broadside array of stacked dipoles results in maximum radiation at a slightly lower angle $\alpha = 9°$ approx.

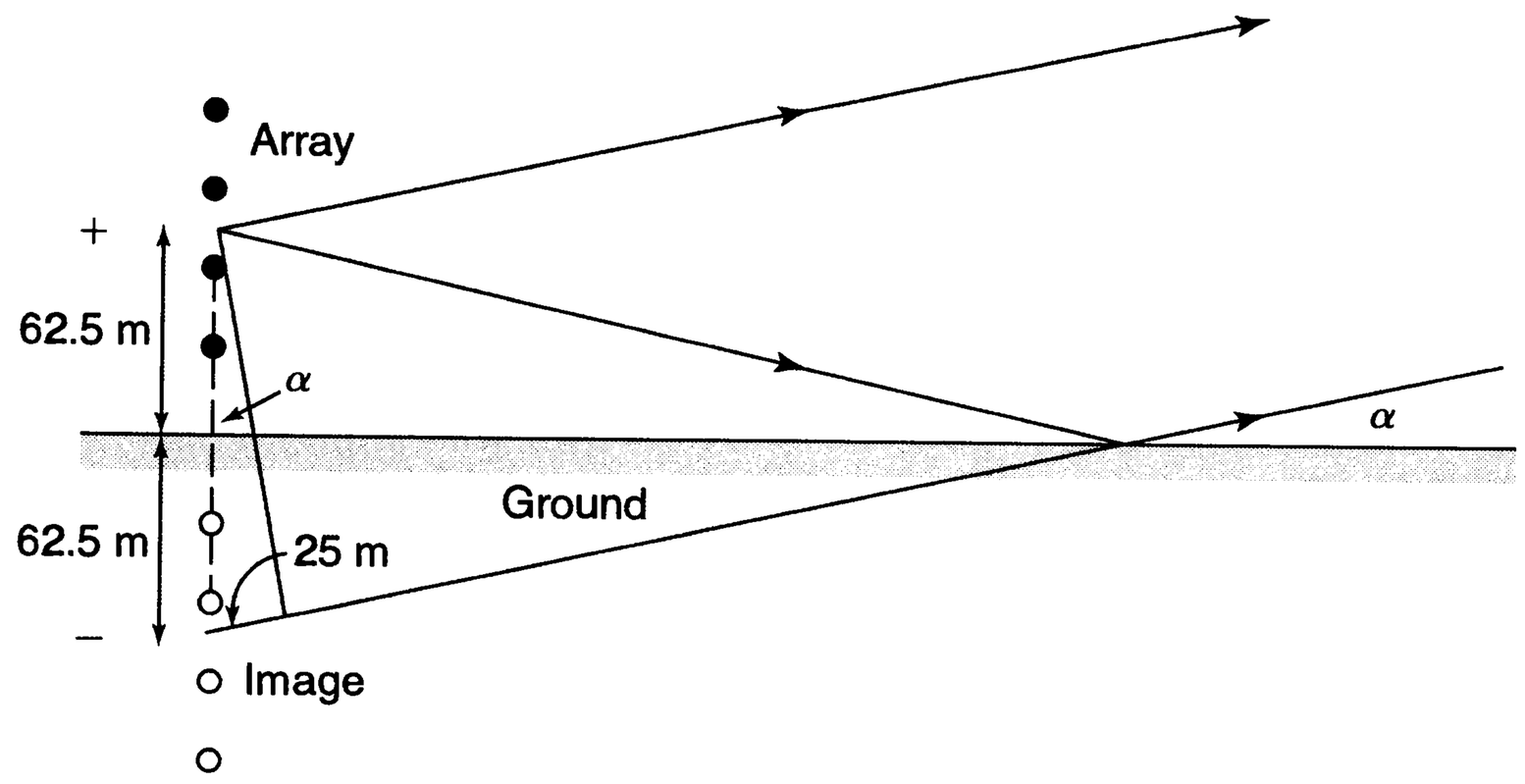


Figure 20–15b
Ground-bounce geometry.

Green Bank. John Findlay's 91-m-diameter meridian-transit radio telescope of the National Radio Astronomy Observatory at Green Bank, West Virginia, had been severely stressed for many years by rocking the dish during right-ascension scans. On November 15, 1998, it collapsed. Senators Robert C. Byrd and Jay Rockefeller of West Virginia immediately obtained a Congressional appropriation of $75 million for a replacement to be built at Green Bank. The National Science Foundation objected. The NSF wanted the replacement constructed in the warmer, drier southwest. U.S. Senators Byrd and Rockefeller wanted it in West Virginia as a symbol of West Virginia high-tech.

The replacement is a 110×100 m completely steerable dish (Figs. 20–16 and 20–17) standing 40 stories tall in a West Virginia mountain valley near Green Bank. Named the Robert C. Byrd telescope, it is the most sophisticated, complex and expensive antenna ever built with a cost as of 2000 of well over $100 million (neck-and-neck in cost with Arecibo).

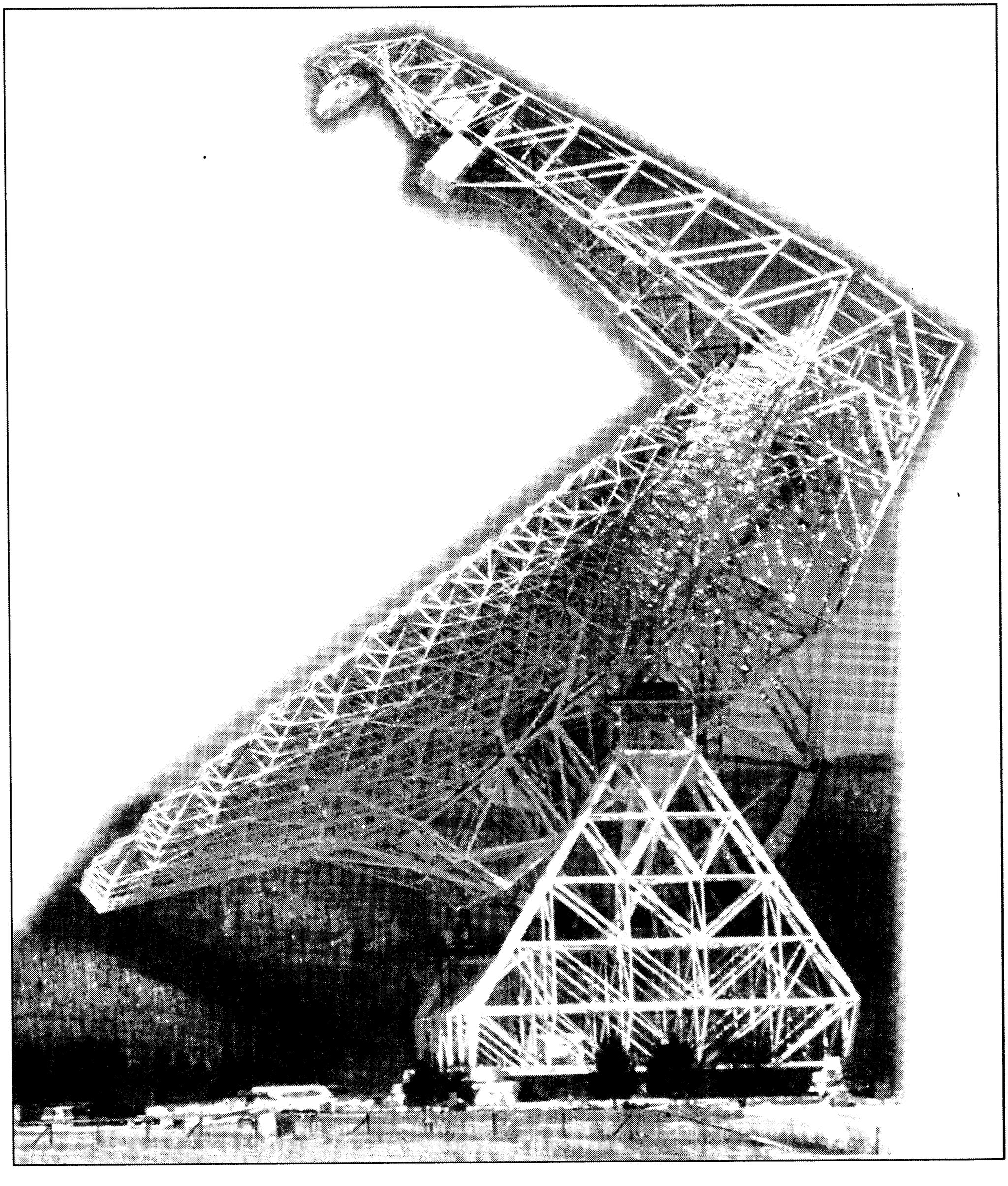


Figure 20–16
The Robert C. Byrd Green Bank Telescope. The alt-azimuth mount provides access to the whole sky down to 5° elevation. The offset feed provides an unblocked aperture. The dish surface has 2004 panels continuously adjusted with laser metrology to a position accuracy of 25 μm. As of 2000 improvements are still in progress with planned frequency coverage from 100 MHz to 100 GHz (3 m to 3 mm). (*NRAO-AUI.*)

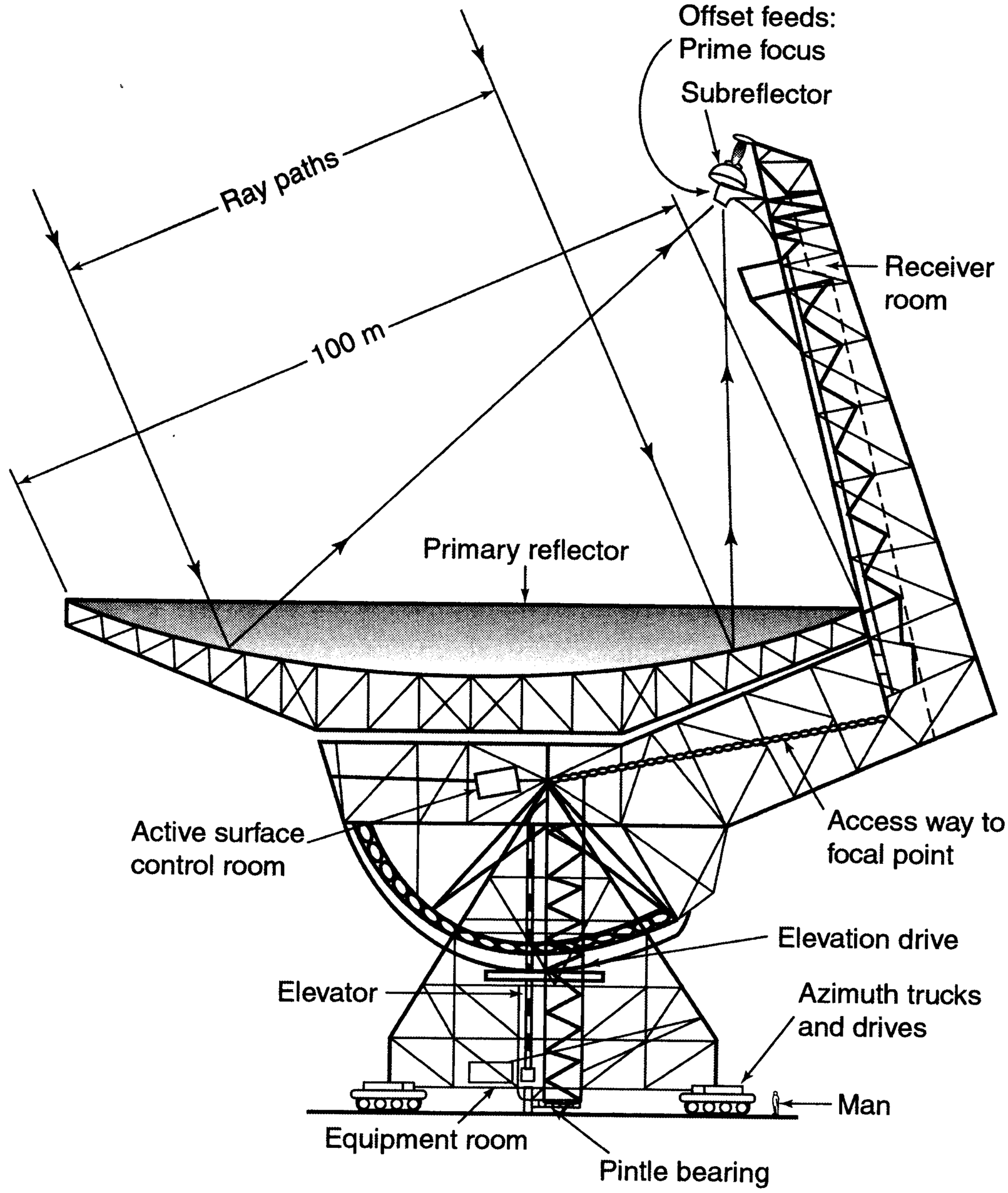


Figure 20–17
Cross section of Robert C. Byrd Green Bank Telescope. Note man at lower right.

REFERENCES（参考文献）

Akabane, K. (1), M. Morimoto, N. Kaifu and M. Ishigro, "The Nobeyama Radio Observatory," *Sky and Tel.,* 495, December 1983.

Chu, T-S. (1), R. W. Wilson, R. W. England, D. A. Gray and W. E. Legg, "The Crawford Hill 7-m Millimeter Wave Antenna," *BSTJ,* **57,** 1257–1288, May-June 1978.

Chu, T-S. (2), personal communication, 1985.

Davis, M. (1), personal communication, Sept. 11, 2000.

Kislyakov, A. G. (1), "Radioastronomical Investigations at Millimetric and Submillimetric Wavelengths," *UFN,* **101,** 607, 1970.

Kraus, J. D. (1), "The Ohio State Radio Telescope," *Sky and Tel.,* **12,** 157–159, April 1953.

Kraus, J. D. (2), "Radio Telescopes," *Sci. Am.,* **192,** 33–43, March 1955.

Kraus, J. D. (3), "Radio Telescopes of Large Aperture," *Proc. IRE,* **46,** 92–97, January 1958.

Kraus, J. D. (4), R. T. Nash and H. C. Ko, "Some Characteristics of the OSU 360-ft Radio Telescope," *IRE Trans. Ants. Prop.,* **AP-9,** 4–8, January 1961.

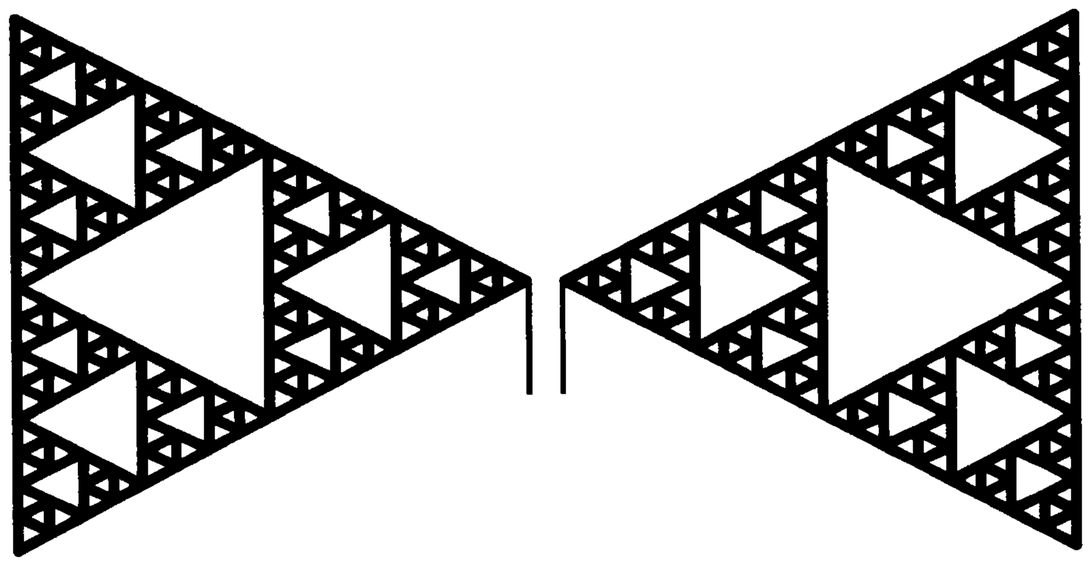
Kraus, J. D. (5), "The Large Radio Telescope of the Ohio State University," *Sky and Tel.,* **26,** 12–16, July 1963.

Kraus, J. D. (6), *Big Ear II,* Cygnus-Quasar, 1995.

CHAPTER 21

Antennas for Special Applications

特殊用途的天线



Topics in this chapter include:

- Electrically small antennas（电小天线）
- Small genetic algorithm antenna（小型遗传算法天线）
- Physically small antennas（物理上的小天线）
- Antenna siting and the effect of imperfect ground（天线架设和非理想地面的影响）
- Ground plane antennas（带接地面的天线）
- Sleeve antennas（套筒天线）
- Turnstile antennas（旋转场天线）
- The superturnstile antenna（超旋转场天线）
- Other omnidirectional antennas（其他全向天线）
- Circularly polarized antennas（圆极化天线）
- The high-gain omni（高增益全向天线）
- Submerged antennas（埋地天线）
- Surface wave and leaky wave antennas（表面波和漏波天线）
- Antenna design for satellite communication（卫星通信用天线的设计）
- Receiving vs. transmitting considerations（用做接收和用做发射的考虑）
- Bandwidth considerations（频带的考虑）
- Architecturally acceptable antennas（建筑上可接受的天线）
- ILS (Instrument Landing System) antennas（仪表着陆系统用天线）
- The Sugar Scoop Antenna and the 3 K Cosmic Sky Background Story（糖铲形天线和 3K 宇宙背景的故事）
- LEO (Low Earth Orbit) satellite antennas（近地轨道卫星线路用天线）
- Asteroid detection antenna（检测小行星的天线）
- Leaky transmission lines as antennas（泄漏传输线用做天线）
- Artistic antennas（艺术性天线）
- Cell phone antennas（蜂窝电话用天线）
- Ground Penetrating Radar (GPR): Pulse bandwidth（探地雷达用天线，脉冲的频带宽度）
- Embedded antennas（植入式天线）
- Ultra-Wide Bandwidth (UWB) antennas for digital applications（数字式应用的特宽频带天线）
- Plasma antennas（等离子态天线）

21-1 INTRODUCTION（引言）

Previous chapters cover the properties of many basic types of antennas which are the mainstream of antenna technology. In this chapter, more specialized, application-oriented antennas are discussed. These include antennas that are electrically small (but sometimes

physically very large) and ones that are physically small (but sometimes electrically large), siting of antennas and the effect of typical ground, feeding and matching arrangements, and antennas for specialized applications from vehicular to satellite communication.

21–2 ELECTRICALLY SMALL ANTENNAS（电小天线）

The *radiation efficiency* of an antenna is given by

$$\text{RE} = \frac{R_r}{R_r + R_L} \qquad \text{(dimensionless)} \tag{1}$$

where

R_r = radiation resistance, Ω
R_L = loss resistance, Ω

From (6–3–15), radiation resistance of a short dipole is given by

$$R_r = 200 l_\lambda^2 \qquad (\Omega) \tag{2}$$

where l_λ = length of dipole in wavelengths, dimensionless

If $l_\lambda = 0.1$, $R_r = 2\ \Omega$; if $l_\lambda = 0.01$, $R_r = 0.02\ \Omega$. With such low values of radiation resistance it is apparent that even small values of loss resistance can result in low radiation efficiency.

EXAMPLE 21–2.1

Referring to Fig. 21–1*a*, a 100-kHz transmitter feeds a 150-m vertical antenna against ground. The loss resistance is 2 Ω. Find (*a*) the effective height, (*b*) the radiation resistance and (*c*) the radiation efficiency.

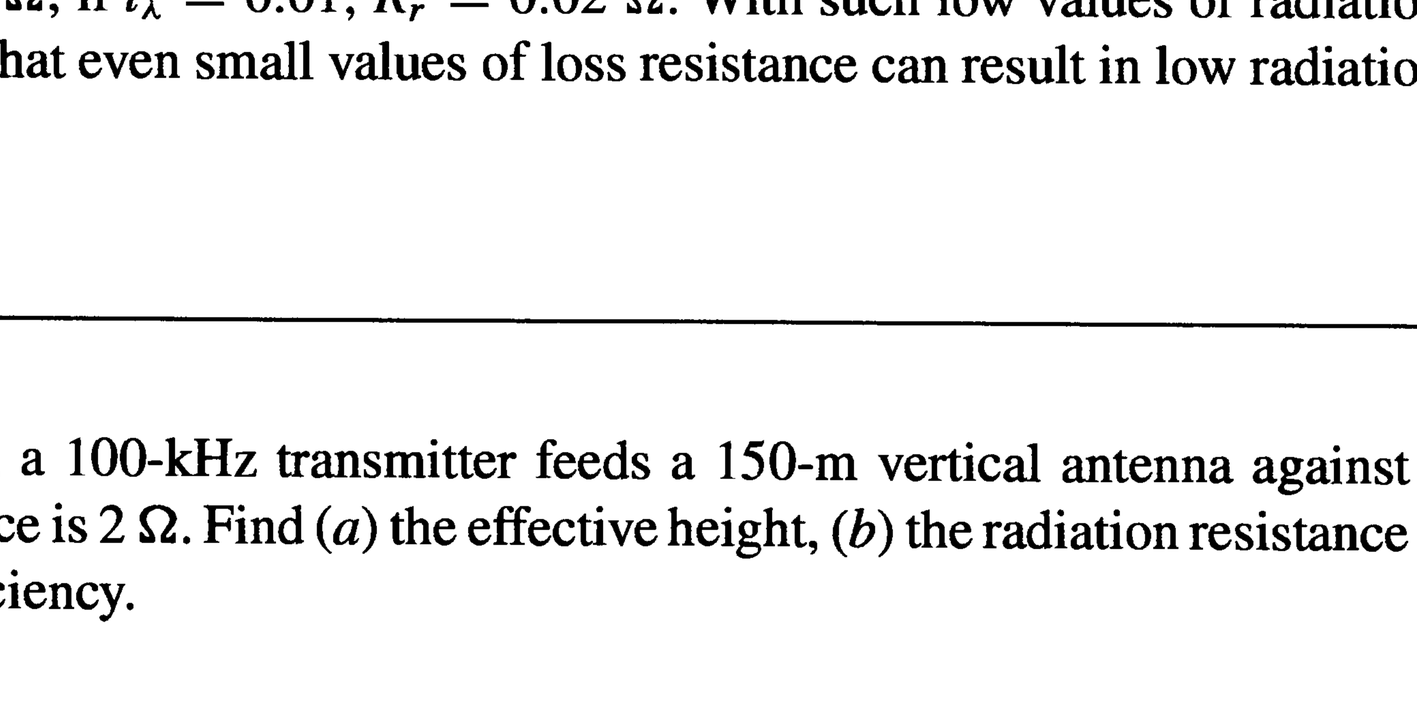


Figure 21–1
Electrically small antennas. (*a*) 150-m tower for operation at $\lambda_0 = 3000$ m. (*b*) Two nonradiating towers supporting flat top for increasing effective height.

■ **Solution**

(*a*) At 100 kHz ($\lambda = 3000$ m), the antenna (physical) height $h_p = 0.05\lambda$ (=150/3000) and with a triangular current distribution (current zero at top increasing linearly to a maximum value I_0 at the base) we have, from (2–10–3), that the effective height $h_e = h_p/2 = 0.025\lambda$.

(*b*) The radiation resistance for a short vertical monopole is given by

$$R_r = 400\left(\frac{h_p}{\lambda}\right)^2 = 1600\left(\frac{h_e}{\lambda}\right)^2 \qquad (\Omega) \tag{3}^1$$

where

h_p = physical height, m
h_e = effective height, m

and for the 150-m vertical radiator, we have

$$R_r = 400 \times 0.05^2 = 1\ \Omega$$

(*c*) From (1) the radiation efficiency

$$\text{RE} = \frac{R_r}{R_r + R_L} = \frac{1}{2+1} = 0.33 \text{ or } 33 \text{ percent}$$

To increase the radiation efficiency requires an increase in the radiation resistance R_r or a decrease in the loss resistance R_L or both. The loss resistance involves losses in the ground system, antenna insulators, tuning coil, conductors (including the tower itself) and corona. These may be reduced but only at a certain cost. On the other hand, doubling the physical height of the antenna quadruples the radiation resistance and increases the radiation efficiency. However, to double the tower height might cost 8 times as much (cost proportional approximately to the cube of the height). Another alternative would be to use 2 towers 150 m tall supporting a vertical wire with flat top as suggested in Fig. 21–1*b*, making the current distribution along the vertical conductor more uniform ($I_{\text{av}} > \frac{1}{2}I_0$) and the effective height more nearly equal to the physical height. This will not quadruple R_r but could increase it by some factor less than 4. The currents on the flat top are in phase opposition and, being separated by a small fraction of a wavelength, have only a small effect on the radiation. What steps, if any, are taken to increase the radiation efficiency involve trade-offs with cost considerations a determining factor.

According to Harold Wheeler, a figure-of-merit for an electrically small antenna is its *radiation power factor:*

$$\text{PF} = \frac{\text{radiated power}}{\text{reactive power}} = \frac{R_r}{X} \ll 1 \tag{4}$$

[1] It is assumed in (2) and (3) that the average antenna current is $\frac{1}{2}$ the terminal current.

where

R_r = radiation resistance
X = reactance

This is not the same as $1/Q$ since

$$Q = \frac{\text{energy stored per unit time}}{\text{energy lost per unit time}} = \frac{X}{R_r + R_L} = \frac{\text{center frequency}}{\text{bandwidth}} \tag{5}$$

Thus, increasing either R_r or R_L, or both, reduces Q and broadens the bandwidth. However, only an increase in R_r increases Wheeler's radiation power factor (Wheeler-1, 2; Hansen-1).

Harold Wheeler defines an electrically small antenna as one which occupies a volume of less than a *radian sphere,* i.e., a sphere of radius $r = \lambda/2\pi$ ($=0.16\,\lambda$). The significance of this definition is that stored energy dominates inside the radian sphere while radiated energy is important outside. Wheeler shows further that the radiation power factor for a small antenna is equal to the ratio of the antenna volume to a radian sphere or

$$\text{PF} = \frac{\text{antenna volume}}{\text{radian sphere}} = \frac{\frac{4}{3}\pi r^3}{\frac{4}{3}\pi(\lambda/2\pi)^3} = \left(\frac{2\pi r}{\lambda}\right)^3 \tag{6}$$

where

r = radius of antenna volume, m
λ = wavelength, m

According to (6), the PF or figure-of-merit of a small antenna varies as the cube of its dimension. If $R_L = 0$ in (5), we have from (4) and (6) that

$$Q = \frac{1}{(2\pi r_\lambda)^3} \tag{7}$$

Thus, an electrically small lossless antenna ($2\pi r_\lambda \ll 1$) is inherently high Q and narrow bandwidth. We note from (5) that losses decrease Q and increase bandwidth.

To reduce the attenuation through salt water, very low frequencies (10 to 50 kHz) are used for transmitting to submerged submarines. At a frequency of 10 kHz ($\lambda = 30$ km) antennas may have flat tops covering many square kilometers and rank as the world's largest antennas (physically), yet they are electrically small ($h_e < 0.01\lambda$). (See Prob. 21–12–1.)

21–2a Short Stub Antenna, Small Genetic Algorithm Antenna and Small Series-Tuned Loop Antenna

（短桩形天线、小型遗传算法天线和小型串联调谐环天线）

A short stub antenna of height $h = \lambda/20$ as in Fig. 21–1–1*a* has a radiation resistance from (21–2–3) given by

$$R_r = 400(h/\lambda)^2 = 400(1/20)^2 = 1\ \Omega$$

If the height is reduced to $\lambda/40$ but the current I is uniform instead of tapered as in Fig. 21–1–1*a*, the radiation resistance from (21–2–3) is

$$R_r = 1600(h/\lambda)^2 = 1600(1/40)^2 = 1\ \Omega$$

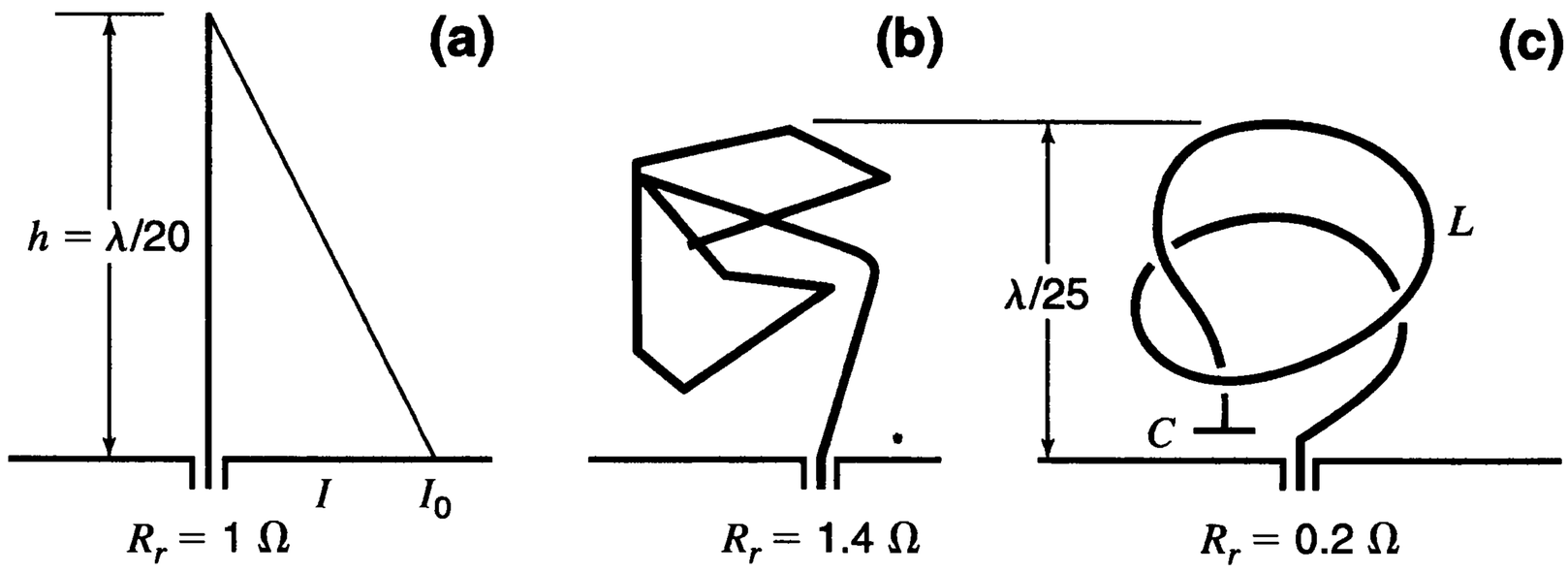


Figure 21–1–1
(*a*) Stub antenna of height $h = \lambda/20$, (*b*) genetic algorithm antenna of height $h = \lambda/25$ and (*c*) resonant series-tuned loop antenna of height $h = \lambda/25$.

A resonant 10-segment genetic algorithm antenna with height $h = \lambda/25$ as in Fig. 21–1–1*b* has a radiation resistance $R_r = 1.4\ \Omega$ (Altshuler-1).

A resonant series-tuned *LC* circuit acting as a small loop antenna of height $h = \lambda/25$, as in Fig. 21–1–1*c*, is analogous to the 10-segment antenna of Fig. 21–1–1*b* (Walter-1). From (7–6–10*a*) the radiation resistance of the loop is

$$R_r = 31{,}200(nA/\lambda^2)^2 = 31{,}200(2\pi\lambda^2/50^2\lambda^2)^2 = 0.2\ \Omega$$

21–3 PHYSICALLY SMALL ANTENNAS（物理上的小天线）

Heinrich Hertz and other radio pioneers used meter and centimeter wavelengths, but the demonstration of long-distance communication with kilometer wavelengths by Guglielmo Marconi and others soon moved interest from the short to the long wavelengths. However, over the intervening years this trend has been reversed and now active radio technology extends down to millimeter and submillimeter wavelengths. One of the bonuses of the short wavelengths is the spectrum space available for many wide-bandwidth video and high-data-rate channels.

Antennas for these short wavelengths include printed and patch antennas and ones with built-in (integrated) active elements (amplifiers and detectors). These active elements can compensate for the increased transmission-line losses at millimeter wavelengths.

The exponential *V* described earlier is well adapted for printed-circuit fabrication, as suggested in Fig. 21–2. Here the *V* takes the form of an exponential notch in the conducting

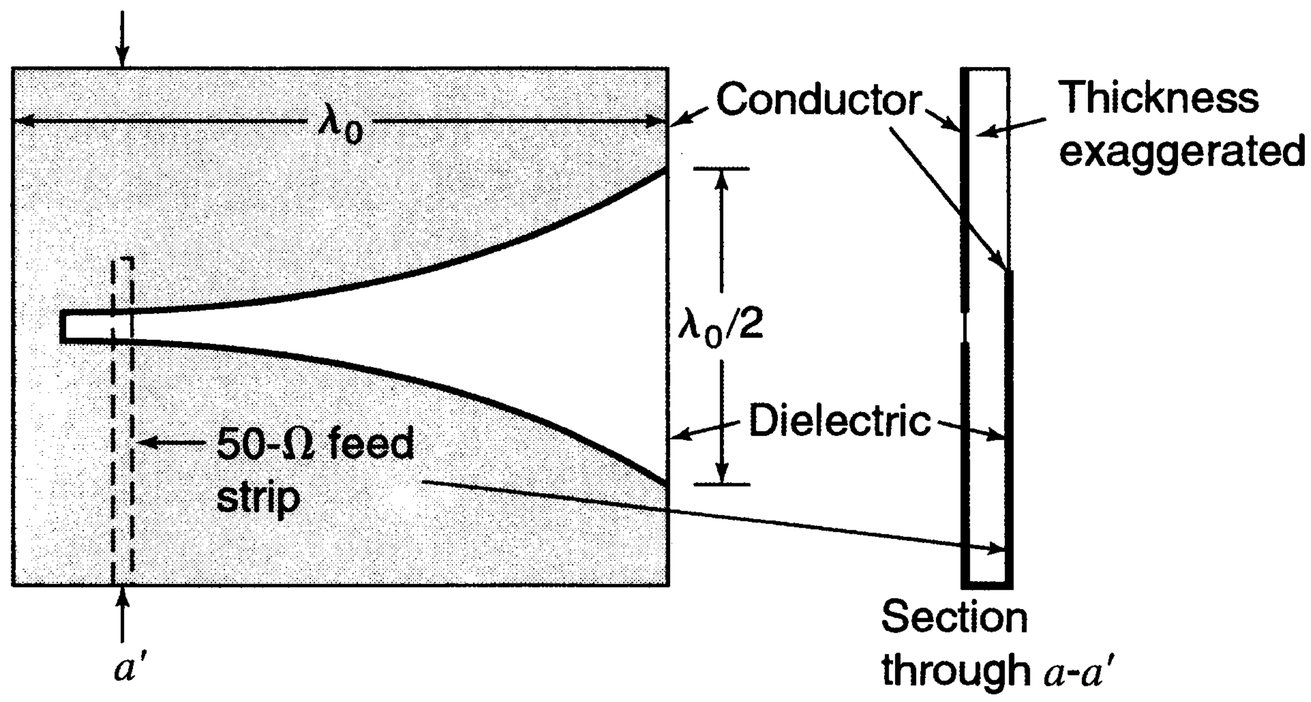


Figure 21–2
Exponential notch antenna with 50-Ω microstrip feed.

surface of a circuit board with coupling from a 50-Ω strip line on the other surface of the board. Bandwidths of 5 to 1 are possible but with only small gain (Prasad-1). However, many such printed notch radiators can be stacked to form highly directional phased arrays (Armitage-1).

Visible wavelengths (blue to red) extend from 400 to 700 nm. Infrared wavelengths extend from 700 nm into the micrometer range where they merge with radio. Whether these micrometer wavelengths are called infrared or radio is arbitrary. With printed-circuit technologies now able to fabricate structures in the micrometer and nanometer range (Robinson-1), it may be possible to construct antennas for even shorter wavelengths (see Chap. 22 on Terahertz Antennas).

21–4 ANTENNA SITING AND THE EFFECT OF TYPICAL (IMPERFECT) GROUND
[天线架设和典型（非理想）地面的影响]

In Secs. 16–7 and 16–8 the vertical plane patterns for horizontal and vertical antennas were calculated assuming that the ground was perfectly conducting ($\sigma = \infty$). Let us consider now the more general situation of a flat ground which is not perfectly conducting, considering first the case for *horizontal polarization*. Referring to Fig. 21–3, the electric field from the horizontal dipole is perpendicular ($\perp$) to the page or plane of incidence. Assuming the ground is nonmagnetic ($\mu = \mu_0$), it may be shown that the reflection coefficient ($\rho_\perp$) is given by

$$\rho_\perp = \frac{\sin\alpha - \sqrt{\varepsilon_r - \cos^2\alpha}}{\sin\alpha + \sqrt{\varepsilon_r - \cos^2\alpha}} \tag{1}$$

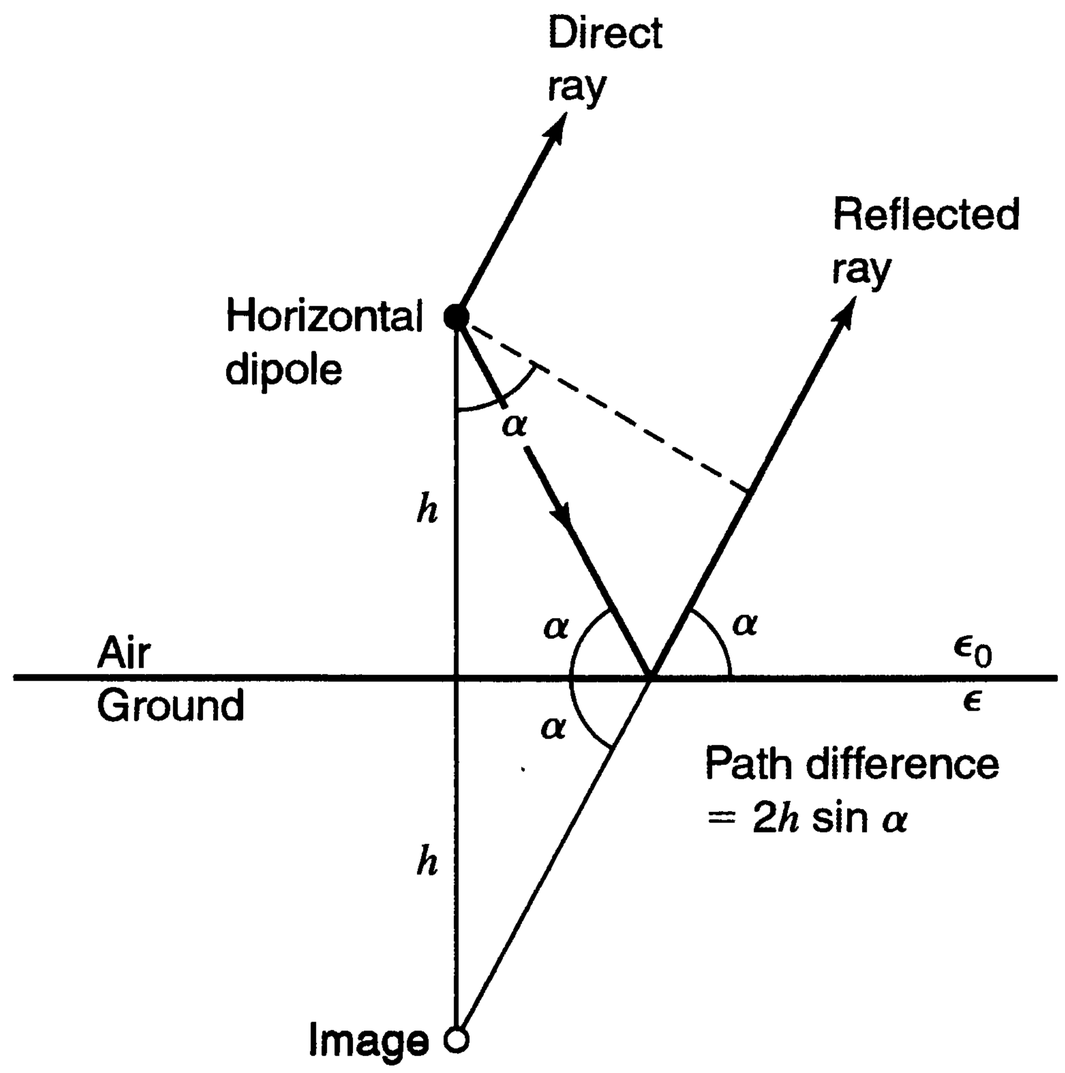


Figure 21–3
Geometry for horizontal dipole at height *h* above a flat earth or ground.

where

ε_r = relative permittivity of the ground = $\varepsilon/\varepsilon_0$ where
ε = ground permittivity
ε_0 = air (or vacuum) permittivity
α = elevation angle (= 90° minus angle of incidence measured from the zenith)

If $\varepsilon_r \gg 1$, (1) reduces to

$$\rho_\perp = \frac{\sin\alpha - n}{\sin\alpha + n} \tag{2}$$

where $n = \sqrt{\varepsilon_r}$ = index of refraction

The relative electric field at a large distance is the resultant of the direct ray from the dipole and the ray reflected from the ground, as suggested in Fig. 21–3, as given by

$$E_\perp = 1 + p_\perp \underline{/2\beta h \sin\alpha} \tag{3}$$

where

$\beta = 2\pi/\lambda$
h = height of horizontal dipole above ground
α = elevation angle
$\rho_\perp$ = reflection coefficient
$2\beta h \sin\alpha$ = path length difference of direct and reflected rays, rad

In its complex form

$$\varepsilon_r = \varepsilon_r' - j\varepsilon_r'' = \varepsilon_r' - j\frac{\sigma}{\omega\varepsilon_0} \tag{4}$$

From (2) we note that if the ground is: (1) perfectly conducting ($\sigma = \infty$) or (2) a lossless dielectric ($\sigma = 0$) of large relative permittivity ($\varepsilon_r' \gg 1$),

$$\rho_\perp = -1 \tag{5}$$

and the patterns for the two cases are the same.

Consider now the situation for *vertical polarization.* Referring to Fig. 21–4, the electric field from the short vertical dipole is parallel (‖) to the page or plane of incidence. Assuming that the ground is nonmagnetic ($\mu = \mu_0$), it may be shown that the reflection coefficient ($\rho_\parallel$) is given by (Kraus (1) and Fleisch)

$$\rho_\parallel = \frac{\varepsilon_r \sin\alpha - \sqrt{\varepsilon_r - \cos^2\alpha}}{\varepsilon_r \sin\alpha + \sqrt{\varepsilon_r - \cos^2\alpha}} \tag{6}$$

If $\varepsilon_r \gg 1$, (6) reduces to

$$\rho_\parallel = \frac{n\sin\alpha - 1}{n\sin\alpha + 1} \tag{7}$$

where $n = \sqrt{\varepsilon_r}$ = index of refraction

As with the horizontal dipole, the field at a large distance is the resultant of the direct ray from the vertical dipole and the ray reflected from the ground, as suggested in Fig. 21–4,

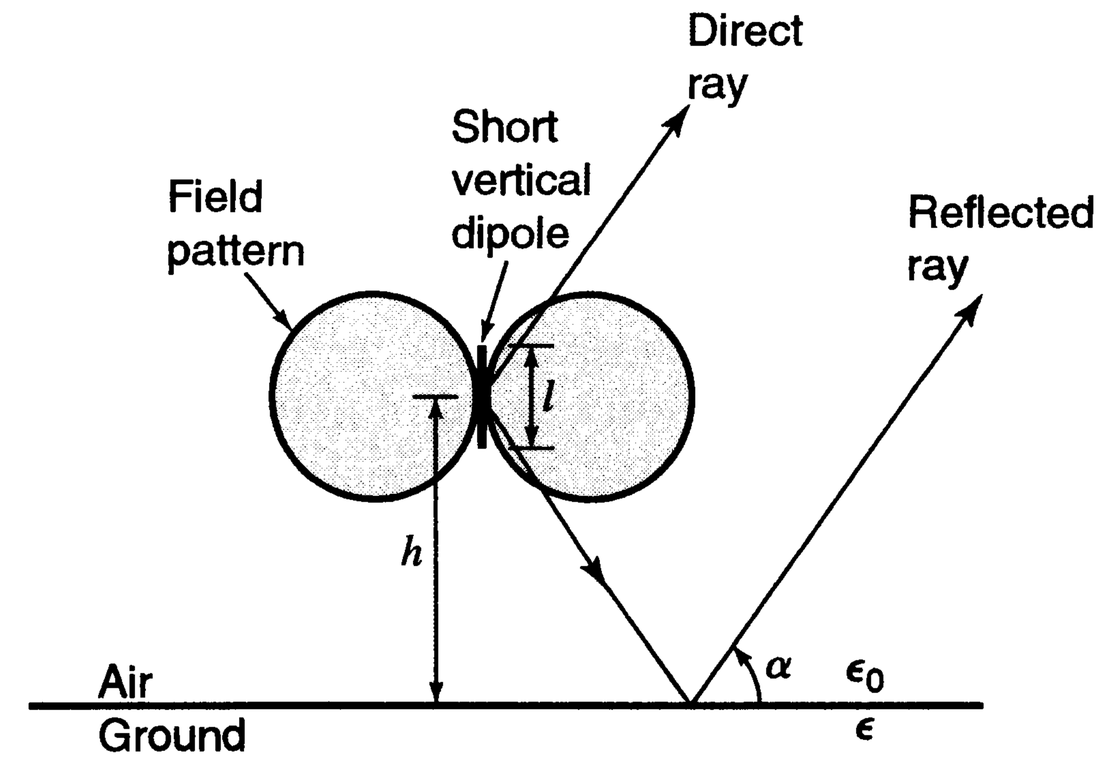


Figure 21–4
Short vertical dipole at height h above a flat ground.

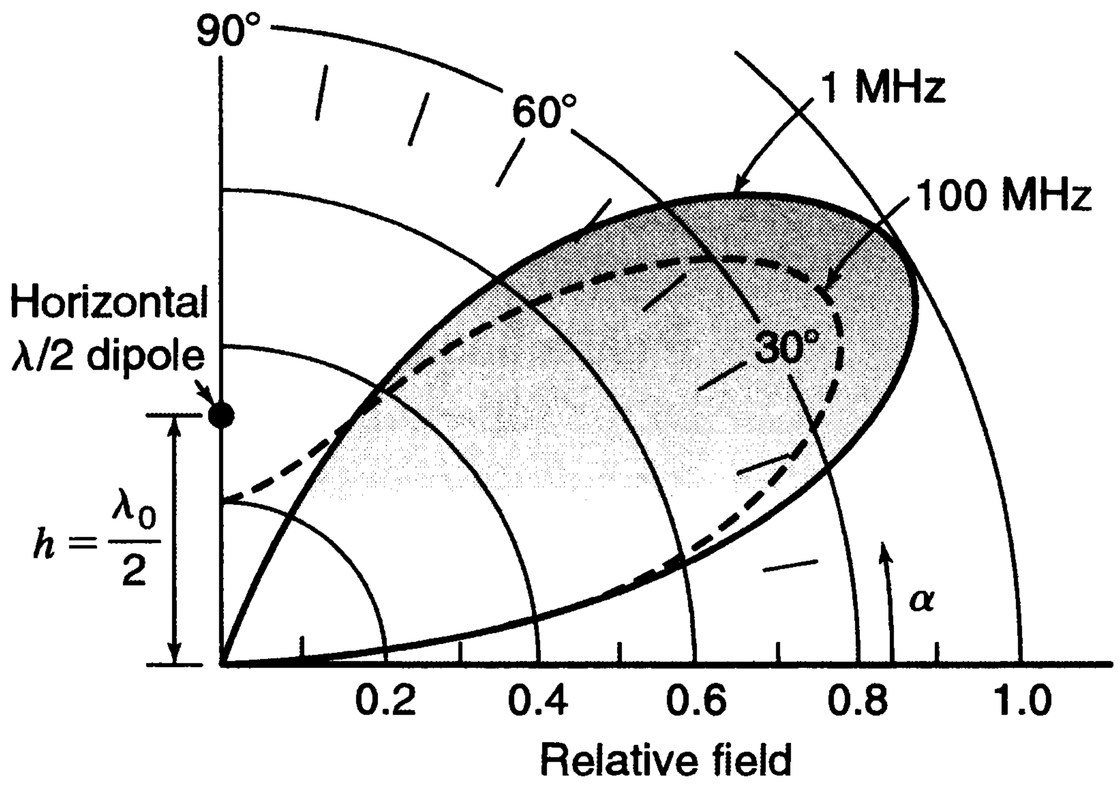


Figure 21–5
Vertical plane patterns for Example 21–4.1 of horizontal $\lambda/2$ dipole $\lambda/2$ above a flat earth with $\varepsilon_r' = 16$ and $\sigma = 10^{-2}$ ℧ m^{-1} at 1 MHz (solid curve) and at 100 MHz (dashed curve). The pattern for perfectly conducting ground ($\sigma = \infty$) is essentially the same as the solid curve. Patterns to left ($90° < \alpha < 180°$) are mirror images.

as given by

$$E_{\parallel} = \cos\alpha[1 + p_{\parallel} \angle 2\beta h \sin\alpha] \tag{8}$$

where

$\beta = 2\pi/\lambda$
h = height of center of short vertical dipole above ground
α = elevation angle
$\rho_{\parallel}$ = reflection coefficient
$2\beta h \sin\alpha$ = path length difference of direct and reflected rays, rad

From (7) we note that if the ground is: (1) perfectly conducting ($\sigma = \infty$) or (2) a lossless dielectric ($\sigma = 0$) of large relative permittivity ($\varepsilon_r' \gg 1$),

$$\rho_{\parallel} = +1 \tag{9}$$

and the patterns for the two cases are similar, except at small values of $\sin\alpha$.

To illustrate the significance of the above relations let us consider several important cases.[1]

EXAMPLE 21–4.1

A horizontal $\lambda/2$ dipole is situated $\lambda/2$ above a homogeneous flat ground of rich, clay soil typical of many midwestern U.S. states ($\varepsilon_r' = 16, \sigma = 10^{-2}$ ℧ m^{-1}). Calculate and plot the vertical-plane electrical field pattern broadside to the dipole at (*a*) 1 MHz and (*b*) 100 MHz.

■ **Solution**

From (4), the loss term of the relative permittivity at 1 MHz is equal to

$$\varepsilon_r'' = \frac{\sigma}{\omega\varepsilon_0} = \frac{10^{-2}}{2\pi \times 10^6 \times 8.85 \times 10^{-12}} = 180$$

which is large compared to the dielectric term $\varepsilon_r' = 16$. However, at 100 MHz, $\varepsilon_r'' = 1.8$ and is small compared to $\varepsilon_r' = 16$.

(*a*) *1 MHz case.* Putting $\varepsilon_r = 16 - j180$ in (2) for $p_\perp$ as a function of the elevation angle α and then in (3) with $h = \lambda/2$, the electric field pattern as a function of α is given by the solid curve in Fig. 21–5. The loss permittivity ε_r'' is sufficiently large at 1 MHz that the pattern is essentially the same as for a perfectly conducting ground.

(*b*) *100 MHz case.* Putting $\varepsilon_r = 16 - j1.8$ in (2) for $\rho_\perp$ as a function of α and then in (3) with $h = \lambda/2$, the pattern is as shown by the dashed curve in Fig. 21–5.

Comparing the 2 cases, we note that at 1 MHz the pattern has a vertical null (at $\alpha = 90°$) which is filled in at 100 MHz and also that the gain is greater at the lower frequency (up 1 dB at $\alpha = 30°$).

EXAMPLE 21–4.2

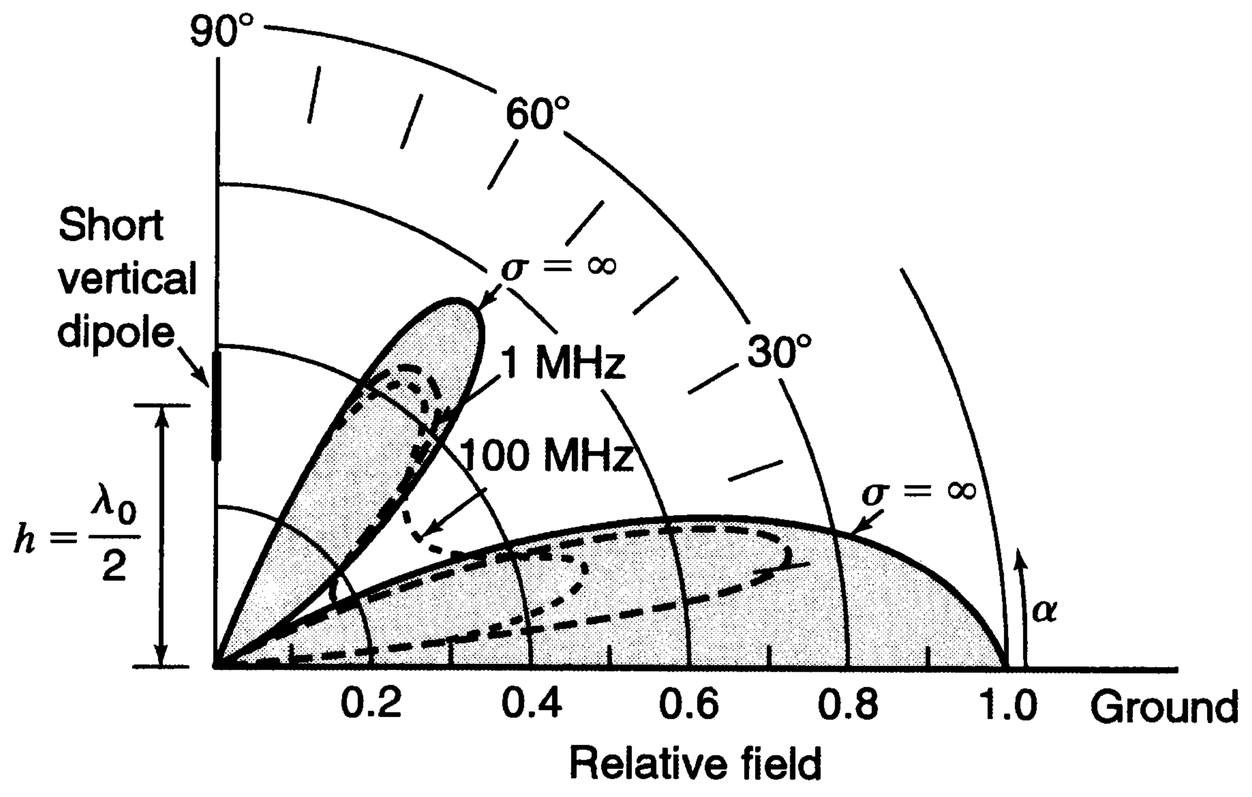


Figure 21–6
Vertical plane patterns for Example 21–4.2 of short vertical dipole $\lambda/2$ above a flat earth with $\varepsilon_r' = 16$ and $\sigma = 10^{-2}$ ℧ m^{-1} at 1 MHz (dashed curve) and at 100 MHz (dotted curve). The pattern for perfectly conducting ground ($\sigma = \infty$), as shown by the solid curve, is the same at all frequencies. Patterns to left ($90° < \alpha < 180°$) are mirror images.

[1]For table of ground and water constants, see App. A, Sec. A–6.

A short vertical dipole ($l \leq \lambda/10$) [$E(\theta) = \cos\alpha$] is located $\lambda/2$ above a ground with the same constants as for Example 21–4.1. Calculate and plot the electric field pattern at (*a*) 1 MHz and (*b*) 100 MHz.

■ **Solution**

(*a*) *1 MHz case.* Putting $\varepsilon_r = 16 - j180$ in (7) for $\rho_\parallel$ as a function of α and then in (8) with $h = \lambda/2$, the pattern is given by the dashed curve in Fig. 21–6.

(*b*) *100 MHz case.* Putting $\varepsilon_r = 16 - j1.8$ in (7) for $\rho_\parallel$ and then in (8) with $h = \lambda/2$, the pattern is as shown by the dotted curve in Fig. 21–6. The solid pattern is for perfectly conducting ground ($\sigma = \infty$) (same at all frequencies).

EXAMPLE 21–4.3

Repeat Example 21–4.2 for the case where the vertical dipole is at the ground (vertical monopole) so that we may set $h = 0$.

■ **Solution**

Since $h = 0$, (8) reduces to

$$E = \cos\alpha(1 + \rho_\parallel) \tag{10}$$

and introducing (7),

$$E = \cos\alpha \frac{2n \sin\alpha}{n \sin\alpha + 1} \tag{11}$$

Evaluating (11) for the 2 frequencies results in the curves shown in Fig. 21–7. The pattern for 1 MHz is given by the dashed curve and at 100 MHz by the dotted curve. The solid pattern is for perfectly conducting ground ($\sigma = \infty$).

At 100 MHz ($\varepsilon_r'' = 1.8$), the field along the ground ($\alpha = 0°$) is zero and the maximum field is down almost 5 dB from the field for perfectly conducting ground ($\sigma = \infty$). At

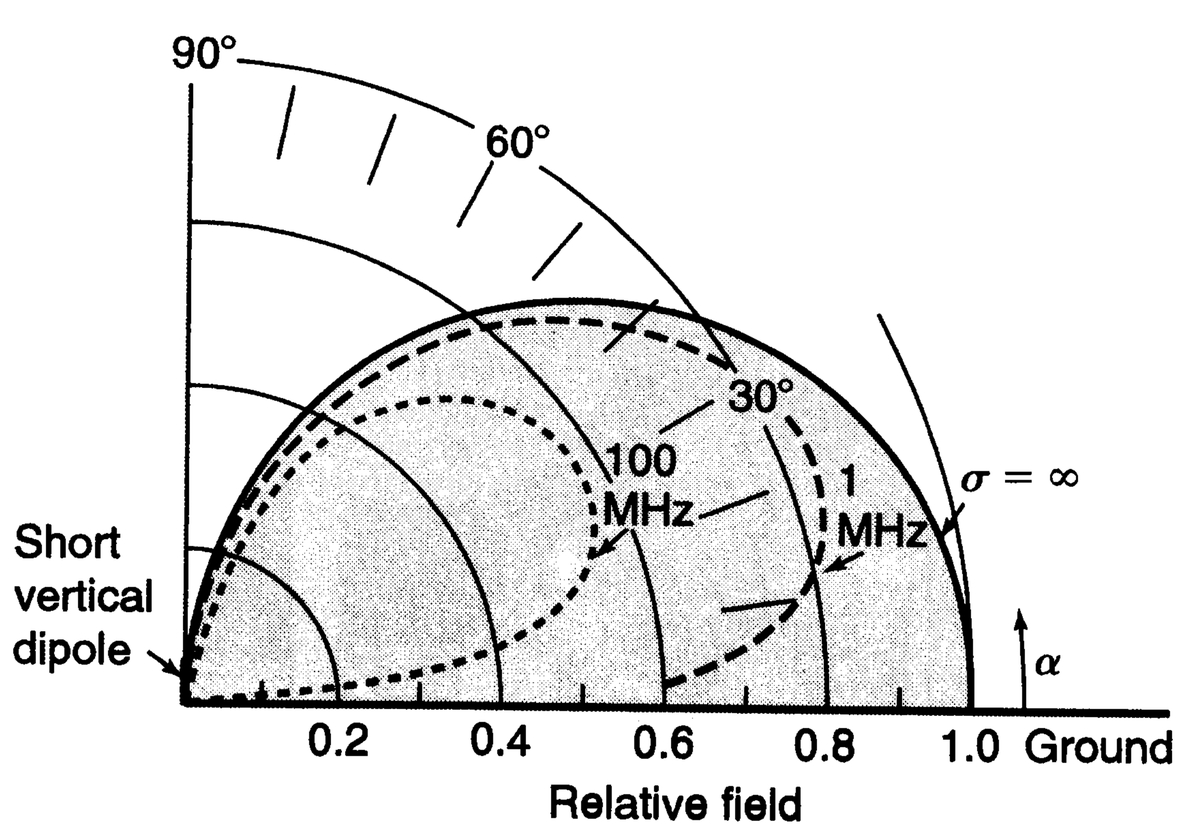


Figure 21–7
Vertical plane patterns for Example 21–4.3 of short vertical dipole at the surface of a flat earth with $\varepsilon_r' = 16$ and $\sigma = 10^{-2}$ ℧ m^{-1} at 1 MHz (dashed curve) and at 100 MHz (dotted curve). The pattern for perfectly conducting ground (same at all frequencies) is shown by the solid curve. Patterns to left ($90° < \alpha < 180°$) are mirror images.

1 MHz ($\varepsilon_r'' = 180$), the situation is intermediate with the maximum field down about 2 dB from the perfectly conducting case. For higher ground conductivity the pattern approaches the $\sigma = \infty$ curve.

The approximate solutions in the above examples involve the assumption that $\varepsilon_r \gg 1$. More rigorous, and necessarily more complex, studies have been conducted, beginning with Arnold Sommerfeld's classic solution of 1909 (Sommerfeld-1). Nevertheless, our examples illustrate some of the principal changes caused by typical ground as compared to perfectly conducting ground (Terman-1; King-1, 2; Lindell-1).

Let us summarize some of the principal differences of nonperfectly versus perfectly conducting ground. With a perfectly conducting ground the reflected-ray amplitude is equal to the direct-ray amplitude (Fig. 21–4), which means that in some directions the 2 fields may add in phase, doubling the field (quadrupling the power) for a 6-dB gain. On the other hand, in some other directions the 2 rays may be out of phase or cancel, resulting in zero field (zero power) for an infinite dB loss. Thus, with a perfectly conducting ground, there is the possibility of anything from a +6 dB to a $-\infty$ dB change from the free-space condition.

With nonperfectly conducting ground the reflected-ray amplitude tends to be *less than* the direct-ray amplitude. Thus, in directions for which the fields add in phase, the maximum gain is less than 6 dB but in directions for which the fields are out of phase there tends not to be complete signal cancellation (a null) except at low frequencies. Referring to Figs. 21–5, 21–6 and 21–7, the above noted trends are apparent, with less gain and filled nulls in Figs. 21–5 and 21–6 and also much reduced field strength along the ground ($\alpha = 0°$) for the vertical antennas in Figs. 21–6 and 21–7.

Comparing Figs. 21–5 and 21–6, we note that for perfectly conducting ground the maximum for horizontal polarization is at an elevation angle of 30° with a null at 0°, while for vertical polarization the null is at 30° with the maximum at 0°. If both the vertical and horizontal antennas are short dipoles, $\lambda/2$ above perfectly conducting ground, and are connected together as a George Brown turnstile to transmit circular polarization, a circularly polarized antenna will receive a constant signal as a function of elevation between 0 and 30° (no nulls, no maxima) but 6 dB below the vertical or horizontal maxima.

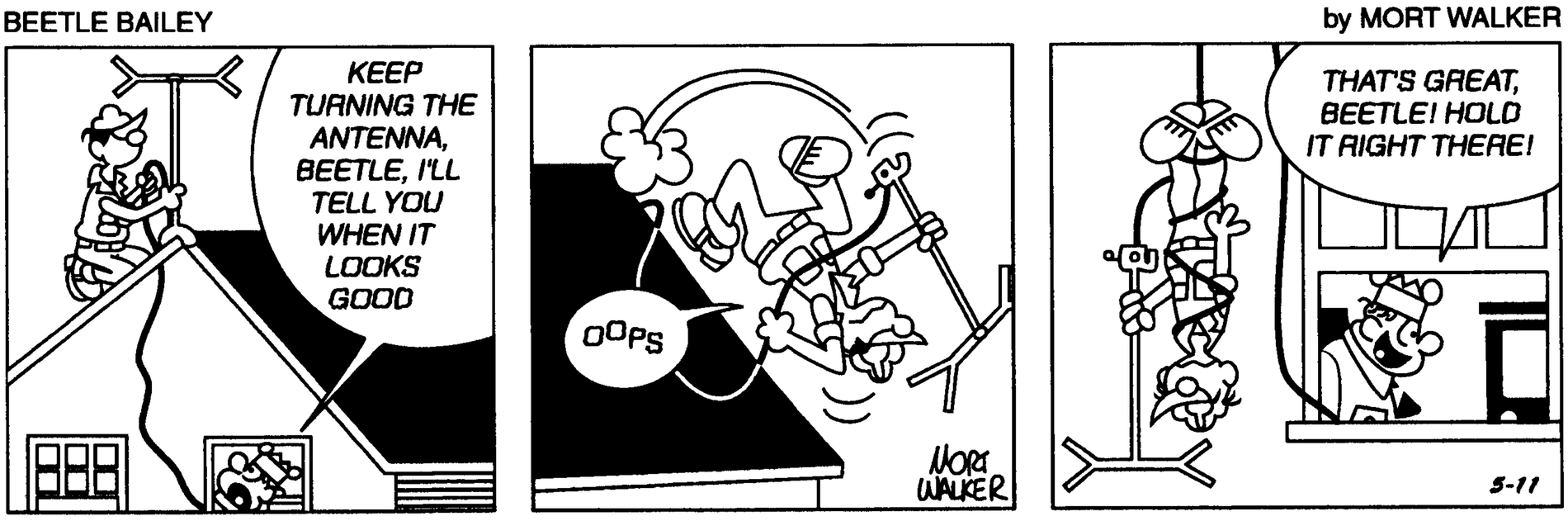


Figure 21–8
Siting the antenna. (*Reprinted with special permission of King Features Syndicate, Inc.*)

In the above discussion it is assumed that the direct and reflected rays are parallel (distance very large). However, if the receiving antenna is closer so that the direct and reflected rays are *not* parallel, the received signal can fluctuate many times between maxima and nulls as a function of height, linear polarization being assumed. In general, avoiding a null with linear polarization may require either raising or *lowering* the receiving antenna (see Prob. 21–10–1, the accompanying figure and solution). Thus, although the siting procedure of Fig. 21–8 is unorthodox, the result has some credibility.

Another siting effect occurs when, for example, a Yagi-Uda array is located less than $\lambda/2$ above ground, the proximity detuning the elements and reducing the gain.

21–5 GROUND-PLANE ANTENNAS（带接地面的天线）

Several types of ground-plane or related antennas are shown in Fig. 21–9. The type at (*a*) has a vertical $\lambda/4$ stub with a circular-sheet ground plane about $\lambda/2$ in diameter. The antenna is fed by a coaxial transmission line with the inner conductor connected to the $\lambda/4$ stub and the outer conductor terminating in the ground plane. In (*b*) the ground plane has been modified to a skirt or cone. By replacing the $\lambda/4$ stub with a disk as in (*c*), a *Kandoian discone* antenna (Kandoian-1, 2) is obtained. The dimensions given are appropriate for the center frequency of operation. In Fig. 21–9*d* the solid-sheet ground plane is replaced by 4 radial conductors. A modification of this antenna is shown at (*e*) in which a short-circuited $\lambda/4$ section of coaxial line is connected in parallel with the antenna terminals.[1] This widens

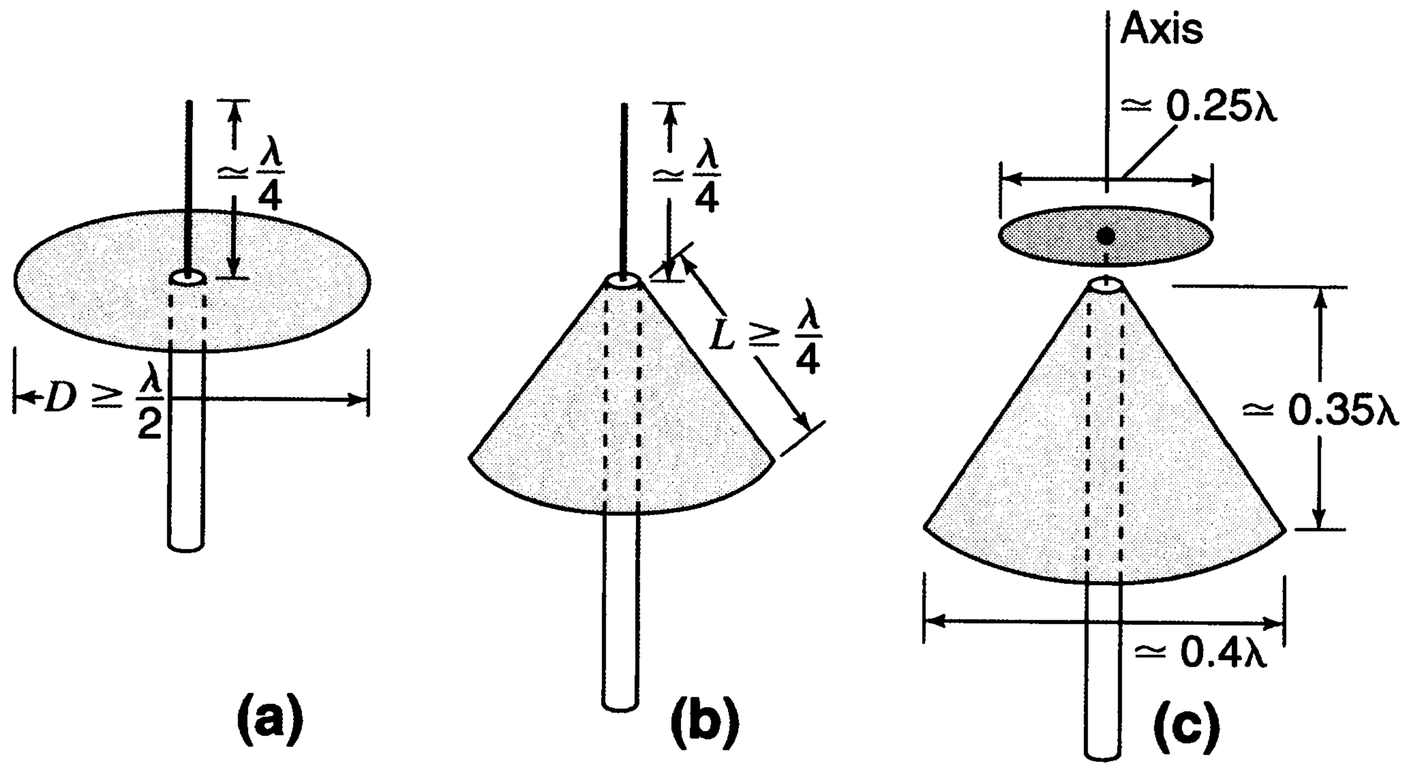


Figure 21–9
(*a*) Stub antenna with flat circular ground plane, (*b*) same antenna with ground plane modified to skirt or cone, (*c*) Kandoian discone antenna, (*d*) stub antenna with 4 radial conductors to simulate ground plane and (*e*) a method of feeding ground-plane antenna.

[1]These radial conductor ground-plane antennas (Fig. 21–9*d* and *e*) were invented in 1938 by George H. Brown, Jess Epstein and Robert Lewis.

the impedance bandwidth and also places the stub antenna at dc ground potential. This is desirable to protect the transmission line from lightning surges.

With reference to solid-sheet ground-plane antennas, it should be noted that the radiation pattern of a vertical $\lambda/4$ stub on a *finite* ground sheet differs appreciably from the pattern with an infinite sheet. This is illustrated by Fig. 21–10. The solid curve is the calculated pattern with a ground sheet of infinite extent. The dashed curve is for a sheet several wavelengths in diameter and the dotted curve for a sheet of the order of 1λ in diameter. With finite solid-sheet ground planes the maximum radiation is generally not in the direction of the ground plane but at an angle α above it. In order that maximum radiation be in the horizontal plane, the ground plane may be modified as in Fig. 21–9*b* or *c*. The maximum radiation from the Kandoian discone antenna is nearly horizontal (normal to axis) over a considerable bandwidth.

By top-loading a vertical stub antenna, it may be modified through the successive stages of Fig. 21–11 to the form in Fig. 21–11*d*. This antenna consists of a circular disk with an

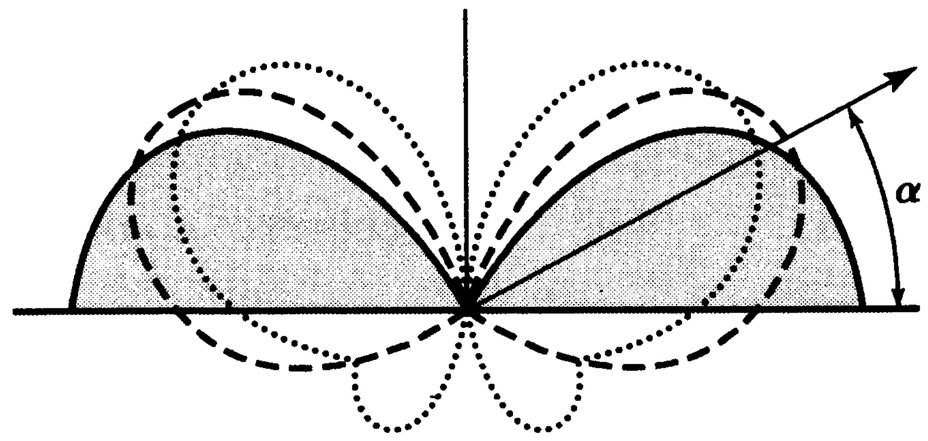
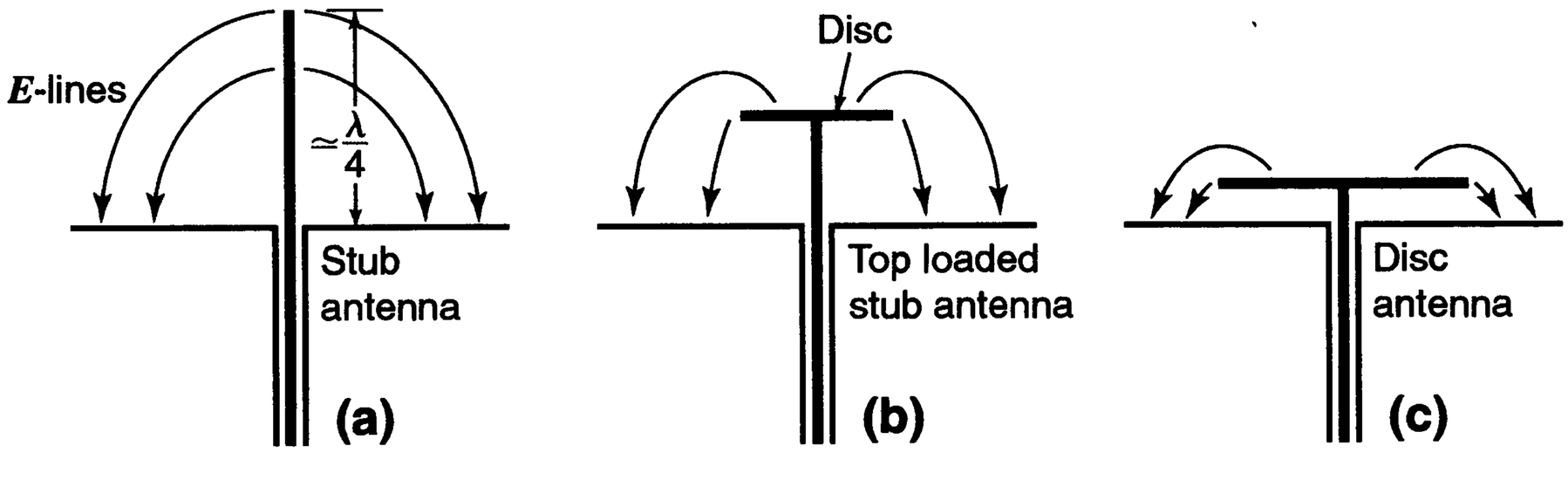


Figure 21–10
Vertical-plane patterns of $\lambda/4$ stub antenna on infinite ground plane (solid), and on finite ground planes several wavelengths in diameter (dashed) and about 1λ in diameter (dotted).



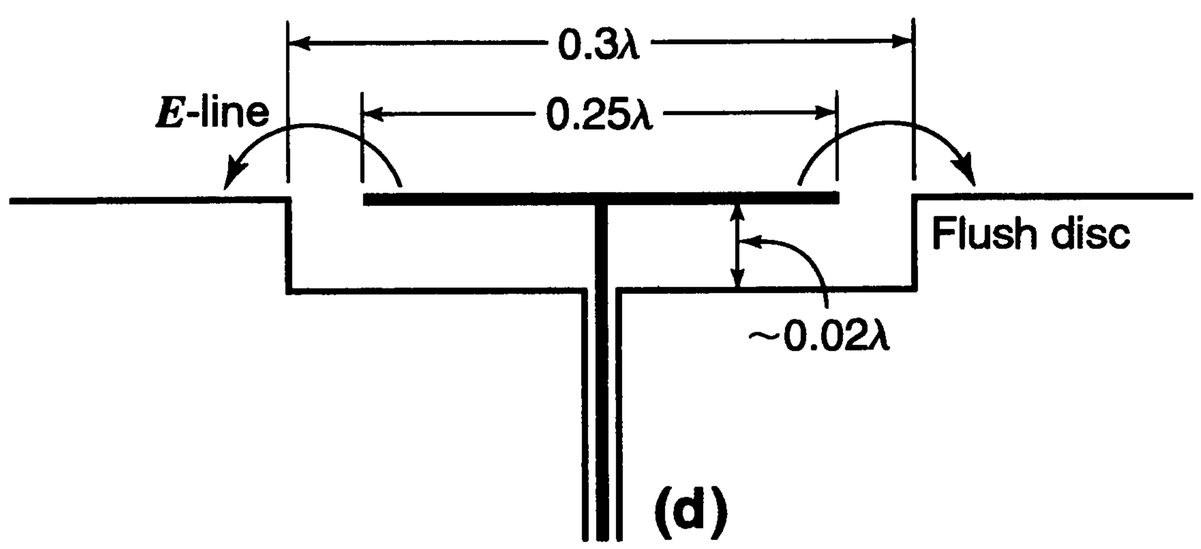


Figure 21–11
Evolution of flush-disk antenna from vertical $\lambda/4$ stub antenna.

annular slot between it and the ground plane. The ground plane is depressed below the disk forming a shallow cavity (Pistolkors-1; Rhodes-1). The radiation pattern of the antenna at (*d*) is quite similar to the pattern for the vertical stub at (*a*).

21–6 SLEEVE ANTENNAS（套筒天线）

Carrying the ground-plane modification of Fig. 21–10*b* a step further results in the vertical $\lambda/2$ sleeve antenna of Fig. 21–12*a*. Here the ground plane has degenerated into a sleeve or cylinder $\lambda/4$ long. Maximum radiation is normal to the axis of this antenna.

Another variety of sleeve antenna is illustrated in Fig. 21–12*b* (Block-1). The antenna is similar to a stub antenna with ground plane but with the feed point moved to approximately the center of the stub. This is accomplished by enclosing the lower end of the stub in a cylindrical sleeve. By varying the characteristic impedance of this $\lambda/8$ section, some control is afforded over the impedance presented to the coaxial line at the ground plane.

A balanced-sleeve dipole antenna corresponding to the sleeve stub type of Fig. 21–12*b* is illustrated in Fig. 21–12*c*. It is shown with a coaxial line feed and balance-to-unbalance

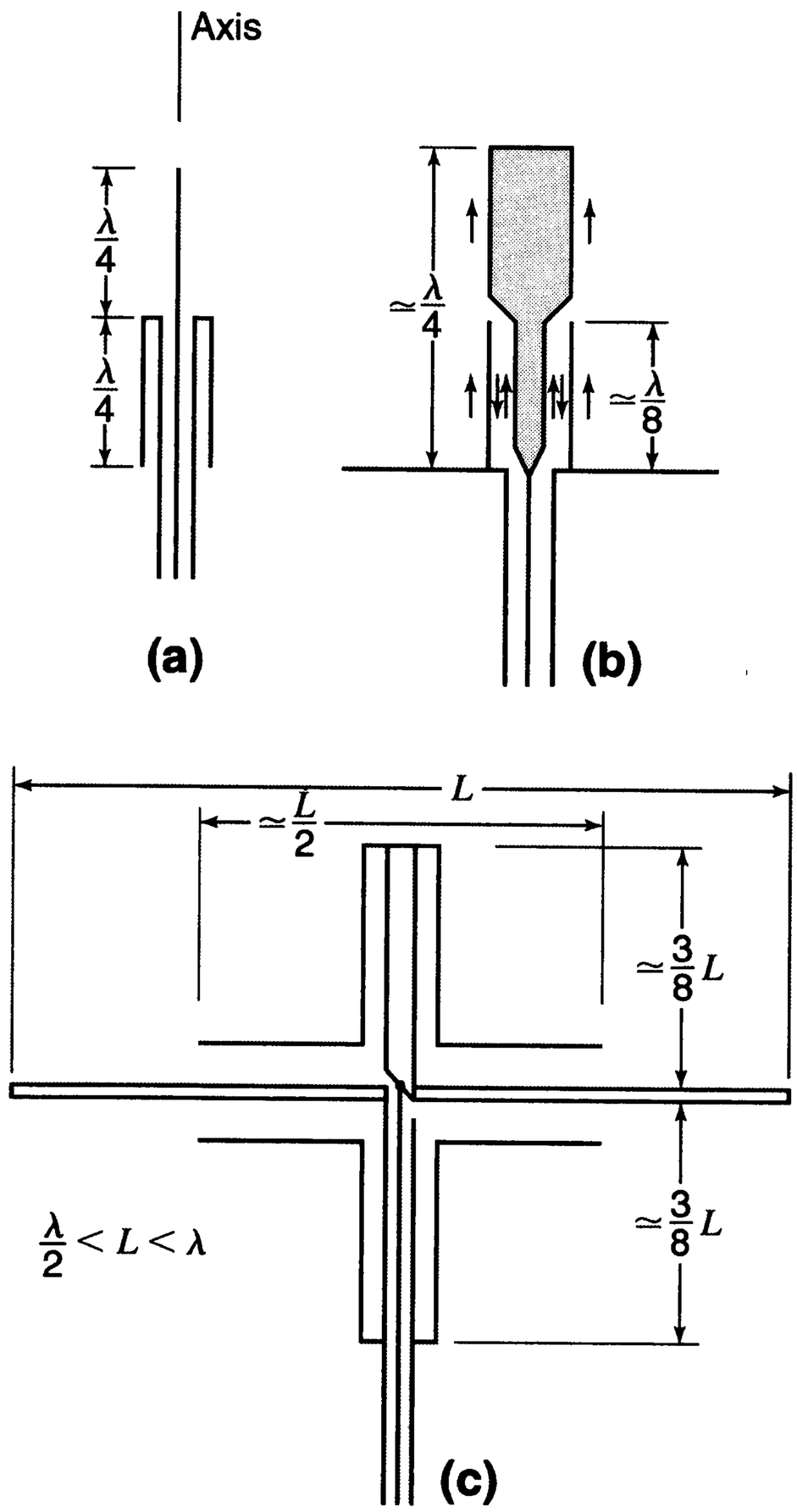


Figure 21–12
(*a*) $\lambda/2$ sleeve antenna, (*b*) sleeve antenna above ground plane and (*c*) balanced-sleeve antenna.

transformer or balun. This antenna may be operated over a frequency range of about 2 to 1 such that L is in the range from about $\frac{1}{2}$ to 1λ.

21–7 TURNSTILE ANTENNA（旋转场天线）

Consider 2 crossed infinitesimal dipoles energized with currents of equal magnitude but in phase quadrature. This arrangement, shown in plan view in Fig. 21–13*a*, produces a circular pattern in the θ plane since the field pattern E as a function of θ and time is given by

$$E = \sin\theta \cos\omega t + \cos\theta \sin\omega t \tag{1}$$

which reduces to

$$E = \sin(\theta + \omega t) \tag{2}$$

At any value of θ the maximum amplitude of E is unity at some instant during each cycle. Hence, the rms field pattern is circular, as shown by the circle in Fig. 21–13*b*. At any instant of time the pattern is a figure-of-eight of the same shape as for a single infinitesimal dipole. An instantaneous pattern is shown in Fig. 21–13*b* for $\omega t = 135°$. As a function of time this pattern rotates completing 1 revolution per cycle. In the case being considered in Fig. 21–13, the pattern rotates clockwise. Thus, the phase of the field as a function of θ is given by $\theta + \omega t =$ constant and, if the constant is zero, by

$$\omega t = -\theta \tag{3}$$

If the field is a maximum in the direction $\theta = 0$ at a given instant, then according to (3) the field is a maximum in the $\theta = -45°$ direction $\frac{1}{8}$-period later.

The above discussion concerns the field in the θ plane (plane of the crossed dipoles). The field in the axial direction (normal to the crossed infinitesimal dipoles) has a constant magnitude given by

$$|E| = \sqrt{\cos^2\omega t + \sin^2\omega t} = 1 \tag{4}$$

Thus, the field normal to the infinitesimal dipoles is circularly polarized. In the case being considered in Fig. 21–13 the field rotates in a clockwise direction.

Replacing the infinitesimal dipoles by $\lambda/2$ dipoles results in a practical type of antenna with approximately the same pattern characteristics. This kind of antenna is a *George Brown turnstile antenna* (Brown-1). Since the pattern of a $\lambda/2$ element is slightly sharper than for an infinitesimal dipole, the θ-plane pattern of the turnstile with $\lambda/2$ elements is not quite

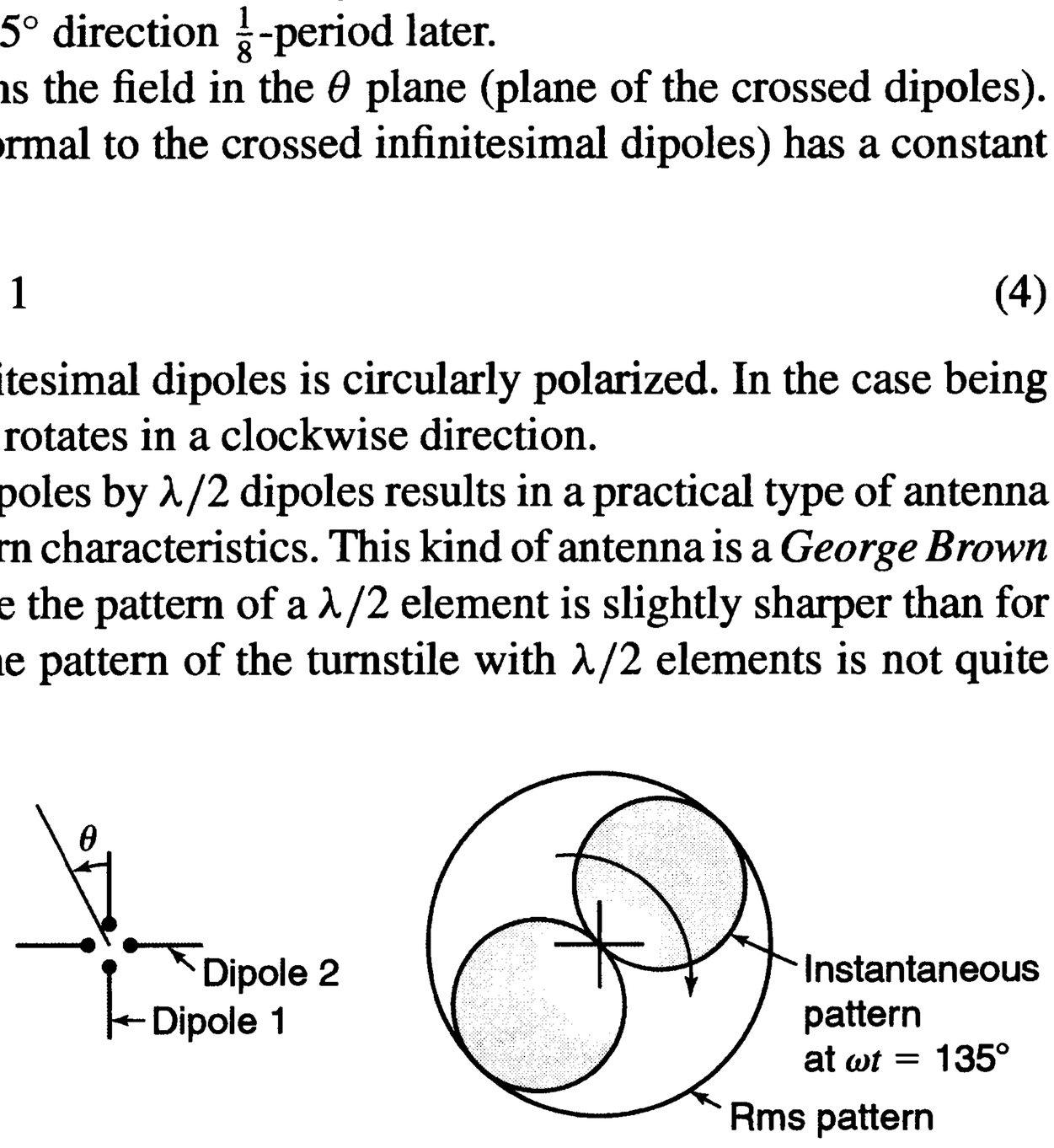


Figure 21–13
George Brown turnstile with short (infinitesimal) dipoles.

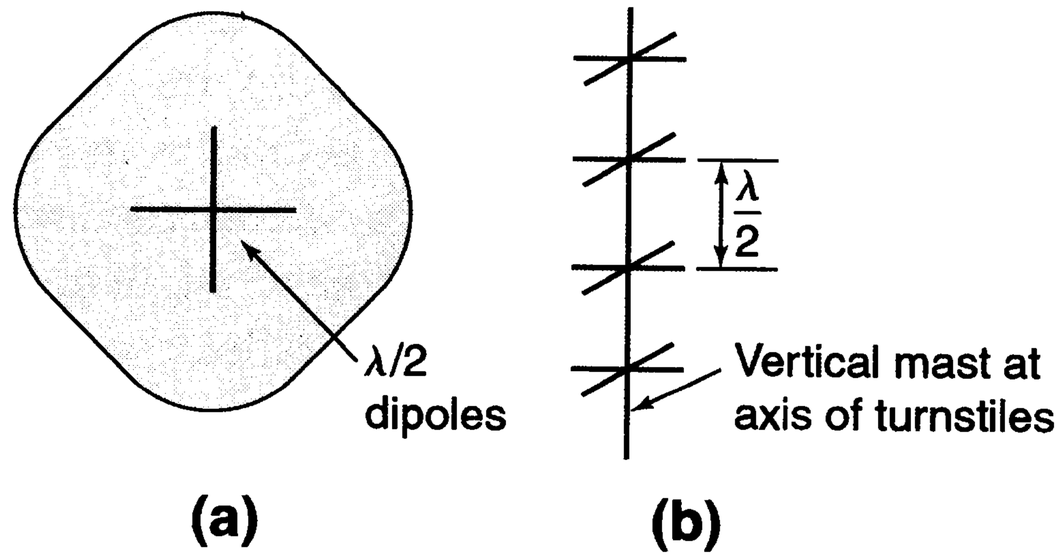


Figure 21–14
George Brown turnstile with $\lambda/2$ dipoles.

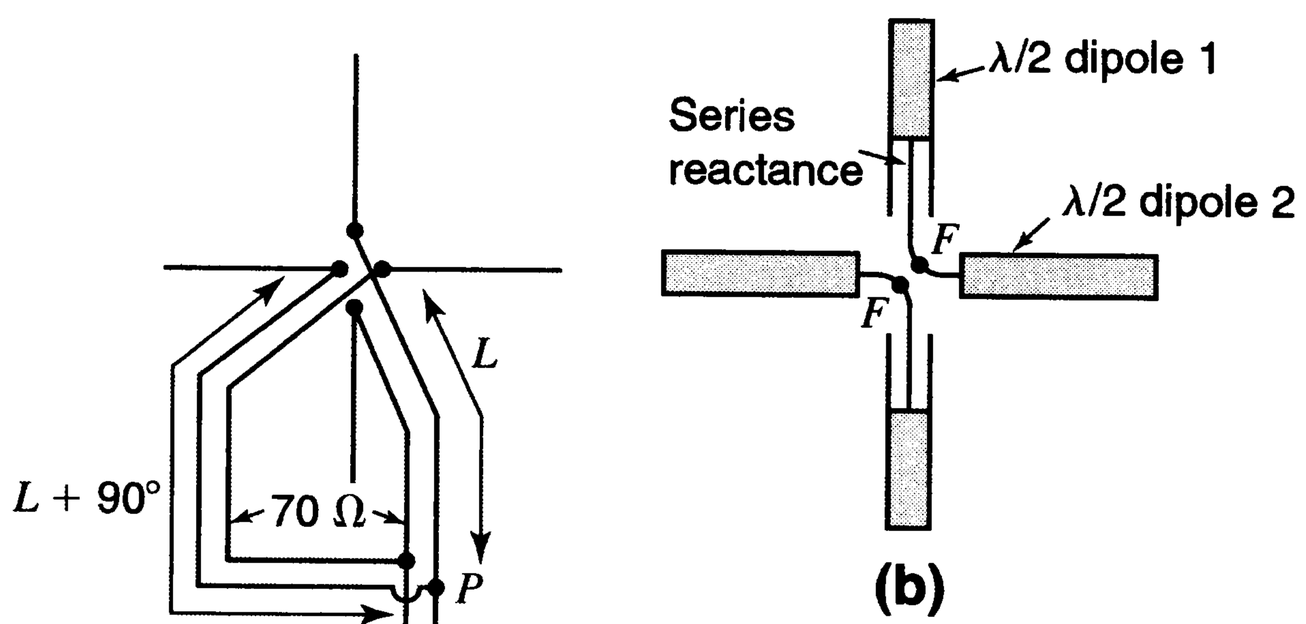


Figure 21–15
Arrangements for feeding turnstile antennas.

circular but departs from a circle by about ±5 percent. The relative pattern is shown in Fig. 21–14*a*. The relative field as a function of θ and time is expressed by

$$E = \frac{\cos(90^\circ \cos\theta)}{\sin\theta} \cos\omega t + \frac{\cos(90^\circ \sin\theta)}{\cos\theta} \sin\omega t \tag{5}$$

Although the θ-plane pattern with $\lambda/2$ elements differs from the pattern with infinitesimal dipoles, the radiation is circularly polarized in the axial direction from the $\lambda/2$ elements provided that the currents are equal in magnitude and in phase quadrature.

A turnstile antenna may be conveniently mounted on a vertical mast. The mast is coincident with the axis of the turnstile. To increase the vertical plane directivity, several turnstile units can be stacked at about $\lambda/2$ intervals as in Fig. 21–14*b*. The arrangement at (*b*) is called a "4-bay" turnstile. It requires 2 bays to obtain a field intensity approximately equal to the maximum field from a single $\lambda/2$ dipole with the same power input.

In order that the currents on the $\lambda/2$ dipoles be in phase quadrature, the dipoles may be connected to separate nonresonant lines of unequal length. Suppose, for example, that the terminal impedance of each dipole in a single-bay turnstile antenna is $70 + j0\ \Omega$. Then by connecting 70-Ω lines (dual coaxial type), as in the schematic diagram of Fig. 21–15*a*, with the length of one line 90 electrical degrees longer than the other, the dipoles will be driven with currents of equal magnitude and in phase quadrature. By connecting a 35-Ω line between the junction point P of the two 70-Ω lines and the transmitter, the entire transmission-line system is matched.

Another method of obtaining quadrature currents is by introducing reactance in series with one of the dipoles (Brown-2 and Epstein). Suppose, for example, that the length and diameter of the dipoles in Fig. 21–15*b* result in a terminal impedance of $70 - j70\ \Omega$. By introducing a series reactance (inductive) of $+j70\ \Omega$ at each terminal of dipole 1 as in Fig. 21–15*b*, the terminal impedance of this dipole becomes $70 + j70\ \Omega$. With the 2 dipoles connected in parallel, the currents are

$$I_1 = \frac{V}{70 + j70}$$

and

$$I_2 = \frac{V}{70 - j70} \tag{6}$$

where

V = impressed emf
I_1 = current at terminals of dipole 1
I_2 = current at terminals of dipole 2

Thus,

$$I_1 = \frac{V}{99}\angle -45^\circ$$

and

$$I_2 = \frac{V}{99}\angle +45^\circ \tag{7}$$

so that I_1 and I_2 are equal in magnitude but I_2 leads I_1 by 90°. The 2 impedances in parallel yield

$$Z = \frac{1}{Y} = \frac{1}{[1/(70 + j70)] + [1/(70 - j70)]} = 70 + j0 \qquad (\Omega) \tag{8}$$

so that a 70-Ω (dual coaxial) line will be properly matched when connected to the terminals FF.

21–8 SUPERTURNSTILE ANTENNA（超旋转场天线）

In order to obtain a very low VSWR over a considerable bandwidth, the turnstile described above has been modified by Masters to the form shown in the photograph of Fig. 21–16. In this arrangement, or *Masters superturnstile,* the simple dipole elements are replaced by flat sheets or their equivalent (Masters-1).

Each "dipole" is equivalent to a slotted sheet about 0.7 by 0.5λ as in Fig. 21–17*a*. The terminals are at FF. As in the slotted cylinder antenna, the length of the slot for resonance is more than $\lambda/2$ (about 0.7λ). The dipole can be mounted on a mast as in Fig. 21–17*b*. To reduce wind resistance, the solid sheet is replaced by a grid of conductors. Typical dimensions for the center frequency of operation are shown. This arrangement gives a VSWR of about 1.1 or less over about a 30 percent bandwidth, which makes it convenient as a mast-mounted television transmitting antenna for frequencies as low as about 50 MHz.

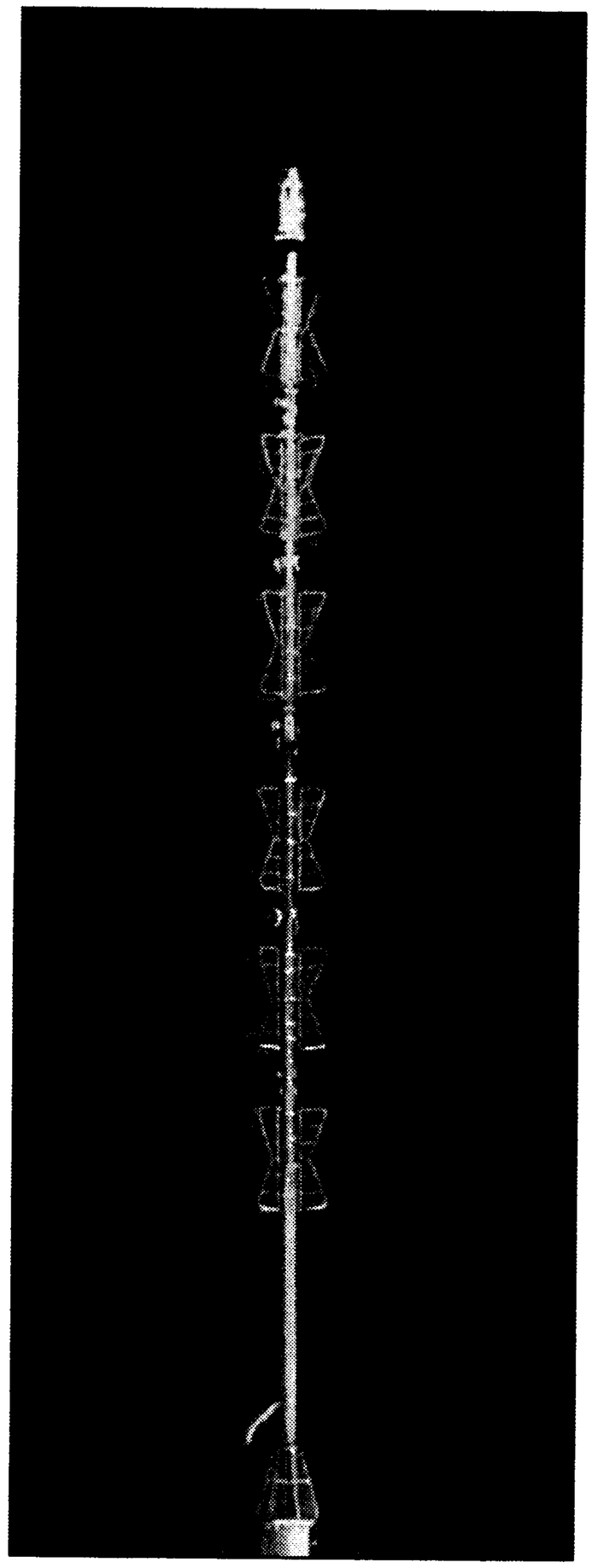


Figure 21–16
Six-bay Masters superturnstile antenna. (*Courtesy RCA.*)

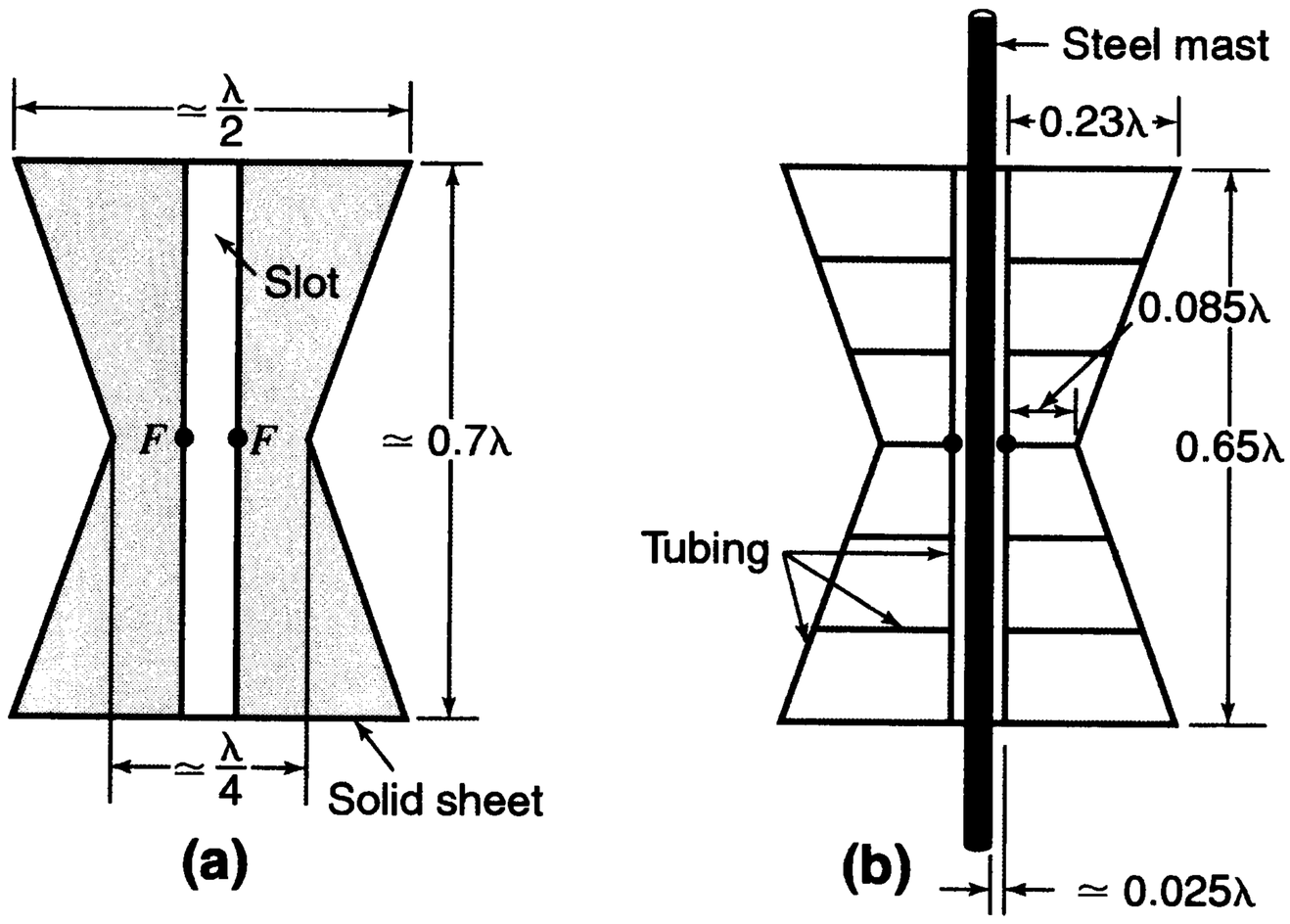


Figure 21–17
Single dipole element of Masters superturnstile antenna. (*a*) Solid-sheet construction, (*b*) tubing construction showing method of mounting on mast.

Unlike the simple turnstile there is relatively little radiation in the axial direction (along the mast), and only one bay is required to obtain a field intensity approximately equal to the maximum field from a single $\lambda/2$ dipole with the same power input. For decreased beamwidth in the vertical plane the superturnstile bays are stacked at intervals of about 1λ between centers. Impedance matching is discussed by Sato *et al.* (Sato-1).

21–9 OTHER OMNIDIRECTIONAL ANTENNAS（其他全向天线）

The radiation patterns of the slotted-cylinder and turnstile antennas are nearly circular in the horizontal plane. Such antennas are sometimes referred to as omnidirectional types, it being understood that "omnidirectional" refers only to the horizontal plane.

As shown in Chap. 7, a circular loop with a uniform current radiates a maximum in the plane of the loop provided that the diameter D is less than about 0.58λ. The pattern is doughnut shaped with a null in the axial direction as suggested by the vertical plane cross section in Fig. 21–18*a*.

One method of simulating the uniform loop is illustrated in Fig. 21–18*b*. Here 4 smaller loops are connected in parallel across a coaxial line. This arrangement is called a "cloverleaf" antenna (Smith-1). Another method is shown in Fig. 21–18*c*, 3 folded dipoles being connected in parallel across a coaxial line (Kandoian-3). A third method utilizing a square loop is illustrated in Fig. 21–18*d* (Alford-1). The terminals are at FF. The side length L

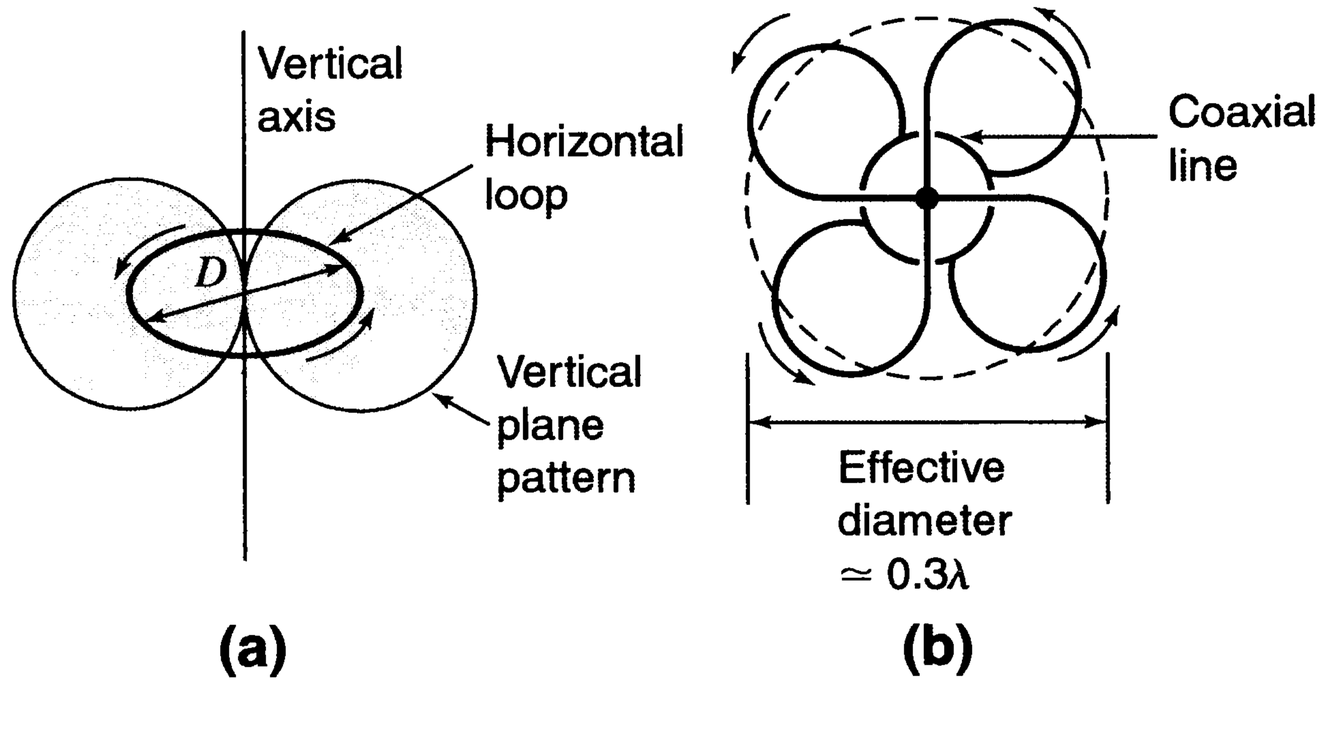


Figure 21–18
(*a*) Circular loop antenna and (*b*) approximately equivalent arrangements of "clover-leaf" type, (*c*) "triangular-loop" type and (*d*) square or Alford loop.

may be of the order of $\lambda/4$. A single equivalent loop or bay of any of these types produces approximately the same field intensity as the maximum field from a single $\lambda/2$ dipole with the same power input. For increased directivity in the vertical plane, several loops may be stacked, forming a multibay arrangement.

21–10 CIRCULARLY POLARIZED ANTENNAS（圆极化天线）

Circularly polarized radiation may be produced with various antennas. The monofilar axial-mode helical antenna (Fig. 21–19*a*) is a simple, effective type of antenna for generating circular polarization. Circular polarization may also be produced in the axial direction from a pair of crossed $\lambda/2$ dipoles with equal currents in phase quadrature (Fig. 21–19*b*) as in a turnstile antenna. If radiation in one axial direction is right-circularly polarized, it is left-circularly polarized in the opposite axial direction.

A third type of circularly polarized antenna consists of 2 in-phase crossed dipoles separated in space by $\lambda/4$ as in Fig. 21–19*c*. With this arrangement the type of circular polarization is the same in both axial directions.

Any of these 3 arrangements can serve as a primary antenna that illuminates a parabolic reflector or they can be placed within a circular waveguide so as to generate a circularly polarized TE_{11} mode wave. By flaring the guide out into a conical horn, a circularly polarized beam can be produced.

Another technique by which a circularly polarized beam may be obtained with a parabolic reflector of large focal length with respect to the diameter is with the aid of a metal grid or grating of parallel wires spaced $\lambda/8$ from the reflector and oriented at 45°

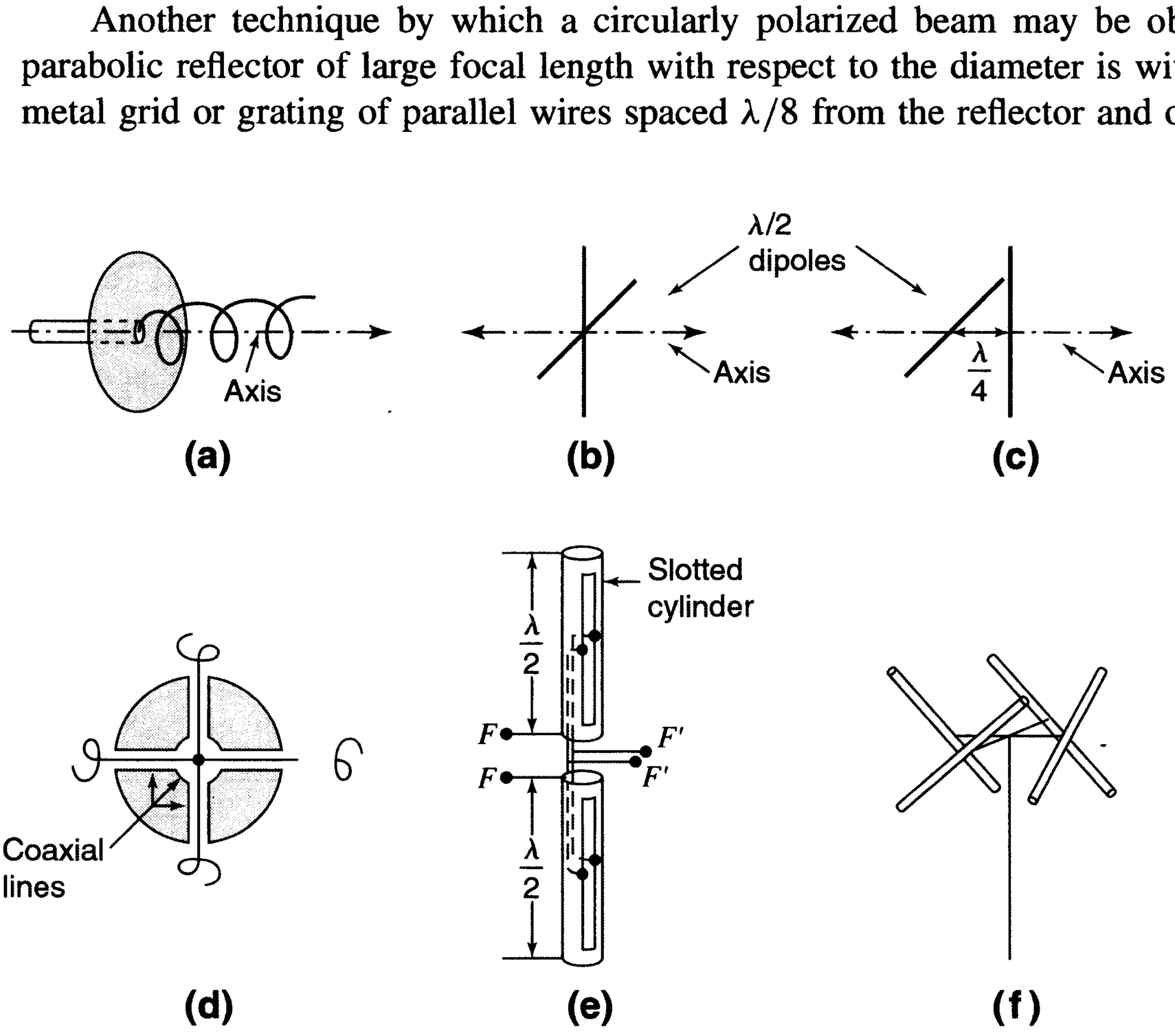


Figure 21–19
Antenna types for circular polarization.

with respect to the plane of polarization of the wave from a linearly polarized primary antenna.

Three arrangements for producing an omnidirectional pattern of circularly polarized radiation are illustrated by Fig. 21–19*d*, *e* and *f*. At (*d*) 4 short axial-mode helices of the same type are disposed around a metal cylinder with axis vertical and fed in phase from a central coaxial line (Kraus-1). In the system at (*e*) vertically polarized omnidirectional radiation is obtained from two vertical $\lambda/2$ cylinders when fed at FF and horizontally polarized omnidirectional radiation is obtained from the slots fed at $F'F'$. By adjusting the power and phasing to the 2 sets of terminals so that the vertically polarized and horizontally polarized fields are equal in magnitude and in phase quadrature, a circularly polarized omnidirectional pattern is produced (Smith-1). At (*f*) 4 in-phase $\lambda/2$ dipoles are mounted around the circumference of an imaginary circle about $\lambda/3$ in diameter (Brown-2). Each dipole is inclined to the horizontal plane as suggested in the figure.

In general, any linearly polarized wave can be transformed to an elliptically or circularly polarized wave, or vice versa, by means of a wave polarizer (Braun-1). For example, assume that a linearly polarized wave is traveling in the negative z direction in Fig. 21–20 and that the plane of polarization is at a 45° angle with respect to the positive x axis. Suppose that this wave is incident on a large grating of many dielectric slabs of depth L with air spaces between. A section of this grating is shown in Fig. 21–20. The slab spacing (in the x direction) is assumed to be a small part of a wavelength.

The incident electric field **E** can be resolved into two components, one parallel to the x axis (E_x) and the other parallel to the y axis (E_y); that is, $\mathbf{E} = \hat{x}E_x + \hat{y}E_y$. The x component (E_x) will be relatively unaffected by the slabs. However, E_y will be retarded (velocity reduced). If the depth L of the slabs is just sufficient to retard E_y by 90° in time phase behind E_x, the wave emerging from the back side of the slabs will be circularly polarized if $|E_x| = |E_y|$. Viewing the approaching wave from a point on the negative z axis, the **E** vector rotates clockwise.

If the depth of the slabs is increased to $2L$, the wave emerging from the back side will again be linearly polarized since E_x and E_y are in opposite phase, but **E** is at a negative angle of 45° with respect to the positive x axis. Increasing the slab depth to $3L$ makes the emerging wave circularly polarized but this time with a counterclockwise rotation direction

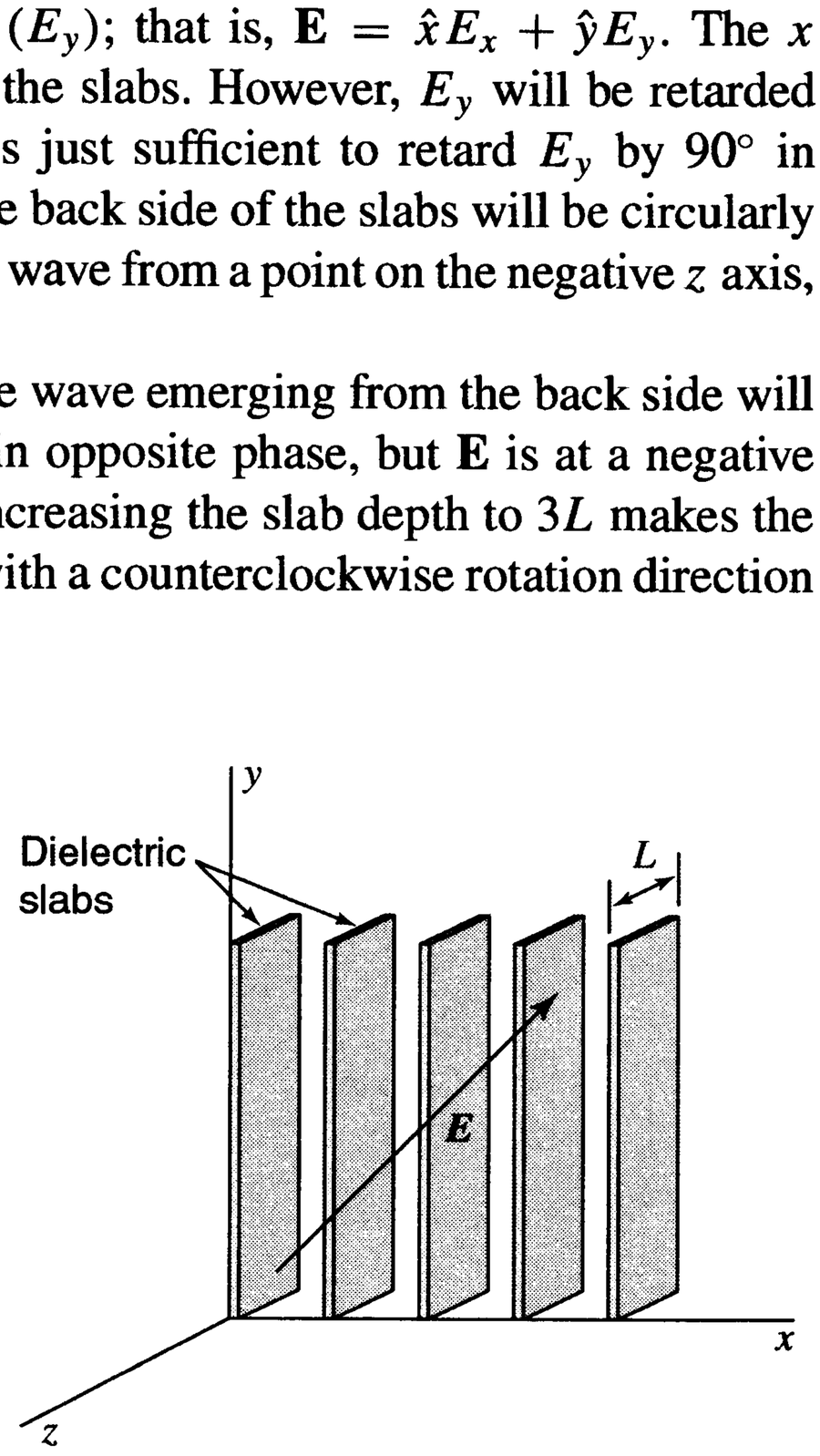


Figure 21–20
Wave polarizer.

for **E** (as viewed from a point on the negative z axis). Finally, if the slab depth is increased to $4L$, the emerging wave is linearly polarized at a slant angle of 45°, the same as the incident wave. The dielectric grating in this example behaves in a similar way to the atomic planes of a uniaxial crystal, such as calcite or rutile, to the propagation of light. For such crystals the velocity of propagation of light, linearly polarized parallel to the optic axis, is different from the velocity for light, linearly polarized perpendicular to the optic axis.

21–11 THE HIGH-GAIN OMNI（高增益全向天线）

A high-gain (high-directivity) antenna which is also omnidirectional involves a contradiction. The directivity D of an antenna is given by

$$D = \frac{4\pi}{\Omega_A} \tag{1}$$

where Ω_A = beam area

A hypothetical isotropic point source has a beam area $\Omega_A = 4\pi$, making it completely omnidirectional. From (1), its directivity $D = 1$, which is the lowest possible directivity. For the directivity (or gain of a lossless radiator) to be more than unity requires that the beam area be less than 4π so that the antenna is no longer omnidirectional. Thus, for a simple antenna, high directivity is incompatible with an omnidirectional (4π sr) pattern. The combination of both is a theoretical and a practical impossibility. However, in the digital signal-processing domain of a large phased array the combination is theoretically possible, giving an arbitrarily large number of simultaneous high-directivity beams but, due to inherent losses, not necessarily ones of high gain.

21–12 SUBMERGED ANTENNAS（埋地天线）

In (21–4–1) for the reflection coefficient $\rho_\perp$, it is assumed that the wave originates in the less-dense medium (air) and is incident on the earth or ground of relative permittivity ε_r. If the wave originates in the denser medium and travels from it into air (21–4–1) becomes

$$\rho_\perp = \frac{\sin\alpha - \sqrt{(1/\varepsilon_r) - \cos^2\alpha}}{\sin\alpha + \sqrt{(1/\varepsilon_r) - \cos^2\alpha}} \tag{1}$$

where ε_r = relative permittivity of denser medium

If $1/\varepsilon_r \leq \cos^2\alpha$, $\rho_\perp$ is complex and $|\rho_\perp| = 1$. Under this condition, the incident wave is totally reflected back into the more-dense medium.[1] When the radical in (1) is zero, $\rho_\perp = 1\angle 0°$, which defines the *critical angle* (see Fig. 21–21)

$$\alpha_c = \cos^{-1}\sqrt{\frac{1}{\varepsilon_r}} \tag{2}$$

For all angles less than α_c, $|\rho_\perp| = 1$, and the wave originating in the denser medium is reflected back from the interface. It may be shown that for $\alpha < \alpha_c$ the electric field in the

[1]This is the case for either perpendicular or parallel polarization.

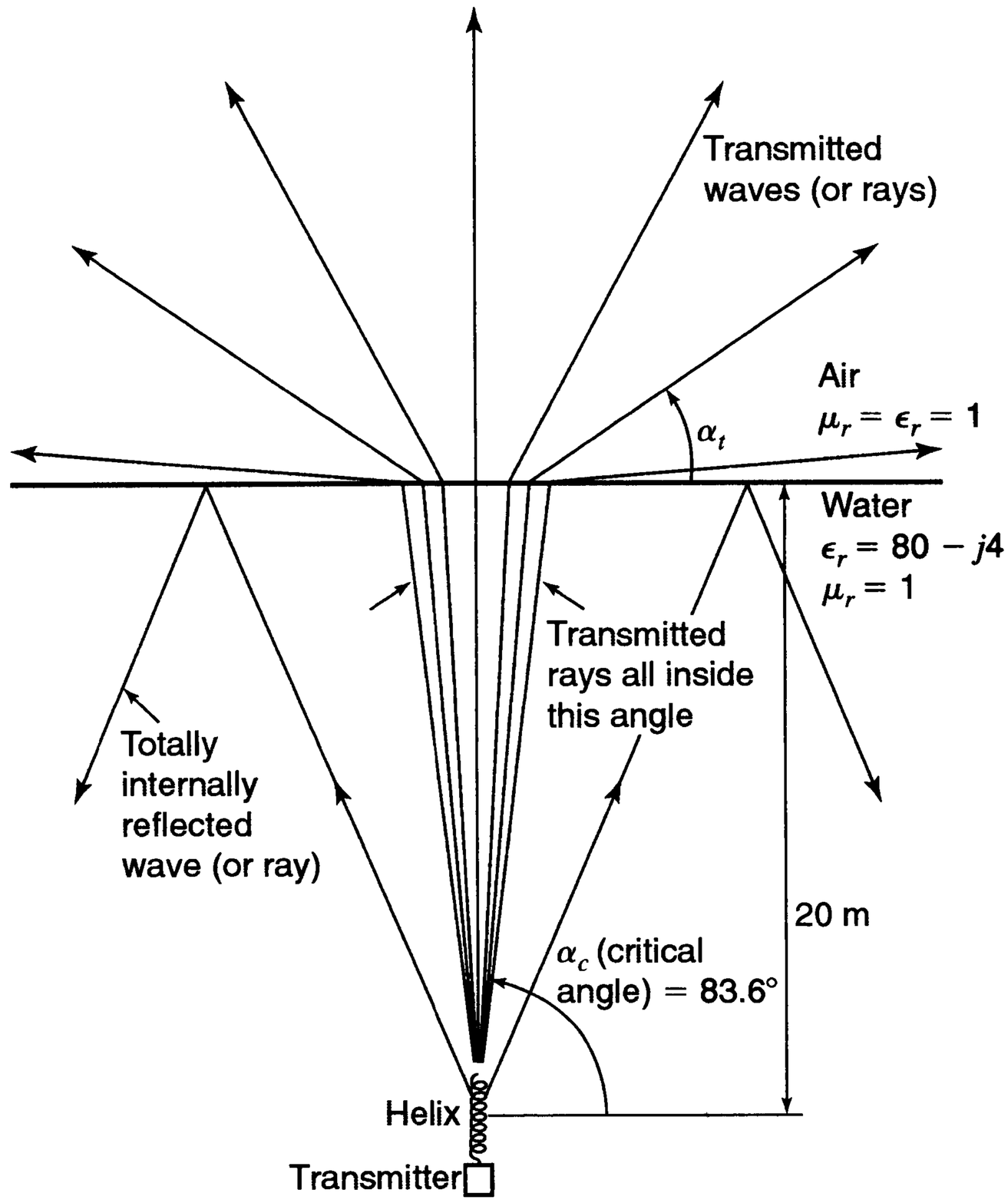


Figure 21–21
Rays from submerged antenna at angles α below 83.6° are totally internally reflected. Rays between 83.6 and 90° are transmitted through the surface into air above and spread out over almost 180°.

less-dense medium decays exponentially away from the interface(evanescent wave) and propagates without loss along the interface (surface wave) (Kraus (1) and Fleisch; King-3; Delogne-1).

EXAMPLE 21–12.1

A 60-MHz ($\lambda_0 = 5$ m) radio transmitter feeding a monofilar axial-mode helical antenna is situated 20 m below the surface of a freshwater lake with constants $\mu_r = 1$, $\varepsilon_r' = 80$ and $\sigma = 1.33 \times 10^{-2}$ ℧ m^{-1}. The helix axis is vertical. See Fig. 21–21. (*a*) For a given transmitter power how many turns should the helix have to maximize the signal at an elevation angle of 10° above the surface? (*b*) What is the relative power density radiated above the surface as a function of the elevation angle above the surface? (*c*) If the transmitter power is 100 W, what is the signal level at an elevation angle of 10° at a distance of 1 km from the submerged antenna site? The receiving antenna is a George Brown turnstile with reflector elements.

■ **Solution**

(*a*) At 60 MHz the loss term of the relative permittivity

$$\varepsilon_r'' = \frac{\sigma}{\varepsilon_0 \omega} = \frac{1.33 \times 10^{-2}}{8.85 \times 10^{-12} \times 2\pi \times 60 \times 10^6} = 4 \tag{3}$$

and the complex relative permittivity

$$\varepsilon_r = \varepsilon_r' - j\varepsilon_r'' = 80 - j4 \cdot \tag{4}$$

The water is not lossless but $\varepsilon_r' \gg \varepsilon_r''$, so neglecting ε_r'' we have from (2) that the *critical angle* is given by

$$\alpha_c = \cos^{-1}\sqrt{\frac{1}{80}} = 83.58^\circ \tag{5}$$

Thus, all rays from the helix at elevation angles less than 83.58° are reflected back into the water, and only rays at elevation angles greater than 83.58° emerge into the air, as suggested in Fig. 21–21. The hole through which the rays emerge is 12.84° wide [$= 2(90^\circ - 83.58^\circ)$] centered on the zenith ($\alpha = 90^\circ$).

From Snell's law we have approximately that

$$\cos\alpha_t \simeq \sqrt{\varepsilon_r'}\cos\alpha = 8.94\cos\alpha \tag{6}$$

where α_t = elevation angle of ray transmitted above the surface

For $\alpha_t = 10^\circ$, we have from (6) that $\alpha = 83.68^\circ$ or just 0.1° ($= 83.68^\circ - 83.58^\circ$) above the critical angle.

For a monofilar axial-mode helical antenna with number of turns $n > 3$, pitch angle $12^\circ < \alpha < 14^\circ$ and circumference $0.8 < C_\lambda < 1.15$, the beamwidth and directivity from (8–3–4) and (8–3–7) are

$$\text{HPBW} \simeq \frac{52^\circ}{C_\lambda\sqrt{nS_\lambda}} \tag{7}$$

and

$$D \simeq 12C_\lambda^2 nS_\lambda \tag{8}$$

Taking $C_\lambda = 1, \alpha = 12.5^\circ$ and $n = 12$, the HPBW $= 32.0^\circ$ and $D = 31.7$.[1] The half-power angle off axis is given by HPBW/2 $= 16.0^\circ$, as compared to 6.32° ($= 90^\circ - 83.68^\circ$) for the ray which emerges at $\alpha_t = 10^\circ$. Thus, with 12 turns the beam is broad enough that at or close to the critical angle the signal level is down from the maximum (on axis) only about 0.3 dB. Doubling the length of the helix ($n = 24$) increases the directivity from (8) by about 3 dB, but narrows the beamwidth sufficiently (by $1/\sqrt{2}$) that at the critical angle the signal level is down about another 0.5 dB. Thus, doubling n results in a net increase in the level of the radiation emerging at an elevation angle $\alpha_t = 10^\circ$ above the surface. However, the narrower beam requires that the helix axis be set within 1° of vertical. Thus, a helix with about 20 turns is a reasonable compromise.

(*b*) The relative power density S of a ray reflected back into the water is given by

$$S_r = \frac{(E_\perp|\rho_\perp|)^2 + (E_\parallel|\rho_\parallel|)^2}{E_\perp^2 + E_\parallel^2} \tag{9}$$

[1]Do not confuse α here for *helix pitch angle* with α in (6) for *elevation angle*.

where

$E_\perp$ = electric field perpendicular to the plane of incidence
$E_\parallel$ = electric field parallel to the plane of incidence
$\rho_\perp$ = reflection coefficient for perpendicular component [Eq. (21–4–1) with ε_r inverted]
$\rho_\parallel$ = reflection coefficient for parallel component [Eq. (21–4–6) with ε_r inverted]

Assuming circular polarization ($E_\perp = E_\parallel$) from the helix within the critical-angle hole, (9) reduces to

$$S_r = \tfrac{1}{2}(|\rho_\perp|^2 + |\rho_\parallel|^2) \tag{10}$$

The relative power density S_t transmitted through the surface and emerging above it is then given by

$$S_t = 1 - S_r \tag{11}$$

Evaluating (10) and (11) for $\varepsilon_r = 80$ as a function of the elevation angle α_t yields the curve of Fig. 21–22*a* and the pattern of Fig. 21–22*b*. Both graphs give the power radiated relative to its value at the zenith ($\alpha_t = 90°$).

(*c*) Note that the dimensions of the helix are reduced to $1/\sqrt{\varepsilon_r} = 0.11$ or 11 percent of the dimensions in air. Thus, the submerged helix diameter $D = \lambda_0/(\sqrt{\varepsilon_r}\pi) = 5\text{ m}/(\sqrt{80}\pi) = 0.178$ m.

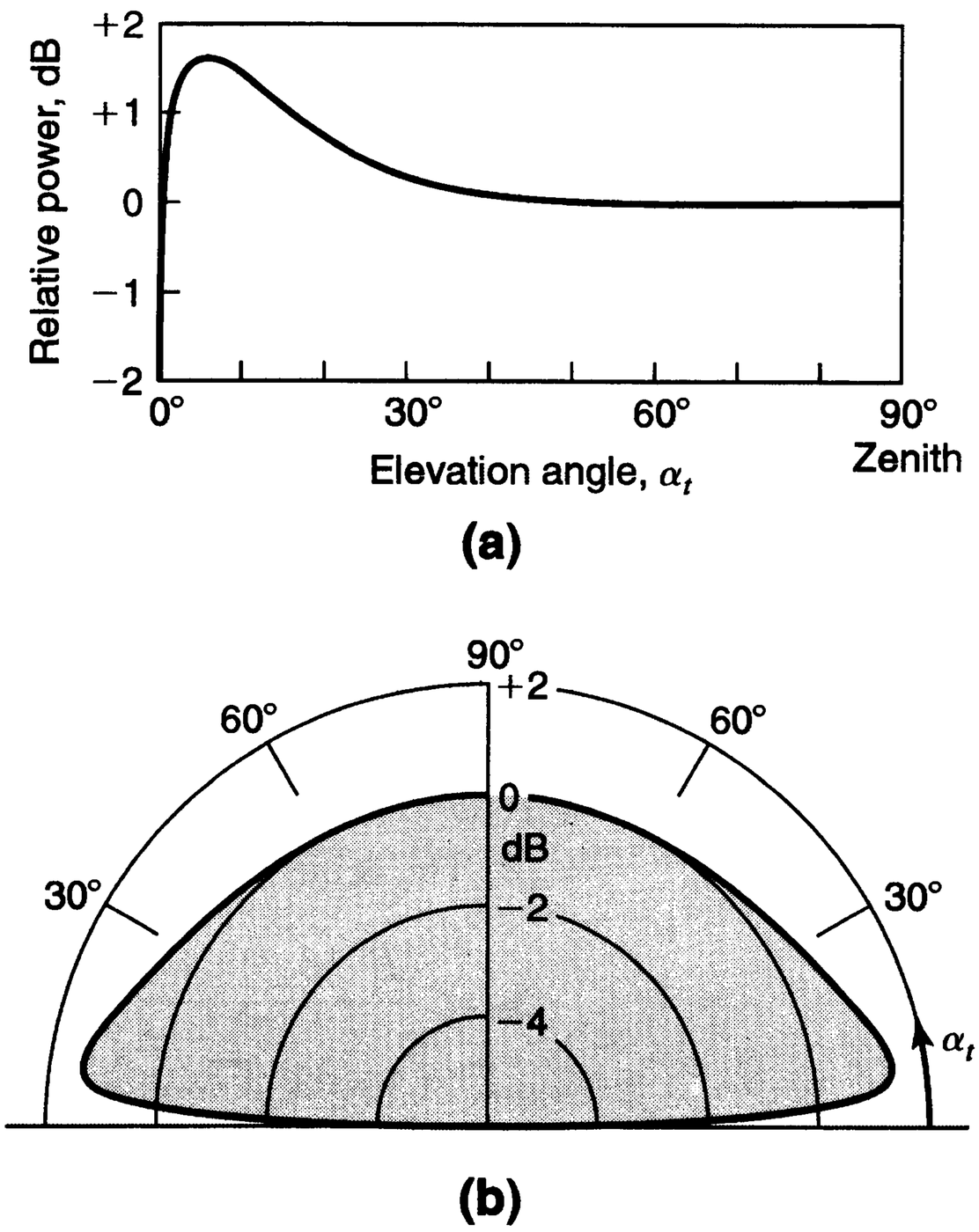


Figure 21–22
(*a*) Ratio of power density at elevation angle α_t to power density at zenith (90°) for waves transmitted above surface of lake from submerged transmitter. (*b*) Pattern of radiation transmitted above surface in polar coordinates.

The loss component of the permittivity $\varepsilon_r'' = 4$ and since $\varepsilon_r'' \ll \varepsilon_r'$ the *attenuation constant* for the water is given by[1]

$$\alpha = \frac{\pi}{\lambda_0}\frac{\varepsilon_r''}{\sqrt{\varepsilon_r'}} = \frac{\pi}{5}\frac{4}{\sqrt{80}} = 0.281 \text{ Np m}^{-1}$$

and the attenuation in the water path d by

$$\text{Attenuation (water)} = 20\log e^{-\alpha d} = 20\log e^{-0.281\alpha 20} = -49 \text{ dB}$$

The loss at the interface is given from (11) for elevation angle α (water) $= 83.68°$ as $10\log 0.51 = -3$ dB.

The directivity of the 20-turn helix is given by

$$D \simeq 12C_\lambda^2 n S_\lambda = 12 \times 20 \times 0.22 = 52.8$$

and its effective aperture by

$$A_e = \frac{D\lambda_0^2}{4\pi} = \frac{52.8 \times 5^2}{4\pi} = 105 \text{ m}^2$$

A George Brown turnstile (Sec. 21–7) with reflector elements has a directivity $D \simeq 3$ and effective aperture at $\lambda_0 = 5$ m of 6 m^2. Taking the distance between transmitter and receiver as 1 km we have from the Friis transmission formula that the received power less water attenuation and surface (interface) loss is

$$P_r = \frac{P_t A_{et} A_{er}}{r^2\lambda_0^2} = \frac{100 \times 105 \times 3}{1000^2 \times 5^2} = 1.25 \text{ mW}$$

The water attenuation and surface (interface) loss total 52 dB (= 49 + 3) so the net (actual) received power is given by

$$P_r = 1.25 \times 10^{-3}/\text{antilog } 5.2 \simeq 8 \times 10^{-9} \text{ W} = 8 \text{ nW}$$

or a bit less than 1 mV on a 100-Ω transmission line.

The transmitted energy-carrying field emerging above the surface goes to zero at the surface (elevation angle $\alpha_t = 0°$), as indicated in Fig. 21–22*a* and *b*. There is also a reactive wave traveling out along (above) the surface that accompanies the totally internally reflected wave. However, it is a reactive wave and carries no energy (**E** and **H** are in time-phase quadrature). Its fields decay exponentially with height above the surface (being simply the matching fields at the surface of the reflected-wave fields below) and are called *evanescent* fields. Due to the difference in transmission of the parallel and perpendicular field components through the surface, the received wave is not necessarily circularly polarized ($AR \neq 1$). However, the receiving antenna should be circularly polarized and of the same hand as the transmitting antenna, an implicit assumption made in solving the example problem.

---

[1]Do not confuse α here as the *attenuation constant* (= real part of propagation constant γ) with α for *helix pitch angle* or α for *elevation angle*. Unfortunately, the English and Greek alphabets do not have enough letters to go around.

It is also assumed in the example that the water surface is smooth. If it is not smooth, as under windy conditions, the situation is different and fluctuations (noise) will occur in the received signal.

The problem of electromagnetic wave transmission from water into air is analogous to that of transmission through the earth's ionosphere to an extraterrestrial point above it. In each case, refraction is from a medium with lower velocity to one with higher velocity of wave propagation with ray-bending and total internal reflection at small enough elevation angles. (See Probs. 12–3–9 and 12–3–10.)

However, there are some fundamental differences in that the ionosphere is inhomogeneous and anisotropic (a magnetized plasma) with polarization changes by Faraday rotation. These effects make circularly polarized antennas essential.

We note that the radio situation differs from the optical one in that $\varepsilon_r \simeq 80$ for water at radio wavelengths but $\varepsilon_r \simeq 1.75$ for water at optical wavelengths. Thus, for light waves the critical angle $\alpha_c = 41°$ as compared to 84° for radio waves.

21–13 SURFACE-WAVE AND LEAKY-WAVE ANTENNAS（表面波和漏波天线）

Traveling-wave antennas discussed in previous chapters include the monofilar axial-mode helix, long-wire and polyrod antennas. *Surface-wave* and *leaky-wave* antennas are also *traveling-wave* antennas but are ones which are adapted to flush or low-profile installations as on the skin of high-speed aircraft. Typically, their bandwidth is narrow (10 percent) and their gain is moderate (~15 dBi).

Consider the plane boundary between air and a perfect conductor as shown in Fig. 21–23*a* with a vertically polarized plane TEM wave traveling to the right along the boundary. From the boundary condition that the tangential component of the electric field vanishes at the surface of a perfect conductor, the electric field of a TEM wave traveling parallel to the boundary must be exactly normal to the boundary, or vertical, as in Fig. 21–23*a*. However, if the conductivity σ of the conductor is not infinite, there will be a tangential electric field E_x at the boundary, as well as the vertical component E_y, so that the total field will have a *forward tilt* as in Fig. 21–23*b*.

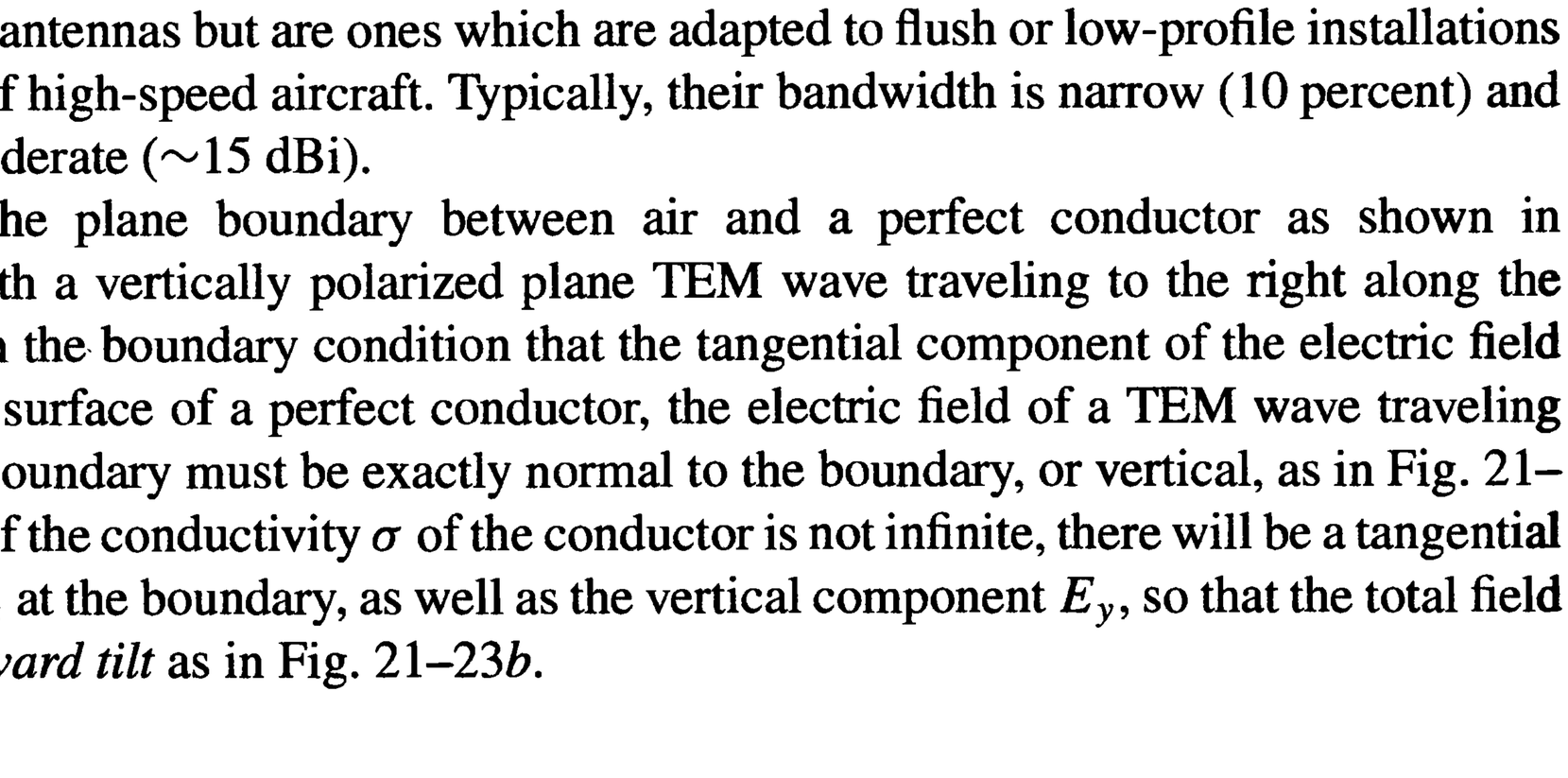


Figure 21–23
Vertically polarized wave traveling to right (*a*) along surface of a perfectly conducting medium and (*b*) along surface of medium with finite conductivity.

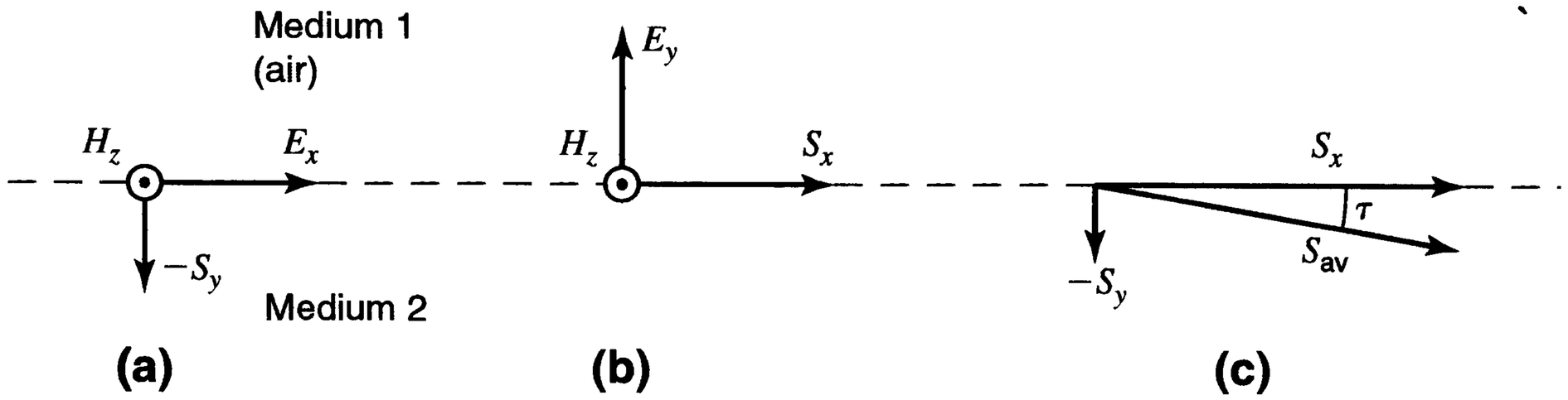


Figure 21–24
Fields and Poynting vector at the surface of a conducting medium with a vertically polarized wave traveling parallel to the surface.

The direction and magnitude of the power flow per unit area are given by the Poynting vector with average value

$$S_{av} = \tfrac{1}{2} \operatorname{Re} \mathbf{E} \times \mathbf{H}^* \qquad (\mathrm{W\ m^{-2}}) \tag{1}$$

At the conducting surface the average power into the conductor ($-y$ direction) is

$$S_y = -\tfrac{1}{2} \operatorname{Re} E_x H_z^* \tag{2}$$

This is power dissipated as heat in the conductor.

The space relation of E_x, H_z (or H_z^*) and S_y is shown in Fig. 21–24*a*. Since

$$\frac{E_x}{H_z} = Z_c \tag{3}$$

where Z_c = intrinsic impedance of the conductor

Then from (2) the power flow into the conductor can be written as

$$S_y = -\tfrac{1}{2} H_z H_z^* \operatorname{Re} Z_c = -\tfrac{1}{2} H_{z0}^2 \operatorname{Re} Z_c \tag{4}$$

where

$H_z = H_{z0}\, e^{-j\xi - \gamma x}$
$H_z^* = H_{z0}\, e^{j\xi + \gamma x}$ = complex conjugate of H_z
ξ = phase lag of H_z with respect to E_x
γ = propagation constant = $\alpha + j\beta$

The power flow parallel to the surface (x direction) is

$$S_x = \tfrac{1}{2} \operatorname{Re} E_y H_z^* \qquad (\mathrm{W\ m^{-2}}) \tag{5}$$

The space relation of E_y, H_z and S_x is shown in Fig. 21–24*b*. Since

$$\frac{E_y}{H_z} = Z_d \tag{6}$$

where Z_d = intrinsic impedance of the dielectric medium (air) above the conductor.

Then from (5) the power flow along the conductor (x-direction) can be written

$$S_x = \tfrac{1}{2} H_{z0}^2 \operatorname{Re} Z_d \qquad (\mathrm{W\ m^{-2}}) \tag{7}$$

The total average Poynting vector is then

$$S_{av} = \hat{x} S_x + \hat{y} S_y = \frac{H_{z0}^2}{2} (\hat{x} \operatorname{Re} Z_d - \hat{y} \operatorname{Re} Z_c) \tag{8}$$

Figure 21–24*c* shows the relation of S_{av} to its x and y components. It is apparent that the average power flow is not parallel to the surface but downward at an angle τ. This angle is the same as the forward tilt angle of the electric field (see Fig. 21–23*b*). If the conductivity of the conductor is infinite ($\sigma = \infty$), the tilt is zero ($\tau = 0$).

From (8) the tilt angle

$$\tau = \tan^{-1} \frac{\operatorname{Re} Z_c}{\operatorname{Re} Z_d} \tag{9}$$

EXAMPLE 21–13.1

Find the tilt angle τ for a vertically polarized 3-GHz wave traveling in air along a flat copper sheet.

■ **Solution**

At 3 GHz, we have for copper that $\operatorname{Re} Z_c = 14.4\ \text{m}\Omega$. The intrinsic impedance of air is real and independent of frequency ($= 377\ \Omega$). From (9) we have

$$\tau = \tan^{-1} \frac{14.4 \times 10^{-3}}{377} = 0.0022^\circ \qquad \textit{Ans.} \tag{10}$$

Although τ is very small it is not zero, indicating some power flow into the copper sheet. If σ is small or if the conductor is replaced by a dielectric, τ may amount to a few degrees. In the Beverage antenna (Sec. 16–16*c*) the horizontal field component (E_x) is parallel to the antenna wire and induces emf's in it.

EXAMPLE 21–13.2

Find the forward tilt angle τ for a vertically polarized 3-GHz wave traveling in air along the smooth surface of a freshwater lake.

■ **Solution**

At 3 GHz the conduction current of fresh water is negligible compared with the displacement current ($\varepsilon_r'' \ll \varepsilon_r'$) so the water may be regarded as a dielectric medium with $\varepsilon_r \simeq \varepsilon_r' = 80$. Thus, from (9),

$$\tau = \tan^{-1} \frac{1}{\sqrt{80}} = 6.4^\circ \qquad \textit{Ans.} \tag{11}$$

The tilt angle τ calculated above is an average value. In general, its instantaneous direction varies with time. For the wave traveling along the copper sheet, E_y and E_x are in phase octature (45° phase difference), so that at one instant the total field **E** may be in the x direction and $\frac{1}{8}$-period later in the y direction. For the 3-GHz wave of Example 21–13.1,

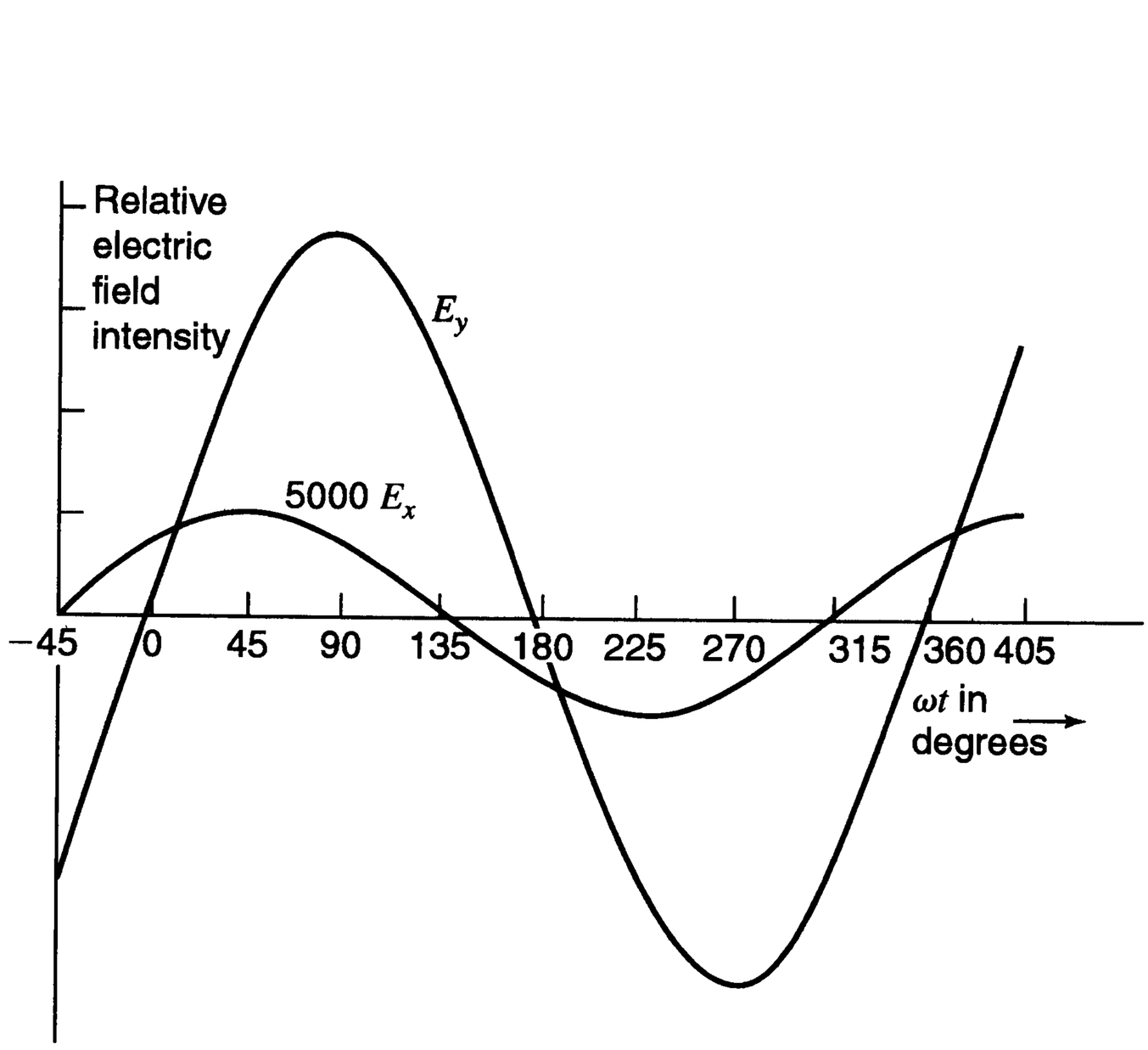


Figure 21–25
(*a*) Magnitude variation with time of E_y and E_x components of **E** in air at the surface of a copper region for a 3-GHz TEM wave traveling parallel to the surface. (*b*) Resultant values of **E** (space vector) at 22.5° intervals over one cycle, illustrating elliptical cross-field at the surface of the copper region. The wave is traveling to the right. Note that although **E** has *x* and *y* components (cross-field) it is linearly polarized as seen from the *x* direction.

E_x and E_y vary with time as in Fig. 21–25*a*. The locus of the tip of **E** describes a cross-field ellipse as in Fig. 21–25*b*, with positions shown as a function of time (ωt). The ellipse is not to scale, the E_x values being magnified by 5000. The variation of the instantaneous Poynting vector is shown in Fig. 21–26 with ordinate values magnified by 5000. It is of note that the tip of the Poynting vector travels around the ellipse *twice* per cycle.

Whereas copper has a complex intrinsic impedance, fresh water at 3 GHz (as in Example 2) has a real intrinsic impedance so that E_x and E_y are essentially in phase and the cross-field ellipse collapses to a straight line with a forward tilt of 6.4°.

The forward tilt of **E** or downward tilt of *S* in the above examples tend to make the wave energy hug the surface, resulting in a *bound wave* or *surface wave*. The phase velocity v of this wave is less than the velocity c of light, i.e., it is a *slow wave* ($v < c$). To initiate a wave along the surface, a launching device, such as a horn, with a height of several wavelengths can be used as in Figs. 21–27 and 21–28.

In 1899 Arnold Sommerfeld showed that a wave could be guided by a round wire of finite conductivity (Sommerfeld-1). Jonathan Zenneck pointed out that for similar reasons a wave traveling along the earth's surface would tend to be guided by it (Zenneck-1).

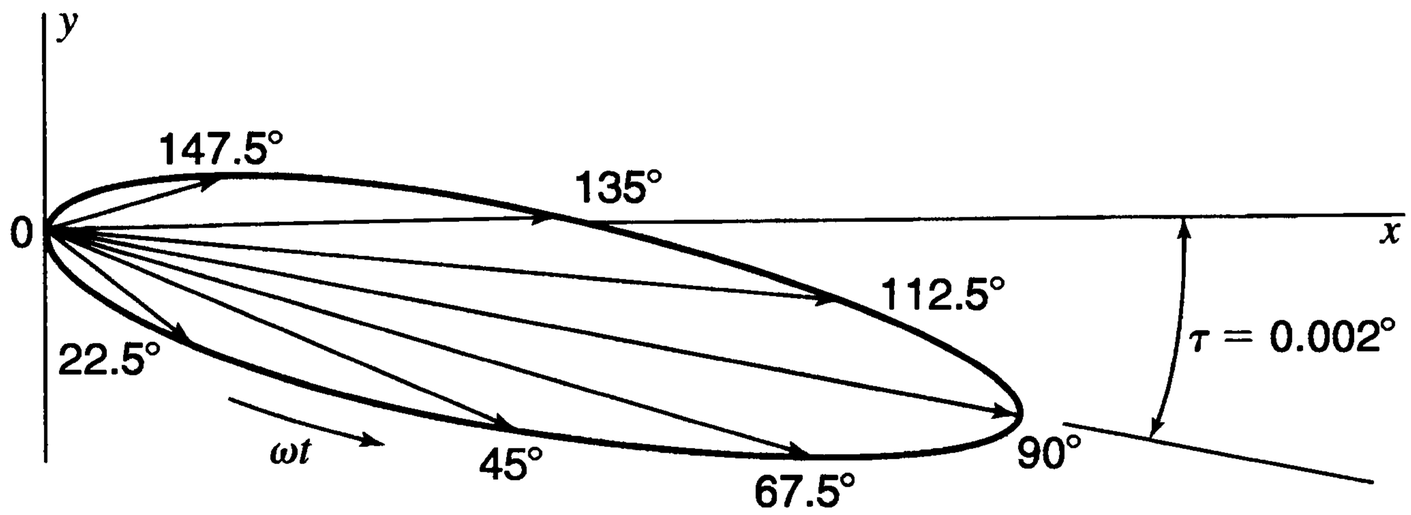


Figure 21–26
Poynting vector in air at a point on the surface of a copper region for a 3-GHz TEM wave traveling along the surface (to right). The Poynting vector is shown at 22.5° intervals over a $\frac{1}{2}$-cycle. The ordinate values are magnified 5000 times compared with the abscissa values. The tip of the Poynting vector travels around the ellipse *twice* per cycle.

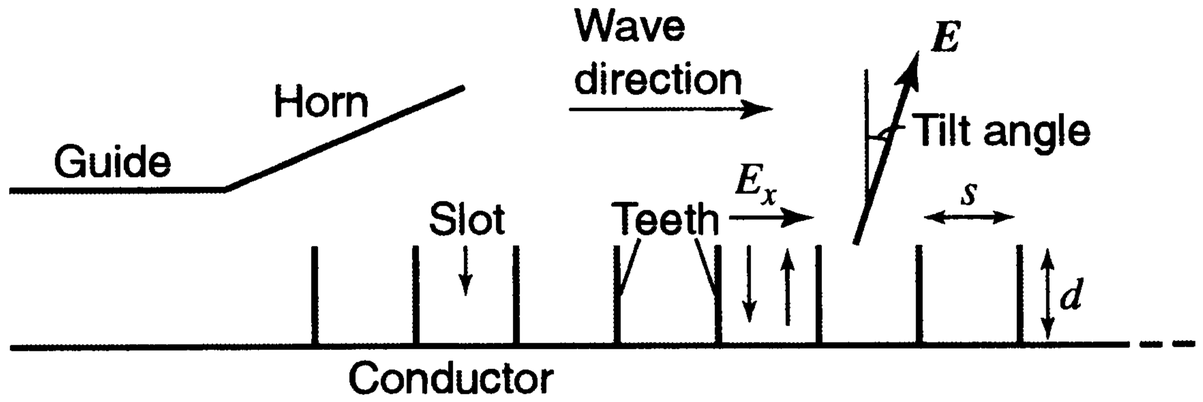


Figure 21–27
Corrugated surface-wave antenna with horn wave-launcher.

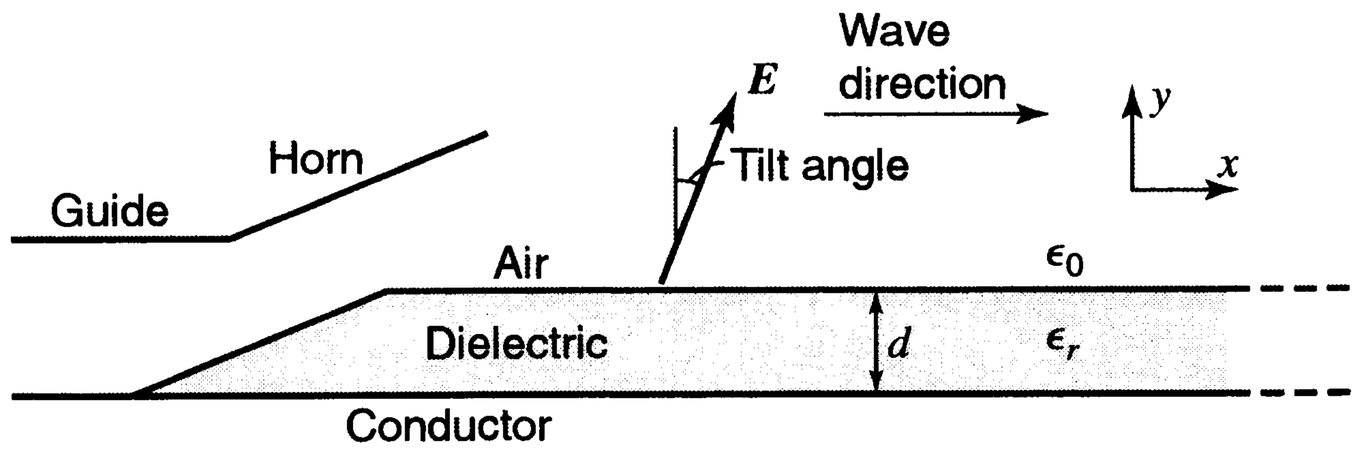


Figure 21–28
Dielectric-slab surface-wave antenna with horn wave-launcher.

The guiding action of a flat conducting surface can be enhanced by adding metal corrugations. These and the horn launcher in Fig. 21–27 form a surface-wave antenna. The corrugations have many teeth per wavelength ($s < \lambda/5$). The slots between the teeth support a TEM wave involving E_x, which travels up and down the depth of the slot (a standing wave). Each slot acts like a short-circuited section of a parallel-plane 2-conductor transmission line of depth d. Assuming lossless materials, the impedance Z presented to a wave traveling down the slots is a pure reactance given by

$$Z = jZ_d \tan \frac{2\pi\sqrt{\varepsilon_r}d}{\lambda_0} \qquad (\Omega) \tag{12}$$

where

Z_d = intrinsic impedance of medium filling the slots, Ω
ε_r = relative permittivity of the medium, dimensionless

λ_0 = free-space wavelength, m
d = depth of slots, m

For air-filled slots ($Z_d = 377\ \Omega$ and $\varepsilon_r = 1$), (12) reduces to

$$Z \simeq j120\pi \tan \frac{2\pi d}{\lambda_0} \qquad (\Omega) \tag{13}$$

The slots store energy from the passing wave. When $d < \lambda_0/4$, the plane along the top of the teeth is inductively reactive. When $d = \lambda_0/4$, $Z = \infty$ and the plane along the top is like an open circuit (nothing below), while when $d = \lambda_0/2$, $Z = 0$ and the plane appears like the conducting sheet below it (a short circuit).

The guiding action of a flat conducting surface can also be enhanced by adding a dielectric coating or slab of thickness d. With a launcher, as in Fig. 21–28, the combination forms another type of surface-wave antenna. The electric field is vertically polarized but has a small forward tilt at the dielectric surface. For a sufficient thickness d, the fields attenuate perpendicular to the surface (y direction) as $e^{-\alpha y}$, where

$$\alpha = \frac{2\pi}{\lambda_0}\sqrt{\varepsilon_r - 1} \qquad (\text{Np m}^{-1}) \tag{14}$$

For $\varepsilon_r = 2$, the attenuation is over 50 dB λ^{-1}. Thus, the fields are confined close to the surface.

Corrugated and dielectric slab surface-wave antennas, as in Figs. 21–27 and 21–28, with a length of several λ and a width of 1λ or more (into page), produce end-fire beams with gain proportional to their length and width. It is assumed that the conducting surface (ground plane) extends beyond the end of the corrugations or slab. If not, the beam direction tends to be off end-fire (elevated). For optimum patterns, the depth of the slots or thickness of the slab may be tapered at both ends.

Although *surface-wave antennas* take many forms, the ones described above are typical. They are traveling-wave antennas carrying a bound wave with the energy flowing *above* the guiding surface and with velocity $v < c$ (*slow wave*). **E** would be perpendicular to the surface except for its forward tilt.

Leaky-wave antennas are also traveling-wave types but ones in which radiating energy leaks continuously or periodically from along the length of the guiding structure, with most of the energy flow *within* the structure. Typically, but not necessarily, the structure carries a *fast wave* ($v > c$). A hollow metal waveguide is an example. With one wall removed, energy can leak out continuously all along the guide.

A leaky-waveguide antenna of this type is shown in Fig. 21–29. Since the wave velocity v in the guide is faster than the velocity c of light, the radiation forms a beam inclined at an angle ϕ with the guide as given by

$$\phi = \cos^{-1}\frac{c}{v} \tag{15}$$

For $v = 1.5c$, $\phi = 48°$. Since v is a function of the frequency, the beam angle ϕ may be scanned by a change in frequency.

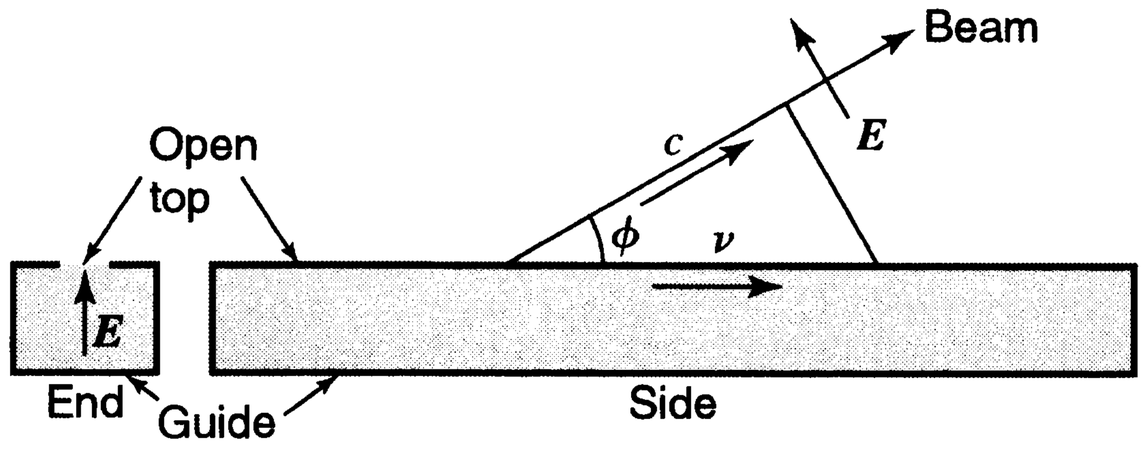


Figure 21–29
Open-top waveguide antenna with continuous energy leakage.

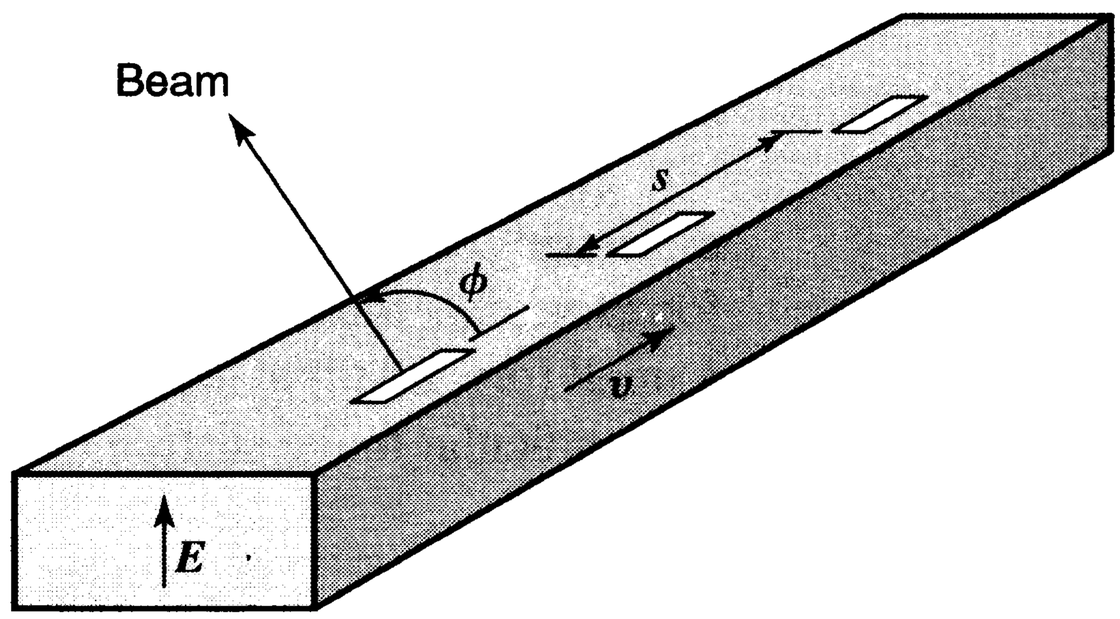


Figure 21–30
Slotted-waveguide antenna with leakage at periodic (discrete) intervals.

A leaky-wave antenna may also be constructed by cutting holes or slots at a regular spacing along the waveguide wall as in Fig. 21–30 (see also the slotted waveguide of Fig. 9–5). Leakage may be controlled by the slot or hole size. For a slot or hole perimeter of $\sim 1\lambda$, leakage is large but decreases rapidly with a decrease in perimeter. This periodic structure radiates at a beam angle ϕ given from (8–14–8) by

$$\phi = \cos^{-1}\left[\frac{\lambda_0}{\lambda_g} + \frac{m}{s/\lambda_0}\right] \tag{16}$$

where

s = hole or slot spacing, m
λ_0 = free-space wavelength, m
λ_g = wavelength in guide, m
m = mode number, $0, \pm\frac{1}{2}, \pm 1, \ldots$

EXAMPLE 21–13.3

Find the beam angle ϕ for $\lambda_g = 1.5\lambda_0$, $s = \lambda_0$ and $m = -1$.

■ **Solution**

From (16),

$$\phi = \cos^{-1}\left[\frac{1}{1.5} - \frac{1}{1}\right] = 109.5° \qquad \text{(in back-fire direction)}$$

If the slots are spaced $\lambda/2$ instead of 1λ, proper phasing requires that alternate ones be placed on opposite sides of the centerline. For this case $m = -\frac{1}{2}$ in (16). The beam angle ϕ may be scanned by a change in frequency. Although described here as a leaky-wave antenna, the antenna of Fig. 21–30 may also be considered as simply an array of waveguide slots.

Consider now a dielectric slab waveguide of thickness d with wave injected at small enough α (below the critical angle) so that the wave is totally internally reflected and propagates by multiple reflections inside the slab. The situation is similar to that for the submerged radio transmitter in Sec. 21–12, except that here the denser medium has upper *and* lower boundaries and a thickness of the order of $1\lambda_0$. Although the energy is transmitted inside the slab, fields exist above and below. These, however, are evanescent and decay exponentially away from the slab. They convey no energy. (Recall the evanescent wave above the water surface in Sec. 21–12.)

It might be supposed that any wave injected into the slab at an angle below the critical value will propagate, but, because of multiple-reflection interference, waves will only propagate at certain angles (eigenvalues) (Kraus (1) and Fleisch; Marcuse-1). Figure 21–31 is an end view of a dielectric slab waveguide carrying a wave with electric field parallel to the surfaces (perpendicular to the plane of incidence). In the notation of Sec. 21–12 this is an $\mathbf{E}_\perp$ field. Propagation is into the page. The evanescent field above and below the slab matches the field inside but carries no energy. However, discontinuities on the slab surface can cause energy leakage from inside and radiation. Thus, a dielectric slab with surface discontinuities can act as a leaky-wave antenna.

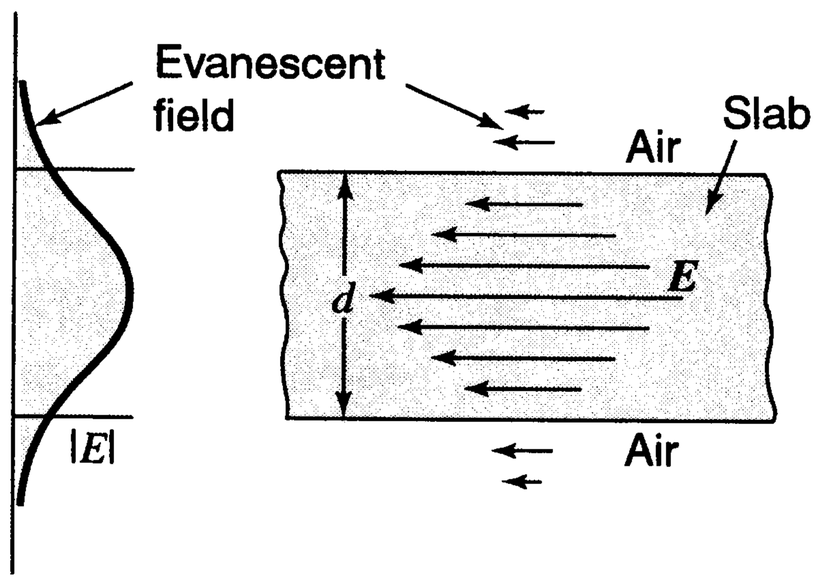


Figure 21–31
End view of dielectric slab waveguide with **E** parallel to the slab surfaces. Propagation is into the page. Arrows and graph (at left) indicate variation in magnitude of **E** across the thickness d of the slab. The field outside the slab is evanescent and transmits no energy.

A dielectric cylinder can also serve as a guide in the same manner as a slab. At or near optical wavelengths the cylinder diameter can be physically small or threadlike. Such guides, or *optical fibers,* consist typically of a transparent glass *core* surrounded by a glass *sheath* or *cladding* of slightly lower index of refraction, with the *sheath* enclosed in an opaque protective jacket. A typical core fiber 25 μm in diameter is as fine as a human hair and can carry one thousand 2-way voice communication channels with an attenuation as small as 1 dB km^{-1} at light or infrared wavelengths ($\frac{1}{2}$ to 1 μm).

The literature on surface- and leaky-wave antennas is extensive. Summary treatments are given by Zucker (1) and by Oliner (1).

21-14 ANTENNA DESIGN CONSIDERATIONS FOR SATELLITE COMMUNICATION[1]（设计卫星通信天线的考虑）

In 1945, while a radar officer in the Royal Air Force, Arthur C. Clarke (now Sir Arthur) published an article in *Wireless World* in which he proposed the use of artificial satellites and, in particular, those in a geostationary orbit, as a solution to the world's communication problem (Clarke-1). The idea then seemed far fetched to many but 12 years later Sputnik went up and only 6 years after that, in 1963, the first successful geostationary satellite, Syncom 2, was put into orbit and Clarke's proposal became a reality. Now there are nearly 1000 satellites in the geostationary or Clarke orbit with more being added at frequent intervals. The satellites move with the earth as though attached to it so that from the earth each appears to remain stationary above a fixed point on the equator. These satellites form a ring around the earth at a height of 36,000 km above the equator, putting the earth in the company of other ringed planets, Saturn and Uranus, with the difference that the earth's ring is man-made and not natural. Arthur Clarke has observed that the ancient superstition that our destinies are controlled by celestial bodies has at last come true, except that the bodies are ones we have put up there ourselves.

Having a transmitting (transponder) antenna on a Clarke-orbit satellite is like having it on an invisible tower 36,000 km high. Since earth-station dishes look upward at the sky, ground reflections are eliminated but the ground is still involved through its effect on the antenna noise temperature via side and back lobes.

The spacing of satellites in the Clarke orbit is closely connected with the design of both satellite and earth-station antennas. Thus, if beamwidths and side-lobe levels are reduced, the spacing can be reduced and more satellites accommodated in orbit.

The full orbit utilization problem involves not only satellite placement and spacing but also the available spectrum. Thus, satellites may be parked closer together than beamwidths would allow if they operate in different frequency bands. However, both *orbit space* and *spectrum space* are limited resources which must be allocated in an optimum manner to meet the tremendous international demand for satellite communications. The problem of maximizing access to the Clarke orbit is under study and planning by the satellite orbit-use (ORB) sessions of the World Administrative Radio Conference (WARC), with its first session (ORB-1) in 1985 and a second session (ORB-2) in 1988.

To permit closer spacings than otherwise, North American satellites sharing the same frequencies use an alternate linear-polarization technique. For example, the even-numbered channels of a satellite may be polarized parallel to the orbit with odd-numbered channels perpendicular to it, while the adjacent satellites on either side have the polarizations reversed (odd-numbered parallel and even-numbered perpendicular to the orbit). Some satellites serving other parts of the world use circular polarizations of opposite hand to do the same thing. Thus, the response level of the earth-station antenna to the opposite state of polarization is a factor (Weiss-1).

It is necessary that the earth-station antenna have gain and pattern properties capable of providing a satisfactory S/N ratio with respect to noise and interference sources (as from adjacent satellites). The satellite transponder power, antenna gain and pattern are

---

[1]Recall Sec. 12–3 regarding SNR.

also factors. At the Clarke-orbit height of 36,000 km, a satellite spacing of 1° amounts to a physical separation of 630 km. Although each satellite wanders about its assigned orbital location (usually in a systematic manner), this "station-keeping" motion is required by regulation to be less than 60 km (0.1°) so that spacing even closer than 1° is possible without danger of satellites colliding.[1]

Some of the antenna design aspects may be illustrated by an example.

EXAMPLE **21–14.1 Earth Station-Satellite Link**

Determine the required parabolic dish diameter of a 4-GHz (C-band) earth-station antenna if its system temperature = 100 K for an S/N ratio of 20 dB and bandwidth = 30 MHz with satellite transponder power = 5 W, satellite parabolic dish diameter = 2 m and spacing between satellites = 2°.

■ **Solution**

From the Friis transmission formula and the Nyquist noise-power equation, we have from (12–3–3) that the S/N (actually C/N or carrier-to-noise) ratio for isotropic antennas and a transmitter power of 1 W is given by

$$\frac{S}{N} = \frac{\lambda^2}{16\pi^2 r^2 k T_{\text{sys}} B} \tag{1}$$

where

λ = wavelength
r = downlink distance, 36,000 km
k = Boltzmann's constant = 1.38×10^{-23} J K^{-1}
T_{sys} = system temperature, K
B = bandwidth, Hz

Introducing the indicated values in (1) we obtain

$$\frac{S}{N} = -61.8 \text{ dB} \tag{2}$$

for the downlink at 1 W isotropic.

For the $D = 2$-m-diameter satellite dish at 50 percent aperture efficiency, the antenna gain G_s is given by

$$G_s = \frac{4\pi A_e}{\lambda^2} = 35.5 \text{ dB} \tag{3}$$

[1]The satellite antenna pointing accuracy in roll, pitch and yaw is also usually of the order of 0.1° or less. Roll, pitch and yaw are of primary concern in the satellite antenna design but of secondary importance in the earth-station antenna design. The Astra satellites have specifications of 0.06° roll, 0.07° pitch, 0.22° yaw and ±0.05° station keeping.

where

$$A_e = \frac{1}{2}\pi\left(\frac{D^2}{4}\right) = 1.6\ \text{m}^2$$
$$\lambda \quad = 0.075\ \text{m}$$

The transponder power of 5 W gives an additional 7 dB (= log 5) gain. The required earth-station antenna gain G_E must then be sufficient to make the system S/N ratio ≥20 dB, or

$$G_E = 20 + 61.8 - 35.5 - 7 = 39.3\ \text{dB} \tag{4}$$

Thus, the required earth-station effective aperture

$$A_e = \frac{G_E\lambda^2}{4\pi} = 3.8\ \text{m}^2 \tag{5}$$

At an assumed aperture efficiency of 50 percent the required physical aperture A_p is twice this, or 7.6 m^2, making the required diameter of the earth-station antenna

$$D_E = 2\sqrt{\frac{A_p}{\pi}} = 3.1\ \text{m} \tag{6}$$

This diameter meets the S/N ratio requirement. To determine if it also meets the 2° spacing requirement we should know the illumination taper across the dish aperture. This is not given. However, we assumed 50 percent aperture efficiency which implies substantial taper. Accordingly, the beamwidths may be estimated as

$$\text{HPBW} \simeq \frac{65^\circ}{D_\lambda} = \frac{65^\circ}{3.1\ \text{m}/0.075\ \text{m}} = 1.6^\circ \tag{7}$$

and

$$\text{BWFN} \simeq \frac{145^\circ}{D_\lambda} = 3.5^\circ \tag{8}$$

Thus, first nulls or minima are approximately 1.75° either side of the satellite or 0.25° less than the 2° spacing of the adjacent satellites, as indicated in Fig. 21–32.

The first side-lobe level of a tapered circular aperture should be at least 20 dB down and since the adjacent satellites are closer to the first nulls or minima than the first side-lobe maxima, the adjacent satellite levels should be at least 25 dB down, which could make the total noise level about 18 dB down. Although this may not be satisfactory, the use of an opposite state of polarization on the adjacent satellites should reduce their interfering signal level to an acceptable level provided the earth-station response to cross-polarization is small.

The station-keeping accuracy of the satellites and the pointing error of the earth-station antenna are also factors. Since these may vary independently, the effects will be random and must be assessed by statistical methods giving upper and lower limits to the overall S/N ratio.

Assuming an earth-station antenna pointing error of ±0.1°, the same as the required satellite station-keeping tolerance, means that the satellite position may vary a maximum

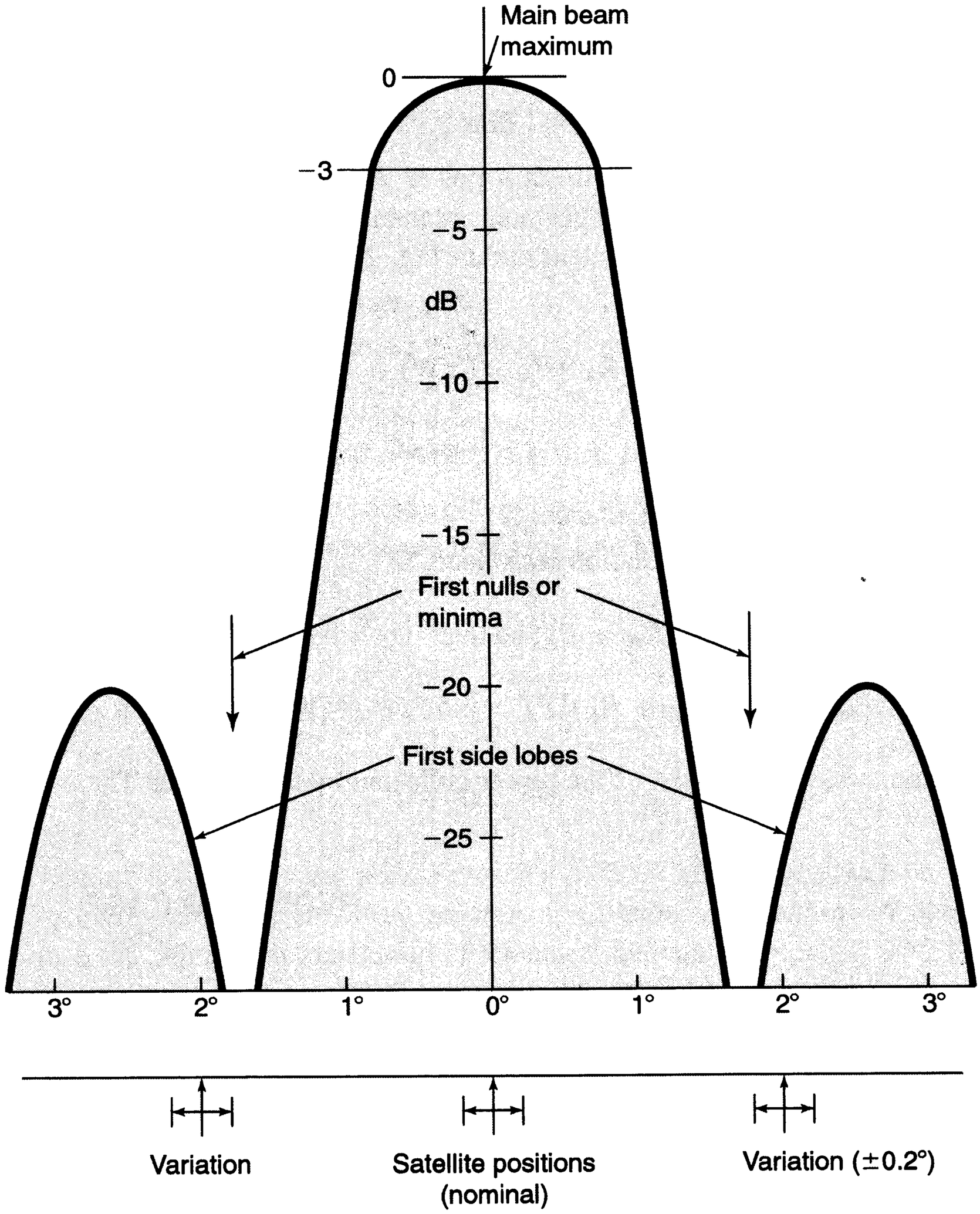


Figure 21–32
Pattern of C-band earth-station dish for worked example. Pattern is shown with respect to satellite positions at 2° spacing in the Clarke orbit.

of ±0.2° with respect to the first null. This is indicated in Fig. 21–32. We note that at one extreme the satellite is almost at the null but at the other extreme is only about 4 dB below the first side-lobe maximum. These and other factors are discussed by Jansky (1) and Jeruchim.

Although we have made a number of assumptions, the example illustrates many of the important factors involved in determining suitable dimensions and pattern characteristics of an earth-station antenna (Fig. 21–32).

21-15 RECEIVING VERSUS TRANSMITTING CONSIDERATIONS
(用做接收和用做发射的考虑)

21-15a Receiving Case (接收状态)

The receiving antenna collects a power P_A from the incoming wave. See Fig. 21–33*a*. Assuming a perfect match between antenna and transmission line, $Z_{\text{line}} = R_A$, and line and load, $R_R = R_A$ (see eqivalent circuit Fig. 21–33*b*) the power delivered to the receiver (or load) is given by

$$P_{\text{rec}} = [R_R/(R_A + R_R)]P_A \qquad \text{(W)} \tag{1}$$

where

R_R = receiver impedance, Ω
R_A = antenna radiation resistance, Ω

For a perfect match, $R_R = R_A$, so that

$$P_{\text{rec}} = [R_R/(R_R + R_R)]P_A = 0.5P_A \qquad \text{(W)} \tag{2}$$

and the receiver gets 1/2 the power collected by the antenna. The other half is reradiated.

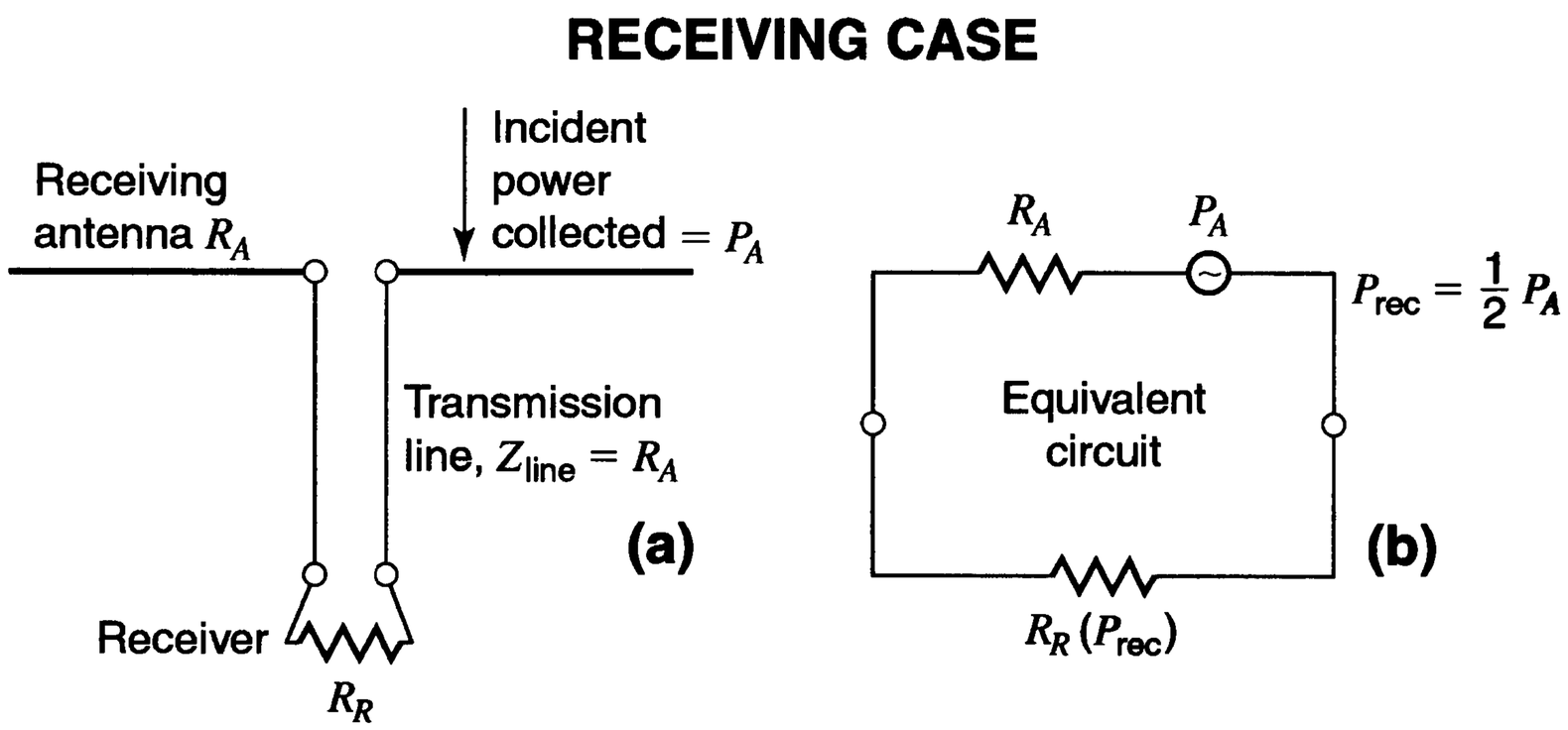


Figure 21–33
In the receiving case, (*a*) and (*b*), maximum power transfer delivers half the power collected by the antenna to the receiver (or load). In the transmitting case, (*c*) and (*d*), the power radiated P_{rad} may be nearly equal to the power P_T generated by the transmitter.

This is the condition of ***maximum power transfer.*** More generally, this condition requires a ***conjugate match,*** that is

$$R_A + jX_A = R_{\text{line}} - jX_{\text{line}} \tag{3}$$

where

X_A = antenna reactance, Ω
X_{line} = line reactance, Ω

21–15b Transmitting Case（发射状态）

Here the transmitter generates a power P_T with a transmitter internal resistance R_T as shown in Fig. 21–33*c* and *d*. Thus, assuming no line losses, the power radiated P_{rad} is given by

$$P_{\text{rad}} = [R_A/(R_A + R_T)]P_T \qquad \text{(W)} \tag{4}$$

If $R_A = 100\ \Omega$ and $R_T = 5\ \Omega$

$$P_{\text{rad}} = [100/(100 + 5)]P_T = 0.95P_T \qquad \text{(W)} \tag{5}$$

Accordingly, *maximum power transfer* at 50% efficiency is the best we can do with a receiving antenna. However, *maximum radiating efficiency* is of paramount importance in a transmitting antenna with efficiencies that can approach 100%. This requires, in the transmitting case, more precise matching of the transmission line and antenna, not only for maximum efficiency but also to eliminate line reflections which could produce ghosts on a TV picture, especially if the line is long.

According to the principle of reciprocity, the field pattern of an antenna is the same for reception as for transmission. However, it does not always follow that because a particular antenna is desirable for a given transmitting application it is also desirable for reception. In transmission the main objective is usually to obtain the largest field intensity possible at the point or points of reception. To this end, high efficiency and gain are desirable. In reception, on the other hand, the primary requirement is usually a large *signal-to-noise ratio*. Thus, although efficiency and gain may be desirable, they are important only insofar as they improve the signal-to-noise ratio. Accordingly, a receiving antenna with low side lobes may be more desirable than one with higher gain or an adaptive antenna may be required to null-out interfering signals. However, by way of contrast, suppose that circuit noise in the receiver is the limiting factor. Then high antenna gain and efficiency would be important in order to raise the signal-to-noise ratio.

In direction-finding antennas, the directional characteristic of the antenna is employed to determine the direction of arrival of the radio wave. If the signal-to-noise ratio is high, a null in the field pattern may be used to find the direction of arrival. With a low signal-to-noise ratio, however, the maximum of the main lobe may provide a more satisfactory indication (Alford-1).

21–16 BANDWIDTH CONSIDERATIONS（频带的考虑）

The useful bandwidth of an antenna depends, in general, on *both its pattern and impedance* characteristics.[1] In thin dipole antennas the bandwidth is usually determined by the impedance variation since the pattern changes less rapidly.[2] However, with very thick cylindrical antennas or biconical antennas of considerable cone angle, the impedance characteristics may be satisfactory over so wide a bandwidth that the pattern variation determines one or both of the frequency limits. The pattern may also determine the useful bandwidth of horn antennas, metal-plate lens antennas or zoned lens antennas.

If the acceptable bandwidth for pattern exceeds that for impedance, the bandwidth can be arbitrarily specified by the frequency limits F_1 and F_2 at which the VSWR on the transmission line exceeds an acceptable value. What is acceptable varies widely depending on the application. In some cases the VSWR must be close to unity. In others it may be as high as 10 to 1 or higher. The bandwidth can be specified as the ratio of $F_2 - F_1$ to F_0 (the center or design frequency) or in percent as

$$\frac{F_2 - F_1}{F_0} \times 100$$

Another definition is the simple ratio F_2/F_1 (or F_2/F_1 to 1) where $F_2 > F_1$.

The bandwidth due to the impedance can also be specified (if the bandwidth is small) in terms of its reciprocal or Q at F_0, where

$$Q = 2\pi \frac{\text{total energy stored by antenna}}{\text{energy dissipated or radiated per cycle}}$$

21–17 ARCHITECTURALLY ACCEPTABLE ANTENNAS（建筑上可接受的天线）

Many municipalities prohibit roof top antennas and antennas in the yard. However, weather-vanes, chimneys and flagpoles are acceptable. Thus, there is a need for antennas that don't look like antennas.

21–17a The Weather-Vane Antenna（气象风标天线）

Two methods of using a weather-vane as an antenna are shown in Fig. 21–34. In (*a*) the mounting rod for the plastic vane acts as a $\lambda/4$ vertical stub antenna and the 4 $\lambda/4$ radials as the ground plane producing an omnidirectional vertically polarized pattern. In (*b*) the radials act as two horizontal $\lambda/2$ dipoles fed in phase quadrature (90°) producing the omnidirectional horizontally polarized pattern of a turnstile antenna (see Sec. 21–7).

---

[1]The pattern is understood to include *polarization* characteristics.

[2]A dipole $\lambda/2$ long has a half-power beamwidth of 78°. If the frequency is reduced so that the dipole length approaches an infinitesimal fraction of a wavelength, the beamwidth only increases from 78 to 90°, while if the frequency is doubled so that the dipole is 1λ long the beamwidth decreases from 78 to about 47°.

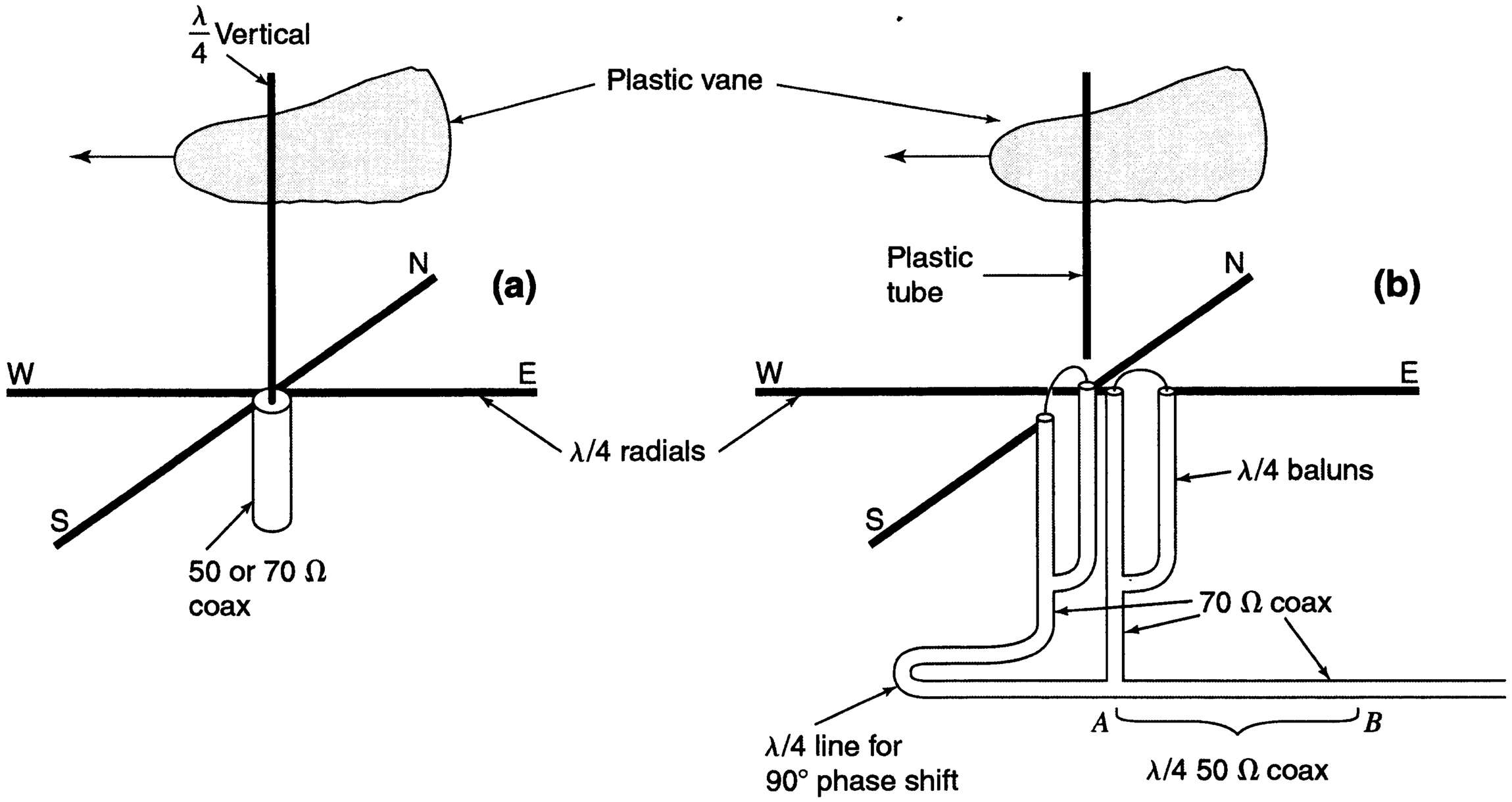


Figure 21–34
(*a*) The weather vane as an omnidirectional vertically polarized antenna and (*b*) as an omnidirectional horizontally polarized antenna turnstile antenna.

The coaxial line impedance at point *A* is about 35 Ω which is transformed by a 50 Ω λ/4 section to about 70 Ω at *B*. Thus, a functioning weather-vane can double as either a vertically polarized or a horizontally polarized antenna.

21–17b The Flagpole Antenna（旗杆天线）

Two ways of using a flagpole as an antenna are shown in Fig. 21–35. In (*a*) the grounded λ/4 metal pole is shunt fed (1/2 of a T-match) while in (*b*) the vertical λ/4 conductor is enclosed in a plastic (PVC) pipe fed directly by a coaxial cable. Thus, a flagpole 10 m tall can function as a vertically polarized omnidirectional antenna for the 40-m wave bands. The plastic pipe could also be used to contain other vertically polarized pole-mounted antennas. Or if the pipe is large enough in diameter it could contain a horizontally polarized slotted-cylinder antenna.

21–17c The Chimney Antenna（烟囱天线）

Patch antennas may be mounted on the sides of an existing chimney with a sheet plastic cover that matches the chimney's brickwork. Or an all-plastic chimney containing a variety of antennas may be added to a roof.

The above examples illustrate just a few of the many possibilities there are for "architecturally acceptable antennas."

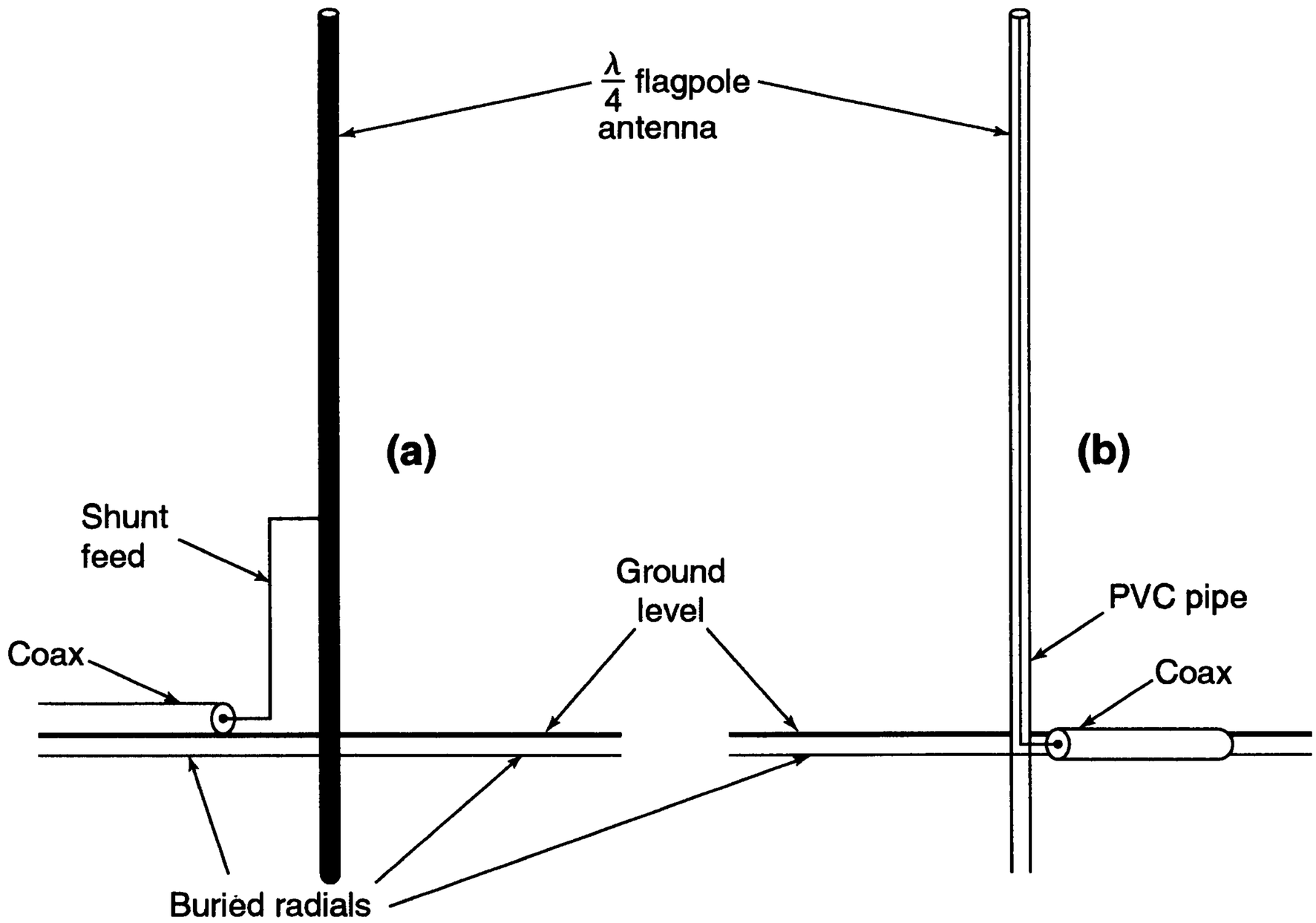


Figure 21–35
(*a*) The grounded shunt fed flagpole antenna and (*b*) the direct fed version.

21–18 ILS (INSTRUMENT LANDING SYSTEM) ANTENNAS[1]
[用于仪表着陆系统（ILS）的天线]

A typical ILS system has 3 components: (1) azimuth or horizontal guidance for aligning with the runway, (2) vertical guidance for aligning with the 3 degree vertical elevation angle of the approach path and (3) range markers for distance to the runway.[2] See Fig. 21–36.

(1) ***Azimuth*** guidance (also called the localizer) may be provided by 6 collinear horizontally polarized corner reflector antennas close to the ground at the far end of the runway. The system operates at 110 MHz. A nose-on beam aligned with the runway is produced with all corners in-phase and a split-beam with null aligned with the runway is produced with the 3 left and the 3 right corners fed in opposite phase. A double-sideband modulated signal provides carrier and sidebands for the nose-on lobe and sideband only modulated signals of opposite phase for the two lobes of the split-beam. A vertical needle in the cockpit moves left to indicate that the aircraft must fly left and to the right to indicate that the aircraft must fly right. See Fig. 21–37.

(2) ***Vertical*** guidance for the glide slope is provided by two corner reflector antennas stacked one above the other on a 10-m tower alongside the runway. See Fig. 21–38. The system operates at 330 MHz. The upper antenna and its image produce a split lobe with null aligned with the approach angle. The lower antenna and its image produce a nose-on lobe

---

[1]Prepared with the assistance of Professor Richard McFarland, Avionics Engineering Center, Ohio University, Athens and Dr. Eric Walton, ElectroScience Laboratory, Ohio State University, Columbus.

[2]Maintenance of ILS Facilities, U.S. Dept. of Transportation, Federal Aviation Administration.

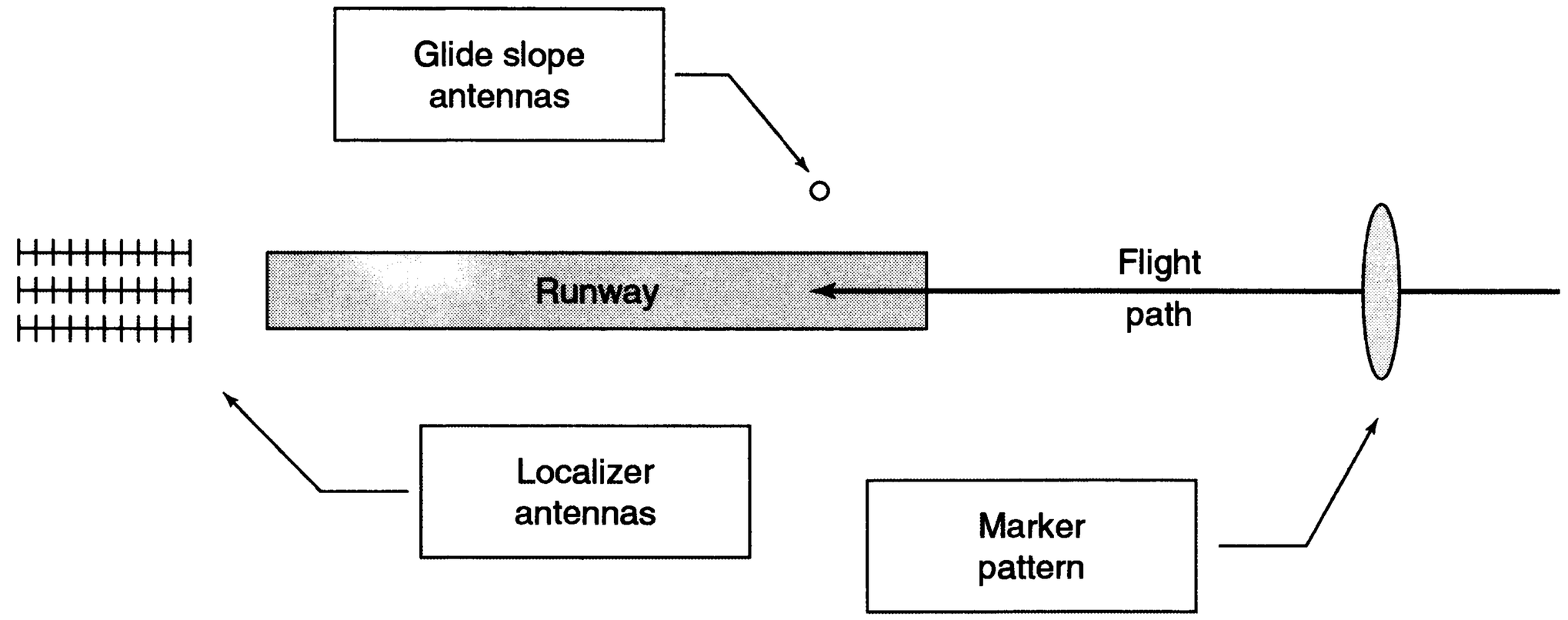


Figure 21–36
ILS antenna locations with azimuth or localizer antennas at the far end of the runway, the vertical guidance or glide-slope antennas alongside the runway and the range makers along the approach path.

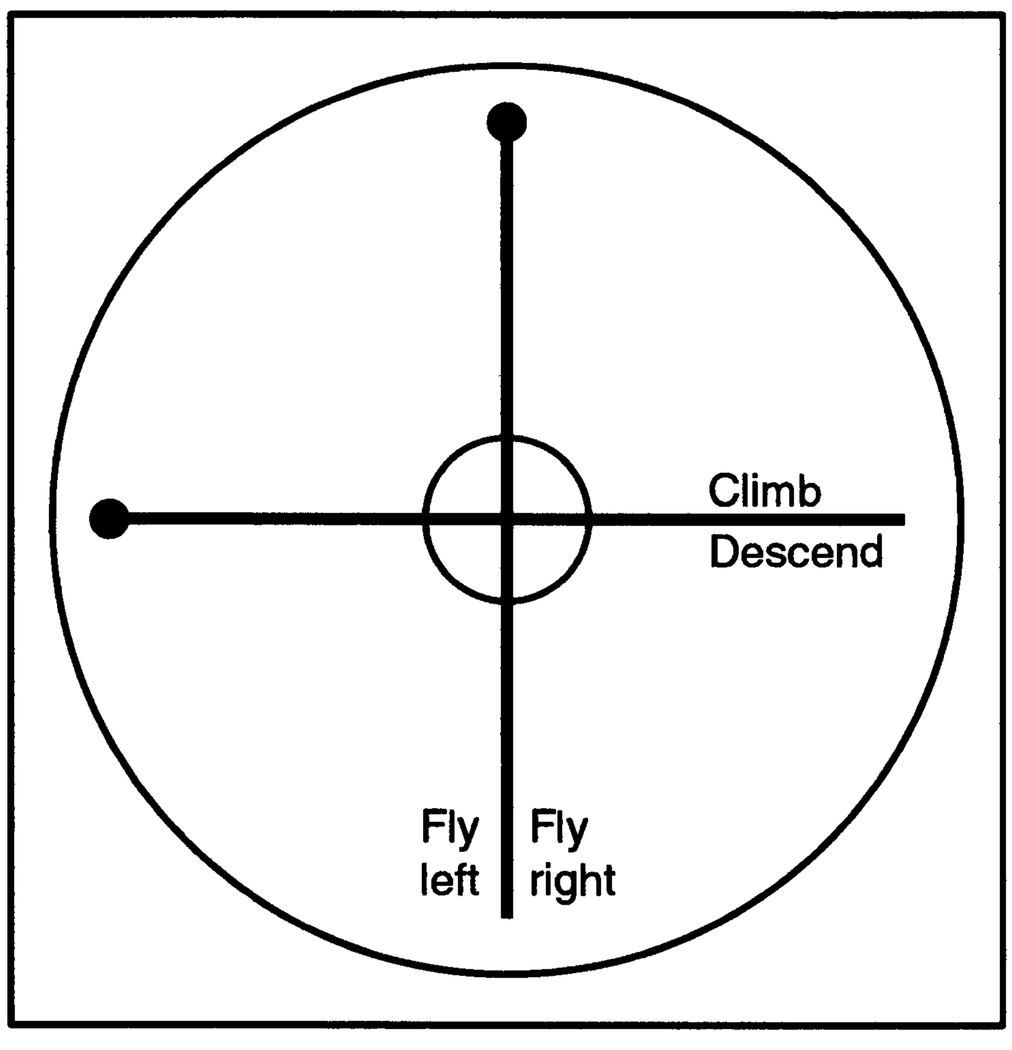


Figure 21–37
ILS cockpit display. With needles as shown, the aircraft is aligned with the runway and is on the proper glide slope. When a needle deflects from normal you "chase the needle" by flying left or right or up or down. In newer systems the needles may be replaced by a computer generated image.

aligned with the approach angle, usually. A double-sideband modulated signal provides carrier and sidebands for the nose-on lobe and sideband only modulated signals of opposite phase for the two lobes of the split beam. A horizontal needle in the cockpit moves up to indicate that the aircraft must climb and down to indicate that it must descend. See Fig. 21–37.

The region of the ground where the reflection takes place is protected as a "critical area" where vehicles and obstructions are not permitted. Deep snow can raise the effective height of the ground which could raise the slope angle an undesirable amount.

(3) ***Distance*** or range to the runway is provided by 75-MHz marker signals at various distances from the runway. As the aircraft passes over a marker beam there is both a visual and audible indication in the cockpit with coding to indicate the distance.

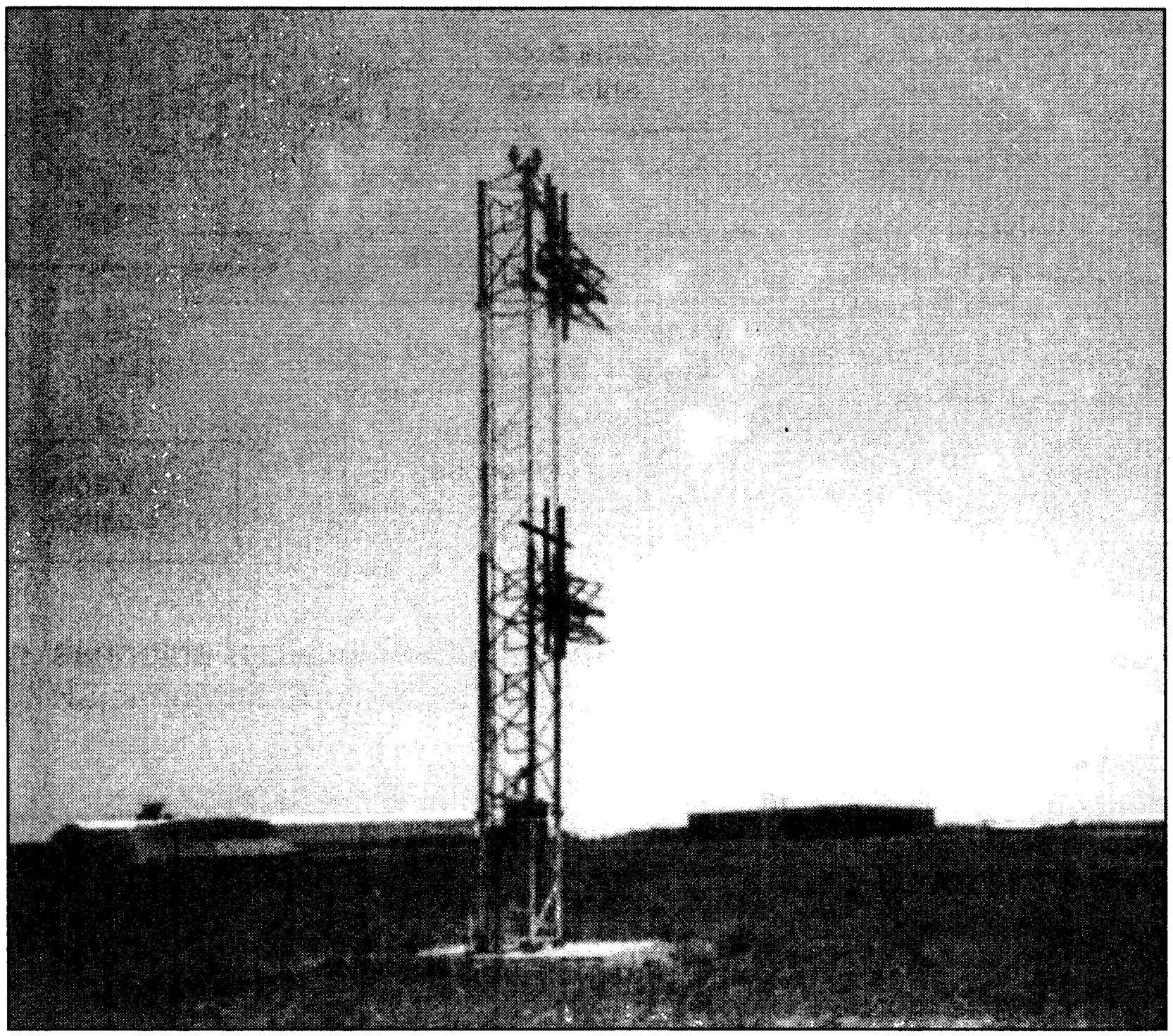


Figure 21–38
Airport installation of two corner reflectors at the heights of Example 21–18.2 to produce a glide slope at 3 degrees elevation angle.

EXAMPLE **21–18.1 Azimuth or Localizer Antennas**

Design an antenna system which fulfills the azimuth guidance requirements of paragraph (1).

■ **Solution**

The beams are provided by 3 horizontally polarized ordinary end-fire arrays each with 12 $\lambda/2$ dipoles as in Fig. 21–39*a*. The arrays are located at the far end of the runway as shown in Fig. 21–36. With ordinary ground there will be adequate signal at low elevation angles. For a null aligned with the runway, arrays 1 and 3 are fed in opposite phase. Array 2 provides the nose-on beam with carrier. The other arrays carry the sideband signals. For a spacing $d = 1.5\ \lambda$ between arrays 1 and 3, the first maximum either side of the null is at an angle $\alpha = \sin^{-1}(0.5\ \lambda/1.5\ \lambda) = 19.5°$ (see insert in Fig. 21–39). This 2 source pattern multiplied by the pattern of an ordinary end-fire array of 12 point sources with $\lambda/4$ spacing and multiplied once more by the pattern of a $\lambda/2$ dipole results in the dashed-line field pattern in Fig. 21–39*b*. *Ans.*

Actual localizers may be more elaborate, but this simple configuration illustrates the basic principles.

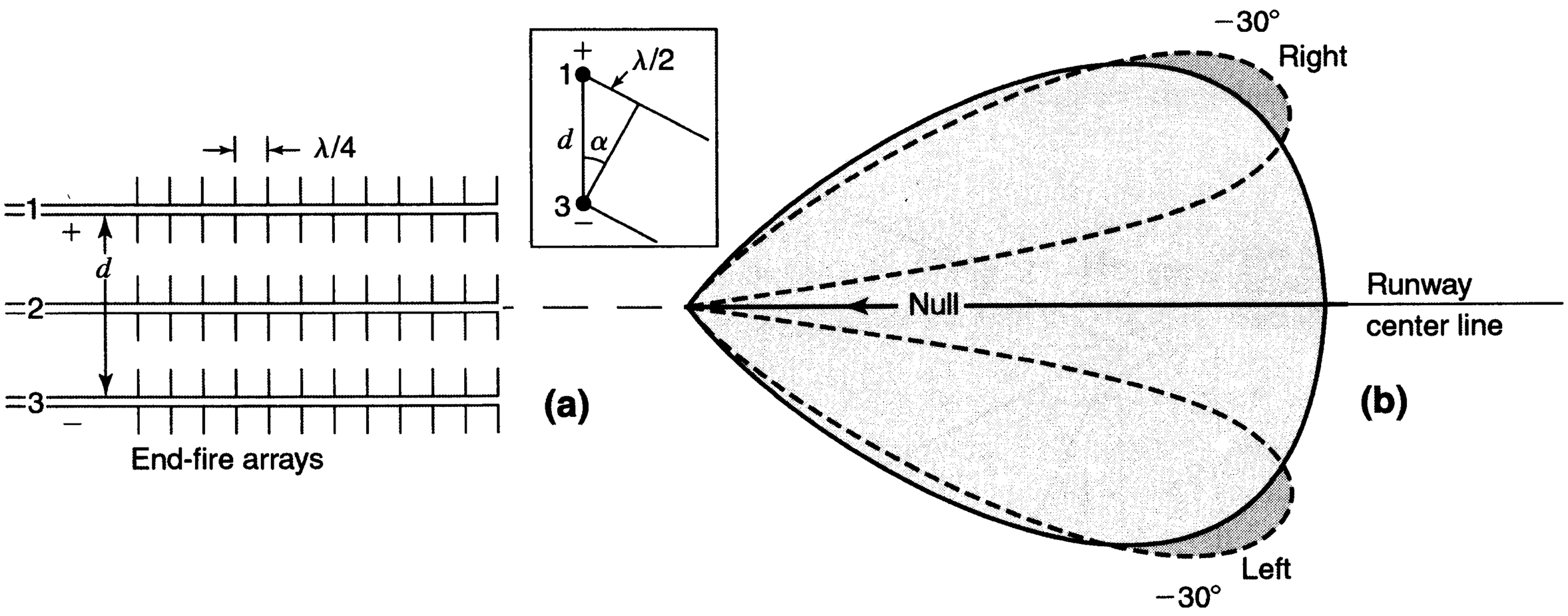


Figure 21–39
(*a*) 12-element end-fire localizer antennas for azimuth guidance and (*b*) their field patterns.

EXAMPLE **21–18.2 Vertical Guidance Antennas**

Design an antenna system which fulfills the vertical guidance requirements of paragraph (2) for a glide slope at an elevation angle of 3°.

■ **Solution**

Using a ground-bounce or image system, a corner reflector with a horizontal dipole is placed at a height h above the ground which is assumed to be perfectly conducting. As suggested in Fig. 21–40, the direct wave from the dipole and its image at a depth h below the surface will reinforce at the angle α and produce a maximum when $d = 0.5\lambda$, 1.5λ, etc., or produce a null when $d = 1\lambda$, 2λ, etc.

Thus, for the split-lobe beam with a null at 3 degrees

$$\sin\alpha = \frac{d}{2h}$$

or

$$h(\text{null}) = \frac{d}{2\sin\alpha} = \frac{\lambda}{2\sin 3^\circ} = 9.6\lambda$$

For the nose-on beam at 3 degrees

$$h(\text{max}) = \frac{0.5\lambda}{2\sin\alpha} = \frac{0.5\lambda}{2\sin 3^\circ} = 4.8\lambda$$

Thus, with one corner at 9.6λ and another at half the height (4.8λ), the requirements are satisfied. See Fig. 21–41. To restrict the beam in azimuth, 3 collinear dipoles are customarily used in the corner reflector. See Fig. 21–38.

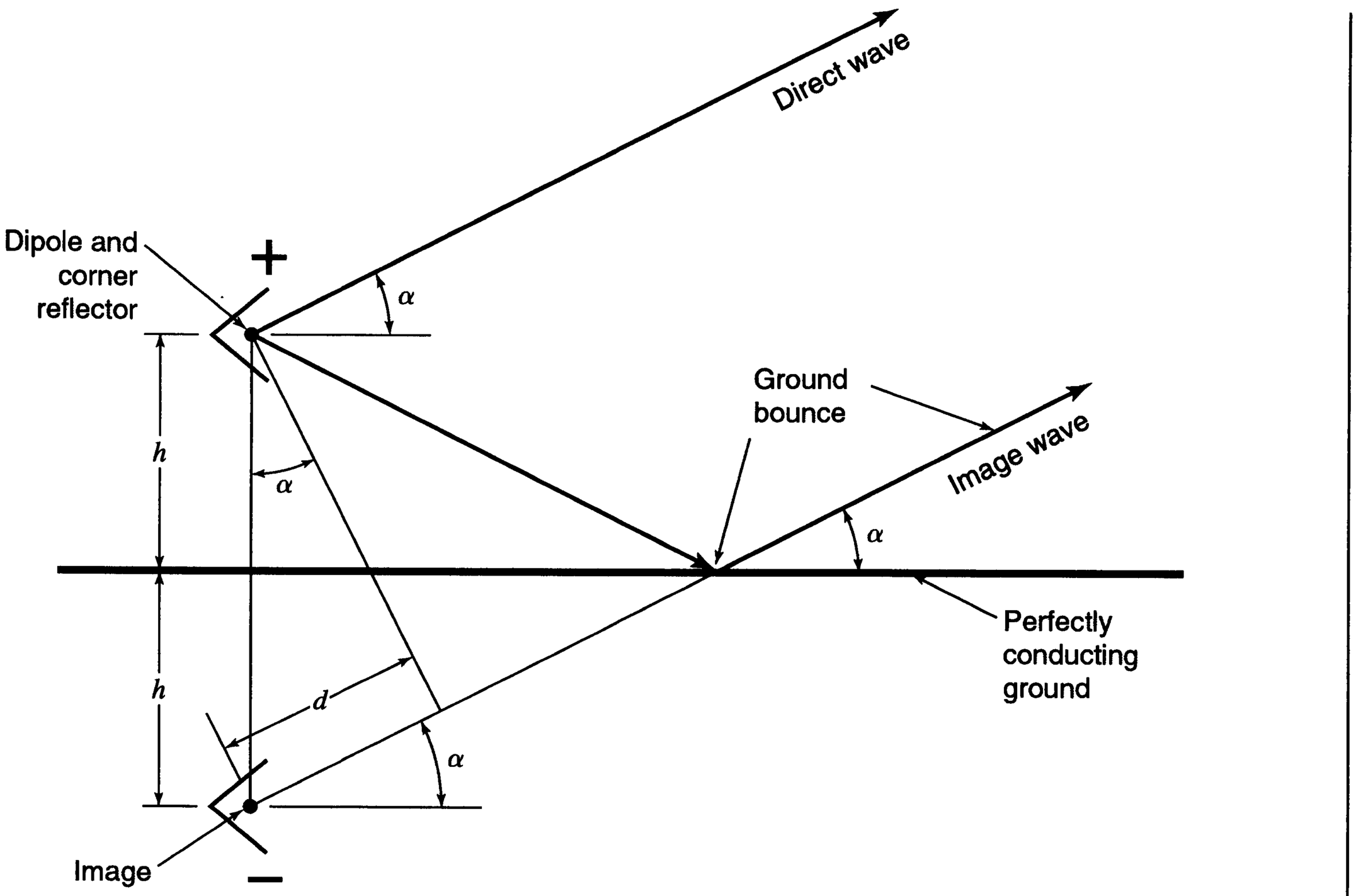


Figure 21–40
Corner reflector antenna and its image.

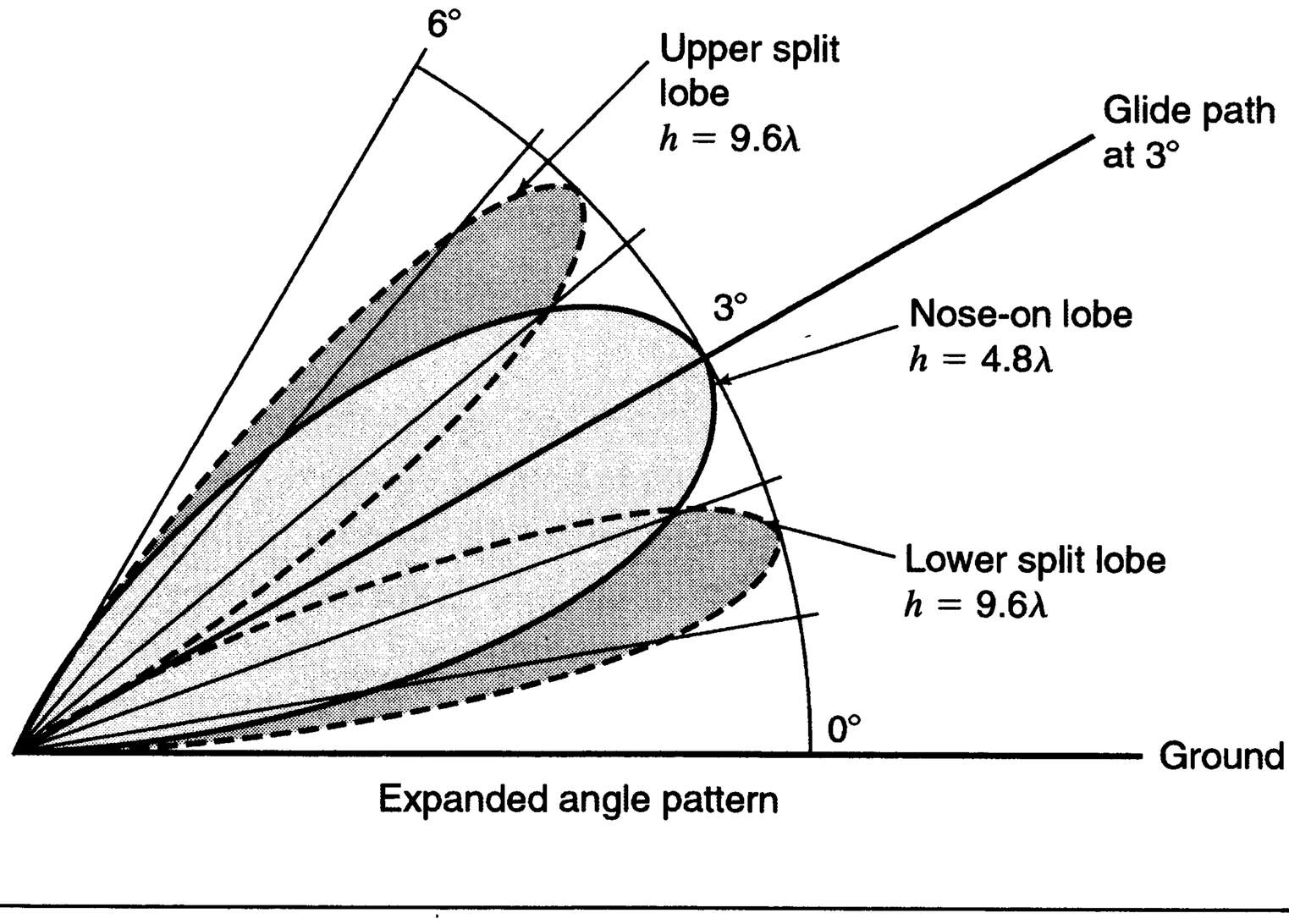


Figure 21–41
Split-lobe and nose-on beams of corner reflector antennas for glide path at elevation angle of 3 degrees in expanded angle display.

EXAMPLE **21–18.3 Range Marker Antennas**

Design an antenna system which fulfills the distance or range requirements of paragraph (3) above.

■ **Solution**

Two in-phase 4-element Yagi-Uda antenna, as in Fig. 21–42*a*, set at 45° from vertical to the left and right of the approach path, produce the pattern shown in Fig. 21–42*b*.

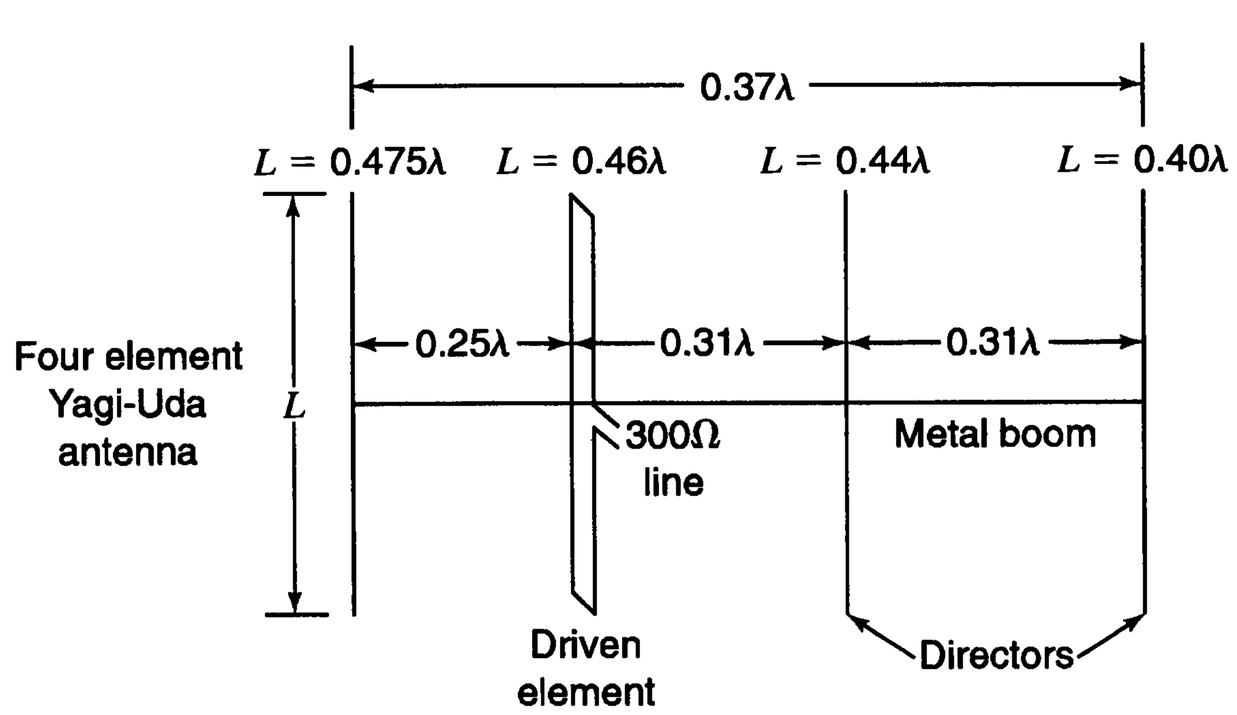


Figure 21–42
(*a*) Four-element Yagi-Uda antennas and (*b*) field pattern of the twin array for the range marker. The glide path is perpendicular to the page.

21–19 THE "SUGAR SCOOP ANTENNA" AND THE 3K COSMIC SKY BACKGROUND STORY（糖铲形天线和 3K 宇宙背景的故事）

The sky temperature is about 25 K at $\lambda = 1$ m but only a few kelvin toward the zenith at $\lambda = 10$ cm (see Fig. 21–44). Before 1965 it was generally thought that the sky temperature at the zenith at centimeter wavelengths would be close to absolute zero (0 K).

The Bell Telephone Laboratory needed more accurate data to determine the SNR of intercontinental radio links via GSO satellites. So in 1965 Arno Penzias and Robert Wilson of the laboratory were comparing the antenna temperature T_A of a large alt-azimuth-mounted

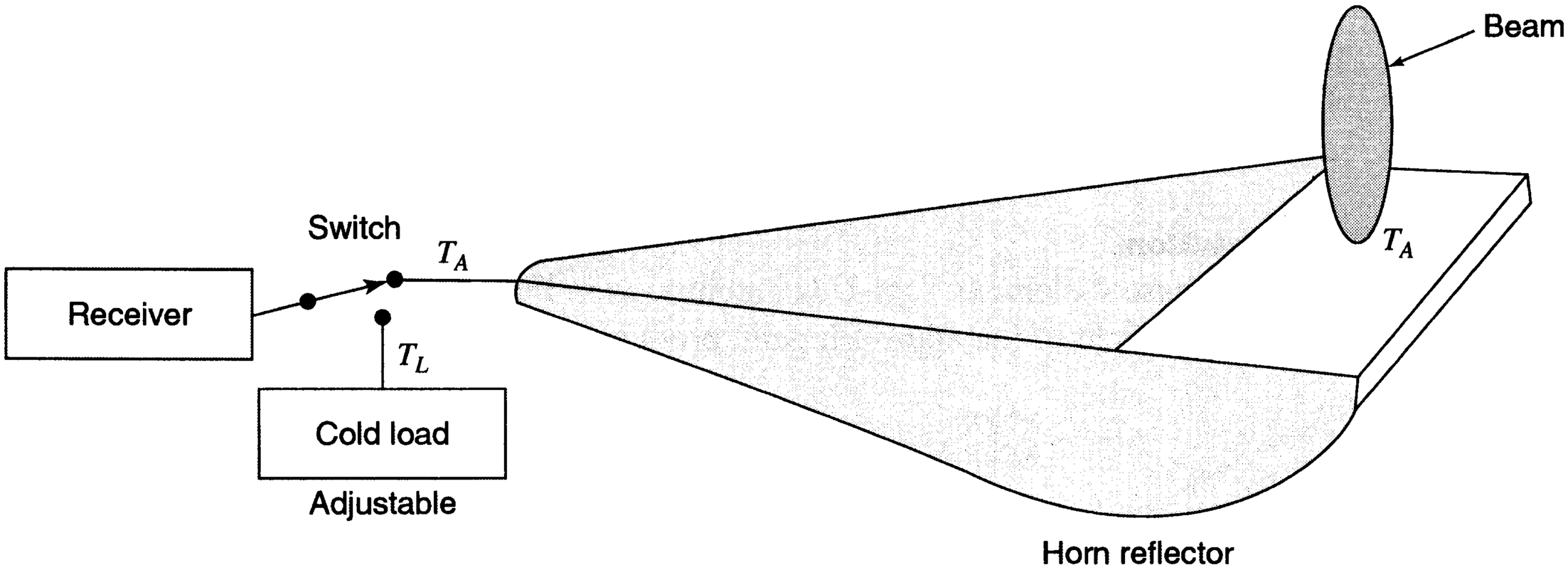


Figure 21–43
The sugar scoop and cold load with switch for null measurements.

Table 21–1 Temperature budget of horn-reflector radio telescope of Penzias and Wilson at 4.08 GHz (see Appendix E)

Temperature measured at zenith		6.7 ± 0.8 K
Temperature accounted for as follows:		
Atmospheric absorption	2.3 ± 0.3	
Ohmic losses	0.8 ± 0.4	
Ground pickup via back lobes	<0.1	
Total accounted for	3.2 ± 0.5	3.2 ± 0.5
Residual (unaccounted-for) temperature		3.5 ± 1.0 K

horn-reflector that resembles a giant sugar scoop with the temperature of a precision calibrated cold load T_L (Fig. 21–43) at $\lambda = 74$ mm. When switching between the two gave a null result, it meant that the sky temperature and load temperature were equal, or $T_A = T_L$. The value Penzias and Wilson measured near the zenith in regions devoid of radio sources was 6.7 K.

Of this 2.3 K could be attributed to radiation from the earth's atmosphere and 0.9 K to losses or spurious pickup by the antenna. See the temperature budget in Table 21–1. This left 3.5 K unaccounted for. This was not large but not the absolute zero expected. Bird droppings and poor connections in the horn were suspected as the source but after the horn was cleaned and the joints were covered with conducting tape the results were unchanged.

Thus, whenever Penzias and Wilson turned the scoop toward apparently empty sky, the 3 K residual persisted. This temperature is believed to be the background radiation from the cosmic fireball of the "Big Bang" that created the universe. It establishes a limit to the sensitivity of all sky scanning antennas. In 1978 Penzias and Wilson shared the Nobel Prize for their discovery. See Fig. 21–45.

Subsequent measurements with the Cosmic Background Explorer satellite (COBE) in 1983 set the temperature at 2.726 + 0.010 K. See Fig. 21–46, also Example 21–19.1.

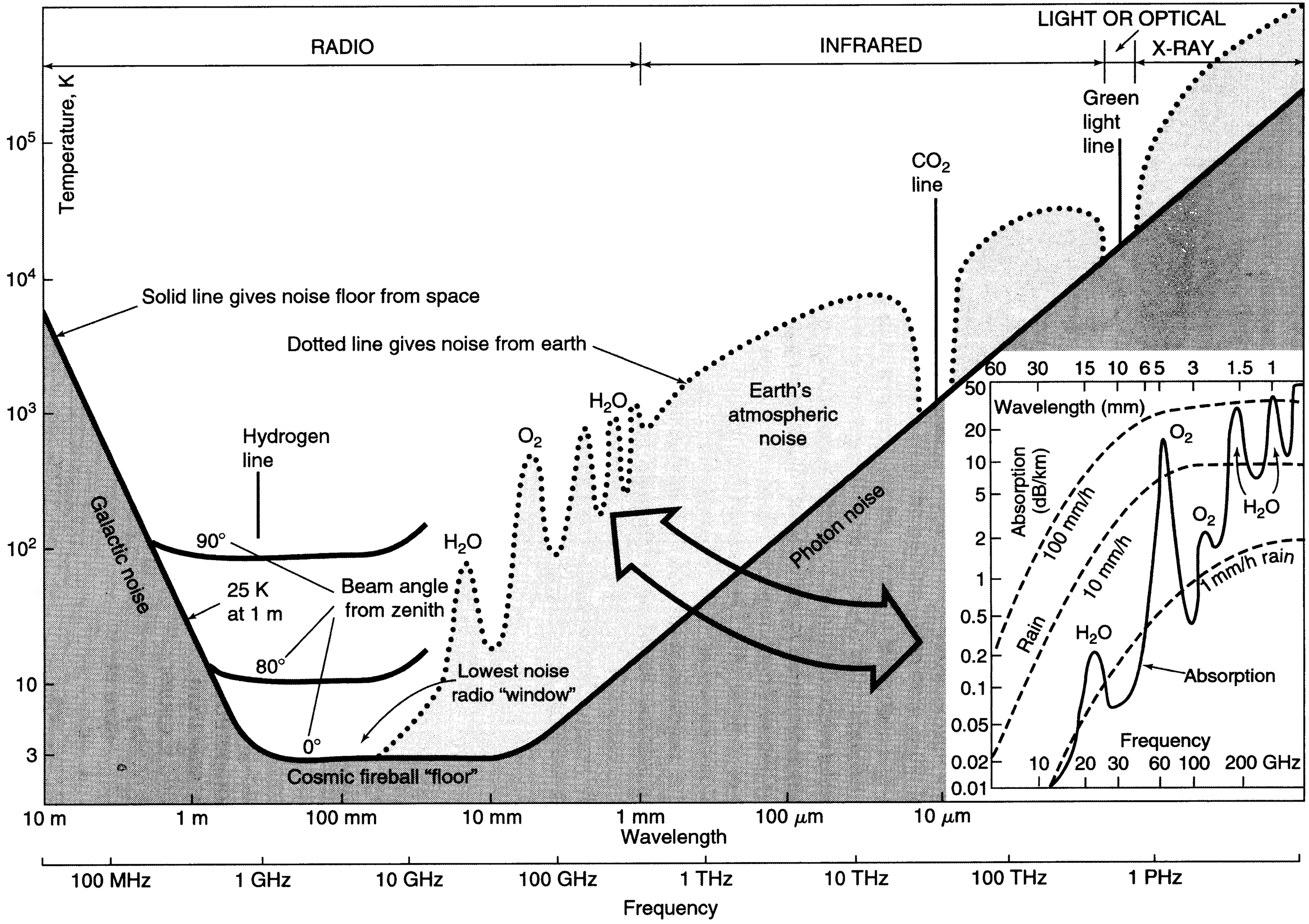


Figure 21–44
Sky noise temperature from radio to x-ray. Inset gives the attenuation by the atmosphere and by rain.

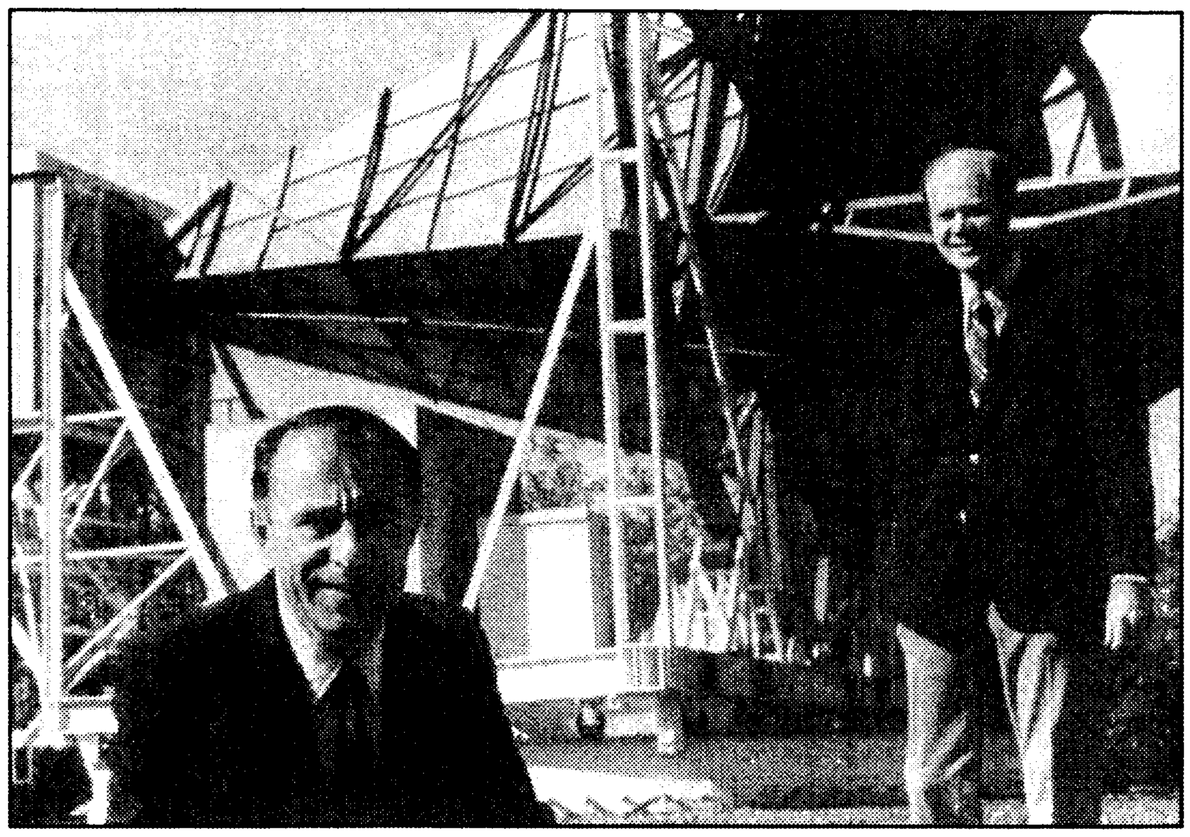


Figure 21–45
Arno Penzias (left) and Robert Wilson with the giant "sugar scoop" horn-reflector antenna with which they found the primordial fireball. A small discrepancy led them to the grandest of all possible answers. (*Courtesy, Bell Telephone Laboratories.*)

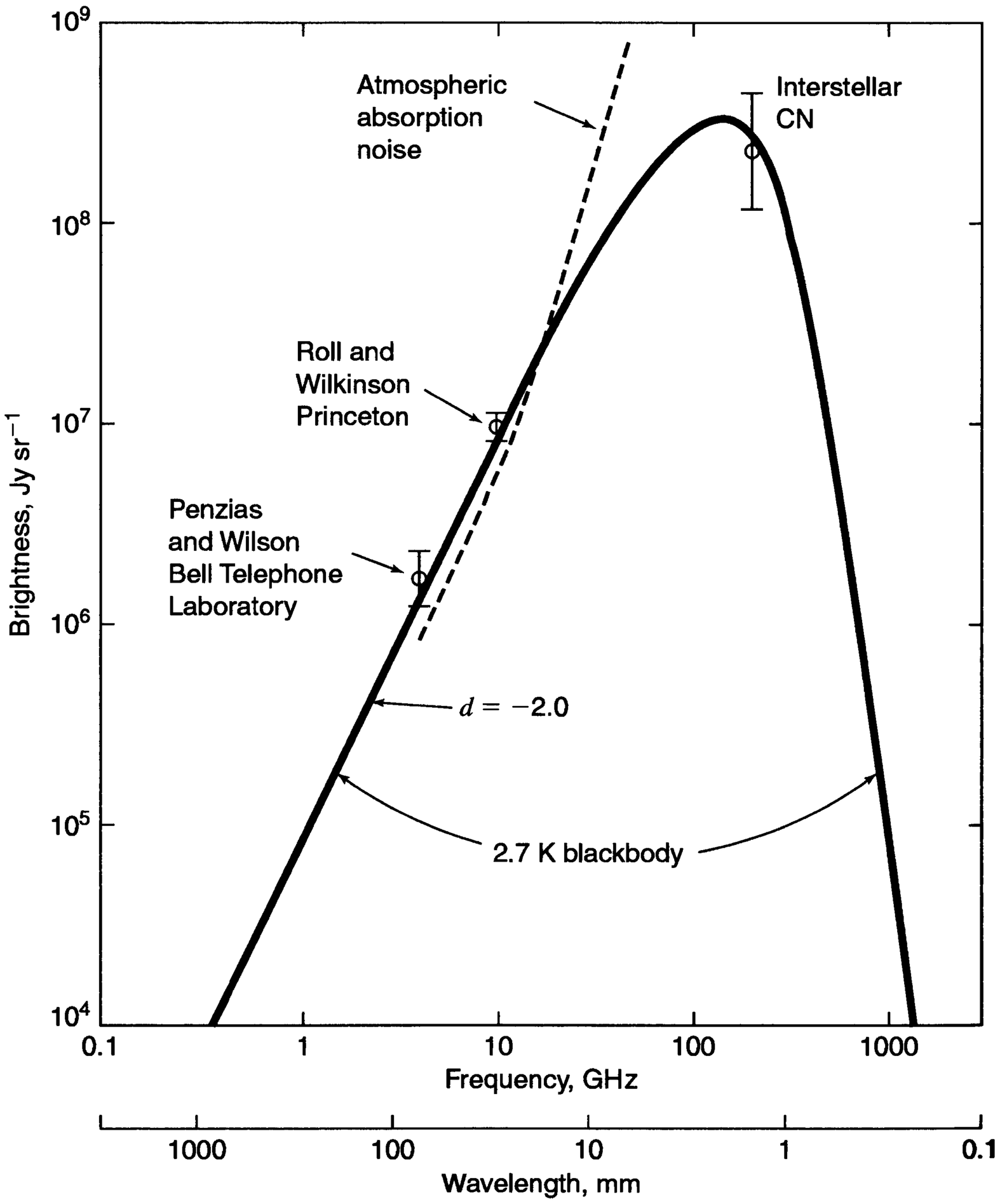


Figure 21–46
Microwave sky background measurements compared with a 2.7 K blackbody spectrum. Cosmic Background Explorer (CODE) measurements (1983) set the temperature at 2.726 ± 0.010 K. (*After Wilkinson (1) and Peebles, and Weiss-1.*)

The microwave background radiation comes from the greatest distance in the universe, even beyond the quasars, and affords the ultimate look-back times—many billions of light-years. For review articles see Wilson (1), Wilkinson (1) and Peebles and Kraus (2).

The sugar-scoop antenna is also referred to as a Hogg horn after D. C. Hogg of the Bell Laboratories (Crawford-1). The horn turned vertical with beam horizontal is widely used on microwave towers for phone and data links.

EXAMPLE **21–19.1 Horn Absorber**

If a perfect absorber with the impedance of space (= 377 Ω/square) is placed so as to completely cover the front of a horn antenna, it will ideally produce an antenna temperature equal to the absorber's thermometer-measured temperature (see Fig. 21–46–1). If the absorber is completely shielded from the outside (open only to the horn) and is cooled, it can provide calibration temperatures for the radio telescope. Thus, if cooled to liquid helium temperature it will give a 4.1 K calibration. Or if the absorber temperature can be controlled so that when the absorber is in front of the horn the radio telescope response is the same as with the absorber removed, the temperature of the object or region being observed by the radio telescope is equal to the absorber temperature. This null type of measurement was used on the Cosmic Background Explorer (COBE) satellite to remeasure the 3-K, 4-GHz sky background originally discovered by Penzias and Wilson. The temperature has now been evaluated more accurately as 2.73 K. This temperature is believed to be from the remnant of the primordial Big Bang and is the lowest possible antenna temperature for a sky-scanning antenna.

If a narrow-beam 4-GHz antenna looking at essentially empty sky at the zenith has an antenna temperature of 4.73 K, how much is due to side lobes or antenna loss?

■ **Solution**

$4.73 - 2.73 = 2°\text{C}$. *Ans.*

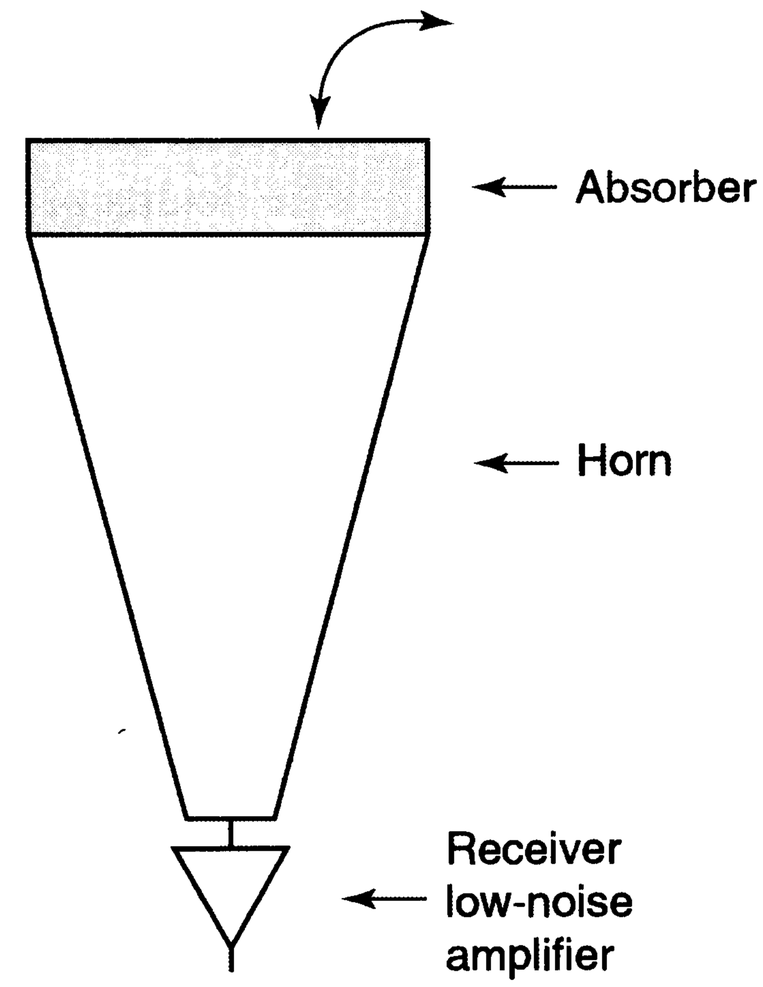


Figure 21–46–1

21-20 LEO SATELLITE LINK ANTENNAS[1] [用于近地轨道（LEO）卫星线路的天线]

The penetration of base-station towers into metropolitan and suburban areas now provides satisfactory service for many. However, the prospector in the Andes Mountain valley or the camel driver in the Sahara Desert may be out of cell-tower range. To serve them and, in fact, anyone else almost anywhere on the earth, there are constellations of satellites in orbit, with more planned (Miller-1). See Fig. 21–47.

At a cost of billions of dollars, Motorola deployed the *Iridium* system (1.65 GHz) with 66 satellites in low-earth orbit (LEO) at 780 km (Schuss-1). These satellites are positioned in six polar-crossing orbits with 11 satellites per orbit.

The *Globalstar* system (1.6 and 2.5 GHz) put up 48 satellites in low-earth orbit at 1414 km (Dietrich-1).

The *Teledesic* system (19 GHz downlink, 29 GHz uplink) is planned to have 288 satellites at 1400 km or less. This system is to provide high-speed data links whereas the other's primary mission was phone service (Wertz-1).

Because of their orbits, LEO satellite signals can come from high angles in the sky in contrast to cell-tower signals at low angles close to the horizon. As a result, cell-tower signals usually penetrate indoors and under trees better than signals from LEO satellites. Furthermore, cell-tower cell phones are lighter and less expensive. The LEO satellite constellations are also expensive to put up and to maintain. The LEO satellites are on the order of 100 times farther from the user, resulting in a 40 dB signal loss. And the 19 and 29 GHz (Ka band) frequencies are subject to attenuation from rain. These factors and the possible limited potential market pose a challenge for the LEO systems. However, the LEO systems are not faced with the NIMBY problem.

Fred J. Dietrich comments as follows: "Satellite antennas vary widely in their gain, beamwidth and coverage performance depending on the application (Dietrich-2). Higher gain always means higher Effective Radiated Power (ERP) for transmit and higher signal to noise for receive, but higher gains involve smaller coverage areas, thus resulting in tradeoffs." Furthermore, he states: "In modern communications, which is almost all digital, methods of calculating RF circuit performance have changed. For example, with forward error correction and methods of demodulation, good communications can take place with lower SNRs than were previously possible, reducing power and antenna gain requirements." The details of the antennas used and all the tradeoffs involved are too extensive to include here; however, it is instructive to calculate power requirements for several antenna system combinations.

---

[1] We thank Fred J. Dietrich of F-D Engineering for assistance on this section.

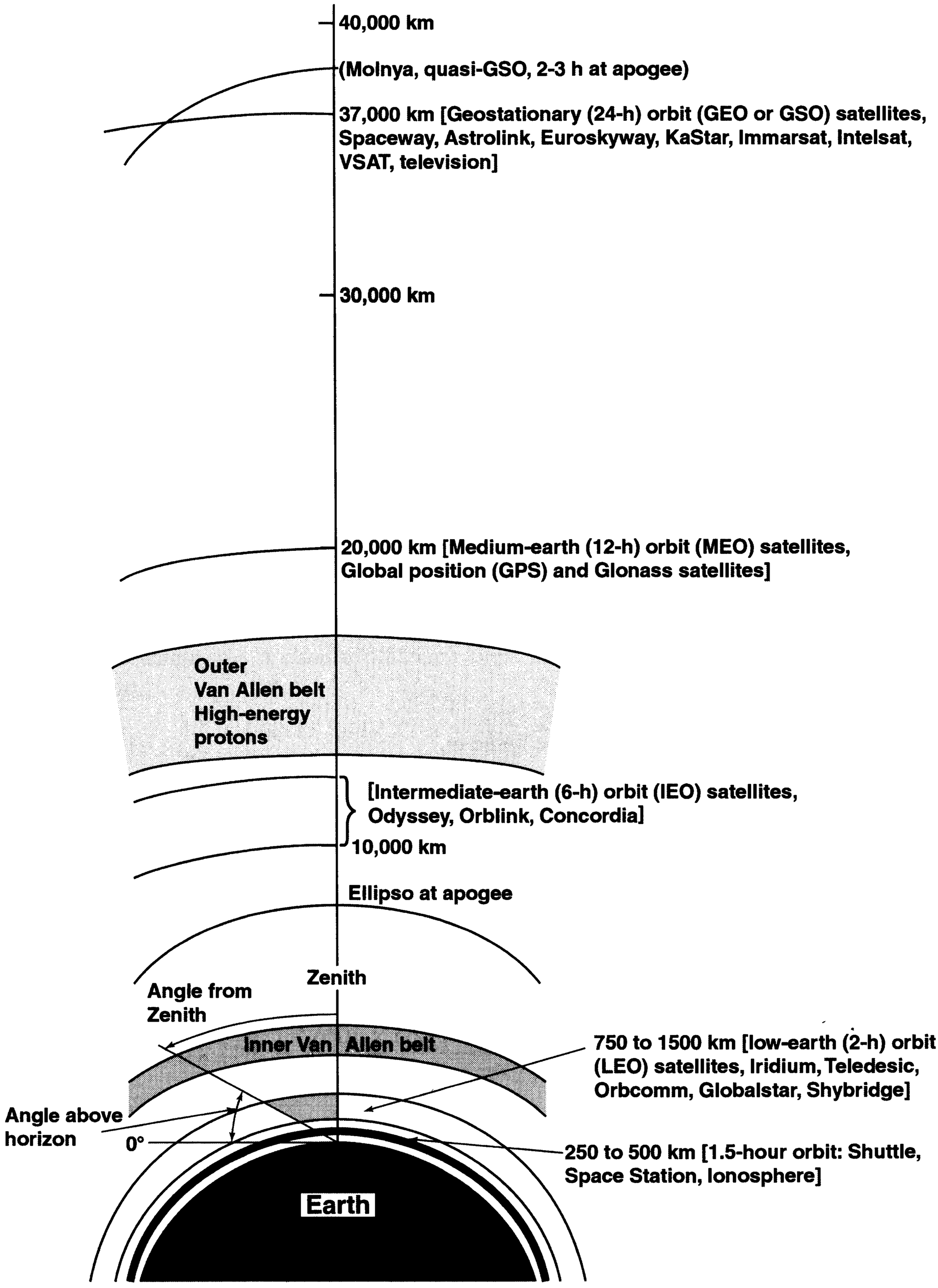


Figure 21–47
The electrons (in the ionosphere), the protons (in the Van Allen belts), and the satellites (LEO to GEO) in space above the earth. Note that the height of Mt. Everest is imperceptible on this figure.

EXAMPLE **21–20.1 Hand-Held Uplink and Downlink Powers with LEO Satellite at 780 km and 1.65 GHz**

The satellite receiver has a temperature $T_R = 45$ K and an RCP antenna with 6 dBi gain. The bandwidth corresponds to a channel of 9.6 kHz.

Assuming that audio communications are used so that an SNR = 10 dB is required and with the satellite 10° off zenith, find the hand-held uplink power and satellite downlink power for hand-held antennas as follows:

(*a*) Uplink power for vertical $\lambda/2$ antenna
(*b*) Downlink power for vertical $\lambda/2$ antenna
(*c*) Uplink power for horizontal $\lambda/2$ antenna
(*d*) Downlink power for horizontal $\lambda/2$ antenna
(*e*) Uplink power for quadrifilar helix antenna directed at the zenith. It has RCP with a gain of 3 dBi.
(*f*) Downlink power for quadrifilar helix directed at zenith.

■ **Solution**

From the Friis transmission formula

$$\frac{P_r}{P_t} = \frac{A_{er}A_{et}}{r^2\lambda^2} = \frac{D_r D_t \lambda^2}{(4\pi)^2 r^2}$$

and since

$$\frac{S}{N} = \frac{P_r}{N}$$

the desired power can be found by

$$P_t = \left(\frac{S}{N}\right)\frac{N}{D_r D_t}\left(\frac{4\pi r}{\lambda}\right)^2$$

working in dB for convenience

$$P_{t\,\mathrm{dB}} = SNR_{\mathrm{dB}} - D_{r\,\mathrm{dB}} - D_{t\,\mathrm{dB}} + N_{\mathrm{dB}} - L_{d\,\mathrm{dB}}$$

where the spreading of the energy as it propagates is being displayed as a spatial loss factor, that is,

$$L_d = \left(\frac{\lambda}{4\pi r}\right)^2 = \left(\frac{3\times 10^8/1.65\times 10^9}{4\pi \times 7.8\times 10^5}\right)^2 = 3.44\times 10^{-16} = -155\ \mathrm{dB}.$$

(a) Uplink power for vertical $\lambda/2$ dipole

The satellite antenna mostly subtends the earth, so the satellite antenna temperature for our purposes will be 300 K and the satellite system temperature is

$$T_{\mathrm{sys}} = T_A + T_R = 300 + 45 = 345\ \mathrm{K}$$

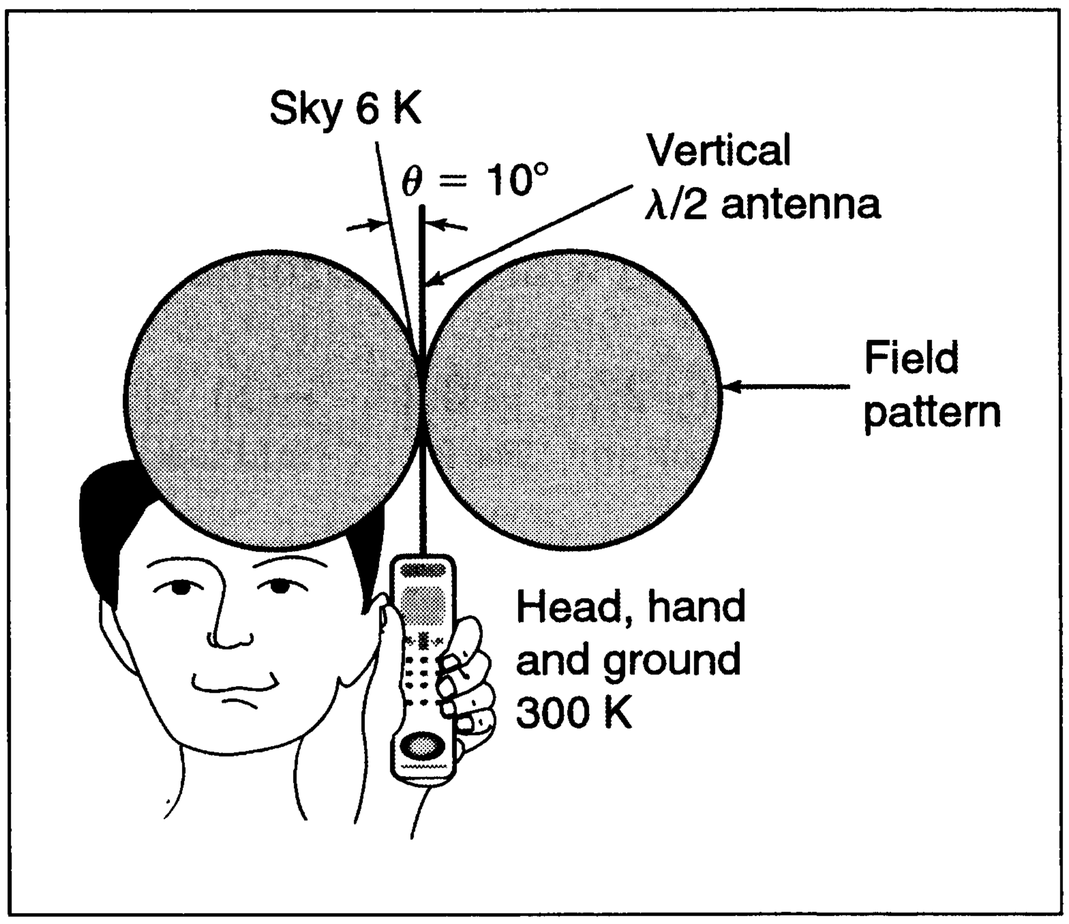
The noise power is

$$N = kT_{\mathrm{sys}}B = 1.4\times 10^{-23}\times 345\times 9.6\times 10^3 = 4.64\times 10^{-17} = -163\ \mathrm{dB}$$

From (6–4–4), the pattern factor for the dipole can be found from

$$E(\text{vert } \lambda/2 \text{ dipole}) = \cos\left(\frac{\pi \cos\theta}{2}\right) \Big/ \sin\theta$$

where θ = zenith angle (see Fig. 21–47, also sketch below).



So for $\theta = 10°$, $E = 0.137$ or -17.2 dB. Also since the dipole is linear polarized and the transmitter is circular polarized, there will be a -3 dB loss in coupling between the two.

$$P_t = \underset{\text{SNR}}{10} - (\underset{\text{Dipole gain}}{+2.15} \underset{\text{Pat. fac.}}{-17.2} \underset{\text{CP to LP loss}}{-3}) - (\underset{\text{Sat. ant.}}{+6}) + (\underset{\text{Noise}}{-163}) - (\underset{\text{Prop. loss}}{-155}) = 14.05 \text{ dB}$$

or ***25.4 W*** required for SNR = 10 dB *Ans.* (*a*)

(b) Downlink power for vertical $\lambda/2$ dipole

About $\frac{1}{2}$ of the beam area of the vertical $\lambda/2$ dipole antenna subtends the sky while the other half subtends the head, hand and ground (300 K); see sketch. Thus, the antenna temperature is

$$T_A = \frac{1}{\Omega_A}\left(\frac{1}{2}\Omega_A 6 + \frac{1}{2}\Omega_A 300\right)\text{K} = 156 \text{ K}$$

$$T_{\text{sys}} = T_A + T_R = 156 + 75 = 231 \text{ K}$$

The noise power is then

$$N = 1.4 \times 10^{-23} \times 231 \times 9.6 \times 10^3 = 3.10 \times 10^{-17} = -165 \text{ dB}$$

so

$$P_t = \underset{\text{SNR}}{10} - (\underset{\text{Dipole gain}}{+2.15} \underset{\text{Pat. fac.}}{-17.2} \underset{\text{CP to LP loss}}{-3}) - (\underset{\text{Sat. ant.}}{+6}) + (\underset{\text{Noise}}{-165}) - (\underset{\text{Prop. loss}}{-155}) = 12.05 \text{ dB}$$

or ***16.0 W*** required power. *Ans.* (*b*)

(c) Uplink power for horizontal $\lambda/2$ dipole

With the $\lambda/2$ dipole being horizontal, the worst case is for $\theta = 80°$ in (6–4–4) for which $E = 0.978$ or -0.2 dB. Thus

$$P_t = \underset{\text{SNR}}{10} \; \underset{\text{Dipole gain}}{-(+2.15} \; \underset{\text{Pat. fac.}}{-0.2} \; \underset{\text{CP to LP loss}}{-3)} \; \underset{\text{Sat. ant.}}{-(+6)} \; \underset{\text{Noise}}{+(-163)} \; \underset{\text{Prop. loss}}{-(-155)} = -2.95 \text{ dB}$$

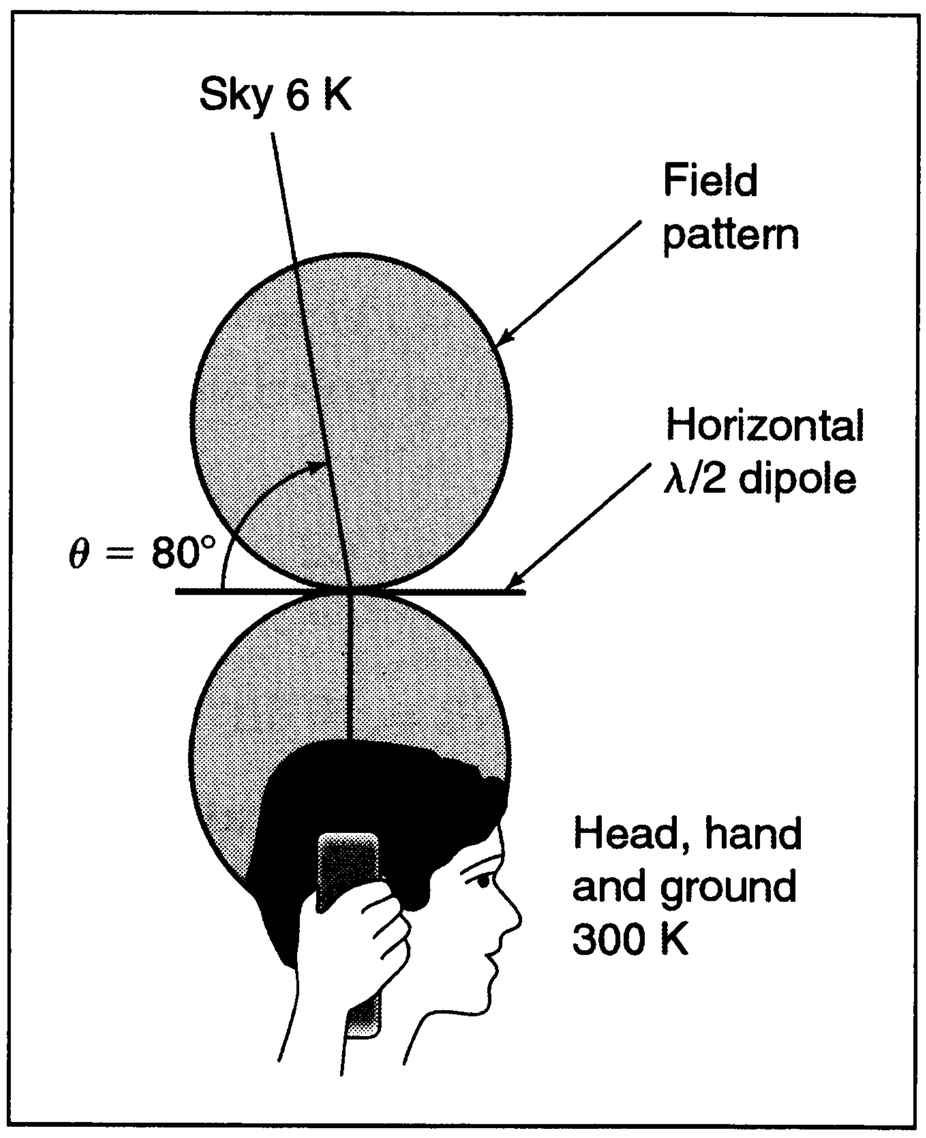
or ***507 mW*** required power for SNR = 10 dB. *Ans.* (*c*)

(d) Downlink power for horizontal $\lambda/2$ dipole

As in Problem (*b*) with the horizontal dipole pattern factor

$$P_t = \underset{\text{SNR}}{10} \; \underset{\text{Dipole gain}}{-(+2.15} \; \underset{\text{Pat. fac.}}{-0.2} \; \underset{\text{CP to LP loss}}{-3)} \; \underset{\text{Sat. ant.}}{-(+6)} \; \underset{\text{Noise}}{+(-165)} \; \underset{\text{Prop. loss}}{-(-155)} = -4.95 \text{ dB}$$

or ***320 mW*** required power for SNR = 10 dB. *Ans.* (*d*)



(e) Uplink power for RCP quadrifilar helix antenna

Proceeding as in Problem (*a*) except with the quadrifilar helix gain and a pattern factor as in Problem (*c*) (see sketch). In addition, since the helix is a circular polarized antenna there is no polarization loss, so

$$P_t = \underset{\text{SNR}}{10} \; \underset{\text{Helix gain}}{-(+3} \; \underset{\text{Pat. fac.}}{-0.2)} \; \underset{\text{Sat. ant.}}{-(+6)} \; \underset{\text{Noise}}{+(-163)} \; \underset{\text{Prop. loss}}{-(-155)} = -6.80 \text{ dB}$$

or ***209 mW*** required power for SNR = 10 dB. *Ans.* (*e*)

(f) Downlink power for RCP quadrifilar helix antenna

About 85% of the beam area of the hand-held helix antenna subtends the sky (6 K). See sketch. The remaining 15% subtends the hand, head and ground (300 K). Thus,

$$T_A = \frac{1}{\Omega_A}(0.85\Omega_A 6 + 0.15\Omega_A 300)\text{K} = 50\ \text{K}$$

The hand-held receiver has a temperature of 75 K so the hand-held system temperature is

$$T_{\text{sys}} = T_A + T_R = 50 + 75 = 125\ \text{K}$$

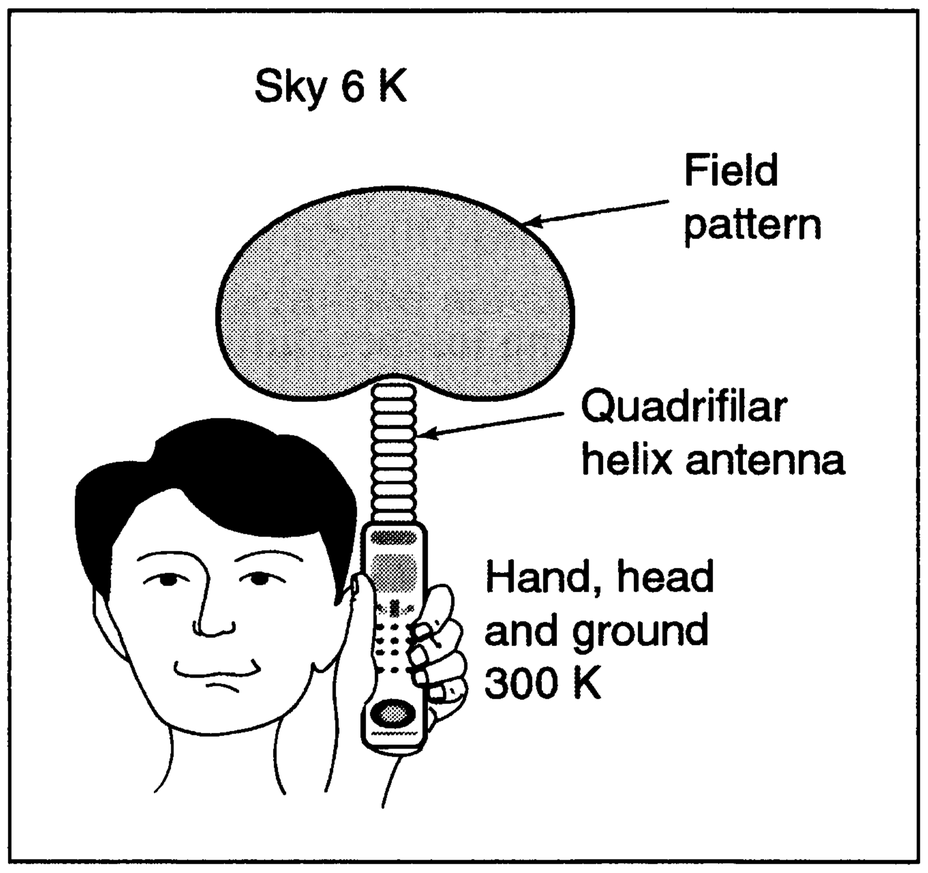
The noise power is then

$$N = 1.4 \times 10^{-23} \times 125 \times 9.6 \times 10^3 = 1.68 \times 10^{-17} = -168\ \text{dB}$$

so

$$P_t = \underset{\text{SNR}}{10} \quad \underset{\text{Helix gain}}{-(+3} \quad \underset{\text{Pat. fac.}}{-0.2)} \quad \underset{\text{Sat. ant.}}{-(+6)} \quad \underset{\text{Noise}}{+(-168)} \quad \underset{\text{Prop. loss}}{-(-155)} = -11.80\ \text{dB}$$

or ***66 mW*** required power per channel for SNR = 10 dB. *Ans.* (*f*)



Summary of results:

(*a*) Uplink power for vertical $\lambda/2$ dipole is 25.4 W.
(*c*) Uplink power for horizontal $\lambda/2$ dipole is 507 mW.
(*e*) Uplink power for RCP quadrifilar helix antenna is 209 mW.
(*b*) Downlink power for vertical $\lambda/2$ dipole is 16.0 W.
(*d*) Downlink power for horizontal $\lambda/2$ dipole is 320 mW.
(*f*) Downlink power for RCP quadrifilar helix antenna is 66 mW.

The type and orientation of antennas play an important role in the power budget in communications. They need to be a part of the engineer's trade-off studies.

EXAMPLE **21–20.2 Hand-Held Uplink and Downlink Power to LEO Satellite at 1400 km and 1.6 GHz**

The conditions are the same as in Example 21–20.1 except that $f = 1.6$ GHz, height = 1400 km, and satellite antenna with 25° × 25° RCP spot beam.

Modern digital communication techniques uses code division multiple access (CDMA) with digital encoding. Signal quality is expressed, not as SNR, but as E_b/N_o, where E_b = energy/bit, N_o = signal power/Hz. A 6 dB E_b/N_o ratio provides satisfactory quality. Both E_b/N_o and SNR are dimensionless. With a bandwidth for an individual communications channel the same as an audio one or 9.6 kHz, find (*a*) the uplink and (*b*) the downlink power for RCP quadrifilar helix hand-held with peak at the spot beam half-power level and the satellite overhead. Find the uplink power for the satellite at 10° from the horizon, which puts the separation distance at 3500 km.

■ **Solution**

From Example 21–20.1 the propagation loss factor for this satellite overhead is

$$L_d = \left(\frac{\lambda}{4\pi r}\right)^2 = \left(\frac{3 \times 10^8/1.6 \times 10^9}{4\pi \times 1.4 \times 10^6}\right)^2 = 1.136 \times 10^{-16} = -159 \text{ dB}$$

$$\text{Satellite gain} = \frac{40{,}000}{25° \times 25°} = 64 \text{ or } 18.1 \text{ dB}$$

(a) Uplink power

The satellite antenna temperature is again 300 K and the satellite system temperature

$$T_{\text{sys}} = T_A + T_R = 300 + 45 = 345 \text{ K}$$

The noise power is

$$N = 1.4 \times 10^{-23} \times 345 \times 9.6 \times 10^3 = 4.64 \times 10^{-17} = -163 \text{ dB}$$

so

$$\underset{\text{SNR}}{P_t =\ 6} \quad \underset{\text{Helix gain}}{-(+3)} \quad \underset{\text{Sat. ant.}}{-(+18.1)} \quad \underset{\text{Noise}}{+(-163)} \quad \underset{\text{Prop. loss}}{-(-159)} = -19.1 \text{ dB}$$

or ***12 mW*** required power for SNR = 6 dB. *Ans.* (*a*)

(b) Downlink power

From (*f*) of Example 21–20.1, the helix hand-held $T_{\text{sys}} = 125$ K so $N = -168$ dB. Thus

$$\underset{\text{SNR}}{P_t =\ 6} \quad \underset{\text{Helix gain}}{-(+3)} \quad \underset{\text{Sat. ant.}}{-(+18.1)} \quad \underset{\text{Noise}}{+(-168)} \quad \underset{\text{Prop. loss}}{-(-159)} = -24.1 \text{ dB}$$

or ***3.9 mW*** required power per channel for SNR = 6 dB. *Ans.* (*b*)

(c) Uplink power with satellite 10° from the horizon

Assume that the helix has about the same pattern near the horizon so the only difference is the separation distance. The propagation loss factor is then

$$L_d = \left(\frac{\lambda}{4\pi r}\right)^2 = \left(\frac{3 \times 10^8/1.6 \times 10^9}{4\pi \times 3.5 \times 10^6}\right)^2 = 1.817 \times 10^{-17} = -167 \text{ dB}$$

So

$$P_t = \underset{\text{SNR}}{6} \quad \underset{\text{Helix gain}}{-(+3)} \quad \underset{\text{Sat. ant.}}{-(+18.1)} \quad \underset{\text{Noise}}{+(-163)} \quad \underset{\text{Prop. loss}}{-(-167)} = -11.1 \text{ dB}$$

or ***78 mW*** required power per channel for SNR = 6. *Ans.* (*c*)

EXAMPLE **21–20.3 Uplink and Downlink to LEO Satellite at 30 GHz**

Satellite receivers have higher noise temperatures at 30 GHz. Thus, $T_R = 300$ K, satellite antenna $T_A = 275$ K, $f = 30$ GHz, height 1400 km, bandwidth per channel = 9.6 kHz, RCP satellite gain = 10 dBi with 10 dB attenuation for rain.

For uplink find (*a*) earth station antenna gain required for SNR = 10 dB; (*b*) the size required for a parabolic dish antenna with 70% aperture efficiency that tracks the satellite; and (*c*) the HPBW of the dish beam; and (*d*) earth station power. (*e*) For downlink find satellite power required with the determined antenna parameters.

■ **Solution**

From Example 21–20.1 the propagation loss factor here is

$$L_d = \left(\frac{\lambda}{4\pi r}\right)^2 = \left(\frac{3 \times 10^8/3.0 \times 10^{10}}{4\pi \times 1.4 \times 10^6}\right)^2 = 3.231 \times 10^{-19} = -185 \text{ dB}$$

(a) Uplink antenna gain

As in Example 21–20.1 the starting point is Friis' transmission formula, which can be rearranged to solve for the dish gain

$$D_r = \frac{1}{D_t}\left(\frac{P_r}{P_t}\right)\left(\frac{4\pi r}{\lambda}\right)^2 = \frac{1}{D_t}\left(\frac{P_r}{N}\right)\left(\frac{N}{P_t}\right)\left(\frac{4\pi r}{\lambda}\right)^2$$

where the noise was factored explicitly into the expression. Noting that $SNR = P_r/N$ and assuming that the transmitted power is assumed to be 1 W and with

$$T_{\text{sys}} = T_A + T_R = 275 + 300 = 575 \text{ K}$$

so that

$$N/(1 \text{ W}) = 1.4 \times 10^{-23} \times 575 \times 9.6 \times 10^3 = 7.7 \times 10^{-17} \text{ or } -161 \text{ dB}$$

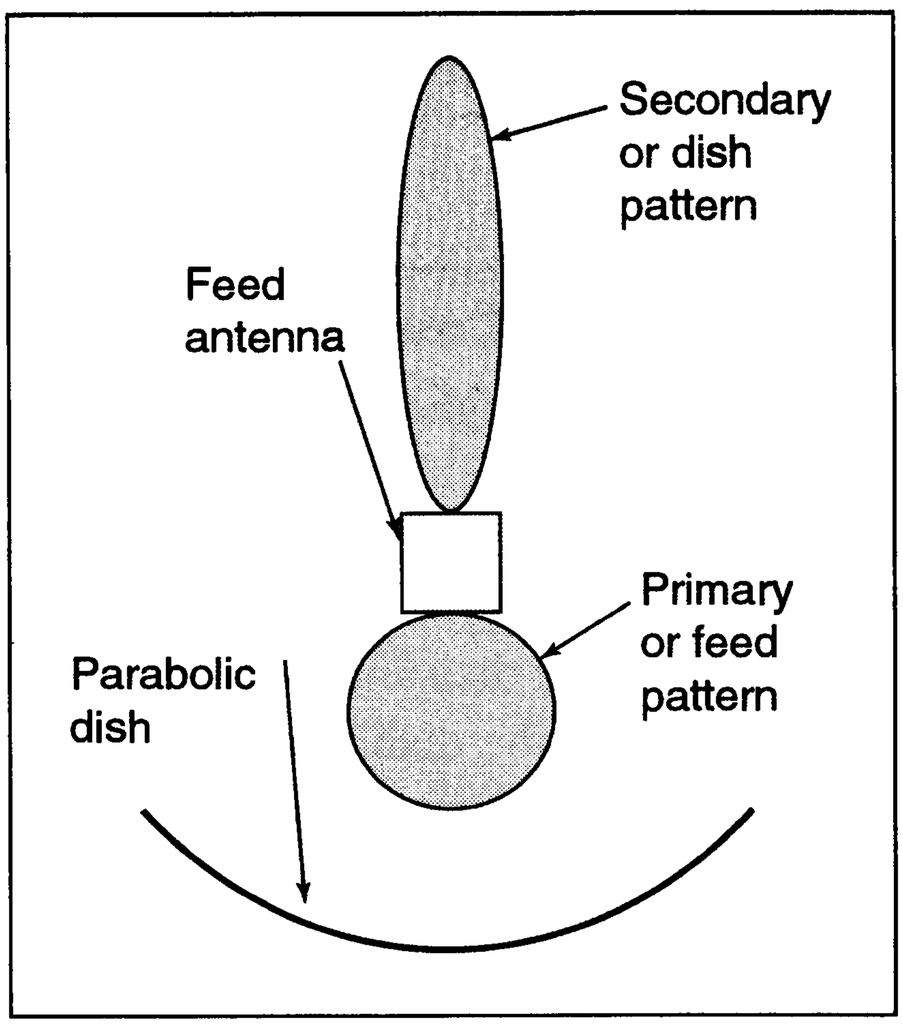
Then using decibels for convenience and adding propagation loss due to rain

$$D_{r\,\mathrm{dB}} = -D_{t\,\mathrm{dB}} + SNR_{\mathrm{dB}} + N_{\mathrm{db}} - L_{d\,\mathrm{dB}} - L_{\mathrm{rain\,dB}}$$

So

$$D_r = \underset{\text{Sat. gain}}{-(+10)} \quad \underset{\text{SNR}}{+(+10)} \quad \underset{\text{Noise}}{-(-185)} \quad \underset{\text{Prop. loss}}{+(161)} \quad \underset{\text{Rain loss}}{-(-10)} = 34\ \mathrm{dB} \ \text{ or } \ 2512$$

But since the parabolic dish aperture efficiency is only 70%, the gain must be increased by a factor of 1/0.7 or therefore should be 3588 or ***35.5 dB*** for SNR = 10 dB. *Ans.* (*a*)



(b) Uplink dish size

From Example 3–7.1 the dish gain is $G = 28\left(\frac{R}{\lambda}\right)^2$ where R is the radius of the dish or

$$\text{Dish diameter} = 2\lambda\sqrt{\frac{G}{28}} = 2 \times 0.01\sqrt{\frac{3588}{28}} = 226\ \mathrm{mm}$$

Commercial dishes for this purpose can be found that are about 450 mm, which would give a nice safety factor over the ***226 mm*** calculated here. *Ans.* (*b*)

(c) Dish HPBW

From (2–7–9) the $HPBW = \sqrt{\dfrac{40{,}000}{G}} = \sqrt{\dfrac{4 \times 10^4}{3588}} = \mathbf{3.3°}$ *Ans.* (*c*)

(d) Earth Temperature Power

With the conditions of (*a*), (*b*), and (*c*) met as above, the required earth terminal power is *1 W* for an SNR = 10 dB. *Ans.* (*d*)

(e) Satellite power

With the antenna parameters determined above and with the dish antenna $T_A = 10$ K and the earth terminal receiver $T_R = 300$ K and using the

$$T_{\text{sys}} = 10 + 300 = 310 \text{ K}$$

or

$$N = 1.4 \times 10^{-23} \times 310 \times 9.6 \times 10^3 = 4.2 \times 10^{-17} \text{ or } -164 \text{ dB}$$

And equation for the transmitted power from Example 21–20.1 with rain attenuation added

$$P_{t\,\text{dB}} = SNR_{\text{dB}} - D_{r\,\text{dB}} - D_{t\,\text{dB}} + N_{\text{dB}} - L_{d\,\text{dB}} - L_{\text{rain dB}}$$

or

$$P_t = \underset{\text{SNR}}{+(+10)} \quad \underset{\text{Dish gain}}{-(+35.5)} \quad \underset{\text{Sat. ant.}}{-(+10)} \quad \underset{\text{Noise}}{+(-164)} \quad \underset{\text{Prop. loss}}{-(-185)} \quad \underset{\text{Rain loss}}{-(-10)} = -4.5 \text{ dB}$$

or *355 mW* for SNR = 10 dB. *Ans. (e)*

Note that the downlink satellite power of 355 mW is about 1/3 the uplink earth terminal power of 1 W because the beam of the earth receiving antenna-subtends more sky on downlink then the satellite receiving antenna on uplink.

Another antenna option is shown in Fig. 21–47–1.

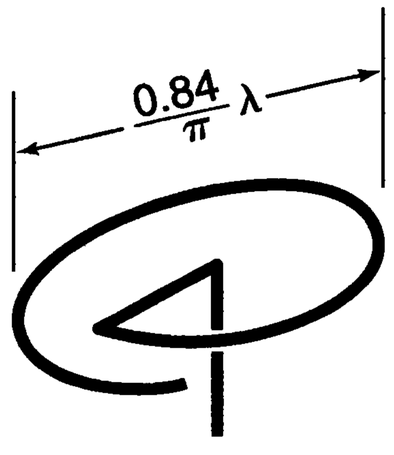


Figure 21–47–1
The curl antenna (Nakano-1) is another candidate for the earth terminal handset antenna.

21–21 ASTEROID DETECTION ANTENNA（检测小行星的天线）

The earth was built up over billions of years by asteroids striking the earth and adding to its mass. Over time the rate of impact has decreased but impacts continue. Millions of years ago an asteroid striking the Yucatan peninsula is believed to have wiped out the dinosaurs. The impact is thought to have kicked up enough dust and debris to form a continuous cloud cover that blocked the sun resulting in a long dark winter all over the earth with catastrophic effects for both animal and plant life.

Less than a hundred years ago, on June 30, 1908, the Tungska fireball slammed into the earth with the force of thousands of Hiroshima bombs, enough to destroy a major city. Luckily, it hit a remote, uninhabited Siberian forest laying waste to an area the size of Los Angeles. There was no advance warning (Mallove-1).

In July 1994, Comet Shoemaker-Levy-9's collision with Jupiter provided a dramatic example of a comet or asteroid-sized object's impact with a planet. And a few years later an asteroid whizzed by the earth too close for comfort.

EXAMPLE **21–21.1 Asteroid Detection Antenna**

Assume an asteroid with diameter 1 km and a power reflection coefficient of 0.4 that of the RCS of a perfectly conducting sphere, which is similar to earth's. A radar transmitting at 4 GHz with a peak power of 1 GW is used. The aperture efficiency, ε_{ap}, of the dish antenna is 75%. Since a typical asteroid's orbit is 1.9 ± 1 AU and the orbit of Mars is 1.5 AU, then it is plausible that when earth and Mars are at their closest to one another we would first need to pick up the errant asteroid at a distance of only 0.4 AU from earth. A typical asteroid's speed is 20 km/s:

(*a*) What diameter dish antenna is required to provide a 10 dB SNR return signal if the system temperature (T_{sys}) is 10 K with a 1 MHz bandwidth?

(*b*) What is the doppler shift of the return signal?

(*c*) What is the time/delay between the transmitted pulse and its reception?

(*d*) How much time is there before the possible impact?

(*e*) What steps do you propose be taken if it appears that the asteroid will strike the earth?

■ **Solution**

The radar cross section of a perfectly conducting sphere is given by

$$\sigma_{PEC} = \pi a^2$$

where a is the radius of the sphere. If the power reflection coefficient for the asteroid is 0.4 and it radius is 500 m, its radar cross section would be

$$\sigma = 0.4 \times \pi(2 \times 10^4 \text{ m})^2 = 1.6\pi \times 10^8 \text{ m}^2$$

Since 1 AU $= 1.5 \times 10^8$ km, then the asteroid is 6×10^{10} m from earth. The wavelength, λ, is 300 Mm s^{-1}/4 GHz = 0.075 m.

(*a*) From the radar equation (12–5–3) and Eq. (2–9–6), we can write

$$P_t = \frac{G^2\lambda^2\sigma P_t}{(4\pi)^3 r^4}$$

For a circular aperture, the gain, or in this case directivity since we will assume a lossless antenna, is given by Eq. (10–8–9). The noise power in the system is given by Eq. (12–1–3). Dividing the above equation by (12–1–3) to obtain the signal to noise ratio SNR, plugging in (10–8–9) and solving for the diameter of the dish d, we obtain

$$d^4 = \frac{64\lambda^2 r^4 k T_{sys} BSNR}{\varepsilon_{ap}\pi\sigma P_t}$$

or

$$d^4 = \frac{64(0.75)^2(6 \times 10^{10})^4(1.38 \times 10^{-23})(10)(1 \times 10^6)(10)}{(0.75)^2(1.6\pi^2 \times 10^8)(1 \times 10^9)} \text{m}^4$$

$$= 7.24 \times 10^9 \text{ m}^4$$

so $d = 292$ m.

(*b*) The doppler shift can be found from

$$v = \frac{c\Delta f}{2f} = \tfrac{1}{2}\lambda\Delta f$$

or

$$\Delta f = \frac{2v}{\lambda} = \frac{2(2 \times 10^4)}{0.075} \text{ s}^{-1} = 533 \text{ kHz}$$

(*c*) The time delay for the radar signal, remembering that we must take into account the travel time to the object and back, is found from

$$\Delta t = \frac{2r}{c} = \frac{2(0.4 \times 1.5 \times 10^{11})}{3 \times 10^8} \text{ s} = 400 \text{ s} = 6.67 \text{ min}$$

(*d*) The time before impact would be

$$t = \frac{r}{v} = \frac{0.4 \times 1.5 \times 10^{11}}{2 \times 10^4} \text{ s} = 3 \times 10^6 \text{ s} = 34.7 \text{ days}$$

(*e*) The size of the dish necessary to give even this short amount of response time is very large. Arecibo is 305 m in diameter. So it would be necessary to utilize the biggest dish in the world to search the orbital plane for just 35 days of lead time.

21-22 LEAKY TRANSMISSION LINES AS ANTENNAS（泄漏传输线用做天线）

There is a need for communication with cellular phones and other wireless devices inside buildings. Two simple methods are to use: (1) a leaky coaxial transmission line which can be installed over the false ceilings of the rooms in a building or (2) a Goubau or "G-string" (Goubau-1).

In a regular flexible coaxial cable the wire strands of the outer conductor are tightly woven to prevent leakage. By using fewer strands and a more open weave the line will leak radiation along its length, providing the desired linkage to wireless devices.

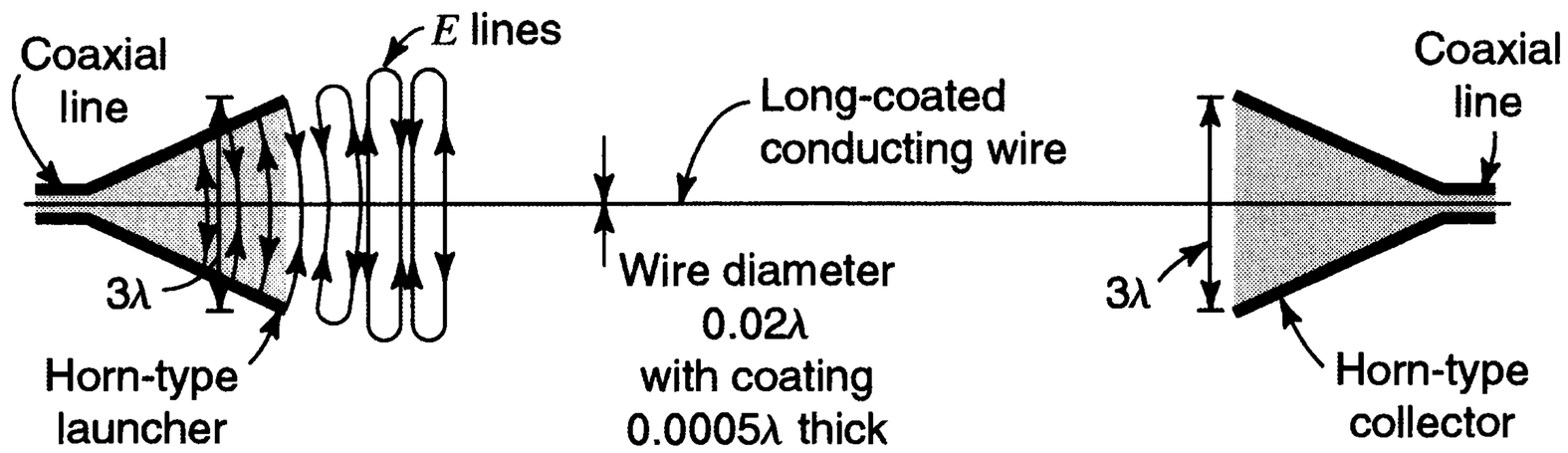


Figure 21–48
Single coated-wire open waveguide, or "G string." (*Goubau-1*)

A dielectric-coated single-wire waveguide is shown in Fig. 21–48 for guiding a wave between launcher and collector. The openness of the system allows radiation which can permit communication with cellular phones or other wireless devices. For example, stringing the wire high above the floor between the ends of a large exhibition hall can permit wireless communication throughout the hall.

21–23 ARTISTIC ANTENNAS (FRACTALS) [艺术性天线（分形）]

Three artistic antennas are shown in Fig. 21–49. The Sierpinsky-triangle bow-tie antenna and the Barnsley-Fern dipole antenna are examples of a large class of *fractal* antennas. Fractals are structures that preserve their shape at different scales. Many are exotic in appearance. They also provide new insights into antenna and array design (Barnsley-1; Prusinkiewicz-1; Mandelbrot-1; Kim-1; and Werner-1).

The promethea moth antenna in Fig. 21–49 is an example of many insect antennae[1] that look like radio antennas. Historically, it is the other way around. What we now call a *radio antenna* was known originally as a *wireless aerial*. Marconi and other early pioneers then began to call them *antennas* because of their resemblance to insect *antennae*.

The moth antenna resembles a twin log-periodic antenna with elements that are like folded dipoles. lnterestingly, the self-repeating different-size structure of a log-periodic antenna qualifies it as a fractal form.

The impedance variation of two Minkowski fractals, a solid plate and square loop are shown in Fig. 21–50.

21–24 CELL-TOWER TREES（蜂窝－塔树）

In urban areas cell-tower antennas can be camouflaged in building cornices, church steeples, and bell towers. In suburban areas the artificial tree has become popular. A steel and aluminium tower with antennas is encased in a plastic tree trunk with protruding plastic limbs. In colder climates the "trees" are made to resemble pine and spruce trees. In warmer climates they are made to resemble palm trees.

---

[1]Note that for insects the plural is *antennae* (the Latin plural) versus *antennas* for radio.

ARTISTIC ANTENNAS (FRACTALS)

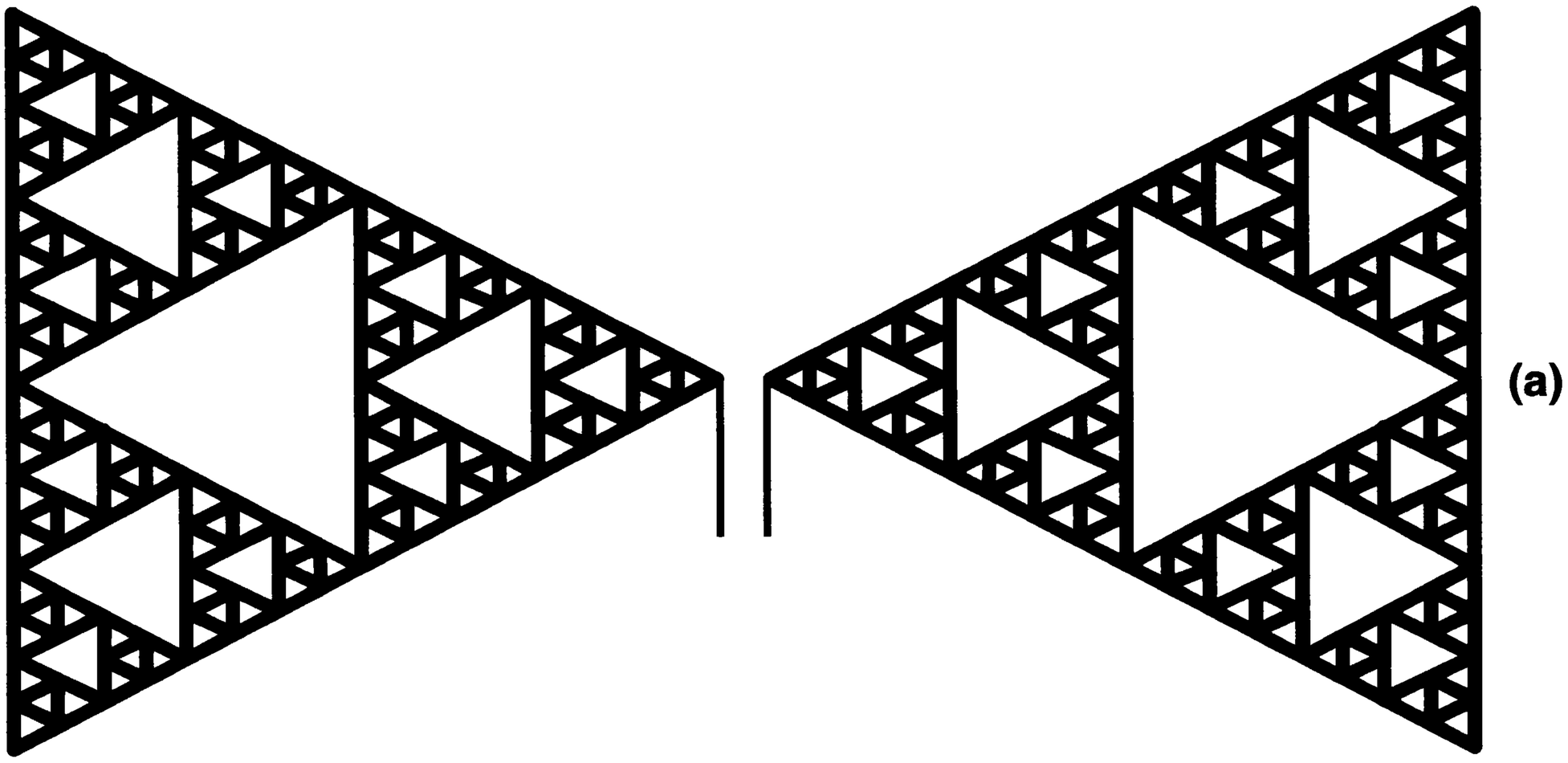
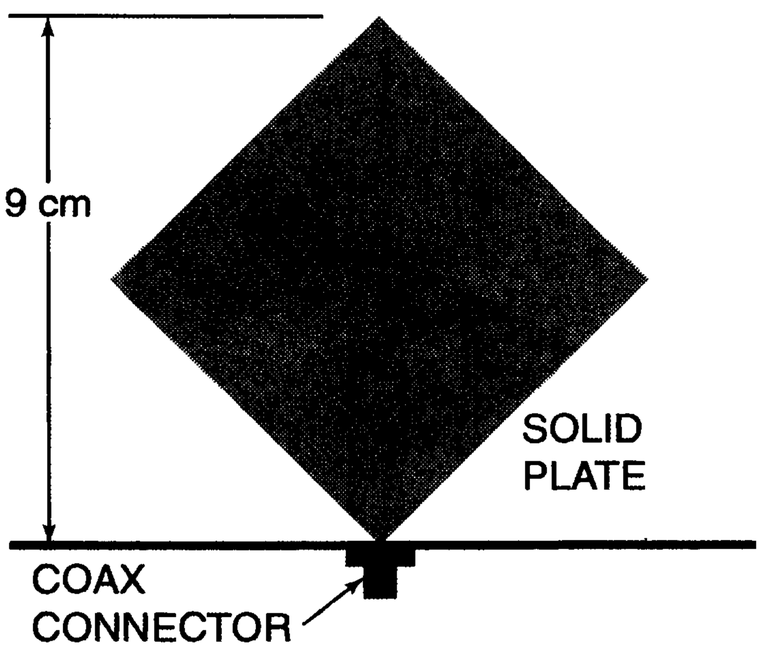
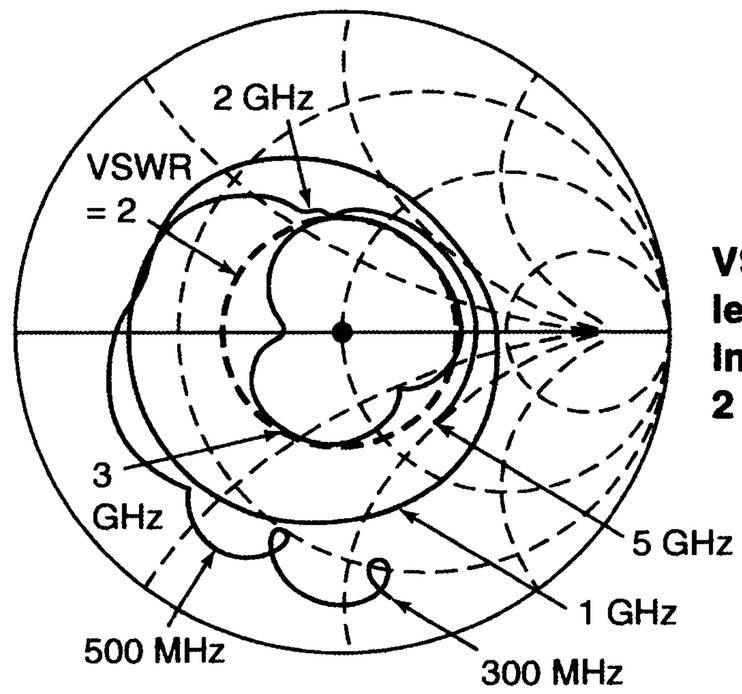
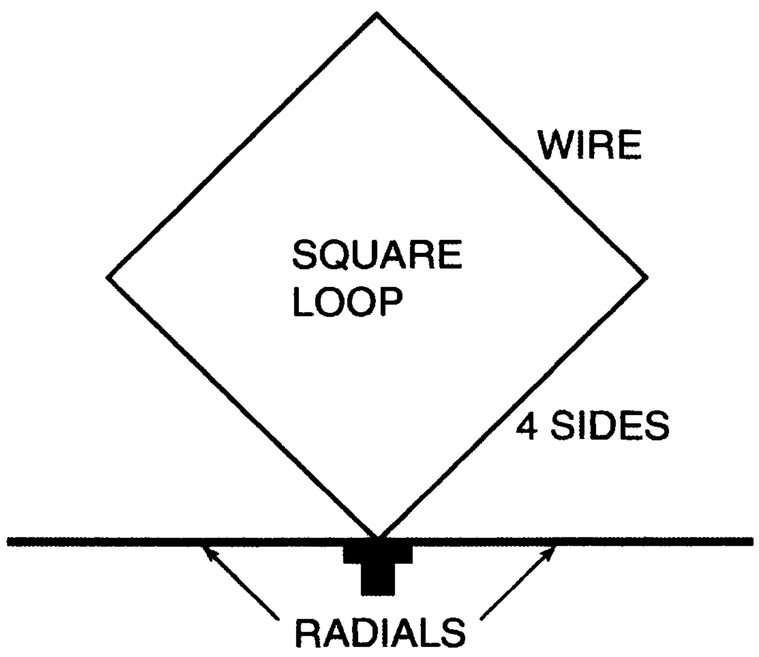


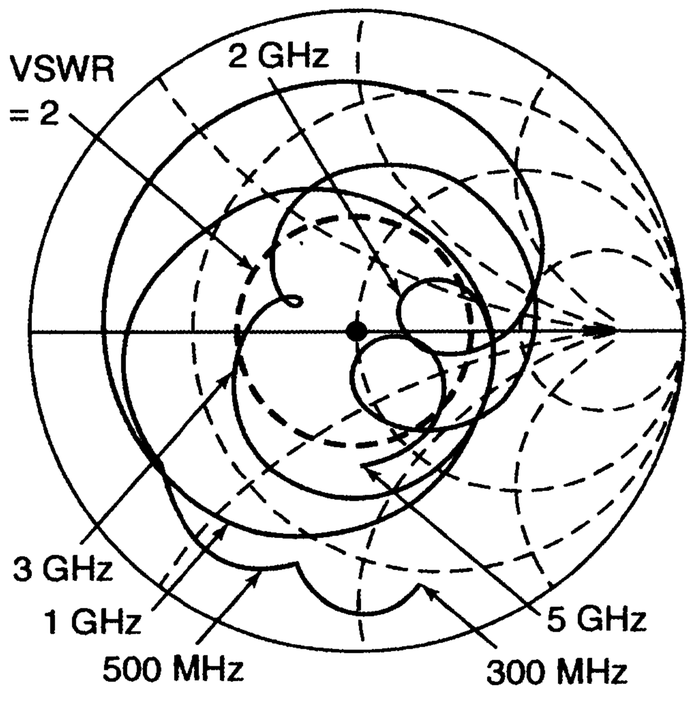
Figure 21–49
Three examples of artistic antennas that are fractal types. The lowest one is the antenna of a moth.



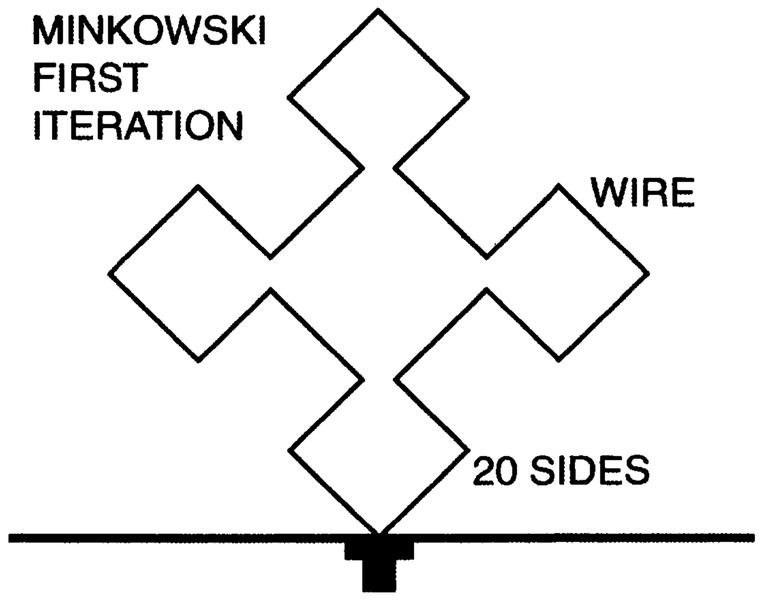


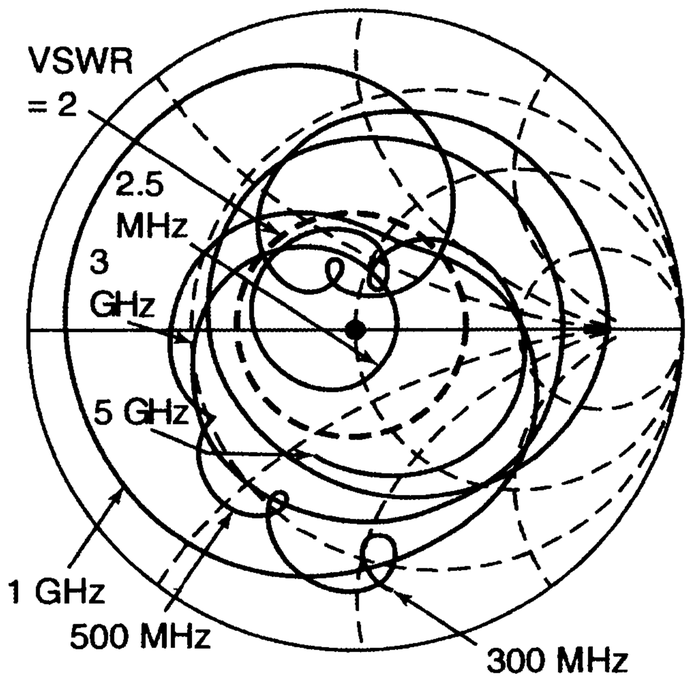
VSWR approximately 2 or less from 2 to 5 GHz. Impedance curve has 2 loops.



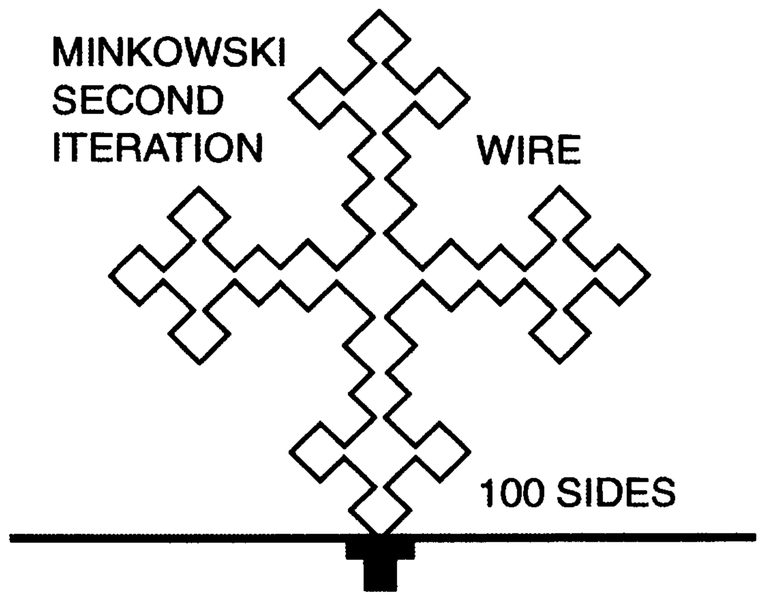


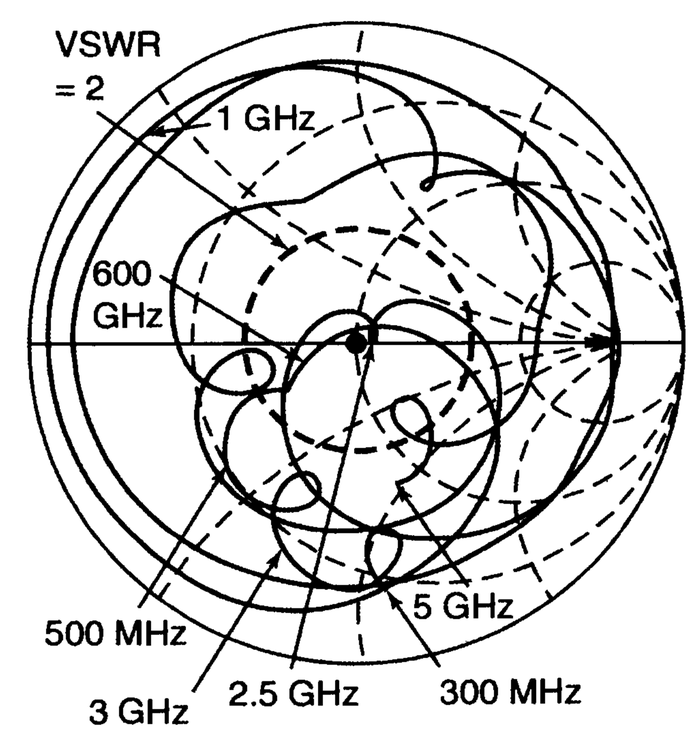
More impedance fluctuation than for plate with curve making 3 loops close to resonance.





Still more impedance fluctuation from non-resonant to resonant conditions making 4 loops.





Large impedance fluctuations making 6 loops with 2 frequencies, 650 MHz and 2.5 GHz, very close to resonance.

Figure 21–50
Impedance variation with respect to 50 Ω for flat plate, square wire loop and Minkowski fractals of first and second iteration. Note the extreme impedance fluctuations of the fractal antennas compared with the plate and square loop. The two radials are 17 cm overall. (*Impedance measurements courtesy of Dr. Daniel Fleisch, Aeroflex-Lintek Corp.*)

21–25 ANTENNAS FOR TERRESTRIAL MOBILE COMMUNICATIONS SYSTEMS

by Pertti Vainikainen[1] （用于陆上移动通信系统的天线）

The rapid development of mobile communication systems has led to the use of novel antennas for base station (BS) and mobile station (MS) vehicular and hand-held applications (Fujimoto (1) and James; Siwiak-1). The frequencies of terrestrial mobile radio systems range from below 200 MHz to over 60 GHz. The most significant of these are the analog and digital cellular radio systems, whose main frequency ranges are around 800–1000 MHz and 1700–2200 MHz, and the Wireless Local Area Network (WLAN) systems at around 2.4–2.5, 5.1–5.8 and 17 GHz. The bandwidths of cellular systems vary from about 8 to 17% with typically two distinct frequency bands for each system due to Frequency Domain Duplexing (FDD). The bandwidths of WLAN systems are below 5% due to higher center frequency and the use of packet-type data transmission.

21–25a Base Station Antennas（基台天线）

The BS antennas should direct signals to the wanted coverage area as effectively as possible. In the case of cellular systems, it is also beneficial that the distribution of the power is restricted accurately in order to minimize the frequency reuse distance in the system.

Nonadaptive Base Station Antennas. The gain of the BS antenna toward the coverage area should be as high as possible. On the other hand the coverage area must be large, so the directivity cannot be increased much in the horizontal plane. This is the case in traditional nonadaptive antennas. Thus, the gain is increased by decreasing the beamwidth in the vertical plane, which results typically in gain values of 5 to 17 dB. The horizontal plane beamwidth varies typically between 50° and 360°. Directive antennas are used at the sectorized BS sites usually having 2 to 3 sectors. In a typical 3-sector configuration the beamwidth of a single antenna is usually about 65° instead of the obvious 120° to optimize the coverage and interference in the network (see Fig. 21–51*a*).

The vertical beamwidth varies typically between 10° and 70°. The beam of BS antennas may be tilted (typically less than 15°) down to reduce the interference level in neighboring cells (Fig. 21–51*b*). Sometimes the beam is shaped so that the vertical plane side lobes above the main beam are minimized (Fig. 21–51*b*).

In addition to the radiation characteristics and bandwidth, important aspects of BS antennas are weight, wind load, size and appearance. Common types are dipoles, corner reflectors, patch arrays (Zürcher-1) and horns. By adjusting the angle of the corner reflector between 60° and 270°, horizontal half-power beamwidths of 60° to 180° can be obtained. Indoor environments can employ leaky lines at the base station. A leaky line is a coaxial cable with a leaky outer conductor (see Sec. 21–22). The radiation from a leaky feeder is characterized by the coupling factor giving the ratio between the power traveling in the cable and the power received by a half-wave dipole located at a certain distance from the cable. Typical norm distance is 3 m and then the coupling factor is typically 70 to 80 dB. With higher coupling giving the coupling factor of 50 dB the attenuation in the cable increases.

---

[1]Radio Laboratory, Helsinki University of Technology.

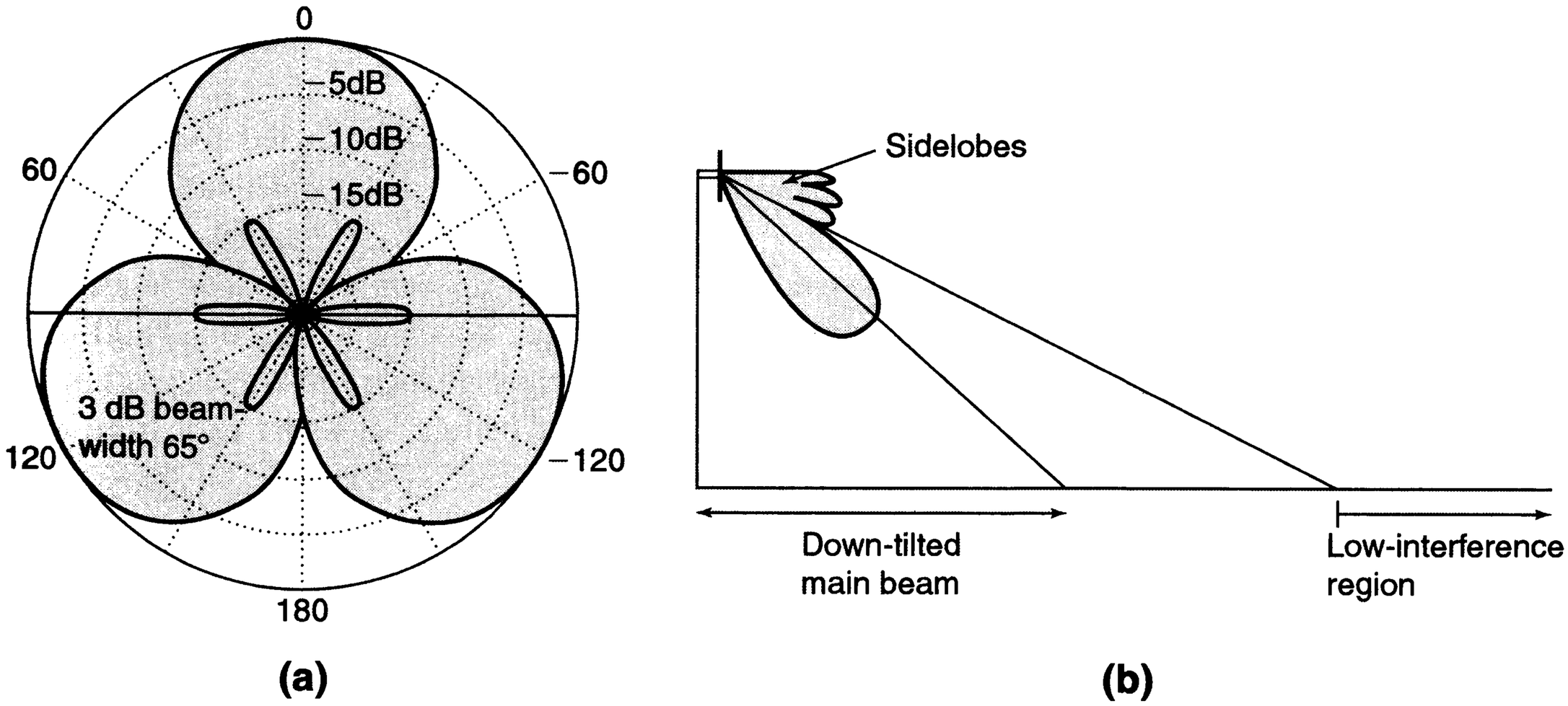


Figure 21–51
Typical beam patterns for base station antennas: (*a*) horizontal plane pattern for a 3-sector base station site, (*b*) vertical plane pattern.

Normally the attenuation is about 5 dB/100 m at 900 MHz for a cable having the outer diameter of 20 mm. The cable can be terminated with a matched load or with an antenna. Base station powers may be over 300 W outdoors and 30 W indoors.

Antenna Diversity for Base Stations. Diversity techniques can be used in base station receivers to reduce the effects of multipath fading. In antenna diversity several receiving antennas are used to obtain independent samples of the incoming field. It is unlikely that all these signals will experience a deep fade simultaneously. The diversity gain compared to single-antenna reception is a statistical feature depending on the probability distribution of fading, the selected probability level, the number of diversity branches, the correlation between the signals in different branches, the relative signal levels, and the diversity combining techniques used (Jakes-1; Vaughan (2) and Andersen). The most common number of diversity branches in mobile systems is 2, which combines low complexity with fairly high system gain of up to 10 dB at 99% cumulative probability level for selection combining of Rayleigh fading channels. If the envelope correlation between the diversity branches is below 0.7 and the difference between the average signal levels below 5 dB, the reduction of diversity gain is fairly small. The diversity reception techniques range from simple selection of the strongest signal in "selection combining" to optimal in-phase combining of signals weighed according to their signal to noise ratios in "maximal ratio combining." In a digital communicator system with substantial signal processing capacity, maximal ratio combining is used at the base station as it provides the highest diversity gain.

The most popular antenna diversity techniques at cellular base stations are space and polarization diversity. In space diversity, low correlation between the signals is achieved by locating the diversity antennas far enough from each other. The required distance is determined by the spatial correlation properties of the incoming field. The correlation properties depend on the angular distribution of the multipath signals, so that wider angular

distribution (larger angular spread) results into shorter correlation distance. Antenna separation of 20 to 40 wavelengths is typically required at traditional elevated base stations, as the scatterers are in this case concentrated close to the hand-held unit. In urban areas small cells ("micro" and "pico" cells) are used to achieve more capacity. In these, the base stations are usually situated below the rooftops or indoors. In this case the angular distribution of the signals at the base station is wide and an antenna separation of less that one wavelength is adequate to ensure low correlation between the diversity branches. The feasibility of polarization diversity is based on the low correlation between vertically and horizontally polarized electric field components especially in urban environments (Vaughan-1). Polarization diversity is useful for hand-helds since these have both vertical and horizontal components due to tilting of the phone. At the base station ±45° slanted linear polarizations can be used to obtain equal signal levels in both diversity branches (Lindmark-1; Ojanpera-1).

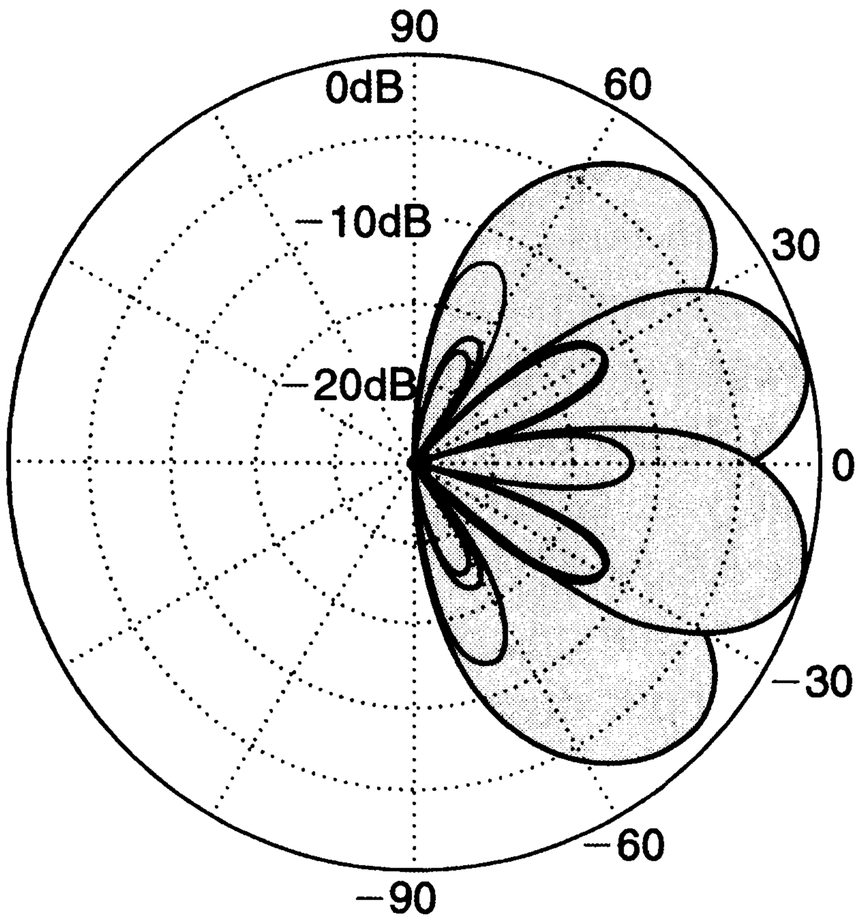
Adaptive Base Station Antennas. In terrestrial mobile communications, the user distribution and propagation channel for each user changes continuously. The basic task of the adaptive BS antenna is to improve the Signal-to-Interference and Noise Ratio (SINR) of single connections and maximize the coupling between the base station and the wanted user while minimizing the coupling with other users (Bach-1). This may be accomplished equally well both in the uplink (from MS to BS) and in the downlink direction. This requirement creates an obvious optimization problem as the complex multipath propagation channel can usually be fully detected only at the base station and thus information required for adaptation is available only for uplink. The respective information for downlink must be either estimated, which is difficult especially for FDD systems where the duplex separation is often larger than the correlation bandwidth of the propagation channel, or feedback control should be used, which increases the required signaling capacity. Therefore, the downlink direction is critical for the system gain obtained through the use of adaptive BS antennas. The benefits obtained by using adaptive BS antennas can be classified as:

- Increased *capacity* due to increase of the SINR
- Increased *coverage* or range due to higher apparent gain of the BS antenna
- Reduced *output power* especially at the MS, where battery lifetime is critical

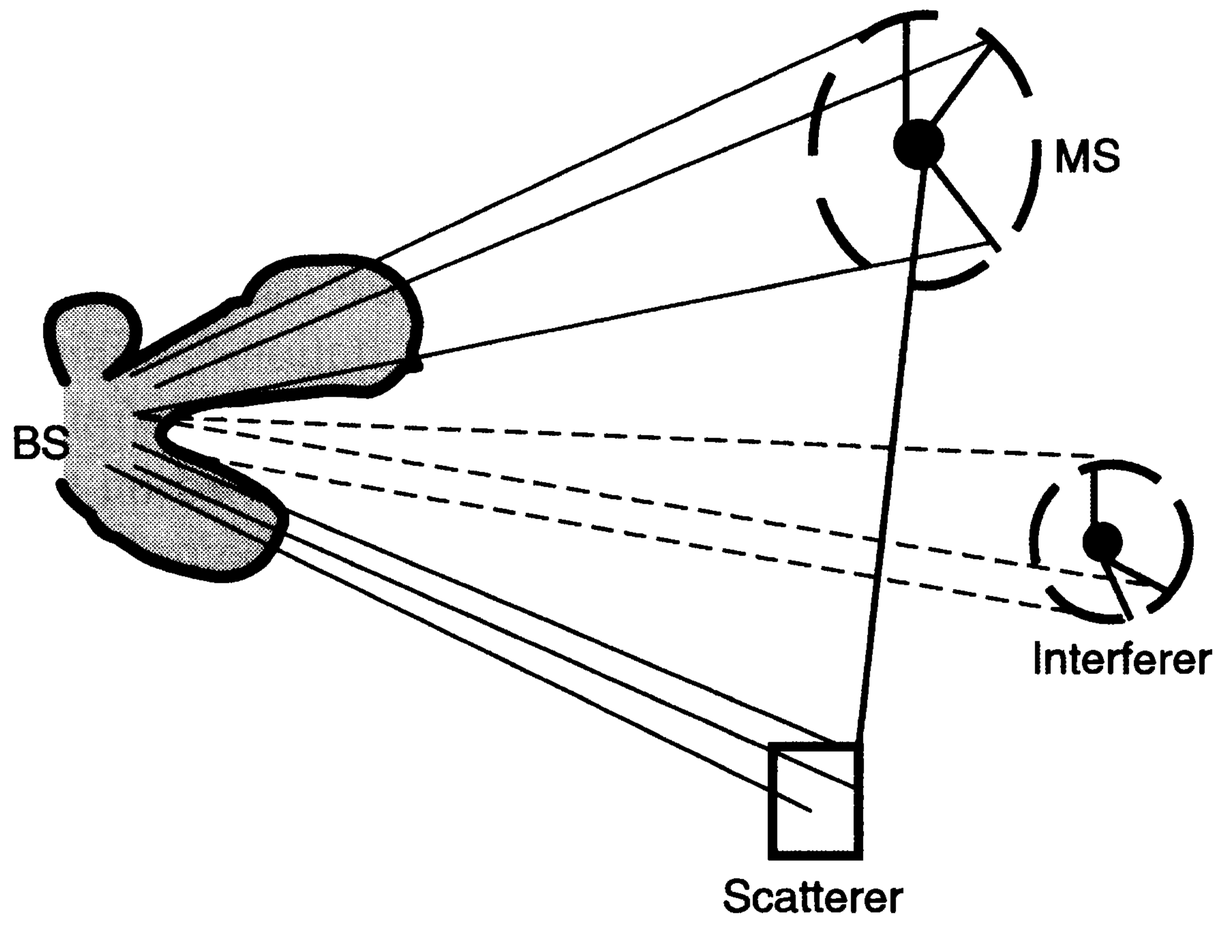
Three Types of Adaptive Base Station Antenna Options.

(1) *Switched beam antennas.* A straightforward means to adapt to the changing mobile station distribution is to use a BS antenna with several selectable beams, of which each covers a certain part of the cell area (see Fig. 21–52*a*). Typical implementation of such a switched beam antenna is based on a Butler matrix, which provides one orthogonal beam per antenna element. The benefit of the switched beam antenna is that only fairly simple RF signal processing is required, which makes it possible to apply the techniques also in existing systems. The drawback is the limited adaptivity.

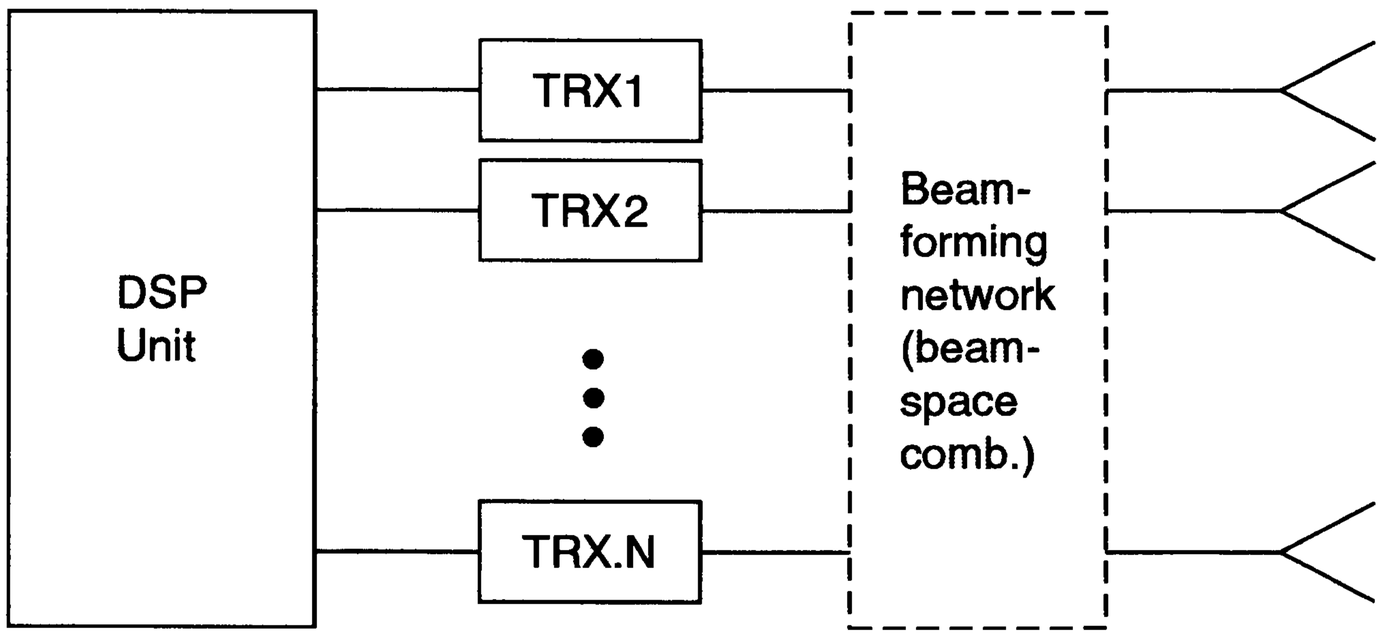
(2) *Beamforming.* Adaptive beamforming implemented at RF or baseband can be used to form pattern maxima to wanted directions and nulls to unwanted directions as shown in Fig. 21–52*b*. However, due to the limited number of antenna elements in practical adaptive BS antennas, only a few maxima and nulls can be realized simultaneously. Furthermore,



(a)



(b)



(c)

Figure 21–52
Adaptive antenna principles. (*a*) Beam patterns of a switched beam antenna realized with a Butler matrix and a linear array of four reflector-backed dipoles, (*b*) beamforming with maxima toward wanted signal paths and a broad minimum toward an interferer, (*c*) adaptive array configuration with optional RF beamforming network like a Butler matrix for beamspace combining.

broad multiple-null minima may be required, because the path between BS and MS is spread in angular domain due to local scattering around the mobile unit (see Fig. 21–52*b*).

(3) *Adaptive arrays.* The most comprehensive and complex configuration for adaptive BS antennas is the fully adaptive array, where each antenna element is connected to a separate transceiver and digital signal processing is used to control the signal weights (see Fig. 21–52*c*). Beamspace combining with, for example, a Butler matrix as the RF beamforming network can also be used to improve signal-to-noise ratio in receiver adaptation. Direction-Of-Arrival (DOA) finding and optimization algorithms are used in reception

(uplink direction) to select the complex weights separately for the multipath propagation channel of each mobile user. The weights can be reused in transmission quite directly for TDD systems. For FDD systems the transmission weights are estimated based on the DOA information.

The use of adaptive arrays at both ends of the radio link would increase the link capacity. The efficiency of the Multiple Input–Multiple Output (MMO) systems is based on the existence of several parallel independent channels in scatter-rich environments (Foschini-1).

21–25b Mobile Station Antennas（移动台天线）

The antenna for a mobile phone should enable connection to the base station in all locations and orientations of the mobile unit. For small cellular hand-helds, the incoming field consists of several multipath signals so the antenna receives several signals with random directions of arrival and polarization (see Sec. 24–6c). Therefore, it is difficult to set any clear requirements for the antenna. In most environments vertical polarization dominates making a vertically polarized omnidirectional antenna, like a vertical dipole, preferable. But the head of the user may block some of the signal. The most critical performance goals especially for the antennas of small portable phones are adequate *bandwidth* (see Sec. 21–25a) and *high efficiency,* which are difficult to achieve simultaneously for small antennas (Hansen-1). As small antennas for cellular systems are usually self-resonant, their performance can be evaluated by using the unloaded quality factor, from which the bandwidth with certain matching criterion at the edge of the band can be calculated in the following way:

$$BW = \frac{S-1}{Q_u\sqrt{S}} \tag{1}$$

where

BW = relative impedance bandwidth, dimensionless
S = *VSWR* at the edge of the frequency band, dimensionless
Q_u = unloaded quality factor of the antenna, dimensionless

Equation (1) holds for the situation where the matching of the antenna is perfect at the center frequency, i.e., with critical coupling of the resonant antenna to the feed circuitry. The typical matching criteria are return loss of 10 dB giving $S = 1.9$ and $BW = 2/(3Q_u)$ and return loss of 6 dB giving $S = 3.0$ and $BW = 1.16/Q_u$. About 10% wider bandwidth can be obtained with slight overcoupling (Hirasawa-1; Wheeler-2).

Antennas on Cellular Handsets. The volume of a typical handset is less than 100 cm^3 and of the antenna is no more than about 5 cm^3. At 800 to 900 MHz, this requirement makes it difficult to achieve system bandwidths of 10% without inducing currents on the entire handset chassis, making the antenna element act more as a coupling structure than a radiating element (Fujimoto-1; Vainikainen-1). Another possibility combines small effective volume and large bandwidth by using a retractable whip antenna.

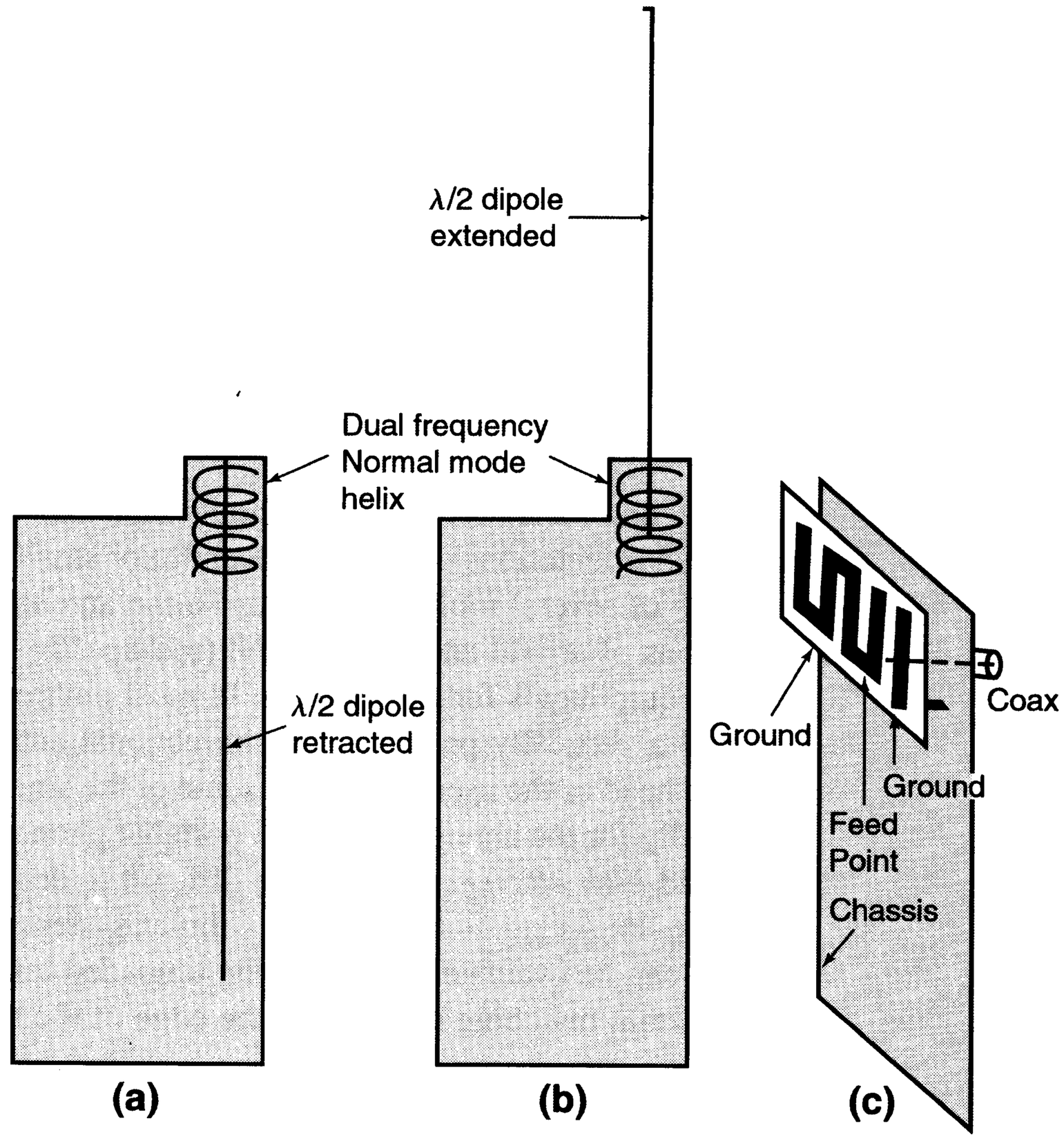


Figure 21–53
(*a*) Handset with dual frequency normal-mode helix and $\lambda/2$ dipole retracted. (*b*) Handset with $\lambda/2$ dipole extended. (*c*) Planar internal multiband antenna.

Dipole and Monopole Antennas. The retractable whip antenna is typical on handsets (see lowest photograph opposite page 1). It may be considered equivalent to either a monopole with the phone chassis as the ground "plane" or an unsymmetric dipole, whose other half is the phone chassis. The length of the whip can be $\lambda/4$ or $3\lambda/8$. The reason for using $3\lambda/8$ whips is to shift the current maximum farther away from the user and reduce the currents on the phone chassis. In this case the antenna works above its resonant frequency, which has to be taken into account in the matching circuitry. The unloaded quality factor of a $\lambda/4$ monopole or $\lambda/2$ dipole is around 5 (see Chap. 6), which means that with them it is fairly easy to obtain the required impedance bandwidth. The whip antenna is usually retractable so that a normal-mode helical antenna can "take over" as the whip is retracted (Fig. 21–53*a*, *b*).

Normal-Mode Helical Antennas. A normal-mode helical antenna may be used for circular polarization (Sec. 8–22). Dual band operation is obtained by using two different pitch angles.

Internal Antennas. The antenna may be enclosed inside the handset. There are two main types of these antennas: planar antennas and chip antennas. A planar antenna is usually a $\lambda/4$ microstrip mounted on the conducting chassis of the handset (Fig. 21–53*c*). Planar antennas are $\lambda/4$ microstrip antennas or Planar Inverted-F Antennas (PIFAs) (Ollikainen-1). The chip antennas are very small and they must be mounted in a certain manner on the circuit board of the phone (Nagumo-1). Miniaturization methods like meandering or high permittivity are often used with internal antennas. Internal antennas may result in a sacrifice in performance, especially if the hand is over the antenna or the unit is close to the head.

Antenna Diversity for Mobile Stations. Antenna diversity is difficult to implement on handsets because of limited space. Therefore, simple diversity combining techniques like selection combining are preferred. The angular distribution of the incoming field for the mobile station is typically quite omnidirectional and thus the correlation distance is on the order of half wavelength. It has been shown that low correlation between the outputs of two antennas can be achieved in cellular handsets with antenna spacing less than 0.2 wavelength. In this case, however, the decorrelation is caused partly by differences in the power patterns and polarization properties of the antennas (Braun-1; Ko-1).

21–26 ANTENNAS FOR GROUND PENETRATING RADAR (GPR): PULSE BANDWIDTH [探地雷达（GPR）的天线:脉冲的频带宽度]

Radar techniques are used for the detection of underground anomalies both natural and man-made. These anomalies include all kinds of buried metallic and nonmetallic objects, voids under highways and earth inhomogeneities. Like above-ground-radar, a GPR receives and displays the radio echo of a short transmitted radio pulse, but with some important differences (Peters-1; Montaya-1).

1. Distances are usually so small that the object is in the near field of the GPR antenna. Thus, the radar equation which assumes far-field conditions cannot be applied without modifications.
2. The ground is a lossy medium with attenuations ranging from 10 dB/m at 3 to 300 MHz to 100s of dB/m at GHz frequencies.
3. There is a large mismatch at the air-ground interface under the GPR antenna.
4. Pulse lengths must be short, of the order of a few nanoseconds with repetition rates of 50 Hz to 1 MHz with resulting spatial resolution of $0.3/\sqrt{\varepsilon_r}$ to $3/\sqrt{\varepsilon_r}$ m where ε_r = relative permittivity of the ground. The short pulse generates a broad spectrum signal. Thus, a 1-ns pulse generates a spectrum measured in GHz.

The antenna is a very critical component of a GPR system. Typically two dipole antennas are used, one for transmitting and the other for receiving, with the pair of dipoles placed parallel to and close to the ground. To reduce reflection of the pulse from the ends of the dipole to avoid "bouncing pulses" and "ringing" effects, the dipole elements may be made of resistance strip as suggested in Fig. 21–54. The wide angle part has low resistance

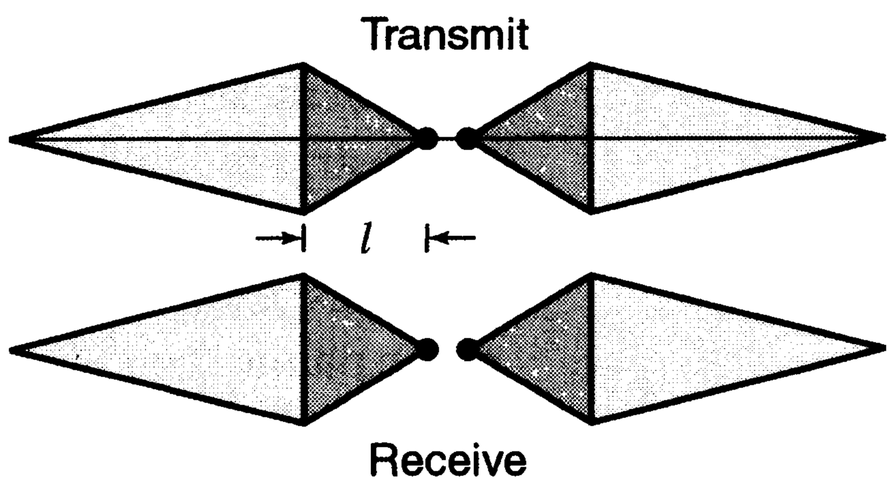


Figure 21–54
Resistance-loaded bow-tie antennas for GPR. Plan view (from above).

and the narrow angle part has high resistance per unit length, forming, in effect, a resistance loaded bow-tie antenna. It is important that the transmission lines be well matched at all junctions. The following example illustrates the principles.

EXAMPLE **21–26.1 Ground Penetrating Radar for Highway Inspection**

Erosion and subsidence under highways can cause pavements to collapse. To detect such conditions before collapse, GPR can be used with a pair of bow-tie antennas, as in Fig. 21–54, towed in a cart along the highway (to right in Fig. 21–55).

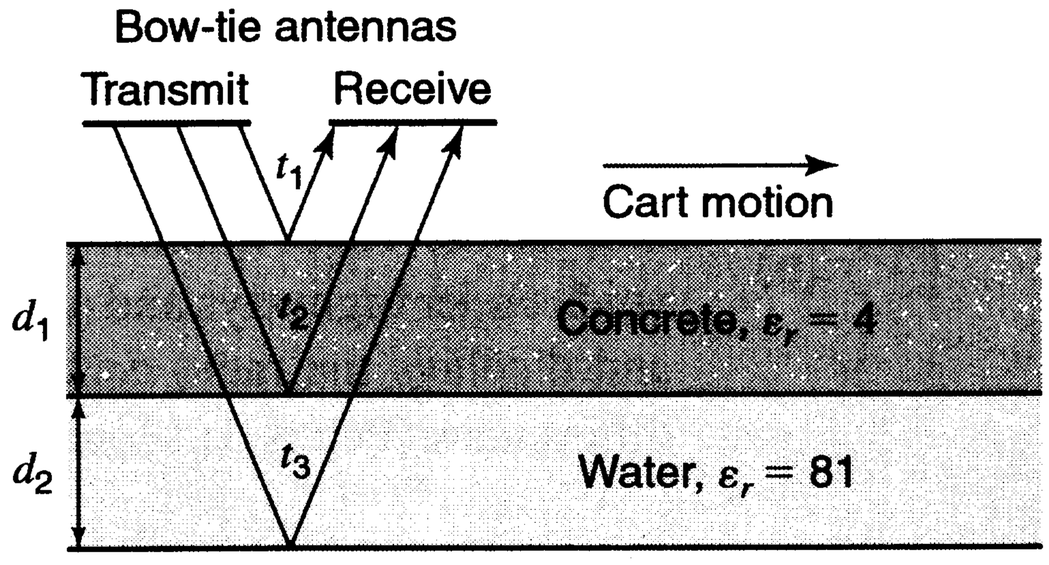


Figure 21–55
Arrangement of bow-tie antennas with respect to pavement. Antennas are perpendicular to the page and are towed to the right.

■ **Problem**

To detect such pockets a short Gaussian pulse is transmitted using a carrier frequency low enough to penetrate the lossy pavement. The pulse must be short enough to resolve the weak echo off the pavement bottom from the strong echo off the top of the pavement as suggested in Fig. 21–55.

Let the echo time off the top of the pavement $= t_1$, off the bottom of the pavement $= t_2$ and off the bottom of the water pocket $= t_3$. If $t_1 = 0.3$ ns, $t_2 = 2.4$ ns and $t_3 = 14.4$ ns, find the thickness of the pavement d_1 and of the water pocket d_2.

■ **Solution**

$$d_1 = (t_2 - t_1)c/2\sqrt{\varepsilon_r} = 2.1 \times 10^{-9} \times 3 \times 10^8/2\sqrt{4} = 0.16 \text{ m} = 16 \text{ cm}$$

$$d_2 = (t_3 - t_2)c/2\sqrt{\varepsilon_r} = 12 \times 10^{-9} \times 3 \times 10^8/2\sqrt{81} = 0.20 \text{ m} = 20 \text{ cm}. \quad \textit{Ans.}$$

EXAMPLE **21–26.2 Design of Bow-tie Antennas for Example 21–26.1**
***by Jonathan D. Young*[1]**

A radar with Gaussian pulse at kHz repetition rate that digitizes the band-limited "impulse response" echo requires an antenna with a wide bandwidth. Assuming that the radar uses a Gaussian baseband pulse whose 3 dB duration is 1/2 the required differential time delay $t_{\text{diff}} = 2$ ns as calculated in Example 21–26.1 or 1 ns. By Fourier transform, the required antenna bandwidth f_b for this short pulse is

$$f_b = 2/t_{\text{diff}}$$

■ **Problem**

(*a*) Calculate the required bandwidth for the antenna.
(*b*) Does a 60° bow-tie antenna have adequate bandwidth for the 1 ns pulse?

■ **Solution**

(*a*) $t_{\text{diff}} = 2.3 - 0.3 = 2$ ns and $f_b = 2/2$ ns $= 1$ GHz. *Ans.* (*a*).
(*b*) Make the bow-tie length such that the antenna second resonance ($X = 0$) occurs at 1 GHz ($\lambda = 0.3$ m). Assume that the pulse generator is matched to the 180 Ω antenna resistance at 1 GHz (see Fig. 11–8–1*a*).

At this "center frequency" f_c, the power transfer P_c is given by $P_c = V^2/4R_c$. At other frequencies f_f the power transfer is given by

$$P_f = V^2 P(f)/[R_c + R(f) + jX(f)]$$

Dividing, we have $P_f/P_c = 4R_cR_f/[R_c + R(f) + jX(f)]$. Using the impedance data of Fig. 11–8–1, the impedance bandwidth of the 60° bow-tie antenna may be estimated by finding the limiting frequencies where

$$|P_{f1,2}/P_c| = 1/2$$

Since the center frequency is 1 GHz and the length is such that $l/\lambda = 0.33$, the bow-tie half-length $l = 10$ cm (see Fig. 21–54). From the impedance curves of Fig. 11–8–1*a* and *b*, the power transfer vs. frequency is as shown in Fig. 21–56.

The lower frequency limit is 400 MHz. The upper frequency limit is well above 1.4 GHz. Thus, the 60° bow-tie antenna will provide ample bandwidth for the pulse. *Ans.* (*b*).

Adding tapered resistance sections to the bow-tie, as in Fig. 21–54, reduces reflections, primarily at lower frequencies, making it easier to match the antenna to the generator. The presence of steel reinforcing bars in the concrete pavement complicates the problem. However, the bar spacing is usually sufficient forming an "open-type" structure or mesh that permits microwave penetration.

In most antenna applications the pattern is of primary concern. In this application bandwidth is the critical factor.

---

[1]ElectroScience Laboratory, Ohio State University.

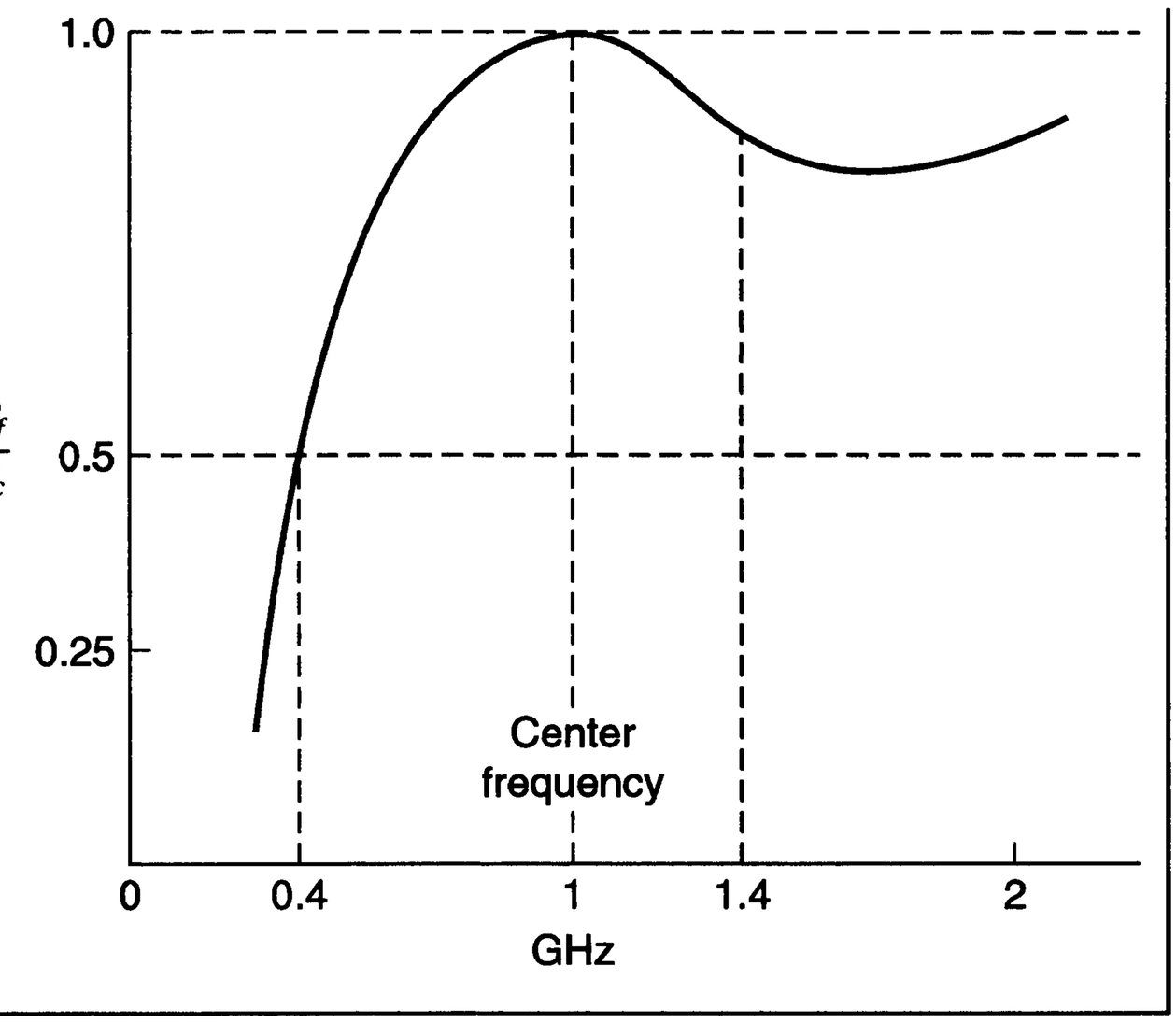


Figure 21–56
Relative power transfer for a 60° bow-tie antenna of half-length $l = 10$ cm using the curves of Fig. 11–8–1*a* and *b*.

21–27 EMBEDDED ANTENNAS（植入式天线）

A $\lambda/2$ dipole embedded in a dielectric medium will resonate when its physical length is $0.5\lambda/\sqrt{\varepsilon_r}$ where ε_r = relative permittivity of dielectric, assuming dielectric of infinite extent. Thus, if $\varepsilon_r = 4$, the $\lambda/2$ dipole will be only half its free-space size. In a dielectric of finite size the reduction is less but it may be an advantage.

The 4-patches of the CP array of Fig. 9–23–1 are embedded in dielectric. They are located on a dielectric substrate and are covered by a dielectric sheet or radome. Thus, the patch conductor is part of a dielectric-conductor-dielectric sandwich. The conductor elements of the frequency-selective surfaces of Chap. 18 are also embedded in dielectric.

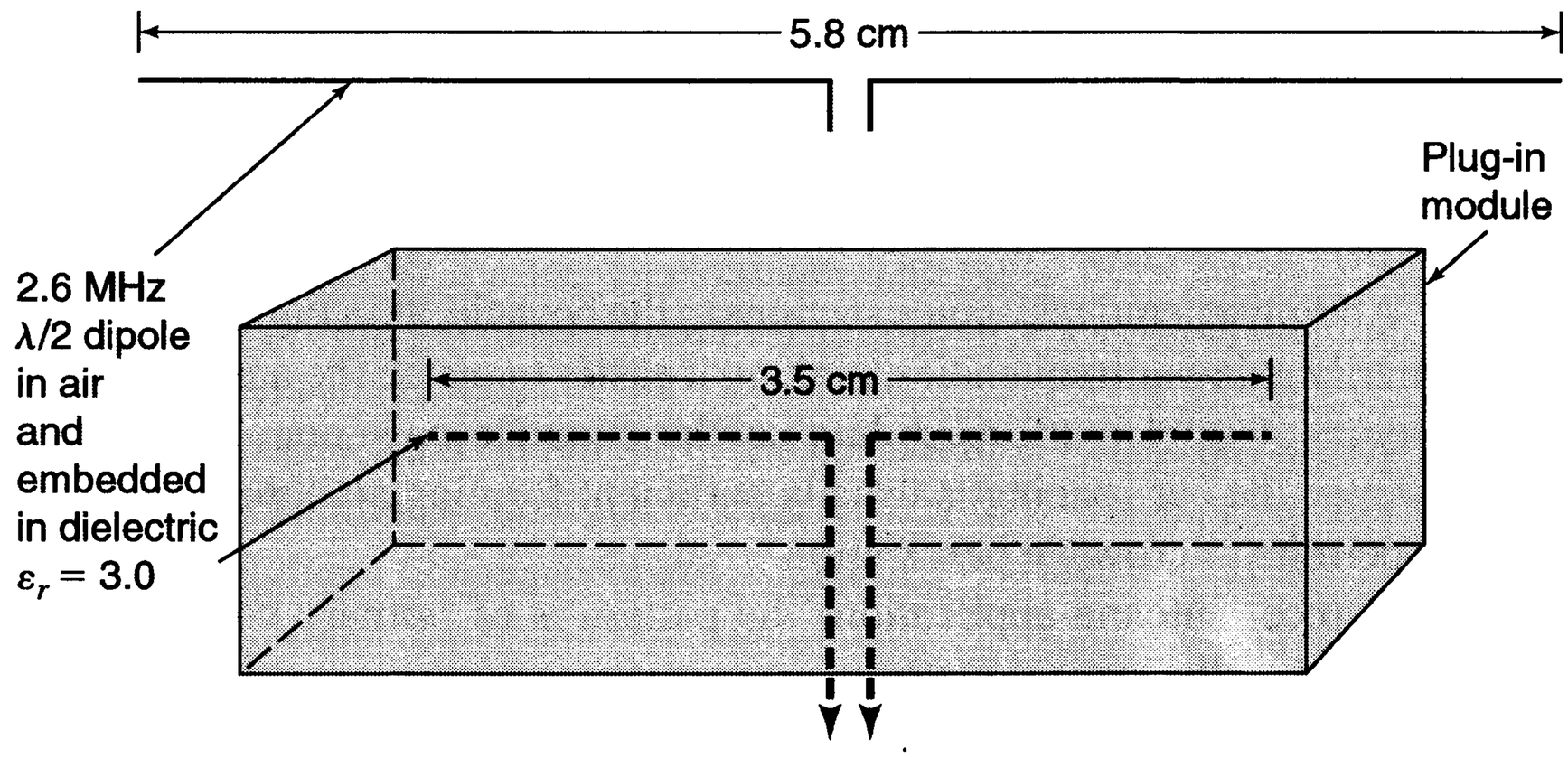


Figure 21–57
Half-wavelength dipole embedded in a dielectric for Bluetooth application.

In Bluetooth[1] technology, wireless links replace coaxial cables for interacting RF networks at short ranges (50 m or less) (Perez-1). Here embedded antennas are useful because of space restrictions. For example, a $\lambda/2$ dipole for 2.6 MHz which would be 5.8 cm long in air is reduced to about 3.5 cm in a dielectric module of $\varepsilon_r = 3.0$ as shown in Fig. 21–57.

21–28 ULTRA-WIDE-BAND (UWB) ANTENNAS FOR DIGITAL APPLICATIONS[2]

（数字式应用的特宽频带天线）

In converting to digital transmission, it was found that many antennas that worked satisfactorily with analog waves worked poorly with digital ones. The reasons are:

1. Digital signals use short pulses and these require wide bandwidths. Thus, digital TV is allocated twice the bandwidths of analog TV.
2. A pulse will radiate from different parts of the antenna at different times causing phase dispersion of the pulse and degrading of the signal.

For example, the vee antenna of two rhombic-shaped plates of Fig. 21–58*a* can radiate from the feed and corners of the plates. These pulses have different time delays as a function of both frequency and angle. Thus, the problem is complex. The following analysis provides insights that are useful for modifying the antenna so that it will transmit a pulse without significant distortion or phase dispersion.

The standard definitions for gain and beamwidth become inadequate for UWB applications, since they are defined on the basis of a single frequency. In order to overcome this situation, modified definitions are proposed by Gwynne and Young (Gwynne-1; Young-1) to define a compact means for determining a UWB antenna's merit. These functions define the normalized spectral pattern and transient functions, as well as gain and beamwidth parameters.

Since the matching of the antenna will normally vary over such large bandwidths, it is necessary to take this into account in the definitions. Transfer functions defining the power transfer between the antenna and the transmitter or receiver are the starting points.

For the transmitting case, the transfer of power $T_b(f, \theta, \phi)$ from the voltage generator V_T with impedance Z_T into radiated power of an antenna with impedance Z_A is

$$\frac{V_T^2}{Z_T + Z_A} T_b^2(f, \theta, \phi) = \frac{E^2(f, \theta, \phi)}{Z_o} \qquad (\text{W m}^{-2}) \tag{1}$$

where $E(f, \theta, \phi)$ = radiated field (V m^{-1}), Z_o = intrinsic impedance of free space (Ω), $T_b(f, \theta, \phi)$ = transmitting transfer function from the total power expended in the system to the total power radiated and is a function of frequency (f) and angle (θ, ϕ) with units of m^{-1}. The total power radiated, P_{rad}, as in Sec. 2–7 can be related to the gain of the antenna

---

[1]Named for "Bluetooth," the 10th century King Harald II of Denmark.

[2]We thank Prof. Jonathon D. Young, ElectroScience Laboratory, Ohio State University for assistance on this section.

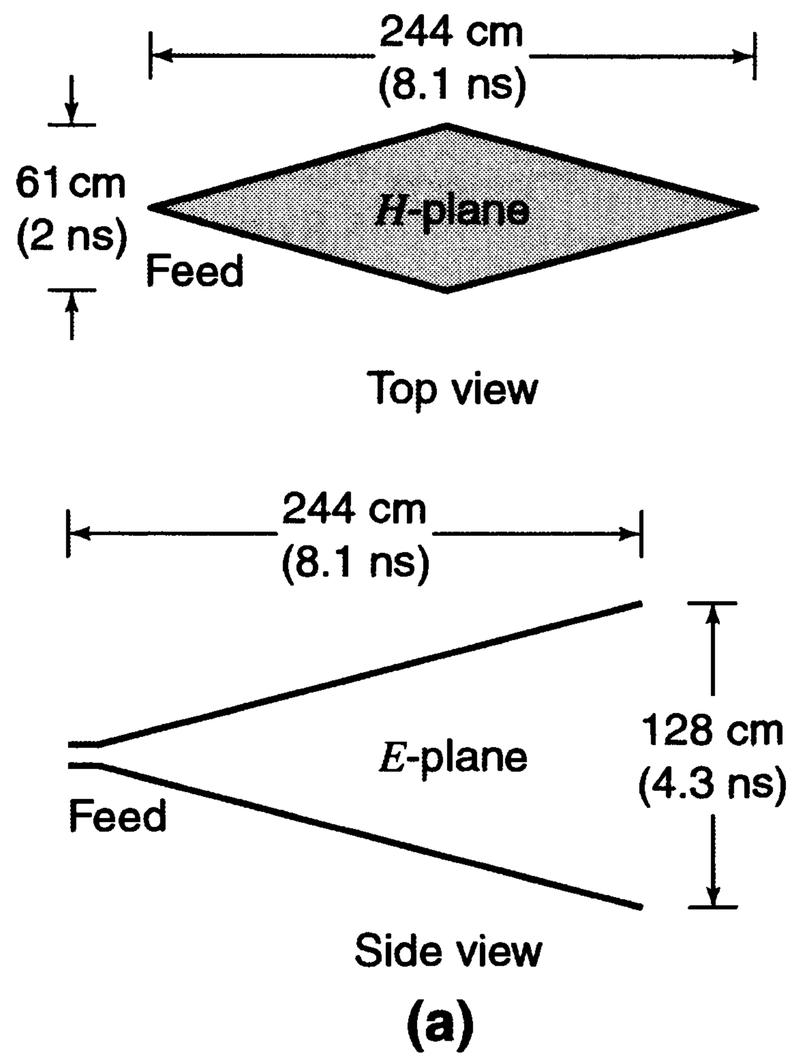


Figure 21–58
Rhombic Vee antenna. (*a*) Top and side view of vee antenna of two rhombic plates and (*b*) its pulse time dispersion (or normalized transient pattern function) versus angle from 0.3 to 2 GHz (100 cm to 15 cm wavelengths).

G such that

$$\frac{V_T^2}{Z_T + Z_A} T_b^2 = \frac{G(f, \theta, \phi)}{4\pi r^2} P_{\text{rad}} = \frac{G}{4\pi r^2} \frac{V_T^2 Z_A}{(Z_T + Z_A)^2} \tag{2}$$

where r = far field distance, m

Thus, the *Transmitting Power Transfer Function* is

$$T_b(f,\theta,\phi) = \sqrt{\frac{Z_A}{Z_T + Z_A}}\frac{\sqrt{G}}{2r\sqrt{\pi}}e^{-j\beta r} \qquad (\mathrm{m}^{-1}) \tag{3}$$

where the phase term has been reinstated into the equation. This can also be related to the effective height h of the antenna using the equations in Sec. 2–10 such that

$$T_b(f,\theta,\phi) = j\beta h\sqrt{\frac{Z_o}{Z_T + Z_A}}\frac{e^{-j\beta r}}{4\pi r} \qquad (\mathrm{m}^{-1}) \tag{4}$$

where the effective height $h = E/V_{oc}$, with units (m), is the parameter that relates the incident field to its open-circuit voltage.

For the receiving case, the transfer of power, $R_b(f,\theta,\phi)$, from the antenna into the terminal load with impedance Z_R is

$$\frac{E^2}{Z_o}R_b^2(f,\theta,\phi) = \frac{V_R^2}{Z_R} \qquad (\mathrm{W}) \tag{5}$$

where $R_b(f,\theta,\phi)$ = the *Receiving Transfer Function* from the power received in the antenna to the power delivered to the load. From Sec. 2–9, the power received is

$$P = \frac{V_R^2}{Z_R} = \frac{E^2}{Z_o}A_e \qquad (\mathrm{W}) \tag{6}$$

where A_e = effective aperture (m^2). Therefore,

$$R_b(f,\theta,\phi) = \sqrt{A_e} \qquad (\mathrm{m}) \tag{7}$$

Again utilizing the definition of the effective height, it can be shown that the Receiving Power Transfer function is

$$R_b(f,\theta,\phi) = \frac{\sqrt{Z_R Z_o}}{(Z_R + Z_A)}h \qquad (\mathrm{m}) \tag{8}$$

A *Normalized Spectral Pattern Function* may now be defined, as a UWB generalization of the single-frequency normalized antenna-pattern. Normalizing the transfer functions to their on-axis responses (or maximums if the on-axis value is a null) deconvolves the antenna on-axis response from the pattern data, which removes the dependence on impedance matching. Thus, the *Normalized Spectral Pattern Function* is

$$P(f,\theta,\phi) = \frac{T_b(f,\theta,\phi)}{T_b(f,\theta_A,\phi_A)} = \frac{R_b(f,\theta,\phi)}{R_b(f,\theta_A,\phi_A)} \qquad \text{(dimensionless)} \tag{9}$$

where the subscripts denote the on-axis angle.

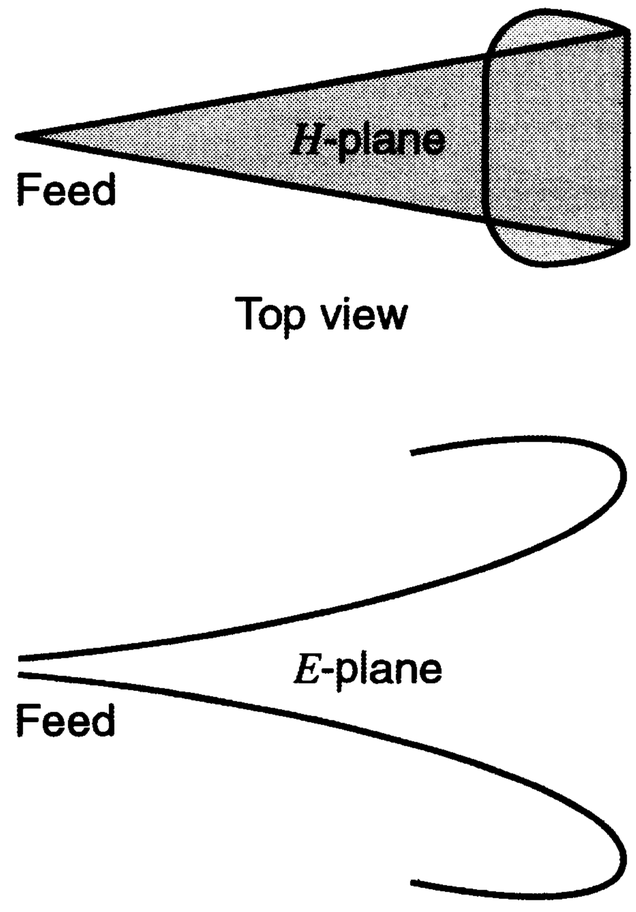


Figure 21–59
Tapered smooth vee antenna in top and side view.

Next a *Normalized Transient Pattern* may be defined. This helps identify and visualize the radiation centers of a broadband antenna through the delay of energy in the resulting plots. It is achieved by Fourier transforming the individual spectra of the Normalized Spectral Patterns into the time domain, such that

$$P(t, \theta, \phi) = \mathcal{I}\{W(f)P(f, \theta, \phi)\} \qquad \text{(dimensionless)} \tag{10}$$

where $W(f)$ = dimensionless windowing function (i.e., Hamming window to reduce Gibb's phenomenon).

To illustrate, the rhombic vee (Fig. 21–58*a*) has a pulse response function, or pulse time dispersion, versus angle as shown in Fig. 21–58*b*. As a consequence of on-axis normalization, the on-axis time response is a "best approximation" of a perfect impulse, given the finite frequency band of measurement. In this case, its 1-ns pulse reflects a measured data range of 0.3 to 2.0 GHz (100 cm to 15 cm wavelength) in 0.1-GHz steps. The pulse travel time "tracks" traced by the impulse energy bursts versus angle can be diagnosed as energy from radiation centers such as the rhombic corners and tips. This leads to pulse distortion off-axis. If this antenna were used for UWB synthetic aperture radar imaging, the echo of a single small off-axis radar object would create a string of pulses, as if several objects were present. As a digital communications antenna, the off-axis coherent signal might get distorted sufficiently to create an error. However, if the antenna is of a gradually tapered smooth design (no corners), like the antenna in Fig. 21–59, the dispersion is significantly reduced both on and off axis.

21–29 THE PLASMA ANTENNA（等离子态天线）

A plasma surface wave can be excited along a column of low-pressure gas (such as argon) by adequate RF power coupled to the column in a glass tube as in Fig. 21–60. The length

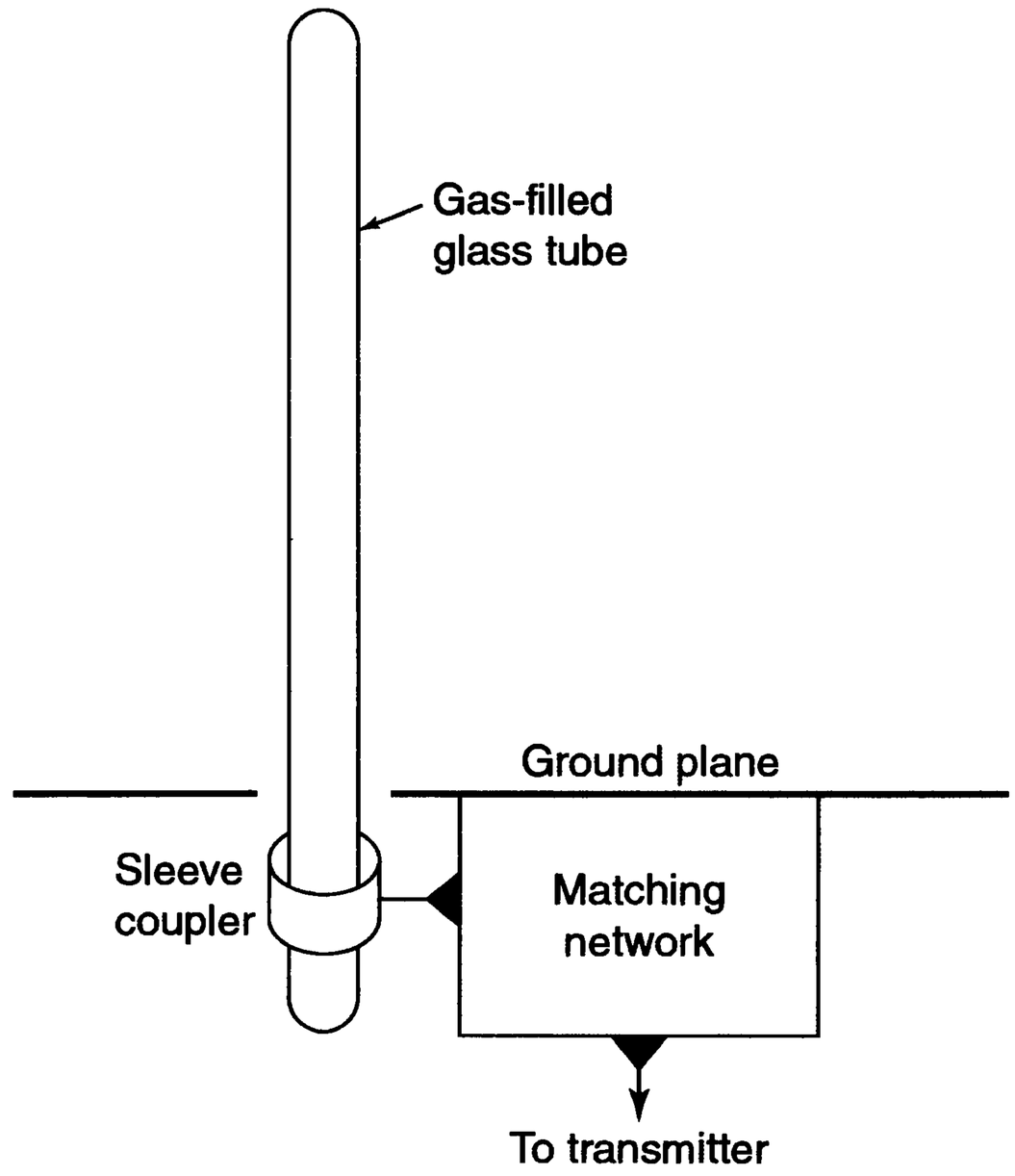


Figure 21–60
Plasma-tube antenna.

of the column that is excited is a function of the power applied. Radiation efficiency is in the range of 25 to 50% (Borg-1).

The plasma antenna has applications where the antenna RCS should be small compared to a metal antenna when no power is applied. When not transmitting the RCS is only that of the thin-wall glass tube. With a *laser beam* producing the plasma column in air in place of the plasma tube, the RCS becomes zero when the laser is off (Anderson (1) and Aiksnoras).

REFERENCES（参考文献）

Alford, A. (1), and A. G. Kandoian, "Ultra-high Frequency Loop Antennas," *Trans. AIEE,* **59,** 843–848, 1940.

Alford, A. (2), J. D. Kraus and E. C. Barkofsky, in *Very High Frequency Techniques,* Radio Research Laboratory Staff, McGraw-Hill, New York, 1947, Chap. 9.

Altshuler, E. E. (1), "Very Small Genetic Antennas," *IEEE Trans. Ants. Prop.*, 2001.

Andersen, J. Bach (1), "Antenna Arrays in Mobile Communications: Gain Diversity and Channel Capacity," *IEEE Ants. Prop. Magazine,* **42,** 12–16, April 2000.

Anderson, T. R. (1) and R. J. Aiksnoras, "Plasma Antennas with Electro-optical Modulation," U.S. Patent 6,087,993, July 11, 2000.

Armitage, J. L. (1), "Electronic Warfare Solid-State Phased Arrays," *Microwave J.,* **29,** 109–122, February 1986.

Barnsley, Michael (1), "Fractals Everywhere," 2d ed., Morgan Kaufman, 1993.

Block, E. L. (1), J. A. Nelson and A. Dorne, in *Very High Frequency Techniques,* Radio Research Laboratory Staff, McGraw-Hill, New York, 1947, Chap. 5.

Borg, G. G. (1), J. H. Harris, D. J. Miljak and N. M. Martin, "Application of Plasma Columns to Radio-frequency Antennas," *Applied Phys. Letters,* Vol. 14, no. 22, 31, May 1999.

Braun, C. (1), G. Engblom and C. Beckman, "Antenna Diversity for Mobile Telephones," *Proc. IEEE AP-S 1998,* 2220–2223, 1998.

Braun, F. (1), "Elektrische Schwingungen und drahtlose Telegraphic," *Jahrbuch der drahtlosen Telegraphie und Telephonie,* **4,** no. 1, 17, 1910.

Brown, G. H. (1), "The Turnstile Antenna," *Electronics,* **9,** 15, April 1936.

Brown, G. H. (2), and J. Epstein, "A Pretuned Turnstile Antenna," *Electronics,* **18,** 102–107, June 1945.

Brown, G. H. (3), and O. M. Woodward, Jr., "Circularly-Polarized OmniDirectional Antenna," *RCA Rev.,* **8,** 259–269, June 1947.

Clarke, A. C. (1), "Extra-terrestrial Relays," *Wireless World,* **51,** 305–308, October 1945.

Crawford, A. B. (1), D. C. Hogg and L. E. Hunt, "A Horn Reflector Antenna for Space Communication," BSTJ, **40,** 1045–1116, July 1961.

Delogne, P. (1), *Leaky Feeders and Subsurface Radio Communication,* IEE, London, 1982.

Dietrich, F. J. (1), P. Metzen, P. Monte, "The Globalstar Cellular Satellite System," *IEEE Trans. Ants. Prop.,* **40,** no. 6, June 1998, pp. 935–942.

Dietrich, F. J. (2), personal communication, 2000.

Foschini, G. J. (1), "Layered Space-Time Architecture for Wireless Communication in a Fading Environment When Using Multi-element Antennas," *Bell Labs. Tech. Journal,* **1,** 41–59, Autumn 1996.

Fujimoto, K. (1), and J. R James, *Mobile Antenna Systems Handbook,* Artech House, Boston, 1994.

Fujimoto, K. (2), A. Henderson, K. Hirasawa and J. R. James, *Small Antennas,* New York, Research Studies Press, 1987.

Goubau, G. (1), "Surface Waves and Their Application to Transmission Lines," *J. Appl. Phys.,* **21,** 1119–1128, November 1950.

Gwynne, J. S. (1), and J. D. Young, "Normalized Transient Pattern Functions of Ultra-Wideband Antennas," submitted to *IEEE Transactions on Antennas and Propagation,* 1999.

Hansen, R. C. (1), "Fundamental Limitations of Antennas," *Proc. IEEE,* **69,** 170–182, February 1981.

Hirasawa, K. (1), and M. Haneishi, *Analysis, Design and Measurement of Small and Low Profile Antennas,* Boston, Artech House, 1992.

Jakes, W. C. (1), *Microwave Mobile Communications,* Wiley, New York, 1974.

Jansky, D. M. (1), and M. C. Jeruchim, *Communication Satellites in the Geostationary Orbit,* Artech House, 1983.

Kandoian, A. G. (1), "Three New Antenna Types and Their Applications," *Proc. IRE,* **34,** 70W–75W, February 1946.

Kandoian, A. G. (2), W. Sichak and G. A. Felsenheld, "High Gain with Discone Antennas," *Proc. Natl. Electronics Conf.,* **3,** 318–328, 1947.

Kandoian, A. G. (3), and R. A. Felsenheld, "Triangular High-Band TV Loop Antenna System," *Communications,* **29,** 16–18, August 1949.

Kim, Y. (1), and D. L. Jaggard, "The Fractal Random Array," *Proc. IEEE,* **74,** 1986, pp. 1278–1280.

King, R. W. P. (1), *Theory of Linear Antennas,* Harvard, 1956, Chap. 7, 75 refs.

King, R. W. P. (2), "Electromagnetic Surface Waves; New Formulas and Applications," *IEEE Trans. Ants. Prop.,* **AP-33,** 1204–1212, November 1985.

King, R. W. P. (3), and G. S. Smith (with M. Owens and T. T. Wu), *Antennas in Matter,* MIT Press, 1980.

Ko, S. (1), and R. Murch, "A Planar Diversity Antenna for Wireless Handsets," *Proc. IEEE AP-S 2000,* 968–971, 2000.

Kraus, J. D. (1), and D. A. Fleisch, *Electromagnetics,* 5th ed., McGraw-Hill, 1999, p. 221.

Kraus, J. D. (2), "Helical Beam Antenna for Wide-Band Applications," *Proc. IRE,* **36,** 1236–1242, October 1948.

Kraus, J. D. (3), and D. A. Fleisch, *Electromagnetics,* 5th ed., McGraw-Hill, 1999, p. 224.

Kraus, J. D. (4), and D. A. Fleisch, *Electromagnetics,* 5th ed., McGraw-Hill, 1999, pp. 483–488.

Kraus, J. D. (5), *Radio Astronomy,* 2d ed., Cygnus-Quasar Books, 1986, pp. 8-9–8-12.

Lindell, I. V. L. (1), E. Alanen and K. Mannersalo, "Exact Image Method for Impedance Computation of Antennas above the Ground," Helsinki University Radio Lab. Rept. S 161, 1984, 23 refs.

Lindmark, B. (1), S. Lundgren, J. Sanford and C. Beckman, "Dual Polarized Array for Signal-Processing Applications in Wireless Communications," *IEEE Trans. Ants. Prop.*, **46,** 758–763, June 1998.

Maltove, E. R. (1), "The Bombarded Earth," *Tech. Rev.*, July 1985.

Mandelbrot, B. B. (1), *The Fractal Geometry of Nature,* W. H. Freeman, 1983.

Marcuse, D. (1), *Theory of Dielectric Optical Waveguides,* Academic Press, 1974.

Masters, R. W. (1), "The Super-turnstile Antenna," *Broadcast News,* **42,** January 1946.

Miller, B. (1), "Satellites Free the Mobile Phone," *IEEE SPECTRUM,* March 1998, pp. 25–26.

Montaya, T. P. (1), and G. S. Smith, "Land-mine Detection Using a Ground-Penetrating Radar Based on Resistively-Loaded Vee Dipoles," *IEEE Trans. Ants. Prop.*, **47,** December 1999, pp. 1795–1806.

Munk, B. (1), *Finite Periodic Structures, FSS,* Chap. 2, Wiley 2002.

Nagumo, S. (1), K. Kawahata, Y. Saitoh, M. Ida and Y. Ishikawa, "Dual Resonance Chip Dielectric Antenna with a Slant Gap for Hand Held Wireless Communication Tools," *Proc. Millennium Conf. Ants. Prop. (AP 2000),* Session 3A9, paper no. 0591, 2000.

Nakano, H. (1), S. Okuzawa, K. Ohishi, H. Mimaki and J. Yamauchi, "A Curl Antenna," *IEEE Ants. Prop.* pp. 1570–1575, November 1993.

Ojanpera, T. (1), and R. Prasad, *Wideband CDMA for Third Generation Mobile Communications,* Artech House, Boston, 1998.

Oliner, A. (1), in *Antenna Engineering Handbook,* R. C. Johnson, (ed.), McGraw-Hill, 1993, Chap. 10.

Ollikainen, J. (1), O. Kivekäs, A Toropainen and P. Vainikainen, "Internal Dual-Band Patch Antenna for Mobile Phones," *Proc. Millennium Conf. Ants. Prop. (AP 2000),* Session 3A9, paper no. 1111, 2000.

Perez, R. (1), "Designing Embedded Antennas for Bluetooth Protocol," *ACES Jour,* **1,** November 2000, pp. 152–158.

Peters, L., Jr. (1), J. J. Daniels and J. D. Young, "Ground Penetrating Radar as a Subsurface Environmental Sensing Tool," *Proc. IEEE,* **82,** December 1994, pp. 1802–1822.

Pistolkors, A. A. (1), "Theory of Circular Diffraction Antenna," *Proc. IRE,* **36,** 56–60, January 1948.

Prasad, S. N. (1), and S. Mahapatra, "A New MIC Slot-Line Aerial," *IEEE Trans. Ants. Prop.*, **AP-31,** 525–527, May 1983.

Prusinkiewicz, P. (1), and A. Lindenmayer, *The Algorithmic Beauty of Plants,* Springer Verlag, 1990.

Rhodes, D. R. (1), "Flush-Mounted Antenna for Mobile Application," *Electronics,* **22,** 115–117, March 1949.

Robinson, A. L. (1), "Cornell Submicron Facility Dedicated," *Science,* **214,** 777–779, 1981.

Sato, G. (1), H. Kawakami, H. Sato and R. W. Masters, "Design Method for Fine Impedance Matching of Superturnstile Antenna and Characteristics of the Modified Batwing Antenna," *Trans. IECE (Japan),* **E65,** 271–278, May 1982.

Schuss, J. J. (1), J. Upton, B. Myers, T. Sikina, A. Rohwer, P. Mokridakas, R. Francois, L. Wardle and R. Smith, "The Iridium Main Mission Antenna Concept," *Trans. Ants. Prop.*, p. 416, March 1999.

Siwiak, K. (1), *Radiowave Propagation and Antennas for Personal Communications,* Artech House, Boston, 1998.

Smith, C. E. (1), and R. A. Fouty, "Circular Polarization in F-M Broadcasting," *Electronics,* **21,** 103–107, September 1948.

Smith, P. H. (1), "Cloverleaf Antenna for FM Broadcasting," *Proc. IRE,* **35,** 1556–1563, December 1947.

Sommerfeld, A. (1), "Über die Ausbreitung der Wellen in der drahtlosen Telegraphic," *Ann. Phys.*, **28,** 665, 1909.

Sommerfeld, A. (2), "Fortpflanzung elektrodynamischer Wellen an einern zylindrischen Leiter," *Ann. Phys. u. Chem.*, **67,** 233, December 1899.

Terman, F. E. (1), *Radio Engineering Handbook,* 1st ed., McGraw-Hill, 1943, sec. 10, 28 refs.

Vainikainen, P. (1), J. Ollikainen, O. Kivekäs and I. Kelander, "Resonator-Based Analysis of the Combination of Mobile Handset Antenna and Chassis," submitted for publication in *IEEE Trans. Ants. Prop.*, 2000.

Vaughan, R. (1), "Polarization Diversity in Mobile Communications," *IEEE Trans. Vehic. Tech.*, **39,** 177–186, August 1990.

Vaughan, R. (2), and J. Bach Andersen, "Antenna Diversity in Mobile Communications," *IEEE Trans. Vehic. Tech.*, **36,** 149–172, November 1987.

Walter, C. H. (1), B. Munk and G. A. Thiele, "Multi-tun Loop Antenna," U.S. Patent No. 3,823,403, July 9, 1974.

Weiss, H. J. (1), "Maximizing Access to the Geostationary-Satellite Orbit," *ITU J.*, **53,** 469–477, August 1986.

Werner, D. H. (1), R. L. Haupt and P. L. Werner, "Fractal Antenna Engineering: The Theory and Design of Fractal Antenna Arrays," *IEEE Ant. and Prop. Mag.*, **41,** October 1999, pp. 37–59.

Wertz, J. (1), and W. Larson, *Space Mission Analysis and Design,* Microcosm Press, 1998.

Wheeler, H. A. (1), "Fundamental Limitations of Small Antennas," *Proc. IRE,* **35,** 1479–1484, December 1947.

Wheeler, H. A. (2), "Small Antennas," *IEEE Trans. Ants. Prop.*, **AP-23,** 462–469, July 1975.

Wilkinson, D. T. (1), and P. J. Peebles, "Discovery of the 3K Radiation," in "Serendipitous Discoveries in Radio Astronomy," pp. 175–184, N.R.A.O., Green Bank, 1983.

Wilson, R. W. (1), "Discovery of Cosmic Microwave Background," in "Serendipitous Discoveries in Radio Astronomy," pp. 185–195, N.R.A.O., Green Bank, 1983.

Young, J. D. (1), and J. S. Gwynne, "Report on Antenna Transient Patterns," Technical Report 732169-3, December 1997, The Ohio State University ElectroScience Laboratory.

Zenneck, J. (1), "Über die Fortpftanzung ebener elektromagnetischer Wellen längs einer ebenen Leiterfläche und ihre Bezichung zur drahtlosen Telegraphie," *Ann. Phys.*, **23,** 846–866, 1907.

Zucker, F. J. (1), in *Antenna Engineering Handbook,* R. C. Johnson (ed.), McGraw-Hill, New York, 1993, Chap. 12.

Zürcher, J-F. (1), and F. E. Gardiol, *Broadband Patch Antennas,* Artech House, Boston, 1995.

PROBLEMS（习题）

21–4–1
（地面上方的水平偶极子）

Horizontal dipole above ground. A thin $\lambda/2$ dipole is parallel to a flat, perfectly conducting ground at a height h above it. (*a*) Calculate and plot the gain of the dipole in the zenith direction as a function of height h for heights from zero to λ. Express the gain with respect to a $\lambda/2$ dipole in free space. Assume zero losses. (*b*) Repeat (*a*) for dipole loss resistance $R_L = 1\ \Omega$.

21–4–2
（非理想地面上方的水平偶极子）

Horizontal dipole above imperfect ground. Calculate the vertical plane field pattern broadside to a horizontal $\lambda/2$ dipole antenna $\lambda/4$ above actual homogeneous ground with constants $\varepsilon_r' = 12$ and $\sigma = 2 \times 10^{-3}\ \Omega^{-1}\ \text{m}^{-1}$ at (*a*) 100 kHz and (*b*) 100 MHz.

21–4–3
（非理想地面上方的铅垂短偶极子）

Short vertical dipole above imperfect ground. The center of a short vertical dipole ($l < \lambda/10$) is located $\lambda/4$ above actual homogeneous ground with the same constants as in Prob. 21–4–2. Calculate the vertical plane field pattern at (*a*) 1 MHz and (*b*) 100 MHz.

21–4–4
（非理想地面上方的天线）

Antennas over imperfect ground. Write and run computer programs for Probs. 21–4–2 and 21–4–3 with a menu for height, frequency and ground constants ε_r' and σ.

21–9–1
（方环）

Square loop. Calculate and plot the far-field pattern in the plane of a loop antenna consisting of four $\lambda/2$ center-fed dipoles with sinusoidal current distribution arranged to form a square $\lambda/2$ on a side. The dipoles are all in phase around the square.

21–9–2
（三角环）

Triangular loop. Calculate and plot the far-field pattern in the plane of a loop antenna consisting of three $\lambda/2$ center-fed dipoles with sinusoidal current distribution arranged to form an equilateral triangle $\lambda/2$ on a side. The dipoles are all in phase around the triangle.

***21–9–3**
（测向与单脉冲）

DF and monopulse. Many direction-finding (DF) antennas consist of small (in terms of λ) loops giving a figure-of-eight pattern as in Fig. P21–9–3*a*. Although the null is sharp the bearing (direction of transmitter signal) may have considerable uncertainty unless the S/N ratio is large. To resolve the 180° ambiguity of the loop pattern, an auxiliary antenna may be used with the loop to give a cardioid pattern with broad maximum in the signal direction and null in the opposite direction.

The maximum of a beam antenna pattern, as in Fig. P21–9–3*b*, can be employed to obtain a bearing with the advantage of a higher S/N ratio but with reduced pattern change per unit angle. However, if 2 receivers and 2 displace beams are used, as in Fig. P21–9–3*c*, a large power-pattern change can be combined with a high S/N ratio. An arrangement of this kind for receiving radar echo signals can give bearing information on a single pulse (*monopulse radar*). If the power received on beam 1 is P_1 and on beam 2 is P_2, then if $P_2 > P_1$ the bearing is to the right. If $P_1 > P_2$ the bearing is to the left and if $P_1 = P_2$ the bearing is on axis (boresight). (With 4 antennas, bearing information left-right *and* up-down can be obtained.) (*a*) If the power pattern is proportional to $\cos^4 \theta$, as in Fig. P21–9–3*c*, determine P_2/P_1 if the interbeam (squint) angle $\alpha = 40°$ for $\Delta\theta = 5$ and 10°. (*b*) Repeat for $\alpha = 50°$. (*c*) Determine the P_0/P_1 of the single power pattern of Fig. P21–9–3*b* for $\Delta\theta = 5$ and 10° if the power pattern is also proportional to $\cos^4 \theta$. (*d*) Tabulate the results for comparison and indicate any improvement of the double over the single beam.

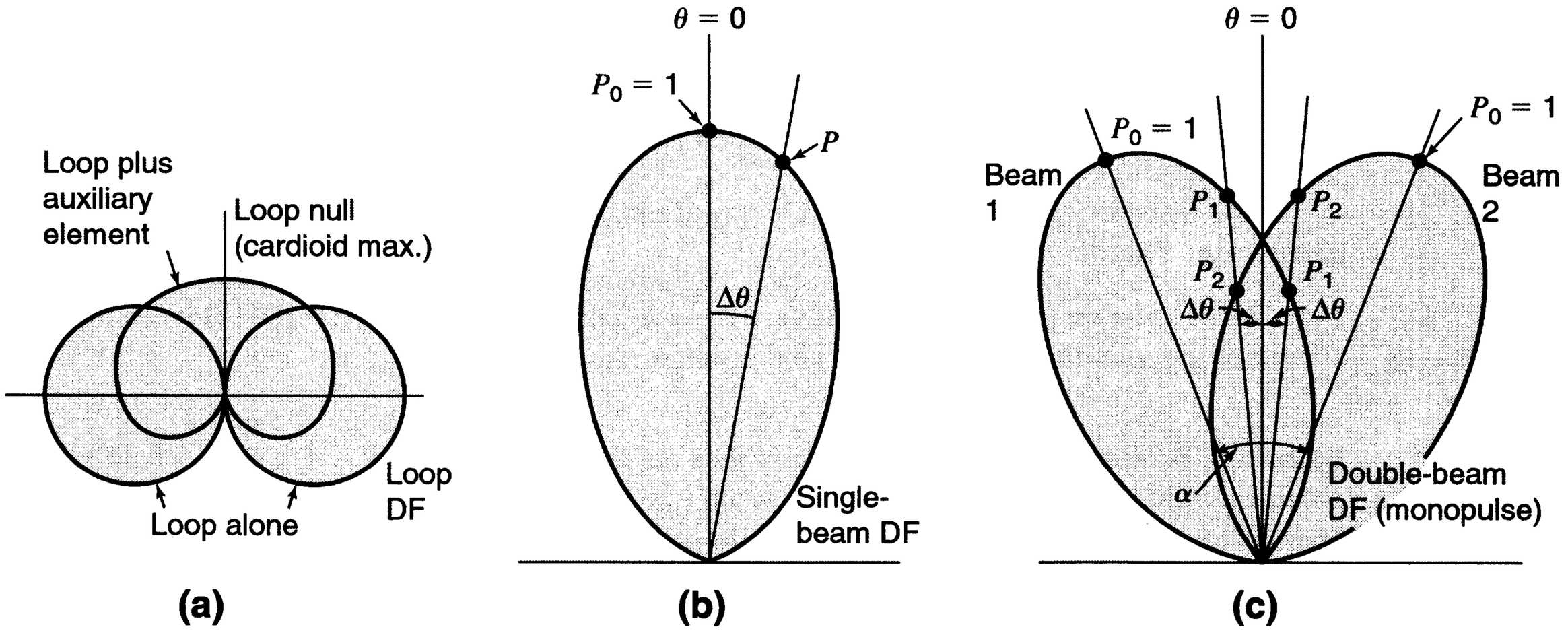


Figure P21–9–3
Direction finding: (*a*) with loop mull, (*b*) with beam maximum and (*c*) with double beam (monopulse).

***21–10–1**
（陆上的用水平、铅垂和圆极化的电视）

Overland TV for HP, VP and CP. (*a*) A typical overland microwave cell-phone communications circuit for AM, FM or TV between a transmitter on a tall building and a distant receiver involves 2 paths of transmission, one direct path (length r_o) and one an indirect path with ground reflection (length $r_1 + r_2$), as suggested in Fig. P21–10–1. Let $h_1 = 300$ m and $d = 5$ km. For a frequency of 100 MHz calculate the ratio of the power received per unit area to the transmitted power as a function of the height h_2 of the receiving antenna. Plot these results in decibels as abscissa versus h_2 as ordinate for 3 cases with transmitting and receiving antennas both (1) vertically polarized, (2) horizontally polarized and (3) right-circularly polarized for h_2 values from 0 to 100 m. Assume that the transmitting antenna is isotropic and that the receiving antennas are also isotropic (all have the same effective aperture). Consider that the ground is flat and perfectly conducting. (*b*) Compare the

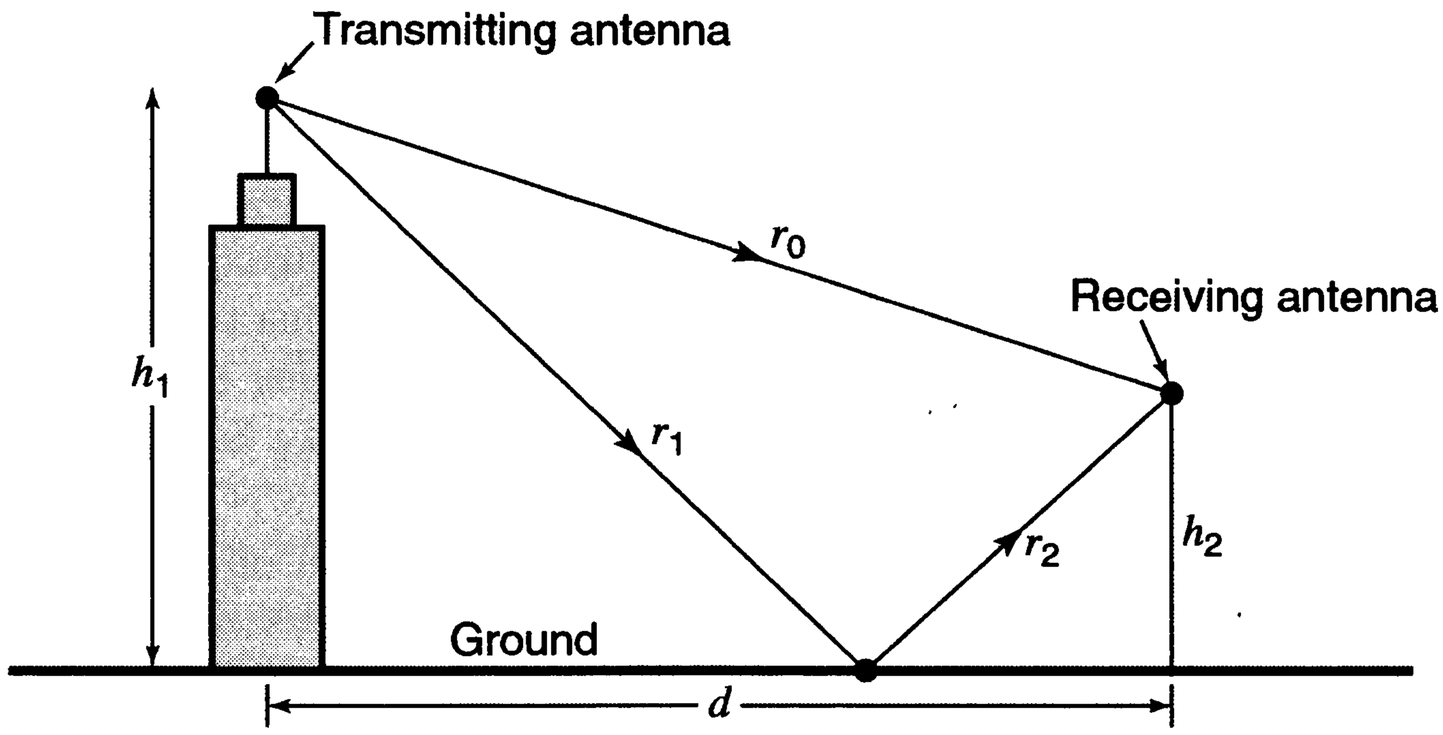


Figure P21–10–1
Overland microwave communication circuit.

results for the 3 types of polarization and show that circular polarization is best from the standpoint of both the noncriticalness of the height h_2 and the absence of echo or ghost signals. Thus, for horizontal or vertical polarization the direct and ground-reflected waves may cancel at certain heights while at other heights, where they reinforce, the images on the TV screen may be objectionable because the time difference via the 2 paths produces a double image (a direct image and its ghost). (*c*) Extend the comparison of (*b*) to consider the effect of other buildings or structures that may produce additional paths of transmission.

Note that direct satellite-to-earth TV downlinks are substantially free of these reflection and ghost image effects.

21–10–2 (陆上无线电线路的路径差) **Path difference on overland radio link.** If $h_1 = h_2$ and $d \gg h_1$ in Prob. 21–10–1 (Fig. P21–10–1), show that the path difference of direct and reflected rays is $2h_1^2/d$.

***21–12–1** (对水下潜艇发信号) **Signaling to submerged submarines.** Calculate the depths at which a 1 μV m^{-1} field will be obtained with E at the surface equal to 1 V m^{-1} at frequencies of 1, 10, 100 and 1000 kHz. What combination of frequency and antennas is most suitable?

***21–13–1** (表面波的功率) **Surface-wave powers.** A 100-MHz wave is traveling parallel to a copper sheet ($|Z_c| = 3.7\times10^{-3}\ \Omega$) with E(=100 V m^{-1} rms) perpendicular to the sheet. Find (*a*) the Poynting vector (watts per square meter) parallel to the sheet and (*b*) the Poynting vector into the sheet.

21–13–2 (表面波的功率) **Surface-wave powers.** A 100-MHz wave is traveling parallel to a conducting sheet for which $|Z_c| = 0.02\ \Omega$. If $\boldsymbol{E}$ is perpendicular to the sheet and equal to 150 V m^{-1} (rms), find (*a*) watts per square meter traveling parallel to the sheet and (*b*) watts per square meter into the sheet.

***21–13–3** (表面波的功率) **Surface-wave power.** A plane 3-GHz wave in air is traveling parallel to the boundary of a conducting medium with $\boldsymbol{H}$ parallel to the boundary. The constants for the conducting medium are $\sigma = 10^7\ \Omega^{-1}$ m^{-1} and $\mu_r = \varepsilon_r = 1$. If the traveling-wave rms electric field $E = 75$ mV m^{-1}, find the average power per unit area lost in the conducting medium.

21–13–4 (表面波的电流片) **Surface-wave current sheet.** A TEM wave is traveling in air parallel to the plane boundary of a conducting medium. Show that if $K = \rho_s v$, where K is the sheet-current density in amperes per meter, ρ_s is the surface charge density in coulombs per square meter and v the velocity of the wave in meters per second, it follows that $K = H$, where H is the magnitude of the $\boldsymbol{H}$ field of the wave.

21–13–5 (介质敷层表面波的功率) **Coated-surface wave power.** Show that for a dielectric-coated conductor, as in Fig. 21–28, the ratio of the power transmitted in the dielectric P_d to the power transmitted in the air P_a is given by

$$\frac{P_d}{P_a} = \frac{\cos \beta d}{\sin^3 \beta d}(\sin 2\beta d - 2\beta d)$$

where d is the thickness of the dielectric coating.

***21–13–6** (介质敷层表面波的截止) **Coated-surface wave cutoff.** A perfectly conducting flat sheet of large extent has a dielectric coating ($\varepsilon_r = 3$) of thickness $d = 5$ mm. Find the cutoff frequency for the TM_0 (dominant) mode and its attenuation per unit distance.

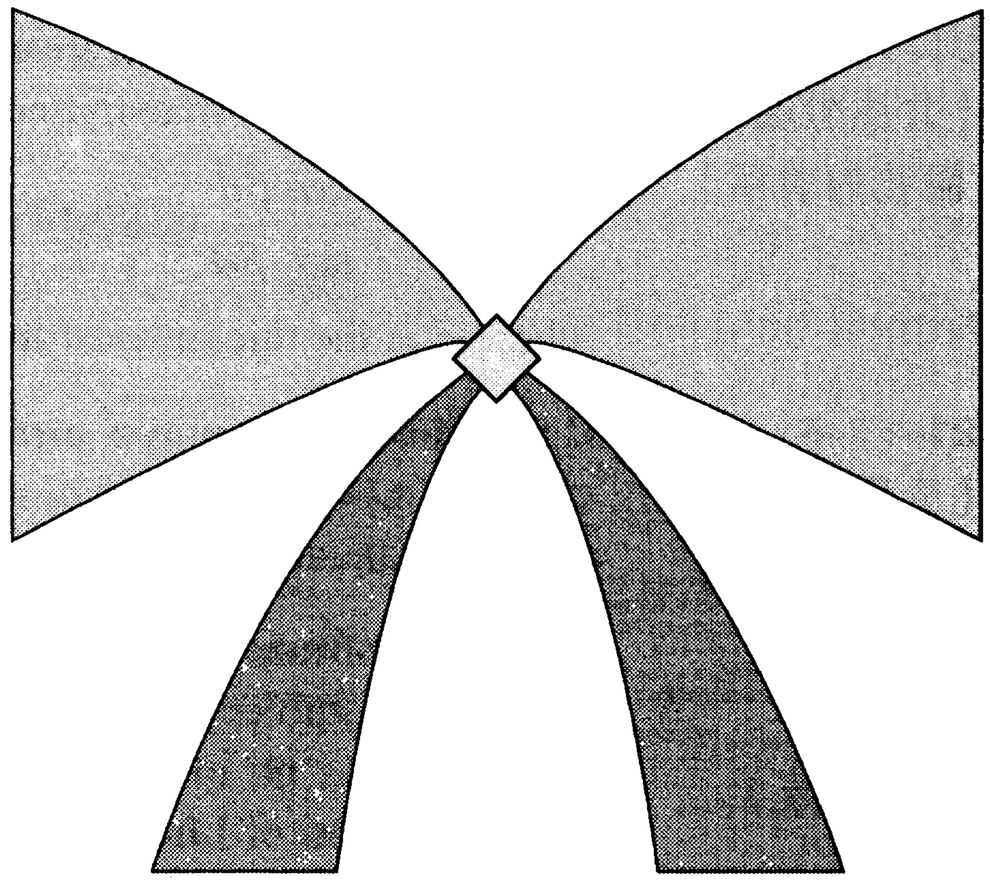
21–13–7 (表面波月球通信) **Lunar communication by surface wave.** Discuss the possibilities of using dielectric-slab surface modes for radio communication around the moon over long distances (1000 km or more). Note that the moon has no ionosphere. See, for example, W. W. Salisbury and D. L. Fernald, "Postoccultation Reception of Lunar Ship Endeavour Radio Transmission," *Nature,* **34,** 95, Nov. 12, 1971; also A. F. Wickersham, Jr., "Generation, Detection and Propagation on the Earth of HF and VHF Radio Surface Waves," *Nature,* **230,** 125–130, Apr. 5, 1971.

21–18–1 (滑翔着陆天线的架高) **Antenna heights for 3° glide path.** What are the 2 antenna heights required for a 3° vertical angle glide slope for an ILS system?

For computer programs, see Appendix C.

CHAPTER 22

Terahertz Antennas
太赫频率的天线



The topics in this chapter include:

- Pyramidal horn cavity antenna（棱锥喇叭腔天线）
- Corner reflector array（夹角反射器阵）
- Bow-tie dipole antenna（领结形偶极子）
- Dielectric lenses（介质透镜）
- Waveguide structures（波导结构）

FROM LONG TO SHORT WAVELENGTHS

Radio communication began in the early 1900s at wavelengths of thousands of meters. Wavelengths of 200 meters or less were considered worthless for commercial purposes and assigned to the amateurs. But beginning in 1921, amateurs demonstrated that wavelengths of 200 meters and less could be used for round-the-world communication with very low powers. Commercial use followed at shorter and shorter wavelengths with the amateurs assigned smaller and smaller frequency segments. By 1994 spectrum space had become so valuable that the U.S. Federal Communication Commission began to auction channels for billions of dollars. At millimeter wavelengths atmospheric absorption limits the useful range but not in space. At micrometer wavelengths we are at the infrared or THz region and at 600 THz or 1/2 micrometer we are into light waves. This short chapter is about antennas for these THz or shortest of "radio" waves.

22–1 INTRODUCTION（引言）

Terahertz frequencies from 1 THz (10^{12} Hz) to 1000 THz (10^{15} Hz) embrace the infrared and visible wavelengths as shown by Table 22–1. These frequencies correspond to wavelengths between 1 mm and 300 nm including the visible spectrum between 400 and 700 nm. With techniques now available very small antenna structures can be fabricated making terahertz antennas feasible. Several are described in the next section.

At frequencies above about 20 GHz atmospheric attenuation may be as much as 20 dB/km (see Fig. 21–44). This attenuation provides a natural shield for radio waves that is useful in many wireless applications. Thus, the same frequency may be employed by many different units in close proximity provided the attenuation is sufficient to prevent mutual interference.

Table 22–1 The Terahertz (THz) Region of the Electromagnetic Spectrum

Frequency, f		Wavelength, λ where $\lambda = c/f$	
	100 GHz	3 mm	
	300	1 mm	
	800	375 μm	← Pyramidal horn cavity with dipole
10^{12} Hz	1 THz	300 μm	
	3	100 μm	← 2.52 THz corner reflector array
10^{13} Hz	10 THz	30 μm	
	30	10 μm	← Bow-tie dipole 28 THz
10^{14} Hz	100 THz	3 μm	
	300	1000 nm	
	600	500 nm	700 nm / 400 nm
10^{15} Hz	1000 THz	300 nm	

Infrared — Terahertz — Visible

Atmospheric windows

22–2a PYRAMIDAL HORN CAVITY WITH DIPOLE（带偶极子的棱锥喇叭腔）

A dipole antenna situated in a pyramidal horn cavity etched in silicon, as in Fig. 22–1, has been operated at frequencies as high as 800 GHz or a wavelength of 375 μm (Fumeaux-1).

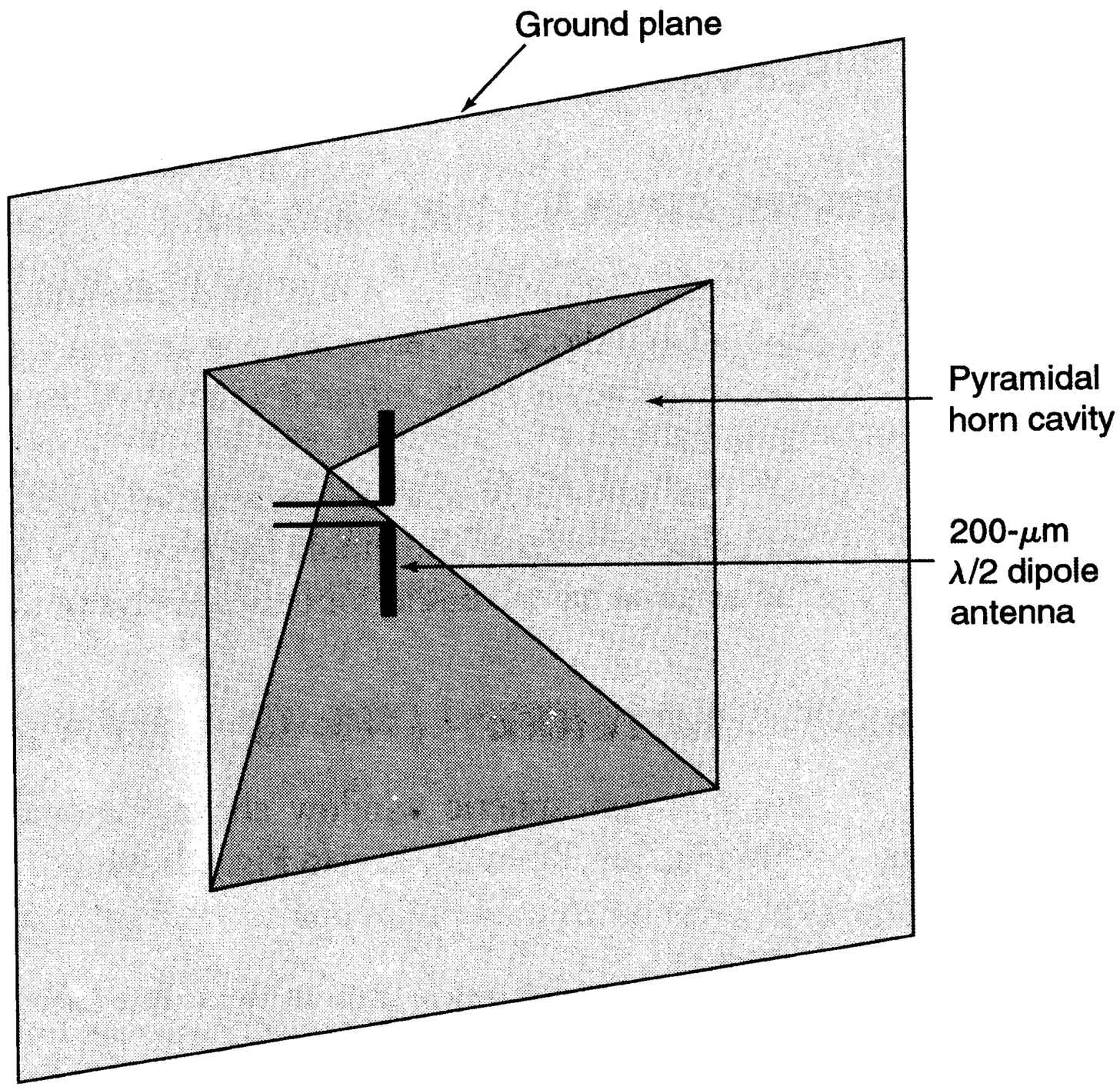


Figure 22–1
Pyramidal horn cavity with $\lambda/2$ dipole for operation at 0.8 THz (800 GHz).

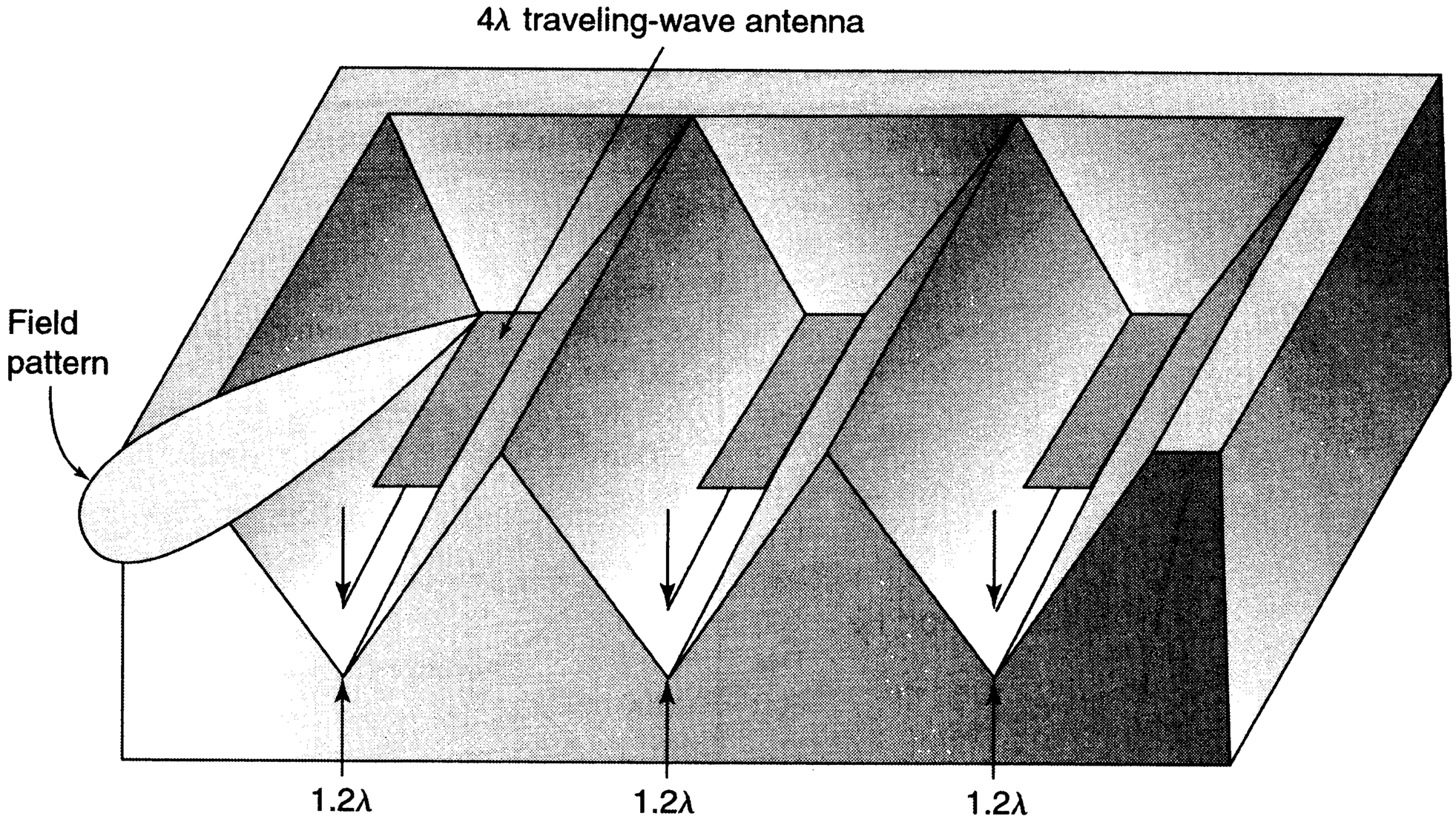


Figure 22–2
Array of three 90°-corner reflectors with traveling-wave antennas 1.2λ from the apex of the corner for operation at 2.5 THz. λ (freespace) = 119 μm.

22–2b CORNER REFLECTOR ARRAY（夹角反射器阵）

These antennas are widely used at THz frequencies. Units with traveling-wave antennas backed by 90-degree corner reflectors have been tested at 119 μm (2.52 THz) (Gearhart-1). The array is shown in Fig. 22–2 with the antenna pattern as a beam at an angle. The antenna is 4 wavelengths long and is supported 1.2 wavelengths from the corner apex.

22–2c BOW-TIE DIPOLE（领结形偶极子）

Figure 22–3 shows a 3/2 λ bow-tie dipole antenna coupled to a MOM (metal-oxide-metal; Ni-NiO-Ni) diode for receiving power from a CO_2 laser at 28 THz (10.6 μm) (Rebeiz-1). This frequency is in the infrared window of the earth's atmosphere. See Fig. 21–44 and also Table 22–1.

The bow-tie is situated on a multilayer (SiO_2-Si-SiO_2) substrate mounted on a flat gold-sheet reflector. The dipole length is only 4.6 μm due to the presence of the dielectric. The antenna has potential for use in arrays of antenna-coupled IR detectors.

22–3 A 600-THz ANTENNA?[1]（一种 600 THz 的天线）

The next atmospheric window above the one at 30 THz is the visible window around 600 THz. See Table 22–1, also Fig. 21–44.

---

[1]We thank Dr. Brian Baertlein of the ElectroScience Laboratory, Ohio State University for assistance on this section.

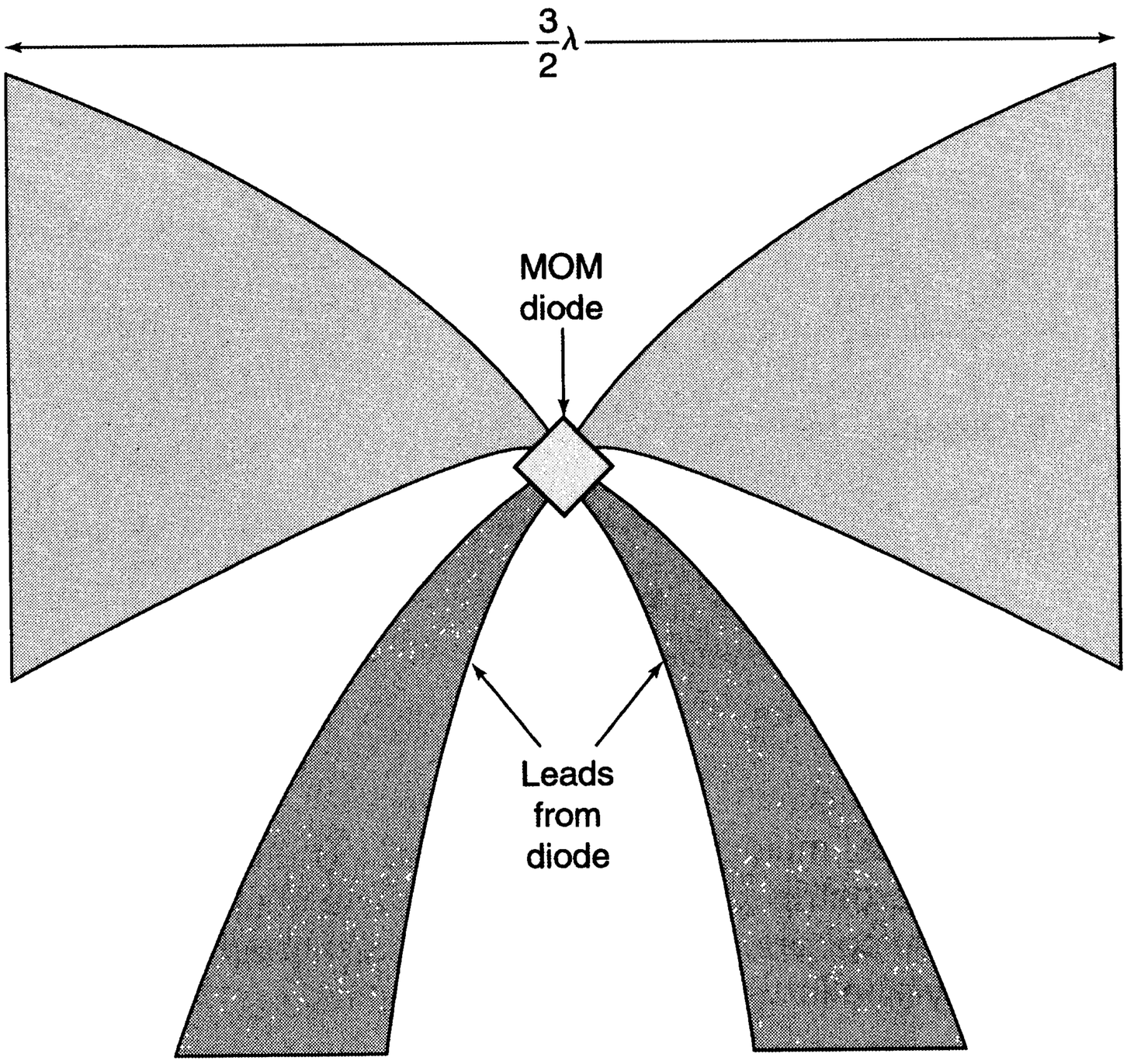


Figure 22–3
3/2 wavelength bow-tie receiving antenna with MOM diode for operation at 28 THz (10.6 μm).

Antennas have been proposed and patented to operate at this frequency for receiving solar radiation and converting it to dc electric power. Since solar radiation is unpolarized, an antenna of any single polarization connected to a 100% efficient rectifier could, in principle, achieve a conversion efficiency of 50% compared to the typical 10 to 20% efficiency of solar cells.

The nominal solar power density at normal incidence is 1 kW per square meter, so that a hypothetical visible-wavelength antenna and rectifier might produce 500 W/m^2 maximum. Unfortunately, it is impractical because at 600 THz the length of one cycle is only

$$1/(6 \times 10^{14})\,\text{s} = 0.17 \times 10^{-14}\,\text{s} = 1.7 \times 10^{-15}\,\text{s} = 1.7 \text{ femtoseconds (fs)}$$

This is less that the electron relaxation time required for a rectifier to function.

The bow-tie MOM diode of Fig. 22–3 operates at 28 THz for which the length of one cycle is 36 fs. This is $36/1.7 = 21$ times as long as the cycle length at 600 THz. The diode limit would appear to be somewhere between 2 and 36 fs.

22–4 PLANAR ANTENNA STRUCTURES ON DIELECTRIC LENSES[1]
(介质透镜的平面天线结构)

At frequencies above 400 GHz quasi-optical mixers have been extensively used in characterizing the performance of mixing devices and in the construction of astronomical receivers.

[1]By Professor Christopher Walker, Astronomy Department, University of Arizona.

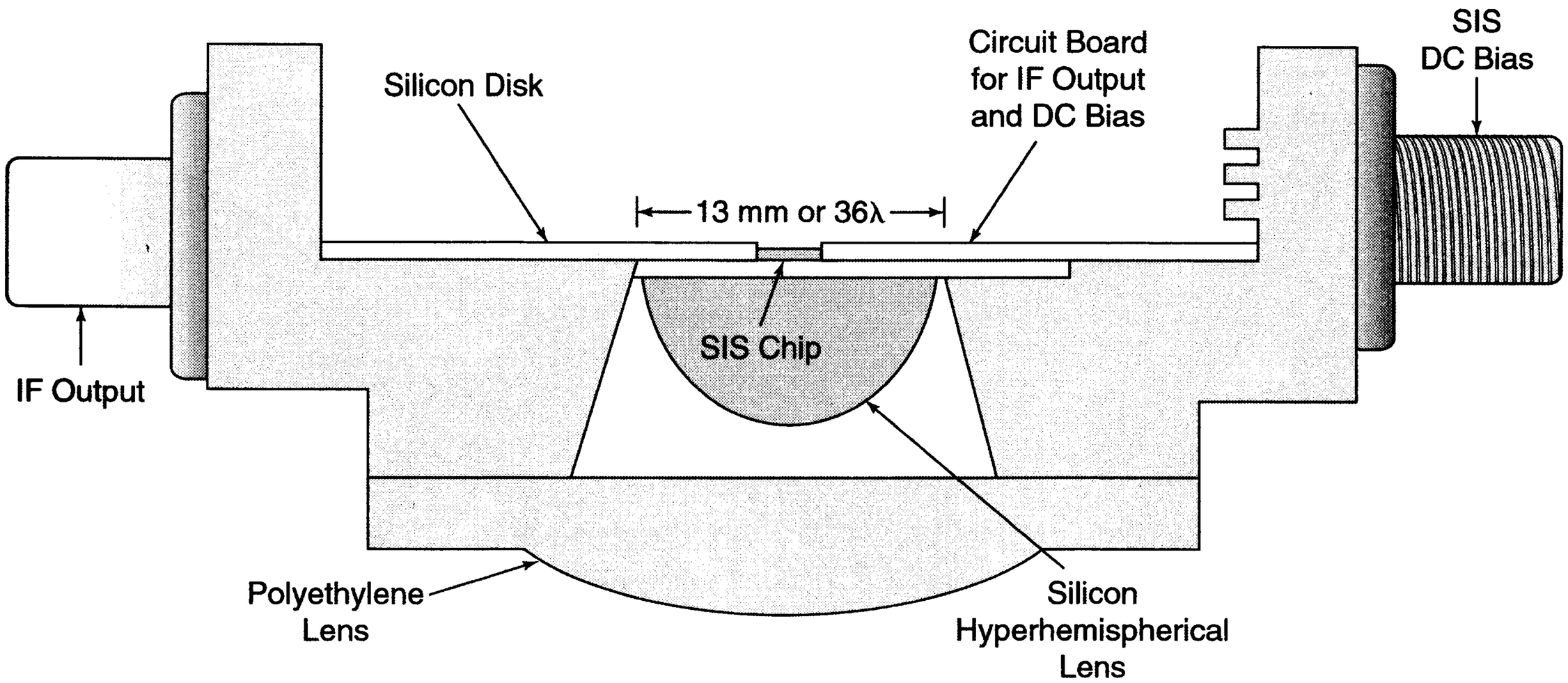


Figure 22–4
Quasioptical mixer block for 850 GHz. The SIS chip contains a twin slot antenna with a microstrip matching network. The chip is bonded to the bottom of a silicon hyperhemispherical lens. The emergent beam is further collimated by a plano-convex polyethylene lens (adapted from Bin-1).

These mixers use a planar antenna mounted on the back of a dielectric lens. The planar antenna can take many forms; log-periodic spirals, twin slots, and twin dipoles (Goldsmith-1). The dielectric lens can be made of quartz or silicon and is plano convex, with the planar antenna mounted on the flat surface. The configuration most commonly used is the twin slot antenna fabricated on a silicon substrate and bonded to a silicon lens. An example of such a mixer is shown in Fig. 22–4 (Bin-1). The great advantage of planar antennas is that they can be fabricated photolithographically along with the mixing device (typically an SIS junction or Hot Electron Bolometer). This property makes planar antenna–dielectric lens beam launchers well suited for characterizing large numbers of experimental mixing devices. Their disadvantages compared to waveguide are that the microstrip tuning structures are intrinsically more lossy and their Gaussian beam coupling efficiencies are >=10% worse than what can be achieved with a corrugated or dual-mode feedhorn (Goldsmith-2). Also, the emergent beam size and direction are often sensitive to the placement of the planar antenna on the dielectric lens.

22–5 THz WAVEGUIDE STRUCTURES[1]（太赫频率的波导结构）

Waveguide-based mixers with feedhorns are relatively low-loss and are excellent beam launchers, with up to 98% of the emergent power in the fundamental Gaussian beam mode. The problem at THz frequencies is achieving the required high machining tolerances (a few

[1]By Professor Christopher Walker, Astronomy Department, University of Arizona.

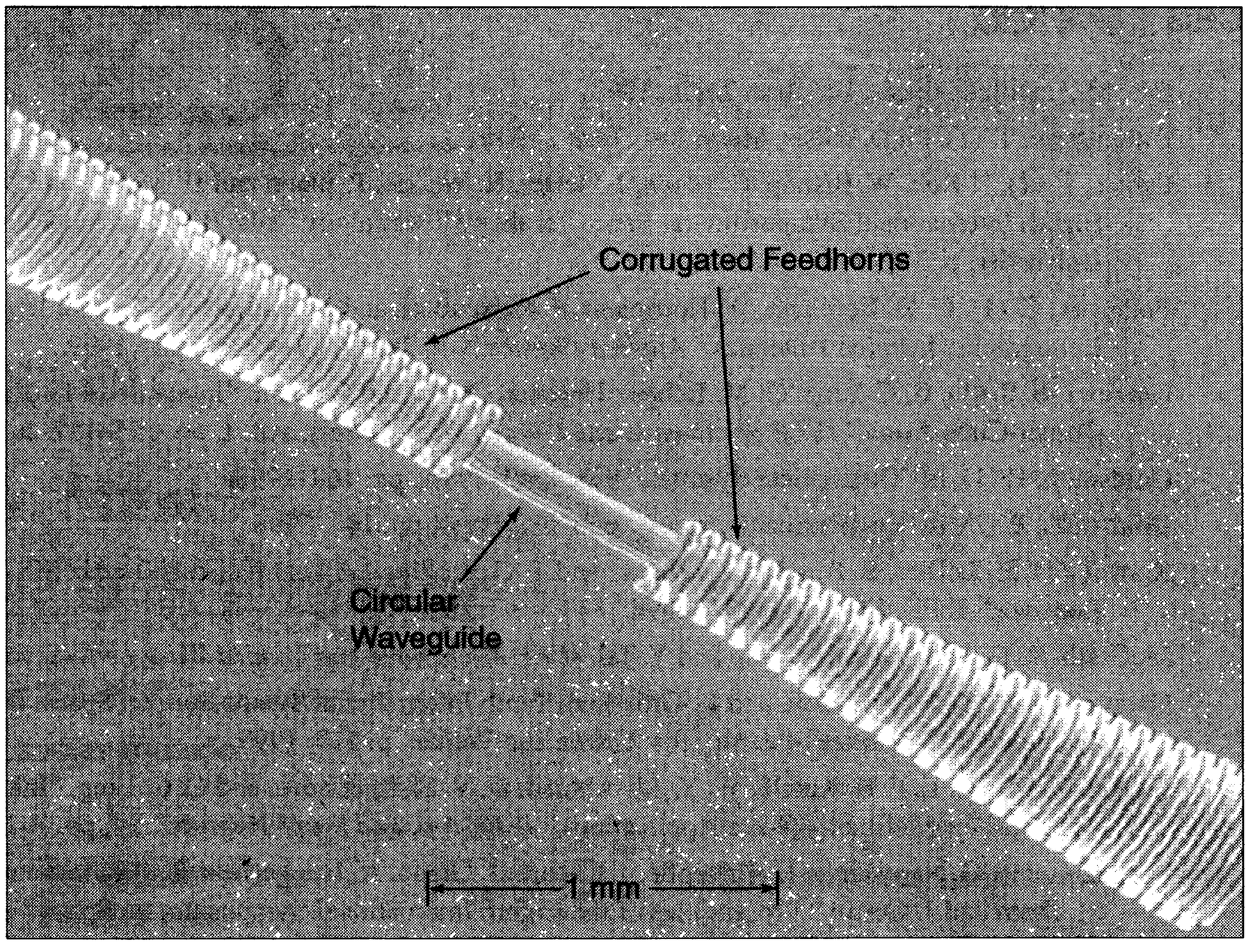


Figure 22–5
Laser micromachined corrugated feedhorns for 2 THz. The figure is an SEM image of one half of a split-block structure. The feedhorns are interconnected with a short section of circular waveguide. In laboratory tests one horn was used to sample the radiation field of a blackbody source. The second horn reradiated the energy onto a bolometer operating at 4K. By scanning the blackbody across the input horn aperture and recording the power detected from the bolometer, the beam pattern could be characterized (Walker-1).

microns) and integrating the mixing devices. High-quality milling machines (Narayanan-1) and several photolithographic techniques for etching silicon and photoresist have been developed to push the usefulness of waveguide into the THz regime (McGrath-1; Crowe-1). However, these techniques are often limited to producing relatively simple structures. One alternative technique is to use a galvo mirror steered laser beam to etch split-block waveguide structures into a silicon wafer (Walker-1; Bloomstein-1). The wafer is then coated with metal and assembled into waveguide devices, such as feedhorns. Figure 22–5 shows a pair of 2-THz feedhorns made with this technology. The excellent performance and repeatability of waveguide feedhorns make them an excellent choice for focal plane arrays. For such applications the mixing devices can be fabricated on a matrix of thin ($\sim$1 μm) dielectric membranes which can then be readily integrated into a feedhorn array. An 810-GHz mixer utilizing a membrane mounted SIS junction in a waveguide mount has been successfully built and tested (Kooi-1).

REFERENCES（参考文献）

Bin, M. (1), Ph.D. thesis, Cal. Inst. Tech., 1997.

Bloomstein, T. (1), Ph.D. thesis, Mass. Inst. Tech., 1996.

Crowe, T. (1), P. Koh, W. Bishop, C. Mann, J. Hesler, R. Weikle, P. Wood and D. Matheson, "Proceedings of the Eighth International Symposium on Space Terahertz Technology," eds. Blundell and Tong, Harvard University, p. 377, 1997.

Fumeaux, C. (1), G. D. Boreman, W. Herrmann, F. K. Neubuhi and H. Rothuizen, "Spatial Impulse Response of Lithographic Infrared Antennas," *Applied Physics,* vol. 38, pp. 37–46, January 1999.

Gearhart, S. S. (1), C. C. Ling, G. M. Rebeiz, Hermant Davee and G. Chin, "Integrated 119-μm Linear Corner-Cube Array," *IEEE Microwave and Guided Wave Letters,* vol. 1, pp. 155–157, July 1991.

Goldsmith, P. (1), in "Quaioptical Systems," pub. IEEE Press, pp. 169, 1998.

Goldsmith, P. (2), in "Quaioptical Systems," pub. IEE Press, pp. 184, 1998.

Kooi, J. (1), H. LeDuc, B. Bumble, C. Walker and T. G. Phillips, "A Low Noise 850-GHz SIS Waveguide Receiver," IEEE, MMT, **46,** 151, 1998.

McGrath, R. (1), C. Walker, M. Yap and Y. Tai, *IEEE Microwave and Guided Wave Letters,* vol. 3, p. 61, 1993.

Narayanan, G. (1), N. Erickson, and R. Grosslein, Tenth International Symposium on Space Terahertz Technology, Charlottesville, Va., eds. Crowe and Weikle, p. 519, 1999.

Rebeiz, G. M. (1), L. F. B. Katehi, W. Y. Ali-Ahmad, G. V. Eleftheriadws and C. C. Ling, "Integrated Horn Antennas for Millimeter-Wave Applications," *IEEE Ant. and Prop. Mag.,* vol. 34, pp. 7–16, February 1992.

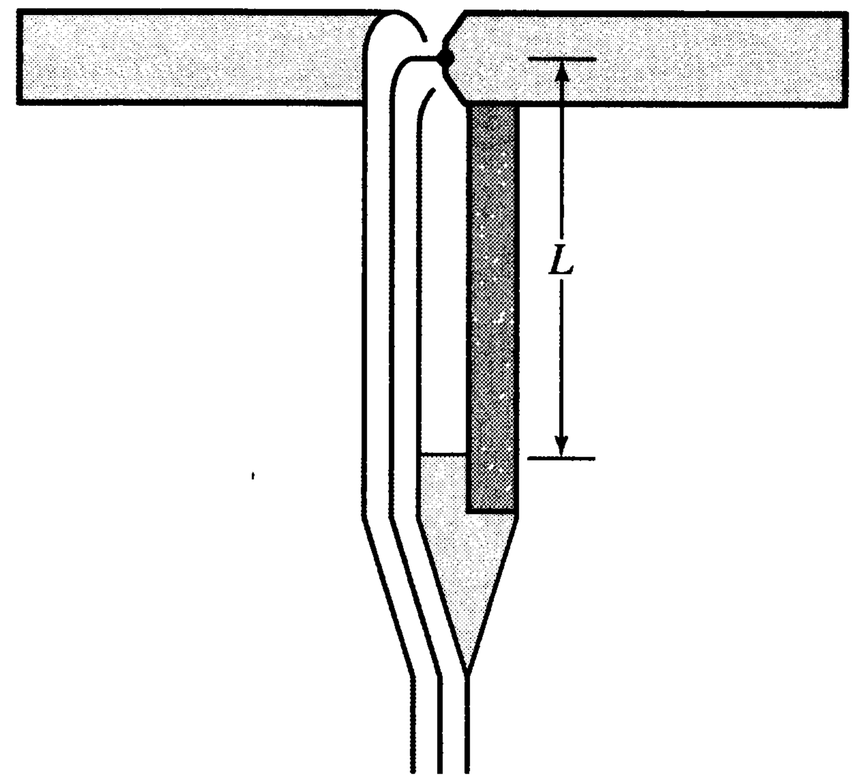
Walker, C. (1), G. Narayanan, H. Knoepfle, J. Capara, J. Glenn, A. Hungerford, T. Bloomstein, S. Palmacci, M. Stern and J. Curtin, "Proceedings of the Eighth International Symposium on Space Terahertz Technology," eds. Blumdell and Tong, Harvard University, p. 358, 1997.

CHAPTER 23

Baluns, etc.

By Ben A. Munk[1]

平衡 – 非平衡转换器等



Topics in this chapter include:

- Balun types: I, II, III and chokes (I型，II型，III型和扼流式巴仑)
- Bandwidth（频带宽度）
- Sleeve-dipole baluns（套筒 – 偶极子天线）
- Modifications of Type II baluns（II型巴仑的变种）
- Mast baluns for dipoles and loops (偶极子和环用的支杆式巴仑)
- Printed baluns（印刷巴仑）
- Bypass balun（旁路式巴仑）
- Balanced transformer（平衡变换器）
- Cutaway baluns（截割式巴仑）
- Natural baluns（天然巴仑）
- Folded dipole to J-match (折合偶极子到J型匹配)
- Matching stubs（匹配短截线）
- Traps（陷波器）

23–1 INTRODUCTION（引言）

The term *balun* is an abbreviation of the words *bal*ance and *un*balance. It is a device that connects a balanced two-conductor line to an unbalanced coaxial line.

Since baluns add complexity and expense to a system, let us consider the consequences of not using one. For example, in Fig. 23–1*a* a horizontal dipole antenna is center fed directly from a coaxial cable. The inner conductor feeds the left half of the dipole while the outer conductor feeds the right half. However, current I_3 will flow down the *outside* of the outer conductor making it part of the radiating system. The result is a mixture of horizontal polarization as in Fig. 23–1*b* and vertical *and* horizontal polarization as in Fig. 23–1*c*. Thus, the pattern is not that of a horizontal dipole. For this, a balun is required. However, in receiving situations where there is an adequate signal-to-noise ratio (SNR) a balun may not be needed.

23–2 BALUN TYPES I, II, AND III AND CHOKE BALUNS (I型，II型，III型和扼流式巴仑)

Figure 23–2 shows five common types of baluns. After Nelson and Stavis (1) the baluns in Fig. 23–2*a*, *b* and *c* are designated Types I, II and III. Type I in (*a*) has a $\lambda/4$ sleeve (shorted at the base) which presents an infinite impedance (ideally) at the top. Type II

[1]ElectroScience Laboratory, Ohio State University, Columbus.

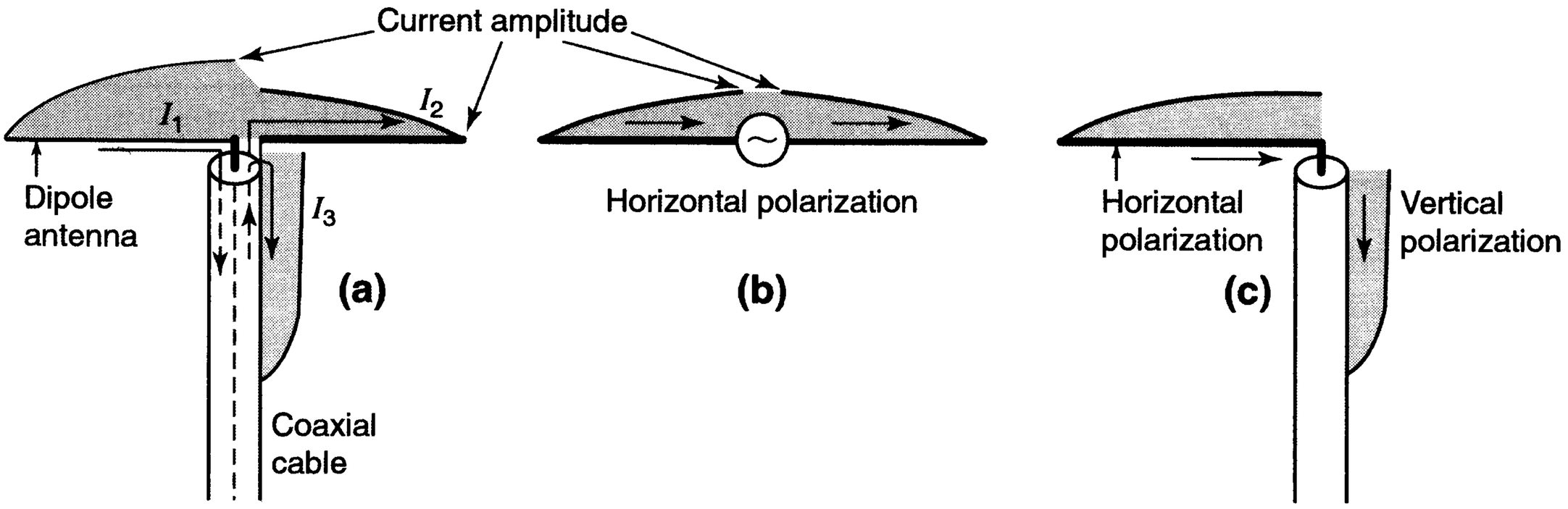
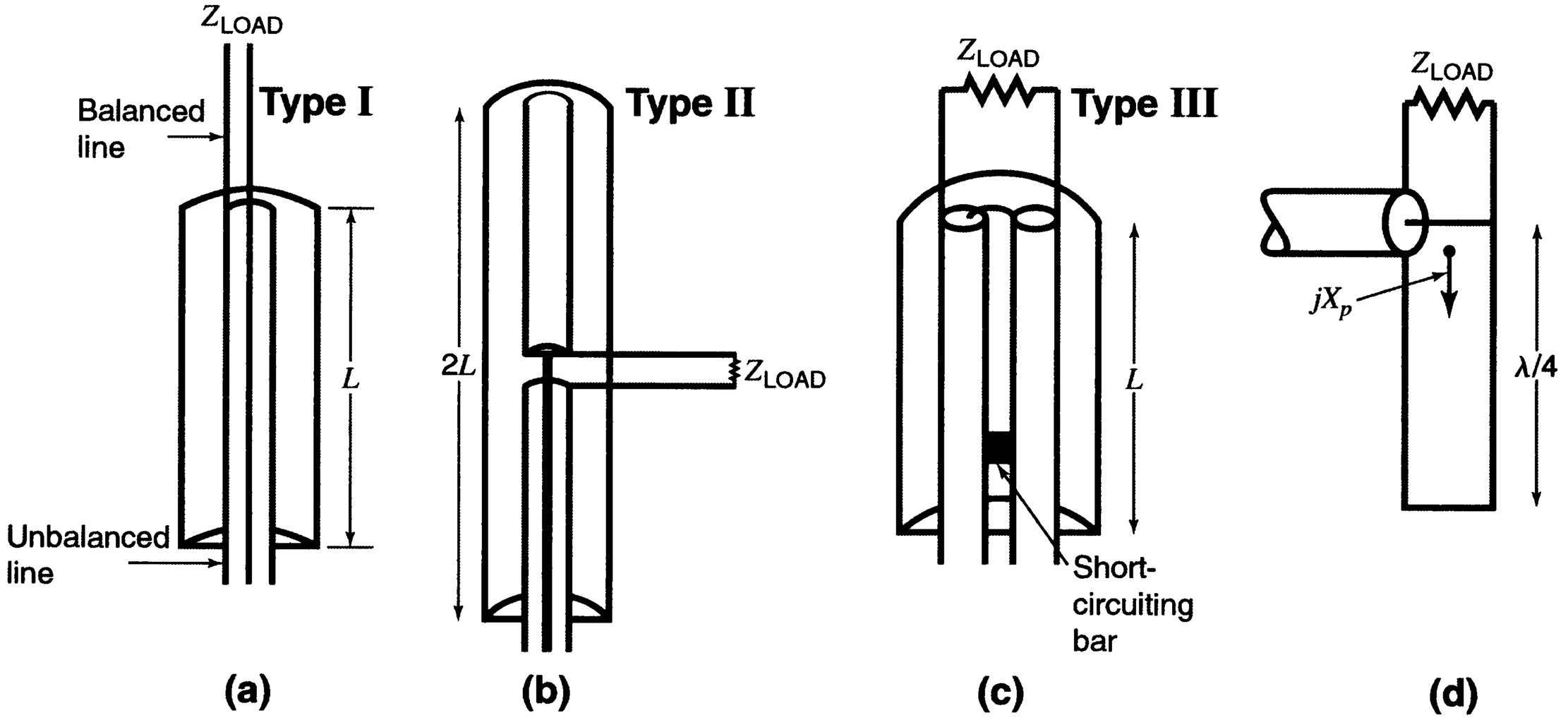


Figure 23–1
A horizontal dipole fed directly from a coaxial line as in (*a*) produces a mixture of balanced horizontal polarization (*b*) and of vertical *and* horizontal polarization (*c*).



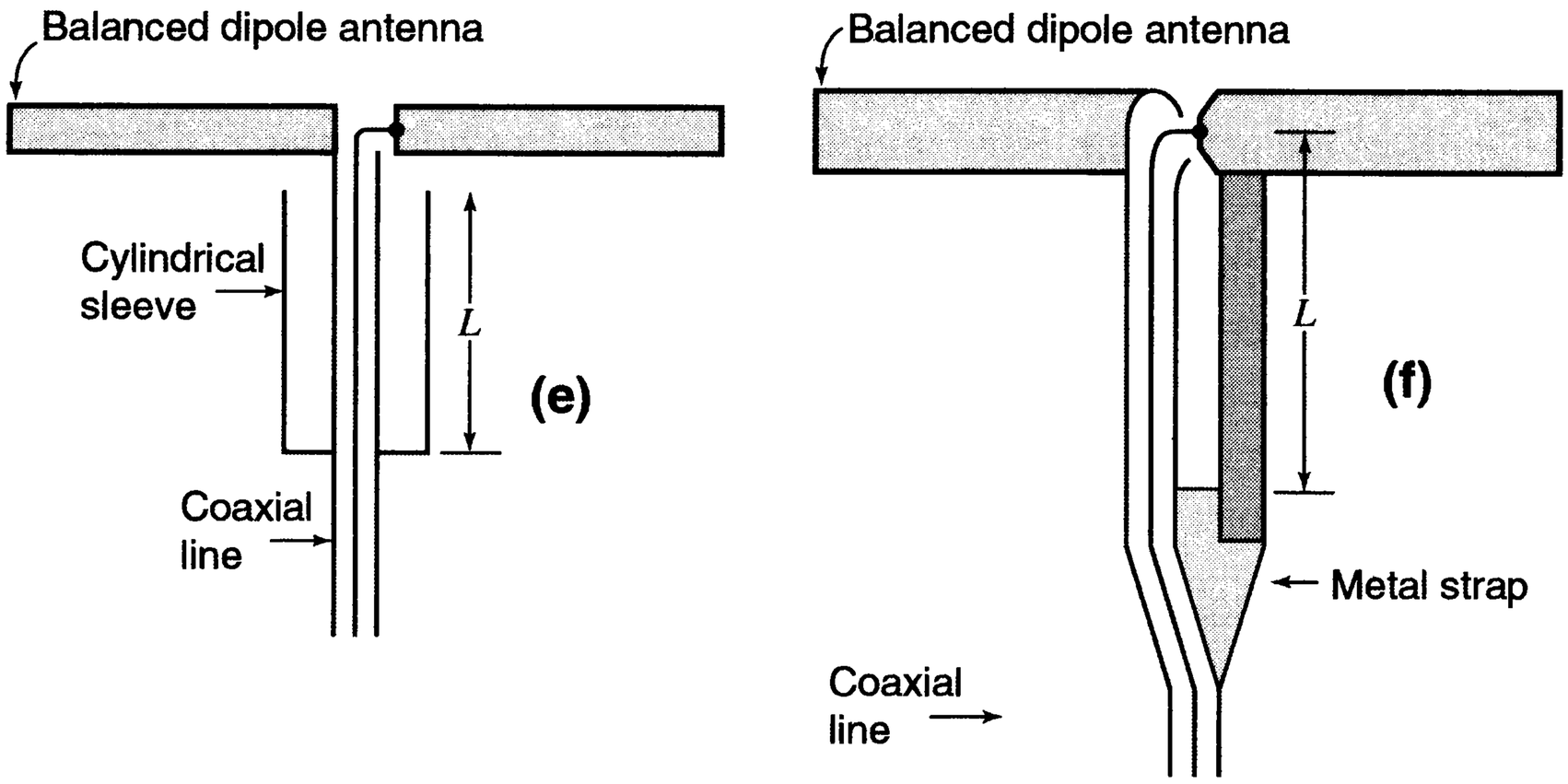


Figure 23–2
(*a*) Type I balun or "bazooka," (*b*) Type II balun, (*c*) Type III balun, (*d*) Type III balun equivalent circuit, (*e*) Type I balun with dipole antenna and (*f*) dipole antenna with Type III balun minus sleeve. [*Figures (a), (b) and (c) adapted from Nelson (1).*]

in (*b*) has two Type I's in series providing more bandwidth and load balance at all frequencies. Type III in (*c*) is a more compact form. The inner conductors form a two conductor $\lambda/4$ line shorted at the base and presenting an infinite impedance at the top. It also features a sliding short-circuit bar for frequency adjustment. See equivalent circuit as shown in (*d*). Figure 23–2*e* has a Type I balun with dipole. Figure 23–2*f* has a dipole fed by a Type III balun minus shielding cavity. The length L of all these baluns is about $\lambda/4$ at the center frequency. In (*e*) and (*f*) a reactive impedance $Z = \pm jZ_0 \tan \beta L$ appears in parallel with the dipole, where Z_0 = characteristic impedance of the balun line of length L.

Two more baluns are shown in Fig. 23–3. These are choke types. The one in Fig. 23–3*a* has the coaxial cable wound into a coil producing a high impedance on the outside of the coil. The coil and its capacitance C form a parallel LC circuit that should resonate at the operating frequency.

A ferrite-bead choke is shown in Fig. 23–3*b* with cylindrical ferrite beads placed on the outside of the coaxial cable. With good-quality ferrite beads large bandwidths may be obtained (an octave or more).

The Type III balun works on the principle that the voltage at point A in Fig. 23–4*a* is zero. Therefore, a cable leaving at this point has zero voltage on its outside. Thus, with the coaxial line connected as in Fig. 23–4*b* balance is maintained. To facilitate obtaining

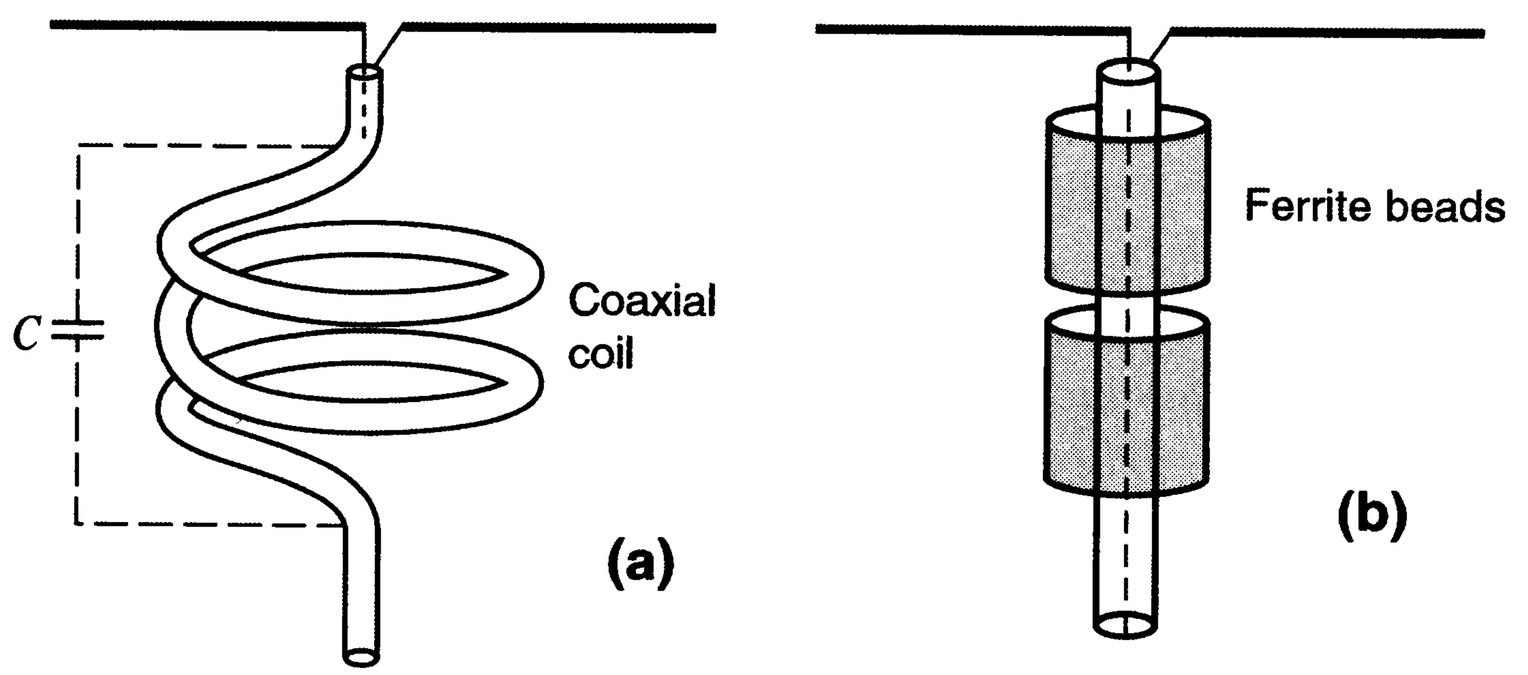


Figure 23–3
Two types of choke balun.

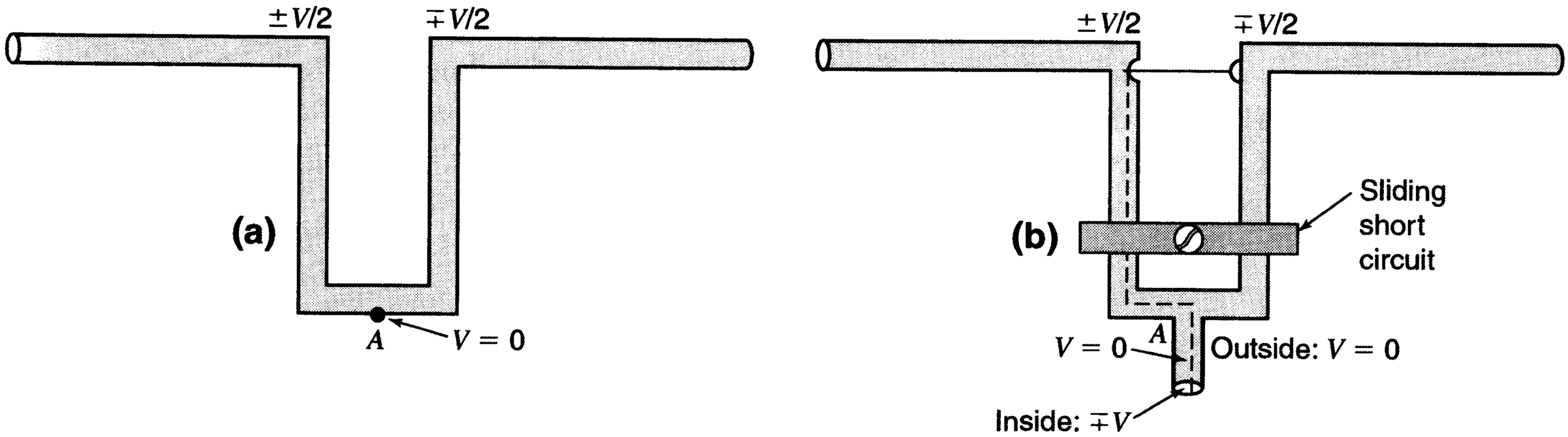


Figure 23–4
(*a*) Basic structure of Type III balun and (*b*) the balun with coaxial line connected.

an infinite impedance across the terminals at the dipole a sliding short may be used as suggested in Fig. 23–4*b*.

23–3 BANDWIDTH（频带宽度）

From the discussion above one might think that the Type III balun is inherently narrow-banded. However, this is not quite so. First of all, because of symmetry, as discussed earlier, a balanced voltage will always result in no current on the outside of the coaxial cable. In other words, it will balance perfectly from DC to terahertz (at least in principle).

The limitation in bandwidth comes from the shunt impedance jX_p (see Fig. 23–2*d*). Since it is purely reactive the parallel combination $Z_A \| jX_p$ will, if plotted in a Smith Chart, be located on a circle going through Z_A and the point (0, 0) as shown in the Fig. 23–5*a*, where Z_A, in this example, is real. At the frequency where $\beta l_p = \pi/2$ the parallel combination $Z_A \| jX_p$ will reduce to Z_A. At lower frequencies jX_p will be inductive and $Z_A \| jX_p$ will consequently be located in the upper (inductive) part of the Smith Chart; Prob. 23–3–1 illustrates how much as a function of frequency. Further, as illustrated in Prob. 23–3–2, if Z_A is a half-wave dipole, where Z_A is capacitive at frequencies below resonance, the addition of the inductive component jX_p can somewhat improve the matching of Z_A to the balun.

The ultimate limitation in bandwidth is realized when $\beta l_p = \pi$. In that case $jX_p = 0\,\Omega$ and Z_A is consequently completely shorted out. However, it is incorrect to assume that the bandwidth is limited to merely ~2 : 1. Typically we may go as low in frequency as $\beta l_p \sim \pi/4$ and as high in frequency as $\beta l_p \sim \frac{3}{4}\pi$. That gives a frequency range of about 3 : 1 or more, all depending on the tolerance for mismatch. See Prob. 23–3–1.

More bandwidth can be obtained by passing the inner conductor through a hole and continuing it inside the right part of the balun as shown in Fig. 23–5*b* (see big arrow). The electrical length l_s inside this arm is $\lambda/4$ at the center frequency and it is terminated in an open circuit. This modification was made by Roberts (1) in 1957.

Let the characteristic of the stub inside the right arm $= Z_{cs}$ and its length $= l_s$. Note that it is terminated in an open circuit; thus, the input impedance is

$$jX_s = jZ_{cs} \cot \beta_s l_s$$

where β_s = propagation constant inside the stub.

Inspection of the equivalent circuit in Fig. 23–5*b* left shows that the parallel combination $Z_A \| jX_p$ is the same as above, as indicated in the Smith Chart, right. Adding the impedance jX_s in series gives circles going through $Z_A \| jX_p$ and we can follow further along the circles to a desired point.

The increase in bandwidth results from by the fact that jX_p and jX_s always have opposite signs (if designed correctly). At lower frequencies, for example, jX_p is inductive, i.e., $Z_A \| jX_p$ will be located in the upper half of the Smith Chart. In contrast, jX_s is capacitive, i.e., it will push $Z_A \| jX_p$ downward toward the lower half of the Smith Chart as indicated in Fig. 23–5*b* right. At higher frequencies the opposite is true while no compensation of Z_A is observed at the center frequency. The net result is that the total impedance observed

How to get more bandwidth

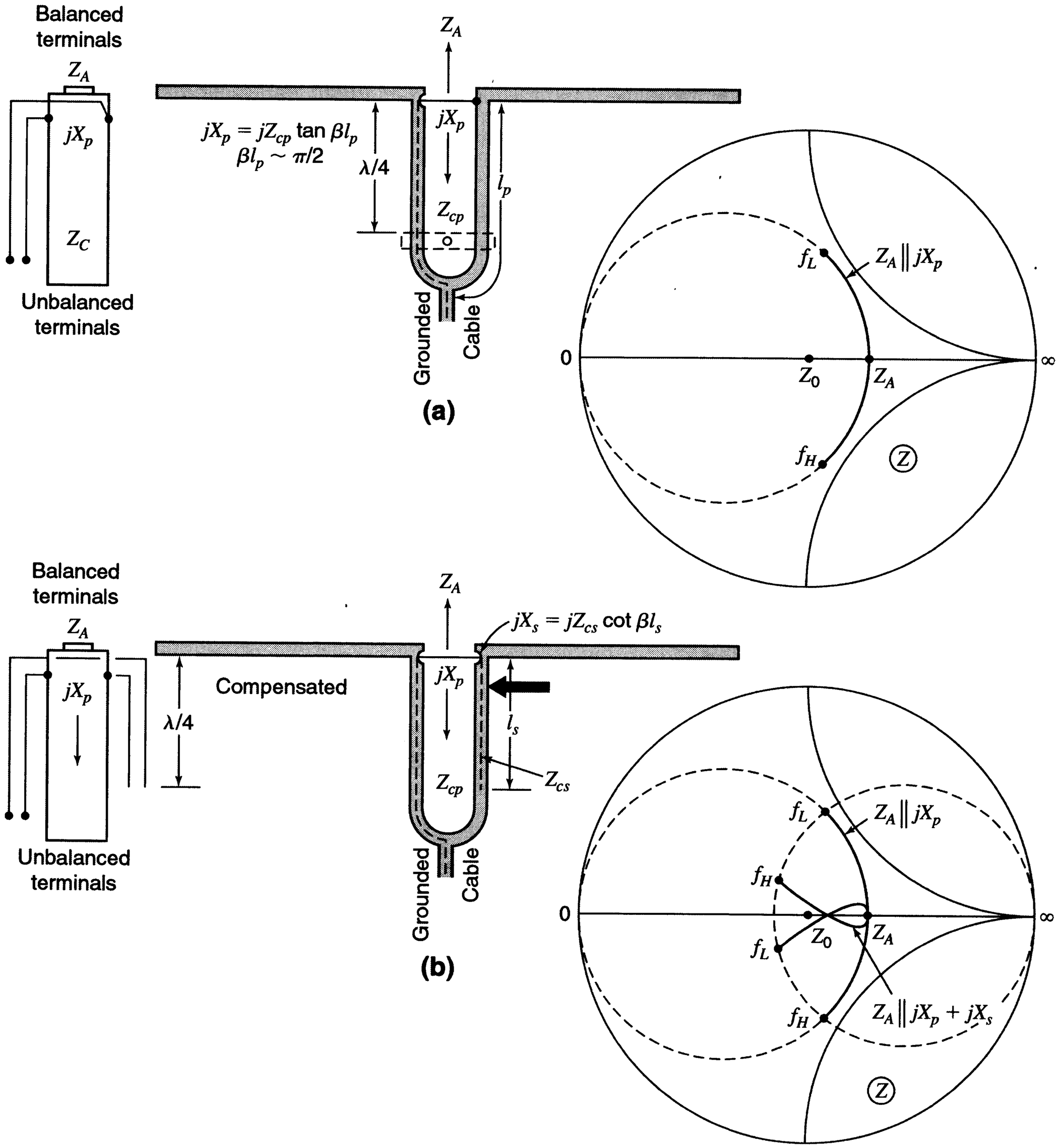


Figure 23–5
(*a*) The antenna impedance Z_A in parallel with the fork impedance jX_p (see equivalent circuit, left, and Smith Chart, right). (*b*) The series reactance jX_s counteracts the parallel impedance jX_p as a function of frequency; see Smith Chart to right.

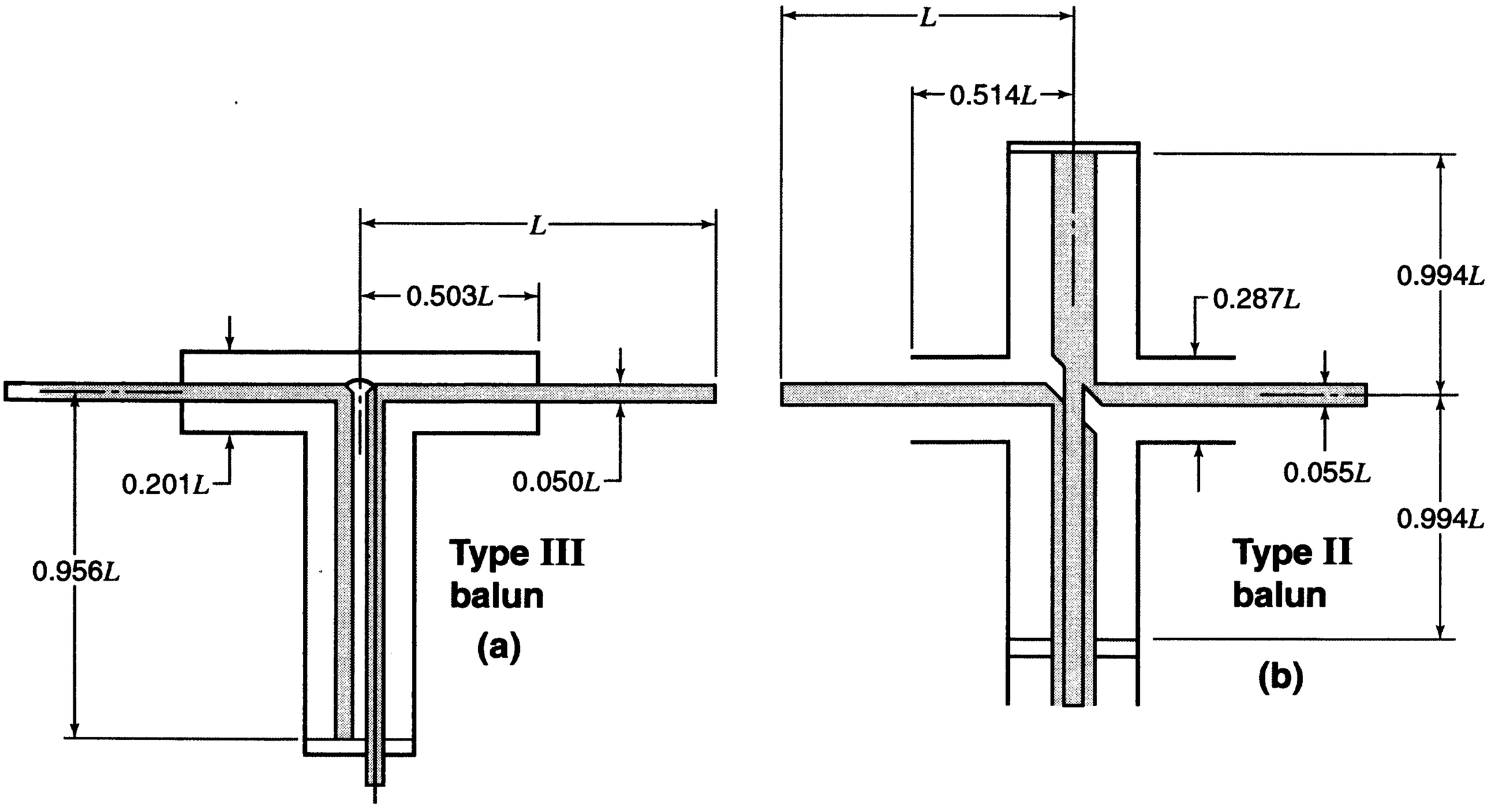


Figure 23–6
(*a*) Cylindrical sleeve dipole with Type III balun and (*b*) cylindrical sleeve dipole with Type II balun. (*Adapted from Bock-1.*)

at the top of the balun is closer to the original impedance Z_A, i.e., we obtain broadband compensation. See Prob. 23–3–3.

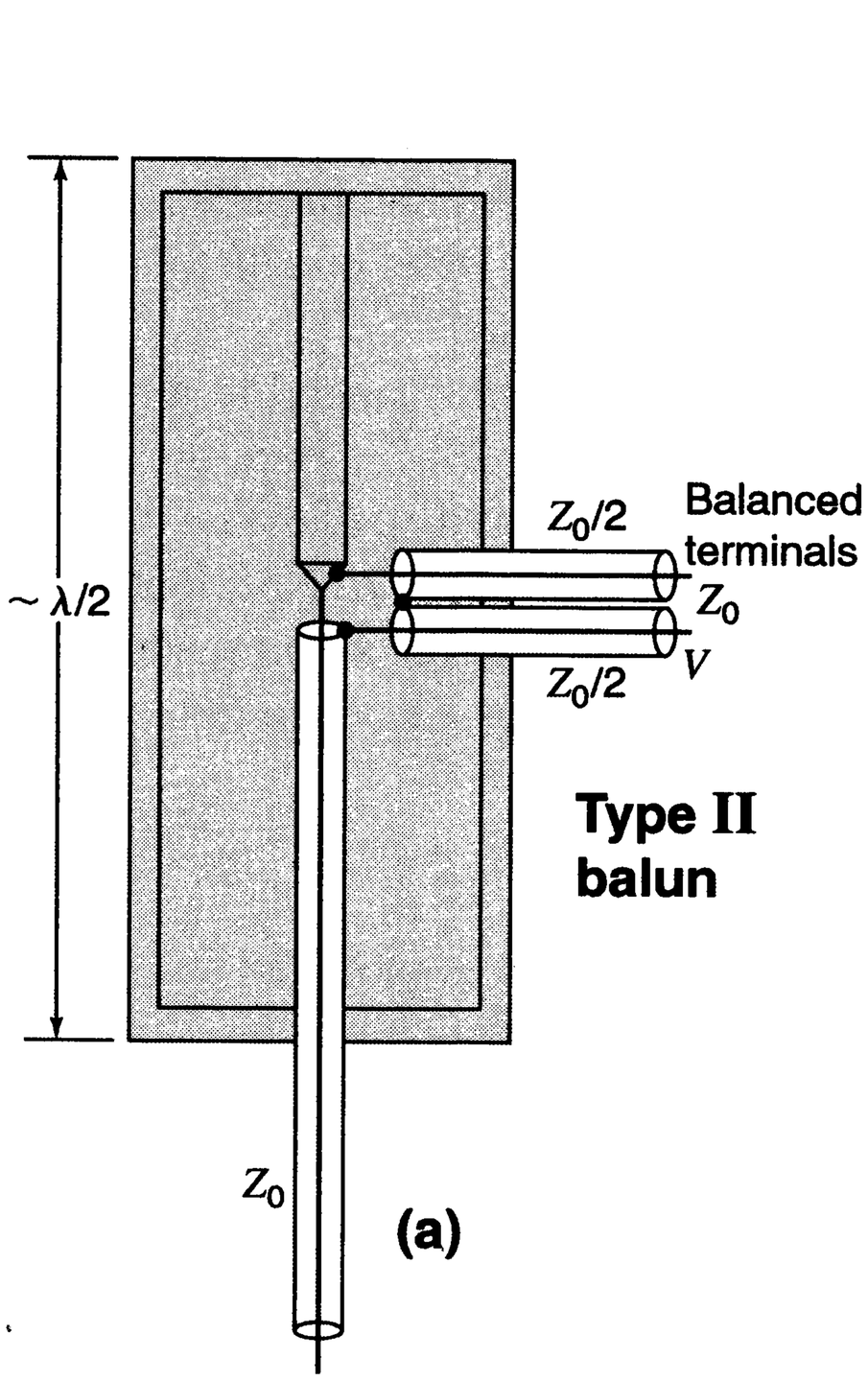
23–4a SLEEVE-DIPOLE BALUNS（套筒－偶极子式巴仑）

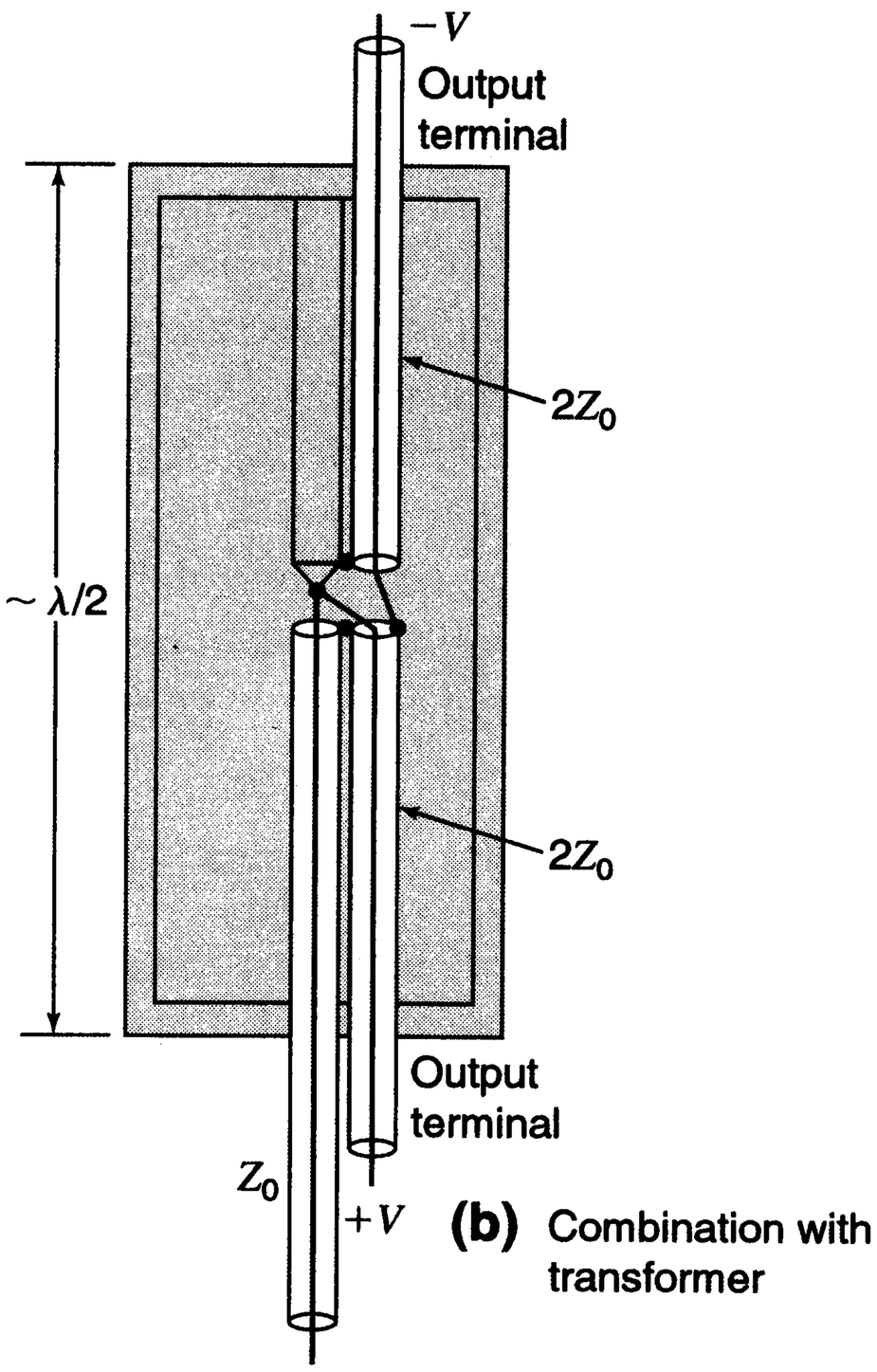
Two examples are shown in Fig. 23–6 of baluns incorporated as part of the antenna. The balun in Fig. 23–6*a* is a Type III while the one in (*b*) is a Type II. The length L is about $\lambda/4$ at the center frequency.

23–4b MODIFICATIONS（变种结构）

The Type III balun has some very desirable features, namely, it always provides a perfect balance and a wide bandwidth. However, it can also be modified in several ways and provide transitions to other types of baluns. Although these could be explained by a direct approach, it is helpful to tie them in with the Type II balun. Also, it will often enable the designer to create a unique type. As a first example Fig. 23–6*a* shows a modification where the single balun at Fig. 23–3 has been replaced by two arranged in series. This doubles the shunt reactance seen at the balanced port and should thus increase the bandwidth.

To obtain a large bandwidth requires that the shunt reactance jX_p of the balun be as large as possible. Quite analogous, the bandwidth of the balun types in Fig. 23–7*a* and *b*





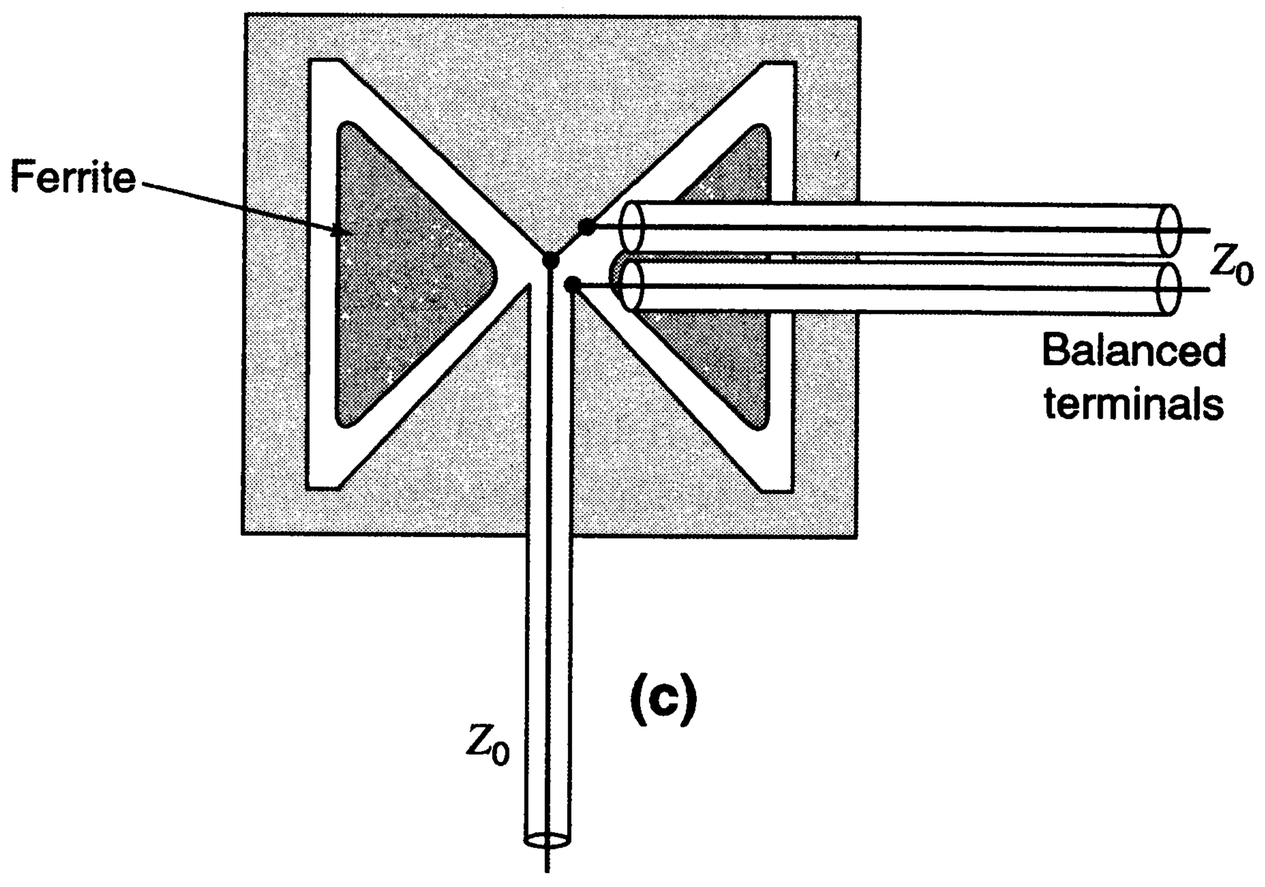


Figure 23–7
(*a*) A Type II balun with two coaxial lines. (*b*) A variation of the balun in (*a*) but with a transformer. (*c*) A variation of the balun in (*a*) where the cavity has been reshaped and filled with ferrite to obtain greater bandwidth.

is determined by the sum of the two coaxial stubs made up of the inside of the enclosure and the outer conductor of the feed cable. However, the characteristic impedance of coaxial lines is smaller than for a two-wire transmission line. This could be remedied by increasing the diameter of the enclosure. However, the characteristic impedance of a coaxial line is a logarithmic function of the ratio between inner and outer conductors, so the benefit of this approach is modest and the enclosure becomes bulky.

Alternatively, the material in the cavity may be changed to a material with a higher intrinsic impedance. An example is shown in Fig. 23–7*c*. Here the cavity has been filled with ferrite with $\mu > \mu_0$. Unfortunately, the permittivity ε of the ferrite will, in general, also be larger than ε_0, which limits the effect of the increase in μ. Note that, the two cavities have been modified to a conical shape. This can increase the shunt reactance and improve the bandwidth. Note also that an increase in μ as well as ε leads to reduced cavity dimensions.

23–5 MAST BALUN（支杆式巴仑）

A Type II balun modification is shown in Fig. 23–8 as mounted on a groundplane with the enclosure removed. The groundplane may be the large conducting cylinder of a mast, making it a "mast balun." The shunt impedance jX_s is considerably higher than for the coaxial versions shown in Fig. 23–7*c*, so we can obtain a larger bandwidth. If a series reactance in the form of an open-ended quarter-wave stub is added additional broadband compensation is obtained precisely in the same manner, as was the case with the compensated Type III balun, shown in Fig. 23–5*b*. This type of design can lead to bandwidth exceeding 4 : 1. Note also the balanced output connected to two pairs of balanced lines in parallel. Each balanced line has two coaxial cables with characteristic impedance Z_0 in series, i.e., total impedance is $2Z_0$. When the left and right sides are connected in parallel as shown, we obtain a total impedance of Z_0. Thus, we have a perfect match at all frequencies.

Note that voltage is present between the balanced output and the mast. Thus, in case of snow or ice the balun needs a dielectric cover or radome.

All the balun types shown in Figs. 23–6, 23–7 and 23–8 are characterized by having the shunt reactance jX_s made up of two reactances in series, namely, one from the top and one from the bottom part. If the two halves are of a little different dimensions they may resonate at somewhat different frequencies. Thus, somewhere between these two resonant frequencies the reactance of one half will be as inductive as the other half is capacitive. Since they are in series they will add up to zero (assuming no losses). This makes the balun unusable at that frequency.

Note that the length of the open-ended stubs used in the balun in Fig. 23–5*b*, and in the mast balun in Fig. 23–8, should be cut approximately 4 to 10% shorter than their mechanical length depending on the diameter. This will compensate for the end capacity of the open-end transmission line.

23–6 PRINTED BALUNS（印刷巴仑）

Figure 23–9*a* shows a printed balun and dipole fed by a coaxial cable. In Fig. 23–9*b*, the situation is basically the same except that the feedline is also printed and an open-ended

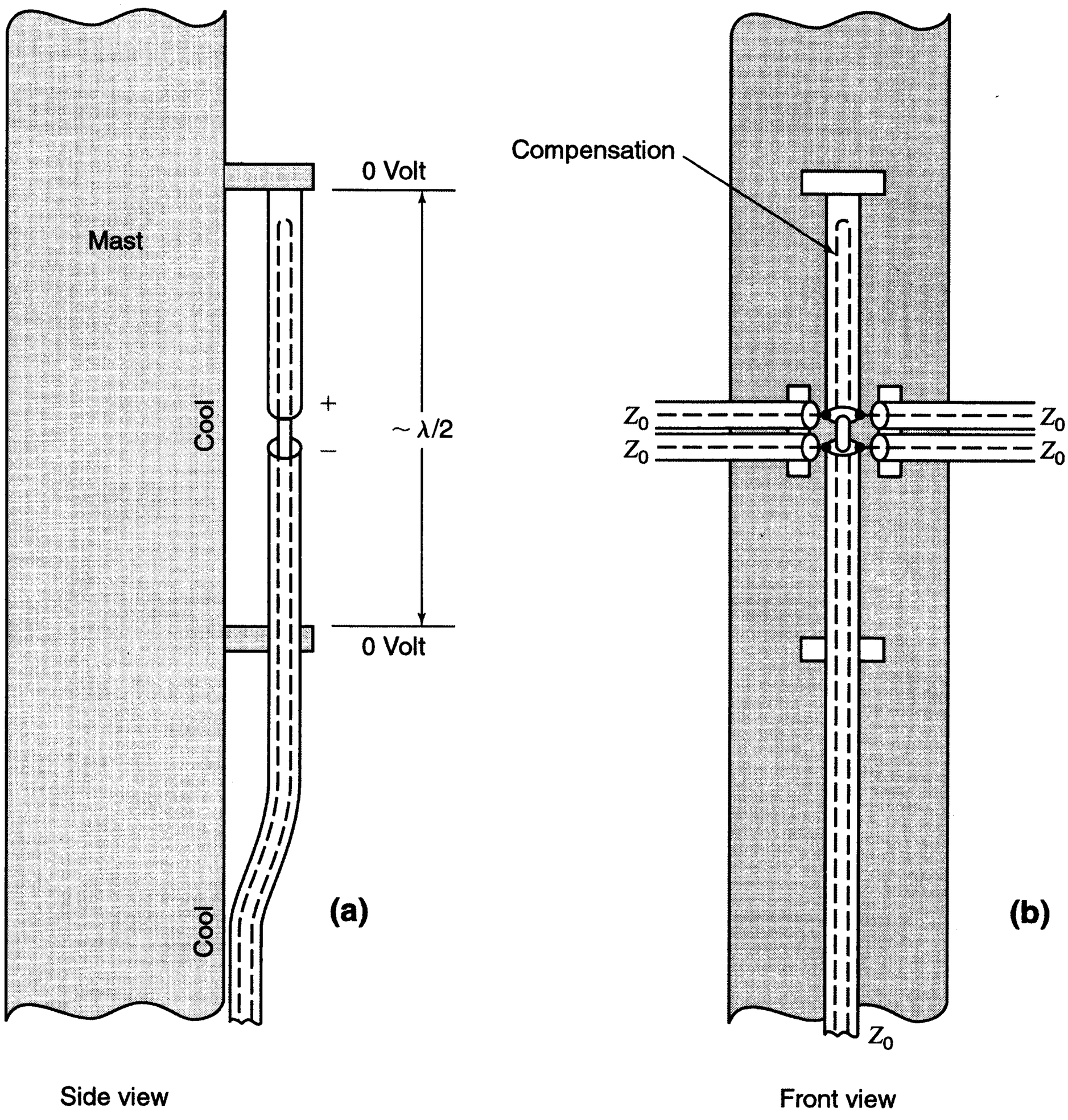


Figure 23–8
The cavity in Fig. 23–7*a* has been essentially removed and substituted by the mast as shown in (*b*). The output cables consist of two pairs of series-connected coaxial lines in parallel yielding the total impedance Z_0 needed for broadband match (>4 : 1).

compensation stub has been added similar to Fig. 23–5*b*. Note that the inner conductor between the dipole gap at the center has been reduced. This suppresses radiation from the feed line.

Finally, Fig. 23–9*c* has a printed version of what is basically the mast balun shown in Fig. 23–8.

Printed circuit version

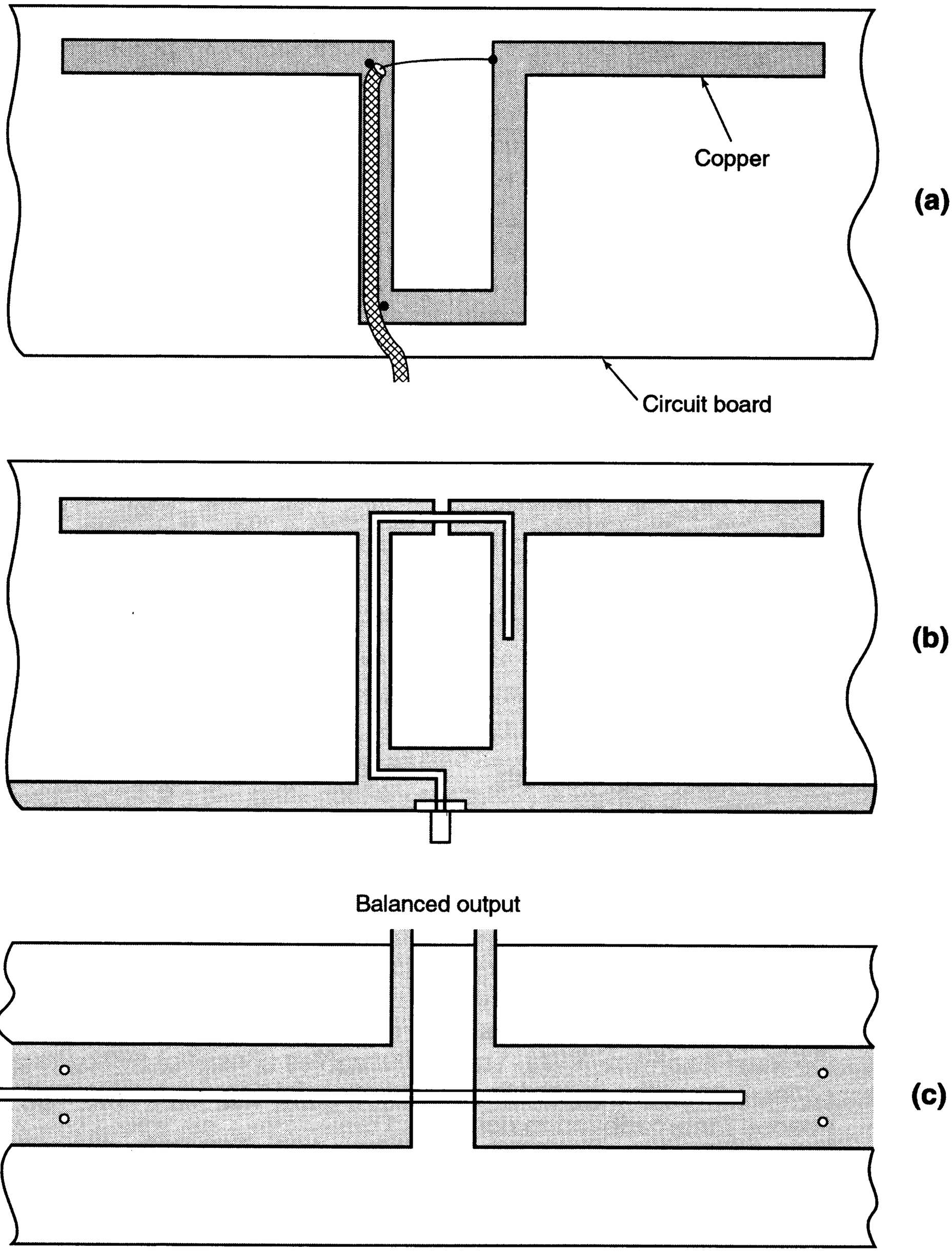


Figure 23–9
Printed versions of various baluns: (*a*) Type III balun and dipole fed by coaxial cable. (*b*) Same as (*a*) but fed by microstrip line and with reactive compensation (*Roberts-1*). (*c*) Printed version of Type II balun. This is covered with a ground plane.

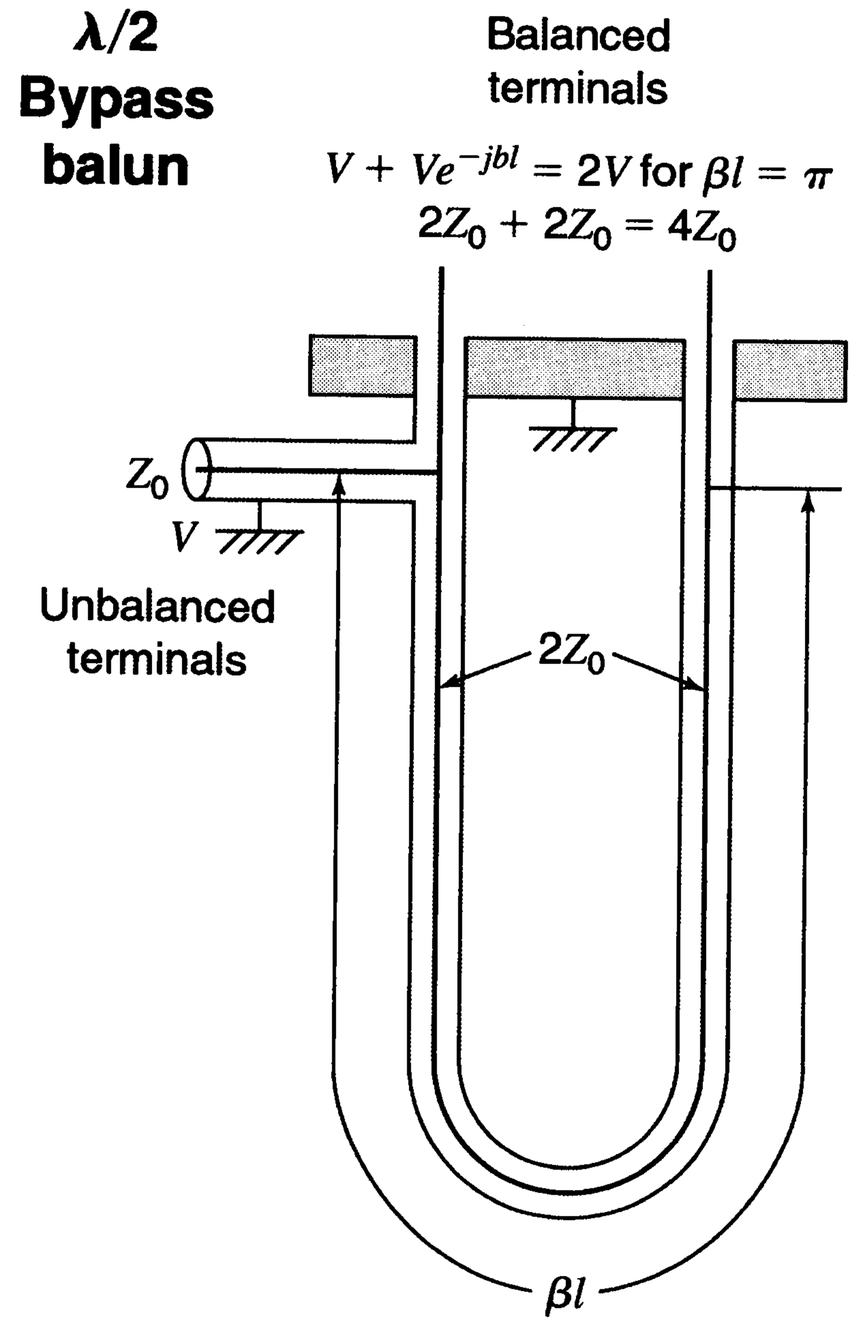


Figure 23–10
The $\lambda/2$ bypass balun. The input cable with characteristic impedance Z_0 splits up into two cables in parallel with characteristic impedance $2Z_0$. One cable is $\lambda/2$ longer than the other producing a balanced voltage difference of $2V$. It is balanced for $\beta l = \pi$ and quadruples the impedance.

23–7 THE $\lambda/2$ BYPASS BALUN[1] (半波长旁路式巴仑)

The balun types considered so far have with the exception of the one in Fig. 23–7*b* merely provide a transition from the balanced to the unbalanced terminals without any impedance transformation. Baluns can also provide impedance transformation.

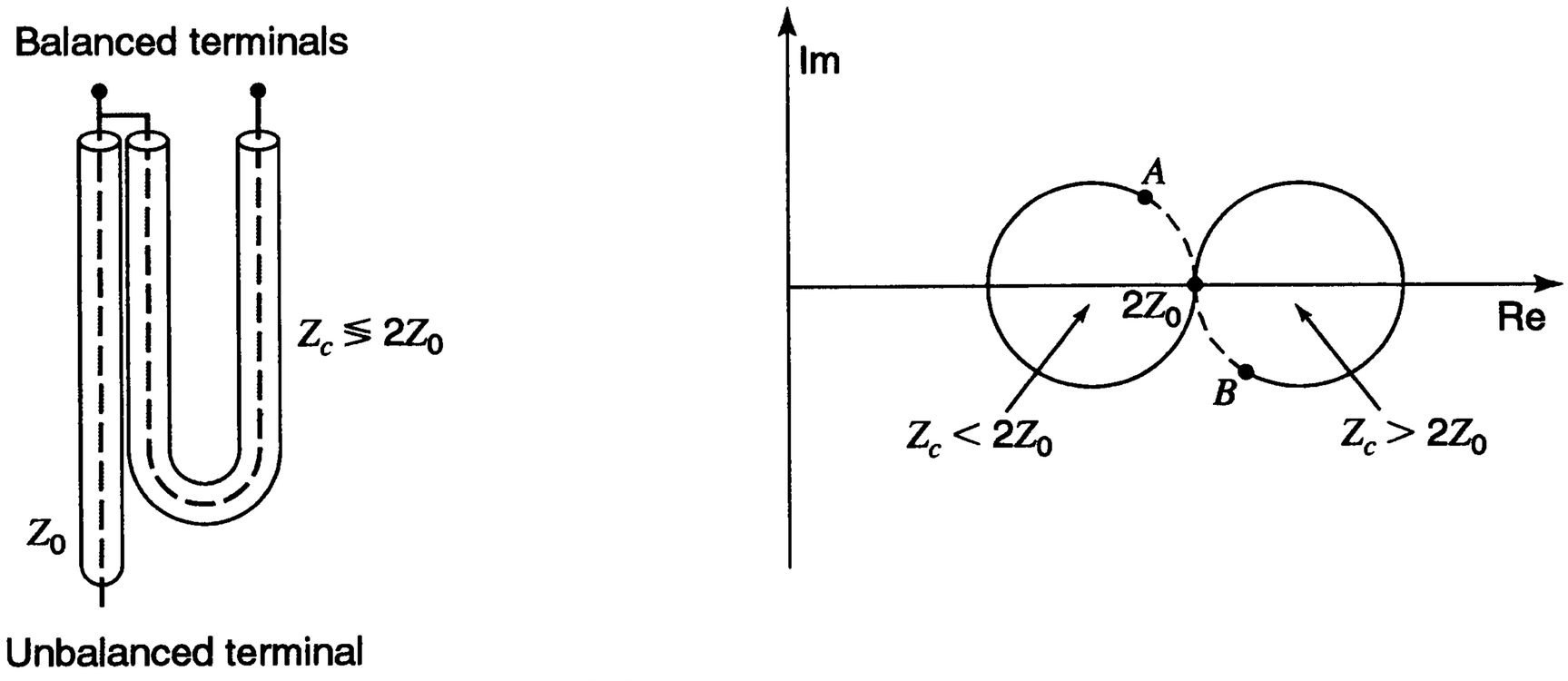
An example is shown in Fig. 23–10. From the unbalanced terminals a coaxial cable with characteristic impedance Z_0 branches out into two coaxial cables with characteristic impedance $2Z_0$. One of these cables is $\lambda/2$ longer than the other. Thus, the voltages will be 180° out of phase with the other cable and the total voltage difference between the two cables will be $2V$; i.e., the impedance change will be 4 : 1. Thus, if the balanced terminals are loaded with a resistance equal to $4Z_0$ the two branch cables are effectively terminated with $2Z_0$, i.e., the input impedance of each is equal to $2Z_0$ and their parallel combination at the branch-point is then equal to Z_0 as required for perfect matching.

The $\lambda/2$-bypass balun is only perfectly balanced when the bypass cable is exactly $\lambda/2$ long. At lower frequencies the voltage through the bypass cable is delayed less than 180° and vice versa at the higher frequencies. Fortunately, this can be compensated to some extent. See Fig. 23–11.

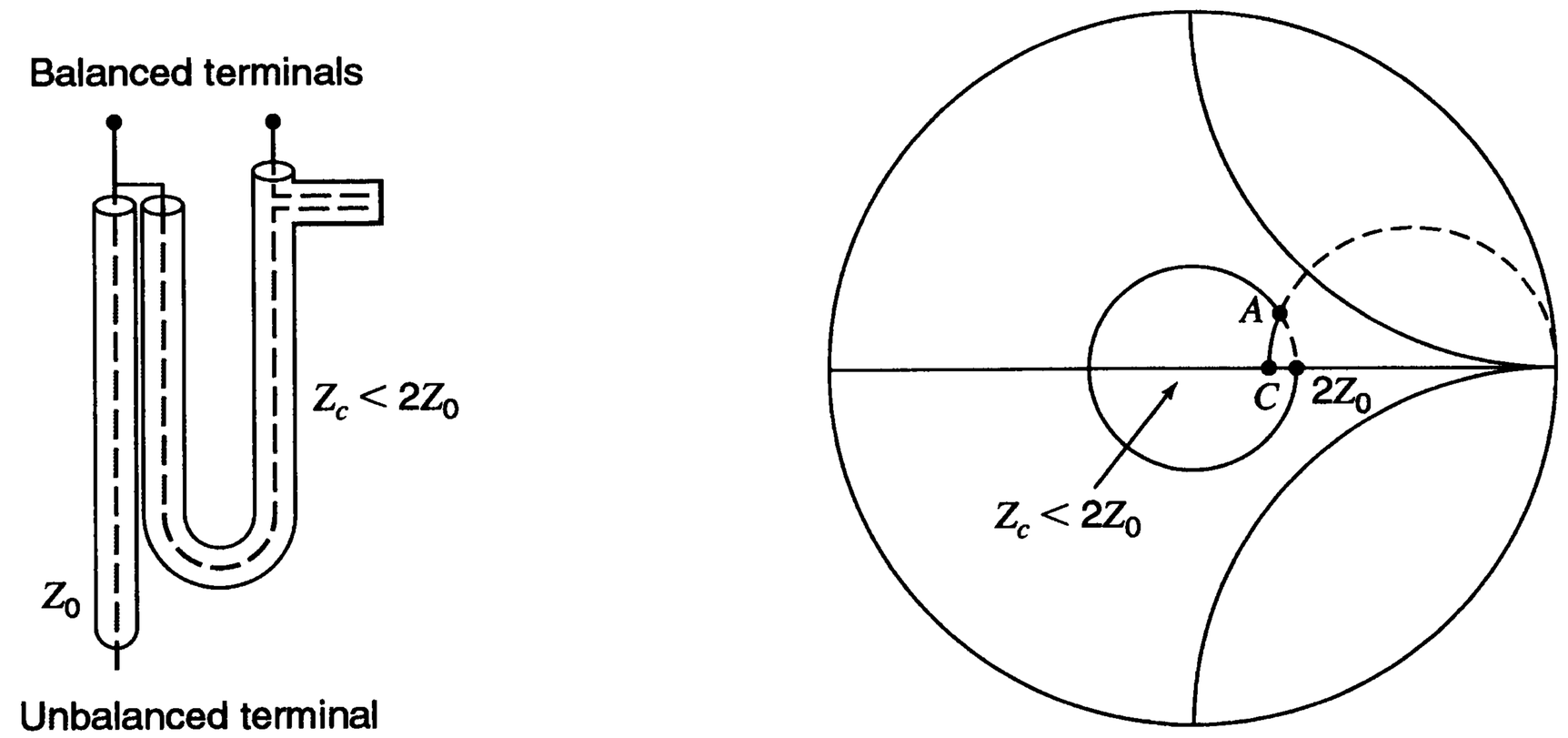
In Fig. 23–11*a* the voltage at the output terminal of the bypass cable is shown for two values of its characteristic impedance, namely, for $Z_c < 2Z_0$ and for $Z_c > 2Z_0$. In the first case, we arrive at point A located in the inductive part of the complex plane and in the

[1]This type of balun was patented in Germany in 1931 (Walter-1). It was called a $\lambda/2$ Umvegleitung.

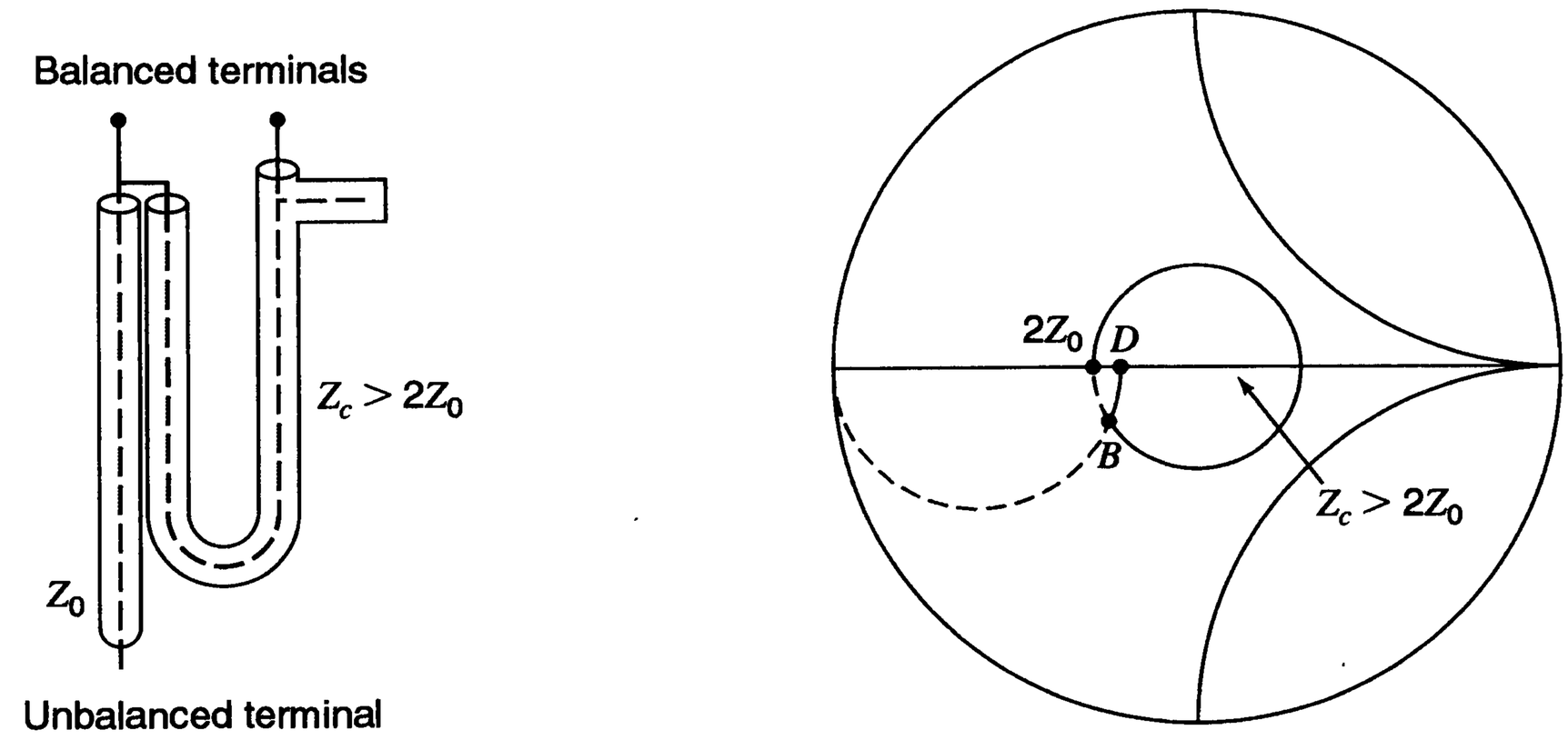
Bypass baluns



(a)



(b)



(c)

Figure 23–11
(*a*) At frequencies below the center frequency the delay on the bypass cable will be less than 180°. For $Z_c < 2Z_0$ it will be in the inductive part of the plane and for $Z_c > 2Z_0$ in the capacitive part. (*b*) For $Z_c < 2Z_0$ adding an open-ended $\sim\lambda/4$ stub increases the delay to ~180°. (*c*) For $Z_c > 2Z_0$ parallel to a short-circuited $\lambda/4$ stub is added to increase the delay to 180°.

second case at point B located in the capacitive half-plane of the Smith Chart. As shown in Fig. 23–11*b* we now for $Z_c < 2Z_0$ add in series the impedance of an open-ended $\lambda/4$ stub (now shown in the Smith Chart). At the lower frequencies this stub impedance is capacitive, i.e., drives the impedance at A downward to point C. This simply corresponds to a further delay and can by proper design be equal to 180° at some of the lower frequencies.

Similarly, as shown in Fig. 23–11*c* for the case $Z_c > 2Z_0$ a short-circuited stub impedance is added in parallel. In this case the stub impedance is inductive at the lower frequencies and will consequently move up toward the inductive half-plane, i.e., obtaining a total phase delay of 180° as desired.

23–8 BALANCED TRANSFORMER（平衡变换器）

It is sometimes desirable to add a transformer to the balanced side of a balun. An example of such a device is shown in Fig. 23–12*a*. It consists of two coaxial lines with characteristic impedance $2Z_0$. At the low-impedance end (lower terminals) the inner and outer conductors of opposing cables are connected in parallel but with the two inner conductors having

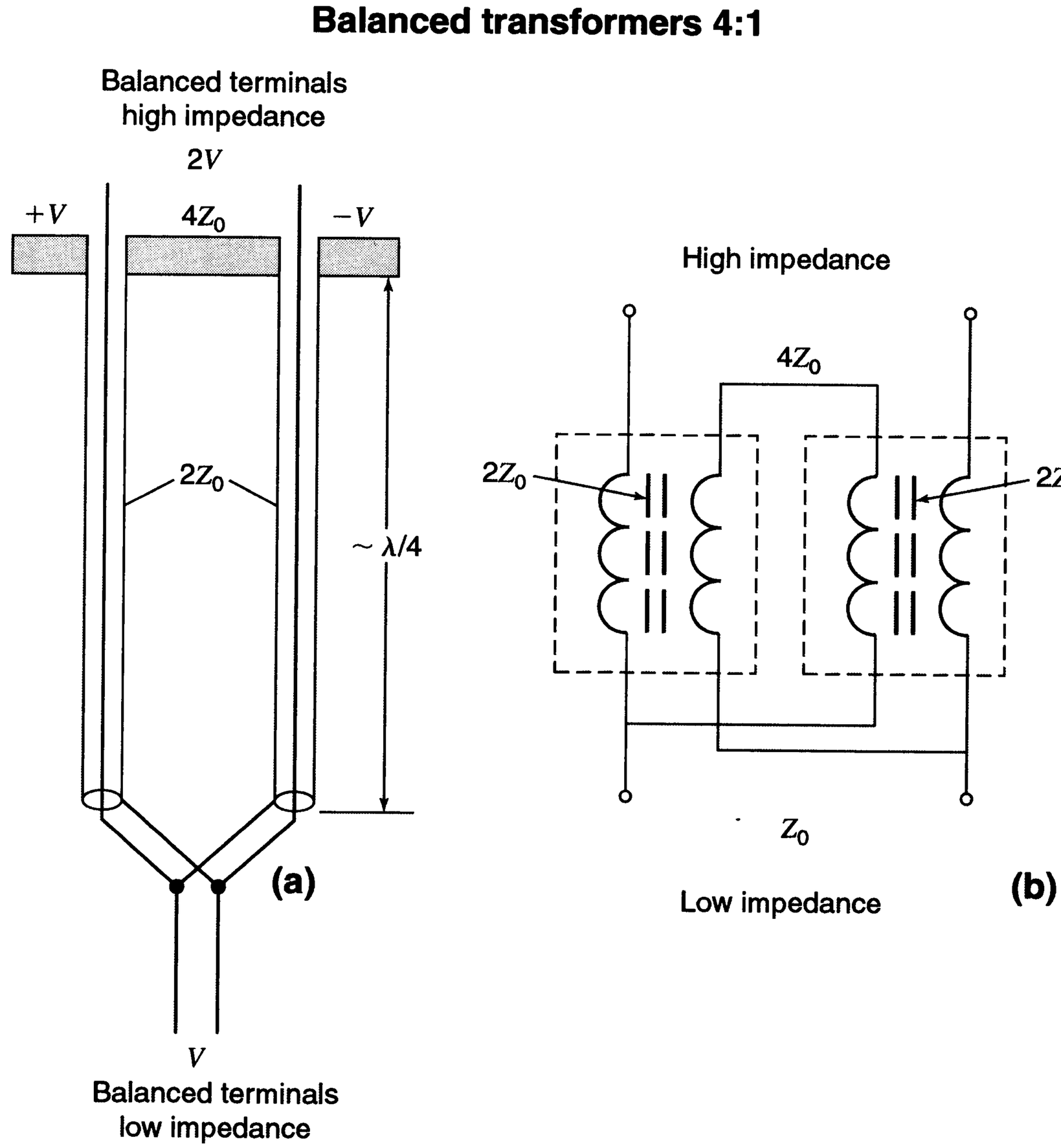
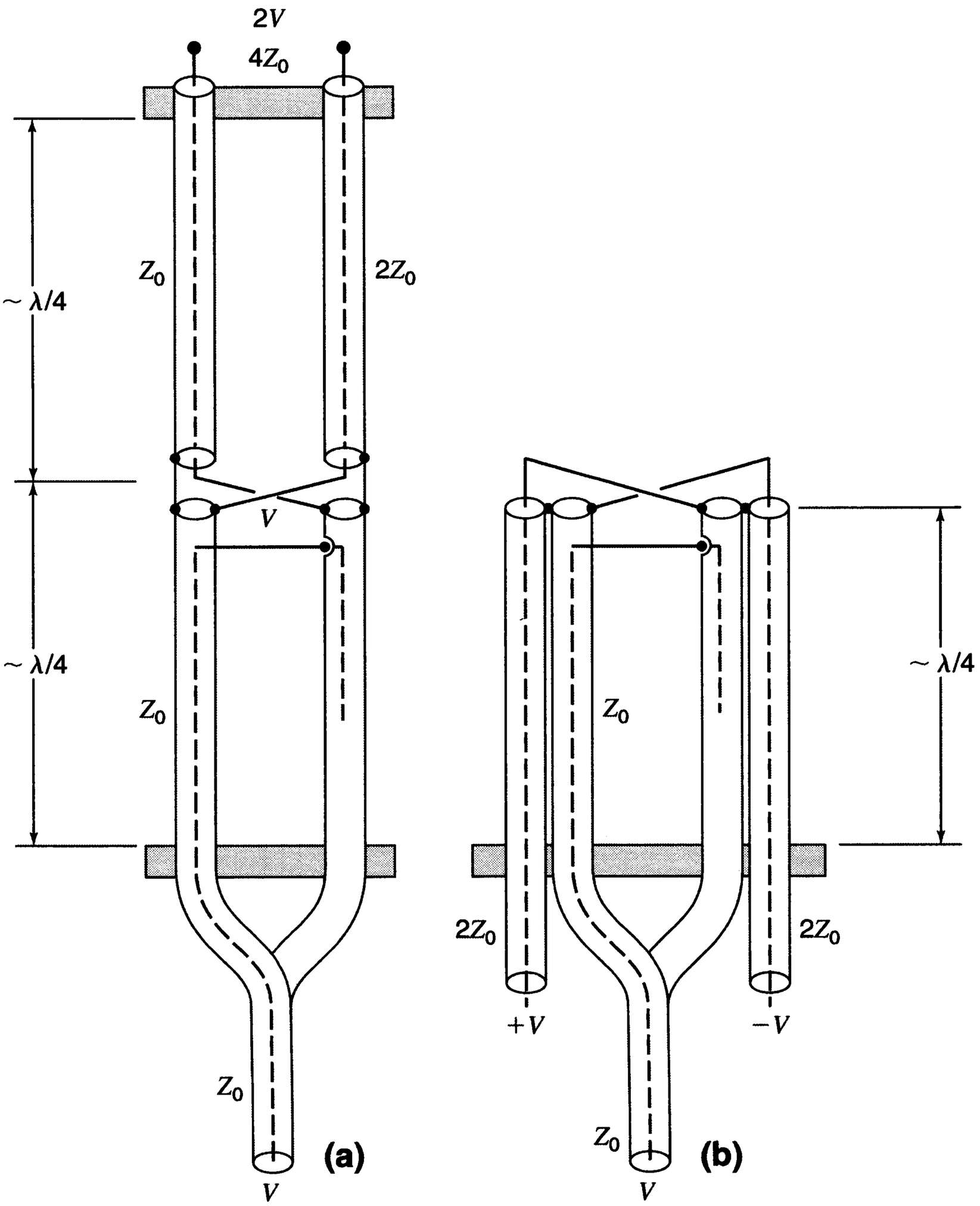


Figure 23–12
By connecting two cables in parallel at the low-impedance end and in series at the high-impedance end as shown we may double the input voltage V. (*a*) A version made of coaxial cables. (*b*) A version made of twin leads wound on ferrite core.

opposite polarity as shown. At the high-impedance terminals the two cables are connected in series and since the inner conductors have opposite polarity their total voltage difference is $2V$; i.e., the impedance level is increased by a factor of 4.

This transformation is relatively independent of the cable length. However, the two outer conductors are connected at the high-impedance terminals. Thus, when seen from the low-impedance terminals the two outer conductors form a fork that has a certain fork impedance in parallel with the input impedance. By making the length of the fork equal to $\lambda/4$ the shunt impedance becomes infinite and will consequently not affect the input impedance.

Combinations of baluns and transformers



Type III balun with transformer

Folded version

Figure 23–13
(*a*) A combination of Type III balun and the transformer shown in Fig. 23–12*a*.
(*b*) The transformer section in (*a*) has been folded down along the fork (*b*).

Instead of coaxial cables twin lead cables may also be used as shown in Fig. 23–12*b*. If these twin lead cables are wound into coils they will exhibit a high resistance for unbalanced but not for balance currents. Thus, they will be effective as a balun as well, similar to the designs shown in Fig. 23–2. In addition, we may apply a ferrite core. It has the potential of increasing the coil impedance and thereby the bandwidth ($>10:1$). They are very popular and inexpensive for low power.

A combination of Type III balun and a transformer is shown in Fig. 23–13*a* left. We observe that the two fork impedances are in parallel; thus the bandwidth will be somewhat reduced (see also Problems 23–3–1, 23–3–2 and 23–3–3). However, compensation can be provided by an open-ended $\lambda/4$ stub; see Fig. 23–5*b*.

A more compact version is shown in Fig. 23–13*b*. Here the transformer lines are folded down along the balun fork (the outer conductors can touch each other everywhere). Besides being shorter, there is now only one fork impedance instead of two in parallel, although the characteristic impedance of the fork has been lowered somewhat due to the effective increase in cable thickness.

Finally, Fig. 23–14 shows a very clever combination of balun and a built-in transformer. By using still more forks connected to a single branch-point, baluns can be designed with a transformation ratio equal to 1 : 9 and even 1 : 16 (Ruhrmann-1 and 2).

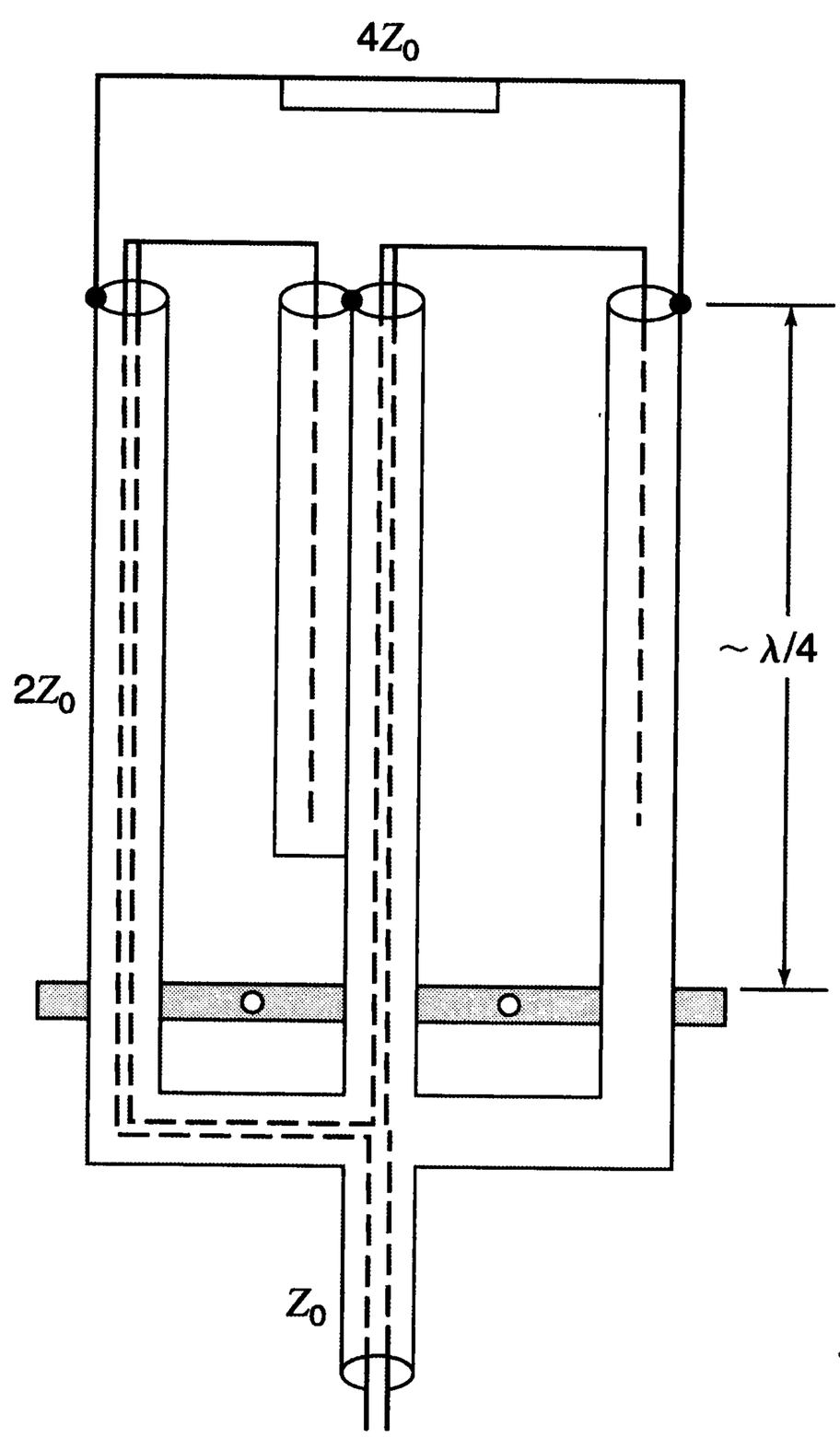


Figure 23–14
A "candelabra" version of Type III balun with transformation ratio 4 : 1. Note the broadband compensation by use of open-ended stubs. By using more arms transformation ratios of 9 : 1 or 16 : 1 can be obtained.

23–9 CUTAWAY BALUNS（截割式巴仑）

This type of balun is shown in Fig. 23–15 and starts with a coaxial cable at the lower end. The outer conductor is then slowly peeled away in a tapered fashion until it becomes a two-wire transmission line at the upper end. It works by the fact that the taper is very gradual thereby making internal reflections very small. This design principle also leads to a rather long design (approximately $\lambda/2$ at the lowest frequency). And it is primarily the long length that leads to a high impedance for the unbalanced currents. In fact, the upper frequency is limited to the excitation of higher-order modes in the coaxial cable (Duncan-1). In contrast to the other transforming baluns such as given in Figs. 23–10 to 23–14, this type of balun can be designed to have transformation ratios that typically range from ~1.5 : 1 to 5 : 1 with a bandwidth approaching 100 : 1, but it must be long!

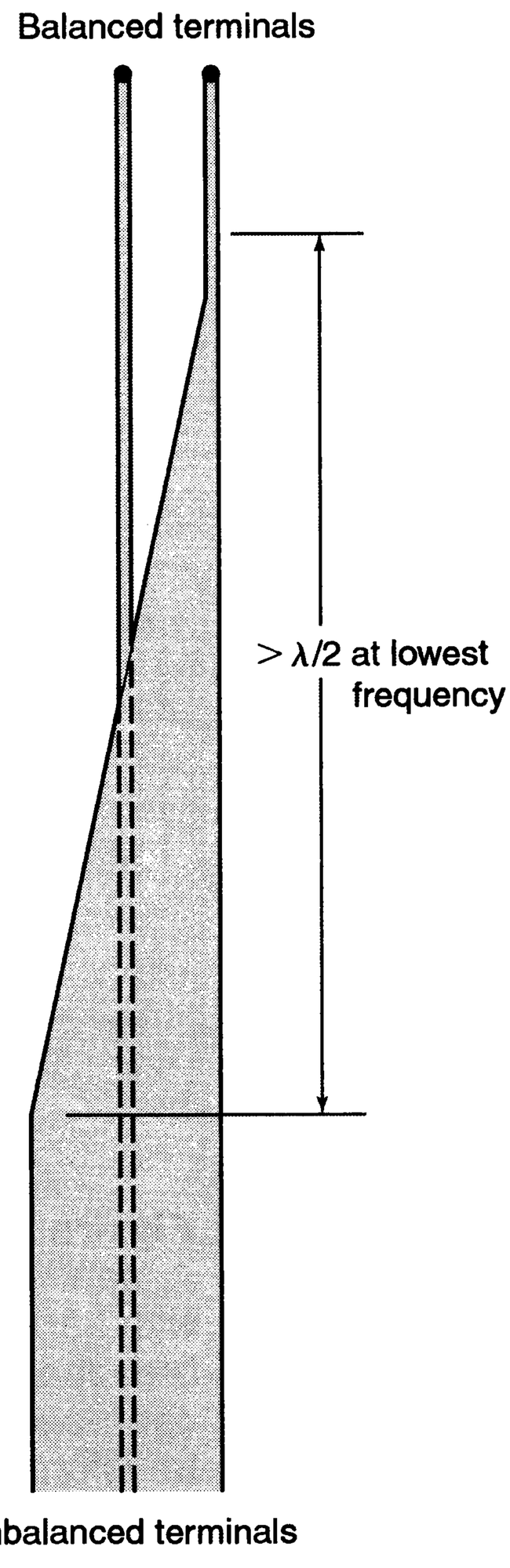


Figure 23–15
A coaxial cable is gradually cut in a tapered fashion and transitioned into a pair of twin leads. Capable of very large bandwidth (100 : 1).

An interesting "mix" is shown in Fig. 23–16. Here the outer conductor has been cut away on two sides while the inner conductor continues to the top where it is connected to the right part of the outer conductor. Thus, it can be considered a cross between the tapered balun above and the Type III balun. However, in contrast to the latter, some impedance transformation will take place. A typical ratio of ~2 : 1 has been my experience as we move from the coaxial input toward the dipole.

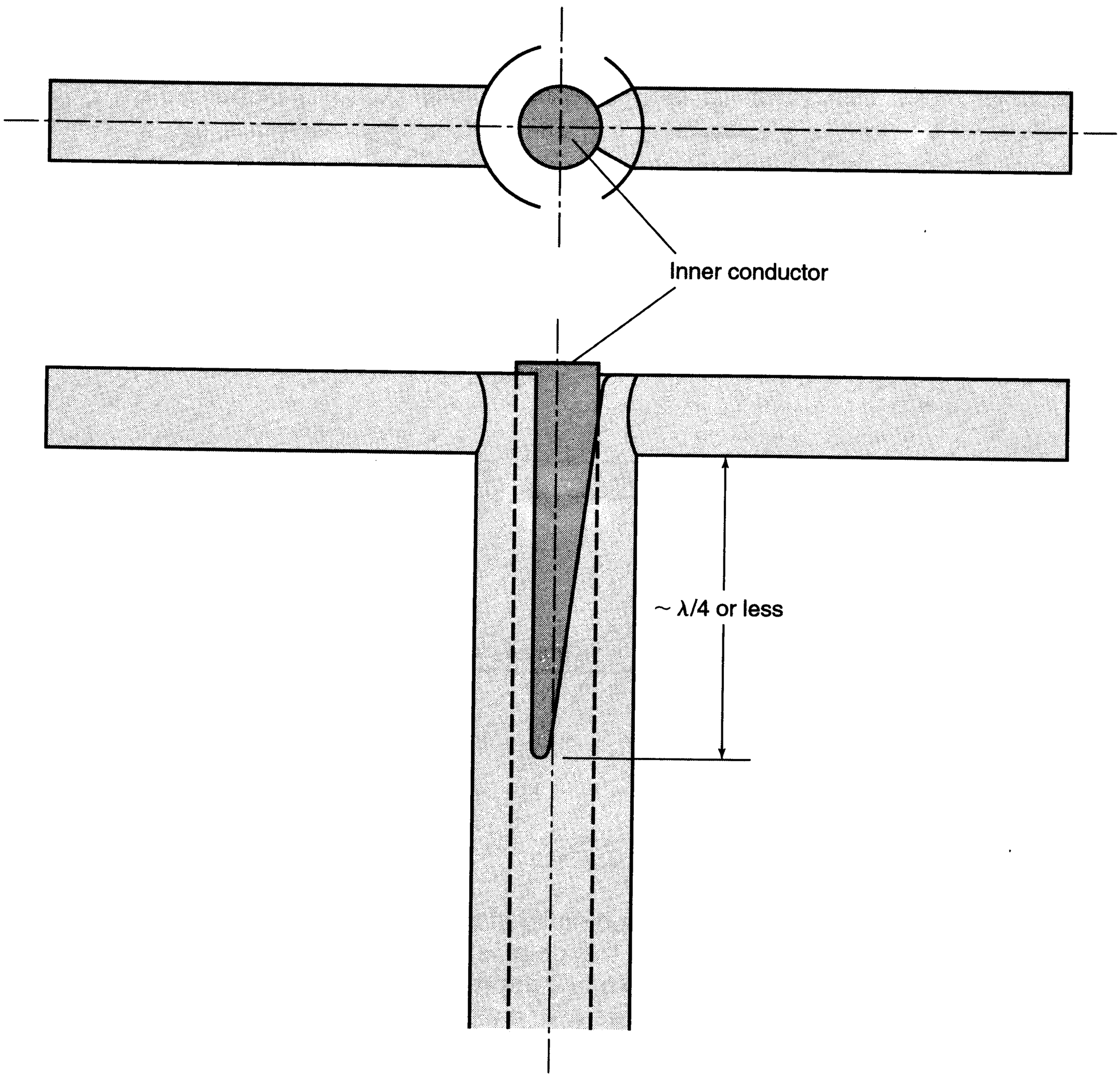


Figure 23–16
A "mix" of a balun: two tapered "slits" are cut at opposite sides in the outer tubing reminiscent of the cutaway balun in Fig. 23–15. However, the inner conductor is connected to one side similar to the Type III balun in Fig. 23–4.

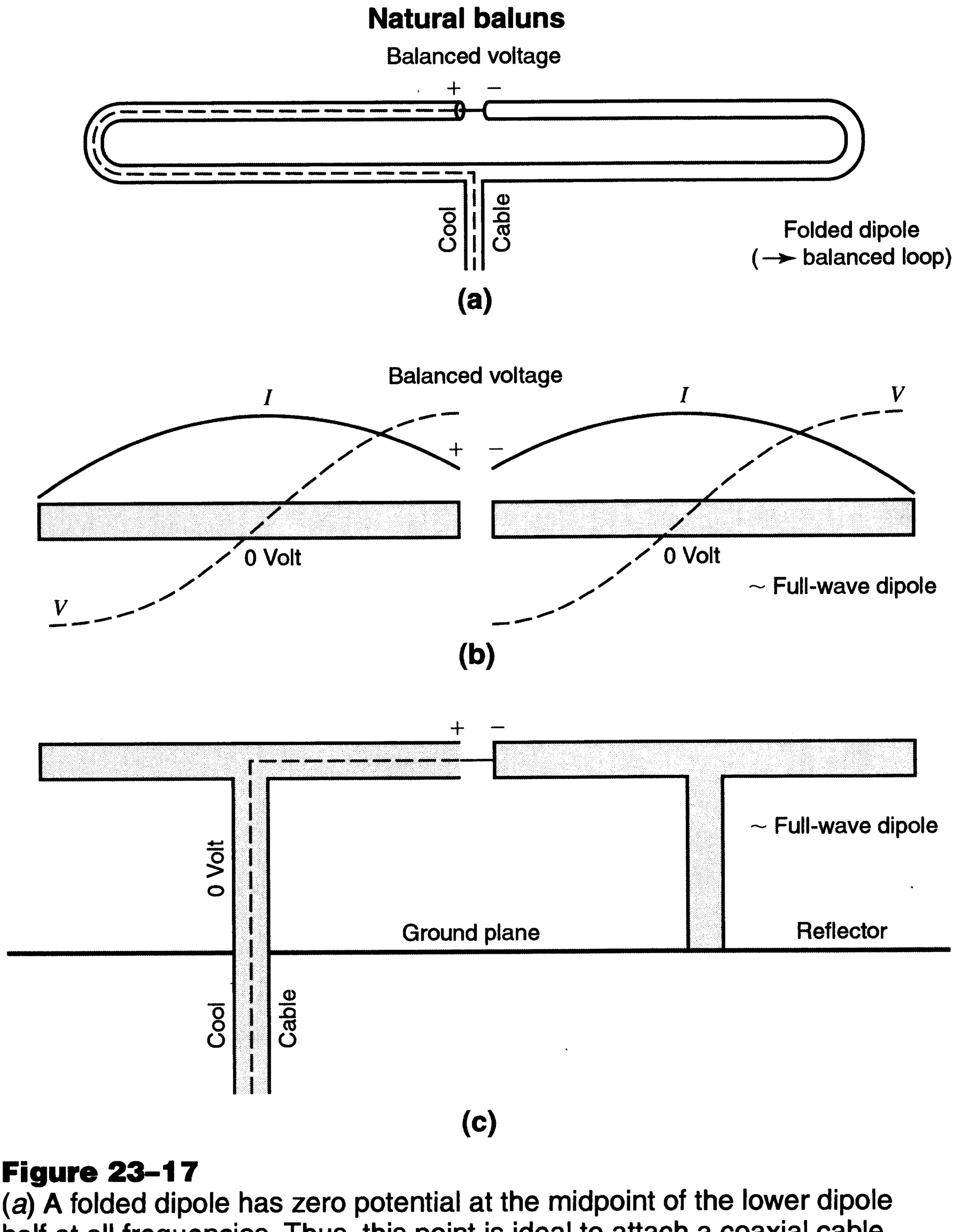


Figure 23–17
(*a*) A folded dipole has zero potential at the midpoint of the lower dipole half at all frequencies. Thus, this point is ideal to attach a coaxial cable. (*b*) Similarly, a full-wave dipole has approximately zero potential near the middle of the two dipole halves over a significant frequency range. (*c*) This point is ideal for support as well as feeding.

23–10 NATURAL BALUNS（天然巴仑）

Some antennas have zero potential somewhere. A classical example is a folded dipole shown in Fig. 23–17*c* where a null is observed in the middle of the unbroken dipole half. Just as was the case with the Type III balun shown in Figs. 23–3 and 23–4, the coaxial cable is attached at that point. The balance is always perfect and only the antenna impedance of the folded dipole is seen at the feed point. In other words, not only a natural but a perfect balun!

The loop baluns shown in Fig. 23–18 are also natural baluns. See also the loop baluns of Figs. 7–8, 7–12 and 7–14.

Another example is shown in Fig. 23–17, namely, a full-wave (or so) dipole. Also shown is the current as well as the voltage distributions. As can be seen, the voltage distribution is zero somewhere in the middle of the two antenna sections. Thus, this is precisely where both antenna halves can be supported without affecting (at least appreciably) the current distribution. The inside of one of these supports (namely, the left) is also used to feed the antenna.

The bandwidth of this arrangement is obviously not infinite but still adequate in practice to work over an octave bandwidth. The zero potential point where the support is attached is best found experimentally. First measure the antenna impedance without support. Then with your hand on the antenna you observe a change in the antenna impedance.

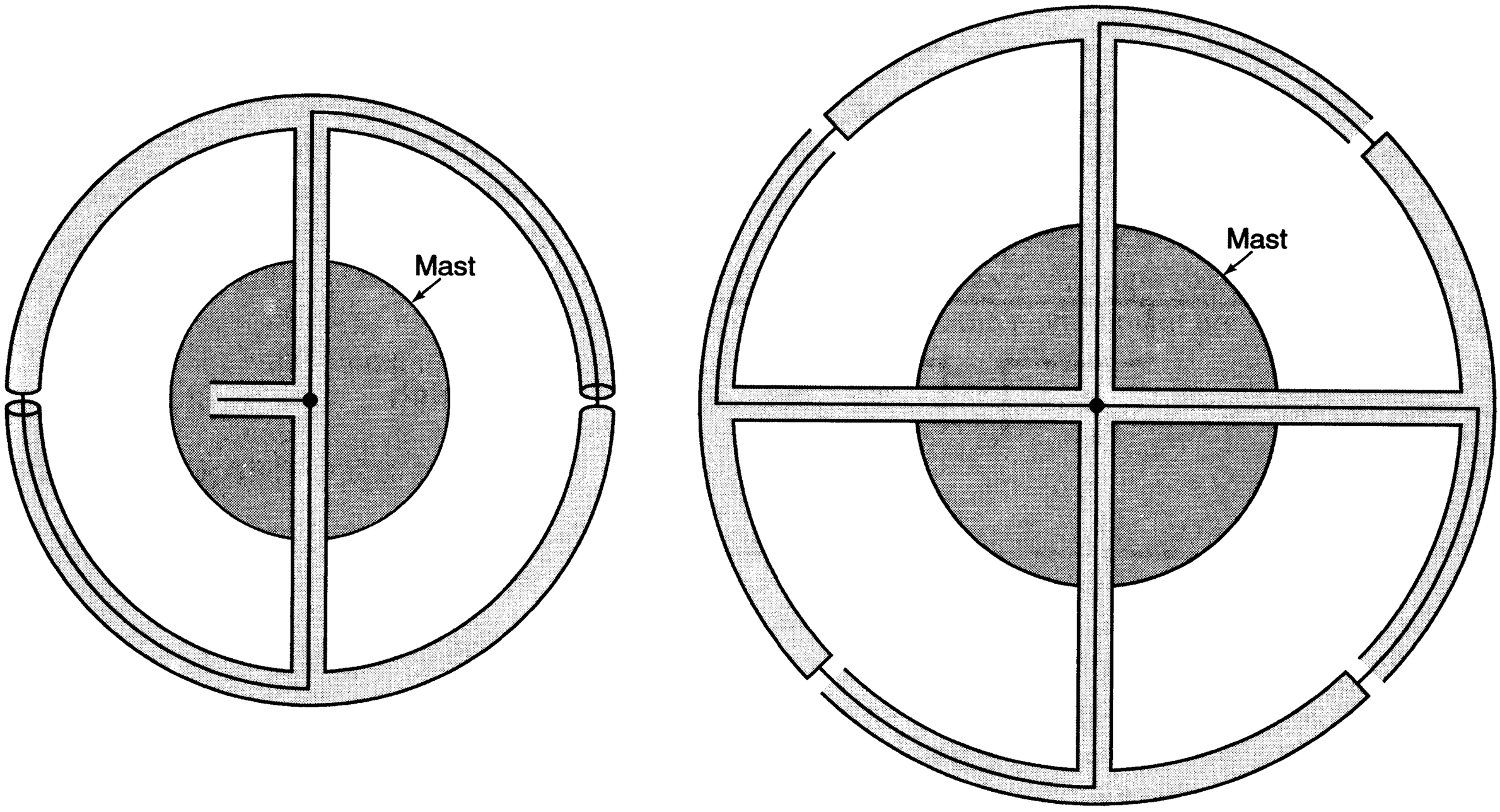
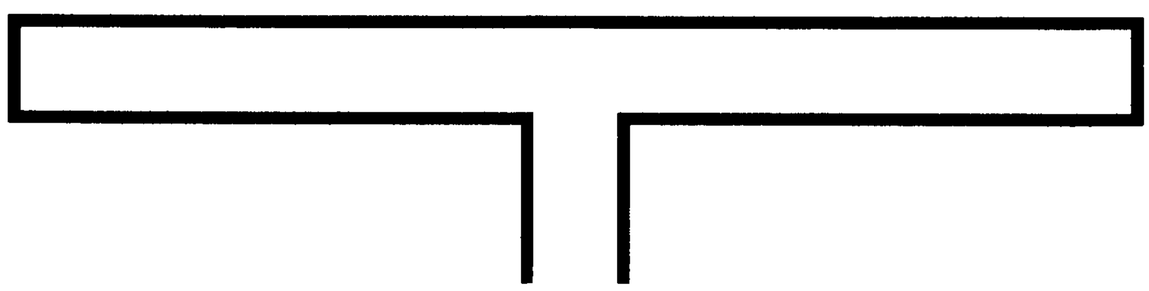


Figure 23–18
Two baluns for feeding loops on a mast.

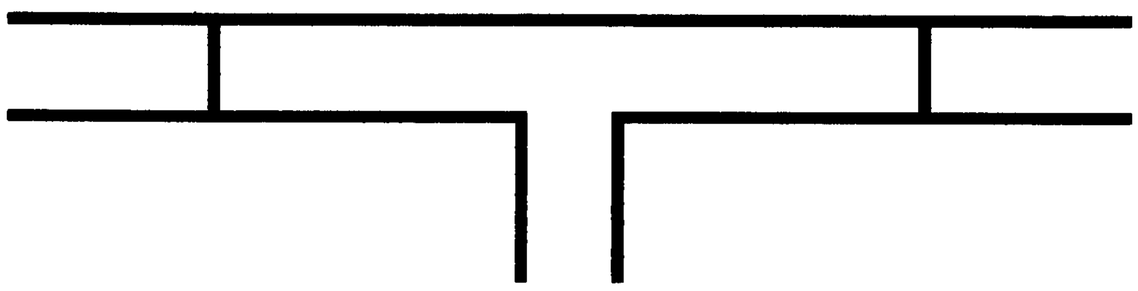
Slide your hand along one of the dipole sections until there is no change in impedance. This indicates the proper support position. (After this exercise you may claim "hands-on experience.")

23–11 FOLDED DIPOLE TO J-MATCH（折合偶极子到J形匹配天线）

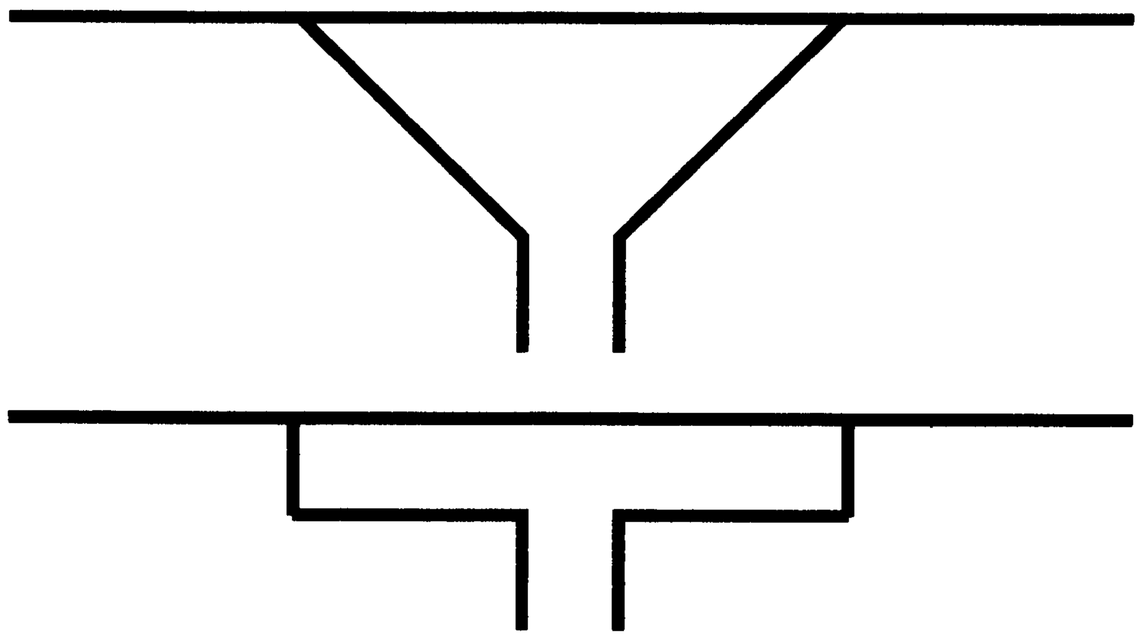
Figure 23–19 shows the five steps in the evolution of a J-match coaxial cable-fed dipole.



(a) Folded dipole

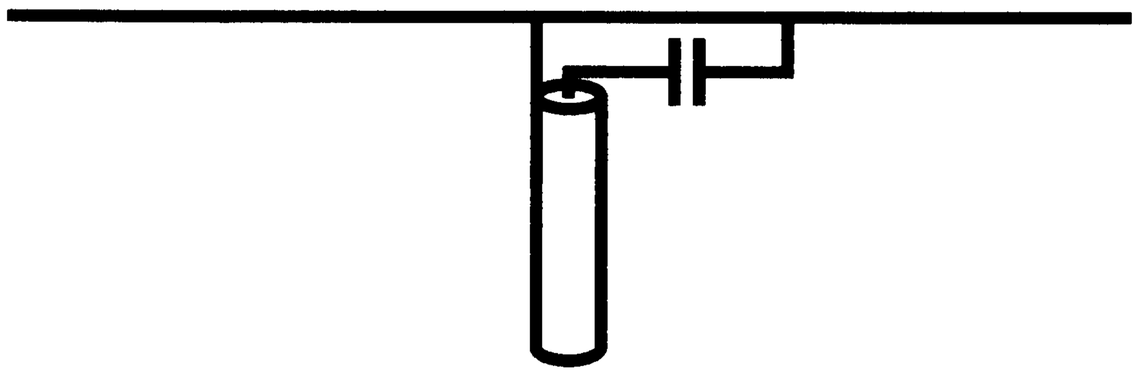


(b) Modified folded dipole



(c) Delta match

(d) T-match



(e) J- or gamma match

Figure 23–19
Five steps in the evolution of a J-match fed dipole from a folded dipole via modified folded dipole, delta-match and T-match (Kraus (1) and Sturgeon).

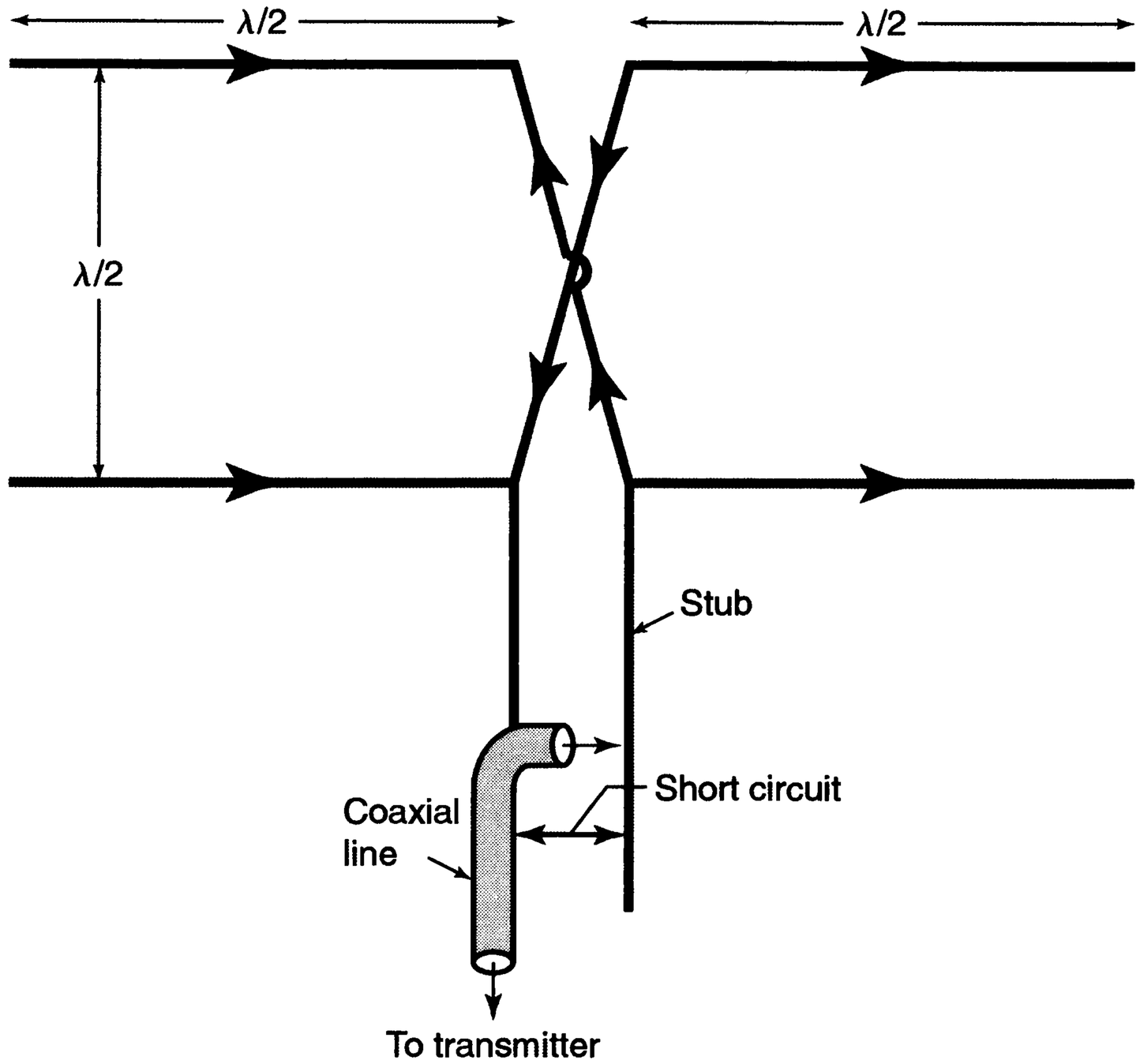


Figure 23–20
Dipole array with matching stub and Type III balun coaxial feed. The spacing of the stub conductors is exaggerated for clarity.

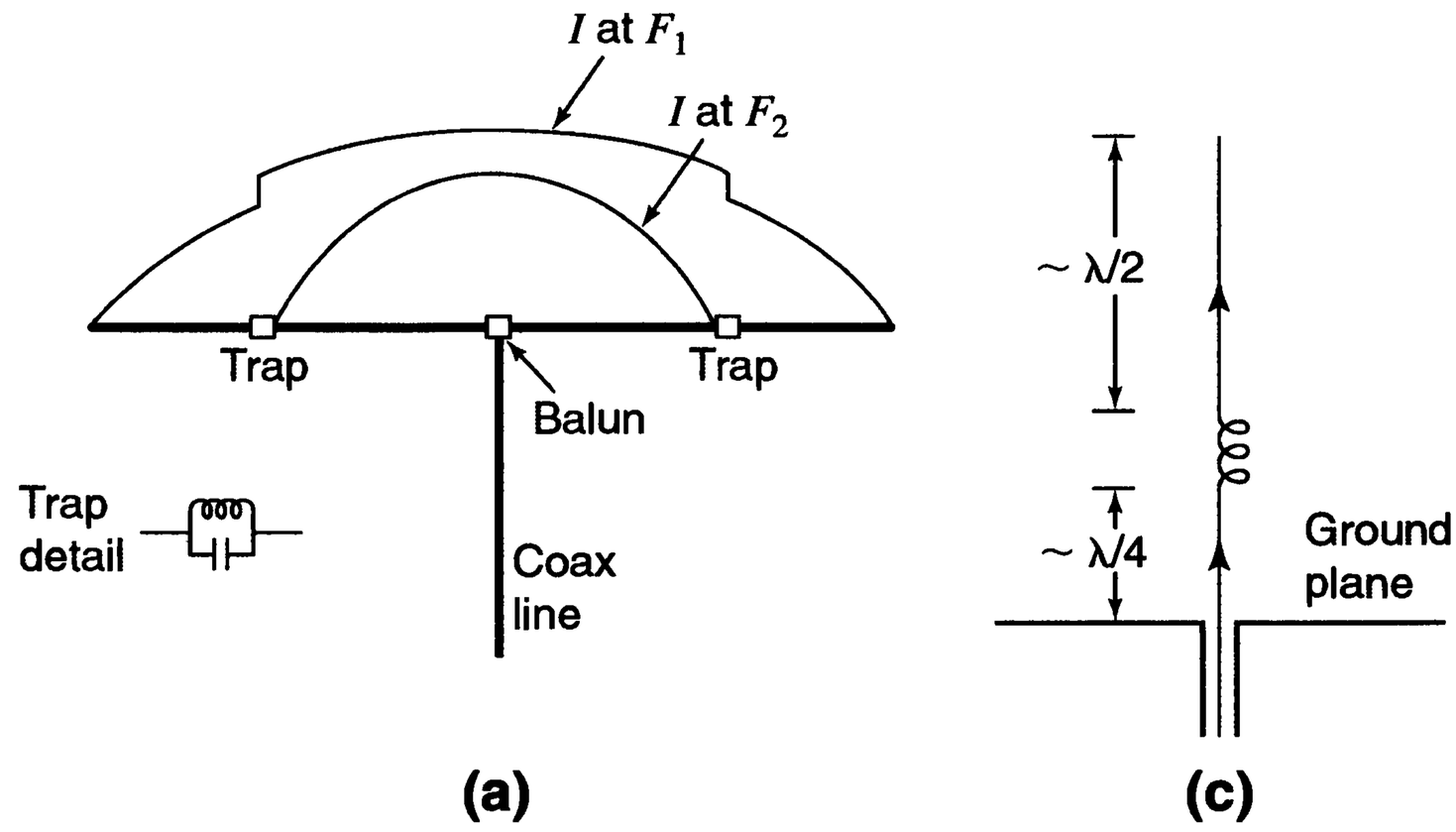
23–12 MATCHING STUBS（匹配短截线）

An array of dipoles with two-conductor transmission line can be fed with a coaxial line as shown in Fig. 23–20. First, the short-circuit on the stub is adjusted for maximum current (or resonance) with the array excited from a nearby transmitting antenna (step 1). The coaxial feed line is then connected and moved up and down the stub for minimum VSWR on the line to the transmitter (step 2).

This balun feed with simple 2-step tune up can be used for a curtain array (as in Fig. 23–20), a W8JK array, a rhombic or other array or a single dipole of any length. The effect of the support structure or other objects in the environment are all accounted for by the tuning of this feed system.

23–13 TRAPS（陷波器）

In many wide-bandwidth applications it is not necessary to have a frequency-independent antenna for continuous spectrum coverage but rather an antenna which can operate at *spot frequencies*. This is possible, for example, with a center-fed dipole by means of tuned traps, as shown in Fig. 23–21*a*, each trap consisting of a parallel-tuned *LC* circuit. At frequency F_1, for which the dipole is $\lambda/2$ long, the traps introduce some inductance so that the resonant length of the dipole is reduced. At twice the frequency F_2 $(=2F_1)$ the traps are resonant $(\omega L = 1/\omega C)$ and the high impedance they introduce effectively isolates the outer sections of the dipole, the inner part becoming a resonant $\lambda/2$ dipole at frequency F_2. Thus, in this simple example, the antenna can perform simultaneously as a matched $\lambda/2$ dipole



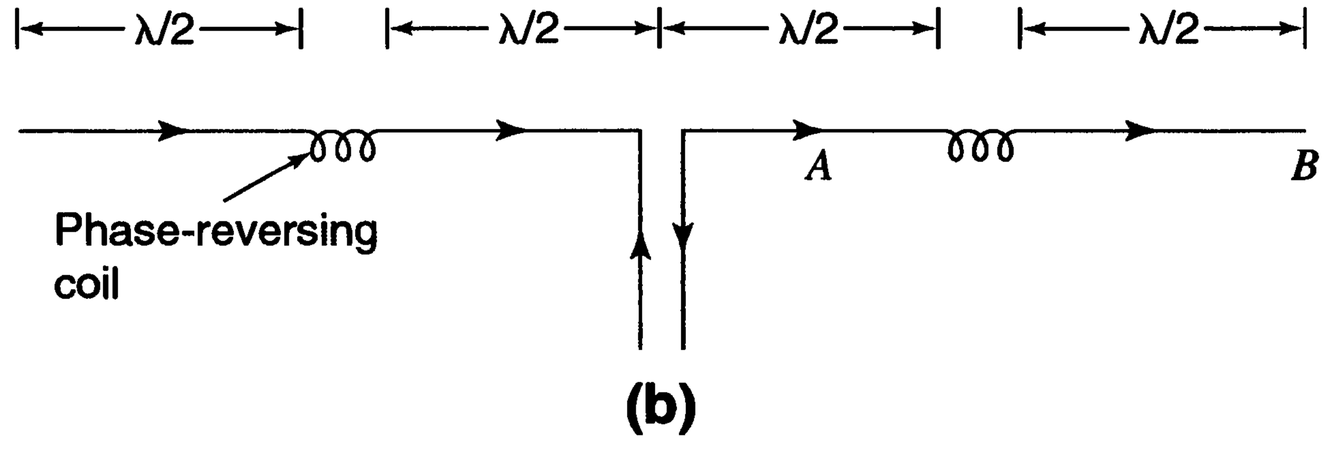


Figure 23–21
(*a*) Dipole with traps for operation at two frequencies separated by an octave ($F_2 = 2F_1$). (*b*) Four in-phase $\lambda/2$ elements with phase-reversing coils. (*c*) Vertical omnidirection in-phase $3\lambda/4$ monopole.

at 2 frequencies, F_1 and F_2, separated by an octave. With more traps, operation may be extended to other frequencies. Although in our example the 2 frequencies are harmonically related, this is not a requirement. Note, however, that the end segment must present a low impedance to the trap to be isolated (i.e., mismatched). In Fig. 23–21*a* the end segment is $\sim\lambda/4$ long at F_2 so this requirement is met.

A coil (or trap) can also act as a 180° phase shifter as in the collinear array of 4 in-phase $\lambda/2$ elements in Fig. 23–21*b*. Here the elements present a high impedance to the coil which may be resonated without an external capacitance due to its distributed capacitance. The coil may also be thought of as a coiled-up $\lambda/2$ element. This 4-element array has a gain of 6.4 dBi as compared to 3.8 and 5.3 dBi gain for 2- and 3-element collinear arrays of $\lambda/2$ elements.

Cutting the antenna of Fig. 23–21*b* at point *A* and turning the section vertical above a ground plane results in the in-phase $3\lambda/4$ omnidirectional monopole antenna of Fig. 23–21*c* with 8.3 dBi gain. To match the approximately 150-Ω terminal resistance to a 50- to 75-Ω coaxial line, a capacitance-inductance network can be used.

23–14 CONCLUSION（结论）

We have considered a wide variety of baluns and transformers. No claim is made as to completeness. However, we have made an attempt to explain the underlying principles and limitations of baluns. It is our hope that it will inspire the reader to produce a unique design that often must be customized for a particular need.

REFERENCES（参考文献）

Bock, E. L. (1), J. A. Nelson and A. Dorne in "Very High-Frequency Techniques," Chap. 5, McGraw-Hill, New York, 1947.

Duncan, J. W. (1), and V. P. Minerva, "100 : 1 Bandwidth Balun Transformer," *Proc. IRE,* pp. 156–164, February 1960.

Kraus, J. D. (1), and S. S. Sturgen, "The T-Matched Antenna," *QST,* pp. 24–25, September 1940.

Nelson, J. A. (1), and G. Stavis, in "Very High-Frequency Techniques," Chap. 3, McGraw-Hill, New York, 1947.

Roberts, W. K. (1), "A New Wide-Band Balun," *Proc. IRE,* pp. 1628–1631, December 1957.

Ruhrmann, A. (1), *Telefunken-Zeitung,* **24,** pp. 237–250, 1951.

Ruhrmann, A. (2), (inventor), (Telefunken), *DRP* 806 446 (1.10.1948).

Walter, L. (1), A. Gothe and H. D. Rosenstein (inventors), (Telefunken), *DRP* 568 559, (18.11.1931).

ADDITIONAL REFERENCES（补充参考资料）

Lindenblad, N. E. (1), "Antennas and Transmission Lines at the Empire State Television Station," *Communications,* Vol. 20, p. 13, May 1940.

Lindenblad, N. E. (2) (inventor), Brit. Patent 487, 708, (28.4.1936).

Meinke, H. (1), and F. W. Gundlach, "Tashenbuch der Hochfrequenztechnik," Sec. 17–24, Springer-Verlag, Berlin/Göttingen/Heidelberg, 1962.

Percival, W. S. (1), and E. C. L. White (inventors), (Electrical Musical Instrument Ltd.), Brit. Patent 438, 506, (15.11.1935).

Radio Research Laboratory (1), Harvard University, "Very High Frequency Techniques," Sec. 3–14, 3–15, 3–16, Boston Technical Publishers, 1965 (originally 1946).

PROBLEMS（习题）

23–3–1
（巴仑 200 Ω，天线 70 Ω）

Balun 200 Ω, antenna 70 Ω. A Type III balun has the characteristic impedance equal to $Z_{cp} = 200\ \Omega$ and the electrical length is equal to $l_p = 7.5$ cm. It is connected to an antenna with impedance $Z_A = 70\ \Omega$. (*a*) Find the balun impedance jX_p at $f = 500$, 1000 and 1500 MHz. (*b*) Calculate the parallel impedances $Z_A \| jX_p$ at 500, 1000 and 1500 MHz and plot them in a Smith Chart normalized to $Z_0 = 50\ \Omega$. Check that all these impedances lie on a circle with a diameter spanning over (0, 0) and $Z_A = 70\ \Omega$. Alternatively, you may determine $Z_A \| jX_p$ graphically in a Smith Chart. (*c*) Explain what effect it would have on the bandwidth if we changed Z_{cp} to 150 Ω or 250 Ω.

23–3–2
（巴仑 200 Ω，天线 80 – *j*20 Ω等）

Balun 200 Ω, antenna 80 – *j*20 Ω, etc. Using the same Type III balun as in Prob. 23–3–1, change the antenna impedance to $Z_A = 80 - j20$, 75 and $80 + j20$ at $f = 500$, 1000 and 1500 MHz, respectively. (*a*) Calculate $Z_A \| jX_p$ at $f = 500$, 1000 and 1500 MHz, and indicate their impedances in a Smith Chart normalized to $Z_0 = 50\ \Omega$. Check that all these impedances are located on the appropriate circles spanning Z_A and (0, 0). A purely graphical approach is satisfactory. (*b*) Repeat the problem for $Z_A = 70 + j20$, 75 and $70 - j20$ at $f = 500$, 1000 and 1500 MHz, respectively. Which of these two cases has the potential for the largest bandwidth?

23–3–3
（带短截线的巴仑）

Balun with stub. We now modify the same Type III balun as in Probs. 23–3–1 and 23–3–2, to the more broadbanded version shown in Fig. 23–5*b*. The characteristic impedance of the coaxial stub is $Z_{cs} = 12.5\ \Omega$ and the dielectric constant is $\varepsilon_1 = 2.25\varepsilon_0$. (*a*) Find the length of the stub for the center frequency $f = 1000$ MHz. (*b*) Calculate the stub impedance jX_s at $f = 500$, 1000 and 1500 MHz, respectively. (*c*) Calculate $Z_A \| jX_p + jX_s$ at the same three frequencies as above. Plot the value in

a Smith Chart normalized to $Z_0 = 50\ \Omega$. (*d*) Compare the potential for large bandwidth for this compensated design with the uncompensated design in Prob. 23–3–1.

Discuss the implications upon bandwidth if the stub characteristic impedance Z_{cs} is changed from $Z_{cs} = 12.5\ \Omega$ to $50\ \Omega$.

This last variation should illustrate that to solve a problem successfully you must not only have the right idea, it must also be *designed* correctly.

23–3–4 (偶极子天线的 VSWR)

VSWR for dipole antenna. Calculate the VSWR on a 65-Ω line connected to the $L/D = 60$ dipole antenna over a 30 percent bandwidth if an open-ended line of 40 Ω characteristic impedance is connected in parallel with the antenna terminals. The line is 180° long at the center frequency of F_0.

23–3–5 (短截线的阻抗)

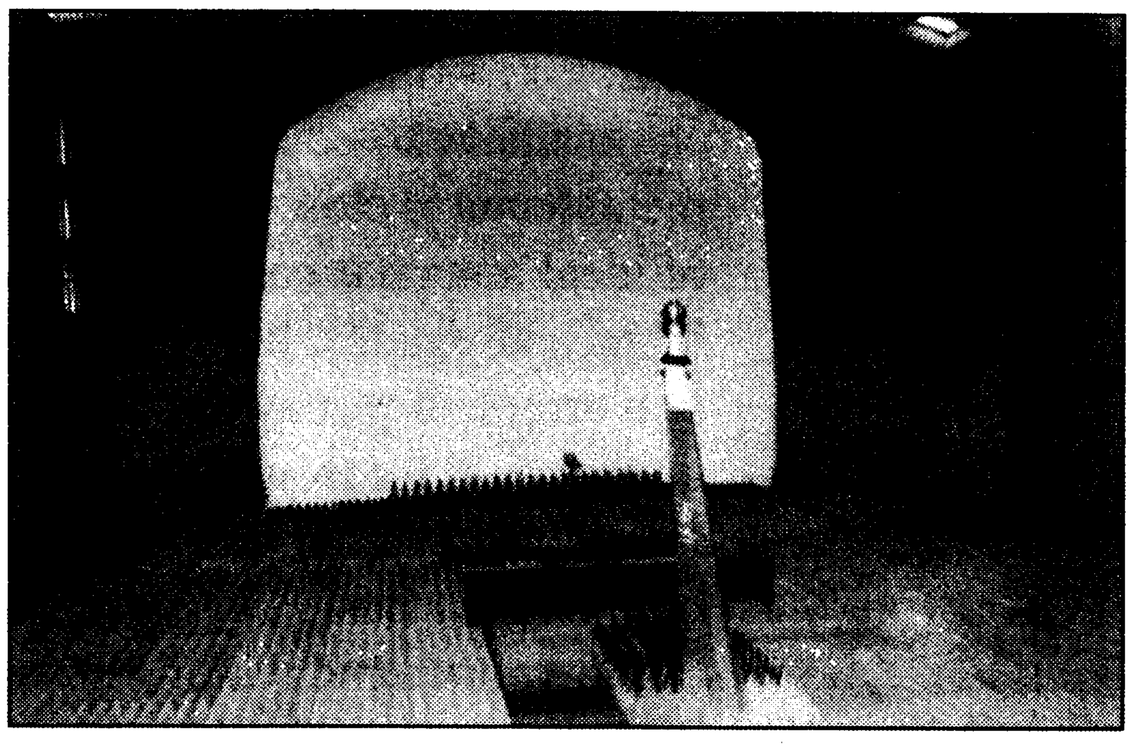
Stub Impedance. (*a*) What is the terminal impedance of a ground-plane mounted stub antenna fed with a 50-Ω air-filled coaxial line if the VSWR on the line is 2.5 and the first voltage minimum is 0.17λ from the terminals? (*b*) Design a transformer so that the VSWR = 1.

CHAPTER 24

Antenna Measurements

By Arto Lehto and Pertti Vainikainen[1]

天线测量



Topics in this chapter include:

- Basic concepts（基本概念）
- Reciprocity（互易性）
- Near field and far field（近场和远场）
- Coordinate system（坐标系）
- Sources of error（误差源）
- Phase error and amplitude taper（相位误差和幅度锥削）
- Reflections（反射）
- Measurement ranges（测量场地）
- Elevated ranges（架高范围）
- Ground-reflection ranges（地面反射范围）
- Anechoic chambers and absorbing materials（吸波室和吸波材料）
- Compact antenna test ranges（紧缩的天线测量场地）
- Near-field ranges（近场场地）
- Testing of ranges（场地的测量）
- Instrumentation（仪器设备）
- Measurement of different antenna parameters（不同天线参量的测量）
- Directional pattern（定向波瓣图）
- Gain（增益）
- Phase（相位）
- Polarization（极化）
- Impedance（阻抗）
- Efficiency（效率）
- Current distribution（电流分布）
- Large antennas（大型天线）
- Electrically small antennas（电小天线）
- Effective gain（有效增益）
- Specific absorption rate（比吸收率）

24–1 INTRODUCTION（引言）

Accurate measurements are necessary to establish the actual performance of antennas: their gain, pattern, polarization, bandwidth, efficiency, etc. Antennas having strict specifications are needed in many applications as in mobile and personal communications, satellite communications, remote sensing, and radar. For example, antennas of a point-to-point radio

[1]Radio Laboratory, Helsinki University of Technology.

link have to fulfill certain gain, side-lobe level, and cross-polarization requirements set by standards to get type approval.

In many cases, antenna properties can be calculated theoretically very accurately. However, for complex antennas this might not be possible—too many idealizations or simplifications have to be made. Often the modeling of the usage environment is difficult, e.g., if the antenna is close to the human head or is installed on an airplane. Even if the properties of ideal antennas can be calculated, performance of real world antennas has to be checked by measurements, because due to fabrication tolerances, and in some cases due to fabrication errors, they may not work as well as predicted. The measurement results give valuable information for troubleshooting.

In this chapter, methods and techniques are discussed for experimental measurements of antennas. Basic concepts, typical sources of measurement errors, measurement ranges and their instrumentation, measurement methods of different antenna parameters, and some miscellaneous topics like measurement of electrically large and small antennas, effective gain, and specific absorption rate are treated.

24–2 BASIC CONCEPTS（基本概念）

The most common antenna measurement is to measure its radiation properties like directional pattern, gain or phase pattern in the far field. In this section we define the basic concepts of such measurements. Typical configuration of the measurement of radiation properties is shown in Fig. 24–1. The basic procedure is to place a transmitting or receiving *source antenna* at different locations with respect to the *Antenna Under Test* (*AUT*) and thus get a number of samples of the pattern. The different locations are normally achieved by rotating the AUT. To ensure the "sharpness" of the pattern sample only one direct signal path should exist between the AUT and the source antenna. This can be achieved in a reflectionless environment like in an anechoic chamber or in free space (see Sec. 24–4).

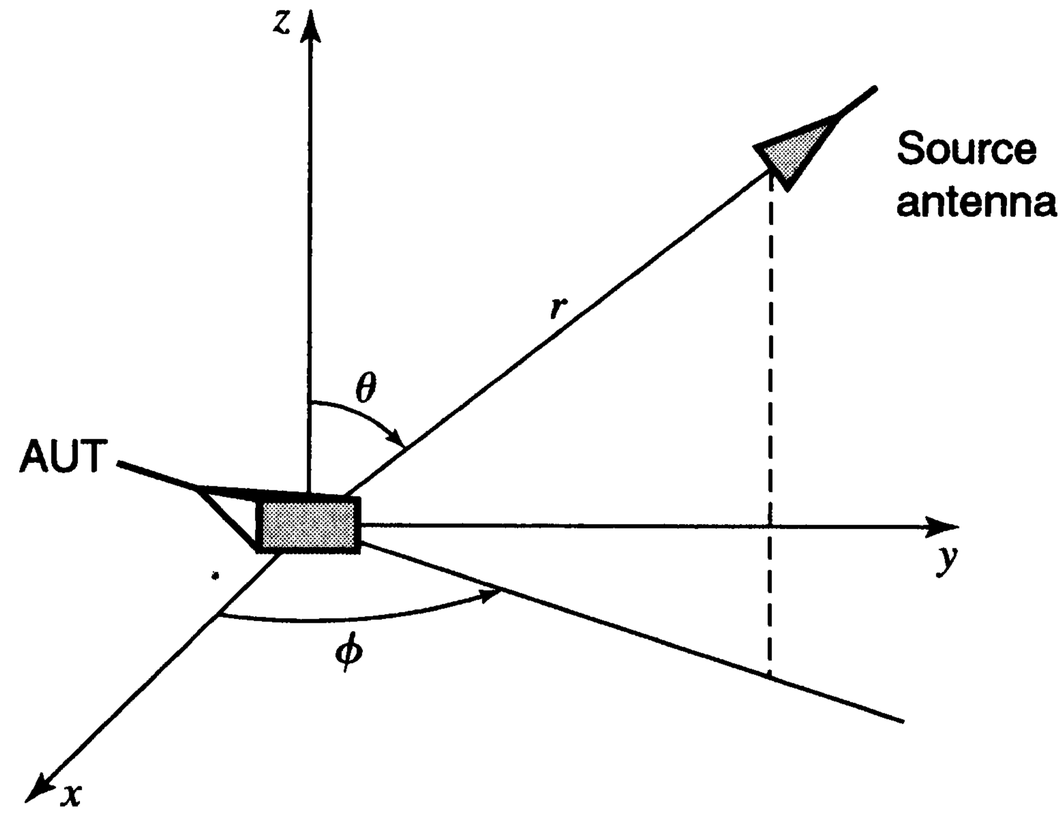


Figure 24–1
Typical configuration for the measurement of the radiation properties of an antenna.

24–2a Reciprocity in Antenna Measurements（天线测量中的互易性）

It was indicated already above that the AUT can act as either a receiving antenna or a transmitting antenna. This is of course due to the reciprocity principle discussed in Sec. 13–2. Two important consequences of the principle from the antenna measurement point of view were given:

- The transmitting and receiving patterns are the same.
- Power flow is the same either way.

Thus it is clear that all radiation parameters of the AUT can be measured in either transmission or reception mode. This is especially useful in cases, where, for example, the AUT is an integral part of a larger device acting as either a receiver or a transmitter thus defining the direction of the signal.

However, in practical antenna measurements one has to be careful in applying the reciprocity principle. In Sec. 13–2 also several important conditions for the validity of the principle were given:

- The emfs in the terminals of the interchanged antennas are of the same frequency.
- The media are linear, passive and isotropic.
- The power flow is equal for matched impedances only.

In standard careful measurements the 1st condition should be met without problems. The 2nd condition should be considered always, when the AUT includes any active or ferrite components as an integral part. The 3rd condition is perhaps the most difficult and may lead into measurement results, which seem to violate the reciprocity principle. The situation is clarified in Fig. 24–2 showing a typical configuration with the instrumentation for the measurement of the radiation properties of an AUT.

We can usually estimate that the multiple reflections between the AUT and source antenna are insignificant due to high free space loss $|1/t_{FS}|$ (see Fig. 24–2). By using, for example, flow graph techniques we can now derive the voltage transmission for the configuration of Fig. 24–2:

$$\frac{V_R}{V_T} = \frac{e^{-\gamma_1 \ell_1}}{1 - \rho_T \rho_{SA} e^{-2\gamma_1 \ell_1}} \cdot t_{FS} \cdot (1 - \rho_{AUT}) \cdot \frac{e^{-\gamma_2 \ell_2}}{1 - \rho_{AUT} \rho_R e^{-2\gamma_2 \ell_2}} \tag{1}$$

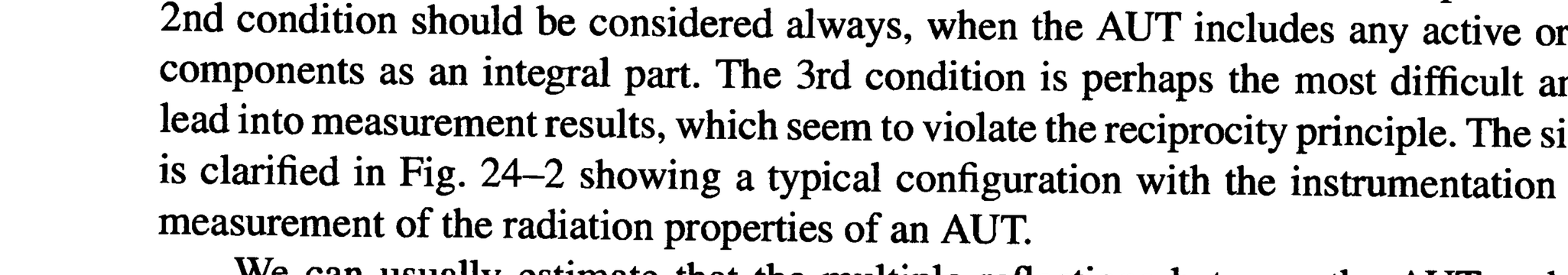


Figure 24–2
Typical instrumentation for the measurement of the radiation properties of Antenna Under Test (AUT).

where

V_R = voltage detected by the receiver from transmission line 2, V
V_T = voltage supplied by the transmitter into transmission line 1, V
γ_1 = complex propagation factor of transmission line 1 between the transmitter and source antenna, m^{-1}
γ_2 = complex propagation factor of transmission line 2 between the AUT and receiver antenna, m^{-1}
ℓ_1 = length of transmission line 1, m
ℓ_2 = length of transmission line 2, m
ρ_T = voltage reflection coefficient of the transmitter output
ρ_{SA} = voltage reflection coefficient of the source antenna
ρ_{AUT} = voltage reflection coefficient of the AUT
ρ_R = voltage reflection coefficient of the receiver input
t_{FS} = voltage transmission coefficient between the antenna terminals

Above it has been assumed that the voltage transmission coefficients of the transmitter output, source antenna input and receiver input are included in the calibration procedures into V_T, V_R and t_{FS}, respectively. Therefore, only the voltage transmission coefficient $1 - \rho_{AUT}$ of the AUT output is taken into account. The first and last term on the right side of (1) are the voltage transmission coefficients of the transmission lines including multiple reflections. Now, according to the reciprocity principle we can exchange the input and the output (V_R and V_T) if we fulfill the conditions above. However, this is not possible normally, because we have to exchange at least the transmitter (generator) and receiver and sometimes also parts of the cabling. In this case the exchange does not change the result significantly, if the numerators of the first and last term on the right side of (1) do not change much, which means that:

1. The reflection coefficients ρ_T and ρ_R and the cables (length, attenuation) are identical, which is seldom the case in practice, or
2. The reflections are small, i.e., $|\rho_T \rho_{SA}|$, $|\rho_R \rho_{AUT}|$, $|\rho_T \rho_{AUT}|$ and $|\rho_R \rho_{SA}| \approx 0$. This depends on the measurement system and can usually be approximated to be true only for a measurement where a calibrated vector network analyzer is used.

It can be seen from the above that problems occur especially when the matching of the antennas or the measurement devices is poor. In this case the exchange of the generator and receiver will change the result and it would appear that the reciprocity principle does not hold.

24–2b Near-Field and Far-Field（近场和远场）

Based on Huygens' principle it does not matter, where the sampling of the field takes place provided enough information is obtained on some surface surrounding the AUT. However, the measuring practices and constrains depend largely on the distance of this surface from the AUT (Hansen-1). It has been defined already in Sec. 2–13 that one can find several regions

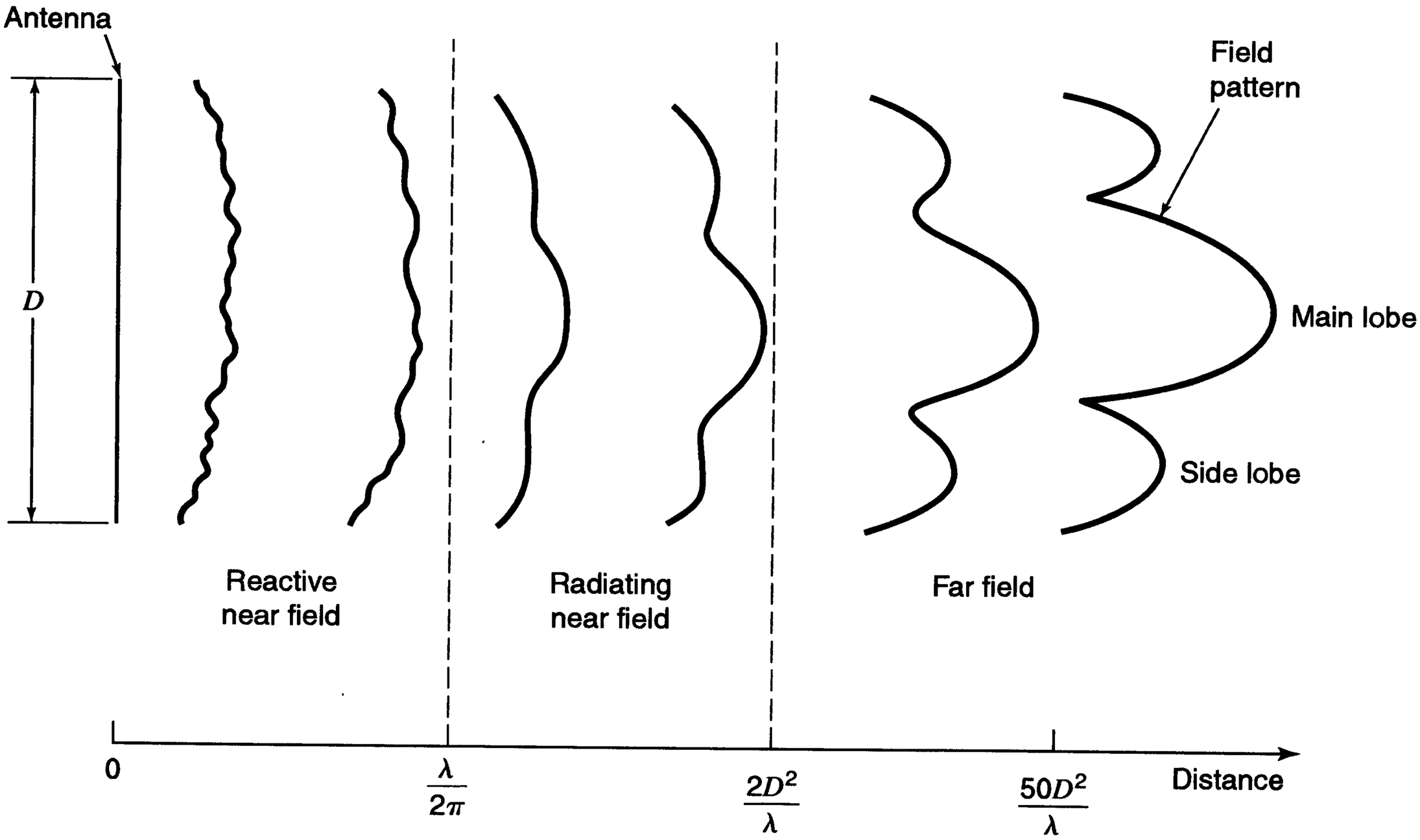


Figure 24–3
Radiation patterns in near-field and far-field regions.

of radiated field in the vicinity of the antenna. These are the reactive near-field region (radiansphere), the radiative near-field or Fresnel region, and the far-field or Fraunhofer region (Fig. 24–3).

As we are almost always interested in the radiation properties in the far field, it is obvious that the measurement usually also takes place in the far field. There are several advantages of the far-field measurement:

- The measured field pattern is valid for any distance in the far-field region; only simple transformation of the field strength according to $1/r$ is required.
- If a power pattern is required, only power (amplitude) measurement is needed.
- The result is not very sensitive to the changes in the location of the phase center of the antennas and thus the rotation of the AUT does not cause significant measurement errors.
- Coupling and multiple reflections between the antennas are not significant.

The main disadvantage of the far-field measurements is the required large distance between the antennas leading to large antenna ranges. The distance can be too large for a measurement in an anechoic chamber or it can result in atmospheric attenuation. In these cases one needs to consider measurements in the radiating near-field region, where so-called near-field measurements take place. The reactive or evanescent near-field region is seldom used for antenna measurements, because it normally is located too close to the AUT

(see below) and because mutual impedance caused by reactive coupling between the antennas makes the measurement complicated.

The borders of the regions around the AUT are defined based on dominances of certain properties of the field. As the dominance is a relative concept also these distances may vary depending on the definition, but below we give the most common values. The outer boundary of the reactive near-field region is typically defined to be at the distance beyond which the amplitude of the far-field component is larger than those of reactive near-field components. For small elementary dipoles, it is at the distance determined by the radius of the radian sphere, that is,

$$r_{rnf} = \frac{\lambda}{2\pi} \qquad \text{(m)} \tag{2}$$

where

r_{rnf} = distance to small AUT at the outer boundary of the reactive near field, m
λ = wavelength, m

For near-field measurements of larger antennas it can be estimated that the evanescent near fields are typically insignificant when the measurement distance is over 3λ (Slater-1).

The inner boundary of the far-field region is most often defined as the distance, where the curvature of the spherical wave front at the edge of the AUT causes a phase difference of $\pi/8$ radians (22.5°) compared to the center of the antenna, which is equal to a curvature of $\lambda/16$. This leads to the well-known Rayleigh distance giving the border between the Fresnel and Fraunhofer regions:

$$r_{ff} = \frac{2D^2}{\lambda} \qquad \text{(m)} \tag{3}$$

where

r_{ff} = distance to AUT at the inner boundary of the far-field region, m
λ = wavelength, m
D = largest dimension of the physical aperture of the antenna, m

In some cases the allowed error due to the curvature is smaller and thus larger far-field distances like $2r_{ff}$ are sometimes applied. For small antennas, the curvature criterion may not be adequate to define the far-field condition. Actually, when $D < \lambda/4$, r_{ff} is smaller than r_{rnf}! Thus in these cases additional criteria for the proper distance for far-field measurement must be used. One such criterion is that the assumed total amplitude of the near-field components should be sufficiently lower than that of the far-field components. If we assume that the near fields add in random phase to the far field and set the peak-to-peak ripple caused by this to ΔL (dB), we get an additional condition for the measurement distance:

$$r'_{ff} = \frac{r_{rnf}}{10^{\Delta L/40} - 1} \qquad \text{(m)} \tag{4}$$

where

r'_{ff} = distance to AUT with certain level of the reactive near fields, m
ΔL = peak-to-peak error caused by the reactive near fields, dB

In (4) we have assumed that the near-field components decay as $1/r^2$, where r is the distance to the AUT. Often the near-field level is required to be at least 35 dB below the far-field component. In this case the peak-to-peak ripple $\Delta L \approx 0.3$ dB and $r'_{ff} = 10\lambda$. Another far-field condition is based on the requirement that the change in the measurement distance r due to rotation of the AUT should not have a large effect on the result obtained. Setting the peak-to-peak limit of this effect to ΔL (dB), we get the condition

$$r''_{ff} = \frac{2D_m}{10^{\Delta L/10} - 1} \quad \text{(m)} \tag{5}$$

where

r''_{ff} = distance to AUT with certain effect of the rotation, m
D_m = largest mechanical dimension of the antenna, m
ΔL = peak-to-peak error caused by the rotation of AUT, dB

In (5) we have estimated that the peak-to-peak change of the distance between the phase center of the AUT and the source antenna is $D_m/2$.

EXAMPLE **24–2.1 Near- and Far-Field Distances for a Normal-Mode Helix on a Handheld Cellular Phone**

A normal-mode helical antenna is located on top of a handset at 900 MHz. The length of the antenna is 25 mm and the length of the handset chassis 110 mm. Find the distances where different near- and far-field criteria are met, when the required peak-to-peak measurement uncertainty due to proximity effects is 0.5 dB.

■ **Solution**

Now the maximum dimension of the antenna should be estimated to be the combined length of the phone chassis and the antenna element, so $D = D_m = 135$ mm. This is due to the fact that for a device smaller than half wavelength typically the whole metallic body is occupied by the currents inducing the radiation.

(*a*) The outer boundary of the reactive near field is at a distance given by Eq. (2):

$$r_{rnf} = (0.333/2\pi)\,\text{m} = 53\,\text{mm} \qquad \textit{Ans. (a)}$$

(*b*) The Fraunhofer region starts at a distance given by Eq. (3):

$$r_{ff} = (2 \cdot (0.135)^2/0.333)\,\text{m} = 110\,\text{mm} \qquad \textit{Ans. (b)}$$

(*c*) Now $\Delta L = 0.5$ dB and we get from Eq. (4) the minimum distance, where the effect of the near fields is small enough:

$$r'_{ff} = ((0.333/2\pi) \cdot 34.2)\,\text{m} = 1.8\,\text{m} \qquad \textit{Ans. (c)}$$

(*d*) Eq. (5) gives the minimum distance, where the effect of the rotation of the AUT is small enough:

$$r''_{ff} = ((2 \cdot 0.135) \cdot 8.20)\,\text{m} = 2.2\,\text{m} \qquad \textit{Ans. (d)}$$

EXAMPLE **24–2.2 Near- and Far-Field Distances for a Pyramidal Standard Gain Horn Antenna at 10 GHz**

The length of the horn is 350 mm and the aperture size 200 mm × 150 mm. For this precision antenna the required peak-to-peak uncertainty due to proximity effects is 0.2 dB.

■ **Solution**

Now $D = 200$ mm and $D_m = 350$ mm. Following the procedure in Example 24–2.1, we get:

$$(a)\ r_{rnf} = 4.8 \text{ mm} \qquad \textit{Ans. } (a)$$
$$(b)\ r_{ff} = 2 \cdot (0.2)^2/0.03 \text{ m} = 2.7 \text{ m} \qquad \textit{Ans. } (b)$$
$$(c)\ r'_{ff} = 0.0048 \cdot 86.4 \text{ m} = 0.41 \text{ m} \qquad \textit{Ans. } (c)$$
$$(d)\ r''_{ff} = 2 \cdot 0.35 \cdot 21.2 \text{ m} = 14.9 \text{ m} \qquad \textit{Ans. } (d)$$

We notice that as we now set a tight limit for ΔL, the measurement distance is determined by (5), if attention is not paid to keeping the movement of the phase center small.

EXAMPLE **24–2.3 Near- and Far-Field Distances for a Reflector Antenna with Diameter $D = 0.5$ m at 300 GHz**

■ **Solution**

Here it is obvious that $r_{ff} = (2 \cdot (0.5)^2/10^{-3})$ m $= 500$ m is determining the measurement distance. At 300 GHz the attenuation of the atmosphere is around 10 dB/km making (among other things) the measurement difficult in full-size ranges. In this case a compact antenna test range (see Sec. 24–4d) is typically applied.

24–2c Coordinate System（坐标系）

The IEEE standard spherical coordinate system is shown in Fig. 24–4 (IEEE-1). The AUT is at the origin. The *elevation* angle θ is measured from the z-axis (zenith). The *azimuth* angle ϕ is measured from the projection of the radius vector to the xy- (horizontal) plane with $\phi = 0$ at the x-axis increasing counterclockwise. Normally the coordinate system is defined based on the mechanical structure of the antenna typically so that the assumed direction of the peak radiation is on the x-axis.

Moving the source antenna along lines of constant ϕ or constant θ results in *conical cuts* or ϕ *cuts,* when θ is constant, and *great-circle cuts* or θ *cuts,* when ϕ is constant. The cut along the equator with $\theta = \pi/2$ belongs to both of these categories. Based on the assumed characteristics of the antenna normally two *principal-plane cuts* are defined. These are orthogonal great-circle cuts through the axis of the main lobe of the antenna. With linearly polarized antennas the cuts are selected to coincide with the assumed direction of the E and H fields in the main lobe and then they are called E- and H-*plane cuts*.

In measuring the polarization of the antenna a standard practice is to establish a local coordinate system for each measurement direction (θ, ϕ). Here the reference direction is

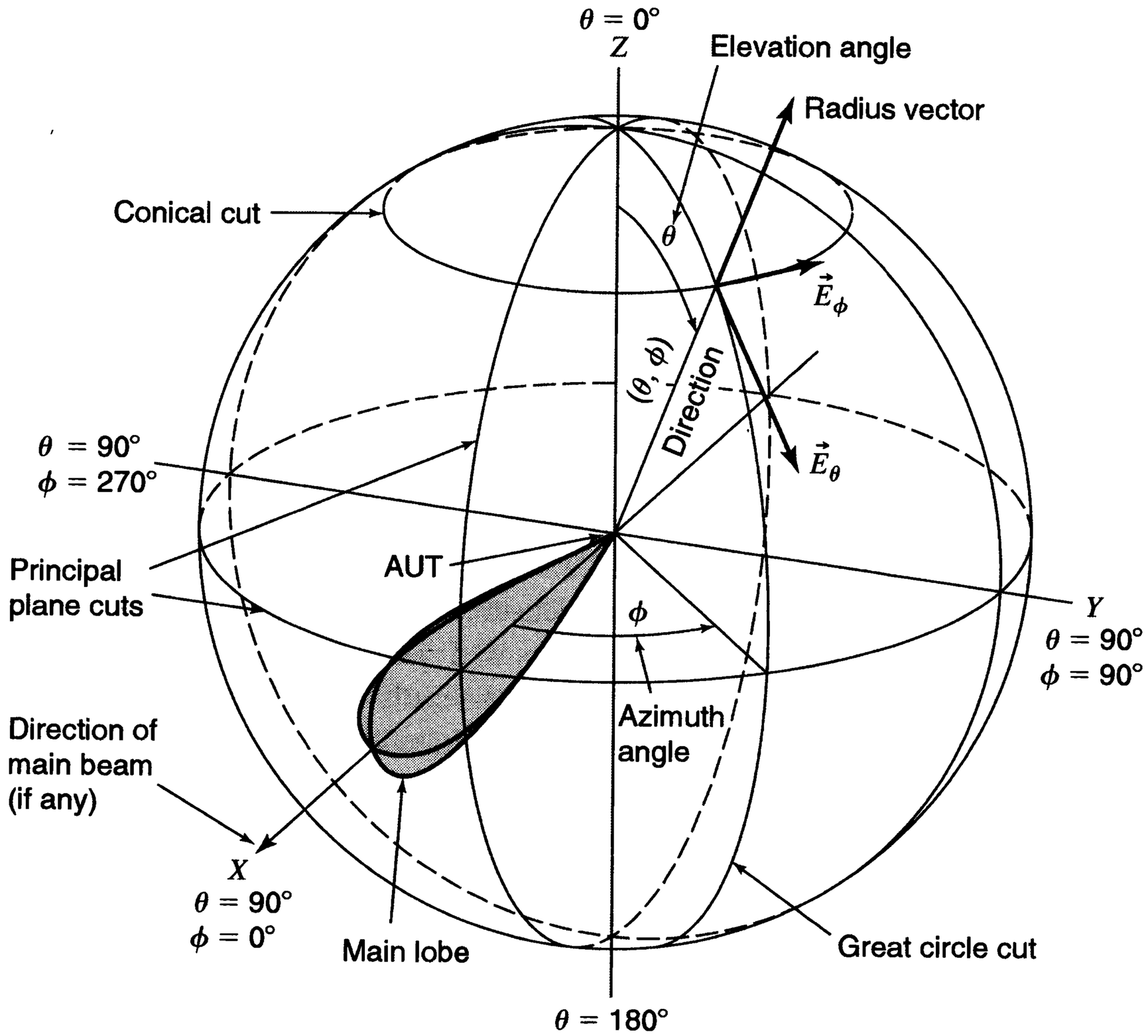


Figure 24–4
Standard coordinate system for antenna measurements showing conical, great circle and principal plane cuts.

normally selected to be along the unit vector u_ϕ of the standard spherical coordinate system of Fig. 24–4. At the z-axis separate definition for the reference direction is required for instance along the positive y-axis. The sense or handedness of the polarization is defined considering the AUT as the transmitter thus having the wave propagation direction (and observation direction) away from the AUT.

24–3 TYPICAL SOURCES OF ERROR IN ANTENNA MEASUREMENTS
(天线测量的典型误差源)

Any measured quantity has a margin of error. Thus, the complete value for the gain of an antenna might be 15 dBi ± 0.5 dB indicating a half decibel uncertainty. To reduce the measurement uncertainty to an acceptable level, the critical sources of error have to be recognized.

A pure plane wave (uniform phase and amplitude) is an ideal test field for the measurement of far-field patterns. However, there are inevitably deviations from the plane wave: insufficient distance between the antennas causes phase curvature and amplitude taper, reflections from surroundings cause amplitude and phase ripple. Phase curvature and amplitude taper can have a significant impact on the main beam, whereas ripples may spoil the

accuracy of side-lobe measurements. These test field imperfections are virtually indistinguishable from antenna errors and therefore, it is worth to spend a lot of expense and labor to reduce them.

Coupling to the reactive near field, alignment errors, interfering signals, effects of the atmosphere, leaking and radiating cables, instrumentation errors, etc., are further sources of error in antenna measurements.

24–3a Phase Error and Amplitude Taper Due to Finite Measurement Distance（有限测试距离所致的相位误差和幅度锥削）

Let us assume that the AUT is a planar antenna, which is receiving a wave coming from the direction of the main beam axis. If the measurement distance is too small, the fields received by different parts of the AUT will not be in phase and there will be a quadratic phase error (Fig. 24–5). At the far-field limit $2D^2/\lambda$, the phase difference between the aperture edge and center is 22.5°. Doubling the measurement distance halves this phase error.

Due to the phase error, the measured gain is smaller and the side lobes are higher than in the ideal plane wave case. Also the nulls of the radiation pattern become filled. For measuring antennas having moderate side-lobe levels (down to about −30 dB), the distance $2D^2/\lambda$ is usually adequate. At this distance, the measured gain is about 0.06 dB smaller than the real far-field gain. If the measurement distance gets shorter, the measurement errors increase rapidly and the near-in side lobes merge with the main beam and either appear as shoulders or disappear altogether.

For low-side-lobe (−30 dB and less) antennas, the $2D^2/\lambda$ distance may be inadequate (Hacker (1) and Schrank). The near side lobes are especially sensitive to phase error. The error on the first side lobe is much larger than the errors on the second and third side lobes. For antennas having very low side lobes, phase errors of 5° or less may be required. However, for measurements of the far-out side lobes larger phase errors may be tolerated.

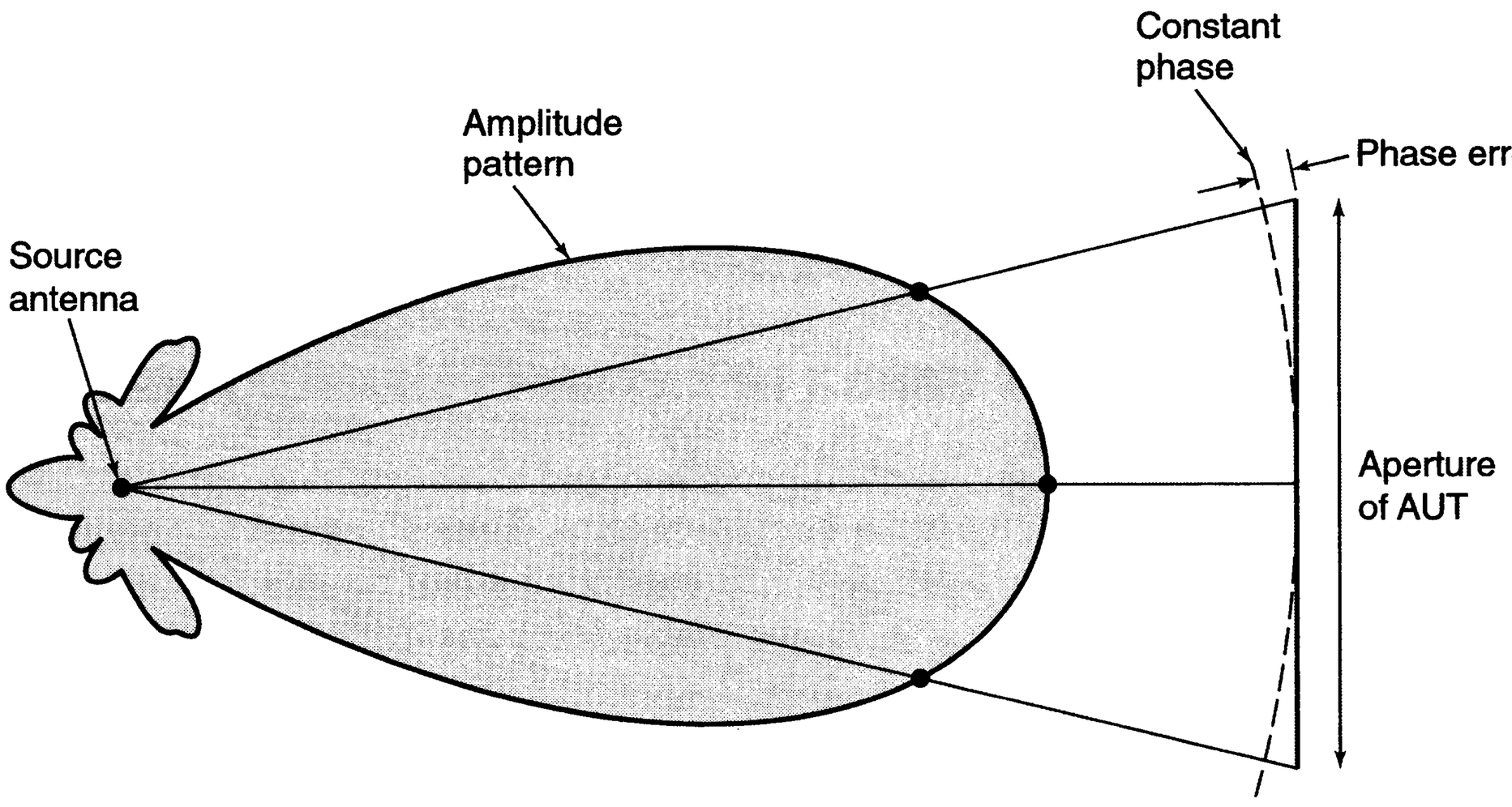


Figure 24–5
Phase error and amplitude taper across the aperture of an AUT.

The finite measurement distance also causes transverse and longitudinal amplitude errors. In the transverse plane, the field amplitude is smaller at the edges of the AUT due to the source antenna pattern and a slightly larger distance to the edge (Fig. 24–5). Usually, an amplitude taper of 0.25 dB is acceptable. This taper decreases the measured gain by about 0.1 dB and causes small errors on the close-in side lobes. In principle, the test field amplitude could be made more uniform by using a source antenna having a broader beamwidth. However, this may not be wise in practice because more power will be reflected from surroundings, which in turn may cause unacceptably large field ripples. The longitudinal amplitude taper causes errors in the measurement of long end-fire antennas.

24–3b Reflections（反射）

Reflections from surroundings produce field variations (amplitude and phase ripples) in the test zone as the direct wave and reflected waves interfere. The ripple length may be of the order of wavelength because the path length difference of the waves changes rapidly as the function of position.

Even small reflected waves may cause large measurement errors, because the fields of the waves are added, not the powers. For example, a reflected wave which has a field 20 dB below the direct wave field (1% of the power density of the direct wave) causes an error of $-0.92 \ldots +0.83$ dB ($-21 \ldots +19\%$ error in power) depending on the phase difference between the direct and reflected field. The error limits in phase measurement are $\pm 5.7°$. If the reflected wave is 40 dB below the direct wave, the error limits in the measured amplitude and phase are ± 0.09 dB and $\pm 0.6°$, respectively.

Reflections are especially harmful in the measurement of low side lobes. A small reflection coupled to the AUT through the main lobe may completely mask the direct wave coupled through the side lobe. If the coupled direct and reflected waves are equally strong, the measured side-lobe level may be 6 dB too high or there may even be a null in the measured pattern.

On outdoor ranges, waves are reflected from the ground and close-by objects like trees and buildings. In anechoic chambers, reflections come from the walls, floor, and ceiling. Also the reflections from the antenna support structures and multiple reflections between the antennas may be significant. Different ways to reduce the reflections or their influence are described in Sec. 24–4. Imaging the sources of reflections is treated in Sec. 24–4f.

24–3c Other Sources of Error（其他误差源）

Phase error, amplitude taper, and ripples due to reflections are deviations from the ideal test field. In addition to these common measurement range imperfections, there are some other possible sources of error:

- Coupling to the reactive near field may be significant at low frequencies. Coupling is considered negligible when the distance is larger than 10λ [see Eq. 24–2–4], but at VHF band and lower frequencies this limit may correspond to a large distance.
- Antenna measurements are three-dimensional vector field measurements. Therefore, many kinds of alignment errors are possible. Careless alignment of the source antenna may increase the amplitude error of the test zone and make the illumination

asymmetrical. Correct source alignment is important also in the measurement of low cross-polar levels. To make proper pattern cuts, the AUT should be aligned correctly with respect to the axes of the positioner, i.e., the coordinate systems of the AUT and positioner should coincide. The phase center of the AUT and the axis of rotation should coincide. Otherwise, the AUT will wander in the test zone during the rotation.

- Man-made interfering signals may couple to the sensitive receiver especially on outdoor ranges. The interfering signal may be at the measurement frequency or at some other frequency at which the receiver has significant response, as at some harmonic sideband frequency of the input mixer.
- At large measurement distances, the effects of the atmosphere may be considerable. Variations of the refractive index can cause scintillation and multipath propagation. At millimeter and submillimeter wave frequencies, the attenuation of the atmosphere is high and, thus, there may be noticeable amplitude variations during the measurement.
- Incorrect use of cables may cause errors. Cables which have insufficient shielding may leak and act as antennas. Also unbalanced transmission lines can radiate as currents may flow on the outer surface of the cable.
- Impedance mismatches between the instruments and antennas may cause errors in gain measurements.
- Imperfections of the transmitter, receiver, and positioner cause measurement errors. However, in most measurements these instrumentation errors are negligible.

24–4 MEASUREMENT RANGES（测量场地）

Generally, one wants to know the far-field performance of an antenna, since the majority of antennas operate in this mode. In a basic far-field measurement, the AUT is separated from the source antenna by a distance that is at least $2D^2/\lambda$. Small antennas can be measured indoors in an anechoic chamber. Often, however, far-field ranges have to be physically large outdoor ranges. At low frequencies, the ground reflection cannot be avoided. In ground-reflection ranges the direct wave and the reflected wave interfere in a controlled manner. Compact antenna test ranges and near-field ranges avoid the problem of large size and can be installed indoors. For the measurement of large millimeter and submillimeter wave antennas having far-field distances of several kilometers or more, a compact antenna test range or a near-field range is a necessity.

Which range type suits best for the measurement of a certain antenna depends mainly on the physical size and frequency of the antenna. Sometimes, the best measurement site is the environment where the antenna is used. For very large fixed antennas, tests using an airborne vehicle can be made. Satellites and celestial radio sources can be used for some tests (Sec. 24–5b).

Radar Cross-Section (RCS) measurements have much in common with antenna measurements: a target illuminated by a plane wave acts like a certain field distribution or an antenna. Therefore, many antenna test ranges, especially compact antenna test ranges, and their instrumentation are used for RCS measurements. Correspondingly, radar techniques are applicable in some antenna measurements (Sec. 24–5b).

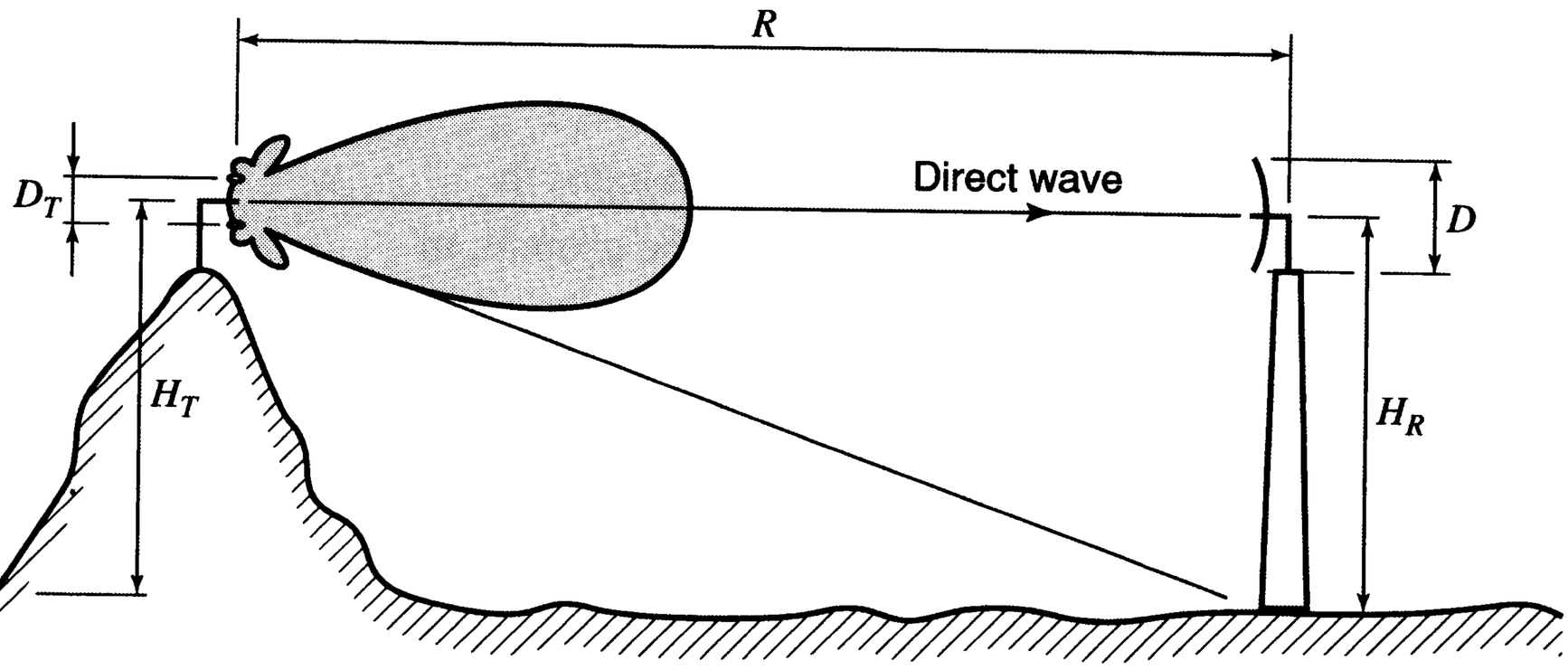


Figure 24–6
Elevated range to minimize reflected waves.

24–4a Elevated Ranges（架高场地）

The basic far-field range is an elevated range (Fig. 24–6). Antennas are placed high on towers, buildings or hills to reduce the effects of environment. In most cases, the AUT is operated as a receiving antenna. If the source antenna is near the ground and the AUT is on a tower, the range is called a slant range.

The range length is usually determined by the far-field criterion. For the measurement of low-side-lobe antennas even longer ranges than $2D^2/\lambda$ may be needed. The range width should be sufficient to keep the main beam of the source antenna within it. However, the land area needed for a long and wide test range may be impractically large and expensive. Then a shorter range must be used.

If the errors due to the phase curvature of the short distance cannot be tolerated, the measurement results can be partly corrected by computer using Fourier transforms. The aperture field of the AUT is calculated from the measured complex (amplitude and phase) pattern, then the aperture phase is corrected, and from the corrected aperture field the corrected far-field pattern is calculated. Defocusing is another technique to obtain far-field patterns when the measurement distance is too short. This technique cannot be applied to all antennas. A reflector antenna can be defocused from infinity to a finite measurement distance by displacing its feed. For ranges as short as $D^2/(8\lambda)$, the measured main lobe is close to the far-field pattern, but side lobes cannot be measured accurately.

The antenna heights and the source antenna pattern should be selected so that the main beam of the source antenna does not illuminate the ground between the antennas. Also low side lobes are a desirable feature for the source antenna to keep ground reflections low. If the measurement distance is large, it may be difficult to place the antennas high enough. The lower limit for the source antenna diameter, D_T, follows from the requirement that the first null of the pattern should fall above the base of the test tower:

$$D_T \geq \frac{1.5\lambda R}{H_R} \qquad \text{(m)} \tag{1}$$

where

λ = wavelength, m
R = separation between the antennas, m
H_R = height of the AUT above the ground, m

A narrow beam of the source antenna leads to a low level of reflections but at the cost of a large-amplitude taper in the test zone. From the maximum accepted amplitude taper of 0.25 dB follows the upper limit for the source antenna diameter:

$$D_T \leq \frac{\lambda R}{3D} \qquad \text{(m)} \tag{2}$$

where

λ = wavelength, m
R = separation between the antennas, m
D = diameter of the AUT, m

Combining the two above-mentioned requirements for the source antenna diameter gives that the AUT should be mounted at least five times as high as its diameter D.

It is not possible to eliminate completely the illumination of the ground. Problems may arise especially in the measurement of elevation cuts when the main beam of the AUT is pointing at the ground. However, it is possible to make the measurement without pointing the AUT below the horizon: the AUT is turned upside down after half of the pattern has been measured. For the measurement of low back lobes, it is advantageous to place the AUT higher than the source antenna.

The level of the reflected field in the test zone can be reduced with diffraction fences which are metallic screens. Fences intercept waves that would be normally reflected from the ground toward the test zone and reflect them skyward and away from the AUT. Diffraction from the edge of the fence to the test zone should be avoided. Often several fences are used. Time-domain techniques to reduce the influence of reflections are discussed in Sec. 24–4g.

EXAMPLE **24–4.1 Amplitude Taper on Test Range**

The source antenna and the AUT are both parabolic reflector antennas, which have a diameter of 1 meter. The operating frequency is 10 GHz and the separation of the antennas is $2D^2/\lambda$. Estimate the amplitude taper.

■ **Solution**

The half-power beamwidth of a parabolic reflector is about $\theta_{3\,\text{dB}} = 70°\lambda/D = 2.1°$. At the measurement distance $2D^2/\lambda = 66.7$ m, this angle corresponds to a transverse distance of $2.1 \cdot \pi \cdot 66.7/180\,\text{m} = 2.44\,\text{m}$. It can be assumed that the shape of the main beam (measured in decibels) is nearly parabolic. Therefore, the pattern level of the source antenna is $[0.5/(2.44/2)]^2 \cdot (-3\,\text{dB}) = -0.50\,\text{dB}$ at the edge of the AUT. It can be verified also from Eq. (2) that the source antenna diameter (1 m) is too large if the maximum allowed edge taper is 0.25 dB.

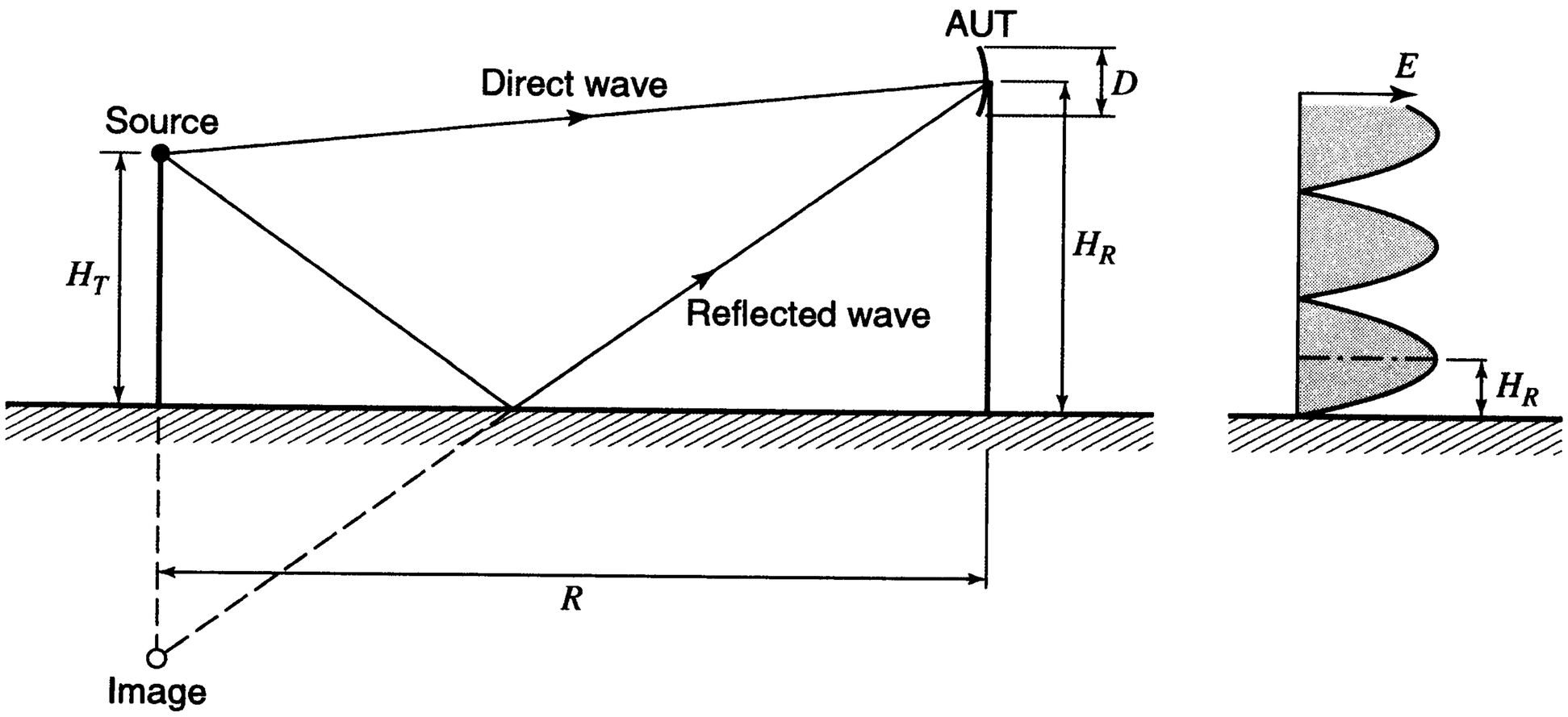


Figure 24–7
Ground-reflection range with direct and reflected waves.

24–4b Ground-Reflection Ranges（地面反射的场地）

At VHF and lower frequencies, ground reflections are difficult to avoid because a directional source antenna is very large and a ground-reflection range (Fig. 24–7) can be used.

Antennas are placed above a flat reflecting surface with (or without) buried ground screen. The specular reflection appears to come from the image source. The real and image source form together an interference pattern. The AUT is placed in the first lobe of the interference pattern. This lobe has the flattest amplitude distribution. Assuming that the image is negative, i.e., the reflection coefficient of the surface is -1, the peak of the first lobe is at the height

$$H_R = \frac{\lambda R}{4H_T} \qquad \text{(m)} \tag{3}$$

where

λ = wavelength, m
R = separation between the antennas, m
H_T = height of the source antenna, m

This equation indicates that the ideal height is a function of wavelength. Therefore, the height of the source antenna or AUT may need to be adjusted for each measurement frequency. The height of the AUT should be at least $3.3D$ to make sure that the amplitude taper is less than 0.25 dB.

In practice, the reflection coefficient at horizontal polarization is about -1 at small grazing angles ψ ($=90°$—angle of incidence). However, with vertical polarization the reflection coefficient may vary rapidly as the function of grazing angle. This polarization dependency of the reflection coefficient may give troubles in polarization measurements.

The surface of a ground-reflection range should be smooth, which requires that the rms height variation of the surface, Δh, is small enough:

$$\Delta h \leq \frac{\lambda}{M \sin \psi} \qquad \text{(m)} \tag{4}$$

where

λ = wavelength, m
M = smoothness factor (16, for example)
ψ = grazing angle

24–4c Anechoic Chambers and Absorbing Materials（吸波室与吸波材料）

The walls, ceiling and floor of an anechoic chamber are covered completely with absorbing material (Emerson-1). An anechoic chamber simulates a reflectionless free space and allows all-weather antenna measurements in a controlled laboratory environment. In an anechoic chamber, the test area is isolated from interfering signals much better than at outdoor ranges. The isolation can be improved further by shielding. Shielded chambers are suitable also for electromagnetic compatibility (EMC) measurements.

Anechoic chambers can be used for the far-field measurements of small antennas. Often the end wall can be opened and the anechoic chamber may be combined with an outdoor range. Large antennas can be measured on compact antenna test ranges and near-field ranges, which are usually installed in anechoic chambers. Complete lining of the chamber with absorbers is then not necessary.

Absorbing Materials.[1] Absorbing materials are an integral part of antenna technology. They are used both in measurement ranges and also as antenna components for reducing side-lobe and back-lobe radiation.

An ideal absorber provides an impedance match for incoming waves at all frequencies and angles of incidence. A tapered impedance transition from free space to the back of the absorber can be obtained by changing the resistivity of the medium gradually or by geometric shaping.

The broadband absorbers used in anechoic chambers are usually made of carbon-loaded polyurethane foam. Pyramids and wedges are widely used shapes (Fig. 24–8). Pyramids work best at normal incidence, and they scatter as a random rough surface if they are large compared to the wavelength (DeWitt-1). At microwave frequencies, the reflection coefficient may be below −50 dB at normal incidence, if the length of the absorber is at least a few wavelengths. At higher frequencies, the reflection coefficient is larger (Lehto-1), whereas at lower frequencies, the absorbers have to be very thick. Near grazing incidence pyramidal absorbers give large backscattered fields. On the other hand, wedges work well at large incident angles with the wedge direction along the plane of incidence but not as well as the pyramids at normal incidence.

---

[1] See Appendix D.

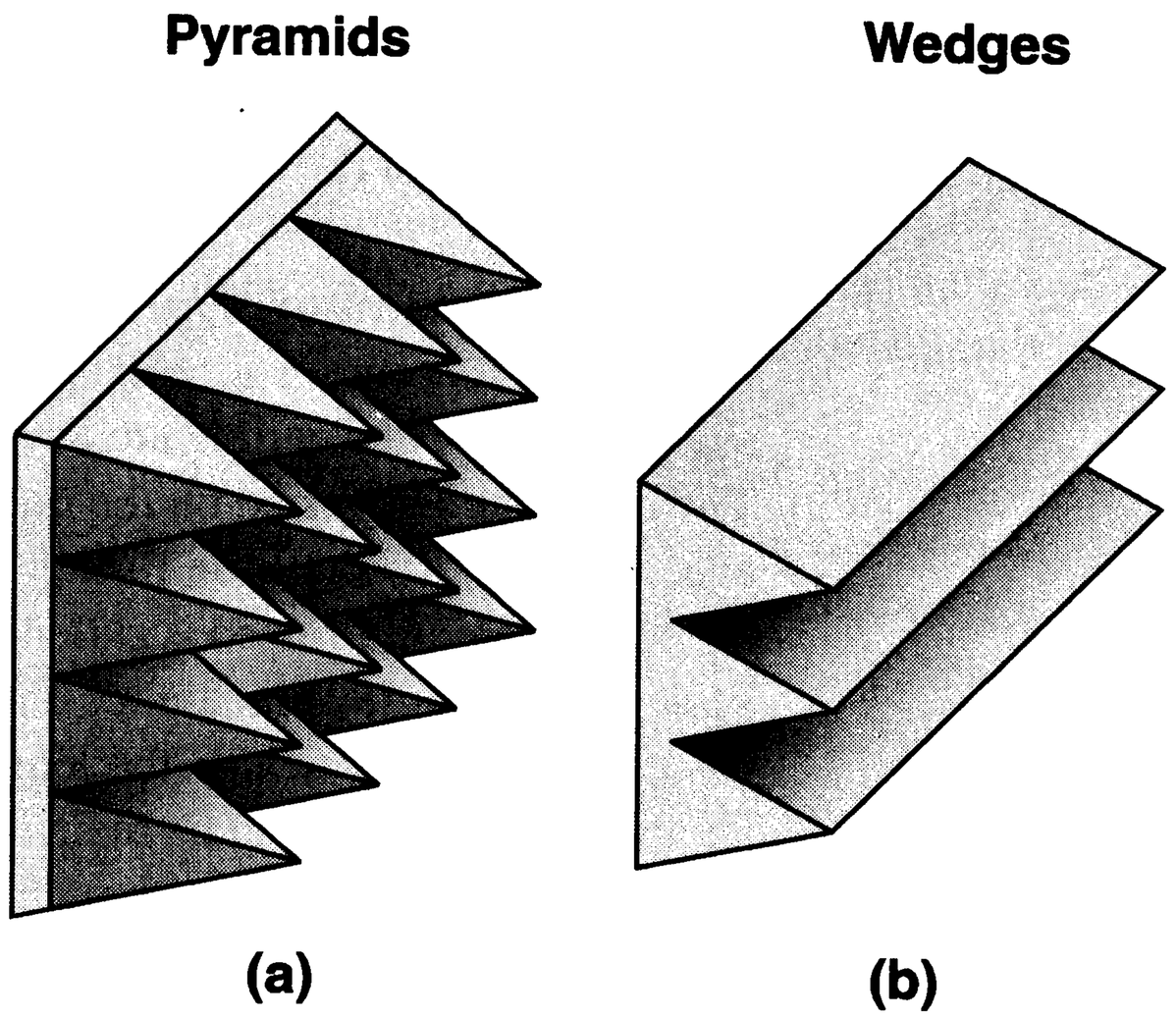


Figure 24–8
Absorbers: (*a*) pyramids, (*b*) wedges.

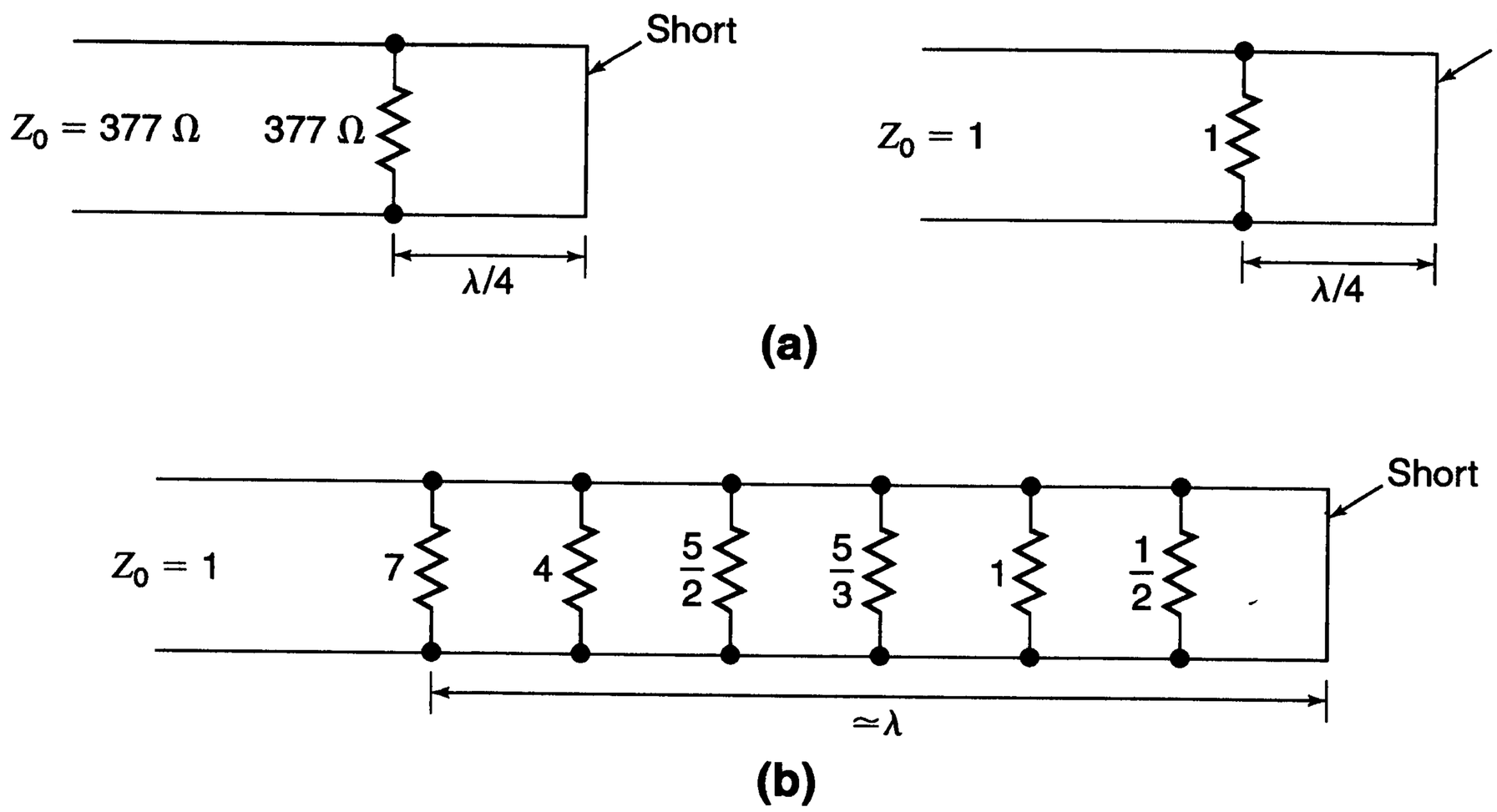


Figure 24–9
Transmission line equivalent of (*a*) Salisbury screen, (*b*) Jauman absorber.

Other ways to make absorbers are magnetic materials with losses, *Salisbury screens,* and *Jauman absorbers*. If a medium has both magnetic and electric losses and the relative permeability and permittivity are equal ($\mu_r = \varepsilon_r$), the wave impedance of the medium is equal to the free space impedance (377 Ω). A wave incident on such medium can enter it without reflection.

The Salisbury screen has a resistive sheet placed $\lambda/4$ from a reflecting plate. If the sheet has a surface impedance of 377 Ω per square, a normally incident wave is completely absorbed. The transmission line equivalent of the Salisbury screen is shown in Fig. 24–9*a*. The reflection coefficient varies rapidly as the function of frequency and angle. The Jauman

absorber has a broader bandwidth than the Salisbury screen. It is made by stacking resistive sheets. The surface impedances and spacings of the sheets increase with distance from the short as shown in the equivalent circuit in Fig. 24–9*b*.

Anechoic Chamber Types. Anechoic chambers are usually rectangular or tapered. Figure 24–10*a* shows a rectangular chamber. The end walls and the center parts of the sidewalls, floor and ceiling are covered with pyramids. Other parts are covered with wedges. The antennas are placed on the middle line of the chamber; the source antenna close to one end wall, the AUT a little further away from the other end wall. The test zone where the reflections are minimized is called the quiet zone. The dimensions of the chamber should be such that the angle of incidence on sidewalls is less than 60°. At larger angles the reflections would be large. Typically, the length to width (or height) ratio is 2 : 1. The source antenna should be chosen so that its main beam does not illuminate the sidewalls, ceiling, and floor.

At frequencies below about 1 GHz, the rectangular chamber having absorbers of reasonable size has a high level of reflections. Then a tapered chamber works better. In a tapered chamber (Fig. 24–10*b*), the source antenna is close to the apex of the tapered section and the specular reflections occur close to the source. The phase difference of the direct wave and the specular reflections changes slowly in the quiet zone which results in a more planar wavefront than in the case of a rectangular chamber. At higher frequencies, the source is moved from the apex closer to the rectangular section and the chamber is used as a normal rectangular chamber. Sec. D–1 has more on absorbing materials with two worked examples.

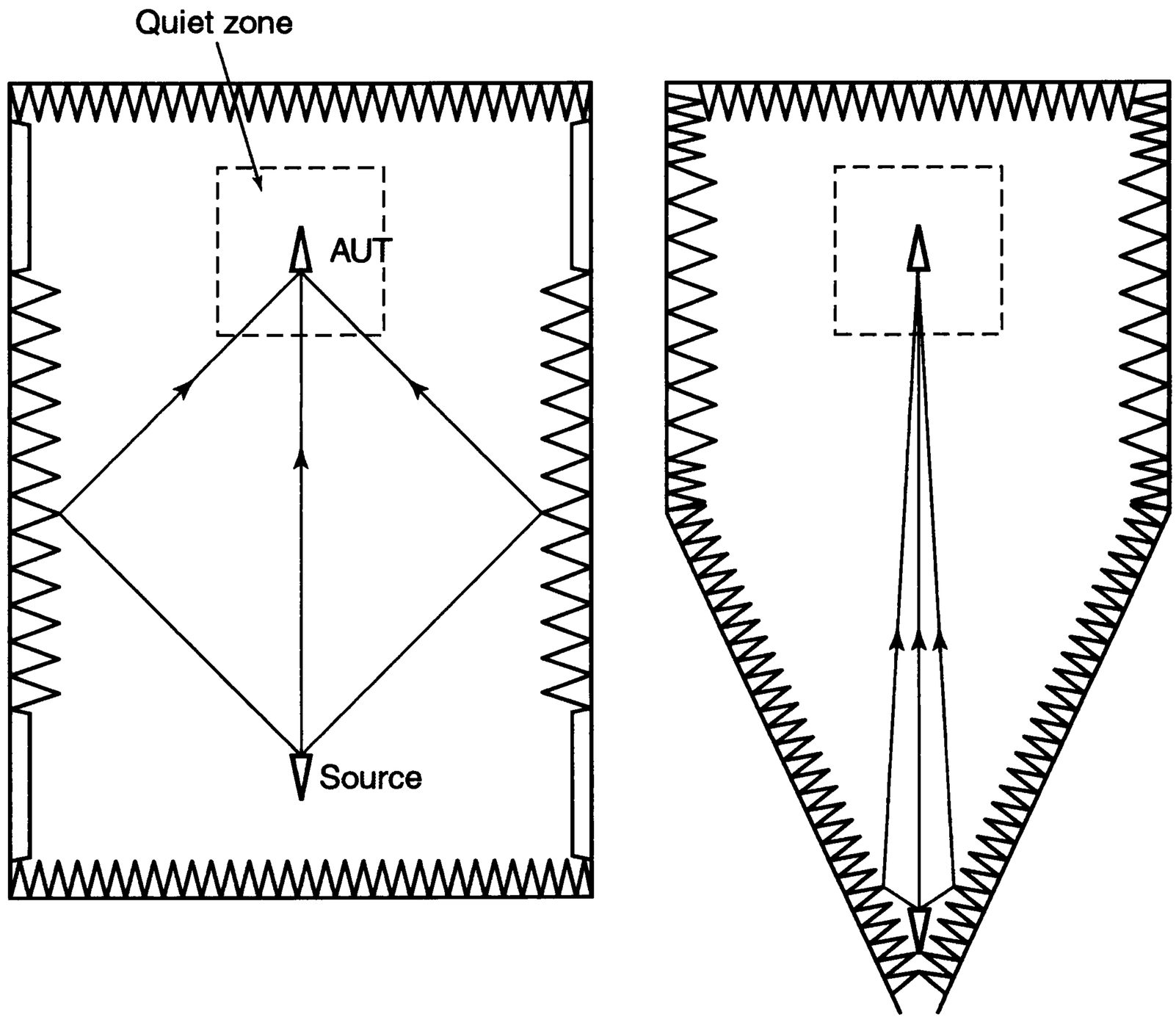


Figure 24–10
Anechoic chambers:
(*a*) rectangular, (*b*) tapered.

24–4d Compact Antenna Test Ranges (CATRs)（紧缩的天线测量场地）

Compact antenna test ranges (CATRs) simulate an infinite range length by producing a flat phase front with a reflector, lens, horn, array, or hologram (Johnson-1). Most CATRs are based on one or more reflectors. A CATR is usually installed indoors in an anechoic chamber but very large CATRs are outdoor ranges. Even some radio telescopes can be used as a CATR. See Fig. 20–8.

Figure 24–11 shows a CATR based on an offset-fed parabolic reflector, which transforms the spherical wave radiated by the feed to a plane wave in front of the reflector. The quiet zone diameter is limited by diffraction and is about a third of the reflector diameter. Because the beam is collimated, the required power is less than on a far-field range.

Room reflections, amplitude taper, edge diffraction, feed leakage, and depolarization inherent for the offset geometry limit the quality of the test field. The direct radiation from the feed to the test zone should be suppressed by placing absorbers between the feed and test zone. At high frequencies, also the surface accuracy of the reflector is of importance. The surface accuracy should be $\lambda/100$ or better.

The waves diffracted from the reflector edge into the test zone may cause large ripples. For low diffraction, the illumination of the reflector edge should be low, but this accompanies a large-amplitude taper. Therefore, the feed antenna should have a shaped pattern. Corrugated horns have symmetric patterns, low side lobes, and low cross-polarization levels, and are used widely as the feed. An array feed is more complicated but allows better control of the beam shape. Beam shaping can be accomplished also by feeding a subreflector with a horn antenna.

In addition to beam shaping, diffraction can be reduced by edge treatment. Three methods are in use: serrating the edge (Fig. 24–12*a*), rolling the edge (Fig. 24–12*b*), and resistive tapering. The serrated edge should be designed to keep the edge diffracted fields outside the quiet zone. The rolled edge illuminates the chamber walls more than the serrated edge but produces better quiet zone field (Lee-1). The transition from the parabola to the rolled edge should be very smooth.

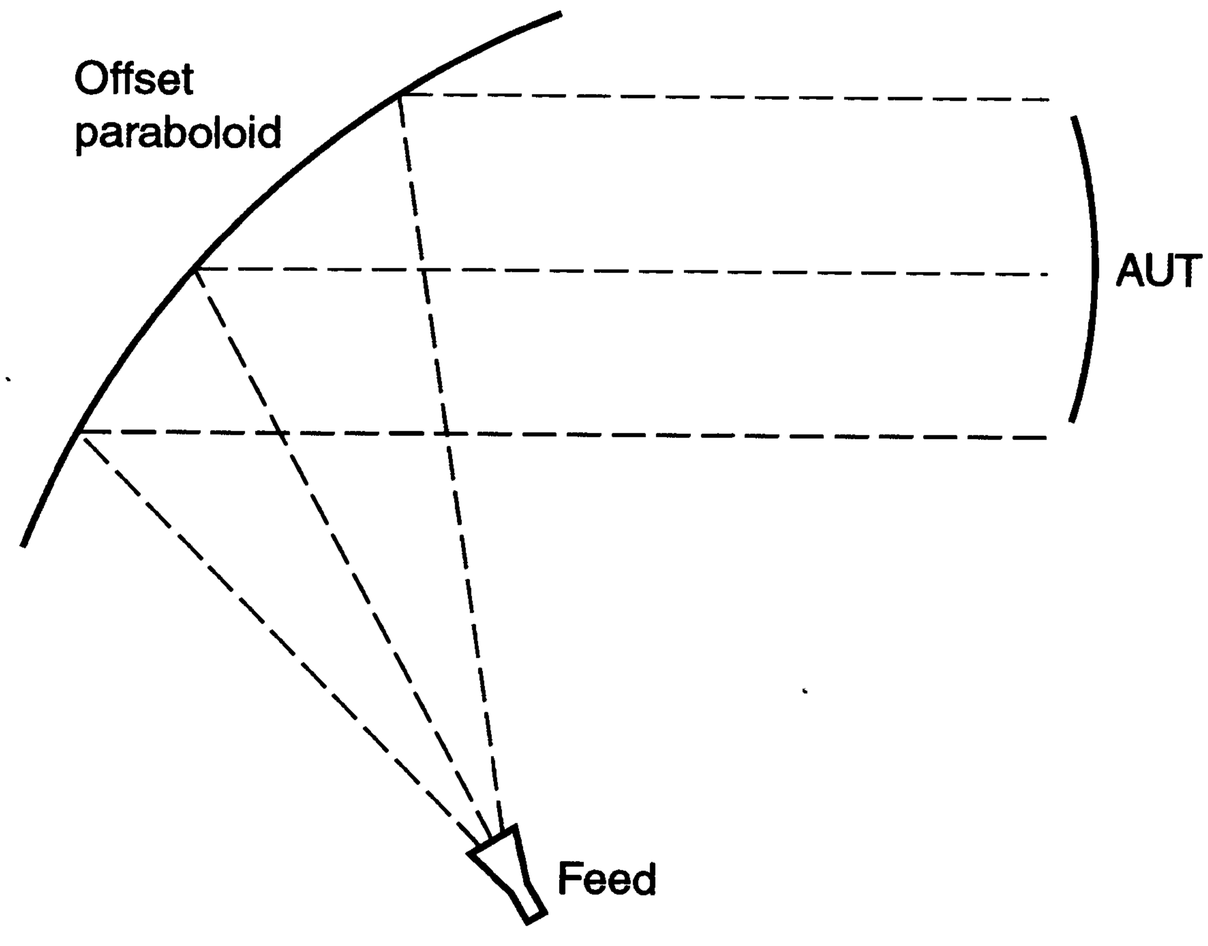


Figure 24–11
Compact antenna test range.

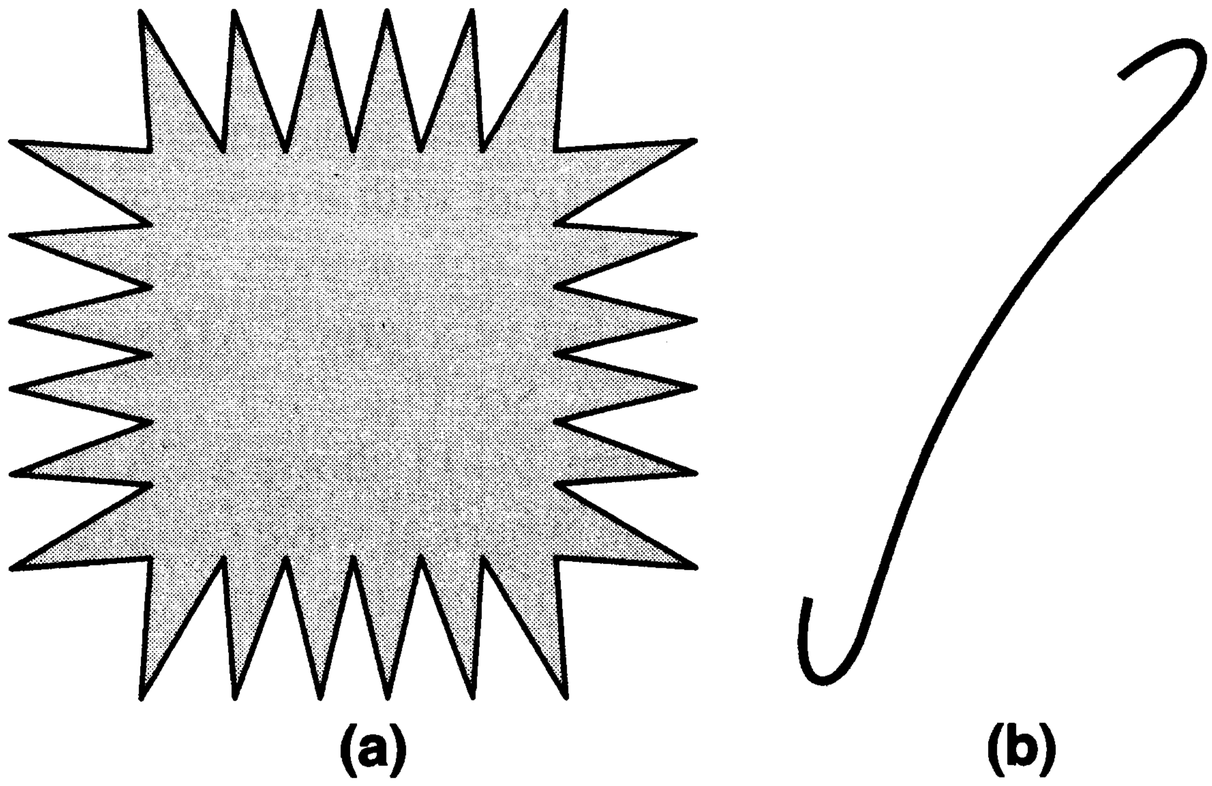


Figure 24–12
Edge treatment of the CATR reflector: (*a*) serrated edge, (*b*) rolled edge.

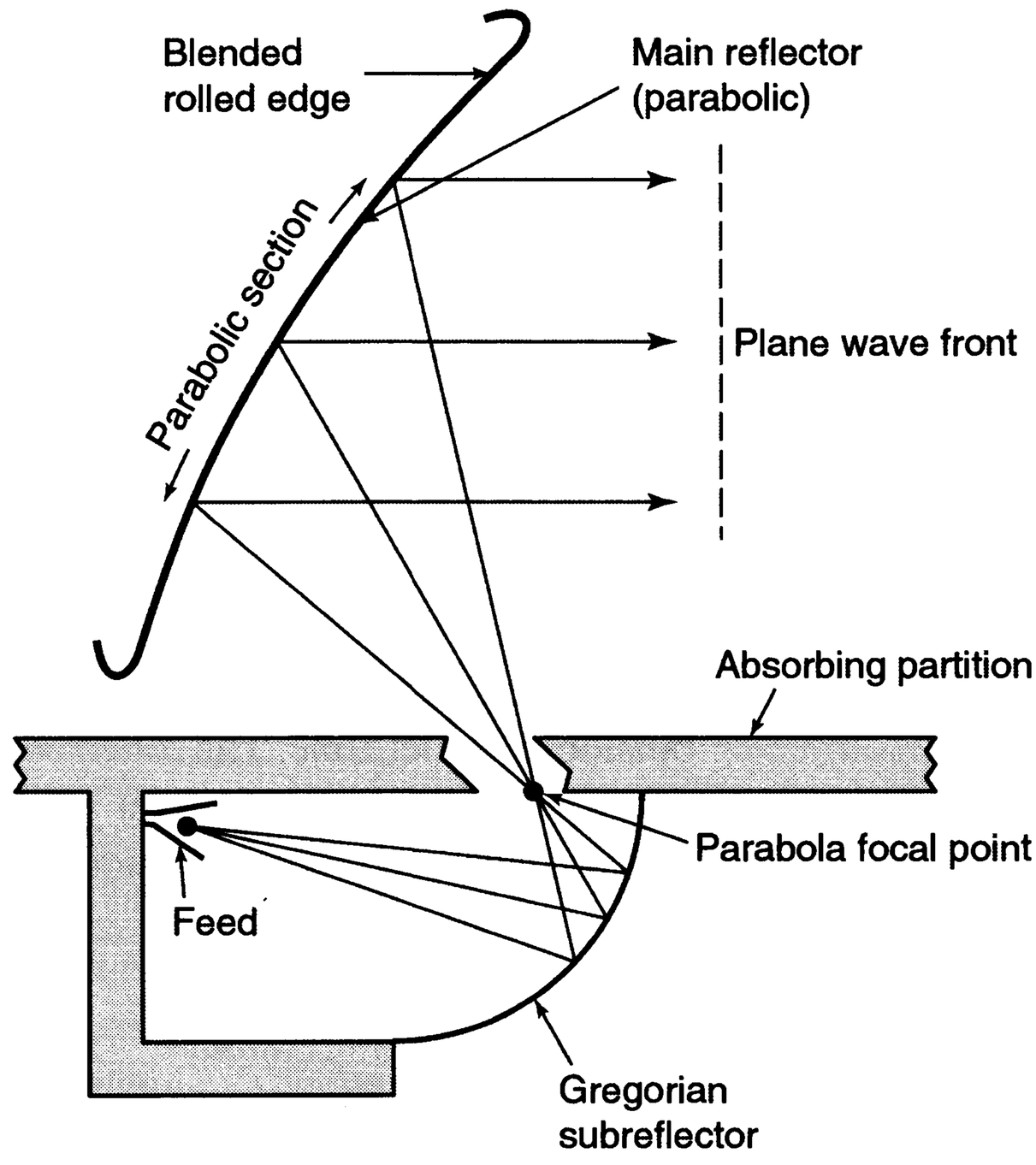


Figure 24–13
Dual-chamber Gregorian-fed CATR.

Dual-reflector CATRs provide some advantages compared to the basic single-reflector CATR: a larger quiet zone for a given reflector size and cancellation of cross-polarization. There are many possible dual-reflector configurations. A CATR may base on two orthogonally placed cylindrical reflectors, one of which collimates in azimuth and the other in elevation. The special feature of the Gregorian type dual-reflector system is that the reflectors can be placed in two chambers having a small opening between them as shown in Fig. 24–13 (Pistorius-1). With this arrangement, the fields diffracted from the edge of the subreflector and the feed spillover can be kept out of the quiet zone.

Figure 24–14 shows a CATR which has a hyperbolic subreflector and parabolic main reflector (Steiner-1). The geometry does not produce any cross-polarization. The size of the main reflector is 7.5 m × 6 m without the serrations, and the diameter of the quiet zone is

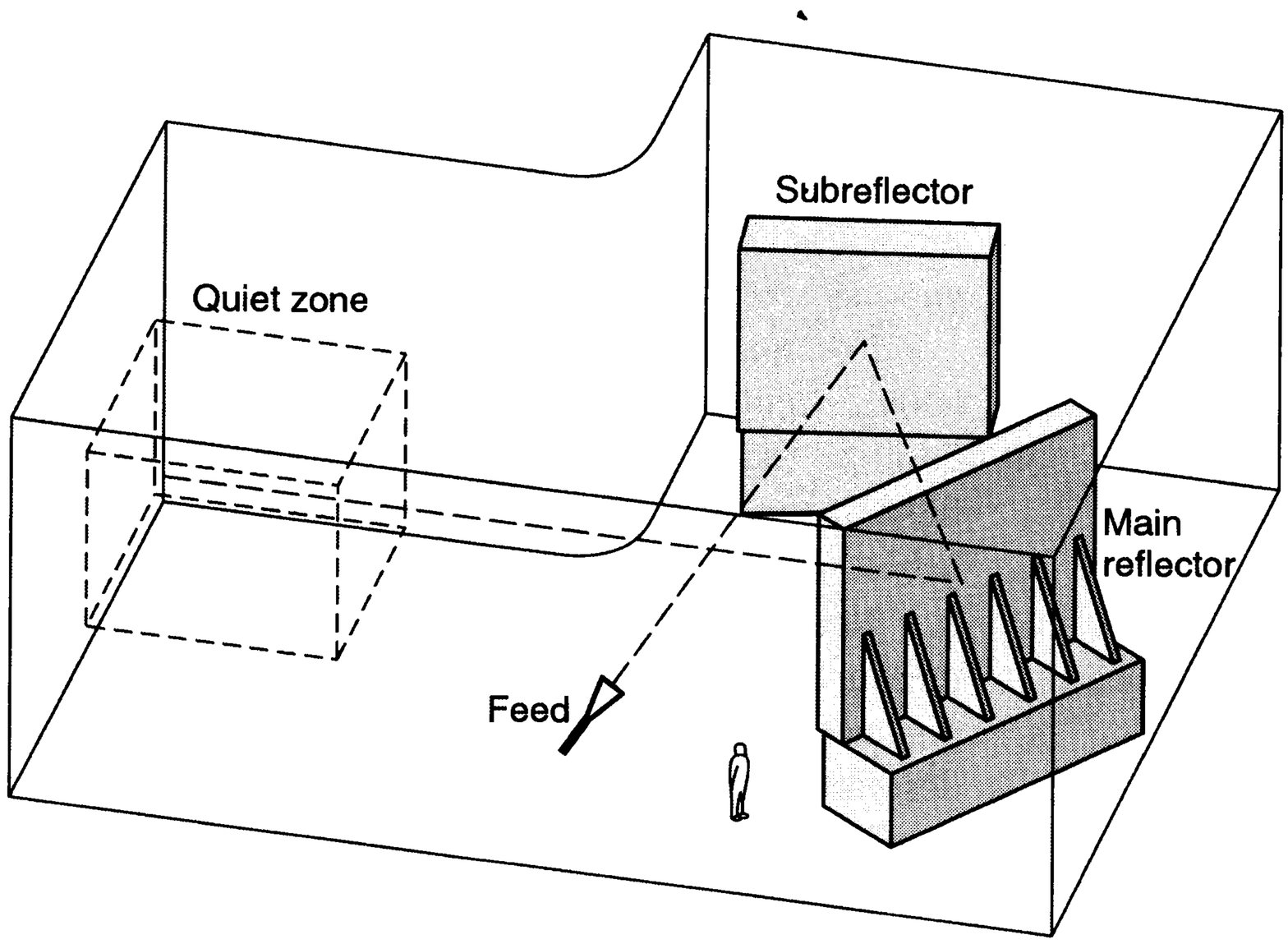
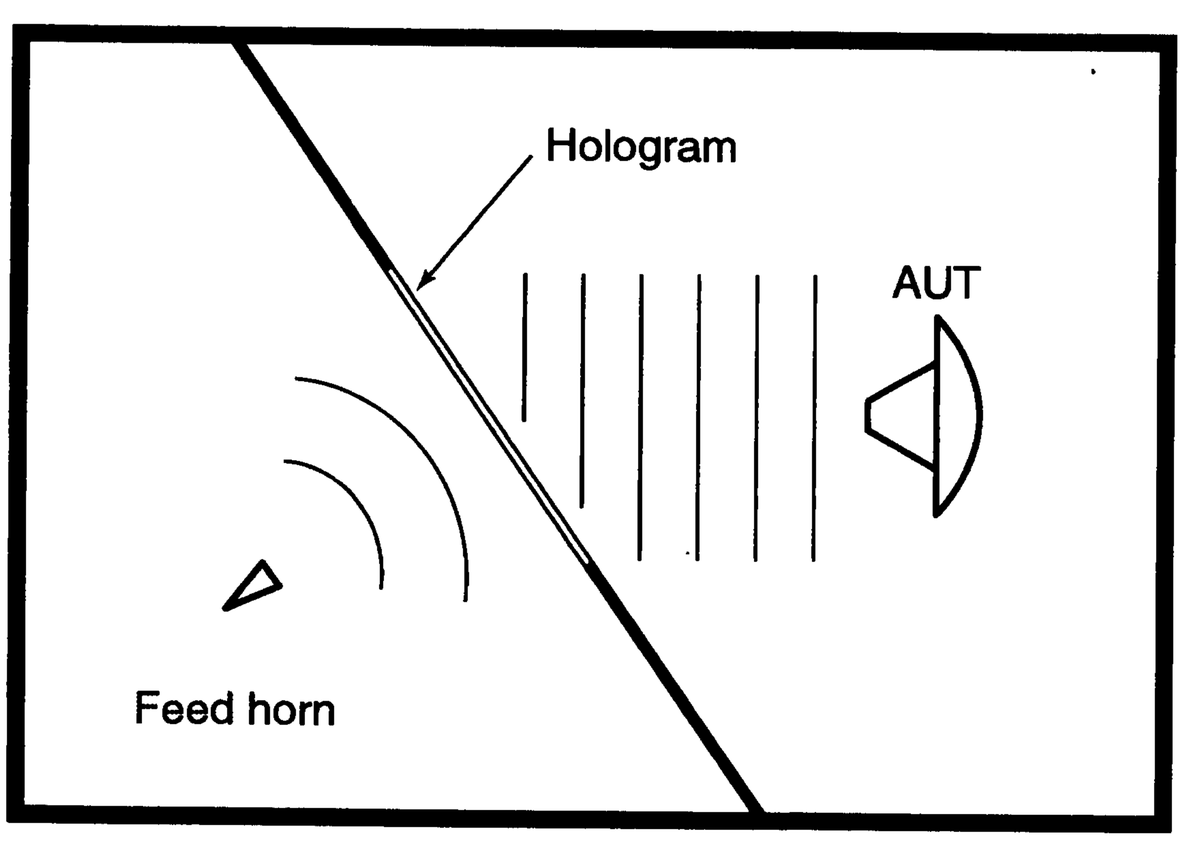


Figure 24–14
Dual-reflector CATR. Serrations and the absorbers of the chamber are not shown.



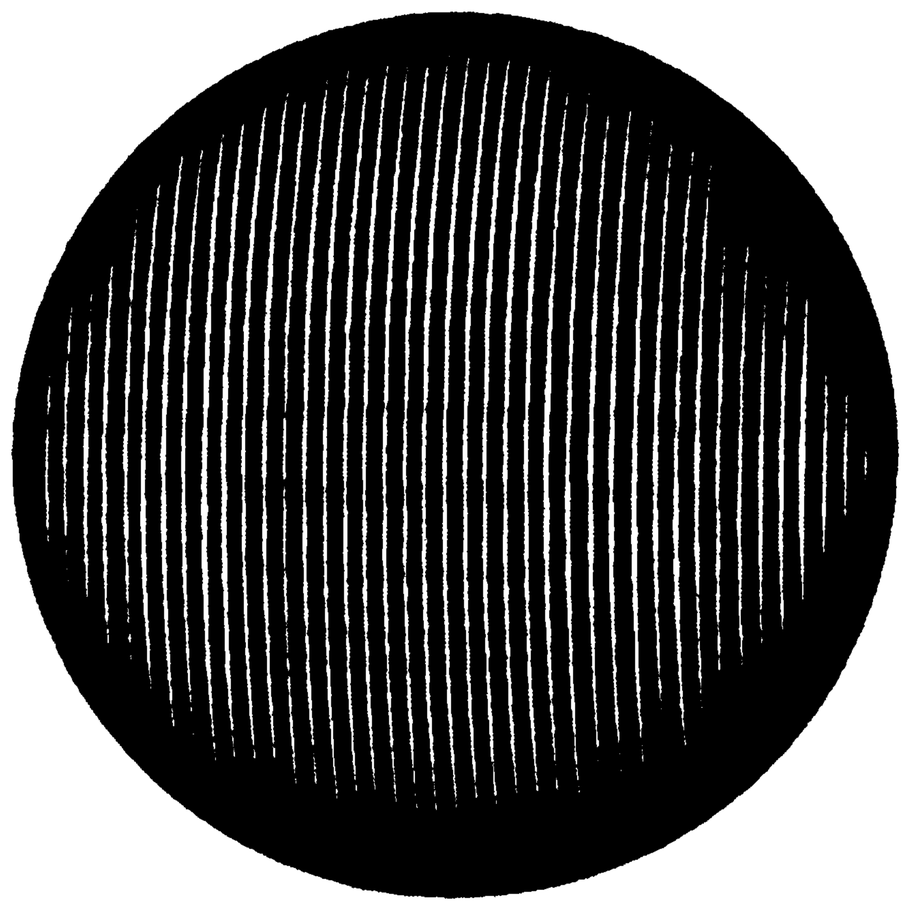


Figure 24–15
CATR based on a hologram.

about 5 m. The rms surface accuracy of the reflectors is better than 20 μm which allows measurements up to 200 GHz. Due to the long effective focal length, the plane wave direction can be scanned accurately by lateral movement of the source.

An approximation of the plane wave can be generated also with a *hologram* (Hirvonen-1). A CATR based on an amplitude hologram is shown in Fig. 24–15. The hologram is made by etching a shaped pattern to a thin metal layer on a dielectric film. The hologram modulates the amplitude of the spherical wave in such a way that a plane wave emanates on the other side. The surface accuracy requirement is about $\lambda/10$ which is much less stringent than in case of the reflector. The strong frequency and polarization dependence are disadvantages of the hologram CATR.

24–4e Near-Field Ranges（近场场地）

If we know the fields radiated by an antenna over a closed surface (with antenna inside), we can calculate the field at any point of space outside that surface. Thus, from the near-field measurements, e.g., the far-field patterns can be computed. This is the basis of *near-field measurements* (Yaghjian-1).

Fortunately, we do not need to measure the field continuously over the surface cited above. According to the sampling theorem, a limited number of samples is enough to describe the field completely. Also the measurement surface need not to be closed, if the field of the AUT is negligible at certain parts of the surface surrounding the AUT, e.g., behind a high-gain reflector antenna.

Near-field measurements have many advantages compared to conventional far-field measurements. Near-field ranges are very compact; thus antennas having large far-field distances can be measured indoors in a small space. Because the measured field is a complete description of the radiation, many kinds of antenna parameters can be computed from the data. Near-field measurements are also a useful diagnostic tool; e.g., defective elements of an array are found easily. If the measurements are performed carefully, far-field patterns of very low-side-lobe (−55 dB) antennas can be determined accurately (Francis-1). However, near-field measurements may be technically very demanding as described below. Because the phase has to be measured accurately, most near-field ranges are limited at frequencies below 60 GHz.

The near-field measurement is realized by scanning the field close to the AUT on a known surface with a known probe antenna. The probe antenna should be small and have a broad beam; e.g., open-ended waveguides are used. The probe is placed in the radiating near field at least a few wavelengths from the AUT. Both the amplitude and phase at two orthogonal polarizations are measured at the sampling points. From the measured field data, the far-field pattern is computed as theoretical patterns are computed from theoretical field distributions.

The basic measurement coordinate systems are planar, cylindrical, and spherical system (Fig. 24–16). For the planar system, the AUT is fixed and the probe moves. Planar measurements are suited for high-gain antennas. The calculations are simple and can be done rapidly with a computer. For the cylindrical system, the AUT rotates and the probe moves on a linear track. Antennas having dipole-type patterns can be measured with the cylindrical system. The calculations are more complicated than in the case of the planar system. Omnidirectional antennas can be measured with the spherical system where the AUT rotates and the probe is fixed. The calculations are quite complex.

Planar near-field measurements are by far the most common. Let us assume that the scan surface is in the xy-plane. Data is measured at equally spaced points in both the x and y directions. The planarity of the scan surface (z-position inaccuracies) should be $\lambda/100$ or better. The positioning accuracy of the probe in the xy-plane is not as demanding. The scan area should be somewhat larger than the area of the AUT. This area determines the maximum angle θ_m to which accurate far-field patterns can be expected. The spacing between sampling points should be slightly less than $\lambda/2$. However, for directive antennas spacings in the order of λ or even more may be sufficient to get accurately the main beam of the far-field pattern.

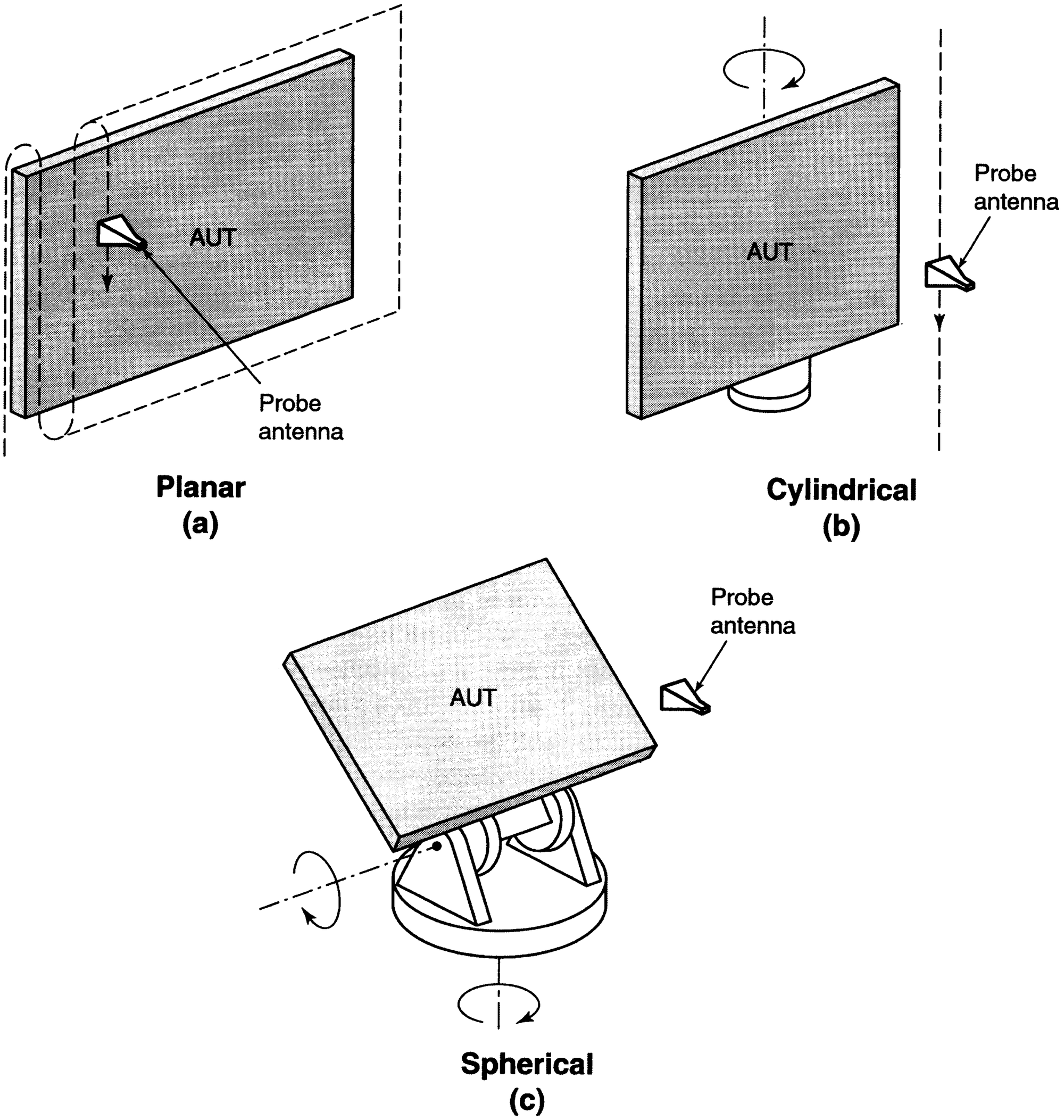


Figure 24–16
Near-field scanning geometries for close-up measurements of an aperture antenna in its near field: (*a*) planar, (*b*) cylindrical, (*c*) spherical.

From the measured data a plane wave spectrum, i.e., an angular spectrum of plane waves (angular spectrum of cylindrical and spherical waves for respective scanning geometries), is computed using two-dimensional Fourier transform. For a rectangular grid of data points, the fast Fourier transform algorithm, FFT, can be used. The plane wave spectrum is distorted because it includes the directive and polarization characteristics of the probe. The spectrum can be corrected with the probe spectrum. This operation is called probe correction. After the probe correction, the far-field pattern, gain, and polarization of the AUT can be calculated.

The error sources are manifold in the near-field techniques: inaccurate probe positioning, reflections, moving cables, receiver nonlinearity, inaccurate probe correction, limited scan area etc. (Newell-1). Accurate positioning requires that the mechanical scanner is a precision instrument whose distortions due to gravitation and temperature are minimized. Lasers can be used to measure the location of the probe. Then the errors caused by inaccurate z-positioning can be reduced by correcting the measured phase. Multiple reflections between the probe and AUT are reduced by covering the scanner and supports with absorbers. The influence of reflections can be reduced by averaging far-field values obtained for several scan planes. Phase errors due to moving cables may be a problem at high frequencies. Because the measurement may take several hours, the stability of equipment is more important than usual.

24–4f Testing of Ranges（场地的检测）

Measurement range quality can be evaluated best by measuring the field uniformity in the test zone. The results of the evaluation can be used either for making error estimates or as a starting point for improving the range.

The measurement range cannot be described with a single figure of merit. Free-space Voltage Standing Wave Ratio (VSWR) method and Antenna Pattern Comparison (APC) method are simple techniques to measure the reflectivity of the range (Appel-Hansen-1). However, the measured reflectivity depends on several parameters, such as frequency, polarization, antenna patterns, and the aspect angle of the probe antenna. More sophisticated evaluation methods can also identify the sources of reflection.

In the VSWR method, a probe antenna having a fixed orientation is moved in the test zone making vertical, horizontal, or raster scans (Fig. 24–17). The probe has a wide beam and thus gathers reflections from a large solid angle. Often it is a small horn antenna. The interference pattern of the direct wave and reflected waves versus location resembles a standing wave pattern. The ratio of the reflected field E_R (sum of all reflected waves) to the direct field E_D, reflectivity R, is calculated from the amplitude ripple S (in dB):

$$R = 20\log\frac{E_R}{E_D} = 20\log\left(\frac{10^{S/20}-1}{10^{S/20}+1}\right) \quad \text{(dB)} \tag{5}$$

where it is assumed that $E_R < E_D$. For example, 1 dB ripple means that the reflected field

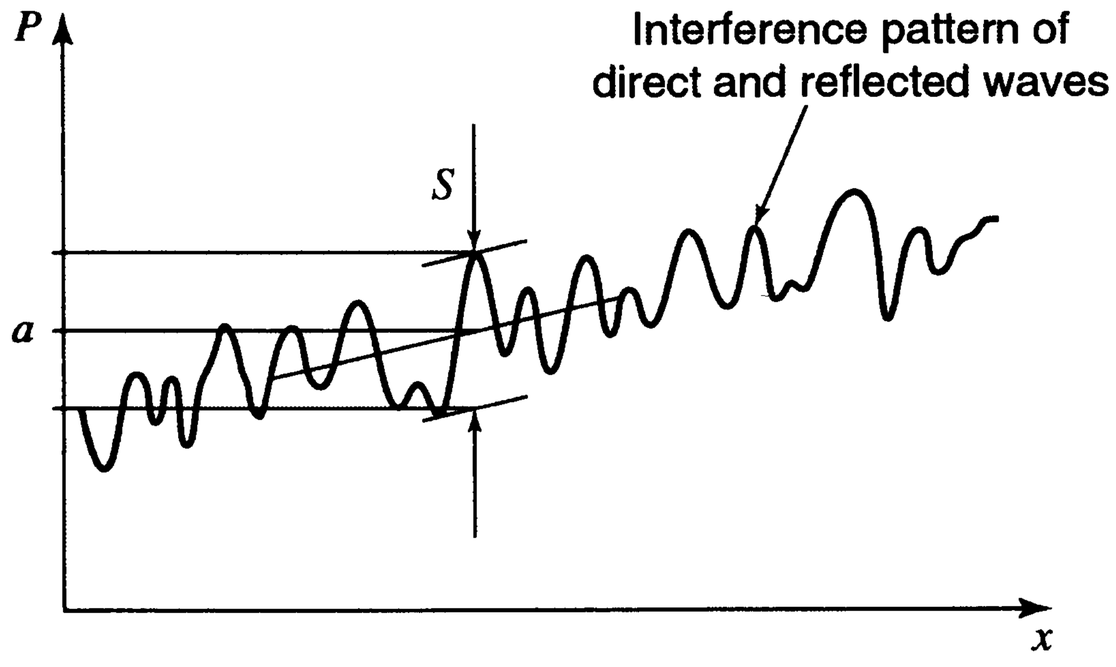


Figure 24–17
Free-space voltage standing wave ratio method of testing a range.

is 24.8 dB below the direct field. It should be noted that the measured fields are those which are coupled to the antenna—they are not the total fields. A directive probe antenna cannot gather all reflections. If the main lobe peak of the probe is not pointed to the source, the coupled direct field is reduced by $10^{a/20}$, where a is the pattern level of the probe toward the source. Then the measured reflectivity is obtained by adding a (dB) to the value calculated from Eq. (10). Often the reflectivity is measured using standard antennas, e.g., 20 dB gain horns, for various orientations of the probe antenna.

EXAMPLE **24–4.2 Testing a Range Using the VSWR Method**

The probe antenna is pointed to the sidewall of an anechoic chamber. The pattern level varies between −22.3 dB and −23.7 dB as the probe is moved. Find the reflectivity.

■ **Solution**

The amplitude ripple $S = 23.7 - 22.3 = 1.4$ dB and the pattern level $a = -(22.3 + 23.7)/2 = -23$ dB. The reflectivity

$$R = a + 20\log\left(\frac{10^{S/20} - 1}{10^{S/20} + 1}\right)$$

$$= -23 + 20\log\left(\frac{10^{0.07} - 1}{10^{0.07} + 1}\right) = -23 - 22 = -45\,\text{dB}$$

The VSWR and APC methods are basically identical. In the APC method, the pattern is measured at several locations within the test zone. The patterns are superimposed on one other so that their main lobe peaks coincide. The maximum variation of the field at a given aspect angle corresponds to the ripple measured with the VSWR method.

The VSWR and APC methods reveal the existence of reflections but cannot identify their sources. If there is only one dominant reflection, the interference pattern is periodic. In case of transverse movement of the probe, a spatial period of $\lambda/\sin\phi$ can be observed, if the angle between the directions of waves is ϕ and the probe moves in the same plane as the waves are propagating. Generally, there are several reflected waves. In principle, the sources of reflections can be mapped with a directive antenna but the coupling of the direct wave prevents the measurement of low reflections.

Reflections can be separated with a Fourier analysis. The Fourier transform of the measured field (amplitude and phase) over a plane surface results in a plane wave spectrum, i.e., the magnitude of reflections versus angle of arrival. This analysis is closely related to the planar near-field measurements. Another approach is the synthetic aperture imaging technique (Seville-1). A focused synthetic aperture array is formed from the measured field samples. This method reveals the directions and distances of scattering objects. Also a monostatic synthetic aperture radar can be used for mapping the sources of reflections.

24–4g Instrumentation（仪器设备）

Figure 24–18 shows a typical instrumentation of an elevated range. This figure applies with some modifications also for ground-reflection ranges, anechoic chambers, and compact

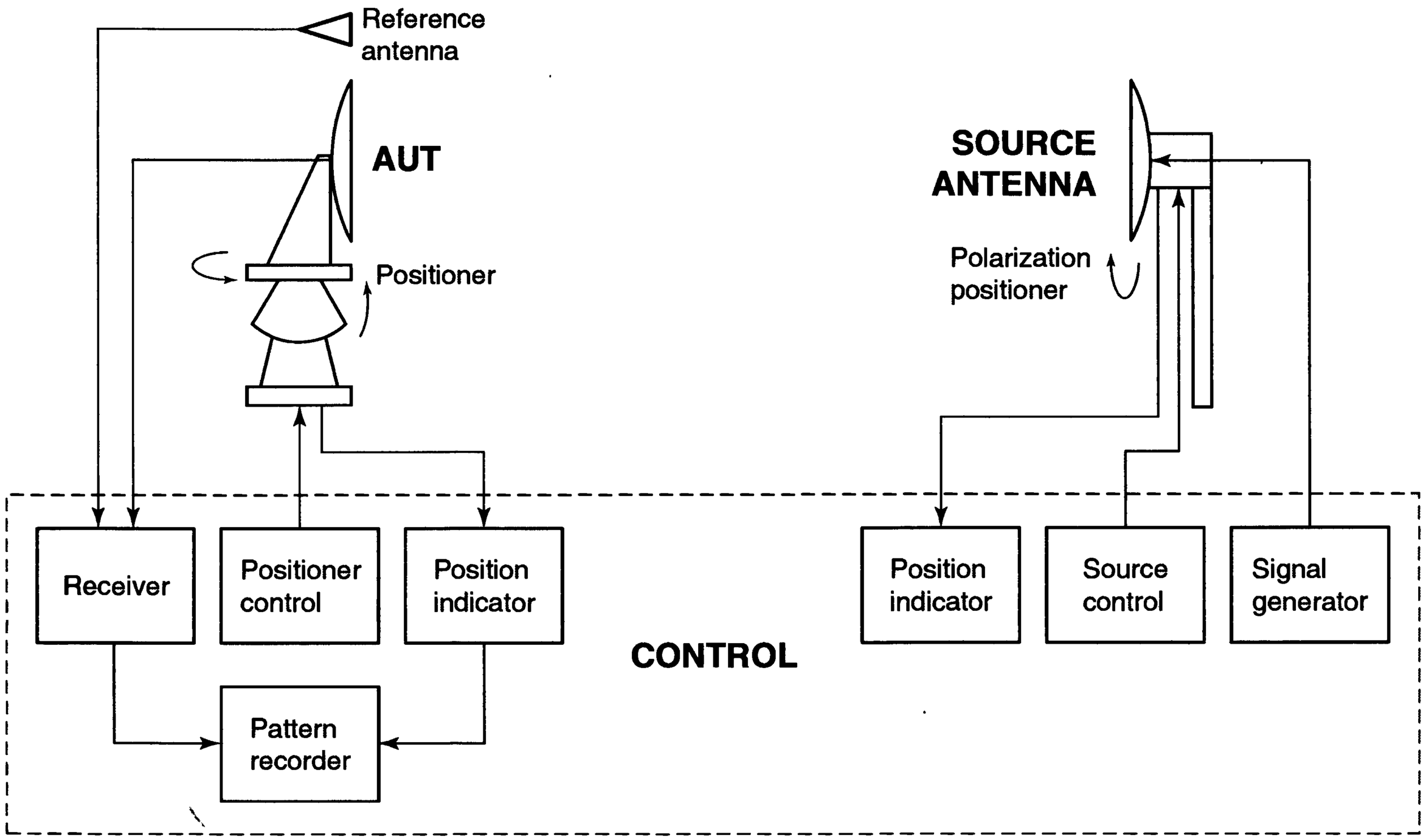


Figure 24–18
Typical instrumentation of an antenna measurement range with Antenna Under Test (AUT) at left, source antenna at right and control equipment below.

ranges. Antenna measurement ranges are general-purpose installations and should allow measurements over a large band of frequencies.

Due to the reciprocity, the direction of signal propagation does not matter—the AUT can be as well the transmitting antenna as the receiving one. The advantage of having the AUT as the receiving antenna is that the data processing and antenna manipulation can occur at one site as in Fig. 24–18. At short ranges there may be RF transmission lines between the towers. The source antenna may be a log-periodic antenna at lower frequencies, a horn or reflector at higher frequencies.

Transmitters and Receivers. To make accurate pattern measurements, a sufficiently powerful transmitter and a good receiver is needed. The transmitter is usually close to the source antenna and is remote controlled. Also the source antenna polarization is remote controlled. The transmitter should have a stable frequency and pure spectrum. Stable signal allows the use of a narrow receiving bandwidth, which is prerequisite for a sensitive receiver. Simple signal generators can be used in many measurements but sophisticated sweeping frequency synthesizers are best for demanding measurement applications.

The receiver should be sensitive, narrow-band to suppress interfering signals, linear, and should have a large dynamic range. Heterodyne receivers dedicated for antenna measurements and Vector Network Analyzer (VNA) systems modified for antenna measurements

are available for demanding measurements. If the measurement bandwidth of a heterodyne receiver is 1 kHz, the noise level may be below −120 dBm and the dynamic range may be more than 80 dB. Similar performance can be obtained with a VNA. At short distances, antenna measurements with a VNA are performed as ordinary measurements of scattering parameter S_{21}. At larger distances, the reference signal has to be regenerated at the receiving site. If the separation between the AUT and receiver is large, the use of an external harmonic mixer mounted close to the AUT eliminates the need for long high-frequency cables. For simple measurements, spectrum analyzers, power meters, or crystal detectors can be used. With these receiving systems only amplitude information is obtained. The dynamic range of power meters and detectors is only about 40 dB.

Often there is a reference antenna at the receiving site to tune the receiver if the signal frequency drifts during the measurement. The reference antenna provides also a phase reference. When measuring large signal level variations the receiver may saturate. A precaution is to insert a known attenuation in the receiver input path when the main beam peak is measured.

The measurement speed or time required for a single measurement depends on the transmitter tuning speed, receiver lock-on time, and measurement bandwidth. Larger bandwidth allows higher data rates but at the cost of sensitivity. Sensitivity can be improved by post-detection averaging but again the measurement speed suffers. The time per measurement point and required angle resolution set also a limit for the speed of rotation.

Positioners. The AUT is placed on a pedestal usually having two orthogonal axes of rotation. Figure 24–19 shows the two basic types: azimuth over elevation and elevation over azimuth positioner. Constant ϕ cuts and constant θ cuts can be measured by using the azimuth over elevation positioner. With the elevation over azimuth positioner the cuts are not along constant θ or ϕ, except on the principal planes. When the positioner has second

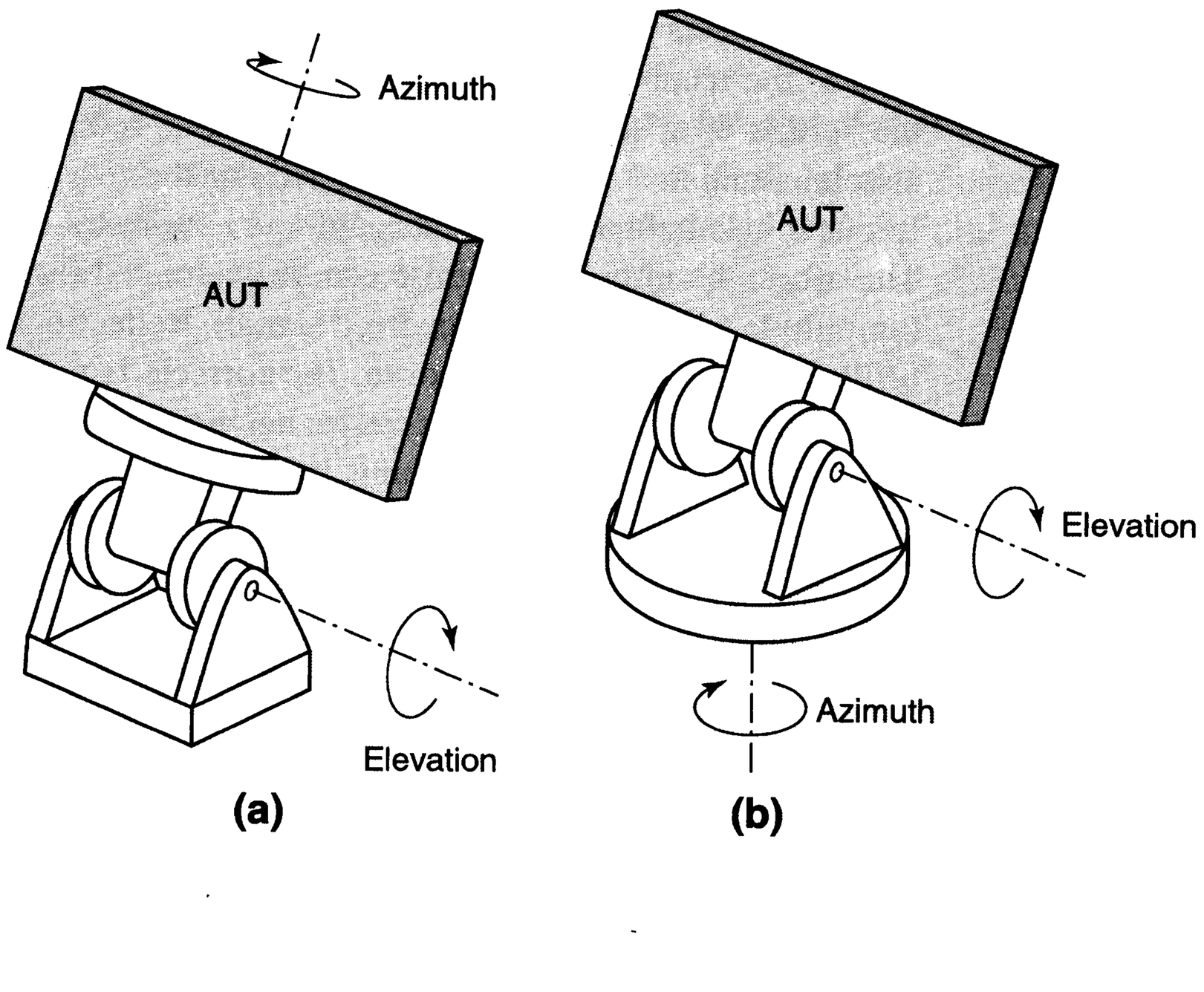


Figure 24–19
Positioners for rotating the Antenna Under Test (AUT): (*a*) azimuth over elevation, (*b*) elevation over azimuth.

azimuth axis it is called upper azimuth over elevation over lower azimuth positioner. Simple azimuth positioners have only one axis. Both ϕ and θ cuts can be measured by mounting the AUT on an azimuth positioner and the source on a carriage that moves along a fixed semicircular arch above the AUT.

The positioning system includes dc or stepping drive motors, synchro systems or digital encoders for shaft position measurement, a position controller, and a position indicator. Rotary joints may be included in the mount.

Load bearing capacity, torque, and speed are the fundamental parameters of the positioner. For the measurement of large and heavy antennas a very robust positioner is needed because the AUT exerts large forces and moments on the mount. Of particular concern is the case when the mass of the antenna must be placed far from the axis. Also the inertial effects must be taken into account when large masses are involved.

Nonorthogonality of the rotation axes, shaft position errors, and deflections due to temperature changes or changes in the forces applied to the positioner cause angle errors. Typical synchro angular accuracy is 0.03°. Backlash is typically 0.1°, but the effects of it can be eliminated by rotating always to the same direction. The speed of rotation should be low enough to avoid distortions in the measured patterns.

Data Processing. Simple measurements can be made manually, but in more complex measurements automation is an essential feature because there are large amounts of data involved. The computer controls the transmitter, receiver, and positioner. The receiver output is fed to a conventional pattern plotter, either a rectangular or polar plotter, or it is converted to digital format and saved to a computer memory. Angle information is obtained from the synchros.

The use of a computer permits many ways for processing and analyzing the data. Different plots, e.g., three-dimensional or constant contour presentations can be produced. The measurements can then be compared to the theoretical results. It is also possible to interpolate between the measured cuts. The power pattern can be integrated to get the directivity.

The effects of reflections can be reduced by transmitting short pulses and keeping the gate of the receiver open as the direct pulses arrive. However, due to the duty factor and the spread of spectrum, the dynamic range is reduced. Time-domain measurements can also be simulated with a computer without the loss of dynamic range. Time domain and frequency domain are related through the Fourier transform and inverse Fourier transform. Therefore, the impulse response can be calculated from the measured frequency response (amplitude and phase of the received signal). Reflections arrive after the direct wave and can be filtered out of the time response. The corrected time response is then transformed back to the frequency domain. Modern vector network analyzers include software to perform these transformations. Time resolution is inversely proportional to the measurement bandwidth, that is for 10 ns resolution (path length difference 3 m) the bandwidth should be about 100 MHz. Thus, this technique is not suited for narrow-band antennas.

24–5 MEASUREMENT OF DIFFERENT ANTENNA PARAMETERS
（不同天线参量的测量）

To describe an antenna completely, several radiation and circuit parameters are needed. In this section, measurement techniques for pattern, gain, phase, polarization, impedance, efficiency, and current distribution are discussed.

24–5a Directional Pattern（定向波瓣图）

The directional pattern, power or amplitude pattern, is usually measured by rotating the AUT on a positioner and by detecting the received power versus angle. The field radiated by the AUT can be divided into two orthogonal components. Correspondingly, the total power is divided between two patterns, the co-polar pattern and cross-polar pattern. It should be noted that the definition of cross-polarization, which applies for all angles, is not self-evident (Ludwig-1). Usually, the co-polar pattern represents the wanted radiation whereas the cross-polar pattern represents "leakage" radiation. Often the patterns are measured only in the principal planes, e.g., in the E and H planes of a linearly polarized antenna. Sometimes several θ or ϕ cuts are measured and a three-dimensional pattern is constructed of them. The use of a frequency synthesizer allows pattern measurements at several frequencies during a single rotation.

The source antenna polarization should match the co-polar and cross-polar fields in the corresponding pattern measurements. For example, for the measurement of a linearly polarized AUT, the polarization vectors of the antennas should be parallel in co-polar measurements, whereas the vectors should be in an angle of 90° in cross-polar measurements. If the AUT has a low cross-polarization level, the angle should be adjusted very carefully. Often, the fine adjustment is performed by minimizing the received power to the direction of the co-polar main lobe peak. If the alignment is not correct, an attenuated co-polar pattern may be obtained instead of the intended cross-polar pattern.

Beamwidth, pattern shape, side-lobe levels and directions, and null directions are parameters which are obtained easily from the measured pattern. The directivity of an antenna, D, cannot be measured directly. However, it can be computed from the normalized power pattern $P_n(\theta, \phi)$ as

$$D = \frac{4\pi}{\iint P_n(\theta, \phi) \sin\theta \, d\theta \, d\phi} \tag{1}$$

EXAMPLE **24–5.1 Alignment Error in Cross-Polar Pattern Measurement**

The level of cross-polarization of an ideal, linearly polarized antenna is measured. The source antenna is also linearly polarized. What is the measured level, if the antennas are aligned carelessly so that the difference of the tilt angles is 88° instead of 90°?

■ **Solution**

Let us denote the field illuminating the AUT by $E_n = 1$. This field can be decomposed to the co-polar component $E_n \sin 88° = 0.9994$ and the cross-polar component $E_n \cos 88° = 0.0349$. Thus, the measured cross-polar level is $20 \log 0.0349 = -29$ dB relative to the co-polar field.

24–5b Gain（增益）

There are many antenna gain definitions depending on which imperfections are included. Traditionally, it is assumed that there are no impedance and polarization mismatch losses.

When the reflection loss due to impedance mismatch is taken into account, the term realized gain is used. Often one wants to know a partial gain, i.e., the gain for a given polarization.

Absolute method and comparison method are the two basic gain measurement methods. Celestial radio sources can be used if the far-field distance is too large for a terrestrial measurement. Also radar techniques and near-field measurements can be used for gain determination (Newell-2).

Absolute Method. The absolute method is based on the Friis transmission formula

$$P_R = P_T G_T G_R \left(\frac{\lambda}{4\pi R}\right)^2 \quad \text{(W)} \tag{2}$$

where

P_R = power received, W
P_T = power accepted by the transmitting antenna, W
G_T = gain of the transmitting antenna
G_R = gain of the receiving antenna
λ = wavelength, m
R = separation between the antennas, m

It is assumed that the polarizations of the antennas are matched, the main lobe peaks are aligned with the line of measurement, and far-field conditions prevail.

In the two-antenna method, two identical (or near identical) antennas are required ($G_T = G_R$). The gain of the antennas is determined from R, λ and the measured powers using Friis transmission formula (2). If two identical antennas are not available, then a third antenna is needed. In the three-antenna method, three sets of measurements are performed, one with each pair of antennas. From the three simultaneous equations, the three unknown gains can be solved straightforwardly.

There are many sources of error in the absolute method: the misalignment of the antennas, polarization mismatch, impedance mismatch both in the transmitting and receiving system, the uncertainty of the power meter or calibrated attenuator used for the measurement of P_R/P_T, and the test field imperfections. The uncertainty in separation R may be significant, if the antenna length is a considerable part of the measurement distance.

Impedance mismatch complicates the power measurement. The power accepted by the transmitting antenna cannot be measured accurately by replacing the antenna with a power meter. The same applies on the receiving side. The efficiency of power transfer between a generator and a load is

$$\eta_{GL} = \frac{(1 - |\rho_G|^2)(1 - |\rho_L|^2)}{|1 - \rho_G \rho_L|^2} \tag{3}$$

where

ρ_G = complex reflection coefficient of the generator
ρ_L = complex reflection coefficient of the load

Thus, to correct the measured powers, the reflection coefficients of the antennas, transmitter, and receiver have to be known. If only the magnitudes of the reflection coefficients are known, the uncertainties for the power measurements can be computed.

Gain reduction due to the phase curvature and amplitude error can be partly corrected: if the measurement distance is $2D^2/\lambda$ and the amplitude taper is 0.25 dB, about 0.15 dB should be added to the measured gain. Due do the test field ripples, the received power may be sensitive to the location. This effect can be reduced by averaging the powers which are received at slightly different locations.

Comparison Method. In the comparison method (gain-transfer method), the powers received with the AUT and with a known reference antenna are compared as shown in Fig. 24–20. This measurement can be performed on either a free-space or a ground-reflection range. The gain of the AUT is

$$G_{AUT} = \frac{P_{AUT}}{P_{ref}} G_{ref} \tag{4}$$

where

P_{AUT} = power received with the AUT, W
P_{ref} = power received with the reference antenna, W
G_{ref} = gain of the reference antenna

G_{ref} has been determined by some other means, e.g., by absolute method or theory. Half-wave dipoles and horn antennas are commonly used reference antennas because they have a predictable gain and pure polarization. The calibration uncertainty of the reference antenna gain is typically ±0.25 dB. The power ratio can be measured simply with a calibrated attenuator: the attenuation is adjusted to give the same output indication with both antennas, and the power ratio is obtained from the attenuator settings.

It is assumed in Eq. (4) that both the AUT and the reference antenna are perfectly matched to the receiving system (transmission line and receiver) and they have the same polarization. Differences in polarization mismatch, impedance mismatch and transmission line losses cause errors. If the AUT and the reference antenna resemble each other, i.e., they have about similar aperture distributions, the errors due to the phase curvature and amplitude taper are about the same for both antennas and thus are partly canceled. If the antennas are mounted close to each other and side by side as in Fig. 24–20, there may be unwanted coupling between them. Often this can be avoided by mounting the antennas on the positioner back-to-back and by rotating the system 180° between the measurements.

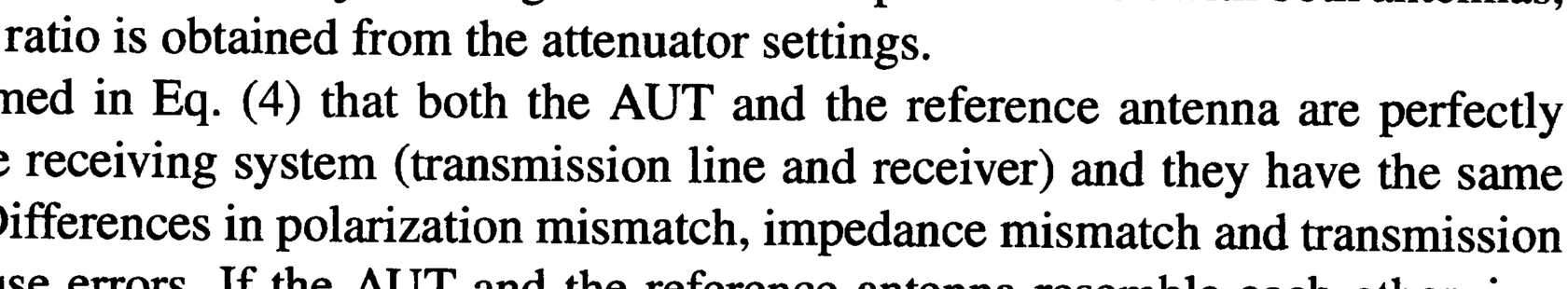


Figure 24–20
Gain measurement by comparing the Antenna Under Test (AUT) with a reference antenna.

The gains of circularly or elliptically polarized antennas are usually determined by measuring the partial gains for two orthogonal linear polarizations. First the polarizations of the linearly polarized source and reference antennas are set horizontally and the gain G_H is measured. Then the measurement is repeated for vertically polarized source and reference antennas and the gain G_V is obtained. The total gain, G_{AUT}, is the sum of the two partial gains:

$$\text{Total gain} = G_{AUT} = G_H + G_V \tag{5}$$

where

G_H = gain of the AUT at horizontal polarization
G_V = gain of the AUT at vertical polarization

Celestial Radio Sources. Celestial radio sources can be used for pattern and gain measurements of very large antennas that cannot be tested any other way. There are many radio "stars," e.g., supernova remnants or radio galaxies, which have an accurately known flux density S (Table 24–1). These sources are broad-band noise sources and have the usual random polarization. However, some radio sources show a significant degree of polarization. Flux density is the measure of power incident on a square meter per hertz. Usually it is given in janskys (1 jansky = 1 Jy = 10^{-26} $\text{Wm}^{-2}\text{Hz}^{-1}$). The spectral index α describes the frequency dependence of the flux density: $S \propto f^{\alpha}$.

If the angular extent of the radio star is much smaller than the half-power beamwidth of the AUT, the power received is

$$P = \frac{SA_eB}{2L} = \frac{S\lambda^2 G_{AUT} B}{8\pi L} \quad \text{(W)} \tag{6}$$

where

S = source flux density, $\text{Wm}^{-2}\text{Hz}^{-1}$
A_e = effective area of the antenna, m^2
B = bandwidth, Hz
L = attenuation of the atmosphere
λ = wavelength, m
G_{AUT} = gain of the AUT

It is assumed that the source is randomly polarized and thus the polarization efficiency is 1/2. The received power can be set equal to the noise power $k\Delta T_A B$. From this the gain of the antenna can be written as

$$G_{AUT} = \frac{8\pi k \Delta T_A}{LS\lambda^2} \tag{7}$$

Table 24–1 Celestial radio sources for antenna measurements*

	Position (epoch 2000.0)								Flux density, Jy						
	Right ascension			Declination											
Source	Hours	Min	Sec	Deg	Arc min	Arc min	Size, arc min	Isolation, deg	38 MHz	178 MHz	750 MHz	1.4 GHz	2.7 GHz	5 GHz	Polarization,
3C 20	00	43	09.21	+52	03	33.5	1	1	112	43	17	11.3	6.4	4.2	1.8
3C 196	08	13	36.09	+48	13	02.5	0.25	1	166	68	23	13.9	7.7	4.4	2.2
3C 273	12	29	06.69	+02	03	08.6	0.35	1	140	63	45	40	42	45	2
3C 295	14	11	20.65	+52	12	09.0	0.3	0.5	94	83.5	35	22.4	12	6.5	0.4
3C 348	16	51	08.25	+04	59	33.8	2	1	1,690	351	84	44.5	22.4	11.9	1.5
3C 380	18	29	31.89	+48	44	46.7	0.25	0.5	211	59	22	14.4	10	7.5	1.2
Cygnus A	19	59	28.45	+40	44	02.0	1.6	†	22,000	8,700	2,980	1,590	785	371	0.2
3C 123	04	37	04.36	+29	40	13.9	0.1		57	189	72	45.8	27	16	1.2
3C 286	13	31	08.36	+30	30	32.9	0.04	0.5	32	24	18	14.6	10	7.5	9
Cassiopeia A	23	23	24	+58	48.9		5	†	37,200	11,600	3,880	2,410	1,470	910	–

*From the second edition of this book.
†In complex galactic plane region.

where

k = Boltzmann's constant = 1.38×10^{-23} J/K
ΔT_A = measured antenna noise temperature due to the source, K
L = attenuation of the atmosphere
S = source flux density, $\mathrm{Wm^{-2}Hz^{-1}}$
λ = wavelength, m

If the source cannot be treated as a point source, a correction factor should be determined. Very large antennas experience structural deviations due to gravity and therefore should be tested as the function of elevation angle.

A sensitive, calibrated radiometer is required for the measurement of small changes of antenna noise temperature. The radiometer is usually calibrated with two loads having accurately known noise temperatures, e.g., one absorber at room temperature, the other immersed in liquid nitrogen (77 K) or liquid helium (4 K). The absorbers should be placed in front of the feed antenna of the radiometer so that they cover the feed pattern completely. Calibrated radiometers can be used for many kinds of remote sensing and radio astronomical applications, e.g., for the measurement of the 3 K (more accurately 2.73 K) cosmic background radiation which is believed to be from the remnant of the primordial Big Bang (see Sec. 21–19).

EXAMPLE **24–5.2 Gain and Aperture Efficiency from Celestial Radio Source Measurements**

Find the gain and aperture efficiency of the Ohio State University 110-m radio telescope antenna at 1.4 GHz if the measured increase in antenna temperature from Cygnus A is 687 K. The physical aperture is 2210 $\mathrm{m^2}$. Assume that there is no atmospheric loss ($L = 1$).

■ **Solution**

From Table 24–1, the flux density of Cygnus A at 1.4 GHz is 1590 Jy. From (7) the gain

$$G = \frac{8\pi k \Delta T_A}{S\lambda^2} = \frac{8\pi \times 1.38 \times 10^{-23} \times 687}{1590 \times 10^{-26} \times 0.214^2} = 3.27 \times 10^5 = 55\,\mathrm{dBi}$$

The effective area

$$A_e = \frac{G\lambda^2}{4\pi} = \frac{3.27 \times 10^5 \times 0.214^2}{4\pi} = 1190\,\mathrm{m^2}$$

The aperture efficiency

$$\varepsilon_{ap} = \frac{A_e}{A_p} = \frac{1190}{2210} = 0.54 \text{ or } 54 \text{ percent}$$

Radar Techniques. The gain of a single antenna can be measured using radar techniques. By placing a sufficiently large flat reflector far enough from the AUT, we have an

arrangement that corresponds to the absolute method with two identical antennas: the AUT sees its image behind the reflector. This measurement may require a very large reflector.

If the AUT illuminates a target with known radar cross section σ, the gain can be solved from the transmitted and received powers using the radar equation. Reflecting spheres are widely used as radar calibration targets because they have a known radar cross section which does not depend on angle. For a perfectly reflecting sphere, σ is equal to its physical cross section πa^2 when its radius $a \gg \lambda$. With a sphere as a target, the gain can be solved from the radar equation:

$$G_{AUT} = \frac{8\pi R^2}{a\lambda}\sqrt{\frac{P_R}{P_T}} \tag{8}$$

where

R = separation of the radar and target, m
a = radius of the target (sphere), m
λ = wavelength, m
P_R = power received, W
P_T = power transmitted, W

The gain of the AUT can be measured also by illuminating it with a calibrated radar. If the AUT is short-circuited, it will reradiate the power which it has received. The radar cross section corresponding to this reradiation is

$$\sigma_r = A_e G_{AUT} = \frac{\lambda^2}{4\pi} G_{AUT}^2 \qquad (\text{m}^2) \tag{9}$$

where

A_e = effective area of the AUT, m^2
G_{AUT} = gain of the AUT
λ = wavelength, m

By moving the position of the short circuit, it is possible to separate σ_r from the scattering of the aperture and support structures.

24–5c Phase（相位）

Knowledge of the antenna phase pattern and phase center is for many applications of importance. In most cases, one wants to know the phase pattern $\psi(\theta, \phi)$ only in the principal planes for co-polar component. For complete description of the field, the phase pattern should be measured to all directions for both the co- and cross-polar components.

Measurement of the phase pattern is equivalent to finding an equiphase surface in the far field. The center of the coordinate system must be clearly defined with respect to the antenna structure within a fraction of a wavelength. If only the position of the phase center is wanted, the AUT is moved with respect to the origin of the coordinate system until a constant phase is observed over the main beam. However, for many antennas there is no unique phase center.

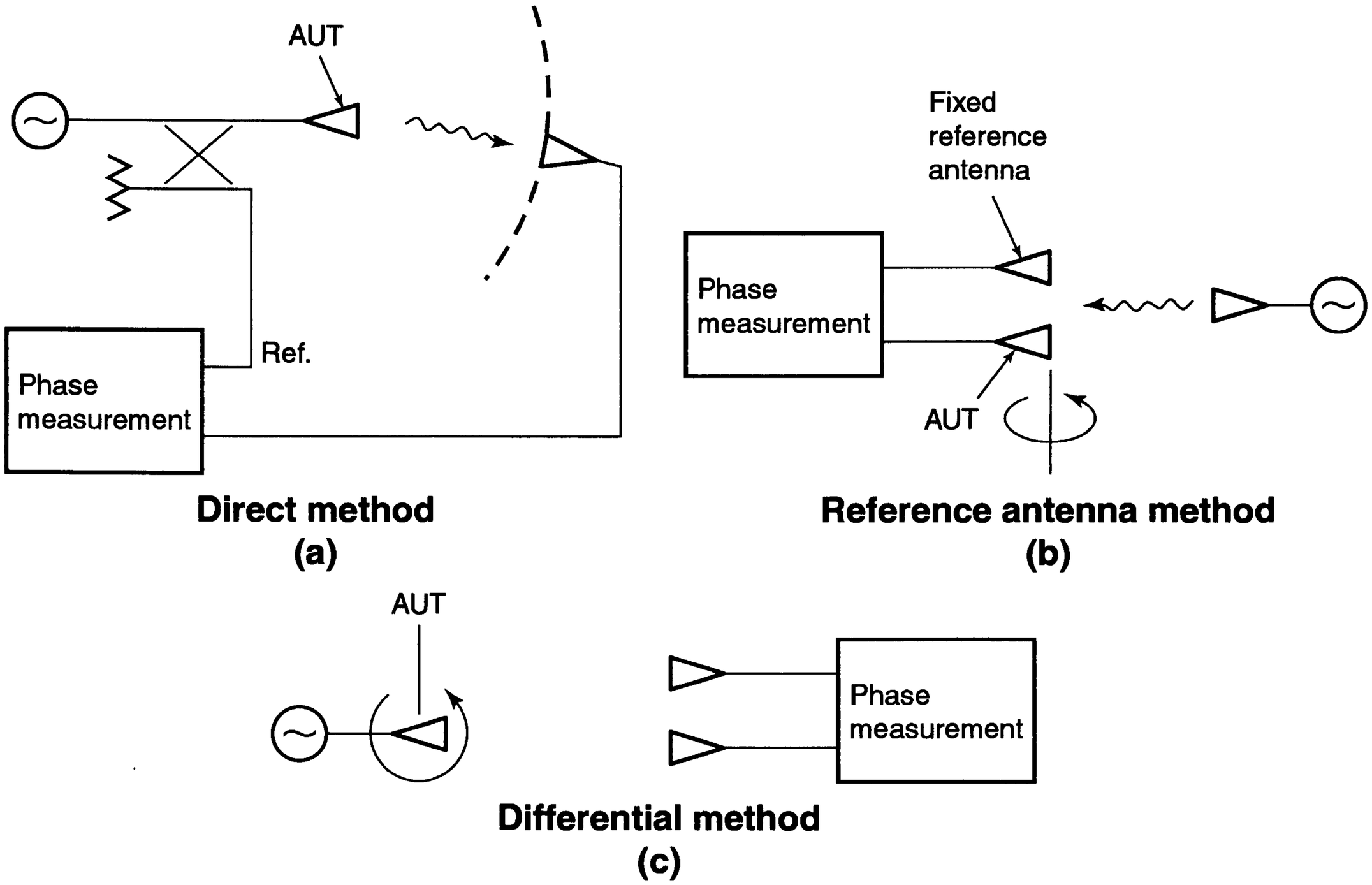


Figure 24–21
Phase measurement methods: (*a*) direct method, (*b*) reference antenna method, (*c*) differential method.

Figure 24–21 shows three arrangements to measure the phase pattern. In the direct method (Fig. 24–21*a*), the phase difference between a sample of the transmitted signal and the received signal is directly measured, as the probe antenna is moved along a circle. If the AUT is rotated, the probe can be fixed. The direct method can be applied at short distances. For larger distances, the reference antenna method may be used. In the reference antenna method (Fig. 24–21*b*), the receiving AUT is rotated and beside it there is a fixed reference antenna, and the phase difference between the received signals is measured.

In the differential phase method (Fig. 24–21*c*), no phase reference is needed. The transmitting AUT is rotated and the phase difference is measured with a two-channel receiver. The phase pattern can be calculated from the measured phase difference pattern (Tuovinen-1). In a modification of this method, the phase difference is calculated from three power readings, which are measured with a single power meter (Mallat-1). The advantage of the differential method is that no rotary joints or flexing cables are needed in the measurement system. At millimeter wave frequencies these components might cause large phase errors.

24–5d Polarization（极化）

As discussed in Secs. 2–15, 16 and 17, any field can be expressed as the sum of two orthogonal components that may be linear, circular, or elliptical. For example, an arbitrary elliptically polarized field **E** can be divided into its linear $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ components, E_x and E_y,

as in (2–15–4), that is

$$\mathbf{E} = \hat{\mathbf{x}} E_x \cos(\omega t) + \hat{\mathbf{y}} E_y \cos(\omega t + \delta) \tag{10}$$

where

$\hat{\mathbf{x}}$ = unit vector
$\hat{\mathbf{y}}$ = unit vector
$\omega = 2\pi f$ = angular frequency, s^{-1}
t = time, s
δ = phase difference of the x and y components, radians

The field could also be divided into its left-hand circular and right-hand circular components.

The polarization ellipse is described in terms of the axial ratio AR, tilt angle τ (direction of the major axis), and the sense of rotation (left or right). A complete description of the radiation pattern requires the measurement of polarization as a function of direction. Often, the complete polarization information is not necessary and only the amplitudes of the co- and cross-polarized components are needed.

There are many polarization measurement methods. In the *polarization-pattern method* (Fig. 24–22), the angle of polarization of a linearly polarized source antenna is rotated. The amplitude versus the tilt angle of the source antenna in polar form is called the polarization pattern. The axial ratio and tilt angle of the polarization ellipse is obtained from this pattern. The sense of rotation can be determined comparing the outputs of two circularly polarized antennas (e.g., axial-mode helical antennas) which have opposite senses of rotation.

If the polarization of a linearly polarized source antenna is rotated rapidly and at the same time the direction of the AUT is changed slowly, a pattern like that in Fig. 24–23 can be obtained. This method is called the *rotating-source method*. The maxima and minima correspond to alignment of the source with the major and minor axes of the polarization

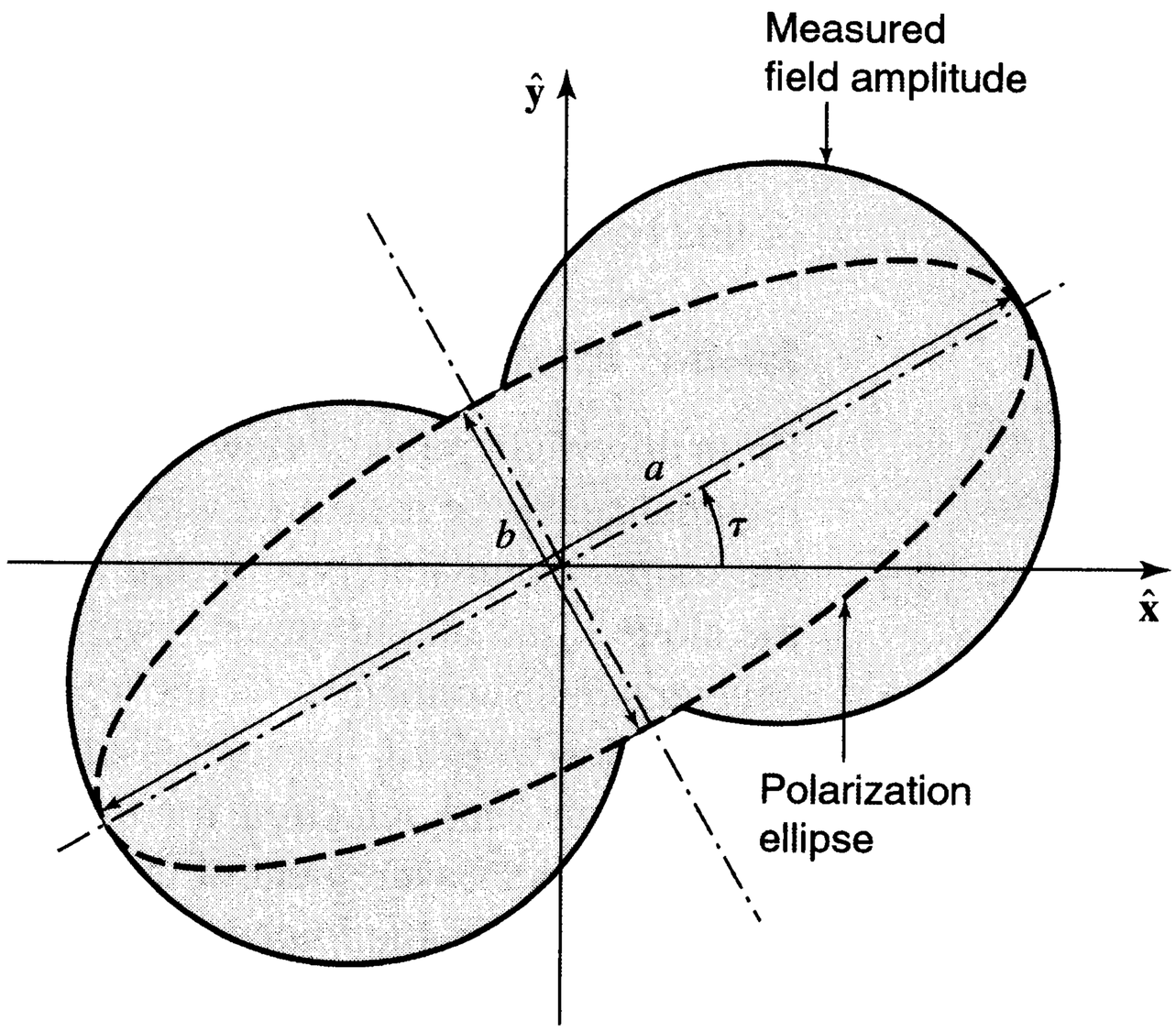


Figure 24–22
Polarization-pattern method. The axial ratio = a/b.

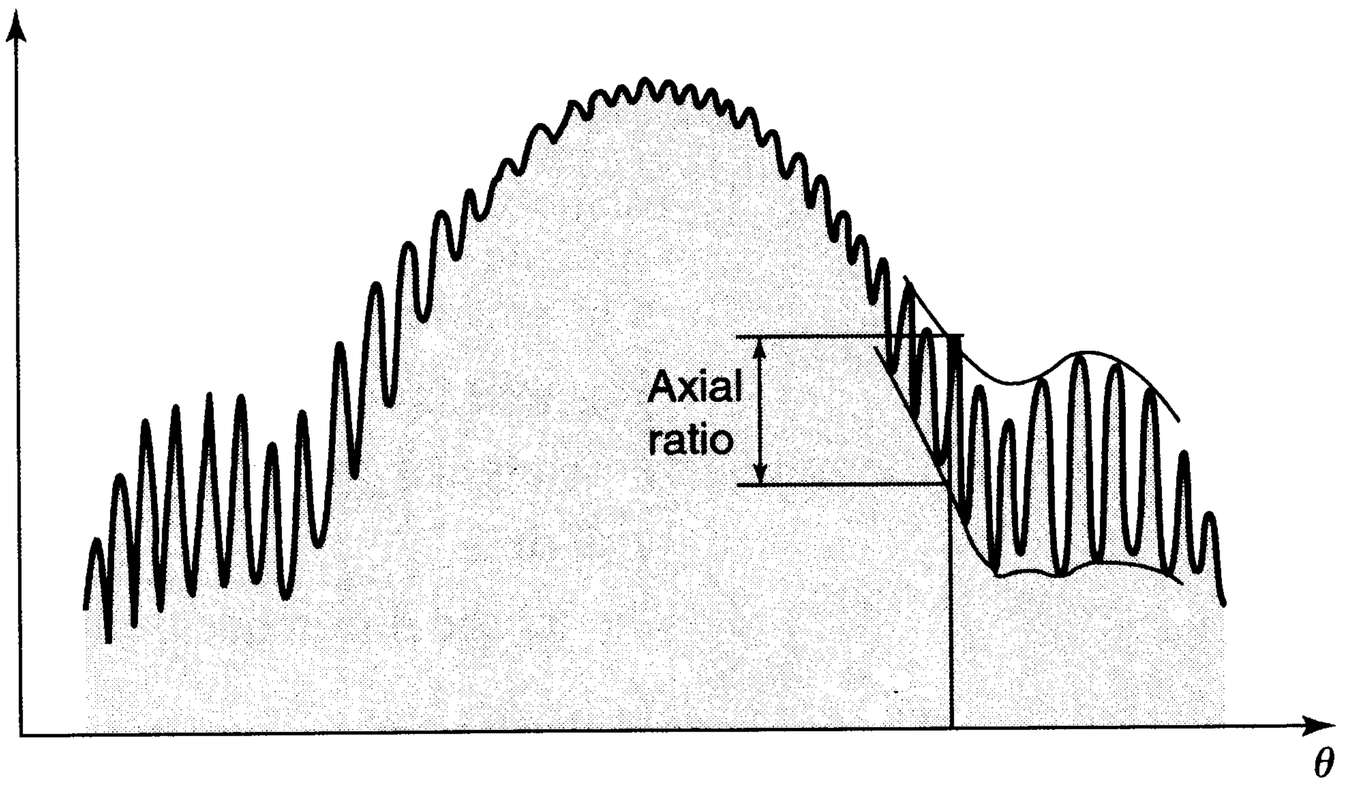


Figure 24–23
Testing of a circularly polarized antenna with the rotating-source method.

ellipse, respectively. Thus, the axial ratio is obtained from the envelopes of the pattern. This method is used for testing nearly circularly polarized antennas.

The parameters E_x, E_y, and δ defining the polarization ellipse can be measured with two fixed linearly polarized antennas which are mounted at right angles. The amplitudes of the fields and their phase difference are measured. Also two circularly polarized antennas of opposite senses of rotation could be used. In this method, the AUT is used as the transmitting antenna.

The polarization ellipse can be determined also from 4 amplitude measurements (or from 3 independent amplitude ratio measurements) with antennas having different, known polarizations. Also the gains of these sampling antennas have to be known. Often 6 antennas are used giving some redundancy: vertically polarized, horizontally polarized, linearly polarized both at 45° and at 135° tilt angle, left-hand and right-hand circularly polarized.

In all the above-mentioned methods, it is assumed that an antenna or antennas having known polarizations are available. The three-antenna absolute method does not have this limitation. A complex response is measured two times for each of the three combinations. In each case, the second antenna is rotated 90° for the second measurement. Polarizations for all three antennas can then be determined.

For a more extensive discussion of polarization measurements, see the second edition of this book, pp. 835–839.

24–5e Impedance（阻抗）

From the circuit point of view the antenna is viewed as an impedance (or one-port). Thus, its impedance can be measured with any impedance measurement method taking into account the transmission line type at the antenna feed. The most common of such methods are the slotted line measurement, swept measurement with directional couplers and network analyzer measurements. As the vector network analyzer (VNA) is commonly found in laboratories working in the area of microwave or millimeter wave engineering, it is also the most obvious selection for the measurement of antenna impedance. It also enables simultaneous measurement of the input impedance (S_{11}) and radiation properties (S_{21}).

Traditionally the impedance properties of an antenna have been connected to its role as the termination of a feed network. Here the most important information required is the VSWR (return loss, magnitude of reflection coefficient) of the AUT often as a function

of frequency. Thus a scalar measurement with one of the methods mentioned above is adequate. Sometimes also the complex impedance of the AUT defined at some reference plane at the antenna feed is necessary. This requirement is usually connected to cases where the AUT is an integrated part of a system as in antenna arrays or with small radio devices.

The requirements for the measurement environment for antenna impedance are not always as stringent as for the radiation properties. If moderate accuracy is adequate as in preliminary checking of the impedance, the measurement can usually be performed in a normal laboratory room if no strong scatterers are situated close to the AUT. The situation can be improved significantly by placing absorbers in the radiating near field of the AUT to cover all significant directions of radiation. In this case the far-field conditions do not have to be fulfilled. For a small antenna with a broad radiation pattern the AUT can be placed inside an "absorber box" with several wavelengths of free space around it. For more directive antennas like horns it is usually adequate to direct the main beam toward an absorber element as in Fig. 24–24. It is also useful to repeat the measurement in a few close-by locations for the AUT and average the result. The most accurate final impedance measurements are, however, best to perform in a proper anechoic chamber, e.g., in conjunction with the radiation pattern measurements.

Specific arrangements are required when the impedance of a single antenna element of a multielement antenna system is measured. A typical example of this is the measurement of the active impedances of antenna array elements including the effect of mutual coupling when all the elements are fed normally. For small arrays or multifeed antennas like quadrifilar helices all the coupling coefficients between the elements or feeds can be measured and the active impedances for all feeding variations can then be obtained by summing the contributions via the couplings to the reflection of the driven element. For an N-element system we get:

$$\rho_{act,n} = \frac{V_n^-}{V_n^+}$$

$$V_n^- = \sum_{m=1}^{N} S_{mn} V_m^+ \qquad \text{(V)} \tag{11}$$

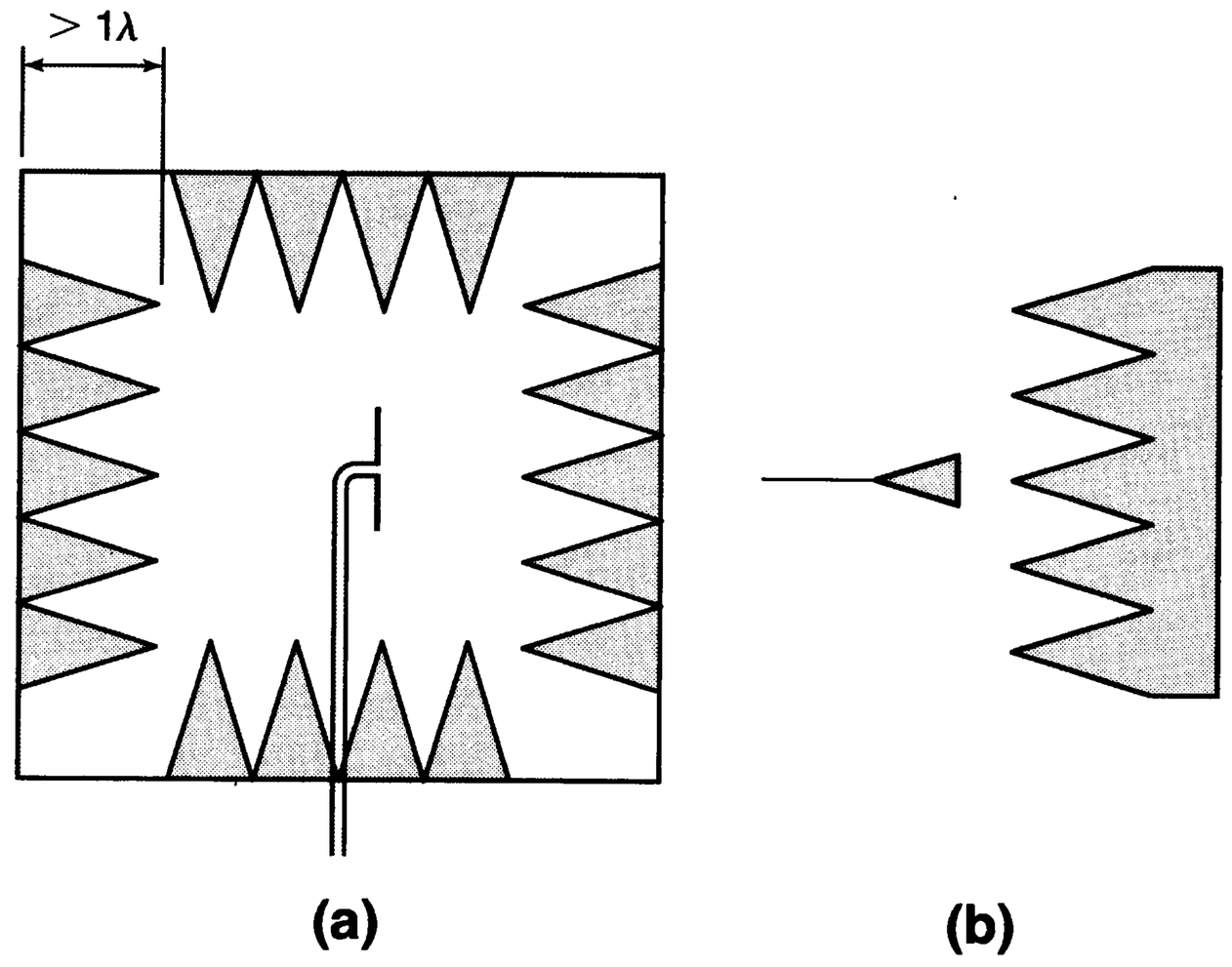


Figure 24–24
Impedance measurement in a laboratory room: (*a*) omnidirectional antenna in an "absorber box," (*b*) directive antenna pointing toward an absorber.

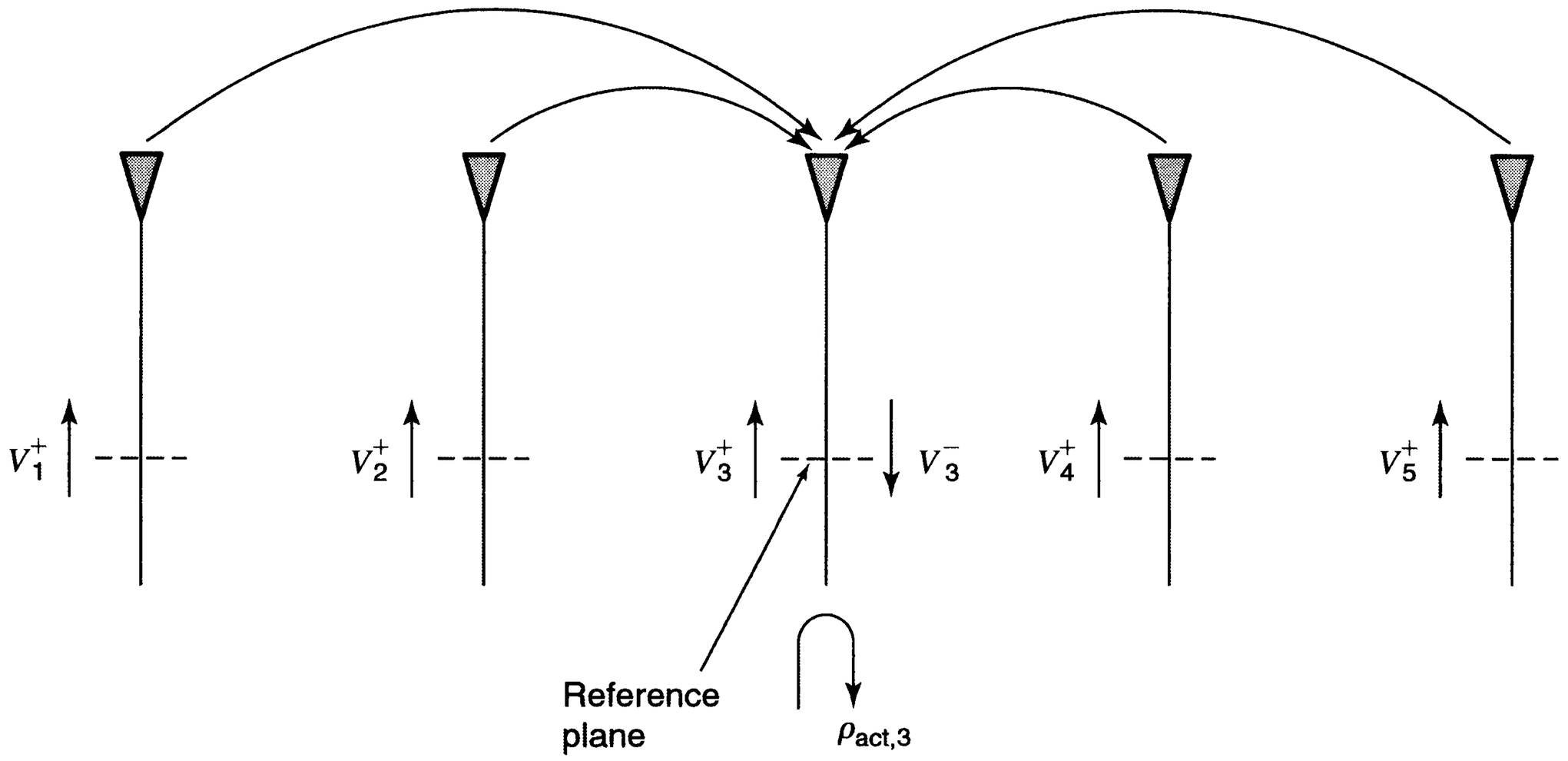


Figure 24–25
Measurement of the active impedance of element 3 in a five-element array. For the sake of clarity, the outward-traveling voltage wave V^- is shown for element 3 only.

where

$\rho_{act,n}$ = the complex active reflection coefficient of element n corresponding to a normalized active impedance $z_{act,n} = (1 + \rho_{act,n})/(1 - \rho_{act,n})$

V^+ = the voltage wave traveling inward, V

V^- = the voltage wave traveling outward, V

S_{mn} = V_m^- / V_n^+ is the S-parameter connected to voltage waves at the reference planes of elements m and n, dimensionless

An example of the impedance measurement for a multielement antenna is shown in Fig. 24–25. For larger arrays determination of all the $N \times N$ S-parameters is cumbersome and thus for them specific simulators with metal side walls to cover a part of the array with a smaller number of elements are usually used. With these certain feed combinations can be tested to find the respective active impedances (Evans-1).

24–5f Efficiency（效率）

The efficiency of an antenna is defined as the ratio between the power radiated by it and the power delivered into the antenna. It can also be defined as the ratio between power gain and directivity in certain directions. Depending on the definition of gain one can obtain several different efficiencies (see Sec. 24–5b). These may include in addition to the internal losses, which are the only ones taken into account in the traditional definition of radiation efficiency η_r, the effect of impedance mismatch with matching efficiency η_m, and the effect of polarization mismatch with the polarization efficiency p. For radiation efficiency one has to also define the interior of the antenna, which is not always self-evident. Often there are losses introduced by the vicinity of the antenna like the matching circuit, the radome, or the head of the user of a cell-phone handset, which are always present in a normal usage situation and can thus be considered as an integral part of the antenna.

Directivity/Gain Method. The most straightforward method to measure the efficiency is to determine directivity and gain as described in Secs. 2–7, 24–5a and 24–5b. As the directivity is often computed based on the integration of the power pattern according to Eq. (1) the method can in these cases be also called pattern integration method. Estimates for the directivity based on beamwidths are also given in Sec. 2–7.

The directivity/gain method is general and simple in principle but is difficult to realize with precision. The problem is associated especially with the full 3D radiation pattern measurement. In addition to the normal error sources of power pattern measurements like reflections in the anechoic chamber one has to cope with specific problems of full 3D measurements like those created by the feed cable of the AUT. The problems are especially difficult with antennas of low directivity as there are no "silent" directions, where the cables can be placed. Taking all the error sources into account it has been estimated that the uncertainty obtained with the pattern integration method for the efficiency (in percent) of microstrip antennas is $\pm 15 \ldots \pm 28$ percent units equivalent to about $\pm 1 \ldots \pm 1.5$ dB uncertainty for gain (Pozar-1). The directivity of the antennas in the estimation ranged from 7.5 to 15 dB. Normally, high directivity can be estimated with lower uncertainty than that of low-gain antennas, because the contribution of the sidewall reflections of the anechoic chamber is smaller. However, especially for small antennas with very low directivity the random errors caused by reflections are partly averaged out in different parts of the radiation pattern and therefore lower uncertainties than that estimated above can be achieved with careful measurements.

Radiometric Method. A lossy antenna can be modeled with a combination of a lossless antenna with similar directional pattern and an attenuator with power loss $L = D/G = 1/\eta_r$. If a lossless antenna is directed toward an absorber, its output power is defined by the temperature of the absorber only. By measuring the output noise power of the AUT with a low-noise receiver like a radiometer the efficiency can be defined according to the following equation:

$$P_{n,out} = G \cdot kB\left(\frac{T_A}{L} + \frac{L-1}{L}T_{AP} + (F-1)T_0\right) \quad \text{(W)} \tag{12}$$

where

$P_{n,out}$ = output noise power of the receiver, W
G = gain of the receiver, dimensionless
k = Boltzmann's constant = 1.38×10^{-23} J/K
B = noise bandwidth of the receiver, Hz
F = noise factor of the receiver, dimensionless
L = power loss of the attenuator describing the losses of the AUT, dimensionless
T_A = antenna noise temperature, K
T_{AP} = antenna physical temperature, K
T_0 = 290 K

Now three independent measurement results of $P_{n,out}$ are required to define L. Typically, two of these can be obtained according to Fig. 24–26 by placing the antenna first to an

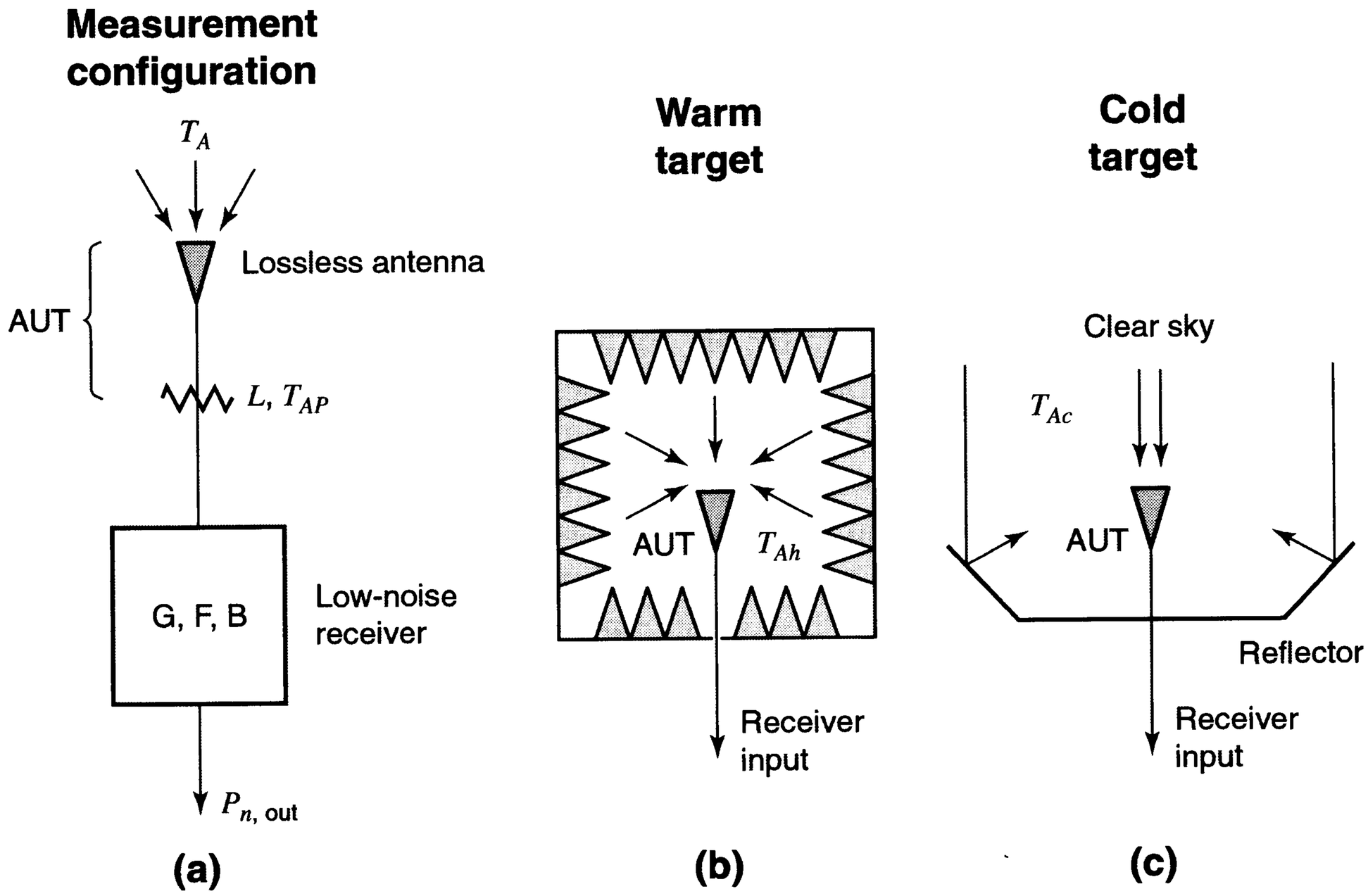


Figure 24–26
Radiometric method for the measurement of the antenna efficiency: (*a*) measurement configuration, (*b*) warm target, (*c*) cold target.

anechoic chamber or smaller enclosure covered with absorbers ($T_A = T_{Ah} \approx 290$ K) and then toward clear sky ($T_A = T_{Ac} = 10$–50 K, see Chap. 12). The warm-target measurement can be replaced by connecting a matched load to the input of the receiver. There is obvious uncertainty associated with the ground radiation in the sky measurement in spite of the reflector (Fig. 24–26*c*). The third measurement can be another cold-target measurement with a lossless standard antenna, with a well-known antenna pattern and antenna noise temperature contributions of the sky and ground. Another option is to connect a calibrated noise source to the receiver input.

From (12) one observes that it is also possible to obtain independent measurements by changing the antenna physical temperature T_{AP}. This can be done by putting the AUT in an anechoic chamber and placing it inside a styrofoam enclosure transparent to radio waves. Again lossless antenna or noise sources can be used for one or two of the three measurements. Here the difference between the power outputs is smaller than with a cold load and thus the method is more sensitive to errors in the measurement system. Also the response time may be long if the antenna is large.

The problems of the radiometric method are that special equipment is needed and that errors may be caused by inaccuracy of the noise sources, mismatch of the AUT and drift of the receiver gain.

Random Field Method. The radiometric method was based on exposing the AUT with an evenly distributed signal source. A similar effect can be achieved in reverberation chambers, which are usually rectangular multimode resonant cavities equipped with 1 to 3 mode

stirrers having typically the form of a propeller. Due to the rotation of the mode stirrers the mode pattern changes as a function of time. Sometimes the stirrers are replaced or accompanied by rotation of the AUT or the source. The technique is similar to that applied in microwave ovens. By investigating the average output power of the AUT over several stirring periods should give the same result as if the AUT would be receiving a noise-like signal with even angular distribution (Corona-1). In this case the output power of an antenna depends on its efficiency only. Thus, the efficiency can be obtained by comparing the AUT to a low-loss reference antenna.

The size of the cavity has to be at least a few wavelengths in each direction to ensure adequate mode density. The direct signal between the source and AUT should be blocked with metallic wall.

Wheeler Cap Method and Other Measurement Methods for Small Antennas. The real part of the impedance of a small antenna may be assumed to consist of loss resistance R_l caused by the internal losses and radiation resistance R_r. In this case the efficiency is

$$\eta_r = \frac{R_r}{R_l + R_r} \tag{13}$$

where

η_r = radiation efficiency of the AUT, dimensionless
R_r = radiation resistance of the AUT, Ω
R_l = loss resistance of the AUT, Ω

The internal losses are defined by the current distribution of the antenna. If the antenna is placed inside a small nonresonant metal shield, which prevents its radiation, but is large enough not to disturb the currents, the real part of the impedance should reduce to R_l. This is the idea of the well-known Wheeler cap (Wheeler-1). Originally Wheeler recommended the use of a spherical shield with the radius $\lambda/2\pi$ of the radiansphere to enclose the AUT. Later the method has been applied mainly to antennas having a groundplane by covering them with a cap and it has been shown that the shape and size of the cap can be chosen quite freely provided the current distribution of the antenna is not disturbed, good contact is maintained between the edge of the cap and the groundplane, and cap resonances are avoided. This limits the maximum size of the AUT approximately to $\lambda/4$ (Newman-1).

The Wheeler cap method can provide results with low uncertainty, estimated to be ± 2 percent units of efficiency (Pozar-1). The problems associated with it are (1) that the small cap may change the current distribution of the AUT and thus the R_l, (2) that the loss mechanisms may include some that cannot be described with one resistor only (both longitudinal and parallel loss mechanisms), and (3) that in measurements of resonant antennas Eq. (13) does not work due to resonance detuning. The changes of the current distribution can be estimated by investigating the change in the reactive components in the equivalent circuit of the AUT due to the cap. As the change of these is normally only on the order of 1 percent, it can be estimated that the change in R_l is not large. The second and third problems appear especially for resonant antennas and can be avoided by employing measurement techniques of resonant circuits (see Example 24–5.3). The AUT is described with a lossy feed structure

EXAMPLE **24–5.3 Wheeler Cap Measurement of Antenna Efficiency**

When measuring a resonant-type antenna in a Wheeler cap, the typical result for the power reflection coefficient is as shown in Fig. 24–27. Far from the resonance one obtains some constant return loss $L_{r,far}$, which is 0 dB, if there are no losses in series with the resonant circuit like those in the feed circuitry. If $L_{r,far} > 0$, then the attenuator in the equivalent circuit model of the AUT in Fig. 24–27 has value $L_s = L_{r,far}/2$ (in dB).

The return loss $L_{r,res}$ at the resonant peak gives the value for the resistor R_s in the series resonant circuit, which we can assume without loss of generality:

$$R_s = Z_c \frac{1 \pm |\rho_{res}|}{1 \mp |\rho_{res}|} \quad (\Omega) \tag{14}$$

$$|\rho_{res}| = 10^{-(L_{r,res} - L_{r,far})/20}$$

where

R_s = resistance of the series resonance circuit (see Fig. 24–27), Ω
Z_c = characteristic impedance of the feed line, Ω
ρ_{res} = voltage reflection coefficient of the resonance circuit, dimensionless
$L_{r,res}$ = return loss at the resonant frequency, dB
$L_{r,far}$ = return loss far from the resonance, dB

In (14) the lower sign is for the overcoupled and the upper sign for the undercoupled case. The coupling can be determined by investigating the size of the resonant circle for the complex reflection coefficient of the AUT. If the diameter multiplied with $10^{L_{r,far}/20}$ is larger than 1, we have the overcoupled case and if <1 the undercoupled case. Now we get the total efficiency by determining the resistor values R_{s1} and R_{s2} with and without the Wheeler cap, respectively, and using Eq. (13) and L_s, which does not change when the cap is placed on top of the antenna:

$$\eta_r = 10^{-L_s/10} \frac{R_{s2} - R_{s1}}{R_{s2}} \tag{15}$$

where

η_r = radiation efficiency of the AUT, dimensionless
R_{s1} = resistance of the series resonance circuit with the Wheeler cap, Ω
R_{s2} = resistance of the series resonance circuit without the Wheeler cap, Ω
$L_s = L_{r,far}/2$ = loss of the series attenuator in the equivalent circuit, dB

Note that the small change of the resonant frequency caused by the cap or the rotation of the impedance locus on the Smith Chart (see Fig. 24–27) had no effect on the result obtained.

ANTENNA MEASUREMENT BY WHEELER CAP METHOD

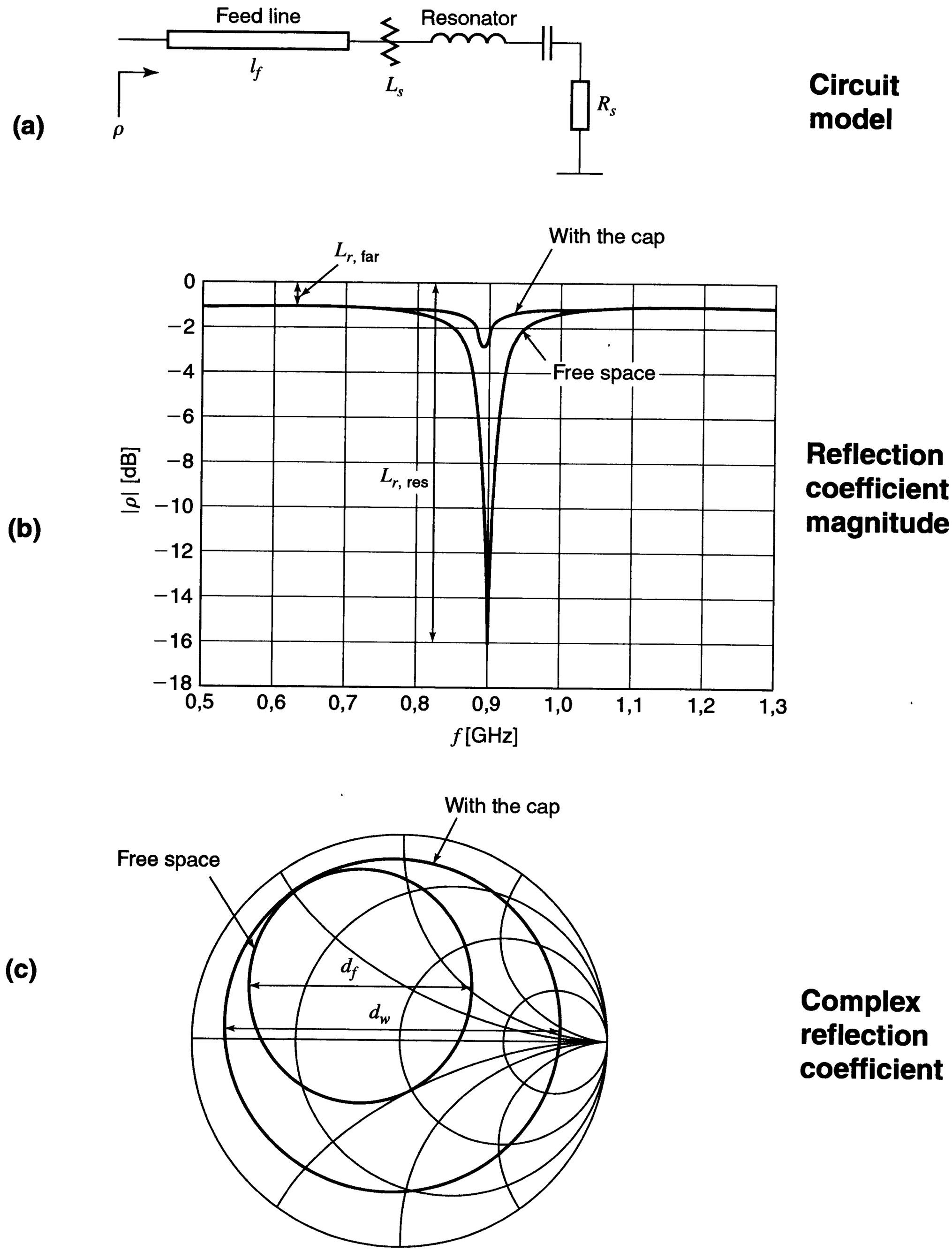


Figure 24–27
Measurement configuration for a resonant-type antenna with the Wheeler cap method. (*a*) Circuit model, (*b*) magnitude of the reflection coefficient, (*c*) complex reflection coefficient.

To avoid some problems associated with the Wheeler cap method it has been proposed to place the AUT inside a large waveguide to collect and measure the power radiated by it (Johnston-1). Here the basic idea is to treat the AUT as a 2-port with the output to free space and define the *S*-parameters of the 2-port with the waveguide measurement setup either with a transmission or reflection measurement. In both cases sliding shorts are used at the ends of the waveguide section to provide proper reflections for each case. The estimated uncertainty of the method is about ± 1 percent units. The result is also quite insensitive to the location of the antenna in the waveguide. When compared to the Wheeler cap method the waveguide method is more general as it does not require any assumption on the equivalent circuit of the AUT. The waveguide has also somewhat larger clearance around the AUT (width $3\lambda/4$, height $3\lambda/8$) than the Wheeler cap and thus the near fields of the AUT are disturbed less in the waveguide. On the other hand, the waveguide method requires somewhat complicated waveguide setup with high-quality sliding shorts.

Both the Wheeler cap method and the waveguide method are limited to the size of the AUT. Also in the case of small antennas one has to often include the environment of the antenna as for a cell phone in the evaluation of the total efficiency of the antenna. This subject is discussed more in Sec. 24–6b.

24–5g Current Distribution[1] (电流分布)

In many cases it is important to know the current distribution along an antenna. For example, if both the magnitude and phase of the current are known at all points along an antenna, the far field can be calculated.

The current can be sampled by a small pickup loop placed close to the antenna conductor. Loop and indicator can be connected to a single unit. The output of the loop probe can be detected and transferred as low-frequency signal with high-impedance lines (Schmid-1). To obtain also the phase information the RF signal can be transferred either directly with a thin coaxial cable or by using a fiber-optic link (Masterson-1). This technology has close connections to measurements of small antennas and the specific absorption rate (see Secs. 24–6b and 24–6d). However, at short wavelengths the indicating instrument may be too large to place near the antenna without disturbing the field. To remove the indicator from the antenna field the arrangement of Fig. 24–28 can be used. Here the loop projects through a longitudinal slot in the hollow antenna conductor. The output cable from the loop is confined within the antenna conductor and is brought out through the end of a grounded stub as shown. The arrangement in Fig. 24–28 permits both amplitude and phase measurements. The phase is measured by comparison with a reference current as suggested by the dashed connections in the figure. The signal picked up by the loop is mixed with a signal of approximately equal amplitude extracted by a probe on a matched slotted line. With the antenna sampling loop fixed, the line probe is moved to give a minimum indication. The phase shift between the line-probe positions then equals the phase shift between the two antenna sampling-loop locations. The phase shift is a linear function of distance on a line with matched termination. Assuming the phase velocity equals that of light in free space, the phase shift θ along the line in degrees per unit length is given by $360^\circ/\lambda_0$, where λ_0 is the free-space wavelength

[1]This section is mostly from the second edition of *Antennas,* by J. D. Kraus, McGraw-Hill, 1988.

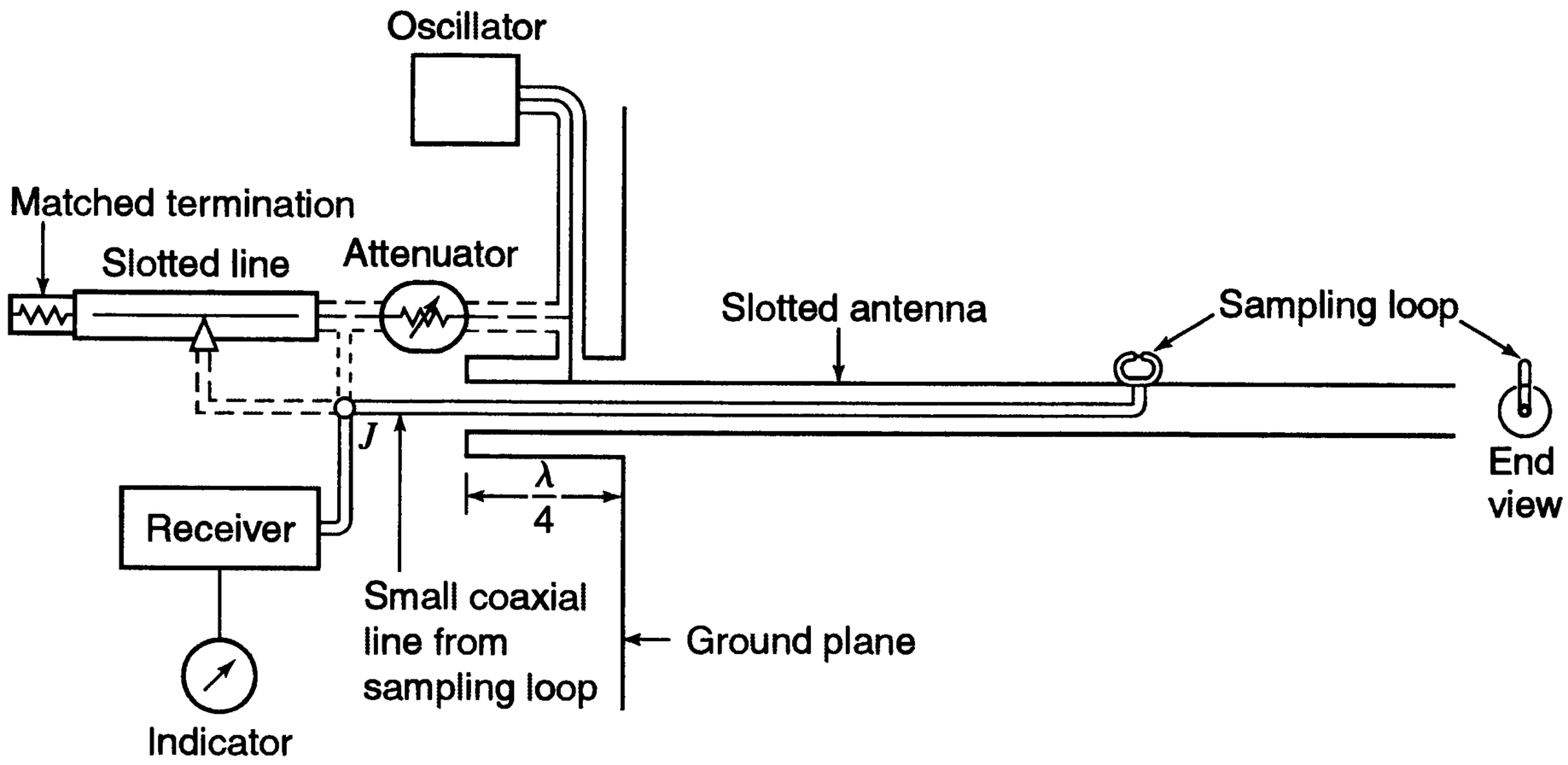


Figure 24–28
Slotted antenna and sampling loop arranged for measurement of current amplitude and phase.

of applied signal. The phase change between two points on the line is then the distance between the points multiplied by θ.

In Chap. 14 on cylindrical antennas Fig. 14–14 shows the current distribution on a 5λ cylindrical monopole antenna 0.2λ in diameter measured using an arrangement similar to that in Fig. 24–28.

24–6 MISCELLANEOUS TOPICS（各种特殊测量）

24–6a Large Millimeter and Submillimeter Wave Antennas
（大型毫米波和亚毫米波天线）

Millimeter and submillimeter wave radiometers up to 3 THz are planned for the future remote sensing and radio astronomy satellites. The antennas of these instruments will be large, of the order of 1 meter (= 10,000 wavelengths at 3 THz), and provide a challenge for the measurement techniques. As the frequency increases, antenna measurements get more and more difficult for many reasons: available power decreases, receiver sensitivity reduces, reflectivity of absorbers increases, dimensional accuracy requirements become tighter, etc. Thus, the upper frequency limits of the existing measurement ranges are at the most 200 GHz.

The far-field distance for a 1 meter antenna operating at 1 THz is 7 km. Water vapor has several strong resonances which make the atmosphere nearly opaque at submillimeter waves. Therefore, a far-field measurement is not practical.

A near-field measurement would be extremely demanding. The number of measurement points would be very large, in the millions. The instrumentation (transmitter, receiver, scanner) should be stable, accurate, and capable of gathering a large number of samples per second.

A compact antenna test range seems to be the most promising alternative. A large, accurate reflector would be needed. The attenuation of air may be significant even at short

distances: close to the resonance frequencies attenuation may be more than 10 dB per meter. Therefore, the measurement chamber should be very dry or nitrogen-filled.

24–6b Electrically Small Antennas（电小天线）

Electrically small antennas, which are used, e.g., in portable radio devices form a specific group when it comes to their measurements. This is especially true for the measurement of their radiation properties. Impedance measurements are less specific, but the strong frequency dependency like the common resonant-type behavior of the impedance of small antennas should be recognized. Here equivalent circuits may form a useful viewpoint (see Example 24–5.3).

The typical feature of electrically small antennas is their low directivity. From the measurement point of view this is beneficial in that sense that measurements of low side-lobe levels are not required. On the other hand, the low directivity means that one has to measure 3D pattern of the antenna or most of it, because there is no specific main lobe or polarization. This is quite difficult and sets specific requirements for the vicinity of the antenna like cabling. Also, the usage environment in the vicinity of the antenna like the user of a handheld phone has to be included in the measurement setup to obtain realistic results in both radiation and impedance measurements. This can be done by using human body phantoms (see Sec. 24–6d) or with test persons (Pedersen-1). An example of this is shown in Fig. 24–29. The number of points required to sample the pattern with

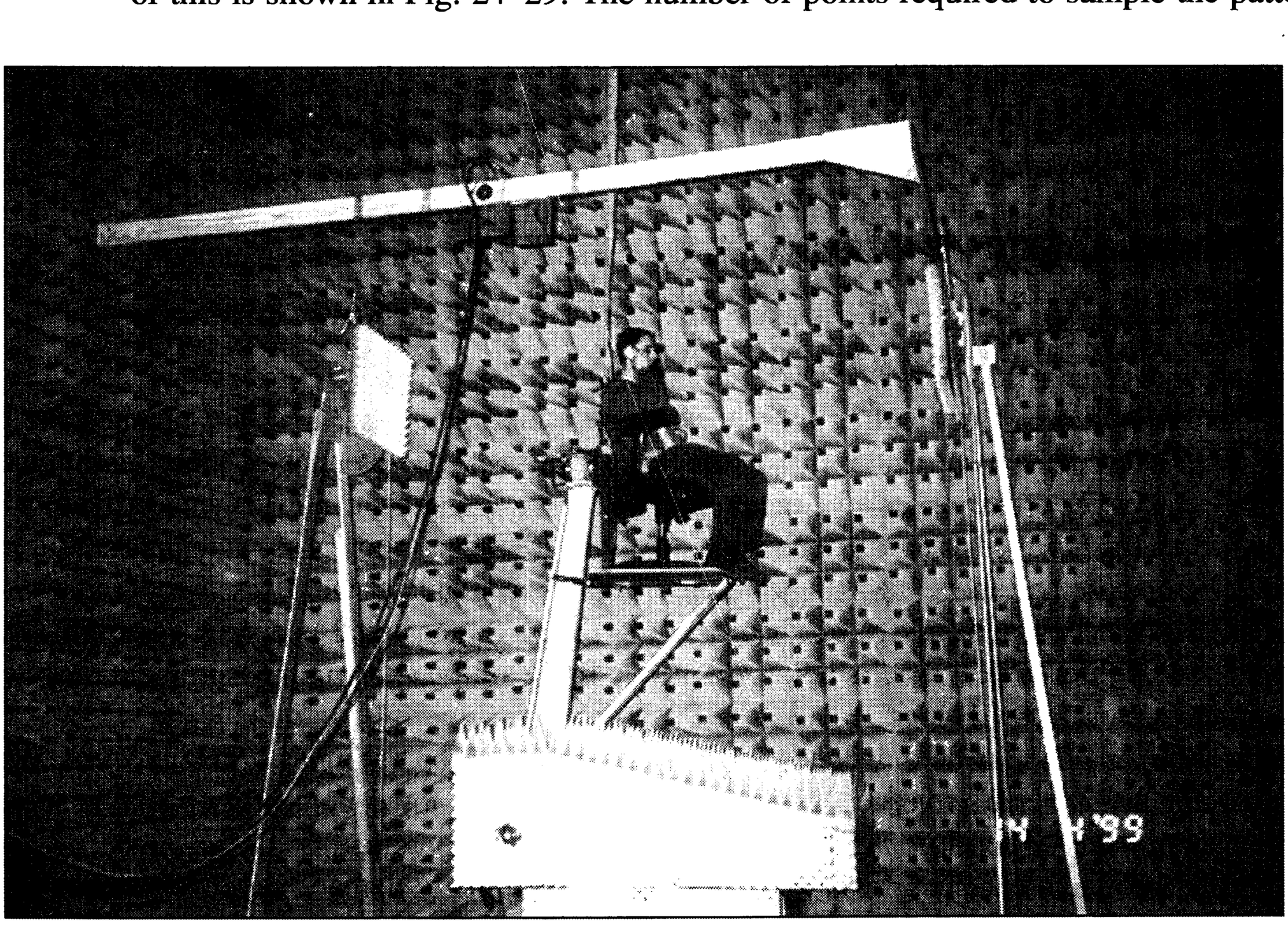


Figure 24–29
3D antenna pattern measurement for a handheld cellphone with a test person in anechoic chamber. (Courtesy of Center for PersonKommunikation, Aalborg University.)

adequate resolution depends on the size of the antenna defining its directivity. An estimation for the number of points is given as $16\pi(D/\lambda)^2$ (Evans-1) by assuming that 2 pattern points are required per 3 dB beamwidth ($\approx \lambda/D$, rad), where D is the largest *dimension* of the physical aperture of the antenna. This issue can also be studied by performing modal analysis of the pattern of the antenna with spherical wavemodes (Hansen-1). Typical grid spacing for a handheld cell-phone antenna measurement with the user is 5 to 10 degrees.

The 3D radiation pattern can be used to estimate the efficiency of the small antenna, which is probably the most significant single factor describing its performance. The necessity to include the effect of the user limits the use of other less complicated efficiency measurement methods described in Sec. 24–5f. The minimum configuration for a handset antenna containing the handset and a head phantom is too large for the Wheeler cap or waveguide methods. The stirred mode chamber has adequate size but the effect of the lossy phantom on the average field intensity in the chamber has to be compensated. Also the radiometric method can be applied with careful consideration of the effect of the changes in impedance matching due to the phantom.

As mentioned in Sec. 24–2b, for electrically small antennas, e.g., the nominal far-field border given by the Rayleigh distance of Eq. (24–2–3) may be closer than the reactive near-field range of Eq. (24–2–2). Thus one has to consider all the Eqs. (24–2–3) to (24–2–5) when defining the measurement range for small antennas. For $\Delta L \leq 1$ dB one can find that if the size D of the antenna is smaller than about 0.3λ, the dominating criterion is the effect of the near fields according to Eq. (24–2–4). For antennas larger than that but still being in the category of small antennas, the most significant criterion for the measurement distance seems to be the effect of the change in the measurement distance according to Eq. (24–2–5).

Another specific problem in measuring a small antenna is the effect of the cable connected to the AUT. As the radiation pattern is more or less isotropic, the cable can be easily situated in some essential part of the pattern. Furthermore, when the cable is connected to the small antenna, it changes the current distribution of it. There are a few options to improve the situation with small antennas in this respect:

1. Use nongalvanic connection to the AUT like optical fiber or high-impedance wire and a diode detector.
2. Reduce currents flowing on the cable with chokes.
3. Place the cable in such a position that it has the least effect on the field.
4. Use a battery-powered small transmitter connected to the AUT.

In all cases except (4) the cable makes the rotation of the AUT in 3D measurements difficult. Thus it is beneficial to use measurement methods, where the AUT can be in a fixed position or possibly rotated around one axis only. Here near-field measurements offer often a good solution but also far-field setups like the one in Fig. 24–29 can be used. Option 1 has certain limitations compared to the use of RF cables. The transformers between optic and RF signals provide phase information and have bandwidth of several GHz, but with their batteries are somewhat bulky. This technique is, however, developing rapidly as applied to mobile communications systems. Another option is to use the passive method with modulation of an optical signal with the RF signal (Masterson-1). The diode

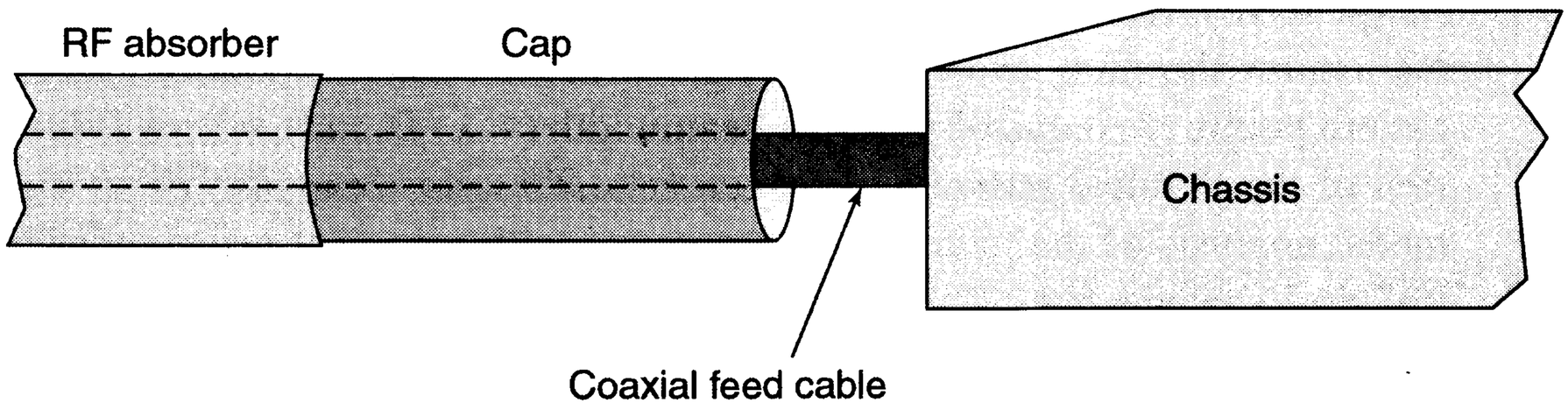


Figure 24–30
Balun-type cap to prevent currents flowing on the surface of the measurement cable.

detectors provide only amplitude information and have the well-known problems with limited dynamic range. The typical solution for option 2 is to use ferrite bead-type common mode chokes on top of the cable shield (Saario-1). However, as the goal is to keep the current distribution in the vicinity of the small AUT unchanged, these are often not the best solution as the ferrite chokes are resistive and thus absorb the current instead of forming an "open circuit" for them. We have proposed a solution utilizing a balun-type cap on the surface of the cable (Fig. 24–30) (Icheln-1). The cap prevents effectively changes on the surface current distribution of the mobile phone chassis. The limitation of this solution is its fairly limited bandwidth of around 20%, which means that one has to change the cap when typical dual-frequency antennas are measured.

Option 3 has been investigated by Saario et al., who have shown that the cable should be positioned perpendicularly to the polarization investigated (Saario-1). This condition, however, makes dual-polarized and full 3D radiation pattern measurements difficult. Option 4 enables normally only amplitude measurements and thus prevents phase pattern or near-field measurements. This limitation can be avoided by using two source antennas and phase comparison (Tuovinen-1).

24–6c Effective Gain（有效增益）

Typically the antenna receives a signal coming from a well-defined source, which is most often a point source as in the case of another antenna. In radiometer measurements of radio astronomy and passive remote sensing the source is a distributed one but signals arriving from different directions are noncoherent and the position of the antenna is known. Thus, it is fairly simple to derive the connection between the source and the received signal from the information in the power pattern of the antenna. The situation becomes complicated if the antenna receives several coherent signals from random directions or the position of the antenna varies. This is the case in mobile communications involving multipath signals. Thus, for mobile terminals a special statistical approach is required to characterize the performance of the antenna. The parameters used here are *effective gain* and *mean effective gain*. Effective gain G_{eff} is the power received by the AUT compared to the power received by a reference antenna like isotropic antenna or half-wave dipole with certain signal distribution. The definition of G_{eff} is (Jakes-1)

$$G_{eff} = \frac{P_r}{P_{i,\theta} + P_{i,\phi}} \tag{1}$$

where

G_{eff} = effective gain of the AUT
$P_{i,\theta}$ = average power received by an isotropic θ-polarized antenna, W
$P_{i,\phi}$ = average power received by an isotropic ϕ-polarized antenna, W

In the numerator of Eq. (1) we can consider the local received power and thus obtain information also on the possible effect the antenna has on the short-term fading. However, to get the reference power in the denominator it is necessary to eliminate the short-term fading by averaging over a distance of at least some wavelengths, because otherwise G_{eff} might become infinite locally.

As the signal distribution in mobile communications is changing continuously, one has to investigate the statistical characteristics of the signal received by the antenna in a number of locations, typically a test measurement route (Andersen-1). Here it is possible to study, e.g., the cumulative distribution of G_{eff} to find certain reliability levels. The mean of G_{eff} is also often used and it is called Mean Effective Gain (MEG) (Taga-1). As the idea of MEG is to describe the performance of an antenna in a certain usage environment (urban, suburban, rural, indoor, etc.), it is defined as an average of G_{eff} over some representative sample of the environment like a test route. The standard procedure is to consider the statistical distribution of the incoming waves and combine that with the information on the 3D radiation properties of the AUT. In this case one obtains based on Eq. (1):

$$\text{MEG} = \int_0^{2\pi}\int_0^{\pi}\left\{\frac{XPR}{1+XPR}G_\theta(\theta,\phi)P_\theta(\theta,\phi) + \frac{1}{1+XPR}G_\phi(\theta,\phi)P_\phi(\theta,\phi)\right\}\sin\theta\, d\theta\, d\phi \tag{2}$$

where

MEG = mean effective gain of the AUT, dimensionless
XPR = $P_{i,\theta}/P_{i,\phi}$ is the cross-polarization ratio of the incident field in the environment, dimensionless
$G_\theta(\theta,\phi)$ = θ component of the gain pattern of the AUT, dimensionless
$G_\phi(\theta,\phi)$ = ϕ component of the gain pattern of the AUT, dimensionless
$P_\theta(\theta,\phi)$ = θ component of the angular density function of the power of the incoming plane waves, dimensionless
$P_\phi(\theta,\phi)$ = ϕ component of the angular density function of the power of the incoming plane waves, dimensionless

The distribution functions are normalized as:

$$\int_0^{2\pi}\int_0^{\pi}\{G_\theta(\theta,\phi)+G_\phi(\theta,\phi)\}\sin\theta\, d\theta\, d\phi = 4\pi\cdot\eta_{tot} \tag{3a}$$

$$\int_0^{2\pi}\int_0^{\pi}P_\theta(\theta,\phi)\sin\theta\, d\theta\, d\phi = \int_0^{2\pi}\int_0^{\pi}P_\phi(\theta,\phi)\sin\theta\, d\theta\, d\phi = 1 \tag{3b}$$

where η_{tot} = total efficiency of the AUT, dimensionless

Table 24–2 Measured parameters of the distributions of the incoming waves as a function of θ in urban environment in Tokyo

Route	θ_{mV}	σ_V	θ_{mH}	σ_H	XPR
Ningyo-cho	71°	20°	58°	64°	5.1 dB
Kabuto-cho	70°	42°	40°	90°	6.8 dB

θ_{mV} = mean value of the Gaussian θ distribution of the incoming VP waves, degrees
θ_{mH} = mean value of the Gaussian θ distribution of the incoming HP waves, degrees
σ_V = standard deviation of the Gaussian θ distribution of the incoming VP waves, degrees
σ_H = standard deviation of the Gaussian θ distribution of the incoming HP waves, degrees
$XPR = P_{i,\theta} / P_{i,\phi}$ is the cross-polarization ratio of the incident field in the environment, dimensionless

The normalization of the gain pattern includes the total efficiency with the effect of both internal losses and mismatch of the antenna. This is due to the purpose of MEG to describe the total behavior of an installed typically small antenna having poor matching often as unavoidable property. The distribution of the incoming waves can be assumed uniform as a function of ϕ in mobile communications. Gaussian distributions were determined as a function of θ for $P_\theta(\theta, \phi)$ and $P_\phi(\theta, \phi)$ in measurements in an urban environment in downtown Tokyo (Taga-1). The parameters obtained are presented in Table 24–2. The results in Table 24–2 show the typical feature in urban environments that the average direction of arrival of the signals is somewhat elevated. The reason is that the waves received by the mobile antenna have first propagated above rooftops and then diffracted down to the street level. The value of *XPR* is also quite typical for urban environments, when the basestation antenna has vertical polarization, and shows that the signal is not totally depolarized in spite of the complex propagation environments. In indoor environments one can expect to obtain wider and in suburban and rural environments narrower distributions as a function of θ than in the urban environment.

Taga calculated also the MEG of a halfwave dipole antenna with different orientations and Gaussian field distributions. It was shown that for a halfwave dipole antenna with an inclination of 55° from vertical the MEG $= -3$ dB practically independent of the *XPR* or the distributions (θ_{mV}, θ_{mH}, σ_V, σ_H) of the incoming waves. Thus, this antenna can be used to define the total incident power $P_{i,\theta} + P_{i,\phi}$ by multiplying the average power obtained with it by 2.

EXAMPLE **24–6.1 Mean Effective Gain (MEG) of a Small Antenna**

Let's consider the MEG of a small lossless antenna having the pattern of an elementary electric dipole. For simplicity, let's assume that the dipole is vertical and the means of the θ distributions of the incoming waves are $\theta_{mV} = \theta_{mH} = 90°$. The other parameters of the considered distributions are:

(*a*) Rural-type distribution: $\sigma_V = 5°$, $\sigma_H = 20°$, $XPR = 15$ dB
(*b*) Urban-type distribution: $\sigma_V = 30°$, $\sigma_H = 50°$, $XPR = 6$ dB
(*c*) Indoor-type distribution: $\sigma_V = 60°$, $\sigma_H = 90°$, $XPR = 3$ dB

■ **Solution**

The normalized gain patterns of the small dipole and the signal distributions are:

$$G_\theta(\theta,\phi) = 1.5\sin^2\theta \qquad G_\phi(\theta,\phi) = 0 \qquad P_\theta(\theta,\phi) = A_\theta e^{-\left[\frac{(\theta-\theta_{mV})^2}{2\sigma_V^2}\right]},$$

where A_θ is a normalization parameter.

We notice from Eq. (2), however, that for this vertical dipole we have to consider P_θ only. Now, in each case (a) to (c) we have to first solve A_θ using Eq. (3) as:

$$A_\theta = \left[2\pi \int_0^\pi e^{-\left[\frac{(\theta-\theta_{mV})^2}{2\sigma_V^2}\right]} \sin(\theta)\, d\theta\right]^{-1}$$

Then we get

$$\text{MEG} = \frac{XPR}{1+XPR} 2\pi \int_0^\pi 1.5\sin^2(\theta) P_\theta(\theta,\phi)\sin(\theta)\, d\theta$$

where the first term gives the average polarization mismatch and the second term describes the match between the antenna pattern and power distribution. The results are:

(a) $A_\theta = 0.730$	$XPR/(1+XPR) = -0.41$ dB	MEG $= 1.3$ dB	*Ans.* (a)
(b) $A_\theta = 0.139$	$XPR/(1+XPR) = -0.97$ dB	MEG $= 0.0$ dB	*Ans.* (b)
(c) $A_\theta = 0.0961$	$XPR/(1+XPR) = -1.8$ dB	MEG $= -1.4$ dB	*Ans.* (c)

The significance of polarization mismatch is obvious especially in the urban and indoor environments, promoting thus the use of polarization diversity or adaptivity in mobile communications terminals.

By determining the 3D gain patterns for antennas of terrestrial or satellite mobile communications systems including the effect of the user or vehicle and combining the result with, e.g., measured distributions of the incoming waves in different environments one can predict the behavior of the mobile antenna without performing tedious test route measurements for each prototype (Pedersen-2; Kalliola-1). Another possibility is to use a test environment having certain random field distribution with typical characteristics like uniform azimuthal distribution. This method, however, gives the performance of the antenna in one average environment only.

24–6d Specific Absorption Rate (SAR) and Exposure Standards
[比吸收率（SAR）和暴露标准]

The quantum energy of radio frequency radiation is so low that the radiation cannot ionize biological materials like human tissues. Thus the harmful biological effects of radio frequency radiation are (according to current scientific information) limited to those caused by the rise of the temperature of the tissues due to absorption of the RF energy. The so-called nonthermal effects on the nervous system, cell membranes or the development of cancer have been studied extensively, but the results are controversial and no conclusive proof of such effects has been found. Due to the vast increase of exposure of the general public to RF energy along with the increased use of portable communication devices, the studies in the area of both thermal and nonthermal effects are continuing. In the investigation of the

health issues of RF transmitters and antennas it is essential to determine the absorption of the energy per unit mass of tissue described with a quantity called the Specific Absorption Rate (SAR) (Kuster-1; USAF-1).

$$\text{Specific Absorption Rate} = \text{SAR} = \frac{dP_a}{dm} = \frac{\sigma E^2}{\rho} \qquad \text{(W/kg)} \tag{4}$$

where

SAR = specific absorption rate, W/kg
dP_a = power absorbed to volume of the tissue, W
dm = the mass of a certain infinitesimal volume, kg
σ = local effective conductivity of the tissue material, S/m (= ℧/m)
ρ = local density, kg/m^3
E = magnitude of the electric field at the inspection point, V/m

Usually this local SAR is averaged over certain volume or mass of tissue and it is applied for localized exposure, for example, in the case of hand-held cell-phones. Traditionally the exposures of potentially harmful RF radiation have mostly occurred either in the near field of low-frequency devices like industrial RF heaters in the 13 MHz and 27 MHz range or in the far field of broadcast or radar transmitters. Thus, the determination of SAR has been mainly performed by measuring the unperturbed electric or magnetic field and studying the absorption in the whole body or parts of it like the head using dosimetric models (USAF-1; Hagmann-1). Due to the increased use of portable radios since the 1980s the localized SAR values close to the transmitter have gained a lot of interest in both research and standardization. As the localized SAR values have the closest connection to antenna design, this section will mainly concentrate on discussion on the determination of those.

The exposure standards for the RF radiation define the SAR limits for persons in both controlled (occupational exposure) and uncontrolled (general public exposure) conditions (ICNIRP-1; prENV-1; IEEE-2; ARIB-1). According to major international and national standards, the limit value for the average whole body SAR for controlled conditions is 0.4 W/kg and for uncontrolled conditions 20% of that, i.e., 0.08 W/kg in the frequency range of 10 MHz to 6 GHz. For localized exposure the average whole body value limit may be met, but still the exposure can be harmful. Thus there are separate values for the local peak SAR. An overview of the peak values for most parts of the body and the parameters

Table 24–3 Local peak SAR values and respective parameters[1]

	USA	ICNIRP, Europe and Japan
Peak SAR	1.6 W/kg	2 W/kg
Averaging period	30 min	6 min
Averaging mass and shape	1 g/cube	10 g/cube*

*ICNIRP has not defined the shape of the volume.

[1]Since the body's heat sensors are in the skin, RF heating can occur internally without much awareness. Thus, safe energy guidelines are needed to prevent undesirable internal heating. Note that [SAR(W/kg)] × [time (s)] = energy/mass (J/kg).

used in their definition is given in Table 24–3. Higher values are applied for hands, wrists, feet and ankles. The estimated temperature increase in the brain due to local SAR = 2 W/kg is around 0.1°C (Anderson-1).

As SAR can be defined in laboratory conditions only, the standards define limits at different frequency ranges for electric and magnetic field strengths, plane wave power density, and induced and contact currents, which, when applied in field tests, should ensure that the whole body SAR values are not exceeded. However, in the determination of peak SAR one has to take into account the local properties of the tissue and the transmitting device. This can be performed either by numerical electromagnetic simulations or by measurements.

In the simulations the tissue can be described in a very detailed manner involving 15 to 30 tissue types with resolution on the order of 1 to 3 mm in the head based on information obtained by Magnetic Resonance Imaging (MRI) (Lazzi-1). Disadvantages of simulations are difficulties in describing the transmitter and antenna with required amount of details and lengthy calculation times. Thus, the simulations can be utilized best at the design phase of mobile handset antennas. The most popular method for SAR calculations is the Finite Difference Time Domain (FDTD) method; see Fig. 24–31. Numerical methods allow the simulation of the fine structure of the tissues but the absolute accuracy is not sufficient for compliance tests.

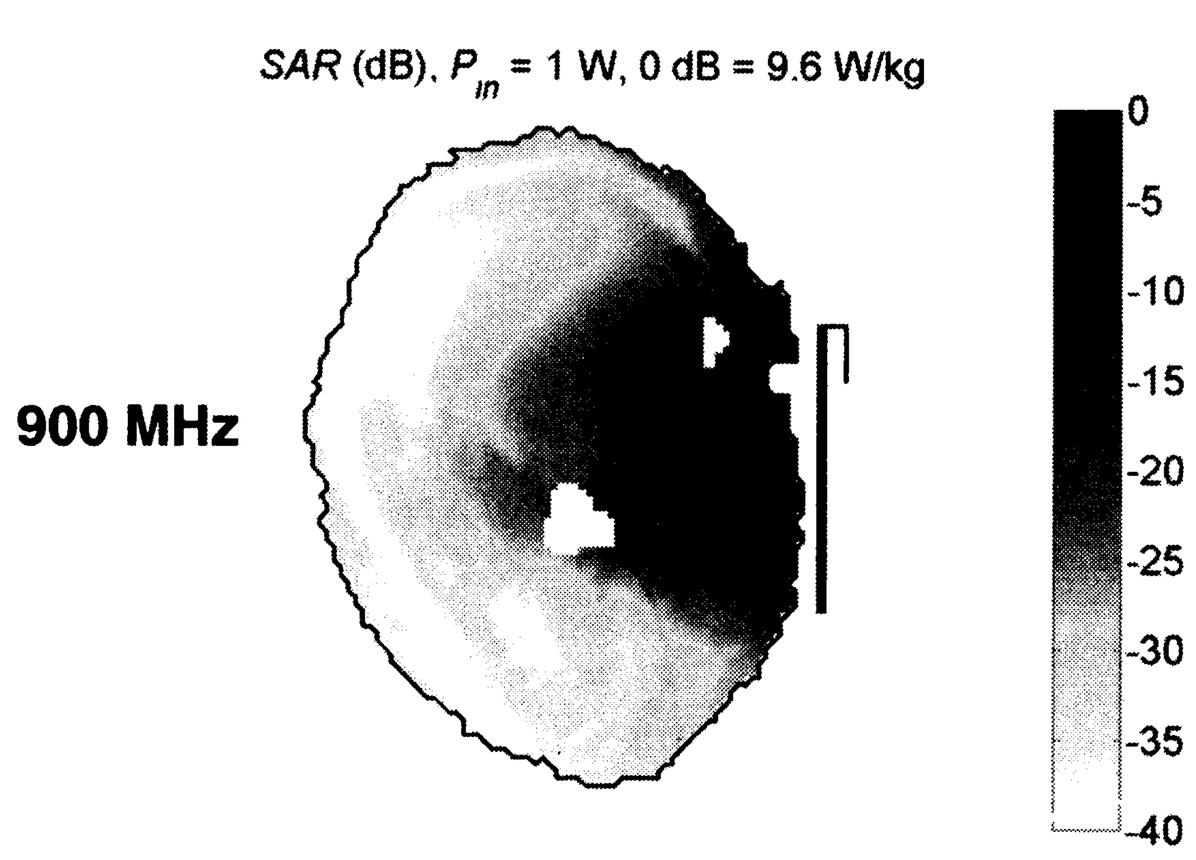
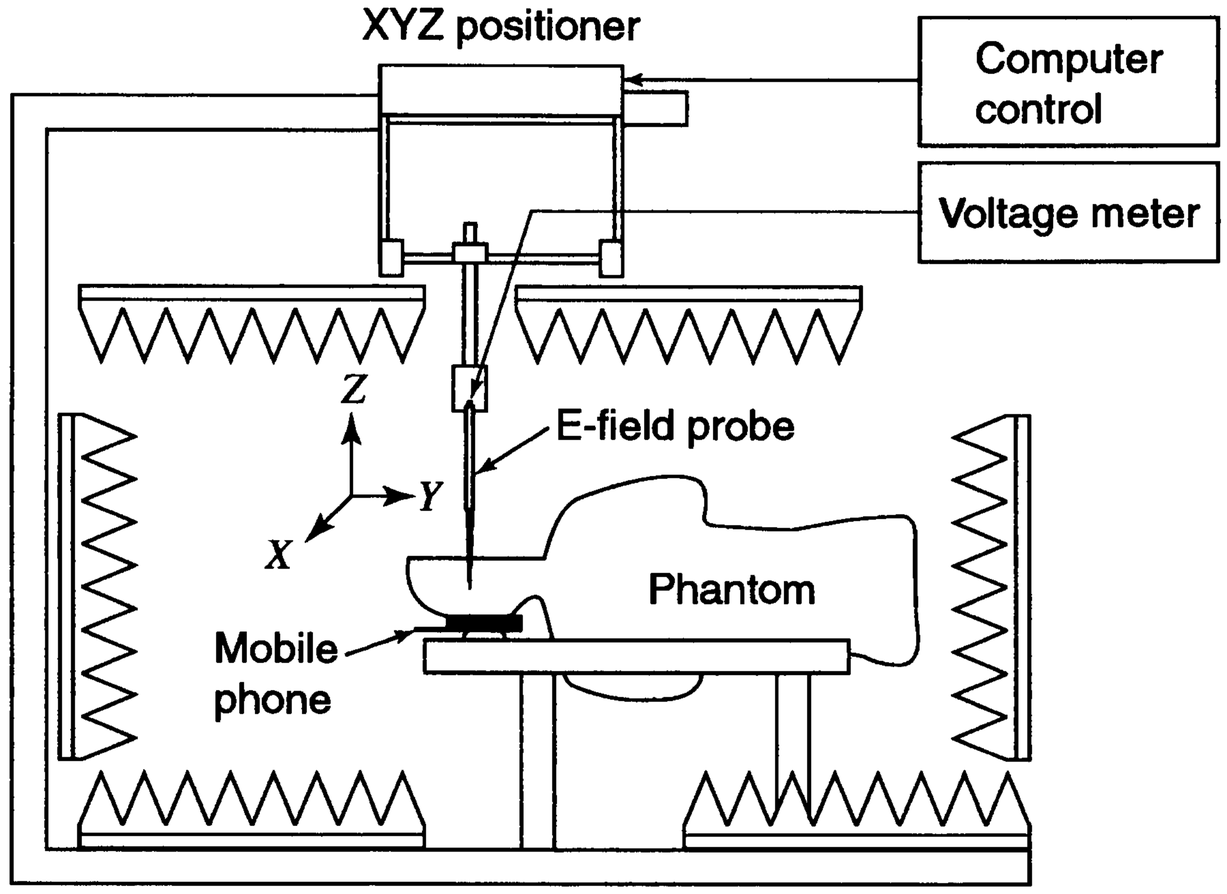


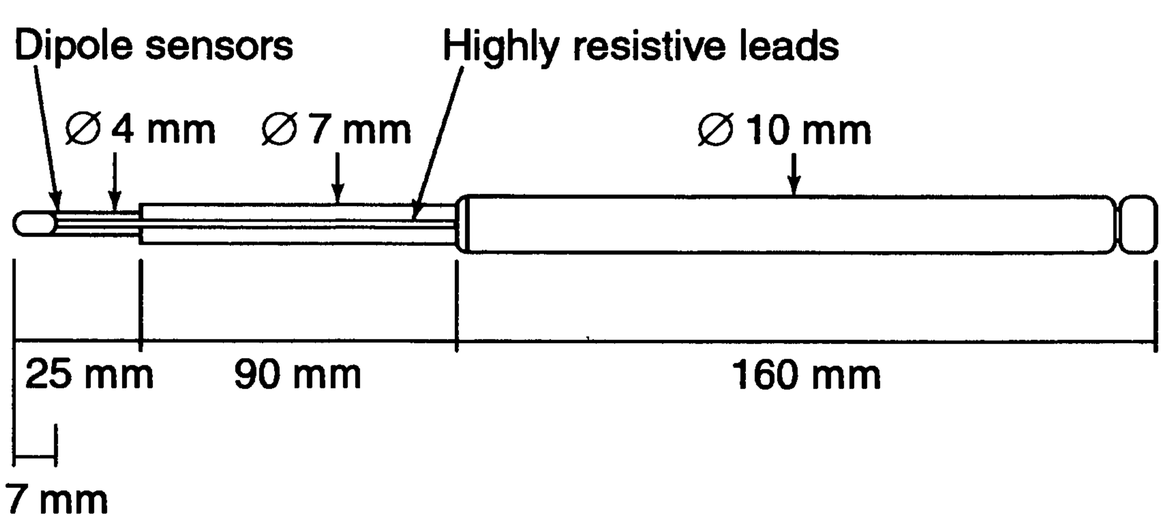
Figure 24–31
An example of calculated SAR distribution inside a head model at 900 MHz (upper) and 1800 MHz (lower).

SPECIFIC ABSORBTION RATES (SAR)



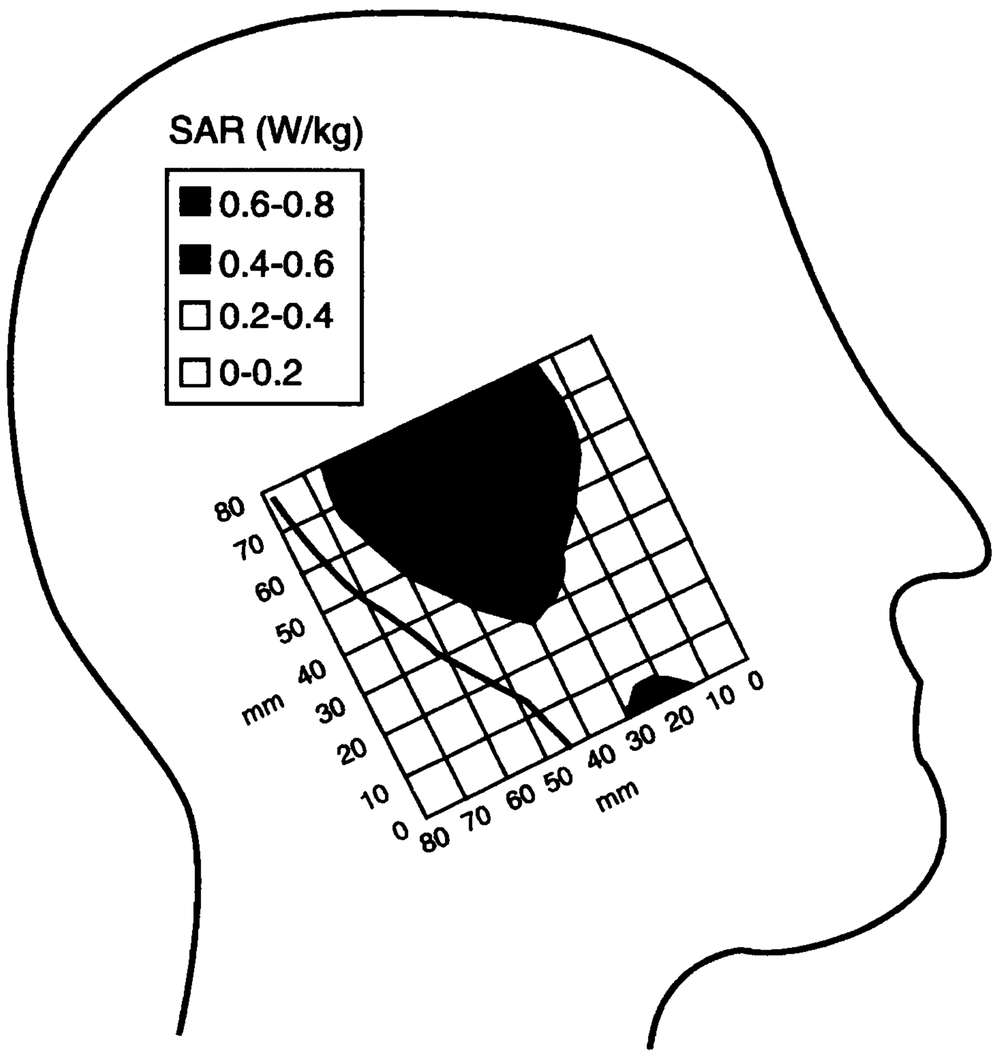
Measurement configuration

(a)



Probe

(b)



Example of measured SAR

(c)

Figure 24–32
SAR measurement: (*a*) measurement configuration of field inside of head and body phantom, (*b*) probe for the measurement of the magnitude of the electric field, (*c*) example of measured SAR. (*Courtesy of STUK Radiation and Nuclear Safety Authority, Finland.*)

In measurements SAR is determined by defining the electric field E emitted by the transmitter and AUT inside a phantom simulating the human body. The SAR is calculated according to Eq. (29) based on the known properties of the tissue equivalent material in the phantom. Several types of phantoms have been used with different levels of details for the shape and the tissues. There are also phantoms for only part of the body like the head or torso and for full body. As the standards require rather detailed determination of SAR, 1 g of brain tissue is equivalent to about 1 cm^3, the most practical solution is to use so-called homogeneous phantom, which has anatomically shaped surface layer and the phantom is filled with tissue-equivalent liquid. The electric field is measured in the liquid using a mechanical 3D scanner and a probe including typically 3 small dipoles and detectors; see Fig. 24–32 (Schmid-1). With measurements the actual product can be tested and automated measurements are also faster than numerical simulations. The disadvantages of the measurement are the limited description of the tissue and difficulties in obtaining low uncertainty due to several error sources.

REFERENCES（参考文献）

Andersen, J. Bach (1), and F. Hansen, "Antennas for VHF/UHF Personal Radio: A Theoretical and Experimental Study of Characteristics and Performance," *IEEE Trans. Vehic. Tech.,* **VT-26,** 349–357, November 1977.

Anderson, V. (1), and K. Joyner, "Specific Absorption Rate Levels Measured in a Phantom Head Exposed to Radio Frequency Transmissions from Analog Hand-Held Mobile Phones," *Bioelectromagnetics,* **16,** 60–69, 1995.

Appel-Hansen, J. (1), "Reflectivity Level of Radio Anechoic Chambers," *IEEE Trans. Ants. Prop.,* **AP-21,** 490–498, July 1973.

ARIB (1), ARIB Std-T56, *Specific Absorption Rate (SAR) Estimation for Cellular Phone,* Association for Radio Industries and Businesses (ARIB), Tokyo, Japan, 1998.

Braun, C., G. Engblom and C. Beckmann, "Evaluation of Antenna Diversity Performance for Mobile Handsets Using 3-D Measurement Data," *IEEE Trans. Ants. Prop.,* **47,** pp. 1736–1738, November 1999.

Corona, P. (1), G. Ferrara and M. Migliaccio, "Reverberating Chambers as Sources of Stochastic Electromagnetic Fields," *IEEE Trans. Electromagnetic Compatibility,* **38,** 348–356, August 1996.

DeWitt, B. T. (1), and W. D. Burnside, "Electromagnetic Scattering by Pyramidal and Wedge Absorber," *IEEE Trans. Ants. Prop.,* **36,** 971–984, July 1988.

Emerson, W. H. (1), "Electromagnetic Wave Absorbers and Aneochoic Chambers through the Years," *IEEE Trans. Ants. Prop.,* **AP-21,** 484–490, July 1973.

Evans, G. E. (1), *Antenna Measurement Techniques,* Artech House, Boston, 1990.

Francis, M. H. (1), A. C. Newell, K. R. Grimm, J. Hoffman and H. E. Schrank, "Planar Near-Field Measurements of Low-Sidelobe Antennas," *J. Res. Natl. Inst. Stand. Technol.,* **99,** 143–167, 1994.

Hacker, P. S. (1), and H. E. Schrank, "Range Distance Requirements for Measuring Low and Ultra-Low Sidelobe Antenna Patterns," *IEEE Trans. Ants. Prop.,* **AP-30,** 956–966, September 1982.

Hagmann, M. (1), O. Gandhi and C. Durney, "Numerical Calculation of Electromagnetic Energy Deposition for Realistic Model of Man," *IEEE Trans. Microwave Th. Tech.,* **27,** 804–809, September 1979.

Hansen, J. E. (ed.) (1), *Spherical Near-Field Antenna Measurements,* Peter Peregrinus Ltd., London, 1988.

Hirvonen, T. (1), J. P. S. Ala-Laurinabo, J. Tuovinen and A. V. Räisänen, "A Compact Antenna Test Range Based on a Hologram," *IEEE Trans. Ants. Prop.,* **45,** 1270–1276, August 1997.

Icheln, C. (1), J. Ollikainen and P. Vainikainen, "Reducing the Influence of Feed Cables on Small Antenna Measurements," *Electronics Letters,* **35,** 1212–1213, July 22, 1999.

ICNIRP (1), International Commission on Non-Ionizing Radiation Protection, "Guidelines for Limiting Exposure to Time-Varying Electric, Magnetic, and Electromagnetic Fields (up to 300 GHz)," *Health Physics,* **74,** 495–522, 1998.

IEEE (1), IEEE Std. 149, *Test Procedures for Antennas,* 1979.

IEEE (2), IEEE Std. 95.1-1999, *IEEE Standard for Safety Levels with Respect to Human Exposure to Radio Frequency Electromagnetic Fields, 3 kHz to 300 GHz,* IEEE, New York, 1999.

Jakes, W. C. (1), *Microwave Mobile Communications,* Wiley, New York, 1974.

Johnson, R. C. (1), H. A. Ecker and R. A. Moore, "Compact Range Techniques and Measurements," *IEEE Trans. Ants. Prop.,* **AP-17,** 568–576, September 1969.

Johnston, R. H. (1), and J. G. McRory, "An Improved Small Antenna Radiation-Efficiency Measurement Method," *IEEE Ants. Prop. Magazine,* **40,** 40–48, October 1998.

Kalliola, K. (1), H. Laitinen, L. Vaskelainen and P. Vainikainen, "Real-Time 3D Spatial-Temporal Dual-Polarized Measurement of Wideband Radio Channel at Mobile Station," *IEEE Trans. Instrum. Meas.,* **49,** April 2000.

Kuster, N. (1), "Review of Exposure Assessment for Handheld Mobile Communications Devices and Antenna Studies for Optimized Performance," Chap. 34 in W. Ross Stone (ed.), *Review of Radio Science,* Oxford University Press, New York, 1999.

Lazzrand, G. (1), and O. Gandhi, "Realistically Tilted Truncated Anatomically Based Models of the Human Head for Dosimetry of Mobile Telephones," *IEEE Trans. EMC,* **EMC-39,** 55–61, 1997.

Lee, T.-H. (1), and W. D. Burnside, "Performance Trade-off between Serrated Edge and Blended Edge Compact Range Reflectors," *IEEE Trans. Ants. Prop.,* **44,** 87–96, January 1996.

Lehto, A. (1), J. Tuovinen and A. Räisänen, "Reflectivity of Absorbers in 100–100 GHz Range," *Electronics Letters,* **27,** 1699–1700, 1991.

Ludwig, A. C. (1), "The Definition of Cross Polarization," *IEEE Trans. Ants. Prop.,* **AP-21,** 116–119, January 1973.

Mallat, J. (1), A. Lehto and J. Tuovinen, "Antenna Phase Pattern Measurements at Millimeter Wave Frequencies Using the Differential Phase Method with One Power Meter," *Int. Journal of Infrared and Millimeter Waves,* **15,** 1497–1506, September 1994.

Masterson, K. D. (1), M. Kanda and D. R. Novotny, "Broadband Electromagnetic-Field Sensors with Optoelectronic Links," Chap. 1 in W. Ross Stone (ed.), *Review of Radio Science,* Oxford University Press, New York, 1999.

Newell, A. C. (1), "Error Analysis Techniques for Planar Near-Field Measurements," *IEEE Trans. Ants. Prop.,* **36,** 754–768, June 1988.

Newell, A. C. (2), R. D. Ward and E. J. McFarlane, "Gain and Power Parameter Measurements Using Planar Near-Field Techniques," *IEEE Trans. Ants. Prop.,* **36,** 792–803, June 1988.

Newman, E. H. (1), P. Bohley and C. H. Walter, "Two Methods for the Measurement of Antenna Efficiency," *IEEE Trans. Ants. Prop.,* **AP-23,** 457–461, July 1975.

Pedersen, G. F. (1), K. Olesen and S. L. Larsen, "Bodyloss for Handheld Phones," *Proc. 1999 IEEE 49th Vehic. Tech Conf.,* Houston, 1580–1584, 1999.

Pedersen, G. Frølund (2), and J. Bach Andersen, "Handset Antennas for Mobile Communications: Integration, Diversity and Performance," Chap. 5 in W. Ross Stone (ed.), *Review of Radio Science,* Oxford University Press, New York, 1999.

Pistorius, C. W. I. (1), G. C. Clerici and W. D. Burnside, "A Dual Chamber Gregorian Subreflector System for Compact Range Applications," *IEEE Trans. Ants. Prop.,* **37,** 305–313, March 1989.

Pozar, D. M. (1), and B. Kaufman, "Comparision of Three Methods for the Measurement of Printed Antenna Efficiency," *IEEE Trans. Ants. Prop.,* **36,** 136–139, June 1988.

prENV (1), European prestandard 50166-2, *Human Exposure to Electromagnetic fields: High Frequency (10 kHz to 300 GHz),* CENELEC, Brussels, Belgium, 1995.

Saario, S. (1), D. V. Thiel, J. W. Lu and S. G. O. Keefe, "An Assessment of Cable Radiation Effects on Mobile Communications Antenna Measurements," *Dig. IEEE Int. Symp. Ants. Prop.,* Montreal, Canada, 550–553, 1997.

Schmid, T. (1), O. Egger and N. Kuster, "Automated E-Field Scanning System for Dosimetric Assessments," *IEEE Trans. Microwave Th. Tech.,* **44,** 105–113, January 1996.

Seville, A. (1), G. G. Cook, J. C. Bennett and E. H. England, "Imaging of Scatterers on Microwave Antenna Test Ranges," *IEEE Trans. Ants. Prop.,* **43,** 766–772, August 1995.

Slater, D. (1), *Near-Field Antenna Measurements,* Artech House, Boston, 1991.

Steiner, H.-J. (1), and N. Kämpfer, "A New Test Facility for EM Field Measuring of Large Antenna Systems up to 200 GHz," *Proc. 19th European Microwave Conference,* 480–485, 1989.

Taga, T. (1), "Analysis for Mean Effective Gain of Mobile Antennas in Land Mobile Radio Environments," *IEEE Trans. Microwave Th. Tech.,* **39,** 117–131, May 1990.

Tuovinen, J. (1), A. Lehto and A. Räisänen, "Phase Measurements of Millimetre Wave Antennas at 105–190 GHz with a Novel Differential Phase Method," *IEE Proc.–H,* **138,** 114–120, April 1991.

USAF (1) School of Aerospace Medicine, Aerospace Medical Division (AFSC), Brooks Air Force Base, TX 78 235-5301, *Radio Frequency Radiation Dosimetry Handbook* (4th ed.), October 1986.

Wheeler, H. A. (1), "The Radiansphere around a Small Antenna," *Proc. IRE,* **47,** 1325–1331, August 1959.

Yaghjian, A. D. (1), "An Overview of Near-Field Antenna Measurements," *IEEE Trans. Ants. Prop.,* **AP-34,** 30–45, January 1986.

PROBLEMS（习题）

24–2–1（电缆对增益测量的影响）

Effect of cabling on gain measurement. Consider the measurement of an AUT, whose VSWR ≈ 2 with a source antenna having VSWR ≈ 1.2. The output return loss of the transmitter in the measurement system is 12 dB and the input return loss of the receiver 22 dB. The lengths of the coaxial cables connecting the measurement system to the AUT and source antenna are 15 m and 5 m, respectively. The permittivity of the dielectrics of the cables is 2.2 and the attenuation 0.25 dB/m. The center frequency in the measurement is 12 GHz. Sketch the gain result obtained around the center frequency with the gain comparison method, where the reference antenna is similar to the source antenna. (*a*) When the AUT is transmitting. (*b*) When the AUT is receiving.

24–3–1（由反射波引起的波瓣图测量的不确定度）

Uncertainty of pattern measurement due to reflected wave. The level of a wave reflected from the ground is 45 dB below the level of the direct wave. How large errors (in dB) are possible in the measurement of: (*a*) main lobe peak; (*b*) −13 dB side lobe; (*c*) −35 dB side lobe?

24–3–2（由允许的口径相位误差确定测量场地的长度）

Range length requirement due to allowed phase curvature. The maximum allowed phase curvature in the measurement of a very low-side-lobe antenna is 5°. The width of the antenna is 8 m and it operates at 5.3 GHz. Find the required separation between the source and AUT.

24–4–1（测量场地的立面设计）

Design of elevated range. Design an elevated range (range length, antenna heights, source antenna diameter) for the measurement of a 1.2 m reflector antenna operating at 23 GHz.

24–4–2（近场扫描的时间要求）

Time required for near-field scanning. Estimate the time needed for a planar near-field measurement of a 2 m antenna at 300 GHz. The sampling speed is 10 samples per second.

24–5–1（实现给定动态范围的功率要求）

Power requirement for certain dynamic range. The AUT has a gain of 40 dBi at 10 GHz. The gain of the source antenna is 20 dBi. The separation between the antennas is 200 m. The receiver sensitivity (signal level that is sufficient for measurement) is −105 dBm. Find the minimum transmitted power that is needed for a dynamic range of 60 dB.

24–5–2 (利用三副未知天线的增益测量) **Gain measurement using three unknown antennas.** Three horn antenna, A, B, and C are measured in pairs at 12 GHz. The separation of antennas is 8 m. The transmitted power is +3 dBm. The received powers are −31 dBm, −36 dBm, and −28 dBm for antennas pairs AB, AC, and BC, respectively. Find the gains of the antennas.

24–5–3 (利用天体射电源的增益测量) **Gain measurement using celestial radio source.** At 2.7 GHz the antenna temperature increases 50 K as a 20 m reflector is pointed to Cygnus A. Find the antenna gain and aperture efficiency.

24–5–4 (实验室中的阻抗) **Impedance in laboratory.** You try to measure the impedance of a horn antenna with 15 dBi gain at 10 GHz in a normal laboratory room by pointing the main lobe of the antenna perpendicularly towards a wall 2 m away. The power reflection coefficient of the wall is 0.3 and it can be assumed to cover practically the whole beam of the AUT. Estimate the uncertainty of the measurement of the reflection coefficient of the AUT due to the reflection of the wall.

24–5–5 (测效率的辐射计法) **Radiometer method for efficiency.** Consider the measurement of antenna efficiency with the radiometer method. In the measurement, first the AUT pointed toward clear sky and then a calibrated noise diode with excess noise ratio $ENR = 30$ dB is connected to the input of the test receiver. The output noise power $P_{n,out}$ is measured with the AUT and with the noise diode switched on ($T_N = (10^{ENR/10} + 1)T_o$) and off ($T_N = T_o$). The power ratios obtained are −1.3 dB between the AUT and the "cold" diode and 25 dB between the "hot" and "cold" diode. The estimated antenna noise temperature in the sky measurement $T_{Ac} = 20$ K taking into account the contribution of the ground. (*a*) Determine the efficiency of the AUT and the noise figure of the test receiver. (*b*) What is the uncertainty of the result, if the uncertainty of ENR is ±0.2 dB, the uncertainty of T_{Ac} is ±5 K and the uncertainty of the gain of the receiver between different measurements (gain fluctuation) ±0.1 dB?

24–5–6 (测效率的惠勒帽法) **Wheeler cap method for efficiency.** Derive Eq. (24–5–14).

24–5–7 (效率与双端口模型) **Efficiency and two-port model.** The antenna can be described as a two-port with a matched output to free space. Present the radiation efficiency of the antenna with its two-port S-parameters.

24–6–1 (平均有效增益) **Mean effective gain.** Derive Eq. (24–6–2). Assume a random propagation environment.

24–6–2 (平均有效增益的实例) **Mean effective gain, practical examples.** (*a*) What is the mean effective gain (MEG) of an antenna in a diffuse signal environment where the polarization and direction of the signal are random? (*b*) Define the MEG for a vertical half-wave dipole in signal conditions of the Kabuto-cho route in Table 24–2.

For computer programs, see Appendix C.

附录A　参考表格

A.1　天线及天线系统的关系式①

口径效率，$\varepsilon_{ap} = \dfrac{A_e}{A_p} = \dfrac{E_{\text{av}}^2}{(E^2)_{\text{av}}}$　（无量纲）

有效口径，$A_e = \dfrac{\lambda^2}{\Omega_A}$　(m^2)

阵因子（n个等幅递差相位δ的源组成的等距d直线阵），$E_n = \dfrac{\sin(n\psi/2)}{n\sin(\psi/2)}$　（无量纲）

式中 $\psi = \beta d\cos\theta + \delta$　（弧度或度）

波束效率，$\varepsilon_M = \dfrac{\Omega_M}{\Omega_A}$　（无量纲）

波束立体角，$\Omega_A = \iint P_n(\theta, \phi)\, d\Omega$　(sr)

波束立体角（近似），$\Omega_A = \theta_{\text{HP}}\, \phi_{\text{HP}}$　$(\text{sr}) = \theta^\circ_{\text{HP}}\, \phi^\circ_{\text{HP}}$（平方度）

电荷-电流连续性，$\dot{I}l = q\dot{v}$

圆口径（均匀分布），$\text{HPBW} = \dfrac{58^\circ}{D_\lambda}$　（D_λ = 直径的波长数）

圆口径（均匀分布），$\text{BWFN} = \dfrac{140^\circ}{D_\lambda}$

圆口径（均匀分布），定向性为$9.9D_\lambda^2$

圆口径（均匀分布），（相对于半波偶极子的）增益 $= 6D_\lambda^2$

（短）偶极子的定向性，$D = 1.5$　$(= 1.76\ \text{dBi})$

（短）偶极子的辐射电阻，$R = 80\pi^2\left(\dfrac{l}{\lambda}\right)^2\left(\dfrac{I_{\text{av}}}{I_0}\right)^2$　(Ω)

（$\lambda/2$）偶极子的定向性，$D = 1.64$　$(= 2.15\ \text{dBi})$

（$\lambda/2$）偶极子的自阻抗，$Z = R_r + jX = 73 + j42.5\ \Omega$

定向性，$D = \dfrac{4\pi A_e}{\lambda^2} = \dfrac{4\pi}{\Omega_A} = \dfrac{P(\theta, \phi)_{\max}}{P_{\text{av}}}$　（无量纲）

定向性（近似），$D \approx \dfrac{4\pi\ (\text{sr})}{\theta_{\text{HP}}\phi_{\text{HP}}\ (\text{sr})} \approx \dfrac{41\ 000\ (\text{平方度})}{\theta^\circ_{\text{HP}}\phi^\circ_{\text{HP}}}$

---

① 术语表中给出了更详细的说明，以及其他关系式的列表。

定向性（较好的近似），$D \approx \dfrac{41\ 000\ \varepsilon_M}{k_P \theta_{\mathrm{HP}} \phi_{\mathrm{HP}}}$

通量密度，$S = \dfrac{2kT_A}{A_e}$ $(\mathrm{W\ m^{-2}\ Hz^{-1}})$

最小可检测通量密度，$\Delta S_{\min} = \dfrac{2k\ \Delta T_{\min}}{A_e}$ $(\mathrm{W\ m^{-2}\ Hz^{-1}})$

弗里斯公式，$\dfrac{P_r}{P_t} = \dfrac{A_{er} A_{et}}{r^2 \lambda^2}$ （无量纲）

增益，$G = kD$ （无量纲）

有效高度，$h_e = \dfrac{V}{E} = \dfrac{I_{\mathrm{av}}}{I_0} h_p = \dfrac{1}{I_0} \displaystyle\int_0^{h_p} I(z)\, dz = \sqrt{\dfrac{2R_r A_e}{Z_0}}$ (m)

单绕轴向模螺旋天线的定向性，$D = 12\left(\dfrac{C}{\lambda}\right)^2 \dfrac{nS}{\lambda}$

（均匀同相的长）直线阵，$\mathrm{HPBW} = \dfrac{1}{L_\lambda}$（弧度）$= \dfrac{57.3^\circ}{L_\lambda}$

（单圈）环的辐射电阻，$R_r = 197 C_\lambda^4$ (Ω)

近场–远场界限，$R = \dfrac{2L^2}{\lambda}$ (m)

接收机噪声功率，$N = kT_{\mathrm{sys}}\ \Delta f$ (W)

奈奎斯特功率，$w = kT$ $(\mathrm{W\ Hz^{-1}})$

雷达方程，$\dfrac{P_r}{P_t} = \dfrac{A^2 \sigma}{4\pi \lambda^2 r^4}$ （无量纲）

辐射功率，$P = \dfrac{\mu^2 q^2 \dot{v}^2}{6\pi Z}$ (W)

辐射电阻，$R_r = \dfrac{S(\theta, \phi)_{\max} r^2 \Omega_A}{I^2}$ (Ω)

矩形口径（均匀分布），$\mathrm{HPBW} = \dfrac{51^\circ}{L_\lambda}$

矩形口径（均匀分布），$\mathrm{BWFN} = \dfrac{115^\circ}{L_\lambda}$

矩形口径（均匀分布），定向性为$12.6 L_\lambda L'_\lambda$

矩形口径（均匀分布），（相当于半波偶极子的）增益为$7.7 L_\lambda L'_\lambda$

角分辨率$\approx \dfrac{\mathrm{BWFN}}{2}$

信噪比，$\dfrac{S}{N} = \dfrac{P_r}{P_n} = \dfrac{P_t A_{et} A_{er}}{r^2 \lambda^2 k T_{\mathrm{sys}}\ \Delta f}$

信噪比（相对于1 W各向同性），$\dfrac{S}{N} = \dfrac{\lambda^2}{16\pi^2 r^2 k T_{\mathrm{sys}}\ \Delta f}$

系统温度，$T_{\mathrm{sys}} = T_A + T_{\mathrm{LP}}\left(\dfrac{1}{\varepsilon} - 1\right) + \dfrac{T_R}{\varepsilon}$ (K)

天线温度（穿过发射-吸收云层），$T_A = T_c(1 - e^{-\tau c}) + T_s e^{-\tau c}$ (K)

最小可检测温度，$\Delta T_{\min} = \dfrac{k' T_{\text{sys}}}{\sqrt{t \Delta f}} = \Delta T_{\text{rms}}$ (K)

（椭圆极化）波的（平均）功率，$S_{\text{av}} = \dfrac{1}{2}\hat{\mathbf{z}}\dfrac{E_1^2 + E_2^2}{Z_0}$ (W m^{-2})

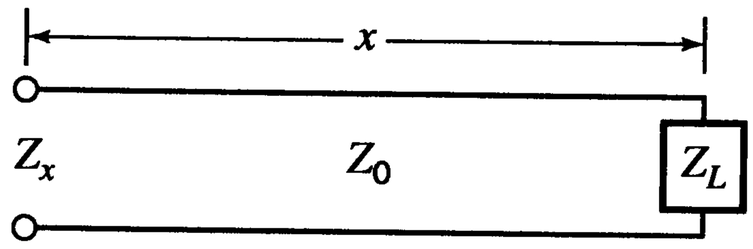
波长与频率（空气或真空中），$\lambda = \dfrac{c}{f}$ (m)

A.2 端接负载传输线的输入阻抗公式

在图A.1所示的特性阻抗为Z_0的传输线上，距离负载（即终端）阻抗Z_L为x处所呈现的输入阻抗公式，按三种负载条件分列于下表：(1) 任意负载Z_L值；(2) 短路线$Z_L = 0$；(3) 开路线$Z_L = \infty$。每种条件下，表的第二列又分两种情况：(a) 线上存在衰减（$\alpha \neq 0$）的一般情况；(b) 线上无耗或有损耗但可忽略（$\alpha = 0$）的情况。

负载条件	一般情况（$\alpha \neq 0$）	无耗情况（$\alpha = 0$）
任意Z_L值	$Z_x = Z_0 \dfrac{Z_L + Z_0 \tanh \gamma x}{Z_0 + Z_L \tanh \gamma x}$	$Z_x = Z_0 \dfrac{Z_L + jZ_0 \tan \beta x}{Z_0 + jZ_L \tan \beta x}$ $Z_x = Z_0^2 / Z_L$†
$Z_L = 0$ 短路线	$Z_x = Z_0 \tanh \gamma x$ $= Z_0 \dfrac{\tanh \alpha x + j \tan \beta x}{1 + j \tanh \alpha x \tan \beta x}$ $Z_x = Z_0 \coth \alpha x$†	$Z_x = jZ_0 \tan \beta x$
$Z_L = \infty$ 开路线路	$Z_x = Z_0 \coth \gamma x$ $= Z_0 \dfrac{1 + j \tanh \alpha x \tan \beta x}{\tanh \alpha x + j \tan \beta x}$ $Z_x = Z_0 \tanh \alpha x$†	$Z_x = -jZ_0 \cot \beta x$

*$\beta x = n\pi/2$, $n = 1, 3, 5, \cdots$。表中的$\gamma = \alpha + j\beta$，式中α为衰减常数，$\beta = 2\pi/\lambda$。



图A.1 特性阻抗为Z_0的传输线。长度为x，负载阻抗为Z_L

A.3 反射系数、传输系数与电压驻波比VSWR

对于特性阻抗为Z_0而终端所接负载阻抗为Z_L的传输线，其电压反射系数ρ_v、电流反射系数ρ_i，电压传输系数或施加于负载的相对电压τ_v、电流传输系数或通过负载的相对电流τ_i，以及电压驻波比VSWR的关系式如下：

电压反射系数 $$\rho_v = \frac{Z_L - Z_0}{Z_L + Z_0}$$

电流反射系数 $$\rho_i = \frac{Z_0 - Z_L}{Z_0 + Z_L} = -\rho_v$$

电压传输系数	$\tau_v = \dfrac{2Z_L}{Z_0 + Z_L} = 1 + \rho_v$
电流传输系数	$\tau_i = \dfrac{2Z_0}{Z_0 + Z_L} = 1 + \rho_i$
电压驻波比 VSWR	$\dfrac{1 + \lvert\rho_v\rvert}{1 - \lvert\rho_v\rvert} = \dfrac{1 + \lvert\rho_i\rvert}{1 - \lvert\rho_i\rvert}$
反射系数的幅度	$\lvert\rho_v\rvert = \lvert\rho_i\rvert = \dfrac{\text{VSWR} - 1}{\text{VSWR} + 1}$

A.4 同轴、双导线及微带传输线的特性阻抗

传输线类型	特性阻抗，Ω
同轴（填充相对介电常数 er 的媒介))	$Z_0 = \dfrac{138}{\sqrt{\varepsilon_r}} \log \dfrac{b}{a}$
同轴（填充空气）	$Z_0 = 138 \log \dfrac{b}{a}$
双导线（相对介电常的媒质），$(D >> a)$	$Z_0 = \dfrac{276}{\sqrt{\varepsilon_r}} \log \dfrac{D}{a}$
双导线（填充空气），$(D >> a)$	$Z_0 = 276 \log \dfrac{D}{a}$
微带$(w \geqslant 2h)$	$Z_0 \approx \dfrac{377}{\sqrt{\varepsilon_r}[(w/h) + 2]}$

表中，b 为外导体的内半径

a 为内导体或导线的半径

D 为导的间距

w 为导带的宽度

h 为介质基片的高度或厚度

这里假设沿线无耗（即 $R << \omega L$ 和 $G << \omega C$）且电流都在各半径对应的导体表面，此条件对于有很小透入深度的高频是近似的。同时，还假设线上只传输 TEM 模。

微带的关系式在比值 w/h 非常大时接近正确。当导带的宽度 w 小于 $2h$ 时，可改用接地平面上方单导线的公式，即

$$Z_0 \approx 138 \log \frac{D}{a} = 138 \log \frac{8}{w/h} \qquad (\Omega)$$

式中，D 为导线与其镜像的间距，$D = 2h$，a 为等效导线半径，$a = w/4$。平的导带被等效成以宽度之半为直径的圆导体。

A.5 用分布参量表示的传输线特性阻抗

下表中给出传输线特性阻抗 Z_0 的三种情况：(1) 有损耗的一般情况；(2) 损耗很小的特殊情况；(3) 无耗情况。表中出现的符号有：

Z_0 ——特性阻抗，Ω

R_0 ——特性电阻，Ω

Z ——串联阻抗，$\Omega\,m^{-1}$

R ——串联电阻，$\Omega\,m^{-1}$

L ——串联电感，$H\,m^{-1}$

Y——并联导纳，$\mho\,m^{-1}$

G——并联电导，$\mho\,m^{-1}$

C ——并联电容，$F\,m^{-1}$

$$Z = R + j\omega L$$

$$Y = G + j\omega C$$

一般情况	$Z_0 = \sqrt{\frac{Z}{Y}} = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$
小损耗	$Z_0 = \sqrt{\frac{L}{C}}\left[1 + j\left(\frac{G}{2\omega C} - \frac{R}{2\omega L}\right)\right]$
无耗情况 * $R = 0, G = 0$	$Z_0 = \sqrt{\frac{L}{C}} = R_0$

* 上列损耗非零的情况，都有近似条件 $\omega L >> R$ 而 $\omega C >> G$。

A.6 材料参数（介电常数、电导率和介质强度）

材料	相对介电常数		电导率	质强度
	ε_r'	ε_r''	$\sigma, \mho\,m^{-1}$	$MV\,m^{-1}$
空气（标准大气压）	1.0006	0	0	3
铝	1	0	3.5×10^7	
胶木（酚醛塑料）	5	0.05	10^{-14}	25
碳			3×10^4	
铜	1	0	5.8×10^7	
玻璃（板）	6	0.03	10^{-13}	30
石墨			10^5	
云母	6	0.2	10^{-15}	200
石（原）油	2.2	0.0002	10^{-14}	15
纸（浆）	3	0.1		50
石蜡	2.1	0.0004	$\sim 10^{-15}$	20
树脂玻璃	3.4			
聚合泡沫	~1.05			
聚苯乙烯	2.7	0.0002	10^{-16}	20
聚氯乙烯(PVC)	2.7			
瓷	5	0.004		
PVC（膨化的）	~1.1			
石英	5	0.001	10^{-17}	35
橡胶（氯丁二烯）	5	0.02	10^{-13}	25
金红石（二氧化钛）	100	0.02		
雪（新鲜的）	1.5	0.5–0.0003		
粘土	14		5×10^{-3}	

（续表）

材料	相对介电常数		电导率	质强度
	ε_r'	ε_r''	σ, $\mho$ m^{-1}	MV m^{-1}
沙土	10		2×10^{-3}	
石灰石			10^{-2}	
板岩石	7			
聚苯乙烯泡沫塑料	1.03			
城市地面	4		2×10^{-4}	
真空	1*	0	0	
凡士林	2.2	0.0003		
聚四氟乙烯	2.1	0.005	10^{-15}	60
蒸馏水	80		10^{-4}	
新鲜水	80		$10^{-2} \sim 10^{-3}$	
海水	80		$4 \sim 5$	
杉木夹板	2	0.04		

* 按定义。

注意，通常ε_r'和ε_r''都是频率的函数，表中所给的是从kHz到GHz范围内的典型值，介电常数还是温度的函数，表中所给的（除了雪之外）都是在25°C左右的典型值。

A.7 介电常数的关系式

$$\varepsilon_r = \varepsilon_r' - j\varepsilon_r''$$

$$\varepsilon_r = \varepsilon_r' - j\frac{\sigma'}{\omega\varepsilon_0}$$

$$\varepsilon_r = \varepsilon_r' - j\left(\frac{\sigma + \omega\varepsilon''}{\omega\varepsilon_0}\right)$$

$$\varepsilon_r = \varepsilon_r' - j\left(\frac{\varepsilon''}{\varepsilon_0} + \frac{\sigma}{\omega\varepsilon_0}\right)$$

$$\varepsilon_r = \varepsilon_r' - j(\varepsilon_{rh}'' + \varepsilon_{rc}'')$$

$$\varepsilon_r \simeq \varepsilon_r' - j\varepsilon_r' \mathrm{PF}, \text{ 对于小的 PF}$$

$$\text{功率损耗} = \sigma E^2 + \omega\varepsilon'' E^2 \qquad (\mathrm{W\ m^{-3}})$$

其中，ε_r——相对介电常数 $= \varepsilon/\varepsilon_0$，无量纲

ε——介电常数，F m^{-1}

ε_0——真空的介电常数，$\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}

ε_r'——有关位移电流的相对介电常数

ε_r''——有关等效传导电流的相对介电常数

σ——直流电导率，$\mho$ m^{-1}

$\sigma'C$——等效电导率，$\mho$ m^{-1}

ε_{rh}''——有关磁滞效应的相对介电常数

ε_{rc}''——有关直流电导率的相对介电常数

PF ——功率因子，$PF=\sigma'/\omega\varepsilon$，$\sigma' \ll \omega\varepsilon$

E ——电场强度，$V\ m^{-1}$

σE^2 ——直流电导率导致的功率损耗，$W\ m^{-3}$

$\omega\varepsilon'' E^2$ ——磁滞效应导致的功率损耗，$W\ m^{-3}$

A.8 麦克斯韦方程组

表1给出微分形式的麦克斯韦方程组，表2给出积分形式的麦克斯韦方程组。表中分别列出了一般情况、自由空间情况、时谐情况、稳态（具有传导电流的静态场）、静态（没有电流的静态场）等的方程组。在表2中，指出了各种方程与电位或电动势 V、磁位或磁动势 U、电流 I、电通量 ψ、磁通量 ψ_m 之间的等效性。

表1 微分形式的麦克斯韦方程组

情况 \ 量纲	安培定律 电流/面积	法拉弟定律 电位/面积	高斯定理 电通量/体积	高斯定理 磁通量/体积
一般	$\nabla\times\mathbf{H}=\mathbf{J}+\frac{\partial\mathbf{D}}{\partial t}$	$\nabla\times\mathbf{E}=-\frac{\partial\mathbf{B}}{\partial t}$	$\nabla\cdot\mathbf{D}=\rho$	$\nabla\cdot\mathbf{B}=0$
自由空间	$\nabla\times\mathbf{H}=\frac{\partial\mathbf{D}}{\partial t}$	$\nabla\times\mathbf{E}=-\frac{\partial\mathbf{B}}{\partial t}$	$\nabla\cdot\mathbf{D}=0$	$\nabla\cdot\mathbf{B}=0$
时谐	$\nabla\times\mathbf{H}=(\sigma+j\omega\varepsilon)\mathbf{E}$	$\nabla\times\mathbf{E}=-j\omega\mu\mathbf{H}$	$\nabla\cdot\mathbf{D}=\rho$	$\nabla\cdot\mathbf{B}=0$
稳态	$\nabla\times\mathbf{H}=\mathbf{J}$	$\nabla\times\mathbf{E}=0$	$\nabla\cdot\mathbf{D}=\rho$	$\nabla\cdot\mathbf{B}=0$
静态	$\nabla\times\mathbf{H}=0$	$\nabla\times\mathbf{E}=0$	$\nabla\cdot\mathbf{D}=\rho$	$\nabla\cdot\mathbf{B}=0$

表2 积分形式的麦克斯韦方程组

情况 \ 量纲（Mks制）	安培定律 电流（A, 安）	法拉弟定律 电位（V, 伏）	高斯定理 电通量（C, 库）	高斯定理 磁通量（Wb, 韦）
一般	$U=\oint\mathbf{H}\cdot d\mathbf{l}=\iint\left(\mathbf{J}+\frac{\partial\mathbf{D}}{\partial t}\right)\cdot d\mathbf{s}=I_{\text{total}}$	$V=\oint\mathbf{E}\cdot d\mathbf{l}=-\iint\frac{\partial\mathbf{B}}{\partial t}\cdot d\mathbf{s}$	$\psi=\iint\mathbf{D}\cdot d\mathbf{s}=\iiint\rho\,d\tau$	$\psi_m=\iint\mathbf{B}\cdot d\mathbf{s}=0$
自由空间	$U=\oint\mathbf{H}\cdot d\mathbf{l}=\iint\frac{\partial\mathbf{D}}{\partial t}\cdot d\mathbf{s}=I_{\text{disp}}$	$V=\oint\mathbf{E}\cdot d\mathbf{l}=-\iint\frac{\partial\mathbf{B}}{\partial t}\cdot d\mathbf{s}$	$\psi=\iint\mathbf{D}\cdot d\mathbf{s}=0$	$\psi_m=\iint\mathbf{B}\cdot d\mathbf{s}=0$
时谐	$U=\oint\mathbf{H}\cdot d\mathbf{l}=(\sigma+j\omega\varepsilon)\iint\mathrm{E}\cdot d\mathbf{s}=I_{\text{total}}$	$V=\oint\mathbf{E}\cdot d\mathbf{l}=-j\omega\mu\iint\mathbf{H}\cdot d\mathbf{s}$	$\psi=\iint\mathbf{D}\cdot d\mathbf{s}=\iiint\rho\,d\tau$	$\psi_m=\iint\mathbf{B}\cdot d\mathbf{s}=0$
稳态	$U=\oint\mathbf{H}\cdot d\mathbf{l}=\iint\mathbf{J}\cdot d\mathbf{s}=I_{\text{cond}}$	$V=\oint\mathbf{E}\cdot d\mathbf{l}=0$	$\psi=\iint\mathbf{D}\cdot d\mathbf{s}=\iiint\rho\,d\tau$	$\psi_m=\iint\mathbf{B}\cdot d\mathbf{s}=0$
静态	$U=\oint\mathbf{H}\cdot d\mathbf{l}=0$	$V=\oint\mathbf{E}\cdot d\mathbf{l}=0$	$\psi=\iint\mathbf{D}\cdot d\mathbf{s}=\iiint\rho\,d\tau$	$\psi_m=\iint\mathbf{B}\cdot d\mathbf{s}=0$

附录B　参考书、录像带和文章

B.1　参考书

Abraham, M., and R. Becker: *Electricity and Magnetism,* Stechert, 1932.

Aharoni, J.: *Antennae,* Oxford, 1946.

American National Standard for Calibration of Antennas Used for Radiated Emission Measurements in Electromagnetic Interference (EMI) Control/Calibration of Antennas (9 kHz to 40 GHz), IEEE, 1998.

Arai, H. *Measurement of Mobile Antenna Systems*, Artech House, 2001.

ARRL Antenna Handbook, American Radio Relay League, 2000.

Bahl, I. J., and P. Bhartia: *Microstrip Antennas,* Artech House, 1980.

Balanis, C. A.: *Antenna Theory: Analysis and Design,* 2d ed., Wiley, 1997.

Biraud, F. (ed.): *Very Long Baseline Interferometry Techniques,* Cepadues, 1983.

Blake, L. V.: Antennas, Artech House, 1984.

Born, M.: *Optik,* Springer, 1933.

Bowman, J. J., T. B. A. Senior and P. L. E. Uslenghi: *Electromagnetic and Acoustic Scattering by Simple Shapes,* North Holland, Amsterdam, 1969.

Bracewell, R. N.: *The Fourier Transform and Its Applications,* McGraw-Hill, 1965.

Bracewell, R. N.: *The Hartley Transform,* Clarendon Press, Oxford, 1986.

Brillouin, L.: *Wave Propagation in Periodic Structures,* McGraw-Hill, 1946.

Brown, George H.: *And Part of Which I Was,* Angus Cupar (117 Hunt Drive, Princeton, NJ 08540).

Brückmann, H.: *Antennen, ihre Theorie und Technik,* Hirzel, 1939.

Burrows, M. L.: *ELF Communications, Antennas,* Peregrinus, 1978.

Cady, W. M., M. B. Karelitz and L. A. Turner: *Radar Scanners and Radomes,* McGraw-Hill, 1948.

Christiansen, W. N., and J. A. Hogbom: *Radio Telescopes,* Cambridge, 1969, 1985.

Clarke, J. (ed.): *Advances in Radar Techniques,* Peregrinus, 1985.

Clarricoats, P. J. B., and A. D. Olver: *Corrugated Horns for Microwave Antennas,* Peregrinus, 1984.

Collin, R. E., and F. J. Zucker (eds.): *Antenna Theory,* McGraw-Hill, 1969.

Collin, R. E.: *Antennas and Radiowave Propagation,* McGraw-Hill, 1985.

Compton, R. T., Jr.,: *Adaptive Antennas; Concepts and Performance,* Prentice-Hall, 1988.

Cornbleet, S.: *Microwave Optics: The Optics of Microwave Antenna Design,* Academic Press, 1984.

Delogne, P.: *Leaky Feeders and Subsurface Radio Communication,* IEE, London, 1982.

Drabowitch, S.: *Modern Antennas,* Ecole Superieure d'Electricite, France.

Elliott, R. S.: *Antenna Theory and Design,* Prentice-Hall, 1981.

Evans, J. V., and T. Hagfors (eds.): *Radar Astronomy,* McGraw-Hill, 1968.

Fanti, R., K. Kellermarm and G. Setti (eds.): *VLBI and Compact Radio Sources,* Reidel, 1984.

Faraday, M.: *Experimental Researches in Electricity,* Quaritch, 1839, 1855.

Fujimoto, K., and J. R. James: *Mobile Antenna Systems Handbook,* Artech House, 2000.

Galeis. J.: *Antennas in Inhomogeneous Media,* Pergamon, 1969.

Garg, R., P. Bhartia, I. Bahl and A. Ittipiboon, *Microstrip Ant. Design Handbook,* Artech House, 2001.

Halim, M. A., *Adaptive Array Measurement and Application in Communication Systems,* Artech House, 2001.

Hall, G. L. (ed.): *ARRL Antenna Book,* American Radio Relay League, 1984.

Hallén, E.: *Teoretisk Electricitetslära,* Skrivbyran Standard, 1945.

Hansen, R. C.: *Microwave Scanning Antennas,* vols. 1, 2, 3, Academic Press, 1966.

Hansen, R. C.: *Phased Array Antennas,* Wiley, 1998.

Harper, A. E.: *Rhombic Antenna Design,* Van Nostrand, 1941.

Harrington, R. F.: *Field Computation by Moment Methods,* Macmillan, 1968.

Hertz, Heinrich, R.: *Electric Waves,* Macmillan, 1893; Dover, 1962.

Hertz, Heinrich: *Memoirs, Letters, Diaries,* San Francisco Press, 1977.

Hord, R. M.: *Remote Sensing, Methods and Applications,* Wiley, 1986.

Hudson, J. E.: *Adaptive Array Principles,* Peregrinus, 1981.

Huygens, C.: *Traite de la Luminière,* Leyden, 1690.

IEEE Standard Methods for Measuring Electromagnetic Field Strength of Sinusoidal Continuous Waves, 30 Hz to 30 GHz, IEEE, 1991.

James, G. L.: *Geometrical Theory of Diffraction for Electromagnetic Waves,* Peregrinus, 1980.

James, J. R., P. S. Hall and C. Wood: *Microstrip Antenna Theory and Design,* Peregrinus, 1982.

Jansky, D. M., and M. C. Jeruchim: *Communication Satellites in the Geostationary Orbit,* Artech House, 1983.

Johnson, R. C., and H. Jasik (eds.): *Antenna Applications Reference Guide,* McGraw-Hill, 1987. (Selected chapters from *Antenna Engineering Handbook.*)

Johnson, R. C., and H. Jasik (eds.): *Antenna Engineering Handbook,* McGraw-Hill, 1984.

Jordan, E. C., and K. G. Balmain: *Electromagnetic Waves and Radiating Systems,* Prentice-Hall, 1968.

Jull, E. V.: *Antennas and Diffraction Theory,* Peregrinus, 1981.

Kiely, D. G.: *Dielectric Aerials,* Methuen, 1953.

King, R. W. P., and G. S. Smith: *Antennas in Matter,* MIT Press, 1980.

King, R. W. P., H. R. Mimno and A. H. Wing: *Transmission Lines, Antennas and Wave Guides,* McGraw-Hill, 1945.

King, R. W. P., R. B. Mack and S. S. Sandier: *Arrays of Cylindrical Dipole Antennas,* Cambridge, 1968.

King, R. W. P.: *The Theory of Linear Antennas,* Harvard, 1956.

Kraus, J. D., and D. A. Fleisch: *Electromagnetics,* 5th ed., McGraw-Hill, 1999.

Kraus, J. D.: *Big Ear II,* Cygnus-Quasar, 1995.

Kraus, J. D.: *Radio Astronomy,* Cygnus-Quasar (PO Box 85, Powell, OH 43065), 1966, 1986.

Kumar, A.: *Antenna Design with Fiber Optics,* Artech House, 1996.

Kuzmin, A. D., and A. S. Solomonovich: *Radio Astronomical Methods for the Measurement of Antenna Parameters,* Soviet Radio, 1974.

Landau, L., and E. Lifshitz: *The Classical Theory of Fields,* Addison-Wesley, 1951.

Laport, E. A.: *Radio Antenna Engineering,* McGraw-Hill, 1952.

Law, P. E., Jr.: *Shipboard Antennas,* Artech House, 1986.

Leonov, A. I., and K. I. Fomichev: *Monopulse Radar,* Artech House, 1986.

Lindell, L. V., A. H. Sihvola, S. A. Tretyakov, and A. J. Viitanen: *Electromagnetic Waves in Chiral and Bi-Isotropic Media,* Artech House, 1994.

Lo, Y. T. (ed.): *Handbook of Antenna Theory and Design,* Van Nostrand Reinhold, 1987.

Love, A. W.: *Electromagnetic Horn Antennas,* IEEE Press, 1976.

Love, A. W.: *Reflector Antennas,* IEEE Press, 1978.

Luneburg, R. K.: *Mathematical Theory of Optics,* Brown University Press, 1944.

Ma, M. T.: *Theory and Application of Antenna Arrays,* Wiley, 1974.

Mailloux, R. J. *Phased Array Antenna Handbook,* Artech House, 1994.

Mar, J. W., and H. Liebowitz (eds.): *Structures Technology for Large Radar and Radio Telescope Systems,* MIT, 1969.

Marconi, Degna: *My Father Marconi,* McGraw-Hill, 1962.

Marcuse, D.: *Theory of Dielectric Optical Waveguides,* Academic Press, 1974.

Maxwell, J. C.: *A Treatise on Electricity and Magnetism,* Oxford, 1873.

Monzingo, R. A., and T. W. Miller: *Introduction to Adaptive Arrays,* Wiley, 1980.

Mortazawi, A, T. Itoh, and James Harvey: *Active Antennas and Quasi-Optical Arrays,* IEEE, 1998.

Moullin, E. B.: *Radio Aerials,* Oxford, 1949.

Nakano, H.: *Helical and Spiral Antennas,* Research Studies Press, Wiley, 1987.

Poincaré, H.: *Theorie Mathematique de la Luminiere,* Carre, 1892.

Popovic, B. D., M. B. Dragovic and A. R. Djordjevic: *Analysis and Synthesis of Wire Antennas,* Wiley, 1982.

Pozar, D., and D. Schaubert: *Microstrip Antennas: The Analysis and Design of Microstrip Antennas and Arrays,* IEEE, 1995.

Pozar, D.: *Antenna Design Using Personal Computers,* Artech House, 1985.

Rappaport, T. S., and J. C. Liberti: *Smart Antennas for Wireless CDMA,* IEEE, 1999.

Rayleigh, Lord: *The Theory of Sound,* Macmillan, 1877,1878, 1929, 1937.

Reich, H. J. (ed.): *Very High Frequency Techniques,* McGraw-Hill, 1947.

Reintjes, J. F. (ed.): *Principles of Radar,* McGraw-Hill, 1946.

Rhodes, D. R.: *Introduction to Monopulse,* McGraw-Hill, 1959.

Rhodes, D. R.: *Synthesis of Planar Antenna Sources,* Clarendon Press, Oxford, 1974.

Rudge, A. W., K. Milne, A. D. Olver and P. Knight (eds.): *Handbook of Antenna Design,* Peregrinus, 1983.

Rumsey, V. H.: *Frequency Independent Antennas,* Academic Press, 1966.

Rusch, W. V. T., and P. D. Potter: *Analysis of Reflector Antennas,* Academic Press, 1970.

Salema, C., C. Fernandes, R. K. Jha: *Solid Dielectric Horn Antennas,* Artech House, 1998.

Schelkunoff, S. A., and H. T. Friis: *Antennas: Theory and Practice,* Wiley, 1952.

Schelkunoff, S. A.: *Advanced Antenna Theory,* Wiley, 1952.

Schelkunoff, S. A.: *Electromagnetic Waves,* Van Nostrand, 1948.

Sherman, S. M.: *Monopulse Principles and Techniques,* Artech House, 1984.

Silver, S.: *Microwave Antenna Theory and Design,* McGraw-Hill, 1949.

Siwiak, K. *Radiowave Properties and Antennas for Personal Communications,* Artech House, 1998.

Skolnik, M. I.: *Introduction to Radar Systems,* McGraw-Hill, 1980.

Slater, J. C., and N. H. Frank: *Introduction to Theoretical Physics,* McGraw-Hill, 1933.

Slater, J. C.: *Microwave Transmission,* McGraw-Hill, 1942.

Smith, C. E.: *Directional Antennas,* Cleveland Institute of Radio Electronics, 1946.

Smith, C. E.: *Theory and Design of Directional Antenna Systems,* National Association of Broadcasters, 1949.

Steinberg, B. D.: *Principles of Aperture and Array System Design,* Wiley, 1976.

Stutzman, W. L., and G. A. Thiele: *Antenna Theory and Design,* 2d ed., Wiley, 1998.

Tai, C-T: *Dyadic Green's Functions in Electromagnetic Theory,* Intext, 1971.

Terman, F. E.: *Radio Engineers' Handbook,* McGraw-Hill, 1943.

Thompson, A. R., J. M. Moran and G. W. Swenson, Jr.: *Interferometry and Synthesis in Radio Astronomy,* Wiley-Interscience, 1986.

Tseitlin, N. M.: *Practical Methods of Radioastronomical and Antenna Techniques,* Soviet Radio, 1966.

Tsoulos, G. V.: *Adaptive Antennas for Wireless Communications,* IEEE, 2001.

Uchida, H.: *Fundamentals of Coupled Lines and Multiwire Antennas,* Sasaki (Sendai), 1967.

Uda, S., and Y. Mushiake: *Yagi-Uda Antenna,* Tohoku University, 1954.

Uda, S.: *Short Wave Projector,* Tohoku University, 1974.

Uda, Shintaro: *On the Wireless Beam of Short Electric Waves.* Series of 11 articles on his wave canal (Yagi-Uda) antenna published in *J. IEE Japan,* between March 1926 and July 1929, plus earlier and later articles on meter wavelength experiments. Privately published bound volume.

Ulaby, F. T., R. K. Moore and A. K. Fung: *Microwave Remote Sensing, Active and Passive,* Addison-Wesley, 1981.

Wait, J. R.: *Antennas and Propagation,* Peregrinus, 1986.

Walter, C. H.: *Traveling Wave Antennas,* Dover, 1972.

Watson, W. H.: *The Physical Principles of Wave Guide Transmission and Antenna Systems,* Oxford, 1947.

Weeks, W. L.: *Antenna Engineering,* McGraw-Hill, 1968.

Wood, P. J.: *Reflector Antenna Analysis and Design,* Peregrinus, 1980.

Zenneck, J.: *Wireless Telegraphy,* McGraw-Hill, 1915.

B.2 录像带

Kraus, J. D.: "Antennas and Radiation," Lecture-demonstration, excellent teaching supplement, 30 min, color, VHS. Cygnus-Quasar, P.O. Box 85, Powell, OH 43065.

Landt, J. A., and E. K. Miller: "Computer Graphics of Transient Radiation and Scattering Phenomena on Antennas and Wire Structures," Fields and currents in slow motion, 20 min, color, VHS. Cygnus-Quasar, P.O. Box 85, Powell, OH 43065.

B.3 进一步选读的文章

Ali, M., M. Okoniewski, M. A. Stuchly and S. S. Stuchly, "Dual-Frequency Strip-Sleeve monopole for Laptop Computers," *IEEE Trans. Ants. Prop.*, **47,** p. 317, February 1999.

Altshuler, E. E., "Design of a Vehicular Antenna for GPS/IRIDIUM Using a Genetic Alogorithm," *IEEE Trans. Ants. Prop.*, pp. 968–972, June 2000.

Anderson, J. B., "Antenna Array in Mobile Communications: Gain, Diversity and Channel Capacity," *IEEE Ants. Prop. Mag.*, pp. 12–16, April 2000.

Auzanneau, F., and R. W. Ziolkowski, "Artificial Composite Materials Consisting of Nonlinearly Loaded Electrically Small Antennas: Operational-Amplifier-Based Circuits with Applications to Smart Skins," *IEEE Trans. Ants. Prop.*, **47,** p. 1330, August 1999.

Baliarda, C. P., J. Romeu and A. Cardama, "The Koch Monopole: A Small Fractal Antenna," *IEEE Trans. Ants. Prop.*, pp. 1773–1781, November 2000.

Barrett, T. W., "History of Ultra-Wide-Band Communication Radar," *Microwave Jour.*, pp. 22–56, January 2001, and pp. 22–52, February 2001.

Brown, A. D., J. L. Volakis, L. C. Kempel, and Y. Y. Botros, "Patch Antennas on Ferromagnetic Substrates," *IEEE Trans. Ants. Prop.*, **47,** p. 26, January 1999.

Chen, H.-M., and K.-L. Wong, "On the Circular Polarization Operation of Annular-Ring Microstrip Antennas," *IEEE Trans. Ants. Prop.*, **47,** p. 1289, August 1999.

Chen, W.-S., K.-L. Wong and C.-K. Wu, "Inset Microstripline-Fed Circularly Polarized Microstrip Antennas, *IEEE Trans. Ants. Prop.*, pp. 1253–1254, August 2000.

Chio, T.-H., and D. H. Schaubert, "Parameter Study and Design of Wide-Band Widescan Dual-Polarized Tapered Slot Antenna Arrays," *IEEE Trans. Ants. Prop.*, pp. 879–886, June 2000.

Chung, S.-J., and K. Chang, "A Retrodirective Microstrip Antenna Array," *IEEE Trans. Ants. Prop.*, pp. 1802–1808, December 1998.

DeMinco, N., "Propagation Prediction Techniques and Antenna Modeling (150 to 1705 kHz) for Intelligent Transportation Systems (ITS) Broadcast Applications," *IEEE Ant. Prop. Mag.*, pp. 9–34, August 2000.

Foltz, H. D., J. S. McLean and G. Crook, "Disk Loaded Monopoles with Parallel Strip Elements," *IEEE Trans. Ants. Prop.*, pp. 1894–1896, December 1998.

Fusco, V. F., and Q. Chen, "Direct-Signal Modulation Using a Silicon Microstrip Patch Antenna," *IEEE Trans. Ants. Prop.*, **47,** p. 1025, June 1999.

Fusco, V. F., R. Roy and S. L. Karode, "Reflector Effects on the Performance of a Retrodirective Antenna Array," *IEEE Trans. Ants. Prop.*, pp. 946–953, June 2000.

George, J., C. K. Aanandan, P. Mohanan and K. G. Noir, "Analysis of New Compact Microstrip Antenna," *IEEE Trans. Ants. Prop.*, pp. 1712–1717, November 1998.

Geyi, W., P. Jarmuszewski and Y. Qi, "The Foster Reactance Theorem for Antennas and Radiation *Q*," *IEEE Trans. Ants. Prop.*, pp. 401–408, March 2000.

Gibson, P. J., "The Vivaldi Aerial," Proc. 9th European Microwave Conf., Brighton, U. K., pp. 101–105, September 1979.

Green, B. M., and M. A. Jensen, "Diversity Performance of Dual-Antenna Handsets Near Operator Tissue," *IEEE Trans. Ants. Prop.*, pp.1017–1024, July 2000.

Hertel, T. W., and G. S. Smith, "Pulse Radiation from an Insulated Antenna: An Analog of Cherenkov Radiation from a Moving Charged Particle," *IEEE Trans. Ants. Prop.*, pp. 165–172, February 2000.

Hertel, T. W., and G. S. Smith, "The Insulated Linear Antenna—Revisited," *IEEE Trans. Ants. Prop.*, pp. 914–920, June 2000.

Hettak, K., G. Delislel and M. Boulmalf, "A Novel Integrated Antenna for Millimeter-Wave Personal Communication Systems," *IEEE Trans. Ants. Prop.*, pp. 1757–1758, November 1998.

Holter, H., T.-H. Chio and D. H. Schaubert, "Elimination of Impedance Anomalies in Single- and Dual-Polarized Endfire Tapered Slot Phased Arrays," *IEEE Trans. Ants. Prop.*, pp. 122–125, January 2000.

Hu, C.-C., C. F. Jou and J.-J. Wu, "An Aperture-Coupled Linear Microstrip Leaky-Wave Antenna Array with Two-Dimensional Dual-Beam Scanning Capability," *IEEE Trans. Ants. Prop.*, pp. 909–913, June 2000.

Huang, C.-Y., J.-Y. Wu and K.-L. Wong, "Cross-Slot-Coupled Microstrip Antenna and Dielectric Resonator antenna for Circular Polarization," *IEEE Trans. Ants. Prop.*, **47,** p. 605, April 1999.

Junkin, G., T. Huang and J. C. Bennett, "Holographic Testing of Terahertz Antennas," *IEEE Trans. Ants. Prop.*, **48,** pp. 409–417, March 2000.

Karmakar, N. C., and M. E. Bialkowski, "Circularly Polarized Aperture-Coupled Circular Microstrip Patch Antennas for *L*-Band Applications," *IEEE Trans. Ants. Prop.*, **47,** p. 933, May 1999.

Kinsey, R. R., "An Edge-Slotted Waveguide Array with Dual-Plane Monopulse," *IEEE Trans. Ants. Prop.*, **47,** p. 474, March 1999.

Kokotoff, D. M., J. T. Aberle, and R. B. Waterhouse, "Rigorous Analysis of Probe-Fed Printed Annular Ring Antennas," *IEEE Trans. Ants. Prop.*, **47,** p. 384, February 1999.

Kyriazidou, C. A., R. E. Diaz and N. G. Alexópoulos, "Novel Material with Narrow-Band Transparency Window in the Bulk," *IEEE Trans. Ants. Prop.*, pp. 107–116, January 2000.

Lee, C. S., and V. Nalbandian, "Planar Circularly Polarized Microstrip Antenna with a Single Feed," *IEEE Trans. Ants. Prop.*, **47,** p. 1005, June 1999.

Lee, J.-I., U.-H. Cho and Y.-K. Cho, "Analysis for a Dielectrically Filled Parallel-Plate Waveguide with Finite Number of Periodic Slots in Its Upper Wall as a Leaky-Wave Antenna," *IEEE Trans. Ants. Prop.*, **47,** p. 701, April 1999.

Leeper, D. G., "Isophoric Arrays—Massively Thinned Phased Arrays with Well Controlled Sidelobes," *IEEE Trans. Ants. Prop.*, **47,** pp. 1825–1835, December 1999.

Leung, K. W., "Analysis of Aperture-Coupled Hemispherical Dielectric Resonator Antenna with a Perpendicular Feed," *IEEE Trans. Ants. Prop.*, pp. 1005–1006, June 2000.

Leung, K. W., K. K. Tse, K. M. Luk and E. K. N. Yung, "Cross-Polarization Characteristics of a Probe-Fed Hemispherical Dielectric Resonator Antenna," *IEEE Trans. Ants. Prop.*, **47,** p. 1238, July 1999.

Lindell, J. V., "The Radiation Operator," *IEEE Trans. Ants. Prop,* pp. 1701–1706, November 2000.

Liu, Z.-F., P.-S. Kooi, L.-W. Li, M.-S. Leong, and T.-S. Yeo, "A Method for Designing Broad-Band Microstrip Antennas in Multilayered Planar Structures," *IEEE Trans. Ants. Prop.*, **47,** p. 1416, September 1999.

Lu, J.-H., C.-L. Tang and K.-L. Wong, "Novel Dual-Frequency and Broad-Band Designs of Slot-Loaded Equilateral Triangular Microstrip Antennas," *IEEE Trans. Ants. Prop.*, pp. 1048–1054, July 2000.

Lu, J.-H., C.-L. Tang, and K.-L. Wong, "Single-Feed Slotted Equilateral-Triangular Microstrip Antenna for Circular Polarization," *IEEE Trans. Ants. Prop.*, **47,** p. 1174, July 1999.

Mak, C. L., K. M. Luk, K. F. Lee and Y. L. Chow, "Experimental Study of a Microstrip Patch Antenna with an *L*-Shaped Probe," *IEEE Trans. Ants. Prop.*, pp. 777–783, May 2000.

Marcano, D., and F. Durán, "Synthesis of Antenna Arrays Using Genetic Algorithms," *IEEE Ants. Prop. Mag.*, pp. 12–20, June 2000.

Mikhnev, V. A., and P. Vainikainen, "Two-Step Inverse Scattering Method for One-Dimensional Permittivity Profiles," *IEEE Trans. Ants. Prop.*, pp. 293–298, February 2000.

Miyashita, H., H. Ohmine, K. Nishizawa, S. Makino and S. Urasaki, "Electromagnetically Coupled Coaxial Dipole Array Antenna," *IEEE Trans. Ants. Prop.*, **47,** pp. 1716–1726, November 1999.

Montoya, T. P., and G. S. Smith, "Land-Mine Detector Using a Grid-Pass Radar Based on Resistively Loaded Vee Dipoles," *IEEE Trans. Ants. Prop.*, **47,** pp. 1795–1806, December 1999.

Muldavin, J. B., and G. M. Rebeiz, "Millimeter-Wave Tapered-Slot Antennas on Synthesized Low Permittivity Substrates," *IEEE Trans. Ants. Prop.*, **47,** p. 1276, August 1999.

Nakano, H., T. Unno, K. Nakayama and J. Yamauchi, "FDTD Analysis and Measurement of Aperture Antennas Based on the Triplate Transmission-Line Structure," *IEEE Trans. Ants. Prop.*, **47,** p. 986, June 1999.

Nishioka, Y., O. Maeshima, T. Uno and S. Adachi, "FDTD Analysis of Resistor-Loaded Bow-Tie Antennas Covered with Ferrite-Coated Conducting Cavity for Subsurface Radar," *IEEE Trans. Ants. Prop.*, **47,** pp. 970–977, June 1999.

Nishirnura, T., N. Ishii and K. Itoh, "Beam Scan Using the Quasi-Optical Antenna Mixer Array," *IEEE Trans. Ants. Prop.*, **47,** p. 1160, July 1999.

Ohtera, I., "Diverging/Focusing of Electromagnetic Waves by Utilizing the Curved Leakywave Structure: Application to Broad-Beam Antenna for Radiating within Specified Wide-Angle," *IEEE Trans. Ants. Prop.*, **47,** p. 1470, September 1999.

Pelosi, G., A. Cocchi and A. Monorchio, "A Hybrid FEM-Based Procedure for the Scattering from Photonic Crystals Illuminated by a Gaussian Beam," *IEEE Trans. Ants. Prop.*, pp. 973–980, June 2000.

Perez, R., "Designing Embedded Antennas for Bluetooth Protocol," *ACES Jour.*, pp. 152–158, November 2000.

Peterson, A. F., and E. O. Rausch, "Scattering Matrix Integral Equation Analysis for the Design of a Waveguide Rotman Lens," *IEEE Trans. Ants. Prop.*, **47,** p. 870, May 1999.

Porath, R., "Theory of Miniaturized Shorting-Post Microstrip Antennas," *IEEE Trans. Ants. Prop.*, pp. 41–47, January 2000.

Pozar, D. M., S. D. Targonski and R. Pokuls, "A Shaped-Beam Microstrip Patch Reflectarray," *IEEE Trans. Ants. Prop.*, **47,** p. 1167, July 1999.

Ramirez, R. R., and N. G. Alexópoulos, "Single Feed Proximity Coupled Circularly Polarized Microstrip Monofilar Archimedean Spiral Antenna Array," *IEEE Trans. Ants. Prop.*, **47,** p. 406, February 1999.

Rao, J. B. C., D. P. Patel and V. Krichevsky, "Voltage-Controlled Ferroelectric Lens Phased Arrays," *IEEE Trans. Ants. Prop.*, **47,** p. 458, March 1999.

Remote Sensing of the Environment, IEEE Proceedings, December 1994 (entire series).

Revster, D. D., and K. J. Cybert, "A High-Efficiency Broadband HF Wire-Antenna System," *IEEE Ants. Prop. Mag.*, pp. 53–61, August 2000.

Romeu, J., and Y. Rahmat-Samii, "Fractal FSS: A Novel Dual-Band Frequency Selective Surface," *IEEE Trans. Ants. Prop.*, pp. 1097–1105, July 2000.

Romley, J. T., and R. B. Waterhouse, "Performance of Shorted Patch Antennas for Mobile Communication Handsets at 1800 MHz," *IEEE Trans. Ants. Prop.*, **47,** pp. 815–822, May 1999.

Rowley, J. T., and R. B. Waterhouse, "Performance of Shorted Microstrip Patch Antennas for Mobile Communications Handsets at 1800 MHz," *IEEE Trans. Ants. Prop.*, **47,** p. 815, May 1999.

Salonen, P., M. Keskilammi and M. Kivikoski, "Single-Feed Dual-Band Planar Inverted-*F* Antenna with U-Shaped Slot," *IEEE Trans. Ants. Prop.*, pp. 1262–1263, August 2000.

Sehm, T., A. Lehto and A. V. Räisänen, "A High-Gain 58-GHz Box-Horn Array Antenna with Suppressed Grating Lobes," *IEEE Trans. Ants. Prop.*, **47,** pp. 1125–1130, July 1999.

Shafai, L. L., W. A. Chamma, M. Barakat, P. C. Strickland and G. Séguin, "Dual-Band Dual-Polarized Perforated Microstrip Antennas for SAR Applications," *IEEE Trans. Ants. Prop.*, **48,** pp. 58–66, January 2000.

Shin, J., and D. H. Schaubert, "A Parameter Study of Stipline-Fed Vivaldi Notch-Antenna Arrays," *IEEE Trans. Ants. Prop.*, **47,** p. 879, May 1999.

Sierra-Garcia, S., and J.-J. Laurin, "Study of a CPW Inductively Coupled Slot Antenna," *IEEE Trans. Ants. Prop.*, **47,** p. 58, January 1999.

Singh, D., C. Kalialakis, P. Gardner and P. S. Hall, "Small *H*-Shaped Antennas for MMIC Applications," *IEEE Trans. Ants. Prop.*, pp. 1134–1141, July 2000.

Stockbroeckx, B., and A. Vander Vorst, "Copolar and Cross-Polar Radiation of Vivaldi Antenna on Dielectric Substrate," *IEEE Trans. Ants. Prop.*, pp. 19–25, January 2000.

Stutzman, W. L., and C. G. Buxton, "Radiating Elements for Wideband Phased Arrays," *Microwave J.*, pp. 130–141, February 2000.

Sze, J.-Y., and K.-L. Wong, "Slotted Rectangular Microstrip Antenna for Bandwidth Enhancement," *IEEE Trans. Ants. Prop.*, pp. 1149–1152, August 2000.

Tai, C. T., "A New Theory of Receiving Antennas," *Radiation Lab. Tech. Rept. RL965,* University of Michigan, February 2001.

Tai, C. T., and R. E. Collin, "Radiation of a Hertzian Dipole Immersed in a Dissipative Medium," *IEEE Trans. Ants. Prop.*, pp. 1501–1506, October 2000.

Tam, M. T. K., and R. D. Murch, "Compact Circular Sector and Annular Sector Dielectric Resonator Antennas," *IEEE Trans. Ants. Prop.*, **47,** p. 837, May 1999.

Tam, M. T. K., and R. D. Murch, "Circularly Polarized Circular Sector Dielectric Resonator Antenna," *IEEE Trans. Ants. Prop.*, **48,** pp. 126–128, January 2000.

Theron, I. P., E. K. Walton, S. Gunawan and L. Cai, "Ultrawide-Band Noise Radar in the VHF-UHF Band," *IEEE Trans. Ants. Prop.*, **47,** pp. 1080–1084, June 1999.

Tseng, W.-J., S.-J. Chung and K. Chang, "A Planar Van Atta Array Reflector with Retrodirectivity in Both *E*-Plane and *H*-Plane," *IEEE Trans. Ants. Prop.*, pp. 173–175, February 2000.

Vaskelainen, L. I., "Phase Synthesis of Conformal Array Antennas," *IEEE Trans. Ants. Prop.*, pp. 987–991, June 2000.

Vaughan, R., "Switched Parasitic Elements for Antenna Diversity," *IEEE Trans. Ants. Prop.*, **47,** p. 399, February 1999.

Vendik, O. G., and Y. V. Yegorov, "The First Phased Array Antenna in Russia," *IEEE Ants. Prop. Mag.*, pp. 46–52, August 2000.

Wang, C.-J., C. F. Jou and J.-J. Wu, "A Novel Two-Beam Scanning Active Leaky-Wave Antenna," *IEEE Trans. Ants. Prop.*, **47,** p. 1314, August 1999.

Wang, H. Y., and M. J. Lancaster, "Aperture-Coupled Thin-Film Superconducting Meander Antennas," *IEEE Trans. Ants. Prop.*, **47,** p. 829, May 1999.

Wang, J. J. H., and V. K. Tripp, "Design of Multioctave Spiral-Mode Microstrip Antennas," *IEEE Trans. Ants. Prop.*, pp. 332–335, March 1991.

Weiss, S. J., and W. K. Kahn, "An Experimental Technique Used to Measure the Unloaded Q of Microstrip Antennas," *IEEE Trans. Ants. Prop.*, pp. 119–122, January 2000.

Werner, D. H., "Radiation and Scattering from Thin Toroidally Knotted Wires," *IEEE Trans. Ants. Prop.*, **47,** pp. 1351–1363, August 1999.

Zaïd, L., G. Kossiavas, J.-Y. Dauvignac, J. Cazajous and A. Papiernik, "Dual-Frequency and Broad-Band Antennas with Stacked Quarter Wavelength Elements," *IEEE Trans. Ants. Prop.*, **47,** p. 654, April 1999.

Zimmerman, R. K., Jr., "Traveling-Wave Analysis of a Bifilar Scanning Helical Antenna," *IEEE Trans. Ants. Prop.*, pp. 1007–1009, June 2000.

附录C　计算机程序（代码）

C.1　引言

电流及其所致辐射场的可视化是学习天线知识过程中的一项挑战，它激发了天线专业的研究者去寻找能使其职业变得简单而又多产的途径。早期的研究者致力于发展测量技术以绘制波瓣图和阻抗图表，目前这仍最好的方法之一，因为其所得的结果来自真实的天线结构。然而，如第24章所述，为此必须了解并控制测量所带来的误差。此外，测量是费时而艰难的，对三维波瓣图尤其如此。

20世纪60年代，计算机的普及为天线设计者提供了另一条途径：开发软件以仿真天线的性能。通常，这类技术既可以借助将问题离散化而用数值法求解麦克斯韦方程组，例如对积分方程用14.11节所述的矩量法（MoM，Moment Methods），或对微分方程用有限元法或时域有限差分法等；也可以借助高频渐近原理近似地求解某些天线问题，例如10.2节所提到的一致性几何绕射理论（UTD，Uniform Geometrical Theory of Diffraction），关于这类方法发展详情的介绍已经超出本书的范围。无论哪种方法，都有着一些对用户相当友好的计算机程序可供有志者进一步探究[①]。

计算机程序领域是一个飞速发展的领域，但本附录只介绍很少几种通用的天线仿真程序。为了避免过多地陈述那些很快就会被淘汰的信息，本书配有网址，提供最即时的计算机程序、习题集以及其他说明材料（详见书后所附“教学支持说明”）。不同型号的计算机和操作系统都可以实时地更新所提供的软件和链接网址。这里无法一一列出全部软件，仅给出少量示例，读者可按需要和兴趣加以选用。

McGraw-Hill为本书提供的万维网址可通过www.mhhe.com.catalogs.sem.engineering接入，并链接于另一网址www.antennas3.com。此外，从上述网址还可以接入其他有用的网址。

C.2　软件建模的原理

在讨论细节之前，先花一点时间来讨论会影响所有软件建模步骤的论题。一位软件工程师应养成良好的习惯，即从用户群体的实践经验中学习，这是非常重要的。各种计算机程序或技术的细节可以互不相同，但它们所遵从的原则是不变的。首先并且首要的就是要去解决尚未由计算机得到正确答案的问题，因此对每项结果都必须仔细评估。深信本书所介绍的基础知识为读者提供了必要的鉴定工具，无论对任何测量或计算的结果来说都是如此。为此，附录A中提供了很多易于记忆的数据和公式。

用户能更多地了解计算机程序背后的理论固然是好事，但典型的工程师往往没有时间去深究正在用着的、或将来会用到的所有解法。然而，借助一个好的说明文件去了解程序所包含技术的基本规定，通

---

① 在14.12节中，矩量法的作者Edward H. Newman写道:数值方法如矩量法（MoM）确实比解析结果更准确，因为它们只含有较少的假设。然而，某些只能得出数值形式而非函数解的数值方法（如有限差分法）的一大缺点是不能获得解的物理解释，以及关于天线的设计信息。因此，数值技术终究不能取代良好（准确而简单）的解析结果。

常总是可能的。例如，矩量法程序需要将表示局部电流幅度的导线模型离散至足够小，使之准确地再现电流分布的形状;高频渐近技术如一致性绕射理论，需要足够大的电尺寸，使之适合于推导绕射系数时采用的大距离近似。因此，即使用户并不完全理解某程序的应用为何存在着限制，但仍能够懂得其物理模型必须设置在这种界限之内，才能维持解的准确性。

通常，用户在建立或分析计算机程序中的模型时，有两种情况会引起建模的不确定性。第一种情况是与物理模型相关联的，用户应明白计算机模型与真实模型应当匹配到什么程度。要为真实天线上的每个螺丝和铆钉都严格建模往往是不可能、也不必要的。如果某物体的尺寸小于波长量级，其效应也是很小的。此外，波瓣图由电流分布的积分得到，它对电流的少量变化是相对不敏感的。然而，馈电点的电压－电流关系随几何上的小改变而发生的变化，足以改变该模型的阻抗。需要根据情况恰当地选择建模技术，或是在建模过程中确定仔细的程度。第二种情况与数学模型相关联，可被特征化为方程误差和解的误差。方程误差是由于数值结果不能严格地满足建模方程（例如，计算机有限数位的舍入误差）而造成，并随着问题的电尺寸增大的积累效应而增加。解的误差的造成原因是数值技术采用了有限个未知量（如MoM），或采用了有限个镜像和（或）彼此相互作用的绕射线（如UTD）。

用户保证所得结果准确而贴切的最重要途径是通过验证。验证虽然耗费时间，但却是一种极其基本的方法。验证可以采取内部和外部两种形式。内部检查由自洽性（self-consistency）测试（如MoM解随加密几何离散化的收敛性、功率守恒性、是否满足边界条件等）组成。此外，用户还应该了解，频带和（或）增益是否对物体尺寸的波长数敏感，波瓣图峰点和零点的个数是否与尺寸相对应，峰值与旁瓣的比值是否对频率敏感？参阅附录A中的数据和关系式。用户可先用尽可能简单的方法开始解题，可利用本书中的各种插图，使结果易懂并可用以比较。随后建立较复杂的问题、直至其更复杂的最终形式，每一步都按此方法检查。即使某一步的检查未获通过，该步骤仍有价值，因为它提供了发现问题出处的机会，并使用户增长了经验与洞察力。

外部检查涉及将所得结果与已知的解或解析的、实验的、或另一种数值方法的解进行比较。解析解只适用于一小类问题，其中多数可在本书中找到。另一种数值方法的解可用来验证类似的理想模型，如果数值解是理论型的，则只能证明两种结果之间的互洽（consistent with one another）而不是它们全都正确。如果作为比较的数值解属不同类型、且它们的解具有可比性，则两个解都正确的概率较大。若能对第24章所指出的众多潜在误差源进行仔细的处理，则用实验解作为比较将是一种非常好的检查方法。如果所得数值解与实验解相匹配，则解的准确性更加可信。

C.3 天线软件说明

如上所述，计算机程序对学生和工程师们来说是非常有用的工具。前面给出的网址所提供的计算机程序直接利用了本书所陈述的经典理论。此外，还可以利用一种基于矩量法的近代导线天线的程序，这将在C.4节中讨论。

下列程序是可供下载的PC版，其他版本则需要凭许可证获取。PC版具有简明、自洽的说明，几何尺寸的输入采用对话框的形式。按用户的选择，辐射波瓣图的绘制可取场强、功率、分贝数的直角坐标或极坐标形式。用户不妨按书中的例子操作一遍，以检测自己使用该程序的能力。

C.3a 计算阵列波瓣图和定向性的程序

ARRAYPATGAIN是应用第5章的理论计算直线阵定向性的计算机程序。指定阵元间距，能求出均匀、二项式、道尔夫－切比雪夫以及其他幅度分布的（各向同性单元）阵因子波瓣图。在有些情况下还需指定附加的信息，例如对道尔夫-切比雪夫分布需指定旁瓣电平，而得出各单元的幅度加权值，对扫瞄波束可给出相位分布。还可得出各种取向的偶极子单元之阵列的定向性。建议用户应用5.10节的知识重复图5.27和图5.34的结果，并结合习题5.8.4至习题5.9.6进行更多的练习。网址上还提供了附加的习题。

C.3b 计算位于平板反射器上方或夹角反射器中的天线波瓣图的程序

CORNEREFLECT是计算位于无限大或有限大平板上方、或夹角反射器中的偶极子波瓣图的计算机程序。对于无限大反射板的情况，运用镜像理论计算合成波瓣图；对于有限大反射板的情况，运用一致性几何绕射理论（Kouyoumjian & Pathak-1）附加来自刃边和两平面接头处的绕射场。该程序补充了第10章所陈述的理论。用户可以指定偶极子至原点的距离（见图10.3）以及反射板间的夹角及其长度（见图10.9）。建议用户重复图10.3，图10.6和图10.12的结果，并结合习题10.2.1至习题10.3.8的练习。网址上还提供了附加的习题。

C.4 学生版Expert MININEC

本书中所介绍的以及实际中所用的许多天线都属于导线天线，且很多口径天线也可应用导线天线程序通过对偶性来建模，因此这类天线模型的计算机程序非常有用。实现导线天线分析的最常用技术是矩量法（MoM），参见14.11节。在工程界广为传播的首批程序之一是适用于PC机的MININEC［Julian（1）Logan & Rockway］，当初是为64 KB存储器的Apple计算机用BASIC语言写成的，此后做了很大的改进，使之在用户友好的Windows环境下扩展了更多的实际应用［Rockway（1，2）&Logan］。

该软件的学生PC版可以通过本书的网址链接到EM Scientific Web的网址而调用。这是非常有力的数值分析工具，包含许多专业版本的有用特点，有利于更好地学习矩量法以及天线设计。C.2节讨论的设计原理，很多就取自MININEC的作者及其用户工程师们的经验。

Expert MININEC网址提供了很多有用的例子，其输入和输出数据组可用做课堂习题。供访问的说明文件有助于用户充分地利用程序。用户界面具有方便的输入模板，可直接为V形偶极子（直偶极子为其特例）、环、三单元八木－宇田等天线建模。程序输出可提供导线上的电流与电荷、阻抗与导纳、辐射波瓣图（dBi或电场、功率或定向增益）等。这些天线模型及其结果有助于第6章至第8章、第13章、第14章和第16章中对应内容的学习。对新用户至为重要的是检验自己使用该程序的能力，且应当核对程序的输出与文件所提供的结果。

C.5 其他有用的软件

上述计算机程序和网址显然是不完全的。有很多采用不同方法设计天线的程序，有些名义上要付费，其他的则属商品软件。很难收集一份完全而又即时的程序清单。但本书网址提供了几个非常有用的链接地址，其中一例就是应用计算电磁学会（Applied Computational Electromagnetic Society）的网址，其中列出了与天线数值分析有关的程序，并链接其他地址。读者可自行搜索这些网址。

参考文献

Julian, A. J. (1), J. C. Logan and J. W. Rockway, "MININEC: A Mini-Numerical Electromagnetics Code," NOSC Technical Document 516, September 1982.

Kouyoumjian, R. G. (1), and P. H. Pathak, "A Uniform Geometrical Theory of Diffraction for an Edge in a Perfectly Conducting Surface," *Proc. IEEE,* **62,** 1448–1461, November 1974.

Rockway, J. W. (1), and J. C. Logan, *MININEC Professional for Windows,* EM Scientific, Inc., Carson City, NV, 1995.

Rockway, J. W. (2), and J. C. Logan, *Expert MININEC Series, Wire Antenna Modeling Code,* EM Scientific, Inc., Carson City, NV, 1999.

关于天线计算机仿真的深入阅读参考资料

Burnside, W. D., and R. J. Marhefka, "Antennas on Aircraft, Ships or Any Large, Complex Environment," Chap. 20 in *Antenna Handbook: Theory, Applications, and Design,* edited by Y. T. Lo and S. W. Lee, Van Nostrand Reinhold, New York, 1988.

Burnside, W. D., and R. J. Marhefka, "Antennas on Complex Platforms," *Proc. IEEE,* **80,** 204–208, January 1992.

Diaz, L., and R. Milligan, *Antenna Engineering Using Physical Optics: Practical CAD Techniques and Software,* Artech House, Massachusetts, 1996.

Harrington, R. F., *Field Computation by Moment Methods,* Krieger Publishing Co., Florida, 1982.

Jin, J., *The Finite Element Method in Electromagnetics,* Wiley, New York, 1993.

Kunz, K. S., and R. J. Luebbers, *The Finite Difference Time Domain Method for Electromagnetics,* CRC Press, Florida, 1993.

Miller E. K., L. Medgyesi-Mitschang and E. H. Newman, *Computational Electromagnetics: Frequency-Domain Method of Moments,* IEEE Press, New Jersey, 1992.

Pathak, P. H., "High-Frequency Techniques for Antenna Analysis," *Proc. IEEE,* **80,** 44–65, January 1992.

Peterson, A. F., S. L. Ray and R. Mittra, *Computational Methods for Electromagnetics,* IEEE Press, New Jersey, 1998.

Sandiku, M. N. O., *Numerical Techniques in Electromagnetics,* CRC Press, Florida, 1992.

Taflove, A., and S. C. Hagness, *Computational Electrodynamics: The Finite-Difference Time-Domain Method,* 2d ed., Artech House, Massachusetts, 2000.

Volakis, J. L., A. Chatterjee and L. C. Kempel, *Finite Element Method for Electromagnetics: Antennas, Microwave Circuits, and Scattering Applications,* IEEE Press, New Jersey, 1998.

Wang, J. J. H., *Generalized Moment Methods in Electromagnetics: Formulation and Computer Solution of Integral Equations,* Wiley, New York, 1991.

附录D　吸波材料

如今吸波材料成了天线技术的一个集成部分，既用于测试现场，又用做天线部件以减少旁瓣与后瓣的辐射，见24.4c节。

第二次世界大战期间Winfield Salisbury（1）在哈佛无线电研究实验室（Harvard Radio Research Laboratory）发明了一种方法，将一片空间织物（$Z = 377$欧每单位面积）置于一反射板前方$\lambda_0/4$处，就能完全吸收其法向入射波。他所用的这种电阻性（碳浆，carbon-impregnated）织物片称为Salisbury片。该结构使法向入射波在1.3：1频带上的反射至少降为-20 dB[①]。

图D.1（a）给出了该结构的特性阻抗为377 Ω的等效传输线。为简便起见，将阻抗与377 Ω相除后成为归一化（无量纲）阻抗，如图D.1（b）中所示。现考虑图D.1（c）的多加载结构，在约1λ的距离内并联一系列电阻，其（归一化）电阻值和间距都随着至反射板的距离而从小变到大[②]。如图D.2所示，归一化线导纳Y从反射板短路位置$Y = \infty$随递增距离以及并联电导而移向Smith圆图的中心点（匹配点）。这种递增锥削终端与图D.1（a）和图D.1（b）所示的单电阻终端相比，具有宽频带上低反射系数的优点。图D.1（c）中的空间等效于图D.1（d），在$1\lambda_0$的距离内插有多层的Salisbury片，它们的每平方阻抗值已标注于图中，且背衬一片反射板。这种在层间填有塑料（电介质）、呈三明治式堆叠状的结构，是由J. Jaumann于二战期间在德国开发的。Emerson（1）曾为吸波材料及其应用撰写过一篇历史札记。

同时增加电阻值和电阻片的数量（且减小它们的间距）至极限情况，即成为连续锥削的媒质材料。设媒质的磁导率和介电常数都含损耗项、且两者各自的相对值相同，即

$$\mu_r = \mu_r' - j\mu_r'' = \varepsilon_r = \varepsilon_r' - j\varepsilon_r'' \tag{1}$$

则该有耗媒质具有与自由空间相同的实阻抗，即

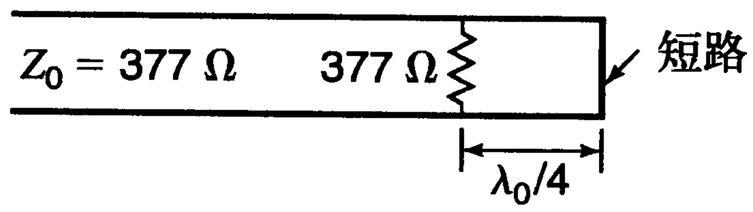
$$Z = \sqrt{\frac{\mu}{\varepsilon}} = \sqrt{\frac{\mu_0}{\varepsilon_0}}\sqrt{\frac{\mu_r}{\varepsilon_r}} = 377\sqrt{\frac{\mu_r}{\varepsilon_r}} = 377\,\Omega \tag{2}$$

原理上，如果入射到一种均匀媒质的波能够无反射地进入媒质的材料内，当该媒质足够厚时波就能被完全吸收。这种均匀媒质与上述锥削媒质恰成对照。

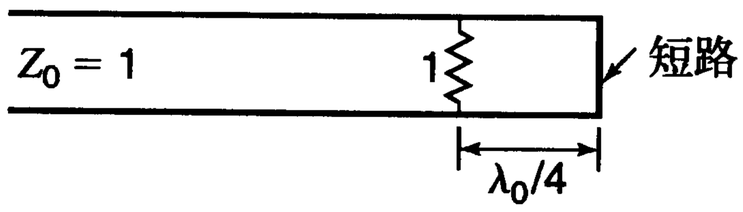
---

① 电阻片与反射板相距为3λ/4或5λ/4等，也能发生波的谐振吸收，但其频带较窄。

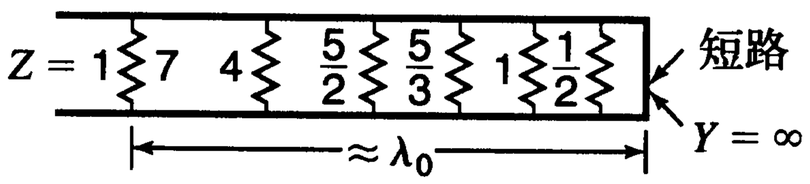
② 电阻值和间距都近似按指数律变化。



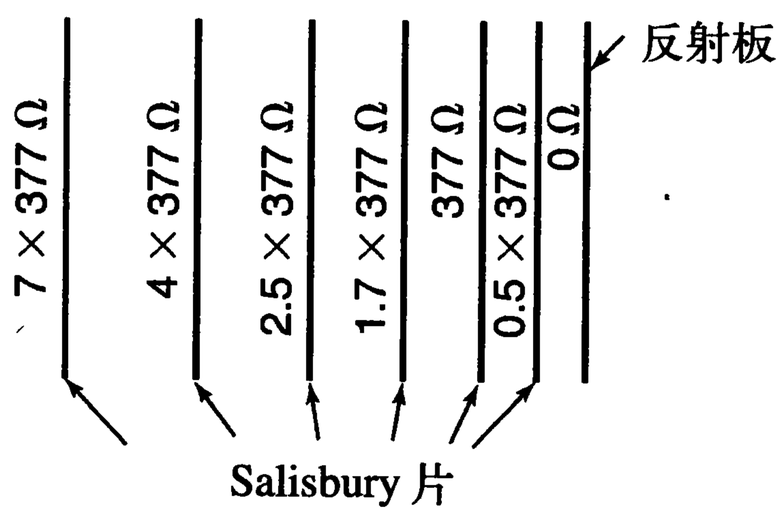
(a) 传输和带有 $\lambda_0/4$ 短截线的单负载作为匹配终端



(b) 结构同(a)，表示成归一化阻抗



(c) 沿线 $1\lambda_0$ 上 6 点加载分布作为宽频带匹配终端



(d) 带有多层 Salisbury 片的空间等效

图 D.1 传输线与空间传播的等效图

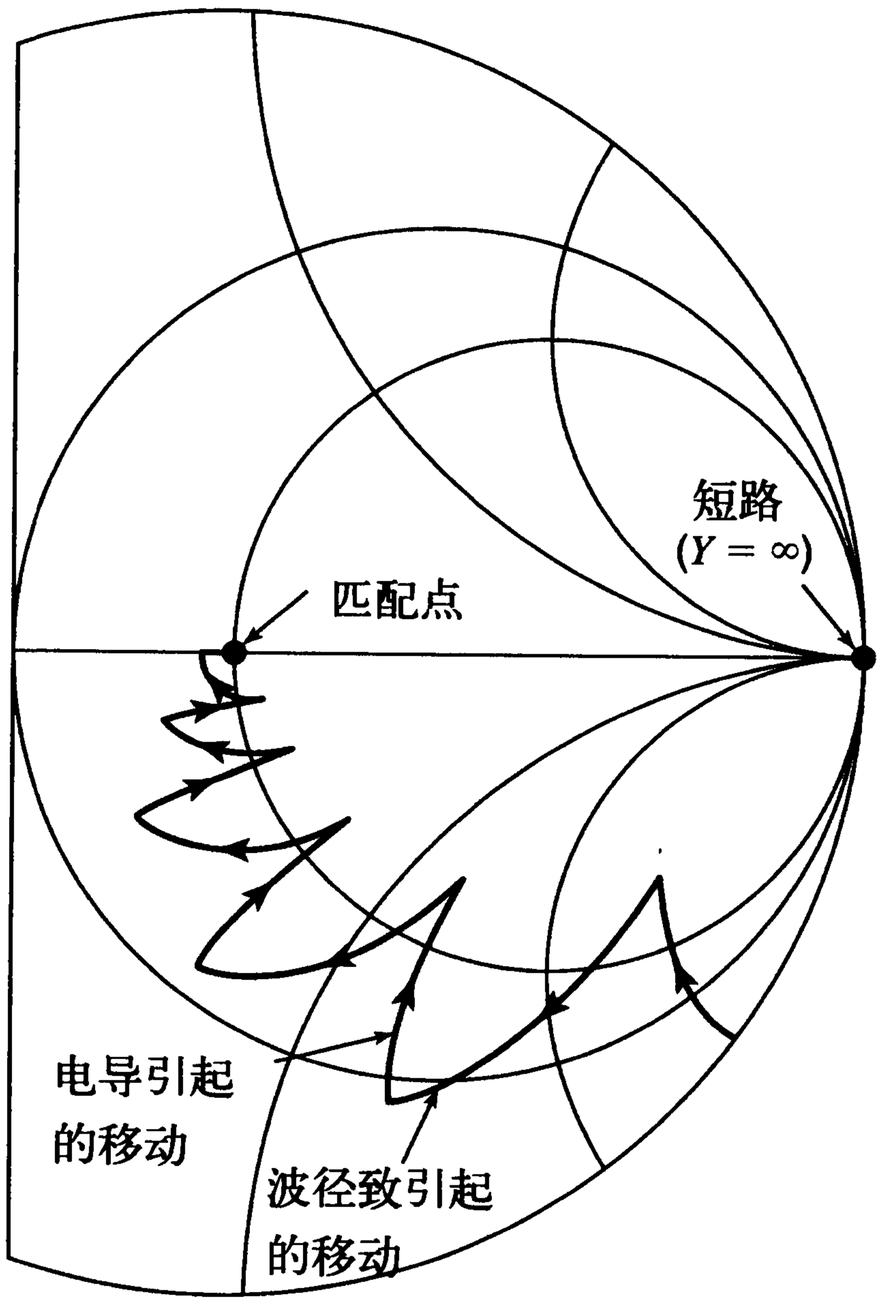


图 D.2 归一化导纳 Y 在 Smith 圆图上的轨迹。从短路点经图 D.1 (c) 的 6 点分布加载，逐步移动到圆图中心的匹配点（在 5 ：1 的频带上反射波降低 −20 dB）

例 D.1

设有 3 mm 厚的吸波片背衬理想导电的金属平板，材料参数为 $\mu_r = \mu_r = 10 - j10$，$\sigma = 0$。求 3 GHz 频率的反射系数。

解：

传播常数

$$\gamma = \alpha + j\beta = j\frac{2\pi}{\lambda_0}(10 - j10) \tag{3}$$

而

$$\lambda_0 = \frac{c}{f} = \frac{3 \times 10^8}{3 \times 10^9} = 0.1\,\text{m} \tag{4}$$

所以

$$\alpha = 2\pi \times 100 = 628\,\text{Np m}^{-1}$$

则波从金属板反射后经吸波片返回的相对场强为

$$\frac{E}{E_0} = |\rho_v| = \text{e}^{-2\alpha x} = \text{e}^{-3.77} = 0.023$$

比入射波场强低 33 dB。

虽然参数 $\mu_r = \varepsilon_r$ 的有耗媒质从原理上说很具有吸引力，但大多数普及型吸波器所具有的典型参数为 $\mu_r = 1$ 而 $\varepsilon_r \approx 2 - j1$。普及的形状则是如图 D.3（a）和图 D.3（b）所示的棱锥形和楔形。棱锥形对于正（对着鼻尖的）入射表现为锥削转换器（如上所述）。然而，DeWitt（1）& Burnside 发现，在大入射角时波的方向对棱锥的侧面几乎是边射的，棱锥吸波器近似于具有随机粗糙表面的散射体，故由于媒质失配而导致较大的反射系数；然而楔形吸波器在波的方向近乎平行于楔脊时的表现要出色得多。

例 D.2

设有高度（从尖顶至底部）为 30 cm 的棱锥阵列，材料参数为 $\mu_r = 1$，$\varepsilon_r = 2 - j1$，$\sigma = 0$。(a) 求（对着鼻尖的）正入射波为 3 GHz 频率（$\lambda = 100$ mm）时的反射系数。(b) 求 10 GHz 频率（$\lambda = 30$ mm）时的反射系数，材料参数同上。

解：

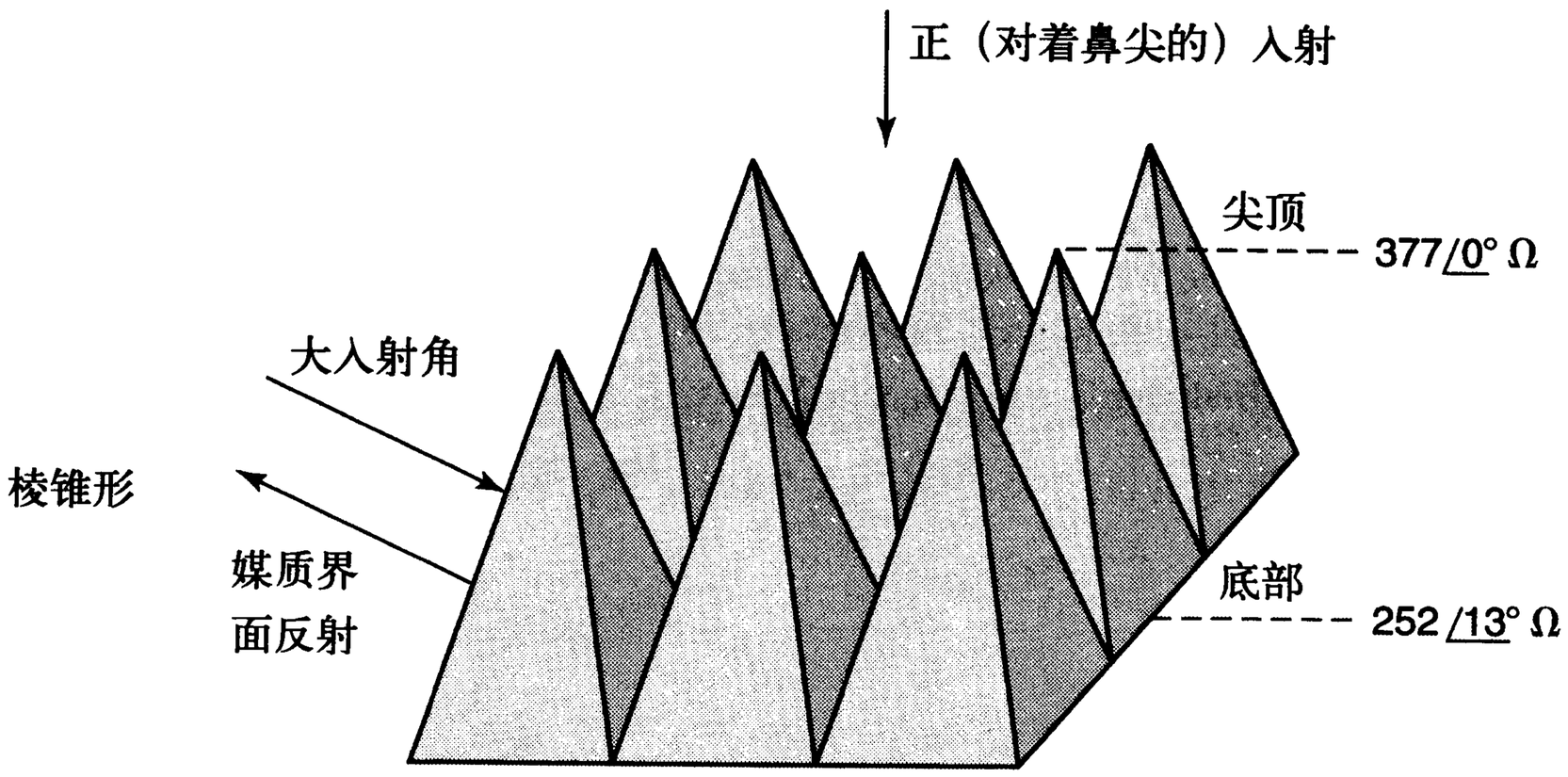
参照图 D.3 (a)，棱锥阵对正入射表现为有效阻抗从尖顶处的 $377\,\angle 0°\,\Omega$ 逐渐提高（3 GHz 时经 3λ 长度、10 GHz 时经 10λ 长度）到底部的

$$Z = \frac{377}{\sqrt{\varepsilon_r}} = \frac{377}{\sqrt{2 - j1}} = 252\,\angle 13.3°\,\Omega$$

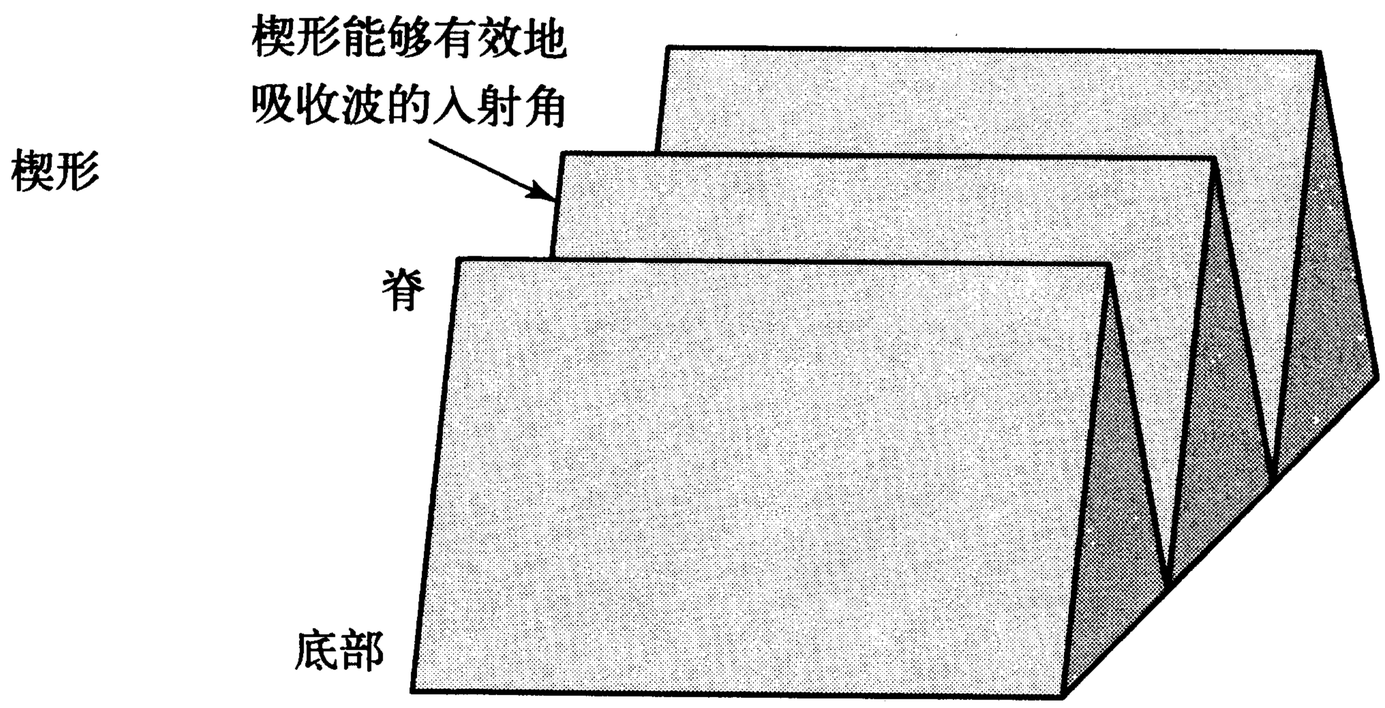
由于锥削段较长，因此足以忽略在媒质界面（空气与棱锥）处的反射。然而，若不存在锥削段，媒质失配的反射系数为

$$|\rho_v| = \left|\frac{Z_L - Z_0}{Z_L + Z_0}\right| = \left|\frac{252\,\angle 13° - 377}{252\,\angle 13° + 377}\right| = 0.24 \qquad -12\,\text{dB}$$

与即将计算的正入射到棱锥的情况相比，这是非常大的反射系数，与以大入射角边射于棱锥侧面[见图 D.3 (a)] 的大反射系数相当。



(a) 棱锥形吸波器



(b) 楔形吸波器

图 D.3 吸波器的形状

作为正入射反射系数的一级近似，假设棱锥体等效为其 1/3 高度的均匀实体。在实体媒质中，3 GHz 波的传播常数为而衰减常数

$$\gamma = j\frac{2\pi}{\lambda_0}\sqrt{\varepsilon_r' - j\varepsilon_r''} = j\frac{2\pi}{0.1}\sqrt{2 - j1} = j\frac{2\pi}{0.1}(1.46 - j0.35)$$

于是，反射系数为

$$\alpha = \frac{2\pi}{0.1} \times 0.35 = 22\,\mathrm{Np\,m^{-1}}$$

故得 (a) 3 GHz 波的反射系数被降至 −38 dB。

$$|\rho_v| = \mathrm{e}^{-2\alpha x} = \mathrm{e}^{-2\times 22\times 0.1} = 0.0123$$

同理有 (b) 10 GHz 波的反射系数被降至 −125 dB。

实际上，10GHz的反射系数确实比3 GHz的反射系数要小的多，但并不像计算所得结果那么悬殊。某些商品吸波材料的非匀质性会增大反射系数和后向散射（DeWitt-1）。

参考文献

DeWitt, B. T. (1), and W. D. Burnside, "Electromagnetic Scattering by Pyramidal and Wedge Absorber," *IEEE Trans. Ants. Prop.*, 1988.

Emerson, W. H. (1), "Electromagnetic Wave Absorbers and Anechoic Chambers through the Years," *IEEE Trans. Ants. Prop.*, **AP-21,** 484–489, July 1973 (49 references).

Salisbury, W. W. (1), "Absorbent Body for Electromagnetic Waves," U.S. Patent 2,599,944, June 10, 1952.

附录E　测量误差

一切被测的量都含有误差。于是，一个球所测出的面积可以是 $2.76 \pm 0.03\ m^2$，这表明误差或不确定性为 $0.03\ m^2$。如果这是均方根（rms，root mean square）偏差或标准偏差，则意味着真值介于此极限值之间、即大于 2.73 而小于 $2.79\ m^2$ 的可能性约为 2：1。对于有限的 n 次读数，有

$$\text{均方根偏差} = \sqrt{\frac{d_1^2 + d_2^2 + \cdots + d_n^2}{n-1}} \tag{1}$$

式中 d_1，d_2，$\cdots d_n$，是相对于一组 n 次观测之中（间）值的测量偏差。

有时给出的误差是概率误差，它是 rms 误差的 0.6745 倍。概率误差指出真值介于极限值之间的可能性恰为 1：1。

凡给出误差，通常意味着已经进行了一组重复性的测量。只进行一遍测量是不规范的，所得到的误差只不过是估计值。

凡在本书中引用的测量值，都应理解为受制于误差的。例如，某天线的增益可以表示为 36.5 dBi，但为了清楚起见，应当包括适当的误差，若误差是 ±0.5 dBi，则增益应为 36.5 ± 0.5dBi。然而，为了表述简便，通常又都省略了误差的标注，除了在 21.19 节中 Penzias & Wilson 所提出的天线温度的测量结果：

2.3 ± 0.3 K	归因于大气层
0.8 ± 0.4 K	归因于欧姆损耗
<0.1 K	归因于指向地面的后瓣
3.2 ± 0.5 K	总和[①]

他们测得的天空温度为 6.7 ± 0.8 K，扣去 3.2 ± 0.5 K，得出剩余量 3.5 ± 1.0 K[①]。

Penzias & Wilson 注意到了这个“误差”，并由此发现了 3K 的天空背景，随后他们因此而荣获了诺贝尔奖。

---

① 注意，该总误差是 rss（rootsumsquare，和方根——无关量的合成）而非 rms（均方根——单个量的误差）。所以上述分项的 0.3 和 0.4 是 rms 而总的 0.5 是 rss。

附录F　带*号习题的答案

第2章

2.7.3　3.8，4.6，6.1。

2.7.4　(a) 2.05×10^4 (43.1 dB)；(b) 1.03×10^4 (40.1 dB)。

2.9.3　$71.6\lambda^2$。

2.11.2　~11 kW。

2.11.5　152 m^2 RCP。

2.16.4　14.9 kW。

2.17.2　(a) AR = 1.5；(b) $\tau = 90°$；(c) CW。

2.17.4　(b) AR = 1.38；(c) $\tau = 45°$。

2.17.6　具有 $\tau = 45°$ 的直线。

2.17.7　(a) AR = −2.33(RH)；(b) $\tau = -45°$；(c) RH。

2.17.9　(a) AR = −5；(b) RH；(c) 34 mW m^{-2}。

2.17.11　(a) AR = 3.0；(b) $\tau = -22.5°$；(c) CW；(d) LH。

第3章

3.2.1　改变(a) $L = 0.5\lambda$，$w \to 0$；(b) $Z_s = 363 - j211\ \Omega$；(c) 1.64。

3.4.2　200 d。

3.5.1　(a) $24.7\lambda^2$；(b) 310(24.9 dBi)；(c) 186(22.7 dBi)。

3.5.2　0.19 m^2。

3.5.3　175 mm。

3.6.1　θ(E 面) = 45°，θ(H 面) = 45°，对于对称的波瓣图。

第4章

4.3.1　(a) 1539 W m^{-2}；(b) 4.29×10^{26} W；(c) 762 V m^{-1}。

4.5.2　(a) 5.1，6，7.07；(b) 见习题 2.7.3。

第5章

5.2.1　(c) 最大方向在 0°，180°，±41.8°，±138.2°

零点方向在 ±19.4°，±90°，±160.6°

半功率方向在±9.6°，±170.4°，±30°，±150°，±56.5°，±123.5°

(d) 最大方向在0°，±90°，180°

零点方向在±30°，±150°

半功率方向在±14.5°，±165.5°，±48.6°，±131.4°

(e) 最大方向在0°，180°，±14.5°，±165.5°，±30°，±150°，±49°，±131°，±90°

零点方向在±7°，±173°±22°，±158°±39°，±141°，±61°，±119°

半功率方向在±3.6°，±176.5°，±11°，±169°，±18.5°，±161.5°，±26°，±154°，±34.5°，±145.5°，±43.5°，±136.5°，±54.5°，±125.5°，±70°，±110°

(f) 最大方向在0°，180°

半功率方向在±90°

5.2.3 (a) 2。

5.6.4 (a)和(b)1个主瓣和5个副瓣；

(c) 常规端射阵D～7，增强定向性端射阵D～12。

5.6.6 (b) 6.6；(c) 6.3。

5.8.4 (a) $E(\phi)=\frac{1}{4}\frac{\sin\left(\frac{5}{2}\pi\sin\phi\right)}{\sin\left(\frac{5}{8}\pi\sin\phi\right)}\angle\frac{15}{8}\pi\sin\phi$

(b) $E(\phi)\cos^3\left(\frac{5}{8}\pi\sin\phi\right)$ 或 $\frac{3}{4}\cos\left(\frac{5}{8}\pi\sin\phi\right)+\frac{1}{4}\cos\left(\frac{15}{8}\pi\sin\phi\right)$

5.9.2 0.61。

5.9.3 (a) 0.52，0.82，1.00，0.82，0.52

(b) 最大方向±39°，±141°，±90°

(c) 零点方向±30°，±54°，±126°，±150°

(d) 24°

5.9.5 R=5：0.93，0.84，1.00，1.00，0.84，0.93

R=7：0.69，0.80，1.00，1.00，0.80，0.69

R=10：0.53，0.78，1.00，1.00，0.78，0.53，旁瓣愈低，则锥削要求愈陡。

5.9.7 (a) $E=\frac{\sin\frac{5}{2}\psi}{\sin\frac{1}{2}\psi}$，式中 $\psi=d_r\cos\phi+\delta$

(b) $\delta=0$,(1)1，1，1，1，1,(2)1，4，6，4，1,(3)1，0，0，0，1。

5.12.1 2500。

5.17.1 (a) 44°；(b) −13.3 dB；(c) 0.17π sr；(d) 0.89；(e) 24；(f) $1.9\lambda^2$。

5.18.1 最大方向0°，180°，±60°，±90°，±120°。

最小方向±41.4°，±75.5°，±104.5°，±138.6°。

第6章

6.2.1 (b) $E_r=\frac{Ql\cos\theta}{2\pi\varepsilon r^3}$，$E_\theta=\frac{Ql\sin\theta}{4\pi\varepsilon r^3}$，$E_\phi=0$。

6.2.2 (a) $2.86\times10^{-2}\angle\underline{-9°}$ V m^{-1}；(b) $8.88\times10^{-2}\angle\underline{-81°}$ V m^{-1}；(c) 2.39 × 10-4 $\angle\underline{-81°}$ A m^{-1}

6.2.4 (a) 一般表达式：282 mV m^{-1}，准静态表达式：121 mV m^{-1}

(b) 一般表达式：242 mV m^{-1}，准静态表达式：61 mV m^{-1}

(c) 一般表达式：784 μA m^{-1}，准静态表达式：338 μA m^{-1}

6.3.1 3.33 Ω。

6.3.2 (a) 2.74 μW；(b) 3.0 A。

6.3.5 (a) π sr；(b) 4。

6.3.7 354 Ω。

6.3.8 (a) 21.9；(b) 1.74λ^2；(c) 6.8 Ω。

6.3.9 (a) 8.16；(b) 654 Ω。

6.3.12 (a) 23.2 Ω。

6.5.1 $E = \tan\theta \sin\left(\frac{\tau}{2}\cos\theta\right)$；(b) 168 Ω；(c) 168，73，197 Ω。

6.8.1 (a) $E = \frac{\sin\theta}{1 - p\cos\theta}\left[\sin\pi\left(\frac{1}{p} - \cos\theta\right)\right]$，四瓣的波瓣图；

(b) 40 瓣的波瓣图。

第 7 章

7.6.1 1890 Ω。

7.8.2 (1) 180 Ω，1.5；(2) 1550 Ω，1.2；(3) 4100 Ω，3.6。

7.9.1 四瓣的波瓣图。

第 8 章

8.3.4 (a) 10；(b) 6 cm；(c) 1.05（95% 纯右旋圆极化）。

8.5.1 (a) 28.8 cm；(b) 27.6 cm；(c) 26.4 cm，26.4 cm，25.8 cm，24.0 cm；(d) 30 cm；(e) 19.1 cm；(f) 465～535 MHz；(g) 11 dBi。

8.8.1 (1) 0.802；(2) 0.763。

8.11.1 (a) $D_\lambda = \sqrt{2H_\lambda}/\pi$；(b) $E = \sin\theta$。

第 9 章

9.5.1 474 Ω。

9.5.2 $779 - j67$。

9.9.1 750。

第 10 章

10.3.4 (a) 73 Ω；(b) 10 dB。

10.3.6 (a) 16 dBi；(b) 12.6°。

10.7.2 76.6 m^2。

第 11 章

11.2.2 $270 + j350\ \Omega$。

第 12 章

12.2.1 14.5 K。

12.2.2 25.6 K。

12.3.2 33 dB。

12.3.3 (a) 10 h；(b) 6.9×10^{-20} W；(c) 50 Hz。

12.3.4 (a) 12.2 dB；(b) 2.3 m。

12.3.5 (a) 0.08 K；(b) 0.09 K；(c) 445 K；(d) 500 mJy。

12.3.7 (a) 12.7 dB； 0.5 dB；(c) ～13 min。

12.3.8 24.2 K。

12.3.9 (a) 15.6 MHz；(b) 26.1 MHz；(c) 41.0 MHz。

12.3.11 (a) 0.06 K；(b) 320 mJy。

12.3.13 1000 光年（LY）。数据率较慢，但当年首条跨大西洋电缆的数据率也并没有快多少。以此 1000 光年为参照，回过头来看：地球的周长为 1/7 光秒，地球至木星的距离为 45 光分、至冥王星的距离为 5 光时、距最近的恒星为 4 光年。1000 光年的距离等于离最近恒星的 250倍。然而，这还仅仅是从我们的银河系到其他几十亿个银河系距离的百分之一，其中有些相距在百亿光年以上。如果地外文明（ETC）发来电讯能被解读并理解，从地球发送一份答复，则地球人要2000年之后才能得到回音。即使这在技术上有可能实现，人们是否有此先见之明和耐心去等待2000年？迄今为止还从未发生过这种遥远距离的对话，人们正在边收听、边怀疑。

12.3.14 650 km 可视距离。

12.3.17 108.4 dB。

12.3.18 (a) 8.16×10^{5} W；(b) 1.0 m；(c) Yes($G/T = 7.4$ dB K^{-1})；(d) 3.2 dB。

12.4.1 23.6 K。

12.4.4 65.1 kW Hz^{-1}。

12.5.3 5.6×10^{-28} m^2。

12.5.5 560 TW 峰值功率。

12.5.7 (a) π m^2；(b) 40π m^2；(c) 0.04π m^2；(d) ～0 m^2。

12.5.8 0.8 m^2。

12.5.9 (a) 79.5 m s^{-1}；(b)46.7 m s^{-1} 前向雷达。

12.5.12 45 m s^{-1} (=162 km h^{-1} = 101 mi h^{-1})。

12.5.13 160 mW。

12.5.14 1.2 kHz。

12.5.18 66 ns。

12.5.20 (a) 3.1×10^{-15} W；(b) 9.0×10^{-17} W；(c) 信噪比 = S/N =31 dB，信杂比 = S/C = 16 dB。

第 13 章

13.4.1 $121 + j46\ \Omega$。

13.6.3 $R_a = 50\ \Omega$，$R_b = 20\ \Omega$，$R_c = 50\ \Omega$。

第 15 章

15.3.1 1.2。

第 16 章

16.2.1 $\sim 0.67\lambda$。

16.3.2 (a) $52 - j21\ \Omega$; (b) G_f（最大）= 1.55。

16.6.1 (a) 1 和 6:$63 + j29\ \Omega$; 2 和 5:$46 - j2\ \Omega$; 3 和 4:$53 + j10\ \Omega$。

16.6.4 11.7°。

16.6.7 (a) 6°22′; (b) −13.15dB; (c) π/4 sr; (d) 0.89; (e) 16; (f) $1.27\lambda^2$。

16.8.6 (a) $0.354\lambda'$; (b) $-\pi/2$。

16.16.1 (b) ~17°。

16.16.2 $H_\lambda = 0.83$，$\phi = 72.5°$，$L_\lambda = 5.5$。

16.16.4 $\phi = 72.5°$，$L_\lambda = 5.14$。

16.16.6 $\phi = 60°$。

第 17 章

17.4.1 (a) 62.5 mm; (c) 28%。

第 19 章

19.1.3 0.81，2036。

19.1.4 0.66，1651。

19.1.6 81.1%，1304 或 31.2 dBi。

19.1.8 (b) 4.2°。

第 21 章

21.9.3 (a) 5°, 1.1 dB; 10°, 2.2 dB

(b) 5°, 1.4dB; 10°, 2.9 dB

(c) 5°, 0.06 dB; 10°, 0.26 dB

21.10.1 (a) 图 P21.10.1 (a) 指出铅垂、水平和圆极化信号随着离地架高的变化。

(b) 注意：若只接收到直射路径（无地面反射）的信号，则用圆极化天线和用铅垂极化 或水平极化天线将得到相同的信号电平。然而，用圆极化天线时的信号电平与架高无关，而用铅垂极化或水平极化天线时相对于无反射情况有 6 dB 的电平变化。

对于电视接收还有一项重要因素，用圆极化天线所接收到的（理想情况）只有直射波，而用铅垂极化或水平极化天线同时接收到直射波与反射波。在合成场强的最大点处，直射与反射信号基本相等，但时间不同，如果此时间差异为微秒量级，就会发生可察觉的重影图像，降低画面质量。

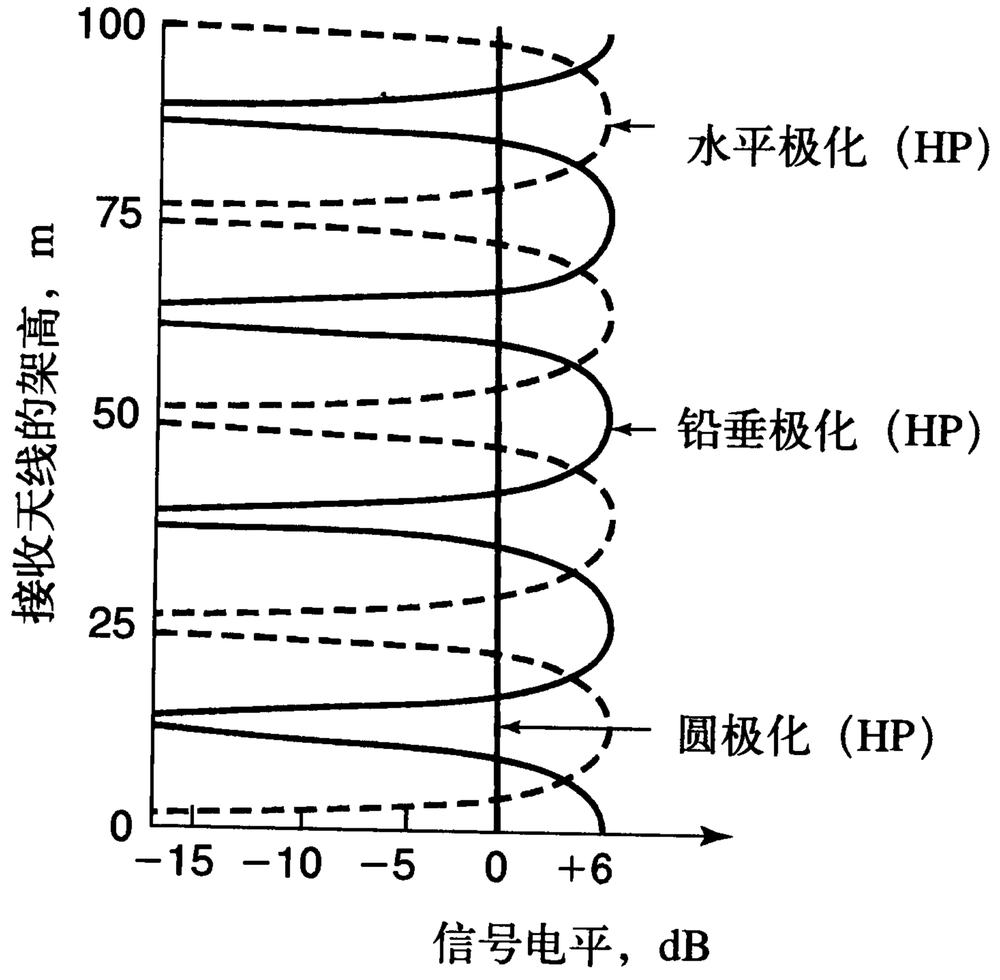


图 P21.10.1 (a) 的解指出信号随着离地架高的变化

21.12.1 106 m，35 m，11 m 和 3.5 m。

21.13.1 (a) 26.5 W m^{-2}; (b) 184 μW m^{-2}。

21.13.3 1.35 nW m^{-2}。

21.13.6 $f_c = 0$，$8.89/\lambda_0$ Np m^{-1}。

术　语　表

Antenna Types　　天线型式

aperture	**口径**
achievement factor	达标因子
complex deviation factor	复偏差因子
aperture distribution	口径分布
Dolph-Tchebyscheff optimum distribution	道尔夫－切比雪夫最优分布
aperture efficiency	口径效率
offset illumination	偏照
Ruze factor	增益损失因子
squint angle	斜视角
surface roughness	表面粗糙度
utilization factor	利用因子
aperture antenna	**口径天线**
aperture plane	**口径平面**
architecturally acceptable antenna	**建筑上可接受的天线**
Argus antenna	**百眼巨人型天线**
array	**阵（列）**
adaptive array	自适应阵
amplitude taper array	幅度锥削（渐变）阵
BC（broadcast）array	广播阵
broadside array	边射阵
chain array	链式阵
Chireix-Mesny array	风琴衣架式阵
circular array	圆形阵
close-spaced array	密距阵
continuous array	连续阵
variable phase velocity continuous array	可变相速连续阵
curtain array	帘幕形阵
dissimilar point sources array	非相似点源的阵
Dolph-Tchebyscheff array	道尔夫－切比雪夫阵

Dipoles array	偶极子阵
end-fire array	端射阵
increased directivity end-fire array	增强定向性端射阵
ordinary end-fire array	常规端射阵
3D end-fire array	三维端射阵
frequency scanning array	频率扫瞄阵
frequency-scanning grid array	频率扫瞄栅格形阵
grid array	栅格形阵
Landsdorfer array	赋形偶极子八木－宇田阵
linear array	直线阵
lobe-sweeping array	波束扫瞄阵
log-periodic dipole array	对数周期偶极子阵
low-side-lobe array	低旁瓣阵
microstrip array	微带阵
multi-aperture array	多口径阵
periodic structure modes	周期性结构模
periodic structures	周期性结构
phased array	相控阵
rotatable helix phased array	可旋转螺旋相控阵
point sources array	点源阵
isotropic point sources array	各向同性点源阵
non-isotropic point sources array	非各向同性点源阵
primary array	初级阵
random array	随机阵
retro array	返回式阵
rectangular array	矩形阵
rotating helix array	旋转螺旋阵
scanning array	扫描阵
secondary array	次级阵
Sterba curtain array	司梯巴帘幕形阵
three-dimensional array	三维阵
triangle array	三角形阵
two-dimensional array	二维阵
Van Atta array	范阿塔阵
verylarge aperture（VLA）array	甚大口径阵
very long baseline（VLBA）array	甚长基线阵
W8JK array	密距端射阵

with missing sources array	缺源阵
Yagi-Uda array	八木－宇田阵
YUCOLP array	八木－宇田－夹角－对数周期阵
asteroid detection antenna	小行星检测天线
antenna diversity	天线分集
antenna under test (AUT)	待测天线
azimuth ILS antenna	方位仪表着陆系统天线
base station antenna	基台天线
billboard antenna	广告牌天线
cell phone antenna	蜂窝电话天线
chimney antenna	烟囱形天线
circularly polarized antenna	圆极化天线
clover-leaf antenna	苜蓿叶形天线
collinear antenna	共线天线
cone antenna	锥形天线
biconical antenna	双锥天线
biconical vee antenna	双锥 V 形天线
bow-tie dipole antenna	领结形偶极子天线
curved biconical vee antenna	弯曲双锥 V 形天线
conical antenna	圆锥天线
discone antenna	盘锥天线
single cone and ground plane	带接地面的单锥天线
square cone antenna	方锥天线
volcano-smoke antenna	火山口－烟嘴形天线
cylindrical antenna	柱形天线，圆柱天线
Deutsche Welle antenna	“德意志之声”天线
digital-beam-forming (DBF) antenna	数字波束形成天线
dipole (antenna)	偶极子（天线）
bow-tie dipole	领结形偶极子
center-fed dipole	中心馈电偶极子
crossed dipoles	交叉偶极子
Delta match antenna	Δ形匹配天线
electric dipole	电偶极子
flagpole	旗杆式天线
folded dipole	折合偶极子
full-wave antenna	全波天线
Gamma match antenna	Γ形匹配天线

half-wave antenna 半波天线
half-wave dipole 半波偶极子
Hertz's dipole 赫兹偶极子
horizontal dipole 水平偶极子
J-match antenna J 形匹配天线
quarter-wave monopole λ/4 单极子
short dipole 短偶极子
stub antenna 短桩形天线
T-match antenna T 形匹配天线
tunable dipole 可调谐偶极子
vertical dipole 铅垂偶极子
whip antenna 鞭状天线
direction-finding antenna 测向天线
earth station antenna 地球站天线
echelon antennas 梯式天线
electrical small antenna 电小天线
embedded antenna 植入式天线
flush disk antenna 嵌入式盘形天线
frequency independent antenna 非频变天线
biconical antenna 双锥天线
fractal antenna 分形天线
artistic antenna 艺术性天线
Barnsley-Fern antenna 班斯莱 – 弗恩天线
Bipectinate antenna 双栉形天线
Minkowski fractal antenna 闵可夫斯基天线
Promethea moth antenna 普罗大蚕蛾天线（触角）
Sierpinsky-triangle antenna 塞平斯基 – 三角形天线
log-periodic antenna 对数周期天线
log-periodic dipole array 对数周期偶极子阵
log-periodic toothed antenna 对数周期齿状天线
log spiral antenna 对数周期螺蜷天线
stacked log periodic antenna 层叠式对数周期天线
self-complementary antenna 自补天线 spiral ant
enna 螺蜷天线
Archimedes spiral antenna 阿基米德螺蜷天线
equiangular spiral antenna 等角螺蜷天线
conical spiral antenna 圆锥螺蜷天线

planar spiral antenna	平面螺蜷天线
twin-alpine horn antenna	双阿尔卑斯喇叭天线
ultra-wide-band（UWB）antenna	特宽频带天线
Galileo antenna	伽利略天线
George Brown turnstile antenna	乔治布朗绕杆式天线
Goubau antenna	G-线天线
ground effect on antenna	天线的地面效应
ground penetrating radar（GPR）antenna	探地雷达天线
ground plane antenna	带接地面的天线
helix（antenna）	螺旋（天线）
axial mode helix	轴向模螺旋
back fire helix	背射螺旋
bifilar helix	双绕螺旋
four-lobed helix	四瓣螺旋
helical beam antenna	螺旋聚束天线
genetic algorithm helix	遗传算法螺旋
monofilar helix	单绕螺旋
Moon helix	登月螺旋
multifilar helix	多绕螺旋
normal mode helix	法向模螺旋
quad-helix	四（支）螺旋
quadrifilar helix	四绕螺旋
rotatable helix	可旋转螺旋
square helix	方形螺旋
tapered helix	锥削螺旋
horizontal antenna	水平天线
horn（antenna）	喇叭（天线）
Alpine horn	阿尔卑斯喇叭
aperture-matched horn	口径匹配喇叭
conical horn	圆锥喇叭
corrugated horn	皱纹喇叭
Hogg horn	糖铲形喇叭
optimum horn	最优喇叭
pyramidal horn	棱锥喇叭
rectangular horn	矩形喇叭
ridge horn	加脊喇叭
rolled-edge horn	卷边喇叭

septum horn	隔膜喇叭
sugar scoop antenna	糖铲形天线
twin-alpine horn	双阿尔卑斯喇叭
internal antenna	**内置式天线**
instrument landing system (ILS)	**仪表着陆系统**
azimuth guidance ILS antenna	方位导引天线
glide slope antenna	滑翔着陆天线
localizer ILS antenna	定位器天线
vertical guidance ILS antenna	铅垂导引天线
isotropic antenna	**各向同性天线**
lens (antenna)	**透镜 (天线)**
artificial dielectric lens	人造介质透镜
of flat metal strips	金属条带的人造介质透镜
of metal spheres	金属球的人造介质透镜
of metal disk	金属盘的人造介质透镜
cylindrical lens	柱形透镜
dielectric lens	介质透镜
Einstein lens	爱因斯坦透镜
E-plane lens	*E*- 面透镜
gravity lens	重力透镜
H-plane lens	*H*- 面透镜
long-focus lens	长焦距透镜
Luneberg lens	龙伯透镜
metal-plate lens	金属板透镜
multiple-helix lens	多螺旋透镜
phase-controlled lens	相位受控透镜
reflector-lens	反射器 – 透镜
short-focus lens	短焦距透镜
tolerance on lens	透镜的容差
unzoned metal-plate lens	未分区金属板透镜
zoned lens	分区透镜
linear antenna	**直天线**
loop (antenna)	**环 (天线)**
Alford loop	埃福特环天线
curl antenna	卷曲天线
ferrite-loaded loop	铁氧体加载环天线
Marconi's square conical loop	马可尼方锥环天线

quad loop	两圈环天线
small loop	小环天线
square loop	方形环天线
triangular loop	三角形环天线
1λ-circumference loop	全波环天线
low—earth orbit (LEO) antenna	近地轨道天线
microstrip antenna	微带天线
mobile communication antenna	移动通信天线
mobile station antenna	移动台天线
Nauen antenna	诺恩大型短波广播天线
notch antenna	凹口天线
omnidirectional antenna	全向天线
patch antenna	贴片天线
physically small antenna	物理小天线
plasma antenna	等离子态天线
prolate spheroidal antenna	长球形天线
range marker antenna	距离标记天线
reflector (antenna)	反射器（天线）
corner reflector	夹角反射器天线
dihedral corner reflector	两面夹角反射器
passive corner reflector	无源夹角反射器
retro-corner reflector	返回式夹角反射器
square-corner reflector	夹直角反射器
trihedral corner reflector	三面夹角反射器
corner-Yagi-Uda hybrid antenna	夹角－八木－宇田混合天线
flat sheet reflector antenna	平板反射器天线
reflector.mirror (antenna)	反射镜（天线）
Cassegrain-type reflector	卡塞格伦型反射镜
cylindrical parabolic reflector	柱形抛物面反射镜
DBS home parabolic dish antenna	直播卫星家用抛物面碟形天线
deep-space dish antenna	深空碟形天线
off-axis reflectors antenna	偏轴反射镜天线
parabolic reflector antenna	抛物面反射镜天线
paraboloid with missing sector	缺损扇片的抛物面
paraboloidal reflector	旋转抛物面反射镜
rolled edge reflector	卷边反射镜
sawtooth edge reflector	锯齿状边缘反射镜

serrated edge reflector	锯齿状边缘反射镜
shaped reflector	赋形反射镜
spherical reflector	球面反射镜
resistance loaded antenna	电阻加载天线
spheroidal antenna	类球体天线
short-wave antenna	短波天线
sleeve antenna	套筒天线
slot antenna	缝隙天线
boxed slot antenna	盒式缝隙天线
complementary antenna	互补天线
slotted cylinders	开缝柱形天线
slotted waveguide	波导缝隙天线
small antenna	小天线
smart antenna	智能天线
smooth vee antenna	光滑 V 形天线
stacked antenna	层叠式天线
submerged antenna	埋地天线
super turnstile antenna	超旋转场天线
surface wave antenna	表面波天线
surveillance antenna	监视天线
T-march antenna	T 形匹配天线
Terahertz antenna	太赫频率的天线
thin cylindrical antenna	细圆柱天线
thin linear antenna	细直天线
traveling wave antenna	行波天线
Beverage antenna	贝弗瑞行波天线
end-fire antenna	端射天线
helical antenna	螺旋天线
polyrod antenna	介质杆天线
cones of retina	视网膜锥
rods of retina	视网膜杆
leaky line antenna	泄漏传输线天线
leaky wave antenna	漏波天线
long wire antenna	长导线天线
rhombic antenna	菱形天线
surface wave antenna	表面波天线
corrugated surface wave antenna	皱纹表面波天线

Vivaldi antenna	韦尔弟天线
Yagi-Uda antenna	八木－宇田天线
turnstile antenna	旋转场天线
ultra-wide-band (UWB) antenna	特宽频带天线
Vee antenna	V 形天线
vertical antenna	铅垂天线
waveguide antenna	波导天线
weather-vane antenna	气象风标天线
Yagi-Uda antenna	八木－宇田天线
Yagi-Uda modifications	变种八木－宇田天线

Antenna Fitting

天线附配件

antenna measurements	天线测量
absorber	吸波器（吸波体）
anechoic chambers	吸波室
horn absorber	喇叭吸波器
Jauman absorber	约曼吸波器
pyramid absorber	棱锥形吸波器
Salisbury screen	索尔兹伯里吸波片
shroud absorber	罩筒式吸波器
space cloth	空间织物
wedges absorber	楔形吸波器
alignment error	对准误差
cable effect on measurement	电缆对测量的影响
compact antenna test range（CATR）	紧缩的天线测量场地
coordinates for measurement	测量用坐标系
elevated range	仰角范围
far field measurement	远场测量
hologram measurement	全息法测量
measurement coordinates	测量坐标系
measurement distance	测量距离
cell phone measurement distance	蜂窝电话测量距离
horn measurement distance	喇叭天线测量距离
reflector measurement distance	反射镜天线测量距离
measurement error	测量误差
measurement range	测量场地
compact measurement range	紧缩的测量场地

dual reflector measurement range	双反射镜测量场地
ground-reflection measurement range	地面反射测量场地
near-field measurement range	近场测量场地
patterns measurement range	波瓣图测量场地
millimeter wave measurement	毫米波测量
near field measurement	近场测量
parameter measurement	参量测量
current distribution measurement	电流分布测量
effective gain measurement	有效增益测量
efficiency measurement	效率测量
directivity.gain method	定向性/增益法
radiometer method	辐射计法
random field method	随机场法
Wheeler cap method	惠勒帽法
gain measurement	增益测量
absolute method	绝对法
comparison method	比较法
using celestial radio source	利用天体射电源
using radar technique	利用雷达技术
using three unknown antennas	利用三个未知天线
impedance measurement	阻抗测量
phase measurement	相位测量
differential method	微分法
direct method	直接法
reference antenna method	参考天线法
polarization measurement	极化测量
rotating source method	旋转源法
pedestal	测试支架，基座
positioner	位置控制器，天线转台
range alignments	测量场地的对准
reciprocity in measurement	测量中的互易性
uncertainty of pattern measurement	波瓣图测量的不确定性
due to reflected wave	反射波引起的测量不确定性
feed of antenna	天线的馈源
Cassegrain feed	卡塞格伦馈源
Gregorian feed	格雷戈里馈源
helix feed	螺旋馈源

horn feed	喇叭馈源
offset feed	偏照馈电
wave launcher	波激励器
feeding network	**馈电网络**
beam forming network	波束形成网络
Butler matrix	巴特勒矩阵
conductor shape equivalence	导体形状的等效
dielectric slab waveguide	介质平板波导
helix frequency shifter	螺旋移频器
helix phase shifter	螺旋移相器
helix terminations	螺旋终端
matching stub	匹配短截线
polarizer	极化器
helix polarizer	螺旋极化器
meanderline polarizer	回折线极化器
wave polarizer	波极化器
traps	陷波器
balun	**巴仑（平衡－非平衡转换器）**
balanced transformers	平衡变换器
“bazooka” balun	火箭筒式巴仑
bypass balun	旁路式巴仑
candelabra balun	烛台式巴仑
choke balun	扼流式巴仑
cutaway balun	截割式巴仑
ferrite balun	铁氧体巴仑
mast balun	支杆式巴仑
natural balun	天然巴仑
printed balun	印刷巴仑
Roberts balun	罗伯特巴仑
sleeve-dipole balun	套筒偶极子式巴仑
two TypeI’s in series balun	两根 I 型串接的巴仑
type I balun	I 型巴仑
type II balun	II 型巴仑
type III balun	III 型巴仑
with stub balun	带短截线巴仑
frequency Selective surface（FSS）	**频率选择表面**
bandwidth of FSS	频率选择表面的频带宽度

band pass FSS 带通频率选择表面
band stop FSS 带阻频率选择表面
complementary surface 互补表面
dichroic surface 分频表面
oblique angle of incidence 斜入射角
periodic slot structure 周期性缝隙结构
periodic wire surface 周期性导线表面
polarization of FSS 频率选择表面的极化
sandwich FSS 夹心式频率选择表面
type elements of FSS 频率选择表面的单元型式
anchor elements 锚型单元
Jerusalem cross elements 耶路撒冷十字型单元
center connected elements 中心连接单元
circle plate elements 圆板单元
dipole elements 偶极子单元
dipole loaded elements 偶极子加载单元
hexagon elements 六边环形单元
hexagon plate elements 六边形板单元
loop elements 环形单元
square plate elements 方板单元
square spiral elements 方蜷线型单元
tripole elements 三极子单元
trislot elements 三缝隙单元
3-legged elements 三腿形单元
4-legged elements 四腿形单元
4-legged loaded elements 四腿形加载单元
N-pole elements *N*极子单元
α-combination elements α型组合单元
β-combination elements β型组合单元
γ-combination elements γ型组合单元
radome 天线罩
dielectric radome 介质天线罩
hybrid radome 混合型天线罩
metal radome 金属天线罩
slotted-metal radome 开缝金属天线罩
stealth radome 隐形天线罩

Antenna Theory

天线理论

Charts

图表

aperture-far-field relations chart
口径 – 远场关系图表

gain-wavelength chart for radio telescopes chart
射电望远镜的增益 – 波长图表

helices chart
螺旋天线图表

RCS values chart
雷达截面值图表

sky noise temperature chart
天空噪音温度图表

codes

计算机程序

ARRAYPATGAIN
直线阵波瓣图和定向性程序

CORNEREFLECT
夹角反射器程序

MININEC
小型数值电磁学程序

software modeling
软件建模

equations&formulae

方程和公式

basic radiation equation
基本辐射方程式

Brown's equation
布朗方程式

Friis transmission formula
弗里斯传输公式

Hallen's integral equation
海伦积分方程

Maxwell's equations
麦克斯韦方程组

transmission line formulas
传输线公式

history stories

历史故事

of antennas
天线的故事

of asteroid
小行星检测天线的故事

of auctioning channels
拍卖信道的故事

of helical beam
螺旋聚束天线的故事

of Hertz
赫兹的故事

of Illinois
伊利诺依的故事

from long to short wavelengths
从长波到短波的故事

of Marconi
马可尼的故事

of Robert C.Byrd telescope
罗伯特·伯德射电望远镜的故事

of Sugar scoop
糖铲形天线的故事

of 3K sky background
3 K 天空背景噪声温的故事

of W8JK
W8JK 的故事

of Yagi-Uda
八木 – 宇田的故事

parameters

天线参量

beam
波束

beam area
波束范围

beam efficiency
波束效率

main-beam efficiency	波束立体角
beamwidth	波束宽度
first null beamwidth（FNBW）	第一零点波束宽度
half-power beamwidth（HPBW）	半功率波束宽度
stray factor	杂散因子
carrier-to-noise ratio	载波噪音比
current distribution	电流分布
directivity	定向性
EP wave power	椭圆极化波的功率
effective aperture	有效口径
effective height	有效高度
effective radiation power（ERP）	有效辐射功率
efficiency	效率
antenna efficiency	天线效率
aperture efficiency	口径效率
radiation efficiency	辐射效率
transmission efficiency	传输效率
field zones	场区
Cornu spiral	考纽螺线
far fields zone	远场区
Fresnel zone	菲涅耳区
Fraunhofer zone	夫琅和费区
near fields zone	近场区
quasi-stationary fields zone	准静态场区
Rayleigh distance	（远.近区分界的）瑞利距离
frequency bandwidth	频带宽度
shape-bandwidth relations	形状－频带的关系
gain	增益
effective gain	有效增益
gain loss factor	增益损失因子
mean effective gain	平均有效增益
impedance	阻抗
characteristic impedance	特性阻抗
driving-point impedance	激励点阻抗
input impedance	输入阻抗
input resistance	输入电阻
intrinsic impedance	本征阻抗

load impedance	负载阻抗
mutual impedance	互阻抗
mutual resistance	互电阻
radiation impedance	辐射阻抗
radiation resistance	辐射电阻
self impedance	自阻抗
self resistance	自电阻
shape-impedance relations	形状－阻抗的关系
space impedance	空间阻抗
terminal impedance	终端阻抗
transfer impedance	转移阻抗
noise temperature	噪声温度
pattern	波瓣图
back lobe	后瓣
decibel pattern	分贝波瓣图
doughnut pattern	饼圈形波瓣图
field pattern	场（强）波瓣图
Fraunhofer pattern	夫琅和费波瓣图
Fresnel pattern	菲涅耳波瓣图
grating lobes	栅瓣
null directions	波瓣图零点方向
main lobe	主瓣
minor-lobes	副瓣
minor-lobe maxima	副瓣最大方向
pattern multiplication	波瓣图乘法
pattern smoothing	波瓣图平滑化
pattern synthesis	波瓣图综合
pencil beam pattern	笔状波束波瓣图
phase pattern	相位波瓣图
polar pattern	极坐标波瓣图
power pattern	功率波瓣图
random side lobes	随机旁瓣
side lobe	旁瓣
side lobe level（SLL）	旁瓣电平
3D pattern	三维波瓣图
phase	相位
polarization	极化

axial ratio	轴比
depolarization	消极化
circular polarization（CP）	圆极化
cross polarization（X-Polar）	交叉极化
elliptical polarization（EP）	椭圆极化
horizontal polarization（HP）	水平极化
linear polarization（LP）	线极化
parallel polarization	平行极化
polarization parameters	极化参量
perpendicular polarization	垂直极化
polarization plane	极化平面
Poincar sphere	庞伽莱球
polarization ratio	极化比
polarization state	极化态
vertical polarization（VP）	铅垂极化
radar cross section	雷达截面
radiation intensity	辐射强度
radiation power	辐射功率
radiation power factor	辐射功率因子
resolution	分辨率
signal-to-clutter ratio	信号杂波比
signal-to-noise ratio（SNR）	信号噪声比
visibility	可见度
complex visibility	复可见度
fringe visibility	条纹可见度
function	可见度函数
standing-wave ratio（SWR）	驻波比
physical quantities & units	**物理量和单位**
ampere（A）	安培
aperture distribution	口径分布
source distributions from visibility functions	由可见度函数重构的源分布
candela（cd）	烛光（光学单位）
charge distribution	电荷分布
conductivities	电导率
coulomb（C）	库伦
dBi	相对于各向同性的分贝
dimensions	量纲

farad（F）	法拉
flux density	通量密度
henry（H）	亨利
hertz（Hz）	赫兹
International System of Units（SI）	国际单位制
joule（J）	焦耳
kelvin（K）	开尔文
ohm（Ω）	欧姆
permeability	磁导率
permittivity	介电常数
Poynting vector	坡印廷矢量
radian（rad）	弧度
radian sphere	弧度球
steradian（sr）	立体弧度
transverse impedance	横向阻抗
volt（V）	伏特
walt（W）	瓦特
weber（Wb）	韦伯
principles & methods	**原理和方法**
Babinet's principle	巴比涅原理
Fermat's principle	费马原理
genetic algorithm	遗传算法
geometric theory of diffraction（GTD）	几何绕射理论
geometric optics（GO）	几何光学法
Huygens principle	惠更斯原理
Image theory	镜像理论
integral-equation（IE）method	积分方程法
moment method（MoM）	矩量法
pattern multiplication	波瓣图乘法
power theorem	功率定理
physical optics（PO）	物理光学法
reciprocity theorem	互易性定理
Rumsey's principle	拉姆塞原理
unified theory of diffraction（UTD）	一致性绕射理论
tables of	**列表**
absorption rates	吸收率表
antenna and array	天线和阵列表

antenna currents	天线电流分布表
antenna relations	天线的关系式表
aperture，directivity，etc.	口径分布与定向性等的表
beamwidth and side-lobe	波束宽度和旁瓣表
binomial and edge distributions	二项式与边缘分布表
celestial radio sources	天空射电源表
circular and rectangular apertures	圆形和矩形口径表
conductivities	电导率表
comer reflector bandwidths	夹角反射器的频带宽度表
comer reflector formulas	夹角反射器公式表
dielectric materials	电介质材料表
dielectric strength	介质强度表
dipole impedance and RCS	偶极子阻抗和雷达截面表
directivities and beamwidths	定向性与波束宽度表
directivities of point source	点源的定向性表
EM spectrum	电磁频谱表
end-fire arrays	端射阵表
Fourier components	傅里叶分量表
helix formulas	螺旋天线公式表
horn beamwidths	喇叭波束宽度表
important antenna relations	重要的天线关系式表
incoming wave angle	波的到达角表
lens tolerances	透镜容差表
loop formulas	环的公式表
material constants	材料参数表
Maxwell's equations	麦克斯韦方程组表
metric prefixes	公制的词头表
minor-lobe maxima	副瓣的最大方向表
mutual impedance	互阻抗表
mutual resistance	互电阻表
null directions	零辐射方向表
permittivities	介电常数表
power factor	功率因子表
rhombic antenna formulas	菱形天线公式表
source amplitudes	源的幅度表
terahertz region	太赫频率范围表

Antenna applications	天线应用
Radar	雷达
axtive ramote sensing	主动遥感
altimeter radar	测高雷达
anti-collision radar	防撞雷达
bistatic radar	双站雷达
cross section（RCS）	雷达截面（RCS）
of aircraft，mosquito，etc.	飞机、蚊子等的 RCS
of comet	慧星的 RCS
of disk，loop，sphere	盘、环、球等的 RCS
of electron	电子的 RCS
of helix	螺旋的 RCS
of space shuttle	航天飞机的 RCS
of stealth aircraft	隐形飞机的 RCS
DF and monopulse radar	测向和单脉冲雷达
Doppler radar	多普勒雷达
radar equation	雷达方程
ground penetrating radar	探地雷达
police radar	警用雷达
Thompson scatter radar	汤姆逊散射雷达
tornado radar	龙卷风雷达
weather radar	气象温度雷达
radio astronomy	射电天文学
interferometer	干涉仪
adding interferometer	加法干涉仪
aperture synthesis	口径合成
clean maps	干净图像
fringe amplitude	条纹幅度
fringe visibility	条纹可见度
grating lobes	栅瓣
phase closure	相位闭合
self-calibration	自校正
simple interferometer	简单干涉仪
two-element interferometer	两单元干涉仪
Max Planck Institute for Radio Astronomy	普朗克射电天文学会
passive remote sensing	被动遥感
radio telescope	射电望远镜

Arecibo telescope	阿利西波望远镜
Bell Telephone Laboratories telescope	贝尔电话实验室望远镜
“Big Ear” telescope	“大耳”望远镜
Bonn telescope	波恩望远镜
Eifel Mountain telescope	埃菲尔山望远镜
Five college Observatory telescope	五大学天文台望远镜
Gorki telescope	高尔基望远镜
Green Bank telescope	绿滩望远镜
Jodrell Bank telescope	乔德雷尔滩望远镜
Lovell telescope	洛弗尔望远镜
Manchester telescope	曼彻斯特望远镜
Nancay telescope	纳西望远镜
Nobeyama telescope	野边山望远镜
Ohio telescope	俄亥俄望远镜
Parkes telescope	帕克斯望远镜
RATAN-600telescope	拉坦 -600 望远镜
SKA array（square kilometer）	平方公里阵
VLA	甚大口径阵
Zelenchukskaya telescope	塞琳楚克斯卡娅望远镜
solar interference	太阳干扰
the outer space	太空
“Big Bang”	宇宙“大爆炸”
cold target	冷目标
coma lobe	慧差波瓣
cosmic background	宇宙背景
Cosmic Background Explorer	宇宙背景探索者卫星
cosmic fireball floor	宇宙火球噪声基底
criterion of detectability	可检测性的判据
Cygnus A radio galaxy	射电星系天鹅座 A
Mars temperature	火星温度
minimum detectable temperature	最小可检测温度
redshifts	红移
SETI institute	地外文明探索协会
Tungska fireball	通古斯火球
Van Allen belts	范·艾伦（辐射）带
3K sky background temperature	3K 天空背景温度

Radio communication link	无线电通信线路
autocorrelation function	自相关函数
bluetooth	蓝牙
cell-tower trees	蜂窝－塔树
drahtlos	无线（德文）
Information capacity	信息容量
LEO satellite link	近地轨道卫星中继
Lunar surface wave communication	月球表面波通信
Mars and Jupiter links	地球－火星或木星的线路
Moon link	地球－月球的线路
sans fils	无线（法文）
Satellite	卫星
Astrolink satellite	航天中继卫星
Clarke-orbit satellite	克拉克（静地）轨道卫星
Concordia satellite	协和卫星
Euroskyway satellite	欧洲天空公路系统卫星
Fleetsatcom satellite	舰队通信卫星
Geostationary orbit（GSO，GEO）satellite	对地静止轨道卫星
Global-position satellite	全球定位卫星
Globalstar satellite	全球星卫星
Glonass satellite	全球导航定位系统卫星
Immarsat satellite	国际海事卫星
Intelsat satellite	国际通信卫星
Intermediate-earth orbit（IEO）satellite	中等地球轨道卫星
Iridium satellite	铱系统卫星
KaStar satellite	Ka 星系统卫星
Low-earth orbit（LEO）satellite	近地轨道卫星
Medium-earth orbit（MEO）satellite	中高度地球轨道卫星
Molnya satellite	闪电卫星
Odyssey satellite	奥德赛卫星
Orbcomm satellite	轨道星系统卫星
Orblink satellite	轨道中继系统卫星
Skybridge satellite	天桥卫星
Spaceway satellite	空间公路卫星
Teledesic satellite	蜂窝通信卫星
VSAT satellite	甚小口径天线终端的卫星
satellite communication	卫星通信

satellites diagram	卫星的图解
satellite footprint	卫星天线波束覆盖图
spacecraft link	航天器间中继
specific absorption rate（SAR）	比吸收率
body SAR distribution	人体比吸收率分布
exposure standards	暴露标准
head phantom	头部模型
head SAR distribution	头部比吸收率分布
transmission	传输
transmission coefficient	传输系数
wave propagation	波的传播
angle of incidence	入射角
atmospheric attenuation	大气层衰减
atmospheric noise	大气层噪音
bound wave	边界波
critical angle	临界角
critical frequency	临界频率
diffraction	绕射
diffraction edge	绕射边缘
diffraction by ground plane	被地面绕射
forest absorption	树林的吸收
forest temperature	树林温度
maximum usable frequency	最高可用频率
minimum usable frequency	最低可用频率
radiation	辐射
rain attenuation	雨致衰减
reflection	反射
reflection coefficient	反射系数
refraction	折射
scattering	散射
sea clutter	海杂波
slow wave	慢波